ÉLÉMENTS DE GÉOMÉTRIE ALGÉBRIQUE IV
COMPLETE CUMULATIVE ENGLISH SOURCE-ALIGNED READER

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Contents

§1. Relative finiteness conditions. Constructible sets in preschemes

In this section, we will resume and complete the exposé of "finiteness conditions" for a morphism of preschemes $f : X \to Y$ given in (**I**, 6.3 and 6.6). There are essentially two notions of "finiteness" of a *global* nature on X, that of *quasi-compact* morphism (defined in (**I**, 6.6.1)) and that of a *quasi-separated* morphism; on the other hand, there are two notions of "finiteness" of a *local* nature on X, that of a morphism *locally of finite type* (defined in (**I**, 6.6.2)) and that of a morphism *locally of finite presentation*. By combining these local notions with the preceding global notions, we obtain the notion of a morphism *of finite type* (defined in (**I**, 6.3.1)) and of a morphism *of finite presentation*. For the convenience of the reader, we will give again in this section the principal properties stated in (**I**, 6.3 and 6.6), referring, of course, to those numbered passages in Chapter I for their proofs.

In n^{os}1.8 and 1.9, we complete, in the context of preschemes, and making use of the previous notions of finiteness, the results on constructible sets given in ($\mathbf{0}_{\mathrm{III}}$, §9).

1.1. Quasi-compact morphisms

Definition (1.1.1). — We say that a morphism of preschemes $f : X \to Y$ is *quasi-compact* if the continuous map f from the topological space X to the topological space Y is quasi-compact (**0**, 9.1.1), in other words, if the inverse image $f^{-1}(U)$ of every quasi-compact open subset U of Y is quasi-compact (cf. (**I**, 6.6.1)).

If $\mathfrak{B}$ is a basis for the topology of Y consisting of affine open sets, then for f to be quasi-compact, it is necessary and sufficient that for all $V \in \mathfrak{B}$, $f^{-1}(V)$ is a *finite union of affine open sets*. For example, if Y is affine and X is quasi-compact, *every* morphism $f : X \to Y$ is quasi-compact (**I**, 6.6.1).

If $f : X \to Y$ is a quasi-compact morphism, then it is clear that for every open subset V of Y, the restriction of f to $f^{-1}(V)$ is a quasi-compact morphism $f^{-1}(V) \to V$. Conversely, if (U_α) is an open cover of Y and $f : X \to Y$ is a morphism such that the restrictions $f^{-1}(U_\alpha) \to U_\alpha$ are quasi-compact, then f is quasi-compact. As a result, if $f : X \to Y$ is an S-morphism of S-preschemes, and **IV-1** | 225
if there exists an open cover (S_λ) of S such that the restrictions $g^{-1}(S_\lambda) \to h^{-1}(S_\lambda)$ of f (where g and h are the structure morphisms) are quasi-compact, then f is quasi-compact.

Proposition (1.1.2). —

(i) *An immersion $X \to Y$ is quasi-compact if it is closed, or if the underlying space of Y is locally Noetherian or if the underlying space of X is Noetherian.*
(ii) *The composition of two quasi-compact morphisms is quasi-compact.*
(iii) *If $f : X \to Y$ is a quasi-compact S-morphism, the same is true of $f_{(S')} : X_{(S')} \to Y_{(S')}$ for every extension $g : S' \to S$ of the base prescheme.*
(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two quasi-compact S-morphisms,*
$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$
is quasi-compact.
(v) *If the composition $g \circ f$ of two morphisms $f : X \to Y$, $g : Y \to Z$ is quasi-compact, and if either g is separated or the underlying space of X is locally Noetherian, then f is quasi-compact.*
(vi) *For a morphism f to be quasi-compact, it is necessary and sufficient that f_{red} be so.*

For the proof, see (**I**, 6.6.4). We note that the assertion (vi) is also a consequence of the following more general proposition:

Proposition (1.1.3). — *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms. If $g \circ f$ is quasi-compact and f is surjective, then g is quasi-compact.*

Proof. Indeed, if V is a quasi-compact open subset of Z, $f^{-1}(g^{-1}(V))$ is quasi-compact by hypothesis, and we have $g^{-1}(V) = f(f^{-1}(g^{-1}(V)))$, since f is surjective; so $g^{-1}(V)$ is quasi-compact. □

Corollary (1.1.4). — *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$; suppose g is quasi-compact and surjective (resp. surjective). Then, for f to be quasi-compact (resp. dominant), it suffices that f' be so.*

Proof. If $g' : X' \to X$ is the canonical projection, g' is a surjective morphism (**I**, 3.5.2), and we have $f \circ g' = g \circ f'$; if f' is quasi-compact (resp. dominant), the same is true of $g \circ f'$ since g is quasi-compact (1.1.2) (resp. surjective); as $f \circ g'$ is quasi-compact (resp. dominant) and g' is surjective, we deduce that f is quasi-compact (1.1.3) (resp. dominant). □

To abbreviate, we will call *maximal point* of a prescheme X every generic point of an irreducible component of X; if $X = \mathrm{Spec}(A)$ is affine, the maximal points of X are the *minimal prime ideals* of A. **IV-1** | 226

Proposition (1.1.5). — *Let $f : X \to Y$ be a quasi-compact morphism. The following conditions are equivalent:*

a) f is dominant;
b) for every maximal point y of Y, we have $f^{-1}(y) \neq \varnothing$;
c) for every maximal point y of Y, $f^{-1}(y)$ contains a maximal point of X.

The equivalence of *a)* and *c)* was proved in (**I**, 6.6.5). It is clear that *c)* implies *b)*; on the other hand, if X' is an irreducible component of X meeting $f^{-1}(y)$, $f(X')$ is irreducible and contains y, so it is contained in the irreducible component $Y' = \overline{\{y\}}$ of Y (**0**, 2.1.6); if x is the generic point of X', we therefore necessarily have $f(x) = y$ (**0**, 2.1.5); so *b)* implies *c)*.

Proposition (1.1.6). — *Let $f' : X' \to Y$, $f'' : X'' \to Y$ be two morphisms, X the coproduct prescheme $X' \coprod X''$, $f : X \to Y$ the morphism equal to f' on X' and to f'' on X''. For f to be quasi-compact, it is necessary and sufficient that f' and f'' be so.*

This follows immediately from the definition.

1.2. Quasi-separated morphisms

Definition (1.2.1). — Let X, Y be two preschemes. We say that a morphism $f : X \to Y$ is *quasi-separated* (or that X is *quasi-separated over* Y) if the diagonal morphism $\Delta_f = (1_X, 1_X)_Y : X \to X \times_Y X$ is quasi-compact. We say that a prescheme X is *quasi-separated* if it is quasi-separated over $\mathrm{Spec}(\mathbf{Z})$.

By definition, every separated morphism is quasi-separated, Δ_f being a closed immersion (1.1.2, (i)); a *scheme* X is a quasi-separated prescheme, being separated over $\mathrm{Spec}(\mathbf{Z})$ (**I**, 5.4.1).

Proposition (1.2.2). —

(i) Every monomorphism of preschemes $f : X \to Y$ (in particular every immersion) is quasi-separated.
(ii) If $f : X \to Y$ and $g : Y \to Z$ are two quasi-separated morphisms, $g \circ f : X \to Z$ is quasi-separated.
(iii) Let X, Y be two S-preschemes, $f : X \to Y$ a quasi-separated S-morphism. Then, for every base change $g : S' \to S$, the morphism $f_{(S')} : X_{(S')} \to Y_{(S')}$ is quasi-separated.
(iv) If $f : X \to Y$, $f' : X' \to Y'$ are two quasi-separated S-morphisms, then

$$f \times_S f' : X \times_S X' \longrightarrow Y \times_S Y'$$

is quasi-separated.
(v) If $f : X \to Y$, $g : Y \to Z$ are two morphisms such that $g \circ f$ is quasi-separated, then f is quasi-separated.
(vi) If $f : X \to Y$ is a quasi-separated morphism, $f_{\mathrm{red}} : X_{\mathrm{red}} \to Y_{\mathrm{red}}$ is quasi-separated.

Proof. By virtue of (**I**, 5.5.12), it suffices to prove (i), (ii) and (iii).

The assertion (i) is trivial, every monomorphism being separated (**I**, 5.5.1). To prove (iii), we reduce immediately to the case where $Y = S$ (**I**, 3.3.11) and the assertion follows from the fact that $\Delta_{f_{(S')}} = (\Delta_f)_{(S')}$ (**I**, 5.3.4) and from (1.1.2, (iii)). To prove (ii), consider the projections p and q of $X \times_Y X$ over X; if $i = (p, q)_Z$, we know that the diagram

$$\begin{tikzcd} X \times_Y X \arrow[r, "i"] \arrow[d, "\pi"'] & X \times_Z X \arrow[d, "f \times_Z f"] \\ Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

(where π is the structure morphism) is commutative and identifies $X \times_Y X$ with the product, as $(Y \times_Z Y)$-preschemes, of Y and $X \times_Z X$ (**I**, 5.3.5). If g is quasi-separated, Δ_g is quasi-compact, and hence so is i (1.1.2, (iii)); if in addition f is quasi-separated, Δ_f is quasi-compact, and hence so is $i \circ \Delta_f$ (1.1.2, (ii)), which is equal to $\Delta_{g \circ f}$. □ **IV-1** | 227

Corollary (1.2.3). —

(i) Let X be a quasi-separated prescheme. Then every morphism $f : X \to Y$ is quasi-separated.

(ii) Let Y be a quasi-separated prescheme. For a morphism $f : X \to Y$ to be quasi-separated, it is necessary and sufficient that the prescheme X be quasi-separated.

This follows immediately from (1.2.2, (ii) and (v)) applied to X, Y and $\mathrm{Spec}(\mathbf{Z})$.

Proposition (1.2.4). — *Let $f : X \to Y$ be a morphism, $g : Y \to Z$ a quasi-separated morphism. If $g \circ f$ is quasi-compact, then so is f.*

Proof. We know that f is the composite morphism $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{\mathrm{pr}_2} Y$ (**I**, 5.3.13); on the other hand, pr_2 is identified with $(g \circ f) \times_Z 1_Y$ (**I**, 3.3.4) and since $g \circ f$ is quasi-compact, so is pr_2 (1.1.2, (iv)). Finally, we have the commutative diagram

(1.2.4.1)
$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] \arrow[d, "f"'] & X \times_Z Y \arrow[d, "f \times 1_Y"] \\ Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

which identifies X with the product, as $(Y \times_Z Y)$-preschemes, of Y and $X \times_Z Y$ (**I**, 5.3.7). As by hypothesis Δ_g is a quasi-compact morphism, so is Γ_f (1.1.2, (iii)); the conclusion then follows from (1.1.2, (ii)). □

Proposition (1.2.5). — *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. If g is quasi-compact and surjective and f' quasi-separated, then f is quasi-separated.*

Proof. We have indeed $(X' \times_{Y'} X') = (X \times_Y X)_{(Y')}$ and $\Delta_{f'} = (\Delta_f)_{(Y')}$ (**I**, 5.3.4); as the projection $X' \times_{Y'} X' \to X \times_Y X$ is quasi-compact (1.1.2, (iii)) and surjective (**I**, 3.5.2), we can, by virtue of (**I**, 3.3.11), apply (1.1.4), which shows that since $\Delta_{f'}$ is a quasi-compact morphism, the same is true of Δ_f. □

Proposition (1.2.6). — *Let $f : X \to Y$ be a morphism, (U_α) a covering of Y by open subsets such that the induced preschemes U_α are quasi-separated. For f to be quasi-separated, it is necessary and sufficient that each of the induced preschemes on the open subsets $f^{-1}(U_\alpha)$ be quasi-separated.*

Proof. The inverse image in $X \times_Y X$ of U_α is $X_\alpha \times_{U_\alpha} X_\alpha$, writing $X_\alpha = f^{-1}(U_\alpha)$, and the restriction $X_\alpha \to X_\alpha \times_{U_\alpha} X_\alpha$ of Δ_f is none other than Δ_{f_α}, denoting by f_α the restriction $X_\alpha \to U_\alpha$ of f; by virtue of the definition (1.2.1) and the local character of the notion of quasi-compact morphism (1.1.1), for f to be quasi-separated, it is necessary and sufficient that each of the morphisms f_α be so. But since by hypothesis the morphism $U_\alpha \to \mathrm{Spec}(\mathbf{Z})$ is quasi-separated, saying that f_α is quasi-separated is equivalent to saying that the composite $X_\alpha \xrightarrow{f_\alpha} U_\alpha \to \mathrm{Spec}(\mathbf{Z})$ is quasi-separated (1.2.2, (ii) and (v)); hence the proposition. □

To verify that a morphism is quasi-separated, we are therefore reduced to giving criteria for a prescheme to be quasi-separated: **IV-1** | 228

Proposition (1.2.7). — *Let X be a prescheme, (U_α) a covering of X formed by quasi-compact open subsets. The following properties are equivalent:*

a) X is quasi-separated.

b) For every quasi-compact open subset U of X, the canonical immersion $U \to X$ is quasi-compact (in other words, U is retrocompact in X).

b') The intersection of two quasi-compact open subsets of X is quasi-compact.

c) For every pair of indices (α, β), the intersection $U_\alpha \cap U_\beta$ is quasi-compact.

Proof. The properties *b)* and *b′)* are trivially equivalent by definition of a quasi-compact morphism. As a quasi-compact open subset is a finite union of affine open subsets, for two quasi-compact open subsets U, V of X, $U \times_{\mathbf{Z}} V$ is a quasi-compact open subset of $X \times_{\mathbf{Z}} X$ (**I**, 3.2.7), whose inverse image under Δ_X is $U \cap V$, so *a)* implies *b′)*. It is trivial that *b′)* implies *c)*; finally, if *c)* is satisfied, the $U_\alpha \times_{\mathbf{Z}} U_\beta$ form a covering of $X \times_{\mathbf{Z}} X$ by quasi-compact open subsets, and we know that for the morphism $\Delta_X : X \to X \times_{\mathbf{Z}} X$ to be quasi-compact, it suffices that the inverse images $U_\alpha \cap U_\beta$ under Δ_X be quasi-compact (1.1.1); so *c)* implies *a)*. □

Corollary (1.2.8). — *Every prescheme X whose underlying space is locally Noetherian (for example a locally Noetherian prescheme) is quasi-separated; every morphism $f : X \to Y$ is then quasi-separated.*

Proposition (1.2.9). — *Let X', X'' be two preschemes, $X = X' \coprod X''$ their coproduct prescheme, $f' : X' \to Y$, $f'' : X'' \to Y$ two morphisms, $f : X \to Y$ the morphism which coincides with f' on X' and with f'' on X''. For f to be quasi-separated, it is necessary and sufficient that f' and f'' be so.*

Proof. Indeed, $X \times_Y X$ is the coproduct of the four preschemes $X' \times_Y X'$, $X' \times_Y X''$, $X'' \times_Y X'$ and $X'' \times_Y X''$, and Δ_f is the morphism which coincides with $\Delta_{f'}$ on X' and with $\Delta_{f''}$ on X''; the proposition then follows immediately from the definitions. □

1.3. Morphisms locally of finite type

(1.3.1). Recall that if A is a ring, a (commutative) A-algebra B is said to be *of finite type* if it is generated by a finite number of elements of B, or, what amounts to the same thing, if it is isomorphic to a quotient of a polynomial algebra $A[T_1, \dots, T_n]$ by an ideal of this algebra. It is clear that for every (commutative) A-algebra A', $B \otimes_A A'$ is then an A'-algebra of finite type. If B is an A-algebra of finite type and C a B-algebra of finite type, C is an A-algebra of finite type, for if C is a quotient of $B[T_1, \dots, T_n]$ and B a quotient of $A[S_1, \dots, S_m]$, $B[T_1, \dots, T_n]$ is isomorphic to $A[S_1, \dots, S_m, T_1, \dots, T_n]/\mathfrak{a}$, where $\mathfrak{a}$ is the kernel of the surjection $A[S_1, \dots, S_m, T_1, \dots, T_n] \to B[T_1, \dots, T_n]$; hence C is also a quotient of $A[S_1, \dots, S_m, T_1, \dots, T_n]$, and so it is indeed an A-algebra of finite type.

Definition (1.3.2). — Let $f : X \to Y$ be a morphism of preschemes, x a point of X, $y = f(x)$. We say that f is *of finite type at x* if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and the ring $A(U)$ is an $A(V)$-algebra of finite type. We say that f is *locally of finite type* if it is of finite type at every point of X.

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Proposition (1.3.3). — *If Y is a locally Noetherian prescheme and $f : X \to Y$ a morphism locally of finite type, then X is locally Noetherian.*

For the proof, see (**I**, 6.3.7).

Proposition (1.3.4). —

(i) *Every local immersion is locally of finite type.*
(ii) *If two morphisms $f : X \to Y$, $g : Y \to Z$ are locally of finite type, then so is $g \circ f$.*
(iii) *If $f : X \to Y$ is an S-morphism locally of finite type, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is locally of finite type for every extension $S' \to S$ of the base prescheme.*
(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms locally of finite type, $f \times_S g$ is locally of finite type.*
(v) *If the composition $g \circ f$ of two morphisms is locally of finite type, f is locally of finite type.*
(vi) *If a morphism f is locally of finite type, the same is true of f_{red}.*

For the proof, see (**I**, 6.6.6).

Corollary (1.3.5). — *Let $f : X \to Y$ be a morphism locally of finite type. For every morphism $Y' \to Y$ such that Y' is locally Noetherian, $X \times_Y Y'$ is locally Noetherian.*

Proof. Indeed, $f_{(Y')} : X \times_Y Y' \to Y'$ is locally of finite type by (1.3.4, (iii)), and it suffices to apply (1.3.3). □

Proposition (1.3.6). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ to be locally of finite type, it is necessary and sufficient that B be an A-algebra of finite type.*

For the proof, see (**I**, 6.3.3), taking into account that f is quasi-compact and using (**I**, 6.6.3).

Proposition (1.3.7). — *For a morphism locally of finite type $f : X \to Y$ to be surjective, it is necessary and sufficient that, for every algebraically closed field Ω, the map $X(\Omega) \to Y(\Omega)$ corresponding to f (**I**, 3.4.1) be surjective.*

For the proof, see (**I**, 6.3.10).

(1.3.8). Let A be a ring. We say that an A-algebra B is *essentially of finite type* if B is A-isomorphic to an A-algebra of the form $S^{-1}C$, where C is an A-algebra *of finite type* and S a multiplicative subset of C.

Proposition (1.3.9). —

(i) *If B is an A-algebra essentially of finite type and C a B-algebra essentially of finite type, then C is an A-algebra essentially of finite type.*
(ii) *Let B be an A-algebra essentially of finite type, A' an A-algebra; then $B' = B \otimes_A A'$ is an A'-algebra essentially of finite type.*

Proof. (i) Let $B = S'^{-1}B'$ and $C = T'^{-1}C'$, where B' (resp. C') is an A-algebra (resp. B-algebra) of finite type and S' (resp. T') a multiplicative subset of B' (resp. C'); then C' is of the form $S'^{-1}C''$, where C'' is also a B'-algebra of finite type, so C is also a ring of fractions of C''; as C'' is an A-algebra of finite type, this proves our assertion.
(ii) If B is a ring of fractions of an A-algebra of finite type C, B' is a ring of fractions of the A'-algebra of finite type $C \otimes_A A'$, which proves (ii).

□

(1.3.10). If B is a *local* A-algebra essentially of finite type, B is of the form $C_{\mathfrak{q}}$, where C is an A-algebra of finite type and $\mathfrak{q}$ a prime ideal of C (Bourbaki, *Alg. comm.*, chap. II, § 5, prop. 11). Let $\mathfrak{p}$ be the inverse image in A of the ideal $\mathfrak{q}$; if we set $S = A - \mathfrak{p}$, $C_{\mathfrak{q}}$ is also the localization of $S^{-1}C$ at a prime ideal; as $S^{-1}C$ is an algebra of finite type over $A_{\mathfrak{p}} = S^{-1}A$, we see that B is also an $A_{\mathfrak{p}}$-algebra *essentially of finite type*, and furthermore the homomorphism $A_{\mathfrak{p}} \to B$ is *local*.

Proposition (1.3.11). — *If B is a local A-algebra essentially of finite type, B is A-isomorphic to a quotient of an A-algebra of the form $C_{\mathfrak{q}}$, where $C = A[T_1, \ldots, T_n]$ is a polynomial algebra over A and $\mathfrak{q}$ a prime ideal of C.*

Proof. Indeed, by definition, B is isomorphic to $C'_{\mathfrak{q}'}$, where C' is an A-algebra of finite type and $\mathfrak{q}'$ a prime ideal of C'; but $C' = C/\mathfrak{b}$, where $C = A[T_1, \ldots, T_n]$ and $\mathfrak{b}$ is an ideal of C; so $\mathfrak{q}' = \mathfrak{q}/\mathfrak{b}$, where $\mathfrak{q}$ is a prime ideal of C, and $C'_{\mathfrak{q}'}$ is isomorphic to $C_{\mathfrak{q}}/\mathfrak{b}C_{\mathfrak{q}}$. □

1.4. Morphisms locally of finite presentation

(1.4.1). Given a ring A, we say that an A-algebra B (commutative) is *of finite presentation* if it is isomorphic to a quotient $A[T_1, \ldots, T_n]/\mathfrak{a}$ of a polynomial algebra over A by an ideal *of finite type* $\mathfrak{a}$ of $A[T_1, \ldots, T_n]$. For every (commutative) A-algebra A', $B \otimes_A A'$ is then an A'-algebra of finite presentation, being isomorphic to the quotient of $A'[T_1, \ldots, T_n]$ by the image of the ideal $\mathfrak{a} \otimes_A A'$ in this ring, which is evidently an $A'[T_1, \ldots, T_n]$-module of finite type. If B is an A-algebra of finite presentation and C a B-algebra of finite presentation, then C is an A-algebra of finite presentation. Indeed, let $B = A[S_1, \ldots, S_m]/\mathfrak{a}$ and $C = B[T_1, \ldots, T_n]/\mathfrak{b}$, where $\mathfrak{a}$ (resp. $\mathfrak{b}$) is an ideal of finite type of $A[S_1, \ldots, S_m]$ (resp. $B[T_1, \ldots, T_n]$); the ring $B[T_1, \ldots, T_n]$ is isomorphic to $A[S_1, \ldots, S_m, T_1, \ldots, T_n]/\mathfrak{a}'$, where $\mathfrak{a}'$ is the canonical image of $\mathfrak{a} \otimes_A A[T_1, \ldots, T_n]$, and is therefore an ideal of finite type of $A[S_1, \ldots, S_m, T_1, \ldots, T_n]$; the ideal $\mathfrak{b}$ of $B[T_1, \ldots, T_n]$ is of the form $\mathfrak{b}'/\mathfrak{a}'$, where $\mathfrak{b}'$ is an ideal of $A[S_1, \ldots, S_m, T_1, \ldots, T_n]$; as $\mathfrak{a}'$ and $\mathfrak{b}'/\mathfrak{a}'$ are modules of finite type over $A[S_1, \ldots, S_m, T_1, \ldots, T_n]$, so is $\mathfrak{b}'$, and C, being isomorphic to $A[S_1, \ldots, S_m, T_1, \ldots, T_n]/\mathfrak{b}'$, is therefore an A-algebra of finite presentation. We note finally that if A is *Noetherian*, then so is $A[T_1, \ldots, T_n]$; hence every A-algebra of finite type is an algebra of finite presentation. **IV-1** | 230

Definition (1.4.2). — Let $f : X \to Y$ be a morphism of preschemes, x a point of X, $y = f(x)$. We say that f is *of finite presentation at* x if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and such that the ring $\mathscr{O}(U)$ is an $\mathscr{O}(V)$-algebra of finite presentation. We say that f is *locally of finite presentation* if it is of finite presentation at every point of X.

If Y is locally Noetherian, it amounts to the same to say that f is locally of finite type or locally of finite presentation.

Proposition (1.4.3). — (i) *Every local isomorphism is locally of finite presentation.*
(ii) *If two morphisms $f : X \to Y$, $g : Y \to Z$ are locally of finite presentation, so is $g \circ f$.*

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(iii) If $f : X \to Y$ is an S-morphism locally of finite presentation, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is locally of finite presentation for every base extension $S' \to S$ of the base prescheme.
(iv) If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms locally of finite presentation, $f \times_S g$ is locally of finite presentation.
(v) If the composition $g \circ f$ of two morphisms is locally of finite presentation and if g is locally of finite type, then f is locally of finite presentation.

Proof. Assertion (i) is trivial. To prove (iii), we may limit to the case where $S = Y$ (**I**, 3.3.11); let x' be a point of $X_{(S')}$, s' and x its projections on S' and X respectively, s the common projection of s' and x on S. By hypothesis, there exists an affine open neighbourhood V (resp. U) of s (resp. x) such that the image of U is contained in V and such that $\mathscr{O}(U)$ is an $\mathscr{O}(V)$-algebra of finite presentation; let V' be an affine open neighbourhood of s' whose image in S is contained in V; then $U \times_V V'$ is an affine open neighbourhood of x' whose image in S' is contained in V' and whose ring $\mathscr{O}(U) \otimes_{\mathscr{O}(V)} \mathscr{O}(V')$ is an $\mathscr{O}(V')$-algebra of finite presentation (1.4.1).

To prove (ii), consider a point $x \in X$, and let $y = f(x)$, $z = g(f(x))$. There is by hypothesis an affine open neighbourhood $W = \operatorname{Spec}(C)$ (resp. $V = \operatorname{Spec}(B)$) of z (resp. y) such that $g(V) \subset W$ and such that B is a C-algebra of finite presentation. On the other hand, there is an affine open neighbourhood $V' = \operatorname{Spec}(B')$ (resp. $U = \operatorname{Spec}(A)$) of y (resp. x) such that $f(U) \subset V'$ and such that A is a B'-algebra of finite presentation. Let $s \in B$ be such that the open set $D(s) = \operatorname{Spec}(B_s)$ in V is a neighbourhood of y contained in V', and let $U' = f^{-1}(D(s)) \subset U$; we have $U' = \operatorname{Spec}(A \otimes_{B'} B_s)$, and $A \otimes_{B'} B_s$ is a B_s-algebra of finite presentation (1.4.1); on the other hand $B_s = B[T]/(1 - sT)$ is a B-algebra of finite presentation by definition; hence by (1.4.1), $A \otimes_{B'} B_s$ is a C-algebra of finite presentation, which proves (ii).

We know that (iv) follows from (i), (ii), and (iii) (**I**, 3.5.1). To prove (v), consider the commutative diagram

$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] & X \times_Z Y \arrow[d, "f \times 1"] \\ Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

which identifies X with the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z Y$ (**I**, 5.3.7), Δ_g being the diagonal morphism. If we know that Δ_g is locally of finite presentation, it will follow by (iii) that so is Γ_f. On the other hand, we have the factorisation of f as $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ (**I**, 5.3.13), where the projection p_2 is equal to $(g \circ f) \times_Z 1_Y$; as $g \circ f$ is assumed locally of finite presentation, so is p_2 by (iv), and it will follow finally from (ii) that f is also locally of finite presentation. We thus see that we are reduced to proving the following. □

Corollary (1.4.3.1). — *Let $g : Y \to Z$ be a morphism locally of finite type; then the diagonal morphism $\Delta_g : Y \to Y \times_Z Y$ is locally of finite presentation.*

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Proof. We may reduce to the case where $Z = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$, B being an A-algebra of finite type; the morphism Δ_g then corresponds to the *augmentation homomorphism* $\delta : B \otimes_A B \to B$ such that $\delta(x \otimes y) = xy$. In view of the definition (1.4.2), it suffices to show that the kernel $\mathfrak{J}$ of δ is an ideal *of finite type*, and this follows from (**0**, 20.4.4). □

Proposition (1.4.4). — *Let A be a ring, B an A-algebra, B' a polynomial algebra $A[T_1, \ldots, T_n]$, $u : B' \to B$ a surjective homomorphism of A-algebras. For B to be an A-algebra of finite presentation, it is necessary and sufficient that the kernel $\mathfrak{J}$ of u be an ideal of finite type of B'.*

Proof. The condition is sufficient by definition (1.4.1). To see that it is necessary, we remark that the morphism $g : \operatorname{Spec}(B') \to \operatorname{Spec}(A)$ is locally of finite type; if B is an A-algebra of finite presentation, the morphism $f : \operatorname{Spec}(B) \to \operatorname{Spec}(B')$ corresponding to u is such that $g \circ f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is locally of finite presentation, and so it follows from (1.4.3, (v)) that f is also locally of finite presentation. Now, to show that the ideal $\mathfrak{J}$ is of finite type, it suffices to prove that $\widetilde{\mathfrak{J}}$ is a $\widetilde{B}'$-module of finite type (**II**, 6.1.4.1). Returning to the definition (1.4.2), we are finally reduced to proving the following. □

Lemma (1.4.4.1). — *Let $\mathfrak{a}$ be an ideal of a ring C, s an element of C, such that $C_s/\mathfrak{a}_s$ is a C-algebra of finite presentation; then $\mathfrak{a}_s$ is an ideal of finite type in the ring C_s.*

Proof. By hypothesis, there exists a polynomial C-algebra $C' = C[T_1, \ldots, T_n]$ and a surjective C-homomorphism $v : C' \to C_s/\mathfrak{a}_s$ whose kernel $\mathfrak{b}$ is of finite type. Let $p : C_s \to C_s/\mathfrak{a}_s$ be the canonical homomorphism; for each i, there is a $t_i \in C$ such that $p(t_i/s^k) = v(T_i)$ (we may assume that the exponent k is the same for all indices i). Consider then the C_s-algebra $D = C_s[T_1, \ldots, T_n]$, and the C_s-algebra homomorphism $w : D \to C_s$ such that $w(T_i) = t_i/s^k$; w is evidently surjective, and so is $v' = p \circ w : D \to C_s/\mathfrak{a}_s$. On the other hand, every polynomial P of D can be written as P/s^m, where $P \in C' = C[T_1, \ldots, T_n]$, and we have $v'(P/s^m) = (1/s^m)v(P)$; as $1/s$ is invertible in C_s, the relation $v'(P/s^m) = 0$ in $C_s/\mathfrak{a}_s$ is equivalent to $v(P) = 0$, and hence the kernel $\mathfrak{b}'$ of v' is generated by the image of $\mathfrak{b}$ in the ring D, and *a fortiori* is *of finite type* in D. But $\mathfrak{b}' = v'^{-1}(0) = w^{-1}(p^{-1}(0)) = w^{-1}(\mathfrak{a}_s)$, and as w is surjective, $\mathfrak{a}_s = w(\mathfrak{b}')$, which proves that $\mathfrak{a}_s$ is an ideal of finite type of C_s. □

Corollary (1.4.5). — *Let X, Y be two preschemes, $j : X \to Y$ an immersion, U an open subset of Y such that $j(X)$ is closed in U, $\mathscr{J}$ the quasi-coherent ideal of $\mathscr{O}_U$ that defines the closed subprescheme of Y associated to j. For j to be locally of finite presentation, it is necessary and sufficient that $\mathscr{J}$ be an $\mathscr{O}_U$-module of finite type.*

Proposition (1.4.6). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ to be locally of finite presentation, it is necessary and sufficient that B be an A-algebra of finite presentation.*

Proof. The condition being trivially sufficient, it remains to prove that it is necessary. If f is locally of finite presentation, it already follows from (1.3.6) that B is an A-algebra of finite type, so there exists **IV-1** | 233
a surjective homomorphism $u : B' = A[T_1, \ldots, T_n] \to B$ of A-algebras; it follows from (1.4.3, (v)) that the immersion $j : \operatorname{Spec}(B) \to \operatorname{Spec}(B')$ is locally of finite presentation; hence by (1.4.5), if $\mathfrak{J}$ is the kernel of u, the $\widetilde{B}'$-module $\widetilde{\mathfrak{J}}$ is of finite type, and hence by (**II**, 6.1.4.1) $\mathfrak{J}$ is an ideal of finite type. □

Proposition (1.4.7). — *Let $\rho : A \to B$ be a homomorphism of rings making B a finite A-algebra. For B to be an A-algebra of finite presentation, it is necessary and sufficient that B be an A-module of finite presentation.*

We will use the following.

Lemma (1.4.7.1). — *If B is a finite A-algebra, there exists a surjective homomorphism of A-algebras $u : B' \to B$, where B' is a finite A-algebra of finite presentation and a free A-module.*

Proof. Indeed, there is a finite system $(a_i)_{1 \leqslant i \leqslant m}$ of generators of the A-algebra B such that each a_i satisfies a relation $F_i(a_i) = 0$, where $F_i \in A[X]$ is a monic polynomial of degree > 0; the A-algebra quotient $B'_i = A[X]/(F_i)$ is therefore free of finite rank over A; let c_i be the class of X in B'_i. There exists an A-homomorphism of algebras $u_i : B'_i \to B$ such that $u_i(c_i) = a_i$; it then suffices to take for B' the tensor product $B'_1 \otimes_A B'_2 \otimes \ldots \otimes_A B'_m$ and to consider the homomorphism

$u_1 \otimes u_2 \otimes \ldots \otimes u_m : B' \to B$, which is surjective by construction. Furthermore, if $B'' = A[T_1, \ldots, T_m]$ and if b'_i denotes the canonical image of c_i in B', the A-homomorphism of algebras $v : B'' \to B'$ such that $v(T_i) = b'_i$ $(1 \leqslant i \leqslant m)$ is surjective, and its kernel $\mathfrak{b}'$ is the ideal of B'' generated by the $F_i(T_i)$, and therefore of finite type; this proves that B' is an A-algebra of finite presentation. □

Proof of (1.4.7). This lemma being proved, let us keep the same notation, and let $w = u \circ v : B'' \to B$.[1] Let $\mathfrak{b}$ (resp. $\mathfrak{a}$) be the kernel of w (resp. u); we have $\mathfrak{a} = v(\mathfrak{b})$ since v is surjective. Now by (1.4.4), if B is an A-algebra of finite presentation, $\mathfrak{b}$ is an ideal of finite type of B'', so $\mathfrak{a}$ is an ideal of finite type of B', hence an A-module of finite type since B' is a finite A-algebra. As B' is a free A-module, B is an A-module of finite presentation. Conversely, if B is an A-module of finite presentation, $\mathfrak{a}$ is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. I, § 2, nº 8, lemme 9) and *a fortiori* an ideal of finite type of B'; hence, B is by definition a B'-algebra of finite presentation, and as B' is an A-algebra of finite presentation, B is an A-algebra of finite presentation. □

1.5. Morphisms of finite type

Proposition (1.5.1). — *Let $f : X \to Y$ be a morphism, (U_α) a covering of Y by open affines. The following conditions are equivalent:*

a) *f is locally of finite type and quasi-compact.*
b) *For all α, $f^{-1}(U_\alpha)$ is a finite union of affine open sets $V_{\alpha i}$ such that each ring $\mathscr{O}(V_{\alpha i})$ is an $\mathscr{O}(U_\alpha)$-algebra of finite type.*
c) *For every affine open subset U of Y, $f^{-1}(U)$ is a finite union of affine open sets V_j such that the rings $\mathscr{O}(V_j)$ are $\mathscr{O}(U)$-algebras of finite type.*

Proof. For the proof, see (**I**, 6.3.2) and (**I**, 6.6.3). □

Definition (1.5.2). — We say that a morphism f is *of finite type* if it satisfies the equivalent conditions of the proposition (1.5.1).

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Proposition (1.5.3). — *Let $f : X \to Y$ be a morphism of finite type; if Y is Noetherian, so is X.*

Proof. For the proof, see (**I**, 6.3.7). □

Proposition (1.5.4). — (i) *Let $j : X \to Y$ be an immersion. If j is quasi-compact (which is the case if j is a closed immersion, or if the underlying space of X is Noetherian, or if the space underlying Y is locally Noetherian), then j is of finite type.*
(ii) *The composition of two morphisms of finite type is of finite type.*
(iii) *If $f : X \to Y$ is an S-morphism of finite type, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is of finite type for every base extension $S' \to S$ of the base prescheme.*
(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms of finite type, $f \times_S g$ is of finite type.*
(v) *If the composition $g \circ f$ of two morphisms is of finite type and if g is quasi-separated or if X is Noetherian, f is of finite type.*
(vi) *If a morphism f is of finite type, so is f_{red}.*

Proof. This follows immediately (except for the case (v) where X is Noetherian, proved in (**I**, 6.3.6)) from the definitions and from (1.1.2), (1.2.4), and (1.3.4). □

Corollary (1.5.5). — *Let $f : X \to Y$ be a morphism of finite type. For every morphism $Y' \to Y$ such that Y' is Noetherian, $X \times_Y Y'$ is Noetherian.*

Proof. This follows from (1.5.4, (iii)) and from (1.5.3). □

Corollary (1.5.6). — *Let X be a prescheme of finite type over a locally Noetherian prescheme S. Then every S-morphism $f : X \to Y$ is of finite type.*

Proof. For the proof, see (**I**, 6.3.9). □

[1] The French source prints $v \circ u$; with the displayed domains and codomains, the typed composite is $u \circ v$.

Proposition (1.5.7). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ to be of finite type, it is necessary and sufficient that φ make B an A-algebra of finite type.*

Proof. For the proof, see (**I**, 6.3.3). □

1.6. Morphisms of finite presentation

Definition (1.6.1). — Let X, Y be two preschemes. We say that a morphism $f : X \to Y$ is *of finite presentation* if it satisfies the following three conditions:

(i) f is locally of finite presentation;
(ii) f is quasi-compact (which, when (i) is satisfied, is equivalent to saying that f is of finite type (1.5.2));
(iii) f is quasi-separated.

We then also say that X is *of finite presentation over* Y, or a Y*-prescheme of finite presentation*.

Condition (iii) is automatically satisfied if f is separated, or if X is locally Noetherian (1.2.8); when Y is locally Noetherian, it amounts to the same to say that f is *of finite presentation* or that f is *of finite type* (the latter condition implying that X is locally Noetherian (1.3.3)).

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Proposition (1.6.2). — (i) *Every quasi-compact immersion that is locally of finite presentation (in particular every quasi-compact open immersion) is of finite presentation.*
(ii) *The composition of two morphisms of finite presentation is of finite presentation.*
(iii) *Let X, Y be two S-preschemes, $f : X \to Y$ an S-morphism of finite presentation. Then, for every morphism $S' \to S$, the morphism $f_{(S')} : X_{(S')} \to Y_{(S')}$ is of finite presentation.*
(iv) *Let $f : X \to Y$, $f' : X' \to Y'$ be two S-morphisms of finite presentation; then $f \times_S f' : X \times_S X' \to Y \times_S Y'$ is of finite presentation.*
(v) *If the composition $g \circ f$ of two morphisms is of finite presentation and if g is quasi-separated and locally of finite presentation, then f is of finite presentation.*

Proof. This follows immediately from (1.1.2), (1.2.2), (1.2.4), and (1.4.3). □

It follows in particular from (1.6.2, (iii)) that if f is a morphism of finite presentation and U an open subset of Y, the restriction $f^{-1}(U) \to U$ of f is a morphism of finite presentation. Conversely, let (U_α) be a covering of Y by open affines, and suppose that the restrictions $f^{-1}(U_\alpha) \to U_\alpha$ of f are morphisms of finite presentation; then f is evidently locally of finite presentation, quasi-compact by virtue of (1.1.1), and quasi-separated by virtue of (1.2.6). We can therefore say that the property for a morphism $f : X \to Y$ of being of finite presentation is *local on* Y.

If X is a quasi-separated prescheme, every morphism $f : X \to Y$ is quasi-separated (1.2.3). Hence, if f is quasi-compact and locally of finite presentation, f is of finite presentation.

In particular:

Corollary (1.6.3). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ to be of finite presentation, it is necessary and sufficient that φ make B an A-algebra of finite presentation.*

Proof. The necessity of the condition follows from (1.4.6). □

Remark (1.6.4). — In the definition (1.6.1), condition (iii) is not a consequence of the two others. Let for example Y be an affine scheme whose underlying space is not Noetherian, and let U be an open subset that is not quasi-compact in Y. Let X be the prescheme obtained by gluing two preschemes Y_1, Y_2 isomorphic to Y along the open subsets U_1, U_2 corresponding to U ($\mathbf{0_I}$, 4.1.7), so that X is the union of two open subsets isomorphic respectively to Y_1 and Y_2 (and that we identify with them), with $Y_1 \cap Y_2 = U$. Furthermore, there is a morphism $f : X \to Y$ that coincides in Y_i with the isomorphism $Y_i \to Y$ ($i = 1, 2$). It is clear that f satisfies condition (i) of (1.6.1), being a local isomorphism; it also satisfies (ii), the inverse image of an open quasi-compact subset of Y being the union of its images in Y_1 and Y_2; but since $Y_1 \cap Y_2$ is not quasi-compact, f is not quasi-separated (1.2.7).

Proposition (1.6.5). — *Let X', X'' be two preschemes, $X = X' \amalg X''$ their sum, $f' : X' \to Y$, $f'' : X'' \to Y$ two morphisms, $f : X \to Y$ the morphism that coincides with f' in X' and with f'' in X''. For f to be of finite presentation, it is necessary and sufficient that f' and f'' be so.*

Proof. It suffices to show that for f to possess one of the three properties of the definition (1.6.1), it is necessary and sufficient that f' and f'' possess it; this is evident for the property of being locally of finite presentation, which is local on X; for the property of being quasi-compact, this was seen in (1.1.6), and for the property of being quasi-separated, in (1.2.9). □ **IV-1** | 236

1.7. Improvements of earlier results We will give in this number a list of propositions proved in the preceding chapters, whose statements can be improved with the aid of the new finiteness conditions introduced above.

(1.7.1). In the statements of (**I**, 6.4.2), (**I**, 6.4.3), and (**I**, 6.4.9), we can, instead of assuming that X and Y are of finite type over the field K, assume only that they are *locally of finite type* over K; for (**I**, 6.4.2), it suffices to observe that every point of a prescheme X that is closed in every affine open subset of X is closed in X, and an affine scheme locally of finite type over K is *ipso facto* of finite type over K (1.5.1). For (**I**, 6.4.9), we use (1.3.7) and (1.3.4, (v)).

(1.7.2). In (**I**, 6.5.1, (ii)), we can, instead of assuming that S is locally Noetherian and Y of finite type over S, assume only that Y is *locally of finite presentation* over S, as one shows immediately by applying the definition (1.4.2). Similarly in (**I**, 6.5.4, (ii)) and (**I**, 6.5.5, (ii)) it suffices to assume that f is an S-morphism, Y being *locally of finite presentation* over S.

(1.7.3). In (**I**, 7.1.11), for the second assertion, instead of assuming S locally Noetherian and Y of finite type over S, we can assume only Y locally of finite presentation over S (the proof depending on (**I**, 6.5.1, (ii))); similarly in (**I**, 7.1.12) to (**I**, 7.1.15).

(1.7.4). In (**I**, 9.2.2), we can replace the three hypotheses by the single hypothesis (implied by each of the others) that f is *quasi-compact and quasi-separated*, by virtue of (1.2.6) and (1.2.7). We note that in (**I**, 9.2.1), the hypothesis means exactly that f is quasi-compact and quasi-separated.

(1.7.5). In (**I**, 9.3.1), (**I**, 9.3.2), and (**I**, 9.3.3), we can replace the two hypotheses on X by the (weaker) hypothesis that X is a prescheme *quasi-compact and quasi-separated*, the proof using these two properties (by virtue of (1.2.7)).

(1.7.6). In (**I**, 9.3.4) and (**I**, 9.3.5), it suffices to assume X quasi-compact, $\mathscr{F}$ and $\mathscr{J}$ quasi-coherent and of finite type.

(1.7.7). In (**I**, 9.4.7), we can weaken hypothesis b) by assuming only X *quasi-separated*, since it suffices that the canonical immersion $U \to X$ be quasi-compact (1.2.7). We can make the same modification in (**I**, 9.4.8), (**I**, 9.4.9), and (**I**, 9.4.10).

(1.7.8). In (**I**, 9.5.1), the conditions indicated in parentheses in the statement can be replaced by the condition that f be quasi-compact and quasi-separated.

(1.7.9). In (**I**, 9.6.5), we can replace the hypotheses on X by the weaker hypothesis that X is *quasi-compact and quasi-separated*, as the proof shows, which uses only (**I**, 9.4.7). In (**I**, 9.6.6), the hypothesis "X is a quasi-compact scheme" may be replaced by "X is a prescheme quasi-compact and quasi-separated". **IV-1** | 237

(1.7.10). In (**II**, 1.7.15), we can suppose only that Y is quasi-compact and quasi-separated, the proof using only (**I**, 9.6.5).

(1.7.11). In the second assertion of (**II**, 3.4.5), we can substitute for the two hypotheses on Y the weaker hypothesis that Y is quasi-compact and quasi-separated. Similarly in (**II**, 3.4.8).

(1.7.12). In (**II**, 3.8.5), we can similarly suppose only that Y is quasi-compact and quasi-separated, this hypothesis entering through (**I**, 9.4.7).

(1.7.13). In (**II**, 4.4.1), (**II**, 4.4.6), and (**II**, 4.4.7), it suffices to assume that Y is *quasi-compact and quasi-separated*, in view of (1.2.3) in the proof of (**II**, 4.4.7). Similarly, in (**II**, 4.4.10.1), it suffices to assume that Z is quasi-compact and quasi-separated.

(1.7.14). In (**II**, 4.5.2) and (**II**, 4.5.5), it suffices to assume X *quasi-compact and quasi-separated*, this hypothesis entering through (**I**, 9.3.1) and (**I**, 9.3.2).

(1.7.15). In (**II**, 4.6.8), it suffices still to assume X *quasi-compact and quasi-separated*, this hypothesis entering through (**I**, 9.4.7).

(1.7.16). In (**II**, 5.1.2), it suffices to assume that X is quasi-compact and quasi-separated (the hypothesis entering through (**II**, 4.5.2) and (**II**, 4.5.5)). Similarly in (**II**, 5.1.9), where we use (**I**, 9.6.5).

(1.7.17). In (**II**, 5.2.1), it suffices always to assume X quasi-compact and quasi-separated (the hypothesis entering through (**II**, 4.5.2)).

(1.7.18). In (**II**, 5.2.2), one must assume f *quasi-compact* and X *quasi-separated*; the current proof is in fact insufficient (with the hypotheses made), for from the fact that for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ we have $\mathrm{R}^1 f_*(\mathscr{F}) = 0$, it does not follow, necessarily, for an affine open subset U of Y, that the same property holds for quasi-coherent $(\mathscr{O}_X|f^{-1}(U))$-modules, if one does not know that such a module is the restriction of a quasi-coherent $\mathscr{O}_X$-module (the same remark applying to conditions b) and c′)); when f is quasi-compact and X quasi-separated, this extension is possible by virtue of (**I**, 9.4.7), modified above (1.7.7).

(1.7.19). In (**II**, 5.3.2), it suffices to assume Y *quasi-compact and quasi-separated* (this hypothesis entering through (**II**, 4.4.6) and (**II**, 4.4.7)). Similarly in (**II**, 5.5.3, (ii)).

(1.7.20). In (**III**, 1.2.2), (**III**, 1.2.3), and (**III**, 1.2.4), we can replace the two hypotheses on X by the weaker hypothesis that X is a prescheme *quasi-compact and quasi-separated*, the proof using only (**I**, 9.3.3).

(1.7.21). In (**III**, 1.4.10) to (**III**, 1.4.14), we can replace "separated" by "quasi-separated", this hypothesis serving only to be able to apply (**I**, 9.2.2) (see above (1.7.4)). Similarly, in (**III**, 1.4.15), it suffices to assume that the morphism u is *quasi-separated and quasi-compact*.

(1.7.22). In (**III**, 2.1.3), we can again replace the hypotheses by the weaker hypothesis that X is a prescheme quasi-compact and quasi-separated, the reasoning being the same as in (**III**, 1.2.2).

1.8. Morphisms of finite presentation and constructible sets **IV-1** | 238

(1.8.1). Let X be a prescheme; we know that every quasi-compact open subset of X is a finite union of affine open subsets, and conversely. For an open subset U of X to be *retrocompact* in X ($\mathbf{0}_{\mathbf{III}}$, 9.1.1), it is necessary and sufficient that for every *open affine* subset V of X, $U \cap V$ be quasi-compact. When X is *quasi-separated* (1.2.1), every quasi-compact open subset of X is retrocompact in X (1.2.7), so every locally constructible part of X is retrocompact in X ($\mathbf{0}_{\mathbf{III}}$, 9.1.13); if *in addition* X is *quasi-compact*, constructible subsets and locally constructible subsets of X coincide ($\mathbf{0}_{\mathbf{III}}$, 9.1.12), as do constructible open subsets and quasi-compact open subsets ($\mathbf{0}_{\mathbf{III}}$, 9.1.5).

Proposition (1.8.2). — *Let X, Y be two preschemes, $f : X \to Y$ a morphism. For every constructible (resp. locally constructible) part Z of Y, $f^{-1}(Z)$ is a constructible (resp. locally constructible) part of X.*

Proof. Suppose Z locally constructible; for every $x \in X$, there will be an affine open neighbourhood V of $f(x)$ such that $Z \cap V$ is constructible in V; if the proposition is proved for constructible parts, $f^{-1}(Z) \cap f^{-1}(V)$ will be constructible in $f^{-1}(V)$, so $f^{-1}(Z)$ will be locally constructible. It therefore suffices to consider the case where Z is constructible, and taking into account ($\mathbf{0}_{\mathbf{III}}$, 9.1.3), we are reduced to the case where Z is open and *retrocompact* in Y, in other words such that the canonical injection $Z \to Y$ is a quasi-compact morphism; it then suffices to see that $f^{-1}(Z)$ is *retrocompact in* X, in other words such that the canonical injection $f^{-1}(Z) \to X$ is a quasi-compact morphism; but this follows from (1.1.2, (iii)), since $f^{-1}(Z) = Z \times_Y X$ (**I**, 4.4.1). □

Lemma (1.8.3). — *Let X be a prescheme quasi-compact and quasi-separated, Z a constructible part of X. There exists then an affine scheme X' and a morphism of finite presentation $f : X' \to X$ such that $f(X') = Z$.*

Proof. Since X is quasi-compact, it is a finite union of affine open subsets X_i ($i \in \mathrm{I}$), and since X is quasi-separated, it follows from (1.2.7) that each of the canonical immersions $g_i : X_i \to X$ is of finite

presentation; one concludes that if $f_i : X'_i \to X_i$ is a morphism of finite presentation, the same is true of $g_i \circ f_i : X'_i \to X$ (1.6.2), and if X' is the prescheme sum of the X'_i, the morphism $h : X' \to X$ which coincides with $g_i \circ f_i$ in each X'_i is also of finite presentation (1.6.5). Since $Z \cap X_i$ is constructible in X_i ($\mathbf{0_{III}}$, 9.1.8), we see that we are reduced to proving the lemma when X_i is *affine* of ring A. Since Z is then a finite union of sets of the form $U \cap \complement V$, where U and V are quasi-compact open subsets **IV-1** | 239
($\mathbf{0_{III}}$, 9.1.3), one sees, by considering yet another suitable sum of preschemes, that one can reduce to the case where $Z = U \cap \complement V$; since moreover U is a finite union of affine open subsets, one can even restrict ourselves to the case where U is *affine*. Since V is quasi-compact, it is a finite union of open subsets of the form X_{f_i}, where $f_i \in A$; let $\mathfrak{J}$ be the ideal of A generated by the f_i, and let Z' be the closed affine subscheme of X that it defines (and which is by construction of finite presentation over X); one has by definition $V = \complement Z'$, so $U \cap \complement V = U \cap Z'$. Consider the affine scheme $X' = Z' \times_X U$ and let $f : X' \to X$ be the structural morphism; we just saw that the canonical immersion $Z' \to X$ is of finite presentation, and it is the same for the open immersion $U \to X$, which is quasi-compact since U and X are affine; one concludes therefore from (1.6.2, (iv)) that f is of finite presentation, and one has $f(X') = Z' \cap U$ (**I**, 3.4.8), which completes the proof. □

Theorem (1.8.4). — (Chevalley). — *Let $f : X \to Y$ be a morphism of finite presentation. For every locally constructible part Z of X, $f(Z)$ is locally constructible in Y.*

Proof. Let $y \in Y$ and V an open affine neighbourhood of y; since f is quasi-compact and quasi-separated, the same is true of its restriction $f^{-1}(V) \to V$, so $f^{-1}(V)$ is a quasi-compact and quasi-separated prescheme (1.2.3); since $Z \cap f^{-1}(V)$ is constructible in $f^{-1}(V)$ (1.8.1), we see that one can restrict ourselves to the case where Y is affine and Z constructible; X is then quasi-compact and quasi-separated, so by (1.8.3) there exists a morphism of finite presentation $g : X' \to X$ such that $Z = g(X')$; since $f \circ g$ is of finite presentation, we see that we are reduced to the case where $Z = X$; in other words, it will suffice to prove the □

Lemma (1.8.4.1). — *Let Y be an affine scheme, $f : X \to Y$ a morphism quasi-compact and locally of finite presentation; then $f(X)$ is a constructible part of Y.*

Proof. Since X is quasi-compact, it is a finite union of affine open subsets; one can therefore restrict ourselves to the case where $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, B being an A-algebra of finite presentation (1.4.6). Now we have the □

Lemma (1.8.4.2). — *Let A_0 be a ring, $(A_\lambda)_{\lambda \in \mathrm{L}}$ an inductive system of A_0-algebras, $A = \varinjlim A_\lambda$; if B is an A-algebra of finite presentation, there exist a λ and an A_λ-algebra of finite presentation B_λ such that B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$.*

Proof. By hypothesis, B is isomorphic to an algebra of the form $A[\mathrm{T}_1, \dots, \mathrm{T}_n]/\mathfrak{J}$, where the T_i are indeterminates and $\mathfrak{J}$ an ideal of finite type; let (F_j) $(1 \leqslant j \leqslant m)$ be a system of generators of $\mathfrak{J}$. Let $\varphi_\lambda : A_\lambda \to A$ be the canonical homomorphism. Since L is filtering, there exists λ such that each of the coefficients of each of the polynomials F_j is the image by φ_λ of an element of A_λ. In other words, there exists a system of polynomials $(\mathrm{F}_{j\lambda})_{1 \leqslant j \leqslant m}$ of $A_\lambda[\mathrm{T}_1, \dots, \mathrm{T}_n]$ such that $\mathrm{F}_j = \varphi_\lambda(\mathrm{F}_{j\lambda})$ for $1 \leqslant j \leqslant m$. If $\mathfrak{J}_\lambda$ is the ideal of $A_\lambda[\mathrm{T}_1, \dots, \mathrm{T}_n]$ generated by the $\mathrm{F}_{j\lambda}$, $\mathfrak{J}$ is the image of $\mathfrak{J}_\lambda \otimes_{A_\lambda} A$ in $A[\mathrm{T}_1, \dots, \mathrm{T}_n] = A_\lambda[\mathrm{T}_1, \dots, \mathrm{T}_n] \otimes_{A_\lambda} A$; the ring $B_\lambda = A_\lambda[\mathrm{T}_1, \dots, \mathrm{T}_n]/\mathfrak{J}_\lambda$ has the required property. □

We apply this lemma here by considering A as the inductive limit of its sub-$\mathbf{Z}$-algebras *of finite type*. One sees that there exist such a sub-$\mathbf{Z}$-algebra A_0 of A, and an A_0-algebra of finite presentation B_0 such that B is isomorphic to $B_0 \otimes_{A_0} A$; let us set $X_0 = \operatorname{Spec}(B_0)$, $Y_0 = \operatorname{Spec}(A_0)$, so that $X = X_0 \times_{Y_0} Y$, f being the projection $X \to Y$; let $f_0 : X_0 \to Y_0$, $g_0 : Y \to Y_0$ be the structural morphisms; it follows from (**I**, 3.4.8) that $f(X) = g_0^{-1}(f_0(X_0))$; taking into account (1.8.2), it therefore suffices to show that $f_0(X_0)$ is constructible. In other words, one is finally reduced to proving the

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Corollary (1.8.5). — *Let Y be a Noetherian prescheme, $f : X \to Y$ a morphism of finite type. For every constructible part Z of X, $f(Z)$ is a constructible part of Y.*

Proof. One reduces as above to proving that $f(X)$ is constructible. Let us apply the criterion ($\mathbf{0_{III}}$, 9.2.3): one must prove that for every irreducible closed part T of Y, $T \cap f(X) = f(f^{-1}(T))$ is rare in T or contains a non-empty open subset of T; taking a subscheme of Y whose underlying space is T (**I**, 5.2.1) and considering its inverse image by f, one sees finally (taking into account (1.5.4, (iii))) that one is reduced to proving that if one supposes Y *irreducible and* f *dominant*, $f(X)$ contains a non-empty open subset of Y. One can moreover suppose Y affine; then X is a finite union of affine open subsets V_i, and since $f(X)$ is dense in Y, the same is true of at least one of the $f(V_i)$. One can therefore also suppose X affine; if (X_j) is the (finite) family of irreducible components of X, one sees as above that at least one of the $f(X_j)$ is dense in Y; one can therefore suppose X irreducible. Finally, by replacing f by f_{red} (1.5.4, (vi)) one can suppose X and Y *integral*. Then $X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$, where A is an integral Noetherian ring, B an integral A-algebra of finite type containing A (**I**, 1.2.7). It suffices to show that there exists $g \in A$ such that, for every $y \in \mathrm{D}(g)$ (that is to say such that $g \notin \mathfrak{p}_y$) the ideal $\mathfrak{p}_y$ is the intersection of A and a prime ideal of B, for this will show that $\mathrm{D}(g) \subset f(X)$. Finally it suffices to prove the □

Lemma (1.8.5.1). — *Let A be an integral ring, B an integral A-algebra of finite type. There exists $g \in A$ such that every homomorphism of A into an algebraically closed field Ω, non-zero on g, extends to a homomorphism of B into Ω.*

Proof. This is a classical result of commutative algebra (Bourbaki, *Alg. comm.*, chap. V, §3, n° 1, cor. 3 of th. 1). □

Corollary (1.8.6). — *Let Y be an irreducible prescheme, η its generic point, $f : X \to Y$ a morphism locally of finite type. If $f^{-1}(\eta)$ is not empty, there exists a neighbourhood U open of η in Y such that $f^{-1}(y)$ is non-empty for every $y \in U$. If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module of finite type, and if $\mathscr{F}_\eta = \mathscr{F} \otimes_{\mathscr{O}_Y} k(\eta)$ is not zero, then there exists a neighbourhood U' of η in Y such that $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$ is not zero for every $y \in U'$.*

Proof. If p_y is the canonical projection of the fibre $f^{-1}(y) = X \times_Y \mathrm{Spec}(k(y))$ to X, one has $\mathscr{F}_y = p_y^*(\mathscr{F})$, so $\mathrm{Supp}(\mathscr{F}_y) = p_y^{-1}(\mathrm{Supp}(\mathscr{F})) = \mathrm{Supp}(\mathscr{F}) \cap f^{-1}(y)$ by virtue of (**I**, 9.1.13.1) and (**I**, 3.6.1), since $\mathscr{F}$ is of finite type; moreover $\mathrm{Supp}(\mathscr{F})$ is closed in X ($\mathbf{0_I}$, 5.2.2), and if Z is a closed subscheme of X having as underlying space $\mathrm{Supp}(\mathscr{F})$ (**I**, 5.2.1), and j the canonical immersion $Z \to X$, $f \circ j$ is locally of finite type (1.3.4); this proves that the first assertion implies the second. To prove the first assertion, let us first note that one can suppose X and Y reduced (1.3.4, (vi)), in other words Y *integral*. Let ξ be a point of $f^{-1}(\eta)$; one can replace X and Y by affine open neighbourhoods of ξ and η respectively, and thus suppose that $X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$, B being an A-algebra of finite type. Moreover, if Z is the reduced closed subscheme of X having $\overline{\{\xi\}}$ as underlying set (**I**, 5.2.1), one can, as above, replace X by Z, in other words suppose B *integral* (**I**, 5.1.4). Since the morphism f is then dominant, the corresponding homomorphism $\varphi : A \to B$ is injective (**I**, 1.2.7); the corollary therefore follows finally from the lemma (1.8.5.1). □

Proposition (1.8.7). — *Let S be a prescheme, $g : X \to S$, $h : Y \to S$ two morphisms of finite presentation, $f : X \to Y$ an S-morphism. For every $s \in S$, let $X_s = g^{-1}(s) = X \times_S \mathrm{Spec}(k(s))$, $Y_s = h^{-1}(s) = Y \times_S \mathrm{Spec}(k(s))$, $f_s = f \times 1 : X_s \to Y_s$. Then the set of $s \in S$ such that f_s is surjective (resp. radicial) is locally constructible.*

Proof. Let E be the set of $s \in S$ such that f_s is surjective; the set $S - E$ is equal to $h(Y - f(X))$; now f is of finite presentation (1.6.2, (v)) so $f(X)$ is locally constructible in Y (1.8.4); since h is of finite presentation, $h(Y - f(X))$ is locally constructible in S, and therefore also E.

To show that the set E' of $s \in S$ such that f_s is radicial is locally constructible, we will use the □

Lemma (1.8.7.1). — *Let U, V be two preschemes; for a morphism $f : U \to V$ to be radicial, it is necessary and sufficient that the diagonal morphism $\Delta_f : U \to U \times_V U$ is surjective. Consequently, every radicial morphism is separated.*

Proof. Indeed, Δ_f being an immersion (**I**, 5.3.9), is a morphism locally of finite type (1.3.4); to say that Δ_f is surjective therefore means that for every algebraically closed field K, the corresponding map $U(K) \to (U \times_V U)(K) = U(K) \times_{V(K)} U(K)$ is surjective (1.3.7) and (**I**, 3.4.2.1); but by definition of a fibre product of sets, this means that the map $U(K) \to V(K)$ corresponding to f is injective, and this is equivalent to saying that f is radicial (**I**, 3.5.5).

This being so, Δ_{f_s} is deduced from $\Delta_f : X \to X \times_Y X$ by the base change $\mathrm{Spec}(k(s)) \to S$ (**I**, 5.3.4). Since f is of finite presentation, it is the same for the structural morphism $X \times_Y X \to Y$ (1.6.2, (iv)), so also for Δ_f (1.6.2, (v)); it therefore suffices to apply the first part of the proposition to Δ_f, using the lemma (1.8.7.1). □

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1.9. Pro-constructible sets and ind-constructible sets

Lemma (1.9.1). — *Let $(A_\alpha)_{\alpha \in \mathrm{I}}$ be an inductive system of rings with index set* I *filtering to the right, and let $A = \varinjlim A_\alpha$. For $A = 0$, it is necessary and sufficient that there exists an index γ such that $A_\gamma = 0$ (and one then has $A_\beta = 0$ for every $\beta \geqslant \gamma$).*

Proof. Indeed, for every $\alpha \in \mathrm{I}$, the canonical homomorphism $\varphi_\alpha : A_\alpha \to A$ sends the unit element to the unit element; to say that $A = 0$ means that $\varphi_\alpha(1) = 0$, or again $\varphi_\alpha(1) = \varphi_\alpha(0)$; one knows that this is equivalent to saying that there exists an index $\gamma \geqslant \alpha$ such that the homomorphism $\varphi_{\gamma\alpha} : A_\alpha \to A_\gamma$ is such that $\varphi_{\gamma\alpha}(1) = \varphi_{\gamma\alpha}(0)$, and this last relation is equivalent to $A_\gamma = 0$, whence the lemma. □

Proposition (1.9.2). — *Let B be a ring, $(A_\alpha)_{\alpha \in \mathrm{I}}$ an inductive system of B-algebras with index set* I *filtering to the right, and let $A = \varinjlim A_\alpha$; set $Y = \mathrm{Spec}(B)$, $X = \mathrm{Spec}(A)$, $X_\alpha = \mathrm{Spec}(A_\alpha)$, and let $u : X \to Y$, $u_\alpha : X_\alpha \to Y$ be the morphisms corresponding to the structural homomorphisms $B \to A$, $B \to A_\alpha$. Then:*

(1.9.2.i). *For $X = \varnothing$, it is necessary and sufficient that there exists $\gamma \in \mathrm{I}$ such that $X_\gamma = \varnothing$, and one then has $X_\alpha = \varnothing$ for $\alpha \geqslant \gamma$.*

(1.9.2.ii). *We have*

(1.9.2.1)
$$u(X) = \bigcap_{\alpha \in \mathrm{I}} u_\alpha(X_\alpha).$$

The assertion (i) is nothing other than the translation of (1.9.1). To prove (ii), note that, since u factors as $X \longrightarrow X_\alpha \xrightarrow{u_\alpha} Y$, the left-hand side is contained in the right-hand side. Conversely, let $y \in Y - u(X)$; one has $u^{-1}(y) = \varnothing$, and $u^{-1}(y)$ is the space underlying the $k(y)$-prescheme $\mathrm{Spec}(A \otimes_B k(y))$; since in the category of B-modules the functor $\varinjlim$ commutes with the tensor product, $A \otimes_B k(y)$ is the inductive limit of the system of rings $A_\alpha \otimes_B k(y)$; the lemma (1.9.1) shows that there exists α such that $A_\alpha \otimes_B k(y) = 0$, that is to say $u_\alpha^{-1}(y) = \varnothing$, so $y \notin u_\alpha(X_\alpha)$.

Proposition (1.9.3). — *Let X be a quasi-compact and quasi-separated prescheme, let E be a part of X, let $(X_\alpha)_{\alpha \in \mathrm{I}}$ be an affine open cover of X, and, for every $\alpha \in \mathrm{I}$, let $E_\alpha = E \cap X_\alpha$. The following conditions are equivalent:*

- *a) There exists a quasi-compact morphism $f : X' \to X$ such that $E = f(X')$.*
- *a′) For every α, there exists a quasi-compact morphism $f_\alpha : X'_\alpha \to X_\alpha$ such that $E_\alpha = f_\alpha(X'_\alpha)$.*
- *a″) For every α, there exist an affine scheme X'_α and a morphism $f_\alpha : X'_\alpha \to X_\alpha$ such that $f_\alpha(X'_\alpha) = E_\alpha$.*
- *b) E is an intersection of constructible parts of X.*
- *b′) $F = X - E$ is a union of constructible parts of X.*

Proof. It is clear that *b*) and *b′*) are equivalent. The condition *a*) implies *a′*) by taking for X'_α the prescheme induced on the open subset $f^{-1}(X_\alpha)$ of X' and for f_α the restriction of f. To see that *a′*) implies *a″*), it suffices to remark that X'_α is a union of affine open subsets $X''_{\alpha\lambda}$ ($\lambda \in \mathrm{L}$); let X''_α be the prescheme *sum* of the $X''_{\alpha\lambda}$ ($\lambda \in \mathrm{L}$), which is an affine scheme; if $g_\alpha : X''_\alpha \to X'_\alpha$ is the morphism coinciding with the identity in each $X''_{\alpha\lambda}$, it is clear that if one sets $f'_\alpha = f_\alpha \circ g_\alpha$, one has $E_\alpha = f'_\alpha(X''_\alpha)$ since g_α is surjective. To show that *a″*) implies *a*), note that there is a finite part J of I such that the X_α for $\alpha \in \mathrm{J}$ cover X; let X' be the prescheme sum of the X'_α for $\alpha \in \mathrm{J}$, and let $f : X' \to X$ be the morphism coinciding with $j_\alpha \circ f_\alpha$ for every $\alpha \in \mathrm{J}$, where $j_\alpha : X_\alpha \to X$ is the canonical injection. Since E is the union of the $E \cap X_\alpha$ for $\alpha \in \mathrm{J}$, one has $E = f(X')$ and it remains to see that f is a quasi-compact morphism; but the hypothesis that X is quasi-separated implies that j_α is quasi-compact (1.2.7), so $j_\alpha \circ f_\alpha$ is quasi-compact (1.1.1) and (1.1.2), and hence so is f (1.1.6).

It remains therefore to prove the equivalence of $a'')$ and $b)$. Let us first prove that $a'')$ implies $b)$: let us consider a finite part J of I such that the X_α for $\alpha \in$ J cover X; it will suffice to show that, for every $\alpha \in$ J, E_α is an intersection of constructible parts $\mathrm{F}_{\alpha\lambda}$ of X_α ($\lambda \in \mathrm{L}_\alpha$); indeed, every $\mathrm{F}_{\alpha\lambda}$ is also a constructible part of X by virtue of ($\mathbf{0_{III}}$, 9.1.8, (ii)), since the hypothesis that X is quasi-separated implies that every X_α is retrocompact in X (1.2.7); since E is the union of the E_α for $\alpha \in$ J, it is the intersection of the finite unions $\bigcup_{\alpha\in \mathrm{J}} \mathrm{F}_{\alpha,\lambda(\alpha)}$, over all choices of $\lambda(\alpha) \in \mathrm{L}_\alpha$; these finite unions are constructible parts of X, which proves our assertion. We may therefore reduce to the case where $X = \mathrm{Spec}(A)$ and $E = f(X')$, with $X' = \mathrm{Spec}(A')$ affine and $f : X' \to X$ corresponding to a homomorphism of rings $A \to A'$. Let us note now that A' is the inductive limit of the family (A'_λ) of its sub-A-algebras *of finite type*, ordered by inclusion; if $X'_\lambda = \mathrm{Spec}(A'_\lambda)$, f factors as $X' \to X'_\lambda \xrightarrow{f_\lambda} X$ and it follows from (1.9.2, (ii)) that $f(X') = \bigcap_\lambda f_\lambda(X'_\lambda)$; if one shows that $f_\lambda(X'_\lambda)$ is an intersection of constructible parts of X, it will be the same for $f(X')$. We may therefore reduce to the case where A' is an A-algebra *of finite type*. Now, we have the following lemma: □ IV-1 | 242 IV-1 | 243

Lemma (1.9.3.1). — *Let A be a ring, A' an A-algebra of finite type. Then A' is A-isomorphic to a filtering inductive limit $\varinjlim A''_\mu$ where the A''_μ are A-algebras of finite presentation* (1.4.1).

Proof. Indeed, one can write $A' = B/\mathfrak{J}$, with $B = A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ and $\mathfrak{J}$ an ideal of B. But $\mathfrak{J}$ is inductive limit of the ideals *of finite type* $\mathfrak{J}'_\mu$ of B, contained in $\mathfrak{J}$, ordered by inclusion; since in the category of B-modules the functor $\varinjlim$ is exact, one concludes that $A' = \varinjlim B/\mathfrak{J}'_\mu$ up to A-isomorphism. □

The same reasoning as above then permits us to reduce to the case where A' is an A-algebra of finite presentation; but then $f(X')$ is constructible in X by virtue of the theorem of Chevalley (1.8.5), which proves $b)$.

Let us prove finally that $b)$ implies $a'')$. If E is an intersection of a family $(G_\lambda)_{\lambda\in\mathrm{L}}$ of constructible parts of X, each E_α is an intersection of the $G_\lambda \cap X_\alpha$, which are constructible in X_α ($\mathbf{0_{III}}$, 9.1.8, (i)), and one is reduced to the case where $X = \mathrm{Spec}(A)$ is affine.

One knows then (1.8.3) that for every $\lambda \in$ L there exists a morphism $f_\lambda : X'_\lambda \to X$, where $X'_\lambda = \mathrm{Spec}(A'_\lambda)$ is affine, such that $f_\lambda(X'_\lambda) = G_\lambda$. It therefore suffices to prove the following lemma:

Lemma (1.9.3.2). — *Let $(A'_\lambda)_{\lambda\in\mathrm{L}}$ be a family of A-algebras, $X = \mathrm{Spec}(A)$, $X'_\lambda = \mathrm{Spec}(A'_\lambda)$, and let $f_\lambda : X'_\lambda \to X$ be the structural morphism. For every finite part* J *of* L, *set $A'_\mathrm{J} = \bigotimes_{\lambda\in\mathrm{J}} A'_\lambda$ (tensor product of A-algebras), $X'_\mathrm{J} = \mathrm{Spec}(A'_\mathrm{J})$, and let $f_\mathrm{J} : X'_\mathrm{J} \to X$ be the structural morphism. One then has $f_\mathrm{J}(X'_\mathrm{J}) = \bigcap_{\lambda\in\mathrm{J}} f_\lambda(X'_\lambda)$. If $A' = \varinjlim A'_\mathrm{J}$,* J *running through the set of finite parts of* L, *$X' = \mathrm{Spec}(A')$ and $f : X' \to X$ is the structural morphism, one has $f(X') = \bigcap_{\lambda\in\mathrm{L}} f_\lambda(X'_\lambda)$.*

Proof. The first assertion follows from (**I**, 3.4.7); the second follows from (1.9.2, (ii)), which gives the relation $f(X') = \bigcap_\mathrm{J} f_\mathrm{J}(X'_\mathrm{J})$. □

Definition (1.9.4). — Let X be a topological space. We say that a part E of X is *pro-constructible* (resp. *ind-constructible*) in X if, for every $x \in X$, there exists an open neighbourhood U of x in X such that $E \cap U$ is an intersection (resp. a union) of locally constructible parts of U.

Taking into account (1.2.7), ($\mathbf{0_{III}}$, 9.1.8), and ($\mathbf{0_{III}}$, 9.1.12), the equivalent conditions of the proposition (1.9.3) are therefore expressed by saying that E is *pro-constructible* in X, the locally constructible parts of X being identical to the constructible parts of X under the hypotheses of (1.9.3).

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Proposition (1.9.5). — *Let X be a prescheme.*

(i) *For a part E of X to be pro-constructible, it is necessary and sufficient that $X - E$ be ind-constructible.*
(ii) *Every finite union and every intersection of pro-constructible parts of X are pro-constructible. Every finite intersection and every union of ind-constructible parts of X are ind-constructible.*
(iii) *Every intersection (resp. union) of locally constructible parts of X is pro-constructible (resp. ind-constructible). Conversely, if X is quasi-compact and quasi-separated, every pro-constructible (resp. ind-constructible) part of X is intersection (resp. union) of constructible parts of X.*

(iv) *Let* (U_α) *be an open cover of* X. *For a part* E *of* X *to be pro-constructible (resp. ind-constructible) in* X, *it is necessary and sufficient that for every* α, $E \cap U_\alpha$ *be pro-constructible (resp. ind-constructible) in* U_α.

(v) *Every pro-constructible part* E *of* X *is retrocompact in* X. *For a locally closed part* E *of* X *to be retrocompact, it is necessary and sufficient that it be pro-constructible in* X.

(vi) *Let* $f : X' \to X$ *be a morphism of preschemes; for every pro-constructible (resp. ind-constructible) part* E *of* X, $f^{-1}(E)$ *is pro-constructible (resp. ind-constructible) in* X'.

(vii) *Let* $f : X' \to X$ *be a quasi-compact morphism; for every pro-constructible part* E' *of* X', $f(E')$ *is pro-constructible in* X; *in particular* $f(X')$ *is pro-constructible in* X.

(viii) *Let* $f : X' \to X$ *be a morphism locally of finite presentation; for every ind-constructible part* E' *of* X', $f(E')$ *is ind-constructible in* X; *in particular* $f(X')$ *is ind-constructible in* X.

(ix) *Suppose that* X *is quasi-compact and quasi-separated; then, for a part* E *of* X *to be pro-constructible, it is necessary and sufficient that there exist an affine scheme* X' *and a (necessarily quasi-compact) morphism* $f : X' \to X$ *such that* $E = f(X')$.

Proof. Properties (i), (ii), (iv) and the first assertion of (iii) follow from the definition (1.9.4) and are valid for any topological space, using the mutual distributivity of intersection and union and the fact that if T is locally constructible in X, $T \cap U$ is locally constructible in U for every open subset U of X. If X is quasi-compact and quasi-separated and E is pro-constructible in X, there is a finite open cover (U_i) of X by affine open subsets such that $E \cap U_i$ is an intersection of constructible parts $\mathrm{M}_{i\lambda}$ of U_i ($\lambda \in \mathrm{L}_i$); the $\mathrm{M}_{i\lambda}$ are also constructible parts of X by virtue of (1.2.7) and ($\mathbf{0_{III}}$, 9.1.8, (ii)), and, since E is the union of the $E \cap U_i$, it is the intersection of the finite unions $\bigcup_i \mathrm{M}_{i,\lambda(i)}$ (where $\lambda(i) \in \mathrm{L}_i$ for every i), which are constructible parts of X; this proves the second assertion of (iii). The assertion (ix) follows from (iii) and (1.9.3). To prove the first assertion of (v), we may restrict ourselves to the case where X is affine, and then prove that E is quasi-compact ($\mathbf{0_{III}}$, 9.1.1); but since X is then quasi-separated, there exists by virtue of (ix) a quasi-compact morphism $f : X' \to X$ such that $E = f(X')$; since X' is quasi-compact, the same is true of E under a continuous map.

To prove (vi), one can restrict ourselves, by virtue of (iv), to the case where X is affine; the conclusion follows then from (iii) and (1.8.2).

To prove (vii) as well, one can restrict ourselves to the case where X is affine; then X' is a finite union of affine open subsets X'_i and $f(E')$ is the union of the $f(E' \cap X'_i)$, so one can also suppose that X' is affine, by virtue of (ii); but then $E' = g(X'')$, where $g : X'' \to X'$ is a quasi-compact morphism, by virtue of (ix), and one has $f(E') = f(g(X''))$, so $f \circ g$ is quasi-compact, and therefore $f(E')$ is pro-constructible.

One deduces from (vii) the proof of the second assertion of (v): indeed, let E be a locally closed and retrocompact part of X, and consider the sub-prescheme of X having E as underlying space (**I**, 5.2.1); the canonical injection $j : E \to X$ is a quasi-compact morphism, and it follows from (vii) that $E = j(E)$ is pro-constructible in X.

Finally, to prove (viii), one can still suppose X affine; moreover, if (X'_α) is an open affine cover of X' such that $E' \cap X'_\alpha$ is union of constructible parts of X'_α ((iii) and (iv)), $f(E')$ is union of the $f(E' \cap X'_\alpha)$, and by virtue of the theorem of Chevalley (1.8.4), each $f(E' \cap X'_\alpha)$ is union of constructible parts, whence the conclusion. □

(1.9.6). Examples (1.9.6). — Every finite part of a prescheme X is pro-constructible: indeed, it suffices to consider the parts $\{x\}$ reduced to a single point (1.9.5, (ii)), and $\{x\}$ is the image of $\mathrm{Spec}(k(x))$ by the canonical morphism $\mathrm{Spec}(k(x)) \to X$, which is quasi-compact; whence the conclusion by (1.9.5, (vii)). On the other hand, a part $\{x\}$ reduced to a point is not necessarily ind-constructible; for example, let A be a Noetherian integral ring having an infinity of maximal ideals, and let x be the generic point of $X = \mathrm{Spec}(A)$; if $\{x\}$ were ind-constructible in X, it would be constructible by virtue of (1.9.5, (iii)) and therefore would contain a non-empty open subset of X ($\mathbf{0_{III}}$, 9.2.2); but this contradicts the hypothesis that A has an infinity of maximal ideals, by virtue of the Artin–Tate theorem (**0**, 16.3.3).

Every closed part of a prescheme X is pro-constructible, by (1.9.5, (v)). Every open part of X is therefore ind-constructible, but an open part U of X is pro-constructible only if it is *retrocompact*, by virtue again of (1.9.5, (v)).

Finally, if X is a *Noetherian* prescheme, therefore quasi-separated (1.2.8), it follows from (1.9.5, (iii)) and ($\mathbf{0}_{\mathbf{III}}$, 9.1.7) that the ind-constructible parts of X are the *unions of locally closed parts* of X.

Theorem (1.9.7). — *Let X be a quasi-compact prescheme, E an ind-constructible part of X, $(\mathrm{F}_\lambda)_{\lambda\in\mathrm{L}}$ a family of pro-constructible parts of X such that $\bigcap_{\lambda\in\mathrm{L}}\mathrm{F}_\lambda \subset E$; then there exists a finite part J of L such that $\bigcap_{\lambda\in\mathrm{J}}\mathrm{F}_\lambda \subset E$.*

Proof. Let us write $F = X - E$; then the sets F and $F\cap\mathrm{F}_\lambda$ are pro-constructible, so one is reduced to the following corollary. □

Corollary (1.9.8). — *Let X be a quasi-compact prescheme, $(\mathrm{F}_\lambda)_{\lambda\in\mathrm{L}}$ a family of pro-constructible parts of X whose intersection is empty; then there exists a finite subfamily $(\mathrm{F}_\lambda)_{\lambda\in\mathrm{J}}$ whose intersection is empty.*

Proof. Covering X by a finite number of affine open subsets, and using (1.9.5, (iv)), one is reduced to the case where X is affine; by virtue of (1.9.5, (ix)), one has $\mathrm{F}_\lambda = f_\lambda(X'_\lambda)$, where $X'_\lambda = \mathrm{Spec}(A'_\lambda)$ **IV-1** | 245
is affine, and f_λ is a morphism $X'_\lambda \to X$; then, with the notations of (1.9.3.2), on the one hand, by hypothesis $f(X') = \varnothing$, hence $X' = \varnothing$; but this implies by (1.9.2, (i)) that there exists a finite part J of L such that $X'_{\mathrm{J}} = \varnothing$, hence $f_{\mathrm{J}}(X'_{\mathrm{J}}) = \varnothing$; but $f_{\mathrm{J}}(X'_{\mathrm{J}}) = \bigcap_{\lambda\in\mathrm{J}}\mathrm{F}_\lambda = \varnothing$. □

Corollary (1.9.9). — *Let X be a quasi-compact prescheme, F a pro-constructible part of X, $(\mathrm{E}_\lambda)_{\lambda\in\mathrm{L}}$ a family of ind-constructible parts of X such that $\mathrm{F} \subset \bigcup_{\lambda\in\mathrm{L}}\mathrm{E}_\lambda$; then there exists a finite part J of L such that $\mathrm{F} \subset \bigcup_{\lambda\in\mathrm{J}}\mathrm{E}_\lambda$. In particular, from every covering of X by ind-constructible parts, one can extract a finite covering of X.*

Proof. This follows from (1.9.7) by passing to complements. □

Proposition (1.9.10). — *Let X be a prescheme. For a part E of X to be ind-constructible, it is necessary that, for every $x \in X$, the intersection $E\cap\overline{\{x\}}$ be a neighbourhood of x in $\overline{\{x\}}$. If X is locally Noetherian, this condition is sufficient.*

Proof. Suppose E ind-constructible, and let Y be the intersection of $\overline{\{x\}}$ and an open affine subset of X containing x; then there is therefore a sub-prescheme of X having Y for underlying space (**I**, 5.2.1), and if $j : Y \to X$ is the canonical injection, $E\cap Y = j^{-1}(E)$ is ind-constructible in Y (1.9.5, (vi)). As Y is quasi-compact and separated, $E\cap Y$ is therefore a union of constructible parts of Y (1.9.5, (iii)); consequently, there are two open subsets U, V in Y such that $x \in U\cap\complement V \subset E\cap Y$ ($\mathbf{0}_{\mathbf{III}}$, 9.1.3). But as x is the generic point of the irreducible space Y, V is necessarily empty and we have $U \subset E\cap Y$.

Suppose now X locally Noetherian, and let E be a part of X satisfying the condition of the statement. By virtue of the definition (1.9.4), one can restrict to the case where X is *Noetherian*. Then, for every $x \in E$, $E\cap\overline{\{x\}}$ contains a non-empty open part of the form $U\cap\overline{\{x\}}$, where U is open in X; this shows that E is a union of locally closed parts of X, and hence is ind-constructible (1.9.6). □

Proposition (1.9.11). — *Let X be a prescheme, E a part of X. The following properties are equivalent:*

a) *E is locally constructible.*
b) *E is ind-constructible and pro-constructible.*
c) *E and $X - E$ are pro-constructible.*
d) *E and $X - E$ are ind-constructible.*

Proof. It clearly suffices to prove that *d*) implies *a*), and one can restrict to the case where X is affine. Then, by (1.9.5, (iii)), E (resp. $X - E$) is a union of a family (E_λ) (resp. (E'_μ)) of constructible parts of X; as the E_λ and the E'_μ form a covering of X, it follows from (1.9.9) that there are finitely many indices λ_i, μ_j such that the E_{λ_i} and the E'_{μ_j} form a covering of X; this implies that E is a union of the E_{λ_i}, and hence is constructible. □

Corollary (1.9.12). — *Let $f : X \to Y$ be a surjective morphism, either quasi-compact, or locally of finite presentation. For a part E of Y to be locally constructible (resp. pro-constructible, resp. ind-constructible), it is necessary and sufficient that $f^{-1}(E)$ be so in X.*

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Proof. We know that the condition is necessary by (1.8.2) and (1.9.5, (vi)); to prove sufficiency, one is reduced to the case where X is affine by virtue of (1.9.5, (iv)); moreover, by virtue of (1.9.11), one can restrict to the case where $f^{-1}(E)$ is pro-constructible, *or* the case where $f^{-1}(E)$ is ind-constructible. If f is surjective and quasi-compact, and $f^{-1}(E)$ is pro-constructible, it follows from (1.9.5, (vii)) that $E = f(f^{-1}(E))$ is pro-constructible. If f is surjective and locally of finite presentation, and $f^{-1}(E)$ is ind-constructible, $E = f(f^{-1}(E))$ is ind-constructible by (1.9.5, (viii)), which completes the proof. □

(1.9.13). Let X be a prescheme, $\mathscr{T}$ its topology; it follows from (1.9.5, (i) and (ii)) that the ind-constructible (resp. pro-constructible) parts of X are the open parts (resp. closed parts) for a topology on X, called the *constructible topology*, which we will denote by $\mathscr{T}^{\mathrm{cons}}$; we will denote by X^{cons} the set X equipped with the topology $\mathscr{T}^{\mathrm{cons}}$.

Proposition (1.9.14). — *Let X be a prescheme, $\mathscr{T}$ its topology, $\mathscr{T}^{\mathrm{cons}}$ the constructible topology on X.*

(i) *The topology $\mathscr{T}^{\mathrm{cons}}$ is finer than $\mathscr{T}$.*
(ii) *The locally constructible parts of X are identical to the parts that are both open and closed in the space X^{cons}.*
(iii) *For every morphism $f : X \to Y$, the underlying map of X^{cons} to Y^{cons} is continuous; we denote it f^{cons}.*
(iv) *If the morphism $f : X \to Y$ is quasi-compact, f^{cons} is a closed map; in particular, if f is quasi-compact and bijective, f^{cons} is a homeomorphism.*
(v) *If the morphism $f : X \to Y$ is locally of finite presentation, f^{cons} is an open map; in particular, if f is bijective and locally of finite presentation, f^{cons} is a homeomorphism.*
(vi) *For every open subset U of X, the topology induced by $\mathscr{T}^{\mathrm{cons}}$ on U is identical to the topology of U^{cons}.*

Proof. Indeed, (i) follows from the fact that every open set for $\mathscr{T}$ is open for $\mathscr{T}^{\mathrm{cons}}$ (1.9.6), and (ii) is a restatement of (1.9.11); (iii), (iv), and (v) are restatements respectively of (1.9.5, (vi)), (1.9.5, (vii)), and (1.9.5, (viii)). Finally, to prove (vi), it suffices to remark that the canonical injection $j : U \to X$ is locally of finite presentation (1.4.3), hence $j^{\mathrm{cons}} : U^{\mathrm{cons}} \to X^{\mathrm{cons}}$ is a continuous open map. □

Proposition (1.9.15). — *Let X be a prescheme.*

(i) *For X^{cons} to be quasi-compact, it is necessary and sufficient that X be quasi-compact.*
(ii) *For X^{cons} to be separated, it is necessary and sufficient that X be quasi-separated, and then X^{cons} is locally compact and totally disconnected.*
(iii) *For X^{cons} to be compact, it is necessary and sufficient that X be quasi-compact and quasi-separated.*
(iv) *Every point of X admits, for the topology $\mathscr{T}^{\mathrm{cons}}$, an open and compact neighbourhood.*
(v) *For a morphism $f : X \to Y$ to be quasi-compact, it is necessary and sufficient that the map $f^{\mathrm{cons}} : X^{\mathrm{cons}} \to Y^{\mathrm{cons}}$ be proper.*

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Proof. (i) As $\mathscr{T}^{\mathrm{cons}}$ is finer than $\mathscr{T}$, it is clear that if X^{cons} is quasi-compact, so is X; the converse follows from (1.9.9).

(ii) Suppose X quasi-separated, and let us show that X^{cons} is totally disconnected: indeed, if x, y are two distinct points of X and if, for example, $x \notin \overline{\{y\}}$, there exists an affine open neighbourhood U of x which does not contain y; as X is quasi-separated, U is retrocompact in X (1.2.7), hence U and $X - U$ are locally constructible in X, and therefore open in X^{cons} by virtue of (1.9.11), whence our assertion. For every open affine subset U of X, the topology induced by $\mathscr{T}^{\mathrm{cons}}$ is that of U^{cons} by virtue of (1.9.14, (vi)); it follows from the above that X^{cons} is locally compact, since U is open in X^{cons} and U^{cons} is compact. It remains to show that if X^{cons} is separated, then X is quasi-separated; consider indeed an open affine subset U of X; the topology induced on U by $\mathscr{T}^{\mathrm{cons}}$ is that of U^{cons} (1.9.14, (vi)), hence U is compact for this induced topology, since U^{cons} is compact by the first part of the reasoning. If V is a second open affine subset in X, $U \cap V$ is therefore the intersection of two compact parts of the separated space X^{cons}; as the topology induced by $\mathscr{T}^{\mathrm{cons}}$ on $U \cap V$ is that of $(U \cap V)^{\mathrm{cons}}$ (1.9.14, (vi)) and the identity $(U \cap V)^{\mathrm{cons}} \to U \cap V$ is continuous, one concludes that

$U \cap V$ is quasi-compact for the topology induced by $\mathscr{T}$; X is therefore quasi-separated by virtue of (1.2.7).

(iii) is an immediate corollary of (i) and (ii).

(iv) For every $x \in X$, an open affine neighbourhood U of x for $\mathscr{T}$ is also a neighbourhood of x for $\mathscr{T}^{\mathrm{cons}}$ and it is compact by virtue of (iii) and (1.9.14, (vi)), the topology induced on U by $\mathscr{T}^{\mathrm{cons}}$ being identical to that of U^{cons}.

(v) Suppose f quasi-compact; then we already know (1.9.14, (iv)) that f^{cons} is a closed map. Let on the other hand y be a point of Y, and let us set

$$Z = f^{-1}(y) = X \times_Y \mathrm{Spec}(k(y));$$

Z is quasi-compact (1.1.2, (iii)) and, as the canonical morphism $p : Z \to X$ is injective, it follows from (i) and from the fact that the map p^{cons} is continuous that the topology induced by that of X^{cons} on $f^{-1}(y)$ makes $f^{-1}(y)$ a quasi-compact space. This proves that f^{cons} is a proper map (Bourbaki, *Top. gén.*, chap. I^{er}, 3^{e} éd., § 10, n° 2, th. 1). Conversely, suppose the continuous map f^{cons} proper, and let V be an open quasi-compact subset of Y; if $h : V \to Y$ is the canonical injection, $h^{\mathrm{cons}} : V^{\mathrm{cons}} \to Y^{\mathrm{cons}}$ is continuous and injective and V^{cons} is quasi-compact by (i), hence the topology induced on V by that of Y^{cons} makes V a quasi-compact space. The hypothesis that f^{cons} is proper implies that the topology induced on $f^{-1}(V)$ by that of X^{cons} makes $f^{-1}(V)$ a quasi-compact space (*loc. cit.*, prop. 6), hence $f^{-1}(V)$ is also a quasi-compact subspace of X, which shows that the morphism f is quasi-compact. □

(1.9.16). We will show later how one can, for every prescheme X, equip the space X^{cons} with a sheaf of rings which makes it a prescheme whose local rings are all fields, identical to the residue fields at the points of X. Such preschemes arise naturally in the translation, into the language of schemes, of the constructions of Néron [32] in his theory of the reduction of abelian varieties.

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1.10. Application to open morphisms

Theorem (1.10.1). — *Let X be a prescheme, E an ind-constructible part of X (1.9.4), x a point of X. For x to be interior to E, it is necessary and sufficient that every generisation* ($\mathbf{0_I}$, 2.1.2) *x' of x belong to E, in other words* (**I**, 2.4.2) *that* $\mathrm{Spec}(\mathscr{O}_{X,x}) \subset E$.

Proof. The condition is evidently necessary, every neighbourhood of x containing the generisations of x. To show that it is sufficient, we may evidently restrict to the case where $X = \mathrm{Spec}(A)$ is affine, x being a prime ideal $\mathfrak{p}$ of A. We know then (1.9.5, (ix)) that there exists an affine scheme $X' = \mathrm{Spec}(A')$ and a morphism $f : X' \to X$ such that $f(X') = X - E$. Let $Y = \mathrm{Spec}(\mathscr{O}_{X,x})$ and $Y' = X' \times_X Y$; the hypothesis means (by virtue of (**I**, 3.6.5)) that $Y' = \varnothing$, in other words $A' \otimes_A A_{\mathfrak{p}} = 0$. As $A_{\mathfrak{p}} = \varinjlim A_t$, where t runs through $A - \mathfrak{p}$ ($\mathbf{0_I}$, 1.4.5), we have $\varinjlim A'_t = 0$, since the functor $\varinjlim$ commutes with the tensor product. Consequently, by (1.9.1), there exists $t \in A - \mathfrak{p}$ such that $A'_t = 0$, and $\mathrm{D}(t)$ is then an open neighbourhood of x in X contained in E. □

Corollary (1.10.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation, and let $y \in Y$. For y to be interior to $f(X)$, it is necessary and sufficient that every generisation y' of y belong to $f(X)$.*

Proof. We know indeed (1.9.5, (viii)) that $f(X)$ is ind-constructible in Y and it suffices to apply (1.10.1). □

We will say that a continuous map $f : X \to Y$ is *open at a point* $x \in X$ if the image under f of *every* neighbourhood of x in X is a neighbourhood of $f(x)$ in Y.

Proposition (1.10.3). — *Let $f : X \to Y$ be a morphism, x a point of X, $y = f(x)$. Consider the following conditions:*

a) *f is open at the point x.*
b) *For every generisation y' of y, there exists a generisation x' of x such that $f(x') = y'$.*
b′) *The image under f of* $\mathrm{Spec}(\mathscr{O}_{X,x})$ *is* $\mathrm{Spec}(\mathscr{O}_{Y,y})$.
c) *For every irreducible closed part Y' of Y containing y, there exists an irreducible component of $X' = f^{-1}(Y')$ containing x and dominating Y'.*

Then we have the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b′) $\Leftrightarrow$ c). *If in addition f is locally of finite presentation, the four conditions are equivalent.*

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Proof. By definition of a generisation, the image under f of $\operatorname{Spec}(\mathscr{O}_{X,x})$ is always contained in $\operatorname{Spec}(\mathscr{O}_{Y,y})$, so b) and b′) are equivalent. It is immediate that c) implies b), since if y' is a generisation of y, $Y' = \overline{\{y'\}}$ is an irreducible closed part of Y containing y; if x' is the generic point of an irreducible component of $f^{-1}(Y')$ containing x and dominating Y', we have $f(x') = y'$ (**0**$_{\mathbf{I}}$, 2.1.5) and x' is a generisation of x. Conversely, b) implies c): indeed, let y' be the generic point of Y' and let x' be a generisation of x such that $f(x') = y'$; let Z' be an irreducible component of $f^{-1}(Y')$ containing x' (and hence x); as $f(x')$ is already dense in Y', so *a fortiori* is $f(Z')$.

To show that a) implies b′), one can evidently restrict to the case where $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$ are affine, x being a prime ideal $\mathfrak{q}$ of B, so that $\mathscr{O}_{X,x} = B_{\mathfrak{q}} = \varinjlim B_t$,

where t runs through $B - \mathfrak{q}$. One has, by (1.9.2), $f(\operatorname{Spec}(\mathscr{O}_{X,x})) = \bigcap_t f(\operatorname{Spec}(B_t)) = \bigcap_t f(\mathrm{D}(t))$, **IV-1** | 250
and by hypothesis, for every $t \in B - \mathfrak{q}$, y is interior to $f(\mathrm{D}(t))$, hence $f(\mathrm{D}(t))$ contains $\operatorname{Spec}(\mathscr{O}_{Y,y})$; whence $\operatorname{Spec}(\mathscr{O}_{Y,y}) \subset f(\operatorname{Spec}(\mathscr{O}_{X,x}))$, which establishes our assertion. Finally, when f is locally of finite presentation, b) implies a) by virtue of (1.10.2) applied to the restriction $U \to Y$ of f to an arbitrary open neighbourhood U of x. □

Corollary (1.10.4). — *Let $f : X \to Y$ be a morphism. Consider the following conditions:*

a) *f is open.*
b) *For every $x \in X$ and every generisation y' of $y = f(x)$, there exists a generisation x' of x such that $f(x') = y'$.*
b′) *For every $x \in X$, the image under f of* $\operatorname{Spec}(\mathscr{O}_{X,x})$ *is* $\operatorname{Spec}(\mathscr{O}_{Y,f(x)})$.
c) *For every irreducible closed part Y' of Y, every irreducible component of $f^{-1}(Y')$ dominates Y'.*

Then we have the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b′) $\Leftrightarrow$ c). *If in addition f is locally of finite presentation, the four conditions are equivalent.*

Proof. To say that f is open means that f is open at every point $x \in X$, hence the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b′) follow from the analogous implications in (1.10.3), as well as the implication b) $\Rightarrow$ a) when f is locally of finite presentation. The implication c) $\Rightarrow$ b) also follows from the analogous implication in (1.10.3); let us finally prove that *b*) implies *c*). Indeed, let x' be the generic point of an irreducible component of $f^{-1}(Y')$, and let us show that $y' = f(x')$ is the generic point of Y'. Let y'' be a generisation of y'; by hypothesis there exists a generisation x'' of x' such that $f(x'') = y''$, and as $x'' \in f^{-1}(Y')$, we have necessarily $x'' = x'$, hence $y'' = y'$, which completes the proof. □

§2. BASE CHANGE AND FLATNESS

This section (unlike § 6) only exceptionally makes use of Noetherian techniques. The n$^{\text{os}}$ 1 and 2 **IV-2** | 5
are essentially nothing more than translations of elementary properties of flatness in commutative Algebra (cf. Bourbaki, *Alg. comm.*, chap. I$^{\text{er}}$), and are included here for the convenience of references. The following numbers are primarily devoted to properties of "descent" by flat or faithfully flat morphisms: if $g : Y' \to Y$ is such a morphism, we want to be able to assert that a subset of Y, or an $\mathscr{O}_Y$-Module, or a morphism $X \to Y$, has a certain property, when we know that its *inverse image* by g possesses this property. We limit ourselves here to properties that do not make use of the general technique of "descent", which will be developed in Chapter V.

2.1. Flat Modules on preschemes

(2.1.1). Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-Module; recall ($\mathbf{0_I}$, 6.7.1) that $\mathscr{F}$ is said to be *f-flat* (or *Y-flat*) *at a point* $x \in X$ if $\mathscr{F}_x$ is a flat $\mathscr{O}_{f(x)}$-module, *f-flat* (or *Y-flat*) if it is f-flat at every point $x \in X$, and finally that the morphism f is said to be *flat at the point* $x \in X$ (resp. *flat*) if $\mathscr{O}_X$ is f-flat at the point x (resp. f-flat). When $f = 1_X$, we will simply say that an $\mathscr{O}_X$-Module $\mathscr{F}$ is *flat at the point* x (resp. *flat*) if it is X-flat at this point (resp. at every point $x \in X$), that is to say if $\mathscr{F}_x$ is a flat $\mathscr{O}_x$-module (resp. if this is the case for every $x \in X$). Recall that we have shown (**III**, 1.4.15.1) the following property:

Proposition (2.1.2). — *Let A, B be two rings, $\varphi : A \to B$ a ring homomorphism, $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$, $f : X \to Y$ the corresponding morphism, and M a B-module. For $\mathscr{F} = \widetilde{M}$ to be f-flat, it is necessary and sufficient that M be a flat A-module.*

Proposition (2.1.3). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

- a) *For every base change $g : Y' \to Y$, if we set $X' = X \times_Y Y'$, the functor $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ from the category of quasi-coherent $\mathscr{O}_{Y'}$-Modules to that of quasi-coherent $\mathscr{O}_{X'}$-Modules is exact.*
- a′) *Condition* a) *is satisfied for all canonical morphisms $g : \operatorname{Spec}(\mathscr{O}_y) \to Y$* (**I**, 2.4.1), *where y ranges over Y.*
- b) *$\mathscr{F}$ is f-flat.*

Proof. The questions being local on X and Y, we can restrict to the case where $Y = \operatorname{Spec}(B)$, $X = \operatorname{Spec}(A)$, $\mathscr{F} = \widetilde{M}$, where M is an A-module. It is clear that *a)* implies *a′)*; condition *a′)* implies that for every $x \in X$, the functor $N \mapsto M_{\mathfrak{n}} \otimes_{B_{\mathfrak{n}}} N$ is exact in the category of $B_{\mathfrak{n}}$-modules, $\mathfrak{n}$ being the ideal $\mathfrak{j}_{f(x)}$ of B; this means that $M_{\mathfrak{n}}$ is a flat $B_{\mathfrak{n}}$-module, and it follows from ($\mathbf{0_I}$, 6.3.3) and from (2.1.2) that $\mathscr{F}$ is f-flat. Finally, to see that *b)* implies *a)*, we can also restrict to the case where $Y' = \operatorname{Spec}(A')$ is affine and $\mathscr{G}' = \widetilde{N'}$, where N' is an A'-module; the conclusion then follows from (2.1.2) and from the definition of flatness, since $(M \otimes_A N')^{\sim} = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$. □

Proposition (2.1.4). — *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms of preschemes, $\mathscr{F}$ a quasi-coherent* **IV-2 | 6**
$\mathscr{O}_X$-Module; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$, and let $g' : X' \to X$ be the canonical projection. Let x' be a point of X', and set $x = g'(x')$, $y' = f'(x')$, and $y = g(y') = f(x)$. If $\mathscr{F}$ is f-flat at the point x, $\mathscr{F}'$ is f'-flat at the point x'; in particular, if $\mathscr{F}$ is f-flat, $\mathscr{F}'$ is f'-flat; if f is flat, f' is flat.

Proof. It suffices to prove the first assertion; applying three times (**I**, 3.6.5), as well as (**I**, 2.4.4), we can reduce to the case where $Y = \operatorname{Spec}(\mathscr{O}_y)$, $X = \operatorname{Spec}(\mathscr{O}_x)$, $Y' = \operatorname{Spec}(\mathscr{O}_{y'})$, $\mathscr{F} = \widetilde{M}$, where $M = \mathscr{F}_x$. The hypothesis and (2.1.2) then imply that $\mathscr{F}$ is f-flat; in other words, we are reduced to proving a special case of the second assertion, and the latter follows immediately from (2.1.3). □

Proposition (2.1.5). — *Consider a commutative diagram of morphisms of preschemes*

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "g'"'] \arrow[d, "f'"] \\ Y & Y' \arrow[l, "g"] \arrow[d, "h"] \\ & Z \end{tikzcd}$$

where $X' = X \times_Y Y'$ and $f' = f_{(Y')}$. Let x' be a point of X', and set $x = g'(x')$, $y' = f'(x')$, $y = f(x) = g(y')$, $z = h(y')$. Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module f-flat at the point x (resp. f-flat), and $\mathscr{G}'$ a quasi-coherent $\mathscr{O}_{Y'}$-Module h-flat at the point y' (resp. h-flat); then $\mathscr{F} \otimes_{\mathscr{O}_{Y'}} \mathscr{G}'$ is a quasi-coherent $\mathscr{O}_{X'}$-Module $(h \circ f')$-flat at the point x' (resp. $(h \circ f')$-flat).

Proof. As in (2.1.4), we reduce to the case where $X = \operatorname{Spec}(\mathscr{O}_x)$, $Y = \operatorname{Spec}(\mathscr{O}_y)$, $Y' = \operatorname{Spec}(\mathscr{O}_{y'})$ and **IV-2 | 7**
$Z = \operatorname{Spec}(\mathscr{O}_z)$, and it suffices then to prove that $\mathscr{F} \otimes_{\mathscr{O}_{Y'}} \mathscr{G}'$ is $(h \circ f')$-flat. Taking into account (2.1.2), the proposition follows from Bourbaki, *Alg. comm.*, chap. I^er, § 2, n^o 7, prop. 8. □

Corollary (2.1.6). — *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-Module. If $\mathscr{F}$ is f-flat at the point $x \in X$ and if g is flat at the point $f(x)$, $\mathscr{F}$ is $(g \circ f)$-flat at the point x. In particular, if f and g are flat morphisms, so is $g \circ f$.*

Proof. This is the special case of (2.1.5) with $Y' = Y$, $\mathscr{G}' = \mathscr{O}_{Y'}$. □

Corollary (2.1.7). — *If $f : X \to X'$, $g : Y \to Y'$ are two flat S-morphisms, $f \times_S g : X \times_S Y \to X' \times_S Y'$ is a flat morphism.*

Proof. This follows from (2.1.4) and (2.1.6) (cf. (**I**, 3.5.1)). □

Proposition (2.1.8). — *Let $f : X \to Y$ be a morphism of preschemes,*

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0$$

an exact sequence of quasi-coherent $\mathscr{O}_X$-Modules, such that $\mathscr{F}''$ is Y-flat.

(i) For every morphism $g : Y' \to Y$ and every quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{G}'$, the sequence $0 \to \mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{G}' \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' \to \mathscr{F}'' \otimes_{\mathscr{O}_Y} \mathscr{G}' \to 0$ of $\mathscr{O}_{X'}$-Modules (where $X' = X \times_Y Y'$) is exact.

(ii) For $\mathscr{F}$ to be Y-flat, it is necessary and sufficient that $\mathscr{F}'$ be so.

Proof. We can evidently suppose X, Y, Y' affine; the conclusion then follows from (2.1.2) and from ($\mathbf{0_I}$, 6.1.2). □

Corollary (2.1.9). — *Let $\mathscr{L}^\bullet$ be a complex of quasi-coherent $\mathscr{O}_X$-Modules, i an index such that if $d^i : \mathscr{L}^i \to \mathscr{L}^{i+1}$ is the derivation operator, $\mathscr{B}^{i+1}(\mathscr{L}^\bullet) = \operatorname{Im}(d^i)$ and $\mathscr{Z}^{i+1}(\mathscr{L}^\bullet) = \operatorname{Coker}(d^i)$ are Y-flat. Then, with the notations of (2.1.8), the canonical homomorphism*

$$\mathscr{H}^i(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}')$$

is bijective.

Proof. As the tensor product is right exact, we have

$$\mathscr{Z}^{i+1}(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' = \operatorname{Coker}(d^i \otimes 1) = \mathscr{Z}^{i+1}(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}')$$

and $\mathscr{Z}^i(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' = \mathscr{Z}^i(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}')$. Moreover, in the exact sequence

$$0 \longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet) \longrightarrow \mathscr{L}^{i+1} \longrightarrow \mathscr{Z}^{i+1}(\mathscr{L}^\bullet) \longrightarrow 0$$

$\mathscr{Z}^{i+1}(\mathscr{L}^\bullet)$ is Y-flat, so it follows from (2.1.8, (i)) that the exact sequence

$$0 \longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{L}^{i+1} \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{Z}^{i+1}(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}') \longrightarrow 0$$

whence $\mathscr{B}^{i+1}(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' = \operatorname{Im}(d^i \otimes 1) = \mathscr{B}^{i+1}(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}')$. Then, as in the exact sequence

$$0 \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet) \longrightarrow \mathscr{Z}^i(\mathscr{L}^\bullet) \longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet) \longrightarrow 0$$

$\mathscr{B}^{i+1}(\mathscr{L}^\bullet)$ is Y-flat, it follows from (2.1.8, (i)) and from what precedes that the sequence

$$0 \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{Z}^i(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}') \longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}') \longrightarrow 0$$

is exact, which proves the corollary. □

Corollary (2.1.10). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module which is Y-flat, $\mathscr{L}_\bullet = (\mathscr{L}_i)$ a left resolution of $\mathscr{F}$ formed of quasi-coherent $\mathscr{O}_X$-Modules which are Y-flat. Then, for every morphism $g : Y' \to Y$ and every quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{G}'$, the complex $\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}' = (\mathscr{L}_i \otimes_{\mathscr{O}_Y} \mathscr{G}')_{i \geqslant 0}$ is a left resolution of $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$. Moreover, if $\mathscr{Z}_i(\mathscr{L}_\bullet) = \operatorname{Ker}(\mathscr{L}_i \to \mathscr{L}_{i-1})$, the $\mathscr{Z}_i(\mathscr{L}_\bullet)$ are Y-flat, and we have $\mathscr{Z}_i(\mathscr{L}_\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' = \mathscr{Z}_i(\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}') = \operatorname{Ker}(\mathscr{L}_i \otimes_{\mathscr{O}_Y} \mathscr{G}' \to \mathscr{L}_{i-1} \otimes_{\mathscr{O}_Y} \mathscr{G}')$. Setting $\mathscr{R}_i = \operatorname{Im}(\mathscr{L}_{i+1} \to \mathscr{L}_i) = \mathscr{Z}_i(\mathscr{L}_\bullet)$; we then have the exact sequences* **IV-2 | 8**

$$0 \leftarrow \mathscr{F} \leftarrow \mathscr{L}_0 \leftarrow \mathscr{R}_0 \leftarrow 0$$

$$0 \leftarrow \mathscr{R}_i \leftarrow \mathscr{L}_{i+1} \leftarrow \mathscr{R}_{i+1} \leftarrow 0 \qquad (i \geqslant 0)$$

and since $\mathscr{F}$ and $\mathscr{L}_i$ are Y-flat, we deduce from (2.1.8, (ii)) by induction that all the $\mathscr{R}_i$ are also Y-flat; using (2.1.8, (i)), we thus have the exact sequences

$$0 \leftarrow \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow \mathscr{L}_0 \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow \mathscr{R}_0 \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow 0$$

$$0 \leftarrow \mathscr{R}_i \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow \mathscr{L}_{i+1} \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow \mathscr{R}_{i+1} \otimes_{\mathscr{O}_Y} \mathscr{G}' \leftarrow 0 \qquad (i \geqslant 0)$$

which prove the corollary.

Proposition (2.1.11). — *Let $f : X \to Y$ be a flat morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_Y$-Module of finite presentation. If $\mathscr{J}$ is the Ideal of $\mathscr{O}_Y$ annihilator of $\mathscr{F}$, $f^*(\mathscr{J})$ is the Ideal of $\mathscr{O}_X$ annihilator of $f^*(\mathscr{F})$.*

Proof. We have indeed by definition an exact sequence ($\mathbf{0_I}$, 5.3.7)

$$0 \longrightarrow \mathscr{J} \longrightarrow \mathscr{O}_Y \longrightarrow \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F}, \mathscr{F})$$

whence, since f is flat, an exact sequence

$$0 \longrightarrow f^*(\mathscr{J}) \longrightarrow \mathscr{O}_X \longrightarrow f^*(\mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F}, \mathscr{F}))$$

and since $\mathscr{F}$ is an $\mathscr{O}_Y$-Module of finite presentation, $f^*(\mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F}, \mathscr{F}))$ canonically identifies with $\mathscr{H}om_{\mathscr{O}_X}(f^*(\mathscr{F}), f^*(\mathscr{F}))$ ($\mathbf{0_I}$, 6.7.6), whence the conclusion. □

Proposition (2.1.12). — *Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module of finite presentation, x a point of X. The following conditions are equivalent:*

a) *$\mathscr{F}_x$ is a flat $\mathscr{O}_x$-module.*
b) *There exists an open neighbourhood U of x such that $\mathscr{F}|U$ is a locally free $(\mathscr{O}_X|U)$-Module.*

Proof. Indeed, $\mathscr{F}_x$ is a module of finite presentation over $\mathscr{O}_x$, and $\mathscr{O}_x$ is a local ring; it amounts to the same thing to say that $\mathscr{F}_x$ is a flat $\mathscr{O}_x$-module or a free $\mathscr{O}_x$-module (Bourbaki, *Alg. comm.*, chap. II, § 3, nº 2, cor. 2 of prop. 5); whence the conclusion, taking into account ($\mathbf{0_I}$, 5.2.7). We note that the proposition is valid for any *ringed space with local rings*. □

Proposition (2.1.13). — *Let $f : X \to Y$ be a morphism of preschemes. If f is flat at a point $x \in X$ and if the ring $\mathscr{O}_x$ is reduced* (*resp. integral and integrally closed*), *then the ring $\mathscr{O}_{f(x)}$ is reduced* (*resp. integral and integrally closed*). *If f is faithfully flat and if X is reduced* (*resp. normal*), *then Y is reduced* (*resp. normal*). **IV-2** | 9

Proof. Set $\mathscr{O}_{f(x)} = A$, $\mathscr{O}_x = B$. If B is a flat A-module, it is also a faithfully flat A-module ($\mathbf{0_I}$, 6.6.2), so A identifies with a subring of B; if B is reduced, so is A. Suppose now that B is integral and integrally closed, and let L be its field of fractions; then $A \subset B$ is integral; denote by $K \subset L$ its field of fractions. The hypothesis implies that $B \cap K = A$ (Bourbaki, *Alg. comm.*, chap. I, § 3, nº 5, prop. 10). If then $t \in K$ is integral over A, it is also integral over B, so belongs to B by hypothesis, and thus $t \in A$, which proves that A is integrally closed. □

Proposition (2.1.14). — *Let $f : X \to Y$ be a faithfully flat morphism. If X is locally integral and if the space underlying Y is locally Noetherian, then Y is locally integral.*

Proof. Indeed, every $y \in Y$ is of the form $f(x)$ for some $x \in X$ and by hypothesis $\mathscr{O}_y$ identifies with a subring of $\mathscr{O}_x$ ($\mathbf{0_I}$, 6.6.1); as $\mathscr{O}_x$ is integral, so is $\mathscr{O}_y$ and this proves the proposition (**I**, 5.1.4). □

2.2. Faithfully flat Modules on preschemes

Proposition (2.2.1). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

a) *For every base change $g : Y' \to Y$, if we set $X' = X \times_Y Y'$, the functor $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$, from the category of quasi-coherent $\mathscr{O}_{Y'}$-Modules to that of quasi-coherent $\mathscr{O}_{X'}$-Modules, is exact and faithful.*
a′) *Condition* a) *is satisfied for all canonical morphisms $g : \mathrm{Spec}(\mathscr{O}_y) \to Y$* (**I**, 2.4.1), *where y ranges over Y.*
a″) *Condition* a) *is satisfied for all canonical immersions $Y' \to Y$, where Y' ranges over the set of affine open sub-preschemes of Y.*
b) *$\mathscr{F}$ is Y-flat and, for every $y \in Y$, if we denote by X_y the fibre $f^{-1}(y)$, the $\mathscr{O}_{X_y}$-Module $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$ is nonzero.*

Proof. It is clear that *a)* implies *a′)* and *a″)*; condition *a′)* implies first that $\mathscr{F}$ is Y-flat (2.1.3); on the other hand it implies that for every $y \in Y$, the functor $N \mapsto \mathscr{F} \otimes_{\mathscr{O}_y} \widetilde{N}$ is faithful in the category of $\mathscr{O}_y$-modules; taking in particular $N = k(y)$, we obtain the second assertion of *b)*. To show that *b)* implies *a)*, we can restrict to the case where Y is affine, since the question is local over Y. Likewise, to prove that *a″)* implies *a)*, we are reduced to proving that, when Y is affine, the fact that $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is an exact and faithful functor implies condition *a)*. In other words, we are reduced to proving the following more precise proposition: □

Proposition (2.2.2). — *Let $Y = \operatorname{Spec}(A)$ be an affine scheme, $f : X \to Y$ a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. Condition* a) *of* (2.2.1) *is equivalent to each of the following:* **IV-2 | 10**

- b′) *$\mathscr{F}$ is Y-flat and, for every closed point y of Y, we have $\mathscr{F}_y \neq 0$.*
- c) *The functor $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ from the category of quasi-coherent $\mathscr{O}_Y$-Modules to that of quasi-coherent $\mathscr{O}_X$-Modules is exact and faithful.*

Proof. If *b′)* holds and y is a closed point of Y, there is at least one $x \in f^{-1}(y)$ such that $(\mathscr{F}_y)_x \neq 0$; let $U = \operatorname{Spec}(B)$ be an affine open neighbourhood of x, and write $\mathscr{F}|_U = \widetilde{M}$, where M is a B-module. Then *b′)* implies that $M/\mathfrak{i}_y M$ is nonzero and hence, since M is a flat A-module (2.1.2), that $M \otimes_A A_y$ is a faithfully flat A_y-module ($\mathbf{0_I}$, 6.4.5). The relation $(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}) \otimes_{\mathscr{O}_Y} \mathscr{O}_y = 0$ for a closed point y of Y implies

$$(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_y) \otimes_{\mathscr{O}_y} \mathscr{G}_y = 0,$$

and therefore $\mathscr{G}_y = 0$. But if $\mathscr{G}_y = 0$ for every closed point y of Y, we conclude that $\mathscr{G} = 0$: indeed, if $\mathscr{G} = \widetilde{N}$, the annihilator of every element of N is contained in no maximal ideal of A, and hence is all of A. The relation $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} = 0$ implies therefore $\mathscr{G} = 0$, in other words the functor $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is faithful; we know moreover that this functor is exact (2.1.3), which shows that *b′)* implies *c)*.

Finally, to see that *c)* implies *a)*, we can restrict to the case where Y' is also affine. The morphism $g : Y' \to Y$ is then affine, and so is the projection $g' : X' \to X$ (**II**, 1.5.5). Moreover, the functor $\mathscr{H}' \mapsto g'_*(\mathscr{H}')$ is exact in the category of quasi-coherent $\mathscr{O}_{X'}$-Modules (**I**, 1.6.4); if $g'_*(\mathscr{H}') = 0$, then $\mathscr{H}' = 0$, so this functor is also faithful. Consequently, to show that $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ is exact and faithful, it suffices to show that the functor

$$\mathscr{G}' \longmapsto g'_*\big(g'^*(\mathscr{F}) \otimes_{\mathscr{O}_{X'}} f'^*(\mathscr{G}')\big)$$

is exact and faithful. The fact that g is affine implies that we have a canonical isomorphism

$$\mathscr{F} \otimes_{\mathscr{O}_X} f^*(g_*(\mathscr{G}')) \xrightarrow{\sim} g'_*(g'^*(\mathscr{F}) \otimes_{\mathscr{O}_{X'}} f'^*(\mathscr{G}')). \tag{2.2.2.1}$$

Indeed, we know (**II**, 1.5.2) that we have a canonical isomorphism

$$f^*(g_*(\mathscr{G}')) \xrightarrow{\sim} g'_*(f'^*(\mathscr{G}')),$$

and on the other hand a canonical homomorphism $\mathscr{F} \to g'_*(g'^*(\mathscr{F}))$ ($\mathbf{0_I}$, 4.4.3.2); composing the homomorphism $\mathscr{F} \otimes_{\mathscr{O}_X} f^*(g_*(\mathscr{G}')) \to g'_*(g'^*(\mathscr{F})) \otimes_{\mathscr{O}_X} g'_*(f'^*(\mathscr{G}'))$ with the canonical homomorphism ($\mathbf{0_I}$, 4.2.2.1), we deduce the homomorphism (2.2.2.1); the verification that it is an isomorphism is immediate by reducing to the case where X is affine. This being so, the functor $\mathscr{G}' \mapsto g_*(\mathscr{G}')$ is exact and faithful, and by hypothesis the same is true of the functor $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} = \mathscr{F} \otimes_{\mathscr{O}_X} f^*(\mathscr{G})$ from quasi-coherent $\mathscr{O}_Y$-Modules to quasi-coherent $\mathscr{O}_X$-Modules. Their composite is therefore exact and faithful, which completes the proof of (2.2.1) and (2.2.2). ☐

Corollary (2.2.3). — *Let $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$ be two affine schemes, $f : X \to Y$ a morphism, $\mathscr{F} = \widetilde{M}$ a quasi-coherent $\mathscr{O}_X$-Module. For $\mathscr{F}$ to satisfy the equivalent conditions of* (2.2.1) *(or* (2.2.2)*), it is necessary and sufficient that the A-module M be faithfully flat.*

Proof. Indeed, condition *c)* of (2.2.2) means that the functor $N \mapsto M \otimes_A N$ from the category of A-modules to that of B-modules is exact and faithful, and the conclusion follows from ($\mathbf{0_I}$, 6.4.1). ☐

Definition (2.2.4). — When the equivalent conditions of (2.2.1) are satisfied, we say that the quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$ is *faithfully flat relative to f* (or to Y).

We note that this notion is local over Y, but *not over* X; in particular we can have $\mathscr{F}_x = 0$ for certain $x \in X$, in other words $\operatorname{Supp}(\mathscr{F})$ is not necessarily equal to X. However, it follows immediately from criterion (2.2.1)*(b)* that for every $y \in Y$, there exists at least one $x \in f^{-1}(y)$ such that $(\mathscr{F}_y)_x = \mathscr{F}_x \otimes_{\mathscr{O}_y} k(y) \neq 0$, and *a fortiori* $\mathscr{F}_x \neq 0$; in other terms: **IV-2 | 11**

Corollary (2.2.5). — *If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-Module, faithfully flat relative to f, then*

$$f(\operatorname{Supp}(\mathscr{F})) = Y.$$

In particular, f is a surjective morphism.

This result admits a partial converse:

Corollary (2.2.6). — *Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module of finite type. For $\mathscr{F}$ to be faithfully flat relative to f, it is necessary and sufficient that $\mathscr{F}$ be f-flat and that $f(\operatorname{Supp}(\mathscr{F})) = Y$.*

Proof. Indeed, by (**I**, 9.1.13) and (**I**, 3.6.1), we have $\operatorname{Supp}(\mathscr{F}_y) = f^{-1}(y) \cap \operatorname{Supp}(\mathscr{F})$, so in this case criterion (2.2.1)*(b)* is precisely the condition of the corollary. In particular, the $\mathscr{O}_X$-Module $\mathscr{O}_X$ is faithfully flat relative to f if and only if f is flat and surjective, that is to say if and only if the *morphism f is faithfully flat* ($\mathbf{0_I}$, 6.7.8). □

Let us spell out the properties intervening in definition (2.2.4):

Proposition (2.2.7). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module faithfully flat relative to f. Then, for a sequence $\mathscr{G}' \to \mathscr{G} \to \mathscr{G}''$ of quasi-coherent $\mathscr{O}_Y$-Modules to be exact, it is necessary and sufficient that the corresponding sequence $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}''$ be so. In particular, for a homomorphism $u : \mathscr{G} \to \mathscr{G}'$ of quasi-coherent $\mathscr{O}_Y$-Modules to be injective* (*resp. surjective, bijective, zero*)*, it is necessary and sufficient that $1_{\mathscr{F}} \otimes f^*(u) : \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ be so. For a quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{G}$ to be zero, it is necessary and sufficient that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} = 0$. For a quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{G}$, the map $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ from the set of quasi-coherent $\mathscr{O}_Y$-submodules of $\mathscr{G}$ to the set of quasi-coherent $\mathscr{O}_X$-submodules of $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is injective.*

Proof. To prove the last assertion, that is to say that for two quasi-coherent sub-$\mathscr{O}_Y$-Modules $\mathscr{G}'$, $\mathscr{G}''$ of $\mathscr{G}$, the relation $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}''$ implies $\mathscr{G}' = \mathscr{G}''$, we can reduce to the case where $\mathscr{G}' \subset \mathscr{G}''$ (by replacing $\mathscr{G}''$ by $\mathscr{G}' + \mathscr{G}''$), and it then suffices to apply the second assertion of the statement to the injection $u : \mathscr{G}' \to \mathscr{G}''$. □

Corollary (2.2.8). — *Let $f : X \to Y$ be a faithfully flat morphism, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Y$-Module. The canonical map*

$$\Gamma(Y, \mathscr{G}) \longrightarrow \Gamma(X, f^*(\mathscr{G})) \tag{2.2.8.1}$$

is injective.

Proof. Indeed, there are canonical identifications ($\mathbf{0_I}$, 5.1.1)

$$\Gamma(Y, \mathscr{G}) \simeq \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{O}_Y, \mathscr{G}),$$
$$\Gamma(X, f^*(\mathscr{G})) \simeq \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{O}_X, f^*(\mathscr{G})).$$

By virtue of (2.2.1) and (2.2.4), the hypothesis implies that the functor $\mathscr{G} \mapsto f^*(\mathscr{G})$ is exact and faithful in the category of quasi-coherent $\mathscr{O}_Y$-Modules, and thus a homomorphism $u : \mathscr{O}_Y \to \mathscr{G}$ is zero if and only if $f^*(u) : f^*(\mathscr{O}_Y) = \mathscr{O}_X \to f^*(\mathscr{G})$ is zero. □

Remark (2.2.9). — The assertions of (2.2.7) and (2.2.8) are still true when the $\mathscr{O}_Y$-Modules $\mathscr{G}', \mathscr{G}, \mathscr{G}''$ appearing there are arbitrary (not necessarily quasi-coherent). Indeed, for every $y \in Y$, there exists $x \in f^{-1}(y)$ such that $\mathscr{F}_x$ is a flat $\mathscr{O}_y$-module faithfully flat, and thus the functor $\mathscr{G}_y \mapsto \mathscr{G}_y \otimes_{\mathscr{O}_y} \mathscr{F}_x$ is **IV-2** | 12
faithful; since moreover for *every* $x \in f^{-1}(y)$ the functor $\mathscr{G}_y \mapsto \mathscr{G}_y \otimes_{\mathscr{O}_y} \mathscr{F}_x$ is exact, we immediately deduce the conclusion.

Proposition (2.2.10). — *Let $f : X \to Y$, $g : Y' \to Y$, $h : Y' \to Z$ be three morphisms of preschemes, $X' = X \times_Y Y'$, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module faithfully flat relative to Y, $\mathscr{G}'$ a quasi-coherent $\mathscr{O}_{Y'}$-Module. For $\mathscr{G}'$ to be flat* (*resp. faithfully flat*) *relative to Z, it is necessary and sufficient that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ be a flat* (*resp. faithfully flat*) *$\mathscr{O}_{X'}$-Module relative to Z.*

Proof. We know already that if $\mathscr{G}'$ is Z-flat, so is $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ (2.1.5). Consider a base change $Z'' \to Z$ and set $X'' = X' \times_Z Z''$; if $\mathscr{G}'$ is faithfully flat relative to Z, the functor

$$\mathscr{H}'' \longmapsto \mathscr{H}'' \otimes_Z \mathscr{G}' \longrightarrow (\mathscr{H}'' \otimes_Z \mathscr{G}') \otimes_{\mathscr{O}_Y} \mathscr{F} = \mathscr{H}'' \otimes_Z (\mathscr{G}' \otimes_{\mathscr{O}_Y} \mathscr{F}) \tag{2.2.10.1}$$

from the category of $\mathscr{O}_{Z''}$-Modules to that of $\mathscr{O}_{X''}$-Modules is composed of two exact and faithful functors, so this functor is also exact and faithful. Conversely, if this composite functor is exact (resp. exact and faithful), the same is true of $\mathscr{H}'' \mapsto \mathscr{H}'' \otimes_Z \mathscr{G}'$, since the functor $\mathscr{M}' \mapsto \mathscr{M}' \otimes_{\mathscr{O}_Y} \mathscr{F}$ from quasi-coherent $\mathscr{O}_{Y'}$-Modules to quasi-coherent $\mathscr{O}_{X'}$-Modules is exact and faithful by hypothesis. □

Corollary (2.2.11). —

(i) *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. If $\mathscr{F}$ is faithfully flat relative to Y, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$ is faithfully flat relative to Y'.*

(ii) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module faithfully flat relative to Y. For g to be a faithfully flat morphism, it is necessary and sufficient that $\mathscr{F}$ be faithfully flat relative to Z.*

(iii) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Y$-Module. Suppose the morphism f is faithfully flat. For $\mathscr{G}$ to be flat* (*resp. faithfully flat*) *relative to Z, it is necessary and sufficient that $f^*(\mathscr{G})$ be flat* (*resp. faithfully flat*) *relative to Z.*

(iv) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, x a point of X. Suppose f is flat at the point x. For $g \circ f$ to be flat at the point x, it is necessary and sufficient that g be flat at the point $f(x)$.*

Proof. To prove (i), we apply (2.2.10), replacing Z by Y' and $\mathscr{G}'$ by $\mathscr{O}_{Y'}$. To prove (ii), we apply (2.2.10) taking $Y' \to Y$ the identity and replacing $\mathscr{G}'$ by $\mathscr{O}_Y$. To prove (iii), we apply (2.2.10), again taking $Y' \to Y$ to be the identity, and replacing $\mathscr{F}$ by $\mathscr{O}_X$ and $\mathscr{G}'$ by $\mathscr{G}$. Finally, (iv) is deduced from (ii) by replacing X by $\operatorname{Spec}(\mathscr{O}_x)$, $\mathscr{F}$ by $\mathscr{O}_X$, Y by $\operatorname{Spec}(\mathscr{O}_{f(x)})$ and Z by $\operatorname{Spec}(\mathscr{O}_{g(f(x))})$, and taking into account ($\mathbf{0_I}$, 6.6.2). ☐

Corollary (2.2.12). — *Let Y be an affine scheme, $f : X \to Y$ a quasi-compact morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. If $\mathscr{F}$ is faithfully flat relative to f, there exists an affine scheme X' and a surjective local isomorphism $g : X' \to X$ such that $g^*(\mathscr{F})$ is faithfully flat relative to $f \circ g$.* **IV-2** | 13

Proof. Indeed, X is quasi-compact, hence a finite union of affine open subsets X_i; it suffices to take for X' the affine scheme given by the disjoint sum of the preschemes induced on the open subsets X_i, and for $g : X' \to X$ the canonical morphism. It is clear that g is a surjective local isomorphism, hence faithfully flat, and the hypothesis implies that $g^*(\mathscr{F})$ is faithfully flat relative to $f \circ g$, by virtue of (2.2.11)(iii). ☐

Corollary (2.2.13). —

(i) *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. If f is a faithfully flat morphism, so is f'.*

(ii) *If $f : X \to X'$, $g : Y \to Y'$ are two faithfully flat S-morphisms,*

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is faithfully flat.

(iii) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms such that f is faithfully flat. For g to be a flat* (*resp. faithfully flat*) *morphism, it is necessary and sufficient that $g \circ f$ be a flat* (*resp. faithfully flat*) *morphism.*

Proposition (2.2.14). — *Let $f : X \to Y$ be a faithfully flat morphism and quasi-compact. If X is locally Noetherian, so is Y.*

Proof. The question being local on Y, we can suppose $Y = \operatorname{Spec}(A)$ affine; as f is quasi-compact, it follows from (2.2.12) that we can also restrict to the case where $X = \operatorname{Spec}(B)$ is affine. Then B is a Noetherian ring by hypothesis, and a faithfully flat A-module (2.2.3); so A is Noetherian ($\mathbf{0_I}$, 6.5.2). ☐

Proposition (2.2.15). — *Let $f : X \to Y$ be a faithfully flat morphism. If $\mathfrak{S}(X)$ and $\mathfrak{S}(Y)$ denote respectively the sets of sub-preschemes of X and of Y, the map $Z \mapsto f^{-1}(Z)$ of $\mathfrak{S}(Y)$ to $\mathfrak{S}(X)$ is injective.*

Proof. Since f is surjective, on the underlying sets of a sub-prescheme Z of Y we have $f(f^{-1}(Z)) = Z$. On the other hand, if U is an open subset of Y containing Z and in which Z is closed, then $f^{-1}(U)$ is open in X, $f^{-1}(Z)$ is closed in $f^{-1}(U)$, and the restriction $f^{-1}(U) \to U$ of f is a faithfully flat morphism. We can therefore restrict to considering only the *closed* sub-preschemes of Y. Now, if Z is a closed sub-prescheme of Y corresponding to a quasi-coherent Ideal $\mathscr{J}$ of $\mathscr{O}_Y$, we know that $f^{-1}(Z)$ corresponds to the quasi-coherent Ideal $f^*(\mathscr{J})\mathscr{O}_X$ of $\mathscr{O}_X$ (**I**, 4.4.5), and since f is flat, $f^*(\mathscr{J})\mathscr{O}_X$ identifies with $f^*(\mathscr{J})$. But the map $\mathscr{J} \mapsto f^*(\mathscr{J})$ from the set of quasi-coherent Ideals of $\mathscr{O}_Y$ to the set of quasi-coherent Ideals of $\mathscr{O}_X$ is injective (2.2.7), whence the conclusion. ☐

Corollary (2.2.16). — *Let X, Y be two S-preschemes; if $S' \to S$ is a faithfully flat morphism, the map $f \mapsto f_{(S')}$ from $\mathrm{Hom}_S(X,Y)$ to $\mathrm{Hom}_{S'}(X_{(S')}, Y_{(S')})$ is injective.*

Proof. We have $X_{(S')} \times_{S'} Y_{(S')} = (X \times_S Y)_{(S')}$ (**I**, 3.3.10), so the morphism projection $X_{(S')} \times_{S'} Y_{(S')} \to X \times_S Y$ is faithfully flat (2.2.13). The elements of $\mathrm{Hom}_S(X,Y)$ correspond bijectively to the sub-preschemes of $X \times_S Y$ that are their graphs (**I**, 5.3.11); if Z_f is the graph of $f \in \mathrm{Hom}_S(X,Y)$, then $Z_{f_{(S')}} = g^{-1}(Z_f)$ (**I**, 5.3.12). It suffices therefore to apply (2.2.15). □

Proposition (2.2.17). — *Let A be a ring, B an A-algebra such that B is a faithfully flat A-module and of finite presentation. Then the structural homomorphism $\varphi : A \to B$ is an isomorphism of the A-module A onto a direct summand of the A-module B. If A is a local ring, B is a free A-module and there exists a basis of this module containing the identity element of B.*

Proof. By virtue of Bourbaki, *Alg. comm.*, chap. II, § 3, nº 3, prop. 12, it suffices to prove the proposition when A is a local ring; we know then (*loc. cit.*, nº 2, cor. 2 of prop. 5) that B is an A-module free of finite type, and the conclusion follows from *loc. cit.*, prop. 5. □ **IV-2** | 14

2.3. Topological properties of flat morphisms

Lemma (2.3.1). — *Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism, $g : Y' \to Y$ a flat morphism; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Then, for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, the canonical homomorphism*

$$g^*(f_*(\mathscr{F})) \longrightarrow f'_*(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}) \tag{2.3.1.1}$$

is bijective.

Proof. This is a special case of (**III**, 1.4.15) (improved in (1.7.21)). □

Proposition (2.3.2). — *Let S be a prescheme, $f : X \to Y$ a quasi-compact and quasi-separated S-morphism, and let Z be the sub-prescheme of Y which is the closed image of X under f* ((**I**, 9.5.3) *and (1.7.8)*). *Let $j : Z \to Y$ be the canonical injection, so that $f = j \circ g$, where $g : X \to Z$ is a morphism* (loc. cit.). *Let $h : S' \to S$ be a flat morphism, and set $f' = f_{(S')} : X_{(S')} \to Y_{(S')}$. Then $j' = j_{(S')} : Z_{(S')} \to Y_{(S')}$ identifies with the canonical injection of the sub-prescheme of $Y_{(S')}$ which is the closed image of $X_{(S')}$ under f'.*

Proof. Since the morphism $Y_{(S')} \to Y$ is flat (2.1.4), we may reduce to the case $S = Y$ (**I**, 3.3.11). If $f = (\psi, \theta)$, then Z is the closed sub-prescheme of S defined by the quasi-coherent Ideal $\mathscr{J} \subset \mathscr{O}_S$ which is the kernel of

$$\theta : \mathscr{O}_S \longrightarrow f_*(\mathscr{O}_X)$$

(**I**, 9.5.2). Since h is flat, the quasi-coherent Ideal $\mathscr{J}\mathscr{O}_{S'}$ identifies with the kernel of

$$h^*(\theta) : \mathscr{O}_{S'} \longrightarrow h^*(f_*(\mathscr{O}_X)).$$

If $f' = (\psi', \theta')$, then $\theta' : \mathscr{O}_{S'} \to f'_*(\mathscr{O}_{X'})$ is the composite of $h^*(\theta)$ with the canonical homomorphism

$$h^*(f_*(\mathscr{O}_X)) \longrightarrow f'_*(\mathscr{O}_{X'})$$

of (2.3.1.1). The conclusion follows from (2.3.1) and (**I**, 9.5.2), since f is quasi-compact and quasi-separated (1.1.2 and 1.2.2). □

(2.3.3). We will say that a morphism $f : X \to Y$ is *quasi-flat* if there exists a quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$ *of finite type* which is f-flat and whose support is equal to X. We will say that f is *quasi-faithfully flat* if it is quasi-flat and surjective. Every flat (resp. faithfully flat) morphism is quasi-flat (resp. quasi-faithfully flat), for then $\mathscr{F} = \mathscr{O}_X$ satisfies the preceding conditions.

It follows immediately from (2.1.4) and from (**I**, 9.1.13), that if f is quasi-flat, then, for every morphism $Y' \to Y$, the morphism $f_{(Y')} : X \times_Y Y' \to Y'$ is quasi-flat. Similarly (taking into account (**I**, 3.5.2)), if f is quasi-faithfully flat, so is $f_{(Y')}$.

Proposition (2.3.4). — *Let $f : X \to Y$ be a quasi-flat morphism* (2.3.3). *Then f possesses the following properties* (*which are equivalent by virtue of (1.10.4)*):

(i) *For every $x \in X$ and every generization y' of $y = f(x)$, there exists a generization x' of x such that $f(x') = y'$.*

(ii) *For every $x \in X$, the image under f of* $\operatorname{Spec}(\mathscr{O}_{X,x})$ *is* $\operatorname{Spec}(\mathscr{O}_{Y,y})$. **IV-2** | 15
(iii) *For every closed irreducible part Y' of Y, every irreducible component of $f^{-1}(Y')$ dominates Y'.*

Proof. It suffices, for example, to prove (ii). By hypothesis, there is an $\mathscr{O}_X$-Module quasi-coherent of finite type $\mathscr{F}$, which is f-flat and such that $\operatorname{Supp}(\mathscr{F}) = X$. For every $x \in X$, $\mathscr{F}_x$ is then an $\mathscr{O}_x$-module of finite type, nonzero, and moreover $\mathscr{F}_x$ is an $\mathscr{O}_y$-module *flat*, for the homomorphism $\rho : \mathscr{O}_y \to \mathscr{O}_x$. As the latter is local and $\mathscr{F}_x \neq 0$, the Nakayama lemma shows that $\mathscr{F}_x \otimes_{\mathscr{O}_y} k(y) \neq 0$, so $\mathscr{F}_x$ is a faithfully flat $\mathscr{O}_y$-module (**0**, 6.4.1). It follows that for every prime ideal $\mathfrak{q}$ of $\mathscr{O}_y$, there exists a prime ideal $\mathfrak{p}$ of $\mathscr{O}_x$ such that $\mathfrak{q} = \rho^{-1}(\mathfrak{p})$ (**0**, 6.5.1), which proves (ii). ☐

Corollary (2.3.5). — *Let $f : X \to Y$ be a morphism satisfying the equivalent conditions (i), (ii), (iii) of* (2.3.4) (*in particular a quasi-flat or open morphism (1.10.4)*).

(i) *Let Z, Z' be two closed irreducible subsets of Y such that $Z \subset Z'$, and let T be an irreducible component of $f^{-1}(Z)$; then there exists an irreducible component T' of $f^{-1}(Z')$ containing T (and dominating Z').*
(ii) *For every irreducible component T of X, $\overline{f(\mathrm{T})}$ is an irreducible component of Y.*
(iii) *Suppose Y irreducible, and, if y is its generic point, suppose that $f^{-1}(y)$ is irreducible. Then X is irreducible.*

Proof. (i) It suffices to apply (2.3.4), (i) by taking for x the generic point of T ($y = f(x)$ is then the generic point of Z) and for y' the generic point of Z'.

(ii) It is clear that $\overline{f(\mathrm{T})} = Z$ is irreducible, and by virtue of (i), Z cannot be strictly contained in a closed irreducible part of Y.

(iii) By virtue of (ii), every irreducible component of X dominates Y, and hence meets $f^{-1}(y)$, where y is the generic point of Y. The conclusion then follows from (**0**, 2.1.8). ☐

Proposition (2.3.6). — *Let Y be a prescheme whose set of irreducible components is locally finite* (*which is the case if the underlying space is locally Noetherian (cf.* (**I**, 6.1.9))).

(i) *Every closed subset W of Y stable by generization* (**0**, 2.1.2) *is open. In particular, every connected component of Y is open.*
(ii) *Let $f : X \to Y$ be a closed morphism satisfying in addition the equivalent conditions (i), (ii), (iii) of* (2.3.4) (*which will be the case if f is quasi-flat or open*). *Then the image under f of every connected component C of X is a connected component of Y.*
(iii) *Let $f : X \to Y$ be a morphism satisfying the equivalent conditions (i), (ii), (iii) of* (2.3.4), *and such that moreover for every $y \in Y$, the set $f^{-1}(y)$ is finite* (*which will be the case if f is quasi-finite or radicial*). *Then the set of irreducible components of X is locally finite.*

Proof. (i) If $y \in W$, the generic points η_i ($1 \leqslant i \leqslant m$) of the irreducible components Y_i of Y containing y belong to W by hypothesis. Since W is closed, these components themselves are contained in W; and since there is by hypothesis a neighbourhood U of y which is the union of the $U \cap Y_i$ (**0**, 2.1.6), we have $U \subset W$, so W is open. Since for every $y \in Y$, $\overline{\{y\}}$ is connected, a connected component of Y is stable by generization, so the second assertion follows immediately from the first. **IV-2** | 16

(ii) Since C is closed in X, $f(C)$ is closed in Y by hypothesis. Moreover, since C is stable by generization, the hypothesis on f implies that $f(C)$ is stable by generization; so, by virtue of (i), $f(C)$ is open and closed, and since it is connected, it is a connected component of Y.

(iii) Let $x \in X$. By hypothesis, there is an open neighbourhood U of $y = f(x)$ which meets only finitely many irreducible components Y_i of Y ($1 \leqslant i \leqslant n$); let y_i be the generic point of Y_i. For every irreducible component Z of X meeting $f^{-1}(U)$, the generic point z of Z is necessarily in one of the sets $f^{-1}(y_i)$ (2.3.4). Since each of these sets is finite by hypothesis, this proves our assertion. ☐

Proposition (2.3.7). — *Let $f : X \to Y$ be a morphism, $g : Y' \to Y$ a quasi-flat morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$.*

(i) *If f is quasi-compact and dominant, so is f'.*
(ii) *If every irreducible component of X dominates an irreducible component of Y, then every irreducible component of X' dominates an irreducible component of Y'.*

Proof. Denote by $g' : X' \to X$ the canonical projection, which is a quasi-flat morphism (2.3.3).

(i) We know already (1.1.2) that f' is quasi-compact; moreover, if y' is a maximal point of Y', $y = g(y')$ is a maximal point of Y, as follows from (2.3.4), (iii). By hypothesis (1.1.5), there exists $x \in X$ such that $f(x) = y$; so (**I**, 3.4.7), there exists $x' \in X'$ such that $f'(x') = y'$.

(ii) Let x' be a maximal point of X', and let $x = g'(x')$; it follows from (2.3.4), (ii) that x is a maximal point of X, and by hypothesis $y = f(x)$ is a maximal point of Y. Set $y' = f'(x')$, so that $g(y') = y$; it remains to show that y' is a maximal point of Y'. By virtue of (**I**, 5.1.7) and (2.3.3), we may reduce to the case where X and Y are reduced, so that $\mathscr{O}_x$ and $\mathscr{O}_y$ are fields. By virtue of (**I**, 3.6.5), applied twice, and (**I**, 2.4.4), we may further reduce to the case where $Y = \operatorname{Spec}(\mathscr{O}_y)$ and $Y' = \operatorname{Spec}(\mathscr{O}_{y'})$. Then f is flat because $\mathscr{O}_y$ is a field (2.1.2), and the same is true of f' (2.1.4); hence (2.3.4), (ii) shows that $y' = f'(x')$ is a maximal point of Y'. □

Corollary (2.3.8). — (Zariski). *Let A, B be two Noetherian local rings, $\mathfrak{m}$, $\mathfrak{n}$ their maximal ideals respectively, $\varphi : A \to B$ a local homomorphism. Suppose the following hypotheses satisfied:*

1° *B is an A-algebra essentially of finite type (1.3.8).*

2° *The $\mathfrak{m}$-adic completion $\widehat{A}$ of A for the $\mathfrak{m}$-adic topology is integral.*

3° *φ is injective.*

Then the $\mathfrak{m}$-adic topology of A is induced by the $\mathfrak{n}$-adic topology of B.

Proof. Set $B' = B \otimes_A \widehat{A}$; by hypothesis 1°, B' is of the form $S^{-1}(C \otimes_A \widehat{A})$, where C is an A-algebra **IV-2** | 17
of finite type and S is a multiplicative subset of C; hence B' is Noetherian. As A identifies with a subring of $\widehat{A}$ (**0**, 7.3.5), A is integral by hypothesis 2°. Hypothesis 3° implies that there exists a prime ideal $\mathfrak{q}$ of B inducing the zero ideal of A (**0**, 1.5.8); consequently, the local homomorphism $A \to B/\mathfrak{q}$ is injective. We can therefore restrict to proving the conclusion of (2.3.8) under the added hypothesis that B is a local ring which is *integral*. Apply (2.3.7), (ii) to $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, $Y' = \operatorname{Spec}(\widehat{A})$, and $X' = \operatorname{Spec}(B')$. Since $Y' \to Y$ is flat and the integral scheme X dominates the integral scheme Y, every irreducible component of X' dominates Y'. If y, x, and y' are the closed points of Y, X, and Y' respectively, there is a point $x' \in X'$ (in fact unique) lying above x and y' (**I**, 3.4.9), and $\operatorname{Spec}(\mathscr{O}_{x'})$ therefore dominates $\operatorname{Spec}(\mathscr{O}_{y'})$. Consequently, there is a commutative diagram of local homomorphisms of Noetherian local rings

$$\begin{tikzcd}
B = \mathscr{O}_x \arrow[r] & \mathscr{O}_{x'} \\
A = \mathscr{O}_y \arrow[u, "\varphi"] \arrow[r, "u"'] & \mathscr{O}_{y'} = \widehat{A} \arrow[u, "v"']
\end{tikzcd}$$

in which u and v are injective (**I**, 1.2.7). Identify A and $\widehat{A}$ with subrings of $\mathscr{O}_{x'}$, and let $\mathfrak{r}$ be the maximal ideal of $\mathscr{O}_{x'}$. Then $\bigcap_k(\mathfrak{r}^k \cap \widehat{A}) = 0$ (**0**, 7.3.5); since $\widehat{A}$ is complete and these ideals are open in $\widehat{A}$, it follows (Bourbaki, *Alg. comm.*, chap. III, § 2, nº 7, prop. 8) that the topology of $\widehat{A}$ is induced by the $\mathfrak{r}$-preadic topology of $\mathscr{O}_{x'}$. *A fortiori*, the same is true of the topology of A (**0**, 7.3.5). Moreover, $\mathfrak{n}^k \cap A \subset \mathfrak{r}^k \cap A$, so the $\mathfrak{n}$-preadic topology of B induces on A a topology finer than the $\mathfrak{m}$-preadic topology; but $\mathfrak{m}^k \subset \mathfrak{n}^k \cap A$, so these two topologies are identical. □

Remark (2.3.9). — We will see further on (7.8.3, (vii)) that for the Noetherian local rings A most commonly used in Algebraic Geometry, the hypothesis that A is integral and integrally closed implies that the same is true of $\widehat{A}$. This is why, in Algebraic Geometry over a base field, one generally states (2.3.8) under the hypothesis that A is *integral and integrally closed*.

Theorem (2.3.10). — *Let $f : X \to Y$ be a quasi-flat morphism (2.3.3). Then, for every pro-constructible part (1.9.4) Z of Y, we have $\overline{f^{-1}(Z)} = f^{-1}(\overline{Z})$.*

Proof. Since f is continuous, we have $\overline{f^{-1}(Z)} \subset f^{-1}(\overline{Z})$ and it amounts to proving that for every $x \in X$ such that $f(x) \in \overline{Z}$, x is adherent to $f^{-1}(Z)$; it is clear that the question is local on Y, so we can suppose Y affine. By virtue of the hypothesis, there exists an affine scheme Y' and a morphism $g : Y' \to Y$ such that $g(Y') = Z$ (1.9.5, (ix)). Let Y_1 be the closed image of Y' under g (**I**, 9.5.3), and let X_1 be the closed sub-prescheme $f^{-1}(Y_1)$ of X; if $f_1 : X_1 \to Y_1$ is the restriction of f to X_1, we know that f_1 is quasi-flat (2.3.3); we can therefore replace X, Y by X_1, Y_1 respectively, in other words suppose that g is *dominant*. Set then $X' = X \times_Y Y'$, and let f' and g' be the projections of X' into Y' and X respectively, so that we have the commutative diagram

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "g'"'] \arrow[d, "f'"] \\ Y & Y' \arrow[l, "g"] \end{tikzcd}$$

Since f is quasi-flat and g is quasi-compact and dominant, it follows from (2.3.7), (i) (with the roles of f and g interchanged) that g' is dominant, which proves the theorem. □

Corollary (2.3.11). — *Let f be a quasi-flat and quasi-compact morphism, F a closed part of X such that $F = f^{-1}(f(F))$; then $F = f^{-1}(\overline{f(F)})$.*

Proof. Let Y' be the reduced sub-prescheme of X having F as underlying space (**I**, 5.2.1), and let $j : Y' \to X$ be the canonical injection. Then $f \circ j$ is quasi-compact (1.1.2), so $Z = f(F)$ is pro-constructible in Y (1.9.5, (vii)), and the corollary follows from the fact that F is closed. **IV-2** | 18

We can further write the result of (2.3.11) in the form $F = f^{-1}(f(X) \cap \overline{f(F)})$. In other words, the closed subsets of the subspace $f(X)$ of Y are the parts $Z \subset f(X)$ such that $f^{-1}(Z)$ is closed in X; this also means that *the topology induced by Y on $f(X)$ is the quotient of that of X by the equivalence relation defined by f*. In particular: □

Corollary (2.3.12). — *Let $f : X \to Y$ be a quasi-faithfully flat morphism* (2.3.3) *and quasi-compact. Then the topology of Y is the quotient of that of X by the equivalence relation defined by f, in other words, for $Z \subset Y$ to be open* (*resp. closed*) *in Y, it is necessary and sufficient that $f^{-1}(Z)$ be open* (*resp. closed*) *in X.*

Proof. Indeed, we then have $f(X) = Y$. □

Corollary (2.3.13). — *Let X, Y be two S-preschemes, $f : X \to Y$ an S-morphism faithfully flat and quasi-compact. For Y to be separated over S, it is necessary and sufficient that the canonical immersion $j : X \times_Y X \to X \times_S X$* (**I**, 5.3.10) *be* closed.

Proof. Let us note for this that we have the commutative diagram (**I**, 5.3.5)

$$\begin{tikzcd} X \times_Y X \arrow[r, "j"] \arrow[d, "\pi"'] & X \times_S X \arrow[d, "f \times_S f"] \\ Y \arrow[r, "\Delta_Y"] & Y \times_S Y \end{tikzcd}$$

identifying $X \times_Y X$ with the product of the $(Y \times_S Y)$-preschemes Y and $X \times_S X$. Since f is surjective, so are π and $f \times_S f$; hence the inverse image of the diagonal $\Delta_Y(Y)$ under $f \times_S f$ is the image $j(X \times_Y X)$ (**I**, 3.4.8). Since $f \times_S f$ is faithfully flat and quasi-compact (1.1.2 and 2.2.13), it suffices to apply (2.3.12) to this morphism. □

Corollary (2.3.14). — *Let $f : X \to Y$ be a quasi-faithfully flat and quasi-compact morphism, and let Z be a part of Y. For Z to be a locally closed part pro-constructible of Y, it is necessary and sufficient that $f^{-1}(Z)$ be a locally closed part pro-constructible of X.*

Proof. We already know (1.9.12) that, for Z to be a pro-constructible part of Y, it is necessary and sufficient that $f^{-1}(Z)$ be a pro-constructible part of X. The condition is evidently necessary. To prove that it is sufficient, consider the closed reduced sub-prescheme Y_1 of Y having $\overline{Z}$ as underlying space and let X_1 be its inverse image, so that $f^{-1}(\overline{Z}) = \overline{f^{-1}(Z)}$ by virtue of (2.3.10). As $f^{-1}(Z)$ is locally closed in X, it is open in $\overline{f^{-1}(Z)}$, so in X_1; now, the morphism $f_1 : X_1 \to Y_1$ deduced from f by restriction is quasi-faithfully flat and quasi-compact (1.1.2 and 2.3.3); we deduce from (2.3.12) that Z is open in $\overline{Z}$, which shows that Z is locally closed in Y. □

Remark (2.3.15). — It does not suffice, in (2.3.12), to suppose only that f is *faithfully flat*. For example, take for Y a reduced irreducible algebraic curve (**II**, 7.4.2), and let X be the prescheme sum **IV-2** | 19
$\coprod_{y \in Y} \operatorname{Spec}(\mathscr{O}_y)$ (**I**, 3.1). Let $f : X \to Y$ be the canonical morphism, which is faithfully flat (**I**, 2.4.2). If η denotes the generic point of Y, then $Z = \{\eta\}$ is not open in Y (**II**, 7.4.3), whereas $f^{-1}(Z) = \operatorname{Spec}(\mathscr{O}_\eta)$ is open in X, since $\mathscr{O}_\eta$ is a field.

2.4. Universally open morphisms and flat morphisms

(2.4.1). We have already defined (**II**, 5.4.9) the notion of *universally closed* morphism; in the same manner, we give the following definitions:

Definition (2.4.2). — We say that a morphism of preschemes $f : X \to Y$ is *universally open* (resp. *universally bicontinuous*, resp. a *universal homeomorphism*) if, for every morphism $g : Y' \to Y$, the morphism $f_{(Y')} : X \times_Y Y' \to Y'$ is open (resp. a homeomorphism onto its image, resp. a homeomorphism onto Y').

We will see further on (14.3.2) that when Y is locally Noetherian, the definition of universally open morphism given here is equivalent to the definition (**III**, 4.3.9) for morphisms of finite type; the reader can verify that we do not use the latter definition before § 14.

Proposition (2.4.3). —

(i) An immersion (resp. an open immersion, resp. a closed immersion) is universally bicontinuous (resp. universally open, resp. universally closed).

(ii) The composite of two universally open morphisms (resp. universally closed morphisms, resp. universally bicontinuous morphisms, resp. universal homeomorphisms) has the same property.

(iii) If $f : X \to Y$ is a universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ after every base change $S' \to S$.

(iv) If the S-morphisms $f : X \to X'$ and $g : Y \to Y'$ are both universally open (resp. universally closed, resp. universally bicontinuous, resp. universal homeomorphisms), then so is

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'.$$

(v) Let $f : X \to Y$ and $g : Y \to Z$ be morphisms, with f surjective. If $g \circ f$ is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), then so is g.

(vi) A morphism f is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism) if and only if f_{red} has the same property.

(vii) Let (U_α) be an open covering of Y. A morphism $f : X \to Y$ is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism) if and only if, for every α, its restriction $f^{-1}(U_\alpha) \to U_\alpha$ has the same property.

Proof. Assertion (i) follows from (**I**, 4.3.2). Assertion (ii) follows immediately from the definitions. The same is true of (iii), after reducing to the case $Y = S$ and $Y' = S'$, as permitted by (**I**, 3.3.11); assertion (iv) follows from (ii), (iii), and (**I**, 3.5.1).

To prove (v), note that after every morphism $Z' \to Z$, the base change

$$f_{(Z')} : X_{(Z')} \longrightarrow Y_{(Z')}$$

is surjective (**I**, 3.5.2). It is therefore enough to use the following purely topological fact: if f is surjective and $g \circ f$ is open (resp. closed, resp. a homeomorphism onto its image, resp. a bijective homeomorphism), then g has the corresponding property. In the open and closed cases this is Bourbaki, *Top. gén.*, chap. I[er], 3[e] éd., § 5, n[o] 1, prop. 1. In the other two cases we may replace Z by $g(Y)$ and thus suppose that $g(f(X)) = g(Y) = Z$. Then $g \circ f$ is a homeomorphism of X onto Z; since f is surjective, g is bijective, and since the preceding case shows that g is open, g is a homeomorphism of Y onto Z.

To prove (vi), note that a morphism g is open (resp. closed, resp. a homeomorphism onto its image, resp. a bijective homeomorphism) if and only if g_{red} has the same property. On the other hand, for every morphism $Y'' \to Y$, we have **IV-2** | 20

$$(X_{\mathrm{red}} \times_{Y_{\mathrm{red}}} Y''_{\mathrm{red}})_{\mathrm{red}} = (X \times_Y Y'')_{\mathrm{red}}$$

by (**I**, 5.1.8); hence the preceding remark shows that if f_{red} is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), then so is f. The converse is proved in the same manner, noting here that for every morphism $Y'' \to Y_{\mathrm{red}}$, we have $(X_{\mathrm{red}} \times_{Y_{\mathrm{red}}} Y'')_{\mathrm{red}} = (X \times_Y Y'')_{\mathrm{red}}$ (**I**, 5.1.8).

Finally, the necessity of condition (vii) follows immediately from (iii). Conversely, suppose the condition satisfied, and let $g : Y' \to Y$ be a morphism; then the $g^{-1}(U_\alpha) = U'_\alpha$ form an open covering of Y' and if one notes f_α the restriction $f^{-1}(U_\alpha) \to U_\alpha$ of f, and f' the morphism $f_{(Y')}$, the restriction $f'^{-1}(U'_\alpha) \to U'_\alpha$ is nothing other than $(f_\alpha)_{(U'_\alpha)}$. We can therefore restrict to proving that f' is open (resp. closed, resp. a homeomorphism onto its image, resp. a homeomorphism onto Y'), which is immediate. □

Proposition (2.4.4). — *A universally bicontinuous morphism $f : X \to Y$ is radicial* (*and hence separated* (**I**, 7.7.1)).

Proof. Indeed, f is universally injective by hypothesis (**I**, 3.5.11). □

Proposition (2.4.5). —

(i) *An integral, surjective, and radicial morphism $f : X \to Y$ is a universal homeomorphism.*

(ii) *Conversely, suppose that Y is locally Noetherian. If a morphism of finite type $f : X \to Y$ is a universal homeomorphism, then it is finite, surjective, and radicial.*

Proof. (i) The three properties of f are preserved by base change (**I**, 3.5.2), (**I**, 3.5.7), and (**II**, 6.1.5). Since an integral morphism is closed (**II**, 6.1.10), f is a homeomorphism of X onto Y.

(ii) The morphism f is of finite type and universally closed by hypothesis, and it is separated by (2.4.4); hence it is proper (**II**, 5.4.1). For every $y \in Y$, the fiber $f^{-1}(y)$ consists of a single point, so (**III**, 4.4.2) implies that f is finite. It is plainly surjective, and it is radicial because it is universally injective (**I**, 3.5.11). □

Theorem (2.4.6). — *Let $f : X \to Y$ be a quasi-flat morphism* (2.3.3) *and locally of finite presentation. Then f is universally open. In particular, a flat morphism locally of finite presentation is universally open.*

Proof. We know that for every base change $Y' \to Y$, $f_{(Y')}$ is quasi-flat (2.3.3) and locally of finite presentation (1.4.3, iii). It suffices therefore to prove that f is an open morphism. But this follows IV-2 | 21
from the criterion (1.10.4) for morphisms locally of finite presentation, conditions *b*), *b'*), and *c*) of (1.10.4) being nothing other than conditions (i), (ii), (iii) of (2.3.4). □

Corollary (2.4.7). — *For every prescheme Y, the structural morphism*

$$\mathbf{V}^n_Y \longrightarrow Y, \qquad \mathbf{V}^n_Y = Y \times_{\mathrm{Spec}(\mathbf{Z})} \mathrm{Spec}\big(\mathbf{Z}[T_1, \ldots, T_n]\big),$$

where $\mathbf{V}^n_Y$ is also denoted by $Y[T_1, \ldots, T_n]$, is universally open.

Proof. Indeed, for $Y = \mathrm{Spec}(A)$, $Y[T_1, \ldots, T_n] = \mathrm{Spec}(A[T_1, \ldots, T_n])$, and $A[T_1, \ldots, T_n]$ is a free A-algebra of finite presentation. □

Remark (2.4.8). —

(i) A faithfully flat and quasi-compact morphism $f : X \to Y$ is *not necessarily open*, even when X and Y are Noetherian. For example, let Y be a reduced irreducible algebraic curve (**II**, 7.4.2) with generic point y, let $X = Y \coprod \mathrm{Spec}(k(y))$, and let f be the canonical morphism. It is clear that f is flat and surjective, hence faithfully flat, and it is quasi-compact. But the image under f of the open subset $\mathrm{Spec}(k(y))$ of X is $\{y\}$, which is not open in Y (**II**, 7.4.3).

(ii) For every prescheme X, the canonical morphism $f : X_{\mathrm{red}} \to X$ is a closed immersion and a *universal homeomorphism* by (2.4.3), (vi). But when X is locally Noetherian, f is flat only if X is reduced, in which case $f = 1_X$ (2.2.17).

Proposition (2.4.9). — *Let Y be a discrete prescheme. Then every morphism $f : X \to Y$ is universally open.*

Proof. The question is local on Y (2.4.3), (vii), so we may assume that the underlying space of Y consists of a single point. Replacing f by f_{red} (2.4.3), (vi), we may further assume that Y is the spectrum of a *field* k. For every base change $Y' \to Y$, the inverse images in $X' = X \times_Y Y'$ of the affine open subsets of X cover X', so we may also suppose that $X = \mathrm{Spec}(B)$ is affine, with B a k-algebra. It suffices then to prove that for every k-algebra A', if we set $B' = B \otimes_k A'$, the image of every open subset of the form $U = D(t)$ ($t \in B'$) is open in $\mathrm{Spec}(A')$. Now B is the filtered direct limit of the increasing family (B_α) of its sub-k-algebras *of finite type*. Since filtered direct limits commute with tensor products, B' is the direct limit of the A'-algebras $B_\alpha \otimes_k A' = B'_\alpha$. There exists α such that t is the image in B' of an element $t_\alpha \in B'_\alpha$, so $D(t)$ is the inverse image under the canonical morphism $u_\alpha : \mathrm{Spec}(B') \to \mathrm{Spec}(B'_\alpha)$ of the open subset $U_\alpha = D(t_\alpha)$ (**I**, 1.2.2.2). But since k is a field, B_α is a k-algebra of finite presentation and a k-module flat, hence B'_α is an A'-algebra of finite presentation and a flat A'-module; we thus conclude from (2.4.6) that the image of U_α by $f'_\alpha : \mathrm{Spec}(B'_\alpha) \to \mathrm{Spec}(A')$ is *open* in $\mathrm{Spec}(A')$. It remains to prove that $f'(U) = f'_\alpha(U_\alpha)$. The inclusion $f'(U) \subset f'_\alpha(U_\alpha)$ is clear, so it remains to show that, for every point $y' \in f'_\alpha(U_\alpha)$, the intersection $V = U \cap f'^{-1}(y')$ is nonempty. Now $V = u_\alpha^{-1}(V_\alpha)$, where $V_\alpha = U_\alpha \cap f'^{-1}_\alpha(y')$; in other words, V_α is a non-empty open subset of the prescheme $\mathrm{Spec}(B'_\alpha \otimes_{A'} k(y')) = \mathrm{Spec}(B_\alpha \otimes_k k(y'))$ and V is its inverse image by the morphism $v : \mathrm{Spec}(B \otimes_k k(y')) \to \mathrm{Spec}(B_\alpha \otimes_k k(y'))$. Since B_α is a subalgebra of B and $k(y')$ is flat over the field k, the homomorphism $B_\alpha \otimes_k k(y') \to B \otimes_k k(y')$ is *injective*. Thus v is dominant (**I**, 1.2.7), which shows that V is nonempty. □

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Corollary (2.4.10). — *Let k be a field, X, Y two k-preschemes; then the morphism projection $X \times_k Y \to X$ is universally open. In particular, for every extension K of k, the morphism projection $X_{(K)} \to X$ is universally open.*

Proof. It suffices to apply (2.4.9) to the structural morphism $Y \to \mathrm{Spec}(k)$. □

Remark (2.4.11). — If $f : X \to Y$ is an *open* morphism, then for every subset E of Y we have $f^{-1}(\overline{E}) = \overline{f^{-1}(E)}$ (Bourbaki, *Top. gén.*, chap. I, 3e éd., § 5, nº 4, prop. 7). This remark applies for example when f is a morphism *flat* locally of finite presentation (2.4.6), or a morphism projection $X \times_k Y \to X$ where X, Y are preschemes over a field k (2.4.10), and generalises then (2.3.10).

2.5. Permanence of properties of Modules by faithfully flat descent

Proposition (2.5.1). — *Let $f : X \to Y$ be a morphism, let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module, and let $g : Y' \to Y$ be faithfully flat. Set*

$$X' = X \times_Y Y', \qquad f' = f_{(Y')} : X' \longrightarrow Y',$$

let $g' : X' \to X$ be the canonical projection, and set $\mathscr{F}' = g'^*(\mathscr{F})$. Then $\mathscr{F}$ is flat (*resp. faithfully flat*) *relative to f if and only if $\mathscr{F}'$ is flat* (*resp. faithfully flat*) *relative to f'.*

Proof. Apply (2.2.10), replacing its X, Y', Z, $\mathscr{F}$, and $\mathscr{G}'$ respectively by Y', X, Y, $\mathscr{O}_{Y'}$, and $\mathscr{F}$. This identifies the required flatness (resp. faithful flatness) condition after base change with the corresponding condition relative to $g \circ f'$. On the other hand, g' is faithfully flat (2.2.13) and

$$g \circ f' = f \circ g'.$$

The assertion therefore follows from (2.2.11, iii). □

Proposition (2.5.2). — *Let $f : X' \to X$ be a morphism faithfully flat and quasi-compact, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module, and $\mathscr{F}' = f^*(\mathscr{F})$. Consider, for a quasi-coherent Module, the property of being:*

(i) of finite type;
(ii) of finite presentation;
(iii) locally free of finite type;
(iv) locally free of rank n.

*If **P** denotes any one of these properties, then $\mathscr{F}$ has property **P** if and only if $\mathscr{F}'$ does.*

Proof. For a quasi-coherent $\mathscr{O}_X$-Module to be locally free of finite type, it is necessary and sufficient that it be flat over X and of finite presentation (Bourbaki, *Alg. comm.*, chap. II, § 5, no. 2, cor. 2 of th. 1, taking (2.1.2) into account). By (2.5.1), applied with the identity morphism for f, $\mathscr{F}$ is flat over X if and only if $\mathscr{F}'$ is flat over X'. Thus case (iii) follows from cases (i) and (ii). The same reduction proves (iv), since $f^*(\mathscr{O}_X^n) = \mathscr{O}_{X'}^n$: if $\mathscr{F}$ and $\mathscr{F}'$ are locally free of finite type and $x = f(x')$, their ranks at x and x' are equal, and f is surjective.

To treat cases (i) and (ii), note that the question is local on X, so we may suppose that X is affine. By (2.2.12), there is an affine scheme X'' and a faithfully flat local isomorphism $g : X'' \to X'$. Consequently, it is equivalent for $f^*(\mathscr{F})$ or $g^*f^*(\mathscr{F})$ to have property **P**. We are therefore reduced to **IV-2 | 23**

$$X = \operatorname{Spec}(\mathrm{A}), \qquad X' = \operatorname{Spec}(\mathrm{A}').$$

In view of (2.2.3) and (**II**, 6.1.4.1), it remains only to prove the following lemma. □

Lemma (2.5.3). — *Let* A *be a ring, let* A' *be a faithfully flat* A*-algebra, and let* M *be an* A*-module. Set* $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$. *Then* M *is of finite type* (*resp. of finite presentation*) *if and only if* M' *is.*

Proof. For the proof, see Bourbaki, *Alg. comm.*, chap. I[er], § 3, no. 6, prop. 11. □

Remark (2.5.4). — The assertions of (2.5.2) for properties (i) and (ii) remain valid if f is assumed only *quasi-faithfully flat* (2.3.3) and quasi-compact. Indeed, one is reduced (Bourbaki, *loc. cit.*) to the following lemma.

Lemma (2.5.4.1). — *Let* $\rho : \mathrm{A} \to \mathrm{A}'$ *be a homomorphism of rings such that the corresponding morphism* $f : \operatorname{Spec}(\mathrm{A}') \to \operatorname{Spec}(\mathrm{A})$ *is surjective. Suppose that there is an* A'*-module* N' *of finite type which is flat over* A *and has support* $\operatorname{Spec}(\mathrm{A}')$. *If* $u : \mathrm{P} \to \mathrm{Q}$ *is a homomorphism of* A*-modules such that*

$$u \otimes 1 : \mathrm{P} \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{Q} \otimes_{\mathrm{A}} \mathrm{A}'$$

is surjective, then u is surjective.

Proof. Tensoring the assumed surjection with N' first shows that

$$u \otimes 1_{\mathrm{N}'} : (\mathrm{P} \otimes_{\mathrm{A}} \mathrm{A}') \otimes_{\mathrm{A}'} \mathrm{N}' \longrightarrow (\mathrm{Q} \otimes_{\mathrm{A}} \mathrm{A}') \otimes_{\mathrm{A}'} \mathrm{N}'$$

is surjective. Let $\mathfrak{q}$ be a prime ideal of A' and put $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$. After localization, the corresponding map is

$$u_{\mathfrak{p}} \otimes 1 : \mathrm{P}_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} \mathrm{N}'_{\mathfrak{q}} \longrightarrow \mathrm{Q}_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} \mathrm{N}'_{\mathfrak{q}}$$

(**0**, 1.5.4). By the support hypothesis $\mathrm{N}'_{\mathfrak{q}} \neq 0$, and $\mathrm{N}'_{\mathfrak{q}}$ is flat over $\mathrm{A}_{\mathfrak{p}}$ (**0**, 6.3.1). Nakayama's lemma gives $\mathfrak{q}\mathrm{N}'_{\mathfrak{q}} \neq \mathrm{N}'_{\mathfrak{q}}$, hence a fortiori $\mathfrak{p}\mathrm{N}'_{\mathfrak{q}} \neq \mathrm{N}'_{\mathfrak{q}}$. Thus $\mathrm{N}'_{\mathfrak{q}}$ is faithfully flat over $\mathrm{A}_{\mathfrak{p}}$ (**0**, 6.4.1), and $u_{\mathfrak{p}}$ is surjective. Since f is surjective, every $\mathfrak{p} \in \operatorname{Spec}(\mathrm{A})$ occurs in this way; therefore u is surjective (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 3, th. 1). □

Proposition (2.5.5). — *Let* $f : X \to Y$ *be a morphism, let* $\mathscr{F}$ *be a quasi-coherent* $\mathscr{O}_X$*-Module of finite type which is flat relative to* f, *and let* $\mathscr{G}$ *be a quasi-coherent* $\mathscr{O}_Y$*-Module of finite type. For* $y \in Y$, *set*

$$\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y).$$

A point $x \in X$ *is a maximal point of* $\operatorname{Supp}(\mathscr{F} \otimes_Y \mathscr{G})$ *if and only if* $y = f(x)$ *is a maximal point of* $\operatorname{Supp}(\mathscr{G})$ *and* x *is a maximal point of* $\operatorname{Supp}(\mathscr{F}_y)$ *in* $f^{-1}(y)$. *When this holds, we have*

(2.5.5.1).

$$\operatorname{length}\big((\mathscr{F} \otimes_Y \mathscr{G})_x\big) = \operatorname{length}(\mathscr{G}_y)\ \operatorname{length}\big((\mathscr{F}_y)_x\big).$$

Proof. It is clear that

$$f\big(\operatorname{Supp}(\mathscr{F} \otimes_Y \mathscr{G})\big) \subset \operatorname{Supp}(\mathscr{G})$$

(**0**, 5.2.2); hence the image under f of every irreducible component of $\operatorname{Supp}(\mathscr{F} \otimes_Y \mathscr{G})$ is contained in an irreducible component of $\operatorname{Supp}(\mathscr{G})$.

We may reduce to the case $\operatorname{Supp}(\mathscr{G}) = Y$. Indeed ($\mathbf{Err}_{\mathbf{III}}$, 30), since $\mathscr{G}$ is of finite type, there is a closed subprescheme $Y' \subset Y$ with underlying space $\operatorname{Supp}(\mathscr{G})$ and a quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{G}_1$ of finite type such that, for the canonical injection $j : Y' \to Y$,

$$\mathscr{G} = j_*(\mathscr{G}_1).$$

Set $X' = X \times_Y Y'$, let $f' : X' \to Y'$ be the projection, and write $\mathscr{F}' = \mathscr{F} \otimes_Y \mathscr{O}_{Y'}$. Then $\mathscr{F}'$ is flat relative to f', and, under the canonical closed immersion $X' \to X$,

$$\mathrm{Supp}(\mathscr{F}' \otimes_{Y'} \mathscr{G}_1) = \mathrm{Supp}(\mathscr{F} \otimes_Y \mathscr{G}).$$

Suppose therefore that $\mathrm{Supp}(\mathscr{G}) = Y$. If Z is an irreducible component of $\mathrm{Supp}(\mathscr{G})$, then

$$f^{-1}(Z) \subset \mathrm{Supp}(\mathscr{F} \otimes_Y \mathscr{G})$$

(**I**, 9.1.13); and (2.3.4) shows that every irreducible component of $f^{-1}(Z)$ dominates Z. Thus, if x is the generic point of an irreducible component of $\mathrm{Supp}(\mathscr{F} \otimes_Y \mathscr{G})$ contained in $f^{-1}(Z)$, then $y = f(x)$ is the generic point of Z. Moreover, by (**0**, 2.1.8), x is the generic point of an irreducible component of

$$f^{-1}(y) = \mathrm{Supp}(\mathscr{F}_y)$$

(**I**, 9.1.13); conversely, every generic point of one of these components is the generic point of an irreducible component of $f^{-1}(Z)$.

It remains to prove (2.5.5.1). We have **IV-2** | 24

$$(\mathscr{F} \otimes_Y \mathscr{G})_x = \mathscr{F}_x \otimes_{\mathscr{O}_{Y,y}} \mathscr{G}_y$$

(**I**, 9.1.12) and $(\mathscr{F}_y)_x = \mathscr{F}_x/\mathfrak{m}_y\mathscr{F}_x$; hence we are reduced to the following lemma. □

Lemma (2.5.5.2). — *Let* A *and* B *be local rings,* $\rho : A \to B$ *a local homomorphism, and* $\mathfrak{m}$ *the maximal ideal of* A*. Let* M *be an* A*-module and* N *a* B*-module which is faithfully flat over* A *and such that* $N/\mathfrak{m}N$ *has finite length as a* B*-module. Then*

(2.5.5.3).

$$\mathrm{length}_B(M \otimes_A N) = \mathrm{length}_A(M) \cdot \mathrm{length}_B(N/\mathfrak{m}N).$$

Proof. If M has infinite length, then so does $M \otimes_A N$. Indeed, every strictly increasing chain of n submodules M_i of M ($1 \leqslant i \leqslant n$) yields pairwise distinct sub-B-modules $M_i \otimes_A N$ of $M \otimes_A N$. Since $N \neq \mathfrak{m}N$, formula (2.5.5.3) holds in this case.

Suppose that M has finite length. If $M = 0$, both sides of (2.5.5.3) vanish, so assume $M \neq 0$. Each $\mathfrak{m}^k M$ has finite length, and if $\mathfrak{m}^k M \neq 0$, Nakayama's lemma gives $\mathfrak{m}^{k+1}M \neq \mathfrak{m}^k M$. Hence some r satisfies $\mathfrak{m}^r M = 0$. The modules $\mathfrak{m}^k M \otimes_A N$ identify with sub-B-modules of $M \otimes_A N$, and

$$\frac{\mathfrak{m}^k M \otimes_A N}{\mathfrak{m}^{k+1} M \otimes_A N} \simeq \frac{\mathfrak{m}^k M}{\mathfrak{m}^{k+1} M} \otimes_{A/\mathfrak{m}} \frac{N}{\mathfrak{m}N}$$

for $0 \leqslant k \leqslant r-1$. Its length as a B-module is the product of $\mathrm{length}_B(N/\mathfrak{m}N)$ and the dimension of the $(A/\mathfrak{m})$-vector space $\mathfrak{m}^k M/\mathfrak{m}^{k+1}M$; the latter dimension is the length of that quotient as an A-module. Summing over $0 \leqslant k \leqslant r-1$ gives (2.5.5.3). □

Remark (2.5.5.4). — When N is an A-module *of finite type*, it is faithfully flat over A if and only if it is flat and $N \neq 0$; indeed, Nakayama's lemma then shows that $\mathfrak{m}N \neq N$.

Lemma (2.5.6). — *Let* B *be a not necessarily commutative ring, let* V *and* W *be isomorphic left* B*-modules, set*

$$C = \mathrm{End}_B(V), \qquad M = \mathrm{Hom}_B(V, W),$$

and equip M *with its canonical right* C*-module structure. Then* M *is isomorphic to the right regular module* C_d*. Moreover, for* $u \in M$*, the following conditions are equivalent:*

a) $\{u\}$ *is a basis of the* C*-module* M*;*
b) u *is an isomorphism from* V *onto* W*.*

Proof. If u is an isomorphism from V onto W, then $v \mapsto u \circ v$ is a bijection from C to M, so *b)* implies *a)*. Conversely, suppose that $\{u\}$ is a basis of M. By hypothesis there is an isomorphism $u' : V \to W$, and $\{u'\}$ is also a basis of M. Hence $u = u' \circ w$ for some invertible $w \in C$, that is, for some automorphism w of V. Thus u is an isomorphism. □

Corollary (2.5.7). — *Under the hypotheses of* (2.5.6)*, suppose in addition that one of the following conditions holds:*

(i) V *and* W *are Noetherian* B*-modules;*

(ii) V *and* W *are modules of finite presentation over a commutative subring of* B.

Then conditions a) *and* b) *of* (2.5.6) *are also equivalent to:*

a′) *u is a generator of the* C*-module* M*;*

b′) *u is an epimorphism of* V *onto* W.

Proof. An epimorphism from an A-module E onto itself is bijective in either of the following cases: 1° E is Noetherian (Bourbaki, *Alg.*, chap. VIII, § 2, no. 2, lemma 3); 2° A is commutative and E is of finite presentation (8.9.3)[2]. Therefore *b)* and b') are equivalent. If u generates M and $\{u'\}$ is a basis, there is $v \in \mathrm{C}$ such that $u' = u \circ v$, so u is surjective. Hence a') implies b'); and *a)* plainly implies a'). This proves the corollary. □

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Proposition (2.5.8). — *Let* A *be a commutative semilocal ring, let* B *be an* A*-algebra* (*not necessarily commutative*), *and let* V *and* W *be* B*-modules. Let* A′ *be a commutative* A*-algebra which is faithfully flat as an* A*-module, and set*

$$\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}', \qquad \mathrm{V}' = \mathrm{V} \otimes_{\mathrm{A}} \mathrm{A}', \qquad \mathrm{W}' = \mathrm{W} \otimes_{\mathrm{A}} \mathrm{A}'.$$

Thus B′ *is an* A′*-algebra and* V′, W′ *are* B′*-modules. Suppose in addition that one of the following conditions holds:*

(i) A *and* A′ *are Noetherian, and* V *and* W *are* A*-modules of finite type;*

(ii) B *is an* A*-module of finite type, and* V *is both a projective* B*-module of finite type and an* A*-module of finite presentation.*

If V′ *and* W′ *are isomorphic as* B′*-modules, then* V *and* W *are isomorphic as* B*-modules.*

Proof. In case (ii), W′, being isomorphic to V′, is an A′-module of finite type. It follows that W is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. Ier, § 3, no. 6, prop. 11). Thus in both cases V and W are A-modules of finite type.

(2.5.8.1). *Under either hypothesis* (i) *or* (ii), $\mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W})$ *is an* A*-module of finite type.*

This is immediate in case (i), because $\mathrm{Hom}_{\mathrm{A}}(\mathrm{V}, \mathrm{W})$ is then an A-module of finite type and $\mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W})$ is an A-submodule. In case (ii), V is a direct factor of a free B-module B^n. Consequently, $\mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W})$ is a direct factor of

$$\mathrm{Hom}_{\mathrm{B}}(\mathrm{B}^n, \mathrm{W}) = \mathrm{W}^n.$$

It is therefore of finite type over A.

Set

$$\mathrm{C} = \mathrm{End}_{\mathrm{B}}(\mathrm{V}), \qquad \mathrm{M} = \mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W}).$$

These are A-modules of finite type in both cases. Under either hypothesis, the canonical homomorphism

(2.5.8.2).

$$\mathrm{Hom}_{\mathrm{A}}(\mathrm{V}, \mathrm{W}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{Hom}_{\mathrm{A}'}(\mathrm{V}', \mathrm{W}')$$

is bijective (Bourbaki, *Alg. comm.*, chap. II, § 2, no. 10, prop. 11). Since A′ is flat over A,

$$\mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W}) \otimes_{\mathrm{A}} \mathrm{A}'$$

identifies canonically with a sub-A′-module of $\mathrm{Hom}_{\mathrm{A}}(\mathrm{V}, \mathrm{W}) \otimes_{\mathrm{A}} \mathrm{A}'$. Its image under (2.5.8.2) lies in $\mathrm{Hom}_{\mathrm{B}'}(\mathrm{V}', \mathrm{W}')$: if $u \in \mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W})$ and $a' \in \mathrm{A}'$, the image of $u \otimes a'$ is the homomorphism $u' : \mathrm{V}' \to \mathrm{W}'$ satisfying $u'(x \otimes 1) = u(x) \otimes a'$, and

$$u'\big((b \otimes 1)(x \otimes 1)\big) = u(bx) \otimes a' = bu(x) \otimes a' = (b \otimes 1)\big(u(x) \otimes a'\big).$$

(2.5.8.3). *Under either hypothesis* (i) *or* (ii), *the homomorphism*

(2.5.8.4).

$$\mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{Hom}_{\mathrm{B}'}(\mathrm{V}', \mathrm{W}')$$

[2] The reader may verify that (2.5.7) and (2.5.8) are not used before § 9.

is bijective.

For $b \in \mathrm{B}$, let $h(b)$ (resp. $h'(b)$) denote the A-endomorphism $x \mapsto bx$ of V (resp. W). Let $(b_\alpha)_{\alpha\in\mathrm{I}}$ be a system of generators of the A-algebra B. The map

$$u \longmapsto (h'(b_\alpha) \circ u - u \circ h(b_\alpha))_{\alpha\in\mathrm{I}}$$

from $\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W})$ to $(\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}))^{\mathrm{I}}$ is A-linear, and its kernel is precisely $\mathrm{Hom}_{\mathrm{B}}(\mathrm{V},\mathrm{W})$. Thus

$$0 \longrightarrow \mathrm{Hom}_{\mathrm{B}}(\mathrm{V},\mathrm{W}) \longrightarrow \mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}) \longrightarrow (\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}))^{\mathrm{I}}.$$

The same reasoning applies after replacing A, B, V, W by A′, B′, V′, W′. Moreover, there is a commutative diagram

(2.5.8.5).

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hom}_{\mathrm{B}}(\mathrm{V},\mathrm{W}) \otimes_{\mathrm{A}} \mathrm{A}' \arrow[r] \arrow[d, "r"] & \mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}) \otimes_{\mathrm{A}} \mathrm{A}' \arrow[r] \arrow[d, "s"] & (\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}))^{\mathrm{I}} \otimes_{\mathrm{A}} \mathrm{A}' \arrow[d, "t"] \\
0 \arrow[r] & \mathrm{Hom}_{\mathrm{B}'}(\mathrm{V}',\mathrm{W}') \arrow[r] & \mathrm{Hom}_{\mathrm{A}'}(\mathrm{V}',\mathrm{W}') \arrow[r] & (\mathrm{Hom}_{\mathrm{A}'}(\mathrm{V}',\mathrm{W}'))^{\mathrm{I}}
\end{tikzcd}$$

Here r is (2.5.8.4), s is (2.5.8.2), and t is the composite

$$(\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}))^{\mathrm{I}} \otimes_{\mathrm{A}} \mathrm{A}' \xrightarrow{w} (\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}) \otimes_{\mathrm{A}} \mathrm{A}')^{\mathrm{I}} \xrightarrow{s^{\mathrm{I}}} (\mathrm{Hom}_{\mathrm{A}'}(\mathrm{V}',\mathrm{W}'))^{\mathrm{I}}$$

where w is the canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3e ed., § 3, no. 7). Since A′ is flat over A, the rows of (2.5.8.5) are exact.

The map s is an isomorphism, hence so is s^{I}. In case (ii) we may take I finite, and then w is **IV-2** | 26
bijective (Bourbaki, *loc. cit.*, prop. 7). In case (i), let B_1 (resp. B_2) be the image of B under h (resp. h') in $\mathrm{End}_{\mathrm{A}}(\mathrm{V})$ (resp. $\mathrm{End}_{\mathrm{A}}(\mathrm{W})$). Both images are A-modules of finite type, so here too I may be chosen finite. Thus t, and consequently r, is bijective in both cases.

(2.5.8.6). It follows from (2.5.8.4) that, if

$$\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}', \qquad \mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$$

then there are canonical bijections

$$\mathrm{C}' \xrightarrow{\sim} \mathrm{End}_{\mathrm{B}'}(\mathrm{V}'), \qquad \mathrm{M}' \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{B}'}(\mathrm{V}',\mathrm{W}')$$

the first an isomorphism of A′-algebras and the second, through the first, an isomorphism of right C′-modules.

(2.5.8.7). *Reduction to the case* $\mathrm{V} = \mathrm{B}_s$. The hypothesis that V′ and W′ are isomorphic B′-modules implies that C'_d and M′ are isomorphic right C′-modules (2.5.6). To prove (2.5.8), it is enough to find $u \in \mathrm{M}$ which generates M as a right C-module. Indeed, $u' = u \otimes 1$ then generates M′ as a right C′-module. In case (i), V′ and W′ are Noetherian A′-modules; in case (ii), they are isomorphic A′-modules of finite presentation. Corollary (2.5.7) therefore shows that u' is a B′-isomorphism $\mathrm{V}' \to \mathrm{W}'$. Since A′ is faithfully flat over A, u is bijective (**0**, 6.4.1).

In case (i), C and M are A-modules of finite type. In case (ii), C is a direct factor of V^n by (2.5.8.1), hence a projective A-module of finite type. After changing notation, we are therefore reduced to proving (2.5.8) when $\mathrm{V} = \mathrm{B}_s$; it is enough to prove that the B-module W has a generator. In this case B is an A-module of finite type.

(2.5.8.8). *Reduction to the case where* A *and* A′ *are fields and* B *is a simple* A*-algebra with centre* A.

Let $\mathfrak{r}$ be the radical of the semilocal ring A. It is enough to prove that $\mathrm{W}/\mathfrak{r}\mathrm{W}$ is a cyclic $(\mathrm{B}/\mathfrak{r}\mathrm{B})$-module. Indeed, a surjection

$$\mathrm{B}_s/\mathfrak{r}\mathrm{B}_s \longrightarrow \mathrm{W}/\mathfrak{r}\mathrm{W}$$

composes with $\mathrm{B}_s \to \mathrm{B}_s/\mathfrak{r}\mathrm{B}_s$ and, since B_s is free as a B-module, lifts to a map $u : \mathrm{B}_s \to \mathrm{W}$. The displayed surjection is $u \otimes 1$ after tensoring over A with $\mathrm{A}/\mathfrak{r}$. Since W is of finite type over A, Nakayama's lemma shows that u is surjective (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 2, cor. 1 of prop. 4).

Put

$$A_1 = A/\mathfrak{r}, \qquad A'_1 = A' \otimes_A A_1 = A'/\mathfrak{r}A',$$
$$B_1 = B/\mathfrak{r}B = B \otimes_A A_1, \quad W_1 = W/\mathfrak{r}W = W \otimes_A A_1.$$

The hypotheses remain valid after replacing A, A′, B, $V = B_s$, and W by A_1, A'_1, B_1, $V_1 = (B_1)_s$, and W_1. Moreover, A'_1 is faithfully flat over A_1, and the corresponding base changes of V_1 and W_1 remain isomorphic. We may therefore suppose that A is a finite product of fields.

The ring B is then Artinian. Let $\mathfrak{N}$ be its radical. It is enough, by the same Nakayama argument, to prove that $W/\mathfrak{N}W$ is cyclic over $B/\mathfrak{N}$. Since

$$B'/\mathfrak{N}B' = (B/\mathfrak{N}) \otimes_A A',$$
$$W'/\mathfrak{N}W' = (W/\mathfrak{N}W) \otimes_A A',$$

and the latter is isomorphic to the right regular module over the former, we may further suppose that $\mathfrak{N} = 0$, that is, that B is a semisimple A-algebra.

Write $A = \prod_{i=1}^n k_i$. Then $A' = \prod_{i=1}^n A'_i$, where each A'_i is a k_i-algebra and is annihilated by the other factors k_j. Faithful flatness implies $A'_i \neq 0$. Choose a field quotient k'_i of each A'_i and set $A'' = \prod_i k'_i$. Then A'' is faithfully flat over A and is a quotient of A′. After the further base change,

$$B'' = B \otimes_A A'', \qquad W'' = W \otimes_A A'',$$

the module W'' remains isomorphic to B''_s. Hence we may also suppose that A′ is a finite product of fields.

Let Z be the centre of B. It is a finite product of fields and an A-module of finite type; B and W are finite, hence projective, Z-modules. Put $Z' = Z \otimes_A A'$. Then

$$B' = B \otimes_Z Z', \qquad W' = W \otimes_Z Z',$$

and Z′ is faithfully flat over Z. We may replace A by Z and thus suppose that A is the centre of B.

If k_i are the field factors of A, then $B = \prod_i B_i$, with B_i a simple ring of centre k_i, and $W = \bigoplus_i W_i$, where the other factors B_j annihilate W_i. We may suppose likewise that $A' = \prod_i k'_i$, where k'_i/k_i is a field extension. For every i,

$$(B_i \otimes_{k_i} k'_i)_s \quad \text{and} \quad W_i \otimes_{k_i} k'_i$$

are isomorphic modules over $B_i \otimes_{k_i} k'_i$. It therefore suffices to treat $n = 1$: A and A′ are fields, and B is a simple algebra with centre A.

(2.5.8.9). *End of the proof.* Every B-module is a direct sum of modules isomorphic to a minimal ideal of B (Bourbaki, *Alg.*, chap. VIII, § 5, props. 6 and 8). Thus two B-modules of finite rank over A are isomorphic if and only if they have the same rank over A. By hypothesis,

$$[W' : A'] = [B'_s : A'].$$

But

$$[W' : A'] = [W : A], \qquad [B'_s : A'] = [B_s : A],$$

so $[W : A] = [B_s : A]$. This completes the proof.

□

2.6. Permanence of set-theoretic and topological properties of morphisms by faithfully flat descent

Proposition (2.6.1). — *Let $f : X \to Y$ be an* S*-morphism of* S*-preschemes, and let $g : S' \to S$ be a surjective morphism. Set*

$$X' = X_{(S')}, \qquad Y' = Y_{(S')}, \qquad f' = f_{(S')} : X' \longrightarrow Y'.$$

Consider, for a morphism, the property of being:

(i) *surjective;*
(ii) *injective;*
(iii) *with finite fibres* (*as sets*)*;*
(iv) *bijective;*
(v) *radicial.*

If **P** *denotes any one of these properties and f' has property* **P**, *then so does f.*

Proof. The projection $Y' \to Y$ is itself surjective (**I**, 3.5.2); hence, by (**I**, 3.3.11), we may reduce to the case $Y = S$ and $Y' = S'$. For $y \in Y$ (resp. $y' \in Y'$), denote by X_y (resp. $X'_{y'}$) the fibre prescheme $f^{-1}(y)$ (resp. $f'^{-1}(y')$) (**I**, 3.6.2). If $y' \in Y'$ and $y = g(y')$, there is a canonical isomorphism

$$X'_{y'} \xrightarrow{\sim} X_y \otimes_{k(y)} k(y')$$

(**I**, 3.6.4). Since $\mathrm{Spec}(k(y')) \to \mathrm{Spec}(k(y))$ is surjective, so is the projection $X'_{y'} \to X_y$ (**I**, 3.5.2). Therefore, if $X'_{y'}$ is nonempty (resp. has at most one point, resp. is a finite set), the same is true of X_y. Since $g : Y' \to Y$ is surjective, this proves the assertion in cases (i), (ii), and (iii); case (iv) follows from (i) and (ii).

Finally, for (v), it is enough to show that if f' is universally injective (**I**, 3.5.11), then so is f. Let $Y_1 \to Y$ be any morphism and set

$$X_1 = X \times_Y Y_1, \qquad f_1 = f_{(Y_1)}.$$

Also set

$$\begin{aligned} Y'_1 &= Y_1 \times_Y Y', \\ X'_1 &= X' \times_{Y'} Y'_1 = X_1 \times_{Y_1} Y'_1, \\ f'_1 &= f'_{(Y'_1)} = (f_1)_{(Y'_1)} : X'_1 \longrightarrow Y'_1. \end{aligned}$$

The projection $g_1 = g_{(Y_1)} : Y'_1 \to Y_1$ is surjective (**I**, 3.5.2); and, since f' is universally injective, f'_1 is injective. Case (ii) then shows that f_1 is injective, which proves the assertion. ☐

Proposition (2.6.2). — *With the notation of (2.6.1), suppose that $g : S' \to S$ is faithfully flat and quasi-compact. Consider, for a morphism, the property of being:*

(i) *open;*
(ii) *closed;*
(iii) *quasi-compact and a homeomorphism onto its image subspace;*
(iv) *a homeomorphism.*

If **P** *denotes any one of these properties and f' has property* **P**, *then so does f.*

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Proof. The morphism $Y' \to Y$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2); hence, by (**I**, 3.3.11), we may reduce to the case $Y = S$, $Y' = S'$. Let $g' : X' \to X$ be the projection. For every subset M of X we have

$$g^{-1}(f(M)) = f'(g'^{-1}(M))$$

(**I**, 3.4.8). If f' is open (resp. closed), then, for every open (resp. closed) subset M of X, the subset $f'(g'^{-1}(M))$ is open (resp. closed) in Y'. Since g is faithfully flat and quasi-compact, (2.3.12) implies that $f(M)$ is open (resp. closed) in Y. This proves cases (i) and (ii).

Let us prove (iii), which also yields (iv) in view of (2.6.1, (iv)). By (2.6.1, (ii)), f is injective; it remains to prove that f, considered as a map from X onto $f(X)$, is quasi-compact and open. Since

f' is quasi-compact, so is f (1.1.4). It is therefore enough to prove, for every closed subset Z of X, that

$$Z = f^{-1}\big(\overline{f(Z)}\big).$$

Since g' is surjective, this is equivalent to

$$\begin{aligned} g'^{-1}(Z) &= g'^{-1}\Big(f^{-1}\big(\overline{f(Z)}\big)\Big) \\ &= f'^{-1}\Big(g^{-1}\big(\overline{f(Z)}\big)\Big). \end{aligned}$$

The composite of the canonical injection $Z \to X$ with f is quasi-compact, where Z denotes here the reduced closed subprescheme of X with underlying space Z. Applying (2.3.10) to the subset $f(Z)$ of Y, the image of the morphism $f|_Z$, gives

$$g^{-1}\big(\overline{f(Z)}\big) = \overline{g^{-1}\big(f(Z)\big)}.$$

If $Z' = g'^{-1}(Z)$, the required formula is consequently equivalent to

$$Z' = f'^{-1}\big(\overline{f'(Z')}\big),$$

which follows from the hypothesis that f' is a homeomorphism from X' onto $f'(X')$. □

Remark (2.6.3). — In cases (i) and (ii), the conclusions of (2.6.2) remain valid if g is assumed only *quasi-faithfully flat* (2.3.3) and quasi-compact. Indeed, by (2.1.4), (**I**, 3.5.2), and (**I**, 9.1.13.1), we may again reduce to the case $Y = S$, $Y' = S'$, and the conclusion follows from (2.3.12). In cases (iii) and (iv), the conclusions remain valid if g is assumed only quasi-faithfully flat, provided that f is already assumed quasi-compact; the proof then uses only (2.3.10) and the surjectivity of g.

Finally, if g is faithfully flat and locally of finite presentation, or if g is surjective and S is discrete, the conclusion of (2.6.2) remains valid when **P** is the property

(iii *bis*) of being a homeomorphism onto its image subspace;

as follows from the proof of (2.6.2) and Remark (2.4.11).

Corollary (2.6.4). — *With the notation of* (2.6.1)*, suppose that* $g : S' \to S$ *is faithfully flat and quasi-compact. Consider, for a morphism, the property of being:*

(i) universally open;
(ii) universally closed;
(iii) quasi-compact and universally bicontinuous;
(iv) a universal homeomorphism;
(v) quasi-compact;
(vi) quasi-compact and dominant.

If **P** *denotes any one of these properties, then* f *has property* **P** *if and only if* f' *does.*

Proof. Properties (v) and (vi) are only given for the record, being consequences of (1.1.4), (1.1.6), and (2.3.7). For the other properties, necessity follows from (2.4.3). Conversely, suppose, for example, that f' is universally open, and let $Y_1 \to Y$ be any morphism. Set

$$X_1 = X \times_Y Y_1, \qquad f_1 = f_{(Y_1)}.$$

Also set

$$\begin{aligned} Y_1' &= Y_1 \times_Y Y', \\ X_1' &= X' \times_{Y'} Y_1' = X_1 \times_{Y_1} Y_1', \\ f_1' &= f'_{(Y_1')} = (f_1)_{(Y_1')} : X_1' \longrightarrow Y_1'. \end{aligned}$$

The projection $g_1 = g_{(Y_1)} : Y_1' \to Y_1$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2). Since f' is universally open, f_1' is open, and (2.6.2) then shows that f_1 is open. Hence f is universally open. The same argument applies in the other cases.

One may again replace "faithfully flat" by "quasi-faithfully flat". Moreover, when g is locally of finite presentation, or when g is surjective and S is discrete, property (iii) may be replaced by

(iii *bis*) universally bicontinuous.

□

2.7. Permanence of various properties of morphisms under faithfully flat descent

Proposition (2.7.1). — *Let $f : X \to Y$ be an* S*-morphism of* S*-preschemes, and let $g : \mathrm{S}' \to \mathrm{S}$ be faithfully flat and quasi-compact. Set*

$$X' = X_{(\mathrm{S}')}, \qquad Y' = Y_{(\mathrm{S}')}, \qquad f' = f_{(\mathrm{S}')} : X' \longrightarrow Y'.$$

Consider, for a morphism, the property of being:

(i) separated;
(ii) quasi-separated;
(iii) locally of finite type;
(iv) locally of finite presentation;
(v) of finite type;
(vi) of finite presentation;
(vii) proper;
(viii) an isomorphism;
(ix) a monomorphism;
(x) an open immersion;
(xi) a quasi-compact immersion;
(xii) a closed immersion;
(xiii) affine;
(xiv) quasi-affine;
(xv) finite;
(xvi) quasi-finite;
(xvii) integral.

If **P** *denotes any one of these properties, then f has property* **P** *if and only if f' does.*

Proof. It was proved in Chapters I and II and in Chapter IV, § 1, that if f has one of the properties **P** above, then so does f' (without any hypothesis on $g : \mathrm{S}' \to \mathrm{S}$). It remains to prove the converse. Since the projection $Y' \to Y$ is
a faithfully flat and quasi-compact morphism (2.2.13) and (1.1.2), we may, by (**I**, 3.3.11), reduce to the case $\mathrm{S} = Y$, $\mathrm{S}' = Y'$.

(i) To say that f is separated means that its diagonal $\Delta_f : X \to X \times_Y X$ is closed. Since $\Delta_{f'} = (\Delta_f)_{(Y')}$ (**I**, 5.3.4), if $\Delta_{f'}$ is closed, then so is Δ_f by (2.6.2); hence f is separated.

(ii) This was already proved under weaker hypotheses (1.2.5).

(iii) and (iv). The question is plainly local on X and Y and, in view of (2.2.12), it is therefore enough to prove the following lemma.

Lemma (2.7.1.1). — *Let A be a ring, B an A-algebra, and A' an A-algebra that is faithfully flat as an A-module; put $B' = B \otimes_A A'$. Then B is an A-algebra of finite type* (*resp. of finite presentation*) *if and only if B' is an A'-algebra of finite type* (*resp. of finite presentation*).

Necessity is already known without any hypothesis on A' (1.3.4), (1.3.6), (1.4.3), and (1.4.6). Suppose that B' is an A'-algebra of finite type. Let $(B_\alpha)_{\alpha \in I}$ be the filtered increasing family of finite-type A-subalgebras of B. Then

$$B = \varinjlim_{\alpha} B_\alpha, \qquad B' = \varinjlim_{\alpha} (B_\alpha \otimes_A A'),$$

because tensor product commutes with inductive limits. If (x'_i) is a finite set of generators of the A'-algebra B', there is an index α such that every x'_i belongs to the subalgebra $B_\alpha \otimes_A A'$ of B'. Hence $B' = B_\alpha \otimes_A A'$, and faithful flatness gives $B = B_\alpha$ (**0**, 6.4.1).

Now suppose that B' is an A'-algebra of finite presentation. What precedes shows that B is an A-algebra of finite type. Thus there are a polynomial A-algebra $C = A[T_1, \ldots, T_m]$ and a surjective A-homomorphism of algebras $C \to B$. Let $\mathfrak{J}$ be its kernel. We have an exact sequence

$$0 \longrightarrow \mathfrak{J} \longrightarrow C \longrightarrow B \longrightarrow 0,$$

and, since A' is flat over A, an exact sequence

$$0 \longrightarrow \mathfrak{J}' \longrightarrow C' \longrightarrow B' \longrightarrow 0,$$

where

$$C' = C \otimes_A A' = A'[T_1, \ldots, T_m], \qquad \mathfrak{J}' = \mathfrak{J} \otimes_A A'$$

and $\mathfrak{J}'$ is identified with an ideal of C'. Since B' is of finite presentation over A', the C'-module $\mathfrak{J}'$ is of finite type (1.4.4). But $\mathfrak{J}' = \mathfrak{J} \otimes_C C'$, and C' is faithfully flat over C (2.2.13) and (2.2.3). It follows that $\mathfrak{J}$ is a C-module of finite type (Bourbaki, *Algèbre commutative*, Chap. I, § 3, no. 6, Prop. 11). Hence B is an A-algebra of finite presentation.

(v) This follows from (iii) and (2.6.4, (v)), by (1.5.2).

(vi) This follows similarly from (iv), (v), and (ii), by (1.6.1).

(vii) This follows from (i), (v), and (2.6.4, (ii)), by (**II**, 5.4.1).

(viii) Since f' is an isomorphism, it is a universal homeomorphism; hence so is f (2.6.4). In particular, f is quasi-compact and separated (2.4.4). Write $f = (\psi, \theta)$, where ψ is therefore a homeomorphism. It remains to prove that

$$\theta : \mathscr{O}_Y \longrightarrow f_*(\mathscr{O}_X)$$

is an isomorphism of $\mathscr{O}_Y$-modules. If $f' = (\psi', \theta')$, then

$$\theta' : \mathscr{O}_{Y'} \longrightarrow f'_*(\mathscr{O}_{X'})$$

is the composite of $g^*(\theta)$ with the canonical homomorphism

$$g^*\big(f_*(\mathscr{O}_X)\big) \longrightarrow f'_*(\mathscr{O}_{X'})$$

(2.3.2). The latter is bijective by (2.3.1). Thus, if θ' is bijective, then so is $g^*(\theta)$; since g is faithfully flat, θ is bijective (2.2.7). This proves (viii).

(ix) This follows from (viii), (**I**, 5.3.4), and (**I**, 5.3.8), which reduces monomorphisms to isomorphisms.

(x) If f' is an open immersion, then $f'(X')$ is open in Y', and

$$f'(X') = g^{-1}\big(f(X)\big)$$

(**I**, 3.4.8). It follows from (2.3.12) that $f(X)$ is open. Replacing Y (resp. Y') by the subprescheme induced on $f(X)$ (resp. $f'(X')$), and using (1.1.2) and (2.2.13), we reduce to the case where f' is an isomorphism. Then f is an isomorphism by (viii), proving (x).

(xi) If f' is a quasi-compact immersion, then it is a quasi-compact and quasi-separated morphism (1.2.2); hence so is f, by (ii) and (2.6.4, (v)). Let Z be the closed subprescheme of Y that is the closed image of X under f (1.7.8), and write $f = j \circ q$, where $j : Z \to Y$ is the canonical injection. Then $f' = j' \circ q'$, with $j' = j_{(Y')}$ and $q' = q_{(Y')}$. Moreover, j' identifies with the canonical injection $Z' \to Y'$ of the closed image Z' of X' under f' (2.3.2). The hypothesis on f' now says that q' is an open immersion (**I**, 9.5.10); hence so is q, by (x), and therefore f is an immersion. **IV-2** | 31

(xii) To say that f (resp. f') is a closed immersion is to say that it is a quasi-compact immersion and a closed morphism. Thus (xii) follows from (xi) and (2.6.2, (ii)).

(xiii) and (xiv). Suppose that f' is affine (resp. quasi-affine). Then f' is quasi-compact and quasi-separated (**II**, 5.1.1), and so is f by (ii) and (2.6.4, (v)). Put

$$\mathscr{A} = f_*(\mathscr{O}_X), \qquad \mathscr{A}' = f'_*(\mathscr{O}_{X'}).$$

By (2.3.1), the canonical homomorphism of $\mathscr{O}_{Y'}$-algebras $g^*(\mathscr{A}) \to \mathscr{A}'$ is bijective. Consequently, if $h : Z = \operatorname{Spec}(\mathscr{A}) \to Y$ is the structure morphism, then the structure morphism $h' : Z' = \operatorname{Spec}(\mathscr{A}') \to Y'$ identifies with $h_{(Y')}$ (**II**, 1.5.2). Let $u : X \to Z$ (resp. $u' : X' \to Z'$) be the canonical Y-morphism (resp. Y'-morphism) corresponding to the identity homomorphism of $\mathscr{A}$ (resp.

$\mathscr{A}'$) (**II**, 1.2.7). We have a commutative diagram

$$\begin{tikzcd} X & X' \arrow[l] \arrow[d, "u'"] \\ Z \arrow[from=u, "u"'] \arrow[d, "h"'] & Z' \arrow[l, "g'"'] \arrow[d, "h'"] \\ Y & Y' \arrow[l, "g"] \end{tikzcd}$$

and $h' \circ u' = f'$. It follows from (**II**, 1.2.7) that $u' = u_{(Z')}$. The projection $g' : Z' \to Z$ is faithfully flat and quasi-compact (1.1.2) and (2.2.13). The hypothesis on f' says that u' is an isomorphism (resp. an open immersion) (**II**, 5.1.6); by (viii) (resp. (x)), u is an isomorphism (resp. an open immersion). This proves (xiii) (resp. (xiv)).

(xv) If f' is finite, it is affine; hence f is affine by (xiii). With the notation used there, $\mathscr{A}'$ is an $\mathscr{O}_{Y'}$-module of finite type and is isomorphic to $g^*(\mathscr{A})$. It follows from (2.5.2) that $\mathscr{A}$ is an $\mathscr{O}_Y$-module of finite type, and therefore f is finite. **IV-2** | 32

(xvi) To say that f is quasi-finite is to say that f is of finite type and that $f^{-1}(y)$ is finite for every $y \in Y$ (**II**, 6.2.2) and (**I**, 6.4.4). The conclusion therefore follows from (v) and (xv), applied to the fibres.

(xvii) As in (xv), one first sees that f is affine. We may reduce to the case

$$Y = \operatorname{Spec}(A), \qquad Y' = \operatorname{Spec}(A'), \qquad X = \operatorname{Spec}(B), \qquad X' = \operatorname{Spec}(B'),$$

where $B' = B \otimes_A A'$. The ring B is the inductive limit of its finite-type A-subalgebras B_α; hence

$$B' = \varinjlim_{\alpha} B'_\alpha, \qquad B'_\alpha = B_\alpha \otimes_A A',$$

and each B'_α is a finite-type A'-algebra. By hypothesis, B' is integral over A', so each B'_α is an A'-module of finite type. Hence B_α is an A-module of finite type (2.5.2), and is therefore integral over A. Since every element of B lies in some B_α, the algebra B is integral over A. □

Corollary (2.7.2). — *Under the hypotheses and notation of* (2.7.1), *suppose that f is quasi-compact. Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module and, if $g' : X' \to X$ is the projection, let $\mathscr{L}' = g'^*(\mathscr{L})$ be its inverse image. Then $\mathscr{L}$ is ample* (*resp. very ample*) *for f if and only if $\mathscr{L}'$ is ample* (*resp. very ample*) *for f'.*

Proof. Necessity holds without any hypothesis on $g : S' \to S$ (**II**, 4.4.10) and (**II**, 4.6.13). To prove sufficiency, we may, as in (2.7.1), reduce to the case $S = Y$, $S' = Y'$. The hypothesis on $\mathscr{L}'$ implies that f' is quasi-compact and separated (**II**, 4.6.1); hence so is f, by (2.6.4, (v)) and (2.7.1, (i)). Put

$$\mathscr{E} = f_*(\mathscr{L}), \qquad \mathscr{E}' = f'_*(\mathscr{L}').$$

By (2.3.1), the canonical homomorphism

$$u : g^*(\mathscr{E}) \longrightarrow \mathscr{E}'$$

is bijective. Suppose first that $\mathscr{L}'$ is very ample for f'. The canonical homomorphism $\sigma' : f'^*(\mathscr{E}') \to \mathscr{L}'$ is then surjective, and

$$r' = r_{\mathscr{L}', \sigma'} : X' \longrightarrow \mathbf{P}(\mathscr{E}')$$

is a necessarily quasi-compact immersion (**II**, 4.4.4, (b)) and (1.1.2, (v)). If

$$h : \mathbf{P}(\mathscr{E}) \longrightarrow Y, \qquad h' : \mathbf{P}(\mathscr{E}') \longrightarrow Y'$$

are the structure morphisms, the bijectivity of u implies that h' identifies with $h_{(Y')}$ (**II**, 4.1.3). On the other hand, the projection $g' : X' \to X$ is faithfully flat (2.2.13), and $f \circ g' = g \circ f'$. Let $\sigma : f^*(\mathscr{E}) \to \mathscr{L}$ be the canonical homomorphism. One readily verifies that $g'^*(\sigma)$ is the composite

$$f'^*(g^*(f_*(\mathscr{L}))) \xrightarrow{f'^*(u)} f'^*(f'_*(\mathscr{L}')) \xrightarrow{\sigma'} \mathscr{L}'.$$

This may be checked, for example, by reducing to the case where Y and Y' are affine. Since σ' is surjective and $f'^*(u)$ is bijective, $g'^*(\sigma)$ is surjective; hence σ is surjective by (2.2.7). Thus the morphism

$$r = r_{\mathscr{L},\sigma} : X \longrightarrow \mathbf{P}(\mathscr{E})$$

is everywhere defined (**II**, 3.7.4). Put $P = \mathbf{P}(\mathscr{E})$ and $P' = \mathbf{P}(\mathscr{E}')$, and let $g'' : P' \to P$ be the projection. Then r' identifies with $r_{(P')}$ (**II**, 4.2.10), while g'' is faithfully flat and quasi-compact (1.1.2) and (2.2.13). By (2.7.1, (xi)), r is an immersion, and therefore $\mathscr{L}$ is very ample (**II**, 4.4.4, (b)).

Now suppose that $\mathscr{L}'$ is ample for f'. To prove that $\mathscr{L}$ is ample for f, we may reduce to the case where Y is affine (**II**, 4.6.4); by (2.2.12) and (**II**, 4.6.13), we may also suppose that Y' is affine. Then X and X' are quasi-compact schemes, and we may apply the criterion of (**II**, 4.6.8, (c)). Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module of finite type, and let

$$\sigma : f^* f_*(\mathscr{F} \otimes \mathscr{L}^{\otimes n}) \longrightarrow \mathscr{F} \otimes \mathscr{L}^{\otimes n}$$

be the canonical homomorphism. As above, $g'^*(\sigma)$ is the composite

$$f'^*\big(g^* f_*(\mathscr{F} \otimes \mathscr{L}^{\otimes n})\big) \xrightarrow{f'^*(u)} f'^* f'_*(\mathscr{F}' \otimes \mathscr{L}'^{\otimes n}) \xrightarrow{\sigma'} \mathscr{F}' \otimes \mathscr{L}'^{\otimes n},$$

where $\mathscr{F}' = g'^*(\mathscr{F})$, using (**0**, 4.3.3.1), and where u is the canonical homomorphism

$$g^* f_*(\mathscr{F} \otimes \mathscr{L}^{\otimes n}) \longrightarrow f'_*(\mathscr{F}' \otimes \mathscr{L}'^{\otimes n}).$$

The homomorphism u is bijective for every n by (2.3.1). Moreover, $\mathscr{F}'$ is quasi-coherent and of finite type, so the ampleness of $\mathscr{L}'$ for f' gives an n_0 such that σ' is surjective for $n \geqslant n_0$. Thus $g'^*(\sigma)$ is surjective for $n \geqslant n_0$; since g' is faithfully flat, σ is surjective for these values of n (2.2.7). This completes the proof. □

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Remarks (2.7.3). — (i) It follows from (2.6.1), (2.6.4), and (2.5.4.1) that the conclusions of (2.7.1) remain valid in cases (i), (iii), (v), (vii), and (xvi) when g is assumed only quasi-compact and *quasi-faithfully flat* (2.3.3). We already noted that case (ii) of (2.7.1) is valid under the sole assumptions that g is surjective and quasi-compact.

(ii) Under the hypotheses and notation of (2.7.1), it can happen that f is proper and f' is projective while f is not quasi-projective. Indeed, Hironaka [34] gave an example of a proper nonprojective morphism $f : X \to Y$, where X and Y are regular algebraic schemes (**0**, 4.1.4) over the same field k, with Y projective over k. Moreover, Y is the union of two affine open subsets Y_i ($i = 1, 2$) such that

$$f_i : X \times_Y Y_i \longrightarrow Y_i$$

is projective for $i = 1, 2$. Let $Y' = Y_1 \amalg Y_2$ be their coproduct as preschemes. The canonical morphism $g : Y' \to Y$, whose restrictions are the canonical injections of Y_1 and Y_2 into Y, is faithfully flat and is quasi-compact by (**I**, 5.5.10). Yet $f' : X \times_Y Y' \to Y'$, whose restriction to each Y_i is f_i, is projective (**II**, 5.5.6), whereas f is not. Consequently there exists an invertible $\mathscr{O}_{X'}$-module $\mathscr{L}'$ that is f'-ample but is not of the form $g'^*(\mathscr{L})$ for an invertible $\mathscr{O}_X$-module $\mathscr{L}$, where $g' : X' \to X$ is the projection, by (2.7.2).

(iii) Under the hypotheses of (2.7.1), it can happen that f' is a local isomorphism while f is not a local immersion. Indeed, let k be a field, let $\bar{k}$ be an algebraic closure of k, and let $K \neq k$ be a finite separable extension of k. The structure morphism $f : X \to Y$, where $X = \operatorname{Spec}(K)$ and $Y = \operatorname{Spec}(k)$, is not a local immersion. But if $Y' = \operatorname{Spec}(\bar{k})$, then $Y' \to Y$ is faithfully flat and quasi-compact, and f' is a local isomorphism, since $X' = X \times_Y Y'$ is a coproduct of finitely many schemes isomorphic to Y'.

2.8. Preschemes over a regular base of dimension 1; closure of a closed subprescheme of the generic fibre

Proposition (2.8.1). — *Let Y be a locally Noetherian, regular, irreducible prescheme of dimension* 1*, with generic point η. Let $f : X \to Y$ be a morphism, let $X_\eta = f^{-1}(\eta)$ be the fibre at the generic point, and let $i : X_\eta \to X$ be the canonical morphism. Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module, put $\mathscr{F}_\eta = i^*(\mathscr{F})$, let $\mathscr{G}'$ be an $\mathscr{O}_{X_\eta}$-module quotient of $\mathscr{F}_\eta$, and let $\overline{\mathscr{G}'}$ be the image of $\mathscr{F}$ under the composite homomorphism* (**0**, 4.4.3.2)

$$\mathscr{F} \xrightarrow{\rho_{\mathscr{F}}} i_*(i^*(\mathscr{F})) \longrightarrow i_*(\mathscr{G}').$$

Then $\overline{\mathscr{G}'}$ is a quasi-coherent $\mathscr{O}_X$-module quotient of $\mathscr{F}$, flat over Y, such that **IV-2 | 34**

$$i^*(\overline{\mathscr{G}'}) = \mathscr{G}',$$

and it is the unique $\mathscr{O}_X$-module quotient of $\mathscr{F}$ having these properties.

Proof. Since i is quasi-compact and quasi-separated (1.1.2) and (1.2.2), it follows from (1.7.4) that, for every quasi-coherent $\mathscr{O}_{X_\eta}$-module $\mathscr{H}'$, the $\mathscr{O}_X$-module $i_*(\mathscr{H}')$ is quasi-coherent. Moreover, for every open subset U of X, by definition (**0**, 3.4.1),

$$i_*(\mathscr{H}')|U = (i|U \cap X_\eta)_*(\mathscr{H}'|U \cap X_\eta).$$

If the proposition is proved when X and Y are affine, the general case follows by gluing, in view of the uniqueness assertion in the affine case. In other words, we are reduced to proving the following lemma.

Lemma (2.8.1.1). — *Let A be a Noetherian regular integral ring of dimension* 1 (**0**, 17.3.6)*, let K be its field of fractions, let M be an A-module, and let N' be a K-module quotient of*

$$M_{(K)} = M \otimes_A K$$

by a sub-K-module P'. Let $\overline{N'}$ be the image of M under the composite homomorphism

$$M \longrightarrow M_{(K)} \longrightarrow N'.$$

Then $\overline{N'}$ is a flat A-module, and it is the unique quotient module N of M that is flat over A and for which the kernel of the surjective homomorphism

$$M_{(K)} \longrightarrow N_{(K)}$$

is equal to P'.

For every maximal ideal $\mathfrak{m}$ of A, the ring $A_\mathfrak{m}$ is a regular local ring of dimension 1, hence a discrete valuation ring. Therefore an A-module N is flat if and only if it is torsion-free (**0**, 6.3.4). Since N' is a K-vector space, it is torsion-free as an A-module, and the same is therefore true of its submodule $\overline{N'}$. Moreover, one immediately verifies that $\overline{N'}_{(K)}$ identifies with N'.

Conversely, let N be an A-module quotient of M having the properties in the statement. Since N is flat over A, the canonical homomorphism

$$N \longrightarrow N_{(K)} = N \otimes_A K$$

is injective. Since $N_{(K)}$ identifies with N', the conclusion follows from the commutativity of the diagram

$$\begin{tikzcd} M \arrow[r] \arrow[d] & N \arrow[d] \\ M_{(K)} \arrow[r] & N_{(K)}. \end{tikzcd}$$

□

Corollary (2.8.2). — *Under the conditions of (2.8.1), the module $\mathscr{F}$ is flat over Y if and only if the canonical homomorphism*

$$\mathscr{F} \longrightarrow i_*(\mathscr{F}_\eta) = i_*(i^*(\mathscr{F}))$$

is injective.

(2.8.3). The formation of the $\mathscr{O}_X$-module $\overline{\mathscr{G}'}$ is *functorial* in $\mathscr{F}$ and $\mathscr{G}'$. More precisely, let $\mathscr{F}_1$ and $\mathscr{F}_2$ be two quasi-coherent $\mathscr{O}_X$-modules, let $u : \mathscr{F}_1 \to \mathscr{F}_2$ be an $\mathscr{O}_X$-homomorphism, let $\mathscr{G}'_i$ be an $\mathscr{O}_{X_\eta}$-module quotient of $(\mathscr{F}_i)_\eta$ for $i = 1, 2$, and let $v : \mathscr{G}'_1 \to \mathscr{G}'_2$ be a homomorphism making the diagram

$$\begin{tikzcd}
(\mathscr{F}_1)_\eta \arrow[r, "i^*(u)"] \arrow[d] & (\mathscr{F}_2)_\eta \arrow[d] \\
\mathscr{G}'_1 \arrow[r, "v"'] & \mathscr{G}'_2
\end{tikzcd}$$

commutative. Such a homomorphism v, when it exists, is uniquely determined by this property. Then the diagram

$$\begin{tikzcd}
\mathscr{F}_1 \arrow[r, "u"] \arrow[d] & \mathscr{F}_2 \arrow[d] \\
i_*(\mathscr{G}'_1) \arrow[r, "i_*(v)"'] & i_*(\mathscr{G}'_2)
\end{tikzcd}$$

is commutative, and consequently there is a unique homomorphism

$$w : \overline{\mathscr{G}'_1} \longrightarrow \overline{\mathscr{G}'_2}$$

making the diagram

$$\begin{tikzcd}
\mathscr{F}_1 \arrow[r, "u"] \arrow[d] & \mathscr{F}_2 \arrow[d] \\
\overline{\mathscr{G}'_1} \arrow[r, "w"'] & \overline{\mathscr{G}'_2}
\end{tikzcd}$$

commutative.

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Proposition (2.8.4). — *Suppose that Y satisfies the hypotheses of (2.8.1). Let X_1 and X_2 be two Y-preschemes, let $\mathscr{F}_i$ be a quasi-coherent $\mathscr{O}_{X_i}$-module, and let $\mathscr{G}'_i$ be an $\mathscr{O}_{(X_i)_\eta}$-module quotient of $(\mathscr{F}_i)_\eta$, for $i = 1, 2$. Then*

(2.8.4.1).

$$\overline{\mathscr{G}'_1} \otimes_{\mathscr{O}_Y} \overline{\mathscr{G}'_2} = \overline{\mathscr{G}'_1 \otimes_{k(\eta)} \mathscr{G}'_2}.$$

Proof. Put $X = X_1 \times_Y X_2$. The left-hand side of (2.8.4.1) is a quasi-coherent $\mathscr{O}_X$-module flat over Y (**0**, 6.2.1); its inverse image on X_η is

$$\mathscr{G}'_1 \otimes_{k(\eta)} \mathscr{G}'_2$$

(**I**, 9.1.5), and it is a quotient of $\mathscr{F}_1 \otimes_{\mathscr{O}_Y} \mathscr{F}_2$. The conclusion therefore follows from the uniqueness assertion of (2.8.1). □

Proposition (2.8.5). — *Suppose that X and Y satisfy the hypotheses of (2.8.1), and let Z' be a closed subprescheme of X_η. Then there exists a unique closed subprescheme $\overline{Z'}$ of X that is flat over Y and satisfies*

$$i^{-1}(\overline{Z'}) = Z'.$$

Proof. Let $\mathscr{J}'$ be the quasi-coherent ideal of $\mathscr{O}_{X_\eta}$ defining Z'. Apply (2.8.1) with

$$\mathscr{F} = \mathscr{O}_X, \qquad \mathscr{G}' = \mathscr{O}_{X_\eta} / \mathscr{J}'.$$

If $\overline{\mathscr{G}'} = \mathscr{O}_X / \mathscr{J}$, then

$$\mathscr{J}' = i^*(\mathscr{J})\mathscr{O}_{X_\eta},$$

and hence $i^{-1}(\overline{Z'}) = Z'$ (**I**, 4.4.5). □

The prescheme $\overline{Z'}$ is the *closed image* of Z' under the composite morphism

$$Z' \longrightarrow X_\eta \xrightarrow{i} X,$$

where the first arrow is the canonical injection (**I**, 9.5.3). Its underlying space is the closure of Z' *in* X (**I**, 9.5.4), which justifies the notation. We also say that $\overline{Z'}$ is the closure subprescheme of X of Z'.

Corollary (2.8.6). — *Let X_1 and X_2 be two Y-preschemes, and let Z'_i be a closed subprescheme of $(X_i)_\eta$, for $i = 1, 2$. Then*

(2.8.6.1).

$$\overline{Z'_1} \times_Y \overline{Z'_2} = \overline{Z'_1 \times_{k(\eta)} Z'_2}.$$

Proof. This follows from (2.8.4) and (2.8.5). □

§3. Associated prime cycles and primary decompositions

In this section, we mainly give the translation of results on modules presented in Bourbaki, *Alg. comm.*, chap. IV, which we follow very closely. The notions that follow seem to be of interest only in the case of *locally Noetherian* preschemes. **IV-2** | 36

3.1. Associated prime cycles of a module

Definition (3.1.1). — Let X be a prescheme and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. We say that a point $x \in X$ is *associated* to $\mathscr{F}$ if the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_x$ is associated to the $\mathscr{O}_x$-module $\mathscr{F}_x$ (in other words, if $\mathfrak{m}_x$ is the annihilator of an element of $\mathscr{F}_x$). We denote by $\operatorname{Ass}(\mathscr{F})$ the set of $x \in X$ associated to $\mathscr{F}$. We say that a closed irreducible subset Z of X is an *associated prime cycle of $\mathscr{F}$* if its generic point is associated to $\mathscr{F}$. When $\mathscr{F} = \mathscr{O}_X$, we also say that the points (resp. *prime cycles*) associated to $\mathscr{F}$ are associated to the *prescheme* X.

We say that an associated prime cycle of $\mathscr{F}$ (resp. X) is *embedded* if it is contained in another associated prime cycle of $\mathscr{F}$ (resp. X) (in other words, if it is not *maximal* in the set of these cycles).

If X is locally Noetherian, the *irreducible components* of X are none other than the *maximal* (or *non-embedded*) prime cycles associated to X.

It is clear that if $x \in \operatorname{Ass}(\mathscr{F})$, then $\mathscr{F}_x \neq 0$, in other words

(3.1.1.1) $$\operatorname{Ass}(\mathscr{F}) \subset \operatorname{Supp}(\mathscr{F}).$$

If $x \in X$ is associated to $\mathscr{F}$, it is evidently associated to $\mathscr{F}|U$ for every open neighbourhood U of x, and conversely, if it is associated to $\mathscr{F}|U$ for one of these neighbourhoods, it is associated to $\mathscr{F}$.

We note finally that for a quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the existence of embedded associated prime cycles is a *local* question, for if y and z are two points of $\operatorname{Ass}(\mathscr{F})$ such that $y \in \overline{\{z\}}$, every neighbourhood of y contains z.

Proposition (3.1.2). — *Let A be a Noetherian ring, M an A-module, $X = \operatorname{Spec}(A)$, $\mathscr{F} = \widetilde{M}$; for a point $x \in X$ to be associated to $\mathscr{F}$, it is necessary and sufficient that the prime ideal $\mathfrak{j}_x$ of A be associated to the module M (in other words, be the annihilator of an element $f \in M$).*

Proof. This follows from the definition (3.1.1) and from Bourbaki, *loc. cit.*, § 1, no 2, cor. of prop. 5, applied to $S = A - \mathfrak{j}_x$. □

Proposition (3.1.3). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module, x a point of X, and Z the reduced closed sub-prescheme of X having $\overline{\{x\}}$ as underlying space* (**I**, 5.2.1). *The following conditions are equivalent:*

a) $x \in \operatorname{Ass}(\mathscr{F})$. **IV-2** | 37
b) *There exist an open neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{F})$ such that $U \cap Z$ is equal to* $\operatorname{Supp}(\mathscr{O}_U \cdot f)$.
b′) *There exist an open neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{F})$ such that $U \cap Z$ is an irreducible component of* $\operatorname{Supp}(\mathscr{O}_U \cdot f)$.
c) *There exist an open neighbourhood U of x and a submodule of $\mathscr{F}|U$ isomorphic to $\mathscr{O}_Z|U$ ($\mathscr{O}_Z$ being identified with a quotient of $\mathscr{O}_X$).*

c′) *There exist an open neighbourhood U of x and a coherent submodule $\mathscr{G}$ of $\mathscr{F}|U$ such that $U \cap Z$ is an irreducible component of* $\operatorname{Supp}(\mathscr{G})$.

Proof. It is clear that *c)* implies *b)*, for it suffices to take for f the element of $\Gamma(U, \mathscr{F})$ which corresponds to the unit section of $\mathscr{O}_Z|U$. As $U \cap Z$ is irreducible (**0**, 2.1.6), *b)* implies *b′)*, and *b′)* implies *c′)* since $\mathscr{O}_X$ is coherent (**0**, 5.3.4). To see that *c′)* implies *a)*, we can reduce to the case where $U = X = \operatorname{Spec}(A)$ is affine, A being Noetherian, and where $\mathscr{F} = \widetilde{M}$, where M is an A-module, and $\mathscr{G} = \widetilde{N}$, where N is a sub-module of M, of finite type. We then know that the minimal elements of $\operatorname{Supp}(\mathscr{G})$ are the maximal points of $V(\operatorname{Ann}(N))$ (**0**, 1.7.4), and these are also the minimal elements of $\operatorname{Ass}(N)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 3, cor. 1 of prop. 7); as $\operatorname{Ass}(N) \subset \operatorname{Ass}(M) = \operatorname{Ass}(\mathscr{F})$, we see that *c′)* implies *a)*. Finally, *a)* implies *c)* by (3.1.2), taking X affine again, $\mathscr{F} = \widetilde{M}$, and Z defined by the ideal $\mathfrak{j}_x A$ (with the notation of (3.1.2)). □

Corollary (3.1.4). — *Let X be a locally Noetherian prescheme and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module. Then the maximal prime cycles associated to $\mathscr{F}$ are the irreducible components of* $\operatorname{Supp}(\mathscr{F})$*, and the generic points of these components are the points $x \in X$ such that $\mathscr{F}_x$ is an $\mathscr{O}_x$-module of finite nonzero length.*

Proof. Indeed, if x is the generic point of an irreducible component Z of $\operatorname{Supp}(\mathscr{F})$, the equivalence of *a)* and *c′)* in (3.1.3) shows that x belongs to $\operatorname{Ass}(\mathscr{F})$, and Z is an associated prime cycle of $\mathscr{F}$, necessarily maximal by (3.1.1.1); the converse follows trivially from (3.1.1.1). Finally, the last assertion is local and follows from Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, cor. 2 of prop. 7. □

Corollary (3.1.5). — *Let X be a locally Noetherian prescheme and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. For $\mathscr{F} = 0$, it is necessary and sufficient that* $\operatorname{Ass}(\mathscr{F}) = \varnothing$.

Proof. The question being local, we reduce to the case where X is affine, and the conclusion follows immediately from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, cor. 1 of prop. 2. □

Proposition (3.1.6). — *Let X be a locally Noetherian prescheme and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module; then* $\operatorname{Ass}(\mathscr{F})$ *is* locally finite *(that is to say, every point of X admits a neighbourhood whose intersection with* $\operatorname{Ass}(\mathscr{F})$ *is finite).*

Proof. It suffices to consider the case where X is affine, hence Noetherian, and then the proposition follows from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, cor. of th. 2. □

Proposition (3.1.7). — *Let X be a prescheme.*

(i) *Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence of quasi-coherent $\mathscr{O}_X$-modules. Then* $\operatorname{Ass}(\mathscr{F}') \subset \operatorname{Ass}(\mathscr{F}) \subset \operatorname{Ass}(\mathscr{F}') \cup \operatorname{Ass}(\mathscr{F}'')$.

(ii) *Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module, $(\mathscr{F}_\alpha)$ a family of quasi-coherent $\mathscr{O}_X$-submodules of $\mathscr{F}$ such that $\mathscr{F}$ is the union of the $\mathscr{F}_\alpha$. Then* $\operatorname{Ass}(\mathscr{F}) = \bigcup_\alpha \operatorname{Ass}(\mathscr{F}_\alpha)$.

(iii) *For every family $(\mathscr{F}_\alpha)$ of quasi-coherent $\mathscr{O}_X$-modules, we have* $\operatorname{Ass}(\bigoplus_\alpha \mathscr{F}_\alpha) = \bigcup_\alpha \operatorname{Ass}(\mathscr{F}_\alpha)$.

Proof. We immediately reduce to the corresponding propositions for modules (Bourbaki, *loc. cit.*, § 1, no 1, formula (1), prop. 3 and cor. 1 of prop. 3). □

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Proposition (3.1.8). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module, U an open subset of X, and $\mathscr{J}$ a coherent ideal of $\mathscr{O}_X$ defining a closed sub-prescheme of X having $X - U$ as underlying space. The following conditions are equivalent:*

a) $\operatorname{Ass}(\mathscr{F}) \subset U$.
b) *For every affine open subset V of X, every section of $\mathscr{F}$ over V whose restriction to $V \cap U$ is zero is itself zero.*
c) *The canonical homomorphism $\mathscr{F} \to \mathscr{H}om_{\mathscr{O}_X}(\mathscr{J}, \mathscr{F})$ is injective.*

Proof. The question being local, we can suppose $X = \operatorname{Spec}(A)$ affine, A being a Noetherian ring, $\mathscr{F} = \widetilde{M}$, where M is an A-module, and $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A. The homomorphism $M \to \operatorname{Hom}_A(\mathfrak{J}, M)$ sends $m \in M$ to the homomorphism $x \mapsto xm$ of $\mathfrak{J}$ into M; to say that it is not injective means that there exists $m \neq 0$ in M such that $\mathfrak{J}m = 0$.

Let us first show that *c)* implies *a)*. If $\operatorname{Ass}(\mathscr{F})$ met $X - U$, there would be a prime ideal $\mathfrak{p} \in \operatorname{Ass}(M)$ containing $\mathfrak{J}$, hence an element $m \neq 0$ of M such that $\mathfrak{p}m = 0$, and therefore $\mathfrak{J}m = 0$, contradicting *c)*.

Next, *b)* implies *c)*. Indeed, if $\mathfrak{J}m = 0$ for some $m \neq 0$ in M, then for every prime ideal $\mathfrak{q}$ not containing $\mathfrak{J}$ there exists $a \in \mathfrak{J}$ with $a \notin \mathfrak{q}$. The relation $am = 0$ then implies that the canonical image of m in $M_{\mathfrak{q}}$ is zero. Thus m would define a nonzero section of $\mathscr{F}$ over X whose restriction to U is zero, contrary to *b)*.

Finally, *a)* implies *b)*. The canonical homomorphism

$$M \longrightarrow \prod_{\mathfrak{p} \in \operatorname{Ass}(M)} M_{\mathfrak{p}}$$

is injective: if N is its kernel, then $\operatorname{Ass}(N) \subset \operatorname{Ass}(M)$; if there existed $\mathfrak{p} \in \operatorname{Ass}(N)$, there would be an element $n \in N$ whose annihilator is $\mathfrak{p}$, while the definition of N would give an $s \notin \mathfrak{p}$ such that $sn = 0$, a contradiction. Hence $\operatorname{Ass}(N) = \varnothing$, and therefore $N = 0$. Under *a)*, a section $m \in M$ whose restriction to U is zero has zero image in $M_{\mathfrak{p}}$ for every $\mathfrak{p} \in \operatorname{Ass}(M)$; consequently $m = 0$. ☐

Corollary (3.1.9). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module, and f a section of $\mathscr{O}_X$ over X. For f to be $\mathscr{F}$-regular* (**0**, 15.2.2)*, it is necessary and sufficient that* $\operatorname{Ass}(\mathscr{F}) \subset X_f$.

Proof. Indeed, it is immediate that to say that f is $\mathscr{F}$-regular means that the canonical homomorphism $\mathscr{F} \to \mathscr{H}om_{\mathscr{O}_X}(f\mathscr{O}_X, \mathscr{F})$ is injective, and it suffices to apply (3.1.8) to the ideal $\mathscr{J} = f\mathscr{O}_X$. ☐

Proposition (3.1.10). — *Let X and Y be locally Noetherian preschemes and $f : X \to Y$ an integral morphism. Then, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we have $f(\operatorname{Ass}(\mathscr{F})) = \operatorname{Ass}(f_*(\mathscr{F}))$.*

Proof. The question being local on Y, and the morphism f being affine, we are immediately reduced to the case where Y is affine, in other words, to the following lemma. ☐

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Lemma (3.1.10.1). — *Let A and B be two Noetherian rings, $\rho : A \to B$ a ring homomorphism making B an integral A-algebra, and M a B-module. Then the prime ideals $\mathfrak{p} \in \operatorname{Ass}(M_{[\rho]})$ are the inverse images by ρ of the prime ideals $\mathfrak{q} \in \operatorname{Ass}(M)$.*

Proof. Indeed, if $\mathfrak{q} \in \operatorname{Ass}(M)$, then $\mathfrak{q}$ is the annihilator in B of an element $x \in M$, and hence $\rho^{-1}(\mathfrak{q})$ is the annihilator in A of x. Conversely, suppose $\mathfrak{p} \in \operatorname{Ass}(M_{[\rho]})$, so that $\mathfrak{p}$ is the inverse image under ρ of the annihilator $\mathfrak{b}$ in B of an element $x \in M$. It follows from the first Cohen–Seidenberg theorem that there exists a prime ideal $\mathfrak{q}$ of B containing $\mathfrak{b}$ and whose inverse image is $\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, no 1, cor. 2 of th. 1). By considering a prime ideal minimal among those contained in $\mathfrak{q}$ and containing $\mathfrak{b}$, we may evidently suppose that $\mathfrak{q}$ itself is one of these minimal primes. Since $B.x \subset M$ is isomorphic to $B/\mathfrak{b}$, we then have $\mathfrak{q} \in \operatorname{Ass}(B/\mathfrak{b}) \subset \operatorname{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, th. 2). ☐

Corollary (3.1.11). — *Under the hypotheses of (3.1.10), for $\mathscr{F}$ to have no embedded associated prime cycle, it suffices that the same be true of $f_*(\mathscr{F})$.*

Proof. Suppose indeed that $f_*(\mathscr{F})$ has no embedded associated prime cycle. Note that if A is an integral algebra over a field k, all prime ideals of A are maximal (Bourbaki, *Alg. comm.*, chap. V, § 2, no 1, prop. 1); it follows from (**I**, 6.2.2) that the fibres of f are *discrete* spaces. If x and x' are two distinct points of $\operatorname{Ass}(\mathscr{F})$, neither can be adherent to the other when $f(x) = f(x')$; and if $f(x) \neq f(x')$, then (3.1.10) and the hypothesis imply that neither of the two points $f(x)$ and $f(x')$ can be adherent to the other, so the same is true of x and x'. ☐

Remark (3.1.12). — Under the hypotheses of (3.1.10), it may happen, on the contrary, that $\mathscr{F}$ has no embedded associated prime cycle whereas $f_*(\mathscr{F})$ does. For example, take $Y = \operatorname{Spec}(k[\mathrm{T}])$, where k is a field (an "affine line"), and let X be the disjoint union of $X_1 = Y$ and $X_2 = \operatorname{Spec}(k)$, the morphism $X_2 \to Y$ corresponding to the canonical homomorphism $k[\mathrm{T}] \to k[\mathrm{T}]/\mathfrak{m}$, where $\mathfrak{m}$ is the maximal ideal (T). It is clear that the morphism $f : X \to Y$ is finite. If one takes $\mathscr{F} = \mathscr{O}_X$, then $\mathscr{F}$ has no embedded associated prime cycle, but $f_*(\mathscr{F}) = \widetilde{M}$, where M is the direct sum of the $k[\mathrm{T}]$-modules $k[\mathrm{T}]$ and k; hence $\operatorname{Ass}(M)$ consists of the generic point (0) of Y and the point $\mathfrak{m}$.

Proposition (3.1.13). — *Let X be a locally Noetherian prescheme, U an open subset of X, and $i : U \to X$ the canonical inclusion. For every quasi-coherent $\mathscr{O}_U$-module $\mathscr{F}$, we have* $\mathrm{Ass}(i_*(\mathscr{F})) = \mathrm{Ass}(\mathscr{F})$.

Proof. Recall that $i_*(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-module (**I**, 2.2) and (**I**, 7.4); as $i_*(\mathscr{F})|U = \mathscr{F}$, we have $\mathrm{Ass}(i_*(\mathscr{F})) \cap U = \mathrm{Ass}(\mathscr{F})$, and it therefore remains to prove that $\mathrm{Ass}(i_*(\mathscr{F})) \subset U$. But by (3.1.8), this relation means that for every affine open subset V of X, every section of $i_*(\mathscr{F})$ over V which is zero on $U \cap V$, is zero, a condition which is trivially satisfied since $\Gamma(V, i_*(\mathscr{F})) = \Gamma(U \cap V, \mathscr{F}) = \Gamma(U \cap V, i_*(\mathscr{F}))$. □

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3.2. Irredundant decompositions

Proposition (3.2.1). — *Let X be a locally Noetherian prescheme and U a dense open subset of X. The following conditions are equivalent:*

a) *X is reduced.*
b) *The induced sub-prescheme on U is reduced, and X has no embedded associated prime cycle.*
c) *X has no embedded associated prime cycle and, for every generic point x of an irreducible component of X, one has* $\mathrm{length}(\mathscr{O}_x) = 1$.

The prime cycles associated with X are then precisely the irreducible components of X.

Proof. It is clear that if X is reduced, then so is the sub-prescheme induced on U. Moreover, since the existence of embedded associated prime cycles is a local question, we may restrict to the case where $X = \mathrm{Spec}(A)$ is affine and A is Noetherian. If A is reduced, its minimal prime ideals form a reduced primary decomposition of (0) (Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, prop. 10) and are precisely the elements of $\mathrm{Ass}(A)$; hence A has no embedded associated prime ideals. Thus *a)* implies *b)*. It is immediate that *b)* implies *c)*, since a generic point x of an irreducible component belongs to U, and consequently $\mathscr{O}_x$ is a field. Finally, *c)* implies *a)*. Indeed, let $\mathscr{N}$ be the nilradical of $\mathscr{O}_X$, which is a coherent ideal. By hypothesis, $\mathrm{Supp}(\mathscr{N})$ contains none of the generic points of the irreducible components of X. If $\mathrm{Supp}(\mathscr{N})$ were nonempty and x were one of its maximal points, criterion (3.1.3), *c′)*, would give $x \in \mathrm{Ass}(\mathscr{O}_X)$; then $\overline{\{x\}}$ would be an embedded associated prime cycle of X, contrary to the hypothesis. Hence $\mathscr{N} = 0$. □

Definition (3.2.2). — Let X be a locally Noetherian prescheme and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module. We say that $\mathscr{F}$ is *reduced* if it satisfies the following two conditions: 1° $\mathscr{F}$ has no embedded associated prime cycle; 2° for every maximal point x of $\mathrm{Supp}(\mathscr{F})$, one has $\mathrm{length}(\mathscr{F}_x) = 1$.

Condition 1° means that the prime cycles associated with $\mathscr{F}$ are the irreducible components of $\mathrm{Supp}(\mathscr{F})$ (3.1.4), and condition 2° means that for every generic point x of such a component one has $\mathrm{length}(\mathscr{F}_x) = 1$.

For an affine scheme X, this definition gives in particular the notion of a *reduced module* over a Noetherian ring A: a finitely generated A-module M is said to be *reduced* if it has no embedded associated prime ideals and if, for every $\mathfrak{p} \in \mathrm{Ass}(M)$, one has $\mathrm{length}_{A_\mathfrak{p}}(M_\mathfrak{p}) = 1$. Returning to a locally Noetherian prescheme X and a coherent $\mathscr{O}_X$-module $\mathscr{F}$, we say that $\mathscr{F}$ is *reduced at a point* $x \in X$ if $\mathscr{F}_x$ is a reduced $\mathscr{O}_x$-module. Equivalently, on the local scheme $\mathrm{Spec}(\mathscr{O}_x)$, the module associated with $\mathscr{F}_x$ is reduced. Thus x belongs to no embedded associated prime cycle of $\mathscr{F}$ and $\mathrm{length}(\mathscr{F}_z) = 1$ *for every maximal point z of* $\mathrm{Supp}(\mathscr{F})$ *such that* $x \in \overline{\{z\}}$. It is clear that if $\mathscr{F}$ is a reduced coherent $\mathscr{O}_X$-module, then it is reduced at every point of X. Conversely, if $\mathscr{F}$ is *reduced at a point* x, there exists an open neighbourhood U of x such that $\mathscr{F}|U$ is a reduced $\mathscr{O}_U$-module: it suffices to take U meeting no embedded associated prime cycle of $\mathscr{F}$ (such a neighbourhood exists because these cycles form a locally finite family of closed subsets of X). To say that $\mathscr{O}_X$ is reduced at x is equivalent to saying that X is *reduced at the point* x.

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Proposition (3.2.3). — *Let X be a locally Noetherian prescheme, U an open subset of X, and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module such that $U \cap \mathrm{Supp}(\mathscr{F})$ is dense in $\mathrm{Supp}(\mathscr{F})$. The following conditions are equivalent:*

a) *$\mathscr{F}$ is reduced.*
b) *$\mathscr{F}|U$ is reduced, and $\mathscr{F}$ has no embedded associated prime cycle.*

c) *There exist a reduced closed sub-prescheme X' of X and a coherent torsion-free $\mathscr{O}_{X'}$-module $\mathscr{F}'$ of rank* 1 *on every irreducible component of X' such that, if $j : X' \to X$ is the canonical inclusion, then $j_*(\mathscr{F}') = \mathscr{F}$.*

Moreover, when these conditions hold, X' is defined by the ideal $\mathscr{J}$ of $\mathscr{O}_X$ that annihilates $\mathscr{F}$.

Proof. To see that *c)* implies *a)*, we may, by (3.1.3), restrict to the case where $X' = X$ is integral, with generic point x. We may further suppose that $X = \operatorname{Spec}(A)$ is affine and $\mathscr{F} = \widetilde{M}$, where A is integral and M is a torsion-free A-module of rank 1. The annihilator of every element of M is then zero, so $\operatorname{Ass}(\mathscr{F}) = \{x\}$ and $\mathscr{F}_x$ is isomorphic to $\mathscr{O}_x$, the field of fractions of A; hence the conditions of Definition (3.2.2) are satisfied. Since the existence of embedded associated prime cycles is a local question, if $\mathscr{F}$ has no such cycles and $U \cap \operatorname{Supp}(\mathscr{F})$ is dense in $\operatorname{Supp}(\mathscr{F})$, then $\operatorname{Ass}(\mathscr{F}) = \operatorname{Ass}(\mathscr{F}|U)$. Thus *a)* and *b)* are equivalent. If *a)* holds, take for X' the reduced closed sub-prescheme of X whose underlying space is $\operatorname{Supp}(\mathscr{F})$ (**I**, 5.2.1), and put $\mathscr{F}' = j^*(\mathscr{F})$. A point x of $\operatorname{Ass}(\mathscr{F})$ is necessarily a maximal point of X', and since $\operatorname{length}(\mathscr{F}_x) = 1$, $\mathscr{F}'_x$ is isomorphic to $\mathscr{F}_x$, hence to the residue field $\boldsymbol{k}(x) = \mathscr{O}_{X',x}$. This proves that $\mathscr{F}'$ is torsion-free and of rank 1 (**I**, 7.4.6) and (**I**, 7.4.2). Finally, the last assertion is immediate, since for every $y \in X'$ the annihilator of the $\mathscr{O}_{X',y}$-module $\mathscr{F}'_y$ is zero. □

Definition (3.2.4). — Let X be a locally Noetherian prescheme and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. We say that $\mathscr{F}$ is *irredundant* if $\operatorname{Ass}(\mathscr{F})$ consists of a single element x; if $\mathscr{F}$ is of finite type, we say that $\mathscr{F}$ is *integral* if, in addition, $\mathscr{F}$ is reduced (in other words, if $\operatorname{length}(\mathscr{F}_x) = 1$). We say that a quasi-coherent $\mathscr{O}_X$-submodule $\mathscr{G}$ of a quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ is *primary in* $\mathscr{F}$ if $\mathscr{F}/\mathscr{G}$ is irredundant.

For an affine scheme X, this definition gives in particular the notion of an *integral module* over a Noetherian ring A: an A-module M is said to be *integral* if it is of finite type, if M is primary (that is, if $\operatorname{Ass}(M)$ consists of a single prime ideal $\mathfrak{p}$), and if, in addition, $\operatorname{length}_{A_\mathfrak{p}}(M_\mathfrak{p}) = 1$. Returning to an arbitrary locally Noetherian prescheme X and a coherent $\mathscr{O}_X$-module $\mathscr{F}$, we say that $\mathscr{F}$ is *integral at a point* $x \in X$ if $\mathscr{F}_x$ is an integral $\mathscr{O}_x$-module. Equivalently, x belongs to only one associated prime cycle of $\mathscr{F}$ (necessarily a nonembedded one), and $\operatorname{length}(\mathscr{F}_z) = 1$ at its generic point z. It is clear that if $\mathscr{F}$ is an integral coherent $\mathscr{O}_X$-module, then it is integral at every point of X. Conversely, if $\mathscr{F}$ is *integral at a point* x, there exists an open neighbourhood U of x such that $\mathscr{F}|U$ is an integral $\mathscr{O}_U$-module: it suffices to take U such that $\mathscr{F}|U$ is a reduced $\mathscr{O}_U$-module (3.2.2).

We say that a locally Noetherian prescheme X is *irredundant* if $\mathscr{O}_X$ is irredundant (which implies that X is *irreducible*). For X to be *integral*, it is necessary and sufficient that the $\mathscr{O}_X$-module $\mathscr{O}_X$ be integral (**I**, 2.1.8) and (3.2.1). If $\mathscr{O}_X$ is integral at a point x, that is, if the ring $\mathscr{O}_x$ is integral, we say that X is *integral at the point* x. We say that a closed sub-prescheme Y of X is *primary in* X if the ideal $\mathscr{J}$ of $\mathscr{O}_X$ defining Y is primary in $\mathscr{O}_X$. **IV-2** | 42

Definition (3.2.5). — Let X be a locally Noetherian prescheme and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module. An *irredundant decomposition* of $\mathscr{F}$ is a family $(\mathscr{F}_\alpha)_{\alpha \in I}$ of quotient $\mathscr{O}_X$-modules of $\mathscr{F}$ such that the $\mathscr{F}_\alpha$ are irredundant, the family $(\operatorname{Supp}(\mathscr{F}_\alpha))_{\alpha \in I}$ is locally finite, and the canonical homomorphism $\mathscr{F} \to \bigoplus_{\alpha \in I} \mathscr{F}_\alpha$ is injective. We say that such a decomposition is *reduced* if the sets $\operatorname{Ass}(\mathscr{F}_\alpha)$ are pairwise distinct and there is no proper subset $J \subsetneq I$ for which the subfamily $(\mathscr{F}_\alpha)_{\alpha \in J}$ is also an irredundant decomposition of $\mathscr{F}$.

If $(\mathscr{F}_\alpha)_{\alpha \in I}$ is an irredundant decomposition (resp. a reduced irredundant decomposition) of $\mathscr{F}$ and $\mathscr{F}_\alpha = \mathscr{F}/\mathscr{G}_\alpha$, we also say that the family $(\mathscr{G}_\alpha)_{\alpha \in I}$ of $\mathscr{O}_X$-submodules of $\mathscr{F}$ is a *primary decomposition of* 0 *in* $\mathscr{F}$. The injectivity of the canonical homomorphism $\mathscr{F} \to \bigoplus_{\alpha \in I} \mathscr{F}_\alpha$ is equivalent to the condition $\bigcap_{\alpha \in I} \mathscr{G}_\alpha = 0$.

If $(\mathscr{F}_\alpha)$ is an irredundant decomposition of $\mathscr{F}$, to say that it is reduced is equivalent to saying that the sets $\operatorname{Ass}(\mathscr{F}_\alpha)$ are pairwise distinct and contained in $\operatorname{Ass}(\mathscr{F})$. If $\operatorname{Ass}(\mathscr{F}_\alpha) = \{x_\alpha\}$ for every $\alpha \in I$, then $\alpha \mapsto x_\alpha$ is a bijection from I onto $\operatorname{Ass}(\mathscr{F})$. These properties are local and therefore follow from Bourbaki, *Alg. comm.*, chap. IV, § 2, no 3, prop. 4.

Proposition (3.2.6). — *Let X be a locally Noetherian prescheme and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module. Then there exists a reduced irredundant decomposition $(\mathscr{F}^{(x)})_{x \in \operatorname{Ass}(\mathscr{F})}$ formed of coherent $\mathscr{O}_X$-modules such that, for* *every $x \in \mathrm{Ass}(\mathscr{F})$, one has $\mathrm{Ass}(\mathscr{F}^{(x)}) = \{x\}$. For every $x \in \mathrm{Ass}(\mathscr{F})$ such that $\overline{\{x\}}$ is not embedded, $\mathscr{F}^{(x)}$ is uniquely determined as the image of the canonical homomorphism $\mathscr{F} \to i_*(i^*(\mathscr{F}))$, where i is the canonical morphism $\mathrm{Spec}(\boldsymbol{k}(x)) \to X$.*

Proof. For every $x \in \mathrm{Ass}(\mathscr{F})$, let U be an affine open neighbourhood of x, with ring A, and write $\mathscr{F}|U = \widetilde{M}$, where M is a finitely generated A-module. We know (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, prop. 4) that there exists a submodule N of M such that, on setting $P = M/N$, one has $\mathrm{Ass}(P) = \{x\}$ and $\mathrm{Ass}(N) = \mathrm{Ass}(M) - \{x\}$. Let $\mathscr{G} = \widetilde{P}$, a quasi-coherent $\mathscr{O}_U$-module, and let $j : U \to X$ be the canonical inclusion. Let $u : j^*(\mathscr{F}) \to \mathscr{G}$ be the surjective homomorphism corresponding to $M \to P$. It induces a homomorphism $j_*(u) : j_*(j^*(\mathscr{F})) \to j_*(\mathscr{G})$, and hence, by composition, a homomorphism

$$v : \mathscr{F} \xrightarrow{\rho_{\mathscr{F}}} j_*(j^*(\mathscr{F})) \xrightarrow{j_*(u)} j_*(\mathscr{G})$$

whose restriction to U is u. We denote by $\mathscr{F}^{(x)}$ the image of $\mathscr{F}$ under this homomorphism; it is a coherent $\mathscr{O}_X$-module (**I**, 6.1.1). By (3.1.13), one has $\mathrm{Ass}(j_*(\mathscr{G})) = \{x\}$, and therefore, by (3.1.7), $\mathrm{Ass}(\mathscr{F}^{(x)}) = \{x\}$, since $\mathscr{F}^{(x)} \neq 0$.

Let $\mathscr{N}^{(x)} = \mathrm{Ker}(v)$. Then $\mathscr{N}^{(x)}|U = \widetilde{N}$, and hence $x \notin \mathrm{Ass}(\mathscr{N}^{(x)})$. It follows that the homomorphism $\mathscr{F} \to \bigoplus_{x \in \mathrm{Ass}(\mathscr{F})} \mathscr{F}^{(x)}$ is injective. Indeed, its kernel $\mathscr{H}$ is contained in every $\mathscr{N}^{(x)}$, so $\mathrm{Ass}(\mathscr{H})$ is contained in the intersection of the sets $\mathrm{Ass}(\mathscr{N}^{(x)})$, which is empty; consequently, by (3.1.5), $\mathscr{H} = 0$. Since $\mathrm{Ass}(\mathscr{F})$ is locally finite (3.1.6), it is clear that $(\mathscr{F}^{(x)})_{x \in \mathrm{Ass}(\mathscr{F})}$ is a reduced irredundant decomposition of $\mathscr{F}$ satisfying the conditions of the statement. Finally, when $\overline{\{x\}}$ is not embedded, the characterization of $\mathscr{F}^{(x)}$ follows from Bourbaki, *Alg. comm.*, chap. IV, § 2, no 3, prop. 5, since the question is local, together with (**I**, 1.6.7). □

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Corollary (3.2.7). — *Under the hypotheses of (3.2.6), if $\mathscr{F}$ has no embedded associated prime cycle, there exists a unique reduced irredundant decomposition of $\mathscr{F}$.*

Corollary (3.2.8). — *Let X be a Noetherian prescheme and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module. There exists a finite filtration $(\mathscr{F}_i)_{0 \leqslant i \leqslant n}$ of $\mathscr{F}$ such that $\mathscr{F}_0 = \mathscr{F}$ and $\mathscr{F}_n = 0$, formed of coherent $\mathscr{O}_X$-modules, and such that the quotients $\mathscr{F}_i/\mathscr{F}_{i+1}$ are either zero or irredundant, with $\mathrm{Ass}(\mathscr{F}_i/\mathscr{F}_{i+1}) \subset \mathrm{Ass}(\mathscr{F})$.*

Proof. Indeed, by (3.2.6), $\mathscr{F}$ is isomorphic to an $\mathscr{O}_X$-submodule of a finite direct sum $\bigoplus_{j=1}^{n} \mathscr{G}_j$, where the $\mathscr{G}_j$ are irredundant and coherent. Since every quasi-coherent $\mathscr{O}_X$-submodule of $\mathscr{G}_j$ is either zero or irredundant (3.1.7), the modules

$$\mathscr{F}_i = \mathscr{F} \cap \left(\bigoplus_{j=1}^{n-i} \mathscr{G}_j \right)$$

satisfy the requirements, since $\mathscr{F}_i/\mathscr{F}_{i+1}$ is isomorphic to a coherent $\mathscr{O}_X$-submodule of $\mathscr{G}_{n-i}$. □

3.3. Relations with flatness

Proposition (3.3.1). — *Let $f : X \to Y$ be a morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module that is f-flat, and $\mathscr{E}$ a quasi-coherent $\mathscr{O}_Y$-module. If, for every $y \in Y$, we set $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} \boldsymbol{k}(y)$, then*

(3.3.1.1) $$\mathrm{Ass}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}) \supset \bigcup_{y \in \mathrm{Ass}(\mathscr{E})} \mathrm{Ass}(\mathscr{F}_y)$$

and the two sides are equal if Y is locally Noetherian.

Proof. (Of course, $\mathscr{F}_y$ is a sheaf on the fibre $f^{-1}(y)$, and this fibre is identified with a subspace of X (**I**, 3.6.1).) The question being local on X and on Y, we reduce to the case where X and Y are affine, and the proposition is then proved in Bourbaki, *Alg. comm.*, chap. IV, § 2, no 6, th. 2. □

Corollary (3.3.2). — *Let Y be a locally Noetherian prescheme with no embedded associated prime cycles, $f : X \to Y$ a morphism, and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module that is f-flat. Then, for every $x \in \mathrm{Ass}(\mathscr{F})$, $f(x)$ is a maximal point of Y.*

Proof. It suffices to apply (3.3.1) with $\mathscr{E} = \mathscr{O}_Y$, since $\mathrm{Ass}(\mathscr{O}_Y)$ is by hypothesis the set of maximal points of Y. □

Corollary (3.3.3). — *Under the hypotheses of (3.3.1), suppose in addition that X and Y are locally Noetherian and that $\mathscr{E}$ and $\mathscr{F}$ are coherent. Then the following conditions are equivalent:*

a) *$\mathscr{E} \otimes_Y \mathscr{F}$ has no embedded associated prime cycle.*
b) *For every point $y \in \mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, $\overline{\{y\}}$ is a nonembedded associated prime cycle of $\mathscr{E}$ and $\mathscr{E}_y \otimes_{\boldsymbol{k}(y)} \mathscr{F}_y$ has no embedded associated prime cycle.*

Proof. Suppose *a)* is satisfied. The hypotheses imply that $\mathscr{E} \otimes_Y \mathscr{F}$ is a coherent $\mathscr{O}_X$-module (**0**, 5.3.11) and (**0**, 5.3.5); its associated prime cycles are therefore the irreducible components of $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$ (3.1.4), and for every maximal point x of $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$, $f(x) = y$ is a maximal point of $\mathrm{Supp}(\mathscr{E})$ and x is a maximal point of $\mathrm{Supp}(\mathscr{F}_y)$ (2.5.5). As, by virtue of (3.3.1) and the fact that $y \in f(\mathrm{Supp}(\mathscr{F}))$ implies $\mathscr{F}_y \neq 0$ (**I**, 9.1.13), every point of $\mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$ is the image by f of a maximal point of $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$, we see that condition *b)* is satisfied. **IV-2 | 44**

Conversely, suppose *b)* is satisfied, and let us show that if z, z' are two distinct points of $\mathrm{Ass}(\mathscr{E} \otimes_Y \mathscr{F})$, neither of the two can be adherent to the other. In the first place, if $f(z) = f(z') = y$, we have $y \in \mathrm{Ass}(\mathscr{E})$ and z and z' belong to $\mathrm{Ass}(\mathscr{F}_y)$ by (3.3.1), whence $y \in f(\mathrm{Supp}(\mathscr{F}))$; as by hypothesis neither of the two points z, z' is adherent to the other *within* $f^{-1}(y)$, neither can be adherent to the other *within* X. If $y = f(z)$ and $y' = f(z')$ are distinct, they belong to $\mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, hence neither of the two can be adherent to the other in Y; it follows from the continuity of f that neither of the points z, z' can be adherent to the other in X. □

Proposition (3.3.4). — *Let X and Y be locally Noetherian preschemes, $f : X \to Y$ a morphism, $\mathscr{E}$ a coherent $\mathscr{O}_Y$-module, and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module that is f-flat. Then the following conditions are equivalent:*

a) *$\mathscr{E} \otimes_Y \mathscr{F}$ is reduced (3.2.2).*
b) *For every point $y \in \mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, $\overline{\{y\}}$ is a nonembedded associated prime cycle of $\mathscr{E}$, $\mathrm{length}(\mathscr{E}_y) = 1$, and $\mathscr{F}_y$ is reduced.*

Proof. Suppose *a)* is satisfied. We already know from (3.3.3) that, for every $y \in \mathrm{Ass}(\mathscr{E})$ lying in $f(\mathrm{Supp}(\mathscr{F}))$, $\overline{\{y\}}$ is a nonembedded associated prime cycle of $\mathscr{E}$ and $\mathscr{F}_y$ has no embedded associated prime cycle. Furthermore, by (2.5.5), for every $x \in \mathrm{Ass}(\mathscr{E} \otimes_Y \mathscr{F}) \cap f^{-1}(y)$, we have

$$1 = \mathrm{length}((\mathscr{E} \otimes_Y \mathscr{F})_x) = \mathrm{length}(\mathscr{E}_y)\,\mathrm{length}((\mathscr{F}_y)_x),$$

hence $\mathrm{length}(\mathscr{E}_y) = \mathrm{length}((\mathscr{F}_y)_x) = 1$, which proves *b)*.

Conversely, suppose *b)* is satisfied; we already know that every point $x \in \mathrm{Ass}(\mathscr{E} \otimes_Y \mathscr{F})$ is a maximal point of $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$, that $y = f(x)$ is a maximal point of $\mathrm{Supp}(\mathscr{E})$ and x a maximal point of $\mathrm{Supp}(\mathscr{F}_y)$ (3.3.1) and (3.3.3); in addition it follows from the hypothesis and from (2.5.5) that $\mathrm{length}((\mathscr{E} \otimes_Y \mathscr{F})_x) = 1$, which proves *a)*. □

Corollary (3.3.5). — *Let X and Y be locally Noetherian preschemes and let $f : X \to Y$ be a flat morphism. If Y is reduced at the points of $f(X)$ and $f^{-1}(y)$ is a reduced $\boldsymbol{k}(y)$-prescheme for every $y \in f(X)$, then X is reduced.*

Proof. Since the nilradical $\mathscr{N}_Y$ is coherent, the set of points where Y is reduced is open (**0**, 5.2.2), and we may restrict to the case where Y is reduced. It suffices then to apply (3.3.4) with $\mathscr{E} = \mathscr{O}_Y$ and $\mathscr{F} = \mathscr{O}_X$. □

Proposition (3.3.6). — *Let $f : X \to S$ and $g : Y \to S$ be two morphisms, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module, and $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Y$-module. Suppose that: 1º $\mathscr{G}$ is g-flat; 2º X is locally Noetherian, and for every $s \in S$, $g^{-1}(s)$ is locally Noetherian (which will hold if Y is also locally Noetherian). Let $Z = X \times_S Y$; for every pair (x, y) such that $x \in X$, $y \in Y$, and $f(x) = g(y) = s$, let $\mathrm{T}_{x,y}$ be the prescheme $\mathrm{Spec}(\boldsymbol{k}(x) \otimes_{\boldsymbol{k}(s)} \boldsymbol{k}(y))$, and let $\mathrm{I}_{x,y}$ be the image of $\mathrm{Ass}(\mathscr{O}_{\mathrm{T}_{x,y}})$ by the canonical monomorphism $\mathrm{T}_{x,y} \to Z$ (**I**, 3.4.9). Then* **IV-2 | 45**

$$\mathrm{Ass}(\mathscr{F} \otimes_S \mathscr{G}) = \bigcup_{x \in \mathrm{Ass}(\mathscr{F})} \left(\bigcup_{y \in \mathrm{Ass}(\mathscr{G}_{f(x)})} \mathrm{I}_{x,y} \right) \tag{3.3.6.1}$$

where for all $s \in S$, $\mathscr{G}_s = \mathscr{G} \otimes_{\mathscr{O}_S} \boldsymbol{k}(s)$.

Proof. Let $p : Z \to X$, $q : Z \to Y$ be the canonical projections, so that we have the commutative diagram

$$\begin{tikzcd} X \arrow[d, "f"'] & Z \arrow[l, "p"'] \arrow[d, "q"] \\ S & Y \arrow[l, "g"] \end{tikzcd}$$

Set $\mathscr{G}' = q^*(\mathscr{G})$, so that $\mathscr{F} \otimes_S \mathscr{G} = \mathscr{F} \otimes_X \mathscr{G}'$; as $\mathscr{G}'$ is p-flat (2.1.4), it follows from (3.3.1) that

(3.3.6.2) $$\operatorname{Ass}(\mathscr{F} \otimes_S \mathscr{G}) = \bigcup_{x \in \operatorname{Ass}(\mathscr{F})} \operatorname{Ass}(\mathscr{G}'_x)$$

with $\mathscr{G}'_x = \mathscr{G}' \otimes_{\mathscr{O}_X} \boldsymbol{k}(x)$. If $s = f(x)$, we have $\mathscr{G}'_x = \mathscr{G}_s \otimes_{\boldsymbol{k}(s)} \boldsymbol{k}(x)$, and $p^{-1}(x) = g^{-1}(s) \otimes_{\boldsymbol{k}(s)} \boldsymbol{k}(x)$; furthermore, as the field $\boldsymbol{k}(x)$ is a flat $\boldsymbol{k}(s)$-module, the morphism $p^{-1}(x) \to g^{-1}(s)$ is flat; applying (3.3.1) to this morphism, we get

(3.3.6.3) $$\operatorname{Ass}(\mathscr{G}'_x) = \bigcup_{y \in \operatorname{Ass}(\mathscr{G}_s)} \operatorname{Ass}(\mathscr{O}_{\mathrm{T}_{x,y}})$$

whence the proposition. □

We note that if, in the statement, we suppress hypothesis 2°, we can still conclude, by virtue of (3.3.1), the relation

(3.3.6.4) $$\operatorname{Ass}(\mathscr{F} \otimes_S \mathscr{G}) \supset \bigcup_{x \in \operatorname{Ass}(\mathscr{F})} \Bigl(\bigcup_{y \in \operatorname{Ass}(\mathscr{G}_{f(x)})} \mathrm{I}_{x,y} \Bigr).$$

Corollary (3.3.7). — *Under the hypotheses of (3.3.6), suppose in addition that S is also locally Noetherian and that $f(\operatorname{Ass}(\mathscr{F})) \subset \operatorname{Ass}(\mathscr{O}_S)$. Then*

(3.3.7.1) $$\operatorname{Ass}(\mathscr{F} \otimes_S \mathscr{G}) = \bigcup_{(x,y) \in C} \mathrm{I}_{x,y}$$

where C is the set of pairs (x, y) such that $x \in \operatorname{Ass}(\mathscr{F})$, $y \in \operatorname{Ass}(\mathscr{G})$ and $f(x) = g(y)$.

Proof. Since $\mathscr{G}$ is g-flat, it follows from (3.3.1) that the relation "$s \in \operatorname{Ass}(\mathscr{O}_S)$ and $y \in \operatorname{Ass}(\mathscr{G}_s)$" is equivalent to "$y \in \operatorname{Ass}(\mathscr{G})$"; the conclusion follows from (3.3.6.1). □

Remarks (3.3.8). — We shall see later (4.2.2) that under the hypotheses of (3.3.6), $\mathrm{T}_{x,y}$ is a prescheme *with no embedded associated prime cycle*; it will follow that if $\mathscr{F}$ and the $\mathscr{G}_s$ have no embedded associated prime cycles, the same is true of $\mathscr{F} \otimes_S \mathscr{G}$.

Corollary (3.3.9). — *Under the conditions of (3.3.7), we have*

(3.3.9.1) $$q(\operatorname{Ass}(\mathscr{F} \otimes_S \mathscr{G})) \subset \operatorname{Ass}(\mathscr{G})$$

(where $q : X \times_S Y \to Y$ is the canonical projection).

Proof. Indeed, if $(x, y) \in Z$, we have $q(\mathrm{I}_{x,y}) = \{y\} \subset \operatorname{Ass}(\mathscr{G})$. □

3.4. Properties of the sheaves $\mathscr{F}/t\mathscr{F}$

Proposition (3.4.1). — *Let X be a locally Noetherian prescheme, t a section of $\mathscr{O}_X$ over X, and Y the closed sub-prescheme of X defined by the ideal $t\mathscr{O}_X$. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module, let S be the reduced closed sub-prescheme of X having $\operatorname{Supp}(\mathscr{F})$ as underlying space, and let (S_i) be the family of reduced closed sub-preschemes of X whose underlying spaces are the irreducible components of S; denote by s_i the generic point of S_i. Finally, let Z be an irreducible component of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F}) = S \cap Y$, z its generic point.*

(i) *For every i such that $Z \subset S_i$, Z is an irreducible component of $S_i \cap Y$.*

(ii) *If Z is not equal to any of the S_i, then*

$$\operatorname{length}((\mathscr{F}/t\mathscr{F})_z) \geqslant \sum_i \operatorname{length}(\mathscr{F}_{s_i}) \tag{3.4.1.1}$$

where the sum on the right-hand side is extended to all i such that $Z \subset S_i$.

(iii) *Suppose that Z is not equal to any of the S_i. For the two sides of (3.4.1.1) to be equal, it is necessary and sufficient that the following two conditions be satisfied:*

α) *t_z is $\mathscr{F}_z$-regular* ($\mathbf{0}$, 15.1.4).

β) *For every i such that $Z \subset S_i$, the canonical image of the germ t_z in $\mathscr{O}_{S_i,z}$ generates the maximal ideal of this ring (which implies that $\mathscr{O}_{S_i,z}$ is a discrete valuation ring and that the image of t_z is a uniformizer of this ring).*

Proof. *(i)* If $j : Y \to X$ is the canonical injection, then

$$\mathscr{F}/t\mathscr{F} = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_Y = j^*\mathscr{F},$$

and hence

$$\operatorname{Supp}(\mathscr{F}/t\mathscr{F}) = j^{-1}(S) = S \cap Y$$

by ($\mathbf{I}$, 9.1.13), which proves the assertion.

(ii) and (iii) The points s_i such that $Z \subset S_i$ are precisely those belonging to $\operatorname{Spec}(\mathscr{O}_z)$. Thus, to prove *(ii)* and *(iii)*, we may replace X by $\operatorname{Spec}(\mathscr{O}_z)$. If $M = \mathscr{F}_z$, we may therefore suppose that $\mathscr{F} = \widetilde{M}$, and then $\mathscr{F}_{s_i} = M_{\mathfrak{p}_i}$, where the $\mathfrak{p}_i$ are the minimal prime ideals of $\mathscr{O}_z$. Moreover, since M is an $\mathscr{O}_z$-module of finite type, we have $S = S'_{\text{red}}$, where

$$S' = \operatorname{Spec}(\mathscr{O}_z/\mathfrak{a})$$

and $\mathfrak{a}$ is the annihilator of M in $\mathscr{O}_z$ ($\mathbf{0}$, 1.7.4). The two sides of (3.4.1.1) have the same values whether M is regarded as an $\mathscr{O}_z$-module or as an $(\mathscr{O}_z/\mathfrak{a})$-module. We may therefore finally replace X by $S' = \operatorname{Spec}(A)$, where A is a Noetherian local ring and M is a faithful A-module. Since Z is closed in S, the hypothesis that $Z \neq S_i$ for every i means that $s_i \notin Z$, and hence that $\dim(A) > 0$. Finally, to say that z is the generic point of Z, an irreducible component of $S \cap V(t)$, means that A/tA has dimension 0 (in other words, that it is a local Artinian ring). We are thus reduced to the following lemma. ☐

Lemma (3.4.1.2). — *Let A be a Noetherian local ring of dimension > 0, let $\mathfrak{p}_i$ be the minimal prime ideals of A, let $\mathfrak{m}$ be its maximal ideal, and let $t \in \mathfrak{m}$ be such that A/tA is Artinian. Then, for every A-module of finite type M,*

$$\operatorname{length}(M/tM) \geqslant \sum_i \operatorname{length}(M_{\mathfrak{p}_i}). \tag{3.4.1.3}$$

Moreover, equality holds if and only if the following two conditions are satisfied:

α) *t is M-regular;*

β) *for every i such that $M_{\mathfrak{p}_i} \neq 0$, the image of t in $A/\mathfrak{p}_i$ generates the maximal ideal of that ring (which implies that $A/\mathfrak{p}_i$ is a discrete valuation ring).*

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Proof. Since A does not have dimension 0 and A/tA is Artinian, necessarily $\dim(A) = 1$ ($\mathbf{0}$, 16.3.4), and $t \notin \mathfrak{p}_i$ for every i. The principal ideal (t) is therefore an ideal of definition of A, and hence contains a power of the maximal ideal $\mathfrak{m}$. Let N be the submodule of M consisting of the elements annihilated by a power of t (or, equivalently in view of what we have just proved, by a power of $\mathfrak{m}$), and put $P = M/N$. Then t is P-regular: if $tx \in N$ for some $x \in M$, then $t^k(tx) = 0$ for some integer k, and hence $x \in N$. We shall use the following elementary lemma.

Lemma (3.4.1.4). — *Let A be a ring and*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

an exact sequence of A-modules. If $t \in A$ is M''-regular, then the sequence

$$0 \longrightarrow M'/tM' \longrightarrow M/tM \longrightarrow M''/tM'' \longrightarrow 0$$

is exact.

Proof. Since $M/tM = M \otimes_A (A/tA)$, it suffices to prove exactness at M'/tM'. Suppose that the image $x \in M$ of an element $x' \in M'$ is of the form $x = ty$, with $y \in M$. For the images x'', y'' of x, y in M'', we have $x'' = ty''$. But $x'' = 0$, so the hypothesis gives $y'' = 0$. Thus y is the image of an element $y' \in M'$, and the relation $x = ty$ gives $x' = ty'$ because $M' \to M$ is injective. □

It follows from this lemma that

$$\text{length}(M/tM) = \text{length}(N/tN) + \text{length}(P/tP). \tag{3.4.1.5}$$

On the other hand, $N_{\mathfrak{p}_i} = 0$ for every i, since $t \notin \mathfrak{p}_i$, and hence $M_{\mathfrak{p}_i} = P_{\mathfrak{p}_i}$. Thus, to prove (3.4.1.3), it suffices to prove it with P in place of M. Furthermore, if the two sides of (3.4.1.3) are equal, then the same inequality for P, together with (3.4.1.5), gives $\text{length}(N/tN) = 0$. Consequently $N/tN = 0$, and Nakayama's lemma gives $N = 0$, since N is of finite type. But $N = 0$ means that t is M-regular. We may therefore reduce to the case $M = P$, that is, suppose from the outset that t is M-regular. This implies that $\mathfrak{m} \notin \text{Ass}(M)$, since t cannot annihilate a nonzero element of M. Because $\dim(A) = 1$, it follows that

$$\text{Ass}(M) \subset \{\mathfrak{p}_i\}_i.$$

We now argue by induction on

$$n = \sum_i \text{length}(M_{\mathfrak{p}_i}).$$

If $n = 0$, then $M_{\mathfrak{p}_i} = 0$ for every i, and hence $M = 0$, since none of the $\mathfrak{p}_i$ belongs to $\text{Ass}(M)$. Both sides of (3.4.1.3) are then zero, and condition *β)* is vacuous. If $n > 0$, the argument at the beginning of the proof of (3.4.1) allows us, moreover, to suppose that M is faithful. It follows that $M_{\mathfrak{p}_i} \neq 0$ for every i (Bourbaki, *Alg. comm.*, chap. II, § 2, no 2, cor. 2 of prop. 4), and consequently

$$\text{Ass}(M) = \{\mathfrak{p}_i\}_i.$$

Suppose first that $n = 1$. There is then only one minimal prime ideal $\mathfrak{p}$ of A, and the assertion that $M_{\mathfrak{p}}$ has length 1 means that it is isomorphic, as an $A_{\mathfrak{p}}$-module, to the residue field **IV-2** | 48

$$k = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

Thus $M_{\mathfrak{p}}$ is annihilated by $\mathfrak{p}A_{\mathfrak{p}}$, and consequently $\mathfrak{p}$ is the annihilator of M (Bourbaki, *Alg. comm.*, chap. II, § 2, no 4, formula (9)). Since M is faithful, $\mathfrak{p} = 0$, and hence A is integral. Because $M \neq 0$, Nakayama's lemma gives $M/tM \neq 0$, so that $\text{length}(M/tM) \geqslant 1$; this proves (3.4.1.3) in this case. Moreover, if $\text{length}(M/tM) = 1$, then M is cyclic (Bourbaki, *Alg. comm.*, chap. II, § 3, no 2, cor. 2 of prop. 4), and hence isomorphic to a quotient $A/\mathfrak{b}$. Since M is faithful, necessarily $\mathfrak{b} = 0$, and M is isomorphic to A. As $\text{length}(A/tA) = 1$, the ideal tA is the maximal ideal $\mathfrak{m}$. Since A is a Noetherian integral local ring, it follows that A is a discrete valuation ring (Bourbaki, *Alg. comm.*, chap. VI, § 3, no 6, prop. 9), with uniformizer t. Conversely, if A is a discrete valuation ring, t is its uniformizer, $\text{length}(M_{\mathfrak{p}}) = 1$, and t is M-regular, then M is torsion-free and therefore isomorphic to a submodule of A (as M is of finite type), hence to A itself. Thus

$$\text{length}(M/tM) = \text{length}(A/tA) = 1.$$

Suppose now that $n \geqslant 2$. There exists an exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

with $M' \neq 0$, $M'' \neq 0$, and

$$\text{Ass}(M) = \text{Ass}(M') \cup \text{Ass}(M'').$$

Indeed, if $\text{Ass}(M)$ has more than one element, this follows from Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, prop. 4. If, on the contrary, $\text{Ass}(M)$ consists of a single prime ideal, that ideal is necessarily the unique minimal prime ideal $\mathfrak{p}$ of A. The hypothesis then gives $\text{length}(M_{\mathfrak{p}}) \geqslant 2$, and it is enough to take for M' the inverse image of a submodule of $M_{\mathfrak{p}}$ distinct from both 0 and $M_{\mathfrak{p}}$. Since t is M-regular, it belongs to none of the prime ideals in $\operatorname{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, cor. 2 of prop. 2). For the same reason, t is both M'-regular and M''-regular. By (3.4.1.4), the sequence

$$0 \longrightarrow M'/tM' \longrightarrow M/tM \longrightarrow M''/tM'' \longrightarrow 0$$

is exact. For every i, so is the sequence

$$0 \longrightarrow M'_{\mathfrak{p}_i} \longrightarrow M_{\mathfrak{p}_i} \longrightarrow M''_{\mathfrak{p}_i} \longrightarrow 0.$$

Consequently

$$\operatorname{length}(M/tM) = \operatorname{length}(M'/tM') + \operatorname{length}(M''/tM''),$$
$$\operatorname{length}(M_{\mathfrak{p}_i}) = \operatorname{length}(M'_{\mathfrak{p}_i}) + \operatorname{length}(M''_{\mathfrak{p}_i}).$$

The induction hypothesis now proves (3.4.1.3). Moreover, the two sides can be equal only if the analogous inequalities for M' and M'' are both equalities. By the induction hypothesis, this is equivalent to condition *β)* for those $\mathfrak{p}_i$ such that $M'_{\mathfrak{p}_i} \neq 0$ or $M''_{\mathfrak{p}_i} \neq 0$; these are precisely the ideals for which $M_{\mathfrak{p}_i} \neq 0$. □

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Corollary (3.4.2). — *Under the general hypotheses of (3.4.1), suppose that Z is not equal to any of the S_i and that* $\operatorname{length}((\mathscr{F}/t\mathscr{F})_z) = 1$*. Then there is a unique S_i containing Z, and for this value of i,* $\operatorname{length}(\mathscr{F}_{s_i}) = 1$*; moreover, $\mathscr{O}_{S_i,z}$ is a discrete valuation ring with uniformizer t_z, and t_z is $\mathscr{F}_z$-regular.*

Proof. This follows from (3.4.1), the two sides of (3.4.1.1) being then equal. □

Proposition (3.4.3). — *Let X be a locally Noetherian prescheme, t a section of $\mathscr{O}_X$ over X, and Y the closed sub-prescheme of X defined by the ideal $t\mathscr{O}_X$. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module, let T be an associated prime cycle of $\mathscr{F}$, let T' be an irreducible component of $T \cap Y$, and let x be the generic point of T'. Suppose that t_x is $\mathscr{F}_x$-regular; then $x \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F})$.*

Proof. As in the proof of (3.4.1), we may reduce to the case $X = \operatorname{Spec}(\mathscr{O}_x)$. Taking (3.1.2) into account, the proposition is then a consequence of (**0**, 16.4.6.3). □

Proposition (3.4.4). — *Let X be a locally Noetherian prescheme, t a section of $\mathscr{O}_X$ over X, and Y the closed sub-prescheme of X defined by the ideal $t\mathscr{O}_X$. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module, and let (S_i) be the family of irreducible components of* $\operatorname{Supp}(\mathscr{F})$*. Let y be a point of Y such that t_y is $\mathscr{F}_y$-regular and no embedded associated prime cycle of $\mathscr{F}/t\mathscr{F}$ contains y. Then the irreducible components of* $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$ *containing y are precisely the irreducible components of the $S_i \cap Y$ containing y, and the associated prime cycles of $\mathscr{F}$ containing y are non-embedded.*

Proof. Let us first prove the last assertion. Let $T \supset T_1$ be two associated prime cycles of $\mathscr{F}$ containing y. If x is the generic point of an irreducible component of $T \cap Y$ containing y, then x is a generization of y, and hence belongs to every neighbourhood of y. The hypothesis that t_y is $\mathscr{F}_y$-regular therefore implies that t_x is $\mathscr{F}_x$-regular (**0**, 15.2.4), so (3.4.3) gives $x \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F})$. Let x_1 be the generic point of an irreducible component of $T_1 \cap Y$ containing y, and let x be the generic point of an irreducible component of $T \cap Y$ containing x_1. The preceding argument shows that x_1 and x both belong to $\operatorname{Ass}(\mathscr{F}/t\mathscr{F})$. Since $x_1 \in \overline{\{x\}}$, the hypothesis implies $x_1 = x$.

Again denote by T and T_1 the integral closed sub-preschemes of X having these spaces as their respective underlying spaces, and set

$$A = \mathscr{O}_{T,x}, \qquad A_1 = \mathscr{O}_{T_1,x}.$$

Then $A_1 = A/\mathfrak{p}$ for a prime ideal $\mathfrak{p}$ of A. By the definitions of x and x_1, the rings A/tA and A_1/tA_1 are Artinian. On the other hand, we saw above that t_x is $\mathscr{F}_x$-regular, so $x \notin \operatorname{Ass}(\mathscr{F})$, and consequently A_1 is not Artinian. Thus $\dim A = \dim A_1 = 1$. This implies $\mathfrak{p} = 0$ and $A = A_1$ (**0**, 16.1.2.2). Since $\operatorname{Spec}(A)$ and $\operatorname{Spec}(A_1)$ are respectively dense in T and T_1, it follows that $T = T_1$.

The S_i containing y are therefore all the associated prime cycles of $\mathscr{F}$ containing y. If x is the generic point of an irreducible component of $S_i \cap Y$ containing y, then (**0**, 15.2.4) again shows that t_x is $\mathscr{F}_x$-regular, and (3.4.3) gives $x \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F})$. This proves the first assertion of (3.4.4). □

Proposition (3.4.5). — *Let X be a locally Noetherian prescheme, t a section of $\mathscr{O}_X$ over X, and Y the closed sub-prescheme of X defined by the ideal $t\mathscr{O}_X$. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module and y a point of Y. Suppose that t_y is $\mathscr{F}_y$-regular and that $\mathscr{F}/t\mathscr{F}$ is integral at y (3.2.4). Then $\mathscr{F}$ is integral at the point y.* **IV-2 | 50**

Proof. Taking into account (3.4.4), it suffices to prove that y is contained in a *unique* irreducible component of $\operatorname{Supp}(\mathscr{F})$, and that if s is the generic point of this component, $\operatorname{length}(\mathscr{F}_s) = 1$. By hypothesis, y belongs to only one irreducible component of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$, and if z is the generic point of this component, then $\operatorname{length}((\mathscr{F}/t\mathscr{F})_z) = 1$. The conclusion follows from (3.4.2). ☐

Proposition (3.4.6). — *The hypotheses being those of (3.4.1), let x be a point of Y. Suppose that Y does not contain any of the S_i containing x, and that $\mathscr{F}/t\mathscr{F}$ is reduced at the point x (3.2.2). Then t_x is $\mathscr{F}_x$-regular and $\mathscr{F}$ is reduced at the point x. Moreover, if z_j is the generic point of an irreducible component of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$ containing x, then z_j is contained in a unique S_i, and $\mathscr{O}_{S_i,z_j}$ is a discrete valuation ring with uniformizer t_{z_j}.*

Proof. The assertion that t_x is $\mathscr{F}_x$-regular follows by applying the following lemma to the ring $\mathscr{O}_x$.

Lemma (3.4.6.1). — *Let A be a Noetherian ring, M an A-module of finite type, $\mathfrak{p}_i$ the minimal elements of $\operatorname{Supp}(M)$, and $t \in A$. Suppose that t belongs to none of the $\mathfrak{p}_i$ and that M/tM is a reduced A-module (3.2.2). Then t is M-regular.*

Proof. Every prime ideal $\mathfrak{p} \in \operatorname{Supp}(M)$ contains one of the $\mathfrak{p}_i$. Since t belongs to none of the $\mathfrak{p}_i$, multiplication by t on $M_\mathfrak{p}$ is not nilpotent (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, cor. of prop. 9). Let N be the submodule of M consisting of the elements annihilated by a power of t, and set $P = M/N$. We shall prove that $N = 0$. Since t is P-regular, (3.4.1.4) gives an exact sequence

$$0 \longrightarrow N/tN \longrightarrow M/tM \longrightarrow P/tP \longrightarrow 0.$$

Since N is of finite type, it is annihilated by a power of t, and it therefore suffices to prove that $N/tN = 0$. As N/tN is a submodule of M/tM, it suffices to prove that $(N/tN)_\mathfrak{p} = 0$ for every $\mathfrak{p} \in \operatorname{Ass}(M/tM)$, or equivalently that

$$u_\mathfrak{p} : (M/tM)_\mathfrak{p} \longrightarrow (P/tP)_\mathfrak{p}$$

is bijective for every $\mathfrak{p} \in \operatorname{Ass}(M/tM)$. Now $(P/tP)_\mathfrak{p} \neq 0$. Indeed,

$$\mathfrak{p} \in \operatorname{Supp}(M/tM) = \operatorname{Supp}(M) \cap V(t)$$

by ($\mathbf{0}$, 1.7.5), so the image of t in $A_\mathfrak{p}$ lies in the maximal ideal $\mathfrak{p}A_\mathfrak{p}$. The equality $P_\mathfrak{p} = tP_\mathfrak{p}$ would therefore imply $P_\mathfrak{p} = 0$ by Nakayama's lemma. We would then have $M_\mathfrak{p} = N_\mathfrak{p}$, and multiplication by t on $M_\mathfrak{p}$ would be nilpotent, contradicting the observation above because $\mathfrak{p} \in \operatorname{Supp}(M)$. Finally, since M/tM is reduced, $\operatorname{length}((M/tM)_\mathfrak{p}) = 1$. As $(P/tP)_\mathfrak{p} \neq 0$, the map $u_\mathfrak{p}$ is necessarily bijective. This proves the lemma. ☐

By hypothesis, no embedded associated prime cycle of $\mathscr{F}/t\mathscr{F}$ contains x. It follows from (3.4.4) that no embedded associated prime cycle of $\mathscr{F}$ contains x. On the other hand, applying (3.4.2) to an irreducible component of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$ containing x, we find that $\operatorname{length}(\mathscr{F}_{s_i}) = 1$ for every S_i containing x. This completes the proof that $\mathscr{F}_x$ is reduced. The final assertions also follow from (3.4.2). ☐

Corollary (3.4.7). — *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, M an A-module of finite type, and $(x_i)_{1\leqslant i\leqslant k}$ a family of elements of $\mathfrak{m}$ forming part of a system of parameters for M ($\mathbf{0}$, 16.3.6). If the A-module* **IV-2 | 51**

$$N = M\Big/\Big(\sum_{i=1}^{k} x_i M\Big)$$

is integral (3.2.4), then M is integral and the sequence $(x_i)_{1\leqslant i\leqslant k}$ is M-regular.

Proof. Induction on k immediately reduces us to the case $k = 1$; write x for x_1. The hypothesis that x forms part of a system of parameters for M implies

$$\dim(N) = \dim(M) - 1$$

by (**0**, 16.3.7). Put $n = \dim(N)$. There is therefore a minimal element $\mathfrak{p}$ of $\mathrm{Supp}(M)$ such that

$$\dim(M/\mathfrak{p}M) = n+1$$

by (**0**, 16.3.4); for every integer $j > 0$ one also has

$$\dim(M/\mathfrak{p}^j M) = n+1$$

by (**0**, 16.3.5). Moreover, x forms part of a system of parameters for $M/\mathfrak{p}^j M$ (**0**, 16.3.5). Set

$$M' = M/\mathfrak{p}^j M, \qquad N' = M'/xM';$$

then $\dim(N') = n$.

There is an evident surjective homomorphism $v : N \to N'$. We claim that it is bijective. Indeed, if $P = \mathrm{Ker}(v)$, then $\mathrm{Ass}(P) \subset \mathrm{Ass}(N)$. Since N is integral, if $P \neq 0$, then $\mathrm{Ass}(P)$ and $\mathrm{Ass}(N)$ would both consist of the unique point $\mathfrak{q}$ of

$$\mathrm{Ass}(N) = \mathrm{Supp}(N).$$

But $\dim(N') = \dim(N)$ implies $N'_{\mathfrak{q}} \neq 0$, and $\mathrm{length}(N_{\mathfrak{q}}) = 1$ implies $\mathrm{length}(N'_{\mathfrak{q}}) = 1$, since $N_{\mathfrak{q}} \to N'_{\mathfrak{q}}$ is surjective. Thus $P_{\mathfrak{q}} = 0$, a contradiction. Hence v is bijective.

It follows that N', being isomorphic to N, is integral. Moreover, the support of N', which is the intersection of $\mathrm{Supp}(M')$ with $V(x)$, cannot contain $\mathrm{Supp}(M')$, and the latter is irreducible by construction. Since M'/xM' is integral, and hence reduced, (3.4.6) shows that x is M'-regular. Consequently the kernel of multiplication by x on M is contained in

$$\mathfrak{p}^j M \subset \mathfrak{m}^j M$$

for every integer j, and this kernel is therefore zero by (**0**, 7.3.5). Thus x is M-regular. Applying (3.4.5), we conclude that M is integral. □

Remark (3.4.8). — The analogue of (3.4.7) obtained by replacing "integral" by "reduced" need not be true. For example, let $C = K[X, Y, Z]$ be the polynomial ring over a field K, let

$$B = C/\mathfrak{p}\mathfrak{q}, \qquad \mathfrak{p} = CZ, \qquad \mathfrak{q} = CX^2 + CY,$$

and let A be the local ring of B corresponding to the maximal ideal that is the image of $CX + CY + CZ$ in B. If x, z are the canonical images of X, Z in A, then $xz \neq 0$ but $x^2z^2 = 0$. On the other hand, A/xA is isomorphic to $K[Y, Z]/(YZ)$, so $\dim(A/xA) = 1$ whereas $\dim(A) = 2$. Thus x belongs to a system of parameters of A (**0**, 16.3.4), and A/xA is reduced, but A is not.

Proposition (3.4.9). — *Let A be a Noetherian ring, M an A-module of finite type, and $f \in A$ an M-regular element such that M/fM has no embedded associated prime ideals. If $\mathfrak{p}_i$ $(1 \leqslant i \leqslant m)$ are the prime ideals associated to M/fM, then, for every integer $n > 0$, $f^n M$ is the intersection of the inverse images of the $f^n M_{\mathfrak{p}_i}$ under the canonical maps $M \to M_{\mathfrak{p}_i}$ $(1 \leqslant i \leqslant m)$.*

Proof. It suffices to prove that the $\mathfrak{p}_i$ are also the prime ideals associated to $M/f^n M$, for then the $\mathfrak{p}_i$-saturations of $f^n M$ are the components of the necessarily unique reduced primary decomposition of $f^n M$ in M. Now, we have

$$\mathrm{Ass}(f^{n-1}M/f^n M) \subset \mathrm{Ass}(M/f^n M) \subset \mathrm{Ass}(M/f^{n-1}M) \cup \mathrm{Ass}(f^{n-1}M/f^n M)$$

by (3.1.7). Since f is M-regular, $f^{n-1}M/f^n M$ is isomorphic to M/fM, and the result follows by induction on n. □

§4. Change of base field for algebraic preschemes

4.1. Dimension of algebraic preschemes We shall develop in § 5 the general theory of the dimension of preschemes. The theory of the dimension of algebraic preschemes, however, can be developed more elementarily and also has many special features; we therefore give here a rapid account independent of the general theory. **IV-2** | 52

If K is a field and L an extension of K, we write $\mathrm{tr.deg}_K L$ for the transcendence degree of L over K.

Definition (4.1.1). — Let k be a field and X a prescheme locally of finite type over k. The *dimension* of X is the number

$$\dim(X) = \sup_x \operatorname{tr.deg}_k \kappa(x), \tag{4.1.1.1}$$

where x runs through the set of maximal points of X.

We shall see in (5.2.2) that Definition (4.1.1) depends only apparently on the base field k over which X is locally of finite type, and that the number $\dim(X)$ so defined agrees with the dimension of the underlying space of X (**0**, 14.1.2). Evidently,

$$\dim(X) = \dim(X_{\text{red}}).$$

Each $\kappa(x)$ is an extension of finite type of k (**I**, 6.3.3), hence has finite transcendence degree over k. If X is of finite type over k and nonempty, it is Noetherian (**I**, 6.3.7), hence has only finitely many irreducible components; consequently $\dim(X)$ is finite and nonnegative. Definition (4.1.1) gives

$$\dim(\varnothing) = -\infty.$$

If (X_α) denotes the family of reduced closed sub-preschemes of X whose underlying spaces are the irreducible components of X (**I**, 5.2.1), then clearly

$$\dim(X) = \sup_\alpha \dim(X_\alpha). \tag{4.1.1.2}$$

The calculation of the dimension is thus reduced to the case of integral preschemes locally of finite type over k. Finally,

$$\dim(X) = \dim(U) \tag{4.1.1.3}$$

for every induced sub-prescheme on an everywhere dense open subset U of X. The notion of dimension is thereby reduced to the case of an affine scheme of finite type over k.

Theorem (4.1.2). — *Let X and Y be two preschemes locally of finite type over a field k, and let $f : X \to Y$ be a k-morphism.*

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(i) If f is quasi-compact and dominant, then

$$\dim(Y) \leqslant \dim(X).$$

(ii) If f is quasi-finite, then

$$\dim(X) \leqslant \dim(Y).$$

(iii) Suppose that X is of finite type over k. For $\dim(X) \geqslant n$ (respectively $\dim(X) \leqslant n$, respectively $\dim(X) = n$), it is necessary and sufficient that there exist an open subset U dense in X and a k-morphism

$$g : U \longrightarrow \mathbf{V}_k^n = \operatorname{Spec}\bigl(k[T_1, \dots, T_n]\bigr)$$

that is surjective (respectively finite, respectively finite and surjective). If X is affine, one may take $U = X$.

Proof. *(i)* Let y be a maximal point of Y. The fibre $f^{-1}(y)$ contains a maximal point x of X (1.1.5); hence $\kappa(x)$ is an extension of $\kappa(y)$, and therefore

$$\operatorname{tr.deg}_k \kappa(y) \leqslant \operatorname{tr.deg}_k \kappa(x) \leqslant \dim(X).$$

This proves (i).

(ii) If x is a maximal point of X, then $\kappa(x)$ is a finite extension of $\kappa(f(x))$ (**II**, 6.2.2), and hence has the same transcendence degree over k. On considering the reduced closed sub-prescheme of Y whose underlying space is $\overline{\{f(x)\}}$, we are reduced to the following corollary. □

Corollary (4.1.2.1). — *Let Y be a k-prescheme locally of finite type. For every sub-prescheme Z of Y, one has*

$$\dim(Z) \leqslant \dim(Y).$$

Suppose in addition that all the irreducible components of Y have the same dimension. Then Z is rare in Y if and only if

$$\dim(Z) < \dim(Y).$$

Proof. Let z be a maximal point of Z. By considering one of the reduced closed sub-preschemes of Y whose underlying space is an irreducible component of Y containing z, we reduce to the case in which Y is integral, with generic point y. Then, by taking in Y an affine open neighbourhood of z, we reduce to the case in which Y is affine and of finite type over k. Write

$$Y = \operatorname{Spec}(A), \qquad Z = \operatorname{Spec}(A/\mathfrak{a}),$$

where A is an integral ring and $\mathfrak{a}$ is an ideal of A distinct from A.

By the normalization lemma (Bourbaki, *Alg. comm.*, chap. V, § 3, no. 1, th. 1), there exists a finite sequence $(x_i)_{1\leqslant i\leqslant n}$ of elements of A, algebraically independent over k, such that A is integral over

$$B = k[x_1, \dots, x_n]$$

and $\mathfrak{a} \cap B$ is generated by a subfamily $(x_i)_{1\leqslant i\leqslant p}$, possibly empty, of $(x_i)_{1\leqslant i\leqslant n}$. Let

$$g : Y \longrightarrow \operatorname{Spec}(B) = \mathbf{V}_k^n$$

be the finite dominant morphism corresponding to the canonical injection $B \to A$. Since

$$C = B/(\mathfrak{a} \cap B) \cong k[x_{p+1}, \dots, x_n],$$

g induces on Z a finite dominant morphism

$$Z \longrightarrow \mathbf{V}_k^{n-p}.$$

Thus

$$\operatorname{tr.deg}_k \kappa(y) = n, \qquad \operatorname{tr.deg}_k \kappa(z) = n - p,$$

because $\kappa(y)$ and $\kappa(z)$ are finite extensions of $\kappa(g(y))$ and $\kappa(g(z))$, respectively. This proves the first assertion.

If $p = 0$, then necessarily $z = y$, since y is the only point of Y whose image under g is the generic point of $\mathbf{V}_k^n$; in this case Z contains a nonempty open subset of Y. If $p > 0$, then $z \neq y$, and $\overline{\{z\}}$ is rare in $\overline{\{y\}}$. This completes the proof. □

Proof of (4.1.2), *continued.* The corollary proves (ii).

(iii) The sufficiency of the stated conditions follows immediately from (i), (ii), and (1.5.4)(v). To prove necessity, choose a dense open subset U of X that is the union of pairwise disjoint irreducible affine open subsets U_i $(1 \leqslant i \leqslant m)$, each containing a maximal point of X (**0**, 2.1.6). We may further suppose that X is reduced; if X is affine, we may of course take $U = X$.

The same argument as in (4.1.2.1) shows that, if $\dim(U_i) = n_i$, there exists a k-morphism **IV-2** | 54

$$g_i : U_i \longrightarrow \mathbf{V}_k^{n_i}$$

that is finite and dominant, hence surjective (**II**, 6.1.10). Set

$$n = \dim(X) = \sup_i n_i.$$

For every n_i there is a morphism

$$h_i : \mathbf{V}_k^{n_i} \longrightarrow \mathbf{V}_k^n$$

which is the closed immersion corresponding to the canonical homomorphism

$$k[T_1, \dots, T_n] \longrightarrow k[T_1, \dots, T_{n_i}]$$

and is the identity when $n_i = n$. Since U is the disjoint union of the U_i, define g on each U_i to be $h_i \circ g_i$. Then g is finite and surjective.

When $n' \geqslant n$, composing g with the canonical closed immersion

$$\mathbf{V}_k^n \longrightarrow \mathbf{V}_k^{n'}$$

gives a finite morphism $U \to \mathbf{V}_k^{n'}$. When $n' \leqslant n$, compose g with the canonical morphism

$$p : \mathbf{V}_k^n \longrightarrow \mathbf{V}_k^{n'}$$

corresponding to the canonical injection

$$k[T_1, \dots, T_{n'}] \longrightarrow k[T_1, \dots, T_n].$$

The morphism p is faithfully flat, hence surjective. □

Remark (4.1.3). — Corollary (4.1.2.1) shows that in formula (4.1.1.1) one may let x run through the set of all points of X.

Corollary (4.1.4). — *Let X be a prescheme locally of finite type over a field k, and let K be an extension of k. Then*

$$\dim(X \otimes_k K) = \dim(X).$$

Proof. It is enough to consider the case in which X is of finite type over k. The morphism

$$u : X \otimes_k K \longrightarrow X$$

is faithfully flat (2.2.13)(i). Thus, if U is an everywhere dense open subset of X, then

$$u^{-1}(U) = U \otimes_k K$$

is dense in $X \otimes_k K$ (2.3.10). If

$$g : U \longrightarrow \mathbf{V}_k^n$$

is finite and surjective, then so is

$$g_{(K)} : U \otimes_k K \longrightarrow \mathbf{V}_K^n$$

(**I**, 3.5.2) and (**II**, 6.1.5). The corollary follows. □

Corollary (4.1.5). — *Let X and Y be two preschemes locally of finite type over a field k. Then*

$$\dim(X \times_k Y) = \dim(X) + \dim(Y).$$

Proof. It is enough to prove the following. Let U and V be affine neighbourhoods of points $x \in X$ and $y \in Y$ such that

$$\operatorname{tr.deg}_k \kappa(x) = m = \dim(U), \qquad \operatorname{tr.deg}_k \kappa(y) = n = \dim(V).$$

Then

$$\dim(U \times_k V) = m + n.$$

Equivalently, by (4.1.2), we may suppose that there exist finite surjective k-morphisms

$$f : X \longrightarrow \mathbf{V}_k^m, \qquad g : Y \longrightarrow \mathbf{V}_k^n.$$

The product morphism

$$f \times g : X \times_k Y \longrightarrow \mathbf{V}_k^{m+n}$$

is finite and surjective (**I**, 3.5.2) and (**II**, 6.1.5), and the result follows. □

4.2. Associated prime cycles on algebraic preschemes

Proposition (4.2.1). — *Let K and L be two extensions of a field k such that $K \otimes_k L$ is Noetherian. Then the prime ideals associated with $K \otimes_k L$ are minimal, and if E is the residue field of the local ring at such an ideal, then*

$$\operatorname{tr.deg}_K E = \operatorname{tr.deg}_k L, \qquad \operatorname{tr.deg}_L E = \operatorname{tr.deg}_k K, \tag{4.2.1.1}$$

whence

$$\operatorname{tr.deg}_k E = \operatorname{tr.deg}_k K + \operatorname{tr.deg}_k L. \tag{4.2.1.2}$$

Proof. It is known that K is an algebraic extension of a purely transcendental extension

$$K' = k(\mathbf{t}), \qquad \mathbf{t} = (t_\alpha)_{\alpha \in I},$$

where $(t_\alpha)_{\alpha\in I}$ is a family of indeterminates. The ring

$$k[\mathbf{t}] \otimes_k L = L[\mathbf{t}]$$

is integral, and hence so is $K' \otimes_k L$, which is a ring of fractions of it; the field of fractions of $K' \otimes_k L$ is $L(\mathbf{t})$. There is a commutative diagram of canonical homomorphisms

(4.2.1.3)
$$\begin{tikzcd} K \arrow[r] & K \otimes_k L \arrow[r] & K \otimes_{K'} L(\mathbf{t}) \\ K' \arrow[u] \arrow[r] & K' \otimes_k L \arrow[u] \arrow[r] & L(\mathbf{t}) \arrow[u] \\ k \arrow[u] \arrow[r] & L \arrow[u] & \end{tikzcd}$$

Since K is faithfully flat over K',

$$K \otimes_k L = K \otimes_{K'} (K' \otimes_k L)$$

is faithfully flat over $K' \otimes_k L$, and therefore $K' \otimes_k L$ is Noetherian ($\mathbf{0_I}$, 6.5.2). Moreover, $K' \otimes_k L$ identifies with a subring of $K \otimes_k L$. The contraction to $K' \otimes_k L$ of a prime ideal $\mathfrak{p}$ associated with $K \otimes_k L$ is the zero ideal, since no nonzero element of $K' \otimes_k L$ is a zero-divisor in $K \otimes_k L$ ($\mathbf{0_I}$, 6.3.4) and Bourbaki, *Alg. comm.*, chap. IV, § 1, no. 1, cor. 3 to prop. 2.

Since K is algebraic over K', the ring $K \otimes_k L$ is integral over $K' \otimes_k L$, and the prime ideals of $K \otimes_k L$ lying over the zero ideal of $K' \otimes_k L$ are necessarily pairwise incomparable (Bourbaki, *Alg. comm.*, chap. V, § 2, no. 1, cor. 1 to prop. 1). This proves the first assertion (Bourbaki, *Alg. comm.*, chap. IV, § 1, no. 3, cor. 1 to prop. 7).

Furthermore, the residue field E of $\mathfrak{p}$ is algebraic over the residue field of the zero ideal of $K' \otimes_k L$, namely $L(\mathbf{t})$. Consequently

$$\operatorname{tr.deg}_L E = \operatorname{tr.deg}_L L(\mathbf{t}) = \operatorname{Card}(I) = \operatorname{tr.deg}_k K.$$

This gives the second equality in (4.2.1.1); interchanging the roles of K and L gives the first, and (4.2.1.2) follows. □

Corollary (4.2.2). — *Under the hypotheses of (3.3.6), if the preschemes $T_{x,y}$ are locally Noetherian, they have no immersed associated prime cycle.*

Corollary (4.2.3). — *Under the hypotheses of (3.3.6) (respectively (3.3.7)), if the $T_{x,y}$ are locally Noetherian and if $\mathscr{F}$ and the $\mathscr{G}_s$ ($s \in S$) (respectively $\mathscr{F}$ and $\mathscr{G}$) have no immersed associated prime cycle, then neither does*

$$\mathscr{F} \otimes_{\mathscr{S}} \mathscr{G}.$$

This follows from (4.2.2), (3.3.2), and the proof of (3.3.6). In particular, since every prescheme over a field k is flat over k, we have thereby proved assertion (i) of the following proposition.

Proposition (4.2.4). — *Let k be a field, and let X and Y be two locally Noetherian k-preschemes such that $X \times_k Y$ is locally Noetherian. Suppose in addition that X and Y are integral. Then:*

(i) $X \times_k Y$ has no immersed associated prime cycle. Each irreducible component of $X \times_k Y$ dominates X and Y, and the set of these components is in bijective correspondence with the set of irreducible components of

$$\operatorname{Spec}\big(\mathrm{R}(X) \otimes_k \mathrm{R}(Y)\big),$$

or equivalently with the set of minimal prime ideals of $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$, where $\mathrm{R}(X)$ and $\mathrm{R}(Y)$ are the fields of rational functions of X and Y, respectively.

(ii) If a maximal point z of $X \times_k Y$ is identified with a minimal prime ideal $\mathfrak{p}$ of $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$, then the local ring

$$\mathscr{O}_{X \times_k Y, z}$$

is isomorphic to the ring of fractions

$$\big(\mathrm{R}(X) \otimes_k \mathrm{R}(Y)\big)_{\mathfrak{p}}.$$

In particular, if $\mathrm{R}(X)$ or $\mathrm{R}(Y)$ is separable over k, then $X \times_k Y$ is reduced.

(iii) If, in addition, X and Y are locally of finite type over k, then every irreducible component of $X \times_k Y$ has dimension

$$\dim(X) + \dim(Y).$$

Proof. Assertion (iii) follows from (4.2.1.2), taking account of (i) and (ii).

To prove (ii), we may plainly reduce to the case in which

$$X = \operatorname{Spec}(A), \qquad Y = \operatorname{Spec}(B)$$

are affine, with A and B integral rings having respective fields of fractions

$$K = \mathrm{R}(X), \qquad L = \mathrm{R}(Y).$$

Assertion (i) shows that every minimal prime ideal $\mathfrak{q}$ of $A \otimes_k B$ is the contraction to $A \otimes_k B$ of a minimal prime ideal $\mathfrak{p}$ of $K \otimes_k L$. Since $K \otimes_k L$ is a ring of fractions of $A \otimes_k B$, the isomorphism

$$(A \otimes_k B)_{\mathfrak{q}} \simeq (K \otimes_k L)_{\mathfrak{p}}$$

follows from ($\mathbf{0_I}$, 1.2.5).

Finally, if, for example, $\mathrm{R}(Y)$ is separable over k, then $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$ is reduced (Bourbaki, *Alg.*, chap. VIII, § 7, no. 3, th. 1). The local rings at the maximal points of $X \times_k Y$ are therefore also reduced, and it follows that $X \times_k Y$ is reduced (3.2.1). □

Proposition (4.2.5). — *Let k be a field, let X and Y be locally Noetherian k-preschemes, and let $\mathscr{F}$ (respectively $\mathscr{G}$) be a quasi-coherent $\mathscr{O}_X$-Module (respectively $\mathscr{O}_Y$-Module). Let (Z'_λ) (respectively (Z''_μ)) be the family of prime cycles associated with $\mathscr{F}$ (respectively $\mathscr{G}$), and denote again by Z'_λ (respectively Z''_μ) the reduced sub-prescheme of X (respectively Y) having Z'_λ (respectively Z''_μ) as its underlying space. If $Z'_\lambda \times_k Z''_\mu$ is locally Noetherian, then its irreducible components $Z_{\lambda\mu\nu}$ dominate Z'_λ and Z''_μ, and $(Z_{\lambda\mu\nu})$ is the family of distinct prime cycles associated with $\mathscr{F} \otimes_k \mathscr{G}$.*

Proof. It suffices to apply (4.2.4) to the product

$$\operatorname{Spec}\bigl(\kappa(x)\bigr) \times_k \operatorname{Spec}\bigl(\kappa(y)\bigr).$$

□

Corollary (4.2.6). — *Let k be a field, and let X and Y be two locally Noetherian k-preschemes such that $X \times_k Y$ is locally Noetherian. Let (Z'_λ) (respectively (Z''_μ)) be the family of reduced sub-preschemes of X (respectively Y) whose underlying spaces are the irreducible components of X (respectively Y). Then the irreducible components $Z_{\lambda\mu\nu}$ of $Z'_\lambda \times_k Z''_\mu$ dominate Z'_λ and Z''_μ, and $(Z_{\lambda\mu\nu})$ is the family of irreducible components of $X \times_k Y$.*

Proof. We may reduce to the case in which X and Y are reduced (**I**, 5.1.8). The irreducible components of X (respectively Y) are then the prime cycles associated with $\mathscr{O}_X$ (respectively $\mathscr{O}_Y$) (3.2.1). Apply (4.2.5) with

$$\mathscr{F} = \mathscr{O}_X, \qquad \mathscr{G} = \mathscr{O}_Y,$$

noting that, by definition,

$$\mathscr{O}_X \otimes_k \mathscr{O}_Y = \mathscr{O}_{X \times_k Y}.$$

The corollary follows because $\mathscr{O}_X \otimes_k \mathscr{O}_Y$ has no immersed associated prime cycles, since by hypothesis the same is true of $\mathscr{O}_X$ and $\mathscr{O}_Y$ (4.2.3). □

We apply the preceding results to the case in which Y is the spectrum of an extension K of k.

Proposition (4.2.7). — *Let k be a field, let X be a k-prescheme, and let K be an extension of k such that $X \otimes_k K$ is locally Noetherian. Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module, let x' be a point of $X \otimes_k K$, and let x be its image in X.*

(i) Let (Z_λ) be the family of reduced sub-preschemes of X whose underlying spaces are the prime cycles associated with $\mathscr{F}$. Then the irreducible components $Z_{\lambda\mu}$ of $Z_\lambda \otimes_k K$ are the prime cycles associated with $\mathscr{F} \otimes_k K$, and $Z_{\lambda\mu}$ dominates Z_λ. Moreover, $Z_{\lambda\mu}$ is immersed if and only if Z_λ is immersed.

(ii) The point x belongs to an immersed prime cycle associated with $\mathscr{F}$ if and only if x' belongs to an immersed prime cycle associated with $\mathscr{F} \otimes_k K$. The Module $\mathscr{F}$ has no immersed associated prime cycle if and only if the same is true of $\mathscr{F} \otimes_k K$.

(iii) *The indices λ for which $x \in Z_\lambda$ are precisely those for which there exists an index μ such that $x' \in Z_{\lambda\mu}$. In particular, if x' belongs to only one prime cycle associated with $\mathscr{F} \otimes_k K$, then x belongs to only one prime cycle associated with $\mathscr{F}$.*

(iv) *If X is locally of finite type over k, then*

$$\dim(Z_{\lambda\mu}) = \dim(Z_\lambda).$$

Proof. The hypothesis implies that X itself is locally Noetherian (2.2.13) and (2.2.14). Assertion (i) follows from (4.2.5) and the proof of (4.2.3), with $\mathscr{G} = \mathscr{O}_Y$ and $Y = \operatorname{Spec}(K)$. Assertions (ii) and (iii) follow from (i) and (2.3.5). Finally, (iv) is a special case of (4.2.4), (iii). ☐

Corollary (4.2.8). — *If X is locally of finite type over k, then the set of dimensions of the prime cycles associated with $\mathscr{F}$ is the same as that of the prime cycles associated with $\mathscr{F} \otimes_k K$. The set of dimensions of the irreducible components is likewise the same for X and $X \otimes_k K$.*

Proposition (4.2.9). — *Suppose that the hypotheses of (4.2.5) hold, and suppose in addition that $\mathscr{F}$ and $\mathscr{G}$ are coherent. Let $(\mathscr{F}_\lambda)_{\lambda \in L}$ and $(\mathscr{G}_\mu)_{\mu \in M}$ be irredundant decompositions of $\mathscr{F}$ and $\mathscr{G}$, respectively. For every pair $(\lambda, \mu) \in L \times M$, let*

$$(\mathscr{H}_{\lambda\mu\nu})_{\nu \in S(\lambda,\mu)}$$

be a reduced irredundant decomposition of $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$, where

$$S(\lambda, \mu) = \operatorname{Ass}(\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu)$$

(3.2.5). Then $(\mathscr{H}_{\lambda\mu\nu})$, over all triples (λ, μ, ν), is an irredundant decomposition of $\mathscr{F} \otimes_k \mathscr{G}$. It is reduced if $(\mathscr{F}_\lambda)$ and $(\mathscr{G}_\mu)$ are reduced.

Proof. The Module $\mathscr{F} \otimes_k \mathscr{G}$ and each $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$ are coherent, and each $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$ identifies with a quotient of $\mathscr{F} \otimes_k \mathscr{G}$. By definition, each $\mathscr{H}_{\lambda\mu\nu}$ identifies with a coherent quotient of $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$, and hence also with a coherent quotient of $\mathscr{F} \otimes_k \mathscr{G}$.

The family of supports of the $\mathscr{H}_{\lambda\mu\nu}$ is locally finite. Indeed,

$$\operatorname{Supp}(\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu) \subset \left|\operatorname{Supp}(\mathscr{F}_\lambda) \times_k \operatorname{Supp}(\mathscr{G}_\mu)\right|,$$

where $\operatorname{Supp}(\mathscr{F}_\lambda)$ and $\operatorname{Supp}(\mathscr{G}_\mu)$ denote the corresponding reduced closed sub-preschemes. For fixed λ and μ, the family of supports of the $\mathscr{H}_{\lambda\mu\nu}$ is locally finite by hypothesis; the assertion therefore follows from the local finiteness of the families $(\operatorname{Supp}(\mathscr{F}_\lambda))$ and $(\operatorname{Supp}(\mathscr{G}_\mu))$.

It remains to show that the canonical homomorphism

$$\mathscr{F} \otimes_k \mathscr{G} \longrightarrow \bigoplus_{\lambda,\mu,\nu} \mathscr{H}_{\lambda\mu\nu}$$

is injective. It is the composite

$$\begin{aligned} \mathscr{F} \otimes_k \mathscr{G} &\longrightarrow \left(\bigoplus_\lambda \mathscr{F}_\lambda\right) \otimes_k \left(\bigoplus_\mu \mathscr{G}_\mu\right) \\ &\simeq \bigoplus_{\lambda,\mu} (\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu) \longrightarrow \bigoplus_{\lambda,\mu,\nu} \mathscr{H}_{\lambda\mu\nu}. \end{aligned}$$

The last homomorphism is injective by definition, and the first is injective because it is the tensor product of the canonical injections

$$\mathscr{F} \longrightarrow \bigoplus_\lambda \mathscr{F}_\lambda, \qquad \mathscr{G} \longrightarrow \bigoplus_\mu \mathscr{G}_\mu,$$

and $\mathscr{F}$ and $\mathscr{G}$ are flat over k.

Finally, if $(\mathscr{F}_\lambda)$ and $(\mathscr{G}_\mu)$ are reduced, we may suppose that

$$L = \operatorname{Ass}(\mathscr{F}), \qquad M = \operatorname{Ass}(\mathscr{G}).$$

The $\mathscr{H}_{\lambda\mu\nu}$ then form a reduced decomposition because, by (4.2.2), the sets

$$\operatorname{Ass}(\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu)$$

partition $\operatorname{Ass}(\mathscr{F} \otimes_k \mathscr{G})$ (3.2.5). ☐

Corollary (4.2.10). — *Under the hypotheses of (4.2.7), suppose in addition that $\mathscr{F}$ is coherent, and let $(\mathscr{F}_\lambda)_{\lambda\in L}$ be an irredundant decomposition of $\mathscr{F}$. For every $\lambda \in L$, let*

$$(\mathscr{F}_{\lambda\mu})_{\mu\in \mathrm{Ass}(\mathscr{F}_\lambda\otimes_k K)}$$

be a reduced irredundant decomposition of $\mathscr{F}_\lambda \otimes_k K$. Then $(\mathscr{F}_{\lambda\mu})$ is an irredundant decomposition of $\mathscr{F} \otimes_k K$. It is reduced if $(\mathscr{F}_\lambda)$ is reduced.

Proof. Apply (4.2.9) with

$$Y = \mathrm{Spec}(K), \qquad \mathscr{G} = \mathscr{O}_Y = K,$$

and with M consisting of a single element. □

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4.3. Reminders on tensor products of fields

For the reader's convenience, we recall here certain properties of tensor products of fields that we shall use in the following numbers; for the proofs, we refer to [1].

(4.3.1). Recall that an extension L of a field k is called *primary* if the largest separable algebraic extension of k contained in L is k itself.

Proposition (4.3.2). — *Let K and L be two extensions of a field k. If L is a primary extension of k, then* $\mathrm{Spec}(L \otimes_k K)$ *is irreducible and, if ζ is its generic point, $\kappa(\zeta)$ is a primary extension of K. Conversely, if* $\mathrm{Spec}(L \otimes_k K)$ *is irreducible for every finite separable extension K of k, then L is a primary extension of k.*

Proof. See [1], pp. 14-03 to 14-06. □

Corollary (4.3.3). — *If k is separably closed (that is, if its algebraic closure is radicial over k), then* $\mathrm{Spec}(L \otimes_k K)$ *is irreducible for every two extensions K and L of k; conversely, if this conclusion holds for every such pair, then k is separably closed.*

Proof. This follows immediately from (4.3.2). □

Corollary (4.3.4). — *Let L be an extension of a field k, and let L_s be the largest separable algebraic extension of k contained in L (sometimes called the separable algebraic closure of k in L). Suppose that L_s is of finite degree over k, and let K be a Galois extension of k containing L_s. Then* $\mathrm{Spec}(L \otimes_k K)$ *has $[L_s : k]$ irreducible components and is their union, and the residue fields at their generic points are primary extensions of K.*

Proof. Indeed,

$$L \otimes_k K = L \otimes_{L_s} (L_s \otimes_k K),$$

and $L_s \otimes_k K$ is isomorphic to a product of $[L_s : k]$ fields, each isomorphic to K (Bourbaki, *Alg.*, chap. VIII, § 7, no. 3, cor. 2 to th. 1). Thus $L \otimes_k K$ is isomorphic to a product of $[L_s : k]$ rings, each isomorphic to $L \otimes_{L_s} K$. Since L is a primary extension of L_s, the conclusion follows from (4.3.2). □

Proposition (4.3.5). — *Let K and L be two extensions of a field k. If L is a separable extension of k, then the ring $L \otimes_k K$ is reduced. Conversely, if $L \otimes_k K$ is reduced for every finite radicial extension K of k, then L is a separable extension of k.*

Proof. See Bourbaki, *Alg.*, chap. VIII, § 7, no. 3, th. 1. □

Corollary (4.3.6). — *If k is perfect, then $K \otimes_k L$ is reduced for every two extensions K and L of k; conversely, if this conclusion holds for every such pair, then k is perfect.*

Proof. This follows immediately from (4.3.5). □

Corollary (4.3.7). — *Let L be a separable extension of k, and let K be an extension of k such that either K or L is finite over k. Then the residue fields of the semilocal ring $L \otimes_k K$ are separable extensions of K.*

Proof. The ring $L \otimes_k K$ is the direct product of its residue fields E_i. For every extension K' of K, the ring $L \otimes_k K'$ is therefore isomorphic to the direct product of the rings $E_i \otimes_K K'$. Since $L \otimes_k K'$ is reduced, the fields E_i are separable over K by (4.3.5). □

Corollary (4.3.8). — *If K and L are two finite separable extensions of k, then $K \otimes_k L$ is a product of finitely many separable extensions of k.*

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Proposition (4.3.9). — *If k is algebraically closed, then the ring $L \otimes_k K$ is an integral domain for every two extensions K and L of k; conversely, if this conclusion holds for every such pair, then k is algebraically closed.*

Proof. This follows from (4.3.3) and (4.3.6), since a field that is both perfect and separably closed is algebraically closed. □

4.4. Irreducible preschemes and connected preschemes over an algebraically closed field

(4.4.1). Let k be a field, let X be a k-prescheme, and let K be an extension of k. Since the morphism

$$\mathrm{Spec}(K) \longrightarrow \mathrm{Spec}(k)$$

is faithfully flat, quasi-compact, and universally open (2.4.9), the same is true of the projection

$$p : X \otimes_k K \longrightarrow X$$

(2.2.13). It therefore follows from (2.3.5) that every irreducible component of $X \otimes_k K$ dominates an irreducible component of X, and, since p is surjective, every irreducible component of X is dominated by an irreducible component of $X \otimes_k K$. Thus p induces a surjective map from the set of irreducible components of $X \otimes_k K$ to the set of irreducible components of X.

Likewise, because p is continuous and surjective, every connected component of X is the image under p of a union of connected components of $X \otimes_k K$. Hence there is a surjective map from the set of connected components of $X \otimes_k K$ to the set of connected components of X. It follows that every irreducible (resp. connected) component Z' of $X \otimes_k K$ is contained in a unique set of the form $p^{-1}(Z)$, where Z is an irreducible (resp. connected) component of X. For every sub-prescheme of X having Z as its underlying space (and again denoted by Z), the space Z' is an irreducible (resp. connected) component of $Z \otimes_k K$.

In particular, suppose that $X \otimes_k K$ has finitely many irreducible (resp. connected) components, say n (resp. n'). This is the case when X is of finite type over k, since then $X \otimes_k K$ is of finite type over K and hence Noetherian. The number of irreducible (resp. connected) components of X is then at most n (resp. at most n'). Equality with n (resp. n') means that, for every reduced sub-prescheme Z of X whose underlying space is an irreducible (resp. connected) component of X, the prescheme $Z \otimes_k K$ is irreducible (resp. connected) (**I**, 5.1.8). Consequently, the number of irreducible (resp. connected) components of $X \otimes_k K$ is independent of K if and only if, for every irreducible (resp. connected) component Z of X, the prescheme $Z \otimes_k K$ is irreducible (resp. connected) for every extension K of k.

In particular, if $X \otimes_k K$ is irreducible (resp. connected), then so is X, as also follows directly from the surjectivity of p. Similarly, if $X \otimes_k K$ is reduced (resp. integral), then so is X, since p is faithfully flat (2.1.13).

In this number and the next, we shall examine more closely what can be said about the irreducible (resp. connected) components of $X \otimes_k K$ as K varies. The preceding observations already show that the number of these components is an increasing function of K.

Lemma (4.4.2). — *Let X' and X be two topological spaces, and let $f : X' \to X$ be a continuous map. Suppose that the following conditions hold:*

(i) *f is open and surjective (resp. the topological space X is canonically identified with the quotient of X' by the equivalence relation defined by f);*

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(ii) *for every $x \in X$, the fibre $f^{-1}(x)$ is irreducible (resp. connected).*

Then X' is irreducible (resp. connected) if and only if X is.

Proof. The assertion concerning connectedness is Bourbaki, *Top. gén.*, chap. I, 3rd ed., § 11, no. 3, prop. 7. We prove the assertion concerning irreducibility. The condition is plainly necessary because f is surjective; let us prove that it is sufficient.

Let X'_1 and X'_2 be two closed subsets of X' such that $X' = X'_1 \cup X'_2$, and, for $i = 1, 2$, let X_i be the set of $x \in X$ such that $f^{-1}(x) \subset X'_i$. Then

$$X - X_i = f(X' - X'_i),$$

and, since f is open, X_i is a closed subset of X. For every $x \in X$, the fibre $f^{-1}(x)$ is irreducible by hypothesis and is the union of the two closed subsets

$$X_1' \cap f^{-1}(x) \quad \text{and} \quad X_2' \cap f^{-1}(x).$$

One of them must equal $f^{-1}(x)$, so $x \in X_1$ or $x \in X_2$. Thus $X = X_1 \cup X_2$. Since X is irreducible, $X = X_1$ or $X = X_2$, and therefore $X' = X_1'$ or $X' = X_2'$. □

Remark (4.4.3). — If the topology of X is not the quotient of that of X' by the equivalence relation defined by f, it can happen that X and all the fibres $f^{-1}(x)$ are irreducible without X' being connected. For example, take for X an irreducible scheme—the affine line $\operatorname{Spec}(k[T])$ over an algebraically closed field k, say—choose a closed point x_0 of X, and take for X' the topological sum of $\{x_0\}$ and the open subspace $X - \{x_0\}$ of X.

Likewise, if the hypothesis that f is open is replaced by the hypothesis that f is closed, or even proper, then it can happen that X and all the fibres $f^{-1}(x)$ are irreducible without X' being irreducible, even though this replacement does ensure that the topology of X is the quotient of that of X' by the relation defined by f. Again take X to be the affine line over k, and put

$$S = X \times_k \mathbf{P},$$

where $\mathbf{P}$ is the projective line over k. Let x_0 be a closed point of X, let t_0 be a closed point of $\mathbf{P}$, and let

$$p : S \longrightarrow X, \qquad q : S \longrightarrow \mathbf{P}$$

be the projections. Let X' be the reduced sub-prescheme whose underlying space is the closed set

$$p^{-1}(x_0) \cup q^{-1}(t_0),$$

and let f be the restriction of p to X'. Then f is proper, but X' is not irreducible.

Theorem (4.4.4). — *Let k be an algebraically closed field and let X be a k-prescheme. If X is irreducible (resp. connected), then $X \otimes_k K$ is irreducible (resp. connected) for every extension K of k.*

Proof. We saw in (4.4.1) that

$$p : X \otimes_k K \longrightarrow X$$

is faithfully flat and open. To apply (4.4.2), it therefore suffices to verify that every fibre $p^{-1}(x)$ is irreducible. The $\kappa(x)$-prescheme $p^{-1}(x)$ is isomorphic to

$$\operatorname{Spec}\big(\kappa(x) \otimes_k K\big)$$

(**I**, 3.6.2), and is therefore integral by the hypothesis on k and (4.3.9). □

Corollary (4.4.5). — *Let k be an algebraically closed field, let X be a k-prescheme, let K be an extension of k, and let*

$$p : X \otimes_k K \longrightarrow X$$

be the canonical projection. If Z is an irreducible (resp. connected) subset of X, then $p^{-1}(Z)$ is irreducible (resp. connected). In particular, if X_0 is an irreducible component (resp. the connected component) of X containing Z, then $p^{-1}(X_0)$ is an irreducible (resp. connected) component of $X \otimes_k K$ containing $p^{-1}(Z)$.

Proof. The second assertion follows from (4.4.1) and the first assertion applied after replacing Z by X_0. To prove the first, let X' be the underlying space of $X \otimes_k K$, and put $Z' = p^{-1}(Z)$.

The equivalence relation R defined by p on X' is open. The induced relation $R_{Z'}$ on the saturated subset Z' is therefore also open, and Z identifies with the quotient space $Z'/R_{Z'}$ (Bourbaki, *Top. gén.*, chap. I, 3rd ed., § 5, no. 2, prop. 4). We may thus apply (4.4.2) to Z and Z', proving the first assertion of (4.4.5). □ **IV-2 | 61**

Corollary (4.4.6). — *Under the hypotheses and notation of (4.4.5), the map*

$$Z \longmapsto p^{-1}(Z)$$

is a bijection from the set of irreducible (resp. connected) components of X to the set of irreducible (resp. connected) components of $X \otimes_k K$. Its inverse is

$$Z' \longmapsto p(Z').$$

4.5. Geometrically irreducible and geometrically connected preschemes

Proposition (4.5.1). — *Let k be a field, let X be a k-prescheme, and let Ω be an algebraically closed extension of k. Let n (resp. n') be the cardinality of the set of irreducible components (resp. connected components) of $X \otimes_k \Omega$. This cardinality is independent of the chosen algebraically closed extension Ω. Moreover, for every extension K of k, the cardinality of the set of irreducible components (resp. connected components) of $X \otimes_k K$ is at most n (resp. at most n').*

Proof. Two algebraically closed extensions Ω and Ω' of k may always be regarded as subextensions of a common extension E of k. Applying (4.4.6) to $X \otimes_k \Omega$ and $X \otimes_k \Omega'$, and to the common extensions E/Ω and E/Ω', proves the first assertion. For the second, take Ω to be an algebraically closed extension of K and use (4.4.1). $\square$

Definition (4.5.2). — The cardinality n (resp. n') of (4.5.1) is called the *geometric number of irreducible components* (resp. *of connected components*) of X relative to k. If $n = 1$ (resp. $n' \leqslant 1$), the k-prescheme X is said to be *geometrically irreducible* (resp. *geometrically connected*).

This recovers the definition given in the special case (**III**, 4.3.4). By (4.5.1), the following properties are equivalent:

(a) X is geometrically irreducible (resp. geometrically connected);
(b) for one algebraically closed extension Ω of k, $X \otimes_k \Omega$ is irreducible (resp. connected);
(c) for every extension K of k, $X \otimes_k K$ is irreducible (resp. connected).

(4.5.3). Let X be a k-prescheme and let Z be a locally closed subset of X. If any sub-prescheme of X having Z as underlying space is again denoted by Z (**I**, 5.2.1), it follows from (**I**, 5.1.8) that, for an extension K of k, the number of irreducible (resp. connected) components of $Z \otimes_k K$ is independent of the chosen sub-prescheme with underlying space Z. The *geometric number of irreducible components* (resp. *of connected components*) of Z is therefore defined to be that of any sub-prescheme of X having Z as underlying space.

Proposition (4.5.4). — *Let X and Y be two k-preschemes, and let $f : X \to Y$ be a surjective k-morphism. If X is geometrically irreducible (resp. geometrically connected), then so is Y.*

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Proof. For every extension K of k, the morphism

$$f_K : X \otimes_k K \longrightarrow Y \otimes_k K$$

is surjective (**I**, 3.5.2). Since $X \otimes_k K$ is irreducible (resp. connected) by hypothesis, the same is true of $Y \otimes_k K$. $\square$

Definition (4.5.5). — A morphism $f : X \to Y$ of preschemes is said to be *irreducible* (resp. *connected*) if, for every $y \in Y$, the $\kappa(y)$-prescheme $f^{-1}(y)$ is geometrically irreducible (resp. geometrically connected).

Proposition (4.5.6). — (i) *Let X be a k-prescheme and let K be an extension of k. The geometric number of irreducible components (resp. connected components) of the K-prescheme $X \otimes_k K$, relative to K, is equal to the geometric number of irreducible components (resp. connected components) of X, relative to k. In particular, $X \otimes_k K$ is geometrically irreducible (resp. geometrically connected) as a K-prescheme if and only if X is so as a k-prescheme.*

(ii) *Let $f : X \to Y$ and $g : Y' \to Y$ be two morphisms, and put*

$$X' = X_{(Y')} = X \times_Y Y', \qquad f' = f_{(Y')} : X' \longrightarrow Y'.$$

If f is irreducible (resp. connected), then so is f'. The converse holds if g is surjective.

Proof. For (i), if Ω is an algebraically closed extension of K, then

$$X \otimes_k \Omega = (X \otimes_k K) \otimes_K \Omega,$$

which proves the assertion.

For (ii), for every $y' \in Y'$, put $y = g(y')$. Then

$$f'^{-1}(y') = f^{-1}(y) \otimes_{\kappa(y)} \kappa(y')$$

(**I**, 3.6.4), so both assertions follow from (i). □

Proposition (4.5.7). — *Let $f : X \to Y$ be a surjective irreducible (resp. connected) morphism, let $g : Y' \to Y$ be a morphism, and put $X' = X \times_Y Y'$. Suppose in addition that Y' is irreducible (resp. connected), and that f is universally open (resp. flat and quasi-compact, or universally open, or universally closed). Then X' is irreducible (resp. connected).*

Proof. All the properties of f under consideration are stable under base change, so it is enough to treat the case $Y' = Y$. The result then follows from (4.4.2), taking (2.3.12) into account. □

Corollary (4.5.8). — *Let X and Y be two k-preschemes.*

(i) *If X is geometrically irreducible (resp. geometrically connected) and Y is irreducible (resp. connected), then $X \times_k Y$ is irreducible (resp. connected).*
(ii) *If X and Y are geometrically irreducible (resp. geometrically connected), then so is $X \times_k Y$.*

Proof. For (i), the structure morphism $X \to \operatorname{Spec}(k)$ is surjective and universally open (2.4.9), and it is also irreducible (resp. connected). It is therefore enough to apply (4.5.7).

For (ii), if Ω is an algebraically closed extension of k, then

$$(X \times_k Y) \otimes_k \Omega = (X \otimes_k \Omega) \times_\Omega (Y \otimes_k \Omega)$$

(**I**, 3.3.10). By hypothesis and (4.5.6), the two factors on the right are geometrically irreducible (resp. geometrically connected) Ω-preschemes, so (i) applies. □

Proposition (4.5.9). — *Let X be a k-prescheme. The following conditions are equivalent:*

(a) *X is geometrically irreducible; in other words, $X \otimes_k K$ is irreducible for every extension K of k;*
(b) *$X \otimes_k K$ is irreducible for every finite separable extension K of k;*
(c) *X is irreducible and, if x is its generic point, $\kappa(x)$ is a primary extension of k.*

Proof. It is clear that (a) implies (b). To see that (b) implies (c), let K be a finite separable extension of k and let

$$p : X \otimes_k K \longrightarrow X$$

be the canonical projection. The maximal points of $X \otimes_k K$ are those of the fibre

$$p^{-1}(x) = \operatorname{Spec}\bigl(\kappa(x) \otimes_k K\bigr)$$

(2.3.4) and $(\mathbf{0}_{\mathrm{I}}, 2.1.8)$. Thus, saying that this fibre is irreducible for every finite separable extension K of k is equivalent, by (4.3.2), to saying that $\kappa(x)$ is a primary extension of k. Conversely, if (c) holds, the same argument, again using (4.3.2), shows that $X \otimes_k K$ is irreducible for every extension K of k. Hence (c) implies (a). □

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Corollary (4.5.10). — *Let X be an irreducible k-prescheme, let x be its generic point, let k' be the separable algebraic closure of k in $\kappa(x)$, and let k'' be a Galois extension of k, of finite degree or not, containing k'. Suppose that k' is finite over k. Then the irreducible components of $X \otimes_k k''$ are geometrically irreducible; their number is $[k' : k]$, and this is also the geometric number of irreducible components of X.*

Proof. As in (4.5.9), it is enough to consider the maximal points of

$$\operatorname{Spec}\bigl(\kappa(x) \otimes_k k''\bigr).$$

By (4.3.4), there are $[k' : k]$ of them, and their residue fields, which are also the residue fields at the maximal points of $X \otimes_k k''$, are primary extensions of k''. The result follows from (4.5.9). □

Corollary (4.5.11). — *Let X be a k-prescheme. Suppose that X has only finitely many maximal points x_i $(1 \leqslant i \leqslant r)$ and that, for each i, the separable algebraic closure k'_i of k in $\kappa(x_i)$ has finite degree over k. Then there is a finite separable extension L of k such that the irreducible components of $X \otimes_k L$ are geometrically irreducible. Their number, equal to*

$$\sum_i [k'_i : k],$$

is the geometric number of irreducible components of X.

Proof. The maximal points of $X \otimes_k L$ are those of the various fibres

$$\operatorname{Spec}\big(\kappa(x_i) \otimes_k L\big),$$

by (2.3.4) and $(\mathbf{0}_{\mathrm{I}}, 2.1.8)$, applied to the irreducible components of X and their inverse images in $X \otimes_k L$. It therefore suffices to take for L a finite Galois extension of k containing all the k_i' and to apply (4.5.10). ☐

Notice that (4.5.11) applies in particular to every k-prescheme X of finite type over k. Indeed, X is then Noetherian, so it has only finitely many irreducible components; moreover, each $\kappa(x_i)$ is an extension of finite type of k, and therefore so is the algebraic closure of k in $\kappa(x_i)$ ([1], p. 6-06, Lemma 5).

Remarks (4.5.12). — (i) The notions defined in (4.5.2) depend on the base field k. For brevity, when it is necessary to name the base field, we may say "k-irreducible" (resp. "k-connected") instead of "geometrically irreducible (resp. connected) relative to k". This terminology is natural in the context of schemes, but is exactly opposite to Weil's. For Weil, "k-irreducible" (resp. "k-connected", "k-normal", and so on) designates an intrinsic property of a prescheme X, independent of the field k chosen as base field (that is, independent of the chosen morphism $X \to \operatorname{Spec}(k)$); we express these properties simply by saying that X is irreducible (resp. connected, normal, and so on). On the other hand, Weil says "absolutely irreducible", "absolutely connected", "absolutely normal", and so on, for the corresponding notions relative to $X \otimes_k \Omega$, where Ω is a suitable algebraically closed extension of k. This terminology is explained by his different point of view: an "algebraic variety" V is first given as a geometric object over an algebraically closed field Ω; the datum of a smaller "field of definition" k (that is, a scheme over k which yields V after extension of the base field to Ω) is an additional and, to some extent, secondary structure. Thus his use of "absolute" or "intrinsic" on the one hand and "relative to k" on the other is opposite to ours.[3] We shall therefore avoid the qualifier "absolute", which may cause confusion; whenever we use the abbreviated terminology introduced above, in conflict with established terminology, we shall refer to (4.5.12) to avoid ambiguity.

(ii) Proposition (4.5.9) gives a "birational" criterion—one depending only on the residue field at the generic point—for a k-prescheme X to be geometrically irreducible. There is no analogous criterion for geometric connectedness. For example, take

$$X = \operatorname{Spec}\big(\mathbf{R}[[T, U]]/(T^2 + U^2)\big),$$

where T and U are indeterminates. Then X is an integral $\mathbf{R}$-scheme, and its field of rational functions $\kappa(x)$ is $\mathbf{C}((T))$, as is readily verified. The scheme $X \otimes_{\mathbf{R}} \mathbf{C}$ is the union of two irreducible components, the two "isotropic lines", which have a point in common: the maximal ideal in $\mathbf{C}[[T, U]]/(T^2 + U^2)$ induced by the maximal ideal $(T) + (U)$ of $\mathbf{C}[[T, U]]$. Thus $X \otimes_{\mathbf{R}} \mathbf{C}$ is connected. Let X' be the open complement of the closed point of X corresponding to the maximal ideal $(T) + (U)$ of $\mathbf{R}[[T, U]]$. Then $X' \otimes_{\mathbf{R}} \mathbf{C}$ is not connected, although X and X' have the same field of rational functions.

Proposition (4.5.13). — *Let X and Y be two k-preschemes and let $f : Y \to X$ be a k-morphism. Suppose that Y is nonempty and geometrically connected, and that X is connected. Then X is geometrically connected.*

Proof. Let Ω be an algebraic closure of k, put

$$X' = X \otimes_k \Omega, \qquad Y' = Y \otimes_k \Omega,$$

and let $f' = f_{(\Omega)} : Y' \to X'$. It is enough to prove that X' is connected. Let U' be a nonempty open-and-closed subset of X'. The projections

$$p : X' \longrightarrow X, \qquad q : Y' \longrightarrow Y$$

[3] Zariski's point of view is already closer to ours, and his terminology is not generally in conflict with that introduced here.

are open (2.4.10), and they are closed because Ω/k is algebraic (**II**, 6.1.10). Consequently $U = p(U')$ is a nonempty open-and-closed subset of X, and hence $U = X$. Since Y is nonempty, it follows from (**I**, 3.4.7) that

$$V' = f'^{-1}(U')$$

is nonempty. It is also open and closed in Y', which is connected by hypothesis, so $V' = Y'$. We conclude that $U' = X'$: otherwise, applying the same argument to the open-and-closed subset $X' \setminus U'$ would give $f'^{-1}(X' \setminus U') = Y'$, which is impossible because Y' is nonempty. □

Corollary (4.5.13.1). — (i) *Let X and Y be two k-preschemes and let $f : Y \to X$ be a k-morphism. If Y is nonempty and geometrically connected, then the connected component X_0 of X containing $f(Y)$ is geometrically connected.*

(ii) *Let X be a k-prescheme. If Y is a geometrically irreducible irreducible component of X, then the connected component X_0 of X containing Y is geometrically connected.*

Proof. For (i), we may suppose that Y is reduced, so that f factors as

$$Y \xrightarrow{g} X_0 \xrightarrow{j} X,$$

where X_0 denotes the reduced sub-prescheme of X whose underlying space is the connected component X_0 (**I**, 5.2.2). Apply (4.5.13) to g.

For (ii), choose a prescheme structure on the underlying space Y. Since a geometrically irreducible prescheme is in particular geometrically connected, (i) applies to the canonical injection $Y \to X$. □

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Corollary (4.5.14). — *Let X be a k-prescheme. If $x \in X$ is such that $\kappa(x)$ is a primary extension of k (in particular, if x is a rational point of X), then the connected component of x in X is geometrically connected.*

Proof. Apply (4.5.13.1) to $Y = \operatorname{Spec}(\kappa(x))$ and to the canonical morphism $Y \to X$. By (4.5.9), the prescheme Y is geometrically irreducible. □

Proposition (4.5.15). — *Let X be a k-prescheme, let $x \in X$, and let k' be the separable algebraic closure of k in $\kappa(x)$. Suppose that:*

(i) *X is connected;*
(ii) *k' is a finite extension of k.*

Then the geometric number of connected components of X is at most $[k' : k]$. Moreover, if k'' is a finite Galois extension of k containing k', the connected components of $X \otimes_k k''$ are geometrically connected.

Condition (ii) is satisfied in particular when X is locally of finite type over k.

Proof. By (ii), there are finite Galois extensions k'' of k containing k'. Put $X'' = X \otimes_k k''$. The projection $p : X'' \to X$ is faithfully flat and finite, hence closed (**II**, 6.1.10). It follows from (2.3.6), (ii), that the image under p of every connected component X''_α of X'' is all of X. Thus X''_α contains a point $x''_\alpha \in p^{-1}(x)$.

The number of points of $p^{-1}(x)$ is the number of irreducible components of

$$\operatorname{Spec}\big(\kappa(x) \otimes_k k''\big)$$

(**I**, 3.4.9), namely $[k' : k]$ by (4.3.4). Therefore the number of connected components of X'' is at most $[k' : k]$. For every $x''_\alpha \in p^{-1}(x)$, the field $\kappa(x''_\alpha)$ is a primary extension of k'', by (4.3.5) and (**I**, 3.4.9). Corollary (4.5.14), applied to x''_α and X'', now shows that every connected component of X'' is geometrically connected. □

Corollary (4.5.16). — *Let X be a k-prescheme, let (X_α) be the family of its connected components, and choose $x_\alpha \in X_\alpha$ for every α. Suppose that:*

(i) *the family (X_α) is finite;*
(ii) *for every α, the separable algebraic closure k'_α of k in $\kappa(x_\alpha)$ is a finite extension of k.*

Then the geometric number of connected components of X is at most

$$\sum_\alpha [k'_\alpha : k],$$

and there exists a finite separable extension k'' of k such that all the connected components of $X \otimes_k k''$ are geometrically connected.

Proof. By (i), the connected components of X are open. Apply (4.5.15) to each sub-prescheme induced on the open subset X_α. To obtain one extension k'' with the asserted property, take a finite Galois extension of k containing all the k'_α. □

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Corollary (4.5.17). — *Suppose that the k-prescheme X contains a point x such that the separable algebraic closure of k in $\kappa(x)$ is finite over k. Then X is geometrically connected if and only if $X \otimes_k K$ is connected for every finite separable extension K of k.*

Remark (4.5.18). — We shall see in § 8 (8.4.5) that the conclusion of (4.5.17) remains valid if, instead of the hypothesis in its statement, one assumes that X is quasi-compact.

Proposition (4.5.19). — *Let X be a k-prescheme, let Z be a subset of X, let k' be an algebraically closed extension of k, put*

$$X' = X \otimes_k k',$$

and let $p : X' \to X$ be the canonical projection. Suppose that $Z' = p^{-1}(Z)$ is contained in a single irreducible component X'_0 of X'. Then

$$X'_0 = p^{-1}(X_0),$$

where $X_0 = p(X'_0)$ is an irreducible component of X containing Z; moreover, X_0 is geometrically irreducible. If, in addition, X'_0 is the only irreducible component of X' which meets Z', then X_0 is the only irreducible component of X which meets Z.

To prove that $X'_0 = p^{-1}(X_0)$, we shall use the following lemma.

Lemma (4.5.19.1). — *Let T and T' be preschemes, let $p : T' \to T$ be a morphism, put*

$$T'' = T' \times_T T',$$

and let $p_1, p_2 : T'' \to T'$ be the canonical projections. A subset U' of T' is of the form $p^{-1}(U)$, for some subset $U \subset T$, if and only if

$$p_1^{-1}(U') = p_2^{-1}(U').$$

Proof. The structure morphism $q : T'' \to T$ is, by definition, both $p \circ p_1$ and $p \circ p_2$, so the condition is necessary. Conversely, suppose the condition holds. Let $u' \in U'$ and let $t' \in p^{-1}(p(u'))$. Since $p(t') = p(u')$, there is a point $t'' \in T''$ such that

$$p_1(t'') = u', \qquad p_2(t'') = t'$$

(**I**, 3.4.7). Thus $t'' \in p_1^{-1}(U') = p_2^{-1}(U')$, and consequently $t' = p_2(t'') \in U'$. This proves the lemma. □

Proof of (4.5.19). Form the fibre product

$$X'' = X' \times_X X'$$

relative to $p : X' \to X$, and let $p_1, p_2 : X'' \to X'$ be the canonical projections. By (4.5.19.1), it is enough to prove that

$$p_1^{-1}(X'_0) = p_2^{-1}(X'_0).$$

Put

$$S = \mathrm{Spec}(k), \qquad S' = \mathrm{Spec}(k'), \qquad S'' = S' \times_S S' = \mathrm{Spec}(k' \otimes_k k').$$

Then X'' may also be written $X' \times_{S'} S''$. If $\varphi'' : X'' \to S''$ is the structure morphism, it is enough to show, for every $s'' \in S''$, that

$$p_1^{-1}(X'_0) \cap \varphi''^{-1}(s'') = p_2^{-1}(X'_0) \cap \varphi''^{-1}(s'').$$

Put

$$T = \operatorname{Spec}(\kappa(s'')), \qquad U = X'' \times_{S''} T = \varphi''^{-1}(s''),$$

and let $v : U \to X''$ be the canonical projection. We must prove that

$$U_1 = v^{-1}\big(p_1^{-1}(X_0')\big), \qquad U_2 = v^{-1}\big(p_2^{-1}(X_0')\big),$$

which are irreducible components of U by (4.4.5), are equal. The relevant morphisms form the commutative diagram

(4.5.19.2)
$$\begin{tikzcd} X \arrow[d] & X' \arrow[l, "p"'] \arrow[d] & X'' \arrow[l, shift right, "p_1"'] \arrow[l, shift left, "p_2"] \arrow[d, "\varphi''"] & U \arrow[l, "v"'] \arrow[d] \\ S & S' \arrow[l] & S'' \arrow[l, shift right] \arrow[l, shift left] & T \arrow[l] \end{tikzcd}$$

By construction, U_1 and U_2 both contain $w^{-1}(Z)$, where

$$w = p \circ p_1 \circ v = p \circ p_2 \circ v.$$

There is a field K which is a common extension of k' and $\kappa(s'')$. If $P = \operatorname{Spec}(K)$, the diagram **IV-2** | 67

$$\begin{tikzcd} S' \arrow[d] & P \arrow[l] \arrow[d] \\ S & T \arrow[l] \end{tikzcd}$$

is commutative. Putting $Q = X \otimes_k K$, the diagram

$$\begin{tikzcd} X' \arrow[d, "p"'] & Q \arrow[l, "q"'] \arrow[d, "r"] \\ X & U \arrow[l, "w"] \end{tikzcd}$$

is likewise commutative. The two irreducible components $r^{-1}(U_1)$ and $r^{-1}(U_2)$ of Q (4.4.5) therefore contain $q^{-1}(Z')$. By (4.4.5), their images

$$q\big(r^{-1}(U_1)\big) \quad \text{and} \quad q\big(r^{-1}(U_2)\big)$$

are irreducible components of X' containing Z'. Both are therefore X_0'. Again by (4.4.5), $r^{-1}(U_1) = r^{-1}(U_2)$, and hence $U_1 = U_2$.

Since p is surjective and X_0' dominates an irreducible component of X, the subset $X_0 = p(X_0')$ is an irreducible component of X containing Z. Since $p^{-1}(X_0) = X_0'$ is an irreducible component of X', the component X_0 is geometrically irreducible.

For the last assertion, let X_1 be another irreducible component of X meeting Z. Choose $z \in Z \cap X_1$. There is an irreducible component of X' which meets Z' and dominates X_1 (2.3.5). By hypothesis X_0' is the only such component, so necessarily $X_1 = X_0$. □

Remark (4.5.20). — The fact that Z' is contained in a single irreducible component X_0' of X' does *not* imply that $X_0 = p(X_0')$ is the only irreducible component of X containing Z. Indeed, take $k = \mathbf{R}$, $k' = \mathbf{C}$, and let X be the k-scheme obtained by gluing[4] the following two schemes at a nongeneric point of each:

$$X_1 = \operatorname{Spec}\big(\mathbf{R}[S,T]/(S^2+T^2+1)\big), \qquad X_2 = \operatorname{Spec}\big(\mathbf{C}[T]\big);$$

the local rings of these two curves at each of their closed points have the same residue field, isomorphic to $\mathbf{C}$. The schemes X_1 and X_2 identify with the two irreducible components of X. If x is their common point, then $p^{-1}(x)$ consists of two distinct points y' and z'; the inverse image $p^{-1}(X_1)$ is irreducible, while $p^{-1}(X_2)$ is the union of two irreducible components Y' and Z' of X', with $y' \in Y'$ and $z' \in Z'$. Thus $p^{-1}(x)$ is contained in only one irreducible component of X'.

[4]The technique of gluing preschemes will be developed in detail in Chapter V. For the case considered here, which concerns algebraic curves, see [38], pp. 68–71.

Proposition (4.5.21). — *Let k be a separably closed field and let k' be an algebraic closure of k. For every k-prescheme X, the canonical projection*

$$X \otimes_k k' \longrightarrow X$$

is a universal homeomorphism. In particular, the irreducible components (resp. connected components) of X are geometrically irreducible (resp. geometrically connected).

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Proof. By definition, k'/k is radicial. The morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is integral, surjective, and radicial, so the first assertion follows from (2.4.5), (i). The second follows from the first and (4.5.1). □

4.6. Geometrically reduced algebraic preschemes

Proposition (4.6.1). — *Let k be a field, let X be a k-prescheme, and let Ω be a perfect extension of k. The following conditions are equivalent:*

(a) *for every reduced k-prescheme S, the prescheme $X \times_k S$ is reduced;*
(b) *for every extension K of k, the prescheme $X \otimes_k K$ is reduced;*
(c) *the prescheme $X \otimes_k \Omega$ is reduced;*
(d) *for every finite radicial extension k' of k, the prescheme $X \otimes_k k'$ is reduced;*
(e) *X is reduced and, for every irreducible component X_α of X, with generic point x_α, the field $\kappa(x_\alpha)$ is a separable extension of k.*

Proof. The implications *(a)*⇒*(b)* ⇒*(c)* are immediate. Since k' may be regarded as a subextension of Ω, (4.4.1) shows that *(c)* implies *(d)*.

Suppose *(d)*. Taking $k' = k$ first shows that X is reduced. To prove that $\kappa(x_\alpha)$ is separable over k, it suffices to show that $\kappa(x_\alpha) \otimes_k k'$ is reduced for every finite radicial extension k' of k (4.3.5). We may reduce to the case $X = \mathrm{Spec}(A)$, where A is a reduced k-algebra. Then $A \otimes_k k'$ is reduced by hypothesis (**I**, 5.1.4), and $\kappa(x_\alpha) \otimes_k k'$ is a ring of fractions of this ring; hence it is reduced ($\mathbf{0}_I$, 1.2.8). Thus *(d)* implies *(e)*.

Finally, suppose *(e)*. We may reduce to the affine case $X = \mathrm{Spec}(A)$ and $S = \mathrm{Spec}(B)$, where A and B are k-algebras. The fields $\kappa(x_\alpha)$ are the fields of fractions of the quotient rings $A/\mathfrak{p}_\alpha$, where $\mathfrak{p}_\alpha = \mathfrak{j}_{x_\alpha}$ are the minimal prime ideals of A. Since $\bigcap_\alpha \mathfrak{p}_\alpha = 0$, the algebra A embeds in $\prod_\alpha A/\mathfrak{p}_\alpha$, and hence in $\prod_\alpha \kappa(x_\alpha)$. Likewise, B may be regarded as a subalgebra of a product $\prod_\beta L_\beta$ of extensions of k. It follows that $A \otimes_k B$ embeds in

$$\left(\prod_\alpha \kappa(x_\alpha)\right) \otimes_k \left(\prod_\beta L_\beta\right),$$

and this tensor product itself embeds in

$$\prod_{\alpha,\beta} (\kappa(x_\alpha) \otimes_k L_\beta)$$

(Bourbaki, *Alg.*, chap. II, 3rd ed., § 7, n° 7, prop. 15). By hypothesis the rings $\kappa(x_\alpha) \otimes_k L_\beta$ are reduced (4.3.5); hence $A \otimes_k B$ is reduced. Thus *(e)* implies *(a)*. □

Definition (4.6.2). — If the equivalent conditions *(a)*–*(e)* of (4.6.1) hold, X is said to be *separable* (or *geometrically reduced*, or *universally reduced*) over k. A k-prescheme X is said to be *geometrically integral* over k if, for every extension K of k, the prescheme $X \otimes_k K$ is integral. By (4.6.1), this is equivalent to saying that X is separable and geometrically irreducible over k.

A commutative k-algebra A is said to be *separable* if $\mathrm{Spec}(A)$ is separable over k; equivalently, for every extension K of k, the ring $A_{(K)} = A \otimes_k K$ is reduced. This definition agrees with that of Bourbaki, *Alg.*, chap. VIII, § 7, n° 5, def. 1, when A has finite rank over k (loc. cit., corollary to **IV-2 | 69**
prop. 7), but not in general: a k-algebra may have nonzero radical even when it is integral.

Corollary (4.6.3). — *Let X be an integral k-prescheme. Then X is geometrically reduced (resp. geometrically integral) over k if and only if its field of rational functions $\mathrm{R}(X)$ is a separable (resp. separable and primary) extension of k.*

Proof. This follows immediately from (4.5.9) and (4.6.1). □

Corollary (4.6.4). — *Let X be a reduced k-prescheme. For every separable extension k' of k, the prescheme $X' = X \otimes_k k'$ is reduced.*

Proof. Apply the equivalence of *(a)* and *(e)* in (4.6.1), replacing X there by $\mathrm{Spec}(k')$ and S by X. □

Proposition (4.6.5). — (i) *Let X be a k-prescheme and let K be an extension of k. Then X is geometrically reduced (resp. geometrically integral) over k if and only if $X \otimes_k K$ is geometrically reduced (resp. geometrically integral) over K.*

(ii) *Let X and Y be two k-preschemes. If X and Y are geometrically reduced (resp. geometrically integral) over k, then so is $X \times_k Y$.*

Proof. Assertion *(i)* is an immediate consequence of the definitions and (4.4.1). For the reduced assertion in *(ii)*, let Ω be an algebraically closed extension of k. Then

$$(X \times_k Y) \otimes_k \Omega = X \times_k (Y \otimes_k \Omega).$$

Since $Y \otimes_k \Omega$ is reduced, the right-hand side is reduced by (4.6.1.a). The remainder of *(ii)* follows from this and (4.5.8.ii). □

Proposition (4.6.6). — *Let X be a k-prescheme of finite type. There exists a finite radicial extension k' of k such that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k'.*

Proof. Since X is covered by finitely many affine open subsets U_i, we may reduce to the affine case. Indeed, suppose for each i that k_i is a finite radicial extension of k such that $(U_i \otimes_k k_i)_{\mathrm{red}}$ is geometrically reduced over k_i. We may choose a common finite radicial extension k' of k containing all the k_i. By (**I**, 5.1.8),

$$(U_i \otimes_k k')_{\mathrm{red}} \simeq (U_i \otimes_k k_i)_{\mathrm{red}} \otimes_{k_i} k',$$

and this is geometrically reduced over k'. Thus the same holds for $(X \otimes_k k')_{\mathrm{red}}$.

Now let $X = \mathrm{Spec}(A)$, where A is a k-algebra of finite type, and let Ω be an algebraic closure of k. Write $\mathfrak{N}'$ for the nilradical of $A \otimes_k \Omega$. Since $A \otimes_k \Omega$ is Noetherian, $\mathfrak{N}'$ is generated by finitely many elements

$$y_i = \sum_j x_{ij} \otimes \xi_{ij}, \qquad x_{ij} \in A, \quad \xi_{ij} \in \Omega.$$

Let K be a finite subextension of Ω containing all the ξ_{ij}, and let $\mathfrak{N}$ be the ideal of $A \otimes_k K$ generated by the y_i, regarding $A \otimes_k K$ as a subring of $A \otimes_k \Omega$. The y_i are nilpotent in $A \otimes_k K$. Moreover, $\mathfrak{N} \otimes_K \Omega = \mathfrak{N}'$, and hence

$$\mathfrak{N}' \cap (A \otimes_k K) = \mathfrak{N}.$$

Thus $\mathfrak{N}$ contains the nilradical of $A \otimes_k K$ and is therefore equal to it. Since

$$(A \otimes_k \Omega)/\mathfrak{N}' = \big((A \otimes_k K)/\mathfrak{N}\big) \otimes_K \Omega,$$

the prescheme $(X \otimes_k K)_{\mathrm{red}} \otimes_K \Omega$ is reduced. Consequently $(X_{(K)})_{\mathrm{red}}$ is geometrically reduced over K by (4.6.1).

Replacing K by the quasi-Galois extension of k that it generates in Ω, we may further suppose, by (**I**, 5.1.8), that K is quasi-Galois over k. It is then a separable extension of a finite radicial extension k' of k (Bourbaki, *Alg.*, chap. V, § 10, n$^{\mathrm{o}}$ 9, prop. 14). By (**I**, 5.1.8), $(X_{(k')})_{\mathrm{red}} \otimes_{k'} K$ is isomorphic to **IV-2** | 70
$(X_{(K)})_{\mathrm{red}}$, and hence is geometrically reduced. It follows from (4.6.5.i) that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced. □

Corollary (4.6.7). — *If K is an extension of finite type of k, there exists a finite radicial extension k' of k such that the residue fields of the semilocal ring $K \otimes_k k'$ are separable over k'.*

Proof. Apply (4.6.6) to $X = \mathrm{Spec}(A)$, where A is a k-algebra of finite type having K as its field of fractions. □

Corollary (4.6.8). — *Let X be a k-prescheme of finite type. There exists a finite extension k' of k such that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k', the irreducible components of $X_{(k')}$ are geometrically irreducible, and its connected components are geometrically connected.*

Proof. This follows immediately from (4.6.6), (4.5.10), and (4.5.15). □

Definition (4.6.9). — Let k be a field, let X be a k-prescheme, and let $x \in X$. We say that X is *geometrically reduced* (or *separable*) (resp. *geometrically punctually integral*) at x over k if, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x, the prescheme X' is reduced (resp. integral) at x'; that is, $\mathscr{O}_{X',x'}$ is reduced (resp. integral).

We say that X is *geometrically punctually integral* if it is geometrically punctually integral at all of its points. Equivalently, for every extension k' of k, all local rings of $X \otimes_k k'$ are integral; in that case $X \otimes_k k'$ is also said to be *punctually integral*.

By (4.6.2), X is geometrically reduced over k if and only if it is geometrically reduced over k at every point $x \in X$.

Proposition (4.6.10). — *Let k be a field, let X be a k-prescheme, let k' be an extension of k, and let x' be a point of $X' = X \otimes_k k'$ with image x in X. Then X is geometrically reduced (resp. geometrically punctually integral) at x over k if and only if X' has the corresponding property at x' over k'. In particular, X is geometrically punctually integral over k if and only if X' is geometrically punctually integral over k'.*

Proof. It is enough to prove the first assertion. Necessity is immediate. Suppose conversely that X' is geometrically reduced (resp. geometrically punctually integral) at x' over k'. Let k'' be an extension of k and let x'' be a point of $X'' = X \otimes_k k''$ above x. There is a point z of $X' \times_X X''$ projecting to x' and x'' (**I**, 3.4.7). Let s be the image of z in $\mathrm{Spec}(k' \otimes_k k'')$ (**I**, 3.4.9), put $K = \kappa(s)$, and put $Z = X \otimes_k K$. Then K may be regarded as a composite extension of k' and k'', and z as a point of Z whose images in X' and X'' are, respectively, x' and x''. By hypothesis, $\mathscr{O}_{Z,z}$ is reduced (resp. integral). Since $\mathscr{O}_{Z,z}$ is a faithfully flat $\mathscr{O}_{X'',x''}$-module, it follows that $\mathscr{O}_{X'',x''}$ is reduced (resp. integral) ($\mathbf{0}_I$, 6.5.1). ☐

Corollary (4.6.11). — *Suppose that k is perfect (resp. algebraically closed). Then X is geometrically reduced (resp. geometrically punctually integral) at x over k if and only if $\mathscr{O}_{X,x}$ is reduced (resp. integral).*

Proof. Necessity is immediate. Conversely, for every extension k' of k, the prescheme $\mathrm{Spec}(\mathscr{O}_{X,x}) \otimes_k k'$ is reduced (4.6.4) (resp. integral (4.4.4)). For every point x' of $X' = X \otimes_k k'$ above x, the local ring $\mathscr{O}_{X',x'}$ is isomorphic to a local ring of $\mathrm{Spec}(\mathscr{O}_{X,x}) \otimes_k k'$, and is therefore reduced (resp. integral). ☐ **IV-2** | 71

Proposition (4.6.12). — *Let k be a field, let X be a k-prescheme, let $x \in X$, and let Ω be a perfect extension of k. The following conditions are equivalent:*

(a) *X is geometrically reduced over k at x; that is, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x, the local ring $\mathscr{O}_{X',x'}$ is reduced;*
(b) *the prescheme $X \otimes_k \Omega$ is reduced at one point above x;*
(c) *for every finite radicial extension k' of k, the prescheme $X' = X \otimes_k k'$ is reduced at the unique point x' above x;*
(d) *the prescheme $\mathrm{Spec}(\mathscr{O}_{X,x})$ is geometrically reduced over k;*
(e) *the local ring $\mathscr{O}_{X,x}$ is reduced and, for every irreducible component Z of X containing x, with generic point z, the field $\kappa(z)$ is a separable extension of k.*

Proof. The implications *(d)⇒(a)⇒(b) ⇒(c)* are immediate, using for the last implication (4.6.10), (4.6.11), and the fact that every radicial extension of k is isomorphic to a subextension of Ω.

Suppose *(c)*. By (4.6.1), it suffices to show that $\mathscr{O}_{X,x} \otimes_k k'$ is reduced for every finite radicial extension k' of k. This ring is the local ring of $X' = X \otimes_k k'$ at the unique point x' above x (**I**, 3.5.8) (**I**, 3.5.7), and is reduced by *(c)*. Thus *(c)* implies *(d)*.

Finally, the equivalence of *(d)* and *(e)* follows from the equivalence of *(b)* and *(e)* in (4.6.1), since the irreducible components of X containing x correspond bijectively to the irreducible components of $\mathrm{Spec}(\mathscr{O}_{X,x})$ (**I**, 2.4.2). ☐

Corollary (4.6.13). — *Under the hypotheses of (4.6.12), suppose in addition that X is locally Noetherian. Then conditions* (a)–(e) *of (4.6.12) are also equivalent to:*

(f) *there exists an open neighbourhood U of x that is geometrically reduced over k.*

Proof. Condition *(f)* plainly implies that X is geometrically reduced at x over k. Conversely, suppose that X is geometrically reduced at x. Since $\mathscr{O}_{X,x}$ is reduced, there is a reduced open neighbourhood U of x in X (**I**, 6.1.13). After shrinking U, we may further suppose that it meets no irreducible component of X that does not contain x. Condition *(e)* of (4.6.12), together with condition *(e)* of (4.6.1), then shows that U is geometrically reduced over k. □

It follows from (4.6.13) that, when X is locally Noetherian, the set of points $x \in X$ at which X is geometrically reduced over k is open; it is the largest open subset of X that is geometrically reduced over k.

(4.6.14). It follows from (4.6.10) and (4.6.11) that, if Ω is an algebraically closed extension of k, saying that X is geometrically punctually integral at x is equivalent to saying that $X \otimes_k \Omega$ is integral at one point above x.

For X to be geometrically punctually integral at x, it is necessary that X be integral at x and that, if z is the generic point of the unique irreducible component of X containing x, the field $\kappa(z)$ be a separable extension of k. This follows from (4.6.10). These conditions are not sufficient, as is shown by (4.5.12.ii). A sufficient but not necessary condition is that X be geometrically reduced at x and belong to only one irreducible component of X, and that this component be geometrically irreducible. If in addition X is locally Noetherian, then x has an open neighbourhood in X that is geometrically integral. **IV-2** | 72

Recall that a locally Noetherian prescheme that is punctually integral is locally integral (**I**, 6.1.13). Thus, if a k-prescheme X locally of finite type over k is geometrically punctually integral over k, then $X \otimes_k k'$ is locally integral for every extension k' of k. In this case X is also said to be *geometrically locally integral*.

Proposition (4.6.15). — (i) *Let k be a field and let X be a k-prescheme of finite type. Then X is geometrically punctually integral if and only if it is geometrically reduced and its geometric number of irreducible components equals its geometric number of connected components.*

(ii) *Let X be a prescheme locally of finite type over k. If X is geometrically punctually integral at x, then x has an open neighbourhood U that is geometrically punctually integral. Equivalently, the set of points $x \in X$ at which X is geometrically punctually integral is open in X.*

Proof. (i) Let Ω be an algebraically closed extension of k. The prescheme $X_{(\Omega)}$ is punctually integral (equivalently, locally integral, since it is Noetherian) if and only if it is reduced and the number of its connected components equals the number of its irreducible components (**I**, 6.1.10). Now apply (4.5.1), (4.6.1), and the first observation in (4.6.14).

(ii) The question is local on X, so we may reduce to the case in which X is of finite type over k. By (4.6.13), x has a geometrically reduced open neighbourhood; hence we may suppose that X itself is geometrically reduced. By (4.6.8), there is a finite extension k' of k such that $X' = X \otimes_k k'$ is reduced and its irreducible components are geometrically irreducible. Let $p : X' \to X$ be the canonical projection, which is finite and surjective, and let x'_j $(1 \leqslant j \leqslant r)$ be the points of $p^{-1}(x)$. By hypothesis X' is integral at each x'_j, so each x'_j has an integral open neighbourhood V'_j in X'. Since p is closed (**II**, 6.1.10), there is an open neighbourhood U of x such that $p^{-1}(U)$ is contained in $\bigcup_j V'_j$. Thus $p^{-1}(U)$ is integral at every point, and hence locally integral. Its irreducible components are consequently open and pairwise disjoint; since they are geometrically irreducible (4.5.9), assertion *(i)* shows that $p^{-1}(U) = U \otimes_k k'$ is geometrically punctually integral. By definition the same is true of U.

□

Proposition (4.6.16). — *Let k be a field, let X be a locally Noetherian k-prescheme, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

(a) *for every extension of finite type k' of k, if $X' = X \otimes_k k'$, then $\mathscr{F}' = \mathscr{F} \otimes_k k'$ is a reduced $\mathscr{O}_{X'}$-Module* (3.2.2)*;*

(b) *for every finite radicial extension k' of k, the Module $\mathscr{F}'$ is a reduced $\mathscr{O}_{X'}$-Module;*

(c) *$\mathscr{F}$ is reduced and, if $\mathscr{J}$ is the Ideal of $\mathscr{O}_X$ annihilating $\mathscr{F}$, the closed sub-prescheme defined by $\mathscr{J}$ is geometrically reduced over k.*

If, moreover, X is locally of finite type over k, these conditions are also equivalent to: **IV-2** | 73

(d) *for every extension k' of k (or for one algebraically closed extension k' of k), the Module $\mathscr{F}'$ is a reduced $\mathscr{O}_{X'}$-Module.*

Proof. It is clear that *(a)* implies *(b)*. Condition *(b)* first implies that $\mathscr{F}$ is reduced. By (3.2.3), this means the following. If Y is the closed sub-prescheme of X defined by the coherent Ideal $\mathscr{J}$ annihilating $\mathscr{F}$, then Y is reduced and there is a coherent torsion-free $\mathscr{O}_Y$-Module $\mathscr{G}$, of rank 1 on every irreducible component of Y, such that $j_*(\mathscr{G}) = \mathscr{F}$, where $j : Y \to X$ is the canonical immersion.

For every extension k' of k, the annihilator of $\mathscr{F}'$ is $\mathscr{J}' = \mathscr{J} \otimes_k k'$, since the projection $X' \to X$ is flat (2.1.11). The closed sub-prescheme of X' defined by $\mathscr{J}'$ is $Y' = Y \otimes_k k'$. If $j' : Y' \to X'$ is the canonical immersion and $\mathscr{G}' = \mathscr{G} \otimes_k k'$, then $\mathscr{F}' = j'_*(\mathscr{G}')$. For every extension k' for which X' is locally Noetherian, the Module $\mathscr{F}'$ has no embedded associated prime cycle (4.2.7). Moreover, every maximal point y' of Y' lies above a maximal point y of Y; since $\mathscr{G}_y \simeq \mathscr{O}_{Y,y}$, one has $\mathscr{G}'_{y'} \simeq \mathscr{O}_{Y',y'}$. Consequently, by (3.2.3), when X' is locally Noetherian, $\mathscr{F}'$ is reduced if and only if Y' is reduced.

It now follows from conditions *(d)* and *(b)* of (4.6.1) that *(b)* implies *(c)* and *(c)* implies *(a)*. In *(a)*, the finite-type hypothesis on k'/k serves only to ensure that X' is locally Noetherian. When X is locally of finite type over k, the prescheme X' is locally Noetherian for every extension k' of k; this proves the equivalence of *(d)* with the other conditions in that case. □

Definition (4.6.17). — When $\mathscr{F}$ satisfies the equivalent conditions of (4.6.16), it is said to be *geometrically reduced* over k, or *separable* over k.

Proposition (4.6.18). — *Let k be a field, let X be a locally Noetherian k-prescheme, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

- (a) *for every extension of finite type k' of k, if $X' = X \otimes_k k'$, then $\mathscr{F}' = \mathscr{F} \otimes_k k'$ is an integral $\mathscr{O}_{X'}$-Module* (3.2.4)*;*
- (b) *for every finite extension k' of k, the Module $\mathscr{F}'$ is an integral $\mathscr{O}_{X'}$-Module;*
- (c) *$\mathscr{F}$ is reduced (or integral) and, if $\mathscr{J}$ is the Ideal of $\mathscr{O}_X$ annihilating $\mathscr{F}$, the closed sub-prescheme defined by $\mathscr{J}$ is geometrically integral over k.*

If, moreover, X is locally of finite type over k, these conditions are also equivalent to:

- (d) *for every extension k' of k (or for one algebraically closed extension k' of k), the Module $\mathscr{F}'$ is an integral $\mathscr{O}_{X'}$-Module.*

Proof. Conditions *(a)*, *(b)*, and *(c)* imply that $\mathscr{F}$ is integral. Hence $\mathrm{Ass}(\mathscr{F})$ consists of one point x, the generic point of $\mathrm{Supp}(\mathscr{F})$. For every extension k' of k for which X' is locally Noetherian, the prime cycles associated to $\mathscr{F}'$ are therefore nonembedded and are the maximal points of $\mathrm{Supp}(\mathscr{F}')$, all lying above x. By (4.6.16), each of conditions *(a)*, *(b)*, and *(c)* also implies that $\mathscr{F}$ is geometrically reduced. It is immediate that *(a)* implies *(b)*, and *(b)* implies that $\mathrm{Supp}(\mathscr{F})$ is geometrically irreducible (4.5.9). Thus *(b)* implies *(c)*, because the closed sub-prescheme Y defined by $\mathscr{J}$ is geometrically reduced and geometrically irreducible, and hence geometrically integral. Conversely, **IV-2** | 74
if Y is geometrically integral, then for every extension k' of k, the support $\mathrm{Supp}(\mathscr{F}')$ is irreducible. Consequently, whenever X' is locally Noetherian, $\mathscr{F}'$ is integral. This proves that *(c)* implies *(a)*, and also that *(c)* implies *(d)* when X is locally of finite type over k. □

Definition (4.6.19). — When $\mathscr{F}$ satisfies the equivalent conditions of (4.6.18), it is said to be *geometrically integral* over k.

Proposition (4.6.20). — *Let k be a field, let X be a locally Noetherian k-prescheme, let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, and let K be an extension of k such that $X \otimes_k K$ is locally Noetherian. If $\mathscr{F}$ is geometrically reduced (resp. geometrically integral) over k, then $\mathscr{F} \otimes_k K$ is geometrically reduced (resp. geometrically integral) over K.*

Proof. For every finite extension K' of K,

$$(X \otimes_k K) \otimes_K K' = X \otimes_k K'$$

is locally Noetherian. The proofs of (4.6.16) and (4.6.18) show that the hypothesis on $\mathscr{F}$ implies that $\mathscr{F} \otimes_k K'$ is reduced (resp. integral), which proves the assertion. □

Proposition (4.6.21). — *Let k be a field, let X and Y be two locally Noetherian k-preschemes such that $X \times_k Y$ is locally Noetherian, let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, and let $\mathscr{G}$ be a coherent $\mathscr{O}_Y$-Module.*

- (i) *If $\mathscr{F}$ is geometrically reduced (resp. geometrically integral) and $\mathscr{G}$ is reduced (resp. integral), then $\mathscr{F} \otimes_k \mathscr{G}$ is reduced (resp. integral).*

(ii) *If $\mathscr{F}$ and $\mathscr{G}$ are both geometrically reduced (resp. geometrically integral), then so is $\mathscr{F} \otimes_k \mathscr{G}$.*

Proof. (i) Using (3.2.3), (4.6.16), and (4.6.18), we may reduce to the case in which

$$\mathrm{Supp}(\mathscr{F}) = X, \qquad \mathrm{Supp}(\mathscr{G}) = Y,$$

X is geometrically reduced (resp. geometrically integral), Y is reduced (resp. integral), $\mathscr{F}$ and $\mathscr{G}$ have no embedded associated prime cycles, and

$$\mathscr{F}_x \simeq \mathscr{O}_x, \qquad \mathscr{G}_y \simeq \mathscr{O}_y$$

at every maximal point x of X and y of Y. Then $\mathscr{F} \otimes_k \mathscr{G}$ has no embedded associated prime cycle (4.2.3), and its associated prime cycles are exactly the irreducible components of $X \times_k Y$ (4.2.5). We may therefore reduce further to the case in which X and Y are irreducible. If X is geometrically reduced and Y is reduced, then $X \times_k Y$ is reduced (4.2.4.ii). If X is geometrically integral and Y is integral, then $X \times_k Y$ is moreover irreducible (4.5.8.i), and hence integral. Finally, every maximal point z of $X \times_k Y$ lies above a maximal point x of X and a maximal point y of Y. The hypotheses on $\mathscr{F}$ and $\mathscr{G}$ therefore imply, by (**I**, 9.1.12), that $(\mathscr{F} \otimes_k \mathscr{G})_z \simeq \mathscr{O}_z$.

(ii) The argument is analogous. After reducing to $X = \mathrm{Supp}(\mathscr{F})$ and $Y = \mathrm{Supp}(\mathscr{G})$, the prescheme $X \times_k Y$ is geometrically reduced (resp. geometrically integral) by (4.6.5.ii). □

The definitions in (4.6.18) and (4.6.19) localize as follows.

Definition (4.6.22). — Let k be a field, let X be a locally Noetherian k-prescheme, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module. We say that $\mathscr{F}$ is *geometrically reduced*, or *separable* (resp. *geometrically punctually integral*), at a point $x \in X$ if, for every finite radicial (resp. finite) extension k' of k, the Module $\mathscr{F} \otimes_k k'$ is reduced (resp. integral) at every point x' of $X \otimes_k k'$ above x; see (3.2.2) and (3.2.4).

If $\mathscr{F}$ is reduced at x, there is an open neighbourhood U of x such that $\mathscr{F}|U$ is reduced (3.2.2). **IV-2** | 75
Arguing as in (4.6.16), one sees that $\mathscr{F}$ is geometrically reduced at x if and only if $\mathscr{F}$ is reduced at x and the closed sub-prescheme Y defined by the annihilator $\mathscr{J}$ of $\mathscr{F}$ is geometrically reduced at x. It follows that x has an open neighbourhood U such that $\mathscr{F}|U$ is geometrically reduced (4.6.11). The arguments of (4.6.16) and (4.6.18) likewise show that, if X is locally of finite type over k, then $\mathscr{F}$ is geometrically punctually integral at x if and only if $\mathscr{F}$ is reduced at x and Y is geometrically punctually integral at x.

4.7. Multiplicities in the primary decomposition on an algebraic prescheme

Lemma (4.7.1). — *Let A and B be two local rings, let $\mathfrak{m}$ be the maximal ideal of A, and let $\rho : A \to B$ be a local homomorphism. Suppose that B is flat as an A-module and that $B/\mathfrak{m}B$ has finite length as a B-module. Then an A-module M has finite length if and only if $M_{(B)}$ has finite length as a B-module, and*

$$\text{(4.7.1.1)} \qquad \mathrm{length}_B(M_{(B)}) = \mathrm{length}_A(M)\ \mathrm{length}_B(B/\mathfrak{m}B).$$

Proof. This is a special case of (2.5.5.2). □

Corollary (4.7.2). — *Let A and B be two Noetherian rings, and let $\rho : A \to B$ be a ring homomorphism such that B is flat as an A-module. Let $\mathfrak{q}$ be a minimal prime ideal of B, put $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$, and let M be an A-module of finite type. Then $\mathfrak{p}$ is a minimal prime ideal of A, $M_\mathfrak{p}$ has finite length as an $A_\mathfrak{p}$-module, and*

$$\text{(4.7.2.1)} \qquad \mathrm{length}_{B_\mathfrak{q}}\big((M_{(B)})_\mathfrak{q}\big) = \mathrm{length}_{A_\mathfrak{p}}(M_\mathfrak{p})\ \mathrm{length}_{B_\mathfrak{q}}\big(B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q}\big).$$

Proof. The ring $B_\mathfrak{q}$ is flat over $A_\mathfrak{p}$ ($\mathbf{0}_I$, 6.3.2), and the homomorphism $A_\mathfrak{p} \to B_\mathfrak{q}$ is local; hence $B_\mathfrak{q}$ is faithfully flat over $A_\mathfrak{p}$ ($\mathbf{0}_I$, 6.6.2). It follows from (2.3.5) that $\mathfrak{p}$ is minimal in A. The ring $A_\mathfrak{p}$ is therefore Artinian (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, prop. 9), so $M_\mathfrak{p}$ has finite length over $A_\mathfrak{p}$. Formula (4.7.2.1) is now a special case of (4.7.1.1). □

Proposition (4.7.3). — *Let k be a field of characteristic exponent p, let X be an integral locally Noetherian prescheme, and let $K = \mathrm{R}(X)$ be the field of rational functions on X. Let k' be an extension of k, let (X'_α) be the family of irreducible components of $X' = X_{(k')}$, and let x'_α be the generic point of X'_α.*

(i) *Suppose that X' is locally Noetherian, as is the case if X is locally of finite type over k or if k' is an extension of finite type of k. Let k'' be any extension of k', set $X'' = X \otimes_k k''$, and let (X''_β) be the family of irreducible components of X'', with generic points x''_β. If x''_β lies above x'_α, set*

$$Z'' = \operatorname{Spec}\bigl(\kappa(x'_\alpha)\bigr) \otimes_{k'} k'',$$

and let z''_β be the point of Z'' corresponding to x''_β. Then

$$\text{length}(\mathscr{O}_{X'',x''_\beta}) = \text{length}(\mathscr{O}_{X',x'_\alpha})\ \text{length}(\mathscr{O}_{Z'',z''_\beta}). \tag{4.7.3.1}$$

In particular, if k'' is a separable extension of k' such that X'' is locally Noetherian, then

$$\text{length}(\mathscr{O}_{X'',x''_\beta}) = \text{length}(\mathscr{O}_{X',x'_\alpha}).$$

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(ii) *If X is locally of finite type over k, or if the extension k'/k is of finite type, then each number*

$$\text{length}(\mathscr{O}_{X',x'_\alpha})$$

is a power of p.

(iii) *Suppose that X is locally of finite type over k. If k' is perfect, all the numbers* $\text{length}(\mathscr{O}_{X',x'_\alpha})$ *are equal and depend only on the extension K/k, not on the particular perfect extension k'/k under consideration. Moreover, there exists a finite radicial extension k_1 of k such that, if x_1 is the generic point of $X_1 = X \otimes_k k_1$, which is irreducible, then*

$$\text{length}(\mathscr{O}_{X',x'_\alpha}) = \text{length}(\mathscr{O}_{X_1,x_1}) \quad \textit{for every } \alpha.$$

Proof. (i) Recall from (4.2.4) that the finitely many irreducible components X'_α correspond bijectively to the minimal prime ideals $\mathfrak{p}_\alpha$ of $K \otimes_k k'$, and that $\mathscr{O}_{X',x'_\alpha}$ is an Artinian ring isomorphic to $(K \otimes_k k')_{\mathfrak{p}_\alpha}$. These rings therefore depend only on the extensions K/k and k'/k; in particular, with k' fixed, they are the same for every integral k-prescheme having K as its field of rational functions.

Formula (4.7.3.1) follows from (4.7.2.1). Indeed, apply that formula to the Noetherian ring A of an affine neighbourhood of x'_α in X' and to the ring B of a sufficiently small affine neighbourhood of x''_β in X'', where $\mathfrak{p}$ and $\mathfrak{q}$ are the minimal prime ideals corresponding respectively to x'_α and x''_β, and take $M = A$. Here $A/\mathfrak{p}$ is the ring of an affine neighbourhood of x'_α in X'_α, while $B = A \otimes_{k'} k''$. If $\mathfrak{q}' = \mathfrak{q}/\mathfrak{p}B$, a minimal prime ideal of $B/\mathfrak{p}B$, then

$$\mathscr{O}_{Z'',z''_\beta} = (B/\mathfrak{p}B)_{\mathfrak{q}'} = B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}},$$

which proves (4.7.3.1). The last assertion of *(i)* follows because Z'' is reduced in this case (4.2.4), and hence $\mathfrak{p}B_{\mathfrak{q}} = \mathfrak{q}B_{\mathfrak{q}}$.

(ii) If k' is of finite type over k, there is a purely transcendental extension k'_0 of k such that k' is finite over k'_0. Since k'_0/k is separable, *(i)*, applied with k' and k'' replaced by k and k'_0, shows that we may replace X by $X \otimes_k k'_0$; thus we may assume that k'/k is finite. There is then a finite quasi-Galois extension k''/k containing k', and (4.7.3.1) shows that it suffices to prove the assertion for k''. But k'' is separable over a finite radicial extension k_1 of k, so *(i)* reduces us to the case $k' = k_1$ radicial. The ring $K \otimes_k k_1$ is then Artinian and has a unique minimal prime ideal (4.3.2). If $\overline{K}$ is an algebraic closure of K, the length of $\overline{K} \otimes_k k_1$ divides the length of the Artinian ring $K \otimes_k k_1$ by (4.7.1.1). The Artinian $\overline{K}$-algebra $\overline{K} \otimes_k k_1$ has a unique prime ideal, necessarily maximal, and its quotient by that ideal is a finite extension of $\overline{K}$, hence is $\overline{K}$. Its length is therefore its rank over $\overline{K}$, namely $[k_1 : k]$, a power of p.

Suppose next that X is locally of finite type over k. Then X'' is locally Noetherian for every extension k'' of k'. Taking k'' algebraically closed, (4.7.3.1) reduces us to the case in which k' is algebraically closed. Since k' is then separable over the algebraic closure $\overline{k}$ of k, another application of *(i)* reduces us to $k' = \overline{k}$. After replacing X by an affine open of finite type over k, there is a finite radicial extension k_1/k such that $(X \otimes_k k_1)_{\text{red}}$ is separable over k_1 (4.6.6). Consequently, $(X \otimes_k k_1)_{\text{red}} \otimes_{k_1} \overline{k}$ is reduced (4.6.4).

By (4.7.3.1), this again reduces us to the case in which k' is a finite radicial extension of k, **IV-2** | 77
and the preceding argument applies.

(iii) Any two extensions k' and k'' of k embed in a common algebraically closed extension of k. To show that the set of numbers $\operatorname{length}(\mathscr{O}_{X',x'_\alpha})$ for perfect k' does not depend on k', it is therefore enough to compare them with those obtained after an algebraically closed extension k''/k'; the assertion then follows from *(i)*, since k''/k' is separable. To prove that all these numbers are equal, we may thus assume $k' = \bar{k}$. For the finite radicial extension k_1/k chosen in *(ii)*, all the points x'_α lie above the unique generic point x_1 of $X_1 = X \otimes_k k_1$. The equality

$$\operatorname{length}(\mathscr{O}_{X',x'_\alpha}) = \operatorname{length}(\mathscr{O}_{X_1,x_1})$$

for every α follows once more from (4.7.3.1) and the choice of k_1.

□

Definition (4.7.4). — Let k be a field, let X be an integral k-prescheme, and let K be the field of rational functions on X. The *radicial multiplicity* of K (or of X) over k is the supremum of the lengths of the Artinian rings $K \otimes_k k_1$, as k_1 ranges over the finite radicial extensions of k. The *separable multiplicity* of K (or of X) over k is the geometric number of irreducible components of X if this number is finite, and $+\infty$ otherwise. Finally, the *total multiplicity* of K (or of X) over k is the product of its radicial and separable multiplicities.

If p is the characteristic exponent of k and the radicial multiplicity of K over k is finite, then it is a power $q = p^f$ of p, by (4.7.3.ii). When $p \neq 1$, the well-defined integer f is called the *inseparability exponent* of K over k; when $p = 1$, the radicial multiplicity is always 1.

To say that the radicial (resp. separable, resp. total) multiplicity of X over k is 1 means that X is geometrically reduced (resp. geometrically irreducible, resp. geometrically integral) over k.

When X is locally of finite type, its radicial multiplicity over k is also the common length of the local rings $A_\mathfrak{q}$ at the minimal prime ideals of the total ring of fractions A of $K \otimes_k k'$, for every perfect extension k'/k. Its separable multiplicity is the number of minimal prime ideals of $A = K \otimes_k \Omega$, for every algebraically closed extension Ω/k. Finally, its total multiplicity is the sum of the lengths of the local rings $A_\mathfrak{q}$ of $A = K \otimes_k \Omega$ at its minimal prime ideals. In this case all three numbers are finite.

Definition (4.7.5). — Let k be a field, let X be a k-prescheme locally of finite type over k, let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, and let $x \in X$ be such that $\mathscr{F}_x$ has finite length as an $\mathscr{O}_{X,x}$-module. The *geometric length* of $\mathscr{F}$ at x relative to k, or the *radicial multiplicity of x for $\mathscr{F}$*, is

$$\lambda_x(\mathscr{F}) = \operatorname{length}_{\mathscr{O}_{X,x}}(\mathscr{F}_x)\, \mu_r(\kappa(x) \mid k),$$

where $\mu_r(\kappa(x) \mid k)$ denotes the radicial multiplicity of $\kappa(x)$ over k.

When X is integral and x is its generic point, $\operatorname{length}(\mathscr{O}_{X,x}) = 1$. Thus the radicial multiplicity of x for $\mathscr{O}_X$ is precisely the radicial multiplicity of X over k defined in (4.7.4).

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Proposition (4.7.6). — *Let*

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0$$

be an exact sequence of coherent $\mathscr{O}_X$-Modules. At a point $x \in X$, the geometric length $\lambda_x(\mathscr{F})$ is defined if and only if both $\lambda_x(\mathscr{F}')$ and $\lambda_x(\mathscr{F}'')$ are defined; in that case

$$\lambda_x(\mathscr{F}) = \lambda_x(\mathscr{F}') + \lambda_x(\mathscr{F}''). \tag{4.7.6.1}$$

Proof. This follows immediately from the definition: the length of $\mathscr{F}_x$ is finite if and only if the lengths of $\mathscr{F}'_x$ and $\mathscr{F}''_x$ are finite, and then

$$\operatorname{length}(\mathscr{F}_x) = \operatorname{length}(\mathscr{F}'_x) + \operatorname{length}(\mathscr{F}''_x).$$

□

(4.7.7). Under the hypotheses of (4.7.5), the only points $x \in X$ for which $\mathscr{F}_x$ is a nonzero $\mathscr{O}_{X,x}$-module of finite length are, by (3.1.2), the maximal points of $\operatorname{Supp}(\mathscr{F})$. This follows from Bourbaki, *Alg. comm.*, chap. IV, § 2, nº 5, cor. 2 to prop. 7, since the question is local on X. There is a closed subprescheme Y with underlying space $\operatorname{Supp}(\mathscr{F})$ and a coherent $\mathscr{O}_Y$-Module $\mathscr{G}$ such that $\mathscr{F} = j_*(\mathscr{G})$, where $j : Y \to X$ is the canonical injection. If x is a maximal point of Y, then the geometric lengths of

$\mathscr{F}$ and $\mathscr{G}$ at x are equal, because $\mathscr{O}_{X,x}$ and $\mathscr{O}_{Y,x}$ have the same residue field. This gives the following characterization of geometric length.

Proposition (4.7.8). — *Under the hypotheses of (4.7.5), let x be a maximal point of* $\operatorname{Supp}(\mathscr{F})$*. Let k' be a perfect extension of k, and set $X' = X \otimes_k k'$ and $\mathscr{F}' = \mathscr{F} \otimes_k k'$. Then $\lambda_x(\mathscr{F})$ equals* $\operatorname{length}_{\mathscr{O}_{X',x'}}(\mathscr{F}'_{x'})$ *at every maximal point x' of* $\operatorname{Supp}(\mathscr{F}')$ *above x. Moreover, there exists a finite radicial extension k_1 of k such that, on setting $X_1 = X \otimes_k k_1$ and $\mathscr{F}_1 = \mathscr{F} \otimes_k k_1$, the number $\lambda_x(\mathscr{F})$ equals* $\operatorname{length}_{\mathscr{O}_{X_1,x_1}}((\mathscr{F}_1)_{x_1})$ *at every maximal point x_1 of* $\operatorname{Supp}(\mathscr{F}_1)$ *above x.*

Proof. The remark in (4.7.7) reduces us to the case in which x is a maximal point of X. Since the question is local on X, suppose $X = \operatorname{Spec}(A)$ is affine Noetherian and $\mathscr{F} = \widetilde{M}$, where M is an A-module of finite type. Put $B = A \otimes_k k'$, and let $\mathfrak{p}$ and $\mathfrak{q}$ be the minimal prime ideals of A and B corresponding respectively to x and x'. Formula (4.7.2.1) gives

$$\operatorname{length}_{B_\mathfrak{q}}((M_{(B)})_\mathfrak{q}) = \operatorname{length}_{A_\mathfrak{p}}(M_\mathfrak{p})\ \operatorname{length}_{B_\mathfrak{q}}(B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q}).$$

Since $\kappa(x) = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$, the argument of (4.7.3) shows that the last factor is the length of $(B/\mathfrak{p}B)_{\mathfrak{q}'}$, where $\mathfrak{q}' = \mathfrak{q}/\mathfrak{p}B$. This is a local ring of $(A/\mathfrak{p}) \otimes_k k'$ at a minimal prime ideal, and therefore also a local ring of the total ring of fractions of $\kappa(x) \otimes_k k'$. By definition its length is $\mu_r(\kappa(x) \mid k)$, proving the first assertion. The second follows in the same way by replacing k' with k_1 and using (4.7.3.iii). □

When x is a maximal point of $\operatorname{Supp}(\mathscr{F})$, the geometric length of $\mathscr{F}$ at x is also called the *radicial multiplicity of the maximal prime cycle* $\overline{\{x\}}$ associated with $\mathscr{F}$.

Corollary (4.7.9). — *Under the hypotheses of (4.7.5), let k' be an extension of k, set $X' = X \otimes_k k'$ and $\mathscr{F}' = \mathscr{F} \otimes_k k'$, and let Z be an irreducible component of* $\operatorname{Supp}(\mathscr{F})$*. If Z' is an irreducible component of* $\operatorname{Supp}(\mathscr{F}')$ *dominating Z (4.2.7), then the radicial multiplicities of Z with respect to $\mathscr{F}$ and of Z' with respect to $\mathscr{F}'$ are equal.*

Proof. Choose a perfect extension k'' of k', set $X'' = X \otimes_k k''$ and $\mathscr{F}'' = \mathscr{F} \otimes_k k''$, and choose a maximal point x'' of $\operatorname{Supp}(\mathscr{F}'')$ above the generic point of Z'. It follows from (4.7.8) that $\operatorname{length}(\mathscr{F}''_{x''})$ equals both multiplicities. □

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Proposition (4.7.10). — *Let k be a field, let X be a k-prescheme locally of finite type over k, let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, let $x \in X$, and let ξ_i be the generic points of the irreducible components of* $\operatorname{Supp}(\mathscr{F})$ *that contain x. Then $\mathscr{F}$ is geometrically reduced at x (4.6.22) if and only if x belongs to no embedded associated prime cycle of $\mathscr{F}$ and $\lambda_{\xi_i}(\mathscr{F}) = 1$ for every i.*

Proof. First suppose the stated conditions hold. Let k'/k be finite, let x' be a point of $X' = X \otimes_k k'$ above x, and put $\mathscr{F}' = \mathscr{F} \otimes_k k'$. By (4.2.7), x' belongs to no embedded associated prime cycle of $\mathscr{F}'$. Moreover, (4.7.8) shows that $\mathscr{F}'$ has geometric length 1 at every generic point ξ' of an irreducible component of $\operatorname{Supp}(\mathscr{F}')$ containing x'; by (2.3.4), such a point lies above one of the ξ_i. A fortiori, $\operatorname{length}(\mathscr{F}'_{\xi'}) = 1$, so $\mathscr{F}'$ is reduced at x'.

Conversely, (4.2.7) first shows that if, for one finite extension k'/k, the Module $\mathscr{F}'$ is reduced at every point x' above x, then x belongs to no embedded associated prime cycle of $\mathscr{F}$. For each ξ_i, take for k' the field k_1 supplied by (4.7.8). The hypothesis that $\mathscr{F}$ is geometrically reduced at x then implies $\lambda_{\xi_i}(\mathscr{F}) = 1$. □

Proposition (4.7.11). — *Let k be a field, let X be a prescheme locally of finite type over k, let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, let k' be an extension of k, set $X' = X \otimes_k k'$ and $\mathscr{F}' = \mathscr{F} \otimes_k k'$, and let $x' \in X'$ lie above $x \in X$. Then $\mathscr{F}$ is geometrically reduced (resp. geometrically punctually integral) at x if and only if $\mathscr{F}'$ is geometrically reduced (resp. geometrically punctually integral) at x'.*

Proof. Every irreducible component of $\operatorname{Supp}(\mathscr{F}')$ containing x' dominates an irreducible component of $\operatorname{Supp}(\mathscr{F})$ containing x, and conversely every such component downstairs is dominated by one containing x' ((4.2.7.i) and (2.3.4)). The assertion concerning geometric reducedness therefore follows from (4.2.7.ii), (4.7.9), and (4.7.10).

Let Y be the closed sub-prescheme of X defined by the annihilator Ideal $\mathscr{J}$ of $\mathscr{F}$. Then $Y' = Y \otimes_k k'$ is the closed sub-prescheme of X' defined by the annihilator Ideal $\mathscr{J}' = \mathscr{J} \otimes_k k'$ of $\mathscr{F}'$. To say that $\mathscr{F}$ (resp. $\mathscr{F}'$) is geometrically punctually integral at x (resp. x') is to say that $\mathscr{F}$ (resp. $\mathscr{F}'$) is geometrically reduced at x (resp. x') and that Y (resp. Y') is geometrically punctually integral at x (resp. x'). By the assertion just proved, either of the two geometric pointwise-integrality hypotheses implies that Y has a separable neighbourhood of x. The conclusion follows from (4.6.10). □

Definition (4.7.12). — Let k be a field, let X be a k-prescheme locally of finite type over k, let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, and let x be a maximal point of $\mathrm{Supp}(\mathscr{F})$.

The *total multiplicity* of x (or of the prime cycle $\overline{\{x\}}$) for $\mathscr{F}$, relative to k, is the product of the radicial multiplicity of x for $\mathscr{F}$ and the separable multiplicity of $\kappa(x)$ over k. **IV-2** | 80

Equivalently, the total multiplicity of x for $\mathscr{F}$ is the product of $\mathrm{length}_{\mathscr{O}_{X,x}}(\mathscr{F}_x)$ and the total multiplicity of $\kappa(x)$ over k (4.7.4). With the notation of (4.7.11), the total multiplicity of x for $\mathscr{F}$ is the sum of the total multiplicities, for $\mathscr{F}'$, of the maximal points x_i' of $\mathrm{Supp}(\mathscr{F}')$ above x. Moreover, there exists a finite extension k'/k for which the total multiplicity of x for $\mathscr{F}$ is

$$\sum_i \mathrm{length}(\mathscr{F}'_{x_i'}).$$

Proposition (4.7.13). — *Under the hypotheses and notation of (4.7.10), it is sufficient, for $\mathscr{F}$ to be geometrically punctually integral at x, that x belong to only one associated prime cycle of $\mathscr{F}$ (necessarily non-embedded), and that the total multiplicity for $\mathscr{F}$ of the generic point ξ of that cycle be* 1.

Proof. Let k'/k be finite, let x' be a point of $X' = X \otimes_k k'$ above x, and put $\mathscr{F}' = \mathscr{F} \otimes_k k'$. The point x' can belong to only one associated prime cycle of $\mathscr{F}'$, since by hypothesis there is only one maximal point of $\mathrm{Supp}(\mathscr{F}')$ above ξ. Moreover, $\mathscr{F}'$ is geometrically reduced at x' by (4.7.10). The conclusion follows. □

4.8. Fields of definition

(4.8.1). Given a prescheme X, throughout this section we denote by $\mathfrak{S}(X)$ the set of sub-preschemes of X, and, as usual, by $\mathfrak{P}(X)$ the set of subsets of the underlying space of X. For every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we denote by $\Phi(\mathscr{F})$ the set of quasi-coherent sub-$\mathscr{O}_X$-Modules of $\mathscr{F}$.

(4.8.2). Let k be a field, let X and Y be two k-preschemes, let $\mathscr{F}$ and $\mathscr{G}$ be two quasi-coherent $\mathscr{O}_X$-Modules, and let $K' \subset K$ be two extensions of k. Corresponding to the morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(K')$, there are canonical maps

$$\Phi(\mathscr{F} \otimes_k K') \longrightarrow \Phi(\mathscr{F} \otimes_k K), \tag{4.8.2.1}$$

$$\mathrm{Hom}(\mathscr{F} \otimes_k K', \mathscr{G} \otimes_k K') \longrightarrow \mathrm{Hom}(\mathscr{F} \otimes_k K, \mathscr{G} \otimes_k K), \tag{4.8.2.2}$$

$$\mathfrak{S}(X_{(K')}) \longrightarrow \mathfrak{S}(X_{(K)}), \tag{4.8.2.3}$$

$$\mathrm{Hom}_{K'}(X_{(K')}, Y_{(K')}) \longrightarrow \mathrm{Hom}_K(X_{(K)}, Y_{(K)}), \tag{4.8.2.4}$$

$$\mathfrak{P}(X_{(K')}) \longrightarrow \mathfrak{P}(X_{(K)}). \tag{4.8.2.5}$$

If $p_X : X_{(K)} \to X_{(K')}$ is the canonical projection, then (4.8.2.5) sends M to $p_X^{-1}(M)$, and similarly (4.8.2.3) sends Z to $p_X^{-1}(Z) = Z \otimes_{K'} K$ (**I**, 4.4.1). Map (4.8.2.1) sends

$$\mathscr{H} \longmapsto \mathscr{H}_{(K)} = \mathscr{H} \otimes_{\mathscr{O}_{X_{(K')}}} \mathscr{O}_{X_{(K)}};$$

(4.8.2.2) sends u to $p_X^*(u)$; and (4.8.2.4) sends f to $f_{(K)}$.

If $s_{K,K'}$ denotes any one of these canonical maps, and if $K'' \subset K' \subset K$ are three extensions of k, then

$$s_{K,K''} = s_{K,K'} \circ s_{K',K''}. \tag{4.8.2.6}$$

Proposition (4.8.3). — *The canonical maps* (4.8.2.1)–(4.8.2.5) *are injective.*

Proof. The morphism p_X is faithfully flat and quasi-compact. The injectivity of (4.8.2.5) follows from the surjectivity of p_X. The injectivity of (4.8.2.1) follows from (2.2.2), that of (4.8.2.2) from (2.2.7), that of (4.8.2.3) from (2.2.15), and that of (4.8.2.4) from (2.2.16). □

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Definition (4.8.4). — With the notation of (4.8.2), a quasi-coherent sub-$\mathscr{O}_{X_{(K)}}$-Module of $\mathscr{F} \otimes_k K$ (resp. a homomorphism $\mathscr{F} \otimes_k K \to \mathscr{G} \otimes_k K$, a sub-prescheme of $X_{(K)}$, a K-morphism $X_{(K)} \to Y_{(K)}$, or a subset of $X_{(K)}$) is said to be *defined over* K' if it belongs to the image of (4.8.2.1) (resp. (4.8.2.2), (4.8.2.3), (4.8.2.4), or (4.8.2.5)). The field K' is then called a *field of definition* of the object in question.

Clearly, K itself is a field of definition for every object in any of the five target sets in (4.8.2). One should note, however, that for a K-prescheme Z, to say that a quasi-coherent $\mathscr{O}_Z$-Module $\mathscr{H}$ is "defined over a subfield K' of K" has no meaning unless Z has been *given* in the form $X_{(K)}$ for a prescheme X over a subfield k of K, and unless $\mathscr{H}$ is a sub-$\mathscr{O}_Z$-Module of an $\mathscr{O}_Z$-Module that has been *given* in the form $\mathscr{F} \otimes_k K$, where $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-Module. Analogous remarks apply to the other notions.

It follows from (4.8.2.6) that if an object in one of the five target sets in (4.8.2) is defined over K', then it is also defined over every intermediate field K'' with $K' \subset K'' \subset K$. Finally, if K_1/K is an extension, then an object in a target set of $s_{K_1,K'}$ is defined over K' if and only if its image under $s_{K_1,K}$ is defined over K'; this follows from

$$s_{K_1,K'} = s_{K_1,K} \circ s_{K,K'}.$$

Proposition (4.8.5). — *With the notation of (4.8.2), let (X_α) be an open covering of X. An element*

$$\mathscr{H} \in \Phi(\mathscr{F} \otimes_k K) \quad (\textit{resp. } u \in \operatorname{Hom}(\mathscr{F} \otimes_k K, \mathscr{G} \otimes_k K),\ Z \in \mathfrak{S}(X_{(K)}),\ \textit{or } M \in \mathfrak{P}(X_{(K)}))$$

is defined over K' if and only if, for every α, its restriction

$$\mathscr{H}|_{(X_\alpha)_{(K)}} \quad (\textit{resp. } u|_{(X_\alpha)_{(K)}},\ Z \cap (X_\alpha)_{(K)},\ \textit{or } M \cap (X_\alpha)_{(K)})$$

is defined over K'.

Proof. This follows immediately from the injectivity of (4.8.2.1), (4.8.2.2), (4.8.2.3), and (4.8.2.5). For example, suppose that for every α there is a quasi-coherent sub-$\mathscr{O}_{(X_\alpha)_{(K')}}$-Module $\mathscr{H}'_\alpha$ such that

$$\mathscr{H}|_{(X_\alpha)_{(K)}} = \mathscr{H}'_\alpha \otimes_{K'} K.$$

For any α, β, injectivity after restriction to $(X_\alpha \cap X_\beta)_{(K')}$ gives

$$\mathscr{H}'_\alpha|_{(X_\alpha \cap X_\beta)_{(K')}} = \mathscr{H}'_\beta|_{(X_\alpha \cap X_\beta)_{(K')}}.$$

The $\mathscr{H}'_\alpha$ therefore glue to a quasi-coherent sub-$\mathscr{O}_{X_{(K')}}$-Module $\mathscr{H}'$ of $\mathscr{F} \otimes_k K'$ such that $\mathscr{H} = \mathscr{H}' \otimes_{K'} K$. The other cases are treated in the same way. □

Lemma (4.8.6). — *Let k be a field, let K/k be an extension, let A be a k-algebra, let M be an A-module, let $\overline{N}$ be an $A_{(K)}$-submodule of $M_{(K)}$, and let K' be an intermediate field of K/k. The following conditions are equivalent:*

(a) *$\overline{N} = N' \otimes_{K'} K$, where N' is an $A_{(K')}$-submodule of $M_{(K')}$;*
(b) *if $X = \operatorname{Spec}(A)$, the quasi-coherent $\mathscr{O}_{X_{(K)}}$-Module $\widetilde{\overline{N}}$ on $X_{(K)} = \operatorname{Spec}(A_{(K)})$ is defined over K';*
(c) *$\overline{N} = N' \otimes_{K'} K$, where N' is a K'-vector subspace of $M_{(K')}$.*

Proof. The equivalence of *(a)* and *(b)* follows immediately from (**I**, 1.6.5), and *(a)* plainly implies *(c)*. **IV-2 | 82** Conversely, if *(c)* holds, then, up to the canonical identification,

$$N' = \overline{N} \cap M_{(K')}.$$

Since $\overline{N}$ and $M_{(K')}$ are $A_{(K')}$-submodules, so is N'. □

Corollary (4.8.7). — *Under the hypotheses of (4.8.6), there is a smallest subfield K' of K containing k for which the equivalent conditions of (4.8.6) hold.*

Proof. It is enough to use condition *(c)*, for which the assertion is Bourbaki, *Alg.*, chap. II, 3rd ed., § 8, n° 6, prop. 6. □

Lemma (4.8.8). — *With the notation of (4.8.2):*

(i) *Let* $\overline{u} : \mathscr{F} \otimes_k K \to \mathscr{G} \otimes_k K$ *be an* $\mathscr{O}_{X_{(K)}}$*-homomorphism, and let*

$$\mathscr{H} \subset (\mathscr{F} \otimes_k K) \oplus (\mathscr{G} \otimes_k K)$$

be its graph. Then $\overline{u}$ *is defined over* K' *if and only if* $\mathscr{H}$ *is.*

(ii) *Let* $\overline{Z}$ *be a closed sub-prescheme of* $X_{(K)}$*, and let* $\overline{\mathscr{J}}$ *be the quasi-coherent Ideal of* $\mathscr{O}_{X_{(K)}}$ *defining* $\overline{Z}$*. Then* $\overline{Z}$ *is defined over* K' *if and only if* $\overline{\mathscr{J}}$ *is.*

(iii) *Let* $\overline{f} : X_{(K)} \to Y_{(K)}$ *be a K-morphism, and let* $\overline{Z}$ *be its graph, regarded as a sub-prescheme of* $X_{(K)} \times_K Y_{(K)}$ (**I**, 5.3.11). *Then* $\overline{f}$ *is defined over* K' *if and only if* $\overline{Z}$ *is.*

Proof. Assertion *(ii)* follows immediately from (**I**, 4.4.5). Necessity in *(iii)* is clear. Conversely, suppose $\overline{Z} = Z'_{(K)}$, where Z' is a sub-prescheme of $(X \times_k Y)_{(K')}$. If $p' : Z' \to X_{(K')}$ is the restriction of the first projection, then $p'_{(K)}$ is an isomorphism, and hence so is p' (2.7.1.viii). Thus Z' is the graph of a K'-morphism $f' : X_{(K')} \to Y_{(K')}$ (**I**, 5.3.11), and $\overline{f} = f'_{(K)}$ (**I**, 5.3.12).

Finally, to prove *(i)*, (4.8.5) permits us to assume $X = \operatorname{Spec}(A)$ affine, with $\mathscr{F} = \widetilde{M}$, $\mathscr{G} = \widetilde{N}$, and $\overline{u} = \widetilde{\overline{v}}$ for an $A_{(K)}$-homomorphism $\overline{v} : M_{(K)} \to N_{(K)}$. Suppose the graph of $\overline{v}$ is $P'_{(K)}$, where P' is an $A_{(K')}$-submodule of $M_{(K')} \oplus N_{(K')}$. If $p' : P' \to M_{(K')}$ is the restriction of the first projection, then $p' \otimes 1_K$ is an isomorphism of $A_{(K)}$-modules. Faithful flatness ($\mathbf{0}_I$, 6.4.1) implies that p' is an isomorphism of $A_{(K')}$-modules. Hence P' is the graph of an $A_{(K')}$-homomorphism $v' : M_{(K')} \to N_{(K')}$, and plainly $\overline{v} = v' \otimes 1_K$. □

Proposition (4.8.9). — *Let k be a field, let K/k be an extension, let X be a k-prescheme, let* $\mathscr{F}$ *be a quasi-coherent* $\mathscr{O}_X$*-Module, and let* $\mathscr{H}$ *be a quasi-coherent sub-*$\mathscr{O}_{X_{(K)}}$*-Module of* $\mathscr{F} \otimes_k K$*. Then* $\mathscr{H}$ *has a smallest field of definition.*

Proof. When X is affine, this is precisely (4.8.7). In general, let (X_α) be an affine open covering of X. For every α, there is a smallest subfield K'_α of K containing k over which $\mathscr{H}|_{(X_\alpha)_{(K)}}$ is defined. By (4.8.5), the subfield of K generated by the K'_α is the smallest field of definition of $\mathscr{H}$. □

Corollary (4.8.10). — *With the notation of* (4.8.9)*, let* $\mathscr{G}$ *be a second quasi-coherent* $\mathscr{O}_X$*-Module and let*

$$\overline{u} : \mathscr{F} \otimes_k K \longrightarrow \mathscr{G} \otimes_k K$$

be an $\mathscr{O}_{X_{(K)}}$*-homomorphism. Then* $\overline{u}$ *has a smallest field of definition.*

Proof. By (4.8.8.i), such a field is also the smallest field of definition of the graph of $\overline{u}$, whose existence follows from (4.8.9). □

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Corollary (4.8.11). — *Let k be a field, let X be a k-prescheme, let K/k be an extension, and let* $\overline{Z}$ *be a closed sub-prescheme of* $X_{(K)}$*. Then* $\overline{Z}$ *has a smallest field of definition.*

Proof. Let $\overline{\mathscr{J}}$ be the quasi-coherent Ideal of $\mathscr{O}_{X_{(K)}}$ defining $\overline{Z}$. By (4.8.8.ii), a field defines $\overline{Z}$ if and only if it defines $\overline{\mathscr{J}}$, and the assertion follows from (4.8.9). □

Corollary (4.8.12). — *Let k be a field, let X and Y be two k-preschemes with Y separated, let K/k be an extension, and let* $\overline{f} : X_{(K)} \to Y_{(K)}$ *be a K-morphism. Then* $\overline{f}$ *has a smallest field of definition.*

Proof. By (4.8.8.iii), the existence of such a field is equivalent to the existence of a smallest field of definition for the graph $\overline{Z}$ of $\overline{f}$. Since $\overline{Z}$ is a closed sub-prescheme of $(X \times_k Y)_{(K)}$ (**I**, 5.4.3), the assertion follows from (4.8.11). □

Proposition (4.8.13). — *Under the hypotheses of* (4.8.11) *(resp.* (4.8.9), (4.8.10), *or* (4.8.12)*), suppose in addition that X is of finite type over k (resp. that X is of finite type over k and* $\mathscr{F}$ *is coherent; that X is of finite type over k and both* $\mathscr{F}$ *and* $\mathscr{G}$ *are coherent; or that X and Y are of finite type over k). Then the smallest field of definition of* $\overline{Z}$ *(resp.* $\mathscr{H}$*,* $\overline{u}$*, or* $\overline{f}$*) is an extension of finite type of k.*

Proof. It is enough to treat (4.8.9), and we may assume that X is affine because it is a finite union of affine opens of finite type over k. With the notation of (4.8.6), it remains to show that if A is a k-algebra of finite type and M an A-module of finite type, then one of the fields K' satisfying the equivalent conditions of (4.8.6) may be chosen of finite type over k.

Let $(m_i)_{1\le i\le n}$ generate M over A. Since A is Noetherian, so is $A_{(K)}$, and the $A_{(K)}$-submodule $\overline{N}$ of $M_{(K)}$ has a finite set of generators n_j $(1 \le j \le r)$. Write

$$n_j = \sum_i (m_i \otimes 1) b_{ij}, \qquad b_{ij} \in A_{(K)},$$

and then

$$b_{ij} = \sum_h a_{ijh} \otimes c_{ijh}, \qquad a_{ijh} \in A, \quad c_{ijh} \in K.$$

The extension K' of k generated by the finitely many c_{ijh} has the required property. □

Proposition (4.8.14). — *Let k be a field, let X be a k-prescheme of finite type, and let K/k be an extension containing a perfect extension of k. Then the smallest field of definition of the closed sub-prescheme $(X_{(K)})_{\mathrm{red}}$ of $X_{(K)}$ is a finite radicial extension of k.*

Proof. If p is the characteristic exponent of k, the hypothesis implies that $k' = k^{p^{-\infty}}$ is contained in K. The prescheme $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k' (4.6.1). Thus

$$(X_{(k')})_{\mathrm{red}} \otimes_{k'} K$$

is reduced and is consequently equal to $(X_{(K)})_{\mathrm{red}}$ (**I**, 5.1.8). In other words, k' is a field of definition of $(X_{(K)})_{\mathrm{red}}$. Since k'/k is algebraic, the conclusion follows from (4.8.13). □

Remarks (4.8.15). — (i) The conclusion of (4.8.14) need not hold if K is not assumed to contain a perfect extension of k.

For example, let k_0 be a field of characteristic $p > 0$, let $k = k_0(s,t)$ be the field of rational functions in two indeterminates, let **IV-2 | 84**

$$L = k(s^{1/p}, t^{1/p}), \qquad X = \mathrm{Spec}(L),$$

let u be a third indeterminate, and put

$$K = k\big(u, s^{1/p} + ut^{1/p}\big).$$

One readily verifies that k is algebraically closed in K and that $L \otimes_k K$ has a nonzero nilradical, so $X_{(K)}$ is not reduced. But $(X_{(K)})_{\mathrm{red}}$ is not defined over k, while K contains no intermediate extension of finite degree over k other than k itself. Hence the smallest field of definition of $(X_{(K)})_{\mathrm{red}}$ cannot be finite over k.

(ii) Given an object in one of the target sets in (4.8.2) and an extension K_1/K, the object has a smallest field of definition if and only if its image under $s_{K_1,K}$ does; the two smallest subfields are necessarily the same, as observed in (4.8.4).

4.9. Fields of definition of subsets of a prescheme

Proposition (4.9.1). — *Let k be a field, let X be a k-prescheme, let K/k be an extension, let $\overline{T}$ be a closed subset of $X_{(K)}$, and let K' be an intermediate field of K/k. The following conditions are equivalent:*

- (a) *$\overline{T}$ is defined over K'.*
- (b) *The open subset $\overline{U} = X_{(K)} - \overline{T}$ of $X_{(K)}$ is defined over K'.*
- (b′) *The sub-prescheme of $X_{(K)}$ induced on the open subset $\overline{U}$ is defined over K'.*
- (c) *There exists a closed sub-prescheme $\overline{Z}$ of $X_{(K)}$, defined over K', whose underlying space is $\overline{T}$.*

Proof. Let $g : X_{(K)} \to X_{(K')}$ be the canonical projection. To say that $\overline{T}$ is defined over K' means that there is a closed subset T' of $X_{(K')}$ such that $\overline{T} = g^{-1}(T')$. Since

$$g^{-1}(X_{(K')} - T') = X_{(K)} - g^{-1}(T')$$

for every subset T' of $X_{(K')}$, conditions (4.9.1.a), (4.9.1.b), and (4.9.1.bprime) are equivalent.

If $\overline{Z}$, whose underlying space is $\overline{T}$, is defined over K', then

$$\overline{Z} = Z' \otimes_{K'} K$$

for a closed sub-prescheme Z' of $X_{(K')}$. If T' is the underlying space of Z', then $\overline{T} = g^{-1}(T')$ (**I**, 4.4.1); thus (4.9.1.c) implies (4.9.1.a). Conversely, if (4.9.1.a) holds, choose a closed sub-prescheme Z' of $X_{(K')}$ whose underlying space is T' (**I**, 5.2.1), and take $\overline{Z} = Z' \otimes_{K'} K$. □

Corollary (4.9.2). — *With the notation of (4.9.1), suppose that K' is a field of definition of $\overline{T}$. Every intermediate field K'' with $k \subset K'' \subset K'$ for which K'/K'' is radicial is also a field of definition of $\overline{T}$.*

Proof. The canonical projection

$$g : X_{(K')} \longrightarrow X_{(K'')}$$

is integral, surjective, and radicial, hence is a homeomorphism (2.4.5). □

Remark (4.9.3). — It follows from (4.9.2) that a closed subset $\overline{T}$ of $X_{(K)}$ need not have a smallest field of definition. For example, let $K = k(t)$, where t is an indeterminate and k is a perfect field of characteristic $p > 0$. A closed subset $\overline{F}$ of $X_{(K)}$ can have a smallest field of definition only if it is already defined over k. Indeed, for every subfield K' of K containing k with $K' \neq k$, one has $(K')^p \neq K'$, the field $(K')^p$ contains k, and $K'/(K')^p$ is radicial. Thus, if K' is a field of definition of $\overline{F}$, then so is $(K')^p$. Nevertheless, one has the following result.

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Proposition (4.9.4). — *With the notation of (4.9.1), suppose that K/K' is separable. Then K' is a field of definition of $\overline{T}$ if and only if it is a field of definition of the reduced sub-prescheme of $X_{(K)}$ whose underlying space is $\overline{T}$.*

Proof. The condition is sufficient by (4.9.1). Conversely, suppose that $\overline{T} = g^{-1}(T')$, where T' is a closed subset of $X_{(K')}$, and let Z' be the reduced sub-prescheme of $X_{(K')}$ whose underlying space is T'. Then

$$\overline{Z} = Z' \otimes_{K'} K = \mathrm{Spec}(K) \times_{K'} Z'$$

is reduced by (4.6.1.e) and the separability of K/K', and its underlying space is $\overline{T}$. □

Corollary (4.9.5). — *Suppose that one of the following hypotheses holds:*

(a) *k has characteristic zero.*
(b) *K/k is algebraic and X is of finite type over k.*

Then every closed subset $\overline{T}$ of $X_{(K)}$ has a smallest field of definition. This field is a separable extension of k, and under (4.9.5.b) it is finite over k.

Proof. Under (4.9.5.a), K is separable over each of its subfields. It follows from (4.9.4) that the fields of definition of $\overline{T}$ are exactly those of the reduced sub-prescheme of $X_{(K)}$ whose underlying space is $\overline{T}$. The assertion therefore follows from (4.8.11).

Under (4.9.5.b), the extension K is radicial over a separable extension K_1 of k. For every intermediate field K' of K/k, the extension $K'/(K' \cap K_1)$ is radicial. By (4.9.2), K' is a field of definition of $\overline{T}$ if and only if $K' \cap K_1$ is. This reduces the question to the case $K' = K_1$, hence to the case of a separable algebraic extension of k, and the argument finishes as under (4.9.5.a). □

Proposition (4.9.6). — *Let k be a field, let X be a k-prescheme, let K/k be an extension, let $\overline{U}$ be a subset of $X_{(K)}$ that is both open and closed, and let $\overline{V}$ be its complement. For an intermediate field K' of K/k, the following conditions are equivalent:*

(a) *K' is a field of definition of the subset $\overline{U}$ of $X_{(K)}$.*
(a′) *K' is a field of definition of the closed sub-prescheme of $X_{(K)}$ induced on the open subset $\overline{U}$.*
(b) *K' is a field of definition of the subset $\overline{V}$ of $X_{(K)}$.*
(b′) *K' is a field of definition of the closed sub-prescheme of $X_{(K)}$ induced on the open subset $\overline{V}$.*

Proof. It is clear that (a′) implies (a), and (a) implies (b′) by (4.9.1). Since $\overline{U}$ and $\overline{V}$ play symmetric roles, (b′) implies (b), and (b) implies (a′). □

Corollary (4.9.7). — *Under the hypotheses of (4.9.6), there is a smallest field of definition K' for the subset $\overline{U}$ of $X_{(K)}$. It is also the smallest field of definition of the subset $\overline{V}$ and of the closed sub-preschemes of $X_{(K)}$ induced on the open subsets $\overline{U}$ and $\overline{V}$. If, in addition, X is of finite type over k, then K'/k is finite and separable.*

Proof. In view of (4.9.6) and (4.8.11), only the last assertion remains to be proved. By (4.8.15.ii), we may reduce to the case in which K is algebraically closed. Let $\bar{k}$ be the algebraic closure of k contained in K, and let W_i $(1 \leq i \leq m)$ be the connected components of $X_{(\bar{k})}$.

If **IV-2 | 86**

$$p : X_{(K)} \longrightarrow X_{(\bar{k})}$$

is the canonical projection, then the $p^{-1}(W_i)$ are the connected components of $X_{(K)}$ (4.4.6). Thus $\overline{U}$ is a union of some of these components and is therefore defined over $\bar{k}$. The conclusion follows from (4.9.5.b). □

§5. Dimension, depth, and regularity in locally Noetherian preschemes

This section merely restates, in geometric language and with various technical complements, the notions and results of commutative algebra set out in Chapter 0, §§ 16 and 17. **IV-2 | 86**

5.1. Dimension of preschemes

(5.1.1). Let A be a ring and let $\mathfrak{J}$ be an ideal of A. Recall (**0**, 16.1) that the definitions in the theory of the dimension of a ring (**0**, 16.1.1) and those in the theory of the combinatorial dimension of topological spaces (**0**, 14.1.2) give the following relations:

(5.1.1.1) $$\dim(\mathrm{Spec}(A)) = \dim(A),$$

(5.1.1.2) $$\dim(\mathrm{V}(\mathfrak{J})) = \dim(A/\mathfrak{J}),$$

(5.1.1.3) $$\mathrm{codim}(\mathrm{V}(\mathfrak{J}), \mathrm{Spec}(A)) = \mathrm{ht}(\mathfrak{J}).$$

(5.1.1.4) $\mathrm{Spec}(A)$ is a catenary space if and only if A is a catenary ring.

(5.1.1.5) $\mathrm{Spec}(A)$ is equidimensional if and only if A is equidimensional, if and only if the minimal prime ideals $\mathfrak{p}_\alpha$ of A are such that the rings $A/\mathfrak{p}_\alpha$ all have the same dimension.

(5.1.1.6) $\mathrm{Spec}(A)$ is equicodimensional if and only if A is equicodimensional, if and only if all maximal ideals of A have the same height.

Recall that a Noetherian ring A is said to be *biequidimensional* if $\mathrm{Spec}(A)$ is biequidimensional; that is, if $\mathrm{Spec}(A)$ is Noetherian and A is equidimensional, equicodimensional, catenary, and of finite dimension.

Proposition (5.1.2). — *Let X be a prescheme, let Y be an irreducible closed subset of X, and let y be the generic point of Y. Then*

(5.1.2.1) $$\mathrm{codim}(Y, X) = \dim(\mathscr{O}_{X,y}).$$

Proof. Indeed, the irreducible closed subsets of X containing y are canonically in bijection with the irreducible closed subsets of $\mathrm{Spec}(\mathscr{O}_{X,y})$ (**I**, 2.4.2), and hence with the prime ideals of $\mathscr{O}_{X,y}$; formula (5.1.2.1) follows from the definitions.

In particular, this recovers the fact that the generic points of the irreducible components of X are the only points $x \in X$ such that $\dim(\mathscr{O}_x) = 0$ (**I**, 1.1.14). □

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Corollary (5.1.3). — *For every closed subset Y of a prescheme X, one has*

(5.1.3.1) $$\mathrm{codim}(Y, X) = \inf_{y \in Y} \dim(\mathscr{O}_{X,y}).$$

Moreover, if X is locally Noetherian, then for every $x \in Y$,

(5.1.3.2) $$\mathrm{codim}_x(Y, X) = \inf_{\substack{y \in Y \\ x \in \overline{\{y\}}}} \dim(\mathscr{O}_{X,y}).$$

Proof. Formula (5.1.3.2) follows from (5.1.3.1) and (**0**, 14.2.6). This corollary allows us to *define*, for any subset Y of a prescheme X, the *codimension* $\operatorname{codim}(Y, X)$ of Y in X to be the right-hand side of (5.1.3.1). □

Proposition (5.1.4). — *For every prescheme X, one has*

$$\dim(X) = \sup_{x \in X} \dim(\mathcal{O}_x). \tag{5.1.4.1}$$

If every irreducible closed subset of X contains a closed point, then one also has

$$\dim(X) = \sup_{x \in F} \dim(\mathcal{O}_x), \tag{5.1.4.2}$$

where F is the set of closed points of X.

Proof. It suffices to observe, by (**I**, 2.4.2), that the chains of irreducible closed subsets of X correspond bijectively to the chains of irreducible closed subsets of a local scheme of X. When every irreducible closed subset of X contains a closed point, one may plainly restrict to the local schemes at closed points of X.

Every irreducible closed subset of X contains a closed point when X is quasi-compact (**0**, 2.1.3); we shall see below (5.1.11) that the same is true when X is locally Noetherian. □

Corollary (5.1.5). — *A prescheme X is catenary if and only if the local ring $\mathcal{O}_x$ is catenary for every $x \in X$. If, in addition, every irreducible closed subset of X contains a closed point, then for X to be catenary it is enough that $\mathcal{O}_x$ be catenary for every closed point x of X.*

Proof. The argument is the same as in (5.1.4), since it is a matter of comparing chains of irreducible closed subsets with the same endpoints. □

(5.1.6). We now examine the more special properties tied to Noetherian hypotheses.

Recall (**0**, 16.2.3) that a Noetherian local ring $A \neq 0$ has finite dimension, equal to the minimum number of generators of an ideal of definition of A. For every prime ideal $\mathfrak{p}$ of A, its height, equal to $\dim(A_{\mathfrak{p}})$, is therefore also finite. Together with (5.1.2.1), (5.1.3.1), and (5.1.4.1), these facts show the following.

Proposition (5.1.7). — *For every nonempty closed subset Y of a locally Noetherian prescheme X, the number* $\operatorname{codim}(Y, X)$ *is finite. If X is Noetherian and affine and Y is an irreducible closed subset of X, then* $\operatorname{codim}(Y, X)$ *is the minimum number of sections s_i of $\mathcal{O}_X$ over X such that Y is an irreducible component of the set of points $x \in X$ for which $s_i(x) = 0$ for every i.*

Corollary (5.1.8). — *Let X be a locally Noetherian prescheme, let $\mathscr{L}$ be an invertible $\mathcal{O}_X$-module, and let f be a section of $\mathscr{L}$ over X. Every irreducible component of the set Z of points $x \in X$ such that $f(x) = 0$ has codimension at most* 1 *in X; it has codimension* 1 *if Z contains no irreducible component of X.*

Proof. We may restrict to the case $\mathscr{L} = \mathcal{O}_X$. If y is a generic point of an irreducible component Y of Z, the ideal (f_y) of $\mathcal{O}_{X,y}$ must be such that $\mathcal{O}_{X,y}/(f_y)$ has only one prime ideal. Thus f_y generates an ideal of definition of the Noetherian local ring $\mathcal{O}_{X,y}$, and $\operatorname{codim}(Y, X) \leqslant 1$ by (5.1.7). If Z contains no irreducible component of X, then $\operatorname{codim}(Y, X)$ cannot be 0 by (**0**, 14.2.1). □

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Proposition (5.1.9). — *Let X be a locally Noetherian prescheme and let Y be a closed subset of X. If $x \in Y$ and $\mathcal{O}_{X,x}$ is a catenary ring, then*

$$\begin{aligned} \operatorname{codim}_x(Y, X) &= \dim(\mathcal{O}_{X,x}) - \operatorname{codim}(\overline{\{x\}}, Y) \\ &= \dim(\mathcal{O}_{X,x}) - \dim(\mathcal{O}_{Y,x}). \end{aligned} \tag{5.1.9.1}$$

Proof. Let Y_i $(1 \leqslant i \leqslant n)$ be the irreducible components of Y containing x; there are only finitely many because Y is locally Noetherian. Let y_i be the generic point of Y_i. Put $A = \mathcal{O}_{X,x}$, and let $\mathfrak{p}_i$ be the prime ideal of A corresponding to $Y_i \cap \operatorname{Spec}(A)$. The irreducible closed subsets of Y containing x correspond bijectively to the prime ideals of A containing one of the $\mathfrak{p}_i$, so

$$\dim(\mathcal{O}_{Y,x}) = \sup_i \dim(A/\mathfrak{p}_i).$$

On the other hand, $\mathscr{O}_{X,y_i} = A_{\mathfrak{p}_i}$, and the hypothesis on A gives

$$\dim(A) = \dim(A/\mathfrak{p}_i) + \dim(A_{\mathfrak{p}_i})$$

by (**0**, 16.1.4). The conclusion follows from

$$\operatorname{codim}_x(Y, X) = \inf_i \dim(\mathscr{O}_{X,y_i})$$

using (5.1.2.1) and (**0**, 14.2.6). □

Proposition (5.1.10). — (i) *In every prescheme X, every nonempty locally constructible subset E contains a point x such that $\{x\}$ is locally closed in X; equivalently, x is isolated in $\overline{\{x\}}$.*

(ii) *Let X be a locally Noetherian prescheme and let $x \in X$ be such that $\{x\}$ is locally closed in X. Then $\dim(\overline{\{x\}}) \leqslant 1$; consequently, every point $y \neq x$ of $\overline{\{x\}}$ is closed in X.*

Proof. (5.1.10.i) is a special case of the following lemma. □

Lemma (5.1.10.1). — *Let X be a topological space with the following property: for every nonempty locally closed subset Z of X, there exist a subset Z' of Z, locally closed in X (or, equivalently, in Z), and a point $x \in Z'$ that is closed in Z'. Then every nonempty locally closed subset Z of X contains a point x isolated in $\overline{\{x\}}$.*

Proof. Let $Z' \subset Z$ be a locally closed subset of Z containing a point x that is closed in Z'. There is an open neighbourhood U of x in X such that x is closed in U; hence

$$U \cap \overline{\{x\}} = \{x\},$$

which says precisely that x is isolated in $\overline{\{x\}}$.

The lemma applies to the underlying space of every prescheme X. Indeed, Z is then itself the underlying space of a prescheme (**I**, 5.2.1), and one may take for Z' an affine open subset of Z, which is a quasi-compact Kolmogorov space and therefore contains a closed point (**0**, 2.1.3). □

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Proof of (5.1.10.ii). Let Z be the reduced sub-prescheme of X whose underlying space is $\overline{\{x\}}$. The hypothesis implies that $\{x\}$ is open in Z. Thus, for every $z \in Z$, the generic point x of $\operatorname{Spec}(\mathscr{O}_{Z,z})$ is isolated in $\operatorname{Spec}(\mathscr{O}_{Z,z})$. The ring $A = \mathscr{O}_{Z,z}$ is a Noetherian local integral domain, and the hypothesis implies that there exists $f \in A$ such that A_f is a field. It follows from (**0**, 16.3.3) that $\dim(A) \leqslant 1$. Since $\dim(\mathscr{O}_{Z,z}) \leqslant 1$ for every $z \in Z$, one has $\dim(Z) \leqslant 1$. □

Corollary (5.1.11). — *If X is a locally Noetherian prescheme, every nonempty closed subset of X contains a closed point.*

Proof. Every closed subset of X is constructible (**0**, 9.1.1) and (**0**, 9.1.5), so it suffices to apply (5.1.10). □

(5.1.12). Let X be a prescheme, let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module of finite type, and let $S = \operatorname{Supp}(\mathscr{F})$ be its support, which is closed in X (**0**, 5.2.2). For every $x \in X$, regard $\operatorname{Supp}(\mathscr{F}_x)$ as a closed subset of the local scheme $\operatorname{Spec}(\mathscr{O}_x)$. By definition (**0**, 16.1.7),

$$\dim(\mathscr{F}_x) = \dim(\operatorname{Supp}(\mathscr{F}_x)).$$

But

$$\operatorname{Supp}(\mathscr{F}_x) = S \cap \operatorname{Spec}(\mathscr{O}_{X,x}) = \operatorname{Spec}(\mathscr{O}_{S,x}),$$

where in the last expression S denotes any closed sub-prescheme of X whose underlying space is S. The irreducible components of $\operatorname{Spec}(\mathscr{O}_{S,x})$ are the intersections of $\operatorname{Spec}(\mathscr{O}_{X,x})$ with the irreducible components of S containing x, and correspond bijectively to the latter. Hence, by (5.1.1.1),

(5.1.12.1) $$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{S,x}).$$

It also follows from (5.1.2.1) that, if X is locally Noetherian,

(5.1.12.2) $$\dim(\mathscr{F}_x) = \operatorname{codim}(\overline{\{x\}}, S).$$

We say that $\mathscr{F}$ is *equidimensional at the point* $x \in X$ if $\mathscr{F}_x$ is an equidimensional $\mathscr{O}_{X,x}$-module; that is (**0**, 16.1.7), if $\operatorname{Supp}(\mathscr{F}_x)$ is equidimensional as a closed subset of $\operatorname{Spec}(\mathscr{O}_{X,x})$. Equivalently, the ring $\mathscr{O}_{S,x}$ is equidimensional.

The *dimension* of $\mathscr{F}$, denoted $\dim(\mathscr{F})$, is the dimension of its support $\operatorname{Supp}(\mathscr{F})$. Equations (5.1.4.1) and (5.1.12.1) give

(5.1.12.3)
$$\dim(\mathscr{F}) = \sup_{x\in X} \dim(\mathscr{F}_x).$$

If $X = \operatorname{Spec}(A)$ is affine and $\mathscr{F} = \widetilde{M}$, where M is a finite A-module, then $\dim(\mathscr{F}) = \dim(M)$ by (**0**, 16.1.7) and (5.1.4.1).

Proposition (5.1.13). — *Let X be a prescheme, let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module of finite type, let $x \in X$, and let x' be a generization of x in X. Then*

(5.1.13.1)
$$\dim(\mathscr{F}_{x'}) \leqslant \dim(\mathscr{F}_x).$$

Proof. This follows immediately from (5.1.12.1) and the definitions. □

5.2. Dimension of an algebraic prescheme

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Proposition (5.2.1). — *Let k be a field, let X be an irreducible prescheme locally of finite type over k, and let ξ be its generic point. Then X is biequidimensional and*

$$\dim(X) = \operatorname{tr.deg}_k \kappa(\xi).$$

Proof. Every local ring $\mathscr{O}_x$ is a local ring of a finite-type k-algebra, hence is a quotient of a regular local ring (**0**, 17.3.9). Such rings are catenary (**0**, 16.5.12); consequently X is a catenary space (5.1.5). Since X is irreducible, it follows from (5.1.1) that it is enough to prove, for every closed point $x \in X$, that

(5.2.1.1)
$$\dim(\mathscr{O}_x) = \operatorname{tr.deg}_k \kappa(\xi).$$

We may plainly suppose that X is reduced and affine, hence integral, with ring A, where A is a finite-type k-algebra. Put

$$n = \operatorname{tr.deg}_k \kappa(\xi), \qquad K = \kappa(\xi) = \operatorname{Frac}(A).$$

There exists a k-subalgebra

$$B = k[t_1, \dots, t_n] \subset A,$$

where the t_i are algebraically independent over k, such that A is a finite B-algebra (Bourbaki, *Algèbre commutative*, chap. V, § 3, n° 1, th. 1). Let $\mathfrak{m} = \mathfrak{j}_x$, which is maximal by hypothesis. Then $\mathfrak{n} = B \cap \mathfrak{m}$ is a maximal ideal of B (ibid., chap. V, § 2, n° 1, prop. 1), and $A_{\mathfrak{m}}$ is a local ring of the finite $B_{\mathfrak{n}}$-algebra $S^{-1}A$, where $S = B - \mathfrak{n}$. Since $B_{\mathfrak{n}}$ is integrally closed and $S^{-1}A$ is integral,

$$\dim(A_{\mathfrak{m}}) = \dim(B_{\mathfrak{n}})$$

by (**0**, 16.1.6).

We may therefore reduce to $A = k[t_1, \dots, t_n]$. The residue field $k' = \kappa(x)$ is then a finite extension of k (**I**, 6.4.2). Put $A' = k'[t_1, \dots, t_n]$. By (**I**, 2.4.6) and (**I**, 3.3.14), there exists a maximal ideal $\mathfrak{m}'$ of A' lying above $\mathfrak{m}$ such that k' is the residue field of $A'_{\mathfrak{m}'}$. Since A' is integral and finite over A, the preceding argument gives

$$\dim(A'_{\mathfrak{m}'}) = \dim(A_{\mathfrak{m}}),$$

so we are reduced to the case $A/\mathfrak{m} = k$.

In that case $\mathfrak{m}$ is generated by the polynomials $t_i - a_i$, where $a_i \in k$ is the image of t_i in $A/\mathfrak{m}$. Replacing t_i by $t_i - a_i$, we are finally reduced to

$$\mathfrak{m} = \sum_{i=1}^{n} At_i.$$

The completion of $A_{\mathfrak{m}}$ is then the formal power-series ring $k[[t_1, \dots, t_n]]$. It has the same dimension as $A_{\mathfrak{m}}$ (**0**, 16.2.4), while $\dim(k[[t_1, \dots, t_n]]) = n$ by (**0**, 17.1.4). This proves (5.2.1.1). □

Corollary (5.2.2). — *For a prescheme X locally of finite type over a field k, $\dim(X)$ agrees with the number defined in (4.1.1).*

Proof. Let X_α be the reduced sub-preschemes whose underlying spaces are the irreducible components of X. Then

$$\dim(X) = \sup_\alpha \dim(X_\alpha)$$

by (**0**, 14.1.2.1), and it suffices to apply (5.2.1) to the X_α. □

Corollary (5.2.3). — *Let X be a prescheme locally of finite type over a field k, and let $x \in X$. Then*

$$\dim_x(X) = \dim(\mathscr{O}_x) + \operatorname{tr.deg}_k \kappa(x). \tag{5.2.3.1}$$

Proof. There is an open neighbourhood U of x in X such that $\dim_x(X) = \dim(U)$ (**0**, 14.1.4.1). We may suppose that the irreducible components of U are the $X_i \cap U$, where the X_i are the irreducible components of X containing x.

Since $U \cap X_i$ is dense in X_i, formula (4.1.1.3) gives $\dim(X_i) = \dim(U \cap X_i)$, and hence **IV-2** | 91

$$\dim_x(X) = \sup_i \dim(X_i).$$

The minimal prime ideals of $\mathscr{O}_x$ correspond to the generic points of the X_i, so

$$\dim(\mathscr{O}_{X,x}) = \sup_i \dim(\mathscr{O}_{X_i,x})$$

by (**0**, 16.1.1.1). We are therefore reduced to the case where X is irreducible. Since X is biequidimensional by (5.2.1),

$$\dim(X) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X)$$

by (**0**, 14.3.5.1). Finally,

$$\dim(\overline{\{x\}}) = \operatorname{tr.deg}_k \kappa(x)$$

by (5.2.1), while

$$\operatorname{codim}(\overline{\{x\}}, X) = \dim(\mathscr{O}_x)$$

by (5.1.2). □

Corollary (5.2.4). — *Let X be a prescheme locally of finite type over a field k, let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module, and let f be a section of $\mathscr{L}$ over X such that the set*

$$Y = \{x \in X \mid f(x) = 0\}$$

is rare in X. Then

$$\dim(Y) \leqslant \dim(X) - 1.$$

Equality holds if Y meets every irreducible component of X having maximum dimension.

Proof. If (X_α) is the family of irreducible components of X, then

$$\dim(Y) = \sup_\alpha \dim(Y \cap X_\alpha).$$

We are therefore reduced to the case where X is irreducible, because $Y \cap X_\alpha$ is rare in X_α and every X_α has nonempty interior in X. We may also suppose that $Y \neq \varnothing$. For every maximal point x of Y, biequidimensionality gives

$$\dim(\overline{\{x\}}) = \dim(X) - \operatorname{codim}(\overline{\{x\}}, X).$$

Since Y is rare in X,

$$\operatorname{codim}(\overline{\{x\}}, X) = 1$$

by (5.1.8), and the result follows. □

Remarks (5.2.5). — (i) Unlike an algebraic k-prescheme, a locally Noetherian prescheme X need not be catenary; see (5.6.11). It is catenary, however, if all its local rings $\mathscr{O}_x$ are quotients of regular local rings (**0**, 16.5.12), and in particular if X is regular (**I**, 4.1.4).

Nevertheless, even an integral scheme $X = \operatorname{Spec}(A)$ with A regular need not be biequidimensional. In other words, the closed points of X need not all have the same codimension (**0**, 14.3.3). For example, let B be a discrete valuation ring, let $\mathfrak{m} = B\pi$ be its maximal ideal, let $k = B/\mathfrak{m}$ be its residue field, let K be its fraction field, and put $A = B[T]$. The ring A has maximal ideals of height 2, such as $(\pi) + (T)$, but also maximal ideals of height 1, such as

the principal ideal $(\pi T - 1)$. Indeed, a nonzero principal prime ideal has height 1 (5.1.8), while
$$A/(\pi T - 1) \simeq B_\pi = K$$
by (**0**, 1.2.3); thus $(\pi T - 1)$ is maximal and has height 1.

(ii) If X is locally of finite type over a field, then $\dim(U) = \dim(X)$ for every dense open subset U of X (4.1.1.3). This does not extend to regular Noetherian preschemes, even when they are biequidimensional. For example, let B be a discrete valuation ring with maximal ideal $\mathfrak{m}$. Then $X = \mathrm{Spec}(B)$ has two points, (0) and $\mathfrak{m}$, the latter being its only closed point. One has $\dim(X) = 1$, whereas the open subset $U = \{(0)\}$ has dimension 0 (see § 10).[5]

5.3. Dimension of the support of a module and Hilbert polynomial This section uses the results of Chapter III; it will not be used in the remainder of this chapter. **IV-2** | 92

Proposition (5.3.1). — *Let A be an Artinian local ring, let X be a projective scheme over* $\mathrm{Spec}(A)$*, let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module that is very ample relative to A, and let $\mathscr{F}$ be a nonzero coherent $\mathscr{O}_X$-module. Put*
$$\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n} \qquad (n \in \mathbf{Z}).$$
Then the degree of the Hilbert polynomial
$$P(n) = \chi_A(\mathscr{F}(n))$$
of $\mathscr{F}$ relative to A (**III**, 2.5.3) *equals* $\dim(\mathrm{Supp}(\mathscr{F}))$.

Proof. We argue by induction on
$$d = \dim(\mathrm{Supp}(\mathscr{F})).$$
There is a closed subprescheme Y of X whose underlying space is $\mathrm{Supp}(\mathscr{F})$, together with a coherent $\mathscr{O}_Y$-module $\mathscr{G}$ such that $\mathscr{F} = j_*(\mathscr{G})$, where $j : Y \to X$ is the canonical injection (**Err$_{\mathbf{III}}$**, 30). The Hilbert polynomials of $\mathscr{F}$ and $\mathscr{G}$ are plainly the same, so we may restrict to the case $X = \mathrm{Supp}(\mathscr{F})$.

Suppose first that $d = 0$. Every point of X is then closed, and X is an Artinian scheme (**I**, 6.2.2); hence $\mathscr{F}(n) = \mathscr{F}$ for every integer n. Consequently, by (**III**, 2.5.3),
$$\chi_A(\mathscr{F}(n)) = \mathrm{length}_A(\Gamma(X, \mathscr{F}))$$
for all sufficiently large n, and the Hilbert polynomial therefore has degree 0.

Suppose now that $d > 0$, and put $Z = \mathrm{Ass}(\mathscr{F})$, a finite set (3.1.6). There are an integer $m > 0$ and a section $f \in \Gamma(X, \mathscr{L}^{\otimes m})$ such that X_f is a neighbourhood of Z (**II**, 4.5.4). Multiplication by f is an injective homomorphism
$$\mu_f : \mathscr{F} \longrightarrow \mathscr{F}(m)$$
by (3.1.8); thus there is an exact sequence

(5.3.1.1) $$0 \longrightarrow \mathscr{F} \xrightarrow{\mu_f} \mathscr{F}(m) \longrightarrow \mathscr{G} \longrightarrow 0,$$

where $\mathscr{G}$ is coherent. By Nakayama's lemma, the points $x \in \mathrm{Supp}(\mathscr{G})$ are exactly those for which $f(x) = 0$. We shall deduce that

(5.3.1.2) $$\dim(\mathrm{Supp}(\mathscr{G})) = d - 1.$$

This follows from the next lemma. □

Lemma (5.3.1.3). — *Let A be an Artinian local ring, let X be a projective scheme over* $\mathrm{Spec}(A)$*, and let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module that is ample relative to A. For every section g of $\mathscr{L}$ over X, the set*
$$X \setminus X_g = \{x \in X \mid g(x) = 0\}$$
meets every irreducible component of X having nonzero dimension.

Proof. Indeed, X_g is an affine open subset of X (**II**, 5.5.7). If it contains an irreducible component X' of X, the reduced closed subprescheme of X whose underlying space is X' is both projective and affine over A, hence finite over A (**III**, 4.4.2). It is therefore an Artinian scheme and has dimension 0. □

[5] The printed French text has $X = \mathrm{Spec}(A)$ here; the ring just specified, and the asserted two-point spectrum, require $X = \mathrm{Spec}(B)$.

Proof of (5.3.1), *continued.* Since Z contains the maximal points of X, the open subset X_f is dense; hence $\mathrm{Supp}(\mathscr{G})$ is rare in X. Equation (5.3.1.2) now follows from the lemma and (5.2.4).

From the exact sequence (5.3.1.1), for every n we obtain the exact sequence **IV-2** | 93

$$0 \longrightarrow \mathscr{F}(n) \longrightarrow \mathscr{F}(n+m) \longrightarrow \mathscr{G}(n) \longrightarrow 0.$$

For all sufficiently large n, there is therefore also an exact sequence (**III**, 2.2.3)

$$0 \longrightarrow \Gamma(X, \mathscr{F}(n)) \longrightarrow \Gamma(X, \mathscr{F}(n+m)) \longrightarrow \Gamma(X, \mathscr{G}(n)) \longrightarrow 0.$$

Taking (**III**, 2.5.3) into account, we have, for all sufficiently large n,

$$\chi_A(\mathscr{G}(n)) = \chi_A(\mathscr{F}(n+m)) - \chi_A(\mathscr{F}(n)).$$

By (5.3.1.2) and the induction hypothesis, the polynomial $\chi_A(\mathscr{G}(n))$ has degree $d-1$. The preceding relation therefore implies that $\chi_A(\mathscr{F}(n))$ has degree d. □

5.4. Dimension of the image of a morphism

Proposition (5.4.1). — *Let X and Y be locally Noetherian preschemes, and let $f : X \to Y$ be a morphism.*

(i) *If f is quasi-finite, then*

$$\dim(X) \leqslant \dim(\overline{f(X)}) \leqslant \dim(Y).$$

(ii) *If f is surjective and open (respectively, surjective and closed), then*

$$\dim(X) \geqslant \dim(Y).$$

Proof. (i) We may replace f by f_{red} (**II**, 6.2.4), and therefore suppose that X and Y are reduced. Let Z be the reduced closed subprescheme of Y whose underlying space is $\overline{f(X)}$. Then $f = j \circ g$, where $j : Z \to Y$ is the canonical injection and $g : X \to Z$ is quasi-finite (**I**, 5.2.2), (**II**, 6.2.4). We may thus restrict to the case in which $f(X)$ is dense in Y.

For every $x \in X$, the ring $\mathscr{O}_x$ is then a quasi-finite $\mathscr{O}_{f(x)}$-module (**II**, 6.2.2); consequently

$$\mathfrak{m}_{f(x)}\mathscr{O}_x$$

is an ideal of definition of $\mathscr{O}_x$ (**0**, 7.4.4). If $A \to B$ is a local homomorphism of Noetherian local rings, with maximal ideal $\mathfrak{m}$ of A, and if $\mathfrak{m}B$ is an ideal of definition of B, then

$$\dim(B) \leqslant \dim(A)$$

by (**0**, 16.3.10). Part (i) now follows from (5.1.4).

(ii) By the definition of dimension (**0**, 14.1.2), it is enough to prove the following. For every sequence $(y_i)_{0 \leqslant i \leqslant n}$ of distinct points of Y such that

$$y_i \in \overline{\{y_{i+1}\}} \qquad (0 \leqslant i \leqslant n-1),$$

there is a sequence $(x_i)_{0 \leqslant i \leqslant n}$ of points of X such that

$$x_i \in \overline{\{x_{i+1}\}} \qquad (0 \leqslant i \leqslant n-1)$$

and $f(x_i) = y_i$ for every i.

Suppose first that f is surjective and open. We construct the x_i by induction on i. Surjectivity gives an $x_0 \in X$ such that $f(x_0) = y_0$. Suppose the x_i have been chosen for $i \leqslant m$ so that $f(x_i) = y_i$ for $i \leqslant m$ and $x_i \in \overline{\{x_{i+1}\}}$ for $i < m$. Since f is open and y_{m+1} is a generization of y_m, there is an $x_{m+1} \in f^{-1}(y_{m+1})$ that is a generization of x_m (1.10.3). The induction therefore continues.

Suppose now that f is surjective and closed. We construct the x_i by descending induction on i. Surjectivity gives an x_n such that $f(x_n) = y_n$. Suppose the x_i have been chosen with the required properties for $i > m$. Because f is closed, **IV-2** | 94

$$f(\overline{\{x_{m+1}\}}) = \overline{\{f(x_{m+1})\}} = \overline{\{y_{m+1}\}}$$

(Bourbaki, *Topologie générale*, chap. I, 3rd ed., § 5, n° 4, prop. 9). There is therefore an $x_m \in \overline{\{x_{m+1}\}}$ such that $f(x_m) = y_m$, and the descending induction continues.

□

Corollary (5.4.2). — *Let X and Y be locally Noetherian preschemes, and let $f : X \to Y$ be a finite morphism, hence a closed morphism. Then*

$$\dim(X) = \dim(\overline{f(X)}).$$

If f is moreover surjective, then $\dim(X) = \dim(Y)$.

Remarks (5.4.3). — (i) We saw in (4.1.2) that if X and Y are preschemes locally of finite type over a field k, and f is a k-morphism, the inequality

$$\dim(Y) \leqslant \dim(X)$$

already holds when f is quasi-compact and dominant. By contrast, if one assumes only that X and Y are locally Noetherian, it is possible to have $\dim(Y) > \dim(X)$ even when f is of finite type, bijective, and a local immersion.

For example, let A be a discrete valuation ring, let K be its fraction field, and let k be its residue field. Put $Y = \operatorname{Spec}(A)$, and denote by a the generic point of Y and by b its closed point; thus $\kappa(a) = K$ and $\kappa(b) = k$. Let

$$X = \operatorname{Spec}(K) \amalg \operatorname{Spec}(k),$$

and let $f : X \to Y$ agree on the two summands with the respective canonical morphisms (**I**, 2.4.5). The morphism f is plainly a bijective local immersion, since $\{a\}$ is open in Y. It is also of finite type: if π is a uniformizer of A, then $K = A[\pi^{-1}]$, so K is a finitely generated A-algebra. Nevertheless,

$$\dim(X) = 0 \qquad \text{and} \qquad \dim(Y) = 1.$$

(ii) Let $A \subset B$ be Noetherian rings, with B a finite A-algebra. Corollary (5.4.2) again gives

$$\dim(B) = \dim(A)$$

(**0**, 16.1.5). Suppose in addition that A is a Noetherian local ring. Then B is a Noetherian semilocal ring (Bourbaki, *Algèbre commutative*, chap. IV, § 2, n$^{\text{o}}$ 5, cor. 3 of prop. 9). If $\mathfrak{n}_i$ $(1 \leqslant i \leqslant r)$ are the maximal ideals of B, then

(5.4.3.1)
$$\dim(A) = \sup_i \dim(B_{\mathfrak{n}_i}).$$

Under these hypotheses one need not have

$$\dim(B_{\mathfrak{n}_i}) = \dim(A)$$

for every i; replacing B by the direct product of B with the residue field k of A gives an immediate example. There are even examples in which A and B are integral domains of dimension 2 and some of the $B_{\mathfrak{n}_i}$ have dimension strictly less than $\dim(A)$ (5.6.11). We shall see later, in (5.6.4) and (5.6.10), that this last phenomenon cannot occur when A is a quotient of a regular local ring.

5.5. Dimension formula for a morphism of finite type

(5.5.1). **IV-2 | 94**

Recall (**0**, 16.3.9) that if A and B are Noetherian local rings, k is the residue field of A, and $\varphi : A \to B$ is a local homomorphism, then

(5.5.1.1)
$$\dim(B) \leqslant \dim(A) + \dim(B \otimes_A k).$$

We deduce the following.

Proposition (5.5.2). — *Let X and Y be locally Noetherian preschemes, let $f : X \to Y$ be a morphism, let $x \in X$, and put $y = f(x)$. Then*

(5.5.2.1)
$$\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y) + \dim\big(\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y)\big).$$

In particular, if x is a maximal point of the fibre $f^{-1}(y)$, then

(5.5.2.2)
$$\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y),$$

because

$$\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y) = \mathscr{O}_x / \mathfrak{m}_y \mathscr{O}_x$$

is the local ring of x in the prescheme $f^{-1}(y)$ and therefore has dimension 0.

We shall now obtain a formula more precise than (5.5.2.1) when f is assumed to be of finite type. **IV-2** | 95

Proposition (5.5.3). — *Let A be a Noetherian ring, let T be an indeterminate, and let $\mathfrak{p}$ be a prime ideal of A. Then*

$$\mathfrak{p}' = \mathfrak{p}A[T]$$

is a prime ideal of $B = A[T]$, and $\mathfrak{p}' \cap A = \mathfrak{p}$. There are infinitely many prime ideals of B, distinct from $\mathfrak{p}'$, whose intersection with A is $\mathfrak{p}$; these prime ideals are pairwise incomparable. Moreover, if $\mathfrak{q}$ is any such ideal, then

$$\dim(B_{\mathfrak{q}}) = \dim(B_{\mathfrak{p}'}) + 1 = \dim(A_{\mathfrak{p}}) + 1. \tag{5.5.3.1}$$

Proof. For the first assertions, replacing A by $A/\mathfrak{p}$ and using

$$A[T]/\mathfrak{p}A[T] = (A/\mathfrak{p})[T]$$

reduces us immediately to the case $\mathfrak{p} = 0$. The prime ideals of $B = A[T]$ whose intersection with A is 0 are exactly those disjoint from the multiplicative set $S = A - \{0\}$ of the domain A. There is an inclusion-preserving bijection from this set of prime ideals to the prime ideals of

$$S^{-1}A[T] = K[T],$$

where K is the fraction field of A (Bourbaki, *Algèbre commutative*, chap. II, § 2, n° 5, prop. 11). The assertions follow.

Furthermore, by (5.5.1.1),[6]

$$\dim(B_{\mathfrak{q}}) \leqslant \dim(A_{\mathfrak{p}}) + \dim(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}).$$

Let k be the fraction field of $A/\mathfrak{p}$. The ring $B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$ is canonically identified with

$$(k[T])_{\overline{\mathfrak{q}}},$$

where $\overline{\mathfrak{q}}$ is a nonzero prime ideal of $k[T]$. It is therefore a discrete valuation ring and has dimension 1. Thus

$$\dim(B_{\mathfrak{q}}) \leqslant \dim(A_{\mathfrak{p}}) + 1.$$

Let

$$\mathfrak{p} = \mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_m$$

be a chain of prime ideals of A of maximum length. The ideals $\mathfrak{p}_j B$ $(0 \leqslant j \leqslant m)$ are prime in B, pairwise distinct, and contained in $\mathfrak{q}$. Hence

$$\dim(B_{\mathfrak{q}}) \geqslant m + 1 = \dim(A_{\mathfrak{p}}) + 1.$$

It follows that

$$\operatorname{ht}(\mathfrak{q}) = \operatorname{ht}(\mathfrak{p}) + 1.$$

Since $\mathfrak{q} \neq \mathfrak{p}'$, we also have

$$\operatorname{ht}(\mathfrak{q}) \geqslant \operatorname{ht}(\mathfrak{p}') + 1 \geqslant \operatorname{ht}(\mathfrak{p}) + 1.$$

All these inequalities are equalities, and (5.5.3.1) follows. □

Corollary (5.5.4). — *For every Noetherian ring A,*

$$\dim(A[T_1, \dots, T_r]) = \dim(A) + r, \tag{5.5.4.1}$$

where the T_i are indeterminates.

By contrast, there are examples of non-Noetherian local rings A such that

$$\dim(A) = 1 \qquad \text{and} \qquad \dim(A[T]) = 3$$

[30].

Corollary (5.5.5). — *For every locally Noetherian prescheme X, the dimension of*

$$X[T_1, \dots, T_r] = X \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \dots, T_r],$$

where the T_i are indeterminates, is $\dim(X) + r$.

[6] The printed French text cites *(5.5.1.2)*, which does not exist. The displayed inequality used here is *(5.5.1.1)*.

Proof. This follows from (5.5.4) and (**0**, 14.1.7). □

Corollary (5.5.6). — *Under the hypotheses of (5.5.3), let $\mathfrak{q}$ be a prime ideal of $B = A[T]$ such that $\mathfrak{q} \cap A = \mathfrak{p}$. If k and k' are the residue fields of $A_\mathfrak{p}$ and $B_\mathfrak{q}$, respectively, then*

(5.5.6.1) $$\dim(A_\mathfrak{p}) + 1 = \dim(B_\mathfrak{q}) + \operatorname{tr.deg}_k k'.$$

Proof. **IV-2 | 96**

If $\mathfrak{q} = \mathfrak{p}B$, then

$$\dim(B_\mathfrak{q}) = \dim(A_\mathfrak{p})$$

by (5.5.3.1), while $k' = k(T)$. Thus (5.5.6.1) holds. If $\mathfrak{q} \neq \mathfrak{p}B$, then

$$\dim(B_\mathfrak{q}) = \dim(A_\mathfrak{p}) + 1,$$

and $\mathfrak{q}$ corresponds to a nonzero prime ideal of $k[T]$. Consequently k' is algebraic over k, and (5.5.6.1) again holds. □

Lemma (5.5.7). — *Let A be a Noetherian local domain, let $\mathfrak{m}$ be its maximal ideal, and let k be its residue field.*

(i) *Let B be a domain containing A such that $B = A[x]$ for some $x \in B$, and let $\mathfrak{q}$ be a prime ideal of B such that $\mathfrak{q} \cap A = \mathfrak{m}$. Then*

(5.5.7.1) $$\dim(A) + \operatorname{tr.deg}_A B \geqslant \dim(B_\mathfrak{q}) + \operatorname{tr.deg}_k k',$$

where k' is the residue field of $B_\mathfrak{q}$ and $\operatorname{tr.deg}_A B$ denotes the transcendence degree of the fraction field of B over that of A.

(ii) *Suppose that for every maximal ideal $\mathfrak{n}$ of $A[T]$ such that $\mathfrak{n} \cap A = \mathfrak{m}$, the ring $(A[T])_\mathfrak{n}$ is catenary. Then the two sides of (5.5.7.1) are equal.*

Proof. (i) If x is transcendental over the fraction field of A, then $\operatorname{tr.deg}_A B = 1$, and the two sides of (5.5.7.1) are equal by (5.5.6).

Suppose instead that x is algebraic over the fraction field of A. Then

$$B = A[T]/\mathfrak{p},$$

where $\mathfrak{p}$ is a nonzero prime ideal of $A[T]$ such that $\mathfrak{p} \cap A = 0$, because B contains A. Thus

$$\operatorname{ht}(\mathfrak{p}) = 1$$

by (5.5.3). The ideal $\mathfrak{q}$ is of the form $\mathfrak{n}/\mathfrak{p}$, where $\mathfrak{n} \supset \mathfrak{p}$ is a prime ideal of $A[T]$ satisfying $\mathfrak{n} \cap A = \mathfrak{m}$, and

$$B_\mathfrak{q} = (A[T])_\mathfrak{n}/\mathfrak{p}(A[T])_\mathfrak{n}.$$

Formula (**0**, 16.1.4.1), applied to

$$X = \operatorname{Spec}((A[T])_\mathfrak{n}) \quad \text{and} \quad Y = \operatorname{Spec}(B_\mathfrak{q}),$$

gives

(5.5.7.2) $$\begin{aligned} \dim((A[T])_\mathfrak{n}) &\geqslant \operatorname{ht}\big(\mathfrak{p}(A[T])_\mathfrak{n}\big) + \dim(B_\mathfrak{q}) \\ &= 1 + \dim(B_\mathfrak{q}). \end{aligned}$$

Indeed,

$$\operatorname{ht}\big(\mathfrak{p}(A[T])_\mathfrak{n}\big) = \operatorname{ht}(\mathfrak{p}) = 1$$

by the bijection between prime ideals of $A[T]$ contained in $\mathfrak{n}$ and prime ideals of $(A[T])_\mathfrak{n}$. Finally, (5.5.6.1) gives

(5.5.7.3) $$\dim((A[T])_\mathfrak{n}) = \dim(A) + 1 - \operatorname{tr.deg}_k k',$$

because $B_\mathfrak{q}$ and $(A[T])_\mathfrak{n}$ have the same residue field. Here $\operatorname{tr.deg}_A B = 0$, and (5.5.7.1) follows.

(ii) If $(A[T])_\mathfrak{n}$ is catenary, the two sides of (5.5.7.2) are equal (**0**, 16.1.4); hence the two sides of (5.5.7.1) are equal as well.

□

Theorem (5.5.8 (dimension formula)). — *Let A be a Noetherian local domain, let B be a domain containing A and finitely generated as an A-algebra, and let $\mathfrak{q}$ be a prime ideal of B such that $\mathfrak{q} \cap A$ is the maximal ideal $\mathfrak{m}$ of A. Let k and k' be the residue fields of A and $B_{\mathfrak{q}}$, respectively. Then*

(5.5.8.1) $$\dim(A) + \operatorname{tr.deg}_A B \geqslant \dim(B_{\mathfrak{q}}) + \operatorname{tr.deg}_k k'.$$

Moreover, the two sides are equal if, for every finitely generated A-subalgebra A' of $B[T]$ and every maximal ideal $\mathfrak{m}'$ of A' such that $\mathfrak{m}' \cap A = \mathfrak{m}$, the local ring $A'_{\mathfrak{m}'}$ is catenary.

Proof. **IV-2** | 97

Write
$$B = A[x_1, \dots, x_n]$$
and argue by induction on n. Put
$$C = A[x_1, \dots, x_{n-1}], \qquad \mathfrak{r} = \mathfrak{q} \cap C.$$
The ring $C_{\mathfrak{r}}$ is a Noetherian local domain. Let
$$S = C - \mathfrak{r}, \qquad B' = S^{-1}B, \qquad \mathfrak{q}' = S^{-1}\mathfrak{q}.$$
Then
$$B_{\mathfrak{q}} = B'_{\mathfrak{q}'}, \qquad B' = C_{\mathfrak{r}}[x_n], \qquad \mathfrak{q}' \cap C_{\mathfrak{r}} = \mathfrak{r}C_{\mathfrak{r}}.$$
Moreover, the fraction fields of B' and $C_{\mathfrak{r}}$ are those of B and C, respectively. If k_1 is the residue field of $C_{\mathfrak{r}}$, then (5.5.7.1) gives

(5.5.8.2) $$\dim(C_{\mathfrak{r}}) + \operatorname{tr.deg}_C B \geqslant \dim(B_{\mathfrak{q}}) + \operatorname{tr.deg}_{k_1} k'.$$

The induction hypothesis gives

(5.5.8.3) $$\dim(A) + \operatorname{tr.deg}_A C \geqslant \dim(C_{\mathfrak{r}}) + \operatorname{tr.deg}_k k_1.$$

Adding (5.5.8.2) and (5.5.8.3) term by term yields (5.5.8.1).

For the second assertion, first note that every finitely generated A-subalgebra of $C[T]$ is also a finitely generated A-subalgebra of $B[T]$. By induction on n, we may therefore suppose that the two sides of (5.5.8.3) are equal.

By (5.5.7), to prove equality in (5.5.8.2), it is enough to show that whenever $\mathfrak{n}$ is a maximal ideal of $C_{\mathfrak{r}}[T]$ satisfying
$$\mathfrak{n} \cap C_{\mathfrak{r}} = \mathfrak{r}C_{\mathfrak{r}},$$
the ring $(C_{\mathfrak{r}}[T])_{\mathfrak{n}}$ is catenary. We have
$$C_{\mathfrak{r}}[T] = S^{-1}C[T].$$
Thus $\mathfrak{n} = S^{-1}\mathfrak{n}'$ for a prime ideal $\mathfrak{n}'$ of $C[T]$ such that $\mathfrak{n}' \cap C = \mathfrak{r}$, and
$$(C_{\mathfrak{r}}[T])_{\mathfrak{n}} = (C[T])_{\mathfrak{n}'}.$$
Choose a maximal ideal $\mathfrak{m}'$ of $C[T]$ containing $\mathfrak{n}'$. Then $\mathfrak{m}' \cap A = \mathfrak{m}$, and $(C[T])_{\mathfrak{m}'}$ is catenary by hypothesis. Since $(C[T])_{\mathfrak{n}'}$ is a localization of this ring, it too is catenary (**0**, 16.1.4). □

5.6. Dimension formula and universally catenary rings **IV-2 | 97**

Proposition (5.6.1). — *Let A be a Noetherian ring. The following properties are equivalent:*

(a) *Every polynomial ring $A[T_1, \ldots, T_n]$ is catenary.*
(b) *Every A-algebra of finite type is catenary.*
(c) *The ring A is catenary, and the following dimension formula holds. Let B' be any integral local A-algebra essentially of finite type (1.3.8); let $\mathfrak{q}$ be its maximal ideal and $\mathfrak{p}$ the inverse image of $\mathfrak{q}$ in A; let A' be the image of $A_{\mathfrak{p}}$ in B'; and let k and k' be the residue fields of A' and B', respectively. Then*

(5.6.1.1)
$$\dim(A') + \operatorname{tr.deg}_{A'} B' = \dim(B') + \operatorname{tr.deg}_k k'.$$

Proof. Suppose first that condition (b) holds. Then B' is a local A'-algebra essentially of finite type (1.3.10); hence

$$B' = B''_{\mathfrak{q}''},$$

where B'' is a finitely generated A'-subalgebra of B' and $\mathfrak{q}''$ is a prime ideal of B'' lying over the maximal ideal of A'. Moreover, B' and B'' have the same fraction field, so

$$\operatorname{tr.deg}_{A'} B' = \operatorname{tr.deg}_{A'} B''.$$

To prove (5.6.1.1), it is enough to show that every finitely generated A'-subalgebra A_1 of $B'[T]$ is catenary. Indeed, the two sides of (5.5.8.1), applied to A', B'', and $\mathfrak{q}''$, are then equal. But Condition (b) implies that every A-algebra essentially of finite type is catenary (**0**, 16.1.4), and A_1 is such an A-algebra (1.3.9). Thus (b) implies (c).

It is immediate that (b) implies (a). Conversely, (a) implies (b), because every A-algebra of finite type is a quotient of a polynomial algebra (**0**, 16.1.4).

It remains to prove that (c) implies (b). Since every quotient by a prime ideal of an A-algebra of finite type is an integral A-algebra of finite type, it is enough to prove the following. Let B be an integral A-algebra of finite type, and let $\mathfrak{q}' \subset \mathfrak{q}$ be prime ideals of B. Then

(5.6.1.2)
$$\dim(B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}) + \dim(B_{\mathfrak{q}'}) = \dim(B_{\mathfrak{q}}).$$

By (**0**, 16.1.4.2), this is precisely the required catenarity condition.

Let $\mathfrak{p}$ and $\mathfrak{p}'$ be the inverse images in A of $\mathfrak{q}$ and $\mathfrak{q}'$, respectively, and let $\mathfrak{n}$ be the kernel of **IV-2 | 98**

$$A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{q}}.$$

The image of $A_{\mathfrak{p}}$ in $B_{\mathfrak{q}}$ is isomorphic to $A_{\mathfrak{p}}/\mathfrak{n}$. Applying (5.6.1.1) to this image and to $B_{\mathfrak{q}}$ gives

(5.6.1.3)
$$\begin{aligned}\dim(A_{\mathfrak{p}}/\mathfrak{n}) &+ \operatorname{tr.deg}_{A_{\mathfrak{p}}/\mathfrak{n}} B_{\mathfrak{q}}\\ &= \dim(B_{\mathfrak{q}}) + \operatorname{tr.deg}_{\kappa(\mathfrak{p})} \kappa(\mathfrak{q}).\end{aligned}$$

The kernel of $A_{\mathfrak{p}'} \to B_{\mathfrak{q}'}$ is $\mathfrak{n}A_{\mathfrak{p}'}$. Applying (5.6.1.1) again gives

(5.6.1.4)
$$\begin{aligned}\dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) &+ \operatorname{tr.deg}_{A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}} B_{\mathfrak{q}'}\\ &= \dim(B_{\mathfrak{q}'}) + \operatorname{tr.deg}_{\kappa(\mathfrak{p}')} \kappa(\mathfrak{q}').\end{aligned}$$

Finally, $B/\mathfrak{q}'$ is an integral A-algebra of finite type. The inverse image of $\mathfrak{q}/\mathfrak{q}'$ in A is $\mathfrak{p}$, and the kernel of

$$A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}$$

is $\mathfrak{p}'A_{\mathfrak{p}}$. Hence

(5.6.1.5)
$$\begin{aligned}\dim(A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}) &+ \operatorname{tr.deg}_{\kappa(\mathfrak{p}')}(B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}})\\ &= \dim(B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}) + \operatorname{tr.deg}_{\kappa(\mathfrak{p})} \kappa(\mathfrak{q}).\end{aligned}$$

Add (5.6.1.4) and (5.6.1.5) term by term. The fields $\kappa(\mathfrak{p}')$ and $\kappa(\mathfrak{q}')$ are the fraction fields of $A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}$ and $B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}$, respectively. Furthermore, $A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}$ and $A_{\mathfrak{p}}/\mathfrak{n}$ have the same fraction field, as do $B_{\mathfrak{q}'}$ and $B_{\mathfrak{q}}$. Since A is catenary, (**0**, 16.1.4.2) gives

$$\dim(A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}) + \dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) = \dim(A_{\mathfrak{p}}/\mathfrak{n}).$$

Taking these observations and (5.6.1.3) into account yields (5.6.1.2). □

Definition (5.6.2). — A Noetherian ring is said to be *universally catenary* if it satisfies the equivalent conditions of (5.6.1).

Remarks (5.6.3). — (i) If A is universally catenary, then so is $S^{-1}A$ for every multiplicative subset S of A. This follows at once from 5.6.1(a) and the fact that every ring of fractions of a catenary ring is catenary.

Conversely, suppose that $A_{\mathfrak{p}}$ is universally catenary for every prime ideal $\mathfrak{p}$ of A. Then A is universally catenary. Indeed, apply 5.6.1(c); if $S = A - \mathfrak{p}$, then $S^{-1}B$ is an $A_{\mathfrak{p}}$-algebra of finite type and

$$B_{\mathfrak{q}} = (S^{-1}B)_{S^{-1}\mathfrak{q}}.$$

(ii) A locally Noetherian prescheme X is said to be *universally catenary* if $\mathscr{O}_x$ is universally catenary for every $x \in X$. By (i), a Noetherian ring A is universally catenary if and only if $\operatorname{Spec}(A)$ is universally catenary.

(iii) The criterion 5.6.1(c) involves only the images of A in integral A-algebras of finite type, hence only integral quotient rings of A.

It follows that if $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ are the minimal prime ideals of A, then A is universally catenary if and only if every $A/\mathfrak{p}_i$ is universally catenary. Likewise, a locally Noetherian prescheme X is universally catenary if and only if X_{red} is universally catenary. **IV-2** | 99

(iv) The criterion 5.6.1(b) and (i) show that every A-algebra essentially of finite type over a universally catenary ring A is universally catenary (1.3.8).

Proposition (5.6.4). — *Every quotient ring of a regular ring is universally catenary.*

Proof. By 5.6.3(iv), it is enough to show that every regular ring A is universally catenary. Each polynomial ring $A[T_1, \dots, T_n]$ is regular (**0**, 17.3.7), hence catenary (**0**, 16.5.12), and the assertion follows from 5.6.1(a). ☐

In particular, Cohen's theorem (**0**, 19.8.8) implies that every complete Noetherian local ring is universally catenary. Likewise, every algebra of finite type over a field is universally catenary, and every prescheme locally of finite type over a field is universally catenary.

We shall see in (6.3.7) that "regular ring" may be replaced by "Cohen–Macaulay ring" in (5.6.4).

Proposition (5.6.5). — *Let Y be an irreducible locally Noetherian prescheme, let X be an irreducible prescheme, and let $f : X \to Y$ be a dominant morphism locally of finite type. Let ξ and η be the generic points of X and Y, respectively, and put*

$$e = \dim(f^{-1}(\eta)) = \operatorname{tr.deg}_{\kappa(\eta)} \kappa(\xi),$$

the dimension of the generic fibre (**0**, 2.1.8), (4.1.1). *For every $x \in X$, with $y = f(x)$, one has*

$$e + \dim(\mathscr{O}_y) \geqslant \operatorname{tr.deg}_{\kappa(y)} \kappa(x) + \dim(\mathscr{O}_x) \tag{5.6.5.1}$$

and

$$\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y)) - \delta(x), \tag{5.6.5.2}$$

where

$$\delta(x) = \dim_x(f^{-1}(y)) - e.$$

If Y is universally catenary, equality holds in (5.6.5.1) and (5.6.5.2). If, in addition, x is closed in $f^{-1}(y)$, then

$$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + e. \tag{5.6.5.3}$$

Proof. We may restrict to the case in which X and Y are affine, and hence f is of finite type, and then replace both preschemes by their reductions (1.5.4). Thus X and Y may be assumed integral.

Since f is dominant, $\mathscr{O}_y$ identifies with a subring of $\mathscr{O}_x$, and their fraction fields are $\kappa(\eta)$ and $\kappa(\xi)$, respectively (**I**, 8.2.7). Moreover, $\mathscr{O}_x$ is the localization at a prime ideal $\mathfrak{q}$ of an integral $\mathscr{O}_y$-algebra B of finite type (**I**, 3.6.5). Applying (5.5.8.1) to $A = \mathscr{O}_y$, B, and $\mathfrak{q}$ gives (5.6.5.1).

Furthermore,

$$\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y) = \mathscr{O}_x / \mathfrak{m}_y \mathscr{O}_x$$

is the local ring of the fibre $f^{-1}(y)$ at x (**I**, 3.6.1). Since the fibre is of finite type over $\kappa(y)$, (5.2.3.1) gives

$$\dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y)) = \dim_x(f^{-1}(y)) - \operatorname{tr.deg}_{\kappa(y)} \kappa(x).$$

Thus (5.6.5.2) is simply another form of (5.6.5.1).

When Y is universally catenary, equality in (5.6.5.1) is exactly (5.6.1.1), applied to $A = \mathscr{O}_y$ and $B' = \mathscr{O}_x$. Finally, if x is closed in the fibre, then in general **IV-2 | 100**

$$\operatorname{tr.deg}_{\kappa(y)} \kappa(x) = \dim(\overline{\{x\}} \cap f^{-1}(y));$$

this follows from (5.2.1), applied to the reduced closed sub-prescheme of $f^{-1}(y)$ whose underlying space is the displayed subspace. Equation (5.6.5.3) follows. □

We shall prove in (13.1.1) that under the hypotheses of (5.6.5) one always has $\delta(x) \geqslant 0$. Thus in this case (5.6.5.2) sharpens (5.5.2.1).

Corollary (5.6.6). — *Under the general hypotheses of (5.6.5),*

(5.6.6.1) $$\dim(X) \leqslant \dim(Y) + e.$$

If Y is universally catenary, equality holds in (5.6.6.1) if and only if

(5.6.6.2) $$\dim(Y) = \sup_{y \in f(X)} \dim(\mathscr{O}_y).$$

In particular, equality holds in (5.6.6.1) when Y is locally of finite type over a field.

Proof. Apply (5.6.5.1) at a closed point x of X, and use (5.1.4.2) and (5.1.11); this gives (5.6.6.1). Conversely, every nonempty fibre $f^{-1}(y)$ contains a point closed in that fibre (5.1.11). If Y is universally catenary, (5.6.5.3) holds at such a point. Taking suprema in (5.6.5.3) as x ranges over X, and observing that every point closed in X is *a fortiori* closed in $f^{-1}(f(x))$, yields the second assertion by (5.1.4.1) and (5.1.4.2).

For the last assertion, choose an affine open neighbourhood U of the generic point of X such that $f|_U$ is of finite type. Then $f(U)$ is constructible (1.8.5) and dense in Y, and therefore contains a nonempty open, hence dense, subset of Y (**0**, 9.2.2). Since Y is locally of finite type over a field, it is universally catenary (5.6.4); moreover, the two sides of (5.6.6.2) are equal by (5.2.2) and (4.1.1.3).

Condition (5.6.6.2) is trivially satisfied when f is surjective. □

Corollary (5.6.7). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a morphism locally of finite type, and let n be an integer. If*

$$\dim(f^{-1}(y)) \leqslant n$$

for every $y \in Y$, then

(5.6.7.1) $$\dim(X) \leqslant \dim(Y) + n.$$

Proof. By (**0**, 14.1.7) and (1.5.4), we may assume that X and Y are affine and reduced, so that f is of finite type. Let X_i $(1 \leqslant i \leqslant m)$ be the integral closed sub-preschemes whose underlying spaces are the irreducible components of X. Then

$$\dim(X) = \sup_i \dim(X_i).$$

Let Z_i be the reduced closed sub-prescheme of Y whose underlying space is $\overline{f(X_i)}$. It is integral (**0**, 2.1.5), and the restriction of f factors as

$$X_i \xrightarrow{g_i} Z_i \xrightarrow{j_i} Y,$$

where j_i is the canonical injection (**I**, 5.2.2), and g_i is dominant and of finite type (1.5.4).

We have $\dim(Z_i) \leqslant \dim(Y)$. If ζ_i is the generic point of Z_i, then **IV-2 | 101**

$$\dim(g_i^{-1}(\zeta_i)) \leqslant n$$

by hypothesis. Thus it is enough to show

$$\dim(X_i) \leqslant \dim(Z_i) + n$$

for every i, and this is (5.6.6.1). □

Corollary (5.6.8). — *Let Y be a locally Noetherian prescheme, let X be irreducible, let $f : X \to Y$ be a morphism locally of finite type, and let $n \geqslant 0$ be an integer. Suppose that Y is universally catenary and that*

$$\dim(f^{-1}(y)) \geqslant n \quad \textit{for every } y \in Y.$$

Then

$$\dim(X) \geqslant \dim(Y) + n. \tag{5.6.8.1}$$

If instead every fibre has dimension exactly n, then

$$\dim(X) = \dim(Y) + n. \tag{5.6.8.2}$$

Proof. Since $n \geqslant 0$, either hypothesis implies that f is surjective, and therefore Y is irreducible. If η is the generic point of Y, then $\dim(f^{-1}(\eta)) \geqslant n$, respectively $\dim(f^{-1}(\eta)) = n$. The conclusion follows from (5.6.6). □

Remarks (5.6.9). —
(i) Even if Y is regular, X is irreducible, and $f : X \to Y$ is dominant and of finite type, the two sides of (5.6.6.1) need not be equal. Take $Y = \mathrm{Spec}(A)$ for a discrete valuation ring A, take $X = \mathrm{Spec}(K)$ for its fraction field K, and let f be the canonical morphism.
(ii) Example 5.4.3(i) shows that the hypothesis that X is irreducible cannot be omitted from (5.6.6) or (5.6.8), even when all the other hypotheses hold. We shall see in (10.6.1) that such phenomena cannot occur under additional hypotheses on Y, satisfied for example when Y is locally of finite type over a field or over a Dedekind ring having infinitely many prime ideals.

Proposition (5.6.10). — *Let A be an integral universally catenary Noetherian local ring, and let B be an integral finite A-algebra containing A. For every maximal ideal $\mathfrak{n}$ of B,*

$$\dim(B_{\mathfrak{n}}) = \dim(A).$$

Proof. We have $\operatorname{tr.deg}_A B = 0$, and the residue field of $B_{\mathfrak{n}}$ is algebraic over the residue field of A.[7] Every maximal ideal of B lies over the maximal ideal of A, so the assertion follows from (5.6.1.1). □

Example (5.6.11). — Let A be an integral, integrally closed Noetherian local ring. Then (**0**, 16.1.6) shows that the conclusion of (5.6.10) holds for every integral finite A-algebra B containing A. We now construct, by contrast, an integral catenary Noetherian local ring A for which that conclusion fails.

Use the construction of 5.2.5(i). Let k_0 be a field, let

$$k = k_0(S_i)_{i \in \mathbf{N}}$$

be a purely transcendental extension of infinite transcendence degree, and let $V = k[S]_{(S)}$ be the discrete valuation ring obtained by localizing $k[S]$ at (S). Put $E = V[T]$. In E, the maximal ideal

$$\mathfrak{m} = (S) + (T)$$

has height 2, whereas the maximal ideal

$$\mathfrak{m}' = (ST - 1)$$

has height 1.

The corresponding residue fields are **IV-2** | 102

$$\kappa(\mathfrak{m}) = k, \qquad \kappa(\mathfrak{m}') = k(S),$$

the latter being the fraction field of V. By the choice of k, these fields are isomorphic. Let

$$\varepsilon : E \longrightarrow E/\mathfrak{m}, \qquad \varepsilon' : E \longrightarrow E/\mathfrak{m}'$$

be the canonical homomorphisms, choose an isomorphism

$$\sigma : \kappa(\mathfrak{m}) \xrightarrow{\sim} \kappa(\mathfrak{m}'),$$

[7] The printed French text says "the fraction field of A". Here the field required by the dimension formula is the residue field of A.

and let C be the subring of E formed by the elements x such that

$$\varepsilon'(x) = \sigma(\varepsilon(x)).$$

This is a special case of the "gluing procedures" to be studied systematically in Chapter V.

It is immediate that

$$\mathfrak{n} = \mathfrak{m}\mathfrak{m}' = \mathfrak{m} \cap \mathfrak{m}'$$

is a maximal ideal of C, and that $C/\mathfrak{n}$ identifies with the subfield of $(E/\mathfrak{m}) \times (E/\mathfrak{m}')$ formed by the pairs $(z, \sigma(z))$. Moreover,

$$E = C + C(ST - 1),$$

so E is a finite C-algebra; it is plainly the integral closure of C. We next prove that C is Noetherian.

Lemma (5.6.11.1). — *Let R be a ring, let S be a subring, and let*

$$\mathfrak{K} = \mathrm{Ann}_S(R/S)$$

be the conductor of S in R, that is, the largest ideal of S which is also an ideal of R.

(i) *For every ideal $\mathfrak{J} \subset \mathfrak{K}$ of R, there is a strictly increasing map from the set of ideals $\mathfrak{a}$ of S such that $R\mathfrak{a} = \mathfrak{J}$ to the set of $(S/\mathfrak{K})$-submodules of $\mathfrak{J}/\mathfrak{K}\mathfrak{J}$.*

(ii) *If $S/\mathfrak{K}$ and R are Noetherian and R is a finite S-module, then every increasing sequence of ideals of S contained in $\mathfrak{K}$ is stationary.*

Proof. For (i), if $R\mathfrak{a} = \mathfrak{J}$, then

$$\mathfrak{K}\mathfrak{J} = \mathfrak{K}(R\mathfrak{a}) = \mathfrak{K}\mathfrak{a} \subset \mathfrak{a}.$$

Thus $\mathfrak{a} \mapsto \mathfrak{a}/\mathfrak{K}\mathfrak{J}$ gives the asserted strictly increasing map.

For (ii), let $(\mathfrak{L}_r)$ be an increasing sequence of ideals of S contained in $\mathfrak{K}$. Since R is Noetherian, the sequence $(R\mathfrak{L}_r)$ is stationary; after discarding finitely many terms, suppose that all its members are the same ideal $\mathfrak{J}$ of R. The module $\mathfrak{J}/\mathfrak{K}\mathfrak{J}$ is finite over R, hence over S, and therefore finite over the Noetherian ring $S/\mathfrak{K}$. It is consequently Noetherian, and (i) shows that $(\mathfrak{L}_r)$ is stationary. □

Apply the lemma with $R = E$ and $S = C$. The ideal $\mathfrak{n}$ is the conductor of C in E. For every ideal $\mathfrak{a}$ of C,

$$\mathfrak{a}/(\mathfrak{n} \cap \mathfrak{a}) \simeq (\mathfrak{a} + \mathfrak{n})/\mathfrak{n}$$

is a submodule of the simple C-module $C/\mathfrak{n}$. It is therefore enough to prove that every ideal $\mathfrak{a} \cap \mathfrak{n}$ is finitely generated; this follows from (5.6.11.1). Thus C is Noetherian.

By (**0**, 16.1.5),

$$\dim(C) = \dim(E) = 2.$$

Set $A = C_{\mathfrak{n}}$ and $U = C - \mathfrak{n}$. Then

$$B = U^{-1}E$$

is the integral closure of A and is a finite A-module. The ideals $\mathfrak{m}$ and $\mathfrak{m}'$ are the only prime ideals of E containing $\mathfrak{n}$. Hence B is semilocal, and its local rings at its two maximal ideals are isomorphic to $E_{\mathfrak{m}}$ and $E_{\mathfrak{m}'}$, of dimensions 2 and 1, respectively. Thus $\dim(B) = 2$ and, by (**0**, 16.1.5), $\dim(A) = 2$.

Since A is an integral local ring of dimension 2, it is catenary: every prime ideal other than 0 and the maximal ideal has height 1. Nevertheless it does not satisfy the conclusion of (5.6.10), and therefore is not universally catenary.

The ideal $\mathfrak{n}$ has height 2 in C. If $\mathfrak{p}$ is a height-1 prime ideal of C, there is a unique prime ideal $\mathfrak{p}'$ of E above $\mathfrak{p}$; it has height 1, and

$$C_{\mathfrak{p}} = E_{\mathfrak{p}'}.$$

Existence follows from integrality, and the Cohen–Seidenberg theorem shows that every prime ideal of E above $\mathfrak{p}$ has height 1.

Moreover, E is a finite C-algebra and $C_{\mathfrak{p}}$ is integrally closed (Bourbaki, *Algèbre commutative*, chap. V, § 1, n° 5, cor. 5 of prop. 16). This proves uniqueness and the displayed equality. **IV-2** | 103

It would be interesting to know whether every integral Noetherian local ring satisfying the conclusion of (5.6.10) is universally catenary. Nagata asserted this [33], but his proof does not appear to be complete.

5.7. Depth and property (S_k) IV-2 | 103

Definition (5.7.1). — Let X be a locally Noetherian prescheme and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. The *depth* (respectively, *codepth*) of $\mathscr{F}$ at a point $x \in X$ is the number $\operatorname{prof}(\mathscr{F}_x)$ (respectively, $\operatorname{coprof}(\mathscr{F}_x)$) (**0**, 16.4.5), (**0**, 16.4.9). The *codepth* of $\mathscr{F}$ is the number

(5.7.1.1) $$\operatorname{coprof}(\mathscr{F}) = \sup_{x \in X} \operatorname{coprof}(\mathscr{F}_x).$$

The module $\mathscr{F}$ is said to be a *Cohen–Macaulay $\mathscr{O}_X$-module at x* if $\mathscr{F}_x$ is a Cohen–Macaulay $\mathscr{O}_x$-module, that is (**0**, 16.5.1), if $\operatorname{coprof}(\mathscr{F}_x) = 0$. It is said to be a *Cohen–Macaulay $\mathscr{O}_X$-module* if it is so at every point, or equivalently if $\operatorname{coprof}(\mathscr{F}) = 0$. A point $x \in X$ such that $\mathscr{O}_x$ is a Cohen–Macaulay ring is also called a *Cohen–Macaulay point* of X.

The *codepth of* X, denoted by $\operatorname{coprof}(X)$, is the number $\operatorname{coprof}(\mathscr{O}_X)$. The prescheme X is said to be *Cohen–Macaulay* if $\mathscr{O}_X$ is a Cohen–Macaulay $\mathscr{O}_X$-module, or equivalently if $\operatorname{coprof}(X) = 0$. Every locally Noetherian prescheme of dimension 0 is evidently Cohen–Macaulay. To say that $\operatorname{Spec}(A)$ is Cohen–Macaulay means that A is a Cohen–Macaulay ring (**0**, 16.5.13).

Definition (5.7.2). — Let X be a locally Noetherian prescheme, let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module, and let k be an integer, positive or negative. The module $\mathscr{F}$ is said to satisfy property (S_k) if, for every $x \in X$,

(5.7.2.1) $$\operatorname{prof}(\mathscr{F}_x) \geqslant \inf\{k, \dim(\mathscr{F}_x)\}.$$

It is said to satisfy property (S_k) *at a point* $x \in X$ if, for every generization x' of x in X,

(5.7.2.2) $$\operatorname{prof}(\mathscr{F}_{x'}) \geqslant \inf\{k, \dim(\mathscr{F}_{x'})\}.$$

The prescheme X is said to satisfy property (S_k) (respectively, property (S_k) at x) if $\mathscr{O}_X$ does so (respectively, does so at x).

The module $\mathscr{F}$ satisfies (S_k) if and only if it satisfies (S_k) at every point of X. If U is an open subset of X and $\mathscr{F}$ satisfies (S_k), then $\mathscr{F}|U$ satisfies (S_k). Conversely, if (U_α) is an open covering of X and $\mathscr{F}|U_\alpha$ satisfies (S_k) for every α, then $\mathscr{F}$ satisfies (S_k).

Remarks (5.7.3). — (i) Recall that $\operatorname{prof}(\mathscr{F}_x) \leqslant \dim(\mathscr{F}_x)$ (**0**, 16.4.5.1) whenever $\mathscr{F}_x \neq 0$. Thus $\mathscr{F}$ satisfies (S_k) precisely when $\operatorname{prof}(\mathscr{F}_x) \geqslant k$ except at points x for which $\dim(\mathscr{F}_x) < k$, while at those latter points

$$\dim(\mathscr{F}_x) = \operatorname{prof}(\mathscr{F}_x).$$

Equivalently (**0**, 16.5.1), $\mathscr{F}$ is Cohen–Macaulay at those points. At points where $\dim(\mathscr{F}_x) = k$, one likewise has $\operatorname{prof}(\mathscr{F}_x) = k$.

Hence $\mathscr{F}$ is also Cohen–Macaulay at those points. To say that $\mathscr{F}$ satisfies (S_k) for every k is therefore the same as to say that $\mathscr{F}$ is a Cohen–Macaulay $\mathscr{O}_X$-module. If $k' \geqslant k$, then $(S_{k'})$ implies (S_k); if $k \leqslant 0$, every coherent $\mathscr{O}_X$-module satisfies (S_k). IV-2 | 104

(ii) To verify (5.7.2.1), it is enough to consider $x \in \operatorname{Supp}(\mathscr{F})$; otherwise $\dim(\mathscr{F}_x) = -\infty$ (**0**, 14.1.2).

(iii) If $X = \operatorname{Spec}(A)$, where A is Noetherian, and $\mathscr{F} = \widetilde{M}$, where M is a finite A-module, then M is said to satisfy (S_k) when $\mathscr{F}$ does.

For an arbitrary locally Noetherian prescheme X, saying that $\mathscr{F}$ satisfies (S_k) at x means the following. Put $Y = \operatorname{Spec}(\mathscr{O}_x)$; then the $\mathscr{O}_Y$-module $\widetilde{\mathscr{F}_x}$ satisfies (S_k). Notice that (5.7.2.1) does not in general imply (5.7.2.2) for every generization x' of x, since there is no general inequality between $\operatorname{prof}(\mathscr{F}_x)$ and $\operatorname{prof}(\mathscr{F}_{x'})$ (**0**, 16.4.6).

(iv) It follows immediately from the definition that if $\mathscr{F}$ satisfies (S_k) at x, then it also satisfies (S_k) at every generization of x.

(v) Property (S_k) is especially important for $k = 1$ and $k = 2$. Serre introduced it for $k = 2$ in order to express his normality criterion; see (5.8.5).

(vi) Let Y be a closed sub-prescheme of X, let $j : Y \to X$ be the canonical immersion, and let $\mathscr{G}$ be a coherent $\mathscr{O}_Y$-module. For every $x \in Y$,

$$\dim(\mathscr{G}_x) = \dim((j_*\mathscr{G})_x), \qquad \operatorname{prof}(\mathscr{G}_x) = \operatorname{prof}((j_*\mathscr{G})_x),$$

and consequently

$$\operatorname{coprof}(\mathscr{G}_x) = \operatorname{coprof}((j_*\mathscr{G})_x).$$

Thus $\mathscr{G}$ satisfies (S_k) if and only if $j_*\mathscr{G}$ does.

Proposition (5.7.4). — *Let X be a locally Noetherian prescheme and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. Put $S = \operatorname{Supp}(\mathscr{F})$, and for every $n \geqslant 0$ let Z_n be the set of points $z \in X$ such that $\operatorname{coprof}(\mathscr{F}_z) > n$; thus $Z_n \subset S$.*

(i) *The module $\mathscr{F}$ satisfies (S_k) if and only if, for every $n \geqslant 0$,*

(5.7.4.1) $$\operatorname{codim}(Z_n, S) > n + k.$$

(ii) *Suppose in addition that every Z_n is closed in X. Then $\mathscr{F}$ satisfies (S_k) at a point $x \in S$ if and only if, for every $n \geqslant 0$,*

(5.7.4.2) $$\operatorname{codim}_x(Z_n, S) > n + k.$$

Proof. (i) By definition (5.1.3),

$$\operatorname{codim}(Z_n, S) = \inf_{z \in Z_n} \dim(\mathscr{O}_{S,z}),$$

and (5.7.4.1) therefore says (5.1.12.2) that, for every $z \in X$ and every $n \geqslant 0$,

$$\dim(\mathscr{F}_z) - \operatorname{prof}(\mathscr{F}_z) \geqslant n + 1$$

implies

$$\dim(\mathscr{F}_z) \geqslant n + k + 1.$$

If $a = \dim(\mathscr{F}_z)$ and $b = \operatorname{prof}(\mathscr{F}_z)$, then $b \leqslant a$; requiring, for every $n \geqslant 0$, that $b \leqslant a - n - 1$ imply $k \leqslant a - n - 1$ is equivalent to $b \geqslant \inf\{k, a\}$. This proves (i).

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(ii) The argument is the same as in (i), except that one restricts to the points z which are generizations of x and uses (5.1.3.2).

□

Proposition (5.7.5). — *Let X be a locally Noetherian prescheme and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. The module $\mathscr{F}$ satisfies (S_k), where $k \geqslant 1$, if and only if the following condition holds: for every integer r with $0 \leqslant r < k$, every open subset U of X, and every $(\mathscr{F}|U)$-regular sequence $(f_i)_{1 \leqslant i \leqslant r}$ of sections of $\mathscr{O}_X$ over U, the module*

$$(\mathscr{F}|U) \Big/ \Big(\sum_{i=1}^{r} f_i(\mathscr{F}|U) \Big)$$

has no embedded associated prime cycle.

Proof. For $k = 1$, the assertion says that $\mathscr{F}$ satisfies (S_1) if and only if $\mathscr{F}$ has no embedded associated prime cycle. Indeed, (S_1) says that $\operatorname{prof}(\mathscr{F}_x) \geqslant 1$ at precisely the points $x \in \operatorname{Supp}(\mathscr{F})$ for which $\dim(\mathscr{F}_x) > 0$; at the other points of the support, $\dim(\mathscr{F}_x) = \operatorname{prof}(\mathscr{F}_x) = 0$. Now $\operatorname{prof}(\mathscr{F}_x) \geqslant 1$ means that the maximal ideal $\mathfrak{m}_x$ is not associated to $\mathscr{F}_x$ (0, 16.4.6(i)), or equivalently that x is not associated to $\mathscr{F}$ (3.1.1). If S' is a sub-prescheme of X whose underlying space is $\operatorname{Supp}(\mathscr{F})$, then

$$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{S',x})$$

(5.1.12.1); saying that this dimension is positive is the same as saying that x is not a maximal point of $\operatorname{Supp}(\mathscr{F})$ (5.1.2). This proves the assertion for $k = 1$.

Let $k > 1$, and first suppose that $\mathscr{F}$ satisfies (S_k). By induction on k and the case just proved, it is enough to show that if f is an $\mathscr{F}$-regular section of $\mathscr{O}_X$, then $\mathscr{F}/f\mathscr{F}$ satisfies (S_{k-1}). At each $x \in X$, the element f_x is $\mathscr{F}_x$-regular. If it is invertible, the quotient stalk is zero. Otherwise $f_x \in \mathfrak{m}_x$, and

$$\operatorname{prof}(\mathscr{F}_x/f_x\mathscr{F}_x) = \operatorname{prof}(\mathscr{F}_x) - 1, \qquad \dim(\mathscr{F}_x/f_x\mathscr{F}_x) = \dim(\mathscr{F}_x) - 1$$

(0, 16.4.6(i)), (**0**, 16.3.4); the assertion follows. Thus the condition is necessary.

Conversely, suppose the stated condition holds. We argue by induction on k. Let $x \in X$, and first suppose that $\dim(\mathscr{F}_x) \geqslant k$. The induction hypothesis gives (S_{k-1}), so $\operatorname{prof}(\mathscr{F}_x) \geqslant k - 1$. By

(**0**, 15.2.4), after replacing X by an open neighbourhood U of x, there is an $(\mathscr{F}|U)$-regular sequence $(f_i)_{1\leqslant i\leqslant k-1}$ with $(f_i)_x \in \mathfrak{m}_x$. Put

$$\mathscr{G} = (\mathscr{F}|U)\Big/\Big(\sum_{i=1}^{k-1} f_i(\mathscr{F}|U)\Big).$$

By hypothesis $\mathscr{G}$ has no embedded associated prime cycle, while

$$\dim(\mathscr{G}_x) = \dim(\mathscr{F}_x) - (k-1) \geqslant 1$$

(**0**, 16.3.4). Hence x is not associated to $\mathscr{G}$, so $\operatorname{prof}(\mathscr{G}_x) \geqslant 1$ (**0**, 16.4.6); since

$$\operatorname{prof}(\mathscr{G}_x) = \operatorname{prof}(\mathscr{F}_x) - (k-1),$$

we obtain $\operatorname{prof}(\mathscr{F}_x) \geqslant k$.

If instead $\dim(\mathscr{F}_x) = r < k$, the induction hypothesis gives

$$\operatorname{prof}(\mathscr{F}_x) \geqslant \inf\{k-1, r\} = r.$$

Thus $\mathscr{F}$ satisfies (S_k), completing the proof. □

Corollary (5.7.6). — *Suppose that $\mathscr{F}$ satisfies (S_k), where $k \geqslant 1$. If $(f_i)_{1\leqslant i\leqslant r}$ is an $\mathscr{F}$-regular sequence of sections of $\mathscr{O}_X$ over X and $r < k$, then*

$$\mathscr{F}\Big/\Big(\sum_{i=1}^{r} f_i\mathscr{F}\Big)$$

satisfies (S_{k-r}).

Proof. This follows immediately from (5.7.5). □

Corollary (5.7.7). — *The module $\mathscr{F}$ satisfies (S_2) if and only if $\mathscr{F}$ has no embedded associated prime cycle and, for every open subset U of X and every $(\mathscr{F}|U)$-regular section f of $\mathscr{O}_X$ over U, the module $(\mathscr{F}|U)/f(\mathscr{F}|U)$ has no embedded associated prime cycle.*

Proof. This follows immediately from (5.7.5). □

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Remarks (5.7.8). — Let X be a locally Noetherian prescheme of dimension 1. The following conditions are equivalent: X is Cohen–Macaulay; X satisfies (S_1); X satisfies (S_n) for some $n \geqslant 1$. This follows from Definitions (5.7.1) and (5.7.2). By (5.7.5), these conditions are also equivalent to the absence of an embedded associated prime cycle. In particular, every reduced locally Noetherian prescheme of dimension 1 is Cohen–Macaulay.

Proposition (5.7.9). — *Let A and B be Noetherian rings, let $\rho : A \to B$ be a ring homomorphism, and let M be a B-module such that the A-module $M_{[\rho]}$ is finite. Let $\mathfrak{p}$ be a prime ideal of A. The prime ideals of B lying over $\mathfrak{p}$ and belonging to* $\operatorname{Supp}(M)$ *are finite in number. If $(\mathfrak{q}_i)_{1\leqslant i\leqslant n}$ is the family of these ideals, then*

(5.7.9.1) $$\dim_{A_\mathfrak{p}}(M_\mathfrak{p}) = \sup_i \dim_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}),$$

(5.7.9.2) $$\operatorname{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \inf_i \operatorname{prof}_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}).$$

Proof. Let $\mathfrak{b}$ be the annihilator of M. The ring $B/\mathfrak{b}$ identifies with an A-submodule of $\operatorname{End}_A(M_{[\rho]})$, and is therefore finite over A. Hence only finitely many prime ideals of B lie over $\mathfrak{p}$ and contain $\mathfrak{b}$; these are precisely the ones belonging to $\operatorname{Supp}(M)$.

Replacing B by $B/\mathfrak{b}$ does not change the right-hand sides of (5.7.9.1) and (5.7.9.2) (**0**, 16.1.9), (**0**, 16.4.8); we may therefore suppose that B is finite over A. Put $S = A - \mathfrak{p}$ and $B' = S^{-1}B$. Then B' is a Noetherian semilocal ring whose maximal ideals are $S^{-1}\mathfrak{q}_i$, and

$$\dim_{A_\mathfrak{p}}(M_\mathfrak{p}) = \dim_{B'}(M_\mathfrak{p})$$

(**0**, 16.1.9). Formula (5.7.9.1) is now the semilocal dimension formula (**0**, 16.1.7.4).

For (5.7.9.2), as in (**0**, 16.4.8), it is enough to treat the case $\mathrm{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) = 0$. The same argument shows that $M_\mathfrak{p}$ contains a B'-submodule of finite length, and hence a simple B'-submodule. Such a module is isomorphic to the residue field of one of the local rings $B_{\mathfrak{q}_i}$. Thus

$$\mathrm{prof}_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}) = 0$$

for at least one i (**0**, 16.4.6), proving the formula. □

Corollary (5.7.10). — *Assume the hypotheses of (5.7.9), and suppose in addition that A is local. The A-module $M_{[\rho]}$ is Cohen–Macaulay if and only if every $M_{\mathfrak{q}_i}$, for $\mathfrak{q}_i$ lying over the maximal ideal of A, is a Cohen–Macaulay $B_{\mathfrak{q}_i}$-module and all the numbers $\dim_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i})$ are equal.*

Proof. By (5.7.9.1) and (5.7.9.2), these conditions are equivalent to

$$\dim_A(M_{[\rho]}) = \mathrm{prof}_A(M_{[\rho]}).$$

□

Corollary (5.7.11). — *Assume the hypotheses of (5.7.9).*

(i) *If $M_{[\rho]}$ satisfies (S_k), then M satisfies (S_k).*

(ii) *Suppose that for every pair of prime ideals $\mathfrak{q}, \mathfrak{q}'$ of B satisfying $\rho^{-1}(\mathfrak{q}) = \rho^{-1}(\mathfrak{q}')$, one has*

$$\dim_{B_\mathfrak{q}}(M_\mathfrak{q}) = \dim_{B_{\mathfrak{q}'}}(M_{\mathfrak{q}'}).$$

If M satisfies (S_k), then $M_{[\rho]}$ satisfies (S_k).

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Proof. This follows immediately from (5.7.9.1), (5.7.9.2), and the definition of (S_k). □

(5.7.12). In accordance with (5.7.1), if A is any Noetherian ring and M is a finite A-module, define

$$\mathrm{coprof}_A(M) = \mathrm{coprof}(\widetilde{M}) = \sup_{x\in X} \mathrm{coprof}_{A_x}(M_x), \qquad X = \mathrm{Spec}(A).$$

We shall see in (6.11.5) that this definition agrees with that of (**0**, 16.4.9) when A is a Noetherian local ring.

Corollary (5.7.13). — *Let A and B be Noetherian rings, let $\rho : A \to B$ be a ring homomorphism, and let M be a B-module such that $M_{[\rho]}$ is a finite A-module. Then*

$$\mathrm{coprof}_A(M_{[\rho]}) \geqslant \mathrm{coprof}_B(M). \tag{5.7.13.1}$$

Proof. This follows from (5.7.12) and (5.7.9.1), (5.7.9.2). □

5.8. Regular preschemes and property (R_k). Serre's normality criterion

(5.8.1). Recall (**0**, 4.1.4) that a ringed space $(X, \mathscr{O}_X)$ is said to be *regular* at a point $x \in X$ if $\mathscr{O}_x$ is a regular local ring. When dealing with preschemes, we shall use this terminology in this chapter only when X is locally Noetherian.

Definition (5.8.2). — A locally Noetherian prescheme X is said to be *regular in codimension* $\leqslant k$, or to satisfy property (R_k), if the set of points where X is not regular has codimension $> k$. In other words (5.1.3), for every $x \in X$,

$$\dim(\mathscr{O}_x) \leqslant k \quad \Longrightarrow \quad \mathscr{O}_x \text{ is regular.}$$

To say that X is regular means that X satisfies (R_k) for every k.

If $X = \mathrm{Spec}(A)$, where A is a Noetherian ring, we say that A satisfies (R_k) if X does; to say that X is regular is to say that the ring A is regular (**0**, 17.3.6). For an arbitrary locally Noetherian prescheme X, we say that X satisfies (R_k) *at a point* $x \in X$ if the local ring $\mathscr{O}_x$ satisfies (R_k). Thus, for every generization x' of x in X, the relation $\dim(\mathscr{O}_{x'}) \leqslant k$ must imply that $\mathscr{O}_{x'}$ is a regular local ring. By (**0**, 17.3.6), saying that X is regular at x is equivalent to saying that X satisfies (R_n) at x for every $n \geqslant 0$.

Proposition (5.8.3). — *Let k be a field and let X be a prescheme locally of finite type over k. For every $x \in X$, there is an open neighbourhood of x in X which is isomorphic to a subscheme of a regular k-scheme.*

Proof. There is an affine open neighbourhood U of x isomorphic to a k-scheme $\operatorname{Spec}(A)$, where A is a k-algebra of finite type. The algebra A is therefore isomorphic to a quotient of a polynomial algebra $k[T_1, \ldots, T_n]$, and the latter is a regular ring (**0**, 17.3.7). □

(5.8.4). By (5.8.2), saying that X satisfies (R_0) means that, at every maximal point x of X, the ring $\mathscr{O}_x$ is a field (**0**, 17.1.4); in other words, X is reduced at x. The set U of points where X is reduced is open, because the nilradical of X is a coherent $\mathscr{O}_X$-module (**I**, 6.1.1), (**0**, 5.2.2). Consequently, saying that X satisfies (R_0) is equivalent to saying that U is everywhere dense.

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Proposition (5.8.5). — *A locally Noetherian prescheme X is reduced if and only if it satisfies* (S_1) *and* (R_0).

Proof. Taking (5.7.5) into account, this follows from the preceding remark and (3.2.1). □

Theorem (5.8.6). — (Serre's criterion). *Let X be a locally Noetherian prescheme. Then X is normal if and only if it satisfies* (S_2) *and* (R_1). *Equivalently, for every $x \in X$, the following conditions hold:*

(i) *If* $\dim(\mathscr{O}_x) \leqslant 1$, *then $\mathscr{O}_x$ is regular; that is, it is a field or a discrete valuation ring* (**0**, 17.1.4).

(ii) *If* $\dim(\mathscr{O}_x) \geqslant 2$, *then* $\operatorname{prof}(\mathscr{O}_x) \geqslant 2$.

Proof. The conditions are necessary. Indeed, saying that X is normal means that, for every $x \in X$, the ring $\mathscr{O}_x$ is an integrally closed Noetherian local ring. If $\dim(\mathscr{O}_x) = 0$, then $\mathscr{O}_x$ is a field because it is a domain; if $\dim(\mathscr{O}_x) = 1$, then $\mathscr{O}_x$ is a discrete valuation ring by (**II**, 7.1.6). Furthermore, for every nonzero element $f_x \in \mathscr{O}_x$, the associated prime ideals of $\mathscr{O}_x/f_x\mathscr{O}_x$ are non-embedded (Bourbaki, *Alg. comm.*, chap. VII, § 1, no. 4, prop. 8). Hence $\mathscr{O}_x$ satisfies (S_2) by (5.7.7).

Conversely, suppose that the two conditions hold. It follows first from (5.8.5) that X is reduced. Since the question is local, we may suppose that $X = \operatorname{Spec}(A)$, where A is a reduced Noetherian ring (**I**, 5.1.4). If R is the total ring of fractions of A, then R is a direct product of finitely many fields; taking (**II**, 6.3.6) into account, it is enough to prove that A is integrally closed in R.

Let $h = f/g \in R$ be integral over A, where $f, g \in A$ and g is a non-zero-divisor. There is a relation

$$f^n + \sum_{i=1}^{n} a_i f^{n-i} g^i = 0, \qquad a_i \in A \quad (1 \leqslant i \leqslant n). \tag{5.8.6.1}$$

Let $\mathfrak{p}$ be a prime ideal of A such that $\dim(A_\mathfrak{p}) = 1$, and write $f_\mathfrak{p}$ and $g_\mathfrak{p}$ for the images of f and g in $A_\mathfrak{p}$. It follows from (5.8.6.1) that $f_\mathfrak{p}/g_\mathfrak{p}$ is integral over $A_\mathfrak{p}$. This quotient belongs to the total ring of fractions of $A_\mathfrak{p}$, since flatness implies that $g_\mathfrak{p}$ is still a non-zero-divisor (**0**, 5.3.1). But $A_\mathfrak{p}$ is regular and hence integrally closed, so $f_\mathfrak{p}/g_\mathfrak{p} \in A_\mathfrak{p}$. Equivalently,

$$(fA)_\mathfrak{p} \subset (gA)_\mathfrak{p}.$$

Since g is a non-zero-divisor in A, condition (S_2) and (5.7.7) imply that A/gA has only non-embedded associated prime ideals $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$. The ideal gA is therefore an intersection of primary ideals $\mathfrak{q}_i$ corresponding to the $\mathfrak{p}_i$; moreover, each $\mathfrak{q}_i$ is the inverse image in A, under $A \to A_{\mathfrak{p}_i}$, of $(gA)_{\mathfrak{p}_i}$ (Bourbaki, *Alg. comm.*, chap. IV, § 2, no. 3, prop. 5). By the Hauptidealsatz (**0**, 16.3.2), $\dim(A_{\mathfrak{p}_i}) = 1$ for every i, so the preceding argument gives

$$(fA)_{\mathfrak{p}_i} \subset (gA)_{\mathfrak{p}_i} \qquad (1 \leqslant i \leqslant n).$$

Now fA is contained in the intersection of the inverse images of the $(fA)_{\mathfrak{p}_i}$. It is therefore contained in the intersection of the inverse images of the $(gA)_{\mathfrak{p}_i}$, which is gA. Thus $fA \subset gA$, and hence $f/g \in A$. □

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5.9. Z-pure and Z-closed modules Some of the notions and results in this section and the next are special cases of notions and results developed in Chapter III in the theory of local cohomology. For the reader's convenience, we give an independent account here.

(5.9.1). Let X be a locally Noetherian prescheme and let Z be a subset of X *stable under specialization*. This means that the closure of every finite subset M of Z is contained in Z. It follows that Z is the union of a filtered increasing family (Z_α) of closed subsets of X; conversely, such a union is plainly stable under specialization.

Put $U_\alpha = X - Z_\alpha$. Then $X - Z$ is the intersection of the filtered decreasing family of open subsets U_α. Let $i_\alpha : U_\alpha \to X$ be the canonical injection, and, whenever $U_\alpha \supset U_\beta$, let $i_{\alpha\beta} : U_\beta \to U_\alpha$ be the canonical injection, so that $i_\beta = i_\alpha \circ i_{\alpha\beta}$.

Let $\mathscr{F}$ be an $\mathscr{O}_X$-module, not necessarily quasi-coherent. We have

$$(i_\beta)_*(\mathscr{F}|U_\beta) = (i_\alpha)_*((i_{\alpha\beta})_*(\mathscr{F}|U_\beta)).$$

The canonical homomorphism (**0**, 4.4.3.2)

$$\mathscr{F}|U_\alpha \longrightarrow (i_{\alpha\beta})_*(\mathscr{F}|U_\beta)$$

therefore gives, after applying $(i_\alpha)_*$, a homomorphism

$$\rho_{\beta\alpha} : (i_\alpha)_*(\mathscr{F}|U_\alpha) \longrightarrow (i_\beta)_*(\mathscr{F}|U_\beta).$$

For $U_\alpha \supset U_\beta \supset U_\gamma$, one immediately checks that $\rho_{\gamma\alpha} = \rho_{\gamma\beta} \circ \rho_{\beta\alpha}$. Thus the modules $(i_\alpha)_*(\mathscr{F}|U_\alpha)$ form an inductive system under the homomorphisms $\rho_{\beta\alpha}$. We put

$$\mathscr{H}^0_{X/Z}(\mathscr{F}) = \varinjlim_{\alpha} (i_\alpha)_*(\mathscr{F}|U_\alpha). \tag{5.9.1.1}$$

This $\mathscr{O}_X$-module does not depend on the increasing family (Z_α) of closed subsets having union Z. Indeed, let V be a Noetherian open subset of X. In the category of $\mathscr{O}_V$-modules, the functor $\mathscr{G} \mapsto \Gamma(V, \mathscr{G})$ commutes with inductive limits (**G**, II.3.10.1); hence (5.9.1.1) gives

$$\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F})) = \varinjlim_{\alpha} \Gamma(V \cap U_\alpha, \mathscr{F}). \tag{5.9.1.2}$$

Let (Z'_λ) be another filtered increasing family of closed subsets of X with union Z. For fixed α, the set $V \cap Z_\alpha$ is the union of the $V \cap Z_\alpha \cap Z'_\lambda$. Now $V \cap Z_\alpha$ is locally closed in X, and every closed irreducible subset of it has a generic point. Since the $V \cap Z_\alpha \cap Z'_\lambda$ form a filtered increasing family of closed subsets of $V \cap Z_\alpha$, there is a λ such that

$$V \cap Z_\alpha \cap Z'_\lambda = V \cap Z_\alpha \qquad (\mathbf{0}, 9.2.4).$$

Equivalently, $V \cap Z_\alpha \subset V \cap Z'_\lambda$. The two filtered decreasing families $V \cap U_\alpha$ and $V \cap U'_\lambda$, where $U'_\lambda = X - Z'_\lambda$, are therefore cofinal. The asserted independence follows from (5.9.1.2).

The subset Z need not be constructible. For example, take $X = \operatorname{Spec}(A)$, where A is a Noetherian domain having infinitely many maximal ideals, and let Z be the complement of the generic point of X. **IV-2** | 110

If Z is closed and $i : X - Z \to X$ is the canonical injection, then

$$\mathscr{H}^0_{X/Z}(\mathscr{F}) = i_*(\mathscr{F}|X - Z). \tag{5.9.1.3}$$

In particular, when $Z = \varnothing$, $\mathscr{H}^0_{X/Z}(\mathscr{F}) = \mathscr{F}$.

Proposition (5.9.2). — (i) *The functor $\mathscr{F} \mapsto \mathscr{H}^0_{X/Z}(\mathscr{F})$ is left exact.*

(ii) *If $\mathscr{F}$ is quasi-coherent, so is $\mathscr{H}^0_{X/Z}(\mathscr{F})$.*

Proof. Assertion (i) follows from (5.9.1.1), from the left exactness of $(i_\alpha)_*$, and from the fact that inductive limits preserve exactness in the category of $\mathscr{O}_X$-modules. Assertion (ii) follows from (**I**, 9.2.2) and from the fact that an inductive limit of quasi-coherent $\mathscr{O}_X$-modules is quasi-coherent (**I**, 1.3.9). □

Remark (5.9.3). — If $\mathscr{F}$ is an $\mathscr{O}_X$-algebra, then $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is one as well (**0**, 4.2.4). In particular, $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is a quasi-coherent $\mathscr{O}_X$-algebra, and, for every $\mathscr{O}_X$-module $\mathscr{F}$, $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is a module over that algebra; it is quasi-coherent when $\mathscr{F}$ is quasi-coherent (**I**, 9.6.1).

More specifically, suppose that $X = \operatorname{Spec}(A)$, where A is a Noetherian domain. Then $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is the $\mathscr{O}_X$-algebra $\widetilde{B}$, where

$$B = \bigcap_{\mathfrak{p} \in X - Z} A_{\mathfrak{p}}. \tag{5.9.3.1}$$

This follows at once from (5.9.1.2) and (**I**, 8.2.1.1).

Proposition (5.9.4). — *Let X be a locally Noetherian prescheme, let $Z \subset X$ be stable under specialization, let X' be a locally Noetherian prescheme, and let $f : X' \to X$ be flat. Then $Z' = f^{-1}(Z)$ is stable under specialization, and, for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, there is a canonical isomorphism*

$$f^*(\mathscr{H}^0_{X/Z}(\mathscr{F})) \xrightarrow{\sim} \mathscr{H}^0_{X'/Z'}(f^*\mathscr{F}). \tag{5.9.4.1}$$

Proof. With the notation of (5.9.1), $Z'_\alpha = f^{-1}(Z_\alpha)$ is closed in X', and Z' is the union of the Z'_α. The base change of i_α to X' is the canonical injection $i'_\alpha : U'_\alpha \to X'$, where $U'_\alpha = X' - Z'_\alpha = f^{-1}(U_\alpha)$. Since f is flat, the canonical homomorphism

$$f^*((i_\alpha)_*(\mathscr{F}|U_\alpha)) \longrightarrow (i'_\alpha)_*(f^*\mathscr{F}|U'_\alpha)$$

is bijective (2.3.1). For $\alpha \leqslant \beta$, the diagram

$$\begin{tikzcd}
f^*((i_\alpha)_*(\mathscr{F}|U_\alpha)) \arrow[r, "\sim"] \arrow[d] & (i'_\alpha)_*(f^*\mathscr{F}|U'_\alpha) \arrow[d] \\
f^*((i_\beta)_*(\mathscr{F}|U_\beta)) \arrow[r, "\sim"] & (i'_\beta)_*(f^*\mathscr{F}|U'_\beta)
\end{tikzcd}$$

is commutative. Passing to the limit gives a canonical isomorphism

$$\varinjlim_\alpha f^*((i_\alpha)_*(\mathscr{F}|U_\alpha)) \xrightarrow{\sim} \mathscr{H}^0_{X'/Z'}(f^*\mathscr{F}).$$

Since f^* commutes with inductive limits (**0**, 4.3.2), this is precisely (5.9.4.1). □ **IV-2** | 111

Corollary (5.9.5). — *Under the hypotheses of (5.9.4), if $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent, then $\mathscr{H}^0_{X'/Z'}(f^*\mathscr{F})$ is coherent. The converse holds when f is faithfully flat and quasi-compact.*

Proof. The first assertion follows from (5.9.4.1) and (**0**, 5.3.11). The second amounts to saying that if $\mathscr{H}^0_{X'/Z'}(f^*\mathscr{F})$ is of finite type, then so is $\mathscr{H}^0_{X/Z}(\mathscr{F})$; this follows from (5.9.4.1) and (2.5.2). □

Corollary (5.9.6). — *Let X be a locally Noetherian prescheme, let $Z \subset X$ be stable under specialization, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module. For every $x \in X$, put $X_x = \operatorname{Spec}(\mathscr{O}_x)$ and $Z_x = Z \cap X_x$. There is a functorial isomorphism*

$$(\mathscr{H}^0_{X/Z}(\mathscr{F}))_x \xrightarrow{\sim} \mathscr{H}^0_{X_x/Z_x}(\widetilde{\mathscr{F}_x}). \tag{5.9.6.1}$$

Proof. Apply (5.9.4) to the canonical flat morphism $X_x \to X$ and use (**I**, 1.6.5). □

(5.9.7). With the notation of (5.9.1), for every α there is a canonical functorial homomorphism

$$\mathscr{F} \longrightarrow (i_\alpha)_*(\mathscr{F}|U_\alpha) \qquad (\mathbf{0}, 4.4.3.2).$$

These homomorphisms form an inductive system, and passage to the limit gives a canonical functorial homomorphism

$$\rho_{X/Z} : \mathscr{F} \longrightarrow \mathscr{H}^0_{X/Z}(\mathscr{F}). \tag{5.9.7.1}$$

Proposition (5.9.8). — *Let X be a locally Noetherian prescheme, let $Z \subset X$ be stable under specialization, and let $\mathscr{F}$ be an $\mathscr{O}_X$-module. The following conditions are equivalent:*

a) *The homomorphism $\rho_{X/Z}$ in (5.9.7.1) is injective (respectively bijective).*

b) *For every Noetherian open subset V of X, the homomorphism*

$$(\rho_{X/Z})_V : \Gamma(V, \mathscr{F}) \longrightarrow \varinjlim_{\alpha} \Gamma(V \cap U_\alpha, \mathscr{F})$$

is injective (respectively bijective).

They are also equivalent to:

a') *For every closed subset $T \subset Z$ of X, the canonical homomorphism* (**0**, 4.4.3.2)

$$\mathscr{F} \longrightarrow i_*(\mathscr{F}|X - T),$$

where $i : X - T \to X$ is the canonical injection, is injective (respectively bijective).

b') *For every closed subset $T \subset Z$ and every Noetherian open subset $V \subset X$, the restriction homomorphism*

$$\Gamma(V, \mathscr{F}) \longrightarrow \Gamma(V \cap (X - T), \mathscr{F})$$

is injective (respectively bijective).

Proof. Taking (5.9.1.2) into account, the equivalence of a) and b), and likewise that of a') and b'), follows from the definition of the functor Γ and its left exactness.

For every closed subset $T \subset Z$, the homomorphism $(\rho_{X/Z})_V$ is the composite

$$\Gamma(V, \mathscr{F}) \longrightarrow \Gamma(V \cap (X - T), \mathscr{F}) \longrightarrow \varinjlim_{\alpha} \Gamma(V \cap U_\alpha, \mathscr{F}). \tag{5.9.8.1}$$

Thus b) implies b') in the injective case, while b') implies b) directly from the definition of an inductive limit.

It remains to show that, if $\rho_{X/Z}$ is bijective, then so is $\Gamma(V, \mathscr{F}) \to \Gamma(V \cap (X - T), \mathscr{F})$. By (5.9.8.1), **IV-2** | 112
it is enough to observe that whenever $U' \subset U$ are open subsets of V containing $V \cap (X - Z)$, the restriction homomorphism $\Gamma(U, \mathscr{F}) \to \Gamma(U', \mathscr{F})$ is injective. This follows from the injectivity of $\rho_{X/Z}$ by replacing V with U and $V \cap (X - T)$ with U'. □

Definition (5.9.9). — Under the hypotheses of (5.9.8), the module $\mathscr{F}$ is called *Z-pure* (respectively *Z-closed*) if $\rho_{X/Z}$ is injective (respectively bijective).

If $X = \operatorname{Spec}(A)$ is affine and $\mathscr{F} = \widetilde{M}$, where M is an A-module, we call M Z-pure (respectively Z-closed) when $\mathscr{F}$ is.

We say that $\mathscr{F}$ is *Z-pure* (respectively *Z-closed*) *at a point* $x \in X$ if, with the notation of (5.9.6), $\widetilde{\mathscr{F}_x}$ is Z_x-pure (respectively Z_x-closed). By (5.9.6), this is equivalent to saying that

$$\mathscr{F}_x \longrightarrow (\mathscr{H}^0_{X/Z}(\mathscr{F}))_x$$

is injective (respectively bijective). At every point $x \in X - Z$, $\mathscr{F}$ is Z-closed by (5.9.8).

Corollary (5.9.10). — (i) *Let (V_λ) be an open covering of X. Then $\mathscr{F}$ is Z-pure (respectively Z-closed) if and only if, for every λ, $\mathscr{F}|V_\lambda$ is $(Z \cap V_\lambda)$-pure (respectively $(Z \cap V_\lambda)$-closed).*

(ii) *Let $Z' \subset Z$ be stable under specialization. If $\mathscr{F}$ is Z-pure (respectively Z-closed), then it is Z'-pure (respectively Z'-closed).*

Proof. This follows immediately from condition b') of (5.9.8). □

Proposition (5.9.11). — *Under the hypotheses of* (5.9.8)*, the supports of* $\ker(\rho_{X/Z})$ *and* $\operatorname{coker}(\rho_{X/Z})$ *are contained in Z, and $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed.*

Furthermore, if $u : \mathscr{F} \to \mathscr{F}'$ is an $\mathscr{O}_X$-module homomorphism and $\mathscr{F}'$ is Z-closed, then u factors uniquely as

$$\mathscr{F} \xrightarrow{\rho_{X/Z}} \mathscr{H}^0_{X/Z}(\mathscr{F}) \xrightarrow{v} \mathscr{F}'.$$

If, in addition, the supports of $\ker(u)$ *and* $\operatorname{coker}(u)$ *are contained in Z, then v is an isomorphism.*

Proof. The first assertion says that, for every $x \in X - Z$,

$$\ker(\rho_{X/Z})_x = \operatorname{coker}(\rho_{X/Z})_x = 0.$$

Equivalently, $(\rho_{X/Z})_x$ is bijective, or $\mathscr{F}$ is Z-closed at x, as noted after (5.9.9).

To prove that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed, it is enough to show, for every Noetherian open $V \subset X$, that

$$\varinjlim_{\beta} \Gamma(V \cap U_\beta, \mathscr{H}^0_{X/Z}(\mathscr{F})) = \Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F})).$$

By definition,

$$\Gamma(V \cap U_\beta, \mathscr{H}^0_{X/Z}(\mathscr{F})) = \varinjlim_{\alpha} \Gamma(V \cap U_\alpha \cap U_\beta, \mathscr{F}).$$

The double family $(U_\alpha \cap U_\beta)$ is filtered decreasing, so the assertion follows from (5.9.1) and the theorem on double inductive limits.

We now prove the second part. For every α, $(i_\alpha)_*(u)$ is the unique homomorphism making the diagram IV-2 | 113

$$\begin{tikzcd} \mathscr{F} \arrow[r, "u"] \arrow[d] & \mathscr{F}' \arrow[d] \\ (i_\alpha)_*(\mathscr{F}|U_\alpha) \arrow[r, "(i_\alpha)_*(u)"] & (i_\alpha)_*(\mathscr{F}'|U_\alpha) \end{tikzcd}$$

commute. Hence there is a unique homomorphism w making all diagrams

$$\begin{tikzcd} (i_\alpha)_*(\mathscr{F}|U_\alpha) \arrow[r, "(i_\alpha)_*(u)"] \arrow[d] & (i_\alpha)_*(\mathscr{F}'|U_\alpha) \arrow[d] \\ \mathscr{H}^0_{X/Z}(\mathscr{F}) \arrow[r, "w"] & \mathscr{H}^0_{X/Z}(\mathscr{F}') \end{tikzcd}$$

commute. Since $\mathscr{F}'$ is Z-closed, it identifies canonically with $\mathscr{H}^0_{X/Z}(\mathscr{F}')$. This proves existence and uniqueness of v.

Suppose finally that the supports of $\ker(u)$ and $\operatorname{coker}(u)$ are contained in Z. We show that, for every Noetherian open $V \subset X$, the homomorphism

$$\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F})) \longrightarrow \Gamma(V, \mathscr{F}')$$

is an isomorphism. If a section $t \in \Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F}))$ maps to zero, then, for some α, it is represented by a section on $V \cap U_\alpha$. By the hypothesis on u, its germ vanishes at every point of $V \cap (X - Z)$. It therefore vanishes on an open neighbourhood of $V \cap (X - Z)$, and so represents zero in $\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F}))$.

Conversely, let $s' \in \Gamma(V, \mathscr{F}')$. For every $x \in V \cap (X - Z)$, there is an open neighbourhood $W^{(x)}$ of x and a section $s^{(x)}$ of $\mathscr{F}$ on $W^{(x)}$ whose image under u is $s'|W^{(x)}$. Hence $s'|W^{(x)}$ is also the image under v of a section $t^{(x)}$ of $\mathscr{H}^0_{X/Z}(\mathscr{F})$. Since v is injective, these sections agree on overlaps. They therefore glue to a section t over an open subset $U \subset V$ containing $V \cap (X - Z)$. As $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed, t extends uniquely over V; its image under v agrees with s' on U, and hence on V by the same argument.

The module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is called the Z-*closure* of $\mathscr{F}$. □

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Remarks (5.9.12). — (i) Let $\mathscr{C}(X)$ be the category of $\mathscr{O}_X$-modules, and let $\mathscr{C}_Z(X)$ be its full subcategory of modules whose support is contained in Z. This is a localizing subcategory in the sense of Gabriel, and $\mathscr{H}^0_{X/Z}$ is precisely Gabriel's localization functor (cf. [27]); this gives another proof of (5.9.11). When Z is closed, the functor $i^* : \mathscr{C}(X) \to \mathscr{C}(X - Z)$, where $i : X - Z \to X$, defines an equivalence

$$\mathscr{C}(X)/\mathscr{C}_Z(X) \simeq \mathscr{C}(X - Z).$$

(ii) By (5.9.11), $\mathscr{H}^0_{X/Z}(\mathscr{F}) = 0$ if and only if $\operatorname{Supp}(\mathscr{F}) \subset Z$. Indeed, the vanishing implies that $\ker(\rho_{X/Z}) = \mathscr{F}$. Conversely, if $\operatorname{Supp}(\mathscr{F}) \subset Z$, apply the second part of (5.9.11) to the unique map $u : \mathscr{F} \to 0$; the corresponding map $v : \mathscr{H}^0_{X/Z}(\mathscr{F}) \to 0$ is an isomorphism.

(iii) The preceding constructions make sense for every locally Noetherian ringed space in which every closed irreducible subset has a unique generic point. In particular, on a locally Noetherian prescheme they apply to arbitrary sheaves of abelian groups, regarded as modules over the sheaf associated with the constant presheaf $\mathbf{Z}$. For every $x \in X$ one still has the canonical isomorphism (5.9.6.1), in which $\widetilde{\mathscr{F}}_x$ denotes the sheaf induced by $\mathscr{F}$ on the subspace X_x of X. A direct proof follows from (5.9.1.2) and the theorem on double inductive limits.

5.10. Property (S_2) and Z-closure

(5.10.1). Let X be a locally Noetherian prescheme and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. For every subset T of X, put

$$\operatorname{prof}_T(\mathscr{F}) = \inf_{z \in T} \operatorname{prof}(\mathscr{F}_z). \tag{5.10.1.1}$$

Proposition (5.10.2). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module. The following conditions are equivalent:*

a) *$\mathscr{F}$ is Z-pure.*
b) *$\operatorname{Ass}(\mathscr{F})$ does not meet Z.*

If, in addition, $\mathscr{F}$ is coherent, these conditions are also equivalent to:

c) $\operatorname{prof}_Z(\mathscr{F}) \geqslant 1$.

Proof. To say that $\mathscr{F}$ is Z-pure means that, for every Noetherian open subset V of X and every open subset $U \supset X - Z$, the restriction homomorphism

$$\Gamma(V, \mathscr{F}) \longrightarrow \Gamma(V \cap U, \mathscr{F})$$

is injective (5.9.8). By (3.1.8), this is equivalent to $V \cap \operatorname{Ass}(\mathscr{F}) \subset U$, which proves the equivalence of a) and b). Further, to say that $x \in \operatorname{Ass}(\mathscr{F})$ means that no element of $\mathfrak{m}_x$ is $\mathscr{F}_x$-regular (3.1.2); when $\mathscr{F}$ is coherent, this is equivalent to $\operatorname{prof}(\mathscr{F}_x) = 0$. The equivalence of b) and c) follows immediately. □

Corollary (5.10.3). — *Let*

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0$$

be an exact sequence of quasi-coherent $\mathscr{O}_X$-modules. If $\mathscr{F}$ is Z-pure, then so is $\mathscr{F}'$. Conversely, if $\mathscr{F}'$ and $\mathscr{F}''$ are Z-pure, then so is $\mathscr{F}$.

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Proof. This follows from the form (5.10.2), b), of the condition for a quasi-coherent $\mathscr{O}_X$-module to be Z-pure, and from (3.1.7). □

Corollary (5.10.4). — *Suppose that $\mathscr{F}$ is coherent. For $\mathscr{F}$ to be Z-pure at a point $x \in X$, it is necessary and sufficient that*

$$\operatorname{prof}_{Z_x}(\widetilde{\mathscr{F}}_x) \geqslant 1,$$

with the notation of (5.9.6).

Proof. This follows immediately from (5.10.2) and (5.9.6). □

Theorem (5.10.5). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. For $\mathscr{F}$ to be Z-closed, it is necessary and sufficient that*

$$\operatorname{prof}_Z(\mathscr{F}) \geqslant 2.$$

Proof. By (5.10.2), we may restrict ourselves to the case in which $\mathscr{F}$ is Z-pure and $\operatorname{prof}_Z(\mathscr{F}) \geqslant 1$. Moreover, to say that $\operatorname{prof}_Z(\mathscr{F}) \geqslant 2$ is equivalent to saying that

$$\operatorname{prof}_{Z_\alpha}(\mathscr{F}) \geqslant 2$$

for every closed subset Z_α of Z; similarly, by (5.9.8), to say that $\mathscr{F}$ is Z-closed is equivalent to saying that it is Z_α-closed for every α. We may therefore first reduce to the case in which Z is closed.

The question is local. For each $x \in Z$, it is enough to prove the theorem for $\mathscr{F}|U$, where U is an affine open neighbourhood of x; we may thus suppose that $X = U$ is affine. Then $\mathrm{Ass}(\mathscr{F})$ is finite (3.1.6), and, since $\mathrm{Ass}(\mathscr{F}) \subset X - Z$, there is a section f of $\mathscr{O}_X$ over X such that

$$\mathrm{Ass}(\mathscr{F}) \subset X_f \subset X - Z \qquad (\mathbf{II}, 4.5.4).$$

It follows that f is $\mathscr{F}$-regular (3.1.9), and for every $y \in Z$ one has $f_y \in \mathfrak{m}_y$, hence

$$\mathrm{prof}(\mathscr{F}_y) = 1 + \mathrm{prof}(\mathscr{F}_y/f_y\mathscr{F}_y) \qquad (\mathbf{0}, 16.4.6).$$

Thus $\mathrm{prof}_Z(\mathscr{F}) \geqslant 2$ is equivalent to $\mathrm{prof}_Z(\mathscr{F}/f\mathscr{F}) \geqslant 1$, or, by (5.10.2), to $\mathscr{F}/f\mathscr{F}$ being Z-pure. It remains to show that this last property is equivalent to $\mathscr{F}$ being Z-closed.

Consider the exact sequence

$$0 \longrightarrow \mathscr{F} \xrightarrow{f} \mathscr{F} \longrightarrow \mathscr{F}/f\mathscr{F} \longrightarrow 0,$$

where multiplication by f is injective by hypothesis. Put $W = X - Z$. We have the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \Gamma(X,\mathscr{F}) \arrow[r, "f"] \arrow[d] & \Gamma(X,\mathscr{F}) \arrow[r] \arrow[d] & \Gamma(X,\mathscr{F}/f\mathscr{F}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \Gamma(W,\mathscr{F}) \arrow[r, "f"] & \Gamma(W,\mathscr{F}) \arrow[r] & \Gamma(W,\mathscr{F}/f\mathscr{F}) &
\end{tikzcd}$$

whose two rows are exact, since X is affine. If the restriction homomorphism

$$\Gamma(X,\mathscr{F}) \longrightarrow \Gamma(W,\mathscr{F})$$

is bijective, the diagram shows that

$$\Gamma(X,\mathscr{F}/f\mathscr{F}) \longrightarrow \Gamma(W,\mathscr{F}/f\mathscr{F})$$

is injective; hence $\mathscr{F}/f\mathscr{F}$ is Z-pure (5.9.8).

Conversely, suppose that $\mathscr{F}/f\mathscr{F}$ is Z-pure, and let s be a section of $\mathscr{F}$ over W. Since $X_f \subset W$, there is an integer $n > 0$ such that $f^n(s|X_f)$ extends to a section t of $\mathscr{F}$ over X (**I**, 1.4.1). The restrictions of t and $f^n s$ to X_f are equal; the relation $\mathrm{Ass}(\mathscr{F}) \subset X_f$ therefore implies, by (5.10.2), that $t|W = f^n s$. Since f is $\mathscr{F}$-regular, it is enough to show that $t = f^n t'$ for some $t' \in \Gamma(X,\mathscr{F})$.

This is equivalent to the image of t in $\Gamma(X,\mathscr{F}/f^n\mathscr{F})$ being zero. For every k, $f^k\mathscr{F}/f^{k+1}\mathscr{F}$ is isomorphic to $\mathscr{F}/f\mathscr{F}$, hence is Z-pure. By induction on n, using (5.10.3), it follows that $\mathscr{F}/f^n\mathscr{F}$ is Z-pure. But the image of $t|W = f^n s$ in $\Gamma(W,\mathscr{F}/f^n\mathscr{F})$ is zero; the assertion follows from Z-purity. ☐ **IV-2** | 116

Corollary (5.10.6). — *Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. For $\mathscr{F}$ to be Z-closed at a point x, it is necessary and sufficient that*

$$\mathrm{prof}_{Z_x}(\widetilde{\mathscr{F}_x}) \geqslant 2.$$

Proof. This follows from (5.9.6) and (5.10.5). ☐

Theorem (5.10.7). — (Hartshorne). *Let X be a locally Noetherian prescheme and let Y be a closed subset of X. Suppose that* $\mathrm{prof}(\mathscr{O}_{X,y}) \geqslant 2$ *for every $y \in Y$. Then, for every connected component C of X, $C - (C \cap Y)$ is connected.*

Proof. We may restrict ourselves to the case in which X is connected. By (5.10.5), the canonical homomorphism

$$\mathscr{O}_X \longrightarrow i_*(\mathscr{O}_X|X - Y),$$

where $i : X - Y \to X$ is the canonical injection, is bijective. Consequently the restriction homomorphism

$$\Gamma(X,\mathscr{O}_X) \longrightarrow \Gamma(X - Y,\mathscr{O}_X)$$

is also bijective. It remains to apply (**III**, 7.8.6.1). ☐

Corollary (5.10.8). — *Let X be a locally Noetherian prescheme and let d be an integer such that, for every $x \in X$,*

$$\dim(\mathscr{O}_x) \geqslant d \quad \Longrightarrow \quad \mathrm{prof}(\mathscr{O}_x) \geqslant 2.$$

Suppose that X is connected. If X' and X'' are two distinct irreducible components of X, there is a sequence $(X_i)_{0 \leqslant i \leqslant n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$, and

$$\mathrm{codim}(X_{i-1} \cap X_i, X) \leqslant d-1 \qquad (1 \leqslant i \leqslant n).$$

In this case one says that X is connected in codimension $\leqslant d-1$.

Proof. If Y is a closed subset of X such that $\mathrm{codim}(Y, X) \geqslant d$, then $\dim(\mathscr{O}_{X,y}) \geqslant d$ for every $y \in Y$ (5.1.3); hence $\mathrm{prof}(\mathscr{O}_{X,y}) \geqslant 2$ for every $y \in Y$, and (5.10.7) shows that $X - Y$ is connected.

For $\mathrm{codim}(Y, X) \geqslant d$, it is necessary and sufficient that, for every $y \in Y$, there be an open neighbourhood V of y such that $\mathrm{codim}(Y \cap V, V) \geqslant d$ (**0**, 14.2.3). Let $\mathfrak{Y}$ be the set of closed subsets Y of X of codimension at least d. The union of two members of $\mathfrak{Y}$ belongs to $\mathfrak{Y}$ (**0**, 14.2.5), and every closed subset of a member of $\mathfrak{Y}$ also belongs to $\mathfrak{Y}$. In other words, $\mathfrak{Y}$ is an *antifilter* of closed subsets of X. The corollary now follows from the following topological lemma. ☐

Lemma (5.10.8.1). — *Let X be a connected locally Noetherian topological space and let $\mathfrak{Y}$ be an antifilter of closed subsets of X. Suppose that a closed subset Y belongs to $\mathfrak{Y}$ whenever, for every $y \in Y$, there exist an open neighbourhood V of y in X and a $Y_y \in \mathfrak{Y}$ such that $V \cap Y = V \cap Y_y$. The following conditions are equivalent:*

a) *For every $Y \in \mathfrak{Y}$, the space $X - Y$ is connected.*
b) *If X' and X'' are two distinct irreducible components of X, there is a sequence $(X_i)_{0 \leqslant i \leqslant n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$, and*

$$X_{i-1} \cap X_i \notin \mathfrak{Y} \qquad (1 \leqslant i \leqslant n).$$

Proof. Suppose b) holds and let $Y \in \mathfrak{Y}$. Put $U = X - Y$. If U' and U'' are two distinct irreducible components of U, there are irreducible components X' and X'' of X such that $X' \cap U = U'$ and $X'' \cap U = U''$ (**0**, 2.1.6).

Choose a sequence (X_i) with the property in b), and put $U_i = X_i \cap U$ for $0 \leqslant i \leqslant n$. Each U_i is an irreducible component of U (**0**, 2.1.6), and $U_i \cap U_{i-1} \neq \varnothing$ for $1 \leqslant i \leqslant n$: otherwise $X_i \cap X_{i-1} \subset Y$, so $X_i \cap X_{i-1} \in \mathfrak{Y}$, contrary to the choice of the X_i. Thus U is connected. **IV-2** | 117

Conversely, suppose a) holds. Let Y be the union of the sets $X_\alpha \cap X_\beta$, where (X_α, X_β) ranges over the pairs of distinct irreducible components of X for which $X_\alpha \cap X_\beta \in \mathfrak{Y}$. Every $y \in Y$ has an open neighbourhood V meeting only finitely many irreducible components of X. It follows both that Y is closed and that $V \cap Y$ is the intersection of V with a member of $\mathfrak{Y}$. By hypothesis, $Y \in \mathfrak{Y}$; hence $U = X - Y$ is connected. Moreover, Y is nowhere dense in X.

Let X' and X'' be distinct irreducible components of X, and let U' and U'' be their respective traces on U. They are distinct irreducible components of U (**0**, 2.1.6). The union of those irreducible components W of U for which there is a sequence $(U_i)_{0 \leqslant i \leqslant n}$ of irreducible components of U satisfying

$$U_0 = U', \qquad U_{i-1} \cap U_i \neq \varnothing \ (1 \leqslant i \leqslant n), \qquad U_n = W,$$

is both open and closed in U, because U is locally Noetherian and its irreducible components form a locally finite family of closed subsets. Since U is connected, there is such a sequence ending at U''. Let X_i be the irreducible component of X such that $X_i \cap U = U_i$ (**0**, 2.1.6). Since $U_{i-1} \neq U_i$, one has $X_{i-1} \neq X_i$. If $X_{i-1} \cap X_i \in \mathfrak{Y}$ for some i, then $X_{i-1} \cap X_i \subset Y$, contrary to $U_{i-1} \cap U_i \neq \varnothing$. This proves b). ☐

The hypothesis on X in (5.10.8) is satisfied when X is a Cohen–Macaulay prescheme and $d \geqslant 2$.

Corollary (5.10.9). — *If a Noetherian local ring A satisfies* (S_2) *and is catenary, then it is equidimensional.*

Proof. The hypothesis of (5.10.8) is satisfied by $X = \mathrm{Spec}(A)$ with $d = 2$. By that corollary, to show that all irreducible components of X have the same dimension, it is enough to consider two such components X' and X'' with

$$\mathrm{codim}(X' \cap X'', X) = 1.$$

There is then an irreducible component Z of $X' \cap X''$ such that $\operatorname{codim}(Z, X) = 1$. Since

$$1 = \operatorname{codim}(Z, X) \geqslant \operatorname{codim}(Z, X') \geqslant 1,$$

one has $\operatorname{codim}(Z, X') = 1$; similarly $\operatorname{codim}(Z, X'') = 1$. Since X is catenary, this implies $\dim(X') = \dim(X'')$. □

Proposition (5.10.10). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. Suppose that the $\mathscr{O}_X$-module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent. Then:*

(i)

$$\operatorname{prof}_Z\big(\mathscr{H}^0_{X/Z}(\mathscr{F})\big) \geqslant 2.$$

(ii) *For every $x \in \operatorname{Ass}(\mathscr{F}) \cap (X - Z)$,*

$$\operatorname{codim}\big(Z \cap \overline{\{x\}}, \overline{\{x\}}\big) \geqslant 2.$$

(iii) *The set U of points $x \in X$ such that*

$$\operatorname{prof}_{Z_x}(\widetilde{\mathscr{F}_x}) \geqslant 2$$

(with the notation of (5.9.6)) is open in X; moreover, $X - U \subset Z$, and U is the largest open subset of X such that $\mathscr{F}|U$ is $(Z \cap U)$-closed.

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Proof. Put, for brevity,

$$\mathscr{F}' = \mathscr{H}^0_{X/Z}(\mathscr{F}).$$

By (5.9.11), $\mathscr{F}'$ is Z-closed, so assertion (i) follows from (5.10.5).

Let $x \in \operatorname{Ass}(\mathscr{F}) \cap (X - Z)$. The restrictions of $\mathscr{F}$ and $\mathscr{F}'$ to $X - Z$ are canonically isomorphic (5.9.11); hence $x \in \operatorname{Ass}(\mathscr{F}')$. For $y \in Z \cap \overline{\{x\}}$, let $\mathfrak{p}$ be the prime ideal of $\mathscr{O}_y$ corresponding to x. It is associated to the $\mathscr{O}_y$-module $\mathscr{F}'_y$. Consequently, by (i) and (**0**, 16.4.6.2),

$$2 \leqslant \operatorname{prof}(\mathscr{F}'_y) \leqslant \dim(\mathscr{O}_y/\mathfrak{p}) = \operatorname{codim}\big(\overline{\{y\}}, \overline{\{x\}}\big),$$

which proves (ii).

Finally, by (5.10.5), U is the set of points x for which $\widetilde{\mathscr{F}_x}$ is Z_x-closed; by (5.9.6), it is equivalently the set of points at which the canonical homomorphism

$$\mathscr{F}_x \longrightarrow \mathscr{F}'_x$$

is bijective. Thus $X - U$ is the union of the supports of $\operatorname{Ker}(\rho_{X/Z})$ and $\operatorname{Coker}(\rho_{X/Z})$. These are coherent $\mathscr{O}_X$-modules by the hypothesis and (**0**, 5.3.4), so their supports are closed (**0**, 5.2.2). Hence U is open. It is plainly the largest open subset on which $\mathscr{F}$ is $(Z \cap U)$-closed, and $X - U \subset Z$ follows from (5.9.11). □

We shall see in (5.11.1) that, in the most important cases, assertion (ii) conversely implies that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent.

(5.10.11). Let X be a locally Noetherian prescheme and let Z be a subset of X stable under specialization. We saw in (5.9.3) that

$$\mathscr{A} = \mathscr{H}^0_{X/Z}(\mathscr{O}_X)$$

is a quasi-coherent $\mathscr{O}_X$-algebra. The X-scheme

$$X' = \operatorname{Spec}(\mathscr{A})$$

(**II**, 1.3.1) is called the *Z-closure* of X.

For every $\mathscr{O}_X$-module $\mathscr{F}$, $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is an $\mathscr{A}$-module, and it is quasi-coherent when $\mathscr{F}$ is quasi-coherent. In that case there is a unique quasi-coherent $\mathscr{O}_{X'}$-module $\mathscr{F}'$ such that, if $g : X' \to X$ denotes the structural morphism, then

(5.10.11.1) $$\mathscr{H}^0_{X/Z}(\mathscr{F}) = g_*(\mathscr{F}').$$

Proposition (5.10.12). — *Use the notation of (5.10.11).*

(i) *Let $x \in X$. The morphism obtained from g by localization,*

$$X' \times_X X_x \longrightarrow X_x = \operatorname{Spec}(\mathscr{O}_{X,x}),$$

is an isomorphism if and only if $\mathscr{O}_X$ is Z-closed at x. This condition holds for every $x \in X - Z$.

(ii) *Put $Z' = g^{-1}(Z)$ and suppose that X' is locally Noetherian. Then $\mathscr{F}'$ is Z'-closed. If, in addition, $\mathscr{F}'$ is a coherent $\mathscr{O}_{X'}$-module, then*

$$\operatorname{prof}_{Z'}(\mathscr{F}') \geqslant 2.$$

(iii) *Suppose that both $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ and $\mathscr{H}^0_{X/Z}(\mathscr{F})$ are coherent. Then $g : X' \to X$ is finite. The set U of points $x \in X$ such that*

$$\operatorname{prof}_{Z_x}(\widetilde{\mathscr{O}_x}) \geqslant 2 \quad \textit{and} \quad \operatorname{prof}_{Z_x}(\widetilde{\mathscr{F}_x}) \geqslant 2$$

is open in X and satisfies $X - U \subset Z$. Moreover, U is the largest open subset of X such that $g^{-1}(U) \to U$ is an isomorphism and the restriction to U of the canonical g-morphism from $\mathscr{F}$ to $\mathscr{F}'$ is an isomorphism.

Proof. Assertion (i) follows from the definitions, and (iii) is an immediate consequence of (5.10.10), (iii).

For (ii), let V be an open subset of X containing $X - Z$, put $V' = g^{-1}(V)$, and let $i : V \to X$ and $i' : V' \to X'$ be the canonical injections.

The canonical homomorphism **IV-2** | 119

$$\rho_{X'/Z'} : \mathscr{F}' \longrightarrow i'_*(\mathscr{F}'|V')$$

has the property that $g_*(\rho_{X'/Z'})$ is the canonical homomorphism

$$\rho_{X/Z} : \mathscr{H}^0_{X/Z}(\mathscr{F}) \longrightarrow i_*\big(\mathscr{H}^0_{X/Z}(\mathscr{F})|V\big),$$

by (**II**, 1.4.2) and (5.10.11.1). Since $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed (5.9.11), $\rho_{X/Z}$ is an isomorphism, and therefore so is $\rho_{X'/Z'}$. The open subset $X' - Z'$ is the intersection of the filtered family of inverse images $g^{-1}(V)$, as V ranges over the open subsets of X containing $X - Z$. Since X' is locally Noetherian, (5.9.1) now shows that $\mathscr{F}'$ is Z'-closed. The final assertion follows from (5.10.5). ☐

(5.10.13). We shall now apply the preceding results when Z is one of the sets $Z^{(n)}(X)$ (or simply $Z^{(n)}$), defined as the set of points $x \in X$ such that

$$\dim(\mathscr{O}_x) \geqslant n.$$

It is clear that $Z^{(n)}$ is stable under specialization. A closed subset T of X is contained in $Z^{(n)}$ if and only if

$$\operatorname{codim}(T, X) \geqslant n.$$

We shall be concerned here with the case $n = 2$.

Proposition (5.10.14). — *Let X be a locally Noetherian prescheme, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module whose support is X.*

(i) *The module $\mathscr{F}$ satisfies (S_1) if and only if it is $Z^{(1)}$-pure.*

(ii) *The module $\mathscr{F}$ satisfies (S_2) if and only if it is $Z^{(2)}$-closed and $Z^{(1)}$-pure, or equivalently, if it is $Z^{(2)}$-closed and has no associated prime cycle of codimension* 1.

Proof. To say that $\mathscr{F}$ has property (S_1) means that it has no embedded associated prime cycle (5.7.5), or equivalently that

$$\dim(\mathscr{F}_x) = 0 \qquad \text{for every } x \in \operatorname{Ass}(\mathscr{F})$$

(3.1.4). Since $\operatorname{Supp}(\mathscr{F}) = X$, this is equivalent to $\dim(\mathscr{O}_x) = 0$ (5.1.12.1); in other words, $\operatorname{Ass}(\mathscr{F})$ does not meet $Z^{(1)}$. Assertion (i) therefore follows from (5.10.2).

To say that $\mathscr{F}$ is $Z^{(2)}$-closed means that

$$\operatorname{prof}_{Z^{(2)}}(\mathscr{F}) \geqslant 2,$$

or equivalently that, for every $x \in X$,

$$\dim(\mathscr{O}_x) \geqslant 2 \quad\Longrightarrow\quad \operatorname{prof}(\mathscr{F}_x) \geqslant 2.$$

Thus (S_2) implies that $\mathscr{F}$ is $Z^{(2)}$-closed. It also implies (S_1), so $\mathscr{F}$ has no embedded associated prime cycle (5.7.5); since its support is X, this says that all its associated prime cycles have codimension 0.

Conversely, suppose that $\mathscr{F}$ is $Z^{(2)}$-closed and has no associated prime cycle of codimension 1. It remains only to consider a point x for which $\dim(\mathscr{O}_x) = 1$. By hypothesis $x \notin \mathrm{Ass}(\mathscr{F})$, and hence $\mathrm{prof}(\mathscr{F}_x) > 0$. Since $\dim(\mathscr{F}_x) = \dim(\mathscr{O}_x) = 1$, it follows that $\mathrm{prof}(\mathscr{F}_x) = 1$. Therefore $\mathscr{F}$ satisfies (S_2). Finally, if $\mathscr{F}$ is $Z^{(1)}$-pure, it satisfies (S_1) by (i); as observed above, all its associated prime cycles then have codimension 0, so the same argument applies.

Notice that a module may be $Z^{(2)}$-closed without satisfying (S_1). For example, if $\dim X = 1$, then $Z^{(2)} = \varnothing$ and every $\mathscr{O}_X$-module is $Z^{(2)}$-closed, even though a module may have embedded associated prime cycles.

We recall that Chapter III gives, by means of local cohomology, a characterization of (S_n) for every $n \geqslant 1$ that generalizes (5.10.14). ☐

Corollary (5.10.15). — *Let X be a locally Noetherian prescheme, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module with support X. Suppose that $\mathscr{F}$ has no associated prime cycle of codimension 1 and that*

$$\mathscr{F}' = \mathscr{H}^0_{X/Z^{(2)}}(\mathscr{F})$$

is coherent. Then:

(i) *$\mathscr{F}'$ satisfies (S_2).*

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(ii) *The set U of points $x \in X$ at which $\mathscr{F}$ satisfies (S_2) (5.7.2) is open in X, and*

$$\mathrm{codim}(X - U, X) \geqslant 2.$$

Proof. By (5.9.11), $\mathscr{F}'$ is $Z^{(2)}$-closed. Moreover, $\mathrm{Supp}(\mathscr{F}') = X$: every maximal point of X lies outside $Z^{(2)}$, and at such a point $\mathscr{F}'_x = \mathscr{F}_x \neq 0$. Thus the support of $\mathscr{F}'$ is dense in X; it is also closed because $\mathscr{F}'$ is coherent, and hence it is all of X.

It remains to see that $\mathscr{F}'$ has no associated prime cycle of codimension 1. If $x \in \mathrm{Ass}(\mathscr{F}')$ and

$$\dim(\mathscr{F}'_x) = \dim(\mathscr{O}_x) = 1,$$

then $x \notin Z^{(2)}$, so $\mathscr{F}'_x = \mathscr{F}_x$ and consequently $x \in \mathrm{Ass}(\mathscr{F})$, contrary to the hypothesis. Assertion (i) now follows from (5.10.14).

For (ii), one has

$$Z^{(n)}(X_x) = Z^{(n)}(X) \cap X_x$$

with the notation of (5.9.6), by (**I**, 2.4.2). The absence of associated prime cycles of codimension 1 for $\mathscr{F}$ implies the same property for $\widetilde{\mathscr{F}}_x$. By (5.10.14), $\widetilde{\mathscr{F}}_x$ satisfies (S_2) if and only if it is $Z^{(2)}(X_x)$-closed. The assertion therefore follows from (5.10.6) and (5.10.10), (iii). ☐

Proposition (5.10.16). — *Let X be a locally Noetherian prescheme, let*

$$X' = \mathrm{Spec}\big(\mathscr{H}^0_{X/Z^{(2)}}(\mathscr{O}_X)\big)$$

be its $Z^{(2)}$-closure, and let $g : X' \to X$ be the structural morphism. Suppose that X has no associated prime cycle of codimension 1.

(i) *At a point $x \in X$, the prescheme X satisfies (S_2) if and only if the morphism*

$$X'_x \longrightarrow X_x$$

deduced from g (with the notation of (5.10.12)) is an isomorphism. This condition always holds if

$$\mathrm{codim}\big(\overline{\{x\}}, X\big) \leqslant 1.$$

(ii) *Suppose in addition that g is finite (see (5.11.2) for sufficient conditions). Then the set U of points at which X satisfies (S_2) is open,*

$$\mathrm{codim}(X - U, X) \geqslant 2,$$

and U is the largest open subset of X for which $g^{-1}(U) \to U$ is an isomorphism.

(iii) *Under the hypotheses of* (ii), X' *satisfies* (S_2). *Moreover, for every* $x' \in X'$ *such that*

$$\operatorname{codim}\big(\overline{\{x'\}}, X'\big) \leqslant 1,$$

the point $x = g(x')$ *satisfies*

$$\operatorname{codim}\big(\overline{\{x\}}, X\big) = \operatorname{codim}\big(\overline{\{x'\}}, X'\big).$$

(iv) *Under the hypotheses of* (ii), *let* $\mathscr{F}$ *be a coherent* $\mathscr{O}_X$*-module with support* X, *and suppose that* $\mathscr{H}^0_{X/Z^{(2)}}(\mathscr{F})$ *is coherent. The* $\mathscr{O}_{X'}$*-module* $\mathscr{F}'$ *defined by*

$$g_*(\mathscr{F}') = \mathscr{H}^0_{X/Z^{(2)}}(\mathscr{F})$$

is coherent and satisfies (S_2), *and its support is a union of irreducible components of* X'.

Proof. Assertions (i) and (ii) have already been proved in substance: (i) follows from (5.10.12), (i), and (5.10.14), while (ii) is the special case of (5.10.15), (ii), obtained by taking $\mathscr{F} = \mathscr{O}_X$.

We prove (iii). Put $x = g(x')$. Since g is finite, so is $X'_x \to X_x$, and therefore

$$\dim(\mathscr{O}_{X',x'}) \leqslant \dim(X'_x) \leqslant \dim(X_x) = \dim(\mathscr{O}_{X,x})$$

by (5.4.1). Suppose first that $\dim(\mathscr{O}_{X',x'}) \leqslant 1$. Then $\dim(\mathscr{O}_{X,x}) \leqslant 1$ as well: otherwise $x \in Z^{(2)}$ and (5.10.12), (ii), applied to $\mathscr{O}_X$, would give

$$\operatorname{prof}(\mathscr{O}_{X',x'}) \geqslant 2,$$

which is impossible. Hence $x \notin Z^{(2)}$, and (5.10.12), (i), gives an isomorphism

$$\mathscr{O}_{X,x} \simeq \mathscr{O}_{X',x'}.$$

In particular their dimensions are equal. Since X has no associated prime cycle of codimension 1, the equality $\dim(\mathscr{O}_{X,x}) = 1$ implies $x \notin \operatorname{Ass}(\mathscr{O}_X)$, so

$$\operatorname{prof}(\mathscr{O}_{X,x}) = \operatorname{prof}(\mathscr{O}_{X',x'}) = 1.$$

If instead $\dim(\mathscr{O}_{X',x'}) \geqslant 2$, then $\dim(\mathscr{O}_{X,x}) \geqslant 2$, so $x \in Z^{(2)}$ and (5.10.12), (ii), gives

$$\operatorname{prof}(\mathscr{O}_{X',x'}) \geqslant 2.$$

This proves all the assertions in (iii).

For (iv), replace $\mathscr{O}_X$ by $\mathscr{F}$ in the preceding argument. Since g is finite and $g_*(\mathscr{F}')$ is coherent, $\mathscr{F}'$ is coherent. The same argument proves that $\mathscr{F}'$ satisfies (S_2) and that, if $\dim(\mathscr{F}'_{x'}) \leqslant 1$, then $\mathscr{F}_x$ and $\mathscr{F}'_{x'}$ are di-isomorphic. In particular, **IV-2** | 121

$$\dim(\mathscr{F}'_{x'}) = 0 \quad\Longrightarrow\quad \dim(\mathscr{F}_x) = 0 \quad\Longrightarrow\quad \dim(\mathscr{O}_{X,x}) = 0,$$

because $\operatorname{Supp}(\mathscr{F}) = X$, and hence also $\dim(\mathscr{O}_{X',x'}) = 0$. Every irreducible component of $\operatorname{Supp}(\mathscr{F}')$ is therefore an irreducible component of X'. Finally, $\operatorname{Supp}(\mathscr{F}')$ is closed because $\mathscr{F}'$ is coherent. □

Proposition (5.10.17). — *Let* A *be a Noetherian integral domain, and denote by* $A^{(1)}$ *the intersection of the local rings* $A_{\mathfrak{p}}$, *where* $\mathfrak{p}$ *ranges over the prime ideals of* A *of height* 1. *Suppose that* $A^{(1)}$ *is a finite* A*-algebra. Then:*

(i) *The ring* $A^{(1)}$ *satisfies* (S_2).

(ii) *The set* U *of primes* $\mathfrak{p} \in \operatorname{Spec}(A)$ *for which the canonical homomorphism*

$$A_{\mathfrak{p}} \longrightarrow (A^{(1)})_{\mathfrak{p}}$$

is bijective is precisely the set of primes for which $A_{\mathfrak{p}}$ *satisfies* (S_2). *The set* U *is open in* $X = \operatorname{Spec}(A)$ *and*

$$\operatorname{codim}(X - U, X) \geqslant 2.$$

(iii) *For every multiplicative subset* S *of* A, $(S^{-1}A)^{(1)}$ *is a finite* $S^{-1}A$*-algebra.*

(iv) *Let* B *be a finite integral* A*-algebra that is an integral domain and contains* A. *Then* $B^{(1)}$ *is a finite* B*-algebra. Moreover, if* $\mathfrak{q}$ *is a prime ideal of* B *of height* 1, *then* $\mathfrak{q} \cap A$ *has height* 1 *in* A.

Proof. Formula (5.9.3.1) shows that

$$X' = \operatorname{Spec}(A^{(1)})$$

is the $Z^{(2)}$-closure of $X = \operatorname{Spec}(A)$. Since A has no embedded associated prime ideals, assertions (i) and (ii) are special cases of (5.10.16), (i)–(iii).

For (iii), observe that

$$(S^{-1}A)^{(1)} = S^{-1}A^{(1)},$$

which is a special case of (5.9.4). Indeed, $S^{-1}A$ is flat over A, its prime ideals are the ideals $S^{-1}\mathfrak{p}$ with $\mathfrak{p} \in \operatorname{Spec}(A)$ disjoint from S, and

$$\operatorname{ht}(S^{-1}\mathfrak{p}) = \operatorname{ht}(\mathfrak{p}).$$

Since $A^{(1)}$ is finite over A, its localization $S^{-1}A^{(1)}$ is finite over $S^{-1}A$.

To prove (iv), put $Y = \operatorname{Spec}(B)$ and let $f : Y \to X$ be the structural morphism. Since f is finite, (5.4.1) gives, for every $y \in Y$,

$$\dim(\mathscr{O}_{Y,y}) \leqslant \dim(\mathscr{O}_{X,f(y)}).$$

Consequently, if

$$T = f^{-1}\big(Z^{(2)}(X)\big),$$

then $T \supset Z^{(2)}(Y)$.

We claim that

$$\mathscr{G} = \mathscr{H}^0_{X/Z^{(2)}(X)}\big(f_*(\mathscr{O}_Y)\big)$$

is coherent. The module $f_*(\mathscr{O}_Y)$ corresponds to B viewed as an A-module. Since B is a finite integral A-algebra and an integral domain, its field of fractions is finite over the field of fractions of A; hence B is contained in a finite free A-module. By (5.9.2), (i), $\mathscr{G}$ is therefore a quasi-coherent $\mathscr{O}_X$-submodule of

$$\big(\mathscr{H}^0_{X/Z^{(2)}(X)}(\mathscr{O}_X)\big)^n$$

for a suitable n. The latter module is coherent by hypothesis, and hence so is $\mathscr{G}$.

By the definition (5.9.1.2), there is an isomorphism

$$\mathscr{G} \simeq f_*\big(\mathscr{H}^0_{Y/T}(\mathscr{O}_Y)\big).$$

It follows, in particular, that $\mathscr{H}^0_{Y/T}(\mathscr{O}_Y)$ is a coherent $\mathscr{O}_Y$-module. Applying (5.10.10), (ii), to $\mathscr{O}_Y$ and the generic point of Y, we obtain

$$\operatorname{codim}(T, Y) \geqslant 2,$$

that is, $T \subset Z^{(2)}(Y)$. Thus $T = Z^{(2)}(Y)$. Formula (5.9.3.1) now identifies the corresponding algebra with $B^{(1)}$, proving that $B^{(1)}$ is finite over B. Finally, if $\mathfrak{q}$ has height 1, then $\mathfrak{q} \notin T$, so $\mathfrak{q} \cap A \notin Z^{(2)}(X)$. Incomparability for the integral extension $A \subset B$ excludes $\mathfrak{q} \cap A = (0)$; hence

$$\operatorname{ht}(\mathfrak{q} \cap A) = 1.$$

□

5.11. Coherence criteria for the modules $\mathscr{H}^0_{X/Z}(\mathscr{F})$

Proposition (5.11.1). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. Denote by (x_α) the family of points of*

$$\mathrm{Ass}(\mathscr{F}) \cap (X - Z),$$

and, for every α, let Y_α be the reduced closed subprescheme of X whose underlying space is $\overline{\{x_\alpha\}}$, and put $Z_\alpha = Z \cap \overline{\{x_\alpha\}}$. The following two conditions are equivalent:

a) *$\mathscr{H}^0_{X/Z}(\mathscr{F})$ is a coherent $\mathscr{O}_X$-module.*
b) *For every α, $\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})$ is a coherent $\mathscr{O}_{Y_\alpha}$-module.*

These conditions imply:

c) *For every α,* $\mathrm{codim}(Z_\alpha, Y_\alpha) \geqslant 2$.

Moreover, conditions a), b), *and* c) *are equivalent if, in addition, either of the following properties holds:*

(i) *Every point of X has an open neighbourhood isomorphic to a subscheme of a regular scheme; one also says in this case that X is* locally immersible in a regular scheme.
(ii) *For every α, Y_α is universally catenary* (5.6.2), *and its normalization $\widetilde{Y}_\alpha$* (**II**, 6.3.8) *is finite over Y_α.*

Proof. All the properties under consideration are local on X. We may therefore restrict ourselves to the case in which $X = \mathrm{Spec}(A)$ is affine, A is Noetherian, and $\mathscr{F} = \widetilde{M}$, where M is a finite A-module. For every α, let $h_\alpha : Y_\alpha \to X$ be the canonical injection. Then

$$(h_\alpha)_*(\mathscr{O}_{Y_\alpha}) = \mathscr{G}_\alpha$$

is the $\mathscr{O}_X$-module corresponding to the quotient $A/\mathfrak{j}_{x_\alpha}$; by the definition of $\mathrm{Ass}(\mathscr{F})$, this quotient is isomorphic to an A-submodule of M. Since $\mathscr{H}^0_{X/Z}(\mathscr{G}_\alpha)$ is a quasi-coherent $\mathscr{O}_X$-submodule of $\mathscr{H}^0_{X/Z}(\mathscr{F})$ (5.9.2), condition a) implies that it is coherent. On the other hand, the definition (5.9.1.2) gives

$$\mathscr{H}^0_{X/Z}(\mathscr{G}_\alpha) \simeq (h_\alpha)_*\big(\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})\big).$$

Thus a) implies b).

To prove the converse, it is enough to find a finite filtration $(\mathscr{F}_i)_{0 \leqslant i \leqslant n}$ by coherent $\mathscr{O}_X$-modules, with $\mathscr{F}_0 = \mathscr{F}$ and $\mathscr{F}_n = 0$, such that

$$\mathscr{H}^0_{X/Z}(\mathscr{F}_i/\mathscr{F}_{i+1})$$

is coherent for every i. Indeed, the assertion then follows by descending induction from the exact sequences

$$0 \longrightarrow \mathscr{F}_{i+1} \longrightarrow \mathscr{F}_i \longrightarrow \mathscr{F}_i/\mathscr{F}_{i+1},$$

the left exactness of $\mathscr{H}^0_{X/Z}$ (5.9.2), and (**0I**, 5.3.3) and (**I**, 6.1.1). In view of (3.2.8), it is therefore enough to treat the case in which $\mathscr{F}$ is irredundant, that is, $\mathrm{Ass}(M) = \{\mathfrak{p}\}$ consists of one element. We use the following lemma.

Lemma (5.11.1.1). — *Let A be a Noetherian ring and let M be a finite A-module such that* $\mathrm{Ass}(M) = \{\mathfrak{p}\}$. *There is a finite filtration $(M_i)_{0 \leqslant i \leqslant r}$ of M, with $M_0 = M$ and $M_r = 0$, such that every quotient M_i/M_{i+1} is isomorphic to a submodule of $A/\mathfrak{p}$.*

Proof. The canonical homomorphism $M \to M_\mathfrak{p} = N$ is injective (Bourbaki, *Algèbre commutative*, Chapter IV, §1, no. 2, Proposition 6). Put

$$B = A_\mathfrak{p}, \qquad \mathfrak{m} = \mathfrak{p}A_\mathfrak{p}.$$

Then $\mathfrak{m}$ is the maximal ideal of B and $\mathrm{Ass}(N) = \{\mathfrak{m}\}$ (*loc. cit.*, Proposition 5).

Since N is a finite B-module, there is an integer s such that $\mathfrak{m}^s N = 0$. The filtration by the powers $\mathfrak{m}^j N$ may be refined to a finite filtration (N_i) of N whose successive quotients are isomorphic to **IV-2** | 123

$$B/\mathfrak{m} = \mathrm{Frac}(A/\mathfrak{p}).$$

Intersecting it with M gives $M_i = M \cap N_i$. Each quotient M_i/M_{i+1} is then isomorphic to a finite $(A/\mathfrak{p})$-submodule of $\mathrm{Frac}(A/\mathfrak{p})$, and every such module is isomorphic to a submodule of $A/\mathfrak{p}$. □

Completion of the proof of Proposition 5.11.1. The lemma and the preceding filtration argument reduce the proof, under condition b), to showing that $\mathscr{H}^0_{X/Z}(\mathscr{G})$ is coherent when

$$\mathscr{G} = (A/\mathfrak{p})^\sim$$

and $\mathfrak{p}$ is an associated prime of M. If $\mathfrak{p} = \mathfrak{j}_y$ with $y \in Z$, the support of $\mathscr{G}$ is a closed subset contained in Z, because Z is stable under specialization. Hence $\mathscr{H}^0_{X/Z}(\mathscr{G}) = 0$ by (5.9.1.2). Otherwise, $\mathfrak{p} = \mathfrak{j}_{x_\alpha}$ for some α, and

$$\mathscr{G} = (h_\alpha)_*(\mathscr{O}_{Y_\alpha}).$$

The displayed isomorphism above and condition b) then show that $\mathscr{H}^0_{X/Z}(\mathscr{G})$ is coherent. This proves the equivalence of a) and b).

Condition a) implies c) by (5.10.10, ii). It remains to prove, under either hypothesis (i) or (ii), that c) implies b). If X satisfies (i), each Y_α does as well. Consequently, in view of (5.9.3.1), the required assertion follows from Corollary 5.11.2 below. □

Corollary (5.11.2). — *Let A be a Noetherian domain satisfying either of the following hypotheses:*

(i) *A is a quotient of a regular Noetherian ring.*
(ii) *A is universally catenary, and its integral closure A' is a finite A-algebra.*

Then

$$A^{(1)} = \bigcap_{\substack{\mathfrak{p}\in\operatorname{Spec}(A)\\ \operatorname{ht}(\mathfrak{p})=1}} A_{\mathfrak{p}}$$

is a finite A-algebra.

Proof. Suppose first that (i) holds, and put $X = \operatorname{Spec}(A)$. The set U of points at which X satisfies (S_2) is open by (6.11.2).[8] Put $Z = X - U$. Then $\operatorname{codim}(Z, X) \geqslant 2$. Indeed, if $x \in X$ and $\dim(\mathscr{O}_x) \leqslant 1$, then $\dim(\mathscr{O}_{x'}) \leqslant 1$ for every generization x' of x. Since $\mathscr{O}_{x'}$ is a domain, it has positive depth whenever its dimension is positive, and therefore X satisfies (S_2) at x.

Thus $Z \subset Z^{(2)}(X)$. If $j : U \to X$ is the canonical injection, (5.9.1.2) gives

$$\mathscr{H}^0_{X/Z^{(2)}(X)}(\mathscr{O}_X) \simeq j_*\big(\mathscr{H}^0_{U/Z^{(2)}(U)}(\mathscr{O}_U)\big).$$

The prescheme U satisfies (S_2), so (5.10.14) identifies the module inside the parentheses with $\mathscr{O}_U$. Since $\operatorname{codim}(Z, X) \geqslant 2$, Chapter III, §9 shows that $j_*(\mathscr{O}_U)$ is coherent. Finally, (5.9.3.1) identifies the corresponding A-module with $A^{(1)}$, proving the assertion under (i).

Now suppose that (ii) holds. The ring A' is a Noetherian normal domain, hence it is the intersection of its local rings $A'_{\mathfrak{p}'}$ as $\mathfrak{p}'$ ranges over the height-one prime ideals of A' (Bourbaki, *Algèbre commutative*, Chapter VII, §1, no. 6, Theorem 4). For such a prime $\mathfrak{p}'$, put

$$\mathfrak{p} = \mathfrak{p}' \cap A, \qquad S = A - \mathfrak{p}.$$

The ring $A'_{\mathfrak{p}'}$ is the localization of the finite $A_{\mathfrak{p}}$-algebra $S^{-1}A'$ at $S^{-1}\mathfrak{p}'$. The latter prime lies above the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ and is therefore maximal. Moreover, $A_{\mathfrak{p}}$ is universally catenary by (5.6.3, i). Formula (5.6.10) consequently gives **IV-2** | 124

$$\dim(A_{\mathfrak{p}}) = \dim(A'_{\mathfrak{p}'}) = 1.$$

It follows that $A^{(1)} \subset A'$. Since A' is a finite A-module and A is Noetherian, its submodule $A^{(1)}$ is finite as well. □

Remark (5.11.3). — The conclusion that $A^{(1)}$ is a finite A-algebra need no longer hold if, in hypothesis (ii) of (5.11.2), one assumes only that A is catenary.

Indeed, consider the catenary local ring A constructed in (5.6.11); its integral closure A' (denoted there by B) is a finite A-algebra. If $A^{(1)}$ were also finite over A, then, being contained in the fraction field of A, it would be contained in A'. On the other hand, in the notation of (5.6.11), every height-one prime ideal of A is of the form $\mathfrak{p}A$, where $\mathfrak{p}$ is a height-one prime of C, and

$$A_{\mathfrak{p}A} = C_{\mathfrak{p}}.$$

[8] The reader may verify that Corollary 5.11.2 is not used in the proof of (6.11.2).

Moreover, $C_{\mathfrak{p}} = E_{\mathfrak{p}'}$, where $\mathfrak{p}'$ is the unique prime of E above $\mathfrak{p}$. Thus $A_{\mathfrak{p}A}$ is integrally closed and consequently contains A'. By definition this gives $A' \subset A^{(1)}$, so finiteness of $A^{(1)}$ would force $A^{(1)} = A'$. This is impossible: one of the two prime ideals of A' above the maximal ideal $\mathfrak{n}A$ of A has height 1, whereas $\mathfrak{n}A$ has height 2, contradicting (5.10.17, iv).

Corollary (5.11.4). — *Let X be a locally Noetherian prescheme, let Z be a closed subset of X, put $U = X - Z$, and let $i : U \to X$ be the canonical injection. Let $\mathscr{F}$ be a coherent $\mathscr{O}_U$-module. A necessary condition for $i_*(\mathscr{F})$ to be a coherent $\mathscr{O}_X$-module is that*

$$\operatorname{codim}\big(\overline{\{x\}} \cap Z, \overline{\{x\}}\big) \geqslant 2 \qquad \text{for every } x \in \operatorname{Ass}(\mathscr{F}).$$

This condition is sufficient in either of the following cases:

(i) *X is locally immersible in a regular scheme.*
(ii) *X is universally catenary and universally Japanese; by definition, the latter means that every point $x \in X$ has an affine open neighbourhood whose ring is universally Japanese* (**0**, 23.1.1).

Proof. By (**I**, 9.4.7), there is a coherent $\mathscr{O}_X$-submodule $\mathscr{G}$ of $i_*(\mathscr{F})$ such that $\mathscr{G}|U = \mathscr{F}$. Plainly,

$$\operatorname{Ass}(\mathscr{G}) \subset \operatorname{Ass}(\mathscr{F}) \subset \operatorname{Ass}\big(i_*(\mathscr{F})\big),$$

and

$$\operatorname{Ass}\big(i_*(\mathscr{F})\big) = \operatorname{Ass}(\mathscr{F})$$

by (3.1.13). Hence $\operatorname{Ass}(\mathscr{G}) = \operatorname{Ass}(\mathscr{F})$. It remains to apply (5.11.1) to $\mathscr{G}$, observing that

$$i_*(\mathscr{F}) = \mathscr{H}^0_{X/Z}(\mathscr{G}),$$

and that, when X is universally catenary and universally Japanese, hypothesis (ii) of (5.11.1) holds by definition. □

Corollary (5.11.5). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module such that $\operatorname{Ass}(\mathscr{F}) \subset X - Z$. Consider the conditions:*

a) *$\mathscr{H}^0_{X/Z}(\mathscr{F})$ is a coherent $\mathscr{O}_X$-module.*
d) *For every subset T of Z closed in X, $i_*(\mathscr{F}|X - T)$ is coherent, where $i : X - T \to X$ is the canonical injection. It is enough here to consider an increasing filtered family of such closed subsets whose union is Z.*

Then a) *implies* d). *If X satisfies either hypothesis* (i) *or* (ii) *of* (5.11.1), *the two conditions are equivalent.*

Proof. The hypothesis and (3.1.13) give

$$\operatorname{Ass}\big(i_*(\mathscr{F}|X - T)\big) = \operatorname{Ass}(\mathscr{F}).$$

It follows from (5.10.2) that the canonical maps

$$i_*(\mathscr{F}|X - T) \longrightarrow \mathscr{H}^0_{X/Z}(\mathscr{F})$$

are injective. Condition a) therefore implies d).

Conversely, condition d) and (5.11.1) imply, in the notation of that proposition, that **IV-2** | 125

$$\operatorname{codim}(T \cap Y_\alpha, Y_\alpha) \geqslant 2.$$

Since Z is the union of its subsets closed in X, it follows that

$$\operatorname{codim}(Z \cap Y_\alpha, Y_\alpha) \geqslant 2.$$

The last assertion now follows from (5.11.1). □

Corollary (5.11.6). — *Let A be a Noetherian ring and put $X = \operatorname{Spec}(A)$. Consider the following properties:*

a) *For every quotient domain B of A, the ring $B^{(1)}$* (5.11.2) *is a finite B-algebra.*
b) *For every coherent $\mathscr{O}_X$-module $\mathscr{F}$ and every subset Z of X stable under specialization such that*

$$\operatorname{codim}\big(\overline{\{x\}} \cap Z, \overline{\{x\}}\big) \geqslant 2$$

for every $x \in \operatorname{Ass}(\mathscr{F}) \cap (X - Z)$, the module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent.

c) *For every closed subset T of X and every coherent $\mathscr{O}_U$-module $\mathscr{G}$, where $U = X - T$, such that*

$$\operatorname{codim}\left(\overline{\{x\}} \cap T, \overline{\{x\}}\right) \geqslant 2 \qquad (x \in \operatorname{Ass}(\mathscr{G})),$$

the module $i_*(\mathscr{G})$ is coherent, where $i : U \to X$ is the canonical injection.

d) *For every quotient domain B of A and every ideal $\mathfrak{J}$ of B of height at least 2, the ring*

$$\bigcap_{\mathfrak{p} \not\supseteq \mathfrak{J}} B_{\mathfrak{p}}$$

is a finite B-algebra.

Then

$$\text{a)} \iff \text{b)} \implies \text{c)} \iff \text{d)}.$$

Moreover, all four properties hold in either of the following cases:

(i) *A is a quotient of a regular ring.*
(ii) *A is universally catenary and universally Japanese.*

Proof. Let $B = A/\mathfrak{q}$, where $\mathfrak{q}$ is a prime ideal of A, so that $Y = \operatorname{Spec}(B) = V(\mathfrak{q})$ is a closed subset of X. Put $Z = Z^{(2)}(Y)$, viewed as a subset of X stable under specialization. Since B is a domain, $\operatorname{Ass}(A/\mathfrak{q})$ consists only of the generic point $\mathfrak{q}$ of Y. If b) holds, apply it to

$$\mathscr{F} = (A/\mathfrak{q})^\sim$$

and to Z. Formula (5.9.3.1) then shows that $B^{(1)}$ is a finite A-module, and hence a finite B-module. Thus b) implies a).

Conversely, suppose a) holds. Let $\mathscr{F}$ and Z satisfy the assumptions in b), and use the notation of (5.11.1). Each $Y_\alpha = \operatorname{Spec}(B_\alpha)$, where B_α is a quotient domain of A. Since $Z_\alpha \subset Z^{(2)}(Y_\alpha)$, condition a), (5.10.2), and the fact that $\operatorname{Ass}(\mathscr{O}_{Y_\alpha})$ consists of the generic point show that

$$\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})$$

is coherent. Property b) follows from (5.11.1).

To prove that c) implies d), argue as above with $\mathscr{G} = (A/\mathfrak{q})^\sim|U$ and $T = V(\mathfrak{J})$. Conversely, d) implies c) by applying the same componentwise argument and (5.11.1). Property c) is plainly a special case of b). Finally, (5.11.2), together with the definitions of universally catenary and universally Japanese rings, proves all four properties under either hypothesis (i) or (ii). □ **IV-2** | 126

Remarks (5.11.7). — (i) It is not known whether condition d) in (5.11.5) implies condition a) without an additional hypothesis on X. We shall see in (7.2.4) that it does when X is a local scheme. Likewise, when A is a Noetherian local ring, the four properties a), b), c), and d) of (5.11.6) are equivalent (7.2.4). It is not known whether this result extends to every Noetherian ring.

(ii) If A satisfies property a) of (5.11.6), then so does every localization $S^{-1}A$ and every finite A-algebra C. Indeed, every quotient domain of $S^{-1}A$ has the form

$$S^{-1}(A/\mathfrak{q}),$$

where $\mathfrak{q}$ is a prime ideal of A disjoint from S. On the other hand, if $\mathfrak{r}$ is a prime ideal of C and $\mathfrak{p}$ is its inverse image in A, then $C/\mathfrak{r}$ is a finite integral $(A/\mathfrak{p})$-algebra containing $A/\mathfrak{p}$. The assertions therefore follow from (5.10.17, iii) and (5.10.17, iv).

5.12. Relations between the properties of a Noetherian local ring A and of a quotient ring A/tA IV-2 | 126
We have already seen in (3.4) relations between the properties of A and those of A/tA concerning associated prime ideals, as well as the properties of being a domain or being reduced. In this subsection we give further relations between the properties of these rings, involving the notions of dimension and depth.

Proposition (5.12.1). — *Let A be a Noetherian local ring, put $X = \mathrm{Spec}(A)$, and let t be an element of A belonging to a system of parameters* ($\mathbf{0}$, 16.3.6). *Let X_0 be the closed subspace $V(t)$ of X, and let $X_1 = X - X_0$. Let Z be a subset of X stable under specialization. Suppose that X is biequidimensional (as is the case when A is equidimensional and is a quotient of a regular local ring* ($\mathbf{0}$, 16.5.12)*). If $Z_0 = Z \cap X_0$ and $Z_1 = Z \cap X_1$, then*

$$\mathrm{codim}(Z_0, X_0) \leqslant \mathrm{codim}(Z, X) \leqslant \mathrm{codim}(Z_1, X_1). \tag{5.12.1.1}$$

Proof. The second inequality follows from the definition (5.1.3). For the first, it is enough to treat the case in which Z is closed. Indeed, by hypothesis Z is a union of closed subsets Z_α, and inequalities

$$\mathrm{codim}(X_0 \cap Z_\alpha, X_0) \leqslant \mathrm{codim}(Z_\alpha, X)$$

for all α give

$$\begin{aligned}\mathrm{codim}(Z_0, X_0) &= \inf_\alpha \mathrm{codim}(X_0 \cap Z_\alpha, X_0) \\ &\leqslant \inf_\alpha \mathrm{codim}(Z_\alpha, X) = \mathrm{codim}(Z, X).\end{aligned}$$

Suppose therefore that Z is closed. Then

$$\mathrm{codim}(Z, X) = \dim(X) - \dim(Z)$$

by ($\mathbf{0}$, 14.3.5.1).

Moreover, X_0 is catenary and equidimensional, and $\dim(X_0) = \dim(X) - 1$ by ($\mathbf{0}$, 16.3.4); hence IV-2 | 127

$$\mathrm{codim}(Z_0, X_0) = \dim(X) - 1 - \dim(Z_0).$$

Since $\dim(Z_0) \geqslant \dim(Z) - 1$ ($\mathbf{0}$, 16.3.4), the first inequality in (5.12.1.1) follows. ☐

Proposition (5.12.2). — *Let A be a catenary Noetherian local ring, let M be a finite A-module, let t be an M-regular element belonging to the maximal ideal $\mathfrak{m}$ of A, and let $k \geqslant 1$. If M/tM is equidimensional and satisfies (S_k), then the same is true of M.*

Proof. In view of ($\mathbf{Err}_{\mathrm{III}}$, 30) and (5.7.3, vi), we may reduce to the case

$$\mathrm{Supp}(M) = \mathrm{Spec}(A) = X.$$

Put $X_0 = V(t) = \mathrm{Spec}(A/tA)$. Since M/tM satisfies (S_1), it has no embedded associated prime ideals (5.7.5). Applying (3.4.4) at the closed point of X (and of every closed subprescheme of X), we see that the irreducible components of $\mathrm{Supp}(M/tM)$ are exactly the irreducible components of $Y_i \cap X_0$, where Y_i $(1 \leqslant i \leqslant r)$ are the irreducible components of X.

Because t is M-regular, $V(t)$ contains no maximal point of X (3.1.8). Every irreducible component of $Y_i \cap X_0$ therefore has codimension 1 in Y_i (5.1.8). Since X is catenary and all irreducible components of $\mathrm{Supp}(M/tM)$ have the same dimension, the Y_i all have the same dimension. Thus X is equidimensional, and hence biequidimensional because A is local and catenary.

It remains to prove (S_k). By the criterion (5.7.4), with its notation, we must show that

$$\mathrm{codim}(Z_n, X) > n + k$$

for every integer $n \geqslant 0$. The hypothesis on M/tM and the same criterion give

$$\mathrm{codim}(Z_n \cap X_0, X_0) > n + k.$$

Every specialization of a point of Z_n again belongs to Z_n, as will be proved in (6.11.5).[9] Finally, t belongs to a system of parameters for M ($\mathbf{0}$, 16.4.1), and hence also to one for A, because the annihilator of M is nilpotent under the hypothesis $\mathrm{Supp}(M) = \mathrm{Spec}(A)$. The required inequality now follows from (5.12.1). ☐

[9]The reader may verify that Proposition 5.12.2 is not used in the proof of (6.11.5).

Remark (5.12.3). — If in Proposition 5.12.2 one assumes only that M/tM is equidimensional, it need not follow that M is equidimensional. For example, let k be a field, let $B = k[T, U, V]$, let C be the local ring of B at the maximal ideal $BT + BU + BV$, and put

$$\mathfrak{p} = CU, \qquad \mathfrak{q} = CV + C(T + U), \qquad A = C/\mathfrak{p}\mathfrak{q}.$$

Geometrically, A is the local ring, at their intersection point x, of the closed subscheme of affine 3-space formed by a plane and a line meeting that plane at x. Let t, u, v be the images of T, U, V in A. The element t is not a zero divisor in A, and

$$A/tA \simeq C_0/\mathfrak{p}_0\mathfrak{q}_0,$$

where C_0 is the local ring of $B_0 = k[U, V]$ at the maximal ideal $B_0U + B_0V$, while

$$\mathfrak{p}_0 = C_0U, \qquad \mathfrak{q}_0 = C_0U + C_0V.$$

Thus $\mathfrak{q}_0$ is the maximal ideal of C_0. It is immediate that $\operatorname{Spec}(A/tA)$ is irreducible of dimension 1, that A/tA does not satisfy (S_1), and that $\operatorname{Spec}(A)$ is not equidimensional.

Corollary (5.12.4). — *Let A be a catenary Noetherian local ring, let t be a regular element belonging to the maximal ideal of A, and let $k \geqslant 1$. If A/tA satisfies (S_k), then so does A.*

Proof. For $k = 1$ this follows from (3.4.4), in view of the interpretation of (S_1) in (5.7.5). For $k \geqslant 2$, Hartshorne's criterion (5.10.9) shows that A/tA is equidimensional, and Proposition 5.12.2 applies. □

Proposition (5.12.5). — *Let A be a catenary Noetherian local ring, let t be a regular element of A belonging to its maximal ideal, and let $k \geqslant 0$. If A/tA is reduced and equidimensional and satisfies (R_k), then A has these same three properties.*

Proof. We already know from (3.4.6) that A is reduced, and Proposition 5.12.2, applied with $k = 1$, shows that A is equidimensional. Put

$$X = \operatorname{Spec}(A), \qquad X_0 = V(t) = \operatorname{Spec}(A/tA),$$

and let Z be the set of points at which X is not regular. Every specialization of a point of Z belongs to Z (**0**, 17.3.2). On the other hand, if $x \in X_0$ and X_0 is regular at x, then X is regular at x (**0**, 17.1.8). Consequently the nonregular locus Z'_0 of X_0 contains $Z_0 = Z \cap X_0$. The hypothesis gives

$$k < \operatorname{codim}(Z'_0, X_0) \leqslant \operatorname{codim}(Z_0, X_0).$$

Since A is equidimensional and catenary, Proposition 5.12.1 gives

$$\operatorname{codim}(Z_0, X_0) \leqslant \operatorname{codim}(Z, X).$$

Thus X satisfies (R_k). □

Remark (5.12.6). — If in Proposition 5.12.5 one assumes only that A/tA is reduced and satisfies (R_k), it need not follow that A satisfies (R_k). For example, let k be a field and let $P(U, V) \in k[U, V]$ be irreducible, with the curve Γ defined by (P) in $\operatorname{Spec}(k[U, V])$ singular at the maximal ideal $(U) + (V)$. One may take, for example,

$$P(U, V) = U(U^2 + V^2) + (U^2 - V^2),$$

which defines a cubic with a double point. Let $B = k[T, U, V, W]$, let C be the local ring of B at the maximal ideal $BT + BU + BV + BW$, and put

$$\mathfrak{p} = CW + CP(U - T, V), \qquad \mathfrak{q} = CU.$$

Consider the local ring $A = C/\mathfrak{p}\mathfrak{q}$. Geometrically, the corresponding closed subscheme X of affine 4-space is the union of a hyperplane H and a 2-dimensional "cylinder" L with base Γ; the "singular line" Y of L is not contained in H, and A is the local ring of X at the point $x = Y \cap H$.

Let t be the image of T in A. The scheme $\operatorname{Spec}(A/tA)$ has x as its only singular point; its local ring there is A/tA, which is reduced and has dimension 2. By contrast, the generic point y of Y, defined by the image in A of

$$C(U - T) + CV + CW,$$

is a singular point of X, and $\mathscr{O}_{X,y}$ has dimension 1. Thus A/tA is reduced and satisfies (R_1), whereas A does not satisfy (R_1).

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Corollary (5.12.7). — *Let A be a catenary Noetherian local ring and let t be a regular element of A belonging to its maximal ideal. If A/tA is a domain and is integrally closed, then so is A.*

Proof. By Serre's criterion (5.8.6) and the fact that A is local, it is enough to prove that $\mathrm{Spec}(A)$ satisfies (S_2) and (R_1). The hypothesis on A/tA implies (R_1) for A by Proposition 5.12.5 and (S_2) for A by Corollary 5.12.4. ☐

Proposition (5.12.8). — (Hironaka's lemma). *Let A be a reduced, equidimensional, catenary Noetherian local ring. Suppose in addition that, for every minimal prime ideal $\mathfrak{q}_i$ of A, the ring $B_i = A/\mathfrak{q}_i$ has the property that $B_i^{(1)}$ (notation of (5.10.17)) is a finite B_i-algebra. This last condition holds, for example, when A has either property* (i) *or property* (ii) *of (5.11.6).*

Let $t \in A$ be a non-zero-divisor satisfying the following conditions:

(i) *The A-module A/tA has a unique nonembedded associated prime ideal $\mathfrak{p}$. (It necessarily has height 1 by the Hauptidealsatz* (**0**, 16.3.2)*, but it is not assumed to be the only associated prime of A/tA.)*

(ii) *The ideal $\mathfrak{p}A_\mathfrak{p}$ is generated by $t/1$.*

(iii) *The ring $A/\mathfrak{p}$ is integrally closed.*

Then A is a domain and is integrally closed, and $\mathfrak{p} = tA$.

Proof. Put $X = \mathrm{Spec}(A)$ and $Z = V(tA)$. Condition (i) says that Z is irreducible, with generic point corresponding to $\mathfrak{p}$. By (ii), the maximal ideal of the Noetherian local ring $A_\mathfrak{p}$ is generated by one element. Thus $A_\mathfrak{p}$ is a discrete valuation ring with uniformizer $t/1$ (Bourbaki, *Algèbre commutative*, chap. VI, §3, no. 6, prop. 9), and $A_\mathfrak{p}/tA_\mathfrak{p}$, its residue field, has length 1 as an $A_\mathfrak{p}$-module.

It follows from (3.4.2) that Z is contained in only one irreducible component, say X_i, of X. For every other irreducible component X_j one consequently has

$$\dim(X_j \cap Z) < \dim(Z) = \dim(X_i \cap Z).$$

Since t is not a zero divisor, it belongs to none of the minimal prime ideals $\mathfrak{q}_j$, and hence

$$\operatorname{codim}(X_i \cap Z, X_i) = \operatorname{codim}(X_j \cap Z, X_j) = 1$$

for every $j \neq i$ (5.1.8). But A is biequidimensional; the displayed strict inequality would therefore force $\dim(X_i) \neq \dim(X_j)$ (**0**, 14.3.3.1), a contradiction. Hence A has only one minimal prime. Since it is reduced, A is a domain.

Let $A^{(1)}$ have the meaning of (5.10.17). It is a finite A-algebra and therefore a Noetherian semilocal ring. Every nonembedded associated prime $\mathfrak{r}$ of $A^{(1)}/tA^{(1)}$ has height 1 by the Hauptidealsatz. Its contraction $\mathfrak{r} \cap A$ also has height 1 (5.10.17, iv), and, since it contains tA, condition (i) forces

$$\mathfrak{r} \cap A = \mathfrak{p}.$$

Let A' be the integral closure of A and put $S = A - \mathfrak{p}$. The integral closure of $A_\mathfrak{p}$ is $S^{-1}A'$ (Bourbaki, *Algèbre commutative*, chap. V, §1, no. 5, prop. 17); it equals $A_\mathfrak{p}$ because this ring is a discrete valuation ring. Consequently there is only one prime $\mathfrak{p}'$ of A' above $\mathfrak{p}$ (*loc. cit.*, §2, no. 1, lemma 1), and therefore only one prime $\mathfrak{p}^{(1)}$ of $A^{(1)}$ above $\mathfrak{p}$. Moreover,

$$A_\mathfrak{p} = A'_{\mathfrak{p}'} = A^{(1)}_{\mathfrak{p}^{(1)}}.$$

The ring $A^{(1)}$ satisfies (S_2) (5.10.17, i). Since t is a non-zero-divisor on $A^{(1)}$, the quotient **IV-2 | 130**

$$A^{(1)}/tA^{(1)}$$

has no embedded associated primes (5.7.7). Thus $\mathfrak{p}^{(1)}$ is its only associated prime, so $tA^{(1)}$ is $\mathfrak{p}^{(1)}$-primary. Equivalently, $tA^{(1)}$ is the inverse image in $A^{(1)}$ of

$$tA^{(1)}_{\mathfrak{p}^{(1)}} = tA_\mathfrak{p},$$

which is the maximal ideal of $A_\mathfrak{p}$ by (ii). It follows that

$$tA^{(1)} = \mathfrak{p}^{(1)}$$

is prime in $A^{(1)}$.

The domain $A^{(1)}/\mathfrak{p}^{(1)}$ is finite over $A/\mathfrak{p}$ and has the same fraction field, namely the residue field of $A^{(1)}_{\mathfrak{p}^{(1)}} = A_{\mathfrak{p}}$. Condition (iii) therefore gives

$$A^{(1)}/\mathfrak{p}^{(1)} = A/\mathfrak{p}.$$

Hence

$$A^{(1)} = A + \mathfrak{p}^{(1)} = A + tA^{(1)}.$$

Since t lies in the Jacobson radical of the Noetherian semilocal ring $A^{(1)}$, Nakayama's lemma yields $A^{(1)} = A$. Thus

$$\mathfrak{p}^{(1)} = \mathfrak{p} = tA.$$

Finally, A is catenary and $A/tA = A/\mathfrak{p}$ is a domain and integrally closed by (iii). Corollary 5.12.7 shows that A is integrally closed. □

Remark (5.12.9). — The conclusion of Hironaka's lemma fails without the hypothesis that A is equidimensional. Let k be a field, put

$$B = k[X,Y,Z], \qquad \mathfrak{r}_1 = BZ, \qquad \mathfrak{r}_2 = BX + BY, \qquad C = B/\mathfrak{r}_1\mathfrak{r}_2.$$

Let $\mathfrak{m} = BX + BY + BZ$, let $\mathfrak{n} = \mathfrak{m}/\mathfrak{r}_1\mathfrak{r}_2$ be its image in C, and let $A = C_{\mathfrak{n}}$. Finally, let t_0 be the image of $Y + Z$ in C, and let t be its image in A. Thus $\operatorname{Spec}(C)$ is the union of a plane P and a line D not contained in P, while $\operatorname{Spec}(C/t_0C)$ is a line D' contained in P and passing through $D \cap P$.

The ring A is reduced and catenary, since it is a quotient of a regular ring, and its minimal prime ideals $\mathfrak{q}_1$ and $\mathfrak{q}_2$ are the images of $\mathfrak{r}_1$ and $\mathfrak{r}_2$. The A-module A/tA has a unique nonembedded associated prime $\mathfrak{p}$, the image of $BY + BZ$; the ideal $\mathfrak{p}A_{\mathfrak{p}}$ is generated by $t/1$, and $A/\mathfrak{p} \simeq k[X]$. Nevertheless, A is not a domain.

Corollary (5.12.10). — *Let A be a Noetherian local domain satisfying one of the following hypotheses:*

a) *A is a quotient of a regular ring;*
b) *A is universally catenary and universally Japanese.*

Let $(x_i)_{1\leqslant i\leqslant n}$ be a sequence of n elements of A, and put

$$\mathfrak{J} = x_1A + \cdots + x_nA.$$

Suppose that:

(i) *$A/\mathfrak{J}$ has a unique nonembedded associated prime $\mathfrak{p}$, and $\mathfrak{p}$ has height n;*
(ii) *the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ is generated by the $x_i/1$ (so $A_{\mathfrak{p}}$ is a regular local ring of dimension n);*
(iii) *$A/\mathfrak{p}$ is integrally closed.*

Then, for every i with $0 \leqslant i \leqslant n$, the quotient

$$A/(x_1A + \cdots + x_iA)$$

is integrally closed and has dimension $\dim(A) - i$. In particular, $\mathfrak{J}$ is prime and equals $\mathfrak{p}$, the ring A is integrally closed, and $(x_i)_{1\leqslant i\leqslant n}$ is an A-regular sequence.

Proof. We argue by induction on n. The assertion is trivial for $n = 0$ and is Hironaka's lemma for $n = 1$, so suppose $n \geqslant 2$. Put

$$\mathfrak{J}' = x_1A + \cdots + x_{n-1}A,$$

and let $\mathfrak{q}$ be a prime minimal among those containing $\mathfrak{J}'$. Then $\operatorname{ht}(\mathfrak{q}) \leqslant n-1$ (**0**, 16.3.1). We first show that $\mathfrak{q} \subset \mathfrak{p}$. If $\mathfrak{p}'$ is minimal among the primes containing $\mathfrak{q} + x_nA$, then $\mathfrak{p}'/\mathfrak{q}$ has height at most 1 by the Hauptidealsatz (**0**, 16.3.2); because A is catenary, $\operatorname{ht}(\mathfrak{p}') \leqslant n$. But $\mathfrak{p}'$ contains $\mathfrak{J}$, and the unique minimal prime of $A/\mathfrak{J}$ is $\mathfrak{p}$. Hence $\mathfrak{p}' = \mathfrak{p}$, and $\mathfrak{q} \subset \mathfrak{p}$.

The prime $\mathfrak{q}$ is therefore induced from a prime of $A_{\mathfrak{p}}$ minimal over $\mathfrak{J}'A_{\mathfrak{p}}$. By (ii) and (**0**, 17.1.7), **IV-2** | 131
the ideal $\mathfrak{J}'A_{\mathfrak{p}}$ is prime. Consequently

$$\mathfrak{q} = A \cap \mathfrak{J}'A_{\mathfrak{p}}$$

is the unique nonembedded associated prime of $A/\mathfrak{J}'$, and has height $n-1$. Moreover,

$$\mathfrak{q}A_{\mathfrak{q}} = \mathfrak{J}'A_{\mathfrak{q}},$$

and, since $A_{\mathfrak{q}} = (A_{\mathfrak{p}})_{\mathfrak{q}A_{\mathfrak{p}}}$, the maximal ideal of $A_{\mathfrak{q}}$ is generated by the images of $x_1, \dots, x_{n-1}$. Thus the shortened sequence already satisfies conditions (i) and (ii).

We verify condition (iii) for it. Put $B = A/\mathfrak{q}$, let $\mathfrak{n} = \mathfrak{p}/\mathfrak{q}$, and let t be the image of x_n in B. Property a) (respectively b)) passes from A to B (5.6.3, iv), and $t \neq 0$ because $x_n \notin \mathfrak{q}$. A prime of B minimal over t comes from a prime of A minimal over $\mathfrak{q} + x_n A$; the only such prime is $\mathfrak{p}$. Hence $\mathfrak{n}$ is the unique nonembedded associated prime of B/tB. Furthermore,

$$B_{\mathfrak{n}} = A_{\mathfrak{p}}/\mathfrak{q}A_{\mathfrak{p}}, \qquad \mathfrak{n}B_{\mathfrak{n}} = \mathfrak{p}A_{\mathfrak{p}}/\mathfrak{q}A_{\mathfrak{p}}.$$

Since $\mathfrak{q}A_{\mathfrak{p}} = \mathfrak{J}'A_{\mathfrak{p}}$ and $\mathfrak{p}A_{\mathfrak{p}}$ is generated by $x_1, \dots, x_n$, the ideal $\mathfrak{n}B_{\mathfrak{n}}$ is generated by t. Finally, $B/\mathfrak{n} = A/\mathfrak{p}$ is integrally closed. Hironaka's lemma therefore gives

$$B = A/\mathfrak{q} \text{ integrally closed}, \qquad \mathfrak{p} = \mathfrak{q} + x_n A.$$

The induction hypothesis applied to $x_1, \dots, x_{n-1}$ now shows that $A/(x_1 A + \dots + x_i A)$ is integrally closed and has dimension $\dim(A) - i$ for $0 \leqslant i \leqslant n-1$, and that $\mathfrak{q} = \mathfrak{J}'$. Hence

$$\mathfrak{p} = \mathfrak{J}' + x_n A = \mathfrak{J},$$

so $A/\mathfrak{J} = A/\mathfrak{p}$ is integrally closed. Finally, $\dim(A_{\mathfrak{p}}) = n$, and biequidimensionality of A gives

$$\dim(A/\mathfrak{p}) = \dim(A) - \dim(A_{\mathfrak{p}}) = \dim(A) - n.$$

This completes the induction. □

5.13. Properties preserved under passage to a direct limit Throughout this subsection, $(A_\alpha, \varphi_{\beta\alpha})$ denotes a direct system of rings whose index set is filtered and preordered. We put **IV-2** | 131

$$A = \varinjlim_{\alpha} A_\alpha$$

and denote by $\varphi_\alpha : A_\alpha \to A$ the canonical homomorphism.

Proposition (5.13.1). — (i) *For every index α, let $\mathfrak{a}_\alpha$ be an ideal of A_α such that*

$$\varphi_{\beta\alpha}(\mathfrak{a}_\alpha) \subset \mathfrak{a}_\beta \qquad (\alpha \leqslant \beta).$$

Then the $\mathfrak{a}_\alpha$, with the restrictions of the $\varphi_{\beta\alpha}$, form a direct system; $\mathfrak{a} = \varinjlim \mathfrak{a}_\alpha$ is canonically identified with an ideal of A, and

$$A/\mathfrak{a} \simeq \varinjlim_{\alpha} (A_\alpha/\mathfrak{a}_\alpha).$$

If, in addition,

$$\mathfrak{a}_\alpha = \varphi_{\beta\alpha}^{-1}(\mathfrak{a}_\beta) \qquad (\alpha \leqslant \beta),$$

then

$$\mathfrak{a}_\alpha = \varphi_\alpha^{-1}(\mathfrak{a}) \qquad \textit{for every } \alpha.$$

(ii) *Conversely, for every ideal $\mathfrak{a}$ of A, if*

$$\mathfrak{a}_\alpha = \varphi_\alpha^{-1}(\mathfrak{a}) \qquad \textit{for every } \alpha,$$

then the $\mathfrak{a}_\alpha$ form a direct system and

$$\mathfrak{a} = \varinjlim_{\alpha} \mathfrak{a}_\alpha.$$

Proof. Assertion (ii) is plainly a special case of (i). In (i), the canonical identification of $\mathfrak{a}$ with an ideal of A, and that of $A/\mathfrak{a}$ with $\varinjlim(A_\alpha/\mathfrak{a}_\alpha)$, follow from the exactness of the direct-limit functor in the category of abelian groups.

Moreover, $\mathfrak{a}$ is the union of the increasing family of images $\varphi_\alpha(\mathfrak{a}_\alpha)$. Thus, if $x_\alpha \in A_\alpha$ and $\varphi_\alpha(x_\alpha) \in \mathfrak{a}$, there are $\beta \geqslant \alpha$ and $y_\beta \in \mathfrak{a}_\beta$ such that

$$\varphi_\alpha(x_\alpha) = \varphi_\beta(y_\beta).$$

For some $\gamma \geqslant \beta$ one then has

$$\varphi_{\gamma\alpha}(x_\alpha) = \varphi_{\gamma\beta}(y_\beta) \in \mathfrak{a}_\gamma.$$

Under the additional inverse-image hypothesis this implies $x_\alpha \in \mathfrak{a}_\alpha$, and proves the last assertion of (i). □

Corollary (5.13.2). — *Let $\mathfrak{N}_\alpha$ be the nilradical of A_α. The nilradical of A is canonically identified with $\varinjlim \mathfrak{N}_\alpha$, and*

$$A_{\mathrm{red}} \simeq \varinjlim_\alpha (A_\alpha)_{\mathrm{red}}.$$

In particular, if every A_α is reduced, then A is reduced.

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Proof. It is clear that

$$\varphi_{\beta\alpha}(\mathfrak{N}_\alpha) \subset \mathfrak{N}_\beta \qquad (\alpha \leqslant \beta),$$

and that $\varinjlim \mathfrak{N}_\alpha$ is a nil ideal of A, hence is contained in the nilradical $\mathfrak{N}$ of A. Conversely, if $x \in \mathfrak{N}$, then $x^h = 0$ for some integer h. Write $x = \varphi_\alpha(x_\alpha)$ with $x_\alpha \in A_\alpha$. The relation $\varphi_\alpha(x_\alpha^h) = 0$ implies that for some $\beta \geqslant \alpha$,

$$\varphi_{\beta\alpha}(x_\alpha^h) = 0.$$

Putting $x_\beta = \varphi_{\beta\alpha}(x_\alpha)$, we obtain $x_\beta^h = 0$, hence $x_\beta \in \mathfrak{N}_\beta$, while $x = \varphi_\beta(x_\beta)$. This proves the assertion, in view of (5.13.1). □

Proposition (5.13.3). — (i) *For every index α, let $\mathfrak{p}_\alpha$ be a prime ideal of A_α such that*

$$\varphi_{\beta\alpha}(\mathfrak{p}_\alpha) \subset \mathfrak{p}_\beta \qquad (\alpha \leqslant \beta).$$

Then $\mathfrak{p} = \varinjlim \mathfrak{p}_\alpha$ is a prime ideal of A. In particular, if all the A_α are domains, then A is a domain.

(ii) *If, in addition,*

$$\mathfrak{p}_\alpha = \varphi_{\beta\alpha}^{-1}(\mathfrak{p}_\beta) \qquad (\alpha \leqslant \beta),$$

then there is a canonical identification

$$A_\mathfrak{p} \simeq \varinjlim_\alpha (A_\alpha)_{\mathfrak{p}_\alpha}.$$

(iii) *Suppose that A is local, with maximal ideal $\mathfrak{p}$, and that the homomorphisms $\varphi_\alpha : A_\alpha \to A$ are injective. If $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, then*

$$A \simeq \varinjlim_\alpha (A_\alpha)_{\mathfrak{p}_\alpha},$$

and every canonical homomorphism $(A_\alpha)_{\mathfrak{p}_\alpha} \to A$ is injective.

Proof. The two assertions in (i) are equivalent, since

$$A/\mathfrak{p} \simeq \varinjlim_\alpha (A_\alpha/\mathfrak{p}_\alpha).$$

Suppose therefore that all the A_α are domains, and let $x, y \in A$ satisfy $xy = 0$. For some α one may write

$$x = \varphi_\alpha(x_\alpha), \qquad y = \varphi_\alpha(y_\alpha),$$

with $x_\alpha, y_\alpha \in A_\alpha$ and $\varphi_\alpha(x_\alpha y_\alpha) = 0$. Hence there is $\beta \geqslant \alpha$ such that, on putting

$$x_\beta = \varphi_{\beta\alpha}(x_\alpha), \qquad y_\beta = \varphi_{\beta\alpha}(y_\alpha),$$

one has $x_\beta y_\beta = 0$. Since A_β is a domain, $x_\beta = 0$ or $y_\beta = 0$, and therefore $x = 0$ or $y = 0$.

For (ii), the hypothesis says that the multiplicative subsets $A_\alpha - \mathfrak{p}_\alpha$ form a direct system whose direct limit is $A - \mathfrak{p}$. The local rings $(A_\alpha)_{\mathfrak{p}_\alpha}$ therefore form a direct system with local transition maps, and the canonical homomorphisms to $A_\mathfrak{p}$ induce a surjection

$$\psi : \varinjlim_\alpha (A_\alpha)_{\mathfrak{p}_\alpha} \longrightarrow A_\mathfrak{p}.$$

To prove injectivity, let $x_\alpha \in A_\alpha$ and $s_\alpha \in A_\alpha - \mathfrak{p}_\alpha$, and suppose that

$$\frac{\varphi_\alpha(x_\alpha)}{\varphi_\alpha(s_\alpha)} = 0 \quad \text{in } A_\mathfrak{p}.$$

There is $s' \in A - \mathfrak{p}$ such that $s'\varphi_\alpha(x_\alpha) = 0$. After replacing α by a larger index, we may write $s' = \varphi_\alpha(s'_\alpha)$ with $s'_\alpha \in A_\alpha - \mathfrak{p}_\alpha$. Then $\varphi_\alpha(s'_\alpha x_\alpha) = 0$, so for some $\beta \geqslant \alpha$,

$$\varphi_{\beta\alpha}(s'_\alpha x_\alpha) = 0 \quad \text{in } A_\beta.$$

It follows that the image of x_α/s_α in the direct limit is zero.

Finally, the first assertion of (iii) follows from (ii) and $A_{\mathfrak{p}} = A$. Every element of $A_\alpha - \mathfrak{p}_\alpha$ maps to a unit of A, and the injectivity of $(A_\alpha)_{\mathfrak{p}_\alpha} \to A$ follows from that of φ_α. ☐

Corollary (5.13.4). — *Suppose that all the A_α are domains and that all the transition homomorphisms $\varphi_{\beta\alpha}$ are injective. If A'_α denotes the integral closure of A_α, then*

$$A' = \varinjlim_\alpha A'_\alpha$$

is the integral closure of A. In particular, if every A_α is integrally closed, then A is integrally closed.

If the A_α are unibranch local rings (respectively, geometrically unibranch local rings) (**0**, 23.2.1)*, and the $\varphi_{\beta\alpha}$ are injective local homomorphisms, then A is unibranch (respectively, geometrically unibranch).*

Proof. Let K_α be the fraction field of A_α. By (5.13.3, ii), applied to $\mathfrak{p}_\alpha = (0)$,

$$K = \varinjlim_\alpha K_\alpha$$

is the fraction field of A. Thus $A' = \varinjlim A'_\alpha$ is identified with a subring of K which is plainly integral over A. **IV-2** | 133

Conversely, let $z \in K$ satisfy an equation of integral dependence

$$z^n + \sum_{i=1}^{n} c_i z^{n-i} = 0, \qquad c_i \in A.$$

There are an index α, an element $z_\alpha \in K_\alpha$, and elements $c_{i\alpha} \in A_\alpha$ whose images are z and the c_i, respectively, and such that the same equation holds in K_α. Hence $z_\alpha \in A'_\alpha$, so $z \in A'$. This also proves at once that a direct limit of the integrally closed rings occurring here is integrally closed.

Suppose now that the A_α are local and that the transition maps are local. Then A is local and, if k_α denotes the residue field of A_α,

$$k = \varinjlim_\alpha k_\alpha$$

is the residue field of A (**0**, 10.3.1.3). If every A'_α is local, then A' is local, and hence A is unibranch. If, moreover, the residue field k'_α of A'_α is a radicial extension of k_α for every α, then

$$k' = \varinjlim_\alpha k'_\alpha,$$

the residue field of A', is a radicial extension of k. Thus A is geometrically unibranch. ☐

(5.13.5). A ring A is called *normal* if $\operatorname{Spec}(A)$ is normal; in other words, if for every prime ideal $\mathfrak{p}$ of A, the ring $A_{\mathfrak{p}}$ is a domain and is integrally closed.

This implies that A is reduced and that $\mathfrak{p}$ can contain only one minimal prime ideal $\mathfrak{q}$ of A; in other words, $\mathfrak{p}$ belongs to only one irreducible component of $\operatorname{Spec}(A)$. Moreover, $\mathfrak{q}$ is the inverse image of (0) under the canonical homomorphism $A \to A_{\mathfrak{p}}$, and hence

$$A_{\mathfrak{p}} \simeq (A/\mathfrak{q})_{\mathfrak{p}/\mathfrak{q}}.$$

It follows that the domain $A/\mathfrak{q}$ is integrally closed, being the intersection of integrally closed rings (**I**, 8.2.1.1).

If A is Noetherian and normal, it is the direct product of finitely many integrally closed domains (**I**, 5.1.4).

Corollary (5.13.6). — *Suppose that all the A_α are normal and that, for $\alpha \leqslant \beta$, every irreducible component of* $\operatorname{Spec}(A_\beta)$ *dominates an irreducible component of* $\operatorname{Spec}(A_\alpha)$*. Then*

$$A = \varinjlim_\alpha A_\alpha$$

is normal.

Proof. Let $\mathfrak{p}$ be a prime ideal of A, and put

$$\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$$

for every α. By hypothesis, $\mathfrak{p}_\alpha$ contains a unique minimal prime ideal $\mathfrak{q}_\alpha$ of A_α. If $\alpha \leqslant \beta$, then $\varphi_{\beta\alpha}^{-1}(\mathfrak{q}_\beta)$ is a minimal prime ideal of A_α contained in $\mathfrak{p}_\alpha$, hence equals $\mathfrak{q}_\alpha$.

Put

$$B_\alpha = A_\alpha / \mathfrak{q}_\alpha.$$

The domains B_α form a direct system under the injective homomorphisms

$$\varphi'_{\beta\alpha} : B_\alpha \longrightarrow B_\beta$$

induced by the $\varphi_{\beta\alpha}$. Each B_α is integrally closed, so

$$B = \varinjlim_\alpha B_\alpha$$

is integrally closed by (5.13.4). If

$$\mathfrak{p}'_\alpha = \mathfrak{p}_\alpha / \mathfrak{q}_\alpha,$$

then

$$(B_\alpha)_{\mathfrak{p}'_\alpha} = (A_\alpha)_{\mathfrak{p}_\alpha}.$$

Consequently,

$$\varinjlim_\alpha (B_\alpha)_{\mathfrak{p}'_\alpha} = \varinjlim_\alpha (A_\alpha)_{\mathfrak{p}_\alpha} = A_\mathfrak{p}$$

is a localization of B at a prime ideal (5.13.3). It is therefore a domain and is integrally closed, proving the corollary. □

Proposition (5.13.7). — *Suppose that every A_α is regular and that, for $\alpha \leqslant \beta$, the homomorphism $\varphi_{\beta\alpha}$ makes A_β a flat A_α-module. Suppose in addition that*

$$A = \varinjlim_\alpha A_\alpha$$

is Noetherian. Then A is regular.

Proof. By definition, it is enough to prove that the Noetherian local ring $A_\mathfrak{p}$ is regular for every prime ideal $\mathfrak{p}$ of A. Put

$$\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p}).$$

By (5.13.3),

$$\mathfrak{p} = \varinjlim_\alpha \mathfrak{p}_\alpha, \qquad A_\mathfrak{p} = \varinjlim_\alpha (A_\alpha)_{\mathfrak{p}_\alpha}.$$

The rings $(A_\alpha)_{\mathfrak{p}_\alpha}$ are regular, and for $\alpha \leqslant \beta$ the ring $(A_\beta)_{\mathfrak{p}_\beta}$ is flat over $(A_\alpha)_{\mathfrak{p}_\alpha}$ (**0**, 6.3.2). We are therefore reduced to the case in which all the A_α, as well as A, are local.

To prove that A is regular, it is enough to show that for any two finite A-modules M and N, there is an integer i_0 such that

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$$\operatorname{Tor}_i^A(M, N) = 0 \qquad (i \geqslant i_0)$$

(**0**, 17.3.1) and (**0**, 17.2.6). We use the following lemma, in which the rings A_α are again arbitrary. □

Lemma (5.13.7.1). — *For every finitely presented A-module M, there are an index α and a finitely presented A_α-module M_α such that*

$$M \simeq M_\alpha \otimes_{A_\alpha} A.$$

Proof. We may write $M = A^r / P$, where P is a finitely generated A-submodule of A^r. Since

$$A^r = \varinjlim_\alpha A_\alpha^r$$

and P is finitely generated, there are an index α and a finitely generated A_α-submodule $P_\alpha \subset A_\alpha^r$ whose canonical image generates P. Right exactness of the tensor product then gives

$$M \simeq (A_\alpha^r / P_\alpha) \otimes_{A_\alpha} A.$$

□

Completion of the proof of Proposition 5.13.7. Since A is Noetherian, the finite modules M and N are finitely presented. After increasing α if necessary, the lemma gives

$$M = M_\alpha \otimes_{A_\alpha} A, \qquad N = N_\alpha \otimes_{A_\alpha} A.$$

The A_α-module A is flat (**0**, 6.1.2), and base change for Tor gives

$$\operatorname{Tor}_i^A(M, N) \simeq \operatorname{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha) \otimes_{A_\alpha} A$$

(**III**, 6.3.9). Since A_α is regular, the right-hand side vanishes for all sufficiently large i. This proves the proposition. □

Corollary (5.13.8). — *Let $k \geqslant 0$. Suppose that the A_α are Noetherian and satisfy (R_k). Suppose also that, for $\alpha \leqslant \beta$, the homomorphism $\varphi_{\beta\alpha}$ makes A_β a flat A_α-module, that $A = \varinjlim A_\alpha$ is Noetherian, and that for every α the induced map*

$$^a\varphi_\alpha : \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(A_\alpha)$$

is a homeomorphism. Then A satisfies (R_k).

Proof. Let $\mathfrak{p}$ be a prime ideal of A, and put

$$\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p}).$$

Then

$$\mathfrak{p} = \varinjlim_\alpha \mathfrak{p}_\alpha, \qquad A_\mathfrak{p} = \varinjlim_\alpha (A_\alpha)_{\mathfrak{p}_\alpha}$$

by (5.13.3). It follows from (**I**, 2.4.2) and the hypothesis that

$$\operatorname{Spec}(A_\mathfrak{p}) \longrightarrow \operatorname{Spec}\big((A_\alpha)_{\mathfrak{p}_\alpha}\big)$$

is a surjective homeomorphism for every α. Hence

$$\dim(A_\mathfrak{p}) = \dim\big((A_\alpha)_{\mathfrak{p}_\alpha}\big) \qquad \text{for every } \alpha.$$

If $\dim(A_\mathfrak{p}) \leqslant k$, then each $(A_\alpha)_{\mathfrak{p}_\alpha}$ is regular. Moreover, for $\alpha \leqslant \beta$, $(A_\beta)_{\mathfrak{p}_\beta}$ is flat over $(A_\alpha)_{\mathfrak{p}_\alpha}$ (**0**, 6.3.2). Proposition (5.13.7) therefore shows that $A_\mathfrak{p}$ is regular. Thus A satisfies (R_k). □

§6. Flat morphisms of locally Noetherian preschemes

Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism. For every $y \in Y$, the fibre $f^{-1}(y) = X \times_Y \operatorname{Spec}(\boldsymbol{k}(y))$ is also a locally Noetherian prescheme, since it suffices to see this when $Y = \operatorname{Spec}(\mathrm{A})$ and $X = \operatorname{Spec}(\mathrm{B})$ are spectra of Noetherian rings, and then $\mathrm{B} \otimes_\mathrm{A} \boldsymbol{k}(y)$ is a ring of fractions of the quotient ring $\mathrm{B}/\mathfrak{J}_y\mathrm{B}$, hence is Noetherian. We propose first of all in this paragraph to relate the properties of X, of Y, and of the fibres $f^{-1}(y)$ (where y ranges over Y), under the hypothesis that the morphism f is *flat*. The questions treated will reduce to the study of relations between the Noetherian local rings $\mathscr{O}_y$, $\mathscr{O}_x$, and $\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$, the homomorphism $\mathscr{O}_y \to \mathscr{O}_x$ being local and making $\mathscr{O}_x$ a flat $\mathscr{O}_y$-module. In n^os^ 6.11 to 6.13 (rather separate from the rest of the paragraph, by their "absolute" rather than "relative" nature), we apply certain preceding results to find criteria allowing us to affirm that the *singular locus* (or certain analogous sets) of certain preschemes are closed sets; these criteria will play an important role in §7.

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6.1. Flatness and dimension

Proposition (6.1.1). — *Let* A, B *be two local Noetherian rings,* $\mathfrak{m}$ *the maximal ideal of* A, $k = \mathrm{A}/\mathfrak{m}$ *its residue field,* $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism. Suppose that for every prime ideal* $\mathfrak{p}$ *of* A*, distinct from* $\mathfrak{m}$*, and for every minimal element* $\mathfrak{q}$ *of the set of prime ideals of* B *containing* $\mathfrak{p}\mathrm{B}$, $\varphi^{-1}(\mathfrak{q})$ *is distinct from* $\mathfrak{m}$ *(in other words, no irreducible component of* $\mathrm{Spec}(\mathrm{B}/\mathfrak{p}\mathrm{B})$ *is contained in the inverse image of the closed point* $\mathfrak{m}$ *of* $\mathrm{Spec}(\mathrm{A})$ *in* $\mathrm{Spec}(\mathrm{B})$*). Then we have*

$$\dim(\mathrm{B}) = \dim(\mathrm{A}) + \dim(\mathrm{B} \otimes_{\mathrm{A}} k). \tag{6.1.1.1}$$

Proof. We argue by induction on $n = \dim(\mathrm{A})$; the assertion is trivial for $n = 0$, $\mathfrak{m}\mathrm{B}$ being then contained in the nilradical of B. We may therefore suppose $n > 0$. Let $\mathfrak{q}_i$ $(1 \leqslant i \leqslant m)$ be the minimal prime ideals of B, and let $\mathfrak{p}_i = \varphi^{-1}(\mathfrak{q}_i)$; for every i, we have $\mathfrak{p}_i \neq \mathfrak{m}$; otherwise (since $n > 0$) there would exist a prime ideal $\mathfrak{p} \neq \mathfrak{m}$ contained in $\mathfrak{p}_i$ and as $\mathfrak{q}_i$ is minimal among the prime ideals of B containing $\mathfrak{p}\mathrm{B}$, we would arrive at a contradiction with the hypothesis. Therefore $\mathfrak{m}$ is distinct from the union of the $\mathfrak{p}_i$ and of the minimal prime ideals $\mathfrak{p}'_j$ $(1 \leqslant j \leqslant r)$ of A (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2) and there exists an $a \in \mathfrak{m}$ not belonging to any of the $\mathfrak{p}_i$ nor any of the $\mathfrak{p}'_j$. Set $\mathrm{A}' = \mathrm{A}/a\mathrm{A}$, $\mathrm{B}' = \mathrm{B}/a\mathrm{B}$; we have (**0**, 16.3.4)

$$\dim(\mathrm{A}') = \dim(\mathrm{A}) - 1, \quad \dim(\mathrm{B}') = \dim(\mathrm{B}) - 1$$

by construction of a, and on the other hand $\mathrm{B}' \otimes_{\mathrm{A}'} k = \mathrm{B} \otimes_{\mathrm{A}} k$, hence

$$\dim(\mathrm{B}' \otimes_{\mathrm{A}'} k) = \dim(\mathrm{B} \otimes_{\mathrm{A}} k)$$

and it suffices to prove (6.1.1) for A' and B'; now, by virtue of the bijective correspondence between ideals of A (resp. B) containing a and ideals of A' (resp. B'), the hypotheses of the statement are also valid for A' and B'; we may therefore apply the induction hypothesis, which completes the proof. □

Corollary (6.1.2). — *Let* A, B *be two local Noetherian rings,* k *the residue field of* A, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* $\mathrm{M} \neq 0$ *an* A*-module of finite type,* $\mathrm{N} \neq 0$ *a* B*-module of finite type. If* N *is a flat* A*-module (resp. if* B *is a flat* A*-module), then we have*

$$\dim_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) = \dim_{\mathrm{A}}(\mathrm{M}) + \dim_{\mathrm{B} \otimes_{\mathrm{A}} k}(\mathrm{N} \otimes_{\mathrm{A}} k) \tag{6.1.2.1}$$

(*resp.* (6.1.1)).

Proof. It suffices to prove the assertion concerning N; on the other hand, if $\mathfrak{b}$ is the annihilator of N, we may replace B by $\mathrm{B}/\mathfrak{b}$, and therefore suppose that $\mathrm{Supp}(\mathrm{N}) = \mathrm{Spec}(\mathrm{B})$; the hypothesis then means that the morphism $f : \mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ corresponding to φ is *quasi-flat* (2.3.3); we conclude (2.3.4) that for every prime ideal $\mathfrak{p} \neq \mathfrak{m}$ of A, every irreducible component of $f^{-1}(\mathrm{V}(\mathfrak{p}))$ **IV-2** | 136
dominates $\mathrm{V}(\mathfrak{p})$, and therefore the condition of (6.1.1) is satisfied, whence the conclusion. □

Corollary (6.1.3). — *Let* A, B *be two local Noetherian rings,* $\mathfrak{m}$ *the maximal ideal of* A, k *its residue field,* $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism. Suppose that* $\dim(\mathrm{B} \otimes_{\mathrm{A}} k) = 0$ *(that is to say* (**0**, 16.2.1) *that* $\mathfrak{m}\mathrm{B}$ *is an* ideal of definition *of* B*). Then we have* $\dim(\mathrm{B}) \leqslant \dim(\mathrm{A})$*. If in addition there exists a* B*-module of finite type* N *that is a flat* A*-module and has support equal to* $\mathrm{Spec}(\mathrm{B})$ *(which holds in particular if* B *is a flat* A*-module), then* $\dim(\mathrm{B}) = \dim(\mathrm{A})$.

Proof. The first assertion follows from (5.5.1) and the second from (6.1.2). □

Corollary (6.1.4). — *Let* X, Y *be two locally Noetherian preschemes,* $f : X \to Y$ *a surjective morphism. For every closed subset* Z *of* Y*, we have*

$$\mathrm{codim}(f^{-1}(Z), X) \leqslant \mathrm{codim}(Z, Y) \tag{6.1.4.1}$$

Moreover, if f *is quasi-flat* (2.3.3)*, the two sides of* (6.1.4) *are equal.*

Proof. Indeed, if y is a maximal point of Z and x a maximal point of the fibre $f^{-1}(y)$, we have $\dim(\mathscr{O}_{X,x}) \leqslant \dim(\mathscr{O}_{Y,y})$ by virtue of (5.5.2); the inequality (6.1.4) then follows from (5.1.2.1) and from (5.1.3.1) and from the fact that f is surjective. If in addition f is quasi-flat, we know (2.3.4) that every irreducible component of $f^{-1}(Z)$ dominates an irreducible component of Z; the maximal points of $f^{-1}(Z)$ are therefore the maximal points of the fibres $f^{-1}(y)$, where y ranges over the set of maximal points of Z (**0**, 2.1.8), and for each of these maximal points x we have $\dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(x)) = 0$; as in addition $\mathscr{O}_x$ is a flat $\mathscr{O}_y$-module, we have $\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y)$ by virtue of (6.1.1), whence the equality in (6.1.4), taking into account that f is surjective. □

The proposition (6.1.1) admits the following partial converse:

Proposition (6.1.5). — *Let* A, B *be two local Noetherian rings, k the residue field of* A, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* M *a* B*-module of finite type. Suppose that:*

1° A *is a regular ring.*
2° M *is a Cohen-Macaulay* B*-module.*
3° $\dim_{\mathrm{B}}(\mathrm{M}) = \dim(\mathrm{A}) + \dim_{\mathrm{B}\otimes_{\mathrm{A}} k}(\mathrm{M} \otimes_{\mathrm{A}} k)$.

Then M *is a flat* A*-module.*

Proof. We argue by induction on $n = \dim(\mathrm{A})$. If $n = 0$, A is a field since it is regular (**0**, 17.1.4) and the assertion is trivial. Suppose $n > 0$; let $\mathfrak{m}$ be the maximal ideal of A, and let $x \in \mathfrak{m}$ be an element not belonging to $\mathfrak{m}^2$; we know then (**0**, 17.1.8) that $\mathrm{A}' = \mathrm{A}/x\mathrm{A}$ is regular and $\dim(\mathrm{A}') = \dim(\mathrm{A}) - 1$. Set

$$\mathrm{B}' = \mathrm{B}/x\mathrm{B}, \quad \mathrm{M}' = \mathrm{M}/x\mathrm{M} = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}';$$

then

$$\mathrm{B}' \otimes_{\mathrm{A}'} k = \mathrm{B} \otimes_{\mathrm{A}} k, \quad \mathrm{M}' \otimes_{\mathrm{A}'} k = \mathrm{M} \otimes_{\mathrm{A}} k$$

and by virtue of (**0**, 16.3.9.2) we have

$$\dim_{\mathrm{B}'}(\mathrm{M}') \leqslant \dim(\mathrm{A}') + \dim_{\mathrm{B}'\otimes_{\mathrm{A}'} k}(\mathrm{M}' \otimes_{\mathrm{A}} k);$$

we conclude therefore from (**0**, 16.3.4) that we have **IV-2 | 137**

$$\dim_{\mathrm{B}}(\mathrm{M}) \leqslant \dim_{\mathrm{B}'}(\mathrm{M}') + 1 \leqslant (\dim(\mathrm{A}') + \dim_{\mathrm{B}\otimes_{\mathrm{A}} k}(\mathrm{M} \otimes_{\mathrm{A}} k)) + 1 = \dim(\mathrm{A}) + \dim_{\mathrm{B}\otimes_{\mathrm{A}} k}(\mathrm{M} \otimes_{\mathrm{A}} k).$$

Since by hypothesis the extreme terms of these inequalities are equal, we have necessarily: (i) $\dim_{\mathrm{B}'}(\mathrm{M}') = \dim_{\mathrm{B}}(\mathrm{M}) - 1$, and as M is by hypothesis a Cohen-Macaulay B-module, this means that x is M-*regular* (**0**, 16.1.9) and (**0**, 16.5.5); (ii) $\dim_{\mathrm{B}'}(\mathrm{M}') = \dim(\mathrm{A}') + \dim_{\mathrm{B}'\otimes_{\mathrm{A}'} k}(\mathrm{M}' \otimes_{\mathrm{A}'} k)$; as M′ is a Cohen-Macaulay B′-module by virtue of (i) and of (**0**, 16.1.9) and (**0**, 16.5.5) and A′ is regular, the induction hypothesis proves that M′ is a flat A′-module; we deduce then from (**0**, 10.2.7) that M is a flat A-module, since x is M-regular by (i). □

6.2. Flatness and projective dimension

Proposition (6.2.1). —

(i) *Let* A, B *be two rings,* $\varphi : \mathrm{A} \to \mathrm{B}$ *a homomorphism such that* B *is a flat* A*-module. Then, for every* A*-module* M*, we have*

(6.2.1.1) $$\dim.\mathrm{proj}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}) \leqslant \dim.\mathrm{proj}_{\mathrm{A}}(\mathrm{M}).$$

(ii) *Suppose in addition that* A *is a Noetherian ring,* B *a faithfully flat* A*-module, and* M *an* A*-module of finite type; then*

(6.2.1.2) $$\dim.\mathrm{proj}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}) = \dim.\mathrm{proj}_{\mathrm{A}}(\mathrm{M}).$$

Proof.

(i) We may restrict to the case where $n = \dim.\mathrm{proj}_{\mathrm{A}}(\mathrm{M})$ is finite; there therefore exists a left resolution

$$0 \longrightarrow \mathrm{P}_n \longrightarrow \mathrm{P}_{n-1} \longrightarrow \ldots \longrightarrow \mathrm{P}_0 \longrightarrow \mathrm{M} \longrightarrow 0$$

of M by projective A-modules; as B is a flat A-module, the sequence

$$0 \longrightarrow \mathrm{P}_n \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow \mathrm{P}_{n-1} \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow \ldots \longrightarrow \mathrm{P}_0 \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow 0$$

is exact; on the other hand, the $\mathrm{P}_i \otimes_{\mathrm{A}} \mathrm{B}$ are projective B-modules; whence the conclusion.

(ii) Suppose that $\dim.\mathrm{proj}_{\mathrm{B}}(\mathrm{M}\otimes_{\mathrm{A}}\mathrm{B}) = m$, and consider an exact sequence

$$0 \longrightarrow \mathrm{R} \longrightarrow \mathrm{P}_{m-1} \longrightarrow \mathrm{P}_{m-2} \longrightarrow \dots \longrightarrow \mathrm{P}_0 \longrightarrow \mathrm{M} \longrightarrow 0$$

where the P_i ($0 \leqslant i \leqslant m-1$) are projective A-modules of finite type; as A is Noetherian, R is also an A-module of finite type. As B is a flat A-module, we also have an exact sequence

$$0 \longrightarrow \mathrm{R}\otimes_{\mathrm{A}}\mathrm{B} \longrightarrow \mathrm{P}_{m-1}\otimes_{\mathrm{A}}\mathrm{B} \longrightarrow \dots \longrightarrow \mathrm{P}_0\otimes_{\mathrm{A}}\mathrm{B} \longrightarrow \mathrm{M}\otimes_{\mathrm{A}}\mathrm{B} \longrightarrow 0$$

and the hypothesis on $\mathrm{M}\otimes_{\mathrm{A}}\mathrm{B}$ implies that $\mathrm{R}\otimes_{\mathrm{A}}\mathrm{B}$ is a projective B-module (**0**, 17.2.1). As B is a faithfully flat A-module, we conclude that R is a projective A-module (Bourbaki, *Alg. comm.*, chap. I$^{\text{er}}$, § 3, n$^{\text{o}}$ 6, prop. 12); hence $\dim.\mathrm{proj}_{\mathrm{A}}(\mathrm{M}) \leqslant m$, which completes the proof. □

Corollary (6.2.2). — *Let $f : X \to Y$ be a flat morphism of preschemes, $\mathscr{E}$ a quasi-coherent $\mathscr{O}_Y$-Module. If* $\dim.\mathrm{proj}(\mathscr{E}) \leqslant n$ (**0**, 17.2.14), *then* $\dim.\mathrm{proj}(f^*(\mathscr{E})) \leqslant n$.

Proposition (6.2.3). — *Let* A, B *be two local Noetherian rings, k the residue field of* A, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* N *a* B*-module of finite type. Suppose that* B *and* N *are flat* A*-modules. Then we have*

(6.2.3.1) $$\dim.\mathrm{proj}_{\mathrm{B}}(\mathrm{N}) = \dim.\mathrm{proj}_{\mathrm{B}\otimes_{\mathrm{A}}k}(\mathrm{N}\otimes_{\mathrm{A}}k).$$

Proof. Consider a left resolution of N by free B-modules of finite type

$$\dots \longrightarrow \mathrm{L}_i \longrightarrow \mathrm{L}_{i-1} \longrightarrow \dots \longrightarrow \mathrm{L}_0 \longrightarrow \mathrm{N} \longrightarrow 0$$

As the L_i and N are flat A-modules (B being a flat A-module) it follows from (2.1.10) that the $\mathrm{Z}_i(\mathrm{L}_\bullet)$ are flat A-modules, that $\mathrm{L}_\bullet\otimes_{\mathrm{A}}k$ is a resolution of $\mathrm{N}\otimes_{\mathrm{A}}k$ and that $\mathrm{Z}_i(\mathrm{L}_\bullet\otimes_{\mathrm{A}}k) = \mathrm{Z}_i(\mathrm{L}_\bullet)\otimes_{\mathrm{A}}k$ for all i. Note that since B is Noetherian, the $\mathrm{Z}_i(\mathrm{L}_\bullet)$ are B-modules of finite type, hence the $\mathrm{Z}_i(\mathrm{L}_\bullet\otimes_{\mathrm{A}}k)$ are $(\mathrm{B}\otimes_{\mathrm{A}}k)$-modules of finite type. Now, to say that a $(\mathrm{B}\otimes_{\mathrm{A}}k)$-module of finite type is flat is equivalent to saying that it is *free* or even *projective* (**0**, 10.1.3). As (where we set C = B) the $\mathrm{Z}_i(\mathrm{L}_\bullet)$ are flat B-modules, it follows on the other hand from (**0**, 10.2.5) that, for $\mathrm{Z}_i(\mathrm{L}_\bullet)$ to be a flat B-module, it is necessary and sufficient that $\mathrm{Z}_i(\mathrm{L}_\bullet)\otimes_{\mathrm{A}}k$ be a $(\mathrm{B}\otimes_{\mathrm{A}}k)$-module that is flat. The smallest integer n such that $\mathrm{Z}_{n-1}(\mathrm{L}_\bullet)$ is a free B-module is therefore also the smallest integer n such that $\mathrm{Z}_{n-1}(\mathrm{L}_\bullet\otimes_{\mathrm{A}}k)$ is a free $(\mathrm{B}\otimes_{\mathrm{A}}k)$-module, which proves the proposition (**0**, 17.2.1). □ **IV-2** | 138

6.3. Flatness and depth

Proposition (6.3.1). — *Let* A, B *be two local Noetherian rings, k the residue field of* A, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* M *an* A*-module of finite type,* N *a* B*-module of finite type. If* N *is a flat* A*-module* $\neq 0$, *then we have*

(6.3.1.1) $$\mathrm{prof}_{\mathrm{B}}(\mathrm{M}\otimes_{\mathrm{A}}\mathrm{N}) = \mathrm{prof}_{\mathrm{A}}(\mathrm{M}) + \mathrm{prof}_{\mathrm{B}\otimes_{\mathrm{A}}k}(\mathrm{N}\otimes_{\mathrm{A}}k).$$

Proof. We may restrict to the case where $\mathrm{M} \neq 0$, without which both sides of (6.3.1) are all equal to $+\infty$. We will argue by induction on the integer n equal to the second member of (6.3.1.1) (which is finite by virtue of the hypotheses, of (**0**, 16.4.6.2) and of the lemma of Nakayama applied to N). Suppose first $n = 0$; then, if $\mathfrak{m}$ and $\mathfrak{n}$ denote the maximal ideals of A and B respectively, $\mathfrak{m}$ is a prime ideal associated to M and $\mathfrak{n}/\mathfrak{m}\mathrm{B}$ a prime ideal of $\mathrm{B}/\mathfrak{m}\mathrm{B} = \mathrm{B}\otimes_{\mathrm{A}}k$ associated to $\mathrm{N}\otimes_{\mathrm{A}}k$. We conclude then from (3.3.1) that $\mathfrak{n}$ is a prime ideal associated to $\mathrm{M}\otimes_{\mathrm{A}}\mathrm{N}$, hence (**0**, 16.4.6), (i)) that $\mathrm{prof}_{\mathrm{B}}(\mathrm{M}\otimes_{\mathrm{A}}\mathrm{N}) = 0$, and we distinguish two cases:

a) Suppose $\mathrm{prof}_{\mathrm{A}}(\mathrm{M}) > 0$. Let $x \in \mathfrak{m}$ be an M-regular element, and set

$$\mathrm{A}' = \mathrm{A}/x\mathrm{A}, \quad \mathrm{B}' = \mathrm{B}/x\mathrm{B}, \quad \mathrm{M}' = \mathrm{M}/x\mathrm{M}, \quad \mathrm{N}' = \mathrm{N}/x\mathrm{N};$$

then

$$\mathrm{B}'\otimes_{\mathrm{A}'}k = \mathrm{B}\otimes_{\mathrm{A}}k, \quad \mathrm{N}'\otimes_{\mathrm{A}'}k = \mathrm{N}\otimes_{\mathrm{A}}k$$

since $x \in \mathfrak{m}$, and

$$\mathrm{M}'\otimes_{\mathrm{A}'}\mathrm{N}' = (\mathrm{M}\otimes_{\mathrm{A}}\mathrm{N})/x(\mathrm{M}\otimes_{\mathrm{A}}\mathrm{N});$$

in addition, as N is a flat A-module, the hypothesis that x is M-regular implies that x is also $(\mathrm{M}\otimes_{\mathrm{A}}\mathrm{N})$-regular. We thus have (**0**, 16.4.6), (ii) and (**0**, 16.4.8)) **IV-2** | 139

$$\mathrm{prof}_{\mathrm{A}'}(\mathrm{M}') = \mathrm{prof}_{\mathrm{A}}(\mathrm{M}) - 1, \quad \mathrm{prof}_{\mathrm{B}'}(\mathrm{M}'\otimes_{\mathrm{A}'}\mathrm{N}') = \mathrm{prof}_{\mathrm{B}}(\mathrm{M}\otimes_{\mathrm{A}}\mathrm{N}) - 1$$

and

$$\operatorname{prof}_{B'\otimes_{A'}k}(N'\otimes_{A'}k)=\operatorname{prof}_{B\otimes_A k}(N\otimes_A k).$$

The equality (6.3.1.1) is therefore a consequence of the same relation for A′, B′, M′ and N′; but as N is a flat A-module, $N' = N\otimes_A A'$ is a flat A′-module (**0**, 6.2.1); we may therefore apply the induction hypothesis, which proves (6.3.1.1) in this case.

b) Suppose $\operatorname{prof}_{B\otimes_A k}(N\otimes_A k) > 0$. Consider an element $y\in\mathfrak{n}$ which is $(N\otimes_A k)$-regular; it follows from (**0**, 10.2.4) that y is N-regular and that we then have

$$N' = N/yN$$

is a flat A-module, since N is supposed to be a flat A-module. Applying (**0**, 6.1.2) to the exact sequence of A-modules

$$0\longrightarrow N\xrightarrow{y}N\longrightarrow N'\longrightarrow 0$$

we conclude that we have isomorphisms

$$(M\otimes_A N)/y(M\otimes_A N)\xrightarrow{\sim}M\otimes_A N'\quad\text{and}\quad N'\otimes_A k\xrightarrow{\sim}(N\otimes_A k)/y(N\otimes_A k)$$

and in addition that y is $(M\otimes_A N)$-regular. We thus have

$$\operatorname{prof}_{B\otimes_A k}(N'\otimes_A k)=\operatorname{prof}_{B\otimes_A k}(N\otimes_A k)-1$$

and

$$\operatorname{prof}_B(M\otimes_A N')=\operatorname{prof}_B(M\otimes_A N)-1;$$

the induction hypothesis shows that the relation (6.3.1.1) is valid for A, B, M and N′, and from what precedes, we deduce (6.3.1.1) for A, B, M and N. □

Corollary (6.3.2). — *Under the hypotheses of (6.3.1), suppose in addition* $M\neq 0$ *and* $N\neq 0$; *then we have*

(6.3.2.1) $$\operatorname{coprof}_B(M\otimes_A N)=\operatorname{coprof}_A(M)+\operatorname{coprof}_{B\otimes_A k}(N\otimes_A k).$$

Proof. This follows immediately from (6.3.1.1) and (6.1.2) and from the definition of the codepth (**0**, 16.4.9). □

Corollary (6.3.3). — *Under the hypotheses of (6.3.1), suppose in addition* $M\neq 0$ *and* $N\neq 0$; *for* $M\otimes_A N$ *to be a Cohen-Macaulay* B*-module, it is necessary and sufficient that* M *be a Cohen-Macaulay* A*-module and that* $N\otimes_A k$ *be a* $(B\otimes_A k)$*-module of Cohen-Macaulay.*

Proof. This follows from the corollary (6.3.2) and from the definition of Cohen-Macaulay modules by the fact that their codepth is zero (**0**, 16.5.1), taking into account that the condition $N\neq 0$ is equivalent to $N\otimes_A k\neq 0$ by virtue of the lemma of Nakayama. □

Corollary (6.3.4). — *Under the hypotheses of (6.3.1), suppose in addition that* $N\otimes_A k$ *is a* B*-module of finite length; then we have*

(6.3.4.1) $$\operatorname{prof}_B(M\otimes_A N)=\operatorname{prof}_A(M).$$

If in addition $M\neq 0$ *and* $N\neq 0$, *then we have* **IV-2** | 140

(6.3.4.2) $$\operatorname{coprof}_B(M\otimes_A N)=\operatorname{coprof}_A(M).$$

Proof. It amounts to the same to say that $N\otimes_A k$ is a B-module of finite length or a $(B\otimes_A k)$-module of finite length, and we know (**0**, 16.2.3) that modules of finite length $\neq 0$ are of dimension 0 and depth 0.

The corollary (6.3.4) is applicable in particular when $B\otimes_A k$ is an Artinian ring, that is to say when $\mathfrak{m}B$ is an *ideal of definition* of the ring B. □

Corollary (6.3.5). — *Let* X, Y *be two locally Noetherian preschemes,* $f : X\to Y$ *a flat morphism.*

(i) *If, at a point* $x\in X$, $\operatorname{coprof}(\mathscr{O}_x)\leqslant n$, *then, setting* $y=f(x)$, *we have* $\operatorname{coprof}(\mathscr{O}_y)\leqslant n$ *and* $\operatorname{coprof}(\mathscr{O}_x\otimes_{\mathscr{O}_y}\boldsymbol{k}(y))\leqslant n$; *in particular, if* $\mathscr{O}_x$ *is a Cohen-Macaulay ring, the same is true of* $\mathscr{O}_y$ *and of* $\mathscr{O}_x\otimes_{\mathscr{O}_y}\boldsymbol{k}(y)$.

(ii) *Conversely, suppose that* $\mathscr{O}_x\otimes_{\mathscr{O}_y}\boldsymbol{k}(y)$ *is a Cohen-Macaulay ring. Then, if* $\mathscr{O}_y$ *is a Cohen-Macaulay ring, we have* $\operatorname{coprof}(\mathscr{O}_x)\leqslant n$ *(resp.* $\mathscr{O}_x$ *is a Cohen-Macaulay ring).*

Proof. It suffices to apply (6.3.2) for $M = \mathcal{O}_y$ and $N = B = \mathcal{O}_x$. □

Corollary (6.3.6). — *Let* A *be a Cohen-Macaulay ring. Then* $A' = A[T_1, \dots, T_n]$ *(*T_i *indeterminates) is a Cohen-Macaulay ring.*

Proof. Indeed, if we set $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(A')$ and if $f : X \to Y$ is the canonical morphism, f is *flat* (since A' is a free A-module (2.1.2)) and for every $y \in Y$, $\boldsymbol{k}(y)[T_1, \dots, T_n]$ is a regular ring (**0**, 17.3.7), hence *a fortiori* a Cohen-Macaulay ring; it therefore suffices to apply (6.3.5). □

Corollary (6.3.7). — *Every quotient ring of a Cohen-Macaulay ring is universally catenary.*

Proof. This follows from (6.3.6), from (5.6.1) and from (**0**, 16.5.12). □

Proposition (6.3.8). — *Let* A *be a local Noetherian ring; suppose that there exists a Cohen-Macaulay* A*-module* M *of finite type whose support is equal to* $\operatorname{Spec}(A)$*. Let* B *be a quotient ring of* A*, and let* $f : \operatorname{Spec}(\widehat{B}) \to \operatorname{Spec}(B)$ *be the canonical morphism; then, for every* $x \in \operatorname{Spec}(B)$*,* $f^{-1}(x)$ *is a Cohen-Macaulay prescheme.*

Proof. As B is an A-module of finite type, one has $\widehat{B} = B \otimes_A \widehat{A}$ (**0**, 7.3.3); hence f is the restriction to $\operatorname{Spec}(\widehat{B})$ of the canonical morphism $\operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}(A)$, and the fibres of these two morphisms at a point of $\operatorname{Spec}(B)$ are the same. We are therefore reduced to proving the proposition when $B = A$. Let $\mathfrak{p} = \mathfrak{j}_x$ be a prime ideal of A. Since $\mathfrak{p} \in \operatorname{Supp}(M)$ by hypothesis, there exists (**0**, 16.5.6) a M-regular sequence $(t_i)_{1 \leqslant i \leqslant r}$ of elements of $\mathfrak{p}$ such that

$$N = M/\Big(\sum_{i=1}^{r} t_i M\Big)$$

is a Cohen–Macaulay A-module, $\mathfrak{p}$ is a minimal associated prime ideal of N, and

$$\dim(N) = \dim(M/\mathfrak{p}M) = \dim(A/\mathfrak{p})$$

(**0**, 16.5.9). The same argument as at the beginning shows that one may replace M by N and A by $A/(\sum_{i=1}^{r} t_i A)$; consequently one may suppose that $\mathfrak{p}$ is a minimal prime ideal of A. Set, to simplify, $A' = \widehat{A}$, $M' = \widehat{M} = M \otimes_A A'$ (**0**, 7.3.3); we know (**0**, 16.5.2) that M' is a Cohen-Macaulay A'-module, which implies that for every prime ideal $\mathfrak{p}'$ of A', $M'_{\mathfrak{p}'}$ is a Cohen-Macaulay $A'_{\mathfrak{p}'}$-module (**0**, 16.5.10). Taking into account (**I**, 3.6.5), one sees that if we set $A'' = A' \otimes_A A_{\mathfrak{p}}$ and $M'' = M' \otimes_A A_{\mathfrak{p}}$, M'' is a Cohen-Macaulay A''-module. For every prime ideal $\mathfrak{p}''$ of A'' above $\mathfrak{p}$, $M''_{\mathfrak{p}''} = M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ is a Cohen-Macaulay $A''_{\mathfrak{p}''}$-module (**0**, 16.5.10); on the other hand, $M_{\mathfrak{p}}$ is by hypothesis a Cohen-Macaulay $A_{\mathfrak{p}}$-module and $A''_{\mathfrak{p}''}$ is a flat $A_{\mathfrak{p}}$-module since A' is a flat A-module (**0**, 7.3.3) and (6.3.2). We conclude from (6.3.3) that, if k is the residue field of $A_{\mathfrak{p}}$, $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ is a Cohen-Macaulay ring. But since $A_{\mathfrak{p}}$ is of dimension 0, the prime ideals of A' correspond bijectively to those of $k \otimes_{A_{\mathfrak{p}}} A''$ (**I**, 3.5.7), and if $\mathfrak{q}''$ is the prime ideal of $k \otimes_{A_{\mathfrak{p}}} A''$ corresponding to $\mathfrak{p}''$, the local rings $(k \otimes_{A_{\mathfrak{p}}} A'')_{\mathfrak{q}''}$ and $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ are isomorphic. Therefore (**0**, 16.5.13), the ring $k \otimes_{A_{\mathfrak{p}}} A''$ is a Cohen-Macaulay ring. □

6.4. Flatness and property (S_k)

Proposition (6.4.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism, $\mathscr{E}$ a coherent $\mathscr{O}_Y$-Module, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module and f-flat.*

(i) Let $x \in \mathrm{Supp}(\mathscr{F})$ be such that $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}$ has property (S_k) at the point x; then $\mathscr{E}$ has property (S_k) at the point $y = f(x)$.

(ii) Suppose that for every $y \in f(\mathrm{Supp}(\mathscr{F}))$, $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ has property (S_k); then, if for a point $y \in f(\mathrm{Supp}(\mathscr{F}))$, $\mathscr{E}$ has property (S_k) at the point y, $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}$ has property (S_k) at every point of $f^{-1}(y)$.

Proof.

(i) We know (**Err$_{\mathbf{III}}$**, 30) that there exists a closed sub-prescheme X' of X having for underlying space $\mathrm{Supp}(\mathscr{F})$ and a coherent $\mathscr{O}_{X'}$-Module $\mathscr{F}'$ such that $\mathscr{F} = j_*(\mathscr{F}')$, where j is the canonical injection $X' \hookrightarrow X$; we may replace X by X' and $\mathscr{F}$ by $\mathscr{F}'$, in other words suppose that $\mathrm{Supp}(\mathscr{F}) = X$. Let y' be a generization of y in Y; we know (2.3.4) that there is a generization x' of x in X such that $y' = f(x')$; we may in addition suppose that x' is the generic point of an irreducible component of $f^{-1}(y')$; we then have $\dim(\mathscr{F}_{x'} \otimes_{\mathscr{O}_y} \boldsymbol{k}(y')) = 0$. By virtue of (6.1.2), we then have, setting $\mathscr{G} = \mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}$,

$$\dim(\mathscr{G}_{x'}) = \dim(\mathscr{E}_{y'})$$

and by virtue of (6.3.1)

$$\mathrm{prof}(\mathscr{G}_{x'}) = \mathrm{prof}(\mathscr{E}_{y'})$$

taking into account that the depth is always at most equal to the dimension. By hypothesis, we have **IV-2 | 142**

$$\mathrm{prof}(\mathscr{G}_{x'}) \geqslant \inf(k, \dim(\mathscr{G}_{x'}))$$

hence

$$\mathrm{prof}(\mathscr{E}_{y'}) \geqslant \inf(k, \dim(\mathscr{E}_{y'}))$$

which proves the first assertion.

(ii) As for every generization x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generization of y, we may restrict to verifying that if $x \in \mathrm{Supp}(\mathscr{G})$ and $f(x) = y$, we have $\mathrm{prof}(\mathscr{G}_x) \geqslant \inf(k, \dim(\mathscr{G}_x))$; it follows from (6.1.2) and (6.3.1) that we have

$$\dim(\mathscr{G}_x) = \dim(\mathscr{E}_y) + \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))$$

$$\mathrm{prof}(\mathscr{G}_x) = \mathrm{prof}(\mathscr{E}_y) + \mathrm{prof}(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)).$$

By hypothesis, we have

$$\mathrm{prof}(\mathscr{E}_y) \geqslant \inf(k, \dim(\mathscr{E}_y))$$

$$\mathrm{prof}(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)) \geqslant \inf(k, \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)))$$

whence, adding term by term,

$$\begin{aligned}\mathrm{prof}(\mathscr{G}_x) &\geqslant \inf(k, \dim(\mathscr{E}_y)) + \inf(k, \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))) \geqslant \\ &\geqslant \inf(k, \dim(\mathscr{E}_y) + \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))) = \inf(k, \dim(\mathscr{G}_x))\end{aligned}$$

which proves (ii).

□

Corollary (6.4.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism, $\mathscr{E}$ a coherent $\mathscr{O}_Y$-Module. Suppose that for every $y \in Y$, the prescheme $f^{-1}(y)$ has property (S_k). Then, for $f^*(\mathscr{E})$ to have property (S_k) at a point x, it is necessary and sufficient that $\mathscr{E}$ have property (S_k) at the point $f(x)$.*

Remark (6.4.3). — Let A, B be two local Noetherian rings, k the residue field of A, $\varphi : \mathrm{A} \to \mathrm{B}$ a local homomorphism, M a B-module of finite type that is a flat A-module, and suppose in addition that the A-module A and the $(\mathrm{B} \otimes_{\mathrm{A}} k)$-module $\mathrm{M} \otimes_{\mathrm{A}} k$ have property (S_n); can one then conclude that the B-module M has property (S_n)? We do not know even when $n = 1$, $\mathrm{M} = \mathrm{B}$ and $\mathrm{B} = \widehat{\mathrm{A}}$, which is to say we do not know whether in general the *completion* of a local Noetherian ring satisfying (S_n) also satisfies (S_n), even for $n = 1$. Setting $Y = \mathrm{Spec}(\mathrm{A})$, $X = \mathrm{Spec}(\mathrm{B})$ and $\mathscr{F} = \widetilde{\mathrm{M}}$, it would suffice, by virtue of (6.4.1), (ii)), to show that for *every* $y \in Y$, $\mathscr{F}_y$ has property (S_k) (and not only when y is the closed point of Y); it would also suffice to show that the set U of $x \in X$ such that $\mathscr{F}_x \otimes_{\mathscr{O}_{f(x)}} \boldsymbol{k}(f(x))$ satisfies (S_k) is *open* in X (since it contains the closed point of X by hypothesis). We will show further on (12.1.4) that U is open when B is a local ring of a finite type A-algebra. On the other hand, for $\mathrm{B} = \widehat{\mathrm{A}}$ and $\mathrm{M} = \mathrm{B}$, we have seen in (6.3.8) that the preschemes $f^{-1}(y)$ are Cohen-Macaulay preschemes (in other words satisfy (S_n)) for *every* n) when we suppose that A is a quotient of a local Cohen-Macaulay ring (or more generally of a local ring satisfying the hypotheses of (6.3.8)), *for* A *to satisfy* (S_n), *it is necessary and sufficient that its completion* $\widehat{\mathrm{A}}$ *satisfy* (S_n). It would remain to see whether this property holds without restriction on A.

6.5. Flatness and property (R_k)

Proposition (6.5.1). — *Let* A, B *be two local Noetherian rings, k the residue field of* A, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism, for which* B *is a flat* A*-module. Then:*

(i) If B *is regular,* A *is regular.*
(ii) If A *and* $\mathrm{B} \otimes_{\mathrm{A}} k$ *are regular,* B *is regular.*

This proposition is given here for the record, having already been proved in (**0**, 17.3.3).

Corollary (6.5.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

(i) If X is regular at a point x, Y is regular at the point $f(x)$.
(ii) If for a $y \in f(X)$, Y is regular at the point y and if $f^{-1}(y)$ is a regular prescheme at a point x, then X is regular at the point x.

Proposition (6.5.3). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

(i) If X has property (R_k) at a point x (resp. has property (R_k)), Y has property (R_k) at the point $f(x)$ (resp. Y has property (R_k) at all points of $f(X)$).
(ii) Suppose that for every $y \in f(X)$, the prescheme $f^{-1}(y)$ has property (R_k); then, if for a point $y \in f(X)$, Y has property (R_k) at the point y, X has property (R_k) at every point of $f^{-1}(y)$.

Proof.

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(i) Set $y = f(x)$ and let y' be a generization of y; as in the proof of (6.4.1), there is a generic point x' of an irreducible component of $f^{-1}(y')$ which is a generization of x; hence $\dim(\mathscr{O}_{x'} \otimes_{\mathscr{O}_{y'}} \boldsymbol{k}(y')) = 0$ and by virtue of the hypothesis and of (6.1.2), $\dim(\mathscr{O}_{x'}) = \dim(\mathscr{O}_{y'})$. The hypothesis implies, provided that $\dim(\mathscr{O}_{x'}) > k$, that $\mathscr{O}_{x'}$ is a regular ring, and then $\mathscr{O}_{y'}$ is regular by virtue of (6.5.1).

(ii) As for every generization x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generization of y, we may restrict to verifying that if $\dim(\mathscr{O}_x) \leqslant k$, then $\mathscr{O}_x$ is a regular ring. Now, by virtue of (6.1.2)

$$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))$$

hence if $\dim(\mathscr{O}_x) \leqslant k$, we have *a fortiori* $\dim(\mathscr{O}_y) \leqslant k$ and $\dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)) \leqslant k$ and the hypothesis implies that $\mathscr{O}_y$ and $\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ are regular rings. We conclude then from (6.5.1) that $\mathscr{O}_x$ is a regular ring.

□

Corollary (6.5.4). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

(i) If X is normal at a point x, Y is normal at the point $f(x)$.
(ii) If, for every $y \in f(X)$, $f^{-1}(y)$ is a normal prescheme and if Y is normal at a point $y \in f(X)$, then X is a normal prescheme at every point of $f^{-1}(y)$.

Proof. This follows immediately from the normality criterion of Serre (5.8.6) and from (6.4.1) applied for $k = 2$ and (6.5.3) applied for $k = 1$. □

Remarks (6.5.5). —

(i) Let A, B be two local Noetherian rings, $\varphi : \mathrm{A} \to \mathrm{B}$ a local homomorphism such that B is a flat A-module. Let k be the residue field of A, and suppose that the two rings A and $\mathrm{B} \otimes_{\mathrm{A}} k$ satisfy property (R_i) (5.8.2); then *it does not necessarily follow* that B satisfies (R_i), even in the particular case where $i = 0$ or $i = 1$ and where B is the *completion* $\widehat{\mathrm{A}}$ of A. Nagata has in fact given an example where A is *normal* (hence satisfies (R_1)) but where $\widehat{\mathrm{A}}$ is not even reduced (hence does not satisfy (R_0)) [30]. We cannot apply the proposition (6.5.3) to this case because the fibres $f^{-1}(y)$ do not necessarily satisfy property (R_i) at the points distinct from the closed point of $Y = \mathrm{Spec}(\mathrm{A})$. We will show further on (7.8.3), (v)) that such phenomena do not arise for the local Noetherian rings that one encounters most often in applications.

(ii) The property for a prescheme of being *integral* does not at all behave like the properties we have just examined and the preceding ones: it can happen that $f : X \to Y$ is a flat morphism of finite type, that all the fibres $f^{-1}(y)$ are geometrically regular (and even geometrically regular (6.7.6)) and that Y is integral, without X being even locally integral. For example, let k be an algebraically closed field of characteristic 0, and let A be the integral ring $k[\mathrm{U},\mathrm{V}]/(\mathrm{UV} - (\mathrm{U}+\mathrm{V})^3)$ (whose spectrum is therefore a "cubic with a double point"); A is not integrally closed, and if u, v are the canonical images of U, V in A, the integral closure of A is the ring $\mathrm{C} = \mathrm{A}[t]$, with $t = (u-v)/(u+v)$, which satisfies the equation $t^2 = 1 + u + v$, whence $u = \frac{1}{2}(t^3 + t^2 - t)$, $v = \frac{1}{2}(-t^3 + t^2 + t)$ and therefore $\mathrm{C} = k[t]$, isomorphic to a polynomial ring in one indeterminate over k. If $\mathfrak{m} = \mathrm{A}u + \mathrm{A}v$, the maximal ideal of A (corresponding to the "double point" of the cubic), there are two maximal ideals $\mathfrak{n}_1$, $\mathfrak{n}_2$ of C, generated respectively by $t-1$ and $t+1$. Let B be the subring of the product $\mathrm{C} \times \mathrm{C}$ formed by the pairs of polynomials (f, g) such that $f(1) = g(-1)$ and $f(-1) = g(1)$ (Spec(B) is the prescheme obtained by "gluing" two copies of Spec(C), the point $\mathfrak{n}_1$ (resp. $\mathfrak{n}_2$) of one of the two copies being "glued" to the point $\mathfrak{n}_2$ (resp. $\mathfrak{n}_1$) of the other; cf. chap. V, where this operation will be discussed in a general way). There are therefore two maximal ideals $\mathfrak{r}_1$, $\mathfrak{r}_2$ of B above the maximal ideal $\mathfrak{m}$ of A. Moreover, as the "gluing" process commutes with localisation and completion, one easily verifies that the canonical homomorphisms $\widehat{\mathrm{A}}_{\mathfrak{m}} \to \widehat{\mathrm{B}}_{\mathfrak{r}_1}$ and $\widehat{\mathrm{A}}_{\mathfrak{m}} \to \widehat{\mathrm{B}}_{\mathfrak{r}_2}$ are bijective; consequently $\mathrm{B}_{\mathfrak{r}_1}$ and $\mathrm{B}_{\mathfrak{r}_2}$ are flat $\mathrm{A}_{\mathfrak{m}}$-modules (Bourbaki, *Alg. comm.*, chap. III, § 3, n° 5, prop. 10). For every other maximal ideal $\mathfrak{p}$ of A, there is a single maximal ideal $\mathfrak{q}_1$, $\mathfrak{q}_2$ of B above $\mathfrak{p}$, and the homomorphisms $\mathrm{A}_{\mathfrak{p}} \to \mathrm{B}_{\mathfrak{q}_1}$ and $\mathrm{A}_{\mathfrak{p}} \to \mathrm{B}_{\mathfrak{q}_2}$ are bijective. We see thus that the morphism $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ is flat and finite, and that all its fibres are geometrically regular (it is even *étale*, as we will see later (17.6.3)); however it is immediate that B is not integral. IV-2 | 144

6.6. Properties of transitivity

Proposition (6.6.1). — *For a locally Noetherian prescheme* Z, *let us denote by* $\boldsymbol{P}(\mathrm{Z})$ *any one of the following properties:*

a) Z *is a Cohen-Macaulay prescheme.*
b) Z *satisfies* (S_n).
c) Z *is regular.*
d) Z *satisfies* (R_n).
e) Z *is normal.*
f) Z *is reduced.*

Let then X, Y, Z *be three locally Noetherian preschemes,* $f : \mathrm{X} \to \mathrm{Y}$, $g : \mathrm{Y} \to \mathrm{Z}$ *two morphisms.*

(i) Suppose that f *and* g *are flat and that for every* $y \in f(\mathrm{X})$ *(resp. every* $z \in g(\mathrm{Y})$*), the prescheme* $f^{-1}(y)$ *(resp.* $g^{-1}(z)$*) satisfies* $\boldsymbol{P}$. *Then* $h = g \circ f$ *is flat,* $h^{-1}(z)$ *satisfies* $\boldsymbol{P}$ *for every* $z \in h(\mathrm{X})$, *and for every* $z \in g(\mathrm{Y})$ *and every* $t \in h(\mathrm{X})$, $h^{-1}(z)$ *satisfies* $\boldsymbol{P}$.

(ii) *Suppose the following conditions are satisfied: f is faithfully flat, $h = g \circ f$ is flat, for every $y \in Y$, the prescheme $f^{-1}(y)$ satisfies* $\boldsymbol{P}$, *and for every $z \in g(Y)$, the prescheme $h^{-1}(z)$ satisfies* $\boldsymbol{P}$. *Then g is flat, and for every $z \in g(Y)$, $g^{-1}(z)$ satisfies* $\boldsymbol{P}$.

Proof. (i) We already know (**0**, 2.1.6) that h is flat; furthermore, for every $z \in Z$, $f_z = f \otimes \mathbf{1}_{\boldsymbol{k}(z)} : h^{-1}(z) \to g^{-1}(z)$ is flat (**0**, 2.1.4), and for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (**I**, 3.6.4). The conclusion follows then respectively from (6.3.5), (ii), (6.4.2), (6.5.2), (ii), (6.5.3), (ii), (6.5.4), (ii) and (3.3.5), (ii). **IV-2** | 145

(ii) We already know that g is flat (2.2.13); in addition, for every $z \in Z$, f_z is faithfully flat (2.2.13). The conclusion follows then respectively from (6.3.5), (i), (6.4.2), (6.5.2), (i), (6.5.3), (i), (6.5.4), (ii) and (2.1.13). □

Remark (6.6.2). — Suppose h flat, f faithfully flat, and suppose that for every $z \in Z$, $h^{-1}(z)$ is of coprofundity $\leqslant n$ (5.7.1); then it follows from the reasoning of (6.6.1), (ii) and (6.3.2) that $g^{-1}(z)$ is of coprofundity $\leqslant n$ for every $z \in g(Y)$ and that for every $y \in Y$, $f^{-1}(y)$ is of coprofundity $\leqslant n$.

6.7. Application to base changes in algebraic preschemes **IV-2** | 146

Proposition (6.7.1). — *Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. Let k' be an extension of k; set $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$ and let $p : X' \to X$ be the canonical projection. We suppose either that X is locally of finite type over k, or that k' is an extension of finite type of k, so that in both cases X' is locally Noetherian. Let x' be a point of X', $x = p(x')$. Then:*

(i) *We have* $\operatorname{coprof}(\mathscr{F}_x) = \operatorname{coprof}(\mathscr{F}'_{x'})$; *in particular, for $\mathscr{F}_x$ to be an $\mathscr{O}_x$-module of Cohen-Macaulay, it is necessary and sufficient that $\mathscr{F}'_{x'}$ be an $\mathscr{O}_{x'}$-module of Cohen-Macaulay.*
(ii) *For $\mathscr{F}$ to satisfy the property (S_n) at the point x, it is necessary and sufficient that $\mathscr{F}'$ satisfy (S_n) at the point x'.*

Proof. We know that p is faithfully flat (2.2.13); the two assertions are then consequences respectively of (6.3.2) and (6.4.2), provided that one proves that $p^{-1}(x) = \operatorname{Spec}(\boldsymbol{k}(x) \otimes_k k')$ is a Cohen-Macaulay prescheme; as one of the two fields $\boldsymbol{k}(x)$, k' is an extension of finite type of k by hypothesis, we are reduced to proving the following: □

Lemma (6.7.1.1). — *Let K, L be two extensions of a field k, one of which is of finite type (so that the ring $K \otimes_k L$ is Noetherian). Then $K \otimes_k L$ is a Cohen-Macaulay ring.*

Proof. Suppose for example that L is an extension of finite type of k, so that L is extension of a pure extension $k(\mathbf{t})$ of k ($\mathbf{t}$ being a system $(t_i)_{1 \leqslant i \leqslant m}$ of indeterminates). We then set $A = K \otimes_k k(\mathbf{t})$, $B = K \otimes_k L$, B is an A-module flat (**0**, 6.2.1) and of finite type; the morphism $h : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is closed and finite, and as every Artinian prescheme is of Cohen-Macaulay, to prove that B is a Cohen-Macaulay ring, it suffices to prove that A is a Cohen-Macaulay ring (6.3.5). Now, if S is the set of nonzero elements of $k[\mathbf{t}]$, then $A = S^{-1}A'$, where $A' = K \otimes_k k[\mathbf{t}] = K[\mathbf{t}]$; by (**0**, 16.5.13), it suffices to prove that A' is regular (**0**, 17.3.7). □

Corollary (6.7.2). — *For $\mathscr{O}_x$ to be a Cohen–Macaulay ring (resp. for X to satisfy (S_n) at the point x), it is necessary and sufficient that $\mathscr{O}_{x'}$ be a Cohen–Macaulay ring (resp. that X' satisfy (S_n) at the point x').*

Corollary (6.7.3). — *Let X and Y be two locally Noetherian k-preschemes, at least one of which is locally of finite type over k. Let $\mathscr{F}$ (resp. $\mathscr{G}$) be a coherent $\mathscr{O}_X$-Module (resp. a coherent $\mathscr{O}_Y$-Module). If $\mathscr{F}$ and $\mathscr{G}$ satisfy property (S_n), then so does $\mathscr{F} \otimes_k \mathscr{G}$.*

Proof. The hypothesis implies that $X \times_k Y$ is locally Noetherian; let $p : X \times_k Y \to X$ and $q : X \times_k Y \to Y$ be the canonical projections, which are flat morphisms. Suppose, for example, that X is locally of finite type over k; we may write $\mathscr{F} \otimes_k \mathscr{G} = p^*(\mathscr{F}) \otimes_k \mathscr{G}$, and since $\mathscr{F}$ is flat relative to the structural morphism $X \to \operatorname{Spec}(k)$, $p^*(\mathscr{F})$ is q-flat (2.1.4). Applying the criterion (6.4.1), (ii) to the morphism q, it suffices to see that, for every $y \in Y$, $p^*(\mathscr{F}) \otimes_{\mathscr{O}_Y} \boldsymbol{k}(y)$ has property (S_n); but $p^*(\mathscr{F}) \otimes_{\mathscr{O}_Y} \boldsymbol{k}(y) = \mathscr{F} \otimes_k \boldsymbol{k}(y)$, and since X is locally of finite type over k, the conclusion follows from (6.7.1), (ii). □

Proposition (6.7.4). — *For a locally Noetherian prescheme Z and a point $z \in Z$, let $\boldsymbol{P}(Z, z)$ be one of the following properties:*

a) Z *is regular at the point* z.
b) Z *satisfies* (R_n) *at the point* z.
c) Z *is normal at the point* z.
d) Z *is reduced at the point* z.

Under the hypotheses and with the notation of (6.7.1), if $\boldsymbol{P}(\mathrm{X}', x')$ *is true, then* $\boldsymbol{P}(\mathrm{X}, x)$ *is true. The converse is true if* k' *is a separable extension of* k.

Proof. The case where $\boldsymbol{P}$ is property d) has been included only for the record, having already been treated in (2.1.13) and (4.6.1). The same reasoning as at the beginning of the proof of (6.7.1) shows that the first assertion follows respectively from (6.5.2), (i), (6.5.3), (i), and (6.5.4), (i); similarly, the second assertion follows from (6.5.2), (ii), (6.5.3), (ii), and (6.5.4), (ii),

provided that one proves that $p^{-1}(x)$ is a regular prescheme; in other words, we are reduced to proving the following lemma. ∎ **IV-2** | 147

Lemma (6.7.4.1). — *Let* K *and* L *be two extensions of a field* k*, one of which is of finite type. If* L *is a separable extension of* k*, then the ring* $\mathrm{K} \otimes_k \mathrm{L}$ *is regular.*

Proof. We distinguish two cases.

A) L is an extension of finite type of k. With the notation of (6.7.1.1), we may then suppose that L is a finite separable extension of $k(\mathbf{t})$. For every $s \in \mathrm{Spec}(\mathrm{A})$, the ring $\boldsymbol{k}(s) \otimes_{k(\mathbf{t})} \mathrm{L}$ is a direct sum of finitely many fields, finite separable extensions of $\boldsymbol{k}(s)$, and is therefore regular; it follows from (6.5.2), (ii) that it suffices to prove that A is regular. Since $\mathrm{A} = \mathrm{S}^{-1}\mathrm{A}'$, it suffices to prove that A' is regular (**0**, 17.3.6), and this was proved in the proof of (6.7.1.1).

B) Let (L_λ) be the filtered family of subextensions of L that are of finite type over k. By A), each ring $\mathrm{C}_\lambda = \mathrm{K} \otimes_k \mathrm{L}_\lambda$ is regular; moreover, if $\mathrm{L}_\lambda \subset \mathrm{L}_\mu$, then L_μ is a flat L_λ-module, so C_μ is a flat C_λ-module (**0**, 6.2.1). Since $\mathrm{C} = \mathrm{K} \otimes_k \mathrm{L} = \varinjlim(\mathrm{K} \otimes_k \mathrm{L}_\lambda)$ is Noetherian, we may apply (5.13.7), and C is therefore regular. ∎

Remarks (6.7.5). — (i) In the proof of the fact that $\boldsymbol{P}(\mathrm{X}, x)$ implies $\boldsymbol{P}(\mathrm{X}', x')$, one cannot dispense with the hypothesis that k' is a *separable* extension of k. We have indeed already seen (4.6.1) when $\boldsymbol{P}$ is the property *d*); but even when X is *geometrically integral* (4.6.2), it can happen that X is *regular* without X' being *normal*.

Take, for example, for X a normal algebraic curve over k (**II**, 7.4.2). Since the local rings of X are integrally closed and of dimension 1, they are discrete valuation rings and hence regular (**II**, 7.1.6); consequently X is a regular k-scheme (and *a fortiori* satisfies (R_n) for every $n \geqslant 0$). To say that X is geometrically integral then means that its field K of rational functions is a separable and primary extension of k (4.6.3).

Now let k be a nonperfect field of characteristic $p > 2$, and let $a \in k$ be an element not belonging to k^p. Let $\mathrm{B} = k[\mathrm{S}, \mathrm{T}]$ be the polynomial ring in two indeterminates. The polynomial

$$\mathrm{P}(\mathrm{S}, \mathrm{T}) = \mathrm{T}^2 - \mathrm{S}^p + a$$

is irreducible in B, since one checks at once that $\mathrm{S}^p - a$ is not a square in $k[\mathrm{S}]$; hence $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, where $\mathrm{A} = \mathrm{B}/\mathrm{PB}$, is an irreducible affine curve over k. To see that X is regular, it suffices to show that it is normal (**II**, 7.4.5). But $\mathrm{A} = k[\mathrm{S}][t]$, where t is a root of P considered as a polynomial in T over $k[\mathrm{S}]$, so the field $\mathrm{K} = \mathrm{R}(\mathrm{X})$ of rational functions of X is the field of fractions $k(\mathrm{S})[\mathrm{T}]/(\mathrm{P})$ of A, a quadratic extension of $k(\mathrm{S})$, hence separable over $k(\mathrm{S})$ and *a fortiori* over k. Since 2 is invertible in k, one checks at once that A is the integral closure of $k[\mathrm{S}]$ in K; thus A is integrally closed, and X is regular. Moreover, if an element $f + tg$ of K (with $f, g \in k(\mathrm{S})$) is algebraic over k, then so are its norm and trace over $k(\mathrm{S})$; since $k(\mathrm{S})$ is a purely transcendental extension of k, it follows readily that $f \in k$ and $g = 0$. Thus k is algebraically closed in K, and *a fortiori* K is a primary extension

of k (4.3.1). Nevertheless, if $k' = k(a^{1/p})$, then $\mathrm{X}' = \mathrm{X} \otimes_k k'$ is not normal: in $k'[\mathrm{S}]$ one has **IV-2** | 148

$$\mathrm{S}^p - a = (\mathrm{S} - a^{1/p})^p,$$

and X' is therefore isomorphic to $\mathrm{Spec}(\mathrm{A}')$, where $\mathrm{A}' = k'[\mathrm{S}][t']$ and t' is a root of $\mathrm{T}^2 - \mathrm{S}^p$. The ring A' is not integrally closed: the element $t'' = t'/\mathrm{S}$ of its field of fractions $\mathrm{K}' = k'(\mathrm{S})[t']$ satisfies the integral-dependence equation

$$(t'')^2 - \mathrm{S}^{p-2} = 0$$

over A', but does not belong to A'. In classical terminology, the "genus" over k' of the field K' of rational functions of X' is *strictly smaller* than that of K over k.

(ii) As recalled in (i), it follows from (4.6.1) that when X is not a geometrically reduced algebraic prescheme over k, there exist finite radicial extensions k' of k such that $X' = X \otimes_k k'$ is not reduced. It is interesting to give an example where X is a regular scheme over k, such that k is *algebraically closed* in the field K of rational functions of X. Let k be a field of characteristic $p > 0$, in which there exist two elements a, b forming a p-free family over k^p. Denoting again by B the ring $k[S, T]$, consider this time the polynomial $P(S, T) = T^p - aS^p - b$; as $aS^p + b$ is not a p-th power in $k(S)$, P is irreducible as a polynomial of $k(S)[T]$, and the scheme $X_0 = \mathrm{Spec}(B/PB)$ is therefore an integral affine curve over k, whose field of rational functions is $K = k(S)[t]$, where t is a root of P; let us show that k is *algebraically closed* in K. Suppose indeed that K contains an element z algebraic over k and not in k; one would then also have $z \notin k(S)$, so $[k(S)[z] : k(S)] > 1$; as $[K : k(S)] = p$, and as K is radicial over $k(S)$, one would have $z^p \in k(S)$, so $c = z^p \in k$ since z is algebraic over k; but then one would have $t^p = aS^p + b \in k^p(S^p)(c)$ hence a and b would belong to $k^p(c)$, which is absurd. So X is the normalisation of the curve X_0 in K, which is therefore a normal curve (and hence *regular*) over k. If $k' = k(a^{1/p}, b^{1/p})$, it is clear that $aS^p + b$ is a p-th power in $k'(S)$, hence $K \otimes_k k'$ is *not reduced*, nor *a fortiori* the scheme $X \otimes_k k'$.

Definition (6.7.6). — Let k be a field, X a k-prescheme locally Noetherian. We say that X is *geometrically regular* at a point x (resp. has the property (R_n) *geometric*) at a point x, resp. is geometrically normal at a point x) if, *for every* finite extension k' of k, $X' = X_{(k')} = X \otimes_k k'$ is regular (resp. has the property (R_n), resp. is normal) at every point x' of X' whose projection in X is x. We say that X is *geometrically regular* (resp. has the property (R_n) *geometric*, or equivalently is geometrically regular in codimension $\leqslant n$), resp. is *geometrically normal*) if X is geometrically regular (resp. has the property (R_n) geometric, resp. is geometrically normal) at every point.

We say that an algebra A over k is a *geometrically regular ring* (resp. geometrically normal, resp. geometrically reduced, resp. having the property (R_n) geometric) if $\mathrm{Spec}(A)$ has the same property.

If $X = \mathrm{Spec}(K)$, where K is an extension of k, then saying that X is geometrically regular, or geometrically normal, or *geometrically reduced*, amounts to saying that K is a *separable* extension of k: this follows from (4.6.1) and the fact that if K is a separable extension of k and k' a finite extension of k, $K \otimes_k k'$ is composed of a direct product of a finite number of fields (Bourbaki, *Alg.*, chap. VIII, § 7, no 3, cor. 1 of th. 1).

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Proposition (6.7.7). — *Let k be a field, X a k-prescheme locally Noetherian, x a point of X; denote by $Q(k')$ the relation "k' is an extension of k, and $\boldsymbol{P}(X_{(k')}, x')$ is true for every point x' of $X_{(k')}$ whose projection in X is x", where $\boldsymbol{P}(Z, z)$ is one of the properties* a), b), c) *of the statement* (6.7.4). *Then the following properties are equivalent:*

a) *$Q(k')$ is true for every finite extension k' of k.*
b) *$Q(k')$ is true for every finite radicial extension k' of k.*
c) *$Q(k')$ is true for every extension of finite type k' of k.*

Suppose in addition that X is locally of finite type over k; then the three preceding properties are also equivalent to the following:

d) *$Q(k')$ is true for every extension k' of k.*
e) *$Q(k')$ is true for a perfect extension k' of k.*
f) *$Q(k')$ is true for every extension $k' = k^{p^{-s}}$, where p is the characteristic exponent of k, and $s > 0$.*

Proof. To prove the equivalence of *a*), *b*) and *c*), it clearly suffices to show that *b*) implies *c*). Let K be then an extension of finite type of k, which is then the field of fractions of a k-algebra of finite type A; set $Y = \mathrm{Spec}(A)$. We know (4.6.6) that there exists a finite and radicial extension k' of k such that $Y' = (Y \otimes_k k')_{\mathrm{red}}$ is separable over k'; if η' is the generic point of Y', $K' = \boldsymbol{k}(\eta')$ is a separable extension of k' by (4.6.1). This being so, $\boldsymbol{P}(X_{(k')}, x')$ is true by hypothesis for every point x' of $X_{(k')}$ above x, so it follows from (6.7.4) that $\boldsymbol{P}(X_{(K')}, x'')$ is true for every point x'' of $X_{(K')}$ above x, since x'' is above a point x' of $X_{(k')}$, itself above x, and that K' is separable over k'. But we also have

$X_{(K')} = (X_{(K)})_{(K')}$ and for every $x_0 \in X_{(K)}$ above x, there exists $x'' \in X_{(K')}$ above x_0; it therefore follows from (6.7.4) that $\boldsymbol{P}(X_{(K)}, x_0)$ is true.

Suppose now that X is locally of finite type over k, so that for *every* extension K of k, $X_{(K)}$ is locally Noetherian. As a radicial extension of k is isomorphic to a sub-extension of a perfect extension of k, it follows immediately from (6.7.4) that $e)$ implies $f)$; also, every finite radicial extension of k is contained in an extension $k^{p^{-s}}$, so $f)$ implies $b)$; $d)$ trivially implies $e)$, and finally $e)$ implies $d)$. Indeed, if K is any extension *whatever* of k, there exists an extension K′ of k containing (up to k-isomorphisms) k' and K; since K′ is a separable extension of k', one deduces from (6.7.4) that $Q(k')$ is true, then that $Q(K)$ is true, reasoning as in the first part of the proof. It therefore suffices to prove that $b)$ *implies* $e)$. We shall take $k' = k^{p^{-\infty}}$ in $e)$.

The question being local on X, one can furthermore limit oneself to the case where X is affine and of finite type over k, by replacing X by a neighbourhood of x; let then $X = \mathrm{Spec}(B)$, where B is a k-algebra of finite type; furthermore, by definition of the property $\boldsymbol{P}$ in the cases considered, one can also replace X by the local scheme of X at the point x, hence suppose that B is a *local* Noetherian ring. Set $B' = B \otimes_k k'$; k' is the inductive limit of its finite sub-extensions k'_λ and if one sets $B'_\lambda = B \otimes_k k'_\lambda$, the morphism $f_\lambda : \mathrm{Spec}(B') \to \mathrm{Spec}(B'_\lambda)$ is a *homeomorphism* over $\mathrm{Spec}(B'_\lambda)$ for every λ;

indeed f_λ is bijective by virtue of (**I**, 3.5.2), (3.5.7) and (3.5.8); on the other hand, it is closed by (**II**, 6.1.10). One can now apply (5.13.6) which proves (by virtue of the hypothesis $b)$) that $b)$ implies $e)$ when $\boldsymbol{P}(Z, z)$ is the property (R_n) at the point z. The case where $\boldsymbol{P}$ is the property of being regular (i.e. of having the property (R_n) for every n) follows trivially. Finally, taking into account Serre's normality criterion (5.8.6), $b)$ implies $e)$ further when $\boldsymbol{P}(Z, z)$ is the property of being normal at the point z, since this is equivalent to saying that Z has at the point z the properties (S_2) and (R_1): one applies what precedes for $n = 1$, and (6.7.2), (ii) for $n = 2$. □ **IV-2** | 150

Corollary (6.7.8). — *Let k be a field; for a k-prescheme locally Noetherian X and a point $x \in X$, denote by $\boldsymbol{P}(X, x)$ any of the following properties:*

(i) $\mathrm{coprof}(\mathscr{O}_x) \leqslant n$.

(ii) $\mathscr{O}_x$ is a Cohen-Macaulay ring.

(iii) X satisfies the property (S_n) at the point x.

(iv) X is geometrically regular at the point x.

(v) X has the property (R_n) geometric at the point x.

(vi) X is geometrically normal at the point x.

(vii) X is geometrically reduced (i.e. separable) at the point x.

Let k' be an extension of k; one supposes either that k' is of finite type, or that X is locally of finite type over k, so that $X' = X \otimes_k k'$ is locally Noetherian. Let x' be a point of X′ whose projection in X is x. Then, for $\boldsymbol{P}(X, x)$ to be true, it is necessary and sufficient that $\boldsymbol{P}(X', x')$ be so.

Proof. This has already been seen for the property (vii) (4.6.11), and for the properties (i), (ii) and (iii) (6.7.1). For (iv), (v) and (vi), this follows from (6.7.7): indeed, the fact that the condition is necessary follows from the equivalence of the criteria $a)$ and $c)$, and the equivalence of $c)$ and $d)$ when X is locally of finite type over k. To see that the condition is sufficient, let k'' be a finite radicial extension of k; one can always consider k' and k'' as sub-extensions of an extension K of k; set $X'' = X \otimes_k k''$, $X_0 = X \otimes_k K$, and note that since k'' is a radicial extension of k, there is only *one* point x'' of X″ above x (**I**, 3.5.7) and (3.5.8). If then x_0 is any point of X_0 above x, it is of the same kind as $\boldsymbol{P}(X_0, x_0)$ by virtue of (6.7.7), $c)$ and $d)$), and if k' is an extension of k of finite type, one can suppose that it is also locally of finite type over k', X′ is also locally of finite type. One deduces then from (6.7.4) that the property $Q(k'')$ is true (with the notations of (6.7.7)), hence $\boldsymbol{P}(X, x)$ is true by (6.7.7), $b)$). □

6.8. Regular, normal, reduced, smooth morphisms

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Definition (6.8.1). — Let X, Y be two preschemes, $f : \mathrm{X} \to \mathrm{Y}$ a morphism such that the fibre $f^{-1}(y)$ is a locally Noetherian prescheme for every $y \in \mathrm{Y}$, x a point of X. We say that f is respectively a morphism:

(i) of coprofundity $\leqslant n$ at the point x;
(ii) of Cohen-Macaulay at the point x; **IV-2 | 151**
(iii) (S_n) at the point x;
(iv) regular at the point x;
(v) (R_n) at the point x;
(vi) normal at the point x;
(vii) reduced at the point x;

if f is flat at the point x, and if in addition the corresponding property $\boldsymbol{P}(f^{-1}(f(x)), x)$ (notation of (6.7.8)) is true.

We say that f is *smooth* at the point x if it is locally of finite presentation in a neighbourhood of x and if it is regular at the point x. We say respectively that f is a morphism: of coprofundity $\leqslant n$, of Cohen-Macaulay, (S_n), regular, (R_n), normal, reduced, smooth, if it has the corresponding property at every point of X.

Proposition (6.8.2). — *Let* X, Y *be two locally Noetherian preschemes,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism,* x *a point of* X*. Denote by* $\boldsymbol{M}(f, x)$ *one of the properties (i) to (vii) of the definition* (6.8.1)*, or the property of being smooth at the point* x*. Let* Y' *be a locally Noetherian prescheme,* $g : \mathrm{Y}' \to \mathrm{Y}$ *a morphism,* $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$*,* $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$*. Suppose that* f *or* g *is locally of finite type. Then, for every* $x' \in \mathrm{X}'$ *above* x*, the property* $\boldsymbol{M}(f, x)$ *implies* $\boldsymbol{M}(f', x')$*.*

Proof. Set $y = f(x)$, $y' = f'(x')$; by transitivity of fibres (**I**, 3.6.4), since either $f^{-1}(y)$ or $\boldsymbol{k}(y')$ is locally of finite type over $\boldsymbol{k}(y)$ (**I**, 6.4.11), it follows from (6.7.8) that the properties $\boldsymbol{P}(f^{-1}(y), x)$ and $\boldsymbol{P}(f'^{-1}(y'), x')$ are equivalent; furthermore, since f is flat at the point x, f' is flat at the point x' (2.1.4), which proves the proposition, taking into account (1.4.3), (iii). ☐

Proposition (6.8.3). — *For a morphism* f *of locally Noetherian preschemes, let* $\boldsymbol{M}(f)$ *be one of the following properties: being Cohen-Macaulay,* (S_n)*, regular,* (R_n)*, normal, reduced.*

(i) Let X, Y, Z *be three locally Noetherian preschemes,* $f : \mathrm{X} \to \mathrm{Y}$*,* $g : \mathrm{Y} \to \mathrm{Z}$ *two morphisms. If* $\boldsymbol{M}(f)$ *and* $\boldsymbol{M}(g)$ *are true,* $\boldsymbol{M}(g \circ f)$ *is true.*
(ii) Conversely, if f *is surjective and if* $\boldsymbol{M}(f)$ *and* $\boldsymbol{M}(g \circ f)$ *are true, then* $\boldsymbol{M}(g)$ *is true.*
(iii) Let X, Y, Y′ *be three locally Noetherian preschemes,* $f : \mathrm{X} \to \mathrm{Y}$*,* $h : \mathrm{Y}' \to \mathrm{Y}$ *two morphisms; set* $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$*,* $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$*. Suppose that* f *or* h *is locally of finite type. If* $\boldsymbol{M}(f)$ *is true, then so is* $\boldsymbol{M}(f')$*; the converse is true when* h *is faithfully flat.*

Proof. The conclusions of (i) and (iii) are also true when $\boldsymbol{M}$ is the property of being smooth and (in (iii)) h is quasi-compact.

(i) We already know that if f and g are flat, so is $u = g \circ f$, and that if f and $g \circ f$ are flat and f is surjective for every $z \in \mathrm{Z}$, the morphism $f_z = f \otimes \mathbf{1}_{\boldsymbol{k}(z)} : u^{-1}(z) \to g^{-1}(z)$ is flat (resp. faithfully flat if f is) and for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (**I**, 3.6.4). If $\boldsymbol{M}(f)$ and $\boldsymbol{M}(g)$ are true, $\boldsymbol{M}(g \circ f)$ is then true for the cases where $\boldsymbol{M}$ is the property of being Cohen-Macaulay or (S_n), by virtue of (6.6.1). On the other hand, let k' be a finite extension of $\boldsymbol{k}(z)$; setting $\mathrm{Y}'_z = g^{-1}(z) \otimes_{\boldsymbol{k}(z)} k'$, $\mathrm{X}'_z = u^{-1}(z) \otimes_{\boldsymbol{k}(z)} k'$, and $f'_z = f_z \otimes \mathbf{1}_{k'} : \mathrm{X}'_z \to \mathrm{Y}'_z$; the morphism f'_z
is flat (resp. faithfully flat) and for every $y' \in \mathrm{Y}'_z$, the fibre $f_z'^{-1}(y')$ is isomorphic to $f^{-1}(y) \otimes_{\boldsymbol{k}(y)} \boldsymbol{k}(y')$, denoting by y the image of y' in $g^{-1}(z)$ (**I**, 3.6.4). When $\boldsymbol{M}$ is the property of being regular, (R_n), normal or reduced, the hypothesis that $\boldsymbol{M}(f)$ and $\boldsymbol{M}(g)$ are true implies that Y'_z and each of the fibres $f_z'^{-1}(y')$ for $y' \in \mathrm{Y}'_z$ have the corresponding property among the properties c), d), e), f) of (6.6.1); one deduces from (6.6.1), (i) that X'_z has the same property, hence $\boldsymbol{M}(g \circ f)$ is true. **IV-2 | 152**

(ii) Conversely, the hypothesis that $\boldsymbol{M}(g \circ f)$ and $\boldsymbol{M}(f)$ are true and that f is surjective implies from (6.6.1), (ii)), f'_z being surjective for every z; hence $\boldsymbol{M}(g)$ is true.

(iii) The first assertion follows immediately from (6.8.2). On the other hand, if h is faithfully flat and f' flat, f is flat (2.4.1); since, with the notations of (6.8.2), the properties $\boldsymbol{P}(f^{-1}(y), x)$ and $\boldsymbol{P}(f'^{-1}(y'), x')$ are equivalent (6.7.8), one sees that $\boldsymbol{M}(f)$ and $\boldsymbol{M}(f')$ are then equivalent.

Finally, the last assertion of the proposition follows from (1.4.3), (iii) and (2.7.1), (iv). □

Remarks (6.8.4). — (i) If f is faithfully flat, g flat and if $g \circ f$ is of coprofundity $\leqslant n$, then g is of coprofundity $\leqslant n$, as follows from the reasoning of (6.6.2).

(ii) When, in (6.8.1), one takes for Y the spectrum of a field k, the notions (iv), (v) and (vi) reduce to those defined in (6.7.5). It is clear that these latter are relative to the base field k. The definition (6.8.1) thus leads to saying that a prescheme X is "regular (resp. (R_n), resp. normal) over k" instead of saying that it is "geometrically regular (resp. (R_n), resp. normal) relative to k"; one will avoid confusing this notion with the property of being regular (resp. (R_n), resp. normal) which is independent of k. One can make in this connection the same remarks as in (4.5.12).

Proposition (6.8.5). — *Let k be a field, X, Y two k-preschemes locally Noetherian, at least one of which is locally of finite type over k. For a k-prescheme Z, denote by $\boldsymbol{P}(\mathrm{Z})$ one of the properties $c)$ to $f)$ of (6.6.1); the property "$\boldsymbol{P}(\mathrm{Z})$ geometric" is then defined in (6.7.6) (resp. (4.6.1)). Then:*

(i) If X has the property $\boldsymbol{P}$, and Y the property $\boldsymbol{P}$ geometric, $\mathrm{X} \times_k \mathrm{Y}$ has the property $\boldsymbol{P}$.

(ii) If X and Y have the property $\boldsymbol{P}$ geometric, so does $\mathrm{X} \times_k \mathrm{Y}$.

Proof. Indeed, let $f : \mathrm{X} \times_k \mathrm{Y} \to \mathrm{X}$, $g : \mathrm{X} \to \operatorname{Spec}(k)$ be the structural morphisms, which are faithfully flat (2.2.13). The hypothesis that Y has the property $\boldsymbol{P}$ geometric implies by virtue of (6.8.2) that $\boldsymbol{M}(f)$ is true, $\boldsymbol{M}$ being the property (6.8.3) which corresponds to $\boldsymbol{P}$; under the hypothesis (ii), $\boldsymbol{M}(g)$ is also true, hence the assertion (ii) follows from (6.8.3), (i). As for the assertion (i), it follows directly from (6.6.1), (i). □

Theorem (6.8.6). — *Let X, Y be two locally Noetherian preschemes, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, x a point of X, $y = f(x)$. The following properties are equivalent:*

a) *f is smooth at the point x.*

b) *f is regular at the point x.* **IV-2 | 153**

c) *$\mathscr{O}_x$ is a formally smooth $\mathscr{O}_y$-algebra* ($\mathbf{0}$, 19.3.1) *for the adic topologies on $\mathscr{O}_x$ and $\mathscr{O}_y$.*

c$'$) *$\mathscr{O}_x$ is a formally smooth $\mathscr{O}_y$-algebra* ($\mathbf{0}$, 19.3.1) *for the discrete topologies on $\mathscr{O}_x$ and $\mathscr{O}_y$.*

Proof. The equivalence of $a)$ and $b)$ follows from the definitions (6.8.1) and from the fact that for locally Noetherian preschemes, morphisms locally of finite type are locally of finite presentation (1.4.2).

Secondly, for $\mathscr{O}_x$ to be a formally smooth $\mathscr{O}_y$-algebra for the adic topologies, it is necessary and sufficient that $\mathscr{O}_x$ be flat over $\mathscr{O}_y$ and that $\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ be formally smooth over $\boldsymbol{k}(y)$ for its adic topology ($\mathbf{0}$, 19.7.1). The latter condition is equivalent to geometric regularity over $\boldsymbol{k}(y)$ ($\mathbf{0}$, 19.6.6); this proves the equivalence of $b)$ and $c)$. Finally, to prove the equivalence of $c)$ and $c')$, one may reduce to the affine case $\mathrm{Y} = \operatorname{Spec}(\mathrm{A})$, $\mathrm{X} = \operatorname{Spec}(\mathrm{C})$, where A is Noetherian and C is a finite-type Noetherian A-algebra, say $\mathrm{C} = \mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_n]/\mathfrak{J}$. Since $\mathfrak{J}/\mathfrak{J}^2$ is a finitely presented C-module, the equivalence follows from ($\mathbf{0}$, 22.6.4). □

Corollary (6.8.7). — *Let X, Y be two locally Noetherian preschemes, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type. Then the set of points $x \in \mathrm{X}$ where f is smooth (or regular) is open in X.*

Proof. Indeed it follows from ($\mathbf{0}$, 22.6.5) that the set of $x \in \mathrm{X}$ satisfying the condition $c')$ of (6.8.6) is open in X, and we conclude by (6.8.6). □

Remark (6.8.8). — In (17.5.1), we shall show that the equivalence of $b)$ and $c')$ in (6.8.6), as well as the corollary (6.8.7), remain valid without the Noetherian hypothesis on X and Y, provided that one restricts to morphisms *locally of finite presentation*.

6.9. The theorem of generic flatness

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Theorem (6.9.1). — *Let* Y *be a locally Noetherian and integral prescheme,* $u : \mathrm{X} \to \mathrm{Y}$ *a morphism of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module. There then exists a non-empty open subset* U *of* Y *such that* $\mathscr{F}|u^{-1}(\mathrm{U})$ *is flat over* U*.*

Proof. One can clearly limit oneself to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ is affine; then X is a finite union of affine opens X_i of finite type over Y; if, for each i, there exists a non-empty open U_i in Y such that $\mathscr{F}|(\mathrm{X}_i \cap u^{-1}(\mathrm{U}_i))$ is flat over U_i, it is clear that taking for U the intersection of the U_i, we shall have answered the question; one can therefore suppose that $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, where B is an A-algebra of finite type. We then have $\mathscr{F} = \widetilde{\mathrm{M}}$, where M is a B-module of finite type; the theorem will follow from the following: ☐

Lemma (6.9.2). — *Let* A *be an integral Noetherian ring,* B *a finite-type* A*-algebra, and* M *a finite* B*-module. Then there exists* $f \neq 0$ *in* A *such that* M_f *is a free* A_f*-module.*

Proof. Denote by K the field of fractions of A; then $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K}$ is an algebra of finite type over K and $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K}$ is a $(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K})$-module of finite type. We shall reason by induction on the *dimension* n of the support of $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K}$, which is either $-\infty$ or finite and $\geqslant 0$. Suppose first $n = -\infty$, i.e. $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K} = 0$; if $(m_i)_{1 \leqslant i \leqslant r}$ is a system of generators of the B-module M, there therefore exists $f \neq 0$ in A such that $f m_i = 0$ for $1 \leqslant i \leqslant r$; hence $\mathrm{M}_f = 0$ and the lemma is true in this case. Suppose now $n \geqslant 0$. We know that there exists a composition series $\mathrm{M} = \mathrm{M}_1 \supset \mathrm{M}_2 \supset \ldots \supset \mathrm{M}_q = 0$ of the B-module M such that each of the quotients $\mathrm{N}_i = \mathrm{M}_i/\mathrm{M}_{i+1}$ is isomorphic to a B-module of the form $\mathrm{B}/\mathfrak{p}_i$, where $\mathfrak{p}_i$ is a prime ideal of B (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, th. 1). If the theorem is true for each of the N_i, there exists for each i an $f_i \neq 0$ in A such that $(\mathrm{N}_i)_{f_i}$ is an A_{f_i}-module free for $1 \leqslant i \leqslant q-1$. But $(\mathrm{N}_i)_f = (\mathrm{M}_i)_f/(\mathrm{M}_{i+1})_f$ (**0**, 1.3.2) and since an extension of free modules is free, we deduce that M_f is free, setting $f = f_1 f_2 \cdots f_{q-1}$, and it therefore suffices (in the case where M = B and B is *integral*) to show that there exists $g \neq 0$ in A and elements t_i $(1 \leqslant i \leqslant m)$ algebraically independent over A and such that B_g is *integral* over $\mathrm{A}_g[t_1, \ldots, t_m]$. One can replace A by A_g, B by B_g and suppose that B is integral over $\mathrm{C} = \mathrm{A}[t_1, \ldots, t_m]$, hence a C-module of finite type and *without torsion*. We know furthermore (4.1.2) that the dimension of $\mathrm{Spec}(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K})$ is equal to m, hence $m = n$.

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This being so, if h is the rank of the C-module without torsion B, there exists a short exact sequence of C-modules

$$0 \longrightarrow \mathrm{C}^h \longrightarrow \mathrm{B} \longrightarrow \mathrm{M}' \longrightarrow 0$$

where M' is a torsion C-module of finite type; the support of M' does not therefore contain the generic point of Spec(C) (**I**, 7.4.6) and hence the support of $\mathrm{M}' \otimes_{\mathrm{A}} \mathrm{K}$ does not contain the generic point of $\mathrm{Spec}(\mathrm{C} \otimes_{\mathrm{A}} \mathrm{K})$ (**I**, 9.1.13.1); one concludes (4.1.2.1) that its dimension is $< n$. By the induction hypothesis, there exists $f \neq 0$ in A such that M'_f is a free A_f-module; moreover C^h_f is a free A_f-module, and therefore so is B_f, which by (**0**, 1.3.2) is an extension of free modules. ☐

Corollary (6.9.3). — *Let* S *be a Noetherian prescheme,* $u : \mathrm{X} \to \mathrm{S}$ *a morphism of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module. There then exists a partition* $(\mathrm{S}_i)_{1 \leqslant i \leqslant n}$ *of* S *into a finite number of locally closed subsets of* S*, such that, if one still denotes by* S_i *the reduced sub-prescheme having* S_i *for underlying space, and if one sets* $\mathrm{X}_i = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}_i$*, the* $\mathscr{O}_{\mathrm{X}_i}$*-Module* $\mathscr{F}_i = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{O}_{\mathrm{S}_i}$ *is flat over* S_i*.*

Proof. We proceed by Noetherian induction (**0**, 2.2.2) on the set of closed subsets T of S such that the reduced closed sub-prescheme with underlying space T satisfies the conclusion of (6.9.3). One can therefore limit oneself to proving the corollary for S by supposing that it is true for every reduced closed sub-prescheme whose underlying space is a closed subset distinct from S. Since the morphism $\mathrm{S}_{\mathrm{red}} \to \mathrm{S}$ is of finite type and surjective, one can replace S by $\mathrm{S}_{\mathrm{red}}$, hence suppose S reduced and non-empty. Since S is Noetherian, the interior T of an irreducible component of S is non-empty, and the induced prescheme on T is integral; there therefore exists by virtue of (6.9.1) a non-empty open $\mathrm{U} \subset \mathrm{T}$

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such that $\mathscr{F}|u^{-1}(\mathrm{U})$ is flat over U. If Y is then the reduced closed sub-prescheme of S with underlying space $\mathrm{S} - \mathrm{U}$, there exists by hypothesis a partition (Y_i) of Y into locally closed subsets of Y (hence of S) such that $\mathscr{F}_i = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{O}_{\mathrm{Y}_i}$ is flat over Y_i for every i; it is clear that the Y_i and U form a partition answering the question. ☐

6.10. Dimension and depth of a module normally flat along a closed sub-prescheme

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(6.10.1). Let X be a locally Noetherian prescheme, $\mathscr{J}$ a quasi-coherent Ideal of $\mathscr{O}_X$, Y the closed sub-prescheme of X defined by $\mathscr{J}$, $j : Y \to X$ the canonical injection. For every integer $k \geqslant 0$, the $\mathscr{O}_X$-Module $\mathscr{J}^k\mathscr{F}/\mathscr{J}^{k+1}\mathscr{F}$ is annihilated by $\mathscr{J}$, hence of the form $j_*(\mathscr{G}_k)$, where $\mathscr{G}_k = j^*(\mathscr{J}^k\mathscr{F}/\mathscr{J}^{k+1}\mathscr{F}) = j^*(\mathscr{J}^k\mathscr{F})$ is a coherent $\mathscr{O}_Y$-Module. By abuse of language, we shall denote by $\mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{F})$ the graded $\mathscr{O}_Y$-Module equal to the direct sum

$$\bigoplus_{k=0}^{\infty}\mathscr{G}_k = j^*\Big(\bigoplus_{k=0}^{\infty}\mathscr{J}^k\mathscr{F}/\mathscr{J}^{k+1}\mathscr{F}\Big);$$

in particular, we have $\mathrm{gr}^0_{\mathscr{J}}(\mathscr{F}) = \mathscr{G}_0 = \mathscr{F}\otimes_{\mathscr{O}_X}\mathscr{O}_Y = j^*(\mathscr{F})$. We shall say (following Hironaka) that $\mathscr{F}$ is *normally flat along* Y if $\mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{F})$ is a flat $\mathscr{O}_Y$-Module. This amounts to the same (2.1.2) and (**0**, 6.1.2) as saying that each of the $\mathscr{O}_Y$-Modules $\mathscr{G}_k = j^*(\mathscr{J}^k\mathscr{F})$ is *locally free*.

Proposition (6.10.2). — *Let* X *be a locally Noetherian prescheme,* Y *a closed integral sub-prescheme of* X*. For every coherent $\mathscr{O}_X$-Module $\mathscr{F}$, there exists an open* U *of* X *such that* $Y\cap U \neq \varnothing$ *and that* $\mathscr{F}|U$ *is normally flat along* $Y\cap U$.

Proof. Indeed, let $\mathscr{J}$ be the coherent ideal of $\mathscr{O}_X$ defining Y. The $\mathscr{O}_Y$-algebra $\mathscr{B} = \mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{O}_X)$ is quasi-coherent and *of finite type*, since it is generated by $\mathrm{gr}^1_{\mathscr{J}}(\mathscr{O}_X)$, the inverse image of $\mathscr{J}/\mathscr{J}^2$. Since $\mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{F})$ is a quasi-coherent $\mathscr{B}$-module generated by $\mathrm{gr}^1_{\mathscr{J}}(\mathscr{F})$, it is a $\mathscr{B}$-Module *of finite type*. If one sets $Z = \mathrm{Spec}(\mathscr{B})$, the structural morphism $u : Z \to Y$ is of finite type, and if $\mathscr{H}$ is the coherent $\mathscr{O}_Z$-Module such that $u_*(\mathscr{H}) = \mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{F})$, it suffices to apply to u and to $\mathscr{H}$ the theorem of generic flatness (6.9.1) to prove the proposition. □

Proposition (6.10.3). — *The notations being those of* (6.10.1), *suppose that $\mathscr{F}$ is normally flat along* Y*. Then:*

(i) $Y\cap\mathrm{Supp}(\mathscr{F})$ *is a part both open and closed of* Y *(in other words, a union of connected components of* Y*).*

(ii) If $\mathscr{J}$ is locally nilpotent, $\mathrm{Supp}(\mathscr{F})$ *is an open and closed part of* X*, and for every* $x \in \mathrm{Supp}(\mathscr{F})$*, we have*

$$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{Y,x}) \tag{6.10.3.1}$$

$$\mathrm{prof}(\mathscr{F}_x) = \mathrm{prof}(\mathscr{O}_{Y,x}) \tag{6.10.3.2}$$

$$\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{O}_{Y,x}). \tag{6.10.3.3}$$

Proof. (i) With the notations of (6.10.1), we have $Y\cap\mathrm{Supp}(\mathscr{F}) = \mathrm{Supp}(\mathscr{G}_0)$ (**I**, 9.1.13), and as by hypothesis $\mathscr{G}_0$ is a locally free $\mathscr{O}_Y$-Module of finite type, its support is at the same time open and closed in Y.

(ii) The hypothesis that $\mathscr{J}$ is locally nilpotent implies that the underlying spaces of X and Y are the same, whence the two first assertions of (ii), taking into account (i); it remains to prove (6.10.3.2) and (6.10.3.3) by difference. Let $(f_i)_{1\leqslant i\leqslant p}$ be a finite family of elements of the maximal ideal of $\mathscr{O}_{X,x}$ whose images in $\mathscr{O}_{Y,x} = \mathscr{O}_{X,x}/\mathscr{J}_x$ form a maximal $\mathscr{O}_{Y,x}$-regular sequence; the hypothesis on $\mathscr{F}$ and $\mathscr{J}$ implies that the $\mathscr{O}_{Y,x}$-module $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{F}_x)$ is free of finite type, hence the suite (f_i) is $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{F}_x)$-regular; one deduces that this suite is also $\mathscr{F}_x$-regular (**0**, 15.1.19). On the other hand, let n be the largest integer such that $\mathscr{F}_x^{(n)} = \mathscr{J}_x^n\mathscr{F}_x \neq 0$. The hypothesis implies also that $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{F}_x/\mathscr{F}_x^{(n)})$ is free of finite type, hence the suite (f_i) is also $(\mathscr{F}_x/\mathscr{F}_x^{(n)})$-regular (*loc. cit.*). Applying the lemma (3.4.1.4), one concludes by induction on i that there is an exact sequence IV-2 | 156

$$0 \longrightarrow \mathscr{F}_x^{(n)}\Big/\Big(\sum_{i=1}^{p} f_i\mathscr{F}_x^{(n)}\Big) \longrightarrow \mathscr{F}_x\Big/\Big(\sum_{i=1}^{p} f_i\mathscr{F}_x\Big).$$

But the hypothesis implies that $\mathscr{F}_x^{(n)}$ is a free $\mathscr{O}_{Y,x}$-module of finite type and $\neq 0$, hence $\mathscr{F}_x^{(n)}/(\sum_{i=1}^p f_i\mathscr{F}_x^{(n)})$ is isomorphic to a module of the form $(\mathscr{O}_{Y,x}/(\sum_{i=1}^p f_i\mathscr{O}_{Y,x}))^m$ with $m > 0$; as $\mathrm{prof}(\mathscr{O}_{Y,x}/(\sum_{i=1}^p f_i\mathscr{O}_{Y,x})) = 0$ (**0**, 16.4.6) one also has, by the characterisation (**0**, 16.4.6), (i) of modules of zero depth, $\operatorname{prof}(\mathscr{F}_x^{(n)}/(\sum_{i=1}^p f_i\mathscr{F}_x^{(n)})) = 0$, then $\operatorname{prof}(\mathscr{F}_x/(\sum_{i=1}^p f_i\mathscr{F}_x)) = 0$; the $(f_i)_x$ belonging to the maximal ideal of $\mathscr{O}_{Y,x}$ and forming a maximal $\mathscr{F}_x$-regular suite, this shows that $\operatorname{prof}(\mathscr{F}_x) = p$ (**0**, 16.4.6), (ii). ☐

Corollary (6.10.4). — *Let* $U_{S_n}(\mathscr{F})$ *(resp.* $U_{C_n}(\mathscr{F})$*) be the set of* $x \in X$ *such that* $\mathscr{F}$ *satisfies* (S_n) *(resp. such that* $\operatorname{coprof}(\mathscr{F}_x) \leqslant n$*). If* $\mathscr{F}$ *is normally flat along* Y*, and if* $\mathscr{J}$ *is locally nilpotent, then* $U_{S_n}(\mathscr{F}) = U_{S_n}(\mathscr{O}_Y)$ *and* $U_{C_n}(\mathscr{F}) = U_{C_n}(\mathscr{O}_Y)$.

Proposition (6.10.5). — *The notations being those of (6.10.1), suppose that* Y *is connected and non-empty, and that* $\mathscr{F}$ *is normally flat along* Y*. For every integer* $n \geqslant 0$*, set*

(6.10.5.1)
$$r(n) = \operatorname{rg}_{\mathscr{O}_Y}(\operatorname{gr}^n_{\mathscr{J}}(\mathscr{F}))$$

(the locally free $\mathscr{O}_Y$*-Module* $\operatorname{gr}^n_{\mathscr{J}}(\mathscr{F})$ *being necessarily of constant rank). Then:*

- *(i) There exists a polynomial* $P \in \mathbf{Q}[T]$ *such that* $P(n) = r(n)$ *for all sufficiently large* n*.*
- *(ii) We have* $Y \cap \operatorname{Supp}(\mathscr{F}) = \varnothing$ *(in other words* $\mathscr{J} \,|\, \mathscr{J}\mathscr{F} = 0$*), or* $Y \cap \operatorname{Supp}(\mathscr{F}) = Y$ *(in other words* $Y \subset \operatorname{Supp}(\mathscr{F})$*).*

In the second case, denote $d-1$ *the degree of* P*; for every maximal point* y *of* Y*, we have*

(6.10.5.2)
$$\dim(\mathscr{F}_y) = d$$

and in particular

(6.10.5.3)
$$\operatorname{codim}(Y, \operatorname{Supp}(\mathscr{F})) = d.$$

- *(iii) Suppose* $Y \subset \operatorname{Supp}(\mathscr{F})$*. For every* $x \in Y$*, we have*

(6.10.5.4)
$$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{Y,x}) + d.$$

More precisely, there exist in $\mathscr{J}_x$ *elements* f_i $(1 \leqslant i \leqslant d)$ *forming part of a system of parameters for* $\mathscr{F}_x$ (**0**, 16.3.6) *and such that, in* $\operatorname{Spec}(\mathscr{O}_{X,x})$*, we have* $V(\mathscr{J}_x) = V(\sum_{i=1}^d f_i\mathscr{O}_{X,x}) \cap \operatorname{Supp}(\mathscr{F})$*.*

Proof. As Y is assumed connected, the first assertion of (ii) follows from (6.10.3), (i). If $Y \cap \operatorname{Supp}(\mathscr{F}) = \varnothing$, assertion (i) is trivial with $P = 0$. Suppose now that $\operatorname{Supp}(\mathscr{F}) \supset Y$, and let y be a maximal point of Y. Since then $Y_y = \operatorname{Supp}(\mathscr{O}_{Y,y})$, the module $\mathscr{F}_y/\mathscr{J}_y\mathscr{F}_y$ has finite length (3.1.4). Setting $s(n) = \operatorname{length}(\operatorname{gr}^n_{\mathscr{J}_y}(\mathscr{F}_y))$, we know that there exists a polynomial $R \in \mathbf{Q}[T]$ such that $s(n) = R(n)$ for n large enough, namely $R(n) = H(n+1) - H(n)$, where H is the Hilbert–Samuel polynomial of $\mathscr{F}_y$ for the $\mathscr{J}_y$-adic filtration (**0**, 16.2.1). Since the $\mathscr{O}_{Y,y}$-modules $\operatorname{gr}^n_{\mathscr{J}_y}(\mathscr{F}_y)$ are free by hypothesis, we have $r(n) = s(n)/m$, denoting by m the length of the $\mathscr{O}_{X,y}$-module $\mathscr{O}_{Y,y} = \mathscr{O}_{X,y}/\mathscr{J}_y$; thus (i) holds with $P = \frac{1}{m}R$. We know furthermore (**0**, 16.2.3) that $\deg(H) = \dim(\mathscr{F}_y)$, whence the relation (6.10.5.2); the relation (6.10.5.3) follows from (5.1.12.2). **IV-2** | 157

It remains to prove (iii) for any point $x \in Y$. Set $A = \mathscr{O}_{X,x}$, $\mathfrak{J} = \mathscr{J}_x$, and $M = \mathscr{F}_x$, so that $\mathscr{O}_{Y,x} = A/\mathfrak{J}$; let $S = \operatorname{gr}^\bullet_{\mathfrak{J}}(A)$, which is a graded $(A/\mathfrak{J})$-algebra of finite type, with positive degrees, such that $S_0 = A/\mathfrak{J}$, and generated by its homogeneous elements of degree 1; let finally $N = \operatorname{gr}^\bullet_{\mathfrak{J}}(M)$, which is a graded S-module of finite type, each homogeneous component N_n being by hypothesis a free $(A/\mathfrak{J})$-module of rank $r(n)$. Let $\mathfrak{m}$ be the maximal ideal of A, $k = A/\mathfrak{m}$ its residue field; $B = S \otimes_A k$ is a graded k-algebra of finite type, with positive degrees, generated by its homogeneous elements of degree 1 and such that $B_0 = k$, so that $\mathfrak{q} = B^+ = \bigoplus_{n \geqslant 1} B_n$ is a maximal ideal of B; $E = N \otimes_A k$ is a graded B-module of finite type such that $\operatorname{rg}_k(E_n) = r(n)$. Applying (**0**, 16.2.7) to B and E, choose homogeneous elements $t_i \in B_{n_i}$ $(1 \leqslant i \leqslant d)$, and lift each t_i to an element $f_i \in \mathfrak{J}^{n_i}$. For $n \geqslant \sup(n_i)$, consider the A-submodule $\sum_{i=1}^d f_i\mathfrak{J}^{n-n_i}M$ of M; as the homogeneous component of degree n of the submodule $\sum_i t_iE$ of E is equal to E_n as soon as n is large enough, one sees that, for n large enough, we have

$$\mathfrak{J}^n M = \sum_{i=1}^d f_i\mathfrak{J}^{n-n_i}M + \mathfrak{m}\mathfrak{J}^n M$$

and as $\mathfrak{J}^n M$ is an A-module of finite type, this implies, by Nakayama's lemma,

$$\mathfrak{J}^n M = \sum_{i=1}^d f_i\mathfrak{J}^{n-n_i}M \subset \sum_{i=1}^d f_i M.$$

If $\mathfrak{a}$ is the annihilator of M, we have therefore (**0**, 1.7.5) in $\mathrm{Spec}(A)$

$$V\Big(\sum_{i=1}^{d} f_i A\Big) \cap V(\mathfrak{a}) \subset V(\mathfrak{J}^n) \cap V(\mathfrak{a}) = V(\mathfrak{J}) \cap V(\mathfrak{a}) = V(\mathfrak{J})$$

since by hypothesis $Y \cap \mathrm{Spec}(A) = V(\mathfrak{J}) \cap \mathrm{Supp}(M) = V(\mathfrak{J})$. Since on the other hand the f_i belong to S, we have $V(\mathfrak{J}) \subset V(\sum_{i=1}^{d} f_i A)$, which proves the last relation of (iii). It remains to show that the f_i form part of a system of parameters for M. Replacing A by $A/\mathfrak{a}$ and the f_i by their images in $A/\mathfrak{a}$, one can limit oneself to the case where $\mathfrak{a} = 0$; one then has to show that we have $\dim(A/(\sum_{i=1}^{d} f_i A)) \leqslant \dim(A) - d$ and it remains therefore to prove (**0**, 16.3.7) that we have

$$\dim(A) \geqslant \dim(A/\mathfrak{J}) + d.$$

Now, let $\mathfrak{p}$ be a prime ideal of A containing $\mathfrak{J}$ and such that $\dim(A/\mathfrak{J}) = \dim(A/\mathfrak{p})$, so that $\mathfrak{p}$ is minimal among the prime ideals containing $\mathfrak{J}$. We have therefore $\mathfrak{p} = \mathfrak{i}_y$ where $y \in \mathrm{Spec}(A)$ is a maximal point of Y. But by virtue of (6.10.5.2) and the hypothesis $\mathrm{Spec}(A) = \mathrm{Supp}(M)$, we have $\dim(A_{\mathfrak{p}}) = d$; the inequality $\dim(A/\mathfrak{p}) + \dim(A_{\mathfrak{p}}) \leqslant \dim(A)$ (**0**, 16.1.4) completes the proof. □

Proposition (6.10.6). — *Let* X *be a locally Noetherian prescheme,* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module,* Y *a closed irreducible sub-prescheme of* X*, of generic point* $y \in \mathrm{Supp}(\mathscr{F})$*. There then exists a neighbourhood* U *open and non-empty of* y *in* X *such that, for every* $x \in U \cap Y$*, we have*

(6.10.6.1) $$\dim(\mathscr{F}_x) = \dim(\mathscr{F}_y) + \dim(\mathscr{O}_{Y,x})$$

(6.10.6.2) $$\mathrm{prof}(\mathscr{F}_x) = \mathrm{prof}(\mathscr{F}_y) + \mathrm{prof}(\mathscr{O}_{Y,x})$$

(6.10.6.3) $$\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{F}_y) + \mathrm{coprof}(\mathscr{O}_{Y,x}).$$

Proof. Let $Y' = Y_{\mathrm{red}}$, which is a closed integral sub-prescheme of Y having the same underlying space, and defined by a locally nilpotent Ideal $\mathscr{J}$ of $\mathscr{O}_Y$ (**I**, 6.1.6). It follows that $\mathrm{gr}^{\bullet}_{\mathscr{J}}(\mathscr{O}_v)$ is a coherent $\mathscr{O}_{v'}$-Module, and as Y' is integral, there exists an open neighbourhood V of y in Y' such that $\mathrm{gr}^{\bullet}_{\mathscr{J}}(\mathscr{O}_v)|V$ is locally free (**0**, 5.2.7); otherwise said, replacing X by a neighbourhood of y, one can suppose that $\mathscr{O}_v$ is *normally flat* along Y'; one then deduces from (6.10.3) that we have **IV-2** | 158

$$\dim(\mathscr{O}_{Y,x}) = \dim(\mathscr{O}_{Y',x}), \qquad \mathrm{prof}(\mathscr{O}_{Y,x}) = \mathrm{prof}(\mathscr{O}_{Y',x})$$

for every $x \in Y$; this allows us to limit ourselves to the case where the closed sub-prescheme Y is *integral*.

This being so, by virtue of (6.10.2), one can, replacing X by an open neighbourhood of y, suppose that $\mathscr{F}$ is *normally flat* along Y. Suppose now $p = \mathrm{prof}(\mathscr{F}_y)$; replacing at need X by an open neighbourhood of y, one can further suppose that there exist p sections f_i ($1 \leqslant i \leqslant p$) of $\mathscr{O}_X$ above X, forming an $\mathscr{F}$-regular suite, such that the $(f_i)_y$ belong to the maximal ideal $\mathfrak{m}_y$ of $\mathscr{O}_X$ at the point y and that $\mathrm{prof}(\mathscr{F}_y/(\sum_{i=1}^{p}(f_i)_y\mathscr{F}_y)) = 0$ (**0**, 16.4.6). If $\mathfrak{J}$ is the Ideal of $\mathscr{O}_X$ defining Y, the hypothesis $\mathrm{prof}(\mathscr{G}_y) = 0$ implies that $\mathscr{G}_y$ contains a sub-module isomorphic to $\mathscr{O}_{X,y}/\mathfrak{m}_y$ (**0**, 16.4.6); reasoning as above (taking into account (**0**, 5.3.8) and (5.2.2)) shows that one can suppose that there exists a sub-$\mathscr{O}_X$-Module $\mathscr{G}'$ of $\mathscr{G}$ *isomorphic to* $\mathscr{O}_X/\mathfrak{J}$ so that $\mathscr{G}'_x = \mathscr{O}_{Y,x}$ for every $x \in Y$. Setting $\mathscr{G}'' = \mathscr{G}/\mathscr{G}'$; replacing X by an open neighbourhood of y, one can, by virtue of (6.10.2), suppose that $\mathscr{G}'$ and $\mathscr{G}''$ are *normally flat* along Y. Let then x be any point of Y, and set $q = \mathrm{prof}(\mathscr{O}_{Y,x}) = \mathrm{prof}(\mathscr{G}'_x)$; let $(g_i)_{1 \leqslant i \leqslant q}$ be a maximal $\mathscr{G}''_x$-regular suite of elements of the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_{X,x}$. Each of the homogeneous components of $\mathrm{gr}^{\bullet}_{\mathscr{J}_x}(\mathscr{G}''_x)$ is a flat $\mathscr{O}_{Y,x}$-module of finite type by hypothesis, hence a free $\mathscr{O}_{Y,x}$-module, and it is therefore also $\mathrm{gr}^{\bullet}_{\mathscr{J}_x}(\mathscr{G}''_x)$-regular, hence it is also $\mathrm{gr}^{\bullet}_{\mathscr{J}_x}(\mathscr{G}''_x)$-regular, and whence the result $\mathrm{gr}^{\bullet}_{\mathscr{J}_x}(\mathscr{G}''_x)$ is $\mathscr{G}''_x$-regular (**0**, 15.1.19). Applying the lemma (**0**, 15.1.18) to the exact sequence

$$0 \longrightarrow \mathscr{G}'_x \longrightarrow \mathscr{G}_x \longrightarrow \mathscr{G}''_x \longrightarrow 0$$

by induction on j, one concludes an exact sequence

$$0 \longrightarrow \mathscr{G}'_x \Big/ \Big(\sum_{j=1}^{q} g_j \mathscr{G}'_x\Big) \longrightarrow \mathscr{G}_x \Big/ \Big(\sum_{j=1}^{q} g_j \mathscr{G}_x\Big).$$

But by hypothesis $\mathrm{prof}(\mathscr{G}'_x/(\sum_{j=1}^q g_j\mathscr{G}'_x)) = 0$ (**0**, 16.4.6); by the characterisation of modules of zero depth, one concludes that $\mathrm{prof}(\mathscr{G}_x/(\sum_{j=1}^q g_j\mathscr{G}_x)) = 0 = \mathrm{prof}(\mathscr{F}_x/(\sum_{i=1}^p f_i\mathscr{F}_x + \sum_{j=1}^q g_j\mathscr{F}_x))$; the suite (g_i) being $\mathscr{G}''_x$-regular and $\mathscr{G}''_x$ being $\mathscr{F}_x$-regular, the suite $((f_i)_x)$ is $\mathscr{F}_x$-regular by hypothesis and is formed of elements of the maximal ideal; one deduces (**0**, 16.4.6) that $\mathrm{prof}(\mathscr{F}_x) = p+q$, which completes the proof. ☐

6.11. Criteria for the sets $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ or $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ to be open

Lemma (6.11.1). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_{\mathrm{X}}$-Module coherent; then the function $x \mapsto \mathrm{dim.\,proj}(\mathscr{F}_x)$ is upper semi-continuous in X.*

Proof. One can restrict to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ is the spectrum of a Noetherian ring and $\mathscr{F} = \widetilde{\mathrm{M}}$, where M is an A-module of finite type. Suppose that for an $x \in \mathrm{X}$, we have $\mathrm{dim.\,proj}(\mathrm{M}_x) = n < +\infty$ (if $n = +\infty$ there is nothing to prove); there exists a resolution

$$\mathrm{L}_{n-1} \longrightarrow \mathrm{L}_{n-2} \longrightarrow \cdots \longrightarrow \mathrm{L}_0 \longrightarrow \mathrm{M} \longrightarrow 0$$

where the L_i are free A-modules of *finite type* (A being Noetherian), whence an exact sequence **IV-2** | 159

$$0 \longrightarrow \mathrm{R} \longrightarrow \mathrm{L}_{n-1} \longrightarrow \mathrm{L}_{n-2} \longrightarrow \cdots \longrightarrow \mathrm{L}_0 \longrightarrow \mathrm{M} \longrightarrow 0$$

where R is an A-module of finite type; one deduces an exact sequence

$$0 \longrightarrow \mathrm{R}_x \longrightarrow (\mathrm{L}_{n-1})_x \longrightarrow \cdots \longrightarrow (\mathrm{L}_0)_x \longrightarrow \mathrm{M}_x \longrightarrow 0$$

where the $(\mathrm{L}_i)_x$ are free A_x-modules of finite type; as by hypothesis $\mathrm{dim.\,proj}(\mathrm{M}_x) = n$, this implies that R_x is a projective A_x-module of finite type (**M**, VI, 2.1) and hence a free A_x-module of finite type ($\mathbf{0}_{\mathrm{III}}$, 10.1.3). We conclude that there exists an open neighbourhood U of x in X such that, for every $x' \in \mathrm{U}$, $\mathrm{R}_{x'}$ is a free $\mathrm{A}_{x'}$-module ($\mathbf{0}_{\mathrm{I}}$, 5.2.7), whence $\mathrm{M}_{x'}$ admits a projective resolution of length n, in other words $\mathrm{dim.\,proj}(\mathrm{M}_{x'}) \leqslant n$, which proves the lemma. ☐

Proposition (6.11.2). — *Let X be a locally Noetherian prescheme and locally immersible in a regular scheme (5.11.1); let $\mathscr{F}$ be an $\mathscr{O}_{\mathrm{X}}$-Module coherent. Then:*

- *(i)* (M. Auslander) *The function $x \mapsto \mathrm{coprof}(\mathscr{F}_x)$ is upper semi-continuous in X (in other words, for every integer n, the set $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ of $x \in \mathrm{X}$ such that $\mathrm{coprof}(\mathscr{F}_x) \leqslant n$ is* open*).*
- *(ii) For every integer n, the set $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ of $x \in \mathrm{X}$ such that $\mathscr{F}$ has the property (S_n) at the point x is open.*

Proof. (i) The question being local on X, one can, by virtue of the hypothesis, restrict to the case where X is a closed sub-prescheme of an affine regular scheme Y. If $j : \mathrm{X} \to \mathrm{Y}$ is the canonical injection, and $\mathscr{G} = j_*(\mathscr{F})$, one knows that one has then, for every $x \in \mathrm{X}$, $\dim(\mathscr{F}_x) = \dim(\mathscr{G}_x)$ and $\mathrm{prof}(\mathscr{F}_x) = \mathrm{prof}(\mathscr{G}_x)$, hence $\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{G}_x)$, and as $\mathrm{coprof}(\mathscr{G}_x) = 0$ for $y \notin \mathrm{X}$—Y, one is reduced to proving the property for $\mathscr{G}$; in other words, one can restrict to the case where X is an affine *regular* scheme. One knows then (**0**, 17.3.4) that one has

$$\mathrm{prof}(\mathscr{F}_x) = \dim(\mathscr{O}_{\mathrm{X},x}) - \mathrm{dim.\,proj}(\mathscr{F}_x).$$

On the other hand, if S is the unique closed sub-prescheme of X having for underlying space $\mathrm{Supp}(\mathscr{F})$, one has $\dim(\mathscr{F}_x) = \mathrm{codim}(\{x\},\mathrm{S})$ (5.1.12.1), and, by virtue of (5.1.9)

$$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{\mathrm{X},x}) - \mathrm{codim}_x(\mathrm{S},\mathrm{X})$$

since $\mathscr{O}_{\mathrm{X},x}$ is a regular ring, and *a fortiori* equidimensional (**0**, 16.5.12). We can therefore write

$$\mathrm{coprof}(\mathscr{F}_x) = \mathrm{dim.\,proj}(\mathscr{F}_x) - \mathrm{codim}_x(\mathrm{S},\mathrm{X}) \tag{6.11.2.1}$$

and the proposition follows then from (6.11.1) and from (**0**, 14.2.6).

(ii) As the $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ are open for every n by (i), the $\mathrm{Z}_n = \mathrm{X} - \mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ are closed in X; furthermore **IV-2** | 160
it is clear that $\mathrm{Z}_{n+1} \subset \mathrm{Z}_n$, and as $\dim(\mathscr{F}_x)$ is finite for every $x \in \mathrm{X}$ and $\mathrm{coprof}(\mathscr{F}_x) \leqslant \dim(\mathscr{F}_x)$, the intersection of the Z_n is *empty*; as one can restrict to the case where X is affine, hence quasi-compact, one can suppose that there exists an m ☐

such that $Z_m = \varnothing$. Now, it follows from (5.7.4) that the relation $x \in U_{S_k}(\mathscr{F})$ is equivalent to the set of relations

$$\operatorname{codim}_x(Z_n, S) > n + k \tag{6.11.2.2}$$

for every $n \geqslant 0$; but for $n \geqslant m$ this relation is automatically satisfied, hence one need only consider the relations (6.11.2.2) for $0 \leqslant n < m$. Now (**0**, 14.2.6) the set $V_{n,k}$ of x satisfying (6.11.2.2) is open, and $U_{S_k}(\mathscr{F})$, intersection of the $V_{n,k}$ for $0 \leqslant n < m$, is also open.

Let us recall that the hypothesis that X is locally immersible in a regular scheme is always satisfied when X is a prescheme *locally of finite type over a field k* (5.8.3).

Corollary (6.11.3). — *Under the hypotheses of (6.11.2), the set* $\mathrm{CM}(\mathscr{F})$ *of points* $x \in X$ *such that* $\mathscr{F}_x$ *is a Cohen-Macaulay module is open in* X.

Proof. Indeed, it is the set $U_{C_0}(\mathscr{F})$. □

Remarks (6.11.4). — (i) The reasoning of (6.11.2), (ii)) proves that (without any hypothesis on X) when $U_{C_n}(\mathscr{F})$ is open for *every* integer n, then $U_{S_n}(\mathscr{F})$ is open for every integer n.

(ii) We have $\mathrm{CM}(\mathscr{F}) = \bigcap U_{S_n}(\mathscr{F})$. If every point $x \in X$ admits an open neighbourhood V of *finite* dimension, the sequence of intersections $V \cap U_{S_n}(\mathscr{F})$ is *stationary* since there exists an m such that $\dim(\mathscr{F}_x) \leqslant m$ for every $x \in V$; consequently, if the $U_{S_n}(\mathscr{F})$ are open in X, then $\mathrm{CM}(\mathscr{F})$ is also open in X.

(iii) We shall write $U_{S_n}(X)$, $U_{C_n}(X)$, $\mathrm{CM}(X)$ instead of $U_{S_n}(\mathscr{O}_X)$, $U_{C_n}(\mathscr{O}_X)$, $\mathrm{CM}(\mathscr{O}_X)$.

Proposition (6.11.5). — *Let* A *be a local Noetherian ring,* M *an* A*-module of finite type. For every prime ideal* $\mathfrak{p}$ *of* A*, we have*

$$\operatorname{coprof}_{A_\mathfrak{p}}(M_\mathfrak{p}) \leqslant \operatorname{coprof}_A(M). \tag{6.11.5.1}$$

Proof. One can restrict to the case where $M_\mathfrak{p} \neq 0$. As $\widehat{A}$ is a faithfully flat A-module ($\mathbf{0}_\mathrm{I}$, 7.3.5), there exists an ideal $\mathfrak{q}$ of $\widehat{A}$ lying over $\mathfrak{p}$ ($\mathbf{0}_\mathrm{I}$, 6.5.1); as $\widehat{M}_\mathfrak{q} = (M \otimes_A \widehat{A})_\mathfrak{q} = M \otimes_A \widehat{A}_\mathfrak{q}$ and $\widehat{A}_\mathfrak{q}$ is a flat A-module, hence also a flat $A_\mathfrak{p}$-module ($\mathbf{0}_\mathrm{I}$, 6.2.1), it follows from (6.3.2), applied to the local homomorphism $A_\mathfrak{p} \to \widehat{A}_\mathfrak{q}$, to $M_\mathfrak{p}$ and $\widehat{M}_\mathfrak{q}$, that $\operatorname{coprof}_{A_\mathfrak{p}}(M_\mathfrak{p}) \leqslant \operatorname{coprof}_{\widehat{A}_\mathfrak{q}}(\widehat{M}_\mathfrak{q})$. On the other part, $X = \operatorname{Spec}(\widehat{A})$ is isomorphic to a sub-prescheme of a regular scheme by virtue of the Cohen structure theorem (**0**, 19.8.8, (i)). One deduces therefore from (6.11.2) that $\operatorname{coprof}_{\widehat{A}_\mathfrak{q}}(\widehat{M}_\mathfrak{q}) \leqslant \operatorname{coprof}_{\widehat{A}}(\widehat{M})$; finally, one knows that $\operatorname{coprof}_{\widehat{A}}(\widehat{M}) = \operatorname{coprof}_A(M)$ (**0**, 16.4.10), which finishes the proof of (6.11.5). This proposition justifies the definition of the coprofundity of an A-module when A is not a local ring, given in (5.7.12). □

Proposition (6.11.6). — *Let* X *be a locally Noetherian prescheme,* n *an integer* > 0, $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module. Suppose that for every closed integral sub-prescheme* Y *of* X*, there exists an open non-empty part* W *of* Y *such that the sub-prescheme of* X *induced on* W *satisfies* (S_n)*. Then the set* $U_{S_n}(\mathscr{F})$ *is open in* X.

Proof. The question being local on X, one can restrict to the case where X is Noetherian. We shall reason by induction on n: for $n = 1$, the set $U_{S_1}(\mathscr{F})$ is open in X, for the set of $x \in X$ where $\mathscr{F}$ does not satisfy (S_1) is the set of x such that $\mathscr{F}_x$ admits immersed associated prime ideals (5.7.5); if (Z_j) is the finite family of the prime cycles associated to $\mathscr{F}$ which are immersed, then $U_{S_1}(\mathscr{F}) = X - \bigcup_j Z_j$, whence our assertion since the Z_j are closed. We shall therefore suppose henceforth that $n > 1$. In the second place, one can restrict to the case where $\operatorname{Supp}(\mathscr{F}) = X$, for there is a closed sub-prescheme T of X having $\operatorname{Supp}(\mathscr{F})$ as underlying space, and an $\mathscr{O}_T$-Module coherent $\mathscr{G}$ such that $\mathscr{F} = j_*(\mathscr{G})$, $j : T \to X$ being the canonical injection ($\mathbf{Err}_\mathbf{III}$, 30); as it amounts to the same to prove that $\mathscr{F}$ or $\mathscr{G}$ satisfies (S_n) at a point $x \in \operatorname{Supp}(\mathscr{F})$, one can restrict to considering the case where $T = X$. Let us note finally that by definition (5.7.2), $U_{S_n}(\mathscr{F})$ is stable by generization. We shall now use the following lemma:

Lemma (6.11.6.1). — *Let* X *be a Noetherian space in which every irreducible closed subset admits a generic point. For* E *to be an open subset of* X*, it is necessary and it suffices that* E *be stable by generization, and that, for every open part* V *of* X *and every irreducible closed subset* Y *of* V*, such that* $V - Y \subset E$ *and that the generic point of* Y *belongs to* E*,* $E \cap Y$ *contains an open non-empty part of* Y.

Proof. We observe that this criterion implies that of ($\mathbf{0}_{\mathrm{III}}$, 9.2.6) when every irreducible closed subset admits a generic point. There is obviously nothing other than to prove the sufficiency of the conditions stated.

Consider the interior U of E; the closed set $X - U$ is a union of its irreducible components, which are finite in number and closed in X. If $E \neq U$, the hypothesis that E is stable by generization would imply that the generic point z of one of the irreducible components of $X - U$ would belong to E. Now, z belongs to only one of the irreducible components of $X - U$; if T is the union of the other irreducible components of $X - U$, $V = X - T$ is open in X, union of U and of the set $Y = \overline{\{z\}} \cap V$ closed in V and irreducible. By hypothesis $E \cap Y$ contains an open part W of Y; we conclude that $U \cup W$ is open *in* V, hence in X, which is absurd since U is supposed to be the interior of E.

By virtue of this lemma, one can suppose that there exists in X an integral sub-prescheme Y of generic point y such that $\mathscr{F}$ satisfies (S_n) at the point y and at every point of $X - Y$, and restrict to verifying that there exists in X an open neighbourhood of y such that $\mathscr{F}$ satisfies (S_n) in this neighbourhood. We distinguish two cases: **IV-2** | 161

1° y is a maximal point of X; as there exists an open neighbourhood of y meeting no irreducible component of X other than $\overline{\{y\}}$, one can suppose that X is irreducible, hence a *fortiori* has the same underlying space as Y, so that Y is defined by the Nilradical of $\mathscr{O}_X$, which is *nilpotent*. On the other hand, on can, replacing X by an open neighbourhood of y, suppose that $\mathscr{F}$ is *normally flat* along Y (6.9.1); it follows then from (6.10.4) that $U_{S_n}(\mathscr{F}) = U_{S_n}(\mathscr{O}_Y)$, and as the latter is by hypothesis an open neighbourhood of y in X, this proves the proposition in this case.

2° y is not a maximal point of X, in other words $\mathrm{Supp}(\mathscr{F}) = X$, $\dim(\mathscr{F}_y) \geqslant 1$, hence also $\mathrm{prof}(\mathscr{F}_y) \geqslant 1$ since $\mathscr{F}$ satisfies by hypothesis (S_n) (and *a fortiori* (S_1)) at the point y. Replacing X by an open neighbourhood of y if necessary, one can therefore suppose that there exists a section f of $\mathscr{O}_X$ over X, $\mathscr{F}$-regular and such that $f_y \in \mathfrak{m}_y$, or equivalently $f(y) = 0$ ($\mathbf{0}$, 15.2.4); one has then also $f(x) = 0$ for every $x \in Y$ ($\mathbf{0}_{\mathrm{I}}$, 5.5.2). One knows that $\mathscr{F}/f\mathscr{F}$ satisfies (S_{n-1}) at the point y (5.7.6). Applying the induction hypothesis, and replacing X by an open neighbourhood of y, one can therefore suppose that $\mathscr{F}/f\mathscr{F}$ satisfies (S_{n-1}) at every point of X. But for every $x \in Y$, the relation $f(x) = 0$ implies that $\mathrm{prof}(\mathscr{F}_x|f_x\mathscr{F}_x) = \mathrm{prof}(\mathscr{F}_x) - 1$ and $\dim(\mathscr{F}_x|f_x\mathscr{F}_x) = \dim(\mathscr{F}_x) - 1$ ($\mathbf{0}$, 16.3.4 and 16.4.6); the relation

$$\mathrm{prof}(\mathscr{F}_x|f_x\mathscr{F}_x) \geqslant \inf(n-1, \dim(\mathscr{F}_x|f_x\mathscr{F}_x))$$

is therefore equivalent to

$$\mathrm{prof}(\mathscr{F}_x) \geqslant \inf(n, \dim(\mathscr{F}_x)).$$

As we supposed that $\mathscr{F}$ satisfies (S_{n-1}) at every point of $X - Y$, this finishes the proof. □

Corollary (6.11.7). — *With the notations being those of (6.11.6):*

(i) The set $U_{S_1}(\mathscr{F})$ *is open in* X.
(ii) For the set $U_{S_n}(\mathscr{F})$ *to be open, it suffices that every maximal point x of* $\mathrm{Supp}(\mathscr{F})$*, belonging to* $U_{S_n}(\mathscr{F})$*, is interior to* $U_{S_n}(\mathscr{F})$.

Proof. Assertion (i) was proved in the course of the proof of (6.11.6); on the other hand, for $n = 2$ the case 2° of the proof of (6.11.6) is valid without any hypothesis on X, since (with the same notations) $U_{S_1}(\mathscr{F})$ and $U_{S_1}(\mathscr{F}/f\mathscr{F})$ are open in X. As for the case 1° of this proof, the hypothesis ensures precisely that it is unnecessary to consider it. □

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Proposition (6.11.8). — *Let* X *be a locally Noetherian prescheme satisfying the following property:*

(CMU) *Every closed integral sub-prescheme* Y *of* X *contains an open non-empty* W *such that the prescheme induced by* X *on* W *is a Cohen-Macaulay prescheme.*

Then, for every coherent $\mathscr{O}_X$*-Module* $\mathscr{F}$*, the function* $x \mapsto \mathrm{coprof}(\mathscr{F}_x)$ *is locally constructible and upper semi-continuous; the sets* $U_{C_n}(\mathscr{F})$ *and* $U_{S_n}(\mathscr{F})$ *are open in* X.

Proof. Indeed, let Y be a closed integral sub-prescheme of X, of generic point y; by virtue of (6.10.6), there is a neighbourhood V of y in Y such that, for every $x \in V \cap Y$, we have

(6.11.8.1) $$\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{F}_y) + \mathrm{coprof}(\mathscr{O}_{Y,x}).$$

But by hypothesis there exists an open non-empty W of Y such that, for $x \in V \cap W$, $\operatorname{coprof}(\mathscr{O}_{Y,x}) = 0$, hence $\operatorname{coprof}(\mathscr{F}_x)$ is constant in a neighbourhood of y in Y, which proves that the function $x \mapsto \operatorname{coprof}(\mathscr{F}_x)$ is locally constructible ($\mathbf{0}_{\text{III}}$, 9.3.2); it follows furthermore from (6.11.5) and from ($\mathbf{0}_{\text{III}}$, 9.3.4) that this function is upper semi-continuous. The last assertion follows from (6.11.4), (i), or from (6.11.6). □

Remarks (6.11.9). — (i) If X satisfies (CMU), then every prescheme X′ locally of finite type over X also satisfies (CMU). Let Y′ be a closed integral subprescheme of X′, let y' be its generic point, put $y = u(y')$, and let Y be the integral closed subprescheme of X with underlying space $\overline{\{y\}}$. Then $u|_{Y'}$ factors as

$$Y' \xrightarrow{v} Y \xrightarrow{j} X,$$

where j is the canonical injection and v is locally of finite type.

Restricting to affine open neighbourhoods of y and y', we may suppose that $X = \operatorname{Spec}(A)$, where A is an integral Cohen–Macaulay ring, and $X' = \operatorname{Spec}(A')$, where A′ is an integral finite-type A-algebra containing A. After replacing A by A_f for some $f \neq 0$, we may suppose that A′ contains a **IV-2** | 163
polynomial ring $A[T_1, \dots, T_n] = A''$ and is finite over A″. The ring A″ is Cohen–Macaulay (6.3.6); hence we may reduce to the case where A′ is finite over A. After one further localisation, A′ is a finite free A-module. It is therefore a Cohen–Macaulay A-module ($\mathbf{0}$, 16.5.1); since it is finite over A, it is also a Cohen–Macaulay A′-module ($\mathbf{0}$, 16.5.3), and hence a Cohen–Macaulay ring.

(ii) Suppose that there exists a coherent $\mathscr{O}_X$-Module $\mathscr{F}$ such that $\operatorname{Supp}(\mathscr{F}) = X$ and that $\mathscr{F}$ is an $\mathscr{O}_X$-Module *of Cohen-Macaulay*. Then X satisfies the condition (CMU): indeed, with the notations of the proof of (6.11.8), the relation (6.11.8.1) shows that $\operatorname{coprof}(\mathscr{O}_{Y,x}) = 0$ in a neighbourhood (with respect to Y) of the generic point y of Y.

It is unknown whether there exist locally Noetherian preschemes of dimension $\geqslant 2$ that do not satisfy (CMU) (if $\dim(X) = 1$, it is immediate that every maximal point of X_{red} admits an open neighbourhood of dimension 1, hence a Cohen-Macaulay ring).

6.12. Nagata's criteria for Reg(X) to be open

(6.12.1). Given a locally Noetherian prescheme X, we call the *singular locus* of X and denote by Sing(X) the set of points $x \in X$ such that X is not regular at the point x (in other words, such that the local ring $\mathscr{O}_x$ is not regular); the complement $X - \operatorname{Sing}(X)$, set of $x \in X$ where X is regular, is denoted Reg(X). We propose in this subsection to give conditions ensuring that Sing(X) is *closed* (i.e. Reg(X) *open*).

Proposition (6.12.2). — *Let* X *be a locally Noetherian prescheme. The following conditions are equivalent:*

a) Reg(X) *is open in* X.
b) *For every* $x \in \operatorname{Reg}(X)$, *there exists an open non-empty part of* $\overline{\{x\}}$ *contained in* Reg(X).

Moreover, these conditions imply the following:

c) *For every* $x \in \operatorname{Reg}(X)$, *if we denote by* Y *the reduced closed sub-prescheme of* X *having* $\overline{\{x\}}$ *as underlying space,* Reg(Y) *is a neighbourhood of* x *in* Y.

Proof. The equivalence of a) and b) follows from the fact that Reg(X) is stable by generization ($\mathbf{0}$, 17.3.2) and from ($\mathbf{0}_{\text{III}}$, 9.2.6). To prove that c) implies b), one can restrict to the case where $X = \operatorname{Spec}(A)$ is the spectrum of an affine ring of x; if $(t_i)_{1 \leqslant i \leqslant n}$ is a regular system of parameters ($\mathbf{0}$, 17.1.6) of the regular local ring A_x, one can suppose (replacing X by an open neighbourhood of x if necessary) that $t_i = (s_i)_x$ for every i, where $s_i \in A$ and where the family $(s_i)_{1 \leqslant i \leqslant n}$ is A-regular ($\mathbf{0}$, 15.2.4). One has then $Y = \operatorname{Spec}(A/\mathfrak{p})$, where $\mathfrak{p} = \mathfrak{j}_x$; as the t_i generate the maximal ideal $\mathfrak{m} = \mathfrak{p}A_x$ of A_x, one can further suppose (replacing X by a smaller neighbourhood of x) that the s_i generate $\mathfrak{p}$. For every $y \in Y$, the $(s_i)_y$ generate therefore $\mathfrak{p}_y$; as they form a suite A_y-regular, they also form a suite $\operatorname{gr}^0_{\mathfrak{p}_y}(\mathscr{O}^Y_y)$-regular, so they form a $\operatorname{gr}_{\mathfrak{p}_y}(\mathscr{O}^Y_y)$-regular sequence and it is thus also $\mathscr{O}^Y_y$-regular, whence the conclusion. □

Corollary (6.12.3). — *Let* X *be a locally Noetherian prescheme. The following conditions are equivalent:*

a) *For every sub-prescheme* Y *of* X, Reg(Y) *is open in* Y. **IV-2** | 164
b) *For every closed integral sub-prescheme* Y *of* X, Reg(Y) *contains an open non-empty part of* Y.

Proof. It is clear that *a*) implies *b*). Conversely, to see that *b*) implies *a*), consider a closed integral sub-prescheme Z of Y with generic point z; if Y' is the integral closed sub-prescheme of X whose underlying space is the closure $\mathrm{Y}' = \overline{\{z\}}$ of z in X, Z is open in Y' and the sub-prescheme Z of Y', being reduced, is induced by Y' on the open subset of Z in Y'. Now the hypothesis implies that $\mathrm{Reg}(\mathrm{Y}')$ is a neighbourhood of z in Y', hence $\mathrm{Reg}(\mathrm{Z})$ is a neighbourhood of z in Z; it suffices then to apply (6.12.2) by replacing X by Y and Y by Z. □

Theorem (6.12.4). — (Nagata). — *Let* A *be a Noetherian ring,* $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$. *The following conditions are equivalent:*

a) *For every prescheme* X' *locally of finite type over* X, $\mathrm{Reg}(\mathrm{X}')$ *is open in* X'.
b) *For every* A*-algebra* A' *finite and integral, there exists an open non-empty subset of* $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$ *contained in* $\mathrm{Reg}(\mathrm{X}')$.
c) *For every prime ideal* $\mathfrak{p}$ *of* A *and every finite radicial extension* K' *of the field of fractions* K *of* $\mathrm{A}/\mathfrak{p}$, *there exists a sub-*A*-algebra* A' *of* K', *finite over* A, *having* K' *as field of fractions, and such that there exists in* $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$ *an open non-empty subset contained in* $\mathrm{Reg}(\mathrm{X}')$.

Proof. It is clear that *a*) implies *b*). To see that *b*) implies *c*), it suffices to note that one can take as generators of the extension K' of K elements integral over $\mathrm{A}/\mathfrak{p}$ (and *a fortiori* over A), and as these elements are finite in number, they generate a finite A-algebra A' whose field of fractions is K'; one can then apply *b*) to A'. It remains therefore to prove that *c*) implies *a*). The question being local on X', one can first suppose that $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$, where A' is an A-algebra of finite type; in view of (6.12.2), it suffices to prove that for every closed integral sub-prescheme Y' of X', $\mathrm{Reg}(\mathrm{Y}')$ contains an open non-empty part of Y'. In other words, one can restrict to proving that if A' is an A-algebra integral *of finite type* and $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{Reg}(\mathrm{X}')$ contains an open non-empty part. Let K' be the field of fractions of A'; if $\mathfrak{p}$ is the canonical image in X of the generic point of X', K' is an extension of the field of fractions K of $\mathrm{A}/\mathfrak{p}$, and $\mathrm{A}/\mathfrak{p}$ is identified with a subring of A', A' being a finite $(\mathrm{A}/\mathfrak{p})$-algebra of finite type. One can obviously restrict in what follows to the proof when $\mathfrak{p} = 0$. We distinguish two cases:

I) K' *is a separable extension* of K. — One is then reduced to proving the

Lemma (6.12.4.1). — *Let* A *be a Noetherian integral ring,* A' *an* A*-algebra integral of finite type, such that the field of fractions* K' *of* A' *is a separable extension of the field of fractions* K *of* A. *If* $\mathrm{Spec}(\mathrm{A})$ *contains an open non-empty subset consisting of regular points, then so does* $\mathrm{Spec}(\mathrm{A}')$.

Proof. Replacing A by a ring of fractions A_f, so that $\mathrm{D}(f) \subset \mathrm{Reg}(\mathrm{Spec}(\mathrm{A}))$, one can already suppose the ring A *regular*. By hypothesis there exists in K' a system $(t_i)_{1 \leqslant i \leqslant n}$ of elements algebraically independent over K and such that K' is a finite separable algebraic extension of $\mathrm{K}(t_1, \dots, t_n)$; considering a finite system of generators t'_j $(1 \leqslant j \leqslant m)$ of K' over $\mathrm{K}(t_1, \dots, t_n)$, which one can suppose **IV-2** | 165
to be integral over $\mathrm{A}_1 = \mathrm{A}[t_1, \dots, t_n]$, one sees that $\mathrm{A}'_1 = \mathrm{A}_1[t'_1, \dots, t'_m]$ is *finite* over A_1 and has K' as field of fractions. If one sets $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}'_1 = \mathrm{Spec}(\mathrm{A}'_1)$, it follows from the fact that the fields of rational functions $\mathrm{R}(\mathrm{X}')$ and $\mathrm{R}(\mathrm{X}'_1)$ are both isomorphic to K' and from the fact that X' and X'_1 are A-preschemes of finite type, that there exists an open subset $\mathrm{U}' \subset \mathrm{X}'$ and an open $\mathrm{U}'_1 \subset \mathrm{X}'_1$ which are A-*isomorphic* (**I**, 6.5.5). One is therefore reduced to proving that $\mathrm{Reg}(\mathrm{X}'_1)$ contains a non-empty open subset, in other words one can suppose that A' is an A_1-algebra *finite*. Now, one knows (**0**, 17.3.7) that A_1 is a regular ring, and one can therefore restrict to the case where A' is an A-algebra *finite* and K' an extension *separable* and *finite* of K. Let ξ be the generic point of X, $\mathrm{A}'_\xi = \mathrm{K}'$ is then a free module over $\mathrm{A}_\xi = \mathrm{K}$, hence (Bourbaki, *Alg., comm.*, chap. II, § 5, no 1, cor. of prop. 2) one can, replacing A by a ring of fractions A_f, suppose that A' is a free A-module of finite type. Let then $(x_h)_{1 \leqslant h \leqslant r}$ be a basis of this A-module, and set

$$d = \det(\mathrm{Tr}_{\mathrm{A}'/\mathrm{A}}(x_h x_k)) = \det(\mathrm{Tr}_{\mathrm{K}'/\mathrm{K}}(x_h x_k)) \in \mathrm{A}.$$

As K' is separable over K, one knows (Bourbaki, *Alg.*, chap. IX, § 2, prop. 5) that $d \neq 0$; replacing A by a ring of fractions A_z, one can therefore suppose d *invertible* in A. But then, for every $z \in \mathrm{Spec}(\mathrm{A})$, if one denotes by $\overline{x}_h$ $(1 \leqslant h \leqslant r)$ the canonical image of x_h in $\mathrm{A}'(z) = \mathrm{A}' \otimes_{\mathrm{A}} \boldsymbol{k}(z)$, one has $\det(\mathrm{Tr}_{\mathrm{A}'(z)/\boldsymbol{k}(z)}(\overline{x}_h \overline{x}_k)) = \overline{d}$, canonical image of d in $\boldsymbol{k}(z) = \mathrm{A}_z/\mathfrak{m}_z$, and as $\overline{d}$ is invertible (hence $\neq 0$) in $\boldsymbol{k}(z)$, one knows (*loc. cit.*) that $\mathrm{A}'(z)$ is a $\boldsymbol{k}(z)$-algebra *separable*, hence a direct product of finite separable field extensions of $\boldsymbol{k}(z)$. Since such an algebra is a *regular* ring, one sees that the morphism $g : X' \to X$ is *flat* and that its fibres $g^{-1}(z)$ are *regular* for every $z \in X$; one concludes then from (6.5.2, (ii)) that X' is *regular*, which finishes the proof in this case. □

II) *General case.* — As A' is an A-module without torsion, $A' \otimes_A K$ is identified with a sub-ring of K', hence $X'' = \mathrm{Spec}(A' \otimes_A K)$ is an integral K-prescheme whose field of rational functions is K'. One knows (4.6.6) that there exists a finite radicial extension K_1 of K such that $X'_1 = \mathrm{Spec}(A' \otimes_A K_1) = X'' \otimes_K K_1$, $(X'_1)_{\mathrm{red}}$ is a K_1-prescheme *geometrically reduced* of finite type; furthermore, the morphism $\mathrm{Spec}(K_1) \to \mathrm{Spec}(K)$ being radicial, finite and surjective, it is a universal homeomorphism (2.4.5), hence X'_1 is homeomorphic to X'', and consequently $(X'_1)_{\mathrm{red}}$ is integral; its field of rational functions K'_1 is a finite radicial extension of K' and a *separable* extension of finite type of K_1 (4.6.1). By virtue of hypothesis c) of the statement, there is a sub-A-algebra *finite* A_1 of K_1, having K_1 as field of fractions and such that if one sets $X_1 = \mathrm{Spec}(A_1)$, $\mathrm{Reg}(X_1)$ contains an open non-empty subset. Let A'_1 be the image of the canonical homomorphism $A' \otimes_A A_1 \to K'_1$ and let $X'_1 = \mathrm{Spec}(A'_1)$; A'_1 is an integral ring which is an A'-algebra *finite* and whose field of fractions is K'_1; furthermore, as the composite homomorphism $A' \to A' \otimes_A A_1 \to A'_1 \to K'_1$ is identical to $A' \to K' \to K'_1$, hence injective, the homomorphism $A' \to A'_1$ is injective; the morphism $g : X'_1 \to X'$ is therefore finite and surjective (**II**, 6.1.10). This being the hypothesis on A_1 and part I) of the proof imply that $\mathrm{Reg}(X'_1)$ contains an open non-empty V'_1; as g is closed (**II**, 6.1.10), one can suppose that **IV-2 | 166**
$V'_1 = g^{-1}(V')$, where V' is an open affine of X'; replacing A' by the ring of V' and A'_1 by that of V'_1, one can therefore suppose that X'_1 is *regular*. Furthermore, the same reasoning as in I) applied to the generic point ξ' of X' allows (replacing X' by a neighbourhood of ξ') to suppose that A'_1 is an A'-module *free*, and hence that the morphism g is *flat*. But it follows then from (6.5.2, (i)) that X' is *regular*. □

Corollary (6.12.5). — (Zariski). — *Let k be a field; for every k-prescheme X locally of finite type over k, the set of $x \in X$ where X is regular (resp.* geometrically regular *over k) is open in* X.

Proof. The assertion of the corollary relative to the property of being regular follows from (6.12.4) by taking $A = k$. The assertion relative to the property of being geometrically regular follows already from (6.8.7); one can also deduce it from (6.12.4) in the following way: set $k' = k^{p^{-\infty}}$ (p being the characteristic exponent of k); as the morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is radicial, integral and surjective, it is a universal homeomorphism (2.4.5), hence the morphism projection $X \otimes_k k' \to X$ is a universal homeomorphism. The projection in X of $\mathrm{Reg}(X \otimes_k k')$ is the set of points of X where X is geometrically regular over k, by virtue of (6.7.7, e)); hence this set is open from what precedes. □

Corollary (6.12.6). — *Let* A *be a ring having one of the following properties:*

(i) A *is a Dedekind ring and its field of fractions* K *is of characteristic* 0.
(ii) A *is a semi-local Noetherian ring of dimension* $\leqslant 1$.

Then, for every prescheme X *locally of finite type over* A, $\mathrm{Reg}(X)$ *is open in* X.

Proof. Let us verify in both cases the condition c) of (6.12.4). In both cases, a prime ideal $\mathfrak{p}$ of A is maximal or minimal; if $\mathfrak{p}$ is maximal, an $(A/\mathfrak{p})$-algebra finite and integral is a field and the condition c) of (6.12.4) is trivially satisfied. Suppose therefore $\mathfrak{p}$ non-maximal, and distinguish the two cases of the statement.

(i) If K is of characteristic 0, there is no radicial extension of K other than K itself; as a Dedekind ring is regular (**0**, 17.1.4), the condition c) of (6.12.4) is trivially satisfied.

(ii) One can then suppose A integral (6.12.2), let K be its field of fractions; if K' is a finite radicial extension of K, A' a sub-A-algebra of K' generated by a finite system of generators of K' over K, integral over A, A' is a semi-local integral ring of dimension 1 (**0**, 16.1.5), and consequently, in $X' = \mathrm{Spec}(A')$, the reduced set at the generic point is open and evidently contained in $\mathrm{Reg}(X')$, which proves the condition c) of (6.12.4) in this case. □

This corollary applies notably when $A = \mathbf{Z}$.

Theorem (6.12.7). — (Nagata). — *Let* A *be a local Noetherian complete ring,* $X = \mathrm{Spec}(A)$*. Then* $\mathrm{Reg}(X)$ *is open in* X.

Proof. In view of (6.12.2), one is reduced to the case where moreover A is *integral*, and to proving that Reg(X) contains an open non-empty part of X. We distinguish two cases:

I) *The field of fractions* K *of* A *is of characteristic* 0. — One knows then (**0**, 19.8.8) that there exists a complete discrete valuation ring C and a subring B of A such that A is a B-algebra *finite* and that B is isomorphic to a formal power series ring $C[[T_1, \dots, T_r]]$. As C is regular (**II**, 7.1.6), so is B (**0**, 17.3.8); furthermore, the field of fractions L of B being of characteristic 0, K is a *separable* extension finite of L, hence one can apply (6.12.4.1) to B, and this proves the proposition. **IV-2** | 167

II) *The field of fractions* K *of* A *is of characteristic* $p > 0$. — Then A contains the prime field $\mathbf{F}_p$, and the theorem has been proved in this case (**0**, 22.7.6). ☐

Corollary (6.12.8). — *Let* A *be a local Noetherian complete ring. Then, for every prescheme* X *locally of finite type over* A, Reg(X) *is open in* X.

Proof. Let us verify the condition *c*) of (6.12.4). If $\mathfrak{p}$ is prime in A, $A/\mathfrak{p}$ is again a local Noetherian complete ring; if K' is a finite extension of the field of fractions K of $A/\mathfrak{p}$, K' is the field of fractions of a sub-A-algebra finite A' of K', generated by a finite system of generators of K' over K, integral over A. One knows then that A' is a semi-local complete ring (Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, cor. 3 of prop. 9), hence a product of local complete rings, and as A' is integral, it is a local complete ring; and by virtue of (6.12.7), if $X' = \mathrm{Spec}(A')$, $\mathrm{Reg}(X')$ is open and non-empty, whence the conclusion. ☐

Proposition (6.12.9). — *Let* X *be a locally Noetherian prescheme such that* Reg(X) *is open; then, for every* $n \geqslant 0$, *the set* $U_{R_n}(X)$ *of points* $x \in X$ *where* X *satisfies* (R_n) *is open.*

Proof. Indeed (5.8.2), $U_{R_n}(X)$ is the union of Reg(X) and of the set of $x \in \mathrm{Sing}(X)$ such that the generic points z_i of the irreducible components of Sing(X) containing x satisfy the relation $\dim(\mathscr{O}_{X,z_i}) \geqslant n+1$ (5.1.2); in other words, these points $x \in \mathrm{Sing}(X)$ are those for which $\mathrm{codim}_x(\mathrm{Sing}(X), X) \geqslant n+1$; as the function $x \mapsto \mathrm{codim}_x(\mathrm{Sing}(X), X)$ is lower semi-continuous (**0**, 14.2.6), the set of these points is open in Sing(X), which finishes the proof. ☐

Let us also note the following elementary result:

Proposition (6.12.10). — *Let* X *be a locally Noetherian prescheme. The set* $U_{R_0}(X)$ *is open in* X. *For* $U_{R_1}(X)$ *to be open in* X, *it is necessary and it suffices that every maximal* $x \in X$ *such that* $x \in U_{R_1}(X)$ *(which means that* X *is reduced at* x*) be interior to* $U_{R_1}(X)$.

Proof. By definition (5.8.2), $U_{R_0}(X)$ is the set $X - \bigcup_\alpha X'_\alpha$, where X'_α ranges over the set of irreducible components of X such that X is not reduced at their generic point; as the set of irreducible components of X is locally finite, $U_{R_0}(X)$ is open. As for $U_{R_1}(X)$, the stated condition is trivially necessary; let us prove that it is sufficient. Let X''_β be the irreducible components of X such that, at their generic point x''_β, the prescheme X is reduced; by hypothesis there exists an open $U \subset U_{R_1}(X)$ containing all the x''_β. Denote then by Z_λ the irreducible components of the closed set $Z = X - U$ whose generic point z_λ is such that $\dim(\mathscr{O}_{X,z_\lambda}) \leqslant 1$ and that $\mathscr{O}_{X,z_\lambda}$ is non-regular. No point belonging to one of the Z_λ can belong to $U_{R_1}(X)$; but conversely, if $x \in Z$ does not belong to any of the Z_λ, then, for every generization x' of x, either x' belongs to U, or $\dim(\mathscr{O}_{X,x'}) \geqslant 2$, or $\dim(\mathscr{O}_{X,x'}) = 1$, and as x' does not belong to any Z_λ, $\overline{\{x'\}}$ is necessarily one of the irreducible components of Z, of codimension 1 in X, distinct from the Z_λ, hence $\mathscr{O}_{X,x'}$ is regular by definition. One concludes that $U_{R_1}(X) = X - \bigcup_\lambda Z_\lambda$; as the set of Z_λ is locally finite in Z, $\bigcup_\lambda Z_\lambda$ is closed, which finishes the proof that $U_{R_1}(X)$ is open in X. ☐ **IV-2** | 168

6.13. Criteria for $\mathrm{Nor}(X)$ to be open

(6.13.1). Given a locally Noetherian prescheme X, we shall denote by $\mathrm{Nor}(X)$ the set of $x \in X$ where X is *normal*; this set contains $\mathrm{Reg}(X)$ and is contained in the (open) set of points x such that $\mathscr{O}_{X,x}$ is integral (i.e. the set of points x where X is reduced, and which belong to only one irreducible component of X).

Proposition (6.13.2). — *Let* X *be a locally Noetherian prescheme reduced,* X' *its normalization* (**II**, 6.3.8). *If the canonical morphism* $f : X' \to X$ *is of finite type,* $\mathrm{Nor}(X)$ *is open in* X.

Proof. Indeed, on $X' = \mathrm{Spec}(f_*(\mathscr{O}_{X'}))$ and $f_*(\mathscr{O}_{X'})$ is a coherent $\mathscr{O}_X$-Algebra by hypothesis (**II**, 6.1.3). To say that X is normal at a point x means that the canonical homomorphism $\mathscr{O}_x \to (f_*(\mathscr{O}_{X'}))_x$ is bijective (**II**, 6.3.4); but the set of these points is *open* since $\mathscr{O}_X$ and $f_*(\mathscr{O}_{X'})$ are coherent ($\mathbf{0}_{\mathrm{I}}$, 5.2.7). □

Corollary (6.13.3). — *If* A *is an integral Noetherian* Japanese *ring,* $\mathrm{Nor}(\mathrm{Spec}(A))$ *is open in* $\mathrm{Spec}(A)$.

Proposition (6.13.4). — *Let* X *be a locally Noetherian prescheme. For* $\mathrm{Nor}(X)$ *to be open, it is necessary and it suffices that every maximal point x of* X *such that* $x \in \mathrm{Nor}(X)$ *(which means that* X *is reduced at the point x) be interior to* $\mathrm{Nor}(X)$.

Proof. There is only to prove the sufficiency of the condition. By virtue of the remark made in (6.13.1), one can restrict (replacing X by an open of X in which X is reduced), i.e. to suppose that the maximal points of X belong to $U_{S_1}(X)$ and to $U_{R_1}(X)$; as $\mathrm{Nor}(X) = U_{S_1}(X) \cap U_{R_1}(X)$ by virtue of Serre's criterion (5.8.6), one deduces from (6.11.7) and (6.12.10) that these two sets are open; hence $\mathrm{Nor}(X)$ is open in X. □

Corollary (6.13.5). — *If* X *is such that* $\mathrm{Reg}(X)$ *is open in* X, *then* $\mathrm{Nor}(X)$ *is open in* X.

Proof. Indeed, every maximal point x where X is reduced (hence $\mathscr{O}_{X,x}$ is a field) belongs to $\mathrm{Reg}(X)$, hence is interior to $\mathrm{Reg}(X)$ by hypothesis, and *a fortiori* is interior to $\mathrm{Nor}(X)$. □

Proposition (6.13.6). — (Nagata). — *Let* A *be a Noetherian integral ring,* K *its field of fractions,* K' *a finite extension of* K, A' *the integral closure of* A *in* K'. *For* A' *to be a finite* A*-algebra, it is necessary and it suffices that the two following conditions be satisfied:*

(i) *There exists* $f \neq 0$ *in* A *such that the integral closure* A'_f *of the ring of fractions* A_f *in* K' *is a finite* A_f*-algebra.*

(ii) *For every prime ideal* $\mathfrak{p} \in \mathrm{Spec}(A)$, *the integral closure* $A'_{\mathfrak{p}}$ *of the local ring* $A_{\mathfrak{p}}$ *in* K' *is a finite* $A_{\mathfrak{p}}$*-algebra.*

Proof. The conditions are necessary, for for every multiplicative subset S of A, $S^{-1}A'$ is the integral closure of $S^{-1}A$ in K' and is by hypothesis a $S^{-1}A$-module of finite type. To see that the conditions are sufficient, consider the filtered increasing family (B_λ) of sub-A-algebras of K' which are A-algebras finite and have K' as field of fractions. Set $Y = \mathrm{Spec}(A)$, $X_\lambda = \mathrm{Spec}(B_\lambda)$, denote by u_λ the morphism $X_\lambda \to Y$ and let $S_\lambda = X_\lambda - \mathrm{Nor}(X_\lambda)$, $T_\lambda = u_\lambda(S_\lambda)$. One can restrict to the B_λ which contain a finite set whose image in A'_f is a system of generators of the A_f-module A'_f; this implies, by virtue of hypothesis (i), that $(B_\lambda)_f = A'_f$ for every λ, or equivalently that $u_\lambda^{-1}(D(f))$ is contained in $\mathrm{Nor}(X_\lambda)$. By virtue of (6.13.2), S_λ is therefore *closed* in Y. But for every $\mathfrak{p} \in \mathrm{Spec}(A)$, it exists by virtue of (ii) a λ such that $(B_\lambda)_{\mathfrak{p}} = A'_{\mathfrak{p}}$, and consequently all points of X_λ lying over $\mathfrak{p}$ belong to **IV-2** | 169
$\mathrm{Nor}(X_\lambda)$; in other words T_λ is *closed* in Y, $\bigcup_\lambda T_\lambda$ is closed, and as u_λ is a finite morphism, T_λ is *closed* in Y. But for every $\mathfrak{p} \in \mathrm{Spec}(A)$, it exists by virtue of (ii) a λ such that $(B_\lambda)_{\mathfrak{p}} = A'_{\mathfrak{p}}$, and consequently all points of X_λ lying over $\mathfrak{p}$ belong to $\mathrm{Nor}(X_\lambda)$; in other words $\mathfrak{p} \notin T_\lambda$; as Y is Noetherian and the T_λ are closed, there exists a λ_0 such that $T_{\lambda_0} = \varnothing$; hence B_{λ_0} is integrally closed, and as its field of fractions is K', one has $B_{\lambda_0} = A'$. □

Proposition (6.13.7). — *Let* A *be a Noetherian ring,* $X = \mathrm{Spec}(A)$. *The following conditions are equivalent:*

a) *For every prescheme* X' *locally of finite type over* X, $\mathrm{Nor}(X')$ *is open in* X'.

b) *For every* A*-algebra finite and integral* A', $\mathrm{Nor}(\mathrm{Spec}(A'))$ *is open in* $\mathrm{Spec}(A')$.

c) *For every prime ideal* $\mathfrak{p}$ *of* A *and every finite radicial extension* K′ *of the field of fractions* K *of* A/$\mathfrak{p}$, *there exists a sub-*A*-algebra* A′ *of* K′, *finite over* A, *having* K′ *as field of fractions and such that* Nor(Spec(A′)) *is open in* Spec(A′).

Proof. The fact that *a*) implies *b*) and that *b*) implies *c*) is proved as in (6.12.4). To show that *c*) implies *a*), one reduces as in (6.12.4) to proving that if A′ is an A-algebra integral and of finite type, the generic point of Spec(A′) is interior to Nor(Spec(A′)). One distinguishes two cases as in (6.12.4) by proving first the

Lemma (6.13.7.1). — *Let* A *be a Noetherian integral ring,* A′ *an* A*-algebra integral of finite type, containing* A *and such that the field of fractions* K′ *of* A′ *is a separable extension of the field of fractions* K *of* A. *If* Nor(Spec(A)) *is open in* Spec(A), Nor(Spec(A′)) *is open in* Spec(A′).

Proof. It suffices to prove (in view of (6.13.2)) that the generic point of Spec(A′) is interior to Nor(Spec(A′)). The proof follows the same approach as that of (6.12.4.1), from which we conserve the notations. We note that when A is integrally closed, and then on takes $A_1 = A[t_1, \dots, t_n]$ and is integrally closed (Bourbaki, *Alg. comm.*, chap. V, § 1, no 3, cor. 2 of prop. 13); one reduces next to the case where A′ is a free A-module of *finite type*; the reasomment then applies (replacing A by an appropriate ring of fractions A_f with $f \neq 0$) the reasoning from case I) of (6.12.4) to see that the fibres $g^{-1}(z)$ of the morphism $g : X' \to X$ are regular and *a fortiori* normal. Furthermore g is flat and X is normal, hence (6.5.4, (ii)) X′ is normal. One then passes to the general case as in (6.12.4); II)), from which we conserve further the notations; applying hypothesis *c*), one sees this time that $\mathrm{Nor}(X'_1)$ is open and one reduces thus to the case where X'_1 is normal and $g : X' \to X'$ is flat and surjective; one concludes by applying (6.5.4, (i)). □

This lemma being proved, one passes to the general case as in (6.12.4; II)), from which we conserve the notations; applying hypothesis *c*), one sees this time that $\mathrm{Nor}(X'_1)$ is open and one reduces thus to the case where X'_1 is normal and $g : X' \to X'$ flat and surjective; one concludes by applying (6.5.4, (i)). □

6.14. Base change and integral closure

Proposition (6.14.1). — *Let* Y, Y′ *be two locally Noetherian preschemes,* $g : Y' \to Y$ *a normal morphism* (6.8.1). *Then, for every normal* Y*-prescheme* X, *the prescheme* $X' = X \times_Y Y'$ *is normal.*

Proof. We note that in this statement one *does not suppose* X locally Noetherian. □

Lemma (6.14.1.1). — *Let* R *be a ring that is a direct product of finitely many fields.*

(i) For a subring A *of* R *having* R *as total ring of fractions to be normal, it is necessary and it suffices that it be integrally closed in* R.
(ii) Let (A_λ) *be a family of normal subrings of* R*; if* $A = \bigcap_\lambda A_\lambda$ *admits* R *as total ring of fractions,* A *is normal.*

Proof. (i) As A is a subring of R, $A_\mathfrak{p}$ is a subring of $R_\mathfrak{p}$ for every prime ideal $\mathfrak{p}$ of A; furthermore $R_\mathfrak{p}$ is a ring of fractions of $A_\mathfrak{p}$; furthermore $R_\mathfrak{p}$ is necessarily a direct product of finitely many fields, hence every non-zero divisor of 0 in $R_\mathfrak{p}$ is invertible; this proves that $R_\mathfrak{p}$ is the total ring of fractions of $A_\mathfrak{p}$. If A is integrally closed in R, $A_\mathfrak{p}$ is therefore integrally closed in $R_\mathfrak{p}$; but if $R_\mathfrak{p}$ is a direct product of at least two rings, the integral closure of $A_\mathfrak{p}$ in $R_\mathfrak{p}$ is a direct product of at least two nonzero rings, which is absurd since $A_\mathfrak{p}$ is a local ring; hence $R_\mathfrak{p}$ is necessarily a field and $A_\mathfrak{p}$ is integral and integrally closed, which by definition means that A is normal. Conversely, if A is normal, $R_\mathfrak{p}$, total ring of fractions of an integral ring $A_\mathfrak{p}$, is a field and $A_\mathfrak{p}$ is integrally closed in $R_\mathfrak{p}$; hence R is an integral element over A, its image in each $R_\mathfrak{p}$ is integral over $A_\mathfrak{p}$, hence belongs to $A_\mathfrak{p}$; we conclude that $x \in A$ (Bourbaki, *Alg. comm.*, chap. II, § 3, no 3, cor. 1 of th. 1), and hence A is integrally closed in R.

(ii) As $A \subset A_\lambda \subset R$ for every λ, R is also the total ring of fractions of each A_λ. In view of the characterisation of normal rings having R as total ring of fractions, the assertion (ii) follows from Bourbaki, *Alg. comm.*, chap. V, § 1, no 3, prop. 12. □ **IV-2** | 170

This lemma being proved, the proof of (6.14.1) proceeds in several steps.

I) *Reduction to the case where* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, A, A′ *being local Noetherian rings,* A integral, B the integral closure of A. — It suffices to prove that for every $x' \in \mathrm{X}'$, the local ring $\mathscr{O}_{\mathrm{X}',x'}$ is integral and integrally closed; let x, y, y' be the canonical images of x' in X, Y, Y′ respectively; if one sets $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$, $\mathrm{A}' = \mathscr{O}_{\mathrm{Y}',y'}$, $\mathrm{B} = \mathscr{O}_{\mathrm{X},x}$, the rings $\mathscr{O}_{\mathrm{X},x}$, $\mathscr{O}_{\mathrm{Y},y}$, $\mathscr{O}_{\mathrm{Y}',y'}$ for x, y, y' fixed, are the local rings at the prime ideals of the ring $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$ (**I**, 3.6.5), and so it suffices to prove that B′ is a *normal* ring. Let us note on the other hand that $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ is a normal morphism by definition (6.8.1 and **I**, 3.6.5); one is reduced to the case where Y and Y′ are local schemes, that is to say to the case where A is an integral and integrally closed local ring ($\mathbf{0}_{\mathrm{I}}$, 6.8.1 and **I**, 3.6.5); one is reduced to the case where Y and Y′ are local schemes. Let us designate by (B_α) the family of *integral closures* of the sub-A-algebras *of finite type* of B; it is clear that B is the union of the filtered increasing family of the B_α. As $\varinjlim$ commutes with the tensor product, B′ is therefore also the union of the filtered increasing family $\mathrm{B}'_\alpha = \mathrm{B}_\alpha \otimes_{\mathrm{A}} \mathrm{A}'$. To prove that B′ is normal, it suffices, by virtue of (5.13.6), to show that the rings B'_α are *normal* and that, for $\alpha \leqslant \beta$, every irreducible component of $\mathrm{Spec}(\mathrm{B}'_\beta)$ *dominates* one irreducible component of $\mathrm{Spec}(\mathrm{B}'_\alpha)$. But this last property follows from the hypothesis that A′ is an A-module flat and from (2.3.7, (ii)).

One can therefore restrict to proving that $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$ is normal when B is the integral closure of a A-algebra integral *of finite type* C; in this case, if one sets $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$, the morphism $\mathrm{Spec}(\mathrm{C}') \to \mathrm{Spec}(\mathrm{C})$ is normal (6.8.2); as $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{C}} \mathrm{C}'$, one sees that one can replace A and A′ by C and C′ respectively, hence suppose that A is integral and that B is the *integral closure* of A. Finally, the procedure from the beginning allows to restrict to the case where A is *local* (taking into account the fact that, if B is the integral closure of A, $\mathrm{B}_\mathfrak{p}$ is the integral closure of $\mathrm{A}_\mathfrak{p}$ for every prime ideal $\mathfrak{p}$ of A).

II) *Reduction to the case where* A *is a local integral ring of dimension* 1, B *a discrete valuation ring, integral closure of* A *and the morphism* $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ *radicial.* — Let K be the field of fractions of A. One knows (**0**, 23.2.7) that B is intersection of a family (V_λ) of discrete valuation rings such that, for every $x \in \mathrm{K}$, one has $x \in \mathrm{V}_\lambda$ except for a finite number of indices λ. One therefore has an exact sequence of A-modules

$$0 \longrightarrow \mathrm{B} \longrightarrow \mathrm{K} \longrightarrow \bigoplus_\lambda (\mathrm{K}/\mathrm{V}_\lambda).$$

Set $\mathrm{K}' = \mathrm{K} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{V}'_\lambda = \mathrm{V}_\lambda \otimes_{\mathrm{A}} \mathrm{A}'$; by flatness, one deduces from the preceding exact sequence a new exact sequence

$$0 \longrightarrow \mathrm{B}' \longrightarrow \mathrm{K}' \longrightarrow \bigoplus (\mathrm{K}'_\lambda/\mathrm{V}'_\lambda)$$

and hence $\mathrm{B}' = \bigcap_\lambda \mathrm{V}'_\lambda$. Furthermore $\mathrm{Spec}(\mathrm{K}')$ is the fibre of the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ **IV-2** | 171
at the generic point of Spec(A), hence K′ is a Noetherian ring, *normal*, and hence a direct product of finitely many integrally closed integral rings; the direct product L′ of the fields of fractions of these latter is the total ring of fractions of A′, hence also that of B′. One can therefore apply the lemma (6.14.1.1), and if one proves that each of the V'_λ is a *normal* ring, then so is B′.

One knows on the other hand (**0**, 23.2.7) that one can take the V_λ such that there is a sub-A-algebra finite C of K such that V_λ is the integral closure of $\mathrm{C}_{\mathfrak{p}_\lambda}$ where $\mathfrak{p}_\lambda$ is a prime ideal of height 1 in C. If one sets $\mathrm{C}'_{\mathfrak{p}_\lambda} = \mathrm{C}_{\mathfrak{p}_\lambda} \otimes_{\mathrm{A}} \mathrm{A}'$, the morphism $\mathrm{Spec}(\mathrm{C}'_{\mathfrak{p}_\lambda}) \to \mathrm{Spec}(\mathrm{C}_{\mathfrak{p}_\lambda})$ is normal (6.8.2) and one has $\mathrm{V}'_\lambda = \mathrm{V}_\lambda \otimes_{\mathrm{C}_{\mathfrak{p}_\lambda}} \mathrm{C}'_{\mathfrak{p}_\lambda}$; one can therefore replace B by V_λ and A by $\mathrm{C}_{\mathfrak{p}_\lambda}$, hence suppose that A is local, integral and *of dimension* 1, B its integral closure and a discrete valuation ring. There is a sub-A-algebra *finite* A_1 of B such that the morphism $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A}_1)$ is radicial (**0**, 23.2.5), which implies in particular that A_1 is also a local ring (obviously of dimension 1); furthermore one can suppose that B and A_1 have the same residue field (*loc. cit.*). One can therefore by the same method replace A by A_1. Applying if necessary the procedure from the beginning of I), one can suppose finally that A′ is also a local ring and that the homomorphism $\mathrm{A} \to \mathrm{A}'$ is *local*.

III) *End of the proof.* — We shall first establish the following lemma: □

Lemma (6.14.1.2). — *Let* A *be a local Noetherian integral ring of dimension* 1, A′ *a local Noetherian integral ring,* $\mathrm{A} \to \mathrm{A}'$ *a local homomorphism such that the corresponding morphism* $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ *is* normal. *Let* K *be the field of fractions of* A, $\mathfrak{m}$ *the maximal ideal of* A, $\boldsymbol{k} = \mathrm{A}/\mathfrak{m}$ *its residue field.*

. (i) If $\mathfrak{q}'_j$ $(1 \leqslant j \leqslant r)$ are the minimal ideals of A', $K' = K \otimes_A A'$ is a direct product of integral rings and integrally closed $K'_j \supset A'/\mathfrak{q}'_j$ $(1 \leqslant j \leqslant r)$, K'_j having as field of fractions the field of fractions L'_j of $A'/\mathfrak{q}'_j$, so that the total ring of fractions L' of A' is identified with the direct product of the L'_j and A' is a subring of K'.

(ii) The ideal $\mathfrak{p}' = \mathfrak{m}A'$ of A' is prime; the ring $A'_{\mathfrak{p}'}$ is of dimension 1 and is identified with a sub-product of the fields L'_j for the indices j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$.

(iii) If $\overline{A'_{\mathfrak{p}'}}$ denotes the subring of L', product of $A'_{\mathfrak{p}'}$ and of the L'_j such that $\mathfrak{q}'_j \not\subset \mathfrak{p}'$, one has

$$A' = K' \cap \overline{A'_{\mathfrak{p}'}}. \tag{6.14.1.3}$$

Proof. Assertion (i) has already been seen in the course of the proof, independent of the hypothesis on the dimension of A. The hypothesis that the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is normal implies that $A' \otimes_A \boldsymbol{k} = A'/\mathfrak{m}A'$ is a normal local ring and *integral*, which shows already that $\mathfrak{p}' = \mathfrak{m}A'$ is prime. One has, by (6.1.2), $\dim(A'_{\mathfrak{p}'}) = \dim(A) + \dim(A'_{\mathfrak{p}'}/\mathfrak{m}A'_{\mathfrak{p}'})$. But $A'_{\mathfrak{p}'}/\mathfrak{m}A'_{\mathfrak{p}'} = A'_{\mathfrak{p}'}/\mathfrak{p}'A'_{\mathfrak{p}'}$ is the residue field of $A'_{\mathfrak{p}'}$, hence $\dim(A'_{\mathfrak{p}'}) = 1$. The fact that $A'_{\mathfrak{p}'}$ is contained in the composed direct of the L'_j for these indices j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$, for these indices j are the minimal prime ideals of $A'_{\mathfrak{p}'}$. **IV-2** | 172

It remains to prove (6.14.1.3). Conversely, let y' be an element of this intersection; let a be a "parameter" for A, i.e. an element such that Aa is $\mathfrak{m}$-primary, and let a' be the image of a in A'; every element of K can be written x/a^n for $x \in A$ and $n > 0$ since Aa contains a power of $\mathfrak{m}$; hence one can write $y' = x'/a'^n$ with $x' \in A'$. Note now that $\mathfrak{p}'$ is the *only* prime ideal associated to $A'/a'^nA' = (A/a^nA) \otimes_A A'$: as A' is a flat A-module, this follows from (3.3.1), $\mathfrak{m}$ being the only prime ideal associated to A/a^nA since A is integral. Hence, a'^nA' is the inverse image in A' of $a'^nA'_{\mathfrak{p}'}$; as by hypothesis $y' \in \widehat{A'_{\mathfrak{p}'}}$, the image of x' in $A'_{\mathfrak{p}'}$ belongs to $a'^nA'_{\mathfrak{p}'}$; whence $x' \in a'^nA'$ and $y' \in A'$, which finishes the proof.

In the situation in which we have arrived at the end of II), B' is *radicial* over A' (**I**, 3.5.7) and is hence also a local ring; furthermore B' is integer over A', hence, if $\mathfrak{q}'$ is the unique prime ideal of B' lying over $\mathfrak{p}'$, $\mathfrak{q}'B'_{\mathfrak{p}'}$ is the unique maximal ideal of $B'_{\mathfrak{p}'}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, no 1, prop. 1), hence $B'_{\mathfrak{q}'} = B'_{\mathfrak{p}'} = B \otimes_{A_\mathfrak{p}} A'_{\mathfrak{p}'}$. We shall first prove that $B'_{\mathfrak{p}'}$ is *Noetherian* and *normal*. $B'_{\mathfrak{p}'}$ contains $A'_{\mathfrak{p}'}$ and is contained in $K'_{\mathfrak{p}'}$, hence in the product L_1 of the L'_j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$. The composed direct L'' of the fields of fractions of these latter is also the total ring of fractions of A', hence also that of $B'_{\mathfrak{p}'}$. For every index j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$, let $\mathfrak{a}'_j$ be the product of the L'_h for these indices h such that $h \neq j$, so that $\mathfrak{a}'_j \cap A'_{\mathfrak{p}'} = \mathfrak{q}'_jA'_{\mathfrak{p}'}$; as every element of $A' - \mathfrak{p}'$ is regular in L_1, one has also

$$\mathfrak{a}'_j \cap A'_{\mathfrak{p}'} = \mathfrak{q}'_jA'_{\mathfrak{p}'}.$$

Let $\mathfrak{r}'_j = B'_{\mathfrak{p}'} \cap \mathfrak{a}'_j$, so that $B'_{\mathfrak{p}'}/\mathfrak{r}'_j$ is isomorphic to the projection of $B'_{\mathfrak{p}'}$ in L'_j; hence $B'_{\mathfrak{p}'}/\mathfrak{r}'_j$ contains the local integral ring of *dimension* 1, $A'_{\mathfrak{p}'}/\mathfrak{q}'_jA'_{\mathfrak{p}'}$, and is contained in its field of fractions L'_j; it is hence a Noetherian ring by virtue of the theorem of Krull-Akizuki (Bourbaki, *Alg. comm.*, chap. VII, § 2, no 5, prop. 5).

As the intersection of the $\mathfrak{r}'_j$ is reduced to 0, one deduces that $B'_{\mathfrak{p}'}$ itself is Noetherian, by virtue of the classical lemma: **IV-2** | 173

Lemma (6.14.1.4). — *Let* R *be a ring,* $\mathfrak{a}$ *and* $\mathfrak{b}$ *two ideals of* R*; if* $R/\mathfrak{a}$ *and* $R/\mathfrak{b}$ *are Noetherian, so is* $R/(\mathfrak{a} \cap \mathfrak{b})$.

Proof. Indeed, let $\mathfrak{c}$ be an ideal of R such that $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{c}$; one has therefore $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{c} \subset \mathfrak{a}$; now $\mathfrak{c}/(\mathfrak{a} \cap \mathfrak{c})$ is an R-module isomorphic to $(\mathfrak{a} + \mathfrak{c})/\mathfrak{a}$, hence an ideal of $R/\mathfrak{a}$, and hence is of finite type; on the other hand $(\mathfrak{a} \cap \mathfrak{c})/(\mathfrak{a} \cap \mathfrak{b})$ is a sub-R-module of $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$, and this latter is isomorphic to $(\mathfrak{a} + \mathfrak{b})/\mathfrak{b}$, hence is of finite type as an ideal of $R/\mathfrak{b}$; hence $(\mathfrak{a} \cap \mathfrak{c})/(\mathfrak{a} \cap \mathfrak{b})$ is also an R-module of finite type, and it follows finally that $\mathfrak{c}/(\mathfrak{a} \cap \mathfrak{b})$ is also an R-module of finite type. ☐

Let us note on the other hand that $\mathrm{Spec}(A)$ is made up of the closed point $\mathfrak{m}$ and of the generic point (o), of residue fields $\boldsymbol{k}$ and K respectively; as in the case where we are placed, the fields of fractions of B and its residue field are respectively isomorphic to K and $\boldsymbol{k}$, the fibres of the morphism $u : \mathrm{Spec}(B'_{\mathfrak{p}'}) \to \mathrm{Spec}(B)$ at the closed point and at the generic point of $\mathrm{Spec}(B)$ are respectively *isomorphic* to the fibres of the morphism $\mathrm{Spec}(A'_{\mathfrak{p}'}) \to \mathrm{Spec}(A)$ at the closed point and at the generic point of $\mathrm{Spec}(A)$ (**I**, 3.6.4), hence are *geometrically normal*; as on the other hand the morphism u is flat, we conclude that it is *normal* (6.8.1). But B and $B'_{\mathfrak{p}'}$ are Noetherian and B is normal, hence (6.5.4) $B'_{\mathfrak{p}'}$ is *normal*.

This being so, B is union of the filtered increasing family of its sub-A-algebras of finite type A_α; by flatness, B' is union of the filtered increasing family of the $A'_\alpha = A_\alpha \otimes_A A'$; if one sets $\mathfrak{p}'_\alpha = \mathfrak{q}' \cap A'_\alpha$, $B'_{\mathfrak{q}'}$ is also union of the filtered increasing family of the $(A'_\alpha)_{\mathfrak{p}'_\alpha}$ (5.13.3). Denote by L'' the direct product of the fields L'_j such that $\mathfrak{q}'_j \not\subset \mathfrak{p}'$; then for every α, $(A'_\alpha)_{\mathfrak{p}'_\alpha}$ is contained in $K'_{\mathfrak{p}'}$, and the ring $\overline{B'_{\mathfrak{q}'}} = L'' \times B'_{\mathfrak{q}'}$ is therefore union of the rings $L'' \times (A'_\alpha)_{\mathfrak{p}'_\alpha} = (\overline{A'_\alpha})_{\mathfrak{p}'_\alpha}$. But each of the A_α is local, Noetherian, integral and of dimension 1, and the morphism $\mathrm{Spec}(A'_\alpha) \to \mathrm{Spec}(A_\alpha)$ is normal (6.8.2), hence one can apply (6.14.1.2) and one has

$$A'_\alpha = K' \cap (\overline{A'_\alpha})_{\mathfrak{p}'_\alpha}$$

for every α; taking the inductive limit of each of the two members, it follows that

$$B' = K' \cap \overline{B'_{\mathfrak{q}'}}.$$

But $\overline{B'_{\mathfrak{q}'}}$, being the direct product of the normal rings L'' and $B'_{\mathfrak{q}'}$, is normal, and as it has the same total ring of fractions as K', the lemma (6.14.1.1) shows that B' is normal. □

Corollary (6.14.2). — *Let k be a field, X a k-prescheme normal (not necessarily locally Noetherian). Then, for every separable extension k' of k, $X_{(k')} = X \otimes_k k'$ is* normal.

Proof. One knows indeed (6.7.6) that the morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is then normal. □

Corollary (6.14.3). — *Let k be a field, X, Y two k-preschemes integral and normal, whose fields of rational functions $K = R(X)$, $L = R(Y)$ are separable extensions of k. Then the prescheme $X \times_k Y$ is* normal.

Proof. The question being local on X and Y, one can suppose that $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, where A and B are two integral rings and integrally closed, of fields of fractions K and L respectively; suppose first that A is a k-algebra of finite type, hence K is a finite type extension of k. By flatness, $A \otimes_k B$ is a sub-k-algebra of $K \otimes_k L$; now $K \otimes_k L$ is a Noetherian normal ring (by virtue of (6.7.1) or (6.14.2)), hence a direct product of finitely many integral rings C_i, so that if E_i is the field of fractions of C_i, the direct product E of the E_i is the total ring of fractions of $K \otimes_k L$; it is also the total ring of fractions of $A \otimes_k B$, since $K \otimes_k L$ is a ring of fractions of $A \otimes_k B$. This being so, one can write $A \otimes_k B = (A \otimes_k L) \cap (K \otimes_k B)$, and it follows from (6.14.2) that $A \otimes_k L$ and $K \otimes_k B$ are normal, hence $A \otimes_k B$ is normal by virtue of (6.14.1.1). **IV-2** | 174

Consider now the general case; A is then union of the filtered increasing family of its sub-k-algebras of finite type A_α, hence $A \otimes_k B$ is the inductive limit of the normal rings $A_\alpha \otimes_k B$. To prove that $A \otimes_k B$ is normal, it suffices hence (5.13.6) to prove that every irreducible component of $\mathrm{Spec}(A_\beta \otimes_k B)$ dominates one irreducible component of $\mathrm{Spec}(A_\alpha \otimes_k B)$ for $A_\alpha \subset A_\beta$, which follows immediately from the fact that B is a k-module flat, from (2.3.7, (ii)) and from the fact that A_α and A_β are integral rings. □

Proposition (6.14.4). — *Let A be a Noetherian ring, A' a Noetherian A-algebra such that the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is normal. Let B be an A-algebra, C the integral closure of A in B; set $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, C' being identified with a subring of B'; then C' is the integral closure of A' in B'.*

Proof. We shall proceed in several steps.

I) *Reduction to the case where B is reduced.* — Set $B_0 = B_{\mathrm{red}} = B/\mathfrak{N}$ where $\mathfrak{N}$ is the nilradical of B, and let C_0 be the integral closure of A in B_0. One has the following lemma:

Lemma (6.14.4.1). — *Let A be a ring, B an A-algebra, $B_0 = B/\mathfrak{n}$ the quotient of B by a nilideal. If C_0 is the integral closure of A in B_0, the inverse image C of C_0 by the canonical homomorphism $\varphi : B \to B_0$ is the integral closure of A in B.*

Proof. Indeed, if $x \in B$ is such that $\varphi(x)$ satisfies an integral dependence equation over A, one deduces that x satisfies a relation of the form $x^n + a_1 x^{n-1} + \ldots + a_n \in \mathfrak{n}$ with $a_k \in A$, whence, by raising to a sufficiently large power, an integral dependence equation for x, with coefficients in A. □

One has therefore the exact sequence of A-modules

$$0 \longrightarrow \mathrm{C} \longrightarrow \mathrm{B} \longrightarrow \mathrm{B}_0/\mathrm{C}_0$$

whence, by flatness, the exact sequence

$$0 \longrightarrow \mathrm{C}' \longrightarrow \mathrm{B}' \longrightarrow \mathrm{B}'_0/\mathrm{C}'_0$$

where one has set $\mathrm{B}'_0 = \mathrm{B}_0 \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{C}'_0 = \mathrm{C}_0 \otimes_{\mathrm{A}} \mathrm{A}'$. If one proves that C'_0 is the integral closure of A' in B'_0, the lemma (6.14.4.1) will prove that C' is the integral closure of A' in B'. One can therefore restrict to the case where B is an A-algebra of finite type; let $\mathfrak{q}_i$ $(1 \leqslant i \leqslant n)$ be its minimal prime ideals; as B is supposed *reduced*, it is identified with a *subring* of the product B_0 of the $\mathrm{B}/\mathfrak{q}_i$; if C_0 is the integral closure of A in B_0, one has $\mathrm{C} = \mathrm{B} \cap \mathrm{C}_0$; if one sets $\mathrm{B}'_0 = \mathrm{B}_0 \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{C}'_0 = \mathrm{C}_0 \otimes_{\mathrm{A}} \mathrm{A}'$, one has, by flatness, $\mathrm{C}' = \mathrm{B}' \cap \mathrm{C}'_0$ $(\mathbf{0}_{\mathrm{I}},\ 6.1.3)$; it suffices therefore to prove that C'_0 is the integral closure of A' in B'_0. But C_0 is the direct product of the C_i, integral closures of A in the $\mathrm{B}_i = \mathrm{B}/\mathfrak{q}_i$; hence C'_0 is the direct product of the $\mathrm{C}'_i = \mathrm{C}_i \otimes_{\mathrm{A}} \mathrm{A}'$ and it suffices to show that C'_i is the integral closure of A' in $\mathrm{B}'_i = \mathrm{B}_i \otimes_{\mathrm{A}} \mathrm{A}'$. One is thus reduced to the case where B is integral and an A-algebra of finite type; if $\mathfrak{p}$ is the kernel of the homomorphism $\mathrm{A} \to \mathrm{B}$, one has also $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}/\mathfrak{p}} (\mathrm{A}'/\mathfrak{p}\mathrm{A}')$; as the morphism $\mathrm{Spec}(\mathrm{A}'/\mathfrak{p}\mathrm{A}') \to \mathrm{Spec}(\mathrm{A}/\mathfrak{p})$ is normal, one can replace A by $\mathrm{A}/\mathfrak{p}$ and A' by $\mathrm{A}'/\mathfrak{p}\mathrm{A}'$, and suppose hence that $\mathrm{A} \subset \mathrm{B}$.

II) *Reduction to the case where* B *is an integral* A*-algebra of finite type containing* A. — Let (B_α) be the filtered increasing family of sub-A-algebras of finite type of B, and let C_α be the integral closure of A in B_α. It follows immediately from the definition of the integral closure that C is union of the C_α; if one sets $\mathrm{B}'_\alpha = \mathrm{B}_\alpha \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{C}'_\alpha = \mathrm{C}_\alpha \otimes_{\mathrm{A}} \mathrm{A}'$, C'_α is contained in B'_α by flatness, and for the same reason B' is union of the filtered increasing family of the B'_α and C' union of the filtered increasing family of the C'_α. If one proves that C'_α is the integral closure of A' in B'_α, it will follow immediately that C' is the integral closure of A' in B'. One can therefore restrict to the case where B is an A-algebra of finite type; let $\mathfrak{q}_i$ $(1 \leqslant i \leqslant n)$ be its minimal prime ideals; as B is supposed *reduced*, it is identified with a *subring* of the product B_0 of the $\mathrm{B}/\mathfrak{q}_i$; if C_0 is the integral closure of A in B_0, one has $\mathrm{C} = \mathrm{B} \cap \mathrm{C}_0$; if one sets $\mathrm{B}'_0 = \mathrm{B}_0 \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{C}'_0 = \mathrm{C}_0 \otimes_{\mathrm{A}} \mathrm{A}'$, one has, by flatness, $\mathrm{C}' = \mathrm{B}' \cap \mathrm{C}'_0$ $(\mathbf{0}_{\mathrm{I}},\ 6.1.3)$; it suffices therefore to prove that C'_0 is the integral closure of A' in B'_0. But C_0 is the direct product of the C_i, integral closures of A in the $\mathrm{B}_i = \mathrm{B}/\mathfrak{q}_i$; hence C'_0 is the direct product of the $\mathrm{C}'_i = \mathrm{C}_i \otimes_{\mathrm{A}} \mathrm{A}'$ and it suffices to show that C'_i is the integral closure of A' in $\mathrm{B}'_i = \mathrm{B}_i \otimes_{\mathrm{A}} \mathrm{A}'$. One is thus reduced to the case where B is integral and A-algebra of finite type; if $\mathfrak{p}$ is the kernel of the homomorphism $\mathrm{A} \to \mathrm{B}$, one has also $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}/\mathfrak{p}} (\mathrm{A}'/\mathfrak{p}\mathrm{A}')$; as the morphism $\mathrm{Spec}(\mathrm{A}'/\mathfrak{p}\mathrm{A}') \to \mathrm{Spec}(\mathrm{A}/\mathfrak{p})$ is normal, one can replace A by $\mathrm{A}/\mathfrak{p}$ and A' by $\mathrm{A}'/\mathfrak{p}\mathrm{A}'$, and suppose hence that $\mathrm{A} \subset \mathrm{B}$. **IV-2** | 175

III) *Case where* A *is a field,* B *a field, finite extension of* A *and such that* A *is algebraically closed in* B. — One has then $\mathrm{C} = \mathrm{A}$, and A' is a Noetherian geometrically normal A-algebra, hence a direct product of integrally closed integral rings A'_i, the fields of fractions K'_i of the A'_i being *separable* extensions of A (4.6.1). The A'-algebra B' is therefore the direct product of the $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'_i = \mathrm{B}'_i$. One can therefore restrict to the case where A' is integral, its field of fractions K' being a separable extension of A. Now, as K' is a separable extension of A, $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K}'$ is reduced (4.3.7), and as furthermore B is a primary extension of A, $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K}'$ is integral (4.3.2); it is hence of the same type as B'. Let L' be the field of fractions of $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K}'$ (which is also that of B'); it is obviously a composed field of $\mathrm{B}(\mathrm{K}')$ of its sub-fields B and K'; as A is algebraically closed in B, that K' is a separable extension of A, and that B and K' are linearly disjoint over A, K' is algebraically closed in $\mathrm{B}(\mathrm{K}') = \mathrm{L}'$ (Bourbaki, *Alg.*, chap. V, § 9, exerc. 2); *a fortiori*, A', being integrally closed, is integrally closed in L', hence in B', which proves the proposition in this case.

IV) *Case where* A *is integral,* B *a field, finite extension of the field of fractions* K *of* A. — Then $\mathrm{B}' = (\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}) \otimes_{\mathrm{K}} \mathrm{B}$ is a B-algebra Noetherian geometrically normal, and hence B' is a direct product of integral rings; the total ring of fractions L' of B' is then a direct product of finitely many fields. This being so, as B is the field of fractions of C, B' is a ring of fractions of C' and L' is therefore also the total ring of fractions of C'. Now $(\mathbf{0},\ 6.14.1)$ as C is a normal ring and $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ a normal morphism of Noetherian rings, C' is a normal ring; but it follows from (6.14.1.1) that C' is then integrally closed in L', and *a fortiori* in B', and hence is the integral closure of A' in B'.

V) *End of the proof.* — By part II), one can suppose that B is an A-algebra integral of finite type containing A. Let K be the field of fractions of A, L that of B, which is a finite type extension of K. Let M be the algebraic closure of K in L, which is a finite algebraic extension of K; let C_0 be the integral closure of A in M, which is also the integral closure of A in L; if one sets $C_0' = C \otimes_A A'$, one has hence $C' = B' \cap C_0'$ by flatness ($\mathbf{0}_I$, 6.1.3). Now, it follows from IV) that C_0' is the integral closure of A' in $M' = M \otimes_A A'$; furthermore, M' is a Noetherian ring and the morphism $\operatorname{Spec}(M') \to \operatorname{Spec}(M)$ is normal (as we have seen in the course of IV)); applying III) to M and M' in place of A and A' and to L in place of B, one deduces that C_0' is the integral closure of A' in L', and $C' = C_0' \cap B'$ is the integral closure of A' in L', and hence C' is the integral closure of A' in B'. □ **IV-2** | 176

Corollary (6.14.5). — *Let* A *be a Noetherian ring,* A' *a Noetherian* A*-algebra such that the morphism* $\operatorname{Spec}(A') \to \operatorname{Spec}(A)$ *is normal. Let* B, C *be two* A*-algebras,* $\varphi : B \to C$ *an* A*-homomorphism,* D *the integral closure of* B *in* C*. If one sets* $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, $D' = D \otimes_A A'$, D' *is the integral closure of* B' *in* C'.

Proof. Let (B_λ) be the filtered increasing family of sub-A-algebras of finite type of B, and set $B'_\lambda = B_\lambda \otimes_A A'$, so that B' is union of the filtered increasing family of the sub-A'-algebras of finite type B'_λ. Let D_λ be the integral closure of B_λ in C, and set $D'_\lambda = D_\lambda \otimes_A A'$, so that D is union of the D_λ and D' union of the D'_λ; if we prove that D'_λ is the integral closure of B'_λ in C', it will follow that D' is the integral closure of B' in C'. Now B_λ and B'_λ are Noetherian and the morphism $\operatorname{Spec}(B'_\lambda) \to \operatorname{Spec}(B_\lambda)$ is normal (6.8.2); one can therefore apply (6.14.4) by replacing A, A' and B by B_λ, B'_λ and C respectively, whence the corollary. □

One can for example apply (6.14.5) when A is its local ring *excellent* and A' its completion $\widehat{A}$, since in this case $\operatorname{Spec}(A') \to \operatorname{Spec}(A)$ is a regular morphism (7.8.2).

6.15. Geometrically unibranch preschemes

(6.15.1). We shall say that a prescheme X is *unibranch* (resp. *geometrically unibranch*) at a point x, or that the point x is unibranch (resp. geometrically unibranch) in X if the local ring $\mathscr{O}_{X,x}$ is unibranch (resp. geometrically unibranch) ($\mathbf{0}$, 23.2.1); we say that X is *unibranch* (resp. *geometrically unibranch*) if it is so at every point. It follows from this definition that, for X to be unibranch (resp. geometrically unibranch) at a point, it is necessary and it suffices that X_{red} be so at this point.

(6.15.2). One must take care that with the definitions of (6.15.1), a *local ring* A can be unibranch (resp. geometrically unibranch) *without* $\operatorname{Spec}(A)$ being so; in other terms, it can happen that there are prime ideals $\mathfrak{p}$ of A such that $A_\mathfrak{p}$ is not unibranch; it amounts to the same to say that on a prescheme, the notion of geometrically unibranch point *is not stable by generization*. We shall see this on the following example.

Let K be an algebraically closed field of characteristic 0, B the integral ring $K[U,V,W]/(U^2(U - W) - V^2(U+W))$ (U, V, W indeterminates) so that $Y = \operatorname{Spec}(B)$ is a "cone with vertex the origin, having a double generatrix". We shall designate by u, v, w the images of U, V, W in B. Let R be the field of fractions of B, and consider in R the element $t = v(u+w)/u$, which does not belong to B; let us show that $C = B[t]$ is the *integral closure* of B. One has indeed $t^2 = u^2 - w^2$, hence t is integral over B, and $v = tu/(u+w)$; the ring $C_1 = K[t,u,w]$ is integrally closed, for it is isomorphic to $K[T,U,W]/(T^2 - U^2 + W^2)$ and is therefore the integral closure of the integrally closed ring $K[U,W]$ in the quadratic extension $K(U,W)(\sqrt{U^2+W^2})$ of its field of fractions (Bourbaki, *Alg. comm.*, chap. V, § 1, no 6, prop. 18). The ring of fractions $K[t,u,w,1/(u+w)]$ of C_1 is hence integrally closed. In the same way, one sees that the ring $C_2 = K[t,v,w]$ is integrally closed, for t is a root of the equation $t(t-v) - w^2(2v-t) = 0$, and hence $K[t,v,w,1/(t-v)] = K[t,v,w,\frac{tu}{u+w},w,\frac{u+w}{tw}]$ is integrally closed. Finally, taking into account the fact that u and w are algebraically independent over K, one proves easily that $C = K[t,u,v,w] = K[t,u,w,1/(u+w)] \cap K[t,v,w,1/(t-v)]$, which finishes showing that C is the integral closure of B. It is immediate that if $\mathfrak{m}_0$ is the maximal ideal of C lying over $\mathfrak{n}_0$, the ideal generated by u, v, w **IV-2** | 177 ("vertex of the cone"), there is only one maximal ideal of C lying over $\mathfrak{m}_0$, namely the ideal generated by t, u, v, w. If one sets $A = B_{\mathfrak{m}_0}$, one deduces easily that $A' = C_{\mathfrak{m}_0}$ is the integral closure of A, which is hence unibranch, and hence also geometrically unibranch since its residue field K is algebraically closed. But in A the prime ideal $\mathfrak{p}$ generated by u and v is such that the integral closure $A_{\mathfrak{p}}[t]$ of $A_{\mathfrak{p}}$ is not a local ring.

We shall see further below (9.7.10) that when X is a locally Noetherian prescheme such that, if X′ is the normalization of X_{red}, the canonical morphism $X' \to X$ is *finite* (which will be the case if X is such that the rings of its affine open subsets are *universally Japanese* (**0**, 23.1.1))), then the set of points $x \in X$ where X is geometrically unibranch is *locally constructible*.

(6.15.3). We shall say for short that a morphism $f : X \to Y$ is *radicial at a point* $y \in Y$, if $f^{-1}(y)$ is empty or reduced to a single point x and if $\boldsymbol{k}(x)$ is a radicial extension of $\boldsymbol{k}(y)$. It amounts to the same to say that f is radicial at every point of Y or that f is radicial (**I**, 3.5.8). If $g : f^{-1}(y) \to \operatorname{Spec}(\boldsymbol{k}(y))$ is the base change inverse image of f by the base change $\operatorname{Spec}(\boldsymbol{k}(y)) \to Y$, it amounts to the same to say that f is radicial at the point $y \in Y$, or that g is *radicial*.

Lemma (6.15.3.1). — *(i) Let $f : X \to Y$, $g : Y \to Z$ be two morphisms. If f is radicial at a point y and g radicial at the point $z = g(y)$, then $g \circ f$ is radicial at the point $g(y)$. The converse is true if f is surjective.*

(ii) Let $f : X \to Y$, $h : Y' \to Y$ be two morphisms, and let $f' = f_{|Y'} : X \times_Y Y' \to Y'$. Let $y' \in Y'$ and $y = h(y')$; then for f to be radicial at the point y, it is necessary and it suffices that f' be radicial at the point y'.

Proof. (i) If f is radicial at the point y and g radicial at the point z, $g^{-1}(z)$ is reduced to y and $f^{-1}(g^{-1}(z)) = f^{-1}(y)$ is empty or reduced to a single point x; furthermore $\boldsymbol{k}(x)$ is a radicial extension of $\boldsymbol{k}(y)$ and $\boldsymbol{k}(y)$ a radicial extension of $\boldsymbol{k}(z)$, hence $\boldsymbol{k}(x)$ is a radicial extension of $\boldsymbol{k}(z)$. Conversely, suppose f surjective; if $g \circ f$ is radicial at the point $z = g(y)$, $g^{-1}(z)$ is reduced to the point y, for otherwise $f^{-1}(g^{-1}(z))$ would have at least two distinct points; furthermore, $f^{-1}(y) = f^{-1}(g^{-1}(z))$ is reduced to a single point x, and by hypothesis one has $\boldsymbol{k}(z) \subset \boldsymbol{k}(y) \subset \boldsymbol{k}(x)$, and $\boldsymbol{k}(x)$ is a radicial extension of $\boldsymbol{k}(z)$, hence $\boldsymbol{k}(y)$ is a radicial extension of $\boldsymbol{k}(z)$ and $\boldsymbol{k}(x)$ radicial over $\boldsymbol{k}(y)$. **IV-2** | 178

(ii) Let $g : f^{-1}(y) \to \operatorname{Spec}(\boldsymbol{k}(y))$, $g' : f'^{-1}(y') \to \operatorname{Spec}(\boldsymbol{k}(y'))$ be the morphisms inverse images of f and f' respectively; as $f'^{-1}(y') = f^{-1}(y) \otimes_{\boldsymbol{k}(y)} \boldsymbol{k}(y')$ (**I**, 3.6.4), g' is the inverse image of g and the assertion reduces to (2.6.1, (v)). □

Suppose then X a prescheme *reduced* not having more than a *finite* number of irreducible components, and let X′ be its *normalization* (**II**, 6.3.8); we say (*loc. cit.*) that the canonical morphism $f : X' \to X$ is surjective. The definition (6.15.1) shows then that for X to be geometrically unibranch at a point, it is necessary and it suffices that f be *radicial at this point*; for X to be geometrically unibranch, it is necessary and it suffices that f be *radicial*.

(6.15.4). Generalizing the definition given in (**I**, 2.2.9), we shall say, for two reduced preschemes X, Y, a morphism $f : X \to Y$ is *birational* if the restriction of f to the set of maximal points of X is a *bijection* of this set onto the set of maximal points of Y, and if, for every maximal point y of Y, the morphism $X \times_Y \operatorname{Spec}(\mathscr{O}_y) \to \operatorname{Spec}(\mathscr{O}_y)$ induced by f is an isomorphism (in other words, if for the fibre $f^{-1}(y)$ reduced to a single point x (maximal in X) and if the homomorphism $\boldsymbol{k}(y) \to \boldsymbol{k}(x)$ induced by f is bijective); this notion reduces to that of (**I**, 2.2.9) when X and Y have only a *finite* number of irreducible components.

Lemma (6.15.4.1). — *Let $f : X \to Y$ be a morphism, $g : Y' \to Y$ a flat morphism, $X' = X \times_Y Y'$, $f' = f_{|Y'} : X' \to Y'$. If f_{red} is birational, so is f'_{red}.*

Proof. Indeed, if y' is a maximal point of Y′, one knows (2.3.4, (ii)) that $y = g(y')$ is a maximal point of Y; there is a unique point x of X in $f^{-1}(y)$ by hypothesis; since moreover $\boldsymbol{k}(x) = \boldsymbol{k}(y)$. It follows therefore from (**I**, 3.4.9) that $f'^{-1}(y')$ is reduced to a single point x' and that $\boldsymbol{k}(x') = \boldsymbol{k}(y')$. We conclude from the reasoning that, noting (2.3.7, (ii)), every irreducible component of X′ dominates one irreducible component of Y′. □

Proposition (6.15.5). — *Let $f : X \to Y$ be a morphism such that f_{red} is entire and birational* (6.15.4), *hence surjective.*

(i) *For* Y *to be geometrically unibranch at a point* y*, it is necessary and it suffices that* f *be radicial at the point* y *and that* X *be geometrically unibranch at the unique point* x *of* $f^{-1}(y)$*.*
(ii) *For* Y *to be geometrically unibranch, it is necessary and it suffices that* X *be geometrically unibranch and that* f *be radicial.*

Proof. One can obviously restrict to the case where X and Y are *reduced*; the fact that f is *surjective* follows from the fact that f is closed (**II**, 6.1.10) and that $f(\mathrm{X})$ contains all the maximal points of Y. This also shows immediately that (i) implies (ii). Using (**I**, 3.6.5) and (**II**, 6.1.5), one can suppose, to prove (i), that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, in other words $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ (with $\mathrm{A} = \mathscr{O}_{\mathrm{X},x}$) is a local **IV-2** | 179
scheme. Note that $\mathrm{A}' = \mathscr{O}_{\mathrm{X}',x'}$ is by hypothesis a faithfully flat A-module, hence containing A, and hence the hypothesis that X is geometrically unibranch at the point x, or the hypothesis that X′ is geometrically unibranch at the point x', both imply that A is a local ring *integral* (since A, being reduced, is also isomorphic to a subring of $\mathrm{A}'_{\mathrm{red}}$). Let K be the field of fractions of A, B the integral closure of A, and set $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$; it amounts to the same to say that X is geometrically unibranch at the point x or that the morphism $g : \mathrm{Y} \to \mathrm{X}$ is radicial at the point x (6.15.3). Let $\mathrm{Y}' = \mathrm{Y} \otimes_k k' = \mathrm{Y} \times_{\mathrm{X}} \mathrm{X}'$ so that one has the commutative diagram

$$\begin{tikzcd} \mathrm{Y} \arrow[r, "h"] & \mathrm{Y}' \\ \mathrm{X} \arrow[u, "g"] \arrow[r, "f"'] & \mathrm{X}' \arrow[u, "g'"'] \end{tikzcd}$$

Note that f is a flat morphism; hence (6.15.4.1) the entire morphism g'_{red} is birational. As Y is normal, Y′ is geometrically unibranch (6.15.6). For X to be geometrically unibranch at the point x, it is necessary and it suffices therefore that g' be radicial at the point x' (6.15.5). Now, x is the projection of x' in X and it is equivalent to say that g is radicial at the point x or that g' is radicial at the point x' (6.15.3.1); finally, for X to be geometrically unibranch at this point, it is necessary and it suffices that g be radicial at this point, whence the proposition. □

Proposition (6.15.6). — *Let* k *be a field,* X *a* k*-prescheme. If* X *is normal, then, for every extension* k' *of* k*,* $\mathrm{X}' = \mathrm{X} \otimes_k k'$ *is* geometrically unibranch.

Proof. One knows that k' is an algebraic extension of a pure extension k_0 of k, and if k'' is the largest separable extension of k_0 contained in k', k' is a radicial extension of k'' and k'' a separable extension of k. One knows (6.14.2) that $\mathrm{X}'' = \mathrm{X} \otimes_k k''$ is normal; as $\mathrm{X} \otimes_k k' = \mathrm{X}'' \otimes_{k''} k'$, one sees that one can restrict to the case where the extension k' of k is *radicial*. Furthermore (**I**, 3.6.5), one can suppose that $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, where A is a *local integral and integrally closed* ring (since X is normal). The projection morphism $f : \mathrm{X}' \to \mathrm{X}$ is a universal homeomorphism, since $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is a universal homeomorphism (2.4.5); as $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$ where $\mathrm{A}' = \mathrm{A} \otimes_k k'$, one sees therefore that A′ is a local ring whose nilradical $\mathfrak{N}'$ is the unique minimal prime ideal, whence $\mathrm{X}'_{\mathrm{red}} = \mathrm{Spec}(\mathrm{A}'_0)$, where $\mathrm{A}'_0 = \mathrm{A}'/\mathfrak{N}'$ is a local integral ring; furthermore, if K is the field of fractions of A, the field of fractions K'_0 of A'_0 is radicial over K, since the morphism f is radicial. But A'_0 is integer over A, its integral closure B is also the integral closure of A, and one knows (Bourbaki, *Alg. comm.*, chap. V, § 2, no 3, lemma 3) that B is the set of $x \in \mathrm{K}'_0$ of which a power p^m-th (for m large enough) belongs to A (p being the characteristic exponent of K); furthermore there exists only a single prime ideal of B lying over every prime ideal of A; in particular B is a local ring and its residue field is a radicial extension of that of A, and *a fortiori* of the same type for A'_0, which proves that A'_0 is geometrically unibranch, and hence so is X′ as well. □

Proposition (6.15.7). — *Let* k *be a field,* X *a* k*-prescheme,* k' *an extension of* k*,* $\mathrm{X}' = \mathrm{X} \otimes_k k'$*. Let* x' *be a point of* X′*,* x *its projection in* X*. For* X *to be geometrically unibranch at the point* x*, it is necessary and it suffices that* X′ *be geometrically unibranch at the point* x'*. For* X *to be geometrically unibranch, it is necessary and it suffices that* X′ *be so.*

Proof. The second assertion follows from the first and from the fact that the projection $f : \mathrm{X}' \to \mathrm{X}$ **IV-2** | 180
is a surjective morphism. To prove the first assertion, one can, by virtue of (**I**, 5.1.8), restrict to the case where X is reduced, and by virtue of (**I**, 3.6.5), to the case $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ (with $\mathrm{A} = \mathscr{O}_{\mathrm{X},x}$) is a

local scheme. Note that $A' = \mathscr{O}_{X',x'}$ is by hypothesis a faithfully flat A-module, hence containing A, and hence the hypothesis that X is geometrically unibranch at the point x, or the hypothesis that X′ is geometrically unibranch at the point x', both imply that A is a local ring *integral* (since A, being reduced, is also isomorphic to a subring of A'_{red}). Let K be the field of fractions of A, B the integral closure of A, and set $Y = \mathrm{Spec}(B)$; it amounts to the same to say that X is geometrically unibranch at the point x or that the morphism $g : Y \to X$ is radicial at the point x (6.15.3). Let $Y' = Y \otimes_k k' = Y \times_X X'$ so that one has the commutative diagram

$$\begin{tikzcd} Y \arrow[r, "h"] & Y' \\ X \arrow[u, "g"] \arrow[r, "f"'] & X' \arrow[u, "g'"'] \end{tikzcd}$$

Note that f is a flat morphism; hence (6.15.4.1) the entire morphism g'_{red} is birational. As Y is normal, Y′ is normal (and *a fortiori* geometrically unibranch) (6.14.1). For X′ to be geometrically unibranch at x', it is necessary and it suffices therefore that g' be radicial at this point (6.15.5); we conclude hence from (6.15.3.1) that f' is radicial at every point $x' \in p^{-1}(x)$, and it follows then from (6.15.5) that X′ is geometrically unibranch (hence integral since it is reduced) at each of these points. □

Lemma (6.15.8). — *Let k be a separably closed field (in other words, such that the algebraic closure of k is* radicial *over k), X a k-prescheme locally of finite type over k, x a closed point of X. If X is unibranch at the point x, it is* geometrically unibranch *at this point.*

Proof. Indeed, one knows (**I**, 6.4.2) that $\boldsymbol{k}(x)$ is an *algebraic* extension of k; as the residue field of the integral closure $(\mathscr{O}_x)_{\mathrm{red}}$ (integral closure which is by hypothesis a local ring) is an algebraic extension of $\boldsymbol{k}(x)$, it is a radicial extension of k by hypothesis, hence also of $\boldsymbol{k}(x)$. □

Corollary (6.15.9). — *Let k be a field, X a k-prescheme locally of finite type over k, k' a separable extension close of k (in other words, such that the algebraic closure of k is radicial over k'); then, for X to be geometrically unibranch, it is necessary and it suffices that $X' = X \otimes_k k'$ be unibranch.*

Proof. In view of (6.15.7), one is reduced to proving that if X is unibranch and k separably closed, then X is geometrically unibranch. Now, X is geometrically unibranch at its closed points (6.15.8); we conclude from (10.4.9) that X is geometrically unibranch at all its points, relying on the fact (pointed out at the end of (6.15.2)) that the set of points where X is geometrically unibranch is constructible (of course, the corollary (6.15.9) will not be used before that point). □

This result justifies, to a certain extent, the terminology of "geometrically unibranch".

Proposition (6.15.10). — *Let Y, Y′ be two locally Noetherian preschemes, $g : Y' \to Y$ a normal morphism (6.8.1), $f : X \to Y$ a morphism; set $X' = X \times_Y Y'$ and let $p : X' \to X$ be the canonical projection, x' a point of X′. Then, if X is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) at the point $x = p(x')$, X′ is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) at the point x'.*

Proof. By virtue of (**I**, 3.6.5), one can restrict to the case where $Y = \mathrm{Spec}(A)$, $Y' = \mathrm{Spec}(A')$, **IV-2** | 181
$X = \mathrm{Spec}(B)$, where A, A′, B are local rings, the homomorphisms $A \to A'$ and $A \to B$ being local, A, A′ Noetherian rings, and x being the closed point of X; it suffices to prove that if B is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) then $B' = B \otimes_A A'$ is also geometrically unibranch at the points of $p^{-1}(x)$, resp. $\mathrm{Spec}(B')$ is integral and geometrically unibranch at these points). Suppose first B reduced; B is union of the filtered increasing family of its sub-A-algebras of finite type B_λ, and by flatness, B′ is union of the filtered increasing family of the sub-A′-algebras $B'_\lambda = B_\lambda \otimes_A A'$. Now, the morphism $\mathrm{Spec}(B'_\lambda) \to \mathrm{Spec}(B_\lambda)$ is normal (6.8.2), and *a fortiori* reduced; the fact that B'_λ is reduced is consequence of (3.3.5); one concludes that B′ itself is reduced by (5.13.2).

Suppose now B geometrically unibranch; in view of (**I**, 5.1.8), one can furthermore suppose B reduced, hence *integral*. Let C be its integral closure, which is by hypothesis a local ring; set

$Z = \mathrm{Spec}(C)$, $C' = C \otimes_A A'$, $Z' = \mathrm{Spec}(C') = Z \times_X X'$, so that one has the commutative diagram

$$\begin{tikzcd} Z \arrow[r] & Z' \\ X \arrow[u, "l"] \arrow[r, "p"'] & X' \arrow[u, "l'"'] \end{tikzcd}$$

By virtue of the first part of the reasoning, X' is *reduced*; on the other hand, as $Z' = Z \times_Y Y'$, it follows from (6.14.1) that Z' is normal (and *a fortiori* geometrically unibranch). As f is entire and birational, so is f' by (6.15.4.1), since p is flat. Finally, the hypothesis that X is geometrically unibranch at the point x implies that f is radicial at this point (6.15.5); we conclude hence from (6.15.3.1) that f' is radicial at every point $x' \in p^{-1}(x)$, and it follows then from (6.15.5) that X' is geometrically unibranch (hence integral since it is reduced) at each of these points. ☐

Remarks (6.15.11). — (i) In the proof of (6.15.10), one cannot dispense with appealing to (6.14.1), even when $X = Y$, for the argument then uses the integral closure of the ring B, which need not be Noetherian even when B is.

(ii) The example (6.5.5, (ii)) shows that in (6.15.10), one cannot replace "geometrically unibranch" by "integral" in the statement, even if the residue field $\boldsymbol{k}(x)$ is algebraically closed and the morphism g is étale.

Nor can "geometrically unibranch" be replaced there by "unibranch". Indeed, let

$$A = \mathbb{R}[[U, V]]/(U^2 + V^2)$$

and let u, v be the images of U, V in A; the maximal ideal of A is $Au + Av$. The integral closure of A is the ring $A[t]$, where $t = u/v$ satisfies $t^2 = -1$, so that $A[t]$ is isomorphic to the local ring $\mathbb{C}[[U]]$. Since the residue field of $\mathbb{C}[[U]]$ is $\mathbb{C}$, A is unibranch but not geometrically unibranch. Yet $A \otimes_{\mathbb{R}} \mathbb{C}$ is not integral: it is isomorphic to

$$\mathbb{C}[[U, V]]/((U - iV)(U + iV)).$$

§7. Formal properties of morphisms; Noetherian approximation

7.1. Formal equidimensionality and formally catenary rings **IV-2** | 183

Definition (7.1.1). — We say that a local Noetherian ring A is *formally equidimensional* if its completion $\widehat{A}$ is equidimensional ($\mathbf{0}$, 16.1.4).

Proposition (7.1.2). — *Let* A *be a local Noetherian ring,* $\mathfrak{p}_i$ *(*$1 \leqslant i \leqslant n$*) its minimal prime ideals. For* A *to be formally equidimensional, it is necessary and sufficient that the* $A/\mathfrak{p}_i$ *be equidimensional (in other words, that the* $A/\mathfrak{p}_i$ *have the same dimension).*

Proof. Indeed, this follows from the fact that for every prime ideal $\mathfrak{p}'$ of $\widehat{A}$, $\mathfrak{p}' \cap A$ contains one of the $\mathfrak{p}_i$, hence $\mathfrak{p}'$ contains one of the $\mathfrak{p}_i\widehat{A}$, and $\widehat{A}/\mathfrak{p}_i\widehat{A} = (A/\mathfrak{p}_i) \otimes_A \widehat{A}$ is the completion of the local ring $A/\mathfrak{p}_i$ ($\mathbf{0}$, 7.3.3), hence has the same dimension. Every maximal chain of prime ideals of $\widehat{A}$ therefore identifies canonically with a maximal chain of prime ideals of one of the $\widehat{A}/\mathfrak{p}_i\widehat{A}$, and conversely, whence the conclusion. ☐

Proposition (7.1.3). — *Let* A, A′ *be two local Noetherian rings,* $\mathfrak{m}$ *the maximal ideal of* A, $\varphi : A \to A'$ *a local homomorphism; suppose that* A′ *is a flat* A*-module, and that* A′ *is equidimensional and catenary. Then:*

i) A *is equidimensional and catenary.*
ii) *Suppose in addition that* $\mathfrak{m}A'$ *is an ideal of definition of* A′*. Then, if* $\mathfrak{a}$ *is an ideal of* A*, for* $A'/\mathfrak{a}A'$ *to be equidimensional, it is necessary and sufficient that* $A/\mathfrak{a}A$ *be so. In particular, for every prime ideal* $\mathfrak{p}$ *of* A, $A'/\mathfrak{p}A'$ *is equidimensional.*

Proof. Let us first prove the second assertion of (ii). Set $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(A')$, $Y = V(\mathfrak{p}) = \mathrm{Spec}(A/\mathfrak{p})$, $Y' = V(\mathfrak{p}A') = \mathrm{Spec}(A'/\mathfrak{p}A')$; if $f : X' \to X$ is the morphism corresponding to φ, Y' is also the closed sub-prescheme $f^{-1}(Y)$ of X'. As $\mathfrak{m}A'$ is an ideal of definition of A', we have

$$\dim(X) = \dim(X') \tag{7.1.3.1}$$

by virtue of (6.1.3); similarly $\mathfrak{m}A'/\mathfrak{p}A'$ is an ideal of definition of $A'/\mathfrak{p}A'$ and Y' is flat over Y (2.1.4), hence (6.1.3)

$$\dim(Y) = \dim(Y'). \tag{7.1.3.2}$$

Let y be the generic point of Y, Y'_i $(1 \leqslant i \leqslant r)$ the irreducible components of Y', y'_i the generic point of Y'_i $(1 \leqslant i \leqslant r)$; we have $f(y'_i) = y$ for all i (2.3.4). On the other hand, y'_i is the maximal point of the fibre $f^{-1}(y)$ (**0**, 2.1.8), hence $\mathscr{O}_{X',y'_i} \otimes_{\mathscr{O}_{X,y}} \boldsymbol{k}(y) = \mathscr{O}_{X',y'_i}/\mathfrak{m}_y\mathscr{O}_{X',y'_i}$ is of dimension 0, in other words $\mathfrak{m}_y\mathscr{O}_{X',y'_i}$ is an ideal of definition of $\mathscr{O}_{X',y'_i}$; we can again apply (6.1.3), which gives

$$\dim(\mathscr{O}_{X',y'_i}) = \dim(\mathscr{O}_{X,y}) \tag{7.1.3.3}$$

that is to say (5.1.2)

$$\operatorname{codim}(Y'_i, X') = \operatorname{codim}(Y, X) \tag{7.1.3.4}$$

for all i. As A' is equidimensional and catenary, we have, by virtue of (**0**, 14.3.5) **IV-2** | 184

$$\dim(X') = \dim(Y'_i) + \operatorname{codim}(Y'_i, X') \tag{7.1.3.5}$$

for all i; by virtue of (7.1.3.1), (7.1.3.2) and (7.1.3.4) and the inequality $\dim(Y'_i) \leqslant \dim(Y')$, we deduce

$$\dim(X) \leqslant \dim(Y) + \operatorname{codim}(Y, X) \tag{7.1.3.6}$$

hence the two sides of this inequality are equal by (**0**, 14.2.2.2); moreover this equality implies

$$\dim(Y'_i) = \dim(Y') = \dim(Y) \tag{7.1.3.7}$$

for all i, which proves the second assertion of (ii).

Let us now prove the first assertion of (ii). Let $\mathfrak{p}_j$ $(1 \leqslant j \leqslant n)$ be the minimal prime ideals of A among those containing $\mathfrak{a}$, and let $\mathfrak{p}'_{jh}$ be the prime ideals of A' minimal among those containing $\mathfrak{p}_jA'$ $(1 \leqslant h \leqslant m_j)$; we know then (2.3.4) that the $\mathfrak{p}'_{jh}$ $(1 \leqslant j \leqslant n, 1 \leqslant h \leqslant m_j$ for all $j)$ are also the minimal prime ideals among those containing $\mathfrak{a}A'$. From what we have just seen, for all j, the dimensions of the rings $A'/\mathfrak{p}'_{jh}$ $(1 \leqslant h \leqslant m_j)$ are all equal and are equal to $\dim(A'/\mathfrak{p}_jA')$, itself equal to $\dim(A/\mathfrak{p}_j)$ by (7.1.3.2). To say that $A'/\mathfrak{a}A'$ is equidimensional therefore means that the dimensions of the $A/\mathfrak{p}_j$ are all equal, that is to say that $A/\mathfrak{a}$ is equidimensional.

Let us finally prove (i). Let $\mathfrak{r}'$ be a prime ideal of A' minimal among those containing $\mathfrak{m}A'$, and set $A'' = A'_{\mathfrak{r}'}$; as A'' is a flat A'-module, it is also a flat A-module; moreover A'' is equidimensional and catenary (**0**, 16.1.4), and $\mathfrak{m}A''$ is an ideal of definition of A''. We can therefore restrict ourselves to the case where $\mathfrak{m}A'$ is an ideal of definition of A'. If then $\mathfrak{p}$ and $\mathfrak{q} \subset \mathfrak{p}$ are two prime ideals of A, we can apply (ii) to $A/\mathfrak{q}$ and $A'/\mathfrak{q}A'$, which is equidimensional and flat over $A/\mathfrak{q}$; if $Z = \operatorname{Spec}(A/\mathfrak{q})$ and $Y = \operatorname{Spec}(A/\mathfrak{p})$, we have $\dim(Z) = \dim(Y) + \operatorname{codim}(Y, Z)$; moreover one can apply (7.1.3.6), which shows that $\operatorname{codim}(Y, X) = \dim X - \dim Y$; as X is equicodimensional, these relations show that it is biequidimensional by virtue of (**0**, 14.3.3). □

Corollary (7.1.4). — *Let A be a local Noetherian ring that is formally equidimensional. Then:*

i) *A is equidimensional and catenary (in other words, biequidimensional).*
ii) *Let $\mathfrak{a}$ be an ideal of A; for $A/\mathfrak{a}$ to be equidimensional, it is necessary and sufficient that $A/\mathfrak{a}$ be formally equidimensional. In particular, for every prime ideal $\mathfrak{p}$ of A, $A/\mathfrak{p}$ is formally equidimensional.*

Proof. If $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}\widehat{A}$ is an ideal of definition of $\widehat{A}$. By hypothesis $\widehat{A}$ is equidimensional, and on the other hand it is known to be catenary (5.6.4); it suffices then to apply (7.1.3) with $A' = \widehat{A}$. □

Corollary (7.1.5). — *Let A be a local Noetherian ring such that there exists a finite type A-module M that is a Cohen-Macaulay A-module and for which $\operatorname{Supp}(M) = \operatorname{Spec}(A)$ (this holds for example if A is a Cohen-Macaulay ring). Then A is formally equidimensional, hence (7.1.4) for a quotient ring B of A to be formally equidimensional, it is necessary and sufficient that it be equidimensional.*

Proof. Indeed, $\widehat{M} = M \otimes_A \widehat{A}$ is a Cohen-Macaulay $\widehat{A}$-module (**0**, 16.5.2), with support equal to **IV-2** | 185
$\operatorname{Spec}(\widehat{A})$; hence (**0**, 16.5.4), $\widehat{A}$ is equidimensional. □

Remark (7.1.6). — We have seen (6.3.8) that under the hypotheses of (7.1.5), for every integral quotient ring B of A, the fibres of the canonical morphism $\operatorname{Spec}(\widehat{B}) \to \operatorname{Spec}(B)$ are preschemes of Cohen-Macaulay; hence (6.4.3) and (5.7.5), *for* B *to have no embedded associated prime ideals, it is necessary and sufficient that the same hold for* $\widehat{B}$.

Lemma (7.1.7). — *Let* A *be a local Noetherian ring,* $\mathfrak{p}_i$ *($1 \leqslant i \leqslant r$) its minimal prime ideals. Suppose that each of the rings* $A/\mathfrak{p}_i$ *is formally equidimensional. Then, for every prime ideal* $\mathfrak{q}$ *of* A *and every i such that* $\mathfrak{p}_i \subset \mathfrak{q}$*, the ring* $A_\mathfrak{q}/\mathfrak{p}_i A_\mathfrak{q}$ *is formally equidimensional.*

Proof. As $A_\mathfrak{q}/\mathfrak{p}_i A_\mathfrak{q}$ is the local ring of $A/\mathfrak{p}_i$ at the prime ideal $\mathfrak{q}/\mathfrak{p}_i$, we can restrict ourselves to the case where A is integral and formally equidimensional. Set $A' = \widehat{A}$, and let $\mathfrak{q}'$ be one of the prime ideals of A' minimal among those containing $\mathfrak{q}A'$; if we set $B = A_\mathfrak{q}$, $B' = A'_{\mathfrak{q}'}$, then B' is a flat B-module (**0**, 6.3.2). Set $C = \widehat{B}$, $C' = \widehat{B'}$; as B' is a flat B-module, we know that C' is a flat C-module (Bourbaki, *Alg. comm.*, chap. III, § 5, nº 4, prop. 4). As A' is catenary (5.6.4) and equidimensional by hypothesis, the same holds for $B' = A'_{\mathfrak{q}'}$ (**0**, 16.1.4); moreover, as A' is isomorphic to a quotient of a regular ring by the theorem of Cohen (**0**, 19.8.8), the same holds for B' (**0**, 17.3.9); we conclude therefore from (7.1.5) that C' is equidimensional. On the other hand, C' is catenary (5.6.4), hence C is equidimensional by virtue of (7.1.3), (i). □

Theorem (7.1.8). — *Let* A *be a local Noetherian ring. The following conditions are equivalent:*

a) *Every integral quotient ring of* A *is formally equidimensional.*
b) *The quotient rings of* A *by its minimal prime ideals are formally equidimensional.*
c) *Every local ring* B *that is an A-algebra essentially of finite type* (1.3.8) *and that is equidimensional, is formally equidimensional.*

Proof. It is trivial that *c*) implies *a*) and that *a*) implies *b*). On the other hand, as every prime ideal of A contains a minimal prime ideal of A, *b*) implies *a*) by virtue of (7.1.4), (ii). It remains to see that *a*) implies *c*).

It suffices to prove that the quotients of B by its minimal prime ideals are formally equidimensional (7.1.2); as every quotient ring of B is also an A-algebra essentially of finite type (1.3.9), we can first suppose B *integral*. If $\mathfrak{q}$ is the inverse image in A of the maximal ideal of B, we know that B is also an $A_\mathfrak{q}$-algebra essentially of finite type (1.3.10), and it follows from *a*) and from (7.1.7) that every integral quotient of $A_\mathfrak{q}$ is formally equidimensional; we can therefore suppose that the homomorphism $A \to B$ is a *local* homomorphism. The kernel $\mathfrak{r}$ of this homomorphism is then a prime ideal of A, and by virtue of *a*), every integral quotient ring of $A/\mathfrak{r}$ is formally equidimensional; as B is an $(A/\mathfrak{r})$-algebra essentially of finite type, we see that we can also suppose that A is *integral* **IV-2** | 186
and is a subring of B. We know (1.3.11) that B is a quotient of a local ring of the form $C_\mathfrak{p}$, where $C = A[T_1, \dots, T_n]$ is a polynomial algebra and $\mathfrak{p}$ is a prime ideal of C above the maximal ideal $\mathfrak{m}$ of A. By virtue of (7.1.7), it suffices to prove that $C_\mathfrak{p}$ is formally equidimensional, in other words one can restrict to the case where $B = C_\mathfrak{p}$.

Set $A' = \widehat{A}$, and $C' = A'[T_1, \dots, T_n] = C \otimes_A A'$; there is a unique prime ideal $\mathfrak{p}'$ of C' above the maximal ideal $\mathfrak{m}A'$ of A', hence above $\mathfrak{p}$; set $B' = C'_{\mathfrak{p}'}$; the homomorphism $B \to B'$ is local and makes B' a flat B-module. We know that $\widehat{B'}$ is a flat $\widehat{B}$-module (Bourbaki, *Alg. comm.*, chap. III, § 5, nº 4, prop. 4); as $\widehat{B'}$ is also (7.1.3), (i)), it will suffice to prove that $\widehat{B'}$ is *equidimensional*, in order to deduce that $\widehat{B}$ is so as well, which will complete the proof.

Now, A' is a quotient of a regular ring by the theorem of Cohen (**0**, 19.8.8), hence by virtue of (7.1.5), to show that B' is formally equidimensional, it suffices to prove that B' is equidimensional, since C' is a quotient of a regular ring, hence catenary (5.6.4). But the minimal prime ideals of C' are the ideals $\mathfrak{q}'C'$, where $\mathfrak{q}'$ ranges over the minimal prime ideals of A' (5.5.3), and we have $C'/\mathfrak{q}'C' = (A'/\mathfrak{q}')[T_1, \dots, T_n]$. As A' is equidimensional by hypothesis (since A is integral), the same holds for C' by (5.5.4). □

Definition (7.1.9). — When the equivalent conditions of (7.1.8) are satisfied, we say that A is a *formally catenary* ring.

For a local Noetherian ring, it is therefore equivalent to say that it is formally equidimensional or formally catenary and equidimensional, by virtue of (7.1.2).

Proposition (7.1.10). — *Every local quotient ring of a local Cohen-Macaulay ring is formally catenary.*

Proof. This follows immediately from (7.1.8) and (7.1.5) applied to the integral quotient rings of A. ☐

Proposition (7.1.11). — i) *A local Noetherian ring that is formally catenary is universally catenary.*

ii) *If* A *is a local Noetherian ring that is formally catenary, every local ring* B *that is an A-algebra essentially of finite type is formally catenary.*

Proof. Assertion (ii) follows immediately from condition *c*) of (7.1.8) and from the fact that if C is a B-algebra essentially of finite type, C is also an A-algebra essentially of finite type (1.3.9). For the proof of (i), let us first note that by virtue of condition *b*) of (7.1.8), a local Noetherian ring that is formally catenary A is catenary, since the quotients of A by its minimal prime ideals are so (7.1.4). If now E is an A-algebra of finite type, it follows from what precedes and from (ii) that every local ring of E at a prime ideal is catenary, hence E is catenary. Therefore A is universally catenary. ☐

Remarks (7.1.12). — *i)* We do not know if, conversely, a local Noetherian ring that is universally catenary is always formally catenary.

ii) A local Noetherian ring A of dimension 1 is necessarily equidimensional (since in this case A would have dimension 0, being its maximal ideal minimal, a case where A is of dimension 0). As $\dim(\widehat{\mathrm{A}}) = 1$, this shows that A is even *formally equidimensional*, and *a fortiori* formally catenary (hence *universally catenary*). By contrast, the local ring of dimension 2 defined in (5.6.11), which is catenary but not universally catenary, is *a fortiori* not formally catenary.

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Corollary (7.1.13). — *Let* Y *be an irreducible prescheme locally Noetherian of dimension* 1, X *an irreducible prescheme,* $f : \mathrm{X} \to \mathrm{Y}$ *a dominant morphism of finite type,* ξ, η_i *the generic points of* X *and* Y *respectively. Then, for all* $y \in f(\mathrm{X})$, *the dimensions of the irreducible components of* $f^{-1}(y)$ *are all equal to* $\deg.\mathrm{tr}_{\boldsymbol{k}(\eta)} \boldsymbol{k}(\xi)$.

Proof. By hypothesis, $f^{-1}(\eta)$ is irreducible of generic point ξ and of dimension $\deg.\mathrm{tr}_{\boldsymbol{k}(\eta)} \boldsymbol{k}(\xi)$ (5.2.1). As Y is irreducible and of dimension 1, we have $\dim(\mathscr{O}_y) = 1$ for all $y \neq \eta$, and $\mathscr{O}_y$ is universally catenary by virtue of (7.1.12), (ii). If $y \in f(\mathrm{X})$ and si z is a generic point of an irreducible component Z of $f^{-1}(y)$, we have therefore, by (5.6.5)

$$\dim(\mathrm{Z}) = 1 + \dim(f^{-1}(\eta)) - \dim(\mathscr{O}_{\mathrm{X},z}).$$

But we have $\dim(\mathscr{O}_{\mathrm{X},z}) \leqslant \dim(\mathscr{O}_y)$ (**0**, 16.3.9), and on the other hand, as z is not the generic point of X, $\dim(\mathscr{O}_{\mathrm{X},z}) > 0$; hence $\dim(\mathscr{O}_{\mathrm{X},z}) = 1$ and $\dim(\mathrm{Z}) = \dim(f^{-1}(\eta))$. ☐

7.2. Strictly equidimensional rings and strictly formally catenary rings

(7.2.1). *Notations.* — For an integral Noetherian ring A, we denote by $\mathrm{A}^{(i)}$ the intersection of the local rings $\mathrm{A}_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1 (5.10.17). If A is a *local* Noetherian integral ring, we denote by $\mathrm{A}^{(a)}$ the intersection of the $\mathrm{A}_{\mathfrak{p}}$ where this time $\mathfrak{p}$ ranges over the set of all the prime ideals of A *distinct from the maximal ideal*.

We will say that a local Noetherian ring A is *strictly equidimensional* if, for every prime ideal $\mathfrak{p} \in \mathrm{Ass}(\mathrm{A})$, we have $\dim(\mathrm{A}/\mathfrak{p}) = \dim(\mathrm{A})$; this amounts to saying that A is equidimensional and *without embedded associated prime ideals*.

(7.2.1.1). *Example.* — Let A be a local Noetherian ring *of dimension* 1. Then the following conditions are equivalent:

a) A has no embedded associated prime ideals.

a′) $\widehat{\mathrm{A}}$ has no embedded associated prime ideals.

b) A is strictly equidimensional.

b′) $\widehat{\mathrm{A}}$ is strictly equidimensional.

c) A is a Cohen-Macaulay ring.

Indeed, a) means that the maximal ideal $\mathfrak{m}$ of A is not associated to A, hence the associated prime ideals to A are the minimal ideals of A, all distinct from $\mathfrak{m}$, and for such an ideal $\mathfrak{p}$, we necessarily have $\dim(A/\mathfrak{p}) = 1$; hence a) implies b). Conversely, b) implies that every associated prime ideal $\mathfrak{p} \in \operatorname{Ass}(A)$ is distinct from $\mathfrak{m}$, hence minimal, and therefore b) implies a). We already know that a) and c) are equivalent for a local ring of dimension 1 (5.7.8). As it comes to the same thing to say that A is a Cohen-Macaulay ring or that $\widehat{A}$ is a Cohen-Macaulay ring (**0**, 16.5.2), and since $\dim(\widehat{A}) = 1$, we see finally that a') and b') are equivalent to c). **IV-2** | 188

Proposition (7.2.2). — *Let* A *be a local Noetherian integral ring; set* $A' = \widehat{A}$, $X = \operatorname{Spec}(A)$, $X' = \operatorname{Spec}(A')$ *and denote by* $f : X' \to X$ *the canonical morphism. Let a be the closed point of* X, $j : X - \{a\} \to X$ *the canonical injection. Let* $\mathscr{F}$ *be a coherent* $\mathscr{O}_X$*-Module,* $\mathscr{F}' = f^*(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$; *then the following conditions are equivalent:*

a) *The* $\mathscr{O}_X$*-Module* $j_*(\mathscr{F}|X - \{a\})$ *is coherent.*
b) *For all* $x' \in \operatorname{Ass}(\mathscr{F}')$, *we have* $\dim(\overline{\{x'\}}) \geqslant 2$.

Proof. Let a' be the closed point of X', which is the unique point of the fibre $f^{-1}(a)$, and let $j' : X' - \{a'\} \to X'$ be the canonical injection. As the morphism f is faithfully flat and quasi-compact, it is sufficient to show that $j_*(\mathscr{F}|X - \{a\})$ is coherent or that $j'_*(\mathscr{F}'|X' - \{a'\})$ is coherent. On the other hand, as A' is isomorphic to a quotient of a regular local ring by the theorem of Cohen (**0**, 19.8.8), it follows from (5.11.4) that condition b) of the statement was equivalent to the fact that $j'_*(\mathscr{F}'|X' - \{a'\})$ is coherent. Whence the proposition.

Instead of applying (5.11.4) and using the theorem of Cohen, which (by (5.11.2)) implicitly appeals to the cohomological results of chap. III, one can also use the fact that A' is universally catenary (5.6.4) and universally Japanese by virtue of (7.6.5) (the proof of this last result not using (7.2.2)). □

Proposition (7.2.3). — *Let* A *be a local Noetherian integral ring. The notations being as in* (7.2.2), *the following conditions are equivalent:*

a) $A^{(1)}$ *is a finite A-algebra.*
b) *For every closed subset* T *of* X *of codimension* $\geqslant 2$, *if we denote by* $i : X - T \to X$ *the canonical injection,* $i_*(\mathscr{O}_{X-T})$ *is a coherent* $\mathscr{O}_X$*-Module.*
c) *For all* $x' \in \operatorname{Ass}(\mathscr{O}_{X'})$ *and every closed subset* T *of* X *of codimension* $\geqslant 2$, *we have* $\operatorname{codim}(f^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geqslant 2$.

Proof. Set $Z = Z^{(2)}(X)$ (5.10.13) and $Z' = f^{-1}(Z)$, which are stable parts under specialisation; the conditions a) and b) are equivalent respectively to the following properties: a_1) $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is coherent; b_1) $\mathscr{H}^0_{X/T}(\mathscr{O}_X)$ is coherent for every closed subset T of X of codimension $\geqslant 2$. Taking into account (5.9.5), these two properties are equivalent respectively to: a'_1) $\mathscr{H}^0_{X'/Z'}(\mathscr{O}_{X'})$ is coherent; b'_1) setting $T' = f^{-1}(T)$, $\mathscr{H}^0_{X'/T'}(\mathscr{O}_{X'})$ is coherent for every closed subset T of X of codimension $\geqslant 2$. Now, every point of $\operatorname{Ass}(\mathscr{O}_{X'})$ projects to the generic point of X since f is a flat morphism (3.3.2); as by definition this point does not belong to Z, we see that $\operatorname{Ass}(\mathscr{O}_{X'})$ does not meet Z'; the equivalence of a'_1 and b'_1 therefore follows from (5.11.5), since A' is isomorphic to a quotient of a regular local ring by the theorem of Cohen (**0**, 19.8.8) (or alternatively because A' is universally Japanese (7.6.5)). **IV-2** | 189
For this reason, conditions b'_1 and c) are also equivalent by virtue of (5.11.4). □

Proposition (7.2.4). — *Let* A *be a local Noetherian ring and set* $X = \operatorname{Spec}(A)$. *The following conditions are equivalent:*

a) *For every integral quotient ring* B *of* A, *the ring* $B^{(1)}$ *is a finite B-algebra.*
b) *For every coherent* $\mathscr{O}_X$*-Module* $\mathscr{F}$ *and every closed subset* Z, *stable under specialisation, and such that for all* $x \in \operatorname{Ass}(\mathscr{F}) \cap (X - Z)$, *we have* $\operatorname{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geqslant 2$, *the* $\mathscr{O}_X$*-Module* $\mathscr{H}^0_{X/Z}(\mathscr{F})$ *is coherent.*
c) *For every closed subset* T *of* X *and every coherent* $\mathscr{O}_U$*-Module* $\mathscr{G}$ *(where* $U = X - T$*) such that, for all* $x \in \operatorname{Ass}(\mathscr{G})$, *we have* $\operatorname{codim}(\overline{\{x\}} \cap T, \overline{\{x\}}) \geqslant 2$, $i_*(\mathscr{G})$ *(where* $i : U \to X$ *is the canonical injection) is a coherent* $\mathscr{O}_X$*-Module.*
d) *For every integral quotient ring* B *of* A, *and every ideal* $\mathfrak{J}$ *of height* $\geqslant 2$ *in* B, *the ring* $\bigcap_{\mathfrak{p} \not\supset \mathfrak{J}} B_{\mathfrak{p}}$ *is a finite B-algebra.*

e) *For every integral quotient ring* B *of* A *and every prime ideal* $\mathfrak{q}$ *of* B*, with* $\mathrm{C} = \mathrm{B}_{\mathfrak{q}}$ *in a prime ideal of* B*, the ring* $\mathrm{C}^{(a)}$ *is a finite C-algebra (or, what amounts to the same, if* $\mathrm{Y} = \mathrm{Spec}(\mathrm{C})$ *and* U *is the complement in* Y *of the closed point of* C *and* $j : \mathrm{U} \to \mathrm{Y}$ *the canonical injection,* $j_*(\mathscr{O}_{\mathrm{U}})$ *is a coherent* $\mathscr{O}_{\mathrm{Y}}$*-Module).*

f) *For every integral quotient ring* B *of* A*, every local ring* $\mathrm{C} = \mathrm{B}_{\mathfrak{q}}$ *in a prime ideal* $\mathfrak{q}$ *of* B*, with* $\dim(\mathrm{C}) \geqslant 2$*, and for every ideal* $\mathfrak{t} \in \mathrm{Ass}(\widehat{\mathrm{C}})$*, we have* $\dim(\widehat{\mathrm{C}}/\mathfrak{t}) \geqslant 2$.

Proof. We already know (5.11.6) that *a*) and *b*) are equivalent, as well as that *c*) and *a*) imply *d*), and that *a*) implies *d*). The equivalence of *a*) and *d*) in the present case follows from (7.2.3) applied to the condition (7.2.3), *b*)), by virtue of (5.9.3.1). The equivalence of *e*) and *f*) is a consequence of (7.2.2) applied to the $\mathscr{O}_{\mathrm{Y}}$-Module coherent $\mathscr{O}_{\mathrm{Y}}$ itself. We have already remarked (5.11.7), (ii) that if A satisfies *a*), the same holds for every A-algebra of finite type and for every ring of fractions of A. The condition *a*) for A therefore implies (with the notations of *e*)) that the ring C satisfies *a*), hence also *c*); but as C is integral and $\dim(\mathrm{C}) \geqslant 2$, one can apply condition *c*) by taking T the closed point of $\mathrm{Y} = \mathrm{Spec}(\mathrm{C})$, which proves that *a*) implies *e*); on the other hand, $\dim(\mathrm{C}) \geqslant 2$, one can apply *c*) the condition *c*) by taking T the closed point of Y, which gives that *a*) implies *a*); we can **IV-2** | 190 obviously restrict ourselves to the case where A is integral and $\mathrm{B} = \mathrm{A}$, and it amounts to showing that condition *e*) implies the condition (7.2.3), *c*)). Let us consider the notations of (7.2.3), *c*)), and let $y' \in f^{-1}(\mathrm{T}) \cap \overline{\{x'\}}$, and set $y = f(y')$ and $\mathrm{C} = \mathscr{O}_{\mathrm{X},y}$; as $y \in \mathrm{T}$, $\dim(\mathrm{C}) \geqslant 2$, and hence (with the notations of *e*)), $j_*(\mathscr{O}_{\mathrm{U}})$ is a coherent $\mathscr{O}_{\mathrm{Y}}$-Module. As the morphism f is flat, $f_z : \mathrm{Y}' \to \mathrm{Y}$; the morphism $f : \mathrm{X}' \to \mathrm{X}$ is flat, being the base change $g = f_{\{\mathrm{Y}\}} : \mathrm{Y}' \to \mathrm{Y}$. The underlying space is identified with $f^{-1}(\mathrm{Y})$ (**I**, 3.6.5), and as A is integral and f flat, $\mathrm{Ass}(\mathscr{O}_{\mathrm{X}'})$ is contained in the fibre of the generic point of X (3.3.2); this last being contained in Y, we have $x' \in \mathrm{Y}'$. Let $\mathrm{U}' = g^{-1}(\mathrm{U})$, and let j' be the canonical injection $\mathrm{U}' \to \mathrm{Y}'$; as $j_*(\mathscr{O}_{\mathrm{U}})$ is a coherent $\mathscr{O}_{\mathrm{Y}}$-Module and g is flat, it follows from (5.9.4) that $j'_*(\mathscr{O}_{\mathrm{U}'})$ is a coherent $\mathscr{O}_{\mathrm{Y}'}$-Module. But we have $x' \in \mathrm{Ass}(\mathscr{O}_{\mathrm{U}'})$, hence we conclude from (5.10.10) that, in Y', $\mathrm{codim}(f^{-1}(\mathrm{T}) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geqslant 2$ in X'. □

Theorem (7.2.5). — *Let* A *be a local Noetherian ring. The following conditions are equivalent:*

a) *For every integral quotient ring* B *of* A*, the completion* $\widehat{\mathrm{B}}$ *is strictly equidimensional (7.2.1).*

b) A *is formally catenary (7.1.9) and the fibres of the canonical morphism* $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$ *satisfy* (S_1) *(in other words, have no embedded associated prime cycle).*

c) A *is universally catenary (5.6.2) and for every integral quotient ring* B *of* A*, the ring* $\mathrm{B}^{(1)}$ *is a finite B-algebra (cf. (7.2.4)).*

d) A *is universally catenary and for every integral quotient ring* B *of* A *and every local ring* $\mathrm{C} = \mathrm{B}_{\mathfrak{q}}$ *in a prime ideal* $\mathfrak{q}$ *of* B*, with* $\dim(\mathrm{C}) \geqslant 2$*, the completed ring* $\mathrm{C}^{(a)}$ *is a finite C-algebra.*

e) A *is universally catenary and for every integral quotient ring* B *of* A *and every local ring* $\mathrm{C} = \mathrm{B}_{\mathfrak{q}}$ *in a prime ideal* $\mathfrak{q}$ *of* B*, with* $\dim(\mathrm{C}) \geqslant 2$*, the completed ring* $\widehat{\mathrm{C}}$ *is such that* $\mathrm{Spec}(\widehat{\mathrm{C}})$ *has no associated prime cycle of dimension* 1.

Moreover, when these conditions are satisfied, for every integral quotient ring B *of* A *that is strictly equidimensional, the completion* $\widehat{\mathrm{B}}$ *is strictly equidimensional.*

Proof. The equivalence of *c*), *d*) and *e*) was proved in (7.2.4). To show that *a*) and *b*) are equivalent, recall (7.1.9) that to say that A is formally catenary means that for every integral quotient ring B of A, $\widehat{\mathrm{B}}$ is equidimensional. On the other hand, for every fibre of the morphism $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$ at a point $x \in \mathrm{Spec}(\mathrm{A})$, this fibre of the morphism $\mathrm{Spec}(\widehat{\mathrm{B}}) \to \mathrm{Spec}(\mathrm{B})$ at the generic point x of $\mathrm{Spec}(\mathrm{B})$, where $\mathrm{B} = \mathrm{A}/\mathfrak{I}_x$; to say that this fibre has no embedded associated prime cycle is equivalent to saying that $\widehat{\mathrm{B}}$ has no embedded associated prime ideals, by virtue of (3.3.3); this proves the equivalence of *a*) and *b*). The same reasoning shows that if *a*) is satisfied, then, for every integral quotient ring B of A that has no embedded associated prime ideals, the completion $\widehat{\mathrm{B}}$ is also without embedded associated prime ideals; on the other hand, the hypothesis that A is formally catenary implies that if B is equidimensional, it is also the case for $\widehat{\mathrm{B}}$ (7.1.8); this establishes the last assertion of the theorem.

Let us show now that *a*) implies *c*). Condition *a*) implies that A is universally catenary (7.1.11); let us show on the other hand that for every integral quotient ring B of A, $\mathrm{B}^{(1)}$ is then a finite B-algebra. Taking into account (7.2.3) applied to B, it amounts to showing that if $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, T is a closed subset of codimension $\geqslant 2$ in X, $X' = \mathrm{Spec}(\widehat{B})$, and $g : X' \to X$ is the canonical morphism, then for all $x' \in \mathrm{Ass}(\mathcal{O}_{X'})$, $\mathrm{codim}(g^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geqslant 2$. But by hypothesis X′ has no associated prime cycle of dimension 1, hence $\inf(\mathrm{codim}(g^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}))$ when x' ranges over $\mathrm{Ass}(\mathcal{O}_{X'})$, is nothing other than $\mathrm{codim}(g^{-1}(T), X')$. Now, as g is a faithfully flat morphism, we have (6.1.4)

$$\mathrm{codim}(g^{-1}(T), X') = \mathrm{codim}(T, X) \geqslant 2.$$

It remains to prove that $c)$ implies $a)$. Let us reason by induction on $n = \dim(A)$, the theorem being trivial for $n = 0$; we can moreover restrict to the case where $A = B$ is integral. We proceed in two steps, n being $\geqslant 1$. **IV-2** | 191

I) *Suppose first that* A *satisfies* (S_2). — Set $X = \mathrm{Spec}(A)$, $A' = \widehat{A}$, $X' = \mathrm{Spec}(A')$ and let $u : X' \to X$ be the canonical morphism; it amounts to showing that for all $x' \in \mathrm{Ass}(\mathcal{O}_{X'})$, we have $\dim(\overline{\{x'\}}) = n$. Let $f \neq 0$ be an element of the maximal ideal of A, and set $C = A/fA$; we know (5.7.6) that A/fA satisfies (S_1); moreover, as the prime ideals of A among those containing f are of height 1 by virtue of the Hauptidealsatz, $C = A/fA$ is *strictly equidimensional* and $\dim(C) = n - 1$. We have $\widehat{C} = C \otimes_A A' = A'/fA'$, and f is A′-regular by flatness (**0**, 6.3.4); if we set $Y' = V(fA') = \mathrm{Spec}(\widehat{C})$, then, for every maximal point y' of $Y' \cap \overline{\{x'\}}$, we have $y' \in \mathrm{Ass}(\mathcal{O}_{Y'})$ by virtue of (3.4.3); on the other hand, we have $y' \neq x'$ since $u(x')$ is the generic point of X (3.3.2) and we conclude (5.1.8) that

(7.2.5.1) $$\dim(\overline{\{y'\}}) = \dim(\overline{\{x'\}}) - 1.$$

But the quotient ring $C = A/fA$ also satisfies hypothesis $c)$ of the statement (5.6.1) and (5.11.7), (ii)), hence by the induction hypothesis, we have $\dim(\overline{\{y'\}}) = \dim(C) = n - 1$; the conclusion therefore follows in this case from (7.2.5.1).

II) *General case.* — As by hypothesis $A^{(1)}$ is a finite A-algebra, it is a semi-local ring, and each of its local rings $(A^{(1)})_{\mathfrak{n}}$ at a maximal ideal $\mathfrak{n}$ and moreover, as A is universally catenary, the rings $(A^{(1)})_{\mathfrak{n}}$ (in finite number) are all of *dimension* n (5.6.10); moreover, these rings satisfy hypothesis $c)$ of the statement (5.6.1) and (5.11.7), (ii). We know that the completion of $A^{(1)}$, equal to $\widehat{A} \otimes_A A^{(1)}$, is composed of the completions of the local rings $(A^{(1)})_{\mathfrak{n}}$. Set $X_1 = \mathrm{Spec}(A^{(1)})$, $X'_1 = \mathrm{Spec}((A^{(1)})^\wedge) = X' \times_X X_1$; for all $x'_1 \in \mathrm{Ass}(\mathcal{O}_{X'_1})$, it follows from what precedes and from case I) that $\dim(\overline{\{x'_1\}}) = n$. Let $u_1 = u_{(X_1)} : X'_1 \to X_1$ be the canonical morphism; as A and $A^{(1)}$ have the same field of fractions, the inverse image of the generic point x of X by the projection $X_1 \to X$ reduces to the generic point x_1 of X_1, the inverse image by the projection $X'_1 \to X'$ of the generic point x' of X′ induces an isomorphism of the prescheme $u_1^{-1}(x_1)$ to the prescheme $u^{-1}(x)$. This being the case, the points of $\mathrm{Ass}(\mathcal{O}_{X'})$ (resp. $\mathrm{Ass}(\mathcal{O}_{X'_1})$) are the generic points of the associated prime cycles to $u^{-1}(x)$ (resp. $u_1^{-1}(x_1)$) (3.3.1). For all $x' \in \mathrm{Ass}(\mathcal{O}_{X'})$, and if Z′ (resp. Z'_1) is the closed sub-prescheme of X′ (resp. X'_1) having $\overline{\{x'\}}$ (resp. $\overline{\{x'_1\}}$) as underlying space, the projection $Z'_1 \to Z'$ is a finite and surjective morphism; we conclude (5.4.2) that $\dim(\overline{\{x'\}}) = \dim(\overline{\{x'_1\}}) = n$. ☐

Definition (7.2.6). — When a local Noetherian ring A satisfies the equivalent conditions of (7.2.5), we say that it is *strictly formally catenary*.

Corollary (7.2.7). — *Let* A *be a local Noetherian ring such that there exists an* A*-module* M *of finite type that is a Cohen-Macaulay* A*-module and for which* $\mathrm{Supp}(M) = \mathrm{Spec}(A)$*; then* A *is strictly formally catenary.*

Proof. Indeed, A is formally catenary (7.1.5) and (7.1.9), and on the other hand the fibres of the canonical morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ are preschemes of Cohen-Macaulay (6.3.8), hence *a fortiori* satisfy (S_1). ☐ **IV-2** | 192

Corollary (7.2.8). — *If* A *is a local Noetherian ring that is strictly formally catenary, every local ring* B *that is an A-algebra essentially of finite type is strictly formally catenary.*

Proof. Indeed, B is formally catenary (7.1.11) and it follows from (7.4.4)[10] that the fibres of $\mathrm{Spec}(\widehat{B}) \to \mathrm{Spec}(B)$ satisfy (S_1), whence the conclusion. ☐

[10]The corollary (7.2.8) is not used to prove (7.4.4).

Corollary (7.2.9). — *Every local Noetherian ring of dimension* 1 *is strictly formally catenary.*

Proof. Indeed, every *integral* quotient ring B of such a ring A is of dimension 0 or 1, hence its completion is strictly equidimensional (7.2.1.1). □

Remarks (7.2.10). — *i)* It follows from (7.2.7) and (7.2.8) that every quotient ring of a Cohen-Macaulay local ring is strictly formally catenary. A local Noetherian ring of dimension 2 satisfying (S_2) is strictly formally catenary, since it is a Cohen-Macaulay ring. Recall however that there exist integral local Noetherian rings of dimension 2 that are not universally catenary, nor *a fortiori* strictly formally catenary (5.6.11).

ii) We do not know if a ring that is formally catenary (7.1.9) is strictly formally catenary; this is because we do not know if for a local Noetherian integral ring B, $\widehat{B}$ has no embedded associated prime ideals (6.4.3). We also do not know if, in the equivalent conditions *c*), *d*), *e*) of (7.2.5), one can replace the hypothesis that A is universally catenary by the hypothesis that A is catenary; one can show that this is the case when A is Henselian (18.9.6).

7.3. Formal fibres of Noetherian local rings

(7.3.1). We will consider in this number and the two following properties $\boldsymbol{P}(Z,k)$ of the following form:

"Z is a locally Noetherian prescheme over a field k,
and for all $z \in Z$, we have $\boldsymbol{Q}(\mathscr{O}_z,k)$"

where $\boldsymbol{Q}(A,k)$ is a property of a local Noetherian ring A that is a k-algebra; one assumes that if k' is a field isomorphic to k, and if A' is a local Noetherian ring that is a k'-algebra di-isomorphic to A, the properties $\boldsymbol{Q}(A,k)$ and $\boldsymbol{Q}(A',k')$ are equivalent.

If X, Y are two locally Noetherian preschemes, we will say that a morphism $f : X \to Y$ is a $\boldsymbol{P}$*-morphism* if:

1° f is *flat*;
2° for all $y \in Y$, the property $\boldsymbol{P}(f^{-1}(y), \boldsymbol{k}(y))$ is true.

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Lemma (7.3.2). — *Let* X, Y *be two locally Noetherian preschemes,* $f : X \to Y$ *a morphism. The following properties are equivalent:*

a) f *is a* $\boldsymbol{P}$*-morphism.*
b) *For all* $x \in X$*, if we set* $y = f(x)$*, then* $\mathscr{O}_x$ *is a flat* $\mathscr{O}_y$*-module and the property* $\boldsymbol{Q}(\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y), \boldsymbol{k}(y))$ *is true.*
c) *For all* $x \in X$*, if we set* $y = f(x)$*, the morphism* $\operatorname{Spec}(\mathscr{O}_x) \to \operatorname{Spec}(\mathscr{O}_y)$ *corresponding to the homomorphism* $\mathscr{O}_y \to \mathscr{O}_x$ *is a* $\boldsymbol{P}$*-morphism.*
c′) *For every closed point* $x \in X$*, the morphism* $\operatorname{Spec}(\mathscr{O}_x) \to \operatorname{Spec}(\mathscr{O}_y)$ *is a* $\boldsymbol{P}$*-morphism.*

Proof. The equivalence of *a*) and *b*) follows from the definitions; the same holds for *b*) and *c*), taking into account (**I**, 2.4.2); finally the equivalence of *b*) and *c*′) follows from the fact that, by (5.1.11), $\overline{\{x\}}$ contains a closed point. □

Corollary (7.3.3). — *Let* A, B *be two Noetherian rings,* $\varphi : A \to B$ *a homomorphism such that the corresponding morphism* $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ *is a* $\boldsymbol{P}$*-morphism. Then, for every multiplicative subset* S *of* A*, the morphism* $\operatorname{Spec}(S^{-1}B) \to \operatorname{Spec}(S^{-1}A)$ *is a* $\boldsymbol{P}$*-morphism.*

Proof. This follows immediately from (7.3.2) and (**I**, 1.6.2). □

(7.3.4). We will suppose throughout what follows that the property $\boldsymbol{Q}$ is such that the following three conditions are satisfied:

(P_I) *(transitivity).* — If $f : X \to Y$ is a regular morphism (6.8.1) and $g : Y \to Z$ is a $\boldsymbol{P}$-morphism, then $g \circ f$ is a $\boldsymbol{P}$-morphism.
(P_{II}) *(descent).* — If $f : X \to Y$ and $g : Y \to Z$ are morphisms of locally Noetherian preschemes such that f is faithfully flat and that $g \circ f$ is a $\boldsymbol{P}$-morphism, then g is a $\boldsymbol{P}$-morphism.
(P_{III}) — For every field k, the property $\boldsymbol{P}(\operatorname{Spec}(k), k)$ is true.

Remarks (7.3.5). — *i)* Let us note that the conditions $(\mathrm{P_I})$ and $(\mathrm{P_{III}})$ imply that *every regular morphism is a* $\boldsymbol{P}$*-morphism.*

ii) Let us note that the hypotheses of $(\mathrm{P_I})$ (resp. $(\mathrm{P_{II}})$) imply that $h = g \circ f$ (resp. g) is flat (2.2.13); the hypotheses of $(\mathrm{P_I})$ or of $(\mathrm{P_{II}})$ imply on the other hand that for all $z \in \mathrm{Z}$, $f_z : h^{-1}(z) \to g^{-1}(z)$ is flat (2.1.4); as, for all $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (**I**, 3.6.4), this shows that it suffices, to verify conditions $(\mathrm{P_I})$ and $(\mathrm{P_{II}})$, to do it only when Z is the *spectrum of a field*.

iii) In certain cases, the property $\boldsymbol{Q}$ will be such that the following condition is also satisfied:
$(\mathrm{P'_I})$ — If $f : \mathrm{X} \to \mathrm{Y}$ and $g : \mathrm{Y} \to \mathrm{Z}$ are two $\boldsymbol{P}$-morphisms, then $g \circ f$ is a $\boldsymbol{P}$-morphism.

(7.3.6). We will say that the property $\boldsymbol{P}$ is *geometric* if it satisfies (in addition to $(\mathrm{P_I})$, $(\mathrm{P_{II}})$ and $(\mathrm{P_{III}})$) the condition:

$(\mathrm{P_{IV}})$ *(finite extension of the base field).* — If $\boldsymbol{P}(\mathrm{Z}, k)$ is true, then, for every finite extension k' of k, $\boldsymbol{P}(\mathrm{Z} \otimes_k k', k')$ is true.

Lemma (7.3.7). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a $\boldsymbol{P}$-morphism of locally Noetherian preschemes, $g : \mathrm{Y}' \to \mathrm{Y}$ a morphism locally of finite type. Then, if $\boldsymbol{P}$ satisfies the condition $(\mathrm{P_{IV}})$, the morphism $f' = f_{(\mathrm{Y}')} : \mathrm{X} \times_\mathrm{Y} \mathrm{Y}' \to \mathrm{Y}'$ is a $\boldsymbol{P}$-morphism.*

Proof. Indeed, for all $y' \in \mathrm{Y}'$, if we set $y = g(y')$, $\boldsymbol{k}(y')$ is an extension of finite type of $\boldsymbol{k}(y)$ (**I**, 6.4.11) and $f'^{-1}(y') = f^{-1}(y) \otimes_{\boldsymbol{k}(y)} \boldsymbol{k}(y')$ (**I**, 3.6.4); it suffices therefore to apply $(\mathrm{P_{IV}})$. □

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(7.3.8). *Examples.* — The following properties $\boldsymbol{P}(\mathrm{Z}, k)$ satisfy the conditions $(\mathrm{P_I})$, $(\mathrm{P_{II}})$ and $(\mathrm{P_{III}})$:

(i) (also denoted (i_n)) Z is of coprofundity $\leqslant n$.
(ii) Z is a prescheme of Cohen-Macaulay.
(iii) (also denoted (iii_n)) Z satisfies (S_n).
(iv) Z is regular.
(v) (also denoted (v_n)) Z satisfies (R_n).
(vi) Z is reduced.
(vii) Z is normal.

For properties (ii) to (vii), this in fact proves the stronger condition $(\mathrm{P'_I})$ which also implies the property (iv) implies all the properties (i) to (vii), and the property (iv') implies all the properties enumerated in (7.3.8). Recall also that (i_0) is equivalent to (ii); the conjunction of (iii_1) and of (v_0) (resp. of (iii_1) and of (v'_0)) is equivalent to (vi) (resp. (vi')); the conjunction of (iii_2) and of (v_1) (resp. of (iii_2) and of (v'_1)) is equivalent to (vii) (resp. (vii')) (5.8.6).

Moreover, the following are also geometric:

(iv') Z is geometrically regular.
(v') (also denoted (v'_n)) Z possesses the property (R_n) geometric.
(vi') Z is geometrically reduced.
(vii') Z is geometrically normal.

Remarks (7.3.9). — *i)* Each of the properties (iv'), (v_n), (vi'), (vii') implies respectively the corresponding property (iv), (v_n), (vi), (vii). The property (iv) implies all the properties (i) to (vii), and the property (iv') implies all the properties enumerated in (7.3.8). Recall also that (i_0) is equivalent to (ii); the conjunction of (iii_1) and of (v_0) (resp. of (iii_1) and of (v'_0)) is equivalent to (vi) (resp. (vi')); the conjunction of (iii_2) and of (v_1) (resp. of (iii_2) and of (v'_1)) is equivalent to (vii) (resp. (vii')) (5.8.6).

ii) In all the examples of (7.3.8), the property $\boldsymbol{Q}(\mathscr{O}_z, k)$ which serves to define $\boldsymbol{P}$ is such that, for every *generization* z' of z in Z, $\boldsymbol{Q}(\mathscr{O}_z, k)$ *implies* $\boldsymbol{Q}(\mathscr{O}_{z'}, k)$: by virtue of (2.3.4), it suffices to verify this for properties (i) to (vii) by (7.3.9), (i) and even (i) to (v) from the remark (7.3.9), (i): but this is included in the definition for (iii) and (v) (5.7.2) and (5.8.2), this follows from (6.3.5), (i), for (i),[11] and finally from (**0**, 16.5.10) and (17.3.2) for (ii) and (iv).

[11]The printed text cites (6.3.9), but §6.3 ends at Proposition (6.3.8); the codepth-under-generization statement used here is Corollary (6.3.5)(i).

(7.3.10). We will say that a property $\boldsymbol{P}(\mathrm{Z},k)$ is of the *first type* if the property $\boldsymbol{Q}(\mathrm{A},k)$ which serves to define it defines a property $\boldsymbol{R}(\mathrm{A})$ *not involving* k and is true when A is a *field*; we will say that $\boldsymbol{P}(\mathrm{Z},k)$ is of the *second type* if the property $\boldsymbol{Q}(\mathrm{A},k)$ is of the form

> "For every finite extension k' of k and every point z' of $\mathrm{Spec}(\mathrm{A}\otimes_k k')$ above the closed point of $\mathrm{Spec}(\mathrm{A})$, we have $\boldsymbol{R}(\mathscr{O}_{z'})$"

$\boldsymbol{R}(\mathrm{A})$ being assumed true when A is a field. It is clear that in the examples of (7.3.8), the properties (i) to (vii) are of the first type, the properties (iv′) to (vii′) of the second type. IV-2 | 195

This being the case, if we take up the reasoning of (6.6.1) and (6.8.3), we observe that when $\boldsymbol{P}$ is of the first or second type for a property $\boldsymbol{R}(\mathrm{A})$, *the conditions* $(\mathrm{P_I})$ *and* $(\mathrm{P_{II}})$ *have the following consequences on* $\boldsymbol{R}$:

$(\mathrm{R_I})$ Let X, Y be two locally Noetherian preschemes, $f:\mathrm{X}\to\mathrm{Y}$ a *regular* morphism; then, for all $x\in\mathrm{X}$, the property $\boldsymbol{R}(\mathscr{O}_{f(x)})$ implies the property $\boldsymbol{R}(\mathscr{O}_x)$.

$(\mathrm{R_{II}})$ Let $\varphi:\mathrm{A}\to\mathrm{B}$ be a homomorphism of local Noetherian rings making B a flat A-module; then $\boldsymbol{R}(\mathrm{B})$ implies $\boldsymbol{R}(\mathrm{A})$.

Moreover, the property $(\mathrm{P_{III}})$ results from the fact that $\boldsymbol{R}(k)$ is true for every field k; finally, when $\boldsymbol{P}$ is of the second type, the condition $(\mathrm{P_{IV}})$ is a consequence of the definition of $\boldsymbol{P}$ and of the transitivity of finite extensions.

Remark (7.3.11). — We leave to the reader the task of formulating the property $\boldsymbol{R}$ corresponding to each of the examples of (7.3.8), and of verifying the conditions $(\mathrm{R_I})$ and $(\mathrm{R_{II}})$ in each case, with the help of the results of § 6. In fact, except for example (i) of (7.3.8), the property $\boldsymbol{R}$ satisfies in the other cases the following condition:

$(\mathrm{R'_I})$ Let X, Y be two locally Noetherian preschemes, $f:\mathrm{X}\to\mathrm{Y}$ a $\boldsymbol{P}$-morphism (for the property $\boldsymbol{P}$ of the first or second type defined by $\boldsymbol{R}$); then, for all $x\in\mathrm{X}$, the property $\boldsymbol{R}(\mathscr{O}_{f(x)})$ implies the property $\boldsymbol{R}(\mathscr{O}_x)$.

The reasoning of (6.6.1) and (6.8.3) proves that if $\boldsymbol{R}$ satisfies the conditions $(\mathrm{R'_I})$ (7.3.4), (iii) and $(\mathrm{P_{II}})$, then $\boldsymbol{P}$ satisfies the conditions $(\mathrm{P'_I})$ (7.3.5), (iii) and $(\mathrm{P_{II}})$.

Proposition (7.3.12). — *Let $\boldsymbol{P}$ be a property of the first or the second type defined from a property $\boldsymbol{R}$ verifying the conditions $(\mathrm{R_I})$ and $(\mathrm{R_{II}})$ (resp. $(\mathrm{R'_I})$ and $(\mathrm{R_{II}})$). If for every locally Noetherian prescheme X, $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X})$ denotes the set of $x\in\mathrm{X}$ such that $\boldsymbol{R}(\mathscr{O}_x)$ is true, then, for every regular morphism (resp. every $\boldsymbol{P}$-morphism) $f:\mathrm{X}\to\mathrm{Y}$ of locally Noetherian preschemes, we have*

$$\mathrm{U}_{\boldsymbol{R}}(\mathrm{X})=f^{-1}(\mathrm{U}_{\boldsymbol{R}}(\mathrm{Y})). \tag{7.3.12.1}$$

Proof. This is an immediate consequence of the definitions. ☐

(7.3.13). Given a semi-local Noetherian ring A, we will call *formal fibres* of A the fibres of the canonical morphism $f:\mathrm{Spec}(\widehat{\mathrm{A}})\to\mathrm{Spec}(\mathrm{A})$; for every prime ideal $\mathfrak{p}$ of A, the formal fibre at $\mathfrak{p}$ is therefore the scheme $\mathrm{Spec}(\widehat{\mathrm{A}}\otimes_{\mathrm{A}}\boldsymbol{k}(\mathfrak{p}))$; as the completion of the local ring $\mathrm{A}/\mathfrak{p}$ is $\widehat{\mathrm{A}}/\mathfrak{p}\widehat{\mathrm{A}}=(\mathrm{A}/\mathfrak{p})\otimes_{\mathrm{A}}\widehat{\mathrm{A}}$, we see that the formal fibre of A at $\mathfrak{p}$ is also the *formal fibre of* $\mathrm{A}/\mathfrak{p}$ *at the generic point* (o) of $\mathrm{Spec}(\mathrm{A}/\mathfrak{p})$.

The property $\boldsymbol{P}$ being defined as in (7.3.1), we will say that *the formal fibres of* A *have the property* $\boldsymbol{P}$, or that A is a $\boldsymbol{P}$-*ring*, if, for all $x\in\mathrm{Spec}(\mathrm{A})$, $\boldsymbol{P}(f^{-1}(x),\boldsymbol{k}(x))$ is true. As the morphism f is *flat*, this amounts to saying that f is a $\boldsymbol{P}$-morphism. IV-2 | 196

Proposition (7.3.14). — *Let A be a semi-local Noetherian ring, $\mathfrak{m}_i$ ($1\leqslant i\leqslant r$) its maximal ideals, and set $\mathrm{A}_i=\mathrm{A}_{\mathfrak{m}_i}$; for A to be a $\boldsymbol{P}$-ring, it is necessary and sufficient that each of the A_i be so.*

Proof. Indeed, $\widehat{\mathrm{A}}$ is the direct product of the $\widehat{\mathrm{A}}_i$, hence the formal fibre of A at a point $x\in\mathrm{Spec}(\mathrm{A})$ is the *sum* of the formal fibres at x of those A_i such that $x\in\mathrm{Spec}(\mathrm{A}_i)$. ☐

Proposition (7.3.15). — *Let A be a semi-local Noetherian ring.*

i) *If A is a $\boldsymbol{P}$-ring, every quotient ring of A is a $\boldsymbol{P}$-ring.*
ii) *If in addition $\boldsymbol{P}$ satisfies the condition $(\mathrm{P_{IV}})$ (7.3.6), then every finite A-algebra is a $\boldsymbol{P}$-ring.*

Proof. For every ideal $\mathfrak{a}$ of A, $\widehat{\mathrm{A}}\otimes_{\mathrm{A}}(\mathrm{A}/\mathfrak{a})$ is the completion of $\mathrm{A}/\mathfrak{a}$, and the formal fibres of $\mathrm{A}/\mathfrak{a}$ are therefore the formal fibres of A at the points of $\mathrm{V}(\mathfrak{a})$, whence (i). On the other hand, if B is a finite A-algebra, B is a semi-local ring, we have $\widehat{\mathrm{B}}=\mathrm{B}\otimes_{\mathrm{A}}\widehat{\mathrm{A}}$, and (ii) follows from (7.3.7). ☐

Proposition (7.3.16). — *Suppose that the property $\boldsymbol{P}$ is of the first (resp. second) type, defined from a property $\boldsymbol{R}$ (7.3.10). When $\boldsymbol{P}$ is of the second type, suppose moreover that the relation*

"for every finite extension k' of k, and every $z' \in \mathrm{Z} \otimes_k k'$, $\boldsymbol{R}(\mathscr{O}_{z'})$ is true"

implies $\boldsymbol{P}(\mathrm{Z}, k)$ (this is verified for the examples (iv') to (vii') of (7.3.8), by virtue of (6.7.7) and (4.6.1)).

Let A be a semi-local Noetherian ring. If the property $\boldsymbol{R}$ satisfies the conditions $(\mathrm{R_I})$ and $(\mathrm{R_{II}})$, the following properties are equivalent:

a) *A is a $\boldsymbol{P}$-ring.*
b) *For every integral quotient ring B of A (resp. every finite integral A-algebra B) and every prime ideal $\mathfrak{q}$ of $\widehat{\mathrm{B}}$ whose inverse image in B is 0, $\boldsymbol{R}(\widehat{\mathrm{B}}_{\mathfrak{q}})$ is true.*

If in addition $\boldsymbol{R}$ satisfies the condition $(\mathrm{R'_I})$, the properties a) and b) are also equivalent to:

c) *For every quotient ring B of A (resp. every finite A-algebra B), if we set $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$, $\mathrm{Y}' = \mathrm{Spec}(\widehat{\mathrm{B}})$, and if $g : \mathrm{Y}' \to \mathrm{Y}$ is the canonical morphism, we have (with the notations of (7.3.12))*

$$\mathrm{U}_{\boldsymbol{R}}(\mathrm{Y}') = g^{-1}(\mathrm{U}_{\boldsymbol{R}}(\mathrm{Y})). \tag{7.3.16.1}$$

Proof. The equivalence of $a)$ and $b)$ is immediate when $\boldsymbol{P}$ is of the first type: indeed, for every prime ideal $\mathfrak{p}$ of A, the formal fibre of $\mathrm{B} = \mathrm{A}/\mathfrak{p}$ at the generic point $\mathfrak{p} \in \mathrm{Spec}(\mathrm{A})$ is nothing other than the formal fibre of A at the point $\mathfrak{p}$ (7.3.13). When $\boldsymbol{P}$ is of the second type, the equivalence of $a)$ and $b)$ follows from the more general lemma below. ☐ **IV-2 | 197**

Lemma (7.3.16.2). — *Let A, A' be two rings, $\varphi : \mathrm{A} \to \mathrm{A}'$ a homomorphism, $f : \mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ the corresponding morphism. In order that for all $x \in \mathrm{Spec}(\mathrm{A})$, every finite extension k of $\boldsymbol{k}(x)$ and every point z of $f^{-1}(x) \otimes_{\boldsymbol{k}(x)} k = \mathrm{Z}$, the property $\boldsymbol{R}(\mathscr{O}_{\mathrm{Z},z})$ be true, it is necessary and sufficient that: for every integral finite A-algebra B, if $g : \mathrm{Spec}(\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{B}) \to \mathrm{Spec}(\mathrm{B})$ is the morphism deduced from f by base extension to B, and if $\mathrm{T} = g^{-1}(\xi)$ is the fibre of the generic point ξ of $\mathrm{Spec}(\mathrm{B})$, then, for all $t \in \mathrm{T}$, $\boldsymbol{R}(\mathscr{O}_{\mathrm{T},t})$ is true.*

Proof. The condition is trivially necessary since x is the point of $\mathrm{Spec}(\mathrm{A})$ above which ξ lies, $\boldsymbol{k}(\xi)$ is a finite extension of $\boldsymbol{k}(x)$. Conversely, let us consider a point $x \in \mathrm{Spec}(\mathrm{A})$ and let k be a finite extension of $\boldsymbol{k}(x)$. If we set $\mathfrak{p} = \mathfrak{j}_x$, then $\boldsymbol{k}(x)$ is the field of fractions of $\mathrm{A}/\mathfrak{p}$, and there is a basis of k over $\boldsymbol{k}(x)$ consisting of elements integral over $\mathrm{A}/\mathfrak{p}$. If B is the subring of k generated by these elements, then B is a finite integral A-algebra and k is the field $\boldsymbol{k}(\xi)$ at the generic point ξ of $\mathrm{Spec}(\mathrm{B})$. Since x is the image of ξ in $\mathrm{Spec}(\mathrm{A})$, the fibre $g^{-1}(\xi)$ is none other than $f^{-1}(x) \otimes_{\boldsymbol{k}(x)} k$, which completes the proof of the lemma. ☐

End of the proof of (7.3.16). That $a)$ implies $c)$ is an immediate consequence of (7.3.15) and (7.3.12). On the other hand, specialise $c)$ to the case where B is integral, and let y denote the generic point of $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$. Then $\mathscr{O}_{\mathrm{Y},y} = \boldsymbol{k}(y)$, so $y \in \mathrm{U}_{\boldsymbol{R}}(\mathrm{Y})$, since $\boldsymbol{R}(k)$ is true for every field k. Expressing that every point $y' \in g^{-1}(y)$ belongs to $\mathrm{U}_{\boldsymbol{R}}(\mathrm{Y}')$ gives statement $b)$; thus $c)$ implies $b)$. ☐

Proposition (7.3.17). — *Suppose that the property $\boldsymbol{R}$ satisfies the conditions $(\mathrm{R'_I})$ and $(\mathrm{R_{II}})$, and that $\boldsymbol{P}$ is the property of the first (resp. second) type defined from $\boldsymbol{R}$ (7.3.10). Then, if A is a $\boldsymbol{P}$-ring, the properties $\boldsymbol{R}(\mathrm{A})$ and $\boldsymbol{R}(\widehat{\mathrm{A}})$ are equivalent.*

Proof. This follows from (7.3.12.1) applied to $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{X} = \mathrm{Spec}(\widehat{\mathrm{A}})$. ☐

Proposition (7.3.18). — *Suppose that $\boldsymbol{P}$ is a property of the first (resp. second) type defined from a property $\boldsymbol{R}$ satisfying the conditions $(\mathrm{R'_I})$ and $(\mathrm{R_{II}})$, and also the condition following:*

$(\mathrm{R_{III}})$ *For every local Noetherian complete ring C, if we set $\mathrm{Z} = \mathrm{Spec}(\mathrm{C})$, the set $\mathrm{U}_{\boldsymbol{R}}(\mathrm{Z})$ (7.3.12) is open in Z.*

Then, if A is a $\boldsymbol{P}$-ring and if we set $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, the set $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X})$ is open in X.

Proof. Indeed, if $\mathrm{X}' = \mathrm{Spec}(\widehat{\mathrm{A}})$ and if $f : \mathrm{X}' \to \mathrm{X}$ is the canonical morphism, $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X}') = f^{-1}(\mathrm{U}_{\boldsymbol{R}}(\mathrm{X}))$ (7.3.12). As f is faithfully flat and quasi-compact (2.3.12), and as by hypothesis $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X}')$ is open in X', the conclusion follows from (2.3.12). ☐

Remarks (7.3.19). — *i)* The property $\boldsymbol{R}$ which serves to define $\boldsymbol{P}$ satisfies the condition $(\mathrm{R_{III}})$ in all the examples enumerated in (7.3.8). For (i), (ii) and (iii), this follows from (6.11.2) and the theorem of Cohen (**0**, 19.8.8); when ***P*** is one of the properties (iv), (iv′), (v), (v′), this follows from (6.12.7) and (6.12.9); when ***P*** is one of the properties (vii), (vii′), it follows from (6.12.7) and (6.13.4); finally, for (vi) and (vi′), the assertion of (7.3.18) is trivial, being true for every locally Noetherian prescheme.

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ii) We have already noted (6.4.3) that when ***P*** is one of the properties (ii) or (iii_1) of (7.3.8), we do not know whether every local Noetherian ring is a ***P***-ring. Recall, however, that when A is a quotient of a Cohen–Macaulay ring, the formal fibres of A are Cohen–Macaulay schemes (6.3.8).

iii) The most interesting instance of the notion of ***P***-ring is that corresponding to the strongest property (iv′) of (7.3.8): the rings whose formal fibres are geometrically regular. Fields, and more generally complete Noetherian local rings, trivially have this property.

iv) Let A be a local Noetherian ring of dimension 1. Then $\mathrm{Spec}(\mathrm{A})$ consists of the closed point a, corresponding to the maximal ideal $\mathfrak{m}$, and the maximal points b_i $(1 \leqslant i \leqslant r)$, corresponding to the minimal prime ideals of A. One has $\dim(\widehat{\mathrm{A}}) = 1$, and the maximal ideal of $\widehat{\mathrm{A}}$ is $\mathfrak{m}\widehat{\mathrm{A}}$. The formal fibre of A at a is therefore $\mathrm{Spec}(k)$, where $k = \mathrm{A}/\mathfrak{m}$ is the residue field of A; the formal fibre at each b_i is the spectrum of an Artinian ring whose residue fields are the residue fields L_{ij} of $\mathrm{Spec}(\widehat{\mathrm{A}})$ at the maximal points b_{ij} lying over b_i. Since an Artinian ring is Cohen–Macaulay, A is a ***P***-ring when ***P*** is property (ii) of (7.3.8). Moreover, since a reduced Artinian ring is a direct sum of fields, the following properties are equivalent (6.7.7):

- *a)* the formal fibres of A are geometrically reduced;
- *b)* the formal fibres of A are geometrically normal;
- *c)* the formal fibres of A are geometrically regular.

When A is reduced, they are also equivalent to:

- *d)* $\widehat{\mathrm{A}}$ is reduced and L_{ij} is a separable extension of K_i for every pair (i, j) (4.6.1).

In particular, if A is a discrete valuation ring, K its field of fractions, and $\widehat{K}$ the completion of K for the valuation corresponding to A (the field of fractions of $\widehat{\mathrm{A}}$), then the formal fibres of A are geometrically regular if and only if $\widehat{K}$ is a separable extension of K. This is always so when K has characteristic 0.

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(7.4.1). Throughout the rest of this number, we suppose that the property ***P*** is of the form defined in (7.3.1), and *satisfies the conditions* $(\mathrm{P_I})$, $(\mathrm{P_{II}})$ *and* $(\mathrm{P_{III}})$ of (7.3.4). Moreover, we suppose that the property ***Q*** which serves to define ***P*** is such that, for every generization z' of z in Z, $\boldsymbol{Q}(\mathscr{O}_z, k)$ implies $\boldsymbol{Q}(\mathscr{O}_{z'}, k)$.

Lemma (7.4.2). — *Let* A, A′ *be two local Noetherian rings,* $\varphi : \mathrm{A} \to \mathrm{A}'$ *a local homomorphism such that* $f = {}^a\varphi : \mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ *is a **P**-morphism. Then, if the formal fibres of* A′ *are geometrically regular,* A *is a **P**-ring.*

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Proof. Consider the completed homomorphism $\widehat{\varphi} : \widehat{\mathrm{A}} \to \widehat{\mathrm{A}'}$ and the corresponding morphism $\widehat{f} = {}^a\widehat{\varphi}$; we have the commutative diagram

$$\begin{tikzcd}
\mathrm{Spec}(\widehat{\mathrm{A}}) & \mathrm{Spec}(\widehat{\mathrm{A}'}) \arrow[l, "\widehat{f}"'] \\
\mathrm{Spec}(\mathrm{A}) \arrow[u, "g"] & \mathrm{Spec}(\mathrm{A}') \arrow[l, "f"] \arrow[u, "g'"']
\end{tikzcd}$$

where g and g' are the canonical morphisms. As by hypothesis f is a ***P***-morphism and g' a regular morphism, it follows from $(\mathrm{P_I})$ that $f \circ g' = g \circ \widehat{f}$ is a ***P***-morphism. On the other hand, the hypothesis that f is a ***P***-morphism implies that f is flat, hence it is the same for $\widehat{f}$ (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, cor. of prop. 3), which is moreover a local homomorphism, hence faithfully flat (**0**, 6.6.2); it follows then from $(\mathrm{P_{II}})$ that g is a ***P***-morphism. □

Corollary (7.4.3). — i) *Let* A *be a* $\boldsymbol{P}$*-ring local Noetherian,* $A' = \widehat{A}$ *its completion. Suppose that for every prime ideal* $\mathfrak{p}'$ *of* A'*, the formal fibres of* $A'_{\mathfrak{p}'}$ *are geometrically regular; then, for every prime ideal* $\mathfrak{p}$ *of* A*,* $A_{\mathfrak{p}}$ *is a* $\boldsymbol{P}$*-ring.*

ii) *Suppose that* $\boldsymbol{P}$ *satisfies the condition* (P_{IV}) (7.3.6)*. Let* A *be a* $\boldsymbol{P}$*-ring local Noetherian,* B *an A-algebra locale essentially of finite type, and such that the homomorphism* $A \to B$ *is local. Set* $A' = \widehat{A}$*, and let* $\mathfrak{n}'$ *be the unique prime ideal of* $B' = B \otimes_A A'$ *above the maximal ideal of* A'*. If the formal fibres of* $B'_{\mathfrak{n}'}$ *are geometrically regular, then* B *is a* $\boldsymbol{P}$*-ring.*

Proof. (i) As by hypothesis $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is a $\boldsymbol{P}$-morphism, the same holds for $\mathrm{Spec}(A'_{\mathfrak{p}'}) \to \mathrm{Spec}(A_{\mathfrak{p}})$ by virtue of (7.3.2), *b*)), for every prime ideal $\mathfrak{p}'$ of A' above $\mathfrak{p}$ and every prime ideal $\mathfrak{p}$ of A. It suffices then to apply the lemma (7.4.2) to this morphism, noting that the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is surjective.

(ii) By virtue of (7.4.2), it suffices to prove that the morphism $\mathrm{Spec}(B'_{\mathfrak{n}'}) \to \mathrm{Spec}(B)$ is a $\boldsymbol{P}$-morphism. Now we have $B = C_{\mathfrak{n}}$, where C is a finite type A-algebra and $\mathfrak{n}$ is a prime ideal of C above the maximal ideal of A. If we set $C' = C \otimes_A A'$, it follows from the hypotheses and from (7.3.7) that $\mathrm{Spec}(C') \to \mathrm{Spec}(C)$ is a $\boldsymbol{P}$-morphism; as $B'_{\mathfrak{n}'}$ is a local ring of C' at a prime ideal of C' above $\mathfrak{n}$, the conclusion follows from (7.3.2), *b*)).

□

Theorem (7.4.4). — *The hypotheses on* $\boldsymbol{P}$ *being those of* (7.4.1)*, let* A *be a* $\boldsymbol{P}$*-ring local Noetherian.*

i) *For every prime ideal* $\mathfrak{p}$ *of* A*,* $A_{\mathfrak{p}}$ *is a* $\boldsymbol{P}$*-ring.*

ii) *Suppose in addition that* $\boldsymbol{P}$ *satisfies the condition* (P_{IV})*. Then every local ring that is an A-algebra essentially of finite type is a* $\boldsymbol{P}$*-ring.*

Proof. (i) Applying (7.4.3), (i), everything reduces to seeing that for every prime ideal $\mathfrak{p}'$ of $A' = \widehat{A}$, $A'_{\mathfrak{p}'}$ has its formal fibres geometrically regular. Now, this has been proved in (**0**, 22.3.3) and (**0**, 22.5.8).

(ii) Let B be an A-algebra essentially of finite type that is a local ring; if $\mathfrak{p}$ is the prime ideal of A above which is the maximal ideal of B, B is also an $A_{\mathfrak{p}}$-algebra essentially of finite type (1.3.8); by virtue of (i), we can therefore suppose that $\mathfrak{p}$ is the maximal ideal $\mathfrak{m}$ of A. We have therefore $B = C_{\mathfrak{q}}$, where C is a finite type A-algebra, $\mathfrak{q}$ a prime ideal of C (necessarily above $\mathfrak{m}$); by virtue of (i), one can therefore suppose that $B = C_{\mathfrak{q}}$.

Let B be an A-algebra essentially of finite type that is a local ring; if $\mathfrak{p}$ is the prime ideal of A above which the maximal ideal of B lies, B is also an $A_{\mathfrak{p}}$-algebra essentially of finite type (1.3.8); by virtue of (i), one can therefore suppose that the homomorphism $A \to B$ is local. **IV-2** | 200

Applying (7.4.3), (ii) with $A' = \widehat{A}$, $B' = B \otimes_A A'$ is a local ring of $C \otimes_A A'$ at a prime ideal above the maximal ideal of A'; as A and A' have the same residue field *k*, the residue field of $C \otimes_A A'$ at this prime ideal is equal to k'. Moreover $C \otimes_A A'$ is a finite type A′-algebra generated by a single element. Hence (7.4.3), (ii) one can restrict to proving (7.4.4.1) when $\boldsymbol{P}$ is the property (iv′) of (7.3.8), that A is *complete* and C is generated by a single element *t*.

□

Lemma (7.4.4.1). — *Let* A *be a* $\boldsymbol{P}$*-ring local Noetherian,* k *its residue field,* C *a finite type A-algebra,* B *a local ring in a prime ideal* $\mathfrak{n}$ *of* C*, such that:* 1° *the homomorphism* $A \to B$ *is local;* 2° *the residue field* k' *of* B *is a finite extension of* k*. If* $\boldsymbol{P}$ *satisfies* (P_{IV})*, then* B *is a* $\boldsymbol{P}$*-ring.*

Proof. Let $(x_i)_{1 \leqslant i \leqslant m}$ be a system of generators of the A-algebra C; let us show first that we can argue by induction on *m*. Let C' be the sub-A-algebra of C generated by $x_1, \ldots, x_{m-1}$, and let $\mathfrak{n}' = \mathfrak{n} \cap C'$. The homomorphism $A \to C_{\mathfrak{n}}$ factors through the local homomorphisms $A \to C'_{\mathfrak{n}'} \to C_{\mathfrak{n}}$; if k'' is the residue field of $C'_{\mathfrak{n}'}$, the extension $k \to k'$ factors through k'', hence k'' is a finite extension of k and k' a finite extension of k''. The induction hypothesis implies that $C'_{\mathfrak{n}'}$ is a $\boldsymbol{P}$-ring. If $S' = C' - \mathfrak{n}'$, then $C_{\mathfrak{n}}$ is a local ring of $S'^{-1}C = C'_{\mathfrak{n}'}[x_m/1]$; the induction hypothesis applied once more shows that $C_{\mathfrak{n}}$ is a $\boldsymbol{P}$-ring. Thus the problem reduces to the case where C is an A-algebra generated by a *single* element *t*.

Applying (7.4.3), (ii), with $A' = \widehat{A}$, the ring $B' = B \otimes_A A'$ is a local ring of $C \otimes_A A'$ at a prime ideal above $\mathfrak{n}$; as A and A' have the same residue field *k*, the residue field of $C \otimes_A A'$ at this prime ideal is equal to k'. Moreover $C \otimes_A A'$ is generated by a single element over A'. It follows therefore from (7.4.3), (ii) that one can restrict to proving (7.4.4.1) when $\boldsymbol{P}$ is the property (iv′) of (7.3.8), that A is *complete* and C generated by a single element t.

To show that the formal fibres of $B = C_{\mathfrak{n}}$ are then geometrically regular, let us apply the criterion (7.3.16), *b*)). Let B_1 be a finite integral B-algebra, hence generated by a finite number of elements integral over B. By multiplying these elements by an element of B, one can suppose that they are integral over C, and we can therefore write $B_1 = S^{-1}C_1$, where C_1 is a sub-C-algebra of B_1 generated by a finite number of integral elements over C, hence a finite integral C-algebra. On the other hand, B_1 is a semi-local ring, and every local ring B_2 of B_1 at a maximal ideal is a local ring of C_1, such that $A \to B_2$ is a local homomorphism; moreover, the residue field of B_2 is a finite extension of k', hence of k. We see therefore (taking into account (7.3.14) and (i)) that we are reduced to the following question: let C be a Noetherian integral ring containing a subring C_0 which is an A-algebra generated by a single element t and such that C is a finite C_0-algebra; if $\mathfrak{n}$ is a maximal ideal of C above the maximal ideal of A, and $B = C_{\mathfrak{n}}$, is the completion $\widehat{B}_{\mathfrak{q}}$ *regular* for every prime **IV-2** | 201
ideal $\mathfrak{q}$ of $\widehat{B}$ such that $\mathfrak{q} \cap A = 0$?

lying over the maximal ideal of A, and $B = C_{\mathfrak{n}}$, the question is to show that for every prime ideal $\mathfrak{q}$ of $\widehat{B}$ whose inverse image in B is 0, the ring $\widehat{B}_{\mathfrak{q}}$ is regular. One may moreover replace A by its image in C, which is a complete local ring (as a quotient of A) and integral. The desired conclusion follows from the more general lemma below. □

Lemma (7.4.4.2). — *Let* A *be a complete Noetherian local integral ring, k its residue field,* C *an integral ring containing* A, *such that there exists $t \in C$ for which* C *is a finite* $A[t]$*-algebra. Let $\mathfrak{n}$ be a maximal ideal of* C *above the maximal ideal $\mathfrak{m}$ of* A*; set* $B = C_{\mathfrak{n}}$, $X = \mathrm{Spec}(B)$, $B' = \widehat{B}$, $X' = \mathrm{Spec}(\widehat{B})$*; if* $U = \mathrm{Reg}(X)$, $U' = \mathrm{Reg}(X')$*, and if $f : X' \to X$ is the canonical morphism, then $f^{-1}(U) \subset U'$.*

Proof. The assertion needed for (7.4.4.1) follows from this lemma because, since B is integral, the generic point of X belongs to U. Since C is an A-algebra of finite type and $\mathfrak{n}$ is a maximal ideal of C, the residue field k' of $C_{\mathfrak{n}} = B$ (hence also of B') is a finite extension of k (**I**, 6.4.11) and (6.4.2).

Set $Y = \mathrm{Spec}(C)$. It follows from (6.12.8) that $\mathrm{Reg}(Y)$ is open in Y; since the local rings of X are local rings of Y (**I**, 2.4.2), we have $U = X \cap \mathrm{Reg}(Y)$, hence U is open in X. On the other hand, (6.12.7) shows that U' is open in X', hence $S' = X' - U'$ is closed. Consequently $S' \cap f^{-1}(U)$ is locally closed in X', and the whole question is to prove that this set is empty. If it were non-empty, (5.1.10) would give a prime ideal $\mathfrak{p}'$ in $S' \cap f^{-1}(U)$ such that $\dim(B'/\mathfrak{p}') \leqslant 1$. The prime ideal $\mathfrak{p}'$ cannot be the maximal ideal $\mathfrak{m}B'$ of B', where $\mathfrak{m}$ is the maximal ideal of B. Indeed, this would mean that $B = B_{\mathfrak{m}}$ is regular, hence so is $B' = \widehat{B}$ (**0**, 17.1.5), and we would have $\mathfrak{m}B' \in U'$, contrary to the hypothesis. We must therefore have $\dim(B'/\mathfrak{p}') = 1$.

Set $\mathfrak{p} = B \cap \mathfrak{p}'$. By hypothesis $B_{\mathfrak{p}}$ is regular, whereas $B'_{\mathfrak{p}'}$ is not. Since $B'_{\mathfrak{p}'}$ is a flat $B_{\mathfrak{p}}$-module, (6.5.2) shows that the fibre Z of f at $\mathfrak{p}$ is not regular at $\mathfrak{p}'$. We may reduce to the case $\mathfrak{p} = 0$. Indeed, in general, set $\mathfrak{q} = \mathfrak{p} \cap C$ and $\mathfrak{r} = \mathfrak{p} \cap A$. Then $C/\mathfrak{q}$ is a finite $(A/\mathfrak{r})[\bar{t}]$-algebra, where $\bar{t}$ is the class of t modulo $\mathfrak{q}$; since $\mathfrak{p} = \mathfrak{q}B$, the ring $B/\mathfrak{p}$ is equal to $(C/\mathfrak{q})_{\mathfrak{n}/\mathfrak{q}}$, and $\mathfrak{n}/\mathfrak{q}$ is a maximal ideal of $C/\mathfrak{q}$ lying over the maximal ideal $\mathfrak{m}/\mathfrak{r}$ of $A/\mathfrak{r}$. Thus the hypotheses of (7.4.4.2) are still satisfied by $A/\mathfrak{r}$, $C/\mathfrak{q}$, and $B/\mathfrak{p}$; since the completion of $B/\mathfrak{p}$ is $B'/\mathfrak{p}B'$, this proves the reduction.

Suppose therefore that $\mathfrak{p} = 0$, so that Z is the generic fibre of f and the homomorphism $B \to B'/\mathfrak{p}'$ is injective. Set $V = B'/\mathfrak{p}'$, and distinguish two cases.

I) V *is a finite* A*-algebra.* Since $B \subset V$, B is *a fortiori* a finite A-algebra; as A is complete, so is B (Bourbaki, *Alg. comm.*, chap. IV, § 2, nº 5, cor. 3 of prop. 9). Hence $B' = B$ and $\mathfrak{p}' = 0$, so $B'_{\mathfrak{p}'}$ is a field, contrary to the hypothesis.

II) V *is not a finite* A*-algebra.* Since the local ring A is complete, V is not a quasi-finite A-algebra (**0**, 7.4.1) and (**0**, 7.4.2); but the residue field k' of V is a finite extension of the residue field k of A, hence (**0**, 7.4.4) the ideal $\mathfrak{m}V$ is not an ideal of definition of V. Since V is a one-dimensional Noetherian local integral ring, 0 is the only ideal of V that is not an ideal of definition, hence $\mathfrak{m}V = 0$. But $A \subset V$ and V is integral, so $\mathfrak{m} = 0$ and $A = k$ is a field. It follows first that $\dim(C) \leqslant 1$ (**0**, 16.1.5); but since $\dim(V) = 1$, the inequalities

$$\dim(V) \leqslant \dim(B') = \dim(B) \leqslant \dim(C)$$

show that $\dim(\mathrm{C}) = \dim(\mathrm{B}) = \dim(\mathrm{B}') = \dim(\mathrm{B}'/\mathfrak{p}') = 1$, and consequently $\mathfrak{p}'$ is a minimal prime ideal of B'. We therefore obtain a contradiction once we prove that $\mathrm{B}'_{\mathfrak{p}'}$ is a field, or equivalently that B' is reduced. Since C is a finite-type k-algebra, its integral closure C_1 is a finite C-algebra (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 2, th. 2). If $\mathrm{S} = \mathrm{C} - \mathfrak{n}$, then $\mathrm{B}_1 = \mathrm{S}^{-1}\mathrm{C}_1$ is the integral closure of B, hence a finite B-algebra and therefore a one-dimensional semi-local Noetherian integral and integrally closed ring (**0**, 16.1.5). If $\mathfrak{m}_j$ $(1 \leqslant j \leqslant h)$ are its maximal ideals, the rings $(\mathrm{B}_1)_{\mathfrak{m}_j}$ are discrete valuation rings (**II**, 7.1.6), and the completion B'_1 of B_1 is the direct product of the completed discrete valuation rings $(\mathrm{B}_1)_{\mathfrak{m}_j}$ (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 13, prop. 18). Thus B'_1 is reduced; since the completion B' of B is a subring of B'_1 (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, cor. 3 of prop. 9), B' is reduced as well. □

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Corollary (7.4.5). — *Suppose that the property* ***P*** *satisfies the conditions* $(\mathrm{P_I})$, $(\mathrm{P_{II}})$, $(\mathrm{P_{III}})$. *Let* A *be a Noetherian ring. The following conditions are equivalent:*

a) *For every prime ideal* $\mathfrak{p}$ *of* A, $\mathrm{A}_{\mathfrak{p}}$ *is a* ***P****-ring.*

b) *For every maximal ideal* $\mathfrak{m}$ *of* A, $\mathrm{A}_{\mathfrak{m}}$ *is a* ***P****-ring.*

If in addition ***P*** *satisfies* $(\mathrm{P_{IV}})$, *the properties a) and b) are also equivalent to:*

c) *For every finite type A-algebra* B *and every prime ideal* $\mathfrak{q}$ *of* B, $\mathrm{B}_{\mathfrak{q}}$ *is a* ***P****-ring.*

Proof. The equivalence of *a*) and *b*) follows from (7.4.4), (i), and that of *a*) and *c*) from (7.4.4), (ii). When condition *a*) of (7.4.5) is satisfied, we say that A is a ***P***-ring; for semi-local Noetherian rings, this definition coincides with that of (7.3.13), when the conditions $(\mathrm{P_I})$, $(\mathrm{P_{II}})$ and $(\mathrm{P_{III}})$ are satisfied. The ring **Z** is a ***P***-ring (7.3.19), (iv); every local Noetherian complete ring is a ***P***-ring. □

Proposition (7.4.6). — *Suppose that the property* ***P*** *satisfies the conditions* $(\mathrm{P_I})$, $(\mathrm{P_{II}})$ *and* $(\mathrm{P_{III}})$. *Let* A *be a Noetherian ring,* $\mathfrak{J}$ *an ideal of* A, $\widehat{\mathrm{A}}$ *the separated completion of* A *for the* $\mathfrak{J}$*-adic topology. Then, if* A *is a* ***P****-ring* (7.4.5), *the canonical morphism* $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$ *is a* ***P****-morphism.*

Proof. Using (7.3.2), c')), it suffices to prove that for every maximal ideal $\mathfrak{n}$ of $\widehat{\mathrm{A}}$ with inverse image $\mathfrak{m}$ in A, the morphism $\mathrm{Spec}((\widehat{\mathrm{A}})_{\mathfrak{n}}) \to \mathrm{Spec}(\mathrm{A}_{\mathfrak{m}})$ is a ***P***-morphism. We know (Bourbaki, *Alg. comm.*, chap. III, § 3, n° 4, prop. 8) that the canonical homomorphism $\mathrm{A}_{\mathfrak{m}} \to (\widehat{\mathrm{A}})_{\mathfrak{n}}$ is injective, that the $\mathfrak{m}\mathrm{A}_{\mathfrak{m}}$-adic topology on $\mathrm{A}_{\mathfrak{m}}$ is induced by the $\mathfrak{n}(\widehat{\mathrm{A}})_{\mathfrak{n}}$-adic topology, and that $\mathrm{A}_{\mathfrak{m}}$ is dense in $(\widehat{\mathrm{A}})_{\mathfrak{n}}$ for this topology, so that the completion of $\mathrm{A}_{\mathfrak{m}}$ for the $\mathfrak{m}\mathrm{A}_{\mathfrak{m}}$-adic topology identifies with that of $(\widehat{\mathrm{A}})_{\mathfrak{n}}$ for the $\mathfrak{n}(\widehat{\mathrm{A}})_{\mathfrak{n}}$-adic topology. We have therefore two morphisms

$$\mathrm{Spec}((\mathrm{A}_{\mathfrak{m}})\widehat{\ }) \xrightarrow{l} \mathrm{Spec}((\widehat{\mathrm{A}})_{\mathfrak{n}}) \xrightarrow{g} \mathrm{Spec}(\mathrm{A}_{\mathfrak{m}})$$

such that l is faithfully flat. Since, by hypothesis, $g \circ l$ is a ***P***-morphism, it follows from $(\mathrm{P_{II}})$ that g is a ***P***-morphism as well. □

Corollary (7.4.7). — *Suppose that* ***P*** *satisfies the conditions* $(\mathrm{P_I})$, $(\mathrm{P'_I})$, $(\mathrm{P_{II}})$, $(\mathrm{P_{III}})$ *and* $(\mathrm{P_{IV}})$. *Then, if* A *is a* ***P****-ring* (7.4.5), *the canonical morphism* $\mathrm{Spec}(\mathrm{A}[[\mathrm{T}_1, \ldots, \mathrm{T}_r]]) \to \mathrm{Spec}(\mathrm{A})$ *is a* ***P****-morphism. In particular, if in addition* A *is integral,* K *its field of fractions, and if* $\mathfrak{p}$ *is a prime ideal of* $\mathrm{B} = \mathrm{A}[[\mathrm{T}_1, \ldots, \mathrm{T}_r]]$ *such that* $\mathfrak{p} \cap \mathrm{A} = 0$, *then the property* $\boldsymbol{P}(\mathrm{B}_{\mathfrak{p}}, \mathrm{K})$ *is true.*

Proof. Indeed, $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ factorises as **IV-2** | 203

$$\mathrm{Spec}(\mathrm{A}[[\mathrm{T}_1, \ldots, \mathrm{T}_r]]) \xrightarrow{f} \mathrm{Spec}(\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_r]) \xrightarrow{g} \mathrm{Spec}(\mathrm{A}).$$

It is clear that the morphism g is regular (**0**, 17.3.7); by virtue of (7.4.5), $\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ is a ***P***-ring, hence it follows from (7.4.6) that f is a ***P***-morphism; as g is regular, hence also a ***P***-morphism (7.3.5), (i), it follows from $(\mathrm{P'_I})$ that $g \circ f$ is a ***P***-morphism.

One will note that the conclusion is still valid if, instead of supposing that ***P*** satisfies $(\mathrm{P'_I})$ and that every regular morphism is a ***P***-morphism, one supposes only that a composite $g \circ f$ is a ***P***-morphism whenever g is regular and f is a ***P***-morphism (a condition in some sense symmetric to $(\mathrm{P'_I})$). □

Remark (7.4.8). — The preceding results pose the following problems:

A) Let A be a Zariski *complete* ring, and let $\mathfrak{J}$ be an ideal of definition of A; if the ring $A/\mathfrak{J}$ is a $\boldsymbol{P}$-ring, is A also a $\boldsymbol{P}$-ring? It would follow that for every Noetherian $\boldsymbol{P}$-ring A and every ideal $\mathfrak{J}$ of A, the separated completion $\widehat{A}$ for the $\mathfrak{J}$-adic topology would also be a $\boldsymbol{P}$-ring.
B) Let k be a valued field, complete and non-discrete; one calls *ring of restricted formal power series* $k\{T_1, \dots, T_n\}$ the subring of the formal power series ring $k[[T_1, \dots, T_n]]$ formed of the series whose coefficients *tend to* 0. Is such a ring a $\boldsymbol{P}$-ring?
C) If A is a linearly topologised $\boldsymbol{P}$-ring, S a multiplicative subset of A, are the rings $A\{S^{-1}\}$ (**0**, 7.6.1) and $A_{\{S\}}$ (**0**, 7.6.15) $\boldsymbol{P}$-rings?

7.5. A criterion for $\boldsymbol{P}$-morphisms

(7.5.0). This number will not be used in the sequel of Chapter IV and may therefore be omitted on a first reading. We shall moreover see later (7.9.8) that the results of the present number can be considerably improved when one has "resolution of singularities" at one's disposal. **IV-2** | 203

In the sequel of this number, we shall consider a property $\boldsymbol{R}(A)$, and we shall denote by $\boldsymbol{P}(Z, k)$ the following property:

"Z is a locally Noetherian prescheme over a field k, and, for every finite extension k' of k, if $Z' = Z \otimes_k k'$, the property $\boldsymbol{R}(\mathscr{O}_{Z',z'})$ is true for every $z' \in Z'$."

Theorem (7.5.1). — *Let* A *and* B *be two complete Noetherian local rings, let* $\mathfrak{m}$ *be the maximal ideal of* A, *let* $k = A/\mathfrak{m}$ *be its residue field, and let* $\varphi : A \to B$ *be a local homomorphism such that:*

(i) the residue field of B *is a finite extension of* k;
(ii) B *is a flat* A*-module.*

Let $\boldsymbol{R}(C)$ *moreover be a property satisfying condition* (R_{III}) (7.3.18) *and the following condition:*

(R_{IV}) *For every local ring* C *at a prime ideal of a complete Noetherian local ring, and every* C*-regular element* t *in the maximal ideal of* C, *the property* $\boldsymbol{R}(C/tC)$ *implies* $\boldsymbol{R}(C)$.

Let $f = {}^a\varphi : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$, *and suppose that* $\boldsymbol{P}(\mathrm{Spec}(B \otimes_A k), k)$ *is true. Then, for every* $x \in \mathrm{Spec}(A)$, *the property* $\boldsymbol{P}(f^{-1}(x), \boldsymbol{k}(x))$ *is true.*

Proof. In other words, property $\boldsymbol{P}$ for the fibre over the closed point of $\mathrm{Spec}(A)$ entails the same property for all fibres of f; that is, f is a $\boldsymbol{P}$-morphism (7.3.1). We proceed in several steps.

I) Reduction to the study of the local rings of the generic fibre. Apply (7.3.16.2). It suffices to show that, for every finite integral A-algebra A' with field of fractions K', if $B' = B \otimes_A A'$, then every local ring of $\mathrm{Spec}(B' \otimes_{A'} K')$ satisfies $\boldsymbol{R}$. The ring A' (resp. B') is complete and semi-local, the latter because it is a finite B-algebra, and is therefore a direct product of complete local rings. Since A' is integral it is local. Every maximal ideal $\mathfrak{n}'$ of B' lies over the maximal ideal $\mathfrak{n}$ of B, hence over $\mathfrak{m}$, and its inverse image in A' is consequently the maximal ideal $\mathfrak{m}'$ of A'. Every local ring of $\mathrm{Spec}(B' \otimes_{A'} K')$ is a local ring of $\mathrm{Spec}(B')$, and hence of one of the $\mathrm{Spec}(B'_{\mathfrak{n}'})$ over the generic point of A'. Thus it is enough to prove that the local rings of

$$\mathrm{Spec}(B'_{\mathfrak{n}'} \otimes_{A'} K')$$

satisfy $\boldsymbol{R}$. Now A' and $B'_{\mathfrak{n}'}$ are complete Noetherian local rings. The residue field of $B'_{\mathfrak{n}'}$ is a finite extension of that of B, hence also of k, and *a fortiori* of the residue field k' of A'. This observation and (2.1.4) show that A' and $B'_{\mathfrak{n}'}$ satisfy conditions (i) and (ii) of the statement. On the other hand, if k'' is a finite extension of k', it is also a finite extension of k, and the local rings of

$$\mathrm{Spec}(B'_{\mathfrak{n}'} \otimes_{A'} k'')$$

are also local rings of $\mathrm{Spec}(B \otimes_A k'')$. The hypothesis $\boldsymbol{P}(\mathrm{Spec}(B \otimes_A k), k)$ therefore implies

$$\boldsymbol{P}(\mathrm{Spec}(B'_{\mathfrak{n}'} \otimes_{A'} k'), k').$$

We are thus reduced to the case where A is integral with field of fractions K, and to proving that the local rings of $\mathrm{Spec}(B \otimes_A K)$ satisfy $\boldsymbol{R}$.

II) The case $\dim(A) = 1$. Let A' be the integral closure of A. By (**0**, 23.1.6), A' is a finite A-algebra and a complete local ring. If $B' = B \otimes_A A'$, then **IV-2** | 204

$$B \otimes_A K = B' \otimes_{A'} K.$$

The same argument as in I) shows that it suffices, for every maximal ideal $\mathfrak{n}'$ of B', to prove that the local rings of $B'_{\mathfrak{n}'} \otimes_{A'} K$ satisfy $\boldsymbol{R}$; it also shows that A' and $B'_{\mathfrak{n}'}$ satisfy hypotheses (i) and (ii), and that

$$\boldsymbol{P}(\operatorname{Spec}(B'_{\mathfrak{n}'} \otimes_{A'} k'), k')$$

is true, where k' is the residue field of A'. Since $\dim(A') = 1$ (**0**, 16.1.5) and A' is integral and integrally closed, it is a complete discrete valuation ring. We may therefore suppose that A itself is a complete discrete valuation ring. If u is a uniformizer of A, then u is A-regular, and the flatness of B over A implies that $t = \varphi(u)$ is B-regular and belongs to the maximal ideal of B. The hypothesis on $\operatorname{Spec}(B \otimes_A k)$ implies in particular that $\boldsymbol{R}(B/tB)$ is true; hence $\boldsymbol{R}(B)$ is true by (R_{IV}). Thus $U_{\boldsymbol{R}}(\operatorname{Spec}(B))$ contains the closed point. It is open by (R_{III}), and Spec(B) is the only open subset of itself containing its closed point. Consequently all local rings of Spec(B), and in particular those of $\operatorname{Spec}(B \otimes_A K)$, satisfy $\boldsymbol{R}$.

III) The general case. The case $\dim(A) = 0$ is trivial, since then $A = k$; we may therefore suppose that $\dim(A) \geqslant 1$. By (6.12.7), Spec(A) contains a non-empty open subset V all of whose points are regular. The intersection V' of V with the complement of the closed point of Spec(A) is non-empty and therefore contains the generic point. It suffices to prove that

$$f^{-1}(V') \subset U_{\boldsymbol{R}}(\operatorname{Spec}(B)).$$

Equivalently, the intersection Z of $f^{-1}(V')$ with the complement of $U_{\boldsymbol{R}}(\operatorname{Spec}(B))$ must be empty. Suppose otherwise. Since $U_{\boldsymbol{R}}(\operatorname{Spec}(B))$ is open by (R_{III}), Z is locally closed in Spec(B); if non-empty, it contains a point x such that $\dim(\overline{\{x\}}) \leqslant 1$ (5.1.10). Since $f(x) \in V'$ is distinct from the closed point of Spec(A), x is distinct from the closed point of Spec(B), and therefore $\dim(\overline{\{x\}}) = 1$.

We show this is impossible. Let $\mathfrak{q}$ be the corresponding prime ideal of B, so that $\dim(B/\mathfrak{q}) = 1$, and put $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$. Then $\mathfrak{p} \neq \mathfrak{m}$ and $A_{\mathfrak{p}}$ is regular. Since $\mathfrak{p} \neq \mathfrak{m}$, the ideal $\mathfrak{m}(B/\mathfrak{q})$ is non-zero, hence is an ideal of definition of the one-dimensional Noetherian local integral ring $B/\mathfrak{q}$. By (i), the residue field of $B/\mathfrak{q}$ is a finite extension of the residue field of $A/\mathfrak{p}$, so $B/\mathfrak{q}$ is a quasi-finite $(A/\mathfrak{p})$-algebra (**0**, 7.4.4). But $A/\mathfrak{p}$ is complete and $B/\mathfrak{q}$ is separated for the $\mathfrak{m}$-preadic topology, which is its local-ring topology; hence $B/\mathfrak{q}$ is finite over $A/\mathfrak{p}$ by (**0**, 7.4.1). The injection $A/\mathfrak{p} \to B/\mathfrak{q}$ consequently gives

$$\dim(A/\mathfrak{p}) = \dim(B/\mathfrak{q}) = 1$$

by (**0**, 16.1.5). Apply II) to $A/\mathfrak{p}$ and $B/\mathfrak{p}B$. Their residue fields are respectively those of A and B, $B/\mathfrak{p}B$ is flat over $A/\mathfrak{p}$, and

$$(B/\mathfrak{p}B) \otimes_{A/\mathfrak{p}} k = B \otimes_A k = B/\mathfrak{m}B.$$

It follows that the local rings of

$$\operatorname{Spec}\big((B/\mathfrak{p}B) \otimes_{A/\mathfrak{p}} \boldsymbol{k}(\mathfrak{p})\big)$$

satisfy $\boldsymbol{R}$. In particular, $B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$ is one of these local rings. Moreover, $B_{\mathfrak{q}}$ is flat over the regular local ring $A_{\mathfrak{p}}$. We now use the following lemma.

Lemma (7.5.1.1). — *Let $\mathscr{C}$ be a full subcategory of the category of Noetherian local rings such that every quotient ring of a ring in $\mathscr{C}$ still belongs to $\mathscr{C}$. Suppose $\boldsymbol{R}(C)$ has the property that, if $C \in \mathscr{C}$ and t is a regular element of the maximal ideal of C such that $\boldsymbol{R}(C/tC)$ is true, then $\boldsymbol{R}(C)$ is true.* **IV-2 | 205**

Let C be a regular local ring with residue field k, let $D \in \mathscr{C}$ be a local ring, and let $\psi : C \to D$ be a local homomorphism making D a flat C-module. If $\boldsymbol{R}(D \otimes_C k)$ is true, then $\boldsymbol{R}(D)$ is true.

Proof. Proceed by induction on $n = \dim(C)$. The assertion is true by hypothesis when $n = 0$, for then $C = k$. Let $t \in \mathfrak{m} - \mathfrak{m}^2$. Then C/tC is regular and $\dim(C/tC) = n - 1$ (**0**, 17.1.8) and (**0**, 16.3.4). Since

$$(D/tD) \otimes_{C/tC} k = D \otimes_C k$$

and $D/tD \in \mathscr{C}$, the induction hypothesis gives $\boldsymbol{R}(D/tD)$. Moreover, t is C-regular, hence D-regular by flatness (**0**, 6.3.4); the assumed property of $\boldsymbol{R}$ therefore gives $\boldsymbol{R}(D)$. □

To finish the proof of (7.5.1), apply (7.5.1.1) with $\mathscr{C}$ the category of local rings of spectra of complete local rings, using (R_{IV}), and take $C = A_{\mathfrak{p}}$ and $D = B_{\mathfrak{q}}$. □

Corollary (7.5.2). — *Let A be a Noetherian local ring, let $\mathfrak{m}$ be its maximal ideal and $k = A/\mathfrak{m}$ its residue field, let B be a Noetherian local ring, and let $\varphi : A \to B$ be a local homomorphism such that:*

(i) the residue field of B *is a finite extension of* k*;*
(ii) B *is a flat* A*-module.*

Let $\boldsymbol{R}(\mathrm{C})$ *be a property satisfying* $(\mathrm{R_{II}})$ (7.3.10), $(\mathrm{R_{III}})$ (7.3.18), *and* $(\mathrm{R_{IV}})$ (7.5.1). *Let* $\boldsymbol{R}'(\mathrm{C})$ *be a property satisfying:*

> $(\mathrm{R_V})$ *If* C *and* D *are Noetherian local rings and* $\psi : \mathrm{C} \to \mathrm{D}$ *is a local homomorphism such that* $g = {}^a\psi : \mathrm{Spec}(\mathrm{D}) \to \mathrm{Spec}(\mathrm{C})$ *is a* $\boldsymbol{P}$*-morphism, then* $\boldsymbol{R}'(\mathrm{C})$ *implies* $\boldsymbol{R}(\mathrm{D})$*.*

Suppose that the canonical morphism $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$ *is a* $\boldsymbol{P}'$*-morphism, where* $\boldsymbol{P}'$ *is defined from* $\boldsymbol{R}'$ *as* $\boldsymbol{P}$ *is defined from* $\boldsymbol{R}$ *in* (7.5.0). *Let* $f = {}^a\varphi : \mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$*, and suppose that*

$$\boldsymbol{P}\big(\mathrm{Spec}(\widehat{\mathrm{B}} \otimes_{\widehat{\mathrm{A}}} k), k\big)$$

is true. Then $\boldsymbol{P}(f^{-1}(x), \boldsymbol{k}(x))$ *is true for every* $x \in \mathrm{Spec}(\mathrm{A})$*; in other words,* f *is a* $\boldsymbol{P}$*-morphism.*

Proof. We proceed in two steps.

I) Reduction to the study of the local rings of the generic fibre. Apply (7.3.16.2) once more. The only difference from part I) of the proof of (7.5.1) is that A' is now only a semi-local integral ring, not necessarily local. Every maximal ideal $\mathfrak{n}'$ of B' lies above a maximal ideal $\mathfrak{m}'$ of A'. Every local ring of $\mathrm{Spec}(\mathrm{B}' \otimes_{\mathrm{A}'} \mathrm{K}')$ is a local ring of $\mathrm{Spec}(\mathrm{B}')$, hence of one of the $\mathrm{Spec}(\mathrm{B}'_{\mathfrak{n}'})$ over the generic point of $\mathrm{Spec}(\mathrm{A}'_{\mathfrak{m}'})$. Thus it is enough to prove that the local rings of

$$\mathrm{Spec}(\mathrm{B}'_{\mathfrak{n}'} \otimes_{\mathrm{A}'_{\mathfrak{m}'}} \mathrm{K}')$$

satisfy $\boldsymbol{R}$. By definition of $\boldsymbol{P}'$, if $\boldsymbol{P}'(\mathrm{Z}, k)$ is true then so is $\boldsymbol{P}'(\mathrm{Z} \otimes_k k', k')$ for every finite extension k' of k; the same argument as in (7.3.7) shows that the formal fibres of $\mathrm{A}'_{\mathfrak{m}'}$ satisfy $\boldsymbol{P}'$. As in part I) of the proof of (7.5.1), the rings $\mathrm{A}'_{\mathfrak{m}'}$ and $\mathrm{B}'_{\mathfrak{n}'}$ satisfy conditions (i) and (ii) above. If k' is the residue field of $\mathrm{A}'_{\mathfrak{m}'}$, then

$$\boldsymbol{P}\big(\mathrm{Spec}(\mathrm{B}'_{\mathfrak{n}'} \otimes_{\mathrm{A}'_{\mathfrak{m}'}} k'), k'\big)$$

is true: indeed, the completion of $\mathrm{A}'_{\mathfrak{m}'}$ is one of the local components of $\widehat{\mathrm{A}}' = \widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{A}'$, and the completion of $\mathrm{B}'_{\mathfrak{n}'}$ is one of the local components of

$$\widehat{\mathrm{B}}' = \widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{B}' = \widehat{\mathrm{B}} \otimes_{\mathrm{A}} \mathrm{A}' = \mathrm{B} \otimes_{\mathrm{A}} \widehat{\mathrm{A}}'.$$

Part I) of the proof of (7.5.1) therefore proves the assertion. We may consequently suppose that A is integral and need only prove, for its field of fractions K, that the local rings of $\mathrm{Spec}(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K})$ satisfy $\boldsymbol{R}$.

II) Reduction to the case where A *and* B *are complete.* Let ξ be the generic point of $\mathrm{Spec}(\mathrm{A})$ and let $y \in f^{-1}(\xi)$. We must prove $\boldsymbol{R}(\mathscr{O}_y)$. By $(\mathrm{R_{II}})$, it is enough to find a point $z \in \mathrm{Spec}(\widehat{\mathrm{B}})$ over y such that $\boldsymbol{R}(\mathscr{O}_z)$ is true. The hypothesis on $\mathrm{Spec}(\widehat{\mathrm{B}} \otimes_{\widehat{\mathrm{A}}} k)$ and (7.5.1) imply that the morphism

$$\widehat{f} : \mathrm{Spec}(\widehat{\mathrm{B}}) \longrightarrow \mathrm{Spec}(\widehat{\mathrm{A}})$$

is a $\boldsymbol{P}$-morphism. For any point $z \in \mathrm{Spec}(\widehat{\mathrm{B}})$ above y—such a point exists because $\widehat{\mathrm{B}}$ is faithfully flat over B—its image x in $\mathrm{Spec}(\widehat{\mathrm{A}})$ lies in the fibre $h^{-1}(\xi)$ of the canonical morphism $h : \mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$. By the hypothesis on A, $\boldsymbol{R}'(\mathscr{O}_x)$ is true; hence $(\mathrm{R_V})$ gives $\boldsymbol{R}(\mathscr{O}_z)$. □

Examples (7.5.3). — Consider the following properties $\boldsymbol{R}(\mathrm{C})$, where C is a Noetherian local ring:

(i) also denoted (i_n), $\mathrm{coprof}(\mathrm{C}) \leqslant n$; **IV-2** | 206
(ii) also denoted (ii_n), C satisfies (S_n);
(iii) C is a Cohen–Macaulay ring;
(iv) C is a regular ring;
(v) also denoted (v_n), C is integral, integrally closed, and satisfies (R_n);
(vi) C is integral and integrally closed;
(vii) C is integral;
(viii) C is reduced.

All these properties satisfy $(\mathrm{R_{III}})$, as was seen in (7.3.19), (i). They also satisfy $(\mathrm{R_{IV}})$. For (i), this follows from (**0**, 16.4.10), (ii); for (ii) and (iii), from (5.12.4) and the fact that a complete Noetherian local ring is catenary (5.6.4); for (iv), it is the special case (**0**, 17.1.8); for (vii), it follows from (3.4.5), and for (viii), from (3.4.6); for (vi), from (5.12.7) and the catenarity just recalled; and finally, for (v), from the preceding facts and (5.12.5).

Theorem (7.5.1) is therefore applicable when $\boldsymbol{R}$ is any one of the above properties. We have also observed in (7.3.11) that properties (i) to (viii) satisfy $(\mathrm{R_{II}})$, except that for (vii) this follows from (2.1.14). As for $(\mathrm{R_V})$, if one takes $\boldsymbol{R}' = \boldsymbol{R}$, it reduces to condition $(\mathrm{R'_I})$ of (7.3.11); properties (ii) to (vi), and (viii), satisfy this condition. For (i), one may take for $\boldsymbol{R}'$ the property of being Cohen–Macaulay, by (6.3.2). For (vii), condition $(\mathrm{R'_I})$ is not satisfied, as shown by (6.15.11), (ii) or (6.5.5), (ii). However, $(\mathrm{R_V})$ is true if $\boldsymbol{R}'$ is the property of being regular: apply (7.5.1.1) with $\mathscr{C}$ the category of all Noetherian local rings and with $\boldsymbol{R}(\mathrm{C})$ the property of being integral; this is possible by (3.4.5).

In particular, when $\boldsymbol{R}$ is any of properties (i) to (viii), the conclusion of (7.5.2) applies if the formal fibres of A are geometrically regular; see (7.8.2).

Remarks (7.5.4). — (i) It would be interesting to know whether (7.5.1) remains true without the finiteness hypothesis (i) on the residue field of B. The answer is affirmative when the residue field k of A has characteristic 0, by Hironaka's results on resolution of singularities; see (7.9.8). A particular case of the question is the following. Let A and B be complete Noetherian rings, let k be the residue field of A, and let $\varphi : \mathrm{A} \to \mathrm{B}$ be a local homomorphism making B a formally smooth A-algebra (**0**, 19.3.1); equivalently, B is flat over A and $\mathrm{B} \otimes_{\mathrm{A}} k$ is geometrically regular over k (**0**, 19.7.1). Are the fibres of ${}^a\varphi : \mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ geometrically regular? This is so when the residue field k' of B is a finite extension of k, by (7.5.1); the answer is also affirmative when k' is an extension of finite type of k. We do not know the answer when $\mathrm{B} \otimes_{\mathrm{A}} k$ is a field equal to a separable closure of k.

(ii) One could state a result analogous to (7.5.1) for a finite-type A-module M and a finite-type B-module N, concerning the properties of

$$(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N})_y \otimes_{\mathscr{O}_{\mathrm{X},x}} \boldsymbol{k}(y),$$

where y runs through Spec(B) and $x = f(y)$, deduced from properties of the same type of M_x and N_y.

(iii) Suppose $\boldsymbol{R}$ satisfies $(\mathrm{R'_I})$, $(\mathrm{R_{II}})$, $(\mathrm{R_{III}})$, and $(\mathrm{R_{IV}})$, and hypotheses (i) and (ii) of (7.5.2) hold. Then $\boldsymbol{R}(\mathrm{A})$ and

$$\boldsymbol{P}\big(\mathrm{Spec}(\widehat{\mathrm{B}} \otimes_{\widehat{\mathrm{A}}} k), k\big)$$

imply $\boldsymbol{R}(\mathrm{B})$. By (7.5.3), this holds for the properties listed there except (i) and (vii). For (vii), it is nevertheless plausible that the following question has an affirmative answer. Let $\varphi : \mathrm{A} \to \mathrm{B}$ be a local homomorphism of Noetherian local rings making B a flat A-module. Suppose A is complete, integral, and geometrically unibranch, and that $\mathrm{Spec}(\mathrm{B} \otimes_{\mathrm{A}} k)$, where k is the residue field of A, is geometrically locally integral. Is B then integral? This is known when k has characteristic 0, by Hironaka's resolution of singularities (7.9.8); it is also true when B is an A-algebra essentially of finite type (1.3.8) (see (11.3.10) and (11.3.11)), or when $\mathrm{Spec}(\mathrm{B} \otimes_{\mathrm{A}} k)$ is geometrically normal, for then (7.5.3) shows that $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ is normal and one concludes by (6.15.10). Even if the residue field of B equals that of A and B is complete, the general answer is unknown.

(iv) Consider the property $\boldsymbol{R}(\mathrm{C})$: "C is reduced, equidimensional, and satisfies (R_n)". It satisfies $(\mathrm{R_{IV}})$ by (5.12.5), but it is not known whether it satisfies $(\mathrm{R'_I})$, $(\mathrm{R_{II}})$, or $(\mathrm{R_{III}})$; the difficulty lies in verifying equidimensionality.

In what follows of this section, we will apply (7.5.1) to the study of *completed tensor products* $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ of local Noetherian rings which are algebras over a field k. **IV-2** | 207

Lemma (7.5.5). — *Let k be a field,* A, B *two local Noetherian complete rings containing k, the residue field of* A *being a finite extension of k. Let* C *be the completed tensor product* $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ (**0**, 7.7.5)*. Then:*

(i) C *is a semi-local Noetherian complete ring.*

(ii) C *is a flat* A*-module and a flat* B*-module.*

(iii) If $\mathfrak{m}$ *is the maximal ideal of* A, $\mathfrak{m}\mathrm{C}$ *is contained in the radical of* C, *and* $\mathrm{C}/\mathfrak{m}\mathrm{C}$ *is isomorphic to* $(\mathrm{A}/\mathfrak{m}) \otimes_k \mathrm{B}$.

(iv) The residue fields of the components of C *are finite extensions of the residue field of* B.

The properties (i), (iii) and the first assertion of (ii) are particular cases of (**0**, 19.7.1.2). *To prove that* C *is a flat* B*-module, note that for every* $h > 0$,

$$\mathrm{C}/\mathfrak{m}^h\mathrm{C} = (\mathrm{A}/\mathfrak{m}^h) \otimes_k \mathrm{B}$$

is a flat B*-module, since* k *is a field; it therefore suffices to apply* (**0**, 10.2.6) *to each local component of* C*. Finally, if* $\mathfrak{n}$ *is the maximal ideal of* B*, the residue fields of the local components of* C *are also those of the local components of the Artinian ring* $(\mathrm{A}/\mathfrak{m}) \otimes_k (\mathrm{B}/\mathfrak{n})$*, and hence are finite extensions of* $\mathrm{B}/\mathfrak{n}$*.*

Proposition (7.5.6). — *Let* k *be a field, and let* A *and* B *be complete Noetherian local rings containing* k*, whose residue fields are finite extensions of* k*. Let* $\boldsymbol{R}$ *be a property satisfying* $(\mathrm{R}'_{\mathrm{I}})$, $(\mathrm{R}_{\mathrm{III}})$ *and* $(\mathrm{R}_{\mathrm{IV}})$*, with* $\boldsymbol{P}$ *defined from* $\boldsymbol{R}$ *as in* (7.5.0)*. Suppose that* $\boldsymbol{R}(\mathrm{A})$ *and* $\boldsymbol{P}(\mathrm{Spec}(\mathrm{B}), k)$ *hold; the latter means that, for every finite extension* k' *of* k *and each complete local ring* B_h *occurring as a component of* $\mathrm{B} \otimes_k k'$*, one has* $\boldsymbol{R}(\mathrm{B}_h)$*. Then* $\boldsymbol{R}(\mathrm{C}_j)$ *holds for every complete local component* C_j *of* $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$*.*

Proof. By (7.5.5), (iii), each homomorphism $\mathrm{A} \to \mathrm{C}_j$ is local; by (7.5.5), (ii), each C_j is flat over A; and by (7.5.5), (iv), its residue field is a finite extension of $\mathrm{A}/\mathfrak{m}$. Moreover,

$$\mathrm{C}/\mathfrak{m}\mathrm{C} = (\mathrm{A}/\mathfrak{m}) \otimes_k \mathrm{B},$$

so the definition of $\boldsymbol{P}$ and the fact that $\mathrm{A}/\mathfrak{m}$ is finite over k imply

$$\boldsymbol{P}\big(\mathrm{Spec}(\mathrm{C}_j \otimes_{\mathrm{A}} \mathrm{A}/\mathfrak{m}), \mathrm{A}/\mathfrak{m}\big).$$

Thus all the hypotheses of (7.5.1) hold for the local homomorphism $\mathrm{A} \to \mathrm{C}_j$, so $\mathrm{Spec}(\mathrm{C}_j) \to \mathrm{Spec}(\mathrm{A})$ is a $\boldsymbol{P}$-morphism. The conclusion follows from $(\mathrm{R}'_{\mathrm{I}})$ and $\boldsymbol{R}(\mathrm{A})$. ☐

Corollary (7.5.7). — *(Chevalley). — Let* k *be a perfect field (resp. algebraically closed),* A, B *two local Noetherian complete rings containing* k *and whose residue fields are finite extensions of* k*. Then, if* A *and* B *are reduced (resp. integral), the completed tensor product* $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ *is reduced (resp. integral).*

Proof. I) Suppose k is perfect and A and B are reduced. Let A_i (resp. B_j) be the quotients of A (resp. B) by its minimal prime ideals. Then A and B identify with subrings of $\prod_i \mathrm{A}_i$ and $\prod_j \mathrm{B}_j$. Since the tensor products are taken over a field, $\mathrm{A} \otimes_k \mathrm{B}$ identifies with a subring of $\prod_{i,j}(\mathrm{A}_i \otimes_k \mathrm{B}_j)$; the tensor-product topology is induced by the product of the corresponding topologies. Hence $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ embeds in $\prod_{i,j}(\mathrm{A}_i\widehat{\otimes}_k\mathrm{B}_j)$, and we are reduced to the case where A and B are integral.

Let A' and B' be their integral closures. By Nagata's finiteness theorem (**0**, 23.1.6), A' (resp. B') is a finite A-module (resp. B-module) and a complete local ring. It suffices to show that $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ embeds in $\mathrm{A}'\widehat{\otimes}_k\mathrm{B}$ and that the latter embeds in $\mathrm{A}'\widehat{\otimes}_k\mathrm{B}'$. This follows from the lemma below.

Lemma (7.5.7.1). — *Let* A, B *be two local Noetherian complete rings containing a field* k *and whose residue fields are finite extensions of* k*. Let* $\mathfrak{m}$ *be the maximal ideal of* A*. For every* A*-module* M *of finite type (equipped with the* $\mathfrak{m}$*-adic topology), the completed tensor product* $\mathrm{M}\widehat{\otimes}_k\mathrm{B}$ (**0**, 7.7.1) *identifies canonically with* $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B})$*.*

Proof. Indeed, one has a canonical isomorphism $\mathrm{M} \otimes_k \mathrm{B} \xrightarrow{\sim} \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A} \otimes_k \mathrm{B})$, hence a canonical composite homomorphism

$$\varphi : \mathrm{M} \otimes_k \mathrm{B} \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A} \otimes_k \mathrm{B}) \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B})$$

and it is immediate that this homomorphism is continuous for the tensor product topologies; furthermore, $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B})$ is separated and complete ((7.5.5) and (**0**, 7.7.8)), hence one also has by completion a continuous homomorphism

$$\widehat{\varphi} : \mathrm{M}\widehat{\otimes}_k\mathrm{B} \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B})$$

It is immediate that this homomorphism is bijective when M is free of finite type; as one has an exact sequence $\mathrm{L}' \to \mathrm{L} \to \mathrm{M} \to 0$, where L and L' are free A-modules of finite type, one deduces a commutative diagram **IV-2 | 208**

$$\begin{tikzcd}
\mathrm{L}'\widehat{\otimes}_k\mathrm{B} \arrow[r] \arrow[d] & \mathrm{L}\widehat{\otimes}_k\mathrm{B} \arrow[r] \arrow[d] & \mathrm{M}\widehat{\otimes}_k\mathrm{B} \arrow[r] \arrow[d] & 0 \\
\mathrm{L}' \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B}) \arrow[r] & \mathrm{L} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B}) \arrow[r] & \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B}) \arrow[r] & 0
\end{tikzcd}$$

where the rows are exact (the first by virtue of the definition of the completed tensor product and of (**0**, 13.2.2)); the first two vertical arrows being bijections, it follows that so is the third. □

The fact that $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ identifies with a subring of $\mathrm{A}' \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B})$ follows then from the fact that $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ is a flat A-module (7.5.5). One can thus further suppose that A and B are *integrally closed*; as moreover k is supposed *perfect*, Spec(A) and Spec(B) are *geometrically normal* over k, by virtue of (6.7.7), *b*)); one can then apply (7.5.6) taking for $\boldsymbol{R}$ the property (vi) of (7.5.3).

II) Suppose k is algebraically closed and A and B are integral. The argument in I) again reduces us to the case where A and B are integrally closed. By (7.5.6), $\mathrm{Spec}(\mathrm{A}\widehat{\otimes}_k\mathrm{B})$ is normal, so $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ is a finite product of integrally closed complete local rings. It remains only to show that it is local. If $\mathfrak{m}$ and $\mathfrak{n}$ are the maximal ideals of A and B, it suffices to show that $(\mathrm{A}/\mathfrak{m}) \otimes_k (\mathrm{B}/\mathfrak{n})$ is local. Since both residue fields are finite extensions of the algebraically closed field k, they are equal to k, and the conclusion follows. □

Remark (7.5.8). — It would be interesting to determine whether, in (7.5.7), one can replace the hypothesis that k is perfect or algebraically closed by the hypothesis that Spec(A) or Spec(B) is *geometrically integral* over k, or at least by the hypothesis that A (for example) is integral and contains a subring A_0 isomorphic to a formal power series ring $k[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$, such that the field of fractions K of A is *separable* over the field of fractions K_0 of A_0 (one can show that this condition entails that A is geometrically integral over k, and that the two properties are equivalent when $[k : k^p]$ is finite (p being the characteristic exponent of k)). Likewise, it would be desirable to develop variants of (7.5.6) and (7.5.7) where one would weaken the hypothesis of finiteness of the residue fields of A and B, by supposing for example that only one of them is a finite extension of k, the other being arbitrary.

7.6. Applications: I. Japanese local rings

Proposition (7.6.1). — *Let* A *be a reduced local Noetherian ring whose formal fibres are geometrically normal. Then the completion* $\widehat{\mathrm{A}}$ *is reduced, the integral closure* A' *of* A *in its total ring of fractions is a finite* A*-algebra (hence a semi-local Noetherian ring), and its completion* $\widehat{\mathrm{A}}'$ *is isomorphic to the integral closure of* $\widehat{\mathrm{A}}$ *in its total ring of fractions.*

Proof. **IV-2 | 209**

The formal fibres of A are *a fortiori* geometrically reduced, so the reducedness of A implies that of $\widehat{\mathrm{A}}$ by (7.3.17). Let $\mathfrak{p}_i$ be the minimal prime ideals of A and put $\mathrm{B}_i = \mathrm{A}/\mathfrak{p}_i$. The formal fibres of the local rings B_i are geometrically normal (7.3.15); hence $\widehat{\mathrm{B}}_i$ is reduced, and (**0**, 23.1.7), (i), shows that the integral closure B'_i of B_i in its field of fractions is finite over B_i, and therefore over A. Since $\mathrm{A}' = \prod_i \mathrm{B}'_i$ (**II**, 6.3.8), A' is a finite A-module.

Let $\mathfrak{m}_j$ be the maximal ideals of the semi-local ring A'. Its completion $\widehat{\mathrm{A}'}$ identifies with the product of the completions $\widehat{\mathrm{A}'_{\mathfrak{m}_j}}$. By the hypothesis and (7.3.15), the formal fibres of each $\mathrm{A}'_{\mathfrak{m}_j}$ are geometrically normal. Since A' is normal, so is each $\mathrm{A}'_{\mathfrak{m}_j}$, and (7.3.17) then shows that each $\widehat{\mathrm{A}'_{\mathfrak{m}_j}}$, and hence $\widehat{\mathrm{A}'}$, is normal.

Since A' is finite over A, one has $\widehat{\mathrm{A}'} = \mathrm{A}' \otimes_{\mathrm{A}} \widehat{\mathrm{A}}$. If R denotes the total ring of fractions of A, then

$$\widehat{\mathrm{A}} \subset \widehat{\mathrm{A}'} \subset \mathrm{R}' = \mathrm{R} \otimes_{\mathrm{A}} \widehat{\mathrm{A}}.$$

Flatness of $\widehat{\mathrm{A}}$ over A implies that every regular element of A remains regular in $\widehat{\mathrm{A}}$; hence R' embeds canonically in the total ring of fractions R'' of $\widehat{\mathrm{A}}$. Since $\widehat{\mathrm{A}'}$ is normal and finite over $\widehat{\mathrm{A}}$, it is precisely the integral closure of $\widehat{\mathrm{A}}$ in R''. □

Corollary (7.6.2). — *Under the hypotheses of* (7.6.1), *there is a bijective correspondence between the maximal ideals* $\mathfrak{m}_j$ *of* A' *(in other words, the set of closed points of* $\mathrm{Spec}(\mathrm{A}')$ *above the closed point of* A*) and the set of minimal prime ideals* $\mathfrak{q}_j$ *of* $\widehat{\mathrm{A}}$ *(in other words, the set of maximal points of* $\mathrm{Spec}(\widehat{\mathrm{A}})$*); in this correspondence, the completion* $\widehat{\mathrm{A}}'_{\mathfrak{m}_j}$ *of* $\mathrm{A}'_{\mathfrak{m}_j}$ *is isomorphic to the integral closure of* $\widehat{\mathrm{A}}/\mathfrak{q}_j$.

Proof. Indeed, one knows that the integral closure of $\widehat{\mathrm{A}}$ in its total ring of fractions is the direct product of the integral closures of the $\widehat{\mathrm{A}}/\mathfrak{q}_j$, which are complete local rings (**0**, 23.1.6). □

Corollary (7.6.3). — *Under the hypotheses of (7.6.1), for $\widehat{\mathrm{A}}$ to be integral, it is necessary and sufficient that A be unibranch; for $\widehat{\mathrm{A}}$ to be geometrically unibranch, it is necessary and sufficient that A be geometrically unibranch.*

Proof. This is a particular case of (7.6.2). □

Theorem (7.6.4). — *(Zariski-Nagata). — Let A be a semi-local Noetherian ring. The following conditions are equivalent:*

a) *For every finite reduced A-algebra C, the completion $\widehat{\mathrm{C}}$ is a reduced ring.*
a′) *For every integral quotient ring B of A, with field of fractions K, the completion $\widehat{\mathrm{B}}$ is reduced, and the fields L_i of the total ring of fractions of $\widehat{\mathrm{B}}$ are separable extensions of K.*
a″) *The formal fibres are geometrically reduced (in other words, for every integral quotient ring B of A, with field of fractions K, the K-algebra $\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K}$ is separable (4.6.2)).*
b) *Every integral quotient ring B of A is a Japanese ring.*

Proof. **IV-2** | 210

To show that a) implies a'), it suffices to verify that the L_i are separable extensions of K, or equivalently (4.6.1) that for every finite extension K' of K, the ring $\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K}'$ is reduced. The field K' is generated by finitely many elements integral over B; these generate a finite B-subalgebra B' of K' whose field of fractions is K'. One has

$$\widehat{\mathrm{B}'} = \widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{B}'$$

(**0**, 7.3.3), and therefore, by associativity of the tensor product,

$$\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K}' = \widehat{\mathrm{B}'} \otimes_{\mathrm{B}'} \mathrm{K}'.$$

Now B' is a finite integral A-algebra, so $\widehat{\mathrm{B}'}$ is reduced by a). Since $\widehat{\mathrm{B}'}$ is flat over B', the nonzero elements of B' are $\widehat{\mathrm{B}'}$-regular; hence $\widehat{\mathrm{B}'} \otimes_{\mathrm{B}'} \mathrm{K}'$ identifies with a subring of the total ring of fractions of $\widehat{\mathrm{B}'}$, and is therefore reduced.

To see that a') implies a''), consider any point x of $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, and let Y be the integral closed subprescheme of X having $\overline{\{x\}}$ as its underlying space. Then $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$, where B is an integral quotient ring of A, and $\mathrm{Spec}(\widehat{\mathrm{B}}) = \mathrm{Y} \times_{\mathrm{X}} \mathrm{Spec}(\widehat{\mathrm{A}})$. Thus the formal fibre of A at x is the same as that of B at its generic point, namely $\mathrm{Spec}(\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K})$. Its local rings are those of $\mathrm{Spec}(\widehat{\mathrm{B}})$ at the points of this fibre, so the hypothesis implies that $\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K}$ is reduced; the remaining condition in a') then says that it is a separable K-algebra (4.6.1).

The condition a'') implies a), for it follows from (7.3.15) that the formal fibres of C are geometrically reduced, and if C is reduced, it follows hence that so is $\widehat{\mathrm{C}}$ (particular case of (7.3.17)).

The fact that a') implies b) is a particular case of (**0**, 23.1.7). It remains to prove that b) implies a). If C is a finite A-algebra and $\mathfrak{q}$ is a prime ideal of C, then the inverse image $\mathfrak{p}$ of $\mathfrak{q}$ in A is prime and $\mathrm{C}/\mathfrak{q}$ is a finite $(\mathrm{A}/\mathfrak{p})$-algebra. Thus b) implies that every integral quotient ring of C is Japanese. We are therefore reduced to proving that, under b), *the completion of every integral quotient ring of* A *is reduced*.

We argue by induction on $n = \dim(\mathrm{A})$, the assertion being trivial for $n = 0$. Replacing A by the quotients by its minimal prime ideals $\mathfrak{p}_i$, we may restrict to the case where A is integral. For every nonzero prime ideal $\mathfrak{p}$, the induction hypothesis already shows that the completion of $\mathrm{A}/\mathfrak{p}$ is reduced; it therefore suffices to prove that $\widehat{\mathrm{A}}$ is reduced.

Moreover, the integral closure A' of A is, by hypothesis, a finite A-module, hence a semi-local Noetherian ring, and $\widehat{\mathrm{A}}$ identifies with a subring of $\widehat{\mathrm{A}'}$ (**0**, 7.3.3). It is therefore enough to prove that $\widehat{\mathrm{A}'}$ is reduced. As observed above, b) also holds for A', and $\dim(\mathrm{A}') = n$ (**0**, 16.1.5); hence we may suppose that A is *integrally closed*. Let $t \neq 0$ belong to the Jacobson radical of A, and let $\mathfrak{q}_j$ $(1 \leqslant j \leqslant r)$ be the prime ideals minimal among those containing $t\mathrm{A}$; then:

(i) t is regular.
(ii) $\mathrm{A}/t\mathrm{A}$ has no immersed associated prime ideals.
(iii) The $\mathrm{A}_{\mathfrak{q}_j}$ are discrete valuation rings.
(iv) The completions of the $\mathrm{A}/\mathfrak{q}_j$ are reduced.

Indeed, (i) is trivial since A is integral and $t \neq 0$. As A is integrally closed, A/tA satisfies (S_1), that is to say (5.7.7) is without immersed associated prime ideals (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 4, prop. 8). Since A is integrally
closed, the $A_{\mathfrak{q}_j}$ are, and one knows (*loc. cit.*) that these rings are of dimension 1, hence they are discrete valuation rings, whence (iii). Finally, (iv) follows from the hypothesis of induction. The proof will be completed if we prove the **IV-2 | 211**

Lemma (7.6.4.1). — *(Zariski). — Let* A *be a semi-local Noetherian ring, t an element of its radical satisfying the conditions (i) to (iv) above; then* $\widehat{A}$ *is reduced.*

Proof. Set for simplicity $A' = \widehat{A}$, and consider t as an element of A'; one has $A'/tA' = (A/tA) \otimes_A A'$, and as A' is a flat A-module, it follows from (3.3.1) that the prime ideals of A' associated to the A'-module A'/tA' are the prime ideals $\mathfrak{q}'_{jh}$ such that $\mathfrak{q}'_{jh}$ lies above $\mathfrak{q}_j$ and is associated to the k_j-module $(A'/tA') \otimes_A k_j$, where k_j is the field of fractions of $A/\mathfrak{q}_j$. Now, $\mathrm{Spec}((A'/tA') \otimes_A k_j)$ is the formal fibre of A at the point $\mathfrak{q}_j$, or equivalently that of $A/\mathfrak{q}_j$ at its generic point. By (iv), the completion $A'/\mathfrak{q}_j A'$ of $A/\mathfrak{q}_j$ is reduced; hence so is its formal fibre at the generic point. This formal fibre therefore has no embedded associated prime, and its irreducible components are fields (3.2.1). It follows from (3.3.3) and (ii) that $\mathfrak{q}_j A'_{\mathfrak{q}'_{jh}}$ is the maximal ideal of $A'_{\mathfrak{q}'_{jh}}$ for every h. Since $A_{\mathfrak{q}_j}$ is a discrete valuation ring, its maximal ideal is principal; therefore the maximal ideal of the Noetherian local ring $A'_{\mathfrak{q}'_{jh}}$ is principal, and this ring is a discrete valuation ring. We have thus verified the relevant hypotheses for the *complete semi-local ring* A'. It therefore suffices to show that if A is complete and satisfies (i) to (iii), then A is reduced.

The hypotheses (i) and (ii) imply that, if $\varphi : A \to \prod_{j=1}^{r} A_{\mathfrak{q}_j}$ is the canonical homomorphism, then for every $m \geqslant 0$ one has $t^m A = \varphi^{-1}(\varphi(t^m A))$ (3.4.9). Since the image of tA in each discrete valuation ring $A_{\mathfrak{q}_j}$ is a power of its maximal ideal, the tA-adic topology on A is the inverse image under φ of the product topology on $\prod_{j=1}^{r} A_{\mathfrak{q}_j}$. As t belongs to the Jacobson radical of A, the tA-adic topology is separated, hence φ is injective. By (iii), the rings $A_{\mathfrak{q}_j}$ are integral; their product is reduced, and therefore so is A. □

□

Corollary (7.6.5). — *Let* A *be a semi-local Noetherian ring satisfying the equivalent conditions of* (7.6.4)*; every semi-local* A*-algebra essentially of finite type also satisfies the conditions of* (7.6.4)*.*

Proof. The condition $a'')$ of (7.6.4) means indeed that A is a $\boldsymbol{P}$-ring, where $\boldsymbol{P}(Z, k)$ is the property "Z is geometrically reduced over k"; the corollary is then a consequence of the general theorem (7.4.4). □

Corollary (7.6.6). — *Let* A *be a semi-local Noetherian integral ring of dimension* 1, K *its field of fractions. For* A *to be a Japanese ring, it is necessary and sufficient that the completion* $\widehat{A}$
be reduced and that, if $\mathfrak{q}_j$ $(1 \leqslant j \leqslant n)$ *are the minimal prime ideals of* $\widehat{A}$*, the fields of fractions of the integral rings* $\widehat{A}/\mathfrak{q}_j$ *be separable extensions of* K. **IV-2 | 212**

Proof. The integral quotients of A are themselves fields, hence the condition $b)$ of (7.6.4) is equivalent to the hypothesis that A is a Japanese ring, and the hypothesis $a')$ to the conditions on $\widehat{A}$, whence the conclusion. □

Remarks (7.6.7). — (i) The equivalent conditions of (7.6.6) mean again that the formal fibres of A are geometrically regular (7.3.19), (iv)). One has already observed (*loc. cit.*) that this condition is satisfied when A is a *discrete valuation ring whose field of fractions is of characteristic* 0.

(ii) The same reasoning as in the first part of the proof of (7.6.4) shows that for a semi-local Noetherian ring A, the two following conditions are equivalent:

a) For every integral quotient ring B of A, the completion $\widehat{B}$ is reduced.

a′) The formal fibres of A are reduced.

Taking into account (**0**, 23.1.7), (i), these two conditions imply the following:

b) For every integral quotient ring B of A, the integral closure of B is a finite B-algebra.

When A is a universally catenary ring, one can prove that the condition $b)$ conversely implies $a)$; we will not need to use this result.

7.7. Applications: II. Universally Japanese rings

(7.7.1). Recall (**0**, 23.1.1) that one calls a ring *universally Japanese* a ring A such that every integral A-algebra *of finite type* is a Japanese ring. This amounts to saying that for every integral A-algebra B of finite type, the integral closure of B is a finite B-algebra.

Theorem (7.7.2). — *(Nagata). — Let* A *be a Noetherian ring. The following conditions are equivalent:*

a) A *is a universally Japanese ring.*
b) *Every integral quotient ring of* A *is a Japanese ring.*
c) *For every maximal ideal* $\mathfrak{m}$ *of* A*, every integral quotient ring of* $A_\mathfrak{m}$ *is a Japanese ring, and for every integral quotient ring* B *of* A*, with field of fractions* K*, every finite radicial extension* K′ *of* K *and every sub-*B*-algebra finite* B′ *of* K′*, with field of fractions* K′*, there exists* $f' \neq 0$ *in* B′ *such that* $B'_{f'}$ *is integrally closed (cf. (6.13.7)).*

Moreover, if A *satisfies these conditions, so does every ring of fractions* $S^{-1}A$ *of* A *and every* A*-algebra of finite type.*

Proof. Let us first show that $b)$ implies $c)$; every integral quotient ring of $A_\mathfrak{m}$ is a ring of fractions of an integral quotient ring of A, hence a Japanese ring (**0**, 23.1.1). On the other hand, every quotient B of A by one of its prime ideals is a Japanese ring (**0**, 23.1.1). One deduces that the set $\mathrm{Nor}(\mathrm{Spec}(B'))$ is open (6.13.3), hence contains a non-empty open set $D(f') = \mathrm{Spec}(B'_{f'})$.

Let us show secondly that, if A satisfies $c)$, so does every A-algebra of finite type B. In the first instance, for every prime ideal $\mathfrak{q}$ of B, $B_\mathfrak{q}$ is a local ring in $S^{-1}B$, where $S = A - \mathfrak{p}$, $\mathfrak{p}$ being the inverse image of $\mathfrak{q}$ in A; as by hypothesis every integral quotient ring of $A_\mathfrak{p}$ is a Japanese ring, so is every integral quotient ring of $B_\mathfrak{q}$, since $S^{-1}B$ is an $A_\mathfrak{p}$-algebra of finite type (7.6.5). On the other hand, if C is an integral quotient ring of B, with field of fractions K, K′ a finite radicial extension of K and C′ a sub-C-algebra finite of K′, with field of fractions K′, C′ is an A-algebra integral of finite type, and in virtue of (6.13.7), the hypothesis $c)$ on A entails that the spectrum of C′ contains a non-empty open part whose points are all normal; in other words, there is a $g' \neq 0$ in C′ such that $C'_{g'}$ is integrally closed, which proves our assertion. **IV-2** | 213

This result shows that, to prove the equivalence of $a)$, $b)$ and $c)$, it suffices to show that $c)$ implies $b)$, and even that for every integral quotient ring B of A, with field of fractions K, the integral closure of B is a B-module of finite type. On the other hand, if A satisfies $b)$, so does every integral quotient ring of A, hence is a Japanese ring, and we have seen above that A verifies the conditions $a)$, $b)$, $c)$; it follows that the same holds for every A-algebra of finite type. Moreover, if A satisfies $b)$, every integral quotient ring of $S^{-1}A$ is an integral quotient ring of A, hence is a Japanese ring by hypothesis, and as $A/\mathfrak{p}$ is by hypothesis a Japanese ring, so is $S^{-1}(A/\mathfrak{p})$ (**0**, 23.1.1). This completes the proof of the last assertion of the statement. □

Corollary (7.7.3). — *Let* A *be a Noetherian ring satisfying the following conditions:*

(i) For every maximal ideal $\mathfrak{m}$ *of* A*, the formal fibres of* $A_\mathfrak{m}$ *are geometrically reduced.*
(ii) The equivalent conditions of (6.12.4) are satisfied.

Then A *is universally Japanese.*

Proof. Indeed, it follows then from (7.6.4) that every integral quotient ring of $A_\mathfrak{m}$ is a Japanese ring; whence the conclusion, by virtue of (7.7.2). □

Corollary (7.7.4). — *A Dedekind ring whose field of fractions is of characteristic* 0 *(in particular* **Z***) is a universally Japanese ring. A local Noetherian ring whose formal fibres are geometrically reduced (in particular a local Noetherian complete ring) is universally Japanese.*

Proof. The first assertion follows from (7.7.3), (7.6.7), (i) and (6.12.6). The second follows from (7.6.4) and (7.7.2). □

7.8. Excellent rings

(7.8.1). The previous paragraphs of this section (as well as (6.11), (6.12) and (6.13)) have been devoted to the study of problems frequently encountered in the use of rings and *Noetherian* preschemes, and which can be grouped into the following types:

A) For a *local* Noetherian ring A, do the properties of A "transfer" to its *completion* $\widehat{\mathrm{A}}$? For **IV-2** | 214 example, if A is reduced (resp. integral, resp. integral and integrally closed), is the same true of $\widehat{\mathrm{A}}$? Most of these questions are related to the properties of the local *formal fibres* of A (recall that these are the fibres of the canonical morphism $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$). An exception is formed by the properties related to the notion of dimension, in particular the notion of being equidimensional; it is then the condition of *chains* and its various refinements that play the essential role.

B) For a locally Noetherian prescheme X (in particular for an affine scheme Spec(A)), does the set of $x \in \mathrm{X}$ at which the local ring $\mathscr{O}_x$ possesses a certain property (for example being integrally closed, or being a Cohen-Macaulay ring, or being regular) form an *open* set?

C) For an *integral* ring A, is the integral closure of A a finite extension of its field of fractions? Of course, this question can be reformulated for Noetherian preschemes (**II**, 6.3).

The problems of type B) or C) may also be posed for *local* rings, but it does not suffice in general that they be solved affirmatively for every local ring $\mathrm{A}_{\mathfrak{p}}$ of a ring A for them to be solved for A (cf. (6.13.6)).

Let us also emphasise that in the study of these problems, we have been systematically concerned with whether the affirmative answer is *stable* for one or the other of the two most important operations in Commutative Algebra: localisation and passage to an algebra of finite type.

The results obtained in the study of these problems lead to identifying a class of Noetherian rings whose behaviour in this regard is the best possible:

Definition (7.8.2). — A ring A is called *excellent* if it is Noetherian and satisfies the following conditions:

(i) A is universally catenary (or, which amounts to the same (5.6.3), (i)), for every prime ideal $\mathfrak{p}$ of A, $\mathrm{A}_{\mathfrak{p}}$ is universally catenary).
(ii) For every prime ideal $\mathfrak{p}$ of A, the formal fibres of $\mathrm{A}_{\mathfrak{p}}$ are geometrically regular.
(iii) For every integral quotient ring B of A and every finite radicial extension K' of the field of fractions K of B, there exists a sub-B-algebra finite B' of K', containing B, having K' as field of fractions, and such that the set of regular points of $\mathrm{Spec}(\mathrm{B}')$ contains a non-empty open set.

With this terminology, the essential content of the results of § 7 and of part of § 6 is summarised as follows:

(7.8.3). (i) In the conditions (i) and (ii) of the definition (7.8.2) one may restrict to the maximal ideals $\mathfrak{p}$ in A. For a local Noetherian ring to be excellent, it is necessary and sufficient that it be universally catenary and that its formal fibres be geometrically regular. **IV-2** | 215

(ii) If A is an excellent ring, so is every ring of fractions $\mathrm{S}^{-1}\mathrm{A}$ and every A-algebra of finite type.

(iii) A local complete ring (in particular a field or a ring of formal power series over a field) is excellent. A Dedekind ring whose field of fractions is of characteristic 0 (in particular **Z**) is excellent.

(iv) Let A be an excellent ring, $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$; the set $\mathrm{Reg}(\mathrm{X})$ (resp. $\mathrm{Nor}(\mathrm{X})$, $\mathrm{U}_{\mathrm{R}_n}(\mathrm{X})$) of points where X is regular (resp. normal, resp. satisfies (R_n)) is open in X. For every coherent $\mathscr{O}_{\mathrm{X}}$-Module $\mathscr{F}$, the set $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ (resp. $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$, $\mathrm{CM}(\mathscr{F})$) of $x \in \mathrm{X}$ where $\mathrm{coprof}(\mathscr{F}_x) \leqslant n$ (resp. the points where $\mathscr{F}$ satisfies (S_n), resp. the points where $\mathscr{F}_x$ is a Cohen-Macaulay $\mathscr{O}_x$-Module) is open in X.

(v) Let A be an excellent ring, $\mathfrak{J}$ an ideal of A, and $\widehat{\mathrm{A}}$ the separated completion of A for the $\mathfrak{J}$-adic topology. Then the canonical morphism $f : \mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$ is regular (in other words, flat and with geometrically regular fibres). If we set $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X}' = \mathrm{Spec}(\widehat{\mathrm{A}})$, use the notation of (iv), and put $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{X}}} \mathscr{O}_{\mathrm{X}'}$, then

(7.8.3.1).

$$\mathrm{Reg}(\mathrm{X}') = f^{-1}(\mathrm{Reg}(\mathrm{X})), \quad \mathrm{Nor}(\mathrm{X}') = f^{-1}(\mathrm{Nor}(\mathrm{X})), \quad \mathrm{U}_{\mathrm{R}_n}(\mathrm{X}') = f^{-1}(\mathrm{U}_{\mathrm{R}_n}(\mathrm{X}))$$
$$\mathrm{U}_{\mathrm{C}_n}(\mathscr{F}') = f^{-1}(\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})), \quad \mathrm{U}_{\mathrm{S}_n}(\mathscr{F}') = f^{-1}(\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})), \quad \mathrm{CM}(\mathscr{F}') = f^{-1}(\mathrm{CM}(\mathscr{F}))$$

In particular, if $\mathfrak{J}$ is contained in the radical of A (for example if A is local and $\mathfrak{J}$ its maximal ideal), for A to be regular (resp. normal, resp. reduced, resp. to satisfy (R_n), resp. its codepth to be $\leqslant n$, resp. to satisfy (S_n), resp. to be a Cohen-Macaulay ring), it is necessary and sufficient that the same hold for $\widehat{A}$; in particular, for A to have no immersed associated primes, it is necessary and sufficient that the same hold for $\widehat{A}$.

(vi) An excellent ring is universally Japanese; in particular, if A is integral, its integral closure in every finite extension of its field of fractions is a finite A-algebra.

(vii) Let A be a reduced excellent local ring. Then the completion $\widehat{A}$ is reduced, the integral closure A' of A in its total ring of fractions is a finite A-algebra (hence a semi-local Noetherian ring), and the integral closure of $\widehat{A}$ in its total ring of fractions is isomorphic to the completion $\widehat{A}'$ of A'. Furthermore, there is a bijective canonical correspondence between the set of maximal points of $\mathrm{Spec}(\widehat{A})$ (in other words, the set of prime minimal ideals of $\widehat{A}$) and the set of closed points of $\mathrm{Spec}(A')$ (in other words, the set of maximal ideals of A'). For the local ring A to be unibranch, it is necessary and sufficient that $\widehat{A}$ be integral; for A to be geometrically unibranch, it is necessary and sufficient that $\widehat{A}$ be.

(viii) If A is an excellent integral ring, the integral closure of A is the intersection of the integral closures of the $A_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1.

(ix) Let A be an excellent ring, $X = \mathrm{Spec}(A)$, Z a closed part of X, $U = X - Z$, $i : U \to X$ the canonical injection, $\mathscr{F}$ an $\mathscr{O}_U$-Module coherent. For the $\mathscr{O}_X$-Module $i_*(\mathscr{F})$ to be coherent, it is necessary and sufficient that, for every $x \in \mathrm{Ass}(\mathscr{F})$, one has $\mathrm{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geqslant 2$. In particular, if A is integral and $\mathscr{F}$ torsion-free, for $i_*(\mathscr{F})$ to be coherent, it is necessary and sufficient that $\mathrm{codim}(Z, X) \geqslant 2$.

(x) Let A be an excellent local ring. For every integral quotient ring B of A, $\widehat{B}$ is equidimensional. **IV-2** | 216
For A to be equidimensional, it is necessary and sufficient that $\widehat{A}$ be.

Proof. Most of these results have already been proved.

(i) The first assertion follows from (5.6.3), (i) and (7.4.4), (i); the second from (7.4.4), (7.3.18), (6.12.7) and (6.12.4).

Assertion (ii) follows from (5.6.1), (5.6.3), (i), (7.4.4) and (6.12.4).

(iii) The first assertion has been seen in (5.6.4) and (7.4.4), taking into account (i). The second has been seen in (5.6.4) and (7.3.19), (iv), taking into account (i).

Assertion (iv) follows from (6.12.4), (6.13.5), (6.12.9) and (6.11.8) (taking into account (ii)).

The first assertion of (v) is a consequence of (7.4.6); the relations follow, taking into account respectively (6.5.1), (6.5.4), (6.5.3), (6.3.5) and (6.4.2). The last assertion of (v) is a particular case corresponding to the property (S_n) for $n = 1$ (5.7.5).

Assertion (vi) follows from (7.7.3), and assertion (vii) from (7.6.1), (7.6.2) and (7.6.3).

For assertion (viii), let us note that the ring $A^{(1)}$, intersection of the $A_{\mathfrak{p}}$ for all prime ideals $\mathfrak{p}$ of height 1 of A, is a finite A-algebra, as it follows from (vi), from (7.8.2), (i) and (5.11.2). As the integral closure A' of A is a finite A-algebra, the prime ideals of height 1 of A' are exactly those which lie above prime ideals of height 1 of A (5.10.17), (iv); the conclusion follows hence from (**0**, 23.2.9).

The first assertion of (ix) is a consequence of (vi) and (5.11.4); the second is a particular case, since, when $\mathscr{F}$ is torsion-free, $\mathrm{Ass}(\mathscr{F})$ is reduced to the generic point of $\mathrm{Spec}(A)$. Finally, to prove (x), it suffices to observe that B is an excellent local ring by (ii). Since B is universally catenary and universally Japanese, the ring $B^{(1)}$ is a finite B-algebra by (vi) and (5.11.2); this shows that B is strictly formally catenary by (7.2.5), *c*)), and completes the proof of (x).[12] □

Remarks (7.8.4). — (i) If one considers only the Noetherian rings A that satisfy the conditions (ii) and (iii) of the definition (7.8.2), one verifies by inspection of the preceding proofs that the assertions (ii), (iii), (iv), (v), (vi) and (vii) of (7.8.3) are still valid, replacing "excellent" by "satisfying (7.8.2), (ii) and (iii)".

(ii) The catenary local ring A constructed in (5.6.11) satisfies the conditions (ii) and (iii) (but not condition (i)) of the definition (7.8.2). Indeed, the formal fibre at the maximal ideal $\mathfrak{n}C_{\mathfrak{n}}$ of A is trivially geometrically regular, and the formal fibre at every other prime ideal $\mathfrak{p}$ of A coincides with

[12]The printed text has A in this last clause; the argument and the first assertion of (x) require B.

that of the ring E at $\mathfrak{p}$. Now E is a V-algebra of finite type, and V is a discrete valuation ring, hence is excellent when its residue field k_0 has characteristic 0 by (7.8.3), (iii). Consequently E is excellent by (7.8.3), (ii), which proves that A satisfies condition (7.8.2), (ii).

On the other hand, the complement of the closed point in $\mathrm{Spec}(A)$ is isomorphic to an open subset of $\mathrm{Spec}(E)$, one of whose affine open subsets is the spectrum of an excellent ring and therefore satisfies condition (6.12.4), *a*)) by (7.8.3), (v). It follows that A itself satisfies condition (6.12.4), *a*)), hence also (6.12.4), *c*)), which is precisely condition (7.8.2), (iii). Nevertheless A does not satisfy condition (7.8.2), (i), since we saw in (5.6.11) that it is not universally catenary. **IV-2** | 217

(iii) One can replace the condition (i) of the definition (7.8.2) by the following (weaker in appearance, taking into account (5.6.10)):

(i′) *For every minimal prime ideal $\mathfrak{p}_i$ of* A $(1 \leqslant i \leqslant r)$, *let* A'_i *be the integral closure of the integral ring* $A_i = A/\mathfrak{p}_i$; *then, for every maximal ideal* $\mathfrak{m}'$ *of* A'_i, *denoting by* $\mathfrak{m}$ *the inverse image of* $\mathfrak{m}'$ *in* A,

(7.8.4.1).
$$\dim(A_{\mathfrak{m}}) = \dim(A'_{i,\mathfrak{m}'}).$$

Indeed, Remark (i) shows that, under only the hypotheses (ii) and (iii) of (7.8.2), the analogues of the properties (ii), (v), (vi) and (vii) of (7.8.3) remain valid. It follows first from (vi) and (ii) that the A'_i satisfy (7.8.2), (ii) and (iii); then (v) shows that the completions $((A'_i)_{\mathfrak{m}'})^\wedge$ are integral (and integrally closed). Let now $\mathfrak{m}$ be an arbitrary maximal ideal of A. For every i such that $\mathfrak{p}_i \subset \mathfrak{m}$, the integral closure of $A_{\mathfrak{m}}/\mathfrak{p}_i A_{\mathfrak{m}}$ is a finite $(A_{\mathfrak{m}}/\mathfrak{p}_i A_{\mathfrak{m}})$-algebra by (vi), hence is semi-local, and its local components are of the form $(A'_i)_{\mathfrak{m}'}$, where $\mathfrak{m}'$ is a prime ideal (necessarily maximal) of A'_i above $\mathfrak{m}/\mathfrak{p}_i$. It then follows from (vii), from what precedes, and from (7.8.4.1) that the quotients of the completion $(A_{\mathfrak{m}}/\mathfrak{p}_i A_{\mathfrak{m}})^\wedge$ by its minimal prime ideals all have the same dimension, equal to $\dim(A_{\mathfrak{m}})$. Consequently, by (7.1.9) and (7.1.8), *b*)), $A_{\mathfrak{m}}$ is formally catenary, and *a fortiori* universally catenary by (7.1.11). The same is therefore true of $A_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A by (5.6.3), (i), hence also of A.

In particular, if A is normal, or more generally if $A_{\mathfrak{m}}$ is unibranch for every maximal ideal $\mathfrak{m}$ of A, the condition (i′) is always satisfied, since $\mathfrak{m}$ can contain only a single prime ideal $\mathfrak{p}_i$ (**0**, 16.1.5); one can then omit (i) from the definition (7.8.2). More particularly, if A is a Noetherian ring, *local unibranch*, it is excellent if and only if its formal fibres are geometrically regular.

(iv) Recall that a quotient ring of a local Cohen–Macaulay ring (and *a fortiori* a quotient ring of a regular local ring) also satisfies the properties (7.8.3), (ix) and (x) by virtue of (7.2.7); but the interest of quotient rings of regular rings lies above all in their cohomological properties (chap. III, 3e Partie).

(v) We will see further on (§ 18) that if k is a field with a complete non-discrete valuation, the ring $k\{\{T_1, \ldots, T_n\}\}$ of *convergent power series* (in a neighbourhood of 0 in k^n) is an *excellent ring*.

(7.8.5). One says that a locally Noetherian prescheme X is *excellent* if for some covering (U_α) of X by affine opens, each of the rings A_α of the U_α is excellent; this property is then true for *every* covering (U_α) of X by affine opens.

Proposition (7.8.6). — *Let* X *be an excellent prescheme.*

(i) If $f : X' \to X$ is a morphism locally of finite type, X′ *is excellent.*

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(ii) If X *is reduced, its normalised* X′ *(**II**, 6.3.8) is finite over* X.

(iii) The sets $\mathrm{Reg}(X)$, $\mathrm{Nor}(X)$, $U_{R_n}(X)$, *as well as* $U_{C_n}(\mathscr{F})$, $U_{S_n}(\mathscr{F})$ *and* $\mathrm{CM}(\mathscr{F})$ *for every coherent* $\mathscr{O}_X$*-Module* $\mathscr{F}$, *are open in* X.

Proof. This follows immediately from (7.8.3), (ii), (vi) and (iv). Let us also note that (7.8.3), (ix) is also valid without modification of the statement when X is an excellent prescheme. □

7.9. Excellent rings and resolution of singularities

(7.9.1). Given a *locally Noetherian reduced* prescheme X, one calls a *resolving morphism* for X a morphism $f : \mathrm{X}' \to \mathrm{X}$ such that X′ is *regular* and that f is *proper* and *birational*. When such a morphism exists, one says that one can *resolve the singularities* of X (or more simply that one can *resolve* X). For example, if A is a *Japanese* ring of dimension 1, one can resolve $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ since the morphism $\mathrm{X}' \to \mathrm{X}$ (where X′ is the *normalised* of X) is finite (hence proper) and birational, and the local rings of X′ are integrally closed rings of dimension 1, hence discrete valuation rings, hence regular.

(7.9.2). It is clear that if one can resolve X, one can also resolve every *open* of X and every local prescheme $\mathrm{Spec}(\mathscr{O}_x)$ of X (**II**, 5.4.2). On the other hand, if one can resolve X, it is clear that there exists in X an open subset *everywhere* dense contained in $\mathrm{Reg}(\mathrm{X})$. These remarks show immediately that if, for every closed *integral* sub-prescheme Y of X and every Y-prescheme Y′ *integral, finite* and *radicial* over Y, one can resolve Y′, then the affine opens of X *satisfy condition* (iii) of (7.8.2).

On the other hand, for local schemes, one has the following proposition:

Proposition (7.9.3). — *Let* A *be a local Noetherian reduced ring, and suppose that one can resolve* $\mathrm{Spec}(\mathrm{A})$*. Then the fibres of the canonical morphism* $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$ *at the maximal points of* $\mathrm{Spec}(\mathrm{A})$ *are regular.*

Proof. Set $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X}' = \mathrm{Spec}(\widehat{\mathrm{A}})$, and let $g : \mathrm{X}' \to \mathrm{X}$ be the canonical morphism. Let $f : \mathrm{Y} \to \mathrm{X}$ be a resolving morphism; set $\mathrm{Y}' = \mathrm{X}' \times_{\mathrm{X}} \mathrm{Y}$, and let $f' : \mathrm{Y}' \to \mathrm{X}'$ and $g' : \mathrm{Y}' \to \mathrm{Y}$ be the canonical projections, so that one has a commutative diagram

$$\begin{tikzcd} \mathrm{X}' \arrow[d, "g"'] & \mathrm{Y}' \arrow[l, "f'"'] \arrow[d, "g'"] \\ \mathrm{X} & \mathrm{Y} \arrow[l, "f"] \end{tikzcd}$$

It will suffice to prove that the prescheme Y′ is *regular*: indeed, as f is birational and of finite type, there is an open U dense in X such that the restriction $f^{-1}(\mathrm{U}) \to \mathrm{U}$ of f is an isomorphism and that $f^{-1}(\mathrm{U})$ is everywhere dense in Y (**I**, 6.5.5); the preschemes induced respectively on the open $g^{-1}(f^{-1}(\mathrm{U}))$ of X′ and the open $g'^{-1}(f^{-1}(\mathrm{U}))$ of Y′ are therefore isomorphic; this proves that $g^{-1}(\mathrm{U})$ is regular, and *a fortiori* so are the fibres of g at the maximal points of X, which are contained in U.

Let us note that the morphism f' is proper (**II**, 5.4.2), and in particular of finite type, hence Y′ **IV-2** | 219
is Noetherian, since it is thus of X′. Let a' be the closed point of X′; to see that Y′ is regular, it will suffice to prove that *for every* $y' \in f'^{-1}(a')$*, the local ring* $\mathscr{O}_{\mathrm{Y}',y'}$ *is regular*. Indeed, one will then have $f'^{-1}(a') \subset \mathrm{Reg}(\mathrm{Y}')$. But as X′ is the spectrum of a complete local ring and f' is of finite type, $\mathrm{Reg}(\mathrm{Y}')$ is *open* in Y′ (6.12.8), and as the morphism f' is *closed* and $f'^{-1}(a') \subset \mathrm{Reg}(\mathrm{Y}')$, there exists a neighbourhood V′ of a' in X′ such that $f'^{-1}(\mathrm{V}') \subset \mathrm{Reg}(\mathrm{Y}')$. As $\widehat{\mathrm{A}}$ is an integral local ring, on necessarily has $\mathrm{V}' = \mathrm{X}'$, hence $\mathrm{Reg}(\mathrm{Y}') = \mathrm{Y}'$ and the proposition will be proved.

Now, if a is the closed point of X, one has $g^{-1}(a) = \{a'\}$ and the residue fields $\boldsymbol{k}(a)$ and $\boldsymbol{k}(a')$ are isomorphic; hence $g'^{-1}(f^{-1}(a)) = f'^{-1}(a')$ and the restriction $f'^{-1}(a') \to f^{-1}(a)$ of g' is an *isomorphism* of these two fibres. Let $y = g'(y')$ and set $\mathrm{B} = \mathscr{O}_{\mathrm{X},y}$; if $\mathfrak{n}$ is the maximal ideal of B, $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \widehat{\mathrm{A}}$ has a single maximal ideal $\mathfrak{n}'$ above $\mathfrak{n}$ of B and of the maximal ideal of $\widehat{\mathrm{A}}$, and one has $\mathscr{O}_{\mathrm{Y}',y'} = \mathrm{B}'_{\mathfrak{n}'}$. To show that the local ring $\mathrm{B}'_{\mathfrak{n}'}$ is regular, it suffices to establish that its *completion* is regular (**0**, 17.1.5), that is, or by hypothesis B is regular, hence so is its completion $\widehat{\mathrm{B}}$. The conclusion will follow from the following lemma:

Lemma (7.9.3.1). — *Let* A, B *be two local Noetherian rings,* $\rho : \mathrm{A} \to \mathrm{B}$ *a local homomorphism such that the ring* $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \widehat{\mathrm{A}}$ *is Noetherian. Then, for every maximal ideal* $\mathfrak{n}'$ *of* B′ *above the maximal ideal* $\mathfrak{n}$ *of* B *and the maximal ideal of* $\widehat{\mathrm{A}}$*, the completions of* B *and of* $\mathrm{B}'_{\mathfrak{n}'}$ *are isomorphic.*

Proof. Let $\mathfrak{m}$ be the maximal ideal of A and set $\mathrm{B}'' = \mathrm{B}'_{\mathfrak{n}'}$. For every integer $h > 0$, one has $\mathrm{B}'/\mathfrak{m}^h\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} (\widehat{\mathrm{A}}/\mathfrak{m}^h\widehat{\mathrm{A}})$; but $\widehat{\mathrm{A}}/\mathfrak{m}^h\widehat{\mathrm{A}}$ is isomorphic to $\mathrm{A}/\mathfrak{m}^h$, hence $\mathrm{B}'/\mathfrak{m}^h\mathrm{B}'$ is isomorphic to $\mathrm{B}/\mathfrak{m}^h\mathrm{B}$, and in particular it is a local ring whose maximal ideal is $\mathfrak{n}'/\mathfrak{m}^h\mathrm{B}'$; consequently $\mathrm{B}''/\mathfrak{m}^h\mathrm{B}''$, isomorphic to $(\mathrm{B}'/\mathfrak{m}^h\mathrm{B}')_{\mathfrak{n}'}$, is B-isomorphic to $\mathrm{B}'/\mathfrak{m}^h\mathrm{B}'$, and finally isomorphic to $\mathrm{B}/\mathfrak{m}^h\mathrm{B}$; in particular, the maximal ideal of B'' is equal to $\mathfrak{n}\mathrm{B}''$, and $\mathrm{B}''/\mathfrak{n}^h\mathrm{B}''$ is isomorphic to $\mathrm{B}/\mathfrak{n}^h$ from what precedes. As $\widehat{\mathrm{B}}'' = \varprojlim_h (\mathrm{B}''/\mathfrak{n}^h\mathrm{B}'')$, the lemma is proved, as was (7.9.3). ☐

☐

Corollary (7.9.4). — *Let* A *be a local Noetherian ring.*

(i) If, for every integral quotient ring B *of* A*, one can resolve* Spec(B)*, then the formal fibres of* A *are regular.*

(ii) Suppose that, for every integral quotient ring B *of* A*, and every integral ring* B′ *containing* B*, which is a finite* B*-algebra and whose field of fractions is radicial over that of* B*, one can resolve* Spec(B′)*; then the formal fibres of* A *are geometrically regular.*

Proof. Every formal fibre of A being a formal fibre of a ring at the generic point of its spectrum, it is clear that (i) is an immediate consequence of (7.9.3), and (ii) follows from (i) and (6.7.7). ☐

Proposition (7.9.5). — *Let* X *be a locally Noetherian prescheme, such that, for every integral closed subprescheme* Y*, finite and radicial over* X*, one can resolve* Y*. Then the rings of the affine opens of* X *satisfy the conditions (ii) and (iii) of* (7.8.2)*; if moreover* X *is universally catenary* (5.6.4)*,*
in particular if X *is* locally immersible in a regular prescheme (5.6.4)*)*, X *is an* excellent prescheme. **IV-2** | 220

Proof. This follows immediately from (7.9.4) and (7.9.2). ☐

Remark (7.9.6). — It is possible that the converse of (7.9.5) be true, that is to say that X is reduced and that the rings of the affine opens of X satisfy the conditions (ii) and (iii) of (7.8.2) implies the possibility of resolving X (and consequently also the possibility of resolving every reduced prescheme locally of finite type over X, by virtue of (7.4.4) and (6.12.4)). This is in any case true when one limits oneself to reduced Noetherian preschemes whose residue fields are of *characteristic* 0, as shown by the recent results of Hironaka [35] (the latter states his results under hypotheses that are too restrictive, the reasoning being in fact valid under the only hypotheses (ii) and (iii) of (7.8.2) for the rings of the affine opens of X, combined with the fact that the residue fields are of characteristic 0). On the contrary, in the general case, it is known so far only how to resolve algebraic schemes of dimension 2 over a perfect field [36]. Given the importance that the resolution of singularities is acquiring in the topological study of schemes (particularly concerning their homological and homotopical properties), this gives a particular interest to the category of excellent rings and excellent preschemes.

One may also note in this regard that the less delicate part of the reasoning of Hironaka (*loc. cit.*) shows independently of all hypothesis on the characteristics of the residue fields, and using only the properties (ii) and (iii) of (7.8.2) for the local rings of X, the problem of resolution of singularities of a prescheme X reduces to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, A being a local Noetherian integral ring and *complete*. Consequently, if subsequent results were to disprove the more advanced conjecture above, and were to lead to formulating more restrictive conditions on the local rings of X, these could only be through the intermediary of restrictive conditions imposed on the local *complete* rings (probably conditions on their residue fields, perhaps the condition $[k : k^p] < +\infty$ (p being the characteristic exponent of k)).

(7.9.7). Consider on the one hand a full subcategory $\mathscr{C}$ of the category of locally Noetherian preschemes, and on the other hand a property $\boldsymbol{R}(\mathrm{A})$, subject to the following conditions:

1° For every $\mathrm{X} \in \mathscr{C}$, every prescheme locally of finite type over X belongs to $\mathscr{C}$. For every Noetherian ring A such that $\mathrm{Spec}(\mathrm{A}) \in \mathscr{C}$ and every multiplicative subset S of A, one has $\mathrm{Spec}(\mathrm{S}^{-1}\mathrm{A}) \in \mathscr{C}$.

2° For every $\mathrm{X} \in \mathscr{C}$, the set $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X})$ of $x \in \mathrm{X}$ such that $\boldsymbol{R}(\mathscr{O}_{\mathrm{X},x})$ is true is open in X.

3° For every local Noetherian ring A such that $\mathrm{Spec}(\mathrm{A}) \in \mathscr{C}$ and every regular element t of the maximal ideal of A, $\boldsymbol{R}(\mathrm{A}/t\mathrm{A})$ implies $\boldsymbol{R}(\mathrm{A})$.

Let us denote as in (7.5.0) by $\boldsymbol{P}(\mathrm{Z}, k)$, for a field k and a k-prescheme $\mathrm{Z} \in \mathscr{C}$, the following property:

"For every finite extension k' of k, all the local rings of $\mathrm{Z} \otimes_k k'$ satisfy the property $\boldsymbol{R}$."

We will further suppose that $\boldsymbol{R}$ satisfies the following condition: **IV-2** | 221

4° If $\boldsymbol{P}(\mathrm{Z}, k)$ is true, then, for every extension *of finite type* k'' of k, all the local rings of $\mathrm{Z} \otimes_k k''$ satisfy the property $\boldsymbol{R}$.

Let us note that these conditions are satisfied when one takes for $\mathscr{C}$ the category of *excellent* preschemes and for $\boldsymbol{R}$ one of the properties (i) to (viii) of (7.5.3); for condition 1°, this follows from (7.8.3), (iii), and for conditions 2° and 3°, from the arguments of (7.5.3), taking into account that every excellent ring is catenary. Finally, condition 4° follows, for (i), (ii), and (iii), from (6.7.1); for (iv), (v), and (vi), from (6.7.7); and for (viii), from (4.6.1). As concerns property (vii) of (7.5.3), the corresponding property $\boldsymbol{P}(\mathrm{Z}, k)$ implies that Z is locally integral (being locally Noetherian) and that each of the integral subpreschemes whose union is Z is geometrically integral, by (4.5.9) and (4.6.1); condition 4° follows in this case as well.

With these notations and hypotheses:

Proposition (7.9.8). — *Let* A *be a local Noetherian ring,* k *its residue field, and* B *a local Noetherian ring such that* $\mathrm{Spec}(\mathrm{B}) \in \mathscr{C}$*. Let* $\varphi : \mathrm{A} \to \mathrm{B}$ *be a local homomorphism making* B *a flat* A*-module; set* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$*,* $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$*, and let* $f : \mathrm{X} \to \mathrm{Y}$ *be the morphism corresponding to* φ*. Suppose that:*

1° the property $\boldsymbol{P}(\mathrm{B} \otimes_{\mathrm{A}} k, k)$ *is true;*

2° for every finite morphism $\mathrm{Y}_1 \to \mathrm{Y}$*, one can resolve* $(\mathrm{Y}_1)_{\mathrm{red}}$*.*

Then, for every $y \in \mathrm{Y}$*, the property* $\boldsymbol{P}(f^{-1}(y), \boldsymbol{k}(y))$ *is true.*

Proof. Let us note that condition 2° of (7.9.8) is still satisfied when one replaces Y by the spectrum of a local ring in a maximal ideal of a finite A-algebra (7.9.2); on the other hand, by virtue of condition 1° of (7.9.7), the spectrum of a ring of fractions of $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$ also belongs to $\mathscr{C}$. The lemma (7.3.16.2) shows then (as in part I) of the reasoning of (7.5.2)) that it suffices to prove that, if Y is integral and if y is the generic point of Y, the local rings of the fibre $f^{-1}(y)$ satisfy $\boldsymbol{R}$.

This being so, there exists by hypothesis a regular prescheme Y' and a proper birational morphism $g : \mathrm{Y}' \to \mathrm{Y}$. Since Y' is Noetherian and locally integral, the fact that g is birational implies that Y' is integral. Put $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, so that one has a commutative diagram

(7.9.8.1).

$$\begin{tikzcd} \mathrm{X} & \mathrm{X}' \arrow[l, "g'"'] \arrow[d, "f'"] \\ \mathrm{Y} \arrow[from=u, "f"'] & \mathrm{Y}' \arrow[l, "g"] \end{tikzcd}$$

where f' and g' are the canonical projections. Taking into account condition 2° of (7.9.7), the same reasoning as at the beginning of (7.9.3) shows that it suffices to prove

that $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X}') = \mathrm{X}'$. If a denotes the closed point of X, it is enough to show that $g'^{-1}(a) \subset \mathrm{U}_{\boldsymbol{R}}(\mathrm{X}')$. **IV-2** | 222

Let $x' \in g'^{-1}(a)$, and set $y' = f'(x')$, so that $b = g(y')$ is the closed point of Y. We have

$$f'^{-1}(y') = f^{-1}(b) \otimes_{\boldsymbol{k}(b)} \boldsymbol{k}(y').$$

Since $\boldsymbol{k}(y')$ is an extension of finite type of k (because g is proper, hence of finite type), the hypothesis that $\boldsymbol{P}(f^{-1}(b), k)$ is true implies, by condition 4° of (7.9.7), that $\boldsymbol{P}(f'^{-1}(y'), \boldsymbol{k}(y'))$ is true. Moreover, $\mathscr{O}_{\mathrm{Y}',y'}$ is regular and $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y}',y'}) \in \mathscr{C}$ by condition 1° of (7.9.7); Lemma (7.5.1.1) therefore proves that $\boldsymbol{R}(\mathscr{O}_{\mathrm{X}',x'})$ is true, completing the proof. □

Corollary (7.9.9). — *Let* A, B *be two local Noetherian rings,* $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism making* B *a flat* A*-module. Suppose that* A *is integral and geometrically unibranch, and that for every* A*-algebra finite integral* A' *one can resolve* $\mathrm{Spec}(\mathrm{A}')$ *(this will be the case if the formal fibres of* A *are geometrically regular and the residue field of* A *is of characteristic* 0*, by virtue of the results of Hironaka [35]). Let* k *be the residue field of* A *and suppose that* $\mathrm{Spec}(\mathrm{B} \otimes_{\mathrm{A}} k)$ *is geometrically punctually integral* (4.6.9)*. Then* B *is integral.*

Proof. We apply (7.9.8), taking for $\mathscr{C}$ the category of all locally Noetherian preschemes and for $\boldsymbol{R}$ the property of being integral. The arguments of (7.5.3) and (7.9.7) show that conditions 1°, 2°, 3° and 4° of (7.9.7) are satisfied in this case. The hypothesis on $\mathrm{B} \otimes_{\mathrm{A}} k$ and Proposition (7.9.8) therefore show (with the notation of (7.9.8)) that the fibres $f^{-1}(y)$ are geometrically punctually integral for $y \in \mathrm{Y}$. In particular the $f^{-1}(y)$ are reduced; since A is integral, and hence reduced, it follows from (3.3.5) that B is reduced. It remains to prove that $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$ is *irreducible*. Now, the proof of

(7.9.8) shows that X' is locally integral (being locally Noetherian). Since $g' : X' \to X$ is surjective, it is therefore enough to show that X' is irreducible, or, since X' is locally integral, that X' is *connected*. For this it is enough to prove that the fibre $g'^{-1}(a)$ is connected. Indeed, if X' were the sum of two non-empty open preschemes X'_1 and X'_2, then, for example, $g'^{-1}(a) \cap X'_2 = \varnothing$; but the restriction $X'_2 \to X$ of g' is proper (since X'_2 is also closed in X'), so $g'(X'_2)$ would be a non-empty closed subset of X not containing the closed point a, a contradiction. Finally,

$$g'^{-1}(a) = g^{-1}(b) \otimes_{\boldsymbol{k}(b)} \boldsymbol{k}(a).$$

The morphism $g : Y' \to Y$ is proper and birational; hence, in its Stein factorisation $Y' \xrightarrow{v} Y'' \xrightarrow{u} Y$ (**III**, 4.3.3), the finite morphism u is also birational, and consequently Y'' is integral. Since A is geometrically unibranch, it follows from (**III**, 4.3.4) that $g^{-1}(b)$ is geometrically connected, completing the proof. □

Remarks (7.9.10). — (i) We will show further on (18.8.11) that for a local Noetherian reduced ring B to be geometrically unibranch, it is necessary and sufficient that for every etale morphism $h : X' \to X = \operatorname{Spec}(B)$ and every point $x' \in X'$ above the closed point of X, $\mathscr{O}_{X',x'}$ is integral. One deduces that if, in (7.9.9) one supposes, not only that $\operatorname{Spec}(B \otimes_A k)$ is geometrically punctually integral, but also geometrically unibranch, then one can conclude that B is *geometrically unibranch*.

(ii) Let X, Y be two excellent preschemes, $f : X \to Y$ a flat morphism, and suppose that for every finite morphism $Y_1 \to Y$, one can resolve $(Y_1)_{\mathrm{red}}$. It follows from (7.9.8), taking for $\boldsymbol{R}$ the property of being *regular*, that the set U of $x \in X$ such that the fibre at the point $f(x)$ of the morphism **IV-2 | 223**

$$\operatorname{Spec}(\mathscr{O}_x \otimes_{\mathscr{O}_{f(x)}} \boldsymbol{k}(f(x))) \longrightarrow \operatorname{Spec}(\boldsymbol{k}(f(x)))$$

is *geometrically regular*, is *stable by generisation*. We do not know if this set is *open* (or, which amounts to the same (**0**, 9.2.5), if it is *constructible*) even in the particular case where Y is the spectrum of **Z** or of a polynomial ring in one indeterminate $k[T]$ over a field k, and where consequently the condition of "resolvability" imposed on Y is trivially satisfied.

§8. Projective limits of preschemes

8.1. Introduction

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(8.1.1). In this paragraph, we will systematically study the following situation. Let I be a filtered preordered increasing set, $(A_\lambda, \varphi_{\mu\lambda})$ an inductive system of rings having I for index set, $A = \varinjlim A_\lambda$ its inductive limit. For every $\alpha \in I$ and every A_α-prescheme X_α, consider the A_λ-preschemes $X_\lambda = X_\alpha \otimes_{A_\alpha} A_\lambda$ for $\lambda \geqslant \alpha$, and the A-prescheme $X = X_\alpha \otimes_{A_\alpha} A$; it is clear that the preschemes X_λ (for $\lambda \geqslant \alpha$) form a *projective system*, and we will see (8.2.5) that X is a *projective limit* of this system in the category of preschemes. We propose to find conditions on X_α or on the A_λ allowing us to prove properties of the following type: for X to have a property **P** (for example, the property of being proper over $S = \operatorname{Spec}(A)$, or irreducible, or connected, etc.), it is necessary and sufficient that there exist an index $\mu \geqslant \alpha$ such that, for every $\lambda \geqslant \mu$, X_λ has (with respect to $S_\lambda = \operatorname{Spec}(A_\lambda)$, as the case may be) the same property **P**. We will obtain analogous statements for properties of $\mathscr{O}_X$-Modules, of A-morphisms of A-preschemes, etc. We will also show (8.9.1) that the datum of an A-prescheme of finite presentation (1.6.1) is essentially equivalent to the datum of an A_λ-prescheme of finite presentation X_λ for some sufficiently large λ, X being then *isomorphic to* $X_\lambda \otimes_{A_\lambda} A$. There are also analogous statements for $\mathscr{O}_X$-Modules of finite presentation, their homomorphisms, the A-morphisms of A-preschemes of finite presentation, etc.

(8.1.2). The utility of such results will appear, for example, in the following questions:

a) Let Y be a prescheme, y a point of Y, (U_λ) the projective system of filtered decreasing open affine neighbourhoods of y in Y; if A_λ is the ring of U_λ, the A_λ form a filtered increasing inductive system, whose inductive limit A is the local ring $\mathscr{O}_y$; moreover, if we denote by $\mathfrak{p}_\lambda$ the prime ideal $\mathfrak{i}_y$ in the ring A_λ, the inductive system (A_λ) is cofinal in every inductive system $(A_\alpha)_f$, where f ranges over $A_\alpha - \mathfrak{p}_\alpha$ (for a fixed α), since the $D(f)$ form a fundamental system of neighbourhoods of y in U_α, hence in Y. The results of the present paragraph will therefore imply that the algebraic geometry of $\mathscr{O}_y$-preschemes of finite presentation (and **IV-3 | 6** the theory of Modules of finite presentation on these preschemes) is essentially equivalent to the algebraic geometry of preschemes of finite presentation on "sufficiently small" open neighbourhoods of y. Thus, the statement (8.10.5), (xii) implies that if a morphism of finite presentation $f : X \to Y$ is such that $X \times_Y \operatorname{Spec}(\mathscr{O}_y)$ is proper over $\operatorname{Spec}(\mathscr{O}_y)$, it is necessary and sufficient that there exist an open neighbourhood U of y in Y such that $f^{-1}(U)$ is proper over U.

An important particular case, classical in a certain sense, is the one where Y is integral and where y is its generic point, so that $\mathscr{O}_y$ is nothing other than *the field* $R(Y) = K$ *of rational functions on* Y. The results of the present paragraph then amount to interpreting the algebraic geometry over K in terms of algebraic geometry over "sufficiently small" non-empty open subsets of Y, that is to say, intuitively, in terms of "families" of geometric objects indexed by the points of such an open set. This point of view has moreover been commonly used for a long time, not only in Algebraic Geometry over algebraically closed fields, but also in the arithmetic study of varieties defined over a *number field* K (a finite extension of **Q**), considering the latter as the field of fractions of its ring of integers A (the "theory of *reduction modulo* $\mathfrak{p}$" ($\mathfrak{p}$ a prime ideal of A); cf. (**I**, 3.7)). The results of §§ 8 and 9 thus provide, among other things, some of the foundations of the language of "reduction modulo $\mathfrak{p}$" in arithmetic.

We note that in the example considered here, the morphisms $S_\mu \to S_\lambda$ (for $\lambda \leqslant \mu$) are the canonical *open immersions* $U_\mu \hookrightarrow U_\lambda$, and *a fortiori* are *flat* morphisms (but not faithfully flat in general), which explains the interest of the statements that appeal to such a restriction.

b) Suppose that the A_λ are *fields*, so that $A = \varinjlim A_\lambda$ is also a field. This case arises generally when one starts from geometric data over an arbitrary field K, which one considers as an extension of a field k (for example the prime subfield of K). There is then interest in considering K as an inductive limit of its sub-extensions that are *of finite type over* k, which allows us to reduce many questions to the case where K is a finite type extension of k. Using also the method sketched in *a)*, one can then reduce generally to the case of a base ring A which is an *integral algebra of finite type over* k.

We note that in this example, the morphisms $S_\mu \to S_\lambda$ are *faithfully flat*.

c) Suppose that we are interested in properties, local on Y, of preschemes of finite presentation over an arbitrary prescheme Y, which we can from now on suppose to be an affine scheme of ring A. There is then interest in considering A as an inductive limit of its subrings that are **Z**-*algebras of finite type*, which allows us to reduce many questions to the case where Y is the spectrum of such an algebra. This is the explication of the "Kroneckerian point of view" according to which Algebraic Geometry reduces to the Algebraic Geometry of preschemes **IV-3** | 7
of finite type over **Z** (which one sometimes qualifies as "Absolute Algebraic Geometry"). This example shows us in particular that in most questions "relative" to a prescheme of base Y, one can reduce to the case where Y is *Noetherian*.

We note that in this example, in contrast to the previous ones, the morphisms $S_\mu \to S_\lambda$ have in general no particular regularity property.

In what follows, when we apply the results that follow to any one of the three particular situations that have just been described, we will not go into detail about the procedure that allows us to make these applications, being content to refer to what precedes.

(8.1.3). In example *a)* of (8.1.2), we saw that if Y is an integral prescheme with generic point y, $f : X \to Y$ a morphism of finite presentation, and if the generic fibre $f^{-1}(y)$ is proper over $\boldsymbol{k}(y)$, then there exists an open neighbourhood U of y in Y such that $f^{-1}(U)$ is proper over U; it follows *a fortiori* that for every $s \in U$, $f^{-1}(s)$ is proper over $\boldsymbol{k}(s)$. It turns out that one needs a converse, ensuring that if $f^{-1}(s)$ is proper over $\boldsymbol{k}(s)$, then $f^{-1}(y)$ is proper over $\boldsymbol{k}(y)$. For example, suppose that X and Y are algebraic preschemes over an algebraically closed field k (one can take for k the field **C** of complex numbers, to fix ideas); one sometimes needs to know that, for every $s \in Y$, *rational over* k, the fibre $f^{-1}(s)$ is proper over $\boldsymbol{k}(s)$, then $f^{-1}(y)$ is proper over $\boldsymbol{k}(y)$, and consequently

$f^{-1}(\mathrm{U})$ is proper over U for a neighbourhood U of y[13]. Now this statement will follow easily from the following: the set E of points $s \in \mathrm{Y}$ such that $f^{-1}(s)$ is proper over $\boldsymbol{k}(s)$ is *constructible* (and therefore identical to all of Y if it contains the closed points of Y, by the theorem of zeros of Hilbert (**I**, 10.4.8)); this amounts again to saying that if $f^{-1}(y)$ *is not proper* over $\boldsymbol{k}(y)$, then there exists an open neighbourhood U of y such that $f^{-1}(s)$ is not proper over $\boldsymbol{k}(s)$ for every $s \in \mathrm{U}$ (cf. (9.6.1), (iv))). This example illustrates the interest of systematically developing *criteria of constructibility*: this is what will be done in § 9.

8.2. Projective limits of preschemes

(8.2.1). Let S_0 be a ringed space, L a filtered preordered increasing set, $(\mathscr{A}_\lambda, \varphi_{\mu\lambda})$ an inductive system of $\mathscr{O}_{\mathrm{S}_0}$-Algebras (not necessarily commutative), having L for index set. We know that, considered as an inductive system of $\mathscr{O}_{\mathrm{S}_0}$-Modules, $(\mathscr{A}_\lambda, \varphi_{\mu\lambda})$ admits an inductive limit $\mathscr{A}$; let us denote by $\varphi_\lambda : \mathscr{A}_\lambda \to \mathscr{A}$ the canonical homomorphism of $\mathscr{O}_{\mathrm{S}_0}$-Modules. Let $m_\lambda : \mathscr{A}_\lambda \otimes \mathscr{A}_\lambda \to \mathscr{A}_\lambda$ be the homomorphism of $\mathscr{O}_{\mathrm{S}_0}$-Modules that defines multiplication in the $\mathscr{O}_{\mathrm{S}_0}$-Algebra $\mathscr{A}_\lambda$; the hypothesis on the $\varphi_{\mu\lambda}$ implies that the m_λ form an inductive system of homomorphisms, and since the functor $\varinjlim$ **IV-3** | 8
commutes with tensor products, $m = \varinjlim m_\lambda$ is a homomorphism $\mathscr{A} \otimes \mathscr{A} \to \mathscr{A}$ of $\mathscr{O}_{\mathrm{S}_0}$-Modules; by passing to the limit on the commutative diagrams expressing the associativity of m_λ and the existence of the identity section in $\mathscr{A}_\lambda$, one sees that m defines on $\mathscr{A}$ a structure of $\mathscr{O}_{\mathrm{S}_0}$-Algebra and that φ_λ is a homomorphism of $\mathscr{O}_{\mathrm{S}_0}$-Algebras for every $\lambda \in \mathrm{L}$. Furthermore, $\mathscr{A}$ is the inductive limit of the system $(\mathscr{A}_\lambda, \varphi_{\mu\lambda})$ *in the category of $\mathscr{O}_{\mathrm{S}_0}$-Algebras*, in other words, for every $\mathscr{O}_{\mathrm{S}_0}$-Algebra $\mathscr{B}$, the canonical map

$$\mathrm{Hom}_{\mathscr{O}_{\mathrm{S}_0}\text{-Alg.}}(\mathscr{A}, \mathscr{B}) \longrightarrow \varprojlim \mathrm{Hom}_{\mathscr{O}_{\mathrm{S}_0}\text{-Alg.}}(\mathscr{A}_\lambda, \mathscr{B}) \tag{8.2.1.1}$$

which to every homomorphism $f : \mathscr{A} \to \mathscr{B}$ of $\mathscr{O}_{\mathrm{S}_0}$-Algebras associates the family $(f \circ \varphi_\lambda)$, is *bijective*. Indeed, we already know that it is injective and identifies $\mathrm{Hom}_{\mathscr{O}_{\mathrm{S}_0}\text{-Alg.}}(\mathscr{A}, \mathscr{B})$ with a subset of $\varprojlim \mathrm{Hom}_{\mathscr{O}_{\mathrm{S}_0}\text{-Mod.}}(\mathscr{A}_\lambda, \mathscr{B})$; it all comes down to seeing that if (f_λ) is an inductive system of $\mathscr{O}_{\mathrm{S}_0}$-Algebra homomorphisms, $f_\lambda : \mathscr{A}_\lambda \to \mathscr{B}$, its inductive limit $f : \mathscr{A} \to \mathscr{B}$, which is by definition a homomorphism of $\mathscr{O}_{\mathrm{S}_0}$-*Modules*, is also a homomorphism of $\mathscr{O}_{\mathrm{S}_0}$-*Algebras*; but this results from passage to the inductive limit in the commutative diagram of $\mathscr{O}_{\mathrm{S}_0}$-Modules

$$\begin{tikzcd}
\mathscr{A}_\lambda \otimes \mathscr{A}_\lambda \arrow[r, "f_\lambda \otimes f_\lambda"] \arrow[d, "m_\lambda"'] & \mathscr{B} \otimes \mathscr{B} \arrow[d] \\
\mathscr{A}_\lambda \arrow[r, "f_\lambda"'] & \mathscr{B}
\end{tikzcd}$$

and from the fact that the functor $\varinjlim$ commutes with tensor products. We note finally that if the $\mathscr{A}_\lambda$ are commutative $\mathscr{O}_{\mathrm{S}_0}$-Algebras, the same holds for $\mathscr{A}$.

(8.2.2). Suppose now that S_0 is a *prescheme*, and that the $\mathscr{A}_\lambda$ are (commutative) *quasi-coherent* $\mathscr{O}_{\mathrm{S}_0}$-Algebras; we know then that $\mathscr{A} = \varinjlim \mathscr{A}_\lambda$ is a quasi-coherent $\mathscr{O}_{\mathrm{S}_0}$-Algebra (**I**, 4.1.1). Let us denote by S_λ (resp. S) the spectrum of the $\mathscr{O}_{\mathrm{S}_0}$-Algebra $\mathscr{A}_\lambda$ (resp. $\mathscr{A}$) (**II**, 1.3.1), and let $u_{\lambda\mu} : \mathrm{S}_\mu \to \mathrm{S}_\lambda$ (for $\lambda \leqslant \mu$) and $u_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ be the S_0-morphisms corresponding to the homomorphisms $\varphi_{\mu\lambda}$ and φ_λ respectively (**II**, 1.2.7); it is clear that $(\mathrm{S}_\lambda, u_{\lambda\mu})$ is a *projective system* in the category of S_0-preschemes. We note that the $u_{\lambda\mu}$ and u_λ are *affine* morphisms (**II**, 1.6.2), hence *quasi-compact* and *separated*.

Proposition (8.2.3). — *With the notations of (8.2.2), the morphisms $u_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ make* S *a projective limit of the projective system $(\mathrm{S}_\lambda, u_{\lambda\mu})$ in the category of preschemes. Furthermore, if $h : \mathrm{S}_0 \to \mathrm{T}$ is a morphism, making every S_0-prescheme a* T*-prescheme,* S *is also a projective limit of the system $(\mathrm{S}_\lambda, u_{\lambda\mu})$ in the category of* T*-preschemes.*

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[13]We note that such a statement is ultimately purely geometric in nature, in the sense that it only appeals to rational points over k, and not to generic points; for example, when $k = \mathbf{C}$, this statement has an obvious topological significance for the analyst, interpreting "proper" in the topological sense of the term, for the underlying analytic spaces formed by the points of X and Y rational over $\mathbf{C}$.

Proof. Let us first prove the second assertion of the statement in the case $\mathrm{T} = \mathrm{S}_0$. It all comes down to proving that if X is any S_0-prescheme, the canonical map **IV-3** | 9

(8.2.3.1) $$\mathrm{Hom}_{\mathrm{S}_0}(\mathrm{X},\mathrm{S}) \longrightarrow \varprojlim \mathrm{Hom}_{\mathrm{S}_0}(\mathrm{X},\mathrm{S}_\lambda)$$

which to every S_0-morphism $v : \mathrm{X} \to \mathrm{S}$ associates the family $(u_\lambda \circ v)$, is *bijective*. Now, if $g : \mathrm{X} \to \mathrm{S}_0$ is the structural morphism and if we set $\mathscr{B} = g_*(\mathscr{O}_\mathrm{X})$, which is an $\mathscr{O}_{\mathrm{S}_0}$-Algebra, the map (8.2.3.1) identifies canonically with (8.2.1.1) (**II**, 1.2.7), and the conclusion follows from what we have seen in (8.2.1).

The other assertions of (8.2.3) are consequences of the following general lemma: □

Lemma (8.2.4). — *Let* **C** *be a category,* T *an object of* **C**, $\mathbf{C}_\mathrm{T}$ *the subcategory of objects of* **C** *over* T. *Let* $(\mathrm{S}_\lambda, u_{\lambda\mu})$ *be a projective system in* $\mathbf{C}_\mathrm{T}$*; then every projective limit of this system in* $\mathbf{C}_\mathrm{T}$ *is also a projective limit in* **C**, *and conversely.*

Proof. Let $f_\lambda : \mathrm{S}_\lambda \to \mathrm{T}$ be the structural morphism. Suppose that S is a projective limit of $(\mathrm{S}_\lambda, u_{\lambda\mu})$ in **C**, and let us denote by $u_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ the canonical corresponding morphisms. Consider a projective system of T-morphisms (in **C**) $w_\lambda : \mathrm{Y} \to \mathrm{S}_\lambda$, where $\mathrm{Y} \in \mathbf{C}_\mathrm{T}$. There exists by hypothesis a unique morphism (in **C**) $w : \mathrm{Y} \to \mathrm{S}$ such that $w_\lambda = u_\lambda \circ w$ for every λ. The hypothesis that the u_λ are T-morphisms implies that the morphisms $f_\lambda \circ u_\lambda : \mathrm{S} \to \mathrm{T}$ are all the same, and this composite f makes S a T-object. If $g : \mathrm{Y} \to \mathrm{T}$ is the structural morphism of Y, one has $f \circ w = f_\lambda \circ u_\lambda \circ w = f_\lambda \circ w_\lambda = g$ for every λ, which proves that w is a T-morphism. Conversely, suppose (with the same notations) that S is a projective limit of $(\mathrm{S}_\lambda, u_{\lambda\mu})$ in $\mathbf{C}_\mathrm{T}$, and consider now a projective system of morphisms (in **C**) $w_\lambda : \mathrm{Y} \to \mathrm{S}_\lambda$. The composites $f_\lambda \circ w_\lambda : \mathrm{Y} \to \mathrm{T}$ are then all the same: indeed, for any two indices λ, μ, there is an index ν such that $\lambda \leqslant \nu$ and $\mu \leqslant \nu$, whence $f_\nu = f_\lambda \circ u_{\lambda\nu} = f_\mu \circ u_{\mu\nu}$; since $w_\lambda = u_{\lambda\nu} \circ w_\nu$ and $w_\mu = u_{\mu\nu} \circ w_\nu$, one has $f_\lambda \circ w_\lambda = f_\lambda \circ u_{\lambda\nu} \circ w_\nu = f_\nu \circ w_\nu$ and $f_\mu \circ w_\mu = f_\mu \circ u_{\mu\nu} \circ w_\nu = f_\nu \circ w_\nu$. If $g : \mathrm{Y} \to \mathrm{T}$ is the unique morphism thus defined, g makes Y a T-object, and the w_λ are T-morphisms; they hence have a projective limit $w : \mathrm{Y} \to \mathrm{S}$ which is a T-morphism, and *a fortiori* a morphism of **C**; moreover, the first part of the argument shows that any projective limit w' (in **C**) of the projective system (w_λ) is also a T-morphism, and as $w' : \mathrm{Y} \to \mathrm{S}'$ in $\mathbf{C}_\mathrm{T}$ gives $p \circ w' = w$ and $p_\lambda \circ w_\lambda = p_\lambda \circ w'_\lambda$, the definition of S'_λ as a fibre product $\mathrm{S}_\lambda \times_\mathrm{T} \mathrm{T}'$ gives $u'_\lambda \circ w' = w'_\lambda$, and it is immediate that w' is the unique T'-morphism satisfying these relations, whence the lemma. □

Proposition (8.2.5). — *Under the conditions of* (8.2.2), *let* α *be an element of* L, X_α *an* S_α*-prescheme. For every* $\lambda \geqslant \alpha$, *set* $\mathrm{X}_\lambda = \mathrm{X}_\alpha \times_{\mathrm{S}_\alpha} \mathrm{S}_\lambda$, *and for* $\alpha \leqslant \lambda \leqslant \mu$, *set* $v_{\lambda\mu} = 1_{\mathrm{X}_\alpha} \times u_{\lambda\mu}$, *so that* $(\mathrm{X}_\lambda, v_{\lambda\mu})$ *is a projective system of* X_α*-preschemes, whose index set is formed of the* $\lambda \geqslant \alpha$ *in* L. *Set likewise* $\mathrm{X} = \mathrm{X}_\alpha \times_{\mathrm{S}_\alpha} \mathrm{S}$ *and* $v_\lambda = 1_{\mathrm{X}_\alpha} \times u_\lambda$. *Then the* X_α*-morphisms* $v_\lambda : \mathrm{X} \to \mathrm{X}_\lambda$ *make* X *a projective limit of the projective system* $(\mathrm{X}_\lambda, v_{\lambda\mu})$ *in the category of* X_α*-preschemes, or in the category of all preschemes.*

Proof. This will also follow from the following general lemma: □

Lemma (8.2.6). — *Let* **C** *be a category in which fibre products exist,* $q : \mathrm{T}' \to \mathrm{T}$ *a morphism of* **C**, $\mathbf{C}_\mathrm{T}$ *(resp.* $\mathbf{C}_{\mathrm{T}'}$*) the category of objects of* **C** *over* T *(resp.* T'*). Let* $(\mathrm{S}_\lambda, u_{\lambda\mu})$ *be a projective system (not necessarily filtered) in* $\mathbf{C}_\mathrm{T}$, *and set* $\mathrm{S}'_\lambda = \mathrm{S}_\lambda \times_\mathrm{T} \mathrm{T}'$, $u'_{\lambda\mu} = u_{\lambda\mu} \times 1_{\mathrm{T}'}$, *so that* $(\mathrm{S}'_\lambda, u'_{\lambda\mu})$ *is a projective system in* $\mathbf{C}_{\mathrm{T}'}$. *Then, if* (S, u_λ) *is a projective limit of* $(\mathrm{S}_\lambda, u_{\lambda\mu})$ *in* $\mathbf{C}_\mathrm{T}$, $(\mathrm{S} \times_\mathrm{T} \mathrm{T}', u_\lambda \times 1_{\mathrm{T}'})$ *is a projective limit of* $(\mathrm{S}'_\lambda, u'_{\lambda\mu})$ *in* $\mathbf{C}_{\mathrm{T}'}$. **IV-3** | 10

Proof. By hypothesis, for every λ, one has a commutative diagram

$$\begin{tikzcd} \mathrm{S}' \arrow[r, "u'_\lambda"] \arrow[d, "p"'] & \mathrm{S}'_\lambda \arrow[r, "h'_\lambda"] & \mathrm{T}' \arrow[d, "q"] \\ \mathrm{S} \arrow[r, "u_\lambda"'] & \mathrm{S}_\lambda \arrow[r, "h_\lambda"'] & \mathrm{T} \end{tikzcd}$$

where we have set $\mathrm{S}' = \mathrm{S} \times_\mathrm{T} \mathrm{T}'$, $u'_\lambda = u_\lambda \times 1_{\mathrm{T}'}$, $h'_\lambda = h_\lambda \times 1_{\mathrm{T}'}$. Let Y be a T'-object, $g' : \mathrm{Y} \to \mathrm{T}'$, and consider a projective system of T'-morphisms $w'_\lambda : \mathrm{Y} \to \mathrm{S}'_\lambda$. Then Y is a T-object for the morphism $g = q \circ g'$, and the $w_\lambda = p_\lambda \circ w'_\lambda$ are T-morphisms since $h_\lambda \circ w_\lambda = h_\lambda \circ p_\lambda \circ w'_\lambda = q \circ h'_\lambda \circ w'_\lambda = q \circ g'$ by hypothesis. Furthermore, one verifies immediately that (w_λ) is a projective system. There thus exists by hypothesis a unique T-morphism $w : Y \to S$ such that $u_\lambda \circ w = w_\lambda$ for every λ. By the definition of the fibre product, there is a unique T′-morphism $w' : Y \to S'$ such that $p \circ w' = w$. Then $u'_\lambda \circ w' = u_\lambda \circ w' = w_\lambda = p_\lambda \circ w'_\lambda$, and since $h'_\lambda \circ u'_\lambda \circ w' = g' = h'_\lambda \circ w'_\lambda$, the definition of S'_λ as the fibre product $S_\lambda \times_T T'$ gives $u'_\lambda \circ w' = w'_\lambda$, and it is immediate that w' is the unique T′-morphism satisfying these relations, whence the lemma. □

Remark (8.2.7). — Given a ringed space S, the inductive limits with respect to a preordered set L (not necessarily filtered) exist in the category of $\mathscr{O}_S$-Algebras, commutative or not, since the inductive limit exists by (8.2.1) and, on the other hand, for any two homomorphisms of $\mathscr{O}_S$-Algebras $\mathscr{A} \to \mathscr{B}$, $\mathscr{A} \to \mathscr{C}$, the tensor product $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{C}$ is the corresponding "amalgamated sum" in this category.

When S is a prescheme, we say that the tensor product $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{C}$ is a quasi-coherent $\mathscr{O}_S$-Algebra when this is so for $\mathscr{A}$, $\mathscr{B}$, and $\mathscr{C}$ (**I**, 1.3.13); from this one concludes that, in the category of quasi-coherent $\mathscr{O}_S$-Algebras, inductive limits for any preordered index set always exist. This allows us to generalise the definition of a projective limit of preschemes and propositions (8.2.3) and (8.2.5) to the case where the preordered set L is not necessarily filtered.

(8.2.8). The notations being those of (8.2.2), set $u_{\lambda\mu} = (\psi_{\lambda\mu}, \theta_{\lambda\mu})$ and $u_\lambda = (\psi_\lambda, \theta_\lambda)$; $(S_\lambda, \psi_{\lambda\mu})$ is thus a projective system of topological spaces and (ψ_λ) a projective system of continuous maps $S \to S_\lambda$.

Proposition (8.2.9). — *With the notations of (8.2.8), the projective limit of the projective system (ψ_λ) of continuous maps is a homeomorphism of the underlying space of* S *onto the projective limit of the projective system $(S_\lambda, \psi_{\lambda\mu})$ of topological spaces.*

Proof. Let T be the topological limit space of the system $(S_\lambda, \psi_{\lambda\mu})$ and set $\psi = \varprojlim \psi_\lambda : S \to T$. We can restrict to the case where $S_0 = S_\alpha$ for some $\alpha \in L$, and $\lambda \geqslant \alpha$. **IV-3** | 11

(i) Let us show first that the topology of S is the inverse image by ψ of the topology of T; in other words, if $\pi_\lambda : T \to S_\lambda$ is the canonical map, it amounts to showing that every open subset of S is a union of open subsets of the form $\psi^{-1}(\pi_\lambda^{-1}(U_\lambda))$, where U_λ is open in S_λ. Now, every open subset of S is by definition a union of open subsets obtained in the following way: one considers an open affine U_0 of S_0, of ring A_0, so that $\psi_\alpha^{-1}(U_0)$ is the open affine subset of S of ring $A = \Gamma(U_0, \mathscr{A})$, then one takes an element $f \in A$ and considers $\psi_\alpha^{-1}(U_0)$, i.e., the open set $D(f)$. These are the open subsets $D(f)$ which form a base of the topology of S (**II**, 1.3.1). Now, if one sets $A_\lambda = \Gamma(U_0, \mathscr{A}_\lambda)$, one has $A = \varinjlim A_\lambda$ (**I**, 1.3.9), hence there exists an index λ such that f is the canonical image of an element $f_\lambda \in A_\lambda$; one has then $D(f) = \psi_\lambda^{-1}(D(f_\lambda))$ (**I**, 1.2.2), and since $\psi_\lambda = \pi_\lambda \circ \psi$, our assertion is proved.

(ii) Let us now show that ψ is *bijective*, which will complete the proof. Since S is a Kolmogoroff space, it already follows from (i) that ψ is injective, and it remains to show that ψ is surjective. We can evidently replace T by an open subset $\pi_\alpha^{-1}(U_0)$, where U_0 is an open affine in $S_\alpha = S_0$, hence we can restrict to the case where the S_λ and S are affine, in other words $\mathscr{A}_\lambda$ is the sheaf of algebras associated to an A_0-algebra A_λ, and $\mathscr{A}$ the sheaf of algebras associated to $A = \varinjlim A_\lambda$; we will still denote $\varphi_{\mu\lambda} : A_\lambda \to A_\mu$ and $\varphi_\lambda : A_\lambda \to A$ the canonical homomorphisms. By definition, an element of T is a family $(\mathfrak{p}_\lambda)_{\lambda \in L}$, where $\mathfrak{p}_\lambda$ is a prime ideal of A_λ and where $\mathfrak{p}_\lambda = \varphi_{\mu\lambda}^{-1}(\mathfrak{p}_\mu)$ for $\lambda \leqslant \mu$. We know ((**0**, 5.13.3) and (5.13.1)) that there exists a prime ideal $\mathfrak{p}$ of A such that $\mathfrak{p}_\lambda = \varphi_\lambda^{-1}(\mathfrak{p})$ for every $\lambda \in L$, which completes the proof. □

In particular, we have thus proved the

Corollary (8.2.10). — *Let $(A_\lambda)_{\lambda \in L}$ be a filtered inductive system of rings, and let $\varphi_\lambda : A_\lambda \to A$ be the canonical homomorphisms. The canonical map $\mathfrak{p} \mapsto (\varphi_\lambda^{-1}(\mathfrak{p}))$ is a homeomorphism of* $\mathrm{Spec}(A)$ *onto the topological space* $\varprojlim \mathrm{Spec}(A_\lambda)$.

Corollary (8.2.11). — *With the notations of (8.2.8), for every quasi-compact open subset* U *of* S, *there exist an index λ and a quasi-compact open subset* U_λ *of* S_λ *such that* $U = \psi_\lambda^{-1}(U_\lambda)$.

Proof. This follows from the fact that, by definition of the projective limit topology, the $\psi_\lambda^{-1}(U_\lambda)$ (U_λ a quasi-compact open subset of S_λ) form a base of the topology of S, and from the fact that the index set L is filtered. □

Corollary (8.2.12). — *With the notations of (8.2.8), the inductive limit of the inductive system of homomorphisms $\theta^*_\lambda : \psi^*_\lambda(\mathscr{O}_{S_\lambda}) \to \mathscr{O}_S$ of sheaves of rings on S is an isomorphism*

(8.2.12.1) $$\varinjlim_\lambda \psi^*_\lambda(\mathscr{O}_{S_\lambda}) \xrightarrow{\sim} \mathscr{O}_S.$$

Proof. We can evidently suppose the S_λ affine; with the notations of the proof of (8.2.9), it all comes down to seeing that the inductive limit of the inductive system of canonical maps $(A_\lambda)_{\mathfrak{p}_\lambda} \to A_\mathfrak{p}$ is an isomorphism, which is nothing other than (**0**, 5.13.3), (ii). □

Proposition (8.2.13). — *Suppose that the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are open immersions, so that S_μ identifies with an open subset of S_λ for $\lambda \leqslant \mu$. Then, for every $\alpha \in L$, the underlying space of the prescheme S identifies with the intersection of the S_λ for $\lambda \geqslant \alpha$, and the structural sheaf $\mathscr{O}_S$ of the sheaf induced* (G, *II*, 1.5) *by $\mathscr{O}_{S_\alpha}$ on this intersection; more generally, for every $\mathscr{O}_{S_\alpha}$-Module $\mathscr{F}_\alpha$, $u^*_\alpha(\mathscr{F}_\alpha)$ identifies with the $\mathscr{O}_S$-Module induced by $\mathscr{F}_\alpha$ on* S.

Proof. The first assertion follows from (8.2.9), in view of the definition of a projective limit of topological spaces; furthermore, all the $\psi^*_\lambda(\mathscr{O}_{S_\lambda})$ are equal to the sheaf induced by $\mathscr{O}_{S_\alpha}$ on S by definition (**0**, 1.3.7.1), and with the notations of the proof of (8.2.9), the homomorphisms $(A_\lambda)_{\mathfrak{p}_\lambda} \to (A_\mu)_{\mathfrak{p}_\mu}$ are bijective for a system $(\mathfrak{p}_\lambda)$ of ideals of primes corresponding to a single point of S; the assertion relative to $\mathscr{O}_S$ is then deduced from (8.2.12). The last assertion then follows from the definition of $u^*_\alpha(\mathscr{F}_\alpha)$ (**0**, 1.4.3.1). □ **IV-3** | 12

Remark (8.2.14). — The results of (8.2.9) and (8.2.12) show that S is a projective limit of the projective system $(S_\lambda, u_{\lambda\mu})$ in the category of *all* ringed spaces (or of all ringed spaces with a local ringed space structure). Indeed, let Y be a ringed space, and consider a projective system of morphisms of ringed spaces $w_\lambda : Y \to S_\lambda$. If one writes $w_\lambda = (\rho_\lambda, \omega_\lambda)$, the ρ_λ form a projective system of continuous maps and, by (8.2.9), their projective limit identifies with a continuous map $\rho : Y \to S$ such that $\rho_\lambda = \psi_\lambda \circ \rho$. On the other hand, the homomorphisms

$$\omega^\sharp_\lambda : \rho^*_\lambda(\mathscr{O}_{S_\lambda}) \longrightarrow \mathscr{O}_Y$$

form an inductive system of homomorphisms of sheaves of rings. Since

$$\rho^*_\lambda(\mathscr{O}_{S_\lambda}) = \rho^*\big(\psi^*_\lambda(\mathscr{O}_{S_\lambda})\big)$$

and the functor ρ^* is exact, the inductive limit of the $\rho^*_\lambda(\mathscr{O}_{S_\lambda})$ is $\rho^*(\mathscr{O}_S)$ by (8.2.12). There is therefore a unique homomorphism

$$\omega^\sharp : \rho^*(\mathscr{O}_S) \longrightarrow \mathscr{O}_Y$$

such that $\omega^\sharp_\lambda = \omega^\sharp \circ \rho^*(\theta^*_\lambda)$ for every λ, which proves the assertion.

8.3. Constructible subsets of a projective limit of preschemes

(8.3.1). In what follows throughout this paragraph, we suppose the conditions of (8.2.2) are satisfied, and we keep the notations.

Theorem (8.3.2). — *For every λ, let E_λ, F_λ be two subsets of S_λ. Set*

(8.3.2.1) $$E = \bigcap_\lambda u_\lambda^{-1}(E_\lambda), \qquad F = \bigcup_\lambda u_\lambda^{-1}(F_\lambda).$$

Suppose the following conditions are satisfied:

(i) For every λ, E_λ is pro-constructible and F_λ is ind-constructible (1.9.4).

(ii) For $\lambda \leqslant \mu$, we have

$$E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda), \qquad F_\mu \supset u_{\lambda\mu}^{-1}(F_\lambda).$$

(iii) There exists $\alpha \in L$ such that S_α is quasi-compact (which implies that S_λ is quasi-compact for $\lambda \geqslant \alpha$).

Then the following properties are equivalent:

a) $E \subset F$.

b) *There exists $\lambda \geqslant \alpha$ such that $u_\lambda^{-1}(E_\lambda) \subset u_\lambda^{-1}(F_\lambda)$ (and then we also have $u_\mu^{-1}(E_\mu) \subset u_\mu^{-1}(F_\mu)$ for $\mu \geqslant \lambda$).* **IV-3** | 13

c) *There exists $\lambda \geqslant \alpha$ such that $E_\lambda \subset F_\lambda$ (and then we also have $E_\mu \subset F_\mu$ for $\mu \geqslant \lambda$).*

The parenthetical remarks in b) *and* c) *follow from (ii). Set*

$$\mathrm{G}_\lambda = \mathrm{E}_\lambda \cap (\mathrm{S}_\lambda - \mathrm{F}_\lambda), \qquad \mathrm{G} = \mathrm{E} \cap (\mathrm{S} - \mathrm{F}).$$

Then G_λ *is a pro-constructible subset of* S_λ (1.9.5), *(i), and by virtue of* (8.3.2.1) *we have*

$$\mathrm{G}_\mu \subset u_{\lambda\mu}^{-1}(\mathrm{G}_\lambda) \qquad \textit{for } \lambda \leqslant \mu$$

$$\mathrm{G} = \bigcap_\lambda u_\lambda^{-1}(\mathrm{G}_\lambda).$$

We are thus reduced to proving the particular case of (8.3.2) *corresponding to* $\mathrm{F}_\lambda = \varnothing$ *for every* λ*:*

Corollary (8.3.3). — *For every* λ*, let* E_λ *be a pro-constructible subset of* S_λ *such that* $\mathrm{E}_\mu \subset u_{\lambda\mu}^{-1}(\mathrm{E}_\lambda)$*. Suppose that there exists* $\alpha \in \mathrm{L}$ *such that* S_α *is quasi-compact. Then the following conditions are equivalent:*

a) $\mathrm{E} = \bigcap_\lambda u_\lambda^{-1}(\mathrm{E}_\lambda) = \varnothing$.
b) *There exists* λ *such that* $u_\lambda^{-1}(\mathrm{E}_\lambda) = \varnothing$ *(and then* $u_\mu^{-1}(\mathrm{E}_\mu) = \varnothing$ *for* $\mu \geqslant \lambda$*).*
c) *There exists* λ *such that* $\mathrm{E}_\lambda = \varnothing$ *(and then* $\mathrm{E}_\mu = \varnothing$ *for* $\mu \geqslant \lambda$*).*

Proof. It is clear that *c)* implies *a)*. Let us prove that *a)* implies *b)*: since S_λ is quasi-compact, E_λ, being pro-constructible, is also quasi-compact, and it is the same for $u_\lambda^{-1}(\mathrm{E}_\lambda)$ (1.9.5), (vi); then the filtered decreasing family of pro-constructible subsets $u_\lambda^{-1}(\mathrm{E}_\lambda)$ has empty intersection, hence (1.9.9) one of them is empty.

Finally, let us show that *b)* implies *c)*. Since S_α is quasi-compact and L is filtered, we can replace S_α by an open affine, hence suppose (8.2.2) that S and the S_λ for $\lambda \geqslant \alpha$ are affine; one has then (1.9.2.1), for $\lambda \geqslant \alpha$,

$$u_\lambda(\mathrm{S}) = \bigcap_{\mu \geqslant \lambda} u_{\lambda\mu}(\mathrm{S}_\mu)$$

whence

$$\mathrm{E}_\lambda \cap u_\lambda(\mathrm{S}) = \bigcap_{\mu \geqslant \lambda} (\mathrm{E}_\lambda \cap u_{\lambda\mu}(\mathrm{S}_\mu)).$$

Now, since u_λ and the $u_{\lambda\mu}$ are quasi-compact, $u_\lambda(\mathrm{S})$ and $u_{\lambda\mu}(\mathrm{S}_\mu)$ are pro-constructible in S_λ (1.9.5), (vii), hence the sets $\mathrm{E}_\lambda \cap u_{\lambda\mu}(\mathrm{S}_\mu)$ for $\mu \geqslant \lambda$ form a filtered decreasing family of pro-constructible subsets of S_λ. Since S_λ is quasi-compact, the hypothesis *b)* implies that the intersection of this family is empty, hence (1.9.9) one of the sets $\mathrm{E}_\lambda \cap u_{\lambda\mu}(\mathrm{S}_\mu)$ is empty, hence $\mathrm{E}_\mu \subset u_{\lambda\mu}^{-1}(\mathrm{E}_\lambda)$ is empty. □

Corollary (8.3.4). — *For every* λ*, let* F_λ *be an ind-constructible subset of* S_λ *such that for* $\lambda \leqslant \mu$ *one has* $u_{\lambda\mu}^{-1}(\mathrm{F}_\lambda) \subset \mathrm{F}_\mu$*. Suppose that there exists* $\alpha \in \mathrm{L}$ *such that* S_α *is quasi-compact. Then the following conditions are equivalent:*

a) *The set* $\mathrm{F} = \bigcup_\lambda u_\lambda^{-1}(\mathrm{F}_\lambda)$ *is equal to* S.
b) *There exists* λ *such that* $u_\lambda^{-1}(\mathrm{F}_\lambda) = \mathrm{S}$ *(and then* $u_\mu^{-1}(\mathrm{F}_\mu) = \mathrm{S}$ *for* $\mu \geqslant \lambda$*).* **IV-3 | 14**
c) *There exists* λ *such that* $\mathrm{F}_\lambda = \mathrm{S}_\lambda$ *(and then* $\mathrm{F}_\mu = \mathrm{S}_\mu$ *for* $\mu \geqslant \lambda$*).*

This follows immediately from (8.3.3) *by passing to complements.*

Corollary (8.3.5). — *For every* λ*, let* E_λ*,* F_λ *be two constructible subsets of* S_λ *such that, for* $\mu \geqslant \lambda$*, one has* $\mathrm{E}_\mu \subset u_{\lambda\mu}^{-1}(\mathrm{E}_\lambda)$ *and* $\mathrm{F}_\mu \supset u_{\lambda\mu}^{-1}(\mathrm{F}_\lambda)$*. Suppose that there exists* α *such that* S_α *is quasi-compact. Then, for* $\mathrm{E} \subset \mathrm{F}$ *(resp.* $\mathrm{E} = \mathrm{F}$*), it is necessary and sufficient that there exist* λ *such that* $\mathrm{E}_\lambda \subset \mathrm{F}_\lambda$ *(resp.* $\mathrm{E}_\lambda = \mathrm{F}_\lambda$*), in which case one also has* $\mathrm{E}_\mu \subset \mathrm{F}_\mu$ *(resp.* $\mathrm{E}_\mu = \mathrm{F}_\mu$*) for* $\mu \geqslant \lambda$*.*

This is a particular case of (8.3.2)*.*

Corollary (8.3.6). — *Suppose that there exists* α *such that* S_α *is quasi-compact. For* $\mathrm{S} = \varnothing$*, it is necessary and sufficient that there exist* λ *such that* $\mathrm{S}_\lambda = \varnothing$*.*

Corollary (8.3.7). — *For every* λ*, we have*

$$u_\lambda(\mathrm{S}) = \bigcap_{\mu \geqslant \lambda} u_{\lambda\mu}(\mathrm{S}_\mu). \tag{8.3.7.1}$$

Proof. It is clear that the first member of (8.3.7.1) is contained in the second. Let s be a point of S_λ and set $X_\lambda = \mathrm{Spec}(\boldsymbol{k}(s))$; consider the projective system $(X_\mu, v_{\mu\nu})$ where $X_\mu = X_\lambda \times_{S_\lambda} S_\mu$ and $v_{\mu\nu} = 1 \times u_{\mu\nu}$ for $\lambda \leqslant \mu \leqslant \nu$; its projective limit is $X = X_\lambda \times_{S_\lambda} S$ (8.2.5). If $s \in \bigcap_{\mu \geqslant \lambda} u_{\lambda\mu}(S_\mu)$, this implies that $X_\mu \neq \varnothing$ for every $\mu \geqslant \lambda$ (**I**, 3.4.8); it follows then from (8.3.6) that $X \neq \varnothing$, hence $s \in u_\lambda(S)$ by (**I**, 3.4.8). □

Proposition (8.3.8). — *(i) For the morphism $u_\lambda : S \to S_\lambda$ to be dominant (resp. surjective), it is necessary and sufficient that for $\mu \geqslant \lambda$ the morphism $u_{\lambda\mu} : S_\mu \to S_\lambda$ be dominant (resp. surjective).*

(ii) If, for some index λ, the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are flat (resp. faithfully flat) for every $\mu \geqslant \lambda$, then the morphism $u_\lambda : S \to S_\lambda$ is flat (resp. faithfully flat).

(iii) Suppose that the morphisms $u_\mu : S \to S_\mu$ are surjective for $\mu \geqslant \lambda$. For u_λ to be an open morphism (resp. universally open), it is necessary and sufficient that, for every $\mu \geqslant \lambda$, $u_{\lambda\mu}$ be an open morphism (resp. universally open).

Proof. (i) Since $u_\lambda(S) \subset u_{\lambda\mu}(S_\mu)$ for every $\mu \geqslant \lambda$, the necessity of the conditions is trivial, and it follows immediately from (8.3.7.1) that if the $u_{\lambda\mu}$ are surjective, then so is u_λ. Suppose now that the $u_{\lambda\mu}$ are dominant for $\mu \geqslant \lambda$, and consider in S_λ an open quasi-compact non-empty subset U_λ; then the $U_\mu = u_{\lambda\mu}^{-1}(U_\lambda)$ for $\mu \geqslant \lambda$ form a projective system whose projective limit is $U = u_\lambda^{-1}(U_\lambda)$ (8.2.5). By hypothesis the U_μ are all non-empty, hence it is the same for U by (8.3.6), which proves that u_λ is dominant.

(ii) By virtue of (i), it suffices to consider the case where the $u_{\lambda\mu}$ are flat; one can then restrict to the case where S_λ is affine, hence also the S_μ and S, and the assertion follows from (2.1.2) and (**0**, 6.2.3).

(iii) By virtue of (8.2.5) and of (**I**, 3.5.2), it suffices to treat the case of open morphisms. Since $u_\lambda = u_{\lambda\mu} \circ u_\mu$ and u_μ is surjective, we know that if u_λ is open then so is $u_{\lambda\mu}$ (Bourbaki, *Top. gén.*, chap. I^er, 3^e ed., § 5, n^o 1, prop. 1). Conversely, to show that u_λ is open when all the $u_{\lambda\mu}$ are open for $\mu \geqslant \lambda$, it suffices to see that for every quasi-compact open subset V of S, $u_\lambda(V)$ is open in S_λ; but there exists then $\mu \geqslant \lambda$ such that $V = u_\mu^{-1}(V_\mu)$, where V_μ is open in S_μ (8.2.11); since u_μ is surjective, one has $V_\mu = u_\mu(V)$ and $u_\lambda(V) = u_{\lambda\mu}(u_\mu(V))$ is therefore open by hypothesis. **IV-3** | 15

One notes that it may happen that all the $u_{\lambda\mu}$ are open without u_λ being open, when the u_μ are not surjective. Indeed, one has an example by considering an integral ring A which is not a field, and its field of fractions K, which is the inductive limit of the A_f, where f ranges over $A - \{0\}$; setting $S_1 = \mathrm{Spec}(A)$, $S_f = \mathrm{Spec}(A_f)$, and $S = \varprojlim S_f = \mathrm{Spec}(K)$, the morphism $S \to S_1$ is not open, even though the morphisms $S_f \to S_1$ are open. □

(8.3.9). For every prescheme X, we will denote as usual by $\mathfrak{P}(X)$ the set of all subsets of the underlying set of X, by $\mathfrak{C}(X)$ (resp. $\mathfrak{Dc}(X)$, $\mathfrak{Fc}(X)$, $\mathfrak{LFc}(X)$) the set of constructible (resp. constructible and open, constructible and closed, and locally closed) subsets of X. It is clear that $(\mathfrak{P}(S_\lambda), u_{\lambda\mu}^{-1})$ is an *inductive system of sets* and that the maps $u_\lambda^{-1} : \mathfrak{P}(S_\lambda) \to \mathfrak{P}(S)$ form an inductive system of maps, whence, by passing to the inductive limit, a canonical map

$$u_{\mathfrak{P}} : \varinjlim \mathfrak{P}(S_\lambda) \longrightarrow \mathfrak{P}(S). \tag{8.3.9.1}$$

Furthermore, it follows from (1.8.2) that $u_{\lambda\mu}^{-1}$ sends $\mathfrak{C}(S_\lambda)$ (resp. $\mathfrak{Dc}(S_\lambda)$, $\mathfrak{Fc}(S_\lambda)$, $\mathfrak{LFc}(S_\lambda)$) into $\mathfrak{C}(S_\mu)$ (resp. $\mathfrak{Dc}(S_\mu)$, $\mathfrak{Fc}(S_\mu)$, $\mathfrak{LFc}(S_\mu)$) and that u_λ^{-1} sends $\mathfrak{C}(S_\lambda)$ (resp. $\mathfrak{Dc}(S_\lambda)$, $\mathfrak{Fc}(S_\lambda)$, $\mathfrak{LFc}(S_\lambda)$) into $\mathfrak{C}(S)$ (resp. $\mathfrak{Dc}(S)$, $\mathfrak{Fc}(S)$, $\mathfrak{LFc}(S)$). We thus have, by restriction of (8.3.9.1), canonical maps

$$\varinjlim \mathfrak{C}(S_\lambda) \longrightarrow \mathfrak{C}(S) \tag{8.3.9.2}$$

$$\varinjlim \mathfrak{Dc}(S_\lambda) \longrightarrow \mathfrak{Dc}(S) \tag{8.3.9.3}$$

$$\varinjlim \mathfrak{Fc}(S_\lambda) \longrightarrow \mathfrak{Fc}(S) \tag{8.3.9.4}$$

$$\varinjlim \mathfrak{LFc}(S_\lambda) \longrightarrow \mathfrak{LFc}(S). \tag{8.3.9.5}$$

(8.3.10). Let $g_\alpha : X_\alpha \to S_\alpha$ be a morphism; with the notations of (8.2.5) one has, on the one hand, a canonical map $v_{\mathfrak{P}} : \varinjlim \mathfrak{P}(S_\lambda) \to \mathfrak{P}(X)$; on the other hand, from the morphisms of projection

$g_\lambda : X_\lambda \to S_\lambda$ for every $\lambda \geqslant \alpha$ and a morphism of projection $g : X \to S$. It is clear that the $g_\lambda^{-1} : \mathfrak{P}(S_\lambda) \to \mathfrak{P}(X_\lambda)$ form an inductive system of maps, and that the diagrams

$$\begin{tikzcd} \mathfrak{P}(S_\lambda) \arrow[r, "u_\lambda^{-1}"] \arrow[d, "g_\lambda^{-1}"'] & \mathfrak{P}(S) \arrow[d, "g^{-1}"] \\ \mathfrak{P}(X_\lambda) \arrow[r, "v_\lambda^{-1}"'] & \mathfrak{P}(X) \end{tikzcd}$$

are commutative; one deduces from this by passing to the inductive limit a commutative diagram **IV-3** | 16

$$\begin{tikzcd} \varinjlim \mathfrak{P}(S_\lambda) \arrow[r, "u_{\mathfrak{P}}"] \arrow[d] & \mathfrak{P}(S) \arrow[d, "g^{-1}"] \\ \varinjlim \mathfrak{P}(X_\lambda) \arrow[r, "v_{\mathfrak{P}}"'] & \mathfrak{P}(X) \end{tikzcd}$$

and it follows from (1.8.2) that one has analogous diagrams replacing $\mathfrak{P}$ by $\mathfrak{C}$, $\mathfrak{Dc}$, $\mathfrak{Fc}$, or $\mathfrak{LFc}$.

(8.3.10.1)

It follows from (8.3.5) that under the hypothesis that for some $\alpha \in L$, S_α is *quasi-compact*, the canonical map (8.3.9.2) is *injective*. Furthermore:

Theorem (8.3.11). — *Suppose that there exists α such that S_α is quasi-compact and quasi-separated. Then the canonical maps (8.3.9.2), (8.3.9.3), (8.3.9.4), and (8.3.9.5) are* bijective.

Proof. By virtue of the preceding remark, it remains to prove that these maps are *surjective*; as every constructible subset of S is a finite union of sets of the form $U \cap \complement V$, where U and V are open and constructible, it will suffice to prove that (8.3.9.3) is surjective for it to be the same for (8.3.9.2) (and also for (8.3.9.4), by passing to complements). Now, since the morphisms $S_\lambda \to S_\alpha$ are affine and separated, the S_λ for $\lambda \geqslant \alpha$ and S are quasi-compact and quasi-separated, and one knows that the constructible open subsets of such a prescheme are none other than the quasi-compact open subsets (1.8.1). The conclusion then follows from (8.2.11) except for the map (8.3.9.5). To prove that this last map is surjective, consider a locally closed and constructible subset Z of S; Z is therefore quasi-compact ($\mathbf{0}_{\rm III}$, 9.1.4). As every point $x \in Z$ admits by hypothesis an open quasi-compact neighbourhood V_x in S such that $Z \cap V_x$ is closed in V_x; we can cover Z by a finite number of the V_x, hence there is a quasi-compact open subset U containing Z such that Z is closed in U; in other words, Z is also constructible in U ($\mathbf{0}_{\rm III}$, 9.1.8). We know (8.2.11) that there exist λ and a quasi-compact open subset U_λ of S_λ such that $U = u_\lambda^{-1}(U_\lambda)$. Applying to U (which is a projective limit of $U_\lambda = u_{\lambda\mu}^{-1}(U_\lambda)$) the map (8.3.9.4), one sees that there exist $\mu \geqslant \lambda$ and a closed constructible subset Z_μ in U_μ such that $Z = u_\mu^{-1}(Z_\mu)$. But since the canonical immersion $U_\mu \to S_\mu$ is quasi-compact by hypothesis (1.2.7), she is of finite presentation (1.6.2), (i), and Z_μ is also a constructible subset *of* S_μ by virtue of (1.8.4) and (1.8.1); since Z_μ is evidently locally closed in S_μ, this completes the proof. □

Corollary (8.3.12). — *Suppose that there exists α such that S_α is quasi-compact, and let, for every λ, Z_λ be a constructible subset of S_λ such that $Z_\mu = u_{\lambda\mu}^{-1}(Z_\lambda)$ for $\mu \geqslant \lambda$. If $Z = u_\lambda^{-1}(Z_\lambda)$, then, for Z to be open (resp. closed, resp. locally closed) in S, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that Z_λ be so in S_λ.* **IV-3** | 17

Proof. Cover S_α by a finite number of open affine subsets $U_\alpha^{(i)}$; then the $U^{(i)} = u_\alpha^{-1}(U_\alpha^{(i)})$ form an open covering of S, and for Z to be open (resp. closed, resp. locally closed) in S, it is necessary and sufficient that each of the $Z \cap U^{(i)}$ be so in $U^{(i)}$. Since L is filtered and each $Z \cap U^{(i)}$ is constructible in $U^{(i)}$ ($\mathbf{0}_{\rm III}$, 9.1.8), one can restrict to proving the corollary when S_α is affine, hence quasi-compact and quasi-separated; but then it follows from (8.3.11). □

Proposition (8.3.13). — *Suppose that the morphisms $u_{\lambda\mu} : \mathrm{S}_\mu \to \mathrm{S}_\lambda$ are flat and that there exists α such that S_α is quasi-compact. For every λ, let Z'_λ, Z''_λ be two pro-constructible subsets of S_λ, such that $\mathrm{Z}'_\mu = u_{\lambda\mu}^{-1}(\mathrm{Z}'_\lambda)$ and $\mathrm{Z}''_\mu = u_{\lambda\mu}^{-1}(\mathrm{Z}''_\lambda)$ for $\mu \geqslant \lambda$; suppose further that $\overline{\mathrm{Z}'_\alpha}$ is constructible in S_α. Set $\mathrm{Z}' = u_\lambda^{-1}(\mathrm{Z}'_\lambda)$, $\mathrm{Z}'' = u_\lambda^{-1}(\mathrm{Z}''_\lambda)$; for $\mathrm{Z}'' \subset \overline{\mathrm{Z}'}$, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $\mathrm{Z}''_\lambda \subset \overline{\mathrm{Z}'_\lambda}$.*

Proof. Indeed, one knows that $u_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ is also a flat morphism for every λ (8.3.8), (i); as Z'_λ is pro-constructible, the closure of Z'_μ for $\mu \geqslant \lambda$ (resp. of Z') in S_μ (resp. S) is equal to $u_{\lambda\mu}^{-1}(\overline{\mathrm{Z}'_\lambda})$ (resp. $u_\lambda^{-1}(\overline{\mathrm{Z}'_\lambda})$) (2.3.10). Since the $u_{\lambda\mu}^{-1}(\overline{\mathrm{Z}'_\lambda})$ and $u_\lambda^{-1}(\overline{\mathrm{Z}'_\lambda})$ are constructible (1.8.2), the conclusion follows from (8.3.2). □

8.4. Criteria of irreducibility and connectedness for projective limits of preschemes

Proposition (8.4.1). — *Suppose that there exists an index α such that S_α is quasi-compact.*

- *(i) If S is not irreducible and if in addition the underlying space of S is Noetherian and S_α is quasi-separated, there exists $\lambda \geqslant \alpha$ such that, for $\mu \geqslant \lambda$, S_μ is not irreducible.*
- *(ii) If S is not connected, there exists $\lambda \geqslant \alpha$ such that, for $\mu \geqslant \lambda$, S_μ is not connected.*

Proof. Suppose that S is a union of two distinct closed subsets S', S'' of S (resp. disjoint non-empty closed subsets). In case (i), S' and S'' are constructible since the space S is Noetherian. By virtue of (8.3.11), there exist $\lambda \geqslant \alpha$ and two constructible closed subsets S'_λ, S''_λ of S_λ such that $\mathrm{S}' = u_\lambda^{-1}(\mathrm{S}'_\lambda)$, $\mathrm{S}'' = u_\lambda^{-1}(\mathrm{S}''_\lambda)$; since $\mathrm{S} = \mathrm{S}' \cup \mathrm{S}''$, it follows from (8.3.11) that one can suppose $\mathrm{S}_\lambda = \mathrm{S}'_\lambda \cup \mathrm{S}''_\lambda$; since S'_λ and S''_λ are distinct from S_λ, this proves that S_λ is not irreducible.

In case (ii), S' and S'' are open and quasi-compact, hence, by virtue of (8.2.11), there exist $\lambda \geqslant \alpha$ and two open quasi-compact subsets S'_λ, S''_λ of S_λ such that $\mathrm{S}' = u_\lambda^{-1}(\mathrm{S}'_\lambda)$, $\mathrm{S}'' = u_\lambda^{-1}(\mathrm{S}''_\lambda)$. Moreover, since S' and S'' are open and closed in S, they are both pro-constructible and ind-constructible (1.9.6), hence constructible (1.9.5), and it follows from (8.3.5) that one can suppose λ taken such that $\mathrm{S}_\lambda = \mathrm{S}'_\lambda \cup \mathrm{S}''_\lambda$ and $\mathrm{S}'_\lambda \cap \mathrm{S}''_\lambda = \varnothing$, which shows that S_λ is not connected. □

Proposition (8.4.2). — *Suppose that the underlying space of S is Noetherian and that one of the two following conditions is satisfied:*

- a) *For $\lambda \leqslant \mu$, $u_{\lambda\mu} : \mathrm{S}_\mu \to \mathrm{S}_\lambda$ is dominant, and there exists α such that S_α is quasi-compact.* **IV-3** | 18
- b) *There exists α such that the underlying space of S_α is Noetherian, and for $\mu \geqslant \lambda$, $u_{\lambda\mu}$ is a homeomorphism of S_μ onto a subspace of S_λ.*

Under these conditions, there exists λ such that, for every $\mu \geqslant \lambda$:

- *(i) For every irreducible component Y_i of S ($1 \leqslant i \leqslant m$), $\overline{u_\mu(\mathrm{Y}_i)}$ is an irreducible component of S_μ, and the map $\mathrm{Y}_i \mapsto \overline{u_\mu(\mathrm{Y}_i)}$ is a bijection of the set of irreducible components of S onto the set of irreducible components of S_μ.*
- *(ii) For every connected component C_j of S ($1 \leqslant j \leqslant n$), $\overline{u_\mu(\mathrm{C}_j)}$ is a connected component of S_μ, and the map $\mathrm{C}_j \mapsto \overline{u_\mu(\mathrm{C}_j)}$ is a bijection of the set of connected components of S onto the set of connected components of S_μ.*

Proof. Let us first establish the

Lemma (8.4.2.1). — *Under condition* a) *or* b) *of (8.4.2), there exists λ such that, for $\mu \geqslant \lambda$, $u_\mu : \mathrm{S} \to \mathrm{S}_\mu$ is dominant.*

In case *a)*, this has already been proved without supposing the space S Noetherian (8.3.8), (i). In case *b)*, set $\mathrm{Z}_\alpha = \overline{u_\alpha(\mathrm{S})}$; as a closed subset of the Noetherian space S_α, Z_α is constructible, and since $u_\alpha^{-1}(\mathrm{Z}_\alpha) = \mathrm{S}$, it follows from (8.3.5) that there exists $\lambda \geqslant \alpha$ such that $u_{\lambda\mu}^{-1}(\mathrm{Z}_\lambda) = \mathrm{S}_\mu$. But since $u_{\alpha\mu}$ is a homeomorphism and $u_\alpha(\mathrm{S})$ is dense in S_α, it follows from $\mathrm{S} \to \mathrm{Z}_\mu = \mathrm{S}_\mu$.

This lemma being proved, one can suppose that for every λ, u_λ is a dominant morphism.

- (i) Each of the S_λ is a union of the $\overline{u_\lambda(\mathrm{Y}_i)}$, which are irreducible. On the other hand, if U_i is the open subset of S complementary to the union of the Y_j of index $j \neq i$ ($1 \leqslant i \leqslant m$), the U_i are pairwise disjoint and $\mathrm{Y}_i = \overline{\mathrm{U}_i}$ (**0**, 2.1.6). Since the underlying space of S is Noetherian, the U_i are quasi-compact, hence there exist an index λ and open subsets $\mathrm{U}_{i\lambda}$ of S_λ such that $U_i = u_\lambda^{-1}(U_{i\lambda})$ for $1 \leqslant i \leqslant m$ (8.2.11). From this one concludes that if one sets $U_{i\mu} = u_{\lambda\mu}^{-1}(U_{i\lambda})$ for $\lambda \leqslant \mu$, the $U_{i\mu}$ are pairwise disjoint, the $U_{i\mu}$ are open and $u_\mu(U_i)$ is dense in $U_{i\mu}$ since u_μ is dominant. By this, none of the closures $\overline{U_{i\mu}}$ is contained in another and $u_\mu(U_i)$ is dense in $U_{i\mu}$ since u_μ is dominant; hence $\overline{U_{i\mu}} = \overline{u_\mu(Y_i)}$, which proves that the $\overline{u_\mu(Y_i)}$ are the irreducible components of S_μ (**0**, 2.1.7) and completes the proof.

(ii) Since the underlying space S is Noetherian, the C_j are open and closed in S and quasi-compact; the same reasoning as in (i) shows that there exist λ and open subsets $V_{j\lambda}$ of S_λ such that $C_j = u_\lambda^{-1}(V_{j\lambda})$ for $1 \leqslant j \leqslant n$. One sees also, as in (i), that if one sets $V_{j\mu} = u_{\lambda\mu}^{-1}(V_{j\lambda})$ for $\mu \geqslant \lambda$, the $V_{j\mu}$ are pairwise disjoint, and $u_\mu(C_j)$ is dense in $V_{j\mu}$. Furthermore, it follows from (8.3.4) that for μ sufficiently large, the union of the $V_{j\mu}$ is S_μ, since every open subset of a Noetherian prescheme is ind-constructible (1.9.6). The $V_{j\mu}$ are thus the connected components of S_μ, this completes the proof.

□

We note that if the morphisms $u_{\lambda\mu}$ are *immersions*, they satisfy in particular condition *b)* of (8.4.2).

Corollary (8.4.3). — *Suppose one of the conditions* a), b) *of* (8.4.2) *is satisfied, the underlying space of* S *being Noetherian; then, for* S *to be irreducible, it is necessary and sufficient that there exist* λ *such that* S_μ *be so for every* $\mu \geqslant \lambda$.

Proposition (8.4.4). — *Suppose that there exists* α *such that* S_α *is quasi-compact and that, for* $\lambda \leqslant \mu$, $u_{\lambda\mu} : S_\mu \to S_\lambda$ *is dominant. Then, for* S *to be connected, it is necessary and sufficient that there exist* λ *such that* S_μ *be so for every* $\mu \geqslant \lambda$. **IV-3** | 19

Proof. The condition is sufficient by virtue of (8.4.1), (i); on the other hand, we have seen (8.3.8), (i) that $u_\lambda : S \to S_\lambda$ is dominant for λ sufficiently large, hence, if S is connected, it is the same for S_λ, since $u_\lambda(S)$ is dense in S_λ and is connected. □

Corollary (8.4.5). — *Let* k *be a field,* X *a quasi-compact* k*-prescheme. For* X *to be geometrically connected* (4.5.2)*, it is necessary and sufficient that, for every finite and separable extension* K *of* k, $X \otimes_k K$ *be connected.*

Proof. The condition is trivially necessary. To see that it is sufficient, we must prove that if Ω is an algebraic closure of k, $X \otimes_k \Omega$ is connected (4.5.1). Now, Ω is a filtered inductive limit of the finite sub-extensions of k, and for $k' \subset k'' \subset \Omega$, the morphism $X \otimes_k k'' \to X \otimes_k k'$ is surjective. We are thus reduced, by virtue of (8.4.4), to proving that $X \otimes_k k'$ is connected for every finite extension k' of k. But if K is the largest separable extension contained in k', the morphism $X \otimes_k k' \to X \otimes_k K$ is finite, surjective, and radicial, hence (2.4.5) a homeomorphism, and since $X \otimes_k K$ is connected by hypothesis, so is $X \otimes_k k'$. □

Remark (8.4.6). — (i) The proof of (8.4.2) shows that the conclusion of this proposition is valid if one supposes that the underlying space of S is Noetherian, that there exists α such that S_α is quasi-compact, and finally that the $u_\lambda : S \to S_\lambda$ are dominant.

(ii) On the other hand, the conclusion of (8.4.2) can be at fault when the u_λ are not dominant for λ sufficiently large, even when the S_λ and S are Noetherian, as shown by the following example. Take **N** for index set, all the S_n equal to $\mathrm{Spec}(A \times K) = \mathrm{Spec}(A) \amalg \mathrm{Spec}(K)$, where K is a field, A any K-algebra, and all the morphisms $u_{n,n+1}$ equal to the same morphism corresponding to the homomorphism $(x, y) \mapsto (j(y), y)$ of $A \times K$ into itself, where $j : K \to A$ is the canonical homomorphism. One easily verifies that the inductive limit of this system of rings is K, the canonical homomorphism $A \times K \to K$ being the second projection. One thus sees that $S = \mathrm{Spec}(K)$ is irreducible even though none of the S_n is connected.

8.5. Modules of finite presentation on a projective limit of preschemes

(8.5.1). We always keep the notations of (8.2.2); we will further restrict ourselves to the case where S_0 is one of the S_λ, to which one can always reduce.

Whenever, in this section, we consider a family $(\mathscr{F}_\lambda)$, *where, for every* $\lambda \in L$, $\mathscr{F}_\lambda$ *is an* $\mathscr{O}_{S_\lambda}$*-Module, it will be understood that this family satisfies the condition*

$$\mathscr{F}_\mu = u^*_{\lambda\mu}(\mathscr{F}_\lambda) \qquad \text{for } \lambda \leqslant \mu. \tag{8.5.1.1}$$

We will then set **IV-3 | 20**

$$\mathscr{F} = u^*_\lambda(\mathscr{F}_\lambda) \tag{8.5.1.2}$$

which is an $\mathscr{O}_S$-Module not depending on the index $\lambda \in L$, by virtue of the hypothesis (8.5.1.1).

Let $(\mathscr{F}_\lambda)$, $(\mathscr{G}_\lambda)$ be two such families of $\mathscr{O}_{S_\lambda}$-Modules. It is clear that the maps $f_\lambda \mapsto u^*_{\lambda\mu}(f_\lambda)$ from $\mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda)$ to $\mathrm{Hom}_{\mathscr{O}_{S_\mu}}(\mathscr{F}_\mu, \mathscr{G}_\mu)$ define an *inductive system* of abelian groups $(\mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda), u^*_{\lambda\mu})$, and that the maps $f_\lambda \mapsto u^*_\lambda(f_\lambda)$ form an inductive system of homomorphisms of abelian groups

$$u^*_\lambda : \mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda) \longrightarrow \mathrm{Hom}_S(\mathscr{F}, \mathscr{G});$$

whence, by passing to the inductive limit, a canonical homomorphism of abelian groups

$$u^*_{\mathscr{F},\mathscr{G}} : \varinjlim \mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda) \longrightarrow \mathrm{Hom}_S(\mathscr{F}, \mathscr{G}). \tag{8.5.1.3}$$

Note that when $\mathscr{F}_\lambda = \mathscr{O}_{S_\lambda}$, the condition (8.5.1.1) is satisfied, and one has $\mathscr{F} = \mathscr{O}_S$; the homomorphism (8.5.1.3) thus gives (**0**, 5.1.1) a canonical homomorphism of abelian groups

$$u_{\mathscr{G}} : \varinjlim \Gamma(S_\lambda, \mathscr{G}_\lambda) \longrightarrow \Gamma(S, \mathscr{G}). \tag{8.5.1.4}$$

Theorem (8.5.2). — *(i) Suppose* S_0 *quasi-compact (resp. quasi-compact and quasi-separated) and that, for some* $\lambda \in L$, $\mathscr{F}_\lambda$ *is quasi-coherent and of finite type (resp. of finite presentation) and* $\mathscr{G}_\lambda$ *quasi-coherent. Then the homomorphism* $u^*_{\mathscr{F},\mathscr{G}}$ *is injective (resp. bijective).*

(ii) Suppose S_0 *quasi-compact and quasi-separated. For every quasi-coherent* $\mathscr{O}_S$*-Module* $\mathscr{F}$ *of finite presentation, there exist* $\lambda \in L$ *and an* $\mathscr{O}_{S_\lambda}$*-Module quasi-coherent of finite presentation* $\mathscr{F}_\lambda$ *such that* $\mathscr{F}$ *is isomorphic to* $u^*_\lambda(\mathscr{F}_\lambda)$.

Proof. (i) We can evidently restrict to the case where $S_0 = S_\lambda$ since the morphisms $u_{0\lambda} : S_\lambda \to S_0$ are affine, hence quasi-compact and separated. Consider first the case where $S_0 = \mathrm{Spec}(A_0)$ is *affine*. Then assertion (i) is equivalent to the

Lemma (8.5.2.1). — *Let* A_0 *be a ring,* $(A_\lambda)_{\lambda \in L}$ *an inductive system of* A_0*-algebras,* $A = \varinjlim A_\lambda$*; let* M_0, N_0 *be two* A_0*-modules, and set* $M_\lambda = M_0 \otimes_{A_0} A_\lambda$, $N_\lambda = N_0 \otimes_{A_0} A_\lambda$, $M = M_0 \otimes_{A_0} A = \varinjlim M_\lambda$, $N = N_0 \otimes_{A_0} A = \varinjlim N_\lambda$. *If* M_0 *is of finite type (resp. of finite presentation), the canonical homomorphism*

$$\varinjlim \mathrm{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \longrightarrow \mathrm{Hom}_A(M, N) \tag{8.5.2.2}$$

is injective (resp. bijective).

We know indeed (Bourbaki, *Alg.*, chap. II, 3^e ed., § 5, n^o 1) that one has canonical functorial isomorphisms

$$\mathrm{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \xrightarrow{\sim} \mathrm{Hom}_{A_0}(M_0, N_\lambda), \qquad \mathrm{Hom}_A(M, N) \xrightarrow{\sim} \mathrm{Hom}_{A_0}(M_0, N)$$

so that the homomorphism (8.5.2.2) is none other, up to canonical isomorphisms, than the canonical homomorphism

$$\varinjlim \mathrm{Hom}_{A_0}(M_0, N_\lambda) \longrightarrow \mathrm{Hom}_{A_0}(M_0, \varinjlim N_\lambda), \tag{8.5.2.3}$$

which, to every inductive system of homomorphisms of A_0-modules $\theta_\lambda : M_0 \to N_\lambda$, associates its inductive limit. **IV-3 | 21**

Now, if M_0 is of finite type (resp. of finite presentation), one has an exact sequence $A_0^m \to M_0 \to 0$ (resp. $A_0^n \to A_0^m \to M_0 \to 0$); since it is clear that (8.5.2.3) is bijective when M_0 is of the form A_0^q, it suffices to use the left-exactness of the functor $M_0 \mapsto \mathrm{Hom}_{A_0}(M_0, P)$ and the exactness of the functor $\varinjlim$ (in the category of abelian groups) to conclude.

Let us pass to the case where S_0 is quasi-compact, and let (U_i) be a finite open affine covering of S_0; for every λ, the $U_{i\lambda} = u_{0\lambda}^{-1}(U_i)$ form a covering of the open affine subsets of S_λ, and the $V_i = u_0^{-1}(U_i)$ a covering of the open affine subsets of S. To show that $u^*_{\mathscr{F},\mathscr{G}}$ is injective, it suffices to show that if $f_\lambda : \mathscr{F}_\lambda \to \mathscr{G}_\lambda$ is such that $f = u^*_\lambda(f_\lambda) = 0$, then there exists $\mu \geqslant \lambda$ such that $u^*_{\lambda\mu}(f_\lambda) = 0$. By virtue of the lemma (8.5.2.1), for each i there exists $\mu_i \geqslant \lambda$ such that $f_\mu|_{U_{i\mu_i}} = 0$. It suffices therefore to take μ greater than all the λ_i.

Suppose further that S_0 is quasi-separated and $\mathscr{F}_0$ is a quasi-coherent $\mathscr{O}_{S_0}$-Module of finite presentation. Note now that S_λ is quasi-separated and $\mathscr{F}_\lambda$ is a quasi-coherent $\mathscr{O}_{S_\lambda}$-Module of finite presentation (**0**, 5.2.5); since, for every pair of indices i, j, $U_{ij\lambda} = U_{i\lambda} \cap U_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.5.2.5) that for every pair of indices i, j, $u^*_\lambda(f^{(i)}_\lambda) = u^*_\lambda(f^{(j)}_\lambda)$ on $U_{ij\lambda}$ by definition, it follows from what we have already seen above that there exists λ_{ij} such that $u^*_{\lambda\mu}(f^{(i)}_\lambda)|_{U_{ij\lambda}} = u^*_{\lambda\mu}(f^{(j)}_\lambda)|_{U_{ij\lambda}}$ for $\mu \geqslant \lambda_{ij}$; taking μ greater than all the λ_{ij}, one sees that $u^*_{\lambda\mu}(f^{(i)}_\lambda)$ and $u^*_{\lambda\mu}(f^{(j)}_\lambda)$ coincide in $U_{i\mu} \cap U_{j\mu}$ for every couple (i, j), and therefore define an homomorphism $f_\mu : \mathscr{F}_\mu \to \mathscr{G}_\mu$ such that $f = u^*_\mu(f_\mu)$.

(ii) Before passing to the proof of (ii), let us note the following corollaries of (i):

□

Corollary (8.5.2.4). — *Suppose* S_0 *quasi-compact,* $\mathscr{F}_\lambda$ *quasi-coherent of finite type,* $\mathscr{G}_\lambda$ *quasi-coherent. Let* $f_\lambda : \mathscr{F}_\lambda \to \mathscr{G}_\lambda$ *be a homomorphism. For* $f = u^*_\lambda(f_\lambda) : \mathscr{F} \to \mathscr{G}$ *to be an isomorphism, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $f_\mu = u^*_{\lambda\mu}(f_\lambda) : \mathscr{F}_\mu \to \mathscr{G}_\mu$ *be so.*

Proof. We can always suppose $S_\lambda = S_0$, and we can further reduce to the case where S_0 is affine, hence also quasi-separated. Then the assertion is equivalent to the lemma (8.5.2.1), for there is by hypothesis an $\mathscr{O}_S$-homomorphism $g : \mathscr{G} \to \mathscr{F}$ such that $g \circ f = 1_{\mathscr{F}}$ and $f \circ g = 1_{\mathscr{G}}$. Since $\mathscr{G}$ is of finite presentation, there exist by virtue of (8.5.2), (i), an index $\nu \geqslant \lambda$ and a homomorphism $g_\nu : \mathscr{G}_\nu \to \mathscr{F}_\nu$ such that $g = u^*_\nu(g_\nu)$ by virtue of (8.5.2), (i); one has therefore $u^*_\nu(g_\nu \circ f_\nu) = 1_{\mathscr{F}}$ and $u^*_\nu(f_\nu \circ g_\nu) = 1_{\mathscr{G}}$, whence the corollary. □

Corollary (8.5.2.5). — *Suppose* S_0 *quasi-compact and quasi-separated. Suppose that* $\mathscr{F}_\lambda$, $\mathscr{G}_\lambda$ *are* $\mathscr{O}_{S_\lambda}$*-Modules quasi-coherent of finite presentation. For* $\mathscr{F}$ *and* $\mathscr{G}$ *to be isomorphic, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $\mathscr{F}_\mu$ *and* $\mathscr{G}_\mu$ *are isomorphic. Moreover, for every isomorphism* $f : \mathscr{F} \xrightarrow{\sim} \mathscr{G}$, *there exist* $\nu \geqslant \mu$ *and an isomorphism* $f_\nu : \mathscr{F}_\nu \xrightarrow{\sim} \mathscr{G}_\nu$ *such that* $f = u^*_\nu(f_\nu)$.

Proof. This follows from (8.5.2.4) and from (8.5.2), (i) since every homomorphism $f : \mathscr{F} \to \mathscr{G}$ is of **IV-3** | 22
the form $u^*_\mu(f_\mu)$ for some $\mu \geqslant \lambda$ and some homomorphism $f_\mu : \mathscr{F}_\mu \to \mathscr{G}_\mu$. □

Proof of (8.5.2), (ii). Consider again first the case where $S_0 = \operatorname{Spec}(A_0)$ is affine. Then the assertion is equivalent to the lemma (5.13.7.1).

In the general case, we start from a finite open affine covering (U_i) of S_0, and for every λ, the $U_{i\lambda} = u_{0\lambda}^{-1}(U_i)$ form an open affine covering of S_λ, and the $V_i = u_0^{-1}(U_i)$ an open affine covering of S. By the affine case, for every i, there exist an index $\lambda(i)$ and an $\mathscr{O}_{U_{i,\lambda(i)}}$-Module quasi-coherent of finite presentation $\mathscr{F}_i$ such that $u^*_{\lambda(i)}(\mathscr{F}_i) = \mathscr{F}|_{V_i}$ (with the notations of (i)). Furthermore, since L is filtered, we can suppose that all the λ_i are equal to a single λ. Note now that S_λ is quasi-separated and $\mathscr{F}_\lambda$ is a quasi-coherent $\mathscr{O}_{S_\lambda}$-Module of finite presentation (**0**, 5.2.5); since, for every pair of indices i, j, $U_{ij\lambda} = U_{i\lambda} \cap U_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.5.2.5) that for every pair (i, j), there exist an index $\lambda_{ij} \geqslant \lambda$ and an isomorphism $\theta_{ij} : \mathscr{F}_i|_{U_{ij\lambda}} \xrightarrow{\sim} u^*_{\lambda,\lambda_{ij}}(\mathscr{F}_j)|_{U_{ij\lambda}}$ such that $u^*_{\lambda_{ij}}(\theta_{ij})$ is the identity automorphism of $\mathscr{F}|_{(V_i \cap V_j)}$; one can further suppose all the λ_{ij} equal to λ. Changing the notations, one can therefore suppose that it holds for every pair (i, j) an isomorphism $\theta_{ij} : \mathscr{F}_i|_{U_{ij\lambda}} \xrightarrow{\sim} \mathscr{F}_j|_{U_{ij\lambda}}$ such that $u^*_\lambda(\theta_{ij})$ is the identity automorphism of $\mathscr{F}|_{(V_i \cap V_j)}$. Finally, for any three indices i, j, k, if one sets $U_{ijk,\lambda} = U_{i\lambda} \cap U_{j\lambda} \cap U_{k\lambda}$, $U_{ijk,\lambda}$ is quasi-compact, and if θ'_{ij}, θ'_{jk}, and θ'_{ik} denote the restrictions of θ_{ij}, θ_{jk}, and θ_{ik} to $U_{ijk,\lambda}$, one has $u^*_\lambda(\theta'_{ij} \circ \theta'_{jk}) = u^*_\lambda(\theta'_{ik})$. There is, hence, by virtue of (i), an index $\mu \geqslant \lambda$ such that $u^*_{\lambda\mu}(\theta'_{ik}) = u^*_{\lambda\mu}(\theta'_{ij} \circ \theta'_{jk})$, hence the isomorphisms $u^*_{\lambda\mu}(\theta'_{ij}) : u^*_{\lambda\mu}(\mathscr{F}_i)|_{U_{ij\mu}} \xrightarrow{\sim} u^*_{\lambda\mu}(\mathscr{F}_j)|_{U_{ij\mu}}$ satisfy the gluing condition, and define on S_μ a quasi-coherent $\mathscr{O}_{\mathrm{S}_\mu}$-Module $\mathscr{F}_\mu$ of finite presentation (**0**, 3.3.1) such that $\mathscr{F}$ is isomorphic to $u_\mu^*(\mathscr{F}_\mu)$. □

(8.5.3). The result of (8.5.2) can be expressed again by saying that if S_0 is quasi-compact and quasi-separated, the category of $\mathscr{O}_\mathrm{S}$-Modules quasi-coherent of finite presentation is determined up to equivalence near the datum of the categories of $\mathscr{O}_{\mathrm{S}_\lambda}$-Modules quasi-coherent of finite presentation, of the functors $u_{\lambda\mu}^*$ between these categories, and of the transition isomorphisms $u_{\mu\nu}^* \circ u_{\lambda\mu}^* \xrightarrow{\sim} u_{\lambda\nu}^*$. In a figurative way, one can say that the datum of a quasi-coherent $\mathscr{O}_\mathrm{S}$-Module of finite presentation $\mathscr{F}$ is "functorially" equivalent to the datum of a quasi-coherent $\mathscr{O}_{\mathrm{S}_\lambda}$-Module of finite presentation $\mathscr{F}'_\lambda$ for some sufficiently large λ and that, if a quasi-coherent $\mathscr{O}_{\mathrm{S}_\mu}$-Module of finite presentation $\mathscr{F}''_\mu$ also has $\mathscr{F}$ as inverse image, then $\mathscr{F}'_\lambda$ and $\mathscr{F}''_\mu$ have the same inverse image in some suitable S_ν ($\nu \geqslant \lambda, \nu \geqslant \mu$).

We will now interpret from this point of view various notions related to quasi-coherent $\mathscr{O}_\mathrm{S}$-Modules.

Corollary (8.5.4). — *Suppose* S_0 *quasi-compact and quasi-separated; then, for every quasi-coherent* $\mathscr{O}_{\mathrm{S}_\lambda}$-*Module* $\mathscr{G}_\lambda$*, the canonical homomorphism* (8.5.1.4) *is bijective.*

Proof. Indeed, it suffices to apply (8.5.2), (i) to the case where $\mathscr{F}_\lambda = \mathscr{O}_{\mathrm{S}_\lambda}$, which is of finite presentation. □

Proposition (8.5.5). — *Suppose* S_0 *quasi-compact, and suppose that* $\mathscr{F}_\lambda$ *is an* $\mathscr{O}_{\mathrm{S}_\lambda}$*-Module quasi-coherent* **IV-3** | 23
of finite presentation. For $\mathscr{F}$ *to be locally free (resp. locally free of rank* n*), it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $\mathscr{F}_\mu$ *be so.*

Proof. The condition being trivially sufficient, let us prove that it is necessary. If $\mathscr{F}$ is locally free (resp. locally free of rank n), there exists a finite open affine covering (V_i) of S such that $\mathscr{F}|_{\mathrm{V}_i}$ is isomorphic to $\mathscr{O}_\mathrm{S}^{n_i}|_{\mathrm{V}_i}$ (resp. $\mathscr{O}_\mathrm{S}^{n}|_{\mathrm{V}_i}$) for every i. By virtue of (8.2.11), there exist $\nu \geqslant \lambda$ and for each i a quasi-compact open subset $\mathrm{U}_{i\nu}$ of S_ν such that $\mathrm{V}_i = u_\nu^{-1}(\mathrm{U}_{i\nu})$. Since S_ν is quasi-compact, each $\mathrm{U}_{i\nu}$ is a finite union of open affines; we are thus reduced to the case where S_0 is affine and $\mathscr{F} = \mathscr{O}_\mathrm{S}^n$; we then know that there exist $\mu \geqslant \lambda$ such that $\mathscr{F}_\mu$ is isomorphic to $\mathscr{O}_{\mathrm{S}_\mu}^n$ (8.5.2.5). □

Proposition (8.5.6). — *Suppose* S_0 *quasi-compact, and consider a sequence*

$$\mathscr{F}_\lambda \longrightarrow \mathscr{G}_\lambda \longrightarrow \mathscr{H}_\lambda \longrightarrow 0$$

of homomorphisms of $\mathscr{O}_{\mathrm{S}_\lambda}$*-Modules quasi-coherent, where* $\mathscr{F}_\lambda$ *and* $\mathscr{G}_\lambda$ *are of finite type and* $\mathscr{H}_\lambda$ *of finite presentation. For the corresponding sequence* $\mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ *to be exact, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that the sequence* $\mathscr{F}_\mu \to \mathscr{G}_\mu \to \mathscr{H}_\mu \to 0$ *be so (in which case it is the same for* $\nu \geqslant \mu$*).*

Proof. The fact that the condition is sufficient and the last assertion result from the fact that the functor u_λ^* (resp. $u_{\mu\nu}^*$) is right-exact. To prove that the condition is necessary, note that it follows from the hypothesis and from (8.5.2), (i) that there exists $\nu \geqslant \lambda$ such that the composite $\mathscr{F}_\nu \to \mathscr{G}_\nu \to \mathscr{H}_\nu$ is zero. If one sets $\mathscr{H}'_\nu = \operatorname{Coker}(\mathscr{F}_\nu \to \mathscr{G}_\nu)$, one has therefore a homomorphism $f_\nu : \mathscr{H}'_\nu \to \mathscr{H}_\nu$; by hypothesis, $u_\nu^*(f_\nu)$ is an isomorphism, and it follows from (8.5.2.4) that there exists $\mu \geqslant \nu$ such that $u_{\nu\mu}^*(f_\nu)$ is an isomorphism, which completes the proof. □

Corollary (8.5.7). — *Suppose* S_0 *quasi-compact,* $\mathscr{F}_\lambda$ *quasi-coherent of finite type, and let* $f_\lambda : \mathscr{F}_\lambda \to \mathscr{G}_\lambda$ *be a homomorphism. For* $f = u_\lambda^*(f_\lambda) : \mathscr{F} \to \mathscr{G}$ *to be surjective, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathscr{F}_\mu \to \mathscr{G}_\mu$ *be so.*

Proof. This is the particular case of (8.5.6) applied to the sequence $0 \to \mathscr{H}_\lambda \to 0 \to 0$, where $\mathscr{H}_\lambda = \operatorname{Coker}(f_\lambda)$, which is quasi-coherent and of finite type (taking into account the fact that one has $\mathscr{H} = \operatorname{Coker}(f)$ and $\mathscr{H}_\mu = \operatorname{Coker}(f_\mu)$, by virtue of the right-exactness of the functors u_λ^* and $u_{\lambda\mu}^*$). □

Corollary (8.5.8). — *Suppose* S_0 *quasi-compact and the morphisms* $u_{\lambda\mu} : \mathrm{S}_\mu \to \mathrm{S}_\lambda$ *flat. Then:*

(i) *Let* $\mathscr{F}_\lambda \xrightarrow{f_\lambda} \mathscr{G}_\lambda \xrightarrow{g_\lambda} \mathscr{H}_\lambda$ *be a sequence of homomorphisms of* $\mathscr{O}_{S_\lambda}$*-Modules quasi-coherent, such that* $\operatorname{Im} f_\lambda$ *and* $\operatorname{Ker} g_\lambda$ *are of finite type. For the corresponding sequence* $\mathscr{F} \xrightarrow{f} \mathscr{G} \xrightarrow{g} \mathscr{H}$ *to be exact, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that the sequence* $\mathscr{F}_\mu \xrightarrow{f_\mu} \mathscr{G}_\mu \xrightarrow{g_\mu} \mathscr{H}_\mu$ *be so (in which case the same holds for* $\nu \geqslant \mu$*).*

(ii) *Let* $f_\lambda : \mathscr{F}_\lambda \to \mathscr{G}_\lambda$ *be a homomorphism of quasi-coherent* $\mathscr{O}_{S_\lambda}$*-Modules, and suppose that* $\operatorname{Ker} f_\lambda$ *is of finite type. For* $f = u_\lambda^*(f_\lambda) : \mathscr{F} \to \mathscr{G}$ *to be injective, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathscr{F}_\mu \to \mathscr{G}_\mu$ *be so.*

Proof. (i) Taking account of (8.3.8), (ii), observe that, by flatness, $\operatorname{Im} f$ and $\operatorname{Ker} g$ (resp. $\operatorname{Im} f_\mu$ and $\operatorname{Ker} g_\mu$ for $\mu \geqslant \lambda$) are the inverse images of $\operatorname{Im} f_\lambda$ and $\operatorname{Ker} g_\lambda$ ($\mathbf{0_I}$, 6.7.2). Suppose that the sequence $\mathscr{F} \xrightarrow{f} \mathscr{G} \xrightarrow{g} \mathscr{H}$ is exact. Since $\operatorname{Im} f_\lambda$ is of finite type, it follows from (8.5.2), (i) that there exists $\mu \geqslant \lambda$ such that the composite $\mathscr{F}_\mu \xrightarrow{f_\mu} \mathscr{G}_\mu \xrightarrow{g_\mu} \mathscr{H}_\mu$ is zero. Changing notation, we may therefore suppose already that $g_\lambda \circ f_\lambda = 0$. Since $\operatorname{Ker} g_\lambda$ is of finite type, it then follows from (8.5.7) that there exists $\mu \geqslant \lambda$ such that the homomorphism $\mathscr{F}_\mu \to \operatorname{Ker} g_\mu$ is surjective, which proves (i).

(ii) This is the special case of (i) applied to the sequence $0 \to \mathscr{F}_\lambda \to \mathscr{G}_\lambda$. □

Lemma (8.5.9). — *Suppose that* S_0 *is quasi-compact and quasi-separated, and that* $\mathscr{F}_\lambda$ *is quasi-coherent and of finite type. Let* $\mathscr{G}'_\lambda$ *and* $\mathscr{G}''_\lambda$ *be two quasi-coherent quotients of* $\mathscr{F}_\lambda$*, with* $\mathscr{G}'_\lambda$ *moreover supposed to be of finite presentation. For* $\mathscr{G}''$ *to be a quotient of* $\mathscr{G}'$*, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $\mathscr{G}''_\mu$ *is a quotient of* $\mathscr{G}'_\mu$*.*

Proof. By hypothesis, there are two surjective homomorphisms $p'_\lambda : \mathscr{F}_\lambda \to \mathscr{G}'_\lambda$ and $p''_\lambda : \mathscr{F}_\lambda \to \mathscr{G}''_\lambda$. By the right exactness of u_λ^* and $u_{\lambda\mu}^*$, the homomorphisms $p' = u_\lambda^*(p'_\lambda)$, $p'' = u_\lambda^*(p''_\lambda)$, $p'_\mu = u_{\lambda\mu}^*(p'_\lambda)$, and $p''_\mu = u_{\lambda\mu}^*(p''_\lambda)$ are also surjective. Moreover, if there exists a homomorphism $f : \mathscr{G}' \to \mathscr{G}''$ (resp. $f_\mu : \mathscr{G}'_\mu \to \mathscr{G}''_\mu$) such that $p'' = f \circ p'$ (resp. $p''_\mu = f_\mu \circ p'_\mu$), that homomorphism is necessarily unique; this shows that the question is local on S_λ, and consequently, since S_0 is quasi-compact and $\mathbf{L}$ is filtered, that we may suppose S_λ affine and therefore quasi-separated. The condition in the statement is plainly sufficient. Conversely, since $\mathscr{G}'_\lambda$ is of finite presentation and S_λ is quasi-compact and quasi-separated, it follows from (8.5.2), (i) that if there exists $f : \mathscr{G}' \to \mathscr{G}''$ such that $p'' = f \circ p'$, then there exist $\mu \geqslant \lambda$ and $f_\mu : \mathscr{G}'_\mu \to \mathscr{G}''_\mu$ such that $f = u_\mu^*(f_\mu)$ and $p''_\mu = f_\mu \circ p'_\mu$, proving the lemma. □

(8.5.10). To finish this section, for every quasi-coherent Module $\mathscr{F}$ on a prescheme, denote by $\mathfrak{Q}(\mathscr{F})$ the set of quotient Modules of $\mathscr{F}$ that are of finite presentation. If $\mathscr{F}_\lambda$ is quasi-coherent and $\mathscr{G}_\lambda \in \mathfrak{Q}(\mathscr{F}_\lambda)$, it follows from the right exactness of $u_{\lambda\mu}^*$ and u_λ^* that $\mathscr{G}_\mu \in \mathfrak{Q}(\mathscr{F}_\mu)$ for $\mu \geqslant \lambda$ and $\mathscr{G} \in \mathfrak{Q}(\mathscr{F})$. It is clear that $(\mathfrak{Q}(\mathscr{F}_\lambda), u_{\lambda\mu}^*)$ is an inductive system of sets and that the $u_\lambda^* : \mathfrak{Q}(\mathscr{F}_\lambda) \to \mathfrak{Q}(\mathscr{F})$ form an inductive system of maps; hence, by passing to the inductive limit, there is a canonical map

$$u_{\mathfrak{Q}} : \varinjlim \mathfrak{Q}(\mathscr{F}_\lambda) \longrightarrow \mathfrak{Q}(\mathscr{F}). \tag{8.5.10.1}$$

Furthermore, if $(\mathscr{F}'_\lambda)$ is a second family of quasi-coherent $\mathscr{O}_{S_\lambda}$-Modules and, for every λ, $\mathscr{F}'_\lambda$ is a quotient of $\mathscr{F}_\lambda$, then $\mathscr{F}'$ is a quotient of $\mathscr{F}$, and one has a commutative diagram

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$$\begin{tikzcd} \varinjlim \mathfrak{Q}(\mathscr{F}_\lambda) \arrow[r, "u_{\mathfrak{Q}}"] & \mathfrak{Q}(\mathscr{F}) \\ \varinjlim \mathfrak{Q}(\mathscr{F}'_\lambda) \arrow[u] \arrow[r, "u_{\mathfrak{Q}}"'] & \mathfrak{Q}(\mathscr{F}') \arrow[u] \end{tikzcd} \tag{8.5.10.2}$$

Proposition (8.5.11). — *Suppose that* S_0 *is quasi-compact (resp. quasi-compact and quasi-separated). Suppose that* $\mathscr{F}_\lambda$ *is quasi-coherent and of finite type (resp. of finite presentation) for every* λ*; then the canonical map* (8.5.10.1) *is injective (resp. bijective).*

Proof. The first assertion follows more precisely from Lemma (8.5.9). For the second, consider a quotient $\mathscr{O}_S$-Module $\mathscr{G}$ of $\mathscr{F}$ that is of finite presentation. It follows from (8.5.2), (ii) that there exist $\lambda' \in \mathbf{L}$ and a quasi-coherent $\mathscr{O}_{S_{\lambda'}}$-Module $\mathscr{G}_{\lambda'}$ of finite presentation such that $\mathscr{G} = u_{\lambda'}^*(\mathscr{G}_{\lambda'})$; since $\mathbf{L}$

is filtered, we may replace λ and λ' by an index majorizing both and suppose that $\lambda' = \lambda$. Consider then the canonical homomorphism $p : \mathscr{F} \to \mathscr{G}$. It follows from (8.5.2), (i) that there exist $\mu \geqslant \lambda$ and a homomorphism $p_\mu : \mathscr{F}_\mu \to \mathscr{G}_\mu$ such that $p = u_\mu^*(p_\mu)$. By (8.5.7), we may moreover suppose that μ is large enough for p_μ to be surjective, which completes the proof. □

8.6. Sub-preschemes of finite presentation of a projective limit of preschemes

(8.6.1). Given a prescheme Y, we denote in this section by $\mathfrak{Spr}(Y)$ the ordered set (**I**, 4.1.10) of sub-preschemes of Y that are *of finite presentation* (**I**, 1.6.1), by $\mathfrak{Spro}(Y)$ (resp. $\mathfrak{Sprf}(Y)$) the subset of $\mathfrak{Spr}(Y)$ formed by the sub-preschemes induced on opens (resp. on closed sub-preschemes) of Y, of finite presentation over Y; it amounts to the same to say that a sub-prescheme Z of Y belongs to $\mathfrak{Spro}(Y)$ (resp. $\mathfrak{Sprf}(Y)$) or that it is induced on an open and that the underlying space is *retrocompact* in Y (resp. that it is closed and that the ideal $\mathscr{J}$ of $\mathscr{O}_Y$ that defines it is *of finite type*, which also means that $j_*(\mathscr{O}_Z) \in \mathfrak{Q}(\mathscr{O}_Y)$ where $j : Z \to Y$ is the canonical injection) (**I**, 1.6.1) and (**I**, 1.4.5).

(8.6.2). We always preserve the notations of (8.2.2) and suppose that S_0 is one of the S_λ. Let Y_λ be a sub-prescheme of S_λ; then $Y_\mu = u_{\mu\lambda}^{-1}(Y_\lambda)$ (resp. $Y = u_\lambda^{-1}(Y_\lambda)$) is a sub-prescheme of S_μ for $\mu \geqslant \lambda$ (resp. of S); it is induced on an open (resp. it is closed) and is of finite presentation on S_μ (resp. S) if Y_λ is of finite presentation on S_λ (resp. S) (**I**, 4.3.2). It follows that $(\mathfrak{Spr}(S_\lambda), u_{\lambda\mu}^{-1})$ (resp. $(\mathfrak{Spro}(S_\lambda), u_{\lambda\mu}^{-1})$, $(\mathfrak{Sprf}(S_\lambda), u_{\lambda\mu}^{-1})$) is an *inductive system* of sets and the maps $u_\lambda^{-1} : \mathfrak{Spr}(S_\lambda) \to \mathfrak{Spr}(S)$ (resp. $\mathfrak{Spro}(S_\lambda) \to \mathfrak{Spro}(S)$, $\mathfrak{Sprf}(S_\lambda) \to \mathfrak{Sprf}(S)$) form an inductive system of maps; whence, by passing to the inductive limit, canonical maps

$$\varinjlim \mathfrak{Spr}(S_\lambda) \longrightarrow \mathfrak{Spr}(S) \tag{8.6.2.1}$$

$$\varinjlim \mathfrak{Spro}(S_\lambda) \longrightarrow \mathfrak{Spro}(S) \tag{8.6.2.2}$$

$$\varinjlim \mathfrak{Sprf}(S_\lambda) \longrightarrow \mathfrak{Sprf}(S). \tag{8.6.2.3}$$

Recall (**I**, 4.1.10) that $\mathfrak{Spr}(X)$, for every prescheme X, is an ordered set by the relation "Z is a sub-prescheme of the sub-prescheme Y" which we write $Z \leqslant Y$. The maps $u_{\lambda\mu}^{-1}$ and u_λ^{-1} are *increasing* for the corresponding order relations in $\mathfrak{Spr}(S_\lambda)$, $\mathfrak{Spr}(S_\mu)$, $\mathfrak{Spr}(S)$. Furthermore, by writing that $\eta \leqslant \eta'$ for two elements of this set whenever there exist λ and two elements Y_λ, Y'_λ of $\mathfrak{Spr}(S_\lambda)$, of which η and η' are the canonical images, and which are such that $Y_\lambda \leqslant Y'_\lambda$; one easily verifies that this does not depend on the choice of representatives, and that one thus has indeed a *relation of order*. The fact that the u_λ^{-1} are increasing immediately implies that the canonical map (8.6.2.1) is *increasing*; the same is evidently true of (8.6.2.2) and (8.6.2.3). **IV-3** | 26

Proposition (8.6.3). — *Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated). Then the maps (8.6.2.1), (8.6.2.2), (8.6.2.3) are injective (resp. bijective).*

Proof. Taking into account the remarks of (8.6.1), the assertions relative to (8.6.2.3) follow from (8.5.11) applied to $\mathscr{F}_\lambda = \mathscr{O}_{S_\lambda}$; similarly, the assertions relative to (8.6.2.2) are special cases of (8.3.5) and (8.3.11), taking into account that the S_λ and S are quasi-compact. It remains to consider the map (8.6.2.1). Let us first prove that it is surjective when S_0 is quasi-compact and quasi-separated. Let Z be a sub-prescheme of S, of finite presentation over S; as S is of the same nature as Z, there exists an open quasi-compact U of S such that Z is a sub-prescheme of U, closed in U, of finite presentation over U. There then exist an index λ and an open quasi-compact U_λ of S_λ such that $U = u_\lambda^{-1}(U_\lambda)$ (8.2.11); as S_λ is quasi-separated, it is of the same nature as U_λ (**I**, 1.2.7), and consequently one can restrict to the case where $U = S$; but in this case, one is reduced to the fact that (8.6.2.3) is surjective.

Finally, to see that (8.6.2.1) is injective when S_0 is quasi-compact, it will suffice to prove the following more precise result: □

Corollary (8.6.3.1). — *Suppose S_0 quasi-compact and let Z'_λ, Z''_λ be two sub-preschemes of S_λ, of finite presentation. For $Z' = u_\lambda^{-1}(Z'_\lambda)$ to be majorised by $Z'' = u_\lambda^{-1}(Z''_\lambda)$* (**I**, 4.1.10), *it is necessary and sufficient that there exist $\mu \geqslant \lambda$ such that $Z'_\mu = u_{\mu\lambda}^{-1}(Z'_\lambda)$ be majorised by $Z''_\mu = u_{\mu\lambda}^{-1}(Z''_\lambda)$.*

Proof. It is trivial that the condition is sufficient. To see that it is necessary, note first that the underlying sets Z'_λ and Z''_λ are locally constructible in S_λ by hypothesis (**I**, 1.8.4), hence the hypothesis implies the existence of a $\nu \geqslant \lambda$ such that $Z'_\lambda \subset Z''_\lambda$ (8.3.5); replacing λ by ν, we can therefore already suppose that $Z'_\lambda \subset Z''_\lambda$. Furthermore, by hypothesis (**I**, 1.6.1), the sub-spaces Z'_λ and Z''_λ of S_λ are quasi-compact; for every point $x \in Z'_\lambda$, there is an open quasi-compact neighbourhood $V(x)$ in S_λ such that $V(x) \cap Z'_\lambda$ and $V(x) \cap Z''_\lambda$ are closed in $V(x)$. By covering S_λ by a finite number of such neighbourhoods, we see that there is an open quasi-compact U_λ of S_λ containing Z'_λ and such that Z'_λ and $Z''_\lambda \cap U_\lambda$ are closed in U_λ. If we denote by Y_λ the sub-prescheme of S_λ induced by Z''_λ on $U_\lambda \cap Z''_\lambda$, it is clear that with the usual notations, Y_μ (resp. Y) is majorised by Z''_μ (resp. Z'') for $\mu \geqslant \lambda$ (**I**, 4.4.1), and as it suffices to prove that Z'_μ is majorised by Y_μ for μ large enough, we see finally that one is reduced (replacing S_λ by U_λ) to the case where Z'_λ and Z''_λ are closed sub-preschemes of S_λ. But then, this has already been proved since (8.6.2.3) is increasing and injective. □ **IV-3 | 27**

Corollary (8.6.4). — *Suppose S_0 quasi-compact, and let Z_λ be a sub-prescheme of S_λ, of finite presentation over S_λ. For $Z = u_\lambda^{-1}(Z_\lambda)$ to be the subprescheme induced on an open subset of S (resp. a closed subprescheme of S), it is necessary and sufficient that there exist $\mu \geqslant \lambda$ such that $Z_\mu = u_{\mu\lambda}^{-1}(Z_\lambda)$ is the subprescheme induced on an open subset of S_μ (resp. a closed subprescheme of S_μ).*

Proof. Let $(U_\lambda^{(i)})$ be a finite affine open covering of S_λ, and set $U_\mu^{(i)} = u_{\mu\lambda}^{-1}(U_\lambda^{(i)})$ for $\mu \geqslant \lambda$ and $U^{(i)} = u_\lambda^{-1}(U_\lambda^{(i)})$. If Z is open (resp. closed) in S, $Z \cap U^{(i)}$ is open (resp. closed) in $U^{(i)}$, and conversely if each of the $Z_\mu \cap U_\mu^{(i)}$ is open (resp. closed) in $U_\mu^{(i)}$, Z_μ is open (resp. closed) in S_μ. Since L is filtered, it suffices to prove the corollary when S_λ is affine, hence quasi-separated, and the result follows from the fact that the maps (8.6.2.1), (8.6.2.2), and (8.6.2.3) are bijective. □

8.7. Criteria for a projective limit of preschemes to be a reduced (resp. integral) prescheme We preserve the hypotheses and notations of (8.2.2) and suppose always that S_0 is one of the S_λ.

Proposition (8.7.1). — *Suppose that S is not reduced. Then there exists λ_0 such that for $\lambda \geqslant \lambda_0$, S_λ is not reduced.*

Proof. The question being local on S_0, we can suppose $S_0 = \operatorname{Spec}(A_0)$ affine, whence $S = \operatorname{Spec}(A)$, where $A = \varinjlim A_\lambda$ is the inductive limit of an inductive system of A_0-algebras (A_λ). We know then (5.13.2) that the nilradical of A is the inductive limit of those of the A_λ; if it is not zero, one of the A_λ contains therefore a nilpotent element $a_\lambda \neq 0$, and the image of a_λ in the A_μ for $\mu \geqslant \lambda$ is therefore a nilpotent element $\neq 0$, and the image of a_λ in A is not zero for $\mu \geqslant \lambda$ since the image in A is not zero. □

Proposition (8.7.2). — *Suppose one of the following hypotheses satisfied:*

a) *S_0 is quasi-compact, the nilradical $\mathscr{N}_0$ of $\mathscr{O}_{S_0}$ is an ideal of finite type (which is satisfied for example when S_0 is Noetherian), and the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are open immersions.*
b) *The morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are faithfully flat.*

Under these conditions, for S to be reduced, it is necessary and sufficient that there exist λ_0 such that S_λ be reduced for $\lambda \geqslant \lambda_0$.

Proof. Furthermore, in case b), if S is reduced, all the S_λ are reduced.

The last assertion follows from the fact that the morphism $S \to S_\lambda$ is then faithfully flat for every λ (8.3.8) and from (2.1.13). On the other hand, (8.7.1) proves that the condition of the statement is sufficient (without hypothesis on S_0 or on the $u_{\lambda\mu}$). It therefore remains to see that the condition is necessary in hypothesis a); then (8.2.13), the underlying space of S is identified with S (the S_λ being identified with sub-preschemes induced on opens of S_0), and the structural sheaf $\mathscr{O}_S$ is identified with the sheaf induced on S by all the $\mathscr{O}_{S_\lambda}$; in particular for every $s \in S$, the local ring $\mathscr{O}_s$ is the same for S and for all the S_λ. If $\mathscr{N}_\lambda$ is the nilradical of $\mathscr{O}_{S_\lambda}$, the nilradical $\mathscr{N}$ of $\mathscr{O}_S$ has at each point of S the same fibre $\mathscr{N}_s$ (induced on S by $\mathscr{N}_\lambda$). The hypothesis that S is reduced implies therefore $u_\lambda^*(\mathscr{N}_\lambda) = 0$; as $\mathscr{N}_0$ is supposed of finite type, the same is true of the $\mathscr{N}_\lambda$, and the conclusion follows then from (8.5.7). □ **IV-3 | 28**

Corollary (8.7.3). — *Suppose one of the following hypotheses satisfied:*

a) *S_0 is a Noetherian prescheme and the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are open immersions.*
b) *The morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are faithfully flat.*

Then, for S to be integral, it is necessary and sufficient that there exist λ_0 such that S_λ be integral for $\lambda \geqslant \lambda_0$.

Proof. To say that a prescheme is integral means that it is both reduced and irreducible; the corollary thus follows from (8.7.2) and from (8.4.3). □

Remark (8.7.4). — If one makes no hypothesis on the $u_{\lambda\mu}$, it can happen that S is integral even though all the S_λ are neither reduced nor connected, as the example (8.4.4) shows, where one takes the ring A non reduced.

8.8. Preschemes of finite presentation over a projective limit of preschemes

(8.8.1). Always preserving the notations and hypotheses of (8.2.2), we shall suppose given in this section two S_α-preschemes X_α, Y_α, which defines (8.2.5) two projective systems of preschemes $(X_\lambda, v_{\lambda\mu})$ and $(Y_\lambda, w_{\lambda\mu})$ by setting $X_\lambda = X_\alpha \times_{S_\alpha} S_\lambda$, $v_{\lambda\mu} = 1_{X_\alpha} \times u_{\lambda\mu}$, $w_{\lambda\mu} = 1_{Y_\alpha} \times u_{\lambda\mu}$ (for $\alpha \leqslant \lambda \leqslant \mu$), whose projective limits are respectively $X = X_\alpha \times_{S_\alpha} S$, $Y = Y_\alpha \times_{S_\alpha} S$, with the canonical morphisms $v_\lambda : X \to X_\lambda$ and $w_\lambda : Y \to Y_\lambda$. For $\alpha \leqslant \lambda \leqslant \mu$, there is a canonical map $\varepsilon_{\mu\lambda} : \mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \to \mathrm{Hom}_{S_\mu}(X_\mu, Y_\mu)$, which to every S_λ-morphism $f_\lambda : X_\lambda \to Y_\lambda$ associates $f_\mu = f_\lambda \times 1_{S_\mu} : X_\lambda \times_{S_\lambda} S_\mu \to Y_\lambda \times_{S_\lambda} S_\mu$, and it is clear that $(\mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda), \varepsilon_{\mu\lambda})$ is an *inductive system* of sets. Similarly, there is a canonical map $\varepsilon_\lambda : \mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \to \mathrm{Hom}_S(X, Y)$ which to f_λ associates $f = f_\lambda \times 1_S : X_\lambda \times_{S_\lambda} S \to Y_\lambda \times_{S_\lambda} S$ and (ε_λ) is an inductive system of maps; whence, by passing to the inductive limit, a canonical map, functorial in S_α, X_α, and Y_α:

$$\varepsilon : \varinjlim \mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \mathrm{Hom}_S(X, Y). \tag{8.8.1.1}$$

Theorem (8.8.2). — *(i) Suppose X_α quasi-compact (resp. quasi-compact and quasi-separated), and Y_α locally of finite type (resp. locally of finite presentation) over S_α. Then the map (8.8.1.1) is injective (resp. bijective).*

(ii) Suppose S_0 quasi-compact and quasi-separated. For every prescheme X of finite presentation over S, there exist a $\lambda \in L$, a prescheme X_λ of finite presentation over S_λ, and an S-isomorphism $X \xrightarrow{\sim} X_\lambda \times_{S_\lambda} S$.

Proof. (i) Consider first the case where $S_0 = \mathrm{Spec}(A_0)$, $X_\alpha = \mathrm{Spec}(B_\alpha)$, and $Y_\alpha = \mathrm{Spec}(C_\alpha)$ are affine; then the $S_\lambda = \mathrm{Spec}(A_\lambda)$ and $S = \mathrm{Spec}(A)$ are also affine, with $A = \varinjlim A_\lambda$, and the assertions of (i) are equivalent to the IV-3 | 29

Lemma (8.8.2.1). — *Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ an inductive system of A_0-algebras, $A = \varinjlim A_\lambda$; let B_α be an A_α-algebra, C_α an A_α-algebra of finite type (resp. of finite presentation). Then the canonical homomorphism*

$$\varinjlim_\lambda \mathrm{Hom}_{A_\lambda\text{-}alg.}(C_\alpha \otimes_{A_\alpha} A_\lambda, B_\alpha \otimes_{A_\alpha} A_\lambda) \longrightarrow \mathrm{Hom}_{A\text{-}alg.}(C_\alpha \otimes_{A_\alpha} A, B_\alpha \otimes_{A_\alpha} A) \tag{8.8.2.2}$$

is injective (resp. bijective).

Proof. We know that there are functorial canonical isomorphisms

$$\mathrm{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, B_\alpha \otimes_{A_\alpha} A_\lambda) \xrightarrow{\sim} \mathrm{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, B_\alpha \otimes_{A_\alpha} A_\lambda)$$

$$\mathrm{Hom}_{A\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A, B_\alpha \otimes_{A_\alpha} A) \xrightarrow{\sim} \mathrm{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, B_\alpha \otimes_{A_\alpha} A)$$

by virtue of the universal property of the tensor product of two algebras. It therefore suffices to prove the □

Lemma (8.8.2.3). — *Let E be a ring, G an E-algebra of finite type (resp. of finite presentation), (F_λ) an inductive system of E-algebras. Then the canonical homomorphism*

$$\varinjlim \mathrm{Hom}_{E\text{-}alg.}(G, F_\lambda) \longrightarrow \mathrm{Hom}_{E\text{-}alg.}(G, \varinjlim F_\lambda)$$

which, to every inductive system of homomorphisms $\theta_\lambda : G \to F_\lambda$ of E-algebras, associates its inductive limit, is injective (resp. bijective).

Proof. Suppose first the E-algebra G of finite type, and let $(t_i)_{1\leqslant i\leqslant n}$ be a system of generators of this E-algebra; let us show that if (θ'_λ), (θ''_λ) are two inductive systems of homomorphisms $G \to F_\lambda$, with $\varinjlim \theta'_\lambda = \varinjlim \theta''_\lambda$, there exists $\mu \geqslant \lambda$ such that $\theta'_\mu = \theta''_\mu$. Indeed, if $\varphi_{\mu\lambda} : F_\lambda \to F_\mu$ and $\varphi_\lambda : F_\lambda \to F = \varinjlim F_\lambda$ are the homomorphisms of the inductive system (F_λ), by hypothesis, for each index i, there exists an index λ_i such that $\varphi_{\lambda_i}(\theta'_{\lambda_i}(t_i)) = \varphi_{\lambda_i}(\theta''_{\lambda_i}(t_i))$, and one can suppose all the λ_i equal to a single λ; it follows in the same way the existence of $\mu \geqslant \lambda$ such that $\varphi_{\mu\lambda}(\theta'_\lambda(t_i)) = \varphi_{\mu\lambda}(\theta''_\lambda(t_i))$ for $1 \leqslant i \leqslant n$, that is to say $\theta'_\mu(t_i) = \theta''_\mu(t_i)$ for $1 \leqslant i \leqslant n$, whence $\theta'_\mu = \theta''_\mu$.

Suppose in the second place G of finite presentation, so that one has

$$G = E[T_1, \dots, T_n]/\mathfrak{J},$$

where $\mathfrak{J}$ is an ideal of finite type, t_i being the class of T_i modulo $\mathfrak{J}$. Let $(P_j)_{1\leqslant j\leqslant m}$ be a system of generators of $\mathfrak{J}$. Suppose given a homomorphism of E-algebras $\theta : G \to F$; set $b_i = \theta(t_i)$; by definition, one has therefore $P_j(b_1, b_2, \dots, b_n) = \theta(P_j(t_1, \dots, t_n)) = 0$ for $1 \leqslant j \leqslant m$. Now, there exist λ and elements $x_1, \dots, x_n$ in F_λ such that $b_i = \varphi_\lambda(x_i)$ for $1 \leqslant i \leqslant n$; one has therefore $\varphi_\lambda(P_j(x_1, \dots, x_n)) = P_j(b_1, \dots, b_n) = 0$, and consequently there exists $\mu \geqslant \lambda$ such that **IV-3 | 30**
$\varphi_{\mu\lambda}(P_j(x_1, \dots, x_n)) = P_j(\varphi_{\mu\lambda}(x_1), \dots, \varphi_{\mu\lambda}(x_n)) = 0$ for $1 \leqslant j \leqslant m$; one therefore deduces for every $\nu \geqslant \mu$ a homomorphism of E-algebras $\theta_\nu = \varphi_{\nu\mu} \circ \theta_\mu$ of G into F_ν, and it is clear that θ is the inductive limit of this inductive system of homomorphisms, which completes the proof of the lemma. □

Let us now pass to the case where X_α is quasi-compact and Y_α locally of finite type over S_α. Set $Z_\alpha = X_\alpha \times_{S_\alpha} Y_\alpha$ and introduce the projective system corresponding to $Z_\lambda = X_\lambda \times_{S_\lambda} Y_\lambda$ and its limit $Z = X \times_S Y$; the canonical bijections (**I**, 3.3.14) give commutative diagrams

$$\begin{tikzcd}
\mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \arrow[r, "\varepsilon_{\mu\lambda}"] \arrow[d, "\wr"] & \mathrm{Hom}_{S_\mu}(X_\mu, Y_\mu) \arrow[r, "\varepsilon_\mu"] \arrow[d, "\wr"] & \mathrm{Hom}_S(X, Y) \arrow[d, "\wr"] \\
\mathrm{Hom}_{X_\lambda}(X_\lambda, Z_\lambda) \arrow[r, "\varepsilon_{\mu\lambda}"] & \mathrm{Hom}_{X_\mu}(X_\mu, Z_\mu) \arrow[r, "\varepsilon_\mu"] & \mathrm{Hom}_X(X, Z)
\end{tikzcd}$$

and consequently we are reduced to proving that (8.8.1.1) is injective in the particular case where $S_\alpha = X_\alpha$ (taking into account (**I**, 1.3.4)). Moreover, as X_α is quasi-compact, hence a finite union of open affines, and as X is a union of the opens $f^{-1}(V_i)$, where the V_i are open affines of Y_α, every point of X has an open affine neighbourhood $W(x)$ in S_α such that f'_α and f''_α both map $W(x)$ into a single open affine $V_{i\alpha}$. As X_α is quasi-compact, it is covered by a finite number of open affines $W(x_k)$; this implies that for each $x \in X_\alpha$, there is an open affine neighbourhood of x such that the restrictions of f'_α and f''_α to $W(x)$ map it into the same open affine $V_{i\alpha}$. As X_α is quasi-compact, it is covered by a finite number of such open affines $W(x_k)$; by virtue of the lemma (8.8.2.1) and of the fact that L is filtered, there exists $\lambda \geqslant \alpha$ such that the restrictions of f'_λ and f''_λ to each of the opens $v_\alpha^{-1}(W(x_k))$ are equal, whence $f'_\lambda = f''_\lambda$.

Suppose now X_α quasi-compact and quasi-separated and Y_α locally of finite presentation over S_α, and let us prove that (8.8.1.1) is surjective. Suppose therefore given an X-morphism $f : X \to Y$. As X is quasi-compact, it is of the same nature as Y, which contains $f(X)$; there exists therefore $\lambda \geqslant \alpha$ and an open quasi-compact Y'_λ in Y_λ such that $Y' = w_\lambda^{-1}(Y'_\lambda)$ (8.2.11). Replacing as needed α by λ and Y_λ by Y'_λ, we can therefore restrict to the case where Y_α is *quasi-compact*, hence also Y and the Y_λ. As Y is quasi-compact, it is a finite union of open affines V_i, and X is a union of the opens $f^{-1}(V_i)$. As X is quasi-compact, we can cover X by a finite number of such open affines. And since X is quasi-compact, for each i, the open quasi-compact sets in X_α may, by replacing α by **IV-3 | 31**
an index $\lambda \geqslant \alpha$, be supposed such that $U_i = v_\alpha^{-1}(U_{i\alpha})$ where the $U_{i\alpha}$ are open quasi-compacts in X_α; moreover, using (8.3.4), on can moreover suppose the $(U_{i\alpha})$ form a *covering* of X_α.

It will then suffice to show that there exist $\lambda \geqslant \alpha$ and for each i a morphism $f_{i\lambda} : U_{i\lambda} \to V_{i\lambda}$ (where $U_{i\lambda} = v_{\alpha\lambda}^{-1}(U_{i\alpha})$ and $V_{i\lambda} = w_{\alpha\lambda}^{-1}(V_{i\alpha})$) such that the morphism $f_i = \varepsilon_\lambda(f_{i\lambda}) : U_i \to V_i$ be equal to the *restriction* of f to U_i. Indeed, if this is so, as X_α is quasi-separated (**I**, 1.2.3), the $U_{i\lambda} \cap U_{j\lambda}$ are *quasi-compact* and the uniqueness result proved above (which applies since Y_λ is locally of finite type over S_λ) proves that there exists $\mu \geqslant \lambda$ such that $f_{i\mu}$ and $f_{j\mu}$ coincide on $U_{i\mu} \cap U_{j\mu}$ for every couple (i, j), and consequently define a S_μ-morphism $f_\mu : X_\mu \to Y_\mu$ such that $\varepsilon_\mu(f_\mu) = f$.

We are thus reduced to the case where Y_α is *affine*, and as in addition we can also suppose that S_α is affine, we can also suppose $S_\alpha = \operatorname{Spec}(A_\alpha)$, $Y_\alpha = \operatorname{Spec}(C_\alpha)$, C_α being an A_α-algebra of finite presentation, $S = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(C)$, with $A = \varinjlim(C_\alpha \otimes_{A_\alpha} A_\lambda) = C_\alpha \otimes_{A_\alpha} A$. One has then

$$\operatorname{Hom}_S(X, Y) = \operatorname{Hom}_{A\text{-alg.}}(C, \Gamma(X, \mathscr{O}_X)) = \operatorname{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, \Gamma(X, \mathscr{O}_X))$$

(**I**, 2.2.4) and similarly

$$\operatorname{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) = \operatorname{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, \Gamma(X_\lambda, \mathscr{O}_{X_\lambda})) = \operatorname{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, \Gamma(X_\lambda, \mathscr{O}_{X_\lambda})).$$

But as X_α is quasi-compact and quasi-separated, one knows (8.5.4) that $\varinjlim \Gamma(X_\lambda, \mathscr{O}_{X_\lambda}) = \Gamma(X, \mathscr{O}_X)$; since C_α is an A_α-algebra of finite presentation, the fact that (8.8.1.1) is bijective then follows from (8.8.2.3).

Before passing to the proof of (ii), let us note the following corollaries of (i): □

Corollary (8.8.2.4). — *Suppose S_0 quasi-compact, X_α of finite presentation over S_α, Y_α of finite type over S_α and quasi-separated over S_α (which is the case, for example, if Y_α is also of finite presentation over S_α).* **IV-3** | 32
Let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism. For $f : X \to Y$ to be an isomorphism, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $f_\lambda : X_\lambda \to Y_\lambda$ is an isomorphism.

Proof. The condition is evidently sufficient. To prove that it is necessary, first observe that, since the question is local on S_0 (because **L** is filtered), we may always suppose S_0 affine and hence quasi-separated. By hypothesis there is an S-morphism $g : Y \to X$ such that $g \circ f = 1_X$ and $f \circ g = 1_Y$. Since X_α is of finite presentation over S_α, and Y_α is quasi-compact and quasi-separated (because S_α is quasi-compact and quasi-separated), it follows from (8.8.2), (ii) that there exist $\mu \geqslant \alpha$ and an S_μ-morphism $g_\mu : Y_\mu \to X_\mu$ such that $g = g_\mu \times 1_S$. It follows moreover from (8.8.2), (i), together with the relations $g \circ f = 1_X$ and $f \circ g = 1_Y$, that there exists $\nu \geqslant \mu$ such that $g_\nu \circ f_\nu = 1_{X_\nu}$ and $f_\nu \circ g_\nu = 1_{Y_\nu}$, since X_α and Y_α are of finite type over S_α and quasi-compact. Thus $f_\nu : X_\nu \to Y_\nu$ is an isomorphism, proving the corollary. □

Corollary (8.8.2.5). — *Suppose S_0 quasi-compact and quasi-separated, X_α and Y_α of finite presentation over S_α. For X and Y to be S-isomorphic, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that X_λ and Y_λ are S_λ-isomorphic. Moreover, for every S-isomorphism $f : X \xrightarrow{\sim} Y$, there exist $\mu \geqslant \lambda$ and an S_μ-isomorphism $f_\mu : X_\mu \xrightarrow{\sim} Y_\mu$ such that $f = f_\mu \times 1_S$.*

Proof. The condition is evidently sufficient. Conversely, if there exists an S-isomorphism $f : X \to Y$, it follows from (8.8.2), (ii) that f is of the form $f_\mu \times 1_S$ for some $\mu \geqslant \alpha$ and some S_μ-morphism $f_\mu : X_\mu \to Y_\mu$. Since f is an isomorphism, it follows from (8.8.2.4) that there exists $\nu \geqslant \mu$ such that $f_\nu : X_\nu \to Y_\nu$ is an isomorphism. □

Proof of (8.8.2), (ii). Consider first the case where $S_0 = \operatorname{Spec}(A_0)$ and $X = \operatorname{Spec}(B)$ are affine; then assertion (ii) is equivalent to Lemma (**I**, 8.4.2).

To prove (ii) in the general case, note that S is quasi-compact and quasi-separated. Since X is of finite presentation over S, and S is affine over S_0, there is a finite open cover (U_i) of S_0 and, if (W_i) denotes the affine open cover of S formed by the inverse images of the U_i under $S \to S_0$, a finite cover (X_r) of X by affine opens such that the image of each X_r under $X \to S$ is contained in some $W_{i(r)}$. The ring $A(X_r)$ is then an algebra of finite presentation over $A(W_{i(r)})$ (**I**, 1.4.6).

By Lemma (**I**, 8.4.2) and the fact that L is filtered, there exist an index $\lambda \in L$ and, for every r, an affine scheme $Z_{r\lambda}$ of finite presentation over the inverse image $W_{i(r),\lambda}$ of $U_{i(r)}$ in S_λ, together with an S-isomorphism

$$g_r : Z_{r\lambda} \times_{S_\lambda} S \xrightarrow{\sim} X_r.$$

Let Z_{rs} be the inverse image under g_r of the sub-prescheme induced on the open $X_r \cap X_s$ of X_r, which is quasi-compact because X is quasi-separated, and let

$$g'_{rs} : Z_{rs} \xrightarrow{\sim} X_r \cap X_s$$

be the restriction of g_r. By (8.8.2.5), there exist $\mu \geqslant \lambda$ and, for every pair (r, s), a quasi-compact open $Z_{rs\mu}$ of $Z_{r\mu} = v_{\lambda\mu}^{-1}(Z_{r\lambda})$ such that Z_{rs} is the inverse image of $Z_{rs\mu}$. Moreover, since S_μ is quasi-separated and $W_{i(r),\mu}$ is a quasi-compact open of S_μ, each $Z_{rs\mu}$ is of finite presentation over S_μ (**I**, 1.6.2).

For every pair (r,s), consider the isomorphism

$$h_{sr} = (g'_{sr})^{-1} \circ g'_{rs} : Z_{rs} \xrightarrow{\sim} Z_{sr}.$$

By (8.8.2.4), there exist $\nu \geqslant \mu$ and, for every pair (r,s), an isomorphism

$$h_{sr\nu} : Z_{rs\nu} \xrightarrow{\sim} Z_{sr\nu}$$

such that $h_{sr} = h_{sr\nu} \times 1_S$. Finally, for every triple (r,s,t), let h'_{sr} denote the restriction of h_{sr} to $Z_{rs} \cap Z_{rt}$. It follows immediately from the definitions that h'_{sr} is an isomorphism from $Z_{rs} \cap Z_{rt}$ onto $Z_{sr} \cap Z_{st}$ and that **IV-3 | 33**

$$h'_{ts} \circ h'_{sr} = h'_{tr}.$$

By (8.8.2), (i), there therefore exists $\rho \geqslant \nu$ such that, for every triple (r,s,t),

$$h'_{ts\rho} \circ h'_{sr\rho} = h'_{tr\rho}.$$

Consequently the isomorphisms $h_{sr\rho}$ satisfy the gluing condition ($\mathbf{0_I}$, 4.1.7) and hence define a prescheme X_ρ such that X is isomorphic to $X_\rho \times_{S_\rho} S$. Moreover, the $Z_{r\rho}$ are of finite presentation over S_ρ, and, identifying them with open subsets of X_ρ, their intersections $Z_{r\rho} \cap Z_{s\rho}$, which are isomorphic to the $Z_{rs\rho}$, are quasi-compact. Thus X_ρ is of finite presentation over S_ρ (**I**, 1.6.3). ☐

(8.8.3). The essential content of (8.8.2) is expressed again by saying that if S_0 is quasi-compact and quasi-separated, the category of S-preschemes of finite presentation is determined up to equivalence by the data of the categories of S_λ-preschemes of finite presentation, the functors $\rho_{\mu\lambda} : X_\lambda \mapsto X_\lambda \times_{S_\lambda} S_\mu$ (for $\lambda \leqslant \mu$) between these categories and the transitivity isomorphisms $\rho_{\mu\lambda} \circ \rho_{\nu\mu} \xrightarrow{\sim} \rho_{\nu\lambda}$. In the form thus stated, one can say that the data of an S_λ-prescheme of finite presentation X is equivalent "functorially" to the data of an S_μ-prescheme of finite presentation X'_μ; if an S_μ-prescheme X''_μ is such that X is S-isomorphic to $X''_\mu \times_{S_\mu} S$, there exists ν such that $\lambda \leqslant \nu$ and $X'_\lambda \times_{S_\lambda} S_\nu$ and $X''_\mu \times_{S_\mu} S_\nu$ are S_ν-isomorphic. The fact that L is filtered implies in addition that if one gives oneself a finite *family* $(X^{(i)})_{i\in I}$ of S-preschemes of finite presentation, and a finite family $(f^{(ij)})_{(i,j)\in J}$ of S-morphisms between these preschemes ($f^{(ij)}$ being therefore of the form $X^{(\sigma(j))} \to X^{(\tau(j))}$ where σ and τ are two maps from J to I), then there is an index $\lambda \in L$, a family $(X^{(i)}_\lambda)_{i\in I}$ of S_λ-preschemes of finite presentation and a family $(f^{(ij)}_\lambda)_{(i,j)\in J}$ of S_λ-morphisms such that $X^{(i)}$ is identified with $X^{(i)}_\lambda \times_{S_\lambda} S$ and $f^{(ij)}$ with $f^{(ij)}_\lambda \times 1_S$; furthermore, if one has a relation $f^{(ij)} = f^{(ik)}$, one can suppose λ chosen such that $f^{(ij)}_\lambda = f^{(ik)}_\lambda$.

Consider in particular three S-preschemes of finite presentation X, Y, Z and two S-morphisms $f : X \to Y$, $g : Y \to Z$, so that the product $X \times_Z Y$ relative to these morphisms is also an S-prescheme of finite presentation (**I**, 1.6.2). It then results from the preceding and from (**I**, 3.3.11) that one can write $X \times_Z Y = (X_\lambda \times_{Z_\lambda} Y_\lambda) \times_{S_\lambda} S$ for a convenient λ, X_λ, Y_λ, Z_λ being S_λ-preschemes of finite presentation; one can therefore say that the determination of S-preschemes of finite presentation by the data of S_λ-preschemes of finite presentation is "*compatible with fibre products*". We have seen on the other hand (8.6.3) that if g is an immersion, we can suppose the same for g_λ; identifying Y (resp. Y_λ) with a sub-prescheme of Z (resp. Z_λ), we see that one can write, for a convenient λ, $f^{-1}(Y) = f_\lambda^{-1}(Y_\lambda) \times_{S_\lambda} S$ (**I**, 4.4.1); there is therefore also "*compatibility with the formation of inverse images of sub-preschemes*". More particularly, if f_1, f_2 are two S-morphisms from X to Y, one calls the *kernel* of this pair of morphisms the inverse image N of the diagonal sub-prescheme of $Y \times_S Y$ by the S-morphism $(f_1, f_2)_S$; one deduces from what precedes that one will have for a convenient λ, $N = N_\lambda \times_{S_\lambda} S$, where N_λ is the kernel of the couple of S_λ-morphisms $(f_{1\lambda}, f_{2\lambda})$ of X_λ into Y_λ. These remarks extend to finite products and to "kernels" of arbitrary finite systems of S-morphisms from an S-prescheme of finite presentation into another; one concludes that in a general fashion the formation of S-preschemes of finite presentation by the data of S_λ-preschemes of finite presentation is "*compatible with finite projective limits*", such a limit being by definition the kernel of a finite system of morphisms from one S-prescheme into a product of S-preschemes (**T**, 1.8). **IV-3 | 33** **IV-3 | 34**

We shall allow ourselves in what follows to freely make the translations implied by the preceding properties (as well as by (8.3.11), (8.5.2), and (8.6.3)) without always striving to justify them as above.

For example, the data of a "group prescheme" G of finite presentation over S is equivalent to the data of a group prescheme on S_λ for a large enough λ; to write indeed the associativity condition for the S-morphism "composition law" $G \times_S G \to G$ of the group prescheme amounts to writing that the kernel of the two S-morphisms from $G \times_S G \times_S G \to G$ is equal to $G \times_S G \times_S G$ (**II**, 8.3.9), and the other conditions intervening in the definition of a group prescheme are interpreted similarly.

8.9. First applications to the elimination of Noetherian hypotheses

Proposition (8.9.1). — *Let A be a ring, X an A-prescheme.*

(i) The following conditions are equivalent:

a) *X is of finite presentation over A.*

b) *There exist a Noetherian ring A_0, a prescheme X_0 of finite type over A_0, a ring homomorphism $A_0 \to A$, and an A-isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*

c) *There exist a subring A_0 of A, which is a $\mathbf{Z}$-algebra of finite type, a prescheme X_0 of finite type over A_0, and an A-isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*

(ii) If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-Module of finite presentation, one can suppose the subring A_0 such that there exists a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ such that $\mathscr{F}$ is isomorphic to $\mathscr{F}_0 \otimes_{A_0} A$; $\operatorname{Supp}(\mathscr{F})$ is constructible and closed in X, and there is a closed sub-prescheme Z of X, having $\operatorname{Supp}(\mathscr{F})$ as underlying space, such that the canonical immersion $Z \to X$ be of finite presentation.

(iii) If Y is a second A-prescheme of finite presentation, and $f : X \to Y$ an A-morphism, one can suppose the subring A_0 of A such that there exists a prescheme Y_0 of finite type over A_0, an A-isomorphism $Y_0 \otimes_{A_0} A \xrightarrow{\sim} Y$, and an A_0-morphism $f_0 : X_0 \to Y_0$ (necessarily of finite type) such that f is identified with $f_0 \otimes 1_A$.

Proof. (i) As A is the inductive limit of its subrings which are of finite type over $\mathbf{Z}$, *a)* implies *c)* by virtue of (8.8.2), (ii); *c)* implies *b)* since a $\mathbf{Z}$-algebra of finite type is a Noetherian ring; finally, if A_0 is Noetherian, an A_0-prescheme of finite type is of finite presentation (**I**, 1.6.1), hence *b)* implies *a)* by virtue of (**I**, 1.6.2), (iii).

The assertion (ii) follows immediately from (8.5.2), (ii), (8.3.11), and (**I**, 1.8.2), and the assertion (iii) from (8.8.2), (i). □

(8.9.2). The proposition (8.9.1) and the results of (8.5.2) and (8.8.2) allow one to extend in numerous cases to morphisms of finite presentation $X \to Y$ the results proved by Noetherian techniques by supposing Y *locally Noetherian*. We shall see numerous examples in what follows and we shall limit ourselves here to giving a few results of this type. **IV-3** | 35

Proposition (8.9.3). — *Let A be a ring, M an A-module of finite presentation. Then every surjective A-endomorphism of M is bijective.*

Proof. Indeed, consider A as an inductive limit of its sub-$\mathbf{Z}$-algebras of finite type. It follows from (8.5.2), (ii) that there exists one of these sub-algebras A_0 and an A_0-module M_0 of finite presentation such that M is A-isomorphic to $M_0 \otimes_{A_0} A$; furthermore, if $f : M \to M$ is a surjective A-endomorphism, one can suppose (8.5.2), (i) that there exists an A_0-endomorphism $f_0 : M_0 \to M_0$ such that $f = f_0 \otimes 1_A$; finally, by (8.5.7), one can suppose that f_0 is *surjective*. But as A_0 is Noetherian and M_0 an A_0-module of finite type, M_0 is a Noetherian module, hence (Bourbaki, *Alg.*, chap. VIII, § 2, n° 2, lemma 3) f_0 is bijective, and it is the same for f by base change. □

Proposition (8.9.4). — *("generic flatness theorem"). — Let Y be an integral prescheme, $u : X \to Y$ a morphism of finite type and locally of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Then there exists an open non-empty U of Y such that $\mathscr{F}|u^{-1}(U)$ be flat over U.*

Proof. The reasoning from the beginning of (6.9.1) leads to proving the

Lemma (8.9.4.1). — *Let A be an integral ring, B an A-algebra of finite presentation, M a B-module of finite presentation. Then there exists $f \neq 0$ in A such that M_f is a free A_f-module.*

Proof. Indeed, by (8.9.1) there is a finitely generated $\mathbf{Z}$-subalgebra A_0 of A, a finite-type A_0-algebra B_0, and a finite B_0-module M_0 such that B is A-isomorphic to $B_0 \otimes_{A_0} A$ and M is B-isomorphic to $M_0 \otimes_{A_0} A$; by (6.9.2), there exists $f_0 \neq 0$ in A_0 such that $(M_0)_{f_0}$ is a free $(A_0)_{f_0}$-module. As $M_{f_0} = (M_0)_{f_0} \otimes_{A_0} A$ and $A_{f_0} = (A_0)_{f_0} \otimes_{A_0} A$, M_{f_0} is a free A_{f_0}-module. □

□

Corollary (8.9.5). — *Let S be a quasi-compact prescheme and quasi-separated, $u : X \to S$ a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Then there exists a partition $(S_i)_{1 \leqslant i \leqslant n}$ of S into finitely many locally closed subsets of S such that, for $1 \leqslant i \leqslant n$, there is a subprescheme S'_i of S, with underlying space S_i, which is of finite presentation over S, and such that, on putting $X_i = X \times_S S'_i$, the $\mathscr{O}_{X_i}$-Module $\mathscr{F}_i = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_i}$ is flat over S'_i.*

Proof. Consider a finite covering $(U_j)_{1 \leqslant j \leqslant m}$ of S by affine open subsets and define, inductively, $T_1 = U_1$ and

$$T_j = U_j \setminus \bigcup_{h<j} (U_j \cap U_h) \qquad (j \geqslant 2).$$

Each T_j is closed in the affine open U_j, and the T_j are pairwise disjoint. Moreover, the $U_j \cap U_h$ are quasi-compact since S is quasi-separated and hence, since S is also quasi-compact, are constructible in S (**I**, 1.8.1); consequently each T_j is constructible as well. It will therefore suffice to prove the corollary for a suitable subprescheme T'_j of S having T_j as underlying space and of finite presentation over S, and for the morphism and the Module deduced respectively from u and $\mathscr{F}$ by the base **IV-3 | 36** change $T'_j \to S$. For this, note that if U_j is given the prescheme structure induced by that of S, the open immersion $U_j \to S$ is quasi-compact because S is quasi-separated (**I**, 1.2.7), and hence is of finite presentation (**I**, 1.6.2), (i). It therefore suffices that T'_j be of finite presentation over U_j; in other words, we may already reduce to the case where $U_j = S$ and $T_j = T$ is closed and constructible in S.

Write $S = \operatorname{Spec}(A)$, and regard A as the inductive limit of its finitely generated **Z**-subalgebras, so that $S = \varprojlim S_\lambda$, where the S_λ are affine and Noetherian. By (8.3.11), there exist λ and a closed (necessarily constructible) subset T_λ of S_λ such that $T = u_\lambda^{-1}(T_\lambda)$, where $u_\lambda : S \to S_\lambda$ is the canonical morphism. Give T_λ a subprescheme structure and take $T' = T_\lambda \times_{S_\lambda} S$. Since the canonical immersion $T_\lambda \to S_\lambda$ is of finite presentation (**I**, 1.6.1), so is the immersion $T' \to S$. As T' is affine, we see finally that it is enough to treat the case where $T' = S$ is affine.

Then, with the notation of (8.9.1), there exist λ, a morphism of finite type $u_\lambda : X_\lambda \to S_\lambda$, and a coherent $\mathscr{O}_{X_\lambda}$-Module $\mathscr{F}_\lambda$ such that X is isomorphic to $X_\lambda \times_{S_\lambda} S$ and $\mathscr{F}$ to $\mathscr{F}_\lambda \otimes_{\mathscr{O}_{X_\lambda}} \mathscr{O}_X$. Applying (6.9.3) to S_λ, X_λ, and $\mathscr{F}_\lambda$, we obtain subpreschemes $S'_{\lambda i}$ of S_λ whose underlying sets form a partition of S_λ and which are such that, on putting $X_{\lambda i} = X_\lambda \times_{S_\lambda} S'_{\lambda i}$, the Module

$$\mathscr{F}_{\lambda i} = \mathscr{F}_\lambda \otimes_{\mathscr{O}_{X_\lambda}} \mathscr{O}_{X_{\lambda i}}$$

is flat over $S'_{\lambda i}$. The subpreschemes $S'_i = S'_{\lambda i} \times_{S_\lambda} S$ of S then have the required properties by (2.1.4). □

8.10. Properties of permanence of morphisms by passage to the projective limit In this section, we preserve the general hypotheses and the notations of (8.8.1).

Proposition (8.10.1). — *If there exists λ such that, for $\mu \geqslant \lambda$, f_μ is an open morphism (resp. universally open), then f is an open morphism (resp. universally open).*

Proof. As $X = X_\lambda \times_{Y_\lambda} Y$, the assertion relative to universally open morphisms is a particular case of (2.4.3), (iii). Let us therefore suppose f_μ open for $\mu \geqslant \lambda$; it suffices to see that for every open *quasi-compact* U of X, $f(U)$ is open in Y. Now, there exists $\mu \geqslant \lambda$ such that $U = v_\mu^{-1}(U_\mu)$, where U_μ is an open quasi-compact in X_μ (8.2.11); one has then $f(v_\mu^{-1}(U_\mu)) = w_\mu^{-1}(f_\mu(U_\mu))$ (**I**, 3.4.8), hence $f(U)$ is open in Y. □

Corollary (8.10.2). — *Let $f : X \to Y$ be a morphism. For f to be universally open, it suffices that, for every integer $n > 0$, if one sets $Y_n = Y \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \ldots, T_n] (= \mathbf{V}_Y^n)$, and $X_n = X \times_Y Y_n$, $f_n = f \times 1_{Y_n} : X_n \to Y_n$, the projection f_n be an open morphism.*

Proof. To prove that f is universally open, it suffices to prove that it remains so for every affine U of Y; by hypothesis, if $U_n = U \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \ldots, T_n]$ is the inverse image of U in Y_n, the morphism $f_n^{-1}(U_n) \to U_n$, restriction of f_n, is open, and one sees that one can restrict to the case where $Y = \operatorname{Spec}(A)$ is affine. Furthermore, it suffices evidently to show that for every morphism $Y' \to Y$, where $Y' = \mathrm{Spec}(A')$ is also *affine*, $f' = f_{|Y'}$ is open. Let us suppose first that A' is an A-algebra of finite type, hence a *quotient* of a polynomial algebra $A[T_1, \dots, T_n]$; then Y' is a closed sub-prescheme of Y_n and f' the restriction of f_n to $f_n^{-1}(Y')$; but for every open V of X_n, one has $f_n(V \cap f_n^{-1}(Y')) =$ **IV-3** | 37 $f_n(V) \cap Y'$, and as by hypothesis $f_n(V)$ is open in Y_n, this shows that f' is also an open morphism. When A' is arbitrary, it can be considered as an inductive limit of its sub-A-algebras of finite type A'_λ, and the fact that f' is an open morphism follows from the preceding and from (8.10.1). □

Proposition (8.10.3). — *Suppose that there exists α such that: 1° S_α is quasi-compact; 2° the morphisms $X_\alpha \to S_\alpha$, $Y_\alpha \to S_\alpha$ are quasi-compact and the morphism $Y_\alpha \to S_\alpha$ quasi-separated; 3° for $\alpha \leqslant \lambda \leqslant \mu$, the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are flat; 4° $f_\alpha(X_\alpha)$ is constructible in Y_α. Then, for f to be dominant, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ be dominant.*

Proof. The hypotheses imply that Y_α is quasi-compact and that the morphism f_α is quasi-compact (**I**, 1.2.4); hence $f_\alpha(X_\alpha) = Z_\alpha$ is pro-constructible (**I**, 1.9.5), (vii) in Y_α. If one sets $Z_\lambda = w_{\alpha\lambda}^{-1}(Z_\alpha)$ for $\lambda \geqslant \alpha$ and $Z = w_\alpha^{-1}(Z_\alpha)$, we have $Z_\lambda = f_\lambda(X_\lambda)$ and $Z = f(X)$ (**I**, 3.4.8). As Z_λ is pro-constructible in Y_λ (8.3.11), the canonical map $\varinjlim \mathfrak{C}(S_\lambda) \to \mathfrak{C}(S)$ is injective, which proves the theorem in the case (vi). It suffices then to apply (8.3.12) by replacing S_λ, Z'_λ, and Z''_λ by Y_λ, Y_λ, and Z_λ respectively. □

Proposition (8.10.4). — *Suppose that there exists α such that Y_α is quasi-compact and f_α is of finite type and quasi-separated. For the morphism f to be separated, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ be separated.*

Proof. The question being local on Y_α (since Y_α is quasi-compact and L filtered), one can restrict to the case where Y_α is affine, hence quasi-separated, and the hypothesis implies that X_α (hence the X_λ and X) are quasi-compact and quasi-separated. Set $X'_\lambda = X_\lambda \times_{Y_\lambda} X_\lambda$ for $\lambda \geqslant \alpha$ and $X' = X \times_Y X$; the first projection $X'_\alpha \to X_\alpha$ is quasi-compact and quasi-separated by hypothesis (**I**, 1.2.3), (iii), hence X'_λ is quasi-compact and quasi-separated. Note now that if we denote by Δ_λ (resp. Δ) the diagonal of $X_\lambda \times_{Y_\lambda} X_\lambda$ (resp. $X \times_Y X$), it results from (**I**, 5.3.4) and (3.4.8) that Δ_λ is the inverse image of Δ_α by the map $v'_{\lambda\alpha} : X'_\lambda \to X'_\alpha$ (resp. $v'_\lambda : X' \to X'_\alpha$). On the other hand, Δ_α is *constructible* in X'_α: indeed, since f_α is quasi-separated, the diagonal immersion $X_\alpha \to X'_\alpha$ is quasi-compact, and locally of finite presentation since f_α is of finite type (**I**, 1.4.3) and (**I**, 1.5.4), (iii); it results then from (**I**, 1.8.4.1) that Δ_λ is locally constructible, hence constructible since X'_λ is quasi-compact. One can now apply (8.3.12) by replacing S_λ and Z_λ by X'_λ and Δ_λ respectively. □

Theorem (8.10.5). — *Suppose S_0 quasi-compact, X_α and Y_α of finite presentation over S_α, and let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism. Consider, for a morphism, the property of being:*

(i) *an isomorphism;*
(i *bis*) *a monomorphism;*
(ii) *an immersion;*
(iii) *an open immersion;*
(iv) *a closed immersion;*
(v) *separated;*
(vi) *surjective;* **IV-3** | 38
(vii) *radicial;*
(viii) *affine;*
(ix) *quasi-affine;*
(x) *finite;*
(xi) *quasi-finite;*
(xii) *proper.*

Then, if **P** *denotes one of the preceding properties, for f to possess the property* **P**, *it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ possess the property* **P** *(in which case f_μ also possesses it for $\mu \geqslant \lambda$).*

If S_0 is furthermore supposed quasi-separated, the same conclusion holds when **P** *is the property of being:*

(xiii) *projective;*
(xiv) *quasi-projective.*

Proof. The case where **P** is one of the properties (i) or (v) was not inserted in the statement for the record; in these cases, the theorem follows from what has been proved respectively in (8.8.2.4) and

(8.10.4). Furthermore, taking into account (**I**, 5.4.1) and (5.3.4), (v) also results from (iv). The case (i *bis*) follows immediately from (i), by using (**I**, 5.3.8) and by noting (as we have already used in (8.10.4)) that the diagonal Δ_f is deduced from Δ_{f_λ} by the base change $S \to S_\lambda$.

We will note on the other hand that (vi), (vii), and (xi) are in fact conditions on the *fibres* $f^{-1}(y)$ of the morphisms considered, taking into account the transitivity of the fibres by base change (**I**, 3.6.4): condition (vi) means indeed that all the fibres must be non-empty, condition (vii) that they must be radicial (**I**, 3.5.8), and condition (xi) that they must be finite (**II**, 6.2.2) and (6.2.3) and (**I**, 6.4.4), taking into account that f and the f_λ are morphisms of finite type by (**I**, 1.5.4), (v). The theorem in these three cases will be yet another consequence of a general result on this type of properties, which will be established in (9.3.3); we therefore postpone until then the proof of the theorem in the case (xi) (of course, the reader will be able to verify that, except for n[os] 8.11 and 8.12, we will make no use of the theorem in these cases until (9.3.3) and that (8.11) and (8.12) will not be used before (9.3.3)).

For the cases that remain to be proved, one can restrict to proving that the condition of the statement is necessary, all the properties **P** considered being invariant by base change (see chap. I[er] and II for the references concerning each of these properties). One can furthermore suppose that $S_\alpha = S_0$ and that $Y_\alpha = S_\alpha$, hence $Y_\lambda = S_\lambda$ for every $\lambda \geqslant \alpha$. Finally, the properties (i) through (xii) are local on S_0, hence as S_0 is a finite union of open affines and L filtered, one can restrict to proving them for $S_0 = \operatorname{Spec}(A_0)$ *affine* (hence quasi-separated). We shall denote by A_λ (resp. A) the ring of S_λ (resp. S).

Cases (ii), (iii), (iv): Suppose that f is an immersion (resp. an open immersion, a closed immersion), and let X' be the sub-prescheme (resp. induced on an open, resp. closed) of S associated to f, which is therefore an S-prescheme of finite presentation. By virtue of (8.6.3), there then exists **IV-3** | 39
$\lambda \geqslant \alpha$ and a sub-prescheme X'_λ of S_λ, of finite presentation over S_λ, such that X' is isomorphic to $X'_\lambda \times_{S_\lambda} S$. For every $\mu \geqslant \lambda$, $X'_\mu = X'_\lambda \times_{S_\lambda} S_\mu$ is therefore a sub-prescheme (resp. induced on an open, resp. closed) of S_μ, and it results therefore from (8.8.2.4) and (8.8.2.5) that there exists $\mu \geqslant \lambda$ and an isomorphism $g_\mu : X_\mu \to X'_\mu$ such that $g = g_\mu \times 1_S$ is the isomorphism $X \to X'$ associated to f; whence the conclusion.

Cases (vi) *and* (vii): We know (**I**, 1.8.1) that $Z_\alpha = f_\alpha(X_\alpha)$ is constructible in S_α; if we set $Z_\lambda = u_{\alpha\lambda}^{-1}(Z_\alpha)$ for $\lambda \geqslant \alpha$ and $Z = u_\alpha^{-1}(Z_\alpha)$, we have $Z_\lambda = f_\lambda(X_\lambda)$ and $Z = f(X)$ (**I**, 3.4.8). As Z_α is pro-constructible in S_α, and by virtue of (8.3.11) the canonical map $\varinjlim \mathfrak{C}(S_\lambda) \to \mathfrak{C}(S)$ is injective, the relation $f(X) = S$ implies the existence of $\lambda \geqslant \alpha$ such that $f_\lambda(X_\lambda) = S_\lambda$, which proves the theorem in the case (vi). To prove it in case (vii), note that the structural morphism $X_\alpha \times_{S_\alpha} X_\alpha \to S_\alpha$ is of finite presentation, since the first projection $X_\alpha \times_{S_\alpha} X_\alpha \to X_\alpha$ is so (1.6.2). It is therefore enough, by (**I**, 1.8.7.1), to apply case (vi) to the diagonal morphism

$$\Delta_{f_\alpha} : X_\alpha \longrightarrow X_\alpha \times_{S_\alpha} X_\alpha,$$

noting that $\Delta_{f_\lambda} = \Delta_{f_\alpha} \times 1_{S_\lambda}$ and $\Delta_f = \Delta_{f_\alpha} \times 1_S$ (**I**, 5.3.4) and (**I**, 3.3.11).

Cases (viii) *and* (ix): As $S = \operatorname{Spec}(A)$ is affine, to say that $f : X \to S$ is affine (resp. quasi-affine) means that there exists an integer r and a closed immersion (resp. an immersion) $j : X \to \mathbf{V}^r_S = \operatorname{Spec}(A[T_1, \ldots, T_r])$ of S-preschemes, since f is of finite type and S quasi-compact (**II**, 5.1.9). As $\mathbf{V}^r_S = \mathbf{V}^r_{S_\lambda} \times_{S_\lambda} S$, and $\mathbf{V}^r_{S_\lambda}$ is an S_0-prescheme of finite presentation, it results from (8.8.2), (i) applied to the S-morphism j, that there exists λ and an S_λ-morphism $j_\lambda : X_\lambda \to \mathbf{V}^r_{S_\lambda}$ such that $j = j_\lambda \times 1_S$; applying (iv) (resp. (ii)) to j, one deduces that there exists $\mu \geqslant \lambda$ such that j_μ is a closed immersion (resp. an immersion); hence f_μ is affine (resp. quasi-affine).

Case (x): By hypothesis, $X = \operatorname{Spec}(B)$, where B is an A-algebra which is an A-module of finite presentation (**I**, 1.4.7); it results therefore from (8.5.2), (ii) that there is λ and an A_λ-module of finite presentation B_λ such that the A-module B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$. The structure of A-algebra of B is defined by an A-homomorphism $m : B \otimes_A B \to B$; as $B \otimes_A B = (B_\lambda \otimes_{A_\lambda} B_\lambda) \otimes_A A$, there exists by (8.5.2), (i) $\mu \geqslant \lambda$ and an A_μ-homomorphism $m_\mu : B_\mu \otimes_{A_\mu} B_\mu \to B_\mu$ such that $m = m_\mu \otimes 1$. Considering the usual diagrams expressing the associativity and the commutativity of m, one sees by applying again (8.5.2), (i) that one can suppose ν taken large enough so that m_ν defines on B_ν an associative and commutative multiplication. In the same way one can suppose ν large enough so that B_ν thus defined admits a unit element. If $X'_\nu = \operatorname{Spec}(B_\nu)$, it is clear then that X is S-isomorphic to $X'_\nu \times_{S_\nu} S$, hence, by virtue of (i), there exists $\rho \geqslant \nu$ such that X_ρ and X'_ρ are S_ρ-isomorphic, which completes the proof in this case.

To prove the theorem in the case (xii), we shall first prove the following proposition: □

Proposition (8.10.5.1). — *(Chow's lemma for morphisms of finite presentation). — Let* A *be a ring, let* X *and* Y *be two* A*-preschemes of finite presentation, and let* $f : X \to Y$ *be a separated* A*-morphism. Then there exist two* A*-preschemes* X' *and* P *of finite presentation, and* A*-morphisms* $p : P \to Y$, $j : X' \to P$, *and* $g : X' \to X$, *such that the diagram* **IV-3 | 40**

$$\begin{tikzcd} X' \arrow[r, "j"] \arrow[d, "g"'] & P \arrow[d, "p"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

is commutative, and such that: 1° p *is projective;* 2° g *is projective and surjective;* 3° j *is an open immersion.*

Proof. Choose $\mathrm{A}_0 \subset \mathrm{A}$, X_0, Y_0, and f_0 as in (8.9.1), with Y_0 Noetherian and f_0 of finite type; by (8.10.4), one may moreover suppose that f_0 is separated. Chow's lemma (**II**, 5.6.1) then gives morphisms of finite type $p_0 : P_0 \to Y_0$, $g_0 : X'_0 \to X_0$, and $j_0 : X'_0 \to P_0$ such that

$$\begin{tikzcd} X'_0 \arrow[r, "j_0"] \arrow[d, "g_0"'] & P_0 \arrow[d, "p_0"] \\ X_0 \arrow[r, "f_0"'] & Y_0 \end{tikzcd}$$

is commutative, p_0 is projective, g_0 is projective and surjective, and j_0 is an open immersion. The assertions follow from the invariance of these properties under base change (**II**, 5.5.5), (iii), (**I**, 3.5.2), and (4.3.2). □

Continuation of the proof of (8.10.5). *Case (xii).* Apply (8.10.5.1) to $f_0 : X_0 \to S_0$. We obtain a commutative diagram

$$\begin{tikzcd} X'_0 \arrow[r, "j_0"] \arrow[d, "g_0"'] & P_0 \arrow[d, "p_0"] \\ X_0 \arrow[r, "f_0"'] & S_0 \end{tikzcd}$$

where p_0 is projective, g_0 is projective and surjective, and j_0 is an open immersion. For each λ one obtains the analogous diagram in which $p_\lambda = p_0 \times 1_{S_\lambda}$, $g_\lambda = g_0 \times 1_{S_\lambda}$, and $j_\lambda = j_0 \times 1_{S_\lambda}$ have the same respective properties, and similarly for the inverse-limit morphisms $p = p_0 \times 1_S$, $g = g_0 \times 1_S$, and $j = j_0 \times 1_S$. Since g is proper (**II**, 5.5.3), so is $p \circ j = f \circ g$ (**II**, 5.4.2); and since p is separated, j is proper, hence a closed immersion. Apply case (iv) to j, noting that X'_0 and P_0 are S_0-preschemes of finite presentation by (8.10.5.1) and (1.6.2). There exists λ such that j_λ is a closed immersion, hence proper (**II**, 5.4.2). Thus

$$f_\lambda \circ g_\lambda = p_\lambda \circ j_\lambda$$

is proper (**II**, 5.5.3) and (**II**, 5.4.2). Since g_λ is surjective and, by the hypothesis on f and case (v), f_λ may be supposed separated, (**II**, 5.4.3) shows that f_λ is proper.

Cases (xiii) and (xiv). By case (xii) and (**II**, 5.5.3), which applies because the S_λ are quasi-compact and quasi-separated by (**I**, 1.7.19), it is enough to consider the case where f is quasi-projective. Suppose there is an invertible $\mathscr{O}_X$-Module $\mathscr{L}$ which is ample for f. Since S_0 is quasi-compact and quasi-separated, so is X_0 (**I**, 1.2.3); hence there are λ and a quasi-coherent $\mathscr{O}_{X_\lambda}$-Module $\mathscr{L}_\lambda$ of finite presentation such that $\mathscr{L} = v_\lambda^*(\mathscr{L}_\lambda)$ (8.5.2), (ii). By (8.5.5), one may moreover suppose $\mathscr{L}_\lambda$ invertible. The assertion now follows from the more precise lemma below.

Lemma (8.10.5.2). — *Suppose that S_0 is quasi-compact, and let $\mathscr{L}_\lambda$ be an invertible $\mathscr{O}_{X_\lambda}$-Module. For $\mathscr{L}$ to be ample for f (resp. very ample for f), it is necessary and sufficient that there exist $\mu \geqslant \lambda$ such that $\mathscr{L}_\mu$ is ample for f_μ (resp. very ample for f_μ).* **IV-3 | 41**

Proof. The condition is plainly sufficient (**II**, 4.4.10) and (**II**, 4.6.13). Conversely, since the S_λ are quasi-compact and the f_λ are of finite type, after replacing $\mathscr{L}$ by a suitable tensor power, one may reduce to the case where $\mathscr{L}$ is very ample (**II**, 4.6.11). The question is local on S_0 by (**II**, 4.4.5) and the fact that L is filtered, so one may suppose that S_0, and hence S, is affine. By (**II**, 4.4.1), (ii) and (**II**, 4.1.2), there is an S-immersion

$$j : X \longrightarrow \mathbf{P}^r_S = P$$

such that $\mathscr{L} \simeq j^*(\mathscr{O}_P(1))$. By (8.8.2), (i), case (ii), and (**II**, 4.1.3), there exist $\mu \geqslant \lambda$ and an immersion $j_\mu : X_\mu \to \mathbf{P}^r_{S_\mu} = P_\mu$ such that $j = j_\mu \times 1_S$. Then (**II**, 4.1.3.2) and (8.5.2.5) give $\nu \geqslant \mu$ such that

$$\mathscr{L}_\nu \simeq j_\nu^*(\mathscr{O}_{P_\nu}(1)),$$

which shows that $\mathscr{L}_\nu$ is very ample for f_ν (**II**, 4.4.2). □

□

8.11. Application to quasi-finite morphisms We propose in this section to prove the two following theorems:

Theorem (8.11.1). — *Let $f : X \to Y$ be a proper morphism, locally of finite presentation, and quasi-finite. Then the morphism f is finite.*

Theorem (8.11.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation, quasi-finite and separated. Then the morphism f is quasi-affine, and a fortiori quasi-projective.*

Remark (8.11.3). — We will see below that one can reduce, for the proof of (8.11.1) and (8.11.2), to the case where Y is locally Noetherian; one will note that in this case one will thus obtain another proof of the theorem of Chevalley (**III**, 4.4.2).

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(8.11.4). The hypotheses and the conclusions of (8.11.1) and (8.11.2) are all local on Y. Indeed, this follows from (**I**, 1.6.1), (**I**, 1.2.6), (**II**, 5.1.1), (**II**, 5.4.1), and (**II**, 6.2.2). Therefore one can suppose $Y = \mathrm{Spec}(A)$ affine. One knows that there then exists a subring A_0 of A, which is a $\mathbf{Z}$-algebra of finite type, and a morphism of finite type $f_0 : X_0 \to \mathrm{Spec}(A_0)$ such that X identifies with $X_0 \otimes_{A_0} A$ and f with $f_0 \times 1$ (8.9.1). Furthermore, A can be considered as inductive limit of its subrings containing A_0 and which are $\mathbf{Z}$-algebras of finite type; using the method of (8.1.2), *c*)) as well as (8.10.5), (v), (xi) and (xii)), one sees that it suffices to prove the theorems (8.11.1) and (8.11.2) for f_0. Suppose henceforth Y Noetherian; using now the method of (8.1.2), *a*)) as well as (8.10.5), (v), (ix), (x), (xi) and (xii)) one can replace Y by $\mathrm{Spec}(\mathscr{O}_y)$, where y is an arbitrary point **IV-3 | 42**
of Y, so one can finally suppose that $Y = \mathrm{Spec}(A)$, where A is a *local Noetherian ring*. Let $\mathfrak{m}$ be the maximal ideal of A, and let $\widehat{A}$ be the completion of A for the $\mathfrak{m}$-adic topology. The ring $\widehat{A}$ is Noetherian local and $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ is faithfully flat and quasi-compact (**0**, 7.3.5); applying (2.7.1), (i), (vii), (xiv), (xv) and (xvi), we may therefore reduce to the case where A is *complete*. It then follows from (**II**, 6.2.6) that $X = X' \sqcup X''$, where X' is a Y-scheme quasi-finite and X'' a Y-scheme quasi-finite such that $X'' \cap f^{-1}(y) = \varnothing$.

Let us first place ourselves in the hypotheses of (8.11.1); as f is proper, X', which is closed in X, is proper over Y (**II**, 5.4.10)), so $f(X'')$ is closed in Y; but y is not contained in $f(X'')$, and moreover $f(X'')$ is adherent to every point of Y, so $f(X'') = \varnothing$, and consequently X'' is empty and f is *finite*.

Let us now place ourselves in the hypotheses of (8.11.2) and, restricting ourselves further (as one can do from what precedes) to the case where $Y = \mathrm{Spec}(A)$ is affine and Noetherian of finite dimension, let us moreover reason by induction on the dimension of Y. Reducing as above to the case where A is furthermore local and complete, one has $\dim(\mathscr{O}_y) = \dim(A) = \dim(Y)$ and for all $z \neq y$, $\dim(\mathscr{O}_z) < \dim(\mathscr{O}_y)$, so $\dim(Y - \{y\}) < \dim(Y)$. Now, by hypothesis one has $f(X'') \subset Y - \{y\}$ and the restriction of f to X'' is evidently a quasi-finite and separated morphism; applying the hypothesis of induction to $Y - \{y\}$ and to X'', one sees that X'' is quasi-affine over

$Y - \{y\}$; but the open subset $Y - \{y\}$ being quasi-affine over Y since Y is Noetherian, X'' is also quasi-affine over Y (**II**, 5.1.10), (ii)); as moreover X' is finite (and *a fortiori* affine) over Y, X is quasi-affine over Y (**II**, 4.6.17) and (**II**, 5.1.2), c')).

Proposition (8.11.5). — *Let $f : X \to Y$ be a morphism of finite presentation. The following properties are equivalent:*

a) *f is a closed immersion.*
b) *f is a proper monomorphism.*
c) *f is proper and for all $y \in Y$, $f^{-1}(y)$ is radicial and geometrically reduced over $k(y)$ (that is to say empty or $k(y)$-isomorphic to* $\operatorname{Spec}(k(y))$*).*

It is clear that *a)* implies *b)*. To see that *b)* implies *c)*, let us note (**I**, 3.3.12) that for all $y \in Y$, the morphism $f^{-1}(y) \to \operatorname{Spec}(k(y))$ deduced from f by extension of the base is a monomorphism, hence injective, and consequently $f^{-1}(y)$ is empty or reduced to a point, and in every case affine. Furthermore, if A is the ring of $f^{-1}(y)$, the canonical homomorphism $A \otimes_k A \to A$ is bijective (**I**, 5.3.8). This evidently implies that $A = k$, for otherwise there would exist an element $a \in A$ not in k and the images of $a \otimes 1$ and of $1 \otimes a$ in A would both be equal to a, so $a \otimes 1 = 1 \otimes a$ since 1 and a form a free system over k.

It remains to prove that *c)* implies *a)*. It follows first from (8.11.1) that f is a *finite* morphism, **IV-3** | 43
so $X = \operatorname{Spec}(\mathscr{A})$, where $\mathscr{A}$ is a finite $\mathscr{O}_Y$-Algebra. It suffices therefore to prove that the canonical homomorphism $\mathscr{O}_Y \to \mathscr{A}$ is *surjective* (**II**, 1.4.10)), or equivalently that for all $y \in Y$, the homomorphism $\mathscr{O}_y \to \mathscr{A}_y$ is surjective. But by hypothesis $f^{-1}(y) = \operatorname{Spec}(\mathscr{A}_y/\mathfrak{m}_y\mathscr{A}_y)$ (**II**, 1.5.5)) is such that the corresponding homomorphism $k(y) = \mathscr{O}_y/\mathfrak{m}_y \to \mathscr{A}_y/\mathfrak{m}_y\mathscr{A}_y$ is bijective; as $\mathscr{A}_y$ is an $\mathscr{O}_y$-module of finite type, the lemma of Nakayama shows that $\mathscr{O}_y \to \mathscr{A}_y$ is surjective, which completes the proof.

Remark (8.11.5.1). — One will note that the preceding reasoning proves that if f is a *monomorphism*, then, for all $y \in Y$, $f^{-1}(y)$ is empty or $k(y)$-isomorphic to $\operatorname{Spec}(k(y))$.

Proposition (8.11.6). — *If a morphism $f : X \to Y$ of finite presentation is a universal homeomorphism, it is finite, surjective and radicial (the converse being true by (2.4.5), (i))).*

Proof. Indeed, f being of finite type, universally closed, and separated by virtue of (2.4.4)), is proper by definition (**II**, 5.4.1)). As it is evidently quasi-finite (**II**, 6.2.3)), it is finite by (8.11.1). One knows moreover that it is radicial (2.4.4)), and evidently surjective. □

8.12. New proof and generalisation of the "Main Theorem" of Zariski

Lemma (8.12.1). — *Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism, $\mathscr{C}$ an $\mathscr{O}_Y$-Algebra quasi-coherent, $Z =$ $\operatorname{Spec}(\mathscr{C})$, which is an affine Y-prescheme over Y. Let $g : X \to Z$ be a Y-morphism, $\varphi = \mathscr{A}(g) : \mathscr{C} \to f_*(\mathscr{O}_X)$ the corresponding $\mathscr{O}_Y$-homomorphism of $\mathscr{O}_Y$-Algebras* (**II**, 1.2.7). *Suppose that g is an immersion. Then, for the closed image of X by g* (**I**, 9.5.3)*) to equal Z, it is necessary and sufficient that φ be injective; g is then an open immersion.*

Proof. The hypothesis implies that $f_*(\mathscr{O}_X)$ is an $\mathscr{O}_Y$-Algebra quasi-coherent (1.7.5)); furthermore, as the canonical morphism $h : Z \to Y$ is affine, hence quasi-compact and quasi-separated (1.2.2) and (1.1.2)), $g_*(\mathscr{O}_X)$ is an $\mathscr{O}_Z$-Algebra quasi-coherent (1.7.5)). This being so, saying that the closed image of X by g equals Z signifies (**I**, 9.5.2)) that the canonical homomorphism $\theta : \mathscr{O}_Z \to g_*(\mathscr{O}_X)$ is injective. But one has $h_*(\mathscr{O}_Z) = \mathscr{C}$ by definition of Z (**II**, 1.3.1)), and $h_*(g_*(\mathscr{O}_X)) = f_*(\mathscr{O}_X)$. As Z is affine over Y, it amounts to the same to say that the homomorphism $\theta : \mathscr{O}_Z \to g_*(\mathscr{O}_X)$ is injective or that the corresponding homomorphism $\varphi = h_*(\theta) : \mathscr{C} \to f_*(\mathscr{O}_X)$ is injective (**I**, 1.3.9)). The fact that g is then an open immersion follows from (**I**, 9.5.10)) and the hypothesis. □

Lemma (8.12.2). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \to Y$ a quasi-separated morphism of finite type, $\mathscr{C}$ an $\mathscr{O}_Y$-Algebra quasi-coherent, $Z =$ $\operatorname{Spec}(\mathscr{C})$. Let $g : X \to Z$ be a Y-morphism, $\varphi : \mathscr{C} \to f_*(\mathscr{O}_X)$ the corresponding $\mathscr{O}_Y$-homomorphism of $\mathscr{O}_Y$-Algebras. Let $(\mathscr{C}_\lambda)$ be the filtered increasing family of sub-$\mathscr{O}_Y$-Algebras quasi-coherent of finite type of $\mathscr{C}$ (where $\mathscr{C}$ is the union* (**I**, 9.6.6) *and (1.7.9))); set $Z_\lambda =$ $\operatorname{Spec}(\mathscr{C}_\lambda)$ and let g_λ be the morphism composed $X \xrightarrow{g} Z \to Z_\lambda$. Then the following conditions are equivalent:*

a) *g is an immersion.*
b) *There exists λ such that g_λ is an immersion.*

Furthermore, when g_λ is an immersion, the same is true of g_μ for $\mu \geqslant \lambda$. IV-3 | 44

It suffices to apply (**II**, 3.8.4)*) by replacing $\mathscr{L}$ by $\mathscr{O}_X$ and $\mathscr{S}$ by $\mathscr{C}[\mathrm{T}]$, and taking into account* (**II**, 3.1.7)*).*

Proposition (8.12.3). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \to Y$ a separated morphism of finite type. Let $\mathscr{B} = f_*(\mathscr{O}_X)$, which is an $\mathscr{O}_Y$-Algebra quasi-coherent* (**I**, 9.2.2)*); let $\mathscr{C}$ be the integral closure of $\mathscr{O}_Y$ in $\mathscr{B}$, which is an $\mathscr{O}_Y$-Algebra quasi-coherent* (**II**, 6.3.4)*); set $Z = \mathrm{Spec}(\mathscr{C})$, and let $g : X \to Z$ be the Y-morphism corresponding to the injection $\mathscr{C} \to \mathscr{B} = f_*(\mathscr{O}_X)$, and for all λ, let $g_\lambda : X \to Z_\lambda = \mathrm{Spec}(\mathscr{C}_\lambda)$ be the Y-morphism corresponding to the injection $\mathscr{C}_\lambda \to \mathscr{B} = f_*(\mathscr{O}_X)$. Then the following conditions are equivalent:*

a) *There exists a factorisation of f as*

$$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

a′) *There exists a factorisation $X \xrightarrow{f'} Y' \xrightarrow{u} Y$ of f, where f' is an open immersion and u a finite morphism.*
b) *The morphism $g : X \to Z$ is an immersion.*
c) *There exists λ such that $g_\lambda : X \to Z_\lambda$ is an immersion.*

Furthermore, when g_λ is an immersion, g is an open immersion, $g(X)$ is dense in Z, and there exists λ such that for $\mu \geqslant \lambda$, g_μ is an open immersion.

where f' is an open immersion and u a finite morphism.

As the homomorphism $\varphi : \mathscr{C} \to f_*(\mathscr{O}_X)$ is injective, it follows from (8.12.1) that if g is an immersion, it is an open immersion and $g(X)$ is dense in Z, and the same is true for g_λ. The fact that *a)* implies *a′)* follows also from (8.12.1), applied by replacing Z by Y' and g by f' (since Y' is finite over Y, hence affine over Y): indeed if Y'' is the closed image of X by f', Y'' is finite over Y and f' factorises as $X \xrightarrow{f''} Y'' \xrightarrow{j} Y'$, where j is the canonical injection, and f'' is an immersion (**I**, 4.1.10)); it follows therefore from (8.12.1) that f'' is an open immersion.

The equivalence of *b)* and *c)* follows from (8.12.2) as well as the fact that g_λ is then an immersion for λ large enough. It is clear that *c)* implies *a)*, since Z_λ is finite over Y. Let us finally show that *a)* implies *c)*. One has seen above that Y' is the closed image of X by f', and it then follows from (8.12.1) that if one sets $\mathscr{B}' = u_*(\mathscr{O}_{Y'})$, the homomorphism $\varphi' : \mathscr{B}' \to \mathscr{B} = f_*(\mathscr{O}_X)$ is *injective*. But as by hypothesis $\mathscr{B}'$ is a finite $\mathscr{O}_Y$-algebra, it identifies by definition of $\mathscr{B}$ with one of the $\mathscr{O}_Y$-subalgebras $\mathscr{C}_\lambda$, which proves *c)*.

We will say that a morphism $f : X \to Y$, where Y is quasi-compact and quasi-separated, is *pseudo-finite*, if it is of finite type and satisfies condition *a)* of (8.12.3) (in which case it is necessarily separated).

Corollary (8.12.4). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \to Y$ a morphism.* IV-3 | 45

(i) Suppose f pseudo-finite. Then, for all morphisms $Y' \to Y$, where Y' is quasi-compact and quasi-separated, $f_{(Y')} : X' = X_{(Y')} \to Y'$ is pseudo-finite.

(ii) Let (U_λ) be a covering of Y by quasi-compact open subsets. For f to be pseudo-finite, it is necessary and sufficient that for all λ, the restriction $f_\lambda : f^{-1}(U_\lambda) \to U_\lambda$ of f be a pseudo-finite morphism.

(iii) Suppose furthermore Y Noetherian, and f of finite type. Then, for f to be pseudo-finite, it is necessary and sufficient that, for all $y \in Y$, the morphism $f_y = f \times 1 : X_y = X \times_Y \mathrm{Spec}(\mathscr{O}_y) \to \mathrm{Spec}(\mathscr{O}_y)$ be pseudo-finite.

Proof. (i) It is clear that $f_{(Y')}$ is of finite type (1.5.4). Moreover, a factorisation $X \xrightarrow{g} Z \xrightarrow{u} Y$, with g an immersion and u finite, induces a factorisation $X' \xrightarrow{g'} Z' \xrightarrow{u'} Y'$ of $f_{(Y')}$, where $Z' = Z_{(Y')}$, $g' = g_{(Y')}$ and $u' = u_{(Y')}$. The morphism g' is an immersion (**I**, 4.3.2) and u' is finite (**II**, 6.1.5); hence $f_{(Y')}$ is pseudo-finite.

(ii) The condition is necessary by virtue of (i), the U_λ being quasi-separated since Y is. To see that it is sufficient, let us observe (with the notations of (8.12.3)) that if one sets $X_\lambda = f^{-1}(U_\lambda)$, we have $\mathscr{B}|U_\lambda = (f_\lambda)_*(\mathscr{O}_{X_\lambda})$, $\mathscr{C}|U_\lambda$ is the integral closure of $\mathscr{O}_{U_\lambda}$ in $\mathscr{B}|U_\lambda$, and consequently, if

$h : Z \to Y$ is the canonical morphism, $Z_\lambda = \mathrm{Spec}(\mathscr{C}_\lambda)$ identifies with $h^{-1}(U_\lambda) = \mathrm{Spec}(\mathscr{C}|U_\lambda)$, the Y-morphism $g_\lambda : X_\lambda \to Z_\lambda = h^{-1}(U_\lambda)$ being the U_λ-morphism corresponding to f_λ. Applying the hypothesis of induction, one sees that X'' is quasi-affine over $Y - \{y\}$; but the open subset $Y - \{y\}$ being quasi-affine over Y since Y is Noetherian, X'' is also quasi-affine over Y (**II**, 5.1.10), (ii)); since on the other hand X' is finite (and *a fortiori* affine) over Y, X is quasi-affine over Y.

(iii) It suffices, in virtue of (ii), to prove that for all $y \in Y$ there exists a neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f be pseudo-finite. Designate by (U_λ) the projective filtering system of affine open neighbourhoods of y, and apply the method of (8.1.2), *a*)). Since Y is Noetherian, the restrictions $f_\lambda : f^{-1}(U_\lambda) \to U_\lambda$ of f are of finite presentation and of the same nature as f_y. By hypothesis f_y factorises as $X_y \xrightarrow{g_y} Z_y \xrightarrow{u_y} \mathrm{Spec}(\mathscr{O}_y)$, where u_y is finite and g_y is an open immersion. As $\mathscr{O}_y$ is Noetherian, it is of the same nature as Z_y, and as Z_y is of finite presentation over $\mathrm{Spec}(\mathscr{O}_y)$, there exists λ and a morphism of finite presentation $u_\lambda : Z_\lambda \to U_\lambda$ such that Z_y identifies with $Z_\lambda \times_{U_\lambda} \mathrm{Spec}(\mathscr{O}_y)$; furthermore, one can suppose λ taken such that g_λ is an open immersion and u_λ a finite morphism (8.10.5), (ii) and (x)), which proves that f_λ is pseudo-finite. ☐

(8.12.5). We can now give the "Main theorem" of Zariski (**III**, 4.4.3)), a proof not using the "global" cohomological results of chap. III, but rather appealing to the finer properties of local Noetherian rings; we will furthermore generalise the statement of the theorem by removing the Noetherian hypotheses:

Theorem (8.12.6). — *("Main theorem" of Zariski). — Let Y be a prescheme quasi-compact and quasi-separated. If a morphism $f : X \to Y$ is quasi-finite, separated and of finite presentation, there exists a factorisation of f*

(8.12.6.1) $$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

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Proof. By virtue of (8.12.4), (ii)) and the local character (on Y) of the notions of quasi-finite morphism, separated and of finite presentation, one can restrict to the case where $Y = \mathrm{Spec}(A)$ is affine. Applying (8.9.1)), one can suppose that there is a subring A_0 of A, which is a **Z**-algebra of finite type, and an A-isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$, f identifying by this isomorphism with $f_0 \times 1$, where $f_0 : X_0 \to \mathrm{Spec}(A_0)$ is a morphism *of finite type*; furthermore (8.10.5), (v) and (xi)) one can suppose that f_0 is separated and quasi-finite; if one proves that f_0 is pseudo-finite, the same will be true of f by (8.12.4), (i)). As A_0 is then Noetherian and the notions of morphism of finite type, separated and quasi-finite are preserved by base change, it follows from (8.12.4), (iii)) that one can even suppose that A is a local ring, *essentially of finite type over* **Z** (1.3.8)). Set $n = \dim(A)$, and let us proceed by induction on n; for $n = 0$, the theorem is evident, A being a field and the morphism f being already finite (**II**, 6.2.2)). Set $B = \Gamma(X, \mathscr{O}_X)$; denote by C the integral closure of A in B, set $Z = \mathrm{Spec}(C)$ and let $g : X \to Z$ be the Y-morphism corresponding to the canonical A-homomorphism injective $C \to B$; by virtue of (8.12.3)), it suffices to show that g is an open immersion. Let a be the closed point of Y, and $U = Y - \{a\}$; U is Noetherian and all its local rings are essentially of finite type over **Z** and of dimension $< n$; taking into account the hypothesis of induction, one sees that the restriction $f^{-1}(U) \to U$ of f is a pseudo-finite morphism. One concludes from (8.12.3)) that, if $h : Z \to Y$ is the structural morphism, the restriction $f^{-1}(U) \to h^{-1}(U)$ of g is an open immersion. Set $A' = \widehat{A}$, $Y' = \mathrm{Spec}(A')$, $X' = X_{(Y')}$, $f' = f_{(Y')} : X' \to Y'$. As the canonical morphism $u : Y' \to Y$ is flat, it follows from (2.3.1)) that $B' = \Gamma(X', \mathscr{O}_{X'})$ identifies with the A'-algebra $B \otimes_A A'$. On the other hand, as A is a local ring excellent (7.8.3)), the morphism $u : Y' \to Y$ is regular, and consequently (6.14.4)) *the integral closure of A' in B' is equal to $C' = C \otimes_A A'$*. One sees therefore that $Z' = \mathrm{Spec}(C')$ is equal to $Z_{(Y')}$ and the morphism $g' : X' \to Z'$ coming from the canonical injection $C' \to B'$ is equal to $g_{(Y')}$. As $u : Y' \to Y$ is faithfully flat and quasi-compact, to prove that g is an open immersion, it suffices to prove that g' is an open immersion (2.7.1), (x)). If $h' : Z' \to Y'$ is the canonical morphism, the fact that the restriction $f'^{-1}(U') \to h'^{-1}(U')$ of g is an open immersion implies that the same is true of the restriction $f'^{-1}(U') \to h'^{-1}(U')$ of g'. Now let us note that $u^{-1}(a)$

is reduced to the closed point a' of Y' and consequently $U' = Y' - \{a'\} = u^{-1}(U)$. If $h' : Z' \to Y'$ is the canonical morphism, it follows from (**II**, 6.2.4)); as A' is *complete*, one deduces from (**II**, 6.2.6)) that X' is Y'-isomorphic to a sum $X'_1 \sqcup X'_2$, where the restriction $f'|X'_1 : X'_1 \to Y'$ is a morphism *finite*, and $X'_2 \subset f'^{-1}(U')$. It follows from this that B' is the direct product of the two A'-algebras $\Gamma(X'_1, \mathscr{O}_{X'_1}) = B'_1$ and $\Gamma(X'_2, \mathscr{O}_{X'_2}) = B'_2$; one concludes immediately that the integral closure C' of A' in B' is the direct product of the integral closures C'_1, C'_2 of A' in B'_1, B'_2 respectively, whence $Z' = Z'_1 \sqcup Z'_2$, where $Z'_i = \operatorname{Spec}(C'_i)$ $(i = 1, 2)$; and the canonical morphism $g' : X' \to Z'$ is such that $g'|X'_i$ is the canonical morphism $g'_i : X'_i \to Z'_i$ $(i = 1, 2)$. But as B'_1 is already a finite A'-algebra, one has $C'_1 = B'_1$, and g'_1 is therefore an isomorphism. On the other hand, as $X'_2 \subset f'^{-1}(U')$ and is open in $f'^{-1}(U')$, one already knows that g'_2 is an open immersion. One concludes that g' is indeed an open immersion. ∎ **IV-3** | 47

Remark (8.12.7). — When, in (8.12.6)), one supposes that Y is an affine *scheme*, the proof by reduction to the Noetherian case shows that, in the factorisation (8.12.6.1)), the morphisms f' and u are also morphisms *of finite presentation* (1.6.2)).

Corollary (8.12.8). — *Let Y be a quasi-compact scheme such that there exists an $\mathscr{O}_Y$-Module ample* (**II**, 4.5.3)), *$f : X \to Y$ a quasi-finite and quasi-projective morphism. Then there exists a factorisation of f as*

$$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

Proof. The hypothesis implies that X identifies with a sub-Y-scheme quasi-compact of a Y-scheme of the form $Z = \mathbf{P}^r_Y$ (**II**, 5.3.3)). There is therefore a quasi-compact open neighbourhood U of X in Z such that X is closed in U; as Z is a scheme, the canonical injection $U \to Z$ is a morphism of finite presentation ((1.2.7) and (1.6.2))), hence the composite morphism $g : U \to Z \to Y$ is also a morphism of finite presentation (the fact that $\mathbf{P}^r_Y$ is of finite presentation over Y following immediately from the definition (**II**, 4.1.1))). Let $\mathscr{J}$ be the quasi-coherent Ideal of $\mathscr{O}_U$ defining the closed sub-prescheme X; as U is a quasi-compact scheme, $\mathscr{J}$ is the filtered inductive limit of its sub-Ideals quasi-coherent of finite type $\mathscr{J}_\lambda$ (**I**, 9.4.9)). If X_λ is the closed sub-prescheme of U defined by $\mathscr{J}_\lambda$, one therefore has $X = \bigcap X_\lambda$. For all $y \in Y$, one therefore has $f^{-1}(y) = \bigcap(X_\lambda \cap g^{-1}(y))$, and as the sets $X_\lambda \cap g^{-1}(y)$ are closed in the Noetherian space $g^{-1}(y)$, there exists for all y an index $\lambda(y)$ such that $f^{-1}(y) = X_{\lambda(y)} \cap g^{-1}(y)$. Denote by E_λ the set of $y \in Y$ such that the fibre $X_\lambda \cap g^{-1}(y)$ of the restriction of g to X_λ is a $k(y)$-prescheme finite. The hypothesis that f is quasi-finite implies, by virtue of what precedes, that $Y = \bigcup_\lambda E_\lambda$. Now, each of the X_λ is, by definition, *of finite presentation* over Y; it follows therefore from (9.2.3) and (9.2.6)) ([1]) that the E_λ are *constructible* sets in the scheme Y; as they form a filtered increasing family, there exists an index λ such that $E_\lambda = Y$ (1.9.9)), and for this λ, the morphism $f_\lambda : X_\lambda \to Y$, restriction of g to X_λ, is of finite presentation, finite and separated, hence one can apply (8.12.6)), and f_λ factorises as

$$X_\lambda \xrightarrow{j_\lambda} Y'_\lambda \xrightarrow{u_\lambda} Y$$

where j_λ is an open immersion and u_λ a finite morphism. As X is a closed sub-prescheme of X_λ, one has thus proved that f possesses the property (8.12.3), *a*)), whence the corollary by virtue of the equivalence of (8.12.3), *a*)) and (8.12.3), *a'*)). ∎

Corollary (8.12.9). — *Let $f : X \to Y$ be a locally quasi-finite morphism* ($\mathbf{Err}_{III}$, 20). *For all $x \in X$ there exist an open neighbourhood U of x in X, an open neighbourhood V of $y = f(x)$ in Y, such that $f(U) \subset V$ and a factorisation* **IV-3** | 48

$$U \xrightarrow{f'} V' \xrightarrow{u} V$$

of the restriction of f to U, where f' is an open immersion and u a finite morphism.

Proof. It evidently suffices to take for V an affine neighbourhood of y in Y, for U an affine neighbourhood of x in X contained in $f^{-1}(V)$ and such that $f|U$ is quasi-finite. The morphism $U \to V$ restriction of f being then affine (hence quasi-projective), one can apply (8.12.8)). ∎

[13]([1]) The reader will verify that the corollaries (8.12.8) to (8.12.11) are not used in § 9.

Corollary (8.12.10). — *Let Y be an integral normal prescheme, X an integral prescheme, and $f : X \to Y$ a birational, locally quasi-finite morphism* ($\mathbf{Err}_{III}$, 20)*. Then f is an open immersion ouverte and furthermore it is necessary and sufficient that f be separated.*

Proof. The second assertion follows immediately from the first and from (**I**, 8.2.8)). To prove the first assertion, one may suppose X and Y affine and f quasi-finite; consider the factorisation $f = u \circ f'$ of (8.12.8), which identifies X, via f', with the subprescheme induced on an open subset of Y'. As X is integral, on can, by virtue of (**I**, 5.2.3)), replace Y' by the reduced sub-prescheme of Y' having underlying space $\overline{X}$, hence one can also suppose that Y' is *integral*. Furthermore, since f is birational, u is also. The conclusion then follows from the following lemma: □

Lemma (8.12.10.1). — *Let Y' be an integral prescheme, Y an integral prescheme and normal; then a morphism $u : Y' \to Y$ finite and birational is an isomorphism.*

Proof. Set $\mathscr{A} = u_*(\mathscr{O}_{Y'})$; then $\mathscr{A}$ is a finite $\mathscr{O}_Y$-algebra and Y' identifies with $\operatorname{Spec}(\mathscr{A})$ (**II**, 1.3.6). If $R(Y)$ is the field of rational functions of Y, one therefore has, for all $y \in Y$, $\mathscr{O}_{Y,y} \subset \mathscr{A}_y \subset R(Y)$; but as the ring $\mathscr{O}_{Y,y}$ is by hypothesis integrally closed and $R(Y)$ a field of fractions, one necessarily has $\mathscr{A}_y = \mathscr{O}_{Y,y}$, whence the lemma. □

Corollary (8.12.11). — *Let Y and X be integral normal preschemes, and let $f : X \to Y$ be a dominant, locally quasi-finite morphism. Let K and L (extension of K) be the fields of rational functions of Y and X respectively; and let Y' be the integral closure of Y relative to L* (**II**, 6.3.4)*); then f factorises in a unique way as $f : X \xrightarrow{f'} Y' \xrightarrow{u} Y$, where f' is birational, and correspond to the identity automorphism of L; f' is then a local isomorphism, and for f' to be an open immersion, it is necessary and sufficient that f be separated.*

Proof. The existence and uniqueness of the factorisation of f follow from (**II**, 6.3.9)). It follows further from (**II**, 6.2.4), (v)), reducing to the affine case, that f' is locally quasi-finite; furthermore, it follows from (**I**, 5.5.1)) that, for f' to be separated, it is necessary and sufficient that f be, since u is affine, hence separated; the last two assertions are therefore consequences of (8.12.10)) applied to f'. □

8.13. Translation in terms of pro-objects The following proposition is essentially equivalent to (8.8.2), (i)): **IV-3 | 49**

Proposition (8.13.1). — *Let S be a prescheme, $(X_\lambda, v_{\lambda\mu})$ a projective filtering system of S-preschemes; suppose that there exists α such that $v_{\alpha\lambda}$ is an affine morphism for all $\lambda \geqslant \alpha$ (which implies* (**II**, 1.6.2)*) that $v_{\lambda\mu}$ is affine for $\alpha \leqslant \lambda \leqslant \mu$), so that the projective limit $X = \varprojlim X_\lambda$ exists in the category of S-preschemes (8.2.3)). Let Y be an S-prescheme, and, for all $\lambda \geqslant \alpha$, let $e_\lambda : \operatorname{Hom}_S(X_\lambda, Y) \to \operatorname{Hom}_S(X, Y)$ be the map which, to every S-morphism $f_\lambda : X_\lambda \to Y$, makes correspond $f = f_\lambda \circ v_\lambda$, where $v_\lambda : X \to X_\lambda$ is the canonical morphism. The family (e_λ) is an inductive system of maps, which therefore defines a canonical map*

$$\varinjlim \operatorname{Hom}_S(X_\lambda, Y) \longrightarrow \operatorname{Hom}_S(X, Y). \tag{8.13.1.1}$$

Suppose X_α quasi-compact (resp. quasi-compact and quasi-separated), and the structural morphism $Y \to S$ locally of finite type (resp. locally of finite presentation). Then the map (8.13.1.1) is injective (resp. bijective).

Proof. Setting, for $\lambda \geqslant \alpha$, $Z_\lambda = Y \times_S X_\lambda$, so that one has $Z_\lambda = Z_\alpha \times_{X_\alpha} X_\lambda$. Setting similarly $Z = Y \times_S X = Z_\alpha \times_{X_\alpha} X$; one knows then (8.2.5)) that, if one also sets $w_{\lambda\mu} = 1 \times v_{\lambda\mu} : Z_\mu \to Z_\lambda$ for $\alpha \leqslant \lambda \leqslant \mu$ and $w_\lambda = 1 \times v_\lambda : Z \to Z_\lambda$ for $\alpha \leqslant \lambda$, Z is the projective limit of the system $(Z_\lambda, w_{\lambda\mu})$ and the w_λ are the canonical morphisms. Let us note furthermore that the morphism $Z_\alpha \to X_\alpha$ is locally of finite type (resp. locally of finite presentation) (1.3.4) and (1.4.3)). Finally, one knows that one has (**I**, 3.3.14))

$$\operatorname{Hom}_S(X_\lambda, Y) = \operatorname{Hom}_{X_\lambda}(X_\lambda, Z_\lambda) \quad \text{and} \quad \operatorname{Hom}_S(X, Y) = \operatorname{Hom}_X(X, Z)$$

It suffices now to apply (8.8.2), (i)) by setting $X_\lambda = S_\lambda$ and replacing Y_λ by Z_λ. □

Corollary (8.13.2). — *With the notations of (8.13.1)), suppose X_α quasi-compact and quasi-separated, and the $v_{\alpha\lambda}$ affine for $\alpha \leqslant \lambda$; suppose furthermore that $Y = \varprojlim Y_\rho$, where $(Y_\rho, t_{\rho o})$ is a projective filtering system of S-preschemes such that, for each ρ, the structural morphism $Y_\rho \to S$ is locally of finite presentation. Then one has a canonical bijection*

$$\mathrm{Hom}_S(X, Y) \xleftarrow{\sim} \varprojlim_\rho \left(\varinjlim_\lambda \mathrm{Hom}_S(X_\lambda, Y_\rho) \right). \tag{8.13.2.1}$$

Proof. Indeed, the fact that Y is the projective limit of the Y_ρ implies in particular that the canonical map $\mathrm{Hom}_S(X, Y) \to \varprojlim \mathrm{Hom}_S(X, Y_\rho)$ is bijective; and on the other hand, the hypotheses imply, for each ρ, the existence of a canonical bijection $\mathrm{Hom}_S(X, Y_\rho) \xleftarrow{\sim} \varinjlim_\lambda \mathrm{Hom}_S(S_\lambda, Y_\rho)$ by virtue of (8.13.1)); whence the conclusion. □

(8.13.3). The preceding results allow one to interpret in the theory of preschemes the notions of "pro-variety" or of "pro-scheme" which arise in certain applications (for example in the theory of the class field body following the ideas of Serre [39] or in the theory of Néron on the reduction of abelian varieties [32]). Let us recall here rapidly the notion of *pro-object* of a category, referring **IV-3** | 50
to chap. V for more ample developments (we will moreover not use before chap. V the interpretation that follows, and the reader can therefore omit it until the end of this number). Given a category $\mathscr{C}$, the category $\mathbf{Pro}(\mathscr{C})$ of *pro-objects* of $\mathscr{C}$ has for objects the *projective systems* (in the universe where one places oneself) $\mathbf{X} = (X_\mu)_{\mu \in M}$ of objects of $\mathscr{C}$ whose index sets (depending on the projective system considered) are supposed filtered preordered. Given two such pro-objects $\mathbf{X} = (X_\mu)_{\mu \in M}$, $\mathbf{X}' = (X'_{\mu'})_{\mu' \in M'}$, the *morphisms* from $\mathbf{X}$ to $\mathbf{X}'$ are by definition the elements of the set $\varprojlim_{\mu'}(\varinjlim_\mu \mathrm{Hom}(X_\mu, X_{\mu'}))$; the verification that one can take these sets as sets of morphisms is immediate, the composition of systems of morphisms $u''_\mu : X_\mu \to X'_{\mu'}$, $u''_{\mu'} : X'_{\mu'} \to X''_{\mu''}$ which are inductive and projective with respect to the lower index and upper index respectively, being carried out "argument by argument", in other words considering the system of $u''^{\mu'}_{\mu''} \circ u''^{\mu}_{\mu'}$.

(8.13.4). Consider then a prescheme quasi-compact and quasi-separated S, and denote by $\mathscr{C}$ the full subcategory of the category $(\mathbf{Sch})_{/S}$ of S-preschemes, formed of the S-preschemes X having the following property: the structural morphism $X \to S$ factorises as $X \xrightarrow{g} X_0 \xrightarrow{f} S$, where $g : X \to X_0$ is affine and $f : X_0 \to S$ is *of finite presentation*; we will say for short that the preschemes of $\mathscr{C}$ are *essentially affine over S*. Consider on the other hand the full subcategory $\mathscr{C}'_0$ of $(\mathbf{Sch})_{/S}$ formed of the S-preschemes *of finite presentation*, and the category $\mathbf{Pro}(\mathscr{C}'_0)$ of pro-objects of $\mathscr{C}'_0$. We will say that an object $\mathbf{X} = (X_\mu)_{\mu \in M}$ of $\mathbf{Pro}(\mathscr{C}'_0)$ is *essentially affine* if there exists a $\gamma \in M$ such that for $\mu \geqslant \gamma$, the transition morphism $v_{\gamma\mu} : X_\mu \to X_\gamma$ is affine (which implies that for $\gamma \leqslant \nu \leqslant \mu$, $v_{\nu\mu} : X_\mu \to X_\nu$ is affine). One will note that an object of $\mathbf{Pro}(\mathscr{C}'_0)$ isomorphic to an essentially affine object *is not* necessarily itself essentially affine. We will denote by $\mathscr{C}'$ the full subcategory of $\mathbf{Pro}(\mathscr{C}'_0)$ formed of pro-objects essentially affine.

This being so, it follows from (8.2.2) and (8.2.3)) that for all objects $\mathbf{X} = (X_\mu)_{\mu \in M}$ of $\mathscr{C}'$, the S-prescheme $X = \varprojlim_\mu X_\mu$ exists; furthermore, as, for μ large enough, the morphism $X \to X_\mu$ is affine (8.2.2)), X is *essentially affine over S* by definition. Set $X = L(\mathbf{X})$; let us show that one has thus defined a canonical *functor*

$$L : \mathscr{C}' \longrightarrow \mathscr{C} \tag{8.13.4.1}$$

One has indeed, for two objects $\mathbf{X} = (X_\mu)$, $\mathbf{X}' = (X'_{\mu'})$ of $\mathscr{C}'$, a canonical map **IV-3** | 51

$$\varprojlim_{\mu'} \varinjlim_{\mu} \mathrm{Hom}_S(X_\mu, X'_{\mu'}) \longrightarrow \mathrm{Hom}_S(\varprojlim_\mu X_\mu, \varprojlim_{\mu'} X'_{\mu'})$$

defined in (8.13.1.1)), and on the other hand, by definition of the projective limit, a canonical bijection

$$\varprojlim_{\mu'} \mathrm{Hom}_S(\varprojlim_\mu X_\mu, X'_{\mu'}) \xrightarrow{\sim} \mathrm{Hom}_S(\varprojlim_\mu X_\mu, \varprojlim_{\mu'} X'_{\mu'})$$

whence a canonical map

$$\varprojlim_{\mu'} \varinjlim_{\mu} \operatorname{Hom}_S(X_\mu, X'_{\mu'}) \longrightarrow \operatorname{Hom}_S(\varprojlim_{\mu} X_\mu, \varprojlim_{\mu'} X'_{\mu'}) \tag{8.13.4.2}$$

evidently functorial in $\mathbf{X}$ and $\mathbf{X}'$, and which completes the definition of the functor L.

Proposition (8.13.5). — *The hypotheses and notations being those of (8.13.4)), the functor L is pleinly faithful. If furthermore S is a Noetherian prescheme (which already implies that S is quasi-compact and quasi-separated (1.2.8))), L is* an equivalence of categories*.*

Proof. To say that L is pleinly faithful signifies that the map (8.13.4.2)) is bijective, which is a particular case of (8.13.2)). Indeed, the structural morphisms $X_\mu \to S$ being of finite presentation, are in particular quasi-compact and quasi-separated, hence the X_μ are quasi-compact and quasi-separated.

To show that when S is Noetherian L is an equivalence of categories, it suffices, since one already knows that L is pleinly faithful, to prove that every S-prescheme essentially affine X is S-*isomorphic* to an object of the form $L(\mathbf{X})$ where $\mathbf{X} \in \mathscr{C}'$ ($\mathbf{0}_{\text{III}}$, 8.1.5). Now, by hypothesis there is a factorisation $X \xrightarrow{g} X_0 \xrightarrow{f} S$ of the structural morphism, f being of finite presentation and g affine. One can therefore write $X = \operatorname{Spec}(\mathscr{A})$, where $\mathscr{A}$ is an $\mathscr{O}_{X_0}$-Algebra quasi-coherent (**II**, 1.3.1)). Now, as X_0 is Noetherian (since it is of finite type over S and S is Noetherian), $\mathscr{A}$ is the filtered inductive limit of the family $(\mathscr{A}_\lambda)$ of its sub-$\mathscr{O}_{X_0}$-Algebras quasi-coherent of finite type (**I**, 9.6.6)). Set $X_\lambda = \operatorname{Spec}(\mathscr{A}_\lambda)$; the morphisms $X_\lambda \to X_0$ are of finite type, hence of finite presentation since X_0 is Noetherian, and consequently the same is true of the composite morphisms $X_\lambda \to X_0 \to S$ (1.6.2)); in other words, the X_λ belong to $\mathscr{C}'_0$ and as the morphisms $X_\lambda \to X_0$ are affine, $\mathbf{X} = (X_\lambda)$ is an object of $\mathscr{C}'$ whose projective limit exists and is S-isomorphic to X by virtue of (8.2.2)). This completes the proof. □

Remark (8.13.6). — It follows from (1.6.2) and (**II**, 1.6.2)) that if X and Y are essentially affine over S, the same is true of $X \times_S Y$. One concludes for example ($\mathbf{0}_{\text{III}}$, 8.2.5) that a $\mathscr{C}$-group is nothing other than a $(\mathbf{Sch})_{/S}$-group which is a prescheme essentially affine over S. On the other hand, the finite products exist in the category $\mathscr{C}'$: indeed, if $\mathbf{X} = (X_\mu)_{\mu \in M}$, $\mathbf{Y} = (Y_\rho)_{\rho \in R}$ are two objects of $\mathscr{C}'$, the products $X_\mu \times_S Y_\rho$ are S-preschemes of finite presentation, and taking for transition morphisms $X_\nu \times_S Y_\sigma \to X_\mu \times_S Y_\rho$ the products of the transition morphisms $X_\nu \to X_\mu$ and $Y_\sigma \to Y_\rho$, one sees immediately that $(X_\mu \times_S Y_\rho)$ is a product of $\mathbf{X}$ and $\mathbf{Y}$ in $\mathbf{Pro}(\mathscr{C}'_0)$; furthermore (**II**, 1.6.2)) the transition morphisms thus defined are affine for μ and ρ large enough, hence the product $\mathbf{X} \times \mathbf{Y}$ also belongs to $\mathscr{C}'$. One concludes then as above that a $\mathscr{C}'$-group is a $\mathbf{Pro}(\mathscr{C}'_0)$-group which is essentially affine. One deduces from (8.13.5)) that the categories of the $\mathscr{C}'$-groups and the $\mathscr{C}$-groups are *equivalent* when S is *Noetherian*. It seems plausible that when S is the spectrum of a field k, the category of $\mathscr{C}$-groups is *equivalent* to that of the ***K**-groups*, where $\boldsymbol{K}$ is the category of S-preschemes *quasi-compact*; in other words, every prescheme in groups quasi-compact over k would be essentially affine. On the other hand, if one denotes by $\mathscr{C}'_0$-$\boldsymbol{gr}$ the category of the $\mathscr{C}'_0$-groups, it **IV-3** | 52
is very likely that the category of the $\mathscr{C}'$-groups is equivalent to the full subcategory of $\mathbf{Pro}(\mathscr{C}'_0\text{-}\boldsymbol{gr})$ formed of the "proalgebraic groups essentially affine", that is to say the projective systems $\mathbf{G} = (G_\mu)_{\mu \in M}$, where the G_μ are algebraic groups over k and the transition morphisms $G_\nu \to G_\mu$ are affine for μ large enough (which is what one can also express by saying that $\mathbf{G}$ is an extension of an algebraic group by a pro-algebraic group affine). The conjunction of these two conjectures is moreover equivalent to the following: every prescheme in groups quasi-compact over k (i.e. a prescheme in groups of finite type over k) is an extension by a prescheme in groups *affine* over k. The only pro-algebraic groups encountered in practice until now being in fact essentially affine, it would therefore be advantageous to substitute for the study of general pro-algebraic groups (introduced and studied by Serre [40]) that of schemes in groups quasi-compact over k, whose definition is conceptually much simpler.

8.14. Characterisation of a prescheme locally of finite presentation over another, in terms of the functor it represents

(8.14.1). Given a prescheme S, we will say still, as in (8.13.4)), that a projective filtering system $(X_\lambda, v_{\lambda\mu})$ of S-preschemes is *essentially affine* if there exists α such that $v_{\alpha\lambda}$ is an affine morphism for $\lambda \geqslant \alpha$.

The following statement, which will be especially useful in chap. V, makes precise (8.8.2), (i)) by providing it with a converse:

Proposition (8.14.2). — *Let S be a prescheme, $f : X \to S$ a morphism. For every S-prescheme T, set*

$$h_X(T) = \mathrm{Hom}_S(T, X)$$

so that h_X is a contravariant functor from the category $(\mathbf{Sch})_{/S}$ of S-preschemes to the category **Ens** *of sets* $(\mathbf{0}_{III}$, 8.1.1), *and X an object representing this functor* $(\mathbf{0}_{III}$, 8.1.8). *The following conditions are equivalent:*

a) *f is locally of finite presentation.*
b) *For every projective filtering system (Z_λ) of S-preschemes, essentially affine* (8.13.4)) *and formed of preschemes quasi-compact and quasi-separated, the canonical map* (8.13.1.1))

$$\varinjlim h_X(Z_\lambda) \longrightarrow h_X(\varprojlim Z_\lambda) \tag{8.14.2.1}$$

is bijective.
c) *For every projective filtering system (Z_λ) of S-preschemes, such that the Z_λ are affine schemes, the map* (8.14.2.1)) *is bijective.*
c′) *For every affine open subset U of S and every projective filtering system (Z_λ) of U-preschemes, such that the Z_λ are affine schemes, the map* (8.14.2.1)) *is bijective.*

Proof. The fact that *a)* implies *b)* is nothing other than (8.13.1)); it is trivial that *b)* implies *c)* and that *c)* implies *c′)*. It remains to see that *c′)* implies *a)*, and as the property *a)* is local on S, one can restrict to the case where S is *affine*. **IV-3** | 53

Suppose first that X is also affine; the assertion to prove is then equivalent to the following:

Corollary (8.14.2.2). — *Let A be a ring, B an A-algebra. In order that, for every inductive filtering system (C_λ) of A-algebras, the canonical map*

$$\varinjlim \mathrm{Hom}_{A\text{-alg.}}(B, C_\lambda) \longrightarrow \mathrm{Hom}_{A\text{-alg.}}(B, \varinjlim C_\lambda) \tag{8.14.2.3}$$

be bijective, it is necessary and sufficient that B be an A-algebra of finite presentation.

It remains only to show that the condition is necessary. First, for (C_λ) take the inductive filtering system of sub-A-algebras of finite type of B, so that $\varinjlim C_\lambda = B$. The fact that (8.14.2.3)) is bijective implies in particular that the identity map $1_B : B \to B$ factorises as $B \to C_\lambda \to B$ for a suitable λ, which implies that $C_\lambda = B$, hence B is an A-algebra *of finite type*. Write then $B = C/\mathfrak{J}$, where $C = A[T_1, \dots, T_n]$ and $\mathfrak{J}$ is an ideal of C; setting $C_\lambda = C/\mathfrak{J}_\lambda$, where $\mathfrak{J}_\lambda$ ranges over the ideals of finite type $\mathfrak{J}_\lambda \subset \mathfrak{J}$ of C; then $\mathfrak{J}$ is the filtered inductive limit of the ideals of finite type of $\mathfrak{J}$, and using the exactness of the inductive limit functor, one sees that B is still isomorphic to the inductive limit of the inductive system (C_λ). There exists therefore a λ and an A-homomorphism $u : B \to C_\lambda$ such that the composite $B \xrightarrow{u} C_\lambda \xrightarrow{p_\lambda} B$ (where p_λ is the canonical homomorphism) is the identity. Let $q_\lambda : C \to C_\lambda$ be the canonical homomorphism, and set $t_i = p_\lambda(q_\lambda(T_i))$. Then $p_\lambda(u(t_i)) = p_\lambda(q_\lambda(T_i))$, in other words $u(t_i) - q_\lambda(T_i) \in \mathfrak{J}/\mathfrak{J}_\lambda$. There is therefore a $\mu \geqslant \lambda$ such that these n elements belong to $\mathfrak{J}_\mu/\mathfrak{J}_\lambda$ $(1 \leqslant i \leqslant n)$. If $p_{\lambda\mu} : C_\lambda \to C_\mu$ is the canonical homomorphism, then $p_{\lambda\mu}(u(t_i)) = p_{\lambda\mu}(q_\lambda(T_i)) = q_\mu(T_i)$. Replacing λ by μ and u by $p_{\lambda\mu} \circ u$, we may therefore suppose that $u(t_i) = q_\lambda(T_i)$ for every i; and if $r = p_\lambda \circ q_\lambda$ is the canonical homomorphism $C \to C/\mathfrak{J} = B$, we may write $u(r(T_i)) = q_\lambda(T_i)$ for every i, whence $q_\lambda = u \circ r$. But this implies necessarily that $\mathfrak{J} = \mathfrak{J}_\lambda$, for if $z \in \mathfrak{J}$, one has $r(z) = 0$; hence we have $B = C_\lambda$.

Let us now pass to the case where S is affine and X arbitrary; what amounts to proving is that an affine open subset V of X is of finite presentation over S, and by virtue of what has just been shown, it suffices to prove that for every projective filtering system (Z_λ) of affine S-preschemes, the map

$$\varinjlim \mathrm{Hom}_S(Z_\lambda, V) \longrightarrow \mathrm{Hom}_S(\varprojlim Z_\lambda, V) \tag{8.14.2.4}$$

is bijective. It is immediate that this map is injective, for if (v_λ), (v'_λ) are two inductive systems of S-homomorphisms $v_\lambda : Z_\lambda \to V$, $v'_\lambda : Z_\lambda \to V$ such that the corresponding morphisms

$$Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v_\lambda} V, \qquad Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v'_\lambda} V$$

are equal (u_λ being the canonical morphism), then the morphisms

$$Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v_\lambda} V \xrightarrow{j} X, \qquad Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v'_\lambda} V \xrightarrow{j} X$$

(where j is the canonical injection) are equal, which implies $j \circ v_\lambda = j \circ v'_\lambda$ by hypothesis for a suitable λ, hence $v_\lambda = v'_\lambda$.

It remains to prove that (8.14.2.4) is surjective. Let $v : Z \to V$ be an S-morphism. By hypothesis there are a λ and an S-morphism $w_\lambda : Z_\lambda \to X$ such that $j \circ v$ factors as

$$Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{w_\lambda} X.$$

It suffices to prove that there is a $\mu \geqslant \lambda$ such that

$$Z_\mu \xrightarrow{w_\lambda \circ u_{\lambda\mu}} X$$

(where $u_{\lambda\mu}$ is the transition morphism) factors as

$$Z_\mu \xrightarrow{v_\mu} V \xrightarrow{j} X.$$

Set $U_\lambda = w_\lambda^{-1}(V)$. Then

$$u_\lambda^{-1}(U_\lambda) = u_\lambda^{-1}(w_\lambda^{-1}(V)) = v^{-1}(V) = Z.$$

The Z_λ are quasi-compact and the U_λ, being open, are ind-constructible (1.9.6); hence (8.3.4) gives a $\mu \geqslant \lambda$ for which $U_\mu = Z_\mu$. ☐

Remark (8.14.3). — The fact that the map (8.14.2.1) is injective when f is locally of finite type (8.8.2), (i) naturally leads one to ask whether this result has a converse. It does not, even when S and X are affine, since there exist monomorphisms $X \to S$ which are not of finite type (**I**, 2.4.2), and these disprove such a conjecture.

§9. Constructible properties

9.1. The finite extension principle Let S be a prescheme, $f : X \to S$ a morphism of *finite presentation* (1.6.1), $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of *finite presentation*. We propose in this paragraph to give criteria ensuring, for example, that the set of $s \in S$ such that the $\mathbf{k}(s)$-prescheme $X_s = f^{-1}(s) = X \times_S \operatorname{Spec}(\mathbf{k}(s))$ has a certain property, or such that the $\mathscr{O}_{X_s}$-Module $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_X} \mathbf{k}(s)$ has a certain property, is *locally constructible*, or at least *ind-constructible* (1.9.4); we will see that this is the case for the majority of properties that arise in algebraic Geometry. Assuming only f *locally of finite presentation*, we will also give (9.9) criteria for the set of points $x \in X$ where the fibre $X_{f(x)}$ (or the $\mathscr{O}_{X_{f(x)}}$-Module $\mathscr{F}_{f(x)}$) has a certain property, to be *locally constructible*. We will see in §12 that these results, together with the additional hypothesis that f is *flat* (resp. *proper and flat*), allow us to prove that the sets considered in X (resp. in S) are even *open* in many cases.

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Proposition (9.1.1). — (Finite extension principle). *Let k be a field, $\mathscr{C}$ a set of extensions of k. Suppose that the following conditions are satisfied:*

- *(i) If $K \in \mathscr{C}$ and if there exists a k-homomorphism $K \to K'$ (where K' belongs to the universe where we have placed ourselves), then $K' \in \mathscr{C}$.*
- *(ii) If $K \in \mathscr{C}$, there exists a sub-extension K' of K, of finite type over k, such that $K' \in \mathscr{C}$.*
- *(iii) If $K \in \mathscr{C}$ is the field of fractions of an integral k-algebra of finite type A, there exists $f \in A - \{0\}$ such that for every maximal ideal $\mathfrak{m}$ of A we have $A_f/\mathfrak{m} \in \mathscr{C}$.*

Let then Ω be an algebraically closed extension of k (in the universe considered). The following conditions are equivalent:

- a) *$\mathscr{C}$ is non-empty.*
- b) *There exists $K' \in \mathscr{C}$ which is a finite extension of k.*

c) *We have* $\Omega \in \mathscr{C}$.

The condition (i) obviously implies that *b)* entails *c)*, and *c)* trivially entails *a)*; let us prove that *a)* entails *b)*. By virtue of (ii) and (iii) there exists an extension $K' \in \mathscr{C}$ which is of the form $A/\mathfrak{m}$, where A is a k-algebra of finite type over k and $\mathfrak{m}$ is a maximal ideal of A. We know, by Hilbert's Nullstellensatz (Bourbaki, *Alg. comm.*, chap. V, §3, n$^{\mathrm{o}}$ 1, cor. 2 of th. 1), that K' is a finite extension of k.

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Corollary (9.1.2). — *Under the hypotheses* (i), (ii), (iii) *of* (9.1.1), *if k is algebraically closed and if $\mathscr{C}$ is non-empty, $\mathscr{C}$ contains all the extensions of k in the universe considered.*

Indeed, as *a)* entails *c)*, and the conclusion follows from the hypothesis (i).

Remark (9.1.3). — In practice, one will verify the condition (ii) of (9.1.1) by noting that K is the inductive limit of its sub-extensions of finite type K_α, and by applying the results of §8, taking into account eventually that for $K_\alpha \subset K_\beta$, K_β is faithfully flat over K_α. Frequently the set $\mathscr{C}$ is formed of *fields* belonging to a set $\mathscr{C}'$ of k-algebras satisfying the following condition:

(i bis) If $A \in \mathscr{C}'$ and if there exists a homomorphism of k-algebras $A \to A'$ (where A' belongs to the universe where we have placed ourselves), then $A' \in \mathscr{C}'$.

Since this is so, the condition (i) is trivially satisfied, and one will verify generally the condition (iii) of (9.1.1) by noting that the field of fractions K of A is the inductive limit of the algebras A_f (for the relation $D(f) \supset D(g)$), and by applying the results of §8, taking into account eventually that the morphisms $D(g) \to D(f)$ are open immersions.

On the other hand, when (i bis) is not satisfied, the proof of (iii) is often more delicate, and is linked to constructibility criteria that will be developed further on.

Here are some typical examples of application of the finite extension principle:

Proposition (9.1.4). — *Let k be a field, Ω an algebraically closed extension of k, X and Y two preschemes of finite type over k. The following conditions are equivalent:*

a) *There exists an Ω-morphism $X_{(\Omega)} \to Y_{(\Omega)}$ (resp. an Ω-morphism possessing one of the determined properties* (i) *to* (xiv) *of* (8.10.5)).

b) *There exists a finite extension k' of k and a k'-morphism $X_{(k')} \to Y_{(k')}$ (resp. a k'-morphism possessing the property considered).*

c) *There exists an extension K of k and a K-morphism $X_{(K)} \to Y_{(K)}$ (resp. a K-morphism possessing the property considered).*

We apply the remark (9.1.3) taking for $\mathscr{C}'$ the set of all k-algebras A (of the universe where we have placed ourselves) such that there exists an A-morphism $X \otimes_k A \to Y \otimes_k A$ (resp. a morphism having the properties of (8.10.5) that one considers). The condition (i bis) of (9.1.3) is then satisfied thanks to the fact that the properties considered in (8.10.5) are all stable under base change. The **IV-3** | 56
condition (iii) of (9.1.1) is thus satisfied by virtue of (8.8.2), (i)) (resp. (8.10.5))), since Spec(k) is quasi-compact and quasi-separated. It remains to verify the condition (ii) of (9.1.1) which also follows from (8.8.2), (i)) and from (8.10.5), considering K as an inductive limit of its sub-extensions of finite type. We conclude therefore by (9.1.1).

In particular, if there exists an extension K of k and a K-isomorphism $X_{(K)} \xrightarrow{\sim} Y_{(K)}$, we say that X and Y are *geometrically isomorphic*.

The following corollary generalises (**II**, 6.6.5):

Corollary (9.1.5). — *Let k be a field, X a k-prescheme. If there exists an extension K of k such that $X_{(K)}$ is projective (resp. quasi-projective) over K, then X is projective (resp. quasi-projective) over k.*

The morphism Spec$(K) \to$ Spec(k) being faithfully flat and quasi-compact, it follows already from (2.7.1), (v)) that X is of finite type over k. The hypothesis means that there exists a closed immersion (resp. an immersion) $X_{(K)} \to \mathbf{P}^n_K = \mathbf{P}^n_k \otimes_k K$ (**II**, 5.5.4, (ii) and 5.5.3); applying (9.1.4) for property (iv) (resp. (ii)) of (8.10.5), we deduce that for some finite extension k' of k there is a closed immersion (resp. an immersion) $X_{(k')} \to \mathbf{P}^n_{k'}$; in other words $X_{(k')}$ is projective (resp. quasi-projective) over k'. We conclude then by (**II**, 6.6.5).

Proposition (9.1.6). — *Let k be a field, Ω an algebraically closed extension of k, X a prescheme of finite type over k, $\mathscr{F}$, $\mathscr{G}$ two coherent $\mathscr{O}_X$-Modules. Suppose that there exists an isomorphism $\mathscr{F} \otimes_k \Omega \xrightarrow{\sim} \mathscr{G} \otimes_k \Omega$. Then there exists a finite extension k' of k and an isomorphism $\mathscr{F} \otimes_k k' \xrightarrow{\sim} \mathscr{G} \otimes_k k'$.*

The reasoning is the same as in (9.1.4), applying (8.5.2), (i)) (using the property (iii) of (9.1.1), the fact that the morphisms $D(g) \to D(f)$ (with the notations of (9.1.3)) are open immersions, and *a fortiori* the morphisms are flat).

9.2. Constructible and ind-constructible properties

Definition (9.2.1). — Let $\mathbf{P}(X, \mathscr{F}, k)$ be a relation. We say that $\mathbf{P}$ is a *constructible* (resp. *ind-constructible*) property of algebraic preschemes if the two following conditions are satisfied:

(i) If k is a field, X an algebraic prescheme over k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, k' an extension of k, then for $\mathbf{P}(X, \mathscr{F}, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, \mathscr{F} \otimes_k k', k')$ be true.

(ii) Let S be an integral Noetherian prescheme, of generic point η, $u : X \to S$ a morphism of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. For every $s \in S$, set $X_s = u^{-1}(s) = X \times_S \operatorname{Spec}(\mathbf{k}(s))$, $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_S} \mathbf{k}(s)$. Let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathscr{F}_s, \mathbf{k}(s))$ is true. Then one of the two sets E, $S - E$ (resp. the set E) contains a non-empty open set (and consequently is a neighbourhood of η) (resp. contains a non-empty open set if it contains η).

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Remarks (9.2.2). — (i) This is of course a convention of meta-mathematical language and not a mathematical definition properly speaking. We have analogous "definitions" for relations between k, one or more algebraic k-preschemes, k-morphisms between these preschemes, coherent Modules on these preschemes or homomorphisms between these Modules.

(ii) We will also need to consider relations involving *constructible parts* of preschemes. For example, let $\mathbf{P}(X, Z, k)$ be a relation; we will say (by abuse of language) that $\mathbf{P}$ is a constructible (resp. ind-constructible) property of the constructible part Z of X if the following two conditions are satisfied:

1° If k is a field, X an algebraic prescheme over k, Z a *constructible part* of X, k' an extension of k, then for $\mathbf{P}(X, Z, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, p^{-1}(Z), k')$ be true ($p : X_{(k')} \to X$ being the canonical projection).

2° Let S be an integral Noetherian prescheme, of generic point η, $u : X \to S$ a morphism of finite type, Z a *constructible part of* X. For every $s \in S$, set $X_s = u^{-1}(s)$, $Z_s = Z \cap X_s$. Let E be the set of $s \in S$ such that $\mathbf{P}(X_s, Z_s, \mathbf{k}(s))$ is true. Then one of the two sets E, $S - E$ (resp. the set E) contains a non-empty open set (and consequently is a neighbourhood of η) (resp. contains a non-empty open set if it contains η).

We will note that in condition 2° it suffices to suppose that Z is a constructible part of X for every s; the first of these two properties entails the second (1.8.2), but not conversely.

(iii) If $\mathbf{P}$ is a constructible property, it is clear that the same is true of "not $\mathbf{P}$". If $\mathbf{P}$, $\mathbf{Q}$ are two constructible properties (resp. ind-constructible), the same is true of the properties "$\mathbf{P}$ or $\mathbf{Q}$" and "$\mathbf{P}$ and $\mathbf{Q}$"; indeed, under the hypotheses of (9.2.1), (ii)), E, E' are two parts of S and if each of E contains a non-empty open set, the same is true of $E \cup E'$, and if $S - E$ and $S - E'$ each contain a non-empty open set, the same is true of $S - (E \cup E') = (S - E) \cap (S - E')$.

(iv) Let $\mathbf{P}(X, \mathscr{F}, k)$ be a relation satisfying the condition (9.2.1), (i)); let S be a prescheme, $u : X \to S$ a morphism of finite type; with the notations of (9.2.1), (ii)), let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathscr{F}_s, \mathbf{k}(s))$ is true. Let on the other hand $g : S' \to S$ be any morphism, and set $X' = X_{(S')}$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$; then it follows from the transitivité of fibres (**I**, 3.6.4) and from the condition (9.2.1), (i)) that *the set of $s' \in S'$ such that $\mathbf{P}(X'_{s'}, \mathscr{F}'_{s'}, \mathbf{k}(s'))$ is true is equal to $g^{-1}(E)$*. This extends immediately to the case where there appear in $\mathbf{P}$ several preschemes, Modules, morphisms of preschemes or homomorphisms of Modules, as well as to properties of the type considered in (ii).

(v) As we will see in the remainder of this paragraph and in the rest of chap. IV, the majority of properties $\mathbf{P}$ satisfying the condition (9.2.1), (i)) also satisfy the condition (9.2.1), (ii)). As possible exceptions, let us note the property of being *projective*, or *quasi-projective*, or *affine*, or *quasi-affine* over the base field (for an algebraic prescheme); we will see (9.6.2) that these properties are *ind-constructible*, but we will give later an example where S is a non-empty open part of $\mathrm{Spec}(\mathbf{Z})$ (or a non-empty open part of a smooth curve over a finite field) and where all fibres X_s

except the generic fibre X_η are projective over $\mathbf{k}(s)$ (all preschemes X_s being of dimension 2). **IV-3** | 58

Proposition (9.2.3). — *Let* $\mathbf{P}$ *be a constructible* (*resp. ind-constructible*) *property of algebraic preschemes, S a prescheme, X a prescheme of finite presentation over S, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. Then the set E of $s \in S$ such that $\mathbf{P}(X_s, \mathscr{F}_s, \mathbf{k}(s))$ is true is locally constructible* (*resp. ind-constructible*). *Furthermore, if S is irreducible of generic point η, one of the two sets E, $S - E$ is a neighbourhood of η* (*resp. E is a neighbourhood of η if it contains this point*).

To prove these assertions, one can restrict to the case where $S = \mathrm{Spec}(A)$ is affine. We know then that there exists a subring A_0 of A which is a $\mathbf{Z}$-algebra of finite type, an A_0-prescheme of finite type X_0 and a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ such that X is isomorphic to $X_0 \otimes_{A_0} A$ (8.9.1). Let $p : S \to S_0 = \mathrm{Spec}(A_0)$ be the morphism corresponding to the injection $A_0 \to A$, and let E_0 be the set of $s_0 \in S_0$ such that

$$\mathbf{P}((X_0)_{s_0}, (\mathscr{F}_0)_{s_0}, \mathbf{k}(s_0))$$

is true; then, by virtue of (9.2.2), (iv)), we have $E = p^{-1}(E_0)$; on the other hand, one can restrict further (1.8.2) to a $\mathbf{Z}$-algebra of finite type, hence a Noetherian scheme. Using the criterion of ind-constructibility ($\mathbf{0}_{\mathrm{III}}$, 9.2.3) (resp. the criterion of constructibility ($\mathbf{0}_{\mathrm{III}}$, 9.2.3)) (9.2.2), (iv)) and replacing S by a closed integral sub-prescheme of S, in the case where S is Noetherian and integral, our assertion then follows from (9.2.3) applied to the ind-constructible part E of S.

We will note that when S is the spectrum of a finite ring of integers of finite type, hence a Noetherian scheme, and it is clear on the other hand that the statement of (9.2.3) also applies when there appear in $\mathbf{P}$ several preschemes, Modules, morphisms of preschemes or homomorphisms of Modules. It applies also when there is a finite number of parts (in finite number) of the preschemes considered, provided one imposes on these parts the condition of being *locally constructible*. Indeed, the restriction to the case where S is affine shows that one can restrict to the case where these parts are *constructible*: applying (8.3.11) then (with the previous notations) one obtains a constructible part of X_0 for a suitable choice of A_0.

Corollary (9.2.4). — *Let* $\mathbf{P}$ *be a constructible* (*resp. ind-constructible*) *property of algebraic preschemes, X, Y two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism. For every $s \in S$, set $X_s = X \times_S \mathrm{Spec}(\mathbf{k}(s))$, $Y_s = Y \times_S \mathrm{Spec}(\mathbf{k}(s))$, $f_s = f \times 1 : X_s \to Y_s$. Then the set E of $s \in S$ such that for every $y \in Y_s$, the property $\mathbf{P}(f_s^{-1}(y), \mathbf{k}(y))$ is true, is locally constructible* (*resp. ind-constructible*).

Indeed, let Z be the set of $y \in Y$ such that $\mathbf{P}(f^{-1}(y), \mathbf{k}(y))$ is true. Since the fibres $f^{-1}(y)$ and $f_s^{-1}(y)$ are isomorphic, we see that E is the set of $s \in S$ such that $Y_s \subset Z$; if $g : Y \to S$ is the structural morphism, we have therefore $E = S - g(Y - Z)$.

Now, f is of finite presentation (1.6.2), (v)), hence it follows from (9.2.3) that Z is locally constructible (resp. ind-constructible) in Y, hence $Y - Z$ is locally constructible (resp. pro-constructible) in Y. As g is of finite presentation, $g(Y - Z)$ is locally constructible (resp. pro-constructible) in S, by virtue of Chevalley's theorem (1.8.4) (resp. of (1.9.5), (vii))); hence E is locally constructible (resp. ind-constructible) in S. **IV-3** | 59

Remark (9.2.5). — We will note that if $\mathbf{P}$ is a property of algebraic preschemes for which prop. (9.2.3) is true, then $\mathbf{P}$ also satisfies the condition (9.2.1), (ii)): this results indeed from the fact that in an irreducible Noetherian space, a constructible set is either rare or contains a non-empty open set ($\mathbf{0}_{\mathrm{III}}$, 9.2.3).

Proposition (9.2.6). — *Let* $\mathbf{P}$ *denote one of the following properties of an algebraic k-prescheme X:*

(i) *X is empty.*
(ii) *X is finite over k.*
(iii) *X is radicial over k.*
(iv) $\dim(X)$ *belongs to a given part Φ of the set $\overline{\mathbf{Z}} = \mathbf{Z} \cup \{-\infty\}$.*

Then **P** *is constructible.*

It is clear that (i) and (ii) are particular cases of (iv), taking for Φ respectively the set $\{-\infty\}$ and the set $\{-\infty, 0\}$. We therefore need only prove (iii) and (iv). In each of these two cases the condition (i) of (9.2.1) is satisfied by virtue of (2.7.1), (xv)) and (4.1.4). On the other hand, in the case (iii), the property **P** satisfies the conclusion of (9.2.3) by virtue of (1.8.7); it therefore remains to check that the same holds in the case (iv). This will follow from the more precise proposition below:

Proposition (9.2.6.1). — *If $f : X \to S$ is a morphism of finite presentation, the function $s \mapsto \dim(f^{-1}(s))$ is locally constructible.*

The question is local on S, so one can suppose that $S = \mathrm{Spec}(A)$ is affine and prove that for every n, the set E of $s \in S$ such that $\dim(X_s) = n$ is constructible. The same reasoning as in (9.2.3) reduces to the case where A is Noetherian and integral, and it suffices then to prove the following

Corollary (9.2.6.2). — *Let S be an integral Noetherian prescheme of generic point η, $f : X \to S$ a morphism of finite type. Then there exists a neighbourhood of η in S such that the function $s \mapsto \dim(X_s)$ is constant in this neighbourhood.*

The images by f of the irreducible components (in finite number) of X that do not meet X_η are contained in closed parts of S not containing η (since S is integral (**0**, 2.1.5)), hence (replacing S by a neighbourhood of η) one can restrict to the case where all irreducible components X_i of X meet X_η. Or, it suffices that there exists a finite covering (U_j) of X by
dense affine open sets, and the numbers $\dim((U_j)_\eta)$ are all equal to $n = \dim(X_\eta)$ (4.1.1.3); **IV-3** | 60
one can therefore restrict to the case where X is affine, hence also X_η. There exists then, by virtue of (4.1.2), a non-empty open set W of X such that $W \supset X_\eta$, and a $\mathbf{k}(\eta)$-morphism of finite type surjective $h : W_\eta \to \mathbf{V}^n_{\mathbf{k}(\eta)} (= \mathrm{Spec}(\mathbf{k}(\eta)[T_1, \ldots, T_n]))$; applying (8.8.2), (i)) and the method of (8.1.2), *a*)), replacing S by a neighbourhood of η if necessary, that $h = g_\eta$, where $g : W \to \mathbf{V}^n_A (= \mathrm{Spec}(A[T_1, \ldots, T_n]))$ is an S-morphism, and one can suppose this morphism of finite type and surjective by virtue of (8.10.5), (vi) and (x)). We therefore conclude that for every $s \in S$, the morphism $g_s : W_s \to \mathbf{V}^n_{\mathbf{k}(s)}$ is of finite type and surjective, hence $\dim(W_s) = n$ (4.1.2).

9.3. Constructible properties of morphisms of algebraic preschemes

Proposition (9.3.1). — *Let $\mathbf{P}(X, k)$ be a constructible* (*resp. ind-constructible*) *property of algebraic preschemes. Let $\mathbf{P}'(f, X, Y, k)$ denote the following relation: $f : X \to Y$ is a k-morphism of algebraic preschemes such that for every $y \in Y$, the property $\mathbf{P}(f^{-1}(y), \mathbf{k}(y))$ is true. Then $\mathbf{P}'$ is a constructible* (*resp. ind-constructible*) *property.*

Indeed, as **P** satisfies the condition (9.2.1), (i)), the same is true of $\mathbf{P}'$ by virtue of the transitivity of fibres (**I**, 3.6.4); furthermore the fact that $\mathbf{P}'$ satisfies the condition (9.2.1), (ii)) follows from (9.2.4), by reason of the remark (9.2.5).

Proposition (9.3.2). — *Let* **P** *denote one of the following properties of a k-morphism $f : X \to Y$ of algebraic preschemes:*

(i) f is surjective.
(ii) f is quasi-finite.
(iii) f is radicial.
(iv) For every $y \in Y$, $\dim(f^{-1}(y))$ belongs to Φ (*notation of* (9.2.6)).

Then **P** *is constructible.*

This follows immediately from (9.3.1) and (9.2.6) if one takes into account that f is of finite type (**I**, 3.5.8), and the characterisation of radicial morphisms (**II**, 6.2.2).

Proposition (9.3.3). — *Suppose that the hypotheses of* (8.8.1) *are satisfied, and let us keep the same notations; suppose furthermore that S_α is quasi-compact, X_α and Y_α of finite presentation over S_α, and let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism. Let* **P** *be an ind-constructible property of morphisms of algebraic preschemes. For every $s \in S$* (*resp. $s_\lambda \in S_\lambda$*) *set $X_s = X \times_S \mathrm{Spec}(\mathbf{k}(s))$, $Y_s = Y \times_S \mathrm{Spec}(\mathbf{k}(s))$, $f_s = f \times 1 : X_s \to Y_s$* (*resp. $X_{\lambda, s_\lambda} = X_\lambda \times_{S_\lambda} \mathrm{Spec}(\mathbf{k}(s_\lambda))$, $Y_{\lambda, s_\lambda} = Y_\lambda \times_{S_\lambda} \mathrm{Spec}(\mathbf{k}(s_\lambda))$, $f_{\lambda, s_\lambda} = f_\lambda \times 1 : X_{\lambda, s_\lambda} \to Y_{\lambda, s_\lambda}$*). *Then,* *in order that for every $s \in S$ the property $\mathbf{P}(f_s)$ be true, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that for every $s_\lambda \in S_\lambda$ we have $\mathbf{P}(f_{\lambda,s_\lambda})$.*

Indeed, let E (resp. E_λ) be the set of $s \in S$ (resp. $s_\lambda \in S_\lambda$) such that the property $\mathbf{P}(f_s)$ (resp. $\mathbf{P}(f_{\lambda,s_\lambda})$) is true; it follows from (9.2.2), (iv)) that $E_\mu = u_{\mu\lambda}^{-1}(E_\lambda)$ for $\lambda \leqslant \mu$, and $E = u_\mu^{-1}(E_\lambda)$; furthermore, by virtue of (9.2.3), E (resp. E_λ) is ind-constructible in S (resp. S_λ); the proposition then follows from (8.3.4) applied to the ind-constructible part E of S.

Remark (9.3.4). — The conjunction of (9.3.3) and of (9.3.2), (ii)) proves the assertion (8.10.5), (xi)).

Proposition (9.3.5). — *Let $\mathbf{P}(X, Y, k)$ be the property: "X and Y are two preschemes of finite type over the field k, and there exists an extension k' of k and a k'-morphism $g : X_{(k')} \to Y_{(k')}$ satisfying $\mathbf{Q}(g)$, or $\mathbf{Q}$ is one of the properties* (i) *to* (xiv) *of* (8.10.5)*". Then $\mathbf{P}$ is an ind-constructible property.*

The definition of $\mathbf{P}$ shows indeed that the condition (9.2.1), (i)) is satisfied, taking into account that the property $\mathbf{Q}(g)$ is stable under base change, and that two extensions of k can always be considered as sub-extensions of a third extension of k. To verify (9.2.1), (ii)), one can restrict to the case where $S = \operatorname{Spec}(A)$ is affine; if $K = \mathbf{k}(\eta)$, the field of fractions of A, there exists by hypothesis and by virtue of (9.1.4) a finite extension K' of K and a K'-morphism $g : (X_\eta)_{(K')} \to (Y_\eta)_{(K')}$, satisfying $\mathbf{Q}(g')$, and K' is evidently the field of fractions of an integral A-algebra of finite type A'. If we set $S' = \operatorname{Spec}(A')$, it follows then from (8.10.5) that there is a neighbourhood U' of the generic point η' of S' such that, if we set $X' = X \otimes_A A'$, $Y' = Y \otimes_A A'$, $Y' = Y \otimes_A A'$, there exists for every $s' \in U'$, a morphism $X'_{s'} \to Y'_{s'}$ having property $\mathbf{Q}$. But the morphism $h : S' \to S$ is of finite type, hence closed, and $h^{-1}(\eta) = \{\eta'\}$, $h(U')$ contains a neighbourhood of η in S; for every $s \in U$, there exists therefore $s' \in U'$ such that $h(s') = s$, and as $X'_{s'} = X_s \otimes_{\mathbf{k}(s)} \mathbf{k}(s')$, $Y'_{s'} = Y_s \otimes_{\mathbf{k}(s)} \mathbf{k}(s')$, the property $\mathbf{P}(X_s, Y_s, \mathbf{k}(s))$ is true for every $s \in U$.

(9.3.6). Example. Take for example for $\mathbf{Q}$ the property of being an isomorphism. Then, combining (9.3.5) and (9.3.3), and the hypotheses being those of (8.8.1), S_α being quasi-compact, X_α and Y_α of finite presentation over S_α, we obtain the following property: using the notations of (9.3.3), in order that, for every $s \in S$, X_s and Y_s be *geometrically isomorphic* (9.1.4), it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that, for every $s_\lambda \in S_\lambda$, X_{λ,s_λ} and Y_{λ,s_λ} be geometrically isomorphic.

We have an analogous result when the preschemes considered are endowed with "composition laws" of a certain kind ($\mathbf{0}_{\text{III}}$, 8.2.1), for example "preschemes in groups", "preschemes in rings", etc. ($\mathbf{0}_{\text{III}}$, 8.2.3). Then the previous statement is still valid when one replaces "isomorphism" by "homomorphisms" for the composition laws considered ($\mathbf{0}_{\text{III}}$, 8.2.2); it suffices here to use not only (8.10.5) but also (8.8.2), (i)) while noting that the notion of homomorphism for a composition law is expressed by saying that diagrams of morphisms of preschemes are commutative (it must be understood that the transition morphisms $X_\mu \to X_\lambda$ and $Y_\mu \to Y_\lambda$ for $\lambda \leqslant \mu$ are homomorphisms for the composition laws considered).

One can also, instead of considering morphisms of preschemes as in (9.3.5), consider homomorphisms of Modules, using (9.1.6) instead of (9.1.5).

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9.4. Constructibility of certain properties of modules

(9.4.1). Notations (9.4.1). In this n° and the following ones until the end of §9, we will systematically use the following notations: given a morphism $f : X \to S$, we will set, for every $s \in S$, $X_s = f^{-1}(s) = X \times_S \operatorname{Spec}(\mathbf{k}(s))$; for every $\mathscr{O}_X$-Module quasi-coherent $\mathscr{F}$, $\mathscr{F}_s$ will denote the $\mathscr{O}_{X_s}$-Module quasi-coherent $\mathscr{F} \otimes_{\mathscr{O}_S} \mathbf{k}(s)$, and for every homomorphism $u : \mathscr{F} \to \mathscr{G}$ of $\mathscr{F}$ into a quasi-coherent $\mathscr{O}_X$-Module $\mathscr{G}$, $u_s : \mathscr{F}_s \to \mathscr{G}_s$ will be the morphism $p^*(u)$, designating by p the canonical projection $X_s \to X$. For every section g of $\mathscr{F}$ over X, we will denote by g_s the image of g by the canonical homomorphism $\Gamma(X, \mathscr{F}) \to \Gamma(X_s, \mathscr{F}_s)$. For every part Z of X, we will denote Z_s the inverse image $p^{-1}(Z) = Z \cap X_s$ (**I**, 3.6.1). Finally, if Y is a second S-prescheme and $h : X \to Y$ an S-morphism, we will denote h_s the morphism $h \times 1 : X_s \to Y_s$.

Proposition (9.4.2). — *Let S be an integral prescheme of generic point η, $f : X \to S$ a morphism of finite presentation, $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ three quasi-coherent $\mathscr{O}_X$-Modules of finite presentation. Let $u : \mathscr{F} \to \mathscr{G}$, $v : \mathscr{G} \to$*

$\mathscr{H}$ be two homomorphisms of $\mathscr{O}_X$-Modules, and suppose that the sequence $\mathscr{F}_\eta \xrightarrow{u_\eta} \mathscr{G}_\eta \xrightarrow{v_\eta} \mathscr{H}_\eta$ is exact. Then there exists a neighbourhood U of η in S such that, for every $s \in U$, the sequence $\mathscr{F}_s \xrightarrow{u_s} \mathscr{G}_s \xrightarrow{v_s} \mathscr{H}_s$ is exact.

Proof. With the general notations of (9.2.1), it is a question here of the relation $\mathbf{P}(X, \mathscr{F}, \mathscr{G}, \mathscr{H}, u, v, k)$: "$X$ is an algebraic prescheme over the field k, $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H}$ is an exact sequence of quasi-coherent $\mathscr{O}_X$-Modules". Since, for every extension k' of k, the canonical projection $X_{(k')} \to X$ is a faithfully flat morphism (2.2.13), the condition (9.2.1), (i)) is satisfied (2.2.7). By virtue of (9.2.3), one can restrict to the case where S is integral and Noetherian, in which case X is Noetherian, and $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ are coherent $\mathscr{O}_X$-Modules. The hypothesis implies that there exists a neighbourhood U of η in S such that $\mathscr{F}|f^{-1}(U) \to \mathscr{G}|f^{-1}(U) \to \mathscr{H}|f^{-1}(U)$ is *exact*, by virtue of (8.5.8), (i)) applied following the general method of (8.1.2), *a*)), and one can therefore already suppose that $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H}$ is exact; it suffices evidently to prove that $\operatorname{Ker}(v_s) = (\operatorname{Ker} v)_s$ and $\operatorname{Im}(u_s) = (\operatorname{Im} u)_s$ for s near η in S. Taking account of the coherence of $\mathscr{F}$, $\mathscr{G}$ and $\mathscr{H}$ and of ($\mathbf{0}_{\mathrm{I}}$, 5.3.4), we are reduced to the particular case where

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{H} \longrightarrow 0$$

is exact. There is then an open neighbourhood U of η in S such that $\mathscr{H}|f^{-1}(U)$ is flat over U (6.9.1); hence (2.1.8) shows that for every $s \in U$ the sequence

$$0 \longrightarrow \mathscr{F}_s \longrightarrow \mathscr{G}_s \longrightarrow \mathscr{H}_s \longrightarrow 0$$

is exact. □

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Corollary (9.4.3). — *Let S be an integral prescheme, of generic point η, $f : X \to S$ a morphism of finite presentation. Let $\mathscr{L}^\bullet = (\mathscr{L}^i)_{i \in \mathbf{Z}}$ be a complex whose terms are quasi-coherent $\mathscr{O}_X$-Modules of finite presentation. For every i, there exists an open neighbourhood U of η in S such that the canonical homomorphisms*

(9.4.3.1) $$(\mathscr{H}^i(\mathscr{L}^\bullet))_s \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet_s)$$

are bijective for every $s \in U$.

Proof. It is enough to consider a three-term complex in degrees $-1, 0, +1$,

$$\mathscr{M} \xrightarrow{u} \mathscr{N} \xrightarrow{v} \mathscr{P},$$

and $i = 0$. The canonical homomorphism in question is then

$$(\operatorname{Ker} v / \operatorname{Im} u)_s \longrightarrow \operatorname{Ker}(v_s) / \operatorname{Im}(u_s).$$

Using (8.9.1) and (8.5.2), (i), we may reduce to the case where S (and hence X) is Noetherian; the Modules $\mathscr{M}$, $\mathscr{N}$ and $\mathscr{P}$ are then coherent. Thus $\operatorname{Im} u$ and $\operatorname{Ker} v$ are coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.4), and (9.4.2) gives a neighbourhood U of η on which

$$\operatorname{Ker}(v_s) = (\operatorname{Ker} v)_s, \qquad \operatorname{Im}(u_s) = (\operatorname{Im} u)_s.$$

The conclusion follows by applying (9.4.2) to the exact sequence

$$0 \longrightarrow \operatorname{Im} u \longrightarrow \operatorname{Ker} v \longrightarrow \operatorname{Ker} v / \operatorname{Im} u \longrightarrow 0,$$

since $\mathscr{O}_\eta = \boldsymbol{k}(\eta)$ and consequently the corresponding sequence over η is exact. □

Proposition (9.4.4). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ three quasi-coherent $\mathscr{O}_X$-Modules of finite presentation. Let $u : \mathscr{F} \to \mathscr{G}$, $v : \mathscr{G} \to \mathscr{H}$ be two homomorphisms of $\mathscr{O}_X$-Modules. Then the set E of $s \in S$ such that the sequence $\mathscr{F}_s \xrightarrow{u_s} \mathscr{G}_s \xrightarrow{v_s} \mathscr{H}_s$ is exact is locally constructible.*

Taking into account (9.2.3), it suffices to establish that the property $\mathbf{P}$ considered in (9.4.2) is constructible. We have already remarked in the proof of (9.4.2) that the condition (9.2.1), (i)) is satisfied for this property, and it remains to verify the condition (9.2.1), (ii)). Let us therefore suppose S integral Noetherian, of generic point η, and let us prove that E or $S - E$ is a neighbourhood of η. If $\eta \in E$, our assertion follows from (9.4.2), and one can therefore restrict to the case where $\eta \notin E$, in other words, the sequence $\mathscr{F}_\eta \xrightarrow{u_\eta} \mathscr{G}_\eta \xrightarrow{v_\eta} \mathscr{H}_\eta$ is not exact. Let us then distinguish two cases:

1° Set $w = v \circ u$, and suppose first that $w_\eta = v_\eta \circ u_\eta \neq 0$. As $\mathscr{F}, \mathscr{G}, \mathscr{H}$ are coherent, the same is true of $\mathscr{N} = \mathrm{Ker}(w)$ ($\mathbf{0}_\mathrm{I}$, 5.3.4); it then follows from (9.4.2) applied to the suite exact $0 \to \mathscr{N} \to \mathscr{F} \xrightarrow{w} \mathscr{H}$ that there exists a neighbourhood U of η in S such that $\mathrm{Ker}(w_s) = \mathscr{N}_s$; restricting S, one can therefore suppose this relation satisfied for every $s \in S$. Let j be the canonical injection $\mathscr{N} \to \mathscr{F}$, and set $\mathscr{M} = \mathrm{Coker}(j)$; the exactness of the right of the functor $\mathscr{F} \to \mathscr{F}_s$ entails that $M_s = \mathrm{Coker}(j_s)$ for every $s \in S$. The hypothesis $w_\eta \neq 0$ means that $M_\eta \neq 0$; as M is coherent ($\mathbf{0}_\mathrm{I}$, 5.3.4), it follows from (1.8.6) that there exists a neighbourhood U of η in S such that $M_s \neq 0$ for every $s \in U$, hence $w_s \neq 0$ for every $s \in U$, and *a fortiori* $S - E$ is a neighbourhood of η.

2° Suppose that $w_\eta = 0$; by virtue of (8.5.2), (i)), applied following the general method of (8.1.2), *a*)), there exists a neighbourhood U of η such that $w|f^{-1}(U) = 0$; replacing S by U, one can already suppose $w = 0$ in X. Then $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H}$ is a complex of three terms, to which one can apply (9.4.3); by hypothesis we have $(\mathscr{H}^0(\mathscr{L}^\bullet))_\eta = \mathscr{H}^0(\mathscr{L}^\bullet_\eta) \neq 0$, and $\mathscr{H}^0(\mathscr{L}^\bullet)$ is coherent ($\mathbf{0}_\mathrm{I}$, 5.3.4 and 5.3.3), hence it follows from (1.8.6) that there exists a neighbourhood U of η such that $(\mathscr{H}^0(\mathscr{L}^\bullet))_s \neq 0$ for every $s \in U$; but as one can suppose that $\mathscr{H}^0(\mathscr{L}^\bullet_s) = (\mathscr{H}^0(\mathscr{L}^\bullet))_s$ for every $s \in U$ by (9.4.3), we see again that $S - E$ is a neighbourhood of η in S.

Corollary (9.4.5). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}, \mathscr{G}$ two quasi-coherent $\mathscr{O}_X$-Modules of finite presentation, $u : \mathscr{F} \to \mathscr{G}$ a homomorphism of $\mathscr{O}_X$-Modules. Then the set of $s \in S$ such that u_s is injective* (*resp. surjective, bijective, zero*) *is locally constructible.*

It suffices to apply (9.4.4) to the sequences $0 \to \mathscr{F} \xrightarrow{u} \mathscr{G}$, $\mathscr{F} \xrightarrow{u} \mathscr{G} \to 0$, $0 \to \mathscr{F} \xrightarrow{u} \mathscr{G} \to 0$, $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{1_\mathscr{G}} \mathscr{G}$.

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Corollary (9.4.6). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. Let h be a section of $\mathscr{F}$ over X; for every $s \in S$, let h_s be the corresponding section of $\mathscr{F}_s$ over X_s* (*for the projection morphism $X_s \to X$*)*. Then the set of $s \in S$ such that $h_s = 0$ is locally constructible.*

It suffices to note that h corresponds to a homomorphism $u : \mathscr{O}_X \to \mathscr{F}$ and h_s to the homomorphism u_s.

Proposition (9.4.7). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. The set E* (*resp. E'*) *of $s \in S$ such that $\mathscr{F}_s$ is a locally free $\mathscr{O}_{X_s}$-module* (*resp. locally free of rank n*) *is locally constructible.*

If X is an algebraic prescheme over a field k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, k' an extension of k, then, for $\mathscr{F}$ to be locally free (resp. locally free of rank n), it is necessary and sufficient that it be so of $\mathscr{F} \otimes_k k'$, since the projection $X_{(k')} \to X$ is a faithfully flat morphism (2.2.7). In other words, the condition (9.2.1), (i)) is satisfied for the properties we wish to prove constructible, and it remains to verify (9.2.1), (ii)); one can therefore still suppose that S is affine, Noetherian and integral. There are then four cases to consider:

1° $\eta \in E$. By virtue of (8.5.5), applied following the general method of (8.1.2), *a*)), there exists a neighbourhood U of η in S such that $\mathscr{F}|f^{-1}(U)$ is locally free (resp. locally free of rank n); *a fortiori* $\mathscr{F}_s$ is locally free (resp. locally free of rank n) for every $s \in U$.

2° $\eta \in E'$. Same reasoning as in 1°.

3° $\eta \in S - E$. As $\mathscr{F}_\eta$ is a coherent $\mathscr{O}_{X_\eta}$-module, to say that it is not locally free means that it is not flat over $\mathscr{O}_{X_\eta}$ (Bourbaki, *Alg. comm.*, chap. II, §5, n° 2, cor. 2 of th. 1). The fact that $S - E$ is a neighbourhood of η will then follow from the more general lemma below:

Lemma (9.4.7.1). — *Let S be an integral Noetherian prescheme, of generic point η, X, Y two S-preschemes of finite type over S, $g : X \to Y$ an S-morphism, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. If $\mathscr{F}_\eta$ is not g_η-flat, then there exists a neighbourhood U of η in S such that for every $s \in U$, $\mathscr{F}_s$ is not g_s-flat.*

Taking into account (2.1.2) and Bourbaki, *Alg. comm.*, chap. Ier, §2, n° 3, Remarque 1, the hypothesis means that there exists a non-empty open V of Y_η, and an injective homomorphism $v : \mathscr{M} \to \mathscr{N}$ of $\mathscr{O}_V$-Modules *coherent*, such that the homomorphism $1 \otimes v : \mathscr{F}_\eta \otimes_{\mathscr{O}_V} \mathscr{M} \to \mathscr{F}_\eta \otimes_{\mathscr{O}_V} \mathscr{N}$ is not injective. We have $V = Y_\eta \cap W$, where W is open in Y ($\mathbf{I}$, 3.6.1), and it follows from (8.5.2), (i) and (ii)), applied following the method of (8.1.2), *a*)), that there exists a neighbourhood U_0 of η in S, two $\mathscr{O}_Z$-Modules coherent $\mathscr{G}$, $\mathscr{H}$ (where $Z = W \cap h^{-1}(U_0)$, $h : Y \to S$ being the structural morphism) and an $\mathscr{O}_Z$-homomorphism $u : \mathscr{G} \to \mathscr{H}$ such that $u_s : \mathscr{G}_s \to \mathscr{H}_s$ is injective (9.4.5). But for every $s \in U_0$, the homomorphism $1 \otimes u_s : \mathscr{F}_s \otimes_{\mathscr{O}_{Z_s}} \mathscr{G}_s \to \mathscr{F}_s \otimes_{\mathscr{O}_{Z_s}} \mathscr{H}_s$ is nothing but $(1 \otimes u)_s$; the hypothesis that $(1 \otimes u)_\eta$ is not injective therefore entails (9.4.5) the existence of a non-empty open $U \subset U_0$ such that for every $s \in U$, $(1 \otimes u)_s$ is not injective, and hence $\mathscr{F}_s$ is not g_s-flat for every $s \in U$. **IV-3** | 65

4° $\eta \in S - E'$. It is clear that $S - E \subset S - E'$, and if $\eta \in S - E$, $S - E'$ is *a fortiori* a neighbourhood of η by 3°. Let us therefore suppose $\eta \in E$, hence $\mathscr{F}_\eta$ is locally free; these hypotheses entail that X_η is non-connected, and that the ranks of the $\mathscr{O}_{X_\eta}$-Module locally free $\mathscr{F}_\eta$ are not the same on the various connected components of X_η. Now, it follows from (8.4.2), applied by the method of (8.1.2), *a*), that one may suppose (after replacing S by a neighbourhood of η) that X and X_η have the same number of connected components, the connected components of X_η being the intersections of X_η with the connected components of X. The conclusion then follows from the reasoning given in 2°, applied to each of the connected components of X (which are finite in number).

Remark (9.4.7.2). — The lemma (9.4.7.1) will be later generalised and freed of Noetherian hypotheses (11.2.8).

Proposition (9.4.8). — *Let S be a locally Noetherian prescheme, $f : X \to S$ a morphism of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Suppose that for every $s \in S$, X_s is a locally integral prescheme. Then the set E (resp. E') of $s \in S$ such that $\mathscr{F}_s$ is a torsion $\mathscr{O}_{X_s}$-module (resp. a torsion-free $\mathscr{O}_{X_s}$-module) is locally constructible.*

Proof. One can evidently suppose $S = \operatorname{Spec}(A)$ affine and Noetherian and prove that E (resp. E') is constructible by using the criterion ($\mathbf{0}_{\text{III}}$, 9.2.3); replacing S by the closed reduced sub-prescheme of S having as underlying space a closed irreducible part Y of S, and noting (**I**, 3.6.4) that for $s \in Y$, the fibre $(X_{(Y)})_s$ is canonically identified with X_s and the sheaf $(\mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_Y)_s$ with $\mathscr{F}_s$, we see that one is reduced to the case where S is integral of generic point η, and to proving that E or $S - E$ (resp. E' or $S - E'$) is a neighbourhood of η in S. Furthermore, X is a finite union of affine open subsets U_i of finite type over S, and each $(U_i)_s$ is an open subprescheme of X_s, hence locally integral. If $(U_i)_\eta$ is empty (resp. nonempty), then $(U_i)_s$ is also empty (resp. nonempty) in a neighbourhood of η (9.2.6). We may therefore suppose that all $(U_i)_\eta$ are nonempty and integral; then $\mathscr{F}_s$ is torsion (resp. torsion-free) if and only if each restriction $\mathscr{F}_s|_{(U_i)_s}$ is torsion (resp. torsion-free).

We are thus reduced to the case where $X = \operatorname{Spec}(B)$ is affine and $\mathscr{F} = \widetilde{M}$, with M a finite B-module. Put $B_s = B \otimes_A \mathbf{k}(s)$ and $M_s = M \otimes_A \mathbf{k}(s)$, and suppose B_η integral. There are four cases.

1° Suppose $\eta \in E$. Then M_η is a finite torsion B_η-module, so there is an $h \neq 0$ in B_η such that $hM_\eta = 0$. By (8.5.2), (i), applied as in (8.1.2), *a*), after shrinking S we may write $h = g_\eta$ for some $g \in \Gamma(X, \mathscr{O}_X)$. Let $u : \mathscr{F} \to \mathscr{F}$ be multiplication by g. Since $u_\eta = 0$, (9.4.5) shows that $u_s = 0$ near η. On the other hand, multiplication by $g_\eta = h$ is injective on $\mathscr{O}_{X_\eta}$ because B_η is integral and $h \neq 0$; again by (9.4.5), multiplication by g_s on $\mathscr{O}_{X_s}$ is injective near η. Thus g_s is $\mathscr{O}_{X_s}$-regular and annihilates $\mathscr{F}_s$, so $\mathscr{F}_s$ is torsion near η.

2° Suppose $\eta \in S - E$. If T is the torsion submodule of M_η, then $M_\eta/T \neq 0$ and, being a finite torsion-free B_η-module, it embeds into B_η^n for some n. Hence there is a nonzero homomorphism $w : M_\eta \to B_\eta^n$. By (8.5.2), (i), after shrinking S it extends to a homomorphism $v : \mathscr{F} \to \mathscr{O}_X^n$. Since $v_\eta \neq 0$, (9.4.5) gives $v_s \neq 0$ near η; as X_s is locally integral, $\mathscr{F}_s$ is not torsion for these s. **IV-3** | 65

3° Suppose $\eta \in E'$. Since M_η is a finite torsion-free B_η-module, there is an injective homomorphism $w : M_\eta \to B_\eta^n$. Using (8.5.2), (i), and (9.4.5) as in 2°, after shrinking S we obtain a homomorphism $v : \mathscr{F} \to \mathscr{O}_X^n$ such that $v_\eta = w$ and v_s is injective near η. Thus $\mathscr{F}_s$ is torsion-free near η.

4° Suppose $\eta \in S - E'$. Let T be the torsion submodule of M_η. Then $T \neq 0$ and is finite. By (8.5.2), (i) and (ii), after shrinking S there are a coherent $\mathscr{O}_X$-module $\mathscr{G}$ and an injective homomorphism $u : \mathscr{G} \to \mathscr{F}$ such that $\mathscr{G}_\eta = \widetilde{T}$ and u_η is the canonical inclusion. By 1° and (**I**, 8.6), near η the module $\mathscr{G}_s$ is a nonzero torsion $\mathscr{O}_{X_s}$-module; by (9.4.5), u_s is injective near η. Therefore the torsion submodule of $\mathscr{F}_s$ is nonzero near η. □

Remark (9.4.9). — The property "X is a locally integral algebraic k-prescheme" does not satisfy condition (9.2.1), (i), and it is therefore not certain that the statement of (9.4.8) remains valid when no hypothesis is imposed on S and one assumes only that f is a morphism of finite presentation and $\mathscr{F}$ is an $\mathscr{O}_X$-Module of finite presentation. Consider, however, the following special case. Let S_0 be a locally Noetherian prescheme, let $f_0 : X_0 \to S_0$ be a morphism of finite type whose fibres $(X_0)_{s_0}$ are locally integral for every $s_0 \in S_0$, and let $\mathscr{F}_0$ be a coherent $\mathscr{O}_{X_0}$-Module. Let $g : S \to S_0$ be any morphism, set $X = X_0 \times_{S_0} S$, $\mathscr{F} = \mathscr{F}_0 \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_S$, and $f = f_0 \times 1_S : X \to S$, and suppose that, for every $s \in S$, the fibre X_s is again locally integral. Then the set E (resp. E') of $s \in S$ such that $\mathscr{F}_s$ is torsion (resp. torsion-free) is again locally constructible. **IV-3** | 66

Indeed, let $s \in S$ and put $s_0 = g(s)$. Taking account of (1.8.2), it suffices to prove that $\mathscr{F}_s$ is torsion (resp. torsion-free) if and only if $(\mathscr{F}_0)_{s_0}$ is so. Now $\operatorname{Supp}(\mathscr{F}_s)$ is the inverse image of $\operatorname{Supp}((\mathscr{F}_0)_{s_0})$ under the projection $p : X_s \to (X_0)_{s_0}$ (**I**, 9.1.13). Since p is faithfully flat and quasi-compact, saying that $\operatorname{Supp}(\mathscr{F}_s)$ contains a maximal point of X_s is equivalent to saying that $\operatorname{Supp}((\mathscr{F}_0)_{s_0})$ contains a maximal point of $(X_0)_{s_0}$ (**I**, 1.1.5) and (2.3.4); this proves the assertion concerning E (**I**, 7.4.6). If the torsion submodule $\mathscr{T}$ of $(\mathscr{F}_0)_{s_0}$ is nonzero, then $\mathscr{T} \otimes_{\boldsymbol{k}(s_0)} \boldsymbol{k}(s)$, which is torsion by what precedes, is nonzero and identifies with a submodule of $\mathscr{F}_s$ (2.2.7); hence the torsion submodule of $\mathscr{F}_s$ is nonzero. Finally, if $(\mathscr{F}_0)_{s_0}$ is torsion-free, then after restricting to an affine open of $(X_0)_{s_0}$ we may suppose that it is isomorphic to a submodule of some $\mathscr{O}^n_{(X_0)_{s_0}}$; consequently $\mathscr{F}_s$ is isomorphic to a submodule of $\mathscr{O}^n_{X_s}$ (2.2.7), which proves the assertion concerning E'.

9.5. Constructibility of topological properties

Proposition (9.5.1). — *Let $f : X \to S$ be a morphism of finite presentation, Z a locally constructible part of X. Then the set E of $s \in S$ such that $Z_s \neq \varnothing$ is locally constructible.*

Proof. Indeed, we have $E = f(Z)$, and it suffices to apply the theorem of Chevalley (**I**, 8.4). □

Corollary (9.5.2). — *If Z', Z'' are two locally constructible parts of X, the set of $s \in S$ such that $Z'_s \subset Z''_s$ (resp. $Z'_s = Z''_s$) is locally constructible.*

Proof. Indeed, the relation $Z'_s \subset Z''_s$ is equivalent to $(Z' \cap (\complement Z''))_s = \varnothing$, and $Z' \cap \complement Z''$ is locally constructible. □

Proposition (9.5.3). — *Let $f : X \to S$ be a morphism of finite presentation, Z, Z' two locally constructible parts of X such that $Z \subset Z'$. Then the set E of $s \in S$ such that Z'_s is dense in Z_s is locally constructible in S.*

Proof. We must verify the two conditions of (9.2.2), (ii). As far as the first is concerned, consider a prescheme X algebraic over a field k, two locally closed constructible parts Z, Z' of X such that $Z \subset Z'$, and an extension k' of k. Then the projection $p : X_{(k')} \to X$ is a faithfully flat and quasi-compact morphism, and we have $p^{-1}(\overline{Z}) = \overline{p^{-1}(Z)}$ and $p^{-1}(\overline{Z'}) = \overline{p^{-1}(Z')}$; by virtue of (2.3.10), as p is surjective, the relation $\overline{Z} = \overline{Z'}$ is equivalent to $\overline{p^{-1}(Z)} = \overline{p^{-1}(Z')}$.

Let us verify the second condition, and suppose therefore S affine, Noetherian, integral, of generic point η. We distinguish two cases:

1° $\eta \in S - E$, in other words, Z_η is not dense in Z'_η; then there exists in X an open V such that $V \cap Z_\eta = \varnothing$ and $V \cap Z'_\eta \neq \varnothing$. As X is Noetherian, V is locally constructible, and by virtue of (9.5.1), there is a neighbourhood U of η in S such that for every $s \in U$, $(V \cap Z)_s = \varnothing$ and $(V \cap Z')_s \neq \varnothing$; this implies that Z_s is not dense in Z'_s for $s \in U$, in other words $U \subset S - E$.

2° $\eta \in E$, so Z_η is dense in Z'_η. Let us first show that we can suppose Z' *closed*. Indeed, Z_η is dense in $\overline{Z'_\eta}$ (closure taken in X_η); setting $V_\eta = X_\eta - \overline{Z_\eta}$, which is open in X_η and does not meet Z'_η, we can suppose V_η of the form $\overline{V} \cap X_\eta$, where V is open (and hence constructible) in X, and the hypothesis $V_\eta \cap Z'_\eta = \varnothing$ implies then $V_s \cap Z'_s = \varnothing$ for every s in a neighbourhood of η in S by virtue of (9.5.1). Replacing S by an open neighbourhood of η, we can therefore suppose $V \cap Z' = \varnothing$, whence $V \cap \overline{Z'} = \varnothing$ (closure taken in X), and consequently $(Z')_\eta = \overline{Z'_\eta}$, whence our assertion. The decomposition of Z' into its irreducible components being a finite union, and restricting further S to a neighbourhood of η, we can suppose that all the irreducible components Z'_i of Z' meet X_η, whence it follows that X_η contains the generic points of the Z'_i (**0**, 2.1.8). To say that Z_s is dense in Z'_s is equivalent to saying that each of the $(Z \cap Z'_i)_s$ is dense in $(Z'_i)_s$, and one is thus reduced to the case where Z' is *irreducible*. Replacing then X by the sub-prescheme having the reduced underlying space of Z', we see that we can suppose $Z' = X$ and that X is *integral* and dominates S. Finally, covering X by a finite number of affine opens W_j and replacing Z by $Z \cap W_j$, we can suppose that $X = \operatorname{Spec}(A)$, where A is a Noetherian integral ring. As X_η is Noetherian integral and Z_η is constructible and dense in X_η, Z_η contains a non-empty open of X_η of the form $(D(t))_\eta$, where t is a nonzero element of A. Replacing S by a neighbourhood of η in S if necessary, by virtue of the relation $(D(t))_\eta \subset Z_\eta$, we can suppose in addition that $D(t) \subset Z$ (9.5.2). Finally, since the homothety of ratio t_s in $\mathscr{O}_{X_s}$ is injective, it follows from (9.4.5) that for s close to η, t_s is $\mathscr{O}_{X_s}$-regular, so $(X_s)_{t_s}$ is dense in X_s, and *a fortiori* it is the same for Z_s which contains $(X_s)_{t_s}$.

□

Corollary (9.5.4). — *Let $f : X \to S$ be a morphism of finite presentation, Z a locally constructible part of X. The set E of $s \in S$ such that Z_s is closed (resp. open, resp. locally closed, resp. locally constructible) in X_s is locally constructible in S.*

Proof. To say that Z_s is open in X_s means that $(X - Z)_s = X_s - Z_s$ is closed in X_s, and as $X - Z$ is locally constructible, we can limit ourselves to considering the set of $s \in S$ such that Z_s is closed and the set of $s \in S$ such that Z_s is locally closed.

Let us verify in each case the two conditions of (9.2.2), (ii). The first results from the fact that $p : X_{(k')} \to X$ is faithfully flat and quasi-compact, and from (2.3.12) and (2.3.14). Let us therefore verify the second condition, S being supposed affine, Noetherian, integral, of generic point η. Set $Z' = \overline{Z}$; the same reasoning as in (9.5.3) shows that Z'_η is equal to the closure of Z_η in X_η; by virtue of (9.5.3), there is thus a neighbourhood U of η such that for $s \in U$, Z_s is dense in Z'_s, the latter being closed. To say that Z_s is closed in X_s means that $Z''_s = \varnothing$, where $Z'' = Z' - Z$; it follows from (9.5.1) that the set E of $s \in S$ where $Z''_s = \varnothing$ is such that E or $S - E$ contains a neighbourhood of η. To say that Z_s is locally closed in X_s means that Z''_s is closed in X_s; it suffices therefore to apply the preceding result replacing Z by Z'', which is locally constructible in X. □

Proposition (9.5.5). — *Let $f : X \to S$ be a morphism of finite presentation, Z a locally constructible part of X such that for every $s \in S$, Z_s is locally closed in X_s. For every $s \in S$, let $D(s) \subset \mathbf{Z} \cup \{-\infty\}$ be the set of dimensions of the irreducible components of Z_s. Then the function $s \mapsto D(s)$ is locally constructible in S.*

Proof. Let Φ be a finite part of $\mathbf{Z} \cup \{-\infty\}$; it is a question of showing that the set of $s \in S$ such that $D(s) = \Phi$ is locally constructible; taking into account (9.2.3), we have once more to verify the two conditions of (9.2.2), (ii).

As far as the first is concerned, it is a question of seeing that if X is a prescheme algebraic over a field k, Z a locally closed part of X, k' an extension of k, $p : X_{(k')} \to X$ the canonical projection, then the set of dimensions of the irreducible components of Z is the same as that of the dimensions of the irreducible components of $p^{-1}(Z)$; taking into account the existence of a sub-prescheme of X having Z as underlying space (4.2.8), this follows from (**I**, 5.2.1).

For the second condition of (9.2.2), (ii), we are in the case where S is Noetherian, integral, of generic point η, and $f : X \to S$ is a morphism of finite type, so that X is Noetherian. The subspace Z_η of the Noetherian space X_η is by hypothesis locally closed, therefore has a finite number of irreducible components $Z_{i\eta}$, which are locally closed in X. It follows for each index i that there exists a locally closed part Z_i of X such that $(Z_i)_\eta = Z_{i\eta}$, so if $Z' = \bigcup_i Z_i$, we have $Z'_\eta = Z_\eta$. But as Z and Z' are locally constructible, we can, replacing S by a neighbourhood of η, suppose that $Z' = Z$ (9.5.2). Furthermore, for $i \neq j$, $Z_{i\eta} \cap Z_{j\eta}$ is rare and *closed* in $Z_{i\eta}$; therefore (9.5.3) and (9.5.4), we can also suppose, restricting S, that for $j \neq i$, $(Z_i)_s \cap (Z_j)_s$ is rare and closed in $(Z_i)_s$. As Z_i is locally closed in X, there is a subprescheme of X having Z_i as underlying space (and still denoted Z_i), which is of finite type over S (**I**, 6.3.5). Set $U_i = Z_i - \bigcup_{j \neq i}(Z_i \cap Z_j)$, which is open in Z_i and such that $(U_i)_s$ is open and everywhere dense in $(Z_i)_s$; furthermore, by construction, the U_i have no two common points. As $(Z_i)_s$ is a $\boldsymbol{k}(s)$-prescheme algebraic, the set of dimensions of the irreducible components of $Z_s = \bigcup_i (Z_i)_s$ is the same as the set of dimensions of the irreducible components of the reunion of the $(U_i)_s$ (4.1.1.3), each of these components being already an irreducible component of one of the $(U_i)_s$. We can therefore limit ourselves to the case where $Z = U_i$; furthermore, as Z_η is then irreducible, there is only a single irreducible component of Z meeting X_η, and one can evidently, restricting Z, suppose Z irreducible. We are thus finally reduced to proving the following particular case of (9.5.5): □

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Corollary (9.5.6). — *Let S be a Noetherian prescheme, integral, of generic point η, X an irreducible prescheme, $f : X \to S$ a dominant morphism of finite type. Then there exists a neighbourhood U of η in S such that, for every $s \in U$, all the irreducible components of X_s are of dimension $n = \dim(X_\eta)$.*

Proof. We can evidently reduce to the case where $S = \operatorname{Spec}(A)$ is affine, A being therefore Noetherian; replacing f by f_{red}, which is of finite type (**I**, 5.4), (vi), we can suppose A integral and X reduced, therefore integral. We know (4.1.2) that there exists an open W dense in X_η and a $\boldsymbol{k}(\eta)$-morphism *finite* and *surjective* $h : W \to \mathbf{V}((\boldsymbol{k}(\eta))^n) = \operatorname{Spec}(B')$, where $B' = \boldsymbol{k}(\eta)[\mathrm{T}_1, \dots, \mathrm{T}_n]$ (T_i indeterminates). If V is an open of X such that $V \cap X_\eta = W$, we know (9.5.3) that for s close to η in S, V_s is an open dense in X_s, and one can therefore (4.1.3) reduce to the case where $V = X$, $W = X_\eta$. Set $B = A[\mathrm{T}_1, \dots, \mathrm{T}_n]$ and $Y = \operatorname{Spec}(B) = \mathbf{V}^n_A$, so that $\operatorname{Spec}(B') = Y_\eta$; it follows from (8.8.2), (i) and (8.10.5), (vi) and (x), applied following the method of (8.1.2), *a*), that replacing S by a neighbourhood of η, we can suppose that $h = g_\eta$, where $g : X \to Y$ is a finite surjective morphism; in other words, we have $X = \operatorname{Spec}(C)$, where C is a finite B-algebra and the homomorphism $B \to C$ corresponding to g is *injective*; as C is an integral ring, C is therefore a B-module of finite type *without torsion*, and $C_\eta = C \otimes_A \boldsymbol{k}(\eta)$ is a module of finite type without torsion over $B_\eta = B' \otimes_A \boldsymbol{k}(\eta) = \boldsymbol{k}(\eta)[\mathrm{T}_1, \dots, \mathrm{T}_n]$ (being a ring of fractions whose denominators lie in $A - \{0\} \subset B - \{0\}$). It follows from (9.4.8) that there is a neighbourhood U of η in S such that for $s \in S$, $C_s = C \otimes_A \boldsymbol{k}(s)$ is a module of finite type *without torsion* over $B_s = B \otimes_A \boldsymbol{k}(s) = \boldsymbol{k}(s)[\mathrm{T}_1, \dots, \mathrm{T}_n]$, and in particular the homomorphism $g_s : B_s \to C_s$ is injective. As no element of B_s is a divisor of zero in C_s, for every prime ideal $\mathfrak{p}_i$ of C_s (the elements being divisors of zero of C_s), we necessarily have $\mathfrak{p}_i \cap B_s = 0$, whence the canonical homomorphism $B_s \to C_s/\mathfrak{p}_i$ is injective. We deduce that for each irreducible component $Z_i = \operatorname{Spec}(C_s/\mathfrak{p}_i)$ of X_s, the restriction to Z_i of g is a finite and dominant morphism $Z_i \to Y_s$, therefore *surjective* (**II**, 6.1.10). We conclude then from (4.1.2) that $\dim Z_i = n$, which completes the proof. □

Remark (9.5.7). — One should note that under the hypotheses of (9.5.6), it may happen that X_s is irreducible for *no* $s \neq \eta$ in a neighbourhood of η, in other words, the property "X is an irreducible k-prescheme algebraic" is *not constructible*. Take for example $S = \operatorname{Spec}(k[\mathrm{T}])$, where k is an algebraically closed field, T an indeterminate; we then have $\boldsymbol{k}(\eta) = K = k(\mathrm{T})$. Let L be a finite separable extension of K of degree > 1, and let X be the *integral closure* of S in L (**II**, 6.3.4); we then have $X = \operatorname{Spec}(B)$, where B is the integral closure of $k[\mathrm{T}]$ in L. We know that B is a Dedekind ring, and that all but finitely many maximal ideals of $k[\mathrm{T}]$ are unramified in B. Moreover, the residue field of every maximal ideal of B is necessarily k, since it is a finite extension of the algebraically closed field k. Thus, for almost all maximal ideals $\mathfrak{j}_s$ of $k[\mathrm{T}]$, the ring $B_s = B/\mathfrak{j}_s B$ is a direct product of $[L : K]$ copies of k; in particular X_s is not irreducible, although $X_\eta = \operatorname{Spec}(L)$ is. The same example shows that the property "X is an integral k-prescheme algebraic" is not constructible. Finally, the property "X is a reduced k-prescheme algebraic" is also not constructible. It suffices for this to take again $S = \operatorname{Spec}(k[\mathrm{T}])$, where this time k is an algebraically closed field of characteristic $p > 0$, and for X the integral closure of S in $L = K^{1/p}$ (where $K = k(\mathrm{T})$), so that $X = \operatorname{Spec}(B)$ with $B = k[\mathrm{T}^{1/p}]$ (k being perfect); every maximal ideal of $k[\mathrm{T}]$ is of the form $(\mathrm{T} - \alpha)$ with $\alpha \in k$; the unique ideal of B over the ideal $(\mathrm{T} - \alpha)$ is the principal ideal $(\mathrm{T}^{1/p} - \alpha^{1/p})$ and it is immediate that the quotient ring $B/(\mathrm{T} - \alpha)B$ has nilpotent elements; in other words, X_s is not reduced for any $s \neq \eta$, whilst $X_\eta = \operatorname{Spec}(L)$ is integral. **IV-3 | 71**

We shall see a little further on (9.7) that one obtains on the other hand constructible properties when one considers the "geometric" notions corresponding to the notions of irreducible, reduced, or integral prescheme (4.5) and (4.6).

9.6. Constructibility of certain properties of morphisms

Proposition (9.6.1). — *Let X, Y be two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism. Let E be the set of $s \in S$ for which f_s has one of the following properties: being*

- *(i) surjective;*
- *(ii) dominant;*
- *(iii) separated;*
- *(iv) proper;*
- *(v) radicial;*
- *(vi) finite;*
- *(vii) quasi-finite;*
- *(viii) an immersion;*
- *(ix) a closed immersion;*
- *(x) an open immersion;*
- *(xi) an isomorphism;*
- *(xii) a monomorphism.*

Then E is locally constructible in S.

Proof. The assertions concerning (i), (v), and (vii) are included only for reference, having already been established in (9.3.2).

(ii): Let us note first that f is of finite presentation (**I**, 6.2), (v), therefore, by virtue of the theorem of Chevalley (**I**, 8.4), $f(X)$ is locally constructible in Y. Furthermore, we have $f_s(X_s) = (f(X))_s$ (**I**, 3.6.1); the set of s such that f_s is dominant is the set of s such that $(f(X))_s$ is dense in Y_s; the conclusion follows therefore from (9.5.3).

(iii): As $f : X \to Y$ is of finite presentation, the diagonal immersion $\Delta_f : X \to X \times_Y X$ is of finite presentation (**I**, 6.2), (iv) and (v); it follows from (1.8.4.1) that $\Delta_f(X) = \Delta_X$ is a locally constructible part of $X \times_Y X$. To say that f_s is separated means (taking into account (**I**, 5.3.4)) that $(\Delta_X)_s$ is closed in $X_s \times_{Y_s} X_s = (X \times_Y X)_s$; the conclusion follows from (9.5.4).

(iv): Let us show that the property "X and Y are preschemes algebraic over a field k, and $f : X \to Y$ a proper k-morphism" is constructible. The condition (9.2.1), (i) is verified by virtue of (2.7.1), (vii). We can therefore reduce to the case where S is affine, Noetherian, integral, of generic point η, and one must prove that E or $S - E$ is a neighbourhood of η. Suppose first that $\eta \in E$; it follows from (8.10.5), (xii), applied following the method of (8.1.2), *a*), that replacing S by a neighbourhood of η, we can suppose that f is itself a proper morphism; one then knows that it is the same for f_s for every $s \in S$ (**II**, 5.4.2), (iii). Suppose on the contrary that $\eta \in S - E$, and we distinguish two cases: **IV-3 | 72**

1° Suppose that f_η is not separated; then it follows from (iii) that f_s is not separated (and *a fortiori* not proper) in a neighbourhood of η.

2° Suppose f_η separated; Y is a finite union of affine opens V_i, and for f_s to be proper, it suffices that each of its restrictions $f_s^{-1}((V_i)_s) \to (V_i)_s$ be so (**II**, 5.4.1); we can therefore reduce to the case where Y is *affine*, hence a scheme. To say that f_η is not proper means (**II**, 5.6.3) that there exists a morphism *of finite type* $h : Z \to Y_\eta$ such that the morphism $(f_\eta)_{(Z)} : X_\eta \times_{Y_\eta} Z \to Z$ is not closed. As Y is a scheme, we deduce from (8.8.2), (i) and (ii) (restricting to S if needed a neighbourhood of η) that there exists a morphism of finite type $g : Y' \to Y$ such that Z is isomorphic to Y'_η and that $g_\eta = h$; we therefore have a closed part M' of X'_η such that $f'_\eta(M')$ is not closed in Y'_η. Now M' is the trace on X'_η of a closed part N' of X'; as X' is Noetherian and f' of finite type, $f'(N')$ is constructible in Y' (**I**, 8.4) and by hypothesis $(f'(N'))_\eta = f'_\eta(N'_\eta) = f'_\eta(M')$ is not closed in Y'_η. We conclude from (9.5.4) that there exists a neighbourhood U of η in S such that for $s \in U$, $(f'(N'))_s = f'_s(N'_s)$ is not closed in Y'_s; in other words, the morphism f'_s is not closed, and *a fortiori* the morphism f_s is not proper.

(vi): The property is a conjunction of properties (iv) and (vii) (8.11.1), therefore the proposition follows from what has already been proved.

(ix): The verification of condition (9.2.1), (i) follows from (2.7.1), (xi). We can therefore reduce to the case where S is affine, Noetherian, integral, of generic point η, and it suffices to prove that E or $S - E$ is a neighbourhood of η. If $\eta \in E$, we can (replacing S by a neighbourhood of η) suppose that

f is a closed immersion by virtue of (8.10.5), (iv), and then it is clear that f_s is a closed immersion for every $s \in S$ (**I**, 4.3.2). Suppose therefore that $\eta \in S - E$, and we distinguish two cases:

1° f_η is not a finite morphism. Then it follows from (vi) that in a neighbourhood of η, f_s is not finite, nor *a fortiori* a closed immersion.

2° f_η is finite; then, by virtue of (8.10.5), (x), we can suppose (restricting S if needed) that f is itself a *finite* morphism. In this case $\mathscr{A} = \mathscr{A}(X) = f_*(\mathscr{O}_X)$ is an $\mathscr{O}_Y$-Module coherent (**II**, 6.1.3) and $f = \mathscr{A}(u)$, where $u : \mathscr{O}_Y \to \mathscr{A}$ is a homomorphism of $\mathscr{O}_Y$-Algebras (**II**, 1.1.2); as f_η is not a closed immersion by hypothesis, u_η is not surjective (**II**, 1.4.10), therefore (9.4.5) there exists a neighbourhood of η in which u_s is not surjective, and consequently (**I**, 4.2.3) f_s is not a closed immersion.

(viii): The verification of condition (9.2.1), (i) is done as in (ix), using this time (2.7.1), (x) and the fact that every immersion of one Noetherian prescheme into another is quasi-compact. We are therefore reduced to the case where S is affine, Noetherian, integral, of generic point η, and it suffices to prove that E or $S - E$ is a neighbourhood of η. If $\eta \in E$, we conclude as in (ix) with the help of (8.10.5), (ii). If $\eta \in S - E$, we distinguish again two cases:

1° $f_\eta(X_\eta)$ is not a locally closed part of Y_η. As $f(X)$ is constructible in Y (**I**, 8.4) and $f_s(X_s) = (f(X))_s$, we deduce from (9.5.4) that for s close to η, $f_s(X_s)$ is not locally closed in Y_s, and *a fortiori* f_s is not an immersion.

2° $f_\eta(X_\eta)$ is locally closed in Y_η. As $f(X)$ is constructible in Y (**I**, 8.4) and since the same holds for $\overline{f(X)}$ since Y is Noetherian, it follows from (8.3.11) that restricting to S if necessary, we can suppose that $f(X)$ is locally closed in Y. There is then an open V of Y containing $f(X)$ and in which $f(X)$ is closed. Since Y is Noetherian, V is of finite type over S; replacing Y by V, we may therefore reduce to the case where $f(X)$ is closed in Y. But then $f_s(X_s) = (f(X))_s$ is closed in Y_s for every $s \in S$, and saying that f_s is an immersion is equivalent to saying that it is a closed immersion. We are therefore reduced to case (ix). **IV-3** | 73

(x): Using this time (2.7.1), (ix) and (8.10.5), (iii), we are reduced to the case where S is affine, Noetherian, integral, of generic point η, and where $\eta \in S - E$. We distinguish three cases:

1° $f_\eta(X_\eta)$ is not open in Y_η. As $f(X)$ is constructible in Y (**I**, 8.4), we deduce from (9.5.4) that $f_s(X_s)$ is not open in Y_s for s close to η, and *a fortiori* f_s is not an open immersion.

2° $f_\eta(X_\eta)$ is open in Y_η but f_η is not an immersion. It follows then from (viii) that for s close to η, f_s is not an immersion, nor *a fortiori* an open immersion.

3° $f_\eta(X_\eta)$ is open in Y_η and f_η is an immersion. As $f(X)$ is constructible in Y (**I**, 8.4) and since the same holds for $\overline{f(X)}$ since Y is Noetherian, it follows from (8.3.11) that restricting to S if necessary, we can already suppose that $f(X)$ is open in Y. But then $f_s(X_s) = (f(X))_s$ is open in Y_s for every $s \in S$, so we can reduce to the case where f is *surjective* and f is a closed immersion, in other words, that X is a closed sub-prescheme of Y defined by a coherent ideal $\mathscr{J}$ of $\mathscr{O}_Y$. By hypothesis f_η is not an isomorphism, hence $\mathscr{J}_\eta \neq 0$; we conclude (9.4.5) that in a neighbourhood of η, $\mathscr{J}_s \neq 0$, and consequently f_s is not an open immersion.

(xi): The property is a conjunction of properties (i) and (x) and therefore follows from what has been proved.

(xii): By virtue of (**I**, 5.3.8), to say that f_s is a monomorphism means that $\Delta_{(f)_s} = (\Delta_f)_s : X_s \to X_s \times_{Y_s} X_s = (X \times_Y X)_s$ is an isomorphism, and as $X \times_Y X$ is an S-prescheme of finite presentation (**I**, 6.2), (iv), the conclusion follows from (xi). □

Proposition (9.6.2). — *Let X, Y be two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism.*

I) *Let E be the set of $s \in S$ for which f_s has one of the following properties: being*
   (i) affine;
   (ii) quasi-affine;
   (iii) projective;
   (iv) quasi-projective.
   Then E is ind-constructible in S.

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II) *Let $\mathscr{L}$ be an $\mathscr{O}_X$-Module invertible. Then the set E' of $s \in S$ such that $\mathscr{L}_s$ is an $\mathscr{O}_{X_s}$-Module ample (resp. very ample) relatively to f_s, is ind-constructible in S.*

Proof. I) Let us verify conditions (i) and (ii) of (9.2.1). As far as the condition (9.2.1), (i) is concerned, it follows, for properties (i) and (ii), from (2.7.1), (xiii) and (xiv); for properties (iii) and (iv), it follows from (9.1.5). Let us then verify condition (9.2.1), (ii), supposing therefore S Noetherian, integral, of generic point η, and that $f_\eta : X_\eta \to Y_\eta$ has one of the properties (i) to (iv) of the statement. Applying (8.10.5), (viii), (ix), (xiii), and (xiv) following the method of (8.1.2), *a*), we see first that there exists a neighbourhood open U of η such that, if V and W are the respective inverse images of U in X and Y, the restriction $V \to W$ of f to the one of the properties (i) to (iv) that one considers. The conclusion follows then from the stability of the properties considered under base change (**II**, 4.6.13) and (4.4.10).

II) We proceed in the same way. For condition (9.2.1), (i) it follows this time from (2.7.2). For condition (9.2.1), (ii), with the same notations as in I), it follows from (8.10.5.2) that for a neighbourhood U of η in S, the restriction $\mathscr{L}|V$ is ample (resp. very ample) relatively to f. The conclusion follows again from the stability of the properties considered under base change (**II**, 4.6.13). □

We can improve (9.6.2), II) under certain conditions:

Proposition (9.6.3). — *Let X, Y be two S-preschemes of finite presentation, $f : X \to Y$ a proper S-morphism, $\mathscr{L}$ an $\mathscr{O}_X$-Module invertible. Then the set E (resp. E') of $s \in S$ such that $\mathscr{L}_s$ is an $\mathscr{O}_{X_s}$-Module ample (resp. very ample) relatively to $f_s : X_s \to Y_s$ is locally constructible in S.*

Proof. Let k be a field, Z, T two preschemes algebraic over k, $\mathscr{M}$ an $\mathscr{O}_Z$-Module invertible, $g : Z \to T$ a k-morphism of finite type; then, if $\mathbf{P}(Z, T, \mathscr{M}, g, k)$ denotes the relation "$\mathscr{M}$ is ample (resp. very ample) relatively to g", we have already noted in (9.6.2) that $\mathbf{P}(Z, T, \mathscr{M}, g, k)$ satisfies condition (9.2.1), (i) by virtue of (2.7.2). We know on the other hand that E and E' are ind-constructible. It remains therefore to see that if S is Noetherian, integral, of generic point η and if $\eta \in S - E$ (resp. $\eta \in S - E'$), then $S - E$ (resp. $S - E'$) contains a neighbourhood of η. We shall separately consider the cases E and E'.

Case of E'. — Since f is separated and Y is quasi-compact, there is an integer h such that **IV-3 | 75**
$R^q f_*(\mathscr{L}) = 0$ for $q > h$ (**III**, 1.4.12). On the other hand, since f is proper and Y is Noetherian, all the $R^q f_*(\mathscr{L})$ are coherent $\mathscr{O}_Y$-Modules (**III**, 3.2.1). They vanish except for finitely many values of q, so the generic flatness theorem (6.9.1) shows that, after restricting S to a neighbourhood of η, we may suppose that $\mathscr{L}$ and all the $R^q f_*(\mathscr{L})$ are flat over S. It then follows from (**III**, 6.9.9) that the canonical homomorphism

$$f_*(\mathscr{L}) \otimes_{\mathscr{O}_S} \boldsymbol{k}(s) \longrightarrow (f_s)_*(\mathscr{L}_s) \tag{9.6.3.1}$$

is an isomorphism.

This amounts to saying that: *either* the canonical homomorphism $(f_s)^*((f_s)_*(\mathscr{L}_s)) \to \mathscr{L}_s$ is not surjective; *or* the preceding homomorphism is surjective and the canonical morphism $r : X_s \to \mathbf{P}((f_s)_*(\mathscr{L}_s))$ is not an immersion. Taking into account the isomorphism (9.6.3.1), these conditions are written respectively in the form: 1° the canonical homomorphism $(f^*(f_*(\mathscr{L})))_s \to \mathscr{L}_s$ is not surjective; 2° the preceding homomorphism is surjective and the canonical morphism $X_s \to \mathbf{P}((f_*(\mathscr{L}))_s)$ is not an immersion.

Suppose first that the canonical homomorphism $(f^*(f_*(\mathscr{L})))_\eta \to \mathscr{L}_\eta$ is not surjective. As $f_*(\mathscr{L})$ is coherent, so is $f^*(f_*(\mathscr{L}))$, and then it follows from (9.4.5) that for every s in a neighbourhood of η, the homomorphism $(f^*(f_*(\mathscr{L})))_s \to \mathscr{L}_s$ is not surjective, which proves in this case that $S - E'$ is a neighbourhood of η.

Suppose next that the canonical homomorphism $(f^*(f_*(\mathscr{L})))_\eta \to \mathscr{L}_\eta$ is surjective but that the morphism $X \to \mathbf{P}((f_*(\mathscr{L}))_\eta)$ is not an immersion. Then the same reasoning as above shows that for s sufficiently close to η, the homomorphism $(f^*(f_*(\mathscr{L})))_s \to \mathscr{L}_s$ is surjective; on the other hand, by virtue of (9.6.1), (viii), the morphism $X_s \to \mathbf{P}((f_*(\mathscr{L}))_s)$ is not an immersion for s in a neighbourhood of η, which completes the proof in the case of E'.

Case of E. — Consider first the particular case where $Y = S$:

Corollary (9.6.4). — *Let $f : X \to S$ be a proper morphism of finite presentation, $\mathscr{L}$ an $\mathscr{O}_X$-Module invertible. Then the set E of $s \in S$ such that $\mathscr{L}_s$ is ample (relatively to f_s) is open in S, and $\mathscr{L}|f^{-1}(E)$ is ample relatively to the restriction $f^{-1}(E) \to E$ of f.*

Proof. As condition (9.2.1), (i) is verified by the property **P** defined above, the result of (9.2.2), (iv) and the reasoning of (9.2.3) show that it suffices to reduce to the case where S is *Noetherian*; but then the result follows from (**III**, 4.7.1) and the stability of amplitude under base change (**II**, 4.6.13). □

Corollary (9.6.5). — *Under the hypotheses of (9.6.4), for $\mathscr{L}$ to be ample relatively to f, it is necessary and sufficient that, for every $s \in S$, $\mathscr{L}_s$ be ample relatively to f_s.*

(9.6.6) End of the proof of (9.6.3). — Let us return to the general case, S being Noetherian, integral, of generic point $\eta \in S - E$. As f is proper, it follows from (9.6.4) that the set V of $y \in Y$ such that $\mathscr{L}_y$ is an $\mathscr{O}_{X_y}$-Module ample, relatively to the morphism $f_y : X_y \to \mathrm{Spec}(\boldsymbol{k}(y))$, is open, and therefore $F = Y - V$ is closed in Y. Since f_s is proper and, for every $y \in Y$ above $s \in S$, one has $\mathscr{L}_y = (\mathscr{L}_s)_y$, it follows from (9.6.5) that s belongs to $S - E$ if and only if $F_s \neq \varnothing$. But F is closed in Y and $F_\eta \neq \varnothing$; hence (9.5.1) shows that the set of $s \in S$ such that $F_s \neq \varnothing$ is a neighbourhood of η in S. □

Remarks (9.6.7). — (i) For each of the properties **P** considered in (9.6.1), proposition (9.3.3) is applicable, and these properties (for the morphisms f_s) are therefore "stable" under passage to a projective limit of a projective system of preschemes X_λ essentially affine (8.13.4) one over another.

(ii) Let Z be a locally constructible part of X such that, for every $s \in S$, $Z \cap X_s$ is *open* in X_s, and denote by Z_s the sub-prescheme of X_s induced on the open $Z \cap X_s$. Then, in propositions (9.6.1) **IV-3 | 76**
and (9.6.2), (I), we can everywhere replace f_s by its restriction $f_s|Z_s : Z_s \to Y_s$ without changing the conclusions. Indeed, the verification of (9.2.1), (i) proceeds as in (9.6.1) and (9.6.2). On the other hand, in the reduction to the Noetherian case, made in (9.2.3), if $Z = q^{-1}(Z_0)$, where Z_0 is the canonical projection and Z_0 a constructible part of X_0 (8.3.11), the fact that $(Z_0)_{s_0}$ is open in $(X_0)_{s_0}$, for $s_0 = p(s)$, follows from (2.4.10) and from the fact that the projection $X_s \to (X_0)_{s_0}$ is surjective. Now, as Z_η is open in X_η, there is an open $Z' \subset X$ such that $Z_\eta = Z'_\eta \cap X_\eta$; as X is Noetherian, Z' is constructible, and since it has the same properties as Z, we conclude from (9.5.2) and from ($\mathbf{0}_{\mathrm{III}}$, 9.2.2) that there is a neighbourhood U of η in S such that $Z_s = Z'_s$ for $s \in U$. Replacing S by U, we can therefore reduce to the case where Z is open in X, and then the result has been proved in (9.6.1) and (9.6.2).

9.7. Constructibility of properties of geometric separability, geometric irreducibility, and geometric connectedness

Lemma (9.7.1). — *Let S be an irreducible prescheme of generic point η, $f : X \to S$ a morphism of finite presentation. If X_η has n irreducible components (resp. n' connected components), there exists a neighbourhood open U of η in S such that for every $s \in U$, X_s has at least n irreducible components (resp. n' connected components).*

Proof. We can reduce to the case where S is affine, so X is quasi-compact and quasi-separated. Let V_i ($1 \leqslant i \leqslant n$) be the interiors of the irreducible components of X_η, which are pairwise disjoint and quasi-compact, and whose reunion is dense in X_η; by virtue of (8.2.11), applied following the method of (8.1.2), *a*), there exists for each i a quasi-compact open W_i in X such that $W_i \cap X_\eta = V_i$; as X is quasi-separated, the intersections $W_i \cap W_j$ are quasi-compact (**I**, 2.7), and therefore, replacing S by a neighbourhood of η, we can suppose that $W_i \cap W_j = \varnothing$ for $i \neq j$ by virtue of (8.3.3). Furthermore, as the W_i are constructible, there is a neighbourhood U of η in S such that for every $s \in U$ the $(W_i)_s$ are non-empty (9.5.1) and (9.2.3) and the reunion of the $(W_i)_s$ is dense in X_s (9.5.3) and (9.2.3). This being the case, as the irreducible components of each $(W_i)_s$ (being finite in number) are also irreducible components of the reunion of the $(W_i)_s$ (**0**, 2.1.7), the closures in X_s of these components are the irreducible components of X_s (**0**, 2.1.6) and their number is evidently $\geqslant n$.

Let now C_j ($1 \leqslant j \leqslant n'$) be the connected components of X_η, which are quasi-compact opens of X_η, pairwise disjoint. If we replace V_i by C_j in the preceding reasoning, we see (using twice (8.3.3)) that we can suppose X a reunion of n' quasi-compact opens M_j ($1 \leqslant j \leqslant n'$), pairwise disjoint, and such that, in a neighbourhood of η, the $(M_j)_s$ are non-empty. As the reunion of the $(M_j)_s$ is X_s, the connected components of the $(M_j)_s$ are the connected components of X_s, and therefore their number is $\geqslant n'$. □

Lemma (9.7.2). — *Let S be an irreducible prescheme of generic point η, $f : X \to S$ a morphism of finite presentation. If X_η is not reduced, there exists a neighbourhood open U of η in S such that, for every $s \in U$, X_s is not reduced.*

Proof. Indeed, let $\mathscr{N}$ be the Nilradical of $\mathscr{O}_X$; it follows from (8.2.13), applied following the method of (8.1.2), *a*), that $\mathscr{N}_\eta$ is the nilradical of $\mathscr{O}_{X_\eta}$, and the hypothesis is that $\mathscr{N}_\eta \neq 0$. We conclude from (9.4.5) and (9.2.3) that there is a neighbourhood U of η such that, for every $s \in U$, $\mathscr{N}_s$ identifies with an Ideal of $\mathscr{O}_{X_s}$ and $\mathscr{N}_s \neq 0$; as $\mathscr{N}$ is evidently contained in the Nilradical of $\mathscr{O}_{X_s}$, we see that X_s is not reduced for $s \in U$. □

(9.7.3) Given a polynomial $F \in A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, where A is a ring and the T_i are indeterminates, for a ring homomorphism $\rho : A \to B$, we denote by $F_{(\rho)}$ or $F_{(B)}$ the polynomial in $B[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ obtained by replacing each coefficient of F by its image under ρ. If k is a field, $F \in k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ a non-constant polynomial, and $X = \operatorname{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n]/(F))$, to say that X is integral (or that the ideal (F) is prime) means that F is *irreducible* (i.e. that in every factorisation $F = F_1F_2$ in polynomials of $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, F_1 or F_2 is of degree 0); this follows from the fact that the polynomial ring $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is factorial. We deduce immediately from (4.6.2) and (4.5.2):

Lemma (9.7.4). — *Let k be a field, Ω an algebraically closed extension of k, F a non-constant polynomial of $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$. The following conditions are equivalent:*

a) *$X = \operatorname{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n]/(F))$ is geometrically integral.*
b) *$F_{(K)}$ is irreducible for every extension K of k.*
c) *$F_{(\Omega)}$ is irreducible.*

We shall say in this case that F is geometrically irreducible*.*

Lemma (9.7.5). — *Let A be an integral ring, K the field of fractions of A, F a non-constant polynomial of $A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ of degree d such that $F_{(K)}$ is geometrically irreducible. Then there exists $f \neq 0$ in A such that for every $x \in D(f)$, $F_{(\boldsymbol{k}(x))}$ is geometrically irreducible.*

Proof. Set $F = \sum_\alpha c_\alpha \mathrm{T}^\alpha$ as usual, with $c_\alpha = c_{\alpha_1\alpha_2\ldots\alpha_n}$, $\mathrm{T}^\alpha = \mathrm{T}_1^{\alpha_1}\mathrm{T}_2^{\alpha_2}\cdots\mathrm{T}_n^{\alpha_n}$, $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n \leqslant d$ (at most the c_α such that $|\alpha| = d$ being not zero). As the nonzero c_α are invertible in K, we can suppose, replacing A by A_g ($g \neq 0$ in A), that the c_α are nonzero and invertible *in* A. It then follows that for every $x \in \operatorname{Spec}(A)$, $F_{(\boldsymbol{k}(x))}$ is of degree d. □

Proof (continued). We shall prove the lemma by means of a preliminary lemma.

Lemma (9.7.5.1). — *Let A be an integral ring, $S = \operatorname{Spec}(A)$, η the generic point of S, B the polynomial ring $A[\mathrm{T}_1, \ldots, \mathrm{T}_m]$, $(P_i)_{i \in I}$ a finite family of elements of B, $\mathfrak{a}$ the ideal of B generated by the P_i. For every $s \in S$, let $\mathfrak{a}_s$ be the ideal of $\boldsymbol{k}(s)[\mathrm{T}_1, \ldots, \mathrm{T}_m]$ generated by the polynomials $(P_i)_{(\boldsymbol{k}(s))}$ ($i \in I$). Then, if $V(\mathfrak{a}_\eta) = \varnothing$, there exists a neighbourhood U of η in S such that $V(\mathfrak{a}_s) = \varnothing$ for every $s \in U$.*

Proof. Indeed, let $Y = \operatorname{Spec}(B)$, and let Z be the closed part $V(\mathfrak{a})$ of Y; as $\mathfrak{a}$ is an ideal of finite type in B, Z is constructible in Y (**0**, 9.1.5) and, with the notations introduced in (9.4.1), we have $V(\mathfrak{a}_s) = Z_s$ for every $s \in S$; as the structural morphism $Y \to S$ is of finite presentation, the conclusion of the lemma follows from (9.5.1) and (9.2.3). □

Let (p, q) be a pair of integers > 0 such that $p + q = d$; we introduce indeterminates T'_β, T''_γ **IV-3** | 78
for all systems of integers β, γ such that $|\beta| \leqslant p$ and $|\gamma| \leqslant q$; for every system of integers α such that $|\alpha| \leqslant d$, consider the polynomial of $B = A[\mathrm{T}'_\beta, \mathrm{T}''_\gamma \mid |\beta| \leqslant p, |\gamma| \leqslant q] : P_\alpha(\mathrm{T}'_\beta, \mathrm{T}''_\gamma) = \sum_{\delta+\gamma=\alpha} \mathrm{T}'_\delta \mathrm{T}''_\gamma - c_\alpha$. Let Ω be an algebraic closure of K; to say that there exist two polynomials F_1, F_2 of $\Omega[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, of respective degrees p and q, and such that $F_1F_2 = F_{(\Omega)}$ means that the system of equations $P_\alpha(t'_\beta, t''_\gamma) = 0$ ($|\alpha| \leqslant d$) admits a solution (t'_β, t''_γ) formed of elements of Ω. Let $\mathfrak{a}$ be the ideal of B generated by the P_α; the preceding interpretation and the theorem of the zeros of Hilbert show that the hypothesis on $F_{(K)}$ implies that $V(\mathfrak{a}_\eta) = \varnothing$, designating by η the generic point of $\operatorname{Spec}(A)$; lemma (9.7.5.1) then proves that in a neighbourhood of η, we have $V(\mathfrak{a}_s) = \varnothing$,

and consequently for these values of x, $F_{(\boldsymbol{k}(x))}$ does not admit any factorisation $F_{(\boldsymbol{k}(x))} = G_1 G_2$, where G_1, G_2 are polynomials of respective degrees p and q, whose coefficients lie in an algebraic closure of $\boldsymbol{k}(x)$. It suffices to apply this result for all pairs of integers (p, q) such that $p > 0$, $q > 0$ and $p + q = d$, to obtain the conclusion of lemma (9.7.5). □

Proposition (9.7.6). — *Let S be a Noetherian prescheme, integral, of generic point η, $f : X \to S$ a morphism of finite type, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. If $\mathscr{F}_\eta$ is without associated prime cycle, there exists a neighbourhood U of η in S such that, for every $s \in U$, $\mathscr{F}_s$ is without associated prime cycle.*

Proof. As X_η is Noetherian, there exists an injective homomorphism $v : \mathscr{F}_\eta \to \bigoplus_{i=1}^m \mathscr{G}_i$, where each $\mathscr{G}_i$ is irredundant and $\operatorname{Ass}(\mathscr{F}_\eta)$ is the set of the x_i, with $\{x_i\} = \operatorname{Ass}(\mathscr{G}_i)$ (3.2.6); if Z_i is the closure of $\{x_i\}$ in X_η, then $Z_i = \operatorname{Supp}(\mathscr{G}_i)$ (3.1.4), and by hypothesis the Z_i are the irreducible components of $Z = \operatorname{Supp}(\mathscr{F}_\eta)$. It follows from (8.5.2), (i) and (ii) (restricting to S if necessary a neighbourhood of η) and (8.5.8) that there exist $\mathscr{O}_X$-Modules coherent $\mathscr{F}_i$ and an injective homomorphism $u : \mathscr{F} \to \bigoplus_{i=1}^m \mathscr{F}_i$ such that $(\mathscr{F}_i)_\eta = \mathscr{G}_i$ for every i and $v = u_\eta$. If $Y_i = \operatorname{Supp}(\mathscr{F}_i)$, we have $(Y_i)_s = \operatorname{Supp}((\mathscr{F}_i)_s)$ for every $s \in S$ (**I**, 9.1.13) and in particular $(Y_i)_\eta = Z_i$ for every i. The hypothesis implies that for $i \neq j$, $Z_i - (Z_i \cap Z_j)$ is open and dense in Z_i; we deduce therefore from (9.5.3) and (9.5.4) that in a neighbourhood of η, $(Y_i)_s - ((Y_i)_s \cap (Y_j)_s)$ is open and dense in $(Y_i)_s$. Suppose that one has proved that each of the $(\mathscr{F}_i)_s$ is without associated prime cycle for $s \in U$. Then the elements of $\operatorname{Ass}((\mathscr{F}_i)_s)$ are the maximal points of $(Y_i)_s$; no two of them can therefore belong to $(Y_j)_s$ for $i \neq j$, and the decomposition is irredundant. We may moreover suppose that $\mathscr{F}_\eta$ is *irredundant*; in addition, we may reduce to the case where $S = \operatorname{Spec}(A)$ is affine and $f : X \to S$ is dominant. Then A is a Noetherian integral ring, B is a finite-type A-algebra containing A, and $\mathscr{F} = \widetilde{M}$ with M a finite B-module. By hypothesis, if K is the field of fractions of A, the $B_{(K)}$-module $M_{(K)}$ is irredundant. Let $\mathfrak{q}$ be the unique element of $\operatorname{Ass}(M_{(K)})$, and let $\mathfrak{p}$ be the prime ideal of B which is the inverse image of $\mathfrak{q}$ by the canonical map $B \to B_{(K)}$. We know (5.11.1.1) that there exists a finite filtration $(N_h)_{0 \leqslant h \leqslant m}$ of $M_{(K)}$ such that $N_m = 0$, and N_h / N_{h+1} is isomorphic to a nonzero $B_{(K)}$-sub-module $\neq 0$ of $B_{(K)}/\mathfrak{q}$. Let M_h be the inverse image of N_h by the canonical map $M \to M_{(K)}$. It follows from (9.4.4) that for s sufficiently close to η, $(M_h)_{(\boldsymbol{k}(s))}$ identifies with a sub-$B_{(\boldsymbol{k}(s))}$-module of $M_{(\boldsymbol{k}(s))}$, and the quotient $(M_h)_{(\boldsymbol{k}(s))}/(M_{h+1})_{(\boldsymbol{k}(s))}$ to a nonzero $B_{(\boldsymbol{k}(s))}$-sub-module $\neq 0$ of $B_{(\boldsymbol{k}(s))}/\mathfrak{p}_{(\boldsymbol{k}(s))}$. Taking into account (3.1.7), with the same notations, it remains to prove, that if B is *integral*, then $B_{(\boldsymbol{k}(s))}$ has no associated prime cycle for s close to η. Now, replacing A by A_g (where g is a nonzero element $\neq 0$ of A), we can suppose that B contains a polynomial sub-algebra $C = A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ such that B is a C-module of finite type (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, cor. 1 of th. 1). As $B_{(K)}$ is a torsion-free $C_{(K)}$-module, we may apply the preceding argument once more, replacing B, M, and $\mathfrak{q}$ by C, B, and (0) respectively; it suffices to show that for s close to η, $C_{(\boldsymbol{k}(s))}$ has no embedded associated prime. But this is obvious since $C_{(\boldsymbol{k}(s))} = \boldsymbol{k}(s)[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is an integral ring. □ **IV-3 | 79**

Theorem (9.7.7). — *Let $f : X \to S$ be a morphism of finite presentation, and let E be the set of $s \in S$ for which X_s has one of the following properties: being*

(i) geometrically irreducible;
(ii) geometrically connected;
(iii) geometrically reduced;
(iv) geometrically integral.

Then E is locally constructible in S.

Proof. To see that the properties considered are constructible, let us note first that they trivially satisfy condition (9.2.1), (i), by virtue of (4.5.6), (i) and (4.6.5), (i). It remains therefore to verify (9.2.1), (ii), so we can suppose $S = \operatorname{Spec}(A)$ affine, Noetherian, integral, of generic point η. By virtue of (4.6.8), there exists a finite extension K' of $K = \boldsymbol{k}(\eta)$ such that $(X_\eta)_{(K')}$ has its irreducible components (resp. connected components, resp. connected components) geometrically irreducible (resp. geometrically connected) and that $((X_\eta)_{(K')})_{\mathrm{red}}$ is geometrically reduced. Furthermore, there exists a base of K' over K formed of elements integral over A, the integral ring A' generated by these elements is finite over A and has K' as field of fractions. Set $S' = \operatorname{Spec}(A')$; the morphism $g : S' \to S$ is finite and dominant, therefore surjective (**II**, 6.1.10), so that $X' = X_{(S')}$; let η' be the generic point of S', $X'_{\eta'} = (X_\eta)_{(K')}$; the set E' of $s' \in S'$ such that $X'_{s'}$ has one of the properties (i), (ii), (iii), or (iv) (E corresponding well to the same property) (9.2.2), (iv); and $S - E = g(S' - E')$ and $g^{-1}(\eta) = \{\eta'\}$ since A' is integral and finite over A, therefore the image by g of every neighbourhood of η' is a neighbourhood of η. The theorem will therefore be proved if one proves that E' or $S' - E'$ is a neighbourhood of η'. In other words, we can henceforth suppose that the irreducible components (resp. connected components) of X_η are geometrically irreducible (resp. geometrically connected) and that $(X_\eta)_{\mathrm{red}}$ is geometrically reduced.

Suppose first that $\eta \in S - E$ for one of the properties (i) to (iv). If X_η is not geometrically irreducible (resp. geometrically connected), it is not irreducible (resp. connected) by virtue of the preceding hypothesis, and therefore it is the same for X_s for s in a neighbourhood of η (9.7.1), and *a fortiori* in this neighbourhood X_s is not geometrically irreducible (resp. geometrically connected). On the other hand, if X_η is not geometrically reduced, it is not reduced (since otherwise it would be equal to $(X_\eta)_{\mathrm{red}}$ which is geometrically reduced by hypothesis); therefore, in a neighbourhood of η, X_s is not reduced (9.7.2) and hence not geometrically reduced. Finally, if X_η is not geometrically integral, either it is not reduced (nor *a fortiori* integral) in a neighbourhood of η; or X_η is reduced (therefore geometrically reduced by hypothesis), and then it is not geometrically irreducible, and we have just seen above that it is then the same for X_s for s close to η; *a fortiori* X_s is not geometrically integral for these values of s. **IV-3** | 80

We shall now suppose that $\eta \in E$ and examine separately each of the properties considered.

1° Suppose that X_η is geometrically integral. Let L be the field of rational functions on X_η; the hypothesis on X_η implies that L is a separable extension of K (4.6.3), therefore a finite separable extension of a pure extension $K(\mathrm{T}_1, \dots, \mathrm{T}_n)$ (T_i indeterminates); there is then an element $x \in L$, integral over the ring $K[\mathrm{T}_1, \dots, \mathrm{T}_n]$, and such that $L = K(\mathrm{T}_1, \dots, \mathrm{T}_n)(x)$; let $G \in K[\mathrm{T}_1, \dots, \mathrm{T}_n, \mathrm{T}_{n+1}]$ be its minimal polynomial. There exists $g \neq 0$ of A such that all the coefficients $\neq 0$ of G (which all belong to K) belong to A_g; replacing A by A_g (which amounts to replacing S by a neighbourhood of η), we can therefore suppose that G has its coefficients in A; designating by F the polynomial G considered as element of $A[\mathrm{T}_1, \dots, \mathrm{T}_n, \mathrm{T}_{n+1}]$, we then have $G = F_{(\boldsymbol{k}(\eta))}$. Set $Y = \mathrm{Spec}(A[\mathrm{T}_1, \dots, \mathrm{T}_{n+1}]/(F))$, so that $Y_\eta = \mathrm{Spec}(K[\mathrm{T}_1, \dots, \mathrm{T}_{n+1}]/(G))$; Y_η is an integral scheme having L as field of rational functions. As X_η and Y_η are Noetherian, there exists a non-empty open $V \subset Y_\eta$ and an immersion (necessarily dominant) $v : V \to X_\eta$ (**I**, 6.5.1), (ii) and (6.5.4), (ii). Let W be an open of Y such that $W \cap Y_\eta = V$; applying (8.8.2), (i) and (8.10.5), (iii) following the method of (8.1.2), *a*), replacing S by a neighbourhood of η, we can suppose that $v = w_\eta$ where $w : W \to X$ is an open immersion.

Since we have seen (4.6.3) that the criterion for a prescheme algebraic integral to be geometrically integral depends only on its field of rational functions; as X_η is geometrically integral by hypothesis, it is the same for Y_η, and the definition of G implies therefore that this polynomial is geometrically irreducible (9.7.4). Applying (9.7.5), we see that there is a neighbourhood U of η in S such that $F_{(\boldsymbol{k}(s))}$ is geometrically irreducible for every $s \in U$, therefore Y_s is geometrically integral for $s \in U$ (9.7.4); furthermore we can also suppose that for $s \in U$, W_s is not empty (9.5.1), and therefore is geometrically integral (4.6.3); finally, we can also suppose that for $s \in U$, $w_s : W_s \to X_s$ is an open immersion (9.6.1), (x), and that its image in X_s is everywhere dense (9.5.3). In other words, for $s \in U$, there is in X_s an open everywhere dense W'_s which is geometrically integral; the criterion (4.5.9), *c*) therefore shows that X_s is already geometrically irreducible for $s \in U$. Finally, as X_η is reduced and without associated prime cycle (3.2.1), we can also suppose that for $s \in U$, X_s has no associated prime cycle (9.7.6); this being the case (for $s \in U$ fixed), let Ω be an algebraically closed extension of $\boldsymbol{k}(s)$, and let $p : (X_s)_{(\Omega)} \to X_s$ be the canonical projection; $p^{-1}(W'_s)$ is an open dense in $(X_s)_{(\Omega)}$ (2.3.10) and is integral by hypothesis; furthermore $(X_s)_{(\Omega)}$ has no associated prime cycle (4.2.7), therefore we conclude from (3.2.1) that $(X_s)_{(\Omega)}$ is reduced; we conclude therefore from (3.2.1) and (4.6.1) that X_s is geometrically integral (4.6.1).

2° Suppose that X_η is geometrically irreducible; as X_{red} is also of finite type over S (**I**, 5.1.8), taking into account (**I**, 5.4), (vi), we can, by replacing X by X_{red}, then X_η is also integral, and as by hypothesis X_η is geometrically irreducible, it is geometrically integral, and we are brought back to the conditions of 1° and we conclude (returning to the initial hypotheses) that X_s is geometrically irreducible for s close to η.

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3° Suppose X_η geometrically connected, and let Z_i ($1 \leqslant i \leqslant n$) be the irreducible components of X_η; there exists (by virtue of (5.10.8.1) applied to $\mathfrak{T} = \{\varnothing\}$) a *surjective* map $j \mapsto \nu(j)$ of an interval $[1, m]$ of $\mathbf{N}$ onto $[1, n]$ such that $Z_{\nu(j)} \cap Z_{\nu(j+1)} \neq \varnothing$ for $1 \leqslant j \leqslant m$. For every i, let X_i be the closure of Z_i in X, and let Y be the reunion of the X_i; as $Y_\eta = X_\eta$ by definition, we can, by virtue of (9.5.1), suppose that $Y_s = X_s$ for every $s \in S$. Now, considering the reduced sub-preschemes of X having the X_i as underlying spaces, we see by 2° that there exists a neighbourhood U of η in S such that for $s \in U$ the $(X_i)_s$ are *geometrically irreducible* (since the $(X_i)_\eta$ can be supposed to be geometrically irreducible, as we saw at the beginning). Furthermore, we can also suppose that for $s \in U$, $(X_{\nu(j)})_s \cap (X_{\nu(j+1)})_s \neq \varnothing$ (9.5.1) for $1 \leqslant j \leqslant m$; we conclude immediately that X_s is connected, therefore (4.5.13.1) geometrically connected for $s \in U$.

4° Suppose X_η geometrically reduced; let Z_i, W_i be the irreducible components of X_η, where W_i is the interior of Z_i in X_η; there is for each i an open V_i of X such that $W_i = V_i \cap X_\eta$ for every i; as the W_i are opens, pairwise disjoint, and their reunion is dense in X_η, (9.5.1), (9.5.3), and (9.5.4) allow us to suppose that for s close to η, the $(V_i)_s$ are opens, pairwise disjoint, in X_s and their reunion is dense in X_s. Furthermore, as the W_i are geometrically reduced and have been supposed at the beginning geometrically irreducible, it follows from 1° that for s close to η, the $(V_i)_s$ are geometrically integral, and *a fortiori* reduced. On the other hand, we deduce from (9.7.6) that X_s has no associated prime cycle, since it is already reduced (3.2.1); we conclude therefore from (3.2.1) and (4.6.1) that X_s is reduced, and from (4.6.1) that X_s is geometrically reduced. □

The parts of the statement of (9.7.7) concerning properties (i) and (ii) generalize as follows.

Proposition (9.7.8). — *Let S be an irreducible prescheme with generic point η, and let $f : X \to S$ be a morphism of finite presentation. Let n (resp. n') be the geometric number of irreducible components (resp. connected components) of X_η (4.5.2). Then there exists a neighborhood U of η in S such that, for every $s \in U$, the geometric number of irreducible components (resp. connected components) of X_s is equal to n (resp. n').* **IV-3 | 82**

Proof. Taking into account that the geometric number of irreducible components (resp. connected components) of an algebraic prescheme is invariant under extension of the base field (4.5.6), one sees by the method of (9.2.3) that one may reduce to the case where S is affine, Noetherian, and integral. Moreover, arguing as at the beginning of the proof of (9.7.7), one sees that one may suppose that the irreducible components (resp. connected components) of X_η are geometrically irreducible (resp. geometrically connected). We already know from (9.7.1) that, for every s near η, X_s has at least n irreducible components and n' connected components.

On the other hand, if Z_i (resp. Z'_j) are the irreducible components (resp. connected components) of X_η, let X_i (resp. X'_j) be the reduced closed subprescheme of X whose underlying space is the closure of Z_i (resp. Z'_j) in X. It follows from (9.5.1) that, in a neighborhood of η, the $(X_i)_s$ (resp. $(X'_j)_s$) cover X_s, and that the $(X'_j)_s$ are pairwise disjoint. Since the $(X_i)_s$ (resp. $(X'_j)_s$) are geometrically irreducible (resp. geometrically connected) by (9.7.7) for s near η, it follows that X_s has at most n irreducible components and at most n' connected components, which proves the proposition. □

Corollary (9.7.9). — *Let $f : X \to S$ be a morphism of finite presentation. For every $s \in S$, let $n(s)$ (resp. $n'(s)$) be the geometric number of irreducible components (resp. connected components) of $f^{-1}(s)$ (4.5.2). Then the function $s \mapsto n(s)$ (resp. $s \mapsto n'(s)$) is locally constructible on S.*

Proof. It is enough to show that the property "X is an algebraic prescheme over a field k and the geometric number of irreducible components (resp. connected components) of X is equal to n (resp. n')" is constructible. It follows from (4.5.6) that this property satisfies condition (9.2.1), (ii); hence one is reduced to the case where S is affine, Noetherian, and integral with generic point η, and the conclusion then follows from (9.7.8). □

Corollary (9.7.10). — *Let X be a locally Noetherian prescheme such that, if X' is the normalization of X_{red}, the canonical morphism $f : X' \to X$ is finite. Then the set of points $x \in X$ such that X is geometrically unibranch at x is locally constructible in X.*

Proof. Indeed, this set is, by definition, the set of points $x \in X$ such that the number of geometric points of $f^{-1}(x)$ is equal to 1. Since f is finite, this number is also the geometric number of irreducible components of the discrete space $f^{-1}(x)$ (taking account of the definition of the normalization (**II**, 6.3.8) and of (4.5.11)); the conclusion therefore follows from (9.7.9). □

Remark (9.7.11). — Let Z be a locally constructible subset of X such that, for every $s \in S$, $Z \cap X_s$ is open in X_s, and write Z_s for the subprescheme of X_s induced on the open subset $Z \cap X_s$. Then, in (9.7.7), one may replace X_s everywhere by Z_s without changing the conclusion; this follows by repeating the argument used in (9.6.7), (ii).

Proposition (9.7.12). — *Let $f : X \to S$ be a morphism of finite presentation, and let $g : S \to X$ be an S-section of X (**I**, 2.5.5). For every $s \in S$, let X_s^0 be the connected component of $f^{-1}(s)$ containing $g(s)$. Then the union X^0 of the X_s^0, for $s \in S$, is a locally constructible subset of X.* **IV-3 | 83**

Proof. We first show that one may reduce to the case where S is affine and Noetherian. One may always suppose that $S = \mathrm{Spec}(A)$ is affine; with the notation of (9.2.3), one has $f = (f_0)_{(S)}$, where $f_0 : X_0 \to S_0$ is a morphism of finite type, and one may moreover suppose that there exists an S_0-section $g_0 : S_0 \to X_0$ such that $g = (g_0)_{(S)}$ (8.9.1). If $p : S \to S_0$ is the canonical morphism, then, for every $s_0 \in S_0$, the connected component $(X_0)^0_{s_0}$ of $f_0^{-1}(s_0)$ containing $g_0(s_0)$ is geometrically connected (4.5.13); consequently, if $s_0 = p(s)$, one has $X_s^0 = q^{-1}((X_0)^0_{s_0})$, where $q : X_s \to (X_0)_{s_0}$ is the canonical projection (4.5.8) and (4.4.1). Our assertion therefore follows from (**I**, 1.8.2).

We now use the constructibility criterion ($\mathbf{0_{III}}$, 9.2.3). Let x be a point of X, let Z be the reduced subprescheme of X whose underlying space is $\overline{\{x\}}$, and let Y be the reduced subprescheme of S whose underlying space is $\overline{f(Z)}$. Since the restriction of f to Z factors as $Z \to Y \to S$ (**I**, 5.2.2), one may replace S by Y, that is, suppose that $f(x) = \eta$, the generic point of the integral prescheme S.

By hypothesis, X_η is the sum of two subpreschemes X_η^0 and X_η^1 induced on complementary open subsets of X_η. By (9.5.4) and (9.5.1), after replacing S, if necessary, by a neighborhood of η, one may suppose that X is the union of two disjoint open subsets X^0 and X^1 such that $(X^0)_\eta = X_\eta^0$ and $(X^1)_\eta = X_\eta^1$. Since $g : S \to X$ is continuous and injective, S is the union of the two disjoint open subsets $g^{-1}(X^0)$ and $g^{-1}(X^1)$; but S is irreducible, and *a fortiori* connected, so one of these two open subsets is empty. Since $g(\eta) \in X_\eta^0$ by definition, one has $g^{-1}(X^1) = \varnothing$.

In other words, g is an S-section of X^0. On the other hand, since $(X^0)_\eta$ is geometrically connected (4.5.13), the same is true of $(X^0)_s$ for every s near η by (9.7.7); and since $g(s) \in (X^0)_s$, one indeed has $(X^0)_s = X_s^0$. □

9.8. Primary decomposition in the neighbourhood of a generic fibre

Proposition (9.8.1). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. Then the set E of $s \in S$ such that $\mathscr{F}_s$ is an $\mathscr{O}_{X_s}$-Module without associated immersed prime cycle is locally constructible.* **IV-3 | 83**

Proof. We know that for quasi-coherent Modules on algebraic preschemes, the property of being without associated immersed prime cycle is invariant by change of base field (4.2.7); we can thus restrict to the case where S is affine, Noetherian and integral, and to proving that in this case E or $S - E$ is a neighbourhood of η. We saw in (9.7.6) that if $\eta \in E$, E is a neighbourhood of η; it remains thus to consider the case where $\mathscr{F}_\eta$ has associated immersed prime cycles. We can restrict to the case where X is affine, and where there is a sub-$\mathscr{O}_{X_\eta}$-Module coherent $\mathscr{H}$ of $\mathscr{F}_\eta$ whose support Z' is non-empty and *rare* with respect to the support Z of $\mathscr{F}$ (3.1.3); then, by virtue of (8.5.2), (i) and (ii)) and (8.5.8), applied following the method of (8.1.2), *a*)), there exists a sub-$\mathscr{O}_X$-Module coherent $\mathscr{G}$ of $\mathscr{F}$ such that $\mathscr{G}_\eta = \mathscr{H}$; if $Y = \mathrm{Supp}(\mathscr{F})$, $Y' = \mathrm{Supp}(\mathscr{G})$, we have by extension $Y_s = \mathrm{Supp}(\mathscr{F}_s)$, $Y'_s = \mathrm{Supp}(\mathscr{G}_s)$ for all $s \in S$ (**I**, 9.1.13) and in particular $Z = Y_\eta$, $Z' = Y'_\eta$; as $Z - Z'$ is dense in Z and $Z' + \varnothing$, there is a neighbourhood U of η such that for $s \in U$, $Y_s - Y'_s$ is dense in Y_s, and $Y'_s \neq \varnothing$ (9.5.1) and (9.5.3); by considering a generic point of an irreducible component of Y'_s and a neighbourhood sufficiently small of this point in X_s, we deduce immediately from (3.1.3) that $\mathscr{F}_s$ admits associated immersed prime cycles. **IV-3 | 84**

(9.8) Let S be a Noetherian prescheme, *integral*, of generic point η, $f : X \to S$ a morphism of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Consider an irredundant reduced decomposition $(\mathscr{G}_i)_{i \in I}$ of $\mathscr{F}_\eta$ (3.2.6); the $\mathscr{G}_i$ are thus quotients of $\mathscr{F}_\eta$ and there is an injective homomorphism $h : \mathscr{F}_\eta \to$

$\bigoplus \mathscr{G}_i$; furthermore $\mathrm{Ass}(\mathscr{G}_i)$ is reduced to a point x_i. Let Z_i be the closure of $\{x_i\}$ in X, so that $(Z_i)_\eta = \mathrm{Supp}(\mathscr{G}_i)$. Denote by $\mathscr{J}_i$ the coherent Ideal of $\mathscr{O}_X$ defining the closed sub-prescheme of X of underlying space Z_i (sub-prescheme still denoted Z_i). By virtue of (8.5.2) and (8.5.8), applied following the method of (8.1.2), *a*)), we can (by restricting S to a neighbourhood of η if needed) suppose that there exist quotients $\mathscr{F}_i$ of $\mathscr{F}$ ($i \in I$) such that $\mathscr{G}_i = (\mathscr{F}_i)_\eta$ for all i, and a homomorphism $g : \mathscr{F} \to \bigoplus \mathscr{F}_i$ such that $h = g_\eta$. Furthermore (**I**, 9.3.5), there exists an integer m such that $(\mathscr{J}_i^m \mathscr{F}_i)_\eta = (\mathscr{J}_i)_\eta^m \mathscr{G}_i = 0$ and by restricting S again, we can suppose that $\mathscr{J}_i^m \mathscr{F}_i = 0$ (8.5.2.5), so that the support of $\mathscr{F}_i$ is contained in Z_i; but as it is closed and contains x_i, it is necessarily equal to Z_i. □

Proposition (9.8.3). — *Under the conditions of* (9.8), *for all $s \in S$, and all $i \in I$, denote by $x_{i\alpha}$ ($\alpha \in J_{s,i}$) the maximal points of $(Z_i)_s$. There exists a neighbourhood U of η in S such that, for all $s \in U$, the $x_{i\alpha}$ (for $i \in I$ and, for each i, $\alpha \in J_{s,i}$) are pairwise distinct and that $\mathrm{Ass}(\mathscr{F}_s)$ is the set of $x_{i\alpha}$ (in other words, the prime cycles associated to $\mathscr{F}_s$ are the irreducible components of $(Z_i)_s$). Furthermore, one can take U such that, for the closure of $\{x_{i\alpha}\}$ in X_s to be a maximal associated prime cycle of $\mathscr{F}_s$, it is necessary and sufficient that $(Z_i)_\eta$ (closure of $\{x_i\}$ in X_η) be a maximal associated prime cycle of $\mathscr{F}_\eta$.*

Proof. It follows from (3.1.3), c')) that for each i, there exists an open W_i in X_η such that $W_i \cap (Z_i)_\eta$ is non-empty, and a coherent $\mathscr{O}_{X_\eta}$-Module $\mathscr{H}_i$, of support $W_i \cap (Z_i)_\eta$, such that there is an injective homomorphism $v_i : \mathscr{H}_i \to \mathscr{F}_\eta | W_i$. Let V_i be an open of X such that $V_i \cap X_\eta = W_i$; applying, as in (9.8.2), the results of (8.5.2) and (8.5.8), on can (by restricting S) suppose that there exists a coherent Module $\mathscr{F}'_i$, of support $V_i \cap Z_i$ and a homomorphism $u_i : \mathscr{F}'_i \to \mathscr{F}|V_i$ such that $\mathscr{H}_i = (\mathscr{F}'_i)_\eta$ and $v_i = (u_i)_\eta$. We will show that there is a neighbourhood U of η such that for $s \in U$, the following properties are satisfied:

(i) The $(\mathscr{F}_i)_s$ are without associated immersed prime cycle.
(ii) The homomorphism $g_s : \mathscr{F}_s \to \bigoplus (\mathscr{F}_i)_s$ is injective.
(iii) $(Z_i)_s \cap (V_i)_s$ is dense in $(Z_i)_s$, $(\mathscr{F}'_i)_s$ has support $(Z_i)_s \cap (V_i)_s$, and $(u_i)_s : (\mathscr{F}'_i)_s \to \mathscr{F}_s | (V_i)_s$ is injective.

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(iv) For $i \neq j$, every irreducible component of $(Z_i)_s$ is distinct from every irreducible component of $(Z_j)_s$.

Now, (i) has already been seen (9.7.6); (ii) is a particular case of (9.4.5); (iii) follows from (9.5.3) and (9.4.5). Finally, if $i \neq j$, $(Z_i)_\eta \cap (Z_j)_\eta = (Z_i \cap Z_j)_\eta$ is rare in $(Z_i)_\eta$ *or* in $(Z_j)_\eta$; suppose for example that $(Z_i \cap Z_j)_\eta$ is rare in $(Z_j)_\eta$; then it follows from (9.5.3) and (9.5.4) that for s in a neighbourhood of η, $(Z_i \cap Z_j)_s = (Z_i)_s \cap (Z_j)_s$ is rare in $(Z_j)_s$, which shows that no irreducible component of $(Z_j)_s$ can be contained in an irreducible component of $(Z_i)_s$, nor *a fortiori* be equal to it.

This being so, it follows from (ii) and (3.1.7) that for $s \in U$, we have

$$\mathrm{Ass}(\mathscr{F}_s) \subset \bigcup \mathrm{Ass}((\mathscr{F}_i)_s),$$

from (i) that $\mathrm{Ass}((\mathscr{F}_i)_s)$ is the set of $x_{i\alpha}$ ($\alpha \in J_{s,i}$). On the other hand, by virtue of (iii) and the criterion (3.1.3), c')), each of the $x_{i\alpha}$ ($\alpha \in J_{s,i}$, $i \in I$) belongs to $\mathrm{Ass}(\mathscr{F}_s)$. Finally (iv) signifies that the $x_{i\alpha}$ are pairwise distinct.

It remains to prove the last assertion of the statement. Now, it follows from (i) that for s and for given i, none of the sets $\{x_{i\alpha}\}$ can be contained in another for $\alpha \in J_{s,i}$. On the other hand, if $(Z_j)_\eta \subset (Z_i)_\eta$, we can suppose U taken such that $(Z_j)_s \subset (Z_i)_s$ for $s \in U$ (9.5.1), so each $x_{j\alpha}$ belongs to the closure of a $x_{i\alpha}$; on the contrary, if $(Z_j)_\eta \cap (Z_i)_\eta$ is rare in $(Z_j)_\eta$, we have seen by proving (iv) that $(Z_j)_s \cap (Z_i)_s$ is rare in $(Z_j)_s$, so none of the $x_{j\alpha}$ is adherent to a $x_{i\alpha}$. In particular, if $(Z_j)_\eta$ is maximal, this is equivalent to saying that $(Z_j)_\eta$ is rare in $(Z_i)_\eta$ for *all* $i \neq j$, and we conclude that $(Z_j)_s \cap (Z_i)_s$ is rare in $(Z_j)_s$ for all $i \neq j$, so that every $x_{j\alpha}$ ($\alpha \in J_{s,j}$) is *maximal*. □

Corollary (9.8.4). — *The notations and hypotheses being those of* (9.8), *there exists a neighbourhood U of η in S such that, for all $s \in U$, each of the $(\mathscr{F}_i)_s$ is without associated immersed prime cycle; furthermore, if $(\mathscr{F}_{i\alpha})_{\alpha \in J_{s,i}}$ is the irredundant reduced decomposition of $(\mathscr{F}_i)_s$, then, for all $s \in U$, the family $(\mathscr{F}_{i\alpha})_{i \in I, \alpha \in J_{s,i}}$ is an irredundant reduced decomposition of $\mathscr{F}_s$.*

Proof. The first assertion follows from (9.8.1) and the definition of $\mathscr{F}_i$. On the other hand, we have seen in (9.8.3) that the homomorphism $\mathscr{F}_s \to \bigoplus(\mathscr{F}_i)_s$ is injective and by definition it is the same for each of the homomorphisms $(\mathscr{F}_i)_s \to \bigoplus \mathscr{F}_{i\alpha}$, so the homomorphism $\mathscr{F}_s \to \bigoplus_{i,\alpha} \mathscr{F}_{i\alpha}$ is injective. As we can suppose (9.4.5) that for $s \in U$ each of the $(\mathscr{F}_i)_s$ is a quotient of $\mathscr{F}_s$, the $\mathscr{F}_{i\alpha}$ are quotients of $\mathscr{F}_s$, and it remains to verify (3.2.5) that the $x_{i\alpha}$ are pairwise distinct and belong to $\mathrm{Ass}(\mathscr{F}_s)$, which has been proved in (9.8.3). $\square$

Proposition (9.8.5). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. For all $s \in S$, let $E(s)$ (resp. $E'(s)$) be the set (finite subset of $\mathbf{Z} \cup \{-\infty\}$) of dimensions of the maximal prime cycles associated to $\mathscr{F}_s$ (resp. of the maximal prime cycles, i.e., the irreducible components of* $\mathrm{Supp}(\mathscr{F}_s)$*). Then the functions $s \mapsto E(s)$ and $s \mapsto E'(s)$ are locally constructible in S.*

Proof. It follows from (4.2.7) and (4.2.8) that the condition (9.2.1), (i)) is satisfied for the properties whose constructibility we wish to show. We are thus reduced to the case where S is affine, Noetherian and integral, and to proving that E and E' are constant in a neighbourhood of the generic point η of S; but this follows from (9.8.3) and (9.8.6). $\square$ **IV-3** | 86

Proposition (9.8.6). — *With the hypotheses and notations of* (9.8)*, let $i \in I$ be such that $(Z_i)_\eta$ is maximal (in other words, an irreducible component of* $\mathrm{Supp}(\mathscr{F}_\eta)$*). Then there exists a neighbourhood U of η in S such that for all $s \in U$ and all $\alpha \in J_{s,i}$, the geometric length of $\mathscr{F}_s$ at $x_{i\alpha}$ (relative to $\mathbf{k}(s)$)* (4.7.5) *is equal to the geometric length of $\mathscr{F}_\eta$ at x_i (relative to $\mathbf{k}(\eta)$).*

Proof. We can evidently restrict to the case where $S = \mathrm{Spec}(A)$ is affine; let us first show that the sub-prescheme $(Z_i)_\eta$ of X_η, which is reduced, is *geometrically integral*. Indeed there is a finite extension K' of $K = \mathbf{k}(\eta)$ such that $(((Z_i)_\eta)_{(K')})_{\mathrm{red}}$ is geometrically reduced and that the irreducible components of $((Z_i)_\eta)_{(K')}$ are geometrically irreducible (4.6.8). We proceed as in the proof of (9.7.7) by considering a sub-A-algebra A' of K' that is a field of fractions and finite over A. We set $S' = \mathrm{Spec}(A')$ and consider the finite surjective morphism $g : S' \to S$; let $X' = X_{(S')}$ and $\mathscr{F}' = \mathscr{F} \otimes_g \mathscr{O}_{S'}$ and let η' be the generic point of S'. For all $s' \in S'$, i.e., $s = g(s')$; if T is an irreducible component of $\mathrm{Supp}(\mathscr{F}_s)$, the irreducible components T'_j of $T_{(\mathbf{k}(s'))}$ are irreducible components of $\mathrm{Supp}(\mathscr{F}'_{s'})$ and dominate T (4.2.7) and the radicial multiplicities of T with respect to $\mathscr{F}_s$ and of each T'_j with respect to $\mathscr{F}'_{s'}$ are the same (4.7.9). The reasoning from the first part of (9.7.7) then shows that we can restrict to proving the proposition for X' and $\mathscr{F}'$; and by virtue of the choice of K', the reduced sub-preschemes having for underlying spaces the irreducible components of $((Z_i)_\eta)_{(K')}$ are geometrically integral (4.6.1).

Suppose now that $(Z_i)_\eta$ is geometrically integral; then (9.7.7), it is the same for (Z_i), for s in a neighbourhood of η; the definition (4.7.5) shows that it will therefore suffice to prove that the length of the $\mathscr{O}_{X_\eta,x_i}$-module $(\mathscr{F}_\eta)_{x_i}$ is equal to that of the $\mathscr{O}_{X_s,x_{i_s}}$-module $(\mathscr{F}_s)_{x_{i_s}}$ (where we have dropped the index α, now unnecessary by hypothesis). The question being evidently local on X, we can suppose (by restricting to a neighbourhood of x_i) that $\mathscr{F} = \mathscr{F}_i$, so that $\mathscr{F}_\eta$ is *irredundant* on $X = \mathrm{Spec}(B)$ affine, and one writes Z instead of Z_i, and x the generic point of Z (and of Z_η). The Noetherian ring B contains A as sub-ring, and $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite type; furthermore, if K is the field of fractions of A, the $B_{(K)}$-module $M_{(K)}$ is $\neq 0$ and irredundant. Let q be the unique element of $\mathrm{Ass}(M_{(K)})$ and let $\mathfrak{p}$ be the prime ideal of B, inverse image of q. Using (5.11.1.1) as in the proof of (9.7.6), we reduce to the case where B is *integral* and M a sub-module $\neq 0$ of B; then $\mathscr{F}_\eta$ is a sub-$\mathscr{O}_{X_\eta}$-Module non-zero of $\mathscr{O}_{Z_\eta}$, and by virtue of (9.4.5), for s in a neighbourhood of η, $\mathscr{F}_s$ is isomorphic to a sub-$\mathscr{O}_{X_s}$-Module non-zero of $\mathscr{O}_{Z_s}$; as Z_s is geometrically integral, the lengths of $(\mathscr{F}_\eta)_x$ and $(\mathscr{F}_s)_{x_s}$ are both equal to 1, which finishes the proof. $\square$

Corollary (9.8.7). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. For all $s \in S$, let $M(s)$ be the set of pairs (d, m) such that there exists an irreducible component of* $\mathrm{Supp}(\mathscr{F}_s)$ *of dimension d and of radicial multiplicity m for $\mathscr{F}_s$* (4.7.8)*. Then the function $s \mapsto M(s)$ is locally constructible in S.* **IV-3** | 87

Proof. It follows from (4.2.7) and (4.7.9) that the condition (9.2.1), (i)) is satisfied for the property whose constructibility we wish to show. We are thus reduced to the case where S is affine, Noetherian and integral, and to proving that M is constant in a neighbourhood of the generic point of S; but this follows from the proposition, namely (9.8.3) and (9.8.6). □

Proposition (9.8.8). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. For all $s \in S$, let $l(s)$ be the sum of the total multiplicities for $\mathscr{F}_s$ (relative to $\mathbf{k}(s)$) of the generic points of the irreducible components of* $\operatorname{Supp}(\mathscr{F}_s)$ *(4.7.12). Then the function $s \mapsto l(s)$ is locally constructible in S.*

Proof. Taking into account (4.7.12), the condition (9.2.1), (i)) is satisfied for the property whose constructibility we wish to show, and we are thus reduced to the case where S is affine, Noetherian and integral, and to showing that l is constant in a neighbourhood of the generic point η, and constant. Using the notations of (9.8.2), this follows from the definition (4.7.12), from the fact that the geometric number of irreducible components of each $(Z_i)_s$ is constant in a neighbourhood of η (9.7.8), that the geometric length of $\mathscr{F}_s$ at $x_{i\alpha}$ is also equal to that of $\mathscr{F}_\eta$ at x_i for each i such that $(Z_i)_\eta$ is maximal (9.8.6) and finally from the fact that the closure of $\{x_{i\alpha}\}$ is a maximal associated prime cycle of $\mathscr{F}_s$ if and only if $(Z_i)_\eta$ is a maximal associated prime cycle of $\mathscr{F}_\eta$ (9.8.3). □

Remark (9.8.9). — One can make precise in various ways the preceding propositions; let us limit ourselves to a statement given as an example. Let us say that a finite subset P of a k-prescheme algebraic X is *saturated* if, for every pair of points x, y of P, the generic points of the irreducible components of $\overline{\{x\}} \cap \overline{\{y\}}$ also belong to P; for every finite subset Q of X, there exists a smallest finite subset P of X containing Q and saturated; we say that P is the *saturate* of Q. For every coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we will call *primary skeleton* of $\mathscr{F}$ the system (P, Q, ω, d, m) where $Q = \operatorname{Ass}(\mathscr{F})$, P is the saturate of Q, ω the ordering relation $\overline{\{x\}} \subset \overline{\{y\}}$ on P, d the function $x \mapsto \dim \overline{\{x\}}$ on P, m the function $x \mapsto \operatorname{length}_\omega \mathscr{F}_x$ defined on the set of maximal elements of Q for the relation ω. We call *virtual skeleton* any system (P, Q, ω, d, m) where P is a set, Q a subset of P, ω a relation of order on P, d a map from P to $\mathbf{N}$, m a map from $\mathbf{N}$ to the set of maximal elements of Q; one defines in an obvious way the notion of *isomorphism* of two virtual skeletons. Finally, with the preceding notations, we call *primary type* of $\mathscr{F}$ the class (for the relation of isomorphism of virtual skeletons) of the primary skeleton of $\mathscr{F}$. It follows from (4.2.6), (4.2.7), (4.2.8), (4.5.1) and (4.7.9) that if, for an algebraically closed extension k' of k, one sets $X' = X \otimes_k k'$ and $\mathscr{F}' = \mathscr{F} \otimes_k k'$, the *primary type* of $\mathscr{F}'$ is independent of the algebraically closed extension k' of k considered; we say that this is the *geometric primary type* of $\mathscr{F}$. Given the definitions, the statement we have in view is the following: **IV-3** | 88

(9.8) Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. For all $s \in S$, let $T(s)$ be the geometric primary type of $\mathscr{F}_s$. Then the function $s \mapsto T(s)$ is locally constructible in S.

Taking into account the remarks that precede, we are reduced as usual to proving that if S is affine, Noetherian and integral, and of generic point η, $T(s)$ is constant in a neighbourhood of η. Reasoning as at the beginning of (9.7.7), we can suppose that all the irreducible components of X_η that intervene are geometrically irreducible, and then the proposition follows from (9.5.1), (9.5.5), (9.8.3) and (9.8.6); we leave the details to the reader. One could generalise by considering several coherent Modules and defining their "simultaneous primary skeleton", etc. The general conclusion of this paragraph is that for all the properties of the type considered (and for an S-irreducible prescheme) *the properties valid on the "generic fibre" are also valid on all neighbouring fibres.*

9.9. Constructibility of local properties of fibres

Proposition (9.9.1). — *Let $f : X \to S$ be a morphism locally of finite presentation, Z a locally constructible subset of X such that for all $s \in S$, Z_s is closed in X_s, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$. Then the following subsets of X are locally constructible:* **IV-3 | 88**

(i) *The set of $x \in X$ such that $\dim_x(Z_{f(x)})$ belongs to Φ.*

(ii) *The set of $x \in X$ such that $\operatorname{codim}_x(Z_{f(x)}, X_{f(x)})$ belongs to Φ.*

(iii) *The set of $x \in X$ such that the local ring $\mathscr{O}_{Z_{f(x)},x}$ is equidimensional.*

We note that properties (i) and (ii) can also be expressed by saying that the functions $x \mapsto \dim_x(Z_{f(x)})$ and $x \mapsto \operatorname{codim}_x(Z_{f(x)}, X_{f(x)})$ are locally constructible *in X (0_{III}, (9.3.1)).*

Proof. The questions being local on X, we can restrict to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine and where f is a morphism of finite presentation; there exists then a sub-ring A_0 of A that is a $\mathbf{Z}$-algebra of finite type, an A_0-prescheme of finite type X_0 and a constructible subset Z_0 of X_0 such that $X = X_0 \otimes_{A_0} A$ and $Z = h^{-1}(Z_0)$, where $h : X \to X_0$ is the canonical projection ((8.9.1) and (8.3.11)). Furthermore, for all $s \in S$, if s_0 is the projection of s in $S_0 = \operatorname{Spec}(A_0)$, we have

$$X_s = (X_0)_{s_0} \otimes_{\mathbf{k}(s_0)} \mathbf{k}(s),$$

and, if h_s is the projection $X_s \to (X_0)_{s_0}$, we have $Z_s = h_s^{-1}((Z_0)_{s_0})$. As the morphism h_s is faithfully flat and quasi-compact, the hypothesis that Z_s is closed in X_s implies that $(Z_0)_{s_0}$ is closed in $(X_0)_{s_0}$ (2.3.12).

This being so, the transitivity of fibres (**I**, 3.6.4) and the prop. (4.2.7) imply that the set of dimensions of the irreducible components of Z_s containing x is the same as the set of dimensions of the irreducible components of $(Z_0)_{s_0}$ containing $x_0 = h_s(x)$. In particular, we have $\dim_x(Z_s) = \dim_{x_0}((Z_0)_{s_0})$. On the other hand, if $Z_s^{(\beta)}$ are the irreducible components of Z_s containing x and $X_s^{(\alpha)}$ the irreducible components of X_s containing x, we have **IV-3 | 89**

$$\operatorname{codim}_x(Z_s, X_s) = \inf_\beta \sup_\alpha \operatorname{codim}(Z_s^{(\beta)}, X_s^{(\alpha)}),$$

where (α, β) ranges over the pairs such that $x \in Z_s^{(\beta)} \subset X_s^{(\alpha)}$ (0, (14.2.6)). As the algebraic preschemes are biequidimensional (5.2.1), we can write, by virtue of (0, (14.3.3.1))

$$\operatorname{codim}_x(Z_s, X_s) = \inf_\beta \sup_\alpha \big(\dim(X_s^{(\alpha)}) - \dim(Z_s^{(\beta)})\big) \tag{9.9.1.1}$$

with the same choice of pairs (α, β). As h_s is faithfully flat and quasi-compact, for every pair formed of an irreducible component $(X_0)_{s_0}^{(\alpha)}$ of $(X_0)_{s_0}$ containing x_0 and an irreducible component $(Z_0)_{s_0}^{(\beta)}$ of $(Z_0)_{s_0}$ containing x_0 and contained in $(X_0)_{s_0}^{(\alpha)}$, there exists a pair $(X_s^{(\alpha)}, Z_s^{(\beta)})$ of the type described above such that $Z_s^{(\beta)}$ dominates $(Z_0)_{s_0}^{(\beta)}$ and $X_s^{(\alpha)}$ dominates $(X_0)_{s_0}^{(\alpha)}$ (2.3.5). The formula (9.9.1.1) (and the analogous formula applied to $(X_0)_{s_0}$) shows then, by virtue of (4.2.7), that we have

$$\operatorname{codim}_x(Z_s, X_s) = \operatorname{codim}_{x_0}((Z_0)_{s_0}, (X_0)_{s_0}).$$

We thus see that if E (resp. E_0) is the set of $x \in X$ (resp. $x_0 \in X_0$) satisfying one of the conditions (i), (ii), (iii) of the statement (resp. the same condition), then $E = h^{-1}(E_0)$; by (1.8.2), we may therefore restrict to the case where A, and hence also B, is Noetherian. Taking account of (0_{III}, (9.2.3)) and of (9.9.1.1), it remains to show that for every $x \in X$ there is a neighbourhood V of x in $\overline{\{x\}}$ such that, for every $x' \in V$, the sets of dimensions of the irreducible components of $X_{f(x')}$ and of $Z_{f(x')}$ containing x' are constant, and likewise the set of pairs

$$\big(\dim Z_{f(x')}^{(\beta)}, \dim X_{f(x')}^{(\alpha)}\big),$$

where $X_{f(x')}^{(\alpha)}$ is an irreducible component of $X_{f(x')}$ and $Z_{f(x')}^{(\beta)}$ is an irreducible component of $Z_{f(x')}$ contained in it and containing x'. For this purpose we may replace S by the reduced closed sub-prescheme S' whose underlying space is $\overline{\{f(x)\}}$, and X by $X' = f^{-1}(S')$, since the fibres of X and X' at points of S' are the same. Thus we may suppose that S is integral and that $\eta = f(x)$ is its generic point.

By hypothesis Z_η is closed in X_η; as Z is constructible, it follows from (8.3.11), applied by the method of (8.1.2), (a), that after replacing S by an open neighbourhood of η we may suppose that Z is *closed* in X. Let X_i (resp. Z_j) be the irreducible components of X (resp. Z) containing x. By (0_I, (2.1.8)), the $X_i \cap X_\eta$ (resp. $Z_j \cap X_\eta$) are the irreducible components of X_η (resp. Z_η) containing x; by (9.5.1), after shrinking S about η, we may further suppose that the X_i (resp. Z_j) are exactly the irreducible components of X (resp. Z) meeting $\overline{\{x\}}$, and that $(X_i)_s \cap \overline{\{x\}}$ and $(Z_j)_s \cap \overline{\{x\}}$ are nonempty for every $s \in S$. This being so, it follows still from (9.7.1) that we can suppose, by restricting S, that the couples (i,j) such that $(Z_j)_s \subset (X_i)_s$ are the same for all $s \in S$. The conclusion follows then from (9.5.6): for all s sufficiently close to η, all the irreducible components of $(X_i)_s$ (resp. $(Z_j)_s$) have the same dimension equal to that of $(X_i)_\eta$ (resp. $(Z_j)_\eta$). Furthermore, if (i,j) is a pair such that $Z_j \not\subset X_i$, $(X_i)_\eta$ does not contain the generic point of $(Z_j)_\eta$, hence $\dim((X_i \cap Z_j)_\eta) <$ **IV-3** | 90
$\dim((Z_j)_\eta)$; consequently, the common dimension of the irreducible components of $(X_i)_s \cap (Z_j)_s$ is, for all $s \in S$, strictly less than the common dimension of the irreducible components of $(Z_j)_s$, which proves that none of the irreducible components of $(Z_j)_s$ is contained in an irreducible component of $(X_i)_s$ for $s \in S$. □

Proposition (9.9.2). — *Let $f : X \to S$ be a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$. The following subsets of X are locally constructible:*

(i) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is equidimensional at the point x (5.1.12).
(ii) The set of $x \in X$ such that the geometric lengths of $\mathscr{F}_{f(x)}$ relative to $\mathbf{k}(f(x))$, at the generic points of the irreducible components of $\operatorname{Supp}(\mathscr{F}_{f(x)})$ *containing x (4.7.5), belong to Φ.*
(iii) The set of $x \in X$ such that the dimensions of the prime cycles associated to $\mathscr{F}_{f(x)}$ and containing x belong to Φ.
(iv) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is geometrically reduced at the point x (4.6.17).
(v) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is geometrically integral at the point x (4.6.22).
(vi) The set of $x \in X$ such that $\dim.\operatorname{proj}((\mathscr{F}_{f(x)})_x) \in \Phi$.
(vii) The set of $x \in X$ such that $\operatorname{coprof}((\mathscr{F}_{f(x)})_x) \in \Phi$.
(viii) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ possesses property (S_k) at the point x (5.7.2).
(ix) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is a Cohen-Macaulay $\mathscr{O}_{X_{f(x)}}$-Module at the point x (5.7.1).

Proof. (i) The support Z of $\mathscr{F}$ is locally constructible and closed in X (8.9.1), and by considering a sub-prescheme of X having Z for underlying space and which is of finite presentation over S (8.9.1), we see that the assertion (i) is a particular case of (9.9.1), (iii)).

(ii) All the properties considered are local on X, and we may therefore suppose that $X = \operatorname{Spec}(B)$ and $S = \operatorname{Spec}(A)$ are affine and that f is of finite presentation. We retain the notation from the beginning of the proof of (9.9.1), and choose A_0 so that there is a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ with $\mathscr{F} \simeq \mathscr{F}_0 \otimes_{A_0} A$. Then (4.2.7) and (4.7.9) show that the set of geometric lengths of $(\mathscr{F}_0)_{f(x_0)}$ at the generic points of the irreducible components of its support containing x_0 is the same as the corresponding set for $\mathscr{F}_{f(x)}$ at x. Thus, if E (resp. E_0) denotes the set of $x \in X$ (resp. $x_0 \in X_0$) satisfying condition (ii), then $E = h^{-1}(E_0)$, and by (1.8.2) we may restrict to the case where A is Noetherian. As in the proof of (9.9.1), it remains to show that for every $x \in X$ there is a neighbourhood V of x in $\overline{\{x\}}$ on which the set of geometric lengths of $\mathscr{F}_{f(x')}$ at the generic points of the irreducible components of its support containing x' is constant. Replacing S by the reduced closed subprescheme with underlying space $\overline{\{f(x)\}}$, and X by its inverse image, we may suppose that S is integral and that $\eta = f(x)$ is its generic point. If $Z = \operatorname{Supp}(\mathscr{F})$ and Z_i are the irreducible components of Z containing x, the argument of (9.9.1) lets us suppose that these are exactly the irreducible components of Z meeting $\overline{\{x\}}$ and that $(Z_i)_s \cap \overline{\{x\}} \neq \varnothing$ for every $s \in S$. The conclusion follows from (9.8.3) and (9.8.6).

(iii) Using (4.2.7), we reduce as in (ii) to the case where S is Noetherian and integral and $\eta = f(x)$ is its generic point. As in the proof of (9.9.1), it remains to find a neighbourhood V of x in $\overline{\{x\}}$ on which the set of dimensions of the prime cycles associated to $\mathscr{F}_{f(x')}$ and containing x' is constant. Let Z'_i be the closures in X of the prime cycles associated to $\mathscr{F}_\eta$ which contain x (cf. (9.8.2)). By (9.8.3) and (9.5.1), for every s sufficiently near η, the prime cycles associated to $\mathscr{F}_s$ which meet $\overline{\{x\}}$ are irreducible components of the $(Z'_i)_s$; by (9.5.6), all these components have dimension $\dim((Z'_i)_\eta)$, which proves the assertion.

(iv) We reason as in (iii), now using (4.7.11). With the same notation, the prime cycles associated to $\mathscr{F}_s$ which are irreducible components of $(Z'_i)_s$ are embedded if and only if $(Z'_i)_\eta$ is an embedded associated prime cycle of $\mathscr{F}_\eta$. It follows that, according as x belongs to an embedded associated prime cycle of $\mathscr{F}_\eta$ or to none, the points $x' \in \overline{\{x\}}$ having the analogous property for $\mathscr{F}_{f(x')}$ form a neighbourhood of x in $\overline{\{x\}}$. The conclusion follows from this observation, the characterization of geometrically reduced Modules in (4.7.10), and (ii).

(v) Taking (4.7.11) into account, we reduce again to the case where S is Noetherian and integral and where $\eta = f(x)$ is its generic point. Let Y be a closed subprescheme of X whose underlying space is $\mathrm{Supp}(\mathscr{F}_\eta)$. As at the beginning of the proof of (9.7.7), there is a finite integral A-algebra A' such that, if $S' = \mathrm{Spec}(A')$, $Y' = Y_{(S')}$, and η' is the generic point of S', the irreducible components of $Y'_{\eta'}$ are geometrically irreducible. If $X' = X_{(S')}$, the projection $h : X' \to X$ is finite and surjective, hence closed; consequently, for a point x' above x, one has $h(\overline{\{x'\}}) = \overline{\{x\}}$, and the image under h of a neighbourhood of x' in $\overline{\{x'\}}$ is a neighbourhood of x in $\overline{\{x\}}$. Writing $\mathscr{F}' = \mathscr{F}_{(S')}$ and using (4.7.11), we are reduced to proving that, according as $\mathscr{F}'_{\eta'}$ is or is not geometrically integral at x', the points $y' \in \overline{\{x'\}}$ where $\mathscr{F}'_{f'(y')}$ is or is not geometrically integral at y' form a neighbourhood of x' in $\overline{\{x'\}}$. We may therefore suppose that $X' = X$ and that the irreducible components of Y_η are geometrically irreducible. Choose closed subsets Z_i of X such that the $Z_i \cap X_\eta$ are the irreducible components of Y_η. By (9.7.7), (9.7.8), and (9.5.3), for every s near η the $(Z_i)_s$ are the irreducible components of Y_s and are geometrically irreducible. At a point $y \in X_s$, the Module $\mathscr{F}_s$ is geometrically integral exactly when it is geometrically reduced there and y belongs to only one of the $(Z_i)_s$ (4.6.22). The conclusion follows from (iv) and from (9.5.1) applied to $\overline{\{x\}} \cap Z_i$ for each i. **IV-3** | 91

(vi) Keeping the notations of (ii), (6.2.1) gives

$$\dim.\mathrm{proj}((\mathscr{F}_s)_x) = \dim.\mathrm{proj}(((\mathscr{F}_0)_{s_0})_{x_0}),$$

so we may again restrict to the case where A is Noetherian. It remains to show that for every $x \in X$ there is a neighbourhood V of x in $\overline{\{x\}}$ such that

$$\dim.\mathrm{proj}((\mathscr{F}_{f(x')})_{x'}) = \dim.\mathrm{proj}((\mathscr{F}_{f(x)})_x)$$

for every $x' \in V$. As above, we may suppose that S is integral and that $\eta = f(x)$ is its generic point, so $(\mathscr{F}_\eta)_x = \mathscr{F}_x$. By generic flatness (6.9.1), after replacing S by an open neighbourhood of η, we may suppose that f and $\mathscr{F}$ are flat over S. Then (6.2.3) gives

$$\dim.\mathrm{proj}((\mathscr{F}_{f(x')})_{x'}) = \dim.\mathrm{proj}(\mathscr{F}_{x'})$$

for every $x' \in X$. By (6.11.1), after replacing X by an open neighbourhood of x, we may suppose that

$$\dim.\mathrm{proj}(\mathscr{F}_{x'}) \leqslant \dim.\mathrm{proj}(\mathscr{F}_x)$$

for every $x' \in X$.

If $\dim.\mathrm{proj}(\mathscr{F}_x) = n$, there is a finite $\mathscr{O}_x$-module M such that $\mathrm{Ext}^n_{\mathscr{O}_x}(\mathscr{F}_x, M) \neq 0$ (**0**, 17.2.4). After replacing X by an open neighbourhood of x, there exists a coherent $\mathscr{O}_X$-module $\mathscr{G}$ such that $M = \mathscr{G}_x$ (**0**, 5.3.8); by virtue of (T, (4.2.2)), we have $(\mathscr{E}xt^n_{\mathscr{O}_X}(\mathscr{F},\mathscr{G}))_x \neq 0$. But $\mathscr{E}xt^n_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})$ is a coherent $\mathscr{O}_X$-Module (0_{III}, (12.3.3)), hence its support Y is closed (0_I, (5.2.2)); as it contains x, it contains also $\overline{\{x\}}$, whence we conclude (by applying again (T, (4.2.2))) that $\dim.\mathrm{proj}(\mathscr{F}_{x'}) \geqslant n$ for all $x' \in \overline{\{x\}}$, which achieves the proof of the assertion in case (vi). **IV-3** | 92

(vii) As B is an A-algebra of finite type, X is S-isomorphic to a *closed sub-scheme* of an S-scheme of the form $Y = \mathrm{Spec}(A[T_1, \ldots, T_r])$; if $j : X \to Y$ is the canonical injection and $\mathscr{G} = j_*(\mathscr{F})$, then $\mathscr{G}_s = (j_s)_*(\mathscr{F}_s)$ for all $s \in S$, and, by virtue of (0, (16.4.11)), we can restrict to proving the assertion for Y and $\mathscr{G}$. In other words, we can suppose that $B = A[T_1, \ldots, T_r]$, so that the schemes $X_s = \mathrm{Spec}(\mathbf{k}(s)[T_1, \ldots, T_r])$ are *regular* (0, (17.3.7)). Let $W = \mathrm{Supp}(\mathscr{F})$ and $W_s = \mathrm{Supp}(\mathscr{F}_s)$ (**I**, 9.1.13); we have, by (6.11.2.1)

(9.9.2.1) $$\mathrm{coprof}((\mathscr{F}_{f(x)})_x) = \dim.\mathrm{proj}((\mathscr{F}_{f(x)})_x) - \mathrm{codim}_x(W_{f(x)}, X_{f(x)}).$$

But as W is constructible (8.9.1) and each W_s is closed, it follows from (vi) and (9.9.1), (ii), that the two functions on the right-hand side of (9.9.2.1) are constructible; hence so is their difference, which proves the assertion in case (vii).

(viii) Let U_n be the set of $x \in X$ such that $\operatorname{coprof}((\mathscr{F}_{f(x)})_x) \leqslant n$, and set $Z_n = X - U_n$; it follows from (vii) that the Z_n are *constructible*. Moreover, since the function $x \mapsto \dim_x(W_{f(x)})$ is constructible by (9.9.1), (i), it takes only finitely many values, and the numbers $\dim(W_{f(x)})$ therefore have a finite upper bound m as x ranges over X. Since $\operatorname{coprof}((\mathscr{F}_{f(x)})_x) \leqslant \dim((\mathscr{F}_{f(x)})_x) \leqslant \dim(W_{f(x)})$, we have $Z_n = \varnothing$ for $n \geqslant m$. Finally, (6.11.2), (i), shows that $(Z_n)_s$ is closed in X_s for every n and every $s \in S$. By (5.7.4), the set of $x \in X$ where $(\mathscr{F}_{f(x)})_x$ possesses property (S_k) is the set of $x \in X$ satisfying all the relations

$$\operatorname{codim}_x((Z_n)_{f(x)}, W_{f(x)}) > n + k \tag{9.9.2.2}$$

for all $n \geqslant 0$; as this relation is automatically satisfied for $n \geqslant m$, we need only consider the relations (9.9.2.2) for $0 \leqslant n < m$. But by virtue of (9.9.1), (ii)), the set $V_{n,k}$ of x satisfying (9.9.2.2) is constructible, and so is the intersection of these sets for $0 \leqslant n < m$.

(ix) The assertion follows immediately from (vii), the set of $x \in X$ where $(\mathscr{F}_{f(x)})_x$ is a Cohen-Macaulay module being defined by the relation $\operatorname{coprof}((\mathscr{F}_{f(x)})_x) = 0$. □

Corollary (9.9.3). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation, $\mathbf{P}$ one of the properties (i) to (ix) of (9.9.2). Then the set of $s \in S$ such that the property $\mathbf{P}$ is true at all the points $x \in X_s$, is locally constructible in S.* **IV-3 | 93**

Proof. Indeed, its complement is the image by f of the complement of the set E of points where $\mathbf{P}$ is true. As E is locally constructible in X, so is $X - E$, and hence $f(X - E)$ is locally constructible in S by virtue of the theorem of Chevalley (1.8.4). □

Proposition (9.9.4). — *Let $f : X \to S$ be a morphism locally of finite presentation. The set of $x \in X$ such that $X_{f(x)}$ has at the point x one of the (determined) following properties:*

- *(i) being geometrically regular;*
- *(ii) possessing property (R_k) geometric;*
- *(iii) being geometrically normal;*
- *(iv) being geometrically reduced (i.e. separable);*
- *(v) being geometrically pointwise integral;*

is a locally constructible subset of X.

Proof. For properties (iv) and (v), this follows from (9.9.2), (iv) and (v)) applied to $\mathscr{F} = \mathscr{O}_X$. For the other properties, taking into account (6.7.8), we reduce, as at the beginning of (9.9.2), (ii)) to the case where S is Noetherian and integral and of generic point $\eta = f(x)$. Furthermore, by virtue of the generic flatness theorem (6.9.1), we can, by replacing S by a neighbourhood open of η, suppose that f is a *flat* morphism. To say that $X_{f(y)}$ is geometrically regular at the point y means that f is *regular* at the point y (6.8.1), and we know that the set L of these points is *open* in X (6.8.7), which proves the proposition in case (i).

To prove case (ii), set $F = X - L$, which is *closed* in X. To say that X_s satisfies the geometric property (R_k) at the point $y \in f^{-1}(s)$ means either that $y \in L$, or that the generic points z_i of the irreducible components of the closed set F_s which contain y satisfy the relation $\dim(\mathscr{O}_{X_{z_i},s_i}) \geqslant k + 1$ (taking into account (4.2.7) and (5.2.3))); in other words, the points $y \in F_s$ where X_s satisfies property (R_k) geometric are those such that $\operatorname{codim}_y(F_s, X_s) \geqslant k + 1$ (5.1.2). The conclusion follows thus from (i) and (9.9.1), (ii)). **IV-3 | 94**

Finally, in case (iii), the conclusion follows from (ii), from (9.9.2), (viii)), from the fact that property (S_k) is stable by extension of the base field (6.7.1) and finally from the criterion of Serre (5.8.6). □

Corollary (9.9.5). — *Let $f : X \to S$ be a morphism of finite presentation, and denote by $\mathbf{P}$ one of the properties (i) to (v) of (9.9.4). Then the set of $s \in S$ such that the property $\mathbf{P}$ is true at all the points $x \in X_s$, is locally constructible in S.*

Proof. The proof from (9.9.4) is the same as that of (9.9.3) from (9.9.2). □

Proposition (9.9.6). — *Let $f : X \to S$ be a morphism locally of finite presentation, $\mathscr{L}_\bullet$ a complex formed of $\mathscr{O}_X$-Modules quasi-coherent of finite presentation; for all $s \in S$, let $(\mathscr{L}_\bullet)_s$ be the complex $\mathscr{L}_\bullet \otimes_{\mathscr{O}_S} \mathbf{k}(s)$ of $\mathscr{O}_{X_s}$-Modules coherent of finite type. Then, for a given integer n, the set of $x \in X$ such that $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = 0$ is locally constructible in X.*

Proof. We can restrict to the case where $\mathscr{L}_i = 0$ except for $i = 0$, 1 or 2, and where $n = 1$. Furthermore, the question being local on X, we can restrict to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, B being a finitely presented A-algebra. There exists then a Noetherian sub-ring A_0 of A, a finite type A_0-prescheme X_0 and a complex $\mathscr{L}_\bullet^{(0)}$ of $\mathscr{O}_{X_0}$-Modules coherent, zero except in dimensions 0, 1 and 2 and such that $X = X_0 \otimes_{A_0} A$ and $\mathscr{L}_\bullet = \mathscr{L}_\bullet^{(0)} \otimes_{A_0} A$. For every $s \in S$, if s_0 is the projection of s in $S_0 = \operatorname{Spec}(A_0)$, then

$$X_s = (X_0)_{s_0} \otimes_{\mathbf{k}(s_0)} \mathbf{k}(s),$$

and the projection $X_s \to (X_0)_{s_0}$ is faithfully flat. Consequently,

$$\mathscr{H}_n((\mathscr{L}_\bullet)_s) = \mathscr{H}_n((\mathscr{L}_\bullet^{(0)})_{s_0}) \otimes_{\mathbf{k}(s_0)} \mathbf{k}(s).$$

If E (resp. E_0) is the set of $x \in X$ (resp. $x_0 \in X_0$) for which $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = 0$ (resp. $(\mathscr{H}_n((\mathscr{L}_\bullet^{(0)})_{f(x_0)}))_{x_0} = 0$), then $E = h^{-1}(E_0)$, where $h : X \to X_0$ is the canonical projection. By (1.8.2), we may therefore restrict to the case where A is Noetherian. It remains to show (0_{III}, (9.2.3)) that if $x \in X$ is such that $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = 0$ (resp. $\neq 0$), there is an open neighbourhood V of x in $\overline{\{x\}}$ such that for every $x' \in V$ one has $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x')}))_{x'} = 0$ (resp. $\neq 0$). Replacing S by the reduced closed subprescheme having $\overline{\{f(x)\}}$ as its underlying space, we may suppose that S is integral and that $f(x) = \eta$ is its generic point. Then, by restricting S to an open neighbourhood of η, we may suppose that for every $s \in S$, $(\mathscr{H}_n(\mathscr{L}_\bullet))_s = \mathscr{H}_n((\mathscr{L}_\bullet)_s)$ (9.4.3); consequently, if Z is the support of $\mathscr{H}_n(\mathscr{L}_\bullet)$, then the support of $\mathscr{H}_n((\mathscr{L}_\bullet)_s)$ is $Z_s = Z \cap X_s$ (**I**, 9.1.13.1). The hypothesis is that $Z_\eta \cap \overline{\{x\}} = \varnothing$ (resp. $\neq \varnothing$). As $Z \cap \overline{\{x\}}$ is closed in the Noetherian space X, we conclude from (9.5.1) that there is a neighbourhood U of η in S such that, for all $s \in U$, we have $Z_s \cap \overline{\{x\}} = \varnothing$ (resp. $\neq \varnothing$). The set $V = f^{-1}(U) \cap \overline{\{x\}}$ thus answers the question. □

§10. Jacobson preschemes

10.1. Very dense subsets of a topological space

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(10.1.1). We say that a subset of a topological space X is *quasi-constructible* if it is a finite union of locally closed subsets of X. We say that a subset T of X is *locally quasi-constructible* if for every $x \in X$, there exists an open neighbourhood V of x such that $T \cap V$ is quasi-constructible in V. It is clear that every quasi-constructible subset of X is locally quasi-constructible; the converse is true if X is quasi-compact. The argument of (**III**, 9.1.3), where one removes the word "retrocompact", shows that the set of quasi-constructible (resp. locally quasi-constructible) subsets of X, which we will denote by $\mathfrak{Qc}(X)$ (resp. $\mathfrak{Lqc}(X)$), is stable under finite union, finite intersection, and passage to complements. If $f : X \to Y$ is a continuous map, it follows immediately from these definitions that, for every quasi-constructible (resp. locally quasi-constructible) subset Z of Y, $f^{-1}(Z)$ is quasi-constructible (resp. locally quasi-constructible) in X.

The constructible (resp. locally constructible) subsets of X are evidently quasi-constructible (resp. locally quasi-constructible); the converse is true when X is Noetherian (resp. locally Noetherian).

In what follows, we will denote respectively by $\mathfrak{O}(X)$, $\mathfrak{F}(X)$, $\mathfrak{Lf}(X)$, $\mathfrak{C}(X)$, $\mathfrak{Lc}(X)$ the set of subsets of X that are respectively open, closed, locally closed, constructible, locally constructible.

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Proposition (10.1.2). — *Let X be a topological space, X_0 a subset of X. The following conditions are equivalent:*

a) *For every locally closed subset $Z \neq \varnothing$ of X, we have $Z \cap X_0 \neq \varnothing$.*

a′) *For every closed subset Z of X, we have $Z = \overline{Z \cap X_0}$.*

b) *For every locally quasi-constructible subset $Z \neq \varnothing$ of X, we have $Z \cap X_0 \neq \varnothing$.*

b′) *For every locally quasi-constructible subset Z of X, we have $Z \subset \overline{Z \cap X_0}$ (in other words, $Z \cap X_0$ is dense in Z).*

c) *The map $U \mapsto U \cap X_0$ of $\mathfrak{O}(X)$ into $\mathfrak{O}(X_0)$ is injective (hence bijective).*

c′) *The map $F \mapsto F \cap X_0$ of $\mathfrak{F}(X)$ into $\mathfrak{F}(X_0)$ is injective (hence bijective).*

c″) *The map $Z \mapsto Z \cap X_0$ of $\mathfrak{Qc}(X)$ into $\mathfrak{Qc}(X_0)$ is injective (hence bijective).*

c‴) *The map $Z \mapsto Z \cap X_0$ of $\mathfrak{Lqc}(X)$ into $\mathfrak{Lqc}(X_0)$ is injective.*

Furthermore, when these conditions are satisfied, the map $Z \mapsto Z \cap X_0$ of $\mathfrak{Lqc}(X)$ into $\mathfrak{Lqc}(X_0)$ is bijective.

Proof. Let us note that the assertions of surjectivity of $c)$ and $c')$ are trivial; they mean that every locally closed subset of X_0 is the trace on X_0 of a locally closed subset of X, and that the map $\mathfrak{Qc}(X) \to \mathfrak{Qc}(X_0)$ defined in $c'')$ is also surjective.

We will prove the implications

$$c''') \Rightarrow c'') \Rightarrow c') \Rightarrow c) \Rightarrow b) \Rightarrow b') \Rightarrow a') \Rightarrow a) \Rightarrow c''').$$

The first three are trivial. To see that $c)$ implies $b)$, let us first note that $c)$ means that X_0 is *dense* in X. Replacing X and X_0 by U and $U \cap X_0$ respectively, one can therefore restrict to the case where Z is locally closed in X, hence $Z = V \cap \complement W$, where V and W are open in X; the hypothesis $Z \neq \varnothing$ means that $V \not\subset W$, or equivalently $V \cup W \neq W$. By virtue of $c)$, we have $(V \cup W) \cap X_0 \neq W \cap X_0$, hence $V \cap X_0 \not\subset W$, and consequently $(V \cap \complement W) \cap X_0 \neq \varnothing$.

To see that $b)$ implies $b')$, it suffices to apply $b)$ to $Z \cap U$, where U is an open neighbourhood of an arbitrary point of Z. As $Z \cap \overline{X_0} \subset \overline{Z}$, it is trivial that $b')$ implies $a')$. To show that $a')$ implies $a)$, let us note that if Z is locally closed in X, we can write $Z = F - F'$ where F and F' are closed in X and $F' \subset F$; whence $Z \cap X_0 = (F \cap X_0) - (F' \cap X_0)$. If $Z \cap X_0 = \varnothing$, then, by virtue of $a')$, we would have $F \cap X_0 = F' \cap X_0$, hence $F = F'$, i.e. $Z = \varnothing$.

To see that $a)$ implies $c''')$, it suffices to show that if $Z \cap X_0 \neq \varnothing$, then $Z' \cap X_0 = Z'' \cap X_0$ is equivalent to $((Z' \cup Z'') - (Z' \cap Z'')) \cap X_0 = \varnothing$. In other words, it suffices to prove that $a)$ implies $b)$; indeed, replacing X and X_0 by U and $U \cap X_0$ respectively, one is reduced to the case where Z is locally closed in X, whence the conclusion.

It remains to show that the map $\mathfrak{Lqc}(X) \to \mathfrak{Lqc}(X_0)$ is surjective. So let Z_0 be a locally quasi-constructible subset of X_0: there is a covering (V_α) of X_0 by open subsets of X_0 such that $Z_0 \cap V_\alpha$ is quasi-constructible in V_α (and hence also in X_0). By virtue of $c)$, there is for each α a unique **IV-3 | 97** quasi-constructible subset Z_α in X such that $Z_0 \cap V_\alpha = X_0 \cap Z_\alpha$. If α and β are any two indices, we have $Z_\beta \cap U_\alpha \cap X_0 = U_\alpha \cap X_0 = V_\alpha \cap Z_0 \cap V_\beta = Z_\alpha \cap X_0 \cap V_\beta \subset Z_0 \cap X_0$; as $Z_\beta \cap U_\alpha$ and Z_α are quasi-constructible in X, it follows from $c'')$ that $Z_\beta \cap U_\alpha \subset Z_\alpha$. If we set $Z = \bigcup_\alpha Z_\alpha$, then $Z \cap U_\alpha = Z_\alpha$ for every α; moreover, since the V_α cover X_0, it follows from $c)$ that the U_α cover X, and we see therefore that Z is locally quasi-constructible in X and $Z_0 = Z \cap X_0$. □

Definition (10.1.3). — When a subset X_0 of a topological space satisfies the equivalent conditions of (10.1.2), we say that X_0 is *very dense* in X.

We have already seen in the course of the proof of (10.1.2) that X_0 is then dense in X.

Corollary (10.1.4). — *If X_0 is very dense in X and U an open subset of X, $U \cap X_0$ is very dense in U. Conversely, if (U_α) is an open covering of X such that $U_\alpha \cap X_0$ is very dense in U_α for every α, then X_0 is very dense in X.*

As every locally closed subset of U is locally closed in X, the first assertion follows from criterion $a)$ of (10.1.2); and the same is true of the second, since if $Z \neq \varnothing$ is locally closed in X, $Z \cap U_\alpha$ is locally closed in U_α for every α, and $Z \cap U_\alpha \neq \varnothing$ for at least one α.

10.2. Quasi-homeomorphisms

Proposition (10.2.1). — *Let X_0, X be two topological spaces, $f : X_0 \to X$ a continuous map. The following conditions are equivalent:*

a) *The map $U \mapsto f^{-1}(U)$ of $\mathfrak{O}(X)$ into $\mathfrak{O}(X_0)$ is bijective.*

a′) *The map $F \mapsto f^{-1}(F)$ of $\mathfrak{F}(X)$ into $\mathfrak{F}(X_0)$ is bijective.*

b) *The topology of X_0 is the inverse image by f of that of X, and $f(X_0)$ is very dense (10.1.3) in X.*

c) *The functor $\mathscr{F} \mapsto f^*(\mathscr{F})$ from the category of sheaves of sets (resp. sheaves of abelian groups) on X to the category of sheaves of sets (resp. sheaves of abelian groups) on X_0 is an equivalence of categories.*

Proof. It is clear that $a)$ and $a')$ are equivalent, and $a)$ implies that the topology of X_0 is the inverse image by f of that of X. On the other hand, if $f(X_0)$ is not very dense in X, there are two distinct open subsets U_1, U_2 of X such that $U_1 \cap f(X_0) = U_2 \cap f(X_0)$, and hence $f^{-1}(U_1) = f^{-1}(U_2)$, which completes showing that $a)$ implies $b)$. Conversely, the condition $b)$ implies that the maps $U \mapsto U \cap f(X_0)$ of $\mathfrak{O}(X)$ into $\mathfrak{O}(f(X_0))$ and $V \mapsto f^{-1}(V)$ of $\mathfrak{O}(f(X_0))$ into $\mathfrak{O}(X_0)$ are bijective, hence so is the composite $U \mapsto f^{-1}(U)$.

To see that $a)$ implies $c)$, it suffices to apply the definition of $f^*(\mathscr{F})$ (**0**, 3.7.1) and the axioms for sheaves: $a)$ means that for every open U of X, the canonical map $\Gamma(U, \mathscr{F}) \to \Gamma(f^{-1}(U), f^*(\mathscr{F}))$ is a functorial bijection in $\mathscr{F}$, whence $c)$. It remains to show that $c)$ implies $b)$.

Suppose first that $f(X_0)$ is not very dense in X; there then exist two distinct closed subsets $Y \supset Y'$ of X such that $Y \cap f(X_0) = Y' \cap f(X_0)$. Let $\mathscr{F}$ (resp. $\mathscr{F}'$) be the sheaf of abelian groups on X that is the direct image by the canonical injection $Y \to X$ (resp. $Y' \to X$) of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y (resp. Y'); the definition of f^* (**0**, 3.7.1) shows that $f^*(\mathscr{F})$ is isomorphic to $f^*(\mathscr{F}')$, but $\mathscr{F}$ is not isomorphic to $\mathscr{F}'$, hence condition $c)$ is not satisfied. (One notes that the functor f^* is not then even *faithful*, for if u and v are the identity automorphism and the zero endomorphism of $\mathscr{F}/\mathscr{F}'$, $f^*(u)$ and $f^*(v)$ are equal.) **IV-3** | 98

Let us now show that $c)$ implies that the topology of X_0 is the inverse image of that of X by f. Let us note that if condition $c)$ is satisfied for the category of sheaves of sets, it is also satisfied for the category of sheaves of abelian groups, for it follows immediately from the definitions (**III**, 8.2.5) that the category of *objects in abelian groups* in the category of sheaves of sets is none other than the category of sheaves of abelian groups. It will therefore suffice to prove our assertion supposing $c)$ satisfied for the categories of sheaves of abelian groups on X and X_0. Now, if $\mathbf{Z}_X$ denotes the simple sheaf on X associated to the constant presheaf $\mathbf{Z}$, there is an isomorphism canonical up to sign $\Gamma(X, \mathscr{F}) \xrightarrow{\sim} \mathrm{Hom}(\mathbf{Z}_X, \mathscr{F})$ functorial in $\mathscr{F}$ (by the same argument as in (**0**, 5.1.1)); as it is clear by definition (**0**, 3.7.1) that $f^*(\mathbf{Z}_X) = \mathbf{Z}_{X_0}$, the hypothesis that f^* is an equivalence means that the canonical homomorphism $\Gamma(X, \mathscr{F}) \to \Gamma(X_0, f^*(\mathscr{F}))$ is *bijective* for every sheaf of abelian groups $\mathscr{F}$ on X. Now, let Y_0 be a closed subset of X_0; let $\mathscr{F}_0$ be the sheaf on X that is the direct image by the canonical injection $Y_0 \to X_0$ of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y_0. As f^* is an equivalence, $\mathscr{F}_0$ is isomorphic to a sheaf of the form $f^*(\mathscr{F})$, where $\mathscr{F}$ is a sheaf of abelian groups on X. Consider then a section s_0 of $\mathscr{F}_0$ over X_0 such that $(s_0)_{x_0} = 1_{x_0}$ if $x_0 \in Y_0$, $(s_0)_{x_0} = 0_{x_0}$ if $x_0 \notin Y_0$. The preceding remarks show that there exists a section s of $\mathscr{F}$ over X such that $(s_0)_{x_0} = s_{f(x_0)}$ for every $x_0 \in X_0$ (the fibres $(\mathscr{F}_0)_{x_0}$ and $\mathscr{F}_{f(x_0)}$ being canonically identified (**0**, 3.7.1)); this means that Y_0 is the inverse image by f of the set Y of $x \in X$ such that $s_x \neq 0_x$, and y is closed in X. □

Definition (10.2.2). — When a continuous map $f : X_0 \to X$ satisfies the equivalent conditions of (10.2.1), we say that f is a *quasi-homeomorphism* of X_0 into X.

By virtue of (10.2.1), $b)$, to say that a subset X_0 of a topological space X is very dense in X means that the canonical injection $X_0 \to X$ is a quasi-homeomorphism.

Corollary (10.2.3). — *The composite of two quasi-homeomorphisms is a quasi-homeomorphism.*

This follows immediately from (10.2.1), $a)$.

Corollary (10.2.4). — *Let $f : X \to Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible subset of Y, $X' = f^{-1}(Y')$; then the restriction $f' = f|X' : X' \to Y'$ is a quasi-homeomorphism.*

Proof. It is clear by virtue of (10.2.1), b) that the topology induced on X' by that of X is the inverse image by f' of the topology induced on Y' by that of Y. On the other hand, let $Z \neq \varnothing$ be a closed subset *in* Y', and $z \in Z$; there is an open neighbourhood U of z in Y such that $U \cap Y'$ is the union of a finite number of closed subsets Y'_j of U; if j is an index such that $z \in Y'_j$, $Z \cap Y'_j$ is therefore closed in U. Since $f(X) \cap U$ is very dense in U (10.1.4), $Z \cap Y'_j \cap f(X)$ is not empty (10.1.2), a); but as $Z \subset Y'$, we have $Z \cap f(X) = Z \cap f'(X')$; the criterion (10.1.2), a) shows that $f'(X')$ is very dense in Y', and we conclude by (10.2.1), b). □ **IV-3** | 99

Corollary (10.2.5). — *Let $f : X \to Y$ be a continuous map, (V_α) an open covering of Y. If, for every α, the restriction $f^{-1}(V_\alpha) \to V_\alpha$ of f is a quasi-homeomorphism, then f is a quasi-homeomorphism.*

This follows immediately from the criterion (10.2.1), b) and from (10.1.4).

Corollary (10.2.6). — *Let $f : X \to Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible subset of Y, $X' = f^{-1}(Y')$. For Y' to be quasi-compact (resp. Noetherian, resp. retrocompact in Y), it is necessary and sufficient that X' be quasi-compact (resp. Noetherian, resp. retrocompact in X).*

Proof. Let us prove the first two assertions; by virtue of (10.2.4), one can restrict to the case where $Y' = Y$. To say that X is quasi-compact (resp. Noetherian) means that for every increasing filtered family (U_α) in $\mathfrak{O}(X)$ (resp. for every increasing filtered family (U_α) in $\mathfrak{O}(X)$) having X as greatest element, there exists γ such that $U_\alpha = U_\gamma$ for $\alpha \geqslant \gamma$. As $U \mapsto f^{-1}(U)$ is a bijection of $\mathfrak{O}(Y)$ onto $\mathfrak{O}(X)$, our assertion follows immediately from the preceding remark.

The quasi-compact open subsets of X are therefore the sets of the form $f^{-1}(U)$ where U is an open quasi-compact subset of Y, by virtue of (10.2.1), a) and of what precedes. For X' to be retrocompact in X, it is necessary and sufficient that for every quasi-compact open U in Y, $f^{-1}(U) \cap X' = f^{-1}(U \cap Y')$ be quasi-compact (**III**, 9.1.1); the first part of the proof shows that this amounts to saying that $U \cap Y'$ is quasi-compact for every open quasi-compact U, i.e. that Y' is retrocompact in Y. □

Proposition (10.2.7). — *Let $f : X \to Y$ be a quasi-homeomorphism. Then the map $Z \mapsto f^{-1}(Z)$ of $\mathfrak{P}(Y)$ into $\mathfrak{P}(X)$ defines by restriction the bijections (cf. (10.1.1) for the notation):*

$$\mathfrak{O}(Y) \xrightarrow{\sim} \mathfrak{O}(X)$$
$$\mathfrak{F}(Y) \xrightarrow{\sim} \mathfrak{F}(X)$$
$$\mathfrak{Lf}(Y) \xrightarrow{\sim} \mathfrak{Lf}(X)$$
$$\mathfrak{Qc}(Y) \xrightarrow{\sim} \mathfrak{Qc}(X)$$
$$\mathfrak{Lqc}(Y) \xrightarrow{\sim} \mathfrak{Lqc}(X)$$
$$\mathfrak{C}(Y) \xrightarrow{\sim} \mathfrak{C}(X)$$
$$\mathfrak{Lc}(Y) \xrightarrow{\sim} \mathfrak{Lc}(X).$$

Proof. For the first two, this is none other than the definition (10.2.2); as the topology of X is the inverse image by f of that of $f(X)$, the five remaining maps, where one replaces Y by $f(X)$, are bijective. It therefore suffices (by (10.2.1), b)) to restrict to the case where X is a very dense subspace of Y, and that the maps $\mathfrak{Lf}(Y) \to \mathfrak{Lf}(X)$, $\mathfrak{Qc}(Y) \to \mathfrak{Qc}(X)$, $\mathfrak{Lqc}(Y) \to \mathfrak{Lqc}(X)$ are bijective has already been proved (10.1.2). Every locally constructible subset being locally quasi-constructible, the maps $\mathfrak{C}(Y) \to \mathfrak{C}(X)$ and $\mathfrak{Lc}(Y) \to \mathfrak{Lc}(X)$ are injective; furthermore, for every open $U \subset Y$, the restriction $f^{-1}(U) \to U$ of f is a quasi-homeomorphism (10.2.4), hence if we show that $\mathfrak{C}(Y) \to \mathfrak{C}(X)$ is surjective, the same will be true of $\mathfrak{Lc}(Y) \to \mathfrak{Lc}(X)$. But by virtue of (10.2.6), every retrocompact open subset Z in X is of the form $f^{-1}(T)$, where T is retrocompact open in Y; this evidently proves the surjectivity of $\mathfrak{C}(Y) \to \mathfrak{C}(X)$. □ **IV-3** | 100

Remarks (10.2.8). — (i) We saw in the proof of (10.2.1) that if f is a quasi-homeomorphism, the canonical map

(10.2.8.1) $$\Gamma(Y, \mathscr{F}) \longrightarrow \Gamma(X, f^*(\mathscr{F}))$$

is a functorial isomorphism in $\mathscr{F}$ in the category of sheaves of abelian groups on Y. As f^* is exact in this category (**0**, 3.7.2), this implies that the canonical homomorphisms $(T, 3.2.2)$

$$\mathrm{H}^i(Y, \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(X, f^*(\mathscr{F}))$$

are *bijective* for every i.

(ii) If $f : X \to Y$ is a continuous map and $\mathscr{F}, \mathscr{G}$ two sheaves of abelian groups on Y, we have $f^*(\mathscr{F} \otimes_{\mathbf{Z}_Y} \mathscr{G}) = f^*(\mathscr{F}) \otimes_{\mathbf{Z}_X} f^*(\mathscr{G})$, an isomorphism canonical up to sign (**0**, 4.3.3). We conclude that if f is a quasi-homeomorphism, f^* is also an *equivalence* of the category of sheaves of rings on Y, and of the category of sheaves of rings on X; the data of a structure of ringed space on Y is therefore equivalent to that of a structure of ringed space on X.

Given two ringed spaces $(X, \mathscr{O}_X)$, $(Y, \mathscr{O}_Y)$, we will say that a morphism of ringed spaces $f = (\psi, \theta) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a *quasi-isomorphism* if ψ is a quasi-homeomorphism of X into Y and $\theta^\sharp : \psi^*(\mathscr{O}_Y) \to \mathscr{O}_X$ an isomorphism of sheaves of rings; when this is so, the ringed space $(X, \mathscr{O}_X)$ is entirely determined, up to isomorphism, by $(Y, \mathscr{O}_Y)$, the space X, and the quasi-homeomorphism ψ (which can be chosen arbitrarily). When f is a quasi-isomorphism of ringed spaces, the functor

$$\mathscr{F} \longmapsto f^*(\mathscr{F})$$

is an *equivalence* of the category of $\mathscr{O}_Y$-Modules and of that of $\mathscr{O}_X$-Modules, since $f^*(\mathscr{F})$ is canonically identified here with $\psi^*(\mathscr{F})$. We conclude for example isomorphisms of bi-∂-functors

$$\mathrm{Ext}^i_{\mathscr{O}_Y}(\mathscr{F}, \mathscr{G}) \xrightarrow{\sim} \mathrm{Ext}^i_{\mathscr{O}_X}(f^*(\mathscr{F}), f^*(\mathscr{G})).$$

In general, one can say that the usual constructions of the theory of sheaves and of homological algebra, carried out on the ringed space Y or the ringed space X, are equivalent.

10.3. Jacobson spaces

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Definition (10.3.1). — We say that a topological space X is a *Jacobson space* if the set X_0 of closed points of X is very dense in X (in other words, if the canonical injection $X_0 \to X$ is a quasi-homeomorphism).

This means therefore (10.1.2) that every locally closed (or only locally quasi-constructible) subset $Z \neq \varnothing$ of X contains a closed point of X, or equivalently that *every closed subset of X is the closure of the set of its closed points*.

Proposition (10.3.2). — *Let X be a Jacobson space, Z a quasi-constructible subset of X; then the subspace Z is a Jacobson space, and for a point of Z to be closed in Z, it is necessary and sufficient that it be closed in X.*

Proof. If X_0 is the set of closed points of X, $Z \cap X_0$ is very dense in Z by virtue of (10.2.4) applied to the injection $i : X_0 \to X$; as the set Z_0 of closed points of Z evidently contains $Z \cap X_0$, it is *a fortiori* very dense in Z, hence Z is a Jacobson space. Let us now show that we have $Z_0 = Z \cap X_0$; let $x \in Z$ be a closed point in Z; let $\overline{\{x\}}$ be its closure in X and $\overline{\{x\}} \cap Z = \{x\}$, which is therefore a locally quasi-constructible subset of X, and as its intersection with X_0 is not empty (10.1.2), we have $x \in X_0$. □

Proposition (10.3.3). — *Let X be a topological space, (U_α) an open covering of X. For X to be a Jacobson space, it is necessary and sufficient that each of the subspaces U_α be a Jacobson space.*

Proof. The condition is necessary by virtue of (10.3.2). Conversely, let us show first that the hypothesis that the U_α are Jacobson spaces implies that for a point $x \in U_\alpha$ to be closed in X, it suffices that it be closed in U_α. It suffices indeed to see that this condition implies that x is also closed in each of the U_β that contain it; but $U_\alpha \cap U_\beta$ is open in U_α, hence x is closed in $U_\alpha \cap U_\beta$, and by virtue of (10.3.2), x is also closed in U_β, which completes the proof. □

10.4. Jacobson preschemes and Jacobson rings

Definition (10.4.1). — We say that a prescheme X is a *Jacobson prescheme* if the underlying topological space of X is a Jacobson space. We say that a ring A is a *Jacobson ring* if $\operatorname{Spec}(A)$ is a Jacobson prescheme.

Every closed subset of $X = \operatorname{Spec}(A)$ is of the form $Z = V(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal equal to its radical, and the set X_0 of closed points of X is the set of maximal ideals of A; to say that $Z \cap X_0$ is dense in Z means therefore that $\mathfrak{a}$ is an *intersection of maximal ideals* (**I**, 1.1.4); as $\mathfrak{a}$ is an intersection of prime ideals, this amounts to saying that every prime ideal of A is an intersection of maximal ideals; by virtue of (10.3.1) and (10.1.2), the usual definition of Jacobson rings (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 4, def. 1) thus coincides with the definition (10.4.1).

Proposition (10.4.2). — *Let X be a prescheme, (U_α) a covering of X formed by affine open subsets. For X to be a Jacobson prescheme, it is necessary and sufficient that the rings $\Gamma(U_\alpha, \mathscr{O}_X)$ of the U_α be Jacobson rings.* **IV-3 | 102**

This follows from (10.3.3) and the definition (10.4.1).

(10.4.3). A discrete prescheme is a Jacobson prescheme; an Artinian ring is therefore a Jacobson ring. Every principal ideal ring having an infinity of maximal ideals (for example **Z**) is a Jacobson ring; a Noetherian local ring A is a Jacobson ring if and only if its maximal ideal is its only prime ideal, i.e. if A is Artinian. Every sub-prescheme of a Jacobson prescheme is a Jacobson prescheme by virtue of (10.3.2).

Proposition (10.4.4). — *Let B be an integral ring. The following conditions are equivalent:*

a) *There exists $f \neq 0$ in B such that B_f is a field.*
b) *The field of fractions of B is a B-algebra of finite type.*
c) *There exists a field K containing B and which is a B-algebra of finite type.*
d) *The generic point of* $\operatorname{Spec}(B)$ *is isolated.*

Proof. It is clear that d) is equivalent to a), since d) means that there exists $f \neq 0$ in B such that $D(f)$ is reduced to the generic point of $\operatorname{Spec}(B)$. It is trivial that a) implies b) and that b) implies c). Finally, c) implies a), by virtue of Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, cor. 2 of th. 1. □

Proposition (10.4.5). — *Let A be a ring. The following conditions are equivalent:*

a) *A is a Jacobson ring.*
b) *For every non-maximal prime ideal $\mathfrak{p}$ of A and every $f \neq 0$ in $B = A/\mathfrak{p}$, B_f is not a field.*
b′) *Every A-algebra of finite type K which is a field is a finite A-algebra.*

Proof. We know that a) implies b') (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 4, cor. 3 of th. 3). Furthermore, the kernel of the homomorphism $A \to K$ is then a *maximal* ideal $\mathfrak{m}$ of A, and K is a finite extension of $A/\mathfrak{m}$ (*loc. cit.*). It is trivial that b') implies b), since $A/\mathfrak{p} = B$ is not a field and B_f is a B-algebra of finite type. It remains to show that b) implies a). We will use the following lemma: □

Lemma (10.4.5.1). — *Let X be a topological space having the following property: for every locally closed subset $Z \neq \varnothing$ of X, there exists a subset Z' of Z, locally closed in X (or in Z, which amounts to the same), and a point $x \in Z'$, closed in Z'. Then, for X to be a Jacobson space, it is necessary and sufficient that no non-closed point x of X be isolated in $\overline{\{x\}}$.*

Proof. If X is a Jacobson space, a non-closed point x of X cannot be isolated in $\overline{\{x\}}$, for this would mean that there exists an open U of X such that $U \cap \overline{\{x\}} = \{x\}$; but $U \cap \overline{\{x\}}$ is locally closed in X and would contain no closed point in X, which contradicts the hypothesis that X is a Jacobson space. Conversely, suppose the conditions of the statement are satisfied; then the set X_0 of closed points of X is identical to the set of $x \in X$ that are isolated in $\overline{\{x\}}$; but it follows from (5.1.10.1) that this set is *very dense* in X, hence X is a Jacobson space by definition. □ **IV-3 | 103**

Let us recall (5.1.10) that the hypothesis made on X in (10.4.5.1) is always satisfied when the underlying space of X is a *prescheme*.

Returning to the proof of (10.4.5), condition $b)$ means that for every non-closed point x of $\mathrm{Spec}(A)$, the generic point x of $\mathrm{Spec}(A/\mathfrak{j}_x) = \overline{\{x\}}$ is not isolated in $\overline{\{x\}}$, hence $b)$ implies $a)$ by the lemma (10.4.5.1) and (10.4.4).

Corollary (10.4.6). — *Every algebra of finite type B over a Jacobson ring A is a Jacobson ring, and the inverse image in A of every maximal ideal of B is a maximal ideal of A. In particular, every algebra of finite type over a field or over* $\mathbf{Z}$ *is a Jacobson ring.*

Proof. A B-algebra of finite type K which is a field is also an A-algebra of finite type, hence an A-module of finite type and *a fortiori* a B-module of finite type, whence the first assertion; the second was proved in the course of the proof of (10.4.5), applied to $K = B/\mathfrak{m}$. □

Corollary (10.4.7). — *If X is a Jacobson prescheme, $f : Y \to X$ a morphism locally of finite type, then Y is a Jacobson prescheme and the image by f of every closed point in Y is a closed point in X.*

Proof. The question being local on X and on Y, one reduces to the case where X and Y are affine, and the corollary then follows from (10.4.6). □

Corollary (10.4.8). — *If k is an algebraically closed field, X a k-prescheme locally of finite type, the set of rational points of X over k is very dense in X.*

Proof. Indeed, X is a Jacobson prescheme (10.4.7) and the closed points are exactly the rational points over k (**I**, 6.4.2). □

(10.4.9). The fact that preschemes locally of finite type over a field or over $\mathbf{Z}$ are Jacobson preschemes is particularly important, because of the possibility of reducing to this case in many questions of algebraic geometry (8.1.2), c). We will give two examples:

(10.4.10). Applications (10.4.10) : I : Proof of (6.15.9)**.** Let k be a separably closed field, X a k-prescheme locally of finite type over k and *unibranch*. We know that the integral closure of a k-algebra of finite type A in an extension of finite type of its field of fractions is a finite A-algebra (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 2, th. 2), hence every k-algebra of finite type is a universally Japanese ring; we conclude that the set of points $x \in X$ where X is geometrically unibranch is locally constructible (9.7.10). But the hypothesis and the lemma (6.15.8) imply that this set contains all the *closed* points of X. The conclusion therefore follows from (10.4.6), (10.3.1) and the bijectivity of the canonical map $\mathfrak{Lc}(X) \to \mathfrak{Lc}(X_0)$ (where X_0 is the set of closed points of X) (10.2.7).

(10.4.11). Applications : II : Proposition (10.4.11). Let S be a prescheme, X an S-prescheme of finite type. Every S-endomorphism of X which is radicial is surjective (hence bijective). **IV-3** | 104

Proof. Let $f : X \to S$ be the structural morphism, $g : X \to X$ the S-endomorphism considered, and, for every $s \in S$, let g_s be the morphism deduced from g by the base change $\mathrm{Spec}(k(s)) \to S$, which is a $k(s)$-endomorphism of the fibre $f^{-1}(s) = X \times_S \mathrm{Spec}(k(s))$.

To prove that g is surjective, it suffices to prove that g_s is surjective for every $s \in S$, hence one can (by virtue of (**I**, 3.5.7)) reduce to the case where $S = \mathrm{Spec}(k)$ is the spectrum of a field k, in which case f is a morphism of finite presentation, since S is Noetherian. Applying (8.9.1) and (8.10.5), (vii), one reduces to the case where $S = \mathrm{Spec}(A)$, where A is a sub-$\mathbf{Z}$-algebra of finite type of k. But then X is a Jacobson prescheme (10.4.7), and $g(X)$ is constructible in X (**I**, 1.8.5), hence, to prove that $g(X) = X$, it suffices to show that $g(X)$ contains all the *closed* points of X (10.3.1).

Lemma (10.4.11.1). — *Let Y be a* $\mathbf{Z}$*-prescheme of finite type.*

(i) *For a closed point $y \in Y$, it is necessary and sufficient that $k(y)$ be a finite field.*
(ii) *For every prime number p and every integer $d \geqslant 1$, the set of points $y \in Y$ such that $k(y)$ is an extension of $\mathbf{F}_p$ of degree dividing d is finite.*

The assertion (i) follows from the fact that the image of a closed point $y \in Y$ in $\mathrm{Spec}(\mathbf{Z})$ is a closed point (10.4.7), in other words a prime number, and from (**I**, 6.4.2). On the other hand, as Y is a finite union of affine open subsets of finite type over $\mathbf{Z}$, one can restrict to proving (ii) in the case where $Y = \mathrm{Spec}(C)$, where C is a $\mathbf{Z}$-algebra of finite type. Now, the points $y \in Y$ such that the degree of $k(y)$ over $\mathbf{F}_p$ divides d correspond bijectively to the homomorphisms $C \to \mathbf{F}_{p^d}$; but if (t_i) is a system of generators of the $\mathbf{Z}$-algebra C, every homomorphism of C is determined by its values on the elements t_i, and there are therefore only a finite number of homomorphisms of C into a finite field.

This being so, since X is a $\mathbf{Z}$-prescheme of finite type, let $T_{p,d}$ be the set of closed points $z \in X$ such that $k(z)$ is an extension of $\mathbf{F}_p$ of degree dividing d; it follows from (10.4.11.1) that the set $T_{p,d}$ is finite and that the set of closed points of X is the union of the $T_{p,d}$. Furthermore, if $z \in T_{p,d}$ and if h is any endomorphism of X, $k(h(z))$ is isomorphic to a subfield of $k(z)$, hence $h(z) \in T_{p,d}$; in other words $T_{p,d}$ is *stable* under every endomorphism of X. As g is by hypothesis injective, its restriction to $T_{p,d}$ is *bijective* since $T_{p,d}$ is finite, which completes proving the proposition.

We will see further on (17.9.6) that when one further supposes that X is an S-prescheme of finite presentation, and on the other hand that g is a monomorphism, then one can assert that g is an *automorphism* of X. □

10.5. Noetherian Jacobson preschemes

Proposition (10.5.1). — *Let B be a Noetherian integral ring. The conditions equivalent to a) through d) of (10.4.4) are then also equivalent to the following:*

e) $\operatorname{Spec}(B)$ *is finite.*
f) *B is a semi-local ring of dimension $\leqslant 1$.*

Proof. It follows indeed from the Artin–Tate theorem (**0**, 16.3.3) that conditions e) and f) are equivalent. Condition f) implies that $\operatorname{Spec}(B)$ is the union of the set of its closed points and of its generic point, hence f) implies e) without assuming A to be Noetherian. Finally, e) implies d) without assuming A to be Noetherian, for the generic point x of X is the complement of the union of the closures $\overline{\{y\}}$, where y ranges over the set of points $y \neq x$, and as these points are finite in number, $\{x\}$ is the complement of a closed set in X. □ **IV-3 | 105**

Corollary (10.5.2). — *For a Noetherian ring A to be a Jacobson ring, it is necessary and sufficient that there exist no prime ideal $\mathfrak{p}$ of A such that $A/\mathfrak{p}$ is a semi-local ring of dimension* 1.

Proof. This follows immediately from (10.5.1) and from condition b) of (10.4.5), the prime ideals $\mathfrak{p}$ of A such that $A/\mathfrak{p}$ is semi-local of dimension 0 being the maximal ideals of A. □

Corollary (10.5.3). — *Let X be an irreducible Noetherian prescheme. The following conditions are equivalent:*

a) *The generic point of X is isolated.*
b) *X is finite.*
c) *X is of dimension $\leqslant 1$ and the set of its closed points is finite.*

Proof. There exists by hypothesis a finite number of irreducible affine open subsets U_i ($1 \leqslant i \leqslant n$) covering X, and each of which contains the generic point of X; it suffices to prove the equivalence of a), b), and c) for each of the $(U_i)_{\mathrm{red}}$ (taking into account (**0**, 14.1.7)). But this equivalence follows from (10.5.1). □

Remark (10.5.4). — A Noetherian prescheme X satisfying the equivalent conditions of (10.5.3) is not necessarily an affine scheme; in fact it can even be *non-separated*, as one sees by replacing, in the example (**I**, 5.5.11) of the "affine line with doubled point", X_1 and X_2 by the spectrum of the ring of the discrete valuation $(K[s])_{(s)}$, U_{12} and U_{21} by the open subset reduced to the generic point in X_1 and X_2 respectively; the non-separated prescheme X that one obtains has exactly 3 points.

Proposition (10.5.5). — *Let X be a Noetherian prescheme.*

(i) *The set X_0 of points $x \in X$ such that $\overline{\{x\}}$ is finite is very dense in X.*
(ii) *For X to be a Jacobson prescheme, it is necessary and sufficient that there exist no sub-prescheme of X isomorphic to the spectrum of a semi-local integral ring of dimension* 1.

Proof. The condition that $\overline{\{x\}}$ is finite is equivalent, here (10.5.3), to the fact that x is isolated in $\overline{\{x\}}$, it being the underlying space of a sub-prescheme (Noetherian) of X; the assertion (i) therefore follows from (10.4.5.1). Similarly, taking into account (10.4.5.1), to show the assertion (ii), let us note that for a non-closed point x of X to belong to X_0, it is necessary and sufficient that the closed sub-prescheme Y of X having $\overline{\{x\}}$ as underlying space be of dimension 1 and finite, hence a finite union of sub-preschemes which are open (in Y) and affine U_i which are spectra of semi-local integral rings of dimension 1. Conversely, if there is a sub-prescheme Z of X which is the spectrum of a semi-local integral ring of dimension 1, Z is not a Jacobson prescheme (10.5.2), hence neither is X (10.3.2). ☐

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Remark (10.5.6). — The assertion (ii) of (10.5.5) is still valid when X is *locally Noetherian*: indeed, if (V_α) is a covering of X by affine open subsets (Noetherian), every sub-prescheme of one of the V_α is a sub-prescheme of X; conversely, if a sub-prescheme Z of X is isomorphic to the spectrum of a semi-local integral ring of dimension 1, there is an α such that $V_\alpha \cap Z$ contains an affine open U of Z which is non-reduced to the generic point of Z, and which is therefore also the spectrum of a semi-local integral ring of dimension 1. We conclude by (10.3.3).

Proposition (10.5.7). — *Let X be a locally Noetherian prescheme, Y a closed subset of X such that every non-empty closed subset of X meets Y. Then the subprescheme induced on the open subset $X - Y$ is a Jacobson prescheme.*

Proof. Applying the criterion (10.5.5), (ii), and supposing that there were in $X - Y$ a sub-prescheme Z which is the spectrum of a semi-local integral ring of dimension 1, the generic point z of Z being isolated in Z (or in the closure $\overline{Z}$ of Z in X), and Z being distinct from $\{z\}$. Let $y \neq z$ be a point of Z; as it does not belong to Y, it is not a closed point in X, and its closure $\overline{\{y\}}$ in X meets Y at a point $x \neq y$, which is therefore a specialisation of y. The existence of the chain $\{x\} \subset \overline{\{y\}} \subset \overline{\{z\}}$ then shows that the dimension of $\overline{\{z\}}$ would be $\geqslant 2$, and the same would be true of the dimension of $U \cap \overline{\{z\}}$, where U is an affine (hence Noetherian) open neighbourhood of x in X. But this contradicts the fact that the generic point z of $U \cap \overline{\{z\}}$ is isolated in $U \cap \overline{\{z\}}$ (10.5.3). ☐

Corollary (10.5.8). — *Let A be a Noetherian ring; for every element f of the radical $\mathfrak{R}$ of A, the ring A_f is a Jacobson ring, and the open subset* $\mathrm{Spec}(A) - V(\mathfrak{R})$ *is a Jacobson scheme.*

Proof. If $X = \mathrm{Spec}(A)$, $Y = V(\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of A, to say that every non-empty closed subset of X meets Y is equivalent (**0**, 2.1.3) to saying that Y contains all the *closed points* of X, or equivalently that $\mathfrak{J}$ is *contained in the radical* $\mathfrak{R}$ of A. If $f \in \mathfrak{R}$, the open $D(f)$ does not meet $V(\mathfrak{R})$, and is a Jacobson space by virtue of (10.5.7). ☐

Corollary (10.5.9). — *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, $X = \mathrm{Spec}(A)$, $Y = X - \{\mathfrak{m}\}$; then Y is a Jacobson scheme, whose closed points are the prime ideals $\mathfrak{p} \in \mathrm{Spec}(A)$ such that* $\dim(A/\mathfrak{p}) = 1$.

Proof. The first assertion is a particular case of (10.5.8); on the other hand the closed points of Y are the prime ideals $\mathfrak{p}$ of A which are maximal elements in the set of prime ideals $\neq \mathfrak{m}$, which, by definition of the dimension, means that $\dim(A/\mathfrak{p}) = 1$. ☐

Proposition (10.5.10). — *Let A be a Noetherian local ring, reduced, complete, and not a field. Then the finite intersections of the kernels of the local homomorphisms of A into discrete valuation rings V, which are A-algebras of finite type, form a basis of filter tending to 0 for the adic topology of A.*

Proof. It suffices (Bourbaki, *Alg. comm.*, chap. III, §2, n° 7, prop. 8) to prove that the intersection of the kernels considered in the statement is reduced to 0. Suppose first that A is integral and of dimension 1; by virtue of the Nagata theorem (**0**, 23.1.5) and (23.1.6), the integral closure A' of A is a complete local integral ring and integrally closed and of dimension 1; it is therefore a discrete valuation ring, and the proposition follows immediately in this case. **IV-3 | 107**

Let us pass to the general case; let x be the closed point of $X = \mathrm{Spec}(A)$, and set $Y = X - \{x\}$; we know (10.5.9) that Y is a Jacobson prescheme. Let us show that the intersection of the prime ideals $\mathfrak{p}$ of A such that $\dim(A/\mathfrak{p}) = 1$ is reduced to 0. But to say that the intersection of these prime ideals is reduced to 0 means that the set of these ideals is dense in X, or equivalently in Y (since Y is dense in X), and this follows immediately from (10.5.9). ☐

10.6. Dimension in Jacobson preschemes The results of this section give precision in certain cases and generalise results from § 5.

Proposition (10.6.1). — *Let S be a locally Noetherian prescheme, satisfying in addition the conditions: 1° S is a Jacobson prescheme; 2° for every $s \in S$, $\mathscr{O}_{S,s}$ is universally catenary (5.6.2); 3° every irreducible component S' of S is equicodimensional (in other words, for every closed point s of S' and every sub-prescheme S' having S' as underlying space, we have $\dim(S') = \dim(\mathscr{O}_{S',s})$). We then have the following properties:*

(i) *For every morphism locally of finite type $g : X \to S$, X satisfies conditions 1°, 2°, and 3° above. In particular, if X is equidimensional (for example if X is irreducible), X is biequidimensional (in other words X is catenary and for every closed point x of X, we have $\dim(\mathscr{O}_{X,x}) = \dim(X)$ (**0**, 14.3.3)).*

(ii) *Let X, Y be two S-preschemes locally of finite type over S, $f : X \to Y$ an S-morphism; suppose X irreducible and f dominant. If ξ (resp. η) is the generic point of X (resp. Y) and $e = \dim(f^{-1}(\eta)) = \deg.\mathrm{tr}_{k(\eta)} k(\xi)$, we have*

$$\dim(X) = \dim(Y) + e. \tag{10.6.1.1}$$

(iii) *Let X, Y be two S-preschemes locally of finite type over S, $f : X \to Y$ an S-morphism, n a non-negative integer such that $\dim(f^{-1}(y)) \geqslant n$ (resp. $\dim(f^{-1}(y)) \leqslant n$) for every $y \in Y$. Then we have*

$$\dim(X) \geqslant \dim(Y) + n \tag{10.6.1.2}$$

(resp.

$$\dim(X) \leqslant \dim(Y) + n). \tag{10.6.1.3}$$

Proof. (i) The property 1° for X follows from (10.4.7). For every $x \in X$, $\mathscr{O}_{X,x}$ is the local ring of an $\mathscr{O}_{S,g(x)}$-algebra of finite type at a prime ideal, and the homomorphism $\mathscr{O}_{S,g(x)} \to \mathscr{O}_{X,x}$ is local; hence (5.6.3), (iv) $\mathscr{O}_{X,x}$ is universally catenary. To show that X satisfies condition 3°, consider several cases: **IV-3 | 108**

a) X is a closed irreducible sub-prescheme of S; let S' be an irreducible component of S containing X, ξ the generic point of X, x a closed point of X; for every $s \in S'$, $\mathscr{O}_{S',s}$, a quotient of $\mathscr{O}_{S,s}$, is catenary (5.6.1), hence conditions 2° and 3° imply that S' is biequidimensional (**0**, 14.3.3). By virtue of (5.1.2) and of (**0**, 14.3.3.2), we therefore have

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{S',x}) - \dim(\mathscr{O}_{S',\xi}) = \dim(S') - \dim(\mathscr{O}_{S',\xi})$$

which proves that $\dim(\mathscr{O}_{X,x})$ does not depend on the closed point x considered, whence the assertion in this case (5.1.4).

b) X is irreducible and g dominant. Then, for every closed point x of X, $g(x)$ is closed in S by (10.4.7); as $\mathscr{O}_{S,g(x)}$ is universally catenary, it follows from (5.6.5.3) that we have

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{S,g(x)}) + e$$

where $e = \dim(g^{-1}(\zeta))$, ζ being the generic point of S. As $\dim(S) = \dim(\mathscr{O}_{S,g(x)})$ by virtue of condition 3° for S, we have $\dim(\mathscr{O}_{X,x}) = \dim(S) + e$ for every closed point $x \in X$; this proves condition 3° for X (5.1.4), and at the same time the formula

$$\dim(X) = \dim(S) + e. \tag{10.6.1.4}$$

c) General case. Considering a reduced sub-prescheme X' of X having for underlying space an irreducible component of X, one reduces to the case where X is integral; using (**I**, 5.2.2) and the case a) proved above, one can then replace S by the reduced sub-prescheme having $g(X)$ as underlying space; one is then brought to case b), and this completes proving (i).

(ii) The morphism f being locally of finite type (**I**, 1.3.4), one can apply the results of (i) replacing S by Y; furthermore, as X is irreducible and f dominant, one can also replace S by Y in (10.6.1.4), which gives (10.6.1.1).

(iii) The relative assertion where $\dim(f^{-1}(y)) \leqslant n$ for every $y \in Y$ has already been proved under more general hypotheses in (5.6.7). Suppose that $\dim(f^{-1}(y)) \geqslant n$ for every $y \in Y$, and consider a generic point η of an irreducible component Y' of Y; there exists at least one irreducible component Z of $f^{-1}(\eta)$ of dimension $\geqslant n$; if ξ is the generic point of Z, ξ is also a generic point of an irreducible component X' of X such that $f(X')$ is dense in Y' (**0**, 2.1.8). Consider the reduced sub-preschemes of X, Y having respectively for underlying spaces X', Y', and the restriction $X' \to Y'$ of

f (**I**, 5.2.2); it then follows from (ii) that $\dim(X') \geqslant \dim(Y') + n$, and *a fortiori* $\dim(X) \geqslant \dim(Y') + n$; this being true for every irreducible component Y' of Y, we conclude the inequality (10.6.1.2). □

Corollary (10.6.2). — *Suppose that X satisfies conditions $1°$, $2°$, and $3°$ of (10.6.1). Then, for every dense open U in X, we have $\dim(U) = \dim(X)$.*

Proof. We know indeed that $\dim(U) \leqslant \dim(X)$ (**0**, 14.1.4); furthermore, as X is a Jacobson prescheme, U contains a closed point of every irreducible component of X, hence $\dim(U) = \dim(X)$ by virtue of (10.6.1) and (**0**, 14.1.2.1). □ **IV-3 | 109**

Proposition (10.6.3). — *Suppose that X satisfies conditions $1°$, $2°$, and $3°$ of (10.6.1), and let Y be a closed subset of X. For every $x \in Y$, and every open neighbourhood U of x in X not meeting the irreducible components of Y that do not contain x, we have*

(10.6.3.1) $$\dim_x(Y) = \dim(U \cap Y) = \dim \overline{\{x\}} + \operatorname{codim}(\overline{\{x\}}, Y) = \sup_i \dim(Y_i)$$

where Y_i ($1 \leqslant i \leqslant m$) are the irreducible components of Y containing x.

Proof. Considering a closed sub-prescheme of X having Y as underlying space, one can restrict, by virtue of (10.6.1), (i), to the case where $Y = X$. By the choice of U, U is the union of the $U \cap X_i$, hence $\dim(U) = \sup_i \dim(U \cap X_i)$; but by (10.6.2) applied to a closed sub-prescheme of X having X_i as underlying space, we have (taking into account (10.6.1), (i)), $\dim(U \cap X_i) = \dim(X_i)$; this proves that the second and the fourth terms of (10.6.3.1) are equal. As $U \cap X_i$ is biequidimensional (10.6.1), (i) (since the immersion $U \cap X_i \to X_i$ is of finite type (**I**, 6.3.5)), we have

(10.6.3.2) $$\dim(X_i) = \dim(U \cap X_i) = \dim(U \cap \overline{\{x\}}) + \operatorname{codim}(U \cap \overline{\{x\}}, U \cap X_i)$$

(**0**, 14.3.5). By (10.6.2) applied to a closed sub-prescheme of X having $\overline{\{x\}}$ as underlying space, $\dim(U \cap \overline{\{x\}}) = \dim(\overline{\{x\}})$; as we also have, for the same reasons,

(10.6.3.3) $$\dim(X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X_i)$$

we obtain

$$\dim(U) = \dim(\overline{\{x\}}) + \sup_i \operatorname{codim}(\overline{\{x\}}, X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X)$$

by definition of the codimension (**0**, 14.2.1). This shows that $\dim(U)$ is *independent* of the open neighbourhood U of x satisfying the conditions of the statement, hence is equal to $\dim_x(X)$, by (**0**, 14.1.4.1). □

Corollary (10.6.4). — *Under the hypotheses of (10.6.3), let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, Y its support. For every $x \in Y$, we have*

(10.6.4.1) $$\dim(\overline{\{x\}}) + \dim(\mathscr{F}_x) = \dim_x Y.$$

Proof. Indeed, this follows from (10.6.3.1) and from the formula $\dim(\mathscr{F}_x) = \operatorname{codim}(\overline{\{x\}}, Y)$ (5.1.12.2). □

10.7. Examples and counter-examples

(10.7.1). Let S be a locally Noetherian prescheme of dimension $\leqslant 1$ and suppose that S is a *Jacobson prescheme*; when S is Noetherian, this amounts to saying that the irreducible components of S of dimension 1 are infinite, for every $x \in S$ which is not a closed point is the generic point of such a component (10.4.5) and (10.5.4). Then S also satisfies conditions 2° and 3° of (10.6.1): indeed, every local ring $\mathscr{O}_{S,s}$ is of dimension 0 or 1, and hence universally catenary (5.6.3), (7.2.9); on the other hand, an irreducible component S' of S is, either reduced to a closed point s of S', or biequidimensional.

From these remarks and from (10.6.1) it follows that every prescheme locally of finite type over S also satisfies the properties 1°, 2°, and 3°; it is in particular the case of preschemes *locally of finite type over a field or over* $\mathbf{Z}$. **IV-3** | 110

(10.7.2). Let A be a Noetherian local ring *universally catenary* and let S be the complementary of *the closed point* a in $X = \operatorname{Spec}(A)$. Then S satisfies conditions 1°, 2°, and 3° of (10.6.1): indeed, one has already seen that S is a Jacobson prescheme (10.5.9); as A is universally catenary, so are all the local rings $A_{\mathfrak{p}}$ at the prime ideals of A (5.6.3). On the other hand, an irreducible component S' of S is the complement of a in an irreducible component X' of X. For every closed point x of S, the closure of x in X is $\{x, a\}$, so $\dim(\overline{\{x\}}) = 1$. Since X' is biequidimensional (**0**, 14.3.3), it follows that $\dim(\mathscr{O}_{X',x}) = \dim(X') - 1$ (**0**, 14.3.2), proving that S' is equicodimensional.

(10.7.3). Let A be a discrete valuation ring; let us show that in $B = A[T_1, \dots, T_n]$, with n indeterminates, there exist two maximal ideals $\mathfrak{m}$, $\mathfrak{n}$ of *respective heights* n and $n+1$. We have seen (5.2.5), (i) for $n = 1$; let us prove it by induction on n. As $A[T_1, \dots, T_{n+1}]$ is a free, hence faithfully flat, $A[T_1, \dots, T_n]$-module, there are maximal ideals $\mathfrak{m}'$ and $\mathfrak{n}'$ of $A[T_1, \dots, T_{n+1}]$ lying respectively over $\mathfrak{m}$ and $\mathfrak{n}$ (**0**, 6.5.1); by (5.5.3), these ideals have respective heights $n+1$ and $n+2$, proving the assertion. Suppose $n \geqslant 2$ in what follows. Let $\mathfrak{J} = \mathfrak{m} \cap \mathfrak{n} = \mathfrak{m}\mathfrak{n}$ and let $R = 1 + \mathfrak{J}$, which is a multiplicative subset of B. If $B' = R^{-1}B$, the ideal $\mathfrak{J}' = R^{-1}\mathfrak{J}$ is contained in the radical of B' (Bourbaki, *Algèbre commutative*, Chap. III, §3, no. 5, Prop. 12). As a topological space, $X' = \operatorname{Spec}(B')$ is identified with a subspace of $X = \operatorname{Spec}(B)$, and at the points x of X' the local rings $\mathscr{O}_{X,x}$ and $\mathscr{O}_{X',x}$ are the same (**I**, 1.6.2). Consider in X' the closed subset $Y' = V(\mathfrak{J}')$, and put $S = X' \setminus Y'$. Then S is a Jacobson prescheme (10.5.7), and is evidently irreducible and Noetherian; moreover, for every $s \in S$, the local ring $\mathscr{O}_{S,s} = \mathscr{O}_{X,s}$ is universally catenary by (5.6.3), since A is universally catenary (5.6.4). Nevertheless, there are two closed points a, b of S such that $\mathscr{O}_{S,a}$ and $\mathscr{O}_{S,b}$ *do not have the same dimension*; in other words, S does not satisfy condition 3° of (10.6.1). To see this, consider the two maximal ideals $\mathfrak{m}' = R^{-1}\mathfrak{m}$ and $\mathfrak{n}' = R^{-1}\mathfrak{n}$ of B', which have heights n and $n+1$, respectively. We have $\mathfrak{J}' = \mathfrak{m}' \cap \mathfrak{n}'$, and hence $\mathfrak{J}'$ is contained in no prime ideal of B' distinct from $\mathfrak{m}'$ and $\mathfrak{n}'$; these are consequently the only maximal ideals of B'. Let a', b' be the only closed points of X', corresponding to $\mathfrak{m}'$ and $\mathfrak{n}'$. There exists in B' a nonmaximal prime ideal $\mathfrak{p}' \subset \mathfrak{m}'$ which is not contained in $\mathfrak{n}'$: it suffices to show that in B there is a nonmaximal prime ideal contained in $\mathfrak{m}$ and not in $\mathfrak{n}$; for this one may, for example, consider the fibre of $\mathfrak{m}$ for the morphism corresponding to the injection $A[T_1] \longrightarrow B = A[T_1, \dots, T_n]$ and apply (6.1.2). By considering a maximal chain of prime ideals between $\mathfrak{p}'$ and $\mathfrak{m}'$, and replacing $\mathfrak{p}'$ by the penultimate ideal in this chain, we may therefore suppose that the point a of X' corresponding to $\mathfrak{p}'$ has closure $\{a, a'\}$ in X'. Since $B'_{\mathfrak{m}'}$ is biequidimensional, $\mathfrak{p}'$ then has height $n-1$. In the same way one constructs a nonmaximal prime ideal $\mathfrak{q}'$ of B' of height n such that, if b is the corresponding point of X', the closure of b in X' is $\{b, b'\}$. Thus a and b belong to S and are closed in S, and consequently answer the question.

10.8. Rectified depth

Definition (10.8.1). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. For every $x \in X$, we call *rectified depth* of $\mathscr{F}$ at the point x and denote $\mathrm{prof}^*_x(\mathscr{F})$ the number (integer $\geqslant 0$ or $+\infty$) equal to

(10.8.1.1) $$\mathrm{prof}^*_x(\mathscr{F}) = \mathrm{prof}(\mathscr{F}_x) + \dim(\overline{\{x\}})$$

where $\overline{\{x\}}$ is the closure of the point x in X. For every closed subset Z of X, we call *rectified depth of $\mathscr{F}$ along Z* and denote $\mathrm{prof}^*_Z(\mathscr{F})$ the number

(10.8.1.2) $$\mathrm{prof}^*_Z(\mathscr{F}) = \inf_{x \in Z} \mathrm{prof}^*_x(\mathscr{F}).$$

In other terms, for every integer n, the relation $\mathrm{prof}^*_Z(\mathscr{F}) \geqslant n$ is equivalent to $\mathrm{prof}^*_x(\mathscr{F}) \geqslant n$ for every $x \in Z$. If $Z = X$, one writes $\mathrm{prof}^*(\mathscr{F})$ instead of $\mathrm{prof}^*_X(\mathscr{F})$.

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Remarks (10.8.2). — (i) At every closed point $x \in X$, the rectified depth equals the depth.

(ii) Let Y be a closed sub-prescheme of X, $j : Y \to X$ the canonical injection. We know (5.7.3), (vi) that $\mathrm{prof}(\mathscr{G}_x) = \mathrm{prof}(j_*(\mathscr{G})_x)$ for every $x \in Y$; we therefore deduce that $\mathrm{prof}^*_x(\mathscr{G}) = \mathrm{prof}^*_x(j_*(\mathscr{G}))$ for every $x \in Y$.

(iii) The notion of rectified depth is of interest only when it is of a *local* character, i.e. when it does not change when one replaces X by a neighbourhood open U of x. This obviously requires that x not be isolated in $\overline{\{x\}}$ when x is not a closed point of X, i.e. most often, that X be a *Jacobson prescheme* (10.4.5.1); furthermore, it will also be necessary to know that $\dim(U) = \dim(X)$ for every dense open U in X, and one must therefore also suppose that X satisfies conditions 2° and 3° of (10.6.1).

Lemma (10.8.3). — *Let X be a regular and biequidimensional prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Then, for every $x \in X$, we have*

(10.8.3.1) $$\mathrm{prof}^*_x(\mathscr{F}) = \dim(X) - \dim.\mathrm{proj}(\mathscr{F}_x).$$

Proof. Indeed, as X is biequidimensional, we have (**0**, 14.3.5.1)

$$\dim(\overline{\{x\}}) = \dim(X) - \mathrm{codim}(\overline{\{x\}}, X) = \dim(X) - \dim(\mathscr{O}_{X,x})$$

by virtue of (5.1.2). On the other hand, as X is regular, we have by (**0**, 17.3.4)

$$\mathrm{prof}(\mathscr{F}_x) = \dim(\mathscr{O}_{X,x}) - \dim.\mathrm{proj}(\mathscr{F}_x)$$

whence the lemma. □

Corollary (10.8.4). — *Under the hypotheses of (10.8.3), the function $x \mapsto \mathrm{prof}^*_x(\mathscr{F})$ is lower semi-continuous.*

Proof. This follows from (10.8.3.1), since $x \mapsto \dim.\mathrm{proj}(\mathscr{F}_x)$ is upper semi-continuous (6.11.1). □

Proposition (10.8.5). — *Let S be a prescheme locally Noetherian, X a prescheme locally of finite type over S, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Suppose that S satisfies the conditions: 1° S is a Jacobson prescheme; 2° S is regular; 3° the irreducible components of S are equicodimensional. Then the function $x \mapsto \mathrm{prof}^*_x(\mathscr{F})$ is semi-continuous inferiorly; in other terms, for every integer n, the set U_n of $x \in X$ such that $\mathrm{prof}^*_x(\mathscr{F}) \geqslant n$ is open.*

Proof. As the local rings $\mathscr{O}_{S,s}$ of S are regular, they are universally catenary (5.6.4); in other words, S satisfies conditions 1°, 2°, and 3° of (10.6.1), hence the same is true of X (10.6.1), (i). The notion of rectified depth being then of local character (10.8.2), (iii), one can restrict to the case where $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, A being a regular ring and B a quotient of A-algebra of finite type, hence a quotient of a polynomial ring $C = A[T_1, \dots, T_n]$, and this last ring is regular (**0**, 17.3.7). One can therefore suppose that X is a closed sub-prescheme of a regular prescheme Y also satisfying conditions 1°, 2°, and 3° of (10.6.1); taking into account the remark (10.8.2), (ii), one is thus brought to the case where X is in addition regular and Noetherian. But as the local rings of X are then integral, the irreducible components of X are open (**I**, 6.1.10), and one can therefore **IV-3** | 112
also suppose X irreducible. Then, as the local rings of X are catenary (**0**, 16.5.12), the hypothesis 3°

of (10.6.1) implies that X is biequidimensional ((5.1.5) and (**0**, 14.3.3)); it therefore suffices to apply (10.8.4). ☐

One notes that if S is the spectrum of a field or of $\mathbf{Z}$, it satisfies the conditions of (10.8.5).

Corollary (10.8.6). — *The hypotheses being those of (10.8.5), for every $x \in X$, the number $\operatorname{prof}^*_x(\mathscr{F})$ is the unique integer n having the following property: there exists an open neighbourhood U of x such that for every point $x' \in U \cap \overline{\{x\}}$, closed in U, we have $\operatorname{prof}(\mathscr{F}_{x'}) = n$. In particular, for every x such that $\operatorname{prof}^*_x(\mathscr{F}) \geqslant m$ (resp. $\operatorname{prof}^*_x(\mathscr{F}) \leqslant m$), it is necessary and sufficient that there exist an open V of X such that, for every point $x' \in V \cap \overline{\{x\}}$, closed in V, we have $\operatorname{prof}(\mathscr{F}_{x'}) \geqslant m$ (resp. $\operatorname{prof}(\mathscr{F}_{x'}) \leqslant m$).*

Proof. Indeed, one can restrict to the case where x is not a closed point; if $\operatorname{prof}^*_x(\mathscr{F}) = n$, the set of $y \in X$ such that $\operatorname{prof}^*_y(\mathscr{F}) \leqslant n$ is closed by virtue of (10.8.5), hence contains $\overline{\{x\}}$, and by virtue of the lower semi-continuity of $\operatorname{prof}^*_y(\mathscr{F})$, there exists an open neighbourhood U of x such that $\operatorname{prof}^*_y(\mathscr{F}) = n$ for every $y \in U \cap \overline{\{x\}}$, hence $\operatorname{prof}(\mathscr{F}_y) = n$ if $x' \in U \cap \overline{\{x\}}$ is closed (since the notion of rectified depth is local). ☐

For preschemes satisfying the hypotheses of (10.8.5), the notion of rectified depth can therefore be defined by means of the values of the depth at the *closed points* of X (the latter forming a set which is very dense in every closed subset of X).

Proposition (10.8.7). — *Let S be a prescheme locally Noetherian satisfying conditions 1°, 2°, and 3° of (10.6.1). Let X be a prescheme locally of finite type over S, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, $Y = \operatorname{Supp}(\mathscr{F})$. Then, for every $x \in Y$, we have*

(10.8.7.1)
$$\operatorname{prof}^*_x(\mathscr{F}) = \dim_x(Y) - \operatorname{coprof}(\mathscr{F}_x).$$

Proof. Indeed, by definition, $\operatorname{coprof}(\mathscr{F}_x) = \dim(\mathscr{F}_x) - \operatorname{prof}(\mathscr{F}_x)$, and it follows from (10.6.4) that $\dim_x(Y) = \dim(\overline{\{x\}}) + \dim(\mathscr{F}_x)$; whence (10.8.7.1) by definition of $\operatorname{prof}^*_x(\mathscr{F})$. ☐

Corollary (10.8.8). — *Under the hypotheses on f and $\mathscr{F}$ being those of (9.9.1), the function $x \mapsto \operatorname{prof}^*_x(\mathscr{F}_{f(x)})$ is constructible.*

Proof. For every $s \in S$, the fibre X_s, being locally of finite type over $k(s)$, is a Jacobson prescheme (10.4.7). Since $\operatorname{Spec}(k(s))$ satisfies the conditions of (10.6.1), one has

$$\operatorname{prof}^*_x(\mathscr{F}_{f(x)}) = \dim_x(\operatorname{Supp}(\mathscr{F}_{f(x)})) - \operatorname{coprof}((\mathscr{F}_{f(x)})_x)$$

by (10.8.7). Now, if $Z = \operatorname{Supp}(\mathscr{F})$, we have $Z_{f(x)} = \operatorname{Supp}(\mathscr{F}_{f(x)})$ (**I**, 9.1.13) and Z is locally constructible (8.9.1); therefore the functions $x \mapsto \dim_x(\operatorname{Supp}(\mathscr{F}_{f(x)}))$ and $x \mapsto \operatorname{coprof}((\mathscr{F}_{f(x)})_x)$ are locally constructible ((9.9.1) and (9.9.3)), which proves the proposition. ☐

10.9. Maximal spectra and ultra-preschemes The results of this section will not be used in what follows.

(10.9.1). Let X be a Jacobson prescheme, and let $S(X)$ be the *ringed space* whose underlying space is the subspace of closed points of X, and the sheaf of rings the sheaf induced on this subspace by $\mathscr{O}_X$, in other words the sheaf $\psi^*(\mathscr{O}_X)$, where $\psi : S(X) \to X$ denotes the canonical injection. As ψ is a quasi- homeomorphism, we have seen (10.2.8), (ii) that if $\theta : \mathscr{O}_X \to \psi_*(\mathscr{O}_{S(X)})$ is the homomorphism **IV-3** | 113
of sheaves of rings such that $\theta^\sharp : \psi^*(\mathscr{O}_X) \to \mathscr{O}_{S(X)}$ is the identity, then $j_x = (\psi, \theta)$ is a *quasi-isomorphism* of ringed spaces, and $\mathscr{F} \mapsto j^*_X(\mathscr{F})$ is an equivalence of the category of $\mathscr{O}_X$-Modules and that of $\mathscr{O}_{S(X)}$-Modules. It is clear that under this equivalence, locally free (resp. coherent) $\mathscr{O}_X$-modules correspond to locally free (resp. coherent) $\mathscr{O}_{S(X)}$-modules. Furthermore, if $\mathscr{L}$ is a locally free $\mathscr{O}_X$-module and U is an open subset of X such that $\mathscr{L}|_U \simeq (\mathscr{O}_X|_U)^n$, then

$$j^*_X(\mathscr{L})|_{U \cap S(X)} \simeq \left(\mathscr{O}_{S(X)}|_{U \cap S(X)}\right)^n.$$

(10.9.2). Let X, Y be two Jacobson preschemes and $f = (p, \lambda) : X \to Y$ a morphism *locally of finite type*. One has seen (10.4.7) that $p(S(X)) \subset S(Y)$ and by restriction of p to $S(X)$, one defines therefore a continuous map $S(p) : S(X) \to S(Y)$. For every open V of Y, one defines by composition a homomorphism of rings

$$\Gamma(V \cap S(Y), \mathscr{O}_{S(Y)}) \longrightarrow \Gamma(V, \mathscr{O}_Y) \xrightarrow{\lambda_V} \Gamma(f^{-1}(V), \mathscr{O}_X) \longrightarrow \Gamma(f^{-1}(V) \cap S(X), \mathscr{O}_{S(X)})$$

where the two extreme isomorphisms have been defined in (10.9.1); it is clear that this defines a homomorphism of sheaves of rings $S(\lambda) : S(\mathscr{O}_{S(Y)}) \to \rho_*(\mathscr{O}_{S(X)})$ (recalling that the open subsets of $S(X)$ (resp. $S(Y)$) correspond bijectively to those of X (resp. Y) (10.2.1)); one obtains thus a morphism of ringed spaces $S(f) = (S(p), S(\lambda)) : (S(X), \mathscr{O}_{S(X)}) \to (S(Y), \mathscr{O}_{S(Y)})$ such that the diagram

$$\begin{tikzcd} S(X) \arrow[r, "S(f)"] \arrow[d, "j_X"'] & S(Y) \arrow[d, "j_Y"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

is commutative; furthermore, if Z is a third Jacobson prescheme and $g : Y \to Z$ a morphism locally of finite type, $S(g \circ f) = S(g) \circ S(f)$. One has thus defined a *covariant functor* $S : \mathbf{C} \to \mathbf{C}'$, where $\mathbf{C}$ is the category of *ringed spaces in local rings*, and $\mathbf{C}'$ the category whose objects are the *Jacobson preschemes* and the morphisms are the morphisms locally of finite type between Jacobson preschemes.

(10.9.3). Let us now determine the subcategory $\mathbf{C}''$ of $\mathbf{C}'$ formed by the ringed spaces isomorphic to $S(X)$. Suppose first that $X = \operatorname{Spec}(A)$, where A is a Jacobson ring; then $S(X)$ is the set of *maximal* ideals of A, equipped: 1° with the topology induced by that of X, so that a base for this topology is formed by the $D^m(h) = D(h) \cap S(X)$, i.e. the set of maximal ideals $\mathfrak{m}$ of A such that $h \notin \mathfrak{m}$; 2° with the sheaf of rings $\mathscr{O}_{S(X)}$ such that $\Gamma(D^m(h), \mathscr{O}_{S(X)}) = A_h$. We will say that this ringed space is the *maximal spectrum* of the Jacobson ring A and denote it $\operatorname{Spm}(A)$.

One notes that $j : D(h) \to X$ is the canonical injection, the ringed space induced on $D^m(h)$ by $S(X)$ and $S(D(h))$ are canonically isomorphic, and that the canonical injection $D^m(h) \to S(X)$ of ringed spaces is equal to $S(j)$.

Let B be a second Jacobson ring, $Y = \operatorname{Spec}(B)$, $\varphi : B \to A$ a ring homomorphism making A a B-algebra of *finite type*, $f = ({}^a\varphi, \widetilde{\varphi}) : X \to Y$ the corresponding morphism of preschemes, and

$$S(f) : \operatorname{Spm}(A) \longrightarrow \operatorname{Spm}(B)$$

the morphism of ringed spaces corresponding to f. It is clear that $S(f) = (\psi, \theta)$ is a morphism of ringed spaces in local rings, i.e. (Err_{11}, 1.8.2) for every $x \in \operatorname{Spm}(A)$, $\theta_x^\sharp$ is a *local* homomorphism. Conversely:

Proposition (10.9.4). — *Let A, B be two Jacobson rings. If $u = (\psi, \theta) : \operatorname{Spm}(A) \to \operatorname{Spm}(B)$ is a morphism of ringed spaces in local rings such that $\Gamma(\theta) : \mathrm{B} \to A$ makes A a B-algebra of finite type, there exists a morphism of preschemes $f : \operatorname{Spec}(A) \to \operatorname{Spec}(B)$ and a unique one such that $u = S(f)$.*

Proof. The uniqueness of f is evident, since if $f = ({}^a\varphi, \widetilde{\varphi})$, one must have $\varphi = \Gamma(\theta)$; it remains to show that $S(f)$ is defined and that $u = S(f)$. Now, the first assertion follows from the fact that φ is supposed to make A a B-algebra of finite type, the fact that $\theta_x^\sharp$ is a local homomorphism for every $x \in \operatorname{Spm}(A)$ allows one to repeat the argument of (**I**, 1.7.3), restricting to points x of $\operatorname{Spm}(A)$, to show thus successively that $\psi(x) = {}^a\varphi(x)$ for every $x \in \operatorname{Spm}(A)$, then that $\theta_x^\sharp = \varphi_x : B_{(x)} \to A_x$, which completes proving that $u = S(f)$. □

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(10.9.5). Consider now a ringed space $(X, \mathscr{O}_X)$; we will say that an open subset U of X is *ultra-affine* if the ringed space induced $(U, \mathscr{O}_X|U)$ is isomorphic to a maximal spectrum $\operatorname{Spm}(A)$, where A is a Jacobson ring. We will say that X is an *ultra-prescheme* if every point of X admits an ultra-affine open neighbourhood. One shows, as in (**I**, 2.1.3) and (2.1.4), that the ultra-affine open subsets form a *base* for the topology of X and that X is a Kolmogoroff space. If Y is a second ultra-prescheme, we will say that a morphism of ringed spaces $f : X \to Y$ is a *morphism of ultra-preschemes* if it satisfies the following conditions: 1° f is a morphism of ringed spaces in local rings; 2° for every $x \in X$, there is an ultra-affine open neighbourhood V of $f(x)$ in Y and an ultra-affine open U of x in X such that $f(U) \subset V$ and the homomorphism $\Gamma(V, \mathscr{O}_Y) \to \Gamma(U, \mathscr{O}_X)$ corresponding to f makes $\Gamma(U, \mathscr{O}_X)$ a $\Gamma(V, \mathscr{O}_Y)$-*algebra of finite type*.

It is immediate that one defines well the composite of two such morphisms, being a third such by the final remark of (10.9.3). It is clear that the category C_0'' thus defined is a subcategory of C' that contains C''; we now propose to show that $C'' = C_0''$, in other words:

Proposition (10.9.6). — *The functor $X \mapsto S(X)$ of $\mathbf{C}$ into $\mathbf{C}_0''$ is an equivalence of categories.*

Proof. 1° Let us first show that the functor $X \mapsto S(X)$ is *fully faithful*, in other words that for X, Y in $\mathbf{C}$, the canonical map

$$\mathrm{Hom}_{\mathbf{C}}(X, Y) \longrightarrow \mathrm{Hom}_{\mathbf{C}_0''}(S(X), S(Y))$$

is *bijective*. In the first place it is *injective*: let f, g be two morphisms locally of finite type of X into Y, and suppose that $S(f) = S(g)$; we have $f^{-1}(V) \cap S(X) = g^{-1}(V) \cap S(X)$ for every open V of Y, hence (10.3.1) and (10.2.7) $f^{-1}(V) = g^{-1}(V)$; it suffices therefore to prove that for every affine open V of Y, f and g coincide in $f^{-1}(V) = g^{-1}(V)$; in other words, one reduces to the case where $Y = \mathrm{Spec}(B)$ is the spectrum of a Jacobson ring B. It suffices (**I**, 2.2.4) to show that the homomorphisms of rings $B \to \Gamma(X, \mathscr{O}_X)$ corresponding to f and g are the same. Now, for every $s \in B$, the images of s under these homomorphisms are two sections of $\mathscr{O}_X$ over X which, by hypothesis, induce the *same* section over $S(X)$; we know (10.2.8) that these sections are identical, whence our assertion. In the second place let us prove that every morphism $h : S(X) \to S(Y)$ (for the category $\mathbf{C}_0''$) is of the form $S(f)$, where $f : X \to Y$ is a morphism locally of finite type. By hypothesis, there exists an ultra-affine open covering (U') (resp. (V_μ')) of $S(X)$ (resp. $S(Y)$) such that for every λ, $h(U_\lambda')$ is contained in one of the V_μ' and that it corresponds to the morphism $h_\lambda : U_\lambda' \to V_\mu'$ making the second ring an algebra of finite type over the first. For every couple of indices α, β, consider the open subset $U_{\alpha\beta}$ of U_α whose trace on $S(X)$ is $U_\alpha' \cap U_\beta'$; by virtue of 1°, the identical self-morphism of $U_\alpha' \cap U_\beta'$ is of the form $S(\theta_{\alpha\beta})$, where $\theta_{\alpha\beta} : U_{\alpha\beta} \to U_{\beta\alpha}$ is an isomorphism of preschemes. We see immediately (by virtue of 1°) that the family $(\theta_{\alpha\beta})$ satisfies the glueing condition of recollement (**0**, 4.1.7), and that this family defines a prescheme X such that we have $X' = S(X)$; it is clear then that the U_α are identified with affine open subsets of X, such that the U_α' are the restrictions of $S(X)$ to the U_α.

2° It remains to prove that every ultra-prescheme X' is of the form $S(X)$ for a Jacobson prescheme X (which will necessarily be unique to isomorphism near, by virtue of 1°). There are ultra-affine open subsets (U_α') of X', each of which is of the form $S(U_\alpha)$, U_α being the spectrum of a Jacobson ring. For every couple of indices α, β, consider the open $U_{\alpha\beta}$ of U_α whose trace U_α' is $U_\alpha' \cap U_\beta'$; by virtue of 1°, the identical self-morphism of $U_\alpha' \cap U_\beta'$ is of the form $S(\theta_{\alpha\beta})$, where $\theta_{\alpha\beta} : U_{\alpha\beta} \to U_{\beta\alpha}$ is an isomorphism of preschemes. One easily verifies (by virtue of 1°) that the family $(\theta_{\alpha\beta})$ satisfies the glueing condition (**0**, 4.1.7), and that this family defines a prescheme X; it is clear then that we have $X' = S(X)$, which completes the proof. □

10.10. Algebraic spaces of Serre

(10.10.1). The language introduced by Serre in (FAC) is sometimes convenient, notably in questions where the main interest is attached to the rational points over the base field k (algebraically closed by hypothesis) of the "algebraic varieties" over k that one considers. We will sketch here this language by linking it to the considerations that precede, to allow the reader to translate the statements of Serre in the language of schemes. It is moreover possible to develop the language of Serre also for preschemes over a non-algebraically closed field (and even over an Artinian ring); but this introduces considerable technical complications, and moreover, on a field of arbitrary base, the advantages (above all psychological ones) of the point of view of Serre disappear; therefore we will confine ourselves to the framework fixed by Serre. The present section, just like the preceding one, will not be used in the sequel of this Treatise, and we will therefore content ourselves with brief indications.

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(10.10.2). Given an ultra-prescheme R, one can naturally (as in the whole category) define the notion of R-*ultra-prescheme*. Consider in particular an algebraically closed field k; $\mathrm{Spm}(k)$ is then identical to $\mathrm{Spec}(k)$; we will say that a $\mathrm{Spm}(k)$-ultra-prescheme X' is a *prealgebraic space over k*: it is a ringed space in local rings whose every point has an open neighbourhood isomorphic to the maximal spectrum of a k-algebra of finite type; it follows that it is the same as saying (by (10.9.6)) that $X' = S(X)$, where X is a k-prescheme locally of finite type. If X, Y, Z are three k-preschemes locally of finite type, it is the same as to say of $X \times_k Y$ (**I**, 1.3.4), hence the same is true of $X' \times_k Y'$, in the category of prealgebraic spaces over k (and also of the "fibre product" $X' \times_{k'} Y'$, where X', Y', Z' are three prealgebraic spaces over k). One can therefore define the diagonal morphism $\Delta_{X'} : X' \to X' \times_k X'$ (which is none other than $S(\Delta_X)$); one says that X' is an *algebraic space over k* if X is a scheme, and it amounts to the same as saying that the image of $\Delta_{X'}$ is a *closed* subset of $X' \times_k X'$.

(10.10.3). The simplifications that come from the hypothesis that k is algebraically closed amount first of all to this: for every k-prescheme X locally of finite type, there is a bijective correspondence between *closed points* of X_k (i.e. the *points of X with values in k* (**I**, 3.4.4)) and *rational points of X over k* (**I**, 3.4.5), by virtue of (**I**, 6.4.2). This shows in particular (**I**, 3.4.3.1) that for two k-prealgebraic spaces X', Y', the underlying set product $X' \times Y'$ *is* identical to the underlying set of the product of the underlying topological spaces of X' and Y' (but it is well understood that the topology of the latter is not the *product* topology of the topologies of the underlying spaces, in general it is strictly finer than the latter).

On the other hand, the local rings $\mathscr{O}_x$ at the points of a prealgebraic space X' over k are k-algebras whose residue field is *isomorphic to k*; if A and B are two such k-algebras, a k-homomorphism $\varphi : A \to B$ is necessarily *local*: indeed, if an element x of the maximal ideal A were such that $\varphi(x)$ is invertible, there would exist $\lambda \in k$ nonzero such that $\varphi(x - \lambda.1)$ belongs to the maximal ideal of B, which is absurd since $x - \lambda.1$ is invertible. We conclude immediately from this that if $X' = S(X)$, $Y' = S(Y)$ are two prealgebraic spaces over k, every morphism $X' \to Y'$ of *k-ringed spaces* is also a morphism of ringed spaces *in local rings*, hence every morphism of k-preschemes $X' \to Y'$ of k-ringed spaces is also, in the notations above, of the form $S(f)$, where $f : X \to Y$ is a morphism of k-preschemes.

Finally, for every open U of X', and every $x \in U$, $s(x)$ (**0**, 5.5.1) is identified with an element of U in k, in other words a map $\overline{s} : x \to s(x)$ of U in k; as the map $h_U : s \to \overline{s}$ is evidently a homomorphism of rings $\Gamma(U, \mathscr{O}_{X'}) \to \Gamma(U, \mathscr{A}(X'))$ (the *germs of maps* of X' *in* k) and commutes with restrictions to a neighbourhood $V \subset U$, the h_U define a homomorphism of *sheaves* of rings $h : \mathscr{O}_{X'} \to \mathscr{A}(X')$. If one takes U to be ultra-affine, $\mathrm{Spm}(A)$, where A is a Jacobson ring, to say that $\overline{s} = 0$ means that for every maximal ideal $\mathfrak{m}$ of A, s belongs to $\mathfrak{m}$, or equivalently that s is in the *radical* of A; but as A is a Jacobson ring, its radical is its nilradical, hence $\overline{s} = 0$ if and only if s is nilpotent.

(10.10.4). One says that the k-prealgebraic space $X' = S(X)$ is *reduced* if it is reduced at each of its points, i.e. the set of points $x \in X$ where X is reduced is open (**0**, 5.2.2), its complement contains at least one closed point x' of X; as each of the local rings $\mathscr{O}_{x'}$ is reduced, one sees in (10.10.3) that for the homomorphism $h : \mathscr{O}_{X'} \to \mathscr{A}(X')$ to be *injective*, it is necessary and sufficient that X' be *reduced*. In (FAC), Serre restricts himself in fact to prealgebraic *reduced* spaces, which allows him to define $\mathscr{O}_{X'}$ as a *sub-sheaf* of $\mathscr{A}(X')$. One notes that if X' and Y' are two k-spaces prealgebraic reduced, it is the same of $X' \times_k Y'$: indeed, this amounts to saying that if A and B are two reduced k-algebras of finite type, $A \otimes_k B$ is also reduced; but we have just seen that the radicals of A and B are 0, and as k is algebraically closed, A and B are "separable" algebras over k in the sense of Bourbaki (*Alg.*, chap. VIII, §7, n° 5, prop. 5); hence $A \otimes_k B$ is without radical (*loc. cit.*, n° 6, cor. 3 of th. 3), and since it is a Jacobson ring, it is reduced. However, if Z' is a third prealgebraic space reduced over k, the "fibre product" $X' \times_{Z'} Y'$ of two prealgebraic reduced spaces over Z' is not in general reduced, which implies that the category of these spaces is insufficient in many questions (notably in the theory of algebraic groups). But as we have seen above, one can keep the language of Serre without restricting oneself, as this latter (who in addition considers only quasi-compact *prealgebraic* spaces), to the case of reduced prealgebraic spaces.

(10.10.5). Finally, one can also consider ultra-preschemes over an arbitrary field k while retaining a language close to Serre's, and introducing, as in Weil, a fixed algebraically closed extension K of k (chosen sufficiently large—for example, of infinite transcendence degree over k—so as to have enough "generic points" in Weil's sense). To every prescheme X locally of finite type over k one then associates the set

$$S_K(X) = S(X \otimes_k K)$$

of points of X with values in K. There is a canonical map $j : S_K(X) \to X$, which is a quasi-homeomorphism when $S_K(X)$ is given the inverse-image topology from X under j; endowing $S_K(X)$ with the sheaf of k-algebras $j^*(\mathscr{O}_X)$ gives a subcategory of the category of k-ringed spaces in local rings, which one might call the category of (k, K)-*prealgebraic spaces*. Products can still be defined there, and the results of (10.10.3) and (10.10.4) can be generalized (with $\mathscr{A}(X')$ replaced here by the sheaf $\mathscr{A}_K(X')$ of germs of maps from X' into K). This point of view, however, assigns an artificial role to an arbitrarily chosen overfield of k, and we mention it only to reject it.

§11. Topological properties of finitely presented flat morphisms. Flatness criteria

11.1. Flatness loci (Noetherian case) IV-3 | 117

Theorem (11.1.1). — *Let* Y *be a locally Noetherian prescheme,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* $\mathscr{F}$ *an* $\mathscr{O}_\mathrm{X}$*-Module coherent. Then the set* U *of* $x \in \mathrm{X}$ *such that* $\mathscr{F}$ *is* f*-flat at the point* x *is open in* X.

Proof. The question being evidently local on X and Y, we can suppose $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, A being Noetherian and B an A-algebra of finite type. We then have $\mathscr{F} = \widetilde{\mathrm{M}}$, where M is a B-module of finite type. Applying the criterion ($\mathbf{0}_{\mathrm{III}}$, 9.2.6): it suffices therefore to prove the following assertion:

(11.1.1.1) *Let* A *be a Noetherian ring,* B *an* A*-algebra of finite type,* M *a* B*-module of finite type,* $\mathfrak{q}$ *a prime ideal of* B, $\mathfrak{p}$ *its inverse image in* A. *Suppose that* $\mathrm{M}_\mathfrak{q}$ *is a flat* $\mathrm{A}_\mathfrak{p}$*-module. Then there exists* $g \in \mathrm{B} - \mathfrak{q}$ *such that, for every prime ideal* $\mathfrak{q}' \supset \mathfrak{q}$ *of* B *such that* $g \notin \mathfrak{q}'$, $\mathrm{M}_{\mathfrak{q}'}$ *is a flat* $\mathrm{A}_{\mathfrak{p}'}$*-module, where* $\mathfrak{p}'$ *denotes the inverse image of* $\mathfrak{q}'$ *in* A *(it amounts to the same* ($\mathbf{0}_\mathrm{I}$, 6.3.1) *to say that* $\mathrm{M}_{\mathfrak{q}'}$ *is a flat* A*-module).*

To this end, regard $\mathrm{B}_{\mathfrak{q}'}$ as an A-algebra; clearly $\mathfrak{p}\mathrm{B}_{\mathfrak{q}'} \subset \mathfrak{q}'\mathrm{B}_{\mathfrak{q}'}$. By ($\mathbf{0}_{\mathrm{III}}$, 10.2.2), the A-module $\mathrm{M}_{\mathfrak{q}'}$ is flat if and only if $\mathrm{M}_{\mathfrak{q}'}/\mathfrak{p}\mathrm{M}_{\mathfrak{q}'}$ is a flat $(\mathrm{A}/\mathfrak{p})$-module and

$$\mathrm{Tor}_1^\mathrm{A}(\mathrm{M}_{\mathfrak{q}'}, \mathrm{A}/\mathfrak{p}) = 0.$$

Now $\mathrm{M}_{\mathfrak{q}'} = \mathrm{M} \otimes_\mathrm{B} \mathrm{B}_{\mathfrak{q}'}$, and $\mathrm{B}_{\mathfrak{q}'}$ is flat over B; hence

$$\mathrm{M}_{\mathfrak{q}'}/\mathfrak{p}\mathrm{M}_{\mathfrak{q}'} = (\mathrm{M}/\mathfrak{p}\mathrm{M})_{\mathfrak{q}'}, \qquad \mathrm{Tor}_1^\mathrm{A}(\mathrm{M}_{\mathfrak{q}'}, \mathrm{A}/\mathfrak{p}) = \big(\mathrm{Tor}_1^\mathrm{A}(\mathrm{M}, \mathrm{A}/\mathfrak{p})\big)_{\mathfrak{q}'}.$$

Here the second equality may be obtained by defining Tor from a projective resolution of $\mathrm{A}/\mathfrak{p}$. For the same reason, since we require $g \notin \mathfrak{q}'$, the ring $\mathrm{B}_{\mathfrak{q}'}$ is flat over B_g, and therefore

$$(\mathrm{M}/\mathfrak{p}\mathrm{M})_{\mathfrak{q}'} = \big((\mathrm{M}/\mathfrak{p}\mathrm{M})_g\big)_{\mathfrak{q}'\mathrm{B}_g}, \qquad \big(\mathrm{Tor}_1^\mathrm{A}(\mathrm{M}, \mathrm{A}/\mathfrak{p})\big)_{\mathfrak{q}'} = \big(\big(\mathrm{Tor}_1^\mathrm{A}(\mathrm{M}, \mathrm{A}/\mathfrak{p})\big)_g\big)_{\mathfrak{q}'\mathrm{B}_g},$$

where $\mathrm{M}/\mathfrak{p}\mathrm{M}$ and $\mathrm{Tor}_1^\mathrm{A}(\mathrm{M}, \mathrm{A}/\mathfrak{p})$ are regarded as B-modules. In view of ($\mathbf{0}_\mathrm{I}$, 6.3.2), we are thus reduced to proving the following lemma.

Lemma (11.1.1.2). — *Under the conditions of* (11.1.1.1), *there exists* $g \in \mathrm{B} - \mathfrak{q}$ *such that:*

i) $(\mathrm{M}/\mathfrak{p}\mathrm{M})_g$ *is a flat* $(\mathrm{A}/\mathfrak{p})$*-module;*
ii) $(\mathrm{Tor}_1^\mathrm{A}(\mathrm{M}, \mathrm{A}/\mathfrak{p}))_g = 0$.

By the generic flatness theorem (6.9.1), applied to the integral ring $\mathrm{A}/\mathfrak{p}$, the finite-type $(\mathrm{A}/\mathfrak{p})$-algebra $\mathrm{B}/\mathfrak{p}\mathrm{B}$, and the finite-type $(\mathrm{B}/\mathfrak{p}\mathrm{B})$-module $\mathrm{M}/\mathfrak{p}\mathrm{M}$, there exists $h \in \mathrm{A} - \mathfrak{p}$ such that, if $\bar{h}$ denotes its canonical image in $\mathrm{A}/\mathfrak{p}$, $(\mathrm{M}/\mathfrak{p}\mathrm{M})_{\bar{h}}$ is a flat $(\mathrm{A}/\mathfrak{p})$-module. On the other hand, $\mathrm{M}_\mathfrak{q}$ is flat over $\mathrm{A}_\mathfrak{p}$ and hence, by ($\mathbf{0}_\mathrm{I}$, 6.3.1), flat over A. Consequently, IV-3 | 118

$$\mathrm{Tor}_1^\mathrm{A}(\mathrm{M}_\mathfrak{q}, \mathrm{A}/\mathfrak{p}) = 0, \qquad \text{or equivalently} \qquad \big(\mathrm{Tor}_1^\mathrm{A}(\mathrm{M}, \mathrm{A}/\mathfrak{p})\big)_\mathfrak{q} = 0.$$

Since A and B are Noetherian, $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/\mathfrak{p})$ is a finite-type B-module. Thus ($\mathbf{0}_{\mathrm{I}}$, 5.2.2) gives an element $g \in \mathrm{B} - \mathfrak{q}$ such that

$$\left(\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/\mathfrak{p})\right)_g = 0.$$

Moreover, $(\mathrm{M}/\mathfrak{p}\mathrm{M})_{\bar{h}} = (\mathrm{M}/\mathfrak{p}\mathrm{M})_h$, where on the right the module is regarded as an A-module. Since $(\mathrm{M}/\mathfrak{p}\mathrm{M})_h$ is also a B-module, $(\mathrm{M}/\mathfrak{p}\mathrm{M})_{hg}$ remains flat over $\mathrm{A}/\mathfrak{p}$: it is $(\mathrm{M}/\mathfrak{p}\mathrm{M})_{\overline{hg}}$, where $\overline{hg}$ is the canonical image of hg in $\mathrm{B}/\mathfrak{p}\mathrm{B}$, and one applies ($\mathbf{0}_{\mathrm{I}}$, 6.3.2). Finally,

$$\left(\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/\mathfrak{p})\right)_{hg} = 0,$$

and $hg \in \mathrm{B} - \mathfrak{q}$ because $h \in \mathrm{A} - \mathfrak{p}$. This proves the lemma and the theorem. □

Corollary (11.1.2). — *Let* Y *be a locally Noetherian prescheme, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, $\mathscr{F}$, $\mathscr{F}'$ two $\mathscr{O}_{\mathrm{X}}$-Modules coherent, $u : \mathscr{F}' \to \mathscr{F}$ a homomorphism of $\mathscr{O}_{\mathrm{X}}$-Modules. Suppose that $\mathscr{F}$ is f-flat. Then the set* U *of $x \in \mathrm{X}$ such that, setting $y = f(x)$, the homomorphism $u_x \otimes 1 : \mathscr{F}'_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y) \to \mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ is injective, is open in* X.

Proof. Let $\mathscr{N}$ (resp. $\mathscr{P}$) be the kernel (resp. cokernel) of u. Apply ($\mathbf{0}_{\mathrm{III}}$, 10.2.4)[14] to the local rings $\mathscr{O}_y$ and $\mathscr{O}_x$, and to the $\mathscr{O}_x$-modules $\mathscr{F}'_x$ and $\mathscr{F}_x$. The injectivity of $u_x \otimes 1$ is equivalent to the two conditions $\mathscr{N}_x = 0$ and that $\mathscr{P}$ be f-flat at x. Since $\mathscr{N}$ and $\mathscr{P}$ are coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.4), the locus where $\mathscr{N}_x = 0$ is open ($\mathbf{0}_{\mathrm{I}}$, 5.2.2), and the locus where $\mathscr{P}$ is f-flat is open by (11.1.1). The result follows.

In particular: □

Corollary (11.1.3). — *Let* Y *be a locally Noetherian prescheme, $f : \mathrm{X} \to \mathrm{Y}$ a flat morphism locally of finite type, and g a section of $\mathscr{O}_{\mathrm{X}}$ over* X. *The set of $x \in \mathrm{X}$ such that $g_x \otimes 1$ is a non-zero-divisor in $\mathscr{O}_x \otimes_{\mathscr{O}_{f(x)}} \boldsymbol{k}(f(x))$ is open in* X.

Proof. It suffices to apply (11.1.2) to the endomorphism of $\mathscr{O}_{\mathrm{X}}$ defined by g. □

Corollary (11.1.4). — *Let* Y *be a locally Noetherian prescheme, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, $\mathscr{F}$ an $\mathscr{O}_{\mathrm{X}}$-Module coherent and f-flat. Let $(g_i)_{1 \leqslant i \leqslant n}$ be a sequence of sections of $\mathscr{O}_{\mathrm{X}}$ above* X. *The set* U *of $x \in \mathrm{X}$ such that the sequence $((g_i)_x \otimes 1)$ is $(\mathscr{F}_x \otimes_{\mathscr{O}_{f(x)}} \boldsymbol{k}(f(x)))$-regular is open in* X.

Proof. Since $\mathscr{F}$ is f-flat, it follows from ($\mathbf{0}$, 15.1.16) that U is also the set of points $x \in \mathrm{X}$ for which **IV-3** | 119
the sequence $((g_i)_x)$ is $\mathscr{F}_x$-regular and the $\mathscr{O}_x$-module

$$\mathscr{G}_x, \qquad \mathscr{G} = \mathscr{F} / \Big(\sum_{i=1}^{n} g_i \mathscr{F}\Big),$$

is f-flat. The modules $\mathscr{F}$ and $\mathscr{G}$ are coherent, so the corollary follows from (11.1.1) and ($\mathbf{0}$, 15.2.4). □

Corollary (11.1.5). — *Let* X, Y, Z *be three locally Noetherian preschemes, $f : \mathrm{X} \to \mathrm{Y}$, $g : \mathrm{Y} \to \mathrm{Z}$ two morphisms of finite type, $\mathscr{F}$ an $\mathscr{O}_{\mathrm{X}}$-Module coherent. Then the set* U *of $y \in \mathrm{Y}$ such that, for every generization y' of y, $\mathscr{F}$ is $(g \circ f)$-flat at all the points of $f^{-1}(y')$ (i.e. such that $\mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$ is flat relative to the morphism $\mathrm{X} \times_{\mathrm{Y}} \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}) \to \mathrm{Z}$) is open in* Y.

[14] Here is a proof of ($\mathbf{0}_{\mathrm{III}}$, 10.2.4) which was not published in N. Bourbaki's *Algèbre commutative*. In view of ($\mathbf{0}_{\mathrm{I}}$, 6.1.2), it is enough to show that b) implies a). Set $\mathrm{P} = \mathrm{Im}(u)$, $\mathrm{Q} = \mathrm{Coker}(u)$, and $\mathrm{R} = \mathrm{Ker}(u)$. The composite $\mathrm{M} \otimes k \to \mathrm{P} \otimes k \to \mathrm{N} \otimes k$ is injective, while $\mathrm{M} \otimes k \to \mathrm{P} \otimes k$ is surjective. Hence $\mathrm{P} \otimes k \to \mathrm{N} \otimes k$ is injective and $\mathrm{M} \otimes k \to \mathrm{P} \otimes k$ is bijective. The exact sequence $0 \to \mathrm{P} \to \mathrm{N} \to \mathrm{Q} \to 0$ gives the exact sequence

$$0 = \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{N}, k) \longrightarrow \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{Q}, k) \longrightarrow \mathrm{P} \otimes k \longrightarrow \mathrm{N} \otimes k \longrightarrow \mathrm{Q} \otimes k \longrightarrow 0$$

and, since $\mathrm{P} \otimes k \to \mathrm{N} \otimes k$ is injective, $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{Q}, k) = 0$. As Q is a finite-type B-module, ($\mathbf{0}_{\mathrm{III}}$, 10.2.2) shows that it is flat over A; then P is also flat over A by ($\mathbf{0}_{\mathrm{I}}$, 6.1.2). From the exact sequence $0 \to \mathrm{R} \to \mathrm{M} \to \mathrm{P} \to 0$, the same result gives an exact sequence $0 \to \mathrm{R} \otimes k \to \mathrm{M} \otimes k \to \mathrm{P} \otimes k \to 0$. Since the last map is bijective, $\mathrm{R} \otimes k = 0$. But B is Noetherian, so R is a finite-type B-module; Nakayama's lemma therefore gives $\mathrm{R} = 0$.

Proof. If V is the set of points $x \in X$ where $\mathscr{F}$ is $(g \circ f)$-flat, then U is the set of $y \in Y$ such that $f^{-1}(y') \subset V$ for every generization y' of y. Now V is open by (11.1.1), hence locally constructible in X, and the set W of $y \in Y$ such that $f^{-1}(y) \subset V$ is

$$W = Y - f(X - V).$$

It is therefore locally constructible in Y by Chevalley's theorem (1.8.4). It follows from ($\mathbf{0}_{\mathrm{III}}$, 9.2.5) that the points of U are the interior points of W, and hence that U is open. ☐

Corollary (11.1.6). — *Let* A *be a Noetherian ring,* B *an* A*-algebra of finite type,* C *a* B*-algebra of finite type,* M *a* C*-module of finite type. Then the set of* $\mathfrak{q} \in \mathrm{Spec}(B)$ *such that* $M_{\mathfrak{q}}$ *is a flat* A*-module is open in* Spec(B).

Proof. Taking into account (2.1.2), this is a consequence of (11.1.5) applied to $Z = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $X = \mathrm{Spec}(C)$, $\mathscr{F} = \widetilde{M}$.

The results of this number will be freed from the Noetherian hypotheses in (11.3). ☐

11.2. Flatness of a projective limit of preschemes

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(11.2.1). Let A be a ring, M, N two A-modules, A′ an A-algebra; set $M' = M \otimes_A A'$, $N' = N \otimes_A A'$. Recall (**III**, 6.3.8) that one defines, for every i, a canonical homomorphism of A-modules

$$\psi_i : \mathrm{Tor}_i^A(M, N) \longrightarrow \mathrm{Tor}_i^{A'}(M', N') \tag{11.2.1.1}$$

in the following way: one considers a left resolution of M by free A-modules

$$\ldots \longrightarrow L_{i+1} \xrightarrow{f_i} L_i \longrightarrow \ldots \longrightarrow L_0 \xrightarrow{\varepsilon} M \longrightarrow 0 \tag{11.2.1.2}$$

whence one deduces by tensoring with A′ a complex of A′-modules

$$\ldots \longrightarrow L'_{i+1} \xrightarrow{f'_i} L'_i \longrightarrow \ldots \longrightarrow L'_0 \xrightarrow{\varepsilon'} M' \longrightarrow 0 \tag{11.2.1.3}$$

where we have set $L'_i = L_i \otimes_A A'$, $f'_i = f_i \otimes 1_{A'}$, $\varepsilon' = \varepsilon \otimes 1_{A'}$. Consider on the other hand a left resolution of M′ by free A′-modules

$$\ldots \longrightarrow L''_{i+1} \xrightarrow{f''_i} L''_i \longrightarrow \ldots \longrightarrow L''_0 \xrightarrow{\varepsilon''} M' \longrightarrow 0 \tag{11.2.1.4}$$

As the L'_i are free A′-modules, we know (M, V, 1.1) that there are A′-homomorphisms u_i forming a commutative diagram IV-3 | 120

$$\begin{tikzcd}
\ldots \arrow[r] & L'_{i+1} \arrow[r, "f'_i"] \arrow[d, "u_{i+1}"] & L'_i \arrow[r] \arrow[d, "u_i"] & \ldots \arrow[r] & L'_0 \arrow[r, "\varepsilon'"] \arrow[d, "u_0"] & M' \arrow[r] \arrow[d, "1_{M'}"] & 0 \\
\ldots \arrow[r] & L''_{i+1} \arrow[r, "f''_i"] & L''_i \arrow[r] & \ldots \arrow[r] & L''_0 \arrow[r, "\varepsilon''"] & M' \arrow[r] & 0
\end{tikzcd}$$

Composing the morphism of complexes $u_\bullet : L'_\bullet \to L''_\bullet$ with the canonical morphism $L_\bullet \to L'_\bullet$ gives a morphism of complexes of A-modules $L_\bullet \to L''_\bullet$. Since

$$(L_i \otimes_A N) \otimes_A A' = L'_i \otimes_{A'} N',$$

we obtain a morphism of complexes

$$L_\bullet \otimes_A N \longrightarrow L''_\bullet \otimes_{A'} N'.$$

Passing to homology yields the canonical homomorphisms (11.2.1, IV.11.2.1.1). Since the u_i are determined up to homotopy (M, V, 1.1), these homomorphisms depend neither on the choice of the u_i nor on the choice of the free resolutions $L_\bullet$ and $L''_\bullet$.

As the $\mathrm{Tor}_i^{A'}(M', N')$ are A′-modules, one deduces canonically from (11.2.1, IV.11.2.1.1) A′-homomorphisms

$$\varphi_i : \mathrm{Tor}_i^A(M, N) \otimes_A A' \longrightarrow \mathrm{Tor}_i^{A'}(M', N'). \tag{11.2.1.5}$$

Consider now two ring homomorphisms

$$\rho : A \longrightarrow A^{(1)}, \quad \sigma : A^{(1)} \longrightarrow A^{(2)},$$

and their composite $\sigma \circ \rho : \mathrm{A} \to \mathrm{A}^{(2)}$; set $\mathrm{M}^{(j)} = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}^{(j)}$, $\mathrm{N}^{(j)} = \mathrm{N} \otimes_{\mathrm{A}} \mathrm{A}^{(j)}$ for $j = 1, 2$. Then the canonical composite homomorphism

$$\mathrm{Tor}_i^{\mathrm{A}}(\mathrm{M},\mathrm{N}) \longrightarrow \mathrm{Tor}_i^{\mathrm{A}^{(1)}}(\mathrm{M}^{(1)},\mathrm{N}^{(1)}) \longrightarrow \mathrm{Tor}_i^{\mathrm{A}^{(2)}}(\mathrm{M}^{(2)},\mathrm{N}^{(2)})$$

is the same as the canonical homomorphism deduced from $\sigma \circ \rho$; this follows from the fact that, if $\mathrm{L}_\bullet^{(j)}$ is a free resolution of $\mathrm{M}^{(j)}$, the diagram

$$\begin{tikzcd}
\mathrm{L}_\bullet \arrow[r] & \mathrm{L}_\bullet \otimes_{\mathrm{A}} \mathrm{A}^{(1)} \arrow[r] \arrow[d] & \mathrm{L}_\bullet^{(1)} \arrow[d] \\
 & \mathrm{L}_\bullet^{(1)} \otimes_{\mathrm{A}^{(1)}} \mathrm{A}^{(2)} \arrow[r] & \mathrm{L}_\bullet^{(2)}
\end{tikzcd}$$

is commutative.

(11.2.2). With the notation of (11.2.1), let $(\mathrm{A}_\alpha, g_{\beta\alpha})$ be a filtered inductive system of A-algebras and set

$$\mathrm{M}_\alpha = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}_\alpha, \qquad \mathrm{N}_\alpha = \mathrm{N} \otimes_{\mathrm{A}} \mathrm{A}_\alpha.$$

For each i, the modules $\mathrm{Tor}_i^{\mathrm{A}_\alpha}(\mathrm{M}_\alpha, \mathrm{N}_\alpha)$, with the canonical maps furnished by (11.2.1), form an inductive system. Put

$$\mathrm{A}' = \varinjlim_\alpha \mathrm{A}_\alpha, \qquad \mathrm{M}' = \varinjlim_\alpha \mathrm{M}_\alpha = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}', \qquad \mathrm{N}' = \varinjlim_\alpha \mathrm{N}_\alpha = \mathrm{N} \otimes_{\mathrm{A}} \mathrm{A}'.$$

If $g_\alpha : \mathrm{A}_\alpha \to \mathrm{A}'$ denotes the canonical homomorphism, we obtain canonical maps

$$\psi_{i\alpha} : \mathrm{Tor}_i^{\mathrm{A}_\alpha}(\mathrm{M}_\alpha, \mathrm{N}_\alpha) \longrightarrow \mathrm{Tor}_i^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}')$$

which again form an inductive system. We complete the result of (M, V, 9.4*) by showing that

$$\psi_i = \varinjlim \psi_{i\alpha} : \varinjlim_\alpha \mathrm{Tor}_i^{\mathrm{A}_\alpha}(\mathrm{M}_\alpha, \mathrm{N}_\alpha) \longrightarrow \mathrm{Tor}_i^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}') \tag{11.2.2.1}$$

are *isomorphisms of* A'*-modules*. To this end, we proceed as in (M, V, 9.5′), associating to each M_α its *canonical free resolution*. It amounts (taking into account the exactness of the functor $\varinjlim$) to proving the

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Lemma (11.2.2.2). — *Let* $(\mathrm{A}_\alpha, g_{\beta\alpha})$ *be a filtered inductive system of rings and* $(\mathrm{M}_\alpha, h_{\beta\alpha})$ *a filtered inductive system of sets. Put* $\mathrm{A}' = \varinjlim \mathrm{A}_\alpha$ *and* $\mathrm{M}' = \varinjlim \mathrm{M}_\alpha$, *and let* $g_\alpha : \mathrm{A}_\alpha \to \mathrm{A}'$ *and* $h_\alpha : \mathrm{M}_\alpha \to \mathrm{M}'$ *be the canonical maps. For each* α, *let* $\mathrm{F}(\mathrm{M}_\alpha)$ *be the* A_α*-module of formal linear combinations of elements of* M_α*; similarly, let* $\mathrm{F}(\mathrm{M}')$ *be the* A'*-module of formal linear combinations of elements of* M'. *If* $h^0_{\beta\alpha} : \mathrm{F}(\mathrm{M}_\alpha) \to \mathrm{F}(\mathrm{M}_\beta)$ *(for* $\alpha \leqslant \beta$*) and* $h'_\alpha : \mathrm{F}(\mathrm{M}_\alpha) \to \mathrm{F}(\mathrm{M}')$ *are the* A_α*-homomorphisms induced by* $h_{\beta\alpha}$ *and* h_α, *respectively, then* $(\mathrm{F}(\mathrm{M}_\alpha), h^0_{\beta\alpha})$ *is an inductive system of* A_α*-modules and* (h'_α) *is an inductive system of homomorphisms. Moreover,*

$$h'' = \varinjlim h'_\alpha : \varinjlim \mathrm{F}(\mathrm{M}_\alpha) \longrightarrow \mathrm{F}(\mathrm{M}') = \mathrm{F}(\varinjlim \mathrm{M}_\alpha)$$

is an isomorphism.

Proof. For the proof, see Bourbaki, *Alg.*, chap. II, 3e éd., § 6, no 6, cor. de la prop. 10. □

(11.2.3). Resume the notation of (11.2.1) and consider the case $i = 1$. Set $\mathrm{T} = \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M},\mathrm{N})$, $\mathrm{T}' = \mathrm{T} \otimes_{\mathrm{A}} \mathrm{A}'$, and $\mathrm{T}'' = \mathrm{Tor}_1^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}')$. Then ψ_1 is the homomorphism

$$\frac{\mathrm{Ker}(f_0 \otimes 1_{\mathrm{N}})}{\mathrm{Im}(f_1 \otimes 1_{\mathrm{N}})} \longrightarrow \frac{\mathrm{Ker}(f''_0 \otimes 1_{\mathrm{N}'})}{\mathrm{Im}(f''_1 \otimes 1_{\mathrm{N}'})}$$

induced on the quotients by the restriction

$$\mathrm{Ker}(f_0 \otimes 1_{\mathrm{N}}) \longrightarrow \mathrm{Ker}(f''_0 \otimes 1_{\mathrm{N}'})$$

of $u_1 \otimes 1 : \mathrm{L}_1 \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{L}''_1 \otimes_{\mathrm{A}'} \mathrm{N}'$.

Set $\mathrm{R} = \mathrm{Ker}(\varepsilon)$, $\mathrm{R}'' = \mathrm{Ker}(\varepsilon'')$, so that one has the exact sequences

$$0 \longrightarrow \mathrm{R} \xrightarrow{j} \mathrm{L}_0 \xrightarrow{\varepsilon} \mathrm{M} \longrightarrow 0 \quad \text{and} \quad 0 \longrightarrow \mathrm{R}'' \xrightarrow{j''} \mathrm{L}''_0 \xrightarrow{\varepsilon''} \mathrm{M}' \longrightarrow 0, \tag{1}$$

whence one deduces the exact homology sequences

$$0 = \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{L}_0, \mathrm{N}) \longrightarrow \mathrm{T} \xrightarrow{\partial} \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N} \xrightarrow{j\otimes 1} \mathrm{L}_0 \otimes_{\mathrm{A}} \mathrm{N} \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} \mathrm{N} \longrightarrow 0 \tag{11.2.3.1}$$

$$0 = \mathrm{Tor}_1^{\mathrm{A}'}(\mathrm{L}''_0, \mathrm{N}') \longrightarrow \mathrm{T}'' \xrightarrow{\partial''} \mathrm{R}'' \otimes_{\mathrm{A}'} \mathrm{N}' \xrightarrow{j''\otimes 1} \mathrm{L}''_0 \otimes_{\mathrm{A}'} \mathrm{N}' \xrightarrow{\varepsilon''\otimes 1} \mathrm{M}' \otimes_{\mathrm{A}'} \mathrm{N}' \longrightarrow 0 \tag{11.2.3.2}$$

On the other hand there is a homomorphism of A'-modules

$$v : \mathrm{R}' = \mathrm{Ker}(\varepsilon) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{Ker}(\varepsilon') \xrightarrow{u_0} \mathrm{Ker}(\varepsilon'') = \mathrm{R}''.$$

Let us show that the diagram **IV-3** | 122

$$\begin{tikzcd} \mathrm{T}' \arrow[r, "\partial\otimes 1"] \arrow[d, "\varphi_1"] & \mathrm{R}' \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] \arrow[d, "v\otimes 1"] & \mathrm{L}'_0 \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] \arrow[d, "u_0\otimes 1"] & \mathrm{M}' \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] \arrow[d, equal] & 0 \\ \mathrm{T}'' \arrow[r, "\partial''"] & \mathrm{R}'' \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] & \mathrm{L}''_0 \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] & \mathrm{M}' \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] & 0 \end{tikzcd} \tag{11.2.3.3}$$

is *commutative*. For this, one verifies immediately (M, IV, 1) that the homomorphism $\partial : \mathrm{T} \to \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}$ comes (in the present case) by passage to the quotient from the homomorphism $w : \mathrm{Ker}(f_0 \otimes 1_{\mathrm{N}}) \to \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}$, restriction of the homomorphism $g_0 \otimes 1_{\mathrm{N}} : \mathrm{L}_1 \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}$, where $g_0 : \mathrm{L}_1 \to \mathrm{R}$ is such that $f_0 = j \circ g_0$; similarly for ∂''. It therefore suffices to see that the diagram

$$\begin{tikzcd} \mathrm{Ker}(f_0 \otimes 1_{\mathrm{N}}) \arrow[r, "w"] \arrow[d] & \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N} \arrow[r] & \mathrm{R}' \otimes_{\mathrm{A}'} \mathrm{N}' = (\mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{A}} \mathrm{A}' \arrow[d] \\ \mathrm{Ker}(f''_0 \otimes 1_{\mathrm{N}'}) \arrow[rr] & & \mathrm{R}'' \otimes_{\mathrm{A}'} \mathrm{N}' \end{tikzcd}$$

is commutative, which follows from the commutativity of the diagram

$$\begin{tikzcd} \mathrm{L}_1 \arrow[r] \arrow[d] & \mathrm{R} \arrow[d] \\ \mathrm{L}''_1 \arrow[r] & \mathrm{R}'' \end{tikzcd}$$

Lemma (11.2.4). — *Let* A *be a ring,* C *an* A*-algebra,* M *an* A*-module,* N *a* C*-module,* A' *an* A*-algebra. Set* $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{N}' = \mathrm{N} \otimes_{\mathrm{A}} \mathrm{A}' = \mathrm{N} \otimes_{\mathrm{C}} \mathrm{C}'$. *Suppose that* $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$ *is a flat* C*-module. Then the canonical homomorphism*

$$\varphi_1 : \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{N}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{Tor}_1^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}')$$

(cf. (11.2.1, IV.11.2.1.5)*)* is surjective.

Proof. Keep the notation of (11.2.3). Right exactness of tensor products shows that

$$\mathrm{L}'_1 \xrightarrow{f'_0} \mathrm{L}'_0 \xrightarrow{\varepsilon'} \mathrm{M}' \longrightarrow 0$$

is exact. Since L'_0 and L'_1 are free A'-modules, we may suppose that $\mathrm{L}''_0 = \mathrm{L}'_0$ and $\mathrm{L}''_1 = \mathrm{L}'_1$, that u_0 and u_1 are the identity maps, and that $f''_0 = f'_0$. Since $\mathrm{Ker}(\varepsilon') = \mathrm{Im}(f'_0)$ and $\mathrm{R}'' = \mathrm{Im}(f''_0) = \mathrm{Im}(f'_0)$, the homomorphism v is surjective, and hence so is $v \otimes 1$. The proof will therefore be complete if the first row of (11.2.3, IV.11.2.3.3) is exact, because $v \otimes 1$ is surjective and $u_0 \otimes 1$ is injective (Bourbaki, *Alg. comm.*, chap. I, § 1, no. 4, Cor. 2 to Prop. 2). For this we use
the hypothesis that $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$ is a flat C-module. Setting $\mathrm{Q} = \mathrm{Ker}(\varepsilon \otimes 1) = \mathrm{Im}(j \otimes 1)$, by the exact **IV-3** | 123
sequence (11.2.3, IV.11.2.3.1), one has the two exact sequences $0 \to \mathrm{Q} \to \mathrm{L}_0 \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{M} \otimes_{\mathrm{A}} \mathrm{N} \to 0$ and $\mathrm{T} \to \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{Q} \to 0$, where the homomorphisms are homomorphisms of *C-modules*; using the hypothesis of flatness, one deduces $(\mathbf{0}_{\mathrm{I}}, 6.1.2)$ the exact sequence

$$0 \longrightarrow \mathrm{Q} \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow (\mathrm{L}_0 \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow (\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow 0$$

and on the other hand, the tensor product being right exact, one has the exact sequence

$$T \otimes_C C' \longrightarrow (R \otimes_A N) \otimes_C C' \longrightarrow Q \otimes_C C' \longrightarrow 0$$

whence finally the exact sequence

$$T \otimes_C C' \longrightarrow (R \otimes_A N) \otimes_C C' \longrightarrow (L_0 \otimes_A N) \otimes_C C' \longrightarrow (M \otimes_A N) \otimes_C C' \longrightarrow 0$$

But by definition, for every C-module P, $P \otimes_C C' = P \otimes_A A'$, whence the conclusion. □

Lemma (11.2.5). — *Let* A *be a ring,* $\mathfrak{J}$ *an ideal of* A, B *an* A*-algebra,* M *a* B*-module, and* $A \to A'$ *a ring homomorphism. Set* $\mathfrak{J}' = \mathfrak{J}A'$, $B' = B \otimes_A A'$, *and* $M' = M \otimes_A A' = M \otimes_B B'$. *Let* $\mathfrak{p}'$ *be a prime ideal of* B′ *containing* $\mathfrak{J}'B'$. *Suppose that one of the following hypotheses holds:*

a) $\mathfrak{J}$ *is nilpotent.*
b) A′ *is Noetherian,* B′ *is an* A′*-algebra of finite type,* M′ *a* B′*-module of finite type.*

Under these conditions, suppose the two following properties are verified:

(i) $M \otimes_A (A/\mathfrak{J})$ *is a flat* $(A/\mathfrak{J})$*-module.*
(ii) *The canonical composite homomorphism*

$$\mathrm{Tor}_1^A(M, A/\mathfrak{J}) \xrightarrow{\psi_1} \mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}') \xrightarrow{\theta} (\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}'))_{\mathfrak{p}'}$$

(where ψ_1 *is the homomorphism* (11.2.1, IV.11.2.1.1) *and* θ *the canonical homomorphism of a* B′*-module localised at* $\mathfrak{p}'$*) is zero.*

Then $M'_{\mathfrak{p}'}$ *is a flat* A′*-module.*

Proof. Under hypothesis b), $M'_{\mathfrak{p}'}$ is a finite-type $B'_{\mathfrak{p}'}$-module, $B'_{\mathfrak{p}'}$ is a Noetherian A′-algebra, and $\mathfrak{J}'B'_{\mathfrak{p}'}$ is contained in the radical $\mathfrak{p}'B'_{\mathfrak{p}'}$ of $B'_{\mathfrak{p}'}$. Under hypothesis a), $\mathfrak{J}'$ is nilpotent. We may therefore apply the flatness criterion $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$ or $(\mathbf{0}_{\mathrm{III}}, 10.2.2)$, according as a) or b) holds. First,

$$\begin{aligned} M'_{\mathfrak{p}'} \otimes_{A'} (A'/\mathfrak{J}') &= (M' \otimes_{A'} (A'/\mathfrak{J}'))_{\mathfrak{p}'} \\ &= \big((M \otimes_A (A/\mathfrak{J})) \otimes_{A/\mathfrak{J}} (A'/\mathfrak{J}')\big)_{\mathfrak{p}'}. \end{aligned}$$

Hypothesis (i) therefore implies, by $(\mathbf{0}_{\mathrm{I}}, 6.2.1 \text{ and } 6.3.2)$, that this is a flat $(A'/\mathfrak{J}')$-module. It remains to prove that

$$\mathrm{Tor}_1^{A'}(M'_{\mathfrak{p}'}, A'/\mathfrak{J}') = 0.$$

By flatness of $B'_{\mathfrak{p}'}$ over B′, this module is

$$\big(\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}')\big)_{\mathfrak{p}'}.$$

Hypothesis (ii) says that $\theta \circ \psi_1$ is zero. On the other hand, (11.2.4), applied with $C = A/\mathfrak{J}$ and $N = C$, shows in view of (i) that φ_1 is surjective (since $A'/\mathfrak{J}' = C \otimes_A A'$). Hence θ is zero. Since the image under θ of $\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}')$ generates its localization as a $B'_{\mathfrak{p}'}$-module, that localization is zero. □

Theorem (11.2.6). — *The notations being those of* (8.5.1) *and* (8.8.1), *suppose* S_α *quasi-compact,* X_α *and* Y_α *of finite presentation over* S_α*; let* $f_\alpha : X_\alpha \to Y_\alpha$ *be an* S_α*-morphism,* $\mathscr{F}_\alpha$ *an* $\mathscr{O}_{X_\alpha}$*-Module quasi-coherent of finite presentation.*

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(i) Let x be a point of X, x_λ its canonical projection in X_λ. For $\mathscr{F}$ to be f-flat at the point x, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $\mathscr{F}_\lambda$ is f_λ-flat at the point x_λ.
(ii) For $\mathscr{F}$ to be f-flat, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $\mathscr{F}_\lambda$ is f_λ-flat.

Proof. One may suppose that $S_\alpha = S_0$; as Y_0 is of finite presentation over S_0, Y_0 is quasi-compact, and $f_0 : X_0 \to Y_0$ is a morphism of finite presentation (1.6.2, (v)), hence one can also restrict to the case where $S_0 = Y_0$. Moreover, by the quasi-compactness of S_0 and of the fact that the set of indices L is filtered, one can restrict by the same reasoning to the case where $S_0 = \mathrm{Spec}(A_0)$ is affine. Furthermore X_0 is also quasi-compact, hence by the same reasoning one can also suppose $X_0 = \mathrm{Spec}(B_0)$ affine; one then has $\mathscr{F}_0 = \widetilde{M}_0$, where M_0 is a B_0-module of finite presentation, and the statement (11.2.6) is in this case equivalent to the following (by $(\mathbf{0}_{\mathrm{I}}, 6.3.1)$):

Corollary (11.2.6.1). — *Let* A_0 *be a ring,* $(A_\lambda)_{\lambda\in L}$ *a filtered inductive system of* A_0*-algebras,* B_0 *an* A_0*-algebra of finite presentation, and* M_0 *a* B_0*-module of finite presentation. Put*

$$B_\lambda = B_0 \otimes_{A_0} A_\lambda, \qquad M_\lambda = M_0 \otimes_{A_0} A_\lambda = M_0 \otimes_{B_0} B_\lambda,$$
$$A = \varinjlim_\lambda A_\lambda, \qquad B = \varinjlim_\lambda B_\lambda = B_0 \otimes_{A_0} A,$$
$$M = \varinjlim_\lambda M_\lambda = M_0 \otimes_{A_0} A = M_0 \otimes_{B_0} B.$$

(i) Let $\mathfrak{p}$ be a prime ideal of B, and for every λ, let $\mathfrak{p}_\lambda$ be its inverse image in B_λ. For $M_\mathfrak{p}$ to be a flat A-module, it is necessary and sufficient that there exist λ such that $(M_\lambda)_{\mathfrak{p}_\lambda}$ be a flat A_λ-module.

(ii) For M to be a flat A-module, it is necessary and sufficient that there exist λ such that M_λ be a flat A_λ-module.

One needs only prove that the conditions are necessary (2.1.4). We will proceed in several steps.

I) *Reduction to the case where the* A_λ *are Noetherian.* — By (8.9.1) there exist a subring $A'_0 \subset A_0$ which is a finite-type **Z**-algebra, a finite-type A'_0-algebra B'_0, and a finite-type B'_0-module M'_0, such that

$$B_0 = B'_0 \otimes_{A'_0} A_0, \qquad M_0 = M'_0 \otimes_{A'_0} A_0.$$

Since $B_\lambda = B'_0 \otimes_{A'_0} A_\lambda$ and $M_\lambda = M'_0 \otimes_{A'_0} A_\lambda$, we may replace A_0, B_0, M_0 by A'_0, B'_0, M'_0, regarding the A_λ as A'_0-algebras. We may therefore already suppose that A_0 is Noetherian.

Let H be the set of pairs (λ, C_λ), where C_λ is a finite-type A_0-subalgebra of A_λ. Order H by declaring $(\lambda, C_\lambda) \leqslant (\mu, D_\mu)$ if $\lambda \leqslant \mu$ and $\varphi_{\mu\lambda}(C_\lambda) \subset D_\mu$. This makes H filtered: two pairs are dominated by choosing $\nu \geqslant \lambda, \mu$ and taking for E_ν the A_0-subalgebra generated by $\varphi_{\nu\lambda}(C_\lambda)$ and $\varphi_{\nu\mu}(D_\mu)$. For $\xi = (\lambda, C_\lambda)$ put $A_\xi = C_\lambda$; the evident restricted transition maps form a filtered inductive system of A_0-algebras. Put $B_\xi = B_0 \otimes_{A_0} A_\xi$ and $M_\xi = M_0 \otimes_{A_0} A_\xi$. The rings A_ξ are Noetherian, and the double-inductive-limit formula (Bourbaki, *Alg.*, chap. II, 3rd ed., § 6, no. 4, Prop. 7) gives

$$\varinjlim_H A_\xi = A, \qquad \varinjlim_H B_\xi = B, \qquad \varinjlim_H M_\xi = M.$$

Suppose (11.2.6.1) proved for the system $(A_\xi)_{\xi\in H}$, and let $\mathfrak{p}$ be a prime ideal of B such that $M_\mathfrak{p}$ is flat over A. Then there exists $\xi \in H$ such that $(M_\xi)_{\mathfrak{p}_\xi}$ is flat over A_ξ, where $\mathfrak{p}_\xi$ is the inverse image **IV-3** | 125
of $\mathfrak{p}$. Write $\xi = (\lambda, C_\lambda)$. Since $\mathfrak{p}_\lambda$ is the inverse image of $\mathfrak{p}$ in B_λ, $\mathfrak{p}_\xi$ is the inverse image of $\mathfrak{p}_\lambda$ in B_ξ, and

$$(M_\lambda)_{\mathfrak{p}_\lambda} = (M_\xi \otimes_{C_\lambda} A_\lambda)_{\mathfrak{p}_\lambda} = \big((M_\xi)_{\mathfrak{p}_\xi} \otimes_{C_\lambda} A_\lambda\big)_{\mathfrak{p}_\lambda}.$$

Thus $(M_\lambda)_{\mathfrak{p}_\lambda}$ is flat over A_λ by ($\mathbf{0}_\mathrm{I}$, 6.2.1 and 6.3.2). Statement (ii) is handled in the same way. Henceforth we may therefore suppose that all A_λ are Noetherian, although A itself need not be.

II) *Reduction of the global statement (ii) to the pointwise statement (i).* — Suppose $\mathscr{F}$ is f-flat. For every λ, let U_λ be the set of $x_\lambda \in X_\lambda$ such that $\mathscr{F}_\lambda$ is f_λ-flat at the point x_λ; we know that U_λ is open in X_λ, since S_λ is Noetherian and f_λ is of finite type (11.1.1); let $V_\lambda = v_\lambda^{-1}(U_\lambda)$ its inverse image in X. By hypothesis, for every $x \in X$, the projection of x in X_λ belongs to U_λ for some λ, which means that $X = \bigcup V_\lambda$. Moreover, by (2.1.4), for $\lambda \leqslant \mu$, $V_\lambda \subset V_\mu$, hence, since X is quasi-compact, there exists an index μ such that $X = V_\mu$. As the X_λ are quasi-compact, it follows from (8.3.4) that there exists a $\nu \geqslant \mu$ such that $v_{\nu\mu}^{-1}(U_\mu) = X_\nu$; but by (2.1.4), this implies that $\mathscr{F}_\nu$ is f_ν-flat.

III) *End of the proof.* — It remains to prove (i), with S_0 affine and Noetherian. Let y_0 be the projection of x in S_0. By (2.1.4) and (**I**, 3.6.5) we may replace S_0 by $\operatorname{Spec}(\mathscr{O}_{S_0,y_0})$; thus we may assume that A_0 is a Noetherian local ring with maximal ideal $\mathfrak{m}_0 = \mathfrak{j}_{y_0}$. By definition, $\mathfrak{m}_0$ is the inverse image of the prime ideal $\mathfrak{p} = \mathfrak{j}_x$ of B, and $M_\mathfrak{p}$ is flat over A. Hence

$$\operatorname{Tor}_1^A(A/\mathfrak{m}_0A, M_\mathfrak{p}) = 0.$$

Since the Tor modules in question are B-modules and $B_\mathfrak{p}$ is flat over B, this is equivalent to

$$\big(\operatorname{Tor}_1^A(A/\mathfrak{m}_0A, M)\big)_\mathfrak{p} = 0.$$

The B_0-module $\mathrm{Tor}_1^{A_0}(A_0/\mathfrak{m}_0, M_0)$ is of finite type. Indeed, it may be computed from a resolution of $A_0/\mathfrak{m}_0$ by finite free A_0-modules; tensoring with M_0 gives a complex of finite-type B_0-modules, whose homology is again of finite type because B_0 is Noetherian. Choose generators t_i^0 $(1 \leqslant i \leqslant m)$ of this module, and let t_i be their canonical images in $\mathrm{Tor}_1^A(A/\mathfrak{m}_0 A, M)$. The preceding localization equality gives an element $h \in B - \mathfrak{p}$ such that $ht_i = 0$ for all i.

By (11.2.2, IV.11.2.2.1),

$$\mathrm{Tor}_1^A(A/\mathfrak{m}_0 A, M) = \varinjlim_{\lambda} \mathrm{Tor}_1^{A_\lambda}(A_\lambda/\mathfrak{m}_0 A_\lambda, M_\lambda).$$

Consequently, for some λ there is an element $h_\lambda \in B_\lambda$ mapping to h such that $h_\lambda t_i^\lambda = 0$ for every i, where t_i^λ is the image of t_i^0. If $\mathfrak{p}_\lambda$ is the inverse image of $\mathfrak{p}$ in B_λ, then $h_\lambda \notin \mathfrak{p}_\lambda$. Thus the images of all the t_i^0 in

$$\left(\mathrm{Tor}_1^{A_\lambda}(A_\lambda/\mathfrak{m}_0 A_\lambda, M_\lambda)\right)_{\mathfrak{p}_\lambda}$$

vanish, and the canonical map from $\mathrm{Tor}_1^{A_0}(A_0/\mathfrak{m}_0, M_0)$ to this localization is zero. The hypotheses of (11.2.5) are therefore satisfied; condition (i) there is automatic because $A_0/\mathfrak{m}_0$ is a field. It follows that $(M_\lambda)_{\mathfrak{p}_\lambda}$ is flat over A_λ, completing the proof.

Corollary (11.2.7). — *Let $S = \mathrm{Spec}(A)$ be an affine scheme, $f : X \to S$ a morphism, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent, x a point of X. The following conditions are equivalent:*

a) *f is a morphism of finite presentation, $\mathscr{F}$ is an $\mathscr{O}_X$-Module of finite presentation and $\mathscr{F}$ is f-flat at the point x (resp. f-flat).* **IV-3** | 126
b) *There exists an affine Noetherian scheme $S_0 = \mathrm{Spec}(A_0)$, a morphism of finite type $f_0 : X_0 \to S_0$, an $\mathscr{O}_{X_0}$-Module coherent $\mathscr{F}_0$, a morphism $S \to S_0$ such that the S-prescheme $X_0 \otimes_{S_0} S$ is S-isomorphic to X and that $\mathscr{F}$ is isomorphic to $\mathscr{F}_0 \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_S$, and $\mathscr{F}_0$ is f_0-flat at the point x_0 the projection of x in X_0 (resp. f_0-flat).*
c) *The conditions of b) are satisfied and in addition A_0 is a sub-$\mathbf{Z}$-algebra of finite type of A, the morphism $S \to S_0$ corresponding to the canonical injection $A_0 \to A$.*

Proof. It is clear that $c)$ implies $b)$ and that $b)$ implies $a)$ by (2.1.4). On the other hand, we may regard A as the filtered direct limit of its finite-type $\mathbf{Z}$-subalgebras, and by (8.9.1) there exist such a subalgebra A_0 and a finite-type morphism $f_0 : X_0 \to S_0$ such that X is S-isomorphic to $X_0 \times_{S_0} S$ and $\mathscr{F}$ is isomorphic to $\mathscr{F}_0 \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_S$. We may therefore write

$$X = \varprojlim_{\lambda} X_\lambda, \qquad X_\lambda = X_0 \otimes_{A_0} A_\lambda,$$

where A_λ runs through the finite-type $\mathbf{Z}$-subalgebras of A containing A_0, and $\mathscr{F} = v_\lambda^*(\mathscr{F}_\lambda)$, where $v_\lambda : X \to X_\lambda$ is the canonical projection and $\mathscr{F}_\lambda = \mathscr{F}_0 \otimes_{A_0} A_\lambda$. It follows from (11.2.6) that there exists λ such that $\mathscr{F}_\lambda$ is f_λ-flat at $x_\lambda = v_\lambda(x)$ (resp. f_λ-flat); the subring A_λ of A then satisfies condition $c)$. □

Proposition (11.2.8). — *Let $g : X \to S$, $h : Y \to S$ be two morphisms of finite presentation, $f : X \to Y$ an S-morphism, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation; for every $s \in S$, let $X_s = g^{-1}(s) = X \times_S \mathrm{Spec}(\boldsymbol{k}(s))$, $Y_s = h^{-1}(s) = Y \times_S \mathrm{Spec}(\boldsymbol{k}(s))$, f_s the morphism $f \times 1 : X_s \to Y_s$, $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_S} \boldsymbol{k}(s)$. Then the set E of $s \in S$ such that $\mathscr{F}_s$ is f_s-flat is locally constructible.*

Proof. The property whose constructibility we wish to prove satisfies condition (9.2.1, (i)), by (2.2.13) and (2.5.1). In view of (9.2.3), we may therefore restrict to the case where S is affine, Noetherian, and integral, with generic point η, and it remains to prove that E or $S - E$ is a neighbourhood of η in S. If $\eta \in S - E$, this follows immediately from Lemma (9.4.7.1). If, on the contrary, $\eta \in E$, it follows first from (11.2.6), applied by the method of (8.1.2, a)), that there is an open neighbourhood U of η in S such that $\mathscr{F}|_{g^{-1}(U)}$ is flat over $h^{-1}(U)$; *a fortiori*, by (2.1.4), $\mathscr{F}_s$ is f_s-flat for every $s \in U$, which completes the proof. □

The following theorem generalizes (11.2.6.1, (ii)).

Theorem (11.2.9). — *(Raynaud). — Let* A_0 *be a ring,* $(\mathrm{A}_\lambda)_{\lambda\in\mathrm{L}}$ *a filtered inductive system of* A_0*-algebras,* B_0 *an* A_0*-algebra of finite presentation,* $\mathfrak{J}_0$ *a finite-type ideal of* B_0*, and* M_0 *a finitely presented* B_0*-module. Set* $\mathrm{B}_\lambda = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_\lambda$*,* $\mathfrak{J}_\lambda = \mathfrak{J}_0\mathrm{B}_\lambda$*,* $\mathrm{M}_\lambda = \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_\lambda = \mathrm{M}_0 \otimes_{\mathrm{B}_0} \mathrm{B}_\lambda$*,* $\mathrm{A} = \varinjlim \mathrm{A}_\lambda$*,* $\mathrm{B} = \varinjlim \mathrm{B}_\lambda = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}$*,* $\mathfrak{J} = \mathfrak{J}_0\mathrm{B}$*, and* $\mathrm{M} = \varinjlim \mathrm{M}_\lambda = \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A} = \mathrm{M}_0 \otimes_{\mathrm{B}_0} \mathrm{B}$*. For* $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ *to be a flat* A*-module, it is necessary and sufficient that there exist* λ *such that* $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda)$ *is a flat* A_λ*-module; the canonical homomorphisms*

$$(11.2.9.1)\qquad \begin{cases} \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda) \otimes_{\mathrm{A}_\lambda} \mathrm{A} \to \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}) \\ \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda) \otimes_{\mathrm{A}_\lambda} \mathrm{A}_\mu \to \mathrm{gr}^*_{\mathfrak{J}_\mu}(\mathrm{M}_\mu) \end{cases} \qquad \text{for } \lambda \leqslant \mu$$

are then bijective and $\mathrm{gr}^*_{\mathfrak{J}_\mu}(\mathrm{M}_\mu)$ *is a flat* A_μ*-module for* $\lambda \leqslant \mu$*.*

Proof. Let us first show that the conditions are sufficient, which amounts to proving the bijectivity of (11.2.9, IV.11.2.9.1). This follows from the following lemma:

Lemma (11.2.9.2). — *Let* A *be a ring,* B *an* A*-algebra,* M *a* B*-module,* $\mathfrak{J}$ *an ideal of* B*. Let* A' *be an* A*-algebra; set* $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$*,* $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}' = \mathrm{M} \otimes_{\mathrm{B}} \mathrm{B}'$*,* $\mathfrak{J}' = \mathfrak{J}\mathrm{B}'$*. If* $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ *is a flat* A*-module, the canonical homomorphism*

$$(\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{gr}^*_{\mathfrak{J}'}(\mathrm{M}')$$

is bijective.

Indeed, by induction on k, the hypothesis that the $\mathfrak{J}^k\mathrm{M}/\mathfrak{J}^{k+1}\mathrm{M}$ are flat A-modules implies first, by $(\mathbf{0}_{\mathrm{I}},\ 6.1.2)$, that $\mathrm{M}/\mathfrak{J}^{k+1}\mathrm{M}$ is a flat A-module; moreover, by $(\mathbf{0}_{\mathrm{I}},\ 6.1.2)$, the sequence **IV-3 | 127**

$$0 \longrightarrow (\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow (\mathrm{M}/\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow 0$$

is then exact, in other words $(\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}'$ identifies with its canonical image $\mathfrak{J}'^{k+1}\mathrm{M}'$ in $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$. On the other hand, always by $(\mathbf{0}_{\mathrm{I}},\ 6.1.2)$, the sequence

$$0 \longrightarrow (\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow (\mathfrak{J}^k\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow (\mathfrak{J}^k\mathrm{M}/\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow 0$$

is exact, which proves the lemma.

To prove the necessity of the conditions of (11.2.9), we proceed in several steps.

I) *Reduction to the case where the* A_λ *are Noetherian.* — One proceeds as in reduction I) of (11.2.6.1), retaining the notation; it is enough to begin by replacing A'_0 by a finite-type sub-A'_0-algebra A''_0 of A_0 such that, if one sets $\mathrm{B}''_0 = \mathrm{B}'_0 \otimes_{\mathrm{A}'_0} \mathrm{A}''_0$, there is in B''_0 a finitely generated ideal $\mathfrak{J}''_0$ such that $\mathfrak{J}''_0\mathrm{B}_0 = \mathfrak{J}_0$. For this, consider the finite-type sub-A'_0-algebras A'_α of A_0, which form a filtered family; one has $\mathrm{B}_0 = \varinjlim \mathrm{B}'_\alpha$, where $\mathrm{B}'_\alpha = \mathrm{B}'_0 \otimes_{\mathrm{A}'_0} \mathrm{A}'_\alpha$. Hence there is an index β such that a finite set of generators of $\mathfrak{J}_0$ consists of the images in B_0 of elements of B'_β. Take $\mathrm{A}''_0 = \mathrm{A}'_\beta$, $\mathrm{B}''_0 = \mathrm{B}'_\beta$, and take for $\mathfrak{J}''_0$ the ideal generated by these elements. We may therefore suppose that A_0 (and hence also B_0) is Noetherian. One then defines, as in *loc. cit.*, the filtered set H and the A_ξ, B_ξ, M_ξ for $\xi \in \mathrm{H}$; set also $\mathfrak{J}_\xi = \mathfrak{J}''_0\mathrm{B}_\xi$ for all $\xi \in \mathrm{H}$. Suppose that one has shown that there exists $\xi = (\lambda, \mathrm{C}_\lambda)$ such that $\mathrm{gr}^*_{\mathfrak{J}_\xi}(\mathrm{M}_\xi)$ is a flat C_λ-module. Since $\mathrm{M}_\lambda = \mathrm{M}_\xi \otimes_{\mathrm{C}_\lambda} \mathrm{A}_\lambda$, it follows from (11.2.9.2) that $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda)$ is a flat A_λ-module. We may therefore suppose from now on that the A_λ are Noetherian for $\lambda \in \mathrm{L}$ (but not necessarily A itself).

II) *Preliminary lemmas.*

Lemma (11.2.9.3). — *Let* A *be a ring, let* $\mathrm{B} = \bigoplus_{i\geqslant 0} \mathrm{B}_i$ *be an* A*-algebra graded in nonnegative degrees, and let* $\mathrm{M} = \bigoplus_{i\in\mathbf{Z}} \mathrm{M}_i$ *be a graded* B*-module. Suppose that* B_0 *is a local ring and that each* M_i *is a finite-type* B_0*-module. For* M *to be a finite-type* B*-module, it is necessary and sufficient that, if* $\mathfrak{q}$ *is the inverse image in* A *of the maximal ideal* $\mathfrak{m}$ *of* B_0*,* $\mathrm{M} \otimes_{\mathrm{A}} \boldsymbol{k}(\mathfrak{q})$ *be a finite-type* $(\mathrm{B} \otimes_{\mathrm{A}} \boldsymbol{k}(\mathfrak{q}))$*-module.*

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Proof. The condition is evidently necessary; let us prove that it is sufficient. Since B_0 (and hence B) is an $\mathrm{A}_{\mathfrak{q}}$-algebra, we may replace A by $\mathrm{A}_{\mathfrak{q}}$; in other words, we may suppose that A is also a local ring with maximal ideal $\mathfrak{q}$. By hypothesis, there is an integer N such that the canonical homomorphism

$$\Big(\bigoplus_{-N\leqslant i\leqslant N} M_i \otimes_{B_0} B\Big) \otimes_A \boldsymbol{k}(\mathfrak{q}) \longrightarrow M \otimes_A \boldsymbol{k}(\mathfrak{q})$$

is surjective. This also means that, for every integer n, the canonical homomorphism of $\boldsymbol{k}(\mathfrak{q})$-modules

$$\Big(\bigoplus_{-N\leqslant i\leqslant N} M_i \otimes_{B_0} B_{n-i}\Big) \otimes_A \boldsymbol{k}(\mathfrak{q}) \longrightarrow M_n \otimes_A \boldsymbol{k}(\mathfrak{q})$$

is surjective. Now M_n is a finite-type B_0-module, B_0 is local, and $\mathfrak{q}B_0 \subset \mathfrak{m}$; Nakayama's lemma therefore shows that each canonical homomorphism

$$\bigoplus_{-N\leqslant i\leqslant N} M_i \otimes_{B_0} B_{n-i} \longrightarrow M_n$$

is surjective, which proves the assertion. □

Corollary (11.2.9.4). — *Under the hypotheses of (11.2.9.3), suppose moreover that each of the* B_i *and each* M_i *is a finitely presented* B_0*-module, and that* M *is a flat* A*-module. For* M *to be a finitely presented* B*-module, it is necessary and sufficient that* $M \otimes_A \boldsymbol{k}(\mathfrak{q})$ *be a finitely presented* $(B \otimes_A \boldsymbol{k}(\mathfrak{q}))$*-module.*

Proof. By (11.2.9.3), if the condition in the statement holds, M is a finite-type B-module. Hence there is a finite-type free graded B-module L (thus having a finite basis of homogeneous elements) and a surjective graded homomorphism of degree 0,

$$u : L \longrightarrow M.$$

Let $R = \mathrm{Ker}(u)$, which is a graded B-module. There are finitely many integers m_j ($1 \leqslant j \leqslant r$) such that, for every integer i, R_i is the kernel of a surjective homomorphism

$$\bigoplus_{1\leqslant j\leqslant r} B_{i+m_j} \longrightarrow M_i.$$

The hypotheses on the M_i and the B_i then imply that R_i is a finite-type B_0-module (Bourbaki, *Alg. comm.*, chap. I, § 2, nº 8, lemma 9). To prove that R is a finite-type B-module, observe that, by the flatness hypothesis and ($\mathbf{0}_{\mathrm{I}}$, 6.1.2), the sequence

$$0 \longrightarrow R \otimes_A \boldsymbol{k}(\mathfrak{q}) \longrightarrow L \otimes_A \boldsymbol{k}(\mathfrak{q}) \longrightarrow M \otimes_A \boldsymbol{k}(\mathfrak{q}) \longrightarrow 0$$

is exact. The hypothesis therefore implies (Bourbaki, *loc. cit.*) that $R \otimes_A \boldsymbol{k}(\mathfrak{q})$ is a finite-type $(B \otimes_A \boldsymbol{k}(\mathfrak{q}))$-module. It is therefore enough to apply (11.2.9.3) to R. □

III) *Reduction to the case where the homomorphisms of transitions* $\varphi_{\mu\lambda} : A_\lambda \to A_\mu$ *(for* $\lambda \leqslant \mu$*) are injective.* — Let A'_λ be the image of A_λ by the canonical homomorphism $\varphi_\lambda : A_\lambda \to A$; it is clear that the A'_λ form an inductive system of sub-rings *Noetherian* of A, and the homomorphisms of transition $A'_\lambda \to A'_\mu$ (for $\lambda \leqslant \mu$) are injective. Set $B'_\lambda = B_0 \otimes_{A_0} A'_\lambda$, $\mathfrak{I}'_\lambda = \mathfrak{I}_0 B'_\lambda$, $M'_\lambda = M_0 \otimes_{A_0} A'_\lambda = M_\lambda \otimes_{A_\lambda} A'_\lambda$; one has still $B = \varinjlim B'_\lambda$, $M = \varinjlim M'_\lambda$. Suppose one has proved that there exists λ such that, for $\mu \geqslant \lambda$, $\mathrm{gr}^*_{\mathfrak{I}'_\mu}(M'_\mu)$ is a flat A'_μ-module. Let a_λ be the kernel of the homomorphism φ_λ, which is an ideal of the Noetherian ring A_λ. It follows from the definition of the inductive limit that there exists an index $\mu \geqslant \lambda$ such that $\varphi_{\mu\lambda}(a_\lambda) = 0$, in other words $\varphi_{\mu\lambda}$ factorises as $\varphi_{\mu\lambda} : A_\lambda \to A'_\lambda \to A_\mu$, and one can write $B_\mu = B'_\lambda \otimes_{A'_\lambda} A_\mu$, $\mathfrak{I}_\mu = \mathfrak{I}'_\lambda B_\mu$ and $M_\mu = M'_\lambda \otimes_{A'_\lambda} A_\mu$; one deduces then from (11.2.9.2) that $\mathrm{gr}^*_{\mathfrak{I}_\mu}(M_\mu)$ is a flat A_μ-module.

IV) *Reduction to the case where* $B_0 = A_0[T_1, \dots, T_n]$ *and* $\mathrm{gr}^*_{\mathfrak{I}_0}(B_0) = B_0$. — By virtue of the hypothesis, there is a system $(t_i)_{1\leqslant i\leqslant n}$ of generators of the finite-type A_0-algebra B_0, such that the t_i for $i \leqslant m$ generate the ideal $\mathfrak{I}_0$ of B_0. Set $B'_0 = A_0[T_1, \dots, T_n]$, a polynomial algebra (hence Noetherian), and let $\mathfrak{I}'_0$ be the ideal of B'_0 generated by the T_i with $i \leqslant m$; there is then a surjective A_0-homomorphism $u_0 : B'_0 \to B_0$ taking each T_i to t_i ($1 \leqslant i \leqslant n$), and hence satisfying $u_0(\mathfrak{I}'_0) = \mathfrak{I}_0$. This homomorphism allows us to regard M_0 as a finite-type B'_0-module. Set

$$B'_\lambda = B'_0 \otimes_{A_0} A_\lambda = A_\lambda[T_1, \dots, T_n], \qquad \mathfrak{I}'_\lambda = \mathfrak{I}'_0 B'_\lambda,$$

and also **IV-3** | 129

$$\mathrm{B}' = \mathrm{B}'_0 \otimes_{\mathrm{A}_0} \mathrm{A} = \mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_n], \qquad \mathfrak{J}' = \mathfrak{J}'_0 \mathrm{B}'.$$

Thus $\mathfrak{J}'$ is the ideal of B' generated by the T_i with $i \leqslant m$. Moreover, for every integer $k \geqslant 0$ one clearly has

$$\mathfrak{J}'^k \mathrm{M} = \mathfrak{J}^k \mathrm{M}, \qquad \mathfrak{J}'_\lambda{}^k \mathrm{M}_\lambda = \mathfrak{J}_\lambda^k \mathrm{M}_\lambda$$

for every λ. Hence

$$\mathrm{gr}^*_{\mathfrak{J}'}(\mathrm{M}) = \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$$

as A-modules, and

$$\mathrm{gr}^*_{\mathfrak{J}'_\lambda}(\mathrm{M}_\lambda) = \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda)$$

as A_λ-modules. We may therefore substitute B', B'_λ, $\mathfrak{J}'$, and $\mathfrak{J}'_\lambda$ for B, B_λ, $\mathfrak{J}$, and $\mathfrak{J}_\lambda$, respectively, in the proof. Note also that, by construction, $\mathrm{gr}^*_{\mathfrak{J}'}(\mathrm{B}')$ identifies with B', and $\mathrm{gr}^*_{\mathfrak{J}'_\lambda}(\mathrm{B}'_\lambda)$ with B'_λ.

V) *Proof that* $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ *is a* $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$*-module of finite presentation.* — We thus suppose from now on that A_0 and the A_λ are Noetherian, that the transition homomorphisms $\mathrm{A}_\lambda \to \mathrm{A}_\mu$ are injective, that $\mathrm{B}_0 = \mathrm{A}_0[\mathrm{T}_1, \dots, \mathrm{T}_n]$, and that $\mathfrak{J}_0$ is generated by the T_i with $i \leqslant m$. The ring $\mathrm{C}_0 = \mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{B}_0)$ therefore identifies with B_0, and $\mathrm{C} = \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$ identifies with B; at first, in what follows, we shall use only the fact that $\mathrm{C} \cong \mathrm{C}_0 \otimes_{\mathrm{A}_0} \mathrm{A}$.

We will need the following variant of (6.9.3):

Lemma (11.2.9.5). — *Let* A_0 *be a Noetherian ring,* B_0 *an* A_0*-algebra of finite type,* $\mathfrak{J}_0$ *an ideal of* B_0, M_0 *a* B_0*-module of finite type. There exists a sequence* $(\mathrm{S}_{0i})_{1 \leqslant i \leqslant m}$ *of sub-preschemes of* $\mathrm{S}_0 = \mathrm{Spec}(\mathrm{A}_0)$ *having the following properties:*

1º *The underlying spaces of the* S_{0i} *are pairwise disjoint and form a covering of* S_0.
2º *For each i, the set* S_{0i} *is open in* $\bigcup_{j \geqslant i} \mathrm{S}_{0j}$.
3º *Each scheme* S_{0i} *is affine.*
4º *If* A_{0i} *is the ring of* S_{0i} *and one sets* $\mathrm{B}_{0i} = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_{0i}$, $\mathfrak{J}_{0i} = \mathfrak{J}_0 \mathrm{B}_{0i}$, *then* $\mathrm{gr}^*_{\mathfrak{J}_{0i}}(\mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_{0i})$ *is a flat* A_{0i}*-module.*

Proof. We proceed as in (6.9.3) by Noetherian induction, assuming the lemma known after replacing A_0 by

$$\mathrm{A}'_0 = \mathrm{A}_0 / \mathfrak{a}_0,$$

where $\mathfrak{a}_0$ is an ideal such that $\mathrm{V}(\mathfrak{a}_0) \neq \mathrm{S}_0$, and replacing B_0, $\mathfrak{J}_0$, and M_0 by

$$\mathrm{B}'_0 = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}'_0, \qquad \mathfrak{J}'_0 = \mathfrak{J}_0 \mathrm{B}'_0, \qquad \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}'_0,$$

respectively. Let $\mathfrak{N}_0$ be the nilradical of A_0. It is plainly enough to prove the lemma after replacing A_0 by $\mathrm{A}_0/\mathfrak{N}_0$ and replacing B_0, $\mathfrak{J}_0$, and M_0 by the corresponding objects just described. In other words, we may suppose that A_0 is *reduced*. Now $\mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{B}_0)$ is a finite-type A_0-algebra and $\mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{M}_0)$ is a finite-type $\mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{B}_0)$-module. By the generic flatness theorem (6.9.1), there exists an open subset $\mathrm{D}(t_0) \subset \mathrm{S}_0$ such that, on setting $\mathrm{A}_{01} = (\mathrm{A}_0)_{t_0}$,

$$\mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{M}_0) \otimes_{\mathrm{A}_0} \mathrm{A}_{01}$$

is a flat A_{01}-module. Since A_{01} is a flat A_0-module, this module identifies with

$$\mathrm{gr}^*_{\mathfrak{J}_{01}}(\mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_{01}),$$

where

$$\mathrm{B}_{01} = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_{01}, \qquad \mathfrak{J}_{01} = \mathfrak{J}_0 \mathrm{B}_{01}.$$

The complement of $\mathrm{D}(t_0)$ in S_0 is then of the form $\mathrm{V}(\mathfrak{a}_0)$, and the conclusion follows by applying the Noetherian-induction hypothesis. □

Apply this lemma to the present situation, keeping the same notations; set $\mathrm{U}_{0i} = \bigcup_{j \leqslant i} \mathrm{S}_{0j}$, which is an open quasi-compact of S_0. There is then a family $(f_{ik})_{1 \leqslant i \leqslant m, 1 \leqslant k \leqslant n}$ of elements of A_0 such that for each i, U_{0i} is the union of the $\mathrm{D}(f_{ik})$ $(1 \leqslant k \leqslant n)$.

The C_0-module $\mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{M}_0)$ is generated by finitely many homogeneous elements of degree 1, so **IV-3** | 130
there is an epimorphism of graded C_0-modules

$$u_0 : \mathrm{C}_0^r \longrightarrow \mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{M}_0).$$

As $\mathrm{C} = \mathrm{C}_0 \otimes_{\mathrm{A}_0} \mathrm{A}$ and the canonical homomorphism $\mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{M}_0) \otimes_{\mathrm{A}_0} \mathrm{A} \to \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ is surjective, one therefore deduces from u_0 an epimorphism of graded C-modules

$$u : \mathrm{C}^r \xrightarrow{u_0 \otimes 1_{\mathrm{A}}} \mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{M}_0) \otimes_{\mathrm{A}_0} \mathrm{A} \longrightarrow \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}),$$

and it remains to show that the graded C-module $\mathrm{Q} = \mathrm{Ker}(u)$ is *of finite type*.

Lemma (11.2.9.6). — *Under the preceding hypotheses, let $\mathfrak{R}_0$ be an ideal of A_0; set*

$$\mathrm{A}'_0 = \mathrm{A}_0/\mathfrak{R}_0, \qquad \mathrm{B}'_0 = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}'_0 = \mathrm{B}_0/\mathfrak{R}_0\mathrm{B}_0, \qquad \mathfrak{J}'_0 = \mathfrak{J}_0\mathrm{B}'_0,$$

and $\mathfrak{R} = \mathfrak{R}_0\mathrm{A}$, $\mathrm{A}' = \mathrm{A}/\mathfrak{R}$, and suppose moreover that

$$\mathrm{gr}^*_{\mathfrak{J}'_0}(\mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}'_0)$$

is a flat A'_0-module. Then there exist finitely many elements q_h ($1 \leqslant h \leqslant s$) of Q such that, for every prime ideal $\mathfrak{p} \supset \mathfrak{R}\mathrm{B}$ of B, the canonical images of the q_h in $\mathrm{Q}_{\mathfrak{p}}$ generate the $\mathrm{C}_{\mathfrak{p}}$-module $\mathrm{Q}_{\mathfrak{p}}$.

Suppose the lemma proved, and note that it can still be applied after replacing A_0 by $(\mathrm{A}_0)_{t_0}$, and B_0, S_0, M_0, A_λ, and A by the corresponding localized objects, for any $t_0 \in \mathrm{A}_0$, since $(\mathrm{A}_0)_{t_0}$ is flat over A_0. Apply the lemma successively after replacing A_0 by each of the rings $(\mathrm{A}_0)_{f_{ik}}$, and $\mathfrak{R}_0$ by the ideal $\mathfrak{R}_{ik}$ defining the closed subscheme of $\mathrm{D}(f_{ik})$ induced by S_{0i} on the open subset $\mathrm{S}_{0i} \cap \mathrm{D}(f_{ik})$ of S_{0i}. Since the flatness hypothesis of (11.2.9.6) is satisfied in each of these cases by (11.2.9.5), for every pair (i,k) one obtains a finite family $(a_{ikh})_{1 \leqslant h \leqslant l}$ of elements of $\mathrm{Q}_{f_{ik}}$ whose images in $\mathrm{Q}_{\mathfrak{p}_{ik}}$ generate this $\mathrm{C}_{\mathfrak{p}_{ik}}$-module for every prime ideal $\mathfrak{p}_{ik}$ of $\mathrm{B}_{f_{ik}}$ containing $\mathfrak{R}_{ik}\mathrm{B}_{f_{ik}}$. For a suitable integer $d > 0$, one may write

$$a_{ikh} = \frac{b_{ikh}}{f_{ik}^d}$$

for all i,k,h, with $b_{ikh} \in \mathrm{Q}$. Since the S_{0i} cover S_0, every prime ideal $\mathfrak{p}$ of B has image in some $\mathrm{S}_{0i} \cap \mathrm{D}(f_{ik})$, and hence its image $\mathfrak{p}_{ik}$ in $\mathrm{B}_{f_{ik}}$ contains $\mathfrak{R}_{ik}\mathrm{B}_{f_{ik}}$. The images of the b_{ikh} ($1 \leqslant h \leqslant l$) in $\mathrm{Q}_{\mathfrak{p}_{ik}} = \mathrm{Q}_{\mathfrak{p}}$ therefore generate this $\mathrm{C}_{\mathfrak{p}}$-module. It follows that the b_{ikh} ($1 \leqslant i \leqslant m$, $1 \leqslant k \leqslant n$, $1 \leqslant h \leqslant l$) generate the C-module Q (Bourbaki, *Alg. comm.*, chap. II, §3, no. 3, th. 1).

It remains to prove Lemma (11.2.9.6). Set

$$\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}' = \mathrm{C}/\mathfrak{R}\mathrm{C}, \qquad \mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}', \qquad \mathfrak{J}' = \mathfrak{J}\mathrm{B}', \qquad \mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'.$$

By hypothesis $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ is a flat A-module, hence $\mathrm{gr}^*_{\mathfrak{J}'}(\mathrm{M}')$ identifies with $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}'$ (11.2.9.2), and one has the exact sequence ($\mathbf{0}_{\mathrm{I}}$, 6.1.2)

$$0 \longrightarrow \mathrm{Q} \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow (\mathrm{C}')^r \xrightarrow{u \otimes 1_{\mathrm{A}'}} \mathrm{gr}^*_{\mathfrak{J}'}(\mathrm{M}') \longrightarrow 0. \tag{11.2.9.7}$$

Set

$$\mathrm{C}'_0 = \mathrm{C}_0 \otimes_{\mathrm{A}_0} \mathrm{A}'_0 = \mathrm{C}_0/\mathfrak{R}_0\mathrm{C}_0$$

and consider the epimorphism of graded C'_0-modules

$$u'_0 : (\mathrm{C}'_0)^r \xrightarrow{u_0 \otimes 1_{\mathrm{A}'_0}} \mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{M}_0) \otimes_{\mathrm{A}_0} \mathrm{A}'_0 \longrightarrow \mathrm{gr}^*_{\mathfrak{J}'_0}(\mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}'_0).$$

Let $\mathrm{Q}_0 = \mathrm{Ker}(u'_0)$. Using the fact that

$$\mathrm{gr}^*_{\mathfrak{J}'_0}(\mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}'_0)$$

is a flat A'_0-module, one sees that its tensor product over A'_0 with A' identifies with $\mathrm{gr}^*_{\mathfrak{J}'}(\mathrm{M}')$ (11.2.9.2), and one has the exact sequence ($\mathbf{0}_{\mathrm{I}}$, 6.1.2)

$$0 \longrightarrow \mathrm{Q}_0 \otimes_{\mathrm{A}'_0} \mathrm{A}' \longrightarrow (\mathrm{C}')^r \xrightarrow{u \otimes 1_{\mathrm{A}'}} \mathrm{gr}^*_{\mathfrak{J}'}(\mathrm{M}') \longrightarrow 0. \tag{11.2.9.8}$$

whence, by comparison with (11.2.9.7), an isomorphism of graded C'-modules **IV-3** | 131

$$\mathrm{Q} \otimes_{\mathrm{A}} \mathrm{A}' \xrightarrow{\sim} \mathrm{Q}_0 \otimes_{\mathrm{A}'_0} \mathrm{A}'.$$

Since C'_0 is Noetherian, Q_0 is a finite-type C'_0-module; hence $\mathrm{Q}_0 \otimes_{\mathrm{A}'_0} \mathrm{A}'$ is a finite-type graded C'-module, and therefore so is $\mathrm{Q} \otimes_{\mathrm{A}} \mathrm{A}'$. Let q_h ($1 \leqslant h \leqslant s$) be homogeneous elements of Q whose images in $\mathrm{Q} \otimes_{\mathrm{A}} \mathrm{A}'$ generate this C'-module.

Now consider an arbitrary prime ideal $\mathfrak{p} \supset \mathfrak{R}B$ of B. By flatness,

$$C_{\mathfrak{p}} = \mathrm{gr}^*_{\mathfrak{J}_{\mathfrak{p}}}(B_{\mathfrak{p}}),$$

and $\mathrm{gr}^0_{\mathfrak{J}_{\mathfrak{p}}}(B_{\mathfrak{p}}) = B_{\mathfrak{p}}/\mathfrak{J}_{\mathfrak{p}}$ is either the zero ring or a local ring; to prove the lemma, it plainly suffices to consider the second case. Each $(B_{\mathfrak{p}}/\mathfrak{J}_{\mathfrak{p}})$-module $\mathrm{gr}^i_{\mathfrak{J}_{\mathfrak{p}}}(B_{\mathfrak{p}})$ is then of finite presentation. On the other hand, for every i,

$$M/\mathfrak{J}^i M \simeq (M_0/\mathfrak{J}_0^i M_0) \otimes_{A_0} A,$$

and is therefore a finitely presented B-module. By induction on i, the hypothesis that $\mathrm{gr}^*_{\mathfrak{J}}(M)$ is flat over A implies, by $(\mathbf{0}_{\mathrm{I}}, 6.1.2)$, that $M/\mathfrak{J}^i M$ is flat over A. Applying (11.2.6.1), with M_0 replaced by $M_0/\mathfrak{J}_0^k M_0$ for $k \leqslant i$, shows that there is an index λ_i such that, for $\mu \geqslant \lambda_i$, every $M_\mu/\mathfrak{J}_\mu^{k+1} M_\mu$ is flat over A_μ; consequently, again by $(\mathbf{0}_{\mathrm{I}}, 6.1.2)$, every $\mathrm{gr}^k_{\mathfrak{J}_\mu}(M_\mu)$ is flat over A_μ for $k \leqslant i$. It follows from (11.2.9.2) that every $(B/\mathfrak{J})$-module $\mathrm{gr}^i_{\mathfrak{J}}(M)$ is of finite presentation, and hence that $\mathrm{gr}^i_{\mathfrak{J}_{\mathfrak{p}}}(M_{\mathfrak{p}})$ is a finitely presented $(B_{\mathfrak{p}}/\mathfrak{J}_{\mathfrak{p}})$-module.

Moreover, the images of the q_h in

$$Q_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}) = (Q \otimes_A A')_{\mathfrak{p}} \otimes_{A'} \boldsymbol{k}(\mathfrak{p})$$

generate this module over $C_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p})$, since

$$C_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}) = C'_{\mathfrak{p}} \otimes_{A'} \boldsymbol{k}(\mathfrak{p}).$$

Since there is an exact sequence

$$Q_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}) \longrightarrow C^r_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}) \longrightarrow \mathrm{gr}^*_{\mathfrak{J}_{\mathfrak{p}}}(M_{\mathfrak{p}}) \otimes_{B_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}) \longrightarrow 0,$$

it follows that $\mathrm{gr}^*_{\mathfrak{J}_{\mathfrak{p}}}(M_{\mathfrak{p}}) \otimes_{B_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p})$ is a finitely presented module over $C_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p})$. Applying (11.2.9.4), with A, B, and M replaced respectively by $B_{\mathfrak{p}}$, $C_{\mathfrak{p}}$, and $\mathrm{gr}^*_{\mathfrak{J}_{\mathfrak{p}}}(M_{\mathfrak{p}})$, we conclude that $Q_{\mathfrak{p}}$ is a finite-type $C_{\mathfrak{p}}$-module. Nakayama's lemma (and the fact that $C = B$) now shows that the images of the q_h in $Q_{\mathfrak{p}}$ generate this $C_{\mathfrak{p}}$-module. This completes the proof of (11.2.9.6), and also proves that $\mathrm{gr}^*_{\mathfrak{J}}(M)$ is a finitely presented C-module.

VI) *End of the proof.* — For brevity, set $C_\lambda = C_0 \otimes_{A_0} A_\lambda$ (in fact equal to B_λ), $N_\lambda = \mathrm{gr}^*_{\mathfrak{J}_\lambda}(M_\lambda)$, and $N = \mathrm{gr}^*_{\mathfrak{J}}(M)$. First note that, for every integer k, $\mathfrak{J}^k M$ identifies with $\varinjlim_\lambda \mathfrak{J}^k_\lambda M_\lambda$ by exactness of the functor $\varinjlim_\lambda$: the image of $\mathfrak{J}^k_\lambda M_\lambda$ in M_μ for $\lambda \leqslant \mu$ (respectively in M) generates $\mathfrak{J}^k_\mu M_\mu$ (respectively $\mathfrak{J}^k M$). Using exactness of $\varinjlim$ once more, we conclude that N identifies canonically with $\varinjlim_\lambda N_\lambda$. With this identification, we shall first prove:

(11.2.9.9) *For λ sufficiently large, the canonical homomorphism $v_\lambda : N_\lambda \otimes_{A_\lambda} A \longrightarrow N$ is bijective.*

Since the C_λ are Noetherian and N_λ is a finite-type C_λ-module, the C-modules $N_\lambda \otimes_{A_\lambda} A$ are of finite presentation and form a filtered inductive system whose inductive limit identifies canonically with N, since filtered colimits commute with tensor products. Moreover, the transition homomorphisms $v_{\mu\lambda} : N_\lambda \otimes_{A_\lambda} A \to N_\mu \otimes_{A_\mu} A$

for $\lambda \leqslant \mu$, as well as the homomorphisms v_λ, are *surjective*. Fix λ, and regard **IV-3** | 132

$$\mathrm{Ker}(v_{\mu\lambda}) = N'_\mu$$

as a C-submodule of $N_\lambda \otimes_{A_\lambda} A$. By the definition of the inductive limit, the N'_μ form an increasing filtered family whose union is $\mathrm{Ker}(v_\lambda)$. We proved in V) that N is a finitely presented C-module; consequently $\mathrm{Ker}(v_\lambda)$ is a finite-type C-module (Bourbaki, *Alg. comm.*, chap. I, § 2, no. 8, lemma 9). Hence there is an index $\mu \geqslant \lambda$ such that $N'_\mu = \mathrm{Ker}(v_\lambda)$. Since v_μ is surjective, this equality says that v_μ is also injective, and therefore proves (11.2.9.9).

After replacing A_0 and M_0 by A_λ and M_λ for a sufficiently large λ, we may therefore assume that the canonical homomorphism v_λ is bijective for *every* λ. Put

$$P_\lambda = N_0 \otimes_{C_0} C_\lambda = N_0 \otimes_{A_0} A_\lambda,$$

so that $N = \varinjlim P_\lambda$. Since C_0 is a finitely presented A_0-algebra and P_0 is a finitely presented C_0-module, (11.2.6.1), applied with $B_0 = C_0$ and $M_0 = P_0$, shows that there is an index λ such that P_μ is a flat A_μ-module for every $\mu \geqslant \lambda$. For such a μ we have a commutative diagram

$$\begin{tikzcd} P_\mu \arrow[r] \arrow[d, "w_\mu"] & N \arrow[d, equal] \\ N_\mu \arrow[r] & N \end{tikzcd}$$

and the definitions show that w_μ is *surjective*. By III), the homomorphism $A_\mu \to A$ is injective; since P_μ is flat over A_μ, the canonical homomorphism

$$P_\mu \longrightarrow N = P_\mu \otimes_{A_\mu} A$$

is therefore *injective*. The preceding diagram now shows that w_μ is injective as well, hence bijective. Thus N_μ is a flat A_μ-module for $\mu \geqslant \lambda$, which completes the proof of (11.2.9). □

Remark (11.2.10). — We do not know whether the analogue of Raynaud's theorem for the generalization of (11.2.6), (i), is valid.

11.3. Application to the elimination of Noetherian hypotheses IV-3 | 132

Theorem (11.3.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. Then the set U of points $x \in X$ at which $\mathscr{F}$ is f-flat is open in X. If, in addition, $\operatorname{Supp}(\mathscr{F}) = X$, the restriction $f|_U$ is an open morphism.*

Proof. The second assertion has already been proved in (2.4.6) and is included here only for reference.

Since the question is local on X and Y, we may assume that both are affine and that f is of finite presentation. Let $x \in X$ be a point at which $\mathscr{F}$ is f-flat. Applying (11.2.7), we may assume that

$$X = X_0 \times_{Y_0} Y, \qquad f = f_0 \times 1, \qquad \mathscr{F} = \mathscr{F}_0 \otimes_{\mathscr{O}_{Y_0}} \mathscr{O}_Y,$$

where Y_0 is Noetherian, $f_0 : X_0 \to Y_0$ is of finite type, and $\mathscr{F}_0$ is a coherent $\mathscr{O}_{X_0}$-Module. Moreover, if x_0 is the canonical projection of x in X_0, we may assume that $\mathscr{F}_0$ is f_0-flat at x_0. By (11.1.1), the set U_0 of points of X_0 at which $\mathscr{F}_0$ is f_0-flat is therefore a neighbourhood of x_0. It follows from (2.1.4) that $\mathscr{F}$ is f-flat at every point of the inverse image of U_0 in X; thus U is a neighbourhood of x. □

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Corollary (11.3.2). — *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Then the set U of $y \in Y$ such that, for every generization y' of y, $\mathscr{F}$ is $(g \circ f)$-flat at all the points of $f^{-1}(y')$ (i.e. such that $\mathscr{F} \otimes_Y \operatorname{Spec}(\mathscr{O}_{Y,y})$ is flat relative to the morphism $X \times_Y \operatorname{Spec}(\mathscr{O}_{Y,y}) \to Z$) is open in Y.*

Proof. The same reasoning as in (11.1.5) shows that if V is the set of $x \in X$ where $\mathscr{F}$ is $(g \circ f)$-flat, U is the set of $y \in Y$ whose every generization belongs to $E = Y - f(X - V)$. As $g \circ f$ is of finite presentation, V is open in X by virtue of (11.3.1), and hence V is ind-constructible (1.9.6), and hence $X - V$ is pro-constructible; but as f is quasi-compact, $f(X - V)$ is pro-constructible in Y (1.9.5, (vii)), and hence E is ind-constructible in Y. It then follows from (1.10.1) that U is the interior of E, and is therefore open in Y. □

Corollary (11.3.3). — *Let A be a ring, B an A-algebra of finite presentation, C a B-algebra of finite presentation, M a C-module of finite presentation. Then the set of $\mathfrak{q} \in \operatorname{Spec}(B)$ such that $M_\mathfrak{q}$ is a flat A-module is open in $\operatorname{Spec}(B)$.*

Proposition (11.3.4). — *Let $f : X \to S$ be a morphism locally of finite presentation, $\mathscr{J}$ a quasi-coherent Ideal of finite type of $\mathscr{O}_X$, Y the closed sub-prescheme of X defined by $\mathscr{J}$, $\mathscr{F}$ an $\mathscr{O}_X$-Module of finite presentation. Suppose that $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{F})$ is f-flat. Under these conditions:*

(i) *$\mathscr{F}$ is f-flat at the points of Y.*
(ii) *If $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is f-flat, $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is an $(\mathscr{O}_X/\mathscr{J})$-Algebra of finite presentation and $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{F})$ is an $(\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X))$-Module of finite presentation.*

(iii) *Suppose* $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ *is* f*-flat (which implies that* $\mathscr{O}_X/\mathscr{J}$ *is* f*-flat). Then the set* W *of* $y \in Y$ *such that* $(\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F}))_y$ *is a flat* $(\mathscr{O}_X/\mathscr{J})_y$*-module is open in* Y.

(iv) *Suppose* $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ *is* f*-flat. Let* $S' \to S$ *be any morphism, let* $X' = X \times_S S'$*, and let* $p : X' \to X$ *be the canonical projection. Set* $Y' = p^{-1}(Y) = Y \times_S S'$*, the closed sub-prescheme of* X' *defined by* $\mathscr{J}' = p^*(\mathscr{J})\mathscr{O}_{X'}$*, and set* $\mathscr{F}' = p^*(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$*. Then, if* $W' = p^{-1}(W)$*, one has* $\mathrm{gr}^*_{\mathscr{J}'}(\mathscr{F}')|_{W'} \cong p^*(\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})|_W)$*, and* $\mathrm{gr}^*_{\mathscr{J}'}(\mathscr{F}')$ *is a flat* $(\mathscr{O}_{X'}/\mathscr{J}')$*-Module at the points of* W'.

Proof. The questions being local on X and S, we may suppose that $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, where B is an A-algebra of finite presentation, that $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite presentation, and that $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of finite type of B. By virtue of (8.9.1) and (8.5.11), there exist a Noetherian subring A_0 of A, an A_0-algebra of finite type B_0 such that $B = B_0 \otimes_{A_0} A$, an ideal $\mathfrak{J}_0$ of B_0 such that $\mathfrak{J} = \mathfrak{J}_0 B$, and a finite-type B_0-module M_0 such that $M = M_0 \otimes_{A_0} A = M_0 \otimes_{B_0} B$. Moreover, A is the inductive limit of its finite-type A_0-subalgebras A_λ. Set $B_\lambda = B_0 \otimes_{A_0} A_\lambda$, $M_\lambda = M_0 \otimes_{A_0} A_\lambda = M_0 \otimes_{B_0} B_\lambda$, and $\mathfrak{J}_\lambda = \mathfrak{J}_0 B_\lambda$, so that $B = \varinjlim B_\lambda$ and $M = \varinjlim M_\lambda$. By hypothesis, $\mathrm{gr}^*_{\mathfrak{J}}(M)$ is a flat A-module; hence it follows from (11.2.9) that there exists λ such that $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(M_\lambda)$ is a flat A_λ-module. To prove that $\mathscr{F}$ is f-flat at the points of Y, we may therefore restrict to the case where S is Noetherian; but then ($\mathbf{0}_{\mathrm{I}}$, 6.1.2), applied by induction on n, proves that the $\mathscr{F}/\mathscr{J}^{n+1}\mathscr{F}$ are f-flat, and one concludes by ($\mathbf{0}_{\mathrm{III}}$, 10.2.6). **IV-3** | 134

The proof of (ii) reduces similarly to the case where S is Noetherian, using (11.2.9); the conclusion is then evident, since in this case $\mathrm{gr}^*_{\mathfrak{J}}(B)$ is a $(B/\mathfrak{J})$-algebra of finite type and $\mathrm{gr}^*_{\mathfrak{J}}(M)$ is an $\mathrm{gr}^*_{\mathfrak{J}}(B)$-module of finite type.

To prove (iii), one reduces as in (i) to the case where S and X are affine. With the same notation, one may suppose, by virtue of (11.2.9), that $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(B_\lambda)$ and $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(M_\lambda)$ are flat A_λ-modules and that

$$\mathrm{gr}^*_{\mathfrak{J}}(B) = \mathrm{gr}^*_{\mathfrak{J}_\lambda}(B_\lambda) \otimes_{A_\lambda} A, \qquad \mathrm{gr}^*_{\mathfrak{J}}(M) = \mathrm{gr}^*_{\mathfrak{J}_\lambda}(M_\lambda) \otimes_{A_\lambda} A.$$

Consequently, $\mathrm{gr}^*_{\mathfrak{J}}(B)$ is a $(B/\mathfrak{J})$-algebra of finite presentation and $\mathrm{gr}^*_{\mathfrak{J}}(M)$ is an $\mathrm{gr}^*_{\mathfrak{J}}(B)$-module of finite presentation. If W^m is the set of $\mathfrak{p} \in \mathrm{Spec}(B/\mathfrak{J})$ such that $\mathrm{gr}^m_{\mathfrak{J}}(M)_{\mathfrak{p}}$ is a flat $(B/\mathfrak{J})_{\mathfrak{p}}$-module, then W^m is open in $\mathrm{Spec}(B/\mathfrak{J})$ by (11.3.1), and $W = \bigcap_m W^m$; thus W is stable under generization. Assertion (iii) now follows from (11.3.2), applied with $X = \mathrm{Spec}(\mathrm{gr}^*_{\mathfrak{J}}(B))$, $Y = Z = \mathrm{Spec}(B/\mathfrak{J})$, and $\mathscr{F} = (\mathrm{gr}^*_{\mathfrak{J}}(M))^{\sim}$.

Finally, (iv) follows immediately from (11.2.9.2).

Generalising the definition of (6.10.1), let Y be a closed sub-prescheme of a prescheme X, defined by a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module. One says that $\mathscr{F}$ is *normally flat along* Y *at a point* y if $(\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F}))_y$ is a flat $\mathscr{O}_{Y,y}$-module. One says that $\mathscr{F}$ is *normally flat along* Y if this is so at every point of Y. □

Corollary (11.3.5). — *Under the general hypotheses of* (11.3.4)*, suppose that* $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ *and* $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ *are* f*-flat. Then the set* W *of* $y \in Y$ *such that* $\mathscr{F}$ *is normally flat along* Y *at the point* y *is open in* Y*, and (with the notations of* (11.3.4)*, (iv)))* $\mathscr{F}'$ *is normally flat along* Y' *at all points of* $W' = p^{-1}(W)$.

Proposition (11.3.6). — *The notations being those of* (8.5.1) *and* (8.8.1)*, suppose* S_α *quasi-compact,* X_α *of finite presentation over* S_α*,* Y_α *a closed sub-prescheme of* X_α *defined by a quasi-coherent Ideal* $\mathscr{J}_\alpha$ *of finite type of* $\mathscr{O}_{X_\alpha}$*; suppose that* $\mathrm{gr}^*_{\mathscr{J}_\alpha}(\mathscr{O}_{X_\alpha})$ *is flat on* S_α*; finally suppose that* $\mathscr{F}_\alpha$ *is an* $\mathscr{O}_{X_\alpha}$*-Module of finite presentation. For* $\mathscr{F}$ *to be normally flat along* Y*, it is necessary and sufficient that there exist* λ *such that* $\mathscr{F}_\lambda$ *be normally flat along* Y_λ.

Proof. Note that Y (resp. Y_λ) is the closed sub-prescheme of X (resp. X_λ) defined by $\mathscr{J} = \mathscr{J}_\alpha \mathscr{O}_X$ (resp. $\mathscr{J}_\lambda = \mathscr{J}_\alpha \mathscr{O}_{X_\lambda}$). By the hypothesis and (11.2.9.2), one has

$$\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X) = \mathrm{gr}^*_{\mathscr{J}_\alpha}(\mathscr{O}_{X_\alpha}) \otimes_{\mathscr{O}_{S_\alpha}} \mathscr{O}_S, \qquad \mathrm{gr}^*_{\mathscr{J}_\lambda}(\mathscr{O}_{X_\lambda}) = \mathrm{gr}^*_{\mathscr{J}_\alpha}(\mathscr{O}_{X_\alpha}) \otimes_{\mathscr{O}_{S_\alpha}} \mathscr{O}_{S_\lambda} \quad (\lambda \geqslant \alpha),$$

and $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ (resp. $\mathrm{gr}^*_{\mathscr{J}_\lambda}(\mathscr{O}_{X_\lambda})$) is flat over S (resp. S_λ). It follows that Y is flat over S (resp. Y_λ is flat over S_λ). If $\mathscr{F}_\lambda$ is normally flat along Y_λ, then $\mathrm{gr}^*_{\mathscr{J}_\lambda}(\mathscr{F}_\lambda)\big|_{Y_\lambda}$ is flat over Y_λ, hence also over S_λ; and since

$$\mathrm{gr}^n_{\mathscr{J}_\lambda}(\mathscr{F}_\lambda)\big|_{X_\lambda \setminus Y_\lambda} = 0 \quad \text{for } n \geqslant 0,$$

$\mathrm{gr}^*_{\mathscr{J}_\lambda}(\mathscr{F}_\lambda)$ is flat over S_λ. By (11.2.9.2),

$$\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F}) = \mathrm{gr}^*_{\mathscr{J}_\lambda}(\mathscr{F}_\lambda) \otimes_{\mathscr{O}_{\mathrm{S}_\lambda}} \mathscr{O}_{\mathrm{S}},$$

so $\mathscr{F}$ is normally flat along Y.

Conversely, one may suppose that S_α and X_α are affine and adopt the notation of (11.2.9). Since $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$ is a flat A-module, the same is true of $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$; hence, by (11.2.9), there exists $\lambda \geqslant \alpha$ such that $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda)$ is a flat A_λ-module and

$$\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}) = \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda) \otimes_{\mathrm{A}_\lambda} \mathrm{A}.$$

Moreover, by (11.3.4), (ii), $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda)$ is an $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{B}_\lambda)$-module of finite presentation and $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{B}_\lambda)$ is a $(\mathrm{B}_\lambda/\mathfrak{J}_\lambda)$-algebra of finite presentation. The conclusion now follows from (11.2.6). ☐

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Proposition (11.3.7). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism locally of finite presentation, $\mathscr{F}$, $\mathscr{F}'$ two $\mathscr{O}_\mathrm{X}$-Modules quasi-coherent of finite presentation, $u : \mathscr{F}' \to \mathscr{F}$ an $\mathscr{O}_\mathrm{X}$-homomorphism, x a point of X and $y = f(x)$; suppose that $\mathscr{F}$ is f-flat at the point x. The following conditions are equivalent:*

a) $(\mathrm{Ker}\, u)_x = 0$ *and* $\mathrm{Coker}\, u$ *is an $\mathscr{O}_\mathrm{X}$-Module f-flat at the point x.*
b) *The homomorphism $(u \otimes 1)_x : (\mathscr{F}' \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))_x \to (\mathscr{F} \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))_x$ is injective.*

If in addition $\mathscr{F}$ is f-flat, the set of points x satisfying the preceding equivalent conditions is open in X.

Proof. Condition $a)$ implies $b)$ by virtue of $(\mathbf{0}_\mathrm{I}$, 6.1.2), without any hypothesis on $\mathscr{F}$. To prove the converse, we may restrict ourselves to the case where X and Y are affine and then, by virtue of (8.9.1) and (11.2.7), suppose that

$$\mathrm{X} = \mathrm{X}_0 \times_{\mathrm{Y}_0} \mathrm{Y}, \qquad f = f_0 \times 1_\mathrm{Y}, \qquad \mathscr{F} = \mathscr{F}_0 \otimes_{\mathscr{O}_{\mathrm{Y}_0}} \mathscr{O}_\mathrm{Y}, \qquad \mathscr{F}' = \mathscr{F}'_0 \otimes_{\mathscr{O}_{\mathrm{Y}_0}} \mathscr{O}_\mathrm{Y}, \qquad u = u_0 \otimes 1_\mathrm{Y},$$

where Y_0 is Noetherian, $f_0 : \mathrm{X}_0 \to \mathrm{Y}_0$ is a morphism of finite type, $\mathscr{F}_0$ and $\mathscr{F}'_0$ are two coherent $\mathscr{O}_{\mathrm{X}_0}$-Modules, and $u_0 : \mathscr{F}'_0 \to \mathscr{F}_0$ is a homomorphism. Moreover, if x_0 is the projection of x in X_0, we may suppose that $\mathscr{F}_0$ is f_0-flat at x_0. Set $y_0 = f_0(x_0)$. By the transitivity of fibres (**I**, 3.6.4), the projection $f^{-1}(y) \to f_0^{-1}(y_0)$ is faithfully flat (2.2.13); and since

$$(u \otimes 1)_x = ((u_0 \otimes 1)_{x_0}) \otimes 1_{\boldsymbol{k}(y)},$$

the hypothesis that $(u \otimes 1)_x$ is injective implies that $(u_0 \otimes 1)_{x_0}$ is injective (2.2.7). Applying $(\mathbf{0}_\mathrm{III}$, 10.2.4) to the Noetherian local rings $\mathscr{O}_{\mathrm{Y}_0,y_0}$ and $\mathscr{O}_{\mathrm{X}_0,x_0}$ and to the $\mathscr{O}_{\mathrm{X}_0,x_0}$-modules $(\mathscr{F}'_0)_{x_0}$ and $(\mathscr{F}_0)_{x_0}$, the second of which is flat over $\mathscr{O}_{\mathrm{Y}_0,y_0}$, we obtain

$$(\mathrm{Ker}\, u_0)_{x_0} = 0 \quad \text{and} \quad \mathrm{Coker}\, u_0 \text{ is } f_0\text{-flat at } x_0.$$

It follows first that $\mathrm{Coker}\, u$ is f-flat at x (2.1.4). Moreover, by right exactness of the tensor product,

$$\mathrm{Coker}\, u = (\mathrm{Coker}\, u_0) \otimes_{\mathscr{O}_{\mathrm{Y}_0}} \mathscr{O}_\mathrm{Y}.$$

Applying $(\mathbf{0}_\mathrm{I}$, 6.1.2) to the exact sequence

$$0 \longrightarrow (\mathscr{F}'_0)_{x_0} \longrightarrow (\mathscr{F}_0)_{x_0} \longrightarrow (\mathrm{Coker}\, u_0)_{x_0} \longrightarrow 0,$$

we conclude that u_x is injective, which proves condition $a)$.

Finally, it follows from (11.1.2) that the set U_0 of points $z_0 \in \mathrm{X}_0$ such that $(u_0 \otimes 1)_{z_0}$ is injective is open. By flatness, for every $z \in \mathrm{X}$ above a point $z_0 \in \mathrm{U}_0$, the homomorphism $(u \otimes 1)_z$ is injective. The set of such points therefore contains the inverse image U of U_0 in X and is consequently a neighbourhood of x, which completes the proof. ☐

Theorem (11.3.8). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism locally of finite presentation, $\mathscr{F}$ an $\mathscr{O}_\mathrm{X}$-Module quasi-coherent of finite presentation, $(g_i)_{1 \leqslant i \leqslant n}$ a sequence of sections of $\mathscr{O}_\mathrm{X}$ above X; set $\mathscr{F}_i = \mathscr{F} / \sum_{j=1}^{i} g_j \mathscr{F}$ for $0 \leqslant i \leqslant n$ (with $\mathscr{F}_0 = \mathscr{F}$). Let x be a point of X, $y = f(x)$, and set $\mathrm{X}_y = f^{-1}(y) = \mathrm{X} \times_\mathrm{Y} \mathrm{Spec}(\boldsymbol{k}(y))$, $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$, which is an $\mathscr{O}_{\mathrm{X}_y}$-Module. Suppose that the $(g_i)_x$ belong to the maximal ideal $\mathfrak{m}_x$. The following conditions are equivalent:*

a) *The sequence $((g_i)_x)$ is $\mathscr{F}_x$-regular and the $\mathscr{F}_i$ $(0 \leqslant i \leqslant n)$ are f-flat at the point x.*
b) *The sequence $((g_i)_x)$ is $\mathscr{F}_x$-regular and $\mathscr{F}_n$ is f-flat at the point x.*

b′) *There exists a neighbourhood* U *of* x *such that the sequence* $(g_i|\mathrm{U})$ *is* $(\mathscr{F}|\mathrm{U})$*-quasi-regular, and* $\mathscr{F}_n$ *is* f*-flat at the point* x. **IV-3** | 136

c) $\mathscr{F}$ *is* f*-flat at the point* x*, and the sequence* $((g_i \otimes 1)_x)$ *of elements of* $\mathscr{O}_{\mathrm{X}_y,x}$ *images of the* $(g_i)_x$ *is* $(\mathscr{F}_y)_x$*-regular.*

d) $\mathscr{F}$ *is* f*-flat at the point* x*, and for every morphism* $\mathrm{Y}' \to \mathrm{Y}$ *and every point* $x' \in \mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$ *above* x*, if one sets* $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$*, the sequence* $(g_i \otimes 1)_{x'}$ *of elements of* $\mathscr{O}_{\mathrm{X}',x'}$ *is* $\mathscr{F}'_{x'}$*-regular.*

Moreover, the set of $x \in \mathrm{X}$ *satisfying these conditions is open in the set* Z *of* x *such that* $g_i(x) = 0$ *for every* i.

Proof. The fact that *a*) implies *d*) follows immediately from (**0**, 15.1.15), and *c*) is a particular case of *d*); moreover, *a*) implies *b*) trivially. Let us show that *b*) or *c*) implies *a*). Since the $\mathscr{F}_i$ are of finite presentation, the fact that *c*) implies *a*) follows immediately from (11.3.7) by induction on n, and this also shows that the set of $x \in \mathrm{X}$ satisfying *c*) is open in Z.

To show that *b*) implies *a*), induction on n reduces us immediately to the case $n = 1$; we write g instead of g_1. Since the question is local on X and Y, we may suppose that $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$ are affine, that B is an A-algebra of finite presentation, and that $\mathscr{F} = \widetilde{\mathrm{M}}$, where M is a B-module of finite presentation. By (8.9.1), we may then write

$$\mathrm{B} = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}, \qquad \mathrm{M} = \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A},$$

where A_0 is a Noetherian subring of A, B_0 is an A_0-algebra of finite type, and M_0 is a B_0-module of finite type. Moreover, A is the filtered inductive limit of its subrings A_λ that are A_0-algebras of finite type (and hence Noetherian). If we set

$$\mathrm{B}_\lambda = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_\lambda, \qquad \mathrm{M}_\lambda = \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_\lambda,$$

then B_λ is an A_λ-algebra of finite type, M_λ is a B_λ-module of finite type, and

$$\mathrm{B} = \varinjlim_\lambda \mathrm{B}_\lambda, \qquad \mathrm{M} = \varinjlim_\lambda \mathrm{M}_\lambda.$$

There therefore exist an index λ and an element $g_\lambda \in \mathrm{B}_\lambda$ such that $g = g_\lambda \otimes 1$. Returning to geometric notation and setting

$$\mathrm{Y}_\lambda = \mathrm{Spec}(\mathrm{A}_\lambda), \qquad \mathrm{X}_\lambda = \mathrm{Spec}(\mathrm{B}_\lambda), \qquad \mathscr{F}_\lambda = \widetilde{\mathrm{M}_\lambda},$$

it will suffice to prove that there exists $\mu \geqslant \lambda$ such that, at the point $x_\mu \in \mathrm{X}_\mu$ that is the projection of x, $(g_\mu)_{x_\mu}$ is $(\mathscr{F}_\mu)_{x_\mu}$-regular and $\mathscr{F}_\mu/g_\mu\mathscr{F}_\mu$ is flat over Y_μ at x_μ. Indeed, it will then follow from (**0**, 15.1.16) that $\mathscr{F}_\mu$ is flat over Y_μ at x_μ, and hence that $\mathscr{F}$ is flat over Y at x.

The fact that *b*) implies *a*) will therefore follow from the following proposition.

Proposition (11.3.9). — *With the notation and general hypotheses of* (8.5.1) *and* (8.8.1)*, suppose that* $f_\alpha : \mathrm{X}_\alpha \to \mathrm{Y}_\alpha$ *is a morphism locally of finite presentation. Let* $\mathscr{F}_\alpha$ *and* $\mathscr{G}_\alpha$ *be two quasi-coherent* $\mathscr{O}_{\mathrm{X}_\alpha}$*-Modules of finite presentation, let* $h_\alpha : \mathscr{F}_\alpha \to \mathscr{G}_\alpha$ *be an* $\mathscr{O}_{\mathrm{X}_\alpha}$*-homomorphism, and let* $\mathscr{H}_\alpha$ *be its cokernel. Finally, let* x *be a point of* X*, and let* x_λ *be its projection in* X_λ *for* $\lambda \geqslant \alpha$*. In order that* h_x *be injective and that* $\mathscr{H} = \mathrm{Coker}\, h$ *be* f*-flat at* x*, it is necessary and sufficient that there exist* $\lambda \geqslant \alpha$ *such that* $(h_\lambda)_{x_\lambda}$ *is injective and* $\mathscr{H}_\lambda = \mathrm{Coker}\, h_\lambda$ *is* f_λ*-flat at* x_λ*. Moreover, the set of* $x \in \mathrm{X}$ *having the preceding properties is open in* X.

Proof. Recall that, by the right exactness of the tensor product,

$$\mathscr{H}_\mu = v^*_{\lambda\mu}(\mathscr{H}_\lambda) \quad \text{for } \lambda \leqslant \mu, \qquad \mathscr{H} = v^*_\lambda(\mathscr{H}_\lambda),$$

which justifies the notation.

The sufficiency of the condition follows from the fact that, if the sequence

$$0 \longrightarrow (\mathscr{F}_\lambda)_{x_\lambda} \longrightarrow (\mathscr{G}_\lambda)_{x_\lambda} \longrightarrow (\mathscr{H}_\lambda)_{x_\lambda} \longrightarrow 0$$

is exact and $(\mathscr{H}_\lambda)_{x_\lambda}$ is a flat $\mathscr{O}_{f_\lambda(x_\lambda)}$-module, then $\mathscr{H}_x$ is a flat $\mathscr{O}_{f(x)}$-module by base change (2.1.4) and the sequence

$$0 \longrightarrow \mathscr{F}_x \longrightarrow \mathscr{G}_x \longrightarrow \mathscr{H}_x \longrightarrow 0$$

is exact (2.1.8). To prove that

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the condition is necessary, note that $\mathscr{H}$ is of finite presentation. Since the question is local on X and Y, we may suppose that X and Y are affine and, by (11.3.1), that $\mathscr{H}$ is f-flat. We now record the following lemma.

Lemma (11.3.9.1). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism locally of finite presentation, $\mathscr{G}$, $\mathscr{H}$ two $\mathscr{O}_{\mathrm{X}}$-Modules quasi-coherent of finite presentation such that $\mathscr{H}$ is f-flat, $p : \mathscr{G} \to \mathscr{H}$ a surjective homomorphism. Then* $\mathrm{Ker}(p)$ *is an $\mathscr{O}_{\mathrm{X}}$-Module of finite presentation.*

Proof. We may indeed suppose that X and Y are affine and that there exist a morphism $\mathrm{Y} \to \mathrm{Y}_0$, where Y_0 is Noetherian, a morphism $f_0 : \mathrm{X}_0 \to \mathrm{Y}_0$ of finite type such that $\mathrm{X} = \mathrm{X}_0 \times_{\mathrm{Y}_0} \mathrm{Y}$ and $f = (f_0)_{(\mathrm{Y})}$, two coherent $\mathscr{O}_{\mathrm{X}_0}$-Modules $\mathscr{G}_0$, $\mathscr{H}_0$, and a homomorphism $p_0 : \mathscr{G}_0 \to \mathscr{H}_0$ such that $\mathscr{G}$, $\mathscr{H}$, and p are obtained from $\mathscr{G}_0$, $\mathscr{H}_0$, and p_0 by base change (8.9.1 and 8.5.2). We may moreover suppose that p_0 is surjective (8.5.7) and that $\mathscr{H}_0$ is f_0-flat (11.2.7). Then, if $\mathscr{K}_0 = \mathrm{Ker}(p_0)$, $\mathscr{K}_0$ is a coherent $\mathscr{O}_{\mathrm{X}_0}$-Module ($\mathbf{0}_{\mathrm{I}}$, 5.3.4), and by ($\mathbf{0}_{\mathrm{I}}$, 6.1.2), $\mathscr{K} = \mathrm{Ker}(p)$ is obtained from $\mathscr{K}_0$ by base change and is therefore of finite presentation. □

This lemma being proved, set $\mathscr{K} = \mathrm{Ker}(\mathscr{G} \to \mathscr{H})$, which is therefore of finite presentation. There is a canonical homomorphism $q : \mathscr{F} \to \mathscr{K}$, and by hypothesis $q_x : \mathscr{F}_x \to \mathscr{K}_x$ is an isomorphism. Hence, by ($\mathbf{0}_{\mathrm{I}}$, 5.2.7), there exists a neighbourhood U of x such that $q|_{\mathrm{U}}$ is an isomorphism; after restricting X, we may therefore suppose that the sequence

$$0 \longrightarrow \mathscr{F} \xrightarrow{h} \mathscr{G} \longrightarrow \mathscr{H} \longrightarrow 0$$

is *exact*. This being so, it follows from (11.2.6) that, for all sufficiently large λ, $\mathscr{H}_\lambda$ is f_λ-flat. Set $\mathscr{K}_\lambda = \mathrm{Ker}(\mathscr{G}_\lambda \to \mathscr{H}_\lambda)$ and $\mathscr{K}_\mu = v^*_{\lambda\mu}(\mathscr{K}_\lambda)$ for $\mu \geqslant \lambda$. It follows from ($\mathbf{0}_{\mathrm{I}}$, 6.1.2) that

$$\mathscr{K}_\mu = \mathrm{Ker}(\mathscr{G}_\mu \longrightarrow \mathscr{H}_\mu) \quad \text{and} \quad \mathscr{K} = v^*_\lambda(\mathscr{K}_\lambda) = \mathrm{Ker}(\mathscr{G} \longrightarrow \mathscr{H}) = \mathscr{F}.$$

On the other hand, for every $\mu \geqslant \lambda$ there is a canonical homomorphism $q_\mu : \mathscr{F}_\mu \to \mathscr{K}_\mu$ such that $q_\mu = v^*_{\lambda\mu}(q_\lambda)$ and $q = v^*_\mu(q_\mu)$. Since q is an isomorphism, it follows from (8.5.2.4) and (11.3.9.1) that there exists $\mu \geqslant \lambda$ for which q_μ is an isomorphism, and hence $u_\mu : \mathscr{F}_\mu \to \mathscr{G}_\mu$ is injective. □

Let us return to the proof of (11.3.8).

Since the set of $x \in \mathrm{X}$ satisfying $b)$ is open in X, it is clear that $b)$ implies $b')$. Let us finally show that $b')$ implies $c)$. First, $\mathscr{F}_n$ is f-flat in a neighbourhood of x (11.3.1); we may therefore reduce to the case where $\mathrm{U} = \mathrm{X}$ and $\mathscr{F}_n$ is f-flat. Since, by definition, $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ is isomorphic to $\mathscr{F}_n \otimes_{\mathscr{O}_{\mathrm{Y}}} \mathscr{O}_{\mathrm{Y}}[\mathrm{T}_1, \dots, \mathrm{T}_n]$ ($\mathbf{0}$, 15.2.2), it is f-flat, and it follows from (11.3.4), (i), that $\mathscr{F}$ itself is f-flat in a neighbourhood of x. On the other hand, if $(\mathscr{J}_y)_x$ is the ideal of $\mathscr{O}_{\mathrm{X}_y,x}$ generated by the $(g_i \otimes 1)_x$, it follows from (11.2.9.2) that, in the diagram

$$\begin{tikzcd}
((\mathrm{gr}^0_{\mathscr{J}_x}(\mathscr{F}_x))[\mathrm{T}_1, \dots, \mathrm{T}_n]) \otimes_{\mathscr{O}_y} \boldsymbol{k}(y) \arrow[r] \arrow[d] & \mathrm{gr}^*_{\mathscr{J}_x}(\mathscr{F}_x) \otimes_{\mathscr{O}_y} \boldsymbol{k}(y) \arrow[d] \\
(\mathrm{gr}^0_{(\mathscr{J}_y)_x}((\mathscr{F}_y)_x))[\mathrm{T}_1, \dots, \mathrm{T}_n] \arrow[r] & \mathrm{gr}^*_{(\mathscr{J}_y)_x}((\mathscr{F}_y)_x)
\end{tikzcd}$$

the vertical arrows are isomorphisms. Since the first horizontal arrow is an isomorphism, so is the second; hence the sequence $((g_i \otimes 1)_x)$ is $(\mathscr{F}_y)_x$-quasi-regular, and therefore regular, since X_y is locally of finite type over $\boldsymbol{k}(y)$. □ **IV-3** | 138

Theorem (11.3.10). — *(Flatness criterion by fibres). — Let* S *be a prescheme, $g : \mathrm{X} \to \mathrm{S}$, $h : \mathrm{Y} \to \mathrm{S}$ two morphisms, $f : \mathrm{X} \to \mathrm{Y}$ an* S*-morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_{\mathrm{X}}$-Module, x a point of* X*, $y = f(x)$, $s = h(y) = g(x)$. Suppose that one of the following two hypotheses holds:*

1º S, X *and* Y *are locally Noetherian and $\mathscr{F}$ is coherent.*

2º g and h are locally of finite presentation and $\mathscr{F}$ is of finite presentation.

Then, with the notation of (9.4.1)*, if $\mathscr{F}_x \neq 0$, the following conditions are equivalent:*

a) *$\mathscr{F}$ is g-flat at the point x and $\mathscr{F}_s$ is f_s-flat at the point x.*

b) *h is flat at the point y and $\mathscr{F}$ is f-flat at the point x.*

Moreover, under hypothesis 2º, the set of $x \in \mathrm{X}$ satisfying condition $b)$ is open in X.

Proof. The last assertion results from (11.3.1) applied to $\mathscr{O}_{\mathrm{Y}}$ and to the morphism h on the one hand, and to $\mathscr{F}$ and to the morphism f (which is locally of finite presentation) on the other.

As $g = h \circ f$, it is clear that $b)$ implies $a)$ without assuming 1° or 2° (2.1.6 and 2.1.4). To prove that $a)$ implies $b)$, we may reduce to the case where S, X, and Y are affine; under hypothesis 2°, by applying (11.2.7), we reduce further to the case where S is *Noetherian*, that is, to the case where hypothesis 1° holds. The assertion is then equivalent to the following lemma, which improves ($\mathbf{0}_{\mathrm{III}}$, 10.2.5):

Lemma (11.3.10.1). — *Let $\rho : \mathrm{A} \to \mathrm{B}$, $\sigma : \mathrm{B} \to \mathrm{C}$ be homomorphisms of local Noetherian rings, k the residue field of* A, M *a* C*-module $\neq 0$ of finite type. Then the following conditions are equivalent:*

a) M *is a flat* A*-module and $\mathrm{M} \otimes_{\mathrm{A}} k$ is a flat $(\mathrm{B} \otimes_{\mathrm{A}} k)$-module.*
b) B *is a flat* A*-module and* M *is a flat* B*-module.*

We will first establish the more general following lemma:

Lemma (11.3.10.2). — *Let* A *be a commutative ring,* B *a commutative* A*-algebra, $\mathfrak{J}$ an ideal of* A*, and* M *a* B*-module. Consider first the following conditions:*

(i) *$\mathfrak{J}$ is nilpotent.*
(ii) B *is Noetherian and* M *is ideally separated for the $\mathfrak{J}\mathrm{B}$-preadic topology* ($\mathbf{0}_{\mathrm{III}}$, 10.2.1).
(iii) *$\mathfrak{J}\mathrm{B}$ is contained in the radical of* B.

Consider, on the other hand, the following four properties:

a) M *is a flat* B*-module.*
b) B *is a flat* A*-module.*
c) M *is a flat* A*-module and $\mathrm{M}/\mathfrak{J}\mathrm{M}$ is a flat $(\mathrm{B}/\mathfrak{J}\mathrm{B})$-module.*
d) *$\mathrm{B}/\mathfrak{J}\mathrm{B}$ is a flat $(\mathrm{A}/\mathfrak{J})$-module and for every maximal ideal $\mathfrak{m} \supset \mathfrak{J}\mathrm{B}$ of* B *one has $\mathfrak{m}\mathrm{M} \neq \mathrm{M}$.*

Then:

1° If one of the conditions (i) or (ii) holds, the conjunction of $a)$ and $b)$ implies $c)$, and $c)$ implies $a)$.
2° If the condition (i) or the conjunction of (ii) and (iii) is verified, the conjunction of $c)$ and $d)$ implies the conjunction of $a)$ and $b)$. **IV-3** | 139

Proof. 1° The first assertion is immediate ($\mathbf{0}_{\mathrm{I}}$, 6.2.1). Suppose therefore $c)$ verified, and let us prove $a)$. Consider the graded rings $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{A})$, $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$ and the graded module $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ (over both $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{A})$ and $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$) relative to the $\mathfrak{J}$-preadic filtrations, as well as the surjective canonical maps ($\mathbf{0}_{\mathrm{III}}$, 10.1.1.2)

$$u : \mathrm{gr}^0(\mathrm{B}) \otimes_{\mathrm{gr}^0(\mathrm{A})} \mathrm{gr}^*(\mathrm{A}) \longrightarrow \mathrm{gr}^*(\mathrm{B})$$
$$v : \mathrm{gr}^0(\mathrm{M}) \otimes_{\mathrm{gr}^0(\mathrm{A})} \mathrm{gr}^*(\mathrm{A}) \longrightarrow \mathrm{gr}^*(\mathrm{M})$$
$$w : \mathrm{gr}^0(\mathrm{M}) \otimes_{\mathrm{gr}^0(\mathrm{B})} \mathrm{gr}^*(\mathrm{B}) \longrightarrow \mathrm{gr}^*(\mathrm{M})$$

It is clear there is a commutative diagram

(11.3.10.3)
$$\begin{tikzcd}
\mathrm{gr}^0(\mathrm{M}) \otimes_{\mathrm{gr}^0(\mathrm{A})} \mathrm{gr}^*(\mathrm{A}) \arrow[rr, "v"] \arrow[dr, "1\otimes u"'] & & \mathrm{gr}^*(\mathrm{M}) \\
& \mathrm{gr}^0(\mathrm{M}) \otimes_{\mathrm{gr}^0(\mathrm{B})} \mathrm{gr}^*(\mathrm{B}) \arrow[ur, "w"'] &
\end{tikzcd}$$

The hypothesis $c)$ implies that v is bijective ($\mathbf{0}_{\mathrm{III}}$, 10.2.1); since the other two maps in the diagram are surjective, they too are bijective. Since, by hypothesis $c)$, $\mathrm{M}/\mathfrak{J}\mathrm{M}$ is a flat $(\mathrm{B}/\mathfrak{J}\mathrm{B})$-module, it follows from ($\mathbf{0}_{\mathrm{III}}$, 10.2.1) that M is a flat B-module.

2° Either of the conditions (i), (iii) implies that every maximal ideal of B contains $\mathfrak{J}\mathrm{B}$. It therefore follows from 1° and the conjunction of $c)$ and $d)$ that M is a *faithfully flat* B-module, and consequently that $\mathrm{gr}^0(\mathrm{M})$ is a *faithfully flat* $\mathrm{gr}^0(\mathrm{B})$-module ($\mathbf{0}_{\mathrm{I}}$, 6.2.1). We saw in 1° that hypothesis $c)$ implies that the three maps in diagram (11.3.10.3) are bijective; the faithful flatness of $\mathrm{gr}^0(\mathrm{M})$ over $\mathrm{gr}^0(\mathrm{B})$ therefore implies that u is bijective ($\mathbf{0}_{\mathrm{I}}$, 6.4.1). On the other hand, conditions (ii) and (iii) imply that B, as an A-module, is ideally separated for the $\mathfrak{J}$-preadic filtration (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, prop. 2). It follows again from ($\mathbf{0}_{\text{III}}$, 10.2.1) that, if condition (i), or the conjunction of (ii) and (iii), holds, B is a flat A-module. □

(11.3.10.4) Lemma (11.3.10.2) being established, one deduces (11.3.10.1) by taking for $\mathfrak{J}$ the maximal ideal of A and observing that conditions (ii) and (iii) of (11.3.10.2) are then satisfied (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, prop. 2). This therefore also completes the proof of (11.3.7). □

Corollary (11.3.11). — *Let $g : X \to S$, $h : Y \to S$ be two morphisms locally of finite presentation, $f : X \to Y$ an S-morphism. The following conditions are equivalent:*

a) *g is flat and $f_s : X_s \to Y_s$ is flat for every $s \in S$.*
b) *h is flat at all the points of $f(X)$ and f is flat.*

Proof. It suffices to apply (11.3.10) for $\mathscr{F} = \mathscr{O}_X$. □

Remark (11.3.12). — It would be interesting to be able to give proofs of (11.3.2) and (11.3.10) not using passage to the inductive limit; it would suffice for this to prove the following criterion:

Let A be a ring, B an A-algebra of finite presentation, M a B-module of finite presentation, $\mathfrak{J}$ an ideal of A, and $\mathfrak{p}$ a prime ideal of B containing $\mathfrak{J}$B. Then the following two conditions are equivalent: **IV-3** | 140

a) $M_{\mathfrak{p}}$ is a flat A-module.
b) $M_{\mathfrak{p}}/\mathfrak{J}M_{\mathfrak{p}}$ is a flat $(A/\mathfrak{J})$-module and $\mathrm{Tor}_1^A(A/\mathfrak{J}, M_{\mathfrak{p}}) = 0$.

When A is *Noetherian*, this follows from ($\mathbf{0}_{\text{III}}$, 10.2.2) applied to the Noetherian A-algebra $B_{\mathfrak{p}}$; can the general statement be deduced from this by passage to the inductive limit?

It should be noted in this connection that such a generalization is certainly not possible if condition *b*) above is replaced by either of conditions *c*), *d*) of ($\mathbf{0}_{\text{III}}$, 10.2.1), with M replaced by $M_{\mathfrak{p}}$. For example, let A be a local ring whose maximal ideal $\mathfrak{J}$ is principal and such that

$$\mathfrak{R} = \bigcap_{n \geqslant 0} \mathfrak{J}^{n+1}$$

is nonzero (for instance, the ring of germs of indefinitely differentiable real-valued functions in a neighbourhood of 0 in **R**). Take $B = A$, $\mathfrak{p} = \mathfrak{J}$, and $M = A/\mathfrak{N}$, where $\mathfrak{N}$ is a cyclic nonzero submodule of $\mathfrak{R}$; M is therefore of finite presentation. It is clear that M is not a flat A-module, for, being of finite presentation, it would then be free (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. 2 to prop. 5), which is absurd since $\mathfrak{N} \neq 0$. Nevertheless,

$$\mathfrak{J}^k M/\mathfrak{J}^{k+n} M \simeq A/\mathfrak{J}^n$$

for all positive integers k and n, so M does satisfy conditions *c*) and *d*) of ($\mathbf{0}_{\text{III}}$, 10.2.1), since $A/\mathfrak{J}$ is a field.

Proposition (11.3.13). — *Let $f : X \to Y$ be a morphism of preschemes and let x be a point of X such that f is flat at x; set $y = f(x)$.*

(i) *If X is reduced (resp. normal) at the point x, then Y is reduced (resp. normal) at the point y.*
(ii) *Suppose in addition that f is of finite presentation at the point x. Then, if Y is reduced (resp. normal) at the point y and if the morphism f is reduced (resp. normal) at the point x* (6.8.1), *X is reduced (resp. normal) at the point x.*

Proof. The first assertion is recalled here only for convenience, having already been proved in (2.1.13). To prove (ii), one may restrict to the case where $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$, where A is a reduced local ring (resp. an integral and integrally closed local ring) and B is an A-algebra of finite presentation. One may then write (8.9.1)

$$B = B_0 \otimes_{A_0} A,$$

where A_0 is a finite type **Z**-subalgebra of A and B_0 is a finite type A_0-algebra. Let (C_α) be the filtered increasing family of finite type A_0-subalgebras of A, which are therefore finite type **Z**-algebras; one has $A = \varinjlim C_\alpha$. We distinguish the two cases:

1° Suppose that A is reduced and that f is reduced at x. If $\mathfrak{m}$ is the maximal ideal of A, let $\mathfrak{p}_\alpha = \mathfrak{m} \cap C_\alpha$ and set $A'_\alpha = (C_\alpha)_{\mathfrak{p}_\alpha}$, so that one also has $A = \varinjlim A'_\alpha$ (5.13.3). Set $Y_\alpha = \mathrm{Spec}(A'_\alpha)$ and $X_\alpha = \mathrm{Spec}(B_0 \otimes_{A_0} A'_\alpha)$, and let $f_\alpha : X_\alpha \to Y_\alpha$, x_α (resp. y_α) be the projection of x (resp. y) in X_α (resp. Y_α). Since f is reduced at x, there exists α_0 such that f_α is reduced at x_α for $\alpha \geqslant \alpha_0$ ((6.7.8) and

(11.2.6)); since Y_α and X_α are Noetherian and A'_α is reduced (because C_α is), it follows from (3.3.5) that X_α is reduced at x_α. But

$$\mathrm{B} = \varinjlim(\mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}'_\alpha)$$

and consequently $\mathscr{O}_{\mathrm{X},x} = \varinjlim \mathscr{O}_{\mathrm{X}_\alpha,x_\alpha}$ (5.13.3); hence $\mathscr{O}_{\mathrm{X},x}$ is reduced (5.13.2).

2° Suppose that A is integrally closed and that f is normal at x. Since C_α is universally Japanese (7.7.4), its integral closure C'_α is a finite C_α-algebra, evidently contained in A. Let $\mathfrak{p}'_\alpha = \mathfrak{m} \cap \mathrm{C}'_\alpha$ and set $\mathrm{A}''_\alpha = (\mathrm{C}'_\alpha)_{\mathfrak{p}'_\alpha}$, so that A''_α is a local Noetherian integral and integrally closed ring, and $\mathrm{A} = \varinjlim \mathrm{A}''_\alpha$ (5.13.3). Set $\mathrm{Y}'_\alpha = \mathrm{Spec}(\mathrm{A}''_\alpha)$ and $\mathrm{X}'_\alpha = \mathrm{Spec}(\mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}''_\alpha)$, and let $f'_\alpha : \mathrm{X}'_\alpha \to \mathrm{Y}'_\alpha$, x'_α (resp. y'_α) be the projection of x (resp. y) in X'_α (resp. Y'_α). Since f is normal at x, there exists α_0 such that, for $\alpha \geqslant \alpha_0$, f'_α is normal at x'_α ((6.7.8) and (11.2.6)); since X'_α and Y'_α are Noetherian and A''_α is integrally closed, it follows from (6.5.4) that X'_α is normal at x'_α. But the morphisms **IV-3 | 141**

$$\mathrm{Spec}(\mathrm{A}''_\beta) \longrightarrow \mathrm{Spec}(\mathrm{A}''_\alpha) \qquad (\alpha \leqslant \beta)$$

are dominant; hence, since by virtue of (11.3.1) we may suppose the f'_α flat for $\alpha \geqslant \alpha_0$, it follows from (2.3.7) that every irreducible component of X'_β dominates an irreducible component of X'_α. We then conclude from (5.13.4), applied to the schemes

$$\mathrm{Spec}\big(\mathscr{O}_{\mathrm{X}_0,x_0} \otimes_{\mathrm{A}_0} \mathrm{A}''_\alpha\big),$$

that X is normal at x. □

Corollary (11.3.14). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism locally of finite presentation, x a point of* X, *and $y = f(x)$. If* Y *is geometrically unibranch at y* (6.15.1) *and if f is normal at x* (6.8.1), *then* X *is geometrically unibranch at x.*

Proof. We may evidently reduce to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, with A a geometrically unibranch local integral domain and y the closed point of Y. Let A' be the integral closure of A; set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ and let y' be the closed point of Y'. Thus the morphism $g : \mathrm{Y}' \to \mathrm{Y}$ is radicial at y (6.15.3), integral, and birational. Set $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, and let $f' : \mathrm{X}' \to \mathrm{Y}'$ and $g' : \mathrm{X}' \to \mathrm{X}$ be the projections. The morphism g' is integral and radicial at x (6.15.3.1). Let x' therefore be the unique point of $g'^{-1}(x)$ lying over y'. The morphism f' is of finite presentation and normal at x' (6.7.8); consequently the local ring $\mathscr{O}_{\mathrm{X}',x'}$ is an integrally closed integral domain (11.3.13). On the other hand, we may suppose f flat (11.3.1), and hence g' is birational (6.15.4.1). It follows that $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ is irreducible; since $\mathscr{O}_{\mathrm{X},x}$ is reduced (11.3.13), it is an integral domain and is geometrically unibranch by (6.15.5). □

Proposition (11.3.15). — *Let* A *be a ring,* B *an* A*-algebra of finite presentation, and* M *a* B*-module of finite presentation which is flat as an* A*-module. Then there exists a finite sequence $(f_i)_{1 \leqslant i \leqslant n}$ of elements of* A *such that the ideal generated by the f_i is* A *and, for $0 \leqslant i < n$,*

$$\frac{\mathrm{M}_{f_{i+1}}}{\sum_{j=1}^{i} f_j \mathrm{M}_{f_{i+1}}}$$

is a free module over

$$\frac{\mathrm{A}_{f_{i+1}}}{\sum_{j=1}^{i} f_j \mathrm{A}_{f_{i+1}}}.$$

Equivalently, the $\mathrm{D}(f_i)$ form an open covering of $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ and, if one sets $\mathrm{Z}_1 = \mathrm{D}(f_1)$ and then, inductively,

$$\mathrm{Z}_{i+1} = \mathrm{D}(f_{i+1}) \cap \mathrm{V}(f_1\mathrm{A} + \cdots + f_i\mathrm{A}),$$

the Z_i form a partition of X into locally closed subsets, while

$$\frac{\mathrm{A}_{f_{i+1}}}{\sum_{j=1}^{i} f_j \mathrm{A}_{f_{i+1}}}$$

is the ring of an affine subscheme of X whose underlying space is Z_{i+1}.

Proof. By virtue of (11.2.7), we may first suppose that there exist a Noetherian subring A_0 of A, a finite type A_0-algebra B_0, and a finite type B_0-module M_0, flat over A_0, such that

$$B = B_0 \otimes_{A_0} A, \qquad M = M_0 \otimes_{A_0} A.$$

It clearly suffices to prove the proposition for A_0, B_0, and M_0: if elements $f_i \in A_0$ satisfy the conditions in the statement in that case, then they also satisfy them for A, B, and M, since

$$\frac{M_{f_{i+1}}}{\sum_{j=1}^{i} f_j M_{f_{i+1}}} = \left(\frac{(M_0)_{f_{i+1}}}{\sum_{j=1}^{i} f_j (M_0)_{f_{i+1}}}\right) \otimes_{A_0} A$$

(Bourbaki, *Alg. comm.*, chap. II, § 2, nº 7, prop. 18). We may therefore henceforth restrict to the case where A is Noetherian.

Let us now note that if C is a Noetherian ring, $\mathfrak{N}$ its nilradical, and P a flat C-module, then it follows from $(\mathbf{0}_{\mathrm{III}}, 10.1.2)$ that, for P to be a free C-module, it is necessary and sufficient that $P \otimes_C (C/\mathfrak{N})$ be a free $(C/\mathfrak{N})$-module. Note also that if C is a reduced Noetherian ring, there exists $g \in C$ such that C_g is an integral domain. **IV-3** | 142

We now apply Lemma (6.9.2). A sequence $(f_i)_{i \geqslant 1}$ of elements of A may be defined inductively as follows:

1° f_1 is chosen so that

$$A_1 = (A_{\mathrm{red}})_{f_1}$$

is an integral domain and $M \otimes_A A_1$ is a free A_1-module;

2° if the ideal $\mathfrak{J}_i$ generated by $f_1, \ldots, f_i$ is equal to A, set $f_{i+1} = f_i$;

3° if, on the contrary, $\mathfrak{J}_i \neq A$, choose $f_{i+1} \notin \mathfrak{J}_i$ so that

$$A_{i+1} = \big((A/\mathfrak{J}_i)_{\mathrm{red}}\big)_{f_{i+1}}$$

is an integral domain and $M \otimes_A A_{i+1}$ is a free A_{i+1}-module.

Since A is Noetherian, the increasing sequence $(\mathfrak{J}_i)$ is stationary. Hence there exists n such that $f_1, \ldots, f_n$ generate the unit ideal of A, and these elements satisfy the required conditions because

$$\big((A/\mathfrak{J}_i)_{\mathrm{red}}\big)_{f_{i+1}} = \big((A/\mathfrak{J}_i)_{f_{i+1}}\big)_{\mathrm{red}}.$$

□

Proposition (11.3.16). — *Let $f : X \to Y$ be a faithfully flat morphism of finite presentation, and let $g : Y \to Z$ be a morphism such that $g \circ f : X \to Z$ is of finite type (resp. of finite presentation). Then g is of finite type (resp. of finite presentation).*

Proof. Since f is surjective and $g \circ f$ is quasi-compact, g is quasi-compact (1.1.3). Let us first show that if $g \circ f$ is of finite presentation, then g is quasi-separated. Indeed, let $W \subset Z$ be an affine open subset and let (V_i) be a finite affine open covering of $g^{-1}(W)$; it is enough to show that the $V_i \cap V_j$ are quasi-compact (1.2.6, 1.2.7). For each i, let (U_{ih}) be a finite affine open covering of $f^{-1}(V_i)$. Since f is of finite presentation, all the intersections $U_{ih} \cap U_{jk}$ are quasi-compact. Since f is surjective, $V_i \cap V_j$ is the union, as h and k vary, of the images $f(U_{ih} \cap U_{jk})$; it is therefore quasi-compact because f is continuous.

The question is thus local on Y, and we may suppose that $Y = \mathrm{Spec}(B)$ and $Z = \mathrm{Spec}(A)$ are affine, while X is the union of finitely many affine open subsets $X_i = \mathrm{Spec}(C_i)$, where the C_i are B-algebras of finite type (resp. of finite presentation). Let X' be the disjoint union of the X_i, and let $p : X' \to X$ be the morphism which on each X_i is the canonical open immersion. The morphism p is faithfully flat and of finite presentation (1.6.5); consequently $g \circ f \circ p$ is of finite type (resp. of finite presentation), and $f \circ p$ is faithfully flat and of finite presentation. We may therefore suppose in addition that $X = \mathrm{Spec}(C)$ is affine, and the proposition is reduced to the following corollary. □

Corollary (11.3.17). — *Let* A *be a ring,* B *an* A*-algebra, and* C *a* B*-algebra of finite presentation which is faithfully flat as a* B*-module. If* C *is an* A*-algebra of finite type (resp. of finite presentation), then* B *is an* A*-algebra of finite type (resp. of finite presentation).*

Proof. Suppose first that C is of finite type over A. Let (B_λ) be the increasing filtered family of finite type A-subalgebras of B. By (8.8.2), there exists an index α such that

$$C = C_\alpha \otimes_{B_\alpha} B,$$

where C_α is a finitely presented B_α-algebra. Moreover, by (8.10.5) and (11.2.6), we may suppose that C_α is faithfully flat as a B_α-module. For $\lambda \geqslant \alpha$, put

$$C_\lambda = C_\alpha \otimes_{B_\alpha} B_\lambda.$$

Then $C = C_\lambda \otimes_{B_\lambda} B$ and C_λ is faithfully flat as a B_λ-module. Since $B_\lambda \to B$ is injective, so is $C_\lambda \to C$. Furthermore, $C = \varinjlim C_\lambda$, so every element of C belongs to some C_λ; since C is of finite type over A, there exists λ for which $C_\lambda \to C$ is bijective. Faithful flatness then implies that $B_\lambda \to B$ is bijective. **IV-3** | 143
Thus $B = B_\lambda$ is of finite type over A.

Suppose now that C is of finite presentation over A. By what has just been proved, B is of finite type over A; write

$$B = A[T_1, \ldots, T_n]/\widetilde{\mathfrak{J}}$$

for an ideal $\widetilde{\mathfrak{J}}$. Let $(\widetilde{\mathfrak{J}}_\lambda)$ be the filtered family of finite type ideals of $A[T_1, \ldots, T_n]$ contained in $\widetilde{\mathfrak{J}}$. Thus

$$\widetilde{\mathfrak{J}} = \varinjlim \widetilde{\mathfrak{J}}_\lambda, \qquad B = \varinjlim B_\lambda, \qquad B_\lambda = A[T_1, \ldots, T_n]/\widetilde{\mathfrak{J}}_\lambda.$$

Applying (8.8.2), (8.10.5), and (11.2.6) as above, we have

$$C = C_\alpha \otimes_{B_\alpha} B,$$

where C_α is a finitely presented B_α-algebra and a faithfully flat B_α-module. For $\lambda \geqslant \alpha$, put

$$C_\lambda = C_\alpha \otimes_{B_\alpha} B_\lambda.$$

By flatness,

$$C_\lambda = C_\alpha \Big/ \Big(C_\alpha \otimes_{B_\alpha} (\widetilde{\mathfrak{J}}_\lambda/\widetilde{\mathfrak{J}}_\alpha)\Big),$$

and similarly

$$C = C_\alpha \Big/ \Big(C_\alpha \otimes_{B_\alpha} (\widetilde{\mathfrak{J}}/\widetilde{\mathfrak{J}}_\alpha)\Big).$$

Since both C and C_α are finitely presented A-algebras, the ideal

$$C_\alpha \otimes_{B_\alpha} (\widetilde{\mathfrak{J}}/\widetilde{\mathfrak{J}}_\alpha)$$

is of finite type in C_α (1.4.4). It is the union of the finite type ideals

$$C_\alpha \otimes_{B_\alpha} (\widetilde{\mathfrak{J}}_\lambda/\widetilde{\mathfrak{J}}_\alpha).$$

Consequently there exists $\lambda \geqslant \alpha$ such that

$$C_\alpha \otimes_{B_\alpha} (\widetilde{\mathfrak{J}}/\widetilde{\mathfrak{J}}_\alpha) = C_\alpha \otimes_{B_\alpha} (\widetilde{\mathfrak{J}}_\lambda/\widetilde{\mathfrak{J}}_\alpha).$$

Faithful flatness then gives $\widetilde{\mathfrak{J}} = \widetilde{\mathfrak{J}}_\lambda$, proving that B is of finite presentation over A. □

11.4. Descent of flatness by arbitrary morphisms: the case of an Artinian base

Theorem (11.4.1). — *Let* A *be a ring,* $\mathfrak{J}$ *a nilpotent ideal of* A*,* $u_\lambda : \mathrm{A} \to \mathrm{B}_\lambda$ $(\lambda \in \mathrm{L})$ *a family of homomorphisms of rings such that the intersection of the kernels of the* u_λ *is reduced to* 0*. Let* M *be an* A*-module such that for all* $\lambda \in \mathrm{L}$*,* $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}_\lambda$ *is a* B_λ*-module free and that* $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ *is an* $(\mathrm{A}/\mathfrak{J})$*-module free. Then* M *is an* A*-module free. If moreover the index set* L *is finite, one can replace everywhere "free" by "flat" in the preceding statement.*

Proof. In both cases it will suffice to prove that M is an A-module *flat*: indeed, when $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ is an $(\mathrm{A}/\mathfrak{J})$-module free, it will follow that M is an A-module free by virtue of $(\mathbf{0}_{\mathrm{III}}, 10.1.2)$.

We will use the following lemma, which generalises $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$:

Lemma (11.4.1.1). — *Let* A *be a ring equipped with a finite filtration* $(\mathfrak{J}_n)_{0 \leqslant n \leqslant \mathrm{N}+1}$ *with* $\mathfrak{J}_0 = \mathrm{A}$*,* $\mathfrak{J}_{\mathrm{N}+1} = 0$*. Let* M *be an* A*-module equipped with the filtration* $(\mathfrak{J}_n\mathrm{M})_{0 \leqslant n \leqslant \mathrm{N}+1}$*, and let us denote by* $\mathrm{gr}(\mathrm{A})$ *and* $\mathrm{gr}(\mathrm{M})$ *the ring and the graded module respectively. Suppose that* $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}_1)$ *is an* $(\mathrm{A}/\mathfrak{J}_1)$*-module flat and that the canonical homomorphism*

$$\mathrm{gr}_0(\mathrm{M}) \otimes_{\mathrm{gr}_0(\mathrm{A})} \mathrm{gr}(\mathrm{A}) \longrightarrow \mathrm{gr}(\mathrm{M}) \tag{11.4.1.2}$$

is injective. Then M *is an* A*-module flat.*

The canonical homomorphism (11.4.1.2) is defined in the same way as that of $(\mathbf{0}_{\mathrm{III}}, 10.1.1)$ as being in degree n the homomorphism

$$(\mathrm{M}/\mathfrak{J}_1\mathrm{M}) \otimes (\mathfrak{J}_n/\mathfrak{J}_{n+1}) = (\mathrm{M} \otimes \mathfrak{J}_n)/(\mathrm{Im}(\mathrm{M} \otimes \mathfrak{J}_{n+1}) + \mathrm{Im}(\mathfrak{J}_1\mathrm{M} \otimes \mathfrak{J}_n)) \longrightarrow \mathfrak{J}_n\mathrm{M}/\mathfrak{J}_{n+1}\mathrm{M}$$

coming from the canonical homomorphism $\mathrm{M} \otimes \mathfrak{J}_n \to \mathfrak{J}_n\mathrm{M}$ by passage to quotients. The lemma is proved by induction on N, since there is nothing to prove for $\mathrm{N} = 0$. The conditions on M imply, by **IV-3** | 144
the induction hypothesis, that $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}_{\mathrm{N}})$ is an $(\mathrm{A}/\mathfrak{J}_{\mathrm{N}})$-module flat. Let us now note that $\mathfrak{J}_{\mathrm{N}}^2 \subset \mathfrak{J}_{\mathrm{N}+1} = 0$ if $\mathrm{N} \geqslant 1$, and $(\mathrm{M}/\mathfrak{J}_1\mathrm{M}) \otimes (\mathfrak{J}_{\mathrm{N}}/\mathfrak{J}_{\mathrm{N}+1}) = (\mathrm{M}/\mathfrak{J}_{\mathrm{N}}\mathrm{M}) \otimes (\mathfrak{J}_{\mathrm{N}}/\mathfrak{J}_{\mathrm{N}+1}) = (\mathrm{M}/\mathfrak{J}_{\mathrm{N}}\mathrm{M}) \otimes \mathfrak{J}_{\mathrm{N}}$; hence the canonical homomorphism

$$(\mathrm{M}/\mathfrak{J}_{\mathrm{N}}\mathrm{M}) \otimes \mathfrak{J}_{\mathrm{N}} \longrightarrow \mathfrak{J}_{\mathrm{N}}\mathrm{M} = \mathfrak{J}_{\mathrm{N}}\mathrm{M}/\mathfrak{J}_{\mathrm{N}}^2\mathrm{M}$$

is injective. Applying $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$ to the $\mathfrak{J}_{\mathrm{N}}$-preadic filtration, we conclude that M is an A-module flat.

To apply this lemma to (11.4.1), we will denote by $\mathfrak{J}_n$ the ideal of A which is the intersection of the inverse images $u_\lambda^{-1}(\mathfrak{J}^n\mathrm{B}_\lambda)$: it is immediate that $\mathfrak{J}_0 = \mathrm{A}$, $\mathfrak{J}_m\mathfrak{J}_n \subset \mathfrak{J}_{m+n}$ for $m \geqslant 0$, $n \geqslant 0$; moreover, if $\mathfrak{J}^{\mathrm{N}+1} = 0$, we also have $\mathfrak{J}_{\mathrm{N}+1} = 0$ since the intersection of the kernels of the u_λ is reduced to 0.

Let us equip A with the filtration $(\mathfrak{J}_n)$, M with the filtration $(\mathfrak{J}_n\mathrm{M})$, and, for each λ, let us equip B_λ and $\mathrm{N}_\lambda = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}_\lambda$ with the $\mathfrak{J}$-preadic filtrations; let us consider for each λ the commutative diagram

$$\begin{tikzcd}
\mathrm{gr}^0(\mathrm{M}) \otimes_{\mathrm{gr}^0(\mathrm{A})} \mathrm{gr}(\mathrm{A}) \arrow[r, "\varphi"] \arrow[d, "f_\lambda"'] & \mathrm{gr}(\mathrm{M}) \arrow[d, "g_\lambda"] \\
\mathrm{gr}^0_{\mathfrak{J}}(\mathrm{N}_\lambda) \otimes_{\mathrm{gr}^0_{\mathfrak{J}}(\mathrm{B}_\lambda)} \mathrm{gr}_{\mathfrak{J}}(\mathrm{B}_\lambda) \arrow[r, "\varphi_\lambda"] & \mathrm{gr}_{\mathfrak{J}}(\mathrm{N}_\lambda)
\end{tikzcd}$$

where the horizontal arrows are the canonical homomorphisms (11.4.1.2) and the vertical arrows are deduced from the canonical homomorphisms $\mathrm{A} \to \mathrm{B}_\lambda$ and $\mathrm{M} \to \mathrm{N}_\lambda$. The hypothesis that N_λ is a B_λ-module flat implies that φ_λ is injective $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$, hence $\mathrm{Ker}(\varphi) \subset \mathrm{Ker}(f_\lambda)$. Setting $\mathrm{A}_0 = \mathrm{A}/\mathfrak{J}_1$, $\mathrm{M}_0 = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}_0$, let us note that

$$\mathrm{gr}^0_{\mathfrak{J}}(\mathrm{N}_\lambda) = \mathrm{N}_\lambda/\mathfrak{J}\mathrm{N}_\lambda = (\mathrm{M}/\mathfrak{J}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{B}_\lambda = (\mathrm{M}/\mathfrak{J}_1\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{B}_\lambda/\mathfrak{J}\mathrm{B}_\lambda)$$

and therefore

$$\mathrm{gr}^0_{\mathfrak{J}}(\mathrm{N}_\lambda) \otimes_{\mathrm{gr}^0_{\mathfrak{J}}(\mathrm{B}_\lambda)} \mathrm{gr}_{\mathfrak{J}}(\mathrm{B}_\lambda) = \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{gr}_{\mathfrak{J}}(\mathrm{B}_\lambda).$$

The homomorphism f_λ can therefore be written **IV-3** | 145

$$1 \otimes \mathrm{gr}(u_\lambda) : \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{gr}(\mathrm{A}) \longrightarrow \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{gr}_{\mathfrak{J}}(\mathrm{B}_\lambda),$$

and since by hypothesis M_0 is an A_0-module *flat*, the kernel of f_λ is equal to $M_0 \otimes_{A_0} R_\lambda$, where R_λ is the kernel of $\mathrm{gr}(u_\lambda) : \mathrm{gr}(A) \to \mathrm{gr}_{\mathfrak{J}}(B_\lambda)$. It therefore suffices to prove that $\bigcap_{\lambda\in L}(M_0 \otimes_{A_0} R_\lambda) = 0$. Now, by definition of the $\mathfrak{J}_n$, the intersection of the kernels of the homomorphisms $\mathfrak{J}_n A/\mathfrak{J}_{n+1}A \to \mathfrak{J}^n B_\lambda/\mathfrak{J}^{n+1}B_\lambda$, as λ ranges over L, is reduced to 0; in other words, $\bigcap_{\lambda\in L} R_\lambda = 0$. This being the case, suppose first that M_0 is an A_0-module free; taking a basis of M_0, we see immediately that $M_0 \otimes_{A_0} (\bigcap_{\lambda\in L} R_\lambda) = \bigcap_{\lambda\in L}(M_0 \otimes_{A_0} R_\lambda)$, whence the proposition in this case. When L is finite, the preceding formula is still true under the sole hypothesis that M_0 is an A_0-module flat ($\mathbf{0}_{\mathrm{I}}$, 6.1.3), which finishes the proof. □

Remark (11.4.2). — The conclusion of (11.4.1) may be inexact if L is infinite and one only supposes that $M \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$-module flat. For example, let V be a discrete valuation ring, K its field of fractions, and let $A = V[T]/(T^2)$ (T an indeterminate); let us take for $\mathfrak{J}$ the image of (T) in A, so that $A/\mathfrak{J} = V$, and let us take $M = K$, which is an $(A/\mathfrak{J})$-module, hence equal to $M \otimes_A (A/\mathfrak{J})$; moreover M is an $(A/\mathfrak{J})$-module flat, but not an A-module flat, since $\mathfrak{J}$ is nilpotent, it would follow from ($\mathbf{0}_{\mathrm{III}}$, 10.1.2) that M would be an A-module free, which is absurd since $\mathfrak{J}K = 0$. On the other hand, consider the maximal ideal $\mathfrak{m}$ of the local Noetherian ring A; we have $\mathfrak{m}K = K$, hence $M \otimes_A (A/\mathfrak{m}^n) = 0$ for every integer n; the $(A/\mathfrak{m}^n)$-modules $M \otimes_A (A/\mathfrak{m}^n)$ are therefore flat for all n, and the intersection of $\mathfrak{m}^n$ is reduced to 0.

Corollary (11.4.3). — *Let* A *be a semi-local ring whose radical* $\mathfrak{J}$ *is nilpotent (for example an Artinian ring),* $u_\lambda : A \to B_\lambda$ $(\lambda \in L)$ *a family of homomorphisms of rings such that the intersection of the kernels of the* u_λ *is reduced to* 0*. For an* A*-module* M *to be flat, it is necessary and sufficient that for all* $\lambda \in L$, $M \otimes_A B_\lambda$ *be a* B_λ*-module flat.*

Proof. Since $A/\mathfrak{J}$ is a direct product of a finite number of fields (Bourbaki, *Alg. comm.*, chap. II, § 3, nº 5, prop. 16) and that $\mathfrak{J}$ is nilpotent, A is a direct product of a finite number of *local* rings A_i whose radical is nilpotent (*loc. cit.*, § 4, nº 3, cor. 1 of prop. 15) and M is therefore a direct sum of A_i-modules M_i, each M_i being annihilated by the A_j of index $j \neq i$; for M to be a flat A-module, it is necessary and it suffices that each of the M_i be a flat A_i-module; moreover the intersection of the kernels of the homomorphisms $A_i \xrightarrow{u_i} B_\lambda$ is reduced to 0, and $M_i \otimes_{A_i} B_\lambda$ is a direct factor of $M \otimes_A B_\lambda$. One can therefore restrict to the case where A is also *local*. Then $A/\mathfrak{J}$ is a field, hence $M \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$-module free, and it suffices to see that for all λ, $M \otimes_A B_\lambda$ is a B_λ-module *free*, by virtue of (11.4.1). But if one sets $\mathfrak{J}_\lambda = \mathfrak{J}B_\lambda$, $(M \otimes_A B_\lambda) \otimes_{B_\lambda} (B_\lambda/\mathfrak{J}_\lambda) = (M \otimes_A (A/\mathfrak{J})) \otimes_{A/\mathfrak{J}} (B_\lambda/\mathfrak{J}_\lambda)$ is a $(B_\lambda/\mathfrak{J}_\lambda)$-module free, and $\mathfrak{J}_\lambda$ is nilpotent. The conclusion therefore follows from the hypothesis that $M \otimes_A B_\lambda$ is a B_λ-module flat and from ($\mathbf{0}_{\mathrm{III}}$, 10.1.2). □

Corollary (11.4.4). — *Let* A *be a ring,* M *an* A*-module; suppose that there exists a nilpotent ideal* $\mathfrak{J}$ *of* A *such that* $M \otimes_A (A/\mathfrak{J})$ *is an* $(A/\mathfrak{J})$*-module free. Then the set of ideals* $\mathfrak{K}$ *of* A *such that* $M \otimes_A (A/\mathfrak{K})$ *is an* $(A/\mathfrak{K})$*-module free admits a smallest element* $\mathfrak{K}_0$ *(which is also the smallest of the ideals* $\mathfrak{K}$ *such that* **IV-3 | 146**
$M \otimes_A (A/\mathfrak{K})$ *is an* $(A/\mathfrak{K})$*-module* flat*).*

Proof. For any homomorphism $u : A \to A'$ such that $M \otimes_A A'$ is an A'-module free (resp. an A'-module flat), it is necessary and it suffices that u factorise through $A \to A/\mathfrak{K}_0 \to A'$ (or equivalently that $\mathfrak{K}_0 A' = 0$).

The fact that the intersection $\mathfrak{K}_0$ of the ideals $\mathfrak{K}$ for which $M \otimes_A (A/\mathfrak{K})$ is an $(A/\mathfrak{K})$-module free is itself the smallest of these ideals follows from (11.4.1) applied to the ring $A/\mathfrak{K}_0$, to a nilpotent ideal $\mathfrak{J}/\mathfrak{K}_0$ and to the homomorphisms $A/\mathfrak{K}_0 \to A/\mathfrak{K}$, whose kernels have 0 for intersection. If A' is an A-algebra such that $M \otimes_A A'$ is an A'-module free and if $\mathfrak{K}$ is the kernel of the homomorphism $A \to A'$, it follows from (11.4.1) applied to the ring $A/\mathfrak{K}$, to the $(A/\mathfrak{K})$-module $M \otimes_A (A/\mathfrak{K})$, to the nilpotent ideal $(\mathfrak{J} + \mathfrak{K})/\mathfrak{K}$ and to the *injective* homomorphism $A/\mathfrak{K} \to A'$, that $M \otimes_A (A/\mathfrak{K})$ is an $(A/\mathfrak{K})$-module free, hence $\mathfrak{K} \supset \mathfrak{K}_0$. The fact that one can replace "free" by "flat" in what precedes (while naturally preserving the hypothesis that $M \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$-module *free*) follows from ($\mathbf{0}_{\mathrm{III}}$, 10.1.2), as we saw at the beginning of the proof of (11.4.1). □

Proposition (11.4.5). — *Let* Y *be an irreducible prescheme,* $f : X \to Y$ *a morphism of finite presentation,* $\mathscr{F}$ *an* $\mathscr{O}_X$*-module of finite presentation. There exists an open non-empty* U *in* Y *and a closed sub-prescheme* Z *of* U*, of finite presentation over* U*, having the following property: for every base change* $U' \to U$*, if we set*

$X' = X \times_Y U'$, $f' = f_{(U')}$ and $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{U'}$, then, for $\mathscr{F}'$ to be f'-flat, it is necessary and sufficient that the morphism $U' \to U$ factorise through $U' \to Z \to U$. Such a scheme Z has the same underlying space as U. Suppose moreover that Y is affine, and let (W_i) be a finite affine open covering of X; then, one can suppose U chosen such that, if $U' = \operatorname{Spec}(A')$ is an affine scheme and $U' \to U$ a morphism that factorises through $U' \to Z \to U$, each of the $\Gamma(W_i \times_Y U', \mathscr{F}')$ is a free A'-module.

Proof. One can evidently restrict to the case where $Y = \operatorname{Spec}(A)$ is affine. Using (**0**, 8.9.1), there is a Noetherian sub-ring A_1 of A, a morphism of finite type $f_1 : X_1 \to Y_1 = \operatorname{Spec}(A_1)$ and a coherent $\mathscr{O}_{X_1}$-module $\mathscr{F}_1$ such that $f = f_{(Y)}$ and $\mathscr{F} = \mathscr{F}_1 \otimes_{\mathscr{O}_{Y_1}} \mathscr{O}_Y$; one can moreover suppose that the W_i are inverse images of affine opens of X_1. Let us note moreover that Y_1 is irreducible, the morphism $Y \to Y_1$ being dominant (**I**, 1.2.7). Suppose the proposition proved for Y_1, X_1 and $\mathscr{F}_1$, and let U_1 and Z_1 be the open set of Y_1 and the closed sub-prescheme of U_1 having the required properties. Then, if $U' \to U$ is a base change such that $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{U'}$ is f'-flat, the morphism $U' \to U_1$ factorises through $U' \to Z_1 \to U_1$; but since $U' \to U \to U_1$ also factorises through $U' \to U \to U_1$, the definition of the product of preschemes shows that $U' \to U$ factorises through $U' \to Z \to U$.

One can therefore restrict to the case where A is *Noetherian*; let $\mathfrak{N}$ be its nilradical, which is nilpotent, and set $A_0 = A_{\text{red}} = A/\mathfrak{N}$, $Y_0 = \operatorname{Spec}(A_0) = Y_{\text{red}}$, $X_0 = X \times_Y Y_0$, $\mathscr{F}_0 = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}$, $W_{i0} = W_i \times_Y Y_0$; if $M_i = \Gamma(W_i, \mathscr{F})$, $M_{i0} = \Gamma(W_{i0}, \mathscr{F}_0)$ is therefore equal to $M_i \otimes_A A_0$. Since A_0 is *integral*, one can, by virtue of the generic flatness theorem (6.9.2), replacing Y by an open non-empty set of Y as needed, suppose that the M_{i0} are free A_0-modules. By virtue of (11.4.4), there is therefore for each i an ideal $\mathfrak{J}_i \subset \mathfrak{N}$ such that the A-algebras A' for which the $M_i \otimes_A A'$ are A'-modules free (or flat) are exactly those for which $\mathfrak{J}_i A' = 0$. Let $\mathfrak{J} = \sum_i \mathfrak{J}_i$, which is an ideal contained in $\mathfrak{N}$; to say that $\mathscr{F} \otimes_{\mathscr{O}_Y} A'$ is A'-flat amounts to saying that the $M_i \otimes_A A'$ are all A'-modules flat, hence that $\mathfrak{J}A' = 0$. It follows that if one takes $Z = V(\mathfrak{J})$, one answers the question, since for any morphism $U' \to U$ having the property of the statement, it is necessary and it suffices that for every affine open $V' \subset U'$, the morphism $V' \to U$ have the same property. ∎ **IV-3 | 147**

Corollary (11.4.6). — *Let A be a ring such that $\operatorname{Spec}(A)$ is irreducible, $\mathfrak{p}$ its unique minimal prime ideal, B an A-algebra of finite presentation, M a B-module of finite presentation. Suppose that $M_{\mathfrak{p}}$ is a flat $A_{\mathfrak{p}}$-module; then there exists $t \in A - \mathfrak{p}$ such that M_t is a free A_t-module.*

Proof. Applying (11.4.5) with $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, $\mathscr{F} = \widetilde{M}$, one can (replacing A by A_h as needed, where $h \in A - \mathfrak{p}$) suppose that there exists an ideal of finite type $\mathfrak{J}$ in A such that the A-algebras A' for which $M \otimes_A A'$ is an A'-module free (or flat) are exactly those such that $\mathfrak{J}A' = 0$. In particular, the hypothesis implies that $M_{\mathfrak{p}}$ is a flat $A_{\mathfrak{p}}$-module, hence $\mathfrak{J}A_{\mathfrak{p}} = 0$, or $\mathfrak{J}_{\mathfrak{p}} = 0$. But since $\mathfrak{J}$ is of finite type, there exists $t \in A - \mathfrak{p}$ such that $\mathfrak{J}_t = 0$, or equivalently $\mathfrak{J}A_t = 0$, and therefore M_t is a free A_t-module. ∎

Proposition (11.4.7). — *Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A, M an A-module ideally separated ($\mathbf{0}_{\text{III}}$, 10.2.1) for the $\mathfrak{J}$-preadic topology. Let $(\mathfrak{p}_i)_{1 \leqslant i \leqslant r}$ be a finite family of prime ideals of A containing $\mathfrak{J}$; for every integer $n \geqslant 0$, let $\mathfrak{p}_i^{(n)}$ be the n-th symbolic power (i.e. the kernel of the homomorphism $A \to A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i}$); set $\mathfrak{J}_n = \bigcap_{i=1}^r \mathfrak{p}_i^{(n)}$, so that $\mathfrak{J}_n \supset \mathfrak{J}^n$, and suppose that the topology defined by the filtration $(\mathfrak{J}_n)$ is identical to the $\mathfrak{J}$-preadic topology (in other words, for every n, there exists m such that $\mathfrak{J}^m \supset \mathfrak{J}_n$). For M to be an A-module flat, it is necessary and sufficient that $M \otimes_A (A/\mathfrak{J})$ be an $(A/\mathfrak{J})$-module flat and that, for all i, $M \otimes_A A_{\mathfrak{p}_i} = M_{\mathfrak{p}_i}$ be a flat $A_{\mathfrak{p}_i}$-module.*

Proof. Since M is ideally separated, it suffices, by virtue of ($\mathbf{0}_{\text{III}}$, 10.2.1), to show that, for every $n \geqslant 0$, $M \otimes_A (A/\mathfrak{J}^n)$ is an $(A/\mathfrak{J}^n)$-module flat; since every $\mathfrak{J}^n$ contains a $\mathfrak{J}_m$, it amounts to the same to prove that for every n, $M \otimes_A (A/\mathfrak{J}_n)$ is an $(A/\mathfrak{J}_n)$-module flat. Now, since $\mathfrak{J}_1 \supset \mathfrak{J}$, $M \otimes_A (A/\mathfrak{J}_1)$ is an $(A/\mathfrak{J}_1)$-module flat; in the ring $A/\mathfrak{J}_n$, the ideal $\mathfrak{J}_1/\mathfrak{J}_n$ is nilpotent and finally the intersection of the kernels of the homomorphisms $A/\mathfrak{J}_n \to A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i}$ is zero in $A/\mathfrak{J}_n$, by definition of $\mathfrak{J}_n$. It therefore suffices, by (11.4.1), to verify that $M \otimes_A (A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i})$ is an $(A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i})$-module flat, which follows from the hypothesis that $M \otimes_A A_{\mathfrak{p}_i}$ is a flat $A_{\mathfrak{p}_i}$-module. ∎

Remark (11.4.8). — The hypothesis made in (11.4.7) on the topology defined by the $\mathfrak{J}_n$ is satisfied if, for every n sufficiently large, $\operatorname{Ass}(A/\mathfrak{J}^n)$ *is contained in the set of the* $\mathfrak{p}_i$. Indeed, $\mathfrak{J}^n$ is then an intersection of primary ideals for the $\mathfrak{p}_i$, each of which contains a symbolic power of $\mathfrak{p}_i$, whence the conclusion. In particular:

Corollary (11.4.9). — *Let* A *be a Noetherian ring,* $\mathfrak{J}$ *a nilpotent ideal of* A, M *an* A*-module. For* M *to be an* A*-module flat (resp. free), it is necessary and sufficient that* $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ *be an* $(\mathrm{A}/\mathfrak{J})$*-module flat (resp. free) and that for every prime ideal* $\mathfrak{p} \in \mathrm{Ass}(\mathrm{A})$, $\mathrm{M}_{\mathfrak{p}}$ *be a flat* $\mathrm{A}_{\mathfrak{p}}$*-module.*

Corollary (11.4.10). — *Let* A *be a Noetherian ring,* $\mathfrak{J}$ *an ideal of* A, M *an* A*-module. Suppose that* M *is ideally separated for the* $\mathfrak{J}$*-preadic topology and that* $\mathrm{gr}_{\mathfrak{J}}(\mathrm{A})$ *is an* $(\mathrm{A}/\mathfrak{J})$*-module flat. For* M *to be an* A*-module flat, it is necessary and sufficient that* $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ *be an* $(\mathrm{A}/\mathfrak{J})$*-module flat and that for every* $\mathfrak{p} \in \mathrm{Ass}(\mathrm{A}/\mathfrak{J})$, $\mathrm{M}_{\mathfrak{p}}$ *be a flat* $\mathrm{A}_{\mathfrak{p}}$*-module.*

Proof. Taking into account (11.4.8), it suffices to show that $\mathrm{Ass}(\mathrm{A}/\mathfrak{J}^n)$ is contained in $\mathrm{Ass}(\mathrm{A}/\mathfrak{J})$ for every n. Now, if $a \in \mathrm{A}$ does not belong to any $\mathfrak{p} \in \mathrm{Ass}(\mathrm{A}/\mathfrak{J})$, the homothety of ratio a is injective in $\mathrm{A}/\mathfrak{J}$; since each of the $\mathfrak{J}^k/\mathfrak{J}^{k+1}$ is an $(\mathrm{A}/\mathfrak{J})$-module flat, a is also a $(\mathfrak{J}^k/\mathfrak{J}^{k+1})$-regular element, hence a is a $(\mathrm{A}/\mathfrak{J}^n)$-regular element for every n (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 1 of th. 1), and therefore it does not belong to any prime ideal associated to $\mathrm{A}/\mathfrak{J}^n$, whence the corollary (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2). □ **IV-3** | 148

Proposition (11.4.11). — *Let* A *be an Artinian local ring,* $\mathfrak{m}$ *its maximal ideal,* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, y *the unique point of* Y, $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-module. Let* (A'_α) *be a family of local rings, and for each* α, $u_\alpha : \mathrm{A} \to \mathrm{A}'_\alpha$ *a homomorphism of rings (necessarily local). Set* $\mathrm{Y}'_\alpha = \mathrm{Spec}(\mathrm{A}'_\alpha)$, $\mathrm{X}'_\alpha = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'_\alpha$, $\mathscr{F}'_\alpha = \mathscr{F} \otimes_{\mathrm{A}} \mathrm{A}'_\alpha$*. Let* x *be a point of* X*, and suppose the following conditions are satisfied:*

- *(i) The intersection of the kernels of the* u_α *is reduced to* 0.
- *(ii) The extension* $k(x)$ *of the residue field* k *of* A *is primary* (4.3.1) *(a condition automatically satisfied if* k *is separably closed).*
- *(iii) For all* α*, there exists a point* $x'_\alpha \in \mathrm{X}'_\alpha$ *whose projections in* X *and* Y'_α *are* x *and the closed point* y'_α *of* Y'_α*, and such that* $\mathscr{F}'_\alpha$ *is* Y'_α*-flat at the point* x'_α.

Under these conditions, $\mathscr{F}$ *is* f*-flat at the point* x.

Proof. The question being local on X, one can evidently restrict to the case where f is a morphism of finite type and suppose $\mathscr{F}_x \neq 0$. We will proceed in several steps.

I) *Reduction to the case where* A'_α *is a local ring of a* Y*-prescheme of finite type and where the residue field* k'_α *of* A'_α *is a finite extension of* k.

Since the reduction is done separately on each A'_α, one can suppress the index α in this part. Let $\mathfrak{m}'$ be the maximal ideal of A'. Let us consider A' as an inductive limit of its sub-A-algebras of finite type B_λ, and set $\mathfrak{n}_\lambda = \mathfrak{m}' \cap \mathrm{B}_\lambda$; A' is also an inductive limit of its local sub-rings $\mathrm{B}'_\lambda = (\mathrm{B}_\lambda)_{\mathfrak{n}_\lambda}$ (5.13.3), and one has evidently $\mathfrak{m}' \cap \mathrm{B}'_\lambda = \mathfrak{n}'_\lambda$, maximal ideal of B'_λ. It follows therefore (11.2.6) that in setting $\mathrm{Z}'_\lambda = \mathrm{Spec}(\mathrm{B}'_\lambda)$, $\mathscr{F} \otimes_{\mathrm{A}} \mathrm{B}'_\lambda$ is Z'_λ-flat at the point x'_λ, projection of x', the projection of x'_λ on Z'_λ being the closed point z'_λ of Z'_λ.

One can therefore suppose that there exists an affine scheme Z' of finite type over Y and a point z' of Z' such that $\mathrm{A}' = \mathscr{O}_{\mathrm{Z}',z'}$, and that, if one sets $\mathrm{T}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Z}'$, there exists a point $t' \in \mathrm{T}'$ whose projections in Z' and X are z' and x, and such that $\mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Z}'$ is Z'-flat at the point t'. Let Z'_1 (resp. X_1) be a closed sub-prescheme of Z' (resp. X) having for underlying space the closure of $\{z'\}$ (resp. $\{x\}$), and set $\mathrm{T}'_1 = \mathrm{X}_1 \times_{\mathrm{Y}} \mathrm{Z}'_1$. The set U of the points of T' where $\mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Z}'$ is Z'-flat is open in T' (11.1.1) and contains t', hence $\mathrm{V} = \mathrm{U} \cap \mathrm{T}'_1$ is a nonempty open subset of T'_1. The ring A, **IV-3** | 149
being Artinian, is a Jacobson ring; hence, by (10.4.6), Z'_1 and T'_1 are Jacobson preschemes. There is therefore a point t'_0 in V that is closed in V, and its image z'_0 in Z'_1 is a *closed* point of Z'_1 (10.4.7). Let f_1 be the restriction of f to X_1, and let $f'_1 : \mathrm{T}'_1 \to \mathrm{Z}'_1$ be the canonical projection. Then $\mathrm{V} \cap f_1'^{-1}(z'_0)$ is a nonempty open subset of $f_1^{-1}(y) \otimes_{k(y)} k(z'_0)$; since this latter prescheme is flat over $f_1^{-1}(y)$, a maximal point t'_1 of $\mathrm{V} \cap f_1'^{-1}(z'_0)$ necessarily lies above the unique maximal point x of X_1 (2.3.4). Finally, $k(z'_0)$ is a *finite* extension of k (**I**, 6.4.2), and the homomorphism $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y} \to \mathrm{A}' = \mathscr{O}_{\mathrm{Z}',z'}$ factors as $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{Z}',z'_0} \to \mathscr{O}_{\mathrm{Z}',z'}$; its kernel is therefore contained in that of $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{Z}',z'_0}$. This completes the announced reduction.

II) *Reduction to the case where there are only finitely many* A'_α *and they are finite* A*-algebras.* — Let $\mathfrak{m}'_\alpha$ be the maximal ideal of A'_α. Since A'_α is a Noetherian local ring, the intersection of the $\mathfrak{m}'^n_\alpha$ is 0 ($\mathbf{0}_\mathrm{I}$, 7.3.5). The intersection of the $u_\alpha^{-1}(\mathfrak{m}'^n_\alpha)$, over all indices n and α, is therefore equal to the intersection of the kernels of the u_α, and hence is reduced to 0 by hypothesis (i). Since A is Artinian, finitely many of these ideals already have intersection 0; denote them by $u_i^{-1}(\mathfrak{m}'^{n_i}_i)$ $(1 \leqslant i \leqslant r)$. Since the A'_i are Noetherian, the $\mathfrak{m}'^h_i/\mathfrak{m}'^{h+1}_i$ are $(A'_i/\mathfrak{m}'_i)$-modules of finite length, and since $A'_i/\mathfrak{m}'_i$ is a finite-dimensional k-vector space, $A''_i = A'_i/\mathfrak{m}'^{n_i}_i$ is a finite A-algebra and an Artinian local ring. The announced reduction therefore follows from (2.1.4).

III) *End of the proof.* — Suppose now that there are only finitely many A'_i $(1 \leqslant i \leqslant r)$ and that they are finite A-algebras. For every i, the residue field k_i of A'_i is a finite extension of k; using hypothesis (ii), one concludes that the inverse image of x in X'_i consists of the *single* point x'_i (4.3.2). Let Y' be the disjoint union of the Y'_i, let $X' = X \times_Y Y'$ be the disjoint union of the X'_i, and put $\mathscr{F}' = \mathscr{F} \otimes_Y Y'$. The hypothesis implies that $\mathscr{F}'$ is Y'-flat at every point of the inverse image of x under the projection $p : X' \to X$. Since p is of finite type, there is therefore an open subset $U' \supset p^{-1}(x)$ such that $\mathscr{F}'$ is Y'-flat at every point of U' (11.1.1). Moreover, the morphism $Y' \to Y$ is finite because the A'_i are finite A-algebras. Thus p is a finite morphism (**II**, 6.1.5), and hence is closed (**II**, 6.1.10); consequently there is an affine neighborhood U of x in X such that $p^{-1}(U) \subset U'$. Let B and A' be the rings of the schemes U and Y', respectively (so A' is the direct product of the A'_i). Replacing X by U, one has $\mathscr{F} = \widetilde{M}$, where M is a B-module, and by hypothesis $M' = M \otimes_A A'$ is a flat A'-module (2.1.2). Since the homomorphism $A \to A'$ is injective by construction, one can apply (11.4.3), which proves that M is a flat A-module. □

Proposition (11.4.12). — *Let* A *be an Artinian local ring with residue field* k, $Y = \mathrm{Spec}(A)$, $f : X \to Y$ *a morphism locally of finite type, and* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-module. Let* (A'_α) *be a family of local rings, and for each* α, $u_\alpha : A \to A'_\alpha$ *a homomorphism of rings. Set* $Y'_\alpha = \mathrm{Spec}(A'_\alpha)$, $X'_\alpha = X \times_Y Y'_\alpha$, $f'_\alpha = f_{(Y'_\alpha)}$, *and* $\mathscr{F}'_\alpha = \mathscr{F} \otimes_A A'_\alpha$. *Let* x *be a point of* X, *and suppose that the following conditions are satisfied:*

(i) The intersection of the kernels of the u_α *is reduced to* 0.

(ii) For all α, $\mathscr{F}'_\alpha$ *is* f'_α*-flat at every point* $x'_\alpha \in X'_\alpha$ *whose projections in* X *and* Y'_α *are* x *and the closed point* y'_α *of* Y'_α.

Then $\mathscr{F}$ *is* f*-flat at the point* x.

Proof. By hypothesis, the intersection of the kernels of the u_α is reduced to 0; since A is Artinian, **IV-3** | 150
there are already a *finite* number of these kernels whose intersection is 0, hence one can restrict to the case where the family (A'_α) is *finite*. Let k' be an algebraic closure of k; one knows ($\mathbf{0}_\mathrm{III}$, 10.3.1) that there exists a local homomorphism $A \to B$ making B an A-module flat, integral over A and such that $B \otimes_A k$ is isomorphic to k'. By flatness, the kernels of the homomorphisms $B \to B'_\alpha = B \otimes_A A'_\alpha$ are deduced from those of u_α by tensorisation with B, and since they are in finite number, their intersection is reduced to 0 ($\mathbf{0}_\mathrm{I}$, 6.1.3). Let us consider the rings $B'_{\alpha\beta}$, localised of B'_α at its *maximal* ideals; one knows (Bourbaki, *Alg. comm.*, chap. II, § 3, nº 3, cor. 2 of th. 1) that the intersection of the kernels of the homomorphisms $B'_\alpha \to B'_{\alpha\beta}$ (for a given α) is reduced to 0; one therefore concludes that the intersection of the kernels of the composite homomorphisms $v_{\alpha\beta} : B \to B'_\alpha \to B'_{\alpha\beta}$ (α and β variable) is reduced to 0. On the other hand, since B is integral over A, B'_α is integral over A'_α, hence its maximal ideals are above the maximal ideal of A'_α. If one sets $Z'_{\alpha\beta} = \mathrm{Spec}(B'_{\alpha\beta})$, $T'_{\alpha\beta} = X'_\alpha \times_{Y'_\alpha} Z'_{\alpha\beta}$, $f'_{\alpha\beta} = f_{(Z'_{\alpha\beta})}$, $\mathscr{G} = \mathscr{F} \otimes_A B$, and $\mathscr{G}'_{\alpha\beta} = \mathscr{G} \otimes_B B'_{\alpha\beta}$, one sees that hypotheses (i) and (ii) are still satisfied when one replaces A, A'_α, u_α, $\mathscr{F}$ and x by B, $B'_{\alpha\beta}$, $v_{\alpha\beta}$, $\mathscr{G}$ and a point t of $T = X \otimes_A B$ above x. Since the residue field of B is separably closed, one deduces from (11.4.11) that $\mathscr{G}$ is flat over B at the point t. But since B is a faithfully flat A-module, it follows from (2.1.4) that $\mathscr{F}$ is flat over A at the point x, which proves (11.4.12). □

Corollary (11.4.13). — *Let* A *be an Artinian local ring,* $Y = \mathrm{Spec}(A)$, $f : X \to Y$ *a morphism locally of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-module. Let* (Y'_α) *be a family of* Y*-preschemes, and for all* α, *set* $X'_\alpha = X \times_Y Y'_\alpha$, $f'_\alpha = f_{(Y'_\alpha)}$, *and* $\mathscr{F}'_\alpha = \mathscr{F} \otimes_Y Y'_\alpha$. *Let* x *be a point of* X *and suppose the following hypotheses satisfied:*

(i) The intersection of the kernels of the homomorphisms $A \to \Gamma(Y'_\alpha, \mathscr{O}_{Y'_\alpha})$ *corresponding to the structural morphisms is reduced to* 0.

(ii) For all α, $\mathscr{F}'_\alpha$ is f'_α-flat at every point $x'_\alpha \in \mathrm{X}'_\alpha$ whose projection in X *is x.*

Then $\mathscr{F}$ is f-flat at the point x.

Proof. Indeed, for every $y'_\alpha \in \mathrm{Y}'_\alpha$, consider the local scheme $\mathrm{Y}''_{\alpha,y'_\alpha} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y}'_\alpha,y'_\alpha})$; by virtue of (2.1.4), $\mathscr{F}'_\alpha \otimes_{\mathrm{Y}'_\alpha} \mathrm{Y}''_{\alpha,y'_\alpha}$ is $\mathrm{Y}''_{\alpha,y'_\alpha}$-flat at the points whose projections on X and $\mathrm{Y}''_{\alpha,y'_\alpha}$ are x and the closed point of $\mathrm{Y}''_{\alpha,y'_\alpha}$. Moreover, the kernel of the homomorphism $\mathrm{A} \to \Gamma(\mathrm{Y}'_\alpha, \mathscr{O}_{\mathrm{Y}'_\alpha})$ is the intersection of the kernels of the homomorphisms $\mathrm{A} \to \Gamma(\mathrm{Y}''_{\alpha,y'_\alpha}, \mathscr{O}_{\mathrm{Y}''_{\alpha,y'_\alpha}})$: indeed, one is immediately reduced to the case where Y'_α is affine, and it then suffices to apply Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, cor. 2 of th. 1. Replacing the family (Y'_α) by the family of the $\mathrm{Y}''_{\alpha,y'_\alpha}$, one is therefore brought back to (11.4.12). ☐

11.5. Descent of flatness by arbitrary morphisms: the general case

Theorem (11.5.1). — *Let* Y *be a locally Noetherian prescheme, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_\mathrm{X}$-module, x a point of* X*, $y = f(x)$. Let (Y'_α) be a family of local* Y*-preschemes $\mathrm{Y}'_\alpha = \mathrm{Spec}(\mathrm{A}'_\alpha)$ such that the images of the closed points y'_α of Y'_α are all equal to y. For all α, let $\mathfrak{m}'_\alpha$ be the maximal ideal of A'_α, and $u_\alpha : \mathscr{O}_y \to \mathrm{A}'_\alpha$ the canonical homomorphism* (**I**, 2.4.4)*; suppose that the finite intersections of the ideals $u_\alpha^{-1}(\mathfrak{m}'^{\,n_\alpha}_\alpha)$ form a fundamental system of neighbourhoods of* 0 *in $\mathscr{O}_y$. Set $\mathrm{X}'_\alpha = \mathrm{X} \times_\mathrm{Y} \mathrm{Y}'_\alpha$, $f'_\alpha = f_{(\mathrm{Y}'_\alpha)}$, $\mathscr{F}'_\alpha = \mathscr{F} \otimes_\mathrm{Y} \mathrm{Y}'_\alpha$, and suppose that one of the following hypotheses holds:* **IV-3** | 151

(i) For all α, $\mathscr{F}'_\alpha$ is f'_α-flat at every point whose projection in X *is equal to x and whose projection in Y'_α is equal to y'_α.*

(ii) For all α, there exists a point $x'_\alpha \in \mathrm{X}'_\alpha$ whose projection on X *is x and whose projection in Y'_α is equal to y'_α, such that $\mathscr{F}'_\alpha$ is f'_α-flat at the point x'_α, and $k(x)$ is a primary extension of $k(y)$.*

Under these conditions, $\mathscr{F}$ is f-flat at the point x.

Proof. Let $\mathfrak{m}_y$ be the maximal ideal of $\mathscr{O}_y$; since $\mathscr{O}_y$ and $\mathscr{O}_x$ are Noetherian and $\mathscr{O}_y \to \mathscr{O}_x$ a homomorphism of local rings, it suffices, by virtue of ($\mathbf{0}_{\mathrm{III}}$, 10.2.2), to prove that for every $n > 0$, $\mathscr{F}_x/\mathfrak{m}_y^n\mathscr{F}_x$ is a flat $(\mathscr{O}_y/\mathfrak{m}_y^n)$-module. Let us denote by $(\mathfrak{J}_\lambda)$ the family of finite intersections of the $u_\alpha^{-1}(\mathfrak{m}'^{\,n_\alpha}_\alpha)$; by hypothesis, there exists λ such that $\mathfrak{J}_\lambda \subset \mathfrak{m}_y^n$, and since $\mathscr{F}_x \otimes_{\mathscr{O}_y} (\mathscr{O}_y/\mathfrak{m}_y^n) = (\mathscr{F}_x \otimes_{\mathscr{O}_y} (\mathscr{O}_y/\mathfrak{J}_\lambda)) \otimes_{\mathscr{O}_y/\mathfrak{J}_\lambda} (\mathscr{O}_y/\mathfrak{m}_y^n)$, it will suffice to prove that $\mathscr{F}_x/\mathfrak{J}_\lambda\mathscr{F}_x$ is a flat $(\mathscr{O}_y/\mathfrak{J}_\lambda)$-module. Now, for each α such that $\mathfrak{J}_\lambda \subset u_\alpha^{-1}(\mathfrak{m}'^{\,n_\alpha}_\alpha)$, passage to quotients gives a homomorphism $u'_\alpha : \mathscr{O}_y/\mathfrak{J}_\lambda \to \mathrm{A}'_\alpha/\mathfrak{m}'^{\,n_\alpha}_\alpha$, and the intersection of the kernels of the u'_α is reduced to 0. Taking into account (**I**, 3.6.1), one sees that one is brought back to (11.4.11) in case (ii), and to (11.4.12) in case (i). ☐

Corollary (11.5.2). — *Let* Y *be a locally Noetherian prescheme, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_\mathrm{X}$-module, x a point of* X*, $y = f(x)$; set $\mathrm{A} = \mathscr{O}_y$. Let $u : \mathrm{A} \to \mathrm{A}'$ be a homomorphism from* A *into a Zariski ring A', such that the inverse image $u^{-1}(\mathfrak{r}')$ of the radical $\mathfrak{r}'$ of A' is the maximal ideal $\mathfrak{m}$ of* A*; suppose moreover that the homomorphism $\widehat{u} : \widehat{\mathrm{A}} \to \widehat{\mathrm{A}}'$ is injective. Set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}' = \mathrm{X} \times_\mathrm{Y} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')}$, $\mathscr{F}' = \mathscr{F} \otimes_\mathrm{Y} \mathrm{Y}'$. For $\mathscr{F}$ to be f-flat at the point x, it is necessary and sufficient that $\mathscr{F}'$ be f'-flat at every point whose projection in* X *is equal to x and whose projection in Y' is equal to a closed point of Y'.*

If moreover A' is a finite A*-algebra, one can in what precedes replace the hypothesis that $\widehat{u}$ is injective by the hypothesis that u is injective.*

Proof. Since A (resp. A') is identified with a subring of $\widehat{\mathrm{A}}$ (resp. $\widehat{\mathrm{A}}'$) (Bourbaki, *Alg. comm.*, chap. III, § 3, n° 3, prop. 6), one sees first that u is itself injective and that $\widehat{u}$ is its extension by continuity to $\widehat{\mathrm{A}}$.

Let $(\mathfrak{m}'_\alpha)$ be the family of maximal ideals of A'; since one has

$$\mathfrak{m}'^{\,n}_\alpha \widehat{\mathrm{A}}' = \widehat{\mathfrak{m}}'^{\,n}_\alpha, \qquad \text{and} \qquad \widehat{\mathfrak{m}}'^{\,n}_\alpha \cap \mathrm{A}' = \mathfrak{m}'^{\,n}_\alpha,$$

and since the $\mathfrak{m}'^{\,n}_\alpha$ are open in A', one has $u^{-1}(\mathfrak{m}'^{\,n}_\alpha) = \mathrm{A} \cap \widehat{u}^{-1}(\widehat{\mathfrak{m}}'^{\,n}_\alpha)$, and it will suffice to prove that in $\widehat{\mathrm{A}}$ the finite intersections of the $\widehat{u}^{-1}(\widehat{\mathfrak{m}}'^{\,n_\alpha}_\alpha)$ form a fundamental system of neighbourhoods of 0. This will permit the application of (11.5.1) to the composite homomorphisms $v_\alpha \circ u : \mathrm{A} \to \mathrm{A}'_{\mathfrak{m}'_\alpha}$, where $v_\alpha : \mathrm{A}' \to \mathrm{A}'_{\mathfrak{m}'_\alpha}$ is the canonical homomorphism. Since $\widehat{\mathrm{A}}$ is complete, it will suffice to prove that the intersection of the $\widehat{u}^{-1}(\widehat{\mathfrak{m}}'^{\,n_\alpha}_\alpha)$ is reduced to 0 (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 7, prop. 8, where in the proof the decreasing sequence may be replaced by an arbitrary filtered set). Now, for every α fixed, the intersection of the ideals $\widehat{\mathfrak{m}}'^{n}_{\alpha}\widehat{A}'_{\widehat{\mathfrak{m}}'_{\alpha}}$ for $n > 0$ is reduced to 0 in the Noetherian local ring $\widehat{A}'_{\widehat{\mathfrak{m}}'_{\alpha}}$. On the other hand the $\widehat{\mathfrak{m}}'_{\alpha}$ are the maximal ideals of $\widehat{A}'$, hence the canonical homomorphism $\widehat{A}' \to \prod_{\alpha} \widehat{A}'_{\widehat{\mathfrak{m}}'_{\alpha}}$ is injective (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, cor. 2 of th. 1), and since by hypothesis $\widehat{u} : \widehat{A} \to \widehat{A}'$ is also injective, this achieves the proof in the general case. The last assertion follows from the fact that $\widehat{A}$ is an A-module faithfully flat ($\mathbf{0}_{\mathrm{I}}$, 7.3.5) and $\widehat{A}' = A' \otimes_A \widehat{A}$ since A' is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. III, § 3, n° 4, th. 3 and chap. IV, § 2, n° 5, cor. 3 of prop. 9). □ **IV-3 | 152**

Proposition (11.5.3). — *Let $f : X \to Y$ be a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module of finite presentation, x a point of X, $y = f(x)$, $g = (\psi, \theta) : Y' \to Y$ a proper morphism of finite presentation. Suppose that:*

(i) The homomorphism $\theta_y : \mathscr{O}_y \to (g_*(\mathscr{O}_{Y'}))_y$ is injective.
(ii) For every $x' \in X' = X \times_Y Y'$ whose projection in X is x, $\mathscr{F}' = \mathscr{F} \otimes_Y Y'$ is Y'-flat at the point x'.

Then $\mathscr{F}$ is f-flat at the point x.

Proof. The question being local on X, one can suppose f of finite presentation. Let $p : X' \to X$ be the canonical projection. Since $f' = f_{|Y'} : X' \to Y'$ is of finite presentation (1.6.2) and $\mathscr{F}'$ is an $\mathscr{O}_{X'}$-module of finite presentation ($\mathbf{0}_{\mathrm{I}}$, 5.2.5), it follows from (11.3.1) that the set U' of points of X' where $\mathscr{F}'$ is f'-flat is open. Since U' contains $p^{-1}(x)$ by hypothesis, and p is proper, hence *closed*, U' contains a set of the form $p^{-1}(U)$, where U is a neighbourhood of x. Replacing X by U, one can therefore suppose that $\mathscr{F}'$ is f'-flat. On the other hand, taking into account (**I**, 3.6.5), (**II**, 5.4.2) and (2.1.4), one can also replace Y by $\mathrm{Spec}(\mathscr{O}_y)$, i.e. suppose that $Y = \mathrm{Spec}(A)$, where A is a local ring. Under these conditions, we shall prove that $\mathscr{F}$ is f-flat. By virtue of (5.13.3), A is the inductive limit of local Noetherian sub-rings A_λ such that the canonical injection $A_\lambda \to A$ is a *local* homomorphism. By virtue of (8.9.1), one can suppose that $X = X_\lambda \otimes_{A_\lambda} A$, $f = f_\lambda \otimes 1_A$, $\mathscr{F} = \mathscr{F}_\lambda \otimes_{A_\lambda} A$, for a suitable λ, where $f_\lambda : X_\lambda \to Y_\lambda = \mathrm{Spec}(A_\lambda)$ is a morphism of finite type and $\mathscr{F}_\lambda$ is a coherent $\mathscr{O}_{X_\lambda}$-module. Similarly, one can suppose that $Y' = Y'_\lambda \otimes_{A_\lambda} A$, $g = g_\lambda \otimes 1_A$, where $g_\lambda : Y'_\lambda \to Y_\lambda$ is a morphism of finite presentation; moreover (8.10.5, (xii)), one can suppose g_λ *proper*. Since by hypothesis the homomorphism $A \to \Gamma(Y', \mathscr{O}_{Y'})$ is injective and that A_λ is a sub-ring of A, the homomorphism $A_\lambda \to \Gamma(Y', \mathscr{O}_{Y'})$ is also injective. Finally, by virtue of (11.2.6), one can suppose λ taken such that $\mathscr{F}'_\lambda = \mathscr{F}_\lambda \otimes_{Y_\lambda} Y'_\lambda$ is f'_λ-flat, since $\mathscr{F}' = \mathscr{F}'_\lambda \otimes_{Y'_\lambda} Y'$. These remarks show that one can henceforth suppose that the ring A is *Noetherian*, the other hypotheses of (11.5.3) being verified. Let us then set $B = \widehat{A}$, $Z = \mathrm{Spec}(B)$; since B is an A-module faithfully flat ($\mathbf{0}_{\mathrm{I}}$, 7.3.5), it amounts to the same to say that $\mathscr{F}$ is f-flat or that $\mathscr{F} \otimes_Y Z$ is Z-flat (2.1.4); similarly, if one sets $Z' = Y' \times_Y Z$, the morphism $Z' \to Y'$ is faithfully flat (2.2.13), hence it amounts to the same to say that $\mathscr{F}'$ is f'-flat or that $\mathscr{F}' \otimes_{Y'} Z'$ is Z'-flat; finally $h = g_{(\mathfrak{m})} : Z' \to Z$ is proper (**II**, 5.4.2) and of finite type (1.5.2), $\widehat{A}$ is Noetherian, and if z is its closed point, the homomorphism $\mathscr{O}_z \to (h_*(\mathscr{O}_{Z'}))_z$ is *injective*, since it follows from (2.3.1) that $h_*(\mathscr{O}_{Z'}) = g_*(\mathscr{O}_{Y'}) \otimes_Y Z$, and our assertion follows from hypothesis (i) and from the definition of flat modules ($\mathbf{0}_{\mathrm{I}}$, 6.1.1). **IV-3 | 153**

One can therefore now suppose the local Noetherian ring A *complete*; the proof will be finished if one proves that the intersection of the kernels of the homomorphisms $A = \mathscr{O}_y \to \mathscr{O}_{y'}$ (where y' ranges over $g^{-1}(y)$) is reduced to 0. Indeed, the $\mathscr{O}_{y'}$ are local Noetherian rings, hence for each y' the intersection of the $\mathfrak{m}^n_{y'}$ ($n \geqslant 0$) is reduced to 0; if $\mathfrak{a}_{n,y'}$ is the inverse image of $\mathfrak{m}^n_{y'}$ in A, the intersections of finitely many of the $\mathfrak{a}_{n,y'}$ are neighbourhoods of 0 in A and the intersection of all the $\mathfrak{a}_{n,y'}$ is reduced to 0; the intersections of finitely many of the $\mathfrak{a}_{n,y'}$ will therefore form a fundamental system of neighbourhoods of 0 in A (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 7, prop. 8, where one can in the proof replace the decreasing sequence by an arbitrary filtered set); one will then be able to apply (11.5.1). Now, let $s \in A$ be an element belonging to the kernel of each of the homomorphisms $A \to \mathscr{O}_{y'}$; the image s' of s in $\Gamma(Y', \mathscr{O}_{Y'})$ is therefore a section of $\mathscr{O}_{Y'}$ above Y' such that $s'_{y'} = 0$ for every $y' \in g^{-1}(y)$; then there is a neighbourhood V' of $g^{-1}(y)$ in Y' such that $s'|V' = 0$. But since g is *closed*, V' contains a set of the form $g^{-1}(V)$, where V is an open neighbourhood of y in Y; but Y is a local scheme and y is its closed point, so $V = Y$ and consequently $V' = Y'$, $s' = 0$, and since $A \to \Gamma(Y', \mathscr{O}_{Y'})$ is injective by hypothesis, $s = 0$. □

The following particular case of (11.5.3) will be useful in chap. V:

Corollary (11.5.4). — *Let $f = (\psi, \theta) : \mathrm{X} \to \mathrm{Y}$ be a proper morphism of finite presentation, and let $p : \mathrm{X} \times_{\mathrm{Y}} \mathrm{X} \to \mathrm{X}$ be the first projection. Suppose that $\theta : \mathscr{O}_{\mathrm{Y}} \to f_*(\mathscr{O}_{\mathrm{X}})$ is injective. Then, for f to be flat, it is necessary and sufficient that p be flat.*

Proposition (11.5.5). — *Let A be a ring, $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_{\mathrm{X}}$-module of finite presentation, x a point of X. Let $\mathrm{A} \to \mathrm{A}'$ be an injective homomorphism making A' an integral algebra over A. Set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f \times 1$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$. Then, if $\mathscr{F}'$ is f'-flat at every point of X' whose projection in X is x, $\mathscr{F}$ is f-flat at the point x.*

Proof. Suppose first that A' is a *finite* A-algebra of *finite presentation*; then the morphism $\mathrm{Y}' \to \mathrm{Y}$ is proper (**II**, 6.1.11) and of finite presentation, hence the hypotheses of (11.5.3) are satisfied, whence the conclusion. In the general case, the proposition results from this particular case, from the fact that A' is the inductive limit of its *finite* sub-A-algebras A'_λ and of the two following lemmas: ☐

Lemma (11.5.5.1). — *Every finite A-algebra A' is the inductive limit of A-algebras A'_λ which are finite and of finite presentation.*

Proof. One reasons as in (1.9.3.1). Indeed $\mathrm{A}' = \mathrm{B}/\mathfrak{J}$, where B is a finite A-algebra that is free as an A-module, and $\mathfrak{J}$ is an ideal of B (1.4.7.1). Now, $\mathfrak{J}$ is the inductive limit of ideals $\mathfrak{J}_\lambda \subset \mathfrak{J}$ of B which are of finite type (and hence, *a fortiori*, are A-modules *of finite type*), hence, by the exactness of the functor $\varinjlim$, A' is the inductive limit of the A-algebras $\mathrm{B}/\mathfrak{J}_\lambda$; now, $\mathrm{B}/\mathfrak{J}_\lambda$ is by definition an A-module of finite presentation, hence also (1.4.7) a finite A-algebra of finite presentation. ☐

Lemma (11.5.5.2). — *Let A be a ring, A' an A-algebra, (A'_λ) an inductive system of A-algebras such that $\mathrm{A}' = \varinjlim \mathrm{A}'_\lambda$; set $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{Y}'_\lambda = \mathrm{Spec}(\mathrm{A}'_\lambda)$. Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_{\mathrm{X}}$-module of finite presentation; set $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{|\mathrm{Y}'}$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$, $\mathrm{X}'_\lambda = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'_\lambda$, $f'_\lambda = f_{|\mathrm{Y}'_\lambda}$, $\mathscr{F}'_\lambda = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'_\lambda$. Let x be a point of X, such that $\mathscr{F}'$ is f'-flat at every point $x' \in \mathrm{X}'$ above x; then there exists λ such that $\mathscr{F}'_\lambda$ is f'_λ-flat at every point $x' \in \mathrm{X}'_\lambda$ above x.*

Proof. Let U' be the set of $x' \in \mathrm{X}'$ such that $\mathscr{F}'$ is f'-flat at the point x'; one knows (11.3.1) that U' is open in X' since f' is of finite presentation (1.6.2); similarly the set U'_λ of points of X'_λ where $\mathscr{F}'_\lambda$ is f'_λ-flat is open in X'_λ, and one knows moreover (11.2.6) that U' is the union of the $v_\lambda^{-1}(\mathrm{U}'_\lambda)$, where $v_\lambda : \mathrm{X}' \to \mathrm{X}'_\lambda$ is the canonical projection. Let us consider the scheme $\mathrm{T} = \mathrm{Spec}(k(x))$; set $\mathrm{T}' = \mathrm{T} \times_{\mathrm{Y}} \mathrm{Y}'$, $\mathrm{T}'_\lambda = \mathrm{T} \times_{\mathrm{Y}} \mathrm{Y}'_\lambda$, and let $w_\lambda : \mathrm{T}' \to \mathrm{T}'_\lambda$, $g' : \mathrm{T}' \to \mathrm{X}'$, $g'_\lambda : \mathrm{T}'_\lambda \to \mathrm{X}'_\lambda$ be the canonical projections. Let us set $\mathrm{V}'_\lambda = g'^{-1}_\lambda(\mathrm{U}'_\lambda)$, $\mathrm{V}' = g'^{-1}(\mathrm{U}') = \bigcup w_\lambda^{-1}(\mathrm{V}'_\lambda)$. By hypothesis (and taking into account (**I**, 3.6.1)), $\mathrm{V}' = \mathrm{T}'$; as T' is quasi-compact, there exists λ such that $w_\lambda^{-1}(\mathrm{V}'_\lambda) = \mathrm{T}'$. One deduces then from (8.3.3) applied to the closed quasi-compact subsets $\mathrm{T}'_\lambda - \mathrm{V}'_\lambda$ of T'_λ, that there exists $\mu \geqslant \lambda$ such that $\mathrm{T}'_\mu = \mathrm{V}'_\mu$; this means that $\mathscr{F}'_\mu$ is f'_μ-flat at every point of X'_μ whose projection to X is x. ☐ **IV-3** | 154

11.6. Descent of flatness by arbitrary morphisms: the case of a unibranch base prescheme

Theorem (11.6.1). — *Let A be an integral geometrically unibranch local ring* (**0**, 23.2.1), $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, *$f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_{\mathrm{X}}$-module of finite presentation. Let A' be a local ring, $u : \mathrm{A} \to \mathrm{A}'$ a local injective homomorphism; set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{|\mathrm{Y}'}$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$. Let x be a point of X whose projection $f(x) = y$ is the closed point of Y, x' a point of X' whose projections in X and Y' are respectively x and the closed point y' of Y'. Then, if $\mathscr{F}'$ is f'-flat at the point x', $\mathscr{F}$ is f-flat at the point x.*

Proof. One can restrict to the case where f is of finite presentation, the question being local on X. We will proceed in several steps.

I) *Reduction to the case where* A *and* A' *are integrally closed local rings.*

Since u is injective and A is integral, there exists a prime ideal $\mathfrak{p}'$ of A' such that $u^{-1}(\mathfrak{p}') = (0)$ ($\mathbf{0}_{\mathrm{I}}$, 1.5.8); the composite homomorphism $\mathrm{A} \to \mathrm{A}' \to \mathrm{A}'' = \mathrm{A}'/\mathfrak{p}'$ is therefore injective and local, and if $\mathrm{Y}'' = \mathrm{Spec}(\mathrm{A}'')$, $\mathrm{X}'' = \mathrm{X}' \otimes_{\mathrm{A}'} \mathrm{A}'' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}''$, $\mathscr{F}'' = \mathscr{F}' \otimes_{\mathrm{A}'} \mathrm{A}''$, $\mathscr{F}''$ is Y''-flat at the points of X'' above x' (2.1.4); replacing A' by A'' as needed and taking into account (**I**, 3.4.7), one can therefore first suppose A' integral. If K' is the field of fractions of A', there is a valuation ring B' in K' that dominates A'; the composite homomorphism $A \xrightarrow{u} A' \to B'$ being injective and local, the same reasoning as the preceding argument allows one to replace A' by B'; one may therefore suppose that the local ring A' is integrally closed, A being a local subring of A' dominated by A'. Let A_1 be the integral closure of A; it is clear that $A \subset A_1 \subset A'$, and by hypothesis A_1 is a *local* ring; if $\mathfrak{m}$, $\mathfrak{m}_1$, $\mathfrak{m}'$ are the maximal ideals of A, A_1, A', one has $\mathfrak{m} \subset \mathfrak{m}_1 \subset \mathfrak{m}'$; indeed, $\mathfrak{m}_1$ is the *sole* prime ideal of A_1 above $\mathfrak{m}$, since A_1 is a local ring (Bourbaki, *Alg. comm.*, chap. V, § 2, nº 1, prop. 1); since $\mathfrak{m}' \cap A = \mathfrak{m}$, one has $\mathfrak{m}' \cap A_1 \cap A = \mathfrak{m}$, hence $\mathfrak{m}' \cap A_1 = \mathfrak{m}_1$. Set $Y_1 = \operatorname{Spec}(A_1)$, $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$, $\mathscr{F}_1 = \mathscr{F} \otimes_Y Y_1$, and let x_1 be the projection of x' in X_1; let us denote furthermore by y_1 the unique closed point of Y_1, so that $y_1 = f_1(x_1)$. By hypothesis the morphism $\operatorname{Spec}(k(y_1)) \to \operatorname{Spec}(k(y))$ is *radicial*, whence one concludes, by the transitivity of fibres (**I**, 3.6.4) and (**I**, 3.5.7), that the morphism $f_1^{-1}(y_1) \to f^{-1}(y)$ is radicial, and in particular x_1 is the *sole* point of X_1 whose projections in X and Y_1 are x and y_1 respectively; moreover, as one has seen, y_1 is the *sole* point of Y_1 whose projection in Y is y, hence x_1 is the *sole* point of X_1 whose projection in X is x. If one proves that $\mathscr{F}_1$ is f_1-flat at the point x_1, one will be able to apply (11.5.5), whence the conclusion. One is therefore brought back to the case where A is also *integrally closed*. **IV-3 | 155**

II) *Reduction to the case where* A *and* A' *are local rings of* **Z***-algebras of finite type integrally closed.*

One may regard A' as the filtered inductive limit of its finite-type sub-**Z**-algebras B'_λ that are integral domains; moreover, since A' is integrally closed and the integral closure of a **Z**-algebra of finite type is also a **Z**-algebra of finite type (7.8.3), one sees that A' is the inductive limit of its sub-**Z**-algebras of finite type B'_λ *integrally closed*; if $\mathfrak{m}'_\lambda = \mathfrak{m}' \cap B'_\lambda$, A' is also the inductive limit of the local subrings $(B'_\lambda)_{\mathfrak{m}'_\lambda} = A'_\lambda$ dominated by A' (5.13.3). For every λ, $B'_\lambda \cap A$ is likewise the inductive limit of its finite-type sub-**Z**-algebras $B_{\alpha\lambda}$, hence $A = \varinjlim_{\alpha,\lambda} B_{\alpha\lambda}$; as above one may replace $B_{\alpha\lambda}$ in this formula by its integral closure (which by hypothesis is contained in $B'_\lambda \cap A$), and then by the local ring $A_{\alpha\lambda} = (B_{\alpha\lambda})_{\mathfrak{m}_{\alpha\lambda}}$, where $\mathfrak{m}_{\alpha\lambda} = \mathfrak{m} \cap B_{\alpha\lambda} = \mathfrak{m}'_\lambda \cap B_{\alpha\lambda}$, so that $A_{\alpha\lambda}$ is dominated by A'_λ and by A. Set $Y_{\alpha\lambda} = \operatorname{Spec}(A_{\alpha\lambda})$; it follows from (8.9.1) that there exist a sufficiently large pair (α, λ), a morphism $f_{\alpha\lambda} : X_{\alpha\lambda} \to Y_{\alpha\lambda}$ of finite type, and a coherent $\mathscr{O}_{X_{\alpha\lambda}}$-module $\mathscr{F}_{\alpha\lambda}$ such that $X = X_{\alpha\lambda} \times_{Y_{\alpha\lambda}} Y$, $f = f_{\alpha\lambda} \times 1_Y$, and $\mathscr{F} = \mathscr{F}_{\alpha\lambda} \otimes_{Y_{\alpha\lambda}} Y$; if $x_{\alpha\lambda}$ is the projection of x in $X_{\alpha\lambda}$, it will suffice to show that $\mathscr{F}_{\alpha\lambda}$ is $f_{\alpha\lambda}$-flat at the point $x_{\alpha\lambda}$. Since A' is the inductive limit of the A'_μ for $\mu \geqslant \lambda$, X' is the projective limit of the $X_{\alpha\lambda} \times_{Y_{\alpha\lambda}} Y'_\mu = X'_\mu$, and one also has $\mathscr{F}' = \mathscr{F}'_{\alpha\mu} \otimes_{Y'_\mu} Y'$, where $\mathscr{F}'_{\alpha\mu} = \mathscr{F}_{\alpha\lambda} \otimes_{Y_{\alpha\lambda}} Y'_\mu$. Applying (11.2.6), one sees that one can take μ large enough for $\mathscr{F}'_{\alpha\mu}$ to be Y'_μ-flat at the point x'_μ, projection of x' in X'_μ, and moreover, by construction of the A'_μ, the projection of x'_μ in Y'_μ is the closed point of Y'_μ.

III) *Reduction to the case where the residue field* k' *of* A' *is a finite extension and radicial of the residue field* k *of* A.

One can first of all repeat the reasoning of part I) of the proof of (11.4.11), taking into account the fact that **Z** is a Jacobson ring; one is thus reduced to the case where k' is a finite extension of k, which we suppose from now on. Let k'' be the largest separable extension of k contained in k', and let k_1 be a finite Galois extension of k containing k'', so that $k'' \otimes_k k_1$ is a direct product of fields isomorphic to k_1. Since k' is a radicial extension of k'', $k' \otimes_k k_1$ is therefore a direct product of radicial extensions of k_1. **IV-3 | 156**

There exists a local ring A_1 which is an A-algebra and a free A-module of finite type, such that $A_1/\mathfrak{m}A_1$ is k-isomorphic to k_1 ($\mathbf{0}_{\text{III}}$, 10.3.1.2). More precisely, write $k_1 = k[T]/(r)$, where r is a monic irreducible separable polynomial of degree n, and let $R \in A[T]$ be a monic polynomial whose canonical image is r; one may take $A_1 = A[T]/(R)$. If K is the field of fractions of A, it is clear that R is an irreducible separable polynomial of K[T]; it follows first that A_1 is an integral ring whose field of fractions $K_1 = K[T]/(R)$ is a separable extension of K. Moreover, if t is the canonical image of T in A_1, the t^j $(0 \leqslant j < n)$ form a basis of the A-module A_1, and their images in k_1 form a basis over k; hence $d = \det\big(\operatorname{Tr}_{A_1/A}(t^{i+j})\big)$ is an element of A whose class in k is nonzero, and which is consequently invertible. The same reasoning as in (6.12.4.1), I) then proves that the morphism $\mathrm{Spec}(A_1) \to \mathrm{Spec}(A)$ is flat and has regular fibres; one concludes from (6.5.4), (ii), that A_1 is *integrally closed*.

Consider now the ring $A'_1 = A' \otimes_A A_1$. It is a free A'-module of finite type, hence a semilocal ring (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, cor. 3 of prop. 9); moreover, the maximal ideals of this finite A'-algebra all lie above the maximal ideal $\mathfrak{m}'$ of A', and *a fortiori* contain $\mathfrak{m}A'_1$. But

$$A'_1/\mathfrak{m}A'_1 = (A'/\mathfrak{m}A') \otimes_k k_1,$$

and, since k_1 is a finite separable extension of k, the radical of $A'_1/\mathfrak{m}A'_1$ is equal to $(\mathfrak{m}'/\mathfrak{m}A') \otimes_k k_1$ (Bourbaki, *Alg.*, chap. VIII, § 7, n° 2, cor. 2 of prop. 3). If $\mathfrak{n}_i$ $(1 \leqslant i \leqslant r)$ are the maximal ideals of A'_1, the fields $A'_1/\mathfrak{n}_i$ are therefore the fields composing the algebra $k' \otimes_k k_1$; in other words, they are *finite radicial extensions of* k_1.

Since $A \to A'$ is injective and A_1 is a flat A-module, the homomorphism $A_1 \to A'_1$ is injective. The canonical homomorphism

$$A'_1 \longrightarrow \prod_{i=1}^{r} (A'_1)_{\mathfrak{n}_i}$$

is also injective (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, th. 1), and consequently so is the composite from A_1 to this product. Now A_1 is integral and there are only finitely many kernels of the homomorphisms $A_1 \to (A'_1)_{\mathfrak{n}_i}$; since their intersection is zero, one of them is already zero. In other words, for some i there is a ring $B_1 = (A'_1)_{\mathfrak{n}_i}$ such that the homomorphism $A_1 \to B_1$ is *injective* and *local*.

Set $Y'_1 = \mathrm{Spec}(B_1)$, $X'_1 = X' \times_{Y'} Y'_1$, and $\mathscr{F}'_1 = \mathscr{F}' \otimes_{Y'} Y'_1$. The module $\mathscr{F}'_1$ is Y'_1-flat at all points of X'_1 above x'; moreover, the maximal ideal of B_1 is the only one above $\mathfrak{m}'$, and hence all these points project to the closed point y'_1 of Y'_1. Let x'_1 be one of these points. On the other hand, set $Y_1 = \mathrm{Spec}(A_1)$, $X_1 = X \times_Y Y_1$, and $\mathscr{F}_1 = \mathscr{F} \otimes_Y Y_1$; if x_1 is the projection of x'_1 in X_1, its projections in X and Y_1 are respectively x and the closed point y_1. If one proves that $(\mathscr{F}_1)_{x_1}$ is a flat $\mathscr{O}_{y_1}$-module, it follows that $\mathscr{F}_x$ is a flat $\mathscr{O}_y$-module: indeed, $\mathscr{O}_{y_1}$ is a flat $\mathscr{O}_y$-module, so $(\mathscr{F}_1)_{x_1}$ is a flat $\mathscr{O}_y$-module ($\mathbf{0}_{\mathrm{I}}$, 6.2.1).

But $(\mathscr{F}_1)_{x_1} = \mathscr{F}_x \otimes_{\mathscr{O}_x} \mathscr{O}_{x_1}$, and $\mathscr{O}_{x_1}$ is a faithfully flat $\mathscr{O}_x$-module; hence (2.2.11), (iii), $\mathscr{F}_x$ is a flat $\mathscr{O}_y$-module. Since $X'_1 = X_1 \times_{Y_1} Y'_1$ and $\mathscr{F}'_1 = \mathscr{F}_1 \otimes_{Y_1} Y'_1$, we have indeed reduced to the situation of (11.6.1), with A and A' replaced respectively by A_1 and B_1. **IV-3** | 157

IV) *End of the proof.* — We are finally reduced to proving (11.6.1) under the following supplementary hypotheses:

(i) A and A' are local rings of **Z**-algebras of finite type (hence *excellent* rings (7.8.3));
(ii) A is integrally closed;
(iii) the residue field k' of A' is a finite radicial extension of the residue field k of A.

It is then known ((7.8.3) and (2.3.8)) that, under conditions (i) and (ii), the $\mathfrak{m}$-adic topology on A is induced by the $\mathfrak{m}'$-adic topology on A'. The homomorphism on completions $\widehat{u} : \widehat{A} \to \widehat{A'}$ is therefore injective. On the other hand, since the morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is radicial, the same holds for $f'^{-1}(y') \to f^{-1}(y)$ (**I**, 3.5.7); there is therefore only one point x' of X' whose projections in X and Y' are respectively x and y'. One can therefore apply (11.5.2). □

Corollary (11.6.2). — *Let* A *be a unibranch integral local ring,* $Y = \mathrm{Spec}(A)$, $f : X \to Y$ *a morphism locally of finite presentation, and* $\mathscr{F}$ *a quasi-coherent* $\mathscr{O}_X$*-module of finite presentation. Let* A' *be a local ring and* $A \to A'$ *an injective local homomorphism; set* $Y' = \mathrm{Spec}(A')$, $X' = X \times_Y Y'$, $f' = f_{(Y')}$, *and* $\mathscr{F}' = \mathscr{F} \otimes_Y Y'$. *Let* x *be a point of* X *whose projection* $f(x) = y$ *is the closed point of* Y, *and suppose that* $\mathscr{F}'$ *is* f'*-flat at all points* x' *of* X' *whose projections in* X *and* Y' *are respectively* x *and the closed point* y' *of* Y'. *Then* $\mathscr{F}$ *is* f*-flat at* x.

Proof. One may repeat part I) of the proof of (11.6.1), which shows (with the same notation) that if $\mathscr{F}_1$ is f_1-flat at *all* points x_1 of X_1 whose projections in X and Y_1 are respectively x and y_1, then $\mathscr{F}$ is f-flat at x. One is thus reduced to the case where A is integrally closed, hence geometrically unibranch, and the conclusion follows from (11.6.1). □

11.7. Counter-examples

(11.7.1). Let us first consider the case where A is an *Artinian* local ring and where the hypotheses of (11.4.11) are satisfied *except* for condition (ii) concerning the residue field k of A. We shall see that the conclusion of (11.4.11) can then fail. Let k be a field admitting a Galois extension k' of degree $[k' : k] > 1$, and let Γ denote its Galois group. Let A be a k-algebra having a basis of three elements $1, a, b$, with multiplication table $a^2 = ab = ba = b^2 = 0$, so that A is an Artinian local ring whose maximal ideal $\mathfrak{m} = ka + kb$ has square zero. Let $\mathrm{A}' = \mathrm{A} \otimes_k k'$, which is a k'-algebra with basis $1, a, b$ and an Artinian local ring with square-zero maximal ideal $\mathfrak{m}' = k'a + k'b = \mathfrak{m}\mathrm{A}'$; the ring A identifies canonically with a subring of A'. Let $\mathfrak{J}$ be the sub-k'-vector space of $\mathfrak{m}'$ generated by $a + \gamma b$, where $\gamma \in k'$ does not belong to k; it is clear that $\mathfrak{J}$ is an ideal of A'. Set $\mathrm{B} = \mathrm{A}'/\mathfrak{J}$; this is an Artinian ring which is a non-flat A-module. Otherwise (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. 2 of prop. 5), B would be a free A-module. Since A' is likewise a free A-module and the canonical homomorphism $\mathrm{A}'/\mathfrak{m}\mathrm{A}' \to \mathrm{B}/\mathfrak{m}\mathrm{B}$ is bijective, the canonical homomorphism $\mathrm{A}' \to \mathrm{B} = \mathrm{A}'/\mathfrak{J}$ would also be bijective (loc. cit., n° 2, cor. of prop. 6), which is absurd. In other **IV-3** | 158
words, if $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, then $\mathscr{O}_\mathrm{X}$ is not Y-flat at the unique point x of X. Now put $\mathrm{A}_1 = \mathrm{B}$ and $\mathrm{Y}_1 = \mathrm{Spec}(\mathrm{A}_1)$, and consider the canonical homomorphism $\mathrm{A} \to \mathrm{A}_1$, which is local and injective since $\mathfrak{J} \cap \mathrm{A} = 0$. If $\mathrm{B}_1 = \mathrm{B} \otimes_\mathrm{A} \mathrm{A}_1 = \mathrm{B} \otimes_\mathrm{A} \mathrm{B}$, we shall see that there is a point of $\mathrm{X}_1 = \mathrm{X} \times_\mathrm{Y} \mathrm{Y}_1 = \mathrm{Spec}(\mathrm{B}_1)$ at which $\mathscr{O}_{\mathrm{X}_1}$ is Y_1-flat. For this, note that

$$\mathrm{B} \otimes_\mathrm{A} \mathrm{B} = \frac{\mathrm{A}' \otimes_\mathrm{A} \mathrm{A}'}{\mathrm{Im}(\mathfrak{J} \otimes_\mathrm{A} \mathrm{A}') + \mathrm{Im}(\mathrm{A}' \otimes_\mathrm{A} \mathfrak{J})}.$$

The structure of $\mathrm{C}' = \mathrm{A}' \otimes_\mathrm{A} \mathrm{A}'$ is obtained easily. Consider the product A-algebra $\mathrm{C} = \prod_{\sigma \in \Gamma} \mathrm{A}'_\sigma$, where all the A'_σ are equal to A', and the canonical map $\varphi : \mathrm{A}' \otimes_\mathrm{A} \mathrm{A}' \to \mathrm{C}$ defined by $\varphi(x \otimes y) = (\sigma(x)y)_{\sigma \in \Gamma}$, where Γ acts canonically on A'. Passing to the quotients gives a homomorphism $\mathrm{C}'/\mathfrak{m}\mathrm{C}' \to \mathrm{C}/\mathfrak{m}\mathrm{C}$ which is none other than the canonical homomorphism $k' \otimes_k k' \to \prod_\sigma k'_\sigma$, with $k'_\sigma = k'$ for all $\sigma \in \Gamma$. This last homomorphism is bijective (Bourbaki, *Alg.*, chap. VIII, § 8, prop. 4), and hence so is φ, since C' and C are free A-modules (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. of prop. 6). It follows that $\mathrm{B} \otimes_\mathrm{A} \mathrm{B}$ is a semilocal A-algebra which is the direct product of the local A-algebras $\mathrm{A}'/(\sigma(\mathfrak{J}) + \mathfrak{J})$. The factor corresponding to the identity element of Γ is isomorphic to $\mathrm{A}'/\mathfrak{J}$, and is therefore a flat A_1-module. Since $\gamma \notin k$, however, there is at least one $\sigma \in \Gamma$ such that $\sigma(\mathfrak{J}) \neq \mathfrak{J}$; then $\sigma(\mathfrak{J}) + \mathfrak{J} = \mathfrak{m}'$, and $\mathrm{A}'/\mathfrak{m}'$ is not a flat $(\mathrm{A}'/\mathfrak{J})$-module, since it is the quotient of $\mathrm{A}'/\mathfrak{J}$ by a nonzero ideal.

(11.7.2). We shall now show that the result of (11.5.4) loses its validity when f is no longer assumed to be proper (and *a fortiori* (11.5.3) ceases to be exact when g is no longer assumed proper). Let k be a field, let $\mathrm{A}_0 = k[\mathrm{S}, \mathrm{T}]$, and let $\mathrm{A} = \mathrm{A}_0/\mathrm{A}_0\mathrm{ST}$. Thus $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ is the reducible curve formed by the two "coordinate axes" in the affine plane $\mathrm{V}_k^2 = \mathrm{Spec}(\mathrm{A}_0)$. The ring A has two minimal prime ideals $\mathfrak{p}_1 = \mathrm{A}_0\mathrm{S}/\mathrm{A}_0\mathrm{ST}$ and $\mathfrak{p}_2 = \mathrm{A}_0\mathrm{T}/\mathrm{A}_0\mathrm{ST}$. Since A is reduced, it embeds canonically into $\mathrm{B} = \mathrm{B}_1 \oplus \mathrm{B}_2$, where $\mathrm{B}_1 = \mathrm{A}/\mathfrak{p}_1$ and $\mathrm{B}_2 = \mathrm{A}/\mathfrak{p}_2$. Moreover, B_1 identifies with $k[\mathrm{T}]$ and B_2 with $k[\mathrm{S}]$; these are integrally closed integral rings, and consequently $\mathrm{Z} = \mathrm{Spec}(\mathrm{B})$ is precisely the normalization of Y relative to $\mathrm{R}(\mathrm{Y})$ (**II**, 6.3.8), the sum of the two schemes $\mathrm{Z}_1 = \mathrm{Spec}(\mathrm{B}_1)$ and $\mathrm{Z}_2 = \mathrm{Spec}(\mathrm{B}_2)$. Denote by y the "double point" of Y corresponding to the maximal ideal $\mathfrak{p}_1 + \mathfrak{p}_2 = \mathfrak{m}$ of A, and by z_1, z_2 the points of Z above y, corresponding respectively to the maximal ideals $\mathfrak{m}_1 = (\mathrm{T})$ and $\mathfrak{m}_2 = (\mathrm{S})$ of B_1 and B_2. Let X be the subprescheme of Z induced on the complement of z_2. Thus $\mathrm{X} = \mathrm{Spec}(\mathrm{B}_1 \oplus (\mathrm{B}_2)_\mathrm{S})$. It is immediate that the homomorphism $\mathrm{A} \to \mathrm{A}' = \mathrm{B}_1 \oplus (\mathrm{B}_2)_\mathrm{S}$ is injective, but the corresponding morphism $f : \mathrm{X} \to \mathrm{Y}$ is not closed, since $f(\mathrm{Z}_2 - \{z_2\})$ is not closed in Y although $\mathrm{Z}_2 - \{z_2\}$ is closed in X; *a fortiori*, f is not proper. We shall now see that f is not flat at z_1. It is enough to show (**0**, 6.6.2) that $\mathscr{O}_{z_1}$ is not a faithfully flat $\mathscr{O}_y$-module, and for this it suffices to observe that the canonical homomorphism $\mathscr{O}_y \to \mathscr{O}_{z_1}$ is not injective. This is immediate, since $\mathscr{O}_{z_1}$ is an integral ring whereas $\mathscr{O}_y$ has zero-divisors. Nevertheless, the first projection $p : \mathrm{X} \times_\mathrm{Y} \mathrm{X} \to \mathrm{X}$ is an isomorphism. Indeed,

$$\mathrm{B}_1 \otimes_\mathrm{A} \mathrm{B}_1 = (\mathrm{A}/\mathfrak{p}_1) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{p}_1) = \mathrm{A}/\mathfrak{p}_1,$$

while $(\mathrm{B}_2)_\mathrm{S} \otimes_\mathrm{A} (\mathrm{B}_2)_\mathrm{S} = (\mathrm{B}_2 \otimes_\mathrm{A} \mathrm{B}_2)_\mathrm{S} = (\mathrm{B}_2)_\mathrm{S}$ for the same reason; finally, $\mathrm{B}_1 \otimes_\mathrm{A} (\mathrm{B}_2)_\mathrm{S} = 0$, since the canonical image of S in B_1 is zero.

(11.7.3). The preceding example can be generalized. Over a field k, consider a reduced algebraic curve Y admitting a single "ordinary double point" y (a notion that will be defined later in general), together with its normalization Z, such that $g : \mathrm{Z} \to \mathrm{Y}$ is finite, the restriction of g to $\mathrm{Z} - g^{-1}(y)$ is an isomorphism onto $\mathrm{Y} - \{y\}$, and $g^{-1}(y)$ consists of two "simple" points z_1, z_2. Moreover, the prescheme $g^{-1}(y)$ is the sum of the two preschemes $\mathrm{Spec}(k(z_1))$ and $\mathrm{Spec}(k(z_2))$, each canonically isomorphic to $\mathrm{Spec}(k(y))$. Let X be the subprescheme of Z induced on the open set $\mathrm{Z} - \{z_2\}$. The morphism $f : \mathrm{X} \to \mathrm{Y}$ obtained by restricting g is not proper; otherwise (**II**, 5.4.3) would imply the same for the canonical injection $j : \mathrm{X} \to \mathrm{Z}$, which is not closed. The morphism f is radicial: for every $y' \in \mathrm{Y}$, the fibre $f^{-1}(y')$ consists of a single point x', and $k(y') \to k(x')$ is an isomorphism. It follows first that the diagonal $\Delta_f : \mathrm{X} \to \mathrm{X} \times_{\mathrm{Y}} \mathrm{X}$ is bijective (**I**, 7.7.1). On the other hand, since f is unramified (17.4.2, (d')), Δ_f is an open immersion (17.4.2, (b')). Consequently Δ_f is an isomorphism, and the first projection $p : \mathrm{X} \times_{\mathrm{Y}} \mathrm{X} \to \mathrm{X}$ is its inverse. Nevertheless, f is not flat at z_1. Otherwise $\mathscr{O}_{z_1}$ would be a faithfully flat $\mathscr{O}_y$-module (**0**, 6.6.2); since $\mathscr{O}_y$ contains two distinct minimal prime ideals $\mathfrak{p}_1, \mathfrak{p}_2$, corresponding to the two branches of Y at y, there would then be two distinct prime ideals of $\mathscr{O}_{z_1}$ whose inverse images under $u : \mathscr{O}_y \to \mathscr{O}_{z_1}$ were $\mathfrak{p}_1$ and $\mathfrak{p}_2$ (**0**, 6.5.1). This is absurd, since $\mathscr{O}_{z_1}$ has only two distinct prime ideals, 0 and its maximal ideal $\mathfrak{m}_1$, while $u^{-1}(\mathfrak{m}_1)$ is the maximal ideal $\mathfrak{m}$ of $\mathscr{O}_y$.

(11.7.4). One will note that in the preceding example the homomorphism u is *injective* when Y is *irreducible* (one may for example take $\mathrm{Y} = \mathrm{Spec}(k[\mathrm{S}, \mathrm{T}]/(\mathrm{S}(\mathrm{S}^2 + \mathrm{T}^2) - (\mathrm{S}^2 - \mathrm{T}^2)))$, a "cubic with a double point"). Indeed, after replacing Y by an affine neighbourhood of y, one may suppose that $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, where A is integral, and hence that $\mathrm{Z} = \mathrm{Spec}(\mathrm{B})$, where B is the integral closure of A. Since B, $\mathscr{O}_y$, and $\mathscr{O}_{z_1}$ then identify with subrings of the field of fractions of A, u is obviously injective. **IV-3** | 159
On the other hand, the homomorphism $\widehat{u} : \widehat{\mathscr{O}}_y \to \widehat{\mathscr{O}}_{z_1}$ is not injective: $\widehat{\mathscr{O}}_{z_1}$ is an integral local ring, since z_1 is a simple point, whereas $\widehat{\mathscr{O}}_y$ has two distinct minimal prime ideals corresponding to the two branches of Y and therefore has zero-divisors. This gives an example showing that in the statement of (11.5.2), the hypothesis that $\widehat{u}$ is injective cannot be replaced by the hypothesis that u itself is injective, even when A' is a local ring. Indeed, with the preceding notation, it suffices to take $\mathrm{A} = \mathscr{O}_y$, $\mathrm{A}' = \mathscr{O}_{z_1}$, $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, and $\mathrm{X} = \mathrm{Spec}(\mathrm{A}')$; the reasoning of (11.7.3) still proves that the first projection $p : \mathrm{X} \times_{\mathrm{Y}} \mathrm{X} \to \mathrm{X}$ is an isomorphism, although f is not flat at z_1.

(11.7.5). The examples of (11.7.2) and (11.7.3) explain the restriction to *unibranch* local rings in (11.6.1) and (11.6.2). We shall now see that in (11.6.1), one cannot weaken the hypothesis on A by supposing only that A is unibranch. Consider the complete local integral ring $\mathrm{A} = \mathbf{R}[[\mathrm{U}, \mathrm{V}]]/(\mathrm{U}^2 + \mathrm{V}^2)$, which is unibranch but not geometrically unibranch (6.15.11). One knows (*loc. cit.*) that if u and v are the images of U and V in A, the integral closure of A is $\bar{\mathrm{A}} = \mathrm{A}[t]$, where $t = v/u$ and $t^2 = -1$, so that $\bar{\mathrm{A}}$ is isomorphic to $\mathbf{C}[[\mathrm{U}]]$. Set $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{X} = \mathrm{Spec}(\bar{\mathrm{A}})$, the normalization of Y (**II**, 6.3.8), and let y and x be the closed points of Y and X respectively. We shall show that for a suitable local A-algebra A', if one sets $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ and $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, and if y' denotes the closed point of Y', then $\mathscr{O}_{\mathrm{X}'}$ is Y'-flat at one point of X' whose projections in X and Y' are respectively x and y', but is not Y'-flat at every point having these projections. It will then follow from (2.1.4) that $\mathscr{O}_{\mathrm{X}}$ is not Y-flat at x (which is moreover trivial *a priori*, since $\bar{\mathrm{A}}$ is not a free A-module).

Let $\mathrm{B} = \mathrm{A} \otimes_{\mathbf{R}} \mathbf{C}$, which is isomorphic to $\mathbf{C}[[\mathrm{U}, \mathrm{V}]]/((\mathrm{U} + i\mathrm{V})(\mathrm{U} - i\mathrm{V}))$. The ring B has two minimal prime ideals $\mathfrak{p}'$ and $\mathfrak{p}''$, generated respectively by $u + iv$ and $u - iv$, and $\mathfrak{n} = \mathfrak{p}' + \mathfrak{p}''$ is the maximal ideal of the complete local ring B. Let $\bar{\mathrm{B}} = \bar{\mathrm{A}} \otimes_{\mathbf{R}} \mathbf{C}$; it is the direct product of two algebras isomorphic to $\mathbf{C}[[\mathrm{U}]]$, generated by the idempotents $e' = (1 \otimes 1 + t \otimes i)/2$ and $e'' = (1 \otimes 1 - t \otimes i)/2$. Since the homomorphism $\mathrm{A} \to \bar{\mathrm{A}}$ is injective, so is $\mathrm{B} \to \bar{\mathrm{B}}$; the images of $u + iv$ and $u - iv$ under this injection are respectively ue' and ue''. It follows immediately that $\bar{\mathrm{B}}$ identifies canonically with $(\mathrm{B}/\mathfrak{p}') \oplus (\mathrm{B}/\mathfrak{p}'')$. Now take for A' the local ring $\mathrm{B}/\mathfrak{p}'$; then $\bar{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{A}'$ identifies with $\bar{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{A}'$. But $(\mathrm{B}/\mathfrak{p}') \otimes_{\mathrm{B}} (\mathrm{B}/\mathfrak{p}') = \mathrm{B}/\mathfrak{p}'$ and $(\mathrm{B}/\mathfrak{p}') \otimes_{\mathrm{B}} (\mathrm{B}/\mathfrak{p}'') = \mathrm{B}/\mathfrak{n}$, so $\bar{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{A}'$ is isomorphic to $\mathrm{A}' \oplus (\mathrm{B}/\mathfrak{n})$. This establishes our assertion, since $\mathrm{B}/\mathfrak{n} = \mathrm{A}'/(\mathfrak{n}/\mathfrak{p}')$ is not a flat A'-module; otherwise it would be a free A'-module (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 2, cor. 2 of prop. 5), which is absurd.

11.8. A valuative criterion for flatness

Theorem (11.8.1). — *Let $f : X \to Y$ be a locally of finite presentation morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation, x a point of X, $y = f(x)$. Suppose that the local ring $\mathscr{O}_y$ is integral (resp. reduced and Noetherian). For $\mathscr{F}$ to be f-flat at the point x, it is necessary and sufficient that, for every valuation ring (resp. every discrete valuation ring) A' and every local homomorphism $\mathscr{O}_y \to A'$, the following condition be satisfied: setting $Y' = \mathrm{Spec}(A')$, $X' = X \times_Y Y'$, $f' = f_{(Y')}$, the $\mathscr{O}_{X'}$-Module $\mathscr{F}' = \mathscr{F} \otimes_Y Y'$ is f'-flat in all the points x' of X' whose projections in X and Y' are respectively x and the closed point y' of Y'.*

Proof. The condition being evidently necessary (2.1.4), it remains to prove that it is sufficient. One can evidently ((**I**, 3.6.5) and (**I**, 2.4.4)) restrict to the case where $Y = \mathrm{Spec}(A)$ is the spectrum of a *local* ring A and y is the closed point of Y.

(i) *Case where* A *is integral.* — Let K be the field of fractions of A, A_1 the integral closure of A; if one sets $Y_1 = \mathrm{Spec}(A_1)$, $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$, it suffices, by virtue of (11.5.5), to show that $\mathscr{F}_1 = \mathscr{F} \otimes_Y Y_1$ is f_1-flat in *all* the points x_1 of X_1 whose projection x is in X. Now, if $f_1(x_1) = y_1$, then $\mathfrak{i}_{y_1} = \mathfrak{m}_1$, where $\mathfrak{m}_1$ is a prime ideal of A_1 whose trace on A is $\mathfrak{m} = \mathfrak{i}_y$ (and $\mathfrak{m}_1$ is moreover necessarily a maximal ideal). Let A' be a valuation ring for K that dominates $\mathscr{O}_{y_1} = (A_1)_{\mathfrak{m}_1}$; since the homomorphism $A \to \mathscr{O}_{y_1}$ is local, so is $A \to A'$. There exists then at least one point $x' \in X'$ **IV-3** | 160
whose projections in X_1 and Y' are respectively x_1 and y' (**I**, 3.4.7); as the hypothesis states that $\mathscr{F}'$ is f'-flat at the point x', and $\mathscr{O}_{y_1}$ is integrally closed, one can apply (11.6.1), and one deduces that $\mathscr{F}_1$ is f_1-flat at the point x_1, whence the theorem in this case.

(ii) *Case where* A *is reduced and Noetherian.* — Let $\mathfrak{p}_i$ ($1 \leqslant i \leqslant m$) be the minimal prime ideals of A; since A is reduced, it identifies canonically with a subring of the product of the $A_i = A/\mathfrak{p}_i$, which are local Noetherian rings; setting $Y_i = \mathrm{Spec}(A_i)$, $X_i = X \times_Y Y_i$, $f_i = f_{(Y_i)}$, it follows from (11.5.2) that it suffices to show that for each i, $\mathscr{F}_i = \mathscr{F} \otimes_Y Y_i$ is f_i-flat in every point x_i of X_i whose projections in X and Y_i are x and the closed point y_i of Y_i respectively. Now, since A_i is a local Noetherian integral ring, there exists in its field of fractions K_i a sub-ring A'_i containing A_i, which is a semi-local Noetherian ring (hence a finite A_i-algebra) and whose local rings are *geometrically unibranch* ((6.15.5) and (**0**, 23.2.5)). Since the maximal ideals of A'_i are necessarily above the maximal ideal of A_i, one deduces further from (**I**, 3.4.7) and (11.5.2) that it suffices, setting $Y'_i = \mathrm{Spec}(A'_i)$, $X'_i = X \times_Y Y'_i$, $f''_i = f_{(Y'_i)}$, to prove that $\mathscr{F}''_i = \mathscr{F} \otimes_Y Y'_i$ is f''_i-flat in every point x''_i of X''_i whose projections in X and Y'_i are x and the closed point y'_i of Y'_i respectively. Now, let A' be a discrete valuation ring for K_i that dominates A'_i, and let x' be a point of X' whose projections in X'_i and Y'_i are x''_i and y' respectively (**I**, 3.4.7); as $\mathscr{O}_{y'_i}$ is geometrically unibranch, one can again apply (11.6.1) and one deduces that $\mathscr{F}''_i$ is indeed f''_i-flat at the point x''_i. □

Remarks (11.8.2). —

(i) In the statement of (11.8.1), one may restrict to assuming that the condition on $\mathscr{F}'$ is satisfied for the *complete* valuation rings A' whose residue field is *algebraically closed*. Indeed, it is known that every such valuation ring A' is dominated by a valuation ring A'' (**II**, 7.1.2), and that if A' is a discrete valuation ring, it is the same for A'' ($\mathbf{0}_{\mathrm{III}}$, 10.3.1).

(ii) The proof of (11.8.1) simplifies when one supposes not only that A is integral and Noetherian, but that its completion $\widehat{A}$ is also *integral*. Replacing X by $X \otimes_A \widehat{A}$ and reasoning as in the proof of (11.5.3), one can in fact reduce to proving (11.8.1) when $A = \mathscr{O}_y$ is integral, Noetherian, and *complete*. Now, one knows (**II**, 7.1.7) that such a ring A is dominated by a discrete complete valuation ring; the conclusion follows therefore directly from (11.5.2).

11.9. Separating and universally separating families of homomorphisms of sheaves of modules

(11.9.1). Let X be a prescheme, $(f_\lambda)_{\lambda\in L}$ a family of morphisms $f_\lambda : Z_\lambda \to X$, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module; for every $\lambda \in L$, suppose given a quasi-coherent $\mathscr{O}_{Z_\lambda}$-Module $\mathscr{G}_\lambda$ and a homomorphism

$$u_\lambda : \mathscr{F} \longrightarrow (f_\lambda)_*(\mathscr{G}_\lambda).$$

We say that the family (u_λ) (or the corresponding family of $u_\lambda^\# : (f_\lambda)^*(\mathscr{F}) \to \mathscr{G}_\lambda$) is *separating* if the intersection of the kernels of the u_λ is *zero*. In other words, this signifies that for every open U of X, and every section t of $\mathscr{F}$ above U, such that, for *every* λ, the section $u_\lambda(t)$ (which, by definition, is a section of $\mathscr{G}_\lambda$ above $f_\lambda^{-1}(U)$) is zero, then t is itself zero. **IV-3** | 161

(11.9.2). With the notations of (11.9.1), let M be a second set of indices; for every $\lambda \in L$, let $(g_{\lambda\mu})_{\mu\in M}$ be a family of morphisms $g_{\lambda\mu} : Z'_{\lambda\mu} \to Z_\lambda$; for every pair (λ, μ), suppose given a quasi-coherent $\mathscr{O}_{Z'_{\lambda\mu}}$-Module $\mathscr{H}_{\lambda\mu}$ and a homomorphism $v_{\lambda\mu} : \mathscr{G}_\lambda \to (g_{\lambda\mu})_*(\mathscr{H}_{\lambda\mu})$; set $h_{\lambda\mu} = f_\lambda \circ g_{\lambda\mu} : Z'_{\lambda\mu} \to X$ and consider the composite homomorphism

$$w_{\lambda\mu} : \mathscr{F} \xrightarrow{u_\lambda} (f_\lambda)_*(\mathscr{G}_\lambda) \xrightarrow{(f_\lambda)_*(v_{\lambda\mu})} (h_{\lambda\mu})_*(\mathscr{H}_{\lambda\mu}).$$

Suppose that, for every $\lambda \in L$, the family $(v_{\lambda\mu})_{\mu\in M}$ is separating: then, it is the same for the family $(f_\lambda)_*(v_{\lambda\mu})$ ($\mu \in M$), as one sees at once. One concludes that, for the family (u_λ) to be separating, it is necessary and sufficient that the family $(w_{\lambda\mu})_{(\lambda,\mu)\in L\times M}$ be so.

(11.9.3). To verify that the family (u_λ) is separating (with the notations of (11.9.1)), one can evidently reduce first to the case where X is *affine*, the property being local on X. One can moreover suppose that $Z_\lambda = X$ *for every* $\lambda \in L$. Indeed, let M be an index set, sum of a family $(M_\lambda)_{\lambda\in L}$, and for every $\lambda \in L$, let $(Y_{\lambda\mu})_{\mu\in M_\lambda}$ be an open affine covering of Z_λ; let $j_{\lambda\mu} : Y_{\lambda\mu} \to Z_\lambda$ be the canonical injection and set $\mathscr{H}_{\lambda\mu} = j_{\lambda\mu}^*(\mathscr{G}_\lambda) = \mathscr{G}_\lambda|Y_{\lambda\mu}$. If one considers the canonical homomorphism $v_{\lambda\mu} : \mathscr{G}_\lambda \to (j_{\lambda\mu})_*(j_{\lambda\mu}^*(\mathscr{G}_\lambda)) = (j_{\lambda\mu})_*(\mathscr{H}_{\lambda\mu})$ relative to $j_{\lambda\mu}$ (**0**, 4.4.3.2), it is immediate that for each $\lambda \in L$, the family $(v_{\lambda\mu})_{\mu\in M_\lambda}$ is separating. By virtue of (11.9.2), one is therefore reduced to proving that the family of composite homomorphisms $((f_\lambda)_*(v_{\lambda\mu})) \circ u_\lambda$, where the f_λ are *affine*, is separating. But then the $(f_\lambda)_*(\mathscr{G}_\lambda)$ are quasi-coherent $\mathscr{O}_X$-Modules (**I**, 1.6.2) and, by virtue of the definition, one can replace the Z_λ by X and the $\mathscr{G}_\lambda$ by $(f_\lambda)_*(\mathscr{G}_\lambda)$, whence our assertion.

One will note further that if L is *finite* and the f_λ are *quasi-compact*, one can, in the preceding reduction, suppose that the M_λ are also *finite*, whence in this case one is brought back to verifying that a *finite* family of homomorphisms of $\mathscr{F}$ in quasi-coherent $\mathscr{O}_X$-Modules is separating.

(11.9.4). Consider therefore the case where $Z_\lambda = X$ for every λ, and where $X = \operatorname{Spec}(A)$ is affine; then $\mathscr{F} = \widetilde{M}$ and $\mathscr{G}_\lambda = \widetilde{N}_\lambda$, where M and N_λ are A-modules, and $u_\lambda = \widetilde{\varphi}_\lambda$, where the $\varphi_\lambda : M \to N_\lambda$ are A-module homomorphisms. To say that the family (u_λ) is separating signifies then that, for every $s \in A$, the intersection of the kernels of $(\varphi_\lambda)_s : M_s \to (N_\lambda)_s$ is zero. One will note that if L is *finite*, it comes down to saying also that the intersection of the kernels of the φ_λ is zero. One says then also that the family (φ_λ) is *separating*. One will note that if L is *finite*, then $\bigcap_{\lambda\in L}\operatorname{Ker}((\varphi_\lambda)_s) = (\bigcap_{\lambda\in L}\operatorname{Ker}(\varphi_\lambda))_s$ (**0**, 1.3.5). But this relation is no longer exact in general when L is *infinite*, and the fact that the intersection of the kernels of the φ_λ is zero *does not therefore entail*, in general, that the family (φ_λ) is separating. For example, suppose A is a discrete valuation ring with maximal ideal $\mathfrak{m}$, and consider the family of homomorphisms $\varphi_k : A \to A/\mathfrak{m}^k$; their kernels are the ideals $\mathfrak{m}^k$, whose intersection is zero. This family is nevertheless not separating, for the fibres of all the $\mathfrak{m}^k$ at the generic point x of $X = \operatorname{Spec}(A)$ (which is open in X) are equal to $\mathscr{O}_x = k(x)$, the field of fractions of A, and hence their intersection is not zero. **IV-3** | 162

(11.9.5). We will mainly be concerned in what follows with the problem of *change of base* for separating families. The notations being those of (11.9.1), consider a morphism $g : X' \to X$ and set $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} = g^*(\mathscr{F})$, and, for every λ, $Z'_\lambda = Z_\lambda \times_X X'$, $f'_\lambda = f_\lambda \times 1$, $\mathscr{G}'_\lambda = \mathscr{G}_\lambda \otimes_{\mathscr{O}_{Z_\lambda}} \mathscr{O}_{Z'_\lambda} = (g'_\lambda)^*(\mathscr{G}_\lambda)$, where $g'_\lambda : Z'_\lambda \to Z_\lambda$ is the projection. For every λ, we denote by $u'_\lambda : \mathscr{F}' \to (f'_\lambda)_*(\mathscr{G}'_\lambda)$ the homomorphism obtained as follows: let

$$\psi : (f_\lambda)_*(\mathscr{G}_\lambda) \longrightarrow (f_\lambda)_*((g'_\lambda)_*((g'_\lambda)^*(\mathscr{G}_\lambda))) = g_*((f'_\lambda)_*(\mathscr{G}'_\lambda))$$

be the homomorphism $(f_\lambda)_*(\rho_{g'_\lambda})$, where $\rho_{g'_\lambda}$ is the canonical homomorphism (**0**, 4.4.3.2) corresponding to g'_λ. Then u'_λ is defined as the composite

$$g^*(\mathscr{F}) \xrightarrow{g^*(u_\lambda)} g^*((f_\lambda)_*(\mathscr{G}_\lambda)) \xrightarrow{g^*(\psi)} g^*g_*((f'_\lambda)_*(\mathscr{G}'_\lambda)) \xrightarrow{\sigma} (f'_\lambda)_*(\mathscr{G}'_\lambda)$$

where $\sigma = \sigma_{(f'_\lambda)_*(\mathscr{G}'_\lambda)}$ is the canonical homomorphism (**0**, 4.4.3.3) corresponding to g. We will say for short that u'_λ is *deduced from* u_λ *by the change of base* g. When $f_\lambda : \mathrm{Z}_\lambda \to \mathrm{X}$ is an *affine* morphism, $(f'_\lambda)_*(\mathscr{G}'_\lambda) = g^*((f_\lambda)_*(\mathscr{G}_\lambda))$, and in this case one has therefore simply $u'_\lambda = g^*(u_\lambda)$.

One can also interpret u'_λ in the following manner: it suffices to know the value of $u'_\lambda(t')$, where t' is a section of $\mathscr{F}'$ above an open U' of X', of the following particular type: t' is the restriction to U' of the canonical image by $\Gamma(\mathrm{U},\mathscr{F}) \to \Gamma(g^{-1}(\mathrm{U}),\mathscr{F}')$ of a section t of $\mathscr{F}$ above an open U of X containing $g(\mathrm{U}')$ (these sections generate the $\mathscr{O}_{\mathrm{X}'}$-Module $\mathscr{F}'$ (**0**, 3.7.1)). Consider the section $u_\lambda(t)$ of $\mathscr{G}_\lambda$ above $f_\lambda^{-1}(\mathrm{U})$, and its canonical image t'' by $\Gamma(f_\lambda^{-1}(\mathrm{U}),\mathscr{G}_\lambda) \to \Gamma(g'^{-1}_\lambda(f_\lambda^{-1}(\mathrm{U})),\mathscr{G}'_\lambda)$; then $u'_\lambda(t')$ is the restriction of t'' to $f'^{-1}_\lambda(\mathrm{U}')$. Consider in particular the case where Z_λ is the sub-prescheme induced on an open subset of X, where $\mathscr{G}_\lambda = j_\lambda^*(\mathscr{F})$, with $j_\lambda : \mathrm{Z}_\lambda \to \mathrm{X}$ the canonical injection, and where u_λ is the canonical homomorphism $\rho_{\mathscr{F}} : \mathscr{F} \to (j_\lambda)_*(j_\lambda^*(\mathscr{F})) = (j_\lambda)_*(\mathscr{G}_\lambda)$ (**0**, 4.4.3.2) corresponding to j_λ. Then Z'_λ is induced on an open subset of X', and the preceding interpretation shows that u'_λ is none other than the canonical homomorphism $\rho_{\mathscr{F}'} : \mathscr{F}' \to (j'_\lambda)_*((j'_\lambda)^*(\mathscr{F}')) = (j'_\lambda)_*(\mathscr{G}'_\lambda)$ corresponding to the canonical injection $j'_\lambda : \mathrm{Z}'_\lambda \to \mathrm{X}'$.

(11.9.6). Under the conditions of (11.9.5), suppose that X and X' are both affine, and that one wishes to prove that every section t' of $\mathscr{F}'$ above X' whose images by all the u'_λ are zero is itself zero. Then, one can also restrict to the case where $\mathrm{Z}_\lambda = \mathrm{X}$ for every $\lambda \in \mathrm{L}$. Indeed, by (11.9.3) and (11.9.5), if one sets $\mathrm{Y}'_{\lambda\mu} = g'^{-1}_\lambda(\mathrm{Y}_{\lambda\mu})$, the homomorphism $v'_{\lambda\mu}$ deduced from $v_{\lambda\mu}$ by the change of base g is none other than the canonical homomorphism $\rho_{\mathscr{G}'_\lambda} : \mathscr{G}'_\lambda \to (j'_{\lambda\mu})_*(j'^*_{\lambda\mu}(\mathscr{G}'_\lambda))$ corresponding to the canonical injection $j'_{\lambda\mu} : \mathrm{Y}'_{\lambda\mu} \to \mathrm{Z}'_\lambda$, as we saw in (11.9.5). The assertion follows then from the reasonings of (11.9.2) and (11.9.3), $\mathrm{Y}_{\lambda\mu}$ and $v_{\lambda\mu}$ being replaced by $\mathrm{Y}'_{\lambda\mu}$ and $v'_{\lambda\mu}$. **IV-3** | 163

One has an analogous reduction when one wishes to prove that the family (u'_λ) is *separating* (X and X' being affine): this follows again from (11.9.2) and (11.9.3).

Proposition (11.9.7). — *With the notations of* (11.9.1) *and* (11.9.5), *suppose* $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ *and* $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$ *affine, and moreover that* A' *is a projective* A*-module. Then, if the family* (u_λ) *is separating, every section* t' *of* $\mathscr{F}'$ *above* X' *whose images by all the* u'_λ *are zero, is itself zero.*

Proof. One has seen in (11.9.6) that one can restrict to the case where all the Z_λ are *equal to* X. The proposition is then a consequence of the following lemma: ☐

Lemma (11.9.7.1). — *Let* A *be a ring,* $(\mathrm{M}_\lambda)_{\lambda\in\mathrm{L}}$ *a family of* A*-modules,* M *an* A*-module, and for each* λ, $u_\lambda : \mathrm{M} \to \mathrm{M}_\lambda$ *a homomorphism. Suppose that the intersection of the kernels of the* u_λ *is reduced to zero. Then, for every projective* A*-module* N*, the intersection of the kernels of the homomorphisms* $u_\lambda \otimes 1 : \mathrm{M}\otimes_\mathrm{A}\mathrm{N} \to \mathrm{M}_\lambda\otimes_\mathrm{A}\mathrm{N}$ *is reduced to zero.*

Proof. Indeed, N is a direct factor of a free A-module L, and it suffices evidently to prove that the intersection of the kernels of the homomorphisms $v_\lambda : \mathrm{M}\otimes_\mathrm{A}\mathrm{L} \to \mathrm{M}_\lambda\otimes_\mathrm{A}\mathrm{L}$ is reduced to zero, since $u_\lambda \otimes 1$ is the restriction of v_λ. But the assertion follows then trivially from the hypothesis. ☐

Remarks (11.9.8). — We do not know whether, under the hypotheses of proposition (11.9.7), the family (u'_λ) is *separating*: indeed, by (11.9.4) one would have to prove that a section t' of $\mathscr{F}'$ above an open subset $\mathrm{D}(h') \subset \mathrm{X}'$ (where $h' \in \mathrm{A}'$) such that all the $u'_\lambda(t')$ are zero is itself zero. Now, one cannot apply proposition (11.9.4) to $\mathrm{D}(h') = \mathrm{Spec}(\mathrm{A}'_{h'})$, for the fact that A' is a projective A-module (or even a free one), does not imply that $\mathrm{A}'_{h'}$ is a projective A-module. For example, one can take for A a discrete valuation ring, for A' a discrete valuation ring that is an A-module free of rank 2, and for $\mathrm{A}'_{h'}$ the fraction field of A'.

One has however the following result:

Corollary (11.9.9). — *Let* X *be an Artinian prescheme,* $g : \mathrm{X}' \to \mathrm{X}$ *a flat morphism (note that these two conditions are satisfied if* X *is the spectrum of a* field *and* g *is an arbitrary morphism). Then, with the notations of* (11.9.1) *and* (11.9.5), *if the family* (u_λ) *is separating, it is the same for the family* (u'_λ).

Proof. One can evidently restrict to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ is the spectrum of an Artinian local ring A (**I**, 6.2.2); one notes then that for every open affine $\mathrm{U}' = \mathrm{Spec}(\mathrm{A}')$ of X', A' is a flat A-module, hence *projective* ($\mathbf{0}_{\mathrm{III}}$, 10.1.3). It suffices therefore to apply (11.9.7) to every open affine of X' to obtain the corollary. □

Theorem (11.9.10). — *Let* X *be a prescheme,* (u_λ) *a family of homomorphisms* (11.9.1.1), $g : \mathrm{X}' \to \mathrm{X}$ *a morphism,* (u'_λ) *the family of homomorphisms deduced from* (u_λ) *by the change of base* g (11.9.5).

(i) If g is a faithfully flat morphism and if the family (u'_λ) is separating, then the family (u_λ) is separating.

(ii) Suppose that g is a flat morphism and moreover that one of the two following conditions is satisfied:
- a) L *is finite and the* f_λ *are quasi-compact.*
- b) *The morphism g is locally of finite presentation.*

Then, if the family (u_λ) is separating, it is the same for the family (u'_λ).

Proof. IV-3 | 164

(i) By virtue of (2.2.8), it suffices to show that if a section t of $\mathscr{F}$ above an open U of X belongs to the kernel of each of the u_λ, its image t' by the canonical homomorphism $\Gamma(\rho) : \Gamma(\mathrm{U}, \mathscr{F}) \to \Gamma(g^{-1}(\mathrm{U}), g^*(\mathscr{F})) = \Gamma(\mathrm{U}, g_*(g^*(\mathscr{F})))$ is zero. Now the images of t' by the $g^*(u_\lambda)$ are the images of $u_\lambda(t)$ by the homomorphism $\Gamma(\mathrm{U}, (f_\lambda)_*(\mathscr{G}_\lambda)) \to \Gamma(g^{-1}(\mathrm{U}), g^*((f_\lambda)_*(\mathscr{G}_\lambda)))$, whence they are zero, and *a fortiori* $u'_\lambda(t') = 0$ for every λ, whence $t' = 0$ by hypothesis, which proves (i).

(ii) The question being local on X and X', one can restrict to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$ are affine, A' being a flat A-module, and it suffices to prove that every section z' of $\mathscr{F}'$ above X' whose images by all the u'_λ are zero is itself zero. We have seen in (11.9.6) that one can also restrict to the case where $\mathrm{Z}_\lambda = \mathrm{X}$ for every $\lambda \in \mathrm{L}$. Let us distinguish now the two cases.

a) If L is finite and the f_λ are quasi-compact, we have seen in (11.9.3) that one can reduce further to the case where $\mathrm{Z}_\lambda = \mathrm{X}$ for every $\lambda \in \mathrm{L}$, and where moreover L is finite. It is then equivalent to say that the intersection of the kernels of the $u_\lambda : \mathscr{F} \to \mathscr{G}_\lambda$ is zero, or that the homomorphism $u = (u_\lambda) : \mathscr{F} \to \mathscr{G} = \bigoplus_\lambda \mathscr{G}_\lambda$ is injective. Since $u' = g^*(u)$ is injective because g is flat, the proposition is proved in this case.

b) Let $\mathscr{F} = \widetilde{\mathrm{M}}$, $\mathscr{G}_\lambda = \widetilde{\mathrm{N}}_\lambda$, and set $\mathrm{M}' = \mathrm{M} \otimes_\mathrm{A} \mathrm{A}'$, $\mathrm{N}'_\lambda = \mathrm{N}_\lambda \otimes_\mathrm{A} \mathrm{A}'$, so that $\mathscr{F}' = \widetilde{\mathrm{M}'}$, $\mathscr{G}'_\lambda = \widetilde{\mathrm{N}'_\lambda}$; by abuse of language, let us still denote by u_λ the homomorphism $\mathrm{M} \to \mathrm{N}_\lambda$, and by u'_λ the homomorphism $u_\lambda \otimes 1 : \mathrm{M}' \to \mathrm{N}'_\lambda$. Given an element $z' \in \mathrm{M}'$ such that $u'_\lambda(z') = 0$ for every λ; it remains to prove that $z' = 0$. Now, the hypothesis that g is flat and of finite presentation implies, by (11.3.15), that there exists a finite sequence $(s_i)_{1 \leqslant i \leqslant n}$ of elements of A, such that, if one sets $\mathfrak{J}_k = s_1\mathrm{A} + \ldots + s_k\mathrm{A}$, and $\mathrm{A}_i = \mathrm{A}_{s_i}/\mathfrak{J}_{i-1}\mathrm{A}_{s_i}$, the ring $\mathrm{A}'_i = \mathrm{A}' \otimes_\mathrm{A} \mathrm{A}_i$ is a *free* A_i-module for $1 \leqslant i \leqslant n$, and $\mathfrak{J}_n = \mathrm{A}$. The proposition will be established if we prove the following assertion for $1 \leqslant i \leqslant n$:

$(*_i)$ *There exists an integer $m_i > 0$ such that $s_j^{m_i} z' = 0$ for $j \leqslant i$.*

Indeed, setting $k = m_n$ and noting that the s_i^k $(1 \leqslant i \leqslant n)$ generate the unit ideal of A, the assertion $(*_n)$ will show that z', as a linear combination of the $s_i^k z'$, is zero.

Let us prove $(*_i)$ by induction on i, the assertion being trivially true for $i = 0$. Suppose $i > 0$ and let m be a common multiple of the m_j for $j \leqslant i-1$. We note (for $1 \leqslant h \leqslant n$) that if $\mathfrak{J}'_h$ is the ideal generated by the s_j^m $(0 \leqslant j \leqslant h)$, then $\mathfrak{J}_h/\mathfrak{J}'_h$ is nilpotent. Replacing the s_j by s_j^m for $1 \leqslant j \leqslant n$ therefore amounts to replacing, for $1 \leqslant h \leqslant n$, A_h by $\mathrm{B}_h = \mathrm{A}_{s_h}/\mathfrak{J}'_{h-1}\mathrm{A}_{s_h}$, so that $\mathrm{A}_h = \mathrm{B}_h/((\mathfrak{J}_{h-1}/\mathfrak{J}'_{h-1})\mathrm{B}_h)$; since A' is a flat A-module, it follows from ($\mathbf{0}_{\mathrm{III}}$, 10.1.2) that $\mathrm{B}'_h = \mathrm{A}' \otimes_\mathrm{A} \mathrm{B}_h$ is again a free B_h-module. One can therefore replace all the s_j $(1 \leqslant j \leqslant n)$ IV-3 | 165
by s_j^m without changing the properties of $\mathfrak{J}_h$ and A'_h, and suppose $m = 1$. Then z', being annihilated by $\mathfrak{J}_{i-1}$, identifies with an element of $\mathrm{Hom}_{\mathrm{A}'}(\mathrm{A}'/\mathfrak{J}_{i-1}\mathrm{A}', \mathrm{M}')$; and since $\mathfrak{J}_{i-1}\mathrm{A}'$ is an ideal of finite type of A', this module of homomorphisms identifies in turn with $\mathrm{Hom}_\mathrm{A}(\mathrm{A}/\mathfrak{J}_{i-1}, \mathrm{M}) \otimes_\mathrm{A} \mathrm{A}'$ ($\mathbf{0}_\mathrm{I}$, 6.2.2). Let $v_\lambda = \mathrm{Hom}(1, u_\lambda) : \mathrm{Hom}_\mathrm{A}(\mathrm{A}/\mathfrak{J}_{i-1}, \mathrm{M}) \to$

$\mathrm{Hom}_{\mathrm{A}}(\mathrm{A}/\mathfrak{J}_{i-1}, \mathrm{N}_\lambda)$, the homomorphism deduced from u_λ; the family (v_λ) is also *separating*. Indeed, for every $t \in \mathrm{A}$, $(\mathrm{Hom}_{\mathrm{A}}(\mathrm{A}/\mathfrak{J}_{i-1}, \mathrm{M}))_t = \mathrm{Hom}_{\mathrm{A}_t}(\mathrm{A}_t/(\mathfrak{J}_{i-1})_t, \mathrm{M}_t)$, and likewise with M replaced by N_λ, since the ideal $\mathfrak{J}_{i-1}$ is of finite type ($\mathbf{0}_{\mathrm{I}}$, 1.3.5); as by hypothesis the intersection of the kernels of the $(u_\lambda)_t$ is zero, the same holds for the intersection of the kernels of the $(v_\lambda)_t = \mathrm{Hom}(1, (u_\lambda)_t)$, whence our assertion by (11.9.4). Replacing A by $\mathrm{A}/\mathfrak{J}_{i-1}$, M by $\mathrm{Hom}_{\mathrm{A}}(\mathrm{A}/\mathfrak{J}_{i-1}, \mathrm{M})$, N_λ by $\mathrm{Hom}_{\mathrm{A}}(\mathrm{A}/\mathfrak{J}_{i-1}, \mathrm{N}_\lambda)$ (which are $(\mathrm{A}/\mathfrak{J}_{i-1})$-modules), u_λ by v_λ, and finally A' by $\mathrm{A}'/\mathfrak{J}_{i-1}\mathrm{A}'$, one sees that one can reduce to the case where, in the initial situation, the element $s = s_i \in \mathrm{A}$ is such that A'_s is a free A_s-module. Now the family $(u_\lambda)_s : \mathrm{M}_s \to (\mathrm{N}_\lambda)_s$ is separating by hypothesis; since $(u'_\lambda)_s(z'/1) = 0$, it follows from (11.9.7) that $z'/1 = 0$ in M'_s, which signifies that there exists an integer r such that $s^r z' = 0$ in M', and this finishes the proof of $(*_i)$ by induction.

□

Remark (11.9.11). — Let us restrict, for simplicity, to the case where $\mathrm{Z}_\lambda = \mathrm{X}$ for every λ. It should be noted that if the family of homomorphisms $u_\lambda : \mathscr{F} \to \mathscr{G}_\lambda$ is separating, it does *not necessarily* follow that, for every $x \in \mathrm{X}$, the intersection of the kernels of the homomorphisms $(u_\lambda)_x : \mathscr{F}_x \to (\mathscr{G}_\lambda)_x$ is reduced to 0. For example, let X be a locally Noetherian prescheme of Jacobson, of dimension $\geqslant 1$, and $\mathscr{F}$ a coherent $\mathscr{O}_{\mathrm{X}}$-Module; for every closed point x of X, and every integer $n \geqslant 0$, $\mathfrak{m}_x^n\mathscr{F}$ is a coherent $\mathscr{O}_{\mathrm{X}}$-Module with support contained in $\{x\}$. The family of canonical homomorphisms $\mathscr{F} \to \mathscr{F}/\mathfrak{m}_x^n\mathscr{F}$ (where x runs through the set X_0 of closed points of X and n through the set of integers $\geqslant 0$) is *separating*: indeed, if t is a section of $\mathscr{F}$ above an open U of X whose images in all the $\Gamma(\mathrm{U}, \mathscr{F}/\mathfrak{m}_x^n\mathscr{F})$ are zero, it follows at once that $t_x = 0$ for every closed point $x \in \mathrm{U}$; as the set of closed points contained in U is very dense in U, this implies $t = 0$ (10.2.1). However, if one takes $\mathscr{F} = \mathscr{O}_{\mathrm{X}}$ and if $y \in \mathrm{X}$ is a *non-closed* point, one has $(\mathscr{O}_{\mathrm{X}}/\mathfrak{m}_x^n\mathscr{O}_{\mathrm{X}})_y = 0$ for every closed point x of X, but $\mathscr{O}_{\mathrm{X},y} \neq 0$, and the intersection of the kernels of the homomorphisms $\mathscr{O}_{\mathrm{X},y} \to (\mathscr{O}_{\mathrm{X}}/\mathfrak{m}_x^n\mathscr{O}_{\mathrm{X}})_y$ is equal to $\mathscr{O}_{\mathrm{X},y}$.

Lemma (11.9.12). — *Let the notations be those of (11.9.1) and (11.9.5), and suppose the family (u_λ) to be separating; suppose in addition that X is an S-prescheme, where $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ is an affine scheme, and that $\mathrm{X}' = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}'$, where $\mathrm{S}' = \mathrm{Spec}(\mathrm{A}')$, A' being an A-algebra; suppose finally that the $\mathscr{G}_\lambda$ are S-flat. Let $(\mathrm{A}'_\alpha)_{\alpha \in \mathrm{I}}$ be the filtered family of sub-A-algebras of A' of finite type, and, for every $\alpha \in \mathrm{I}$, let $\mathrm{S}''_\alpha = \mathrm{Spec}(\mathrm{A}'_\alpha)$, $\mathrm{X}''_\alpha = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}''_\alpha$, $\mathrm{Z}''_{\alpha\lambda} = \mathrm{Z}_\lambda \times_{\mathrm{X}} \mathrm{X}''_\alpha$, and let $f''_{\alpha\lambda} : \mathrm{Z}''_{\alpha\lambda} \to \mathrm{X}''_\alpha$, $\mathscr{F}''_\alpha$, $\mathscr{G}''_{\alpha\lambda}$, and $u''_{\alpha\lambda}$ be the morphisms, Modules, and homomorphisms of Modules deduced from f_λ, $\mathscr{F}$, $\mathscr{G}_\lambda$, and u_λ by means of the change of base $\mathrm{X}''_\alpha \to \mathrm{X}$. Then, if for every $\alpha \in \mathrm{I}$, the family $(u''_{\alpha\lambda})_{\lambda \in \mathrm{L}}$ is separating, it is the same for $(u'_\lambda)_{\lambda \in \mathrm{L}}$.*

Proof. It suffices to prove that if t' is a section of $\mathscr{F}'$ above an open affine U' of X' such that $u'_\lambda(t') = 0$ for every $\lambda \in \mathrm{L}$, then $t' = 0$. If $h_\alpha : \mathrm{X}' \to \mathrm{X}''_\alpha$ is the canonical projection, it follows from (8.2.11) that **IV-3** | 166
there exists an index α and an open quasi-compact $\mathrm{U}''_\alpha \subset \mathrm{X}''_\alpha$ such that $\mathrm{U}' = h_\alpha^{-1}(\mathrm{U}''_\alpha)$; moreover, by virtue of (8.5.2), (i), one can suppose that t' is the canonical image of a section t''_α of $\mathscr{F}''_\alpha$ above U''_α. Replacing, if necessary, S, X, Z_λ, f_λ, $\mathscr{F}$, $\mathscr{G}_\lambda$, and u_λ by S''_α, U''_α, $(f''_{\alpha\lambda})^{-1}(\mathrm{U}''_\alpha)$, and the corresponding restrictions of $\mathscr{F}''_\alpha$, $\mathscr{G}''_{\alpha\lambda}$, and $u''_{\alpha\lambda}$, we may therefore suppose that $\mathrm{U}' = \mathrm{X}'$, that t' is the canonical image of a section t of $\mathscr{F}$ above X, and that the homomorphism $\mathrm{A} \to \mathrm{A}'$ is injective; equivalently, if $p : \mathrm{S}' \to \mathrm{S}$ is the corresponding morphism, the homomorphism $\mathscr{O}_{\mathrm{S}} \to p_*(\mathscr{O}_{\mathrm{S}'})$ occurring in the definition of p is *injective*. It follows at once by virtue of the flatness of $\mathscr{G}_\lambda$ on S (and by reducing for example to the case where Z_λ is affine on S (**I**, 1.6.3) and (**I**, 1.6.5)) that the canonical homomorphism $\rho : \mathscr{G}_\lambda \to (g'_\lambda)_*(\mathscr{G}'_\lambda)$ is also *injective*. But the composite homomorphism

$$\Gamma(\mathrm{X}, \mathscr{F}) \xrightarrow{\Gamma(u_\lambda)} \Gamma(\mathrm{Z}_\lambda, \mathscr{G}_\lambda) \xrightarrow{\Gamma(\rho)} \Gamma(\mathrm{Z}'_\lambda, \mathscr{G}'_\lambda)$$

is equal by definition to the composite homomorphism

$$\Gamma(\mathrm{X}, \mathscr{F}) \xrightarrow{\Gamma(\rho)} \Gamma(\mathrm{X}', \mathscr{F}') \xrightarrow{\Gamma(u'_\lambda)} \Gamma(\mathrm{Z}'_\lambda, \mathscr{G}'_\lambda)$$

whence the image of t by these composite homomorphisms is $u'_\lambda(t') = 0$; by virtue of the injectivity of the homomorphism $\Gamma(\mathrm{Z}_\lambda, \mathscr{G}_\lambda) \xrightarrow{\Gamma(\rho)} \Gamma(\mathrm{Z}'_\lambda, \mathscr{G}'_\lambda)$, one concludes that $u_\lambda(t) = 0$ for every $\lambda \in \mathrm{L}$, whence $t = 0$ by hypothesis, and finally $t' = 0$. □

Proposition (11.9.13). — *Let the notations be those of (11.9.1) and (11.9.5), and suppose that* X *is a prescheme over a field* k*, and that, setting* $\mathrm{S} = \mathrm{Spec}(k)$*, one has* $\mathrm{X}' = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}'$*, where* S' *is an arbitrary* k*-prescheme. Then, if the family* $(u_\lambda)_{\lambda\in\mathrm{L}}$ *is separating, it is the same for* (u'_λ)*.*

Proof. One can restrict to the case where $\mathrm{S}' = \mathrm{Spec}(\mathrm{A}')$ is affine. If A' is a k-algebra of finite type, the morphism $g : \mathrm{X}' \to \mathrm{X}$ is flat and of finite presentation, and one is therefore under the hypotheses of (11.9.10), (ii), b). In the general case, one regards A' as the inductive limit of its sub-k-algebras A'_α of finite type and applies to each A'_α the result of (11.9.10), (ii), b; one then concludes by means of Lemma (11.9.12), since the $\mathscr{G}_\lambda$ are S-flat. □

(11.9.14). Let us keep the notations of (11.9.1) and (11.9.5), and suppose that X is an S-prescheme. If for every change of base $g : \mathrm{X} \times_{\mathrm{S}} \mathrm{S}' \to \mathrm{X}$, where $\mathrm{S}' \to \mathrm{S}$ is an arbitrary morphism, the corresponding family (u'_λ) is separating, we say that the family (u_λ) is *universally separating relatively to* S. When the family (u_λ) is reduced to a single element u, we say further that u is *universally injective*, relatively to S. It is clear then that for every morphism $h : \mathrm{S}' \to \mathrm{S}$, the family (u'_λ) corresponding is universally separating relative to S'; conversely, if h is *faithfully flat* and if (u'_λ) is universally separating relative to S', then (u_λ) is universally separating relative to S, as follows immediately from (11.9.10), (i) and from the fact that for every morphism $\mathrm{S}'' \to \mathrm{S}$, the corresponding morphism $\mathrm{S}'' \times_{\mathrm{S}} \mathrm{S}' \to \mathrm{S}''$ is faithfully flat.

Proposition (11.9.15). — *Let the notations be those of (11.9.1), and suppose that* X *is an* S*-prescheme, the* $\mathscr{G}_\lambda$ *being* S*-flat. Let* S_0 *be a closed sub-prescheme of* S *defined by a nilpotent quasi-coherent Ideal* $\mathscr{J}$ *of* $\mathscr{O}_{\mathrm{S}}$*, such that the* $(\mathscr{O}_{\mathrm{S}}/\mathscr{J})$*-Modules* $\mathscr{J}^k/\mathscr{J}^{k+1}$ *are all locally free. Let* $(u_{\lambda 0})$ *be the family of homomorphisms* **IV-3** | 167
obtained from (u_λ) *by the change of base* $\mathrm{X} \leftarrow \mathrm{X}_0 = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}_0$*. Then, if the family* $(u_{\lambda 0})$ *is separating (resp. universally separating relative to* S_0*), the family* (u_λ) *is separating (resp. universally separating relative to* S*).*

Proof. Let us note that if $\mathrm{S}' \to \mathrm{S}$ is an arbitrary change of base and $\mathrm{S}'_0 = \mathrm{S}' \times_{\mathrm{S}} \mathrm{S}_0$, then S'_0 is a closed sub-prescheme of S' defined by a quasi-coherent nilpotent Ideal $\mathscr{J}'$ of $\mathscr{O}_{\mathrm{S}'}$, such that $\mathscr{J}'^k/\mathscr{J}'^{k+1}$ is a locally free $(\mathscr{O}_{\mathrm{S}'}/\mathscr{J}')$-Module for every k (2.1.8), (i); as moreover the $\mathscr{G}'_\lambda = \mathscr{G}_\lambda \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{O}_{\mathrm{S}'}$ are S'-flat, one sees that the assertion relative to universally separating families is a consequence of that relative to separating families. To prove the latter, one can reduce by (11.9.3) to the case where $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$ are affine, $\mathrm{Z}_\lambda = \mathrm{X}$ for every λ, $\mathscr{F} = \widetilde{\mathrm{M}}$, and $\mathscr{G}_\lambda = \widetilde{\mathrm{N}}_\lambda$, where M and the N_λ are B-modules, the N_λ being flat A-modules. Moreover, the question being local on X and S, it suffices to prove that if $t \in \mathrm{M}$ is such that $u_\lambda(t) = 0$ for every λ, then $t = 0$. One has $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is a nilpotent ideal of A such that the $\mathfrak{J}^k/\mathfrak{J}^{k+1}$ are free $(\mathrm{A}/\mathfrak{J})$-modules, and by hypothesis the homomorphisms $u_{\lambda 0} : \mathrm{M}/\mathfrak{J}\mathrm{M} \to \mathrm{N}_\lambda/\mathfrak{J}\mathrm{N}_\lambda$ form a separating family. Suppose $\mathfrak{J}^{n+1} = 0$ (n an integer $\geqslant 0$) and let us reason by induction on n, the assertion being trivially true for $n = 0$. If $\bar{t}$ is the class of t in $\mathrm{M}/\mathfrak{J}^n\mathrm{M}$, the class of $u_\lambda(t)$ in $\mathrm{N}_\lambda/\mathfrak{J}^n\mathrm{N}_\lambda$ is zero for every λ, whence, by the hypothesis of induction, $\bar{t} = 0$, otherwise said $t \in \mathfrak{J}^n\mathrm{M}$. Now $\mathfrak{J}^n = \mathfrak{J}^n/\mathfrak{J}^{n+1}$ is a free $(\mathrm{A}/\mathfrak{J})$-module; if (e_α) is a basis of this module, one can therefore write $t = \sum_\alpha e_\alpha t^0_\alpha$, with $t^0_\alpha \in \mathrm{M}/\mathfrak{J}\mathrm{M}$, zero for all but a finite number of indices. On the other hand, since N_λ is a flat A-module, $\mathfrak{J}^n \otimes_{(\mathrm{A}/\mathfrak{J})} (\mathrm{N}_\lambda/\mathfrak{J}\mathrm{N}_\lambda)$ identifies with $\mathfrak{J}^n\mathrm{N}_\lambda$ and one can therefore write $u_\lambda(t) = \sum_\alpha e_\alpha u_{\lambda 0}(t^0_\alpha) = \sum_\alpha e_\alpha \otimes u_{\lambda 0}(t^0_\alpha)$. Since by hypothesis $u_\lambda(t) = 0$, one deduces that $u_{\lambda 0}(t^0_\alpha) = 0$ for every α and every λ; whence $t^0_\alpha = 0$ for every α, and hence $t = 0$. □

Theorem (11.9.16). — *Let the notations be those of (11.9.1), and suppose that* X *is a locally Noetherian* S*-prescheme,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module, and that the* $\mathscr{G}_\lambda$ *are* S*-flat. For every* $s \in \mathrm{S}$*, let* $((u_\lambda)_s)_{\lambda\in\mathrm{L}}$ *be the family obtained from* (u_λ) *by the change of base* $\mathrm{X}_s = \mathrm{X} \times_{\mathrm{S}} \mathrm{Spec}(k(s)) \to \mathrm{X}$*. For the family* (u_λ) *to be universally separating relative to* S*, it is necessary and sufficient that for every* $s \in \mathrm{S}$*, the family* $((u_\lambda)_s)$ *be separating.*

Proof. The necessity of the condition follows trivially from the definitions. Conversely, suppose the condition satisfied, and let us prove first that the family (u_λ) is separating. One can by (11.9.3) reduce to the case where $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$ are affine, $\mathrm{Z}_\lambda = \mathrm{X}$ for every λ, $\mathscr{F} = \widetilde{\mathrm{M}}$, $\mathscr{G}_\lambda = \widetilde{\mathrm{N}}_\lambda$, where B is Noetherian, M is a B-module of finite type, and the N_λ are flat A-modules, and restrict to proving that, if $t \in \mathrm{M}$ is such that $u_\lambda(t) = 0$ for every λ, then $t = 0$. To show that $t = 0$, it suffices to prove that the image $t_{\mathfrak{p}}$ of t in $\mathrm{M}_{\mathfrak{p}}$ is zero for every maximal ideal $\mathfrak{p}$ of B (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 3, cor. 1 of th. 1). One may therefore restrict to proving that the intersection of the kernels of the $(u_\lambda)_{\mathfrak{p}} : \mathrm{M}_{\mathfrak{p}} \to (\mathrm{N}_\lambda)_{\mathfrak{p}}$ deduced from (u_λ) by the change of base $\mathrm{Spec}(\mathrm{B}_{\mathfrak{p}}) \to \mathrm{Spec}(\mathrm{B})$ is reduced to 0. Otherwise stated, one is brought back to the case where B is a local Noetherian ring; and, on considering the prime ideal of A which is the inverse image of the maximal ideal of B, one may also suppose that A is a local ring with maximal ideal $\mathfrak{m}$. Then, since $\mathfrak{m}\mathrm{B}$ is contained in the maximal ideal of B, and M is a B-module of finite type, the intersection of the $\mathfrak{m}^n\mathrm{M}$ is reduced to 0 (**0**, 7.3.5), whence it suffices to prove that for every n, the image of t in $\mathrm{M}/\mathfrak{m}^{n+1}\mathrm{M}$ is zero. It suffices therefore to prove that the family deduced from (u_λ) by the change of base $\mathrm{Spec}(\mathrm{B}/\mathfrak{m}^{n+1}\mathrm{B}) \to \mathrm{Spec}(\mathrm{B})$ is separating, which signifies that one can restrict to the case where A is a local ring whose maximal ideal $\mathfrak{m}$ is nilpotent. But then the $\mathfrak{m}^k/\mathfrak{m}^{k+1}$ are $(\mathrm{A}/\mathfrak{m})$-modules free, and, by virtue of the hypothesis on the $(u_\lambda)_s$, one is precisely in the conditions of application of (11.9.15), whence the conclusion announced. **IV-3** | 168

Let now $h : \mathrm{S}' \to \mathrm{S}$ be a change-of-base morphism, and let (u'_λ) be the family obtained from (u_λ) by the change of base $h' : \mathrm{X} \times_{\mathrm{S}} \mathrm{S}' \to \mathrm{X}$; let us prove that (u'_λ) is also separating. Suppose first that h is *locally of finite type*; then the same is true of h', and hence $\mathrm{X}' = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}'$ is locally Noetherian. Moreover, if $s' \in \mathrm{S}'$ lies above $s \in \mathrm{S}$, it follows from (11.9.13), applied to X_s and to $\mathrm{X}_{s'} = \mathrm{X}_s \otimes_{k(s)} k(s')$, that the family $((u'_\lambda)_{s'})$ is separating for every $s' \in \mathrm{S}'$; one may therefore conclude from the first part of the proof that (u'_λ) is separating in this case.

Finally, if h is arbitrary, one can evidently restrict to the case where $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{S}' = \mathrm{Spec}(\mathrm{A}')$ are affine, and consider A' as an inductive limit of its sub-A-algebras of finite type. As the $\mathscr{G}_\lambda$ are S-flat, it suffices to apply what precedes and the lemma (11.9.5) to complete the proof. □

Proposition (11.9.17). — *Let $f : \mathrm{X} \to \mathrm{S}$ be a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_{\mathrm{X}}$-Module of finite presentation and f-flat,* U *an open of* X, *$j : \mathrm{U} \to \mathrm{X}$ the canonical injection, $u : \mathscr{F} \to j_*(j^*(\mathscr{F}))$ the canonical homomorphism* (**0**, 4.4.3.2). *For every $s \in \mathrm{S}$, let X_s be the fibre $\mathrm{X} \times_{\mathrm{S}} \mathrm{Spec}(k(s))$, U_s the open $\mathrm{U} \cap \mathrm{X}_s$ of X_s, $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{S}}} k(s)$, $j_s : \mathrm{U}_s \to \mathrm{X}_s$ the canonical injection, $u_s : \mathscr{F}_s \to (j_s)_*((j_s)^*(\mathscr{F}_s))$ the canonical homomorphism. For u to be universally injective relative to* S, *it is necessary and sufficient that u_s be injective for every $s \in \mathrm{S}$.*

Proof. There is nothing more than to prove the sufficiency of the condition. When X is locally Noetherian, the proposition is an immediate corollary of (11.9.16). We shall now reduce to this case in two stages, restricting, as we clearly may, to the case where $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ and X are affine.

A) Case where U *is quasi-compact.* — We will use the following lemma:

Lemma (11.9.17.1). — *Under the general hypotheses of (11.9.17), and supposing in addition* S *and* X *affine and* U *quasi-compact, the set* E *of the $s \in \mathrm{S}$ such that u_s is injective is constructible.*

Proof. Indeed, the fibres X_s are locally Noetherian preschemes, whence E can also, by virtue of (5.10.2), be defined as the set of $s \in \mathrm{S}$ such that $\mathrm{Ass}(\mathscr{F}_s) \subset \mathrm{U}_s$. Note moreover that U, being quasi-compact in an affine scheme, is constructible. Thus verification of condition (9.2.1), (i) follows immediately from (4.2.7); on the other hand, verification of (9.2.1), (ii) is carried out easily by using the study of associated prime cycles in a neighbourhood of the generic fibre (9.8.3), as well as (9.5.2) and (9.5.3). □ **IV-3** | 169

This lemma being established, one can, by virtue of (8.9.1) and (8.2.11), suppose that there exists a Noetherian sub-ring A_0 of A, a prescheme X_0 of finite type over $\mathrm{S}_0 = \mathrm{Spec}(\mathrm{A}_0)$, an open U_0 of X_0, and a coherent $\mathscr{O}_{\mathrm{X}_0}$-Module $\mathscr{F}_0$, such that $\mathrm{X} = \mathrm{X}_0 \times_{\mathrm{S}_0} \mathrm{S}$ and, if $p_0 : \mathrm{X} \to \mathrm{X}_0$ is the canonical projection, $\mathrm{U} = p_0^{-1}(\mathrm{U}_0)$ and $\mathscr{F} = p_0^*(\mathscr{F}_0)$. Let (A_α) be the filtered family of sub-rings of A that are A_0-algebras of finite type, and set $\mathrm{S}_\alpha = \mathrm{Spec}(\mathrm{A}_\alpha)$, $\mathrm{X}_\alpha = \mathrm{X}_0 \times_{\mathrm{S}_0} \mathrm{S}_\alpha$, $\mathrm{U}_\alpha = \mathrm{U}_0 \times_{\mathrm{S}_0} \mathrm{S}_\alpha$, and $\mathscr{F}_\alpha = \mathscr{F}_0 \otimes_{\mathrm{S}_0} \mathrm{S}_\alpha$; let u_α be the canonical homomorphism relative to $\mathscr{F}_\alpha$ and U_α, defined as in (11.9.17). For every $s \in \mathrm{S}$, the hypothesis that u_s is injective implies that $(u_\alpha)_{s_\alpha}$ is injective (11.9.10), (i), where $s_\alpha = q_\alpha(s)$ is the image of s by the morphism $q_\alpha : \mathrm{S} \to \mathrm{S}_\alpha$. If E_α is the set of $s_\alpha \in \mathrm{S}_\alpha$ such that $(u_\alpha)_{s_\alpha}$ is injective, one therefore has $\mathrm{S} = q_\alpha^{-1}(\mathrm{E}_\alpha)$, and the E_α form a projective system of sets. But the lemma (11.9.17.1), applied to u_α, shows that E_α is constructible in S_α; one therefore deduces from (8.3.4) that there exists an index α such that $\mathrm{E}_\alpha = \mathrm{S}_\alpha$. But then, since X_α is Noetherian, u_α is universally injective by virtue of (11.9.16), whence u is also universally injective.

B) General case. — The open U is a filtered increasing union of quasi-compact opens U_λ; if $j_\lambda : U_\lambda \to X$ is the canonical injection and $u_\lambda : \mathscr{F} \to (j_\lambda)_*(j_\lambda^*(\mathscr{F}))$ the canonical homomorphism corresponding, it follows from the lemma (11.9.17.1) applied to u_λ, that the set E_λ of the $s \in S$ such that $(u_\lambda)_s$ is injective is constructible. On the other hand, for every $s \in S$, to say that u_s (resp. $(u_\lambda)_s$) is injective signifies that $\mathrm{Ass}(\mathscr{F}_s) \subset U \cap X_s$ (resp. $\mathrm{Ass}(\mathscr{F}_s) \subset U_\lambda \cap X_s$) (5.10.2). As $\mathrm{Ass}(\mathscr{F}_s)$ is finite (3.1.6), to say that u_s is injective signifies therefore that there exists λ such that $s \in E_\lambda$. The hypothesis therefore signifies that $S = \bigcup_\lambda E_\lambda$. By virtue of (1.9.9), there exists an index λ such that $S = E_\lambda$, whence one concludes by the first part of the reasoning that u_λ is universally injective. It follows then from the factorisation of u_λ:

$$\mathscr{F} \xrightarrow{u} j_*(\mathscr{F}|U) \longrightarrow (j_\lambda)_*(\mathscr{F}|U_\lambda)$$

that u is also universally injective. □

Remarks (11.9.18). — Let X be a prescheme, $\mathscr{F}, \mathscr{G}$ two quasi-coherent $\mathscr{O}_X$-Modules of finite presentation, $\mathscr{G}$ being moreover supposed *locally free*, $u : \mathscr{F} \to \mathscr{G}$ a homomorphism. The following conditions are equivalent:

a) for every morphism $g : X' \to X$, the homomorphism $g^*(u) : g^*(\mathscr{F}) \to g^*(\mathscr{G})$ is injective;
b) for every $x \in X$, the homomorphism $u \otimes 1_{k(x)} : \mathscr{F} \otimes_{\mathscr{O}_X} k(x) \to \mathscr{G} \otimes_{\mathscr{O}_X} k(x)$ is injective;
c) for every $x \in X$, there exists a neighbourhood U of x such that $u|U : \mathscr{F}|U \to \mathscr{G}|U$ is an isomorphism of $\mathscr{F}|U$ onto a direct factor of $\mathscr{G}|U$.

It is clear that $c)$ implies $a)$, and $a)$ implies $b)$. The fact that $b)$ implies $c)$ follows from (**0**, 19.1.12), (**0**, 5.2.5) and (**0**, 5.5.5).

When the preceding equivalent conditions are satisfied, one says that u is *universally injective*.

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11.10. Schematically dominant families of morphisms and families of schematically dense sub-preschemes

Proposition (11.10.1). — *Let* X *be a prescheme,* $(Z_\lambda)_{\lambda \in L}$ *a family of preschemes, and for every* $\lambda \in L$, *let* $f_\lambda = (\psi_\lambda, \theta_\lambda) : Z_\lambda \to X$ *be a morphism. The following conditions are equivalent:*

a) *The family of homomorphisms* $\theta_\lambda : \mathscr{O}_X \to (f_\lambda)_*(\mathscr{O}_{Z_\lambda})$ *is separating (in other words* (11.9.1), *the intersection of the kernels of the* θ_λ *is* zero*).*

b) *For every open* U *of* X, *every section* t *of* $\mathscr{O}_X$ *above* U *whose images by all the canonical homomorphisms*

$$(\theta_\lambda)_U : \Gamma(U, \mathscr{O}_X) \longrightarrow \Gamma(f_\lambda^{-1}(U), \mathscr{O}_{Z_\lambda})$$

are zero, is itself zero.

c) *For every open* U *of* X *and every closed sub-prescheme* Y *of* U *such that for every* $\lambda \in L$, *there exists a factorisation*

$$f_\lambda^{-1}(U) \xrightarrow{g_\lambda} Y \xrightarrow{j} U$$

of the restriction $f_\lambda^{-1}(U) \to U$ *of* f_λ *(where* j *is the canonical injection), one has* $Y = U$.

If moreover X is an S-prescheme, these conditions are also equivalent to the following:

d) *For every separated morphism* $g : X' \to S$ *and every pair* u, v *of* S*-morphisms from an open* U *of* X *to* X', *such that for every* λ, *the composites of* u *and* v *with the restriction of* f_λ *to* $f_\lambda^{-1}(U)$ *are equal, one has* $u = v$.

Proof. The equivalence of $a)$ and $b)$ follows from the definitions. To see that $b)$ implies $c)$, it suffices to consider the quasi-coherent Ideal $\mathscr{J}$ of $\mathscr{O}_U$ defining Y, and to note that, for every open $V \subset U$, the hypothesis implies that the morphism $(\theta_\lambda)_V$ factorises as

$$\Gamma(V, \mathscr{O}_U) \longrightarrow \Gamma(V, \mathscr{O}_U/\mathscr{J}) \longrightarrow \Gamma(f_\lambda^{-1}(V), \mathscr{O}_{Z_\lambda}).$$

One concludes that every section t of $\mathscr{J}$ above V has image 0 in all the $\Gamma(f_\lambda^{-1}(V), \mathscr{O}_{Z_\lambda})$, whence, by virtue of $b)$, $\mathscr{J} = 0$ and $Y = U$. Conversely, if $c)$ is satisfied, to prove $b)$, it suffices to apply $c)$ to the closed sub-prescheme Y of U defined by the Ideal $t\mathscr{O}_U$: the hypothesis that the images of t by the $(\theta_\lambda)_U$ are all zero implies that one has the factorisations (11.10.1.2) for every λ (**I**, 4.1.9).

To prove that $c)$ implies $d)$, it suffices to apply $c)$ to the closed sub-prescheme Y of U, the inverse image under $(u, v)_{\mathrm{S}}$ of the diagonal X' in $\mathrm{X}' \times_{\mathrm{S}} \mathrm{X}'$, and to use (**I**, 4.4.1). Conversely, one deduces $b)$ from $d)$ by considering the S-scheme $\mathrm{X}' = \mathrm{S}[\mathrm{T}] = \mathrm{S} \otimes_{\mathbf{Z}} \mathbf{Z}[\mathrm{T}]$ (T indeterminate) and noting that the sections of $\mathscr{O}_{\mathrm{U}}$ above U correspond bijectively to the S-morphisms $\mathrm{U} \to \mathrm{S}[\mathrm{T}]$ (**I**, 3.3.15); to say that two sections of $\mathscr{O}_{\mathrm{U}}$ above U have the same images by all the $(\theta_\lambda)_{\mathrm{U}}$ is equivalent to saying that the composites of the two corresponding morphisms with all the morphisms $f_\lambda^{-1}(\mathrm{U}) \to \mathrm{U}$ are equal. ☐

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Definition (11.10.2). — When the equivalent conditions of (11.10.1) are satisfied, one says that the family (f_λ) is *schematically dominant*. When the f_λ are the canonical immersions in X of a family (Z_λ) of sub-preschemes of X, one says also that the family (Z_λ) is *schematically dense*.

Remarks (11.10.3). —

(i) The notion of a schematically dominant family on X, as follows for example from a condition of the form $b)$ of (11.10.1), is *local* on X: if (W_α) is an open covering of X, the family (f_λ) is schematically dominant if and only if each of the families formed of the morphisms $f_\lambda^{-1}(\mathrm{W}_\alpha) \to \mathrm{W}_\alpha$, restrictions of the f_λ, is schematically dominant.

(ii) If Z is the prescheme sum of the Z_λ, $f : \mathrm{Z} \to \mathrm{X}$ the morphism coinciding with f_λ on each Z_λ, it amounts to the same to say that the family (f_λ) is schematically dominant, or that the family reduced to the single element f is.

(iii) Let M be a second set of indices, and, for every $\lambda \in \mathrm{L}$, let $(g_{\lambda\mu})_{\mu \in \mathrm{M}}$ be a family of morphisms $g_{\lambda\mu} : \mathrm{Z}'_{\lambda\mu} \to \mathrm{Z}_\lambda$; if, for every $\lambda \in \mathrm{L}$, the family $(g_{\lambda\mu})_{\mu \in \mathrm{M}}$ is schematically dominant, then, it amounts to the same to say that the family $(f_\lambda)_{\lambda \in \mathrm{L}}$ is schematically dominant, or that the family $(f_\lambda \circ g_{\lambda\mu})_{(\lambda,\mu) \in \mathrm{L} \times \mathrm{M}}$ is schematically dominant (11.9.2).

(iv) Let $f : \mathrm{Z} \to \mathrm{X}$ be a morphism such that $f_*(\mathscr{O}_{\mathrm{Z}})$ is a quasi-coherent $\mathscr{O}_{\mathrm{X}}$-Module (for example a quasi-compact and quasi-separated morphism (1.7.4)). Then, to say that f is schematically dominant signifies that the *closed image* of Z *by* f (**I**, 9.5.3) *is identical to* X.

Proposition (11.10.4). — *If the family of morphisms* $f_\lambda : \mathrm{Z}_\lambda \to \mathrm{X}$ *is schematically dominant, the union of the* $f_\lambda(\mathrm{Z}_\lambda)$ *is dense in* X. *Conversely, if this union is dense in* X *and if moreover* X *is* reduced, *the family* (f_λ) *is schematically dominant.*

Proof. The first assertion follows immediately from (11.10.1), $b)$). On the other hand, if X is reduced, and if the union of the $f_\lambda(\mathrm{Z}_\lambda)$ is dense in X, then, if one has the factorisations (11.10.1.2) for every $\lambda \in \mathrm{L}$, one also has factorisations $(f_\lambda^{-1}(\mathrm{U}))_{\mathrm{red}} \xrightarrow{(g_\lambda)_{\mathrm{red}}} \mathrm{Y}_{\mathrm{red}} \xrightarrow{j_{\mathrm{red}}} \mathrm{U}$, and the hypothesis implies that the underlying space of Y is identical to U, whence $\mathrm{Y} = \mathrm{Y}_{\mathrm{red}} = \mathrm{U}$ since U is reduced. ☐

The results of § 11.9 on separating families translate into results on schematically dominant families:

Theorem (11.10.5). — *Let* (f_λ) *be a family of morphisms* $f_\lambda : \mathrm{Z}_\lambda \to \mathrm{X}$, $g : \mathrm{X}' \to \mathrm{X}$ *a morphism, and set* $\mathrm{Z}'_\lambda = \mathrm{Z}_\lambda \times_{\mathrm{X}} \mathrm{X}'$, $f'_\lambda = (f_\lambda)_{(\mathrm{X}')} : \mathrm{Z}'_\lambda \to \mathrm{X}'$.

(i) If g is faithfully flat and if the family (f'_λ) *is schematically dominant, then the family* (f_λ) *is schematically dominant.*

(ii) Suppose that g is a flat morphism, and moreover that one of the two following conditions is satisfied:

a) L *is finite and the* f_λ *are quasi-compact.*

b) *The morphism g is locally of finite presentation.*

Then, if the family (f_λ) *is schematically dominant, it is the same for the family* (f'_λ).

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Proposition (11.10.6). — *Let the notations be those of* (11.10.5), *and suppose that* X *is a prescheme over a field k, and that, setting* $\mathrm{S} = \mathrm{Spec}(k)$, *one has* $\mathrm{X}' = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}'$, *where* S′ *is an arbitrary k-prescheme. Then, if the family* (f_λ) *is schematically dominant, it is the same for* (f'_λ).

Corollary (11.10.7). — *Let* X *be a prescheme over a field* k, $(Z_\lambda)_{\lambda\in L}$ *a family of* k*-preschemes, geometrically reduced over* k*, and for every* $\lambda \in L$*, let* $f_\lambda : Z_\lambda \to X$ *be a* k*-morphism. Let* Y *denote the reduced sub-prescheme of* X *having for underlying space the closure of* $\bigcup_{\lambda\in L} f_\lambda(Z_\lambda)$*. Let* k' *be an extension of* k, X', Z'_λ, $f'_\lambda : Z'_\lambda \to X'$ *the preschemes deduced from* X, Z_λ *and* f_λ *by extension of the base to* k'*. Then, if* Y' *is the reduced sub-prescheme of* X' *having for underlying space the closure of* $\bigcup_{\lambda\in L} f'_\lambda(Z'_\lambda)$*, one has* $Y' = Y \otimes_k k'$*. In particular,* Y *is geometrically reduced over* k*.*

Proof. Since the Z_λ are reduced, the morphisms f_λ factorise as $Z_\lambda \xrightarrow{g_\lambda} Y \xrightarrow{j} X$, where j is the canonical injection (**I**, 5.2.2). It follows then from (11.10.4) that (g_λ) is a schematically dominant family. Let $Y'_1 = Y \otimes_k k'$, a closed sub-prescheme of X', and let g'_λ be the morphism deduced from g_λ by extension of the base to k', so that f'_λ factorises as $Z'_\lambda \xrightarrow{g'_\lambda} Y'_1 \xrightarrow{j'} X'$, where j' is the canonical injection. It follows from (11.10.6) that the family (g'_λ) is schematically dominant. But by hypothesis the Z'_λ are *reduced*, whence (**I**, 5.2.2) the g'_λ factorise as $Z'_\lambda \to Y' \to Y'_1$; one concludes therefore from (11.10.2) that $Y' = Y'_1$ and $h'_\lambda = g'_\lambda$, which establishes the corollary. ☐

Definition (11.10.8). — Let the notations be those of (11.10.5), and suppose that X is an S-prescheme. One says that the family (f_λ) is *universally schematically dominant* (relative to S) if, for every change of base $S' \to S$, the family (f'_λ) corresponding to the change of base $X' = X \times_S S' \to X$ is schematically dominant.

When S is the spectrum of a field, (11.10.6) then shows that a schematically dominant family is universally schematically dominant (relative to S).

When the f_λ are canonical immersions $Z_\lambda \to X$, one says also that the family of sub-preschemes Z_λ is *universally schematically dense in* X (relative to S) instead of saying that the family (f_λ) is universally schematically dominant (relative to S).

Theorem (11.10.9). — *Let the notations be those of* (11.10.5)*, and suppose that* X *is a locally Noetherian* S*-prescheme and that the* Z_λ *are all* S*-flat. For every* $s \in S$*, let* $X_s = X \times_S \operatorname{Spec}(k(s))$, $(Z_\lambda)_s = Z_\lambda \times_S \operatorname{Spec}(k(s))$, $(f_\lambda)_s = (f_\lambda)_s \times 1 : (Z_\lambda)_s \to X_s$*. For the family* (f_λ) *to be universally schematically dominant relative to* S*, it is necessary and sufficient that for every* $s \in S$*, the family* $((f_\lambda)_s)$ *be schematically dominant.*

Proposition (11.10.10). — *Let* $f : X \to S$ *be a flat locally of finite presentation morphism,* U *an open of* X*. For* U *to be universally schematically dense in* X *relative to* S*, it is necessary and sufficient that, for every* $s \in S$, $U_s = U \cap X_s$ *be schematically dense in* X_s *(or, what amounts to the same, that one has* $\operatorname{Ass}(\mathscr{O}_{X_s}) \subset U_s$*).*

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§12. Study of fibres of flat morphisms of finite presentation

12.0. Introduction *We will use throughout this paragraph the general notation of* (9.4.1).

(12.0.1). Given a morphism *locally of finite presentation* $f : X \to Y$, we have seen (9.9) that for certain *local* properties **P** of preschemes over fields or of Modules over these preschemes, the set E of $x \in X$ such that the property **P** holds, for the *fibre* $X_{f(x)}$, at the point x of this fibre, is *locally constructible in* X. We now propose to show that, for the majority of these properties, if one further supposes that the morphism f is *flat*, then the set E is in fact *open in* X. Similarly ((9.2) to (9.8)) we have shown that if f is *of finite presentation*, and if this time **P** denotes certain *global* properties of preschemes over fields or of Modules over these preschemes, the set F of $y \in Y$ such that the property **P** holds for the fibre X_y is *locally constructible in* Y. We will show that if one further supposes that the morphism f is *proper* and *flat*, F is in fact *open in* Y.

(12.0.2). The general method of proof of the properties in question involves three stages. One first reduces to the case where Y is *affine* and X of finite presentation; then:

A) With the help of (8.9.1) (and possibly other results of § 8) and of (11.2.6), one reduces to the case where X and Y are *Noetherian*.
B) One applies the results of § 9 recalled in (12.0.1) to show that E (resp. F) is *constructible*.

C) To see that E is open, it suffices, by virtue of (**0**, 9.2.5), to show that if $x \in \mathrm{E}$, then every *generization* x' of x also belongs to E. Using (**II**, 7.1.7), one sees that, since Y is *Noetherian*, there exists a spectrum of a *discrete valuation ring* Y_1 and a morphism $h : \mathrm{Y}_1 \to \mathrm{X}$ such that, if y_1 (resp. y'_1) is the closed point (resp. the generic point) of Y_1, one has $h(y_1) = x$, $h(y'_1) = x'$. One then performs the base change $g = f \circ h : \mathrm{Y}_1 \to \mathrm{Y}$; taking into account the results of §§ 4 and 6 on locally Noetherian preschemes over fields and the changes of base field, one is reduced to proving the assertion in question for a point x_1 of $\mathrm{X}_1 = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}_1$ above y_1 and for a generization x'_1 of x_1 above y'_1. Since $\mathrm{Y}_1 = \mathrm{Spec}(\mathrm{A})$, where A is a discrete valuation ring, with uniformiser t, one must in the end, for an A-module M, prove a property of M, knowing that $\mathrm{M}/t\mathrm{M}$ has the same property and that t is M-*regular* (which follows from the flatness hypothesis); one uses for this the results of (3.4) and of (5.12). One proceeds similarly for the set $\mathrm{F} \subset \mathrm{Y}$.

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12.1. Local properties of fibres of a flat morphism locally of finite presentation

Theorem (12.1.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module f-flat and of finite presentation, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$, k an integer. The following subsets of X are open:*

(i) The set of $x \in X$ such that the dimensions of the prime cycles associated to $\mathscr{F}_{f(x)}$ and containing x are elements of Φ.

(ii) The set of $x \in X$ such that all the prime cycles associated to $\mathscr{F}_{f(x)}$ containing x have dimension k.

(iii) The set of $x \in X$ such that x does not belong to any associated immersed prime cycle of $\mathscr{F}_{f(x)}$.

(iv) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is equidimensional at the point x and possesses the property (S_k) *at the point x* (5.7.2).

(v) The set of $x \in X$ such that $\mathrm{coprof}((\mathscr{F}_{f(x)})_x) \leqslant k$ (**0**, 16.4.9).

(vi) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is a Cohen–Macaulay $\mathscr{O}_{X_{f(x)}}$-Module at the point x (5.7.1).

(vii) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is geometrically reduced at the point x (4.6.22).

(viii) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is geometrically punctually integral at the point x (4.6.22).

The questions being local on X, one first reduces to the case where $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$ are affine, f a morphism of finite presentation. There is then a Noetherian subring A_0 of A, an A_0-prescheme of finite type X_0, and a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ such that $\mathscr{F}_0 \otimes_{A_0} A$ is isomorphic to $\mathscr{F}$ (8.9.1)*; in addition, by virtue of* (11.2.6)*, one can suppose that $\mathscr{F}_0$ is Y_0-flat (with $Y_0 = \mathrm{Spec}(A_0)$). If $h : X \to X_0$ is the canonical projection, the set E of points $x \in X$ where one of the properties (i) to (viii) is satisfied is equal to $h^{-1}(E_0)$, where E_0 is the set of $x_0 \in X_0$ where the corresponding property for $\mathscr{F}_0$ is satisfied: this follows, for properties (i) to (iii), from* (4.2.7)*; for properties (iv) to (vi), from* (6.7.1)*; for properties (vii) and (viii), from* (4.7.11).

One is thus reduced to the case where A and B are Noetherian, f of finite type, and $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite type. In virtue of (9.9.2), one knows that the set E is *constructible* for all the properties considered, and it remains in each case the step C) of (12.0.2), where one must prove that E is *stable under generization*.

(12.1.1.1). Let us begin with case (iii), which is the simplest. Let $x' \neq x$ be a generization of x in X. Set $y = f(x)$, $y' = f(x')$; there exists a spectrum of a discrete valuation ring Y_1 and a morphism $u : Y_1 \to X$ such that, if y_1 and y'_1 are the closed point and the generic point of Y_1 respectively, one has $u(y_1) = x$ and $u(y'_1) = x'$ (**II**, 7.1.7); set $g = f \circ u : Y_1 \to Y$; if $X_1 = X \times_Y Y_1$, there exists a Y_1-section $u_1 : Y_1 \to X_1$ such that $u = g_1 \circ u_1$, where $g_1 : X_1 \to X$ is the canonical projection. If $x_1 = u_1(y_1)$, $x'_1 = u_1(y'_1)$, one has $g_1(x_1) = x$ and $g_1(x'_1) = x'$, and x'_1 is a generization of x_1. Applying again (4.2.7), one sees that one is reduced to the case where $Y = \mathrm{Spec}(A)$ is the **IV-3** | 175
spectrum of a discrete valuation ring, y being the closed point and y' the generic point of Y. If t is a uniformiser of A, the hypothesis that $\mathscr{F}$ is f-flat implies that t is $\mathscr{F}$-regular (**0**, 6.3.4). By hypothesis, none of the associated immersed prime cycles of $\mathscr{F}/t\mathscr{F}$ contains x. Then, it follows from (3.4.4) that x does not belong to any associated immersed prime cycle of $\mathscr{F}$, and the same is true of every generization of x. In particular, x' does not belong to any associated immersed prime cycle of $\mathscr{F}$, nor *a fortiori* to any of those associated to $\mathscr{F}_{y'}$ ($X_{y'}$ being a sub-prescheme inducing an open subset of X).

(12.1.1.2). Let us now consider cases (iv) and (v) ((vi) is only a particular case of (v)). One proceeds as above (using (6.7.1) in place of (4.2.7)) and one is reduced to the case where Y is the spectrum of a discrete valuation ring A.

The ring $C = \mathscr{O}_{X,x}$ is then the localisation of an A-algebra of finite type, hence is catenary (5.6.4), and by hypothesis the C-module M_x/tM_x satisfies (S_k) and is equidimensional (resp. one has $\operatorname{coprof}(M_x/tM_x) \leqslant k$); one deduces from (5.12.2) (resp. from (**0**, 16.4.10)) that M_x satisfies (S_k) and is equidimensional (resp. that $\operatorname{coprof}(M_x) \leqslant k$) since t is M_x-regular and belongs to the maximal ideal of C. Whence the conclusions since x' is a generization of x and $(\mathscr{F}_{y'})_{x'} = \mathscr{F}_{x'}$.

(12.1.1.3). Let us pass to cases (vii) and (viii). One can evidently replace Y by the integral sub-prescheme having $\overline{\{y'\}}$ as underlying space, and X by $f^{-1}(\overline{\{y'\}})$, without changing the fibres in y and y', so one can suppose A integral, with field of fractions $K = k(y')$. One knows ((4.5.11) and (4.7.8)) that there exists a finite extension K' of K such that, for the $(\mathscr{O}_X \otimes_A K')$-Module $\mathscr{F}' = \mathscr{F} \otimes_A K'$, the associated prime cycles are geometrically irreducible and the geometric lengths of $\mathscr{F}'$ at the maximal points z'_i of its support are respectively equal to the lengths of $\mathscr{F}'_{z'_i}$ at these points; it amounts therefore to the same to say that $\mathscr{F}$ is geometrically reduced (resp. geometrically punctually integral) at a point x' of $X_{y'} = X \otimes_A K$, or to say that $\mathscr{F}'$ is *reduced* (resp. *integral*) at the points of $X \otimes_A K'$ lying above x', taking into account (4.2.7). Let A' be a finite A-algebra whose field of fractions is K'; it follows from (4.7.11) that at every point of $X \otimes_A A'$ above x, $\mathscr{F} \otimes_A A'$ has property (vii) (resp. (viii)). Replacing A by A' and X by $X \otimes_A A'$, one sees that one can reduce to proving that $\mathscr{F}_{f(x')}$ is *reduced* (resp. *integral*) at every generization x' of x. One proceeds then as in (12.1.1.1), and using this time (4.7.11), in the case where A is a discrete valuation ring, y being the closed point, y' the generic point of Y, and the uniformiser t of A being an M-*regular* element. Since by hypothesis $\mathscr{F}/t\mathscr{F}$ is reduced (resp. integral) at the point x, it follows from (3.4.6) (resp. (3.4.5)) that $\mathscr{F}$ is reduced (resp. integral) at the point x, hence also in a neighbourhood of x, and in particular at the point x', which finishes the proof in cases (vii) and (viii).

(12.1.1.4). It remains to examine cases (i) and (ii). Replacing Y by the integral sub-prescheme having $\overline{\{y'\}}$ as underlying space, one can reduce to the case where Y is irreducible and y' is its generic point. Let z' be a generic point of an associated prime cycle of $\mathscr{F}_{y'}$ containing x', and let Z' be the closure of z' in X, so that one has $x \in Z'$. To treat case (i), it will suffice to prove the **IV-3 | 176**

Proposition (12.1.1.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module and f-flat; for every $y \in Y$, set $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$. Let z' be a point of X, $y' = f(z')$, and suppose that $z' \in \operatorname{Ass}(\mathscr{F}_{y'})$. Let Z' (resp. Y') be the reduced sub-prescheme of X (resp. Y) having as underlying space the closure of $\{z'\}$ in X (resp. of $\{y'\}$ in Y). Then, for every $y \in f(Z')$, the dimensions of all the irreducible components of Z'_y are equal to $\dim(Z'_{y'})$ (dimension of the closure of z' in $X_{y'}$) (which we will express more precisely later (13.2.2) by saying that the restriction $Z' \to Y'$ of f is an equidimensional morphism); furthermore, at every maximal point z of Z'_y, one has $z \in \operatorname{Ass}(\mathscr{F}_y)$.*

For this, we will now reduce to the case where Y is the spectrum of a discrete valuation ring. Let us take a spectrum of a discrete valuation ring Y_1 and a morphism $h : Y_1 \to X$ such that $h(y_1) = z$, $h(y'_1) = z'$, y_1 and y'_1 being the closed point and the generic point of Y_1 respectively (**II**, 7.1.7). Set $g = f \circ h : Y_1 \to Y$, $X_1 = X \times_Y Y_1$, and let $f_1 : X_1 \to Y_1$, $g_1 : X_1 \to X$ be the canonical projections; there is a Y_1-section $h_1 : Y_1 \to X_1$ such that $h = g_1 \circ h_1$ (**I**, 3.3.14). If $Z'_1 = Z'_{y'} \otimes_{k(y')} k(y'_1)$, one knows (4.2.6) that the irreducible components of Z'_1 dominate $Z'_{y'}$; let z'_1 be the generic point of one of these components which contains $h_1(y'_1)$, so that $f_1(z'_1) = y'_1$ and $g_1(z'_1) = z'$. Since $h_1(y_1)$ is a specialisation of $h_1(y'_1)$, it is *a fortiori* a specialisation of z'_1; let z_1 be a generic point of one of the irreducible components of $\overline{\{z'_1\}} \cap (X_1)_{y_1}$ containing $h_1(y_1)$. Then $g_1(z_1)$ is a specialisation of $g_1(z'_1) = z'$ in X, and $z = h(y_1) = g_1(h_1(y_1))$ is a specialisation of $g_1(z_1)$; but since $f(g_1(z_1)) = y$ and z is a generic point of Z'_y, one necessarily has $z = g_1(z_1)$.

Suppose now that one has proved that, if one sets $\mathscr{F}_1 = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_1}$, then $z_1 \in \operatorname{Ass}((\mathscr{F}_1)_{y_1})$; it will follow from (4.2.7) that $z \in \operatorname{Ass}(\mathscr{F}_y)$; furthermore, the dimensions of $\overline{\{z_1\}} \cap (X_1)_{y_1}$ and of $\overline{\{z\}} \cap X_y$ are equal, and likewise the dimensions of $\overline{\{z'_1\}} \cap (X_1)_{y'_1}$ and of $\overline{\{z'\}} \cap X_{y'}$ are equal (4.2.7);

one has therefore indeed reduced, as announced, to the case where Y is the spectrum of a discrete valuation ring A, y and y' being its closed point and its generic point respectively.

Since $z' \in \mathrm{Ass}(\mathscr{F}_{y'})$, one has *a fortiori* $z' \in \mathrm{Ass}(\mathscr{F})$. If t is a uniformiser of A, t is $\mathscr{F}$-regular by flatness, so by (3.4.3) one has $z \in \mathrm{Ass}(\mathscr{F}/t\mathscr{F}) = \mathrm{Ass}(\mathscr{F}_y)$. As for the assertion relative to dimensions, it follows from (7.1.13), applied to a closed sub-prescheme of X having Z' as underlying space.

Let us finally tackle case (ii) of (12.1.1). With the same notation, it is necessary to prove that if all the prime cycles associated to $\mathscr{F}_y$ and containing x have dimension k, the same is true of all the prime cycles associated to $\mathscr{F}_{y'}$ and containing x'. Now, one has just seen that *every* prime cycle associated to $\mathscr{F}_{y'}$ and containing x' has the same dimension as *one* of the prime cycles associated to $\mathscr{F}_y$ and containing x; this shows (ii).

Remarks (12.1.2). — (i) Under the conditions of (12.1.1.5) with $\mathscr{F} = \mathscr{O}_X$ (so that f is *flat*), one cannot in general assert that the restriction $Z' \to Y'$ of f is a flat morphism nor even an *open* morphism. This is what the example (6.5.5), (ii) shows, where one takes for Z' one of the two irreducible components of $X = \mathrm{Spec}(B)$. It is immediate that this restriction morphism is not open at the points of Z' above the "double point" of $Y' = Y$. **IV-3** | 177

(ii) Under the hypotheses of (12.1.1.5), with $\mathscr{F} = \mathscr{O}_X$, one cannot weaken the condition "f is flat" to "f is universally open" (cf. (2.4.6)), as we will see further on with an example (14.4.10), (i)).

Corollary (12.1.3). — *Under the hypotheses of* (12.1.1), *the function* $x \mapsto \mathrm{coprof}((\mathscr{F}_{f(x)})_x)$ *is upper semi-continuous in X and the function* $x \mapsto \mathrm{prof}^*_x(\mathscr{F}_{f(x)})$ (10.8.1) *is lower semi-continuous in X.*

The first assertion is nothing other than an equivalent formulation of (12.1.1), (v). To prove the second, one can first, by taking into account (10.8.7) and (10.8.8), reduce as in (12.1.1) to the case where Y is the spectrum of a discrete valuation ring A of closed point y and of generic point y', $X = \mathrm{Spec}(B)$ being affine, $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite type. Since $\mathscr{F}$ is f-flat, every irreducible component of $\mathrm{Supp}(\mathscr{F})$ which contains $x \in X_y$ dominates Y (2.3.4); in other words, its generic point z' belongs to $X_{y'}$. The irreducible components Z of $\mathrm{Supp}(\mathscr{F})$ which contain a generization x' of x belonging to $X_{y'}$ are therefore exactly those which contain x; furthermore, by (12.1.1.5), one has $\dim(Z_{y'}) = \dim(Z_y)$, whence, if $T = \mathrm{Supp}(\mathscr{F})$, $\dim_x(T_y) = \dim_{x'}(T_{y'})$. Furthermore, for a uniformiser t of A, one has seen in (12.1.1), (v) that $\mathrm{coprof}(M_x) = \mathrm{coprof}(M_x/tM_x)$, and since on the other hand $\mathrm{coprof}(M_{x'}) \leqslant \mathrm{coprof}(M_x)$ (6.11.5), one sees that $\mathrm{coprof}((\mathscr{F}_{y'})_{x'}) \leqslant \mathrm{coprof}((\mathscr{F}_y)_x)$; the relation $\mathrm{prof}^*_x(\mathscr{F}_y) \leqslant \mathrm{prof}^*_{x'}(\mathscr{F}_{y'})$ then follows from (10.8.7).

Remark (12.1.4). — One also deduces from (12.1.1), (i) that the function $x \mapsto \dim_x(X_{f(x)})$ is upper semi-continuous in X, but we will see further on (13.1.3) that this property is true even without supposing f flat.

Corollary (12.1.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module which is f-flat, $\mathscr{G}$ a coherent $\mathscr{O}_Y$-Module, x a point of X, $y = f(x)$. Suppose that $\mathscr{G}$ has the property (S_k) at the point y, and that $\mathscr{F}_y$ has the property (S_k) at the point x and is equidimensional at the point x. Then:*

(i) $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ has the property (S_k) at the point x.

(ii) If furthermore Y is locally immersible in a regular scheme, or is an excellent prescheme (7.8.5), *there exists a neighbourhood V of x in X such that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ has the property (S_k) in this neighbourhood.*

Proof. Indeed, by (12.1.1), (iv), there exists an open neighbourhood V of x such that for every $x' \in V$, $\mathscr{F}_{f(x')}$ satisfies (S_k) at the point x'. On the other hand, $\mathscr{G}$ satisfies (S_k) at all points of $\mathrm{Spec}(\mathscr{O}_y)$ (5.7.2). Replacing Y by $Y' = \mathrm{Spec}(\mathscr{O}_y)$ and X by $V \times_Y Y'$, one is reduced (taking into account (**I**, 3.6.5)) to the case where $\mathscr{G}$ satisfies (S_k) throughout Y and where, for every $y \in f(X)$, $\mathscr{F}_y$ has the property (S_k) at all points of $f^{-1}(y)$; since $\mathscr{F}$ is f-flat, one knows then (6.4.1) that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ has the property (S_k) in X, which proves (i). To prove (ii), it suffices to observe that, under the hypotheses made, $\mathscr{G}$ has the property (S_k) in a neighbourhood U of y in Y (6.11.2) and (7.8.3), (iv)); **IV-3** | 178
one concludes then in the same manner by replacing Y this time by U and X by $V \times_Y U$. □

Theorem (12.1.6). — *Let $f : X \to Y$ be a flat morphism locally of finite presentation, and let k be an integer. The following subsets of X are open:*

(i) The set of points $x \in X$ such that $X_{f(x)}$ has the property (S_k) at the point x.

(ii) The set of $x \in X$ such that $X_{f(x)}$ satisfies the property (R_k) geometrically at the point x, is equidimensional at the point x, and such that, furthermore, x does not belong to any associated immersed prime cycle of $X_{f(x)}$.

(iii) The set of $x \in X$ such that $X_{f(x)}$ is geometrically regular (i.e. smooth) at the point x.

(iv) The set of $x \in X$ such that $X_{f(x)}$ is geometrically normal at the point x.

Steps A) and B) of (12.0.2) are done here as in (12.1.1); for step A), one uses (6.7.8), and also (4.2.7) for (ii); for step B), one uses (9.9.2) and (9.9.4). It remains to examine step C) in each case.

Proof. (i) As in (12.1.1.2), one reduces (using (6.7.8)) to the case where Y is the spectrum of a discrete valuation ring A, $X = \mathrm{Spec}(B)$, where B is an A-algebra of finite type, and x, x' are two points of X such that $f(x) = y$ is the closed point and $y' = f(x')$ is the generic point of Y, with x' in addition a generization of x. As one wishes to prove that X satisfies (S_k) at the point x', one may replace X by $\mathrm{Spec}(\mathscr{O}_{X,x})$; that is, one may suppose that the ring B is local, that the homomorphism $A \to B$ is local, and that B is an A-algebra essentially of finite type (1.3.8). Then, by hypothesis, if t is a uniformiser of A, t is a B-regular element by flatness, and B/tB satisfies (S_k). But since A is a universally catenary ring (5.6.4), B is catenary, so, by (5.12.4), it follows that B satisfies (S_k), which finishes the proof in case (i).

(iii) Here step C) is unnecessary; since f is flat, one knows indeed (6.8.7) that when Y is locally Noetherian and f locally of finite type, the set of $x \in X$ such that $X_{f(x)}$ is geometrically regular at the point x is open in X.

(ii) By reasoning as in (12.1.1.3), to prove that the property considered is stable under generization, one can first, by considering an arbitrary finite extension K' of $k(y')$, and taking into account the definition (6.7.6) of the geometric property (R_k), as well as the invariance under change of base field of the two other properties appearing in (ii), replace in (ii) the geometric property (R_k) by the property (R_k). Proceeding then as in (i), one reduces to the case where Y is the spectrum of a discrete valuation ring A, and since A is catenary, it suffices to apply (5.12.5) to conclude.

(iv) The set of points $x \in X$ such that $X_{f(x)}$ is geometrically normal at the point x is contained in that of $x \in X$ such that $X_{f(x)}$ is geometrically punctually integral and satisfies (S_2) at the point x, and this latter set is open in X by virtue of (12.1.1), (viii). One can therefore already suppose that, for every $x \in X$, $X_{f(x)}$ is geometrically punctually integral and satisfies (S_2) at the point x, and *a fortiori* it is equidimensional. On the other hand, by the Serre criterion (5.8.6), to say that $X_{f(x)}$ is geometrically normal at the point x means that $X_{f(x)}$ satisfies (S_2) and the property (R_1) geometrically at x; but by virtue of (ii) and the preceding remarks, **IV-3** | 179
this set is the intersection of two open subsets of X, hence is open in X.

□

Corollary (12.1.7). — *Let $f : X \to Y$ be a morphism locally of finite presentation. Then the set of points $x \in X$ at which f has one of the following properties (6.8.1):*

(i) satisfies the property (S_k);

(ii) has coprofundity $\leqslant n$;

(iii) is Cohen–Macaulay;

(iv) is regular (or smooth, which amounts to the same);

(v) is normal;

(vi) is reduced;

is open in X.

Proof. Indeed, it follows from (11.3.1) that the set of $x \in X$ where f is *flat* is *open*. One can therefore restrict to the case where f is flat, and then the corollary follows from (12.1.1), (iv), (vi), and (vii), and from (12.1.6), (i), (ii), and (iv). □

Remarks (12.1.8). — (i) In (12.1.6), (ii), one cannot suppress the hypothesis that x does not belong to any associated immersed prime cycle of $X_{f(x)}$. This is shown by the example (5.12.3):

one takes $X = \mathrm{Spec}(A)$, with Y the spectrum of the local ring A_0 of $k[T]$ corresponding to the prime ideal (T), and the morphism $X \to Y$ corresponding to the injective homomorphism $A_0 \to A$ obtained, by localization and passage to the quotient, from the injection $k[T] \to k[T, U, V]$. This morphism is flat since A is a torsion-free A_0-module (**0**, 6.3.4). Here the fibre X_y at the closed point y of Y, equal to $\mathrm{Spec}(A/tA)$, is irreducible, of dimension 1 and satisfies the condition (R_0) geometrically since the local ring at its generic point is a field. In contrast, at the generic point y' of Y, the fibre $X_{y'}$ has two irreducible components of dimensions 0 and 1, and that of dimension 0 is not reduced, so $X_{y'}$ does not satisfy the condition (R_0).

(ii) In (12.1.6), (ii), one cannot suppress the hypothesis that $X_{f(x)}$ is equidimensional at the point x either. This is seen from the example (5.12.6), with $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(A_0)$, where A_0 is defined as in (i); the morphism $X \to Y$ is again obtained from the injection $k[T] \to k[T, U, V, W]$ by localization and passage to the quotient, and is flat since A is a torsion-free A_0-module (**0**, 6.3.4). The fibre X_y at the closed point y of Y is reduced (hence satisfies (S_1)) and satisfies (R_1), but has two irreducible components of dimensions 2 and 1, while the fibre $X_{y'}$ at the generic point y' of Y does not satisfy (R_1).

12.2. Local and global properties of fibres of a proper, flat morphism of finite presentation **IV-3** | 179

Theorem (12.2.1). — *Let $f : X \to Y$ be a* proper *morphism of finite presentation, $\mathscr{F}$ an f-flat $\mathscr{O}_X$-module of finite presentation, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$, and k an integer. The following subsets of Y are open:*

(i) The set of $y \in Y$ such that the set of dimensions of the prime cycles associated to $\mathscr{F}_y$ is contained in Φ.

(ii) The set of $y \in Y$ such that the set of dimensions of the irreducible components of $\mathrm{Supp}(\mathscr{F}_y)$ *contains Φ.*

(iii) The set of $y \in Y$ such that all the prime cycles associated to $\mathscr{F}_y$ have the same dimension, equal to k.

(iv) The set of $y \in Y$ such that $\mathscr{F}_y$ has no associated immersed prime cycle and the set of dimensions of the irreducible components of $\mathrm{Supp}(\mathscr{F}_y)$ *is equal to Φ.* **IV-3** | 180

(v) The set of $y \in Y$ such that $\mathscr{F}_y$ has the property (S_k) and is equidimensional at every point of X_y.

(vi) The set of $y \in Y$ such that $\mathrm{coprof}(\mathscr{F}_y) \leqslant k$.

(vii) The set of $y \in Y$ such that $\mathscr{F}_y$ is a Cohen–Macaulay $\mathscr{O}_{X_y}$-Module.

(viii) The set of $y \in Y$ such that $\mathscr{F}_y$ is geometrically reduced.

(ix) The set of $y \in Y$ such that $\mathscr{F}_y$ is geometrically punctually integral at every point of X_y.

(x) The set of $y \in Y$ such that $\mathscr{F}_y$ is geometrically integral.

(xi) The set of $y \in Y$ such that $\mathscr{F}_y$ has no associated immersed prime cycle and that the sum $l(y)$ of the total multiplicities (4.7.12) of $\mathscr{F}_y$ at the maximal points of $\mathrm{Supp}(\mathscr{F}_y)$ *is $\leqslant k$.*

Proof. With the exception of (ii), (iii), (iv), (x) and (xi), the properties $\mathbf{P}(y)$ considered are of the form: "for every $x \in X_y$, $\mathscr{F}_y$ has the property $\mathbf{Q}(x)$", where $\mathbf{Q}(x)$ is one of the properties (i) to (viii) of (12.1.1). It follows from (12.1.1) that the set U of $x \in X$ such that $\mathbf{Q}(x)$ is true is open, and that the set V of $y \in Y$ such that $\mathbf{P}(y)$ is true is precisely $Y - f(X - U)$; in all these cases, the theorem is therefore already proved under the sole assumption that the morphism f is *closed*. For (iii), one applies (i) with Φ reduced to a single element. It remains to examine cases (ii), (iv), and (xi) separately ((x) follows immediately from (xi) and (4.7.13)),[15] always applying the method described in (12.0.2). □

(12.2.1.1). Case (ii): Steps A) and B) of the proof proceed as at the start of (12.1.1); for step A), one uses (8.9.1), (11.2.6) as well as (4.2.7); for step B), one uses (9.5.5) applied to $\mathrm{Supp}(\mathscr{F})$. It remains to show that if (supposing Y Noetherian) the set of dimensions of the irreducible components of $\mathrm{Supp}(\mathscr{F}_y)$ contains Φ, the same is true of the set of dimensions of the irreducible components of $\mathrm{Supp}(\mathscr{F}_{y'})$, for every *generization* y' of y in Y. Let Y_1 be a spectrum of a discrete valuation ring such that, if y_1 and y'_1 are the closed point and the generic point of Y_1, there exists a morphism $h : Y_1 \to Y$ such that $h(y_1) = y$, $h(y'_1) = y'$ (**II**, 7.1.7). Applying (4.2.7), one sees that one can replace Y by Y_1 and X by $X_1 = X \times_Y Y_1$, in other words, one can restrict to the case where Y is the spectrum of a discrete valuation ring, y its closed point and y' its generic point.

[15]The printed source cites (4.7.14), but Section 4.7 ends with Proposition 4.7.13 before Section 4.8 begins; Proposition 4.7.13 is the relevant geometric-integrality criterion.

Using (**Err**$_{\mathbf{III}}$, 30), one can restrict to the case where $\mathrm{Supp}(\mathscr{F}) = X$, so that f is quasi-flat (2.3.3); the irreducible components Z_i of X then dominate Y (2.3.4), in other words their generic points belong to $X_{y'}$, and the $Z_i \cap X_{y'}$ are the irreducible components of $X_{y'}$. But every irreducible component of X_y is contained in one of the Z_i, hence is an irreducible component of $(Z_i)_y$; now, one knows (7.1.13) that the dimensions of all the irreducible components of $(Z_i)_y$ are equal to $\dim((Z_i)_{y'})$, which finishes the proof of (ii).

(12.2.1.2). Case (iv): The same reasoning as at the start of (12.2.1.1), using (12.1.1), (iii), shows that one can already suppose (replacing Y by an open subset of Y) that for every $y \in Y$, $\mathscr{F}_y$ has no associated immersed prime cycle. The assertion of case (iv) is then an immediate consequence of the assertions of cases (i) and (ii).

(12.2.1.3). Case (xi): For step A) of the reasoning, one uses (8.9.1), (11.2.6), (8.10.5, (xii)) (to keep the hypothesis that f is proper), as well as (4.2.7), (4.5.6) and (4.7.9). For step B), one sees, as in case (iv), that one can suppose that for every $y \in Y$, $\mathscr{F}_y$ has no associated immersed prime cycle, and one applies (9.8.8) which proves that the function $z \mapsto l(z)$ is constructible. It remains therefore to see (supposing Y Noetherian and f proper) that for every generization y' of a point $y \in Y$, $l(y') \leqslant l(y)$. By reasoning as in (12.1.1.3), one sees (with the help of (4.7.8) and (4.5.11)) that one can suppose the irreducible components of $\mathrm{Supp}(\mathscr{F}_{y'})$ are geometrically irreducible and that $l(y')$ is the *sum of the lengths* of $(\mathscr{F}_{y'})_{z'_j}$ at the maximal points z'_j of $\mathrm{Supp}(\mathscr{F}_{y'})$. Applying again (4.2.7), (4.5.6) and (4.7.9), one reduces, as in (12.2.1.1), to the case where Y is a spectrum of a discrete valuation ring, y its closed point and y' its generic point. The fact that $l(y') \leqslant l(y)$ will then be a consequence of the IV-3 | 181

Lemma (12.2.1.4). — *Let Y be a spectrum of a discrete valuation ring, y its closed point, y' its generic point, $f : X \to Y$ a proper morphism, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module and f-flat, z_i (resp. z'_j) the maximal points of* $\mathrm{Supp}(\mathscr{F}_y)$ *(resp.* $\mathrm{Supp}(\mathscr{F}_{y'})$*). Suppose that $\mathscr{F}_y$ has no associated immersed prime cycle. Then one has*

$$\sum_j \mathrm{long}((\mathscr{F}_{y'})_{z'_j}) \leqslant \sum_i \mathrm{long}((\mathscr{F}_y)_{z_i}).$$

One has $Y = \mathrm{Spec}(A)$, where A is a discrete valuation ring, whose uniformiser we will denote by t, so that $\mathscr{F}_y = \mathscr{F}/t\mathscr{F}$. Since $\mathscr{F}$ is f-flat, the z'_j are also the maximal points of $\mathrm{Supp}(\mathscr{F})$ (2.3.4); for every z_i, denote by z'_{ik} those of the z'_j which are generizations of z_i; it follows from (3.4.1.1) that one has

$$\mathrm{long}((\mathscr{F}_y)_{z_i}) \geqslant \sum_k \mathrm{long}((\mathscr{F}_{y'})_{z'_{ik}})$$

whence by summing

$$\sum_i \mathrm{long}((\mathscr{F}_y)_{z_i}) \geqslant \sum_i \sum_k \mathrm{long}((\mathscr{F}_{y'})_{z'_{ik}}).$$

The lemma will therefore be proved if we establish that for every z'_j, there is at least one index i such that z'_j is one of the z'_{ik}. Now, since f is proper (hence closed) and $f(z'_j) = y'$, there exists $x \in X$ which is a specialisation of z'_j and is such that $f(x) = y$, in other words $\overline{\{z'_j\}} \cap X_y$ is non-empty. Since t is $\mathscr{F}$-regular by flatness, one deduces from (3.4.3) that there is at least one point of $\mathrm{Ass}(\mathscr{F}_y)$ of which z'_j is a generization; but since the points of $\mathrm{Ass}(\mathscr{F}_y)$ are, by hypothesis, the z_i, this finishes the proof of the lemma.

Corollary (12.2.2). — *Under the hypotheses of* (12.2.1)*:*

(i) The function $y \mapsto \dim(\mathrm{Supp}(\mathscr{F}_y))$ is continuous (hence locally constant) in Y.
(ii) The function $y \mapsto \mathrm{coprof}(\mathscr{F}_y)$ (5.7.1) *is upper semi-continuous in Y.*
(iii) The function $y \mapsto l(y)$ (12.2.1)*, (xi) is upper semi-continuous in Y when the $\mathscr{F}_y$ have no associated immersed prime cycle.* IV-3 | 182
(iv) The function $y \mapsto \mathrm{prof}^*(\mathscr{F}_y)$ (10.8.1) *is lower semi-continuous in Y.*

Proof. (i) It follows from (12.2.1), (i) applied to Φ equal to the smallest interval of $\mathbf{N}$ containing the dimensions of the prime cycles associated to $\mathscr{F}_y$, that $z \mapsto \dim(\mathrm{Supp}(\mathscr{F}_z))$ is upper semi-continuous at the point y; it follows on the other hand from (12.2.1), (ii) applied to Φ equal to the set of dimensions of the irreducible components of $\mathrm{Supp}(\mathscr{F}_y)$, that this same function is lower semi-continuous at the point y, whence the conclusion. The assertions (ii) and (iii) are nothing but other formulations of (12.2.1), (vi) and (xi). Finally, by (12.1.3), the set of $x \in X$ such that $\mathrm{prof}^*_{f(x)}(\mathscr{F}_{f(x)}) \leqslant k$ is open, and the conclusion of (iv) follows by the same reasoning as at the beginning of (12.2.1).

□

Remarks (12.2.3). — (i) One will observe that for (12.2.1), (ii), one can dispense with the hypothesis that f is *proper*.

(ii) In (12.2.1), (ii), one cannot replace the condition that the set $E(y)$ of dimensions of the irreducible components of $\mathrm{Supp}(\mathscr{F}_y)$ contains Φ, by the condition that $E(y)$ be *equal to* Φ. Indeed, let k be a field, and consider the projective space $P = \mathbf{P}^3_k = \mathrm{Proj}(C)$ of dimension 3, where $C = k[t_0, t_1, t_2, t_3]$ (the t_i being indeterminates). In P, let X_0 be the closed sub-prescheme which is the "union of the line X_1 defined by $t_1 = t_3 = 0$ and the plane X_2 defined by $t_2 - t_3 = 0$". In geometric terms, this says that $X_0 = \mathrm{Proj}(C/\mathfrak{a})$, where the graded ideal $\mathfrak{a} = \mathfrak{b}\mathfrak{c}$, with $\mathfrak{b} = Ct_1 + Ct_3$ and $\mathfrak{c} = C(t_2 - t_3)$. Consider the morphism $f_0 : X_0 \to X_1$ which, geometrically, is the "projection of X_0 onto X_1 from the line at infinity D defined by $t_0 = t_2 = 0$". Algebraically, f_0 corresponds to the homomorphism of graded algebras

$$k[t_0, t_2] \longrightarrow k[t_0, t_1, t_2, t_3]/\mathfrak{a}$$

obtained, by passage to the quotient, from the canonical injection $k[t_0, t_2] \to k[t_0, t_1, t_2, t_3]$. It is clear that f_0 is a projective morphism, hence *proper*; the same is therefore true of its restriction $f : X \to Y$, where $Y = D_+(t_0)$ and $X = f_0^{-1}(Y)$. To see that f is *flat*, it suffices to note that $k[t_2]$ is a principal ideal domain and that the ring

$$C' = k[t_1, t_2, t_3]/(t_2 - t_3)((t_1) + (t_3))$$

is a torsion-free $k[t_2]$-module (**0**, 6.3.4). For every $y \in Y$ distinct from the point y_0 defined by $t_2 = 0$, $f^{-1}(y)$ then has two irreducible components, of dimensions 0 and 1, whereas $f^{-1}(y_0)$ has only one irreducible component, of dimension 1.

(iii) In (12.2.1), (i), one cannot also replace the condition that the set $E'(y) \supset E(y)$ of dimensions of the prime cycles associated to $\mathscr{F}_y$ be contained in Φ by the condition that it be *equal to* Φ. Indeed, let k be a field, let A be the polynomial ring $k[t]$, and let B be the subring $k[t^4, t^3u, tu^3, u^4]$ of the polynomial ring $k[t, u]$ (where t, u are indeterminates). The ring B is not integrally closed: the element t^2u^2 belongs to its integral closure but not to B. If $\mathfrak{m}$ is the maximal ideal of B generated by t^4, t^3u, tu^3, u^4, then $\mathfrak{m}$ is an embedded associated prime ideal of B/Bt^4.[16] Set $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and let $f : X \to Y$ be the morphism corresponding to the homomorphism $A \to B$ which sends t to t^4. Since this homomorphism makes B into an A-module without torsion, f is flat (**0**, 6.3.4). If y is the point of Y corresponding to the maximal ideal tA, the prescheme $f^{-1}(y)$ is irreducible and of dimension 1, but has an associated immersed prime cycle of dimension 0. In contrast, if y' is the generic point of Y, then $f^{-1}(y')$ has no associated immersed prime cycle, since it is the spectrum of an integral $k(y')$-algebra of dimension 1. In this example f is an affine, nonproper morphism; one obtains at once an analogous example in which f is proper and flat by considering X as an everywhere dense open subset of a projective scheme over Y (**II**, 5.3.4) and (**II**, 5.3.2), or by proceeding directly as in Remark (ii).

Remarks (ii) and (iii) show the necessity of including in (12.2.1), (iv) the condition that $\mathscr{F}_y$ have no associated immersed prime cycle.

(iv) In (12.2.1), (xi), one cannot suppress the hypothesis that f be proper. Indeed, let k be a field, let Y be the "affine line", the spectrum of the polynomial ring $k[t]$ (t an indeterminate), and let X be the disjoint union of Y and the open complement of the closed point y of Y defined by $t = 0$. It is clear that the morphism $f : X \to Y$ which is equal to the canonical injections on the two components of X is flat and that, on taking $\mathscr{F} = \mathscr{O}_X$, the Module $\mathscr{F}_z$ has no associated immersed prime cycle for any $z \in Y$. Nevertheless, one has $l(y) = 1$ and $l(z) = 2$ for every $z \neq y$.

[16] The printed source has A/At^4; the construction and the following fibre computation require B/Bt^4.

(v) In (12.2.1), (xi), one cannot also suppress the hypothesis that $\mathscr{F}_y$ has no associated immersed prime cycle. This is shown by the example of Remark (ii): one sees immediately that, with $\mathscr{F} = \mathscr{O}_X$, $l(y_0) = 1$ and $l(y) = 2$ for $y \neq y_0$ in Y.

(vi) We do not know if, in (12.2.1), (xi), one can replace the hypothesis that $\mathscr{F}$ is f-flat by the hypothesis that $\mathrm{Supp}(\mathscr{F}) = X$ and that f is universally open. By taking up the proof of (12.2.1.4), one sees (using also (14.3.6)) that one would have to solve the following question: let A be a local Noetherian ring, M an A-module of finite type, t an element of the maximal **IV-3** | 183
ideal $\mathfrak{m}$ of A, which is not contained in any minimal prime ideal of A; if there exists one of these minimal prime ideals $\mathfrak{p}$ such that $\dim(A/\mathfrak{p}) = 1$, is it true that $\mathfrak{m}$ is an associated prime ideal of M/tM (t not being supposed M-regular)?

One will note, however, that in (12.2.1), (xi) one cannot simply suppress the hypothesis that $\mathscr{F}$ is f-flat. Take here the P of Remark (ii), and in P let X_0 be the closed sub-prescheme which is the union of the three lines X_1, X_2, X_3 defined respectively by $t_1 = t_3 = 0$, $t_1 - t_0 = t_3 = 0$, and $t_2 = t_3 = 0$. Define the projection f_0 of X_0 onto X_1 as above, and its restriction $f : X \to Y$, where $Y = D_+(t_0)$ is the affine line and $X = f_0^{-1}(Y)$. The morphism f is proper but not flat, and, on taking $\mathscr{F} = \mathscr{O}_X$, the Module $\mathscr{F}_y$ has no associated immersed prime cycle for any $y \in Y$. But one has $l(y_0) = 1$ and $l(y) = 2$ for $y \neq y_0$ in Y.

Theorem (12.2.4). — *Let $f : X \to Y$ be a proper, flat morphism of finite presentation, and let k be an integer $\geqslant 1$. The following subsets of Y are open:*

(i) The set of $y \in Y$ such that X_y possesses the property (S_k).

(ii) The set of $y \in Y$ such that X_y satisfies the property (R_k) geometrically, is equidimensional at every point, and has no associated immersed prime cycle.

(iii) The set of $y \in Y$ such that X_y is geometrically regular (i.e. smooth over $k(y)$).

(iv) The set of $y \in Y$ such that X_y is geometrically normal.

(v) The set of $y \in Y$ such that X_y is geometrically reduced.

(vi) The set of $y \in Y$ such that X_y is geometrically reduced and that the number of geometric connected components of X_y is equal to k.

(vii) The set of $y \in Y$ such that X_y is geometrically punctually integral (4.6.9).

(viii) The set of $y \in Y$ such that X_y is geometrically integral.

(ix) The set of $y \in Y$ such that X_y has no associated immersed prime cycle and that the total multiplicity (4.7.4) of X_y over $k(y)$ is $\leqslant k$.

Proof. Except for (vi), these assertions are special cases of assertions of (12.2.1), or are deduced from assertions of (12.1.6) as at the start of the proof of (12.2.1). For (vi), one reduces as always (taking into account the invariance of the geometric number of connected components by base change (4.5.6)) to the case where Y is Noetherian. The set U of $y \in Y$ such that X_y is geometrically reduced is open in Y by virtue of (v); it follows then from (**III**, 7.8.7) and (**III**, 7.8.6) that for every $y \in U$, there is a neighbourhood $V \subset U$ of y and an integer m such that, for *every* $z \in V$, $\Gamma(X_z, \mathscr{O}_{X_z})$ is isomorphic to $(k(z))^m$; but by virtue of (**III**, 4.3.4), m is then the geometric number of connected components of X_z, whence the conclusion. We will give another proof of (vi) in (15.5.9). □

12.3. Local cohomological properties of fibres of a flat morphism, locally of finite presentation

Lemma (12.3.1). — *Let A be a ring, B an A-algebra of finite presentation which is a flat A-module, M a B-module of finite presentation, which is a flat A-module. Then the following conditions are equivalent:*

a) *M is a projective B-module.*

b) *M is a flat B-module.*

c) *For every $y \in \mathrm{Spec}(A)$, $M \otimes_A k(y)$ is a projective $(B \otimes_A k(y))$-module.*

Proof. **IV-3** | 184

The equivalence of *a)* and *b)* follows from Bourbaki, *Alg. comm.*, chap. II, § 5, n° 2, cor. 2 of th. 1. Since *a)* implies *c)* trivially, it remains to prove that *c)* implies *b)*, which follows from the criterion of flatness by fibres (11.3.10), applied with $g = h$, $f = 1_X$. □

Proposition (12.3.2). — *Let A be a ring, B an A-algebra of finite presentation which is a flat A-module, M a B-module of finite presentation which is a flat A-module. Then:*

(i) There exists a left resolution of M by B-modules free of finite type.
(ii) One has

(12.3.2.1) $$\dim.\mathrm{proj}_B(M) = \sup_{y\in\mathrm{Spec}(A)} \dim.\mathrm{proj}_{B\otimes_A k(y)}(M\otimes_A k(y)) = \mathrm{Tor.dim}_B(M).$$

Here $\mathrm{Tor.dim}_B(M)$ *is the smallest integer i such that $\mathrm{Tor}^B_j(M,N) = 0$ for every $j \geqslant i$ and every B-module N (and is $+\infty$ if no such integer i exists).*

Proof. (i) By (8.9.1), there exist a Noetherian subring A_0 of A, an A_0-algebra of finite type B_0, and a B_0-module of finite type M_0, such that $B \simeq B_0 \otimes_{A_0} A$ and $M \simeq M_0 \otimes_{A_0} A$. By (11.2.7), we may moreover suppose that B_0 and M_0 are flat A_0-modules. Let $(L_0)_\bullet$ be a left resolution of M_0 by free B_0-modules of finite type. Since M_0 and the $(L_0)_i$ are flat A_0-modules, $(L_0)_\bullet \otimes_{A_0} A$ is a left resolution of M by free B-modules of finite type (2.1.10).

(ii) By (**0**, 17.2.2), (ii), $\mathrm{Tor.dim}_B(M) \leqslant \dim.\mathrm{proj}_B(M)$; and the definition of projective dimension immediately gives

$$\dim.\mathrm{proj}_{B\otimes_A k(y)}(M\otimes_A k(y)) \leqslant \dim.\mathrm{proj}_B(M)$$

for every $y \in \mathrm{Spec}(A)$. To prove the reverse inequalities, consider a left resolution of M by B-modules free of finite type, and suppose that $\mathrm{Tor.dim}_B(M) = n$ (resp. $\dim.\mathrm{proj}_{B\otimes_A k(y)}(M\otimes_A k(y)) \leqslant n$ for every $y \in \mathrm{Spec}(A)$). Then $R = \mathrm{Im}(L_n \to L_{n-1}) = \mathrm{Ker}(L_{n-1} \to L_{n-2})$ is a B-module of finite type and a flat A-module, by the hypotheses on M and B and by (2.1.10). Moreover,

$$\mathrm{Tor}^B_{n+1}(M,N) = \mathrm{Tor}^B_1(R,N)$$

for every B-module N (M, V, 7). The hypothesis $\mathrm{Tor.dim}_B(M) = n$ therefore implies $\mathrm{Tor}^B_1(R,N) = 0$ for every B-module N; that is, R is a flat B-module, hence projective by (12.3.1). Thus $\dim.\mathrm{proj}_B(M) \leqslant n$. On the other hand, the hypothesis $\dim.\mathrm{proj}_{B\otimes_A k(y)}(M\otimes_A k(y)) \leqslant n$ for every $y \in \mathrm{Spec}(A)$ implies, upon tensoring with $k(y)$, that in each of the following sequences (which are exact by the flatness over A of M, the L_i, and R (2.1.10))

$$0 \longrightarrow R\otimes_A k(y) \longrightarrow L_{n-1}\otimes_A k(y) \longrightarrow \cdots \longrightarrow L_0\otimes_A k(y) \longrightarrow M\otimes_A k(y) \longrightarrow 0$$

the $(B\otimes_A k(y))$-module $R\otimes_A k(y)$ is projective, for every $y \in \mathrm{Spec}(A)$. It follows again from (12.3.1) that R is a projective B-module, and hence that $\dim.\mathrm{proj}_B(M) \leqslant n$.

□

Proposition (12.3.3). — *Let $f : X \to S$ be a morphism locally of finite presentation, and let $\mathscr{L}_\bullet$ be a complex of quasi-coherent $\mathscr{O}_X$-modules of finite presentation; for every $s \in S$, let $(\mathscr{L}_\bullet)_s$ denote the complex $\mathscr{L}_\bullet \otimes_{\mathscr{O}_S} k(s)$ of $\mathscr{O}_{X_s}$-modules of finite type. Suppose that $\mathscr{L}_{n-1}$ is f-flat. Then the set U of $x \in X$ such that $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = 0$ is open in X. If, in addition, $\mathscr{L}_n$ is f-flat, then $\mathscr{H}_n(\mathscr{L}_\bullet)|_U = 0$.*

Proof. It plainly suffices to consider the case $n = 1$ in which the complex $\mathscr{L}_\bullet$ reduces to $0 \to \mathscr{L}_2 \xrightarrow{u} \mathscr{L}_1 \xrightarrow{v} \mathscr{L}_0 \to 0$. We may first reduce to the case where $S = \mathrm{Spec}(A)$ and X are affine, and then to the case where S is *Noetherian*. Indeed, by (8.9.1) and (8.5.2), there exist a Noetherian prescheme $S' = \mathrm{Spec}(A')$, where A' is a subring of A, a morphism of finite type $f' : X' \to S'$, and three coherent $\mathscr{O}_{X'}$-modules $\mathscr{L}'_i$ $(0 \leqslant i \leqslant 2)$, such that $X = X' \times_{S'} S$, $f = f'_{(S)}$, and $\mathscr{L}_i = \mathscr{L}'_i \otimes_{S'} S$, together with two homomorphisms u', v' such that $u = u' \otimes 1$, $v = v' \otimes 1$, and $v' \circ u' = 0$. We may moreover suppose that $\mathscr{L}'_0$ is f'-flat (11.2.7), and that $\mathscr{L}'_1$ is f'-flat whenever $\mathscr{L}_1$ is supposed f-flat. By faithful flatness, the hypothesis $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = 0$ is equivalent to $(\mathscr{H}_n((\mathscr{L}'_\bullet)_{f'(x')}))_{x'} = 0$, where x' is the projection of x in X'. Thus, if U' is the set of $x' \in X'$ such that $(\mathscr{H}_n((\mathscr{L}'_\bullet)_{f'(x')}))_{x'} = 0$, then U is the inverse image of U' under the projection $X \to X'$; this reduces the first assertion to the case where S is Noetherian. **IV-3 | 185**

For the second assertion, note also that A is the filtered colimit of its subrings A_λ which are A'-algebras of finite type. Set $S_\lambda = \mathrm{Spec}(A_\lambda)$, $X_\lambda = X' \times_{S'} S_\lambda$, and $\mathscr{L}^{(\lambda)}_i = \mathscr{L}'_i \otimes_{S'} S_\lambda$; let $u_\lambda = u' \otimes 1 :$

$\mathscr{L}_2^{(\lambda)} \to \mathscr{L}_1^{(\lambda)}$ and $v_\lambda = v' \otimes 1 : \mathscr{L}_1^{(\lambda)} \to \mathscr{L}_0^{(\lambda)}$, so that $\mathscr{H}_1(\mathscr{L}_\bullet^{(\lambda)}) = \operatorname{Ker}(v_\lambda)/\operatorname{Im}(u_\lambda)$. Since filtered colimits are exact in the category of commutative groups,

$$\operatorname{Ker}(v) = \varinjlim_\lambda \operatorname{Ker}(v_\lambda), \qquad \operatorname{Im}(u) = \varinjlim_\lambda \operatorname{Im}(u_\lambda), \qquad \mathscr{H}_1(\mathscr{L}_\bullet) = \varinjlim_\lambda \mathscr{H}_1(\mathscr{L}_\bullet^{(\lambda)}).$$

If $\mathscr{L}_1$ has been supposed f-flat and, after reducing to the case $X = U$, one has proved that $\mathscr{H}_1(\mathscr{L}_\bullet^{(\lambda)}) = 0$ for every λ, it follows that $\mathscr{H}_1(\mathscr{L}_\bullet) = 0$.

I) Suppose henceforth that S is Noetherian. We know (without any flatness hypothesis on $\mathscr{L}_0$) that the set U is *constructible* in X (9.9.6). By ($\mathbf{0}_{\text{III}}$, 9.2.5), it remains to show that every *generization* x' in X of a point $x \in U$ also belongs to U. The method set out in (12.0.2) applies without change (taking into account that, for a prescheme Z over a field k and an extension k' of k, the projection $Z \otimes_k k' \to Z$ is faithfully flat). We may therefore suppose that S is the spectrum of a discrete valuation ring A, with uniformizer t, that x lies over the closed point of S, and that x' lies over the generic point of S. The hypothesis that $\mathscr{L}_0$ is f-flat implies that t is $\mathscr{L}_0$-regular ($\mathbf{0}_{\text{I}}$, 6.3.4). We are thus reduced to proving the following lemma, in which B, M, N, and P will be replaced respectively by $\mathscr{O}_x$, $(\mathscr{L}_2)_x$, $(\mathscr{L}_1)_x$, and $(\mathscr{L}_0)_x$.

Lemma (12.3.3.1). — *Let B be a local Noetherian ring, let M, N, and P be three B-modules, with N of finite type, and let $M \xrightarrow{u} N \xrightarrow{v} P$ be two homomorphisms such that $v \circ u = 0$. Let t be an element of the maximal ideal of B which is P-regular. If the sequence*

$$M/tM \xrightarrow{u \otimes 1_{B/tB}} N/tN \xrightarrow{v \otimes 1_{B/tB}} P/tP \tag{12.3.3.2}$$

is exact, then the sequence $M \xrightarrow{u} N \xrightarrow{v} P$ is exact.

First note that if M is replaced by its image Q in N and u by the injection $j : Q \to N$, the image of $j \otimes 1$ is the same as that of $u \otimes 1$; hence, without changing either the hypothesis or the conclusion, we may suppose that u is injective. On the other hand, if R is the image of v and $\rho : N \to R$ the canonical surjection, then t is plainly R-regular, $\rho \circ u = 0$, and the kernel of $\rho \otimes 1$ is contained in that of $v \otimes 1$. It is therefore equal to it if (12.3.3.2) is exact; since moreover $\operatorname{Ker}(\rho) = \operatorname{Ker}(v)$, we see that, in proving the lemma, we may also suppose that v is surjective. The lemma is then a consequence of the following.

Lemma (12.3.3.3). — *Let B be a ring, let M, N, and P be three B-modules, and let $u : M \to N$, $v : N \to P$ be two homomorphisms such that u is injective, v is surjective, and $v \circ u = 0$. Let t be an element of B which is P-regular. Then one has, up to canonical isomorphism,*

$$\operatorname{Ker}(v \otimes 1)/\operatorname{Im}(u \otimes 1) = (\operatorname{Ker}(v)/\operatorname{Im}(u)) \otimes_B (B/tB) \tag{12.3.3.4}$$

Indeed, the hypothesis that the sequence (12.3.3.2) is exact implies then that $(\operatorname{Ker} v/\operatorname{Im} u) \otimes_B (B/tB) = 0$, and since t belongs to the maximal ideal of B and $\operatorname{Ker}(v)$ is a B-module of finite type (since B is supposed Noetherian in (12.3.3.1)), the Nakayama lemma proves that $\operatorname{Ker}(v) = \operatorname{Im}(u)$.

It remains therefore to prove (12.3.3.3). Set $u' = u \otimes 1$, $v' = v \otimes 1$, $Z = \operatorname{Ker}(v)$, $H = Z/\operatorname{Im}(u)$, so that the sequences

$$0 \longrightarrow M \longrightarrow Z \longrightarrow H \longrightarrow 0$$

$$0 \longrightarrow Z \longrightarrow N \xrightarrow{v} P \longrightarrow 0$$

whence, by tensorising by B/tB and using the lemma (3.4.1.4) and the fact that t is P-regular, the exact sequences **IV-3 | 186**

$$M/tM \xrightarrow{w'} Z/tZ \longrightarrow H/tH \longrightarrow 0$$

$$0 \longrightarrow Z/tZ \longrightarrow N/tN \xrightarrow{v'} P/tP \longrightarrow 0$$

whence $\operatorname{Ker}(v') = Z/tZ$, and since $\operatorname{Im}(w') = \operatorname{Im}(u')$, one obtains (12.3.3.4).

II) Suppose now furthermore that $\mathscr{L}_1$ is f-flat; replacing in addition X by U, one can suppose that $X = U$. It is a question of seeing that for every $x \in X$, $(\mathscr{H}_1(\mathscr{L}_\bullet))_x = 0$. Set $s = f(x)$, and let $\mathfrak{n}$ be the ideal $\mathfrak{m}_s\mathscr{O}_{X,x}$, which is contained in the maximal ideal of $\mathscr{O}_{X,x}$. Since $(\mathscr{H}_1(\mathscr{L}_\bullet))_x$ is an $\mathscr{O}_{X,x}$-module of finite type (X being locally Noetherian), it is separated for the $\mathfrak{n}$-adic topology ($\mathbf{0}_{\mathrm{I}}$, 7.3.5), and it therefore suffices to show that its separated completion for this topology is 0. Now, by virtue of (**III**, 7.4.7.2), this separated completion is equal to $\varprojlim_k \mathscr{H}_1(\mathscr{L}_\bullet \otimes_{\mathscr{O}_S} (\mathscr{O}_{S,s}/\mathfrak{m}_s^{k+1}))$, and it will therefore suffice to prove that each term of this projective system is zero. This is true by hypothesis for $k = 0$; we therefore reason by induction on k. The conclusion will follow from the following more general lemma, which we apply with $A = \mathscr{O}_{S,s}/\mathfrak{m}_s^{k+1}$ and $\mathfrak{J}$ equal to the maximal ideal $\mathfrak{m}_s/\mathfrak{m}_s^{k+1}$ of A.

Lemma (12.3.3.5). — *Let A be a ring, let $\mathfrak{J}$ be a nilpotent ideal of A, and let $L_\bullet : P \xrightarrow{u} Q \xrightarrow{v} R$ be a complex of A-modules. Set $A_0 = A/\mathfrak{J}$ and $L_\bullet^{(0)} = L_\bullet \otimes_A A_0 : P_0 \xrightarrow{u_0} Q_0 \xrightarrow{v_0} R_0$. Suppose that $\mathrm{gr}^\bullet_{\mathfrak{J}}(A)$ is a flat A_0-module, and that Q and R are flat A-modules. Then the relation $\mathrm{H}_1(L_\bullet^{(0)}) = 0$ implies $\mathrm{H}_1(L_\bullet) = 0$.*

Set $L_\bullet^{(k)} = L_\bullet \otimes_A (A/\mathfrak{J}^{k+1}) : P_k \xrightarrow{u_k} Q_k \xrightarrow{v_k} R_k$. Since there exists an integer $k \geqslant 1$ such that $L_\bullet^{(k)} = L_\bullet$, it will suffice to prove, by induction on k, that the sequence $P_k \xrightarrow{u_k} Q_k \xrightarrow{v_k} R_k$ is exact **IV-3** | 187
(this assertion following from the hypothesis for $k = 0$). Let then $x \in Q_k$ be an element such that $v_k(x) = 0$. Since, by the induction hypothesis, the sequence $P_{k-1} \xrightarrow{u_{k-1}} Q_{k-1} \xrightarrow{v_{k-1}} R_{k-1}$ is exact, the canonical image of x in $Q_{k-1} = Q_k/(\mathfrak{J}^k Q/\mathfrak{J}^{k+1}Q)$ belongs to $\mathrm{Im}(u_{k-1})$. There consequently exists $z \in P_k$ such that $x = u_k(z) + x'$ with $x' \in \mathfrak{J}^k Q/\mathfrak{J}^{k+1}Q$. Since $v_k \circ u_k = 0$, the relation $v_k(x) = 0$ implies $v_k(x') = 0$; moreover one plainly has $v_k(x') \in \mathfrak{J}^k R/\mathfrak{J}^{k+1}R$. Everything is therefore reduced to proving that the sequence

$$\mathfrak{J}^k P/\mathfrak{J}^{k+1}P \xrightarrow{u'} \mathfrak{J}^k Q/\mathfrak{J}^{k+1}Q \xrightarrow{v'} \mathfrak{J}^k R/\mathfrak{J}^{k+1}R$$

where u' and v' are induced by u and v by restriction and passage to the quotients, is exact. Now, by hypothesis, $\mathfrak{J}^k/\mathfrak{J}^{k+1}$ is a flat $(A/\mathfrak{J})$-module, so the sequence

$$(P/\mathfrak{J}P) \otimes_{A/\mathfrak{J}} (\mathfrak{J}^k/\mathfrak{J}^{k+1}) \xrightarrow{u''} (Q/\mathfrak{J}Q) \otimes_{A/\mathfrak{J}} (\mathfrak{J}^k/\mathfrak{J}^{k+1}) \xrightarrow{v''} (R/\mathfrak{J}R) \otimes_{A/\mathfrak{J}} (\mathfrak{J}^k/\mathfrak{J}^{k+1})$$

is exact. But $(Q/\mathfrak{J}Q) \otimes_{A/\mathfrak{J}} (\mathfrak{J}^k/\mathfrak{J}^{k+1}) = Q \otimes_A (\mathfrak{J}^k/\mathfrak{J}^{k+1})$ identifies with $\mathfrak{J}^k Q/\mathfrak{J}^{k+1}Q$ by the flatness of Q over A; likewise $(R/\mathfrak{J}R) \otimes_{A/\mathfrak{J}} (\mathfrak{J}^k/\mathfrak{J}^{k+1})$ identifies with $\mathfrak{J}^k R/\mathfrak{J}^{k+1}R$. Finally, the image of u' identifies with that of u'', which completes the proof. □

Corollary (12.3.4). — *Let $f : X \to S$ be a morphism flat and locally of finite presentation, and let $\mathscr{F}$ and $\mathscr{G}$ be two $\mathscr{O}_X$-modules of finite presentation which are f-flat. Let n be an integer $\geqslant 0$, U the set of $x \in X$ such that $(\mathscr{E}xt^n_{\mathscr{O}_{X_{f(x)}}}(\mathscr{F}_{f(x)}, \mathscr{G}_{f(x)}))_x = 0$ (resp. $(\mathscr{T}or_n^{\mathscr{O}_{X_{f(x)}}}(\mathscr{F}_{f(x)}, \mathscr{G}_{f(x)}))_x = 0$). Then U is open, and one has $\mathscr{E}xt^n_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})|_U = 0$ (resp. $\mathscr{T}or_n^{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})|_U = 0$).*

Proof. We may plainly restrict to the case where S and X are affine. Then (12.3.2) gives a left resolution $\mathscr{L}_\bullet = (\mathscr{L}_i)$ of $\mathscr{F}$ formed by $\mathscr{O}_X$-modules *free of finite type*. If we set $\mathscr{M}^\bullet = \mathscr{H}om(\mathscr{L}_\bullet, \mathscr{G})$ (resp. $\mathscr{N}_\bullet = \mathscr{L}_\bullet \otimes_{\mathscr{O}_X} \mathscr{G}$), it follows that each $\mathscr{M}^i$ (resp. $\mathscr{N}_i$) is isomorphic to an $\mathscr{O}_X$-module of the form $\mathscr{G}^m$; hence the $\mathscr{M}^i$ and the $\mathscr{N}_i$ are $\mathscr{O}_X$-modules of finite presentation and are f-flat. Moreover, for every base change $S' \to S$, if we set $X' = X \times_S S'$, $\mathscr{F}' = \mathscr{F} \otimes_S S'$, $\mathscr{G}' = \mathscr{G} \otimes_S S'$, and $\mathscr{L}'_\bullet = \mathscr{L}_\bullet \otimes_S S'$, then $\mathscr{L}'_\bullet$ is still a left resolution of $\mathscr{F}'$ by $\mathscr{O}_{X'}$-modules free of finite type (2.1.10); furthermore, $\mathscr{M}'^\bullet = \mathscr{M}^\bullet \otimes_S S'$ is equal to $\mathscr{H}om(\mathscr{L}'_\bullet, \mathscr{G}')$ (resp. $\mathscr{N}'_\bullet = \mathscr{N}_\bullet \otimes_S S'$ is equal to $\mathscr{L}'_\bullet \otimes_{\mathscr{O}_{X'}} \mathscr{G}'$), by what precedes. In particular, for every $s \in S$, one has $\mathscr{E}xt^n_{\mathscr{O}_{X_s}}(\mathscr{F}_s, \mathscr{G}_s) = \mathscr{H}^n((\mathscr{M}^\bullet)_s)$ (resp. $\mathscr{T}or_n^{\mathscr{O}_{X_s}}(\mathscr{F}_s, \mathscr{G}_s) = \mathscr{H}_n((\mathscr{N}_\bullet)_s)$). Applying (12.3.3) to the f-flat complexes $\mathscr{M}^\bullet$ and $\mathscr{N}_\bullet$, one deduces immediately the corollary. □

§13. Equidimensional morphisms

This section is devoted to the study of the variation of the *dimension* of the fibres of a morphism **IV-3** | 187
locally of finite type $f : X \to Y$ (which has already intervened with respect to the "dimension formula" in (5.5) and (5.6)). We first prove (13.1.3) that the function $x \mapsto \dim_x f^{-1}(f(x))$ is always *upper semi-continuous* in X (Chevalley's semi-continuity theorem). We then study more particularly the morphisms, called "equidimensional", for which this function is *locally constant*. Unfortunately, **IV-3** | 188
the notion of equidimensional morphism is not stable under base change; this is why in many

questions it is more convenient to work with the notion of universally open morphism, the study of which is the subject of §§ 14 and 15.

13.1. Chevalley's semi-continuity theorem

Lemma (13.1.1). — *Let Y be an irreducible locally Noetherian prescheme, X an irreducible prescheme, $f : X \to Y$ a dominant morphism of finite type. Let ξ (resp. η) be the generic point of X (resp. Y) and let $e = \dim(f^{-1}(\eta))$. Then, for every $x \in X$, all the irreducible components of $f^{-1}(f(x))$ are of dimension $\geqslant e$.*

Proof. The assertion is immediate when, for every $y \in f(X)$, $\mathscr{O}_y$ is a *universally catenary* ring: indeed, if x is the generic point of an irreducible component Z of $f^{-1}(y)$, it follows from (5.6.5), combined with (5.2.1), that $e + \dim(\mathscr{O}_y) = \dim(Z) + \dim(\mathscr{O}_x)$; but by (**0**, 16.3.9), one has $\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y)$, whence the conclusion in this case.

We shall now reduce the general case to this particular case. The question is evidently local on Y and, taking into account (4.1.3), it is also local on X; we can therefore restrict to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine and irreducible, B being an A-algebra of finite type. Moreover, by (1.5.4), we may suppose X and Y reduced, so that A and B are integral domains. As f is dominant, A is then a *subring* of B. Consider A as an inductive limit of its sub-**Z**-algebras of finite type; it then follows from (8.9.1) that there exists such a subalgebra A_0 and an A_0-algebra of finite type B_0 such that $B = B_0 \otimes_{A_0} A$. Set $Y_0 = \operatorname{Spec}(A_0)$, $X_0 = \operatorname{Spec}(B_0)$, and let $f_0 : X_0 \to Y_0$ be the morphism corresponding to the homomorphism $A_0 \to B_0$, so that $f = f_0 \times 1_Y$. It is not *a priori* evident that the prescheme X_0 is integral, but we shall see that we can reduce to this case. Let η_0 be the generic point of Y_0, so that, if g denotes the morphism $Y \to Y_0$, one has $g(\eta) = \eta_0$; by transitivity of fibres (**I**, 3.6.4), one has $f^{-1}(\eta) = f_0^{-1}(\eta_0) \otimes_{k(\eta_0)} k(\eta)$, and, since $f^{-1}(\eta)$ is irreducible by hypothesis, so is $f_0^{-1}(\eta_0)$ (4.4.1). Our assertion will then follow from the following lemma:

Lemma (13.1.2). — *Let Y_0, Y be two integral preschemes with generic points η_0, η, let $g : Y \to Y_0$ be a dominant morphism, and let $f_0 : X_0 \to Y_0$ be a dominant morphism such that $f_0^{-1}(\eta_0)$ is irreducible. Let X'_0 be the unique irreducible component of X_0 meeting $f_0^{-1}(\eta_0)$ ($\mathbf{0}_{\mathrm{I}}$, 2.1.8), and also denote by X'_0 the closed reduced sub-prescheme of X_0 having X'_0 as underlying space. Suppose that the prescheme $X = X_0 \times_{Y_0} Y$ is integral; then X is isomorphic to $X'_0 \times_{Y_0} Y$.*

Indeed, if $j_0 : X'_0 \hookrightarrow X_0$ is the canonical injection, which is a closed immersion, $j = j_0 \times 1_Y : X'_0 \times_{Y_0} Y \to X_0 \times_{Y_0} Y = X$ is a closed immersion. On the other hand, X'_0 contains the fibre $f_0^{-1}(\eta_0)$, so $X'_0 \times_{Y_0} Y$ contains $f^{-1}(\eta)$; note also that $f^{-1}(\eta)$ is not empty (**I**, 3.4.7), so it contains the generic point of X; consequently, the image of j is necessarily all of X. But as X is integral, the only closed sub-prescheme of X having X as underlying space is X itself, so j is an isomorphism.

This lemma being established, we can therefore suppose that X_0 is integral; for every $y_0 \in Y_0$, $\mathscr{O}_{y_0}$ is a **Z**-algebra essentially of finite type, hence a universally catenary ring (5.6.4); consequently, **IV-3** | 189
for every $y_0 \in f_0(X_0)$, all the irreducible components of $f_0^{-1}(y_0)$[17] have dimension $\geqslant e$, since we know that $\dim(f_0^{-1}(\eta_0)) = e$ by transitivity of fibres (4.1.4). For every $y \in f(X)$, we then have $y_0 = g(y) \in f_0(X_0)$ and, by transitivity of fibres and (4.2.7), this completes the proof. □

Theorem (13.1.3). — (Chevalley). — *Let $f : X \to Y$ be a morphism locally of finite type. For every integer n, the set $\mathrm{F}_n(X)$ of $x \in X$ such that $\dim_x(f^{-1}(f(x))) \geqslant n$ is closed; in other words, the function $x \mapsto \dim_x(f^{-1}(f(x)))$ is upper semi-continuous in X.*

Proof. I) Suppose first that f is locally of finite presentation. The question is evidently local on X and on Y, and we can therefore suppose $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ affine, B being an A-algebra of finite presentation. We know then (8.9.1) that there exist a Noetherian subring A_0 of A and an A_0-algebra of finite type B_0 such that, on setting $Y_0 = \operatorname{Spec}(A_0)$ and $X_0 = \operatorname{Spec}(B_0)$, we have

$$X = X_0 \times_{Y_0} Y, \qquad f = f_0 \times 1_Y,$$

where $f_0 : X_0 \to Y_0$ is the morphism corresponding to the homomorphism $A_0 \to B_0$. Let $g : Y \to Y_0$ be the morphism corresponding to the canonical injection $A_0 \to A$, let y be a point of Y, and set

[17] The printed source has $f_0^{-1}(y)$ here; the quantified point and the argument require $f_0^{-1}(y_0)$.

$y_0 = g(y)$; we know that $f^{-1}(y) = f_0^{-1}(y_0) \otimes_{k(y_0)} k(y)$, and it follows from (4.2.7) that if p is the canonical projection $f^{-1}(y) \to f_0^{-1}(y_0)$, the irreducible components of $f^{-1}(y)$ are the irreducible components of the spaces $p^{-1}(Z_0)$, where Z_0 runs over the set of irreducible components of $f_0^{-1}(y_0)$, and each irreducible component of $p^{-1}(Z_0)$ dominates Z_0 and has dimension $\dim(Z_0)$. Taking into account (**0**, 14.1.5), we see therefore that if $g' : X \to X_0$ is the canonical projection, x a point of X and $x_0 = g'(x)$, we have $\dim_x(f^{-1}(f(x))) = \dim_{x_0}(f_0^{-1}(f_0(x_0)))$, whence $\mathrm{F}_n(X) = g'^{-1}(\mathrm{F}_n(X_0))$, and we are reduced to proving the theorem when Y is *Noetherian*, which we will henceforth suppose.

We can evidently suppose X and Y reduced (1.5.4). By considering the set of closed subsets Y' of Y such that the theorem is true for the closed sub-prescheme of Y having Y' as underlying space and for $X' = f^{-1}(Y')$, we can reason by Noetherian recurrence ($\mathbf{0}_{\mathrm{I}}$, 2.2.2), and suppose that the theorem is true for every closed subset $Y' \neq Y$ of Y. If X_i ($1 \leqslant i \leqslant m$) are the closed sub-preschemes with reduced underlying spaces that are the irreducible components of X, we have $\mathrm{F}_n(X) = \bigcup \mathrm{F}_n(X_i)$ by virtue of (**0**, 14.1.5), and we can therefore restrict to proving the theorem for each X_i, in other words, we can suppose X *irreducible*. If Z is the closed sub-prescheme of Y having $\overline{f(X)}$ as underlying space, f factorises as $X \xrightarrow{g} Z \xrightarrow{j} Y$, where j is the canonical injection (**I**, 5.2.2) and g is of finite type (1.5.4); it therefore suffices to prove the theorem for Z and g. By the recurrence hypothesis, we are thus reduced to the case $Z = Y$, in other words to the case in which Y is irreducible and f is dominant. Let then η be the generic point of Y and set $e = \dim(f^{-1}(\eta))$; it follows from (13.1.1) that for $n \leqslant e$ we have $\mathrm{F}_n(X) = X$, and consequently we can restrict to the case where $n > e$. But then (9.5.6), there is an open neighbourhood U of η in Y such that $\mathrm{F}_n(X) \subset f^{-1}(Y - U)$; as $Y - U \neq Y$, the hypothesis of recurrence implies that $\mathrm{F}_n(X)$ is closed.

II) We now pass to the general case, still writing $S = \mathrm{Spec}(A)$ for the target and supposing $X = \mathrm{Spec}(B)$ affine, with B an A-algebra of finite type, hence of the form $B = A[T_1, \ldots, T_n]/\mathfrak{J}$. Let **IV-3** | 190
$(\mathfrak{J}_\lambda)$ be the family of finitely generated ideals of $A[T_1, \ldots, T_n]$ contained in $\mathfrak{J}$, so that $\mathfrak{J}$ is the filtered union of the $\mathfrak{J}_\lambda$; if X and the $X_\lambda = \mathrm{Spec}(A[T_1, \ldots, T_n]/\mathfrak{J}_\lambda)$ are considered as closed sub-preschemes of $Z = \mathrm{Spec}(A[T_1, \ldots, T_n])$, we therefore have, for the underlying spaces, $X = \bigcap X_\lambda$. If $f_\lambda : X_\lambda \to S$ is the structural morphism, we deduce that $f^{-1}(s) = \bigcap f_\lambda^{-1}(s)$ for every $s \in S$; and since the sets $f_\lambda^{-1}(s)$ are closed in the Noetherian space Z_s, there exists an index λ (depending on s) such that $f^{-1}(s) = f_\lambda^{-1}(s)$. For every $x \in X$, set $d(x) = \dim_x(f^{-1}(f(x)))$ and $d_\lambda(x) = \dim_x(f_\lambda^{-1}(f_\lambda(x)))$. What precedes shows that $d(x) = \inf_\lambda d_\lambda(x)$; since the functions d_λ are upper semi-continuous by the first part of the proof, so is d. □

Corollary (13.1.4). — *Under the hypotheses of (13.1.3), the set of $x \in X$ such that x is isolated in $f^{-1}(f(x))$ is open in X.*

Proof. Indeed, it is the complement of $\mathrm{F}_1(X)$ (**0**, 14.1.10). □

We note that we thus recover, under more general hypotheses, the consequence (**III**, 4.4.10) of the "Main theorem" of Zariski.

Corollary (13.1.5). — *Under the hypotheses of (13.1.3), suppose in addition that f is a closed morphism. Then, for every integer n, the set of $y \in Y$ such that $\dim(f^{-1}(y)) \geqslant n$ is closed; in other words, the map $y \mapsto \dim(f^{-1}(y))$ is upper semi-continuous; in particular, if y is a specialisation of y', we have $\dim(f^{-1}(y)) \geqslant \dim(f^{-1}(y'))$.*

Proof. Indeed, to say that $\dim(f^{-1}(y)) \geqslant n$ means that $y \in f(\mathrm{F}_n(X))$ (**0**, 14.1.6). □

Corollary (13.1.6). — *Let X, Y be two irreducible preschemes, $f : X \to Y$ a dominant morphism locally of finite type. Let ξ (resp. η) be the generic point of X (resp. Y) and let $e = \dim(f^{-1}(\eta))$. Then, for every $x \in X$, we have $\dim_x(f^{-1}(f(x))) \geqslant e$.*

Proof. Indeed, the set of x such that $\dim_x(f^{-1}(f(x))) < e$ is open by (13.1.3), and as it cannot contain ξ, it is empty. □

Let us note finally the following easy result:

Proposition (13.1.7). — *Let Y be a quasi-compact prescheme, $f : X \to Y$ a morphism of finite type. There exists an integer n such that, for every $y \in Y$, we have $\dim(f^{-1}(y)) \leqslant n$.*

Proof. As there is a finite affine open covering (V_i) of Y such that each $f^{-1}(V_i)$ is a finite union of affine open sets, we are immediately reduced to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, B being an A-algebra of finite type. If B admits a system of n generators, then, for every $y \in Y$, $B \otimes_A k(y)$ is a $k(y)$-algebra admitting n generators, hence $\dim(f^{-1}(y)) \leqslant n$ by virtue of (4.1.1). □

13.2. Equidimensional morphisms: case of dominant morphisms of irreducible preschemes

(13.2.1). Let Y be an irreducible prescheme, X an irreducible prescheme, $f : X \to Y$ a *dominant* morphism locally of finite type; let η be the generic point of Y. We know (13.1.6) that for every $x \in X$, we have $\dim_x(f^{-1}(f(x))) \geqslant \dim(f^{-1}(\eta))$.

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Definition (13.2.2). — Under the hypotheses of (13.2.1), we say that f is *equidimensional at the point* x (or that X is *equidimensional over* Y *at the point* x) if

$$\dim_x(f^{-1}(f(x))) = \dim(f^{-1}(\eta)).$$

We say that f is *equidimensional* (or that X is *equidimensional over* Y) if f is equidimensional at every point $x \in X$.

It follows from Chevalley's theorem (13.1.3) that the set of $x \in X$ where f is equidimensional is *open* and non-empty. Furthermore, if f is equidimensional at the point x, all the irreducible components of $f^{-1}(f(x))$ containing x have the *same dimension*, since each of them has dimension $\geqslant \dim(f^{-1}(\eta))$ (13.1.6) and $\leqslant \dim_x(f^{-1}(f(x)))$ by virtue of (**0**, 14.1.5).

Proposition (13.2.3). — *Let Y be an irreducible locally Noetherian prescheme, X an irreducible prescheme, $f : X \to Y$ a dominant morphism locally of finite type; let η be the generic point of Y, x a point of X, $y = f(x)$, and suppose that we have*

(13.2.3.1) $$\dim(\mathcal{O}_x) = \dim(\mathcal{O}_y) + \dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)).$$

Then f is equidimensional at the point x. The converse holds if equality holds in (5.6.5.2)*, in particular if $\mathcal{O}_y$ is universally catenary.*

Proof. This follows immediately from (5.6.5.2) and from the inequality $\delta(x) \geqslant 0$ (13.1.1). □

(13.2.4). Now, more generally, let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism of finite type, x a point of X, and X' a closed irreducible subset of X containing x; set $y = f(x)$ and $Y' = \overline{f(X')}$, and let η' be the generic point of Y'. Let X' and Y' also denote the closed reduced sub-preschemes of X and Y respectively having X' and Y' as underlying spaces; the restriction $X' \to Y$ of f factorises as $X' \xrightarrow{f'} Y' \xrightarrow{j} Y$, where j is the canonical injection (**I**, 5.2.2), and f' is of finite type (1.5.4). Set, for brevity,

(13.2.4.1) $$A = \mathcal{O}_{Y,y}, \quad B = \mathcal{O}_{X,x}, \quad A' = \mathcal{O}_{Y',y}, \quad B' = \mathcal{O}_{X',x}.$$

The formula (5.6.5.2) applied to the dominant morphism f' and to the irreducible preschemes X', Y', gives

(13.2.4.2) $$\dim(B') \leqslant \dim(A') + \dim(B' \otimes_{A'} k(y)) - (\dim_x(f'^{-1}(y)) - \dim(f'^{-1}(\eta')))$$

On the other hand, the local ring A' (resp. B', $B' \otimes_{A'} k(y)$) is a quotient of A (resp. B, $B \otimes_A k(y)$), hence by (**0**, 16.1.2.1) we have

(13.2.4.3) $$\dim(A') \leqslant \dim(A), \quad \dim(B') \leqslant \dim(B), \quad \dim(B' \otimes_{A'} k(y)) \leqslant \dim(B \otimes_A k(y)).$$

We deduce therefore first from (13.2.4.3) and (**0**, 16.3.9)

(13.2.4.4) $$\dim(B') \leqslant \dim(A') + \dim(B' \otimes_{A'} k(y)) \leqslant \dim(A) + \dim(B \otimes_A k(y)).$$

Furthermore, by virtue of (**0**, 16.3.9), we also have the inequalities

(13.2.4.5) $$\dim(B') \leqslant \dim(B) \leqslant \dim(A) + \dim(B \otimes_A k(y)).$$

Comparing these inequalities therefore gives: **IV-3 | 192**

Lemma (13.2.5). — *With the notations of (13.2.4), the following conditions are equivalent:*

a) $\dim(B') = \dim(A) + \dim(B \otimes_A k(y))$.

b) *The following relations hold simultaneously:*

(i) $\dim(A') = \dim(A)$.

(ii) $\dim(B' \otimes_{A'} k(y)) = \dim(B \otimes_A k(y))$, *in other words*

$$\dim_x(X' \cap f^{-1}(y)) = \dim_x(f^{-1}(y)).$$

(iii) $\dim_x(f'^{-1}(y)) = \dim(f'^{-1}(\eta'))$, *in other words* $f' : X' \to Y'$ *is equidimensional (13.2.2) at the point* x.

(iv) We have the equality

$$\dim(B') = \dim(A') + \dim(B' \otimes_{A'} k(y)) - (\dim_x(f'^{-1}(y)) - \dim(f'^{-1}(\eta')))$$

(a relation which always holds when $A = \mathscr{O}_{Y,y}$ *is a universally catenary ring, by virtue of (5.6.5)).*

c) *The following relations hold simultaneously:*

(i) $\dim(B') = \dim(B)$.

(ii) $\dim(B) = \dim(A) + \dim(B \otimes_A k(y))$.

(13.2.6). Recall now that the irreducible components X_i of X containing x are finite in number and that, by (5.1.2.1) and (**0**, 14.2.1.1),

$$\dim(\mathscr{O}_{X,x}) = \sup_i \dim(\mathscr{O}_{X_i,x}). \tag{13.2.6.1}$$

The equivalence of conditions $b)$ and $c)$ in (13.2.5) therefore implies, taking into account (**0**, 16.3.9) and (**0**, 14.2.1):

Proposition (13.2.7). — *Let* Y *be a locally Noetherian prescheme,* $f : X \to Y$ *a morphism of finite type,* x *a point of* X*,* $f(x) = y$*; we have*

$$\dim(\mathscr{O}_{X,x}) \leqslant \dim(\mathscr{O}_{Y,y}) + \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_{Y,y}} k(y)). \tag{13.2.7.1}$$

For the two sides of (13.2.7.1) to be equal, it is necessary and sufficient that there exist an irreducible closed subset X' *of* X *containing* x *and satisfying simultaneously the following conditions:*

(i) If $Y' = \overline{f(X')}$*, we have* $\dim(\mathscr{O}_{Y',y}) = \dim(\mathscr{O}_{Y,y})$.

(ii) $\dim_x(X' \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$ *(in other words,* X' *contains one of the irreducible components of* $f^{-1}(y)$ *of maximal dimension among those containing* x*). This amounts to saying that* $\dim(\mathscr{O}_{X',x} \otimes_{\mathscr{O}_{Y',y}} k(y)) = \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_{Y,y}} k(y))$.

(iii) X' *is equidimensional over* Y' *at the point* x*, in other words, we have*

$$\dim_x(X' \cap f^{-1}(y)) = \dim(X' \cap f^{-1}(\eta')),$$

where η' *is the generic point of* Y' *(and consequently, all the irreducible components of* $X' \cap f^{-1}(y)$ *containing* x *have the same dimension (13.2.2)).* IV-3 | 193

(iv) We have the equality

$$\dim(\mathscr{O}_{X',x}) = \dim(\mathscr{O}_{Y',y}) + \dim(\mathscr{O}_{X',x} \otimes_{\mathscr{O}_{Y',y}} k(y))$$

(a condition always implied by (iii) when $\mathscr{O}_{Y,y}$ *is a universally catenary ring).*

Moreover, X' *is then an irreducible component of* X*, and* Y' *an irreducible component of* Y.

In the statement, the local rings $\mathscr{O}_{X',x}$ and $\mathscr{O}_{Y',y}$ are relative to the closed reduced sub-preschemes of X, Y respectively having X' and Y' as underlying spaces.

Moreover, this also proves:

Corollary (13.2.8). — *If the two sides of (13.2.7.1) are equal, the irreducible closed subsets* X' *of* X *containing* x *and satisfying conditions (i) to (iv) of (13.2.7) are exactly those for which* $\dim(\mathscr{O}_{X',x}) = \dim(\mathscr{O}_{X,x})$ *(which necessarily implies that* X' *is an irreducible component of* X*).*

Proposition (13.2.7) implies:

Corollary (13.2.9). — *Let* Y *be a locally Noetherian prescheme,* $f : X \to Y$ *a morphism of finite type,* x *a point of* X*,* $f(x) = y$*. Suppose that* $\mathscr{O}_{Y,y}$ *is a universally catenary ring. Then the following conditions are equivalent:*

a) *The two members of (13.2.7.1) are equal.*
b) *There exists an irreducible component Z of $f^{-1}(y)$ containing x, of dimension* $\dim_x(f^{-1}(y))$, *such that, for every x' in a neighbourhood of x in Z, we have*

$$\dim(\mathscr{O}_{X,x'}) = \dim(\mathscr{O}_{Y,y}) + \dim(\mathscr{O}_{X,x'} \otimes_{\mathscr{O}_{Y,y}} k(y)). \tag{13.2.9.1}$$

c) *There exists an irreducible component Z of $f^{-1}(y)$ containing x, of dimension* $\dim_x(f^{-1}(y))$, *such that for the generic point z of Z, we have*

$$\dim(\mathscr{O}_{X,z}) = \dim(\mathscr{O}_{Y,y}). \tag{13.2.9.2}$$

Let us show that $a)$ implies $b)$. Let $\dim_x(f^{-1}(y)) = n$; by virtue of (13.2.7), there exists an irreducible component X' of X satisfying conditions (i) to (iv) of (13.2.7); let Z be an irreducible component of dimension n of $f^{-1}(y)$, containing x and contained in X'. As $f^{-1}(y)$ is locally Noetherian, there exists an open neighbourhood U of x in Z such that U meets no irreducible component of $f^{-1}(y)$ other than those containing x, hence (4.1.1.3) $\dim_{x'}(f^{-1}(y)) = n$ for every $x' \in U$; it is then clear that the conditions (i) to (iii) of (13.2.7) are satisfied when we replace x by any point $x' \in U$, and it follows similarly for condition (iv) since $\mathscr{O}_{Y,y}$ is universally catenary; whence the conclusion by (13.2.7). The condition $b)$ implies trivially $c)$ by (5.1.2). Finally, if $c)$ is satisfied and if X'' is an irreducible component of X containing Z and such that the conditions (i) to (iv) of (13.2.7) are satisfied when we replace X' by X'' and x by z, it is clear that these conditions are also satisfied when we replace X' by X'' and x by x since $\mathscr{O}_{Y,y}$ is universally catenary, hence $c)$ implies $a)$.

Proposition (13.2.10). — *Let Y be an irreducible locally Noetherian prescheme, η its generic point, $f : X \to Y$ a morphism of finite type, y a point of $f(X)$. Let Z_i be the irreducible components of $f^{-1}(y)$, z_i the* **IV-3** | 194
generic point of Z_i, and consider the following conditions:

a) *For every $x \in f^{-1}(y)$, we have the relation*

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{Y,y}) + \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_{Y,y}} k(y)). \tag{13.2.10.1}$$

b) *For every i, we have*

$$\dim(\mathscr{O}_{X,z_i}) = \dim(\mathscr{O}_{Y,y}). \tag{13.2.10.2}$$

c) *For every i, there exists an irreducible component X_i of X containing Z_i and such that* $\dim(X_i \cap f^{-1}(\eta)) = \dim(Z_i)$ *(in other words, the* closed reduced sub-prescheme *X_i of X is equidimensional over Y at the point z_i).*

Then $a)$ implies $b)$ and $b)$ implies $c)$; furthermore, if $\mathscr{O}_{Y,y}$ is universally catenary, the three conditions $a)$, $b)$, $c)$ are equivalent.

The ring $\mathscr{O}_{X,z_i} \otimes_{\mathscr{O}_{Y,y}} k(y)$ being of dimension 0 (5.1.2), $a)$ implies $b)$ evidently; $b)$ implies $c)$ by virtue of (13.2.7) applied at the point z_i. Conversely, suppose that $\mathscr{O}_{Y,y}$ is universally catenary; as the condition $c)$ implies that $f(X_i)$ is dense in Y, it follows from $c)$ that the conditions (i) to (iv) of (13.2.7) are satisfied when we replace X' by X_i and x by z_i, hence $c)$ implies $b)$; finally $b)$ implies $a)$ by (13.2.9).

Corollary (13.2.11). — *With the notation of (13.2.10), suppose that $\mathscr{O}_{Y,y}$ is universally catenary. For every $y' \in Y$, let $E(y')$ be the set of dimensions of the irreducible components of $f^{-1}(y')$ and set $d(y') =$* $\dim(f^{-1}(y')) = \sup(E(y'))$. *Then, if the equivalent conditions $a)$, $b)$, $c)$ of (13.2.10) are satisfied, we have $E(y) \subset E(\eta)$, whence $d(y) \leqslant d(\eta)$.*

Indeed, with the notations of (13.2.10), $X_i \cap f^{-1}(\eta)$ is non-empty and is consequently an irreducible component of $f^{-1}(\eta)$ (**0**, 2.1.8).

Remarks (13.2.12). — (i) Recall from (6.1.2) that the relation (13.2.10.1) is always satisfied when the morphism f is *flat* at the point x.

(ii) With the notations of (13.2.10), suppose that $\mathscr{O}_{Y,y}$ is universally catenary and in addition that the morphism f is *proper*; then, if the equivalent conditions $a)$, $b)$, $c)$ of (13.2.10) are satisfied, we have $d(y) = d(\eta)$, since it follows from (13.1.5) that $d(y) \geqslant d(\eta)$.

(iii) The morphism of (12.2.3), (ii) is *proper* and *flat* and all the local rings of Y are universally catenary; furthermore, the two irreducible components X_1, X_2 of X are *equidimensional* over Y at every point; but $E(y)$ has *two* elements for every $y \neq y_0$, whereas $E(y_0)$ is reduced to a *single* element, hence $E(y)$ is *not constant* in Y.

13.3. Equidimensional morphisms: general case

Proposition (13.3.1). — *Let Y be a prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Let Y_α denote the irreducible components of Y containing y. Then the following conditions are equivalent:*

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a) *There exist an integer e and an open neighbourhood U of x such that the image by f of every irreducible component of U is dense in some Y_α, and such that, for every $x' \in U$, the space $U \cap f^{-1}(f(x'))$ is equidimensional and of dimension e.*

a′) *There exist an integer e and an open neighbourhood U of x such that the image by f of every irreducible component of U is dense in some Y_α, and such that, if we denote by y_α the generic point of Y_α, every irreducible component of the spaces $U \cap f^{-1}(y)$, $U \cap f^{-1}(y_\alpha)$ is of dimension e.*

a″) *There exist an integer e and an open neighbourhood U of x such that, for each of the irreducible components U_λ of U, $f(U_\lambda)$ is dense in some Y_α and such that, for every $x' \in U_\lambda$, the irreducible components of $U_\lambda \cap f^{-1}(f(x'))$ are all of dimension e.*

b) *There exist an integer e, an open neighbourhood U of x and a Y-morphism quasi-finite $g : U \to Y \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \ldots, T_e]$ (a prescheme we will also denote $Y[T_1, \ldots, T_e]$, to abbreviate) such that the image by g of every irreducible component of U is dense in an irreducible component of $Y[T_1, \ldots, T_e]$.*

Proof. It is immediate that a'') implies a), since for every $x \in U$, the irreducible components of $U \cap f^{-1}(f(x))$ are each an irreducible component of one of the $U_\lambda \cap f^{-1}(f(x))$, whence the conclusion by (**0**, 14.1.4). The condition a) trivially implies a''). Let us show that a') implies a''); we can restrict to the case where f is of finite type and X and Y are reduced (1.5.4). Let U_λ be an irreducible component of U, and suppose that $f(U_\lambda)$ is dense in Y_α; then the restriction of f to U_λ factorises as $U_\lambda \xrightarrow{f_\alpha} Y_\alpha \longrightarrow Y$, where f_α is of finite type and dominant (**I**, 5.2.2). Let x_λ be the generic point of U_λ; by virtue of (**0**, 2.1.8), $U_\lambda \cap f_\alpha^{-1}(y_\alpha)$ is the unique irreducible component of $U \cap f^{-1}(y_\alpha)$ containing x_λ and is by hypothesis of dimension e, equal to the dimensions of all the irreducible components of $U_\lambda \cap f^{-1}(y)$, by virtue of the hypothesis and of (13.1.1). But by Chevalley's theorem (13.1.3) and (13.1.1), the set V_λ of $x' \in U_\lambda$ such that $\dim_{x'}(f_\lambda^{-1}(f_\lambda(x'))) = e$ is open and contains x, and it suffices to take the union of the V_λ as an open set satisfying the conditions of a'').

Let us now show that a) implies b); we can restrict to the case where $Y = \operatorname{Spec}(A)$ and $U = X$. Let us first prove the following lemma:

Lemma (13.3.1.1). — *Let $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ be two affine schemes, $f : X \to Y$ a morphism of finite type, y a point of $f(X)$, and set $e = \dim(f^{-1}(y))$. Then there exists a Y-morphism $g : X \to Y[T_1, \ldots, T_e]$ such that (if we set $X_y = X \times_Y \operatorname{Spec}(k(y))$), the morphism $g_y : X_y \to \operatorname{Spec}(k(y)[T_1, \ldots, T_e])$ is finite. Furthermore, for any such morphism g, g_y is necessarily surjective, and there exists an open neighbourhood U of $f^{-1}(y)$ in X such that $g|U$ is quasi-finite.*

Proof. Let $\mathfrak{p} = \mathfrak{j}_y$; the ring $B \otimes_A k(\mathfrak{p})$ is a $k(\mathfrak{p})$-algebra of finite type, hence by the normalisation lemma (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, th. 1) there exist in $B \otimes_A k(\mathfrak{p})$ a finite sequence $(t_i)_{1 \leqslant i \leqslant r}$ of elements algebraically independent over $k(\mathfrak{p})$ and such that if we set $C' = k(\mathfrak{p})[t_1, \ldots, t_r]$, $B \otimes_A k(\mathfrak{p})$ is a C'-algebra *finite*; we therefore have $\dim(B \otimes_A k(\mathfrak{p})) = \dim(C')$ (**0**, 16.1.5) and as $\dim(C') = r$ (5.2.1), we have $r = e$. As $B \otimes_A k(\mathfrak{p}) = (B/\mathfrak{p}B) \otimes_{A/\mathfrak{p}} k(\mathfrak{p})$, we can, by multiplying the t_i by a suitable element of $A/\mathfrak{p}$, suppose that each t_i is the canonical image in $B \otimes_A k(\mathfrak{p})$ of an **IV-3 | 196**
element $s_i \in B$. Let then $u : A[T_1, \ldots, T_e] \to B$ be the homomorphism such that $u(T_i) = s_i$ for every i, and let $g : X \to Y[T_1, \ldots, T_e]$ be the corresponding morphism. It is clear that, by the choice of the s_i, g_y is a finite morphism. For every morphism $g : X \to Y[T_1, \ldots, T_e]$, such that g_y is finite, it follows from (5.4.2) and (4.1.2.1) that g_y is necessarily surjective. On the other hand, by (13.1.4), the set U of $x \in X$ that are isolated in their fibre $g^{-1}(g(x))$ is open in X and contains $f^{-1}(y)$, and by definition the restriction $g|U$ is a quasi-finite morphism. □

This lemma being established, to prove that *a*) implies *b*), it remains to show that if x_λ is a maximal point of U, then $g(x_\lambda) = z_\lambda$ is a maximal point of $Z = Y[T_1, \dots, T_e]$. By hypothesis *a*), we may (after shrinking U if necessary) suppose that $f(x_\lambda)$ is one of the generic points y_α of the irreducible components of Y containing y. If $p : Z \to Y$ is the structural morphism, then $p(z_\lambda) = y_\alpha$, and g consequently induces a quasi-finite $k(y_\alpha)$-morphism

$$h : U \cap f^{-1}(y_\alpha) \longrightarrow p^{-1}(y_\alpha).$$

Now $p^{-1}(y_\alpha) = \operatorname{Spec}(k(y_\alpha)[T_1, \dots, T_e])$ is integral and of dimension e. If z_λ were not a maximal point of Z, it would not be a maximal point of $p^{-1}(y_\alpha)$ (**0**, 2.1.8), and its closure Z'_λ in $p^{-1}(y_\alpha)$ would therefore have dimension $< e$ (4.1.2.1). But since h is quasi-finite, it follows from (4.1.2) and hypothesis *a*) that $\dim(Z'_\lambda) \geqslant e$ (the restriction of h to $\overline{\{x_\lambda\}}$ factorising as

$$\overline{\{x_\lambda\}} \longrightarrow Z'_\lambda \longrightarrow p^{-1}(y_\alpha)$$

by (**I**, 5.2.2)). This is a contradiction, and proves that *a*) implies *b*).

Let us finally prove that *b*) implies *a*). Note that the structural morphism $p : Z = Y[T_1, \dots, T_e] \to Y$ is faithfully flat; hence (2.3.4) the images under p of the maximal points of Z are the maximal points of Y. This already proves that the $f(x_\lambda)$ are generic points of the Y_α. Furthermore, if $f(x_\lambda) = y_\alpha$, the $k(y_\alpha)$-morphism $h : U \cap f^{-1}(y_\alpha) \to p^{-1}(y_\alpha)$ deduced from g is dominant and quasi-finite by hypothesis; hence (4.1.2) we have $\dim(U_\lambda \cap f^{-1}(y_\alpha)) = e$ (U_λ being the irreducible component of U with generic point x_λ). Similarly, for every $x' \in U_\lambda$, the morphism $U_\lambda \cap f^{-1}(f(x')) \to p^{-1}(f(x'))$ deduced from g is a $k(f(x'))$-morphism quasi-finite, hence (4.1.2) we have $\dim(f^{-1}(f(x')) \cap U_\lambda) \leqslant e$; but by (13.1.6) all the irreducible components of $U_\lambda \cap f^{-1}(f(x'))$ are of dimension $\geqslant e$; we see therefore that these components are of exactly dimension e, hence *b*) implies *a*). □

Definition (13.3.2). — Let Y be a prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X. We say that f is *equidimensional at the point* x (or that X is *equidimensional over* Y *at the point* x) if the equivalent conditions of (13.3.1) are satisfied. We say that f is *equidimensional* (or that X is *equidimensional over* Y) if f is equidimensional at every point $x \in X$.

It is clear that, for f to be equidimensional at a point x, it is necessary and sufficient that f_{red} be so. Furthermore, the conditions of (13.3.1) show that the set of points where f is equidimensional is *open* in X.

We note that when X and Y are *irreducible*, to say that f is equidimen- sional at the point x means, **IV-3** | 197
by virtue of (13.3.1), a''), that f is dominant and that $\dim_x(f^{-1}(f(x))) = \dim(f^{-1}(\eta))$ (where η is the generic point of Y); the definition (13.3.2) thus coincides in this case with the definition (13.2.2).

Proposition (13.3.3). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, X_j the irreducible components of X (in finite number) containing x. For f to be equidimensional at the point x, it is necessary and sufficient that, for every j, $Y_j = \overline{f(X_j)}$ is an irreducible component of Y and, denoting by X_j and Y_j the reduced closed sub-preschemes of X and Y with underlying spaces X_j and Y_j, by $f_j : X_j \to Y_j$ the morphism deduced from f* (**I**, 5.2.2), *by y_j the generic point of Y_j, that f_j is equidimensional at the point x and that all the numbers* $\dim(f_j^{-1}(y_j))$ *are equal.*

Proof. This follows immediately from (13.3.1), a')). □

Corollary (13.3.4). — *With the notations of* (13.3.3), *set $y = f(x)$. If $\mathscr{O}_{Y,y}$ is a universally catenary ring and if f is equidimensional at the point x, we have*

$$\dim(\mathscr{O}_{X_j,x}) = \dim(\mathscr{O}_{Y_j,y}) + e - \operatorname{deg.tr}_{k(y)} k(x) \tag{13.3.4.1}$$

where e is the common value of the numbers $\dim(f_j^{-1}(y_j))$.

Proof. As each of the $\mathscr{O}_{Y_j,y}$, a quotient of $\mathscr{O}_{Y,y}$, is a universally catenary ring (5.6.1), the equality is a consequence of (5.6.5). □

Corollary (13.3.5). — *With the notations of* (13.3.4), *suppose that f is equidimensional at the point x and that $\mathscr{O}_{Y,y}$ is a universally catenary ring. If the ring $\mathscr{O}_{Y,y}$ is equidimensional, then so is $\mathscr{O}_{X,x}$; the converse is true if the image under f of the union of the X_j is dense in a neighbourhood of y.*

Proof. Indeed, to say that $\mathscr{O}_{Y,y}$ is equidimensional means that the numbers $\dim(\mathscr{O}_{Y'_i,y})$ are equal for all the irreducible components Y'_i of Y containing y, as follows from (5.1.1.5) applied to the local scheme $\operatorname{Spec}(\mathscr{O}_{Y,y})$; since we have the relation (13.3.4.1), it follows by the same reasoning that $\mathscr{O}_{X,x}$ is then equidimensional. Conversely, if $\mathscr{O}_{X,x}$ is equidimensional, all the numbers $\dim(\mathscr{O}_{Y_j,y})$ are equal by (13.3.4.1); we deduce that $\mathscr{O}_{Y,y}$ is equidimensional if the Y_j are all the irreducible components of Y containing y, which follows from the supplementary hypothesis. □

Proposition (13.3.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that the ring $\mathscr{O}_y$ is equidimensional. Then, if $\mathscr{O}_x$ is equidimensional and if we have the equality*

(13.3.6.1) $$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y))$$

(cf. (13.2.7.1)) f is equidimensional at the point x, and the converse is true if $\mathscr{O}_y$ is a universally catenary ring.

Proof. Keeping the notations of (13.3.3), it follows from (13.2.8) and from the hypothesis that $\mathscr{O}_x$ is equidimensional that each Y_j is an irreducible component of Y and that each f_j is equidimensional at the point x; moreover, by (13.2.8), we have (taking into account (5.6.5))

$$\dim(\mathscr{O}_{X_j,x}) = \dim(\mathscr{O}_{Y_j,y}) + \dim_x(X_j \cap f^{-1}(y)) - \deg.\mathrm{tr}_{k(y)} k(x).$$

Now, since $\mathscr{O}_y$ is assumed equidimensional, this equality can be written **IV-3** | 198

(13.3.6.2) $$\dim_x(X_j \cap f^{-1}(y)) = \dim(\mathscr{O}_{X,x}) - \dim(\mathscr{O}_{Y,y}) + \deg.\mathrm{tr}_{k(y)} k(x).$$

The first member of (13.3.6.2) is therefore independent of j; but as f_j is equidimensional at the point x, we have $\dim_x(X_j \cap f^{-1}(y)) = \dim(f_j^{-1}(y_j))$, hence the criterion of (13.3.3) shows that f is equidimensional at the point x.

Conversely, suppose that $\mathscr{O}_{Y,y}$ is a universally catenary ring and that f is equidimensional at the point x; then (13.3.5) $\mathscr{O}_{X,x}$ is equidimensional, and it follows from (13.3.4) that we have the relation

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{Y,y}) + e - \deg.\mathrm{tr}_{k(y)} k(x)$$

where e is the common value of the numbers $\dim(f_j^{-1}(y_j)) = \dim_x(X_j \cap f^{-1}(y))$; but by definition e is also equal to $\dim_x(f^{-1}(y))$, whence the relation (13.3.6.1), taking into account (5.6.5.2). □

Proposition (13.3.7). — *Let Y be a prescheme, $f : X \to Y$ a morphism locally of finite type and equidimensional, Z a closed subset of X. Then the function*

(13.3.7.1) $$x \longmapsto \operatorname{codim}_x(Z \cap f^{-1}(f(x)), f^{-1}(f(x)))$$

is lower semi-continuous in X.

Proof. As all the irreducible components of $f^{-1}(f(x))$ containing x have the same dimension by hypothesis (13.3.1), we have, by virtue of (5.2.1),

$$\operatorname{codim}_x(Z \cap f^{-1}(f(x)), f^{-1}(f(x))) = \dim_x(f^{-1}(f(x))) - \dim_x(Z \cap f^{-1}(f(x))).$$

But by hypothesis the first term of the right-hand side is a continuous function of x (13.3.1) and the second is an upper semi-continuous function of x by (13.3.3); whence the conclusion. □

Proposition (13.3.8). — *Let Y be a prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X. Let Y' be a prescheme, $g : Y' \to Y$ a flat morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$; if f is equidimensional at the point x, then f' is equidimensional at every point $x' \in X'$ above x.*

Proof. The question being local on X and Y, we can restrict to the case where every irreducible component of X (resp. Y) contains x (resp. y); as the image by f of every irreducible component of X is dense in an irreducible component of Y (13.3.1), we know (2.3.5) that the image by f' of every irreducible component of X' is dense in an irreducible component of Y'. On the other hand, by transitivity of fibres (**I**, 3.6.4), it follows from (4.2.8) and (13.3.1) that, for every $z' \in X'$, the set of dimensions of the irreducible components of $f'^{-1}(f'(z'))$ is the same as the set of dimensions of the irreducible components of $f^{-1}(f(z))$, where z is the projection of z' onto X; whence the conclusion by virtue of (13.3.1), *a*)). □

Remarks (13.3.9). — The property of equidimensionality of a morphism $f : X \to Y$ is *not* stable by arbitrary base change $g : Y' \to Y$, even when g is the canonical injection of an irreducible component Y' of Y into Y (Y' being considered as a reduced closed sub-prescheme of Y). For **IV-3** | 199
example, let k be a field, let $A_0 = k[S, T]$ be the polynomial ring in two indeterminates, and set $\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2$, where $\mathfrak{p}_1$ and $\mathfrak{p}_2$ are the prime ideals A_0S and A_0T of A_0; let $A = A_0/\mathfrak{a}$ and $Y = \operatorname{Spec}(A)$, which has two irreducible components $Y_1 = \operatorname{Spec}(A/\mathfrak{p}_1)$ and $Y_2 = \operatorname{Spec}(A/\mathfrak{p}_2)$; take $X = Y_1$, with $f : X \to Y$ the canonical injection, which is evidently an equidimensional morphism. On the other hand, take $Y' = Y_2$, with $g : Y' \to Y$ the canonical injection; then $X' = X \times_Y Y' = \operatorname{Spec}\big((A/\mathfrak{p}_1) \otimes_A (A/\mathfrak{p}_2)\big) = \operatorname{Spec}\big(A/(\mathfrak{p}_1 + \mathfrak{p}_2)\big)$, and $\mathfrak{p}_1 + \mathfrak{p}_2$ is a maximal ideal of A, so X' is reduced to a point; the morphism $f' : X' \to Y'$ is *not dominant*, the image under f' of the unique point of X' being a closed point of Y'; consequently, f' is not equidimensional.

One can also give a counterexample in which X and Y are *integral*, f is *finite* and *birational* (and *a fortiori* equidimensional by (13.3.1), *b*)), and $g : Y' \to Y$ is *finite* and *dominant*. Let A and $\overline{A}$ be the local rings defined in (**II**, 7.5), and take $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(\overline{A})$; on the other hand, with the notation of (**II**, 7.5), take $Y' = \operatorname{Spec}(\overline{B})$; then $X' = \operatorname{Spec}(\overline{A} \otimes_A \overline{B}) = \operatorname{Spec}(\overline{B} \otimes_B \overline{B})$; but one verifies at once that $\overline{B} \otimes_B \overline{B}$ is the direct product of the rings $B/\mathfrak{p}'$, $B/\mathfrak{p}''$, and two rings isomorphic to $B/\mathfrak{n}$, whose spectra are therefore reduced to a point; as the projections of these points are closed points of Y', here again one sees that f' does not carry an irreducible component of X' onto a dense subset of an irreducible component of Y', and hence f' is not equidimensional.

In this example, the ring A is not geometrically unibranch; we shall see in the following paragraph (14.4.6) that such phenomena cannot occur when the points of Y are *geometrically unibranch*. The lack of stability of the notion of equidimensional morphism greatly restricts its interest, in favour of the notion of universally open morphism, which will be studied in detail in the following paragraph.

§14. Universally open morphisms

Sections §§ 14 and 15 are devoted to the study of the notion of *universally open* morphism (2.4.2). We have already seen (2.4.6) that a *flat* morphism locally of finite presentation is universally open, the converse being false. In § 14, we first examine the properties of the dimensions of the fibres of a universally open morphism $f : X \to Y$; when X and Y are locally Noetherian, f behaves in this regard (14.2.1) like a flat morphism (cf. (6.1.2)), and is in particular *equidimensional* when it is locally of finite type and dominant and X is irreducible (14.2.2). Conversely, an equidimensional morphism locally of finite type $f : X \to Y$ is universally open when one further assumes that Y is *geometrically unibranch*, and in particular when Y is *normal* (Chevalley's criterion, (14.4.4)). We also show that universally open morphisms locally of finite type $f : X \to Y$ (when X and Y are locally **IV-3** | 200
Noetherian and Y irreducible) admit "sufficiently many" *quasi-sections*, i.e. in the neighbourhood of a closed point x of a fibre $f^{-1}(y)$, there is a closed sub-prescheme X' of X containing x and such that the restriction $X' \to Y$ of f is a *quasi-finite* morphism (hence with *discrete* fibres) and dominant (14.5.3).

In § 15 we study various properties of the fibres of universally open morphisms $f : X \to Y$, locally of finite type, notably when X and Y are locally Noetherian. We thus obtain in particular a criterion for a point $x \in X$ to belong to a *single* irreducible component of X, in terms of properties of the *fibre* $f^{-1}(f(x))$ of x: it suffices that Y be geometrically unibranch (for example normal) at the point $f(x)$ and that $f^{-1}(f(x))$ be geometrically punctually integral at the point x (15.3.3); if moreover Y is locally integral at the point $f(x)$, then f is flat at the point x and X is locally integral at the point x. We also study the variation of the *number of geometric connected components* of a fibre $f^{-1}(y)$; for example, if f is universally open and *proper*, and the fibres of f geometrically reduced, this number is *locally constant* (15.5.7). Finally, when f admits a *section* g (which will be the case when X is a *group scheme over* Y) and, for every $y \in Y$, one denotes by X_y^0 the connected component of the fibre $X_y = f^{-1}(y)$ at the point $g(y)$ (the "neutral component" in the case of groups), one studies the union X^0 of the X_y^0 for $y \in Y$, and one shows (15.6.4) that if f is universally open and the fibres X_y are geometrically reduced, then X^0 is an *open* set in X.

14.1. Open morphisms

(14.1.1). Recall (1.10.2) that a continuous map $\psi : X \to Y$ is said to be *open at a point* $x \in X$ if the image by ψ of *every* neighbourhood of x in X is a neighbourhood of $\psi(x)$ in Y.

We note that this does *not* imply that there exists a fundamental system of neighbourhoods of x whose images are *open* in Y.

Proposition (14.1.2). — *Let X, Y be two topological spaces, $\psi : X \to Y$ a continuous map, x a point of X, $y = \psi(x)$.*

(i) If ψ is open at x, then for every subset Y' of Y containing $\psi(x)$ the restriction $\psi^{-1}(Y') \to Y'$ of ψ is open at the point x.

(ii) Suppose that Y is the union of a locally finite family of closed subsets (Y_i) and that for every i such that $\psi(x) \in Y_i$, the restriction $\psi^{-1}(Y_i) \to Y_i$ of ψ is open at the point x; then ψ is open at the point x.

(iii) Let $\gamma : X' \to X$ be a continuous map, x' a point of X'; if the composite map $\psi \circ \gamma : X' \to Y$ is open at the point x', ψ is open at the point $\gamma(x')$.

Proof. If U is a neighbourhood of x, we have $\psi(U \cap \psi^{-1}(Y')) = \psi(U) \cap Y'$; whence (i) follows immediately. To prove (ii), let us note that there is a neighbourhood W of $\psi(x)$ in Y meeting only finitely many of the closed subsets Y_i that *contain* $\psi(x)$; hence W is the union of the $W \cap Y_i$ for these indices. Now, if U is a neighbourhood of x such that $\psi(U) \cap Y_i$ is a neighbourhood of $\psi(x)$ in Y_i, there exists a neighbourhood $V \subset W$ of $\psi(x)$ in Y such that, for every i for which $\psi(x) \in Y_i$, we **IV-3** | 201
have $V \cap Y_i \subset \psi(U) \cap Y_i$ and since the union of the $V \cap Y_i$ for these indices is V, we have $V \subset \psi(U)$, hence $\psi(U)$ is a neighbourhood of $\psi(x)$ in Y. Assertion (iii) is trivial. □

Remarks (14.1.3). — (i) The set Z of points $x \in X$ where a morphism is open is *not necessarily open*. For example, let K be a field, A the polynomial ring $K[S, T]$, V the affine plane $\mathrm{Spec}(A)$, X the closed sub-prescheme of V which is the "union" of the line X_1 defined by $T = 0$ and of the line X_2 defined by $S = 0$, i.e. $\mathrm{Spec}(A/\mathfrak{a})$, where $\mathfrak{a} = AST$; let $Y = X_2 = \mathrm{Spec}(A/AS)$ and let f be the projection corresponding to the canonical injection $K[T] \to A/\mathfrak{a}$; then we have $Z = X_2$, which is not open in X.

(ii) Let X, Y be two Noetherian irreducible preschemes of generic points ξ, η respectively, $f : X \to Y$ a morphism *dominant locally of finite type*; then f *is open at the point* ξ. Indeed, we can evidently restrict to the case where X and Y are reduced (hence integral) (1.5.4) and affine; by virtue of the generic flatness theorem (6.9.1), there exists a non-empty open set V of Y such that the restriction $f^{-1}(V) \to V$ of f is a *flat* morphism, and we conclude from (2.4.6) that this restriction is an open morphism. However, there are dominant morphisms of finite type $f : X \to Y$ (where X and Y are *irreducible*) that are not open (see for example (**II**, 8.1)), the set of points where such a morphism is open is not necessarily *closed* in X.

(iii) We do not know whether or not, when X and Y are locally Noetherian and $f : X \to Y$ is a morphism locally of finite type, the set of points of X where f is open is *locally constructible*.

Proposition (14.1.4). — *Let X, Y be two topological spaces, $\psi : X \to Y$ a continuous map. For every $y \in Y$, the set of $x \in \psi^{-1}(y)$ where ψ is open is a closed subset of $\psi^{-1}(y)$.*

Proof. Indeed, suppose that ψ is not open at a point $x \in \psi^{-1}(y)$; there exists then a neighbourhood V of x in X such that $\psi(V)$ is not a neighbourhood of y; it follows that for every $x' \in V \cap \psi^{-1}(y)$, ψ is not open at the point x'. □

Remark (14.1.5). — Even if X and Y are locally Noetherian and f a *finite* morphism, it can happen that f is open at every point of a fibre $f^{-1}(y)$, without there existing a *neighbourhood* of $f^{-1}(y)$ at every point of which f is open. Take for example K a field, A the polynomial ring $K[\mathrm{T}_1, \mathrm{T}_2, \mathrm{T}_3]$, $V = \mathrm{Spec}(A)$ the affine 3-space over K, $X = \mathrm{Spec}(A/\mathfrak{a})$, where $\mathfrak{a} = \mathfrak{b}\mathfrak{c}$, with $\mathfrak{b} = (\mathrm{T}_3)$ and $\mathfrak{c} = (\mathrm{T}_1) + (\mathrm{T}_2 - \mathrm{T}_3)$ in A, so that X is the union of the plane $X_1 = \mathrm{Spec}(A/\mathfrak{b})$ ("plane with equation $\mathrm{T}_3 = 0$") and of the line $X_2 = \mathrm{Spec}(A/\mathfrak{c})$ ("line with equations $\mathrm{T}_1 = 0$, $\mathrm{T}_2 = \mathrm{T}_3$") which are its irreducible components. Take $Y = X_1$ and let $f : X \to Y$ be the projection corresponding to the canonical injection $K[\mathrm{T}_1, \mathrm{T}_2] \to A/\mathfrak{a}$; if y is the point common to X_1 and X_2, $f^{-1}(y)$ is reduced to y

and f is open at this point but is not open at any point of X_2 in a neighbourhood of y and distinct from y.

(14.1.6). In what follows, the essential role will be played by the criterion (1.10.3) characterising the morphisms *locally of finite presentation* $f : X \to Y$ that are open at a point x by the "lifting of generizations": for every generization y' of $y = f(x)$, there exists $x' \in X$, a generization of x, such that $y' = f(x')$.

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14.2. Open morphisms and the dimension formula

Theorem (14.2.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism, x a point of X, $y = f(x)$. Suppose that f is open at the generic points of the irreducible components of $f^{-1}(y)$ containing x. Then we have the relation*

(14.2.1.1)
$$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)).$$

Proof. Let y' be any generization of y distinct from y, and consider the closed reduced sub-prescheme Y' of Y having $\overline{\{y'\}}$ as underlying space; then no irreducible component X' of $f^{-1}(Y')$ containing x can be contained in $f^{-1}(y)$: indeed, if x' is the generic point of X', we would have $f(x') = y \neq y'$, and as x' is its only generization in $f^{-1}(Y')$, this would contradict the hypothesis that f is open at x', by virtue of the fact that *a*) implies *c*) in (1.10.3). We can therefore apply (6.1.2) to the local homomorphism of Noetherian local rings $\mathscr{O}_y \to \mathscr{O}_x$, whence the conclusion. □

Corollary (14.2.2). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose f is open at the generic points of the irreducible components of $f^{-1}(y)$ containing x. Then f is equidimensional at the point x in each of the two following cases:*

(i) X is irreducible and f is dominant.
(ii) The rings $\mathscr{O}_x$ and $\mathscr{O}_y$ are equidimensional.

Proof. For (i), this follows from (14.2.1) and from (13.2.3). For (ii), this follows from (14.2.1) and from (13.3.6). □

Proposition (14.2.3). — *Let Y be an irreducible locally Noetherian prescheme, η its generic point, $f : X \to Y$ a morphism locally of finite type, y a point of $f(X)$, Z an irreducible component of $f^{-1}(y)$ such that f is open at the generic point of Z. Then Z is contained in an irreducible component X' of X dominant over Y and such that* $\dim(Z) = \dim(X' \cap f^{-1}(y)) = \dim(X' \cap f^{-1}(\eta))$.

Proof. This follows indeed from (14.2.1) applied at the generic point of Z, and from (13.2.7). □

Corollary (14.2.4). — *With the notations of (14.2.3), suppose that f is open at all the points of $f^{-1}(y)$. For every $z \in Y$, let $\mathrm{E}(z)$ be the set of dimensions of the irreducible components of $f^{-1}(z)$ and* $d(z) = \sup \mathrm{E}(z) = \dim(f^{-1}(z))$. *Then $\mathrm{E}(y) \subset \mathrm{E}(\eta)$, whence $d(y) \leqslant d(\eta)$.*

Proof. This follows immediately from (14.2.3). □

Corollary (14.2.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a proper morphism, $y \in Y$ a point such that f is open at all the points of $f^{-1}(y)$. Then the function $z \mapsto$* $\dim(f^{-1}(z))$ *is constant in a neighbourhood of y.*

Proof. Let Y_i ($1 \leqslant i \leqslant n$) be the closed reduced sub-preschemes of Y having as underlying spaces the irreducible components of Y containing y; if $f_i : f^{-1}(Y_i) \to Y_i$ is the restriction of f, we know that each of the f_i is proper (**II**, 5.4.5) and open at all the points of $f^{-1}(y)$ (14.1.2). As the union of the Y_i is a neighbourhood of y in Y, we see that one can restrict to proving the corollary when Y is irreducible; let η be its generic point. It follows from (14.2.4) that $d(y) \leqslant d(\eta)$; on the other hand, as f is proper, we deduce from (13.1.5) that $z \mapsto \dim(f^{-1}(z))$ is upper semi-continuous; in particular $d(y) \geqslant d(\eta)$, hence $d(y) = d(\eta)$. Moreover, there is a neighbourhood V of y such that $d(z) \leqslant d(y)$ for every $z \in V$; but as every z is a specialisation of η, we have on the other hand $d(z) \geqslant d(\eta)$, whence finally $d(z) = d(y)$ for $z \in V$. □ IV-3 | 203

Corollary (14.2.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ an open morphism of finite type, y a point of Y. Let y_i ($1 \leqslant i \leqslant n$) be the generic points of the irreducible components of Y containing y. Then, with the notations of (14.2.4):*

(i) *If for $1 \leqslant i \leqslant n$, we have $\mathrm{E}(y) = \mathrm{E}(y_i)$ (resp. $d(y) = d(y_i)$), the function E (resp. d) is constant in a neighbourhood of y.*

(ii) *There exists an open neighbourhood U of $f^{-1}(y)$ such that the function $z \mapsto \dim(U \cap f^{-1}(z))$ is constant in a neighbourhood of y.*

Proof. (i) The same reasoning as in (14.2.5) shows that one can restrict to considering the case where Y is irreducible, of generic point η. If y' is any generization of y, we have then (applying (14.2.4) to the restriction $f^{-1}(Y') \to Y'$ of f, where $Y' = \overline{\{y'\}}$, and using (14.1.2)) $\mathrm{E}(y) \subset \mathrm{E}(y') \subset \mathrm{E}(\eta)$ and $d(y) \leqslant d(y') \leqslant d(\eta)$; this shows that the relation $\mathrm{E}(y) = \mathrm{E}(\eta)$ (resp. $d(y) = d(\eta)$) implies $\mathrm{E}(y) = \mathrm{E}(y')$ (resp. $d(y) = d(y')$) for every generization y' of y. Now, by virtue of (9.5.5), the functions E and d are locally constructible, hence it follows from ($\mathbf{0}_{\mathrm{III}}$, 9.2.5) applied to the set of z such that $\mathrm{E}(z) = \mathrm{E}(y)$ (resp. $d(z) = d(y)$) that this set is a neighbourhood of y.

(ii) For each of the y_i, $f^{-1}(y_i)$ is a Noetherian space since f is of finite type; let x_{ij} ($1 \leqslant j \leqslant m_i$) be the generic points of its irreducible components, and set $X'_{ij} = \overline{\{x_{ij}\}}$; let us show that the complement U in X of the union of the X'_{ij} that do not meet $f^{-1}(y)$ (which is evidently an open neighbourhood of $f^{-1}(y)$) answers the question. Indeed, the restriction $f|U : U \to Y$ of f is an open morphism of finite type (**I**, 6.3.5); for every pair (i,j) such that $x_{ij} \in U$, x_{ij} is a maximal point of $U \cap f^{-1}(y_i)$, and it follows from (13.1.1) applied to the restriction $X'_{ij} \to Y_i = \overline{\{y_i\}}$ of f (taking into account (**I**, 5.2.2) and (1.5.4)) that all the irreducible components of $X'_{ij} \cap f^{-1}(y)$ have dimension $\geqslant \dim(X'_{ij} \cap f^{-1}(y_i))$. Taking into account (14.2.3) applied to the restriction $f^{-1}(Y_i) \to Y_i$ of f, which is an open morphism (14.1.2), $f^{-1}(y)$ is, for each i, the union of the irreducible components of $X'_{ij} \cap f^{-1}(y)$ ($1 \leqslant j \leqslant m_i$), hence $\dim(f^{-1}(y)) \geqslant \dim(U \cap f^{-1}(y_i))$; but as $f|U$ is open, we also have $\dim(f^{-1}(y)) \leqslant \dim(f^{-1}(y_i) \cap U)$ in virtue of (14.2.4); we see therefore that one can apply (i) to $f|U$, whence the conclusion.

□

Remark (14.2.7). — The example (13.2.12), (iii)) shows that under the hypotheses of (14.2.5) or (14.2.6), (ii)), one cannot, in the conclusion, replace the function $z \mapsto \dim(f^{-1}(z))$ by the function $z \mapsto \mathrm{E}(z)$; indeed, in this example, f is proper and flat (hence universally open (2.4.6)) and every neighbourhood of the unique point of $f^{-1}(y_0)$ contains the generic points of the irreducible components of $f^{-1}(\eta)$.

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14.3. Universally open morphisms

(14.3.1). Recall (2.4.2) that to say that a morphism $f : X \to Y$ is universally open means that for every morphism $g : Y' \to Y$, $f_{(Y')} : X_{(Y')} \to Y'$ is open; it suffices moreover that this be so when $Y' = Y[\mathrm{T}_1, \dots, \mathrm{T}_e]$, for every integer e (8.10.2) (and if Y is locally Noetherian, it suffices therefore that it be so for every Y' locally Noetherian).

Proposition (14.3.2). — *Let $f : X \to Y$ be a morphism of preschemes. If f is universally open, then, for every morphism $g : Y' \to Y$, where Y' is irreducible, the image by $f' = f_{(Y')} : X_{(Y')} \to Y'$ of every irreducible component of $X_{(Y')}$ is dense in Y'. Conversely, if this condition is satisfied for every Y' irreducible and every morphism of finite type $g : Y' \to Y$, and if moreover f is locally of finite presentation, then f is universally open.*

Proof. Indeed, it follows from (1.10.4) that if f is universally open, it satisfies the condition of the statement. Conversely, suppose that f is locally of finite presentation, and let us show that for every integer e, if one sets $Y'' = Y[\mathrm{T}_1, \dots, \mathrm{T}_e]$, $f_{(Y'')}$ is an open morphism. Indeed, let Y' be a closed sub-prescheme of Y'' having as underlying space a closed irreducible subset of Y''; the composite morphism $Y' \to Y'' \to Y$ is of finite type, hence every irreducible component of $X_{(Y')}$ dominates Y' by hypothesis; we deduce therefore from (1.10.4) that $f_{(Y'')}$ is an open morphism. □

This proposition shows that the definition (**III**, 4.3.9) coincides in the case considered with the general definition of universally open morphisms given in (2.4.2).

The notion of an open morphism at a point (14.1.1) gives rise in the same way to the following:

Definition (14.3.3). — Let $f : X \to Y$ be a morphism of preschemes, x a point of X. We say that f is *universally open at the point* x if, for every morphism $g : Y' \to Y$, setting $X' = X \times_Y Y'$, the morphism $f' = f_{(Y')} : X' \to Y'$ is open at every point x' of X' whose projection in X is x.

Remarks (14.3.3.1). — (i) The reasoning of (8.10.1) shows (with the same notations) that if x is a point of X, x_λ its projection in X_λ, then, if f_μ is open at the point x_μ for every $\mu \geqslant \lambda$, f is open at the point x; it suffices to restrict to opens U of X containing x and to remark that the hypothesis implies that $f_\mu(U_\mu)$ is a neighbourhood of $f_\mu(x_\mu)$, hence $f(v_\mu^{-1}(U_\mu))$ is a neighbourhood of $f(x)$. We deduce that the statement (8.10.2) is still exact when one replaces "universally open" by "universally open at the point x", and "open morphism" by "morphism open at every point x_n of X_n whose projection in X is x": it suffices in the proof to limit to opens V containing x_n.

(ii) The result of (14.1.4) is again valid for a morphism $f : X \to Y$, replacing "open" by "universally open". Indeed, suppose that f is not universally open at a point $x \in f^{-1}(y)$; there is then a morphism $Y' \to Y$ and a point $x' \in X'$ projecting to x, such that f' is not open at the point x'.

Now, if $y' = f'(x')$, the projection $f'^{-1}(y') \to f^{-1}(y)$ is an open morphism (2.4.10), and there is, by virtue of (14.1.4), a neighbourhood V' of x' in $f'^{-1}(y')$ where f' is not open; hence f is not universally open at any point of the image of V' in $f^{-1}(y)$, which is a neighbourhood of x in $f^{-1}(y)$. **IV-3** | 205

Proposition (14.3.4). — *(i) Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, x a point of X, $y = f(x)$. If f is universally open at the point x and g universally open at the point y, then $g \circ f$ is universally open at the point x. Conversely, if $g \circ f$ is universally open at the point x, g is universally open at the point y.*

(ii) If $f : X \to Y$ is an S-morphism universally open at the point $x \in X$, then, for every base change $S' \to S$, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is universally open at every point of $X_{(S')}$ above x.

(iii) For $f : X \to Y$ to be universally open at the point $x \in X$, it is necessary and sufficient that f_{red} be so.

(iv) Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$; set $Y_1 = \operatorname{Spec}(\mathscr{O}_y)$, $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$. For f to be universally open at the point x, it is necessary and sufficient that f_1 be so (recall (**I**, 3.6.5) *that X_1 is canonically identified with a subspace of X).*

Proof. Indeed (ii) is an evident consequence of the definition (14.3.3); it also follows from the definition that in order to prove assertion (i), it suffices to do so dropping everywhere the word "universally", and this follows then from (14.1.2). Assertion (iii) follows from (ii) and from the fact that the canonical morphism $X_{\mathrm{red}} \to X$ is surjective. Finally, condition (iv) is trivially necessary. On the other hand, if it is satisfied, and if $g : Y' \to Y$ is any morphism, $X' = X_{(Y')}$, $f' = f_{(Y')}$, x' a point of X' above x, then f' is locally of finite presentation, and to see that it is open at the point x', it suffices to apply the criterion (1.10.3, *b*)). Let $y' = f'(x')$, $Y'_1 = \operatorname{Spec}(\mathscr{O}_{y'})$, $X'_1 = X' \times_{Y'} Y'_1$; since $g(y') = y$, the composite morphism $Y'_1 \to Y' \xrightarrow{g} Y$ factorises as $Y'_1 \to Y_1 \to Y$ (**I**, 2.4.4), hence $X'_1 = X_1 \times_{Y_1} Y'_1$ and $f'_1 = (f_1)_{(Y'_1)}$; the conclusion follows then immediately from the hypothesis that f'_1 is open at the point x' and from (1.10.3). □

Proposition (14.3.5). — *Let X, Y be two preschemes, $f : X \to Y$ a morphism, x a point of X. Let $(Y_i)_{1 \leqslant i \leqslant n}$ be a locally finite family of closed sub-preschemes of Y such that the space Y is the union of the Y_i, and suppose that for every i such that $f(x) \in Y_i$, the restriction $f^{-1}(Y_i) \to Y_i$ of f is universally open at the point x; then f is universally open at the point x.*

Proof. Taking into account the definition, this follows from the analogous proposition (14.1.2, (ii)) for open morphisms at a point. □

Proposition (14.3.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type. For f to be universally open at a point $x \in X$, it is necessary and sufficient that the following condition be satisfied: for every morphism $g : Y' \to Y$, where $Y' = \operatorname{Spec}(A)$ is the spectrum of a discrete valuation ring, such that the image $g(y')$ of the closed point y' of Y' is equal to $y = f(x)$, and for every point $x' \in X' = X \times_Y Y'$ whose projections on X and Y' are x and y' respectively, there exists a generization z' of x' in X' whose projection in Y' is the generic point of Y' (in other words,* there is an irreducible component of X' containing x' and dominating Y'*).*

Moreover, in the preceding condition, one may restrict to the case where A is complete, its residue field is algebraically closed, and x' is rational over $k(y')$. **IV-3** | 206

Proof. If Y' is as in the statement, the necessity of the condition follows from the fact that f' must be open at the point x', and the criterion (1.10.3). To see that the condition is sufficient, consider a morphism of finite type $g : Y'' \to U$, and let $X'' = X \times_Y Y''$, $f'' = f_{(Y'')} : X'' \to Y''$, and x'' a point of X'' above x. Set $y'' = f''(x'')$, and let t be a generization of y'' in Y'', distinct from y''. Since Y'' is locally Noetherian, it follows from (**II**, 7.1.9) and ($\mathbf{0}_{\text{III}}$, 10.3.1) that there exists a scheme $Y' = \operatorname{Spec}(A)$, where A is a complete discrete valuation ring whose residue field is an algebraic closure of $k(x'')$, and a morphism $g : Y' \to Y''$ such that, if s and y' are the generic point and the closed point respectively, we have $g(s) = t$ and $g(y') = y''$. There is then a point x' of $X' = X \times_Y Y' = X'' \times_{Y''} Y'$ whose projections in X'' and Y' are x'' and y' respectively, and which is rational over $k(y')$ (**I**, 3.4.9). The hypothesis implies that there is a generization z' of x' in X' whose projection in Y' is s; if z'' is the projection of z' in X'', z'' is a generization of x'' and its projection in Y'' is t; we conclude therefore from (1.10.3) that f'' is open at the point x'', hence f is universally open at the point x (14.3.3.1, (i)). ☐

Corollary (14.3.7). — *The notations being those of (14.3.6):*

(i) *Given a point $y \in Y$, for f to be universally open at all the points of $f^{-1}(y)$, it is necessary and sufficient that for every morphism $g : Y' \to Y$, where Y' is the spectrum of a discrete valuation ring and where the image $g(y')$ of the closed point y' of Y' is y, every irreducible component of $X' = X \times_Y Y'$ dominates Y'.*

(ii) *For f to be universally open, it is necessary and sufficient that for every morphism $g : Y' \to Y$, where Y' is the spectrum of a discrete valuation ring, every irreducible component of $X' = X \times_Y Y'$ dominates Y'.*

Proof. It is clear that it suffices to prove (i); the necessity of (i) follows from (14.3.2) and its sufficiency from (14.3.6). ☐

Proposition (14.3.8). — *Let Y be a locally Noetherian prescheme, irreducible, regular, and of dimension* 1 *(for example the spectrum of a Dedekind ring), $f : X \to Y$ a morphism locally of finite type, y a point of Y. The following conditions are equivalent:*

a) *f_{red} is flat at every point of $f^{-1}(y)$.*
b) *f is universally open in a neighbourhood of $f^{-1}(y)$.*
c) *f is open in a neighbourhood of $f^{-1}(y)$.*
d) *Every irreducible component of X meeting $f^{-1}(y)$ dominates Y.*

Proof. As f_{red} is locally of finite type (1.3.4), a) implies that f_{red} is flat in a neighbourhood of $f^{-1}(y)$ (11.1.1), and it suffices to apply (2.4.6) in such a neighbourhood to see that a) implies b). The implication $b) \Rightarrow c)$ is trivial, and the implication $c) \Rightarrow d)$ follows from (1.10.4) applied to a neighbourhood of $f^{-1}(y)$. It remains to see that d) implies a). We can evidently, by virtue of (1.3.4), restrict to the case where X is reduced. The question being moreover local on X and on Y, we can suppose $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ affine; if X_i ($1 \leqslant i \leqslant n$) are the closed integral sub-preschemes of X whose underlying spaces are the irreducible components of X, then, for every $x \in X$, $\mathcal{O}_{X,x}$ is a subring of the direct product of the $\mathcal{O}_{X_i,x}$; if $y = f(x)$, it suffices to show that each $\mathcal{O}_{X_i,x}$ is an $\mathcal{O}_y$-module *without torsion*, for the same will then be true of $\mathcal{O}_{X,x}$; since by hypothesis $\mathcal{O}_y$ is a regular local ring of dimension 1, i.e. a discrete valuation ring (**II**, 7.1.6), it then follows from ($\mathbf{0}_1$, 6.3.4) that $\mathcal{O}_{X,x}$ is a *flat* $\mathcal{O}_y$-module. But if $X_i = \operatorname{Spec}(B_i)$, where B_i is an integral ring, the hypothesis implies that the homomorphism $A \to B_i$ is *injective* (**I**, 1.2.7); hence B_i is an A-module without torsion, and *a fortiori* $\mathcal{O}_{X_i,x}$ is an $\mathcal{O}_y$-module without torsion. ☐ **IV-3** | 207

Remarks (14.3.9). — (i) In the statement of (14.3.8), one cannot dispense with the hypothesis that Y is *regular*. With the notations of (11.7.5), take indeed $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(\overline{A})$, so that $f : X \to Y$ is a *finite surjective* morphism; since A is an integral local ring of dimension 1, as is $\overline{A}$, it follows immediately from (1.10.4) that the morphism f is *open*. However f is not universally open (nor *a fortiori* flat) (11.7.5). One could give an analogous example by taking for Y the local scheme at an "ordinary double point" of an algebraic curve, and for X the normalization of Y.

(ii) The example of (14.1.3, (i)) shows that the set of points of X at which a morphism is universally open is not necessarily open, since the morphism f in that example is universally open at every point at which it is open. The example $f : X \to Y$ considered above in (i) likewise shows that the set of points at which a morphism is universally open is not necessarily closed, since it is immediate that f is universally open at all but one point of X (indeed, it is even a local isomorphism there). It would be interesting to know if the set of points where a morphism is universally open is locally constructible.

The following two propositions were pointed out to us by M. Artin:

Lemma (14.3.10). — *Let A be a valuation ring (not necessarily discrete), k its residue field, K its field of fractions, and set $S = \operatorname{Spec}(A)$. Let X be an irreducible prescheme, $f : X \to S$ a dominant morphism of finite type; let $X_0 = X \otimes_A k$ (resp. $X_1 = X \otimes_A K$) be the fibre of f at the closed point (resp. at the generic point) of S. Then, if $X_0 \neq \varnothing$, we have $\dim(X_0) = \dim(X_1)$.*

Proof. We can restrict to the case where X is affine, replacing if necessary X by an open affine containing a generic point of an irreducible component of X_0 of maximal dimension. Let $n = \dim(X_0) \geqslant 0$; using (4.1.1.3), it follows from (13.3.1.1) that there exists a neighbourhood U of X_0 in X and a quasi-finite S-morphism $g : U \to S[\mathrm{T}_1, \dots, \mathrm{T}_n] = Z$ such that the restriction $g_0 : U_0 = U \cap X_0 \to Z_0 = \operatorname{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n])$ is finite and *surjective*. By base change $\operatorname{Spec}(K) \to S$ and restriction to the open subset $U_1 = U \cap X_1$ of X_1, we deduce from g a quasi-finite morphism $g_1 : U_1 \to Z_1 = \operatorname{Spec}(K[\mathrm{T}_1, \dots, \mathrm{T}_n])$. As U_1 is dense in X_1 (4.1.1.3), we have $\dim(U_1) = \dim(X_1)$; if we can prove that the morphism g_1 is *dominant*, the proposition will be established, since then $\dim(U_1) \leqslant \dim(Z_1) = n$ and on the other hand $\dim(X_1) \geqslant n$ by (4.1.2). Suppose the contrary; there would then be a polynomial $\mathrm{F}_1 \neq 0$ in $K[\mathrm{T}_1, \dots, \mathrm{T}_n]$ such that $g_1(U_1) \subset \mathrm{V}(\mathrm{F}_1)$. If ω is a valuation on K associated to A, and if (a_α) is the family of coefficients of F_1, after multiplication of F_1 by a nonzero element of K, we can suppose that $\inf_\alpha(\omega(a_\alpha)) = 0$; in other words, F_1 comes from a polynomial $\mathrm{F} \in A[\mathrm{T}_1, \dots, \mathrm{T}_n]$ (with which it is identified) such that the image F_0 of F in $k[\mathrm{T}_1, \dots, \mathrm{T}_n]$ is *nonzero*. Consider then the closed subset $\mathrm{V}(\mathrm{F})$ of Z; we have $\mathrm{V}(\mathrm{F}) \cap Z_1 = \mathrm{V}(\mathrm{F}_1)$, and as U_1 is dense in U (since it contains the generic point of X), $g(U) \subset \mathrm{V}(\mathrm{F})$; in particular, **IV-3** | 208
$g_0(U_0) \subset \mathrm{V}(\mathrm{F}) \cap Z_0 = \mathrm{V}(\mathrm{F}_0)$; but since $\mathrm{F}_0 \neq 0$, $\mathrm{V}(\mathrm{F}_0)$ is a closed subset of Z_0 distinct from Z_0, and one arrives at a contradiction. □

Proposition (14.3.11). — *Let $f : X \to S$ be a morphism of finite type, $(g_\lambda)_{\lambda \in L}$ a family of universally open morphisms $g_\lambda : Y_\lambda \to S$, and for every $\lambda \in L$, let $u_\lambda : Y_\lambda \to X$ be an S-morphism. For every $s \in S$, set $X_s = X \times_S \operatorname{Spec}(k(s))$, $(Y_\lambda)_s = Y_\lambda \times_S \operatorname{Spec}(k(s))$, and let $(u_\lambda)_s : (Y_\lambda)_s \to X_s$ be the morphism $u_\lambda \times 1$ induced from u_λ by base change. Let $Z(s)$ be the closure in X_s of the union of the sets $(u_\lambda)_s((Y_\lambda)_s)$, and set $d(s) = \dim(Z(s))$. Then, for every generization s' of a point $s \in S$, we have $d(s) \leqslant d(s')$.*

Proof. We know from (**II**, 7.1.4) that there exists a valuation ring A' and a morphism $h : S' = \operatorname{Spec}(A') \to S$ such that, if a (resp. b) is the closed point (resp. the generic point) of S', we have $h(a) = s$, $h(b) = s'$. Furthermore, the projection $p : X_s \otimes_{k(s)} k(a) \to X_s$ is surjective and open (2.4.10), and thus makes X_s a quotient space of $X_s \otimes_{k(s)} k(a)$ by an open equivalence relation; consequently, for every subset M of X_s, $p^{-1}(\overline{M}) = \overline{p^{-1}(M)}$ (Bourbaki, *General Topology*, Chap. I, 4th ed., §5, no. 3, Prop. 7). One reasons in the same way for $X_{s'}$; taking account of (**I**, 3.4.8), (4.2.7), and the fact that the g_λ are universally open, we see that it suffices to prove the proposition after the base change $S' \to S$. Suppose therefore $S' = S$, s being the closed point and s' the generic point of S. The hypothesis that g_λ is open implies that every irreducible component of Y_λ dominates S (1.10.4), hence its generic point is a maximal point of $(Y_\lambda)_{s'}$; if Z denotes the closure in X of $Z(s')$, we have therefore $u_\lambda(Y_\lambda) \subset Z$, and consequently $(u_\lambda)_s((Y_\lambda)_s) \subset Z_s = Z \cap X_s$; in other words,

$Z(s) \subset Z_s$, whence $\dim(Z(s)) \leqslant \dim(Z_s)$. But applying (14.3.10) to a reduced sub-prescheme of X having as underlying space an irreducible component of Z, we obtain $\dim(Z_s) \leqslant \dim(Z_{s'})$ (the inequality coming from the fact that there can be irreducible components of Z not meeting X_s). On the other hand, since $Z(s')$ is closed in $X_{s'}$ by definition, we have $Z_{s'} = Z(s')$, which completes the proof. □

Remark (14.3.12). — The case considered by M. Artin was the case where $Y_\lambda = S$ for every λ, in other words the case where (u_λ) is a family of S-*sections* of X. Another useful case is that where the family (u_λ) is reduced to a single element; one can then always reduce to this case by considering the prescheme Y which is the sum of the Y_λ and the morphisms $g : Y \to S$ and $u : Y \to X$ whose restrictions to each Y_λ are respectively g_λ and u_λ.

Proposition (14.3.13). — *Let $f : X \to Y$ be a morphism locally of finite type, y a point of Y, x a maximal point of the fibre $X_y = X \times_Y \mathrm{Spec}(k(y))$. Consider the following conditions:*

a) *f is universally open at the point x (or even at every point of the irreducible component of X_y with generic point x (14.3.3.1, (ii))).*

b) *For every irreducible component Y_0 of Y containing y, there exists an irreducible component Z of X containing x, dominant over Y_0 and such that $\dim_x(X_y) = \dim_x(Z \cap X_y) = \dim(Z \cap X_\eta)$, where η is the generic point of Y_0 (which implies that Z is* equidimensional *over Y_0 at the point x (13.2.2)).*

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b′) *For every open neighbourhood U of x in X and every generization y' of y, we have $\dim(U \cap X_{y'}) \geqslant \dim_x(U \cap X_y)$.*

Then we have the implications $a) \Rightarrow b) \Leftrightarrow b')$.

Proof. To show that $b)$ implies $b')$, it suffices to remark that y' belongs to an irreducible component Y_0 of Y containing y, of generic point η; taking Z as in $b)$ and noting that the generic point of Z (which is also that of $Z \cap X_\eta$ ($\mathbf{0}_1$, 2.1.8)) is contained in U, we have $\dim(U \cap X_{y'}) \geqslant \dim(U \cap Z \cap X_{y'})$, and, by virtue of (13.1.6), $\dim(U \cap Z \cap X_{y'}) \geqslant \dim(Z \cap X_\eta)$; whence the assertion, since $\dim_x(U \cap X_y) = \dim_x(X_y)$ (4.1.1.3).

To prove that $b')$ implies $b)$, one can first replace X by $f^{-1}(Y_0)$, hence suppose $Y_0 = Y$; we can restrict to the case where X and Y are affine, since $\dim_x(U \cap X_y) = \dim_x(X_y)$ (4.1.1.3). The irreducible components W_i of the Noetherian prescheme X_η are then finite in number, and the complement of the union of the $\overline{W}_i$ (closures *in* X) that do not contain x is an open neighbourhood V of x. Replacing X by V, we can therefore suppose that $x \in \overline{W}_i$ for every i (the hypothesis b') implies $\dim(V \cap X_\eta) \geqslant 0$, hence x belongs to at least one of the $\overline{W}_i$ for at least one i). Furthermore, the $\overline{W}_i$ are exactly the irreducible components of X that dominate Y ($\mathbf{0}_1$, 2.1.8); if one had, for each of these components Z, $\dim(Z \cap X_\eta) < \dim_x(X_y)$, one would conclude $\dim(X_\eta) < \dim_x(X_y)$, contrary to the hypothesis $b')$. The relation $\dim_x(Z \cap X_y) = \dim(Z \cap X_\eta)$ follows then from (13.1.6).

It remains to show that $a)$ implies $b')$. Taking into account (**II**, 7.1.4) and the invariance of the hypotheses and of the conclusion by base change (by virtue of (4.2.7)), we can restrict to the case where Y is the spectrum of a valuation ring, with closed point y and generic point y', and where $U = X$. The hypothesis that f is open at the point x implies that there exists an irreducible component Z of X containing x and dominant over Y (1.10.3). Applying (14.3.10) to a neighbourhood of x in Z, we conclude that $\dim(Z \cap X_{y'}) = \dim_x(Z \cap X_y)$; but as x is maximal in X_y, $Z \cap X_y$ contains the irreducible component of X_y of generic point x, hence $\dim_x(Z \cap X_y) = \dim_x(X_y)$; on the other hand, we have $\dim(X_{y'}) \geqslant \dim(Z \cap X_{y'})$, which completes the proof of $b')$. □

Remark (14.3.14). — We do not know whether, in (14.3.13), the conclusion remains valid when one replaces hypothesis $a)$ by the weaker hypothesis that f is *open* at the point x. One could however easily show that it would suffice to treat the case where Y is the spectrum of an integral local ring, whose generic point is isolated, and where X is a closed sub-prescheme of the vector bundle $Y[\mathrm{T}]$.

14.4. Chevalley's criterion for universally open morphisms

Theorem (14.4.1). — *Let $f : X \to Y$ be a morphism locally of finite type, y a point of Y, x a maximal point of the fibre $X_y = X \times_Y \operatorname{Spec}(k(y))$. Suppose y geometrically unibranch. Then the following conditions are equivalent:*

a) *f is universally open at x (or equivalently at every point of the irreducible component of X_y with generic point x (14.3.3.1, (iii))).* **IV-3** | 210
b) *If Y_0 is the unique irreducible component of Y containing y, η its generic point, there exists an irreducible component Z of X containing x, dominating Y_0 and such that $\dim_x(Z \cap X_y) = \dim(Z \cap X_\eta)$ (which means that Z is equidimensional over Y_0 at the point x (13.2.2)).*
b′) *For every open neighbourhood U of x in X and every generization y' of y, we have $\dim(U \cap X_{y'}) \geqslant \dim_x(U \cap X_y)$.*

If moreover Y is locally Noetherian, these conditions are also equivalent to the following:

c) *f is open at x.*

Let us first note that since $(\mathscr{O}_y)_{\mathrm{red}}$ is integral, y belongs to only one irreducible component Y_0 of Y. The fact that *b)* and *b′)* are equivalent and that *a)* implies *b′)* results from (14.3.13); on the other hand, if Y is locally Noetherian, we have seen in (14.2.3) that *c)* implies *b)*. It remains to show that when y is geometrically unibranch, *b)* implies *a)*.

Lemma (14.4.1.1). — *Let Y be a prescheme, $Y' = Y[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, y' a point of Y', y its image in Y. If Y is geometrically unibranch at the point y, then Y' is geometrically unibranch at the point y'.*

Indeed, for every $y \in Y$, $\operatorname{Spec}(\kappa(y)[\mathrm{T}_1, \ldots, \mathrm{T}_n])$ is geometrically regular over $\kappa(y)$ (**0**, 17.3.7), the structural morphism $Y' \to Y$ is smooth (6.8.1), and it suffices to apply (11.3.14).

Lemma (14.4.1.2). — *Let A be a local integral ring, unibranch, B an integral ring containing and integral over A, and $\mathfrak{n}$ a prime ideal of B above the maximal ideal $\mathfrak{m}$ of A. Then the morphism $\operatorname{Spec}(B_{\mathfrak{n}}) \to \operatorname{Spec}(A)$ is surjective, in other words, for every prime ideal $\mathfrak{p}$ of A, there exists a prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \subset \mathfrak{n}$ and $\mathfrak{q} \cap A = \mathfrak{p}$.*

Let K (resp. L) be the field of fractions of A (resp. B), A' the integral closure of A, B' the subring of L generated by A' and B, so that we have a commutative diagram of canonical injections

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \end{array}$$

As B' is integral over B, there exists a prime ideal $\mathfrak{n}'$ of B' such that $\mathfrak{n}' \cap B = \mathfrak{n}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 1, th. 1), and (for the same reason) $\operatorname{Spec}(A') \to \operatorname{Spec}(A)$ is surjective. On the other hand, as A is unibranch, A' is a *local* ring, so $\mathfrak{n}' \cap A'$, which is above the maximal ideal $\mathfrak{m}$ of A, is necessarily equal to the unique maximal ideal $\mathfrak{m}'$ of A'. By virtue of the second theorem of Cohen–Seidenberg (*loc. cit.*, § 2, n° 4, th. 3) the morphism $\operatorname{Spec}(B'_{\mathfrak{n}'}) \to \operatorname{Spec}(A')$ is surjective, so it is the same for the composite $\operatorname{Spec}(B'_{\mathfrak{n}'}) \to \operatorname{Spec}(A') \to \operatorname{Spec}(A)$; but this morphism is also the composite $\operatorname{Spec}(B'_{\mathfrak{n}'}) \to \operatorname{Spec}(B_{\mathfrak{n}}) \to \operatorname{Spec}(A)$, so the morphism $\operatorname{Spec}(B_{\mathfrak{n}}) \to \operatorname{Spec}(A)$ is surjective.

These lemmas being established, let us return to the proof of the implication $b) \Rightarrow a)$ in (14.4.1). By virtue of (14.3.3.1, (i)), it suffices to prove that, for every integer $n \geqslant 0$ and every point x' of $X' = X[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ above x, the morphism $f' : X' \to Y' = Y[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, deduced from f by base **IV-3** | 211
change, is open at the point x'. Taking into account the lemma (14.4.1.1), (2.3.4) and (4.2.7), we are thus reduced to proving that f is *open* at the point x: furthermore, it plainly suffices, by (14.1.2, (iii)), to show that the restriction of f to a closed sub-prescheme of X having Z as underlying space is open at the point x, so that we can restrict to the case where $X = Z$ is irreducible. Replacing X by an open neighbourhood V of x such that $V \cap X_y$ is irreducible, we may suppose, by virtue of (13.3.1), that the morphism f factorises as $X \xrightarrow{g} Y'' = Y[\mathrm{T}_1, \ldots, \mathrm{T}_m] \to Y$, where g is quasi-finite, dominant and locally of finite type. As the structural morphism $Y'' \to Y$ is open (2.4.6), we are reduced to proving that $g : X \to Y''$ is open at the point x. By (14.4.1.1), $g(x)$ is a geometrically unibranch point of Y''. We are thus reduced to proving the following lemma:

Lemma (14.4.1.3). — *Let X, Y be two irreducible preschemes, $f : X \to Y$ a morphism locally quasi-finite and dominant. If $x \in X$ is such that $y = f(x)$ is a unibranch point of Y, then f is open at the point x.*

It suffices to prove that $f(\operatorname{Spec}(\mathscr{O}_{X,x})) = \operatorname{Spec}(\mathscr{O}_{Y,y})$ (1.10.3). One can thus restrict, by the base change $\operatorname{Spec}(\mathscr{O}_{Y,y}) \to Y$, to the case where $Y = \operatorname{Spec}(A)$, where A is a local ring and y is the closed point of Y (taking into account (**I**, 3.6.5) and (**0**, 1.2.1.8)), which proves that $X \times_Y \operatorname{Spec}(\mathscr{O}_{Y,y})$ is irreducible; replacing f by f_{red}, we may suppose X and Y reduced, hence integral. Replacing X and Y eventually by affine neighbourhoods of x and y respectively, we may suppose (8.12.9) that the morphism f factorises as $X \xrightarrow{j} X_1 \xrightarrow{g} Y$, where j is an *open immersion* and g a *finite* morphism (evidently dominant); as X and X_1 are affine, j is affine, hence separated and quasi-compact, and consequently j factorises as $X \xrightarrow{u} X_2 \xrightarrow{h} X_1$, where X_2 is the closed image of X by j, h the canonical injection and u an open immersion (**I**, 9.5.3). In other terms, one can suppose that X_1 is integral, or even that $X_1 = \operatorname{Spec}(B)$, where B is an integral and finite A-algebra containing A, since g is dominant. If $\mathfrak{n}$ is the prime ideal of B corresponding to the point x, the hypothesis that A is unibranch then implies (14.4.1.2) that the morphism $\operatorname{Spec}(B_{\mathfrak{n}}) \to \operatorname{Spec}(A)$ is surjective, that is, $\operatorname{Spec}(\mathscr{O}_{X,x}) \to \operatorname{Spec}(\mathscr{O}_{Y,y})$ is surjective. Q.E.D.

Corollary (14.4.2). — *Let $f : X \to Y$ be a morphism locally of finite type, y a geometrically unibranch point of Y, η the generic point of the unique irreducible component Y_0 of Y containing y. The following conditions are equivalent:*

a) *f is universally open at all points of X_y (or, what amounts to the same, at the maximal points of X_y (14.3.3.1, (ii)).*

b) *For every $x \in X_y$, there exists an irreducible component Z of X containing x and equidimensional over Y at the point x (13.2.2).*

b′) *For every $x \in X_y$ and every open neighbourhood U of x in X, we have*

$$\dim(U \cap X_\eta) \geqslant \dim_x(U \cap X_y).$$

b″) *For every open set U of X, we have $\dim(U \cap X_\eta) \geqslant \dim(U \cap X_y)$.*

When moreover Y is locally Noetherian, these conditions are also equivalent to the following: **IV-3** | 212

c) f is open at all points of X_y (or, what amounts to the same, at the maximal points of X_y (14.3.3.1, (ii)).

The equivalence of *a)* and *c)* when Y is locally Noetherian results from (14.4.1); the conditions *b)* or *b′)*, applied to the *maximal* points of X_y, imply *a)* by virtue also of (14.3.3.1, (ii)); finally, *b′)* and *b″)* are equivalent, since

$$\dim(U \cap X_y) = \sup_{x \in X_y} (\dim_x(U \cap X_y)).$$

It remains to see that condition *a)* implies *b)* and *b′)* at *every* point $x \in X_y$. Let $d = \dim_x(X_y)$, and let x' be the generic point of an irreducible component of X_y containing x and of dimension d. By virtue of *a)* and of (14.4.1), there is an irreducible component Z of X containing x' and equidimensional over Y at the point x', hence such that $\dim_{x'}(Z \cap X_y) = \dim(Z \cap X_\eta)$. But by construction $\dim_{x'}(Z \cap X_y) = \dim_x(X_y) = d$, and $\dim_x(Z \cap X_y) \leqslant \dim(Z \cap X_y) = d$; taking into account (13.1.6), this proves that Z is equidimensional over Y at the point x; hence *a)* implies *b)*. Moreover, we have $\dim(X_\eta) \geqslant \dim(Z \cap X_\eta) = d = \dim_x(X_y)$. Replacing X by an open neighbourhood U of x, we thus see that *a)* implies *b′)*. Q.E.D.

Corollary (14.4.3). — *Let Y be a geometrically unibranch prescheme, $f : X \to Y$ a morphism locally of finite type. The following conditions are equivalent:*

a) *f is universally open.*

b) *For every open set U of X, every $y \in Y$ and every generization y' of y, we have*

$$\dim(U \cap X_{y'}) \geqslant \dim(U \cap X_y).$$

If moreover Y is locally Noetherian, these conditions are also equivalent to the following:

c) *f is open.*

Corollary (14.4.4). — *(Chevalley's criterion). Let $f : X \to Y$ be a morphism locally of finite type.*

(i) If f is equidimensional at a point $x \in X$ (13.3.2) and if $y = f(x)$ is a geometrically unibranch point of Y, f is universally open at the point x.

(ii) If Y is geometrically unibranch, f is universally open at all the points of X where f is equidimensional, and the set of these points is open in X. In particular, if f is equidimensional, it is universally open.

Assertion (ii) is a trivial consequence of (i), since we already know that the set of points where f is equidimensional is open (13.3.2). As to assertion (i), it follows from the fact that the hypothesis implies that condition *b)* of (14.4.1) is satisfied at the generic point of an irreducible component of X_y containing x, taking into account (13.3.1); it therefore suffices to apply (14.4.1).

Remark (14.4.5). — One can prove that if Y is locally Noetherian, and if all the generizations of $y = f(x)$ are geometrically unibranch points of Y (cf. (6.15.2)), then, if f is equidimensional at the point x, it is universally open *in a neighbourhood of* x.

Corollary (14.4.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type. Let y be a geometrically unibranch point of Y, and suppose furthermore that for every $x \in f^{-1}(y)$, the ring $\mathscr{O}_{X,x}$ is equidimensional. Then the following conditions are equivalent:* **IV-3 | 213**

a) *f is equidimensional (13.3.2) at all points of $f^{-1}(y)$.*
b) *f is open at all points of $f^{-1}(y)$.*
c) *f is universally open at all points of $f^{-1}(y)$.*

Indeed, *c)* implies trivially *b)*, and *a)* implies *c)* by virtue of (14.4.4); finally, because of the hypotheses on $\mathscr{O}_y$ and $\mathscr{O}_x$, *b)* implies *a)* by (14.2.2). More generally:

Proposition (14.4.7). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, such that $y = f(x)$ is geometrically unibranch. The following conditions are equivalent:*

a) *f is equidimensional (13.3.2) at the point x.*
b) *The ring $\mathscr{O}_x$ is equidimensional and f is open at the generic points of the irreducible components of $f^{-1}(y)$ containing x (and hence also at every point of such a component).*
c) *The ring $\mathscr{O}_x$ is equidimensional, and f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x (and hence also at every point of such a component).*

Furthermore, when these conditions are satisfied, for every reduced closed sub-prescheme X_i of X having as underlying space an irreducible component of X containing x, the restriction $f_i : X_i \to Y$ of f is an equidimensional morphism at the point x, and universally open at all the points of the irreducible components of $f^{-1}(y) \cap X_i$ which contain x.

Condition *a)* implies that f is equidimensional at all the generic points of the irreducible components of $f^{-1}(y)$ containing x (13.3.1), and consequently (14.4.4) universally open at these points; the same argument applied to each f_i (taking into account (13.3.3)) proves the last assertion of the proposition, in view of (14.3.3.1, (ii)). Furthermore, by (14.2.1), we have the relations

$$\dim(\mathscr{O}_{X_i,x}) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_{X_i,x} \otimes_{\mathscr{O}_y} \kappa(y)) \tag{14.4.7.1}$$

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_y} \kappa(y)) \tag{14.4.7.2}$$

and since f is equidimensional at the point x, it follows from (13.3.1) that $\dim_x(X_i \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$, hence (5.2.3) $\dim(\mathscr{O}_{X_i,x} \otimes_{\mathscr{O}_y} \kappa(y)) = \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_y} \kappa(y))$. We conclude that $\dim(\mathscr{O}_{X_i,x}) = \dim(\mathscr{O}_{X,x})$ for every i, in other words $\mathscr{O}_{X,x}$ is equidimensional, and this finishes showing that *a)* implies *c)*. It is clear that *c)* implies *b)*; finally, *b)* implies the relation (14.4.7.2) by virtue of (14.2.1); it then follows from (13.3.6) that *b)* implies *a)*.

Proposition (14.4.8). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, y a point of Y, x a maximal point of $X_y = f^{-1}(y)$. The following conditions are equivalent:*

a) *The morphism f is universally open at x, in other words, for every change of base $g : Y' \to Y$, the property $\mathbf{P}(Y')$ holds: for every point x' of $X' = X \times_Y Y'$ above x, the morphism $f' = f_{(Y')} : X' \to Y'$ is open at x'.* **IV-3 | 214**

a′) *The property $\mathbf{P}(Y')$ holds for every finite morphism $g : Y' \to Y$.*
a″) *The property $\mathbf{P}(Y'')$ holds for the normalisation Y'' of Y_{red}* (**II**, 6.3.8).

b) *For every point x'' of $X'' = X \times_Y Y''$ above x, there exists an irreducible component Z'' of X'' containing x'' and equidimensional over Y'' at the point x''.*

It is trivial that *a)* implies *a')*. To show that *a')* implies *a'')*, note that we can write $Y'' = \mathrm{Spec}(\mathscr{B})$, where $\mathscr{B}$ is a quasi-coherent $\mathscr{O}_Y$-Algebra integral over $\mathscr{O}_Y$; as Y is Noetherian, $\mathscr{B}$ is the inductive limit of its sub-$\mathscr{O}_Y$-Algebras $\mathscr{B}_\lambda$, which are quasi-coherent and of finite type (**I**, 9.6.6); but then the $\mathscr{B}_\lambda$ are *finite* $\mathscr{O}_Y$-Algebras (**II**, 6.1.2); we can therefore write $Y'' = \varprojlim Y'_\lambda$, where $Y'_\lambda = \mathrm{Spec}(\mathscr{B}_\lambda)$, and hence $X'' = X \times_Y Y'' = \varprojlim X'_\lambda$, with $X'_\lambda = X \times_Y Y'_\lambda$. By virtue of *a')*, the morphisms $f'_\lambda : X'_\lambda \to Y'_\lambda$ are open at all the points of X'_λ above x; we conclude that f'' is open at all the points of X'' above x, by (8.10.1) and (14.3.3.1, (i)).

As the prescheme Y'' is *normal* by definition, the fact that *b)* implies *a'')* follows from (14.4.4), applied to the equidimensional irreducible component in the statement and to the restriction of f'' to that component. It therefore remains to show that *a'')* implies *a)* and *b)*. Taking into account (1.10.3), one may restrict to the case where $Y = \mathrm{Spec}(\mathscr{O}_y)$, noting that the canonical morphism $\mathrm{Spec}(\mathscr{O}_y) \to Y$ is universally bicontinuous (**I**, 3.6.5), and on the other hand that $Y'' \times_Y \mathrm{Spec}(\mathscr{O}_y)$ is the normalisation of $(\mathrm{Spec}(\mathscr{O}_y))_{\mathrm{red}}$ as follows from the commutativity of the operations of integral closure and localisation (Bourbaki, *Alg. comm.*, chap. V, § 1, nº 5, prop. 16). Suppose therefore that $Y = \mathrm{Spec}(A)$, where A is a local Noetherian ring, and y the closed point of Y; we know (**0**, 23.2.5) that there exists a factorisation

$$Y'' \xrightarrow{u} Y_1 \xrightarrow{v} Y$$

of the structural morphism, such that v is a *finite* and surjective morphism, while u is integral, *radicial* and dominant, hence surjective (**II**, 6.1.10) and consequently a universal *homeomorphism* (2.4.5). If we set $X_1 = X \times_Y Y_1$, the projection $X'' \to X_1$ is therefore a homeomorphism, and the hypothesis *a'')* implies that $f_1 = f_{(Y_1)} : X_1 \to Y_1$ is open at all points of X_1 above x. Furthermore, this shows that in order to prove property *b)*, it suffices to prove the same property where we replace Y'', X'' and x'' by Y_1, X_1 and a point x_1 of X_1 above x. But Y_1 is Noetherian and furthermore geometrically unibranch since Y'' is normal and u is radicial (6.15.1); the property *b)* therefore results from (14.4.1). It remains to show that f is universally open at the point x, which will result from the following lemma:

Lemma (14.4.8.1). — *Let $v : Y_1 \to Y$ be a closed (resp. universally closed) and surjective morphism. For a morphism $f : X \to Y$ to be open (resp. universally open) at a point $x \in X$, it suffices that $f_1 = f_{(Y_1)} : X_1 = X \times_Y Y_1 \to Y_1$ be open (resp. universally open) at all the points of X_1 above x.*

The second assertion results trivially from the first and from the fact that for every base change $Y' \to Y$, the morphism $v_{(Y')} : Y_1 \times_Y Y' \to Y'$ is still surjective and universally closed. To prove the first assertion, consider an open neighbourhood U of x in X; as v is closed and surjective, for $f(U)$ to be a neighbourhood of $y = f(x)$, it is necessary and sufficient that $v^{-1}(f(U))$ be a neighbourhood of $v^{-1}(y)$. But if $p : X_1 \to X$ is the canonical projection, we have $v^{-1}(f(U)) = f_1(p^{-1}(U))$ (**I**, 3.4.8), and the hypothesis implies that $f_1(p^{-1}(U))$ is a neighbourhood of $f_1(p^{-1}(x)) = v^{-1}(y)$ (**I**, 3.4.8). **IV-3** | 215

Corollary (14.4.9). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type. The following conditions are equivalent:*

a) *f is universally open, in other words, for every base change $Y' \to Y$, the morphism $f_{(Y')} : X \times_Y Y' \to Y'$ is open.*

a') *For every finite morphism $Y_1 \to Y$, $f_{(Y_1)}$ is open.*

a'') *If Y'' is the normalisation of Y_{red}, $f_{(Y'')}$ is open.*

b) *For every point x'' of $X'' = X \times_Y Y''$, there exists an irreducible component Z'' of X'' containing x'' and equidimensional over Y'' at the point x'' (cf. (14.4.10, (ii))).*

This results immediately from (14.4.8) and (14.1.4).

Remarks (14.4.10). — (i) The equivalence of conditions *a)* and *b)* in (14.4.8) (resp. (14.4.9)) is also valid for any prescheme Y and any morphism f locally of finite type. Indeed, *a)* implies *b)* by virtue of (14.4.1); conversely, *b)* implies that f'' is universally open at the points of X'' above x (resp. at every point of X'') by virtue of (14.4.1), and we conclude property *a)* by applying the lemma (14.4.8.1) to the integral and surjective morphism $Y'' \to Y$.

It may be that, in (14.4.1), for the equivalence of *a)* and *c)*, the supplementary hypothesis that Y is Noetherian is superfluous (cf. (14.3.14)). If so, the Noetherian hypotheses are also superfluous in (14.4.2), (14.4.3), (14.4.8) and (14.4.9).

(ii) One can give examples of morphisms $f : X \to Y$ with the following properties: Y is Noetherian, regular and of dimension 2, f is *universally open* and of finite type, X has two irreducible components X_1, X_2, but the restriction $X_1 \to Y$ of f to one of them is *not an open morphism*. The principle of the construction relies on the general method of "recollement" which will be presented in chap. V, and which can therefore only be sketched here. Start with a closed point y of Y, and consider the Y-scheme Y_1 obtained by blowing up y (**II**, 8.1.3). If $f_1 : Y_1 \to Y$ is the structural morphism, we know that the restriction of f_1 to $f_1^{-1}(Y - \{y\})$ is an *isomorphism* over $Y - \{y\}$ (*loc. cit.*), while the fibre $f_1^{-1}(y)$ is isomorphic to $\mathrm{Proj}(S)$, where $S = \bigoplus_{k=0}^{\infty} \mathfrak{m}_y^k/\mathfrak{m}_y^{k+1}$ (**II**, 3.5.3), i.e. here $\mathbf{P}^1_{\kappa(y)}$: it follows from (14.4.1) that f_1 is not open at the generic point of $f_1^{-1}(y)$. On the other hand, let us set $Y_2 = \mathbf{P}^1_Y$, and let $f_2 : Y_2 \to Y$ be the structural morphism. It follows from (**II**, 8.4.4) that f_2 is flat, hence universally open (2.4.6); moreover, by (**II**, 3.5.3), $f_2^{-1}(y)$ is isomorphic to $\mathbf{P}^1_{\kappa(y)}$. It then suffices to "glue" Y_1 and Y_2 along the isomorphic fibres $f_1^{-1}(y)$ and $f_2^{-1}(y)$; this gives a morphism $f : X \to Y$ whose irreducible components X_1, X_2 identify canonically with Y_1, Y_2, respectively, and whose restrictions to these components are f_1, f_2.

Let us recall however (12.1.1.5) that if Y is locally Noetherian, $f : X \to Y$ is of finite type and *flat*, then every irreducible component of X is equidimensional over Y (and consequently the restriction of f to such a component is *universally open* if all the points of Y are *geometrically unibranch*).

(iii) Recall (12.1.2, (i)) that there are morphisms $f : X \to Y$ with the following properties: Y is Noetherian (not geometrically unibranch), f is finite and flat (and even étale (17.6.3)), but the restriction of f to an irreducible component of X is *not an open morphism* (although f itself is so by (2.4.6)).

(iv) The Chevalley criterion (14.4.4) explains the importance of the notion of universally open morphism. This notion allows in effect, in numerous results, more or less classical, to replace a hypothesis of *normality* by the hypothesis that a certain morphism is universally open; the more general statement thus obtained will in particular apply to *flat* morphisms (2.4.6), whose importance in algebraic geometry is well known. One can consider that the statements involving the hypothesis that a morphism is universally open are common generalisations of statements involving a hypothesis of normality and of statements involving a hypothesis of flatness. **IV-3** | 216

14.5. Universally open morphisms and quasi-sections

Lemma (14.5.1). — *Let Y be an irreducible locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, such that f is equidimensional* (13.3.2) *at the point x; set $y = f(x)$, $e = \dim_x(f^{-1}(y))$. Let X' be a closed irreducible subspace of X containing x, n an integer such that* $\mathrm{codim}(X', X) \leqslant n$ *and* $\dim_x(X' \cap f^{-1}(y)) \leqslant e - n$. *Then necessarily* $\mathrm{codim}(X', X) = n$, $\dim_x(X' \cap f^{-1}(y)) = e - n$, *and the restriction $X' \to Y$ of f is an equidimensional morphism at x (and a fortiori dominant).*

The question being local on X, we can suppose that f is of finite type and *equidimensional*, so that for *every* $z \in Y$, *all* the irreducible components of $f^{-1}(z)$ are of dimension e (13.3.1, *a''*)). Let x' be the generic point of X', $y' = f(x')$, $Y' = \overline{\{y'\}} = \overline{f(X')}$ and set $Z = f^{-1}(Y')$. By virtue of (**0**, 14.2.2), we have

$$\mathrm{codim}(X', Z) \leqslant n \tag{14.5.1.1}$$

and if the two sides are equal, we necessarily have $\mathrm{codim}(X', X) = n$ and Z contains an irreducible component of X, which implies (13.3.1) that $f(X')$ is dense in Y and hence $Z = X$; thus the equality $\mathrm{codim}(X', Z) = n$ is equivalent to the conjunction of the equality $\mathrm{codim}(X', X) = n$ and the relation $Z = X$.

On the other hand, reasoning in the reduced preschemes of Y and X having Y' and Z respectively as underlying spaces, we deduce from (5.1.2) and from (**I**, 3.6.5) that

$$\mathrm{codim}(X' \cap f^{-1}(y'), f^{-1}(y')) = \mathrm{codim}(X', Z). \tag{14.5.1.2}$$

By hypothesis, $f^{-1}(y')$ is biequidimensional (5.2.1) and of dimension e, hence (**0**, 14.3.5), by virtue of (14.5.1.2) and (14.5.1.1)

(14.5.1.3) $$e' = \dim(X' \cap f^{-1}(y')) = e - \operatorname{codim}(X', Z) \geqslant e - n$$

the equality holding if and only if $\operatorname{codim}(X', X) = n$ and if $f(X')$ is dense in Y. Finally, (13.1.1), we have $\dim_x(X' \cap f^{-1}(y)) \geqslant e'$, whence, by (14.5.1.3),

(14.5.1.4) $$\dim_x(X' \cap f^{-1}(y)) \geqslant e' \geqslant e - n.$$

But, by hypothesis, we also have $\dim_x(X' \cap f^{-1}(y)) \leqslant e - n$, whence the conclusions of the proposition.

Corollary (14.5.2). — *The hypotheses on f, X, Y, x being those of (14.5.1), suppose moreover that x is not a maximal point of $f^{-1}(y)$. Then there exists an affine open neighbourhood U of x in X and a section $g \in \Gamma(U, \mathscr{O}_X)$ such that the set X' of $x' \in U$ such that $g(x') = 0$ contains x and does not contain any maximal point of $f^{-1}(y)$. For every g having these properties, X' is equidimensional over Y at the point x, and we have*

(14.5.2.1) $$\dim_x(X' \cap f^{-1}(y)) = e - 1 \qquad \textit{and} \qquad \operatorname{codim}(X', X) = 1.$$

One can restrict to the case where $X = U$ is an open affine neighbourhood of x such that all the irreducible components of $f^{-1}(y)$ contain x. These components correspond to the minimal prime ideals of $\mathscr{O}_x/\mathfrak{m}_y\mathscr{O}_x$, and by hypothesis these ideals are distinct from $\mathfrak{m}_x/\mathfrak{m}_y\mathscr{O}_x$ (Bourbaki, *Alg. comm.*, chap. II, § 1, nº 1, prop. 2); to obtain a $g \in \Gamma(U, \mathscr{O}_X)$ satisfying the conditions of the statement, it suffices to take $g \in \mathfrak{p}_x$ such that the image of g in $\mathfrak{m}_x$ does not belong to any of the preceding prime ideals. Moreover, as $\operatorname{codim}(X', X) \leqslant 1$ (5.1.8), and as X' does not contain any of the irreducible components of $f^{-1}(y)$ of dimension e, and since these are all the irreducible components of $f^{-1}(y)$, we have (**0**, 14.2.2.2) $\dim_x(X' \cap f^{-1}(y)) \leqslant e - 1$. It then suffices to apply (14.5.1). **IV-3 | 217**

Proposition (14.5.3). — *Let Y be an irreducible locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X. Suppose that f is equidimensional at the point x and that x is closed in $f^{-1}(f(x))$. Then there exists an irreducible subset X' of X, locally closed in X, containing x, such that the restriction $X' \to Y$ of f (where X' is endowed with the reduced subscheme structure induced from X) is a* quasi-finite and dominant *morphism.*

Indeed, with the notation of (14.5.2), the hypothesis that x is closed in $f^{-1}(y)$ implies that x is not a maximal point of $X' \cap f^{-1}(y)$ as long as $e - 1 \geqslant 1$. It therefore suffices to apply (14.5.2) by descending induction on $e = \dim_x(f^{-1}(f(x)))$ until one arrives at $e = 1$; the application of (14.5.2) in this last case gives a subspace X' such that $X' \cap f^{-1}(f(x))$ is Noetherian and of dimension 0, hence finite and discrete; as X' is equidimensional over Y, $X' \cap f^{-1}(f(x'))$ is of dimension 0 for every $x' \in X'$, which implies that the restriction $X' \to Y$ of f is a quasi-finite morphism (**II**, 6.2.2).

Corollary (14.5.4). — *Under the hypotheses of (14.5.3), suppose moreover that $Y = \operatorname{Spec}(A)$, where A is a local Noetherian ring, integral and complete, and that $y = f(x)$ is the unique closed point of Y. Then there exists a local integral ring A', containing A, which is a finite A-algebra, and such that, if we set $Y' = \operatorname{Spec}(A')$ and $X' = X \times_Y Y'$, there exists a Y'-section $h : Y' \to X'$ such that the composite morphism $Y' \xrightarrow{h} X' \to X$ is an immersion whose image contains x.*

Replacing if necessary X by a reduced closed irreducible sub-prescheme of X, we may, by virtue of (14.5.3), restrict to the case where the morphism f is already quasi-finite and dominant. Using (**II**, 6.2.5), we deduce that $\mathscr{O}_x = A'$ is an integral ring which is a finite A-algebra, and that X is a disjoint sum of the closed sub-prescheme $Y' = \operatorname{Spec}(A')$ and of a sub-prescheme Y''; the scheme Y' satisfies the question, the composite morphism $Y' \to X' \to X$ being nothing other than the canonical morphism $\operatorname{Spec}(\mathscr{O}_x) \to X$.

Remark (14.5.5). — If, in the statement of (14.5.4), one does not require that the morphism $Y' \to X$ be an immersion, one can suppose that A' is moreover *integrally closed*: it suffices in effect to replace A' by its *integral closure* A_1, since we know (**0**, 23.1.5) that A_1 is an A-module of finite type.

Proposition (14.5.6). — *Let Y be a locally Noetherian prescheme, irreducible, regular and of dimension* 1*, $f : X \to Y$ a morphism locally of finite type, y a point of Y. The following conditions are equivalent:*

a) *f_{red} is flat at every point of $f^{-1}(y)$.* **IV-3** | 218
b) *f is universally open in a neighbourhood of $f^{-1}(y)$.*
c) *f is open in a neighbourhood of $f^{-1}(y)$.*
d) *Every irreducible component of X meeting $f^{-1}(y)$ dominates Y.*
e) *For every point $x \in f^{-1}(y)$, closed in $f^{-1}(y)$, and every irreducible component X_i of X containing x, there exists an irreducible subset X' of X, locally closed in X, containing x, contained in X_i, and such that the restriction $X' \to Y$ of f is a quasi-finite and dominant morphism.*

The equivalence of *a)*, *b)*, *c)* and *d)* has already been shown (14.3.8). It is clear that *e)* implies *d)*, since for every irreducible component X_i of X meeting $f^{-1}(y)$, there exists in $X_i \cap f^{-1}(y)$ a closed point (5.1.11). It suffices to prove that *c)* implies *e)*: consider a closed point x of $f^{-1}(y)$ and let X_i be an irreducible component of X containing x. As the restriction $f_i : X_i \to Y$ of f to X_i is a dominant morphism, it follows from the equivalence of *c)* and *d)* for f_i that this morphism is open at the generic points of $X_i \cap f^{-1}(y)$, hence from (14.4.2) that f_i is equidimensional at the point x, and we conclude with the aid of (14.5.3).

Remark (14.5.7). — If, in the statement of (14.5.5), we suppose that $Y = \mathrm{Spec}(A)$, where A is a complete discrete valuation ring, and that y is the closed point of Y, we can moreover suppose that $X' = \mathrm{Spec}(A')$, where A' is a discrete valuation ring which is a finite A-algebra, as follows from the proof of (14.5.4) and from the fact that a local integral ring, regular and of dimension 1, is a discrete valuation ring (**II**, 7.1.6).

Proposition (14.5.8). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, y a point of Y. For every Y-prescheme Y', set $X(Y') = \mathrm{Hom}_Y(Y', X)$. Let x be a point of $f^{-1}(y)$; in order that f be universally open at x, it is necessary and sufficient that the following condition be satisfied:*

For every complete discrete valuation ring A with algebraically closed residue field, every morphism $g : Y' = \mathrm{Spec}(A) \to Y$ such that the image under g of the closed point y' of Y' is equal to y, and every element $u_0 \in X(\mathrm{Spec}(\kappa(y')))$ such that $u_0(y') = x$, there exist a discrete valuation ring B, a local homomorphism $A \to B$ making B into a finite A-algebra, and, if we set $Z' = \mathrm{Spec}(B)$, an element $u \in X(Z')$ such that, if z' is the closed point of Z', the diagram

$$\begin{tikzcd} \mathrm{Spec}(\kappa(z')) \arrow[r] \arrow[d, "\ell"] & Z' \arrow[d, "u"] \\ \mathrm{Spec}(\kappa(y')) \arrow[r, "u_0"'] & X \end{tikzcd}$$

is commutative.

Let us note that if A and B satisfy the conditions of the statement, B is a complete discrete valuation ring (Bourbaki, *Alg. comm.*, chap. III, § 3, nº 3, prop. 7 and chap. IV, § 2, nº 2, cor. 3 of prop. 9) whose residue field is isomorphic to that of A, and hence algebraically closed. Moreover, since $u(z') = x$, there is in $X'' = X \times_Y Z'$ a point x'' whose projections in X and Z' are x and z', respectively, and which is rational over $\kappa(z')$; furthermore, since there exists a Z'-section v of X'' such that $v(z') = x''$, the image under v of the generic point s of Z' is a generization t of x'' whose projection in Z' is s. Applying (14.3.6), we see that the condition of the statement is *sufficient*. Let us now prove that it is *necessary*. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. By hypothesis there is a point $x' \in X'$ above x and y' and rational over $\kappa(y')$ (**I**, 3.3.14), hence closed in $f'^{-1}(y')$. By virtue of (14.3.6), there is an irreducible component T' of X' containing x' and dominant over Y'; hence, by (14.3.8) and (14.3.13), the restriction $T' \to Y'$ of f' is equidimensional at x'. It follows from (14.5.5) that there is a finite A-algebra B which is an integrally closed local domain dominating A (and hence is a discrete valuation ring) and, setting $Z' = \mathrm{Spec}(B)$ and $X'' = X \times_Y Z' = X' \times_{Y'} Z'$, a Z'-section $v : Z' \to X''$ such that the image of the closed point z' of Z' under the composite $Z' \xrightarrow{v} X'' \to X'$ is x'; the composite morphism $u : Z' \xrightarrow{v} X'' \to X$ therefore answers the question. **IV-3** | 219

Proposition (14.5.9). — *Let Y be an irreducible locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, and y a geometrically unibranch point of Y. Then the following conditions are equivalent:*

a) *f is universally open at every point of $f^{-1}(y)$.*
b) *f is open at every point of $f^{-1}(y)$.*
c) *For every irreducible component Z of $f^{-1}(y)$, with generic point z, there exists an irreducible component X_i of X containing z and equidimensional over Y at z.*
d) *For every closed point x of $f^{-1}(y)$, there exists an irreducible subset X' of X, locally closed in X, containing x, and such that the restriction $X' \to Y$ of f is a quasi-finite dominant morphism.*

The equivalence of *a)*, *b)* and *c)* has already been shown (14.4.2). To prove that *a)* implies *d)*, note that by (14.4.2), *a)* implies that there exists an irreducible component Z of X containing x and equidimensional over Y at the point x; the existence of X' then comes from (14.5.3) applied to the restriction $Z \to Y$ of f. Conversely, suppose *d)* is satisfied; by the Chevalley criterion (14.4.4), the restriction $X' \to Y$ of f is universally open at x, and *a fortiori* f is open at x. Thus f is open at every closed point of $X_y = f^{-1}(y)$. But X_y is a $\kappa(y)$-prescheme locally of finite type, hence a Jacobson prescheme; the set of closed points of X_y is therefore *dense* in X_y (10.3.1), and it follows from (14.1.4) that f is open at *all* points of X_y, which finishes proving that *d)* implies *b)*.

The following result was brought to light by D. Mumford:

Proposition (14.5.10). — *Let Y be a Noetherian prescheme, $f : X \to Y$ a morphism universally open, surjective and locally of finite type. Then there exists a finite surjective morphism $g : Y' \to Y$ such that, if we set $X' = X \times_Y Y'$ and $f' = f_{(Y')} : X' \to Y'$, every point $y' \in Y'$ admits an open neighbourhood U' such that there exists a U'-section of $f'^{-1}(U')$.* **IV-3** | 220

We will prove the proposition in several steps.

I) Reduction to the case where Y is integral. If we have proved the proposition for each of the reduced sub-preschemes Y_i having as underlying space an irreducible component of Y, and for the inverse images $f^{-1}(Y_i)$, it is clear that the prescheme Y' which is the *sum* of the corresponding Y'_i will answer the question. We can therefore suppose Y *integral* and what follows will be limited to this case. Furthermore, in the conclusion, we will also be able to take Y' integral (up to replacing it by a suitable irreducible component).

II) Local character on Y. We will show that if Y can be covered by finitely many open sets U_j such that, for every j, the conclusion of the proposition holds for the morphism $f^{-1}(U_j) \to U_j$, the restriction of f, then the conclusion also holds for f. Indeed, we can evidently suppose the U_j affine, so that $U_j = \operatorname{Spec}(A_j)$, where A_j is an integral Noetherian ring whose field of fractions $K = R(Y)$ is the field of rational functions on Y. For every j, by hypothesis there is a finite integral A_j-algebra A'_j such that the homomorphism $A_j \to A'_j$ is injective (**I**, 1.2.7) and that the corresponding morphism $g_j : U'_j = \operatorname{Spec}(A'_j) \to \operatorname{Spec}(A_j) = U_j$ satisfies the conditions of the statement (for U_j and $f^{-1}(U_j)$). Let K' be a finite extension of K containing the fields of fractions of all the A'_j (which are finite extensions of K). Consider the normalisation Y'' of Y in K' (**II**, 6.3.8); it is of the form $\operatorname{Spec}(\mathscr{B})$, where $\mathscr{B}$ is an integral quasi-coherent $\mathscr{O}_Y$-Algebra, the integral closure of $\mathscr{O}_Y$ in K' (**II**, 6.3.4). These definitions show that, for every j,

$$A'_j = \Gamma(U'_j, \mathscr{O}_{U'_j}) = \Gamma\big(U_j, (g_j)_*(\mathscr{O}_{U'_j})\big)$$

identifies with a finite sub-A_j-algebra of $\Gamma(U_j, \mathscr{B})$; in other words,

$$(g_j)_*(\mathscr{O}_{U'_j}) = \mathscr{A}'_j$$

is a coherent $\mathscr{O}_{U_j}$-Algebra, a sub-Algebra of $\mathscr{B}|U_j$. There consequently exists a coherent sub-$\mathscr{O}_Y$-Module $\mathscr{C}'_j$ of $\mathscr{B}$ such that $\mathscr{C}'_j|U_j = \mathscr{A}'_j$ (**I**, 9.4.7). If we set $\mathscr{C}' = \sum_j \mathscr{C}'_j$, the sub-$\mathscr{O}_Y$-Algebra $\mathscr{B}'$ of $\mathscr{B}$ generated by $\mathscr{C}'$ is coherent, since $\mathscr{B}$ is an integral $\mathscr{O}_Y$-Algebra. We now show that $Y' = \operatorname{Spec}(\mathscr{B}')$ answers the question. Indeed, it is clear that the morphism $g : Y' \to Y$ is finite and surjective and that Y' is integral; moreover, for every j, $\Gamma(U_j, \mathscr{B}')$ is a finite A'_j-algebra. In other words, the restriction $g^{-1}(U_j) \to U_j$ of g factors as

$$g^{-1}(U_j) \longrightarrow U'_j \longrightarrow U_j,$$

and since the local existence of sections is stable under base change, this proves our assertion, every $y \in Y$ belonging to some U_j.

III) Reduction to the case where Y is integral, local and geometrically unibranch. Suppose first that the proposition has been proved when Y is integral and *local* (with Y' integral), and show that it is valid when Y is integral (Noetherian) affine. Indeed, by virtue of the reduction II), it suffices to prove that for every point $y \in Y$, the proposition holds for an open affine neighbourhood V of y in Y. Let $Y = \operatorname{Spec}(A)$, $\mathfrak{p} = \mathfrak{j}_y$, let $Y_1 = \operatorname{Spec}(A_\mathfrak{p})$, and set $X_1 = X \times_Y Y_1$. By hypothesis, there exists a finite surjective morphism $g_1 : Y_1' \to Y_1$, where $Y_1' = \operatorname{Spec}(B_1)$ and B_1 is a finite integral $A_\mathfrak{p}$-algebra, hence a semilocal ring, such that g_1 satisfies the conditions of the statement for Y_1 and X_1. If y_j' $(1 \leqslant j \leqslant r)$ are the closed points of Y_1', there is therefore a covering of Y_1' by open sets U_j' such that $y_j' \in U_j'$ and there exists a U_j'-section h_j' of $X \times_Y U_j'$ $(1 \leqslant j \leqslant r)$. The $A_\mathfrak{p}$-module B_1 admits a finite system of generators of the form z_i/s (with $s \in A - \mathfrak{p}$, z_i integral over A), which one may **IV-3 | 221**
suppose (multiplying s, if necessary, by an element of A) to be elements of the field of fractions of B_1 integral over A_s, so that if V is the open affine set $D(s) = \operatorname{Spec}(A_s) \subset Y$, Y_1' identifies with $Y_1 \times_V V'$, where V' is the spectrum of the finite A_s-algebra generated by the z_i/s; $g : V' \to V$ is therefore a finite surjective morphism and $g_1 = g_{(Y_1)}$. Moreover, applying the method of (8.1.2), *a*), one may suppose that each U_j' is the inverse image under $p : Y_1' \to V'$ of an open set W_j' of V', that the W_j' cover V' (8.3.11), and that each section h_j' is of the form $(v_j')_{(Y_1)}$, where v_j' is a W_j'-section of $X \times_Y W_j'$ (8.8.2), (i). We are thus indeed reduced to proving the proposition when $Y = \operatorname{Spec}(A)$, where A is a local integral Noetherian ring. We then know that there exists a finite integral A-algebra B, having the same field of fractions as A, such that $A \subset B$ and $\operatorname{Spec}(B)$ is geometrically unibranch (**0**, 23.2.5) and (6.15.5); as the morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective, one may evidently replace Y by $\operatorname{Spec}(B)$ and X by $X \otimes_A B$ to prove the proposition. Arguing as at the beginning of the reduction III), we may therefore suppose that A is local and integral and that $Y = \operatorname{Spec}(A)$ is geometrically unibranch.

IV) Reduction to the case where X is integral, affine, and f quasi-finite, surjective, birational and universally open. Suppose therefore $Y = \operatorname{Spec}(A)$ integral, local and geometrically unibranch. There exists then an irreducible sub-prescheme X_0 of X such that the restriction $f_0 : X_0 \to Y$ of f is a quasi-finite and *dominant* morphism and $f_0(X_0)$ contains the closed point y of Y (14.5.9); as Y is geometrically unibranch, it follows from (14.4.1) that f_0 is still *universally open*. Furthermore, we may suppose X_0 reduced, hence integral, and we see that, by replacing X by X_0, we may suppose X integral, f *quasi-finite and dominant*, and $f(X)$ containing y; since f is *open* and every open subset of Y containing y is equal to Y, f is surjective.

Let ξ, η be the generic points of X and Y respectively; $\kappa(\xi)$ is therefore a finite extension of $K = \kappa(\eta)$. By (4.6.8), there is a *finite* extension K' of K such that $(\operatorname{Spec}(\kappa(\xi) \otimes_K K'))_{\mathrm{red}}$ is geometrically reduced over K and its irreducible components are geometrically irreducible; it follows from (4.5.9) and (4.6.1) that the residue fields of $\operatorname{Spec}(\kappa(\xi) \otimes_K K')$ are finite extensions of K' which are primary and separable, and are therefore equal to K'. Applying once more (**0**, 23.2.5) and (6.15.5), there is a finite integral A-algebra B, having K' as field of fractions, containing A and such that $\operatorname{Spec}(B)$ is geometrically unibranch; as the morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective, one may therefore again replace Y by $\operatorname{Spec}(B)$ and X by $(X \otimes_A B)_{\mathrm{red}}$ in order to prove the proposition; the reduction III) allows then to suppose once more A local, integral and $Y = \operatorname{Spec}(A)$ geometrically unibranch, with closed point y. Furthermore, f is still a quasi-finite, *surjective* and *universally open* morphism; each of the irreducible components X_i of X $(1 \leqslant i \leqslant m)$ dominates Y (1.10.4) and is *birational* over Y. So let x be a point of $f^{-1}(y)$ and let X_i be an irreducible component of X containing x; denote also by X_i the reduced sub-prescheme (hence integral) of X having X_i as underlying space, and let $f_i : X_i \to Y$ be the restriction of f. Applying once more (14.4.1) to the quasi-finite and dominant morphism f_i, we see that f_i is universally open; if U is an affine open set of X_i containing x, $f_i(U)$ is therefore open in Y and contains y, hence is equal to Y. One may thus replace X by U in order to **IV-3 | 222**
prove the proposition.

V) End of the proof. We suppose therefore Y local, integral and geometrically unibranch, X integral and affine, f quasi-finite, surjective, birational and universally open; it follows that f is automatically *separated*. By virtue of the Main theorem (8.12.6), there exists a factorisation $X \xrightarrow{j} Z \xrightarrow{u} Y$, where j is an open immersion and u a *finite* morphism. Replacing moreover Z by the closed image of j (**I**, 9.5.10), we may suppose that Z is *integral*; as u is finite and birational and Y geometrically unibranch, it follows from (**III**, 4.3.5) and (**III**, 4.3.4) that u (and consequently f) is a *radicial* morphism; as f is universally open and surjective, it is therefore a universal *homeomorphism*. Consequently, by (2.4.5), (ii), f is a *finite* morphism. But then we satisfy the conditions of the statement with Y' *integral* by simply taking $Y' = X$ and $g = f$, the Y'-section being the diagonal morphism Δ_f.

This result can be used to develop criteria of "descent" for the properties of universally open and surjective morphisms. Let us point out in particular the following criterion, due to D. Mumford:

Corollary (14.5.11). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, $g : Y' \to Y$ a morphism locally of finite type, universally open and surjective; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Then, for f to be affine, it is necessary and sufficient that f' be so.*

One need only prove that the condition is sufficient. The question being local on Y, we may suppose Y affine, hence Noetherian. By virtue of (14.5.10), there exists a finite surjective morphism $h : Y_1 \to Y$ such that, if we set $Y_1' = Y' \times_Y Y_1$, and $g' = g_{(Y_1)} : Y_1' \to Y_1$, every point of Y_1 admits an open neighbourhood U_1 such that there exists a U_1-section of $g'^{-1}(U_1)$. If we set $X_1 = X \times_Y Y_1$, the canonical projection $p : X_1 \to X$ is a finite surjective morphism, hence, if we prove that X_1 is an affine scheme, it will follow first of all that X is quasi-compact, hence Noetherian, then that X is affine by virtue of the theorem of Chevalley (**II**, 6.7.1). It suffices therefore to prove that the morphism $f_1 = f_{(Y_1)} : X_1 \to Y_1$ is affine. Or, if we set

$$X_1' = X_1 \times_{Y_1} Y_1' = X' \times_{Y'} Y_1' \qquad \text{and} \qquad f_1' = (f_1)_{(Y_1')} = f'_{(Y_1')} : X_1' \longrightarrow Y_1',$$

f_1' is affine by virtue of the hypothesis. We are therefore reduced to proving the corollary by replacing Y, X, f and Y' by Y_1, X_1, f_1 and Y_1'; in other words, it suffices to prove that f is affine when, in the statement of (14.5.11), one assumes in addition that every point of Y admits an open neighbourhood U such that there exists a U-section of $g^{-1}(U)$. The question being local on Y, one may even suppose that there exists a Y-section s of Y'. Now we have the following elementary lemma (valid in any category admitting fibre products):

Lemma (14.5.11.1). — *Let $f : X \to S$, $g : Y \to S$ be two morphisms, $p_1 : X \times_S Y \to X$, $p_2 : X \times_S Y \to Y$ the canonical projections. If $s : S \to Y$ is a section of g, then $s' = (1, s \circ f)_S$ is a section of p_1, and X, equipped with the morphisms $f : X \to S$ and $s' : X \to X \times_S Y$, identifies with the product of the Y-preschemes S and $X \times_S Y$ for the morphisms $s : S \to Y$ and $p_2 : X \times_S Y \to Y$.*

Proof. This is a particular case of (**I**, 3.3.11), where one replaces the diagram by

$$\begin{tikzcd} X \arrow[r, "p_1"] \arrow[d, "f"'] & X \times_S Y \arrow[r, "s'"] \arrow[d, "p_2"] & X \arrow[d, "f"] \\ S \arrow[r, "s"'] & Y \arrow[r, "g"'] & S \end{tikzcd}$$

□

Applying this lemma with S, Y replaced by Y, Y', one sees that one can write $f = (f')_{(Y)}$ for the base change $s : Y \to Y'$, hence f is affine since f' is. Q.E.D.

A variant of this criterion is the following:

Corollary (14.5.12). — *With the notations and general hypotheses of (14.5.11), let $\mathscr{L}$ be an invertible* **IV-3 | 223**
$\mathscr{O}_X$-Module. Suppose that there exists a closed subset Z of X, proper over Y (**II**, 5.4.10)*, and such that the prescheme induced by X on the open subset $X - Z$ is normal. Then, for $\mathscr{L}$ to be ample relative to f, it is necessary and sufficient that $\mathscr{L}' = \mathscr{L} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$ be ample relative to f'.*

Proof. Keeping the notations of the proof of (14.5.11), set $\mathscr{L}_1 = \mathscr{L} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_1}$. By (**III**, 2.6.2), it will suffice to prove that $\mathscr{L}_1$ is ample relative to $f \circ p$; but $f \circ p = h \circ f_1$, and, in view of (**II**, 4.6.13), (v), it will suffice to prove that $\mathscr{L}_1$ is ample relative to f_1. The question being local on Y_1, one may again suppose that g' admits a section s. If $\mathscr{L}_1' = \mathscr{L}' \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y_1'}$, one can then write, by virtue of Lemma (14.5.11.1), $\mathscr{L}_1 = \mathscr{L}_1' \otimes_{\mathscr{O}_{Y_1'}} \mathscr{O}_{Y_1}$ for the base change $s : Y_1 \to Y_1'$; the conclusion follows therefore from two applications of (**II**, 4.6.13), (iii). □

§15. Study of fibres of a universally open morphism

15.1. Multiplicities of fibres of a universally open morphism

Proposition (15.1.1). — *Let* Y *be a locally Noetherian irreducible prescheme, with generic point* η, $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* y *a point of* Y, $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module; we set* $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{Y}}} \mathbf{k}(y)$ *(sheaf of modules on the fibre* $f^{-1}(y)$*). Let* z *be the generic point of an irreducible component of* $\mathrm{Supp}(\mathscr{F}_y)$*, and let* X_i $(1 \leqslant i \leqslant n)$ *be those of the reduced closed sub-preschemes of* X *whose underlying space is an irreducible component of* $\mathrm{Supp}(\mathscr{F})$ *containing* z*, and which are such that the restriction* $\mathrm{X}_i \to \mathrm{Y}$ *of* f *is a universally open morphism at the point* z*; let* z_i *be the generic point of* X_i $(1 \leqslant i \leqslant n)$*. If* $\lambda_x(\mathscr{F}_y)$ *(resp.* $\lambda_t(\mathscr{F}_\eta)$*) denotes the geometric length of* $\mathscr{F}_y$ *at a point* $x \in f^{-1}(y)$ *(resp. that of* $\mathscr{F}_\eta$ *at a point* $t \in f^{-1}(\eta)$*) (4.7.5), then we have* IV-3 | 223

(15.1.1.1)
$$\lambda_z(\mathscr{F}_y) \geqslant \sum_{i=1}^{n} \lambda_{z_i}(\mathscr{F}_\eta).$$

If furthermore we suppose the ring $\mathscr{O}_y$ *regular, then we have*

(15.1.1.2)
$$\mathrm{length}((\mathscr{F}_y)_z) \geqslant \sum_{i=1}^{n} \mathrm{length}((\mathscr{F}_\eta)_{z_i}).$$

Proof. The fibres $f^{-1}(y')$ of f (for all $y' \in \mathrm{Y}$) do not change when we replace f by f_{red}, so we can suppose that X and Y are reduced (2.4.3), (vi), and hence Y integral; we will set $\mathrm{K} = \mathbf{k}(\eta)$. We can furthermore replace f by the restriction $\mathrm{Supp}(\mathscr{F}) \to \mathrm{Y}$ of f to the reduced closed sub-prescheme of X having $\mathrm{Supp}(\mathscr{F})$ as underlying space, and hence restrict to the case where $\mathrm{X} = \mathrm{Supp}(\mathscr{F})$. Finally, if $y = \eta$, there is only a single irreducible component X_i of X containing z (**0**, 2.1.8), and we have $z_i = z$, which renders (15.1.1.1) and (15.1.1.2) trivial (without hypotheses on f). We can thus restrict to the case $y \neq \eta$; it follows then from the hypothesis and from (1.10.3) that the z_i belong to $f^{-1}(\eta)$, so the second members of (15.1.1.1) and (15.1.1.2) are well-defined. Let Z_i be the reduced closed sub-prescheme of $f^{-1}(\eta)$ having $\mathrm{X}_i \cap f^{-1}(\eta)$ as underlying space; we know (4.6.6) that there exists a finite radicial extension K' of K such that, for all i, the K'-prescheme $(\mathrm{Z}_i \otimes_{\mathrm{K}} \mathrm{K}')_{\mathrm{red}}$ is geometrically reduced over K'. Furthermore, the projection morphism $f^{-1}(\eta) \otimes_{\mathrm{K}} \mathrm{K}' \to f^{-1}(\eta)$ is finite, dominant, and radicial (**I**, 3.5.7), hence is a homeomorphism (2.4.5). If z'_i denotes the unique point of $f^{-1}(\eta) \otimes_{\mathrm{K}} \mathrm{K}'$ above z_i, it follows from (4.7.9) and (4.7.5) that we have IV-3 | 224

(15.1.1.3)
$$\lambda_{z_i}(\mathscr{F}_\eta) = \lambda_{z'_i}(\mathscr{F}_\eta \otimes_{\mathrm{K}} \mathrm{K}') = \mathrm{length}((\mathscr{F}_\eta \otimes_{\mathrm{K}} \mathrm{K}')_{z'_i}).$$

Let V be a discrete valuation ring, with field of fractions K', dominating $\mathscr{O}_y$ (**II**, 7.1.7); set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{V})$, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$; if $g : \mathrm{Y}' \to \mathrm{Y}$ is the structural morphism, η' the generic point of Y' and y' its closed point, then we have $g(y') = y$, $g(\eta') = \eta$; the morphism $\mathrm{Spec}(\mathbf{k}(y')) \to \mathrm{Spec}(\mathbf{k}(y))$ being faithfully flat, it is the same for the projection $g' : f'^{-1}(y') \to f^{-1}(y)$ (2.2.13), and hence (2.3.4), there is a generic point z' of an irreducible component of $f'^{-1}(y')$ such that $g'(z') = z$. By construction, we have $\mathbf{k}(\eta') = \mathrm{K}'$, hence $f'^{-1}(\eta') = f^{-1}(\eta) \otimes_{\mathrm{K}} \mathrm{K}'$; on the other hand, if we set $\mathrm{X}'_i = \mathrm{X}_i \times_{\mathrm{Y}} \mathrm{Y}'$, the restriction $\mathrm{X}'_i \to \mathrm{Y}'$ of f' is a universally open morphism at the point z', and hence (1.10.3) there exists a point $z''_i \in f'^{-1}(\eta')$ which is a generization of z', and one can evidently suppose that z''_i is the generic point of an irreducible component of $\mathrm{X}'_i \cap f'^{-1}(\eta')$; as the projection of $\mathrm{X}'_i \cap f'^{-1}(\eta')$ in $f^{-1}(\eta)$ is Z_i, we have $z''_i = z'_i$. By virtue of (4.7.9) and (4.7.5), we have

(15.1.1.4)
$$\lambda_z(\mathscr{F}_y) = \lambda_{z'}(\mathscr{F}'_{y'}) \geqslant \mathrm{length}((\mathscr{F}'_{y'})_{z'})$$

and it follows from this inequality and from (15.1.1.3) that it suffices, in order to establish (15.1.1.1), to prove the inequality

(15.1.1.5)
$$\mathrm{length}((\mathscr{F}'_{y'})_{z'}) \geqslant \sum_{i=1}^{n} \mathrm{length}((\mathscr{F}'_{\eta'})_{z'_i}).$$

Let us now consider the second inequality (15.1.1.2); we will use the following lemma:

Lemma (15.1.1.6). — *Let* A *be a regular local ring that is not a field,* K *its field of fractions, k its residue field. There exists a discrete valuation ring* V *dominating* A*, having* K *as field of fractions, and whose residue field is a pure transcendental extension of k.*

Set $S = \operatorname{Spec}(A)$, and consider the S-scheme P obtained by blowing up the closed point s of S (**II**, 8.1.3); if $\mathfrak{m}$ is the maximal ideal of A, we have by definition $P = \operatorname{Proj}(B)$, where B is the A-algebra graded by $\bigoplus_{n\geqslant 0}\mathfrak{m}^n$. If $h : P \to S$ is the structural morphism, the fibre $h^{-1}(s)$ is isomorphic to $\operatorname{Proj}(B\otimes_A k)$ (**II**, 2.8.10), and by definition $B\otimes_A k$ is the k-algebra graded by $\bigoplus_{n\geqslant 0}\mathfrak{m}^n/\mathfrak{m}^{n+1}$; but as A is regular, this algebra is isomorphic to a polynomial algebra $k[t_1,\dots,t_e]$ (t_i indeterminates) (**0**, 17.1.1) and it follows that $h^{-1}(s)$ is isomorphic to $\mathbf{P}_k^{e-1}$. Let ζ be the generic point of $h^{-1}(s)$; if t'_i ($1 \leqslant i \leqslant e$) are the elements of $\mathfrak{m}$ whose classes $\operatorname{mod}\mathfrak{m}^2$ are the t_i, $B_{(t'_e)}$ is the ring of an affine open neighbourhood of ζ in P, and we have $\mathbf{k}(\zeta) = k(t_1,\dots,t_{e-1})$; on the other hand, as A is integrally closed, one easily verifies that B is as well (Bourbaki, *Alg. comm.*, chap. V, § 1, no 8, cor. 1 of prop. 21), so $B_{(t'_e)}$ is also integrally closed, and hence so is the local ring $\mathscr{O}_{P,\zeta}$. Finally, as $A = \mathscr{O}_s$ is regular, hence a universally catenary ring (5.6.4), we have, by virtue of (5.6.5), $\dim(\mathscr{O}_{P,\zeta}) = \dim(\mathscr{O}_s) - (e-1)$, since h is a birational morphism, $\dim(h^{-1}(s)) = e-1$, and the local ring of the fibre $h^{-1}(s)$ at its generic point ζ is of dimension 0. But $\dim(\mathscr{O}_s) = e$, hence $\dim(\mathscr{O}_{P,\zeta}) = 1$, and $\mathscr{O}_{P,\zeta} = V$ is a discrete valuation ring, Noetherian, of dimension 1 and integrally closed (**II**, 7.1.6); it clearly answers the question. **IV-3 | 225**

This lemma being established, one can resume the reasoning for the inequality (15.1.1.1) with $Y' = \operatorname{Spec}(V)$; this time $\mathbf{k}(y')$ is a *separable* extension of $\mathbf{k}(y)$, and by (4.7.9), we have $\operatorname{length}((\mathscr{F}_y)_z) = \operatorname{length}((\mathscr{F}'_{y'})_{z'})$; as furthermore $f'^{-1}(\eta') = f^{-1}(\eta)$ and $\mathscr{F}'_{\eta'} = \mathscr{F}_\eta$, one is again reduced to proving the inequality (15.1.1.5). Now, if t is a uniformiser of $\mathscr{O}_{y'}$, $f'^{-1}(y')$ is the closed sub-prescheme of X′ defined by the ideal $t\mathscr{O}_{X'}$; on the other hand, we have $\mathscr{F}'_{y'} = \mathscr{F}'/t\mathscr{F}'$; one also verifies immediately (**I**, 9.1.12) that $(\mathscr{F}'_\eta)_{z'} = \mathscr{F}'_{z'_i}$; the inequality to prove (15.1.1.5) is thus nothing but a particular case of (3.4.1.1). □

Corollary (15.1.2). — *The hypotheses being those of (15.1.1), suppose furthermore that $\lambda_z(\mathscr{F}_y) = 1$ (resp. that $\mathscr{O}_y$ is regular and $\operatorname{length}((\mathscr{F}_y)_z) = 1$). Then there exists at most one irreducible component X_i of $\operatorname{Supp}(\mathscr{F})$ containing z and such that the restriction $X_i \to Y$ of f is universally open at the point z, and moreover, if z_i is the generic point of this component, then $\lambda_{z_i}(\mathscr{F}_\eta) = 1$ (resp. $\operatorname{length}((\mathscr{F}_\eta)_{z_i}) = 1$).*

This follows immediately from (15.1.1), noting that one has necessarily, by definition of X_i, $(\mathscr{F}_\eta)_{z_i} \neq 0$ (**I**, 9.1.13).

Corollary (15.1.3). — *Let* Y *be an irreducible locally Noetherian prescheme of generic point η, $f : X \to Y$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-module of support* X*, x a point of* X*, $y = f(x)$. Suppose that: 1° x belongs to a single irreducible component* Z *of $f^{-1}(y)$; 2° $\mathscr{F}_y = \mathscr{F}\otimes_{\mathscr{O}_Y}\mathbf{k}(y)$ is geometrically reduced over $\mathbf{k}(y)$, in other words, if z is the generic point of* Z*, $\lambda_z(\mathscr{F}_y) = 1$ (resp. $\mathscr{O}_y$ is regular and $\operatorname{length}((\mathscr{F}_y)_z) = 1$). Then there exists at most one irreducible component* X′ *of* X *containing x, such that $\dim_x(X'\cap f^{-1}(y)) = \dim_x(f^{-1}(y))$ and that the restriction $X' \to Y$ of f is universally open at the generic points of the irreducible components of $X'\cap f^{-1}(y)$ containing x. Moreover, if such a component* X′ *exists and if z' is its generic point, then $\lambda_{z'}(\mathscr{F}_\eta) = 1$ (resp. $\operatorname{length}((\mathscr{F}_\eta)_{z'}) = 1$).*

Indeed, an irreducible component X′ of X containing x and such that $\dim_x(X'\cap f^{-1}(y)) = \dim_x(f^{-1}(y))$ necessarily contains Z, since an irreducible component of $X'\cap f^{-1}(y)$ containing x must be of the same dimension as the sole irreducible component Z of $f^{-1}(y)$ containing x. One can then apply (15.1.2) to z. **IV-3 | 226**

Remarks (15.1.4). — (i) In the statement of (15.1.1), one cannot replace "universally open" by "equidimensional", as the example (14.4.10), (ii) shows, where we set $\mathscr{F} = \mathscr{O}_X$; the fibres of f are then reduced Artinian schemes, and hence (with the notations introduced *loc. cit.*) we have $\lambda_{x'}(\mathscr{F}_y) = 1$, but there are two irreducible components X_1, X_2 of X containing x', and the right-hand side of (15.1.1.1) is thus equal to 2, even though the restriction of f to each of the components X_1, X_2 is an equidimensional morphism at every point. This same example also shows that in (15.1.2) and (15.1.3), one cannot replace the hypothesis "universally open" by "equidimensional".

(ii) It is plausible that for the validity of the inequality (15.1.1.2), one cannot suppress the hypothesis that $\mathscr{O}_y$ is *regular*.

15.2. Flatness of universally open morphisms with geometrically reduced fibres

(15.2.1). We saw in (2.4.6) that a morphism which is *flat* and locally of finite presentation is *universally open*. We have a partial converse of this result under additional hypotheses:

Theorem (15.2.2). — *Let* Y *be a locally Noetherian prescheme,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-module,* x *a point of* X*,* $y = f(x)$*. Suppose the following conditions are satisfied:*

(i) $\operatorname{Supp}(\mathscr{F}) = \mathrm{X}$.
(ii) f *is universally open at the generic points of the irreducible components of* $f^{-1}(y)$ *containing* x*.*
(iii) $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{Y}}} \mathbf{k}(y)$ *is geometrically reduced at the point* x *(4.6.22).*
(iv) The ring $\mathscr{O}_y$ *is reduced.*

Then $\mathscr{F}$ *is* f*-flat at the point* x*.*

Proof. As $\mathscr{O}_y$ is reduced and Noetherian, we will apply the valuative criterion of flatness (11.8.1). Consider thus a local homomorphism $\mathscr{O}_y \to \mathrm{A}'$, where A' is a discrete valuation ring; set $\mathrm{Y}' = \operatorname{Spec}(\mathrm{A}')$, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$, and let x' be a point of X' whose projections in X and Y' are respectively x and the closed point y' of Y'; if we set $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$, then $\operatorname{Supp}(\mathscr{F}') = \mathrm{X}'$ by (**I**, 9.1.13); every generic point z' of an irreducible component of $f'^{-1}(y')$ containing x' projects in $f^{-1}(y)$ to a generic point z of an irreducible component of $f^{-1}(y)$ containing x, by virtue of (4.2.6); hence f' is universally open at the point z'. Finally, it follows from (4.7.11) that $\mathscr{F}'_{y'}$ is geometrically reduced at the point x'. We see thus that we are reduced to proving the following

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Lemma (15.2.2.1). — *Let* $\mathrm{Y} = \operatorname{Spec}(\mathrm{A})$ *be the spectrum of a discrete valuation ring,* y *its closed point,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* x *a point of* $f^{-1}(y)$*,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-module. Suppose the following conditions are satisfied:*

(i) $\operatorname{Supp}(\mathscr{F}) = \mathrm{X}$.
(ii) f *is open at the generic points of the irreducible components of* $f^{-1}(y)$ *containing* x*.*
(iii) $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_y} \mathbf{k}(y)$ *is reduced at the point* x *(3.2.2).*

Then $\mathscr{F}$ *is* f*-flat at the point* x*.*

Let us denote by t a uniformiser of A, set $\mathrm{B} = \mathscr{O}_x$ and $\mathscr{F}_x = \mathrm{M}$; condition (i) implies that $\operatorname{Supp}(\mathrm{M}) = \operatorname{Spec}(\mathrm{B})$ (**I**, 9.1.13). Condition (ii) implies, by virtue of (14.3.2), that every irreducible component of X containing x dominates Y; in other words, the inverse image in A of every minimal prime ideal $\mathfrak{p}_i$ of B is 0, which means that $\mathrm{M}/t\mathrm{M}$ is a $(\mathrm{B}/t\mathrm{B})$-module *reduced* (3.2.2). It remains to show that under these conditions M is a *torsion-free* A-module (**0**, 6.3.4), and for this it suffices to show that t is an M-*regular* element; but this follows from (3.4.6.1). ☐

Corollary (15.2.3). — *Let* Y *be a locally Noetherian prescheme,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* x *a point of* X*,* $y = f(x)$*. Suppose the following conditions are satisfied:*

(i) f *is universally open at the generic points of the irreducible components of* $f^{-1}(y)$ *containing* x*.*
(ii) $f^{-1}(y)$ *is geometrically reduced over* $\mathbf{k}(y)$ *at the point* x *(4.6.9).*
(iii) The ring $\mathscr{O}_y$ *is reduced.*

Under these conditions f *is flat at the point* x*. It suffices to apply (15.2.2) with* $\mathscr{F} = \mathscr{O}_{\mathrm{X}}$ *(4.6.22).*

Remarks (15.2.4). — (i) The first three conditions of (15.2.2) do not change when we replace f by f_{red}; but if $\mathfrak{N}$ is the nilradical of a Noetherian local ring A, $\mathrm{A}/\mathfrak{N}$ is flat over $\mathrm{A}/\mathfrak{N}$ but not over A itself when $\mathfrak{N} \neq 0$ (Bourbaki, *Alg. comm.*, chap. II, § 3, no 2, cor. 2 of prop. 5). One thus sees that one cannot, in (15.2.2), suppress condition (iv).

(ii) It follows from (2.4.6) that if the conclusion of (15.2.2) is true, and also hypothesis (i), f is universally open in a neighbourhood of x, and in particular at the generic points of the irreducible components of $f^{-1}(y)$ containing x. Even if conditions (i), (iii), and (iv) are satisfied, one cannot replace (ii) by the weaker hypothesis that f is universally open *at the point* x *only*. This is what the example (14.1.3), (i) shows, where f is evidently universally open at every point of X_2, hence in particular at the point x, intersection of X_1 and X_2; conditions (ii) and (iii) of (15.2.3) are also satisfied by this example. However, $\mathscr{O}_x$ is not a torsion-free $\mathscr{O}_y$-module: $\mathscr{O}_y$ identifies with a subring of $\mathscr{O}_x$ that contains zero divisors in $\mathscr{O}_x$; hence f is not flat at the point x. **IV-3** | 228

(iii) In (15.2.3), the conclusion is not necessarily valid when we replace hypothesis (ii) by the weaker hypothesis that $f^{-1}(y)$, considered as a $\mathbf{k}(y)$-prescheme, *has no immersed associated prime cycles*. An example is provided by taking for Y a curve having a cusp, for example $Y = \mathrm{Spec}(K[S,T]/(S^3 - T^2))$, K being an algebraically closed field, and for X its *normalisation* (**II**, 6.3.8). If s and t are the classes of S and T in $A = K[S,T]/(S^3 - T^2)$, one verifies immediately that $X = \mathrm{Spec}(B)$, where $B = K[s,t,u]$, u being the element t/s of the field of fractions of A, and since $s = u^2$, $t = u^3$, we also have $B = K[u]$, isomorphic to a polynomial ring in one indeterminate over K. The only point x of X above the point y corresponding to the maximal ideal $(s) + (t)$ is such that, as the class $\bar{u}$ of u in $\mathscr{O}_x/\mathfrak{m}_y\mathscr{O}_x$ satisfies $\bar{u}^2 = 0$, $f^{-1}(y)$ is not a reduced $\mathbf{k}(y)$-prescheme. Here f is a finite morphism (2.4.5), but it is not flat at the point x, for if $\mathscr{O}_x$ were a flat $\mathscr{O}_y$-module, it would be a free $\mathscr{O}_y$-module (Bourbaki, *Alg. comm.*, chap. II, § 3, no 2, prop. 5), which is absurd.

(iv) Let us finally show that in (15.2.2) or (15.2.3), one cannot replace "geometrically reduced" by "reduced". We will in fact define two rings A, A′ having the following properties:
   1) A is a Noetherian local ring of maximal ideal $\mathfrak{m}$, integral, complete, of dimension 1 and geometrically unibranch.
   2) A′ is the integral closure of A, a finite A-algebra whose maximal ideal is $\mathfrak{m}A'$, and its residue field k' is a non-trivial finite radicial extension of the residue field $k = A/\mathfrak{m}$ of A.

One can then take $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(A')$, $\mathscr{F} = \mathscr{O}_X$, y and x being the closed points of Y and X respectively; the hypotheses (i) and (iv) of (15.2.2) are trivially satisfied; as A is of dimension 1, $f : X \to Y$ is radicial, finite, and surjective, hence a universal homeomorphism (2.4.5), which proves hypothesis (i) of (15.2.3). Finally, it is clear that $f^{-1}(y)$ is reduced. However, f is not flat at the point x, for if A′ were a flat A-module, it would be a free A-module of rank 1 (since it has the same field of fractions as A) generated by the element 1 of A (Bourbaki, *Alg. comm.*, chap. II, § 3, no 2, prop. 5), which is absurd.

To construct the rings A and A′, let us start from an imperfect field k of characteristic $p > 0$; let $K_0 = k((T))$ be the field of formal Laurent series over k, and let A_0 be its valuation ring for the discrete valuation given by the order of a formal Laurent series; the residue field of A_0 is thus k. Let α be an element of k that is not a p-th power, and let us take $k' = k(\alpha^{1/p})$; $K = k'((T))$ is then a finite radicial extension of K_0, and $A' = A_0 \otimes_k k'$ is the discrete valuation ring for K. If $\mathfrak{m}'$ is the maximal ideal of A′, conditions 1) and 2) are satisfied by taking $A = A_0 + \mathfrak{m}'$: indeed, as A_0 is Noetherian and A′ is an A_0-module of finite type, A is a finite A_0-algebra, hence a Noetherian ring; since A′ is finite over A, $\mathfrak{m}'$ is the only maximal ideal of A, which is thus a complete local ring, evidently of dimension 1 (being finite over A_0) and geometrically unibranch, since A′ is integrally closed and has the same field of fractions K as A.

(v) The proof of (15.2.2) shows however that one can weaken condition (iii) by replacing "geometrically reduced" by "reduced" when one can apply the valuative criterion of flatness (11.8.1) using only discrete valuation rings A′ whose residue field k' is *separable* over $\mathbf{k}(y)$; indeed, if x' and z' have the same meanings as in (15.2.2), the radicial multiplicities $\mu_i(\mathbf{k}(z')|k')$ and $\mu_i(\mathbf{k}(z)|\mathbf{k}(y))$ are the same, by virtue of (4.7.3), hence it follows from (4.7.9) that $\mathrm{length}((\mathscr{F}_y)_z)$ and $\mathrm{length}((\mathscr{F}'_{y'})_{z'})$ are equal, and we are again reduced to the case where Y is the spectrum of a discrete valuation ring and y its closed point. For example, one obtains this stronger result when $\mathscr{O}_y$ is *unibranch* and dominated by a discrete valuation ring whose residue field is a separable extension of $\mathbf{k}(y)$, by virtue of (11.6.2).[18] This is the case, for instance, when $\mathscr{O}_y$ is a *regular* ring, by (15.1.1.6). But, as Hironaka has pointed

[18] The printed text cites (11.6.4), which does not occur in §11.6; the applicable unibranch descent result is Corollary (11.6.2).

out, there are Noetherian local rings that are *integrally closed*, of dimension 2 (coming from algebraic local schemes over imperfect fields) that do not satisfy the preceding condition.

15.3. Applications: criteria for reduction and irreducibility

Proposition (15.3.1). — *Let* Y *be a locally Noetherian prescheme,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite* **IV-3** | 228
type, $\mathscr{F}$ *a coherent* $\mathscr{O}_\mathrm{X}$*-module,* x *a point of* X, $y = f(x)$*. Suppose conditions* (i), (ii), *and* (iii) *of* (15.2.2) *are satisfied. Then:*

- *(i) There exists a neighbourhood* U *of* x *in* X *such that* $f|\mathrm{U}$ *is universally open and that, for all* $x' \in \mathrm{U}$, $\mathscr{F}_{f(x')}$ *is geometrically reduced over* $\mathbf{k}(f(x'))$ *at the point* x'.
- *(ii) If furthermore* Y *is reduced at the point* y*, there exists a neighbourhood* $\mathrm{U}_1 \subset \mathrm{U}$ *of* x *such that* $\mathscr{F}|\mathrm{U}_1$ *is* f*-flat and reduced.*

Proof. Taking into account (**I**, 5.1.8), (4.2.7), and (4.7.11), neither the hypotheses nor conclusion (i) of **IV-3** | 229
the statement change when we replace Y by $\mathrm{Y}_{\mathrm{red}}$ and X by $\mathrm{X} \times_\mathrm{Y} \mathrm{Y}_{\mathrm{red}}$, so we may suppose Y *reduced*. It then follows from (15.2.2) that $\mathscr{F}$ is f-flat at the point x, hence, by (11.1.1), in a neighbourhood of x; *a fortiori*, f is universally open in this neighbourhood (2.4.6). The fact that $\mathscr{F}_{f(x')}$ is then geometrically reduced at the point x' for all the points of a neighbourhood of x follows from (12.1.1), (vii). Assertion (ii) follows from this and from (3.3.4). □

Proposition (15.3.2). — *The notations being those of* (15.3.1)*, suppose that conditions* (i), (ii), (iii) *of* (15.2.2) *are satisfied, and furthermore that* $f^{-1}(y)$ *is equidimensional at the point* x*. Then* f *is equidimensional at the point* x*.*

Proof. We have, for all $y' \in \mathrm{Y}$, $\mathrm{Supp}(\mathscr{F}_{y'}) = f^{-1}(y')$. By hypothesis $\mathscr{F}_y$ is reduced (and satisfies (S_1)) and is equidimensional at the point x. By virtue of (12.1.1), (ii), one can thus suppose that $\mathscr{F}_{f(x')}$ is equidimensional and of constant dimension $e = \dim_x(f^{-1}(y))$ at the point x' for all $x' \in \mathrm{U}$, which means that $f^{-1}(f(x'))$ is equidimensional and of dimension e at the point x'. Since moreover $\mathscr{F}|\mathrm{U}$ is f-flat, it follows from (2.3.4) and (13.3.1), a) that f is equidimensional at the point x. □

Corollary (15.3.3). — *The notations being those of* (15.3.1)*, suppose the following conditions are satisfied:*

- *(i)* $\mathrm{Supp}(\mathscr{F}) = \mathrm{X}$.
- *(ii)* f *is universally open at the generic points of the irreducible components of* $f^{-1}(y)$ *containing* x.
- *(iii)* $\mathscr{F}_y$ *is geometrically pointwise integral over* $\mathbf{k}(y)$ *at the point* x (4.6.22).
- *(iv)* Y *is geometrically unibranch at the point* y.

Then x *belongs to a single irreducible component of* X*. If furthermore* Y *is integral at the point* y *and* $\mathscr{F} = \mathscr{O}_\mathrm{X}$*, then* X *is integral at the point* x *and* f *is flat at the point* x*.*

Proof. Note first that (i) and (iii) imply that x belongs to a single irreducible component Z of $f^{-1}(y)$. It follows thus from (14.4.7) and from (15.3.2) that for every irreducible component X_i of X containing x (and hence containing Z), the restriction f_i of f to X_i is equidimensional at the point x and universally open at the generic point of Z. The first assertion of the statement then follows from (15.1.3) and the second from (15.3.1). □

Remarks (15.3.4). — (i) The example (14.4.10), (ii) shows that, in (15.3.3), one cannot suppress the hypothesis that Y is geometrically unibranch at the point y.

(ii) The remark (15.2.4), (vi) shows that the conclusion of (15.3.3) is still true under the following hypotheses: $\mathscr{O}_y$ is regular, $\mathrm{Supp}(\mathscr{F}) = \mathrm{X}$, and $\mathscr{F}_y$ is integral at the point x. It is unknown whether, in this statement, the hypothesis that $\mathscr{O}_y$ is regular can be replaced by the hypothesis that it is geometrically unibranch, or even integral and integrally closed.

15.4. Complements on Cohen-Macaulay morphisms

Proposition (15.4.1). — *Let* Y *be a locally Noetherian prescheme, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_{\mathrm{X}}$-module such that $\mathrm{Supp}(\mathscr{F}) = \mathrm{X}$, x a point of* X, *$y = f(x)$. We set, for all $z \in \mathrm{X}$, $\mathrm{X}_{f(z)} = f^{-1}(f(z))$, $\mathscr{F}_{f(z)} = \mathscr{F} \otimes_{\mathscr{O}_{f(z)}} \mathbf{k}(f(z))$. Suppose that $\mathscr{F}$ is f-flat at the point x and that $\mathscr{F}_y$ is a Cohen-Macaulay $\mathscr{O}_{\mathrm{X}_y}$-module at the point x. Then f is equidimensional at the point x.* **IV-3** | 230
Indeed, by (11.1.1) and (12.1.1), (vi), there exists a neighbourhood U *of x such that $\mathscr{F}|\mathrm{U}$ is f-flat and, for every $x' \in \mathrm{U}$, $\mathscr{F}_{f(x')}$ is a Cohen-Macaulay $\mathscr{O}_{\mathrm{X}_{f(x')}}$-module at the point x'; this latter property implies that $\mathscr{F}_{f(x')}$ is equidimensional at the point x' and that x' belongs to no embedded prime cycle associated with $\mathscr{F}_{f(x')}$* (**0**, 16.5.4)*. Furthermore, by virtue of (12.1.1), (ii), one can suppose that the dimensions of the irreducible components of $\mathrm{Supp}(\mathscr{F}_{f(x')})|(\mathrm{U} \cap \mathrm{X}_{f(x')})$ have a (common) value independent of x'. As $\mathrm{Supp}(\mathscr{F}_{f(x')}) = \mathrm{X}_{f(x')}$ by hypothesis, it follows from (2.3.4) and (13.3.1), a) that f is equidimensional at the point x.*

Proposition (15.4.2). — *Let* Y *be a locally Noetherian prescheme, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_{\mathrm{X}}$-module such that $\mathrm{Supp}(\mathscr{F}) = \mathrm{X}$, x a point of* X, *$y = f(x)$. Suppose that $\mathscr{O}_y$ is regular and that $\mathscr{F}$ is a Cohen-Macaulay $\mathscr{O}_{\mathrm{X}}$-module at the point x. Then (with the notations of (15.4.1)) the following conditions are equivalent:*

a) *$\mathscr{F}$ is f-flat at the point x and $\mathscr{F}_y$ is a Cohen-Macaulay $\mathscr{O}_{\mathrm{X}_y}$-module at the point x.*
b) *$\mathscr{F}$ is f-flat at the point x.*
c) *f is universally open in a neighbourhood of x.*
d) *f is open at the generic points of the irreducible components of X_y containing x.*
e) *$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_y) + \dim((\mathscr{F}_y)_x)$.*
e′) *$\dim(\mathscr{O}_{\mathrm{X},x}) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_{\mathrm{X},x} \otimes_{\mathscr{O}_y} \mathbf{k}(y))$.*

As $\mathrm{Supp}(\mathscr{F}) = \mathrm{X}$, conditions e) and e′) are equivalent by definition (5.1.12). Condition a) trivially implies b); b) implies that $\mathscr{F}$ is f-flat at the points of a neighbourhood of x (11.1.1), hence b) implies c) (2.4.6); c) trivially implies d), and d) implies e′) by (14.2.1). Finally, since $\mathscr{O}_y$ is regular and $\mathscr{F}_x$ is a Cohen-Macaulay $\mathscr{O}_{\mathrm{X},x}$-module, it follows from (6.1.4) that e) implies a).

Proposition (15.4.3). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism locally of finite presentation, $\mathscr{F}$ an $\mathscr{O}_{\mathrm{X}}$-module of finite presentation, such that $\mathrm{Supp}(\mathscr{F}) = \mathrm{X}$, x a point of* X, *$y = f(x)$. Then (with the notations of (15.4.1)) the following conditions are equivalent:*

a) *$\mathscr{F}$ is f-flat at the point x and $\mathscr{F}_y$ is a Cohen-Macaulay $\mathscr{O}_{\mathrm{X}_y}$-module at the point x.*
b) *There exists an open neighbourhood* U *of x in* X *and a quasi-finite* Y*-morphism $g : \mathrm{U} \to \mathrm{Y}' = \mathrm{Y}[\mathrm{T}_1, \dots, \mathrm{T}_n]$, where the T_i are indeterminates, such that $\mathscr{F}|\mathrm{U}$ is g-flat at the point x.*

Proof. Let us first show that *b*) implies *a*). Set $h : \mathrm{Y}' \to \mathrm{Y}$; the $\mathbf{k}(y)$-prescheme $\mathrm{Y}'_y = h^{-1}(y)$ is regular (**0**, 17.3.7); let A be the local ring of this prescheme at the point $y' = g(x)$, k its residue field, and B the local ring of X_y at the point x; set further $(\mathscr{F}_y)_x = \mathrm{M}$, which is a B-module of finite type. The hypothesis that the morphism g is quasi-finite implies the same for $g_y : \mathrm{X}_y \to \mathrm{Y}'_y$ (**II**, 6.2.4), hence $\mathrm{M} \otimes_{\mathrm{A}} k$ is a $(\mathrm{B} \otimes_{\mathrm{A}} k)$-module of finite length (**II**, 6.2.2); since M is a flat A-module by hypothesis and A is a Cohen-Macaulay ring, M is a Cohen-Macaulay B-module (6.3.3). The fact that $\mathscr{F}$ is f-flat at the point x follows from its being g-flat and from the fact that the morphism h is flat (2.1.6).

Let us now show that *a*) implies *b*). The question being local on X and Y, we may suppose that X and Y are affine; using (8.9.1) and (11.2.7), we reduce to the case where X and Y are Noetherian, taking into account (6.7.1). The hypothesis implies that f is equidimensional at the point x (15.4.1), whence there exists a quasi-finite morphism $g : \mathrm{U} \to \mathrm{Y}'$ such that every irreducible component of U dominates Y' (13.3.1), *b*) and such that $g_y : \mathrm{U}_y \to \mathrm{Y}'_y$ is finite and surjective (13.3.1.1). On the other hand, to prove that $\mathscr{F}|\mathrm{U}$ is g-flat at the point x, it suffices, since $h : \mathrm{Y}' \to \mathrm{Y}$ is flat, to show by the fibrewise criterion for flatness (11.3.10) that $\mathscr{F}_y$ is g_y-flat at the point x. But by hypothesis $\mathscr{F}_y$ is a Cohen-Macaulay $\mathscr{O}_{\mathrm{X}_y}$-module at the point x, and Y'_y is a regular prescheme; moreover, since g_y is finite and surjective, it satisfies condition e′) of (15.4.2) by (5.6.10). It therefore suffices, in order to conclude, to apply (15.4.2) to g_y and $\mathscr{F}_y$. □

15.5. Separable rank of fibres of a quasi-finite and universally open morphism. Application to geometric connected components of fibres of a proper morphism

Proposition (15.5.1). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a separated and quasi-finite morphism. For every $z \in Y$, let $n(z)$ be the geometric number of points of $f^{-1}(z)$ (or* separable rank *of $f^{-1}(z)$ over $\kappa(z)$; cf.* (**I**, 6.4.8)*). We have the following properties:*

- *(i) The function $z \mapsto n(z)$ is lower semi-continuous at every point $y \in Y$ such that f is universally open at the points of $f^{-1}(y)$.*
- *(ii) Let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$; if the function $z \mapsto n(z)$ is constant in a neighbourhood of y, then f is proper at the point y (15.7.1).*
- *(iii) Suppose that f is universally open and that every point of Y is geometrically unibranch; then, if the function $z \mapsto n(z)$ is locally constant in Y, the irreducible components of X are pairwise disjoint.*

Proof. (i) Note that since f is quasi-finite, the number $n(z)$ is the geometric number of connected components of $f^{-1}(z)$; we know that the function $z \mapsto n(z)$ is locally constructible (9.7.9); taking into account ($\mathbf{0}_{\mathrm{III}}$, 9.3.4), we are reduced to proving the **IV-3 | 231**

Corollary (15.5.2). — *Under the general hypotheses of (15.5.1), let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$. Then, if y' is a generization of y, we have*

$$n(y) \leqslant n(y'). \tag{15.5.2.1}$$

For every base change $g : Z \to Y$, $f_{(Z)} : X_{(Z)} \to Z$ is separated, quasi-finite, and universally open at the points of $X_{(Z)}$ projecting into $f^{-1}(y)$; moreover, if z, z' are two points of Z such that $g(z) = y$, $g(z') = y'$ and z' is a generization of z, we have $n(z) = n(y)$ and $n(z') = n(y')$ ((**I**, 6.4.12)); it therefore suffices to prove (15.5.2.1) for a suitable g. We can evidently suppose $y \neq y'$. We will use the following lemma:

Lemma (15.5.2.2). — *Under the general hypotheses of (15.5.1), if y is a point of Y and y' is a generization of y distinct from y, there exists a scheme Z that is the spectrum of a discrete valuation ring, with closed point z and generic point z', and a morphism $g : Z \to Y$ such that: 1° $g(z) = y$, $g(z') = y'$; 2° if one sets $f' = f_{(Z)}$, $n(y)$ is the number of points of $f'^{-1}(z)$ and $n(y')$ is the number of points of $f'^{-1}(z')$.*

Indeed, there exists a discrete valuation ring A_1 and a morphism g_1 from $Z_1 = \operatorname{Spec}(A_1)$ to Y such that if z_1 and z'_1 are the closed point and the generic point of Z_1, we have $g_1(z_1) = y$ and $g_1(z'_1) = y'$ ((**II**, 7.1.9)); we can therefore already suppose that Y is the spectrum of a discrete valuation ring A, y its closed point and y' its generic point. For every $x \in f^{-1}(y)$, $\kappa(x)$ is a finite extension of $\kappa(y)$ by hypothesis ((**II**, 6.2.2)); we deduce from (4.5.11) that there exists a finite algebraic extension k of $\kappa(y)$ such that the *number of points* of $f^{-1}(y) \otimes_{\kappa(y)} k$ is equal to $n(y)$. Since there exists a discrete valuation ring A_2 dominating A whose residue field is finite over $\kappa(y)$ and contains k ((**II**, 7.1.2)), we can in the second place suppose that $n(y)$ is the number of points of $f^{-1}(y)$. Finally, the same reasoning shows that there is a *finite* extension K of $\kappa(y')$ such that the number of points of $f^{-1}(y') \otimes_{\kappa(y')} K$ is equal to $n(y')$; if w is a valuation of $\kappa(y')$ associated to A, there is a discrete valuation of K extending w, and the ring A_3 of this valuation dominates A and has K as field of fractions, and satisfies the conditions of the lemma. **IV-3 | 232**

We can therefore henceforth suppose that Y is the spectrum of a discrete valuation ring, y its closed point, y' its generic point, and that $n(y)$ and $n(y')$ are the numbers of points of $f^{-1}(y)$ and $f^{-1}(y')$, so that for every $x \in f^{-1}(y)$ (resp. every $x' \in f^{-1}(y')$) we have $\kappa(x) = \kappa(y)$ (resp. $\kappa(x') = \kappa(y')$).

Denote then by X_i the reduced sub-preschemes of X whose underlying spaces are the irreducible components of X ($1 \leqslant i \leqslant m$). By virtue of (1.10.4), the restriction $X_i \to Y$ of f to each X_i is a *dominant* morphism, and since $f^{-1}(y')$ is discrete, each intersection $X_i \cap f^{-1}(y')$ consists only of the generic point of X_i, whence (denoting by $n_i(z)$ the geometric number of points of $X_i \cap f^{-1}(z)$ for every $z \in Y$), $n_i(y') = 1$ for every i, and $n(y') = \sum_{i=1}^{m} n_i(y') = m$. Furthermore, $f^{-1}(y)$ is the union of the $X_i \cap f^{-1}(y)$, whence

$$n(y) \leqslant \sum_{i=1}^{m} n_i(y), \tag{15.5.2.3}$$

and to prove (15.5.2.1), it suffices to establish that $n_i(y) \leqslant 1$ for every i. In other words, we can suppose X *integral*, and the morphism f birational. But then, for every $x \in f^{-1}(y)$, we have $\mathscr{O}_y \subset \mathscr{O}_x \subset K$, where $K = \kappa(y')$ is the field of fractions of $\mathscr{O}_y$. Since $\mathscr{O}_y$ is a valuation ring and $\mathscr{O}_x$ dominates $\mathscr{O}_y$, we necessarily have $\mathscr{O}_y = \mathscr{O}_x$ ((**II**, 7.1.1)). There exists then a Y-section $h : Y \to X$ such that $h(y) = x$ ((**I**, 2.4.4)); since Y is reduced and X is separated over Y, such a section is *unique* ((**I**, 7.2.2)), so the set $f^{-1}(y)$ contains only a single point, which completes the proof of (i).

(ii) As at the beginning of (i), we are reduced to showing that if the two members of (15.5.2.1) are equal for *every* generization y' of y, f is proper at the point y. Thanks to the criterion of local properness (15.7.5) and to the remarks at the beginning we can further suppose that $Y = \mathrm{Spec}(A)$ is the spectrum of a discrete valuation ring A, y its closed point and y' its generic point, and it remains to show that f is proper.

Let us use (15.5.2.2) and note that if A' is a discrete valuation ring dominating A, A' is a torsion-free A-module, so the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is faithfully flat and quasi-compact, and it therefore amounts to the same to say that f is proper or that the morphism deduced from f by the base change $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is proper (2.7.1), (vii); we can therefore suppose that $n(y)$ and $n(y')$ are the numbers of points of the fibres $f^{-1}(y)$ and $f'^{-1}(y')$ respectively. With the notations of (i), we have then by (15.5.2.3) and (15.5.2.1) **IV-3** | 233

$$n(y) \leqslant \sum_i n_i(y) \leqslant \sum_i n_i(y') = n(y') \tag{15.5.2.4}$$

and for the extreme members to be equal, we need $n_i(y) = n_i(y') = 1$ for every i. As we saw in (i), if $n_i(y) = 1$ for every i, $X_i \cap f^{-1}(y)$ is nonempty and the restriction $X_i \to Y$ of f is an isomorphism; since the X_i are closed sub-preschemes of X, this implies that f is proper ((**II**, 5.4.5)).

(iii) We can restrict to the case where Y is affine and reduced, and since Y is by hypothesis locally integral ((**I**, 5.1.4)), we can suppose Y integral, and the function $y \mapsto n(y)$ constant in Y. Let again X_i ($1 \leqslant i \leqslant m$) be the irreducible components of X. It follows from (1.10.4) that if y' is the generic point of Y, each of the $X_i \cap f^{-1}(y')$ consists of a single point; the restriction $f_i : X_i \to Y$ of f being a quasi-finite and dominant morphism and every point of Y being geometrically unibranch, it follows from the criterion of Chevalley (14.4.4) that f_i is *universally open*. So then for any point y of Y, we have $n_i(y) \leqslant n_i(y')$; on the other hand, since the $X_i \cap f^{-1}(y')$[19] are pairwise disjoint, $\sum_i n_i(y') = n(y')$; we therefore still have the relations (15.5.2.4). But by hypothesis the extreme members are equal, so $n(y) = \sum_i n_i(y)$, which implies that the $X_i \cap f^{-1}(y)$ are pairwise disjoint, and this completes the proof. □

Combining this proposition with the connectivity theorem of Zariski and its consequences ((**III**, 4.3)), we obtain the following results that complement ((**III**, 4.3.7) and (**III**, 4.3.10)):

Proposition (15.5.3). — *Let Y be an irreducible locally Noetherian prescheme, $f : X \to Y$ a proper morphism; for every $y \in Y$, let $n(y)$ be the geometric number of connected components* (4.5.2) *of $f^{-1}(y)$. Suppose that the restriction of f to every irreducible component of X is dominant onto Y. Then, if y_0 is a geometrically unibranch point of Y, the function $y \mapsto n(y)$ is lower semi-continuous at the point y_0.*

Proof. Consider the Stein factorisation $f = g \circ f'$ ((**III**, 4.3.3)), where $g : Y' \to Y$ is a finite morphism and $f' : X \to Y'$[20] is a surjective morphism whose fibres are geometrically connected ((**III**, 4.3.4)). Since f' is surjective, the inverse image under f' of every irreducible component of Y' contains at least one irreducible component of X; therefore each irreducible component of Y' dominates Y. We can therefore apply the criterion of Chevalley (14.4.4) to the restrictions of g to each of these irreducible components, and we conclude that g is *universally open at every point* of $g^{-1}(y_0)$. The conclusion therefore follows from (15.5.2) and from the fact that the number of geometrically

[19] The printed source has $X_i \cap f^{-1}(y)$ here; the following equality $\sum_i n_i(y') = n(y')$ and the generic-fibre argument require y'.

[20] The printed source has $f' : X \to Y$ here; the stated Stein factorisation and the remainder of the proof require target Y'.

connected components of $f^{-1}(y)$ is equal to the geometric number of points of $g^{-1}(y)$ ((**III**, 4.3.4)). ☐

Corollary (15.5.4). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a proper morphism; let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$; then, with the notations of (15.5.3), the function $z \mapsto n(z)$ is lower semi-continuous at the point y.* **IV-3** | 234

Proof. With the notations of (15.5.3), note that in the Stein factorisation $f = g \circ f'$, f' is surjective; since f is universally open at the points of $f^{-1}(y)$, g is universally open at the points of $g^{-1}(y)$ ((14.3.4), (i)); it then suffices to apply to g the proposition (15.5.1), (i) in view of ((**III**, 4.3.4)). ☐

Remarks (15.5.5). — (i) Even if Y is integral and normal, X integral, f finite and surjective (hence universally open by virtue of the criterion of Chevalley (14.4.4)), the function $y \mapsto n(y)$ is not necessarily locally constant. We have an example by taking $Y = \mathrm{Spec}(k[\mathrm{T}])$ where k is algebraically closed, and $X = \mathrm{Spec}(A)$, where $A = k[\mathrm{S}, \mathrm{T}]/(\mathrm{S}^2 + \mathrm{T}^2 - 1)$; at the points y', y'' of Y corresponding to the maximal ideals $(\mathrm{T} - 1)$ and $(\mathrm{T} + 1)$, we have $n(y') = n(y'') = 1$, but $n(y) = 2$ at all other points of Y. We shall give below an additional condition ensuring that $y \mapsto n(y)$ is locally constant (15.5.7).

(ii) The example (14.4.10) (ii) shows that in (15.5.1), (iii), one cannot suppress the hypothesis that the points of Y are geometrically unibranch.

(iii) The example (11.7.5) shows that the conclusion of (15.5.1), (i) is no longer valid if one only supposes that the morphism f is *open* (even if, as is the case in the cited example, f is finite and surjective).

(iv) Finally, in (15.5.1), (iv), one cannot dispense with the hypothesis that the morphism f is *separated*, as shown by the example where X is the affine line in which a point x has been "doubled" ((**I**, 5.5.11)), and Y is the affine line: here f is a local isomorphism (hence flat) and is quasi-finite, but $z \mapsto n(z)$ is not lower semi-continuous.

Lemma (15.5.6). — *Let Y be the spectrum of a discrete valuation ring, a its closed point, b its generic point. Let $f : X \to Y$ be an open morphism locally of finite type. Suppose that X is connected and that the fibre $f^{-1}(a)$ is a reduced prescheme. Then $f^{-1}(b)$ is connected.*

Proof. We will use the following purely topological lemma (a special case of a more general result from Ch. III, 3rd Part):

Lemma (15.5.6.1). — *Let X be a locally Noetherian topological space, every closed irreducible subset of which admits a generic point. Let Z be a nowhere dense closed subset of X having the following property: for every $x \in Z$, if one denotes by T_x the set of generizations of x in X, then $\mathrm{T}_x - \{x\}$ is connected. Under these conditions, if X is connected, $X - Z$ is connected.*

Proof. Let us give an independent proof. Arguing by contradiction, suppose that the open set $U = X - Z$, which is dense in X since Z is nowhere dense, is the union of two disjoint nonempty open sets U' and U'', and denote by X' and X'' the closures in X of U' and U''. Since $X = \overline{U} = X' \cup X''$ and X is connected, we have $X' \cap X'' \neq \varnothing$, and it is clear that $X' \cap X'' \subset Z$. Let x be a generic point of an irreducible component of $X' \cap X''$. Since X is locally Noetherian, there is an open neighbourhood V of x in X such that $V \cap U'$ and $V \cap U''$ have only a finite number of irreducible components; since x is adherent to U' and to U'', it is necessarily in the closure of an irreducible component Z' of $V \cap U'$ and in the closure of an irreducible component Z'' of $V \cap U''$. But $\overline{Z'}$ (resp. $\overline{Z''}$) is closed and irreducible in X, hence admits a generic point z' (resp. z''), and we necessarily have $z' \in U'$ and $z'' \in U''$. This proves that the intersections of $\mathrm{T}_x - \{x\}$ with U' and U'' are nonempty. But by definition $(X' \cap X'') \cap (\mathrm{T}_x - \{x\}) = \varnothing$; therefore the intersections of X' and of X'' with $\mathrm{T}_x - \{x\}$ are two nonempty disjoint closed sets in $\mathrm{T}_x - \{x\}$, whose union is $\mathrm{T}_x - \{x\}$; this contradicts the hypothesis that $\mathrm{T}_x - \{x\}$ is connected. ☐

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Proof of (15.5.6). To prove (15.5.6), we apply the lemma (15.5.6.1) to X and to its nowhere dense closed subset $Z = f^{-1}(a)$, which is nowhere dense because f is open. Note that by hypothesis, taking into account (15.2.2.1), this implies that f is *flat* at the points of $f^{-1}(a)$. For any such point x, $\mathscr{O}_x$ is therefore an $\mathscr{O}_a$-module without torsion ($\mathbf{0}_{\mathrm{I}}$, 6.3.4), and in particular, if t is a uniformiser of $\mathscr{O}_a$, t is an $\mathscr{O}_x$-regular element; moreover $\mathscr{O}_x/t\mathscr{O}_x$ is reduced by hypothesis. If it is of depth 0, it is therefore a field (**0**, 16.4.7), and as $t\mathscr{O}_x$ is then the maximal ideal of $\mathscr{O}_x$, $\mathscr{O}_x$ is a discrete valuation ring (**0**, 17.1.4); hence $\operatorname{Spec}(\mathscr{O}_x) - \{x\}$ consists of a single point, and *a fortiori* is connected. If on the contrary $\operatorname{depth}(\mathscr{O}_x/t\mathscr{O}_x) \geqslant 1$, we have $\operatorname{depth}(\mathscr{O}_x) \geqslant 2$ since t is $\mathscr{O}_x$-regular (**0**, 16.4.6); it then follows from the theorem of Hartshorne (5.10.7) that $\operatorname{Spec}(\mathscr{O}_x) - \{x\}$ is connected, which completes the proof. □

Proposition (15.5.7). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a proper morphism. For every $z \in Y$, let $n(z)$ be the geometric number of connected components of $f^{-1}(z)$. Let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$ and that $f^{-1}(y)$ is geometrically reduced over $\kappa(y)$* (4.6.2). *Then the function $z \mapsto n(z)$ is constant in a neighbourhood of y.*

Proof. We already know from (15.5.4) that under the conditions of the statement, $z \mapsto n(z)$ is lower semi-continuous at the point y; it remains to see that it is also upper semi-continuous, and by the same reasoning as at the beginning of the proof of (15.5.1), (i), it suffices to show that if y' is a generization of y, we have

$$n(y') \leqslant n(y). \tag{15.5.7.1}$$

Using the argument at the beginning of the proof of (15.5.2) and Lemma (15.5.2.2), applied to the finite morphism g in the Stein factorisation $f = g \circ f'$ (recalling that the geometric number of points of $g^{-1}(y)$ is equal to the number of connected components of $f^{-1}(y)$ ((**III**, 4.3.4))), we reduce to the case where $n(y)$ and $n(y')$ are respectively the numbers of connected components of $f^{-1}(y)$ and $f^{-1}(y')$, and where Y is the spectrum of a discrete valuation ring, y is its closed point, and y' is its generic point. We can moreover replace X by any one of its connected components, these being open and closed in X, and proper over Y. Suppose therefore that X is connected. The hypothesis that f is open at the points of $f^{-1}(y)$ implies that if $f^{-1}(y) \neq \varnothing$, then $f^{-1}(y') \neq \varnothing$ ($\mathbf{0}_{\mathrm{I}}$, 1.10.3); the hypothesis that f is closed implies that if $f^{-1}(y') \neq \varnothing$, then $f^{-1}(y) \neq \varnothing$ ((**II**, 7.2.1)). So, if $X \neq \varnothing$ (which we can evidently suppose, the proposition being trivially true in the contrary case), $f^{-1}(y)$ and $f^{-1}(y')$ are both nonempty. Moreover, the hypothesis implies that the prescheme $f^{-1}(y)$ is *reduced* (4.6.1); since f is open at the points of $f^{-1}(y)$, the lemma (15.5.6) implies that $f^{-1}(y')$ is connected, in other words $n(y') = 1$. Since $f^{-1}(y)$ is not empty, this proves (15.5.7.1). □

Remarks (15.5.8). — The relation (15.5.7.1) is not necessarily valid if f is not supposed proper, even if it satisfies the other hypotheses of (15.5.7). With the notations of (15.5.5), (i), it suffices to consider the restriction of f to $X - \{x'\}$, where x' is one of the two points of X above a point y distinct from y' and y''; we then have $n(y) = 1$, whereas if η is the generic point of Y, $n(\eta) = 2$.

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Proposition (15.5.9). — *Let $f : X \to Y$ be a flat and locally of finite presentation morphism; for every $y \in Y$, let $n(y)$ be the geometric number of connected components of $f^{-1}(y)$.*

- *(i) If f is separated and quasi-finite, the function $y \mapsto n(y)$ (then equal to the geometric number of points of $f^{-1}(y)$) is lower semi-continuous on Y; if it is constant in a neighbourhood of y_0, f is proper at y_0.*
- *(ii) If f is proper, the function $y \mapsto n(y)$ is lower semi-continuous on Y; if in addition $f^{-1}(y_0)$ is geometrically reduced over $\kappa(y_0)$, $n(y)$ is constant in a neighbourhood of y_0.*

Proof. In all cases, the questions are local on Y, so we can suppose Y affine and f of finite presentation; using (8.9.1), (8.10.5) and (11.2.7), we reduce to the case where Y is Noetherian (and using moreover (8.2.11) for the second assertion of (i)). Since f is then moreover universally open (2.4.6), it suffices to apply (15.5.1), (15.5.4) and (15.5.7) to conclude. □

Remarks (15.5.10). — We note that we have thus obtained another proof of (12.2.4), (vi). Conversely, one can deduce (15.5.7) from (12.2.4), (vi): indeed, one can restrict to the case where Y is reduced (replacing f by f_{red}), and it then follows from the hypotheses of (15.5.7) and from (15.2.3) that f is flat in an open neighbourhood of $f^{-1}(y)$; since moreover f is proper, we can suppose that this neighbourhood is of the form $f^{-1}(U)$, where U is an open neighbourhood of y in Y. We conclude then by (12.2.4), (vi). The proof of (15.5.7) given above has the advantage of exhibiting the result (15.5.6), which is of independent interest.

15.6. Connected components of fibres along a section

Proposition (15.6.1). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a locally of finite type morphism, $g : Y \to X$ a Y-section of X ((**I**, 2.5.5)). For every $y \in Y$, denote by X_y^0 the connected component of $f^{-1}(y)$ containing the point $g(y)$. Let y be a point of Y, and let y' be a generization of y in Y. Then:*

(i) If there exists an irreducible component Z of $X^0_{y'}$, of dimension equal to $\dim(X^0_{y'})$ and containing $g(y')$ (as will be the case if $X^0_{y'}$ is irreducible), then

$$\dim(X_y^0) \geqslant \dim(X^0_{y'}). \tag{15.6.1.1}$$

(ii) If X_y^0 is moreover irreducible and if the two members of (15.6.1.1) are equal, then $X_y^0 \subset \overline{X^0_{y'}}$ (closure in X).

Proof. (i) Let $\overline{Z}$ be the closure of Z in X; since y is a specialization of y' and g is continuous, $g(y) \in \overline{Z}$. Since $Z = \overline{Z} \cap f^{-1}(y')$, it follows from the semi-continuity theorem of Chevalley ((13.1.3)) applied to the restriction of f to the closed reduced sub-prescheme of X having $\overline{Z}$ as underlying space, that $\dim_{g(y)}(\overline{Z} \cap f^{-1}(y)) \geqslant \dim(Z)$; but the irreducible components of $\overline{Z} \cap f^{-1}(y)$ containing $g(y)$ are evidently contained in X_y^0, whence *a fortiori* the inequality (15.6.1.1).

(ii) The reasoning of (i) shows moreover that if the two members of (15.6.1.1) are equal, we necessarily have $\dim_{g(y)}(\overline{Z} \cap f^{-1}(y)) = \dim_{g(y)}(X_y^0)$; if X_y^0 is irreducible, $\overline{Z} \cap f^{-1}(y) = X_y^0$, so X_y^0 is contained in $\overline{Z}$, and *a fortiori* in $\overline{X^0_{y'}}$. □

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Corollary (15.6.2). — *With the notations of (15.6.1), suppose moreover that for every $y \in Y$, X_y^0 is irreducible. Then the function $z \mapsto \dim(X_z^0)$ is upper semi-continuous on Y. If moreover this function is constant in a neighbourhood of a point $y \in Y$, then $X_y^0 \subset \overline{X^0_{y'}}$ for every generization y' of y.*

Proof. Let y be a point of Y, and let U be an open affine neighbourhood of $g(y)$ in X; then there is an open neighbourhood V of y in Y such that $g(V) \subset U$, and for every $z \in V$, $U \cap X_z^0$ is dense in X_z^0, hence $\dim(X_z^0) = \dim(U \cap X_z^0)$ ((4.1.1.3)). Replacing X by U, we can suppose that f is a morphism of finite type. We then know from (9.7.10) that the function $z \mapsto \dim(X_z^0)$ is locally constructible, and the assertions of the corollary follow from (15.6.1) and from ($\mathbf{0}_{\mathrm{III}}$, 9.3.4). □

Proposition (15.6.3). — *With the notations of (15.6.1), suppose that for every $y \in Y$, X_y^0 is geometrically irreducible (4.5.2). Then, for every $y \in Y$ such that the function $z \mapsto \dim(X_z^0)$ is constant in a neighbourhood of y, f is universally open at the points of X_y^0.*

Proof. Apply the criterion (14.3.7). Let therefore $h : Y' \to Y$ be a morphism, with Y' the spectrum of a discrete valuation ring whose closed and generic points we denote by t and t' respectively, and suppose that $h(t) = y$, so that $y' = h(t')$ is a generization of y. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, $g' = g_{(Y')} : Y' \to X'$. With the same notations as in (15.6.1), the hypothesis on X_y^0 implies that $X'^0_t = X^0_y \otimes_{\kappa(y)} \kappa(t)$ and $X'^0_{t'} = X^0_{y'} \otimes_{\kappa(y')} \kappa(t')$, and that X'^0_t and $X'^0_{t'}$ are irreducible (4.4.1); since $\dim(X_y^0) = \dim(X'^0_t)$ and $\dim(X^0_{y'}) = \dim(X'^0_{t'})$ (4.1.4), we see that it suffices to prove the proposition for f' and X'^0_t. In other words, we can restrict to the case where Y is the spectrum of a discrete valuation ring, y its closed point and y' its generic point; if x' is the generic point of $X^0_{y'}$, it follows from (15.6.2) that every point of X_y^0 is a specialization of x', whence the conclusion. □

Proposition (15.6.4). — *Under the general hypotheses of (15.6.1), let X^0 be the union of the X_y^0 as y runs through Y. If $y \in Y$ is such that X_y^0 is geometrically reduced over $\kappa(y)$ (4.6.2) and if f is universally open at* *every point of X^0_y, then X^0 is a neighbourhood of X^0_y in X. In particular, if f is universally open and if, for every $y \in Y$, X^0_y is geometrically reduced over $\kappa(y)$, X^0 is open in X.*

Proof. Let us first show that we can reduce to the case where f is a morphism of *finite type*. Let x be a point of X^0_y; it follows from (5.10.8.1) (with $\mathfrak{F} = \{\varnothing\}$) that there is a finite sequence $(Z_i)_{0 \leqslant i \leqslant n}$ of irreducible components of $f^{-1}(y)$ such that $g(y) \in Z_0$, $x \in Z_n$ and $Z_i \cap Z_{i+1} \neq \varnothing$ for $0 \leqslant i \leqslant n-1$; the Z_i being irreducible, there is a finite sequence (U_i) ($1 \leqslant i \leqslant n$) of open affine subsets of $f^{-1}(y)$ such that $g(y) \in U_1$, $x \in U_n$, and U_i is an open affine neighbourhood of a point x_i of $Z_i \cap Z_{i-1}$ for $1 \leqslant i \leqslant n$. There exists for each i a quasi-compact open set W_i in X such that $U_i = f^{-1}(y) \cap W_i$; let W be the quasi-compact open set of X which is the union of the W_i. Since g is continuous, there is an open affine neighbourhood V of y in Y such that $g(V) \subset W$; if $X' = W \cap f^{-1}(V)$, the restriction of f to X' is of finite type. Moreover, for every $z \in V$, X'^0_z is the connected component of $X' \cap f^{-1}(z)$ containing $g(z)$, and we have $X'^0_z \subset X^0_z$, while x belongs to X'^0_y. Indeed, each U_i contains the two irreducible sets $U_i \cap Z_{i-1}$ and $U_i \cap Z_i$, which meet, and the irreducible sets $U_i \cap Z_{i-1}$ and $U_{i-1} \cap Z_{i-1}$ **IV-3** | 238
meet for $2 \leqslant i \leqslant n$; hence the union of the $U_i \cap Z_{i-1}$ and the $U_i \cap Z_i$, for $1 \leqslant i \leqslant n$, is connected and contains x and $g(y)$. Since $f|_{X'}$ is universally open at the points of X'^0_y, this completes the proof of our assertion: if the proposition has been proved when f is of finite type, the union X'^0 of the X'^0_z, for $z \in V$, is a neighbourhood of x, and therefore so is X^0.

Suppose therefore that f is of finite type. Then X^0 is locally constructible (9.7.12); it suffices therefore to show that X^0 contains every generization x' of x ($\mathbf{0}_{\text{III}}$, 9.2.5). Set $y' = f(x')$; there exists a discrete valuation ring A and a morphism u from $Y' = \operatorname{Spec}(A)$ to X such that, if z is the closed point and z' the generic point of Y', we have $u(z) = x$ and $u(z') = x'$ ((**II**, 7.1.9)); so $h = f \circ u$, and therefore $h(z) = y$ and $h(z') = y'$. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, which is a morphism of finite type universally open at the points lying over X^0_y,[21] and $g' = g_{(Y')} : Y' \to X'$, which is a Y'-section of X'. If $p_y : f'^{-1}(z) \to f^{-1}(y)$ (resp. $p_{y'} : f'^{-1}(z') \to f^{-1}(y')$) is the canonical projection, $p_y^{-1}(X^0_y)$ (resp. $p_{y'}^{-1}(X^0_{y'})$) is the connected component of $f'^{-1}(z)$ (resp. $f'^{-1}(z')$) containing $g'(z)$ (resp. $g'(z')$), since the connected components X^0_t are geometrically connected (4.5.13) and it suffices to apply (4.4.1). Moreover, the morphism u corresponds to a Y'-section v of X' such that $p_y(v(z)) = x$ and $p_{y'}(v(z')) = x'$, so that $v(z')$ is a generization of $v(z)$; it will therefore suffice to prove that $v(z')$ belongs to $p_{y'}^{-1}(X^0_{y'})$.[22] Finally, it is clear that $p_y^{-1}(X^0_y)$ is geometrically reduced over $\kappa(z)$.

In other words, we are reduced to the case where Y is the spectrum of a discrete valuation ring, y its closed point, y' its generic point. Since $f^{-1}(y) - X^0_y$ is then closed in X, replacing X by the open set $X - (f^{-1}(y) - X^0_y)$, we can suppose that $X^0_y = f^{-1}(y)$; similarly, one may replace X by the open subset that is the connected component of X containing $g(Y)$, this connected component evidently containing X^0; in other words, we can suppose that X is *connected*. The proposition will then be established if we show that $X^0 = X$, or equivalently that $X_{y'}$ is connected; but $f^{-1}(y')$ is geometrically reduced over $\kappa(y')$, and *a fortiori reduced*; moreover, f is open at the points of $f^{-1}(y)$, so we can apply the lemma (15.5.6), whence the conclusion. □

Corollary (15.6.5). — *Let $f : X \to Y$ be a flat morphism of finite presentation, g a Y-section of X; for every $y \in Y$, let X^0_y be the connected component of $f^{-1}(y)$ containing $g(y)$, and let X^0 be the union of the X^0_y as y runs through Y. Then, for every $y \in Y$ such that X^0_y is geometrically reduced over $\kappa(y)$, X^0 is a neighbourhood of X^0_y in X. In particular, if f is reduced (6.8.1), X^0 is open in X.*

Proof. The question being local on Y, we can suppose $Y = \operatorname{Spec}(A)$ affine. There exist therefore a Noetherian subring A_0 and a flat morphism of finite type $f_0 : X_0 \to Y_0 = \operatorname{Spec}(A_0)$ such that $X = X_0 \times_{Y_0} Y$ and $f = (f_0)_{(Y)}$ ((8.9.1) and (11.2.7)); moreover, we can suppose that there exists a Y_0-section g_0 of X_0 such that $g = (g_0)_{(Y)}$ (8.8.2). For every $y \in Y$, let y_0 be its projection in Y_0; it follows from (4.5.13) that $(X_0)^0_{y_0}$ is geometrically connected, so, if $p : f^{-1}(y) \to f_0^{-1}(y_0)$ is the

[21] The printed source says that f' is universally open at the points of $f'^{-1}(z)$; the hypothesis only gives universal openness over the distinguished connected component X^0_y.

[22] The printed source writes $v(y)$ and $v(y')$ in this sentence; the closed and generic points of Y' were denoted by z and z'.

canonical projection, we have $X_y^0 = p^{-1}((X_0)_{y_0}^0)$ (4.4.1); moreover, since f is flat, the hypothesis that X_y^0 is geometrically reduced over $\kappa(y)$ implies that $(X_0)_{y_0}^0$ is geometrically reduced over $\kappa(y_0)$ (4.6.10). We are thus reduced to the case where Y is moreover *Noetherian*; since f is universally open ((11.1.1)), it suffices then to apply (15.6.4). □ **IV-3** | 239

Proposition (15.6.6). — *Let $f : X \to Y$ be a morphism such that Y is locally Noetherian and f is locally of finite type, or such that f is of finite presentation. Let g be a Y-section of X; for every $y \in Y$, let X_y^0 be the connected component of $f^{-1}(y)$ containing $g(y)$; finally, let X^0 be the union of the X_y^0 as y runs through Y. Suppose that, for every $y \in Y$, X_y^0 is geometrically irreducible. Then:*

(i) The function $y \mapsto \dim(X_y^0)$ is upper semi-continuous in Y.

(ii) Let $x_0 \in X^0$, $y_0 = f(x_0)$. If the function $y \mapsto \dim(X_y^0)$ is constant in a neighbourhood of y_0, there exists an open neighbourhood U of y_0 such that f is universally open at all points of $X^0 \cap f^{-1}(U)$. If moreover Y is reduced and $f^{-1}(y_0)$ geometrically reduced over $\kappa(y_0)$ at the point x_0, f is flat at the point x_0.

(iii) Conversely, suppose that f is universally open at the generic point of $X_{y_0}^0$ and in addition that one of the following conditions is satisfied:

α) The fibre $f^{-1}(y_0)$ is geometrically reduced over $\kappa(y_0)$ at the point $g(y_0)$ and the ring $\mathscr{O}_{Y,y_0}$ is Noetherian.

β) For every generic point y' of an irreducible component of Y containing y_0, we have $\dim(f^{-1}(y')) = \dim(X_{y'}^0)$.

Then the function $y \mapsto \dim(X_y^0)$ is constant in a neighbourhood of y_0.

(iv) The set W of points $x \in X^0$ such that the function $y \mapsto \dim(X_y^0)$ is constant in a neighbourhood of $f(x)$ and $X_{f(x)}^0$ is geometrically reduced over $\kappa(f(x))$, is open in X.

Proof. The questions being local on Y, we can suppose that $Y = \operatorname{Spec}(A)$ is affine.

When f is of finite presentation, there exists a Noetherian subring A_1 of A and a morphism of finite type $f_1 : X_1 \to Y_1 = \operatorname{Spec}(A_1)$ such that $X = X_1 \times_{Y_1} Y$ and $f = (f_1)_{(Y)}$ (8.9.1); moreover, we can suppose that there exists a Y_1-section g_1 of X_1 such that $g = (g_1)_{(Y)}$ (8.8.2). For every $y \in Y$, let y_1 be its projection in Y_1; it follows from (4.5.13) that $(X_1)_{y_1}^0$ is geometrically connected, so, if $p : f^{-1}(y) \to f_1^{-1}(y_1)$ is the canonical projection, $X_y^0 = p^{-1}((X_1)_{y_1}^0)$ (4.4.1). Note moreover that one can, by virtue of (9.7.11) and (9.7.12), apply (9.3.3) to the property of being geometrically irreducible; we can therefore suppose A_1 chosen so that $(X_1)_{y_1}^0$ is geometrically irreducible for every $y_1 \in Y_1$. We have $\dim(X_y^0) = \dim((X_1)_{y_1}^0)$ (4.1.4), and, applying (9.3.3) once more to the property of having a prescribed dimension (and using (9.5.5) and (9.7.12)), we can suppose (restricting as needed Y to a neighbourhood of y_0) that if $\dim(X_y^0)$ is constant in Y, then $\dim((X_1)_{y_1}^0)$ is constant in Y_1. If Y is reduced, so is Y_1 since $A_1 \subset A$; moreover, if $f^{-1}(y_0)$ is geometrically reduced over $\kappa(y_0)$ at the point x_0, then $f_1^{-1}(y_{01})$ is geometrically reduced at the point x_{01} (x_{01} and y_{01} being the respective projections of x_0 and y_0 in X_1 and Y_1) (4.6.10).

These remarks show that to prove (i) and (ii), we can restrict to the case where Y is Noetherian and f locally of finite type. But in this case, assertion (i) follows from (15.6.2) and the first assertion of (ii) in (15.6.3). Moreover, if Y is reduced and $f^{-1}(y_0)$ is geometrically reduced over $\kappa(y_0)$ at the point x_0 (with Y still assumed Noetherian), then, since X_y^0 is the unique irreducible component of $f^{-1}(y)$ containing x, we deduce from (15.6.3) and (15.2.3) that if $\dim(X_y^0)$ is constant in a neighbourhood of y_0, f is flat at the point x_0. **IV-3** | 240

To prove (iii), let us first remark that the set X^0 is locally constructible in X (9.7.12), so, by (9.5.5), the set E of $y \in Y$ such that $\dim(X_y^0) = \dim(X_{y_0}^0)$ is locally constructible in Y. Apply then (1.10.1) to E: it suffices to prove, taking (i) into account, that $\dim(X_{y'}^0) = \dim(X_{y_0}^0)$ for the generic point y' of an irreducible component of Y containing y_0; this already allows us to suppose that $Y = \operatorname{Spec}(\mathscr{O}_{Y,y_0})$.

If we are in case α), then, by virtue of (**II**, 7.1.9), and using (4.5.13) and (4.4.1) as in (15.6.4), we reduce immediately to the case where Y is the spectrum of a discrete valuation ring, y_0 its closed point and y' its generic point. But then the hypotheses imply, by virtue of (15.2.2.1), that f is *flat* at the point $g(y_0)$, and also in a neighbourhood V of this point in X ((11.1.1)). To prove our assertion, we can replace X by V, for $V \cap X^0_{y'}$ is nonempty by virtue of (2.3.4), hence is an everywhere dense open set in $X^0_{y'}$ and therefore has the same dimension (4.1.1.3); moreover, since $X^0_{y'}$ is also the connected component of $f^{-1}(y')$ containing $g(y')$, we can suppose V chosen such that it meets no other irreducible component of $f^{-1}(y_0)$ or of $f^{-1}(y')$, in other words, we can suppose that $X^0 = X$; but then the conclusion follows from (12.1.1), (i), since by hypothesis $X^0_{y_0}$ is integral.

Suppose now that we are in the case β). Apply this time ((**II**, 7.1.4)) and ((**II**, 7.1.9)) in the case α): we are then reduced to the case where Y is the spectrum of a discrete valuation ring (not necessarily discrete), y_0 its closed point and y' its generic point. By virtue of (14.3.13), there exists an irreducible component Z of X containing $X^0_{y_0}$ and dominant Y, and such that $\dim(Z \cap f^{-1}(y')) = \dim(X^0_{y_0})$; but the hypothesis β) and the fact that $\dim(Z \cap f^{-1}(y')) \leqslant \dim(f^{-1}(y'))$ show that $\dim(X^0_{y_0}) \leqslant \dim(X^0_{y'})$ whence $\dim(X^0_{y_0}) = \dim(X^0_{y'})$ by virtue of (i).

It remains to prove (iv). Note first that the sets envisaged do not change when we replace f by $f_{(Y_{\text{red}})} : X \times_Y Y_{\text{red}} \to Y_{\text{red}}$, the projection $X \times_Y Y_{\text{red}} \to X$ being a homeomorphism. In other words, we can suppose Y reduced, and then it follows from (ii) that f is *flat* at the points of W. Consider a point x_0 of W and let us prove that W is a neighbourhood of x_0; proceeding as at the beginning of the proof, and using (11.2.7), we can also suppose Y *Noetherian*; it then follows from (ii) and from (15.6.4) that X^0 is a neighbourhood of x_0 in X, and from (12.1.1), (vii) that W is also a neighbourhood of x_0 in X. □

Corollary (15.6.7). — *Assume that the preliminary conditions of (15.6.6) are satisfied by f, and suppose moreover that for every $y \in Y$, X^0_y is geometrically integral over $\kappa(y)$. Then the following conditions are equivalent:*

a) *The function $y \mapsto \dim(X^0_y)$ is locally constant in Y.*

b) *The morphism $f_{(Y_{\text{red}})} : X \times_Y Y_{\text{red}} \to Y_{\text{red}}$ deduced from f by base change, is flat at the points of X^0.*

Moreover, these conditions imply that X^0 is open in X. Finally, when Y is locally Noetherian, a) and b) are also equivalent to **IV-3** | 241

b′) *f is universally open at the points of X^0.*

Proof. The fact that *a*) implies *b*) follows from (15.6.6), (ii), together with the fact that X^0 is then open in X. If *b*) is satisfied, one can restrict to the case where Y is reduced and f flat; then, we reduce, as at the beginning of (15.6.6), and using moreover (11.2.7), to the case where Y is Noetherian, case in which the conclusion follows from (15.6.6), (iii), case α)). The equivalence of *b*) and *b′*) has already been proved when Y is locally Noetherian (15.2.2.1), taking into account the fact that the morphism $X \times_Y Y_{\text{red}} \to X$ is a universal homeomorphism. □

Proposition (15.6.8). — *Let $f : X \to Y$ be a separated morphism of finite presentation, g a Y-section of X; for every $y \in Y$, let X^0_y be the connected component of $f^{-1}(y)$ containing $g(y)$, and let X^0 be the union of the X^0_y as y runs through Y. Then, if $y \in Y$ is such that X^0_y is proper over $\kappa(y)$, X^0 is proper over Y at the point y (i.e., (15.7.1), there exists an open neighbourhood U of y in Y such that $X^0 \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ and proper over U).*

Proof. Proceeding as at the beginning of (15.6.6) (where one uses (8.10.5), (v), and where one replaces (9.7.11) by (9.6.7)), we reduce to the case where Y is Noetherian and f of finite type. We then know from (9.7.12) that X^0 is locally constructible in X; moreover, the X^0_z are geometrically connected (for every $z \in Y$) by virtue of (4.5.13); finally, as $g(Y) \subset X^0$, X^0 is universally submersive over Y (15.7.8). It suffices then to apply to X^0 the criterion (15.7.9). □

Corollary (15.6.9). — *Let $f : X \to Y$ be a separated morphism such that Y is locally Noetherian and f locally of finite type, or that f is of finite presentation. Let g be a Y-section of X, and for every $y \in Y$, let X^0_y be the connected component of $f^{-1}(y)$ containing $g(y)$. Let y be a point of Y such that X^0_y is proper over $\kappa(y)$. Then, for every generization y' of y, $\dim(X^0_y) \geqslant \dim(X^0_{y'})$.*

Proof. Suppose first that f is of finite presentation; then, replacing Y by a neighbourhood of y if necessary, we may suppose that $f|_{X^0}$ is *proper* (15.6.8), and it suffices to apply (13.1.5) to this restriction. If Y is locally Noetherian and f locally of finite type, X is locally Noetherian; moreover, the hypothesis that X^0_y is proper over $\kappa(y)$ implies that X^0_y is Noetherian. There therefore exists a Noetherian open subset $V \subset X$ containing X^0_y; hence there is an open neighbourhood W of y in Y such that $g(W) \subset V$. Moreover, if z is a maximal point of $X^0_{y'}$, we may suppose that $z \in V$. The restriction of f to $V \cap f^{-1}(W)$ is then a morphism of finite type, and $g|_W$ is a W-section; applying to this morphism the result already obtained, we see that if Z is the irreducible component of $X^0_{y'}$ with generic point z, then $\dim(Z) \leqslant \dim(X^0_y)$. Since z is an arbitrary maximal point of $X^0_{y'}$, we indeed have $\dim(X^0_{y'}) \leqslant \dim(X^0_y)$. □

Remarks (15.6.10). — (i) The supplementary hypothesis on the existence of an irreducible component of $X^0_{y'}$ containing $g(y')$ and of dimension equal to $\dim(X^0_{y'})$, made in (15.6.1), is not superfluous, as shown by the following example. Let A be a discrete valuation ring, π a uniformiser of A, K the field of fractions of A, and k its residue field. Set $Y = \operatorname{Spec}(A)$, and denote by y and y' the closed point and the generic point of Y. Consider the two Y-schemes: $X_1 = \operatorname{Spec}(A[t])$, $X_2 = \operatorname{Spec}(K[u,v])$, where t, u, v are indeterminates (we note that the projection of X_2 to Y is therefore $\{y'\}$). In the ring $A[t]$, the principal ideal $(\pi t - 1)$ is *maximal*, and thus corresponds to a closed point x_1 of X_1, which projects to the *generic point* y' of Y and is such that $\kappa(x_1) = K$. Consider also **IV-3** | 242
the closed point x_2 of X_2 corresponding to the maximal ideal $\left(u - \frac{1}{\pi}\right) + (v)$ of $K[u,v]$; we have also $\kappa(x_2) = K$. As one will see in Ch. V, one can therefore *reglue* X_1 and X_2 along two closed subpreschemes Z_1 and Z_2 of X_1 and X_2 having respectively $\{x_1\}$ and $\{x_2\}$ as underlying spaces, by means of the unique Y-isomorphism from Z_1 onto Z_2. Denote by X the Y-prescheme thus obtained, and by x the point of X which is the image of x_1 and x_2. The "null section" g of X_1 ((**II**, 1.7.9)) is then also a Y-section of X; $f^{-1}(y)$ is equal to $(X_1)_y$, whereas $f^{-1}(y')$ has two irreducible components, respectively isomorphic to $(X_1)^0_{y'}$ and $(X_2)^0_{y'}$, having the unique point x in common and having respective dimensions 1 and 2. See, however, Proposition (15.6.9).

(ii) Consider the morphism f defined in (12.2.3), (ii), which is proper and flat; moreover, the restriction of f to $X_1 \cap f_0^{-1}(Y)$ is an isomorphism, so the inverse morphism $g : Y \to X$ is a Y-section of X. We then have $\dim(X_{y_0}) = 1$, whereas $\dim(X^0_y) = 0$ for $y \neq y_0$, although X^0_y is geometrically irreducible for every $y \in Y$ (but $X^0_{y_0}$ is not reduced); one therefore sees that in (15.6.6), (iii), the hypotheses α) and β) cannot be omitted. Moreover, X^0 is not a neighbourhood of $X^0_{y_0}$ in X, which shows that in (15.6.4) one cannot dispense with the hypothesis that $X^0_{y_0}$ is reduced.[23]

(iii) In Ch. VI, we will apply the preceding results to *Y-group preschemes* locally of finite type over a locally Noetherian prescheme Y. If G is such a prescheme, there exists the canonical Y-section g, the "unit section", and one shows that G^0_y is always geometrically irreducible and $\dim(G^0_y) = \dim(G_y)$, setting $G_y = f^{-1}(y)$. It then follows from (15.6.2) and (15.6.3) that the function $z \mapsto \dim(G_z)$ is upper semi-continuous, and that if it is constant in a neighbourhood of a point y, f is universally open at the points of G^0_y. If moreover G_y is geometrically reduced over $\kappa(y)$ at one of its points, one shows that G_y is *smooth* (6.8.1) over $\kappa(y)$ at all its points; it will follow then from (15.6.6) that if moreover $\mathscr{O}_y$ is reduced, then also f is smooth at all the points of G^0_y (though not necessarily at all points of G_y). These results will apply for example in the theory of Picard schemes, where we shall use a general theorem ensuring that, under certain conditions, the function $z \mapsto \dim(G_z)$ is locally constant. Let us remark that it is with applications of this nature in view that the statements such as (15.6.1) are given for morphisms *locally* of finite type, and not only for morphisms of finite type.

We note also that in the case of a Y-group prescheme G, the hypothesis β) of (15.6.6) is always satisfied.

[23] The printed source has X^y here; the preceding sentence and Proposition (15.6.4) require $X^0_{y_0}$.

15.7. Appendix: Valuative criteria for local properness This section gives complements to the valuative criterion of properness proved in ((**II**, 7.3.10)); it is independent of the rest of § 15.

Definition (15.7.1). — Let $f : X \to Y$ be a morphism of preschemes, y a point of Y. We say that f is *proper at the point* y if there exists an open neighbourhood U of y in Y such that the restriction $f^{-1}(U) \to U$ of f is a proper morphism. We say that a subset Z of X is *proper over* Y *at the point* y if there exists an open neighbourhood U of y such that $Z \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ and proper over U ((**II**, 5.4.10)) (which amounts to saying that for every closed sub-prescheme of X having Z as underlying space, the restriction of f to Z is proper at the point y).

Let $g : Y' \to Y$ be any morphism; set $X' = X \times_Y Y'$, $f' = f_{(Y')}$ and, if $p : X' \to X$ is the canonical projection, $Z' = p^{-1}(Z)$. Then, if Z is proper over Y at the point y, Z' is proper over Y' at every point y' above y ((**II**, 5.4.2)).

Proposition (15.7.2). — *Let Y be an integral locally Noetherian prescheme, X an integral prescheme, $f : X \to Y$ a separated, dominant morphism of finite type, and y a point of Y. The following conditions are equivalent:*

a) *f is proper at the point y.*
b) *If $Y_1 = \operatorname{Spec}(\mathscr{O}_y)$, $X_1 = X \times_Y Y_1$, the morphism $f_1 = f_{(Y_1)} : X_1 \to Y_1$ is proper.*
c) *Every discrete valuation ring having $R(X)$ as field of fractions and dominating $\mathscr{O}_y$ dominates a local ring $\mathscr{O}_x$ of X (in which case necessarily $f(x) = y$).*

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Proof. The fact that $a)$ implies $b)$ follows from ((**II**, 5.4.2)), and the implication $b) \Rightarrow c)$ follows from ((**II**, 7.3.10)). It remains therefore to show that $c)$ implies $a)$. The question is local on Y, so we may suppose that Y is affine, hence Noetherian. By Chow's lemma ((**II**, 5.6.1)), there exist an integral prescheme X', a projective morphism $p : P \to Y$, a dominant open immersion $j : X' \to P$, and a projective birational (hence surjective) morphism $g : X' \to X$ such that the diagram

$$\begin{tikzcd} P \arrow[d, "p"'] & X' \arrow[l, "j"'] \arrow[d, "g"] \\ Y & X \arrow[l, "f"] \end{tikzcd}$$

is commutative. Since X' is integral and j is dominant, P is irreducible, and, replacing P by P_{red}, we may suppose that P is integral ((**I**, 5.2.2) and (**II**, 5.5.5), (vi)). It suffices to prove that there exists an open neighbourhood U of y such that $p^{-1}(U) \subset j(X')$, for then the restriction of j to

$$j^{-1}(p^{-1}(U)) = g^{-1}(f^{-1}(U)) = V$$

will be an isomorphism onto $p^{-1}(U)$. Hence the restriction $V \to U$ of $f \circ g$ will be proper, and, since the restriction $V \to f^{-1}(U)$ of g is surjective, the restriction $f^{-1}(U) \to U$ of f will be proper ((**II**, 5.4.3)). Since $T = P - j(X')$ is closed in P, $p(T)$ is closed in Y, and it suffices to show that $y \notin p(T)$, or even that every $z' \in P$ such that $p(z') = y$ *belongs* to $j(X')$. Now, the field of rational functions $R(P)$ is equal to $R(X')$, hence to $R(X)$ by construction; since one can restrict to the case where z' is not the generic point of P, there is a discrete valuation ring A having $R(X)$ as field of fractions and dominating $\mathscr{O}_{z'}$ ((**II**, 7.1.7)); since $p(z') = y$, A dominates $\mathscr{O}_y$, hence by hypothesis there exists $x \in X$ such that $f(x) = y$ and A dominates $\mathscr{O}_x$. Since g is proper, there exists $x' \in X'$ such that $g(x') = x$ and A dominates $\mathscr{O}_{x'}$ ((**II**, 7.3.10)); therefore z' and $j(x')$ are two points of P whose local rings $\mathscr{O}_{z'}$ and $\mathscr{O}_{j(x')} = \mathscr{O}_{x'}$ are related by ((**I**, 8.1.4)); since P is a scheme, this implies $z' = j(x')$ ((**I**, 8.2.2)), which completes the proof. $\square$

Remark (15.7.3). — In (15.7.2), one may omit the hypothesis that Y is locally Noetherian by replacing, in condition $c)$, the discrete valuation rings by arbitrary valuation rings; the proof is then unchanged, taking ((**II**, 7.3.10)) into account.

Corollary (15.7.4). — *Let Y be a Noetherian prescheme, $f : X \to Y$ a separated morphism of finite type, Z a subset of X such that the maximal points of its closure $\overline{Z}$ belong to Z (which is the case when Z is finite, or when Z is constructible ($\mathbf{0}_{\text{III}}$, 9.2.2)). Let y be a point of Y. The following conditions are equivalent:*

a) *The set $\overline{Z}$ is proper over Y at the point y.*
b) *If $Y_1 = \operatorname{Spec}(\mathscr{O}_y)$, $X_1 = X \times_Y Y_1$, and if $g_1 : X_1 \to X$ is the canonical projection and $Z_1 = g_1^{-1}(Z)$, the closure $\overline{Z_1}$ of Z_1 in X_1 is proper over Y_1.*
c) *For every scheme $Y' = \operatorname{Spec}(A')$, where A' is a discrete valuation ring, and every morphism $g : Y' \to Y$, such that the image $g(y')$ of the closed point y' of Y' is y, we have the following property: setting $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, $g' = g_{(X)} : X' \to X$, $Z' = g'^{-1}(Z)$, and denoting by η' the generic point of Y', for every point $x' \in Z' \cap f'^{-1}(\eta')$ rational over $\kappa(\eta')$, there exists a Y'-section of X' containing x'.* **IV-3** | 244

Proof. The fact that a) implies b) follows from the fact that $\overline{Z_1} \subset g_1^{-1}(\overline{Z})$ and from the fact that $g_1^{-1}(\overline{Z})$ is proper over Y_1 ((15.7.1)). The fact that b) implies c) follows from ((**II**, 7.3.3)). It remains therefore to see that c) implies a). It clearly suffices to prove that every irreducible component of $\overline{Z}$ is proper over Y at the point y, and the hypothesis allows us therefore to restrict to the case where Z is reduced to a single point z. We may also, taking into account ((**I**, 5.2.2)) and ((**II**, 5.4.6)), suppose that X and Y are reduced; then, by the definition of a subset proper over Y ((**II**, 5.4.10)), suppose that $\overline{Z} = X = \overline{\{z\}}$ and, applying ((**I**, 5.2.2)) once more, that f is dominant. Thus X and Y are integral and Noetherian, and it remains to prove that f is proper at the point y of Y. For this, let us show that f satisfies condition c) of ((15.7.2)). Let A' be a discrete valuation ring having $R(X) = \kappa(z)$ as field of fractions and dominating $\mathscr{O}_y$; with the notation of c), we have $g(y') = y$, and $g(\eta')$ is the generic point $\eta = f(z)$ of Y. Since by definition $\kappa(\eta') = \kappa(z)$, there exists a $\kappa(\eta)$-morphism $\operatorname{Spec}(\kappa(\eta')) \to \operatorname{Spec}(\kappa(z))$ transforming η' into z, hence a $\kappa(\eta')$-section of $f'^{-1}(\eta')$ ((**I**, 3.3.14)); if z' is the image of η' by this section, z' is therefore rational over $\kappa(\eta')$ and $g'(z') = z$. We conclude from hypothesis c) that there exists a Y'-section h' of X' such that $h'(\eta') = z'$; let $h = g' \circ h' : Y' \to X$ be the corresponding Y-morphism, and let $x = h(y')$. It is clear that $f(x) = y$ and that $A' = \mathscr{O}_{y'}$ dominates $\mathscr{O}_x$, which completes the proof. □

Corollary (15.7.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a separated morphism of finite type, y a point of Y. The following conditions are equivalent:*

a) *f is proper at the point y.*
b) *If $Y_1 = \operatorname{Spec}(\mathscr{O}_y)$ and $X_1 = X \times_Y Y_1$, the morphism $f_1 = f_{(Y_1)} : X_1 \to Y_1$ is proper.*
c) *For every scheme $Y' = \operatorname{Spec}(A')$, where A' is a discrete valuation ring, and every morphism $g : Y' \to Y$ such that the image $g(y')$ of the closed point y' of Y' is y, the following property holds: setting $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, and denoting by η' the generic point of Y', for every $x' \in f'^{-1}(\eta')$ rational over $\kappa(\eta')$, there exists a Y'-section of X' containing x'.*

Corollary (15.7.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism of finite type. The following conditions are equivalent:*

a) *There exists a proper morphism $g : X \to Z$ and an immersion $h : Z \to Y$ such that $f = h \circ g$.*
b) *The subspace $f(X)$ is locally closed in Y; if Z' is the closed-image sub-prescheme of Y determined by X under f ((**I**, 9.5.3) and (**I**, 9.5.1)), and Z'' is the prescheme induced by Z' on the open subset $f(X)$ of Z', so that f factors uniquely as $f = j \circ g$ (where $j : Z'' \to Y$ is the canonical injection), then $g : X \to Z''$ is proper.*
c) *For every $y \in f(X)$, f is proper at the point y.*
d) *For every scheme $Y' = \operatorname{Spec}(A')$, where A' is a discrete valuation ring, and every morphism $g : Y' \to Y$ such that $g(Y') \subset f(X)$, every rational Y'-section ((**I**, 7.1.2)) of $X' = X \times_Y Y'$ extends uniquely to a Y'-section of X'.*

Proof. It is clear that b) implies a). To prove that a) implies c), first note that a) implies that $f(X)$ is locally closed in Y. For every $y \in f(X)$, there is therefore an open neighbourhood U of y in Y such that $f(X) \cap U$ is closed in U; the restriction $h^{-1}(U) \to U$ of h is then a closed immersion, and the restriction

$$f^{-1}(U) = g^{-1}\big(h^{-1}(U)\big) \longrightarrow h^{-1}(U)$$

of g is proper. Consequently the restriction $f^{-1}(U) \to U$ of f is proper ((**II**, 5.4.2)).

To prove that $c)$ implies $b)$, first note that, for every $y \in f(X)$, there exists by $c)$ an open neighbourhood U of y in Y such that $f(X) \cap U$ is closed in U. Thus $f(X)$ is locally closed, and hence open in its closure. Since this closure is the underlying space of Z', and since f factors as $f = j_0 \circ g_0$, where $j_0 : Z' \to Y$ is the canonical injection and $g_0 : X \to Z'$, the equality $g_0(X) = Z''$ (which is open in Z') gives the factorisation in the statement of $b)$. The fact that g is proper then follows from the local character of this property (on Z'') and from (**II**, 5.4.3).

It is clear that $c)$ implies $d)$ by (15.7.5); the uniqueness of the extension of a rational Y'-section follows from the hypothesis that f is separated ((**I**, 7.2.2)). Finally, let us prove that $d)$ implies $c)$. First, it follows from $d)$, taking into account (**II**, 7.2.3) and (**II**, 7.2.4), that f is separated; we may then apply the criterion (15.7.5), $c)$, which shows that f is proper at every point of $f(X)$. □

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Remarks (15.7.7). — (i) In (15.7.4), $c)$, (15.7.5), $c)$, and (15.7.6), $d)$, one may restrict to the case where the discrete valuation ring A' is *complete* and has an algebraically closed residue field. Indeed, in the proof of (15.7.4), consider a complete discrete valuation ring A'' dominating A' and having an algebraically closed residue field ($\mathbf{0}_{\mathrm{III}}$, 10.3.1). Condition $c)$ is then satisfied for $Y'' = \operatorname{Spec}(A'')$, $X'' = X \times_Y Y''$, and $f'' = f_{(Y'')}$; but $X'' = X' \times_{Y'} Y''$, and it follows from the remark (**II**, 7.3.9), (i), that f' is proper. Consequently Y', X', and f' also satisfy condition $c)$ ((**II**, 7.3.8)).

(ii) Taking into account ((**II**, 7.3.8)), condition $c)$ of (15.7.5) can be replaced by the condition that f' is *proper*.

(15.7.8). We say that a morphism of preschemes $f : X \to Y$ is *submersive* if it is surjective and if the topology of Y is equal to the quotient of the topology of X by the equivalence relation defined by f. We say that f is *universally submersive* if for every base change $Y' \to Y$, the morphism $f' = f_{(Y')} : X \times_Y Y' \to Y'$ is submersive. Every surjective morphism that is *universally open*, or *universally closed*, or *faithfully flat and quasi-compact* is universally submersive (2.3.12). Given a morphism $f : X \to Y$, we say that a subset Z of X is *submersive over* $f(Z)$ if the topology of the sub-space $f(Z)$ of Y is equal to the quotient of the topology of the sub-space Z of X by the equivalence relation defined by $f|Z$. We say that Z is *universally submersive over* $f(Z)$ if, for every base change $g : Y' \to Y$, the inverse image $Z' = p^{-1}(Z)$ of Z by the projection $p : X \times_Y Y' \to X$ is submersive over $f'(Z') = g^{-1}(f(Z))$. We note that if the morphism f admits a Y-section $h : Y \to X$, every set Z containing $h(Y)$ is universally submersive over Y.

Proposition (15.7.9). — *Let Y be a locally Noetherian prescheme, y a point of Y, $f : X \to Y$ a separated morphism of finite type. Let Z be a locally constructible subset of X having the following properties:*

- *(i) For every generization y' of y, $Z_{y'} = Z \cap f^{-1}(y')$ is a connected component of $f^{-1}(y')$ and is geometrically connected over $\kappa(y')$* (4.5.2).
- *(ii) Z_y is proper over* $\operatorname{Spec}(\kappa(y))$.
- *(iii) Z is universally submersive over $f(Z)$* (15.7.8).

Then Z is proper over Y at the point y (in other words (15.7.1) *there exists an open neighbourhood U of y such that $Z \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ and proper over U).*

Proof. Using the method of ((8.1.2), a)) and ((8.10.5), (xii)), we are reduced to proving that when **IV-3 | 246**
$Y = \operatorname{Spec}(A)$ is the spectrum of a *local* Noetherian ring, the hypotheses (i), (ii) and (iii) imply that Z is *proper* over Y.

Let us first note that if A' is a local Noetherian ring, $\varphi : A \to A'$ a local homomorphism, $Y' = \operatorname{Spec}(A')$, and $g : Y' \to Y$ the morphism corresponding to φ, the inverse image Z' of Z under the canonical projection $p : X \times_Y Y' \to X$ has, with respect to Y', properties (i), (ii), and (iii) of the statement, taking into account (1.8.2). Using (2.7.1), (vii), it is therefore equivalent to prove that Z' is proper over Y', provided that A' is a faithfully flat A-module. We shall make use of this remark by taking $A' = \widehat{A}$ ($\mathbf{0}_{\mathrm{I}}$, 7.3.5), in other words we shall now limit ourselves to the case where A is *complete*. Since Z_y is a connected component of $f^{-1}(y)$, proper over $\kappa(y)$, it follows from (**III**, 5.5.2) that X is the *sum* of two sub-preschemes X_0, X_1 induced on open and closed subsets of X, such that X_0 is proper over Y and $X_0 \cap f^{-1}(y) = Z_y$. Set $Z_0 = X_0 \cap Z$, $Z_1 = X_1 \cap Z$, so that these sets form a partition of Z into two open and closed subsets. Since the $Z_{y'}$ are connected, we necessarily have $Z_{y'} \subset Z_0$ or $Z_{y'} \subset Z_1$, hence $f(Z_0) \cap f(Z_1) = \varnothing$. But since Z_0 and Z_1 are saturated for the equivalence relation defined by $f|Z$, it follows from (iii) that $f(Z_0)$ and $f(Z_1)$ are both open in $f(Z)$. But $f(Z_0)$ contains y, and every neighbourhood of y in Y is equal to Y, so $f(Z_0) = f(Z)$, and hence $f(Z_1) = \varnothing$, so $Z_1 = \varnothing$ and $Z \subset X_0$.

We can therefore henceforth suppose additionally that f is *proper*, and it remains to prove that Z is closed in X. In other words, it will suffice to prove that a *constructible* subset Z of a prescheme X *proper* over Y, satisfying only conditions (i) and (iii), is *closed* in X. By virtue of ($\mathbf{0}_{\mathrm{III}}$, 9.2.5), it suffices therefore to prove that for every $x' \in Z$ and every specialisation x of x' in X, we have $x \in Z$. Let A_1 be a complete discrete valuation ring, u a morphism of $Y_1 = \operatorname{Spec}(A_1)$ into X such that, if y_1 and y_1' are the closed point and the generic point of Y_1, then $u(y_1) = x$ and $u(y_1') = x'$ ((**II**, 7.1.7)); set $g = f \circ u : Y_1 \to Y$ and $X_1 = X \times_Y Y_1$. There exists a Y_1-section $u_1 : Y_1 \to X_1$ of $f_1 = f_{(Y_1)}$ such that $u = p \circ u_1$, where $p : X_1 \to X$ is the canonical projection. Moreover, $x_1 = u_1(y_1)$, $x_1' = u_1(y_1')$, and then $p(x_1) = x$ and $p(x_1') = x'$, and x_1 is a specialisation of x_1'. Moreover, $x_1' \in Z_1 = p^{-1}(Z)$. Since we have remarked that conditions (i) and (iii) are stable under *every* base change, as is the property of being constructible, we are reduced to the following situation: Y is the spectrum of a complete discrete valuation ring, X is proper over Y, $f(x) = y$ is the closed point of Y, $y' = f(x')$ is its generic point, there exists a Y-section h of X such that $h(y) = x$ and $h(y') = x'$, and finally $x' \in Z$; we must prove that $x \in Z$. Now, Z_y is a connected component of $f^{-1}(y)$, *proper* over $\operatorname{Spec}(\kappa(y))$ since X is proper over Y. Applying once more (**III**, 5.5.2), one sees, as above, that there is an open and closed subset X' of X such that $X' \cap f^{-1}(y) = Z_y$; since Z_y is connected, we may suppose that X' is connected (replacing X', if necessary, by the connected component that contains Z_y). But then the same reasoning as at the beginning proves that necessarily $Z \subset X'$; since $h(Y)$ is connected and contains x', it is necessarily contained in X', hence $x = h(y) \in X' \cap f^{-1}(y) = Z_y$, which completes the proof. □ **IV-3** | 247

Corollary (15.7.10). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a separated morphism of finite type, universally submersive* (15.7.8) *and such that all the fibres $X_y = f^{-1}(y)$ are geometrically connected. Then, if $y \in Y$ is such that X_y is proper over $\kappa(y)$, f is proper at the point y.*

Proof. Taking into account (15.7.5), we are reduced to the case where $Y = \operatorname{Spec}(\mathscr{O}_y)$ since the hypotheses are stable under base change. But in this case it suffices to apply (15.7.9) to $Z = X$. □

Corollary (15.7.11). — *Let $f : X \to Y$ be a separated morphism of finite presentation; suppose that there exists a surjective morphism of finite presentation $g : Y' \to Y$, which is moreover proper or flat, and such that, if one sets $X' = X \times_Y Y'$, there exists a Y'-section of X' (or even a Y-morphism $Y' \to X$ ((**I**, 3.3.14))). Suppose finally that the fibres $X_y = f^{-1}(y)$ are geometrically connected. Then the set U of $y \in Y$ such that X_y is proper over $\kappa(y)$ is open in Y, and the restriction $f^{-1}(U) \to U$ of f is proper.*

Proof. The question being local on Y, we can suppose $Y = \operatorname{Spec}(A)$ affine. Applying (8.9.1) to f and to g, one sees that there exists a Noetherian subring A_0 of A and two morphisms of finite type $f_0 : X_0 \to Y_0 = \operatorname{Spec}(A_0)$, $g_0 : Y_0' \to Y_0$ such that $X = X_0 \times_{Y_0} Y$, $Y' = Y_0' \times_{Y_0} Y$, $f = (f_0)_{(Y)}$ and $g = (g_0)_{(Y)}$; moreover, using (8.10.5), (v), (vi) and (xii) and (11.2.7), we can suppose f_0 separated, and g_0 surjective and proper (resp. flat) if g is proper (resp. flat); using (8.8.2), we can further suppose that there exists a Y_0'-section of X_0.

On the other hand, using (9.7.7), we may apply (9.3.3) to the property of being geometrically connected, and may therefore suppose A_0 chosen so that the $f_0^{-1}(y_0)$ are geometrically connected for every $y_0 \in Y_0$. Moreover, using (2.7.1), (vii), it amounts to the same to say that $f^{-1}(y)$ is proper over $\kappa(y)$ or that $f_0^{-1}(y_0)$ is proper over $\kappa(y_0)$, where y_0 is the projection of y in Y_0. These remarks show that we are reduced to proving the corollary when Y is *Noetherian*. Let us now prove the

Lemma (15.7.11.1). — *Let $f : X \to Y$ be a morphism, and $g : Y' \to Y$ a surjective morphism which is moreover proper or flat and quasi-compact. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Then, if f' is submersive (resp. universally submersive), f is also.*

Proof. We can restrict to the case where f' is submersive. Indeed, suppose it proved in this case, and let us show that if f' is universally submersive, then f is also. Let $Y_1 \to Y$ be any morphism, and set

$Y'_1 = Y' \times_Y Y_1$; if $X'_1 = X' \times_{Y'} Y'_1$, we have $f'_1 = (f')_{(Y'_1)} : X'_1 \to Y'_1$; f'_1 is submersive by hypothesis; moreover, the morphism $g_1 : Y'_1 \to Y_1$ is surjective, and proper (resp. flat and quasi-compact) if g is. Hence, if one sets $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$, it follows from the hypothesis and from the fact that $X'_1 = X_1 \times_{Y_1} Y'_1$, that f_1 is submersive, hence f universally submersive. Suppose therefore only that f' is submersive. It is clear first of all that f is surjective. Let E be a subset of Y such that $f^{-1}(E)$ is closed in X; then, if $p : X' \to X$ is the canonical projection, $p^{-1}(f^{-1}(E)) = f'^{-1}(g^{-1}(E))$ is closed **IV-3** | 248 in X', hence, since f' is submersive, $g^{-1}(E)$ is closed in Y'; but as g is also universally submersive (15.7.8), E is closed in Y, which proves the lemma. □

Since there exists by hypothesis a Y'-section of X', f' is universally submersive (15.7.8), hence f is also by the lemma (15.7.11.1). It suffices then to apply (15.7.10), noting that the set of $y \in Y$ such that f is proper at y is open in Y by definition. □

§16. Differential invariants. Differentially smooth morphisms

In this section we present, in global form, some notions of differential calculus that are particu- **IV-4** | 5 larly useful in algebraic geometry. We pass over many developments classical in differential geometry (connections, infinitesimal transformations associated with a vector field, jets, etc.), although these notions admit particularly natural formulations in the setting of schemes. We likewise pass over here the phenomena special to characteristic $p > 0$ (some of which are studied, in the affine setting, in (**0**, 21)). For further material on the differential formalism for preschemes, the reader may consult Exposés II and VII of [42], as well as later chapters of this treatise.

16.1. Normal invariants of an immersion

(16.1.1). Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two ringed spaces, and let $f = (\psi, \theta) : Y \to X$ be a morphism of ringed spaces (**0**, 4.1.1) such that the homomorphism

$$\theta^\# : \psi^*(\mathscr{O}_X) \longrightarrow \mathscr{O}_Y$$

is surjective, so that $\mathscr{O}_Y$ is identified with a sheaf of quotient rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$. We can then endow $\psi^*(\mathscr{O}_X)$ with the $\mathscr{I}_f$-preadic filtration.

Definition (16.1.2). — The $\mathscr{O}_Y$-augmented sheaf of rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ is called the n-th *normal invariant* of f; the ringed space $(Y, \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1})$ is called the n-th *infinitesimal neighbourhood* of Y for the morphism f and is denoted by $Y_f^{(n)}$ or simply $Y^{(n)}$. The sheaf of graded rings associated to the sheaf of filtered rings $\psi^*(\mathscr{O}_X)$

(16.1.2.1) $$\mathscr{G}r_\bullet(f) = \bigoplus_{n \geqslant 0} (\mathscr{I}_f^n / \mathscr{I}_f^{n+1})$$

is called the sheaf of graded rings *associated with* f. The sheaf $\mathscr{G}r_1(f) = \mathscr{I}_f/\mathscr{I}_f^2$ is called the *conormal sheaf* of f (and is also denoted by $\mathscr{N}_{Y/X}$ when no confusion can result).

It is clear that the $\mathscr{O}_{Y^{(n)}} = \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ (also denoted $\mathscr{O}_{Y_f^{(n)}}$) form a projective system of sheaves **IV-4** | 6 of rings on Y; for $n \leqslant m$, the transition homomorphism $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ identifies $\mathscr{O}_{Y^{(n)}}$ with the quotient of $\mathscr{O}_{Y^{(m)}}$ by the power $(\mathscr{I}_f/\mathscr{I}_f^{n+1})^m$ of the *augmentation ideal* of $\mathscr{O}_{Y^{(n)}}$, the kernel of $\phi_{0n} : \mathscr{O}_{Y^{(n)}} \to \mathscr{O}_Y$. The $Y^{(n)}$ therefore form an inductive system of ringed spaces, all having underlying space Y, and we have canonical morphisms of ringed spaces $h_n : Y^{(n)} \to X$ equal to (ψ, θ_n), where $\theta_n^\#$ is the canonical morphism $\psi^*(\mathscr{O}_X) \to \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$. It is clear that the sheaf $\mathscr{G}r_\bullet(f)$ is a sheaf of graded algebras over the sheaf of rings $\mathscr{O}_Y = \mathscr{G}r_0(f)$, and that the $\mathscr{G}r_k(f)$ are $\mathscr{O}_Y$-modules.

As with every sheaf of filtered rings, we have a *canonical surjective homomorphism* of graded $\mathscr{O}_Y$-algebras

(16.1.2.2) $$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_\bullet(f)$$

which coincides in degrees 0 and 1 with the identities.

Examples (16.1.3). —

(i) Suppose that X is a locally ringed space, Y is reduced to a single point y (endowed with a ring $\mathscr{O}_y$) and that, if $x = \psi(y)$, $\theta^\# : \mathscr{O}_x \to \mathscr{O}_y$ is a *surjective* homomorphism of rings having as kernel the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_x$. So the $\mathscr{O}_{Y^{(n)}}$ are identified with the rings $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$ and $\mathscr{G}r_\bullet(f)$ with the graded ring associated with the local ring $\mathscr{O}_x$ endowed with the $\mathfrak{m}_x$-preadic filtration.

(ii) Suppose that Y is a closed subset of an open subspace U of X and that $\mathscr{O}_Y$ is induced on Y by a quotient sheaf $\mathscr{O}_U/\mathscr{I}$, where $\mathscr{I}$ is an ideal of $\mathscr{O}_U$ such that $\mathscr{I}_x = \mathscr{O}_x$ for every $x \notin Y$; if X is a locally ringed space, suppose further that $\mathscr{I}_x \neq \mathscr{O}_x$ for $x \in Y$, so that $(Y, \mathscr{O}_Y)$ is again a locally ringed space.

Let $\psi_0 : Y \to U$ be the canonical injection and denote by $\theta_0 : \mathscr{O}_U \to (\psi_0)_*(\mathscr{O}_Y)$ the homomorphism such that $\theta_0^\#$ is the canonical homomorphism $\psi_0^*(\mathscr{O}_U) = \mathscr{O}_U|Y \to (\mathscr{O}_U/\mathscr{I})|Y$, so that $j_0 = (\psi_0, \theta_0) : Y \to U$ is a morphism of ringed spaces (and of locally ringed spaces if X is a locally ringed space); if $i : U \to X$ is the canonical injection (morphism of ringed spaces), $j = i \circ j_0$ is the morphism (ψ, θ) of Y to X where $\psi : Y \to X$ is the canonical injection and $\theta : \mathscr{O}_X \to \psi_*(\mathscr{O}_Y)$ is the homomorphism such that $\theta^\# = \theta_0^\#$. Since $\theta^\#$ is surjective we can apply the previous definitions; $\mathscr{O}_{Y^{(n)}}$ is equal to $\psi_0^*(\mathscr{O}_U/\mathscr{I}^{n+1})$, and we have $(\psi_0)_*(\mathscr{O}_{Y^{(n)}}) = \mathscr{O}_U/\mathscr{I}^{n+1}$, and $\mathscr{G}r_n(j) = \mathscr{G}r_n(j_0) = \psi_0^*(\mathscr{I}^n/\mathscr{I}^{n+1}) = j_0^*(\mathscr{I}^n/\mathscr{I}^{n+1})$.

(16.1.4). The example (16.1.3, (ii)) shows that in general the $\mathscr{O}_{Y^{(n)}}$ are *not canonically endowed with a structure of an $\mathscr{O}_Y$-module*, or *a fortiori* with a structure of an $\mathscr{O}_Y$-algebra. The data of such a structure is equivalent to the data of a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation morphism ϕ_{0n}; it is also equivalent to the data of a morphism of ringed spaces $(1_Y, \lambda_n) : Y^{(n)} \to Y$ left inverse to the canonical morphism $(1_Y, \phi_{0n}) : Y \to Y^{(n)}$.

Proposition (16.1.5). — *Let $f = (\psi, \theta) : Y \to X$ be an immersion of preschemes. Then:*

(i) *$\mathscr{G}r_\bullet(f)$ is a quasi-coherent graded $\mathscr{O}_Y$-algebra.* **IV-4** | 7

(ii) *The $Y^{(n)}$ are preschemes, canonically isomorphic to subpreschemes of X.*

(iii) *Every homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation homomorphism ϕ_{0n}, makes $\mathscr{O}_{Y^{(n)}}$ and the $\mathscr{O}_{Y^{(k)}}$ for $k \leqslant n$ quasi-coherent $\mathscr{O}_Y$-algebras; the $\mathscr{O}_Y$-module structures induced from the above structures on the $\mathscr{G}r_k(f)$ for $k \leqslant n$ coincide with the ones defined in (16.1.2).*

Proof. (i) Since the question is local on X and Y, we can reduce to the case where Y is a closed subprescheme of X defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$; since $\mathscr{O}_Y$ is the restriction to Y of $\mathscr{O}_X/\mathscr{I}$, assertion (i) is evident, and $Y^{(n)}$ is the closed subprescheme of X defined by the quasi-coherent ideal $\mathscr{I}^{n+1}$ of $\mathscr{O}_X$. Finally, to prove (iii), note that the data of λ_n makes the ideal $\mathscr{I}/\mathscr{I}^{n+1}$ of the augmentation ϕ_{0n} and its quotients $\mathscr{I}/\mathscr{I}^{k+1}$ $(1 \leqslant k \leqslant n)$ $\mathscr{O}_Y$-modules; it suffices to prove by induction on k that the $\mathscr{I}/\mathscr{I}^{k+1}$ are quasi-coherent $\mathscr{O}_Y$-modules and that the quotient $\mathscr{O}_Y$-module structure thereby induced on $\mathscr{I}^k/\mathscr{I}^{k+1}$ is the same as that defined in (16.1.2). The second assertion is immediate, $\mathscr{I}^k/\mathscr{I}^{k+1}$ being killed by $\mathscr{I}/\mathscr{I}^{n+1}$; the first follows by induction on k, since it is trivial for $k = 1$ and $\mathscr{I}/\mathscr{I}^{k+1}$ is an extension of $\mathscr{I}/\mathscr{I}^k$ by $\mathscr{I}^k/\mathscr{I}^{k+1}$ (1.4.17). ☐

Corollary (16.1.6). — *Under the general hypotheses of (16.1.5), if the immersion f is locally of finite presentation then the $\mathscr{G}r_n(f)$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

Proof. Indeed, with the notation from the proof of (16.1.5), $\mathscr{I}$ is an ideal of finite type of $\mathscr{O}_X$ (1.4.7), therefore the $\mathscr{I}^n/\mathscr{I}^{n+1}$ are $\mathscr{O}_Y$-modules of finite type, hence the conclusion. ☐

Corollary (16.1.7). — *Under the general hypotheses of (16.1.5), let $g : X \to Y$ be a morphism of preschemes, left inverse to f. Then, for every n, the composite morphism $(1, \lambda_n) : Y^{(n)} \xrightarrow{h_n} X \xrightarrow{g} Y$ defines a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$ right inverse to the augmentation ϕ_{0n}, making $\mathscr{O}_{Y^{(n)}}$ a quasi-coherent $\mathscr{O}_Y$-algebra; via these homomorphisms, the transition homomorphisms $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ $(n \leqslant m)$ are homomorphisms of $\mathscr{O}_Y$-algebras. Also, if g is locally of finite type, then the $\mathscr{O}_{Y^{(n)}}$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

Proof. The first assertion follows immediately from the definitions and (16.1.5). On the other hand, if g is locally of finite type, then f is locally of finite presentation (1.4.3, (v)); the $\mathscr{G}r_n(f)$ being then quasi-coherent $\mathscr{O}_Y$-modules of finite type by (16.1.6), the same goes for the $\mathscr{O}_Y$-modules $\mathscr{I}/\mathscr{I}^{n+1}$, being extensions of a finite number of the $\mathscr{G}r_k(f)$ (**III**, 1.4.17). □

Proposition (16.1.8). — *Let X be a locally Noetherian prescheme and let $j : Y \to X$ be an immersion. Then the $Y^{(n)}$ are locally Noetherian preschemes, the $\mathscr{G}r_n(j)$ are coherent $\mathscr{O}_Y$-modules, and $\mathscr{G}r_\bullet(j)$ is a coherent sheaf of rings on the space Y.*

Proof. Everything is local on X and Y, so we reduce to the case where X is affine and j is a closed immersion. All the assertions are then evident except the last; this follows from the fact that, if A is a Noetherian ring and $\mathfrak{I}$ is an ideal of A, then $\mathrm{gr}^\bullet_{\mathfrak{I}}(A)$ is a Noetherian ring, taking into account the exactness of the functor ψ^* and (**0**, 5.3.7). □

Proposition (16.1.9). — *Let X be a prescheme, let $j : Y \to X$ be an immersion locally of finite presentation, and let y be a point of Y. The following conditions are equivalent:* **IV-4 | 8**

(a) *There exists an open neighbourhood U of y in Y such that $j|U$ is a homeomorphism of U onto an open subset of X.*

(b) *There is an integer $n > 0$ such that the canonical homomorphism*

$$(\phi_{n-1,n})_y : \mathscr{O}_{Y^{(n)},y} \longrightarrow \mathscr{O}_{Y^{(n-1)},y}$$

is bijective.

(c) *There is an integer $n > 0$ such that $(\mathscr{G}r_n(j))_y = 0$.*

In addition, if the integer n satisfies (b) *or* (c)*, then there is a neighbourhood V of y in Y such that $\mathscr{G}r_m(j)|V = 0$ for $m \geqslant n$ and that $\phi_{nm}|V : \mathscr{O}_{Y^{(m)}}|V \to \mathscr{O}_{Y^{(n)}}|V$ is bijective for $m \geqslant n$.*

Proof. Since the question is local on Y, we can restrict ourselves to the case where j is a closed immersion, with Y defined by a quasi-coherent ideal *of finite type* $\mathscr{I}$ of $\mathscr{O}_X$. The equivalence of (b) and (c), for a given n, is immediate; also, since $\mathscr{I}^n/\mathscr{I}^{n+1}$ is an $\mathscr{O}_X$-module of finite type, there is an open neighbourhood U of y in X such that $\mathscr{I}^n|U = \mathscr{I}^{n+1}|U$ (**0**, 5.2.2), and hence $\mathscr{I}^n|U = \mathscr{I}^m|U$ for $m \geqslant n$, proving the last assertions. To prove that (a) implies (b), we can restrict ourselves to the case where the underlying space of Y is equal to the underlying space of X and where $\mathscr{I}$ is generated by finitely many sections over X: since $\mathscr{I}$ is contained in the nilradical $\mathscr{N}$ of $\mathscr{O}_X$ (**I**, 5.1.2), it is nilpotent, which proves (b). Finally, to prove that (b) implies (a), we can likewise restrict ourselves to the case where $\mathscr{I}^n = \mathscr{I}^{n+1}$; therefore, for every $y \in Y$, since $\mathscr{I}_y \subset \mathfrak{m}_y$, maximal ideal of $\mathscr{O}_{X,y}$, we must have $\mathscr{I}_y^n = 0$ because of Nakayama's lemma, since $\mathscr{I}_y$ is an ideal of finite type. The set of $x \in X$ such that $\mathscr{I}_x^n = 0$ is therefore an open subset U of X containing Y (**0**, 5.2.2); since on the other hand $\mathscr{I}_x \neq 0$ for $x \notin Y$, we must have $U = Y$. □

Corollary (16.1.10). — *For the restriction of the immersion j to a neighbourhood of y in Y to be an open immersion (in other words, for j to be a* local isomorphism *at the point y), it is necessary and sufficient that $(\mathscr{G}r_1(j))_y = (\mathscr{N}_{Y/X})_y = 0$.*

Proof. The condition is clearly necessary, and the previous reasoning applied to $n = 1$ proves that it is sufficient. □

Remark (16.1.11). —

(i) Under the conditions of the definition (16.1.1), the projective limit of the projective system $(\mathscr{O}_{Y^{(n)}}, \phi_{nm})$ of sheaves of rings on Y is called the *normal invariant of infinite order* of f, and is sometimes denoted by $\mathscr{O}_{Y^{(\infty)}}$. When X is a locally Noetherian prescheme and $j : Y \to X$ is a closed immersion, so that Y is a closed subprescheme of X defined by a coherent ideal $\mathscr{I}$, the sheaf $\mathscr{O}_{Y^{(\infty)}}$ is precisely the *formal completion* of $\mathscr{O}_X$ along Y (**I**, 10.8.4), and $Y^{(\infty)} = (Y, \mathscr{O}_{Y^{(\infty)}})$ is the formal prescheme obtained by *completing* X along Y (**I**, 10.8.5). In all cases, one may say that $Y^{(\infty)}$ is the *formal neighbourhood* of Y in X (for the morphism f). In the particular case we have just considered, it is the formal prescheme that is the inductive limit of the infinitesimal neighbourhoods of order n.

(ii) Note that for a morphism of preschemes $f = (\psi, \theta) : Y \to X$, it can happen that the homomorphism $\theta^\# : \psi^*(\mathscr{O}_X) \to \mathscr{O}_Y$ is surjective without f being a local immersion and without f being injective. We obtain an example by taking Y to be a sum of preschemes Y_λ, all isomorphic to $\mathrm{Spec}(\mathscr{O}_x)$, where $x \in X$, and taking f to be the morphism equal to the canonical morphism on each Y_λ. **IV-4** | 9

16.2. Functorial properties of the normal invariants of an immersion

(16.2.1). Let $f = (\psi, \theta) : Y \to X$ and $f' = (\psi', \theta') : Y' \to X'$ be two morphisms of ringed spaces such that $\theta^\#$ and $\theta'^\#$ are surjective; consider a commutative diagram of morphisms of ringed spaces

(16.2.1.1)
$$\begin{tikzcd} Y \arrow[r, "f"] & X \\ Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \end{tikzcd}$$

Let $u = (\rho, \lambda), v = (\sigma, \mu)$. We have $\rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X))$, and hence a commutative diagram of homomorphisms of sheaves of rings on Y':

$$\begin{tikzcd} \rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X)) \arrow[r, "\psi'^*(\mu^\#)"] \arrow[d, "\rho^*(\theta^\#)"'] & \psi'^*(\mathscr{O}_{X'}) \arrow[d, "\theta'^\#"] \\ \rho^*(\mathscr{O}_Y) \arrow[r, "\lambda^\#"'] & \mathscr{O}_{Y'} \end{tikzcd}$$

from which, if $\mathscr{I}$ and $\mathscr{I}'$ are the kernels of $\theta^\#$ and $\theta'^\#$, we conclude that $\psi'^*(\mu^\#)(\rho^*(\mathscr{I})) \subset \mathscr{I}'$, by the exactness of the functor ρ^*. It follows immediately that, for every integer n, $\psi'^*(\mu^\#)(\rho^*(\mathscr{I}^n)) \subset \mathscr{I}'^n$, which shows that $\psi'^*(\mu^\#)$ defines, on passing to the quotients, a homomorphism of sheaves of rings

(16.2.1.2)
$$\nu_n : \rho^*(\psi^*(\mathscr{O}_X)/\mathscr{I}^{n+1}) \longrightarrow \psi'^*(\mathscr{O}_{X'})/\mathscr{I}'^{n+1}$$

and therefore a morphism of ringed spaces $w_n = (\rho, \nu_n) : Y'^{(n)} \to Y^{(n)}$ (which, for $n = 0$, is none other than u). It follows immediately from this definition that the diagrams

$$\begin{tikzcd} Y^{(n)} \arrow[r, "h_{mn}"] & Y^{(m)} \arrow[r, "h_m"] & X \\ Y'^{(n)} \arrow[u, "w_n"] \arrow[r, "h'_{mn}"'] & Y'^{(m)} \arrow[u, "w_m"] \arrow[r, "h'_m"'] & X' \arrow[u, "v"'] \end{tikzcd} \qquad (n \leqslant m)$$

(where the horizontal arrows are the canonical morphisms (16.1.2)) are commutative.

By passage to the quotients via the morphisms (16.2.1.2), and taking into account the exactness of the functor ρ^*, we obtain a di-homomorphism of graded algebras (relative to the homomorphism $\lambda^\# : \rho^*(\mathscr{O}_Y) \to \mathscr{O}_{Y'}$) **IV-4** | 10

(16.2.1.3)
$$\mathrm{gr}(u) : \rho^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f')$$

(or, if one prefers, a ρ-morphism (**0**, 3.5.1) $\mathscr{G}r_\bullet(f) \to \mathscr{G}r_\bullet(f')$), and, in particular, a di-homomorphism of conormal sheaves

$$\mathrm{gr}_1(u) : \rho^*(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_1(f').$$

It is also immediate that these homomorphisms give rise to a commutative diagram

(16.2.1.4)
$$\begin{tikzcd} \rho^*(\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f))) \arrow[r] \arrow[d, "\mathbf{S}(\mathrm{gr}_1(u))"'] & \rho^*(\mathscr{G}r_\bullet(f)) \arrow[d, "\mathrm{gr}(u)"] \\ \mathbf{S}^\bullet_{\mathscr{O}_{Y'}}(\mathscr{G}r_1(f')) \arrow[r] & \mathscr{G}r_\bullet(f') \end{tikzcd}$$

where the horizontal arrows are the canonical homomorphisms (16.1.2.2).

Finally, if we have a commutative diagram of morphisms of ringed spaces

$$\begin{tikzcd} Y \arrow[r, "f"] & X \\ Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \\ Y'' \arrow[u, "u'"] \arrow[r, "f''"'] & X'' \arrow[u, "v'"'] \end{tikzcd}$$

where $f'' = (\psi'', \theta'')$ is such that $\theta''^{\#}$ is surjective, and if w'_n and w''_n are defined, respectively, from u', v' and from $u'' = u \circ u', v'' = v \circ v'$, then $w''_n = w_n \circ w'_n$, as follows immediately from the definitions and from (**0**, 3.5.5); we have also $\mathrm{gr}(u'') = \mathrm{gr}(u') \circ \rho'^*(\mathrm{gr}(u))$ if $u' = (\rho', \lambda')$. Therefore we can say that $Y^{(n)}$ and $\mathscr{G}r_\bullet(f)$ *depend functorially* on f.

Proposition (16.2.2). — *With the notation and hypotheses of* (16.2.1), *suppose also that f, f', u, and v are morphisms of preschemes. We have:*

(i) *The morphisms $w_n : Y'^{(n)} \to Y^{(n)}$ are morphisms of preschemes.*
(ii) *If $Y' = Y \times_X X'$, with u and f' the canonical projections, and if f is an immersion or v is flat, then $Y'^{(n)} = Y^{(n)} \times_X X'$.*
(iii) *If $Y' = Y \times_X X'$ and if v is flat (resp. if f is an immersion), the homomorphism*

$$\mathrm{Gr}(u) = \mathrm{gr}(u) \otimes 1 : \mathscr{G}r_\bullet(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_\bullet(f')$$

is bijective (resp. surjective).

Proof.

(i) The hypotheses immediately imply that, for every $y' \in Y'$, $\rho^*_{y'}(\theta^{\#}_{\psi'(y')})$ is a *local* homomorphism (**I**, 1.6.2), so w_n is a morphism of preschemes (**I**, 2.2.1). **IV-4** | 11

(ii) and (iii) If f is an immersion, we may restrict to the case where f is a closed immersion, with Y defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$ and $Y^{(n)}$ by the ideal $\mathscr{I}^{n+1}$; the assertions then follow from (**I**, 4.4.5).

Second, suppose that v is flat; we can restrict ourselves to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(A')$ are affines, A' being a flat A-module; so $Y' = \mathrm{Spec}(B')$ where $B' = B \otimes_A A'$; moreover, if $\mathfrak{I}$ is the kernel of the homomorphism $A \to B$, the kernel $\mathfrak{I}'$ of $A' \to B'$ is identified with $\mathfrak{I} \otimes_A A'$ by flatness, and

$$\mathfrak{I}'^n / \mathfrak{I}'^{n+1} = (\mathfrak{I}^n / \mathfrak{I}^{n+1}) \otimes_A A'.$$

It follows at once from (**0**, 4.3.3) that the $\mathscr{O}_{Y'}$-module $\mathscr{I}'^n / \mathscr{I}'^{n+1}$ is equal to

$$\psi'^*(\sigma^*((\mathfrak{I}^n/\mathfrak{I}^{n+1})^\sim) \otimes_{\sigma^*(\mathscr{O}_X)} \mathscr{O}_{X'}) =$$
$$\psi'^*(\sigma^*((\mathfrak{I}^n/\mathfrak{I}^{n+1})^\sim)) \otimes_{\psi'^*(\sigma^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'}) = \rho^*(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^*(\psi^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'})$$

and in particular for $n = 0$, we have

$$\mathscr{O}_{Y'} = \rho^*(\mathscr{O}_Y) \otimes_{\rho^*(\psi^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'})$$

whence we obtain a canonical isomorphism from $\mathscr{I}'^n / \mathscr{I}'^{n+1}$ onto

$$\rho^*(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^*(\mathscr{O}_Y)} \mathscr{O}_{Y'} = (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$$

which proves (iii). Set now $C_n = \Gamma(Y, \mathscr{O}_{Y^{(n)}})$ and $C'_n = \Gamma(Y', \mathscr{O}_{Y'^{(n)}})$. Since $Y^{(n)}$ and $Y'^{(n)}$ are affine schemes (16.1.5), the kernel $\mathfrak{K}_n$ (resp. $\mathfrak{K}'_n$) of the homomorphism $C_n \to C_{n-1}$ (resp.

$C'_n \to C'_{n-1}$) is $\Gamma(Y, \mathscr{I}^n/\mathscr{I}^{n+1})$ (resp. $\Gamma(Y', \mathscr{I}'^n/\mathscr{I}'^{n+1})$); therefore we can deduce from the above results that $\mathfrak{K}'_n = \mathfrak{K}_n \otimes_A A'$. Now, we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathfrak{K}_n \otimes_A A' \arrow[r] \arrow[d, "r"] & C_n \otimes_A A' \arrow[r] \arrow[d, "s_n"] & C_{n-1} \otimes_A A' \arrow[r] \arrow[d, "s_{n-1}"] & 0 \\
0 \arrow[r] & \mathfrak{K}'_n \arrow[r] & C'_n \arrow[r] & C'_{n-1} \arrow[r] & 0
\end{tikzcd}$$

where the left vertical arrow is bijective and both rows are exact (since A' is a flat A-module). It follows by induction that s_n is bijective for every n: this is true by hypothesis for $n = 0$, and the general case follows from the five lemma. This proves the second assertion of (ii).

□

Corollary (16.2.3). — *Let $g : X \to Y$ and $u : Y' \to Y$ be two morphisms of preschemes, let $X' = X \times_Y Y'$, and let $g' : X' \to Y'$ and $v : X' \to X$ be the canonical projections. Let $f : Y \to X$ be a Y-section of X (and hence an immersion), and let $f' = f_{(Y')} : Y' \to X'$ be the Y'-section of X' deduced from f by the base change u. We have:*

(i) *The morphism $w_n : Y'^{(n)}_{f'} \to Y^{(n)}_f$ corresponding to f, f', u, v (16.2.1) and the canonical morphism $h'_n : Y'^{(n)}_{f'} \to X'$ jointly identify $Y'^{(n)}_{f'}$ with the product $Y^{(n)}_f \times_X X'$.*

(ii) *If we endow $\mathscr{O}_{Y^{(n)}_f}$ (resp. $\mathscr{O}_{Y'^{(n)}_{f'}}$) with the structure of an $\mathscr{O}_Y$-algebra defined by g (resp. with the structure of an $\mathscr{O}_{Y'}$-algebra defined by g') (16.1.5, (iii)), then the homomorphism of $\mathscr{O}_{Y'}$-algebras*

$$\rho^*(\mathscr{O}_{Y^{(n)}_f}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{O}_{Y'^{(n)}_{f'}} \tag{16.2.3.1}$$

induced by the homomorphism ν_n (16.2.1.2) is bijective. Also, the homomorphism of $\mathscr{O}_{Y'}$-modules **IV-4** | 12

$$\mathrm{Gr}_1(u) : \mathscr{G}r_1(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_1(f') \tag{16.2.3.2}$$

is bijective.

Proof.

(i) First note that $f' : Y' \to X'$ and $u : Y' \to Y$ jointly identify Y' with the product $Y \times_X X'$ (for the structure morphisms $f : Y \to X$ and $v : X' \to X$) (14.5.11.1).[24] The conclusion of (i) now follows from (16.2.2, (ii)), the morphism g being an immersion.

(ii) The commutative diagram

$$\begin{tikzcd}
Y^{(n)}_f \arrow[d, "h_n"] & Y'^{(n)}_{f'} \arrow[l, "w_n"'] \arrow[d, "h'_n"] \\
X \arrow[d, "g"] & X' \arrow[l, "v"'] \arrow[d, "g'"] \\
Y & Y' \arrow[l, "u"']
\end{tikzcd}$$

identifies $Y'^{(n)}_{f'}$ with the product $Y^{(n)}_f \times_X X'$ and identifies X' with the product $X \times_Y Y'$; hence, by (**I**, 3.3.9), the morphisms $g' \circ h'_n$ and w_n identify $Y'^{(n)}_{f'}$ with the product $Y^{(n)}_f \times_Y Y'$. Since $Y^{(n)}_f$ (resp. $Y'^{(n)}_{f'}$) is the affine prescheme over Y (resp. over Y') associated with the $\mathscr{O}_Y$-algebra $\mathscr{O}_{Y^{(n)}_f}$ (resp. the $\mathscr{O}_{Y'}$-algebra $\mathscr{O}_{Y'^{(n)}_{f'}}$), the bijectivity of the canonical homomorphism (16.2.3.1) follows from (**II**, 1.5.2). Finally, the canonical homomorphism (16.2.3.1) is compatible with the augmentations $\mathscr{O}_{Y^{(n)}_f} \to \mathscr{O}_Y$ and $\mathscr{O}_{Y'^{(n)}_{f'}} \to \mathscr{O}_{Y'}$; since $\mathscr{O}_{Y^{(n)}_f}$ is the direct sum (as an $\mathscr{O}_Y$-module) of $\mathscr{O}_Y$ and the augmentation ideal $\mathscr{I}/\mathscr{I}^{n+1}$, it follows that the

[24]The printed source cites (14.5.12.1); the relevant product lemma is (14.5.11.1).

canonical homomorphism (16.2.3.1), restricted to $(\mathscr{I}/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$, maps this module bijectively onto $\mathscr{I}'/\mathscr{I}'^{n+1}$. For $n = 1$, this shows that $\mathrm{Gr}_1(u)$ is bijective.

□

We note that, under the hypotheses of (16.2.3), the homomorphisms $\mathrm{Gr}_n(u)$ are *surjective* in view of the above, but are not bijective in general for $n \geqslant 2$. However:

Corollary (16.2.4). — *Under the hypotheses of* (16.2.3), *suppose that $u : Y' \to Y$ is a flat morphism (resp. that the $\mathscr{G}r_n(f)$ are flat $\mathscr{O}_Y$-modules for $n \leqslant m$). Then the homomorphism*

$$\mathrm{Gr}_n(u) : \mathscr{G}r_n(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_n(f')$$

is bijective for all n (resp. for $n \leqslant m$).

Proof. If u is flat, then $v : X' \to X$ is flat by base change, and in this case we already know that $\mathrm{Gr}(u)$ is bijective (16.2.2, (iii)). If the $\mathscr{G}r_n(f)$ are flat for $n \leqslant m$, then we first see by induction on n that the same holds for $\mathscr{I}/\mathscr{I}^{n+1}$ for $n \leqslant m$, because of the exact sequences

$$0 \longrightarrow \mathscr{I}^n/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^n \longrightarrow 0$$

(**0**, 6.1.2); in addition, we have the commutative diagram **IV-4** | 13

$$\begin{tikzcd}
0 \arrow[r] & (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^n) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathscr{I}'^n/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^n \arrow[r] & 0
\end{tikzcd}$$

in which the rows are exact (the first by flatness (**0**, 6.1.2)) and the last two vertical arrows are bijective by (16.2.2, (ii)); hence the conclusion. □

Remarks (16.2.5). —

(i) The reasoning of (16.2.2, (i)) still applies when the maps in (16.2.1.1) are morphisms of *locally ringed spaces* ($\mathbf{Err_{III}}$, (**I**, 1.8.2)).

(ii) In (16.2.2, (ii)), the conclusion is no longer necessarily valid if we only suppose that v and f are morphisms of preschemes (f satisfying the condition of (16.1.1)). For example (with the notation of the proof of (16.2.2, (ii))), it can happen that $\mathfrak{I} = 0$ but the kernel $\mathfrak{I}'$ of $A' \to B' = B \otimes_A A'$ is not zero and that $B' \neq 0$, in which case we have $Y^{(n)} = Y$ for all n, but $Y'^{(n)} \neq Y'$. We have an example of this by taking $A = \mathbf{Z}$, $B = \mathbf{Q}$, $A' = \prod_{h=1}^{\infty}(\mathbf{Z}/m^h\mathbf{Z})$ where $m > 1$.

(16.2.6). Consider the special case of diagram (16.2.1.1) in which $X' = X$, v is the identity, X is a prescheme, Y is a subprescheme of X, Y' is a subprescheme of Y, and $f, u, f' = f \circ u$ are the canonical injections; by tensoring with $\mathscr{O}_{Y'}$ over $\rho^*(\mathscr{O}_Y)$, the di-homomorphism (16.2.1.3) gives a homomorphism of graded $\mathscr{O}_{Y'}$-algebras

(16.2.6.1) $$u^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f').$$

On the other hand, we identify $\mathscr{O}_Y$ with $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$ and $\mathscr{O}_{Y'}$ with $\rho^*(\mathscr{O}_Y)/\mathscr{I}_u$; since ρ^* is exact, we have $\rho^*(\mathscr{O}_Y) = \rho^*(\psi^*(\mathscr{O}_X))/\rho^*(\mathscr{I}_f) = \psi'^*(\mathscr{O}_X)/\rho^*(\mathscr{I}_f)$, and since $\mathscr{O}_{Y'}$ is also identified with $\psi'^*(\mathscr{O}_X)/\mathscr{I}_{f'}$, we see that $\mathscr{I}_u = \mathscr{I}_{f'}/\rho^*(\mathscr{I}_f)$. Thus, for every integer n, there is a canonical homomorphism $\mathscr{I}_{f'}^n/\mathscr{I}_{f'}^{n+1} \to \mathscr{I}_u^n/\mathscr{I}_u^{n+1}$, and hence a canonical homomorphism of graded $\mathscr{O}_{Y'}$-algebras

(16.2.6.2) $$\mathscr{G}r_\bullet(f') \longrightarrow \mathscr{G}r_\bullet(u).$$

Proposition (16.2.7). — *Let X be a prescheme, let Y be a subprescheme of X, let Y' be a subprescheme of Y, and let $j : Y' \to Y$ be the canonical injection. We then have an exact sequence of conormal sheaves ($\mathscr{O}_{Y'}$-modules)*

(16.2.7.1) $$j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0$$

where the arrows are the degree 1 components of the canonical homomorphisms (16.2.6.1) *and* (16.2.6.2).

Proof. Since the question is local, we may restrict to the case where $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(A/\mathfrak{J})$, and $Y' = \operatorname{Spec}(A/\mathfrak{K})$, where $\mathfrak{J}$ and $\mathfrak{K}$ are ideals of A such that $\mathfrak{J} \subset \mathfrak{K}$; everything reduces to seeing that the sequence of canonical morphisms $\mathfrak{J}/\mathfrak{K}\mathfrak{J} \to \mathfrak{K}/\mathfrak{K}^2 \to (\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2 \to 0$ is exact, **IV-4** | 14
which is immediate given that the image of $\mathfrak{J}/\mathfrak{K}\mathfrak{J}$ in $\mathfrak{K}/\mathfrak{K}^2$ is $(\mathfrak{J}+\mathfrak{K}^2)/\mathfrak{K}^2$ and that $(\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2$ is identified with $\mathfrak{K}/(\mathfrak{J}+\mathfrak{K}^2)$. ☐

It is easy to give examples where the sequence (16.2.7.1) extended on the left by 0 is not exact; with the above notation, it suffices to take $A = k[T]$, $\mathfrak{J} = AT^2$, $\mathfrak{K} = AT$, because then $(\mathfrak{J}+\mathfrak{K}^2)/\mathfrak{K}^2 = 0$ and $\mathfrak{J}/\mathfrak{K}\mathfrak{J} \neq 0$. See however (16.9.13) and (19.1.5) for some cases where the extended sequence is indeed exact.

16.3. Fundamental differential invariants of morphisms of preschemes

Definition (16.3.1). — Let $f : X \to S$ be a morphism of preschemes, and let $\Delta_f : X \to X \times_S X$ be the corresponding diagonal morphism, which is an immersion ((**I**, 5.3.9) and $\mathbf{Err_{III}}$, 10). We denote by $\mathscr{P}^n_f$ or $\mathscr{P}^n_{X/S}$, and call the *sheaf of principal parts of order n of the S-prescheme X*, the $\mathscr{O}_X$-augmented sheaf of rings that is the n-th normal invariant of Δ_f (16.1.2). We will also write $\mathscr{P}^\infty_f = \mathscr{P}^\infty_{X/S} = \varprojlim_n \mathscr{P}^n_{X/S}$, $\mathscr{G}r_n(\mathscr{P}_f) = \mathscr{G}r_n(\mathscr{P}_{X/S}) = \mathscr{G}r_n(\Delta_f)$ (16.1.2); the $\mathscr{O}_X$-module $\mathscr{G}r_1(\Delta_f)$, the augmentation ideal of $\mathscr{P}^1_{X/S}$, is also denoted by Ω^1_f or $\Omega^1_{X/S}$, and is called the $\mathscr{O}_X$-module of 1-*differentials* of f, or of X with respect to S, or of the S-prescheme X.

It follows from this definition that $\mathscr{P}^0_{X/S}$ is canonically identified with $\mathscr{O}_X$ (16.1.2).
We have (16.1.2.2) a canonical surjective homomorphism of graded $\mathscr{O}_X$-algebras

$$\mathbf{S}^\bullet_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}). \tag{16.3.1.1}$$

It also follows from Definition (16.3.1) that for every open U of X we have $\mathscr{P}^n_{f|U} = \mathscr{P}^n_f|U$, $\mathscr{P}^\infty_{f|U} = \mathscr{P}^\infty_f|U$, $\mathscr{G}r_n(\mathscr{P}_{f|U}) = \mathscr{G}r_n(\mathscr{P}_f)|U$, and $\Omega^1_{f|U} = \Omega^1_f|U$ (in other words, the notions introduced are *local* on X).

(16.3.2). Let p_1, p_2 be the two canonical projections of the product $X \times_S X$; since Δ_f is an X-section of $X \times_S X$ for both p_1 and p_2, *each* of these morphisms defines, for every n, a homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ that is right inverse to the augmentation $\mathscr{P}^n_{X/S} \to \mathscr{O}_X$ (16.1.7); equivalently, this defines on $\mathscr{P}^n_{X/S}$ *two structures of quasi-coherent augmented $\mathscr{O}_X$-algebra*; the corresponding $\mathscr{O}_X$-module structures on $\mathscr{G}r_n(\mathscr{P}_{X/S})$ are the same. We also have, by passing to the limit, two $\mathscr{O}_X$-algebra structures on $\mathscr{P}^\infty_{X/S}$.

(16.3.3). The morphism $s = (p_2, p_1)_S : X \times_S X \to X \times_S X$ is an *involutive automorphism* of $X \times_S X$, called the *canonical symmetry*, such that

$$p_1 \circ s = p_2, \qquad p_2 \circ s = p_1, \qquad s \circ \Delta_f = \Delta_f. \tag{16.3.3.1}$$

If we set $s = (\rho, \lambda)$, $p_i = (\pi_i, \mu_i)$ ($i = 1, 2$), and $\Delta_f = (\delta, \nu)$, then $\lambda^\#$ is an isomorphism from $\rho^*(\pi_1^*(\mathscr{O}_X))$ onto $\pi_2^*(\mathscr{O}_X)$, and $\delta^*(\lambda^\#)$ leaves invariant $\delta^*(\mathscr{O}_{X\times_S X})$ and the kernel $\mathscr{I}$ of the homomorphism $\nu^\# : \delta^*(\mathscr{O}_{X\times_S X}) \to \mathscr{O}_X$. Therefore:

Proposition (16.3.4). — *The homomorphism $\sigma = \delta^*(\lambda^\#)$ induced from s (and also called the* canonical symmetry*) is an involutive automorphism of the projective system $(\mathscr{P}^n_{X/S})$ of $\mathscr{O}_X$-augmented sheaves of* **IV-4** | 15
rings, and as a result also of the projective limit $\mathscr{P}^\infty_{X/S}$. This automorphism interchanges the two $\mathscr{O}_X$-algebra structures on $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$.

(16.3.5). In what follows, the two $\mathscr{O}_X$-algebra structures defined on the $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$ will play very different roles: *we will now agree, unless said otherwise, that when $\mathscr{P}^n_{X/S}$ or $\mathscr{P}^\infty_{X/S}$ is considered as an $\mathscr{O}_X$-algebra, it is the algebra structure induced by p_1.*

For every open U of X and every section $t \in \Gamma(U, \mathscr{O}_X)$, we will simply denote by $t.1$ or even t the image of t under the structure morphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^n_{X/S})$ (resp. $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^\infty_{X/S})$) (that is to say, the homomorphism corresponding to p_1).

Definition (16.3.6). — We denote by d^n_f, or $d^n_{X/S}$ (resp. d^∞_f, or $d^\infty_{X/S}$), or simply d^n (resp. d^∞), the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_f = \mathscr{P}^n_{X/S}$ (resp. $\mathscr{O}_X \to \mathscr{P}^\infty_f = \mathscr{P}^\infty_{X/S}$) induced by p_2. For every open U of X, and every $t \in \Gamma(U, \mathscr{O}_X)$, $d^n t$ (resp. $d^\infty t$) is called the *principal part of order n* (resp. *principal part of infinite order*) of t. We set $dt = d^1 t - t$, and we say that dt is the differential of t (an element of $\Gamma(U, \Omega^1_{X/S})$, also denoted $d_{X/S}(t)$).

It follows immediately [25] from this definition that we have

$$d(t_1 t_2) = t_1 dt_2 + t_2 dt_1 \tag{16.3.6.1}$$

for every t_1, t_2 in $\Gamma(U, \mathscr{O}_X)$; in other words, d is a *derivation* of the ring $\Gamma(U, \mathscr{O}_X)$ with values in the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

In all the notation introduced in (16.3.1) and (16.3.6), we will sometimes replace S by A when $S = \operatorname{Spec}(A)$.

(16.3.7). Suppose in particular that $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine schemes, B then being an A-algebra. Then Δ_f corresponds to the canonical surjective homomorphism $\pi : B \otimes_A B \to B$ such that $\pi(b \otimes b') = bb'$, with kernel $\mathfrak{I} = \mathfrak{I}_{B/A}$ (**0**, 20.4.1); $\mathscr{P}^n_f$ is the structure sheaf of the prescheme $\operatorname{Spec}(P^n_{B/A})$, where

$$P^n_{B/A} = (B \otimes_A B)/\mathfrak{I}^{n+1};$$

$\mathscr{G}r_\bullet(\mathscr{P}_f)$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the graded B-module

$$\operatorname{gr}^\bullet_{\mathfrak{I}}(B \otimes_A B) = \bigoplus_{n \geqslant 0} (\mathfrak{I}^n/\mathfrak{I}^{n+1});$$

in particular $\Omega^1_f = \Omega^1_{X/S}$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the B-module of 1-differentials of B over A, $\Omega^1_{B/A}$ (**0**, 20.4.3). The projection morphisms $p_1 : X \times_S X \to X$ and $p_2 : X \times_S X \to X$ correspond to the two homomorphisms of rings $j_1 : B \to B \otimes_A B$ and $j_2 : B \to B \otimes_A B$ such that $j_1(b) = b \otimes 1$ and $j_2(b) = 1 \otimes b$; thus (by the convention of (16.3.5)), $P^n_{B/A}$ is always considered as a B-algebra via the composite homomorphism $B \xrightarrow{j_1} B \otimes_A B \to P^n_{B/A}$; the ring homomorphism $B \xrightarrow{j_2} B \otimes_A B \to P^n_{B/A}$ is denoted by $d^n_{B/A}$ and corresponds to $d^n_{X/S}$ acting on $\Gamma(X, \mathscr{O}_X)$; for every $t \in B$, dt is equal to $d_{B/A} t$, defined in (**0**, 20.4.6) (cf. **Err$_{\mathbf{IV}}$**, 11).

If $\pi_n : B \otimes_A B \to P^n_{B/A}$ is the canonical homomorphism, then the preceding definitions give

$$\pi_n(b \otimes b') = b \cdot \pi_n(1 \otimes b') = b \cdot d^n_{B/A}(b') \quad \text{for } b \in B, b' \in B. \tag{16.3.7.1}$$

Proposition (16.3.8). — *The image of the canonical homomorphism $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ generates the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$.* **IV-4 | 16**

Proof. We immediately reduce to the case where $X = \operatorname{Spec}(B)$ and $S = \operatorname{Spec}(A)$ are affine; the proposition then follows from (16.3.7.1), since π_n is surjective. We note that in general $d^n_{X/S}$ *is not surjective* (even for $n = 1$). ☐

Proposition (16.3.9). — *Suppose that $f : X \to S$ is a morphism locally of finite type. Then the $\mathscr{P}^n_f$ and the $\mathscr{G}r_n(\mathscr{P}_f)$ are quasi-coherent $\mathscr{O}_X$-modules of finite type.*

Proof. This follows from (16.1.6) and from the fact that Δ_f is locally of finite presentation (**I**, 4.3.1). ☐

[25][Translator's note.] Locally, this is (**0**, 20.1.1).

16.4. Functorial properties of differential invariants

(16.4.1). Consider a commutative diagram of morphisms of preschemes

(16.4.1.1)
$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] \\ S & S' \arrow[l, "w"] \end{tikzcd}$$

We deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & & X' \arrow[ll, "u"'] \arrow[d, "\Delta_{f'}"] \\ X \times_S X & & X' \times_{S'} X' \arrow[ll, "v"] \end{tikzcd}$$

where v is the composite morphism (**I**, 5.3.5) and (**I**, 5.3.15):

(16.4.1.2)
$$X' \times_{S'} X' \xrightarrow{(p_1', p_2')_S} X' \times_S X' \xrightarrow{u \times_S u} X \times_S X.$$

Thus u and v give, as explained in (16.2.1), homomorphisms of augmented sheaves of rings

(16.4.1.3)
$$\nu_n : \rho^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{X'/S'}$$

(where we put $u = (\rho, \lambda)$); these homomorphisms form a projective system and therefore give, in the limit, a homomorphism of augmented sheaves of rings

(16.4.1.4)
$$\nu_\infty : \rho^*(\mathscr{P}^\infty_{X/S}) \longrightarrow \mathscr{P}^\infty_{X'/S'};$$

on the other hand, by passing to the quotients, the homomorphisms ν_n give rise to a di-homomorphism of graded algebras (relative to $\lambda^\#$):

(16.4.1.5)
$$\operatorname{gr}(u) : \rho^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'}).$$

(16.4.2). If we have a commutative diagram **IV-4** | 17

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] & X'' \arrow[l, "u'"'] \arrow[d, "f''"] \\ S & S' \arrow[l, "w"] & S'' \arrow[l, "w'"] \end{tikzcd}$$

we deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & X' \arrow[l, "u"'] \arrow[d, "\Delta_{f'}"] & X'' \arrow[l, "u'"'] \arrow[d, "\Delta_{f''}"] \\ X \times_S X & X' \times_{S'} X' \arrow[l, "v"] & X'' \times_{S''} X'' \arrow[l, "v'"] \end{tikzcd}$$

where v' is defined from u', w', f', f'' as v is from u, w, f, f'. We verify immediately that if $u'' = u \circ u'$ and $w'' = w \circ w'$, then the composite morphism $v \circ v'$ is equal to the morphism v'' deduced from u'', w'', f, f'' as v is deduced from u, w, f, f'. If we set $u' = (\rho', \lambda')$ and $u'' = (\rho'', \lambda'')$, it follows from (16.2.1) that the homomorphism $\nu''_n : \rho''^*(\mathscr{P}^n_{X/S}) \to \mathscr{P}^n_{X''/S''}$ is equal to the composite

$$\rho'^*(\rho^*(\mathscr{P}^n_{X/S})) \xrightarrow{\rho'^*(\nu^\#_n)} \rho'^*(\mathscr{P}^n_{X'/S'}) \xrightarrow{\nu'_n} \mathscr{P}^n_{X''/S''},$$

and there are analogous transitivity properties for the homomorphisms (16.4.1.4) and (16.4.1.5), allowing us to say that $\mathscr{P}^n_{X/S}$, $\mathscr{P}^\infty_{X/S}$, and $\mathscr{G}r_\bullet(\mathscr{P}_{X/S})$ *depend functorially on* f.

(16.4.3). We verify immediately (for example, by reducing to the affine case using (16.3.7)) that, with the notation of (16.4.1), the diagram

(16.4.3.1)
$$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d] & \mathscr{O}_{X'} \arrow[d] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

is commutative, where the vertical arrows define the algebra structures chosen in (16.3.5) (that is, those coming from the first projections); the same goes for the diagram

(16.4.3.2)
$$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d, "\rho^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

the vertical arrows here define the algebra structures coming from the second projections; besides, if σ and σ' are the canonical symmetries corresponding to f and f' (16.3.4), we have **IV-4** | 18

$$\nu_n \circ \rho^*(\sigma) = \sigma' \circ \nu_n$$

which carries one of the preceding diagrams to the other. We therefore deduce from (16.4.3.1) a canonical homomorphism of *augmented $\mathscr{O}_{X'}$-algebras*

(16.4.3.3)
$$P^n(u) : u^*(\mathscr{P}^n_{X/S}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \mathscr{P}^n_{X'/S'}$$

and it follows from (16.4.3.2) that the diagram

(16.4.3.4)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "u^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ u^*(\mathscr{P}^n_{X/S}) \arrow[r, "P^n(u)"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

is commutative. We thus obtain a homomorphism of *graded $\mathscr{O}_{X'}$-algebras*

(16.4.3.5)
$$\mathrm{Gr}_\bullet(u) : u^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'})$$

and in particular a homomorphism of $\mathscr{O}_{X'}$-modules

(16.4.3.6)
$$\mathrm{Gr}_1(u) : \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/S'}$$

giving rise to a commutative diagram

(16.4.3.7)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "d_{X/S}\otimes 1"'] & \mathscr{O}_{X'} \arrow[d, "d_{X'/S'}"] \\ \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \arrow[r] & \Omega^1_{X'/S'} \end{tikzcd}$$

(16.4.4). When $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(A')$, $X = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(B')$ are affine, so that we have a commutative diagram of ring homomorphisms

$$\begin{tikzcd} B \arrow[r] & B' \\ A \arrow[r] \arrow[u] & A' \arrow[u] \end{tikzcd}$$

the image of $\mathfrak{I}_{B/A}$ in $B' \otimes_{A'} B'$ is contained in $\mathfrak{I}_{B'/A'}$, and the homomorphism ν_n corresponds to the homomorphism of rings $P^n_{B/A} \to P^n_{B'/A'}$ induced from the homomorphism $B \otimes_A B \to B' \otimes_{A'} B'$ by passing to quotients. The homomorphism (16.4.3.6) corresponds to the homomorphism defined in (**0**, 20.5.4.1), and the commutative diagram (16.4.3.7) to the diagram (**0**, 20.5.4.2).

Proposition (16.4.5). — *Suppose that $X' = X \times_S S'$, f' and u the canonical projections. Then the canonical homomorphisms $P^n(u)$ (16.4.3.3) and $\mathrm{Gr}_1(u)$ (16.4.3.6) are bijective.* **IV-4 | 19**

Proof. We have $X' \times_{S'} X' = (X \times_S X) \times_S S'$, and it suffices to apply (16.2.3, (ii)), replacing g by the first projection $p_1 : X \times_S X \to X$ and f by the diagonal Δ_f. □

We note that under the hypotheses of (16.4.5) the homomorphism $\mathrm{Gr}_\bullet(u)$ (16.4.3.5) is *surjective*, but not bijective in general. However (16.2.4):

Corollary (16.4.6). — *Under the hypotheses of (16.4.5), suppose in addition that $w : S \to S'$ is flat (resp. that $\mathscr{G}r_n(\mathscr{P}_{X/S})$ are flat $\mathscr{O}_X$-modules for $n \leqslant m$); then the homomorphism*

$$\mathrm{Gr}_n(u) : u^*(\mathscr{G}r_n(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_n(\mathscr{P}_{X'/S'})$$

is bijective for each n (resp. for $n \leqslant m$).

Proof. Indeed, if w is flat, then so is $v : X' \times_{S'} X' \to X \times_S X$, so the conclusion follows from (16.2.4). □

(16.4.7). Let S be a prescheme, let $\mathscr{E}$ be a quasi-coherent $\mathscr{O}_S$-module, and set $X = \mathbf{V}(\mathscr{E})$ (**II**, 1.7.8), the vector bundle associated with $\mathscr{E}$, equal to $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$. Let $f : X \to S$ be the structure morphism. For every open U of S and every section $t \in \Gamma(U, \mathscr{E})$, t is identified with a section of $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ over U; let t' be its image in $\Gamma(f^{-1}(U), \mathscr{O}_X) = \Gamma(U, f_*(\mathscr{O}_X)) = \Gamma(U, \mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$, and set

$$\delta(t) = d^n_{X/S}(t') - t' \in \Gamma(f^{-1}(U), \mathscr{P}^n_{X/S}); \tag{16.4.7.1}$$

it is clear that δ is a di-homomorphism of modules (corresponding to the homomorphism of rings $\Gamma(U, \mathscr{O}_S) \to \Gamma(f^{-1}(U), \mathscr{O}_X)$) from $\Gamma(U, \mathscr{E})$ to $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$, whose image moreover belongs to the augmentation ideal of $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$. We deduce (by varying U) a canonical homomorphism of $\mathscr{O}_X$-algebras

$$f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{P}^n_{X/S} \tag{16.4.7.2}$$

and in view of the preceding remark, if $\mathscr{K}$ is the kernel ideal of the augmentation $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \to \mathscr{O}_S$, the image of $\mathscr{K}^{n+1}$ under (16.4.7.2) is zero; hence, after factoring by $\mathscr{K}^{n+1}$, we finally obtain a canonical homomorphism

$$\delta_n : f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})/\mathscr{K}^{n+1}) \longrightarrow \mathscr{P}^n_{X/S}. \tag{16.4.7.3}$$

Proposition (16.4.8). — *Under the conditions of (16.4.7), the homomorphisms δ_n are bijective and form a projective system of isomorphisms; we deduce an isomorphism of graded $\mathscr{O}_X$-algebras*

$$f^*(\mathbf{S}^\bullet_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}). \tag{16.4.8.1}$$

Proof. The fact that homomorphisms (16.4.7.3) form a projective system follows immediately from their definition. To prove they are isomorphisms, it suffices to prove that (16.4.8.1) is an isomorphism, since the filtrations on both sides of (16.4.7.3) are finite (Bourbaki, *Alg. comm.*, chap. III, §2, no 8, Corollary 3 to Theorem 1). To do this, consider the split exact sequence of $\mathscr{O}_S$-modules **IV-4 | 20**

$$0 \longrightarrow \mathscr{E} \xrightarrow{u} \mathscr{E} \oplus \mathscr{E} \xrightarrow{v} \mathscr{E} \longrightarrow 0 \tag{16.4.8.2}$$

where, for every pair of sections s, t of $\mathscr{E}$ over an open U of S, we take $u(s) = (-s, s)$ and $v(s,t) = s + t$. We have

$$X \times_S X = \mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \otimes_{\mathscr{O}_S} \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) = \mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}))$$

((**II**, 1.4.6) and (**II**, 1.7.11)), and the diagonal morphism $X \to X \times_S X$ corresponds (**II**, 1.2.7) to the homomorphism of $\mathscr{O}_S$-algebras $\mathbf{S}(v) : \mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}) \to \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ (**II**, 1.7.4); thus, if $\mathscr{I}$ is the kernel of this homomorphism, then

$$\mathscr{P}^n_{X/S} = f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E})/\mathscr{I}^{n+1}).$$

The proposition now will be a consequence of the following lemma: □

Lemma (16.4.8.3). — *Let Y be a ringed space, $0 \to \mathscr{F}' \xrightarrow{u} \mathscr{F} \xrightarrow{v} \mathscr{F}'' \to 0$ an exact sequence of $\mathscr{O}_Y$-modules such that each point $y \in Y$ has an open neighbourhood V such that the sequence $0 \to \mathscr{F}'|V \to \mathscr{F}|V \to \mathscr{F}''|V \to 0$ is split. Let $\mathscr{I}$ be the kernel ideal of $\mathbf{S}(v)$:*

$$\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}) \longrightarrow \mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}''),$$

and let $\operatorname{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}))$ be the graded $\mathscr{O}_Y$-algebra associated to the $\mathscr{O}_Y$-algebra $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})$ endowed with the $\mathscr{I}$-preadic filtration. *Then the homomorphism of graded $\mathscr{O}_Y$-algebras*

$$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'') \longrightarrow \operatorname{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})) \tag{16.4.8.4}$$

(where the left-hand side is the graded tensor product of symmetric $\mathscr{O}_Y$-algebras endowed with their canonical gradings (**II**, 1.7.4) *and* (**II**, 2.1.2)*), induced by the canonical injection*

$$\mathscr{F}' \longrightarrow \mathscr{I} = \operatorname{gr}^1_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})),$$

is bijective.

Proof. The injection $\mathscr{F}' \to \mathscr{I}$ indeed canonically gives a homomorphism of graded $\mathscr{O}_Y$-algebras $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \to \operatorname{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}))$, and since the right-hand side is by definition a graded $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$-algebra, we obtain the canonical homomorphism (16.4.8.4) by tensoring the preceding one with $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$. To prove the lemma we can, being a local problem, restrict to the case where $\mathscr{F} = \mathscr{F}' \oplus \mathscr{F}''$, u and v the canonical homomorphisms. Then the graded algebra $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F})$ is canonically identified with the graded tensor product $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$ (**II**, 1.7.4), and it is immediate that $\mathscr{I}$ is therefore the ideal $\mathscr{I}' \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where $\mathscr{I}'$ is the augmentation ideal of $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}')$, that is to say the (direct) sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant 1$. We conclude that $\mathscr{I}^n = \mathscr{I}'^n \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where this time $\mathscr{I}'^n$ is the direct sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant n$; we therefore have $\mathscr{I}^n/\mathscr{I}^{n+1} = \mathbf{S}^n_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, which proves that (16.4.8.4) is bijective. □

Having proved the lemma, it remains to see that the homomorphism (16.4.8.1) is indeed the image under f^* of the homomorphism (16.4.8.4) corresponding to the exact sequence (16.4.8.2); this follows readily from the definitions of u (16.4.8.2) and δ (16.4.7.1), together with the definitions of the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ and of $d^n_{X/S}$ (16.3.5) and (16.4.3.6). **IV-4 | 21**

In particular:

Corollary (16.4.9). — *Under the conditions of (16.4.7), we have a canonical isomorphism*

$$\operatorname{gr}_1(\delta) : f^*(\mathscr{E}) \simeq \Omega^1_{X/S}. \tag{16.4.9.1}$$

Corollary (16.4.10). — *If $S = \operatorname{Spec}(A)$, $\mathscr{E} = \mathscr{O}_S^m$, so that*

$$X = \operatorname{Spec}(A[T_1, \ldots, T_m]),$$

then $\mathscr{P}^n_{X/S}$ is canonically identified with the $\mathscr{O}_X$-algebra corresponding to the quotient $A[T_1, \ldots, T_m]$-algebra
$A[T_1, \ldots, T_m, U_1, \ldots, U_m]/\mathfrak{K}^{n+1}$, where the U_i ($1 \leqslant i \leqslant m$) are m new indeterminates and $\mathfrak{K}$ is the ideal generated by $U_1, \ldots, U_m$.

We thus recover in particular the structure of $\Omega^1_{X/S}$ in this case (**0**, 20.5.13).

Moreover, $d^n_{X/S}$ then sends a polynomial $F(T_1, \ldots, T_m)$ to the class modulo $\mathfrak{K}^{n+1}$ of $F(T_1 + U_1, \ldots, T_m + U_m)$, as follows from the definition (16.4.7.1).

Proposition (16.4.11). — *Let $f : X \to S$ be a morphism, let $g : S \to X$ be an S-section of X, and let $S^{(n)}$ be the n-th infinitesimal neighbourhood of S with respect to the immersion g (16.1.2). Then there exists a unique isomorphism of $\mathscr{O}_S$-algebras*

$$\varpi_n : g^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{O}_{S_g^{(n)}} \tag{16.4.11.1}$$

(via the $\mathscr{O}_S$-algebra structure on $\mathscr{O}_{S_g^{(n)}}$ defined by f (16.1.7)), making the diagram

(16.4.11.2)
$$\begin{tikzcd} \mathscr{O}_S = g^*(\mathscr{O}_X) \arrow[rr, "\lambda_n"] \arrow[dr, "g^*(d^n_{X/S})"'] & & \mathscr{O}_{S_g^{(n)}} \\ & g^*(\mathscr{P}^n_{X/S}) \arrow[ur, "\varpi_n"'] & \end{tikzcd}$$

commutative (where λ_n is the structure morphism).

Proof. In light of (**I**, 5.3.7), where we replace X, Y, S by S, X, S respectively and f by g, the diagrams

(16.4.11.3)
$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\ X \arrow[r, "(g\circ f, 1_X)_S"'] & X \times_S X \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\ X \arrow[r, "(1_X, g\circ f)_S"'] & X \times_S X \end{tikzcd}$$

identify S with the product of the $(X \times_S X)$-preschemes X and X via the morphisms Δ_f and $(g \circ f, 1_X)_S$ (resp. $(1_X, g \circ f)_S$). On the other hand, the diagrams **IV-4** | 22

(16.4.11.4)
$$\begin{tikzcd} X \arrow[r, "(g\circ f, 1_X)_S"] \arrow[d, "f"'] & X \arrow[d, "p_1"] \\ S \arrow[r, "g"'] & X \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "(1_X, g\circ f)_S"] \arrow[d, "f"'] & X \arrow[d, "p_2"] \\ S \arrow[r, "g"'] & X \end{tikzcd}$$

identify X with the product of the X-preschemes S and $X \times_S X$ via the morphisms g and p_1 (resp. p_2) (a special case of the associativity formula (**I**, 3.3.9.1)). We can say that Δ_f, considered as an X-section of $X \times_S X$ (relative to p_1 or p_2) plays the role of a *universal section* for the S-sections of X: each such section g is indeed obtained from it by the *base change* $(g \circ f, 1_X)_S : X \to X \times_S X$. The definition of the homomorphism ϖ_n and the fact that it is bijective therefore follow from these remarks and from (16.2.3, (ii)) applied to the first diagram (16.4.11.4). The commutativity of diagram (16.4.11.2) likewise follows from (16.2.3, (ii)), this time applied to the second diagram (16.4.11.4). To explain ϖ_n, we can restrict ourselves to the case where g is a closed immersion: Indeed, for every $s \in S$, there is an open neighbourhood W of s in S such that $g(W)$ is closed in an open set U of X, and it is clear that $g|W$ is a W-section of the morphism $U \cap f^{-1}(W) \to W$, the restriction of f. We can then suppose that S is a closed subprescheme of X defined by a quasi-coherent ideal $\mathscr{K}$. Then the preceding definitions show that if W is an open of S, t is a section of $\mathscr{O}_X$ over $f^{-1}(W)$, $\varpi_n(d^n t|W)$ is equal to the canonical image of t in $\Gamma(W, (\mathscr{O}_X/\mathscr{K}^{n+1})|W)$. The uniqueness of ϖ_n then follows since the image of $\mathscr{O}_X$ under $d^n_{X/S}$ generates the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$ (16.3.8). □

Corollary (16.4.12). — *Let k be a field, X a k-prescheme, x a point of X* rational over k. *Then $(\mathscr{P}^n_{X/k})_x \otimes_{\mathscr{O}_x} k(x)$ is canonically isomorphic (as an augmented $k(x)$-algebra) to $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$.*

Proof. It suffices to consider the unique k-section g of X such that $g(\mathrm{Spec}(k)) = \{x\}$. □

Corollary (16.4.13). — *Let $f : X \to S$ be a morphism, s a point of S, $X_s = X \times_S \mathrm{Spec}(k(s))$ the fibre of f in s. If $x \in X_s$ is* rational over $k(s)$, *$(\mathscr{P}^n_{X/S})_x \otimes_{\mathscr{O}_s} k(s)$ is canonically isomorphic to $\mathscr{O}_{X_s,x}/\mathfrak{m}'^{\,n+1}_x$, where $\mathfrak{m}'_x$ is the maximal ideal of $\mathscr{O}_{X_s,x}$; more precisely, this isomorphism sends $(d^n t)_x \otimes 1$* (where t is a section of $\mathscr{O}_X$ over an open neighbourhood of x in X) *to the class of $t_x \otimes 1$ modulo $\mathfrak{m}'^{\,n+1}_x$.*

Proof. This follows from (16.4.5) and (16.4.12). □

The preceding corollaries justify the terminology "sheaf of principal parts of order n".

Proposition (16.4.14). — *Let $\rho : A \to B$ be a homomorphism of rings, and let S be a multiplicative subset of B. Then the canonical homomorphisms*

(16.4.14.1) $$S^{-1}P^n_{B/A} \longrightarrow P^n_{S^{-1}B/A}$$

deduced from the canonical homomorphisms $P^n_{B/A} \to P^n_{S^{-1}B/A}$ (16.4.4), form a projective system and are bijective. IV-4 | 23

Proof. It suffices to remark that $S^{-1}((B \otimes_A B)/\mathfrak{I}^{n+1}) = S^{-1}(B \otimes_A B)/(S^{-1}\mathfrak{I})^{n+1}$ by flatness, and that $S^{-1}(B \otimes_A B) = (S^{-1}B) \otimes_A (S^{-1}B)$ (**I**, 1.3.4). ☐

Corollary (16.4.15). — *The notation being that of (16.4.14), let R be a multiplicative subset of A such that $\rho(R) \subset S$. Then we have canonical isomorphisms*

(16.4.15.1) $$S^{-1}P^n_{B/A} \simeq P^n_{S^{-1}B/R^{-1}A}$$

forming a projective system.

Proof. It evidently suffices to define canonical isomorphisms

(16.4.15.2) $$P^n_{S^{-1}B/A} \simeq P^n_{S^{-1}B/R^{-1}A}$$

that is to say that we reduce to the case where $\rho(R)$ consists of *invertible* elements of B. But then the isomorphism (16.4.15.2) is simply induced by the canonical isomorphism $B \otimes_A B \to B \otimes_{R^{-1}A} B$ by passing to quotients (**I**, 1.5.3). ☐

Corollary (16.4.16). — *Let $f : X \to S$ be a morphism of preschemes, x a point of X, $s = f(x)$. Then we have canonical isomorphisms*

(16.4.16.1) $$(\mathscr{P}^n_{X/S})_x \simeq P^n_{\mathscr{O}_x/\mathscr{O}_s}$$

forming a projective system.

We deduce corresponding isomorphisms for the associated graded rings, and in particular a canonical isomorphism

(16.4.16.2) $$(\Omega^1_{X/S})_x \simeq \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}.$$

Corollary (16.4.17). — *Let k be a field, K the field of rational functions $k(T_1, \ldots, T_r)$. Then, for every integer n, the homomorphism from $K[U_1, \ldots, U_r]$ (U_i indeterminates) to $P^n_{K/k}$ which sends U_i to $d^nT_i - T_i.1$ is surjective and induces an isomorphism from the quotient $K[U_1, \ldots, U_r]/\mathfrak{m}^{n+1}$ (where $\mathfrak{m}$ is the ideal generated by the U_i) to $P^n_{K/k}$.*

Proof. This follows from (16.4.8), (16.4.10) and (16.4.14), where we take $A = k$, $B = k[T_1, \ldots, T_r]$ and $S = B - \{0\}$. ☐

We thus recover the fact that the dT_i form a basis of the K-vector space $\Omega^1_{K/k}$ (**0**, 20.5.10).

(16.4.18). Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of preschemes, and consider the canonical homomorphisms of augmented $\mathscr{O}_X$-algebras (16.4.3.3)

(16.4.18.1) $$g_{X/Y/Z} : \mathscr{P}^n_{X/Z} \longrightarrow \mathscr{P}^n_{X/Y}$$

(16.4.18.2) $$f_{X/Y/Z} : f^*(\mathscr{P}^n_{Y/Z}) \longrightarrow \mathscr{P}^n_{X/Z}.$$

Then $g_{X/Y/Z}$ is surjective, and its kernel is the sheaf of ideals generated by the image under $f_{X/Y/Z}$ of the augmentation ideal of $f^*(\mathscr{P}^n_{Y/Z})$.

Proof. First note that $g_{X/Y/Z}$ corresponds to the case in (16.4.3.3) where $X' = X$, $S' = Y$ and $S = Z$, $u = 1_X$, and $f_{X/Y/Z}$ to the case where we replace X', X, S, S' by X, Y, Z, Z respectively and u, f by f, g respectively. IV-4 | 24

We have a commutative diagram (**I**, 3.5)

(16.4.18.3)
$$\begin{tikzcd} X \arrow[r, "\Delta_f"] \arrow[dr, "f"'] & X \times_Y X \arrow[r, "j"] \arrow[d, "p"] & X \times_Z X \arrow[d, "f \times_Z f"] \\ & Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

where $j = (1_X, 1_X)_Z$ is an immersion, $j \circ \Delta_f = \Delta_{g \circ f}$, and p is the structure morphism. Since we can restrict ourselves to the case where X, Y and Z are affine, we can suppose that the immersions Δ_f, Δ_g and j are closed, so that $\mathscr{O}_X$ and $\mathscr{O}_{X \times_Y X}$ are identified respectively with $\mathscr{O}_{X \times_Z X}/\mathscr{I}$ and $\mathscr{O}_{X \times_Z X}/\mathscr{L}$, where $\mathscr{L} \supset \mathscr{I}$ are two quasi-coherent ideals corresponding respectively to the immersions $\Delta_{g \circ f}$ and j. The $\mathscr{O}_X$-algebra $\mathscr{P}^n_{X/Z}$ is identified with $\mathscr{O}_{X \times_Z X}/\mathscr{I}^{n+1}$, and $\mathscr{P}^n_{X/Y}$ is identified with $\mathscr{O}_{X \times_Y X}/(\mathscr{I}/\mathscr{L})^{n+1}$, which is to say with $\mathscr{O}_{X \times_Z X}/(\mathscr{I}^{n+1} + \mathscr{L})$, and therefore with the quotient of $\mathscr{P}^n_{X/Z}$ by $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$. But we know (*loc. cit*) that if p and j make $X \times_Y X$ the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$, so if $\mathscr{O}_Y$ is identified to $\mathscr{O}_{Y \times_Z Y}/\mathscr{K}$, where $\mathscr{K}$ is the ideal corresponding to Δ_g, $\mathscr{L}$ is equal to $(f \times_Z f)^*(\mathscr{K}).\mathscr{O}_{X \times_Z X}$ (**I**, 4.4.5). Since $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$ is the ideal of $\mathscr{P}^n_{X/Z}$ generated by the image of $\mathscr{L}$, we deduce the proposition. ☐

Corollary (16.4.19). — *With the notation of (16.4.18), we have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.19.1)
$$f^*(\Omega^1_{Y/Z}) \xrightarrow{f_{X/Y/Z}} \Omega^1_{X/Z} \xrightarrow{g_{X/Y/Z}} \Omega^1_{X/Y} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 5.7.1).

Proposition (16.4.20). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ a closed immersion, $\mathscr{K}$ the quasi-coherent sheaf of ideals of $\mathscr{O}_Y$ corresponding to j. It follows that $\mathscr{P}^n_{X/Y} = \mathscr{O}_X = \mathscr{O}_Y/\mathscr{K}$, the canonical homomorphism $j_{X/Y/Z} : j^*(\mathscr{P}^n_{Y/Z}) \to \mathscr{P}^n_{X/Z}$ is surjective, and its kernel is the ideal of $j^*(\mathscr{P}^n_{Y/Z})$ generated by $j^*(\mathscr{O}_Y \cdot d^n_{Y/Z}(\mathscr{K}))$ (it should be noted that $d^n_{Y/Z}(\mathscr{K})$ is a subsheaf of abelian groups of $\mathscr{P}^n_{Y/Z}$, but not an $\mathscr{O}_Y$-module in general).*

Proof. We know (**I**, 5.3.8) that the diagonal $\Delta_j : X \to X \times_Y X$ is an isomorphism, from which the first assertion follows. If ϖ_1 and ϖ_2 are the two canonical homomorphisms of algebras $\mathscr{O}_Y \to \mathscr{P}^n_{Y/Z}$ corresponding respectively to the two canonical projections p_1, p_2 of $Y \times_Z Y \to Y$, recall that by definition ((16.3.5) and (16.3.6)) ϖ_1 is the structure homomorphism of the $\mathscr{O}_Y$-algebra $\mathscr{P}^n_{Y/Z}$ and $\varpi_2 = d^n_{Y/Z}$. The $\mathscr{O}_X$-algebra $j^*(\mathscr{P}^n_{Y/Z})$ is therefore identified with $\mathscr{P}^n_{Y/Z}/\varpi_1(\mathscr{K})\mathscr{P}^n_{Y/Z}$ and its quotient by the ideal generated by $j^*(d^n_{Y/Z}(\mathscr{K}))$ to $\mathscr{P}^n_{Y/Z}/(\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K}))\mathscr{P}^n_{Y/Z}$. Now note that we have a commutative diagram **IV-4 | 25**

$$\begin{tikzcd} Y \arrow[d, "\Delta_f"'] & X \arrow[l, "j"'] \arrow[d, "\Delta_{f \circ j}"] \\ Y \times_Z Y & X \times_Z X \arrow[l, "j \times_Z j"] \end{tikzcd}$$

identifying X with the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$ (**I**, 5.3.7). Since $j \times_Z j$ is an immersion, we therefore deduce from this remark and from (16.2.2) that if $\Delta^n_{Y/Z}$ and $\Delta^n_{X/Z}$ denote the infinitesimal neighbourhoods of order n of Y and X by the canonical immersions Δ_f and $\Delta_{f \circ j}$ respectively, then we have a diagram

$$\begin{tikzcd} \Delta^n_{Y/Z} \arrow[d] & \Delta^n_{X/Z} \arrow[l] \arrow[d] \\ Y \times_Z Y & X \times_Z X \arrow[l, "j \times_Z j"] \end{tikzcd}$$

making $\Delta^n_{X/Z}$ the product of the $(Y \times_Z Y)$-preschemes $\Delta^n_{Y/Z}$ and $X \times_Z X$. We can also say that $\mathscr{P}^n_{X/Z}$ is identified with the sheaf of rings $\mathscr{P}^n_{Y/Z} \otimes_{\mathscr{O}_{Y\times_Z Y}} \mathscr{O}_{X\times_Z X}$. But we see immediately (for example, by restricting to the affine case) that $\mathscr{O}_{X\times_Z X} = \mathscr{O}_{Y\times_Z Y}/(p_1^*(\mathscr{K}) + p_2^*(\mathscr{K}))\mathscr{O}_{Y\times_Z Y}$. Therefore $\mathscr{P}^n_{X/Z}$ is identified with the quotient of $\mathscr{P}^n_{Y/Z}$ by the ideal generated by the image in $\mathscr{P}^n_{Y/Z}$ of $p_1^*(\mathscr{K}) + p_2^*(\mathscr{K})$. But by definition this ideal is generated by $\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K})$. □

Corollary (16.4.21). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ an immersion. We have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.21.1)
$$\mathscr{N}_{X/Y} \longrightarrow j^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 20.5.12.1).

Corollary (16.4.22). — *If $f : X \to S$ is a morphism locally of finite presentation, $\mathscr{P}^n_{X/S}$ and $\Omega^1_{X/S}$ are quasi-coherent $\mathscr{O}_X$-modules of finite presentation.*

Proof. We immediately reduce to the case where $S = \operatorname{Spec}(A)$ is affine, $X = \operatorname{Spec}(B)$, where $B = A[T_1, \ldots, T_r]/\mathfrak{K}$, $\mathfrak{K}$ being an ideal of finite type of $C = A[T_1, \ldots, T_r]$. We then apply (16.4.20) with $Z = S$, $Y = \operatorname{Spec}(C)$ and $\mathscr{K} = \widetilde{\mathfrak{K}}$. Then $j^*(\mathscr{P}^n_{Y/Z})$ is a free $\mathscr{O}_X$-module of finite rank (16.4.10) and the hypothesis on $\mathfrak{K}$ implies that $j^*(\mathscr{O}_Y.d^n_{Y/Z}(\mathscr{K}))$ generates a quasi-coherent $\mathscr{O}_X$-module of finite type; hence the conclusion. □

Proposition (16.4.23). — *Let X, Y be two S-preschemes, $Z = X \times_S Y$ their product, $p : X \times_S Y \to X$ and $q : X \times_S Y \to Y$ the canonical projections. Then the canonical homomorphism*

(16.4.23.1)
$$p_{Z/X/S} \oplus q_{Z/Y/S} : p^*(\Omega^1_{X/S}) \oplus q^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{(X\times_S Y)/S}$$

is bijective.

Proof. The commutative diagram **IV-4** | 26

$$\begin{tikzcd}
Y \arrow[d, "g"'] & X \times_S Y \arrow[l, "q"'] \arrow[d, "h"'] & X \times_S Y \arrow[l, "\mathrm{id}"'] \arrow[d, "p"] \\
S & S \arrow[l, "\mathrm{id}"] & X \arrow[l, "f"]
\end{tikzcd}$$

gives us a factorization of the canonical *isomorphism* $P^n(p)$ (16.4.5)

$$p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y}$$

and similarly, switching X with Y, we have a factorization of the *isomorphism* $P^n(q)$

$$q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/X}.$$

This proves that the canonical homomorphism (16.4.18.1)

$$p_{Z/X/S} : p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \quad (\text{resp. } q_{Z/Y/S} : q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S})$$

is *injective*, and that the kernel of the canonical surjective homomorphism (16.4.18.2)

$$\mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y} \quad (\text{resp. } \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/X})$$

is complementary to the image of $p_{Z/X/S}$ (resp. $q_{Z/Y/S}$). On the other hand, by (16.4.18), this kernel is generated by the image under $q_{Z/Y/S}$ (resp. $p_{Z/X/S}$) of the augmentation ideal of $q^*(\mathscr{P}^n_{Y/S})$ (resp. $p^*(\mathscr{P}^n_{X/S})$). We conclude the proposition by considering the case $n = 1$. □

We immediately generalize (16.4.23) to the case of a product of any finite number of S-preschemes.

Remarks (16.4.24). —

(i) We will see (17.2.3) that when the morphism $f : X \to Y$ in (16.4.18) is *smooth*, the homomorphism $f_{X/Y/Z}$ in (16.4.19.1) is locally *left invertible* and in particular injective. Similarly, when the morphism $f \circ j : X \to Z$ of (16.4.20) is *smooth*, the homomorphism on the left in (16.4.21.1) is locally *left invertible* and *a fortiori* injective (17.2.5). In Chapter V, we will also give a variant, in the case of modules over a prescheme, of the "imperfection modules" studied in (**0**, 20.6), and the exact sequences where they occur.

(ii) Let X be a topological space, $\mathscr{A}$ a sheaf of rings over X and $\mathscr{B}$ an $\mathscr{A}$-algebra over X. Then it is clear that

$$U \longmapsto P^n_{\Gamma(U,\mathscr{B})/\Gamma(U,\mathscr{A})} \quad (U \text{ open in } X)$$

is a presheaf of augmented $\Gamma(U,\mathscr{B})$-algebras, and therefore the associated sheaf $\mathscr{P}^n_{\mathscr{B}/\mathscr{A}}$ is an augmented $\mathscr{B}$-algebra. In the particular case where X is a prescheme, $f = (\psi, \theta) : X \to S$ a morphism of preschemes, it follows easily from (16.4.16) and from the exactness of the functor $\varinjlim$ that $\mathscr{P}^n_{X/S}$ is canonically isomorphic to $\mathscr{P}^n_{\mathscr{O}_X/\psi^*(\mathscr{O}_S)}$. It follows that the formalism developed in the present paragraph could be considered as a particular case **IV-4** | 27
of a differential formalism for ringed spaces endowed with a sheaf of algebras over the structure sheaf. However, we did not start with this point of view, which is less intuitive and less convenient for applications. It also seems that, for various kinds of "varieties", the "global" constructions of the $\mathscr{P}^n$ analogous to those we have used here are also better suited for applications.

16.5. Relative tangent sheaves and bundles; derivations.

(16.5.1). Let $f = (\psi, \theta) : X \to S$ be a morphism of ringed spaces. For every $\mathscr{O}_X$-module $\mathscr{F}$, an *S-derivation* (or (X/S)*-derivation*, or f*-derivation*) *from* $\mathscr{O}_X$ *to* $\mathscr{F}$ is any homomorphism of *sheaves of additive groups* $D : \mathscr{O}_X \to \mathscr{F}$ satisfying the following conditions:

(a) for every open V of X, and every pair of sections (t_1, t_2) of $\mathscr{O}_X$ over V, we have

(16.5.1.1) $$D(t_1 t_2) = t_1 D(t_2) + D(t_1) t_2;$$

(b) for every open V of X, every section t of $\mathscr{O}_X$ over V, and every section s of $\mathscr{O}_S$ over an open U of S such that $V \subset f^{-1}(U)$, we have

(16.5.1.2) $$D((s|V)t) = (s|V)D(t).$$

It is clear that this amounts to saying that, for all $x \in X$, the homomorphism of additive groups $D_x : \mathscr{O}_x \to \mathscr{F}_x$ is an $\mathscr{O}_{f(x)}$*-derivation*.

Another interpretation consists of considering the $\mathscr{O}_X$-algebra $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$ as equal to $\mathscr{O}_X \oplus \mathscr{F}$, the algebra structure being defined by the condition that for every open V of X, the product of two sections of $\mathscr{O}_X$ (resp. of a section of $\mathscr{O}_X$ and a section of $\mathscr{F}$) over V is defined by the ring structure of $\Gamma(V, \mathscr{O}_X)$ (resp. the $\Gamma(V, \mathscr{O}_X)$-module structure on $\Gamma(V, \mathscr{F})$), and the product of two sections of $\mathscr{F}$ over V is chosen to be zero; then $\mathscr{F}$ is an ideal of $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, the kernel of the canonical augmentation $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F}) \to \mathscr{O}_X$, and to say that D is an S-derivation from $\mathscr{O}_X$ to $\mathscr{F}$ means that $\iota_{\mathscr{O}_X} + D$ is an $\mathscr{O}_S$*-homomorphism of algebras* from $\mathscr{O}_X$ to $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, which, composed with the augmentation, gives $\iota_{\mathscr{O}_X}$.

The S-derivations from $\mathscr{O}_X$ to $\mathscr{F}$ clearly form a $\Gamma(X, \mathscr{O}_X)$*-module* $\mathrm{Der}_S(\mathscr{O}_X, \mathscr{F})$.

When $\mathscr{F} = \mathscr{O}_X$, an S-derivation of $\mathscr{O}_X$ to itself is simply called an *S-derivation of $\mathscr{O}_X$*.

Proposition (16.5.2). — *Let A be a ring, B an A-algebra, L a B-module; let $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $\mathscr{F} = \widetilde{L}$. Then the map $D \mapsto \Gamma(D)$ which sends every S-derivation D from $\mathscr{O}_X$ to $\mathscr{F}$ to the map $\Gamma(D) : t \mapsto D(t)$ from B to L, is an isomorphism of B-modules from $\mathrm{Der}_S(\mathscr{O}_X, \mathscr{F})$ to $\mathrm{Der}_A(B, L)$ (cf.* (**0**, 20.1.2)*).*

Proof. This follows immediately from the given interpretation of S-derivations in terms of homo- **IV-4** | 28
morphisms of algebras, analogous to the interpretation given in (**0**, 20.1.6), and from the canonical correspondence between homomorphisms of $\mathscr{O}_X$-algebras and homomorphisms of B-algebras ((**I**, 1.3.13) and (**I**, 1.3.8)). ☐

Proposition (16.5.3). — *Let $f = (\psi, \theta) : X \to S$ be a morphism of preschemes.*

(i) *The differential $d_{X/S} : \mathscr{O}_X \to \Omega^1_{X/S}$* (16.3.6) *is an S-derivation.*

(ii) *For every $\mathscr{O}_X$-module $\mathscr{F}$, the map $u \mapsto u \circ d_{X/S}$ is an isomorphism of $\Gamma(X, \mathscr{O}_X)$-modules*

$$\mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \simeq \mathrm{Der}_S(\mathscr{O}_X, \mathscr{F}). \tag{16.5.3.1}$$

Proof. Assertion (i) has already been noted in (16.3.6). On the other hand, it follows immediately (in light of (**0**, 20.4.8)) that $u \mapsto u \circ d_{X/S}$ is injective, by considering the restrictions of both sides to a stalk $\mathscr{O}_x$ and using (16.4.16.2). To see that the homomorphism (16.5.3.1) is surjective, consider an S-derivation $D : \mathscr{O}_X \to \mathscr{F}$; for every affine open $V = \mathrm{Spec}(B)$ of X, such that $f(V)$ is contained in an affine open $U = \mathrm{Spec}(A)$ of S, $D_V : B \to \Gamma(V, \mathscr{F})$ is an A-derivation, and therefore there exists a *unique* B-homomorphism $u_V : \Omega^1_{B/A} \to \Gamma(V, \mathscr{F})$ such that $D_V = u_V \circ d_{B/A}$ (**0**, 20.4.8); in addition, the uniqueness of u_V shows immediately that for an affine open $W \subset V$ we have $u_W = u_V|W$, and therefore the u_V define a homomorphism of $\mathscr{O}_X$-modules $u : \Omega^1_{X/S} \to \mathscr{F}$ answering the question. □

(16.5.4). With the notation of (16.5.1), for every open U of X, $\mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a $\Gamma(U, \mathscr{O}_X)$-module and it is clear that the map $U \mapsto \mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a presheaf; in fact, it is even a *sheaf* (and therefore an $\mathscr{O}_X$-module), in light of the pointwise characterization of S-derivations, seen in (16.5.1). This $\mathscr{O}_X$-module is denoted by $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ and is called the *sheaf of S-derivations from $\mathscr{O}_X$ to $\mathscr{F}$*, and what we have just seen is also expressed by the following corollary:

Corollary (16.5.5). — *For every $\mathscr{O}_X$-module $\mathscr{F}$, the homomorphism of $\mathscr{O}_X$-modules induced by $u \mapsto u \circ d_{X/S}$*

$$\mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \longrightarrow \mathscr{D}er_S(\mathscr{O}_X, \mathscr{F}) \tag{16.5.5.1}$$

is bijective.

Corollary (16.5.6). — (i) *If the morphism $f : X \to S$ is locally of finite presentation and if $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-module.*

(ii) *If in addition S is locally Noetherian and if $\mathscr{F}$ is coherent, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a coherent $\mathscr{O}_X$-module.*

Proof. The assertion (i) follows from the isomorphism (16.5.5.1), from (16.4.22), and (**I**, 1.3.12); the assertion (ii) follows from (**0**, 5.3.5). □

(16.5.7). We set

$$\mathfrak{G}_{X/S} = \mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \mathscr{D}er_S(\mathscr{O}_X, \mathscr{O}_X), \tag{16.5.7.1}$$

and say that it is the *sheaf of S-derivations of $\mathscr{O}_X$*, or even the *tangent sheaf of X relative to S*: it is therefore the *dual* of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. If f is locally of finite presentation, $\mathfrak{G}_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module; if in addition S is locally Noetherian, then $\mathfrak{G}_{X/S}$ is coherent (16.5.6). **IV-4** | 29

(16.5.8). Suppose in particular that $\Omega^1_{X/S}$ is a *locally free* $\mathscr{O}_X$-module (of finite rank) (which will be the case when f is *smooth* (17.2.3)); then $\mathfrak{G}_{X/S}$ is a locally free $\mathscr{O}_X$-module of the same rank as $\Omega^1_{X/S}$ at each point. More specifically, suppose that $\Omega^1_{X/S}$ is of rank n at a point x; then there are n sections s_i ($1 \leqslant i \leqslant n$) of $\mathscr{O}_X$ over an affine neighbourhood U of x such that the canonical images of the ds_i in $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$ form a basis of this $k(x)$-vector space; by virtue of Nakayama's lemma, the germs $(ds_i)_x = d(s_i)_x$ of the ds_i at the point x form a basis of the $\mathscr{O}_x$-module $(\Omega^1_{X/S})_x$, and therefore, by restricting U, we can suppose that the ds_i form a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$. So the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \mathfrak{G}_{X/S})$ is dual to the above; we denote by $(D_i)_{1 \leqslant i \leqslant n}$ or $\left(\frac{\partial}{\partial s_i}\right)_{1 \leqslant i \leqslant n}$ the dual basis of $(ds_i)_{1 \leqslant i \leqslant n}$, so that, by (16.5.3), we have

$$D_i s_j = \langle D_i, ds_j \rangle = \left\langle \frac{\partial}{\partial s_i}, ds_j \right\rangle = \delta_{ij} \quad \text{(Kronecker's symbol).} \tag{16.5.8.1}$$

Every $\Gamma(S, \mathscr{O}_S)$-derivation of the $\Gamma(S, \mathscr{O}_S)$-algebra $\Gamma(U, \mathscr{O}_X)$ is therefore written uniquely as

$$D = \sum_{i=1}^{n} a_i D_i = \sum_{i=1}^{n} a_i \frac{\partial}{\partial s_i},$$

where the a_i ($1 \leqslant i \leqslant n$) are sections of $\mathscr{O}_X$ over U. For every section $g \in \Gamma(U, \mathscr{O}_X)$, if we put $dg = \sum_{i=1}^n c_i ds_i$, then we have $c_i = \langle D_i, dg\rangle = D_i g$ by virtue of (16.5.8.1), in other words,

$$dg = \sum_{i=1}^n (D_i g) ds_i = \sum_{i=1}^n \frac{\partial g}{\partial s_i} ds_i. \tag{16.5.8.2}$$

(16.5.9). Let D_1, D_2 be two S-derivations of $\mathscr{O}_X$. For every open U of X, if D_1^U, D_2^U are the corresponding derivations of the ring $\Gamma(U, \mathscr{O}_X)$, the *bracket*

$$[D_1^U, D_2^U] = D_1^U \circ D_2^U - D_2^U \circ D_1^U$$

is also a derivation of this ring, and therefore the $\psi^*(\mathscr{O}_S)$-endomorphism of $\mathscr{O}_X$

$$[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1 \tag{16.5.9.1}$$

is also an S-derivation; as we immediately check that this bracket satisfies the Jacobi identity, we have thus defined on $\mathrm{Der}_S(\mathscr{O}_X, \mathscr{O}_X)$ a $\Gamma(S, \mathscr{O}_S)$-*Lie algebra* structure. Since the definition of this structure commutes with the restriction to an open of X, we thus see that $\mathfrak{G}_{X/S}$ is canonically equipped with a $\psi^*(\mathscr{O}_S)$-*Lie algebra* structure. Note that the mapping $(D_1, D_2) \mapsto [D_1, D_2]$ is *not* $\Gamma(X, \mathscr{O}_X)$-*bilinear*.

(16.5.10). For every base change $g : S' \to S$, if we set $X' = X \times_S S'$, then we have seen in (16.4.5) that there is a canonical isomorphism

$$\Omega^1_{X/S} \otimes_S S' \simeq \Omega^1_{X'/S'}, \tag{16.5.10.1}$$

from which we deduce, by (16.5.10.1), a canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3) **IV-4** | 30

$$\mathfrak{G}_{X/S} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} \longrightarrow \mathfrak{G}_{X'/S'}, \tag{16.5.10.2}$$

which is neither injective nor surjective in general. However:

Proposition (16.5.11). — (i) *If $g : S' \to S$ is a flat morphism and if f is locally of finite type (resp. locally of finite presentation), then the homomorphism (16.5.10.2) is injective (resp. bijective).*

(ii) *If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module of finite type, then the homomorphism (16.5.10.2) is bijective.*

Proof. The assertion (ii) follows from Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3, prop. 7. The assertion (i) follows similarly from Bourbaki, *Alg. Comm.*, chap. I, §2, n°10, prop. 11 and from the fact that if f is locally of finite type (resp. locally of finite presentation), then $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of finite type (resp. of finite presentation) ((16.3.9) and (16.4.22)). □

(16.5.12). Since $\Omega^1_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module, we can consider the vector bundle over X defined by $\Omega^1_{X/S}$ (**II**, 1.7.8)

$$T_{X/S} = \mathbf{V}(\Omega^1_{X/S}) \tag{16.5.12.1}$$

which is called the *tangent bundle of X relative to S*. We have therefore a canonical bijection (**II**, 1.7.9)

$$\Gamma(T_{X/S}/S) \simeq \mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \Gamma(X, \mathfrak{G}_{X/S})$$

by definition of $\mathfrak{G}_{X/S}$, and we can replace X by an open set U of X in this isomorphism; so we can say that the *tangent sheaf* of X relative to S is isomorphic to the *sheaf of germs of S-sections* of the tangent bundle of X relative to S. If $f : X \to Y$ is an S-morphism, we saw (16.4.19) that we have a canonical homomorphism $f_{X/Y/S} : f^*(\Omega^1_{Y/S}) \to \Omega^1_{X/S}$, which, having in mind that

$$\mathbf{V}(f^*(\Omega^1_{Y/S})) = \mathbf{V}(\Omega^1_{Y/S}) \times_Y X \quad (\mathbf{II}, 1.7.11)\,,$$

gives us an X-morphism $T_{X/S}(f) : T_{X/S} \to T_{Y/S} \times_Y X$. If $g : Y \to Z$ is a second S-morphism, we have $T_{X/S}(g \circ f) = (T_{Y/S}(g) \times 1_X) \circ T_{X/S}(f)$ (**0**, 20.5.4.1).

It follows from (16.5.10.1) and from (**II**, 1.7.11) that for every base change $g : S' \to S$ we have a canonical isomorphism

$$T_{X'/S'} \simeq T_{X/S} \times_S S' = T_{X/S} \times_X X'. \tag{16.5.12.2}$$

(16.5.13). For every point $x \in X$, we define the *tangent space of X at the point x* (relative to S) to be the set of points in the fibre $T_{X/S} \times_X \operatorname{Spec}(k(x))$ that are *rational over* $k(x)$, that is, the set

$$T_{X/S}(x) = \operatorname{Hom}_{k(x)}(\Omega^1_{X/S} \otimes_{\mathscr{O}_x} k(x), k(x)), \tag{16.5.13.1}$$

which is the *dual* of the $k(x)$-vector space $\Omega^1_{\mathscr{O}_x/\mathscr{O}_s}/\mathfrak{m}_x \cdot \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}$. When $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of *finite type*, then $T_{X/S}(x)$ is a vector space of finite rank over $k(x)$, and for every base change $g :$ **IV-4** | 31
$S' \to S$, and every point $x' \in X' = X \times_S S'$ over x, we have a canonical isomorphism

$$T_{X'/S'}(x') \simeq T_{X/S} \otimes_{k(x)} k(x'). \tag{16.5.13.2}$$

If x is *rational over* $k(s)$, where $s = f(x)$ (so that $k(s) \to k(x)$ is an isomorphism), it follows from (16.4.13) that we have a canonical isomorphism

$$T_{X/S}(x) = T_{X_s/k(s)}(x) = \operatorname{Hom}_{k(s)}(\mathfrak{m}'_x/\mathfrak{m}'^2_x, k(x)), \tag{16.5.13.3}$$

where $\mathfrak{m}'_x$ is the maximal ideal of $\mathscr{O}_{X_s,x} = \mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$. In the case where S is the spectrum of a field k, we recover the definition of the Zariski tangent space of a point $x \in X$ *rational over* k, as the dual of $\mathfrak{m}_x/\mathfrak{m}_x^2$.

Let Y be a second S-prescheme and let $g : Y \to X$ be an S-morphism; then we have a canonical homomorphism of $\mathscr{O}_Y$-modules (16.4.19)

$$g_{Y/X/S} : g^*(\Omega^1_{X/S}) \longrightarrow \Omega^1_{Y/S}. \tag{16.5.13.4}$$

Now note that if $y \in Y$ and $x = g(y)$, then we have

$$g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y) = (\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)) \otimes_{k(x)} k(y)$$

and consequently, if $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module *of finite type*, then we can identify

$$\operatorname{Hom}_{k(y)}(g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y), k(y))$$

with $T_{X/S}(x) \otimes_{k(x)} k(y)$. We therefore deduce from the homomorphism (16.5.13.4) a homomorphism of $k(y)$-*vector spaces*

$$T_y(g) : T_{Y/S}(y) \longrightarrow T_{X/S}(x) \otimes_{k(x)} k(y) \tag{16.5.13.5}$$

called the *linear map tangent to g at the point y*. When y is *rational over* $k(s)$, we can identify $k(s)$, $k(y)$, and $k(x)$, and $T_y(g)$ is then a homomorphism of $k(s)$-vector spaces $T_{Y/S}(y) \to T_{X/S}(x)$; also note that in this case, $g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y)$ is identified with $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$, and the above homomorphism is therefore defined without any finiteness conditions on $\Omega^1_{X/S}$ and it is none other than the homomorphism $T_{Y/S}(g)$ (16.5.12) restricted to the fibre of $T_{Y/S}$ at the point y.

(16.5.14). The interpretation of derivations of an A-algebra B to a B-module L, given in (**0**, 20.1.1), translates to the language of preschemes in the following way.

Consider two morphisms of preschemes $f : X \to S$, $g : Y \to S$, and a closed subprescheme Y_0 of Y defined by a *square-zero* ideal $\mathscr{J}$ of $\mathscr{O}_Y$ (so that Y and Y_0 have the *same underlying topological space*). Suppose we are given an S-morphism $u_0 : Y_0 \to X$, so that we have a commutative diagram

$$\begin{tikzcd} X \arrow[d, "f"'] & Y_0 \arrow[l, "u_0"'] \arrow[d, "j"] \\ S & Y \arrow[l, "g"] \arrow[ul, dashed, "u"'] \end{tikzcd} \tag{16.5.14.1}$$

and we seek *S-morphisms* $u : Y \to X$ such that $u_0 = u \circ j$ (in other words, we ask whether the **IV-4** | 32
preceding diagram can be completed by the dotted arrow u while remaining *commutative*).

For that, consider an affine open $U = \operatorname{Spec}(C)$ of Y; its inverse image $j^{-1}(U)$ is the affine open $U_0 = \operatorname{Spec}(C/\mathfrak{L})$, where $\mathfrak{L} = \Gamma(U, \mathscr{J})$, a *square-zero* ideal in C; suppose that U is small enough so that $u_0(U_0)$ is contained in an affine open $V = \operatorname{Spec}(B)$ of X and that $g(U) = f(u_0(U_0))$ is contained in an affine open $W = \operatorname{Spec}(A)$ of S, so that B and C are A-algebras and $u_0|U_0$ corresponds to an A-homomorphism ψ from B to $C/\mathfrak{L}$; Let $P(U_0)$ be the set of restrictions $u|U$ of the sought morphisms; it corresponds canonically to the A-algebra homomorphisms $\phi : B \to C$ such that the *composite* $B \xrightarrow{\phi} C \to C/\mathfrak{L}$ *is equal to* ψ. So we know (**0**, 20.1.1) that the set of such homomorphisms is either empty or of the form $\phi_1 + \mathrm{Der}_A(B, \mathfrak{L})$; when $P(U_0)$ is not empty, the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ acts by addition on $P(U_0)$, which is therefore an *affine space* for the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ (or even a *principal homogeneous space* (or *torsor*) *under* $\mathrm{Der}_A(B, \mathfrak{L})$).

Now notice that, since $\mathfrak{L}$ is equipped with a B-module structure via ψ, we have an *isomorphism* $v \mapsto v \circ d_{B/A}$ of $\mathrm{Hom}_B(\Omega^1_{B/A}, \mathfrak{L})$ onto $\mathrm{Der}_A(B, \mathfrak{L})$ (**0**, 20.4.8). Besides, as $\mathfrak{L}$ is square-zero, therefore a $(C/\mathfrak{L})$-module, every B-homomorphism $v : \Omega^1_{B/A} \to \mathfrak{L}$ can be considered as a $(C/\mathfrak{L})$-homomorphism $\Omega^1_{B/A} \otimes_B (C/\mathfrak{L}) \to \mathfrak{L}$. As $\mathscr{J}$ is square-zero, it can be considered as a quasi-coherent $\mathscr{O}_{Y_0}$-module; let's introduce the $\mathscr{O}_{Y_0}$-module

(16.5.14.2) $$\mathscr{G} = \mathscr{H}om(u_0^*(\Omega^1_{X/S}), \mathscr{J});$$

it follows from the fact that $\Omega^1_{B/A} = \Gamma(V, \Omega^1_{X/S})$ (16.3.7) that we can write $\mathrm{Der}_A(B, \mathfrak{L}) = \Gamma(U_0, \mathscr{G})$.

As $P(U_0)$ is defined as a set of S-morphisms $U \to X$, it is clear that $U_0 \mapsto P(U_0)$ is a *sheaf of sets* $\mathscr{P}$ on Y_0. We can use this fact to prove that the map $h : \Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ defining the torsor structure on $P(U_0)$ is independent of choice of V and W and also that, if $U' \subset U$ is a second affine open of Y, U'_0 its inverse image in Y_0, then the diagram

(16.5.14.3)
$$\begin{tikzcd} \Gamma(U_0, \mathscr{G}) \times P(U_0) \arrow[r, "h"] \arrow[d] & P(U_0) \arrow[d] \\ \Gamma(U'_0, \mathscr{G}) \times P(U'_0) \arrow[r, "h'"'] & P(U'_0) \end{tikzcd}$$

is commutative (the vertical arrows being the restrictions). In light of the above remark, we reduce to proving the commutativity of the above diagram when h is defined as such from affine opens V, W and h' from affine opens $V' \subset V$ and $W' \subset W$. But because of the preceding description of h, **IV-4** | 33
this follows from the commutativity of the diagram (**0**, 20.5.4.2).

The mapping $\Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ therefore defines a homomorphism of *sheaves of sets* $m : \mathscr{G} \times \mathscr{P} \to \mathscr{P}$ such that, for all open sets U_0 for which $\Gamma(U_0, \mathscr{P}) \neq \varnothing$, $m_{U_0} : \Gamma(U_0, \mathscr{G}) \times \Gamma(U_0, \mathscr{P}) \to \Gamma(U_0, \mathscr{P})$ is an external law defining in $\Gamma(U_0, \mathscr{P})$ a torsor structure for the group $\Gamma(U_0, \mathscr{G})$.

(16.5.15). In general, when we are given a sheaf of sets $\mathscr{P}$ over a topological space Z, a sheaf of groups $\mathscr{G}$ (not necessarily commutative), and a homomorphism of sheaves of sets $m : \mathscr{G} \times \mathscr{P} \to \mathscr{P}$ such that, for every open $U \subset Z$ such that $\Gamma(U, \mathscr{P}) \neq \varnothing$, $m_U : \Gamma(U, \mathscr{G}) \times \Gamma(U, \mathscr{P}) \to \Gamma(U, \mathscr{P})$ makes $\Gamma(U, \mathscr{P})$ a *torsor* under the group $\Gamma(U, \mathscr{G})$, then we say that $\mathscr{P}$ is a *pseudo-torsor* (or *formally principal homogeneous sheaf*) under the sheaf of groups $\mathscr{G}$. We say that $\mathscr{P}$ is a *torsor* (or *principal homogeneous sheaf*) under $\mathscr{G}$ [26] if in addition $\Gamma(U, \mathscr{P}) \neq \varnothing$ for every open $U \neq \varnothing$ in a suitable basis for the topology of Z.

For the general theory of torsors, we refer to [42]; we will limit ourselves to recalling the canonical correspondence between isomorphism classes of torsors (for a *given* $\mathscr{G}$) and elements of the cohomology set $\mathrm{H}^1(Z, \mathscr{G})$. Consider a torsor $\mathscr{P}$ under $\mathscr{G}$ and an open cover (U_λ) of Z such that $\Gamma(U_\lambda, \mathscr{P}) \neq \varnothing$ for every λ; denote by p_λ an element of $\Gamma(U_\lambda, \mathscr{P})$. For every pair of indices λ, μ such that $U_\lambda \cap U_\mu \neq \varnothing$, there then exists a unique element $\gamma_{\lambda\mu}$ of $\Gamma(U_\lambda \cap U_\mu, \mathscr{G})$ such that $\gamma_{\lambda\mu} \cdot (p_\mu | U_\lambda \cap U_\mu) = p_\lambda | U_\lambda \cap U_\mu$; in addition, if λ, μ, ν are three indices such that $U_\lambda \cap U_\mu \cap U_\nu \neq \varnothing$, then the restrictions $\gamma'_{\lambda\mu}$, $\gamma'_{\mu\nu}$, $\gamma'_{\lambda\nu}$ of $\gamma_{\lambda\mu}$, $\gamma_{\mu\nu}$, $\gamma_{\lambda\nu}$ to $U_\lambda \cap U_\mu \cap U_\nu$ satisfy the condition $\gamma'_{\lambda\nu} = \gamma'_{\lambda\mu}\gamma'_{\mu\nu}$; in other words, $(\lambda, \mu) \mapsto \gamma_{\lambda\mu}$ is a 1-*cocycle* of the cover (U_λ) with values in $\mathscr{G}$. If, for every λ, p'_λ is a second element of $\Gamma(U_\lambda, \mathscr{P})$, then there exists a unique element $\beta_\lambda \in \Gamma(U_\lambda, \mathscr{G})$ such that $p'_\lambda = \beta_\lambda \cdot p_\lambda$, and the 1-cocycle $(\gamma'_{\lambda\mu})$ corresponding to the family (p'_λ) is given by $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$, that is, it is *cohomologous* to $\gamma_{\lambda\mu}$. Conversely, the data of a 1-cocycle $(\gamma_{\lambda\mu})$ defines, for every pair (λ, μ), an automorphism $\theta_{\lambda\mu}$ of the sheaf *of sets* $\mathscr{G} | U_\lambda \cap U_\mu$, namely the right translation by $\gamma_{\lambda\mu}$, and the fact that it is a cocycle shows that we can *glue* the sheaves of sets $\mathscr{G} | U_\lambda$ via the automorphisms $\theta_{\lambda\mu}$ (**0**, 3.3.1); we thus obtain a torsor under $\mathscr{G}$, denoted $\mathscr{P}$, and if we

[26] [Trans.] This is nowadays more commonly called a $\mathscr{G}$-torsor rather than a torsor *under* $\mathscr{G}$.

take for p_λ the unit section over U_λ, then the corresponding 1-cocycle is none other than the given 1-cocycle $(\gamma_{\lambda\mu})$; in addition, if we replace $(\gamma_{\lambda\mu})$ by a 1-cocycle $\gamma'_{\lambda\mu} = \beta_\lambda\gamma_{\lambda\mu}\beta_\mu^{-1}$ cohomologous to it, then we check immediately that the torsor obtained is isomorphic to $\mathscr{P}$.

In particular, if $(\gamma_{\lambda\mu})$ is a 1-*coboundary*, in other words of the form $\gamma_{\lambda\mu} = \beta_\lambda\beta_\mu^{-1}$, then the torsor $\mathscr{P}$ obtained is *isomorphic to* $\mathscr{G}$ (considered as a torsor under itself by left translations); we say in this case that $\mathscr{P}$ is *trivial*, and the converse is evident.

In particular, it follows from (**III**, 1.3.1) that we have:

Proposition (16.5.16). — *Let Z be an affine scheme, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Z$-module; then every torsor under $\mathscr{G}$ is trivial.*

Returning to the problem considered in (16.5.14), we thus obtain: **IV-4** | 34

Proposition (16.5.17). — *Let X, Y be two S-preschemes, Y_0 a closed subprescheme of Y defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_Y$ such that $\mathscr{I}^2 = 0$, $j : Y_0 \to Y$ the canonical injection. Let $u_0 : Y_0 \to X$ be an S-morphism, and $\mathscr{P}$ the sheaf of sets on Y such that, for every open U of Y, $\Gamma(U, \mathscr{P})$ is the set of S-morphisms $u : U \to X$ such that $u_0|U_0 = u \circ (j|U_0)$, where $U_0 = j^{-1}(U)$. Then $\mathscr{P}$ has the structure of a pseudo-torsor under the $\mathscr{O}_{Y_0}$-module $\mathscr{G} = \mathscr{H}om_{\mathscr{O}_{Y_0}}(u_0^*(\Omega^1_{X/S}), \mathscr{I})$.*

In particular:

Corollary (16.5.18). — *With the notation of (16.5.17), suppose that Y is affine and $\Omega^1_{X/S}$ is of finite presentation; if there is an open cover (U_α) of Y, and, for every index α, an S-morphism $v_\alpha : U_\alpha \to X$ such that, if $U_\alpha^0 = j^{-1}(U_\alpha)$, we have $v_\alpha \circ (j|U_\alpha^0) = u_0|U_\alpha^0$, then there is an S-morphism $u : Y \to X$ such that $u \circ j = u_0$.*

Proof. Indeed, $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_{Y_0}$-module (**I**, 1.3.12); by (16.5.16) and the fact that Y_0 is then affine, the sheaf $\mathscr{P}$, which is by hypothesis a torsor under $\mathscr{G}$, and not only a pseudo-torsor, is *trivial*; if w is an isomorphism from $\mathscr{G}$ to $\mathscr{P}$ (as torsors under $\mathscr{G}$), the image under w of the zero section of $\mathscr{G}$ is the S-morphism sought. ☐

16.6. Sheaf of p-differentials and exterior differentials.

(16.6.1). Let $f : X \to S$ be a morphism of preschemes. We define the *sheaf of p-differentials of X relative to S* (p an integer) to be the *pth exterior power* (**0**, 4.1.5) of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$, denoted by

$$\Omega^p_{X/S} = \bigwedge^p(\Omega^1_{X/S}). \tag{16.6.1.1}$$

So we have $\Omega^0_{X/S} = \mathscr{O}_X$, and $\Omega^p_{X/S} = 0$ for $p < 0$; the $\Omega^p_{X/S}$ are the homogeneous components of the exterior algebra of $\Omega^1_{X/S}$

$$\Omega^\bullet_{X/S} = \bigwedge(\Omega^1_{X/S}) = \bigoplus_{p\in\mathbf{Z}} \bigwedge^p(\Omega^1_{X/S}), \tag{16.6.1.2}$$

which is therefore a quasi-coherent graded anti-commutative $\mathscr{O}_X$-algebra whose elements of degree 1 are square-zero. For every affine U of X, we have $\Gamma(U, \Omega^\bullet_{X/S}) = \bigwedge(\Gamma(U, \Omega^1_{X/S}))$, where $\Gamma(U, \Omega^1_{X/S})$ is considered as a $\Gamma(U, \mathscr{O}_X)$-module.

When $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, so that B is an A-algebra, we have (**0**, 4.1.5) $\Omega^p_{X/S} = (\Omega^p_{B/A})^\sim$, where $\Omega^p_{B/A} = \bigwedge^p \Omega^1_{B/A}$.

Theorem (16.6.2). — *There is one and only one endomorphism d of the sheaf of additive groups $\Omega^\bullet_{X/S}$ with the following properties:*

(i) $d \circ d = 0$.

(ii) *For every open set U of X and every section $f \in \Gamma(U, \mathscr{O}_X)$ we have $df = d_{X/S}f$.* **IV-4** | 35

(iii) *For every open set U of X, every pair of integers p, q and every pair of sections $\omega'_p \in \Gamma(U, \Omega^p_{X/S})$, $\omega''_q \in \Gamma(U, \Omega^q_{X/S})$, we have*

$$d(\omega'_p \wedge \omega''_q) = (d\omega'_p) \wedge \omega''_q + (-1)^p \omega'_p \wedge d\omega''_q. \tag{16.6.2.1}$$

Also, d is an endomorphism of graded $\psi^*(\mathscr{O}_X)$-modules of degree $+1$.

Proof. Suppose that we have proved the existence of an endomorphism d. For every affine open U of X, every section of $\Omega^p_{X/S}$ over U is (by (ii)) a linear combination of finitely many elements of the form $g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)$, where g and the f_i are sections of $\mathscr{O}_X$ over U (**0**, 20.4.7). The conditions (i) and (iii) then show, by induction on p, that we necessarily have

$$d(g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)) = dg \wedge df_1 \wedge df_2 \wedge \cdots \wedge df_p. \tag{16.6.2.2}$$

This therefore proves the *uniqueness* of d and the last claim of the theorem. By virtue of this uniqueness property, to show the existence of d, we can restrict ourselves to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affines. Now (Bourbaki, *Alg.*, chap. III, 3rd ed., §10), to define an A-antiderivation D of degree $+1$ on an exterior algebra $\bigwedge(M)$ (where M is a B-module and B an A-algebra), with values in a *graded anticommutative* A-algebra $C = \bigoplus_{n=0}^{\infty} C_n$ whose elements of degree 1 are square-zero, it suffices to choose arbitrarily an A-derivation D_0 from B to C_1 and an A-homomorphism D_1 from M to C_2; there then exists a unique A-antiderivation D from $\bigwedge(M)$ to C agreeing with D_0 on B and with D_1 on M.

In the present case, D_0 is necessarily equal to $d_{B/A}$ by (ii); we reduce to seeing, having (16.6.2.2) in mind, that there is an A-homomorphism u of $\Omega^1_{B/A}$ in $\Omega^2_{B/A}$ such that

$$u(g.df) = dg \wedge df \tag{16.6.2.3}$$

for all f, g in B; it suffices to show that there is an A-homomorphism $v : B \otimes_A \Omega^1_{B/A} \to \Omega^2_{B/A}$ such that

$$v(g.\omega) = dg \wedge \omega \tag{16.6.2.4}$$

for $g \in B$ and $\omega \in \Omega^1_{B/A}$. Finally, since $\Omega^1_{B/A} = \mathfrak{I}/\mathfrak{I}^2$ (where $\mathfrak{I} = \mathfrak{I}_{B/A}$ is the kernel of the canonical homomorphism $B \otimes_A B \to B$) and $\Omega^1_{B/A}$ is generated by elements of the form $g.df$, it is enough to define an A-homomorphism $w : B \otimes_A (B \otimes_A B) \to \Omega^2_{B/A}$ such that

$$w(g' \otimes g \otimes f) = dg' \wedge (g.df) \tag{16.6.2.5}$$

and such that w is zero on the image of $B \otimes_A \mathfrak{I}^2$. Since the right-hand side of (16.6.2.5) is A-trilinear in g', g, and f, the existence of w satisfying (16.6.2.5) is immediate. Since, on the other hand, $\mathfrak{I}$ is generated by elements of the form $1 \otimes x - x \otimes 1$ ($x \in B$), we reduce to checking that when $z = (1 \otimes x - x \otimes 1)(1 \otimes y - y \otimes 1)$ we have $w(g' \otimes z) = 0$. Now, since $z = 1 \otimes (xy) + (xy) \otimes 1 - x \otimes y - y \otimes x$, formula (16.6.2.4) shows that it is enough to see that we have $d(xy) - x.dy - y.dx = 0$, which is to **IV-4** | 36
say that d is a derivation.

It remains to be shown that d satisfies the condition (i). Now, the square of an antiderivation is a derivation (Bourbaki, *loc. cit.*), and since $\Omega^\bullet_{B/A}$ is generated by $\Omega^1_{B/A}$ as a B-algebra, it is enough to verify that $d(dz) = 0$ when $z \in B$ and when $z \in \Omega^1_{B/A}$; in the first case, this follows from the formula (16.6.2.3) when $g = 1$; for the second, we can restrict ourselves to the case where $z = g.df$ with f, g in B, and then we have, because of (16.6.2.1) and (16.6.2.3),

$$d(d(g.df)) = d(dg \wedge df) = (d(dg)) \wedge (df) - (dg) \wedge (d(df)) = 0.$$

□

Definition (16.6.3). — The antiderivation d defined in (16.6.2) (also denoted by $d_{X/S}$) is called the *exterior differential* on X (relative to S).

Proposition (16.6.4). — *For every base change $g : S' \to S$, if we put $X' = X \times_S S'$, the canonical morphism*

$$\Omega^\bullet_{X/S} \otimes_S S' \longrightarrow \Omega^\bullet_{X'/S'} \tag{16.6.4.1}$$

deduced from the isomorphism (16.5.10.1) is bijective. Also, if s is a section of $\Omega^\bullet_{X/S}$ over an open set U of X, $s \otimes 1$ its inverse image, section of $\Omega^\bullet_{X'/S'}$ over the inverse image U' of U in X', we have $d_{X'/S'}(s \otimes 1) = d_{X/S}(s) \otimes 1$.

Proof. The first claim is immediate, the formation of the exterior algebra of a module commutes with extending the scalar ring. To prove the second, we can, because of (16.6.2.2), restrict ourselves to the case where $s \in \Gamma(U, \mathscr{O}_X)$, and in this case the claim has already been proven (16.4.3.7). □

(16.6.5). Suppose that $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module of rank n at a point x, so that we have n sections $s_i \in \Gamma(U, \mathscr{O}_X)$ such that the ds_i form a basis for the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$ (16.5.8). Then, for every integer $p \geqslant 1$, the p-differentials $ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}$ (for $i_1 < i_2 < \cdots < i_p$ in $[1, n]$) form a basis of $\binom{n}{p}$ elements of $\Gamma(U, \Omega^p_{X/S})$ over $\Gamma(U, \mathscr{O}_X)$. Also the formula (16.6.2.2) shows that for every section $g \in \Gamma(U, \mathscr{O}_X)$, we have

$$(16.6.5.1) \qquad d(g.ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}) = \sum_k (-1)^r \frac{\partial g}{\partial s_k} ds_{i_1} \wedge \cdots \wedge ds_{i_r} \wedge ds_k \wedge ds_{i_{r+1}} \wedge \cdots \wedge ds_{i_p}$$

where, on the right-hand side, k ranges over the $n-p$ indices distinct from the i_h, and i_r is the largest index $< k$.

We note that the relation $d(dg) = 0$ for every section $g \in \Gamma(U, \mathscr{O}_X)$ expresses itself in the form

$$D_i(D_j g) = D_j(D_i g) \quad \text{for } i \neq j;$$

in other words, the derivations D_i defined in (16.5.7) commute with each other.

16.7. The $\mathscr{P}^n_{X/S}(\mathscr{F})$.

(16.7.1). Let $f : X \to S$ be a morphism of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-module. We denote by $X^{(n)}_{\Delta_f}$ the *nth infinitesimal neighbourhood* of X for the diagonal morphism $\Delta_f : X \to X \times_S X$, by $h_n : X^{(n)}_{\Delta_f} \to X \times_S X$ **IV-4 | 37**
the canonical morphism (16.1.2), and consider the two composite morphisms

$$p_1^{(n)} : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_1} X, \quad p_2^{(n)} : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_2} X$$

so that, by definition, $p_1^{(n)}$ corresponds to the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ which we have chosen to define the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ (16.3.5), and $p_2^{(n)}$ to the homomorphism of sheaves of rings $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (16.3.6). Since $X^{(n)}_{\Delta_f}$ and X have the same underlying space, we can write

$$(16.7.1.1) \qquad \mathscr{P}^n_{X/S} = (p_1^{(n)})_*((p_2^{(n)})^*(\mathscr{O}_X)).$$

More generally, we define

$$(16.7.1.2) \qquad \mathscr{P}^n_{X/S}(\mathscr{F}) = (p_1^{(n)})_*((p_2^{(n)})^*(\mathscr{F})).$$

so that $\mathscr{P}^n_{X/S} = \mathscr{P}^n_{X/S}(\mathscr{O}_X)$; by definition, $\mathscr{P}^n_{X/S}(\mathscr{F})$ is an $\mathscr{O}_X$-module.

(16.7.2). Returning to the definition of the inverse image of modules on ringed spaces (**0**, 4.3.1), and bearing in mind that $X^{(n)}_{\Delta_f}$ and X have the same underlying space, we see that definition (16.7.1.2) can be written in the form

$$(16.7.2.1) \qquad \mathscr{P}^n_{X/S}(\mathscr{F}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F},$$

where it must be kept in mind that, in interpreting the symbol $\otimes$, $\mathscr{P}^n_{X/S}$ is endowed with the $\mathscr{O}_X$-module structure defined by *the homomorphism of sheaves of rings* $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$. It follows immediately from such formula (or directly from (16.7.1.2)) that $\mathscr{P}^n_{X/S}(\mathscr{F})$ is canonically equipped with a $\mathscr{P}^n_{X/S}$-module structure.

Proposition (16.7.3). — (i) *The functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$ from the category of $\mathscr{O}_X$-modules to the category of $\mathscr{P}^n_{X/S}$-modules is right exact, and commutes with arbitrary inductive limits; it is exact when $\mathscr{P}^n_{X/S}$ is flat.*

(ii) *If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module (resp. of finite type, resp. of finite presentation), then $\mathscr{P}^n_{X/S}(\mathscr{F})$ is quasi-coherent (resp. of finite type, resp. of finite presentation).*

Proof. The claims from (i) follow from (16.7.2.1) and the consideration of the symmetry of $\mathscr{P}^n_{X/S}$ (16.3.4). The claims from (ii) follow from the right exactness of the functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$. ☐

(16.7.4). The two $\mathscr{O}_X$-module structures on $\mathscr{P}^n_{X/S}$ define two commuting $\mathscr{O}_X$-module structures on $\mathscr{P}^n_{X/S}(\mathscr{F})$, and hence an $\mathscr{O}_X$-bimodule structure. It is convenient to denote the structure coming from the structure homomorphism $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (chosen in (16.3.5)) on the *left* and the one coming from the homomorphism $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ on the *right*. In other words, for every open U of X, and every triple $a \in \Gamma(U, \mathscr{O}_X)$, $b \in \Gamma(U, \mathscr{P}^n_{X/S})$, $t \in \Gamma(U, \mathscr{F})$, we have by definition

(16.7.4.1) $$a(b \otimes t) = (ab) \otimes t, \quad (b \otimes t)a = (b.d^n a) \otimes t = b \otimes (at) = (d^n a).(b \otimes t).$$

The $\mathscr{O}_X$-module structure coming from the definition (16.7.1.2) is therefore, under these conventions, the *left* $\mathscr{O}_X$-module structure. If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then the same is true for **IV-4** | 38 $\mathscr{P}^n_{X/S}(\mathscr{F})$ for any one of its $\mathscr{O}_X$-module structures. If also $\mathscr{F}$ is of finite type (resp. of finite presentation) and $f : X \to S$ is locally of finite type (resp. locally of finite presentation), $\mathscr{P}^n_{X/S}(\mathscr{F})$ is (for any one of its $\mathscr{O}_X$-module structures) of finite type (resp. of finite presentation), which is a consequence of (16.3.9) and (16.4.22).

(16.7.5). The definition (16.7.2.1) entails the existence of a homomorphism of sheaves of commutative groups

(16.7.5.1) $$d^n_{X/S,\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad (\text{also denoted } d^n_{X/S})$$

such that, in the notations of (16.7.4), we have

(16.7.5.2) $$d^n_{X/S,\mathscr{F}}(t) = 1 \otimes t$$

and consequently, because of (16.7.4.1)

(16.7.5.3) $$d^n_{X/S,\mathscr{F}}(at) = (1 \otimes t)a = (d^n_{X/S,\mathscr{F}}(t)).a$$

(16.7.5.4) $$d^n_{X/S,\mathscr{F}}(at) = (d^n_{X/S,\mathscr{F}}(a)).(1 \otimes t) = (d^n_{X/S,\mathscr{F}}(a))(d^n_{X/S,\mathscr{F}}(t)).$$

Therefore it is $\mathscr{O}_X$-linear for the *right* $\mathscr{O}_X$-module structure on $\mathscr{P}^n_{X/S}(\mathscr{F})$, and *semilinear* (relative to the automorphism σ (16.3.4)) for the *left* $\mathscr{O}_X$-module structure.

Proposition (16.7.6). — *The left $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}(\mathscr{F})$ is generated by the image of $\mathscr{F}$ under the homomorphism $d^n_{X/S,\mathscr{F}}$.*

Proof. This is an immediate consequence of (16.7.5.3) and of the particular case $\mathscr{F} = \mathscr{O}_X$ (16.3.8). □

(16.7.7). The canonical homomorphisms of sheaves of rings

$$\phi_{nm} : \mathscr{P}^m_{X/S} \longrightarrow \mathscr{P}^n_{X/S}$$

for $n \leqslant m$ (16.1.2) define, because of (16.7.2.1), canonical homomorphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad (n \leqslant m)$$

which are homomorphisms of $\mathscr{O}_X$-bimodules in light of (16.1.6) and (16.7.4.1); also we have commutative diagrams

$$\begin{tikzcd} \mathscr{P}^m_{X/S}(\mathscr{F}) \arrow[rr] & & \mathscr{P}^n_{X/S}(\mathscr{F}) \\ & \mathscr{F} \arrow[ul, "d^m_{X/S,\mathscr{F}}"] \arrow[ur, "d^n_{X/S,\mathscr{F}}"'] & \end{tikzcd}$$

We have therefore a projective system of $\mathscr{O}_X$-bimodules $(\mathscr{P}^n_{X/S}(\mathscr{F}))$, and we define

(16.7.7.1) $$\mathscr{P}^\infty_{X/S}(\mathscr{F}) = \varprojlim \mathscr{P}^n_{X/S}(\mathscr{F}).$$

Also, this shows that the homomorphisms (16.7.5.1) form a projective system of homomorphisms, and therefore define a canonical homomorphism

(16.7.7.2) $$d^\infty_{X/S,\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{P}^\infty_{X/S}(\mathscr{F}).$$

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(16.7.8). Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules; it follows immediately from the definition (16.7.2.1) that we have a canonical isomorphism of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}) \simeq \mathscr{P}^n_{X/S}(\mathscr{F}) \otimes_{\mathscr{P}^n_{X/S}} \mathscr{P}^n_{X/S}(\mathscr{G}) \tag{16.7.8.1}$$

(Bourbaki, *Alg.*, chap. II, 3rd ed., § 5, no. 1, prop. 3).

We conclude in particular (or see directly from definition (16.7.2.1)) that if $\mathscr{F}$ has an $\mathscr{O}_X$-algebra structure (not necessarily associative), $\mathscr{P}^n_{X/S}(\mathscr{F})$ has a canonical $\mathscr{P}^n_{X/S}$-algebra structure; the latter is associative (resp. commutative, resp. unital, resp. a Lie algebra) if $\mathscr{F}$ is so. Moreover, the canonical homomorphisms $\mathscr{P}^m_{X/S}(\mathscr{F}) \to \mathscr{P}^n_{X/S}(\mathscr{F})$ for $n \leqslant m$ (16.7.7) are then algebra dihomomorphisms; similarly, (16.7.5.1) is then an $\mathscr{O}_X$-algebra homomorphism when $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with the $\mathscr{O}_X$-algebra structure arising from its *right* $\mathscr{O}_X$-module structure.

With the same notation, we also have a canonical homomorphism of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})) \longrightarrow \mathscr{H}om_{\mathscr{P}^n_{X/S}}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{P}^n_{X/S}(\mathscr{G})) \tag{16.7.8.2}$$

(Bourbaki, *Alg.*, chap. II, 3rd ed., § 5, no. 3), which is bijective when $\mathscr{P}^n_{X/S}$ is an $\mathscr{O}_X$-module *locally free of finite type* (*loc. cit.*, prop. 7).

(16.7.9). Suppose we are in the situation described in (16.4.1); then from the canonical homomorphism $P^n(u)$ (16.4.3.3) we deduce immediately a canonical homomorphism of $\mathscr{O}_{X'}$-bimodules

$$u^*(\mathscr{P}^n_{X/S}(\mathscr{F})) \longrightarrow \mathscr{P}^n_{X'/S'}(u^*(\mathscr{F})). \tag{16.7.9.1}$$

We leave it to the reader to extend to this homomorphism the properties established in (16.4) for the case $\mathscr{F} = \mathscr{O}_X$.

Remark (16.7.10). — The definition of $\mathscr{P}^n_{X/S}(\mathscr{F})$ in the form (16.7.1.2) still makes sense when $\mathscr{F}$ is a sheaf of sets (the inverse image of a sheaf of sets by $p_2^{(n)}$ being defined in (**0**, 3.7.1)); a variant of this definition allows one to define the "jet scheme" (relative to S) of an arbitrary X-prescheme.

16.8. Differential operators. [27]

Definition (16.8.1). — Let $f = (\psi, \theta) : X \to S$ be a morphism of preschemes, $\mathscr{F}, \mathscr{G}$ two $\mathscr{O}_X$-modules, n an integer $\geqslant 0$. We say that a morphism $D : \mathscr{F} \to \mathscr{G}$ of sheaves of additive groups is a differential operator of order $\leqslant n$ (relative to S) if there is a homomorphism of $\mathscr{O}_X$-modules $u : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G}$ (where $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with its left $\mathscr{O}_X$-module structure (16.7.4)) such that $D = u \circ d^n_{X/S,\mathscr{F}}$.

It is clear, because of the existence of canonical morphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F})$$

for $n \leqslant m$ (16.7.7), that a differential operator of order $\leqslant n$ is a differential operator of order $\leqslant m$ **IV-4** | 40
for $n \leqslant m$. If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, then, for every open set U of X, $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$.

We say that a homomorphism $D : \mathscr{F} \to \mathscr{G}$ is a *differential operator* (relative to S) if, for every $x \in X$, there is an open neighbourhood U of x and an integer $n \geqslant 0$ such that $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$. The *order* of a differential operator D is the infimum of the integers n such that D is a differential operator of order $\leqslant n$ (and therefore $+\infty$ if there is no such integer); this order is always finite if X is *quasi-compact*. The differential operators of order 0 are exactly the homomorphisms of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G}$; the operators of order < 0 are *zero* by convention. For $n \geqslant 0$, a differential operator of order $\leqslant n$ is not in general a homomorphism of $\mathscr{O}_X$-modules, but is always a homomorphism of $\psi^*(\mathscr{O}_S)$-modules.

When $\mathscr{F} = \mathscr{O}_X$, a differential operator of order $\leqslant 1$ from $\mathscr{O}_X$ to $\mathscr{G}$ can be written uniquely in the form $v + D$, where $v : \mathscr{O}_X \to \mathscr{G}$ is an $\mathscr{O}_X$-homomorphism and D is an S-*derivation* (16.5.1) from $\mathscr{O}_X$ to $\mathscr{G}$; this follows from the structure of $P^1_{B/A}$ (**0**, 20.4.8).

[27] For a more general formalism, see Exposé VII of [42] (due to P. Gabriel).

(16.8.2). To describe in a more precise manner a differential operator of order $\leqslant n$, $D : \mathscr{F} \to \mathscr{G}$, it suffices, for every open set U of X whose image in S is contained in an affine open set V, to characterize the homomorphism $D = D_U : \Gamma(U, \mathscr{F}) \to \Gamma(U, \mathscr{G})$. If we put $\Gamma(V, \mathscr{O}_S) = A$, $\Gamma(U, \mathscr{O}_X) = B$, so that B is an A-algebra, we have $\Gamma(U, \mathscr{P}^n_{X/S}) = (B \otimes_A B)/\mathfrak{I}^{n+1}$, where, for brevity, $\mathfrak{I} = \mathfrak{I}_{B/A}$. Also put $M = \Gamma(U, \mathscr{F})$, $N = \Gamma(U, \mathscr{G})$; then the definition of D means that for every pair (U, V) satisfying the above, the A-homomorphism $D : M \to N$ factors through

$$M \longrightarrow ((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M \xrightarrow{v} N$$

where the first arrow is the canonical homomorphism $t \mapsto 1 \otimes t$, and v is a B-homomorphism, the B-module structure on its source coming from the first factor B (whereas, in forming the tensor product over B, the B-module structure on $(B \otimes_A B)/\mathfrak{I}^{n+1}$ comes from the second factor B). Note also that the B-module $((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M$ is isomorphic to $(B \otimes_A M)/\mathfrak{I}^{n+1}(B \otimes_A M)$, where $B \otimes_A M$ is regarded as a $(B \otimes_A B)$-module and its B-module structure comes from the homomorphism $b \mapsto b \otimes 1$ from B to $B \otimes_A B$. Let D' then be the B-homomorphism from $B \otimes_A M$ to N such that $D'(b \otimes t) = bD(t)$; the factorization condition on D is equivalently that D' vanish on the B-submodule $\mathfrak{I}^{n+1}(B \otimes_A M)$.

(16.8.3). It is clear that the set of differential operators of order $\leqslant n$ from $\mathscr{F}$ to $\mathscr{G}$ forms an additive group, denoted by $\mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G})$; when $\mathscr{F} = \mathscr{G} = \mathscr{O}_X$, we also write $\mathrm{Diff}^n_{X/S}$ instead of $\mathrm{Diff}^n_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$.

We saw in (16.8.1) that for two open sets $U \supset V$ of X, there is a canonical restriction homomorphism

$$\mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U) \longrightarrow \mathrm{Diff}^n_{V/S}(\mathscr{F}|V, \mathscr{G}|V)$$

from which we deduce that $U \mapsto \mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U)$ is a presheaf of additive groups; in fact, it is **IV-4** | 41
actually a *sheaf*, since for an open set U varying in X, the homomorphisms $u \mapsto u \circ d^n_{U/S, \mathscr{F}|U}$ are *isomorphisms* of sheaves of additive groups

(16.8.3.1) $$\mathrm{Hom}_{\mathscr{O}_U}(\mathscr{P}^n_{U/S}(\mathscr{F}|U), \mathscr{G}|U) \simeq \mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U),$$

because of the fact that the image of $\mathscr{F}$ by $d^n_{X/S, \mathscr{F}}$ generates $\mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.6). We denote this sheaf by $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$, and we have:

Proposition (16.8.4). — *The isomorphisms (16.8.3.1) define an isomorphism of sheaves of additive groups*

(16.8.4.1) $$\mathscr{H}\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}) \simeq \mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}).$$

When $\mathscr{F} = \mathscr{G} = \mathscr{O}_X$, we also write $\mathscr{D}\mathit{iff}^n_{X/S}$ instead of $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$; it follows from (16.8.4) that $\mathscr{D}\mathit{iff}^n_{X/S}$ is canonically identified with the *dual* of the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$; so we also write $\langle t, D\rangle$ instead of $u(t)$ if t is a section of $\mathscr{P}^n_{X/S}$ over an open set and if u is a homomorphism from $\mathscr{P}^n_{X/S}$ to $\mathscr{O}_X$ corresponding to D.

(16.8.5). Since $\mathscr{P}^n_{X/S}(\mathscr{F})$ has an $\mathscr{O}_X$-bimodule structure (16.7.4), we canonically obtain an $\mathscr{O}_X$-bimodule structure on $\mathscr{H}\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G})$, and hence on $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ by (16.8.4.1). More precisely, to the left $\mathscr{O}_X$-module structure on $\mathscr{P}^n_{X/S}(\mathscr{F})$ corresponds, because of (16.8.1), the left $\mathscr{O}_X$-module structure on $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ explained as follows: for every open set U of X, every section $a \in \Gamma(U, \mathscr{O}_X)$ and every differential operator $D : \mathscr{F}|U \to \mathscr{G}|U$, aD is the differential operator which, for every section $t \in \Gamma(U, \mathscr{F})$, makes correspond the section

(16.8.5.1) $$(aD)(t) = a(D(t))$$

of $\Gamma(U, \mathscr{G})$. Similarly, the right $\mathscr{O}_X$-module structure on $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ is made explicit as follows: under the same notations as above, Da is the operator which, to every $t \in \Gamma(U, \mathscr{F})$, makes correspond the section

(16.8.5.2) $$(Da)(t) = D(at).$$

Proposition (16.8.6). — *If $f : X \to S$ is a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module of finite presentation and $\mathscr{G}$ a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ is a quasi-coherent $\mathscr{O}_X$-module for either of the structures defined in (16.8.5).*

Proof. The proposition follows from the fact that, under these hypotheses, $\mathscr{P}^n_{X/S}(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-module of finite presentation (16.7.4), together with (**I**, 1.3.12). ☐

(16.8.7). The set of differential operators (of unspecified order (16.8.1)) is denoted by $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$; we also see as in (16.8.3) that $U \mapsto \mathrm{Diff}_{U/S}(\mathscr{F}|U,\mathscr{G}|U)$ is a sheaf of additive groups, which we will denote by $\mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$. It is immediate that $\mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$ is the union of the increasing filtered family of its subsheaves $\mathscr{D}iff^n_{X/S}(\mathscr{F},\mathscr{G})$; if X is quasi-compact, $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$ is similarly the union of its subgroups $\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G})$ (16.8.1). The $\mathscr{O}_X$-bimodule structures on the $\mathscr{D}iff^n_{X/S}(\mathscr{F},\mathscr{G})$ therefore induce an $\mathscr{O}_X$-bimodule structure on $\mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$, again described by (16.8.5.1) and (16.8.5.2). **IV-4** | 42

Note that, for $n \leqslant m$, we have a commutative diagram

(16.8.7.1)
$$\begin{array}{ccc} \mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}) & \xrightarrow{\sim} & \mathscr{D}iff^n_{X/S}(\mathscr{F},\mathscr{G}) \\ \downarrow & & \downarrow \\ \mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^m_{X/S}(\mathscr{F}),\mathscr{G}) & \xrightarrow{\sim} & \mathscr{D}iff^m_{X/S}(\mathscr{F},\mathscr{G}) \end{array}$$

where the horizontal arrows are the isomorphisms (16.8.4.1) and the left vertical arrow comes from the canonical homomorphism $\mathscr{P}^m_{X/S}(\mathscr{F}) \to \mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.7). For every open set U of X, we then equip $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F})) = \varprojlim \Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F}))$ with the projective-limit topology of the discrete topologies on the groups $\Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F}))$. This makes $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F}))$ a topological $\Gamma(U,\mathscr{O}_X)$-bimodule, so that $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ becomes a sheaf with values in the category of topological commutative groups (**0**, 3.2.6). Then, by [I-4, II, 1.11], the inductive limit of the system of sheaves of commutative groups $\big(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G})\big)$ is precisely the sheaf of germs of *continuous* homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$ (the latter equipped with the discrete topology): indeed, the continuous homomorphisms from $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F}))$ to the discrete group $\Gamma(U,\mathscr{G})$ correspond bijectively to the inductive systems of group homomorphisms $\Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F})) \to \Gamma(U,\mathscr{G})$. We can furthermore express (16.8.4) by saying there is a canonical isomorphism

$$\mathscr{H}om.\mathrm{cont}_{\mathscr{O}_X}(\mathscr{P}^\infty_{X/S}(\mathscr{F}),\mathscr{G}) \simeq \mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$$

where the left-hand side denotes the sheaf of germs of continuous homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$.

Proposition (16.8.8). — *Let $\mathscr{F},\mathscr{G}$ be two $\mathscr{O}_X$-modules, $D : \mathscr{F} \to \mathscr{G}$ a homomorphism of $\psi^*(\mathscr{O}_S)$-modules, n an integer $\geqslant 0$. The following conditions are equivalent:*

(a) D is a differential operator of order $\leqslant n$.
(b) For all sections a of $\mathscr{O}_X$ over an open set U, the homomorphism $D_a : \mathscr{F}|U \to \mathscr{G}|U$ such that, for every section t of $\mathscr{F}$ over an open set $V \subset U$, we have

(16.8.8.1)
$$D_a(t) = D(at) - aD(t)$$

is a differential operator of order $\leqslant n-1$.
(c) For every open set U of X, every family $(a_i)_{1\leqslant i\leqslant n+1}$ of $n+1$ sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, we have the identity

(16.8.8.2)
$$\sum_{H\subset I_{n+1}} (-1)^{\mathrm{Card}(H)} \Big(\prod_{i\in H} a_i\Big) D\Big(\Big(\prod_{i\notin H} a_i\Big)t\Big) = 0$$

(where I_{n+1} is the interval $1 \leqslant i \leqslant n+1$ of $\mathbb{N}$).

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Proof. Let us first prove the equivalence of (a) and (c). By definition, to prove that D is a differential operator of order $\leqslant n$, it suffices to prove this for the restriction $D|U : \mathscr{F}|U \to \mathscr{G}|U$ to every affine open subset U of X; on the other hand, property (c) holds for every open subset U of X if it holds for every affine open subset. We may therefore restrict ourselves to the case where $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine. By (16.8.2) (whose notation we retain), condition (a) then means that the A-homomorphism $D' : B \otimes_A M \to N$ defined by $D'(b \otimes t) = bD(t)$ vanishes on $\mathfrak{I}^{n+1}(B \otimes_A M)$; by (**0**, 20.4.4), this is equivalent to saying that D' vanishes on every element of the form

$$(\prod_{i=1}^{n+1}(a_i \otimes 1 - 1 \otimes a_i)).(1 \otimes t)$$

where $a_i \in B$ and $t \in M$. Now this element can be written as $\sum_{H \subset I_{n+1}}(\prod_{i \in H} a_i) \otimes ((\prod_{i \notin H} a_i)t)$, and its image under D' is the left-hand side of (16.8.8.2); this proves the equivalence of (a) and (c).

Let us now prove the equivalence of (b) and (c). We argue by induction on n, the assertion being trivial for $n = 0$. Writing a_{n+1} in place of a in condition (b), we see from the induction hypothesis that condition (b) means that, for every family $(a_i)_{1 \leqslant i \leqslant n}$ of n sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, one has

$$\sum_{H' \subset I_n}(-1)^{\mathrm{Card}(H')}(\prod_{i \in H'} a_i)D_{a_{n+1}}((\prod_{i \notin H'} a_i)t) = 0.$$

But if in this relation we replace $D_{a_{n+1}}$ by its definition (16.8.8.1), we see at once that, up to sign, we obtain the left-hand side of (16.8.8.2); hence the conclusion. □

Proposition (16.8.9). — *If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, and $D' : \mathscr{G} \to \mathscr{H}$ a differential operator of order $\leqslant n'$, then $D' \circ D : \mathscr{F} \to \mathscr{H}$ is a differential operator of order $\leqslant n + n'$.*

Proof. By hypothesis, we can write $D = u \circ d^n_{X/S,\mathscr{F}}$ and $D' = v \circ d^{n'}_{X/S,\mathscr{G}}$, where $u : \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \to \mathscr{G}$ and $v : \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G} \to \mathscr{H}$ are $\mathscr{O}_X$-homomorphisms. It is therefore enough to show that the composite homomorphism of sheaves of additive groups

$$\mathscr{F} \xrightarrow{d^n_{X/S,\mathscr{F}}} \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{d^{n'}_{X/S,\mathscr{G}}} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

factors as

$$\mathscr{F} \xrightarrow{d^{n+n'}_{X/S,\mathscr{F}}} \mathscr{P}^{n+n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{w} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

where w is an $\mathscr{O}_X$-homomorphism. It suffices to prove the following lemma.

Lemma (16.8.9.1). — *There is one and only one $\mathscr{O}_X$-homomorphism*

(16.8.9.2) $$\delta : \mathscr{P}^{n+n'}_{X/S} \longrightarrow \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}) = \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{P}^n_{X/S}$$

making the following diagram commute **IV-4** | 44

(16.8.9.3)
$$\begin{tikzcd} \mathscr{O}_X \arrow[r, "d^{n+n'}_{X/S}"] \arrow[d, "d^n_{X/S}"'] & \mathscr{P}^{n+n'}_{X/S} \arrow[d, "\delta"] \\ \mathscr{P}^n_{X/S} \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}) \end{tikzcd}$$

We will then have, indeed, a commutative diagram deduced from (16.8.9.3) by tensorization with $\mathscr{F}$

$$\begin{tikzcd} \mathscr{F} \arrow[r, "d^{n+n'}_{X/S,\mathscr{F}}"] \arrow[d, "d^n_{X/S,\mathscr{F}}"'] & \mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \arrow[d, "\delta\otimes 1"] \\ \mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F})) \end{tikzcd}$$

and on the other hand, we verify immediately from the definition (16.7.5) that the diagram

$$\begin{tikzcd} \mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "u"] \arrow[d, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{G} \arrow[d, "d^{n'}_{X/S,\mathscr{G}}"] \\ \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F})) \arrow[r, "1\otimes u"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{G}) \end{tikzcd}$$

is commutative. We finish the proof by taking w to be the composite $\mathscr{O}_X$-homomorphism

$$\mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \xrightarrow{\delta\otimes 1} \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F})) \xrightarrow{1\otimes u} \mathscr{P}^{n'}_{X/S}(\mathscr{G}).$$

□

Proof of Lemma 16.8.9.1. It remains to prove Lemma (16.8.9.1). In view of (16.7.6), which proves the uniqueness of δ, we are reduced to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine. Putting $\mathfrak{I} = \mathfrak{I}_{B/A}$, it suffices to define a canonical homomorphism of B-modules

$$\phi : (B \otimes_A B)/\mathfrak{I}^{n+n'+1} \longrightarrow ((B \otimes_A B)/\mathfrak{I}^{n'+1}) \otimes_B ((B \otimes_A B)/\mathfrak{I}^{n+1})$$

the B-module structures on both sides being induced by the first factor B. Recall that in the tensor product on the right, $(B \otimes_A B)/\mathfrak{I}^{n'+1}$ must be regarded as a right B-module by its second B factor, **IV-4** | 45
and $(B \otimes_A B)/\mathfrak{I}^{n+1}$ as a left B-module by its first B factor (16.7.2). This is in turn equivalent to defining a homomorphism of B-modules

$$\phi_0 : B \otimes_A B \longrightarrow ((B \otimes_A B)/\mathfrak{I}^{n'+1}) \otimes_B ((B \otimes_A B)/\mathfrak{I}^{n+1})$$

and proving that it vanishes on $\mathfrak{I}^{n+n'+1}$. Such a homomorphism is defined at once by the condition

$$\phi_0(b \otimes b') = \pi_{n'}(b \otimes 1) \otimes \pi_n(1 \otimes b') \quad \text{for } b, b' \text{ in } B$$

with the notation of (16.3.7). Moreover, it is immediate that ϕ_0 is a homomorphism of *rings*. Now we may write

$$\phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes \pi_n(1\otimes 1) + \pi_{n'}(1\otimes b)\otimes \pi_n(1\otimes 1) - \pi_{n'}(1\otimes 1)\otimes \pi_n(1\otimes b)$$

and we have

$$\pi_{n'}(1\otimes b)\otimes \pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)b\otimes \pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes b\pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes \pi_n(b\otimes 1)$$

whence, finally,

$$\phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes \pi_n(1\otimes 1) + \pi_{n'}(1\otimes 1)\otimes \pi_n(b\otimes 1 - 1\otimes b). \tag{16.8.9.4}$$

A product of $n + n' + 1$ factors of the form (16.8.9.4) is therefore necessarily zero, since this is so for a product of $n + 1$ factors of the form $\pi_n(b \otimes 1 - 1 \otimes b)$ and for a product of $n' + 1$ factors of the form $\pi_{n'}(b \otimes 1 - 1 \otimes b)$. The conclusion therefore follows from (**0**, 20.4.4). □

Corollary (16.8.10). — *The sheaf $\mathscr{D}iff_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$ (also denoted $\mathscr{D}iff_{X/S}$) is canonically endowed with a structure of sheaf of rings, and the $\mathscr{D}iff^n_{X/S}$ form an increasing filtration compatible with this structure.*

In particular, $\mathscr{D}iff^0_{X/S}$ is a sheaf of subrings of $\mathscr{D}iff_{X/S}$, which is canonically identified with $\mathscr{O}_X$ (16.8.1). The formulas (16.8.5.1) and (16.8.5.2) show that the structure of $\mathscr{O}_X$-bimodule of $\mathscr{D}iff_{X/S}$ comes from the multiplication on the left and on the right by sections of $\mathscr{O}_X$ considered as a sheaf of subrings of $\mathscr{D}iff_{X/S}$.

Remarks (16.8.11). — (i) Suppose that $\mathscr{F} = \oplus_{\lambda\in L}\mathscr{F}_\lambda$; then it is clear from (16.7.2.1) that $\mathscr{P}^n_{X/S}(\mathscr{F}) = \oplus_{\lambda\in L}\mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$. Since the functor $\mathscr{F} \mapsto \Gamma(U, \mathscr{F})$ commutes with the formation of arbitrary direct sums, $d^n_{X/S,\mathscr{F}}$ is the homomorphism whose restriction to each $\mathscr{F}_\lambda$ is $d^n_{X/S,\mathscr{F}_\lambda} : \mathscr{F}_\lambda \to \mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$; it follows at once that

$$\operatorname{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\lambda\in L} \operatorname{Diff}^n_{X/S}(\mathscr{F}_\lambda, \mathscr{G}),$$

and therefore also (**0**, 3.2.6)

$$\mathscr{D}iff^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\lambda\in L} \mathscr{D}iff^n_{X/S}(\mathscr{F}_\lambda, \mathscr{G}).$$

Moreover, if $\mathscr{G} = \prod_{\mu\in M} \mathscr{G}_\mu$ (**0**, 3.2.6), we have

$$\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}) = \prod_{\mu\in M} \mathrm{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}_\mu),$$

Every homomorphism u from $\mathscr{P}^n_{X/S}(\mathscr{F})$ to $\mathscr{G}$ corresponds bijectively to the family of its composites $u_\mu : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G} \to \mathscr{G}_\mu$. Thus **IV-4** | 46

$$\mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\mu\in M} \mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G}_\mu),$$

and consequently also

$$\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\mu\in M} \mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}_\mu).$$

(ii) So far, we have encountered differential operators $\mathscr{F} \to \mathscr{G}$ almost exclusively when $\mathscr{F}$ and $\mathscr{G}$ are locally free of finite rank; in that case, by (i), their structure is reduced locally to that of the sheaf $\mathscr{D}\!\mathit{iff}_{X/S}$. The latter will be studied below, in a particular case, in (16.11).

16.9. Regular and quasi-regular immersions

Definition (16.9.1). — Let X be a ringed space. We say that an ideal $\mathscr{I}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular) if, for every point $x \in \mathrm{Supp}(\mathscr{O}_X/\mathscr{I})$, there is an open neighbourhood U of x in X and a regular sequence (**0**, 15.2.2) (resp. quasi-regular sequence (**0**, 15.2.2)) of elements of $\Gamma(U, \mathscr{O}_X)$ which generates $\mathscr{I}|U$.

A regular (resp. quasi-regular) sequence of sections of $\mathscr{O}_X$ over U which generates $\mathscr{I}|U$ is called a *regular system* (resp. *quasi-regular system*) of generators of $\mathscr{I}|U$.

Definition (16.9.2). — Let $j : Y \to X$ be an immersion of preschemes, and let U be an open subset of X such that $j(Y) \subset U$ and j is a closed immersion of Y into U. We say that j is regular (resp. quasi-regular) if the closed subprescheme $j(Y)$ of U associated to j is defined by a regular (resp. quasi-regular) ideal of $\mathscr{O}_U$ (condition independent of the choice of U).

We say that a subprescheme Y of a prescheme X is *regularly immersed* (resp. *quasi-regularly immersed*) if the canonical injection $j : Y \to X$ is a regular immersion (resp. quasi-regular immersion). If Y is a closed subprescheme and $\mathscr{I}$ is the ideal of $\mathscr{O}_X$ that defines Y, it is the same as asking $\mathscr{I}$ to be *regular* (resp. *quasi-regular*).

For example, if A is an *integral* ring and f is a nonzero element of A, the closed subprescheme $V(f)$ of $\mathrm{Spec}(A)$ (isomorphic to $\mathrm{Spec}(A/(f))$) is *regularly immersed* in $\mathrm{Spec}(A)$.

Every regular ideal is quasi-regular (**0**, 15.2.2); every regular immersion is quasi-regular (cf. (16.9.11) for a converse).

Proposition (16.9.3). — *Let X be a ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$, $(f_i)_{1\leqslant i\leqslant m}$ a finite sequence of sections of $\mathscr{O}_X$ over X generating $\mathscr{I}$. For (f_i) to be a quasi-regular sequence* (**0**, 15.2.2), *it is necessary and sufficient that the following conditions be satisfied:*

(i) *The canonical images of f_i in $\mathscr{I}/\mathscr{I}^2$ form a basis of this $\mathscr{O}_X/\mathscr{I}$-module.*
(ii) *The canonical surjective homomorphism* (16.1.2.2)

$$\mathbf{S}^\bullet_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}\!r^\bullet_{\mathscr{I}}(\mathscr{O}_X)$$

is bijective.

Moreover, if this is so, every sequence $(f'_i)_{1\leqslant i\leqslant n}$ of n sections of $\mathscr{I}$ over X which generates $\mathscr{I}$ is quasi-regular.

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Proof. The two conditions in the statement merely express the definition given in (**0**, 15.2.2), taking into account the definition of the canonical homomorphisms (16.2.1.1). The last claim follows from the fact that, if a module M over a commutative ring A admits a basis of n elements, every system of n generators of M is a basis (Bourbaki, *Alg. comm.*, chap. II, §3, cor. 5 of th. 1). □

Corollary (16.9.4). — *Let X be a locally ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$. For $\mathscr{I}$ to be quasi-regular, it is necessary and sufficient that the following conditions hold:*

(i) *$\mathscr{I}$ is of finite type.*
(ii) *$\mathscr{I}/\mathscr{I}^2$ is a locally free $\mathscr{O}_X/\mathscr{I}$-module.*
(iii) *The canonical homomorphism*

(16.9.4.1) $$\mathbf{S}^{\bullet}_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}r^{\bullet}_{\mathscr{I}}(\mathscr{O}_X)$$

is bijective.

Proof. The necessity of the conditions follows immediately from (16.9.3). To see that they are sufficient, it is enough, by (16.9.3), to prove the following: if, at a point $x \in \mathrm{Supp}(\mathscr{O}_X/\mathscr{I})$, there are an open neighbourhood U of x in X and n sections f_i $(1 \leqslant i \leqslant n)$ of $\mathscr{I}$ over U whose canonical images in $\mathscr{I}/\mathscr{I}^2$ form a basis of $(\mathscr{I}/\mathscr{I}^2)|U$ over $(\mathscr{O}_X/\mathscr{I})|U$, then there is an open neighbourhood $V \subset U$ of x such that the $f_i|V$ generate $\mathscr{I}|V$. Now, by hypothesis, $\mathscr{I}_x \neq \mathscr{O}_x$, so $\mathscr{I}_x$ is contained in the maximal ideal of $\mathscr{O}_x$. Since $\mathscr{I}_x$ is an $\mathscr{O}_x$-module of finite type and the classes of $(f_i)_x$ in $\mathscr{I}_x/\mathscr{I}_x^2$ generate this $(\mathscr{O}_x/\mathscr{I}_x)$-module, Nakayama's lemma shows that the $(f_i)_x$ generate $\mathscr{I}_x$. Since $\mathscr{I}$ is of finite type, we conclude by (**0**, 5.2.2). □

Corollary (16.9.5). — *Let X be a locally ringed space, $\mathscr{I}$ a quasi-regular ideal of $\mathscr{O}_X$, $(f_i)_{1\leqslant i\leqslant n}$ a sequence of sections of $\mathscr{I}$ over X, x a point of* $\mathrm{Supp}(\mathscr{O}_X/\mathscr{I})$*. The following conditions are equivalent:*

(a) *There is an open neighbourhood U of x in X such that the $f_i|U$ form a quasi-regular sequence of elements of $\Gamma(U, \mathscr{O}_X)$ generating $\mathscr{I}|U$.*
(b) *The $(f_i)_x$ form a system of generators of $\mathscr{I}_x$ whose size is as small as possible.*
(b′) *The $(f_i)_x$ form a minimal set of generators of $\mathscr{I}_x$.*
(c) *If $\bar{f}_i$ is the canonical image of f_i in $\Gamma(X, \mathscr{I}/\mathscr{I}^2)$, the $(\bar{f}_i)_x$ form a basis for the $(\mathscr{O}_x/\mathscr{I}_x)$-module $\mathscr{I}_x/\mathscr{I}_x^2$.*

Proof. By hypothesis, $\mathscr{O}_x$ is a local ring, $\mathscr{I}_x$ an ideal of finite type of $\mathscr{O}_x$ contained in the maximal ideal of $\mathscr{O}_x$; the equivalence of (b), (b′) and (c) follows from Nakayama's lemma (Bourbaki, *Alg. comm.*, chap. II, §3, n°2, prop. 5). It is clear that (a) implies (c) because of (16.9.3); on the other hand, from (**0**, 5.2.2) it follows that if condition (c) is satisfied (and therefore so is (b)), there is a neighbourhood U of x in X such that $(\mathscr{I}/\mathscr{I}^2)|U$ has constant rank equal to n, and that the $f_i|U$ generate $\mathscr{I}|U$; it suffices now to apply the last assertion of (16.9.3) to U. □

Remarks (16.9.6). — (i) Under the general hypothesis of (16.9.5), for the sequence (f_i) to generate $\mathscr{I}$, it is not enough that the $(\bar{f}_i)_y$ form a basis of the $(\mathscr{O}_y/\mathscr{I}_y)$-module $(\mathscr{I}_y/\mathscr{I}_y^2)$ for all $y \in X$. An example is obtained by taking $X = \mathrm{Spec}(A)$, where A is a Dedekind ring, and **IV-4 | 48**
$\mathscr{I} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is a *non-principal* prime ideal of A. Then $\mathscr{I}_y/\mathscr{I}_y^2 = 0$ at every point y other than the point $x \in X$ corresponding to $\mathfrak{J}$, while $\mathscr{I}_x/\mathscr{I}_x^2$ has rank 1 over the field $\mathscr{O}_x/\mathscr{I}_x$; moreover, $\mathscr{I}$ is plainly a regular ideal.

(ii) In (16.9.5), one cannot replace "quasi-regular" by "regular", even when X is a prescheme (cf. (16.9.12)). Indeed, let B denote the ring of germs at 0 in $\mathbb{R}$ of infinitely differentiable functions. It has a maximal ideal $\mathfrak{m}$ generated by the germ t of the identity map of $\mathbb{R}$ at 0, and the intersection $\mathfrak{n}$ of the $\mathfrak{m}^k$ for $k > 0$ is nonzero. Now let A be the quotient ring $B[T]/\mathfrak{n}TB[T]$, and let f_1, f_2 be the canonical images in A of the elements t and T of $B[T]$. The sequence (f_1, f_2) is *regular* in A: indeed, f_1 is not a zero divisor in A, because the relation $tP[T] \in \mathfrak{n}TB[T]$, for a polynomial $P \in B[T]$, implies that the products of t with the coefficients of P belong to $\mathfrak{n}$; it follows at once that the coefficients themselves belong to $\mathfrak{n}$, and hence $P[T] \in \mathfrak{n}TB[T]$. Since B/tB is isomorphic to $\mathbb{R}$, A/f_1A is isomorphic to the polynomial ring $\mathbb{R}[T]$ and is therefore integral; the image of f_2 in A/f_1A, being equal to T, is consequently not a zero divisor, proving the assertion. However, f_2 is a zero divisor in A, since for every nonzero element $x \in \mathfrak{n}$, the image of x in A is nonzero whereas the image of xT is zero. We conclude that the sequence (f_2, f_1) is *not regular* in A; on the other hand, the ideal $\mathfrak{J} = f_1A + f_2A$ is distinct from A, so the conditions (b), (b′) and (c) of (16.9.5) do not imply the condition (a) when we replace "quasi-regular" by "regular".

(16.9.7). If $X = \mathrm{Spec}(A)$ is an affine scheme, we say that an ideal $\mathfrak{J}$ of A is *regular* (resp. *quasi-regular*) if the ideal $\mathscr{J} = \widetilde{\mathfrak{J}}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular). We note that this notion is *local* and does not imply the existence of a *system of generators* of $\mathfrak{I}$ forming a regular (resp. quasi-regular) sequence in A, as Example (16.9.6, (i)) shows; it does, however, imply this when A is *local* (16.9.5).

The proposition (16.9.4) can be translated in terms of quasi-regular immersions in the following manner:

Proposition (16.9.8). — *Let $j : Y \to X$ be a morphism of preschemes; for j to be a quasi-regular immersion, it is necessary and sufficient that j satisfies the following conditions:*

(i) j is an immersion locally of finite presentation.
(ii) The conormal sheaf $\mathscr{G}r^1(j) = \mathscr{N}_{Y/X}$ (16.1.2) is a locally free $\mathscr{O}_Y$-module.
(iii) The canonical homomorphism

$$\mathbf{S}^{\bullet}_{\mathscr{O}_Y}(\mathscr{G}r^1(j)) \longrightarrow \mathscr{G}r^{\bullet}(j)$$

(16.1.2.2) is bijective.

Proof. Since the question is local on Y, we may restrict ourselves to the case where j is the canonical injection of a closed subprescheme Y of X; the translation of (16.9.4) into (16.9.8) then follows from the description of $\mathscr{G}r^1(j)$ and $\mathscr{G}r^{\bullet}(j)$ in terms of the ideal $\mathscr{I}$ of $\mathscr{O}_X$ defining Y (16.1.3, (ii)). □

Corollary (16.9.9). — *Let Y be a prescheme, X a Y-prescheme, and $j : Y \to X$ a Y-section of X, so that the n-th normal invariant $\mathscr{A}^{(n)}$ of j (16.1.2) is an augmented $\mathscr{O}_Y$-algebra (16.1.7); take $\mathscr{A}^{(\infty)} = \varprojlim \mathscr{A}^{(n)}$. For j to be a quasi-regular immersion, it is necessary and sufficient that j be locally of finite presentation and that every $y \in Y$ admit an affine open neighbourhood U with coordinate ring C such that $\mathscr{A}^{(\infty)}|U$ is isomorphic, as an augmented topological $\mathscr{O}_U$-algebra, to $\mathscr{O}_U[[T_1, \dots, T_n]]$.*

Proof. We may reduce to the case where j is a closed immersion by restricting to a sufficiently small neighbourhood of y (see the argument of (16.4.11)); then $\mathscr{O}_Y$ is identified with the quotient algebra $\mathscr{O}_X/\mathscr{I}$, and the canonical surjective homomorphism $\mathscr{O}_X \to \mathscr{O}_Y$ admits a right inverse (16.1.7). We **IV-4 | 49**
may therefore suppose that $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$ are affine, with B an augmented A-algebra whose augmentation ideal $\mathfrak{I}$ is of finite type. Since $\mathscr{A}^{(n)}$ is then identified with $(B/\mathfrak{I}^{n+1})^{\sim}$, the corollary follows from the equivalence of (b) and (c) in (**0**, 19.5.4), since $B/\mathfrak{I} = A$. □

We note that, in the affine case under consideration, the fact that j is a quasi-regular immersion is also equivalent, by (**0**, 19.5.4), to saying that B is a *formally smooth* A-algebra for the $\mathfrak{I}$-preadic topology.

We also note that the condition that j is locally of finite presentation is always satisfied when $X \to Y$ is locally of finite type (1.4.3, (v)).

Proposition (16.9.10). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, $j : Y \to X$ the canonical injection, y a point of Y.*

(i) *For there to exist an open neighbourhood U of y in X such that the restriction $Y \cap U \to U$ of j be a regular immersion, it is necessary and sufficient that the kernel $\mathscr{I}_y$ of the surjective homomorphism $\mathscr{O}_{X,y} \to \mathscr{O}_{Y,y}$ is generated by a regular sequence of elements of $\mathscr{O}_{X,y}$.*
(ii) *For the immersion j to be regular, it is necessary and sufficient that it is quasi-regular.*

Proof. (i) We can reduce to the case where Y is a subprescheme defined by a *coherent* ideal $\mathscr{I}$ of $\mathscr{O}_X$. The condition is clearly necessary. Conversely, if $\mathscr{I}_y$ is generated by a regular sequence $(s_i)_y$, where the s_i are sections of $\mathscr{I}$ over an open neighbourhood U of y in X, we may suppose that the s_i generate $\mathscr{I}|U$ (**0**, 5.5.2) and form a regular sequence (**0**, 15.2.4), whence the assertion.

(ii) The fact that a quasi-regular immersion is regular follows from (i) and the identification of quasi-regular and regular sequences in $\mathscr{O}_{X,y}$ formed by elements of the maximal ideal (**0**, 15.1.11).

□

When, without any Noetherian hypothesis on X, the kernel $\mathscr{I}_y$ of $\mathscr{O}_{X,y} \to \mathscr{O}_{Y,y}$ is generated by a regular sequence of elements of $\mathscr{O}_{X,y}$, we say that the immersion j is *regular at the point y*.

Corollary (16.9.11). — *Let X be a locally Noetherian prescheme; then every quasi-regular ideal of $\mathscr{O}_X$ is regular.*

Remarks (16.9.12). — (i) We note that a regular immersion is not in general a flat morphism, and hence *a fortiori* is not a regular morphism in the sense of (6.8.1).

(ii) Let A be a local Noetherian ring; it follows immediately from (16.9.4) and from (**0**, 17.1.1) that for A to be *regular*, it is necessary and sufficient that its maximal ideal $\mathfrak{m}$ is *quasi-regular* (or regular, which amounts to the same thing given that A is Noetherian). For an affine Noetherian scheme X to be *regular*, it is necessary and sufficient that for every closed point $x \in X$, the canonical injection $\operatorname{Spec}(k(x)) \to X$ be a *regular* immersion.

Proposition (16.9.13). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, Y' a subprescheme of Y, such that the canonical injection $j : Y' \to Y$ is regular. Then the sequence of $\mathscr{O}_{Y'}$-modules*

(16.9.13.1) $$0 \longrightarrow j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0$$

is exact; furthermore, for every $x \in X$, there is an open neighbourhood U of x such that the restrictions to U of the homomorphisms of (16.9.13.1) form a split exact sequence. **IV-4** | 50

Let us first prove the following lemma:

Lemma (16.9.13.2). — *Let A be a ring, $\mathfrak{I}$ an ideal of A, $A' = A/\mathfrak{I}$, $(f_i)_{1 \leqslant i \leqslant r}$ a sequence of elements of A which is A'-regular, $\mathfrak{K} = \sum_i f_i A$, $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, $\mathfrak{K}' = \sum_i f_i A'$, so that $C = A/\mathfrak{L}$ is isomorphic to $A'/\mathfrak{K}'$. For every integer $n > 0$, and every integer $N \geqslant n$, we have the relation*

(16.9.13.3) $$\mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I}\mathfrak{K}^n + (\mathfrak{I} \cap \mathfrak{K}^N).$$

Proof. It is plainly enough to prove that every element of the left-hand side belongs to the right-hand side, and an induction reduces us to the case $N = n + 1$. An element of the left-hand side of (16.9.13.3), being in $\mathfrak{K}^n$, can be written as $P(f_1, \dots, f_r)$, where $P \in A[T_1, \dots, T_r]$ is homogeneous of degree n. If f'_i is the canonical image of f_i in A', the hypothesis $P(f_1, \dots, f_r) \in \mathfrak{I}$ means that $P(f'_1, \dots, f'_r) = 0$. But $P(f'_1, \dots, f'_r) \in \mathfrak{K}'^n$, so the canonical image of $P(f'_1, \dots, f'_r)$ in $\mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is zero. Now the hypothesis that (f_i) is A'-regular implies that the canonical homomorphism $\mathbf{S}^n_C(\mathfrak{K}'/\mathfrak{K}'^2) \to \mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is bijective (**0**, 15.1.9); we conclude that the coefficients of P are in $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$. It follows at once that $P(f_1, \dots, f_r) \in \mathfrak{I}\mathfrak{K}^n + \mathfrak{K}^{n+1}$, and since $P(f_1, \dots, f_r) \in \mathfrak{I}$, we finally have $P(f_1, \dots, f_r) \in \mathfrak{I}\mathfrak{K}^n + (\mathfrak{I} \cap \mathfrak{K}^{n+1})$, which proves the lemma. □

Taking the quotients of both sides of (16.9.13.3) by $\mathfrak{I}\mathfrak{K}^n$, we see that the relations (16.9.13.3), for $N \geqslant n$, imply

(16.9.13.4) $$(\mathfrak{I} \cap \mathfrak{K}^n)/\mathfrak{I}\mathfrak{K}^n \subset \bigcap_{N \geqslant n} \mathfrak{K}^N \cdot (A/(\mathfrak{I}\mathfrak{K}^n)).$$

We deduce the following corollary.

Corollary (16.9.13.5). — *Suppose that the hypotheses of (16.9.13.2) are satisfied and also that the ring A is Noetherian and $\mathfrak{K}$ is contained in the radical of A. Then for every integer $n > 0$,*

(16.9.13.6) $$\mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I}\mathfrak{K}^n.$$

Proof. Indeed, the second member of (16.9.13.4) is then zero, given that $A/\mathfrak{I}\mathfrak{K}^n$ is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. III, §3, n°3, prop. 6). □

Taking in particular $n = 2$ in (16.9.13.6), and observing that $\mathfrak{L}^2 = \mathfrak{I}^2 + \mathfrak{I}\mathfrak{K} + \mathfrak{K}^2 = \mathfrak{I}\mathfrak{L} + \mathfrak{K}^2$, we have $\mathfrak{I}\mathfrak{L} \subset \mathfrak{I} \cap \mathfrak{L}^2$ and therefore

$$\mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I}\mathfrak{L} + (\mathfrak{I} \cap \mathfrak{K}^2) = \mathfrak{I}\mathfrak{L} + \mathfrak{I}\mathfrak{K}^2 = \mathfrak{I}\mathfrak{L},$$

in other words,

(16.9.13.7) $$\mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I}\mathfrak{L},$$

which may also be expressed by saying that the canonical homomorphism

$$\mathfrak{I}/\mathfrak{I}\mathfrak{L} \longrightarrow (\mathfrak{I} + \mathfrak{L}^2)/\mathfrak{L}^2$$

is bijective.

Proof (16.9.13). Having demonstrated the lemmas, let us prove the first claim of (16.9.13): It is clearly enough to prove that, at a point $x \in Y'$, the sequence of stalks of the sheaves appearing **IV-4** | 51 in (16.9.13.1) is exact. Put $A = \mathscr{O}_{X,x}$. We may write $\mathscr{O}_{Y,x} = A' = A/\mathfrak{I}$, where $\mathfrak{I}$ is an ideal contained in the maximal ideal of A, and then $\mathscr{O}_{Y',x} = A'/\mathfrak{K}'$, where $\mathfrak{K}'$ is generated by an A'-*regular* sequence of elements of A' which are the images of elements of an A-regular sequence in A belonging to the maximal ideal of A. If $\mathfrak{K}$ is the ideal generated by the latter elements and $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, then $\mathscr{O}_{Y',x} = A/\mathfrak{L}$; since we are in the situation of (16.9.13.5), the canonical homomorphism $\mathfrak{I}/\mathfrak{I}\mathfrak{L} \to (\mathfrak{I} + \mathfrak{L}^2)/\mathfrak{L}^2$ is bijective. But this shows that the sequence

$$0 \longrightarrow \mathfrak{I}/\mathfrak{I}\mathfrak{L} \longrightarrow \mathfrak{L}/\mathfrak{L}^2 \longrightarrow (\mathfrak{L}/\mathfrak{I})/(\mathfrak{L}/\mathfrak{I})^2 \longrightarrow 0$$

is exact (see the proof of (16.2.7)), and the modules appearing in this sequence are precisely the stalks at x of the sheaves in (16.9.13.1). The second claim follows from the fact that $\mathscr{N}_{Y'/Y}$ is a locally free $\mathscr{O}_{Y'}$-module (16.9.8) and Bourbaki, *Alg.*, chap. II, 3rd ed., §1, n°11, prop. 21. □

16.10. Differentially smooth morphisms.

Definition (16.10.1). — We say that a morphism of preschemes $f : X \to S$ is differentially smooth (or that X is differentially smooth over S) if it satisfies the following conditions:

(i) $\Omega^1_{X/S}$ is a locally projective $\mathscr{O}_X$-module, which is to say that every point $x \in X$ admits an affine open neighbourhood U such that $\Gamma(U, \Omega^1_{X/S})$ is a projective $\Gamma(U, \mathscr{O}_X)$-module (not necessarily of finite type).

(ii) The canonical morphism (16.3.1.1)

$$\mathbf{S}^{\bullet}_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_{\bullet}(\mathscr{P}_{X/S})$$

is bijective.

In particular, if $\Omega^1_{X/S}$ is locally free of finite rank, the $\mathscr{P}^n_{X/S}$ are locally free of finite rank $\mathscr{O}_X$-modules (being extensions of such modules).

We say that f is *differentially smooth at a point* $x \in X$ if there is an open neighbourhood U of x in X such that $f|U$ is differentially smooth.

We will see later (17.12.4) that a smooth morphism is differentially smooth, which justifies the terminology; but the converse is not true; indeed, a *monomorphism* $f : X \to S$ is differentially smooth, since $\Omega^1_{X/S} = 0$ because of (**I**, 5.3.8), and consequently the surjective homomorphism (16.3.1.1) is clearly bijective; however a monomorphism is not even necessarily flat, neither *a fortiori* smooth. Let us limit ourselves to proving the following proposition:

Proposition (16.10.2). — *Let A be a ring, B a formally smooth A-algebra for the discrete topology* (**0**, 19.3.1). *Then* $\operatorname{Spec}(B)$ *is differentially smooth over* $\operatorname{Spec}(A)$.

Proof. Indeed, $B \otimes_A B$ is then (with the discrete topology) a formally smooth B-algebra (by one or the other canonical homomorphisms $b \mapsto b \otimes 1$, $b \mapsto 1 \otimes b$ of B to $B \otimes_A B$) (**0**, 19.3.5, (iii)); therefore **IV-4** | 52 $B \otimes_A B$ is a formally smooth A-algebra with the discrete topology (**0**, 19.3.5, (ii)). Taking $\mathfrak{I} = \mathfrak{I}_{B/A}$, it follows that $B \otimes_A B$ is a formally smooth A-algebra for the $\mathfrak{I}$-preadic topology (**0**, 19.3.8); since by hypothesis $B = (B \otimes_A B)/\mathfrak{I}$ is a formally smooth A-algebra for the discrete topology, the proposition follows from the equivalence of (a) and (b) in (**0**, 19.5.4). □

Proposition (16.10.3). — *For a morphism $f : X \to S$ to be differentially smooth, it is necessary and sufficient that for every $x \in X$, there is an affine open neighbourhood of x, of ring A, such that $\Gamma(U, \mathscr{P}^\infty_{X/S})$ is an augmented topological A-algebra isomorphic to the completion $\widehat{B}$, where $B = \mathbf{S}_A(V)$, V being a projective A-module and B being endowed with the B^+-preadic topology (where B^+ is the augmentation ideal). If $\Omega^1_{X/S}$ is locally free of finite rank, we can replace $\widehat{B}$ with the ring of formal series $A[[T_1, \ldots, T_n]]$.*

Proof. The notion of a differentially smooth morphism is clearly local on X, so we reduce to the case where $S = \operatorname{Spec}(B)$, $X = \operatorname{Spec}(C)$. Consider $C \otimes_B C$ as a C-algebra (by the first factor); put $\mathfrak{I} = \mathfrak{I}_{C/B}$ and endow $C \otimes_B C$ with the $\mathfrak{I}$-preadic topology; we can apply to the topological C-algebra $C \otimes_B C$ and to the ideal $\mathfrak{I}$ of $C \otimes_B C$ the equivalence of (b) and (c) of (**0**, 19.5.4), given that $(C \otimes_B C)/\mathfrak{I} = C$ is clearly a formally smooth C-algebra for the discrete topologies. The topology of $\Gamma(U, \mathscr{P}^\infty_{X/S})$ is clearly the projective limit topology of this ring (16.1.11). □

We note that the integer n of the proposition (16.10.3) is the *rank* of $\Omega^1_{X/S}$ in the point x. We shall see (17.13.5) that when f is differentially smooth and locally of finite type, n is equal to the dimension of the fibre $f^{-1}(f(x))$.

Proposition (16.10.4). — *Let $f : X \to S$, $g : S' \to S$ be two morphisms, and take $X' = X \times_S S'$, $f' = f_{(S')} : X' \to S'$.*

(i) *If f is differentially smooth, the same is true for f'.*

(ii) *Conversely, if g is faithfully flat and quasi-compact, and if f' is differentially smooth and $\Omega^1_{X'/S'}$ is a finite type $\mathscr{O}_{X'}$-module, f is differentially smooth and $\Omega^1_{X/S}$ is a finite type $\mathscr{O}_X$-module.*

Proof. Indeed, if f is differentially smooth, the $\mathscr{G}r_n(\mathscr{P}_{X/S})$ are *flat* $\mathscr{O}_X$-modules; therefore, by (16.4.6), the homomorphisms $\mathscr{G}r_n(\mathscr{P}_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}_{X'/S'})$ are bijective for every n. In view of the commutativity of diagram (16.2.1.3), it follows from Definition (16.10.1) that f' is differentially smooth. On the other hand, if g is faithfully flat and quasi-compact, it also follows from (16.4.6) that $\mathscr{G}r_n(\mathscr{P}_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}_{X'/S'})$ is bijective for every n. Suppose also that f' is differentially smooth and $\Omega^1_{X'/S'}$ is of finite rank. Since the canonical projection $X' \to X$ is a faithfully flat and quasi-compact morphism, it follows first from (2.5.2) that $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module locally free of finite rank, then from (2.2.7) that the canonical homomorphism (16.3.1.1) is bijective, and therefore f is differentially smooth. □

Proposition (16.10.5). — *For a morphism locally of finite type $f : X \to S$ to be differentially smooth, it is necessary and sufficient that the diagonal immersion $\Delta_f : X \to X \times_S X$ be quasi-regular.*

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Proof. Being a local problem, we can reduce to the case where S and X are affines, and therefore the diagonal subprescheme of $X \times_S X$ is closed. The hypothesis that f is locally of finite type implies that Δ_f is locally of finite presentation (**I**, 4.3.1), therefore the diagonal prescheme of $X \times_S X$ is defined by an ideal $\mathscr{I}$ of finite type, and $\Omega^1_{X/S} = \mathscr{I}/\mathscr{I}^2$ is an $\mathscr{O}_X$-module of finite type. The proposition is now immediate from the comparison of the conditions of (16.10.1) and (16.9.4). □

Remark (16.10.6). — Let $f : X \to S$ be a morphism such that the $\mathscr{O}_X$-module $\Omega^1_{X/S}$ is locally free of finite rank. It follows from (**0**, 20.4.7) that every $x \in X$ has an open neighbourhood such that there is a finite family $(z_\lambda)_{\lambda \in L}$ of sections of $\mathscr{O}_X$ over U for which $(dz_\lambda)_{\lambda \in L}$ forms a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

16.11. Differential operators on a differentially smooth S-prescheme

(16.11.1). Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda \in L}$ a family of sections of $\mathscr{O}_X$ over U such that the dz_λ form a system of generators of $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. Let m be an integer or the symbol ∞, and take, for every λ

(16.11.1.1) $$\zeta_\lambda = \delta z_\lambda = d^m z_\lambda - z_\lambda \in \Gamma(U, \mathscr{P}^m_{X/S}).$$

We will also use the usual notations from analysis; for every $\mathbf{p} = (p_\lambda) \in \mathbb{N}^{(L)}$ (so that $p_\lambda = 0$ except for a finite number of indices), we put

(16.11.1.2) $$|\mathbf{p}| = \sum_\lambda p_\lambda, \quad \mathbf{p}! = \prod_\lambda (p_\lambda !)$$

(16.11.1.3) $$\binom{\mathbf{p}}{\mathbf{q}} = \mathbf{p}!/(\mathbf{q}!(\mathbf{p}-\mathbf{q})!), \quad \text{for } \mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}, \mathbf{q} \leqslant \mathbf{p}$$

and we adopt the convention that $\binom{\mathbf{p}}{\mathbf{q}} = 0$ when $\mathbf{q} \nleqslant \mathbf{p}$,

(16.11.1.4) $$\mathbf{z}^{\mathbf{p}} = \prod_\lambda (z_\lambda)^{p_\lambda}, \quad \boldsymbol{\zeta}^{\mathbf{p}} = \prod_\lambda (\zeta_\lambda)^{p_\lambda}.$$

We therefore have, with this notation

(16.11.1.5) $$d^m(\mathbf{z}^{\mathbf{p}}) = (d^m(\mathbf{z}))^{\mathbf{p}} = (\boldsymbol{\zeta} + \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q} \leqslant \mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}} \mathbf{z}^{\mathbf{p}-\mathbf{q}} \boldsymbol{\zeta}^{\mathbf{q}}$$

(16.11.1.6) $$\boldsymbol{\zeta}^{\mathbf{p}} = (d^m\mathbf{z} - \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q}\leqslant\mathbf{p}} (-1)^{|\mathbf{p}-\mathbf{q}|}\binom{\mathbf{p}}{\mathbf{q}}\mathbf{z}^{\mathbf{p}-\mathbf{q}}d^m(\mathbf{z}^{\mathbf{q}}).$$

Since the dz_λ generate $\Omega^1_{X/S}$, are the images of the δz_λ, and the canonical morphism (16.3.1.1) is surjective, we conclude that, for finite m, the δz_λ generate the $\mathscr{O}_U$-algebra $\mathscr{P}^m_{U/S}$ (Bourbaki, *Alg. comm.*, chap. III, §2, n°8, cor. 2 to th. 1). Therefore the $\boldsymbol{\zeta}^{\mathbf{p}}$ (for $|\mathbf{p}| \leqslant m$) generate the $\mathscr{O}_U$-module $\mathscr{P}^m_{U/S}$. A differential operator $D \in \mathrm{Diff}^m_{U/S}$ is consequently entirely determined by the values $\langle\boldsymbol{\zeta}^{\mathbf{p}}, D\rangle$ for $|\mathbf{p}| \leqslant m$, or, equivalently by (16.11.1.5) and (16.11.1.6), by the values of $\langle d^m(\mathbf{z}^{\mathbf{p}}), D\rangle =$ **IV-4** | 54
$D(\mathbf{z}^{\mathbf{p}})$ for $|\mathbf{p}| \leqslant m$; more precisely, it follows from (16.11.1.5) that

(16.11.1.7) $$D(\mathbf{z}^{\mathbf{p}}) = \langle d^m(\mathbf{z}^{\mathbf{p}}), D\rangle = \sum_{\mathbf{q}\leqslant\mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}}\langle\boldsymbol{\zeta}^{\mathbf{q}}, D\rangle\mathbf{z}^{\mathbf{p}-\mathbf{q}}.$$

Theorem (16.11.2). — *Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda\in L}$ a family of sections of $\mathscr{O}_X$ over U such that the family $(dz_\lambda)_{\lambda\in L}$ generates $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. The following conditions are equivalent:*

(a) $f|U$ is differentially smooth and (dz_λ) is a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$.
(b) There is a family $(D_{\mathbf{p}})_{\mathbf{p}\in\mathbb{N}^{(L)}}$ of differential operators of $\mathscr{O}_U$ to itself verifying the conditions

(16.11.2.1) $$D_{\mathbf{p}}(\mathbf{z}^{\mathbf{q}}) = \binom{\mathbf{q}}{\mathbf{p}}\mathbf{z}^{\mathbf{q}-\mathbf{p}}, \quad (\mathbf{p}, \mathbf{q} \textit{ in } \mathbb{N}^{(L)}).$$

Also, when these conditions are satisfied, the family $(D_{\mathbf{p}})$ is uniquely determined by the conditions (16.11.2.1) and satisfies the relations

(16.11.2.2) $$D_{\mathbf{q}} \circ D_{\mathbf{p}} = D_{\mathbf{p}} \circ D_{\mathbf{q}} = \frac{(\mathbf{p}+\mathbf{q})!}{\mathbf{p}!\mathbf{q}!}D_{\mathbf{p}+\mathbf{q}} \quad (\mathbf{p}, \mathbf{q} \textit{ in } \mathbb{N}^{(L)}).$$

Finally, if L is finite, for every integer m, the $D_{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ form a basis of the $\mathscr{O}_U$-module $\mathscr{D}\!\mathit{iff}^m_{U/S}$, in other words, every differential operator of order $\leqslant m$ on U can be written uniquely as

$$D = \sum_{|\mathbf{p}|\leqslant m} a_{\mathbf{p}}D_{\mathbf{p}}$$

where the $a_{\mathbf{p}}$ are sections of $\mathscr{O}_X$ over U.

Proof. Note first that because of (16.11.1.6) and (16.11.1.5), we verify immediately that the conditions (16.11.2.1) are equivalent to

(16.11.2.3) $$\langle\boldsymbol{\zeta}^{\mathbf{p}}, D_{\mathbf{q}}\rangle = \delta_{\mathbf{pq}} \quad \text{(Kronecker's symbol)}.$$

The existence of the family $(D_{\mathbf{p}})$ verifying these conditions implies first (by taking $|\mathbf{p}| = 1$) that the dz_λ are linearly independent, and therefore form a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$. Then, for every integer $m \geqslant 1$, we deduce similarly from (16.11.2.3) that the $\boldsymbol{\zeta}^{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ are linearly independent; it follows that the canonical homomorphism (16.3.1.1) is injective, and therefore bijective, which proves that (b) implies (a). The converse follows immediately from Definition (16.10.1): the fact that the $\boldsymbol{\zeta}^{\mathbf{p}}$ form a basis of $\mathscr{P}^m_{U/S}$ for $|\mathbf{p}| \leqslant m$ implies the existence and uniqueness of a family of homomorphisms $u_{\mathbf{q},m} : \mathscr{P}^m_{U/S} \to \mathscr{O}_U$ ($|\mathbf{q}| \leqslant m$) such that $\langle\boldsymbol{\zeta}^{\mathbf{p}}, u_{\mathbf{q},m}\rangle = \delta_{\mathbf{pq}}$ for $|\mathbf{p}| \leqslant m$, $|\mathbf{q}| \leqslant m$. For a given value of $\mathbf{q}$, the differential operators corresponding to $u_{\mathbf{q},m}$ for $m \geqslant |\mathbf{q}|$ are identified with the same operator $D_{\mathbf{q}}$. This proves that (a) implies (b), and also that the family $(D_{\mathbf{q}})$ is uniquely determined and that, if L is finite, the $D_{\mathbf{p}}$ with $|\mathbf{p}| \leqslant m$ form a basis of the $\mathscr{O}_U$-module $\mathscr{D}\!\mathit{iff}^m_{U/S}$, the dual of $\mathscr{P}^m_{U/S}$. Finally, the relations (16.11.2.2) follow immediately by evaluating the three operators in question on the $\mathbf{z}^{\mathbf{r}}$ and using the fact that the $\boldsymbol{\zeta}^{\mathbf{r}}$ for $|\mathbf{r}| \leqslant m$ generate $\mathscr{P}^m_{U/S}$. □

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Remarks (16.11.3). — (i) The fact that, by (16.11.2.2), the $D_{\mathbf{p}}$ commute pairwise naturally does not imply that the $\mathscr{O}_U$-algebra $\mathscr{D}\!\mathit{iff}_{U/S}$ is commutative: the $D_{\mathbf{p}}$ do not commute with multiplication by sections of $\mathscr{O}_U$ unless $\mathbf{p} = 0$.

(ii) The indices $\mathbf{p}$ such that $|\mathbf{p}| = 1$ are the $\boldsymbol{\epsilon}_\lambda = (\epsilon_{\lambda\mu})_{\mu\in L}$ where $\epsilon_{\lambda\mu} = 0$ if $\mu \neq \lambda$ and $\epsilon_{\lambda\lambda} = 1$; when L is finite, the operators $D_{\boldsymbol{\epsilon}_\lambda}$ are exactly the S-derivations D_λ introduced in (16.5.7). We note that in general (contrary to what happens in classical analysis), it is not true that a differential operator of any order can be written as a linear combination of powers of D_i (cf. (16.12)).

(iii) For every integer $r \geqslant 1$, we can define the notion of *differentially smooth up to order r* by replacing in (16.10.1) the condition (ii) by the condition that the homomorphisms

$$\mathbf{S}^m_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_m(\mathscr{P}_{X/S})$$

are bijective *for every $m \leqslant r$*. The argument of (16.11.2) proves also that if, in the condition (a), we replace "differentially smooth" by "differentially smooth up to order r", this condition is equivalent to (b) by restricting ourselves to $\mathbf{p} \in \mathbb{N}^{(L)}$, $\mathbf{q} \in \mathbb{N}^{(L)}$ such that $|\mathbf{p}| \leqslant r$, $|\mathbf{q}| \leqslant r$.

16.12. Case of characteristic zero: Jacobian criterion for differentially smooth morphisms.

(16.12.1). We say that a prescheme X is *of characteristic p* (p equal to zero or a prime number) if, for every affine open set U of X, the ring $\Gamma(U, \mathscr{O}_X)$ is of characteristic p (**0**, 21.1.1). It follows from (**0**, 21.1.3) that for X to be of characteristic 0, it is necessary and sufficient that for every *closed* point x of X, the residue field $k(x)$ is of characteristic 0, or even that X can be given a structure of $\mathbb{Q}$-prescheme (necessarily unique).

Theorem (16.12.2). — *Let X be a prescheme of characteristic 0, $f : X \to S$ a morphism. If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module (not necessarily of finite type), f is differentially smooth.*

Proof. The problem being local on X, we can suppose there is a family (z_λ) of sections of $\mathscr{O}_X$ over X such that the (dz_λ) is a basis for the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. Applying the criterion (16.11.2), it is enough for the operators

$$D_{\mathbf{p}} = (\mathbf{p}!)^{-1} \prod_\lambda D_\lambda^{p_\lambda}$$

(where the D_λ are the coordinate forms corresponding to the basis (dz_λ)) to verify the relations (16.11.2.1), which is a consequence of the fact that the D_λ are derivations. □

(16.12.3). The theorem above is not true if we discard the hypothesis that X is of characteristic 0. For example, if $S = \operatorname{Spec}(k)$, where k is a field of characteristic $p > 0$, $X = \operatorname{Spec}(K)$ where $K = k(\alpha)$ where $\alpha \notin k$, $\alpha^p \in k$, we verify immediately that $\Omega^1_{X/S}$ has rank 1, and that the morphism $X \to S$ **IV-4** | 56
is differentially smooth up to order $p - 1$ (16.11.3, (iii)), but not up to order p. However, the proof of (16.12.2) proves that if $\Omega^1_{X/S}$ is locally free, and if $n!1_{\mathscr{O}_X}$ is invertible in $\Gamma(X, \mathscr{O}_X)$, then X is differentially smooth over S up to order n.

§17. Smooth morphisms, unramified (or net) morphisms, and étale morphisms.

In this section, we return to the notions studied in (**0**, 19), expressed in the geometric language of schemes and from the global point of view, for preschemes locally of finite presentation over a given base prescheme.

Most of the results (except those of (17.7), (17.8), (17.9), (17.13), and (17.16)) reduce, moreover, to variants of properties already encountered in (**0**, 19).

For more specific results on étale morphisms, the reader should consult (18).

17.1. Formally smooth morphisms, formally unramified morphisms, formally étale morphisms.

Definition (17.1.1). — Let $f : X \to Y$ be a morphism of preschemes. We say that f is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) if, for all affine schemes Y', all closed subschemes Y'_0 of Y' defined by a nilpotent ideal $\mathscr{J}$ of $\mathscr{O}_{Y'}$, and every morphism $Y' \to Y$, the map

(17.1.1.1) $$\mathrm{Hom}_Y(Y', X) \longrightarrow \mathrm{Hom}_Y(Y'_0, X)$$

induced by the canonical map $Y'_0 \to Y'$, is *surjective* (resp. *injective*, resp. *bijective*).

One also says that X is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) over Y.

It is clear that for f to be formally étale, it is necessary and sufficient for f to be formally smooth and formally unramified.

Remark (17.1.2). —

(i) Suppose that $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, so that f comes from a homomorphism of rings $\phi : A \to B$. According to (**0**, 19.3.1) and (**0**, 19.10.1), saying that f is formally smooth (resp. formally unramified, resp. formally étale) means that, via ϕ, B is a *formally smooth* (resp. *formally unramified*, resp. *formally étale*) A-algebra, for the *discrete* topologies on A and B.

(ii) To verify that f is formally smooth (resp. formally unramified, resp. formally étale), we can, in Definition (17.1.1), restrict to the case where $\mathscr{J}^2 = 0$. To see this, if f satisfies the corresponding condition of Definition (17.1.1) in the particular case $\mathscr{J}^2 = 0$, and if we have $\mathscr{J}^n = 0$, then we consider the closed subscheme Y'_j of Y' defined by the sheaf of ideals $\mathscr{J}^{j+1}$ for $0 \leqslant j \leqslant n-1$, so that Y'_j is a closed subscheme of Y'_{j+1} defined by a square-zero sheaf of ideals; the hypotheses imply that each of the maps

$$\mathrm{Hom}_Y(Y'_{j+1}, X) \longrightarrow \mathrm{Hom}_Y(Y'_j, X) \quad (0 \leqslant j \leqslant n-1)$$

is surjective (resp. injective, resp. bijective); by composition, we conclude that the same holds for (17.1.1.1). **IV** | 57

(iii) Note that the properties of the morphism f defined in (17.1.1) are properties of the *representable functor* ($\mathbf{0}_{\mathbf{III}}$, 8.1.8)

$$Y' \longmapsto \mathrm{Hom}_Y(Y', X)$$

from the category of Y-preschemes to the category of sets; they keep a meaning for *any* contravariant functor with the same domain and codomain, representable or not.

(iv) Assume that the morphism f is formally unramified (resp. formally étale); consider an *arbitrary* Y-prescheme Z and a closed subprescheme Z_0 of Z defined by a *locally nilpotent* sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_Z$. Then the map

(17.1.2.1) $$\mathrm{Hom}_Y(Z, X) \longrightarrow \mathrm{Hom}_Y(Z_0, X)$$

induced by the canonical injection $Z_0 \to Z$, is still injective (resp. bijective). To see this, let (U_α) be an affine open covering of Z such that the sheaves of ideals $\mathscr{J}|U_\alpha$ are nilpotent, and for each α, let U_α^0 be the inverse image of U_α in Z_0, which is the closed subprescheme of U_α defined by $\mathscr{J}|U_\alpha$. Let $f_0 : Z_0 \to X$ be a Y-morphism; by hypothesis, for each α, there is at most one (resp. one and only one) Y-morphism $f_\alpha : U_\alpha \to X$ whose restriction to Z_0 coincides with $f_0|U_\alpha$. We immediately conclude that if f_α and f_β are defined, then, for each affine open $V \subset U_\alpha \cap U_\beta$, we have $f_\alpha|V = f_\beta|V$, as the restrictions of these morphisms to the inverse image V_0 of V in Z_0 coincide. There is therefore at most one (resp. one and only one) Y-morphism $f : Z \to X$ whose restriction to Z_0 coincides with f_0.

Proposition (17.1.3). —

(i) *A monomorphism of preschemes is formally unramified; an open immersion is formally étale.*

(ii) *The composition of two formally smooth (resp. formally unramified, resp. formally étale) morphisms is formally smooth (resp. formally unramified, resp. formally étale).*

(iii) *If $f : X \to Y$ is a formally smooth (resp. formally unramified, resp. formally étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two formally smooth (resp. formally unramified, resp. formally étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.*

(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if $g \circ f$ is formally unramified, then so is f.*

(vi) *If $f : X \to Y$ is a formally unramified morphism, then so is $f_{\mathrm{red}} : X_{\mathrm{red}} \to Y_{\mathrm{red}}$.*

Proof. According to (**I**, 5.5.12), it suffices to prove (i), (ii), and (iii). The assertions in (i) are both trivial. To prove (ii), consider two morphisms $f : X \to Y$, $g : Y \to Z$, an affine scheme Z', a closed subscheme Z'_0 of Z' defined by a nilpotent ideal, and a morphism $Z' \to Z$. Suppose that f and g are formally smooth, and consider a Z-morphism $u_0 : Z'_0 \to X$; the hypothesis on g implies that **IV | 58** there exists a Z-morphism $v : Z' \to Y$ such that $f \circ u_0 = v \circ j$ (where $j : Z'_0 \to Z'$ is the canonical injection); the hypothesis on f then implies that there exists a morphism $u : Z' \to X$ such that $f \circ u = v$ and $u \circ j = u_0$, therefore $(g \circ f) \circ u$ is equal to the given morphism $Z' \to Z$ and $u \circ j = u_0$, which proves that $g \circ f$ is formally smooth; we argue the same way when we suppose that f and g are formally unramified.

Finally, to prove (iii), let $X' = X_{S'}$, $Y' = Y_{S'}$, $f' = f_{S'}$; consider an affine scheme Y'', a closed subscheme Y''_0 defined by a nilpotent sheaf of ideals, and a morphism $g : Y'' \to Y'$ making Y'' a Y'-prescheme; we then know by (**I**, 3.3.8) that $\mathrm{Hom}_{Y'}(Y'', X')$ is canonically identified with $\mathrm{Hom}_Y(Y'', X)$, and $\mathrm{Hom}_{Y'}(Y''_0, X')$ with $\mathrm{Hom}_Y(Y''_0, X)$, and the conclusion follows immediately from Definition (17.1.1). ☐

We note that a *closed immersion* is not necessarily formally smooth.

Proposition (17.1.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is formally unramified. Then, if $g \circ f$ is formally smooth (resp. formally étale), so is f.*

Proof. Let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent sheaf of ideals, $h : Y' \to Y$ a morphism, $j : Y'_0 \to Y'$ the canonical injection, and $u_0 : Y'_0 \to X$ a Y-morphism, so that $f \circ u_0 = h \circ j$. Suppose that $g \circ f$ is formally smooth; then there exists a morphism $u : Y' \to X$ such that $u \circ j = u_0$ and $(g \circ f) \circ u = g \circ h$. But these two relations imply that $f \circ u$ and h are Z-morphisms from Y' to Y such that $(f \circ u) \circ j = h \circ j$; by virtue of the hypothesis that g is formally unramified, we get that $f \circ u = h$, in other words that u is a Y-morphism; thus f is formally smooth. Taking into account (17.1.3, (v)), this proves the proposition. ☐

Corollary (17.1.5). — *Suppose that g is formally étale; then, for $g \circ f$ to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that f is.*

Proof. This follows from (17.1.4) and (17.1.3, (ii) and (v)). ☐

Proposition (17.1.6). — *Let $f : X \to Y$ be a morphism of preschemes.*

(i) *Let (U_α) be an open covering of X and, for each α, let $i_\alpha : U_\alpha \to X$ be the canonical injection. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each $f \circ i_\alpha$ is.*

(ii) *Let (V_λ) be an open covering of Y. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each of the restrictions $f^{-1}(V_\lambda) \to V_\lambda$ of f is.*

Proof. First note that (ii) is a consequence of (i): if $j_\lambda : V_\lambda \to Y$ and $i_\lambda : f^{-1}(V_\lambda) \to X$ are the canonical injections, then the restriction $f_\lambda : f^{-1}(V_\lambda) \to V_\lambda$ of f is such that $j_\lambda \circ f_\lambda = f \circ i_\lambda$; if f is formally smooth (resp. formally unramified), then so is $f \circ i_\lambda$ since i_λ is formally étale (17.1.3); but since j_λ is formally étale, this means that f_λ is formally smooth (resp. formally unramified), by virtue of (17.1.5). Conversely, if all the f_λ are formally smooth (resp. formally unramified), the same applies to $j_\lambda \circ f_\lambda$ (17.1.3), so also to f in virtue of (i).

If we take into account that the i_α are formally étale, everything comes down to proving that if **IV | 59** the $f \circ i_\alpha$ are formally smooth (resp. formally unramified), then the same applies to f.

Therefore let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent ideal $\mathscr{J}$, which we may assume to satisfy $\mathscr{J}^2 = 0$ (17.1.2, (ii)), and finally let $g : Y' \to Y$ be a morphism. Suppose we are given a Y-morphism $u_0 : Y'_0 \to X$; denote by W_α (resp. W^0_α) the prescheme induced by Y' (resp. Y'_0) on the open subset $u_0^{-1}(U_\alpha)$ (we recall that Y' and Y'_0 share the *same underlying topological space*). Let us first suppose that the $f \circ i_\alpha$ are *formally unramified*, and show that, if u' and

u'' are two Y-morphisms from Y' to X whose restrictions to Y'_0 coincide, then we have $u' = u''$. Indeed, taking into account (17.1.2, (iv)), the hypothesis that the $f \circ i_\alpha$ are formally unramified implies that for all α, we have $u'|W_\alpha = u''|W_\alpha$, since the restrictions of both Y-morphisms to W^0_α coincide. Hence the conclusion follows.

Now suppose that the $f \circ i_\alpha$ are *formally smooth* and prove the existence of a Y-morphism $u : Y' \to X$ whose restriction to Y'_0 is u_0. Now, since Y' is an *affine scheme*, we can apply (16.5.17), the hypotheses of which are satisfied, and the conclusion of which precisely proves the existence of u. □

We can therefore say that the notions introduced in (17.1.1) are *local* on X and Y, which always allows, in virtue of (17.1.2, (i)), to be reduced to the study of formally smooth (resp. formally unramified, resp. formally étale) *algebras*.

17.2. General properties of differentials

Proposition (17.2.1). — *For a morphism $f : X \to Y$ to be formally unramified, it is necessary and sufficient that $\Omega^1_f = 0$ (what we still write $\Omega^1_{X/Y} = 0$ (16.3.1)).*

Proof. Taking into account (17.1.6), we reduce to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, and the conclusion then follows from (**0**, 20.7.4) and the interpretation of $\Omega^1_{X/Y}$ in this case (16.3.7). □

Corollary (17.2.2). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms. For f to be formally unramified, it is necessary and sufficient that the canonical morphism (16.4.19)*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is surjective.

Proof. This is an immediate consequence of (17.2.1) and the exact sequence (16.4.19.1). □

Proposition (17.2.3). — *Let $f : X \to Y$ be a formally smooth morphism.*

(i) *The $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is locally projective (16.10.1). If f is locally of finite type, then $\Omega^1_{X/Y}$ is locally free and of finite type.*

(ii) *For all morphisms $g : Y \to Z$, the sequence (16.4.19) of $\mathscr{O}_X$-modules*

(17.2.3.1) $$0 \longrightarrow f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.3.1) form a split *exact sequence.*

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Proof.

(i) We know (16.3.9) that if f is locally of finite type, then Ω^1_f is an $\mathscr{O}_X$-module of finite type. To prove that, in all cases, it is locally projective, we can reduce, by virtue of (17.1.6), to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, and the result follows from the hypothesis on f and from (**0**, 20.4.9) and (**0**, 19.2.1).

(ii) Again, we can restrict to the case where X, Y, and Z are affine (17.1.6), and the conclusion in this case follows from the interpretation of the sheaves of modules in the sequence (17.2.3.1) and from (**0**, 20.5.7).

□

Corollary (17.2.4). — *If $f : X \to Y$ is formally étale, then, for all morphisms $g : Y \to Z$, the canonical homomorphism of $\mathscr{O}_X$-modules*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is bijective.

Proof. This follows from the exactness of the sequence (17.2.3.1) and from the fact that we then have $\Omega^1_{X/Y} = 0$ (17.2.1). □

Proposition (17.2.5). — *Let $f : X \to Y$ be a morphism, X' a subprescheme of X such that the composite morphism $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) is formally smooth. Then the sequence of $\mathscr{O}_{X'}$-modules (16.4.21)*

$$0 \longrightarrow \mathscr{N}_{X'/X} \longrightarrow \Omega^1_{X/Y} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/Y} \longrightarrow 0 \tag{17.2.5.1}$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.5.1) form a split *exact sequence.*

Proof. By virtue of (17.1.6), we reduce to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, and $X' = \mathrm{Spec}(B/\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of B. The conormal sheaf $\mathscr{N}_{X'/X}$ then corresponds to the B-module $\mathfrak{J}/\mathfrak{J}^2$ (16.1.3), and the conclusion follows from (**0**, 20.5.14). □

Proposition (17.2.6). — *Let X and Y be two preschemes, $f : X \to Y$ a morphism locally of finite type. The following conditions are equivalent:*

(a) *f is a monomorphism.*
(b) *f is radicial and formally unramified.*
(c) *For each $y \in Y$, the fibre $f^{-1}(y)$ is empty or $k(y)$-isomorphic to $\mathrm{Spec}(k(y))$ (in other words, it is reduced to a single point z such that $k(y) \to \mathscr{O}_z/\mathfrak{m}_y\mathscr{O}_z$ is an isomorphism).*

Proof. The fact that (a) implies (c) follows from (8.11.5.1). It is clear that (c) implies that f is radicial; let us prove that it also follows from (c) that $\Omega^1_{X/Y} = 0$, which will prove that (c) implies (b) (17.2.1). Note that the $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is quasi-coherent of finite type (16.3.9). It follows from (**I**, 9.1.13.1) that, for $(\Omega^1_{X/Y})_x = 0$, it is necessary and sufficient that if we set $Y_1 = \mathrm{Spec}(k(y))$, $X_1 = f^{-1}(y) = X \times_Y Y_1$, then we have $(\Omega^1_{X_1/Y_1})_x = 0$; but as the morphism $f_1 : X_1 \to Y_1$ induced by f is formally unramified by virtue of the hypothesis (c) (17.1.3), the conclusion follows from (17.2.1). Finally, let us prove that (b) implies (a); for this, consider the diagonal morphism $g = \Delta_f : X \to X \times_Y X$; since f is radicial, g is surjective (1.8.7.1); on the other hand, $\Omega^1_{X/Y}$ is by definition the conormal sheaf $\mathscr{G}r_1(g)$ of the immersion g (16.3.1), and to say that f is formally unramified therefore means that $\mathscr{G}r_1(g) = 0$ (17.2.1). In addition, g is locally of finite presentation (1.4.3.1); therefore the hypothesis **IV | 61**
$\mathscr{G}r_1(g) = 0$ implies that g is an open immersion (16.1.10); being surjective, this immersion is an isomorphism, hence f is a monomorphism (**I**, 5.3.8). □

17.3. Smooth morphisms, unramified morphisms, étale morphisms

Definition (17.3.1). — We say that a morphism $f : X \to Y$ is *smooth* (resp. *unramified*, or *net* [28] resp. *étale*) if it is locally of finite presentation and formally smooth (resp. formally unramified, resp. formally étale).

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over* Y.

We will see later (17.5.2) that this definition of a smooth morphism coincides with the definition already given in (6.8.1); until then, we will exclusively use definition (17.3.1).

It is clear that saying that f is étale means that it is *both* smooth and unramified.

Remarks (17.3.2). —

(i) Note that definition (17.3.1) can be phrased using only the functor

$$Y' \longmapsto \mathrm{Hom}_Y(Y', X)$$

considered in (17.1.2, (iii)) because to say that f is locally of finite presentation is equivalent to saying that the preceding functor *commutes with projective limits of affine schemes* (8.14.2).

(ii) Let A be a ring and B an A-algebra. We say that B is a *smooth* (resp. *unramified*, resp. *étale*) A-algebra if the corresponding morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is smooth (resp. unramified, resp. étale). It is equivalent to saying that B is an A-algebra *of finite presentation* (1.4.6) that is furthermore formally smooth (resp. formally unramified, resp. formally étale) for the discrete topologies.

[28] The words "net" and "formally net" seem preferable to the established terminology "unramified" (resp. "formally unramified") and will be used almost exclusively beginning with Chapter V. In this chapter, we have retained the old terminology so as not to conflict with (**0**, 19.10).

(iii) It follows from (17.1.6) and the definition of a morphism locally of finite presentation (1.4.2) that the notion of a smooth (resp. unramified, resp. étale) morphism is *local on X and on Y*.

Proposition (17.3.3). —

(i) *An open immersion is étale. For an immersion to be unramified, it is necessary and sufficient for it to be locally of finite presentation.*

(ii) *The composition of two smooth (resp. unramified, resp. étale) morphisms is smooth (resp. unramified, resp. étale).*

(iii) *If $f : X \to Y$ is a smooth (resp. unramified, resp. étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are smooth (resp. unramified, resp. étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.* IV | 62

(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if g is locally of finite type and if $g \circ f$ is unramified, then f is unramified.*

Proof. This follows from (1.4.3) and (17.1.3). □

Proposition (17.3.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is unramified. Then, if $g \circ f$ is smooth (resp. unramified, resp. étale), so is f.*

Proof. As g and $g \circ f$ are locally of finite presentation, so is f (1.4.3, (v)); the conclusion thus follows from (17.1.4) and (17.1.3, (v)). □

Corollary (17.3.5). — *Suppose that g is étale; then, for f to be smooth (resp. unramified, resp. étale) it is necessary and sufficient that $g \circ f$ is.*

Proof. This follows from (17.3.4) and (17.3.3, (ii)). □

Proposition (17.3.6). — *Let $g : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation. For an S-morphism $f : X \to Y$ to be unramified, it is necessary and sufficient that the canonical homomorphism* (16.4.19)

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is surjective.

Proof. As f is locally of finite presentation (1.4.3, (v)), the proposition follows from (17.2.2). □

Definition (17.3.7). — Let $f : X \to Y$ be a morphism. We say that f is *smooth* (resp. *unramified*, resp. *étale*) at a point $x \in X$, if there exists an open neighbourhood U of x in X such that the restriction $f|U$ is a smooth (resp. unramified, resp. étale) morphism from U to Y.

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over Y at the point x*.

Taking Remark (17.3.2, (iii)) into account, saying that f is smooth (resp. unramified, resp. étale) is equivalent to saying that it is smooth (resp. unramified, resp. étale) at every point of X.

It is clear that the set of points of X at which the morphism $f : X \to Y$ is smooth (resp. unramified, resp. étale) is *open* in X.

Proposition (17.3.8). — *For all preschemes Y and all locally free $\mathscr{O}_Y$-modules $\mathscr{E}$ of finite type, the vector bundle prescheme $\mathbf{V}(\mathscr{E})$* (**II**, 1.7.8) *associated to $\mathscr{E}$ is a smooth Y-prescheme.*

Proof. By (17.3.2, (iii)), we may restrict to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{V}(\mathscr{E}) = \operatorname{Spec}(A[T_1, \ldots, T_r])$; since $A[T_1, \ldots, T_r]$ is a formally smooth A-algebra for the discrete topologies (**0**, 19.3.2) and is of finite presentation, this proves the proposition (17.3.2, (ii)). □

Corollary (17.3.9). — *Under the hypotheses of* (17.3.8), *the projective prescheme $\mathbf{P}(\mathscr{E})$* (**II**, 4.1.1) *is a smooth Y-prescheme.*

Proof. We can still restrict to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{P}(\mathscr{E}) = \mathbf{P}^r_Y$. We then know (**II**, 2.3.14) that we have a finite open cover of $\mathbf{P}^r_A$ by the $D_+(T_i)$ ($0 \leqslant i \leqslant r$) respectively equal to the spectrum of the ring $S_{(f)}$, where we wrote S for $A[T_1, \ldots, T_r]$ and f for T_i; but it follows immediately from the definition of $S_{(f)}$ (**II**, 2.2.1), that this ring, in this case, is isomorphic to $A[T_0, \ldots, T_{i-1}, T_{i+1}, \ldots, T_r]$; hence the corollary follows by (17.3.8). □

17.4. Characterizations of unramified morphisms.

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Theorem (17.4.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X. The following properties are equivalent:*

(a) f is unramified at the point x.
(b) The diagonal morphism $\Delta_f : X \to X \times_Y X$ is a local isomorphism at the point x.
b') If one sets $\Delta_f = (\psi, \theta)$, $Z = X \times_Y X$ and $z = \psi(x)$, the homomorphism $\theta_z^\# : \mathcal{O}_{Z,z} \to \mathcal{O}_{X,x}$ is bijective.
b") For every morphism $g : Y' \to Y$, every point $y' \in Y'$ above $y = f(x)$, and every Y'-section s' of $X' = X \times_Y Y'$ such that $x' = s'(y')$ lies above x, the section s' is a local isomorphism at y'.
(c) $(\Omega^1_{X/Y})_x = 0$.
(d) The $k(y)$-prescheme $f^{-1}(y)$ is unramified over $k(y)$ at the point x.
*d') The point x is isolated in $X_y = f^{-1}(y)$ (equivalently (**Err**$_{\mathbf{III}}$, 20), the morphism f is quasi-finite at x), and the ring $\mathcal{O}_{X_y,x}$ is a field that is a finite separable extension of $k(y)$.*
d") The ring $\mathcal{O}_{X_y,x} = \mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ is a field that is a finite separable extension of $k(y)$.
(e) The ring $\mathcal{O}_{X,x}$ is a formally unramified $\mathcal{O}_{Y,y}$-algebra for the discrete topologies.

Proof. Since f is locally of finite type, the $\mathcal{O}_X$-module $\Omega^1_{X/Y}$ is of finite type (16.3.9); hence $(\Omega^1_{X/Y})_x = 0$ is equivalent to the existence of an open neighbourhood U of x such that $\Omega^1_{X/Y}|U = 0$. This proves the equivalence of a) and c). On the other hand, if $A = \mathcal{O}_{Y,y}$ and $B = \mathcal{O}_{X,x}$, then $(\Omega^1_{X/Y})_x = \Omega^1_{B/A}$ (16.4.15), and the equivalence of c) and e) follows from (**0**, 20.7.4).

Since d') involves only properties of the morphism $X_y \to \operatorname{Spec}(k(y))$, the equivalence of a) and d') will imply, *ipso facto*, the equivalence of d) and d'). Moreover, d') and d") are equivalent, since, X_y being a $k(y)$-prescheme locally of finite type, saying that $\mathcal{O}_{X_y,x}$ is a *finite* $k(y)$-algebra is equivalent to saying that x is an isolated point of X_y (**I**, 6.4.4).

We now prove the equivalence of b) and b'). We may restrict to the case where $Y = \operatorname{Spec}(R)$ and $X = \operatorname{Spec}(S)$ are affine and f is of finite presentation. Then $Z = \operatorname{Spec}(S \otimes_R S)$, and Δ_f corresponds to the canonical surjective homomorphism $S \otimes_R S \to S$, whose kernel $\mathfrak{J}$ is an ideal of finite type (**0**, 20.4.4). If $\mathscr{J} = \widetilde{\mathfrak{J}}$, the $\mathcal{O}_Z$-module $\psi_*(\mathcal{O}_X) = \mathcal{O}_Z/\mathscr{J}$ is therefore of finite presentation; the hypothesis that $\theta_z^\# : \mathcal{O}_{Z,z} \to (\psi_*(\mathcal{O}_X))_z = \mathcal{O}_{X,x}$ is bijective then implies that, after replacing X if necessary by a suitable open neighbourhood of x, the homomorphism $\theta : \mathcal{O}_Z \to \psi_*(\mathcal{O}_X)$ is itself bijective (**0**, 5.2.7). This shows that b') implies b); the converse is obvious.

On the other hand, the equivalence of b) and b") follows from (**I**, 5.3.7), without any finiteness hypothesis on f: giving a Y'-section $s' : Y' \to X'$ is equivalent to giving a Y-morphism $h = g' \circ s'$: IV | 64
$Y' \to X$ (where $g' : X' \to X$ is the canonical projection), so that $s' = (1_{Y'}, h)_X$, and then the diagram

(17.4.1.1)
$$\begin{tikzcd} Y' \arrow[r, "s'"] \arrow[d, "h"'] & X' = Y' \times_Y X \arrow[d, "h \times 1_X"] \\ X \arrow[r, "\Delta_f"'] & X \times_Y X \end{tikzcd}$$

identifies Y' with the $(X \times_Y X)$-prescheme product of X and X'. Hence ((**I**, 4.3.2)) if Δ_f is an isomorphism local at the point x, s' is an isomorphism local at the point y' (since $x = h(y')$), which proves that b) implies b''). The converse is obtained by applying b'') to the case $Y' = X$, $y' = x$, $g = f$, and $s' = \Delta_f$.

To finish the proof of (17.4.1), it suffices to prove the implications d'') $\Rightarrow c$) $\Rightarrow b$) $\Rightarrow d''$).

d'') $\Rightarrow c$): Since $\Omega^1_{X/Y}$ is an $\mathcal{O}_X$-module of finite type, Nakayama's lemma shows that condition c) is equivalent to $(\Omega^1_{X/Y})_x/\mathfrak{m}_y(\Omega^1_{X/Y})_x = 0$, that is, by (16.4.5), to $(\Omega^1_{X_y/\operatorname{Spec}(k(y))})_x = 0$. We are thus reduced to the case where Y is the spectrum of a field k and X is an algebraic k-prescheme. The hypothesis that $\mathcal{O}_{X,x}$ is a field k', finite extension of k, implies first that x is a *closed* point in X ((**I**, 6.4.2)), then that x is a *maximal* point, hence is an isolated point of X. Replacing X by the open $\{x\}$ of X, one may suppose $X = \operatorname{Spec}(k')$; but then the hypothesis that k' is a finite separable extension of k implies $\Omega^1_{k'/k} = 0$ (**0**, 20.6.20), which proves c).

$c) \Rightarrow b)$: We saw above that one then has $\Omega^1_{X/Y}|U = 0$ for a neighbourhood open U of x in X; the assertion $b)$ follows then from the definition of $\Omega^1_{X/Y}$ (16.3.1) and (16.1.9).

$b) \Rightarrow d'')$: Replacing X by an open neighbourhood of x, we may suppose that Δ_f is an open immersion. If $f_y : X_y \to \mathrm{Spec}(k(y))$ denotes the morphism obtained from f by base change, then Δ_{f_y} is also an open immersion (**I**, 5.3.4). Since condition d″) concerns only X_y, we may reduce to the case where Y is the spectrum of a field k and X is the spectrum of a k-algebra A of finite type. What must be proved is that A is a *finite separable* k-algebra, that is, a finite product of finite separable extensions of k. If K is an algebraically closed extension of k, saying this is equivalent to saying that $A \otimes_k K$ is a finite separable K-algebra (4.6.1); we may therefore suppose that k is algebraically closed. We first show that A is finite over k. It is enough to show that every closed point x of X is isolated: the set of such points will then be open and discrete in X, and hence finite because it is quasi-compact (X being Noetherian), which proves the assertion by (**I**, 6.4.4). But then $k(x) = k$, since k is algebraically closed (**I**, 6.4.3); consequently there is a Y-section s of X such that $s(Y) = \{x\}$. By (17.4.1.1), $\{x\}$ is the inverse image of the diagonal $\Delta_X(X)$ under a morphism $X \to X \times_Y X$, and therefore $\{x\}$ is open in X by hypothesis b). Thus we have shown that A is a *finite* k-algebra. Since a **IV** | 65
finite k-algebra is a finite product of finite local k-algebras, to express that Δ_f is an open immersion we may reduce to the case where A is a finite *local* k-algebra. The residue field of A, being a finite extension of k, is necessarily equal to k; hence (**I**, 3.4.9) the underlying space of $X \times_k X$ consists of a single point, and Δ_f is necessarily an *isomorphism*. But since A is a k-algebra, the canonical homomorphism $A \otimes_k A \to A$ can only be bijective if $A = k$. □

Remark (17.4.1.2). — Suppose only that f is *locally of finite type*. Then $\Omega^1_{X/Y}$ is still an $\mathscr{O}_X$-module of finite type (16.3.9), and Δ_f is an immersion locally of finite presentation (1.4.3.1). The entire proof of (17.4.1) remains valid, provided a) is replaced by: *the restriction of f to a suitable neighbourhood of x is a formally unramified morphism*. One sees moreover that in this case the restriction of f to a suitable neighbourhood of x is a morphism *locally quasi-finite*.

Corollary (17.4.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation. The following properties are equivalent:*

- *(a) f is unramified.*
- *(b) The diagonal morphism $\Delta_f : X \to X \times_Y X$ is an open immersion.*
- *b′) For every morphism $Y' \to Y$, every Y'-section of $X' = X \times_Y Y'$ is an open immersion.*
- *(c) $\Omega^1_{X/Y} = 0$.*
- *(d) For every $y \in Y$, the $k(y)$-prescheme $f^{-1}(y)$ is unramified over $k(y)$.*
- *d′) For every $y \in Y$, the $k(y)$-prescheme $f^{-1}(y)$ is isomorphic to a prescheme of the form $\coprod_{\lambda \in L} \mathrm{Spec}(K_\lambda)$, where each K_λ is a finite separable extension of $k(y)$.*
- *(e) For every $x \in X$, the ring $\mathscr{O}_{X,x}$ is a formally unramified $\mathscr{O}_{Y,f(x)}$-algebra for the discrete topologies.*

Corollary (17.4.3). — *If $f : X \to Y$ is unramified, then f is locally quasi-finite (**Err$_{\mathrm{III}}$**, 20).*

Proof. Indeed, by ((17.4.1), d')), for every $y \in Y$, the fibre $f^{-1}(y)$ is isomorphic to a prescheme of the form $\coprod_{\lambda \in L} \mathrm{Spec}(K_\lambda)$, where the K_λ are finite separable extensions of $k(y)$; hence $f^{-1}(y)$ is a discrete topological space. □

Proposition (17.4.4). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are local Noetherian rings, and let k be the residue field of A. Then the conditions equivalent $a)$ to $e)$ of theorem (17.4.1) are also equivalent to each of the following:*

- *f) $\widehat{B} \otimes_{\widehat{A}} k$ is a field, finite separable extension of k (which implies that $\widehat{B}$ is an $\widehat{A}$-algebra finite).*
- *f′) $\widehat{B}$ is a formally unramified $\widehat{A}$-algebra for the adic topologies.*

If moreover $k(x) = k(y)$, or if k is separably closed, these conditions are also equivalent to:

- *f″) The homomorphism $\widehat{A} \to \widehat{B}$ is surjective.*

Proof. First note, by the same argument as in (**0**, 19.3.6), that saying that B is a formally unramified A-algebra for the preadic topologies is equivalent to saying that $\widehat{B}$ is a formally unramified $\widehat{A}$-algebra for the adic topologies. On the other hand, the hypothesis that f is locally of finite type implies that $\Omega^1_{B/A}$ is a B-module of finite type (16.3.9), and hence is separated for the $\mathfrak{n}$-preadic topology (where $\mathfrak{n}$ is the maximal ideal of B) (**0**, 7.3.5); consequently, $\Omega^1_{B/A} = 0$ is equivalent to $\widehat{\Omega^1_{B/A}} = 0$. By (**0**, 20.7.4), it is therefore equivalent to say that B is a formally unramified A-algebra for the discrete topologies or for the preadic topologies. This proves the equivalence of e) and f′).

If $\mathfrak{m}$ is the maximal ideal of A, then **IV | 66**

$$k = A/\mathfrak{m} = \widehat{A}/\mathfrak{m}\widehat{A}, \qquad \widehat{B} \otimes_{\widehat{A}} k = \widehat{B}/\mathfrak{m}\widehat{B} = \widehat{B} \otimes_B (B/\mathfrak{m}B).$$

It follows from (**0**, 7.3.5) that $\widehat{B}/\mathfrak{m}\widehat{B}$ is the completion of $B/\mathfrak{m}B = B \otimes_A k$ for the $\mathfrak{n}$-preadic topology; this proves the equivalence of d″) and f). Finally, when $k(x) = k(y)$, or when k is separably closed, condition f) implies that the homomorphism $\widehat{A}/\mathfrak{m}\widehat{A} \to \widehat{B}/\mathfrak{m}\widehat{B}$ is bijective. Condition f) also implies that $\widehat{B}$ is a quasi-finite $\widehat{A}$-algebra (**0**, 7.4.4), hence a finite one, since $\widehat{A}$ is complete and $\widehat{B}$ is separated for the $\mathfrak{m}$-preadic topology, $\mathfrak{m}\widehat{B}$ being an ideal of definition of $\widehat{B}$ (**0**, 7.4.1). Nakayama's lemma therefore shows that $\widehat{A} \to \widehat{B}$ is surjective. Thus f) implies f″), and the converse is immediate. □

(17.4.5). Given an S-prescheme Y and two S-morphisms $f : X \to Y$, $g : X \to Y$, one canonically deduces an S-morphism $(f, g)_S : X \to Y \times_S Y$. We call the *prescheme of coincidences* of f and g the inverse image under $(f, g)_S$ of the diagonal $\Delta_{Y|S}$; it is therefore a subprescheme of X, and is a *closed* subprescheme when Y is an S-scheme (**I**, 5.4.1).

Proposition (17.4.6). — *Let $h : Y \to S$ be an unramified morphism, and let $f : X \to Y$, $g : X \to Y$ be two S-morphisms. Then the prescheme C of coincidences of f and g is a sub-prescheme induced on an open of X; if moreover Y is an S-scheme ((**I**, 5.4.1)), C is a closed sub-prescheme of X.*

Proof. Indeed, since $\Delta_h : Y \to Y \times_S Y$ is an open immersion (17.4.2), the inverse image by $(f, g)_S$ of $\Delta_{Y|S}$ is a sub-prescheme induced on an open of X ((**I**, 4.4.1)). The last assertion follows from (17.4.5). □

Corollary (17.4.7). — *Under the hypotheses of (17.4.6), let x be a point of X such that the two composed morphisms* $\operatorname{Spec}(k(x)) \to X \xrightarrow{f} Y$ *and* $\operatorname{Spec}(k(x)) \to X \xrightarrow{g} Y$ *are equal. Then there exists an open neighbourhood U of x such that $f|U = g|U$. If moreover Y is an S-scheme, there exists a closed open neighbourhood X' of x in X such that $f|X' = g|X'$. If finally X is connected, then $f = g$.*

This follows from (17.4.6) and ((**I**, 5.3.17)).

Corollary (17.4.8). — *Under the hypotheses of (17.4.6), suppose that the structural morphism $\varphi = h \circ f = h \circ g$ of X in S is closed. Let s be a point of S; let X_s be the $k(s)$-prescheme $\varphi^{-1}(s)$ and suppose that the two morphisms composed $X_s \to X \xrightarrow{f} Y$ and $X_s \to X \xrightarrow{g} Y$ are equal. Then there exists a neighbourhood open V of s in S such that $f|\varphi^{-1}(V) = g|\varphi^{-1}(V)$. If moreover Y is an S-scheme and φ is open, one can take V open and closed. If finally S is connected, $f = g$.*

Proof. It follows from (17.4.7) that the prescheme C of coincidences of f and g is induced on an open subset of X and contains X_s. Since φ is closed, there exists an open neighbourhood V of s such that $\varphi^{-1}(V) \subset C$. If moreover Y is an S-scheme, then C is closed; hence, when φ is also open, $\varphi(X - C)$ is both open and closed in S, and its complement V in S is therefore an open-and-closed neighbourhood of s such that $\varphi^{-1}(V) \subset C$. If finally S is connected, then $V = S$, and consequently $f = g$. □

Proposition (17.4.9). — *Let Y be a connected prescheme, $f : X \to Y$ an unramified and separated morphism. Then every Y-section g of X is an isomorphism of Y onto a connected component open of X, and the application $g \mapsto g(Y)$ is a bijection of $\Gamma(X/Y)$ onto the set of connected components Z of X (necessarily* open *in X) such that the restriction of f to Z is an isomorphism. In particular, if g' and g'' are two Y-sections of X such that $g'(y) = g''(y)$ for some $y \in Y$, then $g' = g''$.*

Proof. It follows from (17.4.1, b'')) that a Y-section s of X is also an open immersion, and since X is a Y-scheme, s is also a closed immersion ((**I**, 5.4.7)); it follows that s is an isomorphism of Y onto a sub-prescheme of X that is both open and closed, hence a connected component open of X, and since $s(Y)$ is connected, it is necessarily a connected component. The rest of the proposition is immediate. □

Remark (17.4.10). — In view of Remark (17.4.1.2), in statements (17.4.6) through (17.4.9) the word "unramified" may everywhere be replaced by "formally unramified and locally of finite type".

17.5. Characterizations of smooth morphisms. IV | 67

Theorem (17.5.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$. The following conditions are equivalent:*

(a) f is smooth at the point x.

(b) f is flat at the point x and the $k(y)$-prescheme $f^{-1}(y)$ is smooth on $k(y)$ at the point x (6.8.1).

b') f is regular at the point x (6.8.1).

(c) $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-algebra formally smooth for the discrete topologies.

Proof. One may reduce to the case where $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(C)$, where $C = B/\mathfrak{J}$, $B = A[T_1, \ldots, T_n]$ being a polynomial algebra and $\mathfrak{J}$ an ideal *of finite type*. The equivalence of a) and c) follows from (**0**, 22.6.4). On the other hand, applying this result to the morphism locally of finite type $f^{-1}(y) \to \operatorname{Spec}(k(y))$, one sees that the equivalence of b) and b') follows from the equivalence of a) and b) in (6.8.6). It remains to prove the equivalence of a) and b).

Let us first show that a) implies b). Denote by $\mathfrak{p}$ the prime ideal $\mathfrak{i}_x$ of C, and by $\mathfrak{r}$ the prime ideal $\mathfrak{i}_y$ of A; then $\mathfrak{p} = \mathfrak{q}/\mathfrak{J}$, where $\mathfrak{q}$ is a prime ideal of B, and $\mathfrak{r}$ is the inverse image of $\mathfrak{q}$ in A. Hypothesis a) first implies, by (17.3.3), that $f^{-1}(y)$ is smooth over $k(y)$ at x, and it remains to prove that $C_\mathfrak{p} = \mathscr{O}_{X,x}$ is a *flat* $A_\mathfrak{r}$-module. Since $C_\mathfrak{p}$ is a formally smooth $A_\mathfrak{r}$-algebra and B is a formally smooth A-algebra (for the discrete topologies), the Jacobian criterion (**0**, 22.6.4), together with (**0**, 19.1.12), implies that there exist in $\mathfrak{J}$ a system of r polynomials u_i ($1 \leqslant i \leqslant r$) and r indices j_i ($1 \leqslant i \leqslant r$) such that the images of the u_i in $\mathfrak{J}_\mathfrak{q}/\mathfrak{J}_\mathfrak{q}^2$ generate this $B_\mathfrak{q}$-module and

$$\det\left(\frac{\partial u_i}{\partial T_{j_k}}\right) \notin \mathfrak{q}. \tag{17.5.1.1}$$

Now let $g : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ be the structural morphism. The fibre $g^{-1}(y)$ is the spectrum of the regular ring $k(y)[T_1, \ldots, T_n]$ (**0**, 17.3.7); hence the Noetherian local ring $B_\mathfrak{q}/\mathfrak{r}B_\mathfrak{q}$ at a point of this fibre is regular. Condition (17.5.1.1) implies that the canonical images v_i of the u_i in the maximal ideal $\mathfrak{m}$ of $B_\mathfrak{q}/\mathfrak{r}B_\mathfrak{q}$ are *linearly independent* modulo $\mathfrak{m}^2$. Indeed, otherwise there would exist polynomials $w_i \in B$ ($1 \leqslant i \leqslant r$), not all belonging to $\mathfrak{q}$, such that $\sum_{i=1}^r w_i u_i \in \mathfrak{q}^2$. Differentiating with respect to the T_{j_k} would then give $\sum_{i=1}^r w_i(\partial u_i/\partial T_{j_k}) \in \mathfrak{q}$ for $1 \leqslant k \leqslant r$, contradicting (17.5.1.1), since $\mathfrak{q}$ is prime. It follows from (**0**, 17.1.7) that (v_i) is a *regular sequence* in $B_\mathfrak{q}/\mathfrak{r}B_\mathfrak{q}$. But since g is locally of finite presentation and B is a flat A-module, (11.3.8) shows that the canonical images u'_i of the u_i in $B_\mathfrak{q}$ also form a regular sequence and that $B_\mathfrak{q}/(\sum_i u'_i B_\mathfrak{q})$ is a flat $A_\mathfrak{r}$-module. Since the images of the u'_i in $\mathfrak{J}_\mathfrak{q}/\mathfrak{J}_\mathfrak{q}^2$ generate this $B_\mathfrak{q}$-module, Nakayama's lemma gives $\sum_i u'_i B_\mathfrak{q} = \mathfrak{J}_\mathfrak{q}$; hence $C_\mathfrak{p} = B_\mathfrak{q}/\mathfrak{J}_\mathfrak{q}$ is indeed a flat $A_\mathfrak{r}$-module. IV | 68

Finally, let us prove that b) implies a). With the same notation, the hypothesis that $B_\mathfrak{q}/\mathfrak{J}_\mathfrak{q} = C_\mathfrak{p}$ is a flat $A_\mathfrak{r}$-module implies that the canonical homomorphism $\mathfrak{J}_\mathfrak{q}/\mathfrak{r}\mathfrak{J}_\mathfrak{q} \to B_\mathfrak{q}/\mathfrak{r}B_\mathfrak{q}$ is injective ($\mathbf{0_I}$, 6.1.2), so that $\mathfrak{J}_\mathfrak{q}/\mathfrak{r}\mathfrak{J}_\mathfrak{q}$ is identified with an ideal of $B_\mathfrak{q}/\mathfrak{r}B_\mathfrak{q}$. Since $B_\mathfrak{q}/\mathfrak{r}B_\mathfrak{q}$ is a formally smooth $A_\mathfrak{r}$-algebra for the discrete topologies, the Jacobian criterion (**0**, 22.6.4), applied to

$$C_\mathfrak{p} = (B_\mathfrak{q}/\mathfrak{r}B_\mathfrak{q})/(\mathfrak{J}_\mathfrak{q}/\mathfrak{r}\mathfrak{J}_\mathfrak{q}),$$

together with (**0**, 19.1.12), proves, by hypothesis b), the existence of r polynomials $v_i \in k(y)[T_1, \ldots, T_n]$ and r indices j_i ($1 \leqslant i \leqslant r$) such that their images in

$$(\mathfrak{J}_\mathfrak{q}/\mathfrak{r}\mathfrak{J}_\mathfrak{q})/(\mathfrak{J}_\mathfrak{q}/\mathfrak{r}\mathfrak{J}_\mathfrak{q})^2$$

generate this $(B_\mathfrak{q}/\mathfrak{r}B_\mathfrak{q})$-module and

$$\det\left(\frac{\partial v_i}{\partial T_{j_k}}\right) \notin \mathfrak{q}B_\mathfrak{q}/\mathfrak{r}B_\mathfrak{q}. \tag{17.5.1.2}$$

For each i, choose an element $u_i \in \mathfrak{J}$ whose canonical image is v_i. It follows from (17.5.1.2) that the u_i satisfy (17.5.1.1); on the other hand, Nakayama's lemma shows that the images u'_i of the u_i in $\mathfrak{J}_\mathfrak{q}$ generate this $B_\mathfrak{q}$-module. The Jacobian criterion (**0**, 22.6.4), together with (**0**, 19.1.12), then proves that $C_\mathfrak{p} = B_\mathfrak{q}/\mathfrak{J}_\mathfrak{q}$ is a formally smooth $A_\mathfrak{r}$-algebra for the discrete topologies. Q.E.D. □

Corollary (17.5.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be smooth (in the sense of (17.3.1)), it is necessary and sufficient that f be regular (6.8.1), i.e. that f be flat and that for every $y \in Y$, $f^{-1}(y)$ be a $k(y)$-prescheme geometrically regular (6.7.6).*

One has thus established the equivalence of the two definitions of "smooth morphism" given in (6.8.1) and (17.3.1).

Proposition (17.5.3). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are local Noetherian rings. Then the conditions equivalent a) to c) of (17.5.1) are also equivalent to each of the following:*

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d) B is an A-algebra formally smooth for the preadic topologies.
d′) $\widehat{B}$ is an $\widehat{A}$-algebra formally smooth for the adic topologies.

If moreover $k(x) = k(y)$, these conditions are also equivalent to:

d″) $\widehat{B}$ is an $\widehat{A}$-algebra isomorphic to a formal power series algebra $\widehat{A}[[T_1, \dots, T_n]]$.

Proof. The equivalence of condition c) of (17.5.1) and condition d) follows from the equivalence of a) and d) in the Jacobian criterion (**0**, 22.6.4), and the equivalence of d) and d') follows from (**0**, 19.3.6). Moreover, d'') implies d') without any hypothesis on the residue fields (**0**, 19.3.4). Finally, if $\mathfrak{m}$ denotes the maximal ideal of $\widehat{A}$, hypothesis d') implies that $\widehat{B}/\mathfrak{m}\widehat{B}$ is a complete Noetherian local $k(y)$-algebra, formally smooth for its adic topology (**0**, 19.3.5); the hypothesis $k(y) = k(x)$ therefore implies that $\widehat{B}/\mathfrak{m}\widehat{B}$ is $k(y)$-isomorphic to a formal power series algebra $k(y)[[T_1, \dots, T_n]]$ (**0**, 19.6.4). On the other hand, $\widehat{A}[[T_1, \dots, T_n]]$ is a flat $\widehat{A}$-module and a complete Noetherian local $\widehat{A}$-algebra; it follows from (**0**, 19.7.1.5) that this algebra is isomorphic to $\widehat{B}$. Thus d') implies d'') under the additional hypothesis $k(x) = k(y)$. □

Remark (17.5.4). — Suppose that Y is a locally Noetherian prescheme, and $f : X \to Y$ a morphism locally of finite type. The criterion (17.5.3, d)), together with (**0**, 22.1.4) shows that for proving f is smooth, one can apply the definition (17.1.1), *restricting* to the case where the affine scheme Y' is the spectrum of a local Artinian ring.

Proposition (17.5.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that Y is reduced at the point y. Then, for f to be smooth at the point x, it is necessary and sufficient that f be universally open in a neighbourhood of x in $f^{-1}(y)$ and that $f^{-1}(y)$ be a $k(y)$-prescheme geometrically regular at the point x.*

Proof. By (17.5.1), everything reduces to seeing that, if $f^{-1}(y)$ is a $k(y)$-prescheme geometrically regular at the point x, it is equivalent to say that f is flat or universally open in a neighbourhood of x in $f^{-1}(y)$. But if f is flat at the point x, it is in a neighbourhood of x in X (11.1.1) and hence universally open in this neighbourhood (2.4.6). Reciprocally, the hypothesis that f is universally open in a neighbourhood of x in $f^{-1}(y)$ and that $f^{-1}(y)$ is a $k(y)$-prescheme geometrically regular at the point x implies that f is flat at the point x, since $\mathscr{O}_{Y,y}$ is reduced (15.2.2). □

Corollary (17.5.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that Y is reduced and geometrically unibranch (6.15.1) at the point y. Then, for f to be smooth at the point x, it is necessary and sufficient that f be equidimensional at the point x and that $f^{-1}(y)$ be a $k(y)$-prescheme geometrically regular at the point x.*

Observing that the set of points at which f is equidimensional is open (13.3.2), one sees that the corollary follows from (17.5.5) and the Chevalley criterion (14.4.4).

Proposition (17.5.7). — *Let $f : X \to Y$ be a morphism locally of finite presentation, smooth at a point $x \in X$; set $y = f(x)$. Then, for $\mathscr{O}_{X,x}$ to be reduced (resp. integrally closed, resp. geometrically unibranch), it is necessary and sufficient that $\mathscr{O}_{Y,y}$ be so.*

This has been proved in (11.3.13) and (11.3.14) completed by $\mathbf{Err_{IV}}$, 30.

Proposition (17.5.8). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, smooth at a point $x \in X$; set $y = f(x)$. Then:*

(i) $\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{Y,y}) + \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_{Y,y}} k(y))$.

(ii) $\operatorname{coprof}(\mathscr{O}_{X,x}) = \operatorname{coprof}(\mathscr{O}_{Y,y})$.

(iii) For $\mathscr{O}_{X,x}$ to have the property (S_n) (5.7.2) (resp. (R_n) (5.8.2)), it is necessary and sufficient that $\mathscr{O}_{Y,y}$ have it. In particular, for $\mathscr{O}_{X,x}$ to be regular, it is necessary and sufficient that $\mathscr{O}_{Y,y}$ be so.

These are special cases of (6.1.2), (6.3.2), (6.4.1) and (6.5.3).

17.6. Characterizations of étale morphisms. IV | 70

Theorem (17.6.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$. The following conditions are equivalent:*

a) f is étale at the point x.

a′) f is smooth at the point x and non ramified at the point x.

*b) f is smooth at the point x and locally quasi-finite (**Err$_{\mathbf{III}}$**, 20).*

c) f is flat at the point x and non ramified at the point x.

c′) f is flat at the point x and the ring $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is a field, extension finite separable of $k(y)$.

d) $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-algebra formally étale for the discrete topologies.

Proof. The equivalence of $a)$ and $a')$ follows immediately from the definitions; that of $a)$ and $d)$ follows from the equivalence of $a)$ and $c)$ in (17.4.1) and the equivalence of $a)$ and $d)$ in (17.5.1). The equivalence of $c)$ and $c')$ follows from the equivalence of $a)$ and $d')$ in (17.4.1). The fact that $a')$ implies $c')$ follows from (17.5.1); conversely, if $c')$ holds, then f is regular (hence smooth by (17.5.1)) at x, because if K is a finite separable extension of a field k, then, for every extension k' of k, $\operatorname{Spec}(K \otimes_k k')$ is regular, being a finite sum of spectra of fields. The fact that $a')$ implies $b)$ follows from (17.4.1, $d')$) and (17.5.1, $b)$). It remains to show that $b)$ implies $c)$; since one already knows from (17.5.1) that f is flat at x, it is enough to prove that $f^{-1}(y)$ is a $k(y)$-prescheme unramified over $k(y)$. In other words, one is reduced to proving that $b)$ implies $c)$ when $Y = \operatorname{Spec}(k)$ is the spectrum of a field k and $X = \operatorname{Spec}(A)$, where A is a finite local k-algebra ($\mathbf{0_I}$, 7.4.1). By hypothesis $b)$, A is a formally smooth k-algebra for the discrete topologies, which here coincide with the preadic topologies; hence ($\mathbf{0}$, 19.6.5) A is a regular local ring.

It is therefore a field since it is Artinian. It then follows from ($\mathbf{0}$, 19.6.5.1) that A is a finite separable extension of k. ☐ IV | 71

Corollary (17.6.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation. The following conditions are equivalent:*

a) f is étale.

a′) f is smooth and non ramified.

*b) f is smooth and locally quasi-finite (**Err$_{\mathbf{III}}$**, 20).*

c) f is flat and non ramified.

c′) f is flat, and every fibre $f^{-1}(y)$ is a $k(y)$-scheme sum of spectra of fields, extensions finite and separable of $k(y)$.

c″) f is flat, and for every $y \in Y$ and every algebraically closed extension k' of $k(y)$, the "geometric fibre" $f^{-1}(y) \otimes_{k(y)} k'$ is a sum of spectra of fields isomorphic to k'.

Proof. The only point left to prove is the equivalence of $c')$ and $c'')$. It is clear from base change (17.3.3) that $c')$ implies $c'')$. Conversely, since the projection morphism

$$f^{-1}(y) \otimes_{k(y)} k' \longrightarrow f^{-1}(y)$$

is open (2.4.10), hypothesis $c'')$ implies that the space $f^{-1}(y)$ is discrete. Thus, for every point $x \in X_y = f^{-1}(y)$, the local ring $\mathscr{O}_{X_y,x} = A$ is a finite $k(y)$-algebra, hence an Artinian local ring. Moreover, it follows from $c'')$ that $\operatorname{Spec}(A \otimes_{k(y)} k')$ is a sum of spectra of fields isomorphic to k', which is possible only if A is a field that is a finite separable extension of $k(y)$ (4.6.1). ☐

Proposition (17.6.3). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are local Noetherian rings, and let k be the residue field of A. Then the conditions equivalent $a)$ to $d)$ of (17.6.1) are also equivalent to each of the following:*

e) $\widehat{B}$ *is an* $\widehat{A}$*-algebra formally étale for the adic topologies.*

e′) $\widehat{B}$ *is an* $\widehat{A}$*-module free and* $\widehat{B} \otimes_{\widehat{A}} k$ *is a field, extension finite and separable of* k *(which implies that* $\widehat{B}$ *is an* $\widehat{A}$*-algebra finite).*

If moreover $k(x) = k(y)$*, or if* k *is separably closed, these conditions are also equivalent to:*

e″) The canonical homomorphism $\widehat{A} \to \widehat{B}$ *is bijective.*

Proof. The equivalence of e) with each of the conditions of (17.6.1) follows at once from (17.4.4, f')) and (17.5.3, d')). That e) implies e') follows from (17.4.4, f)) and (**0**, 19.7.1), since $\widehat{B}$ is then a finite $\widehat{A}$-algebra (17.4.4), and saying that it is a flat $\widehat{A}$-module is equivalent to saying that it is a free $\widehat{A}$-module ($\mathbf{0_{III}}$, 10.1.3). Conversely, the implication e') $\Rightarrow$ e) follows from (17.4.4) and (**0**, 19.7.1). Finally, e') implies that the homomorphism $\widehat{A} \to \widehat{B}$ is injective; if $k(x) = k(y)$, or if k is separably closed, this homomorphism is surjective by (17.4.4). The converse is immediate. □

Proposition (17.6.4). — *Under the hypotheses of (17.6.3), if* f *is étale at the point* x*, then* $\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{Y,y})$.

This is a particular case of (17.5.8, (i)) since x is isolated in its fibre $f^{-1}(y)$.

17.7. Properties of descent, passage to the limit, and constructibility. IV | 72

Proposition (17.7.1). — *Let* $f : X \to Y$ *be a morphism locally of finite presentation,* $g : Y' \to Y$ *a morphism,* $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$ *and* $g' : X' \to X$ *the canonical projections. Let* x' *be a point of* X' *and set* $x = g'(x')$, $y' = f'(x')$.

(i) If f' *is non ramified at the point* x'*, then* f *is non ramified at the point* x*.*

(ii) Suppose moreover that g *is flat at the point* y'*. Then, if* f' *is smooth (resp. étale) at the point* x'*,* f *is smooth (resp. étale) at the point* x*.*

Proof. Set $y = f(x) = g(y')$, so that $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$; note that f' is locally of finite presentation.

(i) Since the property for a morphism (locally of finite presentation) of being non ramified at a point only involves the fibre of the morphism at that point (17.4.1, d)), one may reduce to the case where Y and Y' are spectra of fields. But then it amounts to the same thing to say that f (resp. f') is non ramified at x (resp. x') or that it is étale at that point (17.6.1, c)); thus (i) is a consequence of (ii).

(ii) Since f' is flat at x' (17.5.1), the hypothesis that g is flat at y' implies that f is flat at x: indeed, the projection $g' : X' \to X$ is flat at x', and $f \circ g' = g \circ f'$ is flat at x', whence the conclusion (2.2.11, (iv)). Whether f is smooth (resp. étale) at x then only involves the fibre $f^{-1}(y)$ (17.5.1 and 17.6.1), and one is again reduced to the case where $Y = \mathrm{Spec}(k)$ and $Y' = \mathrm{Spec}(k')$ are spectra of fields. To say that f' is smooth at x' then means (17.5.1) that X' is geometrically regular over k' at x', and this implies (6.7.8) that X is geometrically regular over k at x; hence f is smooth at x. Suppose moreover that f' is étale at x', so that x' is isolated in $f'^{-1}(y')$ (17.6.1). Since the projection $f'^{-1}(y') \to f^{-1}(y)$ is an open morphism (2.4.10), x is isolated in $f^{-1}(y)$; as f is already known to be smooth at x, it is étale at that point (17.6.1). □

Corollary (17.7.2). — *(i) With the notation of (17.7.1), let* U *(resp.* U'*) be the set of points of* X *(resp.* X'*) where* f *(resp.* f'*) is non ramified; then* $U' = g'^{-1}(U)$*.*

(ii) Suppose moreover that g *is flat, and let* V *(resp.* V'*) be the set of points of* X *(resp.* X'*) where* f *(resp.* f'*) is smooth; then* $V' = g'^{-1}(V)$*.*

Corollary (17.7.3). — *(i) Suppose* g *surjective; then, for* f *to be non ramified, it is necessary and sufficient that* f' *be so.*

(ii) Let $f : X \to Y$ *be an* S*-morphism and* $g : S' \to S$ *a faithfully flat morphism. Suppose that* f *is locally of finite presentation, or that* g *is quasi-compact. Then, for* f *to be smooth (resp. non ramified, resp. étale), it is necessary and sufficient that* f' *be so.*

In (ii), the case where f is locally of finite presentation follows from (17.7.1). If g is quasi-compact and f' is smooth (resp. non ramified, resp. étale), then f' is locally of finite presentation by (2.7.1, (iv)), and one is reduced to the first case.

Proposition (17.7.4). — *Let $f : X \to Y$ be a morphism, $g : Y' \to Y$ a morphism flat and locally of finite presentation, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$ and $g' : X' \to X$ the canonical projections.*

Let V (resp. V') be the set of $x \in X$ (resp. $x' \in X'$) where f (resp. f') has one of the following properties: IV | 73

(i) locally of finite type;
(ii) locally of finite presentation;
(iii) flat;
(iv) non ramified;
(v) smooth;
(vi) étale.

Then $V' = g'^{-1}(V)$ (in other words, for f' to have the property considered at a point x', it is necessary and sufficient that f have this property at x).

Proof. Property (iii) is immediate and does not even require the hypothesis that g be locally of finite presentation (2.2.11, (iv)), since the projection $g' : X' \to X$ is flat. By (17.7.2), the assertions concerning properties (iv), (v), and (vi) are consequences of the assertion concerning property (ii). It is therefore enough to consider cases (i) and (ii). It is clear that $V' \supset g'^{-1}(V)$; it remains to prove the reverse inclusion. The sets V and V' are open, and $W = g'(V')$ is also open in X by (2.4.6). It is thus enough to prove that the morphism $f|W : W \to Y$ is locally of finite type (resp. locally of finite presentation). By hypothesis, the composite

$$V' \xrightarrow{g''} W \xrightarrow{f|W} Y,$$

where g'' is the restriction of g', is locally of finite type (resp. locally of finite presentation), because it is equal to the composite $V' \xrightarrow{f'} Y' \xrightarrow{g} Y$; moreover, g'' is surjective, hence faithfully flat. The assertion is therefore reduced to the following lemma, which strengthens (11.3.16). □

Lemma (17.7.5). — *Let $f : X \to Y$ be a faithfully flat morphism and locally of finite presentation, $g : Y \to Z$ a morphism such that $g \circ f : X \to Z$ has one of the following properties: being*

(i) locally of finite type;
(ii) locally of finite presentation;
(iii) of finite type.

Then g has the same property. If moreover f is quasi-compact or g quasi-separated, the same conclusion holds for the property:

(iv) of finite presentation.

Proof. In cases (i) and (ii), one must show that, for every $y \in Y$, there are an affine open neighbourhood V of y in Y and an affine open neighbourhood W of $z = g(y)$ in Z, containing $g(V)$, such that the morphism $V \to W$ induced by g is of finite type (resp. of finite presentation). By hypothesis, there exist a point $x \in X$ such that $f(x) = y$, an affine open neighbourhood U of x in X, an affine open neighbourhood V' of y in Y containing $f(U)$, and an affine open neighbourhood W of z in Z containing $g(V')$, such that the morphism $f_1 : U \to V'$ induced by f is flat and of finite presentation, while the composite of f_1 with the restriction $g_1 : V' \to W$ of g is of finite type (resp. of finite presentation). Then $f(U)$ is open in Y (2.4.6); if $V \subset f(U)$ is an affine neighbourhood of y, the morphism

$$f_2 : f_1^{-1}(V) \longrightarrow V,$$

obtained by restricting f_1, is still of finite presentation and is moreover faithfully flat. Furthermore, if $g_2 = g_1|V$, then $g_2 \circ f_2$ is of finite type (resp. of finite presentation), while the open set $f_1^{-1}(V)$ is quasi-compact (resp. quasi-compact and quasi-separated). One is thus reduced to the hypotheses of (11.3.16), which proves that g_2 is of finite type (resp. of finite presentation). IV | 74

In case (iii), the question is local on Z, so one may suppose Z affine; then X is quasi-compact, and so is $Y = f(X)$. Case (iii) is therefore a consequence of case (i).

In case (iv), under the additional hypotheses on f or g, one may again suppose Z affine, hence X and Y quasi-compact. Moreover, g is already known to be locally of finite presentation and quasi-compact (I.1.3); thus everything reduces to showing that g is quasi-separated, and it is enough to prove this when f is quasi-compact. But since $g \circ f$ is quasi-separated, the same is true of f

(I.2.2, (v)); as f is quasi-compact and locally of finite presentation, it is of finite presentation (I.6.1). It then suffices to repeat the argument of the first paragraph of the proof of (11.3.16). □

Remark (17.7.6). — Regarding the assertion relative to property (iv), one cannot suppress the hypothesis that f is quasi-compact; without this, it would imply that every morphism $g : Y \to Z$ quasi-compact and locally of finite presentation would be of finite presentation, a conclusion known to be erroneous (**I**, 6.4). Indeed, one can only restrict to the case where Z is affine, hence Y quasi-compact; there is then a finite covering of Y by affine opens U_i such that the restrictions $g|U_i$ are of finite presentation; it would then suffice to take for X the prescheme *sum* of the U_i, for $f : X \to Y$ the canonical morphism, which is evidently faithfully flat and locally of finite presentation (**I**, 6.5); $g \circ f$ would be of finite presentation by virtue of the choice of the U_i and of (**I**, 6.5), whence our assertion.

Proposition (17.7.7). — *Let $f : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$. Suppose that g is flat at the point x. Then, if h is smooth (resp. non ramified, resp. étale) at the point x, f is smooth (resp. non ramified, resp. étale) at the point y.*

Proof. Set $s = h(x) = f(y)$. To say that h is non ramified at x (resp. that f is non ramified at y) amounts to saying that $h^{-1}(s)$ is étale over $k(s)$ at x (resp. that $f^{-1}(s)$ is étale over $k(s)$ at y) (17.4.1 and 17.6.1); since the morphism $g_s : h^{-1}(s) \to f^{-1}(s)$ induced by g is flat at x, one may restrict to proving the proposition when h is smooth or étale at x. Moreover, since h is then flat at x (17.5.1), f is flat at y by (2.2.11, (iv)). It therefore remains to say that $f^{-1}(s)$ is smooth (resp. étale) over $k(s)$ at y, and one is thus reduced to the case where $S = \operatorname{Spec}(k)$ is the spectrum of a field.

(i) Case of smooth morphisms. Since g is locally of finite presentation (I.4.3, (v)), there is an open neighbourhood U of x in X on which g is flat (11.3.1) and h is smooth. Moreover, $g(U)$ is an open neighbourhood of y in Y (2.4.6); replacing X by U and Y by $g(U)$, one may therefore suppose that g is faithfully flat and that h is smooth, and one is reduced to proving that f is smooth. If k' is an algebraically closed extension of k, then $X \otimes_k k'$ is smooth over k'; by (17.7.3, (ii)), it is enough to prove that $Y \otimes_k k'$ is smooth over k' (since $\operatorname{Spec}(k') \to \operatorname{Spec}(k)$ is faithfully flat). Thus one may suppose that k is algebraically closed. The set of k-rational points of Y is then very dense in Y (10.4.8), and the locus where Y is smooth over k is open; it is therefore enough to prove that Y is smooth over k at every k-rational point y. At such a point, saying that Y is smooth over k is equivalent to saying that Y is regular at y (17.5.1 and 6.7.8). Now $y = g(x)$ for some $x \in X$, and by hypothesis X is regular at x (17.5.1); since X and Y are locally Noetherian and g is flat, Y is indeed regular at y (6.5.1, (i)).

(ii) Case of étale morphisms. By (i), one already knows that f is smooth at y; by (17.6.1), it is therefore enough to prove that f is quasi-finite at y, or equivalently that $\mathscr{O}_{Y,y}$ is a finite k-algebra. Since $\mathscr{O}_{X,x}$ is a faithfully flat $\mathscr{O}_{Y,y}$-module ($\mathbf{0}_{\mathbf{I}}$, 6.6.2), $\mathscr{O}_{Y,y}$ is identified with a k-submodule of $\mathscr{O}_{X,x}$; and since $\mathscr{O}_{X,x}$ is, by hypothesis, a finite k-algebra, the same holds for $\mathscr{O}_{Y,y}$. □ **IV** | 75

Proposition (17.7.8). — *With the notation of (8.8.1), suppose that X_α and Y_α are locally of finite presentation over S_α. Let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism and $f : X \to Y$ the corresponding S-morphism.*

(i) Let x be a point of X, x_λ its canonical projection in X_λ. For f to be smooth (resp. non ramified, resp. étale) at x, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ is smooth (resp. non ramified, resp. étale) at x_λ.

(ii) Suppose moreover that X_α is quasi-compact. For f to be smooth (resp. non ramified, resp. étale), it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ is smooth (resp. non ramified, resp. étale).

Proof. (i) If $y = f(x)$ and $y_\lambda = f_\lambda(x_\lambda)$ is the canonical projection of y in Y_λ, then

$$f^{-1}(y) = f_\lambda^{-1}(y_\lambda) \otimes_{k(y_\lambda)} k(y).$$

The assertion concerning non-ramified morphisms therefore follows from (17.7.1, (i)), and it is enough to consider smooth morphisms. Since f and f_λ are locally of finite presentation, (17.5.1) says that it amounts to the same thing to require that $f^{-1}(y)$ be geometrically regular at x or that $f_\lambda^{-1}(y_\lambda)$ be geometrically regular at x_λ (6.7.8). The proposition then follows from (17.5.1) and (11.2.6).

(ii) For every λ, let U_λ be the open set of points $x_\lambda \in X_\lambda$ at which f_λ is smooth (resp. non ramified, resp. étale), and let V_λ be its inverse image in X. By (i), for every $x \in X$ there exists a λ such that f_λ has the required property at x_λ; hence X is the union of the V_λ. Moreover, by (17.3.3), $V_\lambda \subset V_\mu$ for $\lambda \leqslant \mu$; since X is quasi-compact, there is therefore an index μ such that $X = V_\mu$. Since the X_λ are quasi-compact, it follows from (8.3.4) that there exists an index $\nu \geqslant \mu$ such that the inverse image of U_μ in X_ν is all of X_ν; this means that f_ν is smooth (resp. non ramified, resp. étale), by (17.3.3). □

Corollary (17.7.9). — *Let $S = \operatorname{Spec}(A)$ be an affine scheme, $f : X \to S$ a morphism. The following conditions are equivalent:*

a) f is a morphism of finite presentation and smooth (resp. non ramified, resp. étale).
b) There exists an affine Noetherian scheme $S_0 = \operatorname{Spec}(A_0)$, a morphism of finite type $f_0 : X_0 \to S_0$ and a morphism $S \to S_0$ such that the S-prescheme $X_0 \otimes_{S_0} S$ is S-isomorphic to X and f_0 is smooth (resp. non ramified, resp. étale).
c) The conditions of b) are satisfied, and moreover A_0 is a sub-$\mathbf{Z}$-algebra of finite type of A, and the morphism $S \to S_0$ corresponds to the canonical injection $A_0 \to A$.

Proof. The proof from (17.7.8) is the same as that of (11.2.7) from (11.2.6). □

Proposition (17.7.10). — *Let $f : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$. Suppose that h is flat at the point x and that f is non ramified at the point y. Then f is étale at the point y and g is flat at the point x.* **IV | 76**

Proof. (One will see further on (18.4.9) that one can in fact dispense with the hypothesis that h is locally of finite presentation.)

The question being local on S, X, and Y, one may suppose that S, X, and Y are affine, that f, g, and h are of finite presentation, that h is flat, and that f is non ramified. In view of (11.2.7) and (17.7.9), one may moreover suppose that S, X, and Y are Noetherian; finally, one may restrict to the case $S = \operatorname{Spec}(\mathscr{O}_{S,s})$, where $s = f(y) = h(x)$. Since $A = \mathscr{O}_{S,s}$ is a Noetherian local ring, there exist a complete Noetherian local ring B with algebraically closed residue field and a local homomorphism $A \to B$ making B a faithfully flat A-module ($\mathbf{0_{III}}$, 10.3.1). Replacing X and Y by $X \times_S \operatorname{Spec}(B)$ and $Y \times_S \operatorname{Spec}(B)$, one concludes from (2.5.1) and (17.7.1) that one may restrict to the case where A is complete and has algebraically closed residue field. The hypothesis that f is non ramified at y then implies (17.4.4) that $\mathscr{O}_{Y,y}$ is a finite $\mathscr{O}_{S,s}$-algebra and a complete Noetherian local ring ($\mathbf{0_I}$, 7.4.2); since the residue field of $\mathscr{O}_{S,s}$ is algebraically closed, the homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y}$ is surjective (17.4.4). On the other hand, the hypothesis that h is flat at x implies that the composite homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective ($\mathbf{0_I}$, 6.5.1); consequently $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y}$ is bijective, which proves that f is étale at y (17.6.3). Furthermore, $\mathscr{O}_{X,x}$ is a flat $\mathscr{O}_{Y,y}$-module, so g is flat at x. □

Proposition (17.7.11). — *Let S be a prescheme and X, Y two S-preschemes locally of finite presentation over S; let $f : X \to Y$ be an S-morphism. For every $s \in S$, denote X_s, Y_s, f_s the preschemes and morphism deduced from X, Y, f by the change of base $\operatorname{Spec}(k(s)) \to S$. Then:*

(i) The set of $x \in X$ such that, if s is the image of x in S, $f_s : X_s \to Y_s$ is smooth (resp. non ramified, resp. étale, resp. differentially smooth) at x, is locally constructible.
(ii) Suppose moreover that f is of finite presentation. The set of $y \in Y$ such that, if s is the image of y in S, f_s is smooth (resp. non ramified, resp. étale, resp. differentially smooth) at all the points of $f_s^{-1}(y)$, is locally constructible.
(iii) Suppose X and Y of finite presentation on S. The set of $s \in S$ such that f_s is smooth (resp. non ramified, resp. étale, resp. differentially smooth) is locally constructible.

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Proof. Let E be the set of $x \in X$ satisfying the property considered in (i). The set F of points $y \in Y$ satisfying the corresponding property in (ii) is none other than $Y - f(X - E)$; hence (ii) follows from (i) and Chevalley's theorem when f is of finite presentation (1.8.4). Likewise, if $h : Y \to S$ is the structural morphism, the set of $s \in S$ satisfying the corresponding property in (iii) is $S - h(Y - F)$;

thus (iii) again follows from (i) and Chevalley's theorem when f and h are of finite presentation. It therefore remains to prove the assertions in (i).

Let us first prove (i) for the property of being smooth. The question is local on X, so one may restrict to the case where $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and $Y = \mathrm{Spec}(C)$ are affine, with B and C algebras of finite presentation. Reasoning as at the beginning of (9.9.1) and using (17.7.2, (ii)), one is reduced to the case where A is Noetherian. By ($\mathbf{0_{III}}$, 9.2.3), it is enough to show that, if $x \in E$ (resp. $x \notin E$), there is a neighbourhood V of x in $\overline{\{x\}}$ contained in E (resp. in $X - E$).

Denote by $g : X \to S$ and $h : Y \to S$ the structural morphisms. One may first replace S by the reduced sub-prescheme S' having $\overline{\{g(x)\}}$ as underlying space, X by $X' = g^{-1}(S')$, and Y by $Y' = h^{-1}(S')$; the fibres of X and X' (resp. of Y and Y') at the points of S' are the same. Thus one may restrict to the case where S is integral and where $\eta = g(x) = h(y)$, with $y = f(x)$, is its generic point.

1° Suppose first that $x \in E$. The local rings $\mathscr{O}_{X_\eta,x}$ and $\mathscr{O}_{Y_\eta,y}$ are respectively equal to $\mathscr{O}_{X,x}$ and $\mathscr{O}_{Y,y}$. Since smoothness of a morphism of finite presentation at a point depends only on the local ring at that point and the local ring at its image (17.5.1), the hypothesis $x \in E$ means that f is smooth at x. It then has this property at every point of an open neighbourhood of x in X, and (17.3.3, (iii)) yields the conclusion.

2° Suppose next that $x \in X - E$ and that f_η is not flat at x. The conclusion follows from the following lemma, which sharpens (11.2.8):

Lemma (17.7.11.1). — *Let $g : X \to S$ and $h : Y \to S$ be two morphisms locally of finite presentation, $f : X \to Y$ an S-morphism, and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. Then the set E of $x \in X$ such that $\mathscr{F}_{g(x)}$ is $f_{g(x)}$-flat at x is locally constructible.*

Proof of Lemma 17.7.11.1. Reasoning again as at the beginning of (9.9.1) and using (2.5.1), one is reduced to the case where S, X, and Y are Noetherian. Then, as above, one reduces to the case where S is integral, where $\eta = g(x) = h(f(x))$ is its generic point, and where it is enough to prove that, if $x \in E$ (resp. $x \notin E$), there is a neighbourhood V of x in $\overline{\{x\}}$ contained in E (resp. in $X - E$). The case $x \in E$ is an immediate consequence of (11.1.1). To treat the case $x \notin E$, one reasons as in (9.4.7.1): after possibly replacing Y and X by neighbourhoods of $f(x)$ and x, respectively, there exist two coherent $\mathscr{O}_Y$-modules $\mathscr{G}$ and $\mathscr{H}$ and an $\mathscr{O}_Y$-homomorphism $u : \mathscr{G} \to \mathscr{H}$ such that, for every $s \in S$, the homomorphism $u_s : \mathscr{G}_s \to \mathscr{H}_s$ is injective, whereas

$$1 \otimes u_\eta : \mathscr{F} \otimes_{\mathscr{O}_{Y_\eta}} \mathscr{G}_\eta \longrightarrow \mathscr{F} \otimes_{\mathscr{O}_{Y_\eta}} \mathscr{H}_\eta$$

is not injective at x; equivalently, $x \in \mathrm{Supp}(\mathrm{Ker}(1 \otimes u_\eta))$. If $T = \overline{\{x\}}$, then $\mathrm{Supp}(\mathrm{Ker}(1 \otimes u_\eta)) \supset T_\eta$. One may suppose that, for every $s \in S$, base change gives

$$\mathrm{Ker}(1 \otimes u_s) = (\mathrm{Ker}(1 \otimes u))_s$$

(9.4.2), and hence

$$\mathrm{Supp}(\mathrm{Ker}(1 \otimes u_s)) = (\mathrm{Supp}(\mathrm{Ker}(1 \otimes u)))_s$$

(I.9.1.13.1). It follows from (9.5.2) that, for s in a neighbourhood of η, one has $\mathrm{Supp}(\mathrm{Ker}(1 \otimes u_s)) \supset T_s$, which proves the lemma. □

3° Suppose now that $x \in X - E$, that f_η is flat at x, but that f_η is not smooth at x. To say that f_η is flat at x is equivalent to saying that f itself is flat at x; replacing X by a neighbourhood of x, one may therefore suppose that f is flat (11.1.1). It follows that f_s is flat for every $s \in S$; and since, for every $y \in Y$, one has $f^{-1}(y) = f_{h(y)}^{-1}(y)$, it amounts to the same thing to say that $f_{g(x')}$ is smooth at x' or that f is smooth at x'. But the locus where f is smooth is open in X (12.1.7); since it does not contain x by hypothesis, it does not contain $\overline{\{x\}}$ either. This completes the proof for the first property considered in (i).

Let us next prove (i) for the property of being étale. For f_s to be étale at x means that it is both smooth and quasi-finite at x (17.6.1). Now it amounts to the same thing to say that $f_{g(x)}$ is quasi-finite at x or that f itself is quasi-finite at that point; it follows from (13.1.4) that the set of points x at which $f_{g(x)}$ is quasi-finite is open in X, hence a fortiori locally constructible. The conclusion therefore follows from the local constructibility of the set of points x at which $f_{g(x)}$ is smooth. **IV** | 78

Let us pass to the proof of (i) for the property of being differentially smooth. Let $p : X \times_Y X \to X$ be the second canonical projection; for every $s \in S$, the second projection $X_s \times_{Y_s} X_s \to X_s$ is just p_s. It follows from (17.12.5.1)[29] that $f_{g(x)}$ is differentially smooth at x if and only if $p_{g(x)}$ is smooth at $\Delta_f(x)$. Since p is locally of finite presentation, the set of points $z \in X \times_Y X$ at which $p_{g(p(z))}$ is smooth is locally constructible; its intersection with the locally closed subset $\Delta_f(X)$ is therefore locally constructible in $\Delta_f(X)$ (1.8.2). As the restriction of p to $\Delta_f(X)$ is an isomorphism onto X, the set of $x \in X$ at which $f_{g(x)}$ is differentially smooth is locally constructible.

Finally, consider the property of being non ramified. The diagonal morphism $\Delta_f : X \to X \times_Y X$ is an immersion locally of finite presentation (I.4.3.1), and for every $s \in S$ the diagonal $\Delta_{f_s} : X_s \to X_s \times_{Y_s} X_s$ is just $(\Delta_f)_s$. To say that $f_{g(x)}$ is non ramified at x amounts to saying that $(\Delta_f)_{g(x)}$ is a local isomorphism at x (17.4.1); since this is an immersion locally of finite presentation, it amounts to the same thing (17.9.1)[30] to say that $(\Delta_f)_{g(x)}$ is étale at x. It is therefore enough to apply what has just been proved for the property of being étale. ☐

17.8. Criteria for smoothness and non-ramification by fibres. **IV | 79**

Proposition (17.8.1). — *Let $g : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation. For an S-morphism $f : X \to Y$ to be non ramified, it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \to g^{-1}(s)$ deduced from f by the change of base* $\mathrm{Spec}(k(s)) \to S$*, be non ramified.*

Proof. This follows immediately from the fact that, for a morphism locally of finite presentation, being non ramified is a property of the *fibres* of this morphism (17.4.1), *d*)), and from the fact that for every $y \in Y$, one has $f^{-1}(y) = f_s^{-1}(y)$ where $s = g(y)$. ☐

Proposition (17.8.2). — *Let $g : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation; suppose moreover that h is flat. For an S-morphism $f : X \to Y$ to be smooth (resp. étale), it is necessary and sufficient that, for every $s \in S$, the morphism $f_s : h^{-1}(s) \to g^{-1}(s)$ induced from f by the base change* $\mathrm{Spec}(k(s)) \to S$ *be smooth (resp. étale). When this is the case, the morphism g is flat at all points of $f(X)$.*

Proof. Indeed, one knows from (11.3.10) that f is flat if and only if f_s is flat for every $s \in S$, and that in this case g is flat at the points of $f(X)$. But for a flat morphism locally of finite presentation, smoothness is a property of its fibres (17.5.1, *b*)); and for every $y \in Y$, one has $f^{-1}(y) = f_s^{-1}(y)$ when $s = g(y)$. ☐

Remark (17.8.3). — The preceding proofs show (taking (11.3.10) into account) that, under the same hypotheses on g and h, for f to be non ramified (resp. smooth, resp. étale) *at a point* $x \in X$, it is enough that, on setting $s = h(x)$, f_s be non ramified (resp. smooth, resp. étale) at x.

17.9. Étale morphisms and open immersions.

Theorem (17.9.1). — *Let $f : X \to Y$ be a morphism. The following properties are equivalent:*

a) f is an open immersion.
b) f is a flat monomorphism locally of finite presentation.
c) f is étale and radiciel.

Proof. It results from (**I**, 4.3), (i)) that *a*) implies *b*). The condition *b*) implies that for every $y \in Y$, the fibre $f^{-1}(y)$ is either empty or isomorphic to $\mathrm{Spec}(k(y))$ (8.11.5.1), hence *b*) implies *c*), by virtue of (17.6.2), *c*)). It remains to show that *c*) implies *a*). The question being local on X and on Y (since f is injective), one can reduce to the case where Y is affine and f of finite presentation. As f is flat, this is an *open* morphism (2.4.6), hence, replacing Y by $f(X)$, one can suppose that f is *surjective*. For every morphism $Y' \to Y$, $f' = f_{(Y')} : X_{(Y')} \to Y'$ is again étale, radiciel, surjective, and of finite presentation; it is therefore open, hence a homeomorphism. In other words, f is a *universal homeomorphism*; and since it is of finite type and separated (I.8.7.1, (ii)), it is proper. The hypothesis that f is radiciel and of finite type implies that f is quasi-finite; hence (8.11.1) f is a finite morphism.

[29] The printed French text cites (17.12.5.1), but §17.12 has no such item. Proposition (17.12.5), specifically the equivalence of its conditions (a) and (c), is the result used here. The reader may verify that the result of (17.7.11) is not used in the remainder of §17, so there is no vicious circle.

[30] The reader may verify that the result of (17.7.11) is not used in the remainder of §17, so there is no vicious circle.

To prove that f is an isomorphism, one can reduce to the case where $Y = \mathrm{Spec}(A)$, A being a local ring. As f is of finite presentation, one has $X = \mathrm{Spec}(B)$, where B is a flat A-module of finite presentation (I.4.7), hence free (Bourbaki, *Alg. comm.*, chap. II, §5, no. 2, Cor. 2 to Th. 1). Moreover, if $\mathfrak{m}$ is the maximal ideal of A and k its residue field, $B/\mathfrak{m}B$ is by hypothesis a field, which is both radiciel extension and étale extension of k (17.6.1); hence $B/\mathfrak{m}B$ is isomorphic to k, and as B is a free A-module, B is isomorphic to A. C.Q.F.D. □

Corollary (17.9.2). — *Let X be a connected prescheme; if $f : X \to Y$ is a closed étale immersion, then f is an isomorphism of X onto a connected component of Y.*

Proof. Indeed f is étale and radiciel, hence an isomorphism from X onto a sub-prescheme of Y; but by hypothesis $f(X)$ is closed in Y, hence since open and closed, and since $f(X)$ is connected, it is a connected component of Y. □

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Corollary (17.9.3). — *Let $f : X \to Y$ be a morphism étale (resp. étale and separated). Then every Y-section $g : Y \to X$ of X is an open immersion (resp. open and closed); moreover the application $g \mapsto g(Y)$ is a bijection of the set $\Gamma(X/Y)$ of Y-sections of X to the set of open (resp. open and closed) parts Z of X such that the restriction of f to Z is a surjective morphism and radiciel of Z onto Y.*

Proof. Indeed, the fact that g is an open immersion follows already from the fact that f is non ramified (17.4.1), $b'')$), and the restriction of f to the open $g(Y)$ of X is an isomorphism. Conversely, if Z is an open subset of X such that $f|Z$ is surjective and radiciel, then, since f is étale, $f|Z$ is an isomorphism by (17.9.1). If moreover f is separated, then g is a closed immersion (I.5.4.6), which completes the proof. □

Corollary (17.9.4). — *Let Y be a connected prescheme and $f : X \to Y$ an étale and separated morphism. Then every Y-section g of X is an isomorphism of Y onto a connected component of X, and the application $g \mapsto g(Y)$ is a bijection of $\Gamma(X/Y)$ to the set of connected components of X such that the restriction of f to Z is a surjective morphism and radiciel of Z onto Y.*

Corollary (17.9.5). — *Let $g : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation; suppose moreover that h is flat. For an S-morphism $f : X \to Y$ to be an open immersion (resp. isomorphism), it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \to g^{-1}(s)$ deduced from f by the change of base* $\mathrm{Spec}(k(s)) \to S$, *be an open immersion (resp. isomorphism).*

Proof. Indeed, if f_s is an open immersion for every $s \in S$, it results from (17.8.3) that f is étale, and as for every $y \in Y$, $f^{-1}(y) = f_s^{-1}(y)$ with $s = g(y)$, f is radiciel; hence f is an open immersion by virtue of (17.9.1). If moreover f_s is surjective for every $s \in S$, f is surjective, hence an isomorphism. □

The following proposition specifies (10.4.11):

Proposition (17.9.6). — *Let S be a prescheme and X an S-prescheme of finite presentation. Every S-endomorphism of X which is a monomorphism is an automorphism of X.*

Proof. Let $f : X \to S$ be the structural morphism and g the S-endomorphism under consideration. The question being local on S, one can suppose S affine. Using (8.9.1) and (8.10.5), (i *bis*)), one is reduced to the case where $S = \mathrm{Spec}(A)$, where A is a $\mathbf{Z}$-algebra of finite type, and hence X is a $\mathbf{Z}$-prescheme of finite type. It follows already from (10.4.11) that g is a bijective morphism, since a monomorphism is radiciel ((8.11.5.1)); it will thus suffice to show that g is an open immersion, and since g is radiciel, it will suffice, by virtue of (17.9.1), to prove that g is étale. Moreover, as the set of points where g is étale is open and X is a Jacobson prescheme ((10.4.7)), it suffices to show that g is étale at every *closed* point z of X ((10.3.1)). Set $z' = g(z)$; it follows from (10.4.11), (i) that z' is also a closed point of X, and since g is a monomorphism, the map $k(z') \to k(z)$ deduced from g is an isomorphism. For g to be étale at the point z, it is thus necessary and sufficient that the canonical homomorphism $\widehat{\mathscr{O}}_{X,z'} \to \widehat{\mathscr{O}}_{X,z}$ be bijective ((17.6.3), $e'')$). We will show for this purpose that for every integer n, the canonical homomorphism $\mathscr{O}_{X,z'}/\mathfrak{m}_{z'}^{n+1} \to \mathscr{O}_{X,z}/\mathfrak{m}_{z}^{n+1}$ is bijective, from which the conclusion will immediately follow. One can suppose that z belongs to the finite set $T_{p,d}$

of closed points t of X such that $k(t)$ is an extension of $\mathbf{F}_p$ whose degree divides d; in which case the same holds for z'. Denote by $\mathscr{J}$ the coherent Ideal of $\mathscr{O}_X$ such that $\mathscr{J}_x = \mathscr{O}_x$ for $x \notin T_{p,d}$, and $\mathscr{J}_x = \mathfrak{m}_x^{n+1}$ for $x \in T_{p,d}$. Let $T_{p,d,n}$ be the closed sub-prescheme of X defined by $\mathscr{J}$ and $j : T_{p,d,n} \to X$ the canonical injection; the composite $g \circ j : T_{p,d,n} \to X$ maps $T_{p,d}$ into itself, and for every $x \in T_{p,d}$ the homomorphism $\mathscr{O}_{X,g(x)} \to \mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1}$ factors as

$$\mathscr{O}_{X,g(x)} \longrightarrow \mathscr{O}_{X,g(x)}/\mathfrak{m}_{g(x)}^{n+1} \longrightarrow \mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1}.$$

By ((**I**, 4.9)), there therefore exists a unique endomorphism g' of $T_{p,d,n}$ such that $j \circ g' = g \circ j$. As j is a monomorphism, and g is also one by hypothesis, one deduces that g' is also a monomorphism, and the proposition will be proven if one shows that g' is an *automorphism* of $T_{p,d,n}$. But, for every $x \in T_{p,d,n}$, we have seen that $k(x)$ is a finite field, and since $\mathscr{O}_{X,x}$ is Noetherian, each quotient $\mathfrak{m}_x^h/\mathfrak{m}_x^{h+1}$ is a finite-dimensional $k(x)$-vector space. It follows immediately that the local ring $\mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1}$ of $T_{p,d,n}$ at x has finitely many elements and, a fortiori, is a finite-type $\mathbf{Z}$-module. As $T_{p,d,n}$ is the sum of the preschemes $\mathrm{Spec}(\mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1})$ in finite number, it is a finite $\mathbf{Z}$-prescheme; the endomorphism g' is therefore also finite (**II**, 6.1.5, (v)), hence proper. But a proper monomorphism is a closed immersion ((8.11.5)); if $T_{p,d,n} = \mathrm{Spec}(B)$, g' corresponds to a surjective endomorphism $\varphi : B \to B$ of the ring B. But the set B is finite, hence φ is necessarily bijective. □

17.10. Relative dimension of a smooth prescheme over another. IV | 81

Definition (17.10.1). — Let $f : X \to Y$ be a morphism locally of finite type. The *relative dimension* of f at a point $x \in X$ (or the *dimension of X relative to Y at the point x*) is the positive integer $\dim_x f = \dim_x(f^{-1}(f(x)))$.

To say that f is *quasi-finite* at the point x ($\mathbf{Err}_{\mathbf{III}}$, 20) is equivalent to saying that $\dim_x f = 0$. One has seen (13.1.3) that the function $x \mapsto \dim_x f$ is upper semi-continuous. One notes that, even when the morphism f has the property (S_1) (that is to say (6.8.1) is flat and such that its fibres have no embedded associated prime cycle), the function $x \mapsto \dim_x f$ is not necessarily continuous, as shown by the example where $Y = \mathrm{Spec}(k)$, where k is a field, and $X = \mathrm{Spec}(k[\mathrm{U},\mathrm{V},\mathrm{W}]/\mathfrak{p}\mathfrak{q})$, where $\mathfrak{p} = (\mathrm{W})$ and $\mathfrak{q} = (\mathrm{U}) + (\mathrm{V} - \mathrm{W})$, the prime ideals of $k[\mathrm{U},\mathrm{V},\mathrm{W}]$ (X being thus the reunion, in the affine space of 3 dimensions, of a plane and of a line non parallel to this plane).

To say that a morphism locally of finite presentation $f : X \to Y$ is étale at a point $x \in X$ means that f is smooth at x and that $\dim_x f = 0$ (17.6.1).

Proposition (17.10.2). — *Let $f : X \to Y$ be a smooth morphism. For every $x \in X$, the locally free $\mathscr{O}_X$-module Ω^1_f (17.2.3) has rank at x equal to $\dim_x f$ (which implies that $x \mapsto \dim_x f$ is a continuous function on X).*

Proof. Indeed, if $y = f(x)$, then $X_y = f^{-1}(y)$ is smooth over $k(y)$; if $f_y : X_y \to \mathrm{Spec}(k(y))$ is the structural morphism, one has $\Omega^1_{f_y} = \Omega^1_f \otimes_{\mathscr{O}_X} \mathscr{O}_{X_y}$ (16.4.5). Thus one may reduce to the case where $Y = \mathrm{Spec}(k)$ is the spectrum of a field. Moreover, by (16.4.5) and (17.7.1), one may replace k by an algebraically closed extension; in other words, one may suppose that k is algebraically closed. The set of k-rational points of X is then dense in X (10.4.8), so it is enough to consider the case where x is rational over k. Now (16.4.12) identifies $(\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x)$ with $\mathfrak{m}_x/\mathfrak{m}_x^2$; since $\mathscr{O}_{X,x}$ is a regular local ring (17.5.1), one has $\mathrm{rg}_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2) = \dim(\mathscr{O}_{X,x})$ (**0**, 17.1.1). Hence $\mathrm{rg}_{\mathscr{O}_{X,x}}(\Omega^1_{X/k})_x = \dim(\mathscr{O}_{X,x})$. Since $k(x) = k$, one also has $\dim_x f = \dim(\mathscr{O}_{X,x})$ (5.2.3). Q.E.D. □

We will prove a reciprocal of this result further on (17.15.5).

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Corollary (17.10.3). — *Let $f : X \to Y$, $g : Y \to Z$ be two smooth morphisms. Then for every $x \in X$, one has*

(17.10.3.1) $$\dim_x(g \circ f) = \dim_x f + \dim_{f(x)} g.$$

Proof. Indeed, $g \circ f$ is smooth (17.3.3), so the three $\mathscr{O}_X$-modules $\Omega^1_{X/Y}$, $\Omega^1_{X/Z}$, and $f^*(\Omega^1_{Y/Z})$ are locally free (17.2.3 and $\mathbf{0_I}$, 5.4.5); moreover, the rank at x of $f^*(\Omega^1_{Y/Z})$ equals the rank at $y = f(x)$ of

$\Omega^1_{Y/Z}$. Equality (17.10.3.1) is therefore a consequence of (17.10.2) and the exactness of the sequence (17.2.3.1). ☐

Corollary (17.10.4). — *Let $f : X \to Y$ be a smooth morphism and X' a sub-prescheme of X such that the composite $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) is smooth. Then the conormal sheaf $\mathscr{N}_{X'/X}$ is a locally free $\mathscr{O}_{X'}$-module, and for every $x \in X'$ one has*

$$\dim_x f = \dim_x(f \circ j) + \operatorname{rg}_{\mathscr{O}_{X',x}}(\mathscr{N}_{X'/X})_x. \tag{17.10.4.1}$$

Proof. Indeed, $\Omega^1_{X/Y} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$ and $\Omega^1_{X'/Y}$ are locally free, and the exact sequence (17.2.5.1) splits locally in a suitable neighbourhood of every point of X'. Thus $\mathscr{N}_{X'/X}$ is locally free, and relation (17.10.4.1) follows at once from the exactness of (17.2.5.1). ☐

17.11. Smooth morphisms of smooth preschemes.

Theorem (17.11.1). — *Let $f : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X; set $y = g(x)$, $s = f(y) = h(x)$. The following conditions are equivalent:*

a) f is smooth at y and g is smooth at x.
b) g and h are smooth at x.
c) h is smooth at x, and the canonical homomorphism (16.4.18)

$$(g^*(\Omega^1_{Y/S}))_x \longrightarrow (\Omega^1_{X/S})_x \tag{17.11.1.1}$$

is left invertible (in other words is an isomorphism onto a direct factor of $(\Omega^1_{X/S})_x$).
c′) h is smooth at x, and the canonical homomorphism

$$((\Omega^1_{Y/S})_y \otimes_{\mathscr{O}_{Y,y}} k(y)) \otimes_{k(y)} k(x) \longrightarrow (\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x) \tag{17.11.1.2}$$

is injective.

Suppose moreover that the homomorphism $k(y) \to k(x)$ is bijective. Then the preceding conditions are also equivalent to the following:

d) h is smooth at x, and the canonical map $\mathrm{T}_{X/S}(x) \to \mathrm{T}_{Y/S}(y)$ (16.5.12) *from the tangent vector space to X at x to the tangent vector space to Y at y is surjective.*

Proof. That a) implies b) follows at once from (17.3.3), (iii), and b) implies c) by (17.2.3), (ii). To see that c) and c') are equivalent, note that under either condition $\Omega^1_{X/S}$ is locally free of finite type, by (17.2.3), (ii); it is therefore enough to apply (**0**, (19.1.12)) to the local ring $\mathscr{O}_{X,x}$ and to the homomorphism (17.11.1.1) of finite-type $\mathscr{O}_{X,x}$-modules. When $k(x) = k(y)$, the linear tangent map $\mathrm{T}_{X/S}(x) \to \mathrm{T}_{Y/S}(y)$ is, by (16.5.12), the transpose of (17.11.1.2); this proves the equivalence of c') and d) in that case.

It remains to prove that c) implies a). We may restrict ourselves to the case in which $S = \operatorname{Spec}(A)$, IV | 83
$Y = \operatorname{Spec}(B)$, and $X = \operatorname{Spec}(C)$ are affine. The hypothesis implies, by (17.2.3), (i), that $(\Omega^1_{X/S})_x = (\Omega^1_{C/A})_{\mathfrak{m}}$ is a free $C_{\mathfrak{m}}$-module. By (16.10.6), there are elements t_i $(1 \leqslant i \leqslant r)$ of $B_{\mathfrak{p}} = \mathscr{O}_{Y,y}$ such that their differentials generate $(\Omega^1_{B/A})_{\mathfrak{p}}$ and their images in $(\Omega^1_{C/A})_{\mathfrak{m}}$ are part of a basis of that free $C_{\mathfrak{m}}$-module. Since $\Omega^1_{Y/S}$ and $\Omega^1_{X/S}$ are of finite presentation (16.4.22), after replacing X and Y by suitable affine open neighbourhoods of x and y, we may suppose that $\Omega^1_{C/A}$ is free and that the t_i are the images of elements $s_i \in B$ whose differentials $d_{B/A}(s_i)$ generate $\Omega^1_{B/A}$ and whose images in $\Omega^1_{C/A}$ are part of a basis of that C-module (Bourbaki, *Alg. comm.*, chap. II, § 5, nº 1, prop. 2).

Let $\varphi : B' = A[\mathrm{T}_1, \dots, \mathrm{T}_r] \to B$ be the A-homomorphism such that $\varphi(\mathrm{T}_i) = s_i$. The corresponding dihomomorphism $\Omega^1_{B'/A} \to \Omega^1_{B/A}$ (**0**, (20.5.2)) sends the basis elements $d_{B'/A}(\mathrm{T}_i)$ (**0**, (20.4.13)) to the $d_{B/A}(s_i)$, and is therefore surjective. If $Y' = \operatorname{Spec}(B')$ and $u : Y \to Y'$ is the S-morphism corresponding to φ, it follows from (17.2.2) that u is unramified. If $u \circ g : X \to Y'$ is smooth at x, then (17.7.10) shows that u is étale at y, and (17.3.5) then shows that g is smooth at x; moreover, since the structural morphism $f' : Y' \to S$ is smooth (17.3.8), $f = f' \circ u$ is smooth at y. The canonical images of the $d_{B'/A}(\mathrm{T}_i)$ in $\Omega^1_{C/A}$ are the images of the $d_{B/A}(s_i)$ and thus form part of a basis. Consequently, in proving that $c)$ implies $a)$, we may restrict ourselves to the case $Y' = Y$, and hence suppose that f is smooth.

By (17.8.2), we may then reduce to the case in which $S = \mathrm{Spec}(k)$ is the spectrum of a field; by (17.7.1), (ii), we may moreover suppose that k is algebraically closed. After replacing X and Y by suitable open neighbourhoods of x and y, we may suppose that f and h are both smooth and that the canonical homomorphism

$$g^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is left invertible (**0**, (19.1.12)). The set of k-rational points of X is then very dense in X (10.4.8); thus, to prove that g is smooth, it is enough to prove smoothness at every k-rational point $x \in X$, or equivalently that $\mathscr{O}_{X,x}$ is flat over $\mathscr{O}_{Y,y}$ and that $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is regular (17.5.1).

Now $y = g(x)$ is a fortiori k-rational. Since $\mathscr{O}_{Y,y}$ is then a formally smooth k-algebra for the discrete topologies and $\mathscr{O}_{Y,y}/\mathfrak{m}_y = k$, it follows from (**0**, (20.5.14)) that the canonical homomorphism

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \longrightarrow (\Omega^1_{Y/S})_y \otimes_{\mathscr{O}_{Y,y}} k(y)$$

is bijective; likewise the canonical homomorphism

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x)$$

is bijective. Condition $c')$, which is equivalent to $c)$, therefore says here that the canonical homomorphism

$$(\mathfrak{m}_y/\mathfrak{m}_y^2) \otimes_{k(y)} k(x) \longrightarrow \mathfrak{m}_x/\mathfrak{m}_x^2$$

is injective. Since $\mathscr{O}_{X,x}$ is regular, the conclusion follows from (**0**, (17.3.3)). □

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Corollary (17.11.2). — *Under the general hypotheses of (17.11.1), the following conditions are equivalent:*

a) f is smooth at y and g is étale at x.

b) h is smooth at x and g is étale at x.

c) h is smooth at x, and the canonical homomorphism

$$(g^*(\Omega^1_{Y/S}))_x \longrightarrow (\Omega^1_{X/S})_x$$

is bijective.

c') h is smooth at x, and the canonical homomorphism

$$((\Omega^1_{Y/S})_y \otimes_{\mathscr{O}_{Y,y}} k(y)) \otimes_{k(y)} k(x) \longrightarrow (\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x)$$

is bijective.

Suppose moreover that the homomorphism $k(y) \to k(x)$ is bijective. Then the preceding conditions are also equivalent to the following:

d) h is smooth at x, and the canonical map $\mathrm{T}_{X/S}(x) \to \mathrm{T}_{Y/S}(y)$ (16.5.12) is bijective.

Proof. Each of conditions $a)$, $b)$, and $c)$ of (17.11.2) is equivalent to the conjunction of the corresponding condition of (17.11.1) and the assertion that g is unramified at x, taking (17.2.2) into account for condition $c)$; hence $a)$, $b)$, and $c)$ are equivalent. The equivalence of $c)$ and $c')$ follows from the equivalence of the corresponding conditions of (17.11.1) and Nakayama's lemma; when $k(x) = k(y)$, the equivalence of $c')$ and $d)$ follows immediately by transposition (16.5.12). □

Corollary (17.11.3). — *Let $h : X \to S$ be a smooth morphism, let s_i ($1 \leqslant i \leqslant r$) be sections of $\mathscr{O}_X$ over X (and thus also sections of $h_*(\mathscr{O}_X)$ over S), and let $g : X \to S[\mathrm{T}_1, \dots, \mathrm{T}_r] = \mathbf{V}^r_S$ be the S-morphism corresponding to the homomorphism of $\mathscr{O}_S$-modules $\mathscr{O}^r_S \to h_*(\mathscr{O}_X)$ defined by these sections (**II**, (1.2.7)). For g to be smooth (resp. étale) at a point $x \in X$, it is necessary and sufficient that the $(d_{X/S}(s_i))_x$ form part of a basis (resp. form a basis) of the $\mathscr{O}_{X,x}$-module $(\Omega^1_{X/S})_x$.*

Proof. It suffices to apply (17.11.1) (resp. (17.11.2)) taking for f the structural morphism $S[\mathrm{T}_1, \dots, \mathrm{T}_r] \to S$. □

Corollary (17.11.4). — *For a morphism $h : X \to S$ to be smooth at a point $x \in X$, it is necessary and sufficient that there exist an open neighbourhood U of x, an integer r, and an étale S-morphism $U \to S[\mathrm{T}_1, \dots, \mathrm{T}_r]$.*

Proof. The condition is plainly sufficient, since the structural morphism $S[\mathrm{T}_1, \ldots, \mathrm{T}_r] \to S$ is smooth (17.3.8). Conversely, since h is smooth at x, there is an open neighbourhood U of x such that $\Omega^1_{X/S}|U$ is locally free (17.2.3); using (16.10.6) and restricting U if necessary, we obtain sections s_i of $\mathscr{O}_X$ over U such that the $d_{X/S}(s_i)$ form a basis of $\Omega^1_{X/S}|U$. The conclusion follows from (17.11.3). □

Proposition (17.11.5). — *Let $f : Y \to S$, $h : X \to S$ be two smooth morphisms. For an S-morphism $g : X \to Y$ to be an open immersion, it is necessary and sufficient that g be a monomorphism of preschemes and that for every $x \in X$, if one sets $y = g(x)$, one has $\dim_y(f) = \dim_x(h)$ (for a generalisation of this proposition, see (18.10.5)).*

Proof. The condition is evidently necessary; let us prove that it is sufficient. By (17.9.5), we are immediately reduced to the case in which $S = \operatorname{Spec}(k)$ is the spectrum of a field, and by (2.7.1), (x), we may suppose that k is algebraically closed. In view of (17.9.1), it is then enough to prove that g is étale; since the locus where g is étale is open, it is enough to prove this at the closed points of X, or equivalently at its k-rational points (10.4.8).

Let x be such a point and set $y = g(x)$; then y too is k-rational. Put $A = \mathscr{O}_{Y,y}$, a regular local ring with residue field k, let $d = \dim(A)$, and let $(t_i)_{1 \leqslant i \leqslant d}$ be a regular system of parameters of A. Put $B = \mathscr{O}_{X,x}$ and $C = B/\mathfrak{m}_y B$. Since g is a monomorphism, so is the morphism $\operatorname{Spec}(C) \to \operatorname{Spec}(k(y)) = \operatorname{Spec}(k)$ obtained from g by base change (**I**, (3.3.12)). This means that the corresponding homomorphism $u : k \to C$ is surjective, and hence bijective: it has a left inverse $v : C \to k$, while $u \circ v$ and the identity of C, after composition with u, give the same homomorphism $u : k \to C$; the monomorphism property therefore gives $u \circ v = 1_C$.

By hypothesis B is a regular ring of dimension d. The images of the t_i in B consequently form a regular system of parameters of B (**0**, (17.1.7)); condition (17.6.3), e''), is therefore satisfied by g at x (**0**, (17.1.1), and Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 3 to th. 1), which completes the proof. □

17.12. Smooth subpreschemes of a smooth prescheme. Smooth morphisms and differentially smooth morphisms. IV | 85

Theorem (17.12.1). — *Let $f : X \to S$, $h : Y \to S$ be two morphisms locally of finite presentation, $j : Y \to X$ an immersion, y a point of Y, $x = j(y)$. The following conditions are equivalent:*

a) h is smooth at y and f is smooth at x.

b) f is smooth at x, and the canonical homomorphism (16.4.21)

$$(\mathscr{N}_{Y/X})_y \longrightarrow (j^*(\Omega^1_{X/S}))_y$$

is left invertible.

c) h is smooth at y and there exists an open neighbourhood U of y in Y such that $j|U : U \to X$ is a regular immersion (16.9.2).

c′) h is smooth at y and there exists an open neighbourhood U of y in Y such that $j|U : U \to X$ is a quasi-regular immersion; in other words (16.9.8), there is a neighbourhood U of y in Y on which $\mathscr{N}_{Y/X}$ is locally free and the canonical homomorphism

$$\mathbf{S}_{\mathscr{O}_Y}(\mathscr{N}_{Y/X}) \longrightarrow \operatorname{gr}^\bullet(j)$$

is bijective.

Proof. To prove the equivalence of a) and b), we may restrict ourselves to the case in which $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, with $Y = \operatorname{Spec}(B/\mathfrak{J})$, where $\mathfrak{J}$ is a finite-type ideal of B, and where B is a smooth A-algebra (17.3.2), (iii). It is then enough to apply the Jacobian criterion (**0**, (22.6.1)) together with (**0**, (19.1.14)), bearing in mind that $\Omega^1_{X/S}$ is locally free in a neighbourhood of x and that $\mathscr{N}_{Y/X}$ is of finite type (16.1.6).

We next prove that a) implies c'). We may again suppose that S, X, and Y are affine, and that B and $B/\mathfrak{J}$ are smooth A-algebras. It follows from (17.7.9) and (8.10.5), (iv), that there exist a Noetherian subring A' of A, a finite-type A'-algebra B', and an ideal $\mathfrak{J}'$ of B' such that

$$B = B' \otimes_{A'} A, qquad \mathfrak{J} = \mathfrak{J}' B,$$

and such that B' and $B'/\mathfrak{J}'$ are smooth A'-algebras. By (**0**, (19.3.8)), B' is formally smooth when A' is given the discrete topology and B' the $\mathfrak{J}'$-preadic topology. Consequently (**0**, (19.5.4)), $\mathfrak{J}'/\mathfrak{J}'^2$ is a projective $B'/\mathfrak{J}'$-module and the canonical homomorphism

$$\mathbf{S}_{B'/\mathfrak{J}'}(\mathfrak{J}'/\mathfrak{J}'^2) \longrightarrow \mathrm{gr}^{\bullet}_{\mathfrak{J}'}(B')$$

is bijective. Thus the immersion $\mathrm{Spec}(B'/\mathfrak{J}') \to \mathrm{Spec}(B')$ is quasi-regular (16.9.8), and hence regular because B' is Noetherian (16.9.10). Moreover, $B'/\mathfrak{J}'$ is flat over A' (17.5.1) and B' is an A'-algebra of finite presentation; applying (11.3.8) with $\mathfrak{J}'$ in place of $\mathscr{F}$ and then changing base shows that j is a regular immersion, and a fortiori a quasi-regular immersion. The fact that B is an A-algebra of finite presentation and is flat over A also shows, by (11.3.8), that conditions $c')$ and $c)$ are equivalent and are stable under base change.

It remains to prove that $c)$ implies $a)$. Since condition $b)$ of (11.3.8) implies condition $c)$ there, condition $c)$ above already shows that f is flat at x. By (17.5.1), it is therefore enough to prove that the fibre $f^{-1}(s)$, where $s = h(y) = f(x)$, is smooth over $k(s)$ at x. Since condition $c)$ is stable under base change, we reduce to the case in which $S = \mathrm{Spec}(k)$ is the spectrum of a field; in view of (17.7.1), (ii), we may suppose that k is algebraically closed. It is enough to prove that $\mathscr{O}_{X,x}$ is regular (17.5.1). By hypothesis,

$$\mathscr{O}_{Y,y} = \mathscr{O}_{X,x}/\mathfrak{J}_x,$$

where $\mathfrak{J}_x$ is generated by an $\mathscr{O}_{X,x}$-regular sequence (t_i), and $\mathscr{O}_{Y,y}$ is regular. The t_i form part of a system of parameters of $\mathscr{O}_{X,x}$ (**0**, (16.4.1)); the conclusion follows from (**0**, (17.1.7)). ☐

Corollary (17.12.2). — *With the notation of (17.12.1), suppose that f is smooth at x, and let Y be a closed subprescheme of X, defined by an ideal $\mathscr{J}$ of $\mathscr{O}_X$, which is smooth over S at x. Let $(g_i)_{1 \leqslant i \leqslant r}$ be sections of $\mathscr{J}$ over X. The following conditions are equivalent:*

- *a) The $(g_i)_x$ form a system of generators of the $\mathscr{O}_{X,x}$-module $\mathscr{J}_x$ having the smallest possible number of elements.*
- *b) If g'_i denotes the canonical image of g_i in $\Gamma(X, \mathscr{J}/\mathscr{J}^2)$, the $(g'_i)_x$ form a basis of the $(\mathscr{O}_{X,x}/\mathscr{J}_x)$-module $\mathscr{J}_x/\mathscr{J}_x^2$.*
- *c) The images of the $d_{X/S}g_i$ in $(\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x)$ are linearly independent over $k(x)$, and the $(g_i)_x$ generate $\mathscr{J}_x$.*
- *d) There exist an open neighbourhood U of x in X and sections g_j $(r+1 \leqslant j \leqslant n)$ of $\mathscr{O}_X$ over U such that the S-morphism*

  $$u : U \longrightarrow S[\mathrm{T}_1, \dots, \mathrm{T}_n]$$

  *corresponding to the homomorphism $\mathscr{O}_S^n \to f_*(\mathscr{O}_U)$ defined by the sections $g_i|U$ $(1 \leqslant i \leqslant r)$ and g_j $(r+1 \leqslant j \leqslant n)$ (**II**, (1.2.7)) is étale, and the inverse image under this morphism of the subprescheme $Y' = S[\mathrm{T}_{r+1}, \dots, \mathrm{T}_n]$ is the subprescheme induced by $j(Y)$ on the open set $U \cap j(Y)$.*

Proof. Since $(g'_i)_x$ is the canonical image of $(g_i)_x$, the equivalence of $a)$ and $b)$ follows from Nakayama's lemma: $\mathscr{J}_x$ is of finite type and $\mathscr{J}_x/\mathscr{J}_x^2$ is a free $(\mathscr{O}_{X,x}/\mathscr{J}_x)$-module (17.10.4) (Bourbaki, *Alg. comm.*, chap. II, § 3, nº 2, prop. 5). By (17.12.1), $b)$, the module $\mathscr{J}_x/\mathscr{J}_x^2$ is canonically identified with a direct factor of the free $(\mathscr{O}_{X,x}/\mathscr{J}_x)$-module

$$(\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} (\mathscr{O}_{X,x}/\mathscr{J}_x),$$

and the equivalence of $b)$ and $c)$ follows from Bourbaki, *loc. cit.*

If $a)$ holds, $(g'_i)_x$ is identified with $(d_{X/S}g_i)_x \otimes 1$ for $1 \leqslant i \leqslant r$. Since $(\Omega^1_{X/S})_x$ is free of rank n, **IV** | 86
Bourbaki, *loc. cit.*, shows that, after replacing X by an open neighbourhood of x, there exist $n - r$ sections g_i $(r+1 \leqslant i \leqslant n)$ of $\mathscr{O}_X$ over X such that the $(d_{X/S}g_i)_x$ for $1 \leqslant i \leqslant n$ form a basis of $(\Omega^1_{X/S})_x$. The corresponding morphism u is then étale at x by (17.11.3); after a further restriction, we may suppose it is étale. For the same reason, the restriction $u^{-1}(Y') \to Y'$ is étale at x, and we may suppose both morphisms étale near x. Moreover, Y (identified for simplicity with the closed subprescheme $j(Y)$ of X) is plainly a closed subprescheme of $u^{-1}(Y')$; by the choice of the g_i for $i > r$, together with (17.2.5) and (17.11.2), the restriction of u to Y may also be supposed étale. It follows that for every $y' \in Y'$, the immersion $Y \cap u^{-1}(y') \to u^{-1}(y')$ is open (17.9.6); hence $Y \to u^{-1}(Y')$ is an open immersion. This proves that $a)$ implies $d)$. The converse $d) \Rightarrow a)$ follows at once from (17.11.3). ☐

Corollary (17.12.3). — *Let $f : X \to S$ be a morphism locally of finite presentation, $u : S \to X$ an S-section of X. For f to be smooth at a point $x \in X$, it is necessary and sufficient that u be a quasi-regular immersion in a neighbourhood of $s = f(x)$.*

Proof. The conclusion follows from (17.12.1), since $f \circ u = 1_S$ is smooth. ☐

Proposition (17.12.4). — *Every smooth morphism $f : X \to S$ is differentially smooth (in other words (16.10.5) $\Delta_f : X \to X \times_S X$ is a quasi-regular immersion). In particular, the $\mathscr{P}^n_f = \mathscr{P}^n_{X/S}$ and the $\mathrm{gr}_n(\mathscr{P}_{X/S})$ are locally free $\mathscr{O}_X$-modules of finite type.*

Proof. It is enough to note that the structural morphism $p_1 : X \times_S X \to X$ is smooth (17.3.3), (iii), and that Δ_f is an X-section of p_1; the conclusion follows from (17.12.3). ☐

Proposition (17.12.5). — *Let $f : X \to Y$ be a morphism locally of finite presentation. The following conditions are equivalent:*

a) f is differentially smooth.

b) For every morphism $g : Y' \to Y$, if $X' = X \times_Y Y'$ and $f' = f_{(Y')} : X' \to Y'$, then for every Y'-section s' of X', f' is smooth at all points of $s'(Y')$.

c) The second projection $p_2 : X \times_Y X \to X$ is smooth at all points of the diagonal $\Delta_f(X)$.

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Proof. Condition $c)$ is a special case of $b)$: take $Y' = X$ and $g = f$ in $b)$; then f' is the second projection p_2, and Δ_f is an X-section of $X \times_Y X$. Next, $c)$ implies $a)$, since p_2 is locally of finite presentation and, if p_2 is smooth at the points of $\Delta_f(X)$, then Δ_f is a quasi-regular immersion by (17.12.3); hence f is differentially smooth.

Finally, suppose $a)$ and prove $b)$. If Δ_f is quasi-regular, then p_2 is smooth at every point of $\Delta_f(X)$ by (17.12.3). Let $g' : X' \to X$ be the canonical projection and let $v = (g', g')_Y : X' \to X \times_Y X$. Then the diagram

$$\begin{tikzcd} X \times_Y X \arrow[d, "p_2"'] & X' \arrow[l, "v"'] \arrow[d, "f'"] \\ X & Y' \arrow[l, "h = g' \circ s'"] \end{tikzcd}$$

is commutative and identifies X' with the product $(X \times_Y X) \times_X Y'$ (**I**, (3.3.9)). Taking into account that diagram (17.4.1.1) identifies Y' with the product of the $(X \times_Y X)$-preschemes X and X', we conclude from the smoothness of p_2 at every point of $\Delta_f(X)$ that f' is smooth at every point of $s'(Y')$ (17.3.3), (iii). ☐

Corollary (17.12.6). — *With the notations being those of (8.8.1), suppose X_α quasi-compact, and $f_\alpha : X_\alpha \to S_\alpha$ locally of finite presentation. For $f : X \to S$ to be differentially smooth, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $f_\lambda : X_\lambda \to S_\lambda$ is differentially smooth (in which case $f_\mu : X_\mu \to S_\mu$ is differentially smooth for $\mu \geqslant \lambda$).*

Proof. The sufficiency of the condition and the last assertion follow from (16.10.4). To prove necessity, note that $\Delta_f : X \to X \times_S X$ and $p_2 : X \times_S X \to X$ are obtained by base change from $\Delta_{f_\lambda} : X_\lambda \to X_\lambda \times_{S_\lambda} X_\lambda$ and $p_{2,\lambda} : X_\lambda \times_{S_\lambda} X_\lambda \to X_\lambda$. By (17.12.5), for every $z \in \Delta_f(X)$ there is an affine open neighbourhood $U(z)$ of z in $X \times_S X$ on which p_2 is smooth. By (8.2.11) and (17.7.8), there exist an index $\lambda(z)$ and a neighbourhood $U_{\lambda(z)}$ of the projection of z in $X_{\lambda(z)} \times_{S_{\lambda(z)}} X_{\lambda(z)}$ such that $U(z)$ is the inverse image of $U_{\lambda(z)}$ and $p_{2,\lambda(z)}$ is smooth on $U_{\lambda(z)}$. Since X is quasi-compact, $\Delta_f(X)$ is covered by finitely many such neighbourhoods $U(z_i)$. By (8.3.4), there is an index λ greater than all the $\lambda(z_i)$ such that the inverse images of the $U_{\lambda(z_i)}$ in $X_\lambda \times_{S_\lambda} X_\lambda$ cover $\Delta_{f_\lambda}(X_\lambda)$. It follows from (17.12.5) that this λ has the required property. ☐

Example (17.12.7). — Let S be a prescheme, G an S-*prescheme in groups*, locally of finite presentation on S. For G to be differentially smooth over S, it is necessary and sufficient that G be smooth over S at all points of the "identity section" e of G. The condition is indeed necessary by (17.12.5). Conversely, suppose it holds. For every morphism $S' \to S$, $G' = G \times_S S'$ is an S'-prescheme in groups locally of finite presentation over S'; by (17.12.5), it is therefore enough to prove that for every S-section s of G, the structural morphism is smooth at the points of $s(S)$. If $m : G \times_S G \to G$ is the multiplication, then $m \circ (s \times 1_G) : S \times_S G \to G \times_S G \to G$ is an S-isomorphism of G, the "left translation" τ_s, carrying the points of the identity section of G to the points of $s(S)$. Thus G is smooth over S at the points of $s(S)$.

Remark (17.12.8). — An S-prescheme in groups G, of finite type (and even finite) over a locally Noetherian prescheme S, can be differentially smooth over S without being smooth (or even *flat*) over S. For example, take S to be the spectrum of the dual-number algebra $D = k[\mathrm{T}]/\mathrm{T}^2k[\mathrm{T}]$ over a field k, and take G to be the spectrum of the direct-product D-algebra $E = D \oplus k$. Define an S-group-prescheme structure on G by a "diagonal map" $\delta : E \to E \otimes_D E$ as follows. If e' and e'' are the idempotents given by the canonical images of 1 of D in the two factors D and k of E, then

$$\delta(e') = e' \otimes e' + e'' \otimes e'', \qquad \delta(e'') = e' \otimes e'' + e'' \otimes e'.$$

The identity section of the group scheme G corresponds to the homomorphism $E \to D$ that is the identity on D and zero on k. It is an isomorphism of S onto a connected component of G; hence G is differentially smooth over S (17.12.6), whereas it is clear that G is not flat over S.

17.13. Transversal morphisms.

(17.13.1). Let S be a prescheme, let X, Y, and X' be three S-preschemes, let $i : Y \to X$ be an S-immersion, and let $f : X' \to X$ be an S-morphism. Set $Y' = Y \times_X X'$, and let $g : Y' \to Y$, $j : Y' \to X'$ be the canonical projections, so that one has the commutative diagram

(17.13.1.1)
$$\begin{tikzcd} Y & Y' \arrow[l, "g"'] \arrow[d, "j"] \\ X \arrow[from=u, "i"'] & X' \arrow[l, "f"] \end{tikzcd}$$

and j is an S-immersion. There is then a commutative diagram of quasi-coherent $\mathscr{O}_{Y'}$-modules

(17.13.1.2)
$$\begin{tikzcd} g^*(\mathscr{N}_{Y/X}) \arrow[r] \arrow[d, "\mathrm{gr}_1(g)"] & f^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & g^*(\Omega^1_{Y/S}) \arrow[r] \arrow[d] & 0 \\ \mathscr{N}_{Y'/X'} \arrow[r] & \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \arrow[r] & \Omega^1_{Y'/S} \arrow[r] & 0 \end{tikzcd}$$

Here the lower row is the exact sequence (16.4.21) applied to j; the upper row is obtained from the same exact sequence for i by applying the right-exact functor g^*, which therefore preserves its exactness. The homomorphism $\mathrm{gr}_1(g)$ is defined in (16.2.1), and the commutativity of the two squares follows from (**0**, (20.5.7.3)) and (**0**, (20.5.11.3)). Moreover, we saw in (16.2.2), (iii), that $\mathrm{gr}_1(g)$ is surjective here; hence diagram (17.13.1.2) gives the exact sequence

(17.13.1.3)
$$g^*(\mathscr{N}_{Y/X}) \xrightarrow{\alpha} \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \longrightarrow \Omega^1_{Y'/S} \longrightarrow 0.$$

Proposition (17.13.2). — *With the notation of (17.13.1), let x' be a point of Y' and let $x = g(x')$ be its image in Y; suppose that X and Y are smooth over S at x, and that X' is smooth over S at x'. Let m and $m - c$ be the relative dimensions of X and Y over S at x (17.10.1), and let n be that of X' over S at x'. The following conditions are equivalent:*

a) Y' is smooth over S at x' and has relative dimension $n - c$.

b) The homomorphism $\alpha \otimes 1 : g^*(\mathscr{N}_{Y/X}) \otimes_{\mathscr{O}_{Y'}} k(x') \to \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} k(x')$ deduced from the homomorphism α of (17.13.1.3) is injective.

Let s be the canonical image of x in S, and suppose that x' is rational over $k(s)$. Then the conditions a) and b) are also equivalent to the following:

b') The transposed homomorphism of $\alpha \otimes 1$

$$\mathrm{T}_{X'/S}(x') \longrightarrow (\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} k(x'))^\vee = \mathrm{T}_{X/S}(x)/\mathrm{T}_{Y/S}(x)$$

(cf. (16.5.12)) is surjective.

Moreover, when conditions a) and b) hold at x', they hold throughout a neighbourhood of x' in Y'; after replacing X' by a neighbourhood of x', the homomorphism

$$\mathrm{gr}_1(g) : g^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X'}$$

is bijective, and the sequence

$$0 \longrightarrow g^*(\mathscr{N}_{Y/X}) \xrightarrow{\alpha} \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \longrightarrow \Omega^1_{Y'/S} \longrightarrow 0 \tag{17.13.2.1}$$

obtained by adjoining a zero to (17.13.1.3) is exact.

Proof. If the equivalent conditions a) and b) hold at x', they also hold in a neighbourhood of x' in Y': this follows from the openness of the smooth locus (17.3.7) and from (17.10.2).

By (17.12.1), applied to X' and Y', saying that Y' is smooth over S at x' is equivalent to saying that the homomorphism

$$\delta \otimes 1 : \mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} k(x') \longrightarrow \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} k(x')$$

is injective, taking account of (**0**, (19.1.12)) and of the fact that $\Omega^1_{X'/S}$ is locally free at x' (17.2.3). When this holds, $\Omega^1_{Y'/S}$ is locally free in a neighbourhood of x' and the sequence

$$0 \longrightarrow \mathscr{N}_{Y'/X'} \longrightarrow \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \longrightarrow \Omega^1_{Y'/S} \longrightarrow 0 \tag{17.13.2.2}$$

is exact (17.2.5). Thus, by (17.10.2), saying in addition that Y' has relative dimension $n-c$ over S at x' means that $\mathscr{N}_{Y'/X'}$, which is locally free near x', has rank c at x'. By the same reasoning and the hypotheses on X and Y, $\mathscr{N}_{Y/X}$ is locally free of rank c at x; hence $g^*(\mathscr{N}_{Y/X})$ is locally free of rank c at x'. Since $\mathrm{gr}_1(g) : g^*(\mathscr{N}_{Y/X}) \to \mathscr{N}_{Y'/X'}$ is surjective, the preceding conditions are equivalent to its being bijective at x' (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. to prop. 6), and therefore also in a neighbourhood of x' ($\mathbf{0}_{\mathrm{I}}$, (5.2.7)). This plainly implies b) and the final assertion of the statement, by exactness of (17.13.2.2).

Conversely, $\alpha \otimes 1$ factors as

$$g^*(\mathscr{N}_{Y/X}) \otimes k(x') \xrightarrow{\mathrm{gr}_1(g)\otimes 1} \mathscr{N}_{Y'/X'} \otimes k(x') \xrightarrow{\delta\otimes 1} \Omega^1_{X'/S} \otimes k(x').$$

Since $\mathrm{gr}_1(g)$ is surjective, injectivity of $\alpha \otimes 1$ implies both that $\delta \otimes 1$ is injective and that $\mathrm{gr}_1(g) \otimes 1$ is bijective. It follows from (17.12.1) that Y' is smooth over S at x', and $\mathrm{gr}_1(g)$ is bijective in a neighbourhood of x' (Bourbaki, *loc. cit.*). The exact sequence (17.13.2.2) then shows that $\mathscr{N}_{Y'/X'}$, isomorphic to $g^*(\mathscr{N}_{Y/X})$, has rank c at x', while $\Omega^1_{Y'/S}$ has rank $n-c$ there. By (17.10.2), this completes the proof of the equivalence of a) and b).

It remains to prove the equivalence of b) and b') when x' is rational over $k(s)$; this also implies that x is rational over $k(s)$. Then $g^*(\mathscr{N}_{Y/X}) \otimes_{\mathscr{O}_{Y'}} k(x')$ identifies with $\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} k(x')$. Since the sequence

$$0 \longrightarrow \mathscr{N}_{Y/X} \longrightarrow \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_Y \longrightarrow \Omega^1_{Y/S} \longrightarrow 0$$

is exact at x and consists there of locally free $\mathscr{O}_Y$-modules, the dual of $\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} k(x')$ identifies with $\mathrm{T}_{X/S}(x)/\mathrm{T}_{Y/S}(x)$ (16.5.12). Thus injectivity in b) is equivalent to surjectivity of the transposed homomorphism in b'). □

Definition (17.13.3). — With the notation of (17.13.1), we say that the morphism f is *transversal* to Y at x', relative to S, if X and Y are smooth over S at x, X' is smooth over S at x', and the equivalent conditions a) and b) of (17.13.2) hold. When there is no confusion to fear, we suppress the mention of the prescheme S and simply say that f is transversal to Y at the point x'.

Remarks (17.13.4). — (i) Suppose that X, Y and X' are flat and locally of finite presentation on S. For every $s \in S$, denote X_s, Y_s, X'_s, Y'_s the fibres of X, Y, X', Y' at the point s, $f_s : X'_s \to X_s$ the morphism deduced from f by change of base. It follows from (17.5.1) that for f to be transversal to Y at x', it is necessary and sufficient that, if s is the image of x in S, f_s be transversal to Y_s at x', relative to $k(s)$.[31]

[31][Trans.] The original reference (17.5.9) is an erratum in the original; there is no item 17.5.9. The correct reference is (17.5.1).

(ii) We saw in (17.13.2) that the set of points $x' \in Y'$ at which f is transversal to Y is open in Y'. If Y' is moreover *proper* over S, it follows that the set of $s \in S$ such that f is transversal to Y at all points of Y'_s, relative to S, is open in S.

When X, Y, and X' are flat and locally of finite presentation over S, it follows from (i) that the set of $x' \in Y'$ such that f_s is transversal to Y_s at x' (where s is the image of x' in S), relative to $k(s)$, is open in Y'. If, in addition, Y' is proper over S, the set of $s \in S$ such that f_s is transversal to Y_s at all points of Y'_s (relative to $k(s)$) is open in S.

(iii) The property of being smooth over S at a point, as well as the notion of relative dimension over S at a point, is stable under every base change $S' \to S$ ((17.3.3) and (4.2.7)); the same is therefore true of the property that a morphism $f : X' \to X$ be transversal to a subprescheme of X at a point, relative to S.

(iv) Condition b) of (17.13.2) also says that the homomorphism $\alpha : g^*(\mathscr{N}_{Y/X}) \to \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'}$ is *universally injective* relative to Y' (11.9.18).

(17.13.5). Consider a prescheme S, three S-preschemes X, Y, and Z, and two S-morphisms $f : Y \to X$, $g : Z \to X$; put $T = Y \times_X Z$. By (**I**, 5.3.5), there is a commutative diagram IV | 92

(17.13.5.1)
$$\begin{tikzcd} X \arrow[d, "\Delta"] & T = Y \times_X Z \arrow[l] \arrow[d, "j"] \\ X \times_S X & Y \times_S Z \arrow[l, "u"'] \end{tikzcd}$$

which makes T the product of the $(X \times_S X)$-preschemes X and $Y \times_S Z$, where $u = f \times_S g$. Since Δ is an S-immersion (**I**, 5.3.9), we are in the situation of diagram (17.13.1.1). The object corresponding there to $\Omega^1_{X'/S}$ is, by (16.4.23),

$$(\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y \times_S Z}) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_{Y \times_S Z}).$$

On the other hand, the object corresponding to the $\mathscr{O}_Y$-module $\mathscr{N}_{Y/X}$ in (17.13.1) is here, by definition, $\Omega^1_{X/S}$ (16.3.1). Thus (17.13.1.3) becomes the exact sequence

(17.13.5.2)
$$\Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T \xrightarrow{\rho} (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_T) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_T) \xrightarrow{\sigma} \Omega^1_{T/S} \longrightarrow 0$$

where it remains to describe ρ and σ explicitly. First, taking account of (16.4.23) and (**0**, 20.5.2), if $p : T \to Y$ and $q : T \to Z$ are the canonical projections, then, with the notation of (16.4.19),

(17.13.5.3)
$$\sigma = p_{T/Y/S} + q_{T/Z/S}.$$

To evaluate ρ, use the commutativity of the left square in (17.13.1.2). In the present case, this first requires the canonical homomorphism $\rho' : \Omega^1_{X/S} \to \Omega^1_{(X \times_S X)/S} \otimes_{\mathscr{O}_{X \times_S X}} \mathscr{O}_X$ defined in (16.4.21) applied to the immersion Δ. We may reduce to the affine case $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$. Then ρ' corresponds to the homomorphism δ of (**0**, 20.5.11.2), with B there replaced by $B \otimes_A B$ and C by $B = (B \otimes_A B)/\mathfrak{J}_{B/A}$. It sends the class of $x \otimes 1 - 1 \otimes x \pmod{\mathfrak{J}^2_{B/A}}$, for $x \in B$, to the image of

$$(x \otimes 1 - 1 \otimes x) \otimes (1 \otimes 1) - (1 \otimes 1) \otimes (x \otimes 1 - 1 \otimes x)$$

in $\Omega^1_{(B \otimes_A B)/A} \otimes_{B \otimes_A B} B$. The preceding element may be written

$$\begin{aligned} &((x \otimes 1) \otimes (1 \otimes 1) - (1 \otimes 1) \otimes (x \otimes 1)) \\ &\quad - ((1 \otimes x) \otimes (1 \otimes 1) - (1 \otimes 1) \otimes (1 \otimes x)). \end{aligned}$$

Thus ρ' is the difference of the two homomorphisms $\pi^{(1)}$ and $\pi^{(2)}$ from $\Omega^1_{X/S}$ to $\Omega^1_{(X \times_S X)/S} \otimes_{\mathscr{O}_{X \times_S X}} \mathscr{O}_X$ corresponding, by (16.4.3.3), to the first and second projections $X \times_S X \to X$. To obtain ρ, consider the homomorphism $\rho'' : \Omega^1_{(X \times_S X)/S} \otimes_{\mathscr{O}_{X \times_S X}} \mathscr{O}_{Y \times_S Z} \to \Omega^1_{(Y \times_S Z)/S}$ corresponding by (16.4.3.3) to the morphism u of (17.13.5.1); after tensoring with $\mathscr{O}_T$, form the composite $(\rho'' \otimes 1) \circ (\rho' \otimes 1)$. It follows from the preceding computation that, with the notation of (16.4.18),

(17.13.5.4)
$$\rho = (f_{Y/X/S} \otimes 1_{\mathscr{O}_T}) \oplus (-g_{Z/X/S} \otimes 1_{\mathscr{O}_T}).$$

Applying (17.13.2) to diagram (17.13.5.1) (taking account of (17.3.3), (iv), which shows that $X \times_S X$ is smooth over S at (x, x) when X is smooth over S at x) gives the following corollary. IV | 93

Corollary (17.13.6). — *Let S be a prescheme, X, Y, Z three S-preschemes, $f : Y \to X$, $g : Z \to X$ two S-morphisms; set $T = Y \times_X Z$, and let $p : T \to Y$, $q : T \to Z$ be the canonical projections. Let t be a point of T, $y = p(t)$, $z = q(t)$, $x = f(y) = g(z)$. Suppose that X is smooth over S at the point x, of relative dimension m, Y (resp. Z) smooth over S at the point y (resp. z), of relative dimension $m + a$ (resp. $m + b$), a and b being positive or negative integers. Then the following conditions are equivalent:*

a) T is smooth over S at the point t, of relative dimension $m + a + b$.

b) The homomorphism

$$\rho \otimes 1 : \Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(t) \longrightarrow (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} k(t)) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} k(t))$$

where ρ is given by (17.13.5.4), is injective.

c) The morphism $u : Y \times_S Z \to X \times_S X$ is transversal to the diagonal of $X \times_S X$ at the point t.

If t is rational over the residue field of its image s in S, these conditions are also equivalent to the following:

b′) The homomorphism

$$\mathrm{T}_y(f) - \mathrm{T}_z(g) : \mathrm{T}_{Y/S}(y) \oplus \mathrm{T}_{Z/S}(z) \longrightarrow \mathrm{T}_{X/S}(x) \tag{17.13.6.1}$$

(cf. (16.5.13.5)) is surjective.[32]

Moreover, when conditions a) *and* b) *are verified at t, they are so in a neighbourhood of t in T, and restricting T to such a neighbourhood, the sequence*

$$0 \longrightarrow \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T \xrightarrow{\rho} (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_T) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_T) \xrightarrow{\sigma} \Omega^1_{T/S} \longrightarrow 0 \tag{17.13.6.2}$$

(where σ is given by (17.13.5.3)) is exact.

The only point in (17.13.6) still requiring proof is b′); it follows from the fact that, when t is rational over $k(s)$, the homomorphism $\rho \otimes 1$ is the transpose *of $\mathrm{T}_y(f) - \mathrm{T}_z(g)$, by (17.13.5.4).*

When the equivalent conditions of (17.13.6) hold, we say that f and g form a pair of transversal morphisms at t, relative to S; *here again, the mention of S is often omitted.*

(17.13.7). Consider in particular the case where Y and Z are *sub-preschemes* of X, f and g being the canonical injections; T is then the sub-prescheme "intersection" $\inf(Y, Z)$ of X (**I**, 4.4.3), and one has $t = y = z = x$. Instead of saying that the couple (f, g) is transversal at the point x, one says then that Y and Z *cut transversally at the point x* (relatively to S). Let X_s, Y_s, Z_s, and T_s denote the fibres **IV** | 94
of X, Y, Z, and T at s, the image of x in S. Taking account of (5.2.3), (5.1.9), and (**0**, 16.5.12), we see that, when X, Y, and Z are smooth over S at x, they cut transversally at x if and only if T is smooth over S at x and

$$\operatorname{codim}_x(T_s, X_s) = \operatorname{codim}_x(Y_s, X_s) + \operatorname{codim}_x(Z_s, X_s). \tag{17.13.7.1}$$

Proposition (17.13.8). — *Let S be a prescheme, X an S-prescheme locally of finite presentation over S, Y, Z two sub-preschemes of X, $T = \inf(Y, Z)$ the sub-prescheme "intersection" of Y and Z, x a point of T. The following conditions are equivalent:*

a) The canonical injection $i : Y \to X$ is a transversal morphism to Z at the point x, relatively to S (17.13.3).

a′) The canonical injection $j : Z \to X$ is a transversal morphism to Y at the point x, relatively to S.

a″) Y and Z cut transversally at the point x, relatively to S.

b) X, Y, and Z are smooth over S at x, and the homomorphism

$$\rho \otimes 1 : \Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x) \longrightarrow (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} k(x)) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} k(x))$$

is injective.

When x is rational over the residue field $k(s)$ (s image of x in S), these conditions are also equivalent to the following:

b′) The homomorphism

$$\mathrm{T}_x(i) - \mathrm{T}_x(j) : \mathrm{T}_{Y/S}(x) \oplus \mathrm{T}_{Z/S}(x) \longrightarrow \mathrm{T}_{X/S}(x)$$

is surjective.

[32] The printed text cites (16.5.12.5); the tangent map used here is defined in (16.5.13.5).

Moreover, when conditions a) *and* b) *are verified at x, they are so in a neighbourhood of x in T, and restricting X to a neighbourhood of x, the sequence* (17.13.6.2) *is exact, and one has a canonical isomorphism*

(17.13.8.1)
$$\mathscr{N}_{T/X} \xrightarrow{\sim} (\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_T) \oplus (\mathscr{N}_{Z/X} \otimes_{\mathscr{O}_Z} \mathscr{O}_T).$$

Proof. Conditions a), a'), and a'') all imply that X, Y, and Z are smooth over S at x. Let m, $m-a$, and $m-b$ be their respective relative dimensions over S at x. Applying (17.13.2) with Z in place of X' and $T = Y \times_X Z$ in place of Y', conditions a) and a') are both equivalent to saying that T is smooth over S at x and has relative dimension $m-a-b$ there. By (17.13.7), this is precisely condition a''); hence a), a'), and a'') are equivalent. The equivalence of a'') and b) (and of b') when x is rational over $k(s)$), the persistence of these conditions in a neighbourhood of x, and the exactness of (17.13.6.2) there were proved in (17.13.6). It remains to define the canonical isomorphism (17.13.8.1). Denote by α and $-\beta$ the two homomorphisms in the middle term of (17.13.5.4), and by γ and δ the two homomorphisms in the middle term of (17.13.5.2). Exactness of (17.13.6.2), that is, of IV | 95

$$0 \longrightarrow \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T \xrightarrow{\alpha \oplus (-\beta)} (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_T) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_T) \xrightarrow{\gamma+\delta} \Omega^1_{T/S} \longrightarrow 0,$$

means that, in the category of $\mathscr{O}_T$-modules, $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T$ is canonically the fibre product of $\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_T$ and $\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_T$ over $\Omega^1_{T/S}$, for γ and δ. The same reasoning as in (**0**, 18.1.2 and 18.1.3), replacing rings and ideals by modules and submodules, gives the commutative diagram

$$\begin{tikzcd}
 & (\mathscr{N}_{Y/X} \otimes \mathscr{O}_T) \oplus (\mathscr{N}_{Z/X} \otimes \mathscr{O}_T) \arrow[r] \arrow[d] \arrow[dr, "\iota"] & \mathscr{N}_{Z/X} \otimes \mathscr{O}_T \arrow[r, "\sim"] \arrow[d, "\beta"] & \mathscr{N}_{T/Y} \arrow[d] & \\
0 \arrow[r] & \mathscr{N}_{Y/X} \otimes \mathscr{O}_T \arrow[r] \arrow[d] & \Omega^1_{X/S} \otimes \mathscr{O}_T \arrow[r, "\alpha"] \arrow[d, "\beta"] \arrow[dr, "\varepsilon"] & \Omega^1_{Y/S} \otimes \mathscr{O}_T \arrow[r] \arrow[d, "\gamma"] & 0 \\
0 \arrow[r] & \mathscr{N}_{T/Z} \arrow[r] & \Omega^1_{Z/S} \otimes \mathscr{O}_T \arrow[r, "\delta"] \arrow[d] & \Omega^1_{T/S} \arrow[r] \arrow[d] & 0 \\
 & & 0 & 0 &
\end{tikzcd}$$

whose third and fourth rows and columns are exact by the smoothness hypotheses. The identity $\varepsilon = \gamma \circ \alpha = \delta \circ \beta$, and the fact that this is the homomorphism corresponding to the canonical injection $T \to X$, follow from (**0**, 20.5.2.7). Since the diagonal of the preceding diagram is exact, $\ker(\varepsilon)$ identifies canonically with $\mathscr{N}_{T/X}$, taking account of the fact that T is smooth over S (17.2.5). Thus the canonical isomorphism (17.13.8.1) is the inverse of ι. $\square$

(17.13.9). The results of (17.13.5) and (17.13.6) may be generalized to any finite family of S-morphisms $f_i : Y_i \to X$ ($i \in I$). Let X^I be the product of the family $(X_i)_{i \in I}$ of S-preschemes all identical to X (**I**, 3.3.5), and let $p_i : X^I \to X$ be the canonical projections. The diagonal morphism $\Delta : X \to X^I$ is defined, as in (**I**, 5.3.1), by the identities $p_i \circ \Delta = 1_X$ for every $i \in I$; it is an X-section of X^I for p_1, and hence an immersion (**I**, 5.3.11).

Let Y be the product of the S-preschemes Y_i for the composites $Y_i \xrightarrow{f_i} X \to S$, and let T be the product of the X-preschemes Y_i for the morphisms f_i. As in (**I**, 5.3.5), there is a commutative diagram

(17.13.9.1)
$$\begin{tikzcd}
X \arrow[d, "\Delta"] & T \arrow[l, "v"'] \arrow[d, "j"] \\
X^I & Y \arrow[l, "u"]
\end{tikzcd}$$

(where u is the product over S of the f_i), which makes T the product of the X^I-preschemes X and Y. By induction on the number of elements of I, (16.4.23) shows that $\Omega^1_{X^I/S}$ identifies canonically with the direct sum of the $p_i^*(\Omega^1_{X/S})$; similarly, $\Omega^1_{Y/S}$ identifies with the direct sum of the $q_i^*(\Omega^1_{Y_i/S})$, where $q_i : Y \to Y_i$ are the canonical projections. IV | 96

Suppose now that X is smooth over S at a point x, and hence in a neighbourhood of x. Then X^I is smooth at the corresponding point (17.3.3), (iv), and, after restricting X to a neighbourhood of x, (17.2.5) gives an exact sequence of locally free $\mathscr{O}_X$-modules

$$0 \longrightarrow \mathrm{gr}_1(\Delta) \longrightarrow (\Omega^1_{X/S})^I \longrightarrow \Omega^1_{X/S} \longrightarrow 0.$$

Put $\mathscr{J} = \mathrm{gr}_1(\Delta)$. The preceding sequence gives an exact sequence of locally free $\mathscr{O}_T$-modules

$$0 \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_T \longrightarrow (\Omega^1_{X/S})^I \otimes_{\mathscr{O}_X} \mathscr{O}_T \xrightarrow{\sigma} \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T \longrightarrow 0$$

which is the first row of diagram (17.13.1.2), and where σ is the canonical homomorphism sending each family of sections $(t'_i)_{i\in I}$ of $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T$ over an open subset of T to its *sum*. On the other hand, the homomorphism corresponding here to the second vertical arrow of diagram (17.13.1.2) is

(17.13.9.2)
$$\tau : (\Omega^1_{X/S})^I \otimes_{\mathscr{O}_X} \mathscr{O}_T \longrightarrow \bigoplus_{i\in I} (\Omega^1_{Y_i/S} \otimes_{\mathscr{O}_{Y_i}} \mathscr{O}_T)$$

which sends a family $(t'_i \otimes 1)_{i\in I}$, where the t'_i are sections of $\Omega^1_{X/S}$, to the family of the $((f_i)_{Y_i/X/S}(t'_i)) \otimes 1$, with the notation of (16.4.18). The homomorphism ρ corresponding to α of (17.13.1.3) is therefore the restriction of τ to $\mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_T$. With these hypotheses and notation, Proposition (17.13.2) applied to (17.13.9.1) gives the following corollary.

Corollary (17.13.10). — *Under the general hypotheses of (17.13.9), let t be a point of T, y_i ($i \in I$) its projections in the Y_i, x the point of X equal to each of the $f_i(y_i)$. Suppose that X is smooth over S at the point x and of relative dimension d, and that each of the Y_i is smooth over S at the point y_i; let $d + c_i$ (c_i a positive or negative integer) be the relative dimension of Y_i at the point y_i. Then the following conditions are equivalent:*

a) T is smooth over S at the point t, of relative dimension $d + \sum_{i\in I} c_i$.

b) The homomorphism

$$\rho \otimes 1 : \mathscr{J} \otimes_{\mathscr{O}_X} k(t) \longrightarrow \bigoplus_{i\in I} (\Omega^1_{Y_i/S} \otimes_{\mathscr{O}_{Y_i}} k(t))$$

is injective.

When t is rational over the residue field $k(s)$ (s the image of t in S), these conditions are also equivalent to the following: **IV** | 97

b′) The homomorphism transpose of $\rho \otimes 1$

$$\bigoplus_{i\in I} \mathrm{T}_{Y_i/S}(y_i) \longrightarrow (\mathrm{T}_{X/S}(x))^I / \delta(\mathrm{T}_{X/S}(x))$$

(where δ is the diagonal application) is surjective.

It suffices to note that X^I is smooth over S at the point x and of relative dimension $d \cdot \mathrm{Card}(I)$, and Y smooth over S at the point t and of relative dimension $d \cdot \mathrm{Card}(I) + \sum_{i\in I} c_i$.

When the equivalent conditions of (17.13.10) are verified, one says further that the f_i form a family of transversal morphisms at the point t, relatively to S. *One sees further that the set of points $t \in T$ where this holds is* open *in T. The remark ((17.13.4), (ii)) shows moreover that if X and the Y_i are flat and locally of finite presentation over S, and if moreover T is* proper *over S, the set of $s \in S$ such that the f_i form a family of transversal morphisms at* all *the points of T_s, relatively to S, is* open *in S.*

(17.13.11). Consider in particular the case, generalising (17.13.7), where the Y_i ($i \in I$) are *sub-preschemes* of X, the f_i being the canonical injections, so that $T = \inf_{i\in I}(Y_i)$ is also the sub-prescheme "intersection" of the Y_i, and $t = y_i = x$ for every i; instead of saying that the f_i form a family of transversal morphisms at the point x, one says further that the Y_i *cut transversally at the point x* (relatively to S). The condition a) of (17.13.10) is expressed further by the relation that generalises (17.13.7.1):

(17.13.11.1)
$$\mathrm{codim}_x(T_s, X_s) = \sum_{i\in I} \mathrm{codim}_x((Y_i)_s, X_s).$$

Moreover, one has the following property which extends (17.13.8.1), and of which we give another demonstration when I has 2 elements:

Corollary (17.13.12). — *When the Y_i cut transversally at the point x, one has a canonical isomorphism*

$$\mathscr{N}_{T/X} \xrightarrow{\sim} \bigoplus_{i \in I} (\mathscr{N}_{Y_i/X} \otimes_{\mathscr{O}_{Y_i}} \mathscr{O}_T). \tag{17.13.12.1}$$

Proof. One can reduce to the case where the Y_i are closed sub-preschemes of X, defined by quasi-coherent Ideals $\mathscr{J}_i$, so that T is defined by the Ideal $\mathscr{J} = \sum_i \mathscr{J}_i$. By definition of the conormal sheaf of an immersion (16.1.3), the canonical homomorphism $\bigoplus_i \mathscr{J}_i \to \mathscr{J}$ gives, by passage to quotients, a *surjective* homomorphism

$$\bigoplus_{i \in I} (\mathscr{N}_{Y_i/X} \otimes_{\mathscr{O}_{Y_i}} \mathscr{O}_T) \longrightarrow \mathscr{N}_{T/X}. \tag{17.13.12.2}$$

Here the $\mathscr{O}_T$-modules on both sides of (17.13.12.2) are locally free and have the same rank $\sum_i c_i$, **IV** | 98
where c_i is the rank of $\mathscr{N}_{Y_i/X}$, by (17.2.5) and condition *a*) of (17.13.10). Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. to prop. 6, therefore shows that (17.13.12.2) is bijective; (17.13.12.1) is its inverse. □

17.14. Local and infinitesimal characterisations of smooth, unramified, and étale morphisms. **IV** | 98

Proposition (17.14.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$. For f to be smooth (resp. unramified, resp. étale) at x, it is necessary and sufficient that the following condition hold:*

For every local scheme $Y' = \operatorname{Spec}(A')$ with closed point y', every morphism $h : Y' \to Y$ such that $h(y') = y$, every closed subscheme $Y'_0 = \operatorname{Spec}(A'/\mathfrak{J}')$ of Y', where $(\mathfrak{J}')^2 = 0$, and every Y-morphism $g_0 : Y'_0 \to X$ such that $g_0(y') = x$, there exists at least one *(resp.* at most one*, resp.* one and only one*) Y-morphism $g : Y' \to X$ whose restriction to Y'_0 is g_0.*

Proof. In view of Definitions (17.1.1) and (17.3.1), it remains only to show that the stated condition is sufficient in the smooth and unramified cases.

(i) The smooth case. We may restrict ourselves to the affine case $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(C)$, where $C = B/\mathfrak{R}$, $B = A[\mathrm{T}_1, \dots, \mathrm{T}_n]$, and $\mathfrak{R}$ is a finite-type ideal of B. To prove that f is smooth at x, it is enough to prove that $\mathscr{O}_{X,x}$ is formally smooth over $\mathscr{O}_{Y,y}$ for the discrete topologies (17.5.1). Now $\mathscr{O}_{X,x} = B_\mathfrak{q}/\mathfrak{R}B_\mathfrak{q}$, where $\mathfrak{q}$ is a prime ideal of B; if $\mathfrak{p}$ is its inverse image in A, then $B_\mathfrak{q}$ is formally smooth over $A_\mathfrak{p}$ for the discrete topologies (**0**, 19.3.2 and 19.3.5). By the Jacobian criterion (**0**, 22.6.1 and 20.5.12), it is therefore enough to show that $B_\mathfrak{q}/\mathfrak{R}^2B_\mathfrak{q}$ is an $A_\mathfrak{p}$-trivial extension of $B_\mathfrak{q}/\mathfrak{R}B_\mathfrak{q}$ by $\mathfrak{R}B_\mathfrak{q}/\mathfrak{R}^2B_\mathfrak{q}$. But this is precisely the hypothesis applied to

$$A' = B_\mathfrak{q}/\mathfrak{R}^2B_\mathfrak{q}, \qquad \mathfrak{J}' = \mathfrak{R}B_\mathfrak{q}/\mathfrak{R}^2B_\mathfrak{q},$$

and to the morphism g_0 corresponding to the identity automorphism of $A'/\mathfrak{J}' = \mathscr{O}_{X,x}$.

(ii) The unramified case. Put $A = \mathscr{O}_{Y,y}$ and $B = \mathscr{O}_{X,x}$. By (17.4.1), c'), it is enough to prove that $\Omega^1_{B/A} = 0$. With the notation of (**0**, 20.4.1), the ring $C = \mathscr{P}^1_{B/A} = (B \otimes_A B)/\mathfrak{J}^2$ is local, since $C/(\mathfrak{J}/\mathfrak{J}^2)$ is isomorphic to B. Applying the hypothesis to $A' = C$ and $\mathfrak{J}' = \mathfrak{J}/\mathfrak{J}^2$ shows, by definition, that the map $d_{B/A} : B \to \Omega^1_{B/A}$ is zero (**0**, 20.4.6); hence $\Omega^1_{B/A} = 0$ by (**0**, 20.4.7). □

Proposition (17.14.2). — *Let Y be a prescheme locally Noetherian, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. For f to be smooth (resp. unramified, resp. étale) at x, it is necessary and sufficient that the condition of (17.14.1) hold when one additionally requires that A' be Artinian, that $\mathfrak{m}'\mathfrak{J}' = 0$, where $\mathfrak{m}'$ is the maximal ideal of A', and that the residue field $A'/\mathfrak{m}'$ be equal to $k(x)$.*

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Proof. It remains again only to prove sufficiency in the smooth and unramified cases.

(i) The smooth case. Put $A = \mathscr{O}_{Y,y}$ and $B = \mathscr{O}_{X,x}$. In view of (17.5.3), the point is to prove that B is a formally smooth A-algebra for the preadic topologies. By (**0**, 22.1.4), this follows from the hypothesis if, in it, the condition $\mathfrak{m}'\mathfrak{J}' = 0$ is replaced by $\mathfrak{J}' \subset \mathfrak{m}'$. Since $\mathfrak{m}'$ is nilpotent, however, the condition in the statement is already sufficient: consider the rings $A'/\mathfrak{m}'^j\mathfrak{J}'$ and argue by induction on j, as in (**0**, 19.4.3).

(ii) The unramified case. Resume the notation of the proof of (17.14.1), (ii). To prove that the ideal $\mathfrak{J}/\mathfrak{J}^2$ of C is zero, note first that C is a Noetherian local ring: it is the local ring of an affine open

subscheme of $X \times_Y X$, which is locally of finite type over X, and therefore also over Y. If $\mathfrak{r}$ is the maximal ideal of C, it is consequently enough to verify that

$$\mathfrak{J}/\mathfrak{J}^2 \subset \mathfrak{r}^n \quad \text{for every } n \qquad (\mathbf{0}_{\mathrm{I}}, 7.3.5),$$

or, equivalently, that $\mathfrak{r}^n = \mathfrak{r}^n + (\mathfrak{J}/\mathfrak{J}^2)$. With the notation of (**0**, 20.4.6), this says that for every n the two composite maps

$$B \xrightarrow{p_1} C \longrightarrow C/\mathfrak{r}^n, \qquad B \xrightarrow{p_2} C \longrightarrow C/\mathfrak{r}^n$$

are identical. But $C/\mathfrak{r}^n$ is Artinian, and this identity follows by induction on n from the hypothesis applied to

$$A' = C/\mathfrak{r}^n, \qquad \mathfrak{J}' = \big(\mathfrak{r}^n + (\mathfrak{J}/\mathfrak{J}^2)\big)/\mathfrak{r}^n.$$

□

Remark (17.14.3). — Under the hypotheses of (17.14.2), if in addition the residue field $k(x)$ is a *finite* extension of $k(y)$, then the A'-modules $\mathfrak{m}'^j/\mathfrak{m}'^{j+1}$ are finite-dimensional $k(y)$-vector spaces; hence, *a fortiori*, A' is a finite $\mathscr{O}_{Y,y}$-algebra.

17.15. The case of preschemes over a base field. We first recall (6.7.7), (6.7.8), and (6.8.1):

Proposition (17.15.1). — *Let k be a field, X a prescheme locally of finite type over k, and x a point of X. For X to be smooth at x, it is necessary and sufficient that, for every radicial extension k' of k (or only for every finite radicial extension k' of k), the local ring $\mathscr{O}_{X,x} \otimes_k k'$ be regular. If $k(x)$ is a separable extension of k (in particular if k is perfect), this amounts to saying that $\mathscr{O}_{X,x}$ is regular.*

Corollary (17.15.2). — *Under the conditions of (17.15.1), for X to be smooth over k, it is necessary and sufficient that X be geometrically regular over k (6.7.6); this implies that X is reduced. If k is perfect, X is smooth over k if and only if X is regular.*

Proposition (17.15.3). — *Let k be a field, X a prescheme locally of finite type over k, x a point of X, and $n = \dim_x(X)$. Let $(g_i)_{1 \leqslant i \leqslant n}$ be a family of n sections of $\mathscr{O}_X$ over X, and let $f : X \to Y = \operatorname{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n])$ be the morphism corresponding to the k-homomorphism $k[\mathrm{T}_1, \dots, \mathrm{T}_n] \to \Gamma(X, \mathscr{O}_X)$ sending T_i to g_i* (**I**, 2.2.4). *The following conditions are equivalent (and imply that X is smooth over k at x, and therefore regular at x):*

a) f is étale at the point x.
b) The images $d_{X/k}g_i$ form a basis of the $\mathscr{O}_{X,x}$-module $(\Omega^1_{X/k})_x$.
c) The images $d_{X/k}g_i$ generate the $\mathscr{O}_{X,x}$-module $(\Omega^1_{X/k})_x$.

If f is étale at x, then X is smooth over k at x, since $Y = \operatorname{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n])$ is smooth over k (17.3.8); the implication $a) \Rightarrow b)$ is therefore a special case of (17.11.3). As $b)$ trivially implies $c)$, it remains to see that $c)$ implies $a)$.

Proof. First note that hypothesis $c)$ makes the homomorphism $(f^*\Omega^1_{Y/k})_x \to (\Omega^1_{X/k})_x$ surjective. After replacing X by an open neighbourhood of x, we may therefore suppose that $f^*\Omega^1_{Y/k} \to \Omega^1_{X/k}$ is surjective (Bourbaki, *Alg. comm.*, chap. II, § 5, n° 1, prop. 2); hence f is unramified (17.2.2). Let us first show that one can reduce to the case where x is rational over k. Indeed, put $k' = k(x)$, $X' = X \otimes_k k'$, and $Y' = Y \otimes_k k' = \operatorname{Spec}(k'[\mathrm{T}_1, \dots, \mathrm{T}_n])$. There exists a point $x' \in X'$ above x such that $k(x') = k'$. To prove that f is étale at x, it suffices to show that $f' = f_{(k')} : X' \to Y'$ is étale at x' (17.7.1), (ii); moreover, f' is unramified (17.3.3), (iii), and $\dim_{x'}(X') = n$ (4.2.7). In the same way, replacing k' by an algebraically closed extension of k', one can suppose that k is algebraically closed. Now put $y = f(x)$, $A = \mathscr{O}_{Y,y}$, and $B = \mathscr{O}_{X,x}$. Since the residue field of B is k, the same is true of the residue field of A; thus x (resp. y) is a closed point of X (resp. Y) (**I**, 6.4.2), and consequently $\dim A = \dim B = n$. Since f is unramified, the homomorphism $\widehat{A} \to \widehat{B}$ is surjective (17.4.4), f''). But $\dim \widehat{A} = \dim \widehat{B} = n$, and $\widehat{A}$, being regular, is an integral domain; hence $\widehat{A} \to \widehat{B}$ is also injective (**0**, 16.3.10). It is therefore bijective, which completes the proof that f is étale at x (17.6.3), e''). □

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Corollary (17.15.4). — *Under the general hypotheses of (17.15.3), suppose moreover that $k(x)$ is a finite and separable extension of k and that the germs $(g_i)_x$ belong to $\mathfrak{m}_x$. Then the conditions $a)$, $b)$, $c)$ of (17.15.3) are also equivalent to each of the following:*

d) The n germs $(g_i)_x$ generate the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_{X,x}$.

*d′) The ring $\mathcal{O}_{X,x}$ is regular and the $(g_i)_x$ form a regular system of parameters for this ring (**0**, 17.1.6).*

Proof. Clearly $d')$ implies $d)$. Since $k(x)$ is a finite separable extension of k, one has $\Omega^1_{k(x)/k} = 0$ (**0**, 20.6.20), and $k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ is a formally smooth k-algebra for the discrete topologies (**0**, 19.6.1). The exact sequence (**0**, 20.5.14.1), applied with $A = k$, $B = \mathcal{O}_{X,x}$, and $\mathfrak{R} = \mathfrak{m}_x$, therefore gives the canonical isomorphism

$$\delta : \mathfrak{m}_x/\mathfrak{m}_x^2 \xrightarrow{\sim} (\Omega^1_{X/k})_x \otimes_{\mathcal{O}_{X,x}} k(x).$$

It follows first, by Nakayama's lemma, that $c)$ and $d)$ are equivalent. If f is étale at x, the ring $\mathcal{O}_{X,x}$ is regular of dimension n, since x is closed (**I**, 6.4.2); and n elements which generate its maximal ideal necessarily form a regular system of parameters (**0**, 17.1.6). Thus $a)$ implies $d')$. □

Proposition (17.15.5). — *Let k be a field, X a prescheme locally of finite type over k, x a point of X, and $n = \dim_x(X)$. The following conditions are equivalent:*

a) X is smooth over k at x.

b) X is differentially smooth over k at x.

c) $(\Omega^1_{X/k})_x$ is a free $\mathcal{O}_{X,x}$-module of rank n.

d) $(\Omega^1_{X/k})_x$ admits a system of n generators as an $\mathcal{O}_{X,x}$-module. **IV** | 101

e) There exist a perfect extension k' of k and an open neighbourhood U of x in X such that the prescheme $U \otimes_k k'$ is regular.

Proof. If X is smooth over k at x, there is an open neighbourhood U of x which is smooth over k; consequently $U \otimes_k k'$ is regular for every extension k' of k (17.15.2). This proves $a) \Rightarrow e)$. Conversely, $e)$ implies $a)$, because $U \otimes_k k'$ is then smooth over the perfect field k' (17.15.2), and hence U is smooth over k (17.7.1), (ii).

We have already proved $a) \Rightarrow c)$ (17.10.2); clearly $c) \Rightarrow d)$, and $d) \Rightarrow a)$ follows from (**0**, 20.4.7) and (17.15.3), $c)$.

Finally, $a) \Rightarrow b)$ was proved in (17.12.4). Conversely, to prove $b) \Rightarrow a)$, we may reduce to the case where x is rational over k: as in the proof of (17.15.3), take a point x' of $X' = X \otimes_k k'$, above x, with $k(x') = k' = k(x)$, using again (17.7.1), (ii), and the fact that differential smoothness at x implies differential smoothness of X' at x' (16.10.4). When x is rational over k, the smoothness of f at x follows from $b)$ and (17.12.5), applied to the k-section $u : \operatorname{Spec}(k) \to X$ such that $u(\operatorname{Spec}(k)) = \{x\}$. □

Corollary (17.15.6). — *Let X be a prescheme of finite type over a field k. For X to be smooth over k, it is necessary and sufficient that the $\mathcal{O}_X$-module $\Omega^1_{X/k}$ be locally free and that the local rings of X at its maximal points be fields which are separable extensions of k (the latter condition is automatically satisfied if k is perfect and X is* reduced*).*

Proof. The conditions are necessary. Indeed, if X is smooth over k, then $\Omega^1_{X/k}$ is locally free by (17.10.2). Moreover, X is regular and hence, *a fortiori*, reduced; at every maximal point x of X, therefore, $\mathcal{O}_{X,x}$ is a field. This field must be a formally smooth k-algebra for the discrete topologies (17.5.1), and is consequently a separable extension of k (**0**, 19.6.1).

The conditions are sufficient. We may reduce to the case where X is connected; then $\Omega^1_{X/k}$ is locally free of constant rank n. At every maximal point x of X one has $\operatorname{rank}_{k(x)} \Omega^1_{k(x)/k} = n$, since $\mathcal{O}_{X,x} = k(x)$. By hypothesis $k(x)$ is separable over k, so $\Upsilon_{k(x)/k} = 0$ (**0**, 20.6.3), and Cartier's equality gives $\operatorname{deg.tr}_k k(x) = n$ (**0**, 21.7.1). All irreducible components of X therefore have the same dimension n (5.2.1), and the implication $c) \Rightarrow a)$ of (17.15.5) shows that X is smooth over k at every point. □

Corollary (17.15.7). — *If k has characteristic* 0*, then X is smooth over k at x if and only if $(\Omega^1_{X/k})_x$ is a free $\mathcal{O}_{X,x}$-module.*

Proof. This results from (16.12.2) and the equivalence of $b)$ and $a)$ in (17.15.5). □

Proposition (17.15.8). — *Let k be a field, X a prescheme locally of finite type over k, and x a point of X; put $n = \dim_x(X)$ and $r = \dim(\mathcal{O}_{X,x})$, so that $n - r = \operatorname{deg.tr}_k k(x)$ (5.2.3). Let $(g_i)_{1 \leqslant i \leqslant n}$ be a family* *of n sections of $\mathscr{O}_X$ over X, with $(g_i)_x \in \mathfrak{m}_x$ for $1 \leqslant i \leqslant r$; let $f : X \to Y = \mathrm{Spec}(k[T_1, \dots, T_n])$ be the morphism corresponding to the k-homomorphism which sends T_i to g_i. The following conditions are equivalent (and imply that X is smooth over k at x and that $k(x)$ is a separable extension of k):*

a) f is étale at the point x. IV | 102

b) The germs $(g_i)_x$ for $1 \leqslant i \leqslant r$ generate $\mathfrak{m}_x$ (and hence form a regular system of parameters for $\mathscr{O}_{X,x}$), and the images in $\Omega^1_{k(x)/k}$ of the elements $(d_{X/k}g_i)_x$ for $r+1 \leqslant i \leqslant n$ generate $\Omega^1_{k(x)/k}$.

Proof. There is an exact sequence of $k(x)$-modules (**0**, 20.5.12.1)

(17.15.8.1) $$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x) \longrightarrow \Omega^1_{k(x)/k} \longrightarrow 0.$$

Condition b) and Nakayama's lemma therefore imply that the $(d_{X/k}g_i)_x$ generate $(\Omega^1_{X/k})_x$; hence $b) \Rightarrow a)$ by (17.15.3).

Conversely, suppose a) holds. By (17.15.3), the $(d_{X/k}g_i)_x$, $1 \leqslant i \leqslant n$, form a basis of $(\Omega^1_{X/k})_x$. If t_i is the image of $(g_i)_x$ in $k(x)$, it follows that the dt_i generate $\Omega^1_{k(x)/k}$. Since $t_i = 0$ for $1 \leqslant i \leqslant r$, the dt_i for $r+1 \leqslant i \leqslant n$ already generate $\Omega^1_{k(x)/k}$. As $\deg.\mathrm{tr}_k\, k(x) = n-r$, Cartier's equality gives $\Upsilon_{k(x)/k} = 0$; thus $k(x)$ is separable over k (**0**, 20.6.3), and the sequence

(17.15.8.2) $$0 \longrightarrow \mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x) \longrightarrow \Omega^1_{k(x)/k} \longrightarrow 0$$

is exact ((**0**, 20.5.14) and (**0**, 19.6.1)). Moreover the dt_i, $r+1 \leqslant i \leqslant n$, form a basis of $\Omega^1_{k(x)/k}$, so none of the corresponding t_i is zero. It follows that the images of the $(g_i)_x$, $1 \leqslant i \leqslant r$, generate $\mathfrak{m}_x/\mathfrak{m}_x^2$; by Nakayama's lemma the $(g_i)_x$, $1 \leqslant i \leqslant r$, generate $\mathfrak{m}_x$. This proves $a) \Rightarrow b)$. □

Corollary (17.15.9). — *Let X be a prescheme locally of finite type over a field k, and x a point of X; put $n = \dim_x(X)$ and $r = \dim(\mathscr{O}_{X,x})$. The following conditions are equivalent:*

a) The ring $\mathscr{O}_{X,x}$ is regular and $k(x)$ is a separable extension of k.

b) X is smooth over k at x, and the canonical homomorphism

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x)$$

is injective.

c) There exist n sections g_i of $\mathscr{O}_X$ over an open neighbourhood U of x, with $(g_i)_x \in \mathfrak{m}_x$ for $1 \leqslant i \leqslant r$, such that the morphism $U \to \mathrm{Spec}(k[T_1, \dots, T_n])$ corresponding to the g_i (cf. (17.15.8)) is étale at x.

d) There exist n sections g_i $(1 \leqslant i \leqslant n)$ of $\mathscr{O}_X$ over an open neighbourhood U of x, such that the $(g_i)_x$, $1 \leqslant i \leqslant r$, generate $\mathfrak{m}_x$, and the images in $\Omega^1_{k(x)/k}$ of the $(d_{U/k}g_i)_x$, $r+1 \leqslant i \leqslant n$, generate $\Omega^1_{k(x)/k}$.

Proof. The equivalence of c) and d) follows from (17.15.8). Its proof also shows that c) implies that X is smooth over k at x and that the sequence (17.15.8.2) is exact; hence $c) \Rightarrow b)$.

Condition b) implies that $\mathscr{O}_{X,x}$ is regular, so $\mathfrak{m}_x/\mathfrak{m}_x^2$ has rank r; moreover $(\Omega^1_{X/k})_x$ is then free of rank n (17.10.2). Since the sequence (17.15.8.1) is exact by hypothesis, $\Omega^1_{k(x)/k}$ has rank $n-r = \deg.\mathrm{tr}_k\, k(x)$. Cartier's equality therefore gives $\Upsilon_{k(x)/k} = 0$, and $k(x)$ is separable over k (**0**, 20.6.3). Thus $b) \Rightarrow a)$.

Finally, if a) holds, Cartier's equality again shows that $\Omega^1_{k(x)/k}$ has rank $n-r$. Since $\mathscr{O}_{X,x}$ is regular, the existence of sections g_i satisfying d) is immediate. □ IV | 103

Remarks (17.15.10). — (i) For a prescheme X of finite type over k to be smooth over k, it is not enough that $\Omega^1_{X/k}$ be locally free. This is shown by $X = \mathrm{Spec}(K)$, where K is a finite inseparable extension of k.

(ii) When k is not perfect, X may be smooth over k even though $k(x)$ is not separable over k. For example, take $X = \mathrm{Spec}(k[T])$ and let x correspond to the principal prime ideal $(T^p - \lambda)$, where $p > 0$ is the characteristic of k and $\lambda \in k - k^p$.

(iii) Nevertheless, if X is smooth over k, the set of closed points x of X for which $k(x)$ is a separable (and finite) extension of k is *dense* in X. Indeed, let $f : X \to \mathrm{Spec}(k)$ be the structural morphism. For every $x_0 \in X$, there is an open neighbourhood U of x_0 and a factorisation

$$f|U : U \xrightarrow{g} \mathrm{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n]) \xrightarrow{h} \mathrm{Spec}(k),$$

in which g is étale (17.11.4). We may restrict to the case where k is not perfect, and hence is infinite. The open set $g(U)$ then contains a point y rational over k; if $x \in U$ lies above y, then x is closed in X and $k(x)$ is separable over $k(y) = k$ (17.6.2).

Proposition (17.15.11). — *Let X be a prescheme of finite type over a field k. The following conditions are equivalent:*

a) X is étale over k.

b) X is unramified over k.

c) X is isomorphic to $\mathrm{Spec}(A)$, where A is a finite and separable k-algebra.

Proof. This follows from (17.6.2) and (17.4.2), noting that condition b) here implies that X is finite over k. □

Proposition (17.15.12). — *Let k be a field, X a prescheme locally of finite type over k. There exists an everywhere dense open subset U of X which is smooth over k if and only if, at every maximal point x of X, the prescheme X is reduced and $k(x)$ is a separable extension of k. There exists an everywhere dense open subset U of X such that U_{red} is smooth over k if and only if $k(x)$ is a separable extension of k for every maximal point x of X.*

Proof. The second assertion follows immediately from the first. To say that there is an everywhere dense open subset U smooth over k means that X is smooth over k at each of its maximal points x. Necessarily $\mathscr{O}_{X,x}$ is regular (17.15.1), and *a fortiori* reduced; it is therefore the field $k(x)$. Moreover, (17.15.1) requires $k(x) \otimes_k k'$ to be regular for every radicial extension k' of k, so $k(x)$ must be separable over k (4.3.5). The converse follows at once from (17.15.1). □

Corollary (17.15.13). — *Let X be an algebraic prescheme over a field k. Then there exist an everywhere dense open subset U of X and a finite radicial extension k' of k such that $(U_{(k')})_{\mathrm{red}}$ is smooth over k'.*

Proof. There is such an extension k' for which $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k' (4.6.6); **IV** | 104
equivalently (4.6.1), $k(x')$ is separable over k' at every maximal point x' of $X_{(k')}$. It follows from (17.15.12) that there is an everywhere dense open subset U' of $X_{(k')}$ such that U'_{red} is smooth over k'. The morphism $X_{(k')} \to X$ is a homeomorphism (2.4.5), so $U' = U_{(k')}$ for an everywhere dense open subset U of X. □

Proposition (17.15.14). — *Let X be an algebraic prescheme over a field k, of dimension $\leqslant 1$. Then there exists a finite radicial extension k' of k such that the normalization (**II**, 6.3.8) of $(X_{(k')})_{\mathrm{red}}$ is smooth over k'.*

We first prove the following two lemmas.

Lemma (17.15.14.1). — *Let X be a reduced prescheme whose set of irreducible components is locally finite, and let $f : Y \to X$ be a morphism. Then Y is X-isomorphic to the normalization X' of X (**II**, 6.3.8) if and only if the following conditions hold:*

(i) Y is normal;

(ii) f is integral and birational.

If X contains a dense normal open subset (as is always the case when X is excellent (7.8.3), in particular when X is locally of finite type over a field), condition (ii) may be replaced by:

(ii bis) f is integral, and there exists a dense open subset U of X such that $f^{-1}(U)$ is dense in Y and the restriction $f^{-1}(U) \to U$ is an isomorphism.

Proof. The question is local on X, so we may suppose that X has only finitely many irreducible components. Let X_i $(1 \leqslant i \leqslant m)$ be the reduced subpreschemes of X having these components as their underlying spaces. The normalization X' is the sum of the normalizations X'_i of the X_i (**II**, 6.3.6). Each structural morphism $f_i : X'_i \to X_i$ is integral and birational, so f is integral and birational. If X has a dense normal open subset V, condition (ii) plainly implies (ii bis), with $U = V$.

Conversely, suppose (i) and (ii) hold. The prescheme Y has only finitely many irreducible components Y_i $(1 \leqslant i \leqslant m)$, with Y_i dominating X_i for each i (6.15.4). Since Y is normal it is the sum of the Y_i, and the morphism $g_i : Y_i \to X_i$ through which the restriction $Y_i \to X$ of f factors is birational (**I**, 5.2.2). As X_i and Y_i are integral, g_i is integral, and Y_i is normal, for every affine open V of X_i the ring $\Gamma(g_i^{-1}(V), \mathscr{O}_{Y_i})$ is the integral closure of $\Gamma(V, \mathscr{O}_{X_i})$. Thus Y identifies canonically with X' (**II**, 6.3.4).

Finally, suppose (ii bis). We may take U to be a disjoint union of irreducible open subsets U_i ($1 \leqslant i \leqslant m$); then $f^{-1}(U)$ is the disjoint union of the $V_i = f^{-1}(U_i)$. Since the V_i are irreducible and $f^{-1}(U)$ is dense in Y, they are the irreducible components of Y, and f is therefore birational. This proves the lemma. □

Lemma (17.15.14.2). — *Let k be a field and X an algebraic prescheme over k. There exists a finite radicial extension k' of k such that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k' and its normalization is geometrically normal over k'.*

Proof. In view of (4.6.6) and (**I**, 5.1.8), we may reduce to the case where X is already geometrically reduced over k. Let p be the characteristic exponent of k, and let $k_1 = k^{p^{-\infty}}$ be the smallest perfect extension of k; thus k_1 is the filtered direct limit of the finite radicial extensions of k. Put $X_1 = X_{(k_1)}$, which is reduced by hypothesis, and let Y_1 be its normalization. If $f_1 : Y_1 \to X_1$ is the structural morphism, then f_1 is finite (7.8.3), (vi), and surjective.

There is a dense open subset U_1 of X_1 such that $V_1 = f_1^{-1}(U_1)$ is dense in Y_1 and $V_1 \to U_1$ is an isomorphism (17.15.14.1). **IV** | 105

Applying (8.9.1) and (8.10.5), (vi) and (x), we first obtain a finite radicial extension k'' of k and a finite surjective morphism $Y'' \to X'' = X_{(k'')}$ such that

$$Y_1 = Y'' \times_{X''} X_1 = Y'' \otimes_{k''} k_1.$$

Since Y_1 is normal and k_1 is perfect, (6.7.7) shows that Y'' is geometrically normal over k''. The projections $X_1 \to X''$ and $Y_1 \to Y''$ are integral, surjective, and radicial, hence homeomorphisms (2.4.5). If U'' and V'' are the respective images of U_1 and V_1, they are dense open subsets and $V'' = f''^{-1}(U'')$, where $f'' : Y'' \to X''$ is the structural morphism. Since $U_1 = U'' \otimes_{k''} k_1$, (8.10.5), (ii), gives a finite radicial extension k' of k'' such that, after putting

$$X' = X \otimes_k k' = X'' \otimes_{k''} k', \qquad Y' = Y'' \otimes_{k''} k',$$

and writing $f' : Y' \to X'$ for the resulting morphism, the images U' and V' of U'' and V'' satisfy that $V' \to U'$ is an isomorphism. Since Y' is normal and f' is integral and birational, (17.15.14.1) identifies Y' with the normalization of X'. This proves the lemma, because Y' is geometrically normal. □

Proof of Proposition 17.15.14. Apply (17.15.14.2) to k and X. Since $\dim X \leqslant 1$, one also has $\dim X_{(k')} \leqslant 1$ (4.1.4), and the normalization Y' of $X_{(k')}$ has dimension at most 1 ((5.4.2) and **II**, 7.4.6). By the definitions and (**II**, 7.4.5), geometric normality of Y' is then equivalent to geometric regularity over k', and hence to smoothness over k' (17.5.1). □

Proposition (17.15.15). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be smooth at a point $x \in X$, it is necessary and sufficient that f be flat at x and that Ω^1_f be a locally free $\mathscr{O}_X$-module in a neighbourhood of x, of rank at x equal to* $\dim_x f$.

Proof. Necessity follows from (17.5.1) and (17.10.2). Conversely, suppose the conditions hold and put $y = f(x)$. By (17.5.1), it is enough to show that $f^{-1}(y)$ is smooth over $k(y)$ at x; this follows from the definition of $\dim_x f$ (17.10.1), from (16.4.5), and from (17.15.5). □

17.16. Quasi-sections of flat or smooth morphisms. The statements of this number complement those of (14.5), by means of hypotheses of flatness.

Proposition (17.16.1). — *Let $f : X \to S$ be a flat morphism locally of finite presentation. Let $s \in S$, and let x be a closed point of X_s such that $\mathscr{O}_{X_s,x}$ is a Cohen–Macaulay ring. Then there exist an open neighbourhood U of s in S and a subprescheme $X' \subset f^{-1}(U)$ of X, containing x, such that the restriction $X' \to U$ of f is flat, quasi-finite, and of finite presentation.*

Proof. The question is local on X and S, so we may suppose that f is of finite presentation. Let $(t_i)_{1 \leqslant i \leqslant m}$ be a system of parameters of the local ring $\mathscr{O}_{X_s,x}$. There exist an affine open neighbourhood $V = \operatorname{Spec}(A)$ of x in X and m sections g_i of $\mathscr{O}_X$ over V such that the images of the germs $(g_i)_x \in \mathscr{O}_{X,x}$ in **IV | 106**

$$\mathscr{O}_{X_s,x} = \mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$$

are the t_i. Let X' be the closed subprescheme of V defined by the ideal of A generated by the g_i. The sequence (t_i) is regular by hypothesis (**0**, 16.5.7); it therefore follows from (11.3.8), after replacing V if necessary by a smaller open neighbourhood, that the restriction $X' \to S$ of f is flat and of finite presentation. Since the t_i form a system of parameters of $\mathscr{O}_{X_s,x}$, the ring $\mathscr{O}_{X'_s,x}$ is Artinian. As x is closed in X'_s, it is consequently isolated in X'_s. Replacing V once more by a smaller neighbourhood of x, (13.1.4) shows that $X' \to S$ is quasi-finite. □

Corollary (17.16.2). — *Let $f : X \to S$ be faithfully flat and locally of finite presentation. Then there exists a faithfully flat, locally finitely presented, and locally quasi-finite morphism $g : S' \to S$ such that there is an S-morphism $S' \to X$ (equivalently, an S'-section of $X' = X \times_S S'$;* **I***, 3.3.14). If S is quasi-compact (resp. quasi-compact and quasi-separated), S' may be chosen affine (resp. S' may be chosen affine and g of finite presentation and quasi-finite).*

Proof. For every $s \in S$, the fibre X_s is nonempty and is a prescheme locally of finite type over $k(s)$. The set U_s of points x of X_s for which $\mathscr{O}_{X_s,x}$ is Cohen–Macaulay is open in X_s (6.11.3) and nonempty, since it contains the maximal points of X_s (**0**, 16.5.1). It therefore contains a point x_s closed in X_s (10.4.7). Let $X'(s)$ be an affine subprescheme of X, containing x_s, and satisfying the conclusion of (17.16.1).

To obtain a prescheme S' satisfying the assertion, it is enough to take the sum of the $X'(s)$ as s runs through S. Indeed, since $X'(s) \to S$ is flat and locally of finite presentation, its image $U(s)$ is open in S (2.4.6). If S is quasi-compact, it can therefore be covered by finitely many of the $U(s_i)$, and the sum of the corresponding $X'(s_i)$ is still suitable and is affine. If, in addition, S is quasi-separated, the open immersions $U(s_i) \to S$ may be supposed quasi-compact (**I**, 2.7), hence of finite presentation (**I**, 6.2). The morphisms $X'(s_i) \to S$ are then of finite presentation, and so is $S' \to S$. □

Corollary (17.16.3). — *(i) Let $f : X \to S$ be smooth. Let $s \in S$, and let x be a closed point of X_s such that the residue field $k(x)$ is separable over $k(s)$. In the conclusion of (17.16.1), the restriction $X' \to U$ of f may then be chosen étale.*

(ii) Let $f : X \to S$ be smooth and surjective. In the conclusion of (17.16.2), one may moreover suppose that $g : S' \to S$ is étale.

Proof. Assertion (ii) follows from (i) just as (17.16.2) follows from (17.16.1): for every $s \in S$, the nonempty smooth $k(s)$-prescheme X_s has a closed point x for which $k(x)$ is separable over $k(s)$ (17.15.10), (iii). It remains to prove (i). The ring $\mathscr{O}_{X_s,x}$ is regular. Repeat the construction of (17.16.1), taking for (t_i) a regular system of parameters of $\mathscr{O}_{X_s,x}$. The ring $\mathscr{O}_{X'_s,x}$ is then a field isomorphic to $k(x)$, and hence separable over $k(s)$ by hypothesis. The conclusion follows from (17.6.1), c'). □

Proposition (17.16.4). — *Let S be a quasi-compact and quasi-separated prescheme, and let $f : X \to S$ be a surjective morphism locally of finite presentation. Then there exists a finite family $(S_i)_{i \in I}$ of affine subpreschemes of S, of finite presentation over S, pairwise disjoint, and with union S, having the following property: for every $i \in I$, there exist two finite, finitely presented, and surjective morphisms*

$$S''_i \xrightarrow{g_i} S'_i \xrightarrow{h_i} S_i,$$

where h_i is étale and g_i is flat and radicial, and an S-morphism $S''_i \to X$ (in other words, an S''_i-section of $X''_i = X \times_S S''_i$).

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Proof. There is a finite covering $(U_i)_{1\leqslant i\leqslant n}$ of S by affine open subsets. For every i, the intersection

$$W_i = U_i \cap \Big(\bigcup_{j<i} U_j\Big)$$

is quasi-compact (**I**, 2.7); hence there exists a closed subprescheme V_i of U_i whose underlying space is $U_i - W_i$ and which is defined by an ideal of finite type in the ring of U_i. Thus V_i is an affine subprescheme of S of finite presentation over S (**I**, 6.2). It is clear that it is enough to prove the proposition after replacing S by each V_i and X by $X \times_S V_i$. In other words, we may already suppose that S is affine.

On the other hand, X is the union of affine open subsets W_α, and the $f(W_\alpha)$ are constructible in S (**I**, 8.4) and cover S. Consequently (**I**, 9.9), there is a finite subfamily $(W_{\alpha_j})_{1\leqslant j\leqslant m}$ whose images already cover S. Let X' be the sum prescheme of the subpreschemes induced on the open subsets W_{α_j} of X. It is immediate that it is enough to prove the proposition after replacing X by X', since an S-morphism $S''_i \to X'$ gives, by composition, an S-morphism $S''_i \to X' \to X$. We may therefore suppose that X is affine and that f is of finite presentation.

We may then write (8.9.1)

$$X = X_0 \times_{S_0} S,$$

where S_0 is Noetherian and $X_0 \to S_0$ is a morphism of finite type which may be supposed surjective (8.10.5), (vi). If the proposition is proved for this morphism, it follows at once for X by the base change $S \to S_0$. We may thus suppose in addition that S is Noetherian and that f is of finite type.

Let s_h $(1 \leqslant h \leqslant r)$ be the maximal points of S. We show that it is enough to prove the assertion after replacing S by a sufficiently small open neighbourhood U_h of each s_h, and X by the inverse image of U_h in X. Indeed, suppose the proposition established in this case, and argue by Noetherian induction ($\mathbf{0_I}$, 2.2.2), assuming the assertion established for every closed subprescheme of S whose underlying space is not S. We may suppose that the U_h are pairwise disjoint. If T is a closed subprescheme whose underlying space is

$$S - \bigcup_h U_h,$$

the induction hypothesis implies that the assertion is true for T; since it is also true for each U_h, it is plainly true for S.

Since we may clearly suppose S reduced (by replacing S by S_{red} and X by $X \times_S S_{\mathrm{red}}$), each $\mathscr{O}_{S,s_h}$ is a field $k(s_h)$. Suppose that the proposition has been proved when S is the spectrum of a field and f is of finite type. The existence of the U_h then follows from the method of (8.1.2), a), together with (8.8.2), (i) and (ii), (8.10.5), (vi), (vii), and (x), (11.2.6), (ii), and (17.7.8), (ii).

We may therefore suppose that $S = \mathrm{Spec}(k)$, where k is a field, and that X is a nonempty prescheme of finite type over k. Since X is Noetherian, it has a closed point x ($\mathbf{0_I}$, 2.1.3), and $k(x)$ is then a finite extension of k (**I**, 6.4.2). There is consequently a finite separable extension k' of k such that $k'' = k(x)$ is a finite radicial extension of k'. The assertion follows by taking I to have one element, $S'_i = \mathrm{Spec}(k')$, and $S''_i = \mathrm{Spec}(k'')$. □

Corollary (17.16.5). — *Let $f : X \to S$ be a surjective morphism locally of finite presentation. Then there exists a surjective morphism $g : S' \to S$, locally of finite presentation and locally quasi-finite, such that there is an S-morphism $S' \to X$ (in other words, an S'-section of $X' = X \times_S S'$). If S is quasi-compact (resp. quasi-compact and quasi-separated), S' may be supposed affine (resp. S' may be supposed affine and g of finite presentation and quasi-finite).*

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Proof. It is enough to prove the corollary when S is affine. Indeed, S is the union of a family (S_α) of affine open subsets. If, for every α, S'_α is affine and a morphism $g_\alpha : S'_\alpha \to S_\alpha$ satisfies the assertion and is of finite presentation and quasi-finite, take for S' the sum prescheme of the S'_α, with $g : S' \to S$ agreeing with g_α on each S'_α. This morphism has the required properties. Moreover, if S is quasi-compact, the family (S_α) may be supposed finite, so S' is affine; if S is also quasi-separated, the immersions $S_\alpha \to S$ are of finite presentation (**I**, 6.2), and consequently so is g.

Now suppose that S is affine. Form the finite morphisms of finite presentation $S_i'' \to S_i$ supplied by (17.16.4). Since the immersions $S_i \to S$ are of finite presentation (the S_i being affine), the composite morphisms

$$g_i'' : S_i'' \longrightarrow S_i \longrightarrow S$$

are of finite presentation and quasi-finite. The assertion follows by taking S' to be the sum of the S_i'' and g to agree with g_i'' on each S_i''. □

Proposition (17.16.6). — *Let S be a quasi-compact and quasi-separated prescheme, and let $f : X \to S$ be a morphism of finite presentation such that, for every $s \in S$, the fibre $X_s = f^{-1}(s)$ is a prescheme proper over $k(s)$. Then there exists a finite family $(S_i)_{i \in I}$ of affine subpreschemes of S, of finite presentation over S, pairwise disjoint and with union S, such that, for every i, the morphism $X_i = X \times_S S_i \to S_i$ is proper and flat. If, moreover, X_s is finite over $k(s)$ for every $s \in S$, the S_i may be chosen so that each morphism $X_i \to S_i$ factors as*

$$X_i \xrightarrow{u_i} X_i' \xrightarrow{h_i} S_i,$$

where u_i and h_i are finite and locally free (18.2.7), h_i is étale, and u_i is radicial and surjective.

Proof. We follow a course analogous to that of (17.16.4). First reduce to the case in which S is affine and f is flat, using the generic-flatness theorem (8.9.5); note that the subpreschemes S_i' defined in the proof of (8.9.5) are affine. Since f is of finite presentation, we may then write

$$X = X_0 \times_{S_0} S,$$

where S_0 is Noetherian and $X_0 \to S_0$ is a morphism of finite type and flat (11.2.7). Moreover, every fibre $(X_0)_{s_0}$ is still proper over $k(s_0)$, by (2.7.1), (vii), and we are therefore reduced to the case in which S is Noetherian. By Noetherian induction and application of the method of (8.1.2), a) (using also (8.10.5), (xii) on this occasion), the proof of the first assertion is finally reduced to the case in which S is the spectrum of a field; it then follows trivially from the hypothesis, with $S_i = S$.

The case in which X_s is supposed finite over $k(s)$ for every $s \in S$ is treated in the same way, now using (2.7.1), (xv), (8.10.5), (x), (iv), (vi), and (vii), (17.7.8), and (2.1.12). We are reduced to proving the last assertion when $S = \operatorname{Spec}(k)$ is the spectrum of a field. Then $X = \operatorname{Spec}(A)$, where A is a finite k-algebra (**I**, 6.4.4); thus A is a direct product of finite k-algebras A_j which are local rings ($1 \leqslant j \leqslant m$). Since A_j is an Artinian ring and a k-algebra, it contains a subfield K_j' canonically isomorphic to the residue field of A_j (**0**, 19.6.2).

Take $X_i = X$ and take X_i' to be the sum of the $\operatorname{Spec}(K_j')$. This answers the question, since for every j there are homomorphisms **IV** | 109

$$K_j' \longrightarrow A_j \longrightarrow k(A_j)$$

whose composite is an isomorphism. □

§18. Supplement on étale morphisms. Henselian local rings and strictly local rings

In the present section we study various properties peculiar to étale morphisms. Moreover, the notion of an étale morphism makes it possible to develop, in a very natural way, Nagata's theory of Henselian rings, as well as that of strictly local rings. These rings play an important role in many recent developments, appearing whenever one needs a process of "localisation" finer than that furnished by the Zariski topology (cf. for example [43]), pending the appearance of the chapter of our Treatise devoted to the study of the "étale topology."

18.1. A remarkable equivalence of categories.

Proposition (18.1.1). — *Let S be a prescheme, S_0 a closed subprescheme of S, X_0 an S_0-prescheme smooth (resp. étale) over S_0, and x_0 a point of X_0. Then there exist an open neighbourhood U_0 of x_0 in X_0, an S-prescheme U smooth (resp. étale) over S, and an S_0-isomorphism*

$$U \times_S S_0 \xrightarrow{\sim} U_0.$$

Proof. Observe that if X_0 is étale over S_0 at x_0, it is *a fortiori* unramified over S_0 at that point. If an S-prescheme U smooth over S has been constructed such that $U \times_S S_0$ is isomorphic to U_0, then the fibres of the morphisms $U_0 \to S_0$ and $U \to S$ containing x_0 are isomorphic. It follows that U is unramified over S at x_0 (17.4.1), d), hence also in a neighbourhood of x_0; replacing U by this neighbourhood, we conclude that U is étale over S. It is therefore enough to prove the proposition when X_0 is merely supposed smooth over S_0.

The question being local on S and on X_0, we may suppose that $S = \operatorname{Spec}(A)$ and $X_0 = \operatorname{Spec}(C_0)$ are affine, so that $S_0 = \operatorname{Spec}(A_0)$, where A_0 is a quotient ring of A, and

$$C_0 = B_0/\mathfrak{J}_0, qquad B_0 = A_0[T_1, \dots, T_n],$$

where $\mathfrak{J}_0$ is an ideal of finite type in B_0; finally, C_0 is a formally smooth A_0-algebra for the discrete topologies. Let $\mathfrak{p}_0$ be the ideal $\mathfrak{j}_{x_0}$ in C_0; then $\mathfrak{p}_0 = \mathfrak{q}_0/\mathfrak{J}_0$, where $\mathfrak{q}_0$ is a prime ideal of B_0. The Jacobian criterion (**0**, 22.6.4), together with (**0**, 19.1.12), shows that there exist in $\mathfrak{J}_0$ a family of r polynomials u_i $(1 \leqslant i \leqslant r)$ and indices j_h $(1 \leqslant h \leqslant r)$ such that the images of the u_i in

$$(\mathfrak{J}_0)_{\mathfrak{q}_0}/(\mathfrak{J}_0^2)_{\mathfrak{q}_0}$$

generate this $(B_0)_{\mathfrak{q}_0}$-module and

$$\det\left(\frac{\partial u_i}{\partial T_{j_h}}\right) \notin \mathfrak{q}_0. \tag{18.1.1.1}$$

Since $(B_0)_{\mathfrak{q}_0}$ is a local ring, Nakayama's lemma shows that we may suppose that the images of the u_i in $(\mathfrak{J}_0)_{\mathfrak{q}_0}$ generate this $(B_0)_{\mathfrak{q}_0}$-module and then, replacing X_0 if necessary by an affine open neighbourhood of x_0, that the u_i generate $\mathfrak{J}_0$ ($\mathbf{0_I}$, 5.2.2). Put $B = A[T_1, \dots, T_n]$; then B_0 is a quotient ring of B, $\mathfrak{q}_0$ is the image of a prime ideal $\mathfrak{q}$ of B, and $\mathfrak{q}$ is the inverse image of $\mathfrak{q}_0$. For every i, let $v_i \in B$ be an element whose image in B_0 is u_i, and let $\mathfrak{J}$ be the ideal of B generated by the v_i, so that $\mathfrak{J}_0$ is the image of $\mathfrak{J}$ in B_0.

The proposition will be proved by taking for U an open neighbourhood of the point of $\operatorname{Spec}(B/\mathfrak{J})$ **IV** | 110
corresponding to the prime ideal $\mathfrak{p} = \mathfrak{q}/\mathfrak{J}$, provided that we show that $(B/\mathfrak{J})_{\mathfrak{q}/\mathfrak{J}}$ is a formally smooth A-algebra for the discrete topologies. This follows from the Jacobian criterion: the images of the v_i in $\mathfrak{J}_\mathfrak{q}/\mathfrak{J}_\mathfrak{q}^2$ generate this $B_\mathfrak{q}$-module, and (18.1.1.1) gives

$$\det\left(\frac{\partial v_i}{\partial T_{j_h}}\right) \notin \mathfrak{q}.$$

□

Theorem (18.1.2). — *Let S be a prescheme, S_0 a closed sub-prescheme of S whose underlying space is identical to that of S. Then the functor*

$$X \longmapsto X \times_S S_0$$

from the category of S-preschemes étale over S, to the category of S_0-preschemes étale over S_0, is an equivalence of categories.

Proof. Let us first show that this functor is *fully faithful*. Let X, Y be two S-preschemes étale over S, and set $X_0 = X \times_S S_0$, $Y_0 = Y \times_S S_0$. If $Z = X \times_S Y$, the set $\operatorname{Hom}_S(X, Y)$ is in canonical bijective correspondence with the set of X-sections $\Gamma(Z/X)$, and likewise $\operatorname{Hom}_{S_0}(X_0, Y_0)$ is in canonical bijective correspondence with $\Gamma(Z_0/X_0)$, where $Z_0 = Z \times_S S_0 = X_0 \times_{S_0} Y_0$. Now Z is étale over X, Z_0 is étale over X_0, and X_0 (resp. Z_0) is a closed subprescheme of X (resp. Z) having the same underlying space. The open subsets of Z on which the restriction of $Z \to X$ is surjective and radicial are therefore the *same* as the open subsets of Z_0 having the corresponding properties, and our assertion follows from (17.9.3).

To complete the proof, it suffices to show that for every S_0-prescheme X_0 étale over S_0, there exists an S-prescheme X étale over S and an S_0-isomorphism $X_0 \xrightarrow{\sim} X \times_S S_0$. By virtue of Proposition (18.1.1), there exists an open covering (U_α) of X_0 and for each α, an S-prescheme V_α which is étale over S, and finally an S_0-isomorphism $\theta_\alpha : U_\alpha \xrightarrow{\sim} V_\alpha \times_S S_0$. Furthermore, by virtue of the first part of the proof, there exists a unique S_0-isomorphism $\varphi_{\alpha\beta}$ from $\theta_\alpha(U_\alpha \cap U_\beta)$ to $\theta_\beta(U_\alpha \cap U_\beta)$, corresponding to the identity automorphism of $U_\alpha \cap U_\beta$, and it is immediate, for the same reason, that these isomorphisms satisfy the glueing condition ((**0**, 4.1.7)). Hence there exists an S-prescheme X such that the V_α identify canonically with sub-preschemes induced on the open subsets of X, the θ_α identify with S_0-isomorphisms which coincide on the intersections $U_\alpha \cap U_\beta$, and define an S_0-isomorphism $X_0 \xrightarrow{\sim} X \times_S S_0$. It is clear that X is étale over S (17.3.2), which completes the proof. □

Corollary (18.1.3). — *Let S be a prescheme, X an S-prescheme étale over S, S' an S-prescheme, S'_0 a closed sub-prescheme of S' having the same underlying space. Then the canonical map $X(S')_S \to X(S'_0)_S$* (**I**, 3.4.3) *is bijective.*

Proof. Indeed, set $X' = X \times_S S'$, $X'_0 = X \times_S S'_0$, so that $X(S')_S = \Gamma(X'/S')$ and $X(S'_0)_S = \Gamma(X'_0/S'_0)$; the corollary follows from the fact that the functor defined in (18.1.2) (with S and S_0 replaced by S' and S'_0 respectively) is fully faithful (or directly from (17.9.3)). □

18.2. Étale coverings. **IV | 111**

(18.2.1). Given a ring A and a commutative A-algebra B which is *finite* and is a *free* A-module, recall (Bourbaki, *Alg.*, ch. VIII, § 12, no. 2) that one defines on B an *A-linear form* $\mathrm{Tr}_{B/A}$, the "trace form"; from it one obtains a *symmetric A-bilinear form* (also called the "trace form")

$$(x, y) \longmapsto \mathrm{Tr}_{B/A}(xy) \tag{18.2.1.1}$$

whose datum is equivalent to that of the associated A-linear map $\mathrm{astr}_{B/A} : B \to \check{B}$ from the A-module B into its *dual* $\check{B}$; this map is equal to its transpose. When A is a *field*, saying that this bilinear form is *non-degenerate* is equivalent to saying that B is a *separable* A-algebra (Bourbaki, *Alg.*, ch. IX, § 2, prop. 5).

Let $f : A \to A'$ be a ring homomorphism; set $B' = B \otimes_A A'$, and let $g : B \to B'$ be the canonical homomorphism; the image under g of a basis of the A-module B is then a basis of the A'-module B', and it follows from the definitions that, for every $x \in B$,

$$\mathrm{Tr}_{B'/A'}(g(x)) = f(\mathrm{Tr}_{B/A}(x)). \tag{18.2.1.2}$$

(18.2.2). Consider now a *ringed space* $(X, \mathscr{O}_X)$ and let $\mathscr{B}$ be an $\mathscr{O}_X$-algebra which, as an $\mathscr{O}_X$-module, is *locally free of finite rank*. For every open subset $U \subset X$ such that $\mathscr{B}|_U$ is, as an $\mathscr{O}_U$-module, isomorphic to $\mathscr{O}_U^n$ (for some n depending on U), $\Gamma(U, \mathscr{B})$ is a $\Gamma(U, \mathscr{O}_X)$-algebra which, as a $\Gamma(U, \mathscr{O}_X)$-module, is free of finite rank. It therefore defines a $\Gamma(U, \mathscr{O}_X)$-linear form $\mathrm{Tr}_{\Gamma(U,\mathscr{B})/\Gamma(U,\mathscr{O}_X)}$, which we also denote by $\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X,U}$, and hence an associated linear map

$$\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X,U} : \Gamma(U, \mathscr{B}) \longrightarrow \Gamma(U, \mathscr{B})^\vee = \Gamma(U, \check{\mathscr{B}}).$$

Furthermore, it follows from (18.2.1.2) that these linear maps are compatible with restriction from U to a smaller open subset and therefore define, on the one hand, an *$\mathscr{O}_X$-module homomorphism*, again called the *trace homomorphism*,

$$\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X} : \mathscr{B} \longrightarrow \mathscr{O}_X \tag{18.2.2.1}$$

and, on the other hand, an *$\mathscr{O}_X$-module homomorphism*

$$\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X} : \mathscr{B} \longrightarrow \check{\mathscr{B}} \tag{18.2.2.2}$$

called the *homomorphism associated to the trace*, and equal to its transpose. It also follows from (18.2.1.2) that for every $x \in X$, we have

$$(\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X})_x = \mathrm{Tr}_{\mathscr{B}_x|\mathscr{O}_{X,x}} \tag{18.2.2.3}$$

$$(\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X})_x = \mathrm{astr}_{\mathscr{B}_x|\mathscr{O}_{X,x}}. \tag{18.2.2.4}$$

Finally, under the conditions of (18.2.1), if we set $X = \operatorname{Spec}(A)$ and $\mathscr{B} = \widetilde{B}$, the form $\operatorname{Tr}_{\mathscr{B}|\mathscr{O}_X}$ (resp. the homomorphism $\operatorname{astr}_{\mathscr{B}|\mathscr{O}_X}$) corresponds to the form $\operatorname{Tr}_{B/A}$ (resp. the A-module homomorphism $\operatorname{astr}_{B/A}$), as also follows from (18.2.1.2).

Proposition (18.2.3). — *Let $f : X \to Y$ be a finite morphism of preschemes and set $\mathscr{B} = f_*(\mathscr{O}_X)$. The following conditions are equivalent:*

- a) *f is étale.*
- *a′) f is a flat morphism of finite presentation, and for every $x \in X$, if $y = f(x)$, the ring $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is a field which is a finite separable extension of $\kappa(y)$.*
- b) *$\mathscr{B}$ is a locally free $\mathscr{O}_Y$-module and, for every $y \in Y$, $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a finite separable $\kappa(y)$-algebra (hence a direct product of finitely many fields which are finite separable extensions of $\kappa(y)$).*
- c) *$\mathscr{B}$ is a locally free $\mathscr{O}_Y$-module and the homomorphism $\operatorname{astr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \check{\mathscr{B}}$ (18.2.2) is bijective.*

Proof. Since f is quasi-compact, the equivalence of a) and a') has already been shown (17.6.2). To prove the rest of the proposition, we may restrict to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, B being a finite A-algebra and $\mathscr{B} = \widetilde{B}$. To say that f is a morphism of finite presentation is equivalent to saying that B is an A-module of finite presentation (1.4.7). If, in addition, f is flat, then B is a flat A-module; it is known (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 2, cor. 2 to th. 1) that B is then a *projective* A-module. Thus $\mathscr{B}$ is a locally free $\mathscr{O}_Y$-module (*loc. cit.*, no. 2, th. 1), and the converse is immediate. On the other hand, $f^{-1}(y)$ is nothing but the spectrum of the $\kappa(y)$-algebra $\mathscr{B}(y) = \mathscr{B}_y/\mathfrak{m}_y\mathscr{B}_y$, which completes the proof of the equivalence of a') and b). To see that b) is equivalent to c), note that the second assertion of b) is equivalent to the bijectivity of **IV | 112**

$$\operatorname{astr}_{\mathscr{B}(y)|\kappa(y)} : \mathscr{B}(y) \longrightarrow \check{\mathscr{B}(y)}.$$

Since

$$\mathscr{B}(y) = \mathscr{B}_y/\mathfrak{m}_y\mathscr{B}_y, \qquad \check{\mathscr{B}(y)} = \check{\mathscr{B}}_y/\mathfrak{m}_y\check{\mathscr{B}}_y,$$

and $\mathscr{B}_y$ and $\check{\mathscr{B}}_y$ are free $\mathscr{O}_{Y,y}$-modules, it follows from (18.2.2.4) and Bourbaki, *Alg. comm.*, ch. II, § 3, no. 3, cor. to prop. 6, that the homomorphism $\operatorname{astr}_{\mathscr{B}_y|\mathscr{O}_{Y,y}} : \mathscr{B}_y \to \check{\mathscr{B}}_y$ is itself bijective; the converse being evident, this completes the proof.

When an $\mathscr{O}_X$-Algebra $\mathscr{B}$ satisfies the equivalent conditions b) and c) of (18.2.3), we say that $\mathscr{B}$ is a *finite étale $\mathscr{O}_X$-Algebra*. When $X = \operatorname{Spec}(A)$ is affine and $\mathscr{B} = \widetilde{B}$, this amounts, by virtue of (18.2.3), to saying that B is a finite étale A-algebra, or that B is a finite A-algebra that is étale (in the sense of (17.3.2)). ☐

Corollary (18.2.4). — *Let $f : X \to Y$ be a finite morphism of finite presentation, and set $\mathscr{B} = f_*(\mathscr{O}_X)$. Let y be a point of Y. The following conditions are equivalent:*

- a) *There exists an open neighbourhood U of y in Y such that the restriction $f^{-1}(U) \to U$ of f is an étale morphism.*
- b) *$\mathscr{B}_y$ is a free $\mathscr{O}_{Y,y}$-module of finite type and $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a separable $\kappa(y)$-algebra.*

Proof. It is clear that a) implies b) by virtue of (18.2.3). On the other hand, $\mathscr{B}$ is an $\mathscr{O}_Y$-module of finite presentation (1.6.3) and (1.4.7); hence, if $\mathscr{B}_y$ is a free $\mathscr{O}_{Y,y}$-module, there exists an open neighbourhood U of y in Y such that $\mathscr{B}|_U$ is a locally free $\mathscr{O}_U$-module ($\mathbf{0_I}$, 5.2.7). Moreover, since by hypothesis the homomorphism $\operatorname{astr}_{\mathscr{B}_y|\mathscr{O}_{Y,y}} : \mathscr{B}_y \to \check{\mathscr{B}}_y$ is bijective, the same result shows that U may be chosen so that

the homomorphism $\operatorname{astr}_{\mathscr{B}|_U/\mathscr{O}_U}$ is bijective. The fact that b) implies a) then follows from (18.2.3). **IV | 113**
☐

Corollary (18.2.5). — *Let Y be a quasi-compact or locally Noetherian prescheme, and let $f : X \to Y$ be a finite morphism of finite presentation; set $\mathscr{B} = f_*(\mathscr{O}_X)$. Suppose that for every* closed *point y of Y, $\mathscr{B}_y$ is a free $\mathscr{O}_{Y,y}$-module and $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a separable $\kappa(y)$-algebra. Then f is étale.*

Proof. Indeed, it follows from (18.2.4) that every closed point y of Y has an open neighbourhood U such that the restriction $f^{-1}(U) \to U$ of f is étale; the conclusion follows from the fact that in the two cases considered, every non-empty closed subset of Y contains a closed point ((5.1.11) and ($\mathbf{0}$, 2.1.3)). ☐

Corollary (18.2.6). — *If $f : X \to Y$ is a finite étale morphism, and if we set $\mathscr{B} = f_*(\mathscr{O}_X)$, then the $\mathscr{O}_Y$-homomorphism $\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \mathscr{O}_Y$ (18.2.2) (also denoted Tr_f) is surjective.*

Proof. The question being local, we may, by virtue of (18.2.3), suppose that $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, with B a free A-module; as by virtue of (18.2.3) the bilinear form (18.2.1.1) is non-degenerate, this implies in particular that the linear form $\mathrm{Tr}_{B/A}$ is surjective. □

Remarks (18.2.7). — (i) When $f : X \to Y$ is a finite morphism such that $f_*(\mathscr{O}_X)$ is a locally free $\mathscr{O}_Y$-Module (resp. locally free of rank n), we say further that f is a *finite locally free* morphism (resp. *locally free of rank n*). This condition, by virtue of (18.2.3), is satisfied if f is a finite étale morphism, but does not imply by itself that f is étale, as shown by the example where $X = \mathrm{Spec}(K)$ and $Y = \mathrm{Spec}(k)$ are spectra of fields, K being a finite non-separable extension of k. When $f : X \to Y$ is a finite étale morphism, we say further that X is an *étale covering* of Y. One notes that in this case, f is universally open and universally closed, and in particular $f(X)$ is a subset that is both open and closed of Y.

We say that an étale covering X of Y is *trivial* if X is a *sum* of a finite number of preschemes *isomorphic* to Y. We say that an étale covering X of Y is *locally trivial* if the morphism $f : X \to Y$ is such that for every point $y \in Y$ there is an open neighbourhood U for which the covering $f^{-1}(U)$ of U is trivial.

(ii) Let $f : X \to Y$ be a finite morphism, locally free of rank n; set $\mathscr{B} = f_*(\mathscr{O}_X)$ and let $u = \mathrm{astr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \check{\mathscr{B}}$; we deduce from this an n-th exterior power homomorphism $\bigwedge^n u : \bigwedge^n \mathscr{B} \to \bigwedge^n \check{\mathscr{B}} = (\bigwedge^n \mathscr{B})^\vee$ of *invertible* $\mathscr{O}_Y$-Modules, and by (**0**, 5.4.2) an element

(18.2.7.1) $$d_{X/Y} \in \Gamma(Y, (\bigwedge^n \check{\mathscr{B}}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \check{\mathscr{B}}))$$

which we call the *discriminant* of X over Y. Furthermore, since $(\bigwedge^n \check{\mathscr{B}}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \check{\mathscr{B}})$ is the dual of $(\bigwedge^n \mathscr{B}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \mathscr{B})$, $d_{X/Y}$ can also be identified with a homomorphism

(18.2.7.2) $$(\bigwedge^n \mathscr{B}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \mathscr{B}) \longrightarrow \mathscr{O}_Y$$

and we denote by $\mathscr{D}_{X/Y}$ the quasi-coherent ideal of $\mathscr{O}_Y$ of finite type, *image* of the homomorphism (18.2.7.2), also called the *discriminant ideal* of X over Y.

For the homomorphism u to be bijective, it is necessary and sufficient that $\bigwedge^n u$ be bijective, or again that the section $d_{X/Y}$ have an *invertible* germ at every point $y \in Y$, which is also written $d_{X/Y}(y) \neq 0$ for every y ((**0**, 5.5.2)). It amounts to the same to say that the discriminant ideal $\mathscr{D}_{X/Y}$ is *equal* to $\mathscr{O}_Y$. **IV** | 114

The terminology "covering" introduced in (18.2.7), (i), is justified by the following proposition:

Proposition (18.2.8). — *Let $f : X \to Y$ be a morphism étale, separated and of finite type, and for every $y \in Y$, let $n(y)$ be the geometric number of points of $f^{-1}(y)$. Then the function $y \mapsto n(y)$ is lower semi-continuous on Y. For it to be continuous at a point y (and hence constant in a neighbourhood of y), it is necessary and sufficient that there exists an open neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is a* finite *(étale) morphism.*

Proof. Since f is quasi-finite (17.6.1) and locally of finite presentation, it amounts to saying that f is finite or that f is proper (8.11.1): moreover, each fibre $f^{-1}(y)$ is geometrically reduced over $\kappa(y)$. The conclusions therefore follow from (15.5.9), (i) and (ii) and from the fact that f is flat. □

Corollary (18.2.9). — *Let Y be a connected prescheme, $f : X \to Y$ a morphism étale, separated and of finite type. For f to be finite (in other words, for X to be an étale covering of Y), it is necessary and sufficient that all the fibres of f have the same geometric number of points.*

Remarks (18.2.10). — (i) The example of the "affine line with a doubled point" ((**I**, 5.5.11)) shows that a morphism étale of finite type of Noetherian preschemes may not be separated; for this example, the first assertion of (18.2.8) is no longer exact.

(ii) For a separated morphism of finite type and étale $f : X \to Y$ to make X a *locally trivial* covering, it is necessary and sufficient that for every $x \in X$ there exist an open neighbourhood U of $y = f(x)$ and a U-section g of $f^{-1}(U)$ such that $g(y) = x$. Indeed, the condition is obviously necessary; the fact that it is also sufficient follows from the fact that every fibre $f^{-1}(y)$ is finite ((17.6.1)) and (**I**, 6.2.2), from the characterisation of sections of a Y-scheme étale (17.9.3) and from Proposition (18.2.8).

18.3. Finite algebras and étale algebras.

Proposition (18.3.1). — *Let A be a ring, B an A-algebra of finite presentation.*

(i) For B to be an unramified A-algebra, it is necessary and sufficient that B be an A-module of finite presentation and be a projective *$(B \otimes_A B)$-module.*

(ii) Suppose moreover that B is a finite A-algebra. For B to be an étale A-algebra, it is necessary and sufficient that B be a projective A-module and a projective $(B \otimes_A B)$-module.

Proof. The $(B \otimes_A B)$-module structure on B is that coming from the canonical ring homomorphism $B \otimes_A B \to B$, which is surjective and has kernel $\mathfrak{J}_{B/A}$ ((**0**, 20.4.1)). **IV** | 115

(i) To say that the morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is locally of finite presentation is equivalent to saying that B is an A-algebra of finite presentation (1.4.6). To say that B is an unramified A-algebra means (17.4.2) that

$$\operatorname{Spec}\big((B \otimes_A B)/\mathfrak{J}_{B/A}\big)$$

is the subprescheme induced on an open and closed subset of $\operatorname{Spec}(B \otimes_A B)$. This holds if and only if $\mathfrak{J}_{B/A}$ is a *direct-factor ideal* of $B \otimes_A B$ (Bourbaki, *Alg. comm.*, ch. II, § 4, no. 3, prop. 15); but this amounts to saying that the $(B \otimes_A B)$-module quotient $(B \otimes_A B)/\mathfrak{J}_{B/A}$ is projective (Bourbaki, *Alg.*, ch. II, 3e éd., § 2, no. 2, prop. 4).

(ii) Recall that a flat A-module of finite presentation is projective, and conversely (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 2, cor. 2 du th. 1), so the assertion (ii) follows from that of (i) and from (17.6.2). ☐

Proposition (18.3.2). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$-preadic topology, A is separated and complete; set $A_0 = A/\mathfrak{J}$. Then the functor*

$$B \longmapsto B \otimes_A A_0$$

is an equivalence of the category of finite étale A-algebras, to the category of finite étale A_0-algebras.

Proof. We will first prove the following lemma:

Lemma (18.3.2.1). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$-preadic topology, A is separated and complete.*

(i) Every projective A-module of finite type M is separated and complete for the $\mathfrak{J}$-preadic topology, and hence is the projective limit of the projective $(A/\mathfrak{J}^{n+1})$-modules $M/\mathfrak{J}^{n+1}M$.

(ii) Conversely, set $A_n = A/\mathfrak{J}^{n+1}$, and let (M_n) be a projective system of A_n-modules such that, for every n, the homomorphism $M_{n+1} \otimes_{A_{n+1}} A_n \to M_n$ induced by the transition homomorphism $M_{n+1} \to M_n$ is bijective. Suppose moreover that the M_n are projective and M_0 of finite type. Then $M = \varprojlim M_n$ is a projective A-module of finite type such that the canonical homomorphism $M \otimes_A A_0 \to M_0$ is bijective.

Proof. (i) There is a free A-module of finite type L such that M is isomorphic to a direct factor of L; as L is separated for the $\mathfrak{J}$-preadic topology, so is every sub-module N of L since $\mathfrak{J}^{n+1}N \subset \mathfrak{J}^{n+1}L$; in particular M is separated for this topology, and since the surjective homomorphism $f : L \to M$ is continuous for the $\mathfrak{J}$-preadic topology, its kernel N is closed for the topology induced by that of L; since L is complete and f a strict morphism, we conclude that M is complete (Bourbaki, *Top. gén.*, ch. IX, 2e éd., § 3, no. 1, prop. 4).

(ii) It follows from Nakayama's lemma that, if M_0 is generated by a finite family (x_{i0}) of r elements and if, for every n, x_{in} is an element of M_n whose image in M_{n-1} is $x_{i,n-1}$, then (x_{in}) $(1 \leqslant i \leqslant r)$ is a system of generators of M_n (Bourbaki, *Alg. comm.*, ch. II, § 3, cor. 2 to prop. 4). For every n, put $L_n = A_n^r$. If $(e_{in})_{1 \leqslant i \leqslant r}$ is the canonical basis of L_n, let $u_n : L_n \to M_n$ be the A_n-linear map such that $u_n(e_{in}) = x_{in}$ for every i. By hypothesis, we have a split exact sequence

$$0 \longrightarrow N_n \xrightarrow{v_n} L_n \xrightarrow{u_n} M_n \longrightarrow 0$$

and since $L_n = L_{n+1}/\mathfrak{J}^{n+1}L_{n+1}$ and $M_n = M_{n+1}/\mathfrak{J}^{n+1}M_{n+1}$, the vertical arrows in the commutative diagram **IV | 116**

$$\begin{tikzcd} 0 \arrow[r] & N_{n+1} \arrow[r, "v_{n+1}"] \arrow[d] & L_{n+1} \arrow[r, "u_{n+1}"] \arrow[d] & M_{n+1} \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & N_n \arrow[r, "v_n"] & L_n \arrow[r, "u_n"] & M_n \arrow[r] & 0 \end{tikzcd}$$

are all three surjective. Now we have $M = \varprojlim M_n$ and $L = A^r = \varprojlim L_n$; setting $N = \varprojlim N_n$, $u = \varprojlim u_n$, $v = \varprojlim v_n$, by (EGA 0_{III}, 13.2.2) the exact sequence

(18.3.2.2) $$0 \longrightarrow N \xrightarrow{v} L \xrightarrow{u} M \longrightarrow 0$$

is exact. Moreover, since for every n, v_n is left-invertible and M_n is a projective A_n-module, by virtue of ((**0**, 19.1.8)) the exact sequence (18.3.2.2) is split, which proves the lemma. □

This being established, let us first show that the functor of (18.3.2) is *fully faithful*. Set, as in the lemma, $A_n = A/\mathfrak{J}^{n+1}$; let B, C be two finite étale A-algebras, and set, for every n, $B_n = B \otimes_A A_n$, $C_n = C \otimes_A A_n$; by virtue of (18.3.1) and (18.3.2.1), B and C are separated and complete for the $\mathfrak{J}$-preadic topology, and we have $B = \varprojlim B_n$, $C = \varprojlim C_n$; moreover, every A-algebra homomorphism $u : B \to C$ is continuous for the $\mathfrak{J}$-preadic topologies, and hence gives a projective system of A_n-algebra homomorphisms $u_n = u \otimes 1 : B_n \to C_n$, of which it is the projective limit; the converse being evident, we have a canonical bijection

$$\mathrm{Hom}_{A\text{-alg.}}(B, C) \xrightarrow{\sim} \varprojlim \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n).$$

But since B and C are finite étale A-algebras, it follows also from (18.1.2) that the canonical map

$$\mathrm{Hom}_{A_{n+1}\text{-alg.}}(B_{n+1}, C_{n+1}) \longrightarrow \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n)$$

is *bijective* for $n \geqslant 0$, which proves that the canonical map $\mathrm{Hom}_{A\text{-alg.}}(B, C) \to \mathrm{Hom}_{A_0\text{-alg.}}(B_0, C_0)$ is bijective.

To complete the proof of (18.3.2), it suffices to show that for every finite étale A_0-algebra B_0, there exists a finite étale A-algebra B and an A_0-isomorphism $B_0 \xrightarrow{\sim} B \otimes_A A_0$. Now, by (18.1.2), there is a projective system (B_n) such that B_n is an étale A_n-algebra and that the homomorphisms $B_{n+1} \otimes_{A_{n+1}} A_n \to B_n$ are bijective. It follows from (18.3.1) and (18.3.2.1) that the A-algebra $B = \varprojlim B_n$ is a projective A-module of finite type and that B_0 is isomorphic to $B \otimes_A A_0$. To prove that B is a finite étale A-algebra, it suffices, by (18.2.5), to show that for every maximal ideal $\mathfrak{m}$ of A, $B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}}$ is a separable $(A/\mathfrak{m})$-algebra. Now, since $\mathfrak{J}$ is contained in the radical of A ((**0**, 7.1.10)), we have $\mathfrak{J} \subset \mathfrak{m}$, so $\mathfrak{m}_0 = \mathfrak{m}/\mathfrak{J}$, we have $A_0/\mathfrak{m}_0 = A/\mathfrak{m}$ and $B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}} = (B_0)_{\mathfrak{m}_0}/\mathfrak{m}_0(B_0)_{\mathfrak{m}_0}$; the **IV | 117** conclusion therefore follows from the fact that B_0 is a finite étale A_0-algebra (18.2.5). □

(18.3.3). *Example.* — Proposition (18.3.2) applies in particular when A is a *local* ring, separated and complete, $\mathfrak{J}$ being the maximal ideal of A, so that A_0 is a field and the finite étale A_0-algebras are precisely the finite separable A_0-algebras, hence the direct products of finite separable field extensions of A_0. In particular, if the field A_0 is *separably closed*, these extensions are all identical to A_0, and hence every étale covering of $\mathrm{Spec}(A)$ is *trivial* (18.2.7) by virtue of (18.3.2).

Theorem (18.3.4). — *Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is separated and complete for the $\mathfrak{J}$-preadic topology, $A_0 = A/\mathfrak{J}$. Let $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A_0)$. Let X be a proper S-scheme, and set $X_0 = X \times_S S_0$. Then the functor*

$$Z \longmapsto Z \times_X X_0$$

from the category of finite étale X-schemes over X, to the category of finite étale X_0-schemes over X_0, is an equivalence of categories.

Proof. Let us first show that this functor is *fully faithful*. Let Z' and Z'' be two finite étale X-schemes over X. Set $S_n = \mathrm{Spec}(A/\mathfrak{J}^{n+1})$, $X_n = X \times_S S_n$, $Z'_n = Z' \times_S S_n$, $Z''_n = Z'' \times_S S_n$. It follows from ((**III**, 5.4.1)) that we have a canonical bijection $\mathrm{Hom}_X(Z', Z'') \xrightarrow{\sim} \varprojlim \mathrm{Hom}_{X_n}(Z'_n, Z''_n)$. Now, by virtue of (18.1.2), the canonical map $\mathrm{Hom}_{X_{n+1}}(Z'_{n+1}, Z''_{n+1}) \to \mathrm{Hom}_{X_n}(Z'_n, Z''_n)$ is bijective, which proves our assertion.

It remains to show that if $\mathscr{B}_0$ is a finite étale $\mathscr{O}_{X_0}$-algebra, there exists a finite étale $\mathscr{O}_X$-algebra $\mathscr{B}$ and an isomorphism $\mathscr{B}_0 \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_0}$. It follows from (18.1.2) that there is a projective system $(\mathscr{B}_n)$, where $\mathscr{B}_n$ is a finite étale $\mathscr{O}_{X_n}$-algebra, and the second comparison theorem ((**III**, 5.1.4)) proves the existence of a coherent $\mathscr{O}_X$-module $\mathscr{B}$ and of a projective system of isomorphisms $\mathscr{B}_n \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_n}$. The datum of an algebra structure on an $\mathscr{O}_X$-module $\mathscr{F}$ is equivalent to that of a multiplication homomorphism $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{F} \to \mathscr{F}$ making commute certain diagrams in which only tensor powers of $\mathscr{F}$ occur; by (**III**, 5.1.3), (**I**, 10.11.4) and (**I**, 10.11.7), $\mathscr{B}$ is therefore endowed in a natural way with a structure of $\mathscr{O}_X$-algebra for which the isomorphism $\mathscr{B}_0 \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_0}$ is an isomorphism of algebras. Moreover, $\mathscr{B}$ is a *locally free* $\mathscr{O}_X$-module; this also follows from (**III**, 5.1.3), (**I**, 10.11.4), (**I**, 10.11.7) and from the fact that, in the category of coherent $\mathscr{O}_X$-modules, the locally free $\mathscr{O}_X$-modules $\mathscr{F}$ can be characterized as those for which the functors $\mathscr{G} \mapsto \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ are *exact*. Finally, to see that $\mathscr{B}$ is a finite étale $\mathscr{O}_X$-Algebra, it suffices (18.2.5) to show that for every *closed* point $x \in X$, $\mathscr{B}_x \otimes_{\mathscr{O}_{X,x}} \kappa(x)$ is a separable $\kappa(x)$-algebra. But since the structural morphism $f : X \to S$ is *proper*, $f(x)$ is a closed point of S, so it belongs to S_0, since $\mathfrak{J}$ is contained in the radical of A ((**0**, 7.1.10)); the conclusion therefore follows from the fact that $X_0 = f^{-1}(S_0)$ and that $\mathscr{B}_0$ is a finite étale $\mathscr{O}_{X_0}$-algebra. □

18.4. Local structure of unramified morphisms and of étale morphisms. **IV** | 118

Lemma (18.4.1). — *Let A be a ring, B a finite monogenic A-algebra, u a generator of the A-algebra B, $F \in A[\mathrm{T}]$ a polynomial such that $F(u) = 0$, F' the derivative polynomial; set $u' = F'(u)$. Then the annihilator ideal of $\Omega^1_{B/A}$ in B contains $u'B$; it is equal to $u'B$ if the ideal $\mathfrak{J}$ of polynomials $G \in A[\mathrm{T}]$ such that $G(u) = 0$ is generated by F, in other words if the canonical surjection $\varphi : A[\mathrm{T}]/FA[\mathrm{T}] \to B$, which sends the class of* T *to u, is bijective.*

Proof. Set $C = A[\mathrm{T}]$, so that $B = C/\mathfrak{J}$. We have the exact sequence ((**0**, 20.5.12.1))

$$\mathfrak{J}/\mathfrak{J}^2 \longrightarrow \Omega^1_{C/A} \otimes_C B \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

and $\Omega^1_{B/A}$ therefore identifies with the quotient $B/\mathfrak{J}'$, where $\mathfrak{J}'$ is the ideal generated by the elements $G'(u)$, where G ranges over a system of generators of the ideal $\mathfrak{J}$ ((**0**, 20.5.13)); whence the lemma immediately follows. □

Proposition (18.4.2). — *Let the notation be those of* (18.4.1)*, let $\mathfrak{q}$ be a prime ideal of B. Then:*

- *(i) If $\mathfrak{q}$ does not contain u', $B_\mathfrak{q}$ is a formally unramified $A_\mathfrak{p}$-algebra ($\mathfrak{p}$ being the inverse image of $\mathfrak{q}$ in A); in other terms,* $\mathrm{Spec}(B_{u'})$ *is formally unramified over* $\mathrm{Spec}(A)$.
- *(ii) Suppose moreover that F is monic and generates $\mathfrak{J}$. Then, for* $\mathrm{Spec}(B)$ *to be étale over* $\mathrm{Spec}(A)$ *at the point $\mathfrak{q}$, it is necessary and sufficient that $u' \notin \mathfrak{q}$.*

Proof. The hypothesis that $u' \notin \mathfrak{q}$ implies that $\Omega^1_{B_\mathfrak{q}/A_\mathfrak{p}} = 0$ ((**0**, 20.5.9)), hence (i) follows from (17.2.1). Moreover, under the hypotheses of (ii), B is a free A-module by virtue of the Euclidean division; as the annihilator $\mathfrak{J}'$ of $\Omega^1_{B/A}$ is then equal to $u'B$ by virtue of (18.4.1) and $\Omega^1_{B/A}$ is a B-module of finite presentation (16.4.22), the annihilator of $\Omega^1_{B_\mathfrak{q}/A_\mathfrak{p}}$ is equal to $u'B_\mathfrak{q}$ (Bourbaki, *Alg. comm.*, ch. II, § 2, no. 4, formule (g)), and (ii) follows from (i) and from the implication $c) \Rightarrow a)$ in (17.6.1). □

Corollary (18.4.3). — *With the notation of* (18.4.2)*, suppose that F is monic and generates $\mathfrak{J}$. Then, for B to be an étale A-algebra, it is necessary and sufficient that u' be invertible in B (or, what amounts to the same, that the ideal of $A[\mathrm{T}]$ generated by F and F' be equal to $A[\mathrm{T}]$).*

Proof. In view of (18.4.2), (ii), saying that $\mathrm{Spec}(B)$ is étale over $\mathrm{Spec}(A)$ means that u' does not belong to any prime ideal of B, i.e. that it is invertible in B. □

We say that a monic polynomial $F \in A[\mathrm{T}]$ such that the ideal of $A[\mathrm{T}]$ generated by F and F' is equal to $A[\mathrm{T}]$ is *separable*; it is immediate that this definition coincides with the usual definition (Bourbaki, *Alg.*, ch. V, § 7, no. 6) when A is a field.

Lemma (18.4.4). — *Let A be a local ring, B a finite monogenic A-algebra, u a generator of the A-algebra B. Let $\mathfrak{n}_i$ ($1 \leqslant i \leqslant r$) be the maximal ideals of B such that* $\mathrm{Spec}(B)$ *is formally unramified over* $\mathrm{Spec}(A)$ *at* *the points $\mathfrak{n}_i$. Then there exists a monic polynomial $F \in A[\mathrm{T}]$ such that $F(u) = 0$ and $F'(u) \notin \mathfrak{n}_i$ for every i ($1 \leqslant i \leqslant r$). Moreover, if k is the residue field of A and $f \in k[\mathrm{T}]$ is the minimal polynomial of the image of u in $B \otimes_A k$, one can choose F so that its canonical image in $k[\mathrm{T}]$ is f.* **IV** | 119

Proof. The maximal ideal $\mathfrak{m}$ of A is the inverse image of each $\mathfrak{n}_i$ ((**II**, 6.1.10)). Set $L = B \otimes_A k$, a finite k-algebra, and let ξ be the image of u in L; if n is the rank of L over k, the minimal polynomial $f \in k[\mathrm{T}]$ of ξ over k has degree n, and L is isomorphic to $k[\mathrm{T}]/fk[\mathrm{T}]$. If $\mathfrak{n}'_i = \mathfrak{n}_i/\mathfrak{m}B$, the hypothesis implies that $\mathrm{Spec}(L)$ is étale over k at the points $\mathfrak{n}'_i$ ($1 \leqslant i \leqslant r$), by (17.6.1), c); hence $f'(\xi) \notin \mathfrak{n}'_i$ by (18.4.2), (ii). Now $\mathfrak{m}B$ is contained in the radical of B; since $1, \xi, \dots, \xi^{n-1}$ form a basis of L over k, Nakayama's lemma shows that $1, u, \dots, u^{n-1}$ generate the A-module B. Consequently there exists a monic polynomial $F \in A[\mathrm{T}]$ of degree n such that $F(u) = 0$; since ξ is a root of the canonical image of F in $k[\mathrm{T}]$, this image is necessarily f. The image of $F'(u)$ in L is then $f'(\xi)$, and therefore $F'(u) \notin \mathfrak{n}_i$ for every i. □

Proposition (18.4.5). — *Let A be a local ring, k its residue field, and B a finite A-algebra (respectively, a finite A-algebra of finite presentation). Suppose moreover that the residue field k is* infinite, *or else that B is a local ring. Let n be the rank of $L = B \otimes_A k$ over k. For B to be a formally unramified A-algebra (resp. étale), it is necessary and sufficient that there exist a monic separable polynomial $F \in A[\mathrm{T}]$ such that B is isomorphic to a quotient of $A[\mathrm{T}]/F.A[\mathrm{T}]$ (resp. isomorphic to $A[\mathrm{T}]/F.A[\mathrm{T}]$). Moreover, one can suppose that F has degree n (respectively, F necessarily has degree n).*

Proof. The conditions are sufficient by virtue of (18.4.2), without assuming either that k is infinite or that B is local. To see that the conditions are necessary, note that if B is a formally unramified A-algebra, L is a finite separable k-algebra, hence a direct product of finitely many finite separable field extensions k_j of k ($1 \leqslant j \leqslant r$), each generated by an element $\bar{\xi}_j$ with minimal polynomial f_j (Bourbaki, *Alg.*, ch. V, § 11, no. 4, prop. 4). Let us show that by virtue of the hypotheses made on k or B, there exists an element $\bar{\xi}$ of L generating the k-algebra L. This is immediate if B is local, since then $r = 1$. Otherwise, k being assumed infinite, one can suppose that the irreducible polynomials $f_j \in k[\mathrm{T}]$ are all *distinct* (by replacing each $\bar{\xi}_j$ by $\bar{\xi}_j + a_j$, for a suitable element $a_j \in k$); if we set $f = f_1 f_2 \dots f_r$, it is clear that L is isomorphic to $k[\mathrm{T}]/fk[\mathrm{T}]$ in both cases, hence generated by an element $\bar{\xi}$ of minimal polynomial $f \in k[\mathrm{T}]$ of degree n. Choose $u \in B$ whose image in L is $\bar{\xi}$. Since $\mathfrak{m}B$ is contained in the radical of B and $1, \bar{\xi}, \dots, \bar{\xi}^{n-1}$ form a basis of L over k, Nakayama's lemma shows that $1, u, \dots, u^{n-1}$ generate the A-module B; hence there exists a monic polynomial $F \in A[\mathrm{T}]$ of degree n such that $F(u) = 0$; moreover, since $\bar{\xi}$ is a root of the canonical image of F in $k[\mathrm{T}]$, this image is necessarily equal to f. Thus u generates the A-algebra B, which is consequently a quotient of $A[\mathrm{T}]/FA[\mathrm{T}]$. Moreover, B is semilocal and, at each of its maximal ideals $\mathfrak{n}_i$, (18.4.4) gives $F'(u) \notin \mathfrak{n}_i$; hence $F'(u)$ is invertible in B, and F is separable. Finally, if B is an étale A-algebra, then B, being a flat A-module of finite presentation ((1.4.7)), is free, and $1, u, \dots, u^{n-1}$ form a basis of the A-module B (Bourbaki, *Alg. comm.*, ch. II, § 3, no. 2, prop. 5); in other words, the A-algebra B is isomorphic to $A[\mathrm{T}]/FA[\mathrm{T}]$; and any other monic polynomial $G \in A[\mathrm{T}]$ such that $B \simeq A[\mathrm{T}]/GA[\mathrm{T}]$ necessarily has degree n. □ **IV** | 120

Theorem (18.4.6). — *(Chevalley). —*

(i) *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X, $y = f(x)$, and set $A = \mathscr{O}_{Y,y}$. For $\mathscr{O}_{X,x}$ to be a formally étale (respectively, formally unramified) A-algebra, it is necessary and sufficient that there exist a monic polynomial $F \in A[\mathrm{T}]$ and a maximal ideal $\mathfrak{n}$ of $B = A[\mathrm{T}]/FA[\mathrm{T}]$ (respectively, of a quotient algebra B of $A[\mathrm{T}]/FA[\mathrm{T}]$) such that $\mathscr{O}_{X,x}$ is A-isomorphic to $B_{\mathfrak{n}}$ and, if u is the image of T in B, one has $F'(u) \notin \mathfrak{n}$.*
(ii) *Suppose in addition that f is locally of finite presentation. Then, for f to be étale at the point x, it is necessary and sufficient that we can moreover take $B = A[\mathrm{T}]/F.A[\mathrm{T}]$.*

Proof. The conditions are sufficient by virtue of (18.4.2). To see that they are necessary, we may plainly reduce to the case where X and Y are affine and, in view of (17.4.1.2), to the case where f is formally unramified and quasi-finite. It then follows from (8.12.8) that there exist a finite A-algebra C and a maximal ideal $\mathfrak{r}$ of C (necessarily lying above the maximal ideal $\mathfrak{m}$ of A) such that $\mathscr{O}_{X,x}$ is A-isomorphic to $C_{\mathfrak{r}}$. Moreover, by (17.4.1.2), the residue field $C/\mathfrak{r} = C_{\mathfrak{r}}/\mathfrak{r}C_{\mathfrak{r}}$ is a finite separable extension of $k = A/\mathfrak{m}$, and hence is of the form $k[v]$, where v is separable over k. Let $\mathfrak{r}_i$ ($1 \leqslant i \leqslant h$)

be the maximal ideals of the semilocal ring C other than $\mathfrak{r}$. There exists an element $u \in C$ belonging to every $\mathfrak{r}_i$ and whose image in $C/\mathfrak{r}$ is v (Bourbaki, *Alg. comm.*, ch. II, § 1, no. 2, prop. 5). We shall show that the A-subalgebra $B = A[u]$ of C and the ideal $\mathfrak{n} = \mathfrak{r} \cap B$ (necessarily maximal, since C is finite over B) answer the question.

For the case where $\mathscr{O}_{X,x}$ is a formally unramified A-algebra, it suffices to prove that $B_{\mathfrak{n}}$ is isomorphic to $C_{\mathfrak{r}}$; indeed, $B_{\mathfrak{n}}$ will then be formally unramified over A, and the existence of a polynomial $F \in A[\mathrm{T}]$ having the asserted properties will follow from (18.4.4). Since f is formally unramified, $C/\mathfrak{r}$ is isomorphic to the A-algebra $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$ ((17.4.1.2)). We are thus reduced to proving the following lemma.

Lemma (18.4.6.1). — *Let A be a local ring with maximal ideal $\mathfrak{m}$, let C be a finite A-algebra, and let $\mathfrak{r}$ be a maximal ideal of C. Let $u \in C$ belong to every maximal ideal of C distinct from $\mathfrak{r}$, but not to $\mathfrak{r}$, and suppose that $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$ is a monogenic algebra over $k = A/\mathfrak{m}$, generated by the image of u. Set $B = A[u]$ and $\mathfrak{n} = \mathfrak{r} \cap B$. Then the canonical homomorphism $B_{\mathfrak{n}} \to C_{\mathfrak{r}}$ is an isomorphism.*

Proof. Set $R = B \setminus \mathfrak{n}$ and $S = C \setminus \mathfrak{r}$, so that $B_{\mathfrak{n}} = R^{-1}B$ and $C_{\mathfrak{r}} = S^{-1}C$; the canonical homomorphism $B_{\mathfrak{n}} \to C_{\mathfrak{r}}$ factors as

$$R^{-1}B \xrightarrow{g} R^{-1}C \xrightarrow{h} S^{-1}C.$$

It suffices to prove that both homomorphisms are bijective. First, $h : R^{-1}C \to S^{-1}C = C_{\mathfrak{r}}$ is bijective. Indeed, it suffices to see that the images in $R^{-1}C$ of the elements of S are invertible, or equivalently that every maximal ideal $\mathfrak{p}$ of $R^{-1}C$ has inverse image in C disjoint from S, and hence equal to $\mathfrak{r}$. Since $R^{-1}C$ is finite over the local ring $R^{-1}B = B_{\mathfrak{n}}$, the inverse image of $\mathfrak{p}$ in $R^{-1}B$ is its IV | 121
unique maximal ideal $\mathfrak{n}B_{\mathfrak{n}}$; if $\mathfrak{q}$ is the inverse image of $\mathfrak{p}$ in C, then $\mathfrak{q} \cap B = \mathfrak{n}$ and, since $\mathfrak{n}$ is maximal in B and C is finite over B, $\mathfrak{q}$ is one of the maximal ideals of C. Furthermore $u \notin \mathfrak{q}$, since $u \notin \mathfrak{r}$ and $u \in B$; the hypothesis therefore forces $\mathfrak{q} = \mathfrak{r}$.

Since $B \subset C$, the homomorphism $g : R^{-1}B \to R^{-1}C$ is injective (EGA 0_{I}, 1.3.2). To prove that it is surjective, observe that $R^{-1}C$ is an $R^{-1}B$-module of finite type and that $\mathfrak{m}R^{-1}B$ is contained in the maximal ideal of the local ring $R^{-1}B$. By Nakayama's lemma, it suffices to prove that

$$R^{-1}B/\mathfrak{m}R^{-1}B \longrightarrow R^{-1}C/\mathfrak{m}R^{-1}C$$

is surjective. But, by the first part of the proof, $R^{-1}C/\mathfrak{m}R^{-1}C$ identifies with $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$; by hypothesis this k-algebra is generated by the image of u, and hence it is the image of $R^{-1}B/\mathfrak{m}R^{-1}B$. ☐

Consider next the case where f is étale at x. Replacing X by a neighbourhood of x, we may suppose that X is a neighbourhood of $\mathfrak{n}$ in $\mathrm{Spec}(B)$ ((**I**, 7.2)). Set $B' = A[\mathrm{T}]/FA[\mathrm{T}]$ and let $\mathfrak{n}'$ be the inverse image of $\mathfrak{n}$ in B'. Since the image of $F'(\mathrm{T})$ in B' does not belong to $\mathfrak{n}'$, the morphism $\mathrm{Spec}(B') \to \mathrm{Spec}(A)$ is étale at $\mathfrak{n}'$ by (18.4.2), (ii). As $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is étale at $\mathfrak{n}$, (17.3.4) shows that $\mathrm{Spec}(B) \to \mathrm{Spec}(B')$ is étale at $\mathfrak{n}$; but this morphism is an immersion, so it can be étale at a point only if it is a local isomorphism there ((17.9.1)). Thus $B_{\mathfrak{n}}$ and $B'_{\mathfrak{n}'}$ are isomorphic.

Finally, suppose that $\mathscr{O}_{X,x}$ is a formally étale A-algebra. With the preceding notation, (17.1.5) shows that $B_{\mathfrak{n}}$ is a formally étale $B'_{\mathfrak{n}'}$-algebra; since the homomorphism $B'_{\mathfrak{n}'} \to B_{\mathfrak{n}}$ is surjective, this is possible only if it is *bijective* ((**0**, 19.10.3), (i)). This completes the proof of (18.4.6). ☐

Corollary (18.4.7). — *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X. For f to be formally unramified at the point x, it is necessary and sufficient that there exist an open neighbourhood U of x such that $f|U$ factorises as $U \xrightarrow{j} X' \xrightarrow{h} Y$, where h is an étale morphism and j is a closed immersion.*

Proof. We can obviously reduce to the case where $Y = \mathrm{Spec}(R)$ is affine and f is of finite type. If $A = \mathscr{O}_{Y,y}$, with $y = f(x)$, the condition for f to be formally unramified at the point x is equivalent, by virtue of (17.4.1.2), to saying that $\mathscr{O}_{X,x}$ is a formally unramified A-algebra. If it is so, we can apply (18.4.6), (i); replacing Y as needed by an affine neighbourhood of y, we may suppose (with the notation of (18.4.6)) that the polynomial F is the image in $A[\mathrm{T}]$ of a monic unitaire polynomial $G \in R[\mathrm{T}]$. We then set $X' = \mathrm{Spec}(R[\mathrm{T}]/G.R[\mathrm{T}])$; let x' be the image of the point $\mathfrak{n}$ of $\mathrm{Spec}(B)$ by the morphism corresponding to the composite homomorphism $R[\mathrm{T}]/G.R[\mathrm{T}] \to A[\mathrm{T}]/F.A[\mathrm{T}] \to B$. It follows from (18.4.2) that the canonical morphism $h : X' \to Y$ is étale at the point x', hence, restricting X' and Y to open neighbourhoods of x' and y if necessary, we may suppose that h is étale. On the other hand, by virtue of (**I**, 6.5.1), (ii) and (**I**, 7.2), the local homomorphism $\varphi : \mathcal{O}_{X',x'} \to \mathcal{O}_{X,x}$ corresponds to a morphism $j : U \to X'$ from an open neighbourhood U of x in X; in applying (**I**, 6.5.1), (i)), suppose that $h \circ j = f|U$, hence j is of finite type ((**I**, 6.3.4), (v)); finally, the homomorphism φ is surjective by (18.4.6), (i), hence it follows from ((**I**, 6.5.4), (i)) that we can, restricting further U and X', suppose that j is a closed immersion. This proves the necessity of the condition stated; the sufficiency is immediate (17.1.3), (i) and (ii). □ **IV | 122**

Corollary (18.4.8). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be unramified (respectively, étale), it is necessary and sufficient that there exist a family of flat morphisms $g_\alpha : Y'_\alpha \to Y$ and, for each α, an open subset U_α of $X'_\alpha = X \times_Y Y'_\alpha$, such that, if $g'_\alpha : X'_\alpha \to X$ and $f'_\alpha : X'_\alpha \to Y'_\alpha$ are the canonical projections, the $g'_\alpha(U_\alpha)$ cover X, and each composite $U_\alpha \to X'_\alpha \xrightarrow{f'_\alpha} Y'_\alpha$ is a closed immersion (respectively, an open immersion). One can moreover take the g_α to be étale.*

Proof. The necessity of the condition for étale morphisms is immediate: take a single $Y'_\alpha = X$, let $g_\alpha = f$, and take for $U_\alpha \subset X \times_Y X$ the diagonal. When f is unramified, necessity follows from (18.4.7): choose an open covering (V_α) of X such that, for each α, $f|V_\alpha$ factors as $V_\alpha \xrightarrow{j_\alpha} Y'_\alpha \xrightarrow{g_\alpha} Y$, where j_α is a closed immersion and g_α is étale. Then $j_\alpha : V_\alpha \to Y'_\alpha$ factors as $V_\alpha \xrightarrow{s_\alpha} X'_\alpha \xrightarrow{f'_\alpha} Y'_\alpha$, where s_α is a V_α-section of X'_α; since $g'_\alpha : X'_\alpha \to X$ is étale, s_α is an open immersion ((17.4.1)), and it suffices to take $U_\alpha = s_\alpha(V_\alpha)$.

The sufficiency of the conditions follows from (17.7.1); one concludes at each point x of $g'(U_\alpha)$, hence in all of X since the $g'_\alpha(U_\alpha)$ cover X. □

Proposition (18.4.9). — *Let S be a prescheme, $f : X \to S$ a morphism, $h : Y \to S$ a morphism locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X, and $y = g(x)$. The following conditions are equivalent:*

a) *h is étale at the point y and g is flat at the point x.*
b) *h is unramified at the point y and f is flat at the point x.*

Proof. Since $f = h \circ g$, condition a) implies that f is flat at x ((2.1.6)); moreover, an étale morphism is unramified, so a) implies b).

To prove that b) implies a), we may first suppose that h is unramified, after replacing Y by an open neighbourhood of y. Then, after replacing S by an open neighbourhood of $s = h(y) = f(x)$, we may suppose that there exist an étale morphism $u : S' \to S$, a point y' of $Y' = Y_{(S')}$ above y, and an open neighbourhood V of y' in Y' such that, if $h' = h_{(S')} : Y' \to S'$, the restriction $h'|V$ is a closed immersion ((18.4.8)). If one proves that h' is étale at the point y', it will follow that h is étale at the point y (17.7.1), (ii); moreover, $f' = f_{(S')} : X_{(S')} \to S'$ is flat at every point x' above x. As the projections $v : Y' \to Y$, $w : X' \to X$ are étale morphisms (hence flat), if one proves that $g' = g_{(S')} : X_{(S')} \to Y_{(S')}$ is flat at the point x', it will follow from (2.2.11), (iv) that g is flat at the point x. We may therefore reduce to the case where h is a *closed immersion* of finite presentation, with f flat at x. Let $\mathscr{J}$ be the quasi-coherent ideal of finite type in $\mathcal{O}_S$ defining the closed subprescheme Y of S. The hypothesis that f is flat at x implies that the homomorphism $\mathcal{O}_{S,s} \to \mathcal{O}_{X,x}$ is injective (EGA 0_{I}, 6.5.1); *a fortiori* the homomorphism $\mathcal{O}_{S,s} \to \mathcal{O}_{Y,y}$ is injective, which means that $\mathscr{J}_s = 0$, since it is the kernel of the preceding homomorphism. Since $\mathscr{J}$ is of finite type, there exists an open neighbourhood U of s in S such that $\mathscr{J}|U = 0$ ((**0**, 5.2.2)). We can hence suppose that h is an open immersion, and it is clear that g is flat at the point x. □ **IV | 123**

Proposition (18.4.10). — *Let us denote by $\mathbf{P}(f, x)$ a property satisfying the following conditions:*

1° *For every morphism $f : X \to Y$ and every local isomorphism $h : Y \to Z$, $\mathbf{P}(f, x)$ is equivalent to $\mathbf{P}(h \circ f, x)$ for $x \in X$.*
2° *For every morphism $f : X \to Y$, every étale morphism $g : Y' \to Y$, every point $x \in X$, if we set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$ and if $x' \in X'$ is above x, the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(f', x')$ are equivalent ("invariance by étale base change").*

Let then S be a prescheme, $f : X \to S$ and $h : Y \to S$ two morphisms, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$, and suppose h étale at the point y. Then the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(g, x)$ are equivalent.

Proof. With the same notation as in the proof of (18.4.9), and after replacing S (respectively, Y) by a neighbourhood of $s = h(y)$ (respectively, of y), (18.4.8) gives an étale morphism $u : S' \to S$ such that h' is an open immersion. If $x' \in X'$ lies above x, the hypothesis makes $\mathbf{P}(f', x')$ (respectively, $\mathbf{P}(g', x')$) equivalent to $\mathbf{P}(f, x)$ (respectively, $\mathbf{P}(g, x)$). We may therefore reduce to the case where h is an open immersion; condition 1° then makes $\mathbf{P}(g, x)$ equivalent to $\mathbf{P}(h \circ g, x) = \mathbf{P}(f, x)$. □

(18.4.11). *Examples.* — One can take for the property $\mathbf{P}(f, x)$ any one of the following, taking into account (17.7.4), (ii):

(i) f is flat at the point x (2.2.11), (iv);
(ii) f is locally of finite presentation and has codepth $\leqslant n$ at the point x ((6.8.1) and (6.7.8));
(iii) f is locally of finite presentation and Cohen–Macaulay at the point x ((6.8.1) and (6.7.8));
(iv) f is locally of finite presentation and possesses the property (S_n) at the point x ((6.8.1) and (6.7.8));
(v) f is locally of finite presentation and possesses the property (R_n) at the point x ((6.8.1) and (6.7.8));
(vi) f is locally of finite presentation and normal at the point x ((6.8.1) and (6.7.8));
(vii) f is locally of finite presentation and reduced at the point x ((6.8.1) and (6.7.8));
(viii) f is unramified at the point x (17.7.4);
(ix) f is smooth at the point x (17.7.4);
(x) f is étale at the point x (17.7.4).

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Corollary (18.4.12). — *(i) Let $f : X \to Y$ be a morphism locally of finite type, x a point of X. If f is flat and formally unramified at the point x, then, in every factorisation $f|U : U \xrightarrow{j} X' \xrightarrow{h} Y$, where U is an open neighbourhood of x, j a closed immersion and h an étale morphism, the homomorphism $\mathscr{O}_{X',j(x)} \to \mathscr{O}_{X,x}$ corresponding to j is bijective (in particular $\mathscr{O}_{X,x}$ is an essentially étale $\mathscr{O}_{Y,f(x)}$-algebra (18.6.1)).*

(ii) For a morphism $f : X \to Y$ to be étale, it is necessary and sufficient that it be locally of finite type, formally unramified and flat.

Proof. (i) Since the homomorphism $\mathscr{O}_{X',j(x)} \to \mathscr{O}_{X,x}$ is surjective, it suffices to prove that it is injective, and for that it suffices to prove that $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{X',j(x)}$-module *faithfully flat*, or merely *flat* ((**0**, 6.5.1) and (6.6.2)); in other words, it amounts to showing that j is a flat morphism at the point x; but since $h \circ j = f$ is flat at the point x by hypothesis and h is étale, this follows from (18.4.10) and (18.4.11), (i).

(ii) It remains only to prove the sufficiency of the stated conditions. For every $x \in X$, there is an open neighbourhood U of x on which $f|U$ admits a factorization having the properties in (i). Since f is flat and formally unramified at *every* point of U, part (i) applies not only at x but at every $z \in U$. Thus, if $\mathscr{J}$ is the quasi-coherent ideal of $\mathscr{O}_{X'}$ defining the closed subprescheme U associated with j, then $\mathscr{J}_{j(z)} = 0$ for every $z \in U$; consequently j is an open immersion, and f is étale at every point of U, hence at every point of X. □

Corollary (18.4.13). — *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that y admits an open neighbourhood which is a reduced prescheme and has only a finite number of irreducible components. Then, for f to be étale at the point x, it is necessary and sufficient that f be flat and formally non ramified at the point x.*

Proof. There is only the sufficiency of the condition to prove. The question being local on X and Y, one can suppose (18.4.7) that f factorises as $X \xrightarrow{j} X' \xrightarrow{h} Y$ where h is étale and j a closed immersion. Then X' is reduced (17.5.7) and, replacing X' by an open neighbourhood of $j(x)$, one can suppose that X' has only a finite number of irreducible components. Then the maximal points of X' are above the maximal points of Y (2.3.4) and since they are finite, their number is finite. It all amounts to showing, with the notations of the proof of (18.4.12), that $\mathscr{J}_{x'} = 0$ for all points x' in a neighbourhood of $j(x)$ in X', knowing that $\mathscr{J}_{j(x)} = 0$. Now, replacing X' by an affine neighbourhood of $j(x)$, we can suppose that all the irreducible components of X' contain $j(x)$; if $X' = \operatorname{Spec}(A')$, and if $\mathfrak{p}'$ is the prime ideal of A' corresponding to the point $j(x)$, the morphism

$\mathrm{Spec}(A'_{\mathfrak{p}'}) \to \mathrm{Spec}(A')$ is dominant, hence the corresponding homomorphism $A' \to A'_{\mathfrak{p}'}$ is *injective* since A' is reduced (**I**, 1.2.7). If $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A', then $\mathfrak{J}$ identifies with a submodule of $A'_{\mathfrak{p}'}$; since $\mathfrak{J}_{\mathfrak{p}'} = 0$ by hypothesis, it follows that $\mathfrak{J} = 0$. ☐

Corollary (18.4.14). — *Let A be a local ring with maximal ideal $\mathfrak{m}$, residue field k, B a finite A-algebra.* IV | 125

(i) For B to be a formally unramified A-algebra, it is necessary and sufficient that $B \otimes_A k$ be a k-algebra étale.

*(ii) For B to be a finite étale A-algebra, it is necessary and sufficient that $B \otimes_A k$ be an étale k-algebra and that B be a flat A-module (which is equivalent to saying ((**0**, 6.3.3)) that for every maximal ideal $\mathfrak{n}$ of B (necessarily above $\mathfrak{m}$), $B_{\mathfrak{n}}$ is a flat A-module).*

Proof. Only sufficiency needs proof. It is clear that (ii) follows from (i) and (18.4.12), (ii), taking into account (17.1.2), (i). To prove (i), let us note that if $B \otimes_A k$ is an étale k-algebra, we have $\Omega^1_{B\otimes_A k/k} = 0$ (17.2.1). Since $\Omega^1_{B\otimes_A k/k} = \Omega^1_{B/A} \otimes_B k$ ((**0**, 20.5.5)) and since B is an A-algebra of finite type, $\Omega^1_{B/A}$ is a B-module of finite type ((**0**, 20.4.7)). But since B is a finite A-algebra, $\mathfrak{m}B$ is contained in the radical of B (Bourbaki, *Alg. comm.*, ch. V, § 2, no. 1, prop. 1), hence the lemma of Nakayama proves that $\Omega^1_{B/A} = 0$, and hence B is a formally unramified A-algebra (17.2.1). ☐

18.5. Henselian local rings. [33]

(18.5.1). Let X be a prescheme and $\mathscr{E}$ a locally free $\mathscr{O}_X$-module of finite rank. Its dual $\check{\mathscr{E}} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{O}_X)$ is therefore a locally free $\mathscr{O}_X$-module whose rank at every point equals that of $\mathscr{E}$ there, and the canonical homomorphism $\mathscr{E} \to \mathscr{H}om_{\mathscr{O}_X}(\check{\mathscr{E}}, \mathscr{O}_X) = (\check{\mathscr{E}})^\vee$ is *bijective*. For every morphism $X' \to X$, set $\mathscr{E}_{(X')} = \mathscr{E} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$, which is a locally free $\mathscr{O}_{X'}$-module, and consider the *set* $\Gamma(X', \mathscr{E}_{(X')})$ of its sections over X'. We shall see that this defines a *contravariant representable functor*

$$ {}^t V_{\mathscr{E}} : X' \longmapsto \Gamma(X', \mathscr{E}_{(X')}) \tag{18.5.1.1}$$

from the category of X-preschemes to the category of *sets* (EGA 0_{III}, 8.1.8).

First of all, this is indeed a well-defined functor, for if $f : X'' \to X'$ is an X-morphism, we have $\mathscr{E}_{(X'')} = f^*(\mathscr{E}_{(X')})$, whence ((**0**, 4.3.2)) a map $\Gamma(X', \mathscr{E}_{(X')}) \to \Gamma(X'', \mathscr{E}_{(X'')})$ which completes the definition of the functor ${}^tV$. Let us then show that the X-prescheme $\mathbf{V}(\check{\mathscr{E}})$ ((**II**, 1.7.8)) represents the functor ${}^tV$. It is immediate that $(\mathscr{E}_{(X')})^\vee = (\check{\mathscr{E}})_{(X')}$, hence $\mathbf{V}(\check{\mathscr{E}}) \times_X X' = \mathbf{V}((\check{\mathscr{E}})_{(X')})$. Taking (**I**, 3.3.14) into account, it suffices to define a bijection

$$\Gamma(\mathbf{V}(\check{\mathscr{E}})/X) \xrightarrow{\sim} \Gamma(X, \mathscr{E})$$

and to verify that the corresponding bijection $\Gamma(\mathbf{V}((\check{\mathscr{E}})_{(X')})/X') \xrightarrow{\sim} \Gamma(X', \mathscr{E}_{(X')})$ is functorial in X'. There is a canonical bijection from $\Gamma(\mathbf{V}(\check{\mathscr{E}})/X)$ to $\mathrm{Hom}_{\mathscr{O}_X}(\check{\mathscr{E}}, \mathscr{O}_X)$ ((**II**, 1.7.8)), and transposition $u \mapsto {}^tu$ gives a canonical bijection from $\mathrm{Hom}_{\mathscr{O}_X}(\check{\mathscr{E}}, \mathscr{O}_X)$ to $\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{O}_X, \mathscr{E}) = \Gamma(X, \mathscr{E})$, because $(\check{\mathscr{E}})^\vee$ is identified with $\mathscr{E}$. The functoriality is immediate.

Observe also that, in accordance with the general theory (EGA 0_{III}, 8.1.6), the identity automorphism of $\mathbf{V}(\check{\mathscr{E}})$ corresponds canonically to a section c of $\mathscr{E}_{(\mathbf{V}(\check{\mathscr{E}}))}$ over $\mathbf{V}(\check{\mathscr{E}})$. Equivalently ((**II**, 1.4.1)), it corresponds to a homomorphism of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$-modules IV | 126

$$u : \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \longrightarrow \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \otimes_{\mathscr{O}_X} \mathscr{E}.$$

For every affine open $W \subset X$, if $\Gamma(W, \mathscr{O}_X) = A$ and $\Gamma(W, \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}))$ is identified with $C = A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ so that the T_i form a basis of $\Gamma(W, \check{\mathscr{E}})$, and if (e_i) denotes the dual basis of $\Gamma(W, \mathscr{E})$, then u corresponds to the C-module homomorphism satisfying

$$u(1) = \sum_{i=1}^{n} \mathrm{T}_i \otimes e_i.$$

If $X' = \mathrm{Spec}(A)$ is affine and $\mathscr{E}_{(X')} \simeq \mathscr{O}^n_{X'}$, then $\Gamma(X', \mathscr{E}_{(X')})$ is a free A-module of rank n; figuratively, $\mathbf{V}(\check{\mathscr{E}})$ represents "the set of points of the twisted affine space over X defined by $\mathscr{E}$".

[33] The notion of a Henselian local ring is due to Azumaya, and that of Henselization to Nagata, to whom the principal results of this theory are also due.

(18.5.2). Recall ((**II**, 1.7.8)) that by definition $\mathbf{V}(\check{\mathscr{E}}) = \operatorname{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}))$. Let $\mathscr{J}$ be a quasi-coherent ideal of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, so that $\operatorname{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ is a closed subprescheme of $\mathbf{V}(\check{\mathscr{E}})$; we shall interpret it as *representing a functor* from the category of X-preschemes to that of sets. A section $u \in \Gamma(X, \mathscr{E})$ identifies canonically with an $\mathscr{O}_X$-homomorphism $u : \mathscr{O}_X \to \mathscr{E}$, to which correspond by transposition an $\mathscr{O}_X$-homomorphism ${}^t u : \check{\mathscr{E}} \to \mathscr{O}_X$ and hence an $\mathscr{O}_X$-algebra homomorphism $v : \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \to \mathscr{O}_X$. Let $\operatorname{Al}(X, \mathscr{E}, \mathscr{J})$ denote the set of $u \in \Gamma(X, \mathscr{E})$ such that $\mathscr{J}$ is *contained in the kernel* of v; it follows immediately that

$$X' \longmapsto \operatorname{Al}(X', \mathscr{E}_{(X')}, \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'})$$

is represented by $\operatorname{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$. If $X' = \operatorname{Spec}(A)$ is affine, $\mathscr{E}_{(X')} \simeq \mathscr{O}_{X'}^n$, and $\mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} = \widetilde{\mathfrak{J}}$ for an ideal $\mathfrak{J}$ of $A[\mathrm{T}_1, \dots, \mathrm{T}_n]$, then this set identifies with the subset of A^n consisting of the points $(t_1, \dots, t_n)$ such that $P(t_1, \dots, t_n) = 0$ for every $P \in \mathfrak{J}$. Thus $\operatorname{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ represents "the algebraic subset of the twisted affine space defined by $\mathscr{E}$ consisting of the points annihilating $\mathscr{J}$". We also denote this X-prescheme by $\mathbf{Al}(\mathscr{E}, \mathscr{J})$. If $\mathscr{J}$ is an ideal of finite type in $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, then $\mathbf{Al}(\mathscr{E}, \mathscr{J})$ is an X-prescheme of finite presentation, since $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$ is an $\mathscr{O}_X$-algebra of finite presentation.

Lemma (18.5.3). — *Let S be a prescheme, $f : X \to S$ a finite and locally free morphism ((18.2.7)). Consider the contravariant functor from the category of S-preschemes to the category of sets*

$$S' \longmapsto \operatorname{Of}(X \times_S S') \tag{18.5.3.1}$$

where $\operatorname{Of}(X \times_S S')$ denotes the set of subsets that are both open and closed of the underlying space of $X \times_S S'$. Then this functor is representable by an S-prescheme $\mathbf{Of}(X)$, which is affine, étale and of finite presentation over S.

Proof. By hypothesis $X = \operatorname{Spec}(\mathscr{B})$, where $\mathscr{B} = f_*(\mathscr{O}_X)$ is a finite and locally free $\mathscr{O}_S$-algebra. Set $X' = X \times_S S'$, $f' = f_{(S')}$, $\mathscr{B}' = \mathscr{B} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} = f'_*(\mathscr{O}_{X'})$, so that $X' = \operatorname{Spec}(\mathscr{B}')$; there is then a canonical bijection, functorial in S', from the set $\operatorname{Of}(X')$ to the set $\operatorname{Id}(\mathscr{B}')$ of *idempotents* of the ring $\Gamma(S', \mathscr{B}') = \Gamma(X', \mathscr{O}_{X'})$. Indeed, by virtue of the equivalence of the category of affine S'-schemes and the opposite category of quasi-coherent $\mathscr{O}_{S'}$-algebras ((**II**, 1.2.7) and (1.3.1)), there is a canonical bijective correspondence between decompositions of X' as a *disjoint union* $X'_1 \sqcup X'_2$ of two induced open subpreschemes and decompositions of $\mathscr{B}'$ as a *direct sum* of two ideals $\mathscr{B}'_1$, $\mathscr{B}'_2$; these latter in turn form a set in canonical bijective correspondence with $\operatorname{Id}(\mathscr{B}')$. It suffices therefore to prove the lemma for the functor $S' \mapsto \operatorname{Id}(\mathscr{B}')$. IV | 127

For this, we shall show that there exists an ideal $\mathscr{J}$ of finite type in $\mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}})$ such that $\operatorname{Id}(\mathscr{B}')$ is of the form $\operatorname{Al}(S', \mathscr{B}', \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'})$ ((18.5.2)). Since $\mathscr{B}$ is an $\mathscr{O}_S$-algebra, its inverse image $\mathscr{B}_{\mathbf{V}(\check{\mathscr{B}})}$ is an $\mathscr{O}_{\mathbf{V}(\check{\mathscr{B}})}$-algebra, and we may form in this algebra the square c^2 of the canonical section c ((18.5.1)); it corresponds canonically to a homomorphism $u^{(2)} : \mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}}) \to \mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}}) \otimes_{\mathscr{O}_S} \mathscr{B}$, and one verifies immediately that for every affine open subset W of S, with the notations of (18.5.1), $u^{(2)}$ corresponds to the C-module homomorphism such that $u^{(2)}(1) = \sum_k (\sum_{i,j} c_{ijk} \mathrm{T}_i \mathrm{T}_j) \otimes e_k$, where (c_{ijk}) is the table of multiplication of the algebra $\Gamma(W, \mathscr{B})$. Let $\mathscr{J}$ be the ideal of $\mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}})$ locally generated by the coefficients of $(u - u^{(2)})(1)$ with respect to a basis of $\mathscr{B}$; it answers the question. Indeed, $\Gamma(W, \mathscr{J})$ is generated by the polynomials $\mathrm{P}_k(\mathrm{T}_1, \dots, \mathrm{T}_n) = \mathrm{T}_k - \sum_{i,j} c_{ijk} \mathrm{T}_i \mathrm{T}_j$, and the idempotents of $\Gamma(W, \mathscr{B})$ are precisely the elements $\sum_i t_i e_i$ such that $(t_1, \dots, t_n)$ annihilates every P_k $(1 \leqslant k \leqslant n)$. We deduce from (18.5.2) that the S-prescheme affine $\mathbf{Of}(X) = \mathbf{Al}(\mathscr{B}, \mathscr{J})$ does represent the functor (18.5.3.1). It remains to show that $\mathbf{Of}(X)$ is étale over S, or equivalently, that it is *formally étale* over S. But if S' is an S-prescheme, S'_0 a closed sub-prescheme of S' defined by a locally nilpotent Ideal (and having the same underlying space as S'), it is clear that $X' = X \times_S S'$ and $X'_0 = X \times_S S'_0$ have the same underlying space, hence the canonical map $\operatorname{Of}(X') \to \operatorname{Of}(X'_0)$ is *bijective*, which completes the proof (17.1.1). □

Proposition (18.5.4). — *Let S be a prescheme, S_0 a closed sub-prescheme of S; consider the following properties:*

a) *For every finite morphism $g : S' \to S$, the canonical map*

$$\operatorname{Of}(S') \longrightarrow \operatorname{Of}(S' \times_S S_0) \tag{18.5.4.1}$$

is bijective (*cf.* (18.5.3)).

a′) *For every finite and locally free morphism $g : S' \to S$, the map* (18.5.4.1) *is bijective.* **IV** | 128

b) *For every étale and separated morphism $g : S' \to S$, the canonical map*

(18.5.4.2) $$\Gamma(S'/S) \longrightarrow \Gamma(S' \times_S S_0/S_0)$$

is bijective.

Condition b) implies a′); if moreover S is quasi-compact and quasi-separated, condition a) implies b).

Proof. Let us first prove that b) implies a'). Suppose therefore b) satisfied, and let $g : S' \to S$ be a finite and locally free morphism; set $S'_0 = S' \times_S S_0$, so that $g_0 = g_{(S_0)} : S'_0 \to S_0$ is finite and locally free. Then it follows from (18.5.3) that $\mathrm{P} = \mathbf{Of}(S')$ is an S-prescheme *étale and separated*; moreover, by the definition of the functor Of, we see immediately that if we set $\mathrm{P}_0 = \mathbf{Of}(S'_0)$ (for the category of S_0-preschemes), we have $\mathrm{P}_0 = \mathrm{P} \times_S S_0$. This being so, by definition we have the commutative diagram

$$\begin{tikzcd} \Gamma(\mathrm{P}/S) \arrow[r] \arrow[d, "\sim"'] & \Gamma(\mathrm{P}_0/S_0) \arrow[d, "\sim"'] \\ \mathrm{Of}(S') \arrow[r] & \mathrm{Of}(S'_0) \end{tikzcd}$$

where the vertical arrows are the bijections. Since hypothesis b), applied to the morphism $\mathrm{P} \to S$, implies that the first row is a bijection, the same holds for the second, which establishes our assertion.

Before proving that a) implies b) when S is quasi-compact and quasi-separated, we will establish the following lemma.

Lemma (18.5.4.3). — *If S and S_0 satisfy condition a) of* (18.5.4), *then, for every finite morphism $g : S' \to S$, S' is the unique open and closed neighbourhood of $S'_0 = g^{-1}(S_0) = S' \times_S S_0$ in S'.*

Proof. Indeed, it amounts to saying that if T' is a closed subset of S' such that $T' \cap S'_0 = \varnothing$, then $T' = \varnothing$. Now, if we also denote by T' the closed sub-prescheme of S' having T' as underlying space, the composite morphism $h : T' \to S' \xrightarrow{g} S$ is finite and $h^{-1}(S_0)$ is empty; condition a), applied to the morphism h, implies that T' is necessarily empty. □

This lemma being established, let us first prove that, under hypothesis a), the map (18.5.4.2) is *injective*. Indeed, if u', u'' are two S-sections of S', the fact that the morphism $S' \to S$ is unramified implies that the prescheme of coincidences of u' and u'' is induced on an open subset U of S (17.4.6). If the restrictions to S_0 of u' and u'' are equal, then the fact that u' and u'' are open immersions (17.4.1) implies that U contains S_0, and hence is equal to S by virtue of Lemma (18.5.4.3), applied to the case $S' = S$.

It remains to show that, under hypothesis a), the map (18.5.4.2) is *surjective* (when S is quasi-compact and quasi-separated). Let therefore $u_0 : S_0 \to S'_0$ be an S_0-section of S'_0; $u_0(S_0)$ being quasi-compact in S', can be covered by a finite number of open affine subsets V_i such that the restriction $g|V_i$ is a morphism of finite type; if V is the union of the V_i, we conclude that $g|V : V \to S$ is a morphism *of finite presentation*, being separated and locally of finite presentation (1.6.1). Replacing S' by V, one can hence suppose that g is of finite presentation. Since S is quasi-compact and quasi-separated, and g is quasi-finite and separated (17.6.1), it follows from the "Main theorem" (8.12.6) that g factorises as $S' \xrightarrow{j} S'' \xrightarrow{f} S$, where j is an open immersion and f a *finite* morphism. Set $S''_0 = S'' \times_S S_0$, $j_0 = j_{(S_0)} : S'_0 \to S''_0$, which is an open immersion, and $f_0 = f_{(S_0)} : S''_0 \to S_0$, which is a finite morphism. Then u_0 is also an S_0-section of S''_0. Since $g_0 : S'_0 \to S_0$ is étale, u_0 is an open immersion of S_0 into S'_0 (17.4.1), so $u_0(S_0)$ is open in S'_0, and *a fortiori* in S''_0; on the other hand, since f_0 is finite and hence separated, u_0 is a closed immersion of S_0 into S''_0 (**I**, 5.4.6). Thus $X_0 = u_0(S_0)$ is *both open and closed* in S''_0. In virtue of condition a), there exists an open and closed subset X of S'' such that $X \cap S''_0 = X_0$. Let us show that the morphism $f : S'' \to S$ is *étale at the points of* X: indeed, the set of points of X where $f|X$ is étale is open and contains $X_0 \subset S'_0$. But the lemma (18.5.4.3) applied to the finite morphism $f|X$ proves that $U = X$. On the other hand, $S' \cap X$ is open in X

and contains X_0, hence by the same reasoning proves that $S' \cap X = X$, i.e. $X \subset S'$. It remains to show that, for every $s \in S$, the geometric number $n(s)$ of points of $X \cap f^{-1}(s)$ is equal to 1; it will then follow that $f|X$ is radicial and surjective, and, since $f|X$ is étale, we shall have proved (17.9.1) that $f|X$ is an isomorphism from the open subset $X \subset S'$ onto S, whose inverse will be the desired S-section extending u_0. But since $f|X$ is étale and finite, $s \mapsto n(s)$ is continuous on S (18.2.8), and since $X \cap S_0'' = X_0$, we have $n(s) = 1$ on S_0; the set of points $s \in S$ such that $n(s) = 1$ is therefore open and closed in S and contains S_0, so it is equal to S by (18.5.4.3). □

Remark (18.5.4.4). — One can show that the statement (18.5.4) remains valid when, in condition b), one supposes merely the morphism g étale (but not necessarily separated) [SGA 1, exp. XII].

Definition (18.5.5). — We say that a prescheme S and a closed sub-prescheme S_0 of S form a *Henselian couple* if they satisfy condition a) of (18.5.4).

Taking into account (**I**, 5.1.8), it amounts to the same to say that (S, S_0) is a Henselian couple or that $(S_{\mathrm{red}}, (S_0)_{\mathrm{red}})$ is one.

Proposition (18.5.6). — *(i) If (S, S_0) is a Henselian couple, then, for every finite morphism $f : S' \to S$, if S_0' is the closed sub-prescheme $f^{-1}(S_0)$ of S', the couple (S', S_0') is Henselian.*

(ii) Let $S = \coprod_\alpha S^{(\alpha)}$ be a sum of preschemes, S_0 a closed sub-prescheme of S, the sum of sub-preschemes $S_0^{(\alpha)}$ of the $S^{(\alpha)}$. For the couple (S, S_0) to be Henselian, it is necessary and sufficient that each of the couples $(S^{(\alpha)}, S_0^{(\alpha)})$ be Henselian.

Proof. Assertion (i) is an immediate consequence of the definition, since for every finite morphism $g : S'' \to S'$, $f \circ g : S'' \to S$ is a finite morphism. Similarly, under the conditions of (ii), for a morphism $g : S' \to S$ to be finite, it is necessary and sufficient that each of its restrictions $g^{(\alpha)} :$ **IV** | 130
$S'^{(\alpha)} = g^{-1}(S^{(\alpha)}) \to S^{(\alpha)}$ be finite. If we set $S_0' = g^{-1}(S_0)$ and $S_0'^{(\alpha)} = (g^{(\alpha)})^{-1}(S_0^{(\alpha)})$, the open-and-closed subsets U (resp. U_0) of S' (resp. S_0') correspond bijectively to the families $(U^{(\alpha)})$ (resp. $(U_0^{(\alpha)})$), where $U^{(\alpha)}$ (resp. $U_0^{(\alpha)}$) is an open-and-closed subset of $S'^{(\alpha)}$ (resp. $S_0'^{(\alpha)}$); assertion (ii) follows. □

Remark (18.5.7). — Let $S = \mathrm{Spec}(A)$ be an affine scheme, S_0 a closed sub-prescheme defined by an ideal $\mathfrak{J}$ of A. Then, if the couple (S, S_0) is Henselian, the ideal $\mathfrak{J}$ is necessarily contained in the *radical* of A. Indeed, if $\mathfrak{m}$ is a maximal ideal of A, $\mathfrak{m}$ must belong to $V(\mathfrak{J}) = S_0$, by virtue of (18.5.4.3), whence the conclusion. In particular, suppose that S_0 is *reduced to a point*, then $\mathfrak{J}$ must be the radical of A, so A must be a *local* ring, and S_0 the unique closed point of $\mathrm{Spec}(A)$.

Definition (18.5.8). — We say that a ring A is *Henselian* if it is semi-local and if, denoting by $\mathfrak{r}$ the radical of A, the couple $(\mathrm{Spec}(A), \mathrm{Spec}(A/\mathfrak{r}))$ is Henselian. We call a *Henselian local scheme* a scheme isomorphic to the spectrum of a Henselian local ring.

Proposition (18.5.9). — *(i) For a semi-local ring A to be Henselian, it is necessary and sufficient that it be a direct product of Henselian local rings.*

(ii) For a local ring A to be Henselian, it is necessary and sufficient that every finite A-algebra B be isomorphic to a product of local rings.

Proof. (i) Indeed, the definition, applied to $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/\mathfrak{r})$, shows that S_0 is a finite discrete set, closed in S, and S is the union of a finite number of open and closed subsets S_i $(1 \leqslant i \leqslant n)$, pairwise disjoint, each containing exactly one of the maximal ideals $\mathfrak{m}_i$ of A; the conclusion follows from (18.5.6), (ii) and the remark (18.5.7).

(ii) Every finite morphism $S' \to \mathrm{Spec}(A)$ is of the form $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$, where B is a finite A-algebra. If k is the residue field of A, $\mathrm{Spec}(B \otimes_A k)$ is a finite Artinian spectrum, hence finite and discrete. To say that the couple $(\mathrm{Spec}(A), \mathrm{Spec}(k))$ is Henselian means that B is a direct product of rings A_i such that $\mathrm{Spec}(A_i \otimes_A k)$ is reduced to a point, i.e. that each A_i (which is a finite A-algebra) must have only a single maximal ideal (Bourbaki, *Alg. comm.*, ch. V, § 2, no. 1, prop. 1). □

The study of Henselian rings is thus essentially reduced to that of Henselian *local* rings.

Proposition (18.5.10). — *If A is a Henselian ring, every finite A-algebra B is a Henselian ring (hence a direct product of Henselian local rings* (18.5.9)*).*

Proof. Indeed, if $\mathfrak{r}'$ is the radical of B, the inverse image of $\mathfrak{r}'$ in A is the radical $\mathfrak{r}$ of A, and every prime ideal of B lying over a maximal ideal of A is a maximal ideal of B; hence the set $V(\mathfrak{r}')$ in $\operatorname{Spec}(B)$ is the inverse image of the set $V(\mathfrak{r})$ in $\operatorname{Spec}(A)$. The proposition is then a consequence of (18.5.6), (i). □

Theorem (18.5.11). — *Let A be a local ring, $\mathfrak{m}$ its maximal ideal. The following conditions are equivalent:*

a) *A is Henselian, in other words, every finite A-algebra B is isomorphic to a product of local rings.* **IV** | 131

a′) *Condition a) is satisfied for all A-algebras B of the form $A[\mathrm{T}]/F.A[\mathrm{T}]$, where $F \in A[\mathrm{T}]$ is a monic polynomial.*

b) *Let $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(k)$. For every étale morphism $g : S' \to S$, if we set $S'_0 = S' \otimes_A k$, every S_0-section u_0 of S'_0 is the restriction of an S-section u of S'.*

c) *For every morphism locally of finite type $f : X \to S$, separated, and every point $x \in X$ such that $f(x)$ is equal to the closed point s of S and that f is quasi-finite at the point x ((**III**, Err.20)), X is the sum of two preschemes X', X'' such that $X' = \operatorname{Spec}(\mathscr{O}_{X,x})$ and that $f|X' : X' \to S$ is a finite morphism.*

c′) *For every morphism locally of finite type $f : X \to S$, and every point $x \in X$ such that f be quasi-finite at the point x and $f(x)$ equal to the closed point s of S, $\mathscr{O}_{X,x}$ is a finite algebra over $\mathscr{O}_{S,s} = A$.*

c″) *For every morphism locally of finite presentation $f : X \to S$, and every point $x \in X$ such that f be quasi-finite at the point x and $f(x)$ equal to the closed point s of S, $\mathscr{O}_{X,x}$ is a finite algebra and of finite presentation over $\mathscr{O}_{S,s} = A$.*

Proof. Let us first note that c') (resp. c'') is equivalent to the same condition where moreover f is supposed *separated*, the question being local on X. Similarly, condition b) is equivalent to the same condition where moreover g is supposed *separated*: indeed, it suffices to apply the latter to the restriction of g to an open neighbourhood affine of the point $u_0(S_0)$ in S'. Let us henceforth restrict to the case where, in b), c') and c'', the morphisms are separated.

The fact that a) implies b) and that b) implies a') is a particular case of (18.5.4). Let us moreover show that a') implies a). It is a matter of proving that if e_0 is an idempotent of $C = B \otimes_A k$, there is an idempotent $e \in B$ whose image is e_0 in C. If b is an element of B whose image in C is e_0, the sub-A-algebra $A[b] = B'$ of B is finite, and the canonical image C' of $B' \otimes_A k$ in C contains e_0. Now, $B' \otimes_A k$ is a finite k-algebra, hence a direct product of finite local k-algebras, and consequently e_0 is the image in C' of an idempotent e'_0 of $B' \otimes_A k$. We are thus reduced to the case where B is monogenic, and hence by a'), a quotient of an algebra of the form $A[\mathrm{T}]/F.A[\mathrm{T}]$; now, by a'), $A[\mathrm{T}]/F.A[\mathrm{T}]$ is a direct product of local rings, hence so is each of its quotient algebras, which proves the existence of the idempotent e.

It is immediate that c) implies a), as one sees by induction on the number of maximal ideals of the semi-local ring B. To see that a) implies c), we may, by (13.1.4), restrict to the case where f is affine and quasi-finite. Then, by the "Main theorem" (8.12.8), f can be written as the composite

$$X \xrightarrow{j} Y \xrightarrow{g} S,$$

where g is finite and j is an open immersion. Since $Y = \operatorname{Spec}(B)$, where B is a finite A-algebra, condition a) implies that B is a direct product of local rings, which are evidently finite A-algebras, and $\mathscr{O}_{X,x}$ identifies with one of these local rings because $f(x) = s$; moreover, every open subset of Y containing x necessarily contains $\operatorname{Spec}(\mathscr{O}_{X,x})$.

It is trivial that c) implies c'), by virtue of the remark at the beginning. Let us prove that c') implies c''). Suppose c') satisfied and let us prove that, under the hypotheses of c''), the subset **IV** | 132
$Z = \operatorname{Spec}(\mathscr{O}_{X,x})$ identifies with an *open and closed* subset of X, which will establish c'') (**I**, 2.4.2). First, the composite morphism $Z \xrightarrow{j} X \xrightarrow{f} S$, where j is the canonical morphism (**I**, 2.4.1), is finite and of finite presentation by hypothesis; and since f is separated and locally of finite presentation, j is also finite and of finite presentation ((**II**, 6.1.5) and (**I**, 4.3)), hence j is a closed morphism (**II**, 6.1.10), which proves that Z is closed in X. It then follows from (**I**, 2.4.2) and (**I**, 4.2.2) that j is a closed immersion. If $\mathscr{J}$ is the ideal of $\mathscr{O}_X$ defining Z, then by hypothesis $\mathscr{J}_x = 0$, and hence $\mathscr{J}_z = 0$ at every point z of some open neighbourhood V of x in X, since $\mathscr{J}$ is a quasi-coherent $\mathscr{O}_X$-module of finite type ((1.4.7) and (**0**, 5.2.2)). Such a neighbourhood contains Z, so Z is open in X, since $\mathscr{J}|V = 0$.

Finally, $c'')$ implies $a')$: indeed, if $B = A[\mathrm{T}]/F.A[\mathrm{T}]$, the morphism $X = \mathrm{Spec}(B) \to \mathrm{Spec}(A) = Y$ is finite and of finite presentation, and the preceding argument shows that B is a direct product of finite A-algebras $\mathscr{O}_{X,x_i}$, where the x_i are the points of the fibre of the closed point of Y. □

Corollary (18.5.12). — *Let A be a semi-local ring, $\mathfrak{r}$ its radical; set $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A/\mathfrak{r})$. For A to be Henselian, it is necessary and sufficient that, for every finite morphism $f : X \to S$ and every étale and separated morphism $g : Y \to S$, if we set $X_0 = X \times_S S_0$ and $Y_0 = Y \times_S S_0$, the canonical map*

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S_0}(X_0, Y_0)$$

is bijective.

Proof. The sufficiency of the condition follows from the equivalence of $a)$ and $b)$ in (18.5.11), by applying this condition to the case where $f = 1_S$. To see that the condition is necessary, note that if A is Henselian, the couple (X, X_0) is Henselian by (18.5.6), (i); moreover $\mathrm{Hom}_S(X, Y) = \Gamma(X \times_S Y/X)$ and $\mathrm{Hom}_{S_0}(X_0, Y_0) = \Gamma(X_0 \times_{S_0} Y_0/X_0)$; since $X \times_S Y$ is étale and separated over X, the conclusion follows from the fact that $a)$ implies $b)$ in (18.5.4). □

Remark (18.5.13). — The equivalent conditions $a)$, $a')$, $b)$ of Theorem (18.5.11) are also equivalent to the following ("Hensel's lemma"):

a$''$) For every monic polynomial $F \in A[\mathrm{T}]$, with canonical image $F_0 \in k[\mathrm{T}]$, and every decomposition $F_0 = G_0 H_0$ of F_0 into a product of two monic polynomials that are coprime G_0, H_0 in $k[\mathrm{T}]$, there exists a unique couple (G, H) of monic polynomials of $A[\mathrm{T}]$ possessing the following properties: G_0 and H_0 are the canonical images respectively of G and H, we have $F = GH$, and the ideal of $A[\mathrm{T}]$ generated by G and H is equal to $A[\mathrm{T}]$.

Lemma (18.5.13.1). — *Let A be a local ring with residue field k, $F \in A[\mathrm{T}]$ a monic polynomial, B the A-algebra $A[\mathrm{T}]/F.A[\mathrm{T}]$. There exists a canonical bijective correspondence between the decompositions of B into the direct product of two quotient A-algebras B', B'' and the decompositions $F = GH$ of F into a product of two monic polynomials G, H of $A[\mathrm{T}]$, such that the ideal generated by G and H is equal to $A[\mathrm{T}]$; the quotient algebras B', B'' corresponding to such a couple of polynomials G, H are respectively $A[\mathrm{T}]/H.A[\mathrm{T}]$ and $A[\mathrm{T}]/G.A[\mathrm{T}]$.*

Proof. If $F = GH$ and G and H generate the ideal $A[\mathrm{T}]$, there are two polynomials P, Q of $A[\mathrm{T}]$ such that $1 = PG + QH$. We deduce that the intersection of the principal ideals $\mathfrak{a} = G.A[\mathrm{T}]$ and $\mathfrak{b} = H.A[\mathrm{T}]$ is equal to $\mathfrak{c} = F.A[\mathrm{T}]$: indeed, if $R \in G.A[\mathrm{T}] \cap H.A[\mathrm{T}]$, we can write $R = PRG + QRH$; since RG (resp. RH) is a multiple of F since R is a multiple of H (resp. G), therefore $R \in F.A[\mathrm{T}]$. Since $A[\mathrm{T}] = \mathfrak{a} + \mathfrak{b}$, $A[\mathrm{T}]/\mathfrak{c}$ is the direct sum of ideals $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$ and $\mathfrak{b}/(\mathfrak{a} \cap \mathfrak{b})$, canonically isomorphic respectively to $A[\mathrm{T}]/\mathfrak{b}$ and $A[\mathrm{T}]/\mathfrak{a}$. IV | 133

Conversely, suppose given a decomposition of B into a direct product of two A-algebras B', B'', which identify canonically with two ideals $e'B$, $e''B$ of B, corresponding to a decomposition $1 = e' + e''$ into orthogonal idempotents e', e'' of B. Set further $B_0 = B \otimes_A k = k[\mathrm{T}]/F_0.k[\mathrm{T}]$, where F_0 is the canonical image of F in $k[\mathrm{T}]$, of the same degree n as F; if e'_0, e''_0 are the canonical images of e', e'' in B_0, these are two orthogonal idempotents such that $1 = e'_0 + e''_0$, and B_0 is the direct product of $B'_0 = e'_0 B_0$ and $B''_0 = e''_0 B_0$. Let t and t_0 be the canonical images of T in B and B_0; B'_0 (resp. B''_0) is a k-algebra generated by $t'_0 = e'_0 t_0$ (resp. $t''_0 = e''_0 t_0$), and it admits a base of the form $\{e'_0, t'_0, t'^2_0, \dots, t'^{s-1}_0\}$ (resp. $\{e''_0, t''_0, t''^2_0, \dots, t''^{r-1}_0\}$) with $r + s = n$. On the other hand, B being a free A-module (of base $\{1, t, \dots, t^{n-1}\}$), B' and B'' are projective A-modules, hence free since A is a local ring (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 3, cor. de la prop. 5); it follows from the above and from Bourbaki, *Alg. comm.*, ch. II, § 3, no. 3, prop. 5, that if we set $t' = e't$, $t'' = e''t$, $\{e', t', \dots, t'^{s-1}\}$ (resp. $\{e'', t'', \dots, t''^{r-1}\}$) is a base of the A-module B' (resp. B''). There is hence a monic polynomial H (resp. G) of degree s (resp. r) in $A[\mathrm{T}]$ such that $e'H(t') = 0$ and $e''G(t'') = 0$; as $t^h = t'^h + t''^h$ for every integer $h \geqslant 1$, with $t'^h = e't'^h$ and $t''^h = e''t''^h$, we also have $G(t) = e'G(t')$ and $H(t) = e''H(t'')$, whence $G(t)H(t) = 0$; we conclude that the polynomial $G(T)H(T)$ is divisible by $F(T)$; but since these two monic polynomials have the same degree, we have $GH = F$. Moreover, B' (resp. B'') is isomorphic to $A[\mathrm{T}]/H.A[\mathrm{T}]$ (resp. $A[\mathrm{T}]/G.A[\mathrm{T}]$). Finally, there are two polynomials R, S in $A[\mathrm{T}]$ such that $e' = R(t)$ and $e'' = S(t)$; as $R(t) = e'R(t') + e''R(t'')$, we necessarily have $e''R(t'') = 0$, and similarly $e'S(t') = 0$, so that, by definition of G and H, $R = QH$ and $S = PG$, where P, Q belong to $A[\mathrm{T}]$; the relation $1 = R(t) + S(t)$ in B gives $PG + QH = 1 + LF$ for some polynomial $L \in A[\mathrm{T}]$, and since $F = GH$, this proves that the ideal generated by G and H is equal to $A[\mathrm{T}]$, and completes the proof of the lemma. □

This lemma being established, it suffices to apply it, to the ring A on the one hand, to the field k on the other, to see immediately that conditions a') and a'') are equivalent.

Proposition (18.5.14). — *Every semi-local ring, separated and complete for the $\mathfrak{r}$-preadic topology (where $\mathfrak{r}$ is the radical of A) is Henselian.*

Proof. Indeed, A is a direct product of separated and complete local rings (Bourbaki, *Alg. comm.*, ch. III, § 2, no. 13, cor. of prop. 19), so we are reduced to the case where A is a local ring. Let us verify criterion a') of (18.5.11). Since B is a free A-module of finite type, B is separated and complete for the $\mathfrak{r}$-adic topology, which is also the $\mathfrak{s}$-adic topology, where $\mathfrak{s}$ is the radical of the semi-local ring B, because $B/\mathfrak{r}B$ is an Artinian ring with radical $\mathfrak{s}/\mathfrak{r}B$. It is then known (Bourbaki, *Alg. comm.*, ch. III, § 2, no. 13, cor. of prop. 19) that B is a direct product of local rings. □ **IV** | 134

Proposition (18.5.15). — *Let A be a Henselian ring, $\mathfrak{r}$ its radical. Then the functor $B \mapsto B/\mathfrak{r}B$ is an equivalence of the category of finite étale A-algebras with the category of finite étale $(A/\mathfrak{r})$-algebras.*

Proof. The fact that the functor is fully faithful is a particular case of (18.5.12). To show that this functor is an equivalence of categories, we may reduce to the case where A is local; it suffices then to apply (18.1.1) (for étale morphisms), with $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/\mathfrak{r})$, reduced to a single point. □

Remarks (18.5.16). —

(i) We do not know if Proposition (18.5.15) generalises to a Henselian couple (S, S_0), even when S is affine and Noetherian.

(ii) Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is *separated and complete* for the $\mathfrak{J}$-preadic topology. Then the couple $(\mathrm{Spec}(A), \mathrm{Spec}(A/\mathfrak{J}))$ is Henselian: indeed, for every finite A-algebra B, B is separated and complete for the $\mathfrak{J}$-preadic topology ((**0**, 7.3.6)). Replacing B by A and $\mathfrak{J}B$ by $\mathfrak{J}$, everything amounts to seeing that the map that to every idempotent of A associates its class mod $\mathfrak{J}$, is *bijective*. Now $A = \varprojlim A/\mathfrak{J}^n$. Write $\mathrm{Idem}(A)$ for the set of idempotents of A, and, for every ring homomorphism $\varphi : A \to B$, let $\mathrm{Idem}(\varphi)$ denote the map $\mathrm{Idem}(A) \to \mathrm{Idem}(B)$ induced by φ. It follows from the definition of the projective limit that
$$\mathrm{Idem}(A) = \varprojlim_n \mathrm{Idem}(A/\mathfrak{J}^n),$$
for the transition maps $\psi_{nm} : \mathrm{Idem}(A/\mathfrak{J}^m) \to \mathrm{Idem}(A/\mathfrak{J}^n)$ induced by the canonical maps $A/\mathfrak{J}^m \to A/\mathfrak{J}^n$. But since $\mathrm{Spec}(A/\mathfrak{J}^n) \to \mathrm{Spec}(A/\mathfrak{J}^m)$ is a homeomorphism, the ψ_{nm} are bijections (as seen in the proof of (18.5.3)); this proves our assertion.

Theorem (18.5.17). — *Let A be a Henselian local ring, $S = \mathrm{Spec}(A)$, s the closed point. For every smooth morphism $f : X \to S$, the canonical map*
$$\Gamma(X/S) \longrightarrow \Gamma(X_s/\kappa(s))$$
(where $X_s = f^{-1}(s)$) is surjective.

Proof. The datum of a $\kappa(s)$-section of X_s is equivalent to that of a point $x \in X$ above s whose residue field $\kappa(x)$ is isomorphic to $\kappa(s)$, and it is a matter of proving that there exists an S-section $u : S \to X$ such that $u(s) = x$. Taking into account (17.16.3), (i), we can suppose that f is étale. Then the conclusion follows from the criterion (18.5.11), b). (The reader will note that by virtue of this criterion, the validity of (18.5.17) for a local ring is necessary and sufficient for A to be Henselian.) □

Remark (18.5.18). — The conditions of (18.5.11) are also equivalent to the following:

d) For every morphism locally of finite type $f : X \to S$ and every point $x \in X$ such that $f(x)$ be equal to the closed point s of S and that $\mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$ be a field canonically isomorphic to $k = \kappa(s)$, there exists an open neighbourhood U of x in X such that $f|U$ is a closed immersion.

Proof. It is immediate that $d)$ implies condition $b)$ of (18.5.11) since a closed immersion étale is an open immersion (18.9.1). Conversely, suppose the conditions of (18.5.11) satisfied, and let us prove $d)$. The hypothesis of $d)$ implies that f is quasi-finite at the point x ((**III**, Err.20)), by virtue of (13.1.4); hence, by condition $c)$ of (18.5.11), $\mathscr{O}_{X,x}$ is a finite A-algebra and $\mathrm{Spec}(\mathscr{O}_{X,x})$ is an open neighbourhood U of x in X. Moreover, since $\mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$ is isomorphic to $k = A/\mathfrak{m}_s$, the Nakayama lemma proves that the homomorphism $A \to \mathscr{O}_{X,x}$ is surjective, hence $f|U$ is a closed immersion. ☐ **IV | 135**

Proposition (18.5.19). — *Let A be a Noetherian Henselian local ring, $S = \mathrm{Spec}(A)$, s the closed point of S, and $f : X \to S$ a proper morphism; set $X_s = f^{-1}(s)$. Then the map $Y \to Y \cap X_s$ is a bijection from the set of connected components of X to the set of connected components of X_s.*

Proof. By considering a closed sub-prescheme of X having for underlying space a connected component of X, we are reduced to showing that if X is *connected and non-empty*, then X_s is *connected and non-empty*. The fact that X_s is non-empty follows from ((**II**, 7.2.1)); to show that X_s is connected, we argue by contradiction, considering the Stein factorisation $f : X \xrightarrow{f'} S' \xrightarrow{g} S$ of the proper morphism f ((**III**, 4.3.3)); by hypothesis the finite discrete set $g^{-1}(s)$ would contain at least two points. Since A is Henselian and g separated and of finite type, it would then follow from (18.5.11), $c)$ that S' would be the sum of two non-empty preschemes S'_1, S'_2 (since the intersection of any one of them with $g^{-1}(s)$ is reduced to a single point). Since f' is surjective ((**III**, 4.3.1)), we would conclude that X is a sum of two non-empty preschemes, contradicting the hypothesis. ☐

18.6. Henselization.

(18.6.1). Given a local ring A, we shall say that a local A-algebra B is *essentially étale* if there exist an étale A-algebra C and a prime ideal $\mathfrak{n}$ of C such that B is A-isomorphic to $C_{\mathfrak{n}}$ and the composite homomorphism $A \to C \to C_{\mathfrak{n}}$ is local (in other words, $\mathfrak{n}$ lies above the maximal ideal $\mathfrak{m}$ of A). This implies (17.6.1) that $\mathfrak{n}$ is the only prime ideal of B lying above $\mathfrak{m}$, since B is a local ring; if B, B' are two local essentially étale A-algebras, every A-homomorphism $B \to B'$ is therefore local.

If A' is a local essentially étale A-algebra and A'' a local essentially étale A'-algebra, then A'' is a local essentially étale A-algebra. Indeed, by hypothesis $A' = B_{\mathfrak{n}}$, where B is an étale A-algebra and $\mathfrak{n}$ a prime ideal of B above the maximal ideal of A, and $A'' = B'_{\mathfrak{n}'}$, where B' is an étale A'-algebra and $\mathfrak{n}'$ a prime ideal of B' above $\mathfrak{n}B_{\mathfrak{n}}$. If we set $S = B - \mathfrak{n}$, so that $A' = S^{-1}B$, then B' is of the form $S^{-1}C$, where C is a B-algebra of finite presentation. Consequently C is an A-algebra of finite presentation and A'' is of the form $C_{\mathfrak{r}}$, where $\mathfrak{r}$ is a prime ideal of C above the maximal ideal of A. Since A' is formally étale over A and A'' is formally étale over A', A'' is formally étale over A (for the discrete topologies (17.1.3)); the morphism $\mathrm{Spec}(C) \to \mathrm{Spec}(A)$ is therefore étale at the point $\mathfrak{r}$ (17.6.1), and hence there exists an element $g \in C$ such that C_g is an étale A-algebra, which proves that A'' is an essentially étale A-algebra. **IV | 136**

Given a local ring A, there exists a *set* $\mathfrak{E}$ of local essentially étale A-algebras such that every local essentially étale A-algebra is A-isomorphic to an algebra belonging to $\mathfrak{E}$. It suffices indeed to observe that there exists a set $\mathfrak{F}$ of A-algebras of finite type such that every A-algebra of finite type is isomorphic to an algebra belonging to $\mathfrak{F}$; one may take $\mathfrak{F}$ to be the set of quotients of the polynomial algebras $A[\mathrm{T}_1, \dots, \mathrm{T}_n]$ ($n \in \mathbf{N}$).

In what follows, if A and B are two local rings, we denote by $\mathrm{Hom.loc}(A, B)$ the set of *local* homomorphisms from A to B.

Lemma (18.6.2). — *Let A, A' be two local rings, k, k' their residue fields, $\phi : \mathrm{A} \to \mathrm{A}'$ a local homomorphism making A' an A-algebra essentially étale (18.6.1) and such that the corresponding homomorphism $k \to k'$ is bijective. Then, for every local Henselian ring B, the canonical map*

$$\mathrm{Hom}(\phi, 1_{\mathrm{B}}) : \mathrm{Hom.loc}(\mathrm{A}', \mathrm{B}) \longrightarrow \mathrm{Hom.loc}(\mathrm{A}, \mathrm{B})$$

is bijective.

Proof. Let $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$, and let s and y be the closed points of S and Y respectively. By hypothesis, A' is isomorphic to a local ring $\mathscr{O}_{X,x}$ of an S-scheme X étale over S, at a point $x \in \mathrm{X}$ above s. Suppose given a local homomorphism $\psi : \mathrm{A} \to \mathrm{B}$, making Y an S-scheme; it amounts to showing that there exists *one and only one* S-morphism $f : \mathrm{Y} \to \mathrm{X}$ such that $f(y) = x$. Let

$X' = X \times_S Y$, and note that since $k(x) = k(s)$, there exists a unique point $x' \in X'$ above x and y, and that $k(x') = k(y)$. It remains to show that there exists a unique Y-section f' of X′ such that $f'(y) = x'$. Now, the morphism $g : X' \to Y$ is étale and separated, and the fibre $X'_0 = g^{-1}(y)$ at the point x' has the residue field $k(x') = k(y)$. If we set $Y_0 = \mathrm{Spec}(k(y))$, there exists a unique Y_0-section f'_0 of X'_0 such that $f'_0(y) = x'$, and the conclusion follows from the fact that B is assumed Henselian and from (18.5.11), *b*). □

We say that a local A-algebra A′ satisfying the conditions of (18.6.2) is *strictly essentially étale*. Note that the criterion (18.5.11), *b*) means that, *for A to be Henselian, it is necessary and sufficient that every strictly essentially étale A-algebra be A-isomorphic to* A.

Lemma (18.6.3). — *Let* A *be a local ring,* A_1, A_2 *two A-algebras that are local and strictly essentially étale.*

i) *There exists at most one A-homomorphism (necessarily local) from* A_1 *to* A_2.
ii) *There exists a local A-algebra* A_3 *that is strictly essentially étale, and two A-homomorphisms* $A_1 \to A_3$, $A_2 \to A_3$.

Proof. Let $S = \mathrm{Spec}(A)$; by hypothesis there are two S-schemes étale over S, X_1, X_2, and two points $x_1 \in X_1$, $x_2 \in X_2$ above the closed point s of S and such that $A_1 = \mathscr{O}_{X_1,x_1}$, $A_2 = \mathscr{O}_{X_2,x_2}$; Let $X_3 = X_1 \times_S X_2$; the hypotheses $k(x_1) = k(x_2) = k(s)$ (**I**, 3.4.9) imply that there exists a unique point $x_3 \in X_3$ above x_1 and x_2, and that $k(x_3) = k(s)$. Furthermore, X_3 is étale over S (17.3.3), so $A_3 = \mathscr{O}_{X_3,x_3}$ satisfies the conditions of (ii). On the other hand, we have seen that every A-homomorphism **IV | 137**
from A_1 to A_2 is necessarily local; it corresponds to an S-morphism f from $X'_2 = \mathrm{Spec}(A_2)$ to X_1 such that $f(x_2) = x_1$, or, equivalently, on setting $X'_3 = X_1 \times_S X'_2$, to an X'_2-section f' of X'_3 such that $f'(x_2) = x_3$. Since $k(x_3) = k(x_2)$ and X'_2 is connected, the uniqueness of f' follows from (17.4.9), whence (i). □

(18.6.4). Let us denote by $\mathfrak{S}$ the subset of the set $\mathfrak{E}$ defined in (18.6.1) formed by the A-algebras that are strictly essentially étale belonging to $\mathfrak{E}$. It follows from (18.6.3) that the relation "there exists an A-homomorphism from A_1 to A_2" is a *preorder* relation on $\mathfrak{S}$, and makes $\mathfrak{S}$ a *filtered increasing* set. Indexing $\mathfrak{S}$ by itself by means of the identity map, it follows from (18.6.3), (i) that if $\lambda \leqslant \mu$ in $\mathfrak{S}$, there exists a *unique* $\phi_{\mu\lambda} : A_\lambda \to A_\mu$ and $(A_\lambda, \phi_{\mu\lambda})$ is clearly an *inductive system* of local A-algebras, the $\phi_{\mu\lambda}$ being local homomorphisms. Furthermore, it follows from (17.3.5) that for $\lambda \leqslant \mu$, A_μ is an A_λ-algebra that is strictly essentially étale.

Definition (18.6.5). — The *Henselization* of a local ring A is the inductive limit of the inductive system $(A_\lambda, \phi_{\mu\lambda})$, denoted ${}^hA$, defined in (18.6.4).

This definition does not depend on the choice of $\mathfrak{E}$; if $\mathfrak{E}'$ is another set having the same property as $\mathfrak{E}$, and $\mathfrak{S}'$ the subset of $\mathfrak{E}'$ consisting of the strictly essentially étale A-algebras, it follows from (18.6.3), (ii) that, for the preorder relations considered, $\mathfrak{S}$ and $\mathfrak{S}'$ are *cofinal* in $\mathfrak{S}'' = \mathfrak{S} \cup \mathfrak{S}'$, so they give the same inductive limit up to an isomorphism. We shall moreover see below (18.6.6), (i) and (ii) that ${}^hA$ and the canonical homomorphism $A \to {}^hA$ are the solution of a universal problem, and are therefore determined up to a unique isomorphism.

One will observe that if A is a field, then evidently ${}^hA = A$. In the general case, ${}^hA$ is also the Henselization of all the A-algebras that are strictly essentially étale A_λ ($\lambda \in \mathfrak{S}$), since if B is an A_λ-algebra that is strictly essentially étale, B is also an A-algebra that is strictly essentially étale (18.6.1), so (up to an A-isomorphism) one of the A_μ for $\mu \geqslant \lambda$, and the A_μ such that $\mu \geqslant \lambda$ and that A_μ is an A_λ-algebra that is strictly essentially étale form a cofinal set in the preordered set of the A_λ, by virtue of (18.6.1) and (18.6.3).

Theorem (18.6.6). — *Let* A *be a local ring,* ${}^hA$ *its Henselization.*

i) ${}^hA$ *is a Henselian local ring and the structural homomorphism* $A \to {}^hA$ *is local.*
ii) *For every Henselian local ring* B*, the canonical map*
$$\mathrm{Hom\,.\,loc}({}^hA, B) \longrightarrow \mathrm{Hom\,.\,loc}(A, B)$$
is bijective.
iii) ${}^hA$ *is a faithfully flat A-module, and if* $\mathfrak{m}$ *is the maximal ideal of* A, $\mathfrak{m}.{}^hA$ *is the maximal ideal of* ${}^hA$, *and the homomorphism* $A/\mathfrak{m} \to {}^hA/\mathfrak{m}.{}^hA$ *of residue fields is bijective.*

iv) *If* $\widehat{\mathrm{A}}$ *and* $({}^h\mathrm{A})\widehat{}$ *are the separated completions of the local rings* A *and* ${}^h\mathrm{A}$*, the homomorphism* $\widehat{\mathrm{A}} \to ({}^h\mathrm{A})\widehat{}$ *deduced from the structural homomorphism* $\mathrm{A} \to {}^h\mathrm{A}$ *by completion is bijective.*

v) *For* ${}^h\mathrm{A}$ *to be Noetherian, it is necessary and sufficient that A be so.*

vi) *If A is Henselian, the canonical homomorphism* $\mathrm{A} \to {}^h\mathrm{A}$ *is bijective.*

Proof. Let $\mathfrak{m}_\lambda$ be the maximal ideal of A_λ; it follows from the fact that, in (17.6.1), a) implies c'), that for $\lambda \leqslant \mu$ one has $\mathfrak{m}_\mu = \mathfrak{m}_\lambda \mathrm{A}_\mu$, that the homomorphism $\mathrm{A}_\lambda/\mathfrak{m}_\lambda \to \mathrm{A}_\mu/\mathfrak{m}_\mu$ is bijective, and that A_μ is a flat A_λ-module. The fact that ${}^h\mathrm{A}$ is local, that the homomorphism $\mathrm{A} \to {}^h\mathrm{A}$ is local, assertion (iii) and the sufficiency of (v) follow therefore from (**0**, 10.3.1.3); the necessity of (v) follows from the fact that ${}^h\mathrm{A}$ is a flat A-module (6.5.2). IV | 138

To prove that ${}^h\mathrm{A}$ is Henselian, let us apply the criterion (18.5.11), b). Let $\mathrm{S} = \mathrm{Spec}({}^h\mathrm{A})$, $\mathrm{S}_0 = \mathrm{Spec}({}^h\mathrm{A}/\mathfrak{m}.{}^h\mathrm{A})$, let $g : \mathrm{S}' \to \mathrm{S}$ be an étale morphism, set $\mathrm{S}'_0 = g^{-1}(\mathrm{S}_0)$, and let $f_0 : \mathrm{S}_0 \to \mathrm{S}'_0$ be an S_0-section. reasoning as in (18.5.4), one can suppose that g is of finite presentation. It follows then from (8.8.2) and (17.7.5) that there exist an index $\lambda \in \mathfrak{S}$, an étale morphism $g_\lambda : \mathrm{S}'^{(\lambda)} \to \mathrm{S}^{(\lambda)} = \mathrm{Spec}(\mathrm{A}_\lambda)$ and, on setting $\mathrm{S}_0^{(\lambda)} = \mathrm{Spec}(\mathrm{A}_\lambda/\mathfrak{m}\mathrm{A}_\lambda)$ and $\mathrm{S}_0'^{(\lambda)} = g_\lambda^{-1}(\mathrm{S}_0^{(\lambda)})$, an $\mathrm{S}_0^{(\lambda)}$-section $f_0^{(\lambda)} : \mathrm{S}_0^{(\lambda)} \to \mathrm{S}_0'^{(\lambda)}$, such that

$$\mathrm{S}' = \mathrm{S}'^{(\lambda)} \times_{\mathrm{S}^{(\lambda)}} \mathrm{S}, \qquad g = g_\lambda \times 1, \qquad f_0 = f_0^{(\lambda)} \times 1.$$

Let s be the closed point of S, $x = f_0(s)$, let x_λ be the projection of x in $\mathrm{S}'^{(\lambda)}$, and let s_λ be the closed point of $\mathrm{S}^{(\lambda)}$. Since x_λ lies above s_λ, the local ring C_λ of $\mathrm{S}'^{(\lambda)}$ at x_λ is an essentially étale A_λ-algebra; moreover, since $f_0^{(\lambda)}(s_\lambda) = x_\lambda$, we have $k(x_\lambda) = k(s_\lambda)$, in other words C_λ is a local strictly essentially étale A_λ-algebra, and consequently (18.6.1) is A_λ-isomorphic to an A-algebra A_μ with $\mu \geqslant \lambda$. There is therefore an A_λ-homomorphism $\mathrm{C}_\lambda \to {}^h\mathrm{A}$, that is, an $\mathrm{S}^{(\lambda)}$-morphism $h : \mathrm{S} \to \mathrm{S}'^{(\lambda)}$ such that $h(s) = x_\lambda$, and hence an S-section f of S' such that $f(s) = x$; this completes the proof of (i).

To prove (ii), it suffices to note that $\mathrm{Hom}\,.\,\mathrm{loc}({}^h\mathrm{A}, \mathrm{B}) \cong \varprojlim \mathrm{Hom}\,.\,\mathrm{loc}(\mathrm{A}_\lambda, \mathrm{B})$ and that by (18.6.2) the canonical homomorphisms $\mathrm{Hom}\,.\,\mathrm{loc}(\mathrm{A}_\lambda, \mathrm{B}) \leftarrow \mathrm{Hom}\,.\,\mathrm{loc}(\mathrm{A}, \mathrm{B})$ are bijective.

To prove (iv), note that for every λ and every integer $n > 0$, $\mathfrak{m}_\lambda^n = \mathfrak{m}^n \mathrm{A}_\lambda$, and

$$(\mathfrak{m}.{}^h\mathrm{A})^n = \mathfrak{m}^n.{}^h\mathrm{A} = \varinjlim_{\lambda} \mathfrak{m}_\lambda^n.$$

By exactness of the functor $\varinjlim$ in the category of A-modules,

$${}^h\mathrm{A}/(\mathfrak{m}.{}^h\mathrm{A})^n = \varinjlim_{\lambda} (\mathrm{A}_\lambda/\mathfrak{m}_\lambda^n),$$

and it therefore suffices to show that, for every n and every index λ, the homomorphism $\mathrm{A}/\mathfrak{m}^n \to \mathrm{A}_\lambda/\mathfrak{m}_\lambda^n$ is bijective. On the other hand, since A_λ is a flat A-module, one has

$$\mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1} = (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_{\mathrm{A}} \mathrm{A}_\lambda = (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_{\mathrm{A}/\mathfrak{m}} (\mathrm{A}_\lambda/\mathfrak{m}\mathrm{A}_\lambda)$$

and as $\mathrm{A}/\mathfrak{m} \to \mathrm{A}_\lambda/\mathfrak{m}\mathrm{A}_\lambda$ is bijective, the homomorphism $\mathfrak{m}^n/\mathfrak{m}^{n+1} \to \mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1}$ is also bijective; the conclusion then follows from Bourbaki, *Alg. comm.*, chap. III, § 2, no 8, cor. 3 du th. 1.

Finally, if A is Henselian, it follows from the remark preceding (18.6.3) that the canonical homomorphisms $\mathrm{A} \to \mathrm{A}_\lambda$ are bijective, which proves (vi) by definition of ${}^h\mathrm{A}$. □

(18.6.7). Let A be a *semi-local* ring, $\mathfrak{m}_i$ $(1 \leqslant i \leqslant r)$ its maximal ideals. The *Henselization* of A is the ring denoted again ${}^h\mathrm{A}$, defined as the *product* $\prod_i {}^h(\mathrm{A}_{\mathfrak{m}_i})$ of the Henselizations of the local rings $\mathrm{A}_{\mathfrak{m}_i}$. It is a faithfully flat A-module and a semi-local A-algebra, whose maximal ideals are the $\mathfrak{m}_i.{}^h\mathrm{A}$ by virtue of (18.6.6), (iii); furthermore, if $\mathfrak{r} = \bigcap \mathfrak{m}_i$ is the radical of A, it follows from the above that $\mathfrak{r}.{}^h\mathrm{A}$ is the radical of ${}^h\mathrm{A}$ and that the canonical map $\mathrm{A}/\mathfrak{r} \to {}^h\mathrm{A}/\mathfrak{r}.{}^h\mathrm{A}$ is bijective. As the separated completion $\widehat{\mathrm{A}}$ of A for the $\mathfrak{r}$-adic topology is the product of the separated completions $\widehat{\mathrm{A}}_{\mathfrak{m}_i}$ (Bourbaki, *Alg. comm.*, chap. III, § 2, no 13, prop. 18), the canonical homomorphism $\widehat{\mathrm{A}} \to ({}^h\mathrm{A})\widehat{}$ is bijective by (18.6.6), (iv), and it is clear by (18.6.6), (v) that for ${}^h\mathrm{A}$ to be Noetherian, it is necessary and sufficient that A be so. IV | 139

To obtain the analogue of the universal property (18.6.6), (ii), when A and B are two *semi-local* rings of radicals $\mathfrak{r}, \mathfrak{s}$, let us agree to call a *semi-local homomorphism* ϕ from A to B any homomorphism such that $\phi(\mathfrak{r}) \subset \mathfrak{s}$. In the particular case where B is a *product* of local rings B_j $(1 \leqslant j \leqslant n)$, the data of such a homomorphism amounts to the data of its projections $\phi_j : \mathrm{A} \to \mathrm{B}_j$, which are homomorphisms subject only to the condition that $\phi_j(\mathfrak{r}) \subset \mathfrak{n}_j$, where $\mathfrak{n}_j$ is the maximal ideal of B_j. Moreover, since $\mathfrak{m}_i$ $(1 \leqslant i \leqslant m)$ are the maximal ideals of A, the prime ideal $\phi_j^{-1}(\mathfrak{n}_j)$ must contain one of the $\mathfrak{m}_i$, since it must contain their intersection $\mathfrak{r}$, hence it is equal to one of the $\mathfrak{m}_i$ and ϕ_j factorises as $\mathrm{A} \to \mathrm{A}_{\mathfrak{m}_i} \xrightarrow{\psi_{ij}} \mathrm{B}_j$, where ψ_{ij} is a *local* homomorphism. If $\mathrm{Hom.sloc}(\mathrm{A},\mathrm{B})$ denotes the set of semi-local homomorphisms from A to B, one can thus identify this set with $\prod_j \left(\bigcup_i \mathrm{Hom.loc}(\mathrm{A}_{\mathfrak{m}_i}, \mathrm{B}_j)\right)$; in the case considered, since a semi-local Henselian ring is a direct product of Henselian local rings, one sees that the universal property (18.6.6), (ii) is still valid for a semi-local ring A, provided one replaces "local homomorphisms" by "semi-local homomorphisms" (18.6.7).

Proposition (18.6.8). — *Let* A *be a semi-local ring, and let* B *be a semi-local A-algebra integral over* A. *Then* $\mathrm{B} \otimes_{\mathrm{A}} ({}^h\mathrm{A})$ *is a semi-local B-algebra isomorphic to* ${}^h\mathrm{B}$.

Proof. Since $\mathrm{B} \otimes_{\mathrm{A}} ({}^h\mathrm{A})$ is integral over ${}^h\mathrm{A}$, each of its maximal ideals lies above a maximal ideal of ${}^h\mathrm{A}$, hence above a maximal ideal of A. Since B is integral over A, the prime ideals of B lying above a maximal ideal of A are maximal ideals of B. Thus the projections in $\mathrm{Spec}({}^h\mathrm{A})$ and $\mathrm{Spec}(\mathrm{B})$ of a maximal ideal of $\mathrm{B} \otimes_{\mathrm{A}} ({}^h\mathrm{A})$ are both maximal ideals. Taking account of (18.6.6), (iii), and (**I**, 3.4.9), one concludes that $\mathrm{B} \otimes_{\mathrm{A}} ({}^h\mathrm{A})$ is semi-local.

Furthermore, for every Henselian local ring C, $\mathrm{Hom.sloc}(\mathrm{B} \otimes_{\mathrm{A}} ({}^h\mathrm{A}), \mathrm{C})$ is in bijection with the set of pairs (ϕ, ψ) of semi-local homomorphisms $\phi : \mathrm{B} \to \mathrm{C}$ and $\psi : {}^h\mathrm{A} \to \mathrm{C}$ such that the composites

$$\mathrm{A} \longrightarrow \mathrm{B} \xrightarrow{\phi} \mathrm{C}, \qquad \mathrm{A} \longrightarrow {}^h\mathrm{A} \xrightarrow{\psi} \mathrm{C}$$

are equal. By the bijection between $\mathrm{Hom.sloc}(\mathrm{A},\mathrm{C})$ and $\mathrm{Hom.sloc}({}^h\mathrm{A},\mathrm{C})$, for every $\phi \in \mathrm{Hom.sloc}(\mathrm{B},\mathrm{C})$ there exists a unique ψ having this property. Hence the map

$$\mathrm{Hom.sloc}(\mathrm{B} \otimes_{\mathrm{A}} ({}^h\mathrm{A}), \mathrm{C}) \longrightarrow \mathrm{Hom.sloc}(\mathrm{B},\mathrm{C})$$

is bijective, which proves the proposition by the uniqueness of the solution of a universal problem. □

Theorem (18.6.9). — *Let* A *be a semi-local ring.*

i) *For* ${}^h\mathrm{A}$ *to be reduced (resp. normal), it is necessary and sufficient that A be so.*
ii) *Suppose A Noetherian. Then, for every prime ideal* $\mathfrak{p}$ *of A, the ring* $({}^h\mathrm{A})_{\mathfrak{p}}/\mathfrak{p}.({}^h\mathrm{A})_{\mathfrak{p}}$ *is a finite direct product of separable algebraic extensions of* $k(\mathfrak{p})$ *(which implies that the fibres of the canonical morphism* $\mathrm{Spec}({}^h\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$ *are* geometrically regular*).*

Proof. (i) As ${}^h\mathrm{A}$ is a faithfully flat A-module, it follows from (2.1.13) that if ${}^h\mathrm{A}$ is reduced (resp. normal), so is A. To prove the converse, one can restrict to the case where A is local, so that, with the notations of (18.6.4), one has ${}^h\mathrm{A} = \varinjlim \mathrm{A}_\alpha$. Now, it follows from the definition of the A_α and from (17.5.7) that if A is reduced (resp. integral and integrally closed), so is each of the A_α; furthermore, the homomorphisms $\mathrm{A}_\alpha \to \mathrm{A}_\beta$ for $\alpha \leqslant \beta$ are injective by virtue of (**0**, 6.5.1) since A_β is a flat A_α-module; hence the morphisms $\mathrm{Spec}(\mathrm{A}_\beta) \to \mathrm{Spec}(\mathrm{A}_\alpha)$ are dominant, and one concludes from (5.13.2) (resp. (5.13.4)) that ${}^h\mathrm{A}$ is reduced (resp. integral and integrally closed). **IV** | 140

(ii) One can restrict to the case where A is local. By virtue of the fact that the functor $\varinjlim$ commutes with tensor products, the fibre of the morphism $\mathrm{Spec}({}^h\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$ at a point $\mathfrak{p}$ is the projective limit of the fibres of the morphisms $\mathrm{Spec}(\mathrm{A}_\alpha) \to \mathrm{Spec}(\mathrm{A})$ at this point. As ${}^h\mathrm{A}$ is Noetherian, one is reduced to proving the following lemma: □

Lemma (18.6.9.1). — *Let* (B_α) *be an inductive system of Artinian rings,* $\mathrm{B} = \varinjlim \mathrm{B}_\alpha$ *its inductive limit. If* B *is Noetherian, then* B *is Artinian; if moreover, for* $\alpha \leqslant \beta$, *the homomorphisms* $\phi_{\beta\alpha} : \mathrm{B}_\alpha \to \mathrm{B}_\beta$ *are injective, then there exists* λ *such that, for* $\alpha \geqslant \lambda$, *the number of local components* $\mathrm{B}_\alpha^{(i)}$ *of* B_α *is constant, that* $\phi_{\beta\alpha}(\mathrm{B}_\alpha^{(i)}) \subset \mathrm{B}_\beta^{(i)}$ *for every i and that the* $\mathrm{B}^{(i)} = \varinjlim \mathrm{B}_\alpha^{(i)}$ *are the local components of* B.

Proof. Indeed $\mathrm{Y}_\alpha = \mathrm{Spec}(\mathrm{B}_\alpha)$, $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$, which, as a topological space, is equal to $\varprojlim \mathrm{Y}_\alpha$ (8.2.10). The spaces Y_α are finite and discrete; furthermore, if the $\phi_{\beta\alpha}$ are injective, the canonical morphisms $\mathrm{Y}_\beta \to \mathrm{Y}_\alpha$ are dominant, hence surjective. The lemma then follows from the following purely topological lemma: □

Lemma (18.6.9.2). — *Let $(Y_\alpha, \psi_{\alpha\beta})$ be a projective system of finite discrete topological spaces. If $Y = \varprojlim Y_\alpha$ is Noetherian, Y is finite and discrete. If moreover the $\psi_{\alpha\beta}$ are surjective, then there exists λ such that for $\alpha \geqslant \lambda$, the number of elements of Y_α is constant and equal to the number of elements of Y.*

Proof. Indeed, since Y is compact and Noetherian, every subset of Y is compact, hence closed; thus Y is discrete, and therefore finite since it is compact. If moreover the $\psi_{\alpha\beta}$ are surjective, the same holds for $\psi_\alpha : Y \to Y_\alpha$, so $\mathrm{Card}(Y) \geqslant \mathrm{Card}(Y_\alpha)$, and as $\mathrm{Card}(Y_\alpha)$ is an increasing function of α, there exists λ such that for $\alpha \geqslant \lambda$, $\mathrm{Card}(Y_\alpha) = \mathrm{Card}(Y_\lambda)$; the $\psi_{\alpha\beta}$ are then bijective for $\alpha \geqslant \lambda$ and one therefore has $\mathrm{Card}(Y) = \mathrm{Card}(Y_\lambda)$. □

Corollary (18.6.10). — *Let A be a local Noetherian ring. For ^{h}A to possess one of the following properties:*

a) *being a Cohen-Macaulay ring* (**0**, 16.5.3);
b) *satisfying the property* (S_n) (5.7.3);
c) *being regular;*
d) *satisfying the property* (R_n) (5.8.2);

it is necessary and sufficient that A possess the same property.

Proof. It follows immediately from the fact that the fibres of the morphism $\mathrm{Spec}({}^h A) \to \mathrm{Spec}(A)$ are geometrically regular (18.6.9) and from (6.4.1), (6.5.1) and (6.5.3). □ **IV | 141**

Corollary (18.6.11). — *Let A be a semi-local Noetherian ring. For a prime ideal $\mathfrak{p}$ of ^{h}A to belong to $\mathrm{Ass}({}^h A)$, it is necessary and sufficient that $\mathfrak{p} \cap A$ belong to $\mathrm{Ass}(A)$.*

Proof. Taking account of the description of the fibres of the morphism $\mathrm{Spec}({}^h A) \to \mathrm{Spec}(A)$, this also follows immediately from (3.3.1). □

Proposition (18.6.12). — *Let A be a local ring; the following properties are equivalent:*

a) *A is unibranch* (**0**, 23.2.1) *(resp. reduced and unibranch).*
b) *For every A-algebra A′ that is local and strictly essentially étale,* $\mathrm{Spec}(A')$ *is irreducible (resp. integral).*
c) $\mathrm{Spec}({}^h A)$ *is irreducible (resp. integral).*

Proof. First note that ${}^h(A_{\mathrm{red}}) = ({}^h A)_{\mathrm{red}}$: indeed, ${}^h(A_{\mathrm{red}}) = ({}^h A) \otimes_A A_{\mathrm{red}}$ by (18.6.8), and since ${}^h(A_{\mathrm{red}})$ is reduced by (18.6.9), it is equal to $({}^h A)_{\mathrm{red}}$. This allows us, in the proof, to consider only the case where A is reduced; then so are A′ (17.5.7) and ^{h}A (18.6.9).

With the notations of (18.6.4), condition *b*) means that all the A_α are integral; one therefore concludes that ${}^h A = \varinjlim A_\alpha$ is integral (5.13.3), so *b*) implies *c*).

It is clear that if ^{h}A is integral, so are each of the $A_\alpha \subset {}^h A$, hence *c*) implies *b*).

Let us show that *c*) implies *a*). Note first that $A \subset {}^h A$ is integral; let K and L be the fields of fractions of A and ^{h}A respectively. It suffices to show that for every *finite* A-algebra B contained in K, the ring B is local. Now B is semi-local, and its Henselization is $B \otimes_A ({}^h A)$ by (18.6.8). By flatness, $B \otimes_A ({}^h A)$ identifies with a subring of $K \otimes_A ({}^h A)$, and the latter with a subring of L; hence ^{h}B is integral. But since ^{h}B is a semi-local Henselian ring, it is a finite direct product of local rings (18.5.9); it can therefore be integral only if it is local, which indeed implies that B is local.

Let us finally prove that *a*) implies *c*). Let C be the integral closure of the integral ring A. As C is a local ring by hypothesis, $C \otimes_A ({}^h A)$ is the Henselization of C (18.6.8), hence is integral and integrally closed (18.6.9). Since A is a subring of C, flatness identifies ^{h}A with a subring of ^{h}C; it is therefore integral as well, which completes the proof. □

Corollary (18.6.13). — *Let A be a local Henselian ring. Then, for A to be unibranch, it is necessary and sufficient that A_{red} be integral.*

Proposition (18.6.14). —
i) *Every filtered inductive limit of Henselian local rings (where the transition homomorphisms are local) is a Henselian local ring.*
ii) *The functor $A \rightsquigarrow {}^h A$ in the category of local rings (where the morphisms are local homomorphisms) commutes with filtered inductive limits.*
iii) *Every Henselian local ring A is an inductive limit of an inductive filtered system of Henselian and Noetherian local rings A_λ (the transition homomorphisms $A_\lambda \to A_\mu$ being local).*

Proof. (i) Let (A_λ) be a filtered inductive system of Henselian local rings, where the transition homomorphisms are local. Let us show that $A' = \varinjlim A_\lambda$, which is local (**0**, 10.3.1.3), is Henselian. Now, let C be an A'-algebra étale, $\mathfrak{n}$ a prime ideal of C above the maximal ideal $\mathfrak{m}'$ of A', such that $C_\mathfrak{n}$ is strictly essentially étale over A' (18.6.1), and in virtue of (8.8.2), (ii) and (17.7.8), there exist an index λ and an A_λ-algebra C_λ étale such that $C = C_\lambda \otimes_{A_\lambda} A'$ up to an isomorphism. Furthermore, if $\mathfrak{n}_\lambda$ is the inverse image of $\mathfrak{n}$ in C_λ, and $\mathfrak{m}_\lambda$ the maximal ideal of A_λ, one can, by virtue of (8.8.2.4) and of the transitivity of fibres (**I**, 3.6.4), suppose that $(C_\lambda)_{\mathfrak{n}_\lambda}$ is strictly essentially étale over A_λ; since A_λ is Henselian, $(C_\lambda)_{\mathfrak{n}_\lambda}$ is necessarily isomorphic to A_λ (18.5.11), *b*); there is therefore a neighbourhood of $\mathfrak{n}_\lambda$ in $\mathrm{Spec}(C_\lambda)$ isomorphic to $\mathrm{Spec}(A_\lambda)$ (**I**, 6.5.4); one concludes that there is a neighbourhood of $\mathfrak{n}$ in $\mathrm{Spec}(C)$ isomorphic to $\mathrm{Spec}(A')$, hence that $C_\mathfrak{n}$ is isomorphic to A'. This proves that A' is Henselian (18.6.2) and completes the proof. IV | 142

(ii) Suppose that $B = \varinjlim B_\lambda$, where the B_λ are local rings, the transition morphisms $B_\lambda \to B_\mu$ $(\lambda \leqslant \mu)$ being local. Set $A_\lambda = {}^h B_\lambda$; the A_λ are Henselian local rings (18.6.6), (vi), and by virtue of (18.6.6), (ii), the transition morphisms $B_\lambda \to B_\mu$ uniquely determine local homomorphisms $A_\lambda \to A_\mu$, so that (A_λ) is an inductive system. It amounts to showing that $A' = \varinjlim A_\lambda$ is canonically isomorphic to $A = {}^h B$. By virtue of (18.6.6), (ii) and of the definition of inductive limits, one has, for every Henselian local ring E,

$$\mathrm{Hom.loc}(A', E) = \varprojlim \mathrm{Hom.loc}(A_\lambda, E) = \varprojlim \mathrm{Hom.loc}(B_\lambda, E) = \mathrm{Hom.loc}(B, E).$$

But as A' is Henselian by (i), this proves our assertion.

(iii) The ring A is a filtered inductive limit of its sub-local Noetherian rings B_λ, the transition homomorphisms being local (5.13.3), (iii). As the ${}^h B_\lambda$ are Henselian and Noetherian local rings (18.6.6), (v) and that $A = {}^h A$ (18.6.6), (vi), it suffices to apply (ii) to the inductive system (B_λ). □

Corollary (18.6.15). — *Let* A *be a Henselian local ring,* X *an A-prescheme of finite presentation over* A. *Then there exists a Noetherian Henselian local ring* A_0*, a local homomorphism* $A_0 \to A$*, an* A_0*-prescheme* X_0 *of finite type over* A_0 *and an A-isomorphism* $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.

Proof. This follows from (18.6.14) and (8.8.2), (ii). □

18.7. Henselization and excellent rings.

(18.7.1). In this paragraph we denote by $\boldsymbol{P}(Z, k)$ a property of the form considered in (7.3.1), where we further suppose that the property $\boldsymbol{Q}(A, k)$ satisfies the following condition: IV | 142

For every separable algebraic extension k' of k, and every k'-local algebra that is Noetherian, the property $\boldsymbol{Q}(A, k)$ is *equivalent* to $\boldsymbol{Q}(A, k')$.

It is immediate that all the properties $\boldsymbol{P}$ considered in (7.3.8) satisfy the preceding condition, since if K is a finite extension of k, $k' \otimes_k K$ is a finite direct product of separable algebraic extensions of K.

Proposition (18.7.2). — *Let* A *be a local Noetherian ring. For* ${}^h A$ *to be a* $\boldsymbol{P}$*-ring* (7.3.13)*, it is necessary and sufficient that A be so.*

Proof. Indeed, by virtue of (18.6.6), (iv), the completion $\widehat{A}$ of A is also the completion of ${}^h A$ and one therefore has $A \subset {}^h A \subset \widehat{A}$. By (18.6.9), for every $x \in \mathrm{Spec}(A)$, the fibre at x of the morphism $\mathrm{Spec}({}^h A) \to \mathrm{Spec}(A)$ is discrete, finite, and each of the points y_i of this fibre is a separable algebraic extension of $k(x)$. The formal fibre of A at x is therefore the prescheme *sum* of the formal fibres of ${}^h A$ at the points y_i. The conclusion then follows from the hypothesis on $\boldsymbol{P}$ in (18.7.1). □ IV | 143

Corollary (18.7.3). — *Let* A *be a local Noetherian ring. For* ${}^h A$ *to be universally Japanese, it is necessary and sufficient that A be so.*

Proof. Indeed, for a Noetherian local ring B, it amounts to the same to say that B is universally Japanese, or that the formal fibres of B are geometrically reduced (7.6.4) and (7.7.1); the conclusion therefore follows from (18.7.2). □

Corollary (18.7.4). — *Let* A *be a local Noetherian ring. For the formal fibres of* ${}^h A$ *to be geometrically regular, it is necessary and sufficient that those of A be so.*

Proof. This is a particular case of (18.7.2). ☐

Proposition (18.7.5). — *Let* A *be a local Noetherian ring. If A is universally catenary* (5.6.2) *(resp. formally catenary* (7.1.9)*, resp. strictly formally catenary* (7.2.6)*), so is* ^{h}A*.*

Proof. With the notations of (18.6.4), if A is universally catenary, the same holds for the A_α, which are A-algebras essentially of finite type (5.6.3). To prove that in this case ^{h}A is universally catenary, it will therefore suffice to establish the following lemma: ☐

Lemma (18.7.5.1). — *Let* $(B_\alpha, \phi_{\beta\alpha})$ *be an inductive system of Noetherian rings and suppose that, for* $\alpha \leqslant \beta$, $\phi_{\beta\alpha}$ *makes* B_β *a flat* B_α*-module and that* $B = \varinjlim B_\alpha$ *is Noetherian. Then, if the* B_α *are catenary (resp. universally catenary), the same holds for* B*.*

Proof. To say that B is universally catenary is equivalent to saying that $B[T_1, \ldots, T_n]$ is catenary for all n (5.6.2), and it is immediate that the inductive system $(B_\alpha[T_1, \ldots, T_n])$ satisfies the same conditions as (B_α); it therefore suffices to prove that if the B_α are catenary, then B is catenary. If $\mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_r$ is a saturated chain of prime ideals, the inverse images $\mathfrak{p}_{0\alpha} \supset \mathfrak{p}_{1\alpha} \supset \cdots \supset \mathfrak{p}_{r\alpha}$ of these ideals in B_α form a chain of prime ideals for each α sufficiently large, and it suffices to show that for α sufficiently large, this chain is saturated. Now, each $\mathfrak{p}_i$ is an ideal of finite type, hence there exists λ such that, for $\alpha \geqslant \lambda$, one has $\mathfrak{p}_i = \mathfrak{p}_{i\alpha}.B$. One therefore has $1 = \operatorname{codim}(V(\mathfrak{p}_i), V(\mathfrak{p}_{i+1})) = \operatorname{codim}(V(\mathfrak{p}_{i\alpha}), V(\mathfrak{p}_{i+1,\alpha}))$ since B is a flat B_α-module (6.1.4) and (**0**, 6.2.3), which completes the proof of the lemma.

Suppose now A formally catenary. Let $\mathfrak{p}$ be a prime ideal of ^{h}A, and let $\mathfrak{q}$ be its inverse image in A, so that $\mathfrak{q}.{}^hA \subset \mathfrak{p}$ and consequently ${}^hA/\mathfrak{p}$ is a quotient ring of ${}^hA/\mathfrak{q}.{}^hA$. It therefore suffices to prove that ${}^hA/\mathfrak{q}.{}^hA$ is formally catenary for every prime ideal $\mathfrak{q}$ of A (7.1.9). Since ${}^hA/\mathfrak{q}.{}^hA = {}^h(A/\mathfrak{q})$ by (18.6.8), it suffices (7.1.11) to show that if A is *integral* and formally catenary, ^{h}A is formally *equidimensional*. But as the completion of ^{h}A is equal to $\widehat{A}$, this follows from the hypothesis that A is formally catenary.

Finally, suppose that A is strictly formally catenary. We have just seen that ^{h}A is formally catenary, and it remains to prove that the fibres of the morphism $\operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}({}^hA)$ satisfy property (S_1) (7.2.5), *b*). This follows from the hypothesis on A and from (18.7.2), applied to the case where $\boldsymbol{P}(Z, k)$ is property (S_1) for Z. ☐

Corollary (18.7.6). — *If a local Noetherian ring A is excellent* (7.8.2)*, so is* ^{h}A*.*

Remark (18.7.7). — It can happen that ^{h}A is an excellent local ring without A being universally catenary (nor *a fortiori* excellent). To see this, let us take the example of A from (5.6.11), with the same notations. The ring E constructed in (5.6.11) is excellent when k_0 is of characteristic 0 (7.8.3), (ii) and (iii); consider two rings E_1, E_2 isomorphic to E, and "glue" $\operatorname{Spec}(E_1)$ and $\operatorname{Spec}(E_2)$ such that if $\mathfrak{m}_i$ and $\mathfrak{m}'_i$ are the maximal ideals of E_i ($i = 1, 2$), corresponding to $\mathfrak{m}$ and $\mathfrak{m}'$, the point $\mathfrak{m}_1$ is "glued" to $\mathfrak{m}'_2$ and the point $\mathfrak{m}_2$ to $\mathfrak{m}'_1$; to be precise, if ε_i and ε'_i are the canonical homomorphisms from E_i onto $k(\mathfrak{m}_i)$ and $k(\mathfrak{m}'_i)$ ($i = 1, 2$), the scheme obtained is Spec(R), where R is the subring of $E_1 \times E_2$ formed by the pairs (x_1, x_2) such that $\varepsilon'_2(x_2) = \sigma(\varepsilon_1(x_1))$ and $\varepsilon'_1(x_1) = \sigma(\varepsilon_2(x_2))$. One verifies easily (for example with the help of (17.6.3)) that R is a finite étale C-algebra, and that there are two maximal ideals $\mathfrak{r}_1$, $\mathfrak{r}_2$ of R above the maximal ideal $\mathfrak{n}$ of C, the completions of $R_{\mathfrak{r}_1}$ and $R_{\mathfrak{r}_2}$ being canonically isomorphic to that of $A = C_{\mathfrak{n}}$. Thus $R_{\mathfrak{r}_1}$ is an A-algebra that is strictly essentially étale, and consequently its Henselization is equal to ^{h}A. Furthermore, we saw in (7.8.4), (ii), that the formal fibres of A are geometrically regular; the same therefore holds for those of ^{h}A by (18.7.2). Finally, R is universally catenary, since its quotients by its two minimal prime ideals are isomorphic to E, the conclusion following from (5.6.3), (iii). Hence $R_{\mathfrak{r}_1}$ is excellent, and consequently so is ^{h}A by (18.7.6), whereas A is not universally catenary.

18.8. Strictly local rings and strict Henselization. IV | 144

Proposition (18.8.1). — *Let* A *be a local ring. The following conditions are equivalent:*

a) *A is a Henselian ring and its residue field k is separably closed.*
b) *A is a Henselian ring and every étale covering of* $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ *is trivial.*
c) *For every étale morphism* $f : \mathrm{X} \to \mathrm{S}$ *and every point* $x \in \mathrm{X}$ *such that* $f(x)$ *is the closed point s of S, there exists an S-section* $u : \mathrm{S} \to \mathrm{X}$ *of X such that* $u(s) = x$*.*

Proof. The hypothesis c) implies that for every étale morphism $f : \mathrm{X} \to \mathrm{S}$, the residue fields at the points of $f^{-1}(s)$ are all isomorphic to k, and moreover the condition b) of (18.5.11) is satisfied. Hence A is Henselian, and the fact that every étale covering of S is trivial follows from (18.2.10), (ii); hence c) implies b). As étale coverings of $\mathrm{Spec}(k)$ are trivial if and only if k is separably closed, b) implies a) by virtue of (18.5.11), b). Finally, it follows also from (18.5.11), b) that a) implies c). □

Definition (18.8.2). — A local ring is said to be *strictly local* if it satisfies the equivalent conditions of (18.8.1). A scheme isomorphic to the spectrum of a strictly local ring is called a *strictly local scheme.*

Remark (18.8.3). — The conditions of (18.8.1) are also equivalent to the following:

d) For every morphism locally of finite type $f : \mathrm{X} \to \mathrm{S}$ and every point $x \in \mathrm{X}$ such that $f(x)$ is the closed point s of S and that $\mathscr{O}_{\mathrm{X},x}/\mathfrak{m}_s\mathscr{O}_{\mathrm{X},x}$ is a field, a finite separable extension of $k(s) = k$, there exists an open neighbourhood U of x in X such that $f|U$ is a closed immersion.

(One will note that the hypothesis of d) is satisfied if f is *formally unramified* at the point x (17.4.1.2).)

We leave the reader the proof, which is substantially the same as that of (18.5.18).

Lemma (18.8.4). — *Let* A, A′ *be two local rings,* $\phi : \mathrm{A} \to \mathrm{A}'$ *a local homomorphism making* A′ *an A-algebra essentially étale* (18.6.1)*. Then, for every strictly local ring* B*, every local homomorphism* $\psi : \mathrm{A} \to \mathrm{B}$ *and every homomorphism of* $k(\mathrm{A})$*-algebras* $\alpha : k(\mathrm{A}') \to k(\mathrm{B})$*, there exists one and only one local homomorphism* $\psi' : \mathrm{A}' \to \mathrm{B}$ *such that* $\psi = \psi' \circ \phi$ *and that* α *is equal to the homomorphism* $\overline{\psi}'$ *deduced from* ψ' *by passage to quotients.* IV | 145

Proof. With the notations of (18.6.2), the homomorphism α corresponds to a $k(s)$-morphism $\mathrm{Spec}(k(y)) \to \mathrm{Spec}(k(x))$, hence to a point x' well determined in X′ above y. The existence of a Y-section of $f' : \mathrm{X}' \to \mathrm{Y}$ taking y to x' follows from (18.8.1), c), since f' is étale and B is strictly local; its uniqueness follows from the fact that f' is separated and from (17.4.9). □

Lemma (18.8.5). — *Let* A *be a local ring,* A_1*,* A_2 *two A-algebras that are local and essentially étale, K a field that is an extension of* $k(\mathrm{A})$*.*

i) *For every* $k(\mathrm{A})$*-homomorphism* $\gamma : k(\mathrm{A}_1) \to k(\mathrm{A}_2)$*, there exists at most one A-homomorphism (necessarily local)* $\psi : \mathrm{A}_1 \to \mathrm{A}_2$ *such that* γ *is the homomorphism* $\overline{\psi}$ *deduced from* ψ *by passage to quotients.*
ii) *For every pair of local A-homomorphisms* $\beta_1 : \mathrm{A}_1 \to \mathrm{K}$*,* $\beta_2 : \mathrm{A}_2 \to \mathrm{K}$ *there exists a local A-algebra* A_3 *that is essentially étale, two A-homomorphisms* $\phi_1 : \mathrm{A}_1 \to \mathrm{A}_3$*,* $\phi_2 : \mathrm{A}_2 \to \mathrm{A}_3$ *and an A-homomorphism* $\beta : \mathrm{A}_3 \to \mathrm{K}$ *such that* $\beta_1 = \beta \circ \phi_1$ *and* $\beta_2 = \beta \circ \phi_2$*.*

Proof. With the notations of (18.6.3), the homomorphisms β_1, β_2 in (ii) correspond to two S-morphisms $\mathrm{Spec}(\mathrm{K}) \to \mathrm{X}_1$, $\mathrm{Spec}(\mathrm{K}) \to \mathrm{X}_2$, with respective images x_1, x_2; one thus deduces a well-determined $\mathrm{Spec}(\mathrm{K}) \to \mathrm{X}_3$ (**I**, 3.2.1) whose image x_3 lies above x_1 and x_2; the A-algebra $\mathrm{A}_3 = \mathscr{O}_{\mathrm{X}_3,x_3}$ and the homomorphism $\mathrm{A}_3 \to \mathrm{K}$ corresponding to these data then satisfy the conditions of (ii). On the other hand, the data of γ in (i) corresponds to an S-morphism $\mathrm{Spec}(k(x_2)) \to \mathrm{Spec}(k(x_1))$ or also to a point x_3' of X_3 above x_2; the uniqueness of ψ follows from this and from (17.4.9). □

(18.8.6). Let A be a local ring, $\mathfrak{m}$ its maximal ideal, k its residue field, Ω a separable closure of k. Consider the set $\mathfrak{E}$ of essentially étale A-algebras defined in (18.6.1), and denote by L the set of *local A-homomorphisms* $\mathrm{A}' \to \Omega$, where A′ ranges over $\mathfrak{E}$; one will note that such a homomorphism factorises as $\lambda : \mathrm{A}' \to k(\mathrm{A}') \xrightarrow{\omega_\lambda} \Omega$, where $\omega_\lambda : k(\mathrm{A}') \to \Omega$ is a k-homomorphism (since $\mathfrak{m}\mathrm{A}'$ is the maximal ideal of A′); conversely the data of a homomorphism ω_λ uniquely determines a homomorphism $\lambda \in \mathrm{L}$. For every $\lambda \in \mathrm{L}$, we denote by A_λ the A-algebra essentially étale belonging to $\mathfrak{E}$ on which λ is defined. The set L is *preordered* by the relation "there exists a homomorphism $\phi_{\mu\lambda} : \mathrm{A}_\lambda \to \mathrm{A}_\mu$ such that $\lambda = \mu \circ \phi_{\mu\lambda}$", and it follows from (18.8.5), (i) that this homomorphism $\phi_{\mu\lambda}$ is *unique*. Furthermore, L is *filtered increasing* for the preceding preorder relation (which one also denotes $\lambda \leqslant \mu$), by virtue of (18.8.5), (ii). It is clear that $(\mathrm{A}_\lambda, \phi_{\mu\lambda})$ is an *inductive system* of local A-algebras, the $\phi_{\mu\lambda}$ being local homomorphisms; furthermore, it follows from (17.3.5) that for $\lambda \leqslant \mu$, A_μ is an A_λ-algebra that is essentially étale. If $\overline{\phi}_{\mu\lambda} : k(\mathrm{A}_\lambda) \to k(\mathrm{A}_\mu)$ is the k-homomorphism deduced from $\phi_{\mu\lambda}$ by passage to quotients, $(k(\mathrm{A}_\lambda), \overline{\phi}_{\mu\lambda})$ is an inductive system of separable algebraic extensions of k, and the $\omega_\lambda : k(\mathrm{A}_\lambda) \to \Omega$ form an inductive system of homomorphisms of k. Furthermore, the inductive limit IV | 146

$$\omega : \varinjlim k(\mathrm{A}_\lambda) \longrightarrow \Omega \tag{18.8.6.1}$$

is a *k-isomorphism*. It suffices indeed to prove that for every finite separable extension k' of k, there exists an A-algebra A′ such that k' is the residue field of A′ and that the homomorphism $k \to k'$ is deduced from $\mathrm{A} \to \mathrm{A}'$ by passage to quotients. But this follows from (18.1.1) applied to étale morphisms, since Spec(A) is the unique neighbourhood of its closed point.

Note now that if $\mathfrak{m}_\lambda$ is the maximal ideal of A_λ, it follows that $\mathfrak{m}_\mu = \mathfrak{m}_\lambda \mathrm{A}_\mu$ and that A_μ is a flat A_λ-module. Hence it follows from (**0**, 10.3.1.3) that the ring $\varinjlim \mathrm{A}_\lambda$ is *local*, that the homomorphism $w : \mathrm{A} \to \varinjlim \mathrm{A}_\lambda$ is local, and that one has a k-isomorphism

$$\varinjlim k(\mathrm{A}_\lambda) \xrightarrow{\sim} k(\varinjlim \mathrm{A}_\lambda); \tag{18.8.6.2}$$

identifying these two fields, one deduces a k-isomorphism

$$\omega^{-1} : \Omega \xrightarrow{\sim} k(\varinjlim \mathrm{A}_\lambda).$$

Definition (18.8.7). — Let A be a local ring, $k = k(\mathrm{A})$ its residue field, $i : k \to \Omega$ a homomorphism from k into a separable closure Ω of k; the *strict Henselization* of A relative to i, denoted ${}^{hs}\mathrm{A}_{(i)}$ (or ${}^{hs}\mathrm{A}$ when this causes no confusion), is the inductive limit of the inductive system $(\mathrm{A}_\lambda, \phi_{\mu\lambda})$ defined in (18.8.6).

As one has a k-isomorphism $\omega^{-1} : \Omega \xrightarrow{\sim} k({}^{hs}\mathrm{A}_{(i)})$, Ω is endowed canonically with the structure of a ${}^{hs}\mathrm{A}_{(i)}$-algebra by the local homomorphism

$${}^{hs}\mathrm{A}_{(i)} \longrightarrow k({}^{hs}\mathrm{A}_{(i)}) \xrightarrow{\omega} \Omega.$$

As in (18.6.5) one sees that the definition given in (18.8.7) does not depend on the choice of $\mathfrak{E}$; we shall moreover see below (18.8.8), (i) and (ii) that ${}^{hs}\mathrm{A}_{(i)}$ is an object representing a functor entirely defined by the data of A and i (**0**, 8.1.8), and is hence defined as such up to an isomorphism.

Proposition (18.8.8). — *Let* A *be a local ring, k its residue field, $i : k \to \Omega$ a homomorphism from k into a separable closure Ω of k, ${}^{hs}\mathrm{A}_{(i)}$ the strict Henselization of* A *corresponding to i.*

i) *${}^{hs}\mathrm{A}_{(i)}$ is a strictly local ring* (18.8.2) *and the structural homomorphism $w : \mathrm{A} \to {}^{hs}\mathrm{A}_{(i)}$ is local.*
ii) *For every local homomorphism $u : \mathrm{A} \to \mathrm{B}$, where* B *is a strictly local ring, and every k-homomorphism $\psi : \Omega \to k(\mathrm{B})$, there exists one and only one A-homomorphism $v : {}^{hs}\mathrm{A}_{(i)} \to \mathrm{B}$ such that ψ factorises as $\Omega \xrightarrow{\omega^{-1}} k({}^{hs}\mathrm{A}_{(i)}) \xrightarrow{\overline{v}} k(\mathrm{B})$, where $\overline{v}$ is deduced from v by passage to quotients.*
iii) *${}^{hs}\mathrm{A}_{(i)}$ is a faithfully flat A-module, and if $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}.{}^{hs}\mathrm{A}_{(i)}$ is the maximal ideal of ${}^{hs}\mathrm{A}_{(i)}$, and $k({}^{hs}\mathrm{A}_{(i)})$ is a separable closure of k.*
iv) *For ${}^{hs}\mathrm{A}_{(i)}$ to be Noetherian, it is necessary and sufficient that A be so.*
v) *If A is strictly local (hence $\Omega = k$), one has ${}^{hs}\mathrm{A} \cong \mathrm{A}$.*
vi) *If $i' : k \to \Omega'$ is a second homomorphism from k into a separable closure of k, for every k-isomorphism $\sigma : \Omega \xrightarrow{\sim} \Omega'$, the corresponding A-homomorphism (by (ii)) $v_\sigma : {}^{hs}\mathrm{A}_{(i)} \to {}^{hs}\mathrm{A}_{(i')}$ is an isomorphism.*

Proof. We have already seen above that ${}^{hs}\mathrm{A}_{(i)}$ is a local ring and that w is a local homomorphism; the fact that ${}^{hs}\mathrm{A}_{(i)}$ is Henselian (and hence strictly local) is demonstrated as in (18.6.6). The assertions IV | 147 of (iii), as well as the sufficiency of condition (iv), also follow from (**0**, 10.3.1.3); the necessity of condition (iv) follows from the fact that ${}^{hs}\mathrm{A}_{(i)}$ is a faithfully flat A-module (**0**, 6.5.2).

To prove (ii), remark, with the notations of (18.8.6), that for every $\lambda \in \mathrm{L}$, there exists one and only one local homomorphism $v_\lambda : \mathrm{A}_\lambda \to \mathrm{B}$, such that the composite $k(\mathrm{A}_\lambda) \xrightarrow{\omega_\lambda} \Omega \xrightarrow{\psi} k(\mathrm{B})$ is deduced from v_λ by passage to quotients, by virtue of (18.8.4) and the hypothesis that B is strictly local. The uniqueness of v_λ implies moreover that the v_λ form an inductive system, whence the existence of the homomorphism v satisfying the conditions of (ii); its uniqueness follows from (17.4.9). The uniqueness of v immediately implies assertion (vi), by considering the composites $v_\sigma \circ v_{\sigma^{-1}}$ and $v_{\sigma^{-1}} \circ v_\sigma$. Finally, if A is strictly local, and $\mathrm{A}' = \mathrm{B}_\mathfrak{n}$ is an A-algebra local and essentially étale, where B is an A-algebra étale, it follows from (18.8.1) that there exists a Spec(A)-section of Spec(B) taking the value $\mathfrak{n}$ at the closed point of Spec(A), hence by (17.4.1) that the homomorphism $\mathrm{A} \to \mathrm{A}'$ is bijective, which proves (v). ☐

(18.8.8.1). Let $\mathscr{C}$ be the category of strictly local rings, with local homomorphisms as morphisms; for every $\mathrm{B} \in \mathscr{C}$, denote by $\mathrm{F}(\mathrm{B})$ the set of pairs (u, ψ) consisting of a local homomorphism $u : \mathrm{A} \to \mathrm{B}$ and a k-homomorphism $\psi : \Omega \to k(\mathrm{B})$ such that the composite $k(\mathrm{A}) = k \xrightarrow{i} \Omega \xrightarrow{\psi} k(\mathrm{B})$ is deduced from u by passage to quotients. One can say that the object ${}^{hs}\mathrm{A}_{(i)}$ *represents* the covariant functor $\mathrm{F} : \mathscr{C} \to \mathtt{Ens}$ (**0**, 8.1.11).

One will remark that by virtue of (18.8.8), (vi), the strict Henselizations ${}^{hs}\mathrm{A}_{(i)}$ are all A-isomorphic (for the various k-homomorphisms i of k into separable closures of k), but for given i and i', there is in general an infinity of A-isomorphisms ${}^{hs}\mathrm{A}_{(i)} \xrightarrow{\sim} {}^{hs}\mathrm{A}_{(i')}$. To be precise, the group of A-automorphisms of ${}^{hs}\mathrm{A}_{(i)}$ is isomorphic to the Galois group of Ω over k.

(18.8.9). Let now A be a *semi-local* ring, $\mathfrak{m}_j$ $(1 \leqslant j \leqslant r)$ its maximal ideals, and for every j, consider a homomorphism $i_j : k(\mathrm{A}_{\mathfrak{m}_j}) \to \Omega_j$ of the residue field into a separable closure. The *strict Henselization* of A relative to the i_j is the product $\prod_{j=1}^r({}^{hs}(\mathrm{A}_{\mathfrak{m}_j}))$, denoted ${}^{hs}\mathrm{A}$ (when there is no risk of confusion). It is therefore a faithfully flat A-module and a semi-local A-algebra, whose maximal ideals are the $\mathfrak{m}_j.{}^{hs}\mathrm{A}$ and the radical $\mathfrak{r}.{}^{hs}\mathrm{A}$ (if $\mathfrak{r}$ denotes the radical of A). Finally, the universal property (18.8.8), (ii) still holds when replacing "local homomorphism" by "semi-local homomorphism" (18.6.7), and replacing Ω by $\prod_j \Omega_j$. **IV** | 148

Proposition (18.8.10). — *Let* A *be a semi-local ring,* B *a finite A-algebra. Let* $\mathfrak{n}_j$ *$(1 \leqslant j \leqslant r)$ be the maximal ideals of* B*. Then there is, for every j, a maximal ideal* $\mathfrak{q}_j$ *of* $\mathrm{B} \otimes_\mathrm{A} ({}^{hs}\mathrm{A})$ *above* $\mathfrak{n}_j$*, such that* ${}^{hs}\mathrm{B}$ *is isomorphic to the direct product of the local rings* $(\mathrm{B} \otimes_\mathrm{A} ({}^{hs}\mathrm{A}))_{\mathfrak{q}_j}$ *$(1 \leqslant j \leqslant r)$.*

Proof. One can evidently restrict to the case where A is local, replacing A by $\mathrm{A}_{\mathfrak{m}_j}$. On the other hand, let $\mathrm{C} = \mathrm{B}_{\mathfrak{n}_j}$; by virtue of the definition of ${}^{hs}\mathrm{B}$ (18.8.9), everything amounts to seeing that ${}^{hs}\mathrm{C}$ is C-isomorphic to $(\mathrm{C} \otimes_\mathrm{A} ({}^{hs}\mathrm{A}))_\mathfrak{q}$, where $\mathfrak{q}$ is one of the maximal ideals of $\mathrm{C} \otimes_\mathrm{A} ({}^{hs}\mathrm{A})$. Let k and K be the residue fields of A and C, Ω the separable closure of k, and let us suppose K and Ω contained in the same closure of k; as K is a finite extension of k, it is immediate that $\mathrm{K} \otimes_k \Omega$ is a finite direct product of fields, all isomorphic to the separable closure $\mathrm{K}\Omega$ of K. On the other hand, since ${}^{hs}\mathrm{A}$ is Henselian, $\mathrm{C} \otimes_\mathrm{A} ({}^{hs}\mathrm{A})$ is a direct product of the local rings $(\mathrm{C} \otimes_\mathrm{A} ({}^{hs}\mathrm{A}))_{\mathfrak{q}_i}$, where $\mathfrak{q}_i$ $(1 \leqslant i \leqslant s)$ ranges over the maximal ideals of $\mathrm{C} \otimes_\mathrm{A} ({}^{hs}\mathrm{A})$; these local rings are Henselian (18.5.10) and have $\mathrm{K}\Omega$ as residue field, hence they are strictly local (18.8.1). Let $\mathrm{C}' = \mathrm{C} \otimes_\mathrm{A} ({}^{hs}\mathrm{A})$ and $\mathrm{C}'_i = \mathrm{C}'_{\mathfrak{q}_i}$, which is therefore a quotient ring of C'; let $f : \mathrm{C} \to \mathrm{C}'$, $g : {}^{hs}\mathrm{A} \to \mathrm{C}'$, $\phi_i : \mathrm{C}' \to \mathrm{C}'_i$ be the canonical maps.

$$\begin{tikzcd}
 & & {}^{hs}\mathrm{C} \arrow[dl, bend left, "v"'] \\
\mathrm{C} \arrow[r, "f"'] & \mathrm{C}' \arrow[r, "\phi_i"'] & \mathrm{C}'_i \arrow[u, bend left, "w"] \\
\mathrm{A} \arrow[u] \arrow[r] & {}^{hs}\mathrm{A} \arrow[u, "g"] \arrow[ur, "u"'] &
\end{tikzcd}$$

Let $v : \mathrm{C} \to {}^{hs}\mathrm{C}$ be the canonical local homomorphism; there exists a unique $u : {}^{hs}\mathrm{A} \to {}^{hs}\mathrm{C}$ which, by passage to quotients, gives the canonical injection $\Omega \to \mathrm{K}\Omega$ and such that the composite $\mathrm{A} \to \mathrm{C} \xrightarrow{v} {}^{hs}\mathrm{C}$ is equal to the composite $\mathrm{A} \to {}^{hs}\mathrm{A} \xrightarrow{u} {}^{hs}\mathrm{C}$ (18.8.8); hence there exists a unique local homomorphism $w_0 : \mathrm{C}' \to {}^{hs}\mathrm{C}$ such that $w_0 \circ g = u$, $w_0 \circ f = v$, and since ${}^{hs}\mathrm{C}$ is a local ring, this homomorphism factorises as $\mathrm{C}' \to \mathrm{C}'_i \xrightarrow{w} {}^{hs}\mathrm{C}$ for a well-determined index i, w being a local homomorphism. On the other hand, the ring C'_i being strictly local, there exists a local homomorphism $w' : {}^{hs}\mathrm{C} \to \mathrm{C}'_i$ which, by passage to quotients, gives the identity automorphism of $\mathrm{K}\Omega$ and which is such that $w' \circ v = \phi_i \circ f$ (18.8.8). One deduces first that $w \circ w'$ is an endomorphism of ${}^{hs}\mathrm{C}$ which, by passage to quotients, gives the identity automorphism of $\mathrm{K}\Omega$, hence is the identity (18.8.8). On the other hand, $w' \circ w \circ \phi_i \circ f = w' \circ v = \phi_i \circ f$ and $w' \circ w \circ \phi_i \circ g = w' \circ u = \phi_i \circ g$; whence $w' \circ w \circ \phi_i = \phi_i$, which is to say $w' \circ w$ is the identity automorphism of C'_i. ☐ **IV | 149**

Remark (18.8.11). — The proof of (18.8.10) uses the fact that B is integral over A, but uses the fact that B is a *finite* A-algebra only to establish that $\mathrm{K} \otimes_k \Omega$ is a finite direct product of fields. This latter property still holds when B is assumed to be an *integral semi-local* A-algebra whose maximal ideals $\mathfrak{n}_j$ have finite separable residue-field degrees $[k(\mathfrak{n}_j) : k]_s$.

Proposition (18.8.12). — *Let* A *be a semi-local ring.*

i) *For* ${}^{hs}\mathrm{A}$ *to be reduced (resp. normal), it is necessary and sufficient that A be so.*
ii) *Suppose A Noetherian. Then, for every prime ideal* $\mathfrak{p}$ *of A, the ring* $({}^{hs}\mathrm{A})_{\mathfrak{p}}/\mathfrak{p}.({}^{hs}\mathrm{A})_{\mathfrak{p}}$ *is a finite direct product of separable algebraic extensions of* $k(\mathfrak{p})$ *(which implies that the fibres of the canonical morphism* $\mathrm{Spec}({}^{hs}\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$ *are* geometrically regular *and are* discrete *spaces).*

Corollary (18.8.13). — *Let* A *be a local Noetherian ring. For* ${}^{hs}\mathrm{A}$ *to possess one of the following properties:*

a) *being a Cohen-Macaulay ring* (**0**, 16.5.3)*;*
b) *satisfying the condition* (S_n) (5.7.3)*;*
c) *being regular;*
d) *satisfying the condition* (R_n) (5.8.2)*;*

it is necessary and sufficient that A possess the same property.

Corollary (18.8.14). — *Let* A *be a semi-local Noetherian ring. For a prime ideal* $\mathfrak{p}$ *of* ${}^{hs}\mathrm{A}$ *to belong to* $\mathrm{Ass}({}^{hs}\mathrm{A})$*, it is necessary and sufficient that* $\mathfrak{p} \cap \mathrm{A}$ *belong to* $\mathrm{Ass}(\mathrm{A})$.

Proof. The proofs are the same as those of (18.6.9), (18.6.10) and (18.6.11) respectively. ☐

Proposition (18.8.15). — *Let* A *be a local ring; the following properties are equivalent:*

a) *A is geometrically unibranch (resp. reduced and geometrically unibranch).*
b) *For every A-algebra* A' *that is local and essentially étale,* $\mathrm{Spec}(\mathrm{A}')$ *is irreducible (resp. integral).*
c) $\mathrm{Spec}({}^{hs}\mathrm{A})$ *is irreducible (resp. integral).*

Proof. As in (18.6.12), one reduces to the case where A is reduced, using the relation ${}^{hs}(\mathrm{A}_{\mathrm{red}}) = ({}^{hs}\mathrm{A}) \otimes_{\mathrm{A}} \mathrm{A}_{\mathrm{red}}$ (18.8.11); the equivalence of *b*) and *c*) is proved as in (18.6.12). The same is true of the implication *a*) $\Rightarrow$ *c*), taking account of (18.8.11) and of the definition of a geometrically unibranch integral local ring. Finally, to prove that *c*) implies *a*), using the same notations as in (18.6.12), it suffices to see that ${}^{hs}\mathrm{B}$ is integral. But, by (18.8.10), ${}^{hs}\mathrm{B}$ is a localization of the semi-local ring $\mathrm{B} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$. By flatness, the latter again identifies with a subring of the field L and is therefore integral; consequently ${}^{hs}\mathrm{B}$ is integral as well, which completes the proof. ☐

Corollary (18.8.16). — *Let* A *be a local Henselian ring (resp. strictly local). For A to be unibranch (resp. geometrically unibranch), it is necessary and sufficient that* $\mathrm{Spec}(\mathrm{A})$ *be irreducible.*

Proposition (18.8.17). — *Let* A *be a local Noetherian ring. If A is universally catenary, the same holds for* ${}^{hs}\mathrm{A}$.

Proof. The proof is the same as that of (18.7.5). ☐

Proposition (18.8.18). — i) *Every filtered inductive limit of strictly local rings (where the transition homomorphisms are local) is a strictly local ring.*

ii) *The functor* $A \rightsquigarrow {}^{hs}A$ *in the category of local rings (where the morphisms are local homomorphisms) commutes with filtered inductive limits.*
iii) *Every strictly local ring A is an inductive limit of a filtered inductive system of Noetherian strictly local rings (the transition homomorphisms being local).*

Proof. The proofs are modelled on those of (18.6.14). □

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18.9. Formal fibres of Noetherian Henselian rings.

Theorem (18.9.1). — *Let* A *be a local Noetherian Henselian ring, whose formal fibres are geometrically normal. Then the formal fibres of* A *are geometrically integral (hence geometrically connected).*

Proof. The two assertions stated are in fact equivalent, a prescheme that is locally Noetherian connected and normal being integral. As every finite integral A-algebra is a Henselian local ring (18.5.9) and (18.5.10), it follows from (7.3.16.2) that one is reduced to proving that if one further supposes A *integral*, then the fibre of the morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ at the generic point of $\mathrm{Spec}(A)$ is *integral*, which follows from the following corollary. □

Corollary (18.9.2). — *Let* A *be a local Noetherian Henselian integral ring, whose formal fibres are geometrically normal. Then* $\widehat{A}$ *is integral.*

Proof. Indeed, it follows from (18.8.16) that A is unibranch, hence the theorem follows from the hypothesis made on the formal fibres of A and from (7.6.3). □

Corollary (18.9.3). — *Under the hypotheses of (18.9.2), the field of fractions L of* $\widehat{A}$ *is a separable extension of the field of fractions K of A, and K is algebraically closed in L.*

Proof. This follows from the fact that the fibre of the morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ at the generic point is geometrically integral and from (4.3.2) and (4.3.5). □

Corollary (18.9.4). — *Let* A *be a local Noetherian ring, Henselian, and whose formal fibres are geometrically normal (for example a Noetherian Henselian local ring that is excellent), and let* $A' = \widehat{A}$. *Let* X *be an A-prescheme,* $X' = X \otimes_A A'$, $g : X' \to X$ *the canonical projection. Then the map* $U \mapsto g^{-1}(U)$ *is a bijection from the set of subsets of* X *that are both open and closed to the set of subsets of* X′ *that are both open and closed.*

Proof. As g is a faithfully flat and quasi-compact morphism (2.3.12), one knows that the topology of X is the *quotient* of that of X′ by the equivalence relation defined by g. It therefore amounts to seeing that an open and closed subset U′ of X′ is *saturated* for this relation. Now, as the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ has its fibres geometrically connected (18.9.1), the fibres of g are connected, whence our assertion immediately follows.

In particular, if X is locally connected (which will be the case if X is locally Noetherian), then, for every connected component U of X (which is open and closed), $g^{-1}(U)$ is connected (Bourbaki, *Top. gén.*, chap. I^er, 3^e éd., § 11, no 3, prop. 7). If one denotes by $\pi_0(X)$ the set of connected components of X, the map $\pi_0(X') \to \pi_0(X)$ canonically deduced from g is therefore *bijective*. □

Corollary (18.9.5). — *Under the hypotheses of (18.9.4), the functor*

$$Z \longmapsto Z \otimes_A A' \tag{18.9.5.1}$$

from the category of étale preschemes over X*, to that of étale preschemes over* X′*, is fully faithful.*

Proof. Let Z_1, Z_2 be two étale preschemes over X, and set $Z'_i = Z_i \otimes_A A'$ ($i = 1, 2$); it is a matter of proving that every X′-morphism $Z'_1 \to Z'_2$ comes, by base change, from a unique X-morphism $Z_1 \to Z_2$. Suppose first that Z_2 is *separated* over X; then, as $\mathrm{Hom}_X(Z_1, Z_2)$ identifies with $\Gamma((Z_1 \times_X Z_2)/Z_1)$, and that $Z_1 \times_X Z_2$ is étale and separated over Z_1, by virtue of (17.9.3), to the set of open and closed subsets U of $Z_1 \times_X Z_2$ such that the restriction to U of the projection $p : Z_1 \times_X Z_2 \to Z_1$ is a surjective and radicial morphism. The assertion thus follows from (18.9.4) and the fact that $Z'_1 \times_{X'} Z'_2 = (Z_1 \times_X Z_2) \otimes_A A'$. IV | 151

Let us pass to the general case. It will suffice to prove that the graph Γ' of an X'-morphism $Z'_1 \to Z'_2$, which is open in $Z'_1 \times_{X'} Z'_2$ (17.9.3), is of the form $\Gamma \otimes_A A'$, where Γ is induced by an *open* subset of $Z = Z_1 \times_X Z_2$. Indeed, the restriction to Γ of the projection $p : Z \to Z_1$ is then an isomorphism: after the base change $A \to A'$, it becomes the restriction of $p' : Z' \to Z'_1$ to Γ', which is an isomorphism, and it suffices to apply (2.7.1), (viii). The conclusion then follows from the characterisation of graphs (**I**, 5.3).

If $q : Z' = Z \otimes_A A' \to Z$ is the canonical projection, it remains to prove that $\Gamma' = q^{-1}(\Gamma)$ for an open subset Γ of Z. Since q is faithfully flat and quasi-compact, it suffices (2.3.12) to prove that $\Gamma' = q^{-1}(U)$ for an open subset $U \subset Z$. Let $Z'' = Z' \times_Z Z' = Z \times_X X''$ where $X'' = X' \times_X X'$, and let $q_1 : Z'' \to Z'$, $q_2 : Z'' \to Z'$ be the canonical projections. Applying (4.5.19.1), it suffices to show that $q_1^{-1}(\Gamma') = q_2^{-1}(\Gamma')$. But this is a property which is true if and only if it is true after each base change $\operatorname{Spec}(k(x)) \to X$, where x ranges over X. In other words, it suffices to prove the corollary when X is the *spectrum of a field*; but as every X-prescheme that is étale is automatically *separated* over X (17.6.2), c'), one is reduced to the case considered at the beginning of the proof. □

Remarks (18.9.6). — *i)* It is possible that, if the residue field of A is of characteristic 0, the functor (18.9.5.1) induces even an *equivalence* of the category of étale coverings of X and the category of étale coverings of X'. One can indeed prove this when A is *excellent*, using the resolution of singularities of Hironaka (M. Artin). It is plausible that an analogous statement is still true without restriction on the characteristic of the residue field, provided one restricts to the étale coverings whose "principal Galois groups" have order prime to the residual characteristic (cf. [41]).

ii) One does not know whether the formal fibres of A are geometrically connected (or even geometrically irreducible) when A is any Henselian local ring. It would suffice that, for every Noetherian Henselian local *integral* ring A, $\operatorname{Spec}(\widehat{A})$ be irreducible. One does not know if this is always the case.

iii) One does not know whether, when A is an excellent local ring, its strict Henselization ${}^{hs}A$ has geometrically regular formal fibres. To elucidate this question, one can reduce to the case where $A = k[[T_1, \ldots, T_n]]$, k being a field of characteristic $p > 0$. Taking account of (18.8.17), the question is whether the formal fibres of ${}^{hs}A$ are geometrically regular. The answer is affirmative when $n = 1$, but is not known for $n = 2$. One can show that the answer is affirmative whenever, for every scheme Y_1 finite over $Y = \operatorname{Spec}(\widehat{A})$, the singularities of Y_1 can be resolved (7.9.1).

The connectedness assertion made in (18.9.1) generalises in the following way:

Theorem (18.9.7). — *Let* A, B *be two local Noetherian rings,* $\rho : A \to B$ *a local homomorphism,* $f : \operatorname{Spec}(B) = X \to \operatorname{Spec}(A) = Y$ *the corresponding morphism. Suppose the following conditions satisfied:* **IV** | 152

i) f *is flat and all the fibres* $X_y = f^{-1}(y)$ $(y \in Y)$ *are geometrically reduced* (4.6.2).

ii) *Either the ring* A *is geometrically unibranch* (**0**, 23.2.1), *or else the ring A is unibranch* (**0**, 23.2.1) *and* $k(B)$ *is a primary extension* (4.3.1) *of* $k(A)$.

Then, if η *is the generic point of* Y, X_η *is connected, and for every closed subset* T *of X such that* $f(T)$ *is rare in Y,* $X - T$ *is connected.*

Proof. The two conclusions stated are in fact equivalent; indeed, the hypothesis that $f(T)$ is rare implies that $f(T)$ does not contain η. On the other hand, since $\{\eta\}$ is proconstructible (9.6) and f is flat, X_η is dense in X (2.3.10); consequently, if X_η is connected, then so is $X - T$, which contains it. Conversely, as U ranges over the affine open neighbourhoods of η in Y, X_η is the projective limit of the subschemes induced on the open subsets $f^{-1}(U)$ of X (8.1.2), a). If one supposes that $X - T$ is connected for every closed subset T of X such that $f(T)$ is rare in Y, the $f^{-1}(U)$ are connected, and therefore so is X_η (8.4.1).

Let us first show that one can reduce to the case where A is integral and $Y = \operatorname{Spec}(A)$ is geometrically unibranch (6.15.1) (which implies that A is geometrically unibranch, but is not equivalent to this condition (6.15.2)). It is clear that one can first suppose Y reduced, by considering the morphism $X \times_Y Y_{\mathrm{red}} \to Y_{\mathrm{red}}$ obtained from f by base change, which is flat and has the same fibres as f. One can therefore suppose A integral and unibranch. If K is the field of fractions of A, there then exists a finite A-subalgebra A'' of K such that, if A' is the integral closure of A, the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A'')$ is radicial (**0**, 23.2.5); it follows (6.15.5) that $\mathrm{Spec}(A'')$ is geometrically unibranch. Since A is unibranch, A' is a local ring, and hence so is A''. Let us show that the ring $B'' = B \otimes_A A''$ is also local. Indeed, B'' is a finite B-algebra, hence a Noetherian semilocal ring. If k'' is the residue field of A'', the maximal ideals of B'' are the points of $\mathrm{Spec}(B'')$ lying over the closed point of Spec(B), hence the points of $\mathrm{Spec}(k(B) \otimes_{k(A)} k'')$ (**I**, 3.4.9), since the closed point of $\mathrm{Spec}(A'')$ is the only point lying over the closed point of Spec(A). Now, when A is geometrically unibranch, k'' is a radicial extension of $k(A)$, so $\mathrm{Spec}(k(B) \otimes_{k(A)} k'')$ is radicial over $\mathrm{Spec}(k(B))$ and consequently consists of a single point. When A is unibranch and $k(B)$ is a primary extension of $k(A)$, $\mathrm{Spec}(k(B) \otimes_{k(A)} k'')$ is irreducible (4.3.2); since it is a finite discrete space (**I**, 6.4.4), it again consists of a single point. Thus B'' is local in both cases considered. If $Y'' = \mathrm{Spec}(A'')$, $X'' = \mathrm{Spec}(B'') = X \times_Y Y''$, the morphism $f'' = f_{(Y'')} : X'' \to Y''$ is flat and has geometrically reduced fibres. Moreover, if η'' is the generic point of Y'', then $k(\eta'') = k(\eta) = K$ by definition, so the fibres X_η and $X''_{\eta''}$ are isomorphic. It therefore suffices to prove that $X''_{\eta''}$ is connected. □

Remark (18.9.7.1). — When the ring A is *Japanese*, one can, in the case preceding (18.9.7.2), take $A'' = A'$ by definition (**0**, 23.1.1); one sees therefore that in this case one would even be able to prove the theorem when A is *integrally closed*.

(18.9.7.2). Suppose therefore now that Y is integral and geometrically unibranch. Note that if T is a closed subset of X such that $f(T)$ is rare in Y, T contains no maximal point of X since f is flat (2.3.4), and is therefore rare in X. Since X is a local scheme, hence connected, to prove that $X - T$ is connected it suffices, by virtue of (15.5.6.1), to show that for every $x \in X$ such that $f(x) \neq \eta$, $\mathrm{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. Set $y = f(x)$, $A_1 = \mathscr{O}_{Y,y}$, $B_1 = \mathscr{O}_{X,x}$, $Y_1 = \mathrm{Spec}(A_1)$, $X_1 = \mathrm{Spec}(B_1)$, and let $f_1 : X_1 \to Y_1$ be the morphism corresponding to the local homomorphism $A_1 \to B_1$ induced by ρ. It is clear that f_1 is flat, and it follows from (4.6.1) that its fibres are geometrically reduced. Moreover, A_1 is integral and geometrically unibranch (6.15.1), and $\dim(A_1) \geqslant 1$. One is thus reduced to proving the following lemma.

Lemma (18.9.7.3). — *Let* A, B *be two local Noetherian rings,* $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $\rho : A \to B$ *a local homomorphism,* $f : X \to Y$ *the corresponding morphism. Suppose the following conditions satisfied:*

i) f *is flat and all the fibres* X_y ($y \in Y$) *are geometrically reduced.*
ii) *A is integral, geometrically unibranch and* $\dim(A) \geqslant 1$.

Then, if b *is the closed point of* X, $X - \{b\}$ *is connected.*

Proof. Note first that by virtue of the theorem of Hartshorne (5.10.7), $X - \{b\}$ is connected if $\mathrm{prof}(B) \geqslant 2$. Now one has (6.3.1) **IV | 153**

$$\mathrm{prof}(B) = \mathrm{prof}(A) + \mathrm{prof}(B \otimes_A k) \tag{18.9.7.4}$$

where k denotes the residue field of A. On the other hand, since A is integral and $\dim(A) \geqslant 1$, one has $\mathrm{prof}(A) \geqslant 1$, hence $\mathrm{prof}(B) \geqslant 2$ *unless* $\mathrm{prof}(B \otimes_A k) = 0$; moreover, $B \otimes_A k$ is reduced by (i), hence the relation $\mathrm{prof}(B \otimes_A k) = 0$ means that $B \otimes_A k$ is a *field* (**0**, 16.4.7). We shall henceforth suppose that this last condition is satisfied. We shall begin by treating a case where the proof is very simple.

A) *Case where A is integrally closed.* Then, if $\dim(A) \geqslant 2$, one has $\mathrm{prof}(A) \geqslant 2$ (**0**, 16.5.1), hence $\mathrm{prof}(B) \geqslant 2$. One therefore only has to consider the case where $\dim(A) = 1$ and where $B \otimes_A k$ is a field; A is then a discrete valuation ring, hence regular, and since $B \otimes_A k$ is a field and f is flat, B is regular (**0**, 17.3.3). Moreover (6.1.1.1), one has $\dim(B) = \dim(A) + \dim(B \otimes_A k) = \dim(A) = 1$, so B is also a discrete valuation ring. Thus $X - \{b\}$ consists of a single point, which proves the lemma (18.9.7.3) in this case. **IV | 154**

One will observe that this proves the statement of the theorem (18.9.7) when one supposes in addition that, in that statement, the ring A is *Japanese*, for by virtue of the remark (18.9.7.1), one is reduced to the case where A is *integrally closed*, and then, in the preceding reduction (18.9.7.3), the ring $A_1 = \mathscr{O}_{Y,y}$ is also integrally closed and one can apply the result just proved.

B) *Case where* $\dim(A) \geqslant 2$. The result of (18.9.7.3) will follow from the two following lemmas. □

Lemma (18.9.7.5). — *Let* A *be a semi-local Noetherian ring, reduced,* A' *its integral closure in its total ring of fractions,* $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$; *let* S *be the set of closed points of* X *and* $\mathrm{U} = \mathrm{X} - \mathrm{S}$. *Suppose that for every point* $x' \in \mathrm{X}'$ *above a point of* S, $\dim(\mathscr{O}_{\mathrm{X}',x'}) \geqslant 2$; *then* $\mathrm{A}^{(1)} = \Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}})$ *is integral over* A *and is an A-algebra isomorphic to a subalgebra of* A'.

Proof. Recall that if the $\mathfrak{p}_i$ ($1 \leqslant i \leqslant m$) are the minimal prime ideals of A, then A' is the direct product of the integral closures A'_i of the integral rings $\mathrm{A}_i = \mathrm{A}/\mathfrak{p}_i$, so that X' is the sum of the schemes $\mathrm{X}'_i = \mathrm{Spec}(\mathrm{A}'_i)$ ($1 \leqslant i \leqslant m$). If X_0 is the sum of the schemes $\mathrm{X}_i = \mathrm{Spec}(\mathrm{A}_i)$ ($1 \leqslant i \leqslant m$) and U_0 the inverse image of U in X_0, the canonical homomorphism $\Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}}) \to \Gamma(\mathrm{U}_0, \mathscr{O}_{\mathrm{X}_0})$ is injective since X is reduced and the morphism $\mathrm{X}_0 \to \mathrm{X}$ (and hence also $\mathrm{U}_0 \to \mathrm{U}$) is surjective. As $\Gamma(\mathrm{U}_0, \mathscr{O}_{\mathrm{X}_0})$ is the direct product of the $\Gamma(\mathrm{U}_i, \mathscr{O}_{\mathrm{X}_i})$ (where U_i is the inverse image of U in X_i), it suffices to prove that for each i, $\Gamma(\mathrm{U}_i, \mathscr{O}_{\mathrm{X}_i})$ is isomorphic to a sub-A_i-algebra of A'_i. In other words, one is reduced to the case where A is a Noetherian integral semilocal ring. Since X is reduced and the morphism $\mathrm{X}' \to \mathrm{X}$ is surjective, the homomorphism $\Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}}) \to \Gamma(\mathrm{U}', \mathscr{O}_{\mathrm{X}'})$ (where U' is the inverse image of U in X') is injective, and it suffices to verify that the canonical homomorphism $\mathrm{A}' = \Gamma(\mathrm{X}', \mathscr{O}_{\mathrm{X}'}) \to \Gamma(\mathrm{U}', \mathscr{O}_{\mathrm{X}'})$ is *bijective*. Now (**I**, 8.2.1.1), $\Gamma(\mathrm{U}', \mathscr{O}_{\mathrm{X}'})$ is the intersection of the local rings $\mathrm{A}'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over U', and by virtue of the hypothesis, among these local rings are included all those of height 1. But the reasoning of (**0**, 23.2.7) applies also to a Noetherian integral semilocal ring, so A' is a *semilocal Krull ring*, and is therefore the intersection of the local rings $\mathrm{A}'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over the set of prime ideals of height 1 of A' (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 6, th. 4); *a fortiori*, A' is the intersection of the $\mathrm{A}'_{\mathfrak{p}'}$ for $\mathfrak{p}' \in \mathrm{U}'$, which completes the proof of the lemma. □

Remark (18.9.7.6). — We know (**0**, 23.2.5) that there exists a finite sub-A-algebra A'' of K, the field of fractions of A, such that the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A}'')$ is *radicial*; as this morphism is also integral and dominant, hence surjective (**II**, 6.1.10), it is a *homeomorphism* (2.4.5). By virtue of the geometric interpretation of $\dim(\mathscr{O}_{\mathrm{X}',x'})$ (5.1.2), one has therefore, for every point x'' of $\mathrm{X}'' = \mathrm{Spec}(\mathrm{A}'')$ above a point $x \in \mathrm{X}$, $\dim(\mathscr{O}_{\mathrm{X}'',x''}) = \dim(\mathscr{O}_{\mathrm{X}',x'}) \geqslant 2$, where x' is the unique point of X' above x. This being so, it follows first from (**0**, 16.1.5) and (**0**, 16.1.4.1) that the relation $\dim(\mathscr{O}_{\mathrm{X}',x'}) \geqslant 2$ **IV** | 155
implies $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$. Conversely, suppose that $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$ and that A is universally catenary (5.6.2); then so is $\mathscr{O}_{\mathrm{X},x}$ (5.6.3), hence $\dim(\mathscr{O}_{\mathrm{X}'',x''}) = \dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$ by virtue of (5.6.10), and consequently $\dim(\mathscr{O}_{\mathrm{X}',x'}) \geqslant 2$. The same conclusion holds if $\mathscr{O}_{\mathrm{X},x}$ is *geometrically unibranch*, for the fibre of the morphism $\mathrm{X}' \to \mathrm{X}$ at the point x is then reduced to a single point, and $\mathscr{O}_{\mathrm{X}',x'}$ is an *integral* $\mathscr{O}_{\mathrm{X},x}$-algebra; it suffices then to apply (**0**, 16.1.5).

Lemma (18.9.7.7). — *Let* A *be a local Noetherian ring, of dimension* $\geqslant 2$, *integral and geometrically unibranch,* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, y *the closed point of* Y. *Let* X *be a locally Noetherian and connected prescheme and* $f : \mathrm{X} \to \mathrm{Y}$ *a flat morphism. Then, for every closed subset* $\mathrm{T} \subset f^{-1}(y)$, $\mathrm{X} - \mathrm{T}$ *is connected.*

Proof. The same reasoning as at the beginning of (18.9.7.2) reduces to proving that, for every point $x \in f^{-1}(y)$, $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) - \{x\}$ is connected. One can therefore restrict to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, B being a local Noetherian ring and the homomorphism $\mathrm{A} \to \mathrm{B}$ being local; let $\mathrm{U} = \mathrm{Y} - \{y\}$, and note that $\mathrm{V} = f^{-1}(\mathrm{U})$ is dense in X (2.3.10); it suffices therefore to prove that V is *connected*. Keeping the notations of (18.9.7.5), and setting $\mathrm{A}^{(1)} = \Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{Y}})$, $\mathrm{Y}_1 = \mathrm{Spec}(\mathrm{A}^{(1)})$, $\mathrm{X}_1 = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}_1 = \mathrm{Spec}(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}^{(1)})$, we have the commutative diagram

$$\begin{tikzcd} \mathrm{X} \arrow[d, "f"'] & \mathrm{X}_1 \arrow[l, "g_1"'] \arrow[d, "f_1"] \\ \mathrm{Y} & \mathrm{Y}_1 \arrow[l, "g"'] \end{tikzcd}$$

By virtue of (2.3.1) and of the definition of $\mathrm{A}^{(1)}$, one has a canonical isomorphism $\Gamma(\mathrm{V}, \mathscr{O}_{\mathrm{X}}) \xrightarrow{\sim} \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}^{(1)}$, and it suffices therefore to prove that this last ring is *local*, since a local ring cannot be a product of two nonzero rings (**III**, 7.8.6.1). But we have seen (18.9.7.5) that $\mathrm{A}^{(1)}$ is a sub-A-algebra of the integral closure A' of A, since $\dim(\mathrm{A}) \geqslant 2$. The hypothesis that A is geometrically unibranch implies that the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ is radicial at the point y (6.15.3); *a fortiori* g_1 is radicial at the closed point of X; furthermore g_1 is an integral morphism, hence the closed points of X_1 are above x, hence there is only one closed point of X_1, which completes the proof of (18.9.7.7).

It is clear that (18.9.7.7) proves (18.9.7.3) when $\dim(A) \geqslant 2$ (even without supposing the fibres geometrically reduced).

C) *Case where* $\dim(A) = 1$. One has seen at the beginning of the proof of (18.9.7.3) that one can suppose $\dim(B \otimes_A k) = 0$, and by flatness (6.1.1.1), one therefore has

$$\dim(B) = \dim(A) + \dim(B \otimes_A k) = 1.$$

Since A is integral, Y consists of two points, the generic point η and the closed point a. Furthermore, since f is flat, every irreducible component of X dominates Y (2.3.4), hence $X - \{b\}$, which is the set of maximal points ξ_i of X, is equal to the underlying set of $f^{-1}(\eta)$, and the fibre $f^{-1}(\eta)$ is the sum of the $\operatorname{Spec}(k(\xi_i))$. The hypothesis on the X_y and the fact that these fibres are of dimension 0, show that they are *geometrically regular* (6.7.6), and *a fortiori* geometrically normal, in other words f is a *normal morphism*. But since A is integral and geometrically unibranch, it follows then from (6.15.10) that B is also integral and geometrically unibranch, hence $f^{-1}(\eta) = X - \{b\}$ is reduced to a single point, and *a fortiori* connected; this concludes the proof of (18.9.7.3) and of (18.9.7). ☐ **IV | 156**

Remark (18.9.7.8). — In case C) of the proof of (18.9.7.3), one can avoid making appeal to the delicate result (6.15.10) by reasoning as follows: since A is integral and unibranch of dimension 1, it follows from the theorem of Krull-Akizuki (Bourbaki, *Alg. comm.*, chap. VII, § 2, no 5, prop. 5) that its integral closure A' is a *Noetherian* ring. The same reasoning as at the beginning of the proof of (18.9.7) shows that $B' = B \otimes_A A'$ is a local ring; furthermore, by flatness, B' is contained in the total ring of fractions R of the reduced B (3.3.5). Now, the theorem of Krull-Akizuki (Bourbaki, *loc. cit.*) also applies to a local Noetherian *reduced* ring of dimension 1, and shows that for every such ring B, everything contained between B and its total ring of fractions is *Noetherian*. The ring B' being a local Noetherian ring and the morphism $\operatorname{Spec}(B') \to \operatorname{Spec}(A')$ being normal, B' is an integral ring and integrally closed (6.5.4), and one concludes as at the beginning of the proof of (18.9.7).

Corollary (18.9.8). — *With the notations of* (18.9.7), *suppose verified the condition (i) of* (18.9.7) *and one of the two conditions:*

(ii′) *A is a strictly local ring* (18.8.2).
(ii″) *A is a Henselian local ring and* $k(B)$ *is a primary extension of* $k(A)$.

Then all the fibres X_y *(*$y \in Y$*) are geometrically connected.*

Proof. Let $\mathfrak{p}$ be the prime ideal of A corresponding to the point y, and set $A' = A/\mathfrak{p}$ and $B' = B \otimes_A A' = B/\mathfrak{p}B$, which are evidently local Noetherian rings; it is clear that the morphism $f' : \operatorname{Spec}(B') \to \operatorname{Spec}(A')$ satisfies condition (i) of (18.9.7); and as y is the generic point of $\operatorname{Spec}(A')$ and $k(A') = k(A)$, $k(B') = k(B)$, one sees that it suffices to verify that, in the case (ii′) (resp. (ii″)), A' is geometrically unibranch (resp. unibranch). But A' is integral, and in the case (ii′) (resp. (ii″)) it is strictly local (resp. Henselian) by virtue of (18.5.10). One therefore concludes that A' is geometrically unibranch (resp. unibranch) by virtue of (18.8.15) (resp. (18.6.12)). ☐

Corollary (18.9.9). — *Let* A *be a local Noetherian ring, Henselian, universally Japanese (i.e.* (7.6.4) *and* (7.7.2)*), whose formal fibres* (7.3.13) *are geometrically reduced; then all the formal fibres of A are geometrically connected.*

Proof. It suffices to apply (18.9.8) in the case where $B = \widehat{A}$, since B is a flat A-module and $k(B) = k(A)$. ☐

Remark (18.9.10). — The proofs of (18.9.4) and (18.9.5) do not use the fact that the formal fibres of A are geometrically connected. The conclusions of these two propositions are therefore still valid when one supposes the local ring A Noetherian, Henselian and universally Japanese. **IV | 157**

Corollary (18.9.11). — *Let* X, Y *be two locally Noetherian preschemes,* Y *being supposed geometrically unibranch and X connected. Let* $f : X \to Y$ *be a reduced morphism (i.e.* (6.8.1)*), flat and with geometrically reduced fibres. Then, for every closed subset* T *of* X *such that* $f(T)$ *is rare in* Y, $X - T$ *is connected.*

Proof. As f is flat and $f(\mathrm{T})$, being rare, cannot contain a maximal point of Y, T cannot contain a maximal point of X (2.3.4), and hence is rare in X. Utilising (15.5.6.1), it suffices to show that, for every $x \in \mathrm{T}$, $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) - \{x\}$ is connected. Set $y = f(x)$, which is not a maximal point of Y. Since the ring $\mathscr{O}_{\mathrm{Y},y}$ is geometrically unibranch, one can apply (18.9.7) to the morphism

$$f_1 : \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) \longrightarrow \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}),$$

which satisfies conditions (i) and (ii) of this theorem, and to the closed subset $\{x\}$ of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$, since $f_1(\{x\}) = \{y\}$ is rare in $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$. □

18.10. Étale preschemes over a geometrically unibranch or normal prescheme

Theorem (18.10.1). — *Let $f : X \to Y$ be a locally of finite presentation morphism, x a point of X, $y = f(x)$. Suppose that Y is integral and geometrically unibranch at the point y. Then, for f to be étale at the point x, it is necessary and sufficient that the following two conditions be satisfied:*

i) *f is unramified at the point x;*
ii) *the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective.*

Moreover, if this is the case, X is integral and geometrically unibranch at the point x.

Finally, if $\mathscr{O}_{Y,y}$ is Noetherian, one can, in the preceding criterion, replace the condition (ii) *by the condition* (ii bis) $\dim \mathscr{O}_{Y,y} \leqslant \dim \mathscr{O}_{X,x}$.

Proof. The fact that, if f is étale, X is integral and geometrically unibranch at the point x is a particular case of (17.5.7). It is clear on the other hand that the conditions (i) and (ii) are necessary for f to be étale at the point x, since $\mathscr{O}_{X,x}$ is then an $\mathscr{O}_{Y,y}$-module faithfully flat (17.6.1). Let us prove that these conditions are sufficient.

Set $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$, $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$ (18.8.7), $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$. Let y' be the closed point of Y', x' a point of X' above x and y'; since the morphism $g : \mathrm{Y}' \to \mathrm{Y}$ is *flat* (18.8.8), it amounts to showing that f' is étale at the point x' (17.7.1),(ii)). The hypothesis (ii) implies that the morphism $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) \to \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$ is dominant (**I**, 1.2.7); as $\mathscr{O}_{\mathrm{Y},y}$ is integral, $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$ has a single generic point y_1, and there is a generization x_1 of x in X above y_1. The projection morphism $\mathrm{X}' \to \mathrm{X}$ being flat, there exists a generization x_1' of x' above x_1 (2.3.4); set $y_1' = f'(x_1')$. Now, the hypothesis on y implies that A' is *integral* (18.8.15), hence Y' has a single generic point η, necessarily above y_1 since g is flat (2.3.4); on the other hand, η is also the generic point of $g^{-1}(y_1)$ (**0**, 2.1.8), and one knows on the other hand that the fibres of g are discrete spaces (18.8.12), hence $g^{-1}(y_1) = \{\eta\}$, and consequently $y_1' = \eta$. The morphism $\mathrm{Spec}(\mathscr{O}_{\mathrm{X}',x'}) \to \mathrm{Spec}(\mathscr{O}_{\mathrm{Y}',y'})$, restriction of f', is therefore dominant, and since $\mathscr{O}_{\mathrm{Y}',y'} = \mathrm{A}'$ is integral, the corresponding homomorphism $\mathscr{O}_{\mathrm{Y}',y'} \to \mathscr{O}_{\mathrm{X}',x'}$ is injective (**I**, 1.2.7). On the other hand, f' is unramified at the point x'; hence (18.8.3), d), there exists a neighbourhood U of x' in X' such that $f'|\mathrm{U}$ is a *closed immersion* of U into Y'. *A fortiori* the homomorphism $\mathscr{O}_{\mathrm{Y}',y'} \to \mathscr{O}_{\mathrm{X}',x'}$ is surjective, and since it is injective, it is therefore *bijective*. But then $\mathscr{O}_{\mathrm{X}',x'}$ is a flat $\mathscr{O}_{\mathrm{Y}',y'}$-module, and f' is therefore étale at the point x' (17.6.1). **IV | 158**

Finally, when $\mathscr{O}_{\mathrm{Y},y}$ is Noetherian, we know that if f is étale at the point x, we have $\dim \mathscr{O}_{\mathrm{Y},y} = \dim \mathscr{O}_{\mathrm{X},x}$ (17.6.4). Conversely, suppose the conditions (i) and (ii bis) satisfied, and let us show that they imply conditions (i) and (ii). Let $\mathfrak{J}$ be the kernel of the canonical homomorphism $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{X},x}$; restricting Y as needed to a neighbourhood of y, we may suppose that $\mathfrak{J} = \mathscr{J}_y$, where $\mathscr{J}$ is an ideal of $\mathscr{O}_{\mathrm{Y}}$. If Y_1 is the closed subscheme of Y defined by $\mathscr{J}$, one can, restricting further X and Y to neighbourhoods of x and y respectively, suppose that f factorises as $\mathrm{X} \xrightarrow{f_1} \mathrm{Y}_1 \to \mathrm{Y}$ (**I**, 6.5.1). It is clear that f_1 is still unramified at the point x, hence quasi-finite at this point (17.4.1); the proof of (5.4.1), (i) then shows that $\dim(\mathscr{O}_{\mathrm{X},x}) \leqslant \dim(\mathscr{O}_{\mathrm{Y},y}/\mathfrak{J})$. But if one had $\mathfrak{J} \neq 0$, one would conclude that $\dim(\mathscr{O}_{\mathrm{Y},y}/\mathfrak{J}) < \dim(\mathscr{O}_{\mathrm{Y},y})$ since Y is integral (**0**, 16.1.2.2); one would then have $\dim \mathscr{O}_{\mathrm{X},x} < \dim \mathscr{O}_{\mathrm{Y},y}$, contradicting the hypothesis, which proves (ii). □

Remark (18.10.2). —

(i) When $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$ is Noetherian, and one knows that its completion $\widehat{\mathrm{A}}$ is integral (for example if A is regular), one can, in the preceding proof, replace A' by $\widehat{\mathrm{A}}$.

(ii) When Y is locally Noetherian, one can give a faster proof of (18.10.1), without using strict Henselization, but invoking the rather delicate results of §§ 14 and 15. As f is quasi-finite at the point x by hypothesis (17.4.1), one can, replacing X by an open neighbourhood of x, suppose that $f^{-1}(y) = \{x\}$. The hypothesis (ii) implies, as we have seen, the existence of an irreducible component Z of X containing x and dominating the unique irreducible component Y_0 of Y containing y. The morphism $f|Z$ being quasi-finite at the point x, hence equidimensional at this point (13.2.2), it follows from the hypothesis on Y and from the Chevalley criterion (14.4.4) that $g = f|Z$ is universally open at the point x. Furthermore, since f (and hence also g) is unramified at the point x, $g^{-1}(y)$ is geometrically reduced over $k(y)$ (17.4.1); as $\mathscr{O}_{Y,y}$ is integral, one concludes from (15.2.3) that g is *flat* at the point x, hence f is *étale* at this point (18.4.9).

(iii) With the same notations and hypotheses on Y as in (18.10.1), suppose only that f is *locally of finite type* (and not necessarily locally of finite presentation). Then, for $\mathscr{O}_{X,x}$ to be an $\mathscr{O}_{Y,y}$-algebra *essentially étale* (18.6.1), it is necessary and sufficient that f satisfy the two following conditions: **IV** | 159

1° f is formally unramified at the point x;

2° the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective.

There is nothing more than proving the sufficiency of these conditions. Taking into account (18.4.12), it suffices to show that f is *flat* at the point x. Now, by going back to the proof of (18.10.1) and the notations used in this proof, it suffices to show that f' is flat at the point x' (2.5.1); by (17.1.3) furthermore f' is formally unramified at the point x'. One can therefore restrict to carrying out the proof when $Y = \mathrm{Spec}(A)$, where A is a *strictly local* local ring and $y = f(x)$ the closed point of Y. Using then (18.8.3) we see that $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is surjective, hence *bijective* by virtue of the hypothesis, and this suffices to show the flatness of f at the point x.

Suppose furthermore that Y is *locally integral* at the point y (**I**, 2.1.8). Then the conditions 1° and 2° above (together with the fact that Y is geometrically unibranch at the point y and f locally of finite type) already imply that f is *étale at the point* x. Indeed, this follows from what precedes and from (18.4.13).

Corollary (18.10.3). — *Let* Y *be an integral and geometrically unibranch prescheme, η its generic point,* X *a connected prescheme, and $f : X \to Y$ a morphism locally of finite type such that $f^{-1}(\eta)$ is non-empty. Then, if f is formally unramified, f is étale, and* X *is integral and geometrically unibranch.*

Proof. It follows from the hypothesis and from (18.4.13) that f is étale at all the points where it is flat, and in particular at the points of $f^{-1}(\eta)$, since $\mathscr{O}_{Y,\eta} = k(\eta)$ is a field. The open set U of points of X where f is étale is therefore *non-empty*. Let us show on the other hand that for all $x \in U$, the homomorphism $\mathscr{O}_{Y,f(x)} \to \mathscr{O}_{X,x}$ is *injective*. Indeed, let s be a section of $\mathscr{O}_Y$ over an open neighbourhood of $f(x)$, which one may always suppose to be Y itself (the question being local on X and Y), and suppose that its image $t \in \Gamma(Y, f_*(\mathscr{O}_X)) = \Gamma(X, \mathscr{O}_X)$ has zero germ at the point x, hence vanishes in a neighbourhood V of x; then the restriction of t to the non-empty open set $U \cap V$ is zero. But the restriction $f|(U \cap V)$ is étale, hence $W = f(U \cap V)$ is open in Y and consequently contains the generic point η. Since the homomorphism $\mathscr{O}_{Y,\eta} \to \mathscr{O}_{X,z}$ is injective for every point $z \in f^{-1}(\eta)$, the hypothesis that $t|(U \cap V) = 0$ implies that $s_\eta = 0$; and as Y is integral, this implies $s = 0$, proving our assertion. Applying now (18.10.2), (iii)), we conclude that f is *flat* at the point x, in other words $x \in U$, hence U is both open and closed in X; as it is non-empty and X is connected, we have $U = X$.

Note now that since f is étale, hence locally quasi-finite and Y is irreducible, the maximal points of X belong to $f^{-1}(\eta)$ (2.3.4) and for all $x \in X$, there is a neighbourhood of x that contains only a finite number of these maximal points; in other words, the set of irreducible components of X is locally finite. But by (18.10.1), X is integral and geometrically unibranch at every point, hence every point of X belongs to only one irreducible component; and since the irreducible components form a locally finite family, they are open (and closed) in X. Since X is connected, it is integral. ☐

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Remark (18.10.3.1). — If $f : X \to Y$ is locally of finite presentation, dominant and quasi-compact, the fibre $f^{-1}(\eta)$ is non-empty by virtue of (**I**, 1.1.5). On the other hand, if one does not suppose f quasi-compact, *f can be unramified and dominant without being étale* (still assuming that the hypotheses of (18.10.3) are satisfied for X and Y). This is what the following example shows: one takes for Y the affine plane $\operatorname{Spec}(\mathbf{C}[T, U])$; one considers on the other hand a family $(X_i)_{i\in\mathbf{Z}}$ of preschemes isomorphic to the affine line $\operatorname{Spec}(\mathbf{C}[T])$, and one forms the prescheme obtained by "gluing together" X_{2j} and X_{2j+1} at the point -1 and X_{2j+1} and X_{2j+2} at the point $+1$. Let us define on the other hand $f : X \to Y$ as being equal on X_{2j} to the closed immersion whose image is the line of equation $y = 2j$, transforming the point -1 to $(2j-1, 2j)$ and the point 1 to $(2j+1, 2j)$; on X_{2j+1}, f_j is the closed immersion whose image is the line of equation $x = 2j+1$, transforming the point -1 to $(2j+1, 2j)$ and the point 1 to $(2j+1, 2j+2)$. It is clear that f is a local immersion (hence unramified) and dominant, but the fibre over the generic point of Y is empty and f is not étale.

Corollary (18.10.4). — *Let $g : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation, $f : X \to Y$ an S-morphism, x a point of X, $y = f(x)$, and $s = g(y) = h(x)$. Suppose that h is flat at the point x and $g^{-1}(s)$ normal at the point y. Then, for f to be étale at the point x, it is necessary and sufficient that f be unramified at the point x and that $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$.*

Proof. The conditions are necessary by virtue of the fact that if f is étale at the point x, it follows from this that f_s is also flat and quasi-finite at the point x (6.1.2). Conversely, if the conditions of the statement are satisfied, in order to see that f is étale at the point x, it suffices, by virtue of (17.8.3), to see that f_s is so. One can therefore restrict to the case where S is the spectrum of a field. It then follows from the hypothesis $\dim_x X = \dim_y Y$ and from the fact that f is unramified at the point x that one has, by (5.2.3) and (17.4.1), $\dim \mathscr{O}_{X,x} = \dim \mathscr{O}_{Y,y}$. Since f is quasi-finite at the point x and $\mathscr{O}_{Y,y}$ is integral, we deduce from (**0**, 7.4.4) and (**0**, 16.3.10) that the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective; it then suffices to apply (18.10.1). □

Corollary (18.10.5). — *With the notations of (18.10.4), suppose that h is flat and that $g^{-1}(s)$ is normal for all $s \in S$ (which will be the case for example if g is smooth). For f to be an open immersion, it is necessary and sufficient that f be a monomorphism and that, for all $x \in X$, $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$, where $y = f(x)$ and $s = g(y) = h(x)$.*

Proof. It remains only to prove the sufficiency of the conditions. It follows from (17.1.3) that a monomorphism locally of finite presentation is unramified; the hypotheses therefore imply that f is étale by (18.10.4), and one concludes by (17.9.1). □

Lemma (18.10.6). — *Let $f : X \to Y$ be a flat and locally quasi-finite morphism.*

(i) The set $\operatorname{Max}(X)$ of maximal points of X is in canonical bijection with the set $\coprod_{y\in\operatorname{Max}(Y)} f^{-1}(y)$.
(ii) If Y is irreducible with generic point η, the set of irreducible components of X is in canonical bijection with $f^{-1}(\eta)$, and is in particular finite if and only if $f^{-1}(\eta)$ is finite. **IV** | 161
(iii) If the set of irreducible components of Y is locally finite, so is the set of irreducible components of X, and in particular X is locally connected.

Proof. The assertion (i) follows immediately from (2.3.4) and (**0**, 2.1.6) and from the fact that the fibres $f^{-1}(y)$ are discrete sets. It is clear that (ii) is a particular case of (i). Finally, to prove (iii), one can, by virtue of (i), restrict to the case where f is quasi-finite and where Y is irreducible (**0**, 2.1.6) and the conclusion follows from (ii) and from the fact that the fibres of f are finite sets (**II**, 6.2.2). □

Proposition (18.10.7). — *Let Y be a geometrically unibranch and irreducible prescheme (resp. integral), $f : X \to Y$ an étale morphism. Then X is isomorphic to a disjoint union of irreducible (resp. integral) preschemes X_λ, where λ ranges over the fibre $f^{-1}(\eta)$ of the generic point η of Y. The X_λ are geometrically unibranch; if Y is normal, the X_λ are normal.*

Proof. Indeed, X is geometrically unibranch (17.5.7), hence every point of X belongs to only one irreducible component. Moreover, by (18.10.6) and (17.6.1), the family of irreducible components of X is locally finite; hence (**0**, 2.1.6) these are the connected components X_λ of X ($\lambda \in f^{-1}(\eta)$), and X is their disjoint union. By (17.5.7), if Y is integral (resp. normal), the X_λ are integral (resp. normal). □

Proposition (18.10.8). — *Let* Y *be a normal and integral prescheme, of generic point* η, $\mathrm{K} = k(\eta)$ *its field of rational functions,* $f : \mathrm{X} \to \mathrm{Y}$ *an étale morphism. Suppose that* $f^{-1}(\eta)$ *is finite and non-empty, so that the* K*-scheme* $f^{-1}(\eta)$ *is the spectrum of a finite separable* K*-algebra* L*, which is a direct product of finitely many fields* K_i *(*$1 \leqslant i \leqslant n$*), finite separable extensions of* K (17.6.2)*. Let* Y′ *be the integral closure of* Y *in* L*, isomorphic to the sum of the integral closures* Y_i *of* Y *in* K_i (**II**, 6.3.6)*. Then the morphism* f *factorises uniquely as* $\mathrm{X} \xrightarrow{f'} \mathrm{Y}' \xrightarrow{g} \mathrm{Y}$*, such that the restriction* $f'_\eta : f^{-1}(\eta) \to g^{-1}(\eta)$ *is the canonical isomorphism. Moreover* f' *is a local isomorphism; for* f' *to be an open immersion, it is necessary and sufficient that* f *be separated.*

Proof. By virtue of (18.10.7), one can restrict to the case where X is integral and normal. The existence and uniqueness of the factorisation considered for f follow from (**II**, 6.3.9). As f' is birational and locally quasi-finite, the final assertions follow from (8.12.10). □

Corollary (18.10.9). — *Under the hypotheses of* (18.10.8)*, for* f *to be a finite morphism (in other words, for* X *to be an étale covering of* Y*), it is necessary and sufficient that* X *be isomorphic to the integral closure* Y′ *of* Y *in* L*.*

Proof. If f is finite, the canonical injection $j : \mathrm{X} \to \mathrm{Y}'$ (which is an open immersion) is also a finite morphism (**II**, 6.1.5), (v)), hence a closed morphism (**II**, 6.1.10), and as $j(\mathrm{X})$ is dense in Y′ by (18.10.8), we have $j(\mathrm{X}) = \mathrm{Y}'$. Conversely, as Y is normal and L is a composed direct product of finite separable extensions of K, Y′ is finite over Y (Bourbaki, *Alg. comm.*, chap. V, § 1, nº 6, cor. 1 of prop. 18), whence the corollary. □

(18.10.10). Let Y be a normal and integral prescheme, $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ its field of rational functions. We say that a finite-dimensional K-algebra L is *unramified over* Y if: 1° L is a separable K-algebra, hence a direct product of finite separable extensions K_i of K ($1 \leqslant i \leqslant n$); 2° the integral closure Y′ of Y in L (the disjoint union of the integral closures of Y in the K_i) is *unramified over* Y (which, by (18.10.3), is **IV** | 162
equivalent to saying that Y′ is étale over Y). When $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, where A is an integral, integrally closed ring and K its field of fractions, one also says (by a slight abuse of language, and when no confusion with the terminology of (17.3.2), (ii) is to be feared) that L is unramified over A instead of "unramified over Y".

Remark (18.10.11). — Instead of saying that L is unramified over Y, some authors also say that L is "unramified over K"; we shall not follow this potentially confusing usage.

One can then express (18.10.9) in the following form:

Corollary (18.10.12). — *Let* Y *be a normal and integral prescheme,* $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ *its field of rational functions. Then the functor* $\mathrm{X} \mapsto \mathrm{R}(\mathrm{X})$ *is an equivalence between the category of étale coverings of* Y *and the category of finite étale* K*-algebras unramified over* Y*. A quasi-inverse functor sends each finite étale* K*-algebra* L *unramified over* Y *to the integral closure* Y′ *of* Y *in* L*.*

Proposition (18.10.13). — *Let* Y *be a normal and integral prescheme,* $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ *its field of rational functions.*

- *(i) The field* K *is a* K*-algebra unramified over* Y*.*
- *(ii) Let* L *be a finite extension of* K *unramified over* Y*, and let* Z *be the integral closure of* Y *in* L*. Then, if* M *is an* L*-algebra unramified over* Z*,* M *is also a* K*-algebra unramified over* Y *("transitivity" of unramifiedness).*
- *(iii) Let* Y′ *be a normal and integral prescheme,* K′ *its field of rational functions,* $g : \mathrm{Y}' \to \mathrm{Y}$ *a dominant morphism. If* L *is a* K*-algebra unramified over* Y*, then* $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$ *is a* K′*-algebra unramified over* Y′ *("base-change" property). Furthermore if* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ *and* $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ *are affine, and if* C *is the integral closure of* A *in* L*, then* $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ *is the integral closure of* A′ *in* $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$*.*

Proof. The assertion (i) is immediate, Y being its own normalisation in K. To prove (ii), let Z′ be the integral closure of Z in M, which is by hypothesis an étale covering of Z. As Z is étale and separated over Y, so is Z′ (17.3.3), and it is clearly the integral closure of Y in M. As M is a finite separable K-algebra, it follows from (18.10.10) that M is unramified over Y. Finally, to prove (iii), observe that if Z is the integral closure of Y in L, then $\mathrm{Z} \times_{\mathrm{Y}} \mathrm{Y}'$ is étale and separated over Y′ (17.3.3), finite over Y′ since Z is finite over Y, and has $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$ as its ring of rational functions (**I**, 3.4.9). Since Y′ is normal, so is $Z \times_Y Y'$ (17.5.7); it is therefore the integral closure of Y' in $L \otimes_K K'$, which completes the proof. □

Corollary (18.10.14). — *Let* Y *and* Y' *be two normal and integral preschemes,* K *and* K' *their respective fields of rational functions, and* $g : Y' \to Y$ *a dominant morphism, so that* K' *is an extension of* K*. Let* L *be a finite extension of* K *unramified over* Y*, and let* L_1 *be a composite extension (Bourbaki,* Alg.*, chap. VIII, § 8, def. 1) of* L *and* K'*. Then* L_1 *is an extension of* K' *unramified over* Y'*. Furthermore, if* $Y = \mathrm{Spec}(A)$ *and* $Y' = \mathrm{Spec}(A')$ *are affine, and if* C *is the integral closure of* A *in* L*, then the subring* $C_1 = A[C, A']$ *of* L_1 *is the integral closure of* A' *in* L_1*.*

Proof. Indeed, $L \otimes_K K'$ is a direct product of finite separable extensions of K', and L_1 is one of these extensions; hence the integral closure of Y' in $L \otimes_K K'$ is a disjoint union of preschemes, one of which is the integral closure of Y' in L_1. The corollary therefore follows immediately from (18.10.13). **IV** | 163

When, for example, A is the ring of integers of an algebraic number field K, and K' and L are algebraic extensions of K, there are classical examples showing that the relation $C_1 = A[C, A']$ fails when L is not unramified over A. □

(18.10.15). Suppose that $Y = \mathrm{Spec}(A)$ is affine, A being integral and integrally closed; let K be its field of fractions, L a finite separable extension of K, and we suppose that the integral closure C of A in L is a *projective* A-module of finite type (which will be the case for example if A is a Dedekind ring (Bourbaki, *Alg. comm.*, chap. VII, § 4, nº 10, prop. 22)). To say that L is unramified over A means that C is étale (18.10.10); by (18.2.7), (ii), this is equivalent to saying that the *discriminant* $d_{C/A}$ of $\mathrm{Spec}(C)$ over $\mathrm{Spec}(A)$ is *invertible* in A. In the particular case where C is a *free* A-module and $(x_i)_{1 \leqslant i \leqslant n}$ a basis of C over A, this means that $\det(\mathrm{Tr}_{C/A}(x_i x_j))$ is invertible in A.

Theorem (18.10.16). — *Let* Y *be an integral prescheme,* η *its generic point,* $f : X \to Y$ *a separated and locally of finite type morphism. Suppose that every irreducible component of* X *dominates* Y *and that the generic fibre* $X_\eta = f^{-1}(\eta)$ *is a finite prescheme over* $K = k(\eta)$*; then* $(X_\eta)_{\mathrm{red}} = \mathrm{Spec}(L)$*, where* $L = \prod_i L_i$ *is a product of finite extensions* L_i *of* K*. Set*

$$n = [L : K] = \sum_i [L_i : K], \quad n_s = \sum_i [L_i : K]_s \quad \textit{(sum of separable degrees of the } L_i\textit{)}.$$

Let y *be a geometrically unibranch point of* Y*, and denote by* $n(y)$ *the sum of the separable degrees over* $k(y)$ *of the residue fields of the isolated points of* $f^{-1}(y)$*. Then:*

(i) *One has* $n(y) \leqslant n_s \leqslant n$*.*

(ii) *Suppose* X *reduced and the point* y *normal in* Y*. For* $n(y) = n$*, it is necessary and sufficient that there exist a neighbourhood* U *of* y *in* Y *such that the restriction* $f^{-1}(U) \to U$ *of* f *is an étale and finite morphism.*

Proof. A) Reduction to the case where f *is quasi-finite.* — One knows (13.1.4) that the set Z of isolated points $x \in X$ of $f^{-1}(f(x))$ is open in X and by hypothesis Z contains $f^{-1}(\eta)$. Since $f^{-1}(\eta)$ is finite, Z is an increasing filtered union of quasi-compact open sets V_λ containing $f^{-1}(\eta)$ (dense in X by hypothesis). If $n_\lambda(y)$ is the sum of the separable degrees over $k(y)$ of the residue fields of the points of $V_\lambda \cap f^{-1}(y)$, we have $n(y) = \sup n_\lambda(y)$; to prove (i), it will therefore suffice to show that $n_\lambda(y) \leqslant n_s$. On the other hand, if $n(y) = n$, there is an index λ such that $n_\lambda(y) = n$. Suppose that we have shown that there then exists a neighbourhood U of y in Y such that the restriction $V_\lambda \cap f^{-1}(U) \to U$ of f is an étale and finite morphism; since f is separated, the canonical injection $V_\lambda \cap f^{-1}(U) \to f^{-1}(U)$ is then also a finite morphism (**II**, 6.1.5), (v)), hence $V_\lambda \cap f^{-1}(U)$ is closed in $f^{-1}(U)$; but being everywhere dense in $f^{-1}(U)$, it is necessarily equal to it.

B) Reduction to the case where X *is reduced,* f *finite, and* $Y = \mathrm{Spec}(\mathscr{O}_{Y,y})$*.* — We note that the hypothesis on f implies that it is also satisfied for f_{red}, and that the conclusion of (i) concerns only f_{red}. One can therefore replace f everywhere by f_{red} and thus suppose X reduced.

On the other hand, by virtue of (18.12.13) ([1]), one can (after having replaced Y by an open affine neighbourhood V of y and f by its restriction $f^{-1}(V) \to V$) factorise f as $X \xrightarrow{j} X' \xrightarrow{g} Y$, where g is *finite* and j an open immersion. Since X is reduced, one can replace X' by the reduced closed sub-prescheme with underlying space $\overline{j(X)}$ (**I**, 5.2.2), hence suppose X' reduced and such that every **IV** | 164 irreducible component of X' dominates Y. Moreover, $j(X)$ is an open dense set in X' and has only a finite number of irreducible components by virtue of the hypothesis; hence (**0**, 1.2.7) the irreducible components of X' are the closures of those of $j(X)$, and by construction the fibre $g^{-1}(\eta)$ identifies with $f^{-1}(\eta)$. If one supposes the theorem proved for f finite, one can apply it to g, and as $f^{-1}(y)$ identifies with a sub-prescheme of $g^{-1}(y)$, the relation (i) for g implies the same relation for f. Suppose, on the other hand, that $n(y) = n$; then, by what precedes, one can apply the conclusion of (ii) to the morphism g, and after replacing Y if necessary by a neighbourhood of y, one may suppose that g is étale, and consequently so is f; moreover, one then has $f^{-1}(y) = g^{-1}(y)$. As the morphism g is closed and $j(X)$ is open in X', there exists a neighbourhood U of y in Y such that $j(X) \cap g^{-1}(U) = j(f^{-1}(U))$, which proves that (ii) is also true for f.

We are thus reduced to the case where f is moreover a *finite* morphism. It is immediate that, to prove (i), one can restrict to the case where $Y = \mathrm{Spec}(\mathscr{O}_{Y,y})$. Let us show that the same is true for proving (ii). Indeed, suppose it has been proved that the condition $n(y) = n$ implies that $X \times_Y \mathrm{Spec}(\mathscr{O}_{Y,y})$ is étale and finite over $\mathrm{Spec}(\mathscr{O}_{Y,y})$. Since this implies that f is flat and formally unramified at the points of $f^{-1}(y)$, and since Y is *integral*, it follows from (18.4.13) that f is *étale at the points* of $f^{-1}(y)$, hence also on a neighbourhood V of $f^{-1}(y)$ in X; but since f is finite, hence closed, V contains an open set of the form $f^{-1}(U)$, where U is a neighbourhood of y in Y.

C) Reduction to the case where $Y = \mathrm{Spec}(\mathscr{O}_{Y,y})$ *and* $\mathscr{O}_{Y,y}$ *is strictly local.* — Set $A = \mathscr{O}_{Y,y}$, and let $A' = {}^{\mathrm{hs}}A$ be the strict henselisation of A (18.8.7), which is an integral, geometrically unibranch local ring (18.8.16). Let $Y' = \mathrm{Spec}(A')$, and let y' and η' be the closed point and the generic point of Y'; y' is above y and since the morphism $g : Y' \to Y$ is flat (18.8.8), η' is above η (2.3.4). Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$; f' is finite and since g is flat, every irreducible component of X' dominates Y' (2.3.7). If $n'(y')$, n' and n'_s are the numbers defined for f' in the same way as $n(y)$, n and n_s for f, we have $n'_s = n_s$ and $n'(y') = n(y)$ (**I**, 6.4.8), and it therefore amounts to proving the assertion (i) for f or for f'. Moreover, if X is reduced and y is a normal point of Y (hence A is integrally closed), then A' is integrally closed (18.8.12). On the other hand, it follows from the definition (18.8.7), from the remark (8.1.2), *a*)) and from the theorem of the double inductive limit, that A' is the inductive limit of rings A_α such that the morphisms $\mathrm{Spec}(A_\alpha) \to \mathrm{Spec}(A) = Y$ are étale; consequently X' is the projective limit of the $X_\alpha = X \otimes_A A_\alpha$, which are étale over X, hence reduced since X is (17.5.7); by passage to the limit,

we deduce that X' is reduced (8.7.1). This being so, if $n(y) = n$, one has $n_s = n$, and consequently **IV** | 165
the residue fields at the points of X_η are necessarily separable over $k(\eta)$; the residue fields at the points of $X'_{\eta'}$ are then algebraic and separable over $k(\eta')$ (4.6.1), in other words $n'_s = n' = n$. It follows from what precedes that $n'(y') = n'$. The morphism f' therefore has the same properties as f, and if one proves that f' is étale, it will follow from (17.7.1) that f is étale.

D) End of the proof. — Suppose therefore A strictly local. Since A is henselian and the morphism f finite, it follows from (18.5.11) that if x_j ($1 \leqslant j \leqslant m$) are the distinct points of $f^{-1}(y)$, X is the *sum* of open sub-preschemes $\mathrm{Spec}(\mathscr{O}_{X,x_j}) = X_j$, which are the connected components of X. By hypothesis, each of them dominates Y, hence every irreducible component of X meets $f^{-1}(\eta)$, and so the number of points of $f^{-1}(\eta)$ is $\geqslant m$; *a fortiori* we have $n_s \geqslant m$. But since $k(y)$ is separably closed and the $k(x_j)$ are algebraic over $k(y)$, they are necessarily radicial extensions of $k(y)$; hence $m = n(y)$, and this proves (i). Suppose now the hypotheses of (ii) satisfied; the relations $n(y) = n_s = n$ imply first of all, since every irreducible component of X dominates Y, that each of the X_j is *irreducible*; moreover, as the X_j are *reduced*, they are *integral*; finally the relations $m = n_s = n$ imply that for the generic point ξ_j of X_j, $k(\xi_j)$ is a separable extension of $k(\eta)$ of degree 1, hence isomorphic to $k(\eta)$; in other words, the restriction $f|X_j : X_j \to Y$ of f is *finite and birational*; moreover, X_j is *integral* and by hypothesis Y is *normal*; hence (8.12.10.1) $f|X_j$ is an *isomorphism*, which proves that f is étale, and finishes the proof of (18.10.16). □

[33]([1]) The reader will verify that (18.10.16) is not used in the proof of (18.12.13). If f is supposed locally of finite presentation (resp. quasi-projective), one can replace (18.12.13) by (8.12.8) (resp. (8.12.6)).

Remark (18.10.17). — The method of "étale localisation" used in the proof of (18.10.16) also allows one to improve the results of (15.5.1) by eliminating the Noetherian hypothesis. One has first of all the following result which generalises (15.5.2):

(18.10.17.1). Let $f : \mathrm{X} \to \mathrm{Y}$ be a separated and quasi-finite morphism, y a point of Y such that f is universally open at the points of $f^{-1}(y)$. Then, for every generisation y' of y, we have $n(y) \leqslant n(y')$.

Proof. One can replace Y by the reduced closed sub-prescheme $\overline{\{y'\}}$ and X by $f^{-1}(\mathrm{Z})$, hence suppose Y integral with generic point y'. Using (18.12.13) (as in (18.10.16), B)), one can suppose that X is an open dense subset of a prescheme X' and that f is the restriction to X of a finite morphism $g : \mathrm{X}' \to \mathrm{Y}$. Let us show that $g^{-1}(y') = f^{-1}(y')$. Indeed, let $i : g^{-1}(y') \to \mathrm{X}'$ be the canonical injection, which is a flat morphism since y' is the generic point of Y (**I**, 3.6.5); one can write $f^{-1}(y') = i^{-1}(\mathrm{X})$. On the other hand, the canonical injection $j : \mathrm{X} \to \mathrm{X}'$ is a morphism of finite type since the composite $g \circ j = f$ is of finite type and g is separated (1.5.4); hence X is a retrocompact open subset of X', and consequently pro-constructible in X' (**I**, 1.9.5), (v)). One concludes therefore from (2.3.10) that $i^{-1}(\overline{\mathrm{X}}) = \overline{i^{-1}(\mathrm{X})}$, in other words $f^{-1}(y')$ is dense in $g^{-1}(y')$; but as $g^{-1}(y')$ is discrete, we necessarily have $f^{-1}(y') = g^{-1}(y')$.

We are thus reduced to proving that the sum of the numbers $[k(x) : k(y)]_s$ where x ranges over the points of $g^{-1}(y)$ where g is universally open, is at most equal to $n(y')$ (for the morphism g). **IV** | 166
Using then the fact that the property of being universally open at a point is preserved by base change, one shows, as in (18.10.16), C)), that one can reduce to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, with $\mathscr{O}_{\mathrm{Y},y}$ *strictly local*. Applying (18.5.11), *c*)), we see that X is the disjoint union of the open subschemes $\mathrm{X}_j = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j})$, where x_j ranges over $g^{-1}(y)$. The hypothesis that g is open at one of these points x_j implies that the corresponding subscheme X_j dominates Y (1.10.3); one concludes as in (18.10.16), D)).

We then deduce from (18.10.17.1) that *the conclusions of* (15.5.1) *are still valid when one no longer supposes* Y *locally Noetherian, but only that f is separated, quasi-finite, and of finite presentation*. □

Indeed, to prove assertion (i) of (15.5.1), one remarks that, since f is of finite presentation, the set E of points $z \in \mathrm{Y}$ such that $n(z) \geqslant n(y)$ is locally constructible (9.7.9); to show that y is interior to E, it suffices, by virtue of (1.10.1), to see that every generisation y' of y belongs to E, which is nothing other than (18.10.17.1).

To prove assertion (ii) of (15.5.1), one may, since f is of finite presentation, use (8.10.5), (xii) according to the method of (8.1.2), *a*)), and thus first suppose that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$. Applying next (2.7.1), (vii), one sees that after the faithfully flat base change $\mathrm{Y}' \to \mathrm{Y}$, where $\mathrm{Y}' = \mathrm{Spec}({}^{\mathrm{hs}}\mathscr{O}_{\mathrm{Y},y})$, one may suppose that the ring $\mathscr{O}_{\mathrm{Y},y}$ is *strictly local*. Applying (18.5.11), *c*)), we see then that if x_j $(1 \leqslant j \leqslant n)$ are the points of $f^{-1}(y)$, X is the *sum* of n open sub-preschemes $\mathrm{X}_j = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j})$ which are *finite* over Y, and of an open prescheme X''. If one proves that $\mathrm{X}'' = \varnothing$, one will have shown that f is finite, hence proper. Now the fields $k(x_j)$, being algebraic extensions of the separably closed field $k(y)$, are radicial extensions, and consequently $n(y) = n$; since f is open at each point x_j, X_j dominates Y (1.10.3), and the restriction of f to X_j, being finite, is surjective (**II**, 6.1.10). The hypothesis that $z \mapsto n(z)$ is constant on Y therefore implies that $\mathrm{X}'' \cap f^{-1}(z) = \varnothing$ for every $z \in \mathrm{Y}$, that is, $\mathrm{X}'' = \varnothing$.

Finally, the proof of (iii) reproduces without change that given in (15.5.1).

(18.10.17.2). By analogous methods using strict henselisation, one can, in most of the results of §§ 14 and 15, eliminate the Noetherian hypotheses.

Lemma (18.10.18). — *Let* X, Y *be two preschemes, $f : \mathrm{X} \to \mathrm{Y}$ a birational morphism* [1]*, x a point of* X*. For f to be an isomorphism local at the point x, it is necessary and sufficient that f be étale at the point x. For f to be an open immersion, it is necessary and sufficient that f be étale and separated.*

Proof. The conditions stated being trivially necessary, all that is needed is to see that they are sufficient. For the first assertion, the question is local on X and Y; hence one may suppose that f is étale and that X and Y are affine, so that f is *separated*, and one is reduced to proving the second assertion.

33[1] We understand by this the notion defined in (6.15.4), but where we do not suppose X and Y reduced.

By virtue of (17.9.1), it is a question of proving that f is *radicial*. Let $y \in \mathrm{Y}$, and let us prove that $f^{-1}(y)$ is radicial on $k(y)$. The base change $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}) \to \mathrm{Y}$ not changing the fact that f is étale, separated, and birational, one can restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, with $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$. Let $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$, which is a local ring and a faithfully flat A-module (18.8.8), the homomorphism $\mathrm{A} \to \mathrm{A}'$ being local. Set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$, so that f' is étale and separated; moreover, as the morphism $\mathrm{Y}' \to \mathrm{Y}$ is flat, the reasoning of (6.15.4.1) shows that f' is also birational. If y' is the closed point of Y', it suffices, by (2.6.1), (v), to prove that $f'^{-1}(y')$ is radicial over $k(y')$; since f' is étale, it is enough to prove that $f'^{-1}(y')$ contains at most one point (17.6.1), c'). IV | 167

Suppose therefore $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, A strictly local, f separated, étale and birational, and let us show that $f^{-1}(y)$ can contain only a single point. Indeed, if $f^{-1}(y)$ contained two distinct points x_1, x_2, there would exist two Y-sections u_1, u_2 of f such that $u_1(y) = x_1$ and $u_2(y) = x_2$ (18.8.1); but since f is étale and separated, u_1 and u_2 would be isomorphisms of Y onto two connected components (open) of X (17.9.4) and there would then be two maximal points of X at least above every maximal point of Y, which contradicts the hypothesis that f is *birational*. □

Proposition (18.10.19). — *Let* Y *be a reduced prescheme whose set of irreducible components is locally finite,* $f : \mathrm{X} \to \mathrm{Y}$ *a separated morphism, locally of finite type and formally unramified,* g *a* Y*-rational section of* f *(i.e. a* Y*-rational map from* Y *to* X*),* U *the domain of definition of* g (**I**, 7.2.1)*, and let* Z *be the closure of* $g(\mathrm{U})$ *in* X*. Then, for all points* $y \in \mathrm{Y} - \mathrm{U}$ *such that* y *is geometrically unibranch,* $\mathrm{Z}_y = \mathrm{Z} \cap f^{-1}(y) = \varnothing$*. In particular, if* Y *is geometrically unibranch,* $g(\mathrm{U})$ *is a subset of* X *that is both open and closed. If, for every closed irreducible* T *of* X*,* $f(\mathrm{T})$ *is closed in* Y*,* g *is defined at every geometrically unibranch point of* Y*.*

Proof. Since Y is reduced and f separated, it follows from (**I**, 7.2.2) that there exists a U-section u of $f^{-1}(\mathrm{U})$ belonging to the class g; moreover, since f is formally unramified, (17.4.1.2) shows that u is an isomorphism of U onto the subscheme induced on the open set $u(\mathrm{U})$ of X. If one denotes also by Z the reduced sub-prescheme of X having Z for underlying space, $g(\mathrm{U}) = u(\mathrm{U})$, being reduced, is induced also by Z, and by construction $f' = f|\mathrm{Z}$ is a *birational* morphism of Z to Y, moreover formally unramified since f is (17.1.3). The question being local on Y, and since Y is reduced and geometrically unibranch (hence integral) at the point y, one may, after replacing Y by a neighbourhood of y, restrict to the case where Y is *integral*; then $u(\mathrm{U})$ is irreducible, hence so is Z, and consequently Z is integral as well. Since f' is dominant, (**I**, 8.2.7) shows that the homomorphism $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{Z},x}$ is injective for every $x \in \mathrm{Z}_y$, and (18.10.2), (iii) implies that f' is étale. Being separated and birational, it is a local isomorphism at x by (18.10.18); but this would imply that the U-section u extends to an open set strictly larger than U, contrary to the definition of U. We therefore necessarily have $\mathrm{Z}_y = \varnothing$, which establishes the first assertion; the second is an obvious consequence. Finally, under the last hypotheses of the statement, as Z is closed and irreducible, $f(\mathrm{Z})$ is closed in Y, hence equal to Y, i.e. $\mathrm{Z}_y \neq \varnothing$ for all $y \in \mathrm{Y}$, which finishes the proof. □

Remark (18.10.20). — Let Y be a locally Noetherian prescheme. One says that a morphism $f : \mathrm{X} \to \mathrm{Y}$ is *essentially proper* if it is locally of finite type and if, for every base change $\mathrm{Y}'' \to \mathrm{Y}$, where Y'' is *locally Noetherian*, the image by $f_{(\mathrm{Y}'')} : \mathrm{X}_{(\mathrm{Y}'')} \to \mathrm{Y}''$ of every closed irreducible subset of X'' is *closed*. To say that f is essentially proper means that, for every Y-scheme of the form $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A})$, where A is a discrete valuation ring, the canonical map (**II**, 7.3.2.2) relative to the Y'-prescheme $\mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$ is bijective. The reasoning of (**II**, 7.3.8) proves that this is the same as saying that f (supposed locally of finite type) is *separated* and that for every base change $\mathrm{Y}'' \to \mathrm{Y}$, where Y'' is *locally Noetherian*, the image by $f_{(\mathrm{Y}'')} : \mathrm{X}_{(\mathrm{Y}'')} \to \mathrm{Y}''$ of every closed irreducible subset of X'' is *closed*. To say that f is *proper* (for Y locally Noetherian) means therefore that f is *essentially proper and of finite type* (**II**, 7.3.8); but one encounters important examples of essentially proper morphisms that are not of finite type (for example certain "Picard preschemes" or certain "preschemes of Néron-Severi" (chap. VI)). It follows immediately from (18.10.19) that if Y is *locally Noetherian, reduced, and geometrically unibranch*, and if $f : \mathrm{X} \to \mathrm{Y}$ is *essentially proper and unramified*, then every Y-rational section of f is *defined everywhere*. IV | 168

The following proposition generalises the smoothness criterion (17.15.5) for a morphism $\mathrm{X} \to \mathrm{Y}$ when Y is not the spectrum of a field:

Proposition (18.10.21). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$, and suppose that the ring $\mathscr{O}_{\mathrm{Y},y}$ is integral and geometrically unibranch. Let r be the minimum number of generators of the $\mathscr{O}_{\mathrm{X},x}$-module $(\Omega^1_{\mathrm{X}/\mathrm{Y}})_x$ (equal to the rank of the $k(x)$-vector space $(\Omega^1_{\mathrm{X}/\mathrm{Y}})_x \otimes_{\mathscr{O}_{\mathrm{X},x}} k(x)$). Then, for f to be smooth at the point x, it is necessary and sufficient that, if y' is the generic point of the unique irreducible component of Y containing y, there exists a generisation x' of x such that $f(x') = y'$ and $\dim_{x'} f^{-1}(y') \geqslant r$.*

Proof. If f is smooth at the point x, it is also so in an open neighbourhood U of x, hence in every generisation x' of x, and $\dim_{x'}(f^{-1}(f(x'))) = r$ in each of these points, by virtue of (17.10.2). As f is moreover flat in U, there exists a generisation of x above every generisation of y (2.3.4), which proves that the condition is necessary. To show that it is sufficient, let us first note that by (17.5.1), it suffices to prove that the morphism deduced from f by the base change $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}) \to \mathrm{Y}$ is smooth at the point x; one can therefore suppose that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$.

The question is local on X; hence, by (16.4.22) and (**0**, 5.2.2), one may suppose that X is affine and that there exist r sections s_i ($1 \leqslant i \leqslant r$) of $\mathscr{O}_{\mathrm{X}}$ over X such that the sections $d_{\mathrm{X}/\mathrm{Y}}(s_i)$ generate $\Omega^1_{\mathrm{X}/\mathrm{Y}}$. Let

$$g : \mathrm{X} \longrightarrow \mathrm{Z} = \mathrm{Y}[\mathrm{T}_1, \dots, \mathrm{T}_r] = \mathbf{V}_{\mathrm{Y}}$$

be the Y-morphism corresponding to the homomorphism $\mathscr{O}^r_{\mathrm{Y}} \to f_*(\mathscr{O}_{\mathrm{X}})$ of $\mathscr{O}_{\mathrm{Y}}$-modules defined by the s_i (considered as sections of $f_*(\mathscr{O}_{\mathrm{X}})$ over Y) (**II**, 1.2.7). We have the exact sequence (16.4.19)

$$g^*(\Omega^1_{\mathrm{Z}/\mathrm{Y}}) \longrightarrow \Omega^1_{\mathrm{X}/\mathrm{Y}} \longrightarrow \Omega^1_{\mathrm{X}/\mathrm{Z}} \longrightarrow 0.$$

As the $d\mathrm{T}_i$ ($1 \leqslant i \leqslant r$) generate $\Omega^1_{\mathrm{Z}/\mathrm{Y}}$, it follows from the definition of g that the homomorphism $g^*(\Omega^1_{\mathrm{Z}/\mathrm{Y}}) \to \Omega^1_{\mathrm{X}/\mathrm{Y}}$ is surjective, hence $\Omega^1_{\mathrm{X}/\mathrm{Z}} = 0$, and g is consequently *unramified* (17.4.1). We shall now see that g is *étale* at the point x, and taking into account (17.3.8), this will prove that f is smooth at the point x. As $\mathscr{O}_{\mathrm{Y},y}$ is assumed to be integral and geometrically unibranch, so is $\mathscr{O}_{\mathrm{Z},z}$ **IV** | 169
by (17.5.7), where $z = g(x)$, since the structural morphism $\mathrm{Z} \to \mathrm{Y}$ is smooth (17.3.8). By virtue of (18.10.1), it suffices to show that the homomorphism $\mathscr{O}_{\mathrm{Z},z} \to \mathscr{O}_{\mathrm{X},x}$ is injective; it amounts to the same thing to see that the morphism $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) \to \mathrm{Spec}(\mathscr{O}_{\mathrm{Z},z})$ corresponding is *dominant*, since $\mathscr{O}_{\mathrm{Z},z}$ is integral (**I**, 1.2.7). Let y' be the generic point of $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, and let z' be the generic point of the integral scheme $\mathrm{Z} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}[\mathrm{T}_1, \dots, \mathrm{T}_r])$; if $h : \mathrm{Z} \to \mathrm{Y}$ is the structural morphism, then $h(z') = y'$. It suffices to prove that there exists in $f^{-1}(y')$ a generisation x' of x such that the image of x' by the morphism $g_{y'} : f^{-1}(y') \to h^{-1}(y')$ equals z'. Moreover, z' is also the generic point of $h^{-1}(y') = \mathrm{Spec}(k(y')[\mathrm{T}_1, \dots, \mathrm{T}_r])$; since the morphism $g_{y'}$ is unramified, hence quasi-finite (17.4.2), it follows from (5.4.1) that the restriction of $g_{y'}$ to every irreducible component of *dimension* $\geqslant r$ of $f^{-1}(y')$ is dominant. Now, there exists by hypothesis a component containing a generisation x' of x; *a fortiori* its generic point is a generisation of x, whence the conclusion. ☐

18.11. Application to Noetherian local algebras complete over a field The following lemma generalizes (**0**, 21.9.1) and (**0**, 21.9.2):

Lemma (18.11.1). — *Let k be a field, let A be a complete Noetherian local ring that is a k-algebra, and let K be the residue field of A.*

(i) If the K-vector space $\Omega^1_{\mathrm{K}/k}$ is of finite rank, the A-module $\widehat{\Omega}^1_{\mathrm{A}/k}$ is of finite type.

(ii) If, moreover, A is formally smooth as a k-algebra (for the adic topology), $\widehat{\Omega}^1_{\mathrm{A}/k}$ is a free A-module of rank

$$\dim(\mathrm{A}) + \mathrm{rg}_{\mathrm{K}}(\Omega^1_{\mathrm{K}/k}) - \mathrm{rg}_{\mathrm{K}}(\Upsilon_{\mathrm{K}/k}).$$

Furthermore, for every subfield k_0 of k such that Ω^1_{k/k_0} is a finite-dimensional k-vector space, $\widehat{\Omega}^1_{\mathrm{A}/k_0}$ is a free A-module of rank

$$\dim(\mathrm{A}) + \mathrm{rg}_{\mathrm{K}}(\Omega^1_{\mathrm{K}/k_0}) - \mathrm{rg}_{\mathrm{K}}(\Upsilon_{\mathrm{K}/k_0}) + \mathrm{rg}_{\mathrm{K}}(\Omega^1_{k/k_0}).$$

Proof. Assertion (i) is recalled here, having been proved in (**0**, 20.7.15). To prove (ii), note that this assertion was in fact established by the proof of (**0**, 21.9.2); the statement of (**0**, 21.9.2) uses the finiteness of the transcendence degree of K over k only through Cartier's equality (**0**, 21.7.1). ☐

Lemma (18.11.2). — *Let k be a field of characteristic $p > 0$, and let C be an integral complete Noetherian local k-algebra whose residue field is a finite extension of k; let n be its dimension and L its field of fractions. Then $\Omega^1_{\mathrm{L}/k}$ and $\Upsilon_{\mathrm{L}/k}$ are L-vector spaces of finite rank, and one has*

(18.11.2.1) $$\mathrm{rg}_{\mathrm{L}}(\Omega^1_{\mathrm{L}/k}) - \mathrm{rg}_{\mathrm{L}}(\Upsilon_{\mathrm{L}/k}) \geqslant n.$$

Suppose furthermore that $[k : k^p] < +\infty$. Then one has the equality **IV** | 170

(18.11.2.2) $$\mathrm{rg}_{\mathrm{L}}(\Omega^1_{\mathrm{L}/k}) - \mathrm{rg}_{\mathrm{L}}(\Upsilon_{\mathrm{L}/k}) = n.$$

Furthermore, $\Omega^1_{\mathrm{C}/k}$ is then isomorphic to $\widehat{\Omega}^1_{\mathrm{C}/k}$, and hence, by (18.11.1), is a C-module of finite type.

Proof. Indeed, by (**0**, 19.8.9), there exists a k-subalgebra C_0 of C isomorphic to the formal power series algebra $k[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ and such that C is finite over C_0. Consequently, L is a *finite* extension of the field of fractions $\mathrm{L}_0 = k((\mathrm{T}_1, \dots, \mathrm{T}_n))$ of C_0. Applying (**0**, 20.6.17.1) to the prime subfield of k and to the three fields $k \subset \mathrm{L}_0 \subset \mathrm{L}$ (and taking (**0**, 20.6.21), (i), into account) gives the exact sequence of L-vector spaces

$$0 \longrightarrow \Upsilon_{\mathrm{L}_0/k} \otimes_{\mathrm{L}_0} \mathrm{L} \longrightarrow \Upsilon_{\mathrm{L}/k} \longrightarrow \Upsilon_{\mathrm{L}/\mathrm{L}_0} \longrightarrow \Omega^1_{\mathrm{L}_0/k} \otimes_{\mathrm{L}_0} \mathrm{L} \longrightarrow \Omega^1_{\mathrm{L}/k} \longrightarrow \Omega^1_{\mathrm{L}/\mathrm{L}_0} \longrightarrow 0.$$

Since L is a finite extension of L_0, the L-vector spaces $\Omega^1_{\mathrm{L}/\mathrm{L}_0}$ and $\Upsilon_{\mathrm{L}/\mathrm{L}_0}$ have finite and equal ranks, by Cartier's equality (**0**, 21.7.1). Since L_0 is separable over k (**0**, 21.9.6.4), one has $\Upsilon_{\mathrm{L}_0/k} = 0$ (**0**, 20.6.3). The preceding exact sequence therefore shows that $\Upsilon_{\mathrm{L}/k}$ has finite rank and, in every case,

$$\mathrm{rg}_{\mathrm{L}}(\Omega^1_{\mathrm{L}/k}) - \mathrm{rg}_{\mathrm{L}}(\Upsilon_{\mathrm{L}/k}) = \mathrm{rg}_{\mathrm{L}_0}(\Omega^1_{\mathrm{L}_0/k}) - \mathrm{rg}_{\mathrm{L}_0}(\Upsilon_{\mathrm{L}_0/k}).$$

It therefore suffices to prove (18.11.2.1) and (18.11.2.2) after replacing C and L by C_0 and L_0. Since L_0 is separable over k (**0**, 21.9.6.4), one has $\Upsilon_{\mathrm{L}_0/k} = 0$ (**0**, 20.6.3); on the other hand,

$$\Omega^1_{\mathrm{L}_0/k} = \Omega^1_{\mathrm{C}_0/k} \otimes_{\mathrm{C}_0} \mathrm{L}_0 \qquad (\mathbf{0}, 20.5.9).$$

We know from (**0**, 21.9.4) that, if $\mathrm{C}_1 = k[[\mathrm{T}_1^p, \dots, \mathrm{T}_n^p]]$, then $\widehat{\Omega}^1_{\mathrm{C}_0/k}$ identifies with $\Omega^1_{\mathrm{C}_0/\mathrm{C}_1}$. Moreover, $\Omega^1_{\mathrm{C}_0/k} = \Omega^1_{\mathrm{C}_0/k[\mathrm{C}_0^p]}$ (**0**, 21.1.5), and, since $\mathrm{C}_0^p = k^p[[\mathrm{T}_1^p, \dots, \mathrm{T}_n^p]]$, one has $k[\mathrm{C}_0^p] \subset \mathrm{C}_1$, with equality when $[k : k^p] < +\infty$. The exact sequence

$$\Omega^1_{\mathrm{C}_0/k[\mathrm{C}_0^p]} \longrightarrow \Omega^1_{\mathrm{C}_0/\mathrm{C}_1} \longrightarrow 0 \qquad (\mathbf{0}, 20.5.7)$$

therefore yields an exact sequence $\Omega^1_{\mathrm{L}_0/k} \to \widehat{\Omega}^1_{\mathrm{C}_0/k} \otimes_{\mathrm{C}_0} \mathrm{L}_0 \to 0$. Since $\widehat{\Omega}^1_{\mathrm{C}_0/k}$ is a free C_0-module of rank n (**0**, 21.9.3), it follows that $\mathrm{rg}_{\mathrm{L}_0}(\Omega^1_{\mathrm{L}_0/k}) \geqslant n$ in every case, with equality when $[k : k^p] < +\infty$. This proves (18.11.2.1) and (18.11.2.2).

Finally, when $[k : k^p] < +\infty$, to see that $\Omega^1_{\mathrm{C}/k} \to \widehat{\Omega}^1_{\mathrm{C}/k}$ is an isomorphism, it suffices to prove that $\Omega^1_{\mathrm{C}/k}$ is a finite-type C-module, since C is a complete Noetherian local ring (**0**, 7.3.3). Since $\Omega^1_{\mathrm{C}/k} \cong \Omega^1_{\mathrm{C}/k[\mathrm{C}^p]}$ and $k[\mathrm{C}_0^p] \subset k[\mathrm{C}^p]$, this reduces to proving that C is finite over $k[\mathrm{C}_0^p]$ (**0**, 20.4.7); this follows because C is finite over C_0, while C_0 is finite over $k[\mathrm{C}_0^p]$ under the hypothesis $[k : k^p] < +\infty$. □

Lemma (18.11.3). — *Let k be a field, let p be its characteristic exponent, and suppose that $[k : k^p] < +\infty$. Let A be a complete Noetherian local k-algebra whose residue field is a finite extension of k. Let $\mathfrak{p}$ be a prime ideal of A such that $\mathrm{A}_{\mathfrak{p}}$ is geometrically regular over k (6.7.6), so that there is a unique minimal prime ideal $\mathfrak{q}$ of A contained in $\mathfrak{p}$. Then $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is a free $\mathrm{A}_{\mathfrak{p}}$-module of rank* $\dim(\mathrm{A}/\mathfrak{q})$.

Proof. Since A is Noetherian and $\mathrm{Spec}(\mathrm{A})$ is regular (and hence, a fortiori, integral) at $\mathfrak{p}$, the point $\mathfrak{p}$ lies on only one irreducible component of $\mathrm{Spec}(\mathrm{A})$. Thus it contains only one minimal ideal $\mathfrak{q}$ of A, and moreover $\mathfrak{q}_{\mathfrak{p}} = 0$. Put $\mathrm{B} = \mathrm{A}/\mathfrak{q}$. The sequence (**0**, 20.7.20)

$$\mathfrak{q}/\mathfrak{q}^2 \longrightarrow \widehat{\Omega}^1_{\mathrm{A}/k} \widehat{\otimes}_{\mathrm{A}} \mathrm{B} \longrightarrow \widehat{\Omega}^1_{\mathrm{B}/k} \longrightarrow 0$$

is exact. Indeed, $\widehat{\Omega}^1_{\mathrm{A}/k}$ is a finite-type A-module by (18.11.1), and hence

$$\widehat{\Omega}^1_{\mathrm{A}/k} \widehat{\otimes}_{\mathrm{A}} \mathrm{B} = \widehat{\Omega}^1_{\mathrm{A}/k} \otimes_{\mathrm{A}} \mathrm{B} = \widehat{\Omega}^1_{\mathrm{A}/k} / \mathfrak{q} \widehat{\Omega}^1_{\mathrm{A}/k}.$$

As this B-module is of finite type, every one of its submodules is closed (**0**, 7.3.5), taking (**0**, 20.4.5) into account. The image of $\mathfrak{q}/\mathfrak{q}^2$ is dense in the kernel of the displayed homomorphism (**0**, 20.7.20), and is therefore equal to that kernel. Localizing at $\mathfrak{p}$ also shows that the canonical homomorphism $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \to (\widehat{\Omega}^1_{B/k})_{\mathfrak{p}/\mathfrak{q}}$ is bijective. We may consequently reduce to the case $\mathfrak{q} = 0$, that is, to the case where A is integral. We distinguish two cases.

I) $p > 1$. Apply (18.11.2) to the quotient algebra $C = A/\mathfrak{p}$, whose field of fractions K is the residue field of $A_{\mathfrak{p}}$. Thus

$$\text{(18.11.3.1)} \qquad \mathrm{rg}_K(\Omega^1_{K/k}) - \mathrm{rg}_K(\Upsilon_{K/k}) = \dim(A/\mathfrak{p}).$$

Now $A_{\mathfrak{p}}$ is formally smooth as a k-algebra for its $\mathfrak{p}A_{\mathfrak{p}}$-adic topology (**0**, 19.6.6); hence $\Omega^1_{A_{\mathfrak{p}}/k}$ is formally projective (**0**, 20.4.9) for that topology (**0**, 20.4.5). Moreover, $\Omega^1_{A_{\mathfrak{p}}/k} = (\Omega^1_{A/k})_{\mathfrak{p}}$ (**0**, 20.5.9) is of finite type by (18.11.2) applied to A. For every integer j, the module

$$\Omega^1_{A_{\mathfrak{p}}/k}/\mathfrak{p}^{j+1}\Omega^1_{A_{\mathfrak{p}}/k}$$

is therefore projective over $A_{\mathfrak{p}}/\mathfrak{p}^{j+1}A_{\mathfrak{p}}$, of rank $m = \mathrm{rg}_K(\Omega^1_{A_{\mathfrak{p}}/k} \otimes_{A_{\mathfrak{p}}} K)$ (**0**, 19.2.4). It follows from (**0**, 10.2.1) and (**0**, 10.1.3) that $\widehat{\Omega}^1_{A_{\mathfrak{p}}/k}$ is a free $A_{\mathfrak{p}}$-module of rank m. Let $A' = (A_{\mathfrak{p}})^{\wedge}$ be the completion of $A_{\mathfrak{p}}$ for its $\mathfrak{p}A_{\mathfrak{p}}$-adic topology. The algebra A' is again formally smooth over k for its adic topology (**0**, 19.3.6), and (**0**, 20.7.14) and (**0**, 20.4.5) give $\widehat{\Omega}^1_{A'/k} = \widehat{\Omega}^1_{A_{\mathfrak{p}}/k}$. Thus $\widehat{\Omega}^1_{A'/k}$ is a free A'-module of rank m. By (18.11.1) and $\dim(A') = \dim(A_{\mathfrak{p}})$ (**0**, 16.2.4),

$$\text{(18.11.3.2)} \qquad m = \dim(A_{\mathfrak{p}}) + \big(\mathrm{rg}_K(\Omega^1_{K/k}) - \mathrm{rg}_K(\Upsilon_{K/k})\big).$$

By (18.11.3.1), this is $m = \dim(A/\mathfrak{p}) + \dim(A_{\mathfrak{p}})$. Since the complete Noetherian local ring A is a quotient of a regular ring (**0**, 19.8.8), one has $\dim(A) = \dim(A/\mathfrak{p}) + \dim(A_{\mathfrak{p}})$ by (**0**, 16.5.12), which proves the assertion in this case.

II) $p = 1$. As above, $\dim(A) = \dim(A/\mathfrak{p}) + \dim(A_{\mathfrak{p}})$. Put $n = \dim(A)$, $r = \dim(A/\mathfrak{p})$, and $s = \dim(A_{\mathfrak{p}})$. We shall prove that $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$-module of rank n. We first prove the following lemma.

Lemma (18.11.3.3). — *Let* A *be a Noetherian local ring and* $\mathfrak{p}$ *a prime ideal such that* $A_{\mathfrak{p}}$ *is Cohen–Macaulay. For every system of parameters* $(t_i)_{1\leqslant i\leqslant s}$ *of* $A_{\mathfrak{p}}$*, there exists a sequence* $(x_i)_{1\leqslant i\leqslant s}$ *of elements of* $\mathfrak{p}$ *such that the* x_i *form part of a system of parameters of* A *and, for every* i*,* $x_i/1$ *is congruent to* t_i *modulo* **IV | 172**
$t_1A_{\mathfrak{p}} + \cdots + t_{i-1}A_{\mathfrak{p}}$*. In particular, if* $A_{\mathfrak{p}}$ *is regular and the* t_i *form a regular system of parameters of* $A_{\mathfrak{p}}$*, the same is true of the* $x_i/1$*.*

Proof of (18.11.3.3). This lemma is itself a consequence of the following one.

Lemma (18.11.3.4). — *Let* A *be a Noetherian local ring, let* $\mathfrak{p}$ *be a prime ideal of* A*, and let* $x \in A$*. Then there exists* $x' \in A$ *such that* $x'/1 = x/1$ *in* $A_{\mathfrak{p}}$ *and* x' *belongs to none of the minimal prime ideals of* A *not contained in* $\mathfrak{p}$*.*

We first show how (18.11.3.4) implies (18.11.3.3). Multiplying the t_i, if necessary, by units of $A_{\mathfrak{p}}$, we may assume that $t_i = x_i/1$ with $x_i \in \mathfrak{p}$ for $1 \leqslant i \leqslant s$. We argue by induction on s. Since t_1 forms part of a system of parameters of $A_{\mathfrak{p}}$, it belongs to no minimal prime ideal of $A_{\mathfrak{p}}$ (**0**, 16.5.5); hence no $x'_1 \in A$ with $x'_1/1 = t_1$ can belong to a minimal prime ideal of A contained in $\mathfrak{p}$. By (18.11.3.4), we may moreover choose x'_1 such that $x'_1/1 = t_1$ (so $x'_1 \in \mathfrak{p}$) and x'_1 belongs to no minimal prime ideal of A. Thus x'_1 forms part of a system of parameters of A (**0**, 16.3.4) and (16.3.7). We then apply induction to $A' = A/x'_1A$ and $\mathfrak{p}' = \mathfrak{p}/x'_1A$. Since $A'_{\mathfrak{p}'} = A_{\mathfrak{p}}/t_1A_{\mathfrak{p}}$, this ring is Cohen–Macaulay (**0**, 16.5.5), and the images t'_i ($2 \leqslant i \leqslant s$) form a system of parameters of it. The induction hypothesis applied to A' and the t'_i completes the construction. The last assertion follows because, in $A_{\mathfrak{p}}$, a system of parameters that generates the maximal ideal is a regular system of parameters (**0**, 17.1.1).

It remains to prove (18.11.3.4). Let $(\mathfrak{p}'_k)_{1\leqslant k\leqslant r}$ be the minimal prime ideals of A not contained in $\mathfrak{p}$ and containing x, and let $(\mathfrak{p}_j)_{1\leqslant j\leqslant n}$ be the other minimal prime ideals of A; we may assume $r \geqslant 1$. No $\mathfrak{p}'_k$ contains $\bigcap_{1\leqslant j\leqslant n}\mathfrak{p}_j$, so there exists $y \in \bigcap_{1\leqslant j\leqslant n}\mathfrak{p}_j$ belonging to none of the $\mathfrak{p}'_k$ (Bourbaki, *Alg. comm.*, chap. II, § 1, no. 1, prop. 2). Moreover, $y/1$ belongs to every minimal prime ideal of $A_{\mathfrak{p}}$, and is therefore nilpotent. If $(y/1)^h = 0$, the element $x' = x + y^h$ has the required properties: $x'/1 = x/1$; since $y^h \notin \mathfrak{p}'_k$ and $x \in \mathfrak{p}'_k$, one has $x' \notin \mathfrak{p}'_k$ for $1 \leqslant k \leqslant r$; and if $\mathfrak{p}_j$ is a minimal prime ideal not contained in $\mathfrak{p}$ with $x \notin \mathfrak{p}_j$, then $x' \notin \mathfrak{p}_j$ because $y \in \mathfrak{p}_j$. □

We return to the proof of (18.11.3) when $p = 1$. By (18.11.3.3), there exists a regular system of parameters $(t_i)_{1\leqslant i\leqslant s}$ of $A_{\mathfrak{p}}$ such that $t_i = x_i/1$, where the x_i ($1 \leqslant i \leqslant s$) belong to $\mathfrak{p}$ and form part of a system of parameters $(x_j)_{1\leqslant j\leqslant n}$ of A. Put $A_0 = k[[T_1, \ldots, T_n]]$. Since A is a complete k-algebra and the x_j belong to its maximal ideal $\mathfrak{m}$, there is a local k-homomorphism $u : A_0 \to A$ such that $u(T_j) = x_j$ for $1 \leqslant j \leqslant n$ (Bourbaki, *Alg. comm.*, chap. III, § 4, no. 5, prop. 6). If $\mathfrak{n}$ is the ideal generated by the x_j, it is an ideal of definition of A; hence (**0**, 7.4.4) and (7.4.3) show that u makes A finite over A_0.

Put $\mathfrak{p}_0 = \sum_{j=1}^{s} A_0T_j$, $B_0 = (A_0)_{\mathfrak{p}_0}$, and $B = A \otimes_{A_0} B_0$, so that $u_{\mathfrak{p}_0} : B_0 \to B$ makes B finite over B_0. IV | 173
Moreover, if $\mathfrak{p}'$ is the ideal of B generated by $\mathfrak{p}$, then $B_{\mathfrak{p}'} = A_{\mathfrak{p}}$; since $\mathfrak{p}'$ contains $\mathfrak{p}_0B$ by construction, it lies over the maximal ideal $\mathfrak{p}_0B_0$ of B_0. The morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(B_0)$ is unramified at $\mathfrak{p}'$: indeed, $k(\mathfrak{p}')$ is a finite extension of the characteristic-zero field $k(\mathfrak{p}_0)$, hence is necessarily separable, and $B_{\mathfrak{p}'}/\mathfrak{p}_0B_{\mathfrak{p}'} = k(\mathfrak{p}')$ by the choice of the x_j for $1 \leqslant j \leqslant s$ (17.4.1).

We now record the following lemma.

Lemma (18.11.3.5). — *Let k be a field, and let* R *and* S *be complete Noetherian local k-algebras such that, if* K *is the residue field of* S*, $\Omega^1_{K/k}$ has finite rank over* K*. Let $u : R \to S$ be a k-homomorphism making* S *finite over* R*. Then $\widehat{\Omega}^1_{S/R} = \Omega^1_{S/R}$, and the sequence*

$$\widehat{\Omega}^1_{R/k} \otimes_R S \xrightarrow{v} \widehat{\Omega}^1_{S/k} \xrightarrow{w} \Omega^1_{S/R} \longrightarrow 0 \tag{18.11.3.6}$$

(cf. (**0**, 20.7.17.3)*) is exact.*

Proof. The first assertion follows because $\Omega^1_{S/R}$ is a finite-type S-module (**0**, 20.4.7) and by (**0**, 7.3.6). Moreover, (**0**, 20.7.17.3) says that the image of v is dense in $\ker(w)$ and that w is surjective. By (18.11.1), however, $\widehat{\Omega}^1_{S/k}$ is a finite-type S-module, and therefore every one of its submodules is closed (**0**, 7.3.5). The lemma follows. □

Apply the lemma with $R = A_0$ and $S = A$. The module $\widehat{\Omega}^1_{A_0/k}$ is free of rank n over A_0 (**0**, 21.9.3); hence $\widehat{\Omega}^1_{A_0/k}\widehat{\otimes}_{A_0}A = \widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A$, and this is a free A-module of rank n. Since $\operatorname{Spec}(B)$ is unramified over $\operatorname{Spec}(B_0)$ at $\mathfrak{p}'$, one has $(\Omega^1_{A/A_0})_{\mathfrak{p}} = \Omega^1_{B_{\mathfrak{p}'}/B_0} = 0$ (16.4.15) and (17.4.1). Localizing (18.11.3.6), applied to A_0 and A, at $\mathfrak{p}$ therefore gives a surjective homomorphism

$$(\widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A)_{\mathfrak{p}} \longrightarrow (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}.$$

Thus $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ has a system of n generators. On the other hand, $\widehat{\Omega}^1_{A/k}$ has rank n by (**0**, 21.9.5), which applies to the complete integral ring A by Cohen's theorem (**0**, 19.8.8), (ii), and because the field of fractions of A has characteristic zero. Hence $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ also has rank n; its quotient by its torsion submodule, being generated by n elements, is necessarily free. It follows at once that the n generators obtained above form a free system, which proves the assertion. □

Lemma (18.11.4). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k. Let k' be an arbitrary extension of k, and set $A' = A\widehat{\otimes}_k k'$. Then:*

(i) A' is a semi-local Noetherian ring, complete, composed as a direct product of complete local rings A'_i ($1 \leqslant i \leqslant r$) which are A*-modules faithfully flat, and whose residue fields are finite extensions of k'; if $\mathfrak{m}$ is the maximal ideal of* A*, $\mathfrak{m}A'$ is an ideal of definition of A'; and $\dim(A'_i) = \dim(A)$ for all i.*

(ii) For every i, $\widehat{\Omega}^1_{A'_i/k'}$ is canonically isomorphic to $\widehat{\Omega}^1_{A/k} \otimes_A A'_i$.

Proof. The first assertions follow immediately from (7.5.5) and (**0**, 6.6.2); the fact that $\mathfrak{m}A'$ is an ideal of definition of A' also follows from (7.5.5), since K is the residue field of A, $K \otimes_k k'$ is a finite k'-algebra. Finally, $A'_i/\mathfrak{m}A'_i$ is one of the Artinian local rings occurring as direct factors of $K \otimes_k k'$ (7.5.5), and is therefore zero-dimensional. Since A'_i is flat over A, the equality $\dim(A'_i) = \dim(A)$ follows from (6.1.2). IV | 174

(ii) By the hypothesis on K, $\widehat{\Omega}^1_{A/k}$ is an A-module of finite type (18.11.1); hence $\widehat{\Omega}^1_{A/k} \otimes_A A'$ is complete and identifies with the completed tensor product $\widehat{\Omega}^1_{A/k}\widehat{\otimes}_A A'$. By associativity of completed tensor products (**0**, 7.7.4),

$$\widehat{\Omega}^1_{A/k} \otimes_A A' = \widehat{\Omega}^1_{A/k} \otimes_A (A\widehat{\otimes}_k k') \simeq \widehat{\Omega}^1_{A/k}\widehat{\otimes}_k k'.$$

Now $\Omega^1_{A/k} \otimes_k k'$ identifies with $\Omega^1_{A''/k'}$, where $A'' = A \otimes_k k'$ (**0**, 20.5.5); and since A' is by definition the separated completion of A'', the separated completion of $\Omega^1_{A''/k'}$ identifies by construction with that of $\Omega^1_{A'/k'}$ (**0**, 20.7.4). In other words, there is a canonical isomorphism

$$\widehat{\Omega}^1_{A'/k'} \xrightarrow{\sim} \widehat{\Omega}^1_{A/k} \otimes_A A'.$$

The conclusion of (ii) now follows because $\Omega^1_{A'/k'}$ is the direct sum of the $\Omega^1_{A'_i/k'}$ (**0**, 20.4.13). Indeed, if $\mathfrak{r}$ is the radical of A', its $\mathfrak{r}$-adic topology on $\Omega^1_{A'/k'}$ is the product of the $\mathfrak{m}'_i$-adic topologies on the $\Omega^1_{A'_i/k'}$, where $\mathfrak{m}'_i$ is the maximal ideal of A'_i. Thus the separated completion $\widehat{\Omega}^1_{A'/k'}$ is the product of the separated completions $\widehat{\Omega}^1_{A'_i/k'}$, and it remains only to apply (**0**, 20.4.5). □

Proposition (18.11.5). — *Let k be a field, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of A distinct from the maximal ideal $\mathfrak{m}$, such that there exists a unique minimal prime ideal $\mathfrak{q} \subset \mathfrak{p}$ for which $\dim(A/\mathfrak{q}) = \dim(A)$ (which will in particular hold when A is equidimensional), m an integer $\geqslant 0$. The following conditions are equivalent:*

- *a) The $A_\mathfrak{p}$-module $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$ admits a system of m generators.*
- *b) There exists a local k-homomorphism $u : B \to A$, where $B = k[[T_1, \dots, T_m]]$, making A a finite B-algebra, and such that the corresponding morphism $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ is unramified at $\mathfrak{p}$.*

Proof. To prove that *b*) implies *a*), note that by Lemma (18.11.3.5) one has the exact sequence

$$\widehat{\Omega}^1_{B/k} \otimes_B A \longrightarrow \widehat{\Omega}^1_{A/k} \longrightarrow \Omega^1_{A/B} \longrightarrow 0$$

since A is a B-algebra finite; localising at $\mathfrak{p}$ and noting that by hypothesis $(\Omega^1_{A/B})_\mathfrak{p} = 0$ (17.4.1), one obtains a surjective homomorphism $(\widehat{\Omega}^1_{B/k} \otimes_B A)_\mathfrak{p} \to (\widehat{\Omega}^1_{A/k})_\mathfrak{p}$, and the conclusion follows because **IV** | 175
$\widehat{\Omega}^1_{B/k}$ is a free B-module of rank m (**0**, 21.9.3).

To prove that *a*) implies *b*), let us first prove the following lemmas. □

Lemma (18.11.5.1). — *Let k be a field, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of A, $\mathfrak{q}$ a minimal prime ideal of A contained in $\mathfrak{p}$. Then the minimum number of generators of the $A_\mathfrak{p}$-module $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$ is at least equal to $\dim(A/\mathfrak{q})$.*

Proof. Set $n = \dim(A/\mathfrak{q})$, and let m be the minimum number of generators of the $A_\mathfrak{p}$-module $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$, which is equal to $\mathrm{rg}_{k(\mathfrak{p})}((\widehat{\Omega}^1_{A/k})_\mathfrak{p} \otimes_{A_\mathfrak{p}} k(\mathfrak{p}))$ (Bourbaki, *Alg. comm.*, chap. II, § 3, nº 2, cor. 2 of prop. 4). Note that $(\widehat{\Omega}^1_{A/k})_\mathfrak{q} = (\widehat{\Omega}^1_{A/k})_\mathfrak{p} \otimes_{A_\mathfrak{p}} A_\mathfrak{q}$, hence the minimum number of generators of the $A_\mathfrak{q}$-module $(\widehat{\Omega}^1_{A/k})_\mathfrak{q}$ is *at most* equal to m. It suffices therefore to consider the case where $\mathfrak{p} = \mathfrak{q}$ is minimal. In second place, let us show one can restrict to the case where k is *algebraically closed*. Indeed, let k' be an algebraic closure of k, and set $A' = A \widehat{\otimes}_k k'$, which is composed as a direct product of k'-local algebras A'_i $(1 \leqslant i \leqslant r)$, whose residue fields are isomorphic to k', and which are A-modules faithfully flat (18.11.4). Applying (2.3.4) and (6.1.1), one sees that in a given A'_i, there exists a minimal prime ideal $\mathfrak{q}'_i$ above $\mathfrak{q}$ such that $n' = \dim(A'_i/\mathfrak{q}'_i) \geqslant \dim(A/\mathfrak{q}) = n$. On the other hand by (18.11.4), the minimum m' of generators of the $(A'_i)_{\mathfrak{q}'_i}$-module $(\widehat{\Omega}^1_{A'_i/k'})_{\mathfrak{q}'_i}$ is *at most* equal to m; whence our assertion.

Suppose therefore that k is algebraically closed, and let us show that we may further reduce to the case $\mathfrak{q} = 0$. Indeed, $\Omega^1_{(A/\mathfrak{q})/A} = 0$ (**0**, 20.4.12), so the exact sequence (18.11.3.6) gives a surjection

$$\widehat{\Omega}^1_{A/k} \otimes_A (A/\mathfrak{q}) \longrightarrow \widehat{\Omega}^1_{(A/\mathfrak{q})/k}.$$

It follows at once that the minimum number of generators of the K-vector space $\widehat{\Omega}^1_{(A/\mathfrak{q})/k} \otimes_{A/\mathfrak{q}} K$, where K is the fraction field of $A/\mathfrak{q}$, is at most m.

We may thus assume that A is integral. Its fraction field K is separable over the algebraically closed field k, and hence geometrically regular over k (6.7.6). We may therefore apply (18.11.3) with $\mathfrak{p} = 0$; since $k = k^p$, this gives $m = n$. □

Lemma (18.11.5.2). — *Let k be a field, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{m}$ its maximal ideal, $\mathfrak{p} \subset \mathfrak{m}$ a prime ideal of A. Let m be the minimum number* *of generators of the* $\mathrm{A}_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$*, and let* $(x_i)_{1\leqslant i\leqslant m}$ *be a family of elements of* $\mathfrak{m}$ *not belonging to* $\mathfrak{p}$*. Then there exist* m *invertible elements* u_i $(1 \leqslant i \leqslant m)$ *of* A *such that if* $y_i = u_i x_i$*, the canonical images of the* dy_i *in* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ *generate this* $\mathrm{A}_{\mathfrak{p}}$*-module.*

Proof. For all $x \in \mathrm{A}$, denote by $\delta(x)$ the canonical image of dx $(= d_{\mathrm{A}/k}x)$ in $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} k(\mathfrak{p}) = \widehat{\Omega}^1_{\mathrm{A}/k} \otimes_{\mathrm{A}} k(\mathfrak{p})$; as this is a $k(\mathfrak{p})$-vector space of dimension m by hypothesis, it suffices (by virtue of the Nakayama lemma) to prove that one can determine the u_i such that the $\delta(u_i x_i)$ form a free system. We reason by induction and suppose for an integer $r < m$, we have determined the u_i $(1 \leqslant i \leqslant r)$ such that the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$ are linearly independent over $k(\mathfrak{p})$; if $\delta(x_{r+1})$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$, one takes $u_{r+1} = 1$ to continue the induction. In the contrary case, for $u \in \mathrm{A}$ the element $\delta(u x_{r+1})$ is the canonical image of $(du)x_{r+1} + u(dx_{r+1})$. The second term has image in the span of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$, while the image of x_{r+1} in $k(\mathfrak{p})$ is nonzero. It is therefore enough to find a unit $u \in \mathrm{A}$ such that $\delta(u)$ does not belong to that span. Now (**0**, 20.7.15) and (**0**, 7.2.9) show that the $\delta(x)$, for $x \in \mathrm{A}$, generate the $k(\mathfrak{p})$-vector space **IV** | 176 $\widehat{\Omega}^1_{\mathrm{A}/k} \otimes_{\mathrm{A}} k(\mathfrak{p})$. Since $r < m$, there is therefore a $z \in \mathrm{A}$ for which $\delta(z)$ is not in the span of the $\delta(u_i x_i)$. If $z \notin \mathfrak{m}$, take $u = z$; otherwise $1 + z$ is invertible and $\delta(1+z) = \delta(z)$ because $d(1+z) = dz$, so take $u = 1 + z$. This completes the proof of (18.11.5.2). □

Proof of (18.11.5)*, continued.* We now return to the proof of the implication $a) \Rightarrow b)$ in (18.11.5). Let us first note that since $\mathfrak{p} \neq \mathfrak{m}$, there exists a system of parameters $(x_i)_{1\leqslant i\leqslant n}$ of A (with $n = \dim(\mathrm{A})$) such that $x_i \notin \mathfrak{p}$ for all i, without which $\mathrm{A}/\mathfrak{p}$ would be of finite length, and $\mathfrak{p}$ would be maximal, contrary to the hypothesis. But if for example $x_1 \notin \mathfrak{p}$, it suffices, for each index i such that $x_i \in \mathfrak{p}$, to replace x_i by $x_1 + x_i$, to have a system of parameters none of whose elements belong to $\mathfrak{p}$. The relation $\dim(\mathrm{A}/\mathfrak{q}) = n$ implies, by virtue of (18.11.5.1), that $m \geqslant n$; one can therefore consider a family $(x_i)_{1\leqslant i\leqslant m}$ of elements $x_i \notin \mathfrak{p}$, whose n first form a system of parameters of A. Multiplying furthermore the x_i by invertible elements u_i of A, one can, by (18.11.5.2), suppose that the images of dx_i $(1 \leqslant i \leqslant m)$ in $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ generate this $\mathrm{A}_{\mathfrak{p}}$-module, and the multiplication by the u_i has not altered the fact that the x_i for $1 \leqslant i \leqslant n$ form a system of parameters. We shall consider the local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$ such that $u(\mathrm{T}_i) = x_i$ for $1 \leqslant i \leqslant m$ (Bourbaki, *Alg. comm.*, chap. III, § 4, n° 5, prop. 6); as the x_i generate an ideal of definition of A, it follows from (**0**, 7.4.4) and (7.4.3) that u makes A a B-algebra *finite*.

But the $d_{\mathrm{A}} x_i$ are the canonical images by v of the elements $d_{\mathrm{B}}\mathrm{T}_i \otimes 1$ (**0**, 20.5.2.6). Localising the preceding exact sequence at $\mathfrak{p}$, we see that

$$v_{\mathfrak{p}} : (\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}} \longrightarrow (\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$$

is surjective. Consequently

$$0 = (\Omega^1_{\mathrm{A}/\mathrm{B}})_{\mathfrak{p}} = \Omega^1_{\mathrm{A}_{\mathfrak{p}}/\mathrm{B}_{\mathfrak{r}}},$$

where $\mathfrak{r} = u^{-1}(\mathfrak{p})$ is the inverse image of $\mathfrak{p}$ in B; cf. (16.4.15). By (17.4.1), the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is therefore unramified at $\mathfrak{p}$, which proves $b)$. □

Remark (18.11.6). — In the statement of (18.11.5), the hypothesis $\mathfrak{p} \neq \mathfrak{m}$ cannot be omitted. Indeed, for every local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$, where $\mathrm{B} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_r]]$ (r any integer) making A a B-algebra finite, $\mathfrak{m}$ is the only point of $\mathrm{Spec}(\mathrm{A})$ above the maximal ideal $\mathfrak{n}$ of B; the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ cannot be unramified at $\mathfrak{m}$ unless $\mathrm{A}/\mathfrak{n}\mathrm{A}$ is a field, an extension separable of k (17.4.1), which implies that the residue field K of A is a separable extension of k. If this condition is not satisfied, the conclusion of (18.11.5) can never be satisfied for $\mathfrak{p} = \mathfrak{m}$, whatever the integer m.

Corollary (18.11.7). — *Let* k *be a field,* A *a* k*-algebra, local Noetherian and complete, equidimensional, whose residue field is a finite extension of* k*,* $\mathfrak{p}$ *a prime ideal of* A*. Then the minimum number of generators of the* $\mathrm{A}_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ *is at least equal to* $\dim(\mathrm{A})$*.*

Proof. This is a special case of (18.11.5.1). □

Corollary (18.11.8). — *Let* k *be a field,* A *a* k*-algebra, local Noetherian and complete, integral, whose residue field is a finite extension of* k*. If* K *is the field of fractions of* A*, one has* $\mathrm{rg}_{\mathrm{K}}((\widehat{\Omega}^1_{\mathrm{A}/k}) \otimes_{\mathrm{A}} \mathrm{K}) \geqslant \dim(\mathrm{A})$*.*

Proof. It suffices to set $\mathfrak{p} = 0$ in (18.11.7). □

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Corollary (18.11.9). — *Let k be a field, A a k-algebra, local Noetherian and complete, of dimension n, whose residue field is a finite extension of k. Let $\mathfrak{p}$ be a prime ideal of A distinct from the maximal ideal, and containing a minimal prime ideal $\mathfrak{q}$ such that $\dim(\mathrm{A}/\mathfrak{q}) = n$. Suppose that the $\mathrm{A}_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ admits n generators. Then:*

(i) *The $\mathrm{A}_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is free of rank n.*
(ii) *There exists a local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$, where $\mathrm{B} = k[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$, making A a B-algebra finite, and such that the corresponding morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is étale at the point $\mathfrak{p}$.*
(iii) *The k-algebra $\mathrm{A}_{\mathfrak{p}}$ is geometrically regular over k.*

Proof. We first prove (ii). By (18.11.5), there is a local k-homomorphism

$$u : \mathrm{B} = k[[\mathrm{T}_1, \dots, \mathrm{T}_n]] \longrightarrow \mathrm{A}$$

making A a finite B-algebra and such that $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is unramified at $\mathfrak{p}$; put $\mathfrak{r} = u^{-1}(\mathfrak{p})$. The hypothesis on $\mathfrak{q} \subset \mathfrak{p}$, together with the fact that A is a quotient of a regular ring (**0**, 19.8.8), gives

$$\dim(\mathrm{A}_{\mathfrak{p}}) = n - \dim(\mathrm{A}/\mathfrak{p})$$

by (**0**, 16.5.12); likewise $\dim(\mathrm{B}_{\mathfrak{r}}) = n - \dim(\mathrm{B}/\mathfrak{r})$. Since the morphism is unramified at $\mathfrak{p}$, its fibre over $\mathfrak{r}$ has dimension zero, so (**0**, 16.3.9) gives $\dim(\mathrm{A}/\mathfrak{p}) \leqslant \dim(\mathrm{B}/\mathfrak{r})$ and hence $\dim(\mathrm{B}_{\mathfrak{r}}) \leqslant \dim(\mathrm{A}_{\mathfrak{p}})$. It follows from (18.10.1) that the morphism is étale at $\mathfrak{p}$.

Assertion (iii) follows because $\mathrm{A}_{\mathfrak{p}}$ is formally smooth over $\mathrm{B}_{\mathfrak{r}}$, while $\mathrm{B}_{\mathfrak{r}}$ is formally smooth over k, for their preadic topologies (**0**, 19.3.4) and (**0**, 19.3.5). Thus $\mathrm{A}_{\mathfrak{p}}$ is formally smooth over k for its preadic topology, and consequently geometrically regular over k (**0**, 22.5.8).

Finally, we prove (i). Let k' be an algebraically closed extension of k, and consider the semilocal k'-algebra $\mathrm{A}' = \mathrm{A}\widehat{\otimes}_k k'$. Regarding A as a finite B-module via u, there is a canonical isomorphism

$$\mathrm{A}' \simeq \mathrm{A} \otimes_{\mathrm{B}} (\mathrm{B}\widehat{\otimes}_k k')$$

by (7.5.7.1); that result does not require the residue field of B to be finite over k. Put $\mathrm{B}' = \mathrm{B}\widehat{\otimes}_k k'$, which identifies canonically with $k'[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$.

Since $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is finite and étale at $\mathfrak{p}$, the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{B}')$ is finite and étale at every point $\mathfrak{p}'$ over $\mathfrak{p}$ (17.3.3). Moreover, A' is the direct product of local rings A'_i of dimension n (18.11.4), and $\mathfrak{p}'$ identifies with a prime ideal $\mathfrak{p}'_i$ of one of them. The preceding argument shows that $(\mathrm{A}'_i)_{\mathfrak{p}'_i}$ is geometrically regular over k'. Since k' is perfect, (18.11.3) shows that $(\widehat{\Omega}^1_{\mathrm{A}'_i/k'})_{\mathfrak{p}'_i}$ is a free $(\mathrm{A}'_i)_{\mathfrak{p}'_i}$-module of finite type.

But (18.11.4) gives

$$(\widehat{\Omega}^1_{\mathrm{A}'_i/k'})_{\mathfrak{p}'_i} \simeq (\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} (\mathrm{A}'_i)_{\mathfrak{p}'_i},$$

and $(\mathrm{A}'_i)_{\mathfrak{p}'_i}$ is faithfully flat over $\mathrm{A}_{\mathfrak{p}}$. Therefore $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is a free $\mathrm{A}_{\mathfrak{p}}$-module of finite type (2.5.2); and (18.11.5.1), together with $\dim(\mathrm{A}/\mathfrak{q}) = n$, shows that its rank is n. □

Theorem (18.11.10). — *Let k be a field of characteristic exponent p, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of A distinct from the maximal ideal. The following conditions are equivalent:* **IV | 178**

a) *For every extension k' of k, and every prime ideal $\mathfrak{p}'$ of $\mathrm{A}' = \mathrm{A}\widehat{\otimes}_k k'$ above $\mathfrak{p}$, $\mathrm{A}'_{\mathfrak{p}'}$ is a regular ring.*
a′) *There exists a perfect extension k' of k and a prime ideal $\mathfrak{p}'$ of $\mathrm{A}' = \mathrm{A}\widehat{\otimes}_k k'$ above $\mathfrak{p}$ such that $\mathrm{A}'_{\mathfrak{p}'}$ is regular.*
b) *Let n be the largest of the dimensions of the irreducible components of $\mathrm{Spec}(\mathrm{A})$ to which $\mathfrak{p}$ belongs; then $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is an $\mathrm{A}_{\mathfrak{p}}$-module free of rank n.*

Suppose furthermore that $n = \dim(\mathrm{A})$ (which will be the case if A is equidimensional); then the preceding conditions are also equivalent to:

c) *There exists a local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$, where $\mathrm{B} = k[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$, making A a B-algebra finite, and such that the corresponding morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is étale at the point $\mathfrak{p}$.*

Each of the conditions a), a′), and b) implies the following:

d) The ring $A_{\mathfrak{p}}$ is geometrically regular over k.

Furthermore, if $[k : k^p] < +\infty$, condition d) is equivalent to a), a′), and b).

Proof. That $d)$ implies $b)$ when $[k : k^p] < +\infty$ is precisely (18.11.3). It is clear that $a)$ implies $a')$. Let us show that $c)$ implies $a)$ when $\dim(A) = n$. With the notation of $a)$, one then has

$$A' = A \otimes_B B', \qquad B' = B \widehat{\otimes}_k k' = k'[[T_1, \dots, T_n]]$$

by (7.5.7.1). The morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(B')$ is finite and étale at $\mathfrak{p}'$, and the argument proving (18.11.9), (iii), shows that $A'_{\mathfrak{p}'}$ is regular.

That $b)$ implies $c)$ when $\dim(A) = n$ follows from (18.11.9). Let us prove that $a')$ implies $b)$ under the same hypothesis. The ring A' is flat over A (18.11.4), and $\dim(A') = \dim(A) = n$. By (2.3.4) and (6.1.1), there is a minimal prime $\mathfrak{q}'$ of A', contained in $\mathfrak{p}'$ and lying over a minimal prime $\mathfrak{q}$ of A, such that

$$\dim(A'/\mathfrak{q}') = \dim(A/\mathfrak{q}) = n.$$

Moreover, (18.11.4) gives

$$(\widehat{\Omega}^1_{A'/k'})_{\mathfrak{p}'} \simeq (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A'_{\mathfrak{p}'}.$$

Since k' is perfect, the regularity of $A'_{\mathfrak{p}'}$ implies that it is geometrically regular over k' (6.7.7). As $k'^p = k'$, we may apply the implication $d) \Rightarrow b)$ to A' and $\mathfrak{p}'$. Thus $(\widehat{\Omega}^1_{A'/k'})_{\mathfrak{p}'}$ is a free $A'_{\mathfrak{p}'}$-module of rank n; faithful flatness (18.11.4) and (2.5.2) then shows that $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$-module of rank n.

This proves the equivalence of $a)$, $a')$, $b)$, and $c)$ when $\dim(A) = n$. It remains to prove that $a)$, $a')$, and $b)$ are still equivalent in the general case. Let $\mathfrak{J}$ be the kernel of the canonical homomorphism $A \to A_{\mathfrak{p}}$, and put $A_1 = A/\mathfrak{J}$. Necessarily $\mathfrak{J} \subset \mathfrak{p}$; and if $\mathfrak{p}_1 = \mathfrak{p}/\mathfrak{J}$, the canonical homomorphism $A_{\mathfrak{p}} \to (A_1)_{\mathfrak{p}_1}$ is bijective. It follows from (**I**, 6.5.4) that the canonical injection $\mathrm{Spec}(A_1) \to \mathrm{Spec}(A)$ is a local isomorphism at $\mathfrak{p}_1$, and $\mathfrak{J}_{\mathfrak{p}} = 0$. As in (18.11.3), there is an exact sequence

$$\mathfrak{J}/\mathfrak{J}^2 \longrightarrow \widehat{\Omega}^1_{A/k} \otimes_A A_1 \longrightarrow \widehat{\Omega}^1_{A_1/k} \longrightarrow 0.$$

After localising at $\mathfrak{p}$, this yields an isomorphism **IV** | 179

$$(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \xrightarrow{\sim} (\widehat{\Omega}^1_{A_1/k})_{\mathfrak{p}_1}.$$

Thus condition $b)$ for A and $\mathfrak{p}$ is equivalent to condition $b)$ for A_1 and $\mathfrak{p}_1$.

On the other hand, with the notation of $a)$, there is a canonical isomorphism

$$A'_1 = A_1 \widehat{\otimes}_k k' \simeq A'/\mathfrak{J}A'$$

by (7.5.7.1), where the hypothesis on the residue field of B is superfluous. If $\mathfrak{p}'$ is a prime of A' above $\mathfrak{p}$, every element of $\mathfrak{J}A'$ annihilates an element of $A' - \mathfrak{p}'$, and hence $\mathfrak{J}A' \subset \mathfrak{p}'$. Putting $\mathfrak{p}'_1 = \mathfrak{p}'/\mathfrak{J}A'$, the prime $\mathfrak{p}'_1$ lies above $\mathfrak{p}_1$ and $(A'_1)_{\mathfrak{p}'_1}$ identifies canonically with $A'_{\mathfrak{p}'}$. This proves that condition $a)$ (respectively $a')$) for A and $\mathfrak{p}$ is equivalent to the same condition for A_1 and $\mathfrak{p}_1$.

All minimal prime ideals of A contained in $\mathfrak{p}$ contain $\mathfrak{J}$, since $\mathrm{Spec}(A_1) \to \mathrm{Spec}(A)$ is a local isomorphism at $\mathfrak{p}_1$. Moreover, the primes in $\mathrm{Ass}_A(A/\mathfrak{J})$ are precisely those in $\mathrm{Ass}(A)$ which are contained in $\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 2, prop. 6). Hence all minimal primes of A_1 are contained in $\mathfrak{p}_1$, and consequently $\dim(A_1) = n$. It remains only to apply to A_1 and $\mathfrak{p}_1$ what was proved above. □

Remark (18.11.11). — (i) The equivalence of conditions $d)$ and $b)$ in (18.11.10) is no longer valid when one no longer supposes that $[k : k^p] < +\infty$. Indeed, by the example of (**0**, 22.7.7), (ii)), the ring $B = A/\mathfrak{q}$ is integral and of dimension 1; on the other hand, the sequence

$$(\mathfrak{q}/\mathfrak{q}^2) \otimes_A L \xrightarrow{j} \widehat{\Omega}^1_{A/k} \otimes_A L \longrightarrow \widehat{\Omega}^1_{B/k} \otimes_B L \longrightarrow 0$$

is exact (same demonstration as in (18.11.3)), and we know that *j is not injective*, hence $j = 0$ since $(\mathfrak{q}/\mathfrak{q}^2) \otimes_A L$ is of rank 1. We conclude that $\mathrm{rg}_L(\widehat{\Omega}^1_{B/k} \otimes_B L) = 2$; as $B_{\mathfrak{p}}$ is a k-algebra geometrically regular, one sees that the condition $d)$ does not imply here the condition $b)$.

(ii) With the notations of (18.11.10) and supposing that $n = \dim(A)$ and $(x_i)_{1\leqslant i\leqslant n}$ a system of parameters of A not belonging to $\mathfrak{p}$; one deduces a local k-homomorphism $u : B \to A$ such that $u(T_i) = x_i$ for $1 \leqslant i \leqslant n$, making A a B-algebra finite. For the morphism $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ to be *étale* at the point $\mathfrak{p}$, it is necessary and sufficient that the images of $d_A x_i$ in $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ form a *system of generators* of this $A_{\mathfrak{p}}$-module. Indeed, as one has seen in the proof of (18.11.5) that if the morphism $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ is not ramified at $\mathfrak{p}$, the canonical homomorphism $(\widehat{\Omega}^1_{B/k} \otimes_B A)_{\mathfrak{p}} \to (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is surjective and the images of the elements $d_B T_i \otimes 1$, which generate $(\widehat{\Omega}^1_{B/k} \otimes_B A)_{\mathfrak{p}}$, are the $d_A x_i$; whence the *necessity* of the condition. Conversely, the same reasoning shows that if this condition is satisfied, the morphism $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ is not ramified at $\mathfrak{p}$, and the reasoning of (18.11.9) proves in fact that this morphism is étale at $\mathfrak{p}$.

Corollary (18.11.12). — *Let k be a field of characteristic exponent p such that $[k : k^p] < +\infty$, A a k-algebra, local Noetherian and complete and integral, which is not a field, and whose residue field is a finite extension of k. Then there exists a finite radicial extension k' of k such that, setting $A' = A \otimes_k k'$, the k'-algebra A'_{red} and the prime ideal $\mathfrak{o}$ of this algebra satisfy the equivalent conditions a), a′), b), c), and d) of (18.11.10). In particular, if $n = \dim(A) = \dim(A')$, there exists a k'-homomorphism local $B' = k'[[T_1, \dots, T_n]] \to A'_{\mathrm{red}}$* **IV** | 180 *making A'_{red} a B'-algebra finite, and such that the field of fractions K' of A'_{red} is a finite separable extension of the field of fractions $k'((T_1, \dots, T_n))$ of B'.*

Proof. Since $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is radicial and finite, A' is a complete local ring and its nilradical is the unique prime above the zero ideal of A. By flatness, A' identifies with a subring of $K \otimes_k k'$, which is also the total ring of fractions of A'; hence

$$K' = (K \otimes_k k')_{\mathrm{red}}.$$

In view of the equivalence $d) \Leftrightarrow c)$ in (18.11.10), it is therefore enough to find a finite radicial extension k' of k such that $(K \otimes_k k')_{\mathrm{red}}$ is separable over k' (6.7.6).

By (**0**, 19.8.9), under the present hypotheses there is a sub-k-algebra $C = k[[T_1, \dots, T_m]]$ of A such that A is finite over C. Thus K is a finite extension of the fraction field $K_1 = k((T_1, \dots, T_m))$ of C, and K_1 is separable over k (**0**, 21.9.6.4). We may write

$$K \otimes_k k^{p^{-\infty}} = K \otimes_{K_1} (K_1 \otimes_k k^{p^{-\infty}}),$$

and $K_1 \otimes_k k^{p^{-\infty}}$ is a field, radicial over K_1. Consequently $K \otimes_k k^{p^{-\infty}}$ is finite over a field and is therefore Artinian. The conclusion follows from the more general lemma below. ☐

Lemma (18.11.12.1). — *Let k be a field of characteristic $p > 0$, K an extension of k. The following conditions are equivalent:*

a) The ring $K \otimes_k k^{p^{-\infty}}$ is Artinian.
b) There exists an extension k' of k of finite type such that $(K \otimes_k k')_{\mathrm{red}}$ is a k'-algebra separable.
c) There exists a finite radicial extension k' of k such that $(K \otimes_k k')_{\mathrm{red}}$ is a separable extension of k'.

Proof. Let us first show that $c)$ implies $a)$. The ring $A = K \otimes_k k'$ is then local Artinian. If $\mathfrak{N}$ is its nilradical, its residue field $L = A/\mathfrak{N}$ is separable over k', and hence $L \otimes_{k'} k^{p^{-\infty}}$ is a field. Since this field is

$$(A \otimes_{k'} k^{p^{-\infty}})/(\mathfrak{N} \otimes_{k'} k^{p^{-\infty}}),$$

the ideal $\mathfrak{N} \otimes_{k'} k^{p^{-\infty}}$ is the nilradical of $K \otimes_k k^{p^{-\infty}}$. The ideal $\mathfrak{N}$ is finitely generated and nilpotent, so this nilradical is finitely generated; it follows that $K \otimes_k k^{p^{-\infty}}$ is Artinian.

Conversely, let us prove that $a)$ implies $c)$. Let $\mathfrak{N}$ be the nilradical of the local Artinian ring $B = K \otimes_k k^{p^{-\infty}}$. It is generated by finitely many elements

$$y_i = \sum_j x_{ij} \otimes \xi_{ij}, \qquad x_{ij} \in K, \quad \xi_{ij} \in k^{p^{-\infty}}.$$

Let k' be the finite radicial extension of k generated by the ξ_{ij}, and let $\mathfrak{N}_0$ be the ideal of $B_0 = K \otimes_k k'$ generated by the y_i. The y_i are nilpotent in B_0; moreover

$$\mathfrak{N}_0 \otimes_{k'} k^{p^{-\infty}} = \mathfrak{N}, \qquad \mathfrak{N} \cap B_0 = \mathfrak{N}_0.$$

Thus $\mathfrak{N}_0$ is exactly the nilradical of B_0. Since

$$B/\mathfrak{N} = (B_0/\mathfrak{N}_0) \otimes_{k'} k^{p^{-\infty}}$$

is reduced, (4.6.1) shows that $B_0/\mathfrak{N}_0 = (K \otimes_k k')_{\mathrm{red}}$ is separable over k'.

It is clear that *c*) implies *b*). Conversely, suppose *b*) holds. There is a separable extension k_1 of k such that k' is finite radicial over k_1; put $K_1 = K \otimes_k k_1$, which is a field. Applying the equivalence of *a*) and *c*) to the extension K_1/k_1, we see that $K_1 \otimes_{k_1} k_1^{p^{-\infty}}$ is Artinian. But **IV** | 181

$$K \otimes_k k_1^{p^{-\infty}} = (K \otimes_k k^{p^{-\infty}}) \otimes_{k^{p^{-\infty}}} k_1^{p^{-\infty}},$$

so $K \otimes_k k^{p^{-\infty}}$ is Artinian as well (Bourbaki, *Alg. comm.*, chap. I, § 3, n° 5, cor. of prop. 8). Thus *b*) implies *a*), completing the proofs of (18.11.12.1) and (18.11.12). □

18.12. Applications of étale localisation to quasi-finite morphisms (generalisations of earlier results) The results of this section were communicated by P. Deligne.

Theorem (18.12.1). — *Let $f : X \to Y$ be a morphism locally of finite type, let x be a point of X, and put $y = f(x)$. Suppose that x is an isolated point of the space $f^{-1}(y)$. Then there exist an étale morphism $Y' \to Y$, a point x' of $X' = X \times_Y Y'$ above x, and an open neighbourhood V' of x' in X' such that, if $f' = f_{(Y')} : X' \to Y'$, the restriction $f'|V'$ is finite. If, moreover, f is separated, then V' is both open and closed, and X' is therefore the disjoint union of two subschemes induced on open subsets of X', one of which is finite over Y' and contains x'.*

Proof. The last assertion follows because, if f is separated, so is f'. Thus, if $j' : V' \to X'$ denotes the canonical injection, the finiteness of $f'|V' = f' \circ j'$ implies that j' is finite (**II**, 6.1.5); consequently $V' = j'(V')$ is closed in X' (**II**, 6.1.10).

The question is local on X, so we may suppose that $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, with B an A-algebra of finite type. Write $B = C/\mathfrak{J}$, where $C = A[T_1, \dots, T_r]$ and $\mathfrak{J}$ is an ideal of C. Since f is then separated, we may also suppose that $f^{-1}(y) = \{x\}$.

Let $(\mathfrak{J}_\lambda)$ be the filtered family of finitely generated ideals of C contained in $\mathfrak{J}$; then $\mathfrak{J}$ is their union. Regard X and the schemes $X_\lambda = \mathrm{Spec}(C/\mathfrak{J}_\lambda)$ as closed subschemes of $Z = \mathrm{Spec}(C)$. On the underlying spaces, $X = \bigcap_\lambda X_\lambda$. If $f_\lambda : X_\lambda \to Y$ is the structural morphism, then

$$f^{-1}(y) = \bigcap_\lambda f_\lambda^{-1}(y).$$

The sets $f_\lambda^{-1}(y)$ are closed in the Noetherian space $Z_y = \mathrm{Spec}(k(y)[T_1, \dots, T_r])$; hence, for some λ, $f_\lambda^{-1}(y) = f^{-1}(y) = \{x\}$.

We may replace X by such an X_λ. Indeed, if the assertion is proved for X_λ, let x' lie above x in $Z' = Z \times_Y Y'$ and put $X'_\lambda = p^{-1}(X_\lambda)$, where $p : Z' \to Z$ is the projection (so also $X' = p^{-1}(X)$). If V'_λ is an open neighbourhood of x' in X'_λ finite over Y', then the restriction to the closed subscheme $X' \cap V'_\lambda$ is finite as well. We have therefore reduced to the case where f is of finite presentation.

The set of points of X which are isolated in their fibres is open in X (13.1.4); after replacing X by an open neighbourhood of x, we may therefore suppose that f is quasi-finite. Let A'' be the strict henselisation of $\mathscr{O}_{Y,y}$ and put $Y'' = \mathrm{Spec}(A'')$. If $X'' = X \times_Y Y''$ and $f'' = f_{(Y'')} : X'' \to Y''$, then f'' is separated, quasi-finite, and of finite presentation. Since A'' is henselian, every point $x'' \in X''$ above x has an open-and-closed neighbourhood V'' in X'' which is finite over Y'' (18.5.11), c). The immersion $V'' \to X''$ is open and closed, hence quasi-compact and of finite presentation (1.6.2); consequently $f''|V''$ is of finite presentation.

By definition, A'' is the filtered direct limit of strictly essentially étale $\mathscr{O}_{Y,y}$-algebras A_λ (18.6.5). Each A_λ is itself the filtered direct limit of étale $\mathscr{O}_{Y,y}$-algebras $B_{\lambda\mu}$ (18.6.1), (8.1.2), a). Since the $B_{\lambda\mu}$ may be assumed finitely presented over $\mathscr{O}_{Y,y}$, it follows again from (8.1.2), a) and (17.7.8) that they are filtered direct limits of étale A-algebras. The double direct-limit theorem therefore shows that Y'' is the projective limit of a filtered family (Y'_α) of affine schemes étale over Y.

Let x'_α be the projection of x'' in $X'_\alpha = X \times_Y Y'_\alpha$. Since V'' is quasi-compact, after increasing α we may write

$$V'' = V'_\alpha \times_{Y'_\alpha} Y'',$$

where V'_α is an open neighbourhood of x'_α in X'_α (8.2.11). Finally, since V'' is of finite presentation over Y'', (8.10.5), (x) shows that, for a suitable α, V'_α is finite over Y'_α, as required. ☐

Remark (18.12.2). — The preceding proof shows (taking (18.6.2) into account) that Y' may moreover be chosen so that, if $y' = f'(x')$, the homomorphism $k(y) \to k(y')$ is bijective.

Corollary (18.12.3). — *Let $f : X \to Y$ be a separated morphism locally of finite type, and let y be a point of Y such that the space $f^{-1}(y)$ is finite and discrete. Then there exist an étale morphism $Y' \to Y$, a point $y' \in Y'$ above y such that $k(y') = k(y)$, and a decomposition of $X' = X \times_Y Y'$ as a disjoint union of two subschemes X'_1 and X'_2 induced on open subsets of X', such that the restriction of $f' = f_{(Y')} : X' \to Y'$ to X'_1 is finite and $X'_2 \cap f'^{-1}(y') = \varnothing$.*

Proof. Let n be the number of points of $f^{-1}(y)$. We argue by induction on n, the assertion being trivial for $n = 0$. Choose $x \in f^{-1}(y)$. By (18.12.1) and (18.12.2), there are an étale morphism $Y_1 \to Y$, a point $y_1 \in Y_1$ above y with $k(y_1) = k(y)$, and, on putting $S_1 = X \times_Y Y_1$ and $f_1 = f_{(Y_1)}$, a decomposition

$$S_1 = V_1 \amalg X_1$$

into open subschemes, where V_1 is finite over Y_1 and contains a point x_1 above x.

Because $k(y_1) = k(y)$, the fibre $f_1^{-1}(y_1)$ is isomorphic to $f^{-1}(y)$; hence $X_1 \cap f_1^{-1}(y_1)$ is finite, discrete, and has $n-1$ points. Apply the induction hypothesis to $f_1|X_1$. There are an étale morphism $Y' \to Y_1$, a point $y' \in Y'$ above y_1 with $k(y') = k(y_1)$, and, if $p : X' \to S_1$ is the canonical projection, a decomposition **IV | 182**

$$p^{-1}(X_1) = U' \amalg X'_2$$

such that U' is finite over Y' and $X'_2 \cap f'^{-1}(y') = \varnothing$. The open subscheme $V' = p^{-1}(V_1)$ is finite over Y', and

$$X' = U' \amalg V' \amalg X'_2.$$

Taking $X'_1 = U' \amalg V'$ proves the assertion. ☐

Corollary (18.12.4). — *For a morphism $f : X \to Y$ to be finite, it is necessary and sufficient that it be separated, universally closed, locally of finite type, and that $f^{-1}(y)$ be a finite discrete space for every $y \in Y$. In particular, every proper quasi-finite morphism is finite.*

Proof. The necessity is clear. Conversely, apply (18.12.3) at an arbitrary point $y \in Y$. With the notation of that corollary, f' is closed. Since X'_2 is closed in X', the set $f'(X'_2)$ is closed in Y' and does not contain y'. There is therefore an open neighbourhood U' of y' in Y' such that $U' \to Y$ is of finite type and

$$f'^{-1}(U') = X \times_Y U'$$

is finite over U'.

Let U be the image of U' in Y. It is open by (**II**, 3.1), since $Y' \to Y$ is étale, hence flat and locally of finite presentation. Moreover

$$X \times_Y U' = f^{-1}(U) \times_U U'.$$

The morphism $U' \to U$ is faithfully flat and quasi-compact; hence (2.7.1), (xv) implies that $f^{-1}(U) \to U$ is finite. Since y was arbitrary, f is finite. ☐

Remark (18.12.5). — Observe that the proof of (18.12.4) does not use universal closedness in its full form, but only the fact that $f_{(Y')}$ is closed for every étale morphism $Y' \to Y$.

Corollary (18.12.6). — *Let $f : X \to Y$ be a morphism of preschemes. The following conditions are equivalent:*

- *a) f is a closed immersion.*
- *b) f is a proper monomorphism.*
- *c) f is proper and, for all $y \in Y$, the $k(y)$-prescheme $X_y = f^{-1}(y)$ is radicial and geometrically reduced on $k(y)$ (i.e. empty or $k(y)$-isomorphic to $\mathrm{Spec}(k(y))$).*

Proof. This follows from (18.12.4) in the same way that (8.11.5) follows from (8.11.1). ☐

Proposition (18.12.7). — *Let $f : X \to Y$ be a morphism locally of finite type, and let y be a point of* Y. *There exists an open neighbourhood* U *of y such that the restriction $f^{-1}(U) \to U$ is a closed immersion if and only if X_y is a radicial and geometrically reduced $k(y)$-scheme (that is, empty or $k(y)$-isomorphic to* $\mathrm{Spec}(k(y))$*), and there exists an open neighbourhood* V *of y such that $f^{-1}(V) \to V$ is universally closed.*

The following lemma will be used in the proof.

Lemma (18.12.7.1). — *Let $f : X \to Y$ be a closed morphism, and let $y \in Y$ be a point such that every neighbourhood of X_y contains an affine neighbourhood of X_y (a condition which is always satisfied when X_y is empty or consists of a single point). Then there exists an open neighbourhood* U *of y such that $f^{-1}(U) \to U$ is affine.*

Proof. Let U_0 be an affine open neighbourhood of y in Y, and let V be an affine open neighbourhood of X_y contained in $f^{-1}(U_0)$. Since f is closed, there is an affine open neighbourhood $U \subset U_0$ of y such that $f^{-1}(U) \subset V$. The restriction $g : V \to U_0$ of f is affine, and hence so is its restriction $f^{-1}(U) \to U$ (**II**, 1.2.5). □

Proof of Proposition 18.12.7. The conditions are plainly necessary. To prove their sufficiency, we may suppose first that $X_y \neq \varnothing$, and, by the second condition, that f is universally closed. By Lemma 18.12.7.1, after shrinking Y about y, we may also suppose that f is affine, and hence separated.

We may now suppose that X and Y are affine. Since f is of finite type, it is proper. It remains, by (18.12.4) and (18.12.6), to make f quasi-finite over a neighbourhood of y. There is an open neighbourhood V of the one-point set X_y such that $f|V$ is quasi-finite (13.1.4). Since f is closed, there is an open neighbourhood U of y such that $f^{-1}(U) \subset V$. Thus $f^{-1}(U) \to U$ is proper and quasi-finite, hence finite by (18.12.4); its fibres satisfy condition c) of (18.12.6), so it is a closed immersion. The case $X_y = \varnothing$ is immediate by the same closedness argument. □

Proposition (18.12.8). — *Let $f : X \to Y$ be a morphism of preschemes. For f to be integral, it is necessary and sufficient that f be affine and universally closed.*

Proof. The necessity follows from (**II**, 6.1.10). Conversely, suppose that f is affine and universally closed. We may take $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$; it is enough to prove that every $b \in B$ is integral over A (**II**, 6.1.1).

Let $B' = A[b] \subset B$ and set $X' = \mathrm{Spec}(B')$. Then f factors as

$$X \xrightarrow{g} X' \xrightarrow{h} Y,$$

where h is of finite type and g is dominant because $B' \to B$ is injective (**I**, 1.2.7). Since h is separated and $h \circ g$ is universally closed, g is universally closed (**II**, 5.4.3), (**II**, 5.4.9); being dominant, it is therefore surjective. It follows from the same results that h is universally closed as well.

Thus h is proper, since it is of finite type and separated. For every $y \in Y$, the fibre $h^{-1}(y)$ is proper and affine over $k(y)$, hence finite (**III**, 4.4.2). Thus h is quasi-finite, and (18.12.4) shows that h is finite. Consequently B' is a finite A-algebra and b is integral over A. □ **IV** | 183

Remark (18.12.9). — It may be that a morphism $f : X \to Y$ is integral whenever it is separated and universally closed and, for every $y \in Y$, the induced morphism $f^{-1}(y) \to \mathrm{Spec}(k(y))$ is integral. To establish this it would be enough to prove that these conditions imply that f is affine, or equivalently that every fibre X_y is contained in an affine open neighbourhood.

Corollary (18.12.10). — *Every injective universally closed morphism $f : X \to Y$ is integral.*

Proof. It suffices, by (18.12.8), to prove that f is affine, which follows from the lemma (18.12.7.1) and the hypothesis. □

Corollary (18.12.11). — *Let $f : X \to Y$ be a morphism of preschemes (resp. a locally of finite type morphism). For f to be a universal homeomorphism (2.4.2), it is necessary and sufficient that f be integral (resp. finite), radicial, and surjective.*

Proof. One knows that the conditions are sufficient (2.4.5); they are necessary by virtue of (18.12.10) and (18.12.4). The last proposition improves (8.11.2). □

Proposition (18.12.12). — *Every morphism $f : X \to Y$ which is quasi-finite and separated is quasi-affine.*

Proof. Put $\mathscr{A} = f_*(\mathscr{O}_X)$. This is a quasi-coherent $\mathscr{O}_Y$-algebra because f is quasi-compact and separated (1.7.4). Let $Z = \operatorname{Spec}(\mathscr{A})$ (**II**, 1.3.1); then f has the canonical factorisation

$$X \xrightarrow{g} Z \xrightarrow{h} Y,$$

where h is affine and g corresponds to the identity automorphism of $\mathscr{A}$ (**II**, 1.2.7). It is enough to prove that g is an open immersion, or equivalently (17.9.1) that g is étale and radicial. It is plainly enough to establish this at every point of $f^{-1}(y)$, for each $y \in Y$.

Fix $y \in Y$ and apply (18.12.3) to f, retaining its notation. The morphism $Y' \to Y$ is flat, and f is quasi-compact and separated; hence there is a canonical isomorphism

$$f'_*(\mathscr{O}_{X'}) = \mathscr{A} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} =: \mathscr{A}'$$

by (2.3.1). Thus $Z' = Z \times_Y Y'$ identifies with $\operatorname{Spec}(f'_*(\mathscr{O}_{X'}))$, and the canonical factorisation

$$X' \xrightarrow{g'} Z' \xrightarrow{h'} Y'$$

is obtained from the preceding one by base change (**II**, 1.2.7).

The decomposition $X' = X'_1 \amalg X'_2$ yields a direct-product decomposition $\mathscr{A}' = \mathscr{A}'_1 \times \mathscr{A}'_2$ and hence $Z' = Z'_1 \amalg Z'_2$, where $Z'_i = \operatorname{Spec}(\mathscr{A}'_i)$ and $g'(X'_i) \subset Z'_i$. Since X'_1 is finite over Y', it is affine over Y', and therefore $g'|X'_1 : X'_1 \to Z'_1$ is an isomorphism. Moreover $X'_2 \cap f'^{-1}(y') = \varnothing$. It follows that g' is étale and radicial at every point of $f'^{-1}(y')$.

Since $Y' \to Y$ is flat and locally of finite presentation, (17.7.4) shows that g is étale at the projections of the points of $f'^{-1}(y')$, that is, at all points of $f^{-1}(y)$ (**I**, 3.5.2). On the other hand, the induced map

$$g'_{y'} : f'^{-1}(y') \longrightarrow h'^{-1}(y')$$

is radicial; since $k(y') = k(y)$, it is the scalar extension of $g_y : f^{-1}(y) \to h^{-1}(y)$. Hence g_y is radicial as well. Thus g is étale and radicial at every point of X, which proves the proposition. □

Corollary (18.12.13). — *("Main Theorem" of Zariski). Let Y be a quasi-compact and quasi-separated scheme, and let $f : X \to Y$ be a quasi-finite separated morphism. Then f admits a factorisation*

$$X \xrightarrow{g} Z \xrightarrow{u} Y$$

where g is an open immersion (necessarily quasi-compact) and u is finite.

Proof. Indeed, it follows from (18.12.12) that f is quasi-affine, hence quasi-projective. One can therefore apply (8.12.8), where the hypothesis on $\mathscr{O}_Y$-Module ample can in fact be replaced by the sole hypothesis that Y is quasi-compact and quasi-separated: indeed, it follows from (8.12.3) that the existence of the factorisation announced is a *local* property on Y, and it suffices to prove it when Y is affine. □

Remark (18.12.14). — There is also a proof of (18.12.13) analogous to that of (18.12.12). It does not use (8.12.8), but it does use (18.12.3), and hence (18.5.11), which itself uses the "Main Theorem" in its local form (8.12.9).

Retain the notation of the proof of (18.12.12), and let $\mathscr{B}$ be the quasi-coherent $\mathscr{O}_Y$-algebra which is the integral closure of $\mathscr{O}_Y$ in $\mathscr{A}$ (**II**, 6.3.2). Put $T = \operatorname{Spec}(\mathscr{B})$. By (8.12.3), it is enough to prove that the Y-morphism $g : X \to T$ corresponding to the canonical injection $\mathscr{B} \to \mathscr{A}$ is an immersion; for this it is enough, by (17.9.1), to show that g is étale and radicial.

With the notation of (18.12.12), we may suppose that $Y = \operatorname{Spec}(C)$ and $Y' = \operatorname{Spec}(C')$ are affine, with C' an étale C-algebra (17.3.2). Write $\mathscr{A} = \widetilde{A}$, where A is a C-algebra, put $A' = A \otimes_C C'$, and write $\mathscr{B} = \widetilde{B}$, where B is the integral closure of C in A. Recall that $A' \simeq A'_1 \times A'_2$, with A'_1 a finite C'-algebra. It is enough to prove that

$$B' = B \otimes_C C',$$

which identifies by flatness with a C'-subalgebra of A', contains A'_1. Indeed, B' would then decompose as $A'_1 \times A''_2$, where A''_2 is a C'-subalgebra of A'_2. Thus $T' = T \times_Y Y'$ would be the disjoint union of $Z'_1 = \operatorname{Spec}(A'_1)$ and $T''_2 = \operatorname{Spec}(A''_2)$, and $g'|X'_1$ would be an isomorphism from X'_1 onto Z'_1. One would then conclude as in (18.12.12). **IV** | 184

To prove that B' contains A'_1, it plainly suffices to prove the following proposition, which partially extends (6.14.4) when the Noetherian hypotheses are removed.

Proposition (18.12.15). — *Let* C *be a ring, let* C' *be an étale* C*-algebra, let* A *be a* C*-algebra, and let* B *be the integral closure of* C *in* A*. Put* $\mathrm{A}' = \mathrm{A} \otimes_{\mathrm{C}} \mathrm{C}'$ *and* $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{C}} \mathrm{C}'$*, where* B' *is identified with a subalgebra of* A'*. Then* B' *is the integral closure of* C' *in* A'*.*

Proof. Consider the filtered increasing family of finite-type C-subalgebras of A and argue as in (6.14.4), II. We may first suppose that A is a finite-type C-algebra, say $\mathrm{A} = \mathrm{E}/\mathfrak{J}$, where $\mathrm{E} = \mathrm{C}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$. Then $\mathrm{A}' = \mathrm{E}'/\mathfrak{J}\mathrm{E}'$, where $\mathrm{E}' = \mathrm{C}'[\mathrm{T}_1, \ldots, \mathrm{T}_r]$.

Let $(\mathfrak{J}_\lambda)$ be the filtered increasing family of finitely generated ideals of E contained in $\mathfrak{J}$, and let B_λ be the integral closure of C in $\mathrm{E}/\mathfrak{J}_\lambda$. The reasoning of (5.13.4) shows that $\mathrm{B} = \varinjlim_\lambda \mathrm{B}_\lambda$. Likewise, the integral closure of C' in $\mathrm{E}'/\mathfrak{J}\mathrm{E}'$ is the direct limit of the integral closures of C' in $\mathrm{E}'/\mathfrak{J}_\lambda\mathrm{E}'$. If the latter closure equals $\mathrm{B}_\lambda \otimes_{\mathrm{C}} \mathrm{C}'$ for every λ, then

$$\mathrm{B}' = \varinjlim_{\lambda} (\mathrm{B}_\lambda \otimes_{\mathrm{C}} \mathrm{C}')$$

is the integral closure of C' in A'. We are therefore reduced to the case where A is finitely presented over C.

We next reduce to the case where C is Noetherian. Write C as the filtered union of its finite-type $\mathbb{Z}$-subalgebras C_α. By (17.7.8), for some α there is an étale C_α-algebra C'_α such that

$$\mathrm{C}' = \mathrm{C}'_\alpha \otimes_{\mathrm{C}_\alpha} \mathrm{C}.$$

For $\beta \geqslant \alpha$, put $\mathrm{C}'_\beta = \mathrm{C}'_\alpha \otimes_{\mathrm{C}_\alpha} \mathrm{C}_\beta$; then $\mathrm{C}' = \varinjlim_{\beta \geqslant \alpha} \mathrm{C}'_\beta$. Since A is finitely presented over C, after increasing α we may write

$$\mathrm{A} = \mathrm{A}_\alpha \otimes_{\mathrm{C}_\alpha} \mathrm{C},$$

with A_α a finite-type C_α-algebra. Put $\mathrm{A}_\beta = \mathrm{A}_\alpha \otimes_{\mathrm{C}_\alpha} \mathrm{C}_\beta$; then $\mathrm{A} = \varinjlim_{\beta \geqslant \alpha} \mathrm{A}_\beta$ (8.9.1). If B_β is the integral closure of C_β in A_β, the reasoning of (5.13.4) gives $\mathrm{B} = \varinjlim \mathrm{B}_\beta$. Similarly,

$$\mathrm{A}' = \varinjlim_{\beta \geqslant \alpha} \mathrm{A}'_\beta, \qquad \mathrm{A}'_\beta = \mathrm{A}_\beta \otimes_{\mathrm{C}_\beta} \mathrm{C}'_\beta = \mathrm{A}_\alpha \otimes_{\mathrm{C}_\alpha} \mathrm{C}'_\beta.$$

The same argument shows that it is enough to prove that $\mathrm{B}'_\alpha = \mathrm{B}_\alpha \otimes_{\mathrm{C}_\alpha} \mathrm{C}'_\alpha$ is the integral closure of C'_α in A'_α. Thus we may suppose that C is Noetherian.

In that case the proposition is a special case of (6.14.4). One should note, however, that when C' is étale over the Noetherian ring C, the proof of (6.14.4) no longer requires the difficult theorem (6.14.1). Indeed, the successive reductions in that proof lead, without using (6.14.1), to the case in which C is integral and A is its field of fractions. Then B is normal, and since $\mathrm{Spec}(\mathrm{B}') \to \mathrm{Spec}(\mathrm{B})$ is étale, B' is normal by (17.5.7). The argument then uses only the elementary lemma (6.14.1.1), not the difficult part of (6.14.1). ☐

Proposition (18.12.16). — *Let* $g : \mathrm{Y} \to \mathrm{S}$ *be a quasi-compact morphism, and let* $f : \mathrm{X} \to \mathrm{Y}$ *be separated and quasi-finite. For every invertible* $\mathscr{O}_{\mathrm{Y}}$*-module* $\mathscr{L}$ *which is ample relative to* S (**II**, 4.6.1)*, the invertible* $\mathscr{O}_{\mathrm{X}}$*-module* $f^*(\mathscr{L})$ *is ample relative to* S*.*

Proof. By (18.12.12), the morphism f is quasi-affine; hence $\mathscr{O}_{\mathrm{X}}$ is ample for f by (**II**, 5.1.6). It follows from (**II**, 4.6.13), (ii) that

$$\mathscr{O}_{\mathrm{X}} \otimes_{\mathscr{O}_{\mathrm{X}}} f^*(\mathscr{L}^{\otimes n}) = f^*(\mathscr{L}^{\otimes n})$$

is ample relative to S for all sufficiently large n. Since

$$f^*(\mathscr{L}^{\otimes n}) = (f^*\mathscr{L})^{\otimes n},$$

this implies that $f^*(\mathscr{L})$ is ample relative to S (**II**, 4.6.9), (i). ☐

Corollary (18.12.17). — *Let* Z *be an affine scheme, let* $h : \mathrm{X} \to \mathrm{Z}$ *be a morphism of finite type, and let* $\mathscr{L}$ *be an invertible* $\mathscr{O}_{\mathrm{X}}$*-module. With the notation of* (**II**, 4.5.2)*, the conditions in* (**II**, 4.5.2) *(equivalent to the ampleness of* $\mathscr{L}$*) are also equivalent to each of the following:*

b″) *h is separated,* $\mathrm{G}(\varepsilon) = \mathrm{X}$*, and the canonical morphism* $u : \mathrm{G}(\varepsilon) \to \mathrm{Proj}(\mathrm{S})$ *has finite discrete fibres.*

b‴) $\mathrm{G}(\varepsilon) = \mathrm{X}$ *and the canonical morphism u is radicial.*

If $\mathrm{Z} = \mathrm{Spec}(\mathrm{A})$, *then* S *is canonically a graded* A*-algebra; hence there is a structural morphism* $g : \mathrm{Proj}(\mathrm{S}) \to \mathrm{Z}$, *which is separated* (**II**, 2.4.2), *and* $h = g \circ u$. *Since every radicial morphism is separated* (**I**, 1.8.7.1), *condition* $b''')$ *implies condition* $b'')$. *Condition b) of* (**II**, 4.5.2) *plainly implies* $b'')$, *so it remains to prove that* $b'')$ *implies that* $\mathscr{L}$ *is ample.*

Since $h = g \circ u$ *is of finite type and* g *is separated,* u *is of finite type* (**I**, 6.3.4); *condition* $b'')$ *therefore implies that* u *is quasi-finite and separated* (**I**, 5.5.1). *Put* $\mathrm{Y} = \mathrm{Proj}(\mathrm{S})$. *As* X *is quasi-compact, there exist an integer* $n > 0$ *and finitely many elements* $f_j \in \mathrm{S}_n$ *such that the inverse images* $u^{-1}(\mathrm{D}_+(f_j))$ *cover* X. *We may thus regard* u *as a quasi-finite separated morphism from* X *to the open subscheme*

$$\mathrm{Y}' = \bigcup_j \mathrm{D}_+(f_j)$$

of Y.

The invertible $\mathscr{O}_{\mathrm{Y}'}$*-module*

$$\mathscr{L}' = \mathscr{O}_{\mathrm{Y}}(n)|\mathrm{Y}'$$

is very ample relative to Z (**II**, 4.4.3), *and by definition*

$$u^*(\mathscr{L}') = \mathscr{L}^{\otimes n}$$

(**II**, 3.7.9), *(ii). By* (18.12.16), $\mathscr{L}^{\otimes n}$ *is therefore ample relative to* Z *(equivalently, ample* (**II**, 4.6.6)*), and consequently* $\mathscr{L}$ *itself is ample* (**II**, 4.5.6), *(i).*

We leave it to the reader to state, in the same way, conditions equivalent to those of (**II**, 4.6.3) *for relatively ample* $\mathscr{O}_{\mathrm{X}}$*-modules.* **IV** | 185

§19. Regular immersions and normal flatness

This section is devoted, on the one hand, to a study of regular immersions (16.9.2) $Y \to X$ of preschemes, especially when X and Y are flat over the same prescheme S; on the other hand, it gives supplements concerning M-regular sequences and normal flatness, notably generalizing flatness results established by H. Hironaka in his theory of resolution of singularities [35].

19.1. Properties of regular immersions **IV-4** | 185

Proposition (19.1.1). — *Let X be a locally Noetherian prescheme, $j : Y \to X$ an immersion, and y a regular point of Y. For j to be a regular immersion at the point y, it is necessary and sufficient that X be regular at the point $j(y)$.*

Proof. Taking (16.9.10) into account, we are reduced to proving the analogous result for local rings, which follows from (**0**, 17.3.3). ☐

Corollary (19.1.2). — *Let A be a Noetherian local ring and $\mathfrak{J}$ an ideal of A; suppose that the quotient ring $B = A/\mathfrak{J}$ is regular. For A to be regular, it is necessary and sufficient that the ideal $\mathfrak{J}$ be regular.*

Proof. Indeed, if A is regular, then $\mathfrak{J}$ is generated by part of a regular system of parameters of A (**0**, 17.1.9), and hence $\mathfrak{J}$ is regular. Conversely, if $\mathfrak{J}$ is regular and $(x_i)_{1\leqslant i\leqslant r}$ is an A-regular sequence generating $\mathfrak{J}$, then (x_i) is part of a system of parameters of A (**0**, 16.4.1); it therefore follows from (**0**, 17.1.7) that A is regular. ☐

Definition (19.1.3). — Let X be a prescheme, Y a subprescheme of X, and U an open subset of X containing Y such that Y is defined by an ideal $\mathscr{J}$ of $\mathscr{O}_U$; suppose that $\mathscr{J}/\mathscr{J}^2$ is a locally free $(\mathscr{O}_U/\mathscr{J})$-module (as is the case when Y is quasi-regularly immersed in X). For every $y \in Y$, the *transverse codimension* of Y in X at y, denoted by $\mathrm{codim}^*_y(Y, X)$, is the rank of the free $(\mathscr{O}_{U,y}/\mathscr{J}_y)$-module $\mathscr{J}_y/\mathscr{J}_y^2$. If $f : Z \to X$ is an immersion with image Y, the transverse codimension of f at $z \in Z$ is likewise the transverse codimension in X, at $f(z)$, of the subprescheme Y.

Proposition (19.1.4). — *Let X be a locally Noetherian prescheme and Y a subprescheme regularly immersed in X. For every $y \in Y$, one has*

(19.1.4.1) $$\mathrm{codim}^*_y(Y, X) = \mathrm{codim}_y(Y, X).$$

Proof. By virtue of (5.1.3.2), $\operatorname{codim}_y(Y, X)$ is equal to the least value of the dimensions of the local rings $A = \mathscr{O}_{X,z}$, where z runs through the generic points of the irreducible components of Y containing y. Such a point z belongs to every neighbourhood of y in X, and $\mathscr{J}_z/\mathscr{J}_z^2$ is a free $(A/\mathscr{J}_z)$-module of rank

$$n = \operatorname{codim}_y^*(Y, X).$$

It therefore remains to prove that $\dim(A) = n$.

Since z is a maximal point of the subprescheme Y defined by $\mathscr{J}$, one has $\dim(A/\mathscr{J}_z) = 0$; hence $\mathscr{J}_z$ is an ideal of definition of the Noetherian local ring A. Moreover, the quasi-regularity of $\mathscr{J}$ implies that $\mathscr{J}_z^m/\mathscr{J}_z^{m+1}$ is a free $(A/\mathscr{J}_z)$-module of rank $\binom{m+n-1}{n-1}$. If r denotes the length of the Artinian ring $A/\mathscr{J}_z$, the length of $\mathscr{J}_z^m/\mathscr{J}_z^{m+1}$ is therefore **IV-4 | 186**

$$r\binom{m+n-1}{n-1}.$$

Consequently, the Hilbert–Samuel polynomial of A for the $\mathscr{J}_z$-adic filtration has degree n, which proves the proposition by (**0**, 16.2.3). □

By virtue of (19.1.4), henceforth we shall say "codimension" instead of "transverse codimension" when speaking of a subprescheme Y regularly immersed in X, even when X is not locally Noetherian.

Proposition (19.1.5). —

(i) *For an immersion $j : Y \to X$ to be open, it is necessary and sufficient that it be regular and of codimension 0 at every point.*

(ii) *Let $f : Y \to X$ be a regular (resp. quasi-regular) immersion, $g : X' \to X$ a flat morphism, $Y' = Y \times_X X'$; then the morphism $f' = f_{(X')} : Y' \to X'$ is a regular (resp. quasi-regular) immersion, and for every $y' \in Y'$, if y is the projection of y' in Y, the codimension of f' at the point y' is equal to the codimension of f at the point y. Conversely, if f is an immersion and $g : X' \to X$ a faithfully flat and quasi-compact morphism, then, if f' is a quasi-regular immersion, so is f.*

(iii) *If $f : Y \to X$ and $g : Z \to Y$ are regular immersions, then $f \circ g : Z \to X$ is a regular immersion, and for every $z \in Z$, the codimension of $f \circ g$ at the point z is the sum of the codimension of g at the point z and of the codimension of f at the point $g(z)$. Moreover, the sequence of canonical homomorphisms (16.2.7.1)*

$$0 \longrightarrow g^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Z/X} \longrightarrow \mathscr{N}_{Z/Y} \longrightarrow 0$$

(19.1.5.1) *is exact, and for every $z \in Z$, there exists a neighbourhood of z in Z in which the restrictions of the homomorphisms of this sequence form a split exact sequence.*

(iv) *Let X be a locally Noetherian prescheme, $f : Y \to X$ and $g : Z \to Y$ two immersions, z a point of Z. The following conditions are equivalent:*

a) *g is a regular immersion at the point z and f is a regular immersion at the point $g(z)$.*

b) *g and $f \circ g$ are regular immersions at the point z.*

c) *$f \circ g$ is a regular immersion at the point z, f is a regular immersion at the point $g(z)$, and the sequence of canonical homomorphisms (19.1.5.1), restricted to a sufficiently small open neighbourhood of z in Z, is exact and split.*

(v) *Let X be a locally Noetherian prescheme, $f : Y \to X$, $g : Z \to Y$ two immersions, z a point of Z; suppose that $y = g(z)$ is a regular point of Y and $x = f(g(z))$ a regular point of X (so that, by (19.1.1), f is a regular immersion at the point $g(z)$); then, for $f \circ g$ to be a regular immersion at the point z, it is necessary and sufficient that g be a regular immersion at the point z.* **IV-4 | 187**

Proof. Assertion (i) is another way of expressing (16.1.10). To prove the first assertion of (ii), we can restrict ourselves to the case where Y is a closed sub-prescheme of X defined by a regular (resp. quasi-regular) ideal $\mathscr{J}$; then Y' is the closed sub-prescheme of X' defined by the ideal $g^*(\mathscr{J})\mathscr{O}_{X'}$, which is identified here with $g^*(\mathscr{J})$ since g is flat; by replacing X by a sufficiently small affine open, we can suppose that $\mathscr{J}$ admits a regular (resp. quasi-regular) sequence of generators f_i (sections of $\mathscr{O}_X$ over X). If the sequence (f_i) is regular, so is the sequence $(f_i \otimes 1)$ (which generates $g^*(\mathscr{J})\mathscr{O}_{X'}$), by virtue of (**0**, 15.2.5); the analogous result for quasi-regular sequences follows immediately from the flatness of g and from the criterion (16.9.4). Similarly, if g is faithfully flat and quasi-compact, and if $g^*(\mathscr{J})$ is quasi-regular, $\mathscr{J}$ is of finite type by virtue of (2.5.2); by flatness, we have $g^*(\mathscr{J})/g^*(\mathscr{J})^2 = g^*(\mathscr{J}/\mathscr{J}^2)$, hence, since $g^*(\mathscr{J}/\mathscr{J}^2)$ is locally free by hypothesis (16.9.4), it follows from (2.5.2) that $\mathscr{J}/\mathscr{J}^2$ is locally free. Finally, condition (iii) of (16.9.4) relative to $g^*(\mathscr{J})$ implies the same condition for $\mathscr{J}$ by virtue of the faithful flatness of g (2.2.7). We conclude from (16.9.4) that $\mathscr{J}$ is quasi-regular.

To prove (iii), we can likewise suppose that Y and Z are closed sub-preschemes of X, defined respectively by ideals $\mathscr{J}$, $\mathscr{K}$ of $\mathscr{O}_X$ such that $\mathscr{J} \subset \mathscr{K}$; moreover, we can suppose that there exist sections f_i ($1 \leqslant i \leqslant m$) of $\mathscr{J}$ over X generating $\mathscr{J}$ and such that for $1 \leqslant i \leqslant m$, the endomorphism of $\mathscr{O}_X/(\sum_{j=1}^{i-1} f_j\mathscr{O}_X)$ defined by f_i is injective, and sections $\bar{f}_i$ ($m+1 \leqslant i \leqslant n$) of $\mathscr{K}/\mathscr{J}$ over X generating $\mathscr{K}/\mathscr{J}$ and such that for $m+1 \leqslant i \leqslant n$, the endomorphism of

$$(\mathscr{O}_X/\mathscr{J})/\Big(\sum_{j=m+1}^{i-1} \bar{f}_j(\mathscr{O}_X/\mathscr{J})\Big)$$

defined by $\bar{f}_i$ is injective. For every $z \in Z$, let U be an open neighbourhood of z in X such that there exist sections f_i of $\mathscr{K}$ over U, whose classes in $\Gamma(U, \mathscr{K}/\mathscr{J})$ are $\bar{f}_i|U$ for $m+1 \leqslant i \leqslant n$; then the restriction to U of

$$(\mathscr{O}_X/\mathscr{J})/\Big(\sum_{j=m+1}^{i-1} \bar{f}_j(\mathscr{O}_X/\mathscr{J})\Big)$$

is isomorphic to

$$\mathscr{O}_U/\Big((\mathscr{J}|U) + \sum_{j=m+1}^{i-1} f_j\mathscr{O}_U\Big) = \mathscr{O}_U/\Big(\sum_{j=1}^{i-1} f_j\mathscr{O}_U\Big),$$

and we see therefore that the sequence $(f_i)_{1\leqslant i\leqslant n}$ is regular in $\mathscr{O}_U$; as it generates $\mathscr{K}|U$, this proves the first assertion of (iii). To prove the last assertion of (iii), we can restrict ourselves (with the same notation) to the case where the images t_i of the f_i in $\mathscr{K}/\mathscr{K}^2$ (for $1 \leqslant i \leqslant n$) form a basis of this $(\mathscr{O}_X/\mathscr{K})$-module (16.9.5); for the same reason, we can suppose that the t_i for $1 \leqslant i \leqslant m$ are the canonical images of elements of a basis of the $(\mathscr{O}_X/\mathscr{K})$-module $g^*(\mathscr{J}/\mathscr{J}^2)$, and that the canonical images $\bar{t}_i$ of the t_i in $(\mathscr{K}/\mathscr{J})/(\mathscr{K}/\mathscr{J})^2$ for $m+1 \leqslant i \leqslant n$ form a basis of this $(\mathscr{O}_X/\mathscr{K})$-module. This completes the proof of (iii).

Let us pass to the proof of (iv). The fact that *a)* implies *c)* follows from (iii). Let us prove that *c)* implies *a)*. Set again $A = \mathscr{O}_{X,f(g(z))}$, so that $\mathscr{O}_{Y,g(z)} = A/\mathfrak{J}$, $\mathscr{O}_{Z,z} = A/\mathfrak{K}$ with $\mathfrak{J} \subset \mathfrak{K}$. By hypothesis, $\mathfrak{J}$ and $\mathfrak{K}$ are regular ideals of A and we have a split exact sequence **IV-4 | 188**

$$0 \longrightarrow \mathfrak{J}/\mathfrak{J}\mathfrak{K} \longrightarrow \mathfrak{K}/\mathfrak{K}^2 \longrightarrow (\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2 \longrightarrow 0$$

(cf. (16.2.7)). This implies that there exist elements f_j ($1 \leqslant j \leqslant r+s$) of $\mathfrak{K}$ whose images f'_j in $\mathfrak{K}/\mathfrak{K}^2$ form a basis of this $(A/\mathfrak{K})$-module, and such that, in addition, the f_j of index $j \leqslant r$ belong to $\mathfrak{J}$ and are such that their images in $\mathfrak{J}/\mathfrak{J}\mathfrak{K}$ form a basis of this $(A/\mathfrak{K})$-module. The f_j for $1 \leqslant j \leqslant r+s$ generate $\mathfrak{K}$, and the f_j for $1 \leqslant j \leqslant r$ generate $\mathfrak{J}$ since A is Noetherian (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, prop. 5); as $\mathfrak{K}$ is a regular ideal, it follows from (16.9.5) that the sequence $(f_j)_{1\leqslant j\leqslant r+s}$ is regular. By definition, the images of $f_{r+1}, \dots, f_{r+s}$ in $\mathfrak{K}/\mathfrak{J}$ form a regular sequence in this $(A/\mathfrak{J})$-module, which completes the proof that *c)* implies *a)* in (iv).

Finally, taking into account (16.9.10), it is immediate that (v), as well as the equivalence of *a)* and *b)* in (iv), follow from the following corollary. □

Corollary (19.1.6). — *Let* A *be a local Noetherian ring,* $\mathfrak{J}$, $\mathfrak{K}$ *two ideals of* A *contained in its maximal ideal* $\mathfrak{m}$ *and such that* $\mathfrak{J} \subset \mathfrak{K}$, $A' = A/\mathfrak{J}$, $\mathfrak{K}' = \mathfrak{K}/\mathfrak{J}$. *Then:*

(i) If $\mathfrak{J}$ *and* $\mathfrak{K}'$ *are regular ideals, so is* $\mathfrak{K}$.
(ii) If $\mathfrak{K}$ *and* $\mathfrak{K}'$ *are regular ideals, so is* $\mathfrak{J}$.
(iii) Suppose the rings A *and* $A' = A/\mathfrak{J}$ *are regular. For the ideal* $\mathfrak{K}$ *to be regular, it is necessary and sufficient that* $\mathfrak{K}'$ *be so.*

Proof. Assertion (i) is a particular case of (19.1.5), (iii). To prove (ii), consider the canonical surjective homomorphism $u : \mathfrak{K}/\mathfrak{K}^2 \to \mathfrak{K}'/\mathfrak{K}'^2 = \mathfrak{K}/(\mathfrak{K}^2 + \mathfrak{J})$. Since by hypothesis $\mathfrak{K}/\mathfrak{K}^2$ and $\mathfrak{K}'/\mathfrak{K}'^2$ are free $(A/\mathfrak{K})$-modules, whose ranks we denote respectively by $p+q$ and q, the kernel of u is an $(A/\mathfrak{K})$-module *projective* (Bourbaki, *Alg.*, chap. II, 3[e] ed., § 2, n° 2, prop. 4), hence *free* of rank p

since $A/\mathfrak{N}$ is a local Noetherian ring ((**0**, 10.1.3)); in other words, there exists a basis of $\mathfrak{N}/\mathfrak{N}^2$ of which the first p elements form a basis of $(\mathfrak{N}^2 + \mathfrak{J})/\mathfrak{N}^2$. This means (by virtue of Nakayama's lemma) that there exists a system of generators $(f_i)_{1 \leqslant i \leqslant p+q}$ of $\mathfrak{N}$, forming a regular sequence in A, such that the sequence $(f_i)_{1 \leqslant i \leqslant p}$ consists of elements of $\mathfrak{J}$; it remains to show that this last sequence generates $\mathfrak{J}$. For this, denote by $\mathfrak{J}' \subset \mathfrak{J}$ the ideal it generates; considering the ring $A/\mathfrak{J}'$, we see that we are reduced to proving the following lemma: □

Lemma (19.1.6.1). — *Let* B *be a Noetherian local ring with maximal ideal* $\mathfrak{n}$, $(g_j)_{1 \leqslant j \leqslant q}$ *a regular sequence in* B *consisting of elements of* $\mathfrak{n}$, $\mathfrak{b}$ *the ideal it generates, and* $\mathfrak{c} \subset \mathfrak{b}$ *an ideal such that the canonical images of the* g_j *in* $B/\mathfrak{c}$ *still form a regular sequence in this ring. Then* $\mathfrak{c} = 0$.

Proof. The lemma being trivial for $q = 0$, we reason by induction on q. The induction hypothesis, applied to the ring B/g_1B and to the canonical images of the g_j ($2 \leqslant j \leqslant q$) and of $\mathfrak{c}$ in this quotient ring, shows that $\mathfrak{c} \subset g_1B$. Let $\mathfrak{d}$ be the ideal of B consisting of the $x \in B$ such that $g_1x \in \mathfrak{c}$; we thus have $\mathfrak{c} = g_1\mathfrak{d}$; but since by hypothesis g_1 is regular in $B/\mathfrak{c}$, we necessarily have $\mathfrak{d} = \mathfrak{c}$, hence $\mathfrak{c} = g_1\mathfrak{c}$. Since $\mathfrak{c}$ is of finite type and g_1 is contained in the radical of B, Nakayama's lemma shows that $\mathfrak{c} = 0$. **IV-4** | 189

Let us now prove assertion (iii) of (19.1.6). By virtue of (i) and of (19.1.2), it suffices to show that if $\mathfrak{N}$ is regular, so is $\mathfrak{N}'$. Note that $\mathfrak{J}$ is regular (19.1.2), and reason by induction on the rank q of the $(A/\mathfrak{J})$-module $\mathfrak{J}/\mathfrak{J}^2$. If $q = 0$, we have $\mathfrak{J} = 0$ by Nakayama's lemma and the assertion is trivial. Since A and $A/\mathfrak{J}$ are regular, we know ((**0**, 17.1.9)) that $\mathfrak{J}$ is generated by a subset of q elements of a regular system of parameters of A, hence, if f is one of the elements of this system of generators of $\mathfrak{J}$, $A_1 = A/fA$ is regular and $f \notin \mathfrak{m}^2$ ((**0**, 17.1.8)). Let $\mathfrak{J}_1, \mathfrak{N}_1$ be the images of $\mathfrak{J}, \mathfrak{N}$ in A_1; we have $\mathfrak{J}_1 \subset \mathfrak{N}_1$, $A_1/\mathfrak{J}_1 = A/\mathfrak{J}$, hence $A_1/\mathfrak{J}_1$ is regular and consequently $\mathfrak{J}_1$ is a regular ideal (19.1.2). Let us show that $\mathfrak{N}_1$ is a regular ideal in A_1; since $\mathfrak{N}$ is a regular ideal in A, it suffices to show that f is part of a regular system of generators of $\mathfrak{N}$, and for this (16.9.5) it suffices to show that the image of f in $\mathfrak{N}/(\mathfrak{N}^2 + \mathfrak{m}\mathfrak{N}) = \mathfrak{N}/\mathfrak{m}\mathfrak{N}$ is part of a basis of this $(A/\mathfrak{m})$-vector space, in other words that $f \notin \mathfrak{m}\mathfrak{N}$, which indeed follows from $f \notin \mathfrak{m}^2$. Then, as $\mathfrak{J}_1/\mathfrak{J}_1^2$ is a free $(A_1/\mathfrak{J}_1)$-module of rank $q-1$, the induction hypothesis shows that $\mathfrak{N}_1' = \mathfrak{N}_1/\mathfrak{J}_1$ is a regular ideal of $A_1/\mathfrak{J}_1$, and since $\mathfrak{N}'$ is identified with $\mathfrak{N}_1'$ in the canonical isomorphism between $A/\mathfrak{J}$ and $A_1/\mathfrak{J}_1$, $\mathfrak{N}'$ is regular. □

Remarks (19.1.7). —

(i) We do not know whether the composite of two quasi-regular immersions is quasi-regular.

(ii) In a ringed space with local rings X, for every ideal $\mathscr{J}$ of $\mathscr{O}_X$, $\mathscr{O}_X/\mathscr{J}$ is of finite type and therefore has closed support in X ((**0**, 5.2.2)). We can therefore define, as in (**I**, 4.1.3) and (4.2.1), the notions of a (locally closed) ringed subspace of X, locally defined by an ideal $\mathscr{J}$ of a sheaf $\mathscr{O}_U$, where U is an open subset of X, and of an immersion. The definitions of quasi-regular and regular immersions and of transverse codimension then apply without change, and the results of (19.1.5), (i) to (iv) are still valid, [34] provided that, in the second assertion of (i) and in (iv), the sheaf $\mathscr{O}_X$ is coherent and the local rings $\mathscr{O}_{X,x}$ are Noetherian; as for (ii), it is necessary to *define* Y' as the ringed subspace defined by the $\mathscr{O}_{X'}$-ideal $g^*(\mathscr{J})$; the proofs are unchanged.

(19.1.8). We have already noted (in particular (16.9.6), (ii)) that in non-Noetherian rings, the notion of "regular sequence" of elements of the ring does not possess the properties that would make it usable. Recent research seems to show that in this case the notion that should be substituted for that of regular sequence is that of the sequence $\mathbf{f} = (f_1, \ldots, f_n)$ having the property that the complex of the exterior algebra $K_0(\mathbf{f}, M)$ ((**III**, 1.1.3)) has *zero homology in positive degrees* (cf. (**III**, 1.1.4)); see in particular A. Grothendieck, Séminaire de géométrie algébrique, 1966, to appear.

[34] The printed text reads "(19.1.4), (i) to (iv)"; the reference must be to Proposition 19.1.5, the itemized proposition immediately above.

19.2. Transversally regular immersions

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Definition (19.2.1). — Let $u : X \to S$ be a morphism of preschemes and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. We say that a sequence $(f_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{O}_X$ over X is *transversally $\mathscr{F}$-regular relative to S* (or relative to u) if the sequence (f_i) is $\mathscr{F}$-regular and the $\mathscr{O}_X$-modules

$$\mathscr{F}/\Big(\sum_{1\leqslant j\leqslant i} f_j\mathscr{F}\Big)$$

are flat over S for $0 \leqslant i \leqslant n$. We say that a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$ is *transversally regular relative to S* (or relative to u) at a point $x \in \operatorname{Supp}(\mathscr{O}_X/\mathscr{J})$ if there exists an open neighbourhood U of x and a finite sequence (f_i) of sections of $\mathscr{J}$ over U that is transversally $\mathscr{O}_U$-regular relative to S and generates $\mathscr{J}|U$.

In an A-algebra B, we say that an ideal $\mathfrak{J}$ is *transversally regular relative to A* if $\widetilde{\mathfrak{J}}$ is transversally regular relative to $S = \operatorname{Spec}(A)$ at every point of $V(\mathfrak{J})$ in $\operatorname{Spec}(B)$.

Definition (19.2.2). — Let S be a prescheme, X and Y two S-preschemes, and $u : Y \to X$ an S-morphism which is an immersion. We say that u is a *transversally regular immersion relative to S* at a point $y \in Y$ if there exists a neighbourhood V of $u(y)$ in X such that the sub-prescheme induced on $u(Y) \cap V$ by the sub-prescheme of X associated to u is defined by an ideal of $\mathscr{O}_V$ transversally regular relative to S at the point $u(y)$.

It is clear that the set of points of Y where an immersion is transversally regular relative to S is *open* in Y; if it is equal to Y, we say simply that u is a *transversally regular immersion relative to S*.

If $S \to T$ is a flat morphism and $\mathscr{J}$ is an ideal of $\mathscr{O}_X$ transversally regular relative to S at $x \in X$, then it is also transversally regular relative to T ((**0**, 6.2.1)). An S-immersion $Y \to X$ transversally regular relative to S at a point $y \in Y$ is therefore also a T-immersion transversally regular relative to T at this point.

Proposition (19.2.3). — *Let S be a prescheme, X and Y two S-preschemes, and $u : Y \to X$ an S-morphism that is a transversally regular immersion relative to S at a point $y \in Y$. Then, for every base change $S' \to S$, if we set $X' = X \times_S S'$ and $Y' = Y \times_S S'$, the immersion $u' = u_{(S')} : Y' \to X'$ is transversally regular relative to S' at every point $y' \in Y'$ above y.*

Proof. This follows immediately from the definition and from (**0**, 15.1.15). ☐

Proposition (19.2.4). — *Let S be a prescheme, X and Y two S-preschemes, and $u : Y \to X$ an S-morphism that is an immersion. Suppose either that S and X are locally Noetherian, or that the structure morphisms $g : X \to S$ and $h : Y \to S$ are locally of finite presentation. Let y be a point of Y, s its image in S. The following conditions are equivalent:*

a) *u is transversally regular relative to S at the point y.*

b) *The morphisms g and h are flat in a neighbourhood of the points $u(y)$ and y respectively, and if we set $X_s = g^{-1}(s)$, $Y_s = h^{-1}(s)$, the immersion $u_s : Y_s \to X_s$ is regular at the point y.*

b′) *The morphisms g and h are flat in a neighbourhood of the points $u(y)$ and y respectively, and, for every change of base $S' \to S$, if we set $X' = X \times_S S'$, $Y' = Y \times_S S'$, the immersion $u' : Y' \to X'$ is regular at every point of Y' above y.*

c) *The morphism h is flat in a neighbourhood of the point y, and the immersion u is regular in a neighbourhood of the point y.* **IV-4 | 191**

c′) *The morphism h is flat in a neighbourhood of the point y, and the immersion u is quasi-regular in a neighbourhood of the point y.*

Proof. We have already proved in (19.2.3) that *a)* implies *b′)* without the hypothesis of finiteness, and it is trivial that *b′)* implies *b)*. Let us show that *b)* implies *c)*. For this, since the question is local on X and on S, we may suppose that Y is a closed sub-prescheme of X defined by a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$, and that u is the canonical injection. The hypothesis that $\mathscr{O}_X/\mathscr{J}$ is g-flat implies that the sequence

$$0 \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_S} \mathbf{k}(s) \longrightarrow \mathscr{O}_X \otimes_{\mathscr{O}_S} \mathbf{k}(s) \longrightarrow (\mathscr{O}_X/\mathscr{J}) \otimes_{\mathscr{O}_S} \mathbf{k}(s) \longrightarrow 0$$

is exact ((**0**, 6.1.2)); hence Y_s is the closed sub-prescheme of X_s defined by the ideal $\mathscr{J}_s$ of $\mathscr{O}_{X_s} = \mathscr{O}_X \otimes_{\mathscr{O}_S} \mathbf{k}(s)$ that is identified with $\mathscr{J} \otimes_{\mathscr{O}_S} \mathbf{k}(s) = \mathscr{J}/\mathfrak{m}_s\mathscr{J}$. Consider a finite sequence (f_i) of sections of $\mathscr{J}$ over a neighbourhood of y in X whose images in

$$\mathscr{J}_y/(\mathfrak{m}_s\mathscr{O}_{X,y}\mathscr{J}_y + \mathscr{J}_y^2)$$

form a basis of this $(\mathscr{O}_{X,y}/(\mathfrak{m}_s\mathscr{O}_{X,y} + \mathscr{J}_y))$-module, which is free by hypothesis *b)*. Since X_s is locally Noetherian, it follows from (16.9.11), (16.9.5), and hypothesis *b)* that the images $(\overline{f}_i)_y = (f_i)_y \otimes 1$, sections of $\mathscr{J}_s = \mathscr{J} \otimes_{\mathscr{O}_S} \mathbf{k}(s)$ over a neighbourhood of y in X_s, form a regular sequence generating the restriction of $\mathscr{J}_s$ to this neighbourhood. Since $\mathfrak{m}_s\mathscr{O}_{X,y}$ is contained in the radical of $\mathscr{O}_{X,y}$, and since in both cases under consideration $\mathscr{J}_y$ is an ideal of finite type of $\mathscr{O}_{X,y}$, Nakayama's lemma shows that the $(f_i)_y$ also generate $\mathscr{J}_y$; since $\mathscr{J}$ is an ideal of finite type of $\mathscr{O}_X$ in both cases, there is a neighbourhood U of y in X such that the $f_i|U$ generate $\mathscr{J}|U$ ((**0**, 5.2.2)). The implication *b)*$\Rightarrow$*c)* now follows from (**0**, 15.1.16) when S and X are locally Noetherian, and from (11.3.8) when g and h are locally of finite presentation; in the latter case, (11.3.8) also proves the equivalence of *c)* and *c′)*, which is immediate when X and S are locally Noetherian ((**0**, 15.1.11)). Finally, if *c)* holds, then to prove *a)* we may again reduce to the case where Y is a closed sub-prescheme of X defined by $\mathscr{J}$. By hypothesis there is a regular sequence (f_i) of sections of $\mathscr{J}$ over a neighbourhood U of y in X such that the f_i generate $\mathscr{J}|U$ and h is flat on U; to see that *c)* implies *a)*, it again suffices to apply (**0**, 15.1.6) when S and X are locally Noetherian, and (11.3.8) when g and h are locally of finite presentation. □

Remarks (19.2.5). —

(i) If the hypothesis that h is flat in a neighbourhood of y is removed from condition *b)*, the resulting condition no longer implies *a)*. It suffices to take for S the spectrum of a local ring that is not a field, for s the closed point of S, and $Y = X_s$.

(ii) Suppose that Y is a closed sub-prescheme of X defined by a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$. The proof of (19.2.4) shows that, when the equivalent conditions of that proposition hold, then for every system (f_i) of sections of $\mathscr{J}$ over a neighbourhood of y in X whose images in $\mathscr{J}_y/(\mathfrak{m}_s\mathscr{O}_{X,y}\mathscr{J}_y + \mathscr{J}_y^2)$ form a basis of this $(\mathscr{O}_{X,y}/(\mathfrak{m}_s\mathscr{O}_{X,y} + \mathscr{J}_y))$-module, there is an open neighbourhood U of y in X such that the $f_i|U$ satisfy the conditions of Definition (19.2.1).

Corollary (19.2.6). — *Suppose either that S and X are locally Noetherian, or that g and h are locally of finite presentation. Suppose moreover that the fibres X_s and Y_s are regular at the points $u(y)$ and y respectively, and that the morphisms g and h are flat in a neighbourhood of $u(y)$ and y respectively. Then the immersion $Y \to X$ is transversally regular at the point y.*[35] **IV-4** | 192

Proof. Indeed, (19.1.1) shows that the immersion $Y_s \to X_s$ is regular at the point y, and it suffices to apply (19.2.4), *b)*. □

Proposition (19.2.7). —

(i) Every open S-immersion is transversally regular relative to S.

(ii) Let $f : Y \to X$ be an S-immersion, $g : S' \to S$ a morphism and set $X' = X_{(S')}$, $Y' = Y_{(S')}$, $f' = f_{(S')} : Y' \to X'$. If f is transversally regular relative to S, then f' is transversally regular relative to S'. The converse holds if g is faithfully flat and, moreover, X and Y are locally of finite presentation over S.

(iii) If $f : Y \to X$ and $g : Z \to Y$ are transversally regular immersions relative to S, so is $f \circ g : Z \to X$.

(iv) Let X be an S-prescheme, $f : Y \to X$, $g : Z \to Y$ two S-immersions. Suppose either that S and X are locally Noetherian, or that X, Y, and Z are locally of finite presentation over S. Then, if g and $f \circ g$ are transversally regular at the point $z \in Z$ relative to S, f is transversally regular at the point $g(z)$ relative to S.

(v) Under the general conditions of (iv), suppose that X and Y are S-flat, Y_s is regular at the point $g(z)$, and X_s is regular at the point $f(g(z))$; then, if $f \circ g$ is transversally regular at the point z, so is g.

[35] The printed text has "x" here and again in the proof; the point introduced in Proposition 19.2.4 is y.

Proof. Assertion (i) is immediate, and the first assertion of (ii) follows immediately from (19.2.3). To prove the second assertion of (ii), apply (19.2.4). Since X' and Y' are locally of finite presentation over S', this criterion first shows that X' and Y' are flat over S'; hence X and Y are flat over S ((2.5.1)). On the other hand, for every $s \in S$, if $s' \in S'$ lies above s, then $Y'_{s'} \to X'_{s'}$ is a regular immersion by hypothesis; (19.1.5), (ii), therefore shows that the same is true of $Y_s \to X_s$, since these preschemes are locally Noetherian and, in that case, regularity and quasi-regularity of the immersion $Y_s \to X_s$ are equivalent. To prove (iii), reduce to the case where Z is a closed sub-prescheme of Y and Y a closed sub-prescheme of X, and form, as in (19.1.5), (iii), a system of generators of the ideal of $\mathscr{O}_X$ defining Z. For (iv), the hypothesis together with (19.2.4) first shows that X, Y, and Z are S-flat in a neighbourhood of z. If s is the image of z in S, it remains only to prove that $f_s : Y_s \to X_s$ is regular at the point $g(z)$, knowing that $g_s : Z_s \to Y_s$ and $f_s \circ g_s$ are regular at z; this follows from (19.1.5), (iv). Finally, for (v), argue in the same way as for (iv). The hypothesis on $f \circ g$ already implies by (19.2.4), *b)*, that Z is S-flat in a neighbourhood of z, and the same is true of X and Y by hypothesis. We are therefore reduced to proving that g_s is regular at z, knowing that $f_s \circ g_s$ is regular there; this follows from (19.1.5), (v), under the stated hypotheses on X_s and Y_s. □

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Proposition (19.2.8). — *Let X be an S-prescheme, let Y be a closed sub-prescheme of X defined by a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$, and let $u : Y \to X$ be the canonical injection. If u is transversally regular relative to S, the $\mathscr{O}_X$-modules $\mathscr{J}^n$ and $\mathscr{J}^n/\mathscr{J}^m$ $(0 \leqslant n < m)$ are S-flat at the points of Y. For every base change $S' \to S$, if $X' = X_{(S')}$, $Y' = Y_{(S')}$, and $u' = u_{(S')}$, then, up to canonical isomorphism,*

$$\mathscr{G}r_n(u') = \mathscr{G}r_n(u) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$$

for every $n > 0$.

Proof. Indeed, $\mathscr{O}_X$ and $\mathscr{O}_X/\mathscr{J}$ are, by hypothesis, S-flat at the points of Y, while $\mathscr{J}^n/\mathscr{J}^{n+1}$ is a locally free $(\mathscr{O}_X/\mathscr{J})$-module ((16.9.3)), and hence is S-flat at those points. The first assertion then follows from (**0**, 6.1.2), and the second from (11.2.9.2). □

The following proposition generalizes and sharpens (17.16.1).

Proposition (19.2.9). — *Let $f : X \to S$ be a morphism, x a point of X, $s = f(x)$. Suppose that f is of finite presentation at x and is a Cohen–Macaulay morphism at x ((6.8.1)). Then there exists a sub-prescheme Y of X, containing x, with the following properties:*

- *(i) The canonical injection $j : Y \to X$ is transversally regular relative to S.*
- *(ii) The morphism $g = f \circ j : Y \to S$ is Cohen–Macaulay; equivalently, in view of (i), all the fibres of g are Cohen–Macaulay preschemes.*
- *(iii) The point x is a maximal point of $Y_s = g^{-1}(s)$.*

Proof. Set $X_s = f^{-1}(s)$. By hypothesis, the ring $\mathscr{O}_{X_s,x}$ is Cohen–Macaulay; let $(t_i)_{1 \leqslant i \leqslant n}$ be a system of parameters of this ring, and hence a regular sequence ((**0**, 16.5.7)). There is an open neighbourhood U of x in X and sections h_i $(1 \leqslant i \leqslant n)$ of $\mathscr{O}_X$ over U such that the t_i are the canonical images of the germs $(h_i)_x$ in the quotient ring $\mathscr{O}_{X_s,x}$ of $\mathscr{O}_{X,x}$. Let $\mathscr{J}$ be the ideal of $\mathscr{O}_U$ generated by the h_i, and let Y be the closed sub-prescheme of U defined by $\mathscr{J}$. We may suppose that $f|U$ is locally of finite presentation; by the definition of Y, the same is then true of g. The equivalence of *c')* and *a)* in (11.3.8) shows that the sequence $((h_i)_x)$ is regular in $\mathscr{O}_{X,x}$ and that $\mathscr{O}_{Y,x}$ is flat over $\mathscr{O}_{S,s}$. By (11.3.1), f and g are therefore flat in a neighbourhood of x, and (19.2.4) shows that j is a transversally regular immersion relative to S, after replacing Y, if necessary, by $Y \cap V$ for an open neighbourhood V of x in X. Thus (i) is proved, and we may suppose that $g : Y \to S$ is flat. Since $\mathscr{O}_{Y_s,x}$, the quotient of $\mathscr{O}_{X_s,x}$ by the ideal generated by the t_i, is Artinian by construction ((**0**, 16.3.6)), x is a maximal point of Y_s. Finally, every Artinian ring is Cohen–Macaulay ((**0**, 16.5.1)); replacing Y, if necessary, by its intersection with an open neighbourhood of x in X, we may therefore suppose, by (12.1.7), (iii), that g is a Cohen–Macaulay morphism.

In particular, we recover the existence of flat "quasi-sections" Y of a morphism $f : X \to S$ at a point $x \in X_s$ that is closed in X_s, under the hypotheses that f is flat at x and $\mathscr{O}_{X_s,x}$ is Cohen–Macaulay ((17.16.1)); moreover, the immersion $Y \to X$ is transversally regular relative to S. This last refinement also applies to the *étale* quasi-section defined in (17.16.3), (i). □

19.3. Relative complete intersections (flat case)

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Definition (19.3.1). — Let A be a Noetherian local ring. We say that A is a *complete-intersection ring* (or an *absolute complete-intersection ring*, when there is danger of confusion) if its completion $\widehat{\mathrm{A}}$ is isomorphic to the quotient of a complete regular Noetherian local ring B by a regular ideal, that is, by an ideal generated by a regular sequence of elements of B ((16.9.7)).

Every regular local ring is a complete intersection ring.

We say that a locally Noetherian prescheme X is an *(absolute) complete intersection* at a point $x \in X$ if the ring $\mathscr{O}_{X,x}$ is a complete-intersection ring.

Proposition (19.3.2). — *Let* B *be a regular Noetherian local ring and* $\mathfrak{J}$ *an ideal of* B*. For* $\mathrm{A} = \mathrm{B}/\mathfrak{J}$ *to be a complete-intersection ring, it is necessary and sufficient that* $\mathfrak{J}$ *be a regular ideal, that is, be generated by a regular sequence of elements of* B *((16.9.7)).*

Proof. We may write $\widehat{\mathrm{A}} = \widehat{\mathrm{B}}/\mathfrak{J}\widehat{\mathrm{B}}$. Since $\widehat{\mathrm{B}}$ is faithfully flat over B and is Noetherian, $\mathfrak{J}\widehat{\mathrm{B}}$ is regular if and only if $\mathfrak{J}$ is regular ((19.1.5), (ii)). This shows, by (**0**, 17.1.5), both that the condition in the statement is sufficient and that, in proving its necessity, we may reduce to the case where A and B are complete. By hypothesis, there is then a complete regular Noetherian local ring B' such that A is isomorphic to a quotient $\mathrm{B}'/\mathfrak{J}'$, where $\mathfrak{J}'$ is a regular ideal of B'. Consider the fibre product $\mathrm{B}'' = \mathrm{B} \times_{\mathrm{A}} \mathrm{B}'$ for the surjective homomorphisms $\mathrm{B} \to \mathrm{A}$ and $\mathrm{B}' \to \mathrm{A}$ ((**0**, 18.1.2)). We first prove the following lemma.

Lemma (19.3.2.1). — *Let* A, E, F *be three rings,* $f : \mathrm{E} \to \mathrm{A}$, $g : \mathrm{F} \to \mathrm{A}$ *two homomorphisms, and set* $\mathrm{G} = \mathrm{E} \times_{\mathrm{A}} \mathrm{F}$ *((**0**, 18.1.2)).*

- *(i) If f is surjective and if* E *and* F *are Noetherian rings,* G *is Noetherian.*
- *(ii) If* A, E, *and* F *are local rings and f and g are local homomorphisms, then* G *is a local ring and the canonical homomorphisms* $\mathrm{G} \to \mathrm{E}$ *and* $\mathrm{G} \to \mathrm{F}$ *are local.*
- *(iii) If the conditions of (i) and (ii) are satisfied, if g is surjective, if* E *and* F *are complete Noetherian local rings,* G *is complete.*

Proof. (i) In view of (**0**, 18.1.3) and (18.1.5), the projection $\mathrm{G} \to \mathrm{F}$ is surjective. Its kernel $\mathfrak{J}'$ is canonically identified with the kernel $\mathfrak{J}$ of f, and the G-module structure on $\mathfrak{J}'$ corresponds to the E-module structure on $\mathfrak{J}$ via the canonical homomorphism $\mathrm{G} \to \mathrm{E}$. If $\mathfrak{a}$ is any ideal of G, then $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{J}')$ is isomorphic to $(\mathfrak{a} + \mathfrak{J}')/\mathfrak{J}'$, and hence to an ideal of F, so it is finitely generated. On the other hand, $\mathfrak{a} \cap \mathfrak{J}'$ is isomorphic to an ideal of E contained in $\mathfrak{J}$, and is therefore also finitely generated. Thus $\mathfrak{a}$ is finitely generated and G is Noetherian.

(ii) Let $\mathfrak{r}$ and $\mathfrak{s}$ be the maximal ideals of E and F respectively. The set $\mathfrak{m} = \mathfrak{r} \times_{\mathrm{A}} \mathfrak{s}$ is an ideal of G, the inverse image of $\mathfrak{r}$ under $\mathrm{G} \to \mathrm{E}$ and of $\mathfrak{s}$ under $\mathrm{G} \to \mathrm{F}$. Its elements are therefore non-invertible in G. On the other hand, if $(x, y) \notin \mathfrak{m}$ and, for example, $x \notin \mathfrak{r}$, then $f(x)$ does not belong to the maximal ideal of A. Since $g(y) = f(x)$ and g is local, we also have $y \notin \mathfrak{s}$; we conclude that x and y are invertible, hence (x, y) is invertible, which completes **IV-4 | 195** the proof of (ii).

(iii) The image of $\mathfrak{m}$ under the surjective homomorphism $\mathrm{G} \to \mathrm{E}$ is the maximal ideal $\mathfrak{r}$ of E; hence the image of $\mathfrak{m}^n \cap \mathfrak{J}'$ is $\mathfrak{r}^n \cap \mathfrak{J}$. Since $\mathfrak{J}$ is closed, and therefore complete, for the topology induced by the $\mathfrak{r}$-adic topology of E ((**0**, 7.3.5)), $\mathfrak{J}'$ is complete for the topology induced by the $\mathfrak{m}$-preadic topology of G. On the other hand, $\mathrm{G}/\mathfrak{J}'$, endowed with the $\mathfrak{m}$-preadic topology, is isomorphic to F endowed with the $\mathfrak{s}$-adic topology, and is therefore complete for the quotient topology of the $\mathfrak{m}$-preadic topology of G. It follows that G is complete for the $\mathfrak{m}$-preadic topology.

□

The lemma established, Cohen's theorem ((**0**, 19.8.8)) shows that B'' is isomorphic to a quotient of a complete regular Noetherian local ring C. It follows from (19.1.2) that the immersion $\mathrm{Spec}(\mathrm{B}') \to \mathrm{Spec}(\mathrm{C})$ is regular. Since, by hypothesis, the immersion $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B}')$ is regular, (19.1.5), (iii), shows that $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{C})$ is regular. But this immersion is also the composite

$$\mathrm{Spec}(\mathrm{A}) \longrightarrow \mathrm{Spec}(\mathrm{B}) \longrightarrow \mathrm{Spec}(\mathrm{C}).$$

Since B is regular, the immersion $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{C})$ is regular ((19.1.2)); hence $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is regular by (19.1.5), (iv). ☐

Corollary (19.3.3). — *Let X be a locally Noetherian prescheme locally immersible in a regular prescheme. Then the set of $x \in X$ where X is a complete intersection is open in X and contains the set of regular points of X.*

Proof. We can indeed restrict ourselves to the case where $X = \mathrm{Spec}(\mathrm{B}/\mathfrak{J})$, where B is a regular ring and $\mathfrak{J}$ an ideal of B. The set of $\mathfrak{p} \supset \mathfrak{J}$ in $\mathrm{Spec}(\mathrm{B})$ such that $\mathfrak{J}_\mathfrak{p}$ is regular in $\mathrm{B}_\mathfrak{p}$ is open ((16.9.5)), and, by (19.3.2), it is the set of points of X at which X is a complete intersection. ☐

Corollary (19.3.4). — *Let k be a field, X a prescheme locally of finite type over k, k' an extension of k, $X' = X \otimes_k k'$. Let x' be a point of X', x its projection in X. For X to be a complete intersection at x, it is necessary and sufficient that X' be a complete intersection at x'.*

Proof. The question being local on X, we can suppose that $X = \mathrm{Spec}(\mathrm{A})$, where A is a k-algebra of finite type, hence a quotient of a polynomial algebra $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, so X is a closed sub-prescheme of the regular scheme $Y = \mathrm{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n])$ ((**0**, 17.3.7)). If we set $Y' = Y \otimes_k k' = \mathrm{Spec}(k'[\mathrm{T}_1, \ldots, \mathrm{T}_n])$, Y' is also a regular scheme. Now, since $\mathrm{Spec}(k')$ is faithfully flat over $\mathrm{Spec}(k)$, it follows from (19.1.5), (ii), and the fact that these preschemes are Noetherian that the immersion $X \to Y$ is regular at x if and only if the immersion $X' \to Y'$ is regular at x'. The conclusion then follows from the criterion (19.3.2). ☐

Corollary (19.3.5). — *Let* A *and* C *be Noetherian local rings, let $\rho : \mathrm{A} \to \mathrm{C}$ be a local homomorphism, let $\mathrm{B} = \mathrm{C}/\mathfrak{J}$ be a quotient ring of* C*, and let k be the residue field of* A*. Suppose that ρ makes* C *a flat* A*-module and that $\mathrm{C} \otimes_\mathrm{A} k$ is regular. Then the following conditions are equivalent:*

a) *The ideal $\mathfrak{J}$ is transversally regular relatively to* A*.*
b) B *is a flat* A*-module, and $\mathrm{B} \otimes_\mathrm{A} k$ is a complete intersection ring.*

Proof. By virtue of (19.2.4), condition *a)* amounts to saying that B is a flat A-module and that $\mathfrak{J} \otimes_\mathrm{A} k$ **IV-4** | 196
(which is identified with an ideal of $\mathrm{C} \otimes_\mathrm{A} k$ by virtue of the flatness of B over A ((**0**, 6.1.2))) is regular in this ring. Since the ring $\mathrm{C} \otimes_\mathrm{A} k$ is regular, it amounts to the same, by virtue of (19.3.2), to say (when B is a flat A-module) that $\mathfrak{J} \otimes_\mathrm{A} k$ is a regular ideal of $\mathrm{C} \otimes_\mathrm{A} k$, or that $\mathrm{B} \otimes_\mathrm{A} k = (\mathrm{C} \otimes_\mathrm{A} k)/(\mathfrak{J} \otimes_\mathrm{A} k)$ is a complete intersection ring; whence the corollary. ☐

Definition (19.3.6). — Let $f : X \to S$ be a flat morphism locally of finite presentation. We say that X is a *complete intersection relatively to S* at a point x, if the fibre $f^{-1}(f(x))$ is a complete intersection (absolute) at the point x. We say that X is a *complete intersection relatively to S*, or that the morphism f is a *morphism of complete intersection*, if X is a complete intersection relatively to S at each of its points.

Proposition (19.3.7). — *Let $g : X \to S$, $h : Y \to S$ be two flat morphisms locally of finite presentation, $f : Y \to X$ an S-immersion, y a point of Y, $x = f(y)$, $s = g(x) = h(y)$. Suppose that the fibre $X_s = g^{-1}(s)$ is regular at the point x. Then, for f to be a transversally regular immersion at the point y, it is necessary and sufficient that Y be a complete intersection relatively to S at the point y.*

Proof. To say that f is transversally regular at the point y amounts, by virtue of (19.2.4), to saying that $f_s : Y_s \to X_s$ is a regular immersion at the point y. But since X_s is regular at the point $x = f(y)$, this is equivalent, by virtue of (19.3.2), to saying that Y_s is a complete intersection (absolute) at the point y, whence the proposition. ☐

Corollary (19.3.8). — *Let $f : X \to S$ be a flat morphism locally of finite presentation. The set U of $x \in X$ for which X is a complete intersection relatively to S is open in X. If moreover f is proper, the set E of $s \in S$ such that $X_s = f^{-1}(s)$ is a complete intersection at each of its points is an open subset of S.*

Proof. Since $E = S - f(X - U)$,[36] the second assertion follows from the first and from the fact that f is a closed map. The first assertion is local on X, hence we can suppose that X is a closed sub-prescheme of a prescheme of the form $Z = S[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ (T_i indeterminate). Since Z is smooth

[36] The printed text has $Y - f(X - U)$; the ambient set in which E was just defined is S.

over S ((17.3.8)), its fibres Z_s are regular, and, by (19.3.7), the assertion that $x \in U$ is equivalent to the assertion that the immersion $X \to Z$ is transversally regular at x; the result follows from (19.2.2). □

Proposition (19.3.9). —

(i) Every open immersion is a complete-intersection morphism.

(ii) Let $f : X \to S$ be a flat morphism locally of finite presentation that is a complete-intersection morphism. For every change of base $g : S' \to S$, $f' = f_{(S')} : X_{(S')} \to S'$ is a morphism of complete intersection. The converse is true if g is faithfully flat and quasi-compact.

(iii) If $f : X \to Y$ and $g : Y \to Z$ are flat morphisms locally of finite presentation which are complete intersection morphisms, so is $g \circ f : X \to Z$.

Proof. Assertion (i) follows immediately from the definition (19.3.1). To prove (ii), note first that if f is flat and locally of finite presentation, then so is f' ((2.1.4)); conversely, these two properties descend when g is faithfully flat and quasi-compact ((2.5.1) and (2.7.1)). Moreover, for every $s' \in S'$, if $s = g(s')$, the fibre $f^{-1}(s)$ is a complete intersection if and only if $f'^{-1}(s')$ is one ((19.3.4)). This proves (ii), since a faithfully flat morphism is surjective. **IV-4** | 197

Finally, under the hypotheses of (iii), $g \circ f$ is flat and locally of finite presentation. By definition (19.3.6), we are reduced to the case where Z is the spectrum of a field, and must then prove that the local rings $\mathcal{O}_{X,x}$ are complete-intersection rings. This is contained in the following more general result. □

Corollary (19.3.10). — *Let $f : X \to S$ be a flat morphism locally of finite presentation that is a complete-intersection morphism, let x be a point of X, and set $s = f(x)$. If $\mathcal{O}_{S,s}$ is a Noetherian complete-intersection ring, then so is $\mathcal{O}_{X,x}$.*

Proof. We may plainly reduce to the case where $S = \operatorname{Spec}(A)$ with $A = \mathcal{O}_{S,s}$. We first show that we may further reduce to the case where A is complete. Set $A' = \widehat{A}$, $X' = X \otimes_A A'$, and let s' be the unique closed point of $S' = \operatorname{Spec}(A')$, hence the unique point of S' above s. If $f' = f_{(S')} : X' \to S'$, then $f'^{-1}(s')$ is canonically isomorphic to $f^{-1}(s)$ because A and A' have the same residue field ((**I**, 3.6.4)). Thus there is a unique point $x' \in X'$ above x and s', and $\mathcal{O}_{X',x'} = \mathcal{O}_{X,x} \otimes_A A'$. The local rings $\mathcal{O}_{X,x}$ and $\mathcal{O}_{X',x'}$ have isomorphic completions. Indeed, if E is a local ring that is a local A-algebra, then the separated completion $(E \otimes_A \widehat{A})^\wedge$ of $E \otimes_A \widehat{A}$, endowed with the tensor-product topology, is isomorphic to the separated completion of $\widehat{E} \widehat{\otimes}_A \widehat{A} \cong \widehat{E}$, and hence to $\widehat{E}$ itself ((**0**, 7.7.1)). By (19.3.1), it is therefore equivalent for $\mathcal{O}_{X,x}$ or $\mathcal{O}_{X',x'}$ to be a complete-intersection ring.

Suppose, then, that A is complete. We may also suppose that $X = \operatorname{Spec}(B)$, where B is a quotient of the polynomial algebra $C = A[T_1, \ldots, T_r]$. Since polynomial rings over a field are regular ((**0**, 17.3.7)), (19.3.7) shows that the immersion $X \to Z = \operatorname{Spec}(C)$ may be taken transversally regular relative to S. Thus $B = C/\mathfrak{J}$, where $\mathfrak{J}$ is generated by a regular sequence $(g_i)_{1 \leqslant i \leqslant n}$ in C. On the other hand, since A is complete, Cohen's theorem allows us to write $A = A'/\mathfrak{R}$, where A' is a regular local ring and $\mathfrak{R}$ an ideal of A' ((**0**, 19.8.8)). Since A is a complete-intersection ring, $\mathfrak{R}$ is generated by a regular sequence $(f_j)_{1 \leqslant j \leqslant m}$ ((19.3.2)). In $C' = A'[T_1, \ldots, T_r]$, the elements f_j still form a regular sequence, generating $\mathfrak{R}' = \mathfrak{R}[T_1, \ldots, T_r]$ ((**0**, 15.1.4)). Since $C'/\mathfrak{R}' = C$, choose, for each i, an element $g'_i \in C'$ whose image in C is g_i. The sequence consisting of the f_j and the g'_i is regular in C'. If $\mathfrak{J}'$ is the ideal it generates, then $B = C'/\mathfrak{J}'$; the conclusion follows from (19.3.2), since C' is regular ((**0**, 17.3.7)). □

19.4. Application: criteria of regularity and smoothness for blown-up preschemes **IV-4** | 198

(19.4.1). Let X be a prescheme and Y a closed sub-prescheme of X, defined by a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$. Recall ((**II**, 8.1.3)) that the X-scheme X' obtained by blowing up Y is the prescheme

$$X' = \mathrm{Proj}(\mathscr{S}), \quad \text{where} \quad \mathscr{S} = \bigoplus_{n \geqslant 0} \mathscr{J}^n.$$

If $\mathscr{J}$ is of finite type, the structural morphism $f : X' \to X$ is projective. Without hypothesis on $\mathscr{J}$, the closed sub-prescheme

$$Y' = f^{-1}(Y)$$

of X' is defined by the quasi-coherent Ideal $\mathscr{J}\mathscr{O}_{X'}$ of $\mathscr{O}_{X'}$, canonically isomorphic to $\mathscr{O}_{X'}(1)$; more precisely, we have an exact sequence

$$0 \longrightarrow \mathscr{O}_{X'}(1) \xrightarrow{\widetilde{u}_0} \mathscr{O}_{X'} \longrightarrow \mathscr{O}_{Y'} \longrightarrow 0$$

(19.4.1.1) where $\mathscr{J}\mathscr{O}_{X'}$ is the image of $\widetilde{u}_0$ ((**II**, 8.1.7) and (8.1.8)). Since $\mathscr{O}_{X'}(1)$ is free of rank 1 in a neighbourhood of every point of X', $\mathscr{J}\mathscr{O}_{X'}$ is locally generated by a non-zero-divisor of $\mathscr{O}_{X'}$, and $\mathscr{J}\mathscr{O}_{X'}/\mathscr{J}^2\mathscr{O}_{X'}$ is consequently a free $(\mathscr{O}_{X'}/\mathscr{J}\mathscr{O}_{X'})$-module of rank 1. This proves the first assertion of the following lemma:

Lemma (19.4.2). — *The canonical immersion $Y' \to X'$ is regular and of codimension* 1, *and Y' is canonically isomorphic to the Y-scheme* $\mathrm{Proj}(\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X))$.

Proof. The second assertion follows from (**II**, 3.5.3), applied to the canonical injection $Y \to X$, because

$$\mathscr{S} \otimes_{\mathscr{O}_X} \mathscr{O}_Y = \mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$$

by definition: $\mathscr{O}_Y \cong \mathscr{O}_X/\mathscr{J}$ and $\mathscr{J}^n \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J}) \cong \mathscr{J}^n/\mathscr{J}^{n+1}$. ☐

Corollary (19.4.3). — *If the canonical immersion $Y \to X$ is quasi-regular, the restriction $g : Y' \to Y$ of the morphism f is smooth.*

Proof. Indeed, $\mathscr{N}_{Y/X} = \mathscr{J}/\mathscr{J}^2$ is then an $\mathscr{O}_Y$-Module locally free of finite rank and $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is isomorphic to $\mathbf{S}^*_{\mathscr{O}_Y}(\mathscr{N}_{Y/X})$ (16.9.8), hence Y' is Y-isomorphic to $\mathbf{P}(\mathscr{N}_{Y/X})$ and is therefore smooth over Y (17.3.9). ☐

We shall see later (chap. V) that g is smooth also when $Y \to X$ "has only ordinary singularities".

Proposition (19.4.4). — *With the notations of* (19.4.1), *suppose that X is locally Noetherian and that the morphism $g : Y' \to Y$, restriction of f, is smooth. Let x' be a point of Y', $x = g(x')$. For Y to be regular at the point x, it is necessary and sufficient that Y' be regular at the point x'; when this is so, X' is then regular at the point x'.*

Proof. The first assertion follows from (17.5.8) and the second from (19.1.1) and (19.4.2). ☐

Corollary (19.4.5). — *Under the general hypotheses of* (19.4.4), *suppose that Y is regular at the point x. Then, for every generization x_1 of x in X not belonging to Y, X is regular at the point x_1. If* $\mathrm{Reg}(X)$ *is open* (((6.12)), *for example if X is excellent* ((7.8.6), (iii))) *and if Y is regular, then there exists an open neighbourhood U of Y in X such that X is regular at the points of $U - Y$.* **IV-4** | 199

Proof. To prove the first assertion, we may reduce to the case where X is a local scheme with closed point x. The hypothesis then implies that Y is regular ((**0**, 17.3.2)), and hence so is Y' by (19.4.4). The morphism $f : X' \to X$ is projective, hence proper. If x'_1 denotes the unique point of X' above x_1 ((**II**, 8.1.3)), the valuative specialization property ((**II**, 7.2.1)) gives a specialization of x'_1 lying in Y'. Applying (19.4.4) and (**0**, 17.3.2), we conclude that X' is regular at x'_1, and therefore that X is regular at x_1, since $X' - Y'$ is isomorphic to $X - Y$ ((**II**, 8.1.3)). If Y is regular, every point of $X - Y$ which is a generization of a point of Y belongs to $\mathrm{Reg}(X)$; if $\mathrm{Reg}(X)$ is open, $U = Y \cup \mathrm{Reg}(X)$ is then locally constructible and stable under generization, hence open ((**0**, 9.2.5)), and responds to the question. ☐

Proposition (19.4.6). — *Let $g : X \to S$ be a morphism, Y a closed sub-prescheme of X defined by a quasi-coherent Ideal $\mathscr{J}$, X' the X-scheme obtained by blowing up Y, Y' the inverse image of Y in X'.*

(i) The following conditions are equivalent:

a) $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ *is a g-flat* $\mathscr{O}_X$*-algebra.*

b) *For every* $n \geqslant 0$*, the* $\mathscr{O}_X$*-Module* $\mathscr{O}_X/\mathscr{J}^{n+1}$ *is g-flat (in other words, the n-th infinitesimal neighbourhood* $Y^{(n)}$ *of* Y *in* X *(16.1.2) is S-flat).*

c) *For every base change* $S_1 \to S$ *and every integer* $n \geqslant 1$*, if* $X_1 = X \times_S S_1$*, the canonical homomorphism* $\mathscr{J}^n \otimes_{\mathscr{O}_S} \mathscr{O}_{S_1} \to \mathscr{J}^n \mathscr{O}_{X_1}$ *is bijective.*

When this is so, Y' *is S-flat, and for every change of base* $S_1 \to S$*, if* Y_1 *is the inverse image of* Y *in* X_1*, the prescheme* X_1' *obtained by blowing up* Y_1 *in* X_1 *is canonically isomorphic to* $X' \times_S S_1$*.*

(ii) Suppose that the equivalent conditions of (i) are satisfied and moreover that the morphisms $X \to S$ *and* $Y \to S$ *are locally of finite presentation. Then* $\mathscr{S} = \bigoplus_{n \geqslant 0} \mathscr{J}^n$ *is a* $\mathscr{O}_X$*-Algebra of finite presentation, the morphism* $X' \to X$ *is of finite presentation, the canonical immersion* $Y' \to X'$ *is transversally regular of codimension* 1 *relatively to S, and the set of points of X (resp.* X'*) where X (resp.* X'*) is S-flat, is a neighbourhood of Y (resp.* Y'*).*

Proof. **IV-4** | 200

(i) The equivalence of *a)* and *b)* follows immediately from (**0**, 6.1.2). To prove the equivalence of *b)* and *c)*, we may suppose that $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, with $\mathscr{J} = \widetilde{\mathfrak{J}}$. The Tor exact sequence shows that *b)* implies $\mathrm{Tor}_1^A(B/\mathfrak{J}^{n+1}, A') = 0$ for every $n \geqslant 0$ and every A-algebra A'. Taking $A' = A \oplus M$, for an arbitrary A-module M and multiplication $(a,m)(a',m') = (aa', am' + a'm)$, shows that this is equivalent to $\mathrm{Tor}_1^A(B/\mathfrak{J}^{n+1}, M) = 0$, and hence to the flatness of $B/\mathfrak{J}^{n+1}$ over A. If $\mathfrak{J}_1 = \mathfrak{J} \otimes_A A_1$, then

$$\bigoplus_{n\geqslant 0} \mathfrak{J}_1^n = \left(\bigoplus_{n\geqslant 0} \mathfrak{J}^n\right) \otimes_A A_1,$$

and the assertion concerning the blown-up prescheme X_1' follows immediately. Finally, to prove that Y' is S-flat, we may again suppose that S and X are affine. Put $C = \mathrm{gr}^*_{\mathfrak{J}}(B)$. By (19.4.2) and (**II**, 2.3.6), every point of Y' has an affine neighbourhood whose ring is $C_{(t)}$, with t homogeneous of degree 1. Since C is flat over A, so is the localization C_t, and hence its degree-zero direct summand $C_{(t)}$. Thus Y' is S-flat.

(ii) Again suppose that S and X are affine, so that B is a finitely presented A-algebra and $\mathfrak{J}$ is a finitely generated ideal of B. By (8.9.1), (8.6.3), and (11.2.9), there exist a Noetherian subring $A_0 \subset A$, a finite-type A_0-algebra B_0, and a finitely generated ideal $\mathfrak{J}_0 \subset B_0$ such that

$$B = B_0 \otimes_{A_0} A, \qquad \mathfrak{J} = \mathfrak{J}_0 B, \qquad \mathrm{gr}^*_{\mathfrak{J}}(B) = \mathrm{gr}^*_{\mathfrak{J}_0}(B_0) \otimes_{A_0} A,$$

with $\mathrm{gr}^*_{\mathfrak{J}_0}(B_0)$ flat over A_0. By (i),

$$\bigoplus_{n\geqslant 0} \mathfrak{J}^n = \left(\bigoplus_{n\geqslant 0} \mathfrak{J}_0^n\right) \otimes_{A_0} A = \left(\bigoplus_{n\geqslant 0} \mathfrak{J}_0^n\right) \otimes_{B_0} B.$$

The algebra $\bigoplus_{n\geqslant 0} \mathfrak{J}_0^n$ is generated over B_0 by the finitely generated ideal $\mathfrak{J}_0$; it is therefore of finite type, and hence of finite presentation because B_0 is Noetherian. Thus $\bigoplus_{n\geqslant 0} \mathfrak{J}^n$ is a finitely presented B-algebra. Likewise, $\mathrm{Proj}(\bigoplus_{n\geqslant 0} \mathfrak{J}_0^n) \to \mathrm{Spec}(B_0)$ is of finite type ((**II**, 2.7.1)), hence of finite presentation, and its base change $X' \to X$ is of finite presentation. By (19.4.2), the canonical immersion $Y' \to X'$ is regular. Since Y' is S-flat by (i), (19.2.4) shows that $Y' \to X'$ is transversally regular relative to S and that X' is S-flat at the points of Y', hence on an open neighbourhood of Y' by (11.3.1). Finally, since $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is an S-flat $\mathscr{O}_X$-algebra, (11.3.4) shows that X is S-flat at the points of Y, and hence on a neighbourhood of Y by (11.3.1).

□

Corollary (19.4.7). — *Let* $g : X \to S$ *be a morphism locally of finite presentation, Y a closed sub-prescheme of X such that the composite morphism* $Y \to X \to S$ *is locally of finite presentation; suppose moreover that Y is S-flat and that* $\mathscr{O}_X$ *is normally flat along Y ((11.3.4)). Then (with the notations of (19.4.1)), the morphism* $X' \to X$ *is of finite presentation, the canonical immersion* $Y' \to X'$ *is transversally regular of codimension*

1 *relative to S, Y' is flat over Y (and hence over S), and the set of points of X (resp. X') where X (resp. X') is S-flat is a neighbourhood of Y (resp. Y').*

Proof. Indeed, by hypothesis $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is a flat $(\mathscr{O}_X/\mathscr{J})$-module, and hence is S-flat because Y is S-flat. We can therefore apply the results of (19.4.6), (i) and (ii). Since Y' is Y-isomorphic to $\mathrm{Proj}(\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X))$ ((19.4.2)), the same argument as in (19.4.6) shows that Y' is flat over Y. □

Proposition (19.4.8). — *Suppose the hypotheses of (19.4.7) satisfied and moreover that the morphism $Y' \to Y$ is smooth. Let x' be a point of Y', x its image in Y. For Y to be smooth over S at the point x, it is necessary and sufficient that Y' be smooth over S at the point x'; when this is so, X' is then smooth over S at the point x'. The preceding conclusions hold in particular when $X \to S$ is locally of finite presentation and the immersion $Y \to X$ is transversally regular relative to S.*

Proof. Taking into account the compatibility of the formation of a blown-up prescheme with changes of base in the case envisaged (19.4.6), (i) and of ((17.5.1)) and (17.7.1), it suffices to consider the case where S is the spectrum of an algebraically closed field. But then it amounts to the same to say that an S-prescheme locally of finite type is smooth or that it is regular at this point ((17.15.1)), hence the conclusions follow from (19.4.4). **IV-4** | 201

When $X \to S$ is locally of finite presentation and the canonical immersion $Y \to X$ is quasi-regular, that immersion is, by definition, of finite presentation, since the Ideal defining it is of finite type; hence the composite $Y \to X \to S$ is locally of finite presentation. We have already seen in (19.4.3) that under these conditions $Y' \to Y$ is smooth; moreover, $\mathscr{N}_{Y/X}$ is a locally free $\mathscr{O}_Y$-Module and $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is isomorphic to $\mathbf{S}^*_{\mathscr{O}_Y}(\mathscr{N}_{Y/X})$, hence is a flat $\mathscr{O}_Y$-Module. The hypotheses of (19.4.7) are thus all satisfied and we can apply the conclusions of (19.4.8). □

Corollary (19.4.9). — *Under the general hypotheses of (19.4.8), suppose moreover that Y is smooth over S at the point x. Then, for every generization x_1 of x in X not belonging to Y, X is smooth over S at the point x_1. If Y is smooth over S, then there exists an open neighbourhood U of Y in X such that X is smooth over S at the points of $U - Y$.*

Proof. To prove the first assertion, we can restrict ourselves to the case where X is a local scheme with closed point x; the hypothesis then implies that Y is smooth over S, since every neighbourhood in Y of its closed point is necessarily all of Y. We conclude by the same reasoning as in (19.4.5), using the fact that the set of points where X' is smooth over S is open in X'. To prove the second assertion, we can reduce to the case where S is Noetherian, and X of finite type over S, using ((8.9.1)), ((8.6.3)), ((11.2.9)) and (17.7.6); we then conclude as in (19.4.5), using the fact that the set of points where X is smooth over S is open in X. □

Remark (19.4.10). — In (19.4.6), (ii), replace the hypothesis that X and Y are locally of finite presentation over S by the hypothesis that S and X (hence also Y) are *locally Noetherian*. Then it is clear that $\bigoplus_{n\geqslant 0}\mathscr{J}^n$ is an $\mathscr{O}_X$-Algebra of finite type, that the morphism $X' \to X$ is of finite type, and it follows again from (19.2.4) applied to locally Noetherian preschemes, that the immersion $Y' \to X'$ is transversally regular relatively to S and that X' is S-flat at the points of Y'.

Proposition (19.4.11). — *Let X be a prescheme, $\mathscr{E}$ a quasi-coherent $\mathscr{O}_X$-Module, $u : \mathscr{E} \to \mathscr{O}_X$ a homomorphism of $\mathscr{O}_X$-Modules, $\mathscr{J} = u(\mathscr{E})$ the quasi-coherent Ideal of $\mathscr{O}_X$ which is the image of u, and Y the closed sub-prescheme of X defined by $\mathscr{J}$. Let X' be the X-scheme obtained by blowing up Y. On the other hand, let $\mathrm{P} = \mathbf{P}(\mathscr{E})$ be the projective bundle over X defined by $\mathscr{E}$ ((**II**, 4.1.1)), $p : \mathrm{P} \to X$ the structural morphism, $\alpha_1^\# : p^*(\mathscr{E}) \to \mathscr{O}_\mathrm{P}(1)$ the canonical homomorphism ((**II**, 4.1.5.1)) and $\mathscr{H}$ its kernel, so that we have an exact sequence*

$$0 \longrightarrow \mathscr{H} \longrightarrow p^*(\mathscr{E}) \xrightarrow{\alpha_1^\#} \mathscr{O}_\mathrm{P}(1) \longrightarrow 0.$$

Finally, let $\mathscr{K}$ be the quasi-coherent Ideal of $\mathscr{O}_\mathrm{P}$ which is the image of the restriction $v : \mathscr{H} \to \mathscr{O}_\mathrm{P}$ of $p^*(u)$, and let Z be the closed sub-prescheme of P defined by $\mathscr{K}$. **IV-4** | 202

(i) *The image of X' by the closed immersion $j : X' \to \mathbf{P}(\mathscr{E})$ corresponding to the surjective homomorphism of graded $\mathscr{O}_X$-Algebras*

$$\mathbf{S}_{\mathscr{O}_X}(\mathscr{E}) \longrightarrow \mathbf{S}_{\mathscr{O}_X}(\mathscr{J}) \longrightarrow \bigoplus_{n\geqslant 0}\mathscr{J}^n$$

*((**II**, 3.6.2)) is a closed sub-prescheme of Z.*

(ii) If $\mathscr{E}$ is an $\mathscr{O}_X$-Module locally free of rank m, $\mathscr{H}$ is an $\mathscr{O}_P$-Module locally free of rank $m-1$.

(iii) Suppose X is locally Noetherian, that $\mathscr{E}$ is locally free of finite type, that the sub-prescheme Y is regular, and that at every point $x \in Y$, the canonical immersion $i : Y \to X$ is regular and of codimension equal to $\mathrm{rg}_{\mathscr{O}_{X,x}}(\mathscr{E}_x)$ (which we shall express by saying that the homomorphism u is regular at the point x (cf. chap. V)). Then the image of X' by the closed immersion $j : X' \to \mathrm{P}$ is the sub-prescheme Z; at all points of $X' - Y'$ (Y' being the inverse image of Y in X') the immersion j is transversally regular relatively to X; and finally, at all points x' of Y', X' is regular, the immersion j is regular and of codimension $\mathrm{rg}_{\mathscr{O}_{P,z}}(\mathscr{H}_z)$ where $z = j(x')$ (in other words the homomorphism $v : \mathscr{H} \to \mathscr{O}_P$ is regular in z).

Proof. All the questions being local on X, we can restrict ourselves to the case where $X = \mathrm{Spec}(\mathrm{A})$ is affine and $\mathscr{E} = \widetilde{\mathrm{E}}$, where E is an A-module. To verify (i), we can in addition examine what happens in an affine open of P of the form $\mathrm{D}_+(t)$, where $t \in \mathrm{E} = \mathbf{S}^1_{\mathrm{A}}(\mathrm{E})$ ((**II**, 2.3.14)). If we set $\mathrm{S} = \mathbf{S}_{\mathrm{A}}(\mathrm{E})$, then $\Gamma(\mathrm{D}_+(t), \mathscr{O}_P) = \mathrm{S}_{(t)}$ ((**II**, 2.4.1)) and $p^*(\mathscr{E})|_{\mathrm{D}_+(t)} = (\mathrm{E} \otimes_{\mathrm{A}} \mathrm{S}_{(t)})^{\sim}$. Referring to the definition of $\alpha_1^{\#}$ ((**II**, 2.6.2.2)), we see that to every element

$$a = \sum_i x^{(i)} \otimes \frac{x_1^{(i)} x_2^{(i)} \cdots x_n^{(i)}}{t^n}$$

of $\mathrm{E} \otimes_{\mathrm{A}} \mathrm{S}_{(t)}$, where the $x^{(i)}$ and $x_k^{(i)}$ belong to E, it assigns

$$\sum_i \frac{x^{(i)} x_1^{(i)} x_2^{(i)} \cdots x_n^{(i)}}{t^{n+1}} \quad \text{in } (\mathrm{S}(1))_{(t)}.$$

To prove (i), it is enough to show that, if this last element is zero, then the image of

$$a' = v(a) = \sum_i u(x^{(i)}) \frac{x_1^{(i)} x_2^{(i)} \cdots x_n^{(i)}}{t^n}$$

under the canonical homomorphism $\mathrm{S}_{(t)} \to \left(\bigoplus_{n \geqslant 0} u(\mathrm{E})^n\right)_{(u(t))}$ ((**II**, 3.6.2)) is also zero. Its image in $\mathrm{A}_{u(t)}$ is

$$\sum_i \frac{u(x^{(i)}) u(x_1^{(i)}) \cdots u(x_n^{(i)})}{u(t)^n},$$

that is, the product of $u(t)$ and the canonical image of $\sum_i x^{(i)} x_1^{(i)} \cdots x_n^{(i)} / t^{n+1}$ under the algebra homomorphism extending $u : \mathrm{E} \to \mathrm{A}$ from $\mathrm{S}_{(t)}$. This proves (i).

To establish (ii), we may suppose that E is a free A-module of rank m. Since $\alpha_1^{\#} : \mathrm{E} \otimes_{\mathrm{A}} \mathrm{S}_{(t)} \to (\mathrm{S}(1))_{(t)}$ is surjective and $(\mathrm{S}(1))_{(t)}$ is a free $\mathrm{S}_{(t)}$-module of rank 1, the exact sequence

$$0 \longrightarrow \mathrm{H}_{(t)} \longrightarrow \mathrm{E} \otimes_{\mathrm{A}} \mathrm{S}_{(t)} \xrightarrow{\alpha_1^{\#}} (\mathrm{S}(1))_{(t)} \longrightarrow 0$$

splits, and $\mathrm{H}_{(t)}$ is therefore a projective $\mathrm{S}_{(t)}$-module of rank $m-1$, whence (ii).

To prove (iii), let us examine first what happens above a point $x \in X - Y$. The homomorphism $u : \mathscr{E} \to \mathscr{O}_X$, restricted to a neighbourhood of x, is then surjective, so we may restrict to the case $\mathscr{J} = \mathscr{O}_X$, in which X' is canonically identified with X. To see that the closed sub-prescheme of P which is the image of X' is then identical with Z, we may, as at the beginning, suppose $X = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{E} = \mathrm{A}^m$, so that $\mathrm{S} = \mathrm{A}[\mathrm{T}_0, \ldots, \mathrm{T}_{m-1}]$. With the notation above we may suppose $u(t) \neq 0$; the elements $t \in \mathrm{E}$ having this property generate the symmetric algebra $\mathbf{S}_{\mathrm{A}}(\mathrm{E})$. After changing the basis of E if necessary, we may therefore suppose $t = \mathrm{T}_0$ and $u(\mathrm{T}_i) = 0$ for $1 \leqslant i \leqslant m-1$, so that $\mathrm{S}_{(t)}$ is canonically identified with $\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_{m-1}]$. The elements of H are then those of the form

$$\sum_\alpha \mathrm{L}_\alpha(\mathbf{T}) \otimes \mathbf{T}^\alpha$$

in $\mathrm{E} \otimes_{\mathrm{A}} \mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_{m-1}]$, where $\mathrm{L}_\alpha(\mathbf{T}) = \lambda_{\alpha 0} + \lambda_{\alpha 1}\mathrm{T}_1 + \cdots + \lambda_{\alpha, m-1}\mathrm{T}_{m-1}$ and $\sum_\alpha \mathrm{L}_\alpha(\mathbf{T})\mathbf{T}^\alpha = 0$. One checks readily that, subject to $\lambda_{\mathbf{0},0} = 0$, the values of $\lambda_{\alpha 0}$ may be chosen arbitrarily for $|\alpha| \geqslant 1$, after which the $\lambda_{\alpha i}$ for $i \geqslant 1$ may be chosen so as to satisfy the displayed relation. The corresponding elements of the Ideal $\mathscr{K}$ are therefore precisely the polynomials $\sum_\alpha \lambda_{\alpha 0}\mathbf{T}^\alpha$ without constant term. Since $(\mathbf{S}_A(A))_{(t)}$ is identified with A, these polynomials are exactly the kernel of $(\mathbf{S}_A(E))_{(t)} \to (\mathbf{S}_A(A))_{(t)}$. We have thus proved, by (19.2.1), that $j : X' \to P$ is transversally regular relative to X at every point of $X' - Y'$ and that $j(X') = Z$ at those points.

It remains to examine what happens above a point $x \in Y$. The regularity hypothesis on u at x in (iii) is equivalent, by (19.1.1) and ((**0**, 17.1.7)), to saying that X is regular at x and that **IV-4 | 203**

$$u_x \otimes 1 : \mathscr{E}_x \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x) \longrightarrow \mathfrak{m}_x/\mathfrak{m}_x^2$$

is injective. Since $p : P \to X$ is smooth ((17.3.9)), for every $z \in P$ above x, P is regular at z ((17.5.8)); moreover, by ((**0**, 17.3.3)), the canonical homomorphism

$$(\mathfrak{m}_x/\mathfrak{m}_x^2) \otimes_{\mathbf{k}(x)} \mathbf{k}(z) \longrightarrow \mathfrak{m}_z/\mathfrak{m}_z^2 \tag{19.4.11.1}$$

is injective. It follows that the homomorphism

$$(\mathscr{E}_x \otimes_{\mathscr{O}_{X,x}} \mathscr{O}_{P,z}) \otimes_{\mathscr{O}_{P,z}} \mathbf{k}(z) \longrightarrow \mathfrak{m}_z/\mathfrak{m}_z^2$$

induced by $p^*(u)$ is also injective, since it is the composite

$$(\mathscr{E}_x \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x)) \otimes_{\mathbf{k}(x)} \mathbf{k}(z) \longrightarrow (\mathfrak{m}_x/\mathfrak{m}_x^2) \otimes_{\mathbf{k}(x)} \mathbf{k}(z) \longrightarrow \mathfrak{m}_z/\mathfrak{m}_z^2,$$

where the first arrow is injective by flatness and the second is the injective homomorphism (19.4.11.1). Since $\mathscr{H}$ is, in a neighbourhood of z in P, a direct factor of $p^*(\mathscr{E})$, the homomorphism $v_z \otimes 1 : \mathscr{H}_z \otimes_{\mathscr{O}_{P,z}} \mathbf{k}(z) \to \mathfrak{m}_z/\mathfrak{m}_z^2$ is also injective; this establishes the regularity of v at z. It remains to see that the image by j of X' is identical to the closed sub-prescheme Z. It will suffice for this to show that the image by j of the points of $q^{-1}(x)$ (where $q : X' \to X$ is the structural morphism) are exactly those of $Z \cap p^{-1}(x)$ and that on the other hand at each of these points $z = j(x')$, the surjective homomorphism $\mathscr{O}_{Z,z} \to \mathscr{O}_{X',x'}$ is bijective. By the hypothesis and (19.4.3), the restriction $Y' \to Y$ of q is smooth; since $\mathscr{O}_{Y,x}$ is regular, so is $\mathscr{O}_{Y',x'}$ ((17.5.8)). Moreover, $Y' \to X'$ is a regular immersion ((19.4.2)), and hence $\mathscr{O}_{X',x'}$ is regular ((19.1.1)). The regularity of v and of P at z similarly shows that $\mathscr{O}_{Z,z}$ is regular. To prove that $\mathscr{O}_{Z,z} \to \mathscr{O}_{X',x'}$ is bijective, it therefore suffices to show that the dimensions of these two regular rings are equal ((**0**, 17.1.9)). By ((**0**, 17.3.3)), this is equivalent to **IV-4 | 204**
equality of the dimensions of $\mathscr{O}_{X',x'} \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x)$ and $\mathscr{O}_{Z,z} \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x)$. We may therefore replace X' and Z by the fibres $q^{-1}(x)$ and $Z \cap p^{-1}(x)$ respectively, and reduce to proving that the morphism $q^{-1}(x) \to Z \cap p^{-1}(x)$ induced by j is an isomorphism. We may now suppose that $X = \operatorname{Spec}(B)$, where $B = \mathscr{O}_{X,x}$, and that $E = B^m$, so that $S = B[T_1, \dots, T_m]$. The hypothesis means that there is a regular system of parameters $(t_i)_{1 \leqslant i \leqslant n}$ of B such that $u(T_i) = t_i$ for $1 \leqslant i \leqslant m$ ((**0**, 17.1.7)). Let $\mathfrak{J}$ be the image of u, and let $\mathfrak{m}$ be the maximal ideal of B. Then

$$\mathfrak{J}^n \otimes_B (B/\mathfrak{m}) = \mathfrak{J}^n/\mathfrak{m}\mathfrak{J}^n = (\mathfrak{J}^n/\mathfrak{J}^{n+1}) \otimes_B (B/\mathfrak{m}),$$

and consequently, by (16.9.4.1), $(\bigoplus_{n \geqslant 0} \mathfrak{J}^n) \otimes_B \mathbf{k}(x)$ is identified with $\mathbf{k}(x)[T_1, \dots, T_m]$. It follows that the immersion $q^{-1}(x) \to p^{-1}(x)$ is an isomorphism; this implies *a fortiori* that the closed sub-prescheme $Z \cap p^{-1}(x)$ is identical with $j(q^{-1}(x))$, and completes the proof of (19.4.11). □

19.5. Criteria for M-regularity We resume here, in completed form, the criteria for a sequence of elements of a ring A to be M-regular or M-quasi-regular, where M is an A-module, already studied in ((**0**, 15.1)).

Theorem (19.5.1). — *Let* A *be a ring,* $\mathbf{f} = (f_i)_{1 \leqslant i \leqslant n}$ *a sequence of elements of* A, *and* M *an* A*-module. Consider the following properties:*

a) *The sequence* $\mathbf{f}$ *is M-regular.*
b) $H_i(\mathbf{f}, M) = 0$ *for every* $i > 0$ *((***III***, 1.1.3)).*
b') $H_1(\mathbf{f}, M) = 0$.
c) *The sequence* $\mathbf{f}$ *is M-quasi-regular.*

Then we have the implications

$$a) \Longrightarrow b) \Longrightarrow b') \qquad and \qquad a) \Longrightarrow c).$$

Put $\mathfrak{J} = \sum_{i=1}^{n} f_i\mathrm{A}$. If the modules $\mathrm{M}_i = \mathrm{M}/(\sum_{j=1}^{i} f_j\mathrm{M})$ are separated for the $\mathfrak{J}$-preadic topology (respectively, if every quotient module of a submodule of M *is separated for the $\mathfrak{J}$-preadic topology), then we also have the implication c)⇒a) (respectively, b')⇒a)).*

Proof. The implications a)⇒c), and c)⇒a) when the M_i are separated for the $\mathfrak{J}$-preadic topology, were already proved in ((**0**, 15.1.9)) and are recalled only for completeness. We have also shown in ((**III**, 1.1.4)) and ((**III**, 1.1.3.3)) that a) implies b), and b) trivially implies b'). It therefore remains to prove that, when every quotient module of a submodule of M is separated for the $\mathfrak{J}$-preadic topology, b') implies a). We argue by induction on n. For $n = 1$, the assertion follows immediately from the definitions, since $\mathrm{H}_1(\mathbf{f}, \mathrm{M})$ is precisely the kernel in M of multiplication $z \mapsto f_1 z$ ((**III**, 1.1.1), (**III**, 1.1.2)). Suppose then that $n \geqslant 2$, and put $\mathbf{f}' = (f_i)_{1 \leqslant i \leqslant n-1}$. With the notation of ((**III**, 1.1.2)), $\mathrm{K}_\bullet(\mathbf{f}, \mathrm{M}) = \mathrm{K}_\bullet(f_n) \otimes_\mathrm{A} \mathrm{K}_\bullet(\mathbf{f}', \mathrm{M})$. We therefore have ((**III**, 1.1.4.1)) the exact sequence

$$0 \longrightarrow \mathrm{H}_0(f_n, \mathrm{H}_1(\mathbf{f}', \mathrm{M})) \longrightarrow \mathrm{H}_1(\mathbf{f}, \mathrm{M}) \longrightarrow \mathrm{H}_1(f_n, \mathrm{H}_0(\mathbf{f}', \mathrm{M})) \longrightarrow 0. \tag{19.5.1.1}$$

The relation $\mathrm{H}_1(\mathbf{f}, \mathrm{M}) = 0$ implies therefore $\mathrm{H}_0(f_n, \mathrm{H}_1(\mathbf{f}', \mathrm{M})) = 0$ and $\mathrm{H}_1(f_n, \mathrm{H}_0(\mathbf{f}', \mathrm{M})) = 0$. **IV-4** | 205
The first of these two relations means that we have $f_n\mathrm{H}_1(\mathbf{f}', \mathrm{M}) = \mathrm{H}_1(\mathbf{f}', \mathrm{M})$ (**III**, 1.1.3.5). Now, by definition (**III**, 1.1.1), $\mathrm{H}_1(\mathbf{f}', \mathrm{M})$ is isomorphic to a quotient of a submodule N of M^{n-1}; taking on M^{n-1} the filtration of the M^j for $j \leqslant n-1$, on N the induced filtration and on $\mathrm{H}_1(\mathbf{f}', \mathrm{M})$ the quotient filtration from that of N, we obtain on $\mathrm{H}_1(\mathbf{f}', \mathrm{M})$ a finite filtration whose quotients are isomorphic to quotients of submodules of M, hence are separated for the $\mathfrak{J}$-preadic topology by hypothesis. As the relation $f_n\mathrm{H}_1(\mathbf{f}', \mathrm{M}) = \mathrm{H}_1(\mathbf{f}', \mathrm{M})$ implies *a fortiori* $\mathfrak{J}\mathrm{H}_1(\mathbf{f}', \mathrm{M}) = \mathrm{H}_1(\mathbf{f}', \mathrm{M})$, we deduce from the hypothesis and the preceding remarks that $\mathrm{H}_1(\mathbf{f}', \mathrm{M}) = 0$. The induction hypothesis proves then that the sequence $\mathbf{f}'$ is M-regular. On the other hand, the relation $\mathrm{H}_1(f_n, \mathrm{H}_0(\mathbf{f}', \mathrm{M})) = 0$ means that f_n is $(\mathrm{M}/(\sum_{i=1}^{n-1} f_i\mathrm{M}))$-regular, hence the sequence $\mathbf{f}$ is M-regular. □

Corollary (19.5.2). — *Suppose that* A *is a Noetherian ring, that the f_i $(1 \leqslant i \leqslant n)$ belong to the radical of* A*, and that* M *is a finitely generated* A*-module. Then the four conditions* a), b), b'), c) *of* ((19.5.1)) *are equivalent.*

Proof. We know indeed from $(\mathbf{0}_\mathrm{I}, 7.3.5)$ that every finitely generated A-module is separated for the $\mathfrak{J}$-preadic topology. □

Proposition (19.5.3). — *Let*

$$0 \longrightarrow \mathrm{M}' \longrightarrow \mathrm{M} \longrightarrow \mathrm{M}'' \longrightarrow 0$$

be an exact sequence of A*-modules, $\mathbf{f} = (f_i)_{1 \leqslant i \leqslant n}$ an M-regular sequence, $\mathfrak{J} = \sum_{i=1}^{n} f_i\mathrm{A}$. Consider the following properties:*

a) *The sequence* **f** *is* M''*-regular.*
b) *The $\mathfrak{J}$-preadic filtration of* M' *is induced on* M' *by the $\mathfrak{J}$-preadic filtration of* M *(in other words, the canonical homomorphism $\mathrm{gr}^*_\mathfrak{J}(\mathrm{M}') \to \mathrm{gr}^*_\mathfrak{J}(\mathrm{M})$ is injective).*
c) *The canonical homomorphism $\mathrm{M}'/(\sum_{i=1}^{n} f_i\mathrm{M}') \to \mathrm{M}/(\sum_{i=1}^{n} f_i\mathrm{M})$ is injective.*

Then we have the implications a) ⇒ b) ⇒ c)*, and* a) *implies moreover that the sequence* **f** *is* M'*-regular. If every quotient of a submodule of* M'' *is separated for the $\mathfrak{J}$-preadic topology, the conditions* a), b) *and* c) *are equivalent.*

Proof. It is clear that $c)$ is a consequence of $b)$. Let us prove that under all the hypotheses of the last assertion, $c)$ implies $a)$: since **f** is M-regular, by (19.5.1), $\mathrm{H}_1(\mathbf{f}, \mathrm{M}) = 0$, whence an exact sequence, portion of the exact homology sequence

$$0 \longrightarrow \mathrm{H}_1(\mathbf{f}, \mathrm{M}'') \longrightarrow \mathrm{H}_0(\mathbf{f}, \mathrm{M}') \longrightarrow \mathrm{H}_0(\mathbf{f}, \mathrm{M}).$$

Now, condition $c)$ expresses that the homomorphism $\mathrm{H}_0(\mathbf{f}, \mathrm{M}') \to \mathrm{H}_0(\mathbf{f}, \mathrm{M})$ is injective (**III**, 1.1.3.5); it is therefore equivalent to $\mathrm{H}_1(\mathbf{f}, \mathrm{M}'') = 0$, whence, by virtue of the separation hypothesis and (19.5.1), the fact that **f** is M''-regular.

Let us show next that $a)$ implies $c)$, and also that the sequence **f** is M'-regular, proceeding by induction on n. For $n = 1$, f_1 being M-regular is also M'-regular and property $c)$ is precisely the lemma ((3.4.1.4)). For $n > 1$, the induction hypothesis shows that if we set $\mathbf{f}' = (f_1, \ldots, f_{n-1})$, the **IV-4** | 206
sequence $\mathbf{f}'$ is M'-regular and, by virtue of $c)$, we have an exact sequence

$$0 \longrightarrow \mathrm{M}'/(\textstyle\sum_{i=1}^{n-1} f_i\mathrm{M}') \to \mathrm{M}/(\sum_{i=1}^{n-1} f_i\mathrm{M}) \to \mathrm{M}''/(\sum_{i=1}^{n-1} f_i\mathrm{M}'') \to 0.$$

By hypothesis, f_n is $(\mathrm{M}/(\sum_{i=1}^{n-1} f_i\mathrm{M}))$-regular and $(\mathrm{M}''/(\sum_{i=1}^{n-1} f_i\mathrm{M}''))$-regular. The same reasoning shows, on the one hand, that f_n is also $(\mathrm{M}'/(\sum_{i=1}^{n-1} f_i\mathrm{M}'))$-regular and, on the other hand, by ((3.4.1.4)), that the sequence

$$0 \longrightarrow \mathrm{M}'/(\textstyle\sum_{i=1}^{n} f_i\mathrm{M}') \to \mathrm{M}/(\sum_{i=1}^{n} f_i\mathrm{M}) \to \mathrm{M}''/(\sum_{i=1}^{n} f_i\mathrm{M}'') \to 0$$

is exact, whence *c*).

Let us show finally that *a*) implies *b*). As the sequence $\mathbf{f}$ is then M-regular, M-quasi-regular, and M′-quasi-regular, we have canonical isomorphisms

$$\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}') \xrightarrow{\sim} \mathrm{gr}^0_{\mathfrak{J}}(\mathrm{M}')[\mathrm{T}_1,\dots,\mathrm{T}_n], \qquad \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}) \xrightarrow{\sim} \mathrm{gr}^0_{\mathfrak{J}}(\mathrm{M})[\mathrm{T}_1,\dots,\mathrm{T}_n]$$

$(\mathbf{0}, 15.1.7)$, and as we saw above that $\mathrm{gr}^0_{\mathfrak{J}}(\mathrm{M}') \to \mathrm{gr}^0_{\mathfrak{J}}(\mathrm{M})$ is injective, the same holds for $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}') \to \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$.

We note moreover that the equivalence of conditions *a*), *b*), *c*) of (19.5.3) holds in particular when A is Noetherian, M″ is a finitely generated A-module, and the f_i belong to the radical of A. ☐

(19.5.4). In the rest of this number, we preserve the notations A, $\mathbf{f}$, M, $\mathfrak{J}$, and we suppose moreover M endowed with a decreasing filtration (M_k) formed of sub-A-modules, such that $\mathrm{M}_0 = \mathrm{M}$. Recall $(\mathbf{0}, 15.1.5)$ that we then define on M a second decreasing filtration formed of submodules

$$\mathrm{M}'_k = \mathrm{M}_k + \mathfrak{J}\mathrm{M}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{M}_1 + \mathfrak{J}^k\mathrm{M}_0$$

and that, if $\mathrm{gr}_\bullet(\mathrm{M})$ and $\mathrm{gr}'_\bullet(\mathrm{M})$ are the graded A-modules associated to the filtrations (M_k) and (M'_k) respectively, we define a surjective graded homomorphism of degree 0 $(\mathbf{0}, 15.1.5.2)$

$$\psi_{\mathbf{M}} : (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1,\dots,\mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{M})$$

Recall also that we have the canonical surjective homomorphism $(\mathbf{0}, 15.1.1.1)$

$$\varphi_{\mathrm{gr}_\bullet(\mathrm{M})} : (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1,\dots,\mathrm{T}_n] \longrightarrow \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{gr}_\bullet(\mathrm{M}))$$

Theorem (19.5.5). — *With the notations of ((19.5.4)), consider the following properties:*

a) *The sequence* $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{M})$*-regular.*
b) *The canonical homomorphism* $\psi_{\mathbf{M}}$ *is bijective.*
c) *The canonical homomorphism* $\varphi_{\mathrm{gr}_\bullet(\mathrm{M})}$ *is bijective (in other words* $(\mathbf{0}, 15.1.7)$*, the sequence* $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{M})$*-quasi-regular).*

Then we have the implications[37] **IV-4** | 207

$$a) \Rightarrow b) \Rightarrow c).$$

Moreover, if for every integer $k \geqslant 0$*, the quotient modules of submodules of* $\mathrm{gr}_k(\mathrm{M})$ *are separated for the* $\mathfrak{J}$*-preadic topology (which is the case when* A *is Noetherian, the* f_i *belong to the radical of* A*, and the* $\mathrm{gr}_k(\mathrm{M})$ *are finitely generated* A*-modules),* then all conditions a), b) *and* c) are equivalent.

Proof. The implication *a*) ⇒ *b*) has been demonstrated in $(\mathbf{0}, 15.1.8)$; the implication *c*) ⇒ *a*) under the separation hypothesis is a particular case of $(\mathbf{0}, 15.1.9)$. It remains to prove that *b*) implies *c*), which will be done in several steps.

We denote by $\mathfrak{J}$ the ideal $\sum_{i=1}^{n} f_i\mathrm{A}$ generated by the f_i; for every sequence $\mathbf{q} = (q_1,\dots,q_n)$ of n integers $\geqslant 0$, we set $|\mathbf{q}| = \sum_{i=1}^{n} q_i$ and $\mathbf{f}^{\mathbf{q}} = f_1^{q_1} f_2^{q_2} \cdots f_n^{q_n}$.

(19.5.5.1). For the map $\psi_{\mathbf{M}}$ to be injective (and hence bijective), it is necessary and sufficient that the following condition be satisfied:

$(\mathrm{I}_{q,p})$ *For every family* $(x^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ *of elements of* M_p*, the relation*

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x^p_{\mathbf{q}} \in \sum_{i=1}^{q} \mathfrak{J}^{q-i}\mathrm{M}_{p+i} + \mathrm{M}'_{p+q+1}$$

implies $x^p_{\mathbf{q}} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$ *for every* $\mathbf{q}$ *such that* $|\mathbf{q}| = q$*.*

We proceed as in $(\mathbf{0}, 15.1.6)$ by considering the sub-A-module Q_k (resp. Q'_k) as a term of degree k in the first (resp. second) member of $(\mathbf{0}, 15.1.5.2)$. We endow Q_k with the filtration

$$(\mathrm{Q}_k)_i = \textstyle\sum_{j\leqslant k-i} (\sum_{|\mathbf{l}|=j} (\mathrm{gr}_{k-j}(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})))\mathrm{T}^{\mathbf{l}}$$

[37] The implication *b*) ⇒ *c*), without the separation hypothesis on the $\mathrm{gr}_k(\mathrm{M})$, is due to P. Deligne, whose demonstration we reproduce below.

and Q'_k the image filtration formed by $\psi_{\mathbf{M}}((Q_k)_i) = (Q'_k)_i$; it suffices then to prove that the homomorphisms $(\psi_{\mathbf{M}})_{ki} : \mathrm{gr}_i(Q_k) \to \mathrm{gr}_i(Q'_k)$ are injective. Now, we have

$$\mathrm{gr}_i(Q_k) = \textstyle\sum_{|\mathbf{l}|=k-i}\big((M_i/M_{i+1}) \otimes_A (A/\mathfrak{J})\big)T^{\mathbf{l}} = \sum_{|\mathbf{l}|=k-i}\big(M_i/(\mathfrak{J}M_i + M_{i+1})\big)T^{\mathbf{l}}$$

on the other hand, $(Q'_k)_{i+1}$ is the image of $\sum_{h=0}^{k-i-1} \mathfrak{J}^{k-i-h-1}M_{i+h+1}$ in M'_k/M'_{k+1}. To write that $(\psi_{\mathbf{M}})_{ki}$ is injective is therefore equivalent to writing the condition $(I_{k-i,i})$; whence our assertion. **IV-4** | 208

(19.5.5.2). *For the sequence* $\mathbf{f}$ *to be* $\mathrm{gr}_\bullet(M)$*-quasi-regular, it suffices that, for* $p \geqslant 0$ *and* $q \geqslant 0$*, the following condition be satisfied:*

$(S_{q,p})$ *For every family* $(x^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ *of elements of* M_p*, the relation*

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x^p_{\mathbf{q}} \in M_{p+1}$$

implies $x^p_{\mathbf{q}} \in M_{p+1} + \mathfrak{J}M_p$ *for every* $\mathbf{q}$ *such that* $|\mathbf{q}| = q$*.*

By definition, saying that $\mathbf{f}$ is $\mathrm{gr}_\bullet(M)$-quasi-regular means that, for every $p \geqslant 0$ and every $q \geqslant 0$, the relation

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x^p_{\mathbf{q}} \in M_{p+1} + \mathfrak{J}^{q+1}M_p$$

for a family $(x^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ of elements of M_p implies $x^p_{\mathbf{q}} \in M_{p+1} + \mathfrak{J}M_p$ for every $\mathbf{q}$ such that $|\mathbf{q}| = q$. The condition $(S_{q,p})$ means that there exists a family $(y^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ of elements of $\mathfrak{J}M_p$ such that $\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}}(x^p_{\mathbf{q}} - y^p_{\mathbf{q}}) \in M_{p+1}$; it then implies $x^p_{\mathbf{q}} - y^p_{\mathbf{q}} \in M_{p+1} + \mathfrak{J}M_p$, whence $x^p_{\mathbf{q}} \in M_{p+1} + \mathfrak{J}M_p$; hence one sees that the conditions $(S_{q,p})$ imply that the sequence $\mathbf{f}$ is $\mathrm{gr}_\bullet(M)$-quasi-regular.

It suffices therefore to prove the following proposition:

(19.5.5.3). *For every* $r \geqslant 0$*, if the conditions* $(I_{q,p})$ *are satisfied for* $p + q < r$*, then the conditions* $(S_{q,p})$ *are satisfied for* $p + q < r$*.*

Let us introduce the conditions

$(A_{q,p})$ *For every family* $(x^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ *of elements of* M_p*, the relation*

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x^p_{\mathbf{q}} \in M_{p+1}$$

implies

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x^p_{\mathbf{q}} \in \sum_{i=1}^{q} \mathfrak{J}^{q-i}M_{p+i}.$$

It is clear that the conjunction of $(A_{q,p})$ and $(I_{q,p})$ implies $(S_{q,p})$. On the other hand, the conditions $(A_{1,p})$ and $(S_{0,p})$ are trivial. Consequently, (19.5.5.3) will follow from the following proposition:

(19.5.5.4). *For all* $p \geqslant 0$ *and* $q \geqslant 1$*, the condition* $(A_{q+1,p})$ *is implied by the conjunction of the conditions* $(A_{q,p})$*,* $(I_{q-1,p+1})$*,* $(I_{q-2,p+2})$*, …,* $(I_{0,p+q})$*.*

Indeed, once this proposition is proved, the conditions $(I_{q,p})$ supposed satisfied for $p + q < r$ will imply, for each $p \leqslant r$, the condition $(A_{q,p})$ for $1 \leqslant q < r - p$, by induction on q.

Let us prove (19.5.5.4). Let us note that the condition $(A_{q,p})$ is also equivalent to

(19.5.5.5).

$$\mathfrak{J}^qM_p \cap M_{p+1} \subset \textstyle\sum_{i=1}^{q} \mathfrak{J}^{q-i}M_{p+i}.$$

Let therefore $m \in \mathfrak{J}^{q+1}M_p \cap M_{p+1} \subset \mathfrak{J}^qM_p \cap M_{p+1}$. Applying condition $(A_{q,p})$, we see that for each i such that $1 \leqslant i \leqslant q$, there exists a family $(y^{p+i}_{\mathbf{q}})_{|\mathbf{q}|=q-i}$ of elements of M_{p+i} such that

$$m = \textstyle\sum_{i=1}^{q}\big(\sum_{|\mathbf{q}|=q-i} \mathbf{f}^{\mathbf{q}} y^{p+i}_{\mathbf{q}}\big).$$

Suppose that for $1 \leqslant i < j$ ($j \geqslant 1$), we have proved that

$$y^{p+i}_{\mathbf{q}} \in \mathfrak{J}M_{p+i} + M_{p+i+1} \qquad (\text{for } |\mathbf{q}| = q - i).$$

We deduce by definition (19.5.4)

$$m - \textstyle\sum_{i<j}\big(\sum_{|\mathbf{q}|=q-i} \mathbf{f}^{\mathbf{q}} y^{p+i}_{\mathbf{q}}\big) \in M'_{p+q+1},$$

Since by hypothesis $(I_{q-j,p+j})$ is satisfied, we deduce

$$y^{p+j}_{\mathbf{q}} \in \mathfrak{J}M_{p+j} + M_{p+j+1}$$

and, by induction on j, one sees therefore that this condition is satisfied for *every* j such that $1 \leqslant j \leqslant q$. We have therefore **IV-4 | 209**

$$m \in \sum_{i=1}^{q} \mathfrak{J}^{q-i}(\mathfrak{J}\mathrm{M}_{p+i} + \mathrm{M}_{p+i+1}) = \sum_{i=1}^{q+1} \mathfrak{J}^{q-i+1}\mathrm{M}_{p+i}$$

and we have thus proved $(\mathrm{A}_{q+1,p})$ and completed the demonstration of (19.5.5).

□

Remarks (19.5.6). —

(i) The results of this number remain valid when, instead of an A-module M and elements $f_i \in \mathrm{A}$, one takes an object M of any abelian category, and n endomorphisms f_i of M that mutually commute; the demonstrations adapt without difficulty to this more general case. The hypothesis of separation for an A-module N relative to the $\mathfrak{J}$-preadic topology must here be replaced by the hypothesis that every sub-object of N contained in all the $\sum f_i^r(\mathrm{N})$ (r an arbitrary integer) is necessarily zero. The same remark applies to the results of (19.7).

(ii) Suppose that A is a *graded* ring in degrees $\geqslant 0$. If M (resp. each $\mathrm{gr}_p(\mathrm{M})$) is a graded A-module with degrees bounded below, and if the f_i contain no homogeneous component of degree 0, the separation hypothesis of $(\mathbf{0}, 15.1.9)$ is *ipso facto* satisfied for M (resp. each $\mathrm{gr}_p(\mathrm{M})$), and one sees that in this case we have the implication $c) \Rightarrow a)$ in (19.5.1) (resp. (19.5.5)).

(iii) Recall that, without the separation hypothesis, the implication $c) \Rightarrow b)$ is no longer valid even for $n = 1$ $(\mathbf{0}, 15.1.12, \text{(iii)})$.

19.6. Regular sequences relative to a filtered quotient module

(19.6.1). Let A be a ring, $\mathbf{f} = (f_1, \ldots, f_n)$ a finite sequence of elements of A, $\mathfrak{J} = f_1\mathrm{A} + \ldots + f_n\mathrm{A}$ the ideal it generates, and consider an exact sequence of A-modules

$$0 \longrightarrow \mathrm{R} \xrightarrow{j} \mathrm{N} \xrightarrow{p} \mathrm{M} \longrightarrow 0$$

where one supposes that N is endowed with a decreasing filtration (N_k) of sub-A-modules, with $\mathrm{N}_0 = \mathrm{N}$. We set $\mathrm{R}_k = \mathrm{R} \cap \mathrm{N}_k$ (filtration induced by (N_k)), $\mathrm{M}_k = p(\mathrm{N}_k)$ (quotient filtration of (N_k)) and we denote by $\mathrm{gr}_\bullet(\mathrm{N})$, $\mathrm{gr}_\bullet(\mathrm{R})$ and $\mathrm{gr}_\bullet(\mathrm{M})$ the graded A-modules associated with these three filtrations. We set on the other hand for all k,

$$\mathrm{N}'_k = \mathrm{N}_k + \mathfrak{J}\mathrm{N}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{N}_1 + \mathfrak{J}^k\mathrm{N}_0$$

so that $\mathrm{M}'_k = p(\mathrm{N}'_k) = \mathrm{M}_k + \mathfrak{J}\mathrm{M}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{M}_1 + \mathfrak{J}^k\mathrm{M}_0$; we set on the other hand $\mathrm{R}'_k = \mathrm{R} \cap \mathrm{N}'_k$ (filtration *induced* by (N'_k)), and we denote by $\mathrm{gr}'_\bullet(\mathrm{N})$, $\mathrm{gr}'_\bullet(\mathrm{M})$ and $\mathrm{gr}'_\bullet(\mathrm{R})$ the graded A-modules associated with these three filtrations; we thus have a commutative diagram of exact sequences

(19.6.1.1)
$$\begin{tikzcd}
0 \arrow[r] & \mathrm{gr}_\bullet(\mathrm{R}) \arrow[r, "\mathrm{gr}(j)"] \arrow[d] & \mathrm{gr}_\bullet(\mathrm{N}) \arrow[r, "\mathrm{gr}(p)"] \arrow[d] & \mathrm{gr}_\bullet(\mathrm{M}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{gr}'_\bullet(\mathrm{R}) \arrow[r, "\mathrm{gr}'(j)"] & \mathrm{gr}'_\bullet(\mathrm{N}) \arrow[r, "\mathrm{gr}'(p)"] & \mathrm{gr}'_\bullet(\mathrm{M}) \arrow[r] & 0
\end{tikzcd}$$

where the vertical arrows are the canonical homomorphisms from a graded module associated with a filtration to a graded module associated with a coarser filtration. **IV-4 | 210**

We note that if one sets

$$\mathrm{R}''_k = \mathrm{R}_k + \mathfrak{J}\mathrm{R}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{R}_1 + \mathfrak{J}^k\mathrm{R}_0$$

we evidently have $\mathrm{R}''_k \subset \mathrm{R}'_k$, but the filtrations (R''_k) and (R'_k) are in general distinct; we denote by $\mathrm{gr}''_\bullet(\mathrm{R})$ the graded A-module associated with the filtration (R''_k).

(19.6.2). We defined in $(\mathbf{0}, 15.1.5.2)$ the surjective graded homomorphisms of degree 0

$$\psi_\mathrm{N} : (\mathrm{gr}_\bullet(\mathrm{N}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{N})$$
$$\psi_\mathrm{M} : (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{M})$$
$$\psi_\mathrm{R} : (\mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}''_\bullet(\mathrm{R})$$

the first two of which are deduced from the last two vertical arrows of (19.6.1.1).

As the filtration (R''_k) is finer than (R'_k), on R we have a canonical homomorphism $\gamma : \mathrm{gr}''_\bullet(\mathrm{R}) \to \mathrm{gr}'_\bullet(\mathrm{R})$; we will denote by ψ'_{R} the composite homomorphism $\gamma \circ \psi_{\mathrm{R}}$. It then follows immediately from the definitions that we have a commutative diagram

(19.6.2.1)
$$\begin{tikzcd}
& 0 \arrow[d] \\
(\mathrm{gr}_\bullet(\mathrm{R}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \arrow[r, "\psi'_{\mathrm{R}}"] \arrow[d, "j'"] & \mathrm{gr}'_\bullet(\mathrm{R}) \arrow[d, "\mathrm{gr}'(j)"] \\
(\mathrm{gr}_\bullet(\mathrm{N}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \arrow[r, "\psi_{\mathrm{N}}"] \arrow[d, "p'"] & \mathrm{gr}'_\bullet(\mathrm{N}) \arrow[d, "\mathrm{gr}'(p)"] \\
(\mathrm{gr}_\bullet(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \arrow[r, "\psi_{\mathrm{M}}"] \arrow[d] & \mathrm{gr}'_\bullet(\mathrm{M}) \arrow[d] \\
0 & 0
\end{tikzcd}$$

where the columns are exact.

Proposition (19.6.3). — *With the preceding notations, suppose that the sequence* $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{N})$*-regular. Consider the following properties:*

a) $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{M})$*-regular.*
b) ψ_{M} *is bijective.*
c) ψ'_{R} *is surjective (in other words* $\mathrm{gr}'_\bullet(\mathrm{R})$ *is generated as* $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{A})$*-module by* $\mathrm{Im}(\mathrm{gr}_\bullet(\mathrm{R}) \to \mathrm{gr}'_\bullet(\mathrm{R}))$*).*
d) ψ'_{R} *is injective.*
d′) *The homomorphism* $(\psi'_{\mathrm{R}})_0 : \mathrm{gr}_\bullet(\mathrm{R}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}) \to \mathrm{gr}'_\bullet(\mathrm{R})$ *obtained by restricting* ψ'_{R} *to polynomials of degree* 0*, is injective.*
e) ψ'_{R} *is bijective.*
f) j' *is injective.* **IV-4** | 211
f′) *The homomorphism* $\mathrm{gr}_\bullet(\mathrm{R}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}) \to \mathrm{gr}_\bullet(\mathrm{N}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ *obtained by restricting* j' *to polynomials of degree* 0*, is injective.*
g) *The* $\mathfrak{J}$*-preadic filtration of* $\mathrm{gr}_\bullet(\mathrm{R})$ *is induced by that of* $\mathrm{gr}_\bullet(\mathrm{N})$*.*
h) *The filtrations* (R'_k) *and* (R''_k) *on* R *are identical.*

Then we have the following implications:

$$\begin{tikzcd}
a) \arrow[r, Rightarrow] \arrow[dr, Rightarrow] & e) \arrow[r, Rightarrow] \arrow[dr, Rightarrow] & b) \arrow[r, Leftrightarrow] & c) \arrow[r, Leftrightarrow] & h) \\
& g) \arrow[r, Rightarrow] & d) \arrow[r, Leftrightarrow] & d') \arrow[r, Leftrightarrow] & f) \arrow[r, Leftrightarrow] & f')
\end{tikzcd}$$

Moreover, when every quotient module of a submodule of $\mathrm{gr}_k(\mathrm{M})$ *is separated for the* $\mathfrak{J}$*-preadic topology (which is the case when* A *is Noetherian, the* f_i *belong to the radical of* A*, and the* $\mathrm{gr}_k(\mathrm{M})$ *are finitely generated* A*-modules),* then all conditions a) *to* h) are equivalent.

Proof. Let us note that, from (19.5.5), the hypothesis that $\mathbf{f}$ is $\mathrm{gr}_\bullet(\mathrm{N})$-regular implies that ψ_{N} is *bijective*; as $\mathrm{gr}'(j)$ is injective, the equivalence of $d)$ and $f)$ results from the diagram (19.6.2.1). Likewise $d')$ and $f')$ are equivalent, while $f)$ and $f')$ are trivially equivalent; this proves the equivalence of $d)$, $d')$, $f)$ and $f')$.

As one already knows that ψ_{M} is surjective and ψ_{N} bijective, the equivalence of $b)$ and $c)$ also follows from the diagram (19.6.2.1) by a particular case of the five lemma.

The relation $\psi'_{\mathrm{R}} = \gamma \circ \psi_{\mathrm{R}}$ (19.6.2) and the fact that ψ_{R} is surjective show that condition $c)$ is equivalent to saying that γ is surjective, which is also equivalent to condition $h)$; hence we have proved the equivalence of $b)$, $c)$ and $h)$.

It is trivial that $e)$ is equivalent to the conjunction of $c)$ and $d)$, and hence also of $b)$ and $d)$. Let us note now that $a)$ implies $b)$ and that $b)$ implies $a)$ under the separation hypothesis (19.5.5). On the other hand, the implications $a) \Rightarrow g) \Rightarrow f')$, and the implication $f') \Rightarrow a)$ under the separation hypothesis, result from (19.5.3). Hence we have proved $a)$ implies $e)$ (equivalent to the conjunction of $b)$ and $f')$), and also that all conditions are equivalent under the separation hypothesis. □

Corollary (19.6.4). — *Let $\mathfrak{R}$ be an ideal of A, $\mathfrak{L} = \mathfrak{J} + \mathfrak{R}$. Suppose that the filtration (hence also (M_k)) is the $\mathfrak{R}$-preadic filtration, so that the filtrations (M'_k) and (N'_k) are $\mathfrak{L}$-preadic filtrations. Consider $\mathrm{gr}_\bullet(\mathrm{R})$ (resp. $\mathrm{gr}'_\bullet(\mathrm{R})$) as a graded module over $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$ (resp. $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$).*

- (i) *Let S be a part of $\mathrm{gr}_\bullet(\mathrm{R})$ that generates $\mathrm{gr}_\bullet(\mathrm{R})$ as a $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$-module. Then condition* c) *of* ((19.6.3)) *is equivalent to the following condition:*
- c′) *The image of S in $\mathrm{gr}'_\bullet(\mathrm{R})$ by ψ'_{R} generates $\mathrm{gr}'_\bullet(\mathrm{R})$ as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module.*
- (ii) *Suppose condition* e) *of* ((19.6.3)) *is satisfied, and also that the quotient modules of submodules of $\mathrm{gr}_k(\mathrm{R})$ are separated for the $\mathfrak{J}$-preadic topology (which is the case when A is Noetherian, the f_i belong to the radical of A, and the $\mathrm{gr}_k(\mathrm{R})$ are finitely generated A-modules). Then,* for a part S of $\mathrm{gr}_\bullet(\mathrm{R})$ formed of homogeneous elements to generate $\mathrm{gr}_\bullet(\mathrm{R})$ as a $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$-module, it is necessary and sufficient that its image by ψ'_{R} generates $\mathrm{gr}'_\bullet(\mathrm{R})$ as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module.

Proof. **IV-4** | 212

- (i) Consider the homomorphism of degree 0

$$\psi_{\mathrm{A}} : (\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{A}) = \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$$

  ($\mathbf{0}$, 15.1.5.2). The fact that this homomorphism is *surjective* implies that the sub-$\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module of $\mathrm{gr}'_\bullet(\mathrm{R})$ generated by $\mathrm{S}' = \psi'_{\mathrm{R}}(\mathrm{S})$ is none other than $\mathrm{Im}(\psi'_{\mathrm{R}})$, whence the equivalence of $c)$ and $c')$.
- (ii) Since condition $e)$ is satisfied, it follows by (i) that it remains to prove the sufficiency of the condition in the statement. Now, as ψ'_{R} is bijective, to say that S' generates $\mathrm{gr}'_\bullet(\mathrm{R})$ considered as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module amounts to saying that S generates $(\mathrm{gr}_\bullet(\mathrm{R}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \dots, \mathrm{T}_n]$ as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module, and by the surjectivity of ψ_{A}, we can in this statement replace the ring $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$ by $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \dots, \mathrm{T}_n]$. It amounts to saying that S generates $\mathrm{gr}_\bullet(\mathrm{R}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ as a module over the ring $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$, or also as a $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$-module. Now, if T is the graded sub-$\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$-module of $\mathrm{gr}_\bullet(\mathrm{R})$ generated by S, and if we set $\mathrm{T}' = \mathrm{gr}_\bullet(\mathrm{R})/\mathrm{T}$, then by hypothesis $\mathrm{T}'/\mathfrak{J}\mathrm{T}' = 0$, or $\mathfrak{J}\mathrm{T}' = \mathrm{T}'$; but T' is a graded module, and each T'_p is a quotient of a submodule of $\mathrm{gr}_p(\mathrm{R})$, hence separated for the $\mathfrak{J}$-preadic topology. The hypothesis therefore implies $\mathrm{T}' = 0$, which completes the proof of the corollary.

□

19.7. Hironaka's normal flatness criterion

Theorem (19.7.1). — (Hironaka). *Let A be a ring, $\mathfrak{R}$ an ideal of A, $\mathbf{f} = (f_1, \dots, f_n)$ a sequence of elements of A that is $(\mathrm{A}/\mathfrak{R})$-regular, $\mathfrak{J} = f_1\mathrm{A} + \ldots + f_n\mathrm{A}$ the ideal generated by $\mathbf{f}$, $\mathfrak{L} = \mathfrak{J} + \mathfrak{R}$, M an A-module. In this case $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}) = (\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{L}) = (\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L})$, since $(\mathfrak{R}^n\mathrm{M}/\mathfrak{R}^{n+1}\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}) = \mathfrak{R}^n\mathrm{M}/(\mathfrak{R}^{n+1} + \mathfrak{J}\mathfrak{R}^n)\mathrm{M} = \mathfrak{R}^n\mathrm{M}/(\mathfrak{R}^{n+1} + \mathfrak{L}\mathfrak{R}^n)\mathrm{M}$, and the last equality is a result of Bourbaki,* Alg.*, chap. II, 3^e ed., § 3, n^o 7, cor. 2 of prop. 6.*

Consider the following conditions:

- a) $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{R})$*-module.*
- b) $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{L})$*-module and the canonical homomorphism* ($\mathbf{0}$, 15.1.5.2)

$$\psi_{\mathrm{M}} : ((\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \longrightarrow \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$$

  is bijective.
- c) $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L})$ *is a flat* $(\mathrm{A}/\mathfrak{L})$*-module and the sequence* $\mathbf{f}$ *is* $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}))$*-regular.*
- d) $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{L})$*-module, and for every* $\mathfrak{p} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{R}}(\mathrm{A}/\mathfrak{L})$, $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}))_{\mathfrak{p}}$ *is a flat* $(\mathrm{A}/\mathfrak{R})_{\mathfrak{p}}$*-module.*

Then we have the implications

$$\begin{tikzcd} a) \arrow[r, Rightarrow] \arrow[dr, Rightarrow] & c) \arrow[r, Rightarrow] & b) \\ & d) & \end{tikzcd}$$

When every quotient of a submodule of $\mathrm{gr}^k_{\mathfrak{N}}(\mathrm{M})$ $(k \geqslant 0)$ *is ideally separated for* $\mathfrak{J}$ *(Bourbaki,* Alg. comm.*, chap. III, § 5, nº 1), the conditions* a), b), c) *are equivalent.*

Finally, when A *is Noetherian, the* f_i *belong to the radical of* A*, the sequence* $\mathbf{f}$ *is* $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A})$*-regular, and* M *is a finitely generated* A*-module, the conditions* a), b), c), d) *are equivalent.* IV-4 | 213

Proof. The implication $a) \Rightarrow c)$ is immediate $(\mathbf{0}, 15.1.13)$. If $c)$ is satisfied, it results from (19.5.5) that ψ_{M} is bijective; moreover, $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is a flat $(\mathrm{A}/\mathfrak{L})$-module, and hence so is $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$, since ψ_{M} is an $(\mathrm{A}/\mathfrak{L})$-isomorphism; thus $c)$ implies $b)$. The fact that $a)$ also implies $d)$ is immediate.

When every quotient of a submodule of $\mathrm{gr}^k_{\mathfrak{N}}(\mathrm{M})$ is ideally separated for $\mathfrak{J}$, it results from (19.5.5) that condition $b)$ implies that $\mathbf{f}$ is $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}))$-regular; moreover, $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is isomorphic to $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ and hence is flat over $\mathrm{A}/\mathfrak{L}$. Its degree-zero direct summand $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L})$ is therefore flat; this proves that $b)$ implies $c)$. On the other hand, under the same hypothesis, $c)$ implies $a)$ by virtue of $(\mathbf{0}, 15.1.21)$ (where we can replace the Noetherian hypothesis by the hypothesis that M is ideally separated for $\mathfrak{J}$, by virtue of $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$).

It remains therefore to prove that when A is Noetherian, M of finite type, and the f_i belong to the radical of A, $d)$ implies $a)$.

There exists a free A-module N of *finite type* and an exact sequence $0 \to \mathrm{R} \to \mathrm{N} \to \mathrm{M} \to 0$. It suffices to prove that $d)$ implies $b)$, since every quotient of a sub-$\mathrm{gr}^k_{\mathfrak{N}}(\mathrm{M})$ is then ideally separated for $\mathfrak{J}$, by virtue of the fact that $\mathrm{gr}^k_{\mathfrak{N}}(\mathrm{M})$ is a finitely generated A-module, that $\mathfrak{J}$ is contained in the radical of A, and that A is Noetherian (Bourbaki, *Alg. comm.*, chap. III, § 5, nº 1); in other words, it suffices to show that ψ_{M} is *bijective*.

As N is free and the sequence $\mathbf{f}$ is $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A})$-regular by hypothesis, it is also $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{N})$-regular. One can therefore apply (19.6.3), since every quotient module of a sub-$\mathrm{gr}_k(\mathrm{M})$ is separated for the $\mathfrak{J}$-preadic topology, the f_i belonging to the radical of A by hypothesis. The diagram (19.6.2.1) here reads

(19.7.1.1)

$$\begin{tikzcd}
 & 0 \arrow[d] \\
(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{R},\mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \arrow[r, "\psi'_{\mathrm{R}}"] \arrow[d] & \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{R},\mathrm{N}) \arrow[d] \\
(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \arrow[r, "\psi_{\mathrm{N}}"] \arrow[d] & \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{N}) \arrow[d] \\
(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \arrow[r, "\psi_{\mathrm{M}}"] \arrow[d] & \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M}) \arrow[d] \\
0 & 0
\end{tikzcd}$$

where $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{R},\mathrm{N})$ (resp. $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{R},\mathrm{N})$) is the graded associated to R for the filtration $\mathrm{R} \cap \mathfrak{N}^m\mathrm{N}$ (resp. $\mathrm{R} \cap \mathfrak{L}^m\mathrm{N}$); recall that in this diagram the columns are exact and that ψ_{N} is *bijective*. IV-4 | 214

(19.7.1.2). Set $\mathrm{B} = \mathrm{gr}^*_{\mathfrak{N}} \mathrm{A} \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L})$, $\mathrm{P} = \mathrm{gr}^*_{\mathfrak{N}}(\mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L})$, $\mathrm{Q} = \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{R},\mathrm{N})$; since ψ_{N} is bijective, $\mathrm{P}[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ identifies with $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{N})$, and Q is a sub-$\mathrm{B}[\mathrm{T}_1, \ldots, \mathrm{T}_n]$-module of $\mathrm{P}[\mathrm{T}_1, \ldots, \mathrm{T}_n]$; we will first show that we have

(19.7.1.3).

$$\mathrm{Q} = (\mathrm{Q} \cap \mathrm{P})[\mathrm{T}_1, \ldots, \mathrm{T}_n].$$

For this, set $\mathrm{Z} = \mathrm{Q}/((\mathrm{Q} \cap \mathrm{P})[\mathrm{T}_1, \ldots, \mathrm{T}_n])$; this is a sub-$\mathrm{B}[\mathrm{T}_1, \ldots, \mathrm{T}_n]$-module of $\mathrm{P}[\mathrm{T}_1, \ldots, \mathrm{T}_n]/((\mathrm{Q} \cap \mathrm{P})[\mathrm{T}_1, \ldots, \mathrm{T}_n]) = (\mathrm{P}/(\mathrm{Q} \cap \mathrm{P}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$. As an $(\mathrm{A}/\mathfrak{L})$-module, it is therefore isomorphic to a submodule of a direct sum of $(\mathrm{A}/\mathfrak{L})$-modules isomorphic to $\mathrm{P}/(\mathrm{Q} \cap \mathrm{P}) = (\mathrm{P} + \mathrm{Q})/\mathrm{Q}$. But $(\mathrm{P} + \mathrm{Q})/\mathrm{Q}$ is an $(\mathrm{A}/\mathfrak{L})$-module isomorphic to a submodule of $\mathrm{P}[\mathrm{T}_1, \ldots, \mathrm{T}_n]/\mathrm{Q}$, which is none other than $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ by the diagram (19.7.1.1).

But each $\mathrm{gr}^m_{\mathfrak{L}}(\mathrm{M})$ is by hypothesis a *flat* $(\mathrm{A}/\mathfrak{L})$-module of finite type, hence projective, and we have the inclusion $\mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})) \subset \mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$. To see that $\mathrm{Z} = 0$, it suffices by (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 1 of prop. 2 and n° 3, prop. 7) to see that for every $\mathfrak{q} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$, $\mathrm{Z}_{\mathfrak{q}} = 0$. But the ideals of $\mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$ are of the form $\mathfrak{p}/(\mathfrak{L}/\mathfrak{R})$, where $\mathfrak{p} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{R}}(\mathrm{A}/\mathfrak{L})$, and it comes down to the same to see that $\mathrm{Z}_{\mathfrak{p}} = 0$ for every $\mathfrak{p} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{R}}(\mathrm{A}/\mathfrak{L})$. Now, if $\mathfrak{p} = \mathfrak{r}/\mathfrak{R}$, where $\mathfrak{r}$ is a prime ideal of A, then $(\mathrm{A}/\mathfrak{L})_{\mathfrak{p}} = \mathrm{A}_{\mathfrak{r}}/\mathfrak{L}_{\mathfrak{r}}$, and the images of the f_i in $\mathrm{A}_{\mathfrak{r}}$ form a $\mathrm{gr}^*_{\mathfrak{R}_{\mathfrak{r}}}(\mathrm{M}_{\mathfrak{r}})$-regular sequence $(\mathbf{0}, 15.1.14)$; applying to $\mathrm{A}_{\mathfrak{r}}$, $\mathfrak{R}_{\mathfrak{r}}$ and $\mathrm{M}_{\mathfrak{r}}$ the implication $a) \Rightarrow b)$ of the statement, we see from the hypothesis $d)$ that $\psi_{\mathrm{M}_{\mathfrak{r}}}$ is bijective, hence also $\psi'_{\mathrm{R}_{\mathfrak{r}}}$ by virtue of (19.6.3); in other words, $\mathrm{Q}_{\mathfrak{r}} = \mathrm{Q}'_{\mathfrak{r}}[\mathrm{T}_1, \dots, \mathrm{T}_n]$, where we have set $\mathrm{Q}' = \mathrm{gr}^*_{\mathfrak{R}}(\mathrm{R}, \mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L})$, which is here a sub-B-module of P; in particular $\mathrm{Q}'_{\mathfrak{r}} \subset \mathrm{Q}_{\mathfrak{r}} \cap \mathrm{P}_{\mathfrak{r}}$; hence finally $\mathrm{Z}_{\mathfrak{r}} = \mathrm{Z}_{\mathfrak{p}} = 0$, which completes the demonstration of (19.7.1.3).

(19.7.1.5). By virtue of (19.7.1.3), to prove that ψ'_{R} is surjective, it suffices to show that every homogeneous element of degree m of $\mathrm{Q} \cap \mathrm{P}$ is the image of an element of the same degree of $\mathrm{Q}' = \mathrm{gr}^*_{\mathfrak{R}}(\mathrm{R}, \mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L})$. Now, the elements of degree m of P identify (by ψ_{N}) with the elements of $(\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N})/\mathfrak{L}^{m+1}\mathrm{N}$; those which belong to Q identify with the elements of $((\mathrm{R} \cap \mathfrak{L}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N})/\mathfrak{L}^{m+1}\mathrm{N}$; and the images of the elements of degree m of Q' identify with the elements of $((\mathrm{R} \cap \mathfrak{R}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N})/\mathfrak{L}^{m+1}\mathrm{N}$. It will therefore suffice to prove, for $m > 0$, the inclusion

$$((\mathrm{R} \cap \mathfrak{L}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N}) \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}) \subset (\mathrm{R} \cap \mathfrak{R}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N}.$$

As $\mathfrak{R} \subset \mathfrak{L}$, we see immediately that the first member of this relation is equal to $(\mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N})) + \mathfrak{L}^{m+1}\mathrm{N}$, so that everything reduces to proving

(19.7.1.6).

$$\mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}) \subset (\mathrm{R} \cap \mathfrak{R}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N}.$$

We will proceed by induction on m, assuming therefore (19.7.1.6) verified when m is replaced by an integer $m' < m$. We will consider on the other hand, for m fixed and for an integer $d \geqslant 0$, the relation **IV-4 | 215**

$$(*_d) \qquad \mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}) \subset (\mathrm{R} \cap \mathfrak{R}^d\mathrm{N} \cap \mathfrak{L}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N}.$$

It is clear that it suffices to demonstrate $(*_d)$ for $d = m$, and on the other hand $(*_0)$ is trivially true. We will prove $(*_d)$ by *induction on d*; in other words, we suppose, for a $d \geqslant 0$ fixed (with $d < m$), that $(*_d)$ is true, and we wish to prove

$$(*_{d+1}) \qquad \mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}) \subset (\mathrm{R} \cap \mathfrak{R}^{d+1}\mathrm{N} \cap \mathfrak{L}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N}.$$

Consider for this, for an $h \geqslant 0$, the relation

$$(**_{d+1,h}) \qquad \mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^h\mathrm{N}) \subset (\mathrm{R} \cap (\mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^h\mathrm{N}) \cap \mathfrak{L}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N}.$$

The hypothesis $(*_d)$ implies that $(**_{d+1,0})$ is true. We will prove by *induction on h* that $(**_{d+1,h})$ is true for every $h \geqslant 0$. But we have

Lemma (19.7.1.7). — *Let* A *be a Noetherian ring,* N *a finitely generated* A*-module,* $\mathfrak{L}$ *an ideal of* A, E, F *two submodules of* N. *For every $k > 0$ there exists $h > 0$ such that* $\mathrm{E} \cap (\mathrm{F} + \mathfrak{L}^h\mathrm{N}) \subset (\mathrm{E} \cap \mathrm{F}) + \mathfrak{L}^k\mathrm{N}$.

Proof. Indeed, let $\varphi : \mathrm{N} \to \mathrm{N}/(\mathrm{E} \cap \mathrm{F}) = \mathrm{N}_1$ be the canonical homomorphism, and set $\mathrm{E}_1 = \varphi(\mathrm{E})$, $\mathrm{F}_1 = \varphi(\mathrm{F})$, so that $\mathrm{E}_1 \cap \mathrm{F}_1 = 0$ and $\varphi(\mathrm{E} \cap (\mathrm{F} + \mathfrak{L}^h\mathrm{N})) = \mathrm{E}_1 \cap (\mathrm{F}_1 + \mathfrak{L}^h\mathrm{N}_1)$. We know $(\mathbf{0}_{\mathrm{I}}, 7.3.2)$ that there exists h such that $(\mathrm{E}_1 + \mathrm{F}_1) \cap \mathfrak{L}^h\mathrm{N}_1 \subset \mathfrak{L}^k(\mathrm{E}_1 + \mathrm{F}_1)$; if h is thus chosen and if $x_1 \in \mathrm{E}_1$ is such that there exists $y_1 \in \mathrm{F}_1$ for which $x_1 - y_1 \in \mathfrak{L}^h\mathrm{N}_1$, we deduce $x_1 - y_1 \in \mathfrak{L}^k\mathrm{N}_1$, hence $\mathrm{E}_1 \cap (\mathrm{F}_1 + \mathfrak{L}^h\mathrm{N}_1) \subset \mathfrak{L}^k\mathrm{N}_1$; therefore $\mathrm{E} \cap (\mathrm{F} + \mathfrak{L}^h\mathrm{N}) \subset \mathrm{E} \cap \mathrm{F} + \mathfrak{L}^k\mathrm{N}$, which proves the lemma. □

It suffices then to apply this lemma with $\mathrm{E} = \mathrm{R} \cap \mathfrak{L}^m\mathrm{N}$, $\mathrm{F} = \mathfrak{R}^{d+1}\mathrm{N}$ and to take $k = m + 1$ to deduce that $(**_{d+1,h})$ implies $(*_{d+1})$.

We suppose therefore in all that follows that $(**_{d+1,h})$ *is satisfied.*

(19.7.1.8). *Demonstration of $(**_{d+1,h+1})$; First case: $h \leqslant m - d$.*

Start with an element

(19.7.1.9).

$$g \in \mathrm{R} \cap (\mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^h\mathrm{N}) \cap \mathfrak{L}^m\mathrm{N}$$

congruent modulo $\mathfrak{L}^{m+1}\mathrm{N}$ to an element $g_1 \in \mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N})$; we therefore have

(19.7.1.10).
$$g \in \mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}.$$

It will suffice to prove that g is also congruent modulo $\mathfrak{L}^{m+1}\mathrm{N}$ to an element $g_2 \in \mathrm{R} \cap (\mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^{h+1}\mathrm{N})$, for this will imply $g_2 \in \mathfrak{L}^m\mathrm{N}$, since $g \in \mathfrak{L}^m\mathrm{N}$ and $g_2 - g \in \mathfrak{L}^{m+1}\mathrm{N}$. In the case $h \leqslant m-d$ under consideration, it is enough to prove

(19.7.1.11).
$$(\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}) \cap \mathfrak{R}^d\mathrm{N} \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d\mathrm{N} + \mathfrak{R}^{d+1}\mathrm{N}.$$

Indeed, relations (19.7.1.9) and (19.7.1.10) imply that g belongs to the left-hand side of (19.7.1.11); **IV-4** | 216
and since $h \leqslant m-d$, we have $\mathfrak{L}^{m-d+1} \subset \mathfrak{L}^{h+1}$, which will establish $(**_{d+1,h+1})$. Moreover $\mathfrak{R}^m\mathrm{N} \subset \mathfrak{R}^d\mathrm{N}$, and the relation

(19.7.1.12).
$$\mathfrak{L}^{m+1}\mathrm{N} \cap \mathfrak{R}^d\mathrm{N} \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d\mathrm{N}$$

will imply (19.7.1.11). Since N is a free A-module of finite type, it is therefore enough to prove

(19.7.1.13).
$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d.$$

It will suffice to prove that, in general, for $d < m$, one has

(19.7.1.14).
$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d + \mathfrak{R}^{d+1}.$$

Indeed, since $\mathfrak{R} \subset \mathfrak{L}$, hence $\mathfrak{L}^{m-d+1}\mathfrak{R}^d \subset \mathfrak{L}^{m+1}$, the preceding relation, applied successively with $d, d+1, \dots, m$, implies
$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d + \mathfrak{R}^{m+1},$$
whence (19.7.1.13), since $\mathfrak{R}^{m+1} \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d$.

(19.7.1.15). As $\mathfrak{L} = \mathfrak{J} + \mathfrak{R}$, the relation (19.7.1.14) also reads
$$\begin{aligned}&(\mathfrak{J}^{m+1} + \mathfrak{J}^m\mathfrak{R} + \cdots + \mathfrak{J}\mathfrak{R}^m + \mathfrak{R}^{m+1}) \cap \mathfrak{R}^d \\ &\quad\subset \mathfrak{J}^{m-d+1}\mathfrak{R}^d + \mathfrak{J}^{m-d}\mathfrak{R}^{d+1} + \cdots + \mathfrak{J}\mathfrak{R}^m + \mathfrak{R}^{m+1} + \mathfrak{R}^{d+1} \\ &\quad= \mathfrak{J}^{m-d+1}\mathfrak{R}^d + \mathfrak{R}^{d+1}.\end{aligned}$$

We will show that this inclusion is itself a consequence of

(19.7.1.16).
$$\mathfrak{R}^a\mathfrak{J}^b \cap \mathfrak{R}^{a+1} \subset \mathfrak{R}^{a+1}\mathfrak{J}^{b-1}$$

valid for $a \geqslant 0$, $b \geqslant 1$. Indeed, it suffices, to prove (19.7.1.15), to show that for $0 < q \leqslant d$,

(19.7.1.17).
$$(\mathfrak{J}^{m+1} + \mathfrak{J}^m\mathfrak{R} + \ldots + \mathfrak{J}\mathfrak{R}^m + \mathfrak{R}^{m+1}) \cap \mathfrak{R}^d \subset (\mathfrak{J}^{m+1-q}\mathfrak{R}^q + \mathfrak{J}^{m-q}\mathfrak{R}^{q+1} + \ldots + \mathfrak{R}^{m+1}) \cap \mathfrak{R}^d$$

since for $q = d$, this will imply (19.7.1.15). To prove (19.7.1.17), proceed by induction on q, supposing the relation true for $q < d$. An element of the right-hand side corresponding to q can be written $y + z$, with $y \in \mathfrak{J}^{m+1-q}\mathfrak{R}^q$ and $z \in \mathfrak{J}^{m-q}\mathfrak{R}^{q+1} + \cdots + \mathfrak{R}^{m+1}$. Since $y + z \in \mathfrak{R}^d \subset \mathfrak{R}^{q+1}$ and $z \in \mathfrak{R}^{q+1}$, we have $y \in \mathfrak{R}^{q+1}$; by (19.7.1.16), $y \in \mathfrak{J}^{m-q}\mathfrak{R}^{q+1}$, and hence
$$y + z \in \mathfrak{J}^{m-q}\mathfrak{R}^{q+1} + \ldots + \mathfrak{R}^{m+1}, \quad \text{and} \quad y + z \in \mathfrak{R}^d$$
by hypothesis, which proves (19.7.1.17) where q has been replaced by $q+1$.

It remains therefore to establish (19.7.1.16). An element of the left-hand side is written $a_1 f_1 + \ldots + a_n f_n$, where the a_i belong to $\mathfrak{R}^a\mathfrak{J}^{b-1}$. Since the sequence $\mathbf{f}$ is $(\mathfrak{R}^a/\mathfrak{R}^{a+1})$-regular, hence also $(\mathfrak{R}^a/\mathfrak{R}^{a+1})$-quasi-regular $(\mathbf{0}, 15.1.9)$, it follows from the definition $(\mathbf{0}, 15.1.7)$ that the relation $a_1 f_1 + \ldots + a_n f_n \in \mathfrak{R}^{a+1}$, with $a_i \in \mathfrak{R}^a$, necessarily implies $a_i \in \mathfrak{R}^{a+1}$, and hence $a_i \in \mathfrak{R}^a\mathfrak{J}^{b-1} \cap \mathfrak{R}^{a+1}$. It suffices to argue by induction on b for $b \geqslant 1$ (the case $b = 1$ being trivial) to establish (19.7.1.16) and **IV-4** | 217
by extension also $(**_{d+1,h+1})$ for $h \leqslant m-d$.

(19.7.1.18). *Demonstration of* $(**_{d+1,h+1})$; *Second case:* $h > m - d$.

We will first prove that, for every $h \geqslant 0$, we have

(19.7.1.19).
$$\mathrm{R} \cap (\mathfrak{N}^{d+1}\mathrm{N} + \mathfrak{N}^{d}\mathfrak{L}^{h}\mathrm{N}) \subset \mathfrak{J}^{h}(\mathrm{R} \cap \mathfrak{N}^{d}\mathrm{N}) + \mathfrak{L}^{h+1}\mathfrak{N}^{d}\mathrm{N} + \mathfrak{N}^{d+1}\mathrm{N}.$$

Consider an element z of the left-hand side of (19.7.1.19). We note that $\mathfrak{N}^{d+1}\mathrm{N} + \mathfrak{N}^{d}\mathfrak{L}^{h}\mathrm{N} = \mathfrak{N}^{d+1}\mathrm{N} + \mathfrak{N}^{d}\mathfrak{J}^{h}\mathrm{N}$, since $\mathfrak{L} = \mathfrak{J} + \mathfrak{N}$. We consider the class $\bar{z}$ of z in $\mathrm{gr}^{h}_{\mathfrak{J}}(\mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{N}))$, which, by virtue of (19.5.5) and the hypothesis on $\mathbf{f}$, identifies with a sub-$(\mathrm{A}/\mathfrak{L})$-module of $\mathrm{gr}^{h+d}_{\mathfrak{L}}(\mathrm{N})$. Since $z \in \mathrm{R}$, we will show that, under this identification, one even has

(19.7.1.20).
$$\bar{z} \in \mathrm{gr}^{h}_{\mathfrak{J}}(\mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{R},\mathrm{N})).$$

Indeed, we noted in the proof of (19.7.1.3) that, for every prime ideal $\mathfrak{r}$ of A such that $\mathfrak{p} = \mathfrak{r}/\mathfrak{N} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{N}}(\mathrm{A}/\mathfrak{L})$, the map $\psi'_{\mathrm{R}_{\mathfrak{r}}}$ is bijective, and consequently relation (19.7.1.6), with all modules replaced by their localisations at $\mathfrak{r}$, is true. Since z belongs to $\mathrm{R} \cap (\mathfrak{N}^{d}\mathrm{N} + \mathfrak{L}^{d+1}\mathrm{N})$, its image $z/1$ in $\mathrm{R}_{\mathfrak{r}}$ belongs to $(\mathrm{R}_{\mathfrak{r}} \cap \mathfrak{N}^{d}_{\mathfrak{r}}\mathrm{N}_{\mathfrak{r}}) + \mathfrak{L}^{d+1}_{\mathfrak{r}}\mathrm{N}_{\mathfrak{r}}$. Hence the image $\bar{z}/1$ in $\mathrm{gr}^{h}_{\mathfrak{J}}(\mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{N}_{\mathfrak{r}}))$ belongs to $\mathrm{gr}^{h}_{\mathfrak{J}}(\mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{R}_{\mathfrak{r}},\mathrm{N}_{\mathfrak{r}}))$, the localisation at $\mathfrak{q} = \mathfrak{r}/\mathfrak{L}$ of $\mathrm{gr}^{h}_{\mathfrak{J}}(\mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{R},\mathrm{N}))$. In other words, the image of $\bar{z}$ by the canonical map
$$(\mathrm{gr}^{h}_{\mathfrak{J}}(\mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{N})))_{\mathfrak{q}} \longrightarrow (\mathrm{gr}^{h}_{\mathfrak{J}}(\mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{M})))_{\mathfrak{q}}$$
is zero for every $\mathfrak{q} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$. As in the proof of (19.7.1.3), the associated primes of the target are contained in $\mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$; it follows (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 1 of prop. 2 and n° 3, prop. 7) that the image of $\bar{z}$ is zero, which is precisely (19.7.1.20).

(19.7.1.21). This being so, we can write by definition
$$z \equiv \textstyle\sum_{|\mathbf{p}|=h} c_{\mathbf{p}}\mathbf{f}^{\mathbf{p}} \bmod \mathfrak{N}^{d+1}\mathrm{N}$$

where one sets as usual $\mathbf{p} = (p_1, \ldots, p_n)$, $\mathbf{f}^{\mathbf{p}} = f_1^{p_1} f_2^{p_2} \cdots f_n^{p_n}$, $|\mathbf{p}| = p_1 + \ldots + p_n$ and $c_{\mathbf{p}} \in \mathfrak{N}^{d}\mathrm{N}$. Moreover, as ψ_{N} identifies the relevant graded module with $(\mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, the $c_{\mathbf{p}}$ are determined modulo $\mathfrak{N}^{d}\mathfrak{L}\mathrm{N}$; if $\bar{c}_{\mathbf{p}}$ denotes the class of $c_{\mathbf{p}}$ modulo $\mathfrak{N}^{d}\mathfrak{L}\mathrm{N}$, then, under the preceding identification,

(19.7.1.22).
$$\bar{z} = \textstyle\sum_{|\mathbf{p}|=h} \bar{c}_{\mathbf{p}}\mathrm{T}^{\mathbf{p}}.$$

Using (19.7.1.3), we deduce that necessarily, for every $\mathbf{p}$ such that $|\mathbf{p}| = h$ ($\bar{c}_{\mathbf{p}}$ being this time identified by ψ_{N} with an element of $\mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{N})$),

(19.7.1.23).
$$\bar{c}_{\mathbf{p}} \in \mathrm{gr}^{d}_{\mathfrak{N}}(\mathrm{R},\mathrm{N}).$$

But since $d < m$, the hypothesis that $(*_d)$ is true implies that $\bar{c}_{\mathbf{p}}$ belongs to the image of ψ'_{R}, hence we can suppose that $c_{\mathbf{p}} \in (\mathrm{R} \cap \mathfrak{N}^{d}\mathrm{N}) + \mathfrak{N}^{d}\mathfrak{L}\mathrm{N}$. The relation (19.7.1.21) then gives
$$z \in \mathfrak{J}^{h}(\mathrm{R} \cap \mathfrak{N}^{d}\mathrm{N}) + \mathfrak{J}^{h}\mathfrak{N}^{d}\mathfrak{L}\mathrm{N} + \mathfrak{N}^{d+1}\mathrm{N}$$
and as $\mathfrak{J} \subset \mathfrak{L}$, this proves (19.7.1.19).

In particular, we can therefore write IV-4 | 218
$$g = t + g_2$$
with $t \in \mathfrak{J}^{h}(\mathrm{R} \cap \mathfrak{N}^{d}\mathrm{N})$ and $g_2 \in \mathfrak{N}^{d+1}\mathrm{N} + \mathfrak{N}^{d}\mathfrak{L}^{h+1}\mathrm{N}$; moreover, as $t \in \mathrm{R}$ and $g \in \mathrm{R}$, we have $g_2 \in \mathrm{R}$. On the other hand, using the hypothesis $h \geqslant m - d + 1$, one sees that $t \in \mathfrak{L}^{m+1}\mathrm{N}$. The element g_2 hence satisfies all the conditions stated at the start of (19.7.1.8), which completes the demonstration of (19.7.1). □

Remarks (19.7.2). —

(i) When one supposes that $\mathrm{gr}^*_{\mathfrak{N}}(A)$ is an $(A/\mathfrak{N})$-module *flat*, the hypothesis that the sequence $\mathbf{f}$ is $(A/\mathfrak{N})$-regular implies that it is also $\mathrm{gr}^*_{\mathfrak{N}}(A)$-regular $(\mathbf{0}, 15.1.14)$. We note that this will be the case when A and $A/\mathfrak{N}$ *are regular rings* $(\mathbf{0}, 17.3.6)$: indeed, $\mathfrak{N}$ is then a regular ideal of A (19.1.2), hence quasi-regular, and localising at the maximal ideals of A containing $\mathfrak{N}$ and using $(\mathbf{0}, 15.1.7)$, one sees that $\mathrm{gr}^*_{\mathfrak{N}}(A)$ is a *projective* $(A/\mathfrak{N})$-module.

(ii) It would be interesting to be able to prove the last conclusion of (19.7.1) under the sole hypothesis that there exists an A-algebra B which is a Noetherian ring, for which $\mathfrak{J}B$ is contained in the radical of B, and that M is a finitely generated B-module.

Corollary (19.7.3). — *Let* A *be a local Noetherian ring,* $\mathfrak{m}$ *its maximal ideal,* $\mathfrak{N}$ *an ideal of* A *such that the ring* $A/\mathfrak{N}$ *is regular and of dimension* n, M *a finitely generated* A*-module. The following conditions are equivalent:*

a) $\mathrm{gr}^*_{\mathfrak{N}}(M)$ *is a flat* $(A/\mathfrak{N})$*-module.*

b) *If* $P(T)$ *and* $Q(T)$ *are the Poincaré series of the* $(A/\mathfrak{m})$*-modules graded* $\mathrm{gr}^*_{\mathfrak{m}}(M)$ *and* $\mathrm{gr}^*_{\mathfrak{N}}(M) \otimes_{A/\mathfrak{N}} (A/\mathfrak{m})$, *we have*

(19.7.3.1) $$Q(T) = P(T)(1-T)^n.$$

Let $\mathbf{f} = (f_1, \ldots, f_n)$ *be a sequence of elements of* A *such that the images of the* f_i *in* $A/\mathfrak{N}$ *form a regular system of parameters of* $A/\mathfrak{N}$ $(\mathbf{0}, 17.1.6)$, *so that* $\mathbf{f}$ *is an* $(A/\mathfrak{N})$*-regular sequence; moreover, if* $\mathfrak{J}$ *is the ideal generated by* $\mathbf{f}$, $\mathfrak{J} + \mathfrak{N} = \mathfrak{m}$, *and as* $A/\mathfrak{m}$ *is a field,* $\mathrm{gr}^*_{\mathfrak{m}}(M)$ *is a flat* $(A/\mathfrak{m})$*-module. As* A *is Noetherian and* M *a finitely generated* A*-module, one can apply the equivalence of* a) *and* b) *in* (19.7.1)*. Moreover,* $A/\mathfrak{m}$ *being a field, the fact that* ψ_M *is bijective is equivalent to saying that for every* $h \geqslant 0$, $\mathrm{gr}^h_{\mathfrak{m}}(M)$ *and the submodule of elements of degree* h *of* $(\mathrm{gr}^*_{\mathfrak{N}}(M) \otimes_{A/\mathfrak{N}} (A/\mathfrak{m}))[T_1, \ldots, T_n]$ *have the same rank over* $A/\mathfrak{m}$*; but for the second of these modules the rank in question is clearly equal to*

$$\textstyle\sum_{r=0}^{h} \binom{r+n-1}{n-1} \mathrm{rg}(\mathrm{gr}^{h-r}_{\mathfrak{N}}(M) \otimes_{A/\mathfrak{N}} (A/\mathfrak{m})). \quad \textit{As} \quad (1-T)^{-n} = \sum_{r=0}^{\infty} \binom{r+n-1}{n-1} T^r, \quad \textit{this}$$

proves that the relation (19.7.3.1) *is necessary and sufficient for* ψ_M *to be bijective.*

Corollary (19.7.4). — *Let* X *be a locally Noetherian prescheme,* Y *a closed sub-prescheme of* X, Y *regular and connected,* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module normally flat* ((6.10.1)) *along* Y*. Then there exists a formal power series* $R \in \mathbf{Q}[[T]]$ *(independent of* $x \in Y$*) such that, for every* $x \in Y$*, the Poincaré series of* $\mathrm{gr}^*_{\mathfrak{m}_x}(\mathscr{F}_x)$ *is given by the formula*

(19.7.4.1) $$P_x(\mathscr{F})(T) = R(T)(1-T)^{-n} \qquad \textit{where } n = \dim(\mathscr{O}_{Y,x}).$$

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Let $\mathscr{J}$ *be the coherent Ideal of* $\mathscr{O}_X$ *defining* Y*; the hypothesis on* $\mathscr{F}$ *means that* $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ *is a flat* $\mathscr{O}_Y$*-Module. We can therefore, for every* $x \in Y$*, apply* (19.7.3) *taking* $A = \mathscr{O}_{X,x}$, $\mathfrak{N} = \mathscr{J}_x$, $M = \mathscr{F}_x$*; condition* a) *being satisfied, the Poincaré series* $P_x(\mathscr{F})(T)$ *is equal to* $R_x(T)(1-T)^{-n}$*, where* $R_x(T)$ *is the Poincaré series of* $\mathrm{gr}^*_{\mathscr{J}_x}(\mathscr{F}_x) \otimes_{\mathscr{O}_{Y,x}} k(x)$*. Each* $\mathscr{O}_Y$*-Module* $\mathscr{J}^h\mathscr{F}/\mathscr{J}^{h+1}\mathscr{F}$ *is flat and coherent by hypothesis, hence locally free* (2.1.12)*; and since* Y *is connected, its rank is constant on* Y*. This proves that* $R_x(T)$ *is independent of* $x \in Y$*.*

Corollary (19.7.5). — *Let* X *be a locally Noetherian prescheme,* Y *a closed sub-prescheme of* X, Z *a closed sub-prescheme of* Y*; one supposes* Y *and* Z *regular. Let* $\mathscr{F}$ *be a coherent* $\mathscr{O}_X$*-Module.*

(i) *If* $\mathscr{F}$ *is normally flat along* Y *at the points of* Z *(cf.* (11.3.4) *for the definition of normal flatness at a point), then* $\mathscr{F}$ *is normally flat along* Z*.*

(ii) *Suppose moreover that* Z *is connected and that* $\mathscr{O}_X$ *is normally flat along* Y *at the points of* Z *(which is the case in particular if* X *is regular at the points of* Z*). If* $\mathscr{F}$ *is normally flat along* Z *and if there exists a point* $z \in Z$ *at which* $\mathscr{F}$ *is normally flat along* Y*, then* $\mathscr{F}$ *is normally flat along* Y *at all the points of* Z*.*

Let $\mathscr{K}$ *and* $\mathscr{L} \supset \mathscr{K}$ *be the coherent Ideals of* $\mathscr{O}_X$ *defining respectively* Y *and* Z*. For every point* $z \in Z$*, the canonical immersion* $Z \to Y$ *is regular at* z*, since* Y *and* Z *are regular at that point* (19.1.1)*; in other words, there exists a sequence* $(f_i)_{1 \leqslant i \leqslant n}$ *of elements of* $\mathscr{O}_{X,z}$ *which is* $\mathscr{K}_z$*-regular and such that* $\mathscr{L}_z = \mathscr{K}_z + \sum_i f_i \mathscr{O}_{X,z}$*.*

(i) *The hypothesis means that* $\mathrm{gr}^*_{\mathscr{K}_z}(\mathscr{F}_z)$ *is a flat* $\mathscr{O}_{Y,z}$*-module; it therefore implies that* $\mathrm{gr}^*_{\mathscr{L}_z}(\mathscr{F}_z)$ *is a flat* $\mathscr{O}_{Z,z}$*-module, by the implication* a) $\Rightarrow$ b) *of* (19.7.1)*.*

(ii) *The supplementary hypothesis on* X *implies that the sequence* (f_i) *is* $\mathrm{gr}^*_{\mathscr{K}_z}(\mathscr{O}_{X,z})$*-regular at every point* $z \in Z$ (19.7.2). *On the other hand,* Z*, being regular and connected, is integral; if* $\mathrm{gr}^*_{\mathscr{K}_z}(\mathscr{F}_z)$ *is flat over* $\mathscr{O}_{Y,z}$ *at one point* $z \in Z$*, then, since the generic point* ζ *of* Z *is a generization of* z*,* $\mathrm{gr}^*_{\mathscr{K}_\zeta}(\mathscr{F}_\zeta)$ *is flat over* $\mathscr{O}_{Y,\zeta}$ $(\mathbf{0}_{\mathrm{I}}, 6.3.1)$. *If, at every point* z' *of* Z*,* $\mathrm{gr}^*_{\mathscr{L}_{z'}}(\mathscr{F}_{z'})$ *is flat over* $\mathscr{O}_{Z,z'}$*, we then conclude that* $\mathrm{gr}^*_{\mathscr{K}_{z'}}(\mathscr{F}_{z'})$ *is flat over* $\mathscr{O}_{Y,z'}$*, by applying the equivalence of* a) *and* d) *in* (19.7.1).

19.8. Properties of passage to the projective limit In this number, the notations and conventions on projective limits are those of (8.5.1) and (8.8.1).

Proposition (19.8.1). — *Suppose that the transition morphisms* $S_\mu \to S_\lambda$ $(\lambda \leqslant \mu)$ *are flat, and moreover that one of the two following hypotheses is satisfied:*

1° *The preschemes* S_λ *are locally Noetherian.*

2° *The transition morphisms* $S_\mu \to S_\lambda$ *are surjective (and hence faithfully flat).*

Under these conditions: **IV-4 | 220**

(i) *Suppose* S_α *quasi-compact. Let* $\mathscr{F}_\alpha$ *be a quasi-coherent* $\mathscr{O}_{S_\alpha}$*-Module, which one supposes moreover of finite type when hypothesis* 1° *is satisfied. Let* $(f_{i\alpha})_{1\leqslant i\leqslant n}$ *be a finite sequence of sections of* $\mathscr{F}_\alpha$ *above* S_α*, the sequence of sections* f_i *of the* $\mathscr{O}_S$*-Module* $\mathscr{F}$ *above* S*, corresponding to the* $f_{i\alpha}$*. For the sequence of sections* f_i $(1 \leqslant i \leqslant n)$ *to be* $\mathscr{F}$*-regular, it is necessary and sufficient that there exist* $\lambda \geqslant \alpha$ *such that the sequence of sections* $f_{i\lambda}$ *of* $\mathscr{F}_\lambda$ *above* S_λ *is* $\mathscr{F}_\lambda$*-regular.*

(ii) *Suppose* X_α *quasi-compact, and let* $j_\alpha : X_\alpha \to S_\alpha$ *be an immersion, which one supposes locally of finite presentation when hypothesis* 2° *is satisfied. Then, for the corresponding immersion* $j : X \to S$ *to be regular, it is necessary and sufficient that there exist* $\lambda \geqslant \alpha$ *such that* $j_\lambda : X_\lambda \to S_\lambda$ *is regular.*

Proof. (i) If $p_\lambda : S \to S_\lambda$ is the canonical projection, we have $\mathscr{F}/(\sum_{j=1}^{i-1} f_j\mathscr{F}) = p_\lambda^*(\mathscr{F}_\lambda/(\sum_{j=1}^{i-1} f_{j\lambda}\mathscr{F}_\lambda))$, hence we are reduced to the case $n = 1$, in which case we omit the index i. Sufficiency follows from the flatness of p_λ (8.3.8) and $(\mathbf{0}, 15.1.5)$. In case 2°, p_λ is faithfully flat (8.3.8) and necessity follows from $(\mathbf{0}, 15.2.5)$, with $\lambda = \alpha$. In case 1°, let $\mathscr{N}_\lambda$ (resp. $\mathscr{N}$) be the kernel of multiplication by f_λ on $\mathscr{F}_\lambda$ (resp. by f on $\mathscr{F}$); since $\mathscr{F}_\lambda$ is coherent, so is $\mathscr{N}_\lambda$, and the hypothesis $\mathscr{N} = 0$ implies $\mathscr{N}_\lambda = 0$ for some $\lambda \geqslant \alpha$ by (8.5.8), (ii).

(ii) As p_λ is flat, sufficiency follows from (19.1.5), (ii). For necessity, since $j_\alpha(X_\alpha)$ is quasi-compact, it is contained in a quasi-compact open subset of S_α; we may therefore reduce to the case in which S_α is quasi-compact and j_α is a closed immersion. Its image is then defined by a quasi-coherent Ideal $\mathscr{J}_\alpha$ of $\mathscr{O}_{S_\alpha}$, which may moreover be supposed of finite type in cases 1° and 2°. The image of X_λ (resp. X) in S_λ (resp. S) is defined by $\mathscr{J}_\lambda = \mathscr{J}_\alpha\mathscr{O}_{S_\lambda}$ (resp. $\mathscr{J} = \mathscr{J}_\alpha\mathscr{O}_S$), again of finite type. By (8.2.11), we may suppose that $\mathscr{J}$ is generated by a regular sequence $(f_i)_{1\leqslant i\leqslant n}$ of sections over S, and hence by a surjection $u : \mathscr{O}_S^n \to \mathscr{J}$. By (8.5.2), (i), and (8.5.7), there exist $\lambda \geqslant \alpha$ and a surjection $u_\lambda : \mathscr{O}_{S_\lambda}^n \to \mathscr{J}_\lambda$ such that $u = p_\lambda^*(u_\lambda)$; thus the f_i are the canonical images of sections $f_{i\lambda}$ of $\mathscr{J}_\lambda$ generating this Ideal. By (i), there is then $\mu \geqslant \lambda$ for which $(f_{i\mu})$ is $\mathscr{O}_{S_\mu}$-regular, and hence j_μ is a regular immersion. □

Proposition (19.8.2). — *Let* X_α *be an* S_α*-prescheme quasi-compact and locally of finite presentation.*

(i) *Let* $\mathscr{F}_\alpha$ *be a quasi-coherent* $\mathscr{O}_{X_\alpha}$*-Module of finite presentation, and* $(f_{i\alpha})_{1\leqslant i\leqslant n}$ *a finite sequence of sections of* $\mathscr{O}_{X_\alpha}$ *over* X_α*, corresponding to sections* f_i *of* $\mathscr{O}_X$ *over* X*. For the sequence* (f_i) *to be* $\mathscr{F}$*-transversally regular relative to* S ((19.2.1))*, it is necessary and sufficient that there exist* $\lambda \geqslant \alpha$ *such that the sequence* $(f_{i\lambda})$ *over* X_λ *is* $\mathscr{F}_\lambda$*-transversally regular relative to* S_λ*.*

(ii) *Let* Y_α *be an* S_α*-prescheme quasi-compact,* $j_\alpha : Y_\alpha \to X_\alpha$ *an* S_α*-immersion locally of finite presentation. For the corresponding* S*-immersion* $j : Y \to X$ *to be transversally regular relative to* S*, it is necessary and sufficient that there exist* $\lambda \geqslant \alpha$ *such that* $j_\lambda : Y_\lambda \to X_\lambda$ *is transversally regular relative to* S_λ*.*

Proof. (i) The sufficiency of the condition follows from $(\mathbf{0}, 15.1.15)$. To prove necessity, it is enough to show that every point $x \in X$ has an open neighbourhood $V(x)$ for which there exist an index $\lambda = \lambda(x) \geqslant \alpha$ and an open neighbourhood V_λ of the image x_λ of x in X_λ, such **IV-4 | 221** that $\mathrm{V}(x)$ is the inverse image of V_λ and the restrictions of the $f_{i\lambda}$ to V_λ form an $(\mathscr{F}_\lambda|\mathrm{V}_\lambda)$-transversally regular sequence relative to S_λ. Indeed, one then applies (8.3.4) to the open subset U_λ of X_λ defined as the largest open on which the restrictions of the $f_{i\lambda}$ form such a sequence.

Let s be the image of x in S and, for every $\lambda \geqslant \alpha$, let s_λ be the image of s in S_λ. The fibre X_s is the projective limit of the fibres $(\mathrm{X}_\lambda)_{s_\lambda}$ and is locally Noetherian, since $\mathrm{X} \to \mathrm{S}$ is locally of finite presentation. Moreover $(\mathrm{X}_\alpha)_{s_\alpha}$ is quasi-compact and the transition morphisms $(\mathrm{X}_\mu)_{s_\mu} \to (\mathrm{X}_\lambda)_{s_\lambda}$ are flat, since this is so for $\mathrm{Spec}(k(s_\mu)) \to \mathrm{Spec}(k(s_\lambda))$. We may therefore apply (19.8.1), (i), to the sections $f_i \otimes 1$ of $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_\mathrm{S}} k(s)$ over X_s. Thus there exists $\lambda \geqslant \alpha$ such that the sections $f_{i\lambda} \otimes 1$ of $(\mathscr{F}_\lambda)_{s_\lambda} = \mathscr{F}_\lambda \otimes_{\mathscr{O}_{\mathrm{S}_\lambda}} k(s_\lambda)$ form an $(\mathscr{F}_\lambda)_{s_\lambda}$-regular sequence. By (11.2.6), one may moreover suppose that $\mathscr{F}_\lambda$ is flat over S_λ at x_λ. It then follows from (11.3.8) that there exists an open neighbourhood V_λ of x_λ in X_λ satisfying the required conditions.

(ii) Sufficiency follows from (19.2.7), (ii). For necessity, it again suffices to show that every point $x \in j(\mathrm{Y})$ has an open neighbourhood $\mathrm{V}(x)$ for which there exist an index $\lambda = \lambda(x) \geqslant \alpha$ and an open neighbourhood V_λ of the image x_λ of x in X_λ, such that $\mathrm{V}(x)$ is the inverse image of V_λ and the restriction $j_\lambda^{-1}(\mathrm{V}_\lambda) \to \mathrm{V}_\lambda$ is transversally regular relative to S_λ.

Let s be the image of x in S and, for every $\lambda \geqslant \alpha$, let s_λ be the image of s in S_λ. Since X_s (resp. Y_s) is the projective limit of the fibres $(\mathrm{X}_\lambda)_{s_\lambda}$ (resp. $(\mathrm{Y}_\lambda)_{s_\lambda}$), we may use (19.8.1), (ii), as in (i); and since the immersion $\mathrm{Y}_s \to \mathrm{X}_s$ is regular by hypothesis, there exists an index $\lambda(s)$ such that, for $\lambda \geqslant \lambda(s)$, the immersion $(\mathrm{Y}_\lambda)_{s_\lambda} \to (\mathrm{X}_\lambda)_{s_\lambda}$ is regular. Moreover, by (19.2.4), there exists a quasi-compact open neighbourhood $\mathrm{W}(x)$ of x in X such that the structural morphisms $\mathrm{W}(x) \to \mathrm{S}$ and $\mathrm{Y} \cap \mathrm{W}(x) \to \mathrm{S}$ are flat. Applying (11.2.6) and (8.2.11), one may suppose that there exist $\lambda \geqslant \lambda(s)$ and a quasi-compact open neighbourhood $\mathrm{W}(x_\lambda)$ of x_λ in X_λ, such that $\mathrm{W}(x)$ is the inverse image of $\mathrm{W}(x_\lambda)$ and the restrictions to $\mathrm{W}(x_\lambda)$ and $\mathrm{Y}_\lambda \cap \mathrm{W}(x_\lambda)$ of the structural morphisms $\mathrm{X}_\lambda \to \mathrm{S}_\lambda$ and $\mathrm{Y}_\lambda \to \mathrm{S}_\lambda$ are flat. We conclude from (19.2.4) that there exists an open neighbourhood V_λ of x_λ satisfying the required conditions. □

Remark (19.8.3). — The preceding demonstrations show that the statements of (19.8.1) and (19.8.2) subsist when one replaces the conditions concerning regularity or transversal regularity throughout X (resp. X_λ) by these conditions in a neighbourhood of a point $x \in \mathrm{X}$ (resp. in a neighbourhood of x_λ, projection of x).

19.9. $\mathscr{F}$-regular sequences and depth

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(19.9.1). Recall that if X is a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_\mathrm{X}$-Module, T a part of X, one sets (5.10.1)

(19.9.1.1) $$\mathrm{prof}_\mathrm{T}(\mathscr{F}) = \inf_{x \in \mathrm{T}} \mathrm{prof}(\mathscr{F}_x).$$

We set on the other hand, for every point $t \in \mathrm{T}$,

(19.9.1.2) $$\mathrm{prof}_{\mathrm{T},t}(\mathscr{F}) = \inf_{x \in \mathrm{T} \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},t})} \mathrm{prof}(\mathscr{F}_x),$$

and we say that $\mathrm{prof}_{\mathrm{T},t}(\mathscr{F})$ is the *T-depth of $\mathscr{F}$ at the point t*; it is clear that we have

(19.9.1.3) $$\mathrm{prof}_\mathrm{T}(\mathscr{F}) = \inf_{t \in \mathrm{T}} \mathrm{prof}_{\mathrm{T},t}(\mathscr{F}).$$

Lemma (19.9.2). — *Let* A *be a Noetherian ring,* M *a finitely generated* A*-module,* $\mathfrak{J}$ *an ideal of* A*. In order that, for every prime ideal $\mathfrak{p} \supset \mathfrak{J}$, we have $\mathrm{prof}_{\mathrm{A}_\mathfrak{p}}(\mathrm{M}_\mathfrak{p}) \geqslant r$, it is necessary and sufficient that there exist an M-regular sequence of r elements belonging to $\mathfrak{J}$.*

Proof. The sufficiency of the condition follows immediately from $(\mathbf{0}, 16.4.5)$; let us prove its necessity. We argue by induction on r (there is nothing to prove if $r = 0$): since $\mathrm{prof}_{\mathrm{A}_\mathfrak{p}}(\mathrm{M}_\mathfrak{p}) \geqslant r - 1$ for every $\mathfrak{p} \supset \mathfrak{J}$, there exists by the induction hypothesis an M-regular sequence $(g_i)_{1 \leqslant i \leqslant r-1}$ of elements of $\mathfrak{J}$. Set $\mathrm{N} = \mathrm{M}/(g_1\mathrm{M} + \ldots + g_{r-1}\mathrm{M})$; for every $\mathfrak{p} \supset \mathfrak{J}$, the images of the g_i in $\mathrm{A}_\mathfrak{p}$ belong to the maximal ideal $\mathfrak{p}\mathrm{A}_\mathfrak{p}$ and form an $\mathrm{M}_\mathfrak{p}$-regular sequence $(\mathbf{0}, 15.1.14)$. It follows from $(\mathbf{0}, 16.4.6)$ that

$\mathrm{prof}_{\mathrm{A}_\mathfrak{p}}(\mathrm{N}_\mathfrak{p}) = \mathrm{prof}_{\mathrm{A}_\mathfrak{p}}(\mathrm{M}_\mathfrak{p}) - (r-1) \geqslant 1$, so everything reduces to proving the lemma for $r = 1$. The hypothesis then means that, for every $\mathfrak{p} \supset \mathfrak{J}$, $\mathfrak{p}\mathrm{A}_\mathfrak{p}$ is not associated with $\mathrm{M}_\mathfrak{p}$ $(\mathbf{0}, 16.4.6)$; hence (Bourbaki, *Alg. comm.*, chap. IV, § 1, nº 2, prop. 5), $\mathfrak{p}$ is not associated with M. In other words, $\mathfrak{J}$ is contained in none of the prime ideals of Ass(M), and therefore is not contained in their union (Bourbaki, *Alg. comm.*, chap. II, § 1, nº 1, prop. 2). Since this union is the set of elements that are not M-regular (Bourbaki, *Alg. comm.*, chap. IV, § 1, cor. 2 of prop. 2), this proves the lemma. ☐

Proposition (19.9.3). — *Let* X *be a locally Noetherian prescheme,* $\mathscr{F}$ *a coherent* $\mathscr{O}_\mathrm{X}$*-Module,* Y *a closed subset of* X*,* y *a point of* Y*,* n *an integer* > 0*. The following conditions are equivalent:*

a) *There exists an open neighbourhood* U *of* y *in* X *such that* $\mathrm{prof}_{\mathrm{Y}\cap\mathrm{U}}(\mathscr{F}|\mathrm{U}) \geqslant n$.
b) *There exists an open neighbourhood* U *of* y *in* X *and an* $(\mathscr{F}|\mathrm{U})$*-regular sequence of* n *sections* f_i *of* $\mathscr{O}_\mathrm{U}$ *above* U*, such that* $f_i(x) = 0$ $(\mathbf{0}_\mathrm{I}, 5.5.1)$ *for* $1 \leqslant i \leqslant n$ *at every point* $x \in \mathrm{U} \cap \mathrm{Y}$.
c) $\mathrm{prof}_{\mathrm{Y},y}(\mathscr{F}) \geqslant n$.

Moreover, the set of $y \in \mathrm{Y}$ *satisfying these conditions is open in* Y.

Proof. The last assertion follows trivially from *a*). The equivalence of *a*) and *b*) follows from (19.9.2), applied to an affine open neighbourhood of y in X, and from $(\mathbf{0}, 15.1.14)$. It is clear that *a*) implies *c*), since every open neighbourhood of y in X contains $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},y})$; let us prove conversely that *c*) implies *b*). It follows from *c*) and the definition (19.9.1) that we can apply (19.9.2) to the $\mathscr{O}_{\mathrm{X},y}$-module $\mathscr{F}_y$ and the ideal $\mathscr{J}_y$ of $\mathscr{O}_{\mathrm{X},y}$, where $\mathscr{J}$ is an ideal defining Y near y. Hence there exists an $\mathscr{F}_y$-regular sequence $(t_i)_{1\leqslant i\leqslant n}$ of n elements of $\mathscr{J}_y$. There is therefore an open neighbourhood U of y in X and a sequence $(f_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{J}$ above U such that the t_i are the germs of the f_i at y; using $(\mathbf{0}, 15.2.4)$, one deduces that there exists an open neighbourhood $\mathrm{U}' \subset \mathrm{U}$ of y in X such that the $f_i|\mathrm{U}'$ form an $(\mathscr{F}|\mathrm{U}')$-regular sequence. ☐ **IV-4** | 223

Corollary (19.9.4). — *With the notations of* ((19.9.3))*, the function* $y \mapsto \mathrm{prof}_{\mathrm{Y},y}(\mathscr{F})$ *is lower semi-continuous on* Y.

Proposition (19.9.5). — *With the notations of* ((19.9.3))*, let* X′ *be a locally Noetherian prescheme,* $g : \mathrm{X}' \to \mathrm{X}$ *a flat morphism,* $\mathrm{Y}' = g^{-1}(\mathrm{Y})$*,* $\mathscr{F}' = g^*(\mathscr{F})$*. Then, for every point* $y' \in \mathrm{Y}'$ *above* y*, we have* $\mathrm{prof}_{\mathrm{Y}',y'}(\mathscr{F}') = \mathrm{prof}_{\mathrm{Y},y}(\mathscr{F})$.

Proof. Indeed, for every $z' \in \mathrm{Spec}(\mathscr{O}_{\mathrm{X}',y'})$, it follows from (6.3.1) that $\mathrm{prof}(\mathscr{F}'_{z'}) \geqslant \mathrm{prof}(\mathscr{F}_{g(z')})$, and if z'' is a maximal point of the fibre $g^{-1}(g(z'))$ that is a generization of z' (and hence also belongs to $\mathrm{Spec}(\mathscr{O}_{\mathrm{X}',y'})$), we have (*loc. cit.*) $\mathrm{prof}(\mathscr{F}'_{z''}) = \mathrm{prof}(\mathscr{F}_{g(z')})$. The proposition then follows from the fact that $g(\mathrm{Spec}(\mathscr{O}_{\mathrm{X}',y'})) = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},g(y')})$, since g is flat (2.3.4). ☐

Proposition (19.9.6). — *Let* $f : \mathrm{X} \to \mathrm{S}$ *be a morphism locally of finite presentation,* Y *a closed subset of* X*,* $\mathscr{F}$ *a coherent* $\mathscr{O}_\mathrm{X}$*-Module of finite presentation and* f*-flat,* n *an integer* > 0*. For every point* $y \in \mathrm{Y}$*, denoting by* $\mathrm{X}_{f(y)}$ *the fibre of* f *at the point* $f(y)$*, by* $\mathscr{F}_{f(y)}$ *the inverse image of* $\mathscr{F}$ *in* $\mathrm{X}_{f(y)}$*, and setting* $\mathrm{Y}_{f(y)} = \mathrm{X}_{f(y)} \cap \mathrm{Y}$:

a) $\mathrm{prof}_{\mathrm{Y}_{f(y)},y}(\mathscr{F}_{f(y)}) \geqslant n$.
b) *There exists an open neighbourhood* U *of* y *in* X *and a sequence* $(g_i)_{1\leqslant i\leqslant n}$ *of* n *sections of* $\mathscr{O}_\mathrm{X}$ *above* U*, that is transversally* $\mathscr{F}$*-regular relative to* S *(cf.* (19.2.1) *for the terminology) and such that* $g_i(x) = 0$ *for* $1 \leqslant i \leqslant n$ *at every point* $x \in \mathrm{U} \cap \mathrm{Y}$.

The set V_n *of* $y \in \mathrm{Y}$ *satisfying these conditions is open in* Y*; if moreover* Y *is locally constructible in* X*,* V_n *is retrocompact in* X $(\mathbf{0}_{\mathrm{III}}, 9.1.1)$.

Proof. By virtue of (19.9.3), condition *a*) is equivalent to the existence of an open neighbourhood U of y in X and of an $(\mathscr{F}_{f(y)}|(\mathrm{U} \cap \mathrm{X}_{f(y)}))$-regular sequence $(h_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{O}_{\mathrm{X}_{f(y)}}$ above $\mathrm{U} \cap \mathrm{X}_{f(y)}$, such that $h_i(z) = 0$ for $1 \leqslant i \leqslant n$ at every point $z \in \mathrm{U} \cap \mathrm{Y}_{f(y)}$. If we regard Y as a closed sub-prescheme of X having the given closed subset as its underlying space, and denote by $\mathscr{J}$ the quasi-coherent ideal of $\mathscr{O}_\mathrm{X}$ defining it, one can further require that $h_i \in \mathscr{J}_{f(y)}$ for $1 \leqslant i \leqslant n$. Replacing U by a smaller affine open neighbourhood of y, one can suppose that the h_i are the images of a sequence $(g_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{J}$ above U, so that the germs $(g_i)_y$ belong to the maximal ideal $\mathfrak{m}_y$ of $\mathscr{O}_{X,y}$. The equivalence of $a)$ and $b)$ then follows from (11.3.8), which also proves that the set V_n is open in Y.

It remains to prove the last assertion. Since this again concerns properties local on X, we may suppose that $S = \mathrm{Spec}(A)$ is affine and that X is of finite presentation over S. There then exist a Noetherian subring A_0 of A, a prescheme X_0 of finite type over $S_0 = \mathrm{Spec}(A_0)$, and a coherent $\mathscr{O}_{X_0}$-module $\mathscr{F}_0$ such that $X = X_0 \times_{S_0} S$ and $\mathscr{F} = \mathscr{F}_0 \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_X$ (8.9.1). We may moreover suppose that, if $f_0 : X_0 \to S_0$ is the structural morphism, $\mathscr{F}_0$ is f_0-flat ((11.2.6)). Since Y is constructible, we may further suppose (8.3.11) that $Y = p^{-1}(Y_0)$, where $p : X \to X_0$ is the canonical projection. Then, for every $y \in Y$, (19.9.5) and the transitivity of fibres give

$$\mathrm{prof}_{Y_{f(y)},y}(\mathscr{F}_{f(y)}) = \mathrm{prof}_{(Y_0)_{f_0(y_0)},y_0}((\mathscr{F}_0)_{f_0(y_0)}),$$

where $y_0 = p(y)$. If V_n^0 is the set of $y_0 \in Y_0$ for which the right-hand side is at least n, then $V_n = p^{-1}(V_n^0)$. Since X_0 is Noetherian, V_n^0 is a retrocompact open subset of X_0, and consequently (cf. the proof of (I, 8.2)) V_n is retrocompact in X. $\square$

Corollary (19.9.7). — *Under the general hypotheses of (19.9.6), the function $y \mapsto \mathrm{prof}_{Y_{f(y)},y}(\mathscr{F}_{f(y)})$ is lower semi-continuous in Y. If moreover Y is locally constructible, this function is locally constructible.*

Proposition (19.9.8). — *Let $f : X \to S$ be a morphism locally of finite presentation, Z a closed subset of X, $\mathscr{F}$ an $\mathscr{O}_X$-module of finite presentation and f-flat, and $\mathscr{G}$ a quasi-coherent $\mathscr{O}_S$-module. Suppose that, for every $x \in Z$, we have $\mathrm{prof}((\mathscr{F}_{f(x)})_x) \geqslant 1$ (resp. $\mathrm{prof}((\mathscr{F}_{f(x)})_x) \geqslant 2$). Then, if $j : X - Z \to X$ is the canonical injection and if we set $\mathscr{H} = \mathscr{F} \otimes_{\mathscr{O}_X} f^*(\mathscr{G})$, the canonical homomorphism $\rho_{\mathscr{H}} : \mathscr{H} \to j_*(j^*(\mathscr{H}))$ relative to j is injective (resp. bijective).*

Proof. The question is evidently local on X and S, and it suffices to prove the proposition in a neighbourhood of a point $x \in Z$. In other words, we may restrict to the case where X and S are affine. By (19.9.6), we may suppose that there exists a section g_1 of $\mathscr{O}_X$ over X that is transversally $\mathscr{F}$-regular relative to S (resp. a transversally $\mathscr{F}$-regular sequence (g_1, g_2) of two sections of $\mathscr{O}_X$ over X relative to S) such that, if Z' is the set of $x' \in X$ for which $g_1(x') = 0$ (resp. $g_1(x') = g_2(x') = 0$), then $Z \subset Z'$.

This being so, return to the proof of (19.9.8). By (19.9.6), we still have $\mathrm{prof}((\mathscr{F}_{f(x)})_x) \geqslant 1$ (resp. $\mathrm{prof}((\mathscr{F}_{f(x)})_x) \geqslant 2$) for every $x \in Z'$. If the proposition is proved after replacing the pair (X, Z) by each of (X, Z') and $(X - Z, Z' \cap (X - Z))$, it is immediate that it also holds for (X, Z); we may therefore restrict to proving it for X and Z'. In other words, we may restrict to proving (19.9.8) when $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$, where B is an A-algebra of finite presentation, $Z = V(\mathfrak{J})$ for an ideal $\mathfrak{J}$ of finite type of B, $\mathscr{F} = \widetilde{M}$, and $\mathscr{G} = \widetilde{N}$, where N is an A-module and M is a finitely presented B-module that is flat as an A-module. We may moreover reduce to the case where N is a finitely presented A-module. Indeed, N is the filtered inductive limit of a system of finitely presented A-modules N_α (by taking a double inductive limit). If $\mathscr{G}_\alpha = \widetilde{N_\alpha}$, then $\mathscr{H}$ is the inductive limit of the $\mathscr{H}_\alpha = \mathscr{F} \otimes_{\mathscr{O}_X} f^*(\mathscr{G}_\alpha)$; since j_* and j^* commute with inductive limits and $\varinjlim$ is an exact functor in the category of quasi-coherent modules, the proposition for every $\mathscr{H}_\alpha$ implies it for $\mathscr{H}$.

It further suffices to prove that the canonical homomorphism

$$\Gamma(X, \mathscr{H}) \longrightarrow \Gamma(X - Z, \mathscr{H})$$

is injective (resp. bijective). There then exist a Noetherian subring A_0 of A, a finite-type A_0-algebra B_0, an ideal $\mathfrak{J}_0$ of B_0, a finitely generated B_0-module M_0 that is flat as an A_0-module, and an A_0-module N_0, such that $B = B_0 \otimes_{A_0} A$, $\mathfrak{J} = \mathfrak{J}_0 B$, $M = M_0 \otimes_{A_0} A$, and $N = N_0 \otimes_{A_0} A$ (8.9.1), (8.5.11), **IV-4** | 225
and ((11.2.7)). Let (A_λ) be the family of finite-type A_0-subalgebras of A, so that $A = \varinjlim A_\lambda$; set

$$B_\lambda = B_0 \otimes_{A_0} A_\lambda, \qquad \mathfrak{J}_\lambda = \mathfrak{J}_0 B_\lambda, \qquad M_\lambda = M_0 \otimes_{A_0} A_\lambda = M_0 \otimes_{B_0} B_\lambda,$$
$$N_\lambda = N_0 \otimes_{A_0} A_\lambda, \qquad S_\lambda = \mathrm{Spec}(A_\lambda), \quad X_\lambda = \mathrm{Spec}(B_\lambda), \quad Z_\lambda = V(\mathfrak{J}_\lambda).$$

If $p_\lambda : X \to X_\lambda$ is the canonical projection, then $Z = p_\lambda^{-1}(Z_\lambda)$ and $X - Z = p_\lambda^{-1}(X_\lambda - Z_\lambda)$. Since $X_\lambda - Z_\lambda$ is quasi-compact and quasi-separated, it follows from (8.5.2) that, on setting $\mathscr{F}_\lambda = \widetilde{M_\lambda}$ and $\mathscr{H}_\lambda = (M_\lambda \otimes_{A_\lambda} N_\lambda)^\sim$, the homomorphisms

$$\varinjlim \Gamma(X_\lambda, \mathscr{H}_\lambda) \longrightarrow \Gamma(X, \mathscr{H}), \qquad \varinjlim \Gamma(X_\lambda - Z_\lambda, \mathscr{H}_\lambda) \longrightarrow \Gamma(X - Z, \mathscr{H})$$

are bijective. By exactness of $\varinjlim$, it therefore suffices to prove that, for every sufficiently large λ, the canonical homomorphism $\Gamma(X_\lambda, \mathscr{H}_\lambda) \to \Gamma(X_\lambda - Z_\lambda, \mathscr{H}_\lambda)$ is injective (resp. bijective).

For every $x \in Z$, if x_λ is the projection of x in Z_λ, then (4.2.7) and (6.7.1) give

$$\operatorname{prof}((\mathscr{F}_{f(x)})_x) = \operatorname{prof}(((\mathscr{F}_\lambda)_{f_\lambda(x_\lambda)})_{x_\lambda}).$$

Thus, to reduce to proving (19.9.8) when A is Noetherian, it remains to prove that, for every sufficiently large λ, $\operatorname{prof}(((\mathscr{F}_\lambda)_{f_\lambda(x_\lambda)})_{x_\lambda}) \geqslant 1$ (resp. $\geqslant 2$) for every $x_\lambda \in Z_\lambda$. Let Z'_λ be the set of points $x_\lambda \in Z_\lambda$ such that, for every generization z_λ of x_λ in $Z_\lambda \cap (X_\lambda)_{f_\lambda(x_\lambda)}$, one has $\operatorname{prof}(((\mathscr{F}_\lambda)_{f_\lambda(x_\lambda)})_{z_\lambda}) \geqslant 1$ (resp. $\geqslant 2$). If $p_{\lambda\mu} : Z_\mu \to Z_\lambda$ is the canonical projection for $\lambda \leqslant \mu$, the hypothesis and (19.9.5) give $Z'_\mu = p_{\lambda\mu}^{-1}(Z'_\lambda)$ and $Z = p_\lambda^{-1}(Z'_\lambda)$. On the other hand, (19.9.6) shows that Z'_λ is open in Z_λ; the assertion therefore follows from (8.3.4).

Finally, when A is Noetherian, the flatness hypothesis and (6.3.1) give $\operatorname{prof}_Z(\mathscr{H}) \geqslant 1$ (resp. $\operatorname{prof}_Z(\mathscr{H}) \geqslant 2$); in this case the conclusion of (19.9.8) follows from (5.10.2) (resp. (5.10.4)). ☐

Remark (19.9.9). — The same proof, combined with the results of chap. III, 3rd part on depth and local cohomology, allows one to establish the following generalisation of (19.9.8):

Under the general conditions of (19.9.8), suppose that, for every $x \in Z$, we have $\operatorname{prof}((\mathscr{F}_{f(x)})_x) \geqslant k$; then the canonical homomorphism

$$H^i(X, \mathscr{H}) \longrightarrow H^i(X - Z, \mathscr{H}|(X - Z))$$

is bijective for $i \leqslant k-2$ and injective for $i = k-1$ (which, expressed using the general cohomological notion of depth introduced in chap. III, is also written $\operatorname{prof}_Z(\mathscr{H}) \geqslant k$).

§20. Meromorphic functions and pseudo-morphisms

20.0. Introduction Most of the concepts and results of §§20 and 21 relate directly to Chapter I and hardly depend on Chapters II to IV, except for the occasional use of the notion of depth and of a regular local ring (in (20.6), (21.11), (21.13), and (21.15)), of Zariski's "Main Theorem" in (20.4) and (21.12), and of the properties of transversally regular immersions in (20.6) and (21.15).

In §20, we introduce several variants of the concept of a rational map, already studied in (**I**, 7) from a point of view still fairly close to the classical point of view, and for this reason quite ill-suited to the case of not necessarily reduced preschemes. The notions and results of §20 are used in §21 (n^os^(21.1) and (21.7)) to develop the general notion of a divisor and its most elementary properties. This notion is especially convenient when the local rings of the preschemes considered are Noetherian and integrally closed, and especially when they are also *factorial* ((21.6) and (21.7)), because in the latter case it is identified with the notion of a *codimension*-1 *cycle* (a linear combination of irreducible subpreschemes of codimension 1). In (21.9), we determine the divisors on a Noetherian prescheme of dimension 1 but not necessarily normal, which is useful for various applications. Numbers (21.11) and (21.12) give two important theorems, due respectively to Auslander–Buchsbaum and Van der Waerden, and relating to the notion of a factorial ring (numbers (21.9), (21.11), and (21.12) are independent of one another). In numbers (21.13) and (21.14), which are likewise independent of the preceding three, we study a useful variant of the notion of a factorial local ring, namely that of a *parafactorial local ring*; this notion is introduced in particular in [41] in the development of comparison theorems for the Picard group of a projective prescheme X over a field k and of a "hyperplane section". We will see in (21.14.1) (Ramanujam–Samuel theorem) that the local parafactorial rings are much more numerous than one would have expected *a priori*. **IV-4** | 226

In (20.5), (20.6), and (21.15), we review the previous notions but from a point of view "relative" to a fixed base prescheme. For the moment these notions are used only rather rarely; in particular, the notion of a relative divisor is hardly used except for positive divisors, and in that case it can be described advantageously without recourse to relative meromorphic functions, by using the notion of a transversally regular immersion of codimension 1. It is therefore advisable to omit these sections on a first reading.

20.1. Meromorphic functions

(20.1.1). Let $(X, \mathscr{O}_X)$ be a ringed space, and let $\mathscr{S}$ be a subsheaf *of sets* of $\mathscr{O}_X$. For every open U of X, consider the *ring of fractions* $\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ (Bourbaki, *Alg. comm.*, chap. II, §2, n°1). It is immediate that the map $U \mapsto \Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ is a *presheaf of rings* ((**0**, 1.5.1) and (**0**, 1.5.7)). We denote by $\mathscr{O}_X[\mathscr{S}^{-1}]$ the *sheaf of rings* associated to this presheaf and we say that this is the *sheaf of rings of fractions of* $\mathscr{O}_X$ *with denominators in* $\mathscr{S}$; this is a *flat* $\mathscr{O}_X$-module. It is immediate that for every $x \in X$, we have a canonical isomorphism

$$(\mathscr{O}_X[\mathscr{S}^{-1}])_x \simeq \mathscr{O}_{X,x}[\mathscr{S}_x^{-1}], \tag{20.1.1.1}$$

since the reasoning of (**0**, 1.4.5) generalizes immediately in the case where we have an inductive system $(A_\alpha, \phi_{\beta\alpha})$ of rings, and for each index α a subset S_α of A_α such that $\phi_{\beta\alpha}(S_\alpha) \subset S_\beta$ for $\alpha \leqslant \beta$; **IV-4** | 227
we then take S to be the inductive limit, in $A = \varinjlim A_\alpha$, of the inductive system of subsets (S_α).

(20.1.2). Now let $\mathscr{F}$ be an $\mathscr{O}_X$-module. We then set

$$\mathscr{F}[\mathscr{S}^{-1}] = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_X[\mathscr{S}^{-1}], \tag{20.1.2.1}$$

and we say that this is the *sheaf of modules of fractions of* $\mathscr{F}$ *with denominators in* $\mathscr{S}$; it is immediate that it is associated to the presheaf of modules $U \mapsto \Gamma(U, \mathscr{F})[\Gamma(U, \mathscr{S})^{-1}]$, and that for every $x \in X$, we have a canonical isomorphism

$$(\mathscr{F}[\mathscr{S}^{-1}])_x \simeq \mathscr{F}_x[\mathscr{S}_x^{-1}]. \tag{20.1.2.2}$$

(20.1.3). We will focus here on the case where $\mathscr{S}$ is the subsheaf $\mathscr{S}(\mathscr{O}_X)$ of $\mathscr{O}_X$ such that for every open U, $\Gamma(U, \mathscr{S})$ is the *set of regular elements* of the ring $\Gamma(U, \mathscr{O}_X)$; it is immediate that it is a sheaf (and not merely a presheaf), since regularity of a section of $\mathscr{O}_X$ over U can be checked "fibre by fibre" (that is, its germ at x is regular in $\mathscr{O}_{X,x}$ for every $x \in U$); in other words, $\mathscr{S}(\mathscr{O}_X)_x$ is precisely the set of regular elements of $\mathscr{O}_{X,x}$. The corresponding sheaf of rings

$$\mathscr{M}_X = \mathscr{O}_X[\mathscr{S}^{-1}]$$

is called the *sheaf of germs of meromorphic functions on* X, and the sections of $\mathscr{M}_X$ over X are called the *meromorphic functions on* X; they form a ring which we denote by $M(X)$. For every $\mathscr{O}_X$*-module* $\mathscr{F}$,

$$\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_X = \mathscr{F}[\mathscr{S}^{-1}]$$

is also denoted $\mathscr{M}_X(\mathscr{F})$ and called the *sheaf of germs of meromorphic sections of* $\mathscr{F}$; its sections over X form an $M(X)$-module denoted by $M(X, \mathscr{F})$, whose elements are called *meromorphic sections of* $\mathscr{F}$ *over* X. These definitions imply that for every open U of X, we have a canonical isomorphism $\mathscr{M}_X(\mathscr{F})|U \simeq \mathscr{M}_U(\mathscr{F}|U)$, in particular $\mathscr{M}_X|U \simeq \mathscr{M}_U$.

(20.1.3.1). If X is a *reduced prescheme*, then we note that if an element $s \in \Gamma(U, \mathscr{O}_X)$ is such that $s_\xi \neq 0$ for every maximal point ξ of U, then s is *regular*. Indeed, if $st = 0$ for a $t \in \Gamma(U, \mathscr{O}_X)$, then we have $s_\xi t_\xi = 0$, so $t_\xi = 0$ since $\mathscr{O}_{X,\xi}$ is a field, and say that $t_\xi = 0$ for every maximal point ξ of X means that $t = 0$: we are immediately reduced to the case where U is affine, and an element of a reduced ring which belongs to all the minimal prime ideals is zero by definition. The converse is true if the set of irreducible components of X is *locally finite*. We immediately reduce to the case where $X = \operatorname{Spec}(A)$ is affine; if $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ are the minimal prime ideals of A and if $s \in \mathfrak{p}_i$ for an index i, then there exists $t \in A$ such that $t \in \mathfrak{p}_j$ for $j \neq i$ and $t \notin \mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. II, §1, n°1, Prop. 1); so we have $st \in \mathfrak{p}_i$ for all i, and as a result $st = 0$ since A is reduced; s is therefore nonregular.

(20.1.4). For every open U of X, the homomorphism $t \mapsto t/1$ from $\Gamma(U, \mathscr{O}_X)$ to $\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ (which is none other than the *total ring of fractions* of $\Gamma(U, \mathscr{O}_X)$) is injective; these homomorphisms **IV-4** | 228
thus define a *canonical injective homomorphism*

$$i : \mathscr{O}_X \longrightarrow \mathscr{M}_X \tag{20.1.4.1}$$

which allows us to identify $\mathscr{O}_X$ with a subsheaf of $\mathscr{M}_X$. Given a meromorphic function $\phi \in M(X)$, we say that ϕ is *defined* on an open U of X if $\phi|U$ is a *section of* $\mathscr{O}_U$ over U; the axioms of sheaves show that there is, for a given section ϕ, a *largest* open on which ϕ is defined; we call this the *domain of definition* of ϕ and denote it by $\operatorname{dom}(\phi)$.

(20.1.5). For every $\mathscr{O}_X$-module $\mathscr{F}$, we obtain from (20.1.4.1) a di-homomorphism consisting of i and the homomorphism of sheaves of additive groups

$$1_{\mathscr{F}} \otimes i : \mathscr{F} \longrightarrow \mathscr{M}_X(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_X. \tag{20.1.5.1}$$

We note that the latter is not injective in general; when it is injective, we say that $\mathscr{F}$ is *torsion-free*: this means that for every open U of X and every section $s \in \Gamma(U, \mathscr{O}_X)$ which is a regular element in this ring the homothety $z \mapsto sz$ of $\Gamma(U, \mathscr{F})$ is injective; this condition is evidently satisfied if $\mathscr{F}$ is *locally free*.

Proposition (20.1.6). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. For $\mathscr{F}$ to be torsion-free, it is necessary and sufficient that* $\operatorname{Ass}(\mathscr{F}) \subset \operatorname{Ass}(\mathscr{O}_X)$.

Proof. We immediately reduce to the case where $X = \operatorname{Spec}(A)$ is affine, $\mathscr{F} = \widetilde{M}$, and we know that the elements s of A belonging to an ideal of $\operatorname{Ass}(M)$ are exactly those for which the homothety $z \mapsto sz$ is not injective (Bourbaki, *Alg. comm.*, chap. IV, §1, n°1, cor. 2 of prop. 2). ☐

(20.1.7). If u is a section of $\mathscr{M}_X(\mathscr{F})$ over X, then we say that u is *defined* at a point $x \in X$ if there exists an open neighbourhood V of x in X such that $u|V$ is the image of a section of $\mathscr{F}$ over V under the di-homomorphism (20.1.5.1). We say that u is *defined* on an open U of X if it is defined at every point in U; there is still a larger open in which u is defined, called the *domain of definition* of u and denoted $\operatorname{dom}(u)$. When $\mathscr{F}$ is torsion-free, so that $\mathscr{F}$ is identified by (20.1.5.1) with a subsheaf of $\mathscr{M}_X(\mathscr{F})$, saying that u is defined on U means that $u|U$ is a *section of $\mathscr{F}$ over U*.

(20.1.8). In accordance with the general notation of ($\mathbf{0_I}$, 5.4.7), we denote by $\mathscr{M}_X^*$ the sheaf of multiplicative groups such that $\Gamma(U, \mathscr{M}_X^*)$ is (for every open U of X) the group of *invertible elements* of $\Gamma(U, \mathscr{M}_X)$. This sheaf is none other than the sheaf $\mathscr{S}(\mathscr{M}_X)$ defined in (20.1.3): indeed, if $s \in \Gamma(U, \mathscr{S}(\mathscr{M}_X))$, then for every $x \in U$, there exists an open neighbourhood $V \subset U$ of x such that $s|V$ is a regular element in the *total ring of fractions* of $\Gamma(V, \mathscr{O}_X)$, and we know that such an element is necessarily invertible in this ring of fractions. We say that the sections of $\mathscr{M}_X^*$ over X are the *regular meromorphic functions* (note that we are deviating here from the terminology followed by certain authors, who call "regular" meromorphic functions those which are *sections of* $\mathscr{O}_X$, identified with a subsheaf of $\mathscr{M}_X$).

Let $\mathscr{L}$ be an *invertible $\mathscr{O}_X$-module* ($\mathbf{0_I}$, 5.4.1); then it is clear that $\mathscr{M}_X(\mathscr{L}) = \mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{M}_X$ is an *invertible $\mathscr{M}_X$-module*. Let U be an open such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$; as every automorphism of $\mathscr{M}_U$ is multiplication by an invertible element of $\Gamma(U, \mathscr{M}_X)$ ($\mathbf{0_I}$, 5.4.7), it amounts to the same thing to say that a section $s \in \Gamma(U, \mathscr{M}_X(\mathscr{L}))$ has an invertible image in $\Gamma(U, \mathscr{M}_X)$ under *some* isomorphism or under *every* isomorphism onto $\Gamma(U, \mathscr{M}_X)$; we say in this case that s is a *regular meromorphic section of $\mathscr{L}$* over U; a section s of $\mathscr{L}$ over X is called a *regular meromorphic section of $\mathscr{L}$* if, for every open U such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$, $s|U$ is a regular meromorphic section of $\mathscr{L}$ over U. We denote by $(\mathscr{M}_X(\mathscr{L}))^*$ the subsheaf of $\mathscr{M}_X(\mathscr{L})$ such that for every open U, $\Gamma(U, (\mathscr{M}_X(\mathscr{L}))^*)$ is the set of regular meromorphic sections of $\mathscr{L}$ over U. Let s be a meromorphic section of $\mathscr{L}$ over X (i.e. a section of $\mathscr{M}_X(\mathscr{L})$); it defines a homomorphism $h_s : \mathscr{M}_X \to \mathscr{M}_X(\mathscr{L})$ which sends every section t of $\mathscr{M}_X$ over an open U to $(s|U)t$. It follows immediately from the above that for s to be *regular*, it is necessary and sufficient for h_s to be *injective*, and in fact h_s is then a *bijective* homomorphism from $\mathscr{M}_X$ to $\mathscr{M}_X(\mathscr{L})$, and its restriction to $\mathscr{M}_X^*$ is a bijection to $(\mathscr{M}_X(\mathscr{L}))^*$. We conclude that the homothety $t \mapsto ts$ is an isomorphism from $M(X)$ to $M(X, \mathscr{L})$. **IV-4** | 229

(20.1.9). Let s be a regular meromorphic section of an invertible $\mathscr{O}_X$-module $\mathscr{L}$ over X; then for every $\mathscr{O}_X$-module $\mathscr{F}$, s similarly defines a homomorphism $h_s \otimes 1_{\mathscr{F}} : \mathscr{M}_X(\mathscr{F}) \to \mathscr{M}_X(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L})$, which is again *bijective*.

(20.1.10). Let s be a meromorphic section of an invertible $\mathscr{O}_X$-module $\mathscr{L}$ over X; for s to be regular, it is necessary and sufficient for there to exist a meromorphic section s' of $\mathscr{L}^{-1}$ over X such that the canonical image of $s \otimes s'$ in $\mathscr{M}_X$ ($\mathbf{0_I}$, 5.4.3) is the unit section, and this section s' is then unique: indeed, the necessity of the local existence of such a section is evident, and its local uniqueness implies its global (and unique) existence; moreover, the existence of s' is trivially sufficient for s to be regular. We will take $s' = s^{-1}$.

Finally, if $\mathscr{L}'$ is a second invertible $\mathscr{O}_X$-module, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, then $s \otimes s'$ is evidently a regular meromorphic section of $\mathscr{L} \otimes \mathscr{L}'$ over X.

(20.1.11). If $f : X' \to X$ is a morphism of ringed spaces, then there is in general no natural map sending a meromorphic function on X to a meromorphic function on X'. For example, if X is the spectrum of a local integral domain A and X' is the spectrum of its residue field k, then there is no natural homomorphism from the field of fractions K of A to k, and we can only send an element of K to an element of k if it is already in A.

In general, if $f = (\psi, \theta)$, then for every open U of X, denote by $\mathscr{S}_f(U)$ the set of *regular* sections $s \in \Gamma(U, \mathscr{O}_X)$ such that the image of s under

$$\Gamma(\theta^\#) : \Gamma(U, \mathscr{O}_X) \longrightarrow \Gamma(f^{-1}(U), \mathscr{O}_{X'})$$

is a *regular* section. It is immediate that $U \mapsto \mathscr{S}_f(U)$ is a *subsheaf* of the sheaf of sets $\mathscr{S}(\mathscr{O}_X)$, which we denote by $\mathscr{S}_f$. We set $\mathscr{M}_f = \mathscr{O}_X[\mathscr{S}_f^{-1}]$; this is a subsheaf of rings of $\mathscr{M}_X$, and we canonically **IV-4** | 230
obtain from $\theta^\# : \psi^*(\mathscr{O}_X) \to \mathscr{O}_{X'}$ a homomorphism of sheaves of rings $\theta'^\# : \psi^*(\mathscr{M}_f) \to \mathscr{M}_{X'}$ extending $\theta^\#$ (Bourbaki, *Alg. comm.*, chap. II, §2, n°1, prop. 2); hence, recalling that $f^*(\mathscr{M}_f) = \psi^*(\mathscr{M}_f) \otimes_{\psi^*(\mathscr{O}_X)} \mathscr{O}_{X'}$, we get a canonical homomorphism of $\mathscr{O}_{X'}$-algebras

$$f^*(\mathscr{M}_f) \longrightarrow \mathscr{M}_{X'}. \tag{20.1.11.1}$$

For every meromorphic function ϕ on X that *is a section of* $\mathscr{M}_f$, $\Gamma(\theta'^\#)(\phi)$ is a meromorphic function on X', called the *inverse image of ϕ under f*, and denoted by $\phi \circ f$ if there is no cause for confusion.

Similarly, if $\mathscr{F}$ is an $\mathscr{O}_X$-module, then we set $\mathscr{M}_f(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_f$, and we immediately obtain from $\theta'^\#$ a canonical homomorphism (which is also written as $u \mapsto u \circ f$)

$$\Gamma(X, \mathscr{M}_f(\mathscr{F})) \longrightarrow \Gamma(X', \mathscr{M}_{X'}(f^*(\mathscr{F}))).$$

In addition, if $u \in \Gamma(X, \mathscr{M}_f(\mathscr{F}))$ is defined (20.1.7) at a point x, then u coincides, on a neighbourhood U of x, with a section of the form $\sum_i h_i \otimes (t_i/s_i)$, where the h_i belong to $\Gamma(U, \mathscr{F})$, the t_i to $\Gamma(U, \mathscr{O}_X)$, and the s_i to $\Gamma(U, \mathscr{S}_f)$. As by hypothesis the images of the s_i in $\Gamma(f^{-1}(U), \mathscr{O}_{X'})$ are regular, we see that $u \circ f$ is defined at every point of $f^{-1}(U)$; in other words, we have

$$f^{-1}(\mathrm{dom}(u)) \subset \mathrm{dom}(u \circ f). \tag{20.1.11.2}$$

We will see later (20.6.5, (ii)) examples (with $\mathscr{F} = \mathscr{O}_X$) where the two sides of (20.1.11.2) can be different.

Consider in particular the case where $\mathscr{M}_f = \mathscr{M}_X$; then, if $\mathscr{L}$ is an invertible $\mathscr{O}_X$-module, the image in $\mathscr{M}_{X'}(f^*(\mathscr{L}))$, under $\Gamma(\theta'^\#)$, of a *regular* meromorphic section of $\mathscr{L}$ over X (20.1.8) is a *regular* meromorphic section of $f^*(\mathscr{L})$ over X', as it follows immediately from the definition of its sections, and from the fact that a homomorphism of rings sends an invertible element to an invertible element.

Let $f' : X'' \to X'$ be a second morphism of ringed spaces, and suppose that $\mathscr{M}_f = \mathscr{M}_X$ and $\mathscr{M}_{f'} = \mathscr{M}_{X'}$; then, if we set $f'' = f \circ f'$, we also have $\mathscr{M}_{f''} = \mathscr{M}_X$, and we immediately see that for every meromorphic section u of $\mathscr{F}$ over X, we have $u \circ f'' = (u \circ f) \circ f'$.

Proposition (20.1.12). — *If the morphism $f : X' \to X$ is flat* ($\mathbf{0_I}$, 6.7.1)*, then we have $\mathscr{M}_f = \mathscr{M}_X$, and the homomorphism $\phi \mapsto \phi \circ f$ is defined on all of $M(X)$. In addition, if f is a (flat) morphism of locally ringed spaces, then we have* $\mathrm{dom}(\phi \circ f) = f^{-1}(\mathrm{dom}(\phi))$*; if in addition f is surjective (thus faithfully flat), then the homomorphism $\phi \mapsto \phi \circ f$ is injective.*

Proof. The first assertion follows from the fact that, if B is an A-algebra which is a flat A-module, then every element of A not a divisor of 0 in A is not a divisor of 0 in B ($\mathbf{0_I}$, 6.3.4). To prove the other assertions, note that, for every $x' \in X'$, if $x = f(x')$, then $\mathscr{O}_{X',x'}$ is a flat $\mathscr{O}_{X,x}$-module, and as the homomorphism $\mathscr{O}_{X,x} \to \mathscr{O}_{X',x'}$ is *local* by hypothesis, it is injective (($\mathbf{0_I}$, 6.5.1) and ($\mathbf{0_I}$, 6.6.2)); if we set $A = \mathscr{O}_{X,x}$, $B = \mathscr{O}_{X',x'}$, such that A identifies with a subring of B, then $(f^*(\mathscr{M}_X))_{x'}$ is equal to $S^{-1}A \otimes_A B = S^{-1}B$, where S is the set of regular elements of A, $(\mathscr{M}_{X'})_{x'}$ is equal to $T^{-1}B$, where T is the set of regular elements of B, and as we have seen that $S \subset T$, the homomorphism $S^{-1}B \to T^{-1}B$ **IV-4** | 231
is injective; in other words, this proves that the homomorphism (20.1.11.1) $f^*(\mathscr{M}_X) \to \mathscr{M}_{X'}$ is *injective* (hence the last assertion of the statement). The quotient $f^*(\mathscr{M}_X)/\mathscr{O}_{X'}$ identifies with an

$\mathscr{O}_{X'}$-submodule of $\mathscr{M}_{X'}/\mathscr{O}_{X'}$, and $(f^*(\mathscr{M}_X)/\mathscr{O}_{X'})_{x'}$ identifies with $(\mathscr{M}_X/\mathscr{O}_X)_x \otimes_{\mathscr{O}_{X,x}} \mathscr{O}_{X',x'}$. Then suppose that $x \notin \mathrm{dom}(\phi)$; the image of ϕ_x in $(\mathscr{M}_X/\mathscr{O}_X)_x$ is therefore $\neq 0$; by faithful flatness, we deduce that it is the same for the image of $(\phi \circ f)_{x'}$, so $x' \notin \mathrm{dom}(\phi \circ f)$, which finishes the proof. □

Remark (20.1.13). — Let X be a *reduced* complex analytic space; then the notion of a meromorphic function on X defined above coincides with the usual notion. Consider on the other hand a prescheme Y, locally of finite type over the field $\mathbf{C}$; we then know that we can associate to Y an analytic space Y^{an} having the same underlying topological space, and the canonical morphism $f : Y^{\mathrm{an}} \to Y$ is *flat* [37]; by virtue of (20.1.12), the canonical homomorphism $u \mapsto u \circ f$ from $M(Y)$ to $M(Y^{\mathrm{an}})$ is therefore always defined and is injective; but it is not *surjective* in general. For example, when $Y = \mathbf{V}^r_{\mathbf{C}}$ ($\mathbf{Err}_{\mathbf{III}}$, 14) is affine r-space over $\mathbf{C}$, $M(Y)$ canonically identifies with the field $R(Y)$ of rational functions on Y (20.2.13, (i)), while $M(Y^{\mathrm{an}})$ is the field of usual meromorphic functions on $\mathbf{C}^r$. Because of this fact, it is often preferable, in algebraic geometry, to abstain from the terminology introduced in this section, and to use the equivalent terminology of "pseudo-function" which will be defined below.

20.2. Pseudo-morphisms and pseudo-functions The only ringed spaces discussed in this section are *preschemes*.

(20.2.1). Recall (11.10.2) that in a prescheme X, we say that an open U is *schematically dense* if, for every open V of X, the canonical homomorphism $\Gamma(V, \mathscr{O}_X) \to \Gamma(V \cap U, \mathscr{O}_X)$ is *injective*. **IV-4** | 231

Consider two preschemes X, Y, two *schematically dense* opens U, U' of X; we say that two morphisms $u : U \to Y$, $u' : U' \to Y$ are *equivalent* if there exists an open $U'' \subset U \cap U'$, *schematically dense* in X, such that $u|U'' = u'|U''$. As it results immediately from the definition of schematically dense opens that the intersection of two such opens is again one, it is immediate that the preceding relation is indeed an equivalence relation. An equivalence class for this relation is called a *pseudo-morphism* of X in Y, or a *strict rational application* of X in Y.

If S is a prescheme and X, Y two S-preschemes, we call *pseudo-S-morphism* the equivalence class (for the preceding relation) of an *S-morphism* of a schematically dense open of X in Y. A pseudo-morphism is therefore nothing other than a pseudo-S-morphism for $S = \mathrm{Spec}(\mathbf{Z})$.

We denote by $\mathrm{Ps.hom}(X, Y)$ (resp. $\mathrm{Ps.hom}_S(X, Y)$) the set of pseudo-morphisms (resp. pseudo-S-morphisms) of X in Y.

(20.2.2). It follows from the definition recalled above that if U is a schematically dense open in X, then, for every open V of X, $U \cap V$ is schematically dense in V. If two morphisms $u : U \to Y$, $u' : U' \to Y$ of schematically dense opens of X in Y are equivalent, it follows that their restrictions $u|(V \cap U)$ and $u'|(V \cap U')$ are also equivalent morphisms (for the prescheme induced on V); their class is called the *restriction* to V of the pseudo-morphism ω, the class of u, and we denote this pseudo-morphism $\omega|V$. If $W \subset V$ is an open of X, it is clear that $(\omega|V)|W = \omega|W$. This shows that the restriction maps define a *presheaf* of sets $U \mapsto \mathrm{Ps.hom}(U, Y)$; we note that this presheaf *is not* in general a sheaf (cf. (20.2.16)); the associated sheaf is denoted $\mathscr{P}s.\mathrm{hom}(X, Y)$. For pseudo-$S$-morphisms, we see that $U \mapsto \mathrm{Ps.hom}_S(U, Y)$ is a presheaf of sets, whose associated sheaf we denote $\mathscr{P}s.\mathrm{hom}_S(X, Y)$. **IV-4** | 232

If V is a *schematically dense* open in X, then for every open U schematically dense in X, $U \cap V$ is also schematically dense in X, so the map $\omega \mapsto \omega|V$ is a *bijection* of $\mathrm{Ps.hom}(X, Y)$ (resp. $\mathrm{Ps.hom}_S(X, Y)$) onto $\mathrm{Ps.hom}(V, Y)$ (resp. $\mathrm{Ps.hom}_S(V, Y)$).

(20.2.3). Given a pseudo-S-morphism ω of X in Y, we say that it is *defined* at a point $x \in X$ if there exists an open neighbourhood V of x in X, an open $U \subset V$ *containing x and schematically dense in V*, and finally an S-morphism $u : U \to Y$ whose class in $\mathrm{Ps.hom}_S(V, Y)$ is equal to $\omega|V$ (20.2.2); we call the *domain of definition* of ω and denote $\mathrm{dom}_S(\omega)$ (or simply $\mathrm{dom}(\omega)$ if $S = \mathrm{Spec}(\mathbf{Z})$) the set of $x \in X$ where ω is defined; this is evidently an *open* of X. Moreover, for every open W of X, we have

$$\mathrm{dom}_S(\omega|W) = (\mathrm{dom}_S(\omega)) \cap W \tag{20.2.3.1}$$

by virtue of the property of schematically dense opens recalled in (20.2.2).

Proposition (20.2.4). — *Suppose X, Y are S-preschemes such that the structural morphism $Y \to S$ is separated; then, if ω is a pseudo-S-morphism of X in Y, $\mathrm{dom}_S(\omega)$ is the largest of the schematically dense opens U of X such that there exists an S-morphism $u : U \to Y$ belonging to the class ω.*

Proof. It suffices to prove that if U, U' are two schematically dense opens in X such that two S-morphisms $u : U \to Y$ and $u' : U' \to Y$ are equivalent, then the restrictions of u and u' to $U \cap U'$ are equal. But there exists by hypothesis an open $U'' \subset U \cap U'$, schematically dense in X, in which u and u' coincide; as U'' is also schematically dense in $U \cap U'$, our assertion results from (11.10.1, (d)). ☐

Corollary (20.2.5). — *Let S be an S_0-scheme, and let X, Y be two S-preschemes such that the composite morphism $Y \to S \to S_0$ is separated (which implies, by* (**I**, 5.5.1)*, that $Y \to S$ is also separated). For every pseudo-S-morphism ω of X in Y, one then has $\mathrm{dom}_S(\omega) = \mathrm{dom}_{S_0}(\omega)$. In particular, if Y is a prescheme, one has $\mathrm{dom}_S(\omega) = \mathrm{dom}(\omega)$.*

Proof. Indeed, it is clear that $\mathrm{dom}_S(\omega) \subset \mathrm{dom}_{S_0}(\omega)$ without the hypothesis of separation; by virtue of (20.2.4), it suffices to prove that if $u_0 : U_0 \to Y$ is an S_0-morphism of a schematically dense open **IV-4** | 233 U_0 of X in Y such that the composed $v_0 : U_0 \xrightarrow{u_0} Y \to S$ coincides with the restriction of the structural morphism $w_0 : U_0 \to S$ in an open $U \subset U_0$ schematically dense in X, then one has $v_0 = w_0$. But by virtue of the hypothesis that the morphism $S \to S_0$ is separated, this again results from (11.10.1, , d)). ☐

Corollary (20.2.6). — *Under the hypotheses of (20.2.4), the presheaf $U \mapsto \mathrm{Ps.hom}_S(U, Y)$ is a sheaf.*

Proof. Indeed, let U be an open of X, (U_α) a covering of U by opens of X, and suppose given for each α a pseudo-S-morphism ω_α of U_α in Y, such that for every pair of indices α, β, the restrictions (20.2.2) of ω_α and ω_β to $U_\alpha \cap U_\beta$ are equal; by virtue of (20.2.3.1), this implies that $\mathrm{dom}_S(\omega_\alpha) \cap U_\beta = \mathrm{dom}_S(\omega_\beta) \cap U_\alpha$. Let W be the union of the opens $\mathrm{dom}_S(\omega_\alpha)$, and, for each α, let u_α be the unique S-morphism $\mathrm{dom}_S(\omega_\alpha) \to Y$ belonging to the class ω_α (20.2.4); by virtue of (20.2.4), the restrictions of u_α and u_β to $\mathrm{dom}_S(\omega_\alpha) \cap \mathrm{dom}_S(\omega_\beta)$ are equal, so there exists a morphism $u : W \to Y$ whose restriction to each $\mathrm{dom}_S(\omega_\alpha)$ is equal to u_α; it is clear that W is schematically dense in U, so it defines a pseudo-S-morphism ω of U in Y whose restrictions to the U_α are the ω_α. ☐

Remark (20.2.7). — We know (11.10.4) that when X is *reduced*, it amounts to the same to say that an open of X is dense in X or that it is schematically dense in X; the notion of pseudo-morphism (resp. pseudo-S-morphism) of X in Y then coincides with that of *rational application* (resp. *S-rational application*) of X in Y (**I**, 7.1.2). In the general case, the notion of pseudo-morphism seems more useful than that of rational application.

(20.2.8). We call *pseudo-function* on X a pseudo-morphism of X in $\mathrm{Spec}(\mathbf{Z}[T])$ (T indeterminate), or, what amounts to the same, an X-pseudo-morphism of X in $X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$; this amounts also to the same (**I**, 3.3.15) as saying that a pseudo-function on X is an *equivalence class of sections of $\mathscr{O}_X$ over schematically dense opens* of X, two sections $g \in \Gamma(U, \mathscr{O}_X)$, $g' \in \Gamma(U', \mathscr{O}_X)$ over such opens being equivalent if there exists an open $U'' \subset U \cap U'$, schematically dense in X, where g and g' coincide. We can apply (20.2.4) with $S = X$ and $Y = X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$; then, for every pseudo-function φ on X, there is a *largest* schematically dense open $\mathrm{dom}(\varphi)$ in X such that there is a section of $\mathscr{O}_X$ over $\mathrm{dom}(\varphi)$ belonging to the class φ. It is clear that the sheaf $\mathscr{P}s.\mathrm{hom}_X(X, X \otimes_{\mathbf{Z}} \mathbf{Z}[T])$ is here a *sheaf of rings*, and even an *$\mathscr{O}_X$-Algebra*, which we denote $\mathscr{M}'_X$; it results from (20.2.6) that, for every open U of X, $\Gamma(U, \mathscr{M}'_X)$ is equal to the set of pseudo-morphisms of U in $\mathrm{Spec}(\mathbf{Z}[T])$; we set $M'(X) = \Gamma(X, \mathscr{M}'_X)$. To say that a section φ of $\mathscr{M}'_X$ over U is *invertible* in the ring $\Gamma(U, \mathscr{M}'_X)$ means that there exists an open U' schematically dense in $\mathrm{dom}(\varphi)$, thus in U, such that, if g is the unique section of $\mathscr{O}_X$ over $\mathrm{dom}(\varphi)$ belonging to the class φ, $g|U'$ is *invertible* in $\Gamma(U', \mathscr{O}_X)$. It results from (**I**, 3.3.15) that, in the canonical correspondence between $\Gamma(V, \mathscr{O}_X)$ and $\mathrm{Hom}(V, \mathbf{Z}[T])$ (V open of X), the invertible elements of $\Gamma(V, \mathscr{O}_X)$ correspond to the morphisms which *factor through* $V \to \mathrm{Spec}(\mathbf{Z}[T, T^{-1}]) \to$ **IV-4** | 234 $\mathrm{Spec}(\mathbf{Z}[T])$. We conclude that the sheaf $\mathscr{M}'^*_X$ of invertible germs of sections of $\mathscr{M}'_X$ is canonically identified with the sheaf $\mathscr{P}s.\mathrm{hom}_X(X, X \otimes_{\mathbf{Z}} \mathbf{Z}[T, T^{-1}])$.

Lemma (20.2.9). — *Let U be an open of X, s a regular element of the ring $\Gamma(U, \mathscr{O}_X)$; then the open U_s of the $x \in U$ such that $s(x) \neq 0$ is schematically dense in U.*

Proof. Indeed, let V be an open of U, t a section of $\mathscr{O}_X$ over V such that $t|(V \cap U_s) = 0$. For every affine open $W \subset V$, there exists an integer n such that $s^n t|W = 0$ (**I**, 1.4.1); but since s is a *regular* element of $\Gamma(U, \mathscr{O}_X)$, this implies $t|W = 0$, hence $t = 0$. □

(20.2.10). Consider a meromorphic function $f \in M(X)$ (20.1.4); then $\operatorname{dom}(f)$ is *schematically dense* in X: indeed, every point of X admits an open neighbourhood U such that there exists a section $s \in \Gamma(U, \mathscr{O}_X)$ which is a regular element of this ring, and for which $s(f|U) \in \Gamma(U, \mathscr{O}_X)$; as $s|U_s$ is invertible, we conclude that $f|U_s \in \Gamma(U_s, \mathscr{O}_X)$, hence $U_s \subset \operatorname{dom}(f)$ by definition (20.1.4), and our assertion results from the lemma (20.2.9). We can thus associate to f the pseudo-function equal to the class of the section of $\mathscr{O}_X$ over $\operatorname{dom}(f)$, restriction of f; operating in the same way replacing X by an arbitrary open of X, we thus obtain a canonical *homomorphism of $\mathscr{O}_X$-Algebras*

(20.2.10.1) $$\mathscr{M}_X \longrightarrow \mathscr{M}'_X$$

which, by restriction, gives evidently a homomorphism of sheaves of abelian groups

(20.2.10.2) $$\mathscr{M}^*_X \longrightarrow \mathscr{M}'^*_X$$

for the sheaves of invertible germs of sections of $\mathscr{M}_X$ and $\mathscr{M}'_X$.

Proposition (20.2.11). — (i) *The canonical homomorphism* (20.2.10.1) *(and hence also* (20.2.10.2)*) is* injective*.*

(ii) *Suppose either that X is locally Noetherian, or that X is reduced and the set of its irreducible components is locally finite. Then the canonical homomorphism* (20.2.10.1) *(and hence also* (20.2.10.2)*) is* bijective*.*

Proof. The questions being local on X, we can restrict to the case where $X = \operatorname{Spec}(A)$ is affine, and show that the canonical homomorphism $M(X) \to M'(X)$ is injective, and that it is bijective in the cases considered in (ii). Taking into account the definition of the sheaf $\mathscr{M}_X$ (20.1.3), we can in fact remark that (20.2.10.1) comes from a homomorphism of *presheaves*

$$\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}] \longrightarrow \Gamma(U, \mathscr{M}'_X)$$

and it suffices to show that the latter is injective (resp. bijective under the hypotheses of (ii)). Denoting by S the set of regular elements of A (so that $S^{-1}A$ is the *total ring of fractions* of A), we must therefore consider the canonical homomorphism

(20.2.11.1) $$S^{-1}A \longrightarrow M'(X).$$

We can write $S^{-1}A = \varinjlim A_t$, where t runs through the set of *regular* elements of A, ordered by the relation "t is a divisor of t'" (**0**, 1.4.5); we can also write $A_t = \Gamma(\mathrm{D}(t), \mathscr{O}_X)$. On the other hand, by definition $M'(X) = \varinjlim \Gamma(U, \mathscr{O}_X)$, where U runs through the set of schematically dense opens of X (ordered by the relation $\supset$), and the map (20.2.11.1) is none other than the canonical map coming from the fact that the $\mathrm{D}(t)$ constitute a part of the set of schematically dense opens in X (20.2.9). Note that by definition of a schematically dense open, the homomorphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U', \mathscr{O}_X)$ for any two such opens $U' \subset U$ is always *injective*, hence the homomorphisms $\Gamma(U, \mathscr{O}_X) \to M'(X)$ are also injective, and we conclude that (20.2.11.1) is injective. To prove that (20.2.11.1) is bijective, it suffices to show that the $\mathrm{D}(t)$ are *cofinal* in the set of schematically dense opens, i.e., that for such an open U, there exists t regular in A such that $\mathrm{D}(t) \subset U$. Now, when X is reduced and the irreducible components of X form a locally finite set, this set is finite since X has been supposed affine, so A has only a finite number of minimal primes $\mathfrak{p}_i$; as A is reduced, the intersection of the $\mathfrak{p}_i$ is zero, and to say that t is regular is equivalent to saying that t does not belong to any of the $\mathfrak{p}_i$; we conclude by the reasoning of (**I**, 7.1.9.1). When A is Noetherian, to say that $U = X - Y$ (where $Y = V(\mathfrak{J})$ is closed in X) is schematically dense means (5.10.2) that Y *does not meet* $\operatorname{Ass}(\mathscr{O}_X)$, and by virtue of Bourbaki, *Alg. comm.*, chap. IV, §1, nº 4, prop. 8, this implies the existence of a $t \in \mathfrak{J}$ such that t is A-regular, hence $U \supset \mathrm{D}(t)$. IV-4 | 235

We have moreover proved in the course of this demonstration the □

Corollary (20.2.12). — *If $X = \operatorname{Spec}(A)$, where A is Noetherian, or reduced and having only a finite number of minimal prime ideals, the schematically dense opens in X are those which contain an open of the* *form* $\mathrm{D}(t)$, *where* t *is regular in* A, *and* $M(X) = M'(X)$ *is the total ring of fractions* $S^{-1}A$, *where* S *is the set of regular elements of* A.

Remarks (20.2.13). — (i) Let φ be an element of $M(X)$, φ' its image in $M'(X)$; by definition ((20.1.4) and (20.2.8)) $\mathrm{dom}(\varphi) \subset \mathrm{dom}(\varphi')$; but in fact one has the equality $\mathrm{dom}(\varphi) = \mathrm{dom}(\varphi')$, because there exists a section of $\mathscr{O}_X$ over $\mathrm{dom}(\varphi')$, belonging to the class φ', and the corresponding meromorphic function equals φ by (20.2.11, (i)), hence $\mathrm{dom}(\varphi') \subset \mathrm{dom}(\varphi)$.

(ii) We have already noted that when X is *reduced*, one has $\mathscr{M}'_X = \mathscr{R}(X)$ (sheaf of rational functions on X (**I**, 7.3.2)); if moreover the irreducible components of X form a locally finite set, one has $\mathscr{M}_X = \mathscr{M}'_X = \mathscr{R}(X)$. In general, since every schematically dense open is dense, one has a canonical homomorphism $\mathscr{M}'_X \to \mathscr{R}(X)$, but even when X is locally Noetherian, this homomorphism is not necessarily injective. For example, if $X = \mathrm{Spec}(A)$, where A is a Noetherian ring such that $\mathrm{Ass}(A)$ contains embedded prime ideals (which implies that A is not reduced), then $M(X)$ and $M'(X)$ identify with the total ring of fractions $S^{-1}A$, where **IV-4** | 236
S is the set of regular elements of A, the complement of the union of the ideals $\mathfrak{p} \in \mathrm{Ass}(A)$; on the other hand, $R(X)$ identifies with $Q^{-1}A$, where Q is the complement of the union of the *minimal* prime ideals of A (**I**, 7.1.9), and the canonical homomorphism $A \to Q^{-1}A$ (and *a fortiori* $S^{-1}A \to Q^{-1}A$) is therefore not injective, since there exist nonzero elements of $A - Q$ annihilated by elements of Q (Bourbaki, *Alg. comm.*, chap. IV, §1, nº 1, cor. 2 of prop. 1).

(iii) We note also that when X is locally Noetherian, the $\mathscr{O}_X$-Module $\mathscr{M}'_X$ is *not necessarily quasi-coherent*. Consider for example a local Noetherian ring A of dimension $\geqslant 2$, with maximal ideal $\mathfrak{m} \in \mathrm{Ass}(A)$, and if we set $X = \mathrm{Spec}(A)$, the scheme induced on the open U of X, complementary of $\{\mathfrak{m}\}$, is *integral*. The only regular elements of A are the invertible elements, so $\Gamma(X, \mathscr{M}'_X) = M'(X) = A$; if $\mathscr{M}'_X$ were quasi-coherent, it would therefore be equal to $\mathscr{O}_X$; but as U is an integral scheme, $\mathscr{M}'_X|U = R(U)$ is a simple sheaf (**I**, 7.3.5), whereas $\mathscr{O}_U$ is not a simple sheaf since $\dim(A) \geqslant 2$.

It remains to give an example of a ring A having the above properties. It suffices to consider an integral local ring B of dimension $\geqslant 2$ and residue field k, and to take $A = B \oplus k$ with the law of multiplication $(b, x)(b', x') = (bb', bx' + b'x)$.

(iv) If X is locally Noetherian, it results from (5.10.2) that the schematically dense opens in X are those which *contain* $\mathrm{Ass}(\mathscr{O}_X)$.

In particular, if X is reduced and has only finitely many irreducible components, the notion of torsion-free $\mathscr{O}_X$-Module defined in (20.1.5) coincides with that of (**I**, 7.4.1).

(20.2.14). Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent and *torsion-free* (20.1.5), so that $\mathscr{F}$ identifies via (20.1.5.1) with a sub-$\mathscr{O}_X$-Module of $\mathscr{M}_X(\mathscr{F})$. For every meromorphic section u of $\mathscr{F}$ over X, we call the *ideal of denominators* of u the Ideal $\mathscr{J}$ of the section $\overline{u}$, image of u in $\mathscr{M}_X(\mathscr{F})/\mathscr{F}$. The Ideal $\mathscr{J}$ is *quasi-coherent*: indeed, the question being local on X, we can restrict to the case where X is affine, and there exists a section $s \in \Gamma(X, \mathscr{S}(\mathscr{O}_X))$ such that $v = su \in \Gamma(X, \mathscr{F})$; to say that a section $f \in \Gamma(U, \mathscr{O}_X)$ belongs to $\Gamma(U, \mathscr{J})$ means that $f(u|U) \in \Gamma(U, \mathscr{F})$, and since $s|U$ is a regular element of $\Gamma(U, \mathscr{O}_X)$ and $\mathscr{F}$ is torsion-free, the preceding relation is equivalent to $f((su)|U) \in \Gamma(U, s\mathscr{F})$; so if $\overline{v}$ is the section of $\mathscr{F}/s\mathscr{F}$, canonical image of v, we see that $\mathscr{J}$ is the kernel of the homomorphism $\mathscr{O}_X \to \mathscr{F}/s\mathscr{F}$ obtained by multiplication by the section $\overline{v}$. As $\mathscr{F}/s\mathscr{F}$ is quasi-coherent, so is $\mathscr{J}$.

It results immediately from the preceding definition that $\mathrm{dom}(u)$ is the open complement of the closed sub-prescheme of X defined by the Ideal of denominators of u.

Proposition (20.2.15). — *Let* $f : X' \to X$ *be a morphism,* $\mathscr{F}$ *an* $\mathscr{O}_X$*-Module quasi-coherent,* φ *a section of* $\mathscr{M}_f(\mathscr{F})$ *over* X (20.1.11). *Then* $f^{-1}(\mathrm{dom}(\varphi))$ *is schematically dense in* X'.

Proof. The question being evidently local on X and X', we can suppose $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(A')$ affines, $\mathscr{F} = \widetilde{M}$, and $\varphi = h \otimes (1/s)$, where $h \in M$ and s is a regular element of A whose image s' in A' is a regular element. We know then (20.2.9) that $\mathrm{D}(s')$ is a schematically dense open in X', and it is the inverse image by f of $\mathrm{D}(s)$, which is contained in $\mathrm{dom}(\varphi)$. □

Remark (20.2.16). — When Y is not separated, the presheaf $U \mapsto \mathrm{Ps.hom}_S(U,Y)$ on X is not necessarily a sheaf, even when X is Noetherian, as the following example shows. We take for X the spectrum of a semi-local Noetherian ring A of dimension $\geqslant 2$, having exactly two maximal ideals $\mathfrak{m}'$, $\mathfrak{m}''$ (so that X has exactly two closed points x', x''), such that $\mathfrak{m}'$ and $\mathfrak{m}''$ belong to $\mathrm{Ass}(A)$ and that the open $U = X - \{x', x''\}$ is *integral*. Note that the only schematically dense opens in X are those containing x' and x'' (20.2.13, (iv)), and hence they are necessarily equal to X. Let $U' = X - \{x'\}$, $U'' = X - \{x''\}$, so that $U = U' \cap U''$. To have a counterexample, it suffices to define two S-morphisms $u' : U' \to Y$, $u'' : U'' \to Y$ (with $S = X$) whose restrictions to U belong to the same pseudo-morphism, but which for every neighbourhood V' of x' in U' and every neighbourhood V'' of x'' in U'', the restrictions of u' and u'' to $V' \cap V''$ are distinct. For this, let us consider a closed irreducible part Z of X, distinct from X; let Y be the X-prescheme obtained by gluing two copies Y', Y'' of X along the open $X - Z$ (**I**, 2.3.1). We take for u' and u'' the restrictions to U' and U'' respectively of the structural isomorphisms $Y' \xrightarrow{\sim} X$, $Y'' \xrightarrow{\sim} X$. Since V' and V'' contain the generic point of Z, the restrictions of u' and u'' to $V' \cap V''$ are distinct; but the restrictions of u' and u'' to $U - (U \cap Z)$ are equal, and this is a dense open in U, hence schematically dense since U is reduced. **IV-4** | 237

It remains therefore only to define A and Z so as to have the preceding properties. Let $X_0 = \mathrm{Spec}(B)$ be an integral affine scheme (for example the affine plane over a field k), Y an irreducible closed subvariety of X_0 (for example a line) containing two distinct closed points x', x'', corresponding to maximal ideals $\mathfrak{n}'$, $\mathfrak{n}''$ of B. Consider the ring $C = B \oplus (B/\mathfrak{n}' \oplus B/\mathfrak{n}'')$ with the multiplication $(b,z)(b',z') = (bb', bz' + b'z)$. If $X_1 = \mathrm{Spec}(C)$, we have $X_0 = (X_1)_{\mathrm{red}}$ and X_1 is reduced except at the points x', x''. It then suffices to take $A = R^{-1}C$, where R is the complement of the union of the maximal ideals of C at the points x', x'', and for Z the trace of Y on $X = \mathrm{Spec}(A)$.

20.3. Composition of pseudo-morphisms

(20.3.1). Let X, Y, Z be three preschemes, ω a pseudo-morphism of X in Y, $f : Y \to Z$ a morphism. It is clear that if U', U'' are two schematically dense opens in X, $u' : U' \to Y$, $u'' : U'' \to Y$ two morphisms of the class ω, the morphisms fu' and fu'' are equivalent (for the relation defined in (20.2.1)), so, for all morphisms u of the class ω, the morphisms fu belong to a single equivalence class, which we call the pseudo-morphism of X in Z *composed* of f and ω. One has $\mathrm{dom}(f\omega) = \mathrm{dom}(\omega)$. If $g : Z \to T$ is a morphism, it is clear that $g \circ (f\omega) = (g \circ f)\omega$.

(20.3.2). Let us now suppose given a pseudo-S-morphism ω of X in Y, where Y is *separated over S*, so that there exists an S-morphism $u : \mathrm{dom}_S(\omega) \to Y$ of the class ω (20.2.4). Let $f : X' \to X$ be an S-morphism such that the open $V' = f^{-1}(\mathrm{dom}_S(\omega))$ is schematically dense in X'; then we say that the class (for the equivalence relation of (20.2.1)) of the S-morphism $u \circ (f|V')$[38] (a class defined by virtue of the preceding hypothesis) is the pseudo-S-morphism *composed* of ω and f, and we denote it $\omega \circ f$; it is clear that one has

$$f^{-1}(\mathrm{dom}_S(\omega)) \subset \mathrm{dom}_S(\omega \circ f). \tag{20.3.2.1}$$

For $f : X' \to X$ given, we denote by $\mathrm{Ps.hom}_S(X,Y)^f$ the set of pseudo-S-morphisms ω of X in Y which verify the preceding condition. If ω is such a pseudo-S-morphism, it is clear that for every open V of X, **IV-4** | 238

$$f^{-1}(\mathrm{dom}_S(\omega|V)) = f^{-1}(V \cap \mathrm{dom}_S(\omega)) = f^{-1}(V) \cap f^{-1}(\mathrm{dom}_S(\omega))$$

is schematically dense in $f^{-1}(V)$, so, if $f^V : f^{-1}(V) \to V$ is the restriction of f, the composed $(\omega|V) \circ f^V$ is defined and equal to $(\omega \circ f)|f^{-1}(V)$. One thus defines a canonical restriction map $\mathrm{Ps.hom}_S(X,Y)^f \to \mathrm{Ps.hom}_S(V,Y)^{f'}$, which allows to define a *presheaf* of sets $V \mapsto \mathrm{Ps.hom}_S(V,Y)^{f'}$ on X, sub-presheaf of $V \mapsto \mathrm{Ps.hom}_S(V,Y)$, which gives a sheaf of sets that we denote $\mathscr{P}s.\mathrm{hom}_S(X,Y)^f$. Moreover, for every open V of X, we have a map $\omega \mapsto \omega \circ f^V$ in $\mathrm{Ps.hom}_S(f^{-1}(V), Y)$, which defines an f-morphism of sheaves of sets

$$\mathscr{P}s.\mathrm{hom}_S(X,Y)^f \longrightarrow \mathscr{P}s.\mathrm{hom}_S(X',Y).$$

[38] The printed text has $f|U'$ here, but the open introduced in the preceding sentence is $V' = f^{-1}(\mathrm{dom}_S(\omega))$.

(20.3.3). Let now $f' : X'' \to X'$ be an S-morphism such that the open $f'^{-1}(f^{-1}(\mathrm{dom}_S(\omega)))$ is schematically dense in X''; then $\omega \circ (f \circ f')$ is defined and $u \circ f \circ f'$ belongs to this pseudo-S-morphism; moreover, by virtue of (20.3.2.1), $f'^{-1}(\mathrm{dom}_S(\omega \circ f))$ is *a fortiori* schematically dense in X'', so $(\omega \circ f) \circ f'$ is also defined and $u \circ f \circ f'$ belongs to this pseudo-S-morphism, hence $(\omega \circ f) \circ f' = \omega \circ (f \circ f')$. On the other hand, for every S-morphism $g : Y \to Z$, one has $\mathrm{dom}_S(g\omega) = \mathrm{dom}_S(\omega)$ (20.3.1), so $(g\omega) \circ f$ is defined, and $g \circ u \circ f$ belongs to this pseudo-S-morphism, which shows that $(g\omega) \circ f = g \circ (\omega \circ f)$.

(20.3.4). The most important case in the definition (20.3.2) is that where one has $\mathscr{P}s.\mathrm{hom}_S(X, Y)^f = \mathscr{P}s.\mathrm{hom}_S(X, Y)$: it suffices for this that for every open U of X and every open V schematically dense in U, $f^{-1}(V)$ is schematically dense in $f^{-1}(U)$; when this holds, then, for *every* open U of X and *every* pseudo-S-morphism $\omega : U \to Y$, one can define the composed $\omega \circ f^U$, *even if Y is not separated over S*. Indeed, if $u' : U' \to Y$ and $u'' : U'' \to Y$ are two morphisms of the class ω, they coincide in an open U_0 schematically dense in U, so the composed morphisms $f^{-1}(U') \xrightarrow{u'} Y$ and $f^{-1}(U'') \xrightarrow{u''} Y$ coincide in $f^{-1}(U_0)$, which is by hypothesis schematically dense in $f^{-1}(U)$; one can therefore define $\omega \circ f^U$ as the class of any of the morphisms $f^{-1}(U') \xrightarrow{u'} Y$, where u' belongs to ω.

Proposition (20.3.5). — *Let X, X' be two preschemes, $f : X' \to X$ a morphism. Then, in each of the three following cases, for every open U of X and every open V schematically dense in U, $f^{-1}(V)$ is schematically dense in $f^{-1}(U)$:*

(i) *f is flat and locally of finite presentation.*
(ii) *f is flat and the underlying space of X is locally Noetherian.*
(iii) *X' is reduced, the set of irreducible components of X' is locally finite, and every irreducible component of X' dominates an irreducible component of X.*

Proof. Assertion (i) results from (11.10.5, (ii), b)); assertion (ii) results from (11.10.5, (ii), a)), since **IV-4** | 239
then every open V in U is retrocompact, i.e. the canonical injection $j : V \to U$ is a quasi-compact morphism. Finally, to prove (iii), note that since X' is reduced, it suffices to show that for every open U of X and every open V dense in U, $f^{-1}(V)$ is *dense* in $f^{-1}(U)$. Now, for $f^{-1}(V)$ to be dense in $f^{-1}(U)$, it suffices that $f^{-1}(V)$ contain all the maximal points of X' belonging to $f^{-1}(U)$; the conclusion therefore results from the hypothesis that the image under f of every maximal point of X' belonging to $f^{-1}(U)$ is a maximal point of X belonging to U, hence belongs to V since the set of irreducible components of X is locally finite. ☐

(20.3.6). Let X, Y be two S-preschemes; suppose that X satisfies one of the two following hypotheses:

a) X is locally Noetherian;
b) X is reduced and the set of its irreducible components is locally finite.

Then, for every $x \in X$, the canonical S-morphism $j : \mathrm{Spec}(\mathscr{O}_{X,x}) \to X$ is flat and satisfies condition (ii) of (20.3.5) in case a), condition (iii) of (20.3.5) in case b). For every pseudo-S-morphism ω of X in Y, the composite $\omega \circ j$ is therefore defined and is a pseudo-S-morphism of $\mathrm{Spec}(\mathscr{O}_{X,x})$ in Y, called the *restriction* of ω to $\mathrm{Spec}(\mathscr{O}_{X,x})$. Note now that if X satisfies condition a) (resp. b)) of (20.3.6), the same holds for every prescheme induced on an open U of X containing x. By passage to the inductive limit, we deduce from the canonical maps $\mathrm{Ps.hom}_S(U, Y) \to \mathrm{Ps.hom}_S(\mathrm{Spec}(\mathscr{O}_{X,x}), Y)$ thus obtained, a canonical map

$$(\mathscr{P}s.\mathrm{hom}_S(X, Y))_x \longrightarrow \mathrm{Ps.hom}_S(\mathrm{Spec}(\mathscr{O}_{X,x}), Y) \tag{20.3.6.1}$$

where the left-hand side is the stalk at the point x of the sheaf $\mathscr{P}s.\mathrm{hom}_S(X, Y)$, the set of germs at x of pseudo-S-morphisms from open neighbourhoods of x into Y.

Proposition (20.3.7). — *Under the hypotheses of (20.3.6), the canonical map (20.3.6.1) is injective. If moreover Y is locally of finite presentation over S, the map (20.3.6.1) is bijective.*

Proof. By application of the method of (8.1.2, (a)), this proposition will be a particular case of the following: ☐

Proposition (20.3.8). — *The notations being those of (8.8.1), suppose X_α quasi-compact (resp. quasi-compact and quasi-separated) and Y_α locally of finite type (resp. locally of finite presentation) over S_α. Suppose moreover that one of the following conditions is satisfied:*

(i) *The transition morphisms $S_\mu \to S_\lambda$ (for $\lambda \leqslant \mu$) are flat, and the X_λ and X are Noetherian.*

(ii) *The X_λ are reduced, the sets of irreducible components of X and of each X_λ are finite, and, for $\lambda \leqslant \mu$, every irreducible component of X_μ dominates an irreducible component of X_λ.*

Then, the canonical map

$$\varinjlim \operatorname{Ps.hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \operatorname{Ps.hom}_S(X, Y) \tag{20.3.8.1}$$

is injective (resp. bijective).

Proof. Note first that, in case (i), the morphisms $X_\mu \to X_\lambda$ (for $\lambda \leqslant \mu$) and $X \to X_\lambda$ are flat, so by (20.3.4) and (20.3.5) the canonical maps **IV-4** | 240

$$\operatorname{Ps.hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \operatorname{Ps.hom}_{S_\mu}(X_\mu, Y_\mu)$$

for $\lambda \leqslant \mu$, as well as the maps

$$\operatorname{Ps.hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \operatorname{Ps.hom}_S(X, Y),$$

are defined (without any separation hypothesis on the Y_λ or on Y); the same is consequently true of the map (20.3.8.1). The proposition will result from the following lemma.

Lemma (20.3.8.2). — *The notations being those of (8.8.1), suppose X_α quasi-compact, and let U_α be an open in X_α; let U_λ and U be its inverse images in X_λ and X. Suppose that one of the conditions (i), (ii) of (20.3.8) is satisfied. Then, for U to be schematically dense in X, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that U_λ is schematically dense in X_λ, and in this case U_μ is schematically dense in X_μ for $\mu \geqslant \lambda$.*

Suppose first that condition (i) is satisfied. Denote by $j_\lambda : U_\lambda \to X_\lambda$ and $j : U \to X$ the canonical injections, and by $\mathscr{J}_\lambda$ the kernel of the canonical homomorphism $\mathscr{O}_{X_\lambda} \to (j_\lambda)_*(\mathscr{O}_{U_\lambda})$. Since the immersion j_λ is quasi-compact and quasi-separated, $(j_\lambda)_*(\mathscr{O}_{U_\lambda})$ is a quasi-coherent $\mathscr{O}_{X_\lambda}$-module; hence $\mathscr{J}_\lambda$ is a quasi-coherent ideal of $\mathscr{O}_{X_\lambda}$, and, since X_λ is Noetherian, $\mathscr{J}_\lambda$ is coherent and therefore of finite type. On the other hand, since the transition morphism $p_{\lambda\mu} : X_\mu \to X_\lambda$ (resp. $p_\lambda : X \to X_\lambda$) is flat, it follows from (2.3.1) that one may identify $\mathscr{J}_\mu$ with $p_{\lambda\mu}^*(\mathscr{J}_\lambda)$ (resp. $\mathscr{J}$ with $p_\lambda^*(\mathscr{J}_\lambda)$). The assertion results then from the definition of a schematically dense open, and of (8.5.8, (ii)).

To establish (20.3.8.2) when condition (ii) is satisfied, we first prove two lemmas.

Lemma (20.3.8.3). — *Under the hypotheses of (8.2.2), let M_α (resp. M) be the set of maximal points of S_α (resp. S). Suppose that for every λ, the set M_λ is finite, and that the M_λ form a projective system of sets. Then M is the projective limit of the M_λ.*

Let us first show that a point $s \in \varprojlim M_\lambda$ is maximal in S: indeed, if s' is a generalisation of s, the image s'_λ of s' in S_λ is a generalisation of the image s_λ of s, hence equal to s_λ for all λ, which implies $s' = s$, since the set of points underlying S is the projective limit of the sets underlying the S_λ (8.2.9). Inversely, let s be a maximal point of S and let us prove that it belongs to $\varprojlim M_\lambda$. Let s_λ be the image of s in S_λ, M'_λ the set of maximal points of S_λ that are generalisations of s_λ; the M'_λ are non-empty finite sets forming a projective system, hence $M' = \varprojlim M'_\lambda$ is non-empty and the map $M' \to M'_\lambda$ is surjective (Bourbaki, *Ens.*, chap. III, 2$^{\text{e}}$ éd., § 7, n$^{\text{o}}$ 4, Example I). On the other hand, $\operatorname{Spec}(\mathscr{O}_{S,s}) = \varprojlim \operatorname{Spec}(\mathscr{O}_{S_\lambda, s_\lambda})$ by virtue of (8.2.12) and (8.2.9), hence the points of $\varprojlim M'_\lambda$ are also maximal points of $\operatorname{Spec}(\mathscr{O}_{S,s})$ by the first part of the reasoning. Thus $M' = \varprojlim M'_\lambda$ necessarily reduces to the point s; one concludes that M'_λ reduces to the point s_λ and consequently the s_λ are maximal in the S_λ, which completes the proof of the lemma.

Lemma (20.3.8.4). — *The hypotheses being those of (20.3.8.3), suppose moreover that S_α is quasi-compact, and let U_α be an open of S_α, with inverse images U_λ in S_λ for $\lambda \geqslant \alpha$ and U in S.*[39] *If U_α is dense in S_α, then*

[39] The printed statement introduces V_α, V_λ, V but immediately and consistently uses U_α, U_λ, U; the latter notation is used here.

U_λ is dense in S_λ for $\lambda \geqslant \alpha$ and U is dense in S. Inversely, if U is dense in S and if moreover the set M of maximal points of S is finite, there exists $\lambda \geqslant \alpha$ such that U_λ is dense in S_λ.

Indeed, since M_α is finite, the hypothesis that U_α is dense in S_α implies that $M_\alpha \subset U_\alpha$, hence $M_\lambda \subset U_\lambda$ for $\lambda \geqslant \alpha$ and $M \subset U$ by virtue of (20.3.8.3), which proves the first assertion. Inversely, suppose M finite and U dense in S, hence $M \subset U$; note that the U_λ are open, hence ind-constructible, and the M_λ finite, hence pro-constructible (1.9.6). The second assertion then follows from (8.3.2).

20.3.8.5. End of the proof of (20.3.8.2). Suppose condition (ii) is verified, and note that X is then *reduced* by virtue of (8.7.1); it amounts to the same to say that U_λ (resp. U) is schematically dense in X_λ (resp. X) or that it is dense, and the conclusion results from (20.3.8.4) applied to X_λ and X. **IV-4** | 241

20.3.8.6. End of the proof of (20.3.8). To show that the map (20.3.8.1) is injective, consider two morphisms u'_λ, u''_λ of a schematically dense open $U_\lambda \subset X_\lambda$ in Y_λ, and suppose that their images u', u'', morphisms of U of X in Y, coincide in a schematically dense open V of U. Under either hypothesis (i) or (ii), one can suppose V *quasi-compact*; otherwise, since X has only a finite number of maximal points and is reduced, it suffices to replace V by a union of a finite number of affine open neighbourhoods of these maximal points (contained in V by hypothesis). Then there exists $\lambda \geqslant \alpha$ such that V is the inverse image of a quasi-compact open V_λ of X_λ (8.2.11), and by (20.3.8.2) we can suppose that V_λ is schematically dense in X_λ. It results from (8.8.2, (i)) that, on taking λ sufficiently large, u'_λ and u''_λ coincide on V_λ, hence belong to the same pseudo-morphism.

Let us prove finally that the map (20.3.8.1) is surjective. Consider now a morphism u of a schematically dense open U of X in Y; as above, we can suppose U quasi-compact and quasi-separated (U being replaceable, in case (ii), by a union of a finite number of disjoint affine opens). We can furthermore suppose there exists $\lambda \geqslant \alpha$ such that U is the inverse image of a quasi-compact open U_λ of X_λ (8.2.11) which is automatically quasi-separated, and by (20.3.8.2) we can suppose that U_λ is schematically dense in X_λ. Since the Y_λ are supposed locally of finite presentation, it results from (8.8.2, (i)) that there exists λ such that u is the image of a morphism $u_\lambda : U_\lambda \to Y_\lambda$; whence the conclusion.

Remarks (20.3.8.7). — (i) To show that the map (20.3.8.1) is *injective*, it is not necessary, in the hypothesis (i) of (20.3.8), to suppose X Noetherian. Indeed, the lemma (20.3.8.2) does not use this hypothesis. With the notations of (20.3.8.6), let Z_λ be the sub-prescheme of coincidences of u'_λ and u''_λ, and let Z be the sub-prescheme of coincidences of u' and u''; it results from the definition (17.4.5) and from (**I**, 3.3.10.1) that Z is the projective limit of the Z_μ for $\mu \geqslant \lambda$. Now, by hypothesis, Z majorizes a schematically dense open of X; it follows that Z is itself induced on an open of X by virtue of the following lemma:

Lemma (20.3.8.8). — *Let T be a prescheme. Then every sub-prescheme W of T which majorizes a schematically dense open V of T is induced on an open (schematically dense) of T.*

Indeed, the subspace of T underlying W is locally closed, hence open in its closure, hence open in T; the conclusion results from (11.10.1, (c)). **IV-4** | 242

This lemma being established, we conclude that for $\mu \geqslant \lambda$ sufficiently large, Z_μ is induced on an open of X_μ by virtue of (8.6.3); indeed, Z_λ, being a sub-prescheme of a Noetherian prescheme, is of finite presentation over X_λ (1.6.2), and the same is therefore true of Z_μ over X_μ for $\mu \geqslant \lambda$ and of Z over X. One can now apply (20.3.8.2), which shows that for $\mu \geqslant \lambda$ sufficiently large, Z_μ is schematically dense in X_μ, whence the conclusion.

(ii) If, in the hypothesis (i) of (20.3.8), we drop the condition that X is Noetherian, the reasoning of (20.3.8.6) shows yet that the image of (20.3.8.1) is formed of the pseudo-S-morphisms having a representative which is an S-morphism $U \to Y$, where U is schematically dense in X and *quasi-compact and quasi-separated*.

□

Corollary (20.3.9). — *Suppose that either hypothesis a) or b) of* (20.3.6) *is satisfied by X, and let Y be locally of finite presentation over S. Then, for a pseudo-S-morphism ω of X in Y to be defined at a point x* (20.2.3), *it is necessary and sufficient that its restriction to* $\mathrm{Spec}(\mathscr{O}_{X,x})$ *be everywhere defined (in other words, that it be an S-morphism of* $\mathrm{Spec}(\mathscr{O}_{X,x})$ *in Y).*

The following result, which will be used in the proof of (20.3.11), uses the theory of faithfully flat descent from Chapter VI. The reader may verify that the results of (20.3) will not be used in that theory.

Lemma (20.3.10). — *Let $f : X' \to X$ be an S-morphism faithfully flat and quasi-compact, $X'' = X' \times_X X'$, p_1 and p_2 the canonical projections of X'' to X', and let Y be an S-prescheme separated over S. Let U be an open of X, $U' = f^{-1}(U)$, $U'' = p_1^{-1}(U') = p_2^{-1}(U')$, and suppose that U'' is schematically dense in X''. Let $g : U \to Y$ be an S-morphism; then, if $g \circ (f|U')$ extends to an S-morphism $X' \to Y$, g extends to an S-morphism $X \to Y$.*

The hypotheses imply that U (resp. U') is schematically dense in X (resp. X') (11.10.5, (i)). Thus, if ω denotes the pseudo-S-morphism given by the class of g, the assertion of (20.3.10) can equivalently be stated as follows: if $\omega \circ f$ is everywhere defined, then so is ω.

Proof. To prove (20.3.10), denote by g' an S-morphism $X' \to Y$ which extends $g \circ (f|U')$, and set $g''_i = g' \circ p_i : X'' \to Y$ $(i = 1, 2)$. If one sets $f'' = f \circ p_1 = f \circ p_2 : X'' \to X$, it is clear that g''_1 and g''_2 coincide on U'' with $g \circ (f''|U'')$. But as Y is separated over S and U'' is schematically dense in X'', we have $g''_1 = g''_2$ (11.10.1, (d)). As f is faithfully flat and quasi-compact, it results from descent theory (chap. VI) that there exists a unique S-morphism $h : X \to Y$ such that $h \circ f = g'$; as the restriction $U' \to U$ of f is a faithfully flat and quasi-compact morphism and $U'' = U' \times_U U'$, the preceding uniqueness result, applied to U instead of X, shows that $h|U = g$, which proves the lemma. □

Proposition (20.3.11). — *Let Y be an S-prescheme separated on S, ω a pseudo-S-morphism of X in Y, $f : X' \to X$ an S-morphism. Suppose that f is flat, and that one of the following conditions is satisfied:*

(i) *f is an open morphism, or surjective and quasi-compact, and* $\mathrm{dom}_S(\omega)$ *contains an open U schematically dense in X and retrocompact in X.*
(ii) *f is locally of finite presentation.*
(iii) *Y is locally of finite presentation over S, and X satisfies one of the conditions a), b) of (20.3.6).*

Then, $f^{-1}(\mathrm{dom}_S(\omega))$ *is schematically dense in X', so that $\omega \circ f$ is defined (20.3.2) and one has*

$$\mathrm{dom}_S(\omega \circ f) = f^{-1}(\mathrm{dom}_S(\omega)). \tag{20.3.11.1}$$

Proof. Let us first prove that $f^{-1}(\mathrm{dom}_S(\omega))$ is schematically dense in X'. The question being local on X', we can suppose X and X' affines and since f is flat, it suffices to verify, by (11.10.5, (ii), a)), that $\mathrm{dom}_S(\omega)$ contains a retrocompact open U schematically dense in X. This is the hypothesis in case (i), and follows from (20.2.12) in case (iii), since in an affine scheme every open of the form $\mathrm{D}(t)$ is retrocompact; in case (ii), we can directly apply (20.3.5, (i)). **IV-4** | 243

Let us now prove (20.3.11.1), i.e., that, for every point $x' \in \mathrm{dom}_S(\omega \circ f)$, one has $x = f(x') \in \mathrm{dom}_S(\omega)$. Note first that we can restrict to the case where f is *faithfully flat and quasi-compact*. This is already the hypothesis in the second case of (i). In the other cases the question is local on X', so we may suppose X and X' affine, and hence f quasi-compact. In the first case of (i), and in case (ii), f is an open morphism (11.3.1), so, replacing X by the open $f(X')$, we may suppose f surjective, hence faithfully flat. In case (iii), by (20.3.9), it suffices to prove that the restriction of ω to $\mathrm{Spec}(\mathscr{O}_{X,x})$ is everywhere defined, and we can therefore replace X by $\mathrm{Spec}(\mathscr{O}_{X,x})$, X' by $X' \times_X \mathrm{Spec}(\mathscr{O}_{X,x})$ and f by its restriction to this last prescheme, which is a surjective morphism (2.3.4), hence faithfully flat.

Suppose therefore f faithfully flat and quasi-compact; with the notations of the lemma (20.3.10), it suffices to see that U″ is schematically dense in X'', by taking for U an open schematically dense in X, contained in $\mathrm{dom}_S(\omega)$; this will be the case, by virtue of (11.10.5, (ii), a)), if U is taken retrocompact in X (since the morphism $f'' : X'' \to X$ is flat). The existence of such an open U is part of the hypothesis in case (i); in case (iii) it results from (20.2.12) and the fact that in an affine scheme $\mathrm{Spec}(A)$, every open of the form $\mathrm{D}(t)$ is retrocompact. Finally, in case (ii), we take $U = \mathrm{dom}_S(\omega)$ and show directly that U″ is schematically dense in X'': it suffices for this to remark that $f'' : X'' \to X$ is flat and locally of finite presentation and to apply (20.3.5, (i)). □

Corollary (20.3.12). — *Let φ be a pseudo-function on a prescheme X. Then, for every flat morphism locally of finite presentation $f : X' \to X$, the pseudo-function $\varphi \circ f$ is defined and one has* $\mathrm{dom}(\varphi \circ f) = f^{-1}(\mathrm{dom}(\varphi))$.

Remark (20.3.13). — When X satisfies one of the conditions a), b) of (20.3.6), then we have seen (20.2.11) that the pseudo-functions on X are identified with the meromorphic functions on X. By virtue of (20.1.12) and of (20.2.13, (i)), if one only supposes that the morphism $f : X' \to X$ is *flat*, then, for every pseudo-function φ on X, $\varphi \circ f$ is defined and one has $\mathrm{dom}(\varphi \circ f) = f^{-1}(\mathrm{dom}(\varphi))$. **IV-4** | 244

20.4. Properties of domains of definition of rational functions

(20.4.1). Let X, Y be two S-preschemes, ω a pseudo-S-morphism of X in Y. Let u be an S-morphism $U \to Y$ belonging to the class ω, where U is schematically dense in X, and consider the *graph* Γ_u, sub-prescheme of $U \times_S Y$ (**I**, 5.3.11), and therefore sub-prescheme of $X \times_S Y$. Suppose that this sub-prescheme admits an *adherence* (**I**, 9.5.11) in $X \times_S Y$; if $j : \Gamma_u \to X \times_S Y$ is the canonical injection, this will be the case when the $\mathscr{O}_{X \times_S Y}$-Module $j_*(\mathscr{O}_{\Gamma_u})$ is quasi-coherent; it results from the definition of the class of equivalence (20.2.1) that $j_*(\mathscr{O}_{\Gamma_u})$ does not depend on the representative u considered, hence the adherence of this sub-prescheme is well determined by ω; we say that this closed sub-prescheme is the *graph* of the pseudo-S-morphism ω and we denote it Γ_ω. We note that Γ_ω is defined if there exists a morphism $u : U \to Y$ in the class ω such that U is retrocompact in X and if, moreover, Y is quasi-separated over S; indeed, the injection j is then a quasi-compact and quasi-separated morphism (**I**, 2.2,I.7.4). The first hypothesis is always satisfied when X is locally Noetherian. We note also that when X is *reduced*, so is Γ_u, which is isomorphic to U (**I**, 5.3.11); then Γ_ω exists and is none other than the *reduced* closed sub-prescheme of $X \times_S Y$ having for underlying space the closure in $X \times_S Y$ of the underlying space of Γ_u (**I**, 5.2.1,I.5.2.2). Note finally that if Y is *separated* over S, then Γ_u is *closed* in $U \times_S Y$ (**I**, 5.4.3), and hence is induced by Γ_ω (whenever the latter exists) on the open $\Gamma_\omega \cap (U \times_S Y)$ (**I**, 9.5.10); by contrast, the prescheme induced by Γ_ω is in general different from Γ_u when Y is not separated. In particular, if $v : X \to Y$ is an S-*morphism*, the graph Γ_ω of the class ω of v may be distinct from the graph Γ_v. In particular also in what follows, when the question is about the graph of a pseudo-S-morphism, we will restrict to the case where Y *is separated over* S.

(20.4.2). Suppose therefore Γ_ω defined and Y separated on S; denote by p and q the restrictions to Γ_ω of the canonical projections.

$$\begin{tikzcd} & \Gamma_\omega \arrow[dl, "p"'] \arrow[dr, "q"] & \\ X & & Y \end{tikzcd}$$

Then, if $U \subset \mathrm{dom}_S(\omega)$, the restriction $p^{-1}(U) \to U$ of p is an isomorphism (**I**, 5.3.11); reciprocally, **IV-4** | 245
if U is an open of X having this property, and if u is the isomorphism reciprocal of the restriction $p^{-1}(U) \to U$ of p, $q \circ u$ is an S-morphism of U in Y which coincides with a morphism of the class ω in $U \cap \mathrm{dom}_S(\omega)$. We conclude that $\mathrm{dom}_S(\omega)$ is *the largest open* U of X such that the restriction $p^{-1}(U) \to U$ of p is an isomorphism; let $v : \mathrm{dom}_S(\omega) \to \Gamma_\omega$ be the S-morphism inverse to the restriction $p^{-1}(\mathrm{dom}_S(\omega)) \to \mathrm{dom}_S(\omega)$ of p. The pseudo-S-morphism from X to Γ_ω represented by v is sometimes denoted p^{-1}; its associated rational map (20.2.13, (ii)) is then *birational*. Since $p^{-1}(\mathrm{dom}_S(\omega))$ is the graph of the S-morphism $u = q \circ v : \mathrm{dom}_S(\omega) \to Y$, it is schematically dense in Γ_ω (11.10.3, (iv)), so ω can be considered as the *composite* $q \circ p^{-1}$ (20.3.2). For every subset M of the underlying space of X, one sometimes sets $\omega(M) = q(p^{-1}(M))$, which amounts to considering ω as a map from X to the set of subsets of Y (or, as some authors say, a "multiform function"). We note that when $x \in \mathrm{dom}_S(\omega)$, $\omega(\{x\})$ is the singleton $\{u(x)\}$; in general, for any $x \in X$, if one denotes by s the image of x in S, via the structural morphism, $\omega(\{x\})$ is a part of the prescheme Y_s, the fibre at the point s of the structural morphism.

(20.4.3). In all the rest of this n°, we will restrict to the case where X is *reduced*, so that there is then an identity between pseudo-S-morphisms and S-*rational applications* (20.2.7). Moreover, the *graph* of the S-rational application ω of X in Y exists and is none other than the *reduced* closed sub-prescheme of $X \times_S Y$ having for underlying space the adherence of the graph of any representative $u : \mathrm{dom}_S(\omega) \to Y$ in the space underlying $X \times_S Y$ (**I**, 5.2.1) and (5.2.2).

Proposition (20.4.4). — *Let X be an S-prescheme locally integral, Y an S-prescheme locally of finite type and separated on S, ω an S-rational application of X in Y, $p : \Gamma_\omega \to X$ the canonical projection. Then, if $x \in X$ is a normal point such that $p^{-1}(x)$ contains an isolated point, ω is defined at the point x.*

Proof. Indeed, the first projection $p_1 : X \times_S Y \to X$ is then a separated and locally of finite type morphism, and so is its restriction $p : \Gamma_\omega \to X$, which is moreover *birational*, and as Γ_ω is reduced and X integral, Γ_ω is *integral*; it results then from (**Err$_{\mathbf{IV}}$**, 30) that the hypothesis on x implies the existence of an open neighbourhood U of x such that the restriction $p^{-1}(U) \to U$ of p is an isomorphism; whence the conclusion by (20.4.2).

We note that (20.4.4) is the original formulation given by Zariski of his "Main theorem" (up to the fact that it was originally restricted to algebraic schemes over a base field). □

Proposition (20.4.5). — *The hypotheses and notations being those of (20.4.4), suppose moreover X normal, and let X' be a reduced prescheme, $f : X' \to X$ a morphism locally of finite type and universally open. Then $\omega \circ f$ is defined, and one has $\mathrm{dom}_S(\omega \circ f) = f^{-1}(\mathrm{dom}_S(\omega))$ (in other words, if $x' \in X'$ and $x = f(x')$, then for ω to be defined at x it is necessary and sufficient that $\omega \circ f$ be defined at x').*

Proof. As X' is reduced and $f^{-1}(\mathrm{dom}_S(\omega))$ is everywhere dense in X' by virtue of (2.4.11), the composite $\omega \circ f$ is defined. To prove that, whenever $\omega \circ f$ is defined at x', ω is defined at x, we may plainly replace X' by an open neighbourhood of x' and hence suppose $\omega \circ f$ everywhere defined; moreover, since $f(X')$ is open in X, we may replace X by $f(X')$ and thus suppose f surjective. **IV-4** | 246

By the hypothesis on f, the morphism

$$f_{(Y)} : X' \times_S Y \longrightarrow X \times_S Y$$

is then open, and consequently

$$\overline{f_{(Y)}^{-1}(M)} = f_{(Y)}^{-1}(\overline{M})$$

for every subset M of $X \times_S Y$. Taking for M the graph Γ_u, where $u : \mathrm{dom}_S(\omega) \to Y$ is the restriction of ω to its domain, it follows from the preceding relation and from (**I**, 3.3.12) that the underlying space of $\Gamma_{\omega\circ f}$ is $f_{(Y)}^{-1}(\Gamma_\omega)$. Since $\Gamma_{\omega\circ f}$ is a reduced prescheme (20.4.1), the X'-prescheme $\Gamma_{\omega\circ f}$ is therefore equal to

$$(\Gamma_\omega \times_X X')_{\mathrm{red}}.$$

But by hypothesis the composite morphism

$$\Gamma_{\omega\circ f} \longrightarrow \Gamma_\omega \times_X X' \xrightarrow{p^{(X')}} X'$$

is an isomorphism, so $p^{(X')}$ is necessarily radicial. Since f is surjective, the same is true of $p : \Gamma_\omega \to X$ (**I**, 3.4.8); hence $p^{-1}(x)$ consists of a single point (**I**, 3.6.4), and (20.4.4) shows that ω is defined at x. □

Proposition (20.4.6). — *Let S be a prescheme, X, Y two S-preschemes; suppose X locally Noetherian, Y locally of finite type on S. Let U be an open schematically dense in X, $h : U \to Y$ an S-morphism, $x \in X - U$ a normal point of X, $h'_x : \mathrm{Spec}(k(x)) \to Y$ an S-morphism. In order that h can be extended to an S-morphism $h' : U' \to Y$, where U' is an open of X containing U and x, and where the composed morphism $\mathrm{Spec}(k(x)) \to U' \xrightarrow{h'} Y$ is the S-morphism given by h'_x, it is necessary and sufficient that the following condition be satisfied:*

> (P) For every spectrum S_1 of a discrete valuation ring, of closed point a and generic point b, and for every morphism $u : S_1 \to X$ such that $u(a) = x$, $u(b) \in U$, the composed morphism $h_1 = h \circ (u|\{b\}) : \{b\} = u^{-1}(U) \to Y$ extends to a morphism

$h'_1 : S_1 \to Y$ such that the diagram

$$\begin{tikzcd} \mathrm{Spec}(k(a)) \arrow[r] & S_1 \\ \mathrm{Spec}(k(x)) \arrow[r, "h'_x"'] & Y \arrow[u, "h'_1"'] \end{tikzcd}$$

is commutative.

Moreover, if this condition is satisfied, and if $h'' : U'' \to Y$ is a morphism satisfying the same conditions as h', then there exists an open set containing $U \cup \{x\}$, in which h' and h'' coincide.

Proof. Let us first prove the last assertion; we can suppose h' and h'' defined in the same open **IV-4** | 247
$U_0 \supset U \cup \{x\}$. The sub-prescheme Z of coincidences of h' and h'' (17.4.5) contains U and x, so there is an open neighbourhood V of x in U_0 such that the sub-prescheme induced by Z on the open $Z \cap V$ is a *closed* sub-prescheme of the prescheme induced by V on V; as this sub-prescheme $Z \cap V$ majorizes the prescheme induced by V on the open $U \cap V$, and the latter is schematically dense in V, $Z \cap V$ is necessarily equal to V (20.3.8.8), which proves that h' and h'' coincide in $U \cup V$.

The necessity of the condition is evident, let us prove that it is sufficient. By the bijective correspondence between S-morphisms of X in Y and X-sections of $X \times_S Y$ (**I**, 3.3.15), we can restrict to the case where $S = X$ and h is therefore a U-section of Y; we note that then Y is locally Noetherian, and we may plainly (since X is locally integral and locally Noetherian) suppose that $X = S$ is irreducible. Moreover, by virtue of (20.3.7), we can suppose that $X = \mathrm{Spec}(A)$, where A is a local Noetherian ring, integral and integrally closed.

Let $y = h'_x(x)$. For every $x' \in U$, the point y is a specialization of $h(x')$. Indeed, there exists the spectrum S_1 of a discrete valuation ring and a morphism $u : S_1 \to X$ such that $u(a) = x$ and $u(b) = x'$ (**II**, 7.1.9). Applying condition (P), we obtain the assertion at once, since $h(x') = h_1(b) = h'_1(b)$ and $y = h'_1(a)$. If Y' is an affine open neighbourhood of y, it follows that $h(X \cap U) \subset Y'$; we may therefore replace Y by Y' and hence suppose that Y is affine, and thus separated over X. Let ω be the rational X-section of Y to which h belongs, so that its graph has as underlying space the closure of $h(U)$ in Y. Since Y is separated over X, we may apply (20.4.4) to ω; it is enough to prove that, if $p : \Gamma_\omega \to X$ is the canonical projection, then $p^{-1}(x)$ consists of the single point y and $h'_x(x) = y$. Indeed, (20.4.4) then extends h to a section h' over an open U' containing $U \cup \{x\}$, with $h'(x) = y$; and since there is only one X-morphism $\mathrm{Spec}(k(x)) \to Y$ carrying x to y, this proves that h'_x is the composite of h' with $\mathrm{Spec}(k(x)) \to U'$.

Since, for every $x' \in U$, y is a specialization of $h(x')$, we have $y \in p^{-1}(x)$. Now let z be any point of $p^{-1}(x)$. Since Γ_ω is the closure of $h(U)$, and every point of $h(U)$ is adherent to $h(\xi)$, where ξ is the generic point of X, Γ_ω is the closure of $h(\xi)$ in Y. There therefore exist the spectrum S_1 of a discrete valuation ring and a morphism $v : S_1 \to Y$ such that $v(a) = z$ and $v(b) = h(\xi)$ (**II**, 7.1.9); set $u = p \circ v$, so that $u(a) = x$ and $u(b) = \xi$. Applying condition (P) to u, we obtain a morphism $h'_1 : S_1 \to Y$ such that $h'_1(a) = y$ and $h'_1(b) = h(\xi)$. But then $z = y$ by (**II**, 7.2.3), since Y is separated over X and v and h'_1 must coincide, being equal at b. C.Q.F.D. □

Corollary (20.4.7). — *The hypotheses on S, X, Y, U and h being those of (20.4.6), let E be a part of $X - U$ such that X is normal at every point of E, and for each $x \in E$, let $h'_x : \mathrm{Spec}(k(x)) \to Y$ be an S-morphism, such that the condition (P) of (20.4.6) is satisfied. Suppose moreover that the union F of the graphs of the h'_x (identified with parts of $X \times_S Y$) for $x \in E$, is contained in a union of a finite number of opens V_i of $X \times_S Y$ which are* separated *on X (a condition automatically satisfied if Y is* separated *on S, or if X is* quasi-compact *and Y of* finite type *on S). Then there exists an S-morphism $h' : U' \to Y$, where U' is an open of X containing $U \cup E$, such that, for every $x \in E$, the composed*

$$\mathrm{Spec}(k(x)) \longrightarrow U' \xrightarrow{h'} Y$$

is equal to h'_x.

Proof. Note first that, in (20.4.6), if Y is supposed *separated*, there is a largest open $U_0 \supset U$, equal to the domain of the S-rational application corresponding to h, in which h can be extended, and this extension is unique (**I**, 7.2.2); whence the conclusion in this case, by virtue of (20.4.6). In the general case, let E_i be the set of $x \in E$ such that $(x, h'_x(x)) \in V_i$. By virtue of (20.4.6), for every $x \in E$, there is an extension $h^{(x)}$ of h to $U \cup W^{(x)}$, where $W^{(x)}$ is an open neighbourhood of x in X such that the graph of $h^{(x)}|W^{(x)}$ is contained in all the V_i such that $x \in E_i$. Since V_i is separated over X and X is reduced, for two points $x', x'' \in E_i$, the morphisms $h^{(x')}$ and $h^{(x'')}$ coincide on $W^{(x')} \cap W^{(x'')}$, because they coincide on the everywhere dense open $U \cap W^{(x')} \cap W^{(x'')}$. There is therefore a morphism $h_i : U \cup U_i \to Y$ extending h, where U_i is an open of X containing E_i. Moreover, for every pair of indices i, j, the graphs of the restrictions $h_i|(U_i \cap U_j)$ and $h_j|(U_i \cap U_j)$ **IV-4** | 248
are contained in $V_i \cap V_j$; since $V_i \cap V_j$ is separated over X and the preceding morphisms coincide on the everywhere dense open $U \cap U_i \cap U_j$ of $U_i \cap U_j$, they are equal. The morphism h', equal to h on U and to h_i on each U_i, answers the question. □

Remark (20.4.8). — When $E = X - U$, it is clear that if, for every affine open T of X, there exists an S-morphism $h'_T : T \to Y$ extending $h|(U \cap T)$ and such that, for every $x \in T - (U \cap T)$, the composite $\operatorname{Spec}(k(x)) \to T \xrightarrow{h'_T} Y$ is equal to h'_x, then the h'_T are the restrictions of an everywhere-defined S-morphism $h' : X \to Y$ extending h, by the uniqueness assertion in (20.4.6). To prove the existence of h', one is therefore reduced to the case where X is *affine*, and it then suffices that the union F of the graphs of the h'_x, is *quasi-compact* in $X \times_S Y$ so that one can apply (20.4.7). The latter will hold in particular when the h'_x are of the form $\operatorname{Spec}(k(x)) \to Z \xrightarrow{h_0} Y$, where Z is a closed sub-prescheme of X having $X - U$ for underlying space, and h_0 an S-morphism.

Corollary (20.4.9). — *Let S be a locally Noetherian prescheme, X a locally Noetherian prescheme, $f : X \to S$ a flat morphism, and $g : Y \to S$ a morphism locally of finite type. Let U be a dense open of S, $h : f^{-1}(U) \to Y$ an S-morphism, $T = S - U$, Z the reduced closed sub-prescheme of X having $f^{-1}(T)$ as underlying space, and $h_0 : Z \to Y$ an S-morphism. Suppose X normal at all points of Z. In order that there exist a (necessarily unique) S-morphism $h' : X \to Y$ extending h and h_0, it is necessary and sufficient that the following condition be satisfied:*

For every spectrum S_1 of a discrete valuation ring, of closed point a and generic point b, and for every morphism $u : S_1 \to S$ such that $u(a) \in T$ and $u(b) \in U$, there exists an S_1-morphism $h'_1 : X_{(S_1)} \to Y_{(S_1)}$ which extends $h_{(S_1)} : f^{-1}_{(S_1)}(b) \to Y_{(S_1)}$ and is such that the diagram

$$\begin{tikzcd}
\operatorname{Spec}(k(z_1)) \arrow[r] \arrow[d] & Z_{(S_1)} \arrow[d, "(h_0)_{(S_1)}"] \\
X_{(S_1)} \arrow[r, "h'_1"'] & Y_{(S_1)}
\end{tikzcd}$$

is commutative for every $z_1 \in Z_{(S_1)}$.

The hypothesis that f is flat implies indeed that $f^{-1}(U)$ is dense in X (2.3.10), and it then suffices to apply (20.4.8).

Corollary (20.4.10). — *Under the hypotheses of (20.4.6), suppose moreover Y separated and locally quasi-finite on S. Let U be an open schematically dense in X, $h : U \to Y$ an S-morphism, ω the S-rational application corresponding to it, $x \in X - U$ a normal point of X. In order that ω be defined at the point x, it is necessary and sufficient that the following condition be satisfied:* there exists a spectrum S_1 of a discrete valuation ring, of closed point a and generic point b, and a morphism $u : S_1 \to X$ such that $u(a) = x$, $u(b) \in U$, and the S-rational application $\omega \circ u$ is everywhere defined. **IV-4** | 249

Indeed, by hypothesis all the fibres of the projection morphism $X \times_S Y \to X$ consist of isolated points, and to apply (20.4.4) it suffices to know that the fibre $p^{-1}(x)$ is nonempty. Now, if h_1 is the unique morphism in the class $\omega \circ u$, the unique point of $X \times_S Y$ lying over x and over $h_1(a)$ belongs to Γ_ω, whence the conclusion.

Proposition (20.4.11). — *Let X be a locally Noetherian S-prescheme, Y an S-prescheme affine over S, U an open of X, and $Z = X - U$. Suppose that $\operatorname{prof}_Z(\mathscr{O}_X) \geqslant 2$ (5.10.1); then every S-morphism $f : U \to Y$ extends uniquely to an S-morphism of X in Y.*

Proof. We can restrict to the case where S and X are affines, and (by virtue of (**I**, 3.3.14)) to the case $S = X$; so one has $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$, B being a finite type A-algebra. As B is a quotient of a polynomial algebra $A[(T_\lambda)]_{\lambda\in L}$, Y is a closed sub-prescheme of $Y' = X[(T_\lambda)]_{\lambda\in L}$. On the other hand, the hypothesis on Z implies that U is schematically dense in X by virtue of (20.2.13, (iv)) and (5.10.2). If we prove that every X-morphism $u : U \to Y$ extends uniquely to an X-morphism $v' : X \to Y'$, it will result that v' will factorise as $X \xrightarrow{v} Y \to Y'$: indeed, the sub-prescheme $v'^{-1}(Y)$ is closed and contains U (**I**, 4.4.1), and hence is equal to X (20.3.8.8). In these conditions, v will be the unique S-morphism of X in Y extending u. But then there is a bijective correspondence between the X-morphisms of an open $U \subset X$ in Y' and the families $(s_\lambda)_{\lambda\in L}$ of sections of $\mathscr{O}_X$ over U (**II**, 1.7.9); the conclusion results therefore from (5.10.5). □

Corollary (20.4.12). — *Let X be an S-prescheme locally Noetherian reduced and satisfying the condition (S_2) (5.7.2) (for example a locally Noetherian and* normal *prescheme (5.8.6)), Y an S-prescheme affine on S, f an S-rational application of X in Y; then every irreducible component of $X - \operatorname{dom}(f)$ is of codimension 1 in X.*

Proof. This amounts to the same as saying that if $Z^{(2)}$ is the set of $x \in X$ such that $\dim(\mathscr{O}_{X,x}) \geqslant 2$, then, for every closed part $Z \subset Z^{(2)}$ of X, every S-morphism of $X - Z$ in Y extends to an S-morphism of X in Y; but this results from the hypothesis on X (5.7.2) and from (20.4.11). □

20.5. Relative pseudo-morphisms

(20.5.1). Let X, Y be two S-preschemes. It results from the definitions (11.10.8) that the intersection of two opens U, U' of X, *universally schematically dense with respect to S*, possesses this same property. We therefore define an equivalence relation between S-morphisms $u : U \to Y$ by replacing in (20.2.1) "schematically dense" by "universally schematically dense with respect to S". An equivalence class for this relation is called a *pseudo-morphism of X in Y relatively to S*, and the set of these classes is denoted $\operatorname{Ps.hom}_{X/S}(X, Y)$.

(20.5.2). As every open of X universally schematically dense with respect to S is in particular schematically dense, the elements of a pseudo-morphism of X in Y relatively to S are equivalent in the sense of (20.2.1), whence a canonical map **IV-4 | 250**

$$\operatorname{Ps.hom}_{X/S}(X, Y) \longrightarrow \operatorname{Ps.hom}_S(X, Y). \tag{20.5.2.1}$$

Proposition (20.5.3). — *Suppose Y separated on S. Then:*

(i) *The map (20.5.2.1) is injective and identifies $\operatorname{Ps.hom}_{X/S}(X, Y)$ with the subset of $\operatorname{Ps.hom}_S(X, Y)$ formed of the pseudo-S-morphisms ω such that $\operatorname{dom}_S(\omega)$ is universally schematically dense relatively to S.*
(ii) *The presheaf $U \mapsto \operatorname{Ps.hom}_{U/S}(U, Y)$ on X is a sheaf.*

Proof. Assertion (i) is immediate, for two S-morphisms $u : U \to Y$, $u' : U' \to Y$ are equivalent in the sense of (20.2.1) means that they are the restrictions of a single morphism $\operatorname{dom}_S(\omega) \to Y$ (20.2.4), and if U and U' are universally schematically dense with respect to S, the same holds *a fortiori* for $\operatorname{dom}_S(\omega)$. To prove (ii), note that $U \mapsto \operatorname{Ps.hom}_S(U, Y)$ is then a sheaf (20.2.6); on the other hand, given an open covering (U_α) of U and a pseudo-S-morphism ω from U to Y, it follows immediately from the definitions (cf. (20.2.1)) that $\operatorname{dom}_S(\omega)$ is universally schematically dense in U relative to S if and only if each $\operatorname{dom}_S(\omega) \cap U_\alpha = \operatorname{dom}_S(\omega|U_\alpha)$ is universally schematically dense in U_α relative to S; whence (ii). □

When Y is separated, we denote by $\mathscr{P}s.\mathrm{hom}_{X/S}(X, Y)$ the sub-sheaf

$$U \longmapsto \operatorname{Ps.hom}_{U/S}(U, Y)$$

of $\mathscr{P}s.\mathrm{hom}_S(X, Y)$.

When X is *flat and locally of finite presentation* over S and Y separated on S, the pseudo-morphisms of X in Y relatively to S are identified with the pseudo-S-morphisms ω such that, for every fibre X_s of the morphism $X \to S$, $\operatorname{dom}_S(\omega) \cap X_s$ is *schematically dense* in X_s (11.10.10).

(20.5.4). A particularly important case is that where Y is separated on S and $Y = S[T] = S \otimes_{\mathbf{Z}} \operatorname{Spec}(\mathbf{Z}[T])$ (T indeterminate). There is then a bijective correspondence between the pseudo-S-morphisms of X in Y and the *pseudo-functions* on X (20.2.8) by the definition of a product of $\mathbf{Z}$-preschemes. The pseudo-morphisms of X in Y relatively to S are identified then, by virtue of (20.5.3), with the *pseudo-functions* φ *on* X *such that* $\operatorname{dom}(\varphi)$ *is universally schematically dense relatively to* S. The sheaf $\mathscr{P}s.\mathrm{hom}_{X/S}(X, Y)$ is a *sub-sheaf of rings* of $\mathscr{M}'_X$, that we denote $\mathscr{M}'_{X/S}$.

We then set $\mathrm{Ps.\,hom}_{X/S}(X, Y) = M'(X/S)$ and we say that its elements are the *pseudo-functions on* X *relative to* S.

(20.5.5). Let X, Y, Z be three S-preschemes, $f : Y \to Z$ an S-morphism. We can, in the reasoning of (20.3.1), replace everywhere "schematically dense" by "universally schematically dense relatively to S"; for every pseudo-morphism $\omega \in \mathrm{Ps.\,hom}_{X/S}(X, Y)$, the morphisms $f \circ u$, where u runs through ω, belong therefore to a single equivalence class (for the relation defined in (20.5.1)), which we call the pseudo-morphism from X to Z, relative to S, *composite* of f and ω, and denote $f \circ \omega$. If $g : Z \to T$ is a morphism, it is immediate that $g \circ (f \circ \omega) = (g \circ f) \circ \omega$. **IV-4 | 251**

(20.5.6). Suppose Y *separated* on S, and let ω be a pseudo-morphism of X in Y relatively to S. If $f : X' \to X$ is an S-morphism such that $f^{-1}(\operatorname{dom}_S(\omega))$ is universally schematically dense in X' *relative to* S, it follows from (20.3.2) that the pseudo-S-morphism $\omega \circ f$ has a domain universally schematically dense relative to S, and hence, by (20.5.3), may be regarded as a pseudo-morphism *relative to* S. When X' is flat and locally of finite presentation over S, the condition that $f^{-1}(\operatorname{dom}_S(\omega))$ be universally schematically dense in X' relative to S is equivalent to saying that for every $s \in S$, $(f^{-1}(\operatorname{dom}_S(\omega)))_s$ (notation of (11.10.10)) is schematically dense in X'_s. This latter condition will in particular be satisfied if, for every $s \in S$, X'_s and f_s satisfy one of the three conditions (i), (ii), (iii) of (20.3.5).

(20.5.7). Suppose now that X and X' are both S-preschemes *flat and locally of finite presentation* over S, and that $f : X' \to X$ is an S-morphism *flat* (or, what amounts to the same (11.3.10), that for every $s \in S$, $f_s : X'_s \to X_s$ is a flat morphism). Then, for every open U of X and every open $V \subset U$ universally schematically dense in U relatively to S, it results from (11.10.5) and (11.10.9) that $f^{-1}(V)$ is universally schematically dense in $f^{-1}(U)$ relatively to S. For every pseudo-S-morphism ω of X in Y relatively to S, it results from (20.3.4) that the pseudo-S-morphism $\omega \circ f$ is defined and is a pseudo-morphism of X' in Y relatively to S, *even when* Y *is not supposed separated on* S. We deduce from this that for every S-morphism $g : Y \to Z$, $(g \circ \omega) \circ f$ is still defined and equal to $g \circ (\omega \circ f)$ (with the definitions of (20.5.1)), and that it is therefore a pseudo-morphism *relative* to S.

(20.5.8). Let X be an S-prescheme, $S' \to S$ a morphism, $g : X' \to X$ the canonical projection, $X' = X \times_S S'$. Then, for every open U of X and every open $V \subset U$ universally schematically dense in U relatively to S, $V' = g^{-1}(V)$ is universally schematically dense in $U' = g^{-1}(U)$ relatively to S' by definition (11.10.8). Let ω be a pseudo-S-morphism of X in an S-prescheme Y relatively to S; if u_1, u_2 are S-morphisms of the class ω, defined respectively on opens U_1, U_2 of X, universally schematically dense in X relative to S, it follows from what precedes that $u'_1 = u_1 \circ (g|g^{-1}(U_1))$ and $u'_2 = u_2 \circ (g|g^{-1}(U_2))$ coincide on an open U'_3 universally schematically dense relative to S'. Now, if $Y' = Y_{(S')}$ and $h : Y' \to Y$ is the canonical projection, u'_1 and u'_2 factor canonically as $u'_1 = h \circ v_1$ and $u'_2 = h \circ v_2$, where v_1 and v_2 are S'-morphisms into Y' which coincide on U'_3. Thus, as u_1 runs through the class ω, the corresponding S'-morphisms v_1 belong to a single pseudo-morphism relative to S', called the *inverse image* of ω under the base change $S' \to S$ and denoted $\omega_{(S')}$. It is clear that if $S'' \to S'$ is a second morphism, $(\omega_{(S')})_{(S'')} = \omega_{(S'')}$ (for the composed morphism $S'' \to S' \to S$).

20.6. Relative meromorphic functions

(20.6.1). Let S be a prescheme, X an S-prescheme which is *flat and locally of finite presentation* over S; for every $s \in S$, we denote by X_s the fibre at the point s of the structural morphism $X \to S$. In general, if φ is a meromorphic function on X, it is not possible, for each $s \in S$, to associate to it in a "natural" way a meromorphic function "induced" on X_s (20.1.11). For every open U of X, denote by $T_{X/S}(U)$ the set of sections $t \in \Gamma(U, \mathscr{O}_X)$ such that, for every $s \in S$, the image t_s of t by the canonical homomorphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U \cap X_s, \mathscr{O}_{X_s})$ is a *regular* section; this also implies, by the equivalence of a) and b) in (11.3.7), that t is itself a *regular* section. It is clear that $U \mapsto T_{X/S}(U)$ is a *sub-sheaf* of the sheaf of sets $\mathscr{S} = \mathscr{S}(\mathscr{O}_X)$ (notation of (20.1.3)), that we denote $\mathscr{T}_{X/S}$; we set **IV-4 | 252**

$$\mathscr{M}_{X/S} = \mathscr{O}_X[\mathscr{T}_{X/S}^{-1}] \tag{20.6.1.1}$$

(notation of (20.1.1)) and we say that this sheaf of rings is the *sheaf of germs of meromorphic functions on X relative to S*; its sections over X are called *meromorphic functions on X relative to S* and their set is denoted $M(X/S)$. It is clear that $\mathscr{M}_{X/S}$ is a *sub-sheaf* of $\mathscr{M}_X = \mathscr{O}_X[\mathscr{S}^{-1}]$; for every meromorphic function $\varphi \in M(X/S)$ and every $s \in S$, the inverse image of φ by the canonical injection morphism $j_s : X_s \to X$ is then defined (20.1.11), and denoted φ_s.

(20.6.2). Let now $\mathscr{F}$ be an $\mathscr{O}_X$-Module quasi-coherent; we set

$$\mathscr{M}_{X/S}(\mathscr{F}) = \mathscr{F}[\mathscr{T}_{X/S}^{-1}] = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_{X/S}; \tag{20.6.2.1}$$

the sections of $\mathscr{M}_{X/S}(\mathscr{F})$ are called *meromorphic sections of $\mathscr{F}$ over X, relative to S* and their set is denoted $M(X/S, F)$. The canonical homomorphism $\mathscr{F} \to \mathscr{M}_{X/S}(\mathscr{F})$ is not necessarily injective; when it is, we say that $\mathscr{F}$ is *without torsion relatively to S*: this means that for every open U of X and every section $t \in T_{X/S}(U)$, t is $(\mathscr{F}|U)$-regular; this condition is satisfied *a fortiori* when $\mathscr{F}$ is torsion-free (20.1.5). In this last case, it results immediately from the definitions (20.1.2) that the canonical homomorphism of $\mathscr{O}_X$-Modules

$$\mathscr{M}_{X/S}(\mathscr{F}) \longrightarrow \mathscr{M}_X(\mathscr{F}) \tag{20.6.2.2}$$

is *injective*, so that the meromorphic sections of $\mathscr{F}$ relative to S are meromorphic sections of $\mathscr{F}$ in the sense of (20.1.3).

Proposition (20.6.3). — *The image by the injective homomorphism (20.2.10.1) of the sub-sheaf $\mathscr{M}_{X/S}$ of $\mathscr{M}_X$ is the sub-sheaf $\mathscr{M}'_{X/S}$ of pseudo-functions on X relative to S (i.e. of the pseudo-functions whose domain of definition is universally schematically dense relatively to S (20.5.4)).* **IV-4 | 253**

Proof. We can evidently restrict to proving that the image of $M(X/S)$ by the canonical homomorphism $M(X) \to M'(X)$ is equal to $M'(X/S)$; the proposition is then a consequence of the following more general proposition: □

Proposition (20.6.4). — *Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module quasi-coherent of finite presentation and torsion-free. Then, for a meromorphic section φ of $\mathscr{F}$ over X to be a meromorphic section relative to S, it is necessary and sufficient that* $\mathrm{dom}(\varphi)$ *be universally schematically dense relatively to S.*

Proof. The necessity of the condition results from (20.2.15) applied to each canonical injection $X_s \to X$ ($s \in S$), taking into account (11.10.9). To see that the condition is sufficient, it suffices to prove that for every $x \in X$, there exists a neighbourhood U of x and a section of $\mathscr{M}_{X/S}(\mathscr{F})$ over U whose restriction to $U \cap \mathrm{dom}(\varphi)$ coincides with φ in an open schematically dense of $U \cap \mathrm{dom}(\varphi)$. Consider the ideal of denominators $\mathscr{J}$ of φ (20.2.14), which is quasi-coherent and defines the closed sub-prescheme of X whose underlying space is $X - \mathrm{dom}(\varphi)$. By hypothesis, if s is the image of x in S, $\mathrm{dom}(\varphi) \cap X_s$ is schematically dense in the locally Noetherian prescheme X_s, and hence contains $\mathrm{Ass}(\mathscr{O}_{X_s})$ (20.2.13, (iv)); this implies that the Ideal $\mathscr{J}_x$ has an image in $\mathscr{O}_{X_s,x} = \mathscr{O}_{X,x} \otimes_{\mathscr{O}_{S,s}} k(s)$ which is not contained in any of the prime ideals $\mathfrak{p}_i \in \mathrm{Ass}(\mathscr{O}_{X_s,x})$ (finitely many); hence (Bourbaki, *Alg. comm.*, chap. II, §1, nº 1, prop. 2) there exists an element $t_x \in \mathscr{J}_x$ whose image in $\mathscr{O}_{X_s,x}$ does not belong to any of the $\mathfrak{p}_i$, and is therefore *regular* in this Noetherian ring. Let t be a section of $\mathscr{J}$ over an affine neighbourhood U of x, whose germ at x is t_x; since X is flat and locally of finite presentation over S, we can suppose (11.3.8) that t is a regular section of $\mathscr{O}_X$ over U and that, for every $s' \in S$, the image of t in $\Gamma(U \cap X_{s'}, \mathscr{O}_{X_{s'}})$ is also regular; in other words, $t \in T_{X/S}(U)$. But then, since $\mathscr{F}$ is torsion-free, $t(\varphi|(U\cap \mathrm{dom}(\varphi)))$ is a section u of $\mathscr{F}$ over $U\cap\mathrm{dom}(\varphi)$; on the other hand, $U\cap\mathrm{dom}(\varphi)$ contains the open U_t of $x'\in U$ where $t(x')\neq 0$, and this last is schematically dense in U (20.2.9). We see that in U_t, φ coincides with the restriction to U_t of the section u/t of $\mathscr{M}_{X/S}(\mathscr{F})$ over U. C.Q.F.D. □

Remarks (20.6.5). — (i) Let φ be a meromorphic function on X relative to S, so that for every $s\in S$, φ_s is a meromorphic function on X_s (20.6.1); by virtue of (20.1.11.1), one has

(20.6.5.1) $$\mathrm{dom}(\varphi)\cap X_s\subset\mathrm{dom}(\varphi_s).$$

But it is worth noting that even when S is the spectrum of a discrete valuation ring A and $X=S[T]$ (T indeterminate), the two sides of (20.6.5.1) are not necessarily equal: for example, if π is a uniformizer of A, it is immediate that $\varphi=\pi/T$ is a meromorphic function on X relative to S, since if a and b are the closed and generic points of S, T is regular in $\Gamma(X_a,\mathscr{O}_{X_a})=k[T]$ and in $\Gamma(X_b,\mathscr{O}_{X_b})=K[T]$, k being the residue field and K the field of fractions of A. One has $\mathrm{dom}(\varphi)=\mathrm{D}(T)$ in X, but $\mathrm{dom}(\varphi_a)=X_a$ since $\varphi_a=0$.

(ii) For a meromorphic function φ relative to S to be *invertible in the ring* $M(X/S)$, it is necessary and sufficient that for every $s\in S$, φ_s be invertible in $M(X_s)$ (in other words, that φ_s be a regular meromorphic function on X_s (20.1.8)). The condition is trivially necessary. Conversely, if it is satisfied, and if x is any point of X, s its image in S, there exists by hypothesis a neighbourhood U of x and two sections u, t of $\mathscr{O}_X$ over U such that $t\in T_{X/S}(U)$ and $\varphi|U=u/t$; the hypothesis implies that u_s is regular at x in $\Gamma(U\cap X_s,\mathscr{O}_{X_s})$, so u_s is regular at point x. Restricting U, we can therefore suppose, by virtue of (11.3.8), that $u\in T_{X/S}(U)$, whence the conclusion. **IV-4** | 254

When φ is invertible in $M(X/S)$, we also say that φ is a *regular meromorphic function relative to* S. We note that $\varphi\in M(X/S)$ can be invertible in $M(X)$ (in other words, be *regular* in the sense of (20.1.8)) without being invertible in $M(X/S)$, as the above example of (i) shows.

(iii) Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module, and let φ be a regular meromorphic section of $\mathscr{L}$ over X (20.1.8); we say that φ is *regular relatively to* S if, for every open U of X such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$, $\varphi|U$ corresponds to an element of $\Gamma(U,\mathscr{M}_{X/S})$ which is *regular relatively to* S (20.1.8); by virtue of (ii), it is immediate that it is necessary and sufficient that, for every $s\in S$, φ_s be a regular meromorphic section (20.1.8) of the invertible $\mathscr{O}_{X_s}$-Module $\mathscr{L}_s=\mathscr{L}\otimes_{\mathscr{O}_X}k(s)$. If φ' is the inverse of φ in $\mathscr{L}^{-1}$ (20.1.10), φ' is then also regular relatively to S, and if $\mathscr{L}_1$ is a second invertible $\mathscr{O}_X$-module, φ_1 a meromorphic section of $\mathscr{L}_1$ over X, regular relatively to S, then $\varphi\otimes\varphi_1$ is a meromorphic section of $\mathscr{L}\otimes\mathscr{L}_1$ over X, regular relatively to S.

Proposition (20.6.6). — *Let X be an S-prescheme flat and locally of finite presentation over S, and let $\mathscr{F}$ be an $\mathscr{O}_X$-Module locally free of finite type; for every $s\in S$, denote by X_s the fibre at the point s of the structural morphism $f:X\to S$. Let φ be a meromorphic section of $\mathscr{F}$ over X,* relatively to S, *and suppose that φ is defined at all points $x\in X$ such that* $\mathrm{prof}(\mathscr{O}_{X_{f(x)},x})=1$. *Then φ is everywhere defined.*

Proof. By hypothesis, $\mathrm{dom}(\varphi)\cap X_s$ is schematically dense in X_s for every $s\in S$, hence contains the points x of X_s such that $\mathrm{prof}(\mathscr{O}_{X_s,x})=0$ (5.10.2); the hypothesis therefore signifies that if one sets $Z=X-\mathrm{dom}(\varphi)$, one has $\mathrm{prof}(\mathscr{O}_{X_{f(x)},x})\geqslant 2$ for all points x of Z. It suffices then to apply (19.9.8). □

(20.6.7). Let X, X' be two S-preschemes flat and locally of finite presentation, $f=(\psi,\theta):X'\to X$ an S-morphism. For every open U of X, denote by $T_f(U)$ the set of sections $t\in T_{X/S}(U)$ whose image in $\Gamma(f^{-1}(U),\mathscr{O}_{X'})$ belongs to $T_{X'/S}(f^{-1}(U))$; it is immediate that $U\mapsto T_f(U)$ is a sub-sheaf of the sheaf of sets $\mathscr{T}_{X/S}$, that we denote $\mathscr{T}_f$. We set $\mathscr{M}_{X/S,f}=\mathscr{O}_X[\mathscr{T}_f^{-1}]$; this is a sub-sheaf of rings of $\mathscr{M}_{X/S}$, and we canonically obtain from $\theta^{\#}:\psi^*(\mathscr{O}_X)\to\mathscr{O}_{X'}$ a homomorphism of sheaves of rings $\theta'^{\#}:\psi^*(\mathscr{M}_{X/S,f})\to\mathscr{M}_{X'/S}$ extending $\theta^{\#}$. A meromorphic function φ on X, relative to S, that *is a section of* $\mathscr{M}_{X/S,f}$, then has an inverse image $\Gamma(\theta'^{\#})(\varphi)$ which is a meromorphic function on X' relative to S, called the *inverse image of φ by f*, and denoted $\varphi\circ f$ if this causes no confusion. We extend immediately in the same way the definitions of (20.1.11) relative to $\mathscr{O}_X$-Modules.

Proposition (20.6.8). — *With the notations of (20.6.7), if the S-morphism $f : X' \to X$ is* flat, *one has $\mathscr{M}_{X/S,f} = \mathscr{M}_{X/S}$, and the homomorphism $\varphi \mapsto \varphi \circ f$ is defined on all of $M(X/S)$.*

Proof. Indeed, the hypothesis implies, by virtue of (11.3.10), that for every $s \in S$, $f_s : X'_s \to X_s$ is flat; hence, for every section $t \in T_{X/S}(U)$, if t' is its inverse image, t'_s is a *regular* section of $\mathscr{O}_{X'_s}$ over $f^{-1}(U) \cap X'_s$; by virtue of (20.1.12), we conclude that by definition $t' \in T_{X'/S}(f^{-1}(U))$, whence the proposition. ∎ IV-4 | 255

We deduce from this, as in (20.1.12), a canonical homomorphism of $\mathscr{O}_{X'}$-algebras

$$f^*(\mathscr{M}_{X/S}) \longrightarrow \mathscr{M}_{X'/S}. \tag{20.6.8.1}$$

(20.6.9). Consider finally a morphism $S' \to S$, and set $X' = X \times_S S'$, which is flat and locally of finite presentation over S'; let $p : X' \to X$ be the canonical projection. Let U be an open of X, t a section belonging to $T_{X/S}(U)$, t' its image in $\Gamma(p^{-1}(U), \mathscr{O}_{X'})$; for every $s' \in S'$, if s is the image of s', we have $X'_{s'} = X_s \otimes_{k(s)} k(s')$, hence the morphism $X'_{s'} \to X_s$ is flat, and by virtue of (20.1.12) the inverse image $t'_{s'}$ of t_s is regular; this proves that $t' \in T_{X'/S'}(p^{-1}(U))$. We deduce from this, as in (20.1.12), a canonical homomorphism of $\mathscr{O}_{X'}$-Algebras

$$p^*(\mathscr{M}_{X/S}) \longrightarrow \mathscr{M}_{X'/S'}. \tag{20.6.9.1}$$

Using the identification permitted by (20.6.3), this notion of base change for relative meromorphic functions is a particular case of the analogous notion for relative pseudo-morphisms (20.5.8).

§21. Divisors

For the contents of the present section, see the comments in the introduction to ğ20. For the *global* properties of divisors, the reader is referred to the section devoted to them in Chapter V.

21.1. Divisors on a ringed space

(21.1.1). Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{M}_X$ be the sheaf of germs of meromorphic functions on X (20.1.3), and $\mathscr{M}_X^*$ the sheaf (of multiplicative groups) of germs of regular meromorphic functions on X (20.1.8). It is clear that the sheaf $\mathscr{O}_X^*$ (of multiplicative groups) of germs of invertible sections of $\mathscr{O}_X$ is canonically identified with a subsheaf (of multiplicative groups) of $\mathscr{M}_X^*$.

(21.1.2). We call the quotient sheaf (of commutative groups) $\mathscr{M}_X^*/\mathscr{O}_X^*$ the *sheaf of divisors on* X, and denote it by $\mathscr{D}iv_X$. Its sections over X are called the *divisors on* X; they form a commutative group denoted $\mathrm{Div}(X)$. For every section f of $\mathscr{M}_X^*$ over X (that is, every regular meromorphic function on X (20.1.8)), the *divisor of* f, denoted $\mathrm{div}(f)$ (or $\mathrm{div}_X(f)$), is the divisor on X which is the image of f under the canonical homomorphism

$$\Gamma(X, \mathscr{M}_X^*) \longrightarrow \Gamma(X, \mathscr{D}iv_X).$$

The *support* of a divisor D is the closed set of $x \in X$ such that $D_x \neq 0$. We denote it $\mathrm{Supp}(D)$. IV-4 | 256

For every open U of X, we evidently have $\mathscr{M}_X^*|U = \mathscr{M}_U^*$, $\mathscr{O}_X^*|U = \mathscr{O}_U^*$, hence $\mathscr{D}iv_X|U = \mathscr{D}iv_U$, and consequently the sheaf $\mathscr{D}iv_X$ is equal to the presheaf $U \mapsto \mathrm{Div}(U)$.

When $X = \mathrm{Spec}(A)$ is affine, we write $\mathrm{Div}(A)$ instead of $\mathrm{Div}(\mathrm{Spec}(A))$.

(21.1.3). We will always denote *additively* the group $\mathrm{Div}(X)$ of divisors on X. For two regular meromorphic functions f, g on X, we thus have

$$\mathrm{div}(fg) = \mathrm{div}(f) + \mathrm{div}(g), \tag{21.1.3.1}$$

$$\mathrm{div}(f^{-1}) = -\mathrm{div}(f). \tag{21.1.3.2}$$

By definition, for every regular meromorphic function f on X, we have the equivalence

$$\mathrm{div}(f) = 0 \iff f \in \Gamma(X, \mathscr{O}_X^*), \tag{21.1.3.3}$$

whence, for two regular meromorphic functions f, g on X, we have

$$\mathrm{div}(f) = \mathrm{div}(g) \iff fg^{-1} \in \Gamma(X, \mathscr{O}_X^*). \tag{21.1.3.4}$$

(21.1.4). Now let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module, and let s be a *regular meromorphic section* (20.1.8) of $\mathscr{L}$ over X. Every $x \in X$ admits a neighbourhood U such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$, hence $\mathscr{M}_X(\mathscr{L})|U$ is isomorphic to $\mathscr{M}_X|U$; by one of these isomorphisms, $s|U$ corresponds to a section $f \in \Gamma(U, \mathscr{M}_X^*)$, and since two isomorphisms differ only by multiplication by an element of $\Gamma(U, \mathscr{O}_X^*)$ ($\mathbf{0_I}$, 5.4.3), the element $\operatorname{div}_U(f)$ of $\Gamma(U, \mathscr{D}iv_X)$ is *independent* of the isomorphism chosen; it is clear that these elements (for U variable) are the restrictions of a section of $\mathscr{D}iv_X$ over X, which we call the *divisor of s* and denote $\operatorname{div}(s)$ (such a divisor is not necessarily of the form $\operatorname{div}(g)$ for a regular meromorphic function g on X; see (21.2.9)). For $\mathscr{L} = \mathscr{O}_X$, the definition of $\operatorname{div}(s)$ coincides with that of (21.1.2). If $\mathscr{L}, \mathscr{L}'$ are two invertible $\mathscr{O}_X$-Modules, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, we immediately have

(21.1.4.1) $$\operatorname{div}(s \otimes s') = \operatorname{div}(s) + \operatorname{div}(s')$$

(21.1.4.2) $$\operatorname{div}(s^{\otimes n}) = n.\operatorname{div}(s) \quad \text{for all} \quad n \in \mathbf{Z}$$

(s^{-1} being the regular meromorphic section of $\mathscr{L}^{-1}$ over X defined in (20.1.10)) and, for two regular meromorphic sections s, s' of $\mathscr{L}$ over X, we have the relation

(21.1.4.3) $$\operatorname{div}(s) = \operatorname{div}(s') \iff s' = ts \quad \text{with} \quad t \in \Gamma(X, \mathscr{O}_X^*).$$

(21.1.5). The sheaf $\mathscr{S}(\mathscr{O}_X)$ (20.1.3) whose sections over an open U of X are the regular elements of $\Gamma(U, \mathscr{O}_X)$ is a *sub-sheaf of monoids* of $\mathscr{M}_X^*$; we can write

(21.1.5.1) $$\mathscr{S}(\mathscr{O}_X) = \mathscr{O}_X \cap \mathscr{M}_X^*.$$

If we denote by $\mathscr{S}(\mathscr{O}_X)^{-1}$ the sheaf whose sections over U are the inverses in $\Gamma(U, \mathscr{M}_X^*)$ of the **IV-4 | 257**
elements of $\Gamma(U, \mathscr{S}(\mathscr{O}_X))$, it is clear that $\Gamma(U, \mathscr{S}(\mathscr{O}_X)) \cap \Gamma(U, \mathscr{S}(\mathscr{O}_X)^{-1}) = \Gamma(U, \mathscr{O}_X^*)$, hence

(21.1.5.2) $$\mathscr{S}(\mathscr{O}_X) \cap \mathscr{S}(\mathscr{O}_X)^{-1} = \mathscr{O}_X^*.$$

Definition (21.1.6). — The sub-sheaf of sets of $\mathscr{D}iv_X$, the canonical image of the sub-sheaf $\mathscr{S}(\mathscr{O}_X)$ of $\mathscr{M}_X^*$, is denoted $\mathscr{D}iv_X^+$; its sections over X are called *positive divisors* and their set is denoted $\operatorname{Div}^+(X)$.

Since $\mathscr{S}(\mathscr{O}_X)$ is a sheaf of monoids (multiplicative), we have

(21.1.6.1) $$\operatorname{Div}^+(X) + \operatorname{Div}^+(X) \subset \operatorname{Div}^+(X)$$

and on the other hand, by virtue of (21.1.5.2) and (21.1.3.3)

(21.1.6.2) $$\operatorname{Div}^+(X) \cap (-\operatorname{Div}^+(X)) = \{0\}.$$

These two relations show that $\operatorname{Div}^+(X)$ is the set of positive elements for an *order structure* on the group $\operatorname{Div}(X)$, compatible with this group structure; we denote this order relation by $D \leqslant D'$, in other words, we have

(21.1.6.3) $$D \geqslant 0 \iff D \in \operatorname{Div}^+(X).$$

We will always suppose in what follows that $\operatorname{Div}(X)$ is equipped with this order structure.

It is clear that $\mathscr{D}iv_X^+|U = \mathscr{D}iv_U^+$ for every open U of X, and hence that $\Gamma(U, \mathscr{D}iv_X^+) = \operatorname{Div}^+(U)$; thus $\mathscr{D}iv_X^+$ defines on $\mathscr{D}iv_X$ a structure of sheaf of ordered groups. The stalk $(\mathscr{D}iv_X^+)_x$ is a sub-monoid of $(\mathscr{D}iv_X)_x$, consisting of its positive elements. Consequently, for a divisor D on X, to say that $D \geqslant 0$ is equivalent to saying that $D_x \geqslant 0$ for every $x \in X$.

By definition, for every regular meromorphic function f on X, we have the relation

(21.1.6.4) $$\operatorname{div}(f) \geqslant 0 \iff f \in \Gamma(X, \mathscr{S}(\mathscr{O}_X))$$

in other words, $\operatorname{div}(f) \geqslant 0$ means that f is a *regular section* of $\mathscr{O}_X$, or equivalently a section of $\mathscr{O}_X$ *invertible in* $M(X)$.

More generally, given a divisor D on X, the relation $\operatorname{div}(f) \geqslant D$ is equivalent to the following: for every open U of X such that $D|U = \operatorname{div}(g)$, where $g \in \Gamma(U, \mathscr{M}_X^*)$, there exists a regular element t of $\Gamma(U, \mathscr{O}_X)$ such that $f|U = tg$.

(21.1.7). Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module, s a regular meromorphic section of $\mathscr{L}$ over X; we have the relation

$$\text{(21.1.7.1)} \qquad \operatorname{div}(s) \geqslant 0 \iff s \in \Gamma(X, \mathscr{L}) \cap \Gamma(X, (\mathscr{M}_X(\mathscr{L}))^*)$$

as follows immediately from the definitions (21.1.4) and (21.1.6).

Proposition (21.1.8). — *Let X be a locally Noetherian prescheme, D a divisor on X. Suppose that for every $x \in X$ such that* $\operatorname{prof}(\mathscr{O}_{X,x}) = 1$ *we have $D_x \geqslant 0$ (resp. $D_x = 0$). Then $D \geqslant 0$ (resp. $D = 0$).*

Proof. The question being local on X, we can suppose that $D = \operatorname{div}(f)$, f being a regular meromorphic function on X; the hypothesis means that if $T = X - \operatorname{dom}(f)$, then $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 2$ for every $x \in T$ (since $\operatorname{dom}(f)$ contains the maximal points of X). By (5.10.5), the restriction homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(X - T, \mathscr{O}_X)$ is *bijective*, which shows that there exists a section s of $\mathscr{O}_X$ over X such that $f = s|(X - T)$. But by definition of T, this implies $T = \varnothing$, hence $f = s$ and $D \geqslant 0$, and we immediately deduce the conclusion by applying the preceding to $-D$, by virtue of (21.1.6.2). ☐ **IV-4 | 258**

Corollary (21.1.9). — *Let X be a locally Noetherian prescheme, D a divisor on X. Let S be the support of D. Then, for every maximal point x of S,* $\operatorname{prof}(\mathscr{O}_{X,x}) = 1$.

Proof. Set $X_1 = \operatorname{Spec}(\mathscr{O}_{X,x})$; taking into account (20.2.11) and (20.3.6), the sheaf $\mathscr{M}^*_{X_1}$ is induced on X_1 by $\mathscr{M}^*_X$, hence we can restrict to the case where $X = X_1$, in which case $x \in S$ and x is the maximal point of S, so necessarily $S = \{x\}$. If $\operatorname{prof}(\mathscr{O}_{X,x}) \neq 1$, we would conclude, by virtue of (21.1.8), that $D = 0$, which contradicts the definition of S. ☐

Proposition (21.1.10). — *Let A be a local Noetherian ring; in order that* $\operatorname{Div}(A) = 0$, *it is necessary and sufficient that* $\operatorname{prof}(A) = 0$, *in other words, that the maximal ideal $\mathfrak{m}$ of A be associated to A (**0**, 16.4.6).*

Proof. Indeed, to say that $\operatorname{Div}(A) = 0$ means that in A every regular element is invertible, or equivalently that all the elements of $\mathfrak{m}$ are divisors of zero, which means that $\mathfrak{m} \in \operatorname{Ass}(A)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 3 of prop. 2). ☐

21.2. Divisors and invertible fractionary ideals

(21.2.1). Let $(X, \mathscr{O}_X)$ be a ringed space. We call *fractionary Ideal* on X a sub-$\mathscr{O}_X$-Module of the $\mathscr{O}_X$-Module $\mathscr{M}_X$ of germs of meromorphic functions on X. A fractionary Ideal $\mathscr{J}$ on X which is an invertible $\mathscr{O}_X$-Module is called an *invertible fractionary Ideal*.

Proposition (21.2.2). — *For an invertible fractionary Ideal $\mathscr{J}$ on X to be invertible, it is necessary and sufficient that for every $x \in X$, there exists a neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{M}^*_X)$ such that $\mathscr{J}|U = \mathscr{O}_U.f$.*

Proof. The condition is evidently sufficient, since the map $s \mapsto s(f|V)$ of $\Gamma(V, \mathscr{O}_X)$ is evidently bijective for every open $V \subset U$. To see that it is necessary, let us note that there exists by hypothesis a neighbourhood U of x and an isomorphism of $\mathscr{O}_X$-Modules $\mathscr{O}_U \xrightarrow{\sim} \mathscr{J}|U$. If f is the image of the section $1 \in \Gamma(U, \mathscr{O}_X)$ by this isomorphism, we may suppose, after restricting U, that $f = u/s$, where $u \in \Gamma(U, \mathscr{O}_X)$ and $s \in \Gamma(U, \mathscr{S}(\mathscr{O}_X))$, and the isomorphism considered then associates, to every section $v \in \Gamma(V, \mathscr{O}_X)$ (where V is an open contained in U), the section $v(u|V)/(s|V)$; to say that this map is bijective means that $u|V$ is a regular element of $\Gamma(V, \mathscr{O}_X)$, hence $f \in \Gamma(U, \mathscr{M}^*_X)$. ☐

We note that for every open U of X such that $\mathscr{J}|U = \mathscr{O}_U.f$ with $f \in \Gamma(U, \mathscr{M}^*_X)$, the section f is determined *up to multiplication* by an element of $\Gamma(U, \mathscr{O}^*_X)$, since the multiplication by these elements gives all the automorphisms of the $\mathscr{O}_U$-Module $\mathscr{O}_U$ ($\mathbf{0_I}$, 5.4.3). **IV-4 | 259**

Corollary (21.2.3). —
(i) *Let $\mathscr{J}$ be an invertible fractionary Ideal; then the invertible $\mathscr{O}_X$-Module $\mathscr{J}^{-1}$ is canonically identified with the fractionary Ideal $\mathscr{J}'$ (the* transporter *of $\mathscr{J}$ into $\mathscr{O}_X$) defined as follows: for every open U of X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where $f \in \Gamma(U, \mathscr{M}^*_X)$, we have $\mathscr{J}'|U = \mathscr{O}_U.f^{-1}$.*
(ii) *If $\mathscr{J}_1$ and $\mathscr{J}_2$ are two invertible fractionary Ideals, the canonical map $\mathscr{J}_1 \otimes \mathscr{J}_2 \to \mathscr{J}_1\mathscr{J}_2$ is an isomorphism of $\mathscr{O}_X$-Modules.*

Proof. The assertion (ii) follows immediately from (21.2.2). On the other hand, the remark made at the end of (21.2.2) shows that there is indeed a unique fractionary Ideal $\mathscr{J}'$ and a unique isomorphism defined by the condition of the statement; the isomorphism of $\mathscr{J}'$ on $\mathscr{J}^{-1} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{J}, \mathscr{O}_X)$ is obtained by making correspond, for every section $s(f^{-1}|V)$ of $\Gamma(V, \mathscr{J}')$ (where V is an open contained in U and $s \in \Gamma(V, \mathscr{O}_X)$) the homomorphism $t(f|V) \mapsto st$ of $\Gamma(V, \mathscr{J})$ to $\Gamma(V, \mathscr{O}_X)$. □

By virtue of (21.2.3), (i), we will identify the invertible fractionary ideals $\mathscr{J}'$ and $\mathscr{J}^{-1}$ considered as sub-$\mathscr{O}_X$-Modules of $\mathscr{M}_X$.

(21.2.4). It follows from (21.2.3) that the set $\mathrm{Id.\,inv}(X)$ of invertible fractionary Ideals on X is equipped with a structure of *commutative group* for the law of composition $(\mathscr{J}_1, \mathscr{J}_2) \mapsto \mathscr{J}_1\mathscr{J}_2$, the neutral element of this group being $\mathscr{O}_X$. It is clear that for every open U of X, we have $(\mathscr{J}_1\mathscr{J}_2)|U = (\mathscr{J}_1|U)(\mathscr{J}_2|U)$, hence the restriction map $\mathscr{J} \mapsto \mathscr{J}|U$ is a homomorphism of groups $\mathrm{Id.\,inv}(X) \to \mathrm{Id.\,inv}(U)$; we thus define a *presheaf of commutative groups* $U \mapsto \mathrm{Id.\,inv}(U)$; it is immediate that in fact this presheaf is a *sheaf of commutative groups*, which we denote $\mathscr{J}d.inv_X$.

(21.2.5). For every regular meromorphic function $f \in \Gamma(X, \mathscr{M}_X^*)$, it follows from (21.2.2) that $\mathscr{J}(f) = \mathscr{O}_X.f$ is an invertible fractionary Ideal, and we evidently have $\mathscr{J}(f_1f_2) = \mathscr{J}(f_1)\mathscr{J}(f_2)$, in other words the map $f \mapsto \mathscr{J}(f)$ is a homomorphism of the commutative group $\Gamma(X, \mathscr{M}_X^*)$ to the commutative group $\mathrm{Id.\,inv}(X)$. Replacing X by an arbitrary open U of X and noting that the homomorphisms thus obtained are compatible with the restriction operations, we obtain a canonical homomorphism of sheaves of commutative groups:

(21.2.5.1) $$I_0 : \mathscr{M}_X^* \longrightarrow \mathscr{J}d.inv_X.$$

Let us note that if $f \in \Gamma(X, \mathscr{O}_X^*)$, then $\mathscr{J}(f) = \mathscr{O}_X$; we deduce immediately that the homomorphism I_0 factorises as

(21.2.5.2) $$\mathscr{M}_X^* \longrightarrow \mathscr{M}_X^*/\mathscr{O}_X^* = \mathscr{D}iv_X \xrightarrow{I} \mathscr{J}d.inv_X$$

where I is a homomorphism of the sheaf of additive groups $\mathscr{D}iv_X$ to the sheaf of multiplicative groups $\mathscr{J}d.inv_X$; we thus have for every open U of X a homomorphism $\mathscr{J}_U : \mathrm{Div}(U) \to \mathrm{Id.\,inv}(U)$ of commutative groups, such that, for every section $f \in \Gamma(U, \mathscr{M}_X^*)$, we have

(21.2.5.3) $$\mathscr{J}_U(\mathrm{div}_U(f)) = \mathscr{O}_U.f.$$

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We conclude that, for every divisor $D \in \mathrm{Div}(X)$, $\mathscr{J}_X(D)$ is the invertible fractionary ideal defined as follows: for every open U of X such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}_U^*)$, one has $\mathscr{J}_X(D)|U = \mathscr{O}_U.f$. Hence, by virtue of (21.1.6), for every regular meromorphic function $f \in \Gamma(X, \mathscr{M}_X^*)$, we have the relation

(21.2.5.4) $$f \in \Gamma(X, \mathscr{J}_X(D)) \iff \mathrm{div}(f) \geqslant D.$$

Proposition (21.2.6). — *The homomorphism $I : \mathscr{D}iv_X \to \mathscr{J}d.inv_X$ is bijective.*

Proof. We define indeed a homomorphism I'_X of $\mathrm{Id.\,inv}(X)$ to $\mathrm{Div}(X)$ by making correspond to every invertible fractionary Ideal $\mathscr{J}$ on X the divisor $I'_X(\mathscr{J})$ defined as follows: for every open U of X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where $f \in \Gamma(U, \mathscr{M}_X^*)$ (21.2.2), we take $I'_X(\mathscr{J})|U = \mathrm{div}_U(f)$; by virtue of the remark following (21.2.2), this definition is independent of the generator f chosen for $\mathscr{J}|U$, and determines a divisor on X. Moreover, this definition immediately shows that the homomorphisms $\mathscr{J}_X$ and I'_X are inverses of one another. Replacing X by an arbitrary open U, we deduce the definition of the isomorphism $I' : \mathscr{J}d.inv_X \to \mathscr{D}iv_X$, inverse of I, whence the proposition. We set $I'_X(\mathscr{J}) = \mathrm{div}(\mathscr{J})$, so that for every regular meromorphic function f on X

(21.2.6.1) $$\mathrm{div}(\mathscr{O}_X.f) = \mathrm{div}(f).$$

□

(21.2.7). We will often identify the sheaves $\mathscr{D}iv_X$ and $\mathscr{I}d.inv_X$ (resp. the groups $\mathrm{Div}(X)$ and $\mathrm{Id.inv}(X)$) by means of the isomorphisms I and I' (resp. $\mathscr{I}_X$ and I'_X) above. We note that we have the relation

$$D \geqslant 0 \iff \mathscr{I}_X(D) \subset \mathscr{O}_X \quad \text{for} \quad D \in \mathrm{Div}(X) \tag{21.2.7.1}$$

as follows immediately from the definitions (21.1.6) and (21.1.5.1); in other terms, the image $\mathscr{I}_X(\mathrm{Div}^+(X))$ is the set of *Ideals* of $\mathscr{O}_X$ (also sometimes called *integral Ideals*) which are invertible $\mathscr{O}_X$-Modules: such an Ideal $\mathscr{I}$ is further characterised by the fact that for every $x \in X$, there is a neighbourhood U of x in X such that $\mathscr{I}|U = \mathscr{O}_U.f$, where f is a regular element of $\Gamma(U, \mathscr{O}_X)$. The set $\mathscr{I}_X(\mathrm{Div}^+(X))$ of these Ideals is thus a sub-monoid of $\mathrm{Id.inv}(X)$, equal to the set of positive elements for an order relation on $\mathrm{Id.inv}(X)$, and it is immediate that this relation is none other than the relation *opposite* to the inclusion; in other words, we have

$$\mathscr{I}_1 \subset \mathscr{I}_2 \iff \mathrm{div}(\mathscr{I}_1) \geqslant \mathrm{div}(\mathscr{I}_2) \tag{21.2.7.2}$$

for $\mathscr{I}_1$, $\mathscr{I}_2$ in $\mathrm{Id.inv}(X)$.

(21.2.8). For every divisor D on X, we set

$$\mathscr{O}_X(D) = (\mathscr{I}_X(D))^{-1}; \tag{21.2.8.1}$$

$\mathscr{O}_X(D)$ is thus an invertible fractional Ideal, defined as follows: for every open U of X such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, $\mathscr{O}_X(D)|U$ is the frac- tionary Ideal $\mathscr{O}_U.f^{-1}$; by virtue of (21.1.6), for every regular meromorphic function f, we have the relation **IV-4** | 261

$$f \in \Gamma(X, \mathscr{O}_X(D)) \iff \mathrm{div}(f) \geqslant -D. \tag{21.2.8.2}$$

Moreover, it is clear that we have canonical isomorphisms (21.2.3)

$$\begin{cases} \mathscr{O}_X(0) = \mathscr{O}_X, \quad \mathscr{O}_X(D+D') = \mathscr{O}_X(D)\mathscr{O}_X(D') \xrightarrow{\sim} \mathscr{O}_X(D) \otimes \mathscr{O}_X(D') \\ \mathscr{O}_X(nD) = (\mathscr{O}_X(D))^n \xrightarrow{\sim} (\mathscr{O}_X(D))^{\otimes n} \end{cases} \tag{21.2.8.3}$$

for every integer $n \in \mathbf{Z}$, and for any two divisors D, D' on X.

(21.2.9). Let $\mathscr{I}$ be an invertible fractional Ideal on X. The canonical injection $\mathscr{I} \hookrightarrow \mathscr{M}_X$ defines by tensorisation a homomorphism of $\mathscr{O}_X$-Modules

$$\mathscr{M}_X(\mathscr{I}) = \mathscr{I} \otimes_{\mathscr{O}_X} \mathscr{M}_X \longrightarrow \mathscr{M}_X \otimes_{\mathscr{O}_X} \mathscr{M}_X = \mathscr{M}_X \tag{21.2.9.1}$$

which is an *isomorphism*: indeed, if U is an open of X such that $\mathscr{I}|U = \mathscr{O}_U.f$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, the homomorphism (21.2.9.1) restricted to U is none other than the isomorphism which, for every section t of $\mathscr{M}_X(\mathscr{I})|U = \mathscr{M}_X|U$ over $V \subset U$, makes correspond the section $t(f|V)^{-1}$ of the same sheaf. In the isomorphism (21.2.9.1), the regular meromorphic sections of $\mathscr{I}$ over X correspond to the regular meromorphic functions on X.

Consider in particular the case where $\mathscr{I} = \mathscr{O}_X(D)$, where D is a divisor on X; we then have a canonical isomorphism

$$\mathscr{M}_X(\mathscr{O}_X(D)) \xrightarrow{\sim} \mathscr{M}_X \tag{21.2.9.2}$$

and we denote by s_D the regular meromorphic section of $\mathscr{O}_X(D)$ over X which corresponds by this isomorphism to the section 1 of $\mathscr{M}_X$. If U is an open of X such that $\mathscr{O}_X(D)|U = \mathscr{O}_U.f^{-1}$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, then $s_D|U = 1$ in $\Gamma(U, \mathscr{M}_X)$; as $D|U = \mathrm{div}_U(f)$, we deduce from (21.1.4) that

$$\mathrm{div}(s_D) = D. \tag{21.2.9.3}$$

On the other hand, we also deduce from the canonical isomorphisms (21.2.8.3) the formulas

$$s_0 = 1, \quad s_{D+D'} = s_D \otimes s_{D'}, \quad s_{nD} = s_D^{\otimes n} \quad (n \in \mathbf{Z}). \tag{21.2.9.4}$$

(21.2.10). Consider, between two pairs $(\mathscr{L}, s)$, $(\mathscr{L}', s')$, where $\mathscr{L}$ and $\mathscr{L}'$ are two invertible $\mathscr{O}_X$-Modules, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, the relation: "there exists an isomorphism $u : \mathscr{L} \xrightarrow{\sim} \mathscr{L}'$ such that $\overline{u}(s) = s'$", where $\overline{u} : \Gamma(X, \mathscr{M}_X(\mathscr{L})) \xrightarrow{\sim} \Gamma(X, \mathscr{M}_X(\mathscr{L}'))$ is the isomorphism derived from u (we note that the isomorphism u satisfying this condition is then determined in a unique way). It is clear that this is an equivalence relation, and since there exists a set of invertible $\mathscr{O}_X$-Modules such that every invertible $\mathscr{O}_X$-Module is isomorphic to an element of this set ($\mathbf{0_I}$, 5.4.7), we can speak of the *set* $D(X)$ of *equivalence classes* of pairs $(\mathscr{L}, s)$ for the above relation. For every invertible $\mathscr{O}_X$-Module $\mathscr{L}$ and every regular mero- morphic section s of $\mathscr{L}$ over X, we denote by $\mathrm{cl}(\mathscr{L}, s)$ the element of $D(X)$ corresponding to the pair $(\mathscr{L}, s)$. It follows from ($\mathbf{0_I}$, 5.4.3) that if s, s' are two regular meromorphic sections of $\mathscr{L}$ over X, the relation $\mathrm{cl}(\mathscr{L}, s) = \mathrm{cl}(\mathscr{L}, s')$ is equivalent to the existence of a section $t \in \Gamma(X, \mathscr{O}_X^*)$ such that $s' = ts$. **IV-4** | 262

It is immediate that if $(\mathscr{L}, s)$ is equivalent to $(\mathscr{L}_1, s_1)$ and $(\mathscr{L}', s')$ to $(\mathscr{L}'_1, s'_1)$, the pairs $(\mathscr{L} \otimes \mathscr{L}', s \otimes s')$ and $(\mathscr{L}_1 \otimes \mathscr{L}'_1, s_1 \otimes s'_1)$ are equivalent; we thus define in $D(X)$ a law of composition by setting

(21.2.10.1) $$\mathrm{cl}(\mathscr{L}, s)\,\mathrm{cl}(\mathscr{L}', s') = \mathrm{cl}(\mathscr{L} \otimes \mathscr{L}', s \otimes s');$$

it is immediate that this is the law of a *commutative group*, whose neutral element is $\mathrm{cl}(\mathscr{O}_X, 1)$ and where the inverse of $\mathrm{cl}(\mathscr{L}, s)$ is $\mathrm{cl}(\mathscr{L}^{-1}, s^{\otimes(-1)})$.

Proposition (21.2.11). — *The maps*

(21.2.11.1) $$D \longmapsto \mathrm{cl}(\mathscr{O}_X(D), s_D), \qquad \mathrm{cl}(\mathscr{L}, s) \longmapsto \mathrm{div}(s)$$

are inverse isomorphisms of $\mathrm{Div}(X)$ *to* $D(X)$ *and of* $D(X)$ *to* $\mathrm{Div}(X)$ *respectively.*

Proof. Taking into account (21.2.8.3), (21.2.9.4), and (21.2.10.1), it suffices to verify that the composites of these two maps are the identities in $\mathrm{Div}(X)$ and $D(X)$ respectively. The first assertion is none other than (21.2.9.3). On the other hand, let $D = \mathrm{div}(s)$, where s is a regular meromorphic section of $\mathscr{L}$ over X, and let U be an open of X such that there exists an isomorphism of $\mathscr{L}|U$ onto $\mathscr{O}_U$, transforming $s|U$ to $f \in \Gamma(U, \mathscr{M}_X^*)$, so that $D|U = \mathrm{div}_U(f)$, $\mathscr{O}_X(D)|U = \mathscr{O}_U.f^{-1}$ and $s_D|U$ is the unit element of $\Gamma(U, \mathscr{M}_X)$. There is thus an isomorphism $v_U : \mathscr{L}|U \to \mathscr{O}_X(D)|U$ such that $\overline{v}_U$ (notation of (21.2.10)) transforms $s|U$ to $f.f^{-1} = 1$; one sees immediately that these isomorphisms are compatible with the restriction operations, and thus define an isomorphism $v : \mathscr{L} \to \mathscr{O}_X(D)$ such that $\overline{v}(s) = s_D$. □

We can *transport* by the first of the isomorphisms (21.2.11.1) the structure of ordered group of $\mathrm{Div}(X)$ to $D(X)$; the elements $\geqslant 0$ of $D(X)$ are thus the classes $\mathrm{cl}(\mathscr{L}, s)$ such that $\mathrm{div}(s) \geqslant 0$, that is to say (21.1.7.1) such that
$$s \in \Gamma(X, \mathscr{L}) \cap \Gamma(X, (\mathscr{M}_X(\mathscr{L}))^*).$$

(21.2.12). Let D be a *positive* divisor on a *prescheme* X; the fractionary Ideal $\mathscr{J}_X(D)$ is an Ideal of $\mathscr{O}_X$, which is an invertible $\mathscr{O}_X$-Module; let $Y(D)$ be the sub-prescheme of X that it defines. For every $x \in Y(D)$, there is by hypothesis a neighbourhood U of x in X and a regular section $t \in \Gamma(U, \mathscr{O}_X)$ such that $\mathscr{J}_X(D)|U = \mathscr{O}_U.t$ (21.2.7); in other terms, the canonical immersion $Y(D) \to X$ is *regular and of codimension* 1 (19.1.4). Conversely, if Y is a sub-prescheme of X, regularly immersed in X and of codimension 1 at every point of Y, there exists *a unique positive divisor* D such that $Y(D) = Y$, since there is a neighbourhood U of x in X such that $Y \cap U$ is defined by an Ideal of the form $\mathscr{O}_U.t$, where t is a regular element of $\Gamma(U, \mathscr{O}_X)$. We note that $\mathrm{Supp}(D) = Y(D)$, since to say that $D_x \neq 0$ means that t_x is not invertible, that is to say $x \in Y(D)$.

21.3. Linear equivalence of divisors

(21.3.1). We say that a divisor D on X is *principal* if it is of the form $\mathrm{div}(f)$, where f is a regular meromorphic function on X; the meromorphic functions f' such that $\mathrm{div}(f') = D$ are then all of the form tf, where $t \in \Gamma(X, \mathscr{O}_X^*)$ (21.1.3.4). The set of principal divisors is a subgroup of $\mathrm{Div}(X)$, denoted $\mathrm{Div.princ}(X)$, isomorphic to $\Gamma(X, \mathscr{M}_X^*)/\Gamma(X, \mathscr{O}_X^*)$. Two divisors D, D' are said to be *linearly equivalent* if $D - D'$ is a principal divisor; the principal divisors are thus the divisors linearly equivalent to 0.

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(21.3.2). Recall ($\mathbf{0_I}$, 5.4.7) that we can speak of the set of equivalence classes of invertible $\mathscr{O}_X$-Modules for the isomorphism relation; we denote this set by $\mathrm{Pic}(X)$, and for every invertible $\mathscr{O}_X$-Module $\mathscr{L}$, we denote by $\mathrm{cl}(\mathscr{L})$ the equivalence class of $\mathscr{O}_X$-Modules isomorphic to $\mathscr{L}$; moreover, $\mathrm{Pic}(X)$ is a commutative group for the multiplication defined by $\mathrm{cl}(\mathscr{L})\,\mathrm{cl}(\mathscr{L}') = \mathrm{cl}(\mathscr{L} \otimes \mathscr{L}')$. It is clear that the map

(21.3.2.1) $$r : \mathrm{cl}(\mathscr{L}, s) \longmapsto \mathrm{cl}(\mathscr{L})$$

is a homomorphism of the group $D(X)$ (21.2.10) to the group $\operatorname{Pic}(X)$. By composition, we thus deduce a homomorphism

(21.3.2.2) $$l : \operatorname{Div}(X) \xrightarrow{\sim} D(X) \xrightarrow{r} \operatorname{Pic}(X)$$

(also denoted l_X) such that, for every divisor D, we have

(21.3.2.3) $$l(D) = \operatorname{cl}(\mathscr{O}_X(D)).$$

Let us note finally that, if $u : X' \to X$ is a morphism of ringed spaces, $\mathscr{L}_1$, $\mathscr{L}_2$ two invertible isomorphic $\mathscr{O}_X$-Modules, the invertible $\mathscr{O}_{X'}$-Modules $u^*(\mathscr{L}_1)$ and $u^*(\mathscr{L}_2)$ ($\mathbf{0_I}$, 5.4.5) are isomorphic; as moreover, for any two invertible $\mathscr{O}_X$-Modules $\mathscr{L}_1, \mathscr{L}_2$, we have $u^*(\mathscr{L}_1 \otimes \mathscr{L}_2) = u^*(\mathscr{L}_1) \otimes u^*(\mathscr{L}_2)$ up to canonical isomorphism ($\mathbf{0_I}$, 4.3.3), we see that the morphism u canonically defines a homomorphism of commutative groups

(21.3.2.4) $$\operatorname{Pic}(u) : \operatorname{Pic}(X) \longrightarrow \operatorname{Pic}(X').$$

Proposition (21.3.3). — *(i) The kernel of the canonical homomorphism $l : \operatorname{Div}(X) \to \operatorname{Pic}(X)$ is the subgroup* $\operatorname{Div.princ}(X)$*, in other words, for $\mathscr{O}_X(D)$ and $\mathscr{O}_X(D')$ to be isomorphic, it is necessary and sufficient that D and D' be linearly equivalent. There is thus a canonical injective homomorphism*

(21.3.3.1) $$\operatorname{Div}(X)/\operatorname{Div.princ}(X) \longrightarrow \operatorname{Pic}(X)$$

derived from l.

(ii) For an invertible $\mathscr{O}_X$-Module $\mathscr{L}$ to be such that $\operatorname{cl}(\mathscr{L})$ is of the form $l(D)$, or equivalently that $\mathscr{L}$ is isomorphic to an $\mathscr{O}_X$-Module of the form $\mathscr{O}_X(D)$, it is necessary and sufficient that there exists a regular meromorphic section of $\mathscr{L}$.

Proof. The proposition follows immediately from the definitions and from (21.2.10). □ **IV-4** | 264

Proposition (21.3.4). — *Let X be a prescheme satisfying one of the two following hypotheses:*

a) *X is locally Noetherian and $\operatorname{Ass}(\mathscr{O}_X)$ is contained in an affine open of X.*
b) *X is reduced and the set of its irreducible components is locally finite.*

Then the canonical homomorphism $l : \operatorname{Div}(X) \to \operatorname{Pic}(X)$ is surjective, and gives by passage to the quotient an isomorphism

$$\operatorname{Div}(X)/\operatorname{Div.princ}(X) \xrightarrow{\sim} \operatorname{Pic}(X).$$

Proof. It suffices to show that every invertible $\mathscr{O}_X$-Module $\mathscr{L}$ admits a regular meromorphic section over X (21.3.3). In both cases, it will suffice, thanks to (20.2.11), (ii), to define a section s of $(\mathscr{M}_X(\mathscr{L}))^*$ over a *schematically dense* open U of X, or else, in case a), over an open U containing $\operatorname{Ass}(\mathscr{O}_X)$ (20.2.13), (iv). Indeed, let V be an arbitrary open of X such that $\mathscr{L}|V$ is isomorphic to $\mathscr{O}_V$, so that there exists an isomorphism of $\mathscr{M}_X(\mathscr{L})|V$ onto $\mathscr{M}_X|V$, transforming $s|(U \cap V)$ to a section f_V of $\mathscr{M}_X^*$ over $U \cap V$. As $U \cap V$ is schematically dense in V, it follows from (20.2.11) that there exists a unique regular meromorphic function g_V such that $g_V|(U \cap V) = f_V$, and this section corresponds by the isomorphism considered to a regular meromorphic section u_V of $\mathscr{L}$ over V such that $u_V|(U \cap V) = s|(U \cap V)$. Moreover, if V' is a second open of X such that $\mathscr{L}|V'$ is isomorphic to $\mathscr{O}_{V'}$, the restrictions of u_V and $u_{V'}$ to $V \cap V'$ are equal, since they correspond by isomorphism to two meromorphic functions which coincide on an open schematically dense $U \cap V \cap V'$, and we conclude again by (20.2.11); the u_V are thus well the restrictions of a single regular meromorphic section of $\mathscr{L}$ over X.

In case b), we take for each of the maximal points x_λ of X an open U_λ containing x_λ, meeting no irreducible component of X distinct from $\overline{\{x_\lambda\}}$, and such that $\mathscr{L}|U_\lambda$ is isomorphic to $\mathscr{O}_{U_\lambda}$; we take for s the section such that $s|U_\lambda$ is the section of $\mathscr{L}|U_\lambda$ which corresponds by the preceding isomorphism to the unit section of $\mathscr{O}_{U_\lambda}$.

In case a), we can take U to be an affine open (hence Noetherian), in other words, suppose that $X = \operatorname{Spec}(A)$, where A is Noetherian, and $\mathscr{L} = \widetilde{P}$, where P is a projective A-module of rank 1. If S is the set of regular elements of A, we have $\Gamma(X, \mathscr{M}_X) = S^{-1}A$ (20.2.12) and $\Gamma(X, \mathscr{M}_X(\mathscr{L})) = S^{-1}P$; now S is the set of elements not belonging to any of the ideals associated to A, hence $S^{-1}A$ is a semi-local ring whose maximal ideals come from the maximal elements of $\operatorname{Ass}(A)$, and $S^{-1}P$ is a projective $S^{-1}A$-module of rank 1, hence a *free* $S^{-1}A$-module of rank 1 (Bourbaki, *Alg. comm.*, chap. II, § 5, nº 3, prop. 5); an element forming a base of this $S^{-1}A$-module is thus (20.1.8) a regular meromorphic section of $\mathscr{L}$ over X. □

Corollary (21.3.5). — *If X is a Noetherian prescheme such that there exists an ample invertible $\mathscr{O}_X$-Module (**II**, 4.5.3) (for example, a quasi-projective prescheme over the spectrum of a Noetherian ring (**II**, 5.3.1 and 4.6.6)), the canonical homomorphism $l : \mathrm{Div}(X) \to \mathrm{Pic}(X)$ is surjective.*

Proof. Indeed (**II**, 4.5.4), there then exists an affine open neighbourhood of the finite set $\mathrm{Ass}(\mathscr{O}_X)$. □

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Remark (21.3.6). — Recall ($\mathbf{0_I}$, 5.4.7) that there is a canonical isomorphism $\pi : \mathrm{H}^1(X, \mathscr{O}_X^*) \xrightarrow{\sim} \mathrm{Pic}(X)$ defined as follows. We start from a 1-cocycle $(c_{\alpha\beta})$ with values in $\mathscr{O}_X^*$ corresponding to a covering (U_α) of X, $c_{\alpha\beta}$ being an element of $\Gamma(U_\alpha \cap U_\beta, \mathscr{O}_X^*)$, and we associate to it the class of the invertible $\mathscr{O}_X$-Module obtained by gluing the $\mathscr{O}_{U_\alpha}$ along the isomorphisms $\mathscr{O}_{U_\alpha}|(U_\alpha \cap U_\beta) \xrightarrow{\sim} \mathscr{O}_{U_\beta}|(U_\alpha \cap U_\beta)$ defined by multiplication by $c_{\alpha\beta}$. On the other hand, we deduce from the exact sequence of sheaves of commutative groups

$$1 \longrightarrow \mathscr{O}_X^* \longrightarrow \mathscr{M}_X^* \longrightarrow \mathscr{D}iv_X \longrightarrow 0$$

the boundary homomorphism of the exact cohomology sequence

$$\partial : \mathrm{Div}(X) \longrightarrow \mathrm{H}^1(X, \mathscr{O}_X^*). \tag{21.3.6.1}$$

Let us show that the composite homomorphism

$$\mathrm{Div}(X) \xrightarrow{\partial} \mathrm{H}^1(X, \mathscr{O}_X^*) \xrightarrow{\pi} \mathrm{Pic}(X)$$

is the homomorphism l defined in (21.3.2.2). Indeed, we must start from a covering (U_α) of X and a divisor D such that $D|U_\alpha = \mathrm{div}_{U_\alpha}(g_\alpha)$, where g_α is a regular meromorphic function over U_α; $\partial(D)$ is the cohomology class of the cocycle $(c_{\alpha\beta})$, where $c_{\alpha\beta} = g_{\alpha|\beta} g_{\beta|\alpha}^{-1}$, $g_{\alpha|\beta}$ denoting the restriction of g_α to $U_\alpha \cap U_\beta$. It is clear that the image by π of this cohomology class is the class of the invertible fractionary Ideal $\mathscr{L}$ such that for every α, $\mathscr{L}|U_\alpha = \mathscr{O}_{U_\alpha} . g_\alpha^{-1}$, which is none other by definition than $\mathscr{O}_X(D)$ (21.2.8).

21.4. Inverse images of divisors

(21.4.1). Let $f : X' \to X$ be a morphism of ringed spaces; we propose to give conditions allowing us to associate to a divisor D on X a divisor D' on X', the *inverse image* of D by f. Let us first note that for every section $t \in \Gamma(X, \mathscr{O}_X^*)$, the image of t by the canonical homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(X', \mathscr{O}_{X'})$ is still invertible, in other words belongs to $\Gamma(X', \mathscr{O}_{X'}^*)$. Consider then D as given by the class of equivalence of a pair $(\mathscr{L}, s)$, where $\mathscr{L}$ is an invertible $\mathscr{O}_X$-Module and s a regular meromorphic section of $\mathscr{L}$ over X (21.2.11). Form the invertible $\mathscr{O}_{X'}$-Module $f^*(\mathscr{L}) = \mathscr{L}'$; to say that the inverse image $s \circ f$ of s by f exists (20.1.11) and is a regular meromorphic section of $\mathscr{L}'$ over X' amounts to saying that the inverse images by f of s and $s^{\otimes(-1)}$ exist, in other words that $s \in \Gamma(X, \mathscr{M}_f(\mathscr{L}))$ and $s^{\otimes(-1)} \in \Gamma(X, \mathscr{M}_f(\mathscr{L}^{-1}))$; the remark made above shows that if $(\mathscr{L}_1, s_1)$ is a pair equivalent to $(\mathscr{L}, s)$ in the sense of (21.2.10), the inverse image $s_1 \circ f$ exists and is a regular meromorphic section of $\mathscr{L}_1' = f^*(\mathscr{L}_1)$ over X', and the pairs $(\mathscr{L}', s \circ f)$ and $(\mathscr{L}_1', s_1 \circ f)$ are equivalent. We can therefore pose the following definition:

Definition (21.4.2). — Given a morphism $f : X' \to X$ of ringed spaces, we say that the *inverse image by f* of a divisor D on X exists if $s_D \in \Gamma(X, \mathscr{M}_f(\mathscr{O}_X(D)))$ and $s_{-D} \in \Gamma(X, \mathscr{M}_f(\mathscr{O}_X(-D)))$ (cf. (20.1.11)). We call then the *inverse image* of D by f and denote $f^*(D)$ the divisor on X' which corresponds canonically (21.2.11) to the class of the pair $(f^*(\mathscr{O}_X(D)), s_D \circ f)$.

It follows immediately from this definition that if D and D' have inverse images by f, so does $-D$ and $D + D'$ (taking into account (21.2.9.4)) and we have $f^*(D + D') = f^*(D) + f^*(D')$. In **IV-4 | 266**
other words, the set $\mathrm{Div}^f(X)$ of divisors on X whose inverse image by f exists is a *subgroup* of

$\mathrm{Div}(X)$, and the map $D \mapsto f^*(D)$ is an *increasing homomorphism* of the ordered subgroup $\mathrm{Div}^f(X)$ to the ordered group $\mathrm{Div}(X')$, making commutative the diagram

(21.4.2.1)
$$\begin{tikzcd} \mathrm{Div}^f(X) \arrow[r, "l_X"] \arrow[d, "f^*"'] & \mathrm{Pic}(X) \arrow[d, "\mathrm{Pic}(f)"] \\ \mathrm{Div}(X') \arrow[r, "l_{X'}"] & \mathrm{Pic}(X') \end{tikzcd}$$

(21.4.3). The definition (21.4.2) also shows immediately that, for $f^*(D)$ to exist, it is necessary and sufficient that, for every open U of X, the inverse image of $D|U$ by $f^U : f^{-1}(U) \to U$ (the restriction of f) exist. Now, if $D = \mathrm{div}(g)$, where g is a regular meromorphic function on X, to say that the inverse image of s_D by f exists and is a regular meromorphic section of $f^*(\mathscr{O}_X(D))$ means (21.2.9) that the inverse image of g by f exists and is a regular meromorphic function on X'. We deduce also immediately a second description of $\mathrm{Div}^f(X)$ and of $f^*(D)$: we consider the subsheaf of groups of $\mathscr{M}_X^*$, denoted $\mathscr{M}_f^*$, formed of germs of meromorphic functions on an open of X whose inverse image exists and is regular on the inverse image open (20.1.11). Then if $f = (\psi, \theta)$, the canonical homomorphism (20.1.11) $\psi^*(\mathscr{M}_f) \to \mathscr{M}_{X'}$ gives by restriction a homomorphism of sheaves of groups $\psi^*(\mathscr{M}_f^*) \to \mathscr{M}_{X'}^*$. Setting $\mathscr{D}iv'_X = \mathscr{M}_f^*/\mathscr{O}_X^*$, we have $\mathrm{Div}^f(X) = \Gamma(X, \mathscr{D}iv'_X)$, and the map $D \mapsto f^*(D)$ corresponds to the homomorphism of sheaves of groups $\psi^*(\mathscr{M}_f^*)/\psi^*(\mathscr{O}_X^*) \to \mathscr{M}_{X'}^*/\mathscr{O}_{X'}^* = \mathscr{D}iv_{X'}$, derived from the preceding by passage to quotients.

(21.4.4). It follows also from the preceding definitions that if $f' : X'' \to X'$ is a second morphism of ringed spaces, D a divisor on X such that the inverse images $f^*(D)$ and $f'^*(f^*(D))$ exist, then $(f \circ f')^*(D)$ exists and is equal to $f'^*(f^*(D))$.

Proposition (21.4.5). — *Let $f : X' \to X$ be a morphism of ringed spaces. In each of the three following cases, the inverse image by f of every divisor on X is defined:*

(i) f is flat.
(ii) X and X' are locally Noetherian preschemes and $f(\mathrm{Ass}(\mathscr{O}_{X'})) \subset \mathrm{Ass}(\mathscr{O}_X)$.
(iii) X and X' are preschemes, the set of irreducible components of X is locally finite, X' is reduced and every irreducible component of X' dominates an irreducible component of X.

Proof. It suffices indeed to show that in all three cases we have $\mathscr{M}_f = \mathscr{M}_X$; in case (i), this follows from (20.1.12). In case (iii), we can restrict to the case where $X = \mathrm{Spec}(A)$ and $X' = \mathrm{Spec}(A')$ are affine; if $s \in A$ is regular, it does not belong to any minimal prime ideal of A (20.1.3.1), hence the hypothesis implies that its image in A' does not belong to any minimal prime ideal of A', and is therefore a regular element of A' (20.1.3.1). In case (ii) the meromorphic functions on X' are identified with the pseudo-functions on X' (20.2.11), and the hypothesis, combined with (20.2.13), (iv), ensures that the inverse image by f of every schematically dense open in X is a schematically dense open in X'; we conclude therefore by (20.3.12). □ **IV-4 | 267**

Corollary (21.4.6). — *Let X be a prescheme having one of the following properties:*

(i) X is locally Noetherian.
(ii) X is reduced and the set of its irreducible components is locally finite.

Then, for every $x \in X$, we have a canonical isomorphism

(21.4.6.1)
$$(\mathscr{D}iv_X)_x \xrightarrow{\sim} \mathrm{Div}(\mathscr{O}_{X,x}).$$

Proof. This follows indeed from (20.2.11), (20.3.7), and from the fact that $(\mathscr{O}_X^*)_x$ is identified with the group of invertible elements of the ring $\mathscr{O}_{X,x}$. □

(21.4.7). Let X, X' be two preschemes, $f : X' \to X$ a morphism. If D is a *positive* divisor on X such that the inverse image $f^*(D)$ is defined (21.4.2), then the sub-prescheme $Y(f^*(D))$ of X' is none other than the inverse image $f^{-1}(Y(D))$; this follows also from the definitions (21.4.2) and (21.2.12).

Proposition (21.4.8). — *Let X, Y be two preschemes, $f : X \to Y$ a faithfully flat morphism. Then, if D is a divisor on Y such that $f^*(D) \geqslant 0$ (the existence of $f^*(D)$ resulting from (21.4.5)), we have $D \geqslant 0$. In particular, the map $D \mapsto f^*(D)$ of $\mathrm{Div}(Y)$ to $\mathrm{Div}(X)$ is injective.*

Proof. The question being local on Y, we can restrict to the case where $D = \mathrm{div}(w)$, with $w = uv^{-1}$, u and v being two regular sections of $\mathscr{O}_Y$ over Y. By hypothesis we have $u\mathscr{O}_X \subset v\mathscr{O}_X$, hence, for every $x \in X$, setting $y = f(x)$, we have $u_y\mathscr{O}_{X,x} \subset v_y\mathscr{O}_{X,x}$. We conclude that $u_y\mathscr{O}_{Y,y} \subset v_y\mathscr{O}_{Y,y}$ because $\mathscr{O}_{X,x}$ is a faithfully flat $\mathscr{O}_{Y,y}$-module (Bourbaki, *Alg. comm.*, chap. I, § 3, n$^{\text{o}}$ 5, prop. 10, (ii)); since f is surjective, we have $u\mathscr{O}_Y \subset v\mathscr{O}_Y$, and consequently $D \geqslant 0$. □

The following proposition and its corollaries were pointed out to us by M. Raynaud.

Proposition (21.4.9). — *Let X and Y be two locally Noetherian preschemes, and let $f : X \to Y$ be a faithfully flat morphism. Suppose in addition either that f is quasi-compact or that f is open. A divisor D' on X is of the form $f^*(D)$, with D a divisor on Y, if and only if the following condition holds: for every $y \in Y$ such that $\mathrm{prof}(\mathscr{O}_{Y,y}) \leqslant 1$, put*

$$Y_1 = \mathrm{Spec}(\mathscr{O}_{Y,y}), \qquad X_1 = X \times_Y Y_1, \qquad f_1 = f_{(Y_1)} : X_1 \longrightarrow Y_1;$$

then the inverse image D'_1 of D' on X_1 is of the form $f_1^*(D_1)$ for some $D_1 \in \mathrm{Div}(Y_1)$.

Proof. The necessity of the condition is immediate from the transitivity of inverse images of divisors, all the morphisms in question being flat (21.4.4). For every open subset U of Y, write f_U for the restriction $f^{-1}(U) \to U$ of f. By (21.4.8), the map

$$D_U \longmapsto f_U^*(D_U)$$

from $\mathrm{Div}(U)$ to $\mathrm{Div}(f^{-1}(U))$ is injective. Thus, if U_1, U_2 are two open subsets of Y and

$$D'|f^{-1}(U_1) = f_{U_1}^*(D_1), \qquad D'|f^{-1}(U_2) = f_{U_2}^*(D_2),$$

then necessarily $D_1|U_1 \cap U_2 = D_2|U_1 \cap U_2$. It follows at once that there is a largest open subset $U \subset Y$ for which

$$D'|f^{-1}(U) = f_U^*(D_U) \quad \text{with} \quad D_U \in \mathrm{Div}(U).$$

We must show that $U = Y$. Suppose otherwise, and let y be a maximal point of $Y \setminus U$. We shall find an open neighbourhood V of y in Y such that $D'|f^{-1}(V) = f_V^*(D_V)$ for some $D_V \in \mathrm{Div}(V)$, contradicting the maximality of U.

I) The case where f is quasi-compact. Using the quasi-compactness of f and (20.3.8), by the method of (8.1.2, a), we are reduced to proving—in the notation of the statement, but without initially imposing a bound on $\mathrm{prof}(\mathscr{O}_{Y,y})$—that D'_1 is of the form $f_1^*(D_1)$. If $\mathrm{prof}(\mathscr{O}_{Y,y}) \leqslant 1$, this is precisely the hypothesis. Suppose therefore that $\mathrm{prof}(\mathscr{O}_{Y,y}) \geqslant 2$. Since y is a maximal point of $Y \setminus U$, the open subset $W = Y_1 \cap U$ of Y_1 is $Y_1 \setminus \{y\}$. We may therefore restrict to the case $Y = Y_1$ and $U = Y \setminus \{y\}$.

By hypothesis there is a divisor $D_U \in \mathrm{Div}(U)$ such that $f_U^*(D_U) = D'|f^{-1}(U)$. The divisor D_U is the equivalence class of a pair $(\mathscr{L}_U, s)$, where $\mathscr{L}_U$ is an invertible $\mathscr{O}_U$-Module and s a regular meromorphic section of $\mathscr{L}_U$ over U (21.2.11). Hence, by (21.4.1), $D'|f^{-1}(U)$ is the equivalence class of $(\mathscr{L}'_U, s')$, where

$$\mathscr{L}'_U = f_U^*(\mathscr{L}_U), \qquad s' = s \circ f_U.$$

Let $i : U \to Y$ and $j : f^{-1}(U) \to X$ be the canonical injections, and put

$$\mathscr{L} = i_*(\mathscr{L}_U), \qquad \mathscr{L}' = j_*(\mathscr{L}'_U).$$

Since f is flat, $f^*(\mathscr{L})$ is canonically isomorphic to $\mathscr{L}'$ (2.3.1). Moreover, flatness and the inequality $\mathrm{prof}(\mathscr{O}_{Y,y}) \geqslant 2$ imply $\mathrm{prof}(\mathscr{O}_{X,x}) \geqslant 2$ for every $x \in f^{-1}(y)$ (6.3.1).

The divisor D' is the equivalence class of a pair $(\mathscr{L}'', s'')$, where $\mathscr{L}''$ is an invertible $\mathscr{O}_X$-Module and s'' a regular meromorphic section of $\mathscr{L}''$ over X. Consequently $\mathscr{L}'_U$ is isomorphic to $\mathscr{L}''|f^{-1}(U)$, and the preceding observations together with (5.9.8) and (5.10.5) show that $\mathscr{L}'$ is an invertible $\mathscr{O}_X$-Module isomorphic to $\mathscr{L}''$. Since f is faithfully flat and quasi-compact, it follows that $\mathscr{L}$ is an invertible $\mathscr{O}_Y$-Module (2.5.2), and hence isomorphic to $\mathscr{O}_Y$, because Y is a local scheme.

Furthermore, since $\mathrm{prof}(\mathscr{O}_{Y,y}) \geqslant 1$, the open subset $U = Y \setminus \{y\}$ is schematically dense in Y (20.2.13, (iv)). The regular meromorphic section s of $\mathscr{L}_U$ over U therefore extends to a regular meromorphic section s_0 of $\mathscr{L}$ over Y (20.2.11). The equivalence class of $(\mathscr{L}, s_0)$ is thus a divisor D on Y. Plainly $f^*(D)$ and D' induce the same divisor on $f^{-1}(U) = X \setminus f^{-1}(y)$; and since $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 2$ for every $x \in f^{-1}(y)$, (21.1.8) gives $D' = f^*(D)$. This proves case I.

II) The case where f is open. The problem is to find an open neighbourhood V of y in Y such that $D'|f^{-1}(V) = f_V^*(D_V)$ for some $D_V \in \operatorname{Div}(V)$, so we may first suppose Y affine. Because f is open, the images $f(W)$, with W running through the affine open subsets of X, form an open covering of Y. Since Y is quasi-compact, there is a quasi-compact open subset $T \subset X$ such that $f(T) = Y$, and $f|T$ is quasi-compact (I, 1.1.1). A fortiori, for every quasi-compact open $T_\alpha \supset T$ in X, one has $f(T_\alpha) = Y$ and $f|T_\alpha$ is quasi-compact.

Applying case I to $f|T_\alpha$, we obtain a divisor D_α on Y such that

$$(f|T_\alpha)^*(D_\alpha) = D'|T_\alpha.$$

By transitivity of inverse images of divisors under flat morphisms (21.4.4), one also has $(f|T)^*(D_\alpha) = D'|T$. Hence for any α, β,

$$(f|T)^*(D_\alpha) = (f|T)^*(D_\beta),$$

and (21.4.8) gives $D_\alpha = D_\beta$. If D denotes their common value, then $f^*(D)|T_\alpha = D'|T_\alpha$ for every α, and therefore $f^*(D) = D'$. □

Corollary (21.4.10). — *Assume the general hypotheses of (21.4.9), and suppose in addition that Y is normal and that, for every $y \in Y$ with $\dim(\mathscr{O}_{Y,y}) = 1$, the fibre $X_y = f^{-1}(y)$ has no embedded associated prime cycle. A divisor D' on X is of the form $f^*(D)$, with $D \in \operatorname{Div}(Y)$, if and only if the following two conditions hold:*

1° *for every maximal point y of Y, the inverse image of D' in $\operatorname{Div}(X_y)$ is zero;*
2° *for every $y \in Y$ such that $\dim(\mathscr{O}_{Y,y}) = 1$, there is an integer n_y such that, for every maximal point x of X_y, the inverse image of D' in $\operatorname{Div}(\operatorname{Spec}(\mathscr{O}_{X,x}))$ equals $\operatorname{div}(t_y^{n_y})$, where t_y is the image under $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ of a uniformizer of the discrete valuation ring $\mathscr{O}_{Y,y}$.*

Proof. Apply the criterion (21.4.9). At a maximal point y of Y, $\operatorname{Div}(Y_1) = 0$ because $\mathscr{O}_{Y,y}$ is Artinian; condition 1° is therefore equivalent to the criterion of (21.4.9) at such a point. It remains to show that conditions 1° and 2° likewise imply the criterion at a point $y \in Y$ with $\operatorname{prof}(\mathscr{O}_{Y,y}) = 1$. Since Y is normal, these are precisely the points for which $\dim(\mathscr{O}_{Y,y}) = 1$, equivalently those for which $\mathscr{O}_{Y,y}$ is a discrete valuation ring (5.8.6).

We may thus restrict to $Y = \operatorname{Spec}(\mathscr{O}_{Y,y})$. If t is a uniformizer of $\mathscr{O}_{Y,y}$, the map $n \mapsto \operatorname{div}(t^n)$ is an isomorphism from $\mathbb{Z}$ onto $\operatorname{Div}(Y)$. The condition of (21.4.9) says that D' is of the form $\operatorname{div}(t^n \circ f)$ for some $n \in \mathbb{Z}$, and plainly implies condition 2°. Conversely, suppose 2° holds, as does 1°, and write $n = n_y$. It remains to prove $D' = \operatorname{div}(t^n \circ f)$. If η is the generic point of Y, the germs of these two divisors at every point of X_η are zero by condition 1°. Their germs at every maximal point of X_y are equal by condition 2°. At a nonmaximal point $x \in X_y$, one has $\operatorname{prof}(\mathscr{O}_{X_y,x}) \geqslant 1$, because X_y has no embedded associated prime cycle, and hence $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 2$ by (6.3.1). The desired equality follows from (21.1.8). □

Corollary (21.4.11). — *Assume the general hypotheses of (21.4.9), and suppose in addition that Y is normal and that, for every $y \in Y$ with $\dim(\mathscr{O}_{Y,y}) = 1$, the fibre X_y is integral. A divisor D' is of the form $f^*(D)$, with $D \in \operatorname{Div}(Y)$, if and only if, for every maximal point y of Y, the inverse image of D' in $\operatorname{Div}(X_y)$ is zero.*

Proof. It is enough to check that condition 2° of (21.4.10) is then automatic. If $\dim(\mathscr{O}_{Y,y}) = 1$ and X_y is integral, then at the unique maximal point x of X_y one has $\dim(\mathscr{O}_{X,x}) = 1$ by (6.1.3). Moreover, $\mathscr{O}_{Y,y}$ is regular and $\mathscr{O}_{X_y,x}$ is a field, so $\mathscr{O}_{X,x}$ is regular (6.5.1), hence a discrete valuation ring. If t is a uniformizer of $\mathscr{O}_{Y,y}$, its image t_y in $\mathscr{O}_{X,x}$ is a uniformizer of $\mathscr{O}_{X,x}$ (0, 17.3.3). The inverse image of D' in $\operatorname{Div}(\operatorname{Spec}(\mathscr{O}_{X,x}))$ is therefore of the form $\operatorname{div}(t_y^n)$, which is exactly condition 2° of (21.4.10), since X_y has only one maximal point. □

Remark (21.4.12). — If in (21.4.11) one assumes only that the X_y are reduced, one must add to the condition in the statement that the inverse images of D' in each group $\operatorname{Div}(\operatorname{Spec}(\mathscr{O}_{X,x}))$, with x running over the maximal points of X_y, are all of the form $\operatorname{div}((t'_x)^n)$, where t'_x is a uniformizer of $\mathscr{O}_{X,x}$ and the integer n is the same for all these points.

Corollary (21.4.13). — *Assume the hypotheses of (21.4.11). For every invertible $\mathscr{O}_X$-Module $\mathscr{L}$ such that, for every maximal point η of Y, the inverse image*

$$\mathscr{L}_\eta = \mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_\eta}$$

is isomorphic to $\mathscr{O}_{X_\eta}$, there exist an invertible $\mathscr{O}_Y$-Module $\mathscr{M}$ and an isomorphism $f^*(\mathscr{M}) \simeq \mathscr{L}$.

Proof. It is enough to find a regular meromorphic section s of $\mathscr{L}$ over X such that the divisor D' corresponding to $(\mathscr{L}, s)$ has $D'_x = 0$ at every point x of a fibre X_η over a maximal point η of Y. Applying (21.4.11) to D' then gives the result by (21.4.2).

The points of $\mathrm{Ass}(\mathscr{O}_X)$ lie over the maximal points of Y (3.3.1). Since Y is reduced and locally Noetherian, there is a dense open subset V of Y which is the union of a disjoint family of affine open subsets, each containing exactly one maximal point of Y; and $f^{-1}(V)$ is schematically dense in X (5.10.2). We may therefore restrict to the case in which Y is affine and integral.

For the unique maximal point η of Y, the fibre X_η is then integral, so $\mathrm{Ass}(\mathscr{O}_X)$ consists of a single point z, the generic point of X. Let s_η be the section of $\mathscr{L}_\eta$ over X_η which corresponds, under an isomorphism, to the unit section of $\mathscr{O}_{X_\eta}$. Let U be an affine open neighbourhood in X of a point $x \in X_\eta$, and let s be a section of $\mathscr{L}$ over U with the same germ at x as s_η. Since U is schematically dense in X (20.2.13, (iv)), s identifies with a uniquely determined regular meromorphic section of $\mathscr{L}$ over X. It is clear from (20.2.11) and (20.3.5) that this section induces on X_η a regular meromorphic section of $\mathscr{L}_\eta$ equal to s_η, and hence has the required property. □

21.5. Direct images of divisors

(21.5.1). Let X, X' be two preschemes, $f : X' \to X$ a morphism. In this section, we shall give sufficient conditions for one to associate to a divisor D' on X' a divisor on X, the *direct image* of D' by f. We will restrict ourselves to the case where f is a *finite* morphism (for more general conditions, see the chapter of this Treatise devoted to the theory of intersections).

Lemma (21.5.2). — *Let A be a ring, E an A-module free of finite rank. For an endomorphism u of E to be injective, it is necessary and sufficient that* $\det(u)$ *be a regular element of A.*

Proof. This is proved in Bourbaki, *Alg.*, chap. III, 3[e] éd., § 8, n[o] 2, prop. 3. □

(21.5.3). Suppose now that the morphism $f : X' \to X$ is *finite*, and in addition that f satisfies one of the two following properties:

I) f is a finite morphism *locally free*, in other words (18.2.7) the quasi-coherent $\mathscr{O}_X$-Module of finite type $f_*(\mathscr{O}_{X'})$ is *locally free*.

II) X is a locally Noetherian reduced prescheme, the $\mathscr{R}(X)$-Module $f_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is locally free, and for every section $s' \in \Gamma(f^{-1}(U), \mathscr{O}_{X'})$ (U open in X), $\mathrm{N}_{f_*(\mathscr{O}_{X'})|\mathscr{O}_X}(s')$ is a section of $\mathscr{O}_X$ over U (cf. (**II**, 6.5.1)). (We recall that condition (II) is satisfied for every *finite morphism* f when X is a locally Noetherian normal prescheme (*loc. cit.*).) **IV-4** | 268

We know then (**II**, 6.5.5) that for every invertible $\mathscr{O}_{X'}$-Module $\mathscr{L}'$ we define the *norm* $\mathscr{L} = \mathrm{N}_{X'/X}(\mathscr{L}')$ (which we also write $\mathrm{N}(\mathscr{L}')$), which is an invertible $\mathscr{O}_X$-Module. Moreover, for every open U of X and every *regular* section $s' \in \Gamma(f^{-1}(U), \mathscr{O}_{X'})$, the norm $\mathrm{N}_{X'/X}(s') = \mathrm{N}_{f_*(\mathscr{O}_{X'})|\mathscr{O}_X}(s')$ (which we also write $\mathrm{N}(s')$) is a regular element of $\Gamma(U, \mathscr{O}_X)$; we are indeed immediately reduced to the case where $U = X$ is affine, and then the conclusion follows from (21.5.2) in hypothesis (I); on the other hand, in hypothesis (II), the fact that $\mathscr{R}(X)$ is a flat $\mathscr{O}_X$-Module implies that the section $s' \otimes 1$ of $\Gamma(U, f_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X))$ is also regular ($\mathbf{0_I}$, 6.1.2), and the conclusion follows again from (21.5.2) applied to the ring $\Gamma(U, \mathscr{R}(X))$, taking into account the definition of the norm (**II**, 6.5.3). This being so, let u' be a meromorphic section of $\mathscr{L}'$ over X'; the morphism f being affine, every point $x \in X$ admits a neighbourhood U such that $u'|f^{-1}(U)$ is of the form t'/s', where $t' \in \Gamma(f^{-1}(U), \mathscr{L}')$ and s' is a regular section of $\mathscr{O}_{X'}$ over $f^{-1}(U)$; the element $\mathrm{N}(t')/\mathrm{N}(s')$ (where $\mathrm{N}(t')$ is the section of $\mathscr{L}$ defined in (**II**, 6.5.3)) is then a meromorphic section of $\mathscr{L}$ over U, and from what precedes and the multiplicative properties of the norm (**II**, 6.5.3.1) this section depends only on $u'|f^{-1}(U)$ and not on its representation in the form t'/s'; for the same reason, the sections of $\mathscr{L}$

over U thus defined are the restrictions of a single meromorphic section of $\mathscr{L}$ over X, which we call the *norm* of u' and denote $\mathrm{N}_{X'/X}(u')$ (or simply $\mathrm{N}(u')$). The map

(21.5.3.1) $$\mathrm{N}_{X'/X} : \Gamma(X', \mathscr{M}_{X'}(\mathscr{L}')) \longrightarrow \Gamma(X, \mathscr{M}_X(\mathrm{N}_{X'/X}(\mathscr{L}')))$$

extends the norm defined in (**II**, 6.5.5.3); if u' is a *regular* meromorphic section of $\mathscr{L}'$ over X', it follows immediately from what precedes that $\mathrm{N}(u')$ is a regular meromorphic section of $\mathscr{L}$ over X, since (with the same notation) $\mathrm{N}(t')$ is regular if t' is. Finally, if $\mathscr{L}'_1$, $\mathscr{L}'_2$ are two invertible $\mathscr{O}_{X'}$-Modules, s'_1 (resp. s'_2) a regular meromorphic section of $\mathscr{L}'_1$ (resp. $\mathscr{L}'_2$) over X', we have, by virtue of what precedes and of (**II**, 6.5.3.1)

(21.5.3.2) $$\mathrm{N}(s'_1 \otimes s'_2) = \mathrm{N}(s'_1) \otimes \mathrm{N}(s'_2).$$

(21.5.4). Suppose always that f satisfies one of hypotheses I), II) of (21.5.3). If $(\mathscr{L}'_1, s'_1)$, $(\mathscr{L}'_2, s'_2)$ are two pairs each formed of an invertible $\mathscr{O}_{X'}$-Module and a regular meromorphic section of this Module over X', which are moreover *equivalent* in the sense of (21.2.10), then the pairs $(\mathrm{N}(\mathscr{L}'_1), \mathrm{N}(s'_1))$ and $(\mathrm{N}(\mathscr{L}'_2), \mathrm{N}(s'_2))$ are also equivalent, since an isomorphism of invertible $\mathscr{O}_{X'}$-Modules has for norm an isomorphism of their norms (**II**, 6.5.4), and we have already seen above that $\mathrm{N}_{X'/X}$ transforms the sections of $\Gamma(X', \mathscr{O}^*_{X'})$ to sections of $\Gamma(X, \mathscr{O}^*_X)$. We can therefore pose the following definition:

Definition (21.5.5). — Given a finite morphism $f : X' \to X$ of preschemes, satisfying one of conditions I), II) of (21.5.3), we call the *direct image* (or *norm*) of a divisor D' on X' by f, and denote $f_*(D')$ (or $\mathrm{N}_{X'/X}(D')$), the divisor on X which corresponds canonically (21.2.11) to the class of the pair $(\mathrm{N}_{X'/X}(\mathscr{O}_{X'}(D')), \mathrm{N}_{X'/X}(s_{D'}))$.

It follows immediately from this definition, taking into account (21.2.9.4) and (21.5.3.2), that if D'_1, D'_2, D' are divisors on X', we have

(21.5.5.1) $$f_*(D'_1 + D'_2) = f_*(D'_1) + f_*(D'_2)$$

and $D' \geqslant 0$ implies $f_*(D') \geqslant 0$, in other words, $D' \mapsto f_*(D')$ is an *increasing homomorphism* of the ordered group $\mathrm{Div}(X')$ to the ordered group $\mathrm{Div}(X)$. The definition (21.5.5) also shows immediately that for every open U of X, we have $f^U_*(D'|f^{-1}(U)) = f_*(D')|U$ (f^U being the restriction $f^{-1}(U) \to U$ of f), and the homomorphisms f^U_*, for U variable, thus define a *homomorphism of sheaves of ordered groups* **IV-4 | 269**

(21.5.5.2) $$\mathrm{N}_{X'/X} : f_*(\mathscr{D}iv_{X'}) \longrightarrow \mathscr{D}iv_X.$$

Moreover, for every invertible $\mathscr{O}_{X'}$-Module $\mathscr{L}'$ and every regular meromorphic section s' of $\mathscr{L}'$ over $f^{-1}(U)$, we have, from the preceding definitions and (21.1.4)

(21.5.5.3) $$\mathrm{div}_U(\mathrm{N}(s')) = f^U_*(\mathrm{div}_{f^{-1}(U)}(s')).$$

Proposition (21.5.6). — *Let $f : X' \to X$ be a finite morphism locally free and suppose that $f_*(\mathscr{O}_{X'})$ is of constant rank n. Then, for every divisor D on X, $f_*(D)$ is defined and we have*

(21.5.6.1) $$f_*(f^*(D)) = nD.$$

Proof. The first assertion follows from the fact that f is flat (21.4.5), and the second is an immediate consequence of the definitions and of (**II**, 6.5.3.2). □

Proposition (21.5.7). — *Let $f : X' \to X$ be a finite morphism satisfying one of hypotheses I), II) of* (21.5.3), *$f' : X'' \to X'$ a finite morphism locally free of constant rank n. Then $f'' = f \circ f' : X'' \to X$ satisfies the same hypothesis as f, and for every divisor D'' on X'', we have*

(21.5.7.1) $$f''_*(D'') = f_*(f'_*(D'')).$$

Taking into account the definition (21.5.5), it suffices to prove the following result:

Lemma (21.5.7.2). — *Under the hypotheses of* (21.5.7), *we have a functorial isomorphism*

(21.5.7.3) $$\mathrm{N}_{X''/X}(\mathscr{L}'') \xrightarrow{\sim} \mathrm{N}_{X'/X}(\mathrm{N}_{X''/X'}(\mathscr{L}''))$$

in the category of invertible $\mathscr{O}_X$-Modules.

Indeed, taking into account the definition of the norm of a section of $\mathscr{L}''$ (**II**, 6.5.3) and definition (21.5.5), formula (21.5.7.1) follows at once from the lemma. To prove the lemma, it suffices, by the definitions of (**II**, 6.5.2 and 6.5.3), to prove that for every section s of $f''_*(\mathscr{O}_{X''}) = f_*(f'_*(\mathscr{O}_{X''}))$ over an open U of X, we have

$$\mathrm{N}_{f''_*(\mathscr{O}_{X''})|\mathscr{O}_X}(s) = \mathrm{N}_{f_*(\mathscr{O}_{X'})|\mathscr{O}_X}(\mathrm{N}_{f'_*(\mathscr{O}_{X''})|\mathscr{O}_{X'}}(s)) \tag{21.5.7.4}$$

The question is evidently local on X, and we may therefore restrict to the case where $X = \mathrm{Spec}(A)$ is affine; then $X' = \mathrm{Spec}(A')$ and $X'' = \mathrm{Spec}(A'')$, and we may suppose that A'' is a projective A'-module of rank n. When f is locally free, we may suppose that A' is a free A-module of rank m; then A'' is a projective A-module of rank mn, and after restricting X to a suitable open, we may suppose that A'' is a free A-module of rank mn. Formula (21.5.7.4) then follows from the transitivity of the norm (Bourbaki, *Alg.*, chap. VIII, § 12, n° 2, prop. 7). When f satisfies hypothesis II), A is Noetherian and reduced, and if R is its total ring of fractions, $A' \otimes_A R$ is a free R-module of rank m; hence $A'' \otimes_A R = A'' \otimes_{A'} (A' \otimes_A R)$ is a projective R-module of rank mn, and since R is a semi-local ring, this module is free. The proposition then follows again from the transitivity of norms. **IV-4 | 270**

Proposition (21.5.8). — *Let $f : X' \to X$ be a finite morphism, $g : Y \to X$ a morphism; set $Y' = X' \times_X Y$, $f' = f_{(Y)} : Y' \to Y$, $g' = g_{(X)} : Y' \to X'$. Suppose satisfying one of the following conditions:*

(i) f is locally free and g is flat.

(ii) f is locally free, X and Y are locally Noetherian, $g(\mathrm{Ass}(\mathscr{O}_Y)) \subset \mathrm{Ass}(\mathscr{O}_X)$ and $g'(\mathrm{Ass}(\mathscr{O}_{Y'})) \subset \mathrm{Ass}(\mathscr{O}_{X'})$.

(iii) f satisfies hypothesis II) *of (21.5.3), Y is locally Noetherian, Y and Y' are reduced and every irreducible component of Y (resp. Y') dominates an irreducible component of X (resp. X').*

Then, for every divisor D' on X', $g'^*(D')$ is defined, $g^*(f_*(D'))$ is defined and we have

$$g^*(f_*(D')) = f'_*(g'^*(D')). \tag{21.5.8.1}$$

Proof. Indeed, in all cases, it follows from (**II**, 6.5.8) that there is a functorial isomorphism

$$g^*(\mathrm{N}_{X'/X}(\mathscr{L}')) \xrightarrow{\sim} \mathrm{N}_{Y'/Y}(g'^*(\mathscr{L}'))$$

in the category of invertible $\mathscr{O}_Y$-Modules; moreover (**II**, 6.5.4), if s' is a section of $\mathscr{O}_{X'}$ over $f^{-1}(U)$, and s'' the corresponding section of $\mathscr{O}_{Y'}$ over $g'^{-1}(f^{-1}(U))$ (where U is open in X), then $\mathrm{N}_{Y'/Y}(s'')$ is the section of $\mathscr{O}_Y$ over $g^{-1}(U)$ corresponding to the section $\mathrm{N}_{X'/X}(s')$ of $\mathscr{O}_X$ over U. Formula (21.5.8.1) therefore follows from the definitions once it is proved that $g'^*(D')$ and $g^*(D)$ are defined for arbitrary divisors D' on X' and D on X. For D, this follows from the hypotheses and (21.4.5). For D', g' is flat in case (i), while in cases (ii) and (iii) the corresponding hypotheses of (21.4.5) hold; thus $g'^*(D')$ is defined in every case. □

21.6. 1-codimensional cycle associated to a divisor **IV-4 | 270**

(21.6.1). Let X be a locally Noetherian prescheme, and let $\mathfrak{I}(X)$ be the set of *irreducible closed subsets* of X (which is in bijective correspondence with X by the map $x \mapsto \overline{\{x\}}$). In the product group $\mathbf{Z}^X$, consider the subgroup $\mathfrak{Z}(X)$ of elements $(n_x)_{x\in X}$ such that the set of $\overline{\{x\}} \in \mathfrak{I}(X)$ for which $n_x \neq 0$ (or, equivalently, the set of such $x \in X$) is locally finite. It is clear that $\mathfrak{Z}(X)$ is a subgroup of $\mathbf{Z}^X$ containing the direct sum $\mathbf{Z}^{(X)}$ (the free group with basis $\mathfrak{I}(X)$), and equal to it when X is Noetherian. The elements of $\mathfrak{Z}(X)$ are called *cycles* on X, and those of $\mathfrak{I}(X)$ *prime cycles* (they do not in general form a basis of $\mathfrak{Z}(X)$ when X is not Noetherian). We always regard $\mathfrak{Z}(X)$ as ordered by the order induced from the product order on $\mathbf{Z}^X$, and denote by $\mathfrak{Z}^+(X)$ the set of cycles $\geqslant 0$.

For every cycle $Z \in \mathfrak{Z}(X)$, equal to $(n_x)_{x\in X}$, we write, by abuse of language, **IV-4 | 271**

$$Z = \sum_{x\in X} n_x \cdot \overline{\{x\}}.$$

We call *multiplicity of Z at the point x* and denote by $\mathrm{mult}_x(Z)$ the integer n_x (positive or negative). To say that $Z \geqslant 0$ means that $\mathrm{mult}_x(Z) \geqslant 0$ for all $x \in X$. We call the *support* of Z and denote by $\mathrm{Supp}(Z)$ the union of the $\overline{\{x\}}$ such that $\mathrm{mult}_x(Z) \neq 0$; by definition of $\mathfrak{Z}(X)$, this is a closed subset of X, as the union of a locally finite family of closed subsets. We call the *dimension* (resp. *codimension*) of Z and denote by $\dim(Z)$ (resp. $\mathrm{codim}(Z)$) the dimension (resp. codimension in X) of $\mathrm{Supp}(Z)$.

(21.6.2). We say that a closed subspace Y of X is *purely of codimension p* (in X) if *each* of its irreducible components is of codimension p in X. We say that a cycle is *purely of codimension p*, or (by abuse of language) is a *p-codimensional cycle*, if its support is purely of codimension p. We denote by $X^{(p)}$ the set of $x \in X$ such that $\operatorname{codim}(\overline{\{x\}}, X) = p$, or, what amounts to the same (5.1.2), $\dim(\mathscr{O}_{X,x}) = p$: the cycles purely of codimension p form a subgroup $\mathfrak{Z}^p(X)$ of $\mathfrak{Z}(X)$, isomorphic to the sub-group of $\mathbf{Z}^{X^{(p)}}$ consisting of the $(n_x)_{x\in X^{(p)}}$ such that the set of $\overline{\{x\}}$ (or the set of x) for which $n_x \neq 0$ is locally finite; this sub-group contains the free group $\mathbf{Z}^{(X^{(p)})}$ and is identical to it when X is Noetherian. We denote by $\mathfrak{Z}^{p+}(X)$ the set of elements $\geqslant 0$ of $\mathfrak{Z}^p(X)$. It is clear that the ordered group $\mathfrak{Z}(X)$ contains the direct sum of the ordered sub-groups $\mathfrak{Z}^p(X)$, and is identical to this direct sum when X is Noetherian; in the latter case, we consider $\mathfrak{Z}(X)$ as *graded* by the $\mathfrak{Z}^p(X)$ for $p \in \mathbf{N}$.

(21.6.3). Let $Z = \sum_{x\in X} n_x.\overline{\{x\}}$ be a cycle on X, U an open subset of X; we call the *restriction* of Z to U and denote by $Z|U$ the cycle $\sum_{x\in U} n_x.(U \cap \overline{\{x\}})$ on U; we have $\operatorname{Supp}(Z|U) = \operatorname{Supp}(Z) \cap U$. It is clear that $Z \mapsto Z|U$ is a homomorphism of ordered groups from $\mathfrak{Z}(X)$ to $\mathfrak{Z}(U)$ (resp. from $\mathfrak{Z}^p(X)$ to $\mathfrak{Z}^p(U)$), and that if V is an open subset contained in U, we have $Z|V = (Z|U)|V$; the map $U \mapsto \mathfrak{Z}(U)$ (resp. $U \mapsto \mathfrak{Z}^p(U)$) is therefore a *presheaf* on X of ordered commutative groups. In fact, this presheaf is a *sheaf*, as a direct sum of sheaves $(i_x)_*(\mathbf{Z}_{\overline{\{x\}}})$, where x ranges over X (resp. $X^{(p)}$) and for each $x \in X$, i_x is the canonical injection $\overline{\{x\}} \to X$ and $\mathbf{Z}_{\overline{\{x\}}}$ is the simple sheaf associated to the constant sheaf $\mathbf{Z}$ on the space $\overline{\{x\}}$: this follows at once from the description of the sections of a direct sum of sheaves of commutative groups (G, II, 2.7). We denote this sheaf by $\mathscr{Z}_X$ (resp. $\mathscr{Z}_X^p$). We denote by $\mathscr{Z}_X^+$ (resp. $\mathscr{Z}_X^{p+}$) the sub-sheaf of monoids $U \mapsto \mathfrak{Z}^+(U)$ (resp. $U \mapsto \mathfrak{Z}^{p+}(U)$) of $\mathscr{Z}_X$ (resp. $\mathscr{Z}_X^p$). The **IV-4** | 272
sheaf $\mathscr{Z}_X$ is evidently a *direct sum* of sheaves $\mathscr{Z}_X^p$.

Let us note finally that the sheaves $\mathscr{Z}_X^p$ (and therefore also $\mathscr{Z}_X$) are *flasque*. Indeed, suppose given a section Z of $\mathscr{Z}_X^p$ over an open U, so that the set S of $x \in U$ such that $\operatorname{mult}_x(Z) \neq 0$ is locally finite *in* U; this set is also locally finite *in* X, since for all $z \in X$ and every open Noetherian neighbourhood V of z, $U \cap V$ is Noetherian, and thus contains only a finite number of points of S. We thus define a section Z' of $\mathscr{Z}_X^p$ over X by setting $Z' = \sum_{x\in U} \operatorname{mult}_x(Z).\overline{\{x\}}$ and since the closure of x in U is $\overline{\{x\}} \cap U$, we have $Z'|U = Z$, whence our assertion.

(21.6.4). We now propose to define a canonical homomorphism

$$c : \mathscr{D}iv_X \longrightarrow \mathscr{Z}_X^1 \tag{21.6.4.1}$$

of sheaves of commutative groups. It clearly suffices to define a homomorphism of $\mathscr{M}_X^*$ to $\mathscr{Z}_X^1$, which vanishes on $\mathscr{O}_X^*$ (21.1.2); or, by definition, $\mathscr{M}_X^*$ is the *symmetrised* monoid sheaf of $\mathscr{O}_X \cap \mathscr{M}_X^*$, and a homomorphism $\mathscr{M}_X^* \to \mathscr{Z}_X^1$ is uniquely determined by the data of its restriction $\mathscr{S}(\mathscr{O}_X) \to \mathscr{Z}_X^1$, which is a homomorphism of sheaves of *monoids* subject to the sole condition of being zero on $\mathscr{O}_X^*$; it therefore amounts to the same thing to say that, in order to define c, it suffices to define a homomorphism of *sheaves of monoids*

$$c : \mathscr{D}iv_X^+ \longrightarrow \mathscr{Z}_X^{1+}. \tag{21.6.4.2}$$

(21.6.5). Now, we have seen that every positive divisor D on X gives rise to the closed sub-prescheme Y(D) of X, defined by the Ideal $\mathscr{J}_X(D) \subset \mathscr{O}_X$, and which is regularly immersed and of codimension 1. At each of the maximal points x of Y(D), we have ((19.1.4) and (5.1.2)) $\operatorname{codim}(\overline{\{x\}}, X) = \dim(\mathscr{O}_{X,x}) = 1$, in other words $x \in X^{(1)}$; moreover the set of these maximal points is locally finite in X (3.1.6), and $\mathscr{O}_{Y(D),x}$ is an *Artinian* ring. At every point $x \in X^{(1)}$ which is not a maximal point of Y(D), we necessarily have $x \notin Y(D)$, so $\mathscr{O}_{Y(D),x} = 0$. Set

$$\operatorname{cyc}(D) = \sum_{x\in X^{(1)}} \operatorname{length}(\mathscr{O}_{Y(D),x}).\overline{\{x\}} \tag{21.6.5.1}$$

which is thus an element of $\mathfrak{Z}^1(X)$.

Proposition (21.6.6). — *The map* $D \mapsto \operatorname{cyc}(D)$ *is a homomorphism of the monoid* $\operatorname{Div}^+(X)$ *to* $\mathfrak{Z}^{1+}(X)$.

Proof. It amounts to showing that for two positive divisors D, D′, we have

$$\operatorname{cyc}(D + D') = \operatorname{cyc}(D) + \operatorname{cyc}(D').$$

Now, by (21.2.5) $\mathscr{J}_X(D+D') = \mathscr{J}_X(D)\mathscr{J}_X(D')$; it thus amounts to showing that if $x \in X^{(1)}$, setting $A = \mathscr{O}_{X,x}$, and if t and t' are two regular elements of A, then $\mathrm{length}(A/tt'A) = \mathrm{length}(A/tA) + \mathrm{length}(A/t'A)$; but since t is regular, $tA/tt'A$ is isomorphic to $A/t'A$, whence the proposition at once. □

It follows at once from the definitions that for every open $U \subset X$, we have

$$Y(D|U) = Y(D) \cap U,$$

whence $\mathrm{cyc}(D|U) = \mathrm{cyc}(D)|U$, and the homomorphisms $\mathrm{cyc} : \mathrm{Div}^+(U) \to \mathfrak{Z}^1(U)$ define a homomorphism of sheaves of monoids (21.6.4.2), whence the sought-after homomorphism (21.6.4.1) of sheaves of groups. IV-4 | 273

We have $\mathrm{Supp}(\mathrm{cyc}(D)) \subset \mathrm{Supp}(D)$ for every divisor D and

(21.6.6.2).

$$\mathrm{Supp}(\mathrm{cyc}(D)) = \mathrm{Supp}(D) \quad \text{when} \quad D \geqslant 0;$$

we have indeed already seen the second relation (21.2.12); when D is arbitrary, the relation $D_x = 0$ implies the existence of an open neighbourhood U of x such that $D|U = 0$, whence $\mathrm{cyc}(D)|U = 0$, which proves our assertion.

(21.6.7). The homomorphism (21.6.4.1) gives, by passage to groups of sections over X, a *homomorphism of ordered groups* $\mathrm{Div}(X) \to \mathfrak{Z}^1(X)$, which we will also denote $D \mapsto \mathrm{cyc}(D)$; the number $\mathrm{mult}_x(\mathrm{cyc}(D))$ for $x \in X^{(1)}$ is also denoted $\mathrm{mult}_x(D)$ or $\mathrm{mult}_{\overline{\{x\}}}(D)$ and is called the *multiplicity of the divisor* D *at the point* x, or the *multiplicity of the prime cycle* $\overline{\{x\}}$ *in* D; it is a positive or negative integer, equal as we have seen to $\mathrm{length}(\mathscr{O}_{Y(D),x})$ when D is a *positive* divisor; we thus have by definition

$$\mathrm{cyc}(D) = \sum_{x \in X^{(1)}} \mathrm{mult}_x(D).\overline{\{x\}}, \tag{21.6.7.1}$$

and by virtue of the fact that $D \mapsto \mathrm{cyc}(D)$ is a homomorphism, we have

$$\mathrm{mult}_x(-D) = -\mathrm{mult}_x(D), \qquad \mathrm{mult}_x(D+D') = \mathrm{mult}_x(D) + \mathrm{mult}_x(D') \tag{21.6.7.2}$$

for any two divisors D, D′.

For every regular meromorphic function f on X, we set in particular, for $x \in X^{(1)}$

$$o_x(f) = \mathrm{mult}_x(\mathrm{div}(f)) \tag{21.6.7.3}$$

and we say that $o_x(f)$ (a positive or negative integer) is the *order of f at the point* $x \in X^{(1)}$. If $f_x \in \mathscr{O}_{X,x}$ (a regular element of this local ring by hypothesis), then we have

$$o_x(f) = \mathrm{length}(\mathscr{O}_{X,x}/f_x\mathscr{O}_{X,x}); \tag{21.6.7.4}$$

for two regular meromorphic functions f, f' on X, we have

$$o_x(ff') = o_x(f) + o_x(f'), \qquad o_x(f^{-1}) = -o_x(f) \tag{21.6.7.5}$$

for all $x \in X^{(1)}$. The 1-codimensional cycles

$$Z^+(f) = \sum_{x \in X^{(1)}} (o_x(f))^+.\overline{\{x\}}, \qquad Z^-(f) = \sum_{x \in X^{(1)}} (o_x(f))^-.\overline{\{x\}}$$

are called respectively the *cycle of zeros* and the *cycle of poles* (or *polar cycle*) of f, and we evidently have $\mathrm{cyc}(\mathrm{div}(f)) = Z^+(f) - Z^-(f)$. The 1-codimensional cycles of the form $\mathrm{cyc}(\mathrm{div}(f))$ are called *principal* (or *linearly equivalent to zero*) and form a subgroup $\mathfrak{Z}^1_{\mathrm{princ}}(X)$ of $\mathfrak{Z}^1(X)$. The sections over X of the image $c(\mathscr{D}iv_X) \subset \mathscr{Z}^1_X$ are called *locally principal cycles*; such a cycle Z is therefore characterised by the fact that, for all $y \in X$, there is an open neighbourhood U of y in X and a regular meromorphic function f on U such that $Z|U = \sum_{x \in U \cap X^{(1)}} o_x(f)(\overline{\{x\}} \cap U)$. If $Z = \sum_{x \in X^{(1)}} n_x.\overline{\{x\}}$, setting $T_y = \mathrm{Spec}(\mathscr{O}_{X,y})$ and $Z_y = \sum_{x \in X^{(1)} \cap T_y} n_x(\overline{\{x\}} \cap T_y)$, Z_y is *principal*, otherwise said there exists a regular meromorphic function g on T_y such that $Z_y = \mathrm{cyc}(\mathrm{div}(g))$. This follows at once from the definition above and from (20.3.7) which establishes a bijective correspondence between the regular meromorphic functions on T_y and the germs of regular meromorphic functions on the open neighbourhoods of y in X when X is locally Noetherian. We express this further by saying IV-4 | 274 that Z_y is principal in saying that Z is *principal at the point* y. The set of points where Z is principal is evidently *open*, by virtue of what precedes.

We deduce from the canonical homomorphism $\mathrm{cyc} : \mathrm{Div}(X) \to \mathfrak{Z}^1(X)$ a homomorphism $\mathrm{Div}(X)/\mathrm{Div.princ}(X) \to \mathfrak{Z}^1(X)/\mathfrak{Z}^1_{\mathrm{princ}}(X)$, by virtue of the definition of $\mathfrak{Z}^1_{\mathrm{princ}}(X)$. We say that the group $\mathfrak{Z}^1(X)/\mathfrak{Z}^1_{\mathrm{princ}}(X)$ is the *group of classes of* 1-*codimensional cycles* of X and we denote it $\mathfrak{Cl}(X)$. Two elements of $\mathfrak{Z}^1(X)$ having the same image in $\mathfrak{Cl}(X)$ are called *linearly equivalent*.

(21.6.8). Let us consider in particular the case where $X = \mathrm{Spec}(A)$, where A is a *Noetherian integral and integrally closed* ring. Then $X^{(1)}$ is the set of *prime ideals of height* 1 of A, and $\mathfrak{Z}^1(X)$ is identified with the *group of divisors* (in the sense of N. Bourbaki) of the Krull ring A (Bourbaki, *Alg. comm.*, chap. VII, § 1, n° 3, cor. du th. 2 et n° 6, th. 3).

On the other hand, the regular meromorphic functions on X are identified with the nonzero elements of the field of fractions K of A, and the map $f \mapsto \mathrm{cyc}(\mathrm{div}(f))$ from M(X) to $\mathfrak{Z}^1(X)$ is identified with the map denoted $f \mapsto \mathrm{div}(f)$ in Bourbaki (*loc. cit.*, § 1, n° 1); $\mathfrak{Z}^1_{\mathrm{princ}}(X)$ is identified with the *group of principal divisors* of A in the sense of Bourbaki, and $\mathfrak{Cl}(X)$ with the *group of classes of divisors* of A in the sense of Bourbaki (*loc. cit.*, § 1, n° 2 et n° 10).

Theorem (21.6.9). — *Let* X *be a locally Noetherian normal prescheme.*

(i) The canonical homomorphism $\mathrm{cyc} : \mathrm{Div}(X) \to \mathfrak{Z}^1(X)$ *is injective and its image is formed of locally principal cycles.*

(ii) The following conditions are equivalent:

a) *The homomorphism* $\mathrm{cyc} : \mathrm{Div}(X) \to \mathfrak{Z}^1(X)$ *is bijective.*

b) *Every* 1*-codimensional cycle is locally principal.*

c) *For all* $x \in X$*, the local ring* $\mathscr{O}_{X,x}$ *is factorial.*

(We then say that X *is a* locally factorial *prescheme.)*

Proof. (i) It suffices to show that

$$\mathrm{cyc}^{-1}(\mathfrak{Z}^{1+}(X)) = \mathrm{Div}^+(X), \tag{21.6.9.1}$$

since we have $\mathrm{Div}^+(X) \cap (-\mathrm{Div}^+(X)) = 0$ and $\mathfrak{Z}^{1+}(X) \cap (-\mathfrak{Z}^{1+}(X)) = 0$. We are thus reduced to proving that if D is a divisor on X such that $\mathrm{mult}_x(D) \geqslant 0$ for all $x \in X^{(1)}$, then $D \geqslant 0$. Now, for all $x \notin X^{(1)}$, the local ring $\mathscr{O}_{X,x}$ is integral and integrally closed and of dimension 0 or $\geqslant 2$, therefore on the one hand $\mathrm{prof}(\mathscr{O}_{X,x}) = 0$ or $\mathrm{prof}(\mathscr{O}_{X,x}) \geqslant 2$ (**0**, 16.5.1). **IV-4** | 275
On the other hand, for points $x \in X^{(1)}$, the ring $\mathscr{O}_{X,x}$ is a discrete valuation ring ((**II**, 7.1.6)), whence $\mathrm{prof}(\mathscr{O}_{X,x}) = 1$ (**0**, 16.5.1); the only points of X such that $\mathrm{prof}(\mathscr{O}_{X,x}) = 1$ are thus the points of $X^{(1)}$, and the conclusion follows from (21.1.8).

(ii) The equivalence of *a)* and *b)* is clear by virtue of (i). From the characterisation of locally principal cycles given in (21.6.7), and the relation between 1-codimensional cycles on Spec(A) and divisors (in the sense of Bourbaki) of the ring A when A is a Noetherian integrally closed ring (21.6.8), condition *b)* is equivalent to saying that for all $x \in X$, every divisor of the ring $\mathscr{O}_{X,x}$ is principal, in other words that the ring $\mathscr{O}_{X,x}$ is factorial (Bourbaki, *Alg. comm.*, chap. VII, § 3, n° 1), whence the equivalence of *b)* and *c)*.

□

(21.6.9.2). When A is a Noetherian *factorial* ring, it is clear that $X = \mathrm{Spec}(A)$ is locally factorial (Bourbaki, *Alg. comm.*, chap. VII, § 3, n° 4, prop. 3). In this case, if one writes an element $f \neq 0$ of the field of fractions K of A in the form r/s, where r and s are two *coprime* elements of A, whose divisors are well-determined (*loc. cit.*, § 3, n° 3), these divisors are identified respectively with the *divisor of zeros* and the *divisor of poles* of f (21.6.7).

Corollary (21.6.10). — *Let* X *be a locally Noetherian normal prescheme.*

(i) There exists a canonical injective homomorphism

$$\mathrm{Pic}(X) \longrightarrow \mathfrak{Cl}(X). \tag{21.6.10.1}$$

(ii) If X *is locally factorial, the homomorphism* (21.6.10.1) *is bijective, and conversely.*

Proof. We have seen (21.6.7) that the image of $\operatorname{Div.princ}(X)$ by the homomorphism $\mathrm{cyc} : \operatorname{Div}(X) \to \mathfrak{Z}^1(X)$ is $\mathfrak{Z}^1_{\mathrm{princ}}(X)$; we thus deduce from (21.6.9) that the homomorphism $\operatorname{Div}(X)/\operatorname{Div.princ}(X) \to \mathfrak{Z}^1(X)/\mathfrak{Z}^1_{\mathrm{princ}}(X) = \mathfrak{Cl}(X)$ deduced from cyc is injective, and that it is bijective if and only if X is locally factorial. The conclusion follows from the fact that, when X is locally Noetherian and reduced, the canonical homomorphism $\operatorname{Div}(X)/\operatorname{Div.princ}(X) \to \operatorname{Pic}(X)$ (21.3.3.1) is bijective (21.3.4), (ii). □

Corollary (21.6.11). — *Let* X *be a locally Noetherian prescheme which is locally factorial. Then the sheaf* $\mathscr{D}iv_X$ *is flasque, and for every open* U *of* X*, the canonical homomorphism* $\operatorname{Pic}(X) \to \operatorname{Pic}(U)$ *is surjective.*

Proof. Taking into account (21.6.9), (ii), the first assertion is equivalent to saying that the sheaf $\mathscr{Z}^1_X$ is flasque, which was seen above (21.6.3). For every open U of X, the canonical homomorphism $\mathfrak{Z}^1(X) \to \mathfrak{Z}^1(U)$ is thus surjective; as in (21.6.10), the homomorphism $\operatorname{Pic}(X) \to \operatorname{Pic}(U)$ is identified canonically with $\mathfrak{Z}^1(X)/\mathfrak{Z}^1_{\mathrm{princ}}(X) \to \mathfrak{Z}^1(U)/\mathfrak{Z}^1_{\mathrm{princ}}(U)$, it is also surjective. □

Proposition (21.6.12). — *Let* X *be a Noetherian reduced prescheme. Let* $(U_\lambda)_{\lambda\in L}$ *be a decreasing filtered family of opens of* X *satisfying the following conditions:*

1° *If* $Y_\lambda = X - U_\lambda$*, then* $\operatorname{codim}(Y_\lambda, X) \geqslant 2$ *for all* $\lambda \in L$.

2° *For all* $x \in \bigcap_{\lambda\in L} U_\lambda$*, the ring* $\mathscr{O}_{X,x}$ *is factorial.*

Then there exist canonical isomorphisms

$$\varinjlim \operatorname{Div}(U_\lambda) \xrightarrow{\sim} \mathfrak{Z}^1(X), \qquad \varinjlim \operatorname{Pic}(U_\lambda) \xrightarrow{\sim} \mathfrak{Cl}(X) \tag{21.6.12.1}$$

such that, for every open V *of* X*, the diagrams*

$$\begin{tikzcd} \varinjlim \operatorname{Div}(U_\lambda) \arrow[r, "\sim"] \arrow[d] & \mathfrak{Z}^1(X) \arrow[d] \\ \varinjlim \operatorname{Div}(U_\lambda \cap V) \arrow[r, "\sim"] & \mathfrak{Z}^1(V) \end{tikzcd} \qquad \begin{tikzcd} \varinjlim \operatorname{Pic}(U_\lambda) \arrow[r, "\sim"] \arrow[d] & \mathfrak{Cl}(X) \arrow[d] \\ \varinjlim \operatorname{Pic}(U_\lambda \cap V) \arrow[r, "\sim"] & \mathfrak{Cl}(V) \end{tikzcd} \tag{21.6.12.2}$$

are commutative.

Proof. The hypothesis 1° on U_λ implies that for all $\lambda \in L$, we have $X^{(1)} \subset U_\lambda$ (5.1.3.1) and therefore the restriction homomorphism $\mathfrak{Z}^1(X) \to \mathfrak{Z}^1(U_\lambda)$ is bijective and consequently we have an isomorphism $\varinjlim \mathfrak{Z}^1(U_\lambda) \xrightarrow{\sim} \mathfrak{Z}^1(X)$. The canonical homomorphisms $\mathrm{cyc} : \operatorname{Div}(U_\lambda) \to \mathfrak{Z}^1(U_\lambda)$ thus define, by passage to the inductive limit, the first of the canonical homomorphisms (21.6.12.1), and the second is deduced by passage to quotients. Furthermore, it follows from condition 1° that the U_λ are schematically dense in X since X is reduced (11.10.4), and therefore every meromorphic function on X is entirely determined by its restriction to each U_λ; we deduce from this that in the isomorphism $\mathfrak{Z}^1(X) \xrightarrow{\sim} \mathfrak{Z}^1(U_\lambda)$, the image of $\mathfrak{Z}^1_{\mathrm{princ}}(X)$ is $\mathfrak{Z}^1_{\mathrm{princ}}(U_\lambda)$, and therefore we also have a canonical isomorphism $\varinjlim \mathfrak{Cl}(U_\lambda) \xrightarrow{\sim} \mathfrak{Cl}(X)$. The second of the canonical homomorphisms (21.6.12.1) will therefore be an isomorphism if the first is, and it remains to prove this last point, the commutativity of the diagrams (21.6.12.2) being trivial.

Let us first show that the homomorphism $\varinjlim \operatorname{Div}(U_\lambda) \to \mathfrak{Z}^1(X)$ is *injective*. Set $T = \bigcap_\lambda U_\lambda$, and **IV-4 | 276**
note that the U_λ form a *fundamental system of neighbourhoods* of T. Indeed, for every open $V \supset T$, $X - V$ is a closed subspace of X, thus a Noetherian space whose every irreducible closed subspace admits a generic point; as the sets $(X - V) \cap (X - U_\lambda)$ are closed in $X - V$, they form an increasing filtered family and their union is $X - V$; our assertion follows from (**0**, 9.2.4). This being so, it will suffice to show that if $D \in \operatorname{Div}(U_\lambda)$ is such that $\mathrm{cyc}(D) = 0$ in $\mathfrak{Z}^1(U_\lambda)$, then there exists $\mu \geqslant \lambda$ such that $D|U_\mu = 0$. By virtue of what precedes, it will suffice to show that for all $x \in T$, $D_x = 0$; indeed, if this holds, there will be an open neighbourhood $W(x)$ of x in X such that $D|W(x) = 0$, and the union of the $W(x)$ for $x \in T$ contains a U_μ. Taking into account (21.4.6), we are thus reduced to the case where $X = \operatorname{Spec}(\mathscr{O}_{X,x})$ with $x \in T$; but by hypothesis $\mathscr{O}_{X,x}$ is factorial, thus integral and integrally closed, and $\operatorname{Spec}(\mathscr{O}_{X,x})$ is therefore normal; whence the conclusion follows from (21.6.9), (i).

To prove that the homomorphism $\varinjlim \operatorname{Div}(U_\lambda) \to \mathfrak{Z}^1(X)$ is bijective, it will suffice to prove that **IV-4 | 277**
for all $Z \in \mathfrak{Z}^1(X)$ and all $x \in T$, there is a neighbourhood $W(x)$ of x in X and a divisor $D^{(x)}$ on

$W(x)$ such that $\mathrm{cyc}(D^{(x)}) = Z|W(x)$. Indeed, we can then cover the quasi-compact set T by a finite number of the $W(x_i)$, and by the first part of the proof, there will exist an index λ such that in each of the $W(x_i) \cap W(x_j) \cap U_\lambda$, the restrictions of $D^{(x_i)}$ and $D^{(x_j)}$ coincide; since we can moreover assume that U_λ is contained in the union of the $W(x_i)$, we see that the restrictions $D^{(x_i)}|(W(x_i) \cap U_\lambda)$ are the restrictions of a single divisor $D \in \mathrm{Div}(U_\lambda)$ which will be such that $\mathrm{cyc}(D) = Z|U_\lambda$. We are thus reduced again to the case where $X = \mathrm{Spec}(\mathscr{O}_{X,x})$ with $x \in T$; but as $\mathscr{O}_{X,x}$ is factorial, it is Noetherian, integral and integrally closed, and it suffices to apply (21.6.9), (ii). ☐

Corollary (21.6.13). — *Let* A *be a Noetherian local ring which is integral and integrally closed and of dimension* $\geqslant 2$. *Set* $X = \mathrm{Spec}(A)$, *and let* a *be the closed point of* X, $U = X - \{a\}$. *For* A *to be factorial, it is necessary and sufficient that* U *is locally factorial and that* $\mathrm{Pic}(U) = 0$.

Proof. Indeed, to say that A is factorial is equivalent to saying that $\mathfrak{Cl}(X) = 0$ (Bourbaki, *Alg. comm.*, chap. VII, § 1, nº 4, cor. du th. 2 et § 3, th. 1); it therefore suffices to use the existence of the second isomorphism (21.6.12.1), taking the family (U_λ) restricted to the single open U. ☐

Corollary (21.6.14). — *Let* A *be a Noetherian local ring of dimension* $\geqslant 2$, $X = \mathrm{Spec}(A)$, a *the closed point of* X, $U = X - \{a\}$. *The following conditions are equivalent:*

a) A *is factorial.*
b) $\mathrm{Pic}(U) = 0$ *and* $\mathrm{prof}(A) \geqslant 2$ *(conditions which we will express later* (21.13) *by saying that the ring* A *is* parafactorial*), and* U *is locally factorial.*

Proof. It is clear that if A is factorial, it is normal, hence $\mathrm{prof}(A) \geqslant 2$ since $\dim(A) \geqslant 2$ (**0**, 16.5.1), and (21.6.13) shows that *a)* implies *b)*. Conversely, if *b)* is satisfied, it suffices to see that A is integrally closed to conclude by (21.6.13) that *b)* implies *a)*. By virtue of the Serre criterion (5.8.6), it suffices to verify for X the conditions (R_1) and (S_2). Now, U being locally factorial satisfies these conditions, and the hypothesis $\mathrm{prof}(A) \geqslant 2$ implies that X also satisfies them. ☐

21.7. Interpretation of positive 1-codimensional cycles in terms of sub-preschemes

(21.7.1). Let X be a locally Noetherian prescheme, $C = \sum_{x \in X^{(1)}} n_x.\overline{\{x\}}$ a positive 1-codimensional cycle (so that $n_x \geqslant 0$ for all $x \in X^{(1)}$, and $n_x = 0$ except at a locally finite set of points). Let Y(C) denote the closed sub-prescheme of X, *closed image* (**I**, 9.5.3) and (9.5.1) of the sub-prescheme *sum* $Y'(C) = \coprod_{x \in X^{(1)}} \mathrm{Spec}(\mathscr{O}_{X,x}/\mathfrak{m}_x^{n_x})$ by the canonical morphism, and let $\mathscr{J}_X(C)$ (or $\mathscr{J}(C)$) denote the Ideal of $\mathscr{O}_X$ defining Y(C).

Proposition (21.7.2). — *Let* X *be a locally Noetherian prescheme satisfying the condition* (R_1) (5.8.2). *For a closed sub-prescheme* Y *of* X *to be of the form* Y(C), *where* C *is a positive* 1*-codimensional cycle, it is necessary and sufficient that it satisfies the two following conditions:*

(i) Y *is purely of codimension* 1. **IV-4** | 278
(ii) Y *satisfies the condition* (S_1), *in other words* (5.7.5) *it has no embedded associated prime cycle.*

We then have

(21.7.2.1)
$$\mathrm{mult}_x(C) = \mathrm{length}\ \mathscr{O}_{Y,x}.$$

The map $C \mapsto Y(C)$ *is a bijection from* $\mathfrak{Z}^{1+}(X)$ *onto the set of closed sub-preschemes of* X *satisfying conditions (i) and (ii).*

Proof. The conditions are necessary (without assuming that X satisfies (R_1)). Indeed, since the question is local on X, we can assume that $X = \mathrm{Spec}(A)$, where A is a Noetherian ring; we then have $Y(C) = \mathrm{Spec}(A/\mathfrak{J})$, where $\mathfrak{J}$ is, by definition (**I**, 9.5.1), the kernel of the canonical homomorphism $A \to \bigoplus_i A_{\mathfrak{p}_i}/\mathfrak{p}_i^{n_i}A_{\mathfrak{p}_i}$, where the $\mathfrak{p}_i$ are the prime ideals corresponding to points $x \in X^{(1)}$ for which $n_x > 0$, and we have set $n_i = n_x$. The inverse image $\mathfrak{q}_i$ of $\mathfrak{p}_i^{n_i}A_{\mathfrak{p}_i}$ in A is a $\mathfrak{p}_i$-primary ideal and we have $\mathfrak{J} = \bigcap_i \mathfrak{q}_i$. Moreover, as the $x \in X^{(1)}$ are such that $\overline{\{x\}}$ is of codimension 1, no point of $X^{(1)}$ can be adherent to another point of $X^{(1)}$, so the $\mathfrak{p}_i$ are minimal primes of $A/\mathfrak{J}$ and the set of $\mathfrak{p}_i$ is equal to $\mathrm{Ass}(A/\mathfrak{J})$, which proves conditions (i) and (ii).

The conditions are sufficient. Let $\mathscr{J}$ denote the Ideal of $\mathscr{O}_X$ defining Y. Hypothesis (ii) implies that $\mathrm{Ass}(\mathscr{O}_X/\mathscr{J})$ is identical to the set F of maximal points of Y, and hypothesis (i) implies that F is contained in $X^{(1)}$; hence (3.2.6) $\mathscr{O}_X/\mathscr{J}$ is identified with a sub-$\mathscr{O}_X$-Module of $\bigoplus_{x\in F}\mathscr{G}(x)$, where, for every $x \in F$, $\mathscr{G}(x)$ is an irredundant $\mathscr{O}_X$-Module such that $\mathrm{Ass}(\mathscr{G}(x)) = \{x\}$. Now, since X satisfies (R_1), for every $x \in F$ the ring $\mathscr{O}_{X,x}$ is a discrete valuation ring, and consequently the nonzero primary ideals of this ring are the powers $\mathfrak{m}_x^{n_x}$ of its maximal ideal. Supposing again that $X = \mathrm{Spec}(A)$, we conclude that $\mathfrak{J} = \widetilde{\mathfrak{J}}$, where $\widetilde{\mathfrak{J}} = \bigcap_i \mathfrak{q}_i$ and the $\mathfrak{q}_i$ are the inverse images in A of the ideals $\mathfrak{p}_i^{n_i} A_{\mathfrak{p}_i}$, the $\mathfrak{p}_i$ corresponding to the points of F. Thus Y is indeed of the form Y(C). □

Corollary (21.7.3). — *Let* X *be a locally Noetherian prescheme such that* X *satisfies* (R_1). *For every positive divisor* D *on* X *such that* X *satisfies* (R_1) *at the maximal points of the closed sub-prescheme* Y(D) (21.2.12), Y(D) *majorises the closed sub-prescheme* Y(cyc(D)) (21.7.1), *and the two have the same underlying space; for these two sub-preschemes to be equal, it is necessary and sufficient that* Y(D) *satisfies the condition* (S_1), *or else, for every* $z \in Y(D)$ *distinct from a maximal point of* Y(D), *that* $\mathrm{prof}(\mathscr{O}_{X,z}) \geqslant 2$ *(a condition always satisfied when* X *is* normal*).*

Proof. The question being local, we may assume that $\mathscr{J}_X(D) = t\mathscr{O}_X$, where t is a regular section of $\mathscr{O}_X$ over X. At every maximal point x of Y(D), which necessarily belongs to $X^{(1)}$, the ring $\mathscr{O}_{X,x}$ is by hypothesis a discrete valuation ring, and therefore $t_x\mathscr{O}_{X,x} = \mathfrak{m}_x^{n_x}$, where $n_x = \mathrm{mult}_x(D)$. We may suppose that $X = \mathrm{Spec}(A)$; if $\mathscr{J}_X(\mathrm{cyc}(D)) = \widetilde{\mathfrak{J}}$, it follows from what precedes and from (21.7.1) that $\mathfrak{J} = tA$ and $\widetilde{\mathfrak{J}}$ have the same *non-embedded primary ideals*, whence $\widetilde{\mathfrak{J}} \subset \mathfrak{J}$, since $\widetilde{\mathfrak{J}}$ is the intersection of these primary ideals (21.7.2). This proves that Y(D) majorises Y(cyc(D)) and that these two sub-preschemes are equal if and only if Y(D) has no associated immersed prime cycle. For every $x \in Y(D)$, there is by hypothesis an open neighbourhood U of x in X and a regular element $t \in \Gamma(U, \mathscr{O}_X)$ such that $\mathscr{O}_{Y(D)}|(U \cap Y(D))$ is the restriction to $Y(D) \cap U$ of $\mathscr{O}_U/t\mathscr{O}_U$. Thus, to say that Y(D) satisfies (S_1) means that at every non-maximal point $z \in Y(D)$ one has $\mathrm{prof}(\mathscr{O}_{X,z}/t_z\mathscr{O}_{X,z}) \geqslant 1$, or equivalently (**0**, 16.4.6) that $\mathrm{prof}(\mathscr{O}_{X,z}) \geqslant 2$. The assertion for normal X follows immediately, since X satisfies (S_2) and, at every non-maximal point z of Y(D), one has $\dim(\mathscr{O}_{X,z}) \geqslant 2$ (**0**, 16.3.4). □ **IV-4** | 279

(21.7.3.1). We see therefore that when X is *normal*, $\mathrm{Div}^+(X)$ is identified canonically with the set of closed sub-preschemes Y of X satisfying conditions (i) and (ii) of (21.7.2) and *regularly immersed* in X.

Proposition (21.7.4). — *Let* X *be a locally Noetherian prescheme which is reduced at each of its isolated points. The following conditions are equivalent:*

- a) *The canonical homomorphism* $c : \mathscr{D}iv_X \to \mathscr{Z}_X^1$ (21.6.4) *is an isomorphism of sheaves of ordered groups.*
- a′) *Every prime* 1*-codimensional cycle on* X *is the image under* cyc *of a positive divisor, and the canonical homomorphism* $c : \mathscr{D}iv_X \to \mathscr{Z}_X^1$ *is injective.*
- a″) *For every integral closed sub-prescheme* Y *of* X *of codimension* 1, *the canonical immersion* $Y \to X$ *is regular.*
- b) X *is normal and the homomorphism* $\mathrm{cyc} : \mathrm{Div}(X) \to \mathfrak{Z}^1(X)$ *is bijective.*
- c) X *is locally factorial.*

Proof. The equivalence of *b)* and *c)*, as well as the fact that *c)* implies *a)*, was proved in (21.6.9). Moreover, *b)* implies *a″)* by (21.7.3.1), and *a)* trivially implies *a′)*. It remains to show that either *a′)* or *a″)* implies *c)*.

Suppose first that *a′)* holds, and let us show first that X is normal. The condition *a′)* is local, so it also holds for every local scheme $\mathrm{Spec}(\mathscr{O}_{X,x})$. Let $x \in X^{(1)}$, so that $A = \mathscr{O}_{X,x}$ is a Noetherian local ring of dimension 1. Applying *a′)* to Spec(A) and to the prime 1-codimensional cycle formed by its closed point, we see that the maximal ideal of A is generated by a single regular element; hence (**0**, 17.1.1) A is a regular ring. Since the localisations $A_{\mathfrak{p}}$ are then also regular (**0**, 17.3.2), X is regular at all of its non-isolated maximal points; since it is reduced, and therefore regular, at its isolated points by hypothesis, X satisfies (R_1). Let us also show that X satisfies (S_1), that is, for every $x \in X$ such that $\dim(\mathscr{O}_{X,x}) \geqslant 1$, one has $\mathfrak{m}_x \notin \mathrm{Ass}(\mathscr{O}_{X,x})$ (**0**, 16.4.6). Indeed, condition *a′)* applied to $X_1 = \mathrm{Spec}(\mathscr{O}_{X,x})$ shows that this scheme carries at least one nonzero divisor, in other words $\mathscr{M}_{X_1}^* \neq \mathscr{O}_{X_1}^*$, and it suffices to apply (21.1.10). These results already show that X is reduced

(5.8.5). Let us next prove that it satisfies (S_2), which will establish that X is normal by Serre's criterion (5.8.6). Argue by contradiction: suppose that the set E of points $x \in X$ at which X does not satisfy (S_2) is nonempty, and choose $x \in E$ for which $\dim(\mathscr{O}_{X,x})$ is as small as possible. Since X satisfies (S_1), one has $\dim(\mathscr{O}_{X,x}) \geqslant 2$. In $X_1 = \operatorname{Spec}(\mathscr{O}_{X,x})$, the open $U = X_1 - \{x\}$ satisfies (S_2); let us show that X_1 also satisfies (S_2), which gives a contradiction. By (5.10.5), it suffices to show that every section f of $\mathscr{O}_{X_1}$ over U extends to a section over X_1. Since X_1 is reduced and U is dense in X_1, U is schematically dense in X_1, and f is therefore the restriction to U of a regular meromorphic section $g \in M(X_1)$. Moreover, since $\dim(\mathscr{O}_{X,x}) \geqslant 2$, one has $\operatorname{cyc}(\operatorname{div}(g)) \geqslant 0$ because $g|U = f$. As cyc is injective, necessarily $\operatorname{div}(g) \geqslant 0$, so (21.6.4) g is a section of $\mathscr{O}_{X_1}$ over X_1, as required.

To show that *a′)* implies *c)*, observe now that *a′)* implies the surjectivity of the canonical homomorphism $\operatorname{cyc} : \operatorname{Div}(X) \to \mathfrak{Z}^1(X)$; since X is normal, it suffices to apply (21.6.9).

Finally, suppose that *a″)* holds. It follows from (21.6.5) that every prime 1-codimensional cycle on X is the image under cyc of a positive divisor; the first part of the preceding argument again proves that X satisfies (R_1) and (S_1). It remains only to prove that X is normal, the rest of the argument then being the same; equivalently, if $x \in X$ is such that $\dim(\mathscr{O}_{X,x}) \geqslant 2$, we must prove that $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 2$. If y is a generalisation of x such that $\overline{\{y\}}$ has codimension 1 in X, let Y be the reduced closed sub-prescheme of X having $\overline{\{y\}}$ as its underlying space. Then Y is integral and **IV-4** | 280
therefore, by hypothesis, regularly immersed in X. It follows that there is a regular element t of $\mathscr{O}_{X,x}$ such that the ideal $t\mathscr{O}_{X,x}$ is the prime ideal defining the inverse image Y_1 of Y in $X_1 = \operatorname{Spec}(\mathscr{O}_{X,x})$. Since $\mathscr{O}_{X,x}/t\mathscr{O}_{X,x}$ is integral, this proves that $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 2$ and completes the proof of (21.7.4). □

Remark (21.7.5). — We cannot replace in (21.7.4) the condition *a′)* by the weaker condition that every prime 1-codimensional cycle on X is the image by cyc of a positive divisor. This is shown by the example where X is the affine scheme defined in (14.1.5) ("the union of a plane and a line that intersects it") *which is not normal*; with the notation introduced in (14.1.5), the prime 1-codimensional cycles of X are those of the plane X_1 and the closed points of the line X_2 distinct from the point common to X_1 and X_2; if t_1, t_2, t_3 are the images of T_1, T_2, T_3 in $A/\mathfrak{a}$, the 1-codimensional prime ideals of X are defined by the monogenic prime ideals in $A/\mathfrak{a}$ (P irreducible in $K[T_1, T_2]$) and $t_3 - a$, with $a \neq 0$ in K; these elements are indeed regular elements in $A/\mathfrak{a}$, so every 1-codimensional cycle is indeed the canonical image of a positive divisor.

Corollary (21.7.6). — *Let* X *be a locally Noetherian prescheme which is reduced at each of its isolated points. The following conditions are equivalent:*

a) *The canonical homomorphism* $c : \mathscr{D}iv_X \to \mathscr{Z}^1_X$ *is an isomorphism of sheaves of ordered groups, and* $\operatorname{Div}(X) = \operatorname{Div.princ}(X)$.

a′) *The canonical homomorphism* $c : \mathscr{D}iv_X \to \mathscr{Z}^1_X$ *is injective, and every prime* 1*-codimensional cycle is the image under* cyc *of a positive principal divisor, in other words is of the form* $\operatorname{cyc}(\operatorname{div}(f))$*, where* f *is a regular section of* $\mathscr{O}_X$ *over* X.

a″) *For every integral closed sub-prescheme* Y *of* X *of codimension* 1*, there exists a regular section* f *of* $\mathscr{O}_X$ *over* X *such that* Y *is defined by the Ideal* $f\mathscr{O}_X$.

b) X *is normal and every prime* 1*-codimensional cycle on* X *is principal.*

c) X *is locally factorial and* $\operatorname{Pic}(X) = 0$.

d) X *is normal, and for every open* U *of* X*, one has* $\operatorname{Pic}(U) = 0$.

Proof. The equivalence of *a)*, *a′)*, *a″)*, *b)* and *c)* follows at once from (21.7.4) and (21.6.11). Moreover, it follows at once from *a″)* that every nonempty open U of X again satisfies the same conditions; in other words, these conditions imply *d)*. It remains to show that *d)* implies *b)*. Let U_λ be the open complements of finite unions of sets of the form $\overline{\{x\}}$, where $\dim(\mathscr{O}_{X,x}) \geqslant 2$. It is clear that the U_λ form a decreasing filtered family and that, for every $x \in \bigcap_\lambda U_\lambda$, one has $\dim(\mathscr{O}_{X,x}) \leqslant 1$, so $\mathscr{O}_{X,x}$ is a principal ring by hypothesis. We may therefore apply (21.6.12) to the family (U_λ); it shows that $\mathfrak{Cl}(X)$ is isomorphic to $\varinjlim \operatorname{Pic}(U_\lambda)$, hence that $\mathfrak{Cl}(X) = 0$ by hypothesis, which proves *b)* by definition. □

Remark (21.7.7). — When $X = \operatorname{Spec}(A)$, where A is a Noetherian integral ring, the conditions of (21.7.6) are equivalent to saying that A is a *factorial* ring.

21.8. Divisors and normalization

Lemma (21.8.1). — *Let $f : X' \to X$ be an integral morphism of preschemes. For every locally free $\mathscr{O}_{X'}$-Module $\mathscr{E}'$ of constant rank n and every $x \in X$, there exists an open neighbourhood U of x in X such that, if we set $U' = f^{-1}(U)$, $\mathscr{E}'|U'$ is isomorphic to $\mathscr{O}_{U'}^n$.*

Proof. The question being local on X, we can restrict to the case where $X = \operatorname{Spec}(A)$ is affine, in which case $X' = \operatorname{Spec}(A')$, where A' is an A-algebra that is integral. Then A' is the inductive limit of its *finite* sub-A-algebras A'_λ. Setting $X'_\lambda = \operatorname{Spec}(A'_\lambda)$ and denoting by $p_\lambda : X' \to X'_\lambda$ the structural morphism, it follows from (8.5.2) and (8.5.5) that there exists an index λ and an $\mathscr{O}_{X'_\lambda}$-Module locally free $\mathscr{E}'_\lambda$ of rank n such that $\mathscr{E}' = p_\lambda^*(\mathscr{E}'_\lambda)$, and it will clearly suffice to prove the lemma for X'_λ and $\mathscr{E}'_\lambda$; in other words, we may assume that the morphism f is *finite*. Set $B = \mathscr{O}_{X,x}$ and let B' be the ring of the affine scheme $X'_0 = X' \times_X \operatorname{Spec}(\mathscr{O}_{X,x})$; as B is a local ring and B' is a B-algebra which is finite, B' is a semi-local ring (Bourbaki, *Alg. comm.*, chap. V, § 2, nº 1, prop. 3); we conclude that the locally free $\mathscr{O}_{X'_0}$-Module $\mathscr{E}'_0 = \mathscr{E}' \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{X'_0}$ is isomorphic to $\mathscr{O}_{X'_0}^n$ (Bourbaki, *Alg. comm.*, chap. II, § 5, nº 3, prop. 5). Considering X'_0 as the projective limit of the preschemes induced by X' on the opens $f^{-1}(U)$, where U ranges over the set of affine open neighbourhoods of x in X, following the method of (8.1.2), *a*)), and applying (8.5.2.5), we obtain the sought conclusion. ∎ **IV-4 | 281**

Corollary (21.8.2). — *Let $f : X' \to X$ be an integral morphism of preschemes. Then:*

(i) We have $R^1 f_*(\mathscr{O}_{X'}^*) = 0$ (**0**, 12.2.1).

(ii) The group $\operatorname{Pic}(X')$ *is canonically isomorphic to* $H^1(X, f_*(\mathscr{O}_{X'}^*))$.

Proof. (i) $R^1 f_*(\mathscr{O}_{X'}^*)$ is the sheaf associated to the presheaf of commutative groups $U \mapsto H^1(f^{-1}(U), \mathscr{O}_{X'}^*)$ on X (*loc. cit.*); the fibre of this sheaf at a point x can therefore be identified (**0**, 5.4.7) with the commutative group $\varinjlim \operatorname{Pic}(f^{-1}(U))$, where U ranges over the open neighbourhoods of x, the transition homomorphisms $\operatorname{Pic}(f^{-1}(U')) \to \operatorname{Pic}(f^{-1}(U))$ for $U \subset U'$ being defined by (21.3.2.4). Now, for every $x \in X$, every open neighbourhood U' of x in X, and every element $\zeta \in \operatorname{Pic}(f^{-1}(U'))$, it follows from (21.8.1) that there is an open neighbourhood $U \subset U'$ of x such that the image of ζ in $\operatorname{Pic}(f^{-1}(U))$ is zero. Therefore the fibre of $R^1 f_*(\mathscr{O}_{X'}^*)$ at the point x is zero.

(ii) (ii) The Leray spectral sequence for the composed functor $\mathscr{F}' \mapsto \Gamma(X, f_*(\mathscr{F}'))$ of sheaves of commutative groups on X' (T, 2.4) gives the exact sequence of low degree

$$0 \longrightarrow H^1(X, f_*(\mathscr{O}_{X'}^*)) \longrightarrow H^1(X', \mathscr{O}_{X'}^*) \longrightarrow H^0(X, R^1 f_*(\mathscr{O}_{X'}^*))$$

and the conclusion follows from (i) and from the isomorphism of $\operatorname{Pic}(X')$ and $H^1(X', \mathscr{O}_{X'}^*)$ (**0**, 5.4.7). ∎

Proposition (21.8.3). — *Let $f = (\psi, \theta) : X' \to X$ be an integral morphism of preschemes; suppose that, for every open U of X, the homomorphism $\Gamma(\theta) : \Gamma(U, \mathscr{O}_X) \to \Gamma(U, f_*(\mathscr{O}_{X'}))$ sends regular elements to regular elements, which allows us to canonically extend the homomorphism $\theta : \mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ to a homomorphism of sheaves of rings $\theta' : \mathscr{M}_X \to f_*(\mathscr{M}_{X'})$, whence homomorphisms of sheaves of multiplicative groups $\theta^* : \mathscr{O}_X^* \to f_*(\mathscr{O}_{X'}^*)$ and $\theta'^* : \mathscr{M}_X^* \to f_*(\mathscr{M}_{X'}^*)$, giving by passage to quotients a homomorphism $\theta''^* : \mathscr{D}iv_X \to f_*(\mathscr{D}iv_{X'})$. We then have a commutative diagram*

(21.8.3.1)
$$\begin{tikzcd}
1 \arrow[r] & \mathscr{O}_X^* \arrow[r] \arrow[d, "\theta^*"] & \mathscr{M}_X^* \arrow[r] \arrow[d, "\theta'^*"] & \mathscr{D}iv_X \arrow[r] \arrow[d, "\theta''^*"] & 0 \\
1 \arrow[r] & f_*(\mathscr{O}_{X'}^*) \arrow[r] & f_*(\mathscr{M}_{X'}^*) \arrow[r] & f_*(\mathscr{D}iv_{X'}) \arrow[r] & 0
\end{tikzcd}$$

in which the two rows are exact.

Proof. The only point to verify is the exactness of the second row of the diagram, which follows from the application of the exact sequence of cohomology for the functor f_* to the exact sequence of sheaves of commutative groups on X'

$$1 \longrightarrow \mathscr{O}_{X'}^* \longrightarrow \mathscr{M}_{X'}^* \longrightarrow \mathscr{D}iv_{X'} \longrightarrow 0$$

and from (21.8.2). □

Corollary (21.8.4). — *If, in addition to the hypotheses of (21.8.3), the homomorphism θ' is an isomorphism of sheaves of rings, then $\theta^* : \mathscr{O}_X^* \to f_*(\mathscr{O}_{X'}^*)$ is injective, $\theta''^* : \mathscr{D}iv_X \to f_*(\mathscr{D}iv_{X'})$ is surjective, and* $\mathrm{Ker}(\theta''^*)$ *is isomorphic to* $\mathrm{Coker}(\theta^*)$.

Proof. This is an immediate consequence of the snake diagram (Bourbaki, *Alg. comm.*, chap. I, § 2, n° 4, prop. 2) applied to the diagram (21.8.3.1). □

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Proposition (21.8.5). — *Let X, X′ be two locally Noetherian preschemes, $f : X' \to X$ an integral morphism; suppose that there exists a schematically dense open U in X such that $f^{-1}(U)$ is schematically dense in X′ (cf. (20.3.5)), and that the morphism $f^{-1}(U) \to U$, restriction of f, is an isomorphism. Then:*

(i) *The homomorphism $\theta' : \mathscr{M}_X \to f_*(\mathscr{M}_{X'})$ is defined and bijective, the homomorphism*

$$\theta^* : \mathscr{O}_X^* \longrightarrow f_*(\mathscr{O}_{X'}^*)$$

is injective, the homomorphism $\theta''^* : \mathscr{D}iv_X \to f_*(\mathscr{D}iv_{X'})$ is surjective, $\mathrm{Ker}(\theta''^*)$ *is isomorphic to* $\mathscr{N} = \mathrm{Coker}(\theta^*)$*, and the support of the sheaf of multiplicative groups $\mathscr{N}$ is contained in* $S = X - U$.

(ii) *Suppose moreover that the set S is discrete, and put $\mathscr{O}'_X = f_*(\mathscr{O}_{X'})$ for short. Then there is a commutative diagram*

$$\begin{tikzcd}
1 \arrow[r] & \prod_{s\in S} \mathscr{O}'^*_{X,s}/\mathscr{O}^*_{X,s} \arrow[r, "j"] \arrow[d] & \mathrm{Div}(X) \arrow[r] \arrow[d, "k_X"] & \mathrm{Div}(X') \arrow[r] \arrow[d, "k_{X'}"] & 0 \\
1 \arrow[r] & \big(\prod_{s\in S} \mathscr{O}'^*_{X,s}/\mathscr{O}^*_{X,s}\big)/\,\mathrm{Im}\,\Gamma(X', \mathscr{O}^*_{X'}) \arrow[r, "j'"] & \mathrm{Pic}(X) \arrow[r] & \mathrm{Pic}(X') \arrow[r] & 0
\end{tikzcd} \tag{21.8.5.1}$$

whose rows are exact and whose left vertical arrow is the canonical homomorphism.

Proof. (i) The hypothesis gives a canonical isomorphism $\mathscr{M}''_X \xrightarrow{\sim} f_*(\mathscr{M}''_{X'})$ for the sheaves of germs of pseudo-functions (20.2.2); the assertion concerning θ' follows, in view of the canonical isomorphisms $\mathscr{M}_X \xrightarrow{\sim} \mathscr{M}''_X$ and $\mathscr{M}_{X'} \xrightarrow{\sim} \mathscr{M}''_{X'}$ (20.2.11). The rest of assertion (i) follows from (21.8.4), except what concerns the support of $\mathscr{N}$, which follows directly from the hypothesis on U.

(ii) Applying the cohomology exact sequence to the two exact sequences of sheaves of commutative groups

$$\begin{gathered} 1 \longrightarrow \mathscr{N} \longrightarrow \mathscr{D}iv_X \longrightarrow f_*(\mathscr{D}iv_{X'}) \longrightarrow 0, \\ 1 \longrightarrow \mathscr{O}_X^* \longrightarrow f_*(\mathscr{O}_{X'}^*) \longrightarrow \mathscr{N} \longrightarrow 1, \end{gathered} \tag{21.8.5.2}$$

we obtain respectively the exact sequences

$$\begin{gathered} 1 \longrightarrow \Gamma(X, \mathscr{N}) \xrightarrow{j} \mathrm{Div}(X) \longrightarrow \mathrm{Div}(X') \longrightarrow H^1(X, \mathscr{N}), \\ \Gamma(X', \mathscr{O}_{X'}^*) \longrightarrow \Gamma(X, \mathscr{N}) \xrightarrow{\partial} \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(X') \longrightarrow H^1(X, \mathscr{N}), \end{gathered} \tag{21.8.5.3}$$

taking into account the canonical isomorphism $\mathrm{Pic}(X') \xrightarrow{\sim} H^1(X, f_*(\mathscr{O}_{X'}^*))$ (21.8.2). Now, **IV-4 | 283**
since the support of $\mathscr{N}$ is contained in the discrete set S, which is closed in X, we have

$$H^1(X, \mathscr{N}) = H^1(S, \mathscr{N}|S) \quad (G, II, 4.9.2), \qquad H^1(S, \mathscr{N}|S) = \prod_{s\in S} H^1(\{s\}, \mathscr{N}_s) = 0$$

by the definition of cohomology (G, II, 4.4). Similarly, $\Gamma(X, \mathscr{N}) = \prod_{s\in S} \mathscr{N}_s$ and $\mathscr{N}_s = \mathscr{O}'^*_{X,s}/\mathscr{O}^*_{X,s}$, by the second exact sequence (21.8.5.2). It remains to describe the injections j and j'. A section t of $\mathscr{N}$ over X may be described using a covering of X consisting of U and open sets U_i such that $U_i \cap S$ consists of a single point s_i and $U_i \cap U_j \subset U$ for $i \neq j$. For each i, take a section t_i of $\mathscr{N}$ over U_i, necessarily with $(t_i)_{s_i} \in \mathscr{O}'^*_{X,s_i}/\mathscr{O}^*_{X,s_i}$ and $(t_i)_x = 0$ for $x \in U_i - \{s_i\}$. The germ $(t_i)_{s_i}$ comes from a section u_i of $\mathscr{O}^*_{X'}$ over $f^{-1}(U_i)$; this may also be regarded as a section of $\mathscr{M}^*_{X'}$ over $f^{-1}(U_i)$, hence, by the hypothesis, as a section of $\mathscr{M}_X^*$ over U_i. It canonically determines a section $d_i \in \Gamma(U_i, \mathscr{D}iv_X)$. On $f^{-1}(U \cap U_i)$ the restriction of u_i identifies, by the hypothesis, with a section of $\mathscr{O}_X^*$ over $U \cap U_i$; consequently the restrictions of the d_i to $U \cap U_i$ are all zero. Thus the d_i are the restrictions of a single divisor $d \in \mathrm{Div}(X)$, which is the image of t under j. Likewise, the image of t under $\partial : \Gamma(X, \mathscr{N}) \to H^1(X, \mathscr{O}_X^*)$ is represented by the cocycle equal to the restriction of u_i on $U \cap U_i$, and to $(u_i|_{U_i \cap U_j})(u_j|_{U_i \cap U_j})^{-1}$ on $U_i \cap U_j$. This gives the expression for $j'(t)$ and proves the commutativity of (21.8.5.1).

□

Remark (21.8.6). — (i) The hypotheses of (21.8.5), (i) are in particular satisfied when X is a reduced locally Noetherian prescheme with only finitely many irreducible components, X′ is its normalization, and X′ is also locally Noetherian ((**II**, 6.3.8)); the $\mathscr{O}_X$-module $\mathscr{D}iv_X$ is then an *extension* of $f_*(\mathscr{D}iv_{X'})$ by the cokernel $\mathscr{N}$ of $\mathscr{O}_X^* \to f_*(\mathscr{O}_{X'}^*)$, which may be regarded as known. If moreover X is an algebraic curve (reduced) over a field k, we are in the conditions of application of (21.8.5), (ii).

(ii) When the hypotheses of (21.8.5), (ii) hold and, in addition, the canonical homomorphism on global sections $\Gamma(X, \mathscr{O}_X) \to \Gamma(X', \mathscr{O}_{X'})$ is *bijective*, the kernels of the surjective homomorphisms $\mathrm{Div}(X) \to \mathrm{Div}(X')$ and $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$ are *isomorphic*.

(iii) When the hypotheses of (21.8.5), (ii) hold, the homomorphism $\mathrm{Div}(X) \to \mathrm{Div}(X')$ can be injective (and hence bijective) only if $\mathscr{O}'^*_{X,s} = \mathscr{O}^*_{X,s}$ for every $s \in S$. For an $s \in S$ such that $\mathscr{O}'_{X,s}$ is *local*—which, in view of $X' = \mathrm{Spec}(\mathscr{O}'_X)$ (**II**, 1.3), means that there is a single point $s' \in X'$ over s—this necessarily entails that the residue fields $\boldsymbol{k}(s)$ and $\boldsymbol{k}(s')$ are equal and that $1 + \mathfrak{m}_s = 1 + \mathfrak{m}_{s'}$, and hence finally that $\mathscr{O}'_{X,s} = \mathscr{O}_{X,s}$. Equivalently, in view of the hypothesis, there is an open neighbourhood V of s in X such that $f^{-1}(V) \to V$ is an isomorphism. In general, the ring $\mathscr{O}'_{X,s}$ is semilocal (this is evident when f is finite, and in general follows from (**0**, 23.2.6)); the canonical homomorphism $\mathscr{O}^*_{X,s} \to \mathscr{O}'^*_{X,s}$ gives, on passing to quotients, a homomorphism from the multiplicative group $\boldsymbol{k}(s)^*$ to the product of the multiplicative groups $\boldsymbol{k}(s'_j)^*$, where the s'_j are the points (in number > 1) of X′ above s. number > 1) of **IV-4** | 284
X′ above s. It is immediate that this homomorphism can be bijective only if $\boldsymbol{k}(s)$ and all the $\boldsymbol{k}(s'_j)$ are equal to the field $\mathbf{F}_2$ with two elements; moreover, if this condition holds, the multiplicative group $1 + \mathfrak{m}_s$ must have image $1 + \mathfrak{r}$, where $\mathfrak{r}$ is the radical of $\mathscr{O}'_{X,s}$, or equivalently $\mathfrak{m}_s = \mathfrak{r}$. This entails that $\mathscr{O}'_{X,s} \otimes_{\mathscr{O}_{X,s}} \boldsymbol{k}(s)$ is a direct product of fields isomorphic to $\boldsymbol{k}(s)$ (and, when f is finite, that f is unramified at the points s'_j (17.4.1)). Thus, if for example *no* residue field of X is isomorphic to $\mathbf{F}_2$, the canonical homomorphism $\mathrm{Div}(X) \to \mathrm{Div}(X')$ can be bijective only if f is an *isomorphism*. In the case where the canonical homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(X', \mathscr{O}_{X'})$ is bijective, we conclude from this and from (ii) that in the preceding considerations, one can replace the homomorphism $\mathrm{Div}(X) \to \mathrm{Div}(X')$ by the homomorphism $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$.

21.9. Divisors on preschemes of dimension 1

(21.9.1). Let X be a topological space, x a point of X, $i_x : \{x\} \to X$ the canonical injection. If $A(x)$ is a commutative group, we may regard it as a sheaf of commutative groups on the one-point space $\{x\}$ and take its direct image $(i_x)_*(A(x))$, a sheaf of commutative groups on X (**0**, 3.4.1); it follows at once from the definitions that for every open U of X, $\Gamma(U, (i_x)_*(A(x))) = A(x)$ if $x \in U$, and is reduced to 0 in the contrary case, and for $y \in \overline{\{x\}}$, $((i_x)_*(A(x)))_y = A(x)$, and for $y \notin \overline{\{x\}}$, $((i_x)_*(A(x)))_y = 0$.

If now $\mathscr{F}$ is a sheaf of commutative groups on X and Y a part of X, for every open U of X, we have a canonical homomorphism

$$\Gamma(U, \mathscr{F}) \longrightarrow \prod_{x \in U \cap Y} \mathscr{F}_x = \prod_{x \in Y} \Gamma(U, (i_x)_*(\mathscr{F}_x)) \tag{21.9.1.1}$$

and since these homomorphisms commute with the restriction operators, they define a canonical homomorphism of *sheaves*

$$j_Y : \mathscr{F} \longrightarrow \prod_{x \in Y} (i_x)_*(\mathscr{F}_x). \tag{21.9.1.2}$$

Lemma (21.9.2). — *Let* X *be a locally Noetherian topological space,* X_0 *the set of its closed points,* $\mathscr{F}$ *a sheaf of commutative groups on* X*. The following conditions are equivalent:*

a) *The canonical homomorphism* $j_{X_0} : \mathscr{F} \to \prod_{x \in X_0} (i_x)_*(\mathscr{F}_x)$ *is injective and has for image* $\bigoplus_{x \in X_0} (i_x)_*(\mathscr{F}_x)$*.*
a′) *There exists a family of commutative groups* $(A(x))_{x \in X_0}$ *such that* $\mathscr{F}$ *is isomorphic to* $\bigoplus_{x \in X_0} (i_x)_*(A(x))$*.*
b) *Every section of* $\mathscr{F}$ *over an open* U *has a discrete support contained in* X_0*.*

When these conditions hold, for every $x \in X_0$ *the group* $A(x)$ *is determined up to unique isomorphism and is isomorphic to* $\mathscr{F}_x$*. Furthermore, the sheaf* $\mathscr{F}$ *is then flasque.*

Proof. As the points of X_0 are closed in X, we have $((i_x)_*(A(x)))_y = 0$ for $y \neq x \in X_0$; this remark **IV-4** | 285
proves the uniqueness assertion relative to the groups $A(x)$, and it is clear moreover that *a)* implies *a′)*. To see that *a′)* implies *b)*, we may restrict to the case where U is Noetherian; then (G, II, 3.10) we have

$$\Gamma(U, \bigoplus_{x \in X_0} (i_x)_*(A(x))) = \bigoplus_{x \in X_0} \Gamma(U, (i_x)_*(A(x))),$$

and thus every section s of $\mathscr{F}$ over U is a direct sum of a *finite* number of sections $s_k \in \Gamma(U, (i_{x_k})_*(A(x_k)))$ $(1 \leqslant k \leqslant n)$ with $x_k \in X_0$. Since x_k is closed in X, the support of s_k is reduced to $\{x_k\}$, so the support of s is contained in the finite set of the x_k, which is evidently discrete. Let us show that *b)* implies *a)*. For every Noetherian open U, the support of every section of $\mathscr{F}$ over U, being discrete and quasi-compact, is *finite*. It follows from hypothesis *b)* that, for every Noetherian open U, the image of the homomorphism (21.9.1.1) for $Y = X_0$ is contained in $\bigoplus_{x \in X_0} \Gamma(U, (i_x)_*(\mathscr{F}_x))$ and that this homomorphism is injective; this establishes *a)*.

Finally, to show that $\mathscr{F}$ is flasque, consider a section s of $\mathscr{F}$ over an open U of X; as the support of s is a discrete and closed subspace in X, we extend s to a section s' of $\mathscr{F}$ over X by setting $s'_x = 0$ in $X - U$. □

Remark (21.9.3). — (i) In the condition *b)* of (21.9.2), it does not suffice to assume that the support of every section of $\mathscr{F}$ over any open of X is *discrete*; this is shown by the example where we take for X the spectrum of a discrete valuation ring, with a closed point b and a generic point a, and for $\mathscr{F}$ the sheaf of commutative groups such that $\mathscr{F}_b = 0$ and $\mathscr{F}_a = \mathbf{Z}$.

(ii) The conditions of (21.9.2) being supposed satisfied, let E be a discrete part of X_0, and for each $x \in E$, let $a(x)$ be an element of $A(x)$. There then exists a unique section t of $\bigoplus_{x \in X_0} (i_x)_*(A(x))$ over X such that $t_x = a(x)$ for all $x \in E$ and whose support is contained in E. Indeed, for every Noetherian open U, $E \cap U$ is finite, and it suffices to take for t the section $\bigoplus_{x \in E} (i_x)_*(A(x))$ whose restriction to each Noetherian open U is the sum of the $a(x)$ for $x \in E \cap U$.

Proposition (21.9.4). — *Let* X *be a locally Noetherian prescheme of dimension* $\leqslant 1$*, and let* $X^{(1)}$ *be the set of points* $x \in X$ *such that* $\dim(\mathscr{O}_{X,x}) = 1$*. We then have a canonical isomorphism*

$$\mathscr{D}iv_X \xrightarrow{\sim} \bigoplus_{x \in X^{(1)}} (i_x)_*(\mathrm{Div}(\mathscr{O}_{X,x})) \tag{21.9.4.1}$$

and $\mathscr{D}iv_X$ *is flasque.*

Proof. Taking into account the isomorphism (21.4.6.1), the homomorphism (21.9.4.1) is defined by (21.9.1.2): let us prove that it is a bijection. As $\dim(X) \leqslant 1$, the points of $X^{(1)}$ are the *non-isolated closed points* of X. We are thus reduced to proving that: 1° $\mathscr{D}iv_X$ satisfies the condition *b)* of (21.9.2); 2° for every isolated point $x \in X$, we have $(\mathscr{D}iv_X)_x = 0$. The second point follows from the fact that $\mathscr{O}_{X,x}$ is an Artinian ring and that every regular element of $\mathscr{O}_{X,x}$ is therefore invertible. For 1°, it **IV-4** | 286
suffices to note that, for every open U of X and every divisor $D \in \mathrm{Div}(U)$, every maximal point x of the support S of D satisfies $\mathrm{prof}(\mathscr{O}_{X,x}) = 1$ (21.1.9), and hence belongs a fortiori to $X^{(1)} \cap U$. Since $\dim(X) \leqslant 1$, the set of these points is equal to S; thus S is discrete, the set of irreducible components of S being locally finite. □

Corollary (21.9.5). — *Let* X *be a locally Noetherian prescheme of dimension* $\leqslant 1$. *For every discrete part* $\mathrm{E} \subset \mathrm{X}^{(1)}$ *and every family* $(\mathrm{D}(x))_{x\in \mathrm{E}}$ *with* $\mathrm{D}(x) \in \mathrm{Div}(\mathscr{O}_{\mathrm{X},x})$, *there exists a unique divisor* D *on* X *such that the support of* D *is contained in* E *and that* $\mathrm{D}_x = \mathrm{D}(x)$ *for all* $x \in \mathrm{E}$.

Proof. This follows from (21.9.4) and from (21.9.3), (ii). □

Corollary (21.9.6). — *Let* X *be a locally Noetherian prescheme of dimension* $\leqslant 1$. *Every divisor* D *on* X *is of the form* $\mathrm{D}' - \mathrm{D}''$, *where* D', D'' *are two divisors* $\geqslant 0$ *with supports contained in that of* D.

Proof. By virtue of (21.9.5), it is at once reduced to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, where A is a Noetherian local ring; it then suffices to observe that M(X) is the total ring of fractions of A, and that a section of $\mathscr{M}_{\mathrm{X}}^*$ over X is written by definition as the quotient b/a of two regular elements of A. □

Corollary (21.9.7). — *Let* X *be a locally Noetherian prescheme of dimension* $\leqslant 1$, *without isolated point, and* U *a dense open in* X. *There then exists a divisor* $\mathrm{D} \geqslant 0$ *on* X, *with support contained in* U *and meeting each of the irreducible components of* X.

Proof. We can assume that U is the union of disjoint non-empty opens U_α, each of which is contained in a single irreducible component of X; it suffices then to take in each U_α a point x_α closed in X (there exists one since the unique generic point of U_α cannot be isolated by hypothesis), and to apply (21.9.5) to the discrete set of the x_α. □

The interest of this corollary is that it will allow us to prove that a separated algebraic curve X over a field k is quasi-projective, the divisor $\mathrm{D} \geqslant 0$ defined in (21.9.7) being then necessarily *ample* by virtue of the Riemann–Roch theorem for curves (chap. V).

(21.9.8). For locally Noetherian preschemes X of dimension $\leqslant 1$ the proposition (21.9.4) brings back the determination of $\mathscr{D}iv_{\mathrm{X}}$ to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, A being a Noetherian local ring of dimension 1.

When A is a regular local ring of dimension 1 (in other words, a discrete valuation ring), the group Div(A) is canonically identified with **Z** (21.6.8); by (21.9.2):

Proposition (21.9.9). — *Let* X *be a locally Noetherian regular prescheme of dimension* $\leqslant 1$. *Then the sheaf* $\mathscr{D}iv_{\mathrm{X}}$ *is canonically isomorphic to* $\bigoplus_{x\in \mathrm{X}^{(1)}} (i_x)_*(\mathbf{Z}(x))$, *where* $\mathbf{Z}(x)$ *is the additive group* **Z** *considered as a sheaf of groups on the space* $\{x\}$.

(21.9.10). Suppose only that the Noetherian local ring A of dimension 1 is *reduced*; then, if A' is the *normalization* of A (the integral closure of A in its total ring of fractions), A' is Noetherian by virtue of the Krull–Akizuki theorem (Bourbaki, *Alg. comm.*, chap. VII, § 2, nº 5, prop. 5), and we have seen in (21.8.6) how Div(A) can be obtained as an *extension* of $\mathrm{Div}(\mathrm{A}')$; the latter group is of the form $\mathbf{Z}^r$. **IV-4** | 287

Proposition (21.9.11). — *Let* X *be a Noetherian prescheme,* X_0 *a closed sub-prescheme of* X *having the following properties:*

1º $\dim(\mathrm{X}_0) \leqslant 1$.

2º *For every locally closed part* Y *of* X *such that* $\mathrm{Y} \cap \mathrm{X}_0$ *is discrete, there exists a part* Y' *of* Y, *closed in* X *and open in* Y, *containing* $\mathrm{Y} \cap \mathrm{X}_0$.

Under these conditions:

(i) *Let* Z_0 *be the union of the sets* $\overline{\{x\}} \cap \mathrm{X}_0$ *when* x *ranges over the part of* $\mathrm{Ass}(\mathscr{O}_{\mathrm{X}})$ *formed of points such that* $\overline{\{x\}} \cap \mathrm{X}_0$ *is finite. Then, for every divisor* D_0 *on* X_0, *whose support does not meet* Z_0, *there exists a divisor* D *on* X *whose inverse image by the injection* $\mathrm{X}_0 \to \mathrm{X}$ *exists* (21.4) *and is equal to* D_0; *if moreover* $\mathrm{D}_0 \geqslant 0$, *we can assume* $\mathrm{D} \geqslant 0$.

(ii) *Suppose in addition that there exists in* X_0 *an affine open* U_0 *containing* $(\mathrm{Ass}(\mathscr{O}_{\mathrm{X}_0})) \cup \mathrm{Z}_0$ *(a condition automatically satisfied when an ample* $\mathscr{O}_{\mathrm{X}_0}$*-module exists* (**II**, 4.5.4)*). Then the canonical homomorphism* (21.3.2.4) $\mathrm{Pic}(\mathrm{X}) \to \mathrm{Pic}(\mathrm{X}_0)$ *is surjective.*

Proof. (i) In view of (21.9.6), it is enough to prove the assertion for divisors $\mathrm{D}_0 \geqslant 0$. The support T of D_0 is a finite discrete set, closed in X_0, by (21.9.4). We may suppose $\mathrm{D}_0 \neq 0$, that is, $\mathrm{T} \neq \varnothing$; otherwise there is nothing to prove. For each $x \in \mathrm{T}$ there is an element $s_x \in \mathscr{O}_{\mathrm{X}_0,x}$ which is not a zero divisor, belongs to the maximal ideal of that ring, and whose image in

$(\mathscr{D}iv_{X_0})_x$ is $(D_0)_x$. There is an affine open neighbourhood $U^{(x)}$ of x in X and a section $g^{(x)}$ of $\mathscr{O}_X$ over $U^{(x)}$ such that s_x is the image in $\mathscr{O}_{X_0,x}$ of the germ $(g^{(x)})_x \in \mathscr{O}_{X,x}$. After shrinking $U^{(x)}$, we may arrange that $g^{(x)}$ is not a zero divisor in $\Gamma(U^{(x)}, \mathscr{O}_X)$; equivalently (3.1.9), if $V^{(x)}$ denotes the closed subset of $U^{(x)}$ defined by $g^{(x)} = 0$, then $V^{(x)} \cap \mathrm{Ass}(\mathscr{O}_X) = \varnothing$.

Indeed, if $z \in \mathrm{Ass}(\mathscr{O}_X)$, the hypothesis $x \notin Z_0$ implies either that $x \notin \overline{\{z\}}$, or that x is not isolated in $\overline{\{z\}} \cap X_0$. By shrinking $U^{(x)}$, we may suppose that the second alternative holds for every $z \in V^{(x)} \cap \mathrm{Ass}(\mathscr{O}_X)$. But then $V^{(x)}$ would contain an irreducible component of dimension 1 of $X_0 \cap U^{(x)}$ containing x; this would mean that the image $\overline{g}^{(x)}$ of $g^{(x)}$ in $\Gamma(U^{(x)} \cap X_0, \mathscr{O}_{X_0})$ belongs to the nilradical of that ring, and hence that s_x belongs to the nilradical of $\mathscr{O}_{X_0,x}$, a contradiction. Thus $x \notin \overline{\{z\}}$ for every $z \in V^{(x)} \cap \mathrm{Ass}(\mathscr{O}_X)$; since this set is finite, a further shrinking of $U^{(x)}$ gives $V^{(x)} \cap \mathrm{Ass}(\mathscr{O}_X) = \varnothing$.

The choice of $U^{(x)}$ therefore defines a divisor $D'^{(x)} = \mathrm{div}(g^{(x)})$ on $U^{(x)}$. We have also seen that x is necessarily isolated in $V^{(x)} \cap X_0$; shrinking once more, we may suppose $V^{(x)} \cap X_0 = \{x\}$. By condition 2°, there is then a subset $W^{(x)}$ of $V^{(x)}$, closed in X and open in $V^{(x)}$, such that $W^{(x)} \cap X_0 = \{x\}$. Replacing $U^{(x)}$ once more by a smaller open neighbourhood of x, we may suppose $W^{(x)} = V^{(x)}$, or in other words that $V^{(x)}$ is closed in X. We may now define a divisor $D^{(x)}$ on X by requiring $D^{(x)}|U^{(x)} = D'^{(x)}$ and $D^{(x)}|(X - V^{(x)}) = 0$; this is well defined because on $U^{(x)} \cap (X - V^{(x)})$ the section $g^{(x)}$ is invertible. It is then enough to take $D = \sum_{x \in T} D^{(x)}$, a meaningful sum since T is finite.

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(ii) In view of the commutative diagram (21.4.2.1), the conclusion follows from (i) once we prove that every invertible $\mathscr{O}_{X_0}$-module $\mathscr{L}_0$ is isomorphic to one of the form $\mathscr{O}_{X_0}(D_0)$ (21.2.8), where D_0 is a divisor on X_0 whose support does not meet Z_0. Since U_0 is schematically dense in X_0 (20.2.13), (iv), it is enough (3.1.9) to find a section $t \in \Gamma(U_0, \mathscr{L}_0)$ such that $t(z_0) \neq 0$ at every point of $\mathrm{Ass}(\mathscr{O}_{X_0}) \cup Z_0$. We may therefore reduce to the case where $X_0 = \mathrm{Spec}(A_0)$ is affine; then $\mathscr{L}_0$ is ample (**II**, 5.1.4), and since $\mathrm{Ass}(\mathscr{O}_{X_0}) \cup Z_0$ is finite, the conclusion follows from (**II**, 4.5.4).

□

Corollary (21.9.12). — *Let* A *be a Henselian local ring* (18.5.8), $S = \mathrm{Spec}(A)$, s_0 *the closed point of* S, $f : X \to S$ *a separated morphism of finite presentation, and suppose that the set* $X_0 = f^{-1}(s_0)$ *is of dimension* $\leqslant 1$. *Then, for every closed sub-scheme* X'_0 *of* X, *having* X_0 *as underlying set and of finite presentation over* S, *the canonical homomorphism* (21.3.2.4) $\mathrm{Pic}(X) \to \mathrm{Pic}(X'_0)$ *is surjective.*

Proof. Let us first show that it suffices to prove the corollary when the local ring A is moreover Noetherian. Indeed (18.6.15) we can write $A = \varinjlim A_\lambda$, where the A_λ are local Noetherian and Henselian rings, the homomorphisms $A_\lambda \to A$ being local; there exists in addition an index α and a separated morphism of finite presentation $f_\alpha : X_\alpha \to S_\alpha = \mathrm{Spec}(A_\alpha)$ such that X and f are deduced from X_α and f_α by base change $S \to S_\alpha$ (8.10.5), (v); with the usual notation (8.8.1), the morphisms $f_\lambda : X_\lambda \to S_\lambda$ will be separated for $\lambda \geqslant \alpha$ and we will have $X = X_\lambda \times_{S_\lambda} S$ (8.6.3); Moreover, we can assume that if $s_{0\alpha}$ is the unique closed point of S_α, there is a closed sub-scheme $X'_{0\alpha}$ of X_α, such that $X'_0 = X'_{0\alpha} \times_{S_\alpha} S$ (8.6.3); this $X'_{0\alpha}$ has the same underlying set as $f_\alpha^{-1}(s_{0\alpha})$. For $\lambda \geqslant \alpha$, let $X'_{0\lambda}$ denote its base change to S_λ. By transitivity of fibres and (4.1.4), we have $\dim(X'_{0\lambda}) = \dim(X'_0) \leqslant 1$. It remains to see that, if $\mathrm{Pic}(X_\lambda) \to \mathrm{Pic}(X'_{0\lambda})$ is surjective for every $\lambda \geqslant \alpha$, then the same is true for X and X'_0. We evidently have the commutative diagram

$$\begin{tikzcd} \mathrm{Pic}(X_\lambda) \arrow[r] \arrow[d] & \mathrm{Pic}(X) \arrow[d] \\ \mathrm{Pic}(X'_{0\lambda}) \arrow[r] & \mathrm{Pic}(X'_0) \end{tikzcd}$$

For every invertible $\mathscr{O}_{X'_0}$-module $\mathscr{L}'_0$, there exist $\lambda \geqslant \alpha$ and an invertible $\mathscr{O}_{X'_{0\lambda}}$-module $\mathscr{L}'_{0\lambda}$ from which $\mathscr{L}'_0$ is obtained by the base change $S \to S_\lambda$ (8.5.2) and (8.5.5). Choose an invertible $\mathscr{O}_{X_\lambda}$-module $\mathscr{L}_\lambda$ whose class maps to $\mathrm{cl}(\mathscr{L}'_{0\lambda})$, and let $\mathscr{L}$ be its base change to X; by the commutativity of the preceding diagram, $\mathrm{cl}(\mathscr{L}'_0)$ is the image of $\mathrm{cl}(\mathscr{L})$.

Suppose therefore that A is Noetherian, and verify that X and X'_0 satisfy the conditions of (21.9.11), (ii). By hypothesis, $\dim(X'_0) \leqslant 1$. To verify condition 2° of (21.9.11), let Y be the reduced subprescheme of X having a locally closed subset Y as underlying space. The morphism $g : Y \to S$ induced by f is quasi-finite at each point x_i of $Y \cap X_0$ ($1 \leqslant i \leqslant n$). By (18.5.11), c), Y is the sum of the open subpreschemes $Y_i = \mathrm{Spec}(\mathscr{O}_{Y,x_i})$ ($1 \leqslant i \leqslant n$) and a prescheme Y'' such that $Y'' \cap X_0 = \varnothing$; moreover, the canonical injections $j_i : Y_i \to X$ are such that $f \circ j_i$ is finite. Since f is separated, j_i is also finite, and hence Y_i is closed in X. The required subset is therefore $Y' = \bigcup_{1\leqslant i\leqslant n} Y_i$.

It remains to verify the supplementary hypothesis of (21.9.11), (ii). Now X'_0, being a separated curve over the field $\boldsymbol{k}(s_0)$, is a quasi-projective $\boldsymbol{k}(s_0)$-scheme (chap. V)[40]; hence an ample $\mathscr{O}_{X'_0}$-module exists (**II**, 5.3.1), and this completes the proof. □

Remark (21.9.13). — Under the conditions of (21.9.12), supposing in addition f proper, the morphism f is even *projective*: indeed, if $\mathscr{L}_0$ is an ample $\mathscr{O}_{X'_0}$-module, there exists, by virtue of (21.9.12), an invertible $\mathscr{O}_X$-module $\mathscr{L}$ whose inverse image in X'_0 is isomorphic to $\mathscr{L}_0$, hence is ample and, as every neighbourhood of s_0 in S is necessarily the whole of S, we deduce then from (9.6.4) that $\mathscr{L}$ is an ample $\mathscr{O}_X$-module, whence the conclusion ((**II**, 5.3.1) and (**II**, 5.5.3)).

21.10. Inverse images and direct images of 1-codimensional cycles In a later chapter devoted to intersection theory, the notions of inverse image and direct image of cycles will be developed systematically. In the present section we shall confine ourselves to defining these notions in certain useful special cases; for 1-codimensional cycles, the definitions will be chosen so as to be compatible with the corresponding definitions for divisors (21.4) and (21.5), in view of the homomorphism cyc defined in (21.6).

(21.10.1). Let X and X' be two locally Noetherian preschemes, $f : X' \to X$ a morphism, and T a subset of X. Suppose that the image under f of every maximal point of X' is a maximal point of X, and that, for every $x' \in X'^{(1)}$ (that is, $\dim(\mathscr{O}_{X',x'}) = 1$), one of the following three conditions holds for $x = f(x')$: **IV-4 | 289**

(i) $x \notin T$;
(ii) $x \in X^{(1)}$ and $\mathscr{O}_{X',x'}$ is a flat $\mathscr{O}_{X,x}$-module;
(iii) the ring $\mathscr{O}_{X,x}$ is factorial and $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{X',x'})$.

Under these conditions, for every 1-codimensional cycle Z on X whose support is contained in T, we shall define a 1-codimensional cycle $Z' = f^*(Z)$ so that $Z \mapsto f^*(Z)$ is a homomorphism of ordered groups from the subgroup of $\mathfrak{Z}^1(X)$ formed by cycles supported in T to the ordered group $\mathfrak{Z}^1(X')$. Write

$$Z = \sum_{x\in T\cap X^{(1)}} n_x\,\overline{\{x\}} \tag{21.10.1.1}$$

where the family of the $x \in T \cap X^{(1)}$ such that $n_x \neq 0$ is locally finite. For every $x' \in X'^{(1)}$, define an integer $n_{x'}$ as follows, putting $x = f(x')$:

1° if $x \notin T$, we set $n_{x'} = 0$;

2° if $x \in X^{(1)}$ and if $\mathscr{O}_{X',x'}$ is a flat $\mathscr{O}_{X,x}$-module, we know (6.1.1) that $\dim(\mathscr{O}_{X',x'}/\mathfrak{m}_x\mathscr{O}_{X',x'}) = 0$, in other words $\mathscr{O}_{X',x'}/\mathfrak{m}_x\mathscr{O}_{X',x'}$ is an $\mathscr{O}_{X',x'}$-module of finite length $\lambda_{x'}$, and we set $n_{x'} = \lambda_{x'}n_x$;

3° if $\mathscr{O}_{X,x}$ is factorial and $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{X',x'})$, then the canonical homomorphism $\mathrm{cyc} : \mathrm{Div}(\mathscr{O}_{X,x}) \to \mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X,x}))$ is bijective (21.6.9). Furthermore, since $\dim(\mathscr{O}_{X',x'}) = 1$ and $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{X',x'})$, $\mathrm{Ass}(\mathscr{O}_{X',x'})$ consists precisely of the maximal points of $\mathrm{Spec}(\mathscr{O}_{X',x'})$; the hypothesis on f therefore implies $f(\mathrm{Ass}(\mathscr{O}_{X',x'})) \subset \mathrm{Ass}(\mathscr{O}_{X,x})$. It follows from (21.4.5), (ii) that $f^* : \mathrm{Div}(\mathscr{O}_{X,x}) \to \mathrm{Div}(\mathscr{O}_{X',x'})$ is defined. Finally, since x' is the unique closed point of $\mathrm{Spec}(\mathscr{O}_{X',x'})$, $\mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X',x'}))$ is canonically isomorphic to $\mathbf{Z}$.

[40] The reader will verify that (21.9.12) is not used to prove this property in Chapter V.

We have thus defined a canonical composite homomorphism

(21.10.1.2) $$\mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X,x})) \xrightarrow{\mathrm{cyc}^{-1}} \mathrm{Div}(\mathscr{O}_{X,x}) \xrightarrow{f^*} \mathrm{Div}(\mathscr{O}_{X',x'}) \xrightarrow{\mathrm{cyc}} \mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X',x'})) \xrightarrow{\sim} \mathbf{Z}$$

If Z_x is the cycle $\sum_{y\in\mathrm{Spec}(\mathscr{O}_{X,x})\cap X^{(1)}} n_y(\overline{\{y\}}\cap\mathrm{Spec}(\mathscr{O}_{X,x}))$ on $\mathrm{Spec}(\mathscr{O}_{X,x})$, we take $n_{x'}$ to be the *image* of Z_x under (21.10.1.2).

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(21.10.2). We propose to show that:

A) When two of the conditions 1°, 2°, 3° of (21.10.1) are *simultaneously* satisfied, the corresponding values of $n_{x'}$ *coincide*.

B) The set of $x' \in X'^{(1)}$ such that $n_{x'} \neq 0$ is *locally finite* in X'.

To prove A), let us first suppose that $x \notin T$ and that x satisfies one of the conditions 2° or 3° of (21.10.1); then $n_x = 0$ and if one is in case 2°, we have $n_{x'} = 0$; if one is in case 3°, we have $Z_x = 0$ since $\mathrm{Supp}(Z) \subset T$, so again $n_{x'} = 0$. It remains to consider the case where one is simultaneously in cases 2° and 3°; then, since $\dim(\mathscr{O}_{X,x}) = 1$, $\mathscr{O}_{X,x}$ is a discrete valuation ring; if t is a uniformiser of this ring, the divisor corresponding to Z_x is $\mathrm{div}(t^{n_x})$ in $\mathrm{Div}(\mathscr{O}_{X,x})$, and its image in $\mathrm{Div}(\mathscr{O}_{X',x'})$ is the divisor of t'^{n_x}, where t' is the image of t. We may clearly reduce to the case $n_x = 1$; then the definition (21.6.5.1) shows that the image of Z_x under (21.10.1.2) is $\lambda_{x'}$, completing the proof of A).

Let us now prove B). Set $T_0 = \mathrm{Supp}(Z) \subset T$, $T'_0 = f^{-1}(T_0)$; it suffices to prove that the relation $n_{x'} \neq 0$ implies that x' belongs to the set of maximal points of T'_0, this last being locally finite in X'. It is immediate that if x' was not maximal in the closed set T'_0, there would exist a generisation y' IV-4 | 291
of x' in T'_0, distinct from x', and since $x' \in X'^{(1)}$, y' would necessarily be a maximal point of X'; as a result $y = f(y')$ would be a maximal point of X by hypothesis; but this is absurd since $y \in T_0$ and T_0 is purely of codimension 1 in X.

(21.10.3). We can now set

$$f^*(Z) = \sum_{x'\in X'^{(1)}} n_{x'}\overline{\{x'\}},$$

the sum of the second member having a meaning by virtue of what we proved in (21.10.2); we say that the 1-codimensional cycle $f^*(Z)$ is the *inverse image* of Z by f. It is immediate that the map f^* thus defined is a homomorphism of ordered groups. Furthermore, if U is an open of X, V an open of X' such that $f(V) \subset U$ and $f' : V \to U$ the restriction of f, it follows immediately from the definitions that one has

(21.10.3.1) $$f'^*(Z|U) = f^*(Z)|V.$$

Let us denote by $\Gamma_T(\mathscr{Z}^1_X)$ the largest sub-sheaf of commutative groups of $\mathscr{Z}^1_X$ with support contained in T; it follows from the relation (21.10.3.1) that the maps $\Gamma(U, \Gamma_T(\mathscr{Z}^1_X)) \to \Gamma(V, \mathscr{Z}^1_{X'})$ that we have just defined define a homomorphism of sheaves of ordered commutative groups on X'

$$\psi^*(\Gamma_T(\mathscr{Z}^1_X)) \longrightarrow \mathscr{Z}^1_{X'}$$

where ψ is the continuous map underlying the morphism f.

Proposition (21.10.4). — *Suppose satisfied the conditions of* (21.10.1). *Then, for every divisor* D *on* X *such that* $\mathrm{Supp}(D) \subset T$ *and such that* $f^*(D)$ *is defined* (21.4.2), *we have*

(21.10.4.1) $$\mathrm{cyc}(f^*(D)) = f^*(\mathrm{cyc}(D)).$$

Proof. The question being local on X, we can restrict to the case where $X = \mathrm{Spec}(A)$ is affine, where $D = \mathrm{div}(t)$, where t is a regular non-invertible element of A, and where the sub-prescheme Y(D) (21.2.12) has only a single maximal point y, so that $\mathrm{cyc}(D) = n_y.\overline{\{y\}}$, where n_y is the length of $\mathscr{O}_{X,y}/t_y\mathscr{O}_{X,y}$ (21.6.5.1). We saw in the proof of (21.10.2), B) that the points $x' \in X'$ such that $\mathrm{mult}_{x'}(f^*(\mathrm{cyc}(D))) \neq 0$ are maximal points of $f^{-1}(\overline{\{y\}})$. If the point x' is in case 3° of (21.10.1), equality of the multiplicities at x' of the two sides of (21.10.4.1) follows from the definition of $n_{x'}$ by means of the homomorphism (21.10.1.2). Suppose on the contrary that x' is in case 2° of (21.10.1), and let $x = f(x')$; since $x \in X^{(1)} \cap \overline{\{y\}}$, we have necessarily $x = y$. Now $f^*(D) = \mathrm{div}(t')$, where t' is the image of t in $\Gamma(X', \mathscr{O}_{X'})$, and $Y(f^*(D)) = f^{-1}(Y(D))$. The multiplicity of $f^*(D)$ at x' is therefore the length $n'_{x'}$ of the $\mathscr{O}_{X',x'}$-module $\mathscr{O}_{X',x'}/t'_{x'}\mathscr{O}_{X',x'} = (\mathscr{O}_{X,y}/t_y\mathscr{O}_{X,y}) \otimes_{\mathscr{O}_{X,y}} \mathscr{O}_{X',x'}$. It follows from (4.7.1) that $n'_{x'} = \lambda_{x'} n_y$, so the multiplicities at x' of the two sides of (21.10.4.1) are again equal, completing the proof. □

(21.10.5). Suppose now that $f : X' \to X$ is a morphism of locally Noetherian preschemes, transforming every maximal point of X' into a maximal point of X, and suppose furthermore that for every *closed rare* subset T of X, f satisfies the conditions of (21.10.1); this means that for all $x' \in X'^{(1)}$, either $x = f(x')$ is a maximal point of X, or x' satisfies one of the conditions (ii), (iii) of (21.10.1). Since every 1-codimensional cycle has rare support in X, $f^*(Z)$ is therefore defined for *every* 1-codimensional cycle Z on X; by virtue of (21.10.3.1), we have thus defined a homomorphism of sheaves of ordered commutative groups **IV-4 | 292**

$$\psi^*(\mathscr{Z}^1_X) \longrightarrow \mathscr{Z}^1_{X'}.$$

If furthermore, for every divisor D on X, $f^*(D)$ is defined (21.4.5), the fact that the support of D is rare in X (21.6.6) implies that (21.10.4) is applicable, and we therefore have the formula (21.10.4.1) for every divisor D on X. In particular:

Proposition (21.10.6). — *Let* X, X′ *be two locally Noetherian preschemes,* $f : X' \to X$ *a flat morphism. Then* $f^*(Z)$ *is defined for every* 1*-codimensional cycle* Z *on* X, $f^*(D)$ *is defined for every divisor* D *on* X *and we have the relation* (21.10.4.1).

Proof. Indeed, if $x' \in X'^{(1)}$ is such that $x = f(x')$ is not maximal, it follows from (6.1.1) that we have necessarily $x \in X^{(1)}$, so one is in case (ii) of (21.10.1). We can therefore apply (21.10.5), taking into account (21.4.5) and (2.3.4). □

Remark (21.10.7). — The existence of $f^*(Z)$ for every 1-codimensional cycle Z on X already follows from the hypothesis that f is *flat at all points* x' *of* X′ *of codimension* $\leqslant 1$ (i.e. such that $\dim \mathscr{O}_{X',x'} \leqslant 1$); indeed, for every maximal $z' \in X'$, it follows from (6.1.1) that $z = f(z')$ is a maximal point of X since $\dim \mathscr{O}_{X,z} \leqslant \dim \mathscr{O}_{X',z'} = 0$. Moreover, if $x' \in X'^{(1)}$, $x = f(x')$ is maximal or belongs to $X^{(1)}$ by (6.1.1); one can again apply (21.10.5).

Proposition (21.10.8). — *Let* X, X′, X″ *be three locally Noetherian preschemes,* $f : X' \to X$, $g : X'' \to X'$ *two morphisms; suppose that* f *(resp.* g*) is flat at every point of codimension* $\leqslant 1$ *in* X′ *(resp.* X″*). Then* $f \circ g$ *is flat at every point of codimension* $\leqslant 1$ *in* X″ *and for every* 1*-codimensional cycle* Z *on* X, *we have* $(f \circ g)^*(Z) = g^*(f^*(Z))$.

Proof. The first assertion follows from (6.1.1) and (2.1.6). The second follows from the fact that f (resp. g, resp. $f \circ g$) sends maximal points of X′ (resp. X″, resp. X″) to maximal points of X (resp. X′, resp. X), and from the fact that, if $x'' \in X''^{(1)}$ is such that $x' = g(x'') \in X'^{(1)}$ and $x = f(x') \in X^{(1)}$, then

$$\operatorname{length}(\mathscr{O}_{X'',x''}/\mathfrak{m}_x\mathscr{O}_{X'',x''}) = \operatorname{length}(\mathscr{O}_{X'',x''}/\mathfrak{m}_{x'}\mathscr{O}_{X'',x''}) \cdot \operatorname{length}(\mathscr{O}_{X',x'}/\mathfrak{m}_x\mathscr{O}_{X',x'})$$

Indeed, if we set $A = \mathscr{O}_{X,x}$, $A' = \mathscr{O}_{X',x'}$, $A'' = \mathscr{O}_{X'',x''}$, $\mathfrak{m} = \mathfrak{m}_x$, $\mathfrak{m}' = \mathfrak{m}_{x'}$, we have $A''/\mathfrak{m}A'' = (A'/\mathfrak{m}A') \otimes_{A'} A''$, and the formula

$$\operatorname{length}_{A''}(A''/\mathfrak{m}A'') = \operatorname{length}_{A'}(A'/\mathfrak{m}A') \cdot \operatorname{length}_{A''}(A''/\mathfrak{m}'A'')$$

follows from (4.7.1) and the hypothesis of flatness of A″ over A′. □

(21.10.9). Let X be a locally Noetherian prescheme, Λ a commutative group, written additively, and considered as a **Z**-module; we will again denote by **Z** and Λ the simple sheaves on X associated to the constant presheaves respectively equal to **Z** and Λ (**0**, 3.6). We call *sheaf of germs of* 1*-codimensional cycles with coefficients in* Λ the sheaf of commutative groups $\mathscr{Z}^1_X \otimes_{\mathbf{Z}} \Lambda$. If $\Lambda = \mathbf{Q}$, we say that the sections of $\mathscr{Z}^1_X \otimes_{\mathbf{Z}} \mathbf{Q}$ over X are the 1*-codimensional cycles with rational coefficients*. Since **IV-4 | 293**
the fibres of $\mathscr{Z}^1_X$ are torsion-free **Z**-modules (21.6.3), the canonical homomorphism $\mathscr{Z}^1_X \to \mathscr{Z}^1_X \otimes_{\mathbf{Z}} \mathbf{Q}$ is *injective*, so that the 1-codimensional cycles are identified with 1-codimensional cycles with rational coefficients.

(21.10.10). We will now see that under certain conditions, we can enlarge the definition of $f^*(Z)$ given in (21.10.3) for a 1-codimensional cycle Z on X, but to take for $f^*(Z)$ a 1-codimensional cycle *with rational coefficients* on X′. The most general case where we place ourselves is the one where f transforms every maximal point of X′ into a maximal point of X, and where, at every point $x' \in \mathrm{X}'^{(1)}$, we have one of the conditions (i), (ii), (iii) of (21.10.1) *or a fourth condition* (setting $x = f(x')$):

(iv) $x \in \mathrm{X}^{(1)}$, $\mathfrak{m}_x \notin \mathrm{Ass}(\mathscr{O}_{\mathrm{X},x})$, and, in addition, if $\mathrm{A} = \widehat{\mathscr{O}}_{\mathrm{X},x}$, $\mathrm{A}' = \widehat{\mathscr{O}}_{\mathrm{X}',x'}$, and K denotes the total ring of fractions of A, then A′ is a finite A-algebra and $\mathrm{K}' = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}$ is a free K-module.

Let us note then $r_{x'}$ the rank of the free K-module K′, $q_{x'}$ the degree of $\boldsymbol{k}(x')$ over $\boldsymbol{k}(x) = \boldsymbol{k}(\mathrm{A})$, and set $\mu_{x'} = r_{x'}/q_{x'}$. For a 1-codimensional cycle with support contained in T, given by (21.10.1.1) and for all $x' \in \mathrm{X}'^{(1)}$, we define $c_{x'} \in \mathbf{Q}$ to equal $n_{x'}$ when one is in one of the cases 1°, 2°, 3° of (21.10.1); but there remains here a fourth possibility:

4° if x' satisfies condition (iv) above, we set $c_{x'} = \mu_{x'} n_x \in \mathbf{Q}$.

(21.10.11). We must still prove that when condition 4° of (21.10.10) is satisfied *at the same time* as one of the conditions 1°, 2°, 3° of (21.10.1), we have $n_{x'} = c_{x'}$. This is evident if $x \notin \mathrm{T}$ since then $n_x = 0$. To study the other two cases, note that $\mathfrak{m}_x \mathscr{O}_{\mathrm{X}',x'}$ is closed for the $\mathfrak{m}_{x'}$-adic topology; hence the completion of $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x \mathscr{O}_{\mathrm{X}',x'}$ for that topology is $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$. If cases 2° and 4° hold simultaneously, $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x \mathscr{O}_{\mathrm{X}',x'}$ is discrete for the $\mathfrak{m}_{x'}$-adic topology and is therefore isomorphic to $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$. Since A′ is a finite and flat A-algebra ($\mathbf{0}_{\mathrm{III}}$, 10.2.3), it is a free A-module ($\mathbf{0}_{\mathrm{III}}$, 10.1.3). The rank of $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$ over $\mathrm{A}/\mathfrak{m}_x\mathrm{A} = \boldsymbol{k}(x)$ is thus equal to the rank $r_{x'}$ of K′ over K. On the other hand, this rank is the product of the length $\lambda_{x'}$ of the $\mathscr{O}_{\mathrm{X}',x'}$-module $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x \mathscr{O}_{\mathrm{X}',x'}$ by $[\boldsymbol{k}(x') : \boldsymbol{k}(x)] = q_{x'}$. Consequently $\mu_{x'} = r_{x'}/q_{x'} = \lambda_{x'}$ in this case. IV-4 | 294

Suppose finally that one is simultaneously in cases 3° and 4°. Then, since $\dim(\mathscr{O}_{\mathrm{X},x}) = 1$, $\mathscr{O}_{\mathrm{X},x}$ is a discrete valuation ring, hence regular. On the other hand, $\mathscr{O}_{\mathrm{X}',x'}$ is of dimension 1, and since $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{\mathrm{X}',x'})$, $\mathscr{O}_{\mathrm{X}',x'}$ is a Cohen-Macaulay ring ($\mathbf{0}$, 16.4.6); finally, since A′ is an A-module of finite type, $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$ is a $\boldsymbol{k}(x)$-vector space of finite rank; since $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x \mathscr{O}_{\mathrm{X}',x'}$ is contained in $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$, it is also a $\boldsymbol{k}(x)$-vector space of finite rank, hence an Artinian ring. The application of (6.1.5) then shows that $\mathscr{O}_{\mathrm{X}',x'}$ is a flat $\mathscr{O}_{\mathrm{X},x}$-module, so one is also in case 2°, and we conclude by what was seen above.

Accordingly, in the situation under consideration, set

$$f^*(\mathrm{Z}) = \sum_{x' \in \mathrm{X}'^{(1)}} c_{x'} . \overline{\{x'\}}$$

which is thus a 1-codimensional cycle on X′ with *rational* coefficients. We have again defined a homomorphism f^* of ordered groups satisfying (21.10.3.1), and hence a homomorphism of sheaves of ordered commutative groups

$$\psi^*(\Gamma_{\mathrm{T}}(\mathscr{Z}^1_{\mathrm{X}})) \longrightarrow \mathscr{Z}^1_{\mathrm{X}'} \otimes_{\mathbf{Z}} \mathbf{Q}.$$

When f satisfies the preceding conditions for every *closed rare* subset T in X, i.e. that for all $x' \in \mathrm{X}'^{(1)}$, either $x = f(x')$ is a maximal point of X, or x' satisfies one of the conditions (ii), (iii) or (iv), $f^*(\mathrm{Z})$ is then defined for *every* 1-codimensional cycle Z on X, and we have defined a homomorphism of sheaves of commutative ordered groups

$$\psi^*(\mathscr{Z}^1_{\mathrm{X}}) \longrightarrow \mathscr{Z}^1_{\mathrm{X}'} \otimes_{\mathbf{Z}} \mathbf{Q}$$

and hence, by tensoring, a homomorphism of sheaves of ordered $\mathbf{Q}$-vector spaces

$$\psi^*(\mathscr{Z}^1_{\mathrm{X}} \otimes_{\mathbf{Z}} \mathbf{Q}) \longrightarrow \mathscr{Z}^1_{\mathrm{X}'} \otimes_{\mathbf{Z}} \mathbf{Q}. \tag{21.10.11.1}$$

Remark (21.10.12). — When one is in the situation of (21.10.10), one can effectively, for 1-codimensional cycles Z on X, have for $f^*(\mathrm{Z})$ cycles 1-codimensional with *non-integer* coefficients; in other words, the numbers $\mu_{x'}$ can be non-integers. An example is obtained by taking the integral domain A and its integral closure A′ from ($\mathbf{0}_{\mathrm{IV}}$, 6.15.11), (ii): the closed point x' of Spec(A′) satisfies condition (iv) of (21.10.10), and $\mu_{x'} = \frac{1}{2}$.

Lemma (21.10.13). — *Let* A *be a local Noetherian ring of dimension* 1*, t a regular element of* A *belonging to the maximal ideal* $\mathfrak{m}$ *(which implies that* $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$*).*

(i) *For every A-module* M *of finite type, the kernel* $\mathrm{N}_t(\mathrm{M})$ *and the cokernel* $\mathrm{P}_t(\mathrm{M})$ *of the homothety* $t_{\mathrm{M}} : \mathrm{M} \to \mathrm{M}$ *of ratio t are of finite length. We set* $d_t(\mathrm{M}) = \operatorname{length} \mathrm{P}_t(\mathrm{M}) - \operatorname{length} \mathrm{N}_t(\mathrm{M})$.

(ii) *If* $0 \to \mathrm{M}' \to \mathrm{M} \to \mathrm{M}'' \to 0$ *is an exact sequence of A-modules of finite type, we have* $d_t(\mathrm{M}) = d_t(\mathrm{M}') + d_t(\mathrm{M}'')$.

(iii) *We have* $d_t(\mathrm{M}) \geqslant 0$ *for every A-module* M *of finite type; for* $d_t(\mathrm{M}) = 0$*, it is necessary and sufficient that* M *be of finite length.*

(iv) *If* K *is the total ring of fractions of* A *and if* $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K}$ *is a free* K*-module of rank n, we have* $d_t(\mathrm{M}) = n \cdot d_t(\mathrm{A}) = n \cdot \operatorname{length}(\mathrm{A}/t\mathrm{A})$.

(v) *If* M *satisfies the hypotheses of (iv) and if furthermore t is* M*-regular, we have* $\operatorname{length}(\mathrm{M}/t\mathrm{M}) = n \cdot \operatorname{length}(\mathrm{A}/t\mathrm{A})$.

Proof. (i) $\operatorname{Spec}(\mathrm{A})$ is formed of the point $\mathfrak{m}$ and the minimal prime ideals $\mathfrak{p}_i$; since by hypothesis $t \notin \mathfrak{p}_i$ for all i (Bourbaki, *Alg. comm.*, chap. IV, §1, nº 1, cor. 3 of prop. 2), the image of t in each of the $\mathrm{A}_{\mathfrak{p}_i}$ is invertible, and the supports of the A-modules of finite type $\mathrm{N}_t(\mathrm{M})$ and $\mathrm{P}_t(\mathrm{M})$ are therefore empty or reduced to $\mathfrak{m}$; we conclude from this (**0**, 16.1.10) that these modules are of finite length. **IV-4 | 295**

(ii) Since t is regular, we have an exact sequence

$$0 \longrightarrow \mathrm{A} \xrightarrow{t_{\mathrm{A}}} \mathrm{A} \longrightarrow \mathrm{A}/t\mathrm{A} \longrightarrow 0$$

and since $\operatorname{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{A}) = 0$ for $i \geqslant 1$, the Tor exact sequence gives on one hand the exact sequence

$$0 \longrightarrow \operatorname{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{M} \xrightarrow{t_{\mathrm{M}}} \mathrm{M} \longrightarrow \mathrm{M}/t\mathrm{M} \longrightarrow 0$$

and on the other hand, for $i \geqslant 2$,

$$0 = \operatorname{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{A}) \longrightarrow \operatorname{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A}) \longrightarrow \operatorname{Tor}_{i-1}^{\mathrm{A}}(\mathrm{M}, \mathrm{A}) = 0$$

hence $\mathrm{N}_t(\mathrm{M}) = \operatorname{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A})$ and $\operatorname{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A}) = 0$ for $i \geqslant 2$; the Tor exact sequence gives the exact sequence

$$\begin{gathered} 0 \longrightarrow \operatorname{Tor}_1^{\mathrm{A}}(\mathrm{M}', \mathrm{A}/t\mathrm{A}) \longrightarrow \operatorname{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A}) \longrightarrow \operatorname{Tor}_1^{\mathrm{A}}(\mathrm{M}'', \mathrm{A}/t\mathrm{A}) \longrightarrow \\ \mathrm{M}' \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{M}'' \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A}) \longrightarrow 0 \end{gathered}$$

and this sequence can be written, in view of what precedes

$$0 \longrightarrow \mathrm{N}_t(\mathrm{M}') \longrightarrow \mathrm{N}_t(\mathrm{M}) \longrightarrow \mathrm{N}_t(\mathrm{M}'') \longrightarrow \mathrm{P}_t(\mathrm{M}') \longrightarrow \mathrm{P}_t(\mathrm{M}) \longrightarrow \mathrm{P}_t(\mathrm{M}'') \longrightarrow 0$$

which proves (ii).

(iii) To prove (iii), we note that for M there is a composition series (M_h) of M such that the quotients $\mathrm{M}_h/\mathrm{M}_{h+1}$ are isomorphic to $\mathrm{A}/\mathfrak{m}$ or to one of the $\mathrm{A}/\mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. IV, §1, nº 4, th. 1), and that for M to be of finite length, it is necessary and sufficient that all these quotients be isomorphic to $\mathrm{A}/\mathfrak{m}$. It all comes down therefore (by virtue of (ii)) to proving that $d_t(\mathrm{A}/\mathfrak{m}) = 0$ and $d_t(\mathrm{A}/\mathfrak{p}_i) > 0$. Now, the image of t in $\mathrm{A}/\mathfrak{m}$ being 0, we have $\mathrm{N}_t(\mathrm{A}/\mathfrak{m}) = \mathrm{A}/\mathfrak{m}$ and $\mathrm{P}_t(\mathrm{A}/\mathfrak{m}) = \mathrm{A}/\mathfrak{m}$, whence the first assertion; on the other hand, the image of t in $\mathrm{A}/\mathfrak{p}_i$ is regular, so $\mathrm{N}_t(\mathrm{A}/\mathfrak{p}_i) = 0$ and $\mathrm{P}_t(\mathrm{A}/\mathfrak{p}_i) = \mathrm{A}/(t\mathrm{A} + \mathfrak{p}_i)$ which is not reduced to 0, whence the second assertion.

(iv) There is a basis of $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K}$ of the form $(x_j/s)_{1 \leqslant j \leqslant n}$, where s is a regular element of A and $x_j \in \mathrm{M}$. Consider the homomorphism $u : \mathrm{A}^n \to \mathrm{M}$ which transforms the element e_j $(1 \leqslant j \leqslant n)$ of the canonical basis of A^n into x_j and let us show that $\operatorname{Ker}(u)$ and $\operatorname{Coker}(u)$ are of *finite length*; indeed, for each i, the image of s in $\mathrm{A}_{\mathfrak{p}_i}$ is invertible, and since $\mathrm{K}_{\mathfrak{p}_i} = \mathrm{A}_{\mathfrak{p}_i}$, the images of the x_j/s in $\mathrm{M}_{\mathfrak{p}_i}$ form a basis of this $\mathrm{A}_{\mathfrak{p}_i}$-module; hence $u_{\mathfrak{p}_i} : \mathrm{A}_{\mathfrak{p}_i}^n \to \mathrm{M}_{\mathfrak{p}_i}$ is bijective. As in (i), we conclude that the supports of $\operatorname{Ker}(u)$ and of $\operatorname{Coker}(u)$ are contained in $\{\mathfrak{m}\}$, so these modules are of finite length. This being so, it follows from (ii) and (iii) that $d_t(\mathrm{M}) = d_t(\mathrm{A}^n) = n \cdot d_t(\mathrm{A}) = n \cdot \operatorname{length}(\mathrm{A}/t\mathrm{A})$.

Finally, it is clear that (v) follows immediately from (iv), since then $\mathrm{N}_t(\mathrm{M}) = 0$. □

This lemma allows us to generalise (21.10.4):

Proposition (21.10.13*). — *Suppose that f satisfies the conditions of* (21.10.10)*. Then, for every divisor* D *on* X *such that* $\operatorname{Supp}(\mathrm{D}) \subset \mathrm{T}$ *and such that* $f^*(\mathrm{D})$ *is defined, we have*

(21.10.13.1) $$\operatorname{cyc}(f^*(\mathrm{D})) = f^*(\operatorname{cyc}(\mathrm{D})).$$

Proof. Reasoning as in (21.10.4), it all amounts to seeing (with the same notation) that if x' is in case 4° of (21.10.10) and if n_y is the length of $\mathscr{O}_{X,y}/t_y\mathscr{O}_{X,y}$, $\mu_{x'}$ being the rational number defined in (21.10.10), then the length $n_{x'}$ of $\mathscr{O}_{X',x'}/t'_{x'}\mathscr{O}_{X',x'}$ is equal to $\mu_{x'}n_y$. Since $\mathscr{O}_{X',x'}/t'_{x'}\mathscr{O}_{X',x'}$ is of finite length, it has the same length as its $\mathfrak{m}_{x'}$-preadic completion $A'/t'_{x'}A'$, which can also be written A'/t_yA'; moreover, since $t'_{x'}$ is regular by hypothesis in $\mathscr{O}_{X',x'}$, it is also regular in A' by flatness ($\mathbf{0}_{\mathrm{I}}$, 6.3.4), and when A' is considered as an A-module, we can also say that t_y is A'-regular. Since A has dimension 1 and A' is a finite A-module such that $A' \otimes_A K$ is a free K-module of rank $r_{x'}$, we may apply (21.10.13), (v) to $M = A'$ and t_y; the length of A'/t_yA' as an A-module is therefore $r_{x'}n_y$. Since $\boldsymbol{k}(x')$ is a $\boldsymbol{k}(x)$-vector space of rank $q_{x'}$, the length of A'/t_yA' as an A'-*module* is therefore $r_{x'}n_y/q_{x'} = \mu_{x'}n_y$, which finishes the proof. ☐ IV-4 | 296

(21.10.14). Let now X, X′ be two locally Noetherian preschemes, $f : X' \to X$ a morphism having the *two* following properties:

a) f is finite;

b) the image by f of every maximal point of X′ is a maximal point of X.

For all $x \in X^{(1)}$, the points $x' \in f^{-1}(x)$ all belong to $X'^{(1)}$, as a consequence of hypothesis *b*) and of the inequality (**0**, 16.3.9.1), the fibre $f^{-1}(x)$ being discrete. Let then

$$Z' = \sum_{x' \in X'^{(1)}} n_{x'}.\overline{\{x'\}}$$

be a 1-codimensional cycle on X′. For all $x \in X^{(1)}$, we define an integer n_x by the formula

$$n_x = \sum_{x' \in f^{-1}(x)} n_{x'} \cdot [\boldsymbol{k}(x') : \boldsymbol{k}(x)]$$

which has a meaning, the points of $f^{-1}(x)$ being in finite number and $\boldsymbol{k}(x')$ being a field of finite degree over $\boldsymbol{k}(x)$ (**I**, 6.4.4). Furthermore, the set of $x \in X^{(1)}$ such that $n_x \neq 0$ is *locally finite* in X, since it is contained in the image by f of the set of $x' \in X'^{(1)}$ such that $n_{x'} \neq 0$, and the conclusion follows from the fact that the morphism f is quasi-compact. We can therefore define a 1-codimensional cycle on X by setting

(21.10.14.1) $$f_*(Z') = \sum_{x \in X^{(1)}} n_x.\overline{\{x\}}$$

and we say that $f_*(Z')$ is the *direct image* of Z′ by f. It is clear that the map $f_* : \mathfrak{Z}^1(X') \to \mathfrak{Z}^1(X)$ thus defined is a homomorphism of ordered groups. Furthermore, if U is an open of X, $f_U : f^{-1}(U) \to U$ the restriction of f, it follows immediately from the definitions that we have

(21.10.14.2) $$(f_U)_*(Z'|f^{-1}(U)) = f_*(Z')|U$$

hence, denoting by ψ the continuous map underlying the morphism f, the maps $\Gamma(f^{-1}(U), \mathscr{Z}^1_{X'}) \to \Gamma(U, \mathscr{Z}^1_X)$ that we have just defined define a homomorphism of sheaves of ordered commutative groups on X

$$\psi_*(\mathscr{Z}^1_{X'}) \longrightarrow \mathscr{Z}^1_X.$$

Proposition (21.10.15). — *Let* X, X′, X″ *be three locally Noetherian preschemes,* $f : X' \to X$, $g : X'' \to X'$ *two morphisms satisfying conditions* a) *and* b) *of* (21.10.14)*. Then* $f \circ g$ *satisfies the same conditions, and for every* 1*-codimensional cycle* Z″ *on* X″*, we have* $(f \circ g)_*(Z'') = f_*(g_*(Z''))$. IV-4 | 297

Proof. This follows immediately from the definitions. ☐

Proposition (21.10.16). — *Let* X, X′, X_1 *be three locally Noetherian preschemes,* $f : X' \to X$ *a morphism satisfying conditions* a) *and* b) *of* (21.10.14)*,* $g : X_1 \to X$ *a flat morphism. Set* $X'_1 = X' \times_X X_1$ *(so that* X'_1 *is locally Noetherian) and let* $f_1 : X'_1 \to X_1$ *and* $g_1 : X'_1 \to X'$ *be the canonical projections. Then* f_1 *satisfies conditions* a) *and* b) *of* (21.10.14)*, and for every* 1*-codimensional cycle* Z′ *on* X′*, we have*

(21.10.16.1) $$g^*(f_*(Z')) = (f_1)_*(g_1^*(Z')).$$

Proof. It is clear that f_1 is finite, and it satisfies condition b) of (21.10.14) by virtue of (2.3.7). To prove (21.10.16.1), we are immediately reduced to the case where X, X' and X_1 are spectra of local Noetherian rings of dimension 1, A, A' and A_1, with A' a finite A-module and A_1 a flat A-module. Denoting by x, x' and x_1 the closed points of X, X' and X_1 respectively, it is a matter of seeing that we have

(21.10.16.2) $$\sum_{x_1'} \lambda_{x_1'} [\boldsymbol{k}(x_1') : \boldsymbol{k}(x_1)] = \lambda_{x_1} [\boldsymbol{k}(x') : \boldsymbol{k}(x)]$$

where x_1' ranges over the closed points of X_1' (i.e. the points lying above both x' and x_1) and where $\lambda_{x_1'} = \text{length}(\mathscr{O}_{\mathrm{X}_1',x_1'}/\mathfrak{m}_{x'}\mathscr{O}_{\mathrm{X}_1',x_1'})$ and $\lambda_{x_1} = \text{length}(\mathrm{A}_1/\mathfrak{m}_x\mathrm{A}_1)$. Since

$$[\boldsymbol{k}(x_1') : \boldsymbol{k}(x')]\,[\boldsymbol{k}(x') : \boldsymbol{k}(x)] = [\boldsymbol{k}(x_1') : \boldsymbol{k}(x_1)]\,[\boldsymbol{k}(x_1) : \boldsymbol{k}(x)],$$

the left-hand side of (21.10.16.2) may also be written

$$\frac{[\boldsymbol{k}(x') : \boldsymbol{k}(x)]}{[\boldsymbol{k}(x_1) : \boldsymbol{k}(x)]} \text{length}_{\mathrm{A}'}(\mathrm{A}_1'/\mathfrak{m}_{x'}\mathrm{A}_1') = \text{length}_{\mathrm{A}_1}(\mathrm{A}_1'/\mathfrak{m}_{x'}\mathrm{A}_1'),$$

where $\mathrm{A}_1' = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{A}_1$. Now $\mathrm{A}_1'/\mathfrak{m}_{x'}\mathrm{A}_1' = (\mathrm{A}'/\mathfrak{m}_{x'}\mathrm{A}') \otimes_{\mathrm{A}} \mathrm{A}_1$, and since A_1 is a flat A-module, (4.7.1) gives

$$\text{length}_{\mathrm{A}_1}(\mathrm{A}_1'/\mathfrak{m}_{x'}\mathrm{A}_1') = \text{length}_{\mathrm{A}'}(\mathrm{A}'/\mathfrak{m}_{x'}\mathrm{A}')\,\text{length}_{\mathrm{A}_1}(\mathrm{A}_1/\mathfrak{m}_x\mathrm{A}_1) = [\boldsymbol{k}(x') : \boldsymbol{k}(x)]\lambda_{x_1},$$

which completes the proof. □

Proposition (21.10.17). — *Let* X, X' *be two locally Noetherian preschemes,* $f : \mathrm{X}' \to \mathrm{X}$ *a finite locally free morphism. Then* f *satisfies condition* b) *of* (21.10.14)*, and for every divisor* D' *on* X'*, we have*

(21.10.17.1) $$f_*(\text{cyc}(\mathrm{D}')) = \text{cyc}(f_*(\mathrm{D}')).$$

Proof. Since f is flat and finite, condition b) of (21.10.14) and the relation $f(\mathrm{X}'^{(1)}) \subset \mathrm{X}^{(1)}$ follow from (6.1.1). The definition (21.10.14.1) shows that we can reduce to the case where $\mathrm{X} = \text{Spec}(\mathrm{A}) = \text{Spec}(\mathscr{O}_{\mathrm{X},x})$ for an $x \in \mathrm{X}^{(1)}$; then $\mathrm{X}' = \text{Spec}(\mathrm{A}')$, where A' is a free A-module of finite rank; furthermore, we can suppose that $\mathrm{D}' = \text{div}(t')$, where t' is a regular element of A'; we then have $f_*(\mathrm{D}') = \text{div}(t)$, where $t = \mathrm{N}_{\mathrm{A}'/\mathrm{A}}(t')$ is a regular element of A (21.5.2). We can restrict to the case where t' is not invertible in A', which is equivalent to t not being invertible in A; the ring $\mathrm{A}/t\mathrm{A}$ is then nonzero and of finite length, and A' is a finite direct product of local rings A_i' $(1 \leqslant i \leqslant r)$, whose residue fields are $\boldsymbol{k}(x_i')$, where the x_i' are the points of X' lying above x. If t_i' is the image of t' in A_i', $\mathrm{A}'/t'\mathrm{A}'$ is the direct product of the $\mathrm{A}_i'/t_i'\mathrm{A}_i'$; since the product $\text{length}_{\mathrm{A}_i'}(\mathrm{A}_i'/t_i'\mathrm{A}_i') \cdot [\boldsymbol{k}(x_i') : \boldsymbol{k}(x)]$ is equal to $\text{length}_{\mathrm{A}}(\mathrm{A}_i'/t_i'\mathrm{A}_i')$, we see that the multiplicity at point x of the left-hand side of (21.10.17.1) is $\text{length}_{\mathrm{A}}(\mathrm{A}'/t'\mathrm{A}')$, so the formula to prove reduces to

(21.10.17.2) $$\text{length}_{\mathrm{A}}(\mathrm{A}'/t'\mathrm{A}') = \text{length}_{\mathrm{A}}(\mathrm{A}/t\mathrm{A}).$$

This relation follows from the following more general lemma: □

Lemma (21.10.17.3). — *Let* A *be a local Noetherian ring of dimension* 1, M *a free A-module of finite rank,* u *an injective endomorphism of* M*. Then we have* **IV-4** | 298

(21.10.17.4) $$\text{length}_{\mathrm{A}}(\text{Coker}\, u) = \text{length}_{\mathrm{A}}(\mathrm{A}/(\det u)\mathrm{A}).$$

Proof. We distinguish several cases.

I) A *is a discrete valuation ring*. Let π be a uniformiser of A, and note $\text{length}_{\mathrm{A}}(\mathrm{A}/\pi^k\mathrm{A}) = k$; if n is the rank of M and if π^{m_i} $(1 \leqslant i \leqslant n)$ are the invariant factors of u, $\text{Coker}(u)$ is a direct sum of A-modules $\mathrm{A}/\pi^{m_i}\mathrm{A}$, so has length $m = \sum_{i=1}^n m_i$, and $\det(u)$ is the product of an invertible element and of π^m, whence the conclusion in this case (Bourbaki, *Alg.*, chap. VII, §4, nº 5, cor. 1 of prop. 4).

II) A *is a complete integral ring (of dimension* 1*)*. We know then (**0**, 19.8.8, (ii)) that there is a subring B of A, which is a discrete valuation ring, such that $\mathrm{B} \to \mathrm{A}$ is a local homomorphism making A a B-module of finite type; since this B-module is evidently torsion-free, it is *free* (Bourbaki, *Alg. comm.*, chap. VI, §3, nº 6, lemma 1). Let M' denote the set M equipped with its structure of B-module (free), by u' the endomorphism u considered as a B-endomorphism. It follows from I) that

(21.10.17.5) $$\text{length}_{\mathrm{B}}(\text{Coker}\, u') = \text{length}_{\mathrm{B}}(\mathrm{B}/(\det u')\mathrm{B}).$$

But $\det u' = \mathrm{N}_{\mathrm{A}/\mathrm{B}}(\det u)$, therefore, by applying (21.10.17.4) to the homothety $x \mapsto (\det u)x$ of the free B-module A, we get

$$\mathrm{length}_{\mathrm{B}}(\mathrm{B}/(\det u')\mathrm{B}) = \mathrm{length}_{\mathrm{B}}(\mathrm{A}/(\det u)\mathrm{A}),$$

whence, substituting in (21.10.17.5) and dividing by $[\boldsymbol{k}(\mathrm{A}) : \boldsymbol{k}(\mathrm{B})]$, the length of the B-module $\boldsymbol{k}(\mathrm{A})$, we obtain (21.10.17.4) in the case considered.

III) A *is a complete ring*. Let us note that we can further suppose that $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$; indeed, since $\det(u)$ is a regular element of A, if we had $\mathfrak{m} \in \mathrm{Ass}(\mathrm{A})$, then $\det(u)$ would be invertible, u an automorphism of M, and the formula (21.10.17.4) would then be trivial, both members being zero. **IV-4** | 299

In what follows, for an endomorphism v of a module N over a ring R, such that $\mathrm{Ker}\, v$ and $\mathrm{Coker}\, v$ are of *finite length*, we set

$$\chi(\mathrm{N}, v) = \mathrm{length}_{\mathrm{R}}(\mathrm{Ker}(v)) - \mathrm{length}_{\mathrm{R}}(\mathrm{Coker}(v)).$$

We note that the hypothesis on v amounts to saying that the complex

$$\mathrm{K}^0 : 0 \longrightarrow \mathrm{N} \xrightarrow{v} \mathrm{N} \longrightarrow 0$$

has its cohomology modules of finite length and that $\chi(\mathrm{N}, v) = \chi(\mathrm{H}^0(\mathrm{K}^0))$ ($\mathbf{0}_{\mathrm{III}}$, 11.10). We deduce that if $\mathrm{N}', \mathrm{N}, \mathrm{N}''$ are three R-modules, and if there is a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{N}' \arrow[r, "r"] \arrow[d, "v'"] & \mathrm{N} \arrow[r, "s"] \arrow[d, "v"] & \mathrm{N}'' \arrow[r] \arrow[d, "v''"] & 0 \\ 0 \arrow[r] & \mathrm{N}' \arrow[r, "r"'] & \mathrm{N} \arrow[r, "s"'] & \mathrm{N}'' \arrow[r] & 0 \end{tikzcd}$$

whose rows are exact, and if two of the three numbers $\chi(\mathrm{N}, v)$, $\chi(\mathrm{N}', v')$ and $\chi(\mathrm{N}'', v'')$ are defined, then so is the third and one has

(21.10.17.6) $$\chi(\mathrm{N}, v) = \chi(\mathrm{N}', v') + \chi(\mathrm{N}'', v'').$$

This follows indeed immediately from the cohomology exact sequence.

Finally, if N is an R-module of *finite length*, we have $\chi(\mathrm{N}, v) = 0$.

With this notation, we have the following lemma: □

Lemma (21.10.17.7). — *Let* A *be a local Noetherian ring of dimension* 1 *whose maximal ideal* $\mathfrak{m}$ *is such that* $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$; *let* $\mathfrak{p}_i$ *(*$1 \leqslant i \leqslant n$*) be the minimal prime ideals of* A. *Let* M *be a free A-module of finite type, u an endomorphism of* M *such that* $\chi(\mathrm{M}, u)$ *is defined; for each i, set* $\mathrm{M}_i = \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{p}_i)$ *and let* u_i *be the endomorphism* $u \otimes 1_{\mathrm{A}/\mathfrak{p}_i}$ *of* M_i; *then if* $\chi(\mathrm{M}_i, u_i)$ *is defined for each i, we have*

(21.10.17.8) $$\chi(\mathrm{M}, u) = \sum_{i=1}^{n} \mathrm{length}(\mathrm{A}_{\mathfrak{p}_i}) \cdot \chi(\mathrm{M}_i, u_i).$$

Proof. Since $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$, we have a reduced primary decomposition $(0) = \bigcap_{1 \leqslant i \leqslant n} \mathfrak{q}_i$, where the ideal $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary for $1 \leqslant i \leqslant n$. If we set $\mathrm{M}'_i = \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{q}_i)$, we have therefore an exact sequence **IV-4** | 300

$$0 \longrightarrow \mathrm{M} \longrightarrow \bigoplus_i \mathrm{M}'_i \longrightarrow \mathrm{M}'' \longrightarrow 0$$

of A-modules, where M'' is of *finite length*: indeed by localising the preceding exact sequence at each of the $\mathfrak{p}_i$, we obtain $\mathrm{M}''_{\mathfrak{p}_i} = 0$, since $(\mathfrak{q}_i)_{\mathfrak{p}_i} = 0$ and $(\mathfrak{q}_j)_{\mathfrak{p}_i} = \mathrm{A}_{\mathfrak{p}_i}$ for $j \neq i$ (Bourbaki, *Alg. comm.*, chap. IV, §2, n$^{\mathrm{o}}$ 4, prop. 6), so $(\mathrm{M}'_i)_{\mathfrak{p}_i} = \mathrm{M}_{\mathfrak{p}_i}$ and $(\mathrm{M}'_j)_{\mathfrak{p}_i} = 0$; the support of M'' is therefore reduced to $\mathfrak{m}$, hence M'' is of finite length ($\mathbf{0}$, 16.1.10). If we set $u'_i = u \otimes 1_{\mathrm{A}/\mathfrak{q}_i}$, and if u'' is the endomorphism of M'' deduced from $\bigoplus u'_i$ by passing to quotients, we deduce from (21.10.17.6) that $\chi(\mathrm{M}, u) + \chi(\mathrm{M}'', u'') = \sum_i \chi(\mathrm{M}'_i, u'_i)$, and since M'' is of finite length, $\chi(\mathrm{M}'', u'') = 0$. To prove (21.10.17.8), we can therefore reduce to the case where $n = 1$. We will then denote by $\mathfrak{p}$ the unique minimal prime ideal of A; if $\mathrm{A}_0 = \mathrm{A}/\mathfrak{p}$, $\mathrm{M}_0 = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}_0$, $u_0 = u \otimes 1_{\mathrm{A}_0}$, it is a matter of proving that if $\chi(\mathrm{M}_0, u_0)$ is defined, then

(21.10.17.9) $$\chi(\mathrm{M}, u) = \mathrm{length}\, \mathrm{A}_{\mathfrak{p}} \cdot \chi(\mathrm{M}_0, u_0).$$

Let $\mathfrak{n}_j$ $(0 \leqslant j \leqslant r)$ be the "j-th symbolic power" of $\mathfrak{p}$, the inverse image of $(\mathfrak{p}A_{\mathfrak{p}})^j$ in A $(1 \leqslant j \leqslant r)$, with $\mathfrak{n}_0 = A$ and $\mathfrak{n}_r = 0$; set

$$M_j = \mathfrak{n}_j M/\mathfrak{n}_{j+1}M \qquad (0 \leqslant j \leqslant r-1),$$

and denote by v_j the endomorphism of M_j deduced from u by restriction to $\mathfrak{n}_j M$ and passage to quotients. We shall first show that each of the numbers $\chi(M_j, v_j)$ is defined and that

$$\chi(M, u) = \sum_j \chi(M_j, v_j). \tag{21.10.17.10}$$

The first assertion implies the second, by applying (21.10.17.6) to each of the exact sequences

$$0 \longrightarrow \mathfrak{n}_j M/\mathfrak{n}_{j+1}M \longrightarrow M/\mathfrak{n}_{j+1}M \longrightarrow M/\mathfrak{n}_j M \longrightarrow 0.$$

To prove the first assertion, we note that if m is the rank of the free A-module M, M_j is isomorphic to $(\mathfrak{n}_j/\mathfrak{n}_{j+1})^m$, or since $\mathfrak{p}$ annihilates each of the quotients (since $\mathfrak{p} \cdot \mathfrak{n}_j \subset \mathfrak{n}_{j+1}$), M_j is an A_0-module isomorphic to $M_0 \otimes_{A_0} (\mathfrak{n}_j/\mathfrak{n}_{j+1})$. Let us denote by l_j the rank of the A_0-module $\mathfrak{n}_j/\mathfrak{n}_{j+1}$; since the field of fractions K_0 of A_0 is the residue field of $A_{\mathfrak{p}}$, l_j is also the length of the $A_{\mathfrak{p}}$-module $(\mathfrak{p}A_{\mathfrak{p}})^j/(\mathfrak{p}A_{\mathfrak{p}})^{j+1}$. There is a system of generators of the A_0-module M_j which contains a basis of $M_j \otimes_{A_0} K_0$; hence there is an A_0-homomorphism

$$w_j : M_0^{l_j} \longrightarrow M_j$$

whose localisation at the ideal (0) of A_0 is an isomorphism, so that the support of $\operatorname{Ker}(w_j)$ and of $\operatorname{Coker}(w_j)$ is reduced to the maximal ideal $\mathfrak{m}/\mathfrak{p}$ of A_0; $\operatorname{Ker}(w_j)$ and $\operatorname{Coker}(w_j)$ are therefore A_0-modules of finite length (**0**, 16.1.10). Since by hypothesis $\chi(M_0, u_0)$ is defined, it follows from the same of $\chi(M_0^{l_j}, u_0^{l_j}) = l_j \chi(M_0, u_0)$ and by virtue of (21.10.17.6) and the fact that $\operatorname{Ker}(w_j)$ and $\operatorname{Coker}(w_j)$ are of finite length, we see that $\chi(M_j, v_j)$ is defined and equal to $l_j \chi(M_0, u_0)$; the relation (21.10.17.10) then gives

$$\chi(M, u) = (\sum_j l_j)\chi(M_0, u_0),$$

and by virtue of a preceding remark, $\sum_j l_j$ is none other than the length of $A_{\mathfrak{p}}$, which finishes the proof of the lemma (21.10.17.7). □

To apply this lemma when A is a complete ring and $\mathfrak{m} \notin \operatorname{Ass}(A)$, we observe that if u is injective, **IV-4** | 301
so is each of the u_i (with the notation of the lemma): indeed, $\det u$ is then a regular element of A, so is not contained in any of the $\mathfrak{p}_i$, and its image $\det u_i$ in $A/\mathfrak{p}_i$ is therefore a nonzero element of this integral ring, which proves that u_i is injective. Since $\operatorname{Coker}(u_i)$ is an image of $\operatorname{Coker}(u)$, it too has finite length, so $\chi(M_i, u_i)$ is defined for every i; formula (21.10.17.8) therefore applies. On the other hand, since $\det(u)$ is a regular element of A, it is not contained in any of the $\mathfrak{p}_i$; the ideal $(\det u)A$ is therefore $\mathfrak{m}$-primary and the quotient $A/(\det u)A$ is of finite length. Applying the same reasoning above to the injective homothety $t : \xi \to (\det u)\xi$ of A and to its images $t_i = \det u_i$ in the $A/\mathfrak{p}_i$, we get

$$\chi(A, t) = \sum_i \operatorname{length}(A_{\mathfrak{p}_i}) \cdot \chi(A/\mathfrak{p}_i, t_i).$$

But all the rings $A/\mathfrak{p}_i$ are integral and complete, and by applying the result of II) to each of them, one obtains again the relation (21.10.17.4) for M and u.

IV) *General case*. Set $A' = \widehat{A}$, $M' = M \otimes_A A'$, $u' = u \otimes 1_{A'}$; we have $\det(u') = \det(u)$, and by flatness, $\operatorname{Coker}(u') = (\operatorname{Coker}(u))_{(A')}$ and $A'/(\det u)A' = (A/(\det u)A)_{(A')}$; since the formula (21.10.17.4) is true for A' and u' by virtue of III), it is also true for A and u.

This finishes the proof of (21.10.17).

Proposition (21.10.18). — *Let* X, X′ *be two locally Noetherian preschemes,* $f : X' \to X$ *a finite morphism, transforming every maximal point of* X′ *into a maximal point of* X*, and satisfying for all* $x' \in X'$ *one of the conditions (ii), (iii), (iv) of* (21.10.10)*. Suppose furthermore that there exists an integer* n *such that, for every maximal point* x *of* X*,* $(f_*(\mathscr{O}_{X'}))_x$ *is a free* $\mathscr{O}_{X,x}$*-module of rank* n*. Then, for every* 1*-codimensional cycle* Z *on* X*, we have*

$$f_*(f^*(Z)) = n \cdot Z \tag{21.10.18.1}$$

("projection formula").

Proof. By virtue of the definitions, we are immediately reduced to the case where X is the spectrum **IV-4** | 302
of a local Noetherian ring A of dimension 1, with closed point x, and $\mathrm{Z} = \overline{\{x\}}$; write $\mathrm{X}' = \operatorname{Spec}(\mathrm{A}')$, where A' is a finite A-algebra and, for every minimal prime ideal $\mathfrak{p}_i$ of A, $\mathrm{A}'_{\mathfrak{p}_i}$ is a free $\mathrm{A}_{\mathfrak{p}_i}$-module of rank n. Let us show that we may moreover reduce to the case where A is *complete*. Make the base change $h : \mathrm{Y} \to \mathrm{X}$, where $\mathrm{Y} = \operatorname{Spec}(\mathrm{B})$ and $\mathrm{B} = \widehat{\mathrm{A}}$, and set

$$\mathrm{Y}' = \mathrm{X}' \times_{\mathrm{X}} \mathrm{Y} = \operatorname{Spec}(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'), \qquad g = f_{\{\mathrm{Y}\}} : \mathrm{Y}' \longrightarrow \mathrm{Y}.$$

The morphism g is finite. Since h is flat, the maximal points of Y lie above those of X; above each $\mathfrak{p}_i$ there are only finitely many minimal prime ideals $\mathfrak{p}_{ij}$ of B, and $(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}')_{\mathfrak{p}_{ij}}$ is a free $\mathrm{B}_{\mathfrak{p}_{ij}}$-module of rank n. Finally, if f satisfies one of conditions (ii), (iii), (iv) of (21.10.10) at every point x' of $f^{-1}(x)$, then g satisfies the corresponding condition at the unique point y' of $g^{-1}(y)$ lying above x', where y is the closed point of Y. This is immediate for (ii) and (iv); for (iii), it follows because A is then a discrete valuation ring, hence so is B, while $\mathfrak{m}_{y'} \notin \operatorname{Ass}(\mathscr{O}_{\mathrm{Y}',y'})$ follows by flatness from $\mathfrak{m}_{x'} \notin \operatorname{Ass}(\mathscr{O}_{\mathrm{X}',x'})$ ($\mathbf{0}$, 3.3.1). Thus g satisfies the same hypotheses as f; if (21.10.18.1) holds for g and $\overline{\{y\}}$, it holds for f and $\overline{\{x\}}$ by (21.10.16.1).

We may therefore suppose that A is complete. Then A' is complete as well and is a finite direct product of complete local rings, so we may further reduce to the case where A' is local, and verify (21.10.18.1) separately in cases (ii), (iii), and (iv), for the closed point x' of X'.

In case (ii), A' is a finite flat A-module, hence a free A-module ($\mathbf{0}_{\mathrm{III}}$, 10.1.3), of rank n by hypothesis. By definitions (21.10.1) and (21.10.3),

$$f^*(\mathrm{Z}) = \lambda_{x'} \cdot \overline{\{x'\}},$$

where $\lambda_{x'}$ is the length of the A'-module $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$. Hence

$$f_*(f^*(\mathrm{Z})) = \lambda_{x'}[\boldsymbol{k}(x') : \boldsymbol{k}(x)] \cdot \overline{\{x\}}.$$

But the coefficient is the length of $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$ as an A-module, equivalently its dimension as a $\boldsymbol{k}(x)$-vector space, and is therefore equal to n.

In case (iii), A is a discrete valuation ring, hence regular, and the hypothesis $\mathfrak{m}_{x'} \notin \operatorname{Ass}(\mathscr{O}_{\mathrm{X}',x'})$ implies that A' is Cohen–Macaulay ($\mathbf{0}$, 16.4.6). Since $\dim(\mathrm{A}') = \dim(\mathrm{A}) = 1$ and $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$ is Artinian, (6.1.5) shows that A' is flat over A, reducing this to case (ii).

In case (iv), if K is the total ring of fractions of A, then $\mathrm{K}' = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}$ is by hypothesis a free K-module of rank n and, by definition (21.10.10),

$$f^*(\mathrm{Z}) = \frac{n}{[\boldsymbol{k}(x') : \boldsymbol{k}(x)]} \cdot \overline{\{x'\}},$$

whence $f_*(f^*(\mathrm{Z})) = n \cdot \overline{\{x\}}$.

We note that the formula (21.10.18.1) is applicable in particular when the morphism f is *finite* and *flat* and such that for every maximal point x of X, $(f_*(\mathscr{O}_{\mathrm{X}'}))_x$ is a free $\mathscr{O}_{\mathrm{X},x}$-module of rank n. □

Corollary (21.10.19). — *Under the hypotheses of* (21.10.18), *let* D *be a divisor on* X *such that* $f^*(\mathrm{D})$ *is defined* (21.4.5); *then we have*

$$f_*(\operatorname{cyc}(f^*(\mathrm{D}))) = n \cdot \operatorname{cyc}(\mathrm{D}). \tag{21.10.19.1}$$

Proof. This follows from (21.10.18) and (21.10.13bis). □

21.11. Factoriality of regular local rings

Theorem (21.11.1). — (Auslander-Buchsbaum) — *A regular local Noetherian ring is factorial.* **IV-4** | 302

Proof. The proof that follows is due to I. Kaplansky.

Let A be a regular local ring of dimension n; we will reason by induction on n. For $n = 0$, A is a field, and for $n = 1$, a discrete valuation ring, hence principal (and *a fortiori* factorial). Suppose therefore $n \geqslant 2$ and the theorem proved for regular rings of dimension $< n$. Set $X = \mathrm{Spec}(A)$, denote by a the closed point of X and set $U = X - \{a\}$. At every point $y \in U$, we have $\dim(\mathscr{O}_{X,y}) \leqslant n-1$, and since A is regular, the rings $\mathscr{O}_{X,y}$ are also regular (**0**, 17.3.2), hence by the induction hypothesis they are factorial. Moreover, since $\mathrm{prof}(A) = \dim(A) \geqslant 2$ since A is regular, hence Cohen-Macaulay (**0**, 17.1.3). Using (21.6.14), we are reduced to proving that $\mathrm{Pic}(U) = 0$. Consider **IV-4** | 303 therefore an invertible $\mathscr{O}_U$-Module $\mathscr{L}$, and let us prove that it is isomorphic to $\mathscr{O}_U$. It follows from (**I**, 9.4.5) that there exists a *coherent* $\mathscr{O}_X$-Module $\mathscr{F}$ such that $\mathscr{F}|U = \mathscr{L}$. Since A is regular, hence of finite cohomological dimension (**0**, 17.3.1), there exists a *finite* free resolution of $\mathscr{F}$:

$$0 \leftarrow \mathscr{F} \leftarrow \mathscr{O}_X^{n_1} \leftarrow \mathscr{O}_X^{n_2} \leftarrow \ldots \leftarrow \mathscr{O}_X^{n_h} \leftarrow 0$$

(**0**, 17.2.8 and **0**, 17.2.2, (iii)). By restriction to U, we have therefore a finite resolution

$$0 \leftarrow \mathscr{L} \leftarrow \mathscr{O}_U^{n_1} \leftarrow \mathscr{O}_U^{n_2} \leftarrow \ldots \leftarrow \mathscr{O}_U^{n_h} \leftarrow 0. \tag{21.11.1.1}$$

□

The theorem will follow from the following general considerations. On a ringed space X, let $\mathscr{E}$ be a locally free $\mathscr{O}_X$-module of finite rank; we will denote by $\Lambda^{\max}\mathscr{E}$ the invertible $\mathscr{O}_X$-module which, in a neighbourhood of each point of X, is equal to $\Lambda^p\mathscr{E}$, denoting by p the rank of $\mathscr{E}$ in this neighbourhood (which can vary with the connected component). With this notation, we have the

Lemma (21.11.2). — *Let* X *be a ringed space in local rings, and*

$$0 \leftarrow \mathscr{E}_0 \xleftarrow{u_0} \mathscr{E}_1 \leftarrow \ldots \leftarrow \mathscr{E}_h \leftarrow 0$$

an exact sequence of $\mathscr{O}_X$-Modules locally free of finite rank; then the invertible $\mathscr{O}_X$-Module $\bigotimes_{0\leqslant i\leqslant h}(\Lambda^{\max}\mathscr{E}_i)^{\otimes(-1)^i}$ is isomorphic to $\mathscr{O}_X$.

Proof. Let us show how this lemma will allow us to conclude the proof of (21.11.1). It suffices to note that, for every integer n, $\Lambda^{\max}(\mathscr{O}_U^n) = \Lambda^n\mathscr{O}_U^n$ is isomorphic to $\mathscr{O}_U$, and on the other hand $\Lambda^{\max}\mathscr{L} = \mathscr{L}$ for every invertible $\mathscr{O}_U$-Module $\mathscr{L}$; the lemma (21.11.2), applied to the exact sequence (21.11.1.1), shows that $\mathscr{L}$ is isomorphic to $\mathscr{O}_U$.

It remains to prove (21.11.2); we proceed by induction on h, the lemma being trivial for $h = 1$. For $h > 1$, $\mathscr{N} = \mathrm{Ker}(u_0)$ is an $\mathscr{O}_X$-Module locally free of finite rank ($\mathbf{0}_{\mathrm{I}}$, 5.5.5), and we have the two exact sequences

$$0 \leftarrow \mathscr{E}_0 \leftarrow \mathscr{E}_1 \leftarrow \mathscr{N} \leftarrow 0$$
$$0 \leftarrow \mathscr{N} \leftarrow \mathscr{E}_2 \leftarrow \ldots \leftarrow \mathscr{E}_h \leftarrow 0$$

By virtue of the induction hypothesis, $(\Lambda^{\max}\mathscr{N}) \otimes (\bigotimes_{2\leqslant i\leqslant h}(\Lambda^{\max}\mathscr{E}_i)^{\otimes(-1)^{i-1}})$ is isomorphic to $\mathscr{O}_X$; it will therefore suffice to define a canonical isomorphism $(\Lambda^{\max}\mathscr{N}) \otimes (\Lambda^{\max}\mathscr{E}_0) \xrightarrow{\sim} \Lambda^{\max}\mathscr{E}_1$ to finish the proof. Now, there exists an open covering (U_α) of X such that in each U_α, $\mathscr{E}_1|U_\alpha$ is a direct sum of $\mathscr{N}|U_\alpha$ and of a free $\mathscr{O}_{U_\alpha}$-Module $\mathscr{M}_\alpha$ ($\mathbf{0}_{\mathrm{I}}$, 5.5.5), whence a canonical isomorphism $v_\alpha : \mathscr{M}_\alpha \xrightarrow{\sim} \mathscr{E}_0|U_\alpha$. Since we have a canonical isomorphism

$$r_\alpha : (\Lambda^{\max}\mathscr{N}|U_\alpha) \otimes (\Lambda^{\max}\mathscr{M}_\alpha) \xrightarrow{\sim} (\Lambda^{\max}\mathscr{E}_1)|U_\alpha$$

we deduce, by means of v_α, an isomorphism

$$u_\alpha : (\Lambda^{\max}\mathscr{N}|U_\alpha) \otimes (\Lambda^{\max}\mathscr{E}_0|U_\alpha) \xrightarrow{\sim} \Lambda^{\max}\mathscr{E}_1|U_\alpha$$

and it is a matter of showing that u_α and u_β coincide in $U_\alpha \cap U_\beta$ for any two indices α, β. Now, if **IV-4** | 304 v'_α and v'_β are the restrictions to $U_\alpha \cap U_\beta$ of v_α and v_β respectively, we have $v'_\alpha = v'_\beta \circ w_{\beta\alpha}$, where $w_{\beta\alpha} : \mathscr{M}_\alpha|(U_\alpha \cap U_\beta) \xrightarrow{\sim} \mathscr{M}_\beta|(U_\alpha \cap U_\beta)$ is the "parallel projection" such that for every section $s \in \Gamma(U_\alpha \cap U_\beta, \mathscr{M}_\alpha)$, $w_{\beta\alpha}(s) = s + t$ with $t \in \Gamma(U_\alpha \cap U_\beta, \mathscr{N})$; the identity of u_α and u_β follows immediately from this fact and from the definition of the canonical isomorphism r_α (Bourbaki, *Alg.*, chap. III, 3^e ed.). □

21.12. The Van der Waerden purity theorem for the ramification locus of a birational morphism

(21.12.1). Let X and U be two preschemes, $f : \mathrm{U} \to \mathrm{X}$ a quasi-compact and quasi-separated morphism, so that $f_*(\mathscr{O}_\mathrm{U})$ is a quasi-coherent $\mathscr{O}_\mathrm{X}$-Algebra (1.7.4). We call *affine envelope* of the X-prescheme U the X-prescheme affine over X

$$\mathrm{U}^0 = \mathrm{Aff}(\mathrm{U}/\mathrm{X}) = \mathrm{Spec}(f_*(\mathscr{O}_\mathrm{U})) = \mathrm{Spec}(\mathscr{A}(\mathrm{U})) \quad (\mathbf{II},\ 1.1.1).$$

If $f^0 : \mathrm{U}^0 \to \mathrm{X}$ is the structural morphism, we have therefore by definition

$$\mathscr{A}(\mathrm{U}^0) = f^0_*(\mathscr{O}_{\mathrm{U}^0}) = f_*(\mathscr{O}_\mathrm{U}),$$

and at the identity isomorphism of $f_*(\mathscr{O}_\mathrm{U})$, there corresponds by (**II**, 1.2.7) a canonical X-morphism

$$i_\mathrm{U} : \mathrm{U} \longrightarrow \mathrm{U}^0. \tag{21.12.1.1}$$

For i_U to be an *isomorphism*, it is necessary and sufficient that the morphism $f : \mathrm{U} \to \mathrm{X}$ be *affine*. For every affine X-prescheme V, the map $u \mapsto u \circ i_\mathrm{U}$ is a *bijection*

$$\mathrm{Hom}_\mathrm{X}(\mathrm{U}^0, \mathrm{V}) \xrightarrow{\sim} \mathrm{Hom}_\mathrm{X}(\mathrm{U}, \mathrm{V})$$

functorial in V: this follows from the existence of canonical bijections

$$\mathrm{Hom}_\mathrm{X}(\mathrm{U}, \mathrm{V}) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_\mathrm{X}}(\mathscr{A}(\mathrm{V}), \mathscr{A}(\mathrm{U}))$$

and $\mathrm{Hom}_\mathrm{X}(\mathrm{U}^0, \mathrm{V}) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_\mathrm{X}}(\mathscr{A}(\mathrm{V}), \mathscr{A}(\mathrm{U}^0))$ (**II**, 1.2.7). We can therefore say that U^0 *represents* the covariant functor $\mathrm{V} \mapsto \mathrm{Hom}_\mathrm{X}(\mathrm{U}, \mathrm{V})$ in the category of X-preschemes affine over X. We deduce from this ($\mathbf{0}_{\mathrm{III}}$, 8.1.7) that for a fixed X, $\mathrm{U} \mapsto \mathrm{Aff}(\mathrm{U}/\mathrm{X})$ is a covariant functor from the category of X-preschemes quasi-compact and quasi-separated over X, into the category of X-preschemes affine over X; more precisely, if U_1, U_2 are two X-preschemes quasi-compact and quasi-separated over X, to every X-morphism $g : \mathrm{U}_1 \to \mathrm{U}_2$ corresponds the unique X-morphism $g^0 : \mathrm{U}_1^0 \to \mathrm{U}_2^0$ making commutative the diagram

$$\begin{tikzcd} \mathrm{U}_1 \arrow[r, "g"] \arrow[d, "i_{\mathrm{U}_1}"'] & \mathrm{U}_2 \arrow[d, "i_{\mathrm{U}_2}"] \\ \mathrm{U}_1^0 \arrow[r, "g^0"'] & \mathrm{U}_2^0 \end{tikzcd}$$

More generally, consider a commutative diagram **IV-4** | 305

$$\begin{tikzcd} \mathrm{U} \arrow[r, "v"] \arrow[d, "f"'] & \mathrm{U}' \arrow[d, "f'"] \\ \mathrm{X} \arrow[r, "u"'] & \mathrm{X}' \end{tikzcd}$$

where the morphisms f, f' are quasi-compact and quasi-separated and the morphism u *affine*. If $h = u \circ f$, we have $h_*(\mathscr{O}_\mathrm{U}) = u_*(f_*(\mathscr{O}_\mathrm{U})) = u_*(f^0_*(\mathscr{O}_{\mathrm{U}^0}))$ and $u \circ f^0$ is an affine morphism; hence $\mathrm{U}^0 = \mathrm{Aff}(\mathrm{U}/\mathrm{X}')$ (relative to the morphism h), whence a unique X′-morphism $v^0 : \mathrm{U}^0 \to \mathrm{U}'^0$ making commutative the diagram

$$\begin{tikzcd} \mathrm{U} \arrow[r, "v"] \arrow[d, "i_\mathrm{U}"'] & \mathrm{U}' \arrow[d, "i_{\mathrm{U}'}"] \\ \mathrm{U}^0 \arrow[r, "v^0"'] & \mathrm{U}'^0 \end{tikzcd}$$

Proposition (21.12.2). — *Let $f : \mathrm{U} \to \mathrm{X}$ be a quasi-compact and quasi-separated morphism, $h : \mathrm{X}' \to \mathrm{X}$ a flat morphism; set $\mathrm{U}' = \mathrm{U} \times_\mathrm{X} \mathrm{X}'$, $f' = f_{\{\mathrm{X}'\}} : \mathrm{U}' \to \mathrm{X}'$. Then there is a canonical X′-isomorphism*

$$\mathrm{Aff}(\mathrm{U}'/\mathrm{X}') \xrightarrow{\sim} \mathrm{Aff}(\mathrm{U}/\mathrm{X}) \times_\mathrm{X} \mathrm{X}'. \tag{21.12.2.1}$$

Proof. Indeed, we have $\mathrm{Aff}(\mathrm{U}'/\mathrm{X}') = \mathrm{Spec}(f'_*(\mathscr{O}_{\mathrm{U}'}))$, and $\mathrm{Aff}(\mathrm{U}/\mathrm{X}) \times_\mathrm{X} \mathrm{X}' = \mathrm{Spec}(h^*(f_*(\mathscr{O}_\mathrm{U})))$; the isomorphism of the assertion comes from the canonical isomorphism $h^*(f_*(\mathscr{O}_\mathrm{U})) \xrightarrow{\sim} f'_*(\mathscr{O}_{\mathrm{U}'})$ (2.3.1). ∎

Corollary (21.12.3). — *For every quasi-compact and quasi-separated morphism $f : \mathrm{U} \to \mathrm{X}$ and every $x \in \mathrm{X}$, there is, up to canonical isomorphism, an equality*

$$\mathrm{U}^0 \times_{\mathrm{X}} \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) = (\mathrm{U} \times_{\mathrm{X}} \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}))^0.$$

It also follows from (21.12.2) that, up to canonical isomorphism, for every open V of X

(21.12.4) $$(f^{-1}(\mathrm{V}))^0 = (f^0)^{-1}(\mathrm{V}).$$

(21.12.5). We will consider in particular the case where $f : \mathrm{U} \to \mathrm{X}$ is an *open immersion*, U thus being identified with an open of X. Since the morphism $f^{-1}(\mathrm{U}) \to \mathrm{U}$ is the identity, it follows from (21.12.4) that the morphism $(f^0)^{-1}(\mathrm{U}) \to \mathrm{U}$ restriction of f^0 is an isomorphism, $i_{\mathrm{U}} : \mathrm{U} \to \mathrm{U}^0$ being thus an *open immersion* allowing us to identify U with an open of U^0. More generally, for an open V of X, the restriction $(f^0)^{-1}(\mathrm{V}) \to \mathrm{V}$ of f^0 is an isomorphism of $(f^0)^{-1}(\mathrm{V})$ onto the prescheme induced on the open $\mathrm{V} \cap \mathrm{U}$ if and only if the open immersion $\mathrm{U} \cap \mathrm{V} \to \mathrm{V}$ is an *affine* morphism. It is clear (**II**, 1.2.1) that the union of these opens V is the *largest* of them, U_1, which contains U; by virtue of what precedes, U_1 is also the largest open not meeting the set **IV-4** | 306

$$\mathrm{Daf}(\mathrm{U}/\mathrm{X}) = f^0(\mathrm{U}^0) - \mathrm{U}$$

("affinity defect" of the open U relative to X, which is empty if and only if U is affine over X); in other words, the closed set $\mathrm{Z} = \mathrm{X} - \mathrm{U}_1$ is the *closure* of $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$.

We note that for every *flat* morphism $h : \mathrm{X}' \to \mathrm{X}$, if we set $\mathrm{U}' = h^{-1}(\mathrm{U})$, we have

(21.12.5.1) $$\mathrm{Daf}(\mathrm{U}'/\mathrm{X}') = h^{-1}(\mathrm{Daf}(\mathrm{U}/\mathrm{X}))$$

as follows immediately from (21.12.2.1) and from (**I**, 3.4.8). In particular, for every open V of X, we have

(21.12.5.2) $$\mathrm{Daf}((\mathrm{U} \cap \mathrm{V})/\mathrm{V}) = \mathrm{Daf}(\mathrm{U}/\mathrm{X}) \cap \mathrm{V}$$

and for all $x \in \mathrm{X}$,

(21.12.5.3) $$\mathrm{Daf}((\mathrm{U} \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}))/\,\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})) = \mathrm{Daf}(\mathrm{U}/\mathrm{X}) \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}).$$

When X is locally Noetherian, we shall give more precise information on the nature of $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$; it will show, for example, that when U is everywhere dense in X, $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$ is not just an arbitrary rare closed subset:

Theorem (21.12.6). — *Let X be a locally Noetherian prescheme, U a non-empty open, $f : \mathrm{U} \to \mathrm{X}$ the canonical injection. Then:*

(i) The closure $\mathrm{Z} = \mathrm{X} - \mathrm{U}_1$ of $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$ is of codimension $\geqslant 2$ in X.
(ii) If $\mathrm{T} = \mathrm{X} - \mathrm{U} \supset \mathrm{Z}$ is of codimension $\geqslant 2$, the morphism $f^0 : \mathrm{U}^0 \to \mathrm{X}$ is surjective.

Proof. (i) Let us first show that for every point $x \in \mathrm{Daf}(\mathrm{U}/\mathrm{X})$ we have necessarily $\dim(\mathscr{O}_{\mathrm{X},x}) > 1$. Indeed, x plainly cannot be a maximal point of X, since it lies in $\overline{\mathrm{U}} - \mathrm{U}$; it remains to see that $\dim(\mathscr{O}_{\mathrm{X},x})$ cannot equal 1. That equality would imply, by (21.12.5.3), that $x \in \mathrm{Daf}(\mathrm{U}^{(x)}/\,\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}))$, where $\mathrm{U}^{(x)} = \mathrm{U} \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$. But the only open subsets of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ are $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ itself and the parts of the (finite) set of maximal points of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$. Now, an open set reduced to a single maximal point of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ is affine by the definition of preschemes; we conclude that *all* open subsets of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ are affine, hence $\mathrm{Daf}(\mathrm{U}^{(x)}/\,\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})) = \varnothing$, contradicting the hypothesis.

To prove (i), it remains to prove the stronger assertion that if $x \in \mathrm{X}$ satisfies $\dim(\mathscr{O}_{\mathrm{X},x}) = 1$, then there exists an open neighbourhood W of x in X which does not meet $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$, i.e. for which the canonical injection $f_{\mathrm{W}} : \mathrm{U} \cap \mathrm{W} \to \mathrm{W}$ is affine. But, with the same notation as above, the injection $f^{(x)} : \mathrm{U}^{(x)} \to \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ is affine. We may plainly reduce to the case where X is Noetherian; since f is then of finite presentation, the conclusion follows from (**IV**, 8.10.5, (viii)), by the method of (**IV**, 8.1.2, a)).

(ii) It remains to prove that $x \in f^0(\mathrm{U}^0)$ for every $x \in \mathrm{T}$. We first reduce to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, with A a complete local Noetherian ring and x the closed point of X. Indeed, make the base change $h : \mathrm{X}' = \mathrm{Spec}(\widehat{\mathscr{O}}_{\mathrm{X},x}) \to \mathrm{X}$, set $\mathrm{U}' = h^{-1}(\mathrm{U})$, and let $f' = f_{\{\mathrm{X}'\}} : \mathrm{U}' \to \mathrm{X}'$ be the **IV-4** | 307 canonical injection. Since h is flat, (21.12.2) shows that if the closed point belongs to $f'^{,0}(\mathrm{U}'^{,0})$, then $x \in f^0(\mathrm{U}^0)$; by (6.1.1), this gives the desired reduction.

Let X_1 be the reduced closed sub-prescheme of X whose underlying space is an irreducible component of X of *maximal* dimension among those containing an irreducible component of T, and set

$$\mathrm{U}_1 = \mathrm{U} \cap \mathrm{X}_1, \qquad \mathrm{T}_1 = \mathrm{T} \cap \mathrm{X}_1 = \mathrm{X}_1 - \mathrm{U}_1.$$

Then $\operatorname{codim}(\mathrm{T}_1, \mathrm{X}_1) \geqslant 2$ (**0**, 14.2.1), and the pair $(\mathrm{U}_1, \mathrm{X}_1)$ satisfies the same hypotheses as (U, X); indeed, X_1 is the spectrum of a quotient of A and is therefore local Noetherian and complete. We also have a commutative diagram

$$\begin{tikzcd} \mathrm{U}_1 \arrow[r, "j"] \arrow[d, "f_1"'] & \mathrm{U} \arrow[d, "f"] \\ \mathrm{X}_1 \arrow[r, "i"'] & \mathrm{X}, \end{tikzcd}$$

where i and j are the canonical injections, and i is therefore *affine*. From (21.12.1) we obtain a morphism $j^0 : \mathrm{U}_1^0 \to \mathrm{U}^0$ making commutative the diagram

$$\begin{tikzcd} \mathrm{U}_1^0 \arrow[r, "j^0"] \arrow[d, "f_1^0"'] & \mathrm{U}^0 \arrow[d, "f^0"] \\ \mathrm{X}_1 \arrow[r, "i"'] & \mathrm{X}. \end{tikzcd}$$

Consequently, to prove that $x \in f^0(\mathrm{U}^0)$, it suffices to prove that $x \in f_1^0(\mathrm{U}_1^0)$. We may therefore replace X by X_1 and suppose in addition that A is *integral*. By the Cohen structure theorem (**0**, 19.8.8, (i)), A is a quotient of a regular Noetherian ring; since it is integral, we may apply (**IV**, 5.11.1) with the family (x_α) reduced to the single maximal point of X. As $\operatorname{codim}(\mathrm{T}, \mathrm{X}) \geqslant 2$, it follows that $f_*(\mathscr{O}_{\mathrm{U}})$ is a *coherent* $\mathscr{O}_{\mathrm{X}}$-Module, so $f^0 : \mathrm{U}^0 \to \mathrm{X}$ is *finite*. Since it is dominant (U being everywhere dense), it is surjective (**II**, 6.1.10), and therefore $x \in f^0(\mathrm{U}^0)$. □

Corollary (21.12.7). — *Let* X *be a locally Noetherian prescheme,* U *a sub-prescheme induced on an open of* X*,* $j : \mathrm{U} \to \mathrm{X}$ *the canonical immersion; suppose that* j *is an affine morphism. Then every irreducible component of* $\mathrm{T} = \mathrm{X} - \mathrm{U}$ *is of codimension* $\leqslant 1$ *(and hence of codimension* 1 *if* U *is everywhere dense).*

Proof. Suppose indeed that one of the irreducible components T_1 of T is of codimension $\geqslant 2$ in X. Replacing X if needed by an open neighbourhood of the generic point of T_1, we can suppose T irreducible and of codimension $\geqslant 2$. But then the hypothesis that $j : \mathrm{U} \to \mathrm{X}$ is affine implies that **IV-4** | 308
U^0 is identified with U and j^0 with j, in other words $j^0(\mathrm{U}^0) = \mathrm{U}$; but this contradicts the conclusion of (21.12.6) which, under the hypothesis of dimension, says that j^0 must be surjective. □

(21.12.8). Let A be a local Noetherian ring, $\mathrm{Y} = \operatorname{Spec}(\mathrm{A})$, y the unique closed point of Y, $\mathrm{Y}' = \mathrm{Y} - \{y\}$. Consider the following condition:

- (W) *For every open* U *of* Y *contained in* Y'*, containing no irreducible component of* Y' *and such that the canonical injection* $\mathrm{U} \to \mathrm{Y}'$ *is affine,* U *is itself an affine open.*

This condition is implied by the following:

- (W′) *For every closed subset* T *of* Y *whose every irreducible component is of codimension* 1 *and such that for every irreducible component* Y_i *of* Y*,* $\mathrm{Y}_i \cap \mathrm{T} \neq \{y\}$*, the open* $\mathrm{U} = \mathrm{Y} - \mathrm{T}$ *is affine.*

Indeed, if (W′) holds and U satisfies the hypotheses of (W), (21.12.7) shows that no irreducible component of $\mathrm{T} = \mathrm{Y} - \mathrm{U}$ can have codimension $\geqslant 2$; moreover, U contains none of the irreducible components $\mathrm{Y}_i - \{y\}$ of Y', so (W′) shows that U is affine.

We note that condition (W′) simplifies when Y is *irreducible* and is then equivalent to the following:

- (W″) *For every closed irreducible subset* T *of* Y*, of codimension* 1 *in* Y*,* $\mathrm{Y} - \mathrm{T}$ *is an affine open.*

Indeed, it is clear that (W′) implies (W″) when Y is irreducible, and the converse is true by considering the irreducible components of T and of Y and noting that the intersection of a finite number of affine open subsets is an affine open (**I**, 5.5.6).

(21.12.9). *Examples.* — If A is a local Noetherian *factorial* ring, it satisfies (W′) (and *a fortiori* (W)), since every prime ideal of height 1 is principal (**I**, 1.3.6). There are nevertheless nonfactorial local Noetherian rings satisfying (W″), for example in dimension $\leqslant 1$: indeed, as remarked in the proof of (21.12.6), (i), *all* open subsets of Y are then affine. On the other hand, the theory of local duality (chap. III, Part 3) shows that every Noetherian local ring of dimension 2 satisfies (W′).

The interest of condition (W) resides in the following result:

Proposition (21.12.10). — *Let* X, Y *be two locally Noetherian preschemes,* $g : X \to Y$ *a morphism,* y *a closed point of* Y, $Y' = Y - \{y\}$, *and* $X' = g^{-1}(Y')$; *suppose that the restriction* $g' : X' \to Y'$ *is an open immersion, and that the local ring* $\mathscr{O}_{Y,y}$ *satisfies condition* (W) *of* (21.12.8). *Then, for every irreducible component* Z *of* $X_y = g^{-1}(y)$, *either* Z *has codimension* $\leqslant 1$ *in* X, *or its generic point is isolated in* Z. *If* Z *is locally of finite type over* $\boldsymbol{k}(y)$, *the second alternative implies that* Z *is reduced to a single point.*

Proof. The last assertion of the statement results from the fact that Z is then a prescheme of Jacobson (**I**, 10.4.7), and since the set of closed points of Z is then dense in Z, the generic point of Z cannot be isolated in Z unless Z is reduced to a single point.

Suppose that X_y has an irreducible component Z whose generic point z is not isolated in Z and for which $\operatorname{codim}(Z, X) \geqslant 2$. The question is local on X and Y; since z is not isolated in Z, after replacing X by an open neighbourhood of z we may suppose that X and Y are affine, that $X_y = Z$ is irreducible and not reduced to a single point, and that $\operatorname{codim}(X_y, X) \geqslant 2$. The image $g(X - X_y) = U$ is an open subset of Y′ isomorphic to $X - X_y$ by hypothesis. We claim that, after shrinking X around z if necessary, U contains none of the irreducible components of Y′ whose closure contains y. **IV-4** | 309

We may first suppose that every maximal point of X is a generisation of z; there are only finitely many such points. The maximal points of U are then the images $y_i = g(x_i)$ of the maximal points x_i of X (no maximal point of X can lie in X_y, since $\dim(\mathscr{O}_{X,z}) \geqslant 2$). Put $X_i = \overline{\{x_i\}}$. By hypothesis $z \in X_i$, and because z is not isolated in X_y, there is a point $x \in X_y$ with $x \neq z$. Since $X_i \neq Z$, there exists a point $t_i \in X_i$ such that, if $T_i = \overline{\{t_i\}}$, then $\dim(\mathscr{O}_{T_i,x}) = 1$ (**I**, 10.5.9). Hence $t_i \notin Z$, so t_i is not a generisation of z. Replacing X by $X' = X - \bigcup_i T_i$, the image under g of $X' - X_y$ contains none of the points $g(t_i)$, and consequently none of the irreducible components of Y′ whose closure contains y.

Since X and Y are affine, $g : X \to Y$ is affine, and so is its restriction $g' : X' \to Y'$. Because g' is also an open immersion, the canonical immersion $U \to Y'$ is affine. Set $Y_1 = \operatorname{Spec}(\mathscr{O}_{Y,y})$, $Y'_1 = Y_1 - \{y\} = Y' \cap Y_1$, and $U_1 = U \cap Y_1$. The preceding paragraph shows that $U_1 \to Y'_1$ is affine, hence (W) says that U_1 is an affine open of Y_1, or equivalently that $U_1 \to Y_1$ is affine. This immersion is of finite presentation (**I**, 6.2); applying (**IV**, 8.10.5, (viii)) by the method of (**IV**, 8.1.2, a)), and shrinking Y to an affine open neighbourhood of y if necessary, we may suppose that $U \to Y$ is affine. It would follow that the open $X - X_y$ of X, isomorphic to U, is affine, contradicting (21.12.7). This proves the proposition. □

Theorem (21.12.12). — (van der Waerden) — *Let* X, Y *be two locally Noetherian integral preschemes,* $g : X \to Y$ *a birational morphism and locally of finite type. Suppose furthermore that* Y *is normal and that for all* $y \in Y$, *the local ring* $\mathscr{O}_{Y,y}$ *satisfies the condition* (W) *of* (21.12.8) *(conditions satisfied in particular when* Y *is* locally factorial (21.12.9)*). If* V *is the largest open of* X *such that the restriction* $g|V : V \to Y$ *is a local isomorphism, all the irreducible components of* $T = X - V$ *are of codimension* 1 *in* X. **IV-4** | 310

Proof. We note that since g is birational, the open V is not empty; it suffices therefore to prove that a maximal point z of T cannot be isolated in X_y, where $y = g(z)$. But, after restricting to an open neighbourhood of z, we may reduce to the case $T = X_y$; then every fibre $g^{-1}(y')$ ($y' \in Y$) would be empty or reduced to a single point, and the "Main theorem" (8.12.10) would imply that g is a local isomorphism, contradicting $z \in T$. □

Corollary (21.12.13). — *Suppose the hypotheses of (21.12.12) hold and, in addition, that g is quasi-finite at every point of $X^{(1)}$ (recall that these are the points x for which $\dim(\mathscr{O}_{X,x}) = 1$). Then g is a local isomorphism; if moreover g is separated, it is an open immersion.*

Proof. It suffices to prove the first assertion, the second being a consequence of the first and of (**I**, 8.2.8). It all comes down to proving, with the notations of (21.12.12), that $T = \varnothing$; in the contrary case, a generic point z of an irreducible component of T would belong to $X^{(1)}$ by virtue of (21.12.12), and by hypothesis it would be isolated in X_y with $y = g(z)$; but as we saw in the proof of (21.12.12) this is not possible. □

Remarks (21.12.14). — (i) The conclusion of (21.12.12) applies when Y is regular and integral, by virtue of the Auslander-Buchsbaum theorem (21.11.1). On the contrary, one knows examples where X and Y are normal algebraic schemes of dimension 3, over an algebraically closed field of any characteristic, and where the conclusion of (21.12.12) is no longer valid.

(ii) The set T of (21.12.12) is also the set of points where g is *ramified*. Indeed, if g is unramified at a point $x \in X$, it is also unramified in an open neighbourhood of x (17.3.7), hence $g|U$ is a separated, quasi-finite and birational morphism. Since Y is normal, we conclude from the "Main theorem" (**III**, 4.4.9) that g is an isomorphism local at the point x, hence $x \in X - T$. Conversely, g is evidently unramified at every point where it is a local isomorphism. This justifies the title given to this section.

(iii) Without assuming that Y is locally factorial, but assuming instead that X is a complete inter- **IV-4** | 311
section over Y (chap. V), we shall see in chap. V a result more precise than (21.12.12), expressing T as the 1-codimensional cycle of an invertible $\mathscr{O}_X$-Module canonically associated with g. This applies notably when X and Y are both regular, or when g is a S-morphism, X and Y being smooth preschemes over S.

(iv) One may ask whether (21.12.10) has a converse. In other words, let A be a local Noetherian ring, set $Y = \mathrm{Spec}(A)$, and let y be its closed point. Suppose that, for every locally Noetherian prescheme X and every morphism $g : X \to Y$ for which $g' : X' \to Y'$ is an open immersion, every irreducible component of X_y has codimension $\leqslant 1$ in X or is reduced to a single point. Must A then satisfy condition (W) of (21.12.8)?

(v) Let Y be an integral *regular* prescheme, X a locally Noetherian *normal* prescheme, $g : X \to Y$ a morphism locally of finite type; suppose furthermore that, if η is the generic point of Y, the fibre X_η is *étale* over $\boldsymbol{k}(\eta)$, and let T' be the complement of the largest open V' of X such that $g|V' : V' \to Y$ is *étale*; is it true that all the irreducible components of T' are of codimension 1? This is indeed the case as well when one moreover supposes that g is *locally quasi-finite* ("purity theorem" of Zariski-Nagata). One can show that the preceding conjecture would follow from the following statement (and would even be equivalent to it if the conjecture in (iv) held): if a regular local ring A is contained in a local integral and integrally closed ring B which is finite as an A-module, then the open subset U of Spec(B) at whose points Spec(B) is unramified over Spec(A)—or, equivalently by (**IV**, 18.10.1), étale over Spec(A)—is affine.

Proposition (21.12.15). — *Let S be a prescheme, $g : X \to S$ a flat morphism locally of finite presentation, whose fibres $X_s = g^{-1}(s)$ are geometrically irreducible (4.5.2), $h : Y \to S$ a smooth morphism; denote by $Y_s = h^{-1}(s)$ the fibres of this morphism. Let $f : X \to Y$ be a proper S-morphism such that, for every maximal point η of S, the morphism $f_\eta : X_\eta \to Y_\eta$ is an isomorphism. Then f is an isomorphism.*

Proof. Since h is flat, every maximal point of Y lies above a maximal point of S (2.3.4), hence the **IV-4** | 312
union of the Y_η, as η ranges over the set of maximal points of S, is dense in Y; since the f_η are isomorphisms, f is dominant and hence *surjective* since it is proper. We must therefore show that f is an *open immersion*. In view of (17.9.5), it suffices to prove that $f_s : X_s \to Y_s$ is an open immersion for every $s \in S$.

Every $s \in S$ is a generisation of a maximal point η. Making the base change $\mathrm{Spec}(\mathscr{O}_{S',s}) \to S$, where S' is the reduced sub-prescheme of S with underlying space $\overline{\{\eta\}}$, we may first reduce to the case where S is a local integral scheme with closed point s and generic point η; the fibres at s and η and the relevant properties of g, h, and f are unchanged by this base change. The question is local on Y. Replacing Y by an affine open neighbourhood of a point of Y_s, and X by its inverse image (which is quasi-compact because f is proper), we may also suppose that X and Y are of finite presentation over S. There then exist a local Noetherian subring A_0 of $A = \mathscr{O}_{S,s}$, with $A_0 \to A$ local, finite-type morphisms

$$g_0 : X_0 \longrightarrow S_0 = \operatorname{Spec}(A_0), \qquad h_0 : Y_0 \longrightarrow S_0,$$

and an S_0-morphism $f_0 : X_0 \to Y_0$ from which g, h, and f are obtained by base change (**IV**, 8.9.1 and 5.13.3). We may moreover suppose that g_0 is flat (**IV**, 11.2.7), h_0 smooth (**IV**, 17.7.0), and f_0 proper (**IV**, 8.10.5, (xii)). The image in S_0 of the generic point η of S is the generic point η_0 of S_0, and $(f_0)_{\eta_0} : (X_0)_{\eta_0} \to (Y_0)_{\eta_0}$ is an isomorphism (**IV**, 2.7.1, (viii)). The closed point s of S lies above the closed point s_0 of S_0, and $(X_0)_{s_0}$ is geometrically irreducible (**IV**, 4.5.6). Thus we may reduce to the case where S is the spectrum of a Noetherian local integral ring, while weakening the fibre hypothesis to require only that X_s and X_η be geometrically irreducible; under these hypotheses we must prove that f_s is an open immersion.

There exists a discrete valuation ring A' and a morphism $S' = \operatorname{Spec}(A') \to S$ carrying the closed point a of S' to s and the generic point b to η (**II**, 7.1.9). Let g', h', and f' be obtained from g, h, and f by this base change. They again satisfy the hypotheses of (21.12.15); if $f'_a : X'_a \to Y'_a$ is an open immersion, then so is $f_s : X_s \to Y_s$ by (**IV**, 2.7.1, (x)). We may therefore finally suppose that S is the spectrum of a discrete valuation ring.

Since $h : Y \to S$ is smooth, Y is regular (**IV**, 17.5.8); as the question is local on Y, we may suppose that Y is integral. Because $g : X \to S$ is flat, the maximal points of X lie in X_η (**IV**, 2.3.4); since f_η is an isomorphism, X is irreducible and the local ring at its generic point is a field. Moreover, flatness (**IV**, 3.3.2) shows that X has no embedded prime cycles, so X is reduced (**IV**, 3.2.1) and hence integral, while f is birational and separated. We may therefore apply (21.12.13); it remains only to check that f is quasi-finite at the points of $X^{(1)}$. This is clear for those lying in X_η. By (6.1.1), the only point of $X^{(1)}$ lying in X_s is the generic point x of X_s. By (2.4.6) and (14.2.2), g and h are equidimensional; since X_η and Y_η are isomorphic, the irreducible components of X_s and Y_s all have the same dimension. The scheme X_s is irreducible by hypothesis, and f_s is surjective because f is; hence Y_s is also irreducible. If y is its generic point, then $f(x) = y$, and $\dim(X_s) = \dim(Y_s)$ gives $\dim(f_s^{-1}(y)) = 0$ by (5.6.6). Thus f_s, and hence f, is quasi-finite at x, completing the proof. □

Corollary (21.12.16). — *Let $g : X \to S$ be a proper, flat morphism of finite presentation, $h : Y \to S$ a smooth, proper morphism of finite presentation, $f : X \to Y$ an S-morphism. Suppose that the fibres $X_s = g^{-1}(s)$ of g are geometrically irreducible. Let* U *be the set of $s \in S$ such that $f_s : X_s \to Y_s$ is an isomorphism. Then* U *is open and closed in* S*, and the morphism $g^{-1}(U) \to h^{-1}(U)$, restriction of f, is an isomorphism.*

Proof. The last assertion follows from the first and from (17.9.5). We already know (9.6.1), (xi) that U is locally constructible in S. By virtue of (1.10.1), it suffices to prove that U is stable by *specialisation* and by *generisation*. To prove these assertions, one can, by a change of base $\operatorname{Spec}(\mathscr{O}_{S,s}) \to S$, reduce to the case where S is a local integral scheme of closed point s and generic point η.

Stability under specialisation follows from (21.12.15). **IV-4** | 313

To prove stability under generisation, we argue as in the proof of (21.12.15) (noting, with the same notation, that the image of the closed point of S is the closed point of S_0) and reduce to the case where S is moreover *Noetherian*; the conclusion then follows from (**III**, 4.6.7, (ii)). □

Remarks (21.12.17). — (i) The conclusion of the proposition (21.12.15) is no longer valid if we eliminate the hypothesis that the fibres X_s are irreducible. Take indeed $S = \operatorname{Spec}(A)$, where A is a discrete valuation ring, $Y = \mathbb{P}^1_S$, which is proper and smooth over S (17.3.9). Let s and η denote the closed and generic points of S; let z be a closed point of Y_s, for example a k-rational point where $k = \boldsymbol{k}(s)$, and let X be the Y-prescheme obtained by blowing up z. Since A[T] is regular of dimension 2, the argument of the proof of (15.1.1.6) shows that, if $f : X \to Y$ is the structural morphism, $f^{-1}(z)$ is isomorphic to $\mathbb{P}^1_k$; moreover, the complement of $f^{-1}(z)$ in X_s is isomorphic to $Y_s - \{z\}$. Thus $Z_1 = f^{-1}(z)$ and the closure Z_2 in X_s of its complement are the two irreducible components of X_s. The morphism f is plainly proper, and $g = h \circ f$ is flat: X is integral (**II**, 8.1.4) and, for every affine open U of X, the homomorphism $A \to \Gamma(U, \mathscr{O}_X)$ is injective (**I**, 1.2.7), so the **IV-4** | 313 latter is a torsion-free A-module and hence flat because A is a discrete valuation ring ($\mathbf{0}_{\mathrm{I}}$, 6.3.4). It is clear that $f_\eta : X_\eta \to Y_\eta$ is an isomorphism, whereas f_s is not.

(ii) The conclusion of (21.12.15) also fails if, under the hypotheses, we suppress the hypothesis that f is *proper*. Take S and Y as in (i), but replace X by the complement X' of Z_2 in the prescheme constructed there, and replace f by its restriction $f' : X' \to Y$. The restriction $g' : X' \to S$ of g is still flat, and now X'_s is geometrically irreducible. Moreover, X'_s is an affine scheme isomorphic to the complement of a closed point in $\mathbb{P}^1_k$, hence to $\mathbb{A}^1_k$; since its image under f'_s is reduced to the closed point z, f' is not proper and f'_s is not an isomorphism, although f'_η is.

(iii) It is possible that the statement of the proposition (21.12.15) remains valid when we replace the word "isomorphism" by "étale morphism" (cf. (21.12.14), (v)). It will then also be the same of (21.12.16).

21.13. Parafactorial pairs. Parafactorial local rings

Definition (21.13.1). — Let X be a ringed space, Y a closed subset of X, $U = X - Y$. We say that the pair (X, Y) is *parafactorial* if, for every open V of X, the restriction functor $\mathscr{L} \mapsto \mathscr{L}|(U \cap V)$, from the category of $\mathscr{O}_V$-Modules invertibles into that of $\mathscr{O}_{U\cap V}$-Modules invertibles, is an *equivalence of categories*.

To say that the pair (X, Y) is parafactorial means therefore that, for every open V of X, the following conditions are satisfied:

1° the functor $\mathscr{L} \mapsto \mathscr{L}|(U \cap V)$ is *fully faithful*, in other words, for two $\mathscr{O}_V$-Modules invertibles $\mathscr{L}, \mathscr{L}'$, the map of restriction

$$\mathrm{Hom}_{\mathscr{O}_V}(\mathscr{L}, \mathscr{L}') \longrightarrow \mathrm{Hom}_{\mathscr{O}_{U\cap V}}(\mathscr{L}|(U \cap V), \mathscr{L}'|(U \cap V))$$

is *bijective*;

2° the functor $\mathscr{L} \mapsto \mathscr{L}|(U \cap V)$ is *essentially surjective*, in other words, for every invertible $\mathscr{O}_{U\cap V}$-module $\mathscr{L}_0$, there exists an invertible $\mathscr{O}_V$-module $\mathscr{L}$ such that $\mathscr{L}_0$ is isomorphic to **IV-4** | 314
$\mathscr{L}|(U \cap V)$; this is again expressed by saying that the canonical homomorphism (21.3.2.4)

$$\mathrm{Pic}(V) \longrightarrow \mathrm{Pic}(U \cap V)$$

is *surjective*.

Lemma (21.13.2). — *Let $f : X' \to X$ be a morphism of ringed spaces; for every open V of X, denote $f_V : f^{-1}(V) \to V$ the restriction of f. The following conditions are equivalent:*

a) *For every open V of X, the functor $\mathscr{E} \mapsto f_V^*(\mathscr{E})$ from the category of $\mathscr{O}_V$-Modules locally free of finite rank, into the category of $\mathscr{O}_{f^{-1}(V)}$-Modules locally free of finite rank, is faithful (resp. fully faithful).*

a') *For every open V of X, the functor $\mathscr{L} \mapsto f_V^*(\mathscr{L})$ from the category of $\mathscr{O}_V$-Modules locally free of rank 1 into the category of $\mathscr{O}_{f^{-1}(V)}$-Modules locally free of rank 1 is faithful (resp. fully faithful).*

b) *For every open V of X, the canonical homomorphism ($\mathbf{0}_{\mathrm{I}}$, 4.4.3.2) $\mathscr{E} \to (f_V)_*(f_V^*(\mathscr{E}))$ is injective (resp. an isomorphism) for every $\mathscr{O}_V$-Module locally free of finite rank $\mathscr{E}$.*

b') *The canonical homomorphism $\mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ is a monomorphism (resp. an isomorphism).*

Suppose that the canonical homomorphism $\mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ is bijective. Then, for every $\mathscr{O}_{X'}$-Module locally free of finite rank $\mathscr{F}$, $\mathscr{F}$ is of the form $f^*(\mathscr{E})$, where $\mathscr{E}$ is an $\mathscr{O}_X$-Module locally free of finite rank, if and only if the two following conditions are satisfied:

(i) *$f_*(\mathscr{F})$ is an $\mathscr{O}_X$-Module locally free;*

(ii) *the canonical homomorphism ($\mathbf{0}_{\mathrm{I}}$, 4.4.3.3) $f^*(f_*(\mathscr{F})) \to \mathscr{F}$ is an isomorphism.*

When these two conditions are satisfied, $\mathscr{E}$ is isomorphic to $f_*(\mathscr{F})$.

Proof. To say that the functor $\mathscr{E} \mapsto f_V^*(\mathscr{E})$ is faithful (resp. fully faithful) means that, for two $\mathscr{O}_V$- **IV-4** | 315
Modules $\mathscr{E}_1, \mathscr{E}_2$ locally free of finite rank, the map

$$\mathrm{Hom}_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2) \longrightarrow \mathrm{Hom}_{\mathscr{O}_{f^{-1}(V)}}(f_V^*(\mathscr{E}_1), f_V^*(\mathscr{E}_2))$$

is injective (resp. bijective). Since this must hold after replacing V by an open $W \subset V$ and $\mathscr{E}_1, \mathscr{E}_2$ by $\mathscr{E}_1|W, \mathscr{E}_2|W$, and since

$$\mathrm{Hom}_{\mathscr{O}_W}(\mathscr{E}_1|W, \mathscr{E}_2|W) = \Gamma(W, \mathscr{H}om_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2)),$$

the condition also means that the canonical homomorphism of sheaves

(21.13.2.1) $$\mathscr{H}om_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2) \longrightarrow (f_V)_*(f_V^*(\mathscr{H}om_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2)))$$

is injective (resp. bijective). But since $\mathscr{E}_1$ and $\mathscr{E}_2$ are locally free of finite rank, $\mathscr{H}om_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2)$, isomorphic to $\mathscr{E}_1^\vee \otimes_{\mathscr{O}_V} \mathscr{E}_2$ ($\mathbf{0}_{\mathrm{I}}$, 5.4.2), is also locally free of finite rank. This already shows that b) implies a); conversely, b) is a particular case of a), in view of the isomorphism $\mathscr{E} \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_V}(\mathscr{O}_V, \mathscr{E})$. It is clear that a') is a particular case of a) and that b') is a particular case of a'). Conversely, since b) is a local property, one may, in order to verify it, restrict to the case $\mathscr{E} = \mathscr{O}_V^n$; this shows that b') implies b).

If the canonical homomorphism $\mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ is bijective, and if conditions (i) and (ii) are satisfied, it is clear that $\mathscr{F} = f^*(\mathscr{E})$ with $\mathscr{E} = f_*(\mathscr{F})$, up to canonical isomorphism. Conversely, suppose that $\mathscr{F} = f^*(\mathscr{E})$, with $\mathscr{E}$ locally free. Since the question is local on X, we may suppose that $\mathscr{E} = \mathscr{O}_X^n$, whence $\mathscr{F} = \mathscr{O}_{X'}^n$; conditions (i) and (ii) then follow from the hypothesis that $\mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ is an isomorphism. □

In this paragraph, we will apply the preceding lemma to a canonical injection $j : \mathrm{U} \to \mathrm{X}$, where U is the ringed space induced by X on an *open* of X. With these notations, (21.13.2) translates as:

Corollary (21.13.3). — *Let* X *be a ringed space,* Y *a closed subset of* X, $\mathrm{U} = \mathrm{X} - \mathrm{Y}$, $j : \mathrm{U} \to \mathrm{X}$ *the canonical injection. The following conditions are equivalent:*

- a) *For every open* V *of* X*, the functor restriction* $\mathscr{E} \mapsto \mathscr{E}|(\mathrm{U} \cap \mathrm{V})$ *from the category of* $\mathscr{O}_V$*-Modules locally free of finite rank into the category of* $\mathscr{O}_{U\cap V}$*-Modules locally free of finite rank is faithful (resp. fully faithful).*
- a′) *For every open* V *of* X*, the restriction functor* $\mathscr{L} \mapsto \mathscr{L}|(\mathrm{U} \cap \mathrm{V})$ *from the category of* $\mathscr{O}_V$*-Modules locally free of rank* 1 *into the category of* $\mathscr{O}_{U\cap V}$*-Modules locally free of rank* 1 *is faithful (resp. fully faithful).*
- b) *For every open* V *of* X*, the canonical homomorphism* $\mathscr{E} \to (j_V)_*(\mathscr{E}|(\mathrm{U} \cap \mathrm{V}))$ *is injective (resp. bijective) for every* $\mathscr{O}_V$*-Module locally free of finite rank* $\mathscr{E}$.
- b′) *The canonical homomorphism* $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ *is injective (resp. bijective).*

Suppose that the canonical homomorphism $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ *is bijective. Then, for an* $\mathscr{O}_U$*-Module locally free of finite rank* $\mathscr{F}$ *to be of the form* $\mathscr{E}|\mathrm{U}$*, where* $\mathscr{E}$ *is an* $\mathscr{O}_X$*-Module locally free of finite rank, it is necessary and sufficient that* $j_*(\mathscr{F})$ *is an* $\mathscr{O}_X$*-Module locally free of finite rank, and in this case, we can take* $\mathscr{E} = j_*(\mathscr{F})$.

Lemma (21.13.4). — *Let* X *be a locally Noetherian prescheme,* Y *a closed subset of* X, $\mathrm{U} = \mathrm{X} - \mathrm{Y}$*, and* $j : \mathrm{U} \to \mathrm{X}$ *the canonical injection. For the canonical homomorphism* $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ *to be injective (resp. bijective), it is necessary and sufficient that* $\operatorname{prof}_Y(\mathscr{O}_X) \geqslant 1$ *(resp.* $\operatorname{prof}_Y(\mathscr{O}_X) \geqslant 2$*) (in other words, that* $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 1$ *(resp.* $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 2$*) for every* $x \in \mathrm{Y}$*).*

Proof. These assertions are indeed particular cases of (5.10.2) and (5.10.5). □

Proposition (21.13.5). — *Let* X *be a ringed space,* Y *a closed subset of* X, $\mathrm{U} = \mathrm{X} - \mathrm{Y}$, $j : \mathrm{U} \to \mathrm{X}$ *the canonical injection. For the pair* (X, Y) *to be parafactorial, it is necessary and sufficient that the canonical homomorphism* $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ *is bijective, and that for every open* V *of* X *and every invertible* $\mathscr{O}_{U\cap V}$*-module* $\mathscr{L}$, $(j_V)_*(\mathscr{L})$ *is an invertible* $\mathscr{O}_V$*-module (notation of (21.13.3)).*

Proof. This is an immediate consequence of the definition (21.13.1) and of (21.13.3). □

Corollary (21.13.6). — *Let* X *be a ringed space,* Y *a closed subset of* X.

- *(i) If the pair* (X, Y) *is parafactorial, then so is* $(\mathrm{W}, \mathrm{Y} \cap \mathrm{W})$ *for every open* W *of* X*. Conversely, if* (W_α) *is an open covering of* X *such that each of the pairs* $(\mathrm{W}_\alpha, \mathrm{Y} \cap \mathrm{W}_\alpha)$ *is parafactorial, then the pair* (X, Y) *is parafactorial.*
- *(ii) If the pair* (X, Y) *is parafactorial, then so is* (X, Y′) *for every closed subset* Y′ *of* Y.
- *(iii) Suppose that* X *is a prescheme, and let* X′ *be a prescheme,* $f : \mathrm{X}' \to \mathrm{X}$ *a faithfully flat and quasi-compact morphism; set* $\mathrm{Y}' = f^{-1}(\mathrm{Y})$*. Suppose that the pair* (X′, Y′) *is parafactorial and that the open* $\mathrm{U} = \mathrm{X} - \mathrm{Y}$ *is retrocompact in* X ($\mathbf{0}_{\mathrm{III}}$, 9.1.1)*. Then the pair* (X, Y) *is parafactorial.*

Proof. (i) Since the bijectivity of the canonical homomorphism $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is a local property on X, it remains only to see, by virtue of (21.13.5), that the following property holds. Denoting **IV-4** | 316 by $j_\alpha : U \cap W_\alpha \to W_\alpha$ the canonical injection, if, for every α and every invertible $\mathscr{O}_U$-Module $\mathscr{L}$, $(j_\alpha)_*(\mathscr{L}|(U \cap W_\alpha))$ is an invertible $\mathscr{O}_{W_\alpha}$-Module, then $j_*(\mathscr{L})$ is an invertible $\mathscr{O}_X$-Module. But being an invertible $\mathscr{O}_X$-Module is local on X, and (W_α) is an open covering of X; since

$$j_*(\mathscr{L})|W_\alpha = (j_\alpha)_*(\mathscr{L}|(U \cap W_\alpha)),$$

this proves the assertion.

(ii) Set $U' = X - Y'$ and let $j' : U' \to X$ be the canonical injection. To say that the canonical homomorphism $\mathscr{O}_X \to j'_*(\mathscr{O}_{U'})$ is bijective amounts to saying that, for every open V of X, the homomorphism $\Gamma(V, \mathscr{O}_X) \to \Gamma(V \cap U', \mathscr{O}_X)$ is bijective. But the composite

$$\Gamma(V, \mathscr{O}_X) \longrightarrow \Gamma(V \cap U', \mathscr{O}_X) \longrightarrow \Gamma(V \cap U, \mathscr{O}_X)$$

is bijective by hypothesis, and, on replacing V by $V \cap U'$, one sees that the second arrow is bijective; hence the first arrow is bijective.

Next let $j'' : U \to U'$ be the canonical injection, and let $\mathscr{L}'$ be an invertible $\mathscr{O}_{U'}$-Module. Then $\mathscr{L}'|U$ is an invertible $\mathscr{O}_U$-Module, so by hypothesis

$$j_*(\mathscr{L}'|U) = j'_*(j''_*(\mathscr{L}'|U))$$

is an invertible $\mathscr{O}_X$-Module. Since the pair $(U', U' \cap Y)$ is parafactorial by (i), one has $j''_*(\mathscr{L}'|U) \cong \mathscr{L}'$, and therefore $j'_*(\mathscr{L}')$ is an invertible $\mathscr{O}_X$-Module. To conclude, it suffices to replace X, U, and U′ in the preceding argument by V, $V \cap U$, and $V \cap U'$, where V is an open subset of X.

(iii) Set $U' = X' - Y' = f^{-1}(U)$ and note that, since we are dealing with preschemes, one may write $U' = U \times_X X'$. Let $f_U : f^{-1}(U) \to U$ be the restriction of f, and let $j' : U' \to X'$ be the canonical injection, also denoted $j_{(X')}$. Let us first show that the canonical homomorphism $\rho : \mathscr{O}_X \to j_*(\mathscr{O}_U)$ is bijective. By hypothesis, the canonical homomorphism $\mathscr{O}_{X'} \to j'_*(\mathscr{O}_{U'})$ is bijective. Since j is quasi-compact and separated by hypothesis, and f is flat, $j'_*(\mathscr{O}_{U'}) = j'_*(f_U^*(\mathscr{O}_U))$ identifies canonically with $f^*(j_*(\mathscr{O}_U))$ by (2.3.1). Since $\mathscr{O}_{X'} = f^*(\mathscr{O}_X)$, the homomorphism $f^*(\rho) : f^*(\mathscr{O}_X) \to f^*(j_*(\mathscr{O}_U))$ is therefore bijective; hence ρ is bijective because f is faithfully flat (2.2.7).

Now let $\mathscr{L}$ be an invertible $\mathscr{O}_U$-Module, and set $\mathscr{L}' = f_U^*(\mathscr{L})$, which is an invertible $\mathscr{O}_{U'}$-Module. By hypothesis, $j'_*(\mathscr{L}') = j'_*(f_U^*(\mathscr{L}))$ is an invertible $\mathscr{O}_{X'}$-Module; as above, it is isomorphic by (2.3.1) to $f^*(j_*(\mathscr{L}))$. Since f is faithfully flat and quasi-compact, it follows that $j_*(\mathscr{L})$ is a locally free, hence invertible, $\mathscr{O}_X$-Module (2.5.2). To complete the proof, it suffices to replace X by an open V and X′ by $f^{-1}(V)$ in the preceding argument. □

Definition (21.13.7). — We say that a *local ring* A is parafactorial if, setting $X = \operatorname{Spec}(A)$ and denoting by a the closed point of X, the pair $(X, \{a\})$ is parafactorial.

Proposition (21.13.8). — *The notations being those of* (21.13.7), *set* $U = X - \{a\}$. *For* A *to be parafactorial, it is necessary and sufficient that it satisfies the following conditions:*

(i) The canonical homomorphism $A = \Gamma(X, \mathscr{O}_X) \to \Gamma(U, \mathscr{O}_X)$ *is bijective.*

(ii) $\operatorname{Pic}(U) = 0$.

If furthermore A *is Noetherian, condition (i) is equivalent to* **IV-4 | 317**

(i bis) $\operatorname{prof}(A) \geqslant 2$.

Proof. Indeed, the only open of X containing a is X itself, and hence every invertible $\mathscr{O}_X$-module is isomorphic to $\mathscr{O}_X$, i.e. $\operatorname{Pic}(X) = 0$; the first assertion follows therefore from the definition (21.13.1) and from (21.13.3). The second assertion is nothing but a particular case of (21.13.4). □

(21.13.9). *Examples.* — (i) A local Noetherian parafactorial ring is necessarily of dimension $\geqslant 2$ by virtue of (21.13.8); in other words a local Noetherian ring of dimension $\leqslant 1$ is not parafactorial.

(ii) A local Noetherian *factorial* ring A of dimension $\geqslant 2$ is parafactorial, as follows from (21.13.8) and (21.6.14).

(iii) If A is a local Noetherian ring of dimension $\geqslant 3$ and parafactorial, it is not necessarily normal nor even reduced. Consider a regular local ring B of dimension $\geqslant 3$, and let $A = D_B(B)$ (**0**, 18.2.3), isomorphic to $B[T]/(T^2)$; we see immediately that $\operatorname{prof}(A) = \operatorname{prof}(B) \geqslant 3$. To show that A is parafactorial, it suffices therefore, with the notations of (21.13.8), to prove that $\operatorname{Pic}(U) = 0$. Let $\mathfrak{J}$ be the ideal of the augmentation $A \to B$, which is such that $\mathfrak{J}^2 = 0$ and which, as a B-module, is isomorphic to B. Since $B = A/\mathfrak{J}$, $X = \operatorname{Spec}(A)$ and $X_0 = \operatorname{Spec}(B)$ have the same underlying space;

if $\mathscr{J} = \widetilde{\mathfrak{J}}$, X_0 is the sub-prescheme defined by $\mathscr{J}$; we will denote U_0 the prescheme induced by X_0 on the open U. For every $z \in \mathfrak{J}$, set $\varphi(z) = 1 + z$; since $(1+z)(1-z) = 1$, $1+z$ is invertible in A, and $\varphi(\mathfrak{J})$ is the kernel of the canonical surjective homomorphism of multiplicative groups $A^* \to B^*$; in other words, we have an exact sequence of commutative groups

$$0 \longrightarrow \mathfrak{J} \xrightarrow{\varphi} A^* \longrightarrow B^* \longrightarrow 1$$

(the three last groups being multiplicative, the two first being additive). For the same reason, for every $t \in A$, we have the exact sequence

$$0 \longrightarrow \mathfrak{J}_t \xrightarrow{\varphi_t} (A_t)^* \longrightarrow (B_{t_0})^* \longrightarrow 1$$

denoting by t_0 the image of t in B, since $B_{t_0} = A_t/\mathfrak{J}_t$; in other words, we have an exact sequence of sheaves of commutative groups on the topological space X

$$0 \longrightarrow \mathscr{J} \xrightarrow{u} \mathscr{O}_X^* \longrightarrow \mathscr{O}_{X_0}^* \longrightarrow 1$$

whence by restriction to the open U, an exact sequence

$$0 \longrightarrow \mathscr{J}|U \longrightarrow \mathscr{O}_U^* \longrightarrow \mathscr{O}_{U_0}^* \longrightarrow 1.$$

By the cohomology exact sequence, we deduce an exact sequence

(21.13.9.1) $$H^1(U, \mathscr{J}) \longrightarrow H^1(U, \mathscr{O}_U^*) \xrightarrow{u_1} H^1(U, \mathscr{O}_{U_0}^*).$$

But since $\mathfrak{J}$ is a B-module isomorphic to B, we have $H^1(U, \mathscr{J}) = H^1(U_0, \mathscr{O}_{U_0})$, and it follows from chap. III, Part 3 (see also [41, III, Example III-1]) that $H^1(U_0, \mathscr{O}_{U_0}) = 0$ by reason of the relation **IV-4** | 318
$\operatorname{prof}(B) \geqslant 3$. On the other hand, since B is factorial, $H^1(U, \mathscr{O}_{U_0}^*) = \operatorname{Pic}(U_0) = 0$; we obtain therefore from the preceding exact sequence $\operatorname{Pic}(U) = H^1(U, \mathscr{O}_U^*) = 0$.

(iv) There also exist local Noetherian rings of dimension 3 which are integral, integrally closed and parafactorial, but not factorial. Let indeed B be a local Noetherian ring, complete, integral, integrally closed, of dimension $\geqslant 2$ and not factorial (for example the completion of the localised polynomial algebra $k[U, V, W]/(W^2 - UV)$ at the ideal image of $(U) + (V) + (W)$). Then it follows from what we will see below (21.14.2) that the local ring $A = B[[T]]$ of formal power series in T over B is parafactorial, but it is not factorial, without which B would be so by virtue of (21.13.12) below.

(v) One can show that an absolute complete-intersection local ring (19.3.1) of dimension $\geqslant 4$ is parafactorial (cf. [41, XI, 3.13 (i)]).

(vi) We have seen (Remark (ii)) that every local Noetherian factorial ring A of dimension 2 is parafactorial. But there are local Noetherian rings of dimension 2 which are parafactorial and not factorial. One can show that, for a local Noetherian ring A of dimension 2 to be parafactorial and not factorial, it is necessary and sufficient that it satisfy the following three conditions:

1° A is a Cohen-Macaulay ring (in other words $\operatorname{prof}(A) = 2$).

2° A is integral and if A' is its integral closure, A' is factorial and is a finite A-algebra.

3° Let $\mathfrak{J}$ be the conductor of A in A' (the annihilator of the A-module A'/A, or equivalently the largest ideal of A' contained in A); set $B = A/\mathfrak{J}$, $B' = A'/\mathfrak{J}$. Then $\dim(B) = 1$ (which implies $A' \neq A$, i.e. A is not integrally closed), and the canonical map $\operatorname{Div}(B) \to \operatorname{Div}(B')$ (21.4.5) is *surjective*.

One can furthermore show that these conditions imply the following property:

4° The ring B' (and, a fortiori, $B \subset B'$) is reduced, and the morphism $\operatorname{Spec}(B') \to \operatorname{Spec}(B)$ is bijective.

If we set $X = \operatorname{Spec}(A)$, $X' = \operatorname{Spec}(A')$, $Y = \operatorname{Spec}(B)$, and $Y' = \operatorname{Spec}(B')$, so that Y (resp. Y') is defined by the ideal $\mathfrak{J}$ of A (resp. A'), the structural morphism $f : X' \to X$ is an isomorphism of $X' - Y'$ onto $X - Y$ (Bourbaki, *Alg. comm.*, chap. V, §1, no. 5, Cor. 5 of Prop. 16). By 4°, f is a bijective morphism; in other words, X is a Noetherian local unibranch prescheme (6.15.1) (and in particular A' is a Noetherian local ring). In general, X is not geometrically unibranch.

The space Y, of dimension 1, consists of the closed point x of X and the maximal points y_i ($1 \leqslant i \leqslant r$) of Y; the unique point y_i' of Y' above y_i is likewise a maximal point of Y'. Since B and B' are reduced, it follows that

$$\mathfrak{m}_{y_i}\mathscr{O}_{X',y_i'} = \mathfrak{m}_{\mathscr{O}_{X',y_i'}}.$$

Thus the relation $\mathfrak{m}_z\mathscr{O}_{X',z'} = \mathfrak{m}_{z'}$ holds for $z = y_i$ and $z' = y_i'$, and it is plainly also true for $z \notin Y$ and the unique point z' of $f^{-1}(z)$; hence it holds for every $z \in U = X - \{x\}$ and the unique point z' of $f^{-1}(z)$. In particular, if A has characteristic 0 (21.1.1), then U′ is *unramified* over U (but not étale in general).

We shall confine ourselves here to proving that conditions 1°, 2°, and 3° above are sufficient for A to be parafactorial. By condition 1° and (21.13.8), it suffices to show that $\mathrm{Pic}(U) = 0$. Since A′ is factorial by 2°, one has $\mathrm{Pic}(U') = 0$, and (21.8.5), (ii) gives an exact sequence

$$1 \longrightarrow \left(\prod_{i=1}^{r} \mathscr{O}^*_{U',y_i'}/\mathscr{O}^*_{U,y_i}\right)/\operatorname{Im}\Gamma(U',\mathscr{O}^*_{U'}) \longrightarrow \mathrm{Pic}(U) \longrightarrow \mathrm{Pic}(U') \longrightarrow 0.$$

It therefore remains to show that the second term of this sequence is reduced to the neutral element. Here we take $S = \{y_i \mid 1 \leqslant i \leqslant r\}$ and note that $f_*(\mathscr{O}_{X'})$ is the $\mathscr{O}_X$-algebra $\widetilde{A'}$. Let $\mathfrak{p}_i$, $\mathfrak{p}_i'$ be the prime ideals of A and A′ corresponding to y_i, y_i', and note that $\mathfrak{p}_i' \cap A = \mathfrak{p}_i$. For each i, choose an invertible element a_i'/s_i of $A'_{\mathfrak{p}_i'}$, with $a_i' \in A'$ and $s_i \notin \mathfrak{p}_i'$. Since only the quotient groups $(A'_{\mathfrak{p}_i'})^*/(A_{\mathfrak{p}_i})^*$ are at issue, we may suppose that $s_i = 1$ for every i. Let $b_i' \in B' = A'/\mathfrak{J}$ be the canonical image of a_i', so that $b_i'/1$ is a nonzero element of $\kappa(y_i') = \mathscr{O}_{Y',y_i'}$.

The open $U \cap Y$, consisting of the points y_i, is an affine open of the form $D(t)$ in Y, with $t \in \mathfrak{m}$, the maximal ideal of A. There is therefore an invertible element $b' \in B'_t$ whose canonical images are the $b_i'/1$. Since t is regular in the integral ring A′, we may write $b' = b''/t^m$, where $b'' \in B'$ is invertible. Let $a'' \in A'$ represent the class b''; it is necessarily invertible. Then $a' = a''/t^m$ is invertible in A'_t, and for every i one has

$$u_i = a'' - a_i't^m \in \mathfrak{J}, \qquad t^m a_i' = a'' - u_i = a''(1 - a''^{-1}u_i).$$

But $a''^{-1}u_i \in \mathfrak{J} \subset \mathfrak{m}$, so $1 - a''^{-1}u_i$ is invertible in A; hence the classes of a' and a_i' in $(A'_{\mathfrak{p}_i'})^*/(A_{\mathfrak{p}_i})^*$ are the same, which completes the proof.

For an explicit example of a parafactorial ring of dimension 2 obtained in this way and not factorial, consider **IV-4** | 319

$$E = \mathbb{R}[[U,V]]/(U^2+V^2),$$

whose integral closure E′ identifies with $\mathbb{C}[[U]]$ (6.15.11). If $A = E[[T]]$, its integral closure is $A' = E'[[T]]$ (Bourbaki, *Alg. comm.*, chap. V, §1, no. 4, Prop. 14). One verifies at once that the conductor of E in E′ is the maximal ideal $\mathfrak{n}$ of E, hence that the conductor $\mathfrak{J}$ of A in A′ is $\mathfrak{n}[[T]]$, and one has $B = A/\mathfrak{J} \cong \mathbb{R}[[T]]$, $B' = A'/\mathfrak{J} \cong \mathbb{C}[[T]]$. Conditions 1°, 2°, and 3° above are therefore satisfied, although A is not even integrally closed.

One may vary this example: if k is an algebraically closed field, the localisation at the maximal ideal generated by the images of U, V, and W of

$$k[U,V,W]/(U^2 - WV^2)$$

also satisfies conditions 1°, 2°, and 3° above.

It may be that these three conditions even imply that each of the rings $B'/\mathfrak{p}_i'$ is a discrete valuation ring, which would imply that A′ itself is a regular ring.

Proposition (21.13.10). — *Let* X *be a locally Noetherian prescheme,* Y *a closed subset of* X*. For the pair* (X, Y) *to be parafactorial, it is necessary and sufficient that, for all* $y \in Y$*, the local ring* $\mathscr{O}_{X,y}$ *is parafactorial.*

Proof. Set $U = X - Y$ and let $j : U \to X$ be the canonical injection. **IV-4** | 320

By (21.13.4), to say that the canonical homomorphism $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is bijective amounts to saying that each of the local rings $\mathscr{O}_{X,y}$, for $y \in Y$, satisfies condition (i bis) of (21.13.8).

Let us first show that if the pair (X, Y) is parafactorial, each of the rings $\mathscr{O}_{X,y}$ ($y \in Y$) is parafactorial. Taking into account the preceding remark, it all amounts to seeing that if $T_y = \mathrm{Spec}(\mathscr{O}_{X,y})$ and $U_y = T_y - \{y\}$, every invertible $\mathscr{O}_{U_y}$-module $\mathscr{L}_0$ is isomorphic to $\mathscr{O}_{U_y}$. Now, when V ranges over the open neighbourhoods of y in X, it follows from (8.2.13) and (**I**, 2.4.2) that the prescheme U_y is projective limit of the preschemes induced by X on the opens $V - (V \cap \overline{\{y\}})$, which are quasi-compact since X is Noetherian. Since the opens $V - (V \cap \overline{\{y\}})$ are separated, it follows from (8.5.2), (ii) and (8.5.5) that there exist an affine open neighbourhood V of y in X and an invertible module $\mathscr{L}'_V$ on $V - (V \cap \overline{\{y\}})$ which induces $\mathscr{L}_0$ on U_y. But the hypothesis implies, by (21.13.6), (i), that the pair $(V, V \cap \overline{\{y\}})$ is parafactorial. Hence there exists an invertible $\mathscr{O}_V$-Module $\mathscr{L}_V$ inducing $\mathscr{L}'_V$ on $V - (V \cap \overline{\{y\}})$. Replacing V if necessary by a smaller open neighbourhood of y, we may suppose that $\mathscr{L}_V \cong \mathscr{O}_V$; it follows that $\mathscr{L}_0 \cong \mathscr{O}_{U_y}$.

Conversely, let us prove that if all the $\mathscr{O}_{X,y}$ ($y \in Y$) are parafactorial, the pair (X, Y) is parafactorial. This is evidently reduced, taking into account the remark at the beginning, to showing that for every invertible $\mathscr{O}_U$-module $\mathscr{L}$, $j_*(\mathscr{L})$ is an invertible $\mathscr{O}_X$-module. The set of $x \in X$ such that the restriction of $j_*(\mathscr{L})$ to an open neighbourhood of x in X is invertible is evidently open in X; it is a matter of showing that $V = X$. For this, suppose the contrary and set $Z = X - V \neq \varnothing$. Let y be a *maximal* point of Z; since $Z \subset Y$, $\mathscr{O}_{X,y}$ is by hypothesis a parafactorial ring. Replacing X if necessary by an open neighbourhood of y, we may suppose that the restriction of $j_*(\mathscr{L})$ to $V - (V \cap \overline{\{y\}})$ is invertible. With the notation introduced above, the invertible $\mathscr{O}_{U_y}$-Module $\mathscr{L}_0$ induced by $j_*(\mathscr{L})$ is therefore invertible. Since $\mathscr{O}_{X,y}$ is parafactorial, $\mathscr{L}_0$ is induced by an invertible $\mathscr{O}_{T_y}$-Module $\mathscr{L}'$.

As T_y is the projective limit of the open neighbourhoods W of y in X, another application of (8.5.2), (ii) and (8.5.5) yields such a neighbourhood W and an invertible $\mathscr{O}_W$-Module $\mathscr{L}''$ inducing $\mathscr{L}'$ on T_y, and therefore inducing $\mathscr{L}_0$ on U_y. By (8.5.2.5) and (8.5.2), (i), after replacing W by a smaller open neighbourhood of y, we may suppose that the restrictions of $j_*(\mathscr{L})$ and $\mathscr{L}''$ to $W - (W \cap \overline{\{y\}})$ are equal.

Set $V_1 = V \cup W$, and let $\mathscr{L}_1$ be the invertible $\mathscr{O}_{V_1}$-Module equal to $\mathscr{L}''$ on W and to $j_*(\mathscr{L})|V$ on V. Then $U \subset V_1$ and $\mathscr{L}_1|U = \mathscr{L}$. Since $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is bijective, (21.13.2), *b*) shows that $j_*(\mathscr{L})|V_1 \cong \mathscr{L}_1$, hence is invertible. But this contradicts the definition of V, and concludes the proof of (21.13.10). □

Corollary (21.13.11). — *Let* X *be a locally Noetherian prescheme,* Y *a closed subset of* X, $U = X - Y$. *Suppose that the pair* (X, Y) *is parafactorial and that* U *is locally factorial; in other terms* (21.13.10), *for all* $x \in X$, *the local ring* $\mathscr{O}_{X,x}$ *is:* 1° *parafactorial if* $x \in Y$; 2° *factorial if* $x \notin Y$. *Then* X *is locally factorial (in other words,* $\mathscr{O}_{X,x}$ *is in fact factorial for* all $x \in X$).

Proof. Suppose indeed the contrary, and let y be a point of Y such that $\mathscr{O}_{X,y}$ is not factorial and has a minimal dimension among all the points of Y having this property. Then, if we set $T_y = \mathrm{Spec}(\mathscr{O}_{X,y})$, $U_y = T_y - \{y\}$, the hypothesis that $\mathscr{O}_{X,y}$ is parafactorial implies $\mathrm{Pic}(U_y) = 0$ and $\mathrm{prof}(\mathscr{O}_{X,y}) \geqslant 2$ (21.13.8); moreover, the choice of y implies that U_y is locally factorial. But then (21.6.14) proves that $\mathscr{O}_{X,y}$ is factorial, contrary to the hypothesis. □

Proposition (21.13.12). — *Let* A, B *be two local Noetherian rings,* $\rho : A \to B$ *a local homomorphism making* B *a flat A-module. Then, if* B *is factorial, so is* A.

Proof. We already know that A is integral and integrally closed (6.5.4); we may therefore restrict to the case $\dim(A) \geqslant 2$ and argue by induction on $n = \dim(A)$. Let $X = \mathrm{Spec}(A)$, a the closed point of X, $X' = \mathrm{Spec}(B)$, $f : X' \to X$ the structural morphism, $U = X - \{a\}$, $U' = f^{-1}(U)$. The local rings $\mathscr{O}_{X',x'}$ at points $x' \in f^{-1}(a)$ are factorial and of dimension $\geqslant 2$ (6.1.2), hence parafactorial (21.13.9), (ii); we conclude from (21.13.10) that the pair $(X', f^{-1}(a))$ is parafactorial. By virtue of (21.13.6), (iii), this shows that the ring A is parafactorial, hence $\mathrm{prof}(A) \geqslant 2$ and $\mathrm{Pic}(U) = 0$ (21.13.8). Finally, the local rings of U being of dimension $< n$, the induction hypothesis proves that U is *locally factorial*; the conclusion follows then from (21.6.14). □

(21.13.12.1). We note that the statement of (21.13.12) where one replaces "factorial" by "parafactorial" is no longer exact, as shown by the example where $A = k$ is a field and B the local factorial ring $k[[T_1, T_2]]$. However, it follows from (21.13.6), (iii) that, under the conditions of (21.13.12), if $\mathfrak{m}$ denotes the maximal ideal of A and if $\mathfrak{m}B$ is an *ideal of definition* of B (i.e. $\dim(B/\mathfrak{m}B) = 0$), then, when B is parafactorial, so is A. In particular, if the completion $\widehat{A}$ of a local Noetherian ring is parafactorial, so is A. IV-4 | 321

Remark (21.13.13). — Some of the preceding definitions and results belong to more general considerations on the cohomology of sheaves of commutative groups, developed in chap. III, Part 3, for which we have given an independent account here for the convenience of the reader. Indeed, if X is a ringed space, there is a canonical *equivalence* between the category of invertible $\mathscr{O}_X$-modules and that of *principal homogeneous sheaves under the sheaf of groups* $\mathscr{O}_X^*$ (16.5.15). This equivalence associates with every invertible $\mathscr{O}_X$-module $\mathscr{L}$ the sheaf $\mathscr{L}^* = \mathscr{I}som_{\mathscr{O}_X}(\mathscr{O}_X, \mathscr{L})$ of germs of isomorphisms of $\mathscr{O}_X$ onto $\mathscr{L}$; it is immediate that $\mathscr{L}^*$ is canonically equipped with a structure of principal homogeneous sheaf under $\mathscr{O}_X^* \cong \mathscr{I}som_{\mathscr{O}_X}(\mathscr{O}_X, \mathscr{O}_X)$. The verification of this equivalence of categories is immediate. This being said, let X be a topological space, $\mathscr{G}$ a sheaf of groups on X, Y a closed subset of X, and i an integer such that $0 \leqslant i \leqslant 2$. We say that the pair (X, Y) is *i-pure* for $\mathscr{G}$ if, for every open V of X, the restriction functor $\mathscr{P} \mapsto \mathscr{P}|(U \cap V)$, from the category of principal homogeneous sheaves under the sheaf of groups $\mathscr{G}|V$ into the analogous category under $\mathscr{G}|(U \cap V)$, is *faithful* ($i = 0$), resp. *fully faithful* ($i = 1$), resp. an *equivalence of categories* ($i = 2$). In the case where X is a ringed space and $\mathscr{G} = \mathscr{O}_X^*$, to say that (X, Y) is i-pure therefore means, for $i = 0$, that the canonical homomorphism $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is *injective*; for $i = 1$, that this homomorphism is *bijective*; and finally, for $i = 2$, that the pair (X, Y) is *parafactorial* (21.13.3).

Returning to the general case, recall that the morphisms of the category of principal sheaves are by definition isomorphisms. To say that the pair (X, Y) is 0-pure means therefore that for every open V of X, the canonical homomorphism $H^0(V, \mathscr{G}) \to H^0(U \cap V, \mathscr{G})$ is injective; to say that the pair (X, Y) is 1-pure means that the canonical homomorphism $H^0(V, \mathscr{G}) \to H^0(U \cap V, \mathscr{G})$ is bijective (and so also the canonical homomorphism $H^1(V, \mathscr{G}) \to H^1(U \cap V, \mathscr{G})$ is injective); finally, one can show that for the pair (X, Y) to be 2-pure, it is necessary and sufficient that the homomorphisms $H^i(V, \mathscr{G}) \to H^i(U \cap V, \mathscr{G})$ be bijective for $i = 0$ and $i = 1$. When $\mathscr{G}$ is a sheaf of commutative groups, introduce the cohomology sheaves $\mathscr{H}_Y^p(\mathscr{G})$ defined in chap. III, Part 3. To say that (X, Y) is i-pure for $i \leqslant 2$ then means that $\mathscr{H}_Y^p(\mathscr{G}) = 0$ for $p \leqslant i$; in this form, the notion generalises immediately to every integer i.

Proposition (21.13.14). — *Let* X *be a locally Noetherian* normal *prescheme,* $(U_\lambda)_{\lambda \in L}$ *a decreasing filtered family of opens of* X*. The following conditions are equivalent:*

a) *Every* 1*-codimensional cycle* Z *on* X *such that there exists an index* λ *for which* $Z|U_\lambda$ *is locally principal* (21.6.7)*, is locally principal.*
b) *For all* $x \in X$ *such that* $\dim(\mathscr{O}_{X,x}) \geqslant 2$ *and such that* x *does not belong to* $\bigcap_{\lambda \in L} U_\lambda$*, the local ring* $\mathscr{O}_{X,x}$ *is parafactorial.*

Proof. Condition b) may also be expressed as follows: **IV-4** | 322

b') *For every closed subset* Y *of* X *such that* $\operatorname{codim}(Y, X) \geqslant 2$ *and* $Y \subset X - U_\lambda$ *for some* λ*, the pair* (X, Y) *is parafactorial.*

Property a) may in turn be expressed as follows. If the closed set N of points $x \in X$ at which Z is not principal is contained in one of the sets $X - U_\lambda$, then Z is locally principal. Since X is normal, the condition $\dim(\mathscr{O}_{X,x}) \leqslant 1$ implies that $\mathscr{O}_{X,x}$ is a field or a discrete valuation ring, and therefore that Z_x is principal; thus necessarily $\operatorname{codim}(N, X) \geqslant 2$ (5.1.3). Consequently, a) is equivalent to:

a') *If there exists a closed subset* Y *contained in one of the sets* $X - U_\lambda$*, such that* $\operatorname{codim}(Y, X) \geqslant 2$ *and* $Z|(X - Y)$ *is locally principal, then* Z *is locally principal.*

Condition b) implies b') by (21.13.10). Conversely, if b') holds and $x \notin U_\lambda$, then $Y = \{x\}$ is contained in $X - U_\lambda$ and $\operatorname{codim}(Y, X) \geqslant 2$; it follows again from (21.13.10) that $\mathscr{O}_{X,x}$ is parafactorial. This proves the equivalence of b) and b').

We are thus reduced to proving the equivalence of a') and b'). Since X is normal, for every closed subset Y of X such that $\operatorname{codim}(Y, X) \geqslant 2$, one has $\operatorname{prof}_Y(\mathscr{O}_X) \geqslant 2$ (5.8.6). If $U = X - Y$ and $j : U \to X$ is the canonical injection, the homomorphism $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is therefore bijective (21.13.4). Since a') and b') are local conditions, it suffices to prove, for X Noetherian and normal and Y closed with $\operatorname{codim}(Y, X) \geqslant 2$, the equivalence of:

a'') *For every* 1*-codimensional cycle* Z *on* X*, if* $Z|U$ *is locally principal, then* Z *is locally principal.*

b'') *The canonical homomorphism* $\operatorname{Pic}(X) \to \operatorname{Pic}(U)$ *is surjective.*

First suppose a''), and let $\zeta_0 \in \operatorname{Pic}(U)$. Its image in $\mathfrak{Cl}(U)$ (21.6.11) is the class of a locally principal 1-codimensional cycle Z_0 on U. The hypothesis $\operatorname{codim}(Y, X) \geqslant 2$ implies that the restriction homomorphism $\mathscr{Z}^1(X) \to \mathscr{Z}^1(U)$ is bijective; hence $Z_0 = Z|U$ for a 1-codimensional cycle Z on X. By a''), Z is locally principal. It follows from (21.6.11) that the image of Z in $\mathfrak{Cl}(X)$ is the image of a unique element $\zeta \in \operatorname{Pic}(X)$, and it is then clear that ζ_0 is the image of ζ.

Conversely, suppose $b'')$, and let Z be a 1-codimensional cycle on X such that $\mathrm{Z}|\mathrm{U}$ is locally principal. The image of $\mathrm{Z}|\mathrm{U}$ in $\mathfrak{Cl}(\mathrm{U})$ is the image of a unique element $\zeta_0 \in \mathrm{Pic}(\mathrm{U})$ (21.6.11). By hypothesis, there exists $\zeta \in \mathrm{Pic}(\mathrm{X})$ whose image in $\mathrm{Pic}(\mathrm{U})$ is ζ_0; the image of ζ in $\mathfrak{Cl}(\mathrm{X})$ is therefore the class of a locally principal 1-codimensional cycle Z' on X such that $\mathrm{Z}|\mathrm{U}$ and $\mathrm{Z}'|\mathrm{U}$ are linearly equivalent. But U is schematically dense in X, and the isomorphism $\mathscr{Z}^1(\mathrm{X}) \to \mathscr{Z}^1(\mathrm{U})$ carries $\mathscr{Z}^1_{\mathrm{princ}}(\mathrm{X})$ onto $\mathscr{Z}^1_{\mathrm{princ}}(\mathrm{U})$; hence Z and Z' are linearly equivalent. Since Z' is locally principal, so is Z. □

Corollary (21.13.15). — *Let X be a locally Noetherian normal prescheme, S a subset of X. The following conditions are equivalent:*

a) *Every 1-codimensional cycle on X which is principal at the points of S is locally principal.* **IV-4** | 323
b) *For all $x \in \mathrm{X}$ which is not a generisation of a point of S (in other words, such that $\overline{\{x\}} \cap \mathrm{S} = \varnothing$) and such that $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$, the local ring $\mathscr{O}_{\mathrm{X},x}$ is parafactorial.*

Proof. It is clear that the set S' of generisations in X of points of S is the intersection of the open neighbourhoods of S. We may restrict to the case where X is Noetherian, since properties a) and b) are local on X. Every 1-codimensional cycle on X which is principal at the points of S is also principal at the points of an open neighbourhood of S; condition a) of (21.13.15) is therefore condition a) of (21.13.14) applied to the family (U_λ) of open neighbourhoods of S. The corollary now follows from the equivalence of a) and b) in (21.13.14). □

Remark (21.13.16). — Under the general conditions of (21.13.14), suppose moreover $\varinjlim \mathrm{Pic}(\mathrm{U}_\lambda) = 0$. Then condition a) of (21.13.14) is also equivalent to the following:

c) *Every 1-codimensional cycle on X whose support is contained in one of the sets $\mathrm{X} - \mathrm{U}_\lambda$ is locally principal.*

We can restrict to the case where X is irreducible and all the U_λ are non-empty. It is clear that a) implies c), since if the 1-codimensional cycle Z on X has its support in $\mathrm{X} - \mathrm{U}_\lambda$, $\mathrm{Z}|\mathrm{U}_\lambda = 0$, hence $\mathrm{Z}|\mathrm{U}_\lambda$ is locally principal. Inversely c) implies a): let Z be a 1-codimensional cycle on X such that $\mathrm{Z}|\mathrm{U}_\lambda$ is locally principal; since X is normal, $\mathrm{Z}|\mathrm{U}_\lambda = \mathrm{cyc}(\mathrm{D}_\lambda)$, where D_λ is a divisor on U_λ. The hypothesis $\varinjlim \mathrm{Pic}(\mathrm{U}_\lambda) = 0$ implies, by (21.3.4), that there is an index μ with $\mathrm{U}_\mu \subset \mathrm{U}_\lambda$ such that $\mathrm{D}_\mu = \mathrm{D}_\lambda|\mathrm{U}_\mu$ is equivalent to 0. If $\mathrm{D}_\mu = \mathrm{div}(f_\mu)$, f_μ may be considered as a regular rational function on X, regular on U_μ; hence $\mathrm{Z}' = \mathrm{cyc}(\mathrm{div}(f_\mu))$, we see therefore that $\mathrm{Z}'' = \mathrm{Z} - \mathrm{Z}'$ has its support in $\mathrm{X} - \mathrm{U}_\mu$; by hypothesis, Z'' is locally principal, whence the conclusion.

The condition $\varinjlim \mathrm{Pic}(\mathrm{U}_\lambda) = 0$ is satisfied in particular when (U_λ) is the family of open neighbourhoods of a point $z \in \mathrm{X}$: for every invertible $\mathscr{O}_{\mathrm{U}_\lambda}$-Module $\mathscr{L}_\lambda$, there exists by definition an open neighbourhood $\mathrm{U}_\mu \subset \mathrm{U}_\lambda$ of z such that $\mathscr{L}_\lambda|\mathrm{U}_\mu \cong \mathscr{O}_{\mathrm{U}_\mu}$. We see therefore that, in (21.13.15), if $\mathrm{S} = \{z\}$, conditions a) and b) are equivalent also to the following:

c) *Every 1-codimensional cycle on X whose support does not contain z is locally principal.*

If moreover $\mathrm{Pic}(\mathrm{X}) = 0$, it follows that this condition implies that every 1-codimensional cycle on X whose support does not contain z is principal.

21.14. The Ramanujam-Samuel theorem

Theorem (21.14.1). — (Ramanujam-Samuel) — *Let A be a local Noetherian ring with maximal ideal $\mathfrak{m}$,* **IV-4** | 323 *such that its completion $\widehat{\mathrm{A}}$ is integral and integrally closed. Let B be a local Noetherian ring and $\rho : \mathrm{A} \to \mathrm{B}$ a local homomorphism making B a formally smooth A-algebra (for the preadic topologies* ($\mathbf{0}$, 19.3.1)*) and such that the residue field of B is finite over that of A. Then every 1-codimensional cycle on $\mathrm{Spec}(\mathrm{B})$ which is principal at the point $\mathfrak{p} = \mathfrak{m}\mathrm{B}$, is a principal 1-codimensional cycle.*

Proof. If $k = \mathrm{A}/\mathfrak{m}$ is the residue field of A, then $\mathrm{B}/\mathfrak{m}\mathrm{B}$ is a formally smooth k-algebra for its preadic **IV-4** | 324 topology ($\mathbf{0}$, 19.3.5), hence regular and in particular integral. Thus $\mathfrak{p} = \mathfrak{m}\mathrm{B}$ is indeed a prime ideal of B, which justifies the statement. It evidently all amounts to proving that every prime ideal $\mathfrak{q}$ of B not contained in $\mathfrak{m}\mathrm{B}$ is *principal*.

Let $\widehat{\mathrm{A}}$, $\widehat{\mathrm{B}}$ be the completions of A and B respectively, so that the maximal ideal of $\widehat{\mathrm{A}}$ is $\mathfrak{m}\widehat{\mathrm{A}}$; we know ($\mathbf{0}$, 19.3.6) that $\widehat{\mathrm{B}}$ is a formally smooth $\widehat{\mathrm{A}}$-algebra for the adic topologies. Let k' be the residue field of $\widehat{\mathrm{B}}$, a finite extension of k. There exists a local homomorphism $\widehat{\mathrm{A}} \to \mathrm{C}$, where C is a local

Noetherian ring which is finite and flat (hence free) as an $\widehat{\mathrm{A}}$-module and such that $\mathrm{C}/\mathfrak{m}\mathrm{C} \cong k'$ ($\mathbf{0}_{\mathrm{III}}$, 10.3.1). It follows that C is complete, and then from (7.5.1), (7.5.3), and (6.5.4), (ii) that C is *integral* and *integrally closed*.

Moreover,

$$\mathrm{D} = \widehat{\mathrm{B}}\widehat{\otimes}_{\widehat{\mathrm{A}}}\mathrm{C}$$

is a complete semilocal ring, a direct product of complete local rings, one of which, D_0, has residue field k' (since $k' \otimes_k k'$ is a direct product of local rings, one of them isomorphic to k'). Since D is formally smooth over C, so is D_0; consequently $\mathrm{D}_0/\mathfrak{m}\mathrm{D}_0$ is a formally smooth k'-algebra with residue field k', and is therefore k'-isomorphic to a formal power-series algebra $k'[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ ($\mathbf{0}$, 19.6.4). It follows from ($\mathbf{0}$, 19.7.1.5) that D_0 is C-isomorphic to $\mathrm{C}[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$, and hence is integral and integrally closed (Bourbaki, *Alg. comm.*, chap. V, §1, no. 4, Prop. 14). Since the morphisms $\mathrm{Spec}(\mathrm{D}_0) \to \mathrm{Spec}(\widehat{\mathrm{B}})$ and $\mathrm{Spec}(\mathrm{D}_0) \to \mathrm{Spec}(\mathrm{B})$ are faithfully flat, it follows that B and $\widehat{\mathrm{B}}$ are likewise integral and integrally closed (2.1.13).

The ideal $\mathfrak{q}\mathrm{D}_0$ is therefore divisorial (Bourbaki, *Alg. comm.*, chap. VII, §1, no. 10, Prop. 15), and it is not contained in $\mathfrak{m}\mathrm{D}_0$; otherwise faithful flatness would give

$$\mathfrak{q} = (\mathfrak{q}\mathrm{D}_0) \cap \mathrm{B} \subset (\mathfrak{m}\mathrm{D}_0) \cap \mathrm{B} = \mathfrak{m}\mathrm{B} = \mathfrak{p}$$

($\mathbf{0}_{\mathrm{I}}$, 6.5.1). Hence none of the height-one prime ideals $\mathfrak{r}_h$ of D_0 containing $\mathfrak{q}\mathrm{D}_0$ is contained in $\mathfrak{m}\mathrm{D}_0$. If these ideals are principal, then $\mathfrak{q}\mathrm{D}_0$ is principal, since the divisors (in Bourbaki's sense) of $\mathfrak{q}\mathrm{D}_0$ and of a product of powers of the $\mathfrak{r}_h$ are equal (Bourbaki, *Alg. comm.*, chap. VII, §1, no. 4, Prop. 5). As $\mathfrak{q}\mathrm{D}_0 \cap \mathrm{B} = \mathfrak{q}$, faithful flatness then shows that $\mathfrak{q}$ is principal (2.5.2), using the fact that B is local.

We are thus reduced to proving the theorem in the particular case where A is complete and $\mathrm{B} = \mathrm{A}[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$. Let us first show that we may reduce to $n = 1$. We argue by induction on n, and choose $f \in \mathfrak{q}$ with $f \notin \mathfrak{p} = \mathfrak{m}\mathrm{B}$. It is enough to show that there exists an A-automorphism σ of B such that $\sigma(f)$ does not belong to

$$\mathfrak{p}' = \mathfrak{p} + \mathrm{B}\mathrm{T}_1 + \dots + \mathrm{B}\mathrm{T}_{n-1}.$$

Indeed, if $\mathrm{B}' = \mathrm{A}[[\mathrm{T}_1, \dots, \mathrm{T}_{n-1}]]$, then B' is complete and has maximal ideal $\mathfrak{m}\mathrm{B}' + \mathrm{B}'\mathrm{T}_1 + \dots + \mathrm{B}'\mathrm{T}_{n-1} = \mathfrak{m}'$. Since $\mathrm{B} = \mathrm{B}'[[\mathrm{T}_n]]$ and $\mathfrak{p}' = \mathfrak{m}'\mathrm{B}$, the induction hypothesis, taking into account that B' is integral and integrally closed, shows that $\sigma(\mathfrak{q})$ is principal in B, and hence so is $\mathfrak{q}$. Now, if $\overline{f} \in k[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ is the image of f, then $\overline{f} \neq 0$; one knows (Bourbaki, *Alg. comm.*, chap. VII, §3, no. 7, Lemma 3) that there is an A-automorphism σ of B such that $(\overline{\sigma(f)})(0, \dots, 0, \mathrm{T}_n) \neq 0$, which plainly implies $\sigma(f) \notin \mathfrak{p}'$.

Suppose therefore $n = 1$, and write T instead of T_1, so that $\mathrm{B} = \mathrm{A}[[\mathrm{T}]]$. It is enough to show that the ideal $\mathfrak{q} \cap \mathrm{A}[\mathrm{T}]$ of the polynomial ring $\mathrm{A}[\mathrm{T}]$ is principal. Choose $f_0 \in \mathfrak{q}$ with $f_0 \notin \mathfrak{m}\mathrm{B}$; it is a formal power series whose reduced series $\overline{f}_0$ is nonzero. By the preparation theorem (Bourbaki, *Alg. comm.*, chap. VII, §3, no. 8, Prop. 5), for every $f \in \mathfrak{q}$ there exist $g \in \mathrm{B}$ and $r \in \mathrm{A}[\mathrm{T}]$ such that $f = gf_0 + r$, so $r \in \mathfrak{q} \cap \mathrm{A}[\mathrm{T}]$. On the other hand (*loc. cit.*, Prop. 6), there exist a nonconstant distinguished polynomial $\mathrm{F}_0 \in \mathrm{A}[\mathrm{T}]$ and a unit $u \in \mathrm{B}$ such that $f_0 = u\mathrm{F}_0$; hence $\mathrm{F}_0 \in \mathfrak{q} \cap \mathrm{A}[\mathrm{T}]$. Thus, putting $\mathfrak{q}_1 = \mathfrak{q} \cap \mathrm{A}[\mathrm{T}]$, one has $\mathfrak{q} = \mathfrak{q}_1\mathrm{B}$. Since B is flat over $\mathrm{A}[\mathrm{T}]$ ($\mathbf{0}_{\mathrm{I}}$, 7.3.3), it follows from Bourbaki, *Alg. comm.*, chap. VII, §1, no. 10, Prop. 15 that $\mathfrak{q}_1$ is a height-one prime ideal of $\mathrm{A}[\mathrm{T}]$.

Necessarily $\mathfrak{q}_1 \cap \mathrm{A} = 0$. Otherwise $\mathfrak{q}_1 \cap \mathrm{A}$ would have height at least 1, and (5.5.3) would give $\mathfrak{q}_1 = (\mathfrak{q}_1 \cap \mathrm{A})\mathrm{A}[\mathrm{T}]$; since $\mathfrak{q}_1 \cap \mathrm{A} \subset \mathfrak{m}$, this would imply $\mathfrak{q}_1 \subset \mathfrak{m}\mathrm{A}[\mathrm{T}]$, contrary to the hypothesis on $\mathfrak{q}$.

Let K be the field of fractions of A. Then $\mathfrak{q}_1\mathrm{K}[\mathrm{T}]$ is a prime ideal distinct from both 0 and $\mathrm{K}[\mathrm{T}]$, hence is of the form $h\mathrm{K}[\mathrm{T}]$, where

$$h(\mathrm{T}) = \mathrm{T}^m + a_1\mathrm{T}^{m-1} + \dots + a_m, \qquad m \geqslant 1,$$

the a_i belonging to K and h being irreducible in $\mathrm{K}[\mathrm{T}]$. We saw above that $\mathfrak{q}_1$ contains a nonconstant distinguished polynomial $\mathrm{F}_0 \in \mathrm{A}[\mathrm{T}]$. If t is the class of T in $\mathrm{K}[\mathrm{T}]/\mathfrak{q}_1\mathrm{K}[\mathrm{T}]$, then t is a root of F_0 in an extension of K, and therefore h divides F_0 in $\mathrm{K}[\mathrm{T}]$. Since h and F_0 are monic, the coefficients a_i of h are integral over A (Bourbaki, *Alg. comm.*, chap. V, §1, no. 3, Prop. 11), and thus belong to A because A is integrally closed. In other words,

$$h \in \mathrm{A}[\mathrm{T}] \cap \mathfrak{q}_1\mathrm{K}[\mathrm{T}] = \mathfrak{q}_1$$

(Bourbaki, *Alg. comm.*, chap. II, §1, no. 5, Prop. 11). Since every polynomial $g \in \mathfrak{q}_1$ is divisible by h in $\mathrm{K}[\mathrm{T}]$ and h is monic, the coefficients of g/h belong to A; hence $\mathfrak{q}_1 = h\mathrm{A}[\mathrm{T}]$. This completes the proof. □

Corollary (21.14.2). — *Under the hypotheses of (21.14.1) on* A, B *and* ρ, *for every prime ideal* $\mathfrak{q}$ *of* B *not contained in* $\mathfrak{p} = \mathfrak{m}\mathrm{B}$ *and such that* $\dim(\mathrm{B}_{\mathfrak{q}}) \geqslant 2$, *the ring* $\mathrm{B}_{\mathfrak{q}}$ *is parafactorial. In particular, if* $\dim(\mathrm{B} \otimes_{\mathrm{A}} k) > 0$ *(i.e.* $\dim \mathrm{B} > \dim \mathrm{A}$*)* $((\mathbf{0}, 19.7.1)$ *and* $(\mathbf{0}, 6.1.1)))$, *the ring* B *is parafactorial.*

Proof. Indeed, we saw in the proof of (21.14.1) that the conditions of the statement imply that B is integral and integrally closed; and the equivalence of the statements (21.14.1) and (21.14.2) follows then from (21.13.15) applied to $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$ and $\mathrm{S} = \{\mathfrak{p}\}$. □

Proposition (21.14.3). — *Let* S *be a normal prescheme,* $f : \mathrm{X} \to \mathrm{S}$ *a smooth morphism.*

(i) *If* S *is locally Noetherian (hence also* X*), then, for all* $x \in \mathrm{X}$ *such that* $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$ *and such that* x *is not a maximal point of its fibre* $f^{-1}(f(x))$, *the ring* $\mathscr{O}_{\mathrm{X},x}$ *is parafactorial. Every* 1*-codimensional* **IV-4 | 326** *cycle* Z *on* X *such that* $\mathrm{Supp}(\mathrm{Z})$ *contains no irreducible component of any fibre* X_s, $f^{-1}(s) = \mathrm{X}_s$, *is locally principal.*

(ii) *Let* Y *be a closed subset of* X *containing no irreducible component of any fibre* X_s, *and such that, for every maximal point* η *of* S, $\mathrm{Y}_\eta = \mathrm{Y} \cap \mathrm{X}_\eta$ *has codimension* $\geqslant 2$ *in* X_η. *Then the pair* (X, Y) *is parafactorial.*

Proof. We note that the conditions of (ii) are satisfied if Y is a closed subset of X such that, for all $s \in \mathrm{S}$, $\mathrm{Y}_s = \mathrm{Y} \cap \mathrm{X}_s$ is of codimension $\geqslant 2$ in X_s.

To prove (i), suppose first that $\mathrm{Y} = \overline{\{x\}}$. There is an open neighbourhood V of x in X such that $\mathrm{Y} \cap \mathrm{V}$ and V satisfy the conditions of (ii). Indeed, the hypothesis and (9.5.3) allow us to choose V so that $\mathrm{Y} \cap \mathrm{V}$ contains no irreducible component of a fibre X_s. Moreover, for every maximal point η of S for which $\mathrm{Y} \cap \mathrm{X}_\eta \neq \varnothing$, the points z of $\mathrm{Y} \cap \mathrm{X}_\eta$ are specializations of x; hence $\dim(\mathscr{O}_{\mathrm{X},z}) \geqslant 2$. Since $\mathscr{O}_{\mathrm{X},z} = \mathscr{O}_{\mathrm{X}_\eta,z}$ (the prescheme S being reduced at η, by (2.1.13)), $\mathrm{Y} \cap \mathrm{X}_\eta$ has codimension $\geqslant 2$ in X_η. Thus (ii) makes the pair $(\mathrm{V}, \mathrm{Y} \cap \mathrm{V})$ parafactorial, and (21.13.10) shows that $\mathscr{O}_{\mathrm{X},x}$ is parafactorial. This proves the first assertion of (i).

For the second assertion, it is enough to consider the case $\mathrm{Z} = \overline{\{z\}}$, where $z \in \mathrm{X}^{(1)}$. Since $\mathscr{O}_{\mathrm{X},z}$ is an integrally closed Noetherian local ring of dimension 1, it is a discrete valuation ring, and Z is therefore principal at z. At every other point $x \in \mathrm{Supp}(\mathrm{Z})$, one has $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$, and by hypothesis x is not a maximal point of its fibre; hence $\mathscr{O}_{\mathrm{X},x}$ is parafactorial. Apply (21.13.15), taking for S the set formed by z and by the maximal points of the fibres of f. The cycle Z is principal at the points of this set, since z is the only point of it belonging to $\mathrm{Supp}(\mathrm{Z})$; and condition (b) of (21.13.15) is plainly satisfied. It follows that Z is locally principal.

It remains to prove (ii). Put $\mathrm{U} = \mathrm{X} - \mathrm{Y}$, and let $j : \mathrm{U} \to \mathrm{X}$ be the canonical injection. Since the hypotheses and the conclusion are local on S and on X by (21.13.6), (i), we may assume that S and X are affine; then f is of finite presentation. Since the fibres of f are regular, the hypothesis on Y implies that, for every $x \in \mathrm{Y}$,

$$\mathrm{prof}(\mathscr{O}_{\mathrm{X}_{f(x)},x}) = \dim(\mathscr{O}_{\mathrm{X}_{f(x)},x}) \geqslant 1.$$

It follows from (19.9.8) that the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ is injective. Moreover, replacing Y by a larger closed subset by the method used in the proof of (19.9.8), and using (21.13.6), (ii), we may suppose that Y is defined by an ideal of finite type in the ring of X. Consequently $\mathrm{U} = \mathrm{X} - \mathrm{Y}$ is quasi-compact and quasi-separated, and Y is constructible.

For every invertible $\mathscr{O}_{\mathrm{U}}$-module $\mathscr{L}$ and every $x \in \mathrm{Y}$, it is enough to establish the existence of an open neighbourhood V of x in X such that: 1° the canonical homomorphism $\mathscr{O}_{\mathrm{V}} \to (j_{\mathrm{V}})_*(\mathscr{O}_{\mathrm{U}\cap\mathrm{V}})$ is surjective; and 2° there is an invertible $\mathscr{O}_{\mathrm{V}}$-module $\mathscr{L}'_{\mathrm{V}}$ whose restriction to $\mathrm{U} \cap \mathrm{V}$ is isomorphic to $\mathscr{L}|(\mathrm{U} \cap \mathrm{V})$ ((21.13.5) and (21.13.3)).

Set $s = f(x)$. We first reduce to the case $\mathrm{S} = \mathrm{S}_0 = \mathrm{Spec}(\mathscr{O}_{\mathrm{S},s})$. Let (S_ν) be a fundamental system of affine open neighbourhoods of s in S, and put $\mathrm{X}_\nu = f^{-1}(\mathrm{S}_\nu)$, $\mathrm{Y}_\nu = \mathrm{Y} \cap \mathrm{X}_\nu$, $\mathrm{U}_\nu = \mathrm{X}_\nu - \mathrm{Y}_\nu$, with canonical injection $j_\nu : \mathrm{U}_\nu \to \mathrm{X}_\nu$. Then $\mathrm{X}_0 = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}_0$ is the projective limit of the X_ν $(\mathbf{8.1.2}, a))$. Put $\mathrm{Y}_0 = \mathrm{Y} \cap \mathrm{X}_0$, $\mathrm{U}_0 = \mathrm{X}_0 - \mathrm{Y}_0$, and let $j_0 : \mathrm{U}_0 \to \mathrm{X}_0$ be the canonical injection. If $p_\nu : \mathrm{X}_0 \to \mathrm{X}_\nu$ denotes the affine projection, then

$$(j_0)_*(\mathscr{O}_{U_0}) = p_\nu^*((j_\nu)_*(\mathscr{O}_{U_\nu})) \qquad (\mathbf{II},\ 1.5.2),$$

and the canonical homomorphism $\rho_0 : \mathscr{O}_{X_0} \to (j_0)_*(\mathscr{O}_{U_0})$ is $p_\nu^*(\rho_\nu)$. Since ρ_0 is surjective by hypothesis, ρ_ν is surjective for sufficiently large ν (8.5.7).

Let $\mathscr{L}_0$ be the restriction of $\mathscr{L}$ to U_0. The X_ν are affine, hence quasi-compact and quasi-separated. By hypothesis there exists an invertible $\mathscr{O}_{X_0}$-module $\mathscr{L}_0'$ extending $\mathscr{L}_0$. For sufficiently large ν there is a quasi-coherent $\mathscr{O}_{X_\nu}$-module $\mathscr{L}_\nu'$ of finite presentation such that $\mathscr{L}_0' \simeq p_\nu^*(\mathscr{L}_\nu')$ (8.5.2); by (8.5.5) we may suppose that $\mathscr{L}_\nu'$ is invertible. Finally, since the U_ν are quasi-compact and quasi-separated and $\mathscr{L}_0'|U_0 \simeq \mathscr{L}_0$, for sufficiently large ν the modules $\mathscr{L}_\nu'|U_\nu$ and $\mathscr{L}|U_\nu$ are isomorphic (8.5.2.5). This proves the reduction.

We are thus reduced to the case $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, where A is local. Since X is normal **IV-4** | 327
and f smooth, S is normal (17.5.7); hence A is integral and integrally closed. Regard A as the filtered direct limit of its **Z**-subalgebras of finite type. The integral closure of such a subalgebra is a subring of A and is again a **Z**-algebra of finite type (**0**, 7.8.3, (ii), (iii), and (vi)). Thus A is the filtered direct limit of integrally closed **Z**-subalgebras A_λ of finite type. By (1.8.4.2), for some λ there is an A_λ-algebra B_λ of finite type such that, up to an A-isomorphism,

$$B = B_\lambda \otimes_{A_\lambda} A.$$

Put $S_\lambda = \mathrm{Spec}(A_\lambda)$ and $X_\lambda = \mathrm{Spec}(B_\lambda)$. If $p_\lambda : X \to X_\lambda$ is the canonical projection, we may also suppose, since Y is constructible, that $Y = p_\lambda^{-1}(Y_\lambda)$ for a closed subset Y_λ of X_λ (8.3.11). Write $f_\lambda : X_\lambda \to S_\lambda$ for the structural morphism. By transitivity of fibres and (4.2.6), after increasing λ we may suppose that Y_λ contains no irreducible component of any fibre of f_λ. Since f is smooth, we may likewise suppose that f_λ is smooth (17.7.8). Finally, the image η_λ in S_λ of the generic point η of S is the generic point of S_λ, and

$$\mathrm{codim}((Y_\lambda)_{\eta_\lambda}, (X_\lambda)_{\eta_\lambda}) = \mathrm{codim}(Y_\eta, X_\eta) \geqslant 2$$

by (6.1.4). Thus S_λ, X_λ, and Y_λ satisfy all the hypotheses of (ii), as do their base changes S_μ, X_μ, and Y_μ for $\lambda \leqslant \mu$.

If (X_μ, Y_μ) is parafactorial for all sufficiently large μ, then so is (X, Y). Indeed, put $U = X - Y$, $U_\mu = X_\mu - Y_\mu$, and let j and j_μ be the corresponding open immersions. Since $p_\mu : X \to X_\mu$ is affine,

$$j_*(\mathscr{O}_U) = p_\mu^*((j_\mu)_*(\mathscr{O}_{U_\mu})) \qquad (\mathbf{II},\ 1.5.2),$$

so the canonical map $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is the pullback of the corresponding map on X_μ, and is therefore bijective. Moreover, for every invertible $\mathscr{O}_U$-module $\mathscr{L}$, there is, for sufficiently large μ, an invertible $\mathscr{O}_{U_\mu}$-module $\mathscr{L}_\mu$ such that $\mathscr{L} \simeq (p_m'u)^*(\mathscr{L}_\mu)$ (8.5.2) and (8.5.5); here we use that U_λ is quasi-compact because X_λ is Noetherian. By parafactoriality, $\mathscr{L}_\mu$ extends to an invertible $\mathscr{O}_{X_\mu}$-module $\mathscr{L}_\mu'$, whose pullback extends $\mathscr{L}$. Hence it remains only to prove (ii) when A is an integrally closed **Z**-algebra of finite type.

The local rings $\mathscr{O}_{S,s}$ are then excellent integrally closed rings, and their completions are integrally closed (**0**, 7.8.3, (ii), (iii), and (vii)). By (21.13.10), it is enough to show that $\mathscr{O}_{X,x}$ is parafactorial for every $x \in Y$. Choose a closed point y of $Y_{f(x)}$ which is a specialization of x (5.1.11), and put $s = f(x) = f(y)$. The completion of $\mathscr{O}_{S,s}$ is integrally closed, and $\mathscr{O}_{X,y}$ is a formally smooth $\mathscr{O}_{S,s}$-algebra for the preadic topologies (17.5.3). Since Y contains no irreducible component of X_s,

$$\dim(\mathscr{O}_{X,y}) > \dim(\mathscr{O}_{S,s}) \qquad \text{by (17.5.8).}$$

If $s \neq \eta$, then $\dim(\mathscr{O}_{S,s}) \geqslant 1$; if $s = \eta$, the hypothesis gives $\dim(\mathscr{O}_{X,x}) \geqslant 2$. In either case $\dim(\mathscr{O}_{X,y}) \geqslant 2$. Furthermore, the prime ideal of $\mathscr{O}_{X,y}$ corresponding to x is not contained in $\mathfrak{m}_s\mathscr{O}_{X,y}$, because Y contains no irreducible component of X_s. Finally y is closed in X_s, so $k(y)$ is a finite extension of $k(s)$ (**I**, 6.4.2). We may therefore apply (21.14.2) with $A = \mathscr{O}_{S,s}$, $B = \mathscr{O}_{X,y}$, and $B_\mathfrak{q} = \mathscr{O}_{X,x}$. This completes the proof. □

(21.14.4). *Remarks.* — (i) It is possible that the statements (21.14.1) and (21.14.2) remain valid *without* **IV-4** | 328
hypothesis on the residue field of B. The statement with this generality would be equivalent, by virtue of (21.13.15) and (21.13.12.1), to the following: let A be a complete, integral, integrally closed

Noetherian local ring, and let B be a Noetherian local ring which is a formally smooth A-algebra, such that $\dim(B) > \dim(A)$; then B is parafactorial.

(ii) We will see in chap. VI, using a technique of "finite descent" applied to the morphism $Y' \to Y = \operatorname{Spec}(A)$, where Y' is the normalization of Y, that the conclusion of (21.14.1) (or of (21.14.2)) remains valid when the hypothesis that $\widehat{A}$ is integrally closed is replaced by the hypothesis that $\widehat{A}$ is reduced, provided that $\dim B \geqslant \dim A + 2$. If this last condition is replaced by $\dim B \geqslant \dim A + 3$, the hypothesis that $\widehat{A}$ is reduced may even be omitted. Similarly, the conclusion of (21.14.3) remains valid by replacing the hypothesis that X is normal by the hypothesis that X is reduced, provided that $\dim(\mathscr{O}_{X,x}) \geqslant \dim(\mathscr{O}_{S,f(x)}) + 2$.

(iii) In chap. III, Part 3, we prove that, if $f : X \to S$ is a smooth morphism and Y is a closed locally constructible subset of X such that, for every $s \in S$, $\operatorname{codim}(Y_s, X_s) \geqslant 3$ (with the notation of (21.14.3)), then the pair (X, Y) is parafactorial. This conclusion is no longer valid when we only suppose $\operatorname{codim}(Y_s, X_s) \geqslant 2$ for all $s \in S$ and when we do not suppose X reduced. For example, let k be a field, let $A = k[T]/(T^2)$ be the algebra of dual numbers over k, and put $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(A[T_1, T_2]) = \mathbb{V}_S^2$, where T_1 and T_2 are indeterminates; let Y be the "zero section" of this fibration. If s is the unique closed point of S, then $X_s = \mathbb{V}_k^2$ and Y_s is the "zero section" of X_s. To see that the pair (X, Y) is not parafactorial, it suffices to show that the ring $B = A[T_1, T_2]_{\mathfrak{m}}$, where $\mathfrak{m}$ is the ideal $(T_1) + (T_2)$ which defines Y, is not parafactorial (21.13.10). Let $B_0 = B_{\mathrm{red}}$, and let U and U_0 be the complementary open subsets of the closed point in $\operatorname{Spec}(B)$ and $\operatorname{Spec}(B_0)$; reasoning as in (21.13.9), we have the exact sequence, extension of (21.13.9.1)

$$\Gamma(U, \mathscr{O}_U)^* \longrightarrow \Gamma(U_0, \mathscr{O}_{U_0})^* \longrightarrow H^1(U_0, \mathscr{O}_{U_0}) \longrightarrow \operatorname{Pic}(U) \longrightarrow \operatorname{Pic}(U_0).$$

Now, one has $\Gamma(U, \mathscr{O}_U) = B$, $\Gamma(U_0, \mathscr{O}_{U_0}) = B_0$ since $\operatorname{prof}(B) = \operatorname{prof}(B_0) = 2$ (5.10.5). On the other hand $\operatorname{Pic}(U_0) = 0$ since B_0 is factorial, and $H^1(U_0, \mathscr{O}_{U_0}) \neq 0$ [41, 3, Example III-1]; since the homomorphism $B^* \to B_0^*$ is surjective, we conclude that $\operatorname{Pic}(U) \neq 0$, hence that B is not parafactorial.

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(21.14.4bis). (iv) The result of (21.14.3) was first proved by C. Seshadri [44] in the particular case where S is an algebraic normal scheme over an algebraically closed field k and $X = S \times_k T$, where T is a k-algebraic prescheme smooth over k. The proof of Seshadri [44, pp. 188–189] is of global nature and uses the theory of Picard schemes. It gives moreover (*loc. cit.*) results such as the following (for which one does not currently possess a proof by local methods). Let S and T be two preschemes locally of finite type over a field k, put $X = S \times_k T$, and let Z be a 1-codimensional cycle on X (regarded as an S-prescheme). Suppose that the following conditions hold:

1° S and T are geometrically normal over k (6.7.6);

2° For every maximal point η of S, the 1-codimensional cycle Z_η on the fibre X_η, having the same multiplicity as Z at every point of $X^{(1)} \cap X_\eta$, is locally principal (in other words, it is the image of a divisor of X_η, since X_η is normal);

3° For all $s \in S$, Z is principal at the maximal points of the fibre X_s.

Then Z is locally principal. In other words, since X is normal (6.14.1), for every $x \in X$ which is not maximal in its fibre X_s and which belongs to none of the "generic fibres" X_η (which implies $\dim(\mathscr{O}_{X,x}) \geqslant 2$ by (6.1.1)), the local ring $\mathscr{O}_{X,x}$ is parafactorial, by (21.12.15).[41]

[41] In the cited paper Seshadri in fact assumes that k is algebraically closed and that T is separated and "semi-complete" (that is, $\Gamma(T, \mathscr{O}_T) \simeq k$), and replaces condition 3° by the stronger hypothesis that Supp(Z) contains no fibre X_s, $s \in S$. Since the statement is local on S, however, condition 3° suffices. The resulting conclusion, interpreted as above in terms of parafactoriality of the rings $\mathscr{O}_{X,x}$, is local on both S and T; this removes altogether the assumptions that T is semi-complete and that k is algebraically closed. Indeed, after first passing to an algebraic closure of k, one may assume that T is affine, embed it as an open subscheme of a normal projective prescheme over k, and apply Seshadri's result. This reduction also shows that it is enough to prove Seshadri's theorem when T is projective (rather than merely semi-complete), the case in which the Picard-scheme theory used by Seshadri is contained in chap. VI of this Treatise.

21.15. Relative divisors

(21.15.1). Let S be a prescheme, $f : X \to S$ a *flat* morphism and locally of finite presentation. We defined in (20.6.1) the sheaf of rings $\mathscr{M}_{X/S}$ of the *germs of meromorphic functions on* X *relative to* S, sub-sheaf of $\mathscr{M}_X$; it is clear that the canonical injection $\mathscr{O}_X \hookrightarrow \mathscr{M}_X$ (20.1.4.1) maps $\mathscr{O}_X$ into a sub-sheaf of $\mathscr{M}_{X/S}$, with which we identify it. Let $\mathscr{M}^*_{X/S}$ be the sheaf (of multiplicative groups) of germs of invertible sections of $\mathscr{M}_{X/S}$; it is therefore a sub-sheaf of $\mathscr{M}^*_X$ and contains $\mathscr{O}^*_X$ as a sub-sheaf.

Definition (21.15.2). — We call *sheaf of divisors on* X *relative to* S, or *sheaf of divisors on* X *transversal to* f, and denote $\mathscr{D}iv_{X/S}$ the quotient sheaf (of commutative groups) $\mathscr{M}^*_{X/S}/\mathscr{O}^*_X$; the sections of this sheaf over X are called *divisors relative to* S, or *divisors on* X *transversal to* f; they form a commutative group denoted $\mathrm{Div}(X/S)$.

It is clear that $\mathscr{D}iv_{X/S}$ identifies canonically with a sub-sheaf of $\mathscr{D}iv_X$, and hence $\mathrm{Div}(X/S)$ with a subgroup of $\mathrm{Div}(X)$, which we denote also *additively*.

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For every open U of X, we have $\mathscr{M}^*_{X/S}|U = \mathscr{M}^*_{U/S}$, hence $\mathscr{D}iv_{X/S}|U = \mathscr{D}iv_{U/S}$, and the sheaf $\mathscr{D}iv_{X/S}$ is therefore equal to the presheaf $U \mapsto \mathrm{Div}(U/S)$.

Since $\mathscr{M}^*_{X/S}$ is a sub-sheaf of $\mathscr{M}^*_X$, the definitions, notations and formulas relative to divisors of sections of $\mathscr{M}_X$ over X (21.1.3) apply without change to sections of $\mathscr{M}_{X/S}$ over X.

(21.15.3). The structure of sheaf of ordered groups on $\mathscr{D}iv_X$ (21.1.6) induces on the sub-sheaf $\mathscr{D}iv_{X/S}$ a structure of sheaf of ordered groups, for which the sheaf of monoids of germs of positive sections is $\mathscr{D}iv^+_{X/S} = \mathscr{D}iv^+_X \cap \mathscr{D}iv_{X/S}$, which we denote $\mathscr{D}iv^+_{X/S}$. We have $\Gamma(X, \mathscr{D}iv^+_{X/S}) = \mathrm{Div}^+(X) \cap \mathrm{Div}(X/S)$; we denote this sub-monoid of $\mathrm{Div}(X/S)$ by $\mathrm{Div}^+(X/S)$, and it consists of elements $\geqslant 0$ with respect to the ordered group structure on $\mathrm{Div}(X/S)$. It follows from (21.5.1) that $\mathscr{D}iv^+_{X/S}$ is the image in $\mathscr{D}iv_{X/S}$ of the sub-sheaf of monoids

$$\mathscr{O}^{\mathrm{reg}}_{X/S} = \mathscr{O}_X \cap \mathscr{M}^*_{X/S} \tag{21.15.3.1}$$

For every open U of X, $\Gamma(U, \mathscr{O}^{\mathrm{reg}}_{X/S})$ is the set of sections t of $\mathscr{O}_X$ over U such that t is regular and that $1/t$ belongs to $\Gamma(U, \mathscr{M}_{X/S})$, which means, with the notations of (20.6.1), that $t \in T_{X/S}(U)$, so that the sheaf $\mathscr{O}^{\mathrm{reg}}_{X/S}$ is none other than the sheaf denoted $\mathscr{T}_{X/S}$ in (20.6.1). We can therefore write

$$\mathscr{D}iv^+_{X/S} = \mathscr{O}^{\mathrm{reg}}_{X/S}/\mathscr{O}^*_X \tag{21.15.3.2}$$

up to canonical isomorphism.

(21.15.3bis). Let $D \in \mathrm{Div}^+(X/S)$ and consider the closed sub-prescheme $Y(D)$ defined by the ideal $\mathscr{J}_X(D)$ of $\mathscr{O}_X$ (21.6.5); by virtue of what precedes, for all $x \in Y(D)$, there is an open neighbourhood U of x in X and a section $t \in T_{X/S}(U)$ such that $\mathscr{J}_X(D)|U = \mathscr{O}_U \cdot t$; since $x \in Y(D)$, the image $t_{f(x)}$ of t in $\Gamma(U \cap X_{f(x)}, \mathscr{O}_{X_{f(x)}})$ belongs to the maximal ideal of $\mathscr{O}_{X_{f(x)},x}$, and furthermore, by definition, for all $s \in S$, the image t_s of t in $\Gamma(U \cap X_s, \mathscr{O}_{X_s})$ is a regular section. We deduce therefore from (11.3.8) and (19.2.4) that the canonical injection $Y(D) \to X$ is *transversely regular relative to* S *and of codimension* 1. The converse being immediate, we see that one can *identify* canonically the positive divisors on X relative to S with the closed sub-preschemes Y of X such that the canonical injection $Y \to X$ is a transversely regular immersion relative to S and of codimension 1. We will ordinarily make this identification.

Proposition (21.15.4). — *Let* D *be a divisor on* X, *and let* $\mathscr{J}_X(D)$ *be the corresponding invertible fractional ideal* (21.2.5). *For* $D \in \mathrm{Div}(X/S)$, *it is necessary and sufficient that, for all* $x \in X$, $(\mathscr{J}_X(D) \cap \mathscr{M}^*_{X/S})_x \neq 0$ *(or equivalently, that* $\mathscr{J}_X(D) \subset \mathscr{M}_{X/S}$ *and* $\mathscr{J}_X(D)^{-1} \subset \mathscr{M}_{X/S}$*).*

Proof. Indeed, to say that $D \in \mathrm{Div}(X/S)$ means that for all $x \in X$, there exists an open neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{M}^*_{X/S})$ such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}^*_X)$; since $\mathscr{J}_X(D)|U$ is the fractional ideal $\mathscr{O}_U \cdot f$, the proposition follows immediately. □

Proposition (21.15.5). — ***IV-4 | 331***

Let D *be a divisor on* X, *let* $\mathscr{O}_X(D)$ *be the invertible fractional ideal, and let* s_D *be the regular meromorphic section of* $\mathscr{O}_X(D)$ *canonically defined by* D (21.2.8) *and* (21.2.9). *For* $D \in \mathrm{Div}(X/S)$, *it is necessary and sufficient that* $s_D \in \Gamma(X, \mathscr{M}_{X/S}(\mathscr{O}_X(D)))$.

Proof. Indeed, if U is an open of X such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, to say that $s_D \in \Gamma(U, \mathscr{M}_{X/S}(\mathscr{O}_X(D)))$ means, by virtue of the definitions (20.6.2), that $f^{-1} \in \Gamma(U, \mathscr{M}_{X/S})$, whence the proposition. □

The interpretation of the divisors on X by means of the classes $\mathrm{cl}(\mathscr{L}, s)$ (21.2.11) allows us therefore to interpret the elements of $\mathrm{Div}(X/S)$ as the pairs $(\mathscr{L}, s)$ (up to isomorphism) such that $\mathscr{L}$ is an invertible $\mathscr{O}_X$-module and s is a meromorphic section of $\mathscr{L}$ regular relative to S (20.6.5), (iii)).

Proposition (21.15.6). — *Let* D *be a divisor on* X *relative to* S, *and suppose that for all* $x \in X$ *such that* $\mathrm{prof}(\mathscr{O}_{X_{f(x)},x}) = 1$, *we have* $D_x \geqslant 0$ *(resp.* $D_x = 0$*). Then* $D \geqslant 0$ *(resp.* $D = 0$*).*

Proof. As in (21.1.8), we may restrict to the case where $D = \mathrm{div}(\varphi)$, with φ a meromorphic function regular relative to S. It is then enough to show that, if $D_x \geqslant 0$ at every point $x \in X$ for which $\mathrm{prof}(\mathscr{O}_{X_{f(x)},x}) = 1$, then φ is defined everywhere on X. This hypothesis means that, if $T = X - \mathrm{dom}(\varphi)$, then $\mathrm{prof}(\mathscr{O}_{X_{f(x)},x}) \geqslant 2$ for every $x \in T$; the conclusion follows from (20.6.6). □

(21.15.7). Let X′ be a second S-prescheme flat and locally of finite presentation over S, and let $g : X' \to X$ be an S-morphism. If the S-morphism g is *flat*, we know (21.4.5) that the inverse image by g of every divisor on X is defined; if furthermore $D \in \mathrm{Div}(X/S)$, it follows from the definition (21.15.2) and from (20.6.8) that $g^*(D) \in \mathrm{Div}(X'/S)$.

(21.15.8). Let X′ be a second S-prescheme flat and locally of finite presentation over S, and let $g : X' \to X$ be an S-morphism *finite* and *flat*. We note that g is also necessarily of finite presentation (1.4.3), (1.4.6) and (1.6.3), hence $g_*(\mathscr{O}_{X'})$ is a $\mathscr{O}_X$-Module *locally free* (2.1.12), in other words g is a locally free morphism (18.2.7); for all $s \in S$, the corresponding morphism $g_s : X'_s \to X_s$ is also finite and locally free. We deduce from (21.5.2) and (20.6.1) that, for every open U of X and every section $t' \in \Gamma(g^{-1}(U), \mathscr{T}_{X'/S})$, the norm $N_{X'/X}(t')$ belongs to $\Gamma(U, \mathscr{T}_{X/S})$. The argument of (21.5.3) then shows that, for every invertible $\mathscr{O}_{X'}$-module $\mathscr{L}'$ and every meromorphic section u' of $\mathscr{L}'$ over X′ which is regular relative to S, the norm $N_{X'/X}(u')$ is a meromorphic section of $\mathscr{L} = N_{X'/X}(\mathscr{L}')$ over X, regular relative to S. The interpretation of divisors relative to S given in (21.15.5), together with the definition of the direct image of a divisor (21.5.5), therefore proves that *for every divisor* $D' \in \mathrm{Div}(X'/S)$, one has $g_*(D') \in \mathrm{Div}(X/S)$.

(21.15.9). Consider finally an arbitrary morphism $S' \to S$, and (under the hypotheses of (21.15.1)) set $X' = X \times_S S'$, which is flat and locally of finite presentation over S′; if $p : X' \to X$ is the canonical projection, we have seen (20.6.9) that there is a canonical homomorphism $p^*(\mathscr{M}_{X/S}) \to \mathscr{M}_{X'/S'}$, which transforms every section of $\mathscr{M}_{X/S}$ over U, *regular relative to* S, into a section of $\mathscr{M}_{X'/S'}$ regular relative to S′ (20.6.5), (iii); we conclude then from the definition (21.15.2) and the right exactness of the functor p^*, that the preceding homomorphism defines by passage to quotients a canonical homomorphism **IV-4 | 332**

$$\mathrm{Div}(X/S) \longrightarrow \mathrm{Div}(X'/S') \tag{21.15.9.1}$$

which evidently transforms elements of $\mathrm{Div}^+(X/S)$ into elements of $\mathrm{Div}^+(X'/S')$. We also set $\mathrm{Div}(X'/S') = \mathrm{Div}_{X/S}(S')$ (resp. $\mathrm{Div}^+(X'/S') = \mathrm{Div}^+_{X/S}(S')$), and we see immediately that we have thus defined two contravariant functors

$$\mathrm{Div}_{X/S} : \mathbf{Sch}_{/S} \longrightarrow \mathbf{Ab}, \qquad \mathrm{Div}^+_{X/S} : \mathbf{Sch}_{/S} \longrightarrow \mathbf{Ens}$$

from the category of S-preschemes into that of commutative groups (resp. sets). We will see further (chap. VI) the important cases where the functor $\mathrm{Div}^+_{X/S}$ is *representable* ($\mathbf{0}_{\mathrm{III}}$, 8.1.8).

For every divisor $D \in \mathrm{Div}(X/S)$, the image of D by the homomorphism (21.15.9.1) is none other than the *inverse image* $p^*(D)$ (in the sense of (21.4.2)): the existence of this inverse image and the preceding assertions are immediate consequences of (20.6.5), (iii) and (20.6.9).

The elements of $\mathrm{Div}(X'/S')$ are often called, by abuse of language, "*families of divisors on* X *relative to* S, *parametrised by the S-prescheme* S′"; this terminology is used especially when it concerns positive divisors.

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The printed Part 4 bibliography gives the initials "G. S. Seshadri." The bibliographic identity of the cited paper establishes the author as C. S. Seshadri; the corrected initials are used above.

Entries [32] and [34] reproduce the continued bibliographies printed in Parts 1 and 2. The EGA I and SGA 1 entries are included so that the two explicit cross-volume citations resolve in this standalone cumulative reader.

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ERRATA AND ADDENDA

(List 3)

A) Typographical errors

(II, 3.3.2). On line 1 of p. 58, replace $\mathscr{S}$ by $\mathscr{M}$. On line 6 of p. 58, replace $S(n)$ by $M(n)$.
(II, 4.1.1). On line 8 of p. 71, replace $\mathscr{O}$-P-Module by $\mathscr{O}_P$-Module.
(II, 5.4.5). On line 6 of p. 102, replace u by $f \circ u$.
(II, 5.4.8). Before the fourth line from the bottom of p. 102, replace $f \times 1_{S'}$ by $f_{(Y')}$ in the diagram. On the fourth line from the bottom of p. 102, replace $f \times 1_{S'}$ by $f_{(Y')}$.
(II, 6.4.4). On the first line at the top of p. 122, replace "normal" by "integrally closed."
(II, 6.4.7). On the fourteenth line from the bottom of p. 122, replace "dimension" by "rank."
(II, 6.5.2). On line 5 of p. 127, replace $(\mathscr{B}|U_\lambda)$ by $(B|U_\lambda)$. On line 15 of p. 127, delete the first occurrence of $(N_{\mathscr{B}/\mathscr{A}}\omega_{\lambda_\mu})$.
(0_{III}, 8.1.11). On the eighth line from the bottom of p. 8, replace $h_X(u)$ by $h'_X(u)$.
(0_{III}, 9.1.2). On the tenth line from the bottom of p. 12, replace $\mathscr{F}$ by X.
(0_{III}, 10.2.1). On the sixth line from the bottom of p. 18, replace "in addition if" by "if in addition."
(III, 4.4.12). On line 10 of p. 138, replace "chap. V" by "chap. IV (8.11.2)."
(III, 5.2.5). On line 12 of p. 153, replace (I, 5.5.4) by (II, 5.5.4).
(III, 7.4.7). On line 15 of p. 57, replace "every A-module" by "every A-module M."
(III, 7.8.5). On the eighteenth line from the bottom of p. 74, replace F by $\mathscr{F}$.
(III, 7.8.9). On the ninth line from the bottom of p. 75, replace M by $\mathscr{M}$.
(III, 7.9.11). On the seventeenth line from the bottom of p. 79, replace chap. V by chap. VI.
(0_{IV}, 16.2.2.2). On line 13 of p. 26, replace $M' \cap \mathfrak{q}^n M'$ by $M' \cap \mathfrak{q}^n M$.
(0_{IV}, 17.3.7). On the sixteenth line from the bottom of p. 50, replace (17.1.10) by (17.3.3).
(0_{IV}, 18.1.3). In diagram (18.1.3.1), p_2 must be beside the arrow $G \to F$.
(0_{IV}, 19.8.7). On the seventh, eighth, and ninth lines from the bottom of p. 111, replace (21.5.5) by (21.5.3).
(0_{IV}, 19.8.8). On the second line from the bottom of p. 112, replace A-module by B-module.
(0_{IV}, 20.5.4). On the fifth line from the bottom of p. 130, replace $\Omega_{B'/A'}$ by $\Omega^1_{B'/A'}$.
(0_{IV}, 20.7.8). On the ninth line from the bottom of p. 149, replace (18.4.3) by (18.5.4). On the fourth line from the bottom, replace (18.4.1) by (18.5.1).
(0_{IV}, 21.5.3). On the last line of p. 166, replace the feminine form of "ramified" by the masculine form.
(0_{IV}, 21.7.1). On line 4 of p. 170, replace (20.6.15.1) by (20.6.17.1).
(0_{IV}, 22.1.1). On line 12 of p. 183, replace (20.6.20) by (20.6.10).

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(0_{IV}, 22.3.3). On the thirteenth line from the bottom of p. 192, replace $\widehat{B}_{\mathfrak{p}}$ by $\widehat{B}_{\mathfrak{q}}$.
(0_{IV}, 22.7.3). On the twenty-first line from the bottom of p. 212, replace $\mathrm{Ker}(i)$ by $\mathrm{Ker}(i_0)$. On the twelfth line from the bottom of p. 212, replace $(\mathfrak{p}/\mathfrak{p}^2) \otimes_L L$ by $(\mathfrak{p}/\mathfrak{p}^2) \otimes_A L$.
(0_{IV}, 23.1.3). On line 4 of p. 215, replace $x'^q \in K'$ by $x' \in K'$. On line 9, replace $x'^q \in A'$ by $x'^q \in A$.

(IV, 2.1.1). On the fourth line from the bottom of p. 5, replace (III, 1.4.15.1) by (III, 1.4.15.5).
(IV, 2.4.3). On line 12 of p. 20, replace $(X_{\mathrm{red}} \times_{Y_{\mathrm{red}}} Y_{\mathrm{red}})_{\mathrm{red}}$ by $(X_{\mathrm{red}} \times_{Y_{\mathrm{red}}} Y'_{\mathrm{red}})_{\mathrm{red}}$.
(IV, 2.6.1). On line 10 of p. 27, replace $X' \to X_{(S')}$ by $X' = X_{(S')}$.
(IV, 4.2.1). On the twentieth line from the bottom of p. 55, replace $\operatorname{tr.deg}_k E$ by $\operatorname{tr.deg}_L E$ and $\operatorname{tr.deg}_k L(t)$ by $\operatorname{tr.deg}_L L(t)$; on the nineteenth line from the bottom, interchange "first" and "second."
(IV, 4.4.2). On the last line of p. 59, replace X by X'.
(IV, 4.7.3). On the tenth line from the bottom of p. 75, replace "prescheme" by "k-prescheme."
(IV, 5.6.1). On line 16 of p. 98, replace the subscript $k(\mathfrak{p}($ by $k(\mathfrak{p})$.
(IV, 5.6.5). On line 20 of p. 99, delete the second closing parenthesis in $k(\xi))$.
(IV, 5.6.11.1). On line 16 of p. 102, replace K by $\mathfrak{K}$.
(IV, 5.11.2). In the footnote on p. 123, replace "5.11.2)" by "(5.11.2)."
(IV, 6.1.5). On the second line from the bottom of p. 136, replace (5.5.1.2) by (0, 16.3.9.2).
(IV, 6.3.1). On line 9 of p. 139, replace $y \in \mathfrak{m}$ by $y \in \mathfrak{n}$.
(IV, 6.7.2). On line 18 of p. 146, replace $\mathscr{O}'_{x'}$ by $\mathscr{O}_{x'}$.
(IV, 6.10.1). On line 13 of p. 155, replace $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_Y$ by $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_Y$.
(IV, 6.14.1). On line 21 of p. 170, replace "no. 3, prop. 12" by "no. 2, prop. 8."
(IV, 6.15.2). On line 13 of p. 177, replace $t(t-v)$ by $t^2(t-v)^2$.
(IV, 6.15.7). On the sixth line from the bottom of p. 179, replace $X \otimes_A k'$ by $X \otimes_k k'$.
(IV, 7.5.4). On the fourteenth line from the bottom of p. 206, replace $\operatorname{Spec}(\widehat{B} \otimes k)$ by $\operatorname{Spec}(\widehat{B} \otimes_{\widehat{A}} k)$.
(IV, 7.8.4). On the tenth line from the bottom of p. 216, move the opening parenthesis so that the phrase reads "satisfying (7.8.2, (ii) and (iii))."
(IV, Part 2, Bibliography). On line 6 of p. 224, replace *Amer. J. of Math.* by *Annals of Math.*
(IV, 8.5.8). On line 6 of p. 24, replace $\operatorname{Ker} g_\mu$ at the end of the line by $\operatorname{Ker} g_\lambda$.
(IV, 8.10.1). On the thirteenth line from the bottom of p. 36, replace (2.3.11) by (8.2.11).
(IV, 10.2.1). On the fourteenth line from the bottom of p. 98, replace y by Y.
(IV, 10.4.11). On the tenth line from the bottom of p. 104, replace (17.9.7) by (17.9.6).
(IV, 11.2.9). On the third line from the bottom of p. 126, replace A_λ by A_μ.
(IV, 11.3.4). On line 16 of p. 134, replace F by $\mathscr{F}$.
(IV, 11.3.10). On the ninth line from the bottom of p. 139, replace (11.3.7) by (11.3.10).
(IV, 11.7.5). On line 11 of p. 159, replace (6.5.11) by (6.15.11).
(IV, 12.1.1). On line 13 of p. 174, add at the end: "for $k \geqslant 1$."
(IV, 14.2.1). On line 11 of p. 202, replace (6.1.2) by (6.1.1).

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(IV, 14.3.8). On the tenth line from the bottom of p. 206, replace $(f^{-1}y)$ by $f^{-1}(y)$.
(IV, 14.5.7). On line 16 of p. 218, replace (14.5.5) by (14.5.6).
(IV, 14.5.11). In the diagram at the top of p. 223, every horizontal arrow must point from right to left.
(IV, 15.4.2). On line 23 of p. 230, replace (6.1.4) by (6.1.5).
(IV, 15.5.2). On line 16 of p. 233, replace $f^{-1}(y)$ by $f^{-1}(y')$.
(IV, 15.6.10). On line 6 of p. 242, replace $^{-1}(y')$ by $f^{-1}(y')$. On line 11, replace "α) and β)" by "α) or β)."

B) Textual modifications

($\mathrm{Err}_{\mathrm{IV}}$, 1). In ($0_{\mathrm{III}}$, 11.1.6), the statement (which reproduces a statement of (G, I, 4.4.1)) is false when $q(n)$ is an arbitrary integer; it is correct only when $q(n) = 0$ or $q(n) = n$. Indeed, in the first case the kernel of $d_r^{p,0}$ is $E_r^{p,0}$, since the image of $d_r^{p,0}$ lies in $E_r^{p+r,-r+1} = 0$ by hypothesis; likewise the image of $d_r^{p-r,r-1}$ is zero since $E_r^{p-r,r-1} = 0$. Hence $E_{r+1}^{p,0} = E_r^{p,0}$. The case $q(n) = n$ for every n is handled in the same way.

($\mathrm{Err}_{\mathrm{IV}}$, 2). In (III, 3.4), delete lines 2 and 3 of p. 119.

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($\mathrm{Err}_{\mathrm{IV}}$, 3). In (III, 4.4.9), on the fifteenth line from the bottom of p. 137, replace "finite" by "finite as sets."

($\mathrm{Err}_{\mathrm{IV}}$, 4). In (III, 7.8.10), on line 9 of p. 76, replace "simple" by "smooth."

($\mathrm{Err}_{\mathrm{IV}}$, 5). In ($0_{\mathrm{IV}}$, 16.1.4), on the eighth line from the bottom of p. 23, replace "To say that A is catenary" by "To say that an *integral* ring A is catenary."

($\mathrm{Err}_{\mathrm{IV}}$, 6). In ($0_{\mathrm{IV}}$, 16.1.9), the hypothesis that A is Noetherian may be deleted. Indeed, for every $u \in B_M$, $A[u]$ is an A-subalgebra of C, so u is integral over A (Bourbaki, *Alg. comm.*, chap. V, §1, no. 1, corollary to prop. 1), and hence also over A_M; one concludes again by (16.1.5).

($\mathrm{Err}_{\mathrm{IV}}$, 7). In ($0_{\mathrm{IV}}$, 16.5.12), note that relation (16.5.11.1) is equivalent to the same relation after replacing A by the quotient rings $A/\mathfrak{r}$, where $\mathfrak{r}$ ranges over the minimal prime ideals of A. Thus in the proof of (16.5.12) one may reduce to the case in which A is integral and apply (16.1.4.2).

($\mathrm{Err}_{\mathrm{IV}}$, 8). Delete ($0_{\mathrm{IV}}$, 19.3.12), whose proof is erroneous (the sets W_λ, U_λ, and V_λ are not linear varieties).

($\mathrm{Err}_{\mathrm{IV}}$, 9). In ($0_{\mathrm{IV}}$, 19.8.3), on the sixth and eighth lines from the bottom of p. 110, replace "homomorphism" by "local homomorphism"; make the same replacement in (0_{IV}, 19.8.4), line 1 of p. 111.

($\mathrm{Err}_{\mathrm{IV}}$, 10). In ($0_{\mathrm{IV}}$, 20.1.2), on the second line from the bottom of p. 117, replace $a \in A$ by $b \in B$ and aD by bD; on the last line, replace A-module by B-module.

($\mathrm{Err}_{\mathrm{IV}}$, 11). In ($0_{\mathrm{IV}}$, 20.2.1), on line 8 of p. 119, replace A-modules by B-modules. On lines 21 and 22, replace "di-homomorphisms of modules (relative to u)" by "homomorphism of B-modules."

($\mathrm{Err}_{\mathrm{IV}}$, 12). In ($0_{\mathrm{IV}}$, 20.4.6.1), replace $p_2 - p_1$ by $p_1 - p_2$; on the tenth line from the bottom of p. 125, replace $x \otimes 1 - 1 \otimes x$ by $1 \otimes x - x \otimes 1$.

($\mathrm{Err}_{\mathrm{IV}}$, 13). In ($0_{\mathrm{IV}}$, 20.4.8), on line 14 of p. 126, replace $p_2 = p_1 - d_{B/A}$ by $p_1 = p_2 - d_{B/A}$.

($\mathrm{Err}_{\mathrm{IV}}$, 14). In ($0_{\mathrm{IV}}$, 20.4.13), replace the sixth through eleventh lines from the bottom of p. 127 by the following text:

(iv) Let A and B be integral rings such that $A \subset B$, with B integral over A, and suppose that the fraction field of B is a separable extension of that of A. Then the B-module $\Omega^1_{B/A}$ is a torsion module. Indeed, for every $x \in B$, let f be the minimal polynomial of x over the fraction field K of A; there is a nonzero $a \in A$ such that $af[T] \in A[T]$. Since x is separable over K, $f'(x) \neq 0$, while the relation $af(x) = 0$ gives $af'(x)d_{B/A}x = 0$; the assertion follows from (20.4.7).

($\mathrm{Err}_{\mathrm{IV}}$, 15). In ($0_{\mathrm{IV}}$, 22.2.7), on line 5 of p. 189, replace "formally smooth" by "formally smooth for the preadic topologies."

($\mathrm{Err}_{\mathrm{IV}}$, 16). In the Summary of Chapter IV, on the fourth line from the bottom of p. 222, add "and strictly local rings." Replace the last three lines by:

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§20. Meromorphic functions and pseudo-morphisms.
§21. Divisors.

On line 16 of p. 223, replace "of finite presentation" by "locally of finite presentation"; on line 19, replace "as" by "as being."

($\mathrm{Err}_{\mathrm{IV}}$, 17). In (IV, 3.1.10), add after line 16 of p. 39:

Remark (3.1.10.2). — Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module, so that $f_*(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_Y$-Module (1.7.4). Then

$$\mathrm{Ass}(f_*(\mathscr{F})) \subset f(X), \tag{3.1.10.3}$$

$$\mathrm{Ass}(f_*(\mathscr{F})) \subset f(\mathrm{Ass}(\mathscr{F})) \quad \text{if } X \text{ is locally Noetherian.} \tag{3.1.10.4}$$

Indeed, let $y \in \mathrm{Ass}(f_*(\mathscr{F}))$. By (2.3.1) and the flatness of $\mathrm{Spec}(\mathscr{O}_{Y,y}) \to Y$, $(f_*(\mathscr{F}))_y$ equals $(f'_*(\mathscr{F}'))_y$, where

$$f' : X \times_Y \mathrm{Spec}(\mathscr{O}_{Y,y}) \longrightarrow \mathrm{Spec}(\mathscr{O}_{Y,y})$$

is obtained from f by base change and $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y,y}$. In view of (I, 3.6.5) and the definition of $\mathrm{Ass}(\mathscr{F}')$, we may therefore replace Y by $\mathrm{Spec}(\mathscr{O}_{Y,y})$, f by f', and $\mathscr{F}$ by $\mathscr{F}'$.

By definition there is a nonzero element $t_y \in (f_*(\mathscr{F}))_y$ annihilated by $\mathfrak{m}_y$. Since Y is local, $(f_*(\mathscr{F}))_y$ identifies with $\Gamma(Y, f_*(\mathscr{F}))$, and hence t_y identifies with a section $s \in \Gamma(X, \mathscr{F})$. The annihilator Ideal of s in $\mathscr{O}_X$ contains $\mathfrak{m}_y\mathscr{O}_X$, so the support of s is contained in $f^{-1}(y)$; if this inverse image were empty then $s = 0$, a contradiction. If moreover X is locally Noetherian, the coherent sub-$\mathscr{O}_X$-Module $\mathscr{O}_X s$ of $\mathscr{F}$ is nonzero, and therefore $\mathrm{Ass}(\mathscr{O}_X s) \neq \varnothing$ (3.1.5). But

$$\mathrm{Ass}(\mathscr{O}_X s) \subset \mathrm{Supp}(\mathscr{O}_X s) \subset f^{-1}(y), \qquad \mathrm{Ass}(\mathscr{O}_X s) \subset \mathrm{Ass}(\mathscr{F})$$

by (3.1.7), and hence $y \in f(\mathrm{Ass}(\mathscr{F}))$.

Remark (3.1.10.5). — Even when f is proper, one need not have $f(\mathrm{Ass}(\mathscr{F})) \subset \mathrm{Ass}(f_*(\mathscr{F}))$. Take Y to be the spectrum of a field and X a projective space over k, for example $\mathbf{P}^1_k$; then $\mathscr{F} = \mathscr{O}_X(-1)$ is a nonzero coherent $\mathscr{O}_X$-Module with $\Gamma(X, \mathscr{F}) = 0$ (III, 2.1.13), so $\mathrm{Ass}(f_*(\mathscr{F})) = \varnothing$ whereas $\mathrm{Ass}(\mathscr{F}) \neq \varnothing$.

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($\mathrm{Err}_{\mathrm{IV}}$, 18). After (IV, 3.2.8), insert after line 17 of p. 43:

Remark (3.2.9). — For every point s of a prescheme X, write $i_s : \mathrm{Spec}(k(s)) \to X$ for the canonical morphism, which is a quasi-compact monomorphism. For every $k(s)$-module $G(s)$, regarded as a Module on $\mathrm{Spec}(k(s))$, $(i_s)_*(G(s))$ is therefore a quasi-coherent $\mathscr{O}_X$-Module (1.7.4). Its fibre is zero at every point $x \notin \overline{\{s\}}$ and is isomorphic to $G(s)$ otherwise. If X is locally Noetherian and $G(s)$ has finite length, then $\mathrm{Ass}((i_s)_*(G(s)))$ consists of the single point s; in other words, $(i_s)_*(G(s))$ is irredundant.

Now suppose that X is locally Noetherian and that $\mathscr{F}$ is a coherent $\mathscr{O}_X$-Module. Let S be a discrete subset of X, and for every $s \in S$ put $\mathscr{F}(s) = i_s^*(\mathscr{F})$, a $k(s)$-module. For each $s \in S$, choose a finite-length quotient $G(s)$ of $\mathscr{F}(s)$. There is a canonical homomorphism

$$u : \mathscr{F} \longrightarrow \bigoplus_{s \in S} (i_s)_*(\mathscr{F}(s)) \longrightarrow \bigoplus_{s \in S} (i_s)_*(G(s))$$

of coherent $\mathscr{O}_X$-Modules, since a discrete subset of X is locally finite. If $\mathscr{H} = \mathrm{Ker}\, u$ and $\mathscr{G} = \mathscr{F}/\mathscr{H}$, then $\mathscr{G}$ is coherent and u induces an injection

$$v : \mathscr{G} \longrightarrow \bigoplus_{s \in S} (i_s)_*(G(s)).$$

If $\mathscr{G}^{(s)}$ denotes the image of $\mathscr{G}$ in $(i_s)_*(G(s))$, the family $(\mathscr{G}^{(s)})_{s \in S}$ is an irredundant decomposition, reduced when all the $G(s)$ are nonzero, and $\mathrm{Ass}(\mathscr{G}) = S$; thus $\mathscr{G}$ has no embedded associated prime cycle.

This gives a bijection $(G(s))_{s \in S} \leftrightarrow \mathscr{G}$, where S ranges over the discrete subsets of X for which $\mathscr{F}(s) \neq 0$ for every $s \in S$, and, for each S, the $G(s)$ range over the nonzero finite-length quotients of $\mathscr{F}(s)$; on the other side, $\mathscr{G}$ ranges over the coherent quotient $\mathscr{O}_X$-Modules of $\mathscr{F}$ without embedded associated prime cycles. The inverse bijection associates to $\mathscr{G}$ its reduced irredundant decomposition.

($\mathrm{Err}_{\mathrm{IV}}$, 19). In (IV, 4.2.1), before the seventeenth line from the bottom of p. 55, insert:

Remark (4.2.1.4). — Under the hypotheses of (4.2.1), one also has

$$\dim(K \otimes_k L) = inf(\mathrm{tr.deg}_k K, \mathrm{tr.deg}_k L). \tag{4.2.1.5}$$

If both fields have infinite transcendence degree (which may be an arbitrary cardinal), the left-hand side is to be understood as $+\infty$; the dimension here is that of (0, 16.1.1).

Suppose, for example, that $\mathrm{tr.deg}_k K \leqslant \mathrm{tr.deg}_k L$, and first that $\mathrm{tr.deg}_k K = n < +\infty$. In the proof of (4.2.1) we saw that $K \otimes_k L$ is an algebraic extension of $K' \otimes_k L$, and therefore has the same dimension (0, 16.1.5). We may reduce to $K = k[T_1, \dots, T_n]$; then $K \otimes_k L$ is a ring of fractions of

$$k[T_1, \dots, T_n] \otimes_k L = L[T_1, \dots, T_n],$$

which has dimension n (5.5.4), so $\dim(K \otimes_k L) \leqslant n$ (0, 16.1.3.1). To prove equality, it is enough to find in $\mathrm{Spec}(L[T_1, \dots, T_n])$ an L-rational point belonging to $\mathrm{Spec}(K \otimes_k L)$ (5.2.3), or equivalently (I, 1.7.4) an L-homomorphism $u : K \otimes_k L \to L$. By the definition of the tensor product this is equivalent to a k-homomorphism $v : K \to L$, whose existence follows from the inequality of transcendence degrees.

If the two transcendence degrees are infinite, the same argument gives a rational point of $\mathrm{Spec}(L[(t_\alpha)_{\alpha \in I}])$ belonging to $\mathrm{Spec}(K \otimes_k L)$. The kernel of an L-homomorphism from this polynomial ring to L is a maximal ideal generated by a family $(t_\alpha - a_\alpha)_{\alpha \in I}$ with $a_\alpha \in L$. Such a maximal ideal contains an infinite descending chain of distinct prime ideals; the corresponding local ring is non-Noetherian and infinite-dimensional. Since it is a local ring of $K \otimes_k L$, the latter is also non-Noetherian and infinite-dimensional.

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($\mathrm{Err}_{\mathrm{IV}}$, 20). After (IV, 4.5.19), before the twelfth line from the bottom of p. 67, insert:

Corollary (4.5.19.3). — *Let k be a field and X a finite-type k-prescheme which is irreducible but not geometrically irreducible. There is a closed subset Y of X, nowhere dense in X, containing every $x \in X$ for which $k(x)$ is radicial over k (and in particular every k-rational point of X).*

Let k' be an algebraic closure of k, let $X' = X \otimes_k k'$, and let $p : X' \to X$ be the canonical projection. If $k(x)$ is radicial over k, there is only one point $x' \in X'$ over x (I, 3.4.9); this point must belong to two distinct irreducible components of X', since otherwise X would be geometrically irreducible by (4.5.19). Let X'_i $(1 \leqslant i \leqslant n)$ be the irreducible components of the Noetherian prescheme X', and for $1 \leqslant i < j \leqslant n$ put $X'_{ij} = X'_i \cap X'_j$. Then

$$Y = p\left(\bigcup_{i<j} X'_{ij}\right)$$

has the required properties. Since p is integral and the X'_{ij} are closed in X', Y is closed in X. If ξ is the generic point of X, $p^{-1}(\xi)$ is discrete and consists of the generic points of the X'_i (4.5.11), so it meets none of the X'_{ij}; hence Y is nowhere dense.

($\mathrm{Err}_{\mathrm{IV}}$, 21). In (IV, 5.6.11.1), on the seventh line from the bottom of p. 102, add "of dimension 2" after "integral."

($\mathrm{Err}_{\mathrm{IV}}$, 22). After (IV, 6.2.3), after line 18 of p. 138, insert:

Proposition (6.2.4). — *Under the hypotheses of (6.2.3), let in addition M be a finite A-module, and suppose that $M \neq 0$ and $N \neq 0$. Then*

(6.2.4.1) $$\mathrm{pd}_B(M \otimes_A N) = \mathrm{pd}_A(M) + \mathrm{pd}_{B\otimes_A k}(N \otimes_A k).$$

Consider the left resolution $L_\bullet$ of N introduced in (6.2.3), and a left resolution $K_\bullet$ of M by finite free A-modules:

$$\cdots \longrightarrow K_i \longrightarrow K_{i-1} \longrightarrow \cdots \longrightarrow K_0 \longrightarrow M \longrightarrow 0.$$

We saw that the L_i, N, and the $Z_i(L_\bullet)$ are flat A-modules. The same is true of the $B_i(L_\bullet)$, which equal $Z_i(L_\bullet)$ except for $i = 0$; in that case $B_0(L_\bullet)$ is flat by (0_{I}, 6.1.2). Finally the $H_i(L_\bullet)$ vanish except for $H_0(L_\bullet) = N$, which is flat over A. Under these conditions one knows that

$$H_n(K_\bullet \otimes_A L_\bullet) \simeq \bigoplus_{p+q=n} H_p(K_\bullet) \otimes_A H_q(L_\bullet)$$

(G, I, 5.5). Thus $K_\bullet \otimes_A L_\bullet$, with the usual total grading, is a resolution of $M \otimes_A N$ by free B-modules.

Put

$$r = \mathrm{pd}_A(M), \qquad s = \mathrm{pd}_B(N) = \mathrm{pd}_{B\otimes_A k}(N \otimes_A k).$$

There are free resolutions $K_\bullet$ and $L_\bullet$ of lengths r and s, so their tensor product is a free resolution of length $r + s$; this proves $\mathrm{pd}_B(M \otimes_A N) \leqslant r + s$. It remains, by (0, 17.2.6), to show

$$\mathrm{Tor}^B_{r+s}(M \otimes_A N, k(B)) \neq 0.$$

Using the preceding resolution, the left-hand side is

$$H_{r+s}((K_\bullet \otimes_A L_\bullet) \otimes_B k(B)).$$

But

$$(K_\bullet \otimes_A L_\bullet) \otimes_B k(B) = (K_\bullet \otimes_A k) \otimes_k (L_\bullet \otimes_B k(B)).$$

This is a tensor product of complexes of vector spaces over k, so the Künneth formula gives

$$H_{r+s} \simeq H_r(K_\bullet \otimes_A k) \otimes_k H_s(L_\bullet \otimes_B k(B)),$$

since the homology of the first, respectively second, complex vanishes above r, respectively s. Hence, up to canonical isomorphism,

$$\mathrm{Tor}^B_{r+s}(M \otimes_A N, k(B)) = \mathrm{Tor}^A_r(M, k) \otimes_k \mathrm{Tor}^B_s(N, k(B)),$$

and both factors are nonzero by hypothesis; so is their tensor product.

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($\mathrm{Err}_{\mathrm{IV}}$, 23). In (IV, 6.10.1), on line 8 of p. 155, add after "canonical": "$\mathscr{F}$ a coherent $\mathscr{O}_X$-Module."

($\mathrm{Err}_{\mathrm{IV}}$, 24). In (IV, 6.14.4), the reference on the twelfth line from the bottom of p. 175 is insufficient; argue as follows. First, if M is a transcendence basis of K' over A, then $A(M)$ is algebraically closed in $B(A(M))$ (Bourbaki, *Alg.*, chap. V, §6, exercise 8(c)). We may therefore reduce to the case in which K' is an algebraic separable extension of A.

Let p be the characteristic exponent of A. It follows from Bourbaki, *loc. cit.*, §9, exercise 2, that the algebraic closure of K' in $B(K')$ is radicial over K'. We show that every $x \in B(K')$ such that $x^{p^r} \in K'$ already belongs to K'. Choose a basis (e_λ) of K' over A. Since K'/A is algebraic separable, the $e_\lambda^{p^r}$ also form a basis of K' over A (Bourbaki, *loc. cit.*, §8, no. 3, corollary to prop. 4). Write $x = \sum_\lambda e_\lambda b_\lambda$ with $b_\lambda \in B$. By hypothesis there are $a_\lambda \in A$ such that

$$\sum_\lambda e_\lambda^{p^r} b_\lambda^{p^r} = \sum_\lambda e_\lambda^{p^r} a_\lambda.$$

Therefore $b_\lambda^{p^r} = a_\lambda$ for every λ. Since A is algebraically closed in B, each b_λ lies in A, and hence $x \in K'$.

($\mathrm{Err}_{\mathrm{IV}}$, 25). In (IV, 7.4.8, B), p. 203, the question was answered affirmatively by R. Kiehl, "Ausgezeichnete Ringe der nichtarchimedischen analytischen Geometrie," forthcoming in *Inventiones Mathematicae*.

($\mathrm{Err}_{\mathrm{IV}}$, 26). In (IV, 7.8.4), on the ninth line from the bottom of p. 217, replace "We shall see later (§18) that if k is a complete nondiscrete valued field…" by:

> R. Kiehl proved (in the article cited in ($\mathrm{Err}_{\mathrm{IV}}$, 25)) that if k is a complete nondiscrete valued field…

($\mathrm{Err}_{\mathrm{IV}}$, 27). In (IV, 7.8.4), insert before the sixth line from the bottom of p. 217:

(vi) Let A be a one-dimensional Noetherian local integral ring. It is then universally catenary (7.2.9), and hence, by (7.3.19), it is excellent if and only if its formal fibres are geometrically reduced; equivalently, if and only if its completion $\widehat{A}$ is reduced and the component fields of the total ring of fractions of $\widehat{A}$ are separable extensions of the fraction field K of A. By (7.6.4) and (7.7.2), this is also equivalent to saying that A is universally Japanese. When A is a discrete valuation ring, so is $\widehat{A}$; to say that A is excellent then means that the fraction field of $\widehat{A}$ is a separable extension of K, a condition which is always satisfied when K has characteristic zero.

($\mathrm{Err}_{\mathrm{IV}}$, 28). After (IV, 8.10.3), following line 14 of p. 37, insert:

Remark (8.10.3.1). — The hypothesis in (8.10.3) that the sets $\overline{f_\alpha(X_\alpha)}$ be constructible cannot be omitted. For example, take $S = \mathrm{Spec}(A)$, where A is a ring such that S is compact, totally disconnected, and has no isolated point, every prime ideal $\mathfrak{p}$ of A is maximal, and $A_{\mathfrak{p}}$ is a field (Bourbaki, *Alg. comm.*, chap. II, §4, exercises 16 and 17). Let $s \in S$, and take for $Y_\alpha = S_\alpha$ the affine open neighbourhoods of s in S. For every α, take $X_\alpha = \mathrm{Spec}(k(s))$, with f_α and f the canonical morphisms. It is clear that f, which is the identity, is dominant, whereas none of the f_α is.

($\mathrm{Err}_{\mathrm{IV}}$, 29). In (IV, 8.12.8), on line 7 of p. 47, replace "such that there exists an ample $\mathscr{O}_Y$-Module" by "and quasi-separated." On line 12 of p. 47, replace "The hypothesis" by:

> It follows from (8.12.3) that the existence of the factorisation in the statement is a property local on Y; we may therefore suppose Y affine, which

($\mathrm{Err}_{\mathrm{IV}}$, 30). After (IV, 8.12.11), add at the end of p. 48:

Corollary (8.12.12). — *Let X and Y be two integral preschemes and let $f : X \to Y$ be a separated, birational morphism locally of finite type. Let $y \in Y$ be a normal point of Y, and suppose that a point $x \in f^{-1}(y)$ is isolated in $f^{-1}(y)$. Then there exists an open neighbourhood U of y in Y such that the restriction $f^{-1}(U) \to U$ of f is an isomorphism.*

Proof. It is enough to prove that there exist an open neighbourhood U of y in Y and a U-section $g : U \to f^{-1}(U)$ of $f^{-1}(U)$ such that $g(y) = x$. Indeed, U contains the generic point η of Y and $f^{-1}(U)$ contains the generic point ξ of X; necessarily $g(\eta) = \xi$, since f is birational (6.15.4). As g is an immersion, $g(U)$ is locally closed in $f^{-1}(U)$, hence open in its closure there; the preceding observation shows that this closure is $f^{-1}(U)$. The subprescheme of $f^{-1}(U)$ associated with g is isomorphic to U, hence reduced, and $f^{-1}(U)$ is reduced, so (I, 5.2.1) shows that g is an open immersion. On the other hand, since f is separated, g is a closed immersion (I, 5.4.6). Thus $g(U)$ is both open and closed in $f^{-1}(U)$. Since $f^{-1}(U)$ is integral, this gives $g(U) = f^{-1}(U)$, as required.

We consider two cases.

I) First suppose Y normal, and let V be the set of points of X at which f is quasi-finite; this is open in X (13.1.4).[42] By (8.12.10), the restriction $f|V : V \to Y$ is an open immersion. Hence there is an open neighbourhood U of y in Y such that $f|V$ is an isomorphism from V onto the subprescheme induced by Y on U; its inverse g is a U-section of $f^{-1}(U)$ satisfying $g(y) = x$.

II) The construction of a U-section g with $g(y) = x$ is local on X and Y. We may therefore suppose that X and Y are affine and that f is of finite type, so that X is a closed subprescheme of a Y-prescheme Z of finite presentation over Y (for example one of the form $\mathbb{A}^r_Y$). Put

$$Y' = \mathrm{Spec}(\mathscr{O}_{Y,y}), \qquad X' = X_{(Y')}, \qquad Z' = Z_{(Y')};$$

[42] The reader will verify that neither (8.12.12) nor any of its consequences is used in the proof of (13.1.4).

thus X' is a closed subprescheme of Z'. If $f' = f_{(Y')}$, then $f'^{-1}(y) = f^{-1}(y)$ (I, 3.6.5), so x is isolated in $f'^{-1}(y)$. Since Y' is normal, case I applies to f'; and since Y' is the only neighbourhood of y in Y', there is a Y'-section g' of X' with $g'(y) = x$, which may also be regarded as a Y'-section of Z'.

The structural morphism $h : Z \to Y$ is of finite presentation. Applying (8.8.2, (i) and (ii)) by the method of (8.1.2, a), we obtain an open neighbourhood U of y in Y and a U-section g of $h^{-1}(U)$ such that $g(y) = x$. It remains to see that g is in fact a U-section of $f^{-1}(U)$. Since $f^{-1}(U)$ is a closed subprescheme of $h^{-1}(U)$ and U is reduced, it is enough by (I, 5.2.2) to show that $g(U) \subset f^{-1}(U)$. As U is integral and $f^{-1}(U)$ is closed in $h^{-1}(U)$, it is enough to check this at the generic point η of Y (and of U). But the restriction of g to Y' is a Y'-section of $X' \subset f^{-1}(U)$, and $\eta \in Y'$. □

(Err$_{\text{IV}}$, 31). In (IV, 10.4), after Proposition (10.4.11), add:

Corollary (10.4.12). — *Let S be a prescheme, let X be an S-prescheme of finite presentation, and let f be a radicial S-endomorphism of X.*

(i) *The endomorphism f is bijective and finite (and hence is a universal homeomorphism (2.4.5)).*

(ii) *Suppose in addition that X is reduced, that the local rings of S are Noetherian and universally Japanese, and that f is birational (a condition automatically satisfied if $k(x)$ is perfect for every maximal point x of X). Then f is an S-automorphism.*

Proof. (i) The question is local on S, so we may restrict to the case $S = \operatorname{Spec}(A)$ affine. There is a finitely generated $\mathbb{Z}$-subalgebra A_0 of A such that, on putting $S_0 = \operatorname{Spec}(A_0)$, one has $X = X_0 \times_{S_0} S$, where X_0 is a prescheme of finite type over S_0, and such that f is obtained from an S_0-endomorphism f_0 of X_0 by the base change $S \to S_0$ (8.9.1). We may moreover suppose f_0 radicial (8.10.5, (vii)). It is enough to prove that f_0 is bijective and finite, so from now on we may suppose that A is a finitely generated $\mathbb{Z}$-algebra. It follows that S and X are excellent preschemes ((7.8.3) and (7.8.6)), and in particular that X is universally Japanese ((5.11.4) and (7.8.3, (vi))).

We may also suppose X reduced. Indeed, $f_{\text{red}} : X_{\text{red}} \to X_{\text{red}}$ is a radicial S_{red}-endomorphism (I, 5.1.6). If f_{red} is finite, then so is f: because f is radicial, it is separated (1.8.7.1); since f_{red} is proper, f is proper as well (II, 5.4.6), and since f is plainly quasi-finite, it is finite (III, 4.4.2).

Let X' be the normalization of X (II, 6.3.8). If X_i ($1 \leqslant i \leqslant n$) are the reduced subpreschemes of X whose underlying spaces are its irreducible components, then X' is the disjoint sum of the normalizations X'_i of the integral preschemes X_i. Write x_i (respectively x'_i) for the generic point of X_i (respectively X'_i). The canonical morphism $p : X' \to X$ is finite because X is universally Japanese, and $p^{-1}(x_i) = \{x'_i\}$, with $k(x_i) \to k(x'_i)$ bijective.

Write $f = (\psi, \theta)$. We already know from (10.4.11) that ψ is a continuous bijection of the space X onto itself. Put $y_i = \psi(x_i)$. Then y_i is again a maximal point of X: otherwise a generalization $y'_i \neq y_i$ of y_i would make x_i a generalization, different from itself, of $\psi^{-1}(y'_i)$. The set M of maximal points of X is finite, so $\psi|M$ is a permutation; write $\psi(x_i) = x_{\sigma(i)}$ for a permutation σ of $\{1, \ldots, n\}$. For each i, the restriction of f to X_i therefore factors as

$$X_i \xrightarrow{f_i} X_{\sigma(i)} \longrightarrow X,$$

the second arrow being the canonical injection (I, 5.2.2). Since the X_i are integral, there is by (II, 6.3.9) a unique morphism $f'_i : X'_i \to X'_{\sigma(i)}$ making the diagram

$$\begin{array}{ccc} X'_i & \xrightarrow{f'_i} & X'_{\sigma(i)} \\ \downarrow p_i & & \downarrow p_{\sigma(i)} \\ X_i & \xrightarrow{f_i} & X_{\sigma(i)} \end{array}$$

commute. Taking f' to agree with f'_i on every X'_i gives a commutative diagram

(10.4.12.1)
$$\begin{array}{ccc} X' & \xrightarrow{f'} & X' \\ \downarrow p & & \downarrow p \\ X & \xrightarrow{f} & X. \end{array}$$

Since f is separated, so is $f \circ p = p \circ f'$, and hence f' is separated (I, 5.5.1). Since p is finite and f bijective, the diagram (10.4.12.1) also shows that f' is quasi-finite. The X'_i being integral and normal, (8.12.11) gives a factorisation

$$X'_i \xrightarrow{f''_i} X''_{\sigma(i)} \xrightarrow{u_{\sigma(i)}} X'_{\sigma(i)},$$

where $X''_{\sigma(i)}$ is the integral closure of $X'_{\sigma(i)}$ relative to the extension $k(x'_i)$ of $k(x'_{\sigma(i)})$, and f''_i is an open immersion. As $k(x'_i)/k(x'_{\sigma(i)})$ is radicial, $u_{\sigma(i)}$ is radicial (Bourbaki, *Alg. comm.*, chap. V, §2, no. 3, lemma 4). Consequently f' is radicial, and (10.4.11) shows that it is surjective. Since the $u_{\sigma(i)}$ are injective, each f''_i is bijective and hence an isomorphism. It follows that f' is finite, because the $u_{\sigma(i)}$ are finite, X' being universally Japanese (7.8.3). Thus $p \circ f' = f \circ p$ is finite; as p is surjective and f separated, f is proper (II, 5.4.3), and being quasi-finite it is finite (8.11.1).

(ii) By (8.10.5, (i)), applied by the method of (8.1.2, a), we may restrict to the case in which S is a local scheme. The preliminary reductions made in the proof of (i) are therefore unnecessary here. In the same notation, one obtains the commutative diagram (10.4.12.1), in which X is an S-prescheme of finite type, p is finite, f' is an isomorphism, and the canonical homomorphism $\mathscr{O}_X \to p_*(\mathscr{O}_{X'})$ is injective. In Chapter VI, in the theory of quotients of preschemes by equivalence relations, we shall prove that under these conditions f itself is an isomorphism (the reader will verify that (10.4.12) is not used in that theory). $\square$

Corollary (10.4.13). — *Let k be a field and X an algebraic prescheme over k. Every radicial k-endomorphism f of X is finite and bijective, hence a universal homeomorphism. If, in addition, X is reduced and f is birational (for example if k has characteristic zero), then f is a k-automorphism.*

Remark (10.4.14). —

(i) In (10.4.12) and (10.4.13), the hypothesis that X is reduced cannot be omitted from the second conclusion. This is shown by the example of the B-endomorphism corresponding to the augmentation $D_B(L) \to B$ of the ring $D_B(L) = A$ defined in (0, 18.2.3), where B is identified with a subring of A.

(ii) Let X be a Noetherian prescheme. We do not know whether every radicial endomorphism f of X is necessarily surjective and finite. If f is assumed surjective and X is Japanese, the argument of (10.4.12) shows that f is finite. If moreover X is reduced and f is birational—for example, if $k(x)$ is perfect for every maximal point $x \in X$—the same argument shows that f is an automorphism.

(iii) There are examples of nonsurjective radicial endomorphisms of non-Noetherian integral affine schemes. Let k be a ring, let $A = k[(T_\lambda)_{\lambda \in L}]$ be a polynomial ring with L countably infinite, and let $\mathfrak{J}$ be the ideal generated by a subfamily $(T_\lambda)_{\lambda \in H}$, where $L \setminus H$ is infinite and $H \neq \varnothing$. Then $A/\mathfrak{J}$ is isomorphic to $k[(T_\lambda)_{\lambda \in L\setminus H}]$, hence to A. The endomorphism of $\mathrm{Spec}(A)$ corresponding to the composite

$$A \xrightarrow{p} A/\mathfrak{J} \xrightarrow{u} A,$$

where p is the canonical projection and u an isomorphism, is such an example: its image is the closed subprescheme of $\mathrm{Spec}(A)$ defined by $\mathfrak{J}$.

($\mathrm{Err}_{\mathrm{IV}}$, 32). In (IV, 11.3.14), on line 7 of p. 141, insert "(i)" before "Let." After line 9 of p. 141, add:

(ii) Conversely, let $f : X \to Y$ be a flat morphism. If X is unibranch (respectively geometrically unibranch) at a point x, then Y is unibranch (respectively geometrically unibranch) at the point $y = f(x)$.

At the beginning of line 10 of p. 141, insert "(i)." After line 20 of p. 141, add:

(ii) Put $Y_1 = Y_{\mathrm{red}}$ and $X_1 = X \times_Y Y_1$. The underlying space of X_1 is that of X, and the morphism $f_1 : X_1 \to Y_1$ obtained from f by base change is flat. We may therefore restrict to the case $Y = \mathrm{Spec}(A)$, with A a reduced local ring and y the closed point of Y. Put $B = \mathscr{O}_{X,x}$. By hypothesis B_{red} is unibranch, so $\mathrm{Spec}(B)$ has only one maximal point; flatness (2.3.4) then gives the same conclusion for $\mathrm{Spec}(A)$. Since A is reduced, it is therefore integral.

Let K be the fraction field of A, and let A' be its integral closure. By flatness, $B \otimes_A A'$ identifies with a subring of $B \otimes_A K$ containing B. Since the nonzero elements of A are B-regular by flatness, $B \otimes_A K$ identifies with a subring of the total ring of fractions R of B. To show that A is unibranch, it is enough to show that A' cannot have two distinct maximal ideals (necessarily lying over y;

Bourbaki, *Alg. comm.*, chap. V, §2, no. 1, prop. 1). If it did, then $B' = B \otimes_A A'$ would have two distinct prime ideals over x (I, 3.4.7), and we need only show that this is impossible.

Let $\mathfrak{N}$ be the nilradical of B. If S is the set of regular elements of B, then $R = S^{-1}B$, and $S^{-1}\mathfrak{N} = R\mathfrak{N}$ is the nilradical of R. Hence

$$R/R\mathfrak{N} = S^{-1}B/S^{-1}\mathfrak{N} = S^{-1}(B/\mathfrak{N})$$

is contained in the fraction field L of the integral local ring $B/\mathfrak{N} = B_{\mathrm{red}}$. The same is therefore true of

$$B'_{\mathrm{red}} = B'/(B' \cap R\mathfrak{N}),$$

which is integral over B_{red}. Since B'_{red} is contained in the integral closure of B_{red}, that ring is local by hypothesis; hence B'_{red}, and therefore B', is local. This proves the assertion.

If B is moreover geometrically unibranch, the residue field of B'_{red}, being contained in the residue field of the integral closure of B_{red}, is a radicial extension of $k(x)$. But the residue field of B'_{red} is also that of $B' = B \otimes_A A'$, hence is $k(x) \otimes_{k(y)} k'$, where k' is the residue field of A'. It follows from (2.6.1) that $k'/k(y)$ is radicial, so A is geometrically unibranch as well.

($\mathrm{Err}_{\mathrm{IV}}$, 33). The example in (IV, 12.1.2, (i)) is incorrect: the morphism considered there is open. An example of a flat morphism $f : X \to Y$ whose restriction to an irreducible component of X is not open may be constructed as follows. Let $Z = \mathrm{Spec}(\mathbb{C}[U, V])$ be the affine plane over the complex numbers, and let Y be the scheme obtained by identifying two distinct closed points a, b of Z. Thus $Y = \mathrm{Spec}(B)$, where B is the subring of $\mathbb{C}[U, V]$ formed by the polynomials taking the same value at a and b (the set of closed points of Z being identified with $\mathbb{C}^2$). Plainly B is a complex vector subspace of codimension one in $\mathbb{C}[U, V]$, so $\mathbb{C}[U, V]$ is a finite B-module. Consequently $Z \to Y$ is finite, and the inverse image of the point c of Y obtained by identifying a and b is $\{a, b\}$.

Now form X by gluing two copies Z_1, Z_2 of Z so that, if a_i, b_i ($i = 1, 2$) are the points corresponding to a, b, then a_1 is glued to b_2 and b_1 to a_2 (cf. (6.5.5, (ii))). It is easy to see (*loc. cit.*) that the resulting morphism $X \to Y$ is flat and finite (indeed, étale), and that Z_1 and Z_2 identify with the two irreducible components of X. If D is a line in Z_1 through a_1 but not through b_1, the image in Y of the open set $Z_1 \setminus D$ contains the image of b_1, namely c, but does not contain the image of the generic point of D. That image is therefore not a neighbourhood of c.

($\mathrm{Err}_{\mathrm{IV}}$, 34). In the title of (IV, 13.2), on the fifth line from the bottom of p. 190, replace "equidimensional" by "locally equidimensional." Likewise, in (IV, 13.2.2), on line 4 of p. 191, replace both occurrences of "equidimensional" by "locally equidimensional." On line 5 of p. 191, add:

> If, in addition, the fibres $f^{-1}(f(x))$ of f are equidimensional (0_{IV}, 14.1.3), then f is said to be *equidimensional*.

On the sixth line from the bottom of p. 194, replace "equidimensional" by "locally equidimensional."

($\mathrm{Err}_{\mathrm{IV}}$, 35). In the title of (IV, 13.3), on the fourth line from the bottom of p. 94, replace "equidimensional" by "locally equidimensional." In (IV, 13.3.2), on the sixth line from the bottom of p. 196, make the same replacement twice. On the fifth line from the bottom of p. 196, add:

> If, in addition, the fibres $f^{-1}(f(x))$ of f are equidimensional (0_{IV}, 14.1.3), then f is said to be *equidimensional*.

In (IV, 13.3.7), on line 14 of p. 198, replace "equidimensional" by "locally equidimensional." In (IV, 13.3.9), on the third line from the bottom of p. 198, replace "equidimensionality" by "local equidimensionality." On lines 10, 12, 20, and 24 of p. 199, replace "equidimensional" by "locally equidimensional."

($\mathrm{Err}_{\mathrm{IV}}$, 36). In the introduction to §14, on the sixth line from the bottom of p. 199, replace "equidimensional" by "locally equidimensional."

($\mathrm{Err}_{\mathrm{IV}}$, 37). In (IV, 14.4.1), on the fifth line from the bottom of p. 209, replace "locally of finite type" by "locally of finite presentation." The present statement of (IV, 14.4.1) is false even when $f : X \to Y$ is a closed immersion, X consists of a single closed point y of Y, and $\mathscr{O}_{Y,y}$ is a field. Indeed, there exist rings A such that $Y = \mathrm{Spec}(A)$ is compact, has no isolated points, and is totally disconnected, while every prime ideal $\mathfrak{p}$ of A is maximal and $A_{\mathfrak{p}}$ is a field (Bourbaki, *Alg. comm.*, chap. II, §4, exercises 16 and 17). Clearly the closed immersion $\{y\} \to Y$ is not open, although it satisfies conditions b) and b') of (IV, 14.4.1).

($\mathrm{Err}_{\mathrm{IV}}$, 38). In (IV, 14.4.1.3), on line 14 of p. 211, replace "locally quasi-finite and dominant" by "locally quasi-finite, locally of finite presentation, and dominant."

($\mathrm{Err}_{\mathrm{IV}}$, 39). In (IV, 14.4.2), on the tenth line from the bottom of p. 211, replace "locally of finite type" by "locally of finite presentation."

($\mathrm{Err}_{\mathrm{IV}}$, 40). In (IV, 14.4.3), on line 18 of p. 212, replace "locally of finite type" by "locally of finite presentation."

($\mathrm{Err}_{\mathrm{IV}}$, 41). In (IV, 14.4.4), on the fifteenth and sixteenth lines from the bottom of p. 212, replace "locally of finite type" by "locally of finite presentation." On the tenth and eleventh lines from the bottom, replace "equidimensional" by "locally equidimensional."

($\mathrm{Err}_{\mathrm{IV}}$, 42). In (IV, 14.4.8), after line 1 of p. 214, add:

> If, in addition, the set of irreducible components of Y is locally finite, these two conditions are equivalent to the following ones:

($\mathrm{Err}_{\mathrm{IV}}$, 43). In (IV, 14.4.10), on line 15 of p. 215, replace "arbitrary" by "whose set of irreducible components is locally finite." On line 16 of p. 215, replace "of finite type" by "of finite presentation."

($\mathrm{Err}_{\mathrm{IV}}$, 44). In (IV, 15.5.6.1), on the fifteenth line from the bottom of p. 234, after the first sentence, add:

> It is enough to consider the case in which every irreducible closed subset of X has a unique generic point. Otherwise, consider the space $Y = X/R$, the quotient of X by the equivalence relation
>
> $$\overline{\{x\}} = \overline{\{x'\}}.$$
>
> It follows immediately from this definition that every open subset of X is saturated for R; hence the topology of X is the inverse image of that of Y under the canonical map $\pi : X \to X/R$. A dense open subset of X is of the form $\pi^{-1}(U)$, where U is dense and open in Y; consequently a nowhere dense closed subset Z of X is of the form $\pi^{-1}(S)$, where $S = \pi(Z)$ is nowhere dense and closed in Y. Thus $X \setminus Z$ is connected if and only if $Y \setminus S$ is connected.
>
> Similarly, $\pi(T_x)$ is the set of generalizations of $\pi(x)$, and $T_x \setminus \{x\}$ is connected if and only if $\pi(T_x) \setminus \{\pi(x)\}$ is connected. Finally, by construction, Y is a locally Noetherian Kolmogorov space, so every irreducible closed subset of Y has a unique generic point. Proving the lemma for X is therefore equivalent to proving it for Y, and we may suppose that every irreducible closed subset of X has a unique generic point.

($\mathrm{Err}_{\mathrm{IV}}$, 45). In (IV, 15.6.1), on the eighteenth line from the bottom of p. 236, delete "locally Noetherian" (this hypothesis is not used in the proof).

($\mathrm{Err}_{\mathrm{IV}}$, 46). In (IV, 15.6.3), on line 11 of p. 237, replace "suppose that for every $y \in Y$" by "suppose that f is locally of finite presentation and that for every $y \in Y$." Replace the proof (lines 14–24 of p. 237) by the following:

> By (1.10.3), it is enough to prove that for every $x \in X_y^0$ and every generalization y' of y, there exists a generalization x' of x such that $f(x') = y'$. By (15.6.2), it is enough to take for x' the generic point of $X_{y'}^0$.

($\mathrm{Err}_{\mathrm{IV}}$, 47). In (IV, 15.6.7), on line 2 of p. 241, after "Noetherian," add "or if condition β) of (15.6.6, (iii)) is satisfied." After line 9 of p. 241, add:

> When condition β) of (15.6.6, (iii)) is satisfied, the implication a)$\Rightarrow$b') follows from (15.6.6, (ii)), and the implication b')$\Rightarrow$a) follows from (15.6.6, (iii)).

($\mathrm{Err}_{\mathrm{IV}}$, 48). In (IV, 20.1.5), line 13 of p. 228; (IV, 20.1.6), line 17 of p. 228; and (IV, 20.1.7), the thirteenth line from the bottom of p. 228, delete "strictly."

($\mathrm{Err}_{\mathrm{IV}}$, 49). In (IV, 20.2.13), after line 6 of p. 236, add:

> In particular, if X is reduced and has only finitely many irreducible components, the notion of torsion-free $\mathscr{O}_X$-Module defined in (IV, 20.1.5) coincides with that of (I, 7.4.1).

($\mathrm{Err}_{\mathrm{IV}}$, 50). In (IV, 20.2.14), on lines 18 and 25 of p. 236, delete "strictly."

($\mathrm{Err}_{\mathrm{IV}}$, 51). In (IV, 20.6.2), on the thirteenth line from the bottom of p. 252, delete "strictly."

($\mathrm{Err}_{\mathrm{IV}}$, 52). In (IV, 20.6.4), on lines 23 and 22 of p. 253, delete "strictly."

($\mathrm{Err}_{\mathrm{IV}}$, 53). In (IV, 21.4), after line 23 of p. 267, add:

The following proposition and its corollaries were pointed out to us by M. Raynaud.

Proposition (21.4.9). — *Let X and Y be two locally Noetherian preschemes, and let $f : X \to Y$ be a faithfully flat morphism. Suppose in addition either that f is quasi-compact or that f is open. A divisor D' on X is of the form $f^*(D)$, with D a divisor on Y, if and only if the following condition holds: for every $y \in Y$ such that* $\mathrm{prof}(\mathscr{O}_{Y,y}) \leqslant 1$, *put*

$$Y_1 = \mathrm{Spec}(\mathscr{O}_{Y,y}), \qquad X_1 = X \times_Y Y_1, \qquad f_1 = f_{(Y_1)} : X_1 \longrightarrow Y_1;$$

then the inverse image D'_1 of D' on X_1 is of the form $f_1^*(D_1)$ for some $D_1 \in \mathrm{Div}(Y_1)$.

Proof. The necessity of the condition is immediate from the transitivity of inverse images of divisors, all the morphisms in question being flat (21.4.4). For every open subset U of Y, write f_U for the restriction $f^{-1}(U) \to U$ of f. By (21.4.8), the map

$$D_U \longmapsto f_U^*(D_U)$$

from $\mathrm{Div}(U)$ to $\mathrm{Div}(f^{-1}(U))$ is injective. Thus, if U_1, U_2 are two open subsets of Y and

$$D'|f^{-1}(U_1) = f_{U_1}^*(D_1), \qquad D'|f^{-1}(U_2) = f_{U_2}^*(D_2),$$

then necessarily $D_1|U_1 \cap U_2 = D_2|U_1 \cap U_2$. It follows at once that there is a largest open subset $U \subset Y$ for which

$$D'|f^{-1}(U) = f_U^*(D_U) \quad \text{with} \quad D_U \in \mathrm{Div}(U).$$

We must show that $U = Y$. Suppose otherwise, and let y be a maximal point of $Y \setminus U$. We shall find an open neighbourhood V of y in Y such that $D'|f^{-1}(V) = f_V^*(D_V)$ for some $D_V \in \mathrm{Div}(V)$, contradicting the maximality of U.

I) The case where f is quasi-compact. Using the quasi-compactness of f and (20.3.8), by the method of (8.1.2, a), we are reduced to proving—in the notation of the statement, but without initially imposing a bound on $\mathrm{prof}(\mathscr{O}_{Y,y})$—that D'_1 is of the form $f_1^*(D_1)$. If $\mathrm{prof}(\mathscr{O}_{Y,y}) \leqslant 1$, this is precisely the hypothesis. Suppose therefore that $\mathrm{prof}(\mathscr{O}_{Y,y}) \geqslant 2$. Since y is a maximal point of $Y \setminus U$, the open subset $W = Y_1 \cap U$ of Y_1 is $Y_1 \setminus \{y\}$. We may therefore restrict to the case $Y = Y_1$ and $U = Y \setminus \{y\}$.

By hypothesis there is a divisor $D_U \in \mathrm{Div}(U)$ such that $f_U^*(D_U) = D'|f^{-1}(U)$. The divisor D_U is the equivalence class of a pair $(\mathscr{L}_U, s)$, where $\mathscr{L}_U$ is an invertible $\mathscr{O}_U$-Module and s a regular meromorphic section of $\mathscr{L}_U$ over U (21.2.11). Hence, by (21.4.1), $D'|f^{-1}(U)$ is the equivalence class of $(\mathscr{L}'_U, s')$, where

$$\mathscr{L}'_U = f_U^*(\mathscr{L}_U), \qquad s' = s \circ f_U.$$

Let $i : U \to Y$ and $j : f^{-1}(U) \to X$ be the canonical injections, and put

$$\mathscr{L} = i_*(\mathscr{L}_U), \qquad \mathscr{L}' = j_*(\mathscr{L}'_U).$$

Since f is flat, $f^*(\mathscr{L})$ is canonically isomorphic to $\mathscr{L}'$ (2.3.1). Moreover, flatness and the inequality $\operatorname{prof}(\mathscr{O}_{Y,y}) \geqslant 2$ imply $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 2$ for every $x \in f^{-1}(y)$ (6.3.1).

The divisor D' is the equivalence class of a pair $(\mathscr{L}'', s'')$, where $\mathscr{L}''$ is an invertible $\mathscr{O}_X$-Module and s'' a regular meromorphic section of $\mathscr{L}''$ over X. Consequently $\mathscr{L}'_U$ is isomorphic to $\mathscr{L}''|f^{-1}(U)$, and the preceding observations together with (5.9.8) and (5.10.5) show that $\mathscr{L}'$ is an invertible $\mathscr{O}_X$-Module isomorphic to $\mathscr{L}''$. Since f is faithfully flat and quasi-compact, it follows that $\mathscr{L}$ is an invertible $\mathscr{O}_Y$-Module (2.5.2), and hence isomorphic to $\mathscr{O}_Y$, because Y is a local scheme.

Furthermore, since $\operatorname{prof}(\mathscr{O}_{Y,y}) \geqslant 1$, the open subset $U = Y \setminus \{y\}$ is schematically dense in Y (20.2.13, (iv)). The regular meromorphic section s of $\mathscr{L}_U$ over U therefore extends to a regular meromorphic section s_0 of $\mathscr{L}$ over Y (20.2.11). The equivalence class of $(\mathscr{L}, s_0)$ is thus a divisor D on Y. Plainly $f^*(D)$ and D' induce the same divisor on $f^{-1}(U) = X \setminus f^{-1}(y)$; and since $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 2$ for every $x \in f^{-1}(y)$, (21.1.8) gives $D' = f^*(D)$. This proves case I.

II) The case where f is open. The problem is to find an open neighbourhood V of y in Y such that $D'|f^{-1}(V) = f_V^*(D_V)$ for some $D_V \in \operatorname{Div}(V)$, so we may first suppose Y affine. Because f is open, the images $f(W)$, with W running through the affine open subsets of X, form an open covering of Y. Since Y is quasi-compact, there is a quasi-compact open subset $T \subset X$ such that $f(T) = Y$, and $f|T$ is quasi-compact (I, 1.1.1). A fortiori, for every quasi-compact open $T_\alpha \supset T$ in X, one has $f(T_\alpha) = Y$ and $f|T_\alpha$ is quasi-compact.

Applying case I to $f|T_\alpha$, we obtain a divisor D_α on Y such that

$$(f|T_\alpha)^*(D_\alpha) = D'|T_\alpha.$$

By transitivity of inverse images of divisors under flat morphisms (21.4.4), one also has $(f|T)^*(D_\alpha) = D'|T$. Hence for any α, β,

$$(f|T)^*(D_\alpha) = (f|T)^*(D_\beta),$$

and (21.4.8) gives $D_\alpha = D_\beta$. If D denotes their common value, then $f^*(D)|T_\alpha = D'|T_\alpha$ for every α, and therefore $f^*(D) = D'$. □

Corollary (21.4.10). — *Assume the general hypotheses of (21.4.9), and suppose in addition that Y is normal and that, for every $y \in Y$ with $\dim(\mathscr{O}_{Y,y}) = 1$, the fibre $X_y = f^{-1}(y)$ has no embedded associated prime cycle. A divisor D' on X is of the form $f^*(D)$, with $D \in \operatorname{Div}(Y)$, if and only if the following two conditions hold:*

1° *for every maximal point y of Y, the inverse image of D' in $\operatorname{Div}(X_y)$ is zero;*

2° *for every $y \in Y$ such that $\dim(\mathscr{O}_{Y,y}) = 1$, there is an integer n_y such that, for every maximal point x of X_y, the inverse image of D' in $\operatorname{Div}(\operatorname{Spec}(\mathscr{O}_{X,x}))$ equals $\operatorname{div}(t_y^{n_y})$, where t_y is the image under $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ of a uniformizer of the discrete valuation ring $\mathscr{O}_{Y,y}$.*

Proof. Apply the criterion (21.4.9). At a maximal point y of Y, $\operatorname{Div}(Y_1) = 0$ because $\mathscr{O}_{Y,y}$ is Artinian; condition 1° is therefore equivalent to the criterion of (21.4.9) at such a point. It remains to show that conditions 1° and 2° likewise imply the criterion at a point $y \in Y$ with $\operatorname{prof}(\mathscr{O}_{Y,y}) = 1$. Since Y is normal, these are precisely the points for which $\dim(\mathscr{O}_{Y,y}) = 1$, equivalently those for which $\mathscr{O}_{Y,y}$ is a discrete valuation ring (5.8.6).

We may thus restrict to $Y = \operatorname{Spec}(\mathscr{O}_{Y,y})$. If t is a uniformizer of $\mathscr{O}_{Y,y}$, the map $n \mapsto \operatorname{div}(t^n)$ is an isomorphism from $\mathbb{Z}$ onto $\operatorname{Div}(Y)$. The condition of (21.4.9) says that D' is of the form $\operatorname{div}(t^n \circ f)$ for some $n \in \mathbb{Z}$, and plainly implies condition 2°. Conversely, suppose 2° holds, as does 1°, and write $n = n_y$. It remains to prove $D' = \operatorname{div}(t^n \circ f)$. If η is the generic point of Y, the germs of these two divisors at every point of X_η are zero by condition 1°. Their germs at every maximal point of X_y are equal by condition 2°. At a nonmaximal point $x \in X_y$, one has $\operatorname{prof}(\mathscr{O}_{X_y,x}) \geqslant 1$, because X_y

has no embedded associated prime cycle, and hence $\operatorname{prof}(\mathscr{O}_{X,x}) \geqslant 2$ by (6.3.1). The desired equality follows from (21.1.8). □

Corollary (21.4.11). — *Assume the general hypotheses of (21.4.9), and suppose in addition that Y is normal and that, for every $y \in Y$ with $\dim(\mathscr{O}_{Y,y}) = 1$, the fibre X_y is integral. A divisor D' is of the form $f^*(D)$, with $D \in \operatorname{Div}(Y)$, if and only if, for every maximal point y of Y, the inverse image of D' in $\operatorname{Div}(X_y)$ is zero.*

Proof. It is enough to check that condition 2° of (21.4.10) is then automatic. If $\dim(\mathscr{O}_{Y,y}) = 1$ and X_y is integral, then at the unique maximal point x of X_y one has $\dim(\mathscr{O}_{X,x}) = 1$ by (6.1.3). Moreover, $\mathscr{O}_{Y,y}$ is regular and $\mathscr{O}_{X_y,x}$ is a field, so $\mathscr{O}_{X,x}$ is regular (6.5.1), hence a discrete valuation ring. If t is a uniformizer of $\mathscr{O}_{Y,y}$, its image t_y in $\mathscr{O}_{X,x}$ is a uniformizer of $\mathscr{O}_{X,x}$ (0, 17.3.3). The inverse image of D' in $\operatorname{Div}(\operatorname{Spec}(\mathscr{O}_{X,x}))$ is therefore of the form $\operatorname{div}(t_y^n)$, which is exactly condition 2° of (21.4.10), since X_y has only one maximal point. □

Remark (21.4.12). — If in (21.4.11) one assumes only that the X_y are reduced, one must add to the condition in the statement that the inverse images of D' in each group $\operatorname{Div}(\operatorname{Spec}(\mathscr{O}_{X,x}))$, with x running over the maximal points of X_y, are all of the form $\operatorname{div}((t'_x)^n)$, where t'_x is a uniformizer of $\mathscr{O}_{X,x}$ and the integer n is the same for all these points.

Corollary (21.4.13). — *Assume the hypotheses of (21.4.11). For every invertible $\mathscr{O}_X$-Module $\mathscr{L}$ such that, for every maximal point η of Y, the inverse image*

$$\mathscr{L}_\eta = \mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_\eta}$$

is isomorphic to $\mathscr{O}_{X_\eta}$, there exist an invertible $\mathscr{O}_Y$-Module $\mathscr{M}$ and an isomorphism $f^*(\mathscr{M}) \simeq \mathscr{L}$.

Proof. It is enough to find a regular meromorphic section s of $\mathscr{L}$ over X such that the divisor D' corresponding to $(\mathscr{L}, s)$ has $D'_x = 0$ at every point x of a fibre X_η over a maximal point η of Y. Applying (21.4.11) to D' then gives the result by (21.4.2).

The points of $\operatorname{Ass}(\mathscr{O}_X)$ lie over the maximal points of Y (3.3.1). Since Y is reduced and locally Noetherian, there is a dense open subset V of Y which is the union of a disjoint family of affine open subsets, each containing exactly one maximal point of Y; and $f^{-1}(V)$ is schematically dense in X (5.10.2). We may therefore restrict to the case in which Y is affine and integral.

For the unique maximal point η of Y, the fibre X_η is then integral, so $\operatorname{Ass}(\mathscr{O}_X)$ consists of a single point z, the generic point of X. Let s_η be the section of $\mathscr{L}_\eta$ over X_η which corresponds, under an isomorphism, to the unit section of $\mathscr{O}_{X_\eta}$. Let U be an affine open neighbourhood in X of a point $x \in X_\eta$, and let s be a section of $\mathscr{L}$ over U with the same germ at x as s_η. Since U is schematically dense in X (20.2.13, (iv)), s identifies with a uniquely determined regular meromorphic section of $\mathscr{L}$ over X. It is clear from (20.2.11) and (20.3.5) that this section induces on X_η a regular meromorphic section of $\mathscr{L}_\eta$ equal to s_η, and hence has the required property. □